Graphon Mean Field Games

A Physicist's Path from the Mean-Field Limit to Network Equilibria

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Preface

This book is an attempt to carry a reader trained in theoretical physics — and in particular one comfortable with the language of statistical mechanics, stochastic processes, and field theory — across the bridge into the modern mathematics of *mean field games* and their network-structured generalization, *graphon mean field games*. The destination is a body of theory that sits at the confluence of stochastic analysis, partial differential equations, optimal control, game theory, and the theory of graph limits. The premise is that a physicist already possesses many of the load-bearing intuitions — the passage to a thermodynamic limit, the notion of a self-consistent field, conservation laws and continuity equations, and the effect of disorder and broken homogeneity — and that the most efficient route into the subject is to name those intuitions explicitly and then build the rigorous mathematics on top of them.

A word on how that premise is used. Physical analogies appear throughout, but they are kept deliberately light and are always optional: a physical picture motivates a definition or makes a theorem memorable, but it never carries mathematical weight. If every *Physics note* in the book were deleted, the mathematics would stand unchanged. The standard of rigor is what we call "physicist's rigor, honestly kept": every object is defined precisely and every theorem is stated with its full hypotheses, but proofs are given at the level of a complete and honest sketch when a full argument would consume more space than it illuminates, always with an explicit pointer to where the missing details live. We never write "it can be shown" and leave it there.

The book is long by design and contains no exercises. In place of exercises, two running examples — a linear–quadratic–Gaussian mean field game, and its graphon generalization — are carried from the foundational chapters all the way to the numerical and learning-based frontiers, acquiring one new layer of structure in each relevant chapter. Watching a single, fully explicit problem pass through every stage of the theory is, we believe, worth more than any number of detached drills.

The roadmap is organized in five parts. Part I assembles exactly the measure theory, functional analysis, stochastic calculus, and optimal control that the rest of the book consumes — no more, no less. Part II develops the classical theory of mean field games for homogeneous populations: the coupled forward–backward system, its probabilistic counterpart, existence and uniqueness, the master equation, and the convergence of large finite games to the mean field limit. Part III is a self-contained introduction to dense graph limits and graphons. Part IV is the synthesis: mean field games played over a graphon, where each agent's coupling to the population is mediated by a kernel $W(x, y)$ indexed by a continuum of types. Part V turns to active research — sparse network limits, numerical methods, and reinforcement-learning approaches to computing equilibria.

A note on conventions. The single most important page in this book for day-to-day reading is Appendix A, which fixes notation and resolves every clash between physical and mathematical convention used in the text. The reader is encouraged to skim it once before Part II and to return to it whenever a symbol surprises them.

Part I

Mathematical Foundations

Chapter 1

Measure, Integration, and the Passage to a Continuum

This is the first chapter of the book and it imports nothing: it builds, from the ground up, the language in which every later statement is made precise. The reader who already manipulates N-body sums, Dirac combs, occupation densities, and the "thermodynamic limit" $N \to \infty$ has, in fact, already been doing measure theory – informally. Nothing in this chapter asks the reader to take up a foreign subject; it asks only that a handful of habits already in daily use – summing point masses, smearing a Dirac comb into a profile, letting a particle count run to infinity – be given names sharp enough to be trusted under a limit. Our task here is to install the formal apparatus so that the limits the physicist takes by reflex become *theorems* with hypotheses one can check, rather than gestures one hopes are legal. The single object that organizes the whole chapter is the *empirical measure*

$$\mu^N \; = \; \frac{1}{N}\sum_{i=1}^{N} \delta_{X_i},$$

the normalized occupation profile of a finite population of agents located at $X_1, \ldots, X_N$. The central result, proved in section 1.7, is that when the X_i are independent draws from a common law m, the random measures μ^N converge – almost surely, in a precise sense we will define – to the deterministic measure m. This is the cleanest rigorous incarnation of the statement "fluctuations are suppressed in the large-population limit," and it is the prototype of every passage-to-the-continuum argument in mean field game theory.

We develop exactly the toolbox that the rest of the book consumes and no more, and the order of development is dictated by that single destination: each section exists because the previous one left something we still need. First, the limiting profile m and the empirical measure μ^N need a *home* – a family of subsets we are entitled to assign a size to, and a rule that does the assigning – so we build σ-algebras and measures (section 1.2), motivated in section 1.1 by the failures that make them unavoidable. Then we need an *integral* compatible with limits, one that survives the monotone and concentrating limits on which the elementary integral founders, so we construct the Lebesgue integral (section 1.3) and check that $\int f \; \mathrm{d}\mu^N = \frac{1}{N}\sum_i f(X_i)$ is the literal bridge between the discrete average and the continuum one. Then we need *permits* to move a limit through that integral, so we prove the three convergence theorems – monotone, Fatou, dominated (section 1.4) – which are exactly the licenses that every "take the limit inside the average" step silently invokes. Then, because expectations factor over independent sources and the graphon chapters integrate a kernel $W(x, y)$ against two population variables at once, we need a *product theory*, supplied by the Fubini–Tonelli theorem (section 1.5). Then we need *completeness* of the spaces of integrable functions, both as a structural backbone and for the spectral chapter to come, so we build the L^p spaces and prove they are Banach (section 1.6). Only with every part in hand do we assemble the empirical-measure theorem in the closing worked computation. Each section is thus the next thing one is forced to want, and we flag at every step where the tool will be reused later.

The functional-analytic life of these spaces – duality, weak topologies, the spectral theorem – is the subject of chapter 2; the full theory of weak convergence of probability measures, with the Portmanteau theorem and Prokhorov's compactness criterion, is developed in chapter 3; the Wasserstein geometry in which the mean field actually lives is chapter 4. We use only as much of those subjects here as the empirical-measure statement strictly requires. The lone result of pure measure theory we accept without proof is the Carathéodory extension theorem (named where it is used, in theorem 1.11); every other borrowed fact is flagged on the spot.

Notation watch

Throughout this chapter the symbol x denotes a generic point of the *state space* $\mathbb{R}^d$ – the position, velocity, wealth, or opinion of a single agent – and the integration variable ranges over $\mathbb{R}^d$. This is *not* yet the graphon *type* variable $x \in [0, 1]$, an exchangeable label that carries no metric meaning of its own: it is first put to work in chapter 2 as the domain of the graphon integral operator (the Hilbert–Schmidt example there) and is formally introduced as a graph limit in chapter 15. The two never appear together in this chapter. When, in section 1.5, we foreshadow the graphon double integral, we will write the type variables explicitly and point to where they are defined.

1.1 Why a physicist needs measure theory

The preamble made a promise: the limits a physicist takes by reflex – replacing a normalized sum of point masses by a smooth profile, replacing a population average by an integral – are to become theorems with hypotheses one can check. The purpose of this section is to show that the promise has teeth. Taken literally, the reflex poses three questions, and the elementary calculus the reader already owns cannot answer any of them; worse, the obstruction is not at the margin but at the precise joint where analysis meets limits. By the end of the section the need for new machinery should be felt as a concrete pressure rather than accepted on authority, and the shape of that machinery – a structured family of "sizeable" sets and a notion of size stable under countable limits – should already be visible in outline. That is exactly the object the next section builds.

Begin where the physicist begins: with an average. Given N agents at positions $X_1, \ldots, X_N \in \mathbb{R}^d$ – positions, velocities, wealths, opinions – the population average of an observable f is the finite sum $\frac{1}{N}\sum_{i=1}^{N} f(X_i)$. One would like to say that as the population grows this settles down to an integral $\int f(x)\, m(\mathrm{d}x)$ against some limiting "density profile" m, and – the genuinely powerful claim – that the entire population is thereafter summarized by the single object m rather than by the ever-lengthening list of positions. This is the move that makes a continuum theory possible at all: it trades N coordinates for one profile. But to make it precise we must answer three questions, and each is, in a sharp sense, unanswerable inside elementary calculus.

1. What *is* the object m? The population may concentrate on a lower-dimensional set, develop atoms, or spread to infinity, and in none of these cases is m a Riemann-integrable density function.

2. In what *sense* does $\frac{1}{N}\sum_i \delta_{X_i} \to m$? The left-hand side is a sum of infinitely sharp spikes and the right-hand side may be perfectly smooth, so the convergence cannot be the pointwise convergence of densities.

3. When may we *exchange* the limit $N \to \infty$ with the integral $\int f(\cdots)$? Every thermodynamic-limit argument performs this interchange, almost always silently, and – as we will see – it is sometimes simply false.

Consider the first question concretely, because its difficulty is easy to underestimate. Suppose the agents' positions, as N grows, pile up onto a constraint surface – particles pinned to a manifold, a population of opinions collapsing to a single consensus value, a wealth distribution in which a finite fraction of agents holds exactly zero. The limiting profile then assigns *positive* mass to a set that has *zero* length, area, or volume: a single point, a curve, a lower-dimensional surface. No function $\rho(x)$ with $m(\mathrm{d}x) = \rho(x)\,\mathrm{d}x$ can describe this, for the integral of any honest function over a set of zero volume is zero. A condensate – a macroscopic fraction of the population sitting at one location – is an *atom*, and atoms are invisible to densities. Even in the benign case where m does turn out to have a smooth density, m is not handed to us *as* a density: it is handed to us as the rule "how much population sits in the region A," an assignment of a number to each set A. The right primitive notion is therefore not a function on points but an assignment of mass to sets – a *measure* – and only sometimes is this assignment representable by a density.

The second question is the mirror image of the first on the side of the approximants. The empirical measure $\frac{1}{N}\sum_i \delta_{X_i}$ places mass $1/N$ at each of N points and nothing in between; it is as far from a smooth density as an object can be, concentrated entirely on a finite set of zero total volume. There is no sense in which the "heights" of these spikes converge to the values of a limiting density, because the spikes have no heights – they are point masses, not tall thin bumps. Whatever convergence we mean cannot live at the level of values at points. The only thing that *does* behave well is what we get by testing the spikes against a smooth observable: the number $\frac{1}{N}\sum_i f(X_i)$ is a perfectly ordinary average, and it is this number, for each fixed reasonable f, that we can hope to control. Convergence of the population, then, must mean convergence of these test averages – $\int f\,\mathrm{d}\mu^N \to \int f\,\mathrm{d}m$ for a rich enough class of f – which is precisely the notion of *weak convergence* we develop in chapter 3. The heuristic below records the shift in viewpoint this forces, and the third question exposes why even the test averages must be handled with care.

Example 1.1 (Riemann integration is not closed under monotone limits)**.** Enumerate the rationals in $[0,1]$ as $\mathbb{Q} \cap [0,1] = \{q_1, q_2, \dots\}$ and set $f_n = \mathbf{1}_{\{q_1,\dots,q_n\}}$, the indicator of the first n of them. Each

f_n vanishes except at the finitely many points $q_1, \ldots, q_n$, so on any partition of $[0,1]$ its lower Darboux sum is 0, while its upper Darboux sum can be made smaller than any $\varepsilon > 0$ by covering those n points with cells of total length $< \varepsilon$; hence f_n is Riemann integrable with $\int_0^1 f_n = 0$. The sequence increases pointwise, $f_n \uparrow f$, to the indicator $f = \mathbf{1}_{\mathbb{Q}\cap[0,1]}$ of all the rationals. But f is *not* Riemann integrable, and the reason is worth seeing in full. Fix any partition $0 = x_0 < x_1 < \cdots < x_m = 1$. On each cell $[x_{k-1}, x_k]$ both the rationals and the irrationals are dense, so f takes both the value 1 (at some rational) and the value 0 (at some irrational) inside the cell; thus $\inf_{[x_{k-1},x_k]} f = 0$ and $\sup_{[x_{k-1},x_k]} f = 1$. The lower Darboux sum is therefore $\sum_k 0 \cdot (x_k - x_{k-1}) = 0$ and the upper Darboux sum is $\sum_k 1 \cdot (x_k - x_{k-1}) = 1$, *for every partition.* The lower integral (sup over partitions of the lower sums) is 0 and the upper integral (inf of the upper sums) is 1; they disagree, so $\int_0^1 f$ does not exist.

The example does more than exhibit a monotone limit of integrable functions that fails to be integrable; it shows that the desired interchange statement cannot even be *written down* in Riemann's theory. The left-hand side $\lim_n \int_0^1 f_n = \lim_n 0 = 0$ is a perfectly good number. The right-hand side $\int_0^1 (\lim_n f_n) = \int_0^1 f$ is a symbol that *denotes nothing*, because the function it would integrate is not Riemann integrable. The proposed identity $\lim_n \int f_n = \int \lim_n f_n$ is thus not false – it is ill-formed, an equation one side of which is undefined. A theory in which the natural question cannot be posed is the wrong theory for taking limits inside integrals, which is the only thing we will ever do.

The Lebesgue theory, built in sections 1.2 to 1.3, repairs exactly this. There the rationals are a *null set* – a countable set, coverable by intervals of arbitrarily small total length – the function $\mathbf{1}_{\mathbb{Q}\cap[0,1]}$ is measurable and integrable with integral 0, and the identity $\lim_n \int f_n = 0 = \int \lim_n f_n$ now has both sides defined and equal, as a special case of the monotone convergence theorem (theorem 1.20). The repair is not cosmetic. The whole business of this book – limits of densities, weak limits of empirical measures, the exchange of $N \to \infty$ with an integral – requires an integral that is *stable under the limits we actually take*, and that stability is precisely what Riemann's construction lacks and Lebesgue's supplies.

Heuristic (non-rigorous)

The physicist's instinct "a sum of many sharp spikes becomes a smooth profile" is correct, but it is a statement about *measures*, not about *functions*. The empirical measure $\frac{1}{N}\sum_i \delta_{X_i}$ is never a function – it has no density with respect to Lebesgue measure, being a sum of point masses sitting on a set of zero volume – yet it can converge to a measure that has a perfectly smooth density. The two sides of "spikes $\to$ profile" therefore live in different worlds if read as functions, and in the *same* world, the world of measures, if read correctly: a measure is what both a sum of Diracs and a smooth density already are. This is why the right category for the passage to the continuum is the category of measures, with convergence tested by integrating against smooth observables rather than by comparing values at points. Measure theory is the price of admission to that category, and the rest of this chapter pays it.

The third question – when may we move the limit inside the integral – has its own canonical warning, and it is even sharper than the rationals example because here both sides are perfectly well-defined and simply *unequal.* Consider the sequence $f_n = n\,\mathbf{1}_{(0,1/n)}$ on $[0,1]$ with ordinary length: a rectangle of height n and width $1/n$ pressed against the origin. Its integral is the area of that rectangle,

$$\int_0^1 f_n \,\mathrm{d}x \;=\; n \cdot \tfrac{1}{n} \;=\; 1 \qquad \text{for every } n.$$

Now fix a point $x > 0$ and let n grow: as soon as $n > 1/x$ the point x lies outside the interval $(0, 1/n)$, so $f_n(x) = 0$ from then on; and $f_n(0) = 0$ for all n. Hence $f_n(x) \to 0$ for *every* $x \in [0,1]$, and the pointwise limit is the zero function, with $\int_0^1 \big(\lim_n f_n\big)\,\mathrm{d}x = 0$. The limit and the integral disagree outright:

$$\lim_n \int_0^1 f_n \,\mathrm{d}x \;=\; 1 \;\neq\; 0 \;=\; \int_0^1 \lim_n f_n \,\mathrm{d}x.$$

The unit of mass did not vanish; it *concentrated.* Where did it go? Test against any bounded continuous g:

$$\int_0^1 g\, f_n \,\mathrm{d}x \;=\; n \int_0^{1/n} g(x)\,\mathrm{d}x \;\xrightarrow[n\to\infty]{}\; g(0),$$

because the average of g over the shrinking interval $(0, 1/n)$ tends to the value $g(0)$ by continuity. But $g(0) = \int g \,\mathrm{d}\delta_0$. So in the language we are about to build, the measures $f_n\,\mathrm{d}x$ converge *weakly* to the Dirac mass δ_0: the mass survives, relocated entirely to the origin, and it is the *pointwise* limit – not the measures – that threw it away. This is the same phenomenon as the empirical-measure problem read in reverse, and it shows once more that the robust notion of convergence is the weak one, tested against observables.

This example is not merely a warning; it plants a seed we will harvest twice. First, it pinpoints the hypothesis that makes the interchange legal. The monotone, Fatou, and dominated convergence theorems of section 1.4 are the three permits for moving a limit inside an integral, and the dominated convergence theorem (theorem 1.23) demands exactly one thing the present sequence cannot supply: a single integrable function g with $|f_n| \leq g$ for all n. Here the smallest function dominating every f_n is $\sup_n f_n(x)$, which behaves like $1/x$ near the origin and is *not* integrable on $[0,1]$; the dominating-function hypothesis fails, and that failure is the precise mechanism by which mass escapes the pointwise limit. The hypothesis of theorem 1.23 is therefore not bureaucratic fine print – it is the wall that holds the mass in. Second, the very phenomenon "mass leaks out of every bounded picture – to a point here, to infinity in other examples" is, at the level of populations, controlled by the condition called *tightness*, developed in chapter 3; tightness is what guarantees that the limit of a sequence of probability measures is still a probability measure and has not silently shed mass. Recognizing when the interchange is legal, and recognizing the exact way it fails when it does, is thus not pedantry: it is the difference between a correct continuum limit and a lost order parameter.

We have now turned the physicist's reflex into three precise demands and shown that elementary calculus meets none of them. Each failure points to the same missing structure. To say what m is, to test convergence against observables, and to keep mass from leaking through a limit, we need two objects that Riemann's world does not provide. We need, first, a structured family of subsets of the state space that we are entitled to assign a size to – and the family must be closed under the countable unions and intersections that limits perform, since every object above (the rationals, the increasing f_n, the concentrating bumps) was produced by a countable limiting operation. That family is the *σ-algebra.* We need, second, a notion of size defined on that family – a *measure* – that is stable under exactly those countable limits, so that continuity-of-measure can replace the broken continuity of the Riemann integral. With those two objects in hand we will at last be able to say what the limiting profile m and the empirical measure μ^N *are*, as honest mathematical objects sharing a common home. Constructing them, and the maps between such homes, is the work of the next section.

1.2 σ-algebras, measures, and measurable functions

The previous section closed with a precise shopping list rather than a vague unease. To turn the physicist's reflex into theorems we need two things Riemann's world withholds: a structured family of subsets of the state space that we are entitled to assign a size to, *closed under the countable unions and intersections that limits perform*, and a notion of size on that family that is *stable under exactly those countable limits*. This section delivers both, and with them the first concrete payoff – an honest answer to the first of the three questions of section 1.1, "what *is* the object m." The answer will be: m is a *measure*, an assignment of mass to sets, living on a *measurable space*; the empirical measure μ^N is another such assignment on the same space; and the law of a single random agent is manufactured from a third by a structure-respecting map. Building those three notions – a family of sets, a size on them, and the maps between such homes – is the whole of the section, and the order in which we build them is forced by how they depend on one another.

The arc runs through five movements, each the minimal next thing the previous one requires. We begin with *measurable spaces* (definition 1.2 below): one cannot speak of the size of a set before declaring which sets are eligible to have a size, so the very first object is the family $\mathcal{F}$ of admissible sets, and its defining axiom – closure under countable operations – is the direct answer to the limit-incompatibility diagnosed in section 1.1. Onto that arena we then place a *measure* (definition 1.5): the size itself, a single rule $\mu\colon \mathcal{F} \to [0, \infty]$ whose one nontrivial axiom, countable additivity, immediately yields the continuity-under-limits properties that the broken Riemann integral could not supply. Because we will need to *construct* specific measures by prescribing their values on a small, convenient class of sets and then trusting that the prescription determines them everywhere, we pause for the uniqueness machinery of *generating classes* – π-systems and Dynkin's lemma (theorem 1.9) – the combinatorial engine that propagates agreement from a generating class to the whole σ-algebra, and which we reuse three times in this chapter. Armed with it we construct the one canonical concrete measure every density is implicitly measured against, *Lebesgue measure* (theorem 1.11), the rigorous form of length, area, and volume – and the background whose sets of zero volume are exactly the atoms that defeated the naïve density picture. Finally we turn to the *maps*: *measurable functions* (definition 1.12), the morphisms that respect the structure, which are both the observables we will integrate and the device – via the *pushforward* (definition 1.14) – that produces m as the law of a single agent and μ^N as a sum of pushed-forward point masses; and we prove the decisive stability fact that a pointwise limit of measurable functions is again measurable, without which none of the limiting objects built later in the book would be legal to integrate.

The ordering is not a matter of taste. A measure is a function defined *on* a σ-algebra, so the σ-algebra must exist first; a measurable function is defined by *pulling back* measurable sets of its target, so again the σ-algebra must come first; and integrating such a function against such a measure – the construction of the next section – presupposes both. Measurable spaces precede measures precede measurable functions because each later notion is literally built out of the earlier ones. We start, then, with the family of sizeable sets.

1.2.1 Measurable spaces

The previous section ended with a shopping list whose first item was a *home* for the limiting profile m and the empirical measure μ^N: a structured family of subsets of the state space that we are entitled to assign a size to. We build that family now, before any notion of size is placed on it, because one cannot speak of the size of a set before declaring which sets are eligible to have a size. The single design constraint is the one section 1.1 forced on us. Every object that broke the elementary theory – the rationals $\mathbb{Q} \cap [0,1]$, the increasing indicators $f_n \uparrow \mathbf{1}_{\mathbb{Q}}$, the concentrating bumps $n\mathbf{1}_{(0,1/n)}$ – was manufactured by a *countable limiting operation*, and the family of sizeable sets must be stable under exactly those operations or it will not survive the limits we intend to take. The family is the *σ-algebra*, and its axioms are nothing but the closure properties this stability demands, neither more nor less.

Before writing the axioms down, it is worth asking what a limiting operation actually does to sets, because the axioms are reverse-engineered from the answer. Three moves recur. We pass to a *complement*: “the agents *not* in region A” is as legitimate a region as A itself, and the size we will eventually assign should satisfy $\mu(A^c) = \mu(\Omega) - \mu(A)$, which is meaningless unless A^c is also in the family. We form a *countable union*: an increasing sequence of regions $A_1 \subseteq A_2 \subseteq \cdots$ swept out by a limit – think of the first n rationals as $n \to \infty$, or the sublevel sets that a convergence argument unions together – must have its limit region $\bigcup_n A_n$ in the family, or the whole point of building a limit-stable theory is lost. And, dually, *countable intersection*, the shrinking counterpart. These three are precisely the operations the convergence theorems of section 1.4 will perform on the sublevel sets $\{f_n > a\}$ of a sequence of functions; a family closed under them is exactly a family on which those theorems can run. This motivates the definition.

Definition 1.2 (σ-algebra)**.** Let Ω be a set. A collection $\mathcal{F} \subseteq 2^\Omega$ is a *σ-algebra* on Ω if (i) $\Omega \in \mathcal{F}$; (ii) $A \in \mathcal{F} \implies \Omega \setminus A \in \mathcal{F}$ (closure under complements); and (iii) $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_n A_n \in \mathcal{F}$ (closure under countable unions). The pair $(\Omega, \mathcal{F})$ is a *measurable space* and the members of $\mathcal{F}$ are the *measurable sets*.

Three axioms, and each earns its place. Axiom (i) merely fixes the total space as sizeable – the population as a whole is a region. Axiom (ii), closure under complements, is what makes “everywhere except A” admissible whenever A is, and – combined with (iii) – it silently delivers closure under countable *intersections*: by De Morgan, $\bigcap_n A_n = \left(\bigcup_n A_n^c\right)^c$, a complement of a countable union of complements, so the shrinking limits are covered without a separate axiom. (Taking $A = \Omega$ in (ii) also gives $\emptyset = \Omega^c \in \mathcal{F}$, so the empty region is sizeable, as it must be with size zero.) Axiom (iii), closure under *countable* unions, is the crux, and the adjective is doing all the work. It is strong enough to absorb a limit – a limit is assembled from a *sequence*, an indexed-by-$\mathbb{N}$, hence countable, collection of approximants – which is exactly the operation Riemann’s theory had no stable counterpart for, the gap diagnosed in section 1.1.

Why not demand more – closure under *arbitrary* unions, which would look cleaner? Because that demand is self-defeating: it is too strong to leave any room for a notion of size. Every single point $\{x\}$ of $\mathbb{R}^d$ will turn out to be measurable (it is closed, hence Borel below), and an *arbitrary* subset $E \subseteq \mathbb{R}^d$ is the union $E = \bigcup_{x\in E}\{x\}$ of its points. Closure under arbitrary unions would therefore force $\mathcal{F} = 2^\Omega$, the full power set – and on the full power set of $\mathbb{R}$ there is *no* translation-invariant way to assign a length to every set consistently (the Vitali construction exhibits a set whose translates partition the line into countably many congruent pieces, which no countably additive, translation-invariant length can accommodate). Arbitrary unions collapse the family to everything and leave nothing for a measure to be. Countable unions are the Goldilocks closure: wide enough to contain every set a limit can build, narrow enough that a consistent size still exists on the family. The single word "countable" is thus the operative hinge of the entire subject, the precise place where measure theory parts company with naïve set theory and gains the limits.

Generated σ-algebras. In practice one never writes down a σ-algebra by listing its members; one names a small, intuitive family of sets one insists be measurable – the open sets, say, or the boxes – and asks for the smallest σ-algebra that contains them. That such a smallest one exists is a one-line consequence of a closure fact worth isolating.

Remark 1.3 (Intersections of σ-algebras, and the generated σ-algebra)**.** *An arbitrary intersection of σ-algebras on Ω is again a σ-algebra.* Indeed, let $\{\mathcal{F}_i\}_{i\in I}$ be σ-algebras and put $\mathcal{F} = \bigcap_{i\in I}\mathcal{F}_i$. Then $\Omega \in \mathcal{F}_i$ for every i, so $\Omega \in \mathcal{F}$; if $A \in \mathcal{F}$ then $A \in \mathcal{F}_i$ for every i, whence $A^c \in \mathcal{F}_i$ for every i and so $A^c \in \mathcal{F}$; and if $A_1, A_2, \dots \in \mathcal{F}$ then for each fixed i all the A_n lie in $\mathcal{F}_i$, so $\bigcup_n A_n \in \mathcal{F}_i$, and as this holds for every i, $\bigcup_n A_n \in \mathcal{F}$. All three axioms pass through the intersection because each is a condition checked one index at a time. Consequently, for any family $\mathcal{E} \subseteq 2^\Omega$ the intersection of *all* σ-algebras containing $\mathcal{E}$ – a nonempty collection, since the power set 2^Ω is one such σ-algebra – is itself a σ-algebra, it contains $\mathcal{E}$, and it is contained in every σ-algebra that contains $\mathcal{E}$. It is therefore the *smallest* such, denoted $\sigma(\mathcal{E})$ and called the σ-algebra *generated* by $\mathcal{E}$. One should read $\sigma(\mathcal{E})$ not as an explicit list but as "everything forced to be measurable once the sets in $\mathcal{E}$ are," the closure of $\mathcal{E}$ under the three axioms.

A deliberately tiny example fixes the idea. Take $\Omega = \{1, 2, 3\}$ and let $\mathcal{E} = \{\{1\}\}$ insist only that the singleton $\{1\}$ be measurable. Closure under complements forces $\{2, 3\}$ in; the axioms put $\emptyset$ and Ω in; and one checks that $\sigma(\mathcal{E}) = \{\emptyset, \{1\}, \{2, 3\}, \Omega\}$ is already closed under all three operations, so nothing further is forced. This four-element family is strictly smaller than the eight-element power set $2^{\{1,2,3\}}$: the points 2 and 3 are never separated, because nothing we demanded distinguishes them. The generated σ-algebra is exactly the record of what our chosen "observable" sets can resolve and no finer.

The Borel σ-algebra. The state space $\mathbb{R}^d$ carries a topology, and the only sets we ever need to measure are those buildable from the open sets by the countable operations – which is to say, the σ-algebra generated by the open sets. This is the canonical choice on any space where a notion of nearness already lives.

Definition 1.4 (Borel σ-algebra)**.** On a topological space (X, τ) the *Borel σ-algebra* is $\mathcal{B}(X) := \sigma(\tau)$, the σ-algebra generated by the open sets. On $\mathbb{R}^d$ it is generated equally by the open balls, by the half-open boxes $\prod_j (a_j, b_j]$, or by the half-spaces; we write $\mathcal{B}(\mathbb{R}^d)$ throughout. Every set one can name by a countable construction from open sets – closed sets, countable intersections of open sets, and so on – is Borel.

The claim that the open balls, the half-open boxes, and the half-spaces all generate the *same* σ-algebra is not obvious and deserves the short argument that makes it routine. The principle is a single one: $\sigma(\mathcal{E}_1) = \sigma(\mathcal{E}_2)$ *as soon as each family lies inside the σ-algebra generated by the other*, because if $\mathcal{E}_1 \subseteq \sigma(\mathcal{E}_2)$ then $\sigma(\mathcal{E}_2)$ is a σ-algebra containing $\mathcal{E}_1$, hence contains the smallest such, $\sigma(\mathcal{E}_1)$, and the reverse inclusion is symmetric. So it suffices, in each case, to write the members of one family as countable set-operations on members of the other. The open balls are open, so $\sigma(\text{balls}) \subseteq \mathcal{B}(\mathbb{R}^d)$; conversely every open set $U \subseteq \mathbb{R}^d$ is the *countable* union of the open balls with rational center and rational radius that it contains (around each point of U sits such a ball lying inside U, and there are only countably many candidate balls), so $U \in \sigma(\text{balls})$ and thus $\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\text{balls})$. The half-open box $\prod_j (a_j, b_j]$ is a *countable intersection* of open boxes, $\prod_j (a_j, b_j] = \bigcap_n \prod_j (a_j, b_j + \frac{1}{n})$, hence Borel, giving $\sigma(\text{boxes}) \subseteq \mathcal{B}(\mathbb{R}^d)$; conversely every open U is a countable union of half-open boxes with rational corners, so $\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\text{boxes})$. And the half-open box is even a *finite* intersection of half-spaces and their complements, $\prod_j (a_j, b_j] = \bigcap_j (\{x_j \leq b_j\} \setminus \{x_j \leq a_j\})$, so the half-spaces generate the boxes (hence all of $\mathcal{B}(\mathbb{R}^d)$), while each half-space $\{x_j \leq c\}$ is closed and so Borel. The three families therefore generate one and the same $\mathcal{B}(\mathbb{R}^d)$, and one is free to use whichever is convenient – a freedom we exploit immediately, since the boxes are the natural domain on which volume is prescribed.

The closing sentence of definition 1.4 is the working intuition to carry forward: *anything nameable by a countable construction starting from open sets is Borel.* To see this is not empty, take the set that broke Riemann's theory in example 1.1, the rationals $\mathbb{Q} \cap [0, 1]$. Each singleton $\{q\}$ is closed – it is the complement of the open set $\mathbb{R} \setminus \{q\}$, or directly $\{q\} = \bigcap_n (q - \frac{1}{n}, q + \frac{1}{n})$, a countable intersection of open intervals – hence Borel; and $\mathbb{Q} \cap [0, 1] = \bigcup_{q \in \mathbb{Q} \cap [0,1]} \{q\}$ is a *countable* union of Borel sets, hence Borel. The very set on which the Darboux sums disagreed is, in the new framework, a perfectly ordinary measurable set, and we will see in the next subsection that it is one a measure can size (at zero). More generally the same two moves – a closed set is the complement of an open set, a countable family of measurable sets unions and intersects to a measurable set – certify every set that arises in this book.

This completes the home. We have a measurable space $(\Omega, \mathcal{F})$ – concretely $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ – a family of sets closed under exactly the countable operations a limit performs, generated economically from the boxes. The next subsection places a notion of *size* on this family, a measure $\mu\colon \mathcal{F} \to [0, \infty]$, whose one substantive axiom, countable additivity, is the size-level counterpart of the countable-union closure we just built and the direct source of the continuity-under-limits properties that Riemann's integral lacked. One feature of the present construction is worth flagging before we go, because the measure machinery will lean on it hard: the half-open boxes are closed under finite intersection (the intersection of two boxes is a box), and they generate $\mathcal{B}(\mathbb{R}^d)$. A generating family with this intersection-closure – a *π-system* (definition 1.8) – is exactly the lever that the uniqueness machinery of section 1.2.3 and the construction of Lebesgue measure in theorem 1.11 will pull: a measure is pinned down everywhere on $\mathcal{B}(\mathbb{R}^d)$ once its values on the boxes are fixed. The boxes, introduced here as a convenient generator, will return there as the small class on which every concrete measure of this chapter is actually prescribed.

1.2.2 Measures

We now own a home: a measurable space $(\Omega, \mathcal{F})$ whose family $\mathcal{F}$ of admissible sets is closed under exactly the countable unions and intersections a limit performs. What that family conspicuously lacks is a *size*. The previous subsection declared which sets are eligible to be measured but assigned no number to any of them; the limiting profile m and the empirical measure μ^N still have nowhere to live as actual objects. We supply the missing ingredient here – a rule $\mu\colon \mathcal{F} \to [0,\infty]$ that reads off "how much population sits in the region A" for each admissible A. The design of that rule is forced by the same diagnosis that shaped $\mathcal{F}$. In section 1.1 the elementary theory broke because size did not survive countable limits; the Riemann integral could not be slid through the increasing sequence $f_n \uparrow \mathbf{1}_{\mathbb{Q}}$. We are about to impose a single axiom on μ – countable additivity – and the entire payoff of this subsection is that this one axiom is logically equivalent to a package of *continuity-under-limits* properties, the exact properties whose absence sank Riemann's construction. Where the σ-algebra gave us a family of sets stable under limits, the measure gives us a size stable under limits; the two halves of the section 1.1 shopping list are then both in hand.

What should the rule demand? Physically, size is mass, and mass has one defining feature: it *adds over disjoint pieces*. Split a region into parts that do not overlap and the masses of the parts sum to the mass of the whole; there is no double counting and nothing is lost in the seams. Finite additivity, $\mu(A \sqcup B) = \mu(A) + \mu(B)$ for disjoint A, B, is the bare statement of this, and any honest notion of length, charge, probability, or population count obeys it. But finite additivity alone is too weak for our purpose, because every object we care about is assembled by a *limit*, i.e. from a countable sequence of pieces. A density profile is built by refining a partition without end; the empirical measure is a sum of N atoms and we let $N \to \infty$; the rationals are a countable union of singletons. We therefore promote additivity from finite to *countable*: mass adds not just over two disjoint pieces but over a whole sequence of them at once. This single strengthening – from finitely many disjoint pieces to countably many – is precisely the axiom that will let us exchange a limit with a sum, and it is the entire content of the definition.

Definition 1.5 (Measure)**.** A *measure* on $(\Omega, \mathcal{F})$ is a function $\mu\colon \mathcal{F} \to [0,\infty]$ with $\mu(\emptyset) = 0$ that is *countably additive*: for any sequence $(A_n)_{n\geq 1}$ of pairwise disjoint sets in $\mathcal{F}$,

$$\mu\Big(\bigsqcup_{n\geq 1} A_n\Big) \;=\; \sum_{n\geq 1} \mu(A_n). \tag{1.1}$$

The triple $(\Omega, \mathcal{F}, \mu)$ is a *measure space*. If $\mu(\Omega) = 1$ it is a *probability space*, μ a *probability measure*, and we usually write $(\Omega, \mathcal{F}, \mathbb{P})$. If $\mu(\Omega) < \infty$, μ is *finite*; if $\Omega = \bigcup_n \Omega_n$ with $\mu(\Omega_n) < \infty$, it is *σ-finite*.

A few words on the axiom as written. The values land in $[0,\infty]$, the extended nonnegative reals: a measure may legitimately assign infinite size to a set (the whole line has infinite length), so ∞ is an allowed output, and the sum on the right of (1.1) is a sum of nonnegative terms, which always converges in $[0, \infty]$ – either to a finite number or to $+\infty$ – with no question of conditional convergence or reordering, since the terms are nonnegative and the left side does not depend on how we indexed the pieces. The normalization $\mu(\emptyset) = 0$ rules out the degenerate $\mu \equiv \infty$ and is exactly what one expects of "the mass of nothing." The three named special cases are the ones we will actually meet: *probability measures*, total mass one, are the laws m and the empirical measures μ^N at the center of the chapter; *finite* measures are probability measures up to a harmless rescaling; and σ*-finite* measures – those exhausted by a countable family of finite-mass pieces – are the natural habitat of Lebesgue measure (the line is the countable union of the bounded intervals $(-n, n]$, each of finite length) and the precise hypothesis under which Fubini's theorem and the uniqueness machinery later run. The gradation finite $\subset$ σ-finite is not pedantry; σ-finiteness is the exact amount of tameness those theorems need, as we will see when we invoke it.

Countable additivity (1.1) is the entire content of the definition, and its consequences are precisely the continuity properties limits need. We now extract those consequences. The four properties below are not a miscellany; they are the four ways a measure interacts with the set operations a limit performs – inclusion, union, and the two monotone limits $\uparrow$ and $\downarrow$ – and each one is invoked by name later in the chapter.

Proposition 1.6 (Elementary properties)**.** *Let μ be a measure on $(\Omega, \mathcal{F})$ and $A, B, A_n \in \mathcal{F}$. Then:*

1. (Monotonicity) $A \subseteq B \implies \mu(A) \le \mu(B)$.
2. (Countable subadditivity) $\mu\big(\bigcup_n A_n\big) \le \sum_n \mu(A_n)$.
3. (Continuity from below) *If $A_n \uparrow A$ (increasing, $\bigcup_n A_n = A$), then $\mu(A_n) \uparrow \mu(A)$.*
4. (Continuity from above) *If $A_n \downarrow A$ and $\mu(A_1) < \infty$, then $\mu(A_n) \downarrow \mu(A)$.*

Proof. Monotonicity. If $A \subseteq B$ then B is the disjoint union $B = A \sqcup (B \setminus A)$ of two measurable sets (recall $B \setminus A = B \cap A^c \in \mathcal{F}$). Finite additivity – the two-term case of (1.1), obtained by padding the pair $A, B \setminus A$ with $\emptyset, \emptyset, \ldots$ – gives $\mu(B) = \mu(A) + \mu(B \setminus A)$, and since $\mu(B \setminus A) \ge 0$ this is $\ge \mu(A)$. Size respects inclusion; nothing more is happening than "the whole is at least the part."

Subadditivity. The sets A_n may overlap, so we cannot apply (1.1) to them directly – additivity is a statement about *disjoint* pieces only. The remedy is a device we will reuse throughout the book, so it is worth naming: the *disjointification trick*. Replace the overlapping A_n by the disjoint "new mass" sets

$$B_1 = A_1, \qquad B_n = A_n \setminus \bigcup_{k<n} A_k \quad (n \ge 2),$$

where B_n collects the points that enter the union at stage n and not before. Three facts hold by construction. The B_n are pairwise disjoint, because a point's first entry index is unique. They have the same union as the A_n, $\bigcup_n B_n = \bigcup_n A_n$, since stripping off earlier sets removes only points already counted. And $B_n \subseteq A_n$, since B_n is A_n with some points deleted. Now apply countable additivity to the disjoint family (B_n) and then monotonicity term by term:

$$\mu\Big(\bigcup_n A_n\Big) = \mu\Big(\bigsqcup_n B_n\Big) = \sum_n \mu(B_n) \le \sum_n \mu(A_n).$$

The inequality, not equality, is the price of overlap: the right side recounts the shared mass once per set that contains it.

Continuity from below. Here the A_n increase, $A_1 \subseteq A_2 \subseteq \cdots$ with union A, and we use the *same* disjointification to convert a nested union into a disjoint one. With $B_n = A_n \setminus A_{n-1}$ (and $B_1 = A_1$) the B_n are disjoint, the finite union telescopes, $A_N = \bigsqcup_{n \le N} B_n$, and the full union is $A = \bigsqcup_{n \ge 1} B_n$. Countable additivity applied to A, and finite additivity applied to A_N, give

$$\mu(A_N) = \sum_{n \le N} \mu(B_n) \xrightarrow[N \to \infty]{} \sum_{n \ge 1} \mu(B_n) = \mu(A),$$

where the limit is simply the definition of the infinite sum as the limit of its partial sums. Because each new term $\mu(B_n) \ge 0$, the partial sums increase, so $\mu(A_n) \uparrow \mu(A)$ – the convergence is monotone, matching the monotone sweep of the sets. This is the property the monotone convergence theorem (theorem 1.20) rests on: applied to the sublevel sets of an increasing sequence of functions, it is what lets the integral follow the functions up.

Continuity from above. Now the A_n *decrease*, $A_1 \supseteq A_2 \supseteq \cdots$ with intersection A, and we are given $\mu(A_1) < \infty$. The natural move is to complement into the increasing case, but the complement must be taken *inside the finite-mass set* A_1 to keep every quantity finite. Set

$$C_n \;:=\; A_1 \setminus A_n.$$

As A_n shrinks, C_n grows: $A_n \supseteq A_{n+1}$ gives $C_n \subseteq C_{n+1}$, and since $\bigcap_n A_n = A$ we have $\bigcup_n C_n = A_1 \setminus \bigcap_n A_n = A_1 \setminus A$. Thus $C_n \uparrow A_1 \setminus A$, an increasing sequence to which continuity from below applies:

$$\mu(A_1 \setminus A_n) \;=\; \mu(C_n) \;\uparrow\; \mu(A_1 \setminus A).$$

It remains to turn these differences back into the quantities we want, and this is the step that consumes the hypothesis. Because $A_n \subseteq A_1$, additivity on the disjoint splitting $A_1 = A_n \sqcup (A_1 \setminus A_n)$ reads $\mu(A_1) = \mu(A_n) + \mu(A_1 \setminus A_n)$; and since $\mu(A_1) < \infty$ every term here is finite, so we may legitimately subtract,

$$\mu(A_1 \setminus A_n) \;=\; \mu(A_1) - \mu(A_n),$$

and likewise $\mu(A_1 \setminus A) = \mu(A_1) - \mu(A)$. Substituting into the displayed convergence, $\mu(A_1) - \mu(A_n) \uparrow \mu(A_1) - \mu(A)$, and cancelling the finite constant $\mu(A_1)$ from both sides gives $\mu(A_n) \downarrow \mu(A)$. The whole argument is a reflection of continuity from below across the fixed finite-mass background A_1; the finiteness of that background is what makes the reflection $\mu(A_1 \setminus \cdot) = \mu(A_1) - \mu(\cdot)$ a genuine subtraction rather than the meaningless $\infty - \infty$. □

The finiteness hypothesis in (iv) is not a technical convenience; without it the conclusion is simply false, and the failure is instructive. On $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with Lebesgue measure take the half-lines $A_n = (n, \infty)$. They decrease, $A_1 \supseteq A_2 \supseteq \cdots$, and their intersection is empty – no real number exceeds every n – so $A_n \downarrow \emptyset$ and continuity from above would predict $\mu(A_n) \to \mu(\emptyset) = 0$. Yet each A_n is a ray of infinite length, $\mu(A_n) = \infty$ for all n, so $\mu(A_n) \equiv \infty \not\to 0$. Here $\mu(A_1) = \infty$, the hypothesis fails, and with it the conclusion. What went wrong is physically transparent: the sets are shrinking to nothing, but the mass they carry does not shrink with them – it *escapes to infinity*, sliding rightward out of every bounded window while keeping its infinite total. The subtraction step in the proof would have read $\infty - \infty$, which is exactly the algebra refusing to track mass that has run off the edge of the space. This little example is the elementary ancestor of *tightness*, the condition developed in chapter 3 that forbids precisely this leakage: a family of probability measures is tight when no mass can escape to infinity in the limit, and the mechanism we will use to verify tightness – pin a measure's mass inside a large bounded set and let the set grow – is nothing but continuity from above, now always legitimate because a *probability* measure has $\mu(\Omega) = 1 < \infty$ and the hypothesis holds automatically. The finiteness that fails here is the finiteness that tightness guarantees there.

With the abstract properties in hand we can name the concrete measures the book runs on. They are a small cast, and each will reappear: the Dirac mass is the atom from which the empirical measure is built, Lebesgue measure is the backdrop against which densities are defined, and the empirical measure is the protagonist.

Example 1.7 (The measures we will actually use)**.**

- *Dirac measure.* For $a \in \Omega$, the *Dirac measure* δ_a is defined by $\delta_a(A) = \mathbf{1}_A(a)$: it places unit mass at the point a. It is the building block of the empirical measure.
- *Counting measure.* $\#(A) =$ the number of elements of A, with the value ∞ for infinite A.
- *Lebesgue measure* Leb on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, the unique measure assigning to each box $\prod_j (a_j, b_j]$ its elementary volume $\prod_j (b_j - a_j)$ – constructed in theorem 1.11 below.
- *Empirical measure.* For points $X_1, \ldots, X_N \in \mathbb{R}^d$, $\mu^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_i}$, a probability measure on $\mathbb{R}^d$.

Each of these warrants a sentence on why it satisfies the definition, since a definition is only as good as the examples one can certify against it. The *Dirac measure* δ_a is countably additive for a trivial but telling reason: for disjoint sets $A_1, A_2, \ldots$ the point a lies in at most one of them, so exactly one indicator $\mathbf{1}_{A_n}(a)$ is 1 and the rest are 0; both sides of (1.1) equal 1 if $a \in \bigsqcup_n A_n$ and 0 otherwise. It is a probability measure, $\delta_a(\Omega) = 1$, concentrating all its mass at the single point a – the purest possible *atom*, the very object that defeated the density picture in section 1.1, since no function ρ with $\delta_a = \rho \, \mathrm{d}x$ can put positive mass on the volumeless set $\{a\}$. The *counting measure* $\#$ is countably additive because the size of a disjoint union is the sum of the sizes, finite or infinite; it is finite only on finite sets and is the measure with respect to which sums *are* integrals, a fact we exploit in section 1.3. *Lebesgue measure* is the one entry whose existence is not self-evident – assigning a consistent volume to *every* Borel set, not just to boxes, is a genuine construction – and we have forward-referenced it to theorem 1.11, where it is built and shown unique; that pointer is deliberate and we keep it, because the present subsection is not the place to construct Lebesgue measure, only to record where it sits in the cast.

The *empirical measure* $\mu^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_i}$ is the reason the Dirac mass leads the list. It is a finite *average* of Dirac measures, and one checks directly that any nonnegative combination $\sum_i c_i \, \nu_i$ of measures is again a measure – evaluate both sides of (1.1) and interchange the two sums, which is legal because all terms are nonnegative (a nonnegative double sum may be summed in either order). With $c_i = 1/N$ and $\nu_i = \delta_{X_i}$ the total mass is $\frac{1}{N}\sum_{i=1}^N 1 = 1$, so μ^N is a bona fide *probability* measure on $\mathbb{R}^d$: it assigns to a region A the fraction

$$\mu^N(A) \;=\; \frac{1}{N}\sum_{i=1}^N \delta_{X_i}(A) \;=\; \frac{1}{N}\#\{i : X_i \in A\}$$

of the population lying in A. This is the honest mathematical object behind the physicist's "occupation profile," and it is exactly the input the next section integrates against to recover the population average. We hand it forward in that role.

Notation watch

The *Dirac measure* δ_a defined above is an honest measure: it inputs a set and returns a number $\delta_a(A) \in \{0, 1\}$, and it obeys (1.1). The physicist's "Dirac delta function" $\delta(x - a)$ is a different kind of object – or rather, it is not an object of the same type at all. There is no function $\mathbb{R} \to \mathbb{R}$ that vanishes off a single point yet integrates to 1; the symbol $\delta(x - a)$ acquires meaning only *under an integral*, through the rule $\int f(x)\,\delta(x - a)\,\mathrm{d}x = f(a)$, which is a *prescription of how the symbol acts on test functions*, not a statement about the values of a function at points. That action is precisely integration against the Dirac *measure*: in our notation the same content reads $\int f\,\mathrm{d}\delta_a = f(a)$ (a special case of eq. (1.7) below), with the measure on the right carrying all of the information and none of the abuse. We therefore never write $\delta(x)$ as a standalone function. The payoff of this discipline is sharpest for the empirical measure. A *sum* of Dirac measures, $\mu^N = \frac{1}{N}\sum_i \delta_{X_i}$, is again a perfectly well-defined measure – we just verified it – assigning $\mu^N(A) = \frac{1}{N}\#\{i : X_i \in A\}$ to every Borel set. But "a sum of delta functions," $\frac{1}{N}\sum_i \delta(x - X_i)$ read as a function of x, has *no pointwise meaning*: it is 0 at every $x \notin \{X_i\}$ and undefined ("∞") at the X_i, which is not a function one can integrate or take limits of. The measure-theoretic reading is the one that survives the limit $N \to \infty$ – the empirical measures converge weakly to m, as we will prove in section 1.7 – whereas the pointwise reading converges to nothing. This is the spikes-versus-profile heuristic of section 1.1, now sharpened into a notational rule we hold to for the rest of the book.

We have placed a size on the home and read off its behavior under the four set operations a limit can perform. Two threads now run forward from here. The continuity properties (iii) and (iv) are the workhorses of the limit arguments to come: continuity from below drives the monotone convergence theorem (theorem 1.20) and reappears in the uniqueness corollary of the next subsection and in the section-measurability lemma (lemma 1.25), while continuity from above – the finite-mass reflection just proved – is exactly what delivers tightness in the empirical-measure analysis of section 1.7. The Dirac mass δ_a and the empirical measure μ^N, together with the transfer identity for δ_a, are handed to section 1.3, where $\int f\,\mathrm{d}\mu^N = \frac{1}{N}\sum_i f(X_i)$ becomes the literal bridge between the discrete average and the continuum integral. But there is an immediate gap to close first. We have *named* Lebesgue measure and asserted that its box values determine it uniquely, yet a measure is a rule on the *whole* of $\mathcal{F}$, an uncountable family, while we only ever get to prescribe its values on a small, convenient class – the boxes. Why should agreement on so meager a class force agreement everywhere? Answering that is the business of the next subsection: we extract, once and reusably, the principle that two measures agreeing on a sufficiently rich generating class agree on all of $\sigma(\mathcal{P})$. The continuity properties just proved are precisely the fuel that argument burns.

1.2.3 Generating classes and the uniqueness of measures

The previous subsection closed by naming Lebesgue measure and asserting – without yet earning it – that its values on the boxes determine it everywhere. That assertion is not a luxury but a structural necessity, and not only for Lebesgue measure. Look ahead at how every concrete measure in this chapter is actually going to be *produced*. We never get to specify a measure by writing down $\mu(A)$ for each of the uncountably many Borel sets A at once; that would be an impossible amount of data, most of it pathological. Instead we prescribe μ on a small, hands-on, intuitive class of sets – the boxes for Lebesgue measure (theorem 1.11), the measurable rectangles for the product measure (theorem 1.26) – and then *trust* that the prescription has pinned the measure down on all of $\sigma(\mathcal{P})$. The same trust is what lets the Fubini argument (theorem 1.27) identify a second iterated-integral measure with the first by checking them only on rectangles, and what makes the section-measurability lemma (lemma 1.25) go through. Three later constructions, one tool: rather than reprove the propagation principle three times in three disguises, we extract it once here, before any of them needs it, as a clean black box. This is the deliberate logic of the chapter's ordering – the reusable lever is forged in advance of the loads it will bear.

The principle we want is a *uniqueness* statement: if two measures agree on a class of sets that is rich enough, they agree on everything those sets generate. "Rich enough" has to be made precise, and the right notion of richness is the weakest one that does the job. Agreement on a single set, or on a few unrelated sets, plainly propagates nowhere. What turns out to be exactly sufficient is that the class be closed under finite intersection: if A and B are in the class then so is $A \cap B$. A family with this closure is a *π-system*. The boxes are a π-system – the intersection of two boxes is a box (or empty) – and so are the rectangles, which is why this is the right hypothesis to isolate. The combinatorial engine that converts "agree on a π-system" into "agree on the generated σ-algebra" is Dynkin's π–λ theorem, and we prove it from the ground up because the whole chapter, and much of the book, will lean on it.

Definition 1.8 (π-system and λ-system)**.** A nonempty family $\mathcal{P} \subseteq 2^{\Omega}$ is a *π-system* if it is closed under finite intersections. A family $\mathcal{D} \subseteq 2^{\Omega}$ is a *λ-system* (or *Dynkin system*) if (i) $\Omega \in \mathcal{D}$; (ii) $A, B \in \mathcal{D}$ with $A \subseteq B \implies B \setminus A \in \mathcal{D}$ (closure under proper differences); and (iii) $A_n \in \mathcal{D}$ with $A_n \uparrow A \implies A \in \mathcal{D}$ (closure under increasing countable unions).

The two notions look ad hoc until one sees where they come from, and the source is a single observation that organizes the entire proof to follow: *a σ-algebra is exactly a family that is both a π-system and a λ-system.* One direction is immediate from the axioms of definition 1.2 – a σ-algebra is closed under complements and countable unions, hence under finite intersections (by De Morgan), so it is a π-system, and it trivially satisfies the three λ-system clauses. The converse is the useful half: suppose $\mathcal{D}$ is both. It contains Ω, and for any $A \in \mathcal{D}$ the proper difference $\Omega \setminus A = A^c$ lies in $\mathcal{D}$ (clause (ii) with $A \subseteq \Omega$), so $\mathcal{D}$ is closed under complements. Given $A, B \in \mathcal{D}$, the π-system property gives $A \cap B \in \mathcal{D}$, hence $A \cup B = (A^c \cap B^c)^c \in \mathcal{D}$, so $\mathcal{D}$ is closed under finite unions; and a countable union $\bigcup_n A_n$ is the *increasing* union of its finite partial unions $\bigcup_{k \le n} A_k$, each of which is in $\mathcal{D}$, so clause (iii) promotes finite-union closure to countable-union closure. Thus $\mathcal{D}$ is a σ-algebra. The payoff of splitting the σ-algebra axioms into these two weaker halves is strategic: the two halves are easy to verify in *different* places. The λ-system clauses (closure under proper differences and increasing unions) are exactly the operations under which a statement of the form "$\mu(A) = \nu(A)$" is preserved – because they correspond to *subtraction* and to *continuity from below*, both of which a measure respects – so the set of A on which two measures agree is automatically a λ-system. The π-system clause, meanwhile, is the single piece of structure we can demand of the small generating class we actually prescribe. Dynkin's theorem is the statement that having the two halves in these two separate places is enough to force them together.

Theorem 1.9 (Dynkin's π–λ theorem)**.** *If $\mathcal{P}$ is a π-system and $\mathcal{D}$ is a λ-system with $\mathcal{P} \subseteq \mathcal{D}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{D}$.*

Before the formal argument, here is the strategy in one breath, because the proof is dense and its moving parts only make sense once their goal is fixed. We pass to $\mathcal{D}_0$, the *smallest* λ-system containing $\mathcal{P}$ – smaller than the given $\mathcal{D}$, hence enough to control. The plan is to show this minimal λ-system $\mathcal{D}_0$ is in fact closed under finite intersections; by the boxed observation above, a λ-system that is also a π-system is a σ-algebra, and a σ-algebra containing $\mathcal{P}$ must contain $\sigma(\mathcal{P})$. So the whole proof reduces to one combinatorial task: *upgrade $\mathcal{D}_0$ from a λ-system to one closed under intersection.* The device for that upgrade is to fix a set A and collect all B that intersect A back into $\mathcal{D}_0$; this collection is the auxiliary family $\mathcal{G}_A$, and the engine of the proof is that $\mathcal{G}_A$ is itself a λ-system, so once it contains $\mathcal{P}$ it must contain all of $\mathcal{D}_0$.

Proof. Let $\mathcal{D}_0$ be the smallest λ-system containing $\mathcal{P}$. This exists for the same reason generated σ-algebras exist (remark 1.3): an arbitrary intersection of λ-systems is again a λ-system – each clause (i)–(iii) is checked one member-family at a time, exactly as in remark 1.3 – and the power set 2^Ω is a λ-system containing $\mathcal{P}$, so the intersection of all λ-systems containing $\mathcal{P}$ is the smallest one. Since the given $\mathcal{D}$ is one such, $\mathcal{D}_0 \subseteq \mathcal{D}$, and it therefore suffices to prove $\sigma(\mathcal{P}) \subseteq \mathcal{D}_0$. By the observation preceding the theorem, this follows the moment we know $\mathcal{D}_0$ is closed under finite intersections: a λ-system that is also a π-system is a σ-algebra, and being a σ-algebra containing $\mathcal{P}$ it contains the smallest such, $\sigma(\mathcal{P})$. Everything now rests on proving that closure.

The auxiliary families. For any $A \subseteq \Omega$ define

$$\mathcal{G}_A = \{B \subseteq \Omega : A \cap B \in \mathcal{D}_0\},$$

the collection of sets whose intersection with A lands back in $\mathcal{D}_0$. Closure of $\mathcal{D}_0$ under finite intersection is precisely the assertion that $\mathcal{D}_0 \subseteq \mathcal{G}_A$ for every $A \in \mathcal{D}_0$, so this family is built to package exactly the statement we are chasing.

Each $\mathcal{G}_A$ is a λ-system, provided $A \in \mathcal{D}_0$. We check the three clauses, and each one inherits directly from the corresponding clause for $\mathcal{D}_0$ because the map $B \mapsto A \cap B$ commutes with the set operations involved. (i) $\Omega \in \mathcal{G}_A$, since $A \cap \Omega = A \in \mathcal{D}_0$ by hypothesis on A. (ii) If $B \subseteq B'$ both lie in $\mathcal{G}_A$, then $A \cap B \subseteq A \cap B'$ are both in $\mathcal{D}_0$, and intersecting a proper difference distributes,

$$A \cap (B' \setminus B) = (A \cap B') \setminus (A \cap B),$$

a proper difference of two members of $\mathcal{D}_0$, hence in $\mathcal{D}_0$ by clause (ii) for $\mathcal{D}_0$; so $B' \setminus B \in \mathcal{G}_A$. (iii) If $B_n \uparrow B$ with $B_n \in \mathcal{G}_A$, then $A \cap B_n \uparrow A \cap B$ is an increasing union of members of $\mathcal{D}_0$, hence in $\mathcal{D}_0$ by clause (iii) for $\mathcal{D}_0$; so $B \in \mathcal{G}_A$. Thus $\mathcal{G}_A$ is a λ-system whenever $A \in \mathcal{D}_0$.

First pass: anchor on $\mathcal{P}$. Fix $A \in \mathcal{P}$. For any $B \in \mathcal{P}$ the π-system property gives $A \cap B \in \mathcal{P} \subseteq \mathcal{D}_0$, so $B \in \mathcal{G}_A$; that is, $\mathcal{P} \subseteq \mathcal{G}_A$. Since $A \in \mathcal{P} \subseteq \mathcal{D}_0$, the family $\mathcal{G}_A$ is a λ-system, and a λ-system containing $\mathcal{P}$ contains the smallest one, so $\mathcal{D}_0 \subseteq \mathcal{G}_A$. Unwinding the definition of $\mathcal{G}_A$, we have proved: for every $A \in \mathcal{P}$ and every $B \in \mathcal{D}_0$, the intersection $A \cap B \in \mathcal{D}_0$.

The symmetry step: second pass with the roles exchanged. The conclusion just reached is symmetric in form but was proved with A restricted to $\mathcal{P}$; we now bootstrap it to let A range over all of $\mathcal{D}_0$. Read the boxed conclusion the other way: it says that for every fixed $B \in \mathcal{D}_0$, every $A \in \mathcal{P}$ satisfies $A \cap B \in \mathcal{D}_0$, i.e. $\mathcal{P} \subseteq \mathcal{G}_B$. But $B \in \mathcal{D}_0$, so $\mathcal{G}_B$ is again a λ-system by the second step, and a λ-system containing $\mathcal{P}$ contains $\mathcal{D}_0$; hence $\mathcal{D}_0 \subseteq \mathcal{G}_B$. Unwinding once more: for every $B \in \mathcal{D}_0$ and every $A \in \mathcal{D}_0$, the intersection $A \cap B \in \mathcal{D}_0$. That is exactly closure of $\mathcal{D}_0$ under finite intersection. The two passes are the same lemma applied twice – first to push agreement from $\mathcal{P} \times \mathcal{P}$ out to $\mathcal{P} \times \mathcal{D}_0$, then, exploiting the symmetry $A \cap B = B \cap A$, from $\mathcal{P} \times \mathcal{D}_0$ out to $\mathcal{D}_0 \times \mathcal{D}_0$ – and the λ-system structure of the $\mathcal{G}$'s is what powers each push. By the reduction of the first paragraph, $\mathcal{D}_0$ is a σ-algebra and $\sigma(\mathcal{P}) \subseteq \mathcal{D}_0 \subseteq \mathcal{D}$. □

The theorem is abstract, but the corollary we actually use is concrete and is the whole reason the machinery was built: it says that pinning a measure down on a π-system pins it down everywhere, provided the space can be exhausted by finite-mass members of that π-system.

Corollary 1.10 (Uniqueness of measures from a π-system)**.** *Let μ, ν be measures on $(\Omega, \sigma(\mathcal{P}))$ that agree on a π-system $\mathcal{P}$. Suppose there exist $\Omega_n \in \mathcal{P}$ with $\Omega_n \uparrow \Omega$ and $\mu(\Omega_n) = \nu(\Omega_n) < \infty$. Then $\mu = \nu$ on all of $\sigma(\mathcal{P})$.*

Proof. The idea is to feed Dynkin's theorem the set on which μ and ν agree, after localizing to a finite-mass window Ω_n so that subtraction is legal. Fix n and define

$$\mathcal{D}_n = \{A \in \sigma(\mathcal{P}) : \mu(A \cap \Omega_n) = \nu(A \cap \Omega_n)\},$$

the sets on which the two measures agree once everything is intersected with the window Ω_n. We claim $\mathcal{D}_n$ is a λ-system, and we verify each clause, taking care to see which measure property delivers which closure and exactly where the finiteness $\mu(\Omega_n) = \nu(\Omega_n) < \infty$ is spent.

Clause (i): $\Omega \in \mathcal{D}_n$. Here $\Omega \cap \Omega_n = \Omega_n$, and $\mu(\Omega_n) = \nu(\Omega_n)$ by hypothesis, so $\Omega \in \mathcal{D}_n$.

Clause (ii): closure under proper differences. Let $A \subseteq B$ with $A, B \in \mathcal{D}_n$. Intersecting with Ω_n preserves the inclusion, $A \cap \Omega_n \subseteq B \cap \Omega_n$, and additivity on the disjoint splitting $B \cap \Omega_n = (A \cap \Omega_n) \sqcup \big((B \setminus A) \cap \Omega_n\big)$ gives, for each of μ and ν,

$$\mu\big((B \setminus A) \cap \Omega_n\big) = \mu(B \cap \Omega_n) - \mu(A \cap \Omega_n),$$

and likewise for ν. This subtraction is where finiteness earns its keep: because $A \cap \Omega_n \subseteq \Omega_n$ has finite mass $\leq \mu(\Omega_n) < \infty$, the difference is an honest real number rather than the meaningless $\infty - \infty$. Since the two right-hand sides agree term by term ($A, B \in \mathcal{D}_n$), so do the left-hand sides, and $B \setminus A \in \mathcal{D}_n$. This is the clause that forces the λ-system to be the right vehicle: agreement of measures is preserved under *differences*, which is precisely the proper-difference closure, and it is preserved only because of finiteness.

Clause (iii): closure under increasing unions. Let $A_k \uparrow A$ with $A_k \in \mathcal{D}_n$. Intersecting with the fixed window preserves the monotone sweep, $A_k \cap \Omega_n \uparrow A \cap \Omega_n$, so continuity from below (proposition 1.6(iii)) applied to each measure gives

$$\mu(A \cap \Omega_n) = \lim_k \mu(A_k \cap \Omega_n) = \lim_k \nu(A_k \cap \Omega_n) = \nu(A \cap \Omega_n),$$

the middle equality because every $A_k \in \mathcal{D}_n$. Hence $A \in \mathcal{D}_n$. This is the clause that needs continuity from below – the very property handed forward from proposition 1.6 for exactly this purpose – and note it needs no finiteness, since continuity from below holds for all measures.

So $\mathcal{D}_n$ is a λ-system, and it contains $\mathcal{P}$: for $A \in \mathcal{P}$, the π-system property gives $A \cap \Omega_n \in \mathcal{P}$ (an intersection of two members of $\mathcal{P}$), and μ, ν agree on $\mathcal{P}$, so $\mu(A \cap \Omega_n) = \nu(A \cap \Omega_n)$. Dynkin's theorem (theorem 1.9) now applies with this π-system inside this λ-system: $\sigma(\mathcal{P}) \subseteq \mathcal{D}_n$, and since $\mathcal{D}_n$ was a subfamily of $\sigma(\mathcal{P})$ to begin with, $\mathcal{D}_n = \sigma(\mathcal{P})$. In words,

$$\mu(A \cap \Omega_n) = \nu(A \cap \Omega_n) \qquad \text{for every } A \in \sigma(\mathcal{P}) \text{ and every } n.$$

Finally let $n \to \infty$. Because $\Omega_n \uparrow \Omega$ we have $A \cap \Omega_n \uparrow A$ for each fixed measurable A, and one last application of continuity from below to each measure passes the equality to the limit:

$$\mu(A) = \lim_n \mu(A \cap \Omega_n) = \lim_n \nu(A \cap \Omega_n) = \nu(A).$$

Thus $\mu = \nu$ on all of $\sigma(\mathcal{P})$. The two appearances of continuity from below – once to close clause (iii) inside the fixed window, once to remove the window at the end – are the two places where the limit-stability of measure, built in proposition 1.6, does the real work; finiteness of $\mu(\Omega_n)$ is spent in exactly one place, the subtraction of clause (ii). □

We now have the black box the rest of the chapter ordered in advance. To prove two measures equal it suffices to exhibit a π-system on which they agree and a finite-mass exhaustion drawn from it – a far more tractable task than confronting all of $\sigma(\mathcal{P})$ directly. The next subsection cashes this in immediately: it constructs Lebesgue measure and must show the construction is *unique*, and the boxes – already identified in definition 1.4 as a π-system generating $\mathcal{B}(\mathbb{R}^d)$, exhausted by the finite-volume boxes $(-n, n]^d \uparrow \mathbb{R}^d$ – are precisely the generating class corollary 1.10 asks for. The same corollary will be invoked again on the rectangles when we build the product measure (theorem 1.26), identify the second iterated integral in Fubini's theorem (theorem 1.27), and run the section-measurability lemma (lemma 1.25). Each of those later steps will simply name its π-system and cite this corollary; the combinatorial work is done once, here.

1.2.4 Lebesgue measure

The previous subsection forged a tool and left it idle: corollary 1.10 says that a measure is pinned down everywhere on $\mathcal{B}(\mathbb{R}^d)$ once its values are fixed on a generating π-system that admits a finite-mass exhaustion. We promised, in the cast of characters back at example 1.7, one specific measure on which every density in this book is implicitly weighed – the rigorous form of "length, area, volume" – and forwarded its construction to here. Now is when the debt comes due, and the uniqueness engine is exactly what discharges it. The generating π-system will be the very one introduced in definition 1.4: the half-open boxes $\prod_j (a_j, b_j]$, which we noted there are closed under intersection and generate $\mathcal{B}(\mathbb{R}^d)$. On those boxes the prescription is not in doubt for a moment – a box *should* have the volume $\prod_j (b_j - a_j)$, the product of its side lengths, and nothing else. The whole content of the subsection is therefore split cleanly in two: that this box-prescription *extends* to a genuine measure on all of $\mathcal{B}(\mathbb{R}^d)$ (existence), and that it does so in only one way (uniqueness). The first half we import; the second half, and a characterization that pins down Lebesgue measure by a symmetry, we prove.

It is worth being explicit about where the seam between imported and proved material falls, because honoring the self-containedness doctrine here means naming a mechanism precisely and then using only its conclusion. Existence rests on one foundational construction – the Carathéodory extension of a set function from a small algebra to a full σ-algebra – and this is the *single* result of pure measure theory that the entire book accepts without proof. It is standard, its proof is long and bookkeeping-heavy, and reproducing it would teach the reader nothing they will reuse; so we state exactly what it delivers, cite a clean reference for the reader who wants the details, and move on. Every other measure-theoretic fact in this book is proved in these pages. Uniqueness, by contrast, is short and illuminating and follows directly from the machinery we just built, so we give it in full.

Theorem 1.11 (Existence and uniqueness of Lebesgue measure)**.** *There is a unique measure* Leb *on* $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ *such that* $\mathrm{Leb}\big(\prod_{j=1}^d (a_j, b_j]\big) = \prod_{j=1}^d (b_j - a_j)$ *for all boxes. It is σ-finite, translation invariant, and* – among measures that are finite on bounded (equivalently, compact) sets – *characterized up to a positive multiplicative constant by translation invariance.*

Proof. Existence (the accepted import). The construction begins not on the Borel sets but on *all* subsets of $\mathbb{R}^d$ at once, by way of an auxiliary set function that overestimates size. For an arbitrary $E \subseteq \mathbb{R}^d$ define the *Lebesgue outer measure*

$$\mathrm{Leb}^*(E) \;=\; \inf\Big\{ \sum_k \mathrm{vol}(Q_k) \;:\; E \subseteq \bigcup_k Q_k,\ Q_k \text{ boxes} \Big\}, \tag{1.2}$$

the cheapest total box-volume of any countable cover of E by boxes. This is defined for every set, with no measurability question, and it is the honest meaning of "E takes up at most this much room": cover E as economically as boxes allow and add up their volumes. The trouble is that Leb^* is only *sub*additive, not additive – covering two sets together can be cheaper than covering them apart when their boundaries interlock – so it is not yet a measure. Carathéodory's insight is to let the outer measure itself decide which sets are well behaved: call A *Carathéodory measurable* if A cuts every test set cleanly, in the sense that

$$\mathrm{Leb}^*(E) \;=\; \mathrm{Leb}^*(E \cap A) + \mathrm{Leb}^*(E \setminus A) \qquad \text{for all } E \subseteq \mathbb{R}^d,$$

so that no room is lost or double-counted at the cut between the part of E inside A and the part outside. The *Carathéodory extension theorem* then asserts three things at once: the Carathéodory-measurable sets form a σ-algebra; Leb^* restricted to that σ-algebra is countably additive, i.e. an honest measure; and that σ-algebra contains every box – hence, being a σ-algebra, contains all of $\sigma(\text{boxes}) = \mathcal{B}(\mathbb{R}^d)$ – on which the measure returns the elementary volume. That is precisely a measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ with the prescribed box values, which is existence. We take this theorem as the book's one external measure-theoretic import; the full construction, including the verification that boxes satisfy the splitting condition, is carried out in [123, Ch. 1], cited here for attribution and further reading, never as a stand-in for an argument we rely on.

Uniqueness (proved here). This is where the previous subsection pays off. The half-open boxes $\mathcal{P} = \{\prod_j (a_j, b_j]\}$ form a π-system – the intersection of two boxes is again a box (intersect coordinatewise) or empty – and they generate $\mathcal{B}(\mathbb{R}^d)$, both facts recorded at definition 1.4. The cubes $\Omega_n = (-n, n]^d$ are themselves members of $\mathcal{P}$, they increase to $\mathbb{R}^d$ as $n \to \infty$, and each has the finite prescribed volume $\mathrm{Leb}(\Omega_n) = (2n)^d < \infty$. So $\mathcal{P}$ is a generating π-system equipped with a finite-mass exhaustion: exactly the hypotheses of corollary 1.10. Any measure ν with the same box values as Leb agrees with Leb on all of $\mathcal{P}$ (that is what "same box values" says) and in particular on the Ω_n, so the corollary forces $\nu = \mathrm{Leb}$ on all of $\mathcal{B}(\mathbb{R}^d)$. The construction is therefore unique, and σ-finiteness is visible in the same breath: $\mathbb{R}^d = \bigcup_n \Omega_n$ exhibits the whole space as a countable union of the finite-volume cubes Ω_n, which is the very definition of σ-finite. Translation invariance, $\mathrm{Leb}(A+t) = \mathrm{Leb}(A)$, is equally quick: for fixed $t \in \mathbb{R}^d$ the set function $A \mapsto \mathrm{Leb}(A+t)$ is a measure (translation is a bijection that carries disjoint unions to disjoint unions) agreeing with Leb on boxes, because translating a box $\prod_j (a_j, b_j]$ to $\prod_j (a_j + t_j, b_j + t_j]$ leaves every side length $b_j - a_j$ unchanged; so the same uniqueness forces $\mathrm{Leb}(\cdot + t) = \mathrm{Leb}(\cdot)$ everywhere. Already this is a small lesson in how corollary 1.10 is meant to be used: to prove two measures equal one never wrestles with all Borel sets, one checks agreement on boxes and invokes the corollary.

The characterization (proved here). We claim that translation invariance *alone*, together with local finiteness, already determines the measure up to a constant – that Leb is, essentially, the unique notion of size that does not prefer one location to another. Let μ be any measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ that is translation invariant and finite on bounded sets. Set

$$c \;:=\; \mu\big((0,1]^d\big),$$

the μ-mass of the unit cube, which is finite because the cube is bounded. We show $\mu = c\,\mathrm{Leb}$. The engine is the same corollary, fed by a subdivision argument that turns translation invariance into proportionality on boxes, in three graded steps – unit cube, then dyadic and rational boxes, then all Borel sets.

Step 1: subdivide the unit cube. Fix an integer $k \geq 1$ and chop $(0,1]^d$ into k^d congruent small cubes of side $1/k$, the boxes $\prod_j (\frac{m_j - 1}{k}, \frac{m_j}{k}]$ indexed by $(m_1, \ldots, m_d)$ with each $m_j \in \{1, \ldots, k\}$. These k^d boxes are pairwise disjoint and their union is $(0,1]^d$, and – crucially – each is a *translate* of the single small cube $(0, \frac{1}{k}]^d$. Translation invariance makes them all carry the same μ-mass, call it s, and countable (here finite) additivity sums them to the whole:

$$c \;=\; \mu\big((0,1]^d\big) \;=\; \sum_{k^d \text{ boxes}} s \;=\; k^d\, s, \qquad \text{so} \qquad s \;=\; \mu\big((0, \tfrac{1}{k}]^d\big) \;=\; \frac{c}{k^d}.$$

Since $\mathrm{Leb}\big((0, \frac{1}{k}]^d\big) = k^{-d}$, this reads $\mu\big((0, \frac{1}{k}]^d\big) = c\,\mathrm{Leb}\big((0, \frac{1}{k}]^d\big)$: the two measures already agree, up to the factor c, on every small dyadic-type cube.

Step 2: pass to boxes with rational corners. Let $B = \prod_j (a_j, b_j]$ have rational endpoints, and choose a common denominator k so that every ka_j and kb_j is an integer. Then B tiles exactly into translates of the small cube $(0, \frac{1}{k}]^d$: along the j-th axis the interval $(a_j, b_j]$ is partitioned into $k(b_j - a_j)$ subintervals of length $1/k$, so B is the disjoint union of

$$\prod_{j=1}^{d} k(b_j - a_j) \;=\; k^d \prod_{j=1}^{d} (b_j - a_j) \;=\; k^d \,\mathrm{Leb}(B)$$

such small cubes, each of μ-mass c/k^d by Step 1. Additivity then gives

$$\mu(B) \;=\; \big(k^d \,\mathrm{Leb}(B)\big) \cdot \frac{c}{k^d} \;=\; c\,\mathrm{Leb}(B),$$

the desired proportionality on every rational-corner box. (The factors of k^d cancel; the subdivision was only a scaffold for transporting the unit-cube value to an arbitrary rational box by translation.)

Step 3: pass to all Borel sets. The rational-corner boxes again form a π-system – their pairwise intersections are rational-corner boxes – and they still generate $\mathcal{B}(\mathbb{R}^d)$, since every open set is a countable union of them. They are exhausted by the integer cubes $\Omega_n = (-n, n]^d$, which have rational (indeed integer) corners and finite mass under both μ and $c\,\mathrm{Leb}$. Steps 1–2 show μ and $c\,\mathrm{Leb}$ agree on this π-system, so corollary 1.10, invoked one last time, propagates the agreement to all of $\mathcal{B}(\mathbb{R}^d)$: $\mu = c\,\mathrm{Leb}$. This completes the characterization. □

The local-finiteness hypothesis in the characterization is not decoration; drop it and the conclusion collapses, and the counterexample is instructive enough to keep in view. Counting measure # on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ – which assigns to A its number of points, with the value $+\infty$ for infinite A – is perfectly translation invariant, since shifting a set by t neither creates nor destroys points: $\#(A + t) = \#(A)$. Yet it is wildly far from a multiple of Leb. Every nonempty box contains uncountably many points and so has #-mass $+\infty$, while its Lebesgue volume is finite and positive; no single constant c can satisfy $+\infty = c \cdot (\text{finite})$. What rescued the characterization above was the finite value $c = \mu((0, 1]^d)$, and that is exactly what counting measure refuses to supply: it is not finite on any bounded set with infinitely many points. Worse, # is not even σ-finite – a set of finite counting measure is a *finite* set, a countable union of finite sets is countable, but $\mathbb{R}^d$ is uncountable, so $\mathbb{R}^d$ cannot be written as a countable union of finite-mass pieces. Local finiteness is therefore the precise dividing line: among translation-invariant measures it separates the σ-finite, geometrically meaningful ones (the constant multiples of Leb) from pathologies like # that measure cardinality rather than volume.

This closes the construction and hands two concrete bequests to the sections that follow. To the integral of section 1.3 we hand a definite, σ-finite reference measure Leb: it is the background against which a density ρ means the measure $\rho\,\mathrm{Leb}$, and its σ-finiteness – the exhaustion $\mathbb{R}^d = \bigcup_n (-n, n]^d$ – is the standing hypothesis under which the Fubini–Tonelli theorem of section 1.5 will be allowed to run. To the L^p density theorem of section 1.6 we hand the box-cover formula (1.2) itself: because the Lebesgue measure of a Borel set is approached from outside by countable box covers, any integrable indicator can be approximated by simple combinations of box-indicators and thence by continuous functions, which is exactly the mechanism by which $C_c(\mathbb{R}^d)$ will be shown dense in L^p. The outer measure is not scaffolding discarded after existence; it is a quantitative tool the later density argument reuses.

One last piece of hygiene sets up a convention we will lean on for the rest of the book. The Carathéodory construction actually produces a measure on a σ-algebra slightly larger than $\mathcal{B}(\mathbb{R}^d)$:

the *Lebesgue σ-algebra*, obtained by adjoining to $\mathcal{B}(\mathbb{R}^d)$ all subsets of Borel sets of measure zero (every subset of a null set is declared measurable, with measure zero). This *completion* matters in some treatments, but it will not matter for us, because every function we ever integrate in this book is Borel measurable, and on the Borel sets the two agree. What we *do* keep from it is the notion it formalizes: a property that holds outside a set of Lebesgue measure zero is said to hold *almost everywhere* (a.e.), and a property holding outside a set of probability zero holds *almost surely* (a.s.). Null sets are negligible for every purpose involving integration – changing a function on a null set changes no integral, as we are about to see – and the "a.e." qualifier, which from here on attaches silently to nearly every statement about functions, draws its meaning from exactly the Lebesgue null sets this construction identifies. With a home, a notion of size, and now a canonical reference measure in hand, the one structure still missing is the maps between such homes – the functions we will integrate – and it is to those that we turn next.

1.2.5 Measurable functions

We now have a home and a size: a measurable space $(\Omega, \mathcal{F})$ – concretely $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ – and on it a notion of mass, with Lebesgue measure as the canonical reference against which densities are read. Two of the three objects the chapter needs are in place. The third is the class of *maps*. Sets acquired a size in definitions 1.2 and 1.5; the natural next question is which functions *respect* that structure, and the answer supplies at once the observables we will integrate, the device that manufactures the law m of a single agent, and – the point that closes the circle opened in section 1.1 – the guarantee that the limiting objects the book keeps producing are still legitimate things to integrate. Recall the third worry of that opening section: every interesting object (the limit profile, the empirical limit, the solution of an existence argument) is built by a countable limiting operation, and an integral theory is worthless if the result of such a limit falls outside the class of integrable functions. The decisive proposition of this subsection – that a pointwise limit of measurable functions is again measurable – is exactly the answer to that worry, and it is the reason the trio sets/measures/maps is now complete rather than merely two-thirds built.

Begin with the definition, and with why it is phrased through *preimages* rather than images. A continuous map between topological spaces is the one whose preimages of open sets are open: it is the morphism that respects nearness, and crucially the condition runs backward, from target to source. The measure-theoretic morphism is the exact analogue with "open" replaced by "measurable." The reason the direction is backward is the same in both settings: the structure we wish to preserve lives on *sets* of the target, and the only way a function f can transport a question about a target set A' – "how much source mass lands in A'?" – into a question the source can answer is to look at $f^{-1}(A')$, the set of source points that f sends into A'. If that preimage is measurable, the question has an answer; if it is not, the question is meaningless. So measurability is precisely the demand that f never turn a legitimate target question into an illegitimate source one.

Definition 1.12 (Measurable function)**.** A map $f\colon (\Omega, \mathcal{F}) \to (\Omega', \mathcal{F}')$ between measurable spaces is *measurable* if $f^{-1}(A') \in \mathcal{F}$ for every $A' \in \mathcal{F}'$. When the target is $\mathbb{R}^d$ (or $\mathbb{R}$, or $[-\infty, \infty]$) with its Borel σ-algebra we speak of a *Borel measurable* or simply *measurable* function; it suffices that $f^{-1}((a, \infty)) \in \mathcal{F}$ for every $a \in \mathbb{R}$ in the scalar case, since such sets generate $\mathcal{B}(\mathbb{R})$. On a probability space a measurable function is exactly a *random variable.*

The generating-set shortcut in the last sentence is what makes the definition usable, and it deserves to be seen, because verifying the preimage condition against *every* Borel set would be hopeless – there is no list of the Borel sets to run through. The shortcut rests on one structural fact: the family of target sets whose preimage is measurable is *itself a σ-algebra.* Write $\mathcal{G} := \{A' \subseteq \Omega' : f^{-1}(A') \in \mathcal{F}\}$. Preimage commutes with all the set operations – $f^{-1}(\Omega') = \Omega$, $f^{-1}((A')^c) = f^{-1}(A')^c$, and $f^{-1}(\bigcup_n A'_n) = \bigcup_n f^{-1}(A'_n)$ – so each of the three σ-algebra axioms for $\mathcal{G}$ is inherited from the corresponding axiom for $\mathcal{F}$: $\mathcal{G}$ is a σ-algebra. Now suppose $f^{-1}(E) \in \mathcal{F}$ merely for every E in some generating family $\mathcal{E}$ with $\sigma(\mathcal{E}) = \mathcal{F}'$. Then $\mathcal{E} \subseteq \mathcal{G}$, and since $\mathcal{G}$ is a σ-algebra it contains the smallest σ-algebra over $\mathcal{E}$, namely all of $\mathcal{F}' = \sigma(\mathcal{E})$. So checking the preimage condition on a generating family checks it everywhere – the same propagation-from-a-generating-class principle that powered remark 1.3 and the equivalence of generators for $\mathcal{B}(\mathbb{R}^d)$. For a scalar target the rays (a, ∞) generate $\mathcal{B}(\mathbb{R})$ (definition 1.4), so one need only confirm that each sublevel-type set $\{f > a\} = f^{-1}((a, \infty))$ is measurable – a single inequality to test, exactly the form in which measurability is checked in practice. A one-line instance: every *continuous* $f\colon \mathbb{R}^d \to \mathbb{R}$ is Borel measurable, since $f^{-1}((a, \infty))$ is open (a preimage of an open set under a continuous map) and open sets are Borel; continuity is more than enough, and measurability is the strictly weaker demand that lets in the discontinuous indicators and limits we actually need.

Before the central stability result, we dispose of the routine closure properties, since the observables we integrate are built from simpler ones by arithmetic and composition. If f, g are measurable and $\varphi\colon \mathbb{R} \to \mathbb{R}$ (or $\mathbb{R}^2 \to \mathbb{R}$) is continuous, then $\varphi(f)$ and more generally $\varphi(f, g)$ are measurable. The mechanism is the preceding one-liner run once more: the pair $(f, g)\colon \Omega \to \mathbb{R}^2$ is measurable because its preimage of a box $(a, b] \times (c, d]$ is $\{a < f \leq b\} \cap \{c < g \leq d\}$, an intersection of two measurable sets, and the boxes generate $\mathcal{B}(\mathbb{R}^2)$; composing with the continuous (hence Borel) map φ preserves measurability because $(\varphi \circ (f, g))^{-1}(B) = (f, g)^{-1}(\varphi^{-1}(B))$ and $\varphi^{-1}(B)$ is Borel. Taking $\varphi(u, v) = u + v$ and $\varphi(u, v) = uv$ shows that sums and products of measurable functions are measurable; taking φ any continuous scalar map covers compositions. These facts we use without further comment from here on; the only one that is not a routine corollary – and the only one the limits of this book genuinely depend on – is stability under *pointwise limits*, to which we now turn.

Proposition 1.13 (Stability of measurability under limits)**.** *If $f_n\colon \Omega \to [-\infty, \infty]$ are measurable, then* $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$, *and* $\liminf_n f_n$ *are measurable. In particular, if $f_n \to f$ pointwise, then f is measurable.*

Proof. Everything follows from a single set identity, read off the definition of a supremum. A point ω satisfies $\sup_n f_n(\omega) > a$ if and only if at least one $f_n(\omega)$ exceeds a – if the supremum beats a, some term must already beat a, and conversely a single term beating a forces the supremum past it. In symbols,

$$\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\},$$

a *countable* union of the measurable sets $\{f_n > a\} = f_n^{-1}((a, \infty))$, hence measurable; and as a ranges over $\mathbb{R}$ these sets generate the relevant σ-algebra, so $\sup_n f_n$ is measurable by the generating-set criterion. The infimum is the mirror image: a point lies below a in infimum iff some term lies below a, giving $\{\inf_n f_n < a\} = \bigcup_n \{f_n < a\}$, again a countable union of measurable sets. (Equivalently, $\inf_n f_n = -\sup_n(-f_n)$.) The two one-sided limits are now built from these by the very formulas that define them as iterated extrema,

$$\limsup_n f_n = \inf_N \sup_{n \geq N} f_n, \qquad \liminf_n f_n = \sup_N \inf_{n \geq N} f_n,$$

each a countable infimum of countable suprema of the f_n, hence measurable by two applications of what we have just shown. Finally, wherever the sequence converges its limit equals its lim sup (and its lim inf); so if $f_n \to f$ pointwise everywhere, $f = \limsup_n f_n$ is measurable. The whole proof is the observation that a limit is assembled from sup and inf, and that sup and inf over a *sequence* cost only countable unions – exactly the operations $\mathcal{F}$ was built to absorb. □

It is worth pausing on the word "countable" here, because it is the same hinge the whole chapter turns on. The identity $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$ would be useless if the index ran over an uncountable set – an uncountable union of measurable sets need not be measurable, and indeed the supremum of an uncountable family of measurable functions can fail to be measurable. It is precisely because a limit is the limit of a *sequence* that everything stays inside the σ-algebra. This proposition is the unglamorous engine of the entire subject. Every convergence theorem in section 1.4 hands back a function defined as a pointwise limit, and every existence proof in this book – the fixed-point constructions of the mean field game system, the limits of empirical measures, the weak solutions assembled by compactness – produces its object as a limit of approximants and must then *integrate that object* or apply a further estimate to it. Each such step silently invokes proposition 1.13 to know the limit is a legitimate measurable function in the first place. Without it, the constructions would produce objects one is not entitled to integrate, and the analysis would have nothing to stand on. This is the precise discharge of the third worry of section 1.1: limiting objects remain legal.

With measurable functions in hand we can describe how a measure is *transported* along one. This is the construction that produces m in the first place. A random agent is a measurable map $X : \Omega \to \mathbb{R}^d$ from an abstract sample space carrying a probability measure $\mathbb{P}$; we never see Ω directly, only where X lands, and the right object to record is "how likely is the agent to land in the region A?" – the number $\mathbb{P}(X^{-1}(A))$. Reading this as a function of A defines a new measure, on the state space, that forgets the sample space entirely.

Definition 1.14 (Pushforward measure)**.** If $f : (\Omega, \mathcal{F}, \mu) \to (\Omega', \mathcal{F}')$ is measurable, the *pushforward* (or *image*) measure $f_\#\mu$ on $\mathcal{F}'$ is $(f_\#\mu)(A') = \mu(f^{-1}(A'))$. For a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$, the pushforward $\mathcal{L}(X) := X_\#\mathbb{P}$ is its *law* or *distribution*, a probability measure on $\mathbb{R}^d$.

That $f_\#\mu$ really is a measure is a direct inheritance: $f_\#\mu(\emptyset) = \mu(f^{-1}(\emptyset)) = \mu(\emptyset) = 0$, and for disjoint A'_n the preimages $f^{-1}(A'_n)$ are disjoint (a point maps into at most one A'_n), so countable additivity of $f_\#\mu$ is countable additivity of μ read through the preimage. Measurability of f is exactly what makes the right-hand side $\mu(f^{-1}(A'))$ defined for every $A' \in \mathcal{F}'$ – the definition would be empty without it. For a random variable X the total mass is $X_\#\mathbb{P}(\mathbb{R}^d) = \mathbb{P}(X^{-1}(\mathbb{R}^d)) = \mathbb{P}(\Omega) = 1$, so the law $\mathcal{L}(X)$ is a probability measure on $\mathbb{R}^d$, a point of $\mathcal{P}(\mathbb{R}^d)$.

The pushforward is engineered so that integrating against it is the same as integrating the pulled-back integrand against the original measure. This *transfer* (or *change-of-variables*) identity is the bridge we use again and again – it is what turns an expectation $\mathbb{E}[g(X)]$ computed over the unseen sample space into an honest integral $\int g \, \mathrm{d}\mathcal{L}(X)$ against the law on the state space, and it is in particular the single load-bearing equality of the empirical-measure computation in section 1.7. One must be careful about exactly what definition 1.14 has and has not delivered. The definition fixes $f_{\#}\mu$ on *sets* only: it says how to push forward the mass of a region, nothing more. The statement we want is about *integrands* – arbitrary measurable g, not just indicators of sets – and that statement does not come for free from the set-level definition. It is a separate lemma, and the work of the lemma is precisely to climb from the set-level identity up to the function-level one through the standard ladder of the integral.

Lemma 1.15 (Transfer of integrals under a pushforward)**.** *Let $f\colon (\Omega, \mathcal{F}, \mu) \to (\Omega', \mathcal{F}')$ be measurable and let $g\colon \Omega' \to [-\infty, \infty]$ be $\mathcal{F}'$-measurable. Then $g \circ f$ is $\mathcal{F}$-measurable, and*

$$\int_{\Omega'} g \, \mathrm{d}(f_{\#}\mu) \;=\; \int_{\Omega} g \circ f \, \mathrm{d}\mu \tag{1.3}$$

holds whenever $g \geq 0$ (both sides in $[0, \infty]$), and also whenever $g \in L^1(f_{\#}\mu)$ – equivalently $g \circ f \in L^1(\mu)$ – with both sides finite and equal.

Proof. The proof is the prototype of an argument we will run many times: *verify an integral identity on indicators, extend by linearity to simple functions, pass to nonnegative functions by monotone approximation, and split a signed function into its positive and negative parts.* Each of the four rungs is the natural next case once the previous is settled.

First, $g \circ f$ is $\mathcal{F}$-measurable, so both sides of (1.3) are even defined: for any Borel B, $(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B))$, which lies in $\mathcal{F}$ because $g^{-1}(B) \in \mathcal{F}'$ (measurability of g) and f^{-1} carries $\mathcal{F}'$ into $\mathcal{F}$ (measurability of f). This is just the closure of measurability under composition, applied to the specific composite we integrate.

Rung 1: indicators. Take $g = \mathbf{1}_{A'}$ with $A' \in \mathcal{F}'$. A point ω satisfies $g(f(\omega)) = 1$ exactly when $f(\omega) \in A'$, i.e. $\omega \in f^{-1}(A')$, so $g \circ f = \mathbf{1}_{f^{-1}(A')}$. The integral of an indicator is the measure of its set ((1.4)), and so

$$\int_{\Omega} g \circ f \, \mathrm{d}\mu = \mu(f^{-1}(A')) = (f_{\#}\mu)(A') = \int_{\Omega'} g \, \mathrm{d}(f_{\#}\mu),$$

where the middle equality is nothing but the *definition* of the pushforward. So (1.3) for indicators *is* the set-level definition of $f_{\#}\mu$, restated under an integral sign – this rung is where the definition enters, and it is the only rung that uses it.

Rung 2: simple functions. Both sides of (1.3) are linear in g over nonnegative coefficients, and $(\sum_k c_k \mathbf{1}_{A'_k}) \circ f = \sum_k c_k \mathbf{1}_{f^{-1}(A'_k)}$, so summing Rung 1 over the pieces of a nonnegative simple $g = \sum_k c_k \mathbf{1}_{A'_k}$ gives (1.3) for every nonnegative simple g.

Rung 3: nonnegative functions. Let $g \geq 0$ be measurable. Choose simple functions $s_n \uparrow g$ increasing to g pointwise – the canonical sequence constructed in section 1.3 (the explicit dyadic truncation s_n used there to define the integral of a nonnegative function). Composing with f

preserves the monotone convergence, $s_n \circ f \uparrow g \circ f$, because composition acts pointwise and the s_n increase at every point. Rung 2 gives $\int s_n \circ f \, \mathrm{d}\mu = \int s_n \, \mathrm{d}(f_\# \mu)$ for each n, and the monotone convergence theorem (theorem 1.20) lets us pass to the limit on *both* sides at once:

$$\int_\Omega g \circ f \, \mathrm{d}\mu = \lim_n \int_\Omega s_n \circ f \, \mathrm{d}\mu = \lim_n \int_{\Omega'} s_n \, \mathrm{d}(f_\# \mu) = \int_{\Omega'} g \, \mathrm{d}(f_\# \mu).$$

This is (1.3) for all nonnegative g, with both sides possibly $+\infty$ but always equal.

Rung 4: signed functions. For general measurable g write $g = g^+ - g^-$ with $g^\pm \geq 0$. Composition respects the splitting, $(g \circ f)^\pm = g^\pm \circ f$ (because f only relabels the point at which g is evaluated, it cannot change the sign of the value), so applying Rung 3 to g^+ and to g^- and adding gives $\int |g| \, \mathrm{d}(f_\# \mu) = \int |g \circ f| \, \mathrm{d}\mu$. Hence $g \in L^1(f_\# \mu)$ if and only if $g \circ f \in L^1(\mu)$ – the two integrability conditions are literally the same finite number – and when this common value is finite, subtracting the (now finite) Rung-3 identities for g^+ and g^- yields (1.3) for the signed g. □

Remark 1.16 (A mild and honest forward dependency)**.** Two ingredients of Rung 3 – the canonical simple approximation $s_n \uparrow g$ and the monotone convergence theorem – are objects we have not yet built; they live in section 1.3 and theorem 1.20, a few pages ahead. We deliberately use them here only as named black boxes and do *not* reproduce their construction or proof, because the logical order is not circular: the integral and the monotone convergence theorem are developed without any reference to the pushforward or to transfer, so importing them forward into this lemma creates no loop. The transfer identity is placed in this subsection because it belongs to the story of measurable maps, not because it could be proved before the integral exists; the reader meeting it should carry the two forward references as promissory notes redeemed in the next section.

A first sanity check makes the identity concrete. Take $\mu = \delta_a$, the Dirac mass at a point $a \in \Omega$, and any measurable f. The pushforward is $f_\# \delta_a = \delta_{f(a)}$, since $(f_\# \delta_a)(A') = \delta_a(f^{-1}(A')) = \mathbf{1}[f(a) \in A'] = \delta_{f(a)}(A')$; and (1.3) reads $\int g \, \mathrm{d}\delta_{f(a)} = \int g \circ f \, \mathrm{d}\delta_a$, i.e. $g(f(a)) = g(f(a))$ – the transfer identity for a point mass is just the statement that pushing a spike forward moves it to its image, which is reassuring rather than surprising, and is exactly the $N = 1$ atom of the empirical measure.

The law $\mathcal{L}(X)$ produced by this machinery is the object the rest of the book treats as the fundamental unknown. In mean field game theory the entire population is summarized by a single point $m \in \mathcal{P}(\mathbb{R}^d)$, and an individual agent's randomness enters the theory only through its law: two agents with the same law are interchangeable, which is the precise content of the exchangeability that underlies the whole continuum description. We adopt the global convention of appendix A – restated here on first use – that $m = m$ denotes such a population measure. In the present chapter m is simply a *fixed* probability law on $\mathbb{R}^d$, the common law from which a population is sampled; from chapter 8 onward it acquires a time index and *evolves* forward in time as the density of a continuum of agents, the forward Fokker–Planck partner of the backward value function. It is the recurring protagonist of everything that follows, and this subsection has given it both its definition (as a pushforward) and the one identity through which it will be computed.

That identity is the precise thing we hand to the next two stops. To the integral construction of section 1.3 we hand the class of measurable functions it was built to feed – the integrand g in (1.3) has nowhere to be integrated until that section says what $\int g \, \mathrm{d}\nu$ means – together with the assurance, from proposition 1.13, that the monotone and dominated limits the integral will be asked to survive land back inside this same class. And to the closing worked computation of section 1.7 we hand its single most-used equality: for an i.i.d. agent X_i with law $\mathcal{L}(X_i) = m$ and a bounded test function f_k, transfer gives

$$\mathbb{E}[f_k(X_i)] \;=\; \int_{\mathbb{R}^d} f_k \,\mathrm{d}\,\mathcal{L}(X_i) \;=\; \int_{\mathbb{R}^d} f_k \,\mathrm{d}m,$$

the conversion of a population average into an integral against the law that the entire empirical-measure theorem is built to exploit. With the maps in hand, the trio is complete; we now build the integral they were designed to feed.

1.3 The Lebesgue integral

The previous subsection handed us the two ingredients an integral pairs together: the *measurable functions* – the observables f we wish to average – and, from the subsection before it, the *measures* – the sizes μ against which we average them. What is still missing is the pairing itself, the rule that turns a function and a measure into a number $\int f \,\mathrm{d}\mu$. We build that rule now. Two demands govern its design, and both come straight from section 1.1. First, the rule must reproduce the physicist's average exactly: integrating an observable against the empirical measure must return the ensemble mean $\frac{1}{N}\sum_i f(X_i)$, the literal bridge (1.8) between the discrete population and its continuum summary. Second – and this is the demand Riemann's construction could not meet – the rule must *survive the limits we actually take.* The increasing sequence $f_n \uparrow \mathbf{1}_{\mathbb{Q}}$ of example 1.1 broke the Riemann integral precisely because its value did not pass to the monotone limit; we engineer the Lebesgue integral so that the monotone limit is not an afterthought but the very mechanism of the construction. Each of the three strokes below – simple functions, then nonnegative functions, then general ones – is defined so that the convergence theorems of section 1.4 come out cleanly, repairing exactly the defect diagnosed in section 1.1. Fix throughout a measure space $(\Omega, \mathcal{F}, \mu)$.

First stroke: simple functions. We begin where the measure already tells us everything. The only functions whose integral needs no analysis at all are the *simple* ones – finite combinations of indicators of measurable sets – because for them "area under the graph" is literally a finite sum of "height × size of base," and the measure supplies the size. A *simple function* is a finite combination

$$s = \sum_{k=1}^{m} c_k \,\mathbf{1}_{A_k}, \qquad c_k \in [0,\infty),\;\; A_k \in \mathcal{F},$$

taking only finitely many nonnegative values. Its integral is *declared* – there is nothing to compute, only a convention to fix – to be

$$\int_\Omega s \,\mathrm{d}\mu \;:=\; \sum_{k=1}^{m} c_k \,\mu(A_k) \;\in [0,\infty]: \tag{1.4}$$

the value c_k weighted by the measure $\mu(A_k)$ of the set where s takes it, summed over the values. The measure does all the work; no limiting process appears. This is the entire reason simple functions come first: they are the level at which integration is just arithmetic with the numbers $\mu(A_k)$ that the previous subsections taught us to assign.

There is one thing to check before (1.4) is even well posed, and it is the kind of check that is easy to wave away and worth doing once carefully, because the argument is the prototype of several to come. A simple function does not come with a canonical representation: the same s can be written as $\sum_k c_k \mathbf{1}_{A_k}$ and as $\sum_\ell d_\ell \mathbf{1}_{B_\ell}$ with completely different sets and coefficients (merge two sets sharing a value, or split one in two, and the formula changes while the function does not). For (1.4) to define a number attached to s rather than to an accidental way of writing it, the two sums $\sum_k c_k \mu(A_k)$ and $\sum_\ell d_\ell \mu(B_\ell)$ must agree. They do, by passing to the *common refinement*. There is no loss in assuming each representation partitions Ω – adjoin the set where $s = 0$ with coefficient 0 if needed – so that $\{A_k\}_k$ and $\{B_\ell\}_\ell$ are each partitions of Ω into disjoint measurable pieces. Their common refinement $\{A_k \cap B_\ell\}_{k,\ell}$ is a single partition refining both. On a nonempty cell $A_k \cap B_\ell$ the function s has one value, read off either representation, so $c_k = d_\ell$ there. Finite additivity of μ – the two-set case of countable additivity (1.1) – splits each base set along the refinement, $\mu(A_k) = \sum_\ell \mu(A_k \cap B_\ell)$ and $\mu(B_\ell) = \sum_k \mu(A_k \cap B_\ell)$, and therefore

$$\sum_k c_k\, \mu(A_k) = \sum_{k,\ell} c_k\, \mu(A_k \cap B_\ell) = \sum_{k,\ell} d_\ell\, \mu(A_k \cap B_\ell) = \sum_\ell d_\ell\, \mu(B_\ell),$$

the middle equality holding cell by cell because $c_k = d_\ell$ on every nonempty $A_k \cap B_\ell$ (and the empty cells contribute $0 \cdot \mu(\emptyset) = 0$ to either side). The integral of a simple function is thus a property of the function, not of its bookkeeping, and – the point to carry forward – it is the *additivity of the measure* that makes the representation drop out. This same common-refinement move will reappear in a moment as the proof that the integral is additive on simple functions.

Second stroke: nonnegative measurable functions. A general nonnegative observable is not simple; it takes a continuum of values. The way to integrate it is to approach it from below by the simple functions we can already integrate, and to take the integral to be as large as those lower approximants force it to be. For measurable $f\colon \Omega \to [0,\infty]$ we set

$$\int_\Omega f \; \mathrm{d}\mu \;:=\; \sup\Big\{ \int_\Omega s \; \mathrm{d}\mu \;:\; s \text{ simple},\ 0 \le s \le f \Big\} \;\in [0,\infty]. \tag{1.5}$$

Why the supremum over simple functions *below* f, rather than some other recipe? Because it is forced. Whatever the integral of f is to be, monotonicity demands $\int s \le \int f$ for every simple $s \le f$, so $\int f$ can be no smaller than the supremum on the right; and if the integral is to be compatible with monotone limits at all – the property we are building the whole theory to secure – it can be no larger, since (as we verify next) the simple functions below f climb all the way up to f. The supremum in (1.5) is thus not one choice among many but the *unique* extension of the simple-function rule (1.4) that respects monotone limits. Everything downstream – the monotone convergence theorem above all – is the cashing-out of this design choice.

That the simple functions below f genuinely exhaust it is made concrete by an explicit approximating sequence, and the formula repays a careful reading because it encodes the single idea that distinguishes Lebesgue's integral from Riemann's. Define

$$s_n \;=\; \sum_{k=0}^{n2^n-1} \frac{k}{2^n}\, \mathbf{1}_{\{\, k/2^n \le f < (k+1)/2^n \,\}} \;+\; n\, \mathbf{1}_{\{f \ge n\}}.$$

Read it as a slicing not of the domain but of the *range*. Riemann chops the x-axis into small intervals and asks for the height of f over each; Lebesgue chops the y-axis – the values of f – into the horizontal bands $[k/2^n, (k+1)/2^n)$ of width 2^{-n}, and for each band asks *how much of the space* the measure sees with f-value in that band. The set $\{k/2^n \le f < (k+1)/2^n\} = f^{-1}([k/2^n, (k+1)/2^n))$ is precisely the preimage of one band; on it s_n takes the band's floor value $k/2^n$, contributing $\frac{k}{2^n}\,\mu(\{k/2^n \le f < (k+1)/2^n\})$ to the integral – the height of the band times the size of the part of the space living in it. Above the height n, where f is too large for any of the finitely many bands to reach, the last term truncates s_n at the value n on the set $\{f \ge n\}$. Here is exactly where measurability earns its keep: every set the formula names is the preimage of a Borel set under f, hence lies in $\mathcal{F}$ by the very definition of a measurable function (definition 1.12), so each $\mu(\cdots)$ in $\int s_n \,\mathrm{d}\mu$ is defined. Were f not measurable, the slicing sets would have no size and the construction would not even start.

The sequence climbs to f, $s_n \uparrow f$, and seeing why pins down what each refinement does. Increasing n to $n+1$ does two things at once: it *halves* every band, replacing a band of height 2^{-n} by two bands of height $2^{-(n+1)}$, so that the floor value attached to a point can only rise (a point formerly assigned the lower edge of a wide band is now assigned the lower edge of the narrower band it actually sits in, which is at least as high); and it *raises the ceiling* from n to $n+1$, capturing more of the large-value region. Hence $s_n \le s_{n+1}$ everywhere. As for the limit, fix a point x. If $f(x) < \infty$, then for all $n > f(x)$ the value $f(x)$ lies below the ceiling and inside some band of width 2^{-n}, so $0 \le f(x) - s_n(x) < 2^{-n} \to 0$; if $f(x) = \infty$, then $x \in \{f \ge n\}$ for every n and $s_n(x) = n \to \infty = f(x)$. Either way $s_n(x) \uparrow f(x)$. Thus every nonnegative measurable f is a monotone increasing pointwise limit of simple functions, and (1.5) is the value that limit is forced to have. The doubling of resolution as n grows is the hallmark of the range-slicing picture: refinement adds horizontal detail, and the measure converts each new sliver of detail into a definite contribution.

Third stroke: general functions. A signed observable is split into its nonnegative parts and integrated piece by piece. A measurable $f\colon \Omega \to [-\infty, \infty]$ has *positive and negative parts* $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$, both nonnegative and measurable, with the two identities $f = f^+ - f^-$ and $|f| = f^+ + f^-$. We call f *integrable* (and write $f \in L^1(\mu)$) when the total mass of $|f|$ is finite,

$$\int |f| \,\mathrm{d}\mu = \int f^+ \,\mathrm{d}\mu + \int f^- \,\mathrm{d}\mu < \infty,$$

which is the same as asking that f^+ and f^- each have finite integral; and for such f we define

$$\int_\Omega f \,\mathrm{d}\mu \;:=\; \int_\Omega f^+ \,\mathrm{d}\mu - \int_\Omega f^- \,\mathrm{d}\mu. \tag{1.6}$$

The integrability requirement is exactly what makes the subtraction in (1.6) meaningful: with both terms finite there is no risk of the undefined $\infty - \infty$, the same algebraic catastrophe that sank continuity-from-above for an infinite-mass set in proposition 1.6. For a vector-valued $f\colon \Omega \to \mathbb{R}^d$ integrability means $\int |f| \,\mathrm{d}\mu < \infty$ with $|f|$ the Euclidean norm, and the integral is taken componentwise.

The basic algebra Riemann lacked. The integral now exists; we must check it obeys the rules that make it usable – linearity, monotonicity, the triangle inequality – and one of these rules is the precise statement whose failure under limits doomed the Riemann theory. Linearity has two halves. Positive homogeneity $\int(cf) = c\int f$ for $c \geq 0$ is immediate: scaling a simple function scales its integral by c directly from (1.4), and the supremum (1.5) scales with it. *Additivity*, $\int(f+g) = \int f + \int g$, is the substantive half, and it is exactly the kind of statement that fails for the Riemann integral under limits – recall that in example 1.1 the obstruction was precisely an additive/limit interchange that Riemann's construction could not perform. We prove it now, because the chapter uses it repeatedly: to split $\int f \, \mathrm{d}\mu^N$ over the sum of Diracs in (1.8), and to manipulate $\int(2g - |f_n - f|)$ in the proof of dominated convergence.

For *simple* $s = \sum_k c_k \mathbf{1}_{A_k}$ and $t = \sum_\ell d_\ell \mathbf{1}_{B_\ell}$, additivity is the common-refinement argument again, now used to add rather than to compare. Refine both to the partition $\{A_k \cap B_\ell\}_{k,\ell}$; on each cell $s+t$ takes the constant value $c_k + d_\ell$, so $s+t$ is itself simple with this representation, and (1.4) gives $\int(s+t)\,\mathrm{d}\mu = \sum_{k,\ell}(c_k + d_\ell)\,\mu(A_k \cap B_\ell)$, which splits into $\sum_{k,\ell} c_k\,\mu(A_k \cap B_\ell) + \sum_{k,\ell} d_\ell\,\mu(A_k \cap B_\ell) = \int s\,\mathrm{d}\mu + \int t\,\mathrm{d}\mu$ by the same finite-additivity bookkeeping as before. At the simple level additivity is therefore pure arithmetic with the measure – no limit anywhere.

For general nonnegative measurable f, g the limit finally enters, and this is the heart of the matter. Take the canonical dyadic sequences $s_n \uparrow f$ and $t_n \uparrow g$ built above. Their sum increases to the sum, $s_n + t_n \uparrow f + g$, and we pass to the limit through the integral three times – once on each summand and once on the sum:

$$\int (f+g)\,\mathrm{d}\mu \overset{(1)}{=} \lim_n \int (s_n + t_n)\,\mathrm{d}\mu \overset{(2)}{=} \lim_n \Big(\int s_n\,\mathrm{d}\mu + \int t_n\,\mathrm{d}\mu \Big) \overset{(3)}{=} \int f\,\mathrm{d}\mu + \int g\,\mathrm{d}\mu.$$

Step (2) is just simple-function additivity, applied to each n. Steps (1) and (3) are each an instance of the monotone convergence theorem (theorem 1.20): the integral of a monotone increasing sequence of nonnegative functions equals the limit of the integrals. This is exactly the interchange of limit and integral that Riemann could not perform; here it is licensed, because (1.5) was built to make it so.

Remark 1.17 (The forward reference to theorem 1.20 is not circular)**.** The argument just given invokes the monotone convergence theorem, which is stated and proved only in the next section, section 1.4. This forward dependence is deliberate and it is *benign*, not a hidden circle. The proof of theorem 1.20 rests on two things alone: the definition (1.5) of the integral as a supremum over simple minorants, and the continuity-from-below of the measure (proposition 1.6(iii)). It never uses additivity of the integral. So the logical order is sound – monotone convergence is proved first from the definition, and additivity then follows from it – even though, for narrative reasons, we want additivity in hand here, before we reach the convergence theorems. We keep the construction in this order and simply signpost the pointer: when reading theorem 1.20, note that its proof leans on nothing established in the present paragraph.

With additivity secured for nonnegative functions, the rest of the algebra unwinds mechanically. Splitting a signed $f = f^+ - f^-$ and applying nonnegative additivity extends $\int(f+g) = \int f + \int g$ to all of $L^1(\mu)$. The triangle inequality is then a one-line consequence of $|f| = f^+ + f^-$,

$$\Big|\int f\,\mathrm{d}\mu\Big| = \Big|\int f^+\,\mathrm{d}\mu - \int f^-\,\mathrm{d}\mu\Big| \leq \int f^+\,\mathrm{d}\mu + \int f^-\,\mathrm{d}\mu = \int |f|\,\mathrm{d}\mu,$$

and monotonicity is the observation that for $f \le g$ the difference $g - f \ge 0$ has $\int (g - f)\ d\mu \ge 0$, so by additivity $\int g\ d\mu - \int f\ d\mu \ge 0$. A fuller account of these manipulations is in [123, Ch. 2]. We have what the next section needs: a *linear, monotone* integral, additive on $L^1(\mu)$, with the triangle inequality. Before handing it on, we record two facts that fix the integral's relationship to the rest of the chapter – its agreement with the elementary integral, and its blindness to null sets.

Remark 1.18 (Agreement with Riemann; the role of null sets)**.** The Lebesgue integral does not overturn the elementary one; it *extends* it. If $f : [a, b] \to \mathbb{R}$ is Riemann integrable, then it is Lebesgue measurable and the two integrals coincide. The mechanism is the Darboux squeeze, run through the convergence theorems. Fix a sequence of partitions refining to mesh zero, and on each form the lower and upper Darboux *step functions* $\underline{\varphi}_n \le f \le \overline{\varphi}_n$ – constant on the partition cells, hence *simple* – with $\underline{\varphi}_n$ increasing and $\overline{\varphi}_n$ decreasing as the partition refines. By the very definition (1.4), the Lebesgue integrals $\int \underline{\varphi}_n$ dLeb and $\int \overline{\varphi}_n$ dLeb are exactly the lower and upper Darboux sums, and Riemann integrability is the statement that these share a common limit, the Riemann integral $\int_a^b f$. Everything on the bounded interval $[a, b]$ is dominated by the constant $\sup |f| < \infty$, so monotone (equivalently dominated) convergence applies to both monotone sequences: the limits $\underline{\varphi} = \lim_n \underline{\varphi}_n$ and $\overline{\varphi} = \lim_n \overline{\varphi}_n$ are measurable and

$$\int \underline{\varphi}\ \mathrm{dLeb} = \int \overline{\varphi}\ \mathrm{dLeb} = \int_a^b f.$$

Now the squeeze. The difference $\overline{\varphi} - \underline{\varphi} \ge 0$ is nonnegative with $\int (\overline{\varphi} - \underline{\varphi})$ dLeb $= 0$, and a nonnegative measurable function whose integral vanishes must vanish almost everywhere. (This little principle is used twice in the book, so here is its one-line proof: if $h \ge 0$ and $\int h\ d\mu = 0$, then for each k the simple minorant $\frac{1}{k}\mathbf{1}_{\{h \ge 1/k\}} \le h$ gives $\frac{1}{k}\mu(\{h \ge 1/k\}) \le \int h\ d\mu = 0$, so $\mu(\{h \ge 1/k\}) = 0$; and $\{h > 0\} = \bigcup_k \{h \ge 1/k\}$ is then a countable union of null sets, hence null by subadditivity, proposition 1.6(ii).) Applied here, $\overline{\varphi} = \underline{\varphi}$ Lebesgue-a.e., and since $\underline{\varphi} \le f \le \overline{\varphi}$ this forces $\underline{\varphi} = f = \overline{\varphi}$ a.e. Thus f agrees almost everywhere with the Borel function $\underline{\varphi}$, is itself measurable, and $\int f$ dLeb $= \int_a^b f$. The Lebesgue integral agrees with the elementary one wherever the latter is defined, and – crucially – remains defined on the monotone limits (example 1.1) where the latter breaks.

The squeeze just used reveals a structural feature that will organize the entire book. The integral *never sees a null set*: two functions equal μ-almost everywhere – agreeing outside a set of μ-measure zero – have the same integral, because their difference is supported on a null set and the integral of anything supported there is zero (it is dominated by constants times the indicator of a measure-zero set, whose integral (1.4) vanishes). This is why, throughout the book, identities between integrable functions are asserted "almost everywhere" rather than pointwise: the pointwise values on a null set are genuinely invisible to every integral we will ever form, so insisting on them would be insisting on information the theory discards by design. It is also the precise reason the L^p spaces of section 1.6 are spaces of *equivalence classes* – functions identified when they agree a.e. – rather than spaces of functions: the quotient by null sets is not a technical nuisance but the honest record of what the integral can resolve. The "a.e." convention and the L^p quotient both descend from this single remark.

We close the section by collecting the dividend the whole construction was built to pay: the identity that turns a population average into an integral. It is the simplest possible computation in the new machinery, and it is the chapter's destination in miniature.

Example 1.19 (Integration against a Dirac mass and the empirical average)**.** Take first a single point mass δ_a. For a simple $s = \sum_k c_k \mathbf{1}_{A_k}$, the definition (1.4) together with $\delta_a(A_k) = \mathbf{1}_{A_k}(a)$ gives $\int s \, \mathrm{d}\delta_a = \sum_k c_k \mathbf{1}_{A_k}(a) = s(a)$: integration against δ_a is just evaluation at a. For a nonnegative measurable f this passes to the supremum (1.5): every simple $s \le f$ has $s(a) \le f(a)$, while the dyadic minorants $s_n \le f$ achieve $s_n(a) \uparrow f(a)$, so the supremum of $\int s \, \mathrm{d}\delta_a = s(a)$ over simple $s \le f$ is exactly $f(a)$. Hence

$$\int_\Omega f \, \mathrm{d}\delta_a \;=\; f(a). \tag{1.7}$$

Only the value of f at the atom a survives – which is the rigorous content of the physicist's sifting property $\int f(x)\,\delta(x-a)\,\mathrm{d}x = f(a)$, now read as a statement about the Dirac *measure* rather than a fictional function. Summing (1.7) over the atoms of the empirical measure $\mu^N = \frac{1}{N}\sum_{i=1}^N \delta_{X_i}$ and using the linearity just proved, integration against μ^N is the population average:

$$\int_{\mathbb{R}^d} f \, \mathrm{d}\mu^N \;=\; \frac{1}{N}\sum_{i=1}^N \int_{\mathbb{R}^d} f \, \mathrm{d}\delta_{X_i} \;=\; \frac{1}{N}\sum_{i=1}^N f(X_i). \tag{1.8}$$

This identity is the bridge between the discrete and the continuum that the whole chapter is built to cross: the left side is an integral against a measure, the right side is the physicist's ensemble average over a finite population, and they are *equal*, not merely analogous. With it, every statement about the empirical average becomes a statement about an integral against μ^N, and conversely. The empirical-measure theorem of section 1.7 is precisely the assertion that, for f bounded continuous, the right side converges to $\int f \, \mathrm{d}m$ as $N \to \infty$ – that the population summary μ^N settles onto the law m. To prove it we will need to move the limit $N \to \infty$ across the integral sign, and that is the one permit the integral we have just built does not yet carry. Acquiring it – the monotone, Fatou, and dominated convergence theorems – is the work of the next section, which receives from us a linear, monotone integral, additive on $L^1(\mu)$, ready to be tested against limits, and the identity (1.8) whose limiting behavior is the goal.

1.4 Exchanging limits and integrals

The previous section closed by naming the one permit the freshly built integral does not yet carry. We have a linear, monotone integral $f \mapsto \int f \, \mathrm{d}\mu$ that agrees with the elementary integral wherever the latter exists and survives the monotone limits where it breaks, and we have the bridge identity (1.8) that turns the population average $\frac{1}{N}\sum_i f(X_i)$ into the single integral $\int f \, \mathrm{d}\mu^N$. What we do *not* yet have is the right to push a limit through the integral sign. This is exactly the third of the three questions raised in section 1.1 – *when may we exchange the limit $N \to \infty$ with the integral $\int f\,(\cdots)$?* – the one we flagged there as "sometimes simply false." The first two questions are now answered: m and μ^N are measures (section 1.2), and the integral against them is built (section 1.3). The third is the business of this section.

It is worth being precise about why this last permit is the hard one. The interchange $\lim_n \int f_n = \int \lim_n f_n$ is the formal content of every "take the thermodynamic limit inside the average" step a physicist writes, almost always silently. And section 1.1 already exhibited the canonical way it fails: the escaping-mass sequence $f_n = n\mathbf{1}_{(0,1/n)}$, for which $\int f_n = 1$ for every n while the pointwise limit is the zero function, so $\lim_n \int f_n = 1 \neq 0 = \int \lim_n f_n$. The mass did not vanish; it concentrated, and the pointwise limit threw it away. That example was left there as a warning with a promised resolution: there is a precise hypothesis whose presence makes the interchange legal and whose absence is the exact mechanism of the failure. We now redeem the promise. The resolution comes in three theorems – monotone convergence, Fatou's lemma, and dominated convergence – each supplying a *different* sufficient hypothesis, and each tailored to a different way the integrand can behave under the limit. Read together they form a complete answer: the escaping-mass sequence violates the one hypothesis (a fixed integrable envelope) that dominated convergence demands, and that is precisely why it escapes. By the end of the section the warning of section 1.1 will be a closed loop, and we will have, as a free corollary, the right to differentiate under the integral sign – a manipulation used silently in nearly every later chapter.

We continue to work on a fixed measure space $(\Omega, \mathcal{F}, \mu)$. The order of the three theorems is not arbitrary: monotone convergence is proved first and directly from the construction of the integral; Fatou's lemma falls out of it in one line; and dominated convergence is then deduced from Fatou by a cancellation trick. Each is the lever for the next.

The first theorem treats the most structured situation – an integrand that only *increases*. This is the case in which no mass can ever escape, because nothing is ever taken away: the functions climb monotonically toward their limit, and one expects the integrals to climb with them. It is also the case the integral was *built* to handle, since the very definition (1.5) of $\int f \,\mathrm{d}\mu$ as a supremum over simple minorants is itself a statement about an increasing approximation. Monotone convergence says that this internal monotone structure of the integral propagates to *any* monotone approximating sequence, not just the dyadic one used to define it.

Theorem 1.20 (Monotone convergence, MCT)**.** *Let $f_n \colon \Omega \to [0, \infty]$ be measurable with $f_n \uparrow f$ pointwise ($0 \le f_1 \le f_2 \le \cdots$ and $f_n(\omega) \to f(\omega)$ for every ω). Then f is measurable and*

$$\int_\Omega f_n \,\mathrm{d}\mu \;\uparrow\; \int_\Omega f \,\mathrm{d}\mu.$$

Proof. Measurability of the limit f is exactly the stability fact proved in proposition 1.13: a pointwise limit of measurable functions is measurable, so $\int f \,\mathrm{d}\mu$ is at least *defined*. We must show the integrals converge to it.

The easy inequality. The integral is monotone – $0 \le u \le v$ implies $\int u \le \int v$, immediate from (1.5) since every simple minorant of u is one of v. Applying this along the chain $f_1 \le f_2 \le \cdots \le f$ gives $\int f_1 \le \int f_2 \le \cdots \le \int f$. The left-hand side is therefore an increasing sequence in $[0, \infty]$, so it has a limit

$$L \;:=\; \lim_{n\to\infty} \int f_n \;=\; \sup_n \int f_n \;\in [0, \infty],$$

and every term, hence the limit, is bounded above by $\int f$: thus $L \le \int f$. The content of the theorem is the reverse inequality $L \ge \int f$, for then $L = \int f$ and the monotone convergence $\int f_n \uparrow \int f$ is exactly the assertion.

The reverse inequality – setup. By the definition (1.5) of $\int f \, \mathrm{d}\mu$ as the supremum of $\int s \, \mathrm{d}\mu$ over simple functions $0 \le s \le f$, it suffices to show $L \ge \int s \, \mathrm{d}\mu$ for *each* fixed such s; taking the supremum over s then yields $L \ge \int f$. So fix one simple $0 \le s \le f$. The obstacle is that we cannot compare f_n to s directly – it is entirely possible that $f_n < s$ on a large set for every finite n, since f_n only reaches up to s in the limit. The device that circumvents this is to relax the comparison by a multiplicative slack $c < 1$ and chase the sets where it holds. Fix a constant $c \in (0,1)$ and define the *capture sets*

$$E_n \ := \ \{\omega \in \Omega : f_n(\omega) \ge c\, s(\omega)\},$$

the region where f_n has already climbed to within a factor c of s. Each E_n is measurable (it is a sublevel-type set of the measurable function $f_n - cs$), and they increase: since $f_n \le f_{n+1}$, the inequality $f_n \ge cs$ forces $f_{n+1} \ge cs$, so $E_n \subseteq E_{n+1}$.

Why $E_n \uparrow \Omega$. This is the step the original proof compressed into a parenthesis, and it is the crux, so we give it in full by splitting on the value of s at a fixed point ω. There are two cases.

- If $s(\omega) = 0$, then $c\, s(\omega) = 0$, and since $f_n(\omega) \ge 0$ always, the defining inequality $f_n(\omega) \ge c\, s(\omega) = 0$ holds for *every* n; thus $\omega \in E_n$ for all n, in particular for $n = 1$.
- If $s(\omega) > 0$, then because $c < 1$ we have the *strict* inequality $c\, s(\omega) < s(\omega) \le f(\omega) = \lim_n f_n(\omega)$. The increasing sequence $f_n(\omega)$ converges to a value strictly above the threshold $c\, s(\omega)$, so it must eventually exceed that threshold: there is some N with $f_n(\omega) \ge c\, s(\omega)$ for all $n \ge N$, i.e. $\omega \in E_n$ for all large n.

In both cases ω belongs to E_n from some index on, so $\bigcup_n E_n = \Omega$. Together with $E_n \subseteq E_{n+1}$ this is precisely $E_n \uparrow \Omega$. The role of the slack $c < 1$ is visible here: it is what converts the possibly non-strict $s(\omega) \le f(\omega)$ into the strict gap $c\, s(\omega) < f(\omega)$ that lets the increasing sequence *overtake* the threshold in finite time. With $c = 1$ the argument would stall at points where $f(\omega) = s(\omega)$ is approached but never attained.

Cashing it in via continuity from below. On E_n we have $f_n \ge cs$ by definition, so restricting the integral to E_n and using monotonicity twice,

$$\int_\Omega f_n \, \mathrm{d}\mu \ \ge \ \int_{E_n} f_n \, \mathrm{d}\mu \ \ge \ \int_{E_n} c\, s \, \mathrm{d}\mu \ = \ c \int_{E_n} s \, \mathrm{d}\mu.$$

Now the simple function $s = \sum_k a_k \mathbf{1}_{A_k}$ turns the set map $A \mapsto \int_A s \, \mathrm{d}\mu = \sum_k a_k \, \mu(A \cap A_k)$ into a finite nonnegative combination of the measures $A \mapsto \mu(A \cap A_k)$, hence *itself a measure* ν_s. The increasing sweep $E_n \uparrow \Omega$ is exactly the hypothesis of continuity from below for ν_s (proposition 1.6), which therefore gives

$$\int_{E_n} s \, \mathrm{d}\mu \ = \ \nu_s(E_n) \ \uparrow \ \nu_s(\Omega) \ = \ \int_\Omega s \, \mathrm{d}\mu.$$

This is the precise point at which the continuity-under-limits property of measures, built back in proposition 1.6, is spent: the monotone build-up of the *sets* E_n drives the monotone build-up of the *integral* of s over them.

The two final passages. Combine the last two displays and let $n \to \infty$. The left side tends to L and the right side tends to $c \int_\Omega s \, \mathrm{d}\mu$, so

$$L \ \ge \ c \int_\Omega s \, \mathrm{d}\mu \qquad \text{for every } c \in (0,1).$$

The constant c was an artifact of the argument, not of the statement, so we now retire it: letting $c \uparrow 1$ – legitimate because $\int_\Omega s \, \mathrm{d}\mu$ does not depend on c – gives $L \geq \int_\Omega s \, \mathrm{d}\mu$. Finally, this holds for every simple $0 \leq s \leq f$, and the supremum over such s is by definition $\int_\Omega f \, \mathrm{d}\mu$; hence $L \geq \int f$. With the easy inequality this closes the proof. □

A word on what was really used, since the same skeleton recurs. The only substantive ingredient was *continuity from below* of the measure ν_s, and that in turn was the one-axiom payoff of countable additivity isolated in proposition 1.6. Monotone convergence is thus not a new analytic miracle; it is countable additivity, viewed through the integral. In its native habitat MCT is the statement that the defining approximation of the integral is robust: applied to the dyadic minorants $s_n \uparrow f$ of (1.5) it merely returns $\int s_n \uparrow \int f$, the definition itself, but applied to *any* increasing sequence reaching f it says one may use whichever monotone approximation is convenient and the integral will follow it up. We will exploit exactly this freedom – replace an awkward sequence by a convenient monotone one with the same limit – repeatedly, beginning in the very next section.

Example 1.21 (MCT recovers the rationals computation)**.** The monotone limit that broke Riemann's theory in example 1.1 is now trivial. There $f_n = \mathbf{1}_{\{q_1,\ldots,q_n\}} \uparrow \mathbf{1}_{\mathbb{Q}\cap[0,1]} = f$, each f_n is simple and nonnegative with $\int f_n \, \mathrm{dLeb} = 0$ (a finite set has Lebesgue measure 0), and the sequence increases to f. MCT applies verbatim and gives $\int f \, \mathrm{dLeb} = \lim_n \int f_n \, \mathrm{dLeb} = 0$. The symbol that "denoted nothing" in Riemann's theory now denotes the number 0, and the interchange $\lim_n \int f_n = \int \lim_n f_n$ that could not even be *written down* there is a special case of MCT here. This is the repair promised in section 1.1 made literal.

Monotone convergence assumed the integrand climbs. Most sequences do not; they oscillate, spike, or wander. For those we cannot expect an equality, but we can ask for the safe *one-sided* statement that survives in full generality: whatever the sequence does, the integral of its eventual floor cannot exceed the eventual size of its integrals. "Mass can be lost in the limit but never spuriously gained" is the slogan, and the precise form is Fatou's lemma. It is the workhorse-of-the-workhorse: it costs no hypothesis at all beyond nonnegativity, and the dominated convergence theorem will be wrung out of it.

The right object to compare against is the *liminf* of the sequence, $\liminf_n f_n = \sup_n \inf_{k\geq n} f_k$ – the largest function the f_n are eventually above. Its appeal is that the inner infima $g_n = \inf_{k\geq n} f_k$ *increase* in n (an infimum over a shrinking index set can only grow), which hands us exactly the monotone structure MCT consumes. That single observation is the whole proof.

Lemma 1.22 (Fatou)**.** *For any measurable $f_n \colon \Omega \to [0,\infty]$,*

$$\int_\Omega \liminf_{n\to\infty} f_n \, \mathrm{d}\mu \;\leq\; \liminf_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mu.$$

Proof. Set $g_n = \inf_{k\geq n} f_k$. Three facts hold by construction. First, g_n is measurable, as a countable infimum of measurable functions (proposition 1.13). Second, the g_n increase and converge upward to the liminf: $g_n \uparrow \sup_n g_n = \liminf_n f_n$ by the very definition of lim inf. Third, g_n is a lower bound for every tail term: $g_n = \inf_{k\geq n} f_k \leq f_k$ for each $k \geq n$, so integrating and taking the infimum over those k, $\int g_n \, \mathrm{d}\mu \le \inf_{k\ge n} \int f_k \, \mathrm{d}\mu$. Now apply monotone convergence (theorem 1.20) to the increasing sequence $g_n \uparrow \liminf_n f_n$ and let $n \to \infty$:

$$\int_\Omega \liminf_n f_n \, \mathrm{d}\mu \;=\; \lim_n \int_\Omega g_n \, \mathrm{d}\mu \;\le\; \lim_n \inf_{k\ge n} \int_\Omega f_k \, \mathrm{d}\mu \;=\; \liminf_n \int_\Omega f_n \, \mathrm{d}\mu,$$

the last equality being the definition of the liminf of the number sequence $\left(\int f_n\right)_n$. This is the claim. $\square$

The inequality is genuinely one-directional, and seeing *why* it can be strict is what finally closes the loop opened in section 1.1. Return to the escaping-mass sequence $f_n = n\mathbf{1}_{(0,1/n)}$ on $[0,1]$ with Lebesgue measure – the rectangle of height n and width $1/n$ pinned to the origin. We computed there that $\int_0^1 f_n \, \mathrm{d}x = 1$ for every n, so the right-hand side of Fatou is $\liminf_n \int f_n = \lim_n 1 = 1$. On the other side, every fixed point $x \in [0,1]$ has $f_n(x) = 0$ for all $n > 1/x$ (and $f_n(0) = 0$ always), so the pointwise liminf is the zero function, and the left-hand side is $\int_0^1 \liminf_n f_n \, \mathrm{d}x = \int_0^1 0 \, \mathrm{d}x = 0$. Fatou's lemma here reads

$$0 \;=\; \int_\Omega \liminf_n f_n \, \mathrm{d}\mu \;<\; \liminf_n \int_\Omega f_n \, \mathrm{d}\mu \;=\; 1,$$

a *strict* inequality. The lemma holds – $0 \le 1$ – but with room to spare, and the gap is precisely the unit of mass that concentrated at the origin and was discarded by the pointwise limit. This is the same escaping mass diagnosed in section 1.1, now read through the integral: Fatou correctly refuses to certify that the lost mass reappears, which is exactly the safe one-sided guarantee it advertises. The example also tells us what is missing for *equality*. Nothing prevented the mass from sliding off to a point because nothing held the functions down: the smallest function dominating all the f_n is $\sup_n f_n(x) \sim 1/x$ near the origin, which is not integrable. To upgrade Fatou's inequality to an equality we must forbid exactly this – we must supply a single integrable function that caps every f_n from above. That hypothesis is the entire content of the dominated convergence theorem, the workhorse of the entire book, and the failure we have just witnessed is the precise failure its hypothesis rules out.

Theorem 1.23 (Dominated convergence, DCT)**.** *Let $f_n \colon \Omega \to \mathbb{R}$ (or $\mathbb{R}^d$) be measurable with $f_n \to f$ pointwise μ-almost everywhere. Suppose there is an integrable* dominating function $g \in L^1(\mu)$*, $g \ge 0$, with $|f_n| \le g$ μ-a.e. for every n. Then $f \in L^1(\mu)$,*

$$\lim_{n\to\infty} \int_\Omega |f_n - f| \, \mathrm{d}\mu = 0, \qquad \textit{and in particular} \qquad \lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mu = \int_\Omega f \, \mathrm{d}\mu.$$

Proof. Reduction to a clean pointwise setting. The hypotheses are stated "μ-a.e.": the convergence $f_n \to f$ may fail on a null set, and each bound $|f_n| \le g$ may fail on a null set. There are countably many such exceptional sets – one for the convergence, one for each n – and a countable union of null sets is null (countable subadditivity, proposition 1.6). Off that single null set N_0 every hypothesis holds at every point. Since the integral never sees a null set (the a.e. principle of section 1.3), we may freely redefine all functions to be 0 on N_0 without changing any integral; after this harmless surgery $f_n \to f$ *everywhere* and $|f_n| \le g$ *everywhere*. For the vector-valued case ($f_n \colon \Omega \to \mathbb{R}^d$) we argue one coordinate at a time: each component $f_n^{(j)}$ converges to $f^{(j)}$ and is bounded by $|f_n| \le g$, so it falls under the scalar statement, and finite linear combinations of the coordinate conclusions give the vector conclusion. So assume f_n, f scalar with the bounds holding pointwise.

The limit is integrable. Letting $n \to \infty$ in $|f_n| \le g$ and using $f_n \to f$ gives $|f| = \lim_n |f_n| \le g$ pointwise. Hence $\int |f|\, \mathrm{d}\mu \le \int g\, \mathrm{d}\mu < \infty$, so $f \in L^1(\mu)$ and the symbol $\int f\, \mathrm{d}\mu$ is legitimate.

The Fatou trick. The quantity we want to drive to zero is $\int |f_n - f|\, \mathrm{d}\mu$. The obstruction is that $|f_n - f|$ is not monotone, so MCT is unavailable; but it *is* nonnegative and uniformly capped, since

$$|f_n - f| \;\le\; |f_n| + |f| \;\le\; g + g \;=\; 2g.$$

This is the cue to apply Fatou to the nonnegative functions $2g - |f_n - f| \ge 0$, which are legitimate integrands and converge pointwise to $2g$ (because $|f_n - f| \to 0$). Fatou (lemma 1.22) gives

$$\int 2g\, \mathrm{d}\mu \;=\; \int \liminf_n \left(2g - |f_n - f|\right) \mathrm{d}\mu \;\le\; \liminf_n \int \left(2g - |f_n - f|\right) \mathrm{d}\mu.$$

Now $2g$ is a *fixed* function, so $\int 2g\, \mathrm{d}\mu$ pulls out of the liminf as a constant, and a $\liminf$ of a difference with a fixed term is that term minus the corresponding $\limsup$:

$$\liminf_n \Big(\int 2g\, \mathrm{d}\mu - \int |f_n - f|\, \mathrm{d}\mu \Big) \;=\; \int 2g\, \mathrm{d}\mu - \limsup_n \int |f_n - f|\, \mathrm{d}\mu.$$

Chaining the two displays, $\int 2g\, \mathrm{d}\mu \le \int 2g\, \mathrm{d}\mu - \limsup_n \int |f_n - f|\, \mathrm{d}\mu$.

The cancellation – and where integrability of g is spent. Here is the one place the hypothesis $g \in L^1(\mu)$ does irreplaceable work. Because $\int 2g\, \mathrm{d}\mu < \infty$, it is a *finite* number and may be subtracted from both sides of the last inequality without producing the indeterminate $\infty - \infty$. Cancelling it leaves

$$0 \;\le\; -\limsup_n \int |f_n - f|\, \mathrm{d}\mu, \qquad \text{i.e.} \qquad \limsup_n \int |f_n - f|\, \mathrm{d}\mu \;\le\; 0.$$

But $\int |f_n - f|\, \mathrm{d}\mu \ge 0$ for every n, so its $\limsup$ is also ≥ 0; squeezed between 0 and itself, the sequence has limit 0: $\lim_n \int |f_n - f|\, \mathrm{d}\mu = 0$. Had g failed to be integrable, $\int 2g\, \mathrm{d}\mu = \infty$, the cancellation would have read $\infty \le \infty - (\cdots)$ and concluded nothing – which is exactly the loophole through which the escaping-mass sequence (whose minimal envelope $\sup_n f_n \sim 1/x$ is not integrable) drives its mass away. The finiteness of the envelope is the wall that holds the mass in, precisely as foretold in section 1.1.

From L^1 convergence to the interchange. Finally, the convergence of the integrals is a one-line consequence of the L^1 convergence just established, by linearity and the triangle inequality for the integral:

$$\Big| \int_\Omega f_n\, \mathrm{d}\mu - \int_\Omega f\, \mathrm{d}\mu \Big| \;=\; \Big| \int_\Omega (f_n - f)\, \mathrm{d}\mu \Big| \;\le\; \int_\Omega |f_n - f|\, \mathrm{d}\mu \;\xrightarrow[n\to\infty]{}\; 0,$$

so $\int f_n\, \mathrm{d}\mu \to \int f\, \mathrm{d}\mu$. This is the permit we set out to obtain. □

We can now collect the payoff of the entire section 1.1 motivation in a single picture. That section turned the physicist's reflex into three demands and warned that the third – exchanging a limit with an integral – is sometimes false; the three theorems just proved are the precise resolution, and it is worth seeing them side by side as three different licenses for the *same* maneuver, each matched to a different physical regime.

Heuristic (non-rigorous)

Every thermodynamic-limit manipulation of the form "$\lim_{N\to\infty} \frac{1}{N}\sum_i \varphi(X_i^{(N)}) = \int \varphi \, \mathrm{d}\big(\lim_N \mu^N\big)$" is an interchange of a limit with an integral, and the three theorems are its three permits, distinguished by what they assume about the integrand. *Monotone convergence* is the permit when the integrand only *increases* – a free energy or partition function assembled by adding modes one at a time, a sublevel content swept upward – and it gives an exact equality with no envelope required, because a monotone build-up can lose nothing. *Fatou* asks for nothing at all beyond nonnegativity and pays for that generality with a mere one-sided inequality; it is the right tool exactly when one needs only that mass is *not spuriously gained* in the limit (a lower bound on a limiting energy, say), and it is honest about the possibility that mass is *lost*, which is what the strict escaping-mass case records. *Dominated convergence* is the generic workhorse: once an a priori bound furnishes a single integrable envelope g with $|f_n| \le g$, fluctuations of either sign are uniformly controlled and the interchange holds as an equality. The order MCT $\to$ Fatou $\to$ DCT is also the order of increasing applicability and decreasing structure – from monotone, to nonnegative, to merely dominated. When *none* applies the interchange is genuinely false, and the escaping-mass example $f_n = n\mathbf{1}_{(0,1/n)}$ of section 1.1 is the canonical warning of how: its minimal envelope $\sup_n f_n \sim 1/x$ is not integrable, the DCT hypothesis fails, and the unit of mass concentrates at a point that the pointwise limit cannot see. The hypothesis of DCT is thus not fine print – it is the wall that holds the mass in. In the empirical-measure theorem of section 1.7 we will sit squarely in the tamest corner of this picture: the observable φ is bounded and the measures are probability measures (total mass $1 < \infty$), so the constant envelope $g \equiv \|\varphi\|_\infty$ is integrable and DCT applies in its simplest guise. This special case – a uniformly bounded sequence on a finite measure space – is so common it is named separately, the *bounded convergence theorem* (BCT), and it is the engine of the chapter's closing computation.

One immediate dividend of dominated convergence deserves to be recorded as a named tool, because the rest of the book uses it constantly and almost always without comment: differentiation under the integral sign. Whenever we write $\partial_t \int \cdots \mathrm{d}\mu = \int \partial_t(\cdots)\,\mathrm{d}\mu$ – differentiating a Riccati functional in time, varying a value function with respect to a parameter, computing the linear response of a free energy – we are silently invoking the corollary below. It is nothing but DCT applied to difference quotients, with the mean value theorem manufacturing the dominating function.

Corollary 1.24 (Differentiation under the integral)**.** *Let $(t,\omega) \mapsto f(t,\omega)$ with t in an open interval $I \subseteq \mathbb{R}$ satisfy: $f(t,\cdot) \in L^1(\mu)$ for each t; the partial derivative $\partial_t f(t,\omega)$ exists for μ-a.e. ω and all $t \in I$; and $|\partial_t f(t,\omega)| \le g(\omega)$ for some $g \in L^1(\mu)$ independent of t. Then $t \mapsto \int f(t,\cdot)\,\mathrm{d}\mu$ is differentiable on I with $\frac{\mathrm{d}}{\mathrm{d}t}\int f(t,\cdot)\,\mathrm{d}\mu = \int \partial_t f(t,\cdot)\,\mathrm{d}\mu$.*

Proof. Fix $t \in I$ and any sequence $h_n \to 0$ with $h_n \neq 0$ and $t + h_n \in I$; it suffices to show the difference quotients of $t \mapsto \int f(t,\cdot)\,\mathrm{d}\mu$ converge to $\int \partial_t f(t,\cdot)\,\mathrm{d}\mu$ along every such sequence, since that pins the derivative. Put

$$q_n(\omega) \;:=\; \frac{f(t+h_n,\omega) - f(t,\omega)}{h_n}.$$

Two things hold for μ-a.e. ω. First, $q_n(\omega) \to \partial_t f(t,\omega)$ as $n \to \infty$, by the definition of the partial derivative, which exists a.e. by hypothesis. Second, each q_n is dominated by the fixed integrable g: the mean value theorem applied to the differentiable function $\tau \mapsto f(\tau,\omega)$ on the interval between t and $t+h_n$ produces an intermediate point $\xi_n(\omega)$ with $q_n(\omega) = \partial_t f(\xi_n(\omega),\omega)$, and the uniform bound $|\partial_t f(\tau,\omega)| \le g(\omega)$ gives $|q_n(\omega)| \le g(\omega)$ for every n. The hypotheses of DCT (theorem 1.23) are met – a.e. pointwise convergence and a single integrable envelope – so the integrals pass to the limit:

$$\lim_{n\to\infty} \frac{\int f(t+h_n,\cdot)\,\mathrm{d}\mu - \int f(t,\cdot)\,\mathrm{d}\mu}{h_n} \;=\; \lim_{n\to\infty} \int_\Omega q_n\,\mathrm{d}\mu \;=\; \int_\Omega \partial_t f(t,\cdot)\,\mathrm{d}\mu,$$

where the first equality is the linearity of the integral pulling the constant $1/h_n$ inside. As the sequence $h_n \to 0$ was arbitrary, the difference quotient of $t \mapsto \int f(t,\cdot)\,\mathrm{d}\mu$ converges to the displayed value, which is therefore its derivative. □

The third question of section 1.1 is now fully answered. The three convergence theorems are the exact licenses for "take the limit inside the integral," the escaping-mass warning has hardened into the failure of a checkable hypothesis, and we have, for free, the right to differentiate under the integral. What we have really installed, though, is a reusable *architecture*, and it is worth naming because it recurs almost everywhere ahead: *monotone control, then dominated passage.* One first builds or bounds an object by a monotone approximation – where MCT and Fatou apply without any envelope – and then, once a uniform integrable bound is in hand, upgrades the one-sided estimate to an equality by DCT. This two-step pattern is the skeleton of nearly every limit argument in the book.

The very next section is the first to lean on it. To define the product measure $\mu_1 \otimes \mu_2$ and prove the Fubini–Tonelli theorem we will verify countable additivity of a candidate set function by a *monotone* passage (continuity from below, hence MCT), and Tonelli's theorem is exactly the statement that for nonnegative integrands the two iterated integrals agree – proved by the monotone-class machinery and MCT, with no envelope – after which Fubini's theorem handles signed integrands by the dominated upgrade, splitting into positive and negative parts controlled by an integrable bound. The same architecture returns in section 1.6: the completeness of L^p (Riesz–Fischer) is proved by first extracting a *monotone* control on a rapidly convergent subsequence (an increasing sum of norms, handled by MCT) and then passing the limit through with a dominating function (DCT). The density of smooth compactly supported functions C_c^∞ in L^p, also established there, leans on corollary 1.24 to differentiate a mollified integrand under the integral sign. And the bounded strong law that powers the empirical-measure theorem of section 1.7 is the same DCT, specialized to the bounded regime flagged in the heuristic above. We hand the next section a working set of interchange permits and the monotone-control-then-dominated-passage pattern that will organize its proofs.

1.5 Product measures and the Fubini–Tonelli theorem

The previous section built the single-variable integral and equipped it with the three permits – monotone, Fatou, dominated – for moving a limit through it, and it named the recurring rhythm of the subject: *monotone control, then dominated passage.* Every object we have integrated so far, though, has lived over a single space and been written against a single variable. The book is full of objects that do not. An expectation that factors over independent sources – the state of an agent and the Brownian noise driving it, say – is naturally an integral over a *pair* of variables, one for each source; and the graphon coupling that organizes Parts III–IV is the double integral of a kernel $W(x, y)$ against two copies of the population at once. To compute either we must integrate over two variables in turn, and we must know that the answer does not depend on which variable we integrate first – that the iterated integrals agree, and agree with the single integral over the product. That is the content of the Fubini–Tonelli theorem, and the business of this section is to build the product measure it lives on and then prove it. The section generalizes the one-variable integral to two variables exactly as far as those two applications demand and no further.

The construction reuses, rather than reinvents, the two engines already forged in this chapter. The monotone convergence theorem from section 1.4 is what lets us pass from indicators to general nonnegative integrands, in the same "monotone control" pattern just named; and the π–λ uniqueness machinery from section 1.2.3 – forged there precisely so that we would not have to reprove it here – is what pins the product measure down from its values on rectangles and, in one decisive lemma, certifies a measurability we could not otherwise reach. This is the third and last time in the chapter that the Dynkin engine is loaded, and it is the place the brief promised it would be indispensable.

The product space and its sections. Let $(\Omega_1, \mathcal{F}_1, \mu_1)$ and $(\Omega_2, \mathcal{F}_2, \mu_2)$ be σ-finite measure spaces. We want to measure subsets of the Cartesian product $\Omega_1 \times \Omega_2$, and the sets whose size is beyond dispute are the *measurable rectangles* $A_1 \times A_2$ with $A_1 \in \mathcal{F}_1$ and $A_2 \in \mathcal{F}_2$: a rectangle ought to have area $\mu_1(A_1)\, \mu_2(A_2)$, length times width, just as in elementary geometry. But the rectangles do not themselves form a σ-algebra – the complement of a rectangle is an L-shaped region, not a rectangle – so we take the smallest σ-algebra that contains them. The *product σ-algebra* $\mathcal{F}_1 \otimes \mathcal{F}_2$ on $\Omega_1 \times \Omega_2$ is $\sigma(\{A_1 \times A_2\})$, generated by the measurable rectangles in the sense of remark 1.3. It is the smallest family of subsets of the product that contains every rectangle and is closed under the countable operations; equivalently, on $\mathbb{R}^d \times \mathbb{R}^{d'}$ one checks (by the now-familiar "each family lies in the σ-algebra of the other" argument of definition 1.4) that it coincides with the Borel σ-algebra $\mathcal{B}(\mathbb{R}^{d+d'})$ of the product, so nothing exotic is being introduced.

The one piece of notation the whole section turns on is the *section*. For a set $E \subseteq \Omega_1 \times \Omega_2$ and a fixed first coordinate $\omega_1 \in \Omega_1$, the ω_1-*section* of E is

$$E_{\omega_1} \;=\; \{\omega_2 \in \Omega_2 :\; (\omega_1, \omega_2) \in E\} \;\subseteq\; \Omega_2,$$

the slice of E sitting above ω_1, viewed as a subset of the second factor alone. Geometrically, if one draws E in the plane $\Omega_1 \times \Omega_2$ and fixes a vertical line at horizontal position ω_1, then E_{ω_1} is the set of heights at which that line meets E. Sections are the device that turns a two-variable integral into a one-variable one: integrating a function over E "column by column" means, for each fixed ω_1, integrating over the section E_{ω_1} and then integrating the result over ω_1. For this two-step recipe even to be *writable*, two things must hold: each section must be a measurable subset of Ω_2 (so that $\mu_2(E_{\omega_1})$ makes sense), and the resulting function $\omega_1 \mapsto \mu_2(E_{\omega_1})$ – the "height of the column above ω_1" – must itself be $\mathcal{F}_1$-measurable (so that it can be integrated against μ_1). Neither is obvious for a general $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$, which may be a wildly complicated set built from rectangles by countably many operations. The whole theory of product measures hinges on this one lemma, and it is the single place in the construction where a π–λ argument is genuinely indispensable; we isolate it and prove it in full.

Lemma 1.25 (Measurability of sections)**.** *Let μ_2 be σ-finite. For every $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$ each section E_{ω_1} is in $\mathcal{F}_2$, and the map $\omega_1 \mapsto \mu_2(E_{\omega_1})$ is $\mathcal{F}_1$-measurable.*

Proof. Each section is measurable. This first claim costs nothing and uses no π–λ machinery; it is a pure closure argument. Let $\mathcal{S} = \{E \subseteq \Omega_1 \times \Omega_2 : \ E_{\omega_1} \in \mathcal{F}_2 \text{ for every } \omega_1 \in \Omega_1\}$ be the family of sets all of whose sections are measurable. It contains every rectangle, because the ω_1-section of $A_1 \times A_2$ is simply A_2 when $\omega_1 \in A_1$ and $\emptyset$ otherwise – in either case a member of $\mathcal{F}_2$. And it is a σ-algebra, because sectioning commutes with all the set operations: taking the section at a fixed ω_1 is the preimage under the inclusion $\omega_2 \mapsto (\omega_1, \omega_2)$, so $(E^c)_{\omega_1} = (E_{\omega_1})^c$ and $\left(\bigcup_n E^{(n)}\right)_{\omega_1} = \bigcup_n (E^{(n)})_{\omega_1}$, whence membership in $\mathcal{S}$ is preserved under complements and countable unions, and $\Omega_1 \times \Omega_2 \in \mathcal{S}$ trivially. Thus $\mathcal{S}$ is a σ-algebra containing the rectangles, so it contains the σ-algebra they generate, $\mathcal{F}_1 \otimes \mathcal{F}_2 \subseteq \mathcal{S}$. Every $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$ therefore has all sections in $\mathcal{F}_2$, and in particular $\mu_2(E_{\omega_1})$ is a well-defined number in $[0, \infty]$ for each ω_1.

The column function is measurable: the finite case. The second claim – that $\omega_1 \mapsto \mu_2(E_{\omega_1})$ is $\mathcal{F}_1$-measurable – is the substance, and here a bare closure argument no longer suffices, because measurability of this function is *not* preserved under complements without some finiteness to license a subtraction. This is exactly the situation the π–λ theorem was built for: the property we want is stable under the weaker λ-system operations but not under all σ-algebra operations. Assume first that μ_2 is *finite*, $\mu_2(\Omega_2) < \infty$, and collect the good sets,

$$\mathcal{D} = \{E \in \mathcal{F}_1 \otimes \mathcal{F}_2 : \ \omega_1 \mapsto \mu_2(E_{\omega_1}) \text{ is } \mathcal{F}_1\text{-measurable}\}.$$

The strategy is the standard one of section 1.2.3: exhibit a π-system inside $\mathcal{D}$ that generates the whole σ-algebra, show $\mathcal{D}$ is a λ-system, and let Dynkin's theorem do the rest.

The generating π-system is the rectangles $\mathcal{P} = \{A_1 \times A_2\}$, which form a π-system because the intersection of two rectangles is a rectangle, $(A_1 \times A_2) \cap (A_1' \times A_2') = (A_1 \cap A_1') \times (A_2 \cap A_2')$, and which generate $\mathcal{F}_1 \otimes \mathcal{F}_2$ by definition. They lie in $\mathcal{D}$: for $E = A_1 \times A_2$ the section computation above gives $E_{\omega_1} = A_2$ on A_1 and $\emptyset$ off it, so

$$\mu_2(E_{\omega_1}) \ = \ \mathbf{1}_{A_1}(\omega_1)\, \mu_2(A_2),$$

which is $\mu_2(A_2)$ times the indicator of a measurable set, hence an $\mathcal{F}_1$-measurable function of ω_1. So $\mathcal{P} \subseteq \mathcal{D}$.

Now verify that $\mathcal{D}$ is a λ-system in the sense of definition 1.8, and notice as we go that this is precisely where the finiteness of μ_2 is spent. (i) $\Omega_1 \times \Omega_2 \in \mathcal{D}$: its section is all of Ω_2, so $\mu_2(E_{\omega_1}) \equiv \mu_2(\Omega_2)$ is a (finite) constant, trivially measurable. (ii) Closure under proper differences: if $E \subseteq F$ both lie in $\mathcal{D}$, then $E_{\omega_1} \subseteq F_{\omega_1}$ for every ω_1, and the section of the difference is the difference of the sections, $(F \setminus E)_{\omega_1} = F_{\omega_1} \setminus E_{\omega_1}$, so that

$$\mu_2((F \setminus E)_{\omega_1}) \ = \ \mu_2(F_{\omega_1}) - \mu_2(E_{\omega_1}).$$

This is a difference of two measurable functions of ω_1, hence measurable – but the *subtraction itself is only meaningful because μ_2 is finite*: with $\mu_2(\Omega_2) < \infty$ every term $\mu_2(E_{\omega_1}) \le \mu_2(\Omega_2)$ is a finite real number, and $\mu_2(F_{\omega_1}) - \mu_2(E_{\omega_1})$ is an honest difference of reals rather than the indeterminate $\infty - \infty$. Strip away finiteness and this clause collapses; it is the load-bearing place. (iii) Closure under increasing unions: if $E_n \uparrow E$ in $\mathcal{D}$, then sectioning preserves the monotone union, $(E_n)_{\omega_1} \uparrow E_{\omega_1}$, and continuity from below of μ_2 (proposition 1.6(iii)) gives

$$\mu_2(E_{\omega_1}) \;=\; \lim_{n\to\infty} \mu_2\big((E_n)_{\omega_1}\big) \qquad \text{for every } \omega_1,$$

exhibiting $\omega_1 \mapsto \mu_2(E_{\omega_1})$ as the pointwise limit of the measurable functions $\omega_1 \mapsto \mu_2((E_n)_{\omega_1})$, hence measurable by proposition 1.13. So all three λ-system clauses hold. By Dynkin's theorem (theorem 1.9), the λ-system $\mathcal{D}$ contains the σ-algebra generated by the π-system $\mathcal{P}$, that is, $\mathcal{D} \supseteq \mathcal{F}_1 \otimes \mathcal{F}_2$; and since $\mathcal{D} \subseteq \mathcal{F}_1 \otimes \mathcal{F}_2$ by definition, $\mathcal{D} = \mathcal{F}_1 \otimes \mathcal{F}_2$. Every E in the product σ-algebra has measurable column function. This is the indispensable π–λ argument the section was built around.

Removing finiteness: the σ-finite extension. It remains to drop "μ_2 finite" to the standing hypothesis "μ_2 σ-finite," and this is a routine exhaustion. Write $\Omega_2 = \bigcup_m B_m$ with $B_m \uparrow \Omega_2$ and each $\mu_2(B_m) < \infty$ – possible exactly because μ_2 is σ-finite – and define the finite measures $\mu_2^{(m)}(\cdot) = \mu_2(\cdot \cap B_m)$. The finite case just proved, applied to each $\mu_2^{(m)}$, shows $\omega_1 \mapsto \mu_2^{(m)}(E_{\omega_1}) = \mu_2(E_{\omega_1} \cap B_m)$ is measurable for every $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$ and every m. As $B_m \uparrow \Omega_2$ we have $E_{\omega_1} \cap B_m \uparrow E_{\omega_1}$, so continuity from below gives $\mu_2^{(m)}(E_{\omega_1}) \uparrow \mu_2(E_{\omega_1})$ pointwise in ω_1; the limit of measurable functions is measurable (proposition 1.13), so $\omega_1 \mapsto \mu_2(E_{\omega_1})$ is measurable. This is again the monotone-passage pattern of section 1.4, now run over the exhausting sets rather than over an integrand. □

With the lemma in hand the product measure builds itself: the formula that *should* define the area of E – integrate the column heights – is now a legal expression, and the only work left is to check it is a measure and that it is the only one with the right values on rectangles. That uniqueness is again the Dynkin engine, invoked through the corollary corollary 1.10 that we extracted in section 1.2.3 for exactly this purpose.

Theorem 1.26 (Product measure)**.** *There is a unique σ-finite measure $\mu_1 \otimes \mu_2$ on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$ with $(\mu_1 \otimes \mu_2)(A_1 \times A_2) = \mu_1(A_1)\, \mu_2(A_2)$ for all $A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2$.*

Proof. Existence via the section-integral formula. Lemma 1.25 guarantees that for each $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$ the column function $\omega_1 \mapsto \mu_2(E_{\omega_1})$ is a nonnegative $\mathcal{F}_1$-measurable function, so its integral against μ_1 is a well-defined element of $[0, \infty]$, and we may *define*

$$(\mu_1 \otimes \mu_2)(E) \;:=\; \int_{\Omega_1} \mu_2(E_{\omega_1})\, \mu_1(\mathrm{d}\omega_1), \qquad E \in \mathcal{F}_1 \otimes \mathcal{F}_2.$$

This is the "area = integral of column heights" prescription made rigorous; the lemma is exactly what was needed to license it. We must check it is a measure. It vanishes on the empty set, since $\emptyset_{\omega_1} = \emptyset$ and $\mu_2(\emptyset) = 0$. For countable additivity, let $E^{(1)}, E^{(2)}, \ldots$ be pairwise disjoint members of $\mathcal{F}_1 \otimes \mathcal{F}_2$. For each fixed ω_1 the sections $E^{(j)}_{\omega_1}$ are pairwise disjoint subsets of Ω_2 (a point (ω_1, ω_2) lies in at most one $E^{(j)}$, so ω_2 lies in at most one section), and $\big(\bigsqcup_j E^{(j)}\big)_{\omega_1} = \bigsqcup_j E^{(j)}_{\omega_1}$, so the countable additivity of μ_2 itself gives, pointwise in ω_1,

$$\mu_2\Big(\big(\textstyle\bigsqcup_j E^{(j)}\big)_{\omega_1}\Big) \;=\; \textstyle\sum_{j\geq 1} \mu_2(E^{(j)}_{\omega_1}).$$

Integrating this identity against μ_1 and pulling the sum out through the integral is the one analytic step, and it is precisely a monotone-passage: the partial sums $\sum_{j\leq J} \mu_2(E^{(j)}_{\omega_1})$ are nonnegative and increase to the full sum as $J \to \infty$, so the monotone convergence theorem (theorem 1.20) permits the interchange,

$$\int_{\Omega_1} \sum_{j\geq 1} \mu_2(E^{(j)}_{\omega_1})\ \mathrm{d}\mu_1 \;=\; \sum_{j\geq 1} \int_{\Omega_1} \mu_2(E^{(j)}_{\omega_1})\ \mathrm{d}\mu_1 \;=\; \sum_{j\geq 1} (\mu_1 \otimes \mu_2)(E^{(j)}).$$

The left side is $(\mu_1 \otimes \mu_2)\big(\bigsqcup_j E^{(j)}\big)$, so the prescription is countably additive: it is a measure. On a rectangle $E = A_1 \times A_2$ we computed $\mu_2(E_{\omega_1}) = \mathbf{1}_{A_1}(\omega_1)\mu_2(A_2)$, whose μ_1-integral is $\mu_1(A_1)\mu_2(A_2)$, the required value. Finally σ-finiteness transfers from the factors: if $\Omega_1 = \bigcup_m \Omega_1^{(m)}$ and $\Omega_2 = \bigcup_m \Omega_2^{(m)}$ with finite-measure pieces, the products $\Omega_1^{(m)} \times \Omega_2^{(m)}$ increase to $\Omega_1 \times \Omega_2$ and each has product measure $\mu_1(\Omega_1^{(m)})\mu_2(\Omega_2^{(m)}) < \infty$, exhausting the product space by finite-mass sets.

Uniqueness via corollary 1.10. Suppose ν is any σ-finite measure on $\mathcal{F}_1 \otimes \mathcal{F}_2$ agreeing with $\mu_1 \otimes \mu_2$ on rectangles. The rectangles are a π-system generating $\mathcal{F}_1 \otimes \mathcal{F}_2$, and the exhausting products $\Omega_1^{(m)} \times \Omega_2^{(m)}$ are rectangles of finite measure increasing to the whole space. These are exactly the hypotheses of the uniqueness corollary corollary 1.10: two measures that agree on a generating π-system and on an exhausting sequence of finite-measure sets in it agree on the entire generated σ-algebra. Hence $\nu = \mu_1 \otimes \mu_2$ on $\mathcal{F}_1 \otimes \mathcal{F}_2$. This is the same propagation principle that fixed Lebesgue measure from the boxes; here it fixes the product from the rectangles. A fuller account is in [123, Ch. 6]. □

We could, by sectioning in the *other* variable, equally well have defined the area of E as $\int_{\Omega_2} \mu_1(E_{\omega_2})\, \mu_2(\mathrm{d}\omega_2)$ – integrating row heights rather than column heights. Both prescriptions assign $\mu_1(A_1)\mu_2(A_2)$ to a rectangle, so by the uniqueness just proved *they are the same measure*. Already at the level of sets, then, the order of integration does not matter; the Fubini–Tonelli theorem is the upgrade of this set-level fact to arbitrary integrands, and its proof is the systematic climb from indicators, where it is the statement just made, up to all integrable functions.

Theorem 1.27 (Tonelli and Fubini). *Let $(\Omega_1, \mathcal{F}_1, \mu_1)$ and $(\Omega_2, \mathcal{F}_2, \mu_2)$ be σ-finite and let $f\colon \Omega_1 \times \Omega_2 \to [-\infty, \infty]$ be $\mathcal{F}_1 \otimes \mathcal{F}_2$-measurable.*

1. (Tonelli.) *If $f \geq 0$, then $\omega_1 \mapsto \int f(\omega_1, \cdot)\ \mathrm{d}\mu_2$ and $\omega_2 \mapsto \int f(\cdot, \omega_2)\ \mathrm{d}\mu_1$ are measurable, and*

$$\int_{\Omega_1\times\Omega_2} f\ \mathrm{d}(\mu_1 \otimes \mu_2) = \int_{\Omega_1} \Big(\int_{\Omega_2} f\ \mathrm{d}\mu_2\Big)\,\mathrm{d}\mu_1 = \int_{\Omega_2} \Big(\int_{\Omega_1} f\ \mathrm{d}\mu_1\Big)\,\mathrm{d}\mu_2. \tag{1.9}$$

   The common value may be $+\infty$; all three integrals are always equal.

2. (Fubini.) *If f is integrable for the product measure – equivalently, by Tonelli applied to $|f|$, if* one *of the iterated integrals of $|f|$ is finite – then the inner integrals are finite a.e., are integrable, and (1.9) holds with all integrals finite.*

Proof. Tonelli, by the indicator $\to$ simple $\to$ nonnegative ladder. The proof is the standard three-rung ascent through the construction of the integral, each rung licensed by a tool already in hand. *Rung one, indicators.* Take $f = \mathbf{1}_E$ for $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$. Then for each fixed ω_1 the inner integrand $f(\omega_1, \cdot) = \mathbf{1}_{E_{\omega_1}}$ is the indicator of the section, so $\int_{\Omega_2} f(\omega_1, \cdot)\,\mathrm{d}\mu_2 = \mu_2(E_{\omega_1})$, which lemma 1.25 certifies is a measurable function of ω_1; and by the very construction of the product measure (theorem 1.26),

$$\int_{\Omega_1} \Big(\int_{\Omega_2} \mathbf{1}_E \,\mathrm{d}\mu_2 \Big) \mathrm{d}\mu_1 = \int_{\Omega_1} \mu_2(E_{\omega_1})\,\mathrm{d}\mu_1 = (\mu_1 \otimes \mu_2)(E) = \int_{\Omega_1 \times \Omega_2} \mathbf{1}_E \,\mathrm{d}(\mu_1 \otimes \mu_2),$$

which is the first equality of (1.9). The second equality – the genuine order-swap – is the row-versus-column remark made just above the theorem: the iterated integral taken in the opposite order, $E \mapsto \int_{\Omega_2} \mu_1(E_{\omega_2})\,\mathrm{d}\mu_2$, defines a measure assigning $\mu_1(A_1)\mu_2(A_2)$ to every rectangle, so by the uniqueness corollary corollary 1.10 it *is* $\mu_1 \otimes \mu_2$, and the two iterated integrals agree. Thus (1.9) holds for indicators. *Rung two, simple functions.* A nonnegative simple function is a finite combination $\sum_k c_k \mathbf{1}_{E_k}$ with $c_k \geq 0$, and all three integrals in (1.9) are linear in f, so the identity extends from indicators to nonnegative simple functions by adding finitely many copies of the indicator case. *Rung three, general $f \geq 0$, by monotone passage.* For an arbitrary measurable $f \geq 0$ take simple functions $s_n \uparrow f$ pointwise – the canonical increasing approximation built in section 1.3. For each fixed ω_1 the slices increase, $s_n(\omega_1, \cdot) \uparrow f(\omega_1, \cdot)$, so the monotone convergence theorem (theorem 1.20) gives $\int_{\Omega_2} s_n(\omega_1, \cdot)\,\mathrm{d}\mu_2 \uparrow \int_{\Omega_2} f(\omega_1, \cdot)\,\mathrm{d}\mu_2$; the left sides are measurable in ω_1 by the simple case, so their increasing pointwise limit – the inner integral of f – is measurable in ω_1 by proposition 1.13. Applying the monotone convergence theorem a second time, now to the outer μ_1-integral and to the product integral, and using the simple-case identity at each finite stage, lets us pass $n \to \infty$ in all three integrals of (1.9) simultaneously and preserve the equalities. (Should the common value be $+\infty$, the identity reads $\infty = \infty = \infty$ and is still true: for nonnegative integrands there is no integrability hypothesis to impose, which is the whole convenience of Tonelli.) This proves part (i).

Fubini, by the signed splitting $f = f^+ - f^-$ guarded by absolute integrability. Now f may change sign. Decompose it into its nonnegative and nonpositive parts, $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$, so that $f = f^+ - f^-$, $|f| = f^+ + f^-$, and both $f^\pm \geq 0$ are measurable. The hypothesis is that one iterated integral of $|f|$ is finite; applying Tonelli (part (i)) to the nonnegative function $|f|$, all three of its integrals are equal, so this single finiteness forces

$$\int_{\Omega_1 \times \Omega_2} |f| \,\mathrm{d}(\mu_1 \otimes \mu_2) = \int_{\Omega_1} \Big(\int_{\Omega_2} |f| \,\mathrm{d}\mu_2 \Big) \mathrm{d}\mu_1 < \infty,$$

i.e. f is product-integrable. Tonelli also applies to f^+ and f^- separately, giving (1.9) for each with possibly infinite value – and the point of absolute integrability is to rule that possibility out. Here is the step the skeleton left terse, spelled out. Since $f^\pm \leq |f|$ pointwise, monotonicity of the integral gives, at the level of inner integrals,

$$\int_{\Omega_2} f^\pm(\omega_1, \cdot)\,\mathrm{d}\mu_2 \;\leq\; \int_{\Omega_2} |f|(\omega_1, \cdot)\,\mathrm{d}\mu_2 \qquad \text{for every } \omega_1,$$

and integrating over ω_1,

$$\int_{\Omega_1} \Big(\int_{\Omega_2} f^\pm \,\mathrm{d}\mu_2 \Big) \mathrm{d}\mu_1 \;\leq\; \int_{\Omega_1} \Big(\int_{\Omega_2} |f| \,\mathrm{d}\mu_2 \Big) \mathrm{d}\mu_1 < \infty.$$

So the outer integral of the nonnegative measurable function $\omega_1 \mapsto \int_{\Omega_2} f^{\pm}(\omega_1, \cdot)\, \mathrm{d}\mu_2$ is finite. A nonnegative measurable function with finite integral is finite almost everywhere – for if it were equal to $+\infty$ on a set of positive measure, that set alone would already force the integral to be $+\infty$. Hence for μ_1-almost every ω_1 *both* inner integrals $\int_{\Omega_2} f^{+}(\omega_1, \cdot)\, \mathrm{d}\mu_2$ and $\int_{\Omega_2} f^{-}(\omega_1, \cdot)\, \mathrm{d}\mu_2$ are finite real numbers, and only then is their difference – the inner integral of f itself, $\int_{\Omega_2} f(\omega_1, \cdot)\, \mathrm{d}\mu_2 = \int_{\Omega_2} f^{+}\, \mathrm{d}\mu_2 - \int_{\Omega_2} f^{-}\, \mathrm{d}\mu_2$ – a legitimate subtraction of two finite numbers rather than the indeterminate $\infty - \infty$. *This is what absolute integrability buys: it makes both columns finite a.e., so the difference defining the signed inner integral is well-posed.* Subtracting the two finite Tonelli identities for f^+ and f^- – each a true instance of (1.9) with all integrals finite – term by term now yields (1.9) for f, with every integral finite. Running the identical argument with the roles of the two factors exchanged gives the other iterated integral the same value, so all three agree. A fuller account is in [123, Ch. 6]. ☐

The two halves of the theorem divide the labor cleanly, and it is worth saying exactly how, for this is the rule one applies in practice. Tonelli is the workhorse for *nonnegative* integrands: it asks nothing beyond σ-finiteness of the two spaces and lets one swap the order of integration freely, with the understanding that the common value may be $+\infty$. Fubini handles *signed* (or complex) integrands but exacts a price – absolute integrability – and the standard way to pay it is to run Tonelli on $|f|$ first: if either iterated integral of $|f|$ comes out finite, Fubini's swap is licensed; if both come out infinite, it may not be. The famous counterexample makes the division vivid. On $\mathbb{N} \times \mathbb{N}$ with counting measure, the array $a_{m,n}$ equal to $+1$ on the diagonal, -1 just below it, and 0 elsewhere has iterated sums $\sum_m \sum_n a_{m,n} = 0$ but $\sum_n \sum_m a_{m,n} = 1$ (or, in the continuous guise, an integrand on $(0,1)^2$ whose two iterated integrals are $+1$ and -1). The two orders disagree – but this is no contradiction with the theorem, because $\sum_{m,n} |a_{m,n}| = \infty$, so the absolute-integrability hypothesis of Fubini *fails*. The counterexample violates the hypothesis; it never violates the conclusion under that hypothesis. The lesson to carry into every later interchange is the discipline: *before swapping the order on a signed integrand, check that the integral of the absolute value is finite*, and one almost always checks it by a Tonelli computation on the modulus.

Remark 1.28 (Foreshadowing the graphon double integral)**.** The single most important double integral in this book has not appeared yet, but it is what the whole machinery of this section was built to make rigorous, and naming it now – while the theorem is fresh – means it will arrive in Part III as a settled object rather than a leap of faith. The kernel coupling at the center of Parts III–IV is the integral operator

$$(\mathbb{W}f)(x) \;=\; \int_0^1 W(x,y)\, f(y)\, \mathrm{d}y, \qquad x, y \in [0,1],$$

with $W\colon [0,1]^2 \to [0,1]$ the graphon kernel and x, y the *type* labels. We keep these type variables explicit and we do not conflate them with the state x of this chapter: as the chapter's opening notation watch stressed, $x, y \in [0,1]$ are exchangeable network labels carrying no metric meaning, put to work in chapter 2 (the Hilbert–Schmidt example there) and formally introduced as a graph limit in chapter 15, not the spatial state over which we have been integrating. Under the standing regularity hypothesis **(A-Reg-W)** – W symmetric, measurable, and bounded – the kernel lies in $L^2([0,1]^2, \mathrm{Leb})$, since a bounded measurable function on the finite-measure square $[0,1]^2$ is automatically square-integrable, $\iint |W|^2 \le \|W\|_\infty^2 < \infty$. Theorem 1.27 is then exactly what makes the energy double integral

$$\iint_{[0,1]^2} W(x,y)\, g(x)\, f(y)\; \mathrm{d}x\; \mathrm{d}y$$

well defined and *order-free*: for $f, g \in L^2([0,1])$ the product $W(x,y)g(x)f(y)$ is product-integrable on $[0,1]^2$ – one checks $\iint |W\, g\, f| < \infty$ by Tonelli on the modulus, bounding it by $\|W\|_\infty \|g\|_{L^1} \|f\|_{L^1}$ – so Fubini licenses computing it as either iterated integral, $\int g(x)(\mathbb{W}f)(x)\; \mathrm{d}x$ or $\int f(y)(\mathbb{W}g)(y)\; \mathrm{d}y$, the order-freedom that will become the self-adjointness of $\mathbb{W}$ for symmetric W. The same theorem, applied to $|W|^2$, is what shows in chapter 2 (proposition 2.24) that $\mathbb{W}$ is a Hilbert–Schmidt operator on $L^2([0,1])$ – the finiteness of the Hilbert–Schmidt norm $\|\mathbb{W}\|_{\mathrm{HS}}^2 = \iint |W|^2$ is a Tonelli statement about a nonnegative integrand – the property recorded in **(A-Reg-W)**; and its spectral payoff – eigenvector centrality, the spectral gap, and low-rank reductions – is the subject of chapter 18. We flag the double integral now so that, when it arrives, the right to write it as an iterated integral and to swap the order is a known quantity, owned by theorem 1.27, rather than a new act of faith.

This closes the integration theory proper. We began the chapter needing an integral stable under limits; we now have, in addition, an integral stable under the *reordering* of independent integrations, with two clean licenses – Tonelli for nonnegative integrands, Fubini under absolute integrability – and the Dynkin engine of section 1.2.3 reused one final time to forge them. The product structure built here is not a sidebar: it is what every later double integral in the book runs on – the independent-noise expectations of the stochastic chapters, the graphon energy just foreshadowed for chapters 2 to 18, and, more immediately, the Minkowski-integral and convolution arguments that the next section meets when it manufactures the L^p spaces. With the integral now complete – limits and products both under control – the remaining structural question is geometric: do the integrable functions, packaged as a normed space, form a *complete* space, a Banach space in which Cauchy sequences converge? That completeness is the property the whole measure-theoretic core has been driving toward, and assembling it from the integral and the three convergence theorems is the work of the next section.

1.6 The L^p spaces and completeness

The previous section completed the integration theory: we now own an integral stable under limits and stable under the reordering of independent integrations. The natural next move is the one a physicist makes without comment – to stop treating integrable functions one at a time and instead regard them as the *points of a space*, equipped with a notion of length (a norm) and hence of distance, so that one may speak of two functions being close, of a sequence of functions converging, and of a limit. This packaging is not cosmetic. Every existence proof in this book – a density solving a Fokker–Planck equation, a fixed point of a best-response map, a minimizer of an energy – is built the same way: one writes down a sequence of approximate objects, shows the sequence is *Cauchy* in some norm, and then needs to know that it actually *converges to a limit that is itself a member of the space*. That last guarantee is *completeness*, and it is the decisive structural fact about the spaces we now build. A normed space in which every Cauchy sequence converges is a *Banach space*; the rationals $\mathbb{Q}$ are the cautionary non-example – the sequence 3, 3.1, 3.14, 3.141, . . . is Cauchy yet escapes $\mathbb{Q}$, converging to nothing rational – and the real line $\mathbb{R}$ is the completion that repairs the hole. The L^p spaces are the function-space analogues of $\mathbb{R}$: complete by a limiting argument, and complete *because* of the three convergence theorems of section 1.4, which is exactly why they could not be assembled until now. Completeness is the analytic counterpart of the compactness arguments of chapters 2 and 3: where compactness extracts a convergent subsequence, completeness certifies that the limit it produces lands back inside the space. Fix a measure space $(\Omega, \mathcal{F}, \mu)$ throughout.

We build a whole one-parameter family at once, indexed by an exponent $1 \le p \le \infty$, because the book uses several values and it is wasteful to construct them separately. The exponent $p = 2$ is the Hilbert space of the spectral chapter; $p = 1$ is the home of densities and of the integral itself; $p = \infty$ houses bounded coefficients; and $p < 2$ will be forced on us by sparse graphon kernels in chapter 27. Intuitively the exponent dials *how a function is required to be small*: a large p penalizes tall spikes severely (a height-h spike contributes h^p, which explodes), a small p is lenient about height but unforgiving about spread. The norm is built by raising $|f|$ to the p-th power, integrating, and taking the p-th root – the final root restoring the homogeneity $\|\lambda f\|_p = |\lambda| \, \|f\|_p$ that any honest notion of length must possess.

Definition 1.29 (L^p spaces)**.** For $1 \le p < \infty$, let $\mathcal{L}^p(\mu)$ be the measurable $f \colon \Omega \to \mathbb{R}$ (or $\mathbb{R}^d$) with $\|f\|_p := \left(\int_\Omega |f|^p \, \mathrm{d}\mu\right)^{1/p} < \infty$. For $p = \infty$, let $\mathcal{L}^\infty(\mu)$ be the measurable f with $\|f\|_\infty := \operatorname{ess\,sup}_\Omega |f| < \infty$, the essential supremum being the smallest M with $|f| \le M$ μ-a.e. The space $L^p(\mu)$ is the quotient of $\mathcal{L}^p(\mu)$ by the relation "equal μ-a.e."; its elements are equivalence classes, and $\|\cdot\|_p$ is a genuine norm on it (the quotient is exactly what makes $\|f\|_p = 0 \implies f = 0$ true).

Two features of this definition need the motivation that a bare statement withholds. The first is the passage to *equivalence classes*, which is not fastidiousness but a forced repair. The candidate norm $\|f\|_p = \left(\int |f|^p\right)^{1/p}$ fails one axiom of a genuine norm on the raw function space $\mathcal{L}^p(\mu)$: a norm must satisfy $\|f\|_p = 0 \Rightarrow f = 0$, yet $\int |f|^p \, \mathrm{d}\mu = 0$ forces only $|f|^p = 0$ μ-almost everywhere, hence $f = 0$ *off a μ-null set* – not everywhere. A function supported on a single point carries Leb-integral zero without being the zero function; the indicator $\mathbf{1}_\mathbb{Q}$ has $\|\mathbf{1}_\mathbb{Q}\|_p = 0$ without vanishing anywhere on $\mathbb{Q}$. The integral simply cannot *see* a null set, so it cannot distinguish two functions that differ only there. The honest conclusion is that the integral was never measuring functions in the first place; it was measuring them *up to almost-everywhere equality*. We therefore declare $f \sim g$ when $f = g$ μ-a.e. – the same null-set quotient that ran through section 1.3, now promoted from a convenience to the very definition of the objects – and let $L^p(\mu)$ be the set of resulting equivalence classes. On classes $\|f\|_p = 0$ does mean $f = 0$ (the functions vanishing a.e. are precisely the zero class), so $\|\cdot\|_p$ becomes a genuine norm. In practice one writes f for both a function and its class and never suffers for it, because every operation we ever perform on an element of L^p – integration against a test function, the norm, the integral of any pointwise expression in f – is blind to null-set changes and so descends unambiguously to classes. The cost is one discipline, worth stating once: a statement about an element of L^p is legitimate only if it is invariant under modification on a null set. "The value $f(x_0)$" is meaningless for $f \in L^p$; "the integral $\int_A f$" is meaningful.

The second feature is the definition for $p = \infty$, where the recipe "integrate the p-th power and take a root" has no integral to take and is replaced by the right notion of "largest value," the *essential supremum*. The ordinary supremum is the wrong quantity, for the same null-set reason:

$\sup|\mathbf{1}_{\mathbb{Q}}| = 1$ even though $\mathbf{1}_{\mathbb{Q}}$ is a.e. zero, so the ordinary sup is *not* blind to null sets and would fail to descend to classes. The essential supremum $\operatorname{ess\,sup}|f| = \inf\{M : |f| \le M \ \mu\text{-a.e.}\}$ is the smallest ceiling $|f|$ respects except possibly on a negligible set; it ignores values attained only on a null set, giving $\operatorname{ess\,sup}|\mathbf{1}_{\mathbb{Q}}| = 0$, and on a finite-measure space it is exactly the limit of $\|f\|_p$ as $p \to \infty$ (whence the notation). With this reading $L^\infty(\mu)$ is the space of *essentially bounded* functions – the natural home of bounded coefficients and of the kernels $W \in L^\infty([0,1]^2)$ met in the previous section.

We have a candidate norm; what is not yet established is that it *is* one, and the single nonobvious axiom is the triangle inequality $\|f+g\|_p \le \|f\|_p + \|g\|_p$. For $p = 1$ it follows at once from $|f+g| \le |f| + |g|$ and monotonicity of the integral, and for $p = \infty$ from the same bound applied to essential suprema; but for intermediate p the p-th power entangles f and g, and the inequality – here named *Minkowski's* – is a genuine theorem. It rests on a companion, *Hölder's inequality*, the most-used estimate in analysis, which bounds the integral of a product by the product of norms taken in *conjugate* exponents $\frac{1}{p} + \frac{1}{q} = 1$. The conjugacy is not arbitrary bookkeeping: it is exactly the condition that makes the product of a "height in L^p" and a "height in L^q" have total degree one, so the estimate is homogeneous under rescaling. Both inequalities descend from a single elementary convexity fact, *Young's inequality*, and we prove the chain Young $\Rightarrow$ Hölder $\Rightarrow$ Minkowski.

Proposition 1.30 (Hölder and Minkowski inequalities)**.** *Let $1 \le p, q \le \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$ (conjugate exponents). For measurable f, g,*

$$\|fg\|_1 \le \|f\|_p \|g\|_q \qquad \textit{(Hölder)}, \qquad\qquad \|f+g\|_p \le \|f\|_p + \|g\|_p \qquad \textit{(Minkowski)}.$$

In particular $\|\cdot\|_p$ satisfies the triangle inequality and $L^p(\mu)$ is a normed vector space.

Proof. Everything flows from *Young's inequality*: for $a, b \ge 0$ and conjugate exponents $1 < p, q < \infty$,

$$ab \ \le \ \frac{a^p}{p} + \frac{b^q}{q}.$$

If $a = 0$ or $b = 0$ the inequality is trivial; assume $a, b > 0$. It is then a one-line consequence of the *concavity of the logarithm*. Read $\frac{1}{p}$ and $\frac{1}{q}$ as two weights summing to 1 – this is precisely what conjugacy $\frac{1}{p} + \frac{1}{q} = 1$ supplies – and apply the defining inequality of a concave function (the function of a weighted average dominates the weighted average of the values) at the two points a^p and b^q:

$$\log\Big(\tfrac{1}{p}a^p + \tfrac{1}{q}b^q\Big) \ \ge \ \tfrac{1}{p}\log a^p + \tfrac{1}{q}\log b^q \ = \ \log a + \log b \ = \ \log(ab).$$

Since log is increasing, exponentiating preserves the inequality and gives $\frac{1}{p}a^p + \frac{1}{q}b^q \ge ab$, which is Young's. (Equality holds iff $a^p = b^q$, the degenerate case where the two evaluation points coincide; we shall not need it.)

Hölder. Take first $1 < p < \infty$. If $\|f\|_p = 0$ then $f = 0$ a.e. and both sides vanish; if $\|f\|_p = \infty$ the right side is infinite and there is nothing to prove; so assume $0 < \|f\|_p, \|g\|_q < \infty$. The trick is to *normalize away the scales*. Put

$$a(\omega) = \frac{|f(\omega)|}{\|f\|_p}, \qquad b(\omega) = \frac{|g(\omega)|}{\|g\|_q},$$

so that $\int a^p = \int |f|^p / \|f\|_p^p = 1$ and likewise $\int b^q = 1$; the normalization is chosen exactly so the right-hand side of Young will integrate to a clean constant. Applying Young pointwise and integrating,

$$\int a\, b\, \mathrm{d}\mu \;\le\; \int \Big(\frac{a^p}{p} + \frac{b^q}{q}\Big)\, \mathrm{d}\mu \;=\; \frac{1}{p}\int a^p + \frac{1}{q}\int b^q \;=\; \frac{1}{p} + \frac{1}{q} \;=\; 1.$$

Multiplying through by $\|f\|_p \|g\|_q$ and recognizing $\|f\|_p \|g\|_q \int ab = \int |f||g| = \|fg\|_1$ yields $\|fg\|_1 \le \|f\|_p \|g\|_q$. The endpoint cases $p \in \{1, \infty\}$ need no convexity at all: there $|fg| \le \|g\|_\infty |f|$ pointwise a.e. (the essential bound on g), and integrating gives $\|fg\|_1 \le \|f\|_1 \|g\|_\infty$.

Minkowski. The triangle inequality is now a short manoeuvre built on the algebraic identity $|f+g|^p = |f+g|\,|f+g|^{p-1}$. Bounding the first factor by $|f+g| \le |f| + |g|$ and integrating,

$$\|f+g\|_p^p \;=\; \int |f+g|^p \;\le\; \int |f|\,|f+g|^{p-1} \;+\; \int |g|\,|f+g|^{p-1}.$$

Each integral on the right is a product of two functions, ready for Hölder with exponent p on the first factor and the conjugate $q = p/(p-1)$ on the second. The point of choosing that exponent is that $(p-1)q = p$, so the q-norm of the second factor is

$$\Big(\int |f+g|^{(p-1)q}\Big)^{1/q} = \Big(\int |f+g|^p\Big)^{1/q} = \|f+g\|_p^{p/q},$$

and the two applications of Hölder combine to give

$$\|f+g\|_p^p \;\le\; \big(\, \|f\|_p + \|g\|_p \,\big)\; \|f+g\|_p^{p/q}.$$

If $\|f+g\|_p = 0$ the triangle inequality is trivial. Otherwise $\|f+g\|_p$ is finite and nonzero – finite because the pointwise convexity bound $|f+g|^p \le 2^{p-1}(|f|^p + |g|^p)$ keeps $f+g$ in L^p – so we may divide both sides by $\|f+g\|_p^{p/q}$. Using $p - \frac{p}{q} = p(1 - \frac{1}{q}) = p \cdot \frac{1}{p} = 1$, the left side becomes $\|f+g\|_p^{p-p/q} = \|f+g\|_p$, and Minkowski's inequality $\|f+g\|_p \le \|f\|_p + \|g\|_p$ drops out. With the triangle inequality established, $\|\cdot\|_p$ is a norm and $L^p(\mu)$ a normed vector space. A fuller development is in [31, Ch. 1]. $\square$

We now reach the structural heart of the section. A normed space is only as useful as its limits are trustworthy, and the guarantee we need is that the space has *no holes*: every sequence whose terms crowd arbitrarily close together – every Cauchy sequence – actually converges, and to a limit inside the space. This is the Riesz–Fischer theorem. It is the single fact licensing the standard existence strategy of the entire book: manufacture approximants, prove they are Cauchy, invoke completeness to produce the limit, identify it. Its proof is a showcase application of the convergence theorems just built, and it exhibits an architecture we will meet again and again, worth naming in advance: *monotone control via MCT, then dominated passage via DCT.* One first tames the sequence by passing to a rapidly-converging subsequence whose increments are summable; one packs those increments into a single dominating function G, finite because MCT converts their summability into an L^p bound; and one then runs DCT against G to upgrade the resulting almost-everywhere limit to a norm limit.

Theorem 1.31 (Riesz–Fischer: completeness of L^p)**.** *For every $1 \le p \le \infty$, $L^p(\mu)$ is a Banach space: every Cauchy sequence converges in $\|\cdot\|_p$ to an element of $L^p(\mu)$.*

Proof. We give the argument for $1 \le p < \infty$ first, as it is short, illuminating, and uses precisely the convergence theorems of section 1.4; the case $p = \infty$ is elementary and handled separately at the end. Let (f_n) be Cauchy in $L^p(\mu)$. A Cauchy sequence converges iff it has a convergent subsequence, so choose $n_1 < n_2 < \cdots$ with $\|f_{n_{k+1}} - f_{n_k}\|_p \le 2^{-k}$. Define the nonnegative measurable function

$$G \;=\; |f_{n_1}| + \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}|.$$

By Minkowski applied to partial sums and then MCT (theorem 1.20) on the increasing partial sums of G^p,

$$\|G\|_p \;\le\; \|f_{n_1}\|_p + \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_p \;\le\; \|f_{n_1}\|_p + \sum_{k=1}^{\infty} 2^{-k} \;<\; \infty.$$

Hence $G \in L^p(\mu)$, so $G < \infty$ μ-a.e.; on that full-measure set the telescoping series $f_{n_1} + \sum_k (f_{n_{k+1}} - f_{n_k})$ converges absolutely to a finite limit, which we call f. Thus $f_{n_k} \to f$ pointwise a.e., $|f| \le G$, and $|f - f_{n_k}|^p \le (2G)^p \in L^1(\mu)$; by DCT (theorem 1.23), $\|f - f_{n_k}\|_p \to 0$. Finally, a Cauchy sequence with a convergent subsequence converges to the same limit: given $\varepsilon > 0$, pick N with $\|f_n - f_m\|_p < \varepsilon$ for $m, n \ge N$ and k with $n_k \ge N$ and $\|f_{n_k} - f\|_p < \varepsilon$; then $\|f_n - f\|_p < 2\varepsilon$ for $n \ge N$.

For $p = \infty$ the argument is even shorter. Let (f_n) be Cauchy in $\|\cdot\|_\infty$. For each pair (m, n) the inequality $|f_n - f_m| \le \|f_n - f_m\|_\infty$ holds off a μ-null set $N_{m,n}$, and $|f_n| \le \|f_n\|_\infty$ off a null set N_n; let $N = \bigcup_{m,n} N_{m,n} \cup \bigcup_n N_n$, a countable union of null sets, hence null. Off N the sequence (f_n) is *uniformly* Cauchy – $\sup_{\omega \notin N} |f_n - f_m| \le \|f_n - f_m\|_\infty \to 0$ – so it converges uniformly there to a bounded limit f (set $f = 0$ on N). Then f is measurable, $\|f\|_\infty \le \sup_n \|f_n\|_\infty < \infty$, and $\|f_n - f\|_\infty \le \sup_{\omega \notin N} |f_n - f| \to 0$. Thus $L^\infty(\mu)$ is complete as well. □

The proof is worth dwelling on because its architecture recurs: build a dominating object (G) with MCT, extract an almost-everywhere limit, and upgrade to norm convergence with DCT. That pattern – monotone control, then dominated passage – is the analytic skeleton of nearly every limit taken in this book.

Two further facts about L^p structure the way it is used. First, the *density of nice functions*, which is what lets one prove an L^p estimate by checking it on smooth functions and passing to the limit. We prove it in full, as the book consumes it later (mollification and testing arguments, e.g. in chapters 8 and 22).

Proposition 1.32 (Density of simple, C_c, and C_c^∞ functions)**.** *Let $1 \le p < \infty$.*

1. *Integrable simple functions are dense in $L^p(\mu)$ for every measure space $(\Omega, \mathcal{F}, \mu)$.*
2. *On $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mathrm{Leb})$ the compactly supported continuous functions $C_c(\mathbb{R}^d)$ are dense in L^p.*
3. *On $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mathrm{Leb})$ the smooth compactly supported functions $C_c^\infty(\mathbb{R}^d)$ are dense in L^p.*

Proof. (i) Let $f \in L^p(\mu)$ and split $f = f^+ - f^-$. For the canonical simple sequences $s_n \uparrow f^+$ and $t_n \uparrow f^-$ of (1.5), each s_n, t_n is dominated by $|f| \in L^p$, hence integrable simple, and $|f - (s_n - t_n)|^p \to 0$

pointwise with $|f-(s_n-t_n)|^p \le (2|f|)^p \in L^1(\mu)$; dominated convergence (theorem 1.23) gives $\|f-(s_n-t_n)\|_p \to 0$.

(ii) By (i) it suffices to approximate a single indicator $\mathbf{1}_A$, with A Borel and $\mathrm{Leb}(A)<\infty$, by a C_c function in L^p. By the very definition of the outer measure used to build Leb (theorem 1.11), given $\varepsilon>0$ there are boxes Q_k with $A \subseteq \bigcup_k Q_k$ and $\sum_k \mathrm{vol}(Q_k) \le \mathrm{Leb}(A)+\varepsilon$; choosing M with $\sum_{k>M}\mathrm{vol}(Q_k)<\varepsilon$ and setting $U=\bigcup_{k\le M} Q_k$, a bounded finite union of boxes, we get $\mathrm{Leb}(A\setminus U) \le \sum_{k>M}\mathrm{vol}(Q_k)<\varepsilon$ and $\mathrm{Leb}(U\setminus A) \le \sum_k \mathrm{vol}(Q_k)-\mathrm{Leb}(A) \le \varepsilon$, so $\|\mathbf{1}_A-\mathbf{1}_U\|_p^p=\mathrm{Leb}(A\triangle U)\le 2\varepsilon$. It remains to approximate $\mathbf{1}_U$. For a single box $Q=\prod_j (a_j,b_j]$ and $\eta>0$ let $\varphi_{Q,\eta}\in C_c(\mathbb{R}^d)$ be the product of one-dimensional continuous trapezoids that equal 1 on $[a_j,b_j]$, drop linearly to 0 over a collar of width η, and vanish outside $[a_j-\eta,b_j+\eta]$; then $0\le\varphi_{Q,\eta}\le 1$, $\varphi_{Q,\eta}=1$ on Q, and $\varphi_{Q,\eta}\ne\mathbf{1}_Q$ only on a collar of Lebesgue measure $O(\eta)$, so $\|\mathbf{1}_Q-\varphi_{Q,\eta}\|_p^p=O(\eta)$. Taking $\varphi=\max_{k\le M}\varphi_{Q_k,\eta}\in C_c(\mathbb{R}^d)$ and η small makes $\|\mathbf{1}_U-\varphi\|_p$ as small as desired (the maximum of finitely many continuous functions is continuous, and $\mathbf{1}_U=\max_k \mathbf{1}_{Q_k}$). Combining the two estimates gives a C_c function within $3\varepsilon^{1/p}$ of $\mathbf{1}_A$ in L^p, as required.

(iii) By (ii) it suffices to approximate a fixed $g\in C_c(\mathbb{R}^d)$ by C_c^∞ functions in L^p. Fix the standard *mollifier*: the function $\rho(x)=c\exp\big(-1/(1-|x|^2)\big)$ for $|x|<1$ and $\rho(x)=0$ for $|x|\ge 1$ is smooth, nonnegative, and supported in the unit ball; choose c so that $\int_{\mathbb{R}^d}\rho=1$, and set $\rho_\varepsilon(x)=\varepsilon^{-d}\rho(x/\varepsilon)$, so $\int\rho_\varepsilon=1$ and $\operatorname{supp}\rho_\varepsilon\subseteq\overline{B_\varepsilon}$. Define the convolution

$$(g*\rho_\varepsilon)(x)=\int_{\mathbb{R}^d} g(x-y)\,\rho_\varepsilon(y)\;\mathrm{d}y.$$

Differentiation under the integral sign (corollary 1.24, with dominating function supported in a fixed compact set since g and all x-derivatives of ρ_ε are bounded and compactly supported) shows $g*\rho_\varepsilon\in C^\infty$, and $\operatorname{supp}(g*\rho_\varepsilon)\subseteq\operatorname{supp}g+\overline{B_\varepsilon}$ is compact, so $g*\rho_\varepsilon\in C_c^\infty(\mathbb{R}^d)$. Finally, since $\int\rho_\varepsilon=1$,

$$(g*\rho_\varepsilon)(x)-g(x)=\int_{\mathbb{R}^d}\big(g(x-y)-g(x)\big)\rho_\varepsilon(y)\;\mathrm{d}y,$$

and g, being continuous with compact support, is uniformly continuous, so $\sup_x|g(x-y)-g(x)|\le\omega_g(|y|)$ with the modulus $\omega_g(\delta)\to 0$ as $\delta\to 0$; hence $\|g*\rho_\varepsilon-g\|_\infty\le\omega_g(\varepsilon)\to 0$. As all the functions $g*\rho_\varepsilon$ are supported in the fixed compact set $K=\operatorname{supp}g+\overline{B_1}$ for $\varepsilon\le 1$, $\|g*\rho_\varepsilon-g\|_p\le\mathrm{Leb}(K)^{1/p}\,\|g*\rho_\varepsilon-g\|_\infty\to 0$. This proves $C_c^\infty(\mathbb{R}^d)$ is dense. □

The density statement is one place corollary 1.24 (differentiation under the integral) and the box-cover construction of Lebesgue measure earn their keep; a comparison treatment is in [31, Ch. 4]. Second, the case $p=2$ is special: $\|f\|_2^2=\int|f|^2\;\mathrm{d}\mu$ comes from the inner product $\langle f,g\rangle=\int f\,\overline{g}\;\mathrm{d}\mu$, so $L^2(\mu)$ is a *Hilbert* space. The geometry this buys – orthogonal projection, self-adjoint operators, the spectral theorem – is developed in chapter 2, which is also the setting in which the graphon operator $\mathbb{W}$ is constructed and shown to be Hilbert–Schmidt (proposition 2.24); its spectral consequences are taken up in chapter 18. The home of unbounded, sparse kernels is instead L^p for $p<2$, treated in chapter 27.

1.7 The empirical measure and its continuum limit

With the last of the toolbox in place – the complete function spaces of the previous section join the measures, the integral, and the three interchange permits assembled before them – every part the chapter set out to build is now in hand, and the empirical-measure theorem announced in the opening is their assembly. Drawing on the core of that toolbox – measures to house the spikes and their limit, the bridge identity (1.8), and the continuity-from-above that keeps mass from leaking to infinity – we can at last make rigorous the passage from a finite population to a continuum. The result is the prototype every later convergence theorem refines: McKean–Vlasov propagation of chaos (chapter 7), convergence of N-player Nash equilibria (chapter 13), and the finite-network-to-graphon limit (chapter 25) are all, at their core, statements that an empirical measure converges to a deterministic law.

1.7.1 Weak convergence, lightly

The empirical measure is a sum of spikes and its limit may be smooth, so convergence cannot be tested pointwise or in any L^p norm of densities. The correct notion tests against smooth observables.

Definition 1.33 (Weak convergence of probability measures)**.** Let $\nu_N, \nu \in \mathcal{P}(\mathbb{R}^d)$. We say ν_N *converges weakly* to ν, written $\nu_N \Rightarrow \nu$, if

$$\int_{\mathbb{R}^d} f \, \mathrm{d}\nu_N \longrightarrow \int_{\mathbb{R}^d} f \, \mathrm{d}\nu \qquad \text{for every bounded continuous } f \in C_b(\mathbb{R}^d).$$

This is the single most-used mode of convergence in the book. Its full theory – the equivalent formulations of the Portmanteau theorem, metrizability by the Lévy–Prokhorov distance, and Prokhorov's compactness criterion – is developed in chapter 3; the finer Wasserstein metric, which additionally controls second moments, is chapter 4. None of that machinery is needed here. The one structural input the empirical-measure argument requires – that weak convergence can be tested against a fixed *countable* family of observables rather than all of $C_b(\mathbb{R}^d)$ – we prove from scratch in the next lemma, by an elementary construction using only the continuity-from-above of measures (proposition 1.6). The chapter therefore borrows nothing forward.

Lemma 1.34 (A countable convergence-determining family)**.** *There exists a countable family $\mathcal{F} = \{f_1, f_2, \dots\} \subseteq C_b(\mathbb{R}^d)$ with the following property: if $\nu_N, \nu \in \mathcal{P}(\mathbb{R}^d)$ satisfy $\int f_k \, \mathrm{d}\nu_N \to \int f_k \, \mathrm{d}\nu$ for* every *k, then $\nu_N \Rightarrow \nu$.*

Proof. Write $C_0(\mathbb{R}^d)$ for the continuous functions g that vanish at infinity, i.e. with $|g(x)| \to 0$ as $|x| \to \infty$; such a g is automatically *uniformly* continuous (it is continuous, and outside a large ball it is uniformly small), and bounded. We proceed in three steps: build a countable family; show convergence against it forces convergence against all of $C_0(\mathbb{R}^d)$; then upgrade that to weak convergence using that the measures are probability measures.

Step 1: an explicit countable family $\mathcal{F} \subseteq C_0(\mathbb{R}^d)$. For $n \in \mathbb{N}$ consider the dyadic lattice $2^{-n}\mathbb{Z}^d$. To each assignment of rational values at the lattice points lying in a ball B_R of rational radius R

centred at a rational point, with the value 0 assigned to all lattice points outside B_R, associate the function on $\mathbb{R}^d$ obtained by *multilinear interpolation* inside each dyadic cube from its 2^d corner values. The result is continuous, compactly supported (it vanishes outside the bounded region where nonzero corner values were prescribed), hence lies in $C_0(\mathbb{R}^d)$. Let $\mathcal{F} = \{f_1, f_2, \dots\}$ be the collection of all such functions over all n, all admissible balls, and all rational corner assignments; it is a countable union of finite (countable) families, hence countable.

This family is sup-norm dense in $C_0(\mathbb{R}^d)$. Indeed, fix $g \in C_0(\mathbb{R}^d)$ and $\varepsilon > 0$. Choose R with $|g| < \varepsilon$ outside B_R, and using uniform continuity choose n so large that $|g(x) - g(x')| < \varepsilon$ whenever $|x - x'| \le 2^{-n}\sqrt{d}$ (the diameter of a dyadic cube). Pick $f \in \mathcal{F}$ that interpolates rational values within ε of g at the lattice points in B_{R+1} and vanishes at lattice points outside. Because a multilinear interpolant on a cube takes values between the minimum and maximum of its corner values, on each cube meeting B_R we get $|g - f| \le 3\varepsilon$ (the corner error ε, the interpolation oscillation ε, and the rounding ε), while outside B_R both $|g| < \varepsilon$ and $|f| \le \varepsilon$. Hence $\|g - f\|_\infty \le 3\varepsilon$, proving density.

Step 2: convergence against $\mathcal{F}$ implies convergence against $C_0(\mathbb{R}^d)$. Assume $\int f_k \,\mathrm{d}\nu_N \to \int f_k \,\mathrm{d}\nu$ for every k. Let $g \in C_0(\mathbb{R}^d)$ and $\varepsilon > 0$; choose $f \in \mathcal{F}$ with $\|g - f\|_\infty < \varepsilon$. Since all ν_N, ν are probability measures, $\left|\int g \,\mathrm{d}\rho - \int f \,\mathrm{d}\rho\right| \le \|g - f\|_\infty < \varepsilon$ for $\rho \in \{\nu_N, \nu\}$, so

$$\left|\int g \,\mathrm{d}\nu_N - \int g \,\mathrm{d}\nu\right| \;\le\; 2\varepsilon + \left|\int f \,\mathrm{d}\nu_N - \int f \,\mathrm{d}\nu\right|,$$

whence $\limsup_N \left|\int g \,\mathrm{d}\nu_N - \int g \,\mathrm{d}\nu\right| \le 2\varepsilon$. As ε was arbitrary, $\int g \,\mathrm{d}\nu_N \to \int g \,\mathrm{d}\nu$ for every $g \in C_0(\mathbb{R}^d)$.

Step 3: from C_0 to all of C_b. The family $\mathcal{F}$ contains, for every rational $R > 0$, a *cutoff* χ_R with $0 \le \chi_R \le 1$, $\chi_R \equiv 1$ on B_R, and $\operatorname{supp} \chi_R \subseteq B_{2R}$ (prescribe corner value 1 at lattice points in B_R and 0 at lattice points outside B_{2R}, with n large; multilinear interpolation of values in $[0, 1]$ stays in $[0, 1]$). The pointwise inequalities $\mathbf{1}_{B_{2R}^c} \le 1 - \chi_R \le \mathbf{1}_{B_R^c}$ hold by construction. Now fix $f \in C_b(\mathbb{R}^d)$; by homogeneity we may assume $\|f\|_\infty \le 1$. Split

$$\int f \,\mathrm{d}\nu_N - \int f \,\mathrm{d}\nu \;=\; \underbrace{\int f\chi_R \,\mathrm{d}(\nu_N - \nu)}_{\text{(I)}} \;+\; \underbrace{\int f(1 - \chi_R) \,\mathrm{d}\nu_N}_{\text{(II)}} \;-\; \underbrace{\int f(1 - \chi_R) \,\mathrm{d}\nu}_{\text{(III)}}.$$

The function $f\chi_R$ is continuous with support in B_{2R}, hence in $C_0(\mathbb{R}^d)$, so (I) $\to 0$ by Step 2. For (III), $|\text{(III)}| \le \int (1 - \chi_R) \,\mathrm{d}\nu \le \nu(B_R^c)$. For (II), $|\text{(II)}| \le \int (1 - \chi_R) \,\mathrm{d}\nu_N$, and since $1 - \chi_R \in C_0(\mathbb{R}^d)$ Step 2 gives $\int (1 - \chi_R) \,\mathrm{d}\nu_N \to \int (1 - \chi_R) \,\mathrm{d}\nu \le \nu(B_R^c)$. Therefore

$$\limsup_{N\to\infty} \left|\int f \,\mathrm{d}\nu_N - \int f \,\mathrm{d}\nu\right| \;\le\; 2\,\nu(B_R^c).$$

Because ν is a probability measure and $B_R^c \downarrow \emptyset$ as $R \uparrow \infty$, continuity from above (proposition 1.6, applicable since $\nu(\mathbb{R}^d) = 1 < \infty$) gives $\nu(B_R^c) \to 0$. Letting $R \to \infty$ through rational values forces $\int f \,\mathrm{d}\nu_N \to \int f \,\mathrm{d}\nu$. As $f \in C_b(\mathbb{R}^d)$ was arbitrary, $\nu_N \Rightarrow \nu$. □

Remark 1.35 (The role of total mass)**.** Step 3 is exactly where the hypothesis "probability measures" does its work. Convergence against $C_0(\mathbb{R}^d)$ alone is *vague* convergence, which does not see mass escaping to infinity – the sequence $\nu_N = \delta_{N\mathbf{e}_1}$ converges vaguely to the zero measure yet is not weakly convergent. What rules this out is that the total mass is pinned at 1 and the limit ν is itself a probability measure, so no mass can leak past every ball; the cutoff estimate turns that into the tightness bound $\nu(B_R^c) \to 0$. This is the elementary, finite-dimensional shadow of Prokhorov's theorem (chapter 3).

1.7.2 The strong law for bounded observables

The probabilistic engine of the theorem is the strong law of large numbers. We do not import it as a black box: for *bounded* summands – the only case the empirical-measure argument needs, since we test against bounded f – it has a short, fully self-contained proof via a fourth-moment estimate and the Borel–Cantelli lemma. This keeps section 1.7 independent of the probability machinery built later in chapter 3.

Lemma 1.36 (Borel–Cantelli, first lemma)**.** *Let (A_n) be events in a probability space with $\sum_n \mathbb{P}(A_n) < \infty$. Then $\mathbb{P}(\limsup_n A_n) = 0$; that is, almost surely only finitely many A_n occur.*

Proof. $\limsup_n A_n = \bigcap_N \bigcup_{n\geq N} A_n$, so for every N, $\mathbb{P}(\limsup_n A_n) \leq \mathbb{P}(\bigcup_{n\geq N} A_n) \leq \sum_{n\geq N} \mathbb{P}(A_n)$ by countable subadditivity (proposition 1.6). The right side is the tail of a convergent series, hence $\to 0$ as $N \to \infty$. $\square$

Lemma 1.37 (Strong law for bounded i.i.d. variables)**.** *Let $(Y_i)_{i\geq 1}$ be i.i.d. real random variables with $|Y_i| \leq M$ a.s. for some $M < \infty$, and write $m_Y = \mathbb{E}[Y_1]$. Then $\frac{1}{N}\sum_{i=1}^N Y_i \to m_Y$ almost surely.*

Proof. Replacing Y_i by $Z_i = Y_i - m_Y$ we may assume $\mathbb{E}[Y_i] = 0$; then $|Z_i| \leq 2M$, $\mathbb{E}[Z_i] = 0$, and the Z_i are i.i.d.. Let $S_N = \sum_{i=1}^N Z_i$. Expanding the fourth power and using independence and $\mathbb{E}[Z_i] = 0$, every term $\mathbb{E}[Z_iZ_jZ_kZ_l]$ vanishes unless the indices coincide in pairs; there are N terms of the form $\mathbb{E}[Z_i^4]$ and $3N(N-1)$ of the form $\mathbb{E}[Z_i^2]\,\mathbb{E}\Big[Z_j^2\Big]$, so

$$\mathbb{E}\Big[S_N^4\Big] \;=\; N\,\mathbb{E}\Big[Z_1^4\Big] + 3N(N-1)(\mathbb{E}\Big[Z_1^2\Big])^2 \;\leq\; C\,N^2, \qquad C := \mathbb{E}\Big[Z_1^4\Big] + 3(\mathbb{E}\Big[Z_1^2\Big])^2 \leq 4(2M)^4.$$

Fix $\varepsilon > 0$. We use *Markov's inequality*: for a nonnegative random variable Z and $a > 0$, $a\,\mathbf{1}_{\{Z\geq a\}} \leq Z$ pointwise, so taking expectations (monotonicity of the integral) gives $\mathbb{P}(Z \geq a) = \mathbb{E}\Big[\mathbf{1}_{\{Z\geq a\}}\Big] \leq \mathbb{E}[Z]/a$. Applied to $Z = S_N^4 \geq 0$ with $a = N^4\varepsilon^4$,

$$\mathbb{P}\Big(\Big|\tfrac{S_N}{N}\Big| \geq \varepsilon\Big) \;=\; \mathbb{P}(S_N^4 \geq N^4\varepsilon^4) \;\leq\; \frac{\mathbb{E}[S_N^4]}{N^4\varepsilon^4} \;\leq\; \frac{C}{N^2\varepsilon^4},$$

which is summable in N. By Borel–Cantelli (lemma 1.36), a.s. only finitely many of the events $\{|S_N/N| \geq \varepsilon\}$ occur, i.e. $\limsup_N |S_N/N| \leq \varepsilon$ a.s.Taking the intersection over $\varepsilon = 1/j$, $j \in \mathbb{N}$ – a countable intersection of full-measure events, hence of full measure – gives $S_N/N \to 0$ almost surely. $\square$

1.7.3 The worked computation

We now state and prove the chapter's self-contained model computation: the empirical measure of N independent draws converges, almost surely, to their common law. This is the cleanest "thermodynamic limit" statement in the book. It is not one of the two carried threads of appendix A (LQG and graphon-LQG), which begin only once optimization enters; it is the foundational prototype on which both, and every later propagation-of-chaos result, rest.

Worked computation 1.38 (Almost-sure weak limit of the empirical measure)**.** Let $m \in \mathcal{P}(\mathbb{R}^d)$ and let $X_1, X_2, \dots$ be i.i.d. random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with common law $\mathcal{L}(X_i) = m$. Form the random empirical measures

$$\mu^N \;=\; \frac{1}{N}\sum_{i=1}^{N} \delta_{X_i} \;\in\; \mathcal{P}(\mathbb{R}^d), \qquad N \geq 1.$$

Then, almost surely, $\mu^N \Rightarrow m$ as $N \to \infty$. Equivalently, there is an event $\Omega_0 \subseteq \Omega$ with $\mathbb{P}(\Omega_0) = 1$ such that for every $\omega \in \Omega_0$ and every $f \in C_b(\mathbb{R}^d)$,

$$\frac{1}{N}\sum_{i=1}^{N} f\big(X_i(\omega)\big) \;\longrightarrow\; \int_{\mathbb{R}^d} f \,\mathrm{d}m.$$

Proof. Let $\mathcal{F} = \{f_1, f_2, \dots\} \subseteq C_b(\mathbb{R}^d)$ be the countable convergence-determining family of lemma 1.34. Fix k. The random variables $Y_i := f_k(X_i)$ are i.i.d. (measurable functions of the i.i.d. X_i) and bounded, $|Y_i| \leq \|f_k\|_\infty < \infty$, with mean

$$\mathbb{E}[Y_i] \;=\; \mathbb{E}[f_k(X_i)] \;=\; \int_{\mathbb{R}^d} f_k \,\mathrm{d}\,\mathcal{L}(X_i) \;=\; \int_{\mathbb{R}^d} f_k \,\mathrm{d}m,$$

the middle equality being the transfer identity (1.3) for the pushforward $\mathcal{L}(X_i) = (X_i)_{\#}\mathbb{P}$ (lemma 1.15), applied to the bounded continuous $g = f_k$, and the last equality the hypothesis $\mathcal{L}(X_i) = m$. By the strong law for bounded variables (lemma 1.37), there is an event Ω_k with $\mathbb{P}(\Omega_k) = 1$ on which, recalling (1.8),

$$\int_{\mathbb{R}^d} f_k \,\mathrm{d}\mu^N \;=\; \frac{1}{N}\sum_{i=1}^{N} f_k(X_i) \;\longrightarrow\; \int_{\mathbb{R}^d} f_k \,\mathrm{d}m.$$

Set $\Omega_0 = \bigcap_{k\geq 1} \Omega_k$. As a countable intersection of full-measure events, $\mathbb{P}(\Omega_0) = 1$ (its complement is a countable union of null sets, hence null by subadditivity). On Ω_0 the convergence $\int f_k \,\mathrm{d}\mu^N \to \int f_k \,\mathrm{d}m$ holds *simultaneously for every* k, so the defining hypothesis of lemma 1.34 is met – with $\nu_N = \mu^N(\omega)$ and $\nu = m$ – and we conclude $\mu^N(\omega) \Rightarrow m$ for every $\omega \in \Omega_0$. This is the claim. □

Three remarks fix the meaning and the limitations of this computation.

Remark 1.39 (What is and is not claimed)**.** The order of quantifiers is the whole point. Worked computation 1.38 produces a *single* full-measure event Ω_0 on which weak convergence holds against *all* test functions at once – this is stronger than, and follows from, having a full-measure event for each fixed f separately, precisely because the countable family $\mathcal{F}$ determines weak convergence and a countable intersection of full-measure events is again full measure. Note also what is *not* asserted: the empirical measure μ^N has no density and never converges to m in any L^p sense of densities; convergence is weak, tested by integration, exactly as the heuristic of section 1.1 predicted.

Remark 1.40 (Generality and the role of moments)**.** The statement requires no moment assumption on m: it holds for every probability measure, because we only ever apply the strong law to the *bounded* variables $f_k(X_i)$. The companion quantitative theory – rates of convergence and convergence in the Wasserstein metric $\mathcal{W}_2$ – does require moments and is genuinely deeper; it is taken up in chapter 4. Two distinct scales must not be conflated here. The *metric* rate $\mathbb{E}\Big[\mathcal{W}_2(\mu^N, m)\Big]$ is dimension-dependent: for i.i.d. samples it scales like $N^{-1/d}$ in dimension $d > 4$ (with borderline logarithmic corrections at $d = 2, 4$), the Fournier–Guillin estimate developed in chapter 4 [69], and is *not* $O(N^{-1/2})$ in general. The $N^{-1/2}$ scale is the size of the Gaussian *fluctuations of a smooth linear statistic* $\int f \, \mathrm{d}\mu^N$ around its mean – the central limit theorem of chapter 3 – and it is this linear-statistic structure, not the full $\mathcal{W}_2$ distance, that drives the $O(N^{-1/2})$ propagation-of-chaos couplings of chapters 7 and 13. The present result is the qualitative backbone all of these quantitative refinements stand on.

Physics note

Worked computation 1.38 is the rigorous form of the thermodynamic-limit statement that a macroscopic system is described by a density profile rather than by the microstate. The empirical measure μ^N is the occupation measure of N identical, independently prepared agents; its almost-sure weak limit m is the deterministic single-particle distribution – the analog of the one-body density to which a noninteracting ensemble self-averages. The almost-sure character is the precise meaning of "self-averaging": a *single* realization of the large system already reproduces the ensemble law, with no need to average over disorder. The fluctuations the pointwise limit discards are of order $N^{-1/2}$ by the central limit theorem (chapter 3); they vanish in the limit, which is why the leading-order description is deterministic. Everything genuinely new in mean field *game* theory begins one layer up: there the law m is not given by a preparation postulate but is the self-consistent output of a Nash fixed point, so that the agents whose empirical measure converges to m are simultaneously *optimizing against* m. That strategic closure has no thermodynamic analogue and is the subject of Part II.

Rigor ledger

Proved.

- The basic apparatus: σ-algebras and measures with their continuity properties (definitions 1.2 and 1.5 and proposition 1.6); measurable functions, stable under pointwise limits (proposition 1.13); the Lebesgue integral and its identity $\int f \, \mathrm{d}\mu^N = \frac{1}{N}\sum_i f(X_i)$ (section 1.3, eq. (1.8)).

- The three interchange theorems – monotone convergence, Fatou, dominated convergence – with full proofs and the differentiation-under-the-integral corollary (theorems 1.20 and 1.23, lemma 1.22, and corollary 1.24).

- Dynkin's π–λ theorem and the resulting uniqueness of a measure from a generating π-system (theorem 1.9 and corollary 1.10); the section-measurability lemma (lemma 1.25); existence *and* uniqueness of the product measure (theorem 1.26) and uniqueness of

Lebesgue measure with its local-finiteness characterization (theorem 1.11), all proved in-book.

- Tonelli–Fubini for σ-finite spaces (theorem 1.27); additivity of the integral and Young's inequality (used to prove Hölder/Minkowski, proposition 1.30); completeness of L^p for all $1 \le p \le \infty$ – including the $p = \infty$ case – via the monotone-control/dominated-passage architecture (theorem 1.31); and density of simple, C_c, and C_c^∞ functions in L^p (proposition 1.32).

- The transfer identity for pushforward integrals (lemma 1.15), Markov's inequality (in the proof of lemma 1.37), and the existence of a countable convergence-determining family on $\mathbb{R}^d$ (lemma 1.34), all proved from scratch within the chapter's tools.

- The self-contained model computation: the almost-sure weak limit of the empirical measure of N i.i.d. draws (worked computation 1.38), together with its two elementary engines, Borel–Cantelli (lemma 1.36) and the bounded strong law (lemma 1.37), both proved from scratch.

Assumed.

- The *Carathéodory extension theorem* – the existence half of theorem 1.11 (extending the box premeasure to a countably additive measure on $\mathcal{B}(\mathbb{R}^d)$). This is the book's single accepted external measure-theoretic import; it is named explicitly at the point of use, and everything else is proved in-book. Reference: [123, Ch. 1].

Deferred.

- The full theory of weak convergence – Portmanteau, Lévy–Prokhorov metrizability, Prokhorov tightness/compactness – to chapter 3.

- Hilbert-space structure of L^2, duality, weak/weak-$*$ topologies, the spectral theorem, and the construction of the graphon operator with its Hilbert–Schmidt property under **(A-Reg-W)** (remark 1.28, proposition 2.24) to chapter 2; the spectral payoff of $\mathbb{W}$ (eigenvector centrality, spectral gap, low-rank reductions) to chapter 18.

- Quantitative empirical-measure convergence and Wasserstein rates to chapters 4 and 7.

Open.

- Nothing in this chapter is open: it is curated classical analysis. The genuinely open questions of the book begin once the limiting law m is determined by a strategic fixed point rather than by an i.i.d. preparation; see the open ledgers of chapters 11, 24 and 31.

Bibliographic notes

The measure-theoretic core of this chapter – σ-algebras, the construction of Lebesgue measure, the Lebesgue integral, the monotone, Fatou, and dominated convergence theorems, product measures and Fubini–Tonelli, and the L^p spaces with the Riesz–Fischer completeness theorem – follows the development in Stein and Shakarchi's *Real Analysis* [123, Ch. 1, 2, 6], the roadmap text for this chapter; the L^p inequalities, density statements, and the $p = \infty$ completeness are as in Brézis [31, Ch. 1, 4]. The Carathéodory extension and π–λ arguments underlying theorems 1.11 and 1.26 are standard and may be read in either source.

The empirical-measure theorem (worked computation 1.38) is the almost-sure weak law of large numbers for empirical measures, often attributed to Varadarajan; a textbook treatment, together with the convergence-determining-class machinery of lemma 1.34, is in Billingsley [23, Ch. 1]. The elementary fourth-moment route to the strong law (lemma 1.37) and the Borel–Cantelli lemma are classical; see Durrett [62, Ch. 1–2], which also contains the central limit theorem behind the $N^{-1/2}$ fluctuation scaling mentioned in the physics note. The reading of the empirical measure as the fundamental object of the mean-field limit, and the program of refining worked computation 1.38 into quantitative propagation-of-chaos statements, is developed probabilistically in Carmona and Delarue [40, Vol. I, Ch. 1] and is the direct ancestor of chapters 7 and 13. The role of L^2 kernels and Hilbert–Schmidt operators foreshadowed in remark 1.28 is the operator-theoretic side of the graphon formalism of Lovász [99], taken up in chapter 18. The Wasserstein geometry in which the limiting law m actually lives, and the quantitative form of empirical-measure convergence, is the subject of chapter 4 and of Villani [131].

Chapter 2

Functional Analysis for Fields and Operators

The previous chapter built measures and the Lebesgue spaces L^p as the rigorous home of densities and population profiles (chapter 1). Those spaces are, so far, only *sets of functions with an addition law*: we can add two densities and rescale one, but we cannot yet say how large a field is, how far apart two fields are, whether a sequence of fields is converging, or—most consequentially for what is to come—whether two fields point in the "same direction." This chapter supplies the missing structure, and it arrives in two flavors that the reader should hold apart from the outset. The first is *geometry*: a norm makes "size" and "distance" meaningful, and in the Hilbert case an inner product makes "angle," "orthogonality," and "projection" meaningful as well, so that a space of functions becomes a place where Euclidean reasoning is legal. The second is *operators*: linear maps that carry one field into another—the verbs to the nouns that geometry describes. Geometry tells us what the objects are and how they sit relative to one another; operators tell us what acts on them. Nearly every technical object in this chapter is one or the other, and the chapter as a whole is the deliberate assembly of a single toolbox out of these two kinds of part.

Almost every object we will eventually call by a more exotic name is, underneath, one of the structures assembled here, and it is worth naming the three that matter most before they arrive, so that the reader later meets them as old acquaintances rather than surprises. The graphon coupling $\mathbb{W}$ of appendix A.5—the kernel through which a continuum of agent types influence one another—will turn out to be nothing more exotic than a self-adjoint Hilbert–Schmidt integral operator, and that single identification is what later lets us diagonalize the interaction into the network's normal modes. The value function of an optimal control problem (chapter 6) will live not in a space of smooth functions but in a Sobolev space, precisely because the Hamilton–Jacobi–Bellman equation that governs it only ever differentiates it *weakly*, against test functions, so we must first build a space in which such a derivative is legal at all. And every existence theorem in the book—for a single Nash equilibrium (chapter 10), for a graphon equilibrium (chapter 24)—is, stripped to its core, a fixed-point theorem applied to a best-response map on one of these spaces: an equilibrium is a population behavior that reproduces itself under everyone's optimal response, and proving one exists is proving that such a self-reproducing point exists. Geometry, operators, fixed points: the chapter exists to put these three in the reader's hands.

We assume from chapter 1 the construction of $L^p(\Omega, \mathcal{F}, \mu)$ for a σ-finite measure space, their completeness (the Riesz–Fischer theorem theorem 1.31), the density of simple and of smooth functions, and the Fubini–Tonelli theorem (theorem 1.27); these are the load-bearing imports, and we will call on each by name at the moment it does work rather than stockpile them here. The *weak convergence of probability measures* enters this chapter only by name: its lightweight definition is already in hand from definition 1.33 (testing against bounded continuous functions, used there in the empirical-measure computation worked computation 1.38), and its full theory—the Portmanteau theorem, Lévy–Prokhorov metric, and Prokhorov's compactness theorem—is developed in chapter 3. We mention it now only to wall it off, because the weak and weak-$*$ *operator* topologies introduced in this chapter are its functional-analytic namesakes and must be kept deliberately distinct from it: this is the first appearance of a "three weak words" bookkeeping we make explicit in remark 2.15. Beyond that one borrowed definition, nothing in this chapter uses probability at all—everything here is L^2 geometry and the linear algebra of operators—so the reader may safely set randomness aside until chapter 3 takes it up again.

The plan is linear. Section 2.1 fixes Banach and Hilbert spaces and the inner-product geometry we exploit relentlessly. Section 2.2 introduces bounded operators, dual spaces, and adjoints. Section 2.3 introduces the weak and weak-$*$ topologies and the Banach–Alaoglu theorem—the abstract compactness that powers existence proofs. Section 2.4 treats compact and Hilbert–Schmidt operators, where the graphon first appears. Section 2.5 proves the spectral theorem for compact self-adjoint operators; this is staged deliberately, because in chapter 18 the graphon's spectrum becomes the network's "band structure." Section 2.6 develops Sobolev spaces to exactly the depth that PDE existence needs. Section 2.7 collects the three fixed-point theorems—Banach, Schauder, Kakutani—that are the engines of every existence proof to come.

Throughout, scalars are real unless a statement explicitly needs the complex field; we flag the distinction where the spectral theory makes it matter. We write $\|\cdot\|_X$ for the norm of a space X, dropping the subscript when context is unambiguous, and $\langle \cdot, \cdot \rangle_H$ for the inner product of a Hilbert space H.

2.1 Banach and Hilbert spaces

A field is a function, and chapter 1 has already handed us spaces of such functions, the Lebesgue spaces L^p. What those spaces still lack is the geometry that turns a vector space into a place where analysis can be carried out: a way to measure how big a function is, and through that a way to decide when a sequence of functions has settled down to a limit. This section installs exactly that, in two graded steps the reader should keep distinct. First we equip L^p with a *norm*, which supplies size and, with it, a notion of convergence; demanding that the resulting metric be *complete*—that every Cauchy sequence actually converges to a point still inside the space—promotes a normed space to a *Banach* space, and it is this completeness that licenses the move repeated throughout the book, building a desired object as the limit of an approximating sequence and then concluding that the limit is really there. Next, for the single exponent $p = 2$, we add a second and stronger structure, an *inner product*, which records not merely the length of a function but the angle between two of them, and brings with it orthogonality, projection, and a privileged basis. Completeness and inner-product geometry are the two ingredients being layered on, and the order is deliberate: completeness is the property *every* space we use must have, whereas the inner product is the luxury that *only* L^2 enjoys and around which, as we will see again and again, much of the book is engineered. We begin with the norm.

2.1.1 Normed and Banach spaces

Both capabilities the section preamble promised—a measure of how big a function is and, through it, a test for when a sequence of them has settled down to a limit—issue from a single piece of equipment, the norm: it assigns to each function one nonnegative number, its size, and a notion of size is automatically a notion of distance—two functions are close when their difference is small—and a notion of distance is automatically a notion of convergence. So the norm is the whole of the first capability and the seed of the second. We begin by writing down what we demand of it.

Definition 2.1 (Normed and Banach space)**.** A *normed vector space* is a real (or complex) vector space X equipped with a map $\|\cdot\| : X \to [0, \infty)$ such that, for all $x, y \in X$ and scalars λ,

1. $\|x\| = 0 \iff x = 0$ (positive definiteness),
2. $\|\lambda x\| = |\lambda| \, \|x\|$ (homogeneity),
3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

A normed space is a *Banach space* if it is complete: every Cauchy sequence converges to a limit in X.

None of the three axioms is decorative; each rules out a way the notion of "size" could fail to behave like the lengths and distances we reason with by reflex. Positive definiteness says that the only function of size zero is the function that is identically zero—without it "$\|x\| = 0$" would no longer mean "x has vanished," the implication on which every uniqueness argument secretly runs (to prove two solutions x, y coincide one proves $\|x - y\| = 0$, and that is worth nothing unless it forces $x = y$). It is exactly here that the passage from chapter 1 leaves a fingerprint: on L^p a function that is nonzero only on a set of measure zero must be declared equal to zero, for otherwise it would have $\int |f|^p \, \mathrm{d}\mu = 0$ yet $f \neq 0$ and axiom (1) would fail; this is why the elements of L^p are equivalence classes modulo null sets, not literal functions. Homogeneity says size scales the way length does—doubling a field doubles its size, and the sign of a scalar cannot change a magnitude—so that the norm is compatible with the linear structure rather than indifferent to it. The triangle inequality is the substantive one: it says detours do not shorten distances, $\|x + y\| \leq \|x\| + \|y\|$, and it is what makes the induced distance $d(x, y) = \|x - y\|$ an honest metric and, just as importantly, what makes the norm a *continuous* function of its argument, so that estimates can be chained. Nearly every inequality in this book is, after enough unwrapping, the triangle inequality applied to a shrewdly chosen decomposition $x = (x - x_n) + x_n$; without it "small errors accumulate controllably" would be false and analysis would have no purchase.

A norm thus delivers convergence for free: $x_n \to x$ means $\|x_n - x\| \to 0$. But it does *not*, by itself, deliver the one thing we most want from convergence—a guarantee that a sequence which is *trying* to converge actually has somewhere to converge to. A sequence is Cauchy when its terms get and stay arbitrarily close to one another, $\|x_n - x_m\| \to 0$, which is exactly the situation in which the terms behave as though they were homing in on a limit; completeness is the demand that in such a situation the limit genuinely exists *inside the space*. The rationals under the usual absolute value show how this can fail: the truncated decimals of $\sqrt{2}$ are Cauchy but escape to a point ($\sqrt{2}$ itself) that is not rational, so $\mathbb{Q}$ is incomplete. A Banach space is, by fiat, a normed space in which no such escape is possible.

This is not a technicality to be noted and forgotten. Completeness is the property that licenses the central move of the entire subject, the move executed in one form or another by every existence theorem in this book: *build the object you want as the limit of a sequence of approximations you can construct, and then invoke completeness to conclude that the limit is a genuine member of the space and hence is the object you were after.* The sequence is always something concrete and constructible; the limit is the thing whose existence was in doubt. Three later chapters cash this single principle out in three different currencies, and it is worth seeing in advance that they are the same coin. First, in the fixed-point arguments of section 2.7, an equilibrium is produced as the limit of the iterates $x_{n+1} = F(x_n)$ of a best-response map: the contraction property forces the iterates to be Cauchy, and completeness is precisely what turns "the iterates bunch up" into "the iterates converge to an actual fixed point," the equilibrium itself. Second, in the compactness arguments of section 2.3, a solution is produced as the limit of a subsequence extracted from a bounded family of approximate solutions; extracting the subsequence is one job, but knowing its limit lives in the space—and is therefore a candidate solution rather than a phantom—is completeness doing its work again. Third, in the Galerkin and Lax–Milgram constructions of section 2.6, the solution of a partial differential equation is assembled as the limit of its projections onto finite-dimensional subspaces, exact finite computations whose limit exists only because the ambient Sobolev space is complete. Iteration, extraction, projection: three procedures, one underlying guarantee. Completeness is what each of them spends at the decisive step, and it is the reason the whole book insists on working in Banach (and, better, Hilbert) spaces rather than in the bare vector spaces of chapter 1.

With the abstract demand in place, it pays to fix the handful of concrete spaces that will recur, so that the names below arrive later as familiar rooms rather than fresh definitions.

Example 2.2 (The working spaces)**.** The following Banach spaces recur.

1. $\mathbb{R}^d$ with any norm; all norms on a finite-dimensional space are equivalent and complete.

2. For a measure space $(\Omega, \mathcal{F}, \mu)$ and $1 \leq p < \infty$, the Lebesgue space $L^p(\Omega, \mu)$ with $\|f\|_{L^p} = \left(\int_\Omega |f|^p \, \mathrm{d}\mu\right)^{1/p}$, and $L^\infty(\Omega, \mu)$ with the essential-supremum norm. Completeness is the Riesz–Fischer theorem theorem 1.31. For the graphon we will overwhelmingly use $\Omega = [0, 1]$ with Lebesgue measure, written $L^p[0, 1]$, and the product $L^p([0, 1]^2)$.

3. For a compact metric space K, the space $C(K)$ of continuous functions with the uniform norm $\|f\|_\infty = \sup_{x\in K} |f(x)|$.

These three are worth walking through rather than merely listing, because each plays a distinct and recurring role. The finite-dimensional space $\mathbb{R}^d$ is the trivial case, but trivial in a way we will lean on: because all norms on it are equivalent and it is automatically complete, we never have to worry about *which* norm we put on the finite-dimensional state space of an agent, and the Galerkin method of section 2.6 will reduce an infinite-dimensional PDE to a sequence of problems posed on exactly such spaces, where existence is elementary linear algebra. The Lebesgue spaces L^p are the main characters: they are the home of the densities and population profiles built in chapter 1, their completeness is not free but is bought by the Riesz–Fischer theorem (theorem 1.31)—which is precisely the statement that an L^p-Cauchy sequence of functions converges to an L^p function—and the special instances $L^2[0,1]$ and $L^2([0,1]^2)$ will host, respectively, the fields the graphon operator acts on and the graphon kernel itself. Finally $C(K)$, the continuous functions on a compact set under the sup norm, is the space in which we will phrase uniform estimates: it is where the value functions live before we have learned to differentiate them weakly, and its sup-norm completeness is the familiar fact that a uniform limit of continuous functions is continuous. The reader should keep these three in view as the recurring settings; almost every later inequality is posed in one of them.

Of these working spaces, one is singled out for special treatment in the next subsection, and the reason is worth stating now because it organizes much of the book. The norm of L^p, for any p, faithfully measures the size of a function, and that is all a general Banach space ever provides: a length for each vector and a distance between each pair. What it does not provide is any sense of *direction*—no way to say that two fields are aligned, or opposed, or at right angles. For a single exponent, $p = 2$, the norm turns out to arise from a richer object that does record direction, and that single piece of extra structure is responsible for nearly every clean theorem in the chapters to come. That structure is the *inner product*, and the next subsection installs it—and with it the angle, the Pythagorean decomposition, and the projection onto preferred directions that the bare norm of L^p can never supply.

2.1.2 Inner products and Hilbert spaces

The previous subsection closed on a deliberate complaint, and this one answers it. The norm of L^p, for any exponent, faithfully reports the size of a function and the distance between two functions, and for $p \neq 2$ that is all it can ever do: it knows nothing of *angle*. There is no way to declare two fields orthogonal, no Pythagorean theorem to split a field into independent components whose sizes add in quadrature, and no operation that drops a field perpendicularly onto a subspace of preferred directions. These are not exotic luxuries; they are the everyday moves of Euclidean geometry, the ones that make linear algebra in $\mathbb{R}^d$ feel concrete, and a space that lacks them is a space in which one can measure but cannot resolve. What restores all three at once is a single richer object from which the norm descends—an *inner product*, recording not merely the length of a vector but the relative orientation of two of them. Exactly one Lebesgue exponent admits such an object: $p = 2$, where $\|f\|_{L^2}^2 = \int |f|^2$ is visibly a quadratic form and so can come from a symmetric pairing of f with itself. That single accident is why the entire book is engineered to live in L^2, and it is what makes the spectral theory of section 2.5 and the variational PDE theory of section 2.6 as clean as they are. We install the inner product now, record the one inequality it forces, and then extract from it the three structural theorems—projection, Riesz representation, Fourier expansion—that a general Banach space cannot supply at any price.

Definition 2.3 (Inner product, Hilbert space)**.** An *inner product* on a real vector space H is a map $\langle\cdot,\cdot\rangle\colon H \times H \to \mathbb{R}$ that is bilinear, symmetric ($\langle x, y\rangle = \langle y, x\rangle$), and positive definite ($\langle x, x\rangle > 0$ for $x \neq 0$). It induces the norm $\|x\| = \langle x, x\rangle^{1/2}$. A *Hilbert space* is an inner-product space complete in this norm. Over the complex field one requires instead $\langle x, y\rangle = \overline{\langle y, x\rangle}$ (Hermitian symmetry) and linearity in the first argument; then $\langle\cdot,\cdot\rangle$ is conjugate-linear in the second.

Notation watch

A physicist writes the inner product as the bra–ket $\langle\psi|\phi\rangle$, which is conjugate-linear in the *first* (bra) slot and linear in the second (ket). The mathematics convention used in this book—and in definition 2.3—is the opposite: $\langle\cdot,\cdot\rangle$ is linear in the *first* argument and conjugate-linear in the second. For the real Hilbert spaces that carry the entire graphon theory the distinction evaporates, since the inner product is then symmetric and bilinear and $\langle f, g\rangle = \int fg$. We flag the convention only because the complex spectral theorem (theorem 2.26) is occasionally invoked, and there the placement of the conjugate matters for which operator one calls self-adjoint.

Example 2.4 (L^2 as a Hilbert space)**.** $L^2(\Omega, \mu)$ is a Hilbert space with $\langle f, g\rangle = \int_\Omega f\,\overline{g}\,\mathrm{d}\mu$. On $[0, 1]$ this is $\langle f, g\rangle = \int_0^1 f(x)\,\overline{g(x)}\,\mathrm{d}x$. The Cauchy–Schwarz inequality $|\langle f, g\rangle| \le \|f\|\,\|g\|$ here is the statement that the correlation of two square-integrable fields is controlled by their energies; it is used so often it ceases to be visible.

Before extracting the deep theorems it is worth isolating the one algebraic identity that drives all of them, because it is the precise sense in which an inner-product norm is more rigid than a bare norm. Expanding $\|x \pm y\|^2 = \langle x \pm y, x \pm y\rangle = \|x\|^2 \pm 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2$ and adding the two signs cancels the cross term and leaves the *parallelogram law*

$$\|x + y\|^2 + \|x - y\|^2 = 2\,\|x\|^2 + 2\,\|y\|^2\,,$$

the statement that the squared diagonals of a parallelogram equal the sum of the squares of its four sides. No general Banach norm satisfies it—the L^p norm fails it for $p \neq 2$, which is the analytic fingerprint of "no angle"—and conversely a norm that does satisfy it comes from an inner product, recoverable by the polarization identity that subtracting the two signs produces, $4\,\mathrm{Re}\langle x, y\rangle = \|x + y\|^2 - \|x - y\|^2$. The parallelogram law is the engine of the projection theorem below; the Cauchy–Schwarz inequality of example 2.4, itself a one-line consequence of positive-definiteness applied to $x - \frac{\langle x,y\rangle}{\|y\|^2}y$, is the engine of almost everything else. With these two in hand, the inner product buys three structural theorems that no general Banach space enjoys. We state them because each is used directly later, and we prove them in full: each is short, and the proofs are exactly where the Hilbert geometry earns its keep. We follow [123, Ch. 4] and [31, Ch. 5] for attribution.

The first of the three is the one that makes a Hilbert space a place where optimization has guaranteed answers. We will repeatedly need to find, inside a set of admissible objects, the one closest to a given target: the conditional expectation of chapter 3 is the closest σ-algebra-measurable random variable to a given one in the L^2 sense; the finite-element and Galerkin solutions of section 2.6 are the best approximations in energy to a PDE solution from a finite-dimensional trial space; a best response in an equilibrium computation is the minimizer of a cost over an admissible set. In a general Banach space such a nearest point may fail to exist or fail to be unique. The next theorem says that in a Hilbert space, against a closed convex target, it always exists and is always unique, and it identifies the minimizer by a checkable inequality rather than by the unworkable infimum itself.

Theorem 2.5 (Projection onto a closed convex set)**.** *Let H be a Hilbert space and $C \subseteq H$ a nonempty closed convex set. For every $x \in H$ there is a unique point $P_C x \in C$ of minimal distance, $\|x - P_C x\| = \inf_{c\in C} \|x - c\|$, characterized by the variational inequality $\langle x - P_C x, c - P_C x\rangle \leq 0$ for all $c \in C$. When $C = M$ is a closed subspace, P_M is linear, $\|P_M\| \leq 1$, and $H = M \oplus M^\perp$ orthogonally, where $M^\perp = \{y : \langle y, m\rangle = 0 \ \forall m \in M\}$.*

Proof. Existence. Write $d = \inf_{c\in C} \|x - c\|$. By definition of the infimum we may pick a *minimizing sequence* $c_n \in C$ with $\|x - c_n\|^2 \to d^2$; the content of the theorem is that this sequence, chosen only to make the distances tend to the infimum, automatically converges. This is the place the parallelogram law does its work. Applying it to the two vectors $u = x - c_n$ and $v = x - c_m$, and noting $u - v = c_m - c_n$ while $u + v = 2x - (c_n + c_m) = 2\big(x - \frac{c_n+c_m}{2}\big)$, gives

$$\|c_n - c_m\|^2 = 2\|x - c_n\|^2 + 2\|x - c_m\|^2 - 4\Big\|x - \tfrac{c_n+c_m}{2}\Big\|^2.$$

Now convexity enters, and it enters in exactly one place: it guarantees that the midpoint $\frac{c_n+c_m}{2}$ again lies in C, so its distance to x is at least the infimum d, and the subtracted term is at least $4d^2$. Hence $\|c_n - c_m\|^2 \leq 2\|x - c_n\|^2 + 2\|x - c_m\|^2 - 4d^2$, and as $n, m \to \infty$ the two leading terms each tend to $2d^2$, so the right-hand side tends to $2d^2 + 2d^2 - 4d^2 = 0$. Thus (c_n) is Cauchy. Here the second hypothesis does its work: *completeness* of H supplies a limit and *closedness* of C guarantees that limit is still in C, so $P_C x := \lim c_n \in C$, and by continuity of the norm $\|x - P_C x\| = d$. Drop convexity and the midpoint can leave C, so the estimate collapses; drop closedness or completeness and the Cauchy sequence can converge to a forbidden or nonexistent point—each hypothesis blocks one specific failure.

Uniqueness. The same identity settles it without any further work. If $c, c' \in C$ both achieve the distance d, take the minimizing "sequence" $c, c', c, c', \ldots$; the displayed identity with $c_n = c$, $c_m = c'$ reads $\|c - c'\|^2 \leq 2d^2 + 2d^2 - 4d^2 = 0$, so $c = c'$.

The variational characterization. The infimum is unusable as a test for whether a candidate is the minimizer; we convert it to a checkable inequality. Fix any $c \in C$. By convexity the segment $(1-t)P_C x + tc$ lies in C for $t \in [0, 1]$, so the scalar function

$$g(t) = \|x - (1-t)P_C x - tc\|^2 = \|(x - P_C x) - t(c - P_C x)\|^2$$

is minimized over $[0, 1]$ at $t = 0$. Expanding the square, $g(t) = \|x - P_C x\|^2 - 2t\langle x - P_C x, c - P_C x\rangle + t^2\|c - P_C x\|^2$, so $g'(0) = -2\langle x - P_C x, c - P_C x\rangle$. A minimum at the left endpoint forces $g'(0) \geq 0$, i.e. $\langle x - P_C x, c - P_C x\rangle \le 0$, which is the asserted variational inequality. Conversely, if $P_C x \in C$ satisfies it for all $c \in C$, then for any competitor c,

$$\|x-c\|^2 = \|(x-P_Cx)+(P_Cx-c)\|^2 = \|x-P_Cx\|^2+\|P_Cx-c\|^2-2\langle x-P_Cx, c-P_Cx\rangle \ge \|x-P_Cx\|^2,$$

because the last inner product is ≤ 0; so $P_C x$ is indeed the minimizer and the characterization is exact.

The subspace case. When $C = M$ is a closed subspace the inequality upgrades to an equality, because a subspace contains lines, not just segments. Given $m \in M$, both $P_M x + tm$ and $P_M x - tm$ lie in M for every $t > 0$; feeding each into the variational inequality gives $\mp t\,\langle x - P_M x, m\rangle \le 0$ for both signs, which is only possible if $\langle x - P_M x, m\rangle = 0$. Thus $x - P_M x \in M^\perp$, and the trivial decomposition $x = P_M x + (x - P_M x)$ exhibits x as a member of M plus a member of $M^\perp$. The decomposition is unique: if $x = m_1 + m_1^\perp = m_2 + m_2^\perp$ with the obvious memberships, then $m_1 - m_2 = m_2^\perp - m_1^\perp$ lies in $M \cap M^\perp$, and any vector orthogonal to itself has $\langle v, v\rangle = 0$, hence $v = 0$ by positive-definiteness. This is the orthogonal direct sum $H = M \oplus M^\perp$. Linearity of P_M now follows for free from this uniqueness: for scalars a, b the vector $aP_M x + bP_M y$ lies in M and the remainder $a(x - P_M x) + b(y - P_M y)$ lies in $M^\perp$, so by uniqueness applied to $ax + by$ we read off $P_M(ax + by) = aP_M x + bP_M y$. Finally, since $x - P_M x \perp P_M x$, the Pythagorean identity gives $\|x\|^2 = \|P_M x\|^2 + \|x - P_M x\|^2 \ge \|P_M x\|^2$, so $\|P_M x\| \le \|x\|$ and $\|P_M\| \le 1$ (with equality unless $M = \{0\}$, since P_M fixes M pointwise). $\square$

The orthogonal decomposition $H = M \oplus M^\perp$ is not a formality; it is the geometric fact behind two constructions the book leans on constantly. The first is conditional expectation, which chapter 3 builds precisely as the L^2-orthogonal projection of a random variable onto the closed subspace of variables measurable with respect to a sub-σ-algebra—the best mean-square predictor is literally a foot of perpendicular. The second is the “best approximation in energy” principle of variational PDE theory: when the energy norm of section 2.6 is the relevant inner product, the Galerkin approximation to a solution from a finite-dimensional trial space M is P_M of the true solution, and the error $u - P_M u$ is orthogonal in energy to the whole trial space. Both are this one theorem wearing different clothes.

The second theorem identifies what the bounded linear functionals on a Hilbert space *are.* A linear functional is a scalar-valued linear measurement of a field—an “observable,” in the physicist’s idiom—and the dual space H^* of all bounded ones is generally an abstract object one knows only through its defining property. Riesz’s theorem collapses this abstraction: on a Hilbert space every bounded functional is just inner product against a fixed vector, so H^* is a faithful copy of H itself. This is far from automatic; it is exactly the payoff of having geometry. We will spend it twice over. It is what lets weak convergence in H (section 2.3) be read concretely as $\langle x_n, y\rangle \to \langle x, y\rangle$ for every y, against vectors rather than against unknown functionals, and it is the representation step that powers the Lax–Milgram theorem (theorem 2.34), where a PDE in weak form is recast as “find the vector that represents a given functional” and then solved by inverting a coercive bilinear form. The proof is a single application of the projection theorem just proved: the kernel of a nonzero functional is a closed hyperplane, and its one-dimensional orthogonal complement supplies the representing vector.

Theorem 2.6 (Riesz representation). *Let H be a Hilbert space. For every bounded linear functional $\phi\colon H \to \mathbb{R}$ there is a unique $y_\phi \in H$ with $\phi(x) = \langle x, y_\phi\rangle$ for all $x \in H$, and $\|\phi\| = \|y_\phi\|$. The map $\phi \mapsto y_\phi$ identifies H with its continuous dual H^* isometrically.*

Proof. If $\phi = 0$ the representing vector is $y_\phi = 0$, so assume $\phi \neq 0$. The kernel $M = \ker\phi = \{x : \phi(x) = 0\}$ is a subspace; it is *closed* because ϕ is continuous (boundedness was used here, and is the only place it is used) and *proper* because $\phi \neq 0$. By the projection theorem $H = M \oplus M^\perp$ with $M \neq H$, so $M^\perp$ contains a nonzero vector, which we normalize to a unit vector $z \in M^\perp$, $\|z\| = 1$.

The trick is to manufacture, for an arbitrary x, a vector that ϕ kills and that therefore must be orthogonal to z. Consider $w = \phi(x)z - \phi(z)x$. Applying ϕ and using linearity, $\phi(w) = \phi(x)\phi(z) - \phi(z)\phi(x) = 0$, so $w \in M$ and hence $w \perp z$. Writing out $\langle w, z\rangle = 0$ with $\langle z, z\rangle = \|z\|^2 = 1$,

$$0 = \langle \phi(x)z - \phi(z)x, z\rangle = \phi(x)\langle z, z\rangle - \phi(z)\langle x, z\rangle = \phi(x) - \phi(z)\langle x, z\rangle,$$

which rearranges to $\phi(x) = \phi(z)\langle x, z\rangle$. To absorb the scalar $\phi(z)$ into the second slot of the inner product, recall it is conjugate-linear there, so $\phi(z)\langle x, z\rangle = \langle x, \overline{\phi(z)}\, z\rangle$; thus $y_\phi = \overline{\phi(z)}\, z$ represents ϕ on all of H (in the real case $\overline{\phi(z)} = \phi(z)$ and the conjugate is invisible). *Uniqueness.* If $\langle x, y\rangle = \langle x, y'\rangle$ for every x then $\langle x, y - y'\rangle = 0$ for every x; taking $x = y - y'$ gives $\|y - y'\|^2 = 0$, so $y = y'$ by positive-definiteness. *Isometry.* On one hand Cauchy–Schwarz gives $|\phi(x)| = |\langle x, y_\phi\rangle| \le \|x\|\,\|y_\phi\|$, so taking the supremum over $\|x\| \le 1$ yields $\|\phi\| \le \|y_\phi\|$. On the other hand evaluating at the specific argument $x = y_\phi$ gives $\phi(y_\phi) = \langle y_\phi, y_\phi\rangle = \|y_\phi\|^2$, so $\|\phi\| \ge |\phi(y_\phi)| \,/\, \|y_\phi\| = \|y_\phi\|$ (when $y_\phi \neq 0$; the case $y_\phi = 0$ is $\phi = 0$). The two bounds give $\|\phi\| = \|y_\phi\|$, and since $\phi \mapsto y_\phi$ is linear (immediate from uniqueness) and norm-preserving, it is the asserted isometric identification of H^* with H. □

This self-duality—a Hilbert space is, via Riesz, its own continuous dual—is the structural reason two later constructions are so much more concrete in H than in a general Banach space. It is what makes weak convergence in a Hilbert space the transparent statement $\langle x_n, y\rangle \to \langle x, y\rangle$ of section 2.3 rather than a quantifier over abstract functionals, and it is the abstract engine of the Lax–Milgram theorem (theorem 2.34) for elliptic PDE, where solving the equation is exactly producing the Riesz vector of a prescribed functional with respect to the energy inner product. It also gives, for free, the construction of the adjoint in definition 2.12: the functional $x \mapsto \langle Tx, y\rangle$ is bounded, so Riesz hands us a unique vector to call T^*y.

The third theorem is the one that makes a Hilbert space computable. The projection theorem gives best approximations and Riesz identifies functionals, but neither yet lets us write a generic field down in coordinates. In $\mathbb{R}^d$ one expands a vector in an orthonormal basis and manipulates the d coordinates; we want the same in infinite dimensions—a countable coordinate system in which any field is reconstructed from its inner products with the basis vectors, and in which the norm is computed from those coordinates alone. The obstruction is that "basis" can no longer mean finite linear combinations: we must allow infinite sums and must control their convergence. The right notion replaces "every vector is a finite combination" with "finite combinations are *dense*." We also need the space to admit a *countable* such system at all, which is the separability hypothesis. Both ideas are packaged in the next definition.

Definition 2.7 (Orthonormal basis)**.** A family $\{e_n\}_{n\in\mathbb{N}}$ in H is *orthonormal* if $\langle e_n, e_m\rangle = \delta_{nm}$. It is an *orthonormal basis* if its linear span is dense. A Hilbert space is *separable* if it has a countable dense subset; equivalently (by Gram–Schmidt) if it has a countable orthonormal basis.

The equivalence asserted at the end deserves a word, since it is what makes separability the right hypothesis. Given a countable dense set, the Gram–Schmidt process—orthonormalize the vectors one at a time, discarding any that are already in the span of their predecessors—produces a countable orthonormal family with the same (hence dense) span, i.e. an orthonormal basis; conversely a countable orthonormal basis has, as a countable dense subset, the rational (or rational-complex) finite combinations of its members. So "countable dense subset" and "countable orthonormal basis" are interchangeable, and the next theorem shows the second is as good as a coordinate system on $\mathbb{R}^d$.

Theorem 2.8 (Fourier expansion and Parseval)**.** *If $\{e_n\}$ is an orthonormal basis of a separable Hilbert space H, then every $x \in H$ has the norm-convergent expansion $x = \sum_n \langle x, e_n\rangle\, e_n$, and $\|x\|^2 = \sum_n |\langle x, e_n\rangle|^2$ (Parseval). Conversely any ℓ^2-summable sequence of coefficients defines an element of H.*

Proof. The argument is the projection theorem applied along the basis, run in four short steps: truncate, bound (Bessel), converge (completeness), and identify the limit (density).

Truncate. Let $S_N = \sum_{n\le N}\langle x, e_n\rangle e_n$ be the N-th partial sum. By orthonormality it is exactly the orthogonal projection of x onto the finite-dimensional subspace $\operatorname{span}\{e_1,\dots,e_N\}$: indeed $\langle x - S_N, e_k\rangle = \langle x, e_k\rangle - \langle x, e_k\rangle = 0$ for each $k \le N$, so $x - S_N$ is orthogonal to that subspace, which characterizes the projection by theorem 2.5. The Pythagorean identity for the orthogonal split $x = S_N + (x - S_N)$ reads

$$\|x\|^2 = \|S_N\|^2 + \|x - S_N\|^2 = \sum_{n\le N} |\langle x, e_n\rangle|^2 + \|x - S_N\|^2 ,$$

where $\|S_N\|^2 = \sum_{n\le N} |\langle x, e_n\rangle|^2$ again by orthonormality.

Bound. Dropping the nonnegative remainder term gives, for every N, $\sum_{n\le N} |\langle x, e_n\rangle|^2 \le \|x\|^2$. This is *Bessel's inequality*; letting $N \to \infty$ it says the coefficient sequence $(\langle x, e_n\rangle)_n$ lies in ℓ^2 with ℓ^2-norm at most $\|x\|$.

Converge. Square-summability of the coefficients makes the partial sums Cauchy: for $M > N$, again by orthonormality, $\|S_M - S_N\|^2 = \sum_{N<n\le M} |\langle x, e_n\rangle|^2$, which is a tail of a convergent series and so tends to 0. By completeness of H the Cauchy sequence converges, $S_N \to S$ for some $S \in H$. Note that up to here we have used only orthonormality; we have *not* yet used that the span is dense, and indeed without that the limit S need not equal x.

Identify. This is where density—the defining property of a *basis* as opposed to a mere orthonormal family—enters. For each fixed k, continuity of the inner product gives $\langle S, e_k\rangle = \lim_N \langle S_N, e_k\rangle = \langle x, e_k\rangle$, so $\langle x - S, e_k\rangle = 0$ for every k. Thus $x - S$ is orthogonal to every basis vector, hence to their entire linear span, hence—by continuity of $\langle x - S, \cdot\rangle$—to the closure of that span, which is all of H. A vector orthogonal to all of H is orthogonal to itself, so $\|x - S\|^2 = 0$ and $x = S$. This proves the expansion $x = \sum_n \langle x, e_n \rangle e_n$. Returning to the Pythagorean identity and letting $N \to \infty$, the remainder $\|x - S_N\|^2 \to 0$ and Bessel's inequality becomes the *equality* $\|x\|^2 = \sum_n |\langle x, e_n \rangle|^2$, which is Parseval's theorem: the norm is computed from the coordinates exactly as in $\mathbb{R}^d$.

Converse. The same Cauchy estimate, read forward, shows every square-summable coefficient sequence is realized. If $(c_n) \in \ell^2$, set $S_N = \sum_{n \le N} c_n e_n$; then $\|S_M - S_N\|^2 = \sum_{N<n\le M} |c_n|^2 \to 0$, so (S_N) is Cauchy and converges by completeness to some $x \in H$, whose coefficients are $\langle x, e_n \rangle = c_n$. Hence the coordinate map $x \mapsto (\langle x, e_n \rangle)_n$ is an isometric isomorphism $H \cong \ell^2$: a separable Hilbert space is, up to this identification, the single space ℓ^2. □

It is worth dwelling on the closing remark, because it is the hinge between this section and the spectral machinery built later, and it sets up the chapter's central object. Every Hilbert space this book uses is separable, so the apparatus just proved always applies. Concretely, $L^2[0,1]$—the home of the fields the graphon operator acts on—is separable, with the trigonometric system $\{e^{2\pi i n x}\}_{n\in\mathbb{Z}}$ (or, in the real setting, the sines and cosines, or the Legendre polynomials, or any complete orthonormal system) as a ready-made orthonormal basis; and $L^2([0,1]^2)$, the home of the kernel itself, is separable with the products $e_n(x)e_m(y)$ as a basis, a fact we will use directly in identifying Hilbert–Schmidt operators with L^2 kernels (theorem 2.22). So far these are *generic* bases: fixed once and for all, indifferent to whatever operator we later study. The decisive move of the spectral theorem (section 2.5) is to trade a generic basis for a *problem-adapted* one. When the operator in question is compact and self-adjoint, its own eigenfunctions form an orthonormal basis, and in *that* basis the operator is diagonal—it merely rescales each coordinate by an eigenvalue. For the graphon operator $\mathbb{W}$ this is the whole game: Parseval expresses any field in eigen-coordinates, $\mathbb{W}$ acts on each coordinate by multiplication by λ_n, and the resulting mode-by-mode picture is exactly the network's "band structure" exploited in chapter 18. The Fourier/Parseval theorem is thus not the end of the geometric story but the template the spectral theorem will fill in with the right, operator-specific, basis.

With the inner-product geometry installed, we have in hand the three objects the rest of the chapter will act on and through. We have the *bounded linear functionals*—the scalar observables on a field—and, via Riesz, the knowledge that on a Hilbert space they are abundant and concrete, each one inner product against a vector; in a general Banach space their abundance is not free and must be bought from the Hahn–Banach theorem instead, a distinction the next section confronts head-on. We have, through Riesz, an explicit grip on the *dual space* H^*, identified isometrically with H itself. And we have the seed of the *adjoint*: Riesz representation is precisely what will let every operator cast a shadow on the dual side, the construction from which self-adjointness—the defining property of the graphon operator and the hypothesis of the spectral theorem—is born. What we have not yet done is let anything *act* on a field. A space of fields, however richly geometric, becomes a setting for dynamics only once we introduce the maps that carry one field into another; and the right notion of such a map, as the next section takes up, is the continuous linear operator.

2.2 Bounded operators, dual spaces, and adjoints

The previous section ended by handing us three things and a promise. The things were the bounded linear functionals on a Hilbert space, the dual space H^* they fill out, and—through Riesz representation (theorem 2.6)—the assurance that on a Hilbert space these functionals are both abundant and concrete, each one inner product against a fixed vector. The promise was that a space of fields, however richly geometric, becomes a setting for dynamics only once we let something *act* on a field. A functional is the humblest such action: it reads a field and returns a single number, a scalar observable. But the actions that drive the rest of the book return not a number but another field. The graphon coupling $\mathbb{W}$ averages a population profile against its kernel and hands back a new profile; the generator of a diffusion turns one function into another; the solution operator of a partial differential equation turns data into a solution. Each is a linear map from a space of fields into a space of fields, and this section settles which such maps we admit and what structure the admissible ones carry. The short answer is the *continuous* linear map. The work is to show that continuity has a clean quantitative face—the operator norm; that the continuous maps from a space to itself form an algebra under composition whose one structural inequality is exactly what later makes iteration converge; and that the scalar-valued special case, the dual space, is rich enough to deserve the name, a richness that off a Hilbert space is not free but must be bought from the Hahn–Banach theorem.

What must be pinned down is not linearity, which every operator in the book respects because superposition of fields is built into the physics, but the analytic side: which linear maps are tame enough to commute with limits, so that from $x_n \to x$ one may conclude $Tx_n \to Tx$. For linear maps this continuity has a strikingly simple equivalent—the map may not stretch any vector by more than some fixed factor—and that factor is the natural measure of the operator's size.

Definition 2.9 (Bounded linear operator)**.** Let X, Y be normed spaces. A linear map $T\colon X \to Y$ is *bounded* if

$$\|T\|_{X\to Y} \;:=\; \sup_{x\neq 0} \frac{\|Tx\|_Y}{\|x\|_X} \;=\; \sup_{\|x\|_X\leq 1} \|Tx\|_Y \;<\; \infty.$$

For linear maps, boundedness is equivalent to continuity. The bounded operators $X \to Y$ form a normed space $\mathcal{B}(X, Y)$ under the operator norm; if Y is Banach so is $\mathcal{B}(X, Y)$. We write $\mathcal{B}(X) = \mathcal{B}(X, X)$.

The single clause in the definition that asserts "boundedness is equivalent to continuity" carries more than it shows, and since the operator norm's claim to be the right notion of size rests on it, we prove it. Suppose first that T is bounded, so $\|Tx\|_Y \leq \|T\| \, \|x\|_X$ for all x. Then by linearity $\|Tx - Tx'\|_Y = \|T(x - x')\|_Y \leq \|T\| \, \|x - x'\|_X$, so T is Lipschitz with constant $\|T\|$ and *a fortiori* continuous everywhere. Conversely, suppose T is merely continuous at the single point 0. Then for $\varepsilon = 1$ there is a $\delta > 0$ with $\|Tx\|_Y \leq 1$ whenever $\|x\|_X \leq \delta$. Given any $x \neq 0$, the rescaled vector $\delta x / \|x\|_X$ has norm exactly δ, so $\|T(\delta x / \|x\|_X)\|_Y \leq 1$, and pulling the scalar out by homogeneity gives $\|Tx\|_Y \leq \delta^{-1} \|x\|_X$. Thus T is bounded with $\|T\| \leq \delta^{-1}$. For a linear map, then, three notions collapse into one: bounded, continuous, and continuous at a single point are the same property, and the operator norm is precisely the best Lipschitz constant. Linearity is the whole of the magic here—it is what lets a lone estimate valid in a small ball around the origin be inflated, by rescaling, to a bound at every scale. Drop linearity and the shortcut evaporates: the continuous map $x \mapsto x^2$ on $\mathbb{R}$ has no global Lipschitz bound at all, which is exactly why "bounded equals continuous" is a theorem about linear maps and not a tautology.

The operator norm earns its name as a size because it is the *gain* of the map: the smallest constant C for which $\|Tx\| \leq C\,\|x\|$ holds universally, hence the largest factor by which T can amplify a vector. Two micro-examples fix the picture. On $L^2(\Omega)$ the multiplication operator $(M_a f)(x) = a(x)f(x)$ by a bounded function a has $\|M_a\| = \|a\|_{L^\infty}$: it cannot stretch any field by more than the largest value of $|a|$, and a field concentrated where $|a|$ nearly attains its essential supremum is stretched by nearly that much, so the bound is sharp. The graphon operator itself is the example we are building toward: on $L^2[0,1]$ the map $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y$ is bounded, since Cauchy–Schwarz in the inner integral gives $|(\mathbb{W}f)(x)| \leq \big(\int_0^1 W(x,y)^2\,\mathrm{d}y\big)^{1/2}\,\|f\|_{L^2}$, and integrating the square over x yields $\|\mathbb{W}f\|_{L^2} \leq \|W\|_{L^2([0,1]^2)}\,\|f\|_{L^2}$, so $\|\mathbb{W}\| \leq \|W\|_{L^2([0,1]^2)}$. A square-integrable kernel thus automatically defines a bounded operator—the first installment of the identification, completed in section 2.4, of the graphon with a Hilbert–Schmidt operator.

It is worth recording why $\mathcal{B}(X,Y)$ is itself complete when Y is, since this is the same completeness principle the chapter keeps spending, now lifted to the level of operators. If (T_n) is Cauchy in the operator norm, then for each fixed x the vectors $T_n x$ are Cauchy in Y, because $\|T_n x - T_m x\| \leq \|T_n - T_m\|\,\|x\|$; completeness of Y furnishes a limit, which we name Tx, and passing the linear relations to the limit shows T is linear. The Cauchy condition is uniform over $\|x\| \leq 1$, so the convergence $T_n \to T$ is in operator norm and the limit T is bounded. The dual space defined below is precisely the instance $Y = \mathbb{R}$ of this fact, and it is the reason every dual is a Banach space.

When the domain and codomain coincide we may compose operators, and composition meets the norm in the one inequality that organizes everything downstream. If $S, T \in \mathcal{B}(X)$ then for every x, $\|STx\| \leq \|S\|\,\|Tx\| \leq \|S\|\,\|T\|\,\|x\|$, and taking the supremum over $\|x\| \leq 1$ gives the *submultiplicative* bound $\|ST\| \leq \|S\|\,\|T\|$. A complete normed space carrying an associative, submultiplicative product with unit is a *Banach algebra*, and $\mathcal{B}(X)$ is its prototype, with composition for product and the identity Id for unit. The reason this one-line inequality matters out of all proportion to its proof is that it controls *powers*: iterating gives $\|T^n\| \leq \|T\|^n$, so the size of the n-fold composite decays geometrically the instant $\|T\| < 1$. Two payoffs follow that the book collects repeatedly. First, the Neumann series $\sum_{n\geq 0} T^n$ then converges in operator norm—its tail is dominated by the convergent geometric series $\sum_n \|T\|^n$, and completeness of $\mathcal{B}(X)$ supplies the limit—and a direct telescoping shows it sums to $(\mathrm{Id} - T)^{-1}$, so $\mathrm{Id} - T$ is invertible whenever $\|T\| < 1$; this is the workhorse that solves a linear fixed-point equation $(\mathrm{Id} - T)x = b$ by iteration. Second, and equivalently, the affine map $x \mapsto Tx + b$ is then a contraction of ratio $\|T\|$, whose unique fixed point is the limit of the iterates $x_{n+1} = Tx_n + b$, with $\|T^n\| \leq \|T\|^n$ giving the exact geometric rate of approach. This is the linear shadow of the contraction mapping theorem (theorem 2.36), and it is no accident that it looks like the self-consistent-field iteration that defines an equilibrium: when a best-response map is a contraction, this very submultiplicative estimate is what controls its iterates and certifies their convergence. Submultiplicativity, in short, is the precise algebraic fact that lets a bound on a single step govern an entire trajectory.

We now specialize the codomain to the scalar field, where the operators become measurements rather than transformations of a field, and a structure of independent importance emerges.

Definition 2.10 (Dual space)**.** The (continuous) *dual* of a normed space X is $X^* = \mathcal{B}(X, \mathbb{R})$, the space of bounded linear functionals, with the norm $\|\phi\|_{X^*} = \sup_{\|x\| \le 1} |\phi(x)|$. It is always a Banach space, even when X is not complete.

The dual is always a Banach space, even when X is not, and this costs nothing: it is the case $Y = \mathbb{R}$ of the completeness argument above, with the scalar field standing in for the complete codomain. So even a defective, incomplete space of fields possesses a complete space of observables—a small but genuine asymmetry we will exploit when we put topologies on X^* in section 2.3.

What is *not* free, and not even obvious, is that X^* holds enough functionals to be useful. The definition guarantees that the bounded functionals form a Banach space, but a Banach space can be the trivial space $\{0\}$; nothing said so far excludes a normed X on which the only bounded linear functional is the zero map. On such a space X^* would distinguish no two distinct fields, and every dual-based construction in the book—weak convergence, adjoints, convex separation—would be vacuous. On a Hilbert space Riesz representation (theorem 2.6) already dispels the worry, manufacturing for each direction a functional that points along it. Off a Hilbert space we have no inner product to build functionals from, and the abundance must be secured by other means. That bounded functionals nonetheless exist in profusion—enough to separate any two points, enough to expose any convex body along a supporting hyperplane—is the content of the Hahn–Banach theorem, which we state in the two forms we use.

Theorem 2.11 (Hahn–Banach, extension and separation)**.** *Let X be a normed space.*

1. (Extension.) *Every bounded linear functional on a subspace $M \subseteq X$ extends to a bounded linear functional on all of X with the same norm. Consequently, for every $x_0 \neq 0$ there is $\phi \in X^*$ with $\|\phi\| = 1$ and $\phi(x_0) = \|x_0\|$, so X^* separates points of X.*

2. (Separation.) *If $A, B \subseteq X$ are disjoint nonempty convex sets with A closed and B compact, there exist $\phi \in X^*$ and $\alpha \in \mathbb{R}$ with $\phi(a) \le \alpha < \phi(b)$ for all $a \in A,\ b \in B$.*

Both forms are proved in full in appendix B.2, following [31, Ch. 1]; here we expose the mechanism, because the reason the theorem is *true* is more illuminating than the bookkeeping that makes it airtight. Everything rests on a single one-dimensional extension step. Suppose a linear functional ℓ defined on a subspace M is dominated by a *sublinear* gauge p—a function with $p(x+y) \le p(x)+p(y)$ and $p(\lambda x) = \lambda p(x)$ for $\lambda \ge 0$—in the sense that $\ell \le p$ on M. We try to extend ℓ across one fresh direction $x_0 \notin M$, which amounts to choosing the single new value $c = \ell(x_0)$. For the enlarged functional on $M \oplus \mathbb{R}x_0$ to stay dominated by p we need $\ell(m) + tc \le p(m + tx_0)$ for all $m \in M$ and all real t. Splitting into the cases $t > 0$ and $t < 0$ and dividing by $|t|$—legitimate by the positive homogeneity of p—collapses this pair of requirements into the single sandwich

$$\sup_{m \in M} \left[\ell(m) - p(m - x_0)\right] \ \le\ c \ \le\ \inf_{m' \in M} \left[p(m' + x_0) - \ell(m')\right].$$

The crux is that this sandwich is never empty. For any $m, m' \in M$, subadditivity of p gives $\ell(m) + \ell(m') = \ell(m + m') \le p(m + m') \le p(m - x_0) + p(m' + x_0)$, which rearranges to exactly “the left-hand bracket $\leq$ the right-hand bracket"; hence the supremum on the left lies at or below the infimum on the right, and any c between them yields a dominated extension one dimension larger. That is the entire analytic content of the theorem; everything else is reach. To extend all the way to X one orders the dominated partial extensions by inclusion, notes that the union of any chain of them is again such an extension and so is an upper bound for the chain, and invokes Zorn's lemma to obtain a maximal element. Were its domain not all of X, the one-dimensional step would enlarge it, contradicting maximality—so the maximal extension is defined everywhere. Specializing the gauge to $p(x) = \|\ell\|_M \|x\|$ recovers the norm-preserving extension form, since domination by this p is precisely the bound $|\ell(x)| \leq \|\ell\|_M \|x\|$.

The geometric separation form is the same theorem read through the *Minkowski gauge*. To an open convex set C containing the origin one attaches $p_C(x) = \inf\{t > 0 : x/t \in C\}$, the factor by which x must be shrunk to fit inside C; convexity makes p_C sublinear, and openness makes $\{p_C < 1\} = C$. A point $x_0 \notin C$ then has $p_C(x_0) \geq 1$, so the functional $\ell(\lambda x_0) = \lambda$ defined on the line $\mathbb{R}x_0$ satisfies $\ell \leq p_C$ there; extending it by the step just described produces $\phi \in X^*$ with $\phi \leq p_C \leq 1$ on C and $\phi(x_0) \geq 1$—a hyperplane with C on one side and x_0 on the other. The two-set statement of the theorem follows by applying this to the translated difference set $A - B$, which is convex and avoids the origin precisely because A and B are disjoint, the compactness of B against the closedness of A being what upgrades weak separation to the strict inequality $\phi(a) \leq \alpha < \phi(b)$.

These two forms divide the labor cleanly between the two halves of the book. The separation form is the analytic root of convex duality: it manufactures the supporting functional that exposes a convex set along a face, and it is exactly the device that later represents a constraint by a Lagrange multiplier and underwrites the primal–dual formulations of the variational and optimal-transport arguments. The extension form is the one we lean on immediately. Its corollary that for every $x_0 \neq 0$ some unit functional attains $\phi(x_0) = \|x_0\|$ says that X^* *separates points*: distinct fields are told apart by some observable. That single fact is what makes the weak topology of section 2.3 Hausdorff, so that weak limits are unique and the compactness we extract there is worth having.

There remains the handoff to the next subsection, and Riesz representation is the bridge. Hahn–Banach secured the abundance of functionals on a general Banach space; on a Hilbert space Riesz did better, attaching to every functional a concrete representing vector. That representation is exactly the tool that lets an operator cast a shadow on the dual side, and the shadow it casts is the *adjoint*—where self-adjointness, the hypothesis of the spectral theorem and the defining symmetry of the graphon operator, is born. We turn to it now.

2.2.1 Adjoints

The previous subsection ended by naming the bridge. Hahn–Banach guaranteed that a general Banach space carries enough functionals to separate its points, but on a Hilbert space Riesz representation (theorem 2.6) did something sharper: it pinned each bounded functional to a concrete representing vector, so that the abstract dual H^* became a faithful copy of H itself. We now collect the dividend that self-duality was always pointing toward. On a Hilbert space every operator T casts a shadow on the dual side—a second operator T^* that records how T looks when probed by inner products—and the construction of that shadow is nothing but Riesz applied one functional at a time. The shadow is interesting in its own right, but its real importance is that it lets us ask a question that has no analogue for a bare bounded operator: when does an operator *coincide* with its own shadow? That coincidence is self-adjointness, and it is the single hypothesis on which the entire spectral theory of section 2.5 turns and the exact structural feature that will make the graphon operator diagonalizable. So this short subsection is where self-adjointness is born; the rest of the chapter is where it is spent.

The mechanism is worth seeing before the definition formalizes it. Fix an operator $T \in \mathcal{B}(H)$ and a vector $y \in H$. Pairing the output Tx against y produces, as x ranges over H, a scalar-valued map

$$\phi_y \colon x \longmapsto \langle Tx, y\rangle,$$

and this map is linear in x (because T is linear and the inner product is linear in its first slot) and bounded, since Cauchy–Schwarz together with boundedness of T gives $|\phi_y(x)| = |\langle Tx, y\rangle| \le \|Tx\|\,\|y\| \le \|T\|\,\|y\|\,\|x\|$. So $\phi_y \in H^*$, and Riesz representation (theorem 2.6) hands back a *unique* vector representing it—a vector that depends on y, which we name T^*y. By construction $\phi_y(x) = \langle x, T^*y\rangle$, which is precisely the defining relation below. Running Riesz once for every y assembles the map $y \mapsto T^*y$; this is the adjoint.

Definition 2.12 (Adjoint, self-adjoint operator)**.** Let H be a Hilbert space and $T \in \mathcal{B}(H)$. The *adjoint* $T^* \in \mathcal{B}(H)$ is the unique bounded operator with

$$\langle Tx, y\rangle = \langle x, T^*y\rangle \qquad \text{for all } x, y \in H.$$

Its existence and uniqueness follow from Riesz representation applied to the functional $x \mapsto \langle Tx, y\rangle$, and $\|T^*\| = \|T\|$. The operator T is *self-adjoint* if $T^* = T$, i.e. $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all x, y.

The definition asserts three things that the construction above does not by itself supply, and each deserves its line. First, that T^* is *linear*: for vectors y_1, y_2 and scalars a, b we have $\langle x, T^*(ay_1+by_2)\rangle = \langle Tx, ay_1 + by_2\rangle = \bar{a}\langle Tx, y_1\rangle + \bar{b}\langle Tx, y_2\rangle = \langle x, aT^*y_1 + bT^*y_2\rangle$ for every x, and a vector determined by its inner products against all x is unique, so $T^*(ay_1 + by_2) = aT^*y_1 + bT^*y_2$. (Over the reals the conjugates are invisible; we keep them so the statement is correct over $\mathbb{C}$, where it matters for the spectral theorem.) Second, that T^* is *bounded with the same norm*, $\|T^*\| = \|T\|$. To see it, bound T^*y by testing it against the worst x: for any unit x, $|\langle x, T^*y\rangle| = |\langle Tx, y\rangle| \le \|T\|\,\|y\|$, and taking the supremum over $\|x\| \le 1$ identifies that supremum, by Riesz isometry, with $\|T^*y\|$ itself, giving $\|T^*y\| \le \|T\|\,\|y\|$ and hence $\|T^*\| \le \|T\|$. The reverse inequality costs nothing once one notices that the whole construction is symmetric in T and T^*: the defining relation $\langle Tx, y\rangle = \langle x, T^*y\rangle$ read the other way, $\langle T^*y, x\rangle = \langle y, Tx\rangle$ (conjugate both sides of the original), exhibits T as the adjoint of T^*, so $(T^*)^* = T$ and the bound just proved applied to T^* gives $\|T\| = \|(T^*)^*\| \le \|T^*\|$. The two inequalities close to the equality $\|T^*\| = \|T\|$: passing to the adjoint never changes an operator's size. Third, that T^* is *unique*, which is immediate—two operators satisfying the defining relation would agree in every inner product $\langle x, \cdot\rangle$ and so agree as vectors for each y.

It pays to anchor all of this in the finite-dimensional case the reader already owns, because the adjoint is the exact infinite-dimensional lift of a construction from linear algebra. Take $H = \mathbb{R}^d$ with the Euclidean inner product $\langle x, y\rangle = x^\top y$, and let T act as a matrix A. Then

$\langle Ax, y\rangle = (Ax)^\top y = x^\top(A^\top y) = \langle x, A^\top y\rangle$, so the adjoint is the transpose, $A^* = A^\top$; over $\mathbb{C}$ with $\langle x, y\rangle = \sum_i x_i \bar{y}_i$ the same computation produces the conjugate transpose $A^* = \bar{A}^\top$. The defining relation $\langle Tx, y\rangle = \langle x, T^* y\rangle$ is thus the coordinate-free statement of "move the matrix to the other side of the inner product and transpose it," and self-adjointness $T^* = T$ is exactly the condition $A = A^\top$ that the matrix be *symmetric* (Hermitian over $\mathbb{C}$). The whole point of the Riesz construction is that this move survives the passage to infinite dimensions, where there is no matrix to transpose: the inner product, not a list of entries, is what carries the structure, and Riesz is what guarantees the transposed object still exists as an honest bounded operator.

Why single out the self-adjoint operators? Because symmetry is precisely the hypothesis that makes spectral analysis behave. For a real symmetric matrix the reader already knows the two facts that matter: its eigenvalues are all real, and one can choose its eigenvectors to form an orthonormal basis of $\mathbb{R}^d$, in which basis the matrix is diagonal. Neither is true for a general matrix—eigenvalues wander into the complex plane and eigenvectors may fail to span—and neither is an accident of finite dimension. Self-adjointness is exactly the structural property responsible for both, and section 2.5 will show that it continues to deliver them in infinite dimensions, with one extra hypothesis. The eigenvalues of a bounded self-adjoint operator are real, and—once we also assume *compactness*, the substitute for finite-dimensionality that keeps the spectrum discrete—its eigenvectors again form an orthonormal basis in which the operator is diagonal. The reason to isolate self-adjointness *here*, several sections before it is used, is that it is the working hypothesis of that later theorem and we want the reader to meet it as a familiar object: the abstract avatar of the real symmetric matrix, carrying the same guarantee of a real spectrum and a clean eigenbasis.

This is also the moment to plant the thread the chapter is laid out to pay off, because the graphon operator is the self-adjoint operator the book actually cares about. Recall from definition 2.9 that the graphon coupling acts by $(\mathbb{W}f)(x) = \int_0^1 W(x, y) f(y)\,\mathrm{d}y$ on $L^2[0, 1]$. Compute its adjoint directly from the definition, using the L^2 inner product $\langle f, g\rangle = \int fg$ (real-valued here) and Fubini (theorem 1.27) to exchange the order of integration:

$$\langle \mathbb{W}f, g\rangle = \int_0^1\int_0^1 W(x, y)\, f(y)\, g(x)\ \mathrm{d}y\ \mathrm{d}x = \int_0^1 f(y)\Big(\int_0^1 W(x, y)\, g(x)\ \mathrm{d}x\Big)\,\mathrm{d}y = \langle f, \mathbb{W}^* g\rangle,$$

which exhibits the adjoint as the integral operator with the *transposed* kernel, $(\mathbb{W}^* g)(y) = \int_0^1 W(x, y)\, g(x)\ \mathrm{d}x$—the kernel $W(x, y)$ read with its two arguments swapped, exactly as a matrix transpose swaps row and column indices. Therefore $\mathbb{W}$ is self-adjoint if and only if its kernel is symmetric, $W(x, y) = W(y, x)$. This is not a technical imposition but a modeling truth: a graphon encodes an *undirected* network of agent types, in which the influence of type y on type x equals the influence of x on y, so its kernel is symmetric by construction and $\mathbb{W}$ is self-adjoint automatically. That single sentence is the hinge of the chapter's design. It is the reason the compact-operator section (section 2.4) will be able to place $\mathbb{W}$ in the Hilbert–Schmidt class and the spectral section (section 2.5) will be able to diagonalize it into the mode-by-mode "band structure" that chapter 18 reads off as the network's normal modes. Self-adjointness, born here from a one-line Fubini computation, is what every one of those later payoffs cashes.

With the adjoint in hand we have completed the operator-theoretic toolkit this part of the chapter set out to assemble: bounded operators and their norm, the dual space, and now the map $T \mapsto T^*$ that ties an operator to its action on the dual side. Two distinct strands lead out of here.

The immediate one is the weak-topology section that follows, which puts operators and their duals to work on a problem we have so far only flagged: existence proofs need to extract convergent subsequences from bounded families, finite-dimensional Heine–Borel fails to provide them in infinite dimensions, and the repair is to weaken the topology until compactness returns. The more pointed strand is the pair of sections after it. Self-adjointness has been defined here in the abstract, but it does no work until it is married to compactness; the compact and Hilbert–Schmidt section supplies that partner and shows the graphon operator owns it, and the spectral section then collects the reward—the real eigenvalues and orthonormal eigenbasis previewed above. The reader should carry forward a single expectation: every time the phrase "self-adjoint" reappears, it is this construction being invoked, and the symmetry of the graphon kernel is why it applies.

2.3 Weak and weak-$*$ topologies and Banach–Alaoglu

The previous section closed by handing the operator and dual machinery to a problem we have so far only named, and the time has come to cash the promise. Every existence theorem in this book follows the same arc: one builds a sequence of approximate solutions, controls their size by an a priori estimate, extracts a convergent subsequence, and christens its limit the solution sought. The hinge of that arc is the extraction step, and it is a statement about *compactness*—about when a bounded family is forced to have a convergent subsequence. In a finite-dimensional space the Heine–Borel theorem settles the matter outright: a set is compact precisely when it is closed and bounded, so a bounded sequence in $\mathbb{R}^d$ always accumulates and a convergent subsequence can always be pulled out, full stop. The whole difficulty of infinite-dimensional analysis is that this clean equivalence fails, and fails badly: the closed unit ball, the most basic bounded set there is, is *never* compact in the norm topology once the dimension is infinite. Before we can repair the situation we must see exactly how it breaks, because the repair is shaped to the break.

The obstruction is geometric and completely concrete. In $\mathbb{R}^d$ a bounded sequence cannot run away to infinity and cannot disperse forever inside a bounded region, so it must pile up somewhere; in infinite dimensions there is always "another direction" to escape into, and one can build a bounded sequence whose terms stay rigidly far apart, so that no subsequence is even Cauchy. The device that manufactures such an escape is a single geometric fact—that a unit vector can always be found almost perpendicular to any proper closed subspace—which goes by the name of Riesz's lemma. We isolate it, because the half-separated sequence it produces is the cleanest possible witness to non-compactness, and because the lemma's mechanism (a residual vector, normalized) is one we will reuse.

Lemma 2.13 (Riesz's lemma)**.** *Let X be a normed space and $Y \subsetneq X$ a proper closed subspace. For every $\theta \in (0,1)$ there is a unit vector $x \in X$, $\|x\| = 1$, with $\operatorname{dist}(x,Y) = \inf_{y\in Y} \|x-y\| \geq \theta$.*

Proof. Since Y is a proper subspace, choose a vector $z \in X \setminus Y$. Because Y is *closed*, z sits at a strictly positive distance from it: writing $d := \operatorname{dist}(z,Y) = \inf_{y\in Y} \|z-y\|$, we have $d > 0$, for if d were 0 there would be $y_k \in Y$ with $z = \lim y_k$, forcing $z \in Y$ by closedness and contradicting the choice of z. The number d is an infimum, so it is approached but need not be attained; crucially we do not need it attained, only approached to within a controlled factor. Fix $\theta \in (0,1)$. Since $d/\theta > d$, the value d/θ is not a lower bound for the distances, so the definition of infimum supplies a $y \in Y$ with

$$d \;\le\; \|z-y\| \;\le\; \frac{d}{\theta}.$$

Normalize the residual: set $x = (z-y)/\,\|z-y\|$, a unit vector. We claim x is the vector we want. To see it, estimate the distance from x to an arbitrary $w \in Y$. Because Y is a subspace, the combination $y + \|z-y\|\; w$ again lies in Y, so its distance from z is at least d; therefore

$$\|x-w\| = \Big\|\frac{z-y}{\|z-y\|} - w\Big\| = \frac{\|z-(y+\|z-y\|\; w)\|}{\|z-y\|} \;\ge\; \frac{d}{\|z-y\|} \;\ge\; \frac{d}{d/\theta} = \theta.$$

Since $w \in Y$ was arbitrary, $\operatorname{dist}(x,Y) \ge \theta$. Each hypothesis paid its way exactly once: closedness of Y is what made $d > 0$ (without it z could sit on the boundary of Y at distance zero, and no separated vector could exist), and properness is what let us find a z outside Y to begin with. $\square$

The lemma now drives a short induction that exhibits the unit ball's non-compactness. The key observation feeding it is that in an infinite-dimensional space the spans of finitely many vectors are always both proper and closed, so Riesz's lemma can be applied to each of them in turn. We build unit vectors $x_1, x_2, \ldots$ one at a time. Take x_1 to be any unit vector. Suppose $x_1, \ldots, x_n$ have been chosen, and let $Y_n = \operatorname{span}\{x_1, \ldots, x_n\}$. This Y_n is finite-dimensional, hence complete—all norms on a finite-dimensional space are equivalent and complete (example 2.2)—hence closed in X; and it is a *proper* subspace, because an infinite-dimensional X can never be exhausted by the span of finitely many vectors. Riesz's lemma with $\theta = \frac{1}{2}$ therefore furnishes a unit vector x_{n+1} with $\operatorname{dist}(x_{n+1}, Y_n) \ge \frac{1}{2}$. In particular, since every earlier x_m (for $m \le n$) lies in Y_n, we get $\|x_{n+1} - x_m\| \ge \frac{1}{2}$. Carrying this out for all n produces an infinite sequence of unit vectors with $\|x_n - x_m\| \ge \frac{1}{2}$ for all $n \ne m$.

This sequence lives entirely inside the closed unit ball, yet no two of its terms ever come within $\frac{1}{2}$ of one another, so it has no Cauchy subsequence and *a fortiori* no convergent subsequence. The closed unit ball of an infinite-dimensional normed space is thus not sequentially compact—and since the ball is a metric space, where sequential compactness and compactness coincide, it is not compact at all. This is the precise sense in which Heine–Borel dies in infinite dimensions, and it is the obstacle the rest of the section is built to circumvent.

The diagnosis points straight at its own cure. Norm convergence demands that $x_n - x$ shrink to zero *as measured by the norm*, and that is simply too severe a demand for bounded sequences to meet—the half-separated sequence above never makes its increments small. So we ask for less. The only linear instruments we possess for probing a vector are the bounded functionals of X^*, and the weakest reasonable sense in which "x_n approaches x" is that no such instrument can tell them apart in the limit: $\phi(x_n) \to \phi(x)$ for every ϕ. This is a genuine loosening—it tests convergence one scalar reading at a time rather than all at once in the norm—and the whole payoff, to be delivered by Banach–Alaoglu, is that bounded sets recover under it the compactness they lost. The same loosening applies in two guises according to whether the sequence in question lives in X (test it against X^*) or already in the dual X^* (test it against X), and we name both.

Definition 2.14 (Weak and weak-$*$ convergence)**.** Let X be a Banach space with dual X^*.

- A sequence $x_n \in X$ *converges weakly* to x, written $x_n \rightharpoonup x$, if $\phi(x_n) \to \phi(x)$ for every $\phi \in X^*$.
- A sequence $\phi_n \in X^*$ *converges weakly-$*$* to ϕ, written $\phi_n \overset{*}{\rightharpoonup} \phi$, if $\phi_n(x) \to \phi(x)$ for every $x \in X$.

These are the topologies of pointwise convergence on X^* and on X respectively. In a Hilbert space, by Riesz representation, weak convergence reads $\langle x_n, y\rangle \to \langle x, y\rangle$ for every $y \in H$, and weak and weak-$*$ coincide.

The coincidence in the Hilbert case deserves a moment, because it is the only case the book's existence proofs ever invoke and it erases a distinction that is otherwise essential. In general X and X^* are different spaces, so weak convergence (of vectors in X, tested against X^*) and weak-$*$ convergence (of functionals in X^*, tested against X) are different notions living on opposite sides of the duality; they can be compared only inside the double dual X^{**}, and they agree exactly when X is reflexive. A Hilbert space is its own dual through Riesz representation (theorem 2.6), so the two sides fold onto one another and weak convergence becomes the single transparent statement $\langle x_n, y\rangle \to \langle x, y\rangle$ for every fixed field y. It pays to see at once that this is *strictly* weaker than norm convergence, and the cleanest witness is an orthonormal sequence. If $\{e_n\}$ is orthonormal, then for any fixed y Bessel's inequality (theorem 2.8) makes $\sum_n |\langle y, e_n\rangle|^2 \le \|y\|^2$ finite, so its terms must tend to zero, $\langle e_n, y\rangle \to 0$; thus $e_n \rightharpoonup 0$. Yet $\|e_n\| = 1$ for every n, so e_n does *not* converge to 0 in norm—indeed $\|e_n - e_m\| = \sqrt{2}$ for $n \neq m$, the very half-separation Riesz's lemma produced. The weak limit 0 has strictly smaller norm than every term of the sequence: under weak convergence mass can leak away. This one example already carries the warning that will organize the rest of the section—weak limits are slippery, the norm is only *lower* semicontinuous rather than continuous along them, and nonlinear quantities need not pass to the limit at all—and we will return to it when we distill the existence motif at the section's end.

Remark 2.15 (Three distinct "weak" words)**.** The single word "weak" is about to do triple duty across this book, and because all three of its uses are load-bearing and recur to the very end, we fix the bookkeeping once and post it as a permanent signpost. There are three arenas. *(i) The weak and weak-$*$ operator topologies* just defined: convergence of vectors or of functionals tested against the dual, written $x_n \rightharpoonup x$ and $\phi_n \overset{*}{\rightharpoonup} \phi$, the subject of the present section. *(ii) The weak convergence of probability measures* of chapter 3, written $\mu_n \Rightarrow \mu$: convergence of measures tested against bounded continuous functions, $\int f \, \mathrm{d}\mu_n \to \int f \, \mathrm{d}\mu$, the notion already met in definition 1.33 and used there to make sense of empirical measures collapsing onto a limit law. *(iii) The weak derivative* of section 2.6, written with ∂^{w}: a function's derivative defined not pointwise but through integration against smooth, compactly supported test functions, the move that lets us differentiate a value function the Hamilton–Jacobi–Bellman equation refuses to differentiate classically. All three are instances of one organizing idea—replace a strong, direct requirement by its *tested* action against a class of probe objects, trading a demand that is hard to verify head-on for a family of scalar checks that is easy—which is exactly why the same adjective attached itself to each. But they live on different spaces and probe with different test classes (bounded functionals, bounded continuous functions, smooth bumps), and they are emphatically not interchangeable: a sequence can converge in one of these senses while diverging in another. We therefore keep the symbols $\rightharpoonup$, $\Rightarrow$, and ∂^{w}

rigidly distinct throughout, so the reader can always read off which arena is in play, and chapter 3 will credit definition 1.33 when it resumes the measure-theoretic strand.

With the weak-$*$ topology in hand we can state the theorem the whole section was built to reach. It delivers exactly what Riesz's lemma denied us in the norm topology: that the closed unit ball of a dual space, though never norm-compact, becomes compact the moment we pass to the weak-$*$ topology. The trade is now visible in full. Compactness was lost when the topology was strong enough to keep the half-separated sequence apart; it is bought back by making the topology coarse enough that those same vectors are forced to accumulate. The coarser the topology, the easier it is for a set to be compact—fewer open covers to survive—and weak-$*$ is precisely coarse enough.

Theorem 2.16 (Banach–Alaoglu)**.** *Let X be a Banach space. The closed unit ball $B_{X^*} = \{\phi \in X^* : \|\phi\| \leq 1\}$ is compact in the weak-$*$ topology. If X is separable, B_{X^*} is in addition weak-$*$* sequentially *compact and metrizable.*

Proof. The two claims call for two different arguments, and we give each in full because the book uses them differently. The general statement rests on Tychonoff's theorem—the one ingredient in this chapter that genuinely sits above our prerequisite line, which we flag honestly and quarantine—while the separable refinement, the only form the book's existence proofs ever call on, needs nothing beyond a diagonal extraction the reader already half knows.

The general statement, via Tychonoff. The idea is to regard a functional as nothing more than the table of its values, and to recognize that table as a single point in a product of compact intervals. To each $\phi \in B_{X^*}$ associate the tuple of all its readings, $\Phi(\phi) = (\phi(x))_{x\in X}$. Because $\|\phi\| \leq 1$, each reading is confined to its own compact interval, $\phi(x) \in [-\|x\|, \|x\|]$, so Φ carries B_{X^*} into the product cube $P = \prod_{x\in X}[-\|x\|, \|x\|]$. This P is a product of compact spaces and is therefore compact in the product topology—and this is the sole place the argument exceeds the prerequisites, for an *uncountable* product of compacta is compact only by Tychonoff's theorem, whose proof runs through the Alexander subbase lemma and the axiom of choice. *We quote Tychonoff as a black box and record its proof in appendix B.3.* Two things now remain, and both are elementary. First, the product topology on P—the topology of coordinatewise convergence—restricts on the image $\Phi(B_{X^*})$ to *exactly* the weak-$*$ topology, because a net of functionals converges coordinatewise precisely when $\phi_\alpha(x) \to \phi(x)$ for every x, which is verbatim the definition of weak-$*$ convergence. Second, the image $\Phi(B_{X^*})$ is closed in P: a point $(t_x)_{x\in X}$ of P is the table of a *linear* functional if and only if it satisfies, for every pair x, x' and scalars a, b, the relation $t_{ax+bx'} = a\,t_x + b\,t_{x'}$, and each such relation cuts out a closed subset of P (it is the zero set of a continuous combination of three coordinate projections); intersecting all of these closed conditions—together with the norm bound already baked into the cube—leaves exactly the tables of functionals in B_{X^*}. A closed subset of a compact space is compact, so B_{X^*} is weak-$*$ compact, which is the first claim. We stress once more that the separable case below avoids Tychonoff entirely, so a reader unwilling to grant the axiom of choice loses nothing the book actually uses.

The separable statement, via diagonal extraction. Suppose now X is separable and fix a countable dense set $\{x_k\}_{k\geq 1} \subseteq X$. We prove sequential compactness by hand, first building one subsequence that converges against every x_k, then promoting that to convergence against every $x \in X$. Let

$\phi_n \in B_{X^*}$ be any sequence. Probe it first at x_1: the scalars $\phi_n(x_1)$ all lie in the compact interval $[-\|x_1\|, \|x_1\|]$, so by Bolzano–Weierstrass some subsequence, call it $\phi_n^{(1)}$, has $\phi_n^{(1)}(x_1)$ convergent. Probe *that* subsequence at x_2: its values $\phi_n^{(1)}(x_2)$ are again bounded, so a further subsequence $\phi_n^{(2)}$ makes $\phi_n^{(2)}(x_2)$ converge while—being a subsequence of the previous one—keeping $\phi_n^{(2)}(x_1)$ convergent as well. Continuing, we obtain nested subsequences $\phi_n^{(1)} \supseteq \phi_n^{(2)} \supseteq \cdots$, the k-th of which converges when tested against each of $x_1, \ldots, x_k$. No single one of these subsequences handles *all* the x_k at once; the standard remedy is to slide down the diagonal. Set $\psi_j := \phi_j^{(j)}$, the j-th term of the j-th subsequence. For each fixed k, the tail $(\psi_j)_{j \geq k}$ is a subsequence of $\phi_n^{(k)}$ (this is exactly what the nesting buys us), so $\psi_j(x_k)$ converges—and this holds for every k simultaneously.

We now have convergence on the dense set; the boundedness $\|\psi_j\| \leq 1$ is what lets us spread it to all of X, by a three-ε argument. Fix any $x \in X$ and $\varepsilon > 0$, and use density to pick x_k with $\|x - x_k\| < \varepsilon$. For any two indices i, j, split the difference through the nearby point x_k:

$$|\psi_i(x) - \psi_j(x)| \ \leq \ |\psi_i(x) - \psi_i(x_k)| + |\psi_i(x_k) - \psi_j(x_k)| + |\psi_j(x_k) - \psi_j(x)| \, .$$

The two outer terms are each at most $\|\psi\| \ \|x - x_k\| \leq \varepsilon$ by the operator-norm bound, and the middle term tends to 0 as $i, j \to \infty$ because $\psi(x_k)$ converges; hence $\limsup_{i,j\to\infty} |\psi_i(x) - \psi_j(x)| \leq 2\varepsilon$. As $\varepsilon > 0$ was arbitrary, $(\psi_j(x))_j$ is Cauchy in $\mathbb{R}$ and so converges; call its limit $\phi(x)$. The limit functional inherits both of the properties that matter from the ψ_j: passing the linearity relation $\psi_j(ax + bx') = a\psi_j(x) + b\psi_j(x')$ to the limit shows ϕ is linear, and passing $|\psi_j(x)| \leq \|x\|$ to the limit shows $|\phi(x)| \leq \|x\|$, so $\phi \in B_{X^*}$ and $\psi_j \xrightarrow{*} \phi$. This is weak-∗ sequential compactness.

Finally, metrizability on the ball. Define $d(\phi, \psi) = \sum_{k\geq 1} 2^{-k} \min\bigl(1, |\phi(x_k) - \psi(x_k)|\bigr)$. The series converges (it is dominated termwise by $\sum_k 2^{-k}$), and d is a genuine metric: it is symmetric and satisfies the triangle inequality coordinatewise, and it separates points because $\{x_k\}$ is dense, so two functionals agreeing at every x_k agree everywhere. Convergence in d is exactly coordinatewise convergence against all the x_k, which—by the same three-ε argument, now run on the bounded set B_{X^*}—is weak-∗ convergence there. So on B_{X^*} the weak-∗ topology is induced by a metric, and the sequential compactness just established is honest compactness. Full details are in [31, Ch. 3] and [113, Ch. IV]. □

The separable case is the workhorse: it converts a uniform bound into an honest convergent subsequence, and that conversion is the engine of every compactness argument to come. We package the consequence the book leans on relentlessly as a named corollary.

Corollary 2.17 (Bounded sequences have weakly convergent subsequences)**.** *If H is a separable Hilbert space, every bounded sequence has a weakly convergent subsequence. More generally this holds in any reflexive Banach space X (one for which the canonical embedding $X \hookrightarrow X^{**}$ is onto). The reader may keep in mind the example L^p, $1 < p < \infty$, which is reflexive (this rests on $(L^p)^* = L^{p'}$ and uniform convexity, [31, Ch. 4]); we record it only as an aside, since the chapter's load-bearing use is exclusively the separable-Hilbert case.*

Proof. We take the Hilbert case first, since it is immediate and is the only one the book actually spends. A Hilbert space H is reflexive—it is its own dual by Riesz representation (theorem 2.6), so the canonical embedding $H \hookrightarrow H^{**}$ is onto. Given a bounded sequence (x_n) in a *separable* H, view it inside $H = (H^*)^*$ as a bounded sequence of *functionals* on the separable space H^*. The separable case of theorem 2.16 now applies verbatim: a subsequence converges weak-$*$ in $(H^*)^*$, which—because $H^* = H$ via Riesz—is exactly weak convergence in H. That is the entire content of the Hilbert statement.

For the general reflexive version we reduce to this core, and the reduction is worth seeing, because it makes plain that the hypothesis doing the real work is *separability*, not reflexivity by itself. Let (x_n) be bounded in a reflexive Banach space X, which we do not assume separable; we instead manufacture a separable subspace that carries the whole sequence. Put $X_0 = \overline{\operatorname{span}}\,\{x_n\}$, the closed linear span of the terms. This X_0 is *separable*, because the finite combinations of the x_n with rational coefficients form a countable dense subset; and it is *reflexive*, because a closed subspace of a reflexive space is reflexive. A separable reflexive space has *separable dual* X_0^* (reflexivity transports separability from $X_0 = X_0^{**}$ back down to X_0^*). Now repeat the Hilbert paragraph with X_0 in the role of H: the sequence (x_n), sitting inside $X_0 = (X_0^*)^*$, is a bounded family of functionals on the separable space X_0^*, so by the separable case of theorem 2.16 a subsequence x_{n_k} converges weak-$*$ in $(X_0^*)^*$, that is, weakly in X_0, to some x_*. It remains to lift this from the subspace to all of X: every functional $\phi \in X^*$ restricts to an element of X_0^*, so the weak convergence in X_0 gives $\phi(x_{n_k}) \to \phi(x_*)$ for *every* $\phi \in X^*$, i.e. $x_{n_k} \rightharpoonup x_*$ in X. No appeal to Hahn–Banach is needed at this step: each $\phi \in X^*$ is simply evaluated on the points $x_{n_k}, x_* \in X_0$ through its restriction $\phi|_{X_0}$, so $\phi(x_{n_k}) = \phi|_{X_0}(x_{n_k}) \to \phi|_{X_0}(x_*) = \phi(x_*)$; surjectivity of the restriction map plays no role here. The fully general reflexive statement, with *no* passage to a separable subspace, is the Eberlein–Šmulian theorem [31, Ch. 3]; we never need it, because every space the book actually uses—separable Hilbert spaces, and L^p over a separable measure space—is separable, and there the reduction just carried out is exact. □

Heuristic (non-rigorous)

We can now state the motif that every compactness-based existence proof in this book instantiates, and that the reader should carry forward as a fixed template. It has three beats. *(i) Bound.* Produce an a priori estimate: show that the approximating sequence x_n—Galerkin truncations of a PDE, iterates of a best-response map, regularized solutions—stays inside a fixed ball, $\|x_n\| \le C$, with C independent of n. This is invariably the analytically substantive beat; it is some energy inequality, maximum principle, or entropy bound, and producing it is where the structure of the particular problem enters. It is worth saying once and remembering: *every* compactness in this book is, at bottom, such an a priori estimate wearing a topological costume. We will meet the same fact dressed three more ways—as Hilbert–Schmidt summability of a kernel (section 2.4), as the Rellich–Kondrachov embedding (section 2.6), and as Schauder's hypothesis that a map sends a set into a compact one (theorem 2.38)—and each time the compactness will turn out to be a bound in disguise.
(ii) Extract. Feed the bound into Banach–Alaoglu (corollary 2.17): the ball is weak-$*$ compact—in a Hilbert space, weakly sequentially compact—so a subsequence $x_{n_k} \rightharpoonup x_*$ converges to some candidate limit. This beat is now automatic; it is the entire content of the theorem just proved, and it costs nothing beyond the bound from beat (i). The limit x_* exists and lives in the right space. What it *is* remains to be seen.

(iii) Identify. Show that the weak limit x_* actually solves the equation. Here the difficulty migrates, and this is exactly why the first two beats, easy as they have become, do not finish the job. Weak convergence is treacherous under anything nonlinear: from $x_{n_k} \rightharpoonup x_*$ one cannot in general conclude $f(x_{n_k}) \rightharpoonup f(x_*)$ for a nonlinear f, nor that a product $x_{n_k} y_{n_k}$ tends to $x_* y_*$; and even the norm is only lower semicontinuous, $\|x_*\| \leq \liminf_k \|x_{n_k}\|$, never guaranteed to be preserved—the orthonormal sequence $e_n \rightharpoonup 0$ with $\|e_n\| = 1$ from earlier in the section is the standing reminder that mass can vanish in the limit. So passing the equation to the limit requires the maps appearing in it to be *weakly* (sequentially) continuous, or, where only an inequality is at stake, weakly lower semicontinuous. For bounded *linear* maps weak continuity is automatic; the obstruction is precisely the nonlinear terms and the merely-bounded bilinear ones, and overcoming it is the genuine labor of every existence proof in the book.

The decisive question is therefore which maps upgrade the weak convergence handed to us in beat (ii) back into the strong, norm convergence that nonlinear terms need before they will pass to the limit. There is a single class of operators with exactly this property—they turn weakly convergent sequences into strongly convergent ones—and they are important enough that the next section is built around them. They are the *compact* operators, and the fact that the graphon operator $\mathbb{W}$ is one of them is what will make beat (iii) tractable wherever it appears, and what makes Schauder's fixed-point theorem (theorem 2.38) applicable at all.

2.4 Compact and Hilbert–Schmidt operators

The heuristic that closed the last section ended on a precise promise, and this section is where we keep it. The three-beat template—bound, extract, identify—left us stranded at the third beat: Banach–Alaoglu hands us a weak limit for free, but weak convergence is too weak to survive the nonlinear and bilinear terms that every equation in the book contains, so to finish an existence proof we must upgrade that weak convergence back into the strong, norm convergence those terms require. The heuristic named the maps that perform exactly this upgrade, the ones that send a weakly convergent sequence to a strongly convergent image, and promised a section built around them. Here it is. The operators in question are the *compact* operators, and the burden of this section is twofold: to prove that they really do convert weak convergence to strong—so the qualitative property the existence proofs lean on becomes a theorem rather than a slogan—and then to locate, inside the compact operators, the concrete computable subclass that the integral operators of this book actually inhabit.

That two-part burden is worth making explicit before either half is discharged, because it explains why the section needs two subsections rather than one. The two notions answer two different questions. *Compactness* is the *qualitative* property existence demands: it is defined purely by what the operator does to bounded sets—it crushes them into relatively compact ones—and that single abstract feature is precisely what licenses the weak-to-strong upgrade and, downstream, makes Schauder's fixed-point theorem (theorem 2.38) applicable. It asks nothing about how the operator is presented and gives nothing one could compute. *Hilbert–Schmidt* is the *quantitative* answer to the complementary question: of the compact operators, which ones do our integral kernels actually furnish, and can we certify membership by a number we can evaluate? It is a strictly smaller class, carved out by a finite sum-of-squares condition that, for an integral operator, turns out to be nothing more than square-integrability of its kernel—the very condition $W \in L^2([0,1]^2)$ that the graphon satisfies by construction. So compactness is the property we *need* and Hilbert–Schmidt is the property we can *check*; the first subsection establishes the former and the inclusion that every Hilbert–Schmidt operator is compact, and the second pins down the latter and uses it to place the graphon operator $\mathbb{W}$ in the class where the spectral theorem of section 2.5 can finally act on it.

We take the two in that order. The compact-operator subsection makes the weak-to-strong upgrade a theorem; the Hilbert–Schmidt subsection identifies L^2 kernels with a computable subclass of compact operators and exhibits the graphon as its first concrete inhabitant.

2.4.1 Compact operators

The compact operators are those that turn a weakly convergent sequence into a strongly convergent one—named in the heuristic but not yet defined, and the present subsection pays that promissory note. We give the definition the upgrade rests on, show that the two ways of phrasing it coincide, and then prove the upgrade itself as a theorem—so that "compact operators convert weak to strong," the slogan we leaned on to motivate the whole weak-topology detour, becomes a result with a proof. Along the way we record the two structural facts the rest of the chapter consumes: that the compact operators are closed under operator-norm limits, and that they form a two-sided ideal. The first of these is what the next subsection uses to certify that Hilbert–Schmidt operators are compact; the second is part of the standing furniture of operator theory.

The definition isolates a single demand on what an operator does to bounded sets. A bounded operator is, by definition, one that does not blow boundedness up—it sends bounded sets to bounded sets. A *compact* operator does strictly more: it sends bounded sets to sets that are not merely bounded but *precompact*, sets whose closure is compact. In infinite dimensions this is a genuine strengthening, because the closed unit ball is bounded but *never* compact (that failure was the whole reason the weak topology had to be invented in section 2.3); demanding that T crush every bounded set into a compact one is therefore a real restriction, and it is exactly the restriction that buys back the sequential extraction that strong topology denies.

Definition 2.18 (Compact operator)**.** Let X, Y be Banach spaces. An operator $T \in \mathcal{B}(X, Y)$ is *compact* if it maps bounded sets to relatively compact sets: the closure of $T(B_X)$ is compact in Y, where B_X is the closed unit ball of X. Equivalently, for every bounded sequence x_n the image Tx_n has a norm-convergent subsequence.

The two formulations in the definition deserve a word, because the equivalence is what lets us move freely between a topological statement about sets and a sequential statement about extracting limits—and it is the sequential form that all the analysis below actually uses. The bridge is the metric-space fact that compactness and sequential compactness coincide: in a metric space a set is compact iff every sequence in it has a subsequence converging to a point of the set.

Suppose first that $\overline{T(B_X)}$ is compact, and let (x_n) be bounded, say $\|x_n\| \le R$; then each Tx_n lies in $T(RB_X) = RT(B_X)$, a scaling of a precompact set and hence itself precompact, so the sequence (Tx_n) lives in a compact closure and has a norm-convergent subsequence. Conversely, if every bounded sequence has an image subsequence that converges, take any sequence (y_k) in $T(B_X)$; writing $y_k = Tx_k$ with $\|x_k\| \le 1$, the hypothesis gives a subsequence Ty_{k_j} converging in Y, so every sequence in $T(B_X)$ has a convergent subsequence, which is precisely sequential—hence topological—compactness of the closure (completeness of Y guarantees the limit exists in Y and lands in the closure). The two phrasings are therefore interchangeable, and we use whichever is convenient, most often the sequential one.

We can now keep the promise of the previous section: a compact operator is exactly a machine that repairs the defect of weak convergence, returning strong convergence on the other side.

Proposition 2.19 (Compact operators upgrade weak to strong)**.** *Let X, Y be Banach spaces and $T \in \mathcal{B}(X, Y)$ compact. If $x_n \rightharpoonup x$ in X, then $Tx_n \to Tx$ in the norm of Y.*

Proof. The argument has two moving parts, and the whole subtlety is that they pull in opposite directions until compactness reconciles them: weak convergence pins down what the strong limit *must* be if one exists, while compactness guarantees that one *does* exist along subsequences.

The weak limit pins the only candidate. A weakly convergent sequence is norm-bounded—this is the uniform-boundedness consequence recorded in section 2.3—so $\sup_n \|x_n\| < \infty$. Moreover bounded linear maps are automatically continuous from the weak topology of X to the weak topology of Y: for any functional $\phi \in Y^*$ the composition $\phi \circ T$ lies in X^*, so $x_n \rightharpoonup x$ gives $\phi(Tx_n) = (\phi \circ T)(x_n) \to (\phi \circ T)(x) = \phi(Tx)$, which is exactly the statement $Tx_n \rightharpoonup Tx$. Thus *if* some subsequence Tx_{n_k} converges in norm to a limit y, then it also converges weakly to y (norm convergence implies weak convergence), while the line just proved forces $Tx_{n_k} \rightharpoonup Tx$; weak limits are unique, so $y = Tx$. Every norm-convergent subsequence of (Tx_n) therefore has the *same* limit, namely Tx.

Compactness supplies a strong limit. Because (x_n) is bounded, compactness of T (definition 2.18, sequential form) guarantees that every subsequence of (Tx_n) has a further subsequence converging in norm—necessarily to Tx by the previous paragraph. Now invoke the elementary metric-space principle that a sequence converges to L iff every subsequence has a further subsequence converging to L: were $Tx_n \not\to Tx$, some $\varepsilon > 0$ and some subsequence would keep $\|Tx_{n_k} - Tx\| \ge \varepsilon$, yet that subsequence still has a sub-subsequence converging to Tx, a contradiction. Hence $Tx_n \to Tx$ in norm. □

This is the property the existence machinery of the book actually consumes. Whenever a self-consistent-field iteration produces only a weak limit at beat (ii), feeding that limit through a compact operator restores the norm convergence that the nonlinear and bilinear terms demand at beat (iii); and it is the same compactness, in its set-theoretic guise, that will make the image of Schauder's map relatively compact and so make theorem 2.38 applicable. The graphon operator $\mathbb{W}$ will be compact, so this upgrade is available everywhere it appears.

Two structural facts about the class of compact operators remain, and we prove the one we will soon cite. The compact operators form a *closed two-sided ideal* in $\mathcal{B}(X)$: they are a subspace closed under operator-norm limits, and pre- or post-composing a compact operator with any bounded operator leaves it compact. Closedness is the fact the Hilbert–Schmidt subsection needs, so we give its $\varepsilon/3$ proof in full rather than as a slogan.

Proposition 2.20 (Compact operators are operator-norm closed)**.** *If $T_n \in \mathcal{B}(X,Y)$ are compact and $T_n \to T$ in operator norm, then T is compact.*

Proof. Let (x_k) be any sequence with $\|x_k\| \le R$; we must produce a norm-convergent subsequence of (Tx_k). We build it in three steps.

Step 1 (diagonal extraction). Since T_1 is compact, (T_1x_k) has a convergent subsequence; record its index set and pass to it. Inside that subsequence (T_2x_k) again has a convergent subsequence, by compactness of T_2; pass to it. Continuing and taking the diagonal subsequence, which we relabel (x_k), yields a single subsequence along which T_nx_k converges, hence is Cauchy in k, *for every fixed* n simultaneously.

Step 2 (the $\varepsilon/3$ split). Fix $\varepsilon > 0$. Choose n so large that $\|T - T_n\| < \varepsilon/(3R)$, possible because $T_n \to T$ in operator norm. For this n and any indices k, k', insert $\pm T_nx_k$ and $\pm T_nx_{k'}$ and apply the triangle inequality:

$$\|Tx_k - Tx_{k'}\| \le \|(T-T_n)x_k\| + \|T_nx_k - T_nx_{k'}\| + \|(T_n - T)x_{k'}\| \le 2R\,\|T - T_n\| + \|T_nx_k - T_nx_{k'}\|\,.$$

The first and third terms together are at most $2R\,\|T - T_n\| < \frac{2}{3}\varepsilon$.

Step 3 (the middle term). With n now fixed, Step 1 makes $(T_nx_k)_k$ Cauchy, so there is N with $\|T_nx_k - T_nx_{k'}\| < \varepsilon/3$ for all $k, k' \ge N$. Combining the two bounds, $\|Tx_k - Tx_{k'}\| < \varepsilon$ for $k, k' \ge N$. Hence (Tx_k) is Cauchy along the diagonal subsequence, and since Y is complete it converges. So T maps the bounded sequence (x_k) to a sequence with a convergent subsequence, which is compactness. $\square$

The ideal property is quicker and uses no completeness. If T is compact and $S, U \in \mathcal{B}$, then STU is compact: U sends a bounded set to a bounded set, T sends that to a relatively compact set, and the continuous map S sends a relatively compact set to a relatively compact one (a continuous image of a compact set is compact). Taking U or S to be the identity covers both one-sided cases, so the compacts absorb bounded operators from either side—a two-sided ideal. Finally, *finite-rank operators are compact*: if $\dim T(X) < \infty$, then $T(B_X)$ is a bounded subset of a finite-dimensional space, hence relatively compact by Heine–Borel. This last fact, combined with the closedness just proved, is precisely what the next subsection needs: any operator that is an operator-norm limit of finite-rank operators is compact.

One honest caveat closes the subsection. On a Hilbert space the converse of that last observation also holds—*every* compact operator is a norm-limit of finite-rank ones [113, Ch. VI]—which makes "compact" and "finite-rank-approximable" the same notion there. That converse is genuinely deeper, and we do not need it: every operator we will actually certify as compact, the graphon operator chief among them, arrives already exhibited as a finite-rank limit, so the easy direction suffices throughout. We therefore take only what we have proved—compactness upgrades weak to strong, the class is a closed ideal, and finite-rank limits are compact—into the next subsection, where these qualitative facts are made quantitative: the compact operators we meet in this book are integral operators, and almost all of them belong to a strictly smaller, computable class whose membership is certified by a single finite number.

2.4.2 Hilbert–Schmidt operators and integral kernels

The previous subsection left compactness as a qualitative property: a compact operator crushes bounded sets into precompact ones, it upgrades weak convergence to strong, and it is certified compact the moment it is exhibited as an operator-norm limit of finite-rank maps. That is exactly the structural inheritance the graphon operator will need. But it is a *yes/no* property, and the operators this book actually meets carry more information than a yes/no answer can record. They are not abstract compact maps produced by some existence argument; they are *integral operators*, given by an explicit kernel $K(s,t)$ that one can write down, square, and integrate. For such operators compactness is not the sharpest thing one can say. There is a finer, quantitative class—the *Hilbert–Schmidt* operators—that sits strictly inside the compact operators, that contains essentially every integral operator we will form, and whose membership is certified not by an extraction argument but by a single finite number computed from the kernel. The promise of this subsection is to make that number, and the class it defines, completely concrete: we will see that the Hilbert–Schmidt operators on $L^2(\Omega)$ are *precisely* the integral operators with square-integrable kernels, and that the correspondence is an exact isometry. Once that identification is in hand the graphon operator falls into place with no further work—it is the integral operator of a bounded, hence square-integrable, symmetric kernel—and the chapter's central object is finally certified compact and self-adjoint, which is all the spectral theorem of the next section will ask of it.

We begin with the definition that attaches the certifying number to an operator. The idea is the most naive one imaginable: in finite dimensions the size of a matrix can be measured not only by its largest singular value (the operator norm) but by the root-sum-of-squares of all its entries, the Frobenius norm $\big(\sum_{ij}|a_{ij}|^2\big)^{1/2}$. The Hilbert–Schmidt norm is nothing but the Frobenius norm carried verbatim into infinite dimensions, with the entries of T in an orthonormal basis playing the role of the matrix entries. The only subtlety—and it is the one the definition must immediately confront—is that an infinite matrix has infinitely many entries, so the root-sum-of-squares may diverge; finiteness is therefore a genuine condition, and it is exactly the condition that names the class.

Definition 2.21 (Hilbert–Schmidt operator)**.** Let H be a separable Hilbert space with orthonormal basis $\{e_n\}$. An operator $T \in \mathcal{B}(H)$ is *Hilbert–Schmidt* if

$$\|T\|_{\mathrm{HS}}^2 \;:=\; \sum_n \|Te_n\|^2 \;<\; \infty.$$

The quantity $\|T\|_{\mathrm{HS}}$ is independent of the chosen basis, dominates the operator norm, $\|T\| \le \|T\|_{\mathrm{HS}}$, and is itself the norm induced by the inner product $\langle S,T\rangle_{\mathrm{HS}} = \sum_n \langle Se_n, Te_n\rangle$, under which the Hilbert–Schmidt operators form a Hilbert space. Every Hilbert–Schmidt operator is compact.

Three claims are bundled into that definition, and each is worth drawing out, because together they are what make the Hilbert–Schmidt class so pleasant to work in—a Hilbert space of operators, sitting inside the Banach algebra of all bounded operators, on which the size of an operator is computed coordinate by coordinate exactly as Parseval computes the size of a vector.

The first claim is *basis-independence*: the number $\|T\|_{\mathrm{HS}}$ does not depend on which orthonormal basis $\{e_n\}$ is used to compute it, even though the individual summands $\|Te_n\|^2$ manifestly do. This is what entitles us to call $\|T\|_{\mathrm{HS}}$ a property of T rather than of T together with a choice of coordinates. The reason is a double application of Parseval. Expand each image Te_n in a *second* orthonormal basis $\{f_m\}$; Parseval (theorem 2.8) gives $\|Te_n\|^2 = \sum_m |\langle Te_n, f_m\rangle|^2$, so

$$\|T\|_{\mathrm{HS}}^2 = \sum_n \|Te_n\|^2 = \sum_{n,m} |\langle Te_n, f_m\rangle|^2 .$$

Every term in this double sum is nonnegative, so by Tonelli the order of summation is irrelevant and we may sum over n first. Moving the inner product onto the adjoint, $\langle Te_n, f_m\rangle = \langle e_n, T^* f_m\rangle$, and summing over n by Parseval again gives $\sum_n |\langle e_n, T^* f_m\rangle|^2 = \|T^* f_m\|^2$. Hence

$$\|T\|_{\mathrm{HS}}^2 = \sum_m \|T^* f_m\|^2 ,$$

which on the one hand is manifestly the Hilbert–Schmidt sum of T^* computed in the basis $\{f_m\}$, and on the other hand makes no reference to the original basis $\{e_n\}$. Taking $\{f_m\} = \{e_m\}$ shows $\|T\|_{\mathrm{HS}} = \|T^*\|_{\mathrm{HS}}$, and taking a general $\{f_m\}$ shows the value is the same in any basis. The same computation thus simultaneously proves basis-independence and the symmetry between T and T^*—a fact we will use the moment the graphon's self-adjointness is in play.

The second claim is *domination of the operator norm*, $\|T\| \le \|T\|_{\mathrm{HS}}$. This is what places the Hilbert–Schmidt operators inside the bounded operators and gives the certifying number teeth: controlling it controls the gain of T as well. The proof is one line of Parseval. For any unit vector x, expand it as $x = \sum_n \langle x, e_n\rangle e_n$ and apply T termwise; the triangle inequality followed by Cauchy–Schwarz on the coefficient sequence gives

$$\|Tx\| = \Big\| \sum_n \langle x, e_n\rangle Te_n \Big\| \le \sum_n |\langle x, e_n\rangle| \; \|Te_n\| \le \Big(\sum_n |\langle x, e_n\rangle|^2 \Big)^{1/2} \Big(\sum_n \|Te_n\|^2 \Big)^{1/2} = \|x\| \; \|T\|_{\mathrm{HS}} ,$$

using Parseval $\sum_n |\langle x, e_n\rangle|^2 = \|x\|^2 = 1$ in the last step. Taking the supremum over unit x yields $\|T\| \le \|T\|_{\mathrm{HS}}$. The inequality can be strict and badly so—the identity operator on an infinite-dimensional H has $\|\mathrm{Id}\| = 1$ but $\|\mathrm{Id}\|_{\mathrm{HS}}^2 = \sum_n \|e_n\|^2 = \sum_n 1 = \infty$, so the identity is bounded but emphatically *not* Hilbert–Schmidt. This is the quantitative shadow of the qualitative fact from the previous subsection that the identity is not compact in infinite dimensions: the Hilbert–Schmidt norm sees the failure as an outright divergence.

The third claim packages the Hilbert–Schmidt operators into a Hilbert space of their own. The expression $\langle S, T\rangle_{\mathrm{HS}} = \sum_n \langle Se_n, Te_n\rangle$ is, by the polarization identity applied to the basis-independent norm, itself basis-independent and is a genuine inner product (it is the ℓ^2 inner product of the two operators' coordinate arrays); its induced norm is $\|T\|_{\mathrm{HS}}$; and completeness in this norm transfers from the completeness of ℓ^2 once one knows—which the kernel theorem below will make transparent—that the coordinate array $(\langle Te_m, e_n\rangle)_{n,m}$ ranges over all of $\ell^2(\mathbb{N} \times \mathbb{N})$. So the Hilbert–Schmidt operators are not merely a subset of $\mathcal{B}(H)$ but a Hilbert space, with their own geometry: one can speak of orthogonality of operators, of best approximation within the class, of the angle between two operators. This is more structure than $\mathcal{B}(H)$ itself carries (the operator norm comes from no inner product), and it is exactly the structure that the kernel identification will reveal to be a perfect copy of the geometry of $L^2(\Omega \times \Omega)$.

The final assertion—that every Hilbert–Schmidt operator is compact—is the bridge back to the previous subsection, and it is worth seeing precisely how the finite-rank machinery cashes it out, because this is the argument that ultimately certifies the graphon compact. Order the basis and let P_N be the orthogonal projection onto $\operatorname{span}\{e_1, \dots, e_N\}$. The truncated operator $T_N := TP_N$ has range inside $T(\operatorname{span}\{e_1, \dots, e_N\})$, a space of dimension at most N, so T_N is *finite-rank*, hence compact by the last fact of the previous subsection. We claim $T_N \to T$ in operator norm. Indeed $T - T_N = T(\mathrm{Id} - P_N)$ annihilates $e_1, \dots, e_N$ and agrees with T on the orthogonal tail, so by the very domination inequality just proved, applied to the operator $T(\mathrm{Id} - P_N)$ whose only nonzero basis images are Te_n for $n > N$,

$$\|T - T_N\|^2 \leq \|T - T_N\|_{\mathrm{HS}}^2 = \sum_{n>N} \|Te_n\|^2 ,$$

and the right-hand side is the tail of the convergent series $\sum_n \|Te_n\|^2 = \|T\|_{\mathrm{HS}}^2 < \infty$, so it tends to 0 as $N \to \infty$. Thus T is an operator-norm limit of finite-rank operators, and by the closedness of the compact ideal (proposition 2.20) T is compact. The Hilbert–Schmidt condition is, in this light, simply a *quantitative* sufficient condition for compactness—a convergent root-sum-of-squares—and the explicit finite-rank truncations it supplies are precisely the kind of concrete witness the previous subsection promised we would always have in hand.

With the class understood intrinsically, we come to the identification that is the real engine of the subsection, the one that turns "Hilbert–Schmidt operator" and "integral operator with L^2 kernel" into two names for the same thing. This is what lets us read every property of the graphon operator straight off its kernel.

Theorem 2.22 (Hilbert–Schmidt operators are L^2 kernels)**.** *Let (Ω, μ) be a σ-finite measure space and $H = L^2(\Omega, \mu)$. For $K \in L^2(\Omega \times \Omega, \mu \otimes \mu)$ define, writing s, t for the two integration variables of Ω,*

$$(T_K f)(s) \;=\; \int_\Omega K(s,t)\, f(t)\; \mathrm{d}\mu(t).$$

Then T_K is a well-defined Hilbert–Schmidt operator on H with

$$\|T_K\|_{\mathrm{HS}} \;=\; \|K\|_{L^2(\Omega\times\Omega)} ,$$

and the map $K \mapsto T_K$ is an isometric isomorphism from $L^2(\Omega \times \Omega)$ onto the Hilbert–Schmidt operators on $L^2(\Omega)$. Moreover $T_K^* = T_{K^*}$ where $K^*(s,t) = \overline{K(t,s)}$; in particular T_K is self-adjoint iff $K(s,t) = \overline{K(t,s)}$ for $\mu \otimes \mu$-almost every (s,t).

The statement has four parts, and we prove them in the order in which each becomes possible: T_K is well-defined and bounded; its Hilbert–Schmidt norm equals the L^2 norm of the kernel (so the map $K \mapsto T_K$ is an isometry into the Hilbert–Schmidt class); the map is onto (every Hilbert–Schmidt operator arises this way); and the adjoint is read off by transposing-and-conjugating the kernel. The reader should keep the payoff in view throughout: an *exact* dictionary, with no constants lost, between a space of *functions of two variables* and a space of *operators*.

Proof. Well-definedness and boundedness. Fix $f \in L^2(\Omega)$. For almost every s the function $t \mapsto K(s,t)$ lies in $L^2(\Omega)$—this is Tonelli applied to the finite double integral $\iint |K|^2$, which guarantees the inner slice is square-integrable for a.e. s—so the defining integral $\int K(s,t)f(t)\,\mathrm{d}\mu(t)$ is the inner product of two L^2 functions and Cauchy–Schwarz bounds it: $|(T_K f)(s)| \le \big(\int |K(s,t)|^2\,\mathrm{d}\mu(t)\big)^{1/2}\,\|f\|$. Squaring and integrating over s,

$$\|T_K f\|^2 \le \int_\Omega \Big(\int_\Omega |K(s,t)|^2\,\mathrm{d}\mu(t)\Big)\,\|f\|^2\,\mathrm{d}\mu(s) = \|K\|^2_{L^2(\Omega\times\Omega)}\,\|f\|^2,$$

where the last equality is Fubini–Tonelli (theorem 1.27) re-reading the iterated integral of the nonnegative integrand $|K|^2$ as the double integral $\iint |K|^2$. Thus $T_K f \in L^2$, T_K is linear, and $\|T_K\| \le \|K\|_{L^2}$: a square-integrable kernel already gives a bounded operator, recovering the bound previewed in section 2.2.

The isometry $\|T_K\|_{\mathrm{HS}} = \|K\|_{L^2}$. Here the geometry of $L^2(\Omega\times\Omega)$ does the work. Fix an orthonormal basis $\{e_n\}$ of $L^2(\Omega)$ and form the two-variable products

$$E_{nm}(s,t) \;=\; e_n(s)\,\overline{e_m(t)}.$$

These are orthonormal in $L^2(\Omega\times\Omega)$, by Fubini and the separation of variables: $\langle E_{nm}, E_{n'm'}\rangle = \big(\int e_n\overline{e_{n'}}\big)\big(\int \overline{e_m}e_{m'}\big) = \delta_{nn'}\delta_{mm'}$. The crux—and the densest point of the whole proof—is that they are also *complete*, i.e. form an orthonormal basis of $L^2(\Omega\times\Omega)$. We establish this by a short density chain, each link of which reduces an unfamiliar density to one already in hand.

1. *Simple functions on the product are dense in $L^2(\Omega\times\Omega)$.* This is the Chapter-1 fact (proposition 1.32) that simple functions are L^2-dense in any L^2 space, applied to the product measure space $(\Omega\times\Omega, \mu\otimes\mu)$. A simple function here is a finite combination of indicators $\mathbf{1}_E$ of measurable sets $E \subseteq \Omega\times\Omega$.

2. *Product indicators are dense in product simple functions, hence in L^2.* Each measurable E in the product σ-algebra is approximated in measure (theorem 1.26) by finite disjoint unions of measurable rectangles $A\times B$, so $\mathbf{1}_E$ is approximated in L^2 by finite combinations of $\mathbf{1}_{A\times B} = \mathbf{1}_A(s)\mathbf{1}_B(t)$. Composing with the previous link, the *product simple functions*—finite sums $\sum_k g_k(s)h_k(t)$ with $g_k, h_k \in L^2(\Omega)$—are dense in $L^2(\Omega\times\Omega)$.

3. *Each factor expands in $\{e_n\}$, so $\{E_{nm}\}$ captures product functions.* It now suffices to approximate a single product $g(s)h(t)$ by finite combinations of the E_{nm}, since these products span a dense set by link 2. Because $\{e_n\}$ is an orthonormal basis of $L^2(\Omega)$, the partial sums $g_N = \sum_{n\le N}\langle g, e_n\rangle e_n \to g$ and $h_N = \sum_{m\le N}\langle h, e_m\rangle e_m \to h$ in $L^2(\Omega)$. Their product $g_N(s)\,h_N(t)$ is a finite linear combination of the functions $e_n(s)\,\overline{e_m(t)} = E_{nm}(s,t)$ (the conjugation of the second factor being absorbed into the coefficients, and harmless for the real orthonormal systems—sines and cosines, Legendre polynomials—that we actually use on $[0,1]$), and it converges to $g(s)h(t)$ in $L^2(\Omega\times\Omega)$ because

$$\|g\otimes h - g_N\otimes h_N\|_{L^2(\Omega\times\Omega)} \le \|(g-g_N)\otimes h\| + \|g_N\otimes(h-h_N)\| = \|g-g_N\|\,\|h\| + \|g_N\|\,\|h-h_N\| \to 0,$$

using the multiplicativity $\|u \otimes v\|_{L^2(\Omega\times\Omega)} = \|u\|_{L^2} \|v\|_{L^2}$ (again Fubini) and the boundedness of $\|g_N\| \leq \|g\|$.

Chaining the three links, finite combinations of the E_{nm} are dense in $L^2(\Omega \times \Omega)$, so $\{E_{nm}\}$ is an orthonormal basis. Now Parseval (theorem 2.8) in this basis computes the kernel norm coordinate by coordinate,

$$\|K\|_{L^2}^2 = \sum_{n,m} |\langle K, E_{nm}\rangle|^2 .$$

On the operator side, the same coefficients are matrix entries of T_K. Unwinding the definition with Fubini,

$$\langle K, E_{nm}\rangle = \iint K(s,t)\,\overline{e_n(s)}\,e_m(t)\,\mathrm{d}\mu(s)\,\mathrm{d}\mu(t) = \int \overline{e_n(s)}\Big(\int K(s,t)e_m(t)\,\mathrm{d}\mu(t)\Big)\,\mathrm{d}\mu(s) = \langle T_K e_m, e_n\rangle,$$

where the last step recognizes the inner integral as $(T_K e_m)(s)$ and the outer integral as the inner product $\langle T_K e_m, e_n\rangle$ (the conjugate sitting on e_n, matching our convention). Therefore, summing over n first by Parseval in $L^2(\Omega)$,

$$\|T_K\|_{\mathrm{HS}}^2 = \sum_m \|T_K e_m\|^2 = \sum_m \sum_n |\langle T_K e_m, e_n\rangle|^2 = \sum_{n,m} |\langle K, E_{nm}\rangle|^2 = \|K\|_{L^2}^2 .$$

This is the asserted isometry: the certifying number of the operator is exactly the L^2 norm of its kernel. In particular T_K is Hilbert–Schmidt (so, by definition 2.21, compact) precisely because $K \in L^2$.

Surjectivity. The isometry already shows $K \mapsto T_K$ is injective with closed range; we check the range is everything. Let T be any Hilbert–Schmidt operator and set $t_{nm} = \langle T e_m, e_n\rangle$. By the basis-independence computation above its array is square-summable, $\sum_{n,m} |t_{nm}|^2 = \|T\|_{\mathrm{HS}}^2 < \infty$, so $(t_{nm}) \in \ell^2(\mathbb{N} \times \mathbb{N})$. Since $\{E_{nm}\}$ is an orthonormal basis, the converse half of Parseval (theorem 2.8) assembles these coefficients into a genuine L^2 kernel $K = \sum_{n,m} t_{nm} E_{nm} \in L^2(\Omega \times \Omega)$, with $\|K\|_{L^2} = \|T\|_{\mathrm{HS}}$. By the matrix-entry identity just derived, $\langle T_K e_m, e_n\rangle = \langle K, E_{nm}\rangle = t_{nm} = \langle T e_m, e_n\rangle$ for all n, m, and two bounded operators with the same matrix in an orthonormal basis are equal. Hence $T = T_K$: every Hilbert–Schmidt operator is the integral operator of an L^2 kernel, and the map is onto. Together with the isometry, $K \mapsto T_K$ is the asserted isometric isomorphism of Hilbert spaces—$L^2(\Omega \times \Omega)$ *is* the Hilbert–Schmidt class, relabelled.

The adjoint. Finally we read off the adjoint. For $f, g \in L^2$, Fubini lets us move the kernel across the inner product:

$$\langle T_K f, g\rangle = \iint K(s,t) f(t)\,\overline{g(s)}\,\mathrm{d}\mu(t)\,\mathrm{d}\mu(s) = \int f(t)\,\overline{\Big(\int \overline{K(s,t)}\,g(s)\,\mathrm{d}\mu(s)\Big)}\,\mathrm{d}\mu(t) = \langle f, T_{K^*} g\rangle,$$

where $K^*(s,t) = \overline{K(t,s)}$ is the conjugate transpose of the kernel and the bracketed integral is $(T_{K^*} g)(t)$. By the defining property of the adjoint (definition 2.12), $T_K^* = T_{K^*}$. Self-adjointness $T_K = T_K^*$ is then, through the isometry's injectivity, equivalent to equality of the kernels $K = K^*$, i.e. $K(s,t) = \overline{K(t,s)}$ for $\mu \otimes \mu$-a.e. (s,t): an operator is self-adjoint exactly when its kernel is Hermitian-symmetric. We have followed the line of [113, Ch. VI, Thm. VI.23]. □

The theorem is worth pausing over, because it is the precise sense in which the analysis of integral operators *is* the analysis of two-variable L^2 functions. Everything one knows about $L^2(\Omega \times \Omega)$—it is a separable Hilbert space, its elements are approximable by simple functions, its norm is computed by Parseval—transfers verbatim to the Hilbert–Schmidt operators, and conversely the operator-theoretic content (compactness, the spectral structure to come) becomes a statement about kernels. The self-adjointness criterion is the link the chapter has been building toward: *symmetry of the kernel is self-adjointness of the operator*, with no analysis required beyond the one-line Fubini computation above. That is precisely the property the spectral theorem will demand, and a graphon supplies it by fiat.

We can now meet the central object of the book's operator theory and watch every clause of the theorem discharge at once.

Definition 2.23 (The graphon operator)**.** A *graphon* is a symmetric measurable function $W\colon [0,1]^2 \to \mathbb{R}$; in the basic theory it is bounded, $W\colon [0,1]^2 \to [0,1]$ (assumption (A-Reg-W) of appendix A.6). Its *integral operator* $\mathbb{W}\colon L^2[0,1] \to L^2[0,1]$ is

$$(\mathbb{W}f)(x) \;=\; \int_0^1 W(x,y)\, f(y)\, \mathrm{d}y.$$

Read against the theorem, this definition is just the instance $\Omega = [0,1]$ with μ the Lebesgue measure and the kernel $K = W$. Two features of that instance do real work and deserve to be flagged before we use them: what the integration variable *means*, and which W this is.

Notation watch

This is the first place in the body where the type variable x does mathematical work, so we recall (appendix A.5) that $x, y \in [0,1]$ are exchangeable *type / label* coordinates—an agent's kind or network position—*not* spatial positions and *not* momenta; in particular $|x-y|$ carries no intrinsic meaning, only what W assigns it. Concretely, $(\mathbb{W}f)(x)$ is the average of the field f over all labels y, weighted by the connection strength $W(x,y)$ between type x and type y; reordering the labels by any measure-preserving bijection of $[0,1]$ leaves the operator unitarily equivalent, which is the analytic face of exchangeability and the reason no metric on $[0,1]$ is built in. The letter W here is the graphon kernel, and no Brownian motion is present in this chapter, so the W-versus-W clash flagged in appendix A.5 does not arise: the disambiguation (write Brownian motion as B once a graphon is in scope) becomes binding only from Part III onward, where the two coexist. Until then W unambiguously denotes the kernel. The type variable and the operator $\mathbb{W}$ previewed here are formally introduced as a graph limit in chapter 15; here we need only that $\mathbb{W}$ is the integral operator of the kernel W, so that the kernel theorem applies to it directly.

Proposition 2.24 (The graphon operator is self-adjoint Hilbert–Schmidt)**.** *Under* (A-Reg-W) *the operator* $\mathbb{W}$ *is a self-adjoint Hilbert–Schmidt operator on* $L^2[0,1]$*, hence compact, with*

$$\|\mathbb{W}\|_{\mathrm{HS}} = \|W\|_{L^2([0,1]^2)} \le 1, \qquad \|\mathbb{W}\| \le \|\mathbb{W}\|_{\mathrm{HS}} \le 1.$$

Its spectrum is real and accumulates only at 0.

Proof. Every hypothesis the kernel theorem asks for is supplied by (A-Reg-W) and the finiteness of the type interval. First, $W \in L^2([0,1]^2)$: the kernel is bounded by 1 and $[0,1]^2$ has finite Lebesgue measure, so $\|W\|^2_{L^2([0,1]^2)} = \iint W(x,y)^2 \, \mathrm{d}x \, \mathrm{d}y \le \iint 1 \, \mathrm{d}x \, \mathrm{d}y = 1 < \infty$. By theorem 2.22, $\mathbb{W} = T_W$ is therefore Hilbert–Schmidt with $\|\mathbb{W}\|_{\mathrm{HS}} = \|W\|_{L^2([0,1]^2)} \le \|W\|_\infty \le 1$, and the operator norm is dominated by the Hilbert–Schmidt norm (definition 2.21), giving the chain $\|\mathbb{W}\| \le \|\mathbb{W}\|_{\mathrm{HS}} \le 1$. Second, self-adjointness: a graphon is by definition symmetric and real, $W(x,y) = W(y,x) = \overline{W(y,x)}$, which is exactly the Hermitian-kernel condition $K = K^*$ of the theorem, so $\mathbb{W} = \mathbb{W}^*$. Third, Hilbert–Schmidt implies compact (definition 2.21). The remaining two claims—that the spectrum is real and accumulates only at 0—are precisely the content of the spectral theorem for compact self-adjoint operators, proved in the next section; we have here established exactly the two hypotheses (compact, self-adjoint) that theorem consumes. □

It is worth dwelling on the inequality $\|\mathbb{W}\| \le \|\mathbb{W}\|_{\mathrm{HS}} = \|W\|_{L^2([0,1]^2)} \le 1$, because it is more than bookkeeping: it is the first *a priori estimate* the graphon contributes to the book's existence machinery. The bound $\|W\|_{L^2} \le 1$ is a direct quantitative consequence of the qualitative regularity assumption (A-Reg-W)—boundedness of the kernel into $[0,1]$ on a domain of unit measure—and through the kernel theorem it hands us, for free, control of the *operator* norm $\|\mathbb{W}\| \le 1$. That operator-norm control is exactly the quantity the coupling assumption (A-Coupling) of appendix A.6 regulates: the contraction-based uniqueness arguments of the later mean-field-game chapters require $\|\mathbb{W}\|$ to lie below a critical threshold, and the present proposition is what guarantees the norm is finite and computable from the kernel in the first place, so that (A-Coupling) is a condition one can actually check. The recurring pattern of the chapter is visible in miniature here: a regularity hypothesis becomes, via Hilbert–Schmidt, a hard bound on an operator, and that bound is what a downstream fixed-point argument will spend.

This proposition is the reason section 2.5 exists. The entire spectral apparatus—the network's eigenvector centrality, its spectral gap, its low-rank reductions—rests on the single fact that $\mathbb{W}$ is compact and self-adjoint [99], and the payoff is collected in chapter 18. We have now placed $\mathbb{W}$ exactly where the spectral theorem can act on it: it is the infinite-dimensional counterpart of a real symmetric matrix, compact so that its eigenvalues form a discrete sequence rather than a continuum, and self-adjoint so that those eigenvalues are real and its eigenvectors orthogonal. The next section makes good on the two claims this proof deferred—real spectrum, accumulation only at 0—by proving the spectral theorem itself, the infinite-dimensional analogue of the orthogonal diagonalization of a symmetric matrix, and reads off from it the orthonormal basis of eigenfunctions in which $\mathbb{W}$ becomes the network's mode-by-mode band structure.

2.5 The spectral theorem for compact self-adjoint operators

The previous section did its work and stopped one step short of the payoff. Proposition 2.24 placed the graphon operator $\mathbb{W}$ exactly where we wanted it: it is compact and self-adjoint, the infinite-dimensional counterpart of a real symmetric matrix, and the proof there deferred two precise claims to "the spectral theorem" without proving them—that the spectrum is *real*, and that it can *accumulate only at* 0. This section makes good on both, and in doing so converts the abstract operator into the concrete object the rest of the book reads off it.

The template is one the reader already knows from finite dimensions. A real symmetric matrix is orthogonally diagonalizable: it has a complete orthonormal set of eigenvectors, its eigenvalues are real, and in the eigenbasis the matrix is simply a list of rescalings, one per coordinate. The spectral theorem is the statement that a compact self-adjoint operator on a Hilbert space enjoys the *same* structure, with a single unavoidable amendment forced by infinite dimensionality: the eigenvalues no longer form a finite list but a sequence, and that sequence must tend to zero. The "tend to zero" is not a blemish—it is exactly compactness leaving its fingerprint on the spectrum, and it is what later licenses approximating the full coupling by a finite-rank truncation. Once proved, the theorem hands us, for $T = \mathbb{W}$, an orthonormal basis of eigenfunctions in which $\mathbb{W}$ acts diagonally—each mode rescaled by its eigenvalue—and that mode-by-mode picture is precisely the network's "band structure" that chapter 18 exploits.

Everything rests on building the eigenvalues, and we build them not by solving a characteristic equation—there is none in infinite dimensions—but by a *variational principle*: the largest eigenvalue is the largest value the quadratic form $\langle Tx, x\rangle$ takes on the unit sphere, the next is the largest value on the orthogonal complement of the first eigenvector, and so on. For this strategy to find eigenvalues at all, the maximum of the quadratic form must *equal* the operator norm; otherwise maximizing $\langle Tx, x\rangle$ would tell us nothing about how large T really is. That identity is false for a general bounded operator and true for a self-adjoint one, and it is the single algebraic fact the whole construction turns on. We isolate it first.

Lemma 2.25 (Norm via the quadratic form)**.** *If $T \in \mathcal{B}(H)$ is self-adjoint, then $\|T\| = \sup_{\|x\|=1} |\langle Tx, x\rangle|$.*

Proof. Write $M = \sup_{\|x\|=1} |\langle Tx, x\rangle|$ for the quantity on the right. One inequality is free and uses nothing about T beyond boundedness: for a unit vector x, Cauchy–Schwarz gives $|\langle Tx, x\rangle| \le \|Tx\|\,\|x\| = \|Tx\| \le \|T\|$, so taking the supremum yields $M \le \|T\|$. The whole content of the lemma is the reverse inequality $\|T\| \le M$, and this is where self-adjointness is indispensable.

First record what M buys us by homogeneity: for *every* x (not just unit vectors), $|\langle Tx, x\rangle| \le M\,\|x\|^2$, since rescaling x to unit length and back multiplies both sides by $\|x\|^2$. Now expand the quadratic form on the sum and difference of two vectors. Using bilinearity and then self-adjointness, $\langle Ty, x\rangle = \langle y, Tx\rangle = \overline{\langle Tx, y\rangle}$, the cross terms combine into twice a real part:

$$\langle T(x+y), x+y\rangle - \langle T(x-y), x-y\rangle = 2\langle Tx, y\rangle + 2\langle Ty, x\rangle = 4\,\mathrm{Re}\langle Tx, y\rangle.$$

This is the polarization identity, and *self-adjointness is exactly what made the four cross terms collapse to* $4\,\mathrm{Re}\langle Tx, y\rangle$ rather than an uncontrolled combination. Bounding each quadratic form on the left by M times the square of its argument and then applying the parallelogram law $\|x+y\|^2 + \|x-y\|^2 = 2\,\|x\|^2 + 2\,\|y\|^2$,

$$4\;|\mathrm{Re}\langle Tx, y\rangle| \le M\big(\,\|x+y\|^2 + \|x-y\|^2\,\big) = 2M\big(\,\|x\|^2 + \|y\|^2\,\big),$$

so $|\mathrm{Re}\langle Tx, y\rangle| \le \frac{M}{2}(\|x\|^2 + \|y\|^2)$. Specialize to $\|x\| = \|y\| = 1$, where the right-hand side is just M. The bound holds for every unit y, so we are free to choose the one that makes the left-hand side as large as possible: if $Tx \neq 0$, take $y = Tx/\|Tx\|$, the unit vector aligned with Tx, for which $\mathrm{Re}\langle Tx, y\rangle = \|Tx\|$. That choice turns the inequality into $\|Tx\| \leq M$ for every unit x (trivially true when $Tx = 0$), and the supremum over unit x gives $\|T\| \leq M$. The two inequalities together are the claimed equality. □

The lemma is the lever for the entire construction: it says that to locate the largest-magnitude eigenvalue of a self-adjoint operator it suffices to maximize the real scalar quantity $\langle Tx, x\rangle$ over the unit sphere—a genuine optimization problem, the kind compactness is built to solve. The theorem now turns that single maximization into a complete diagonalization by iterating it.

Theorem 2.26 (Spectral theorem, compact self-adjoint operators)**.** *Let H be a separable real (or complex) Hilbert space and $T \in \mathcal{B}(H)$ a nonzero compact self-adjoint operator. Then there is a finite or countably infinite orthonormal system $\{e_n\}$ of eigenvectors of T, with corresponding real eigenvalues $\{\lambda_n\}$ satisfying*

$$|\lambda_1| \geq |\lambda_2| \geq \cdots > 0, \qquad \lambda_n \to 0 \ \text{ if the system is infinite,}$$

such that

$$Tx \;=\; \sum_n \lambda_n \langle x, e_n\rangle\, e_n \qquad \text{for all } x \in H,$$

the series converging in H. Each nonzero eigenvalue has finite-dimensional eigenspace, the only possible accumulation point of the eigenvalues is 0, and $H = \ker T \;\oplus\; \overline{\operatorname{span}}\{e_n\}$ orthogonally. Equivalently, H has an orthonormal basis of eigenvectors of T (adjoining any orthonormal basis of $\ker T$).

Proof sketch. The mechanism is iterated maximization of the quadratic form, with compactness supplying the maximizers at each stage. We give the argument in full in the real case; the only piece genuinely deferred is the complex case, noted at the end.

Step 1: produce the first eigenvalue and eigenvector. By lemma 2.25, $\|T\| = \sup_{\|x\|=1} |\langle Tx, x\rangle|$, so there is a maximizing sequence x_k with $\|x_k\| = 1$ and $|\langle Tx_k, x_k\rangle| \to \|T\|$. Each $\langle Tx_k, x_k\rangle$ is a real number in $[-\|T\|, \|T\|]$ (real by self-adjointness), and its absolute value approaches the endpoint $\|T\|$; hence one of the two values $+\|T\|$, $-\|T\|$ is approached along an infinite subsequence. Pass to that subsequence and call the limit λ_1, so $\langle Tx_k, x_k\rangle \to \lambda_1$ with $|\lambda_1| = \|T\| > 0$ (the operator is nonzero by hypothesis).

The sequence (x_k) is bounded, living on the unit sphere, so corollary 2.17—the Hilbert-space form of Banach–Alaoglu—lets us extract a weakly convergent subsequence $x_k \rightharpoonup e_1$. This is where compactness is spent: a compact operator turns weak convergence into strong convergence of images, so $Tx_k \to Te_1$ *strongly.* Weak convergence alone would leave Tx_k only weakly convergent and the argument would stall; compactness is exactly the upgrade that makes it run. Now the self-adjoint computation that pins down e_1. Expanding the square and using $\|x_k\| = 1$,

$$\|Tx_k - \lambda_1 x_k\|^2 = \|Tx_k\|^2 - 2\lambda_1\langle Tx_k, x_k\rangle + \lambda_1^2.$$

The first term obeys $\|Tx_k\|^2 \leq \|T\|^2 = \lambda_1^2$, so the right-hand side is at most $2\lambda_1^2 - 2\lambda_1\langle Tx_k, x_k\rangle = 2\lambda_1(\lambda_1 - \langle Tx_k, x_k\rangle) \to 0$ as $\langle Tx_k, x_k\rangle \to \lambda_1$. Hence $Tx_k - \lambda_1 x_k \to 0$ strongly. Combining with

$Tx_k \to Te_1$ gives $\lambda_1 x_k = Tx_k - (Tx_k - \lambda_1 x_k) \to Te_1$, and since $\lambda_1 \neq 0$ we may divide: $x_k \to \lambda_1^{-1} Te_1$ *strongly.* A strong limit must agree with the weak limit e_1, so $e_1 = \lambda_1^{-1} Te_1$, that is $Te_1 = \lambda_1 e_1$; and strong convergence preserves norm, so $\|e_1\| = \lim \|x_k\| = 1$. Thus e_1 is a genuine unit eigenvector and its eigenvalue λ_1 has the largest possible modulus, $|\lambda_1| = \|T\|$.

Step 2: peel off and iterate. The subspace $e_1^\perp$ is *invariant* under T, and this is again self-adjointness at work: if $\langle x, e_1\rangle = 0$ then $\langle Tx, e_1\rangle = \langle x, Te_1\rangle = \lambda_1 \langle x, e_1\rangle = 0$, so $Tx \in e_1^\perp$. The restriction $T|_{e_1^\perp}$ is the restriction of a compact self-adjoint operator to a closed invariant subspace, hence is itself compact and self-adjoint, and Step 1 applies to it verbatim, producing a unit eigenvector $e_2 \in e_1^\perp$ (so $e_2 \perp e_1$) with eigenvalue $|\lambda_2| = \left\|T|_{e_1^\perp}\right\| \leq \|T\| = |\lambda_1|$. Iterating—at each stage restricting to $\{e_1, \dots, e_n\}^\perp$ and maximizing again—manufactures an orthonormal system $\{e_n\}$ with eigenvalues of nonincreasing modulus, $|\lambda_1| \geq |\lambda_2| \geq \cdots > 0$. The process halts at a finite stage exactly when the restricted operator becomes the zero operator (then everything left lies in $\ker T$); otherwise it runs forever and yields an infinite orthonormal sequence.

Step 3: the eigenvalues tend to zero. Suppose the system is infinite. We claim $\lambda_n \to 0$, and here is where compactness pays off a second time, now obstructing the eigenvalues from staying large. If $\lambda_n \not\to 0$, then since $|\lambda_n|$ is nonincreasing it is bounded below by some $\delta > 0$ for all n. The eigenvectors e_n are orthonormal, hence bounded, so by compactness their images $Te_n = \lambda_n e_n$ must have a convergent subsequence. But for $n \neq m$, orthonormality gives

$$\|Te_n - Te_m\|^2 = \|\lambda_n e_n - \lambda_m e_m\|^2 = \lambda_n^2 + \lambda_m^2 \geq 2\delta^2,$$

so no two of the Te_n are within $\delta\sqrt{2}$ of each other and *no* subsequence can be Cauchy. This contradiction forces $\lambda_n \to 0$. The same orthogonality-versus-compactness mechanism settles the two structural fine points the previous section had deferred. *Finite multiplicity:* were a nonzero eigenvalue λ to have an infinite-dimensional eigenspace, it would contain an infinite orthonormal sequence (f_j) with $Tf_j = \lambda f_j$, and $\|Tf_j - Tf_i\|^2 = 2\lambda^2 > 0$ for $i \neq j$ would again deny the bounded sequence (f_j) any convergent image, contradicting compactness; so each nonzero eigenspace is finite-dimensional. *No nonzero accumulation:* if distinct eigenvalues λ_{n_j} converged to some $\mu \neq 0$, their orthonormal eigenvectors would produce the identical contradiction. Hence 0 is the only possible accumulation point of the spectrum—precisely the "accumulation only at 0" that proposition 2.24 promised the spectral theorem would supply, and the reality of every λ_n is the other deferred claim, delivered by self-adjointness in Step 1.

Step 4: assemble the expansion and identify the kernel. Fix any $x \in H$ and let $r_N = x - \sum_{n \leq N} \langle x, e_n\rangle e_n$ be the Fourier remainder after N terms; by construction $r_N \perp e_1, \dots, e_N$, so r_N lies in the subspace $\{e_1, \dots, e_N\}^\perp$ on which T has norm exactly $|\lambda_{N+1}|$ (that was the defining maximization of Step 2). Therefore $\|Tr_N\| \leq |\lambda_{N+1}|\,\|r_N\| \leq |\lambda_{N+1}|\,\|x\| \to 0$, using Bessel's inequality $\|r_N\| \leq \|x\|$ and $\lambda_{N+1} \to 0$. But $Tr_N = Tx - \sum_{n \leq N} \lambda_n \langle x, e_n\rangle e_n$ by linearity and $Te_n = \lambda_n e_n$, so the partial sums $\sum_{n \leq N} \lambda_n \langle x, e_n\rangle e_n$ converge to Tx: this is the asserted expansion $Tx = \sum_n \lambda_n \langle x, e_n\rangle e_n$. Finally we identify the kernel. Let $M = \overline{\operatorname{span}}\,\{e_n\}$. If $x \in M^\perp$ then $\langle x, e_n\rangle = 0$ for all n and the expansion collapses to $Tx = 0$, so $M^\perp \subseteq \ker T$. Conversely, if $Tx = 0$ then for each n, $\lambda_n \langle x, e_n\rangle = \langle x, \lambda_n e_n\rangle = \langle x, Te_n\rangle = \langle Tx, e_n\rangle = 0$, and since $\lambda_n \neq 0$ this forces $\langle x, e_n\rangle = 0$, i.e. $x \in M^\perp$; so $\ker T \subseteq M^\perp$. The two inclusions give $\ker T = M^\perp$ and hence the orthogonal decomposition $H = \ker T \oplus \overline{\operatorname{span}}\,\{e_n\}$. Adjoining any orthonormal basis of $\ker T$ (each of whose vectors is an eigenvector with eigenvalue 0) to the $\{e_n\}$ produces an orthonormal basis of all of H consisting of eigenvectors of T, which is the diagonalization claimed.

This is the complete argument in the real case; we follow [123, Ch. 6], [31, Ch. 6], and [113, Thm. VI.16]. The only genuinely deferred piece is the *complex* case—where "self-adjoint" is read with the Hermitian conjugate of definition 2.3, and one first checks that $\langle Tx, x\rangle$ is real so that the eigenvalues come out real—which is carried out in appendix B.4. (No measurable selection enters: extracting a maximizing subsequence is ordinary sequential compactness, not a measurable-selection problem; the latter appears only later, for a *parametrized family* of operators in chapter 18.) $\square$

The variational construction does more than locate the eigenvalues one at a time; it endows each eigenvalue with a characterization that never mentions the eigenvectors. This is the comparison principle we will need in chapter 18 to read off the network's leading modes—its spectral gap, the size of its dominant band—without ever solving an eigenproblem exactly: bounding a Rayleigh quotient on a cleverly chosen subspace bounds an eigenvalue. We state it for the positive part of the spectrum; the negative part is identical with signs reversed (apply the result to $-T$).

Corollary 2.27 (Courant–Fischer min–max)**.** *Let T be compact self-adjoint with positive eigenvalues $\lambda_1^+ \geq \lambda_2^+ \geq \cdots > 0$ listed with multiplicity, and with orthonormal eigenvectors $e_1^+, e_2^+, \ldots$ from theorem 2.26. Then for each n not exceeding the number of positive eigenvalues (so that λ_n^+ exists),*

$$\lambda_n^+ \;=\; \max_{\substack{V\subseteq H\\ \dim V = n}} \; \min_{\substack{x\in V\\ \|x\|=1}} \langle Tx, x\rangle \;=\; \min_{\substack{V\subseteq H\\ \dim V = n-1}} \; \max_{\substack{x\in V^\perp\\ \|x\|=1}} \langle Tx, x\rangle .$$

In particular $\lambda_1^+ = \max_{\|x\|=1}\langle Tx, x\rangle$, attained at the leading eigenvector.

Proof. The single tool is the spectral expansion of the quadratic form. Diagonalizing T by theorem 2.26 and writing the coordinates of a unit vector x in the full eigenbasis, $\langle Tx, x\rangle = \sum_k \lambda_k \,|\langle x, e_k\rangle|^2$, a weighted average of eigenvalues with weights $|\langle x, e_k\rangle|^2$ that sum to at most $\|x\|^2 = 1$ (Parseval). Each identity has two halves, and they have the same logic: to bound the min–max from *below* we exhibit one good subspace on which the inner extremum is already large enough, and to bound it from *above* we show that *every* candidate subspace contains one bad direction that drags the inner extremum down. The bridge between the two is a dimension count.

First identity, lower bound (a good subspace). Test the particular n-dimensional subspace $V_0 = \operatorname{span}\left\{e_1^+, \ldots, e_n^+\right\}$ spanned by the top n positive eigenvectors. Any unit $x \in V_0$ expands as $x = \sum_{k\leq n} c_k e_k^+$ with $\sum_{k\leq n} |c_k|^2 = 1$, so $\langle Tx, x\rangle = \sum_{k\leq n} \lambda_k^+ |c_k|^2 \geq \lambda_n^+ \sum_{k\leq n} |c_k|^2 = \lambda_n^+$, because each eigenvalue in play satisfies $\lambda_k^+ \geq \lambda_n^+$ for $k \leq n$. Thus the inner minimum over V_0 is $\geq \lambda_n^+$, and the outer maximum over *all* n-dimensional subspaces is therefore $\geq \lambda_n^+$.

First identity, upper bound (a bad direction in every subspace). Let V be *any* n-dimensional subspace. Consider the closed subspace $\operatorname{span}\left\{e_1^+, \ldots, e_{n-1}^+\right\}^\perp$, the orthogonal complement of the top $n-1$ positive eigenvectors, which is cut out by $n-1$ linear conditions. Asking a vector of the n-dimensional V to satisfy those $n-1$ conditions leaves at least $n-(n-1) = 1$ dimension of solutions, so there is a unit vector $x \in V \cap \operatorname{span}\left\{e_1^+, \ldots, e_{n-1}^+\right\}^\perp$. Being orthogonal to $e_1^+, \ldots, e_{n-1}^+$ kills exactly the $n-1$ largest weights, so its quadratic form runs only over the remaining spectrum:

$\langle Tx, x\rangle = \sum_{k\notin\{1,\ldots,n-1\}} \lambda_k \left|\langle x, e_k\rangle\right|^2$. Every eigenvalue still present is $\leq \lambda_n^+$—the surviving positive eigenvalues are $\lambda_n^+ \geq \lambda_{n+1}^+ \geq \cdots$, and all zero and negative eigenvalues are $\leq 0 < \lambda_n^+$ a fortiori—so $\langle Tx, x\rangle \leq \lambda_n^+ \sum_k \left|\langle x, e_k\rangle\right|^2 \leq \lambda_n^+$. Hence the inner minimum over V is $\leq \lambda_n^+$ for every V, and the outer maximum is $\leq \lambda_n^+$. Combining the two halves, the maximum equals λ_n^+, attained at V_0.

Second identity. This is the same argument with the roles of "test a good subspace" and "find a bad direction" exchanged, now optimizing over $(n-1)$-dimensional subspaces V and looking inside their complements $V^\perp$. For the *upper* bound, test $V = \operatorname{span}\left\{e_1^+, \ldots, e_{n-1}^+\right\}$: its complement $V^\perp$ contains no e_k^+ with $k < n$, so by the computation just performed $\langle Tx, x\rangle \leq \lambda_n^+$ for every unit $x \in V^\perp$, with equality at $x = e_n^+ \in V^\perp$; thus the inner maximum over this $V^\perp$ is exactly λ_n^+, and the outer minimum is $\leq \lambda_n^+$. For the *lower* bound, let V be any $(n-1)$-dimensional subspace and intersect $V^\perp$ with $U = \operatorname{span}\left\{e_1^+, \ldots, e_n^+\right\}$: a vector $u \in U$ lies in $V^\perp$ precisely when it is orthogonal to the $n-1$ basis vectors of V, again $n-1$ conditions on the n-dimensional U, so $\dim(U \cap V^\perp) \geq n - (n-1) = 1$ and there is a unit $x \in U \cap V^\perp$. Since $x \in U$ puts all its weight on the top n positive eigenvalues, $\langle Tx, x\rangle = \sum_{k\leq n} \lambda_k^+ \left|\langle x, e_k^+\rangle\right|^2 \geq \lambda_n^+$. So the inner maximum over $V^\perp$ is $\geq \lambda_n^+$ for every V, and the outer minimum is $\geq \lambda_n^+$. The two halves again pinch the value to λ_n^+. □

Physics note

Theorem 2.26 is the statement that a compact self-adjoint operator has *normal modes.* Reading $T = \mathbb{W}$ as the (self-generated, type-indexed) coupling of a network, the eigenfunctions e_n are the collective modes and the eigenvalues λ_n their coupling strengths; the expansion $\mathbb{W}f = \sum_n \lambda_n \langle f, e_n\rangle e_n$ diagonalizes the interaction exactly as a normal-mode transform diagonalizes a quadratic Hamiltonian. The leading eigenfunction e_1, carrying the largest $|\lambda_1| = \|\mathbb{W}\|$, is the network's dominant, most strongly coupled collective mode—it is the *eigenvector centrality* of chapter 18, the profile that the population settles into under repeated averaging by $\mathbb{W}$. The decay $\lambda_n \to 0$ is the statement that high-index modes are increasingly weakly coupled and contribute increasingly little, which is what makes finite-rank truncations of the coupling accurate. This picture is genuinely useful; but the proof above uses no physics, and every assertion stands on the variational argument alone.

Abstract diagonalization is only as convincing as the examples one can carry it out on, and the operator we care about admits three that we can solve in closed form. Each is a member of Thread B (appendix A.7), and the three are arranged in increasing structural richness: a constant kernel that collapses to a single mode and recovers Thread A, a rank-one kernel $f(x)f(y)$ that is the cleanest graphon reduction, and a block kernel that bridges to the multi-population games of chapter 30. In every case the spectral theorem is not invoked abstractly—we exhibit the eigenfunctions and eigenvalues explicitly and check them against the Hilbert–Schmidt identity $\sum_n \lambda_n^2 = \|W\|^2_{L^2([0,1]^2)}$ of proposition 2.24, which the spectral expansion must satisfy term by term.

Worked computation 2.28 (Spectrum of the graphon operator: constant and rank-one kernels)**.** Throughout, $H = L^2[0,1]$ with $\langle f, g\rangle = \int_0^1 fg$, and $\mathbb{W}$ is the operator of definition 2.23.

(a) Constant graphon (Erdős–Rényi / Curie–Weiss closure). Let $W(x,y) \equiv c$ with $c \in (0,1]$.

Then

$$(\mathbb{W}f)(x) = c\int_0^1 f(y)\ \mathrm{d}y = c\,\langle f, \mathbf{1}\rangle,$$

where $\mathbf{1}$ is the constant function 1, which has $\|\mathbf{1}\| = 1$. Thus $\mathbb{W}f$ is always a multiple of $\mathbf{1}$: the operator has rank one, with the single nonzero eigenvalue $\lambda_1 = c$ and normalized eigenfunction $e_1 = \mathbf{1}$, while every function with zero mean lies in $\ker \mathbb{W}$. Here $\|\mathbb{W}\| = \|\mathbb{W}\|_{\mathrm{HS}} = c$, consistent with $\|W\|^2_{L^2([0,1]^2)} = c^2$. The operator collapses the entire population field to its mean and rescales it—this is precisely the global mean-field coupling of Thread A (appendix A.7), recovered as the constant-graphon special case, with the coupling strength c playing the role of the smallness parameter in (A-Coupling).

(b) Rank-one graphon (the clean GMFG reduction). Let $W(x,y) = f(x)f(y)$ for a fixed $f \in L^2[0,1]$, symmetric by construction and (for W to be a graphon) with values in $[0,1]$. The integral now factors through a single scalar pairing:

$$(\mathbb{W}g)(x) = f(x)\int_0^1 f(y)g(y)\ \mathrm{d}y = \langle g, f\rangle\, f(x).$$

This is the structural heart of the reduction. Whatever field g we feed in, the output points in the *one fixed direction* f, and all the information $\mathbb{W}$ retains about g is the single number $\langle g, f\rangle$. It is this number that Thread B names the scalar *order parameter* $\varrho = \langle g, f\rangle$ (appendix A.7), so that the entire action of the operator compresses to $\mathbb{W}g = \varrho\, f$. Spectrally, $\mathbb{W}$ has rank one: its only nonzero eigenvalue is $\lambda_1 = \|f\|^2_{L^2} = \int_0^1 f^2$—read off by applying $\mathbb{W}$ to f itself, $\mathbb{W}f = \langle f, f\rangle f = \|f\|^2_{L^2} f$—with normalized eigenfunction $e_1 = f/\|f\|_{L^2}$, while every field orthogonal to f is annihilated, $\ker\mathbb{W} = \{f\}^{\perp}$. The Hilbert–Schmidt check is immediate and confirms the spectral bookkeeping: $\|W\|^2_{L^2([0,1]^2)} = \int_0^1\!\int_0^1 f(x)^2 f(y)^2\ \mathrm{d}x\ \mathrm{d}y = \big(\int_0^1 f^2\big)^2 = \lambda_1^2$, which is exactly $\sum_n \lambda_n^2$ with its single surviving term. The payoff is the cleanest reduction in the book: because $\mathbb{W}g = \varrho f$ collapses the coupling onto one scalar, a mean-field game played through this kernel is no longer an infinite-dimensional fixed-point problem in the field but a *one-parameter* self-consistent equation for the scalar ϱ—each agent best-responds to the single aggregate ϱ, which in turn is the f-weighted average of the resulting population profile. We take this up dynamically, with ϱ promoted to a time-dependent order parameter ϱ_t, in chapter 22.

(c) Block graphon (the bridge to multi-population games). Let $\{P_1, \ldots, P_K\}$ partition $[0,1]$ into measurable sets of masses $\pi_k = |P_k| > 0$, $\sum_k \pi_k = 1$, and let $W(x,y) = \mathsf{B}_{k\ell}$ be constant on $P_k \times P_\ell$ for a symmetric matrix $\mathsf{B} = (\mathsf{B}_{k\ell}) \in [0,1]^{K\times K}$. The operator $\mathbb{W}$ then maps everything into the K-dimensional subspace of functions that are constant on each block, and it annihilates every function whose average over each block vanishes: if $\int_{P_\ell} f = 0$ for all ℓ then $(\mathbb{W}f)(x) = \sum_\ell \mathsf{B}_{k\ell}\int_{P_\ell} f = 0$ for $x \in P_k$, so the block-mean-zero functions lie in $\ker\mathbb{W}$. On the block-constant subspace, writing $f = \sum_\ell f_\ell \mathbf{1}_{P_\ell}$,

$$(\mathbb{W}f)(x) = \sum_\ell \mathsf{B}_{k\ell}\,\pi_\ell\, f_\ell \qquad (x \in P_k),$$

so in the coordinates (f_ℓ) the operator acts as the $K \times K$ matrix $\mathsf{B}D$ with $D = \mathrm{diag}(\pi_1, \ldots, \pi_K)$. This matrix is not symmetric, but its nonzero eigenvalues—which are the nonzero spectrum of $\mathbb{W}$—are those of the *symmetric* matrix $D^{1/2}\mathsf{B}D^{1/2}$, to which $\mathsf{B}D$ is similar via conjugation by $D^{1/2}$: indeed $D^{1/2}(\mathsf{B}D)D^{-1/2} = D^{1/2}\mathsf{B}D^{1/2}$. Similarity preserves eigenvalues, and the symmetric matrix on the right has real ones, so the spectrum of $\mathbb{W}$ is real exactly as self-adjointness demands—the finite-dimensional shadow of the reality clause in theorem 2.26. The natural inner product making $\mathbb{W}$ self-adjoint here is not the bare Euclidean one on $\mathbb{R}^K$ but the mass-weighted one $\langle u, v\rangle_D = \sum_k \pi_k u_k v_k$, inherited from $L^2[0,1]$ restricted to block-constant fields, and in that inner product $\mathsf{B}D$ is genuinely self-adjoint. The Hilbert–Schmidt check again closes the loop: with W constant $\mathsf{B}_{k\ell}$ on $P_k \times P_\ell$,

$$\|W\|^2_{L^2([0,1]^2)} = \sum_{k,\ell} \mathsf{B}^2_{k\ell}\, \pi_k \pi_\ell = \operatorname{tr}\left[(D^{1/2}\mathsf{B}D^{1/2})^2\right] = \sum_n \lambda_n^2,$$

the trace of the square of the symmetric matrix being the sum of the squares of its eigenvalues, in exact agreement with the spectral theorem. This is precisely the finite multi-population coupling: the blocks are sub-populations, the π_k their mass fractions, and B the between-group interaction matrix; we take it up as the bridge from graphon to multi-population MFG in chapter 30.

The three reductions share one feature worth naming, because it is what chapter 18 will read off the spectrum. In each, the leading eigenfunction e_1 is the profile to which repeated application of $\mathbb{W}$ drives any input—iterating $\mathbb{W}$ amplifies the λ_1-mode relative to all others by the ratio $(\lambda_2/\lambda_1)^n$, so the dominant mode wins, and that dominant mode is the network's eigenvector centrality. And in each, $\|\mathbb{W}\| = \lambda_1$ is precisely the operator norm constrained by the smallness condition (A-Coupling) when a contraction argument is needed for uniqueness in chapter 24: the spectral theorem is what makes that abstract norm a concrete, computable number—c, $\int f^2$, or the top eigenvalue of $D^{1/2}\mathsf{B}D^{1/2}$, respectively.

With the spectral theorem proved and exhibited on the three diagnostic kernels, the operator side of the chapter is complete. We have taken the graphon W from a measurable kernel (chapter 1), through the Hilbert–Schmidt class (proposition 2.24), to a fully diagonalized object: an orthonormal eigenbasis $\{e_n\}$, real eigenvalues $\lambda_n \to 0$, and the mode-by-mode action $\mathbb{W}f = \sum_n \lambda_n \langle f, e_n\rangle e_n$ that chapter 18 interprets as band structure, eigenvector centrality, spectral gap, and the low-rank truncations the decay $\lambda_n \to 0$ justifies. Threads B's later chapters draw on this directly: the rank-one order parameter ϱ of chapter 22, the operator-norm threshold of chapter 24, and the block bridge to multi-population games of chapter 30 all read their structure off the spectrum computed here.

That closes the book's work with operators acting on L^2. But L^2 is not the only space the book needs, and the remaining sections turn from operators to the *other* spaces—the homes of the partial differential equations that the mean-field game couples together—beginning with the Sobolev spaces, in which differentiation itself must be re-founded.

2.6 Sobolev spaces and weak derivatives

The previous section closed the book's work with operators on L^2 and pointed past it: the value function u of an optimal control problem and the density m of a population both solve partial differential equations—the Hamilton–Jacobi–Bellman and Fokker–Planck equations of chapters 6 and 8, the backward and forward halves of every mean-field game—and the preamble already warned that those solutions are generally *not* classically differentiable. A value function built as the infimum of many smooth cost-to-go functions typically has kinks where the optimal control switches; a density transported by a rough drift need not be smooth either. At such points the classical derivative simply does not exist, and an equation written in terms of it has no solution to look for. The repair is the same move the chapter has made twice already. Weak convergence rescued compactness by testing against functionals instead of demanding norm convergence; the weak derivative will rescue differentiation by testing against smooth functions instead of demanding a pointwise limit of difference quotients. This is the third and last of the chapter's "three weak words," and it is the one that builds the spaces the PDEs actually live in.

The mechanism is integration by parts read backwards. For a genuinely C^1 function the derivative is characterized—not just computed—by how it pairs against test functions, and that characterization survives even when the pointwise derivative does not. The Sobolev spaces are the function spaces in which this surviving characterization is taken as the definition of a derivative, and in which the resulting objects can be measured in L^2. We develop them to exactly the depth the existence theory of the MFG PDEs requires and no further: the weak derivative itself, the H^k and H^1_0 scales, the density–Rellich–Poincaré trio that supplies PDE-side compactness, the fractional scale and Gelfand triple that house right-hand sides and parabolic problems, and the Lax–Milgram theorem that solves each linear step inside a mean-field iteration. The heavier elliptic *regularity* theory—that these weak solutions are secretly smooth—is genuinely needed later but not here, and we defer it honestly to chapter 6 and its sources, for a reason remark 2.35 makes precise.

Definition 2.29 (Weak derivative)**.** Let $\Omega \subseteq \mathbb{R}^d$ be open and $u \in L^1_{\mathrm{loc}}(\Omega)$. A function $v \in L^1_{\mathrm{loc}}(\Omega)$ is the *weak (distributional) partial derivative* $\partial_i^{\mathrm{w}} u$ if

$$\int_\Omega u\,\partial_i\varphi \,\mathrm{d}x \;=\; -\int_\Omega v\,\varphi\,\mathrm{d}x \qquad \text{for all } \varphi \in C_c^\infty(\Omega).$$

This is the integration-by-parts identity with the boundary term legislated to vanish; when u is classically C^1 the weak derivative coincides with the classical one. Weak derivatives, when they exist, are unique up to a null set.

Three things in that definition deserve to be unpacked, because together they are the whole content of "weak differentiation." The first is *why this is the right identity at all.* For a C^1 function u on Ω and a test function $\varphi \in C_c^\infty(\Omega)$, ordinary integration by parts gives $\int_\Omega (\partial_i u)\varphi \,\mathrm{d}x = -\int_\Omega u\,\partial_i\varphi \,\mathrm{d}x +$ (boundary term), and the boundary term is the surface integral of $u\varphi$ over $\partial\Omega$. The point of insisting that φ have *compact support* inside the open set Ω is precisely to kill that boundary term: φ vanishes identically near $\partial\Omega$, so there is no surface to integrate over. What remains, $\int_\Omega u\,\partial_i\varphi = -\int_\Omega (\partial_i u)\varphi$, is an identity that mentions $\partial_i u$ only *under an integral against a smooth weight*—never at a point. That is the loophole the definition exploits: it asks for a function v that plays the role $\partial_i u$ would play in this identity, without ever asking u to be differentiable anywhere. The boundary term has been "legislated away" not by fiat but by the choice of test class.

The second is the *micro-check* that this generalizes rather than replaces the classical derivative. If $u \in C^1(\Omega)$, then by the computation just made $v = \partial_i u$ (the genuine, classical partial derivative) satisfies the defining identity for every φ, so the classical derivative *is* a weak derivative; the two notions agree wherever the classical one exists. The gain is entirely at the points and sets where the classical derivative fails. The cleanest one-dimensional witness is $u(x) = |x|$ on $\Omega = (-1, 1)$, which has no derivative at 0: integrating $\int_{-1}^{1} |x|\ \varphi'(x)\ \mathrm{d}x$ by parts on each half $(-1, 0)$ and $(0, 1)$ separately (where u *is* smooth, with derivative ∓ 1) and adding, the two boundary contributions at 0 cancel because $|x|$ is continuous there, leaving $\int_{-1}^{1} |x|\ \varphi' = -\int_{-1}^{1} \operatorname{sgn}(x)\, \varphi$. So $\partial^{\mathrm{w}} |x| = \operatorname{sgn}(x)$, an honest L^1_{loc} function and exactly the "derivative" a physicist would write. The same cancellation is what *fails* for the Heaviside step $H(x) = \mathbf{1}_{x>0}$: integrating by parts leaves a leftover $\varphi(0)$ that no L^1_{loc} function can produce against every φ, so H has no weak derivative inside L^1_{loc}—its derivative is the Dirac mass, a distribution but not a function. The boundary between "has a weak derivative in L^1_{loc}" and "does not" is the boundary a Sobolev space draws, and it is exactly the kink-versus-jump distinction: kinks are allowed, jumps are not.

The third is *uniqueness up to a null set*, which is what makes "the" weak derivative a legitimate phrase. If v_1 and v_2 both satisfy the identity, then $\int_\Omega (v_1 - v_2)\varphi = 0$ for every $\varphi \in C_c^\infty(\Omega)$; a function in L^1_{loc} that integrates to zero against every compactly supported smooth weight is zero almost everywhere (the fundamental lemma of the calculus of variations, itself a density statement about C_c^∞). So $v_1 = v_2$ a.e., and the weak derivative is unique as an element of L^1_{loc}—which, since all our spaces identify functions agreeing a.e., is exactly the right sense of uniqueness. With one notion of derivative now defined for non-smooth functions, we can build the spaces of functions whose weak derivatives are not merely locally integrable but square-integrable, the regime where Hilbert-space geometry returns.

Definition 2.30 (Sobolev space H^1 and H^k)**.** For an open $\Omega \subseteq \mathbb{R}^d$ and integer $k \geq 1$, the Sobolev space $H^k(\Omega)$ consists of all $u \in L^2(\Omega)$ whose weak derivatives $D^\alpha u$ exist in $L^2(\Omega)$ for every multi-index $|\alpha| \leq k$. It is a Hilbert space under

$$\langle u, v \rangle_{H^k} = \sum_{|\alpha| \leq k} \int_\Omega D^\alpha u\, D^\alpha v\ \mathrm{d}x.$$

The case $k = 1$, $\langle u, v \rangle_{H^1} = \int_\Omega (uv + \nabla u \cdot \nabla v)\ \mathrm{d}x$, is the one the variational theory uses. We write $H_0^1(\Omega)$ for the closure of $C_c^\infty(\Omega)$ in $H^1(\Omega)$—morally, the functions in H^1 that vanish on $\partial\Omega$.

The inner product is engineered so that the H^k norm measures a function *together with* all of its weak derivatives up to order k in a single Hilbertian quantity, $\|u\|_{H^k}^2 = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^2}^2$; controlling $\|u\|_{H^k}$ is exactly controlling u and each $D^\alpha u$ in L^2 at once. The distinguished subspace H_0^1 encodes a homogeneous Dirichlet boundary condition without ever evaluating u on the measure-zero set $\partial\Omega$ (where an L^2 function has no canonical value): a function lies in H_0^1 iff it is an H^1-limit of test functions, all of which vanish near the boundary, so "$u = 0$ on $\partial\Omega$" is enforced in the only way an L^2 theory can enforce it—by approximation. This is the space in which the homogeneous boundary-value problems of chapter 6 are posed, and the space on which Poincaré's inequality below will hold.

That H^k is complete—so that it is a genuine Hilbert space and not merely an inner-product space—is the small argument that makes the whole construction worth having, and it is worth seeing in full because it is where the weak derivative earns its keep. Let (u_n) be Cauchy in H^k. By the form of the norm, each sequence $(D^\alpha u_n)_n$, for $|\alpha| \leq k$, is Cauchy in $L^2(\Omega)$; since L^2 is complete (chapter 1), each converges to some $u^\alpha \in L^2$, and in particular $u_n \to u^{(0)} =: u$ in L^2. The only thing left to verify is that the limits are consistent—that u^α really is the weak derivative $D^\alpha u$, not just an unrelated L^2 function. Here is the step that "passes the identity to the limit." For each n and each test function φ, the weak-derivative identity holds, $\int_\Omega u_n\, D^\alpha\varphi = (-1)^{|\alpha|}\int_\Omega (D^\alpha u_n)\varphi$. Both sides are L^2 inner products against fixed φ (or $D^\alpha\varphi$), and inner products are continuous, so letting $n\to\infty$ and using $u_n \to u$ and $D^\alpha u_n \to u^\alpha$ in L^2 gives $\int_\Omega u\, D^\alpha\varphi = (-1)^{|\alpha|}\int_\Omega u^\alpha\varphi$ for every φ. That is precisely the statement $u^\alpha = D^\alpha u$. Hence $u \in H^k$ and $u_n \to u$ in H^k. Completeness was almost free, but only because the weak-derivative identity is preserved under L^2 limits—which is the whole reason the weak derivative, not the classical one, is the right notion to build a Banach space around. The same argument shows H^1_0 is complete (it is a closed subspace of H^1 by definition).

Two further facts turn these complete spaces from formal objects into working tools, and they are exactly the two the PDE existence theory will call on: smooth functions are *dense* in them, so the weak theory is nothing but the completion of the classical one, and on a bounded domain the inclusion of H^1 into L^2 is *compact*, which is the PDE-side source of the compactness the chapter's existence scheme runs on. We collect both, with a third—Poincaré—that they naturally travel with, as a single motivated theorem.

Theorem 2.31 (Density and the compact embedding)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open, bounded, with Lipschitz boundary.*

1. (Meyers–Serrin / density.) *$C^\infty(\overline{\Omega})$ is dense in $H^1(\Omega)$, and $C^\infty_c(\Omega)$ is dense in $H^1_0(\Omega)$; thus the weak theory is the completion of the classical one.*
2. (Rellich–Kondrachov.) *The inclusion $H^1(\Omega) \hookrightarrow L^2(\Omega)$ is* compact*: every H^1-bounded sequence has an L^2-convergent subsequence.*
3. (Poincaré.) *There is $C = C(\Omega)$ with $\|u\|_{L^2} \le C\,\|\nabla u\|_{L^2}$ for all $u \in H^1_0(\Omega)$, so on H^1_0 the gradient norm alone is an equivalent norm.*

The three are deliberately bundled because each answers a different need of the variational method, and reading them in order tells the story. The *density* statement is the bridge between the classical and the weak worlds: it says H^1 is nothing more exotic than the completion of ordinary smooth functions in the H^1 norm, so any identity one can verify for smooth functions and which is continuous in that norm—integration by parts, the product rule, chain rules—extends to all of H^1 by approximation. Without density, the weak theory would be a separate universe from the classical one; with it, the two are the same object seen at two resolutions, and every manipulation we will perform on weak solutions is justified by passing to a smooth approximant and back.

The *Rellich–Kondrachov* statement is the one this chapter has been pointing at since section 2.4, and it deserves to be named for what it is: a *compact embedding.* The inclusion map $\iota\colon H^1(\Omega) \to L^2(\Omega)$, $u \mapsto u$, is a bounded operator between two Hilbert spaces—bounded because $\|u\|_{L^2} \le \|u\|_{H^1}$ trivially—and Rellich says this particular bounded operator is *compact* in the exact sense of definition 2.18: it sends the H^1 unit ball to a precompact subset of L^2, equivalently it maps H^1-bounded sequences to sequences with L^2-convergent subsequences. This is not a restatement of Banach–Alaoglu. Alaoglu only gives a *weakly* convergent subsequence in L^2 from an L^2-bounded sequence; Rellich upgrades that to *strong* (norm) L^2 convergence, and the price of the upgrade is the extra information that the gradients are also bounded—the sequence is bounded not merely in L^2 but in H^1. That extra derivative bound is exactly an *a priori estimate*, and the moral the chapter has repeated in every guise reappears here in its sharpest form: compactness is an a priori estimate in disguise, and Rellich is the disguise worn on the PDE side. It is the engine that converts the bound–extract–identify scheme into a working existence proof for the HJB and Fokker–Planck equations: bound the approximate solutions in H^1 by an energy estimate, extract an L^2-strongly convergent subsequence by Rellich, identify the limit as a weak solution by passing the equation to the limit (the manoeuvre completeness above already rehearsed). What weak compactness did for measures and for L^2, Rellich does for Sobolev functions; it is the same beat played in a new register.

The *Poincaré* statement is the small inequality that makes the scheme self-supporting on H^1_0. It says that on $H^1_0(\Omega)$—functions vanishing at the boundary—the function cannot be large unless its gradient is, so the gradient seminorm $\|\nabla u\|_{L^2}$ already controls the full H^1 norm and is therefore an *equivalent* norm. The hypothesis that u vanish on the boundary is essential: on all of H^1 the constant functions have zero gradient but nonzero L^2 norm, so no such inequality can hold; pinning u to zero at $\partial\Omega$ removes exactly the constant (and near-constant) directions that would break it, and a bounded domain keeps the constant $C(\Omega)$ finite. What Poincaré buys, concretely, is *coercivity*: it is the reason the Dirichlet form $a(u,v) = \int_\Omega \nabla u \cdot \nabla v$ is coercive on H^1_0—because $a(u,u) = \|\nabla u\|^2_{L^2} \geq C^{-2}\,\|u\|^2_{H^1}$—which is the single hypothesis Lax–Milgram below will demand. Density justifies the calculus, Rellich supplies the compactness, Poincaré supplies the coercivity: the trio is the complete toolkit for solving one linear elliptic problem, and we shall use all three within a page.

All three are proved in appendix B.8, and we keep the proofs there rather than here because nothing the chapter *exports* depends on their internals, only on their statements. For orientation: Poincaré and the Meyers–Serrin density go through directly, while Rellich–Kondrachov goes through the Fréchet–Kolmogorov compactness criterion in L^2 (equicontinuity of translates, which an H^1 bound supplies). The cleanest instance of the mechanism is the torus, where it falls straight out of the Fourier definition of definition 2.32: there an H^1 bound is a bound on $\sum_\xi (1+|\xi|^2)\,|\hat{u}(\xi)|^2$, the weight $(1+|\xi|^2)^{-1}$ decays, and tails in frequency are uniformly small—which is precisely L^2-precompactness. We keep [64, Ch. 5] and [31, Ch. 8–9] for attribution and further reading. That Fourier picture is no accident, and it is the natural way to interpolate between the integer orders H^k to the fractional scale where the precise regularity of HJB solutions is measured; we turn to it now.

Definition 2.32 (Fractional Sobolev space $H^s(\mathbb{R}^d)$)**.** For $s \in \mathbb{R}$ and $u \in L^2(\mathbb{R}^d)$ with Fourier transform $\hat{u}$, set

$$H^s(\mathbb{R}^d) = \left\{ u : \ \|u\|^2_{H^s} = \int_{\mathbb{R}^d} \left(1 + |\xi|^2\right)^s \, |\hat{u}(\xi)|^2 \, \mathrm{d}\xi < \infty \right\}.$$

For integer $s = k \geq 0$ this norm is equivalent to that of definition 2.30, since multiplication by ξ^α on the Fourier side is differentiation. Increasing s means increasing smoothness; $s < 0$ gives spaces of distributions, and $(H^s)^* = H^{-s}$ under the L^2 pairing.

The Fourier definition does two things at once. First, it makes “half a derivative” meaningful: where the integer space H^k weights $|\hat{u}(\xi)|^2$ by the polynomial $(1+|\xi|^2)^k$, the fractional space simply allows any real exponent s, and since each factor of $|\xi|$ on the Fourier side is one derivative in space, s counts derivatives on a continuous dial. Increasing s demands faster decay of $\hat{u}$ and hence more smoothness; this is the scale on which one says, precisely, "the HJB value function has s derivatives" even when s is not an integer. Second, and decisively for the variational theory, allowing $s < 0$ produces honest spaces of *distributions*—objects rougher than functions, like the Dirac mass, whose Fourier transform does not decay but whose negative-order weighted norm is still finite. The pairing that ties the two ends together is the duality $(H^s)^* = H^{-s}$ under the L^2 inner product: a distribution $f \in H^{-s}$ acts on a test function $u \in H^s$ by the extension of $\int fu$, and the Cauchy–Schwarz split $\int \hat{f}\,\overline{\hat{u}} = \int \big[(1+|\xi|^2)^{-s/2}\hat{f}\big]\big[(1+|\xi|^2)^{s/2}\overline{\hat{u}}\big]$ shows this pairing is bounded by $\|f\|_{H^{-s}}\|u\|_{H^s}$, with H^{-s} realizing *every* bounded functional on H^s this way. The negative exponent is not a curiosity: it is the home of the right-hand sides of elliptic equations, which after one integration by parts naturally lose a derivative and land in a negative-order space. The duality $(H^s)^* = H^{-s}$ thus organizes the variational solution of elliptic equations into a single clean statement, and the way to package it for use is a triple of spaces.

Definition 2.33 (Gelfand triple)**.** A *Gelfand triple* (or rigged Hilbert space) is a continuous dense embedding $V \hookrightarrow H \hookrightarrow V^*$ of a Hilbert space V into a pivot Hilbert space H (identified with its own dual) and thence into V^*, with $\langle\cdot,\cdot\rangle_H$ extending to the duality pairing of V and V^*. The model case is $H_0^1(\Omega) \hookrightarrow L^2(\Omega) \hookrightarrow H^{-1}(\Omega)$.

The triple is not idle bookkeeping, and it is worth saying concretely what later chapters do with it, because the abstraction is doing real work in three distinct places. The first is the one visible on the next page. The datum $\ell \in V^*$ of the Lax–Milgram theorem below, for the model choice $V = H_0^1(\Omega)$, lives in exactly the endpoint $V^* = H^{-1}(\Omega)$ of this triple. That is why "a right-hand side in H^{-1}" is the natural generality for a weak elliptic problem: a source term need not be a function at all—it may be a derivative of an L^2 function, hence a genuine distribution—and the triple is what makes $-\Delta\colon H_0^1 \to H^{-1}$ an isomorphism onto the right space rather than a map with a domain mismatch. The second is time-dependence. In the kernel HJB–Fokker–Planck system of chapter 22, the unknowns are not single functions but *trajectories* of functions, curves $t \mapsto u(t) \in V$; their time derivatives $\dot{u}(t)$, formed weakly, fall a notch in regularity and live in V^*, while the functions themselves live in V and their natural "energy" is measured in the pivot H. The Gelfand triple is precisely the staircase $V \hookrightarrow H \hookrightarrow V^*$ that lets a solution and its time derivative inhabit adjacent rungs and still be paired—$\langle \dot{u}(t), u(t)\rangle_{V^*\times V} = \frac{1}{2}\frac{d}{dt}\|u(t)\|_H^2$, the identity behind every parabolic energy estimate. The third is the compactness that closes those parabolic existence proofs: the Aubin–Lions lemma, which the graphon FBSDE well-posedness of chapter 23 and the parabolic systems of chapter 22 both invoke, is stated exactly relative to a Gelfand triple—it says that a family bounded in $L^2(0,T;V)$ with time derivatives bounded in $L^2(0,T;V^*)$ is precompact in $L^2(0,T;H)$, the time-dependent analogue of Rellich, and it is unstatable without the three named spaces. So the triple is not notation we will carry idly: it is the minimal structure in which the book's parabolic problems can even be posed, and the home of the only compactness argument strong enough to solve them.

Theorem 2.34 (Lax–Milgram)**.** *Let V be a Hilbert space and $a\colon V\times V \to \mathbb{R}$ a bilinear form that is* bounded*, $|a(u,v)| \le M\|u\|_V\|v\|_V$, and* coercive*, $a(u,u) \ge \theta\|u\|_V^2$ for some $\theta > 0$. Then for every*

$\ell \in V^*$ there is a unique $u \in V$ with $a(u,v) = \ell(v)$ for all $v \in V$, and $\|u\|_V \leq \theta^{-1} \|\ell\|_{V^*}$.

Proof. The whole theorem is a translation of the abstract weak equation into a single operator equation, which the two hypotheses then make uniquely solvable; the translation is Riesz representation, used twice. Fix $u \in V$. The map $v \mapsto a(u,v)$ is, by boundedness, a bounded linear functional on V (its norm is at most $M\|u\|_V$), so Riesz representation (theorem 2.6) hands back a unique vector we call Au with $a(u,v) = \langle Au, v\rangle_V$ for all v. Linearity of a in its first slot makes $u \mapsto Au$ linear, and $\|Au\|_V \leq M\|u\|_V$ makes it bounded: $A \in \mathcal{B}(V)$. Applying Riesz once more to the functional ℓ produces $f \in V$ with $\ell(v) = \langle f, v\rangle_V$. The weak problem "find u with $a(u,v) = \ell(v)$ for all v" now reads $\langle Au, v\rangle_V = \langle f, v\rangle_V$ for all v, i.e. simply $Au = f$. Everything reduces to showing A is a bijection of V with a quantitative inverse.

Coercivity is what supplies the bijection, in two stages. First it gives a *lower bound* on A. Putting $v = u$ in the defining relation, $\langle Au, u\rangle_V = a(u,u) \geq \theta\|u\|_V^2$, and by Cauchy–Schwarz the left side is at most $\|Au\|_V \|u\|_V$; cancelling one factor of $\|u\|_V$ (trivial if $u = 0$) yields

$$\|Au\|_V \;\geq\; \theta\,\|u\|_V \qquad \text{for all } u \in V.$$

This single estimate does most of the work. It makes A *injective* ($Au = 0 \Rightarrow u = 0$), and it makes the range of A *closed*: if $Au_n \to g$ in V, then (Au_n) is Cauchy, so by the lower bound $\|u_n - u_m\|_V \leq \theta^{-1}\|Au_n - Au_m\|_V$ shows (u_n) is Cauchy too; completeness of V gives $u_n \to u$, boundedness of A gives $Au_n \to Au$, and uniqueness of limits gives $Au = g$, so g lies in the range. The range $R = A(V)$ is therefore a closed subspace.

The second stage shows that closed subspace is *all* of V—this is where the adjoint enters. Suppose, for contradiction, $R \neq V$. Then R is a proper closed subspace, so by the projection theorem its orthogonal complement contains a nonzero vector w: $\langle Av, w\rangle_V = 0$ for every $v \in V$. Read through the adjoint of definition 2.12, $\langle Av, w\rangle_V = \langle v, A^*w\rangle_V$, this says $A^*w = 0$—the leftover direction is exactly a null vector of the adjoint. But now test coercivity against w itself: taking $v = w$ in $\langle Av, w\rangle_V = 0$ gives $\langle Aw, w\rangle_V = 0$, while coercivity demands $\langle Aw, w\rangle_V = a(w,w) \geq \theta\|w\|_V^2 > 0$ since $w \neq 0$. The two are incompatible, so no such w exists and $R = V$: A is surjective. (When a is symmetric, $A = A^*$ and this last step is automatic; the adjoint is what handles the general, non-symmetric forms that the convection terms of Fokker–Planck produce.)

Injective, surjective, and bounded below, A is a bijection with $\|A^{-1}\| \leq \theta^{-1}$ directly from $\|Au\|_V \geq \theta\|u\|_V$. Hence $u = A^{-1}f$ is the unique solution, and $\|u\|_V \leq \theta^{-1}\|f\|_V = \theta^{-1}\|\ell\|_{V^*}$, the last equality because Riesz is an isometry. That is the stated estimate. A streamlined account along the same lines is in [31, Ch. 5]. □

It is worth seeing the role this theorem plays before leaving it, because it is the linchpin of the chapter's whole division of labour. A mean-field game is nonlinear—the coefficients of the HJB equation depend on the population density m, which is itself the output of solving the Fokker–Planck equation, which depends on the value function u, and round again. Lax–Milgram does not solve that nonlinear loop; nothing linear could. What it solves is each *frozen* step: hold the population (hence the coefficients) fixed, and the HJB and Fokker–Planck equations become *linear* elliptic or parabolic problems with given coefficients, exactly of the form $a(u,v) = \ell(v)$ with a bounded and coercive (boundedness from the regularity of the coefficients, coercivity from nondegeneracy of the diffusion and Poincaré). Lax–Milgram produces the unique solution of that frozen step, with the quantitative bound $\|u\|_V \leq \theta^{-1} \|\ell\|_{V^*}$ that will later be reused as the a priori estimate feeding compactness. The nonlinearity is then absorbed by iterating the frozen-step solver and seeking a population that reproduces itself—which is precisely the fixed-point problem of the next section. Lax–Milgram solves each link; fixed points close the chain.

The coercive form carries a bonus when it is also *symmetric*, $a(u, v) = a(v, u)$, and it is the one a physicist's instinct reaches for first. Then the solution of $a(u, v) = \ell(v)$ is exactly the minimizer of the quadratic energy $J(u) = \frac{1}{2}a(u, u) - \ell(u)$. To see the equivalence is one line of the calculus of variations: for any direction v and scalar t, $J(u + tv) = J(u) + t\big(a(u, v) - \ell(v)\big) + \frac{t^2}{2}a(v, v)$, a strictly convex parabola in t (strict because $a(v, v) \geq \theta \|v\|^2 > 0$), whose stationarity at $t = 0$ for every v is the condition $a(u, v) = \ell(v)$. So "solve the weak equation" and "minimize the energy" are the same problem, and coercivity is what guarantees the energy is bounded below and the minimum attained. This is the Dirichlet principle the reader expects—for $a(u, v) = \int \nabla u \cdot \nabla v$ and $\ell(v) = \int fv$ it is the statement that the solution of $-\Delta u = f$ minimizes Dirichlet energy—and the variational structure of certain MFG systems (chapter 8), where an entire equilibrium is characterized as the minimizer of a convex functional on the space of densities, is the nonlinear elaboration of exactly this picture.

Remark 2.35 (How far we go)**.** We deliberately stop here. Elliptic and parabolic *regularity*—that weak solutions are in fact smooth when the data are, the Sobolev and Morrey embedding theorems giving $H^s \hookrightarrow C^k$, and the Schauder and Calderón–Zygmund estimates—is the machinery that upgrades the weak solutions Lax–Milgram produces to classical ones. We invoke it as a black box in chapters 6 and 8 with explicit pointers to [64, Ch. 6], rather than developing it here, because none of the *functional-analytic* structure the rest of this chapter exports depends on it.

This completes the analytic apparatus the chapter set out to build, and it is worth standing back to see the shape of what is now in hand, because the final section depends on all of it. Two compactness theorems have been assembled, one on each side of a mean-field game. On the PDE side, Rellich–Kondrachov (theorem 2.31) turns an H^1 energy bound into an L^2-convergent subsequence; on the measure side, the weak-$*$ compactness of section 2.3—and the tightness criterion that will accompany it—turns a bound on a family of population measures into a weakly convergent subsequence. These are the two inputs the next section's existence theorems consume: every fixed-point argument for an equilibrium needs the best-response map to land in a set that is compact in a usable topology, and that compactness is always one of these two estimates in disguise, exactly as the chapter's recurring motif promised. And Lax–Milgram has supplied the other half: it solves each frozen-coefficient linear step that the best-response map is built from, with a quantitative bound that is itself the a priori estimate feeding the compactness. The division of labour is now fully drawn. Lax–Milgram solves one linear problem; fixed points glue the solved problems into a self-consistent equilibrium. It remains only to supply the gluing, and that is the business of the three fixed-point theorems—Banach, Schauder, Kakutani—to which we turn, recognizing the self-consistent loop they govern as the same self-reproduction picture the mean-field idea has worn since the start.

2.7 Fixed-point theorems: the engines of existence

Every equilibrium in this book is a fixed point. An agent best-responds to the population; the population is the aggregate of best responders; an equilibrium is a population that reproduces itself under this loop. Formalizing "reproduces itself" as $\Phi(m) = m$ for a best-response map S on a suitable space, existence of an equilibrium becomes existence of a fixed point. Three theorems supply fixed points under three different trade-offs between what one assumes about the map and what one concludes.

Heuristic (non-rigorous)

The physical reflex is the self-consistent field: guess a mean field, compute the best response to it, read off the new mean field it generates, and iterate $m^{(0)} \mapsto m^{(1)} = \mathsf{S}(m^{(0)}) \mapsto m^{(2)} = \mathsf{S}(m^{(1)}) \mapsto \cdots$, hoping for a fixed point. The three theorems below are precisely the three rigorous guarantees one can attach to this iteration. *Banach* (contraction) says: if the map shrinks distances, the iteration converges geometrically to the unique fixed point—this is convergence *and* uniqueness, but it demands a smallness/contraction hypothesis (in the graphon theory, smallness of $\|\mathbb{W}\|$, (A-Coupling)). *Schauder* says: if the map is merely continuous and lands in a compact convex set, a fixed point exists—this is existence without uniqueness and without convergence of the naive iteration, and it is what one uses when no smallness is available. *Kakutani* extends this to *set-valued* best responses, for when the best response is not unique and one must allow a convex set of optimal replies.

Theorem 2.36 (Banach contraction principle)**.** *Let (X, d) be a nonempty complete metric space and $\mathsf{S}\colon X \to X$ a* contraction*: there is $\kappa \in [0, 1)$ with $d(\mathsf{S}x, \mathsf{S}y) \le \kappa\, d(x, y)$ for all x, y. Then S has a unique fixed point x_*, and for every x_0 the iterates $x_{n+1} = \mathsf{S}x_n$ converge to it with $d(x_n, x_*) \le \dfrac{\kappa^n}{1-\kappa}\, d(x_1, x_0)$.*

Proof. The iterates are Cauchy: $d(x_{n+1}, x_n) \le \kappa^n d(x_1, x_0)$, and summing the geometric series bounds $d(x_{n+p}, x_n) \le \frac{\kappa^n}{1-\kappa} d(x_1, x_0)$. Completeness gives a limit x_*; continuity of the contraction (it is Lipschitz) gives $\mathsf{S}x_* = \lim \mathsf{S}x_n = \lim x_{n+1} = x_*$. If x_*, x_*' are both fixed, $d(x_*, x_*') = d(\mathsf{S}x_*, \mathsf{S}x_*') \le \kappa\, d(x_*, x_*')$ forces $d(x_*, x_*') = 0$. □

This is the only one of the three that yields uniqueness and a constructive, convergent algorithm, which is why every *uniqueness* theorem in the book that is not monotonicity-based (chapters 11 and 24) ultimately reduces best response to a contraction under a smallness hypothesis. Its limitation is exactly that hypothesis: when the coupling is not small the map need not contract, and one must trade uniqueness for the weaker existence guarantees that compactness provides. The bridge is the finite-dimensional fixed-point theorem of Brouwer, which we state as the seed.

Theorem 2.37 (Brouwer)**.** *Every continuous map of a nonempty compact convex subset of $\mathbb{R}^d$ into itself has a fixed point.*

Brouwer's theorem sits above our prerequisite line, so we do not take it on faith: an elementary and fully self-contained proof, via Sperner's combinatorial lemma, is given in theorem B.13, with [79, Ch. 1] for the degree-theoretic and simplicial-approximation routes as further reading. Its infinite-dimensional extension is the workhorse of MFG existence.

Theorem 2.38 (Schauder)**.** *Let C be a nonempty closed convex subset of a Banach space X and $\mathsf{S}\colon C \to C$ a continuous map whose image $\mathsf{S}(C)$ is relatively compact. Then S has a fixed point. (Equivalently: any continuous self-map of a nonempty compact convex set has a fixed point.)*

Proof sketch. The mechanism is to approximate the infinite-dimensional problem by finite-dimensional ones to which Brouwer applies, then pass to the limit using compactness. Since $K = \overline{\mathsf{S}(C)}$ is compact, for each $\varepsilon > 0$ it has a finite ε-net $\{y_1, \ldots, y_N\} \subseteq K$. The Schauder projection $P_\varepsilon z = \frac{\sum_j \theta_j(z)\, y_j}{\sum_j \theta_j(z)}$, with $\theta_j(z) = \max(0, \varepsilon - \|z - y_j\|)$, maps K continuously into the finite-dimensional convex hull $C_\varepsilon = \operatorname{conv}\{y_1, \ldots, y_N\}$ and moves each point by less than ε. Because every $y_j \in K = \overline{\mathsf{S}(C)} \subseteq C$ and C is convex, the hull $C_\varepsilon \subseteq C$; hence $P_\varepsilon \circ \mathsf{S}$ is a continuous self-map of the finite-dimensional compact convex set C_ε (S maps $C_\varepsilon \subseteq C$ into $\mathsf{S}(C) \subseteq K$, and P_ε maps K back into C_ε), so Brouwer (theorem 2.37) gives a fixed point $x_\varepsilon \in C_\varepsilon$ with $\|\mathsf{S}x_\varepsilon - x_\varepsilon\| = \|\mathsf{S}x_\varepsilon - P_\varepsilon \mathsf{S}x_\varepsilon\| < \varepsilon$. Letting $\varepsilon \to 0$, compactness of K extracts a convergent subsequence $\mathsf{S}x_\varepsilon \to x_*$, whence $x_\varepsilon \to x_*$ as well, and continuity of S gives $\mathsf{S}x_* = x_*$. The argument is carried out in full in theorem B.14; see also [79, Ch. II] and [31, Ch. 11]. □

Schauder is the theorem behind the existence proofs of chapters 10 and 24: the best-response map is shown continuous, and the relatively compact image is produced by an a priori estimate (tightness in chapter 3 on the measure side, the Rellich–Kondrachov compactness of theorem 2.31 on the PDE side). It gives existence with no uniqueness and no algorithm—which is honest, because mean field games genuinely can have multiple equilibria (chapter 11). When the best response is not single-valued—when an agent is indifferent among several optimal replies—one needs a set-valued refinement.

Definition 2.39 (Upper hemicontinuity)**.** A set-valued map $\mathsf{S}\colon C \rightrightarrows C$ (assigning to each x a subset $\mathsf{S}(x) \subseteq C$) is *upper hemicontinuous with closed graph* if its graph $\{(x, y) : y \in \mathsf{S}(x)\}$ is closed in $C \times C$; concretely, $x_n \to x$, $y_n \in \mathsf{S}(x_n)$, $y_n \to y$ imply $y \in \mathsf{S}(x)$.

A word on the name. Having a closed graph and being upper hemicontinuous in the standard sense (preimages of open sets pull back to open sets under the upper inverse) are not equivalent for arbitrary correspondences. They *do* coincide in the only setting we use, where the target C is compact and the values $\mathsf{S}(x)$ are closed (hence compact): there a closed graph is exactly upper hemicontinuity. We verify and invoke the closed-graph form throughout because it is what the fixed-point argument actually checks, and we keep the name "upper hemicontinuous" to match the literature.

Theorem 2.40 (Kakutani–Fan–Glicksberg)**.** *Let C be a nonempty compact convex subset of a locally convex space (e.g. a Banach space, or a convex weak-$*$ compact set of measures), and $\mathsf{S}\colon C \rightrightarrows C$*

an upper hemicontinuous set-valued map with closed graph such that every value $\mathsf{S}(x)$ *is nonempty, convex, and closed. Then* S *has a fixed point: some* $x_* \in C$ *with* $x_* \in \mathsf{S}(x_*)$*.*

The proof reduces the set-valued problem to the single-valued Schauder theorem by an approximate-selection construction. For each $\varepsilon > 0$, cover the compact C by finitely many balls $B(x_i, \varepsilon)$, take a continuous partition of unity $\{\psi_i\}$ subordinate to this cover, choose one value $z_i \in \mathsf{S}(x_i)$ at each centre, and set $f_\varepsilon = \sum_i \psi_i\, z_i$. This $f_\varepsilon : C \to C$ is continuous and single-valued (a convex combination of points of C, which is convex), and it is an ε-approximate selection of S in the sense that its graph lies within ε of the graph of S. Schauder (theorem 2.38) applied to f_ε on the compact convex C yields a fixed point $x_\varepsilon = f_\varepsilon(x_\varepsilon)$; letting $\varepsilon \to 0$, compactness extracts $x_\varepsilon \to x_*$, and the closed graph together with convexity of the values forces $x_* \in \mathsf{S}(x_*)$ in the limit. The full argument—including the approximate-selection lemma and the closed-graph passage to the limit—is carried out in theorem B.16; see also [79, Ch. II]. Kakutani is what licenses *relaxed* or *mixed* equilibria in chapter 10, where the optimal control need not be unique and the best response is the (convex) set of all optimal replies. The convexity of the values is the crucial hypothesis and the reason relaxed controls—probability measures over actions—are introduced: convexifying the action set is what makes the best-response correspondence convex-valued.

Notation watch

The three theorems sit in a strict logical hierarchy of hypothesis strength, and it is worth fixing which is which to avoid over-claiming. Banach needs a metric *contraction* and delivers the most (unique fixed point, convergent iteration). Schauder needs only *continuity plus compactness* and delivers existence of a (possibly non-unique) fixed point with no algorithm. Kakutani needs *upper hemicontinuity plus convex values plus compactness* and delivers existence for set-valued maps. A common error is to invoke “Schauder” for a map that is continuous but whose image is not relatively compact; without the compactness input the theorem is false in infinite dimensions (the identity map on the unit ball has no fixed point). The compactness input is never free—it is always an a priori estimate in disguise.

Rigor ledger

Rigor ledger

Proved (in full or by complete sketch with the mechanism named).

- Hilbert-space geometry: projection onto closed convex sets, Riesz representation, Parseval (theorems 2.5, 2.6 and 2.8); stated with proofs cited.
- Banach–Alaoglu and the extraction of weakly convergent subsequences from bounded sequences in reflexive/Hilbert spaces (theorem 2.16 and corollary 2.17), proved via Tychonoff.
- Identification of Hilbert–Schmidt operators with L^2 kernels and the self-adjoint Hilbert–Schmidt structure of the graphon operator (theorem 2.22 and proposition 2.24).

- The spectral theorem for compact self-adjoint operators and the Courant–Fischer min–max (theorem 2.26 and corollary 2.27), with a complete variational-construction sketch; the explicit spectra of the constant and rank-one graphon operators (worked computation 2.28).
- The Banach contraction principle in full (theorem 2.36); Lax–Milgram by a complete sketch (theorem 2.34).

Imported from chapter 1 (proved there, used here).

- Completeness of L^p (theorem 1.31), density of simple and smooth functions, and Fubini–Tonelli (theorem 1.27); the lightweight definition of weak convergence of probability measures (definition 1.33).
- Standing assumption (A-Reg-W) of appendix A.6, used to make $\mathbb{W}$ self-adjoint Hilbert–Schmidt.

Proved in-book in appendix B (stated here with body sketch where given).

- Hahn–Banach extension and separation (theorem 2.11), via Zorn's lemma and the Minkowski gauge: appendix B.2.
- Tychonoff's theorem for the product of compact intervals underlying Banach–Alaoglu (theorem 2.16), via the Alexander subbase lemma: appendix B.3. The separable case used throughout the book needs no Tychonoff and is proved in the body.
- Brouwer's theorem (theorem 2.37) via Sperner's lemma (theorem B.13); the Schauder and Kakutani–Fan–Glicksberg theorems (theorems 2.38 and 2.40), with body sketches and the full arguments in theorems B.14 and B.16.
- Meyers–Serrin density, the Rellich–Kondrachov compact embedding, and the Poincaré inequality (theorem 2.31): appendix B.8.

Deferred (named destination).

- The complex case of the spectral theorem: appendix B.4 (the real case is proved in full in the body).
- Elliptic/parabolic regularity (Sobolev/Morrey embeddings, Schauder estimates) that upgrades weak to classical solutions: invoked in chapters 6 and 8, sources [64, Ch. 6].
- The spectral payoff—eigenvector centrality, spectral gap, operator-norm thresholds, low-rank reductions of $\mathbb{W}$: chapter 18 (static games chapter 20, GMFG existence and the critical coupling chapter 24).

Open (genuinely, beyond this book's scope).

- No claim in this chapter is open; it is classical functional analysis. The open questions it *feeds* are downstream: sharp operator-norm thresholds for GMFG uniqueness (chapter 24) and the spectral theory of *unbounded* (L^p, $p < \infty$) graphon operators where $\mathbb{W}$ is no longer Hilbert–Schmidt (chapter 27).

Bibliographic notes

The Hilbert-space material of sections 2.1 and 2.2—inner products, projection, Riesz representation, orthonormal bases—follows [123, Ch. 4], the roadmap's designated source, supplemented by [31, Ch. 5–6] and the physicist-friendly [113, Ch. II]. The weak and weak-$*$ topologies and Banach–Alaoglu (section 2.3) are treated in [31, Ch. 3] and [113, Ch. IV]; [50] is a good alternative reference for the duality theory and the Hahn–Banach theorem. Compact and Hilbert–Schmidt operators and the spectral theorem (sections 2.4 and 2.5) are in [123, Ch. 6], [31, Ch. 6], and [113, Ch. VI]; the variational construction of the eigenvalues and the min–max principle are classical. That the graphon is exactly a self-adjoint Hilbert–Schmidt operator, and the spectral reading of network structure that this enables, is the operator-theoretic core of graph-limit theory [99, Ch. 7.5], collected in chapter 18. The Sobolev-space theory of section 2.6—weak derivatives, the H^s scale, Rellich–Kondrachov, Lax–Milgram—follows [31, Ch. 8–9] and [64, Ch. 5], which are also the sources for the deferred elliptic regularity. The three fixed-point theorems of section 2.7 are developed, with the finite-dimensional-approximation proofs sketched here given in full, in [79, Ch. I–II]; their use as the engines of mean-field-game existence follows [36] and [40, Vol. I], and in the graphon setting [34, 108].

Chapter 3

Probability and Weak Convergence of Measures

The previous two chapters built the two pillars on which this one stands, and it is worth recalling not merely their contents but their purpose, because every instrument assembled here is welded out of them. From chapter 1 we import the measure-theoretic apparatus: σ-algebras as the bookkeeping of events, the Lebesgue integral as the universal averaging operation, the monotone and dominated convergence theorems (theorem 1.20, theorem 1.23) that license the exchange of a limit with an integral, the Fubini–Tonelli theorem (theorem 1.27) that disentangles a joint integral into iterated ones, and the picture of the empirical measure $\mu^N = \frac{1}{N}\sum_{i=1}^{N} \delta_{X_i}$ as the rigorous occupation profile of a finite ensemble — the histogram that records *where* the agents are without recording *which* agent is where. From chapter 2 we import Hilbert-space geometry: closed subspaces, orthogonal projection onto a closed convex set as the unique nearest-point map, and the duality between a Banach space and its continuous dual. The first pillar tells us how to integrate; the second tells us how to project. This chapter welds the two into the single instrument the rest of the book uses more than any other: a *topology on the space of probability measures* in which the empirical measure of a large population converges to a deterministic limit, a precise accounting of the residual fluctuations around that limit, and — the projection pillar's contribution — the conditional expectation the stochastic chapters demand.

The reason this matters can be stated before any formalism, and it is the whole motivation for the chapter. A mean field game confronts an N-player problem in which every agent's optimal behaviour depends on what all the others do; with the coupling fully retained the problem is intractable as N grows. The mean field idea is to replace it, in the limit $N \to \infty$, by a problem with a single *representative* agent who best-responds not to the other agents individually but to a *measure* m summarizing the population's state, with the closing consistency requirement that when every agent plays this best response the population they generate is again m. Two phrases in this story must be given precise meaning, and supplying that meaning is what the chapter does. First, "the population converges to a measure": the object that converges is the empirical measure μ^N, itself a *random point in the space of probability measures*, and to say it settles down to a deterministic m we need a notion of *convergence of measures*. Second, "the finite game is well approximated by the limit game": this demands control of the discrepancy between μ^N and m, and an elementary variance computation (carried out at the end of section 3.1) already shows that discrepancy is of typical size $N^{-1/2}$, with — as the central limit theorem will sharpen — a Gaussian profile.

Concretely, four tools do this work, and the reader should keep each tool's downstream consumer in view from the start. *(i) Weak convergence*, with the Portmanteau theorem as its working characterization, is the right notion of "the population converges to a measure"; it is the topology in which the existence arguments operate. *(ii) Tightness together with Prokhorov's theorem* is the compactness theory that extracts a limit from a sequence of measures when no limit is given in advance; it is the engine of the existence proofs in chapter 10 and chapter 24, where an equilibrium is obtained as a subsequential limit of approximate ones. *(iii) The characteristic-function route to the central limit theorem* pins the $N^{-1/2}$ Gaussian fluctuation scale, the quantity that chapter 7 (propagation of chaos) and chapter 13 (fluctuation analysis) make their business. *(iv) Conditional expectation as an orthogonal projection* — where the second pillar finally pays off — is the prerequisite for the filtrations and backward stochastic equations of chapter 5 and chapter 9. These are not a generic probability refresher; they are exactly the pieces of probability the mean field and graphon mean field theories consume, curated to that end. And, honouring the self-containedness demanded in chapter 1, *every named theorem among them is proved here, not cited*: Portmanteau, Prokhorov, the Skorokhod representation, the central limit theorem, and the projection theorem for conditional expectation all receive complete in-book proofs.

A word on the setting before we begin. Throughout, the state space of a single agent is a complete separable metric (*Polish*) space, most often $\mathbb{R}^d$; Polishness is precisely what makes tightness and the Skorokhod representation available, and we flag each place it is used. The type variable $x \in [0,1]$ of the graphon chapters plays no role here — this chapter concerns the law of a *single* agent's state, not the network of types — and no standing assumption from the thread specification is invoked, because this is a foundations chapter and its theorems hold in the generality stated. The order of march follows the order of the four tools, and each later section opens by saying where in this march we stand: we begin (section 3.1) with the objects — probability spaces, random variables, expectation, independence; build weak convergence and prove Portmanteau, the metrization, and the Skorokhod representation that make it a usable topology (section 3.2); develop tightness and prove Prokhorov (section 3.3); quantify the $N^{-1/2}$ fluctuations through characteristic functions and the central limit theorem (section 3.4); pause to record two concentration tools — the Borel–Cantelli lemmas and the Azuma–Hoeffding inequality — that the later chapters draw on (section 3.5); and close with conditional expectation as projection (section 3.6).

3.1 Probability spaces, expectation, and independence

With the four-tool itinerary fixed, we start at the bottom, with the objects themselves, and in doing so redeem the preamble's central promise: that this chapter makes good on the slogan "every probabilistic quantity is an integral" by welding the measure theory of chapter 1 into probability. The recollection that follows is short, but it turns on a single conceptual move — the one a physicist most needs to internalize. A random variable is not a number equipped with a "spread"; it is a *measurable function on a sample space.* And every probabilistic quantity built from it — the probability of an event, a mean, a variance, a correlation — is an *integral* of such a function against a measure. Once that identification is in hand, the whole convergence apparatus of chapter 1 (the monotone and dominated convergence theorems, Fubini–Tonelli) transfers to probability with no further work, and the remainder of this section is the harvest.

Definition 3.1 (Probability space)**.** A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a set (the sample space), $\mathcal{F}$ is a σ-algebra of subsets of Ω (the events), and $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a measure with $\mathbb{P}(\Omega) = 1$. A property that holds for all ω outside a set $N \in \mathcal{F}$ with $\mathbb{P}(N) = 0$ is said to hold *almost surely* (a.s.).

The structure is *exactly* that of chapter 1, specialized to total mass one; nothing is added, and one thing is fixed. The normalization $\mathbb{P}(\Omega) = 1$ is the entire difference between a general measure and a probability measure, and it is what licenses the two probabilistic readings we will lean on: $\mathbb{P}(A)$ is the "chance" of the event A, a number in $[0, 1]$ rather than an arbitrary nonnegative quantity, and $\mathbb{E}[X]$ (defined below) is a genuine *average* — each outcome weighted by how likely it is — rather than an unnormalized total. Everything underneath, from countable additivity to the monotone and dominated convergence theorems, is the measure theory already in hand, inherited verbatim. The almost-sure language of definition 3.1 is the probabilistic name for "up to a $\mathbb{P}$-null set"; it is the equivalence under which we tacitly identify random variables throughout, exactly as chapter 1 identifies functions that agree almost everywhere.

Definition 3.2 (Random variable and its law)**.** Let $(S, \mathcal{S})$ be a measurable space. An *S-valued random variable* is a measurable map $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$, i.e. $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{S}$. Its *law* (or *distribution*) is the push-forward measure

$$\mathcal{L}(X) \;=\; \mathbb{P} \circ X^{-1} \;:=\; \mu_X, \qquad \mu_X(B) = \mathbb{P}(X \in B), \quad B \in \mathcal{S},$$

a probability measure on $(S, \mathcal{S})$. When S is a metric space we always take $\mathcal{S} = \mathcal{B}(S)$, the Borel σ-algebra.

The law μ_X *forgets the sample space entirely.* Nothing in μ_X remembers which $\omega \in \Omega$ produced which value; it records only how the unit mass of $\mathbb{P}$ is spread over the state space S. This forgetting is not a loss but the whole point: Ω is an auxiliary scaffold, chosen for convenience and never canonical, whereas μ_X is the invariant content. It is the law, not the sample space, that survives into every limit theorem of this chapter, and it is the law on which the entire mean field theory is built — the mean field m *is* a law, a point of $\mathcal{P}_2(\mathbb{R}^d)$ in the sense of chapter 4, and the empirical occupation profile μ^N is a (random) law as well. Because the sample space is disposable, two random variables with the same law are interchangeable for every purpose this book cares about; we write $X \overset{d}{=} Y$, call them *equal in distribution*, and allow them to live on entirely different probability spaces. This is precisely why weak convergence, the subject of section 3.2, will be phrased as a convergence of laws rather than of the variables themselves: the variables may not even share a home.

The single quantity we extract from a real random variable more than any other is its average. A physicist reads $\mathbb{E}[X]$ as "the value of X averaged over the ensemble of outcomes, each weighted by its probability," and the next definition makes exactly this precise by declaring the average to *be* the Lebesgue integral of chapter 1: built first for $X \geq 0$, where the integral is the supremum of the integrals of simple functions beneath X and so is always defined in $[0, \infty]$, and then for signed X by subtracting the contributions of its positive and negative parts.

Definition 3.3 (Expectation as integration)**.** For a real random variable $X \geq 0$ the *expectation* is the Lebesgue integral $\mathbb{E}[X] = \int_\Omega X \, \mathrm{d}\mathbb{P} \in [0, \infty]$. For general real X, X is *integrable* if $\mathbb{E}[|X|] < \infty$, and then $\mathbb{E}[X] = \int_\Omega X^+ \, \mathrm{d}\mathbb{P} - \int_\Omega X^- \, \mathrm{d}\mathbb{P}$. We write $X \in L^p(\Omega, \mathcal{F}, \mathbb{P})$, $p \geq 1$, when $\mathbb{E}[|X|^p] < \infty$.

Because the expectation is *literally* an integral against $\mathbb{P}$, every theorem of chapter 1 applies to it verbatim — this is the dividend of insisting that a random variable is a measurable function rather than a number with a spread. Two consequences are used on nearly every page that follows: the change-of-variables formula, which moves a computation off the abstract space Ω and onto the concrete state space S where the law lives, and the convergence theorems, which let us exchange a limit with an expectation. We record both here in probabilistic dress.

Lemma 3.4 (Change of variables; convergence under the integral)**.** *Let X be an S-valued random variable with law μ_X and let $f : S \to \mathbb{R}$ be Borel measurable.*

1. (Transport of the integral.) *If $f \geq 0$ or $f(X)$ is integrable, then*

$$\mathbb{E}[f(X)] \;=\; \int_S f \, \mathrm{d}\mu_X. \tag{3.1}$$

2. (Dominated convergence.) *If $X_n \to X$ a.s. and $|X_n| \leq Y$ a.s. with $\mathbb{E}[Y] < \infty$, then $\mathbb{E}[X_n] \to \mathbb{E}[X]$. If instead $0 \leq X_n \uparrow X$ a.s., then $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$ (monotone convergence).*

Proof. Part (1) is the pushforward transfer formula of chapter 1 (lemma 1.15), re-derived here in probabilistic notation by the *standard machine*: establish the identity for indicators, extend it by linearity to simple functions, by monotone convergence to nonnegative functions, and by a difference to integrable ones. We run the four steps because the same ladder reappears, unannounced, in almost every later proof.

Indicators. Let $f = \mathbf{1}_B$ with $B \in \mathcal{S}$. Then $f(X) = \mathbf{1}\{X \in B\} = \mathbf{1}_{X^{-1}(B)}$, an indicator on Ω, so by the definition of the integral of an indicator and of the law μ_X (definition 3.2),

$$\mathbb{E}[\mathbf{1}_B(X)] = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B) = \mu_X(B) = \int_S \mathbf{1}_B \, \mathrm{d}\mu_X.$$

Thus (3.1) holds when f is an indicator.

Simple functions. Let $f = \sum_{k=1}^r c_k \, \mathbf{1}_{B_k}$ with $c_k \geq 0$ and $B_k \in \mathcal{S}$. Both sides of (3.1) are linear in f — the left side by linearity of the integral over Ω, the right side by linearity of the integral of a simple function over S (eq. (1.4)) — so the indicator case applied termwise gives $\mathbb{E}[f(X)] = \sum_k c_k \, \mu_X(B_k) = \int_S f \, \mathrm{d}\mu_X$.

Nonnegative functions. Let $f \geq 0$ be Borel. By the standard approximation of measurable functions in chapter 1 there are simple functions s_n with $0 \leq s_n \uparrow f$ pointwise on S; composing with X preserves the monotone increase, $0 \leq s_n(X) \uparrow f(X)$ pointwise on Ω. Applying the monotone convergence theorem (theorem 1.20) on $(\Omega, \mathcal{F}, \mathbb{P})$ to the left and on $(S, \mathcal{S}, \mu_X)$ to the right, with the simple-function identity in the middle,

$$\mathbb{E}[f(X)] = \lim_n \mathbb{E}[s_n(X)] = \lim_n \int_S s_n \, \mathrm{d}\mu_X = \int_S f \, \mathrm{d}\mu_X.$$

Integrable functions. For a general Borel f write $f = f^+ - f^-$ with $f^\pm \geq 0$. The nonnegative case applied to $|f| = f^+ + f^-$ gives $\mathbb{E}[|f(X)|] = \int_S |f| \, \mathrm{d}\mu_X$, so the hypothesis $\mathbb{E}[|f(X)|] < \infty$ is exactly $\int_S |f| \, \mathrm{d}\mu_X < \infty$ and renders both $\int_S f^+ \, \mathrm{d}\mu_X$ and $\int_S f^- \, \mathrm{d}\mu_X$ finite. Subtracting the nonnegative identity for f^- from that for f^+ yields (3.1) for f.

Part (2) is the convergence machinery of chapter 1 read on the probability space, with no new idea. If $X_n \to X$ a.s. and $|X_n| \leq Y$ a.s. with $\mathbb{E}[Y] < \infty$, then Y is an integrable dominating function, and the dominated convergence theorem (theorem 1.23) applied to (X_n) on $(\Omega, \mathcal{F}, \mathbb{P})$ gives $\mathbb{E}[X_n] = \int_\Omega X_n \, \mathrm{d}\mathbb{P} \to \int_\Omega X \, \mathrm{d}\mathbb{P} = \mathbb{E}[X]$. If instead $0 \leq X_n \uparrow X$ a.s., the monotone convergence theorem (theorem 1.20) delivers $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$. The almost-sure qualifiers are harmless: altering each X_n on a $\mathbb{P}$-null set changes none of the integrals, by the very definition of the Lebesgue integral. □

Formula (3.1) is the bridge we cross constantly, and it is worth saying exactly what it buys. Its left-hand side lives on the abstract, inaccessible space Ω, which we never describe; its right-hand side lives on the concrete state space S — usually $\mathbb{R}^d$ — where f is an explicit function and μ_X an explicit distribution. The identity therefore lets every later computation be carried out on S against the law, never needing to mention Ω at all. A tiny instance fixes the idea: with $f(x) = x$ the formula reads $\mathbb{E}[X] = \int_S x \, \mathrm{d}\mu_X(x)$, the familiar statement that the mean is the integral of the value against the distribution; and if X takes finitely many values $x_1, \ldots, x_r$ with probabilities $p_k = \mathbb{P}(X = x_k)$, then $\mu_X = \sum_k p_k \, \delta_{x_k}$ and the integral collapses, by the Dirac evaluation (1.7), to $\mathbb{E}[X] = \sum_k x_k \, p_k$ — the elementary weighted-average formula, recovered as a special case rather than posited. The same bridge with $f(x) = x^2$, with $f(x) = e^{itx}$, and with $f(x) = \mathbf{1}_{\{x>a\}}$ produces, respectively, the second moment, the characteristic function of section 3.4, and a tail probability, each as a single integral against μ_X; these are the three computations the rest of the chapter actually performs.

Two such integrals earn their own names, because they are the L^2 data of a random variable and the L^2 geometry of chapter 2 will return for them in force when we reach conditional expectation. For $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ the *variance* and *covariance* are the special cases

$$\operatorname{Var}(X) = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big], \qquad \operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Read geometrically, these are nothing but the squared norm and the inner product of the *centered* variables in the Hilbert space L^2: $\operatorname{Var}(X) = \|X - \mathbb{E}[X]\|_{L^2}^2$ and $\operatorname{Cov}(X, Y) = \langle X - \mathbb{E}[X]\,, Y - \mathbb{E}[Y] \rangle_{L^2}$. They are finite precisely because $X, Y \in L^2$ — this is where the square-integrability hypothesis is spent — and the Cauchy–Schwarz inequality (proposition 1.30 with $p = q = 2$) both guarantees the covariance exists and gives the bound $|\operatorname{Cov}(X, Y)| \leq \sqrt{\operatorname{Var} X}\,\sqrt{\operatorname{Var} Y}$. Reading variance and covariance as L^2 geometry is not a flourish: it is the very viewpoint under which conditional expectation becomes an orthogonal projection in section 3.6, and it is why "uncorrelated" will turn out to mean "orthogonal."

Expectation is now pinned down as integration against the law, and with it the L^2 bookkeeping of fluctuations. The next structural ingredient the limit theorems require is the encoding of "no interaction" between random variables. It is this notion — independence — that makes the law of large numbers and the central limit theorem run, and, fittingly, it is expressed through the very integral pairing just established: independence will be the statement that the expectation of a product *factorizes* into the product of expectations.

3.1.1 Independence

Expectation is now the integral pairing $\mathbb{E}[f(X)] = \int_S f \, \mathrm{d}\mu_X$ of (3.1), and the L^2 data of a random variable read off as the geometry of that pairing. With this in hand we can give precise meaning to the one structural relation between several random variables that the limit theorems require: that they do *not* interact. "No interaction" is a physical phrase, and the task is to translate it into a statement about the integral pairing just established. The translation is the cleanest possible: two variables fail to interact when the expectation of their product *splits* into the product of their expectations, so that knowing one carries no information about the average of the other. This is independence, and the whole engine of the law of large numbers and the central limit theorem turns on it. It is, moreover, the exact structural fact at stake in the mean field programme: a mean field limit *manufactures* independence asymptotically — this is propagation of chaos (chapter 7), the statement that any fixed finite group of agents becomes independent as $N \to \infty$ — while the coupling in an N-player game *destroys* it at finite N, every agent's state depending on every other's through the shared population. The residual dependence is precisely what section 3.4 will weigh, and the $N^{-1/2}$ that measures it is already visible, as we will see, in a one-line variance computation.

We give the definition in its most structural form, at the level of σ-algebras, and then unfold it into the two working forms one actually computes with.

Definition 3.5 (Independence)**.** Sub-σ-algebras $\mathcal{F}_1, \ldots, \mathcal{F}_k \subseteq \mathcal{F}$ are *independent* if for all $A_i \in \mathcal{F}_i$,

$$\mathbb{P}\Big(\bigcap_{i=1}^{k} A_i\Big) = \prod_{i=1}^{k} \mathbb{P}(A_i).$$

Random variables $X_1, \ldots, X_k$ are independent if the generated σ-algebras $\sigma(X_1), \ldots, \sigma(X_k)$ are. An infinite family is independent if every finite subfamily is. The variables are *independent and identically distributed* (i.i.d.) if they are independent and share a common law. We write $X \perp\!\!\!\perp \mathcal{G}$ when $\sigma(X)$ and the σ-algebra $\mathcal{G}$ are independent.

The definition is phrased through σ-algebras because that is the form that composes cleanly — $\sigma(X_i)$ records *everything* measurable about X_i, so factorization of probabilities across the $\sigma(X_i)$ says no event decided by one variable shifts the odds of any event decided by another. But it is not the form one verifies or computes with. There are two equivalent working forms, and tracing the equivalence is worthwhile because the bridge between them is exactly the Fubini–Tonelli theorem of chapter 1, the same disentangling of a joint integral into iterated ones that the chapter preamble flagged as imported machinery.

The first working form is a statement about the joint law on the product space. The variables $X_1, \ldots, X_k$ assemble into a single S^k-valued variable $(X_1, \ldots, X_k)$, whose law $\mu_{(X_1,\ldots,X_k)}$ is a probability measure on S^k; its one-dimensional marginals are the individual laws μ_{X_i}. Then $X_1, \dots, X_k$ are independent *iff* this joint law is the product measure

$$\mu_{(X_1,\dots,X_k)} \;=\; \mu_{X_1} \otimes \cdots \otimes \mu_{X_k}, \tag{3.2}$$

the unique measure on S^k assigning mass $\prod_i \mu_{X_i}(B_i)$ to each rectangle $B_1 \times \cdots \times B_k$. That this is the same condition as definition 3.5 is immediate on rectangles — the event $\{X_i \in B_i \ \forall i\} = \bigcap_i \{X_i \in B_i\}$ has probability $\prod_i \mathbb{P}(X_i \in B_i) = \prod_i \mu_{X_i}(B_i)$ exactly when the $\sigma(X_i)$ factorize — and rectangles form an intersection-stable generating class for the product σ-algebra, so agreement there forces agreement everywhere by the uniqueness lemma for measures (corollary 1.10). This is the cleanest formulation, but it lives on S^k and speaks of sets.

The second working form lives back on the level of expectations, where we actually integrate. When the relevant products are integrable, $X_1, \dots, X_k$ are independent iff

$$\mathbb{E}\Big[\textstyle\prod_{i=1}^k f_i(X_i)\Big] = \displaystyle\prod_{i=1}^k \mathbb{E}[f_i(X_i)] \qquad \text{for all bounded measurable } f_i : S \to \mathbb{R}. \tag{3.3}$$

This is the form the limit theorems consume, and it is *exactly* the product-measure form (3.2) read through Fubini–Tonelli. Indeed, transporting the expectation to the product space by the change-of-variables formula (3.1) and then, *because* the joint law is a product, iterating the integral via Fubini–Tonelli (theorem 1.27),

$$\mathbb{E}\Big[\textstyle\prod_{i=1}^k f_i(X_i)\Big] = \displaystyle\int_{S^k} \prod_{i=1}^k f_i(x_i) \; \mathrm{d}(\mu_{X_1} \otimes \cdots \otimes \mu_{X_k})(x_1, \dots, x_k) = \prod_{i=1}^k \int_S f_i \; \mathrm{d}\mu_{X_i} = \prod_{i=1}^k \mathbb{E}[f_i(X_i)]\,.$$

The middle equality is the entire content: Fubini–Tonelli is the theorem that licenses splitting an integral against a product measure into a product of one-dimensional integrals, and it is precisely the integrability of $\prod_i |f_i(X_i)|$ — guaranteed here by boundedness of the f_i — that its hypotheses demand. Conversely, taking $f_i = \mathbf{1}_{B_i}$ in (3.3) returns the rectangle identity and hence definition 3.5, so the three forms are genuinely one notion wearing three costumes: σ-algebra factorization for structure, product law for cleanliness, expectation factorization for computation, with Fubini the hinge between the last two.

The single scalar consequence we extract is the one that already encodes the fluctuation scale. Specializing (3.3) to $k = 2$ and to the *centered* test functions $f_1(x) = x - \mathbb{E}[X_i]$, $f_2(x) = x - \mathbb{E}[X_j]$ (legitimate whenever $X_i, X_j \in L^2$, where these products are integrable) gives, for $i \neq j$,

$$\mathrm{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])] = \mathbb{E}[X_i - \mathbb{E}[X_i]]\; \mathbb{E}[X_j - \mathbb{E}[X_j]] = 0,$$

since each centered factor has mean zero. So independent variables are *uncorrelated*: in the L^2 geometry of section 3.1 their centered versions are mutually orthogonal. The converse fails, and it is worth one concrete witness rather than a bare warning. Let X be standard Gaussian and set $Y = X^2$. Then $\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\,\mathbb{E}[X^2] = 0 - 0 \cdot 1 = 0$ by the vanishing of the odd moments of a symmetric law, yet Y is a deterministic function of X — about as dependent as two variables can be, since X determines Y outright. Vanishing covariance probes only the second-order, linear relationship and is blind to the quadratic dependence here; the pair (X, X^2) is of course not jointly Gaussian, and the one setting where uncorrelated does force independent is precisely a *jointly* Gaussian vector, whose dependence is entirely carried by its covariance matrix.

The reason uncorrelatedness is worth isolating is that it is all one needs for the identity that seeds the entire fluctuation story: the additivity of variance. If $X_1, \dots, X_N$ are independent — indeed merely pairwise uncorrelated — with finite variances, then

$$\operatorname{Var}\Big(\sum_{i=1}^N X_i\Big) = \sum_{i=1}^N \operatorname{Var}(X_i). \tag{3.4}$$

The computation is short and the cross-term cancellation is the whole point, so we show it in full. Writing $\bar{X}_i = X_i - \mathbb{E}[X_i]$ for the centered variables, so that $\sum_i X_i - \mathbb{E}[\sum_i X_i] = \sum_i \bar{X}_i$, and expanding the square of the sum,

$$\operatorname{Var}\Big(\sum_{i=1}^N X_i\Big) = \mathbb{E}\Bigg[\Big(\sum_{i=1}^N \bar{X}_i\Big)^2\Bigg] = \mathbb{E}\Bigg[\sum_{i=1}^N\sum_{j=1}^N \bar{X}_i\bar{X}_j\Bigg] = \sum_{i=1}^N\sum_{j=1}^N \mathbb{E}\Big[\bar{X}_i\bar{X}_j\Big] = \sum_{i=1}^N\sum_{j=1}^N \operatorname{Cov}(X_i, X_j),$$

the third equality by linearity of expectation across the N^2 terms. Now split the double sum into its diagonal $i = j$ and off-diagonal $i \neq j$ parts:

$$\sum_{i,j} \operatorname{Cov}(X_i, X_j) = \underbrace{\sum_{i=1}^N \operatorname{Cov}(X_i, X_i)}_{=\sum_i \operatorname{Var}(X_i)} + \underbrace{\sum_{i\neq j} \operatorname{Cov}(X_i, X_j)}_{=0},$$

and the off-diagonal block vanishes term by term because each $\operatorname{Cov}(X_i, X_j) = 0$ for $i \neq j$ — this is exactly where independence (through uncorrelatedness) is spent, and without it the discarded cross-terms would survive and could be of either sign. What remains is the diagonal $\sum_i \operatorname{Var}(X_i)$, which is (3.4).

Innocuous as it looks, (3.4) already contains the seed of the $N^{-1/2}$ law that organizes the entire fluctuation analysis of this book. Take the X_i i.i.d. with common variance σ^2. The sum has variance $\operatorname{Var}(\sum_i X_i) = N\sigma^2$, growing *linearly* in N: the random contributions, being uncorrelated, neither systematically reinforce nor cancel but accumulate additively. The empirical average $\bar{X}_N = \frac{1}{N}\sum_i X_i$ then has variance

$$\operatorname{Var}(\bar{X}_N) = \frac{1}{N^2}\operatorname{Var}\Big(\sum_{i=1}^N X_i\Big) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N},$$

the factor N^{-2} coming from pulling the constant $1/N$ out of the variance twice. So the variance of the average shrinks like $1/N$, and its standard deviation — the typical size of a fluctuation of $\bar{X}_N$ about its mean — like $\sigma/\sqrt{N}$. This $N^{-1/2}$ is the rate at which the empirical average of a non-interacting ensemble concentrates on its mean, and read on the empirical measure μ^N it is the rate at which the population profile settles onto the deterministic mean field m. At this stage it is only a statement about the second moment; section 3.4 promotes it, through the characteristic function, from variance bookkeeping into the full distributional statement that the rescaled fluctuation $\sqrt{N}(\bar{X}_N - \mathbb{E}[X_1])$ is asymptotically Gaussian — the central limit theorem.

With independence installed and the fluctuation scale already in view, the static catalogue of probabilistic objects is complete: we have sample spaces, laws, expectation-as-integration, the L^2

geometry of fluctuations, and the encoding of non-interaction. Every one of these is an object attached to a *fixed* law. But the mean field story is not about a fixed law; it is about a *sequence* of them — the empirical measures μ^N as N grows — and the assertion that this sequence converges to m. What survives into that limit is, as we stressed for the law itself, not the sample space but the law, and so the question that now organizes the rest of the chapter is dynamic rather than static: in *what sense* does a sequence of laws converge? Answering it requires choosing a topology on the space of probability measures, and the next section builds exactly that topology, together with the theorem — Portmanteau — that makes it usable.

3.2 Weak convergence and the Portmanteau theorem

The previous section closed by isolating what survives the passage to a limit. A random variable's sample space is disposable scaffolding; the invariant content, the thing that reappears in every limit theorem and on which the entire mean field theory is built, is its *law*. The mean field assertion "the population converges to a measure" is therefore, stripped to its mathematical core, the assertion that a sequence of laws — the empirical measures μ^N as N grows — converges to a deterministic law m. To make that assertion mean something we must say in *what sense* a sequence of probability measures converges, and that is a choice of topology on the space of all of them. Choosing and mastering that topology is the business of this section and the two that follow; it is the single piece of apparatus the rest of the book leans on more than any other, because every existence proof, every propagation-of-chaos statement, and every fluctuation theorem is ultimately a convergence statement in it.

We fix the setting once and for all. Let (S, d) be a metric space carrying its Borel σ-algebra $\mathcal{B}(S)$ — the reader may keep the concrete case $S = \mathbb{R}^d$ in mind throughout, since that is the single agent's state space in nearly every application, though separability and, later, completeness are the only structural facts we will actually use. Write $C_b(S)$ for the space of bounded continuous real-valued functions on S, $U_b(S) \subseteq C_b(S)$ for the bounded *uniformly* continuous ones, and $\mathcal{P}(S)$ for the set of Borel probability measures on S. These function classes are the test objects against which we will probe a measure: a probability measure μ is examined not by listing the μ-mass of every set — there are far too many sets — but by recording the averages it assigns to a rich enough family of functions, and $C_b(S)$ and $U_b(S)$ are the two families that turn out to be rich enough. We abbreviate that averaging by the *integral pairing*

$$\mu(f) \;:=\; \int_S f \,\mathrm{d}\mu, \qquad f \in C_b(S),\ \mu \in \mathcal{P}(S),$$

which is exactly the expectation $\mathbb{E}[f(X)]$ of (3.1) when μ is the law of X. This pairing $\mu \mapsto \mu(f)$ is the lens through which a measure is viewed for the remainder of the chapter, and weak convergence will be nothing other than convergence of every such reading. The notation $\mu(f)$ for $\int f \,\mathrm{d}\mu$ is used without further comment from here on.

The development proceeds in three movements, and it is worth naming them in advance so the reader sees the shape of the argument rather than meeting each piece cold. The *first* movement (section 3.2.1) steps back to the random variables themselves and sorts out the four standard senses in which a sequence X_n can converge — almost surely, in probability, in L^p, and in distribution — arranging them into a strict hierarchy. This is the orientation that prevents the most common confusions, and it earns its place here because the fourth and weakest mode, convergence in distribution, is precisely convergence of *laws*: it is the one mode that compares random variables through their laws alone, demands no common sample space, and is therefore the only one that can survive the mean field limit, where the coupling between agents makes almost-sure or L^p statements about individual trajectories hopeless. Locating it among its stronger cousins is what motivates singling it out for deep study. The *second* movement (section 3.2.2) makes that mode precise as the *weak convergence* of measures — $\mu_n(f) \to \mu(f)$ for every $f \in C_b(S)$ — and immediately confronts the awkwardness of the definition: testing against *all* bounded continuous functions is clean to state but unwieldy to verify and to exploit, since one cannot test directly against the indicator of an event. The Portmanteau theorem is the resolution, assembling five mutually equivalent characterizations — in terms of continuous tests, uniformly continuous tests, closed sets, open sets, and continuity sets — each tuned to a different use, and we prove it in full because it is the workhorse of every later chapter. The same movement shows that, when S is separable, weak convergence is not merely a convergence class but a genuine *metric* topology, exhibited through one explicit metric; this metrization is what licenses the constant use of sequential and subsequence arguments that the compactness theory of section 3.3 will demand. The *third* movement (section 3.2.3) supplies the bridge back from measures to random variables: Skorokhod's representation theorem realizes any weakly convergent sequence of laws as a sequence of random variables on one common probability space that converges *almost surely*. This is the device that lets us import the powerful pointwise convergence theorems of chapter 1 into the weak-convergence world, and it is the technical engine behind many of the later limit arguments.

We begin, then, with the first movement: before topologizing the laws we ask how the random variables underneath them converge, so as to place weak convergence correctly among the modes it is the weakest of.

3.2.1 Modes of convergence

We have just fixed the two test classes $C_b(S)$ and $U_b(S)$ and announced that a measure will be probed through the integral pairing $\mu \mapsto \mu(f)$; the plan is to declare a sequence of laws convergent when every such reading converges. Before we topologize the laws, however, it pays to drop back one level and ask the more elementary question the whole programme rests on: in what sense can the *random variables* themselves converge? There is not one answer but four, and they are genuinely different. Sorting them out now is not a digression but the orientation that fixes the target: the weakest of the four will turn out to be convergence of laws — the very thing the pairing $\mu(f)$ is built to detect — and seeing it sit at the bottom of a strict hierarchy is what explains why it, and not its stronger cousins, is the notion the mean field limit forces upon us. Keeping the four straight also heads off the most common errors, because the implications between them run one way only and the reverse implications fail in ways worth seeing concretely.

Let us name the four and say in words what each demands, since the formal statement that follows compresses them. *Almost-sure* convergence is the strongest and the most intuitive: for almost every outcome ω the numerical sequence $X_n(\omega)$ converges to $X(\omega)$ in the ordinary sense, so the trajectories settle down individually. *Convergence in probability* relaxes this to a statement about the bulk: the probability that X_n strays from X by more than a fixed tolerance ε tends to zero, but on the small bad set the discrepancy is uncontrolled and may be enormous, and that bad set is allowed to *move* with n rather than shrinking onto one fixed null set. *Convergence in L^p* measures closeness by an average of the p-th power of the error, so it controls the typical magnitude of the discrepancy but is sensitive to rare events carrying large values — a vanishing probability multiplying a huge value can keep an L^p distance from converging. Finally, *convergence in distribution* asks only that the laws converge: the histograms $\mathcal{L}(X_n)$ approach $\mathcal{L}(X)$, with no requirement that the variables be close, or even defined on the same probability space. The first three compare X_n to X *pathwise*, outcome by outcome, and so presuppose a common sample space; the fourth compares them only through their laws and so does not. This last distinction is the one that will matter most, and we return to it once the hierarchy is established.

Lemma 3.6 (Hierarchy of modes)**.** *Let X_n, X be S-valued random variables on a common probability space. Consider:* (a) $X_n \to X$ *almost surely;* (b) $X_n \to X$ *in probability, i.e.* $\mathbb{P}(d(X_n, X) > \varepsilon) \to 0$ *for every $\varepsilon > 0$;* (c) $X_n \to X$ *in L^p (for real-valued X_n), i.e.* $\mathbb{E}[|X_n - X|^p] \to 0$*;* (d) $X_n \to X$ *in distribution, i.e.* $\mathcal{L}(X_n) \to \mathcal{L}(X)$ *in the sense of definition 3.7 below. Then* (a)⇒(b)*,* (c)⇒(b)*, and* (b)⇒(d)*. Moreover* (b)⇒(a) along a subsequence*. None of the reverse implications holds in general; but if the limit X is a.s. a constant, then* (d)⇒(b)*.*

The lemma asserts a single chain, (a) ⇒ (b) ⇒ (d) with (c) ⇒ (b) feeding in from the side, so that both almost-sure and L^p convergence are strictly stronger than convergence in probability, which is in turn strictly stronger than convergence in distribution. Two of the four arrows are one-line consequences of machinery already in hand; the third, (b) ⇒ (d), is the only substantive one and is where the function classes of the previous section earn their keep; and the partial converses — recovery of almost-sure convergence along a subsequence, and the full converse when the limit is constant — are the refinements the later chapters actually invoke. We take the arrows in that order.

Proof. (a)⇒(b). Fix $\varepsilon > 0$ and consider the indicator $\mathbf{1}\{d(X_n, X) > \varepsilon\}$. If $X_n \to X$ almost surely then for almost every ω one has $d(X_n(\omega), X(\omega)) \to 0$, so eventually $d(X_n(\omega), X(\omega)) \le \varepsilon$ and the indicator is 0 from some n on; thus $\mathbf{1}\{d(X_n, X) > \varepsilon\} \to 0$ a.s.. These indicators are bounded by the constant 1, an integrable dominating function on a probability space, so dominated convergence (lemma 3.4) lets us pass the limit through the expectation: $\mathbb{P}(d(X_n, X) > \varepsilon) = \mathbb{E}[\mathbf{1}\{d(X_n, X) > \varepsilon\}] \to 0$. As ε was arbitrary, this is (b). The mechanism is worth registering: it is exactly the dominated convergence theorem of chapter 1, applied to indicators, that converts a pathwise statement into a statement about probabilities.

(c)⇒(b). This is Markov's inequality and nothing more. The event $\{|X_n - X| > \varepsilon\}$ is contained in the event where $|X_n - X|^p > \varepsilon^p$, and on that event the integrand $|X_n - X|^p$ exceeds ε^p, so

$$\varepsilon^p\, \mathbb{P}(\, |X_n - X| > \varepsilon) \le \mathbb{E}[|X_n - X|^p\ \mathbf{1}\{|X_n - X| > \varepsilon\}] \le \mathbb{E}[|X_n - X|^p]\,,$$

whence $\mathbb{P}(|X_n - X| > \varepsilon) \le \varepsilon^{-p}\mathbb{E}[|X_n - X|^p] \to 0$. Convergence of the p-th-power average therefore forces convergence in probability, with an explicit rate.

(b)⇒(d). This is the one arrow that is not a formality, and the reason is structural, so we pause on it. The Portmanteau theorem (theorem 3.9), which we prove in the next subsection, characterizes weak convergence of two *deterministic* sequences of measures; every one of its five equivalent forms speaks of the measures $\mathcal{L}(X_n)$ and $\mathcal{L}(X)$ alone. Convergence in probability, by contrast, is a statement about the *joint* behaviour of X_n and X on a common space — it measures how close the two variables are pathwise — and there is simply no way to extract a joint-space statement from a list of measure-level equivalences. The random variables must be brought in by hand, and the device that brings them in is the modulus of continuity. Let $f \in U_b(S)$, say $|f| \le M$, with modulus of continuity

$$\omega_f(\delta) \;=\; \sup\{|f(x) - f(y)| : d(x, y) \le \delta\},$$

which tends to 0 as $\delta \downarrow 0$ *precisely because f is uniformly continuous.* This is the crux, and it is why we test against $U_b(S)$ rather than all of $C_b(S)$: $\omega_f(\delta)$ is a *single* number bounding the oscillation of f over every pair of nearby points at once, so it can be pulled outside an expectation. A merely continuous f would supply only a local modulus, one that depends on *where* the two points sit, and such a thing cannot be factored out of $\mathbb{E}[\cdot]$. Now split the error according to whether X_n is within δ of X or not. On $\{d(X_n, X) \le \delta\}$ the integrand $|f(X_n) - f(X)|$ is at most $\omega_f(\delta)$ by the very definition of the modulus; on the complement it is at most $2M$ since $|f| \le M$. Therefore

$$|\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]| \le \mathbb{E}[|f(X_n) - f(X)|] \le \omega_f(\delta) \cdot \mathbb{P}(d(X_n, X) \le \delta) + 2M\, \mathbb{P}(d(X_n, X) > \delta),$$

and bounding the first probability by 1 gives the cleaner $|\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]| \le \omega_f(\delta) + 2M\, \mathbb{P}(d(X_n, X) > \delta)$. The estimate now does its work in two stages, in the right order. Fix $\varepsilon > 0$ and *first* choose δ small enough that $\omega_f(\delta) < \varepsilon$ — possible by uniform continuity, and this choice does not depend on n. *Then* let $n \to \infty$: hypothesis (b) makes $\mathbb{P}(d(X_n, X) > \delta) \to 0$, killing the second term, so $\limsup_n |\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]| \le \varepsilon$. Since ε was arbitrary the lim sup is 0, i.e. $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ for every $f \in U_b(S)$. In the integral-pairing notation this reads $\mathcal{L}(X_n)(f) \to \mathcal{L}(X)(f)$ for all $f \in U_b(S)$, which is exactly hypothesis (ii) of Portmanteau; that theorem then upgrades it to convergence against all of $C_b(S)$, i.e. $\mathcal{L}(X_n) \Rightarrow \mathcal{L}(X)$, which is (d). We are invoking theorem 3.9 ahead of its proof, but there is no circularity: the Portmanteau theorem is established in the next subsection by a self-contained argument that never refers to this lemma, so the forward reference is one of exposition only, and the reader may take it on credit until section 3.2.2 discharges it.

(b)⇒(a) along a subsequence. Convergence in probability cannot give almost-sure convergence of the *whole* sequence — the counterexample below shows the bad set roaming forever — but it always gives it back along a suitably thinned subsequence, and this partial recovery is used repeatedly later when one wants to pass to a pointwise limit. The idea is to choose the subsequence so fast that the bad probabilities become *summable*, at which point Borel–Cantelli forbids the bad events from recurring infinitely often. Concretely, apply (b) with tolerance $\varepsilon = 2^{-k}$: for each k the probability $\mathbb{P}(d(X_n, X) > 2^{-k}) \to 0$ as $n \to \infty$, so we may pick an index n_k — and, increasing it if necessary, arrange $n_1 < n_2 < \cdots$ — with $\mathbb{P}(d(X_{n_k}, X) > 2^{-k}) \le 2^{-k}$. The bound 2^{-k} was engineered to be the k-th term of a convergent geometric series, so $\sum_k \mathbb{P}(d(X_{n_k}, X) > 2^{-k}) \le \sum_k 2^{-k} = 1 < \infty$. The first Borel–Cantelli lemma (theorem 3.27(i), also proved later in this chapter, by an elementary argument independent of the present lemma — again no circularity) then says that summability

of the probabilities forces $\mathbb{P}(d(X_{n_k}, X) > 2^{-k}$ for infinitely many $k) = 0$. Outside that null set, $d(X_{n_k}, X) \le 2^{-k}$ for all sufficiently large k, and since $2^{-k} \to 0$ this gives $X_{n_k} \to X$ almost surely.

Constant-limit converse. The reverse arrow (d)⇒(b) is false in general — again, see below — but it holds in the one case the law of large numbers and the propagation-of-chaos arguments care about: when the limit is a deterministic constant, $X \equiv c$, so that $\mathcal{L}(X) = \delta_c$. Here convergence in distribution to a point mass already pins the variables down in probability, and the reason is that the only place δ_c puts mass is the single point c, which sits comfortably in the interior of every ball around it. Let us make this precise, expanding the sphere argument the statement compresses. Fix $\varepsilon > 0$ and let $\overline{B}(c, \varepsilon) = \{x : d(x, c) \le \varepsilon\}$ be the closed ε-ball about c. Its topological boundary lies in the sphere $\Sigma_\varepsilon = \{x : d(x, c) = \varepsilon\}$, because a boundary point is a limit of points inside and of points outside the ball and so must sit exactly on the level $d(\cdot, c) = \varepsilon$. Now observe two things. First, the spheres Σ_ε for distinct radii $\varepsilon > 0$ are pairwise disjoint, since a point has exactly one distance from c. Second, δ_c is a probability measure and so has finite total mass, and *any* finite measure can charge at most countably many members of a disjoint family with positive mass — if uncountably many of the Σ_ε each had mass $> 1/k$ for some fixed k, their disjoint union would have infinite mass, a contradiction, and the union over k of these countable sets is still countable. Hence all but countably many radii ε give $\delta_c(\Sigma_\varepsilon) = 0$, and for every such ε the ball $\overline{B}(c, \varepsilon)$ has δ_c-null boundary and is therefore a δ_c-continuity set in the sense of Portmanteau (v). (For the particular measure δ_c one can see this even more directly: the centre c lies in the open ball, hence in the interior, hence never on the boundary, so $\delta_c(\partial\overline{B}(c, \varepsilon)) = 0$ for *every* $\varepsilon > 0$; but the disjoint-spheres mechanism is the general one, and it recurs verbatim when we select continuity level-sets in the proof of Portmanteau.) For such a continuity radius, Portmanteau (v) applied to the hypothesis $\mathcal{L}(X_n) \Rightarrow \delta_c$ gives

$$\mathbb{P}\big(d(X_n, c) \le \varepsilon\big) = \mathcal{L}(X_n)\big(\overline{B}(c, \varepsilon)\big) \;\longrightarrow\; \delta_c\big(\overline{B}(c, \varepsilon)\big) = 1,$$

the limit being 1 because $c \in \overline{B}(c, \varepsilon)$. Equivalently $\mathbb{P}(d(X_n, c) > \varepsilon) \to 0$. Finally, the at-most-countably-many exceptional radii cause no trouble: given *any* $\varepsilon > 0$, pick a continuity radius $\varepsilon' < \varepsilon$, and monotonicity gives $\mathbb{P}(d(X_n, c) > \varepsilon) \le \mathbb{P}(d(X_n, c) > \varepsilon') \to 0$. Thus $X_n \to c$ in probability, which is (b). $\square$

Each failing reverse implication deserves a concrete witness, because the counterexamples are short and they fix the intuition far better than the warning "does not hold in general." Work on the unit interval $[0, 1]$ with Lebesgue measure as the probability $\mathbb{P}$. *Convergence in probability does not imply almost-sure convergence:* the standard "travelling bump" does it. Enumerate the dyadic subintervals $I_{k,j} = [\, j2^{-k}, (j{+}1)2^{-k}\,]$, $0 \le j < 2^k$, in a single list $A_1, A_2, \ldots$ sweeping across $[0, 1]$ at each scale k before passing to scale $k{+}1$, and set $X_n = \mathbf{1}_{A_n}$. Since the length of A_n tends to 0, $\mathbb{P}(|X_n| > \varepsilon) \le |A_n| \to 0$, so $X_n \to 0$ in probability; but every point of $[0, 1]$ lies in infinitely many of the sweeping intervals and outside infinitely many others, so $X_n(\omega)$ equals 1 infinitely often and 0 infinitely often, and converges for *no* ω. The bad set never settles; it merely keeps moving, exactly as the informal description of mode (b) warned. The same example, incidentally, converges to 0 in every L^p as well, since $\mathbb{E}[|X_n|^p] = |A_n| \to 0$, so it simultaneously shows that L^p convergence does not imply almost-sure convergence. *Almost-sure (or L^p) convergence and the other do not coincide either:* the spike $X_n = n\,\mathbf{1}_{[0,1/n]}$ converges to 0 pointwise everywhere except possibly at 0, hence almost surely and in probability, yet $\mathbb{E}[|X_n|] = n \cdot \frac{1}{n} = 1 \not\to 0$, so it fails to converge in L^1 — a vanishing-probability event carrying a diverging value, precisely the sensitivity that distinguishes L^p from convergence in probability. *Convergence in distribution does not imply convergence in probability:* let X be any symmetric non-degenerate variable (standard Gaussian, say) and set $X_n = -X$ for every n. By symmetry $\mathcal{L}(X_n) = \mathcal{L}(X)$, so trivially $X_n \Rightarrow X$; but $X_n - X = -2X$, and $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(2\,|X| > \varepsilon)$ is a fixed positive number, bounded away from 0. The laws agree perfectly while the variables stay resolutely far apart — the cleanest illustration that mode (d) sees only the law and is blind to the joint coupling between X_n and X. That blindness is not a defect to be lamented; in the mean field setting it is exactly the feature we need.

This last point is the throughline payoff, and it is why this entire subsection was worth its place. In the N-player game the agents are coupled — each one's state depends, through the shared population, on every other's — so at finite N there is no fixed limiting random variable that an individual trajectory approaches pathwise, and almost-sure or L^p statements about "the" limit have nothing to attach to. What *does* stabilize as $N \to \infty$ is the population's statistical profile, the empirical measure μ^N, and what we can hope to prove is that its law settles onto a deterministic mean field m. That is a statement in mode (d), and the example $X_n = -X$ shows precisely why no stronger mode is available: convergence in distribution is the one notion that compares objects through their laws alone, demanding neither a common sample space nor any pathwise proximity, and so it is the one mode robust to the coupling that the stronger modes cannot survive. Mode (d) is, in short, the survivor — the only sense of convergence that the mean field limit leaves standing.

Having located convergence in distribution at the foot of the hierarchy and argued that it is the notion the rest of the book must develop, we now give it its proper name and its working machinery. The next subsection states the formal definition of weak convergence of measures — the precise meaning of the $\mathcal{L}(X_n) \to \mathcal{L}(X)$ written as item (d) above — and proves the Portmanteau theorem that the arrow (b)$\Rightarrow$(d) has already leaned on, turning the bare definition into the five interchangeable characterizations every later chapter will use.

3.2.2 The definition and its working form

The previous subsection ran the four modes of convergence to ground and left us with a single survivor: convergence in distribution, mode (d), the one notion that compares random variables through their laws alone and so is the only one the mean field coupling cannot destroy. We promised it its proper name, and we now deliver it. Stripped of the random variables, the assertion "the laws $\mathcal{L}(X_n)$ approach $\mathcal{L}(X)$" is a statement purely about a sequence of probability measures, and the name for that statement is *weak convergence*. Its definition is the one already foreshadowed by the integral pairing $\mu \mapsto \mu(f)$ of section 3.2: a measure is probed not by listing the mass of every set but by recording the averages it assigns to a rich family of test functions, and a sequence of measures is declared convergent when every one of those readings converges.

Definition 3.7 (Weak convergence of probability measures)**.** A sequence $(\mu_n)_{n\geq 1}$ in $\mathcal{P}(S)$ *converges weakly* to $\mu \in \mathcal{P}(S)$, written $\mu_n \Rightarrow \mu$, if

$$\mu_n(f) \;\longrightarrow\; \mu(f) \qquad \text{for every } f \in C_b(S).$$

If X_n, X are random variables we write $X_n \Rightarrow X$ to mean $\mathcal{L}(X_n) \Rightarrow \mathcal{L}(X)$, and call this convergence in distribution.

The test class in definition 3.7 is $C_b(S)$, the bounded *continuous* functions, and neither qualifier is idle. Boundedness keeps every pairing $\mu(f) = \int f \, \mathrm{d}\mu$ finite for every probability measure at once, so the definition never stumbles over an integrability question; continuity makes the pairing insensitive to the fine, measure-zero structure of μ and so lets it detect only the genuine "shape" of the measure. One might ask why we do not simply demand $\mu_n(B) \to \mu(B)$ for every Borel set B, the most literal reading of "the measures converge". The answer, which the whole section will make vivid, is that this demand is far too strong: it fails already for the most basic approximation one wants to permit. If $\mu_n = \delta_{1/n}$ is a unit mass sliding toward the origin, then for the half-open set $B = (0, \infty)$ one has $\mu_n(B) = 1$ for every n while the natural limit $\mu = \delta_0$ has $\mu(B) = 0$, so the set-wise demand is violated even though, on any reasonable reading, the masses are converging to a point at the origin. Testing against continuous f repairs exactly this: $\mu_n(f) = f(1/n) \to f(0) = \mu(f)$ for every $f \in C_b(S)$, because a continuous function cannot tell the difference between sitting *at* a boundary point and sitting just inside it. The price of this robustness is that the test class is enormous, and removing that awkwardness is the whole purpose of the Portmanteau theorem below.

Notation watch

The probabilist's "weak convergence" is testing against bounded *continuous* functions, $C_b(S)$. The functional analyst's "weak-$*$ convergence" in the dual pairing $\left(C_b(S)\right)^*$ would test against the *entire* dual and need not preserve total mass. On a compact metric space, where the continuous dual of $C(S)$ is the space of finite signed Borel measures $\mathcal{M}(S)$ — this is the Riesz–Markov representation theorem, classical and not used as a tool below [61, Ch. 7] — weak convergence of probability measures coincides with weak-$*$ convergence restricted to the probability simplex. On a non-compact S such as $\mathbb{R}^d$ they genuinely differ: mass can escape to infinity under weak-$*$ limits, producing a sub-probability measure, whereas definition 3.7 insists the limit again be a probability measure. Tightness (section 3.3) is exactly the condition that rules out this escape. We always mean definition 3.7 when we say "weak convergence"; the macro is `\weakto` ($\Rightarrow$).

The escape-of-mass phenomenon flagged in the box is worth picturing concretely, because it is the precise gap the compactness theory of section 3.3 exists to close. On $S = \mathbb{R}$ take $\mu_n = \delta_n$, a unit mass marching off to $+\infty$. For any $f \in C_b(\mathbb{R})$ that vanishes at infinity, $\mu_n(f) = f(n) \to 0$, while for the constant $f \equiv 1$ one has $\mu_n(f) = 1$ for all n. There is no single probability measure μ on $\mathbb{R}$ with $\mu_n(f) \to \mu(f)$ for every $f \in C_b$: the would-be limit has lost all of its mass over the horizon, and any honest limiting object is the zero measure, which is not a probability measure at all. So a sequence in $\mathcal{P}(S)$ need not converge weakly to anything in $\mathcal{P}(S)$ — the space is not compact when S is non-compact. The functional analyst tolerates this by admitting sub-probability limits in the weak-$*$ sense; we cannot, because the mean field m must be an honest probability measure. *Tightness* — a uniform control that forbids the mass from escaping to infinity — is exactly the hypothesis that restores a limit inside $\mathcal{P}(S)$, and Prokhorov's theorem (theorem 3.16) will show it is the *only* obstruction. For the present section we set the escape problem aside and study weak convergence where a limit is already given; we will, however, twice need to borrow tightness and Prokhorov before they are proved, and we flag each borrowing as it occurs.

Testing against *all* of $C_b(S)$ is convenient for stating the definition but awkward for verifying it and for deducing consequences. In practice one rarely has the whole of $C_b(S)$ under control at once: one would prefer to test against a manageable *sub*family — say the uniformly continuous functions, which can be built by hand — or against the indicator $\mathbf{1}_A$ of an explicit event A, and that indicator is not continuous, so it is not even a legal test function. The Portmanteau theorem is the resolution. It shows that the single clause of definition 3.7 is equivalent to four others, assembling *five* mutually interchangeable characterizations — continuous tests, uniformly continuous tests, a closed-set inequality, an open-set inequality, and exact convergence on sets whose boundary is null — each tuned to a different use, so that one may verify weak convergence through whichever clause is cheapest and then exploit it through whichever is sharpest. It is, without exaggeration, the single most-used technical tool in this book; we therefore prove it in full.

The engine of the whole equivalence is one approximation lemma: a quantitative, constructive form of Urysohn's lemma that holds in *any* metric space, with no appeal to partitions of unity or to compactness. Given a closed set F, it manufactures an explicit sequence of Lipschitz functions decreasing to the indicator $\mathbf{1}_F$, and thereby builds the bridge from continuous tests to statements about sets that every step of Portmanteau will cross.

Lemma 3.8 (Continuous approximation of a closed set)**.** *Let $F \subseteq S$ be closed and define, for $k \in \mathbb{N}$,*

$$f_k(x) \;=\; \big(1 - k\, d(x,F)\big)^+, \qquad d(x,F) = \inf_{y\in F} d(x,y).$$

Then each f_k is k-Lipschitz, hence $f_k \in U_b(S)$ with $0 \le f_k \le 1$; $f_k = 1$ on F; and $f_k(x) \downarrow \mathbf{1}_F(x)$ for every $x \in S$ as $k \to \infty$.

Proof. First, $x \mapsto d(x,F)$ is 1-Lipschitz. Indeed, for any $x, x' \in S$ and any $y \in F$ the triangle inequality gives $d(x,y) \le d(x,x') + d(x',y)$; taking the infimum over $y \in F$ on both sides yields $d(x,F) \le d(x,x') + d(x',F)$, i.e. $d(x,F) - d(x',F) \le d(x,x')$, and by symmetry $|d(x,F) - d(x',F)| \le d(x,x')$. Consequently $x \mapsto 1 - k\,d(x,F)$ is k-Lipschitz, and its positive part is too, since $t \mapsto t^+$ is 1-Lipschitz and a 1-Lipschitz function of a k-Lipschitz map is k-Lipschitz; thus $f_k \in U_b(S)$ with $0 \le f_k \le 1$. If $x \in F$ then $d(x,F) = 0$, so $f_k(x) = 1$. The sequence $f_k(x)$ is non-increasing in k because $d(x,F) \ge 0$ makes $1 - k\,d(x,F)$ decrease in k while $t \mapsto t^+$ is non-decreasing. Finally we identify the pointwise limit. If $x \in F$ then $f_k(x) = 1 = \mathbf{1}_F(x)$ for every k. If $x \notin F$ then — and this is the one place closedness of F is spent — the complement $S \setminus F$ is open, so x sits in it with room to spare and $d(x,F) > 0$; hence $k\,d(x,F) \to \infty$, $1 - k\,d(x,F) \to -\infty$, and $f_k(x) = (1 - k\,d(x,F))^+ = 0$ for all sufficiently large k, matching $\mathbf{1}_F(x) = 0$. Therefore $f_k \downarrow \mathbf{1}_F$ pointwise. □

Two features of this construction will be used again and again. First, because the f_k decrease to $\mathbf{1}_F$ and are dominated by the constant 1 (integrable against any probability measure), dominated convergence gives $\mu(f_k) \downarrow \mu(F)$ for *every* probability measure μ: a closed set's mass is the decreasing limit of the integrals of explicit continuous functions. Second, the f_k lie in $U_b(S)$, the smaller of our two test classes, so any characterization controlling $\mu(f)$ for $f \in U_b$ already controls the masses of closed sets. These are precisely the two facts the first nontrivial implication of Portmanteau exploits. We now state the theorem and prove it by chasing a single closed cycle of implications, (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Leftrightarrow$ (iv) $\Rightarrow$ (v) $\Rightarrow$ (i), so that every one of the five clauses is reachable from every other.

Theorem 3.9 (Portmanteau)**.** *Let S be a metric space and $\mu_n, \mu \in \mathcal{P}(S)$. The following are equivalent.*

(i) $\mu_n(f) \to \mu(f)$ *for every* $f \in C_b(S)$ *(i.e.* $\mu_n \Rightarrow \mu$*).*

(ii) $\mu_n(f) \to \mu(f)$ *for every* $f \in U_b(S)$*.*

(iii) $\limsup_n \mu_n(F) \le \mu(F)$ *for every closed* $F \subseteq S$*.*

(iv) $\liminf_n \mu_n(G) \ge \mu(G)$ *for every open* $G \subseteq S$*.*

(v) $\mu_n(A) \to \mu(A)$ *for every Borel* continuity set A*, i.e. every* A *with* $\mu(\partial A) = 0$*.*

Proof. We prove (i)$\Rightarrow$(ii)$\Rightarrow$(iii)$\Leftrightarrow$(iv)$\Rightarrow$(v)$\Rightarrow$(i), a closed loop through which any clause implies any other.

(i)$\Rightarrow$(ii). Immediate, since $U_b(S) \subseteq C_b(S)$: testing against fewer functions is a weaker demand, so it follows from the stronger one. This trivial arrow is worth one comment, because its *converse* — that controlling only the uniformly continuous tests already forces control of all continuous tests — is the genuinely useful content buried in the loop, and it is delivered only once the full circuit closes at (v)$\Rightarrow$(i).

(ii)$\Rightarrow$(iii). Here the Urysohn functions do their work. Fix a closed set F and let f_k be as in lemma 3.8. Since $\mathbf{1}_F \le f_k$ pointwise, monotonicity of the integral gives $\mu_n(F) = \mu_n(\mathbf{1}_F) \le \mu_n(f_k)$ for every n and every k. Take $\limsup_n$ with k held fixed: because $f_k \in U_b(S)$, hypothesis (ii) supplies $\mu_n(f_k) \to \mu(f_k)$, so

$$\limsup_n \mu_n(F) \;\le\; \limsup_n \mu_n(f_k) \;=\; \mu(f_k).$$

The order of the two limits is the crux and must not be reversed: we send n to infinity *first*, at fixed k, so that (ii) applies to the single fixed test function f_k, and only then send $k \to \infty$. By lemma 3.8, $f_k \downarrow \mathbf{1}_F$, and each $f_k \le 1$ is dominated by the constant 1, which is μ-integrable since μ is a probability measure; dominated convergence therefore gives $\mu(f_k) \downarrow \mu(F)$. Hence $\limsup_n \mu_n(F) \le \mu(F)$, which is (iii). The inequality can be strict — mass can pile onto the boundary of F in the limit, as $\mu_n = \delta_{1/n} \Rightarrow \delta_0 = \mu$ with $F = \{0\}$ shows ($\mu_n(F) = 0$ for all n, yet $\mu(F) = 1$): a closed set can only *gain* mass in the limit, never lose it, and that one-sidedness is exactly what (iii) records.

(iii)$\Leftrightarrow$(iv). The two are the same statement read through complementation. A set G is open iff its complement $F = S \setminus G$ is closed, and for any probability measure ν one has $\nu(G) = 1 - \nu(F)$. Therefore

$$\liminf_n \mu_n(G) = \liminf_n \big(1 - \mu_n(F)\big) = 1 - \limsup_n \mu_n(F),$$

using $\liminf(1-a_n) = 1 - \limsup a_n$. Comparing with $\mu(G) = 1 - \mu(F)$, the open-set inequality $\liminf_n \mu_n(G) \geq \mu(G)$ is literally the closed-set inequality $\limsup_n \mu_n(F) \leq \mu(F)$ with both sides subtracted from 1. The two clauses are thus interchangeable, and we may use whichever is convenient; (iv) is the form that propagates lower semicontinuity, as the remark below explains.

(iii) & (iv)⇒(v). Recall the topological anatomy of a Borel set A: its interior A° (the largest open set contained in A), its closure $\overline{A}$ (the smallest closed set containing A), and its boundary $\partial A = \overline{A} \setminus A^\circ$. A *continuity set* is one whose boundary is μ-null, $\mu(\partial A) = 0$ — the name signalling that μ puts no mass on the very places where "inside A" and "outside A" touch. From $A^\circ \subseteq A \subseteq \overline{A}$ and monotonicity of each μ_n we have the sandwich $\mu_n(A^\circ) \leq \mu_n(A) \leq \mu_n(\overline{A})$; applying (iv) to the open set A° on the left and (iii) to the closed set $\overline{A}$ on the right,

$$\mu(A^\circ) \leq \liminf_n \mu_n(A^\circ) \leq \liminf_n \mu_n(A) \leq \limsup_n \mu_n(A) \leq \limsup_n \mu_n(\overline{A}) \leq \mu(\overline{A}).$$

Now the continuity hypothesis is spent: since $\mu(\overline{A}) - \mu(A^\circ) = \mu(\partial A) = 0$ and $A^\circ \subseteq A \subseteq \overline{A}$, all three of $\mu(A^\circ), \mu(A), \mu(\overline{A})$ coincide, so the two outer terms of the display are equal. They squeeze the $\liminf$ and $\limsup$ of $\mu_n(A)$ together, and equal to the common value $\mu(A)$, so the genuine limit $\mu_n(A) \to \mu(A)$ exists. Without $\mu(\partial A) = 0$ the outer bounds would straddle a gap of width $\mu(\partial A)$ and $\mu_n(A)$ could oscillate within it — which is precisely what happens for $A = (0, \infty)$ under $\mu_n = \delta_{1/n} \Rightarrow \delta_0$, whose boundary $\{0\}$ carries the entire limit mass.

(v)⇒(i). This closes the loop, upgrading convergence on continuity sets back to convergence against every bounded continuous test. Let $f \in C_b(S)$. Replacing f by $(f - \inf f)/(\sup f - \inf f)$ — an affine rescaling that changes both sides of $\mu_n(f) \to \mu(f)$ by the same affine map and is harmless if f is constant — we may assume $0 \leq f < 1$. The tool is the *layer-cake* (or distribution-function) representation: for any nonnegative bounded measurable f and any probability measure ν,

$$\nu(f) = \int_S f \, \mathrm{d}\nu = \int_0^1 \nu(\{f > t\}) \, \mathrm{d}t. \tag{3.5}$$

This is itself a Fubini–Tonelli computation: write the value as an integral over its own height, $f(x) = \int_0^1 \mathbf{1}\{f(x) > t\} \, \mathrm{d}t$ (the length of the interval of levels below $f(x)$), integrate in x against ν, and apply the Tonelli theorem (chapter 1) to the nonnegative integrand $\mathbf{1}\{f(x) > t\}$ to swap the order of the x- and t-integrals, giving the right-hand side. Now fix a level $t \in (0,1)$. Because f is continuous, the superlevel set $\{f > t\}$ is open and $\{f \geq t\}$ is closed, so the boundary of the former is squeezed,

$$\partial\{f > t\} \;\subseteq\; \{f \geq t\} \setminus \{f > t\} \;\subseteq\; \{f = t\},$$

the closure of $\{f > t\}$ lying inside the closed set $\{f \geq t\}$ and its interior containing the open set $\{f > t\}$. The level sets $\{f = t\}$ for distinct t are pairwise disjoint — a point x has the single value $f(x)$ and so lies in exactly one of them — and μ, being a finite measure, can assign positive mass to at most countably many members of a disjoint family (if uncountably many had mass exceeding $1/m$ for some fixed m, a finite sub-collection would already exceed the total mass 1, and the union over m of these countable sets is still countable). Hence for all but countably many $t \in (0,1)$ we have $\mu(\{f = t\}) = 0$, so $\mu(\partial\{f > t\}) = 0$ and $\{f > t\}$ is a μ-continuity set; for each such t, clause (v) gives $\mu_n(\{f > t\}) \to \mu(\{f > t\})$. The exceptional t form a Lebesgue-null set and do not affect the t-integral. The integrands $t \mapsto \mu_n(\{f > t\})$ are bounded by 1, an integrable dominating function on the finite interval $(0, 1)$, so dominated convergence in t passes the limit through (3.5):

$$\mu_n(f) = \int_0^1 \mu_n(\{f > t\})\ \mathrm{d}t \ \longrightarrow\ \int_0^1 \mu(\{f > t\})\ \mathrm{d}t = \mu(f).$$

This is (i), closing the cycle and, with it, establishing the mutual equivalence of all five clauses. □

Remark 3.10 (Which form to use)**.** The value of Portmanteau is that one rarely uses the same clause twice in a row: each is the natural language for a different task, and the theorem lets us switch between them for free. Form (i) is the definition and the thing one ultimately wants to conclude. Form (ii) is what one actually *verifies*, because uniformly continuous test functions can be built and bounded by hand — the proof of (b)⇒(d) in lemma 3.6 produced exactly such functions, and the central limit theorem of section 3.4.2 will establish weak convergence by checking convergence against the single smooth family $x \mapsto e^{itx}$, which lies in U_b: one verifies (ii) for that family and Portmanteau upgrades it to (i). Forms (iii) and (iv) are the currency of compactness and *lower-semicontinuity* arguments. Many functionals of a measure that the later chapters minimize — the relative entropy, the running cost of a mean field game — are suprema of pairings $\mu(f)$ over families of test functions, hence lower semicontinuous under weak convergence, and it is precisely the one-sided inequality (iv) (or (iii) for an upper bound) that survives a weak limit; this is what makes the direct method of the calculus of variations run in chapter 10. Form (v) is the one used to pass to the limit in the probability of an *explicit* event: if $\mu_n \Rightarrow \mu$ and the event A has μ-null boundary, then literally $\mu_n(A) \to \mu(A)$, no continuity of an indicator required — this is how the constant-limit converse in lemma 3.6 concluded $\mathbb{P}(d(X_n, c) \le \varepsilon) \to 1$. The mental habit to form is: *verify* through (ii), *transport functionals* through (iii)/(iv), *read off event probabilities* through (v).

A frequently used consequence — needed whenever a law is pushed forward through a map, as when an equilibrium control is applied to a state or a trajectory is read out through an observable — is the continuous mapping theorem: weak convergence survives the application of a map that is continuous except on a set the limit measure ignores. One might hope to prove it in a single line by testing $h_\#\mu_n$ against $g \in C_b(S')$, since $(h_\#\mu_n)(g) = \mu_n(g \circ h)$; but $g \circ h$ is continuous only where h is, so when h has discontinuities $g \circ h \notin C_b(S)$ and is *not a legal test function* for $\mu_n \Rightarrow \mu$. The naive argument thus fails exactly at the discontinuities, and the repair is to abandon the continuous-test form (i) and work instead through the closed-set form (iii), which never mentions a test function and tolerates the bad set as long as μ does not charge it.

Corollary 3.11 (Continuous mapping theorem)**.** *Let (S, d), (S', d') be metric spaces, $h : S \to S'$ Borel measurable, and let $D_h = \{x \in S : h \text{ is discontinuous at } x\}$ (a Borel set). If $\mu_n \Rightarrow \mu$ in $\mathcal{P}(S)$ and $\mu(D_h) = 0$, then $h_\#\mu_n \Rightarrow h_\#\mu$ in $\mathcal{P}(S')$, where $h_\#\nu = \nu \circ h^{-1}$ is the push-forward.*

Proof. First note that the construction makes sense: h is Borel, so each push-forward $h_\#\nu = \nu \circ h^{-1}$ is a Borel probability measure on S', and D_h is a Borel set (it is an F_σ, the countable union over rational thresholds of the closed sets where the oscillation of h exceeds the threshold). By Portmanteau (theorem 3.9) applied on S', it suffices to verify the closed-set inequality (iii) for $h_\#\mu_n$ and $h_\#\mu$. Let $F' \subseteq S'$ be closed and set $A = h^{-1}(F') \subseteq S$, so that $(h_\#\nu)(F') = \nu(A)$ for every

ν. The set A need not be closed — h is not continuous — so we cannot apply (iii) to A directly; instead we control its closure. We claim

$$\overline{A} \;\subseteq\; A \cup D_h.$$

Indeed, take $x \in \overline{A}$ and suppose $x \notin D_h$, i.e. h is continuous at x. Choose $x_k \in A$ with $x_k \to x$ (possible since x lies in the closure of A). Continuity at x gives $h(x_k) \to h(x)$; each $h(x_k) \in F'$ because $x_k \in A = h^{-1}(F')$, and F' is closed, so the limit satisfies $h(x) \in F'$, i.e. $x \in A$. Thus any point of $\overline{A}$ outside D_h already lies in A, which is the claim, and in particular $\overline{A} \setminus A \subseteq D_h$. Since $\mu(D_h) = 0$ by hypothesis, $\mu(\overline{A}) = \mu(A)$. Now $\overline{A}$ *is* closed, so (iii) applies to it:

$$\limsup_n (h_{\#}\mu_n)(F') = \limsup_n \mu_n(A) \le \limsup_n \mu_n(\overline{A}) \le \mu(\overline{A}) = \mu(A) = (h_{\#}\mu)(F').$$

This is the closed-set inequality on S', valid for every closed F', so $h_{\#}\mu_n \Rightarrow h_{\#}\mu$. □

The hypothesis $\mu(D_h) = 0$ cannot be dropped, and one line shows why. With $S = S' = \mathbb{R}$, take $h = \mathbf{1}_{(0,\infty)}$, discontinuous only at the origin so that $D_h = \{0\}$, and $\mu_n = \delta_{1/n} \Rightarrow \delta_0 = \mu$. The pushed-forward laws are $h_{\#}\mu_n = \delta_{h(1/n)} = \delta_1$ for every n, whereas $h_{\#}\mu = \delta_{h(0)} = \delta_0$, so $h_{\#}\mu_n \not\Rightarrow h_{\#}\mu$. The obstruction is precisely that μ places its entire mass on the lone discontinuity $D_h = \{0\}$ — exactly the situation the hypothesis forbids.

definition 3.7 presents weak convergence as a bare *convergence class* — a rule declaring which sequences converge to which limits — and a priori such a rule need not come from any metric. It matters a great deal that here it does, because the entire compactness machinery of the next sections, and the routine "pass to a subsequence" manoeuvres of the existence proofs, are licensed only in a *metric* (or at least sequential) topology: in a general topological space compactness and sequential compactness can diverge, and a convergence class that is not topological at all can behave pathologically. We now show that on a separable S weak convergence *is* induced by an honest metric, and on a Polish S that metric is itself complete and separable — so $\mathcal{P}(S)$ is Polish and weak compactness reduces to the extraction of convergent subsequences, exactly the form the tightness arguments will use. We exhibit one such metric explicitly, the *bounded Lipschitz* (or Fortet–Mourier) metric, chosen because its test class is at once rich enough to separate measures and rigid enough — a uniform Lipschitz bound — to be precompact in the sense Arzelà–Ascoli needs. For $f : S \to \mathbb{R}$ write

$$\mathrm{Lip}(f) = \sup_{x \neq y} \frac{|f(x) - f(y)|}{d(x,y)}, \qquad \|f\|_{\mathrm{BL}} = \|f\|_\infty + \mathrm{Lip}(f),$$

the latter finite precisely for the bounded Lipschitz functions, and define the distance between two probability measures as the largest gap their pairings can exhibit over the unit ball of this norm:

$$d_{\mathrm{BL}}(\mu,\nu) = \sup\Big\{\, |\mu(f) - \nu(f)| : \; \|f\|_{\mathrm{BL}} \le 1 \Big\} \qquad (\mu,\nu \in \mathcal{P}(S)). \tag{3.6}$$

Read as a supremum over a unit ball of test functions, d_{BL} is a dual-norm-type distance: the smaller it is, the more nearly μ and ν assign the same average to every reasonably regular function at once.

Theorem 3.12 (Metrization of weak convergence)**.** *Let (S,d) be a Polish space. Then d_{BL} is a metric on $\mathcal{P}(S)$, and for $\mu_n, \mu \in \mathcal{P}(S)$,*

$$d_{\mathrm{BL}}(\mu_n,\mu) \to 0 \qquad \Longleftrightarrow \qquad \mu_n \Rightarrow \mu.$$

Moreover $(\mathcal{P}(S), d_{\mathrm{BL}})$ *is itself Polish: separable and complete.* (*The metric and separability statements use only separability of S; the* $\Rightarrow\Rightarrow d_{\mathrm{BL}}$ *direction and completeness use the completeness of S through tightness.*)

Proof. The proof has five movements: d_{BL} is a metric; it metrizes weak convergence (the two directions of the displayed equivalence); $\mathcal{P}(S)$ is separable; and $\mathcal{P}(S)$ is complete. Two of these movements — the harder direction $\Rightarrow\Rightarrow d_{\mathrm{BL}} \to 0$ and completeness — borrow tightness, Prokhorov's theorem (theorem 3.16) and Ulam's single-measure tightness (lemma 3.15) from the *next* section. There is no circularity in this: those results are proved there by arguments that never invoke the present theorem, so we may use them here on credit, and we flag each use as it arises.

d_{BL} *is a metric.* Symmetry is built into the absolute value in (3.6), and the triangle inequality is inherited termwise — for each admissible f, $|\mu(f) - \lambda(f)| \le |\mu(f) - \nu(f)| + |\nu(f) - \lambda(f)| \le d_{\mathrm{BL}}(\mu, \nu) + d_{\mathrm{BL}}(\nu, \lambda)$, and taking the supremum over f on the left preserves the bound. Clearly $d_{\mathrm{BL}}(\mu, \mu) = 0$. Finiteness: the constraint $\|f\|_{\mathrm{BL}} \le 1$ forces in particular $\|f\|_\infty \le 1$, so $|\mu(f) - \nu(f)| \le \|f\|_\infty (\mu(S) + \nu(S)) \le 2$ and $d_{\mathrm{BL}} \le 2$ always. Only definiteness has content. Suppose $d_{\mathrm{BL}}(\mu, \nu) = 0$, so $\mu(f) = \nu(f)$ for every bounded Lipschitz f. The Urysohn functions f_k of lemma 3.8 attached to a closed set F are bounded and Lipschitz, hence admissible after scaling by their BL-norm, so $\mu(f_k) = \nu(f_k)$ for all k; letting $k \to \infty$ with $f_k \downarrow \mathbf{1}_F$ and dominated convergence on both sides gives $\mu(F) = \nu(F)$ for every closed F. The closed sets form a π-system (they are stable under finite intersection) that generates $\mathcal{B}(S)$, so two probability measures agreeing on all of them agree everywhere, by the uniqueness lemma corollary 1.10 imported from chapter 1. Thus $\mu = \nu$, and d_{BL} separates points.

$d_{\mathrm{BL}} \to 0 \Rightarrow\Rightarrow$. This direction is elementary and uses only the Urysohn functions. Each f_k from lemma 3.8 is bounded Lipschitz, say with $\|f_k\|_{\mathrm{BL}} = c_k < \infty$; then f_k/c_k is admissible in (3.6), so for each fixed k, $|\mu_n(f_k) - \mu(f_k)| \le c_k\, d_{\mathrm{BL}}(\mu_n, \mu) \to 0$ as $n \to \infty$. Running the argument of step (ii)⇒(iii) of theorem 3.9 with these f_k — $\mu_n(F) \le \mu_n(f_k) \to \mu(f_k)$, then $k \to \infty$ with $\mu(f_k) \downarrow \mu(F)$ — gives $\limsup_n \mu_n(F) \le \mu(F)$ for every closed F, which is Portmanteau (iii); hence $\mu_n \Rightarrow \mu$.

$\Rightarrow\Rightarrow d_{\mathrm{BL}} \to 0$. This is the substantive direction, and it is where tightness enters. Argue by contradiction. If $d_{\mathrm{BL}}(\mu_n, \mu) \not\to 0$, then along some subsequence (not relabelled) there are $\varepsilon > 0$ and admissible functions f_n, $\|f_n\|_{\mathrm{BL}} \le 1$, with $|\mu_n(f_n) - \mu(f_n)| \ge \varepsilon$ for all n. The difficulty is that the offending f_n *change with* n, so we cannot quote $\mu_n \Rightarrow \mu$ against any single one of them; we must first replace the moving family by a fixed test function, and that requires an Arzelà–Ascoli compactness together with a tightness control of where the mass lives.

Step 1 (tightness localizes the mass). The set $\{\mu_n\}_n \cup \{\mu\}$ is weakly convergent, hence relatively compact in $\mathcal{P}(S)$, hence tight by the necessity half of Prokhorov's theorem (theorem 3.16(b)). So, given $\delta > 0$, there is a compact $K \subseteq S$ with $\sup_n \mu_n(K^c) \le \delta$ and $\mu(K^c) \le \delta$ simultaneously: all the mass, uniformly in n, sits on one fixed compact set up to δ.

Step 2 (Arzelà–Ascoli extracts a fixed limit on K). On the compact K the f_n are uniformly bounded, $|f_n| \le 1$, and uniformly equicontinuous, since $\mathrm{Lip}(f_n) \le 1$ gives $|f_n(x) - f_n(y)| \le d(x, y)$ for all n at once. The Arzelà–Ascoli theorem (a prerequisite of undergraduate analysis) therefore furnishes a further subsequence along which $f_n \to f_0$ *uniformly on* K, the limit $f_0 : K \to \mathbb{R}$ inheriting $|f_0| \le 1$ and $\mathrm{Lip}(f_0) \le 1$.

Step 3 (McShane extends the limit to all of S). We need f_0 as a global test function, so we extend it off K without enlarging either its Lipschitz constant or its bound, by the McShane formula

$$f(x) = \Big(\sup_{y\in K}\big(f_0(y) - d(x,y)\big)\Big) \vee (-1) \wedge 1.$$

Each $x \mapsto f_0(y) - d(x,y)$ is 1-Lipschitz; a supremum of 1-Lipschitz functions is 1-Lipschitz; and truncation into $[-1,1]$ neither raises the Lipschitz constant nor breaks continuity, so $f \in C_b(S)$ with $|f| \le 1$. Moreover $f = f_0$ on K: for $x \in K$ the choice $y = x$ gives the value $f_0(x) \in [-1,1]$, and any other $y \in K$ gives $f_0(y) - d(x,y) \le f_0(x)$ because $\mathrm{Lip}(f_0) \le 1$, so the supremum is attained at $y = x$ and the truncation is inert. (Only $f \in C_b(S)$ and the uniform approximation $\sup_K |f_n - f| = \sup_K |f_n - f_0| \to 0$ are used below; the precise form of the extension is immaterial.)

Step 4 (assemble the contradiction). Split the offending quantity through the fixed test function f:

$$|\mu_n(f_n) - \mu(f_n)| \le \underbrace{|\mu_n(f_n - f)|}_{\text{(I)}} + \underbrace{|\mu_n(f) - \mu(f)|}_{\text{(II)}} + \underbrace{|\mu(f - f_n)|}_{\text{(III)}}.$$

For (I), separate the integral over K from its complement: on K the integrand $|f_n - f|$ is at most $\sup_K |f_n - f|$, while off K it is at most $|f_n| + |f| \le 2$, so (I) $\le \sup_K |f_n - f| + 2\mu_n(K^c) \le \sup_K |f_n - f| + 2\delta$, which tends to 2δ as $n \to \infty$ by Step 2. The same bound with μ in place of μ_n gives (III) $\le \sup_K |f_n - f| + 2\mu(K^c) \to 2\delta$. And (II) $\to 0$ because $f \in C_b(S)$ is a single fixed function and $\mu_n \Rightarrow \mu$. Therefore $\limsup_n |\mu_n(f_n) - \mu(f_n)| \le 4\delta$. Since $\delta > 0$ was arbitrary the $\limsup$ is 0, flatly contradicting $|\mu_n(f_n) - \mu(f_n)| \ge \varepsilon$. Hence $d_{\mathrm{BL}}(\mu_n, \mu) \to 0$.

Separability. We exhibit a countable dense subset of $\mathcal{P}(S)$. Fix a countable dense set $\{x_i\} \subseteq S$ (here separability of S is used), and let $\mathcal{D}$ be the collection of finitely-supported measures $\sum_i q_i \delta_{x_i}$ with rational weights $q_i \ge 0$ summing to 1; this set is countable. Given $\mu \in \mathcal{P}(S)$ and $\eta > 0$, single-measure tightness (lemma 3.15) supplies a compact K with $\mu(K^c) < \eta$. Cover K by finitely many balls $B(x_i, \eta)$ (possible by compactness), and disjointify them into Borel pieces $E_1, \dots, E_r$, each of diameter $< 2\eta$ and each containing a centre $x_{i_j} \in E_j$. The measure $\nu = \sum_j \mu(E_j)\delta_{x_{i_j}}$ then approximates μ: for any admissible f (so $\mathrm{Lip}(f) \le 1$ and $\|f\|_\infty \le 1$),

$$|\mu(f) - \nu(f)| \le \sum_j \int_{E_j} \left|f(x) - f(x_{i_j})\right| \,\mathrm{d}\mu(x) + \|f\|_\infty\, \mu(K^c) \le \sum_j \int_{E_j} d(x, x_{i_j})\,\mathrm{d}\mu + \eta \le 2\eta + \eta = 3\eta,$$

using $d(x, x_{i_j}) \le \mathrm{diam}(E_j) < 2\eta$ on each piece and bounding $\sum_j \mu(E_j) \le 1$. Hence $d_{\mathrm{BL}}(\mu, \nu) \le 3\eta$; rounding the weights $\mu(E_j)$ to nearby rationals (a further $O(\eta)$ error spread over finitely many atoms) lands the approximant in $\mathcal{D}$. Thus $\mathcal{D}$ is dense and $\mathcal{P}(S)$ is separable.

Completeness on Polish S. Let (μ_n) be d_{BL}-Cauchy. The strategy is to prove (μ_n) tight, extract a weak limit through Prokhorov, and then upgrade that subsequential weak limit to a full d_{BL}-limit. Granting tightness for the moment, Prokhorov (theorem 3.16(a)) gives a subsequence $\mu_{n_j} \Rightarrow \mu$, hence $d_{\mathrm{BL}}(\mu_{n_j}, \mu) \to 0$ by the metrization direction just proved; and a Cauchy sequence with a convergent subsequence converges to the same limit, so $d_{\mathrm{BL}}(\mu_n, \mu) \to 0$ with $\mu \in \mathcal{P}(S)$. It remains to prove the Cauchy sequence is tight, and this is the delicate construction the statement compressed.

Fix $\varepsilon > 0$; we work one length-scale at a time. At the scale set by a parameter $\eta > 0$, introduce the *bump function* sitting on the first k dense centres,

$$\psi_k(x) = \Big(1 - \eta^{-1} d\big(x, \textstyle\bigcup_{i\le k} \overline{B}(x_i, \eta)\big)\Big)^+,$$

which by the Urysohn computation of lemma 3.8 (applied to the closed set $\bigcup_{i\le k}\overline{B}(x_i,\eta)$ with slope η^{-1}) is bounded Lipschitz with $\|\psi_k\|_{\mathrm{BL}} \le 1+\eta^{-1}$, obeys the squeeze

$$\mathbf{1}_{\bigcup_{i\le k} B(x_i,\eta)} \;\le\; \psi_k \;\le\; \mathbf{1}_{\bigcup_{i\le k} B(x_i,2\eta)}$$

(equal to 1 on the inner union, where $d(x,\cdot)=0$, and to 0 outside the 2η-enlargement, since a point at distance $\ge \eta$ from every η-ball is at distance $\ge 2\eta$ from every centre), and increases to 1 as $k\to\infty$ because the centres are dense and so every x eventually lands within η of some x_i. Now use the Cauchy property to reduce to finitely many measures: choose N so large that $d_{\mathrm{BL}}(\mu_n,\mu_N) \le \varepsilon\eta/(2(\eta+1))$ for all $n\ge N$. By Ulam's tightness (lemma 3.15) each of the finitely many $\mu_1,\dots,\mu_N$ is individually tight, so we may pick k large enough that, for all $j\le N$ at once,

$$\mu_j(\psi_k) \;\ge\; \mu_j\Big(\bigcup_{i\le k} B(x_i,\eta)\Big) \;>\; 1-\varepsilon/2.$$

For the tail $n\ge N$ we transport this lower bound off μ_N using the dual bound $|\mu_n(\psi_k)-\mu_N(\psi_k)| \le \|\psi_k\|_{\mathrm{BL}}\, d_{\mathrm{BL}}(\mu_n,\mu_N)$ together with $\|\psi_k\|_{\mathrm{BL}} \le 1+\eta^{-1} = (\eta+1)/\eta$:

$$\mu_n(\psi_k) \;\ge\; \mu_N(\psi_k) - \frac{\eta+1}{\eta}\, d_{\mathrm{BL}}(\mu_n,\mu_N) \;>\; (1-\varepsilon/2) - \frac{\eta+1}{\eta}\cdot\frac{\varepsilon\eta}{2(\eta+1)} \;=\; 1-\varepsilon.$$

Combining with the upper squeeze $\psi_k \le \mathbf{1}_{\bigcup_{i\le k} B(x_i,2\eta)}$ gives $\mu_n\big(\bigcup_{i\le k} B(x_i,2\eta)\big) \ge \mu_n(\psi_k) > 1-\varepsilon$ for all $n\ge N$; the finitely many indices $n<N$ are absorbed by enlarging k if necessary (each μ_n is tight on its own). Writing $k=k(\eta,\varepsilon)$ for the resulting index, we have shown

$$\text{for every } \eta>0, \qquad \sup_n \mu_n\Big(\bigcup_{i\le k(\eta,\varepsilon)} B(x_i,2\eta)\Big) \;>\; 1-\varepsilon. \tag{3.7}$$

Finally, assemble the scales into one compact set. Apply (3.7) with $\eta = 2^{-m}$ and tolerance $\varepsilon\, 2^{-m}$ in place of ε, obtaining for each $m\ge 1$ a finite union $U_m = \bigcup_{i\le k_m}\overline{B}(x_i,2^{1-m})$ of closed balls with $\sup_n \mu_n(U_m^c) \le \varepsilon\, 2^{-m}$, and set

$$K = \bigcap_{m\ge 1} U_m.$$

Then K is closed, being an intersection of closed sets, and *totally bounded*: for any prescribed accuracy 2^{1-m} it is covered by the finitely many balls of radius 2^{1-m} comprising U_m. A closed, totally bounded subset of a *complete* metric space is compact — and this is the one place completeness of S (its Polishness) is spent, exactly as the theorem's parenthetical foretold. The mass outside K is controlled by summing the per-scale leaks:

$$\sup_n \mu_n(K^c) = \sup_n \mu_n\Big(\bigcup_{m\ge 1} U_m^c\Big) \le \sum_{m\ge 1}\sup_n \mu_n(U_m^c) \le \sum_{m\ge 1} \varepsilon\, 2^{-m} = \varepsilon.$$

Thus for every $\varepsilon>0$ there is a compact K with $\sup_n \mu_n(K^c)\le\varepsilon$: the Cauchy sequence (μ_n) is tight, and the proof is complete. □

The metric d_{BL} is not the only one inducing the weak topology; the Lévy–Prokhorov metric does too, and in chapter 4 the Wasserstein distance $\mathcal{W}_2$ refines all of them by additionally controlling second moments. What we will use repeatedly is only the qualitative content of theorem 3.12: "the topology of weak convergence" is a genuine separable metric topology (Polish when S is), so weak compactness coincides with sequential compactness and a weakly convergent sequence is pinned down by countably many test functions — the fact exploited in worked computation 3.26. This is exactly the foothold the existence proofs need: to produce a mean field equilibrium one builds approximate ones, shows the family tight, and extracts a convergent *subsequence*; metrizability is what makes "extract a convergent subsequence" a legitimate move.

There remains, however, a gap between what weak convergence asserts and what the limit machinery of chapter 1 is equipped to exploit, and naming it now sets up the next subsection precisely. Weak convergence $\mu_n \Rightarrow \mu$ is a statement about *laws*: it compares the measures through their integrals and says nothing whatever about how any underlying random variables behave pointwise — indeed the variables may live on unrelated probability spaces, or fail to be jointly defined at all, as the example $X_n = -X$ of section 3.2.1 made plain. Yet the most powerful tools we own for passing a limit through a nonlinear functional — dominated and monotone convergence, Fatou's lemma — are pointwise, almost-sure statements that presuppose a single common space on which $X_n(\omega) \to X(\omega)$. The natural question, and the one the next subsection answers, is whether the gap can be bridged: given only that the laws converge weakly, can we *realize* them as genuine random variables on one common probability space that converge almost surely? Skorokhod's representation theorem says yes — and, fittingly, its proof leans on exactly the completeness-and-tightness machinery just assembled, so the metrization built here is itself the springboard for what comes next.

3.2.3 Almost-sure realization: Skorokhod's representation theorem

The metrization of the previous subsection finally placed $\mathcal{P}(S)$ on the same footing as any other metric space: "a sequence of laws converges" now has a bona fide topological meaning, with subsequences, Cauchy sequences, and completeness all available. But that very success threw the limitation flagged at its close into relief. Weak convergence is a statement that lives *entirely on the measures*. It compares μ_n and μ through the readings $\mu_n(f) \to \mu(f)$ and is indifferent to whether the underlying random variables share a probability space, or are jointly defined at all — recall the toy sequence $X_n = -X$ of section 3.2.1, whose laws converge while the variables themselves do nothing of the kind. The trouble is that the sharpest instruments we own for pushing a limit through a nonlinear functional — dominated convergence, monotone convergence, and Fatou's lemma — are *pointwise* theorems. Each presupposes a single sample space on which $X_n(\omega) \to X(\omega)$ for almost every ω, precisely the structure weak convergence declines to provide. So the natural question, already posed at the end of section 3.2.2, is whether the gap can be bridged from the measure side: given only that $\mu_n \Rightarrow \mu$, can we *manufacture* random variables with those laws, on one common space, that converge almost surely? Skorokhod's representation theorem answers yes, and the construction is wholly explicit.

It is worth dwelling on *why* one wants this before building it, because the motivation dictates exactly the form of the conclusion. The recurring need is to move an inequality *upward* in strength: from an inequality that is trivial for almost-surely convergent variables to the same inequality at the level of weak convergence, where it is not trivial at all. The cleanest example, and the one chapter 4 consumes verbatim, is lower semicontinuity of a moment or a cost. Take a continuous, nonnegative

$g \geq 0$ on S — think of $g(x) = |x|^p$, an energy — and ask whether the functional $\mu \mapsto \mu(g) = \int_S g \, \mathrm{d}\mu$ is lower semicontinuous along weak convergence, i.e. whether

$$\mu_n \Rightarrow \mu \quad \Longrightarrow \quad \int_S g \, \mathrm{d}\mu \, \leq \, \liminf_{n\to\infty} \int_S g \, \mathrm{d}\mu_n. \tag{3.8}$$

For an unbounded g this is not handed to us by the definition of weak convergence, which tests only against *bounded* continuous functions, so g is not a legal test function. But if we had random variables $Y_n \to Y$ almost surely with $\mathcal{L}(Y_n) = \mu_n$ and $\mathcal{L}(Y) = \mu$, the inequality would be immediate: $g \geq 0$ and continuity give $g(Y_n) \to g(Y)$ almost surely, and Fatou's lemma (chapter 1) applied to the nonnegative sequence $g(Y_n)$ yields $\mathbb{E}[g(Y)] \leq \liminf_n \mathbb{E}[g(Y_n)]$, which by the change-of-variables formula (3.1) is exactly (3.8). The hard, measure-level statement collapses to a one-line application of a pointwise theorem — *once the variables are realized on a common space.* Skorokhod's theorem is the device that performs that realization, and (3.8) is the template: weak convergence in, an a.s.-convergent coupling out, the desired inequality recovered by Fatou or dominated convergence. Lower semicontinuity of the cost in an existence proof, and completeness of the Wasserstein metric in chapter 4, are both instances of this single move.

How could such a coupling possibly be built? The blueprint is already familiar from the real line, and the general theorem is its faithful generalization. On $S = \mathbb{R}$, if the distribution functions satisfy $F_n \to F$ at every continuity point of F — which is what weak convergence amounts to there — one fixes a *single* uniform random variable U on $[0, 1)$ and sets $Y_n = F_n^{-1}(U)$, $Y = F^{-1}(U)$ through the quantile (generalized-inverse) transform. Each Y_n then has law μ_n by construction, all of them are driven by the *same* U, and one checks that $F_n^{-1}(u) \to F^{-1}(u)$ at every continuity point of F^{-1}, i.e. for Lebesgue-almost every u, so $Y_n \to Y$ almost surely. The common randomness U is what couples the laws; the monotone inversion is what turns convergence of the F_n into pointwise convergence of the Y_n. On a general Polish space there is no order to invert, but there is a substitute: a sequence of ever-finer *partitions* into sets whose boundaries the limit measure does not charge, coded consistently into nested subintervals of $[0, 1)$. Reading off which nested intervals a point $\omega \in [0, 1)$ falls through plays the role of $F^{-1}(\omega)$, and the boundary-free (continuity-set) property of the partitions is what lets Portmanteau (v) transfer the convergence. That is the entire idea; the proof is its careful execution, and its only inputs are Portmanteau (v), the Borel–Cantelli lemma, and the completeness and separability of S.

Theorem 3.13 (Skorokhod representation)**.** *Let S be a Polish space and let $\mu_n \Rightarrow \mu$ in $\mathcal{P}(S)$. Then on the probability space $([0, 1), \mathcal{B}([0, 1)), \lambda)$, with λ Lebesgue measure, there exist S-valued random variables Y_n $(n \geq 1)$ and Y with*

$$\mathcal{L}(Y_n) = \mu_n, \qquad \mathcal{L}(Y) = \mu, \qquad Y_n \xrightarrow{a.s.} Y.$$

Proof. Step 0 (the sole distributional input: Portmanteau (v)). Call $B \in \mathcal{B}(S)$ a *μ-continuity set* if its topological boundary is μ-null, $\mu(\partial B) = 0$. These are exactly the sets on which weak convergence is guaranteed to act numerically: by the continuity-set form Portmanteau (v) (theorem 3.9), $\mu_n(B) \to \mu(B)$ for every μ-continuity set B. This single implication is the *only* fact about $\mu_n \Rightarrow \mu$ that the construction will use; everything else is bookkeeping on $[0, 1)$. Two closure properties we lean on repeatedly: a finite intersection or union of μ-continuity sets is again one, because $\partial(A \cap B)$

and $\partial(A \cup B)$ are both contained in $\partial A \cup \partial B$ (a boundary point of the intersection or union is a boundary point of one of the pieces), and a μ-null set has μ-null subsets; likewise a *set difference* $A \setminus B = A \cap B^c$ is a continuity set, since $\partial B^c = \partial B$. We refer back to this paragraph as "Step 0" throughout.

Step 1 (nested continuity-set partitions of small mesh). The geometric scaffold is a sequence of countable Borel partitions $\mathcal{Q}_m = \{A_{m,1}, A_{m,2}, \dots\}$ of S, $m \geq 1$, with three properties: every cell is a μ-continuity set, every cell has diameter $\leq 1/m$, and $\mathcal{Q}_{m+1}$ *refines* $\mathcal{Q}_m$ (each cell of $\mathcal{Q}_{m+1}$ sits inside a single cell of $\mathcal{Q}_m$). The continuity-set property is what makes Step 0 applicable cell by cell; the shrinking diameter is what will pin down a limit point; the nesting is what will make the codings at successive levels consistent.

Fix m and build a partition $\mathcal{Q}'_m$ of mesh $\leq 1/m$ first. Here is where *separability* is spent: a separable metric space is covered by countably many open balls of any prescribed radius, so fix a countable dense set and cover S by balls centred at its points with radii in $(1/(2m), 1/m)$. The radius is not yet determined — we must still arrange that each ball is a continuity set — and this is where the disjoint-spheres argument used earlier in the chapter (for δ_c, in the constant-limit converse of lemma 3.6) reappears in full generality. For a fixed centre x, the spheres $\Sigma_r = \{y : d(y,x) = r\}$ over distinct radii $r > 0$ are pairwise disjoint, since a point has exactly one distance from x. A finite measure can assign positive mass to at most countably many members of a disjoint family — if uncountably many Σ_r each carried mass exceeding $1/k$, their disjoint union would have infinite mass — so all but countably many radii r give $\mu(\Sigma_r) = 0$. The boundary of the open ball $B(x,r)$ lies in Σ_r, so for all but countably many r the ball $B(x,r)$ is a μ-continuity set; in particular we may pick, for each centre, a radius in the nonempty interval $(1/(2m), 1/m)$ that works. Enumerate the chosen continuity-set balls $U_1, U_2, \dots$ and disjointify in the standard way,

$$A'_{m,i} = U_i \setminus (U_1 \cup \dots \cup U_{i-1}),$$

so the $A'_{m,i}$ are pairwise disjoint and cover S, hence partition it. Each is a continuity set, being a finite set difference of continuity sets (Step 0), and each has diameter $\leq 1/m$ because it sits inside the ball U_i of radius $< 1/m$. This is the partition $\mathcal{Q}'_m$.

The partitions $\mathcal{Q}'_m$ have small mesh but need not be nested across m. We force nesting by taking common refinements: let $\mathcal{Q}_m$ be the partition whose cells are the nonempty intersections $A'_{1,i_1} \cap \dots \cap A'_{m,i_m}$, one factor drawn from each $\mathcal{Q}'_1, \dots, \mathcal{Q}'_m$. Each such cell is a *finite* intersection of continuity sets, hence a continuity set by Step 0; it has diameter $\leq 1/m$ because one of its factors comes from $\mathcal{Q}'_m$; and by construction $\mathcal{Q}_m$ refines $\mathcal{Q}_{m-1}$, since dropping the last factor A'_{m,i_m} sends a $\mathcal{Q}_m$-cell into the $\mathcal{Q}_{m-1}$-cell with the same first $m-1$ factors. The three required properties now hold simultaneously.

Step 2 (consistent interval coding: the CDF-inversion analogue). We now turn each measure into a map $[0,1) \to S$ by stacking its cell masses into nested subintervals — the multidimensional replacement for inverting a distribution function. Order the cells of every $\mathcal{Q}_m$ so that the children of any one $\mathcal{Q}_{m-1}$-cell are listed *contiguously*, in one consecutive block; this contiguous ordering is the linchpin that will make the interval coding respect the nesting. For a probability measure $\rho \in \mathcal{P}(S)$, assign to the j-th cell $A_{m,j}$ the half-open "stacking" interval

$$I^\rho_{m,j} = \Big[\textstyle\sum_{i<j} \rho(A_{m,i}),\ \sum_{i\leq j} \rho(A_{m,i})\Big) \subseteq [0,1),$$

of length exactly $\rho(A_{m,j})$. At a fixed level m these intervals are disjoint and tile $[0,1)$ (up to the single point 1, which has λ-measure zero), because the cell masses sum to $\rho(S)=1$. The key consistency is that they nest *across* levels. Suppose $A_{m,j} \subseteq A_{m-1,\ell}$. Because the children of $A_{m-1,\ell}$ are listed contiguously and their masses sum to the parent's mass, $\sum_{i:\, A_{m,i}\subseteq A_{m-1,\ell}} \rho(A_{m,i}) = \rho(A_{m-1,\ell})$, the block of level-$m$ intervals belonging to the children of $A_{m-1,\ell}$ is exactly the parent interval $I^{\rho}_{m-1,\ell}$ resubdivided; hence $I^{\rho}_{m,j} \subseteq I^{\rho}_{m-1,\ell}$. Were the children listed non-contiguously, the child intervals would be scattered across $[0,1)$ and this containment would fail — which is the whole reason the ordering was imposed.

Now read off a point of S from a point $\omega \in [0,1)$. For each m there is a unique cell index $j_m(\rho,\omega)$ with $\omega \in I^{\rho}_{m,j_m}$, and by the nesting just proved the cells $A_{m,j_m(\rho,\omega)}$ *descend*: each is contained in its predecessor. This is precisely a nested sequence of sets whose diameters tend to zero. Pick any representative $y_m \in A_{m,j_m(\rho,\omega)}$ from each. For $m' \geq m$ the cell at level m' lies inside the cell at level m, so y_m and $y_{m'}$ both lie in a set of diameter $\leq 1/m$, giving $d(y_m, y_{m'}) \leq 1/m \to 0$: the sequence (y_m) is *Cauchy.* Here is where *completeness* of S is spent — a Cauchy sequence in a Polish space converges — and the limit is independent of which representatives were chosen, since two competing choices $y_m, \tilde{y}_m$ at level m also lie in a common cell of diameter $\leq 1/m$, so their limits coincide. Define $R_\rho(\omega)$ to be this common limit. The map $R_\rho : [0,1) \to S$ is Borel, being the pointwise limit of the cell-valued step maps $\omega \mapsto y_m$. Finally R_ρ has law ρ: for a cell $A_{m,j}$, the set $\{R_\rho \in A_{m,j}\}$ agrees with $I^{\rho}_{m,j}$ up to the λ-null overlap of adjacent interval endpoints, so $\lambda(R_\rho \in A_{m,j}) = \lambda(I^{\rho}_{m,j}) = \rho(A_{m,j})$; the cells of all the $\mathcal{Q}_m$ generate $\mathcal{B}(S)$ (their diameters shrink to zero, so they separate points and generate the Borel sets), whence $\mathcal{L}(R_\rho) = \rho$ by the uniqueness lemma for measures. Apply this to each law in sight: set $Y = R_\mu$ and $Y_n = R_{\mu_n}$ for every n, all defined on the *single* space $[0,1)$ under λ, with $\mathcal{L}(Y) = \mu$ and $\mathcal{L}(Y_n) = \mu_n$ as required. The common variable ω is the coupling — the Polish-space stand-in for the shared uniform U of the one-dimensional quantile construction.

Step 3 (almost-sure convergence via Borel–Cantelli). It remains to show $Y_n \to Y$ almost surely, and here is where the convergence $\mu_n \to \mu$ on continuity sets (Step 0) finally enters. Fix a level m. Since the cell masses $\mu(A_{m,j})$ sum to 1, we may choose a cutoff K_m so that the first K_m cells already carry most of the mass: $\mu(E_m) > 1 - 2^{-m}$, where $E_m = \bigcup_{j \leq K_m} A_{m,j}$ is a finite union of continuity sets. Consider the level-m interval endpoints for the laws μ_n. The j-th endpoint is the partial sum $\sum_{i<j} \mu_n(A_{m,i})$, a *finite* sum (at most K_m terms for $j \leq K_m + 1$) of masses of continuity sets. By Step 0 each term $\mu_n(A_{m,i}) \to \mu(A_{m,i})$, and a finite sum of convergent sequences converges to the sum of the limits, so every such endpoint of $I^{\mu_n}_{m,j}$ converges, as $n \to \infty$, to the corresponding endpoint of $I^{\mu}_{m,j}$. This is exactly why we restricted to the first K_m cells: it keeps the relevant partial sums finite, so that Step 0 applies term by term without any uniformity in j.

Convergence of the endpoints converts into pointwise agreement of the codings on the interiors of the limit intervals. Fix $j \leq K_m$ and take any ω in the *open* interval $\operatorname{int} I^{\mu}_{m,j}$. Because the endpoints of $I^{\mu_n}_{m,j}$ converge to those of $I^{\mu}_{m,j}$ and ω sits strictly inside the limit interval, ω falls inside $I^{\mu_n}_{m,j}$ for all sufficiently large n. For those n, the level-m cell index of ω is the same j for both μ_n and μ, so $Y_n(\omega)$ and $Y(\omega)$ are both limits of representatives drawn from (nested subcells of) the single cell $A_{m,j}$, a set of diameter $\leq 1/m$; hence $d(Y_n(\omega), Y(\omega)) \leq 1/m$ for all large n. Collect the good points at level m:

$$G_m = \bigcup_{j \le K_m} \operatorname{int} I^{\mu}_{m,j}, \qquad \lambda(G_m) = \sum_{j \le K_m} \mu(A_{m,j}) = \mu(E_m) > 1 - 2^{-m},$$

the middle equality because the intervals are disjoint with lengths $\mu(A_{m,j})$. Thus $\lambda(G_m^c) < 2^{-m}$, and these bad-set measures are summable: $\sum_m \lambda(G_m^c) < \sum_m 2^{-m} = 1 < \infty$. The first Borel–Cantelli lemma (theorem 3.27(i)) — which is stated and proved later in this chapter, by an elementary argument that nowhere invokes the present theorem, so the forward reference carries no circularity — then says that summability of the $\lambda(G_m^c)$ forces $\lambda(\, G_m^c \text{ for infinitely many } m\,) = 0$. Equivalently, almost every $\omega \in [0,1)$ lies in G_m for all sufficiently large m. For such an ω and each large m we showed $\limsup_n d(Y_n(\omega), Y(\omega)) \le 1/m$; letting $m \to \infty$ drives the lim sup to 0, so $Y_n(\omega) \to Y(\omega)$. Hence $Y_n \to Y$ λ-almost surely, completing the proof. □

Step back and notice how economical the inputs were. The distributional hypothesis $\mu_n \Rightarrow \mu$ entered *only* through Step 0 — the continuity-set form Portmanteau (v), used to push masses of finitely many cells to the limit; the limit point of the codings was produced by *completeness* of S closing up a Cauchy sequence; the countable scaffolding of small-mesh partitions came from *separability*; and the upgrade from “most ω are good at each level” to “almost every ω is eventually always good” was a single application of the *Borel–Cantelli lemma* proved in section 3.5.1. No further machinery — no metrization, no compactness, no abstract existence principle — was needed, so the theorem is fully self-contained within this chapter, exactly as the chapter’s self-containedness pledge demands. That same frugality is why the metrization argument of theorem 3.12 could lean on this result without circularity: Skorokhod’s representation rests on Portmanteau and Borel–Cantelli alone.

With Skorokhod’s theorem in hand the bridge between the two worlds is complete: weak convergence of laws can always be *realized* as almost-sure convergence of suitably coupled variables, so every pointwise limit theorem of chapter 1 transfers to the weak-convergence setting through the template (3.8), and chapter 4 will invoke precisely this to prove completeness of the Wasserstein space and to push lower-semicontinuity and Fatou-type inequalities up to the level of measures. But notice what the theorem takes for granted: it begins with a sequence that *already converges weakly*. It realizes a limit that is handed to it; it does not produce one. The existence proofs of the mean field theory face the opposite situation. There one starts with a sequence of approximate equilibria — approximate population measures — with *no* limit given in advance, and the entire task is to extract one. That is a *compactness* question, and weak convergence has so far supplied no compactness at all: as the toy escape $\mu_n = \delta_n$ already warned in section 3.2.2, a sequence of probability measures can simply leak its mass off to infinity and converge to nothing in $\mathcal{P}(S)$. The next section identifies the condition that forbids this leakage — tightness — and proves, in Prokhorov’s theorem, that it is the *only* obstruction to extracting a weakly convergent subsequence, turning the realization device of this section into the manufacturing device the existence proofs require.

3.3 Tightness and Prokhorov's theorem

Skorokhod's theorem closed the previous section by completing a bridge in one direction: a sequence of laws that *already converges weakly* can be realized on a single probability space as variables that converge almost surely, so the pointwise machinery of chapter 1 transfers to the world of measures. But, as we were careful to note there, that device takes its convergent sequence as a gift; it realizes a limit, it does not manufacture one. The existence proofs of the mean field theory stand on the other side of exactly this gap. There one is handed a sequence of approximate equilibria — approximate population measures produced by some iteration or relaxation — with no limit given in advance, and the whole problem is to *produce* a limit to which the candidate could converge. That is a question of *compactness*, and weak convergence, for all that the Portmanteau theorem and the metric of section 3.2.2 did for it, has so far supplied none. Worse, it cannot supply it unconditionally. The toy escape $\mu_n = \delta_n$ flagged back in section 3.2.2 already shows why: a sequence of probability measures can simply let its unit of mass slide off to infinity and converge to nothing in $\mathcal{P}(S)$. Concretely, for any $f \in C_b(\mathbb{R}^d)$ that has a limit $f(\infty)$ at infinity one finds $\mu_n(f) = f(n) \to f(\infty)$, so the integral readings *do* settle down — but they settle on the functional $f \mapsto f(\infty)$, which is not $\mu(f)$ for any probability measure μ; the limiting "measure" would have to place its entire mass at the point at infinity, which $\mathbb{R}^d$ does not contain. The mass is not redistributed, it is lost, and no $\mu \in \mathcal{P}(\mathbb{R}^d)$ catches it.

This section identifies the one condition that forbids the leak — *tightness* — and proves, in Prokhorov's theorem, that ruling out the leak is the *only* thing standing between a sequence of measures and a convergent subsequence. The payoff is precisely the manufacturing device the existence proofs need: an a priori estimate, of exactly the moment-bound type a physicist produces by an energy argument, certifies tightness, and tightness is then converted by Prokhorov into a weakly convergent subsequence whose limit is the equilibrium. The realization device of the previous section and the compactness theory of this one are the two halves of how weak convergence is actually used downstream: one turns a limit into almost-sure statements, the other conjures the limit in the first place.

Definition 3.14 (Tightness)**.** A family $\Pi \subseteq \mathcal{P}(S)$ is *tight* if for every $\varepsilon > 0$ there is a compact set $K_\varepsilon \subseteq S$ with

$$\mu(K_\varepsilon) \;\geq\; 1 - \varepsilon \qquad \text{for every } \mu \in \Pi.$$

A family is *relatively compact* (in the weak topology) if every sequence in Π has a subsequence converging weakly to some element of $\mathcal{P}(S)$.

Read the definition slowly, because every word of it is load-bearing. The word "compact" is what forbids the escape to infinity: a compact set in $\mathbb{R}^d$ is bounded, so demanding that μ place at least $1-\varepsilon$ of its mass on K_ε is demanding that at most ε of it sit beyond a fixed horizon. The word that does the real work, though, is the quantifier "for every $\mu \in \Pi$": the *same* compact K_ε must absorb $1-\varepsilon$ of the mass of *every* member of the family at once. It is this uniformity, not the inner-mass condition on any individual measure, that turns into compactness — indeed, by Ulam's theorem below, every *single* probability measure already satisfies the inner-mass condition, so a family that failed to be tight would do so only because no single compact works for all of its members simultaneously, the horizon needed to corral the mass receding without bound as one runs through the family. That is exactly the pathology of $\mu_n = \delta_n$: each δ_n is by itself perfectly concentrated, but the point n at which it concentrates marches off to infinity, so no fixed compact catches all of them.

On $\mathbb{R}^d$ this becomes wholly concrete, because there the compact sets are exactly the closed bounded sets (Heine–Borel), and a closed ball $\overline{B}(0,R)$ can be made to contain any prescribed compact by taking R large. Tightness of Π is therefore equivalent to the statement that the mass living *outside* large balls goes to zero *uniformly* over the family,

$$\sup_{\mu\in\Pi} \mu\big(\{|x|>R\}\big) \;\longrightarrow\; 0 \qquad \text{as } R\to\infty,$$

which is the form we actually check. (One direction: given ε, pick R with the supremum below ε and take $K_\varepsilon = \overline{B}(0,R)$. The other: given the compacts K_ε, enclose each in a ball.) This is the shape every verification of tightness in the book will take — show that no fixed fraction of the population can wander arbitrarily far out — and there is one sufficient condition for it that recurs so often it deserves to be isolated, because it is precisely what an energy estimate delivers. Suppose the family obeys a *uniform moment bound*: for some exponent $p>0$,

$$\sup_{\mu\in\Pi} \int_{\mathbb{R}^d} |x|^p \,\mathrm{d}\mu \;:=\; M \;<\; \infty,$$

so that no member has too much mass far from the origin, with the bound M holding across the whole family. Then Markov's inequality — the same one-line device $\mathbb{P}(|Y|>R) \le R^{-p}\mathbb{E}[|Y|^p]$ that powered the L^p arrow of lemma 3.6, applied here to the identity $Y=|x|$ under μ — gives, for every $\mu\in\Pi$ at once,

$$\mu(\{|x|>R\}) \;=\; \int_{\{|x|>R\}} \mathrm{d}\mu \;\le\; \int_{\{|x|>R\}} \frac{|x|^p}{R^p}\,\mathrm{d}\mu \;\le\; \frac{1}{R^p}\int_{\mathbb{R}^d} |x|^p \,\mathrm{d}\mu \;\le\; \frac{M}{R^p},$$

the first inequality because $|x|^p/R^p \ge 1$ on the region $\{|x|>R\}$. The right-hand side M/R^p does not depend on μ and tends to 0 as $R\to\infty$, so the supremum over Π does too, and Π is tight. This implication — "a single uniform p-th moment bound $\Rightarrow$ tightness" — is the workhorse that converts the output of a mean field existence argument into compactness: one proves, by an a priori energy or moment estimate on the population (an estimate uniform in the approximation parameter), that $\sup\int |x|^p\,\mathrm{d}\mu \le M$, and tightness follows mechanically. Note how little is asked: *any* fixed positive moment, bounded *uniformly*, suffices; one does not need a sharp constant, only finiteness that does not blow up across the family.

There is one preliminary fact to settle before the family statement, and it is the reassurance that the inner-mass half of the tightness condition is never itself the obstruction. On a Polish space *a single* probability measure is automatically tight: it always concentrates $1-\varepsilon$ of its mass on some compact, however spread out it may look. This is Ulam's theorem. We need it for two distinct reasons, and it is worth flagging both. First, it is the inner-regularity fact that keeps the limit measure produced in Prokhorov's theorem honest — without it one could not be sure the constructed limit is genuinely a probability measure rather than a defective content that has quietly shed mass. Second, it is one of the results that the metrization theorem of the previous section borrowed forward: the completeness half of theorem 3.12 invoked Ulam (and Prokhorov direction (b)) on credit, promising they would be proved here by arguments that never refer back to metrization. We are now discharging that promise, and the reader can verify the non-circularity directly: the proof below uses only a countable dense set and continuity from below, nothing about the topology of $\mathcal{P}(S)$.

Lemma 3.15 (Ulam: tightness of a single measure)**.** *Let S be Polish and $\mu \in \mathcal{P}(S)$. Then μ is tight: for every $\varepsilon > 0$ there is a compact $K \subseteq S$ with $\mu(K) \geq 1 - \varepsilon$. Consequently every finite subset of $\mathcal{P}(S)$ is a tight family.*

Proof. The idea is to approximate S from inside, at every resolution scale simultaneously, by a finite union of small balls, and then intersect those finite unions across all scales to produce a set that is closed and totally bounded — the two properties that, on a complete space, force compactness. Polishness enters twice and we mark each entrance: separability supplies the countable dense set that makes "finitely many balls" meaningful, and completeness is what upgrades total boundedness to compactness at the end.

Fix a countable dense set $\{x_i\}_{i\geq 1} \subseteq S$ — it exists because S is separable — and fix $\varepsilon > 0$. Work one resolution scale $1/m$ at a time. The open balls $B(x_i, 1/m)$, $i \geq 1$, cover S, since every point lies within $1/m$ of some x_i by density. Their finite unions $\bigcup_{i\leq k} B(x_i, 1/m)$ increase to all of S as $k \to \infty$, so by continuity from below of the measure μ (chapter 1; this is the import we lean on) their μ-masses increase to $\mu(S) = 1$. Hence we may choose an index k_m large enough that the truncated cover already captures all but an m-dependent sliver of the mass:

$$\mu\Big(\bigcup_{i\leq k_m} B(x_i, 1/m)\Big) \;>\; 1 - \varepsilon\, 2^{-m}.$$

The budget $\varepsilon\, 2^{-m}$ is allotted to scale m precisely so that the slivers across all scales sum to ε, by the geometric series $\sum_m 2^{-m} = 1$; this is the standard device for spending a finite error budget over countably many constraints.

Now close up and intersect. Set $D_m = \bigcup_{i\leq k_m} \overline{B}(x_i, 1/m)$, the union of the corresponding *closed* balls (which only enlarges the set, so the mass bound is inherited: $\mu(D_m) \geq \mu(\bigcup_{i\leq k_m} B(x_i, 1/m)) > 1 - \varepsilon 2^{-m}$), and let $K = \bigcap_{m\geq 1} D_m$. We check K is compact. It is closed, as an intersection of closed sets. It is totally bounded: given any $\eta > 0$, pick m with $1/m < \eta$; then $K \subseteq D_m$ is covered by the finitely many balls $\overline{B}(x_i, 1/m)$, $i \leq k_m$, each of radius $< \eta$, which is exactly the definition of total boundedness. A closed and totally bounded subset of a *complete* metric space is compact — this is where completeness is spent — so K is compact. Finally, the mass left out is controlled by summing the per-scale slivers, using countable subadditivity on the complement:

$$\mu(K^c) = \mu\Big(\bigcup_{m\geq 1} D_m^c\Big) \leq \sum_{m\geq 1} \mu(D_m^c) \leq \sum_{m\geq 1} \varepsilon\, 2^{-m} = \varepsilon,$$

so $\mu(K) \geq 1 - \varepsilon$, as claimed. The final sentence is the bridge to families: a finite collection $\{\mu_1, \ldots, \mu_r\}$ is tight because, taking the compact furnished for each μ_j at level ε and forming their union, one gets a single compact set (a finite union of compacts is compact) that carries $1 - \varepsilon$ of every μ_j at once. It is only when the family is *infinite* that this union may fail to stay compact, and that failure is exactly the content tightness is designed to rule out. $\square$

Theorem 3.16 (Prokhorov). *Let S be a metric space and $\Pi \subseteq \mathcal{P}(S)$.*

(a) *If Π is tight, then Π is relatively compact.*

(b) *If S is Polish and Π is relatively compact, then Π is tight.*

In particular, on a Polish space tightness and relative compactness are equivalent.

The two directions are not of equal importance to us, and it helps to say which one does the work. Direction (a) — tight $\Rightarrow$ relatively compact — is the engine: it is the implication that takes an a priori estimate, repackaged as tightness, and *produces* a weakly convergent subsequence whose limit can then be identified as the equilibrium. Direction (b) — relatively compact $\Rightarrow$ tight, valid on Polish spaces — is the converse that certifies tightness is not merely *a* sufficient condition for compactness but the *exact* one, nothing weaker will do; it is also, as just recalled, the result the metrization proof borrowed forward, so proving it here closes that loop. We prove both in full for S Polish, which is the only case the book ever needs (a general metric space reduces to it by restricting to the closure of a separable support, on which the measures live). The reader should be warned in advance that the proof of (a) is the single longest and most tightly compressed argument in the chapter, and the reason is structural: weak convergence is defined by testing against continuous functions, but a *limit* measure is not handed to us as a functional — it must be *built*, set by set, as an honest Borel measure, before there is anything for the continuous tests to converge to. The construction is the classical one: from the compacts supplied by tightness we manufacture an inner-regular *content* γ, promote it to a metric outer measure, and invoke the Carathéodory extension theorem of chapter 1 to crystallize a genuine Borel probability measure, which a final Portmanteau check identifies as the weak limit. Each of these moves is a step in the proof below, and we expand the terse justifications — the compactness-driven subadditivity, the subordinated finite cover, the separation additivity — that the skeleton draft compressed to a line apiece.

Proof of (a), *S Polish.* The strategy is to read off, from the sequence, the *values* a limiting measure would have to take on a rich-but-countable supply of compact sets; to extract a subsequence along which all those values converge (a diagonal trick, legitimate only because the supply is countable); and then to grow those values into a genuine Borel measure by the inner-content/outer-measure construction. Tightness enters once, at the start, to confine the action to a σ-compact part of S where the construction has compacts to feed on.

Let (μ_n) be a sequence in the tight family Π; we must find a weakly convergent subsequence. By tightness, for each m there is a compact carrying $1 - 1/m$ of every measure's mass, and replacing each by the union of the first m of them we may take the compacts increasing, $K_1 \subseteq K_2 \subseteq \cdots$, with $\mu_n(K_m) \geq 1 - 1/m$ for all n simultaneously (the uniformity in n is exactly what tightness provides). Their union $K = \bigcup_m K_m$ is σ-compact, hence separable; fix a countable dense set $D \subseteq K$. Everything from here lives on K, where mass 1 has been corralled in the limit.

Step 1 (a countable family of compacts, and diagonal extraction). We need a countable supply of compact sets, closed under finite unions enough to do combinatorics, on which to read the limiting values. Let $\mathcal{K}$ be the collection of all sets of the form $C = F \cap K_m$, $m \geq 1$, where F ranges over finite unions of *closed* balls $\overline{B}(d,q)$ with centre $d \in D$ and rational radius $q \in \mathbb{Q}_{>0}$. There are countably many such F (finite subsets of the countable index set $D \times \mathbb{Q}_{>0}$) and countably many m, so $\mathcal{K}$ is countable. Each $C = F \cap K_m$ is a closed subset of the compact K_m, hence itself compact. Now the diagonal extraction: each real sequence $\big(\mu_n(C)\big)_n$ lives in the compact interval $[0,1]$, so it has a convergent subsequence; because $\mathcal{K}$ is countable we may enumerate it as $C^{(1)}, C^{(2)}, \ldots$, extract a subsequence along which $\mu_n(C^{(1)})$ converges, a further subsequence for $C^{(2)}$, and so on, and take the diagonal sequence, which from its j-th term on is a subsequence of the j-th extraction and hence makes $\mu_n(C^{(j)})$ converge for *every* j at once. (This is the one place countability of $\mathcal{K}$ is indispensable: an uncountable family of conditions cannot be met by a single subsequence this way.) Relabel this diagonal subsequence again as (μ_n). Along it,

$$\gamma(C) := \lim_n \mu_n(C) \in [0,1] \qquad \text{exists for every } C \in \mathcal{K}.$$

The limit γ inherits from the μ_n exactly the two algebraic properties we will use. It is *monotone*: if $C \subseteq C'$ then $\mu_n(C) \le \mu_n(C')$ for every n, and the inequality passes to the limit. It is *finitely subadditive*: $\mu_n(C \cup C') \le \mu_n(C) + \mu_n(C')$ for every n (a property of any measure), so $\gamma(C \cup C') \le \gamma(C) + \gamma(C')$ in the limit. And it retains the mass $\gamma(K_m) = \lim_n \mu_n(K_m) \ge 1 - 1/m$. These three facts are all we know about γ, and the rest of the proof manufactures a measure out of them.

Step 2 (inner content and outer measure). A content lives on compacts; a measure must live on all Borel sets. We bridge the gap in the standard two stages, working outward from $\mathcal{K}$ to open sets and then inward to arbitrary sets. For an open $U \subseteq S$ define the *inner content*

$$\tau(U) = \sup\{\, \gamma(C) : C \in \mathcal{K},\ C \subseteq U \,\},$$

the most mass our compacts can certify inside U, and then for an arbitrary $A \subseteq S$ the *outer value*

$$\mu^*(A) = \inf\{\, \tau(U) : U \supseteq A \text{ open} \,\},$$

the least open content that can be wrapped around A. We claim μ^* is a *metric outer measure*, meaning: it is monotone with $\mu^*(\emptyset) = 0$; it is countably subadditive; and it is *additive across positively separated sets*, $\mu^*(A \sqcup B) = \mu^*(A) + \mu^*(B)$ whenever $d(A,B) > 0$. Monotonicity and $\mu^*(\emptyset) = 0$ are immediate from the definitions (a larger A admits fewer enclosing U, raising the infimum; the empty set is enclosed by ever-smaller opens containing no C of positive content). The two substantive claims are where the geometry of S is used, and we give each in full.

Countable subadditivity, and why compactness is exactly what makes it work. This is the only step that touches compactness, and the reason is that subadditivity must defeat an *infinite* cover using only the *finite* subadditivity of γ — compactness is precisely the tool that reduces the infinite cover to a finite one. Let $U = \bigcup_k U_k$ be a countable union of open sets, and let $C \in \mathcal{K}$ with $C \subseteq U$ be any compact competing in the supremum defining $\tau(U)$. Since C is compact and $\{U_k\}$ is an open cover of it, *finitely many* of them, say $U_1, \ldots, U_J$, already cover C. We now subordinate C to this finite cover: we write C as a finite union $C = \bigcup_{k \le J} C_k$ of members of $\mathcal{K}$ with $C_k \subseteq U_k$. To build the C_k, observe that each point $x \in C$ lies in some U_k, and since U_k is open and D is dense there is a closed ball $\overline{B}(d,q)$ with $d \in D$, $q \in \mathbb{Q}_{>0}$, small enough that $x \in \overline{B}(d,q) \subseteq U_k$; the interiors $B(d,q)$ of such balls form an open cover of the compact C, so finitely many suffice, and grouping the chosen closed balls according to which U_k contains them and intersecting with the relevant K_m

yields sets $C_k = F_k \cap K_m \in \mathcal{K}$ with $C_k \subseteq U_k$ and $C \subseteq \bigcup_{k \le J} C_k$ (we may intersect each C_k further with C to keep it inside C, still a member of $\mathcal{K}$ since C itself is). Finite subadditivity of γ from Step 1, applied to this finite decomposition, gives

$$\gamma(C) \le \sum_{k \le J} \gamma(C_k) \le \sum_{k \le J} \tau(U_k) \le \sum_k \tau(U_k),$$

where the middle inequality is just $\gamma(C_k) \le \tau(U_k)$ from the definition of the inner content (each $C_k \subseteq U_k$ is one of the compacts competing in $\tau(U_k)$). The right-hand side no longer mentions C, so taking the supremum over all admissible C gives $\tau(U) \le \sum_k \tau(U_k)$. Finally, for arbitrary sets A_k with $A = \bigcup_k A_k$, enclose each A_k in an open U_k with $\tau(U_k) \le \mu^*(A_k) + \eta\, 2^{-k}$; then $U = \bigcup_k U_k \supseteq A$ is open and $\mu^*(A) \le \tau(U) \le \sum_k \tau(U_k) \le \sum_k \mu^*(A_k) + \eta$, and letting $\eta \downarrow 0$ yields $\mu^*(\bigcup_k A_k) \le \sum_k \mu^*(A_k)$.

Additivity across separated sets, via a separating pair of opens. Suppose $d(A, B) = \rho > 0$; we must show μ^* does not over-count their union. Subadditivity already gives $\mu^*(A \sqcup B) \le \mu^*(A) + \mu^*(B)$, so only the reverse inequality is at issue. The positive separation lets us *physically separate* any enclosing open. Let $U \supseteq A \sqcup B$ be open, and carve out of it the two disjoint open collars

$$U_A = U \cap \{x : d(x, A) < \rho/2\}, \qquad U_B = U \cap \{x : d(x, B) < \rho/2\};$$

they are disjoint because a point within $\rho/2$ of A is, by the triangle inequality and $d(A, B) = \rho$, more than $\rho/2$ from B, so it cannot lie in both. Now take any compacts $C_A \subseteq U_A$ and $C_B \subseteq U_B$ from $\mathcal{K}$ competing in the inner contents $\tau(U_A)$ and $\tau(U_B)$ respectively. Their union $C_A \cup C_B$ lies in U and, being a disjoint union of two members of $\mathcal{K}$, is again a member of $\mathcal{K}$ with $\gamma(C_A \cup C_B) = \gamma(C_A) + \gamma(C_B)$ — here the limit of the exact additivity $\mu_n(C_A \cup C_B) = \mu_n(C_A) + \mu_n(C_B)$ on the disjoint pair is used. Hence $\tau(U) \ge \gamma(C_A) + \gamma(C_B)$, and taking suprema over C_A, C_B gives $\tau(U) \ge \tau(U_A) + \tau(U_B) \ge \mu^*(A) + \mu^*(B)$. Taking the infimum over enclosing U yields $\mu^*(A \sqcup B) \ge \mu^*(A) + \mu^*(B)$, and with subadditivity, equality.

Step 3 (a Borel probability measure). The point of verifying the *metric* outer measure property is that it unlocks an import from chapter 1: for a metric outer measure every Borel set is Carathéodory measurable, so the restriction $\mu := \mu^*|_{\mathcal{B}(S)}$ is a countably additive Borel measure. This is the Carathéodory extension theorem (chapter 1), and it is the precise machine that turns the separation-additive outer measure μ^* into a bona fide measure on the Borel σ-algebra — exactly the role flagged in the preamble to the proof. It remains to check that no mass leaked away in the construction, i.e. $\mu(S) = 1$. Each K_m is itself a member of $\mathcal{K}$: it is compact, so the interiors of the closed balls $\overline{B}(d, q)$ ($d \in D$, $q \in \mathbb{Q}_{>0}$) meeting it cover it, finitely many suffice, and $K_m = F \cap K_m$ for the resulting finite union F of closed balls. Therefore $\gamma(K_m)$ is defined and competes in $\tau(S)$, so

$$\mu(S) = \mu^*(S) = \tau(S) \ge \gamma(K_m) = \lim_n \mu_n(K_m) \ge 1 - \tfrac{1}{m} \qquad \text{for every } m,$$

forcing $\mu(S) \ge 1$. The reverse bound is automatic: every competing $\gamma(C) \le 1$ (as γ is a limit of probabilities), so $\tau(S) \le 1$. Hence $\mu(S) = 1$ and $\mu \in \mathcal{P}(S)$ — a genuine probability measure, with Ulam's tightness retroactively guaranteeing it could not have been a defective content. This is where the construction would have failed without the mass control of Step 1: had tightness not pinned $1 - 1/m$ inside compacts, the limit μ could carry mass < 1, which is the analytic form of the escape-to-infinity the whole section is built to prevent.

Step 4 (identification of the weak limit). A measure has been built; we now verify it is the weak limit of the subsequence, and the natural certificate is the open-set form of Portmanteau, since open sets are exactly where the inner content τ was defined. Let $U \subseteq S$ be open. For any $C \in \mathcal{K}$ with $C \subseteq U$, monotonicity of each μ_n gives $\mu_n(U) \geq \mu_n(C)$, hence $\liminf_n \mu_n(U) \geq \liminf_n \mu_n(C) = \gamma(C)$; the limit on the right is an honest equality because $\gamma(C) = \lim_n \mu_n(C)$ exists. Taking the supremum over all such C replaces $\gamma(C)$ by $\tau(U)$, and $\tau(U) \geq \mu^*(U) = \mu(U)$ by the very definition of μ^* as an infimum over enclosing opens (with U itself one of them). Chaining these,

$$\liminf_n \mu_n(U) \;\geq\; \tau(U) \;\geq\; \mu(U) \qquad \text{for every open } U,$$

which is precisely characterization (iv) of the Portmanteau theorem (theorem 3.9), the lower-semicontinuity of mass on open sets. That theorem then upgrades it to $\mu_n \Rightarrow \mu$. We have thus extracted from the original sequence a weakly convergent subsequence with limit in $\mathcal{P}(S)$. Since the argument applies verbatim to *any* subsequence of the tight sequence — a subsequence of a tight family is tight — every subsequence has a further weakly convergent sub-subsequence, which is the definition of relative compactness of Π. □

Proof of (b)*, S Polish.* This is the converse: relative compactness forces tightness, so that tightness is not just sufficient but necessary for weak compactness on a Polish space. The proof runs the construction of (a) in reverse — it builds the same kind of compact set, an intersection of finite ball-unions across all scales, but now it must show the mass outside it is uniformly small over the *whole* family, and the only lever available is relative compactness. The heart of the matter is a single uniform covering claim, after which the assembly mimics Ulam.

Fix a countable dense set $\{x_i\} \subseteq S$ and a scale $\delta > 0$. The claim is that finitely many δ-balls already capture, uniformly over Π, all but an arbitrarily small fraction of the mass:

$$\forall \varepsilon > 0 \;\exists\, k : \quad \inf_{\mu \in \Pi} \; \mu\Big(\bigcup_{i \leq k} B(x_i, \delta)\Big) \geq 1 - \varepsilon. \tag{3.9}$$

The content of (3.9) is the *uniformity*: for one fixed μ the finite union $\bigcup_{i \leq k} B(x_i, \delta)$ increases to S and its mass tends to 1 as $k \to \infty$, so a single measure poses no problem; the claim is that a single k works for the entire family at once. Suppose it failed. Then for some $\varepsilon > 0$, no finite k suffices: for every k there is a witness $\mu_k \in \Pi$ that evades the bound, $\mu_k\big(\bigcup_{i \leq k} B(x_i, \delta)\big) < 1 - \varepsilon$. Here is where relative compactness is spent: the sequence of witnesses (μ_k), living in the relatively compact Π, has a weakly convergent subsequence $\mu_{k_j} \Rightarrow \nu$ with limit $\nu \in \mathcal{P}(S)$. We now show this ν cannot be a probability measure, the desired contradiction. Fix any m and consider the open set $G_m = \bigcup_{i \leq m} B(x_i, \delta)$. For every index $k_j \geq m$ the nesting $G_m \subseteq \bigcup_{i \leq k_j} B(x_i, \delta)$ holds (more balls, larger union), so by monotonicity $\mu_{k_j}(G_m) \leq \mu_{k_j}\big(\bigcup_{i \leq k_j} B(x_i, \delta)\big) < 1 - \varepsilon$. Applying the open-set Portmanteau bound (iv) to the weak convergence $\mu_{k_j} \Rightarrow \nu$ on the fixed open G_m,

$$\nu(G_m) \;\leq\; \liminf_j \mu_{k_j}(G_m) \;\leq\; \liminf_j \mu_{k_j}\Big(\bigcup_{i \leq k_j} B(x_i, \delta)\Big) \;\leq\; 1 - \varepsilon.$$

This bound is uniform in m. But the sets G_m increase to S as $m \to \infty$, because the dense points $\{x_i\}$ put a δ-ball within reach of every point, so continuity from below (chapter 1) gives $\nu(S) = \lim_m \nu(G_m) \leq 1 - \varepsilon < 1$, contradicting $\nu(S) = 1$. The contradiction establishes (3.9).

With the uniform covering claim in hand the compact set is assembled exactly as in Ulam's lemma, with (3.9) doing the job that continuity from below did there, but now uniformly over Π. Fix $\varepsilon > 0$. Apply (3.9) at scale $\delta = 1/m$ with tolerance $\varepsilon 2^{-m}$ to get an index k_m for which $A_m := \bigcup_{i \le k_m} B(x_i, 1/m)$ satisfies $\inf_{\mu\in\Pi} \mu(A_m) \ge 1 - \varepsilon 2^{-m}$, i.e. $\mu(A_m^c) \le \varepsilon 2^{-m}$ for *every* $\mu \in \Pi$ at once. Set $K = \bigcap_{m\ge1} \overline{A_m}$. As in Ulam, K is closed and totally bounded — for each m it sits inside the finitely many closed balls $\overline{B}(x_i, 1/m)$, $i \le k_m$, of radius $1/m$ — hence compact by completeness of S. And the mass outside it is controlled uniformly:

$$\mu(K^c) = \mu\Big(\bigcup_{m\ge1} \overline{A_m}^{\,c}\Big) \le \sum_{m\ge1} \mu(A_m^c) \le \sum_{m\ge1} \varepsilon 2^{-m} = \varepsilon \qquad \text{for every } \mu \in \Pi,$$

using $\overline{A_m}^{\,c} \subseteq A_m^c$. Thus the single compact K carries $\mu(K) \ge 1 - \varepsilon$ for every $\mu \in \Pi$ simultaneously, which is tightness. Notice where Polishness was spent: completeness, in the step "closed + totally bounded $\Rightarrow$ compact." On a non-Polish space that implication can fail, and indeed relative compactness then need not imply tightness — so direction (b), unlike (a), genuinely requires the Polish hypothesis. This also retroactively licenses the forward use made in section 3.2.2: the completeness half of the metrization theorem (theorem 3.12) invoked precisely this direction (b) together with Ulam (lemma 3.15), and both have now been proved by arguments that never refer to that theorem, so there was no circularity in borrowing them ahead. □

We can now state, in one place, the recipe that every existence proof in the book follows, because the abstract development of this section was curated to exactly that recipe and it is worth seeing the pieces snap together.

Remark 3.17 (How Prokhorov is used downstream)**.** The pattern, instantiated in chapter 10 for classical MFG and in chapter 24 for graphon MFG, has four moves. *(1) Candidate map.* One sets up a candidate equilibrium as a fixed point: either an iteration $m \mapsto \Phi(m)$ that maps a population law to the law induced by the best response against it, or a relaxed family of approximate-equilibrium laws indexed by a regularization parameter. *(2) Uniform bound $\Rightarrow$ tightness.* One proves an a priori moment or energy estimate that is *uniform* in the iteration index or regularization parameter — typically $\sup \int |x|^p \, \mathrm{d}\mu \le M$ — which, by the Markov argument at the head of this section, certifies tightness of the family of candidates. *(3) Prokhorov.* Direction (a) converts that tightness into a weakly convergent subsequence of candidates, $\mu_{n_j} \Rightarrow m_\star$, manufacturing a limit object out of thin air. *(4) Portmanteau identification.* Continuity of the relevant functionals — the cost, the coupling, the best-response map — combined with Portmanteau (iv), shows the limit $m_\star$ is a fixed point of Φ, hence an equilibrium. The nondegeneracy standing assumption (A-Nondeg) of the thread specification enters at move (2) in the diffusive case: it is the uniform parabolic smoothing that produces the a priori bound in the first place, so the whole machine turns on it; we flag the dependence there, where the estimate is actually carried out, not here, where only its consequence is consumed.

This completes the compactness theory and, with it, the deterministic side of the mean field limit. Tightness plus Prokhorov is the law-of-large-numbers half of the story: it guarantees that the population profile μ^N does not escape to infinity but settles — along subsequences, and in the equilibrium analyses genuinely — onto a deterministic limit measure m. What it does *not* tell us is how the population fluctuates *around* that limit while N is large but finite. That residual jitter is not noise to be ignored; it is the $N^{-1/2}$ fluctuation whose variance bookkeeping we already glimpsed in (3.4), where the additivity of variance over an uncorrelated ensemble made the standard deviation of an empirical average scale like $\sigma/\sqrt{N}$. Prokhorov delivered the limit; the complementary task is to quantify the spread about it, and to show that the spread is not merely of size $N^{-1/2}$ but *Gaussian* in profile. That is the business of the next section, which sharpens the second-moment estimate into a full distributional statement by passing to the characteristic function and proving the central limit theorem.

3.4 Characteristic functions, the central limit theorem, and the $N^{-1/2}$ fluctuation scale

Tightness and Prokhorov, in the section just closed, settled the law-of-large-numbers half of the mean field limit: they guarantee that the population profile μ^N does not run off to infinity but condenses — along subsequences, and in the equilibrium analyses genuinely — onto a deterministic limit measure m. That is the *deterministic* skeleton of the limit. It says nothing, however, about the flesh on the skeleton: how the population trembles *about* m while N is large but still finite. This residual trembling is the complementary object, and quantifying it is the task of the present section. We are not starting from nothing. The variance bookkeeping of the independence section already pinned the *size* of the trembling: the additivity of variance over an uncorrelated ensemble, (3.4), forced the standard deviation of an empirical average to scale like $\sigma/\sqrt{N}$, so a fluctuation of typical magnitude $N^{-1/2}$ was visible at the level of the second moment alone. What the second moment cannot see is the *shape* of that fluctuation — whether the $N^{-1/2}$-sized deviation is distributed in one way or another. The slogan a physicist carries, "mean field suppresses fluctuations, which are of order $N^{-1/2}$ and Gaussian," contains two claims, and only the first has so far been earned. This section earns the second. It promotes the variance estimate into a full distributional statement: the rescaled fluctuation $\sqrt{N}(\bar{X}_N - \mathbb{E}[X])$ does not merely have bounded variance, it converges in law to a Gaussian. That is the central limit theorem, and it is the precise content of the suppression slogan.

The variance computation got us this far because variance is *additive* over independent summands; to reach the full law of a sum of independent variables we need an instrument that turns the entire distribution, not just its second moment, into something additive. That instrument is the *characteristic function*, the Fourier transform of a law, $\varphi_X(t) = \mathbb{E}\left[e^{i\langle t,X\rangle}\right]$. Its decisive virtue is a single algebraic fact: *independence becomes multiplication*. If X and Y are independent, the expectation of the product $e^{itX}e^{itY} = e^{it(X+Y)}$ factorizes by (3.3), so $\varphi_{X+Y} = \varphi_X\,\varphi_Y$. Adding independent random variables, which on the level of laws is the operation of *convolution* — $\mathcal{L}(X + Y) = \mathcal{L}(X) * \mathcal{L}(Y)$, an integral entangling the two distributions — is thereby converted into the elementary pointwise *product* of their transforms. The Fourier transform diagonalizes convolution; the hard nonlinear bookkeeping of combining many independent contributions collapses into multiplying numbers. A sum of N independent variables, whose law is an N-fold convolution and is essentially uncomputable directly, has a characteristic function that is simply the N-fold product of the factors, and the asymptotics of a product are read off from the asymptotics of its logarithm. This is exactly why the Fourier route, and not a direct attack on the convolution, proves the central limit theorem.

The parallel with equilibrium statistical mechanics is genuine and worth one sentence, kept light. There the partition function of a system assembled from *non-interacting* subsystems factorizes, $Z = \prod_i Z_i$, so that the free energies $-\log Z_i$ add and become extensive; here the characteristic function of a sum of *independent* variables factorizes in the same way, so that the *cumulant generating function* $\log \varphi$ adds over the summands. The $N^{-1/2}$ scaling and the Gaussian shape are then the statement that, after centering and rescaling, the additive log-transform of a non-interacting ensemble flows to a quadratic — the universal fixed point whose inverse Fourier transform is the Gaussian. Independence is, in both settings, precisely the hypothesis that makes the generating function multiplicative and hence its logarithm additive; the coupling in a finite N-player game spoils the factorization, and the corrections to it are the very fluctuations chapter 7 and chapter 13 will track.

Three subsections carry this out, in the only order the logic permits, and it is worth seeing the dependency chain whole before meeting its links. The Fourier route rests on the transform being a *faithful* encoding of the law and a *continuous* one, and these two properties must be established before any limit theorem can use them. The first subsection (section 3.4.1) builds the transform and proves both. Faithfulness is the *Lévy inversion formula* (lemma 3.20), which reconstructs the measure from its characteristic function by an explicit integral and so shows the transform loses no information; its immediate corollary is *uniqueness* — equal characteristic functions mean equal laws (proposition 3.19(4)) — the property that will let us *identify* a limit once we recognize its transform. Continuity is the *Lévy continuity theorem* (theorem 3.21), which says that pointwise convergence of characteristic functions is equivalent to weak convergence of laws, provided the limit transform is continuous at the origin — a proviso that is, as we will see, a tightness condition wearing a Fourier disguise, tying this pillar back to the previous one. The chain is therefore inversion $\Rightarrow$ uniqueness $\Rightarrow$ continuity $\Rightarrow$ central limit theorem: inversion makes the transform invertible, uniqueness makes the limiting transform name a unique law, continuity converts convergence of transforms into convergence of laws, and only then can the second subsection (section 3.4.2) prove the central limit theorem itself — by the strategy already advertised, expanding $\log \varphi$ to second order and watching the rescaled product converge to the Gaussian transform $e^{-\sigma^2 t^2/2}$. The third subsection (section 3.4.3) spends the theorem on the object this chapter exists to serve: a mean field observable averaged over the empirical measure, where the central limit theorem makes the $N^{-1/2}$ Gaussian fluctuation about the mean field m a precise distributional statement rather than a variance estimate.

We begin, then, with the transform itself and the two properties — inversion and continuity — on which everything downstream rests.

3.4.1 Characteristic functions

The section preamble made a promise this subsection must now redeem: that the Fourier transform of a law converts independence into multiplication, and that two analytic facts about that transform — that it is *faithful* (uniqueness) and *continuous* (Lévy's theorem) — are between them all the central limit theorem will need. We proceed in the order the logic forces. First we write the transform down and establish the elementary properties that surround it; among them, as property (3), sits the promised conversion of independence into a product, made concrete and proved on the spot. Then we build the two heavier facts on this elementary base: the inversion formula behind uniqueness, and the continuity theorem behind the limit. Nothing here is cited in place of an argument — both named theorems are proved in full, honouring the self-containedness rule imported from chapter 1.

Definition 3.18 (Characteristic function)**.** The *characteristic function* of an $\mathbb{R}^d$-valued random variable X with law μ is

$$\varphi_X(t) = \varphi_\mu(t) = \mathbb{E}\Big[e^{\,i\langle t,X\rangle}\Big] = \int_{\mathbb{R}^d} e^{\,i\langle t,x\rangle}\,\mathrm{d}\mu(x), \qquad t \in \mathbb{R}^d.$$

A characteristic function is thus nothing more exotic than the expectation of the bounded oscillating observable $x \mapsto e^{i\langle t,x\rangle}$, computed by the change-of-variables bridge (3.1) as an integral of that observable against the law. It is the law viewed in Fourier space: one complex number $\varphi_X(t)$ for each frequency vector t, recording how the unit mass of μ interferes when wound around the unit circle at rate t. Because the integrand has modulus one and is smooth, this encoding is automatically tame, and the next proposition collects the five facts about it that the rest of the development consumes. Read them as a checklist of what the central limit theorem will ask for: that the transform is bounded and normalized, so a limit of transforms cannot escape the unit disc; that it is uniformly continuous, so pointwise limits are well behaved; that it turns independence into multiplication, the algebraic engine of the whole argument; that it determines the law uniquely, so that a limiting transform names a single limit measure; and that its derivatives at the origin read off the moments, so that a finite-variance law is captured near $t = 0$ by its mean and variance alone — the exact datum the central limit theorem expands.

Proposition 3.19 (Elementary properties)**.** *For any $\mathbb{R}^d$-valued X: (1) $\varphi_X(0) = 1$ and $|\varphi_X(t)| \le 1$; (2) φ_X is uniformly continuous on $\mathbb{R}^d$; (3) if $X \perp\!\!\!\perp Y$ then $\varphi_{X+Y} = \varphi_X\varphi_Y$; (4)* (uniqueness) *$\varphi_X = \varphi_Y$ implies $X \overset{d}{=} Y$; (5) if $\mathbb{E}[|X|^p] < \infty$ for an integer p, then $\varphi_X \in C^p$ and $\partial^\alpha\varphi_X(0) = i^{|\alpha|}\mathbb{E}[X^\alpha]$ for multi-indices $|\alpha| \le p$. In particular for scalar $X \in L^2$,*

$$\varphi_X(t) = 1 + it\,\mathbb{E}[X] - \tfrac{1}{2}t^2\,\mathbb{E}\Big[X^2\Big] + o(t^2) \qquad (t \to 0). \tag{3.10}$$

Proof. We take the five in turn; the first three fall straight out of the definition and the unit modulus $\left|e^{i\langle t,x\rangle}\right| = 1$, while the last two carry the analytic weight.

(1) Normalization and the unit bound. At $t = 0$ the integrand is $e^0 = 1$, so $\varphi_X(0) = \int_{\mathbb{R}^d} 1\,\mathrm{d}\mu = 1$. For general t the integrand has modulus $\left|e^{i\langle t,x\rangle}\right| = 1$, so by the triangle inequality for integrals $|\varphi_X(t)| = \left|\int e^{i\langle t,x\rangle}\,\mathrm{d}\mu\right| \le \int \left|e^{i\langle t,x\rangle}\right|\,\mathrm{d}\mu = \int 1\,\mathrm{d}\mu = 1$. The transform thus lives in the closed unit disc and is pinned to the value 1 at the origin — the normalization on which the entire continuity theory below will key.

(2) Uniform continuity. Subtracting two values and factoring out the common phase $e^{i\langle t,x\rangle}$,

$$\varphi_X(t+h) - \varphi_X(t) = \int_{\mathbb{R}^d} e^{i\langle t,x\rangle}\left(e^{i\langle h,x\rangle} - 1\right)\,\mathrm{d}\mu(x),$$

so, since $\left|e^{i\langle t,x\rangle}\right| = 1$, $|\varphi_X(t+h) - \varphi_X(t)| \le \int_{\mathbb{R}^d}\left|e^{i\langle h,x\rangle} - 1\right|\,\mathrm{d}\mu(x) = \mathbb{E}\Big[\left|e^{i\langle h,X\rangle} - 1\right|\Big]$. The decisive feature of this bound is that it does *not* depend on t: it controls the oscillation of φ_X over a step of size h uniformly across all of $\mathbb{R}^d$ at once, which is exactly the difference between uniform and merely pointwise continuity. It remains to see the bound tends to 0 with h. The integrand $\left|e^{i\langle h,x\rangle}-1\right|$ is bounded by the constant 2, an integrable dominating function on the probability space, and for each fixed x it tends to 0 as $h \to 0$ by continuity of the exponential; dominated convergence (lemma 3.4) therefore sends $\mathbb{E}\left[\left|e^{i\langle h,X\rangle}-1\right|\right] \to 0$. Uniform continuity follows.

(3) Factorization under independence — the promised conversion. This is the property the section was built to deliver, and it is a single line once the expectation-factorization form of independence is invoked. If $X \perp\!\!\!\perp Y$ then $e^{i\langle t,X\rangle}$ and $e^{i\langle t,Y\rangle}$ are bounded functions of the two independent variables, so the factorization (3.3) applies to their product:

$$\varphi_{X+Y}(t) = \mathbb{E}\Big[e^{\,i\langle t,(X+Y)\rangle}\Big] = \mathbb{E}\Big[e^{\,i\langle t,X\rangle}e^{\,i\langle t,Y\rangle}\Big] = \mathbb{E}\Big[e^{\,i\langle t,X\rangle}\Big]\,\mathbb{E}\Big[e^{\,i\langle t,Y\rangle}\Big] = \varphi_X(t)\,\varphi_Y(t).$$

Addition of independent variables — which on the level of laws is the convolution $\mathcal{L}(X+Y) = \mathcal{L}(X)*\mathcal{L}(Y)$, an integral entangling the two distributions — has become the pointwise multiplication of two numbers. This is the diagonalization of convolution the section preamble advertised, and it is the entire reason the Fourier route, rather than a direct attack on the N-fold convolution, will deliver the central limit theorem.

(4) Uniqueness. This is the substantive analytic claim, the assertion that the transform forgets nothing: $\varphi_X = \varphi_Y$ forces $X \overset{d}{=} Y$. Its proof is the content of the Lévy inversion formula, which reconstructs μ from φ by an explicit integral; we establish it in full as lemma 3.20 below and borrow the conclusion here. The mechanism, previewed: inversion recovers the μ-measure of every interval whose endpoints are atom-free, and those intervals form an intersection-stable class generating the Borel σ-algebra, so two laws with the same transform agree on a generating π-system and hence everywhere.

(5) Smoothness and moments. Suppose $\mathbb{E}[|X|^p] < \infty$ for an integer $p \ge 1$. We claim $\varphi_X \in C^p$ with $\partial^\alpha\varphi_X(0) = i^{|\alpha|}\mathbb{E}[X^\alpha]$ for every multi-index $|\alpha| \le p$, and the whole content is the legitimacy of *differentiating under the integral sign.* Differentiating the integrand $e^{i\langle t,x\rangle}$ in t_j brings down a factor ix_j, and after α such differentiations

$$\partial_t^\alpha\, e^{\,i\langle t,x\rangle} = i^{|\alpha|}\,x^\alpha\,e^{\,i\langle t,x\rangle}, \qquad x^\alpha := \prod_j x_j^{\alpha_j}.$$

To pull ∂^α inside the integral we need a single integrable function dominating this derivative for all t at once, and the unit modulus supplies one: since $\left|e^{i\langle t,x\rangle}\right| = 1$,

$$\left|\partial_t^\alpha e^{\,i\langle t,x\rangle}\right| = |x^\alpha| \le |x|^{|\alpha|} \le 1+|x|^{\,p} \qquad (|\alpha| \le p),$$

and $1+|X|^{\,p}$ is integrable by the hypothesis $\mathbb{E}[|X|^p] < \infty$. The differentiation-under-the-integral theorem of chapter 1 — itself dominated convergence applied to the difference quotients $h^{-1}(e^{i\langle(t+he_j),x\rangle} - e^{i\langle t,x\rangle})$, which are dominated by $|x_j| \le 1+|x|^p$ uniformly in small h by the mean value theorem — therefore licenses exchanging each ∂_{t_j} with $\int \mathrm{d}\mu$. The resulting integrand $i^{|\alpha|}x^\alpha e^{i\langle t,x\rangle}$ is continuous in t and dominated as above, so $\partial^\alpha\varphi_X$ exists and is continuous: $\varphi_X \in C^p$. Evaluating at $t=0$ kills the exponential and leaves $\partial^\alpha\varphi_X(0) = i^{|\alpha|}\int x^\alpha\,\mathrm{d}\mu = i^{|\alpha|}\mathbb{E}[X^\alpha]$, the announced formula — the derivatives of the transform at the origin *are* the moments, up to powers of i. Now specialize to a scalar $X \in L^2$, where $p = 2$: then $\varphi_X \in C^2$ with $\varphi_X(0) = 1$, $\varphi'_X(0) = i\mathbb{E}[X]$, and $\varphi''_X(0) = i^2\mathbb{E}[X^2] = -\mathbb{E}[X^2]$. Taylor's theorem with the second-order Peano remainder, valid because φ_X is twice continuously differentiable at 0, then gives

$$\varphi_X(t) = \varphi_X(0) + t\,\varphi'_X(0) + \tfrac{1}{2}t^2\,\varphi''_X(0) + o(t^2) = 1 + it\,\mathbb{E}[X] - \tfrac{1}{2}t^2\,\mathbb{E}\Big[X^2\Big] + o(t^2),$$

which is (3.10). This small expansion is the single fact the central limit theorem will plug in: near the origin of Fourier space a finite-variance law is determined to second order by its mean and variance alone, and everything beyond is the $o(t^2)$ that the $N^{-1/2}$ rescaling will wash out. □

The uniqueness asserted in (4) is consumed downstream more than once — it identifies the Gaussian limit in the central limit theorem and it is the device that *closes* Lévy's continuity theorem by pinning all subsequential limits to one measure — so, the self-containedness rule of chapter 1 forbidding any bare appeal to "it is known," we prove the inversion formula it rests on rather than cite it. The proof needs exactly one analytic input, and we isolate it now and take it explicitly as a *prerequisite* of elementary analysis: the *Dirichlet integral*, together with the uniform boundedness of its truncations,

$$\mathsf{S}(\theta) := \int_0^\infty \frac{\sin(\theta t)}{t}\,\mathrm{d}t = \frac{\pi}{2}\operatorname{sgn}(\theta), \qquad \sup_{R>0,\ \theta\in\mathbb{R}} \Big|\int_0^R \frac{\sin(\theta t)}{t}\,\mathrm{d}t\Big| =: C_0 < \infty. \tag{3.11}$$

Both halves are standard facts of undergraduate analysis that we do not reprove here: the convergence of the improper integral is conditional (the integrand is not absolutely integrable), its value is the classical $\pm\pi/2$, and the truncated integrals $\int_0^R \frac{\sin\theta t}{t}\,\mathrm{d}t$ stay bounded by a single constant C_0 uniformly in the cutoff R and the parameter θ — the overshoot familiar from the Gibbs phenomenon. These two facts are all we borrow; everything else in the inversion proof is built here.

The shape of the argument is worth stating before the computation, because the formula (3.12) looks more forbidding than it is. We want to recover $\mu((a, b))$ from φ. We substitute the definition of φ into the right-hand side, swap the order of integration by Fubini so that the t-integral is done *first*, and discover that the inner t-integral is a smooth function $\Psi_R(x)$ that — thanks to the Dirichlet integral — converges as $R \to \infty$ to the indicator of the interval (a, b) (with a half-weight exactly at the two endpoints a, b). Integrating that limiting indicator against μ returns $\mu((a, b))$, provided the half-weighted endpoints carry no mass — which is precisely the atom-free hypothesis. The entire proof is the careful execution of those four steps.

Lemma 3.20 (Lévy inversion formula)**.** *Let $\mu \in \mathcal{P}(\mathbb{R})$ with characteristic function φ. For any real $a < b$ that are atoms-free, i.e. $\mu(\{a\}) = \mu(\{b\}) = 0$,*

$$\mu((a, b)) = \lim_{R\to\infty} \frac{1}{2\pi} \int_{-R}^{R} \frac{e^{-ita} - e^{-itb}}{it}\,\varphi(t)\,\mathrm{d}t. \tag{3.12}$$

Consequently a probability measure on $\mathbb{R}$ is determined by its characteristic function, and the same holds on $\mathbb{R}^d$.

Proof. Step 1: substitute and prepare for the swap. Fix the cutoff R and call the truncated integral I_R. Writing $\varphi(t) = \int_{\mathbb{R}} e^{itx}\,\mathrm{d}\mu(x)$ inside it and combining the two exponentials,

$$I_R := \frac{1}{2\pi} \int_{-R}^{R} \frac{e^{-ita} - e^{-itb}}{it}\,\varphi(t)\,\mathrm{d}t = \frac{1}{2\pi} \int_{-R}^{R}\int_{\mathbb{R}} \frac{e^{it(x-a)} - e^{it(x-b)}}{it}\,\mathrm{d}\mu(x)\,\mathrm{d}t.$$

Before swapping the two integrals we must check the joint integrand is absolutely integrable over the product domain, because the kernel $1/(it)$ is singular at $t=0$ and a careless swap is exactly the kind of step the Fubini hypotheses guard against. The singularity is illusory: the numerator vanishes at $t=0$ at the same rate, and in fact writing the difference of exponentials as an integral,

$$\frac{e^{it(x-a)}-e^{it(x-b)}}{it}=\int_a^b e^{\,it(x-s)}\ \mathrm{d}s,$$

exhibits the whole quotient as an average of unit-modulus terms, so its absolute value is at most $b-a$ for *every* t, including $t=0$. Hence the joint integrand is bounded by $b-a$ on the product $[-R,R]\times\mathbb{R}$, and since μ is a probability measure and $[-R,R]$ has finite length, that product carries finite measure $2R$ and the integrand is absolutely integrable on it. Fubini–Tonelli (chapter 1) therefore applies, and we may do the t-integral first:

$$I_R=\int_{\mathbb{R}}\Psi_R(x)\ \mathrm{d}\mu(x),\qquad \Psi_R(x)=\frac{1}{2\pi}\int_{-R}^{R}\frac{e^{it(x-a)}-e^{it(x-b)}}{it}\ \mathrm{d}t.$$

Step 2: compute the inner kernel Ψ_R. This is where the Dirichlet integral enters. Expand each exponential as $e^{i\theta}=\cos\theta+i\sin\theta$ and divide by $it=\frac{-i}{t}\cdot(-1)\cdots$ — concretely, using $\frac{1}{it}=\frac{-i}{t}$, the integrand becomes

$$\frac{e^{it(x-a)}-e^{it(x-b)}}{it}=\underbrace{\frac{\sin t(x-a)-\sin t(x-b)}{t}}_{\text{even in }t}-i\,\underbrace{\frac{\cos t(x-a)-\cos t(x-b)}{t}}_{\text{odd in }t}.$$

The second (imaginary) piece is an odd function of t, so its integral over the symmetric interval $[-R,R]$ vanishes; the first (real) piece is even, so its integral is twice the integral over $[0,R]$. With the notation $D_R(\theta):=\int_0^R\frac{\sin(\theta t)}{t}\ \mathrm{d}t$ — the truncated Dirichlet integral of (3.11) — this gives

$$\Psi_R(x)=\frac{1}{2\pi}\cdot 2\int_0^R\frac{\sin t(x-a)-\sin t(x-b)}{t}\ \mathrm{d}t=\frac{1}{\pi}\Big(D_R(x-a)-D_R(x-b)\Big).$$

So Ψ_R is built entirely out of two copies of the one elementary object we took as a prerequisite.

Step 3: pass to the limit $R\to\infty$. The uniform bound in (3.11) gives $|D_R(\theta)|\le C_0$ for all R,θ, hence $|\Psi_R(x)|\le 2C_0/\pi$ uniformly in R and x — this is the constant dominating function we will need. Pointwise, $D_R(\theta)\to\frac{\pi}{2}\operatorname{sgn}(\theta)$ as $R\to\infty$ by the Dirichlet integral, so

$$\Psi_R(x)\to\Psi(x):=\tfrac12\big(\operatorname{sgn}(x-a)-\operatorname{sgn}(x-b)\big)=\begin{cases}1, & a<x<b,\\ \tfrac12, & x\in\{a,b\},\\ 0, & x<a\text{ or }x>b.\end{cases}$$

This limiting Ψ is the indicator of the open interval (a,b) everywhere except at the two endpoints, where the sign function jumps and the average lands on the midpoint value $\frac12$ — the source of the half-weights that the atom-free hypothesis is about to discard. We now exchange limit and integral. The dominating function $2C_0/\pi$ is constant, hence μ-integrable because μ is finite, so dominated convergence (lemma 3.4) gives

$$I_R=\int_{\mathbb{R}}\Psi_R\ \mathrm{d}\mu\ \longrightarrow\ \int_{\mathbb{R}}\Psi\ \mathrm{d}\mu=\mu\big((a,b)\big)+\tfrac12\mu(\{a\})+\tfrac12\mu(\{b\}),$$

the last equality reading off the three values of Ψ on the open interval, the two endpoints, and the exterior. *Here* the hypothesis is spent: $\mu(\{a\}) = \mu(\{b\}) = 0$ kills the two half-weights, leaving exactly $\mu((a, b))$. This is (3.12), and we see why atom-free endpoints are needed — at an atom the half-weight would survive and the formula would overcount.

Step 4: uniqueness from inversion. The formula expresses $\mu((a, b))$ as an explicit functional of φ alone, for every interval with atom-free endpoints. Two laws sharing a characteristic function therefore agree on every such interval. That this suffices to force the laws equal is the standard π-system argument. The atoms of μ are at most countable — distinct atoms carry disjoint chunks of positive mass and a finite measure can host only countably many of those — so the atom-free points are dense, hence the intervals (a, b) with atom-free endpoints form an intersection-stable family (the intersection of two such intervals is again one) generating the Borel σ-algebra $\mathcal{B}(\mathbb{R})$. Two probability measures agreeing on a generating π-system agree on the whole σ-algebra by the uniqueness lemma for measures (corollary 1.10), so φ determines μ on $\mathcal{B}(\mathbb{R})$. In $\mathbb{R}^d$ nothing changes but the bookkeeping: the same computation with the product kernel $\prod_{j=1}^d (e^{-it_j a_j} - e^{-it_j b_j})/(it_j)$ recovers μ on every rectangle $\prod_j (a_j, b_j)$ with atom-free faces, and rectangles are again a generating π-system, so uniqueness holds in every dimension. This proves property (4) of proposition 3.19. $\square$

Inversion delivered the *faithfulness* of the transform; the second pillar of the Fourier route is its *continuity*, and that is Lévy's theorem. It is the practical reason the Fourier route proves limit theorems at all: it converts a question about weak convergence of measures, which is awkward to verify directly, into a question about pointwise convergence of their transforms, which the central limit theorem will settle by a one-line Taylor expansion. The theorem has two directions. Direction (a) is the easy half — weak convergence implies convergence of transforms — and is little more than the observation that each $e^{i\langle t,x\rangle}$ is a bounded continuous test function. Direction (b) is the substantive half and the one the limit theorems consume: pointwise convergence of the transforms, *plus* continuity of the limit at the origin, implies weak convergence. The role of that innocuous-looking proviso is the throughline point of this entire section, so we state it before proving it. Pointwise convergence of characteristic functions, on its own, does *not* force a weak limit; it does so only when the candidate limit is continuous at $t = 0$. And continuity at the origin is not a technical decoration — it is exactly a *tightness* condition wearing a Fourier disguise. The proof of (b) makes this literal: it extracts from the continuity of ψ at 0 a uniform tail bound on the measures μ_n, which is tightness, and only tightness then licenses Prokhorov to produce a limit. Without the proviso the mass can leak to infinity and the "limit" transform fails to be the transform of any probability measure. This is the precise sense, promised at the top of the chapter, in which tightness is indispensable to the Fourier route.

Theorem 3.21 (Lévy continuity theorem)**.** *Let $\mu_n \in \mathcal{P}(\mathbb{R}^d)$ with characteristic functions φ_n.*

(a) *If $\mu_n \Rightarrow \mu$ then $\varphi_n(t) \to \varphi_\mu(t)$ for every t, uniformly on compact sets.*

(b) *If $\varphi_n(t) \to \psi(t)$ pointwise for some function ψ that is continuous at $t = 0$, then $\psi = \varphi_\mu$ is the characteristic function of a probability measure $\mu \in \mathcal{P}(\mathbb{R}^d)$ and $\mu_n \Rightarrow \mu$.*

Proof. The argument is complete given the in-book Prokhorov theorem (theorem 3.16) and the

Fourier uniqueness (lemma 3.20); the external citations at the end are attribution only, not a substitute for any step.

(a) Weak convergence implies convergence of transforms. For each fixed t the map $x \mapsto e^{i\langle t,x\rangle}$ has bounded real and imaginary parts $\cos\langle t,x\rangle$ and $\sin\langle t,x\rangle$, both continuous, so it belongs to $C_b(\mathbb{R}^d)$ (read componentwise). Weak convergence is by definition convergence of the integral pairing against every such test function, so $\varphi_n(t) = \mu_n(e^{i\langle t,\cdot\rangle}) \to \mu(e^{i\langle t,\cdot\rangle}) = \varphi_\mu(t)$ for each t. That is the pointwise claim. For uniformity on a compact set of frequencies, we upgrade pointwise to uniform via equicontinuity. A weakly convergent sequence is in particular relatively compact, so the family $\{\mu_n\} \cup \{\mu\}$ is tight by the converse half of Prokhorov (theorem 3.16(b)): given $\varepsilon > 0$ there is a radius R with $\sup_n \mu_n(\{|x| > R\}) \le \varepsilon$. Splitting the bound $\left|e^{i\langle (t-s),x\rangle} - 1\right|$ according to whether $|x| \le R$ or not — on the ball it is at most $|\langle (t-s), x\rangle| \le |t-s|\ R$ by the elementary estimate $\left|e^{i\theta} - 1\right| \le |\theta|$, off the ball it is at most 2 — gives, for all s, t,

$$|\varphi_n(t) - \varphi_n(s)| \le \int_{\mathbb{R}^d} \left|e^{i\langle (t-s),x\rangle} - 1\right| \,\mathrm{d}\mu_n(x) \le |t-s|\ R + 2\varepsilon.$$

The right-hand side is independent of n and small when $|t-s|$ and ε are small, so $\{\varphi_n\}$ is equicontinuous; and a uniformly equicontinuous sequence that converges pointwise converges uniformly on compact sets. This gives (a).

(b) Convergence of transforms plus continuity at 0 *implies weak convergence.* As advertised, the crux is that continuity of ψ at the origin forces tightness of (μ_n), after which Prokhorov and uniqueness do the rest. The quantitative bridge from the transform to a tail bound is the elementary inequality, valid for any $\nu \in \mathcal{P}(\mathbb{R})$ and any $u > 0$,

$$\nu\Big(\Big\{\ |x| > \tfrac{2}{u}\Big\}\Big) \le \frac{1}{u}\int_{-u}^{u} \left(1 - \varphi_\nu(t)\right)\,\mathrm{d}t, \tag{3.13}$$

which bounds the mass ν puts far out by how far the transform departs from 1 near the origin. We derive it by computing the right-hand side. Fix $u > 0$ and start from the inner t-average of $1 - e^{itx}$ at a single point x:

$$\frac{1}{u}\int_{-u}^{u} \left(1 - e^{itx}\right)\,\mathrm{d}t = \frac{1}{u}\Big(2u - \int_{-u}^{u} e^{itx}\,\mathrm{d}t\Big) = \frac{1}{u}\Big(2u - \frac{e^{iux} - e^{-iux}}{ix}\Big) = 2\Big(1 - \frac{\sin ux}{ux}\Big),$$

using $\int_{-u}^{u} e^{itx}\,\mathrm{d}t = \frac{2\sin ux}{x}$ and simplifying. Now compare this with the indicator of the far tail. Writing $v = ux$, the claim is

$$2\Big(1 - \frac{\sin v}{v}\Big) \ \ge\ \mathbf{1}\{|v| > 2\} \qquad \Longleftrightarrow \qquad 2\Big(1 - \frac{\sin ux}{ux}\Big) \ge \mathbf{1}\Big\{\ |x| > \tfrac{2}{u}\Big\}.$$

This holds because $\frac{\sin v}{v} \le 1$ always, so the left side is ≥ 0 and the inequality is automatic where the indicator is 0; and where the indicator is 1, i.e. $|v| > 2$, we have $\left|\frac{\sin v}{v}\right| \le \frac{1}{|v|} < \frac{1}{2}$, hence $1 - \frac{\sin v}{v} \ge \frac{1}{2}$ and the left side is ≥ 1. So the elementary fact "$1 - \frac{\sin v}{v} \ge \frac{1}{2}$ for $|v| > 2$" is exactly what makes the inner average dominate the tail indicator pointwise in x. Integrating both sides against ν and exchanging the $\mathrm{d}t$- and $\mathrm{d}\nu$-integrals — legitimate by Fubini, since the integrand $1 - e^{itx}$ is bounded by 2 on the finite-measure product $[-u, u] \times \mathbb{R}$ — turns the pointwise inequality into

$$\frac{1}{u}\int_{-u}^{u} \left(1 - \varphi_\nu(t)\right)\,\mathrm{d}t = \int_{\mathbb{R}} 2\Big(1 - \frac{\sin ux}{ux}\Big)\,\mathrm{d}\nu(x) \ \ge\ \int_{\mathbb{R}} \mathbf{1}\Big\{\ |x| > \tfrac{2}{u}\Big\}\,\mathrm{d}\nu(x) = \nu\Big(\Big\{\ |x| > \tfrac{2}{u}\Big\}\Big),$$

which is (3.13); the left equality used $\frac{1}{u}\int_{-u}^{u}(1-\varphi_\nu)\,\mathrm{d}t = \int \frac{1}{u}\int_{-u}^{u}(1-e^{itx})\,\mathrm{d}t\,\mathrm{d}\nu$ and the single-point computation above.

With the inequality in hand the tightness is immediate. Apply (3.13) to each μ_n and let $n\to\infty$ on the right: the integrand $1-\varphi_n$ is bounded by 2 and converges pointwise to $1-\psi$, so bounded convergence on the finite interval $[-u,u]$ gives

$$\limsup_{n\to\infty}\mu_n\Big(\Big\{\,|x|>\tfrac{2}{u}\Big\}\Big)\le\frac{1}{u}\int_{-u}^{u}\big(1-\psi(t)\big)\,\mathrm{d}t.$$

Now the proviso pays off. Since $\psi(0)=1$ (as the pointwise limit of the φ_n, each equal to 1 at 0) and ψ is continuous at 0, the integrand $1-\psi(t)$ is small for t near 0; concretely, given $\varepsilon>0$ choose u small enough that $|1-\psi(t)|<\varepsilon/2$ on $[-u,u]$, whence the right-hand side is at most $\frac{1}{u}\cdot 2u\cdot\frac{\varepsilon}{2}=\varepsilon$. Thus $\limsup_n\mu_n(\{|x|>2/u\})\le\varepsilon$, and enlarging u slightly absorbs the finitely many early terms, so $\sup_n\mu_n(\{|x|>2/u\})\le\varepsilon$ for this u. As ε was arbitrary, (μ_n) is tight — and one sees plainly that had ψ been discontinuous at 0 this bound would have failed and the mass could have escaped to infinity, which is the throughline point made concrete.

The endgame is Prokhorov plus uniqueness. By the forward half of Prokhorov (theorem 3.16(a)) tightness makes (μ_n) relatively compact, so every subsequence has a further subsequence converging weakly to some probability measure. Let μ be any such subsequential limit. By part (a) its characteristic function is the pointwise limit of the φ_n along that subsequence, namely ψ; so *every* subsequential limit has the same characteristic function ψ. By the inversion-based uniqueness (lemma 3.20) a characteristic function determines its law, so all these subsequential limits are one and the same measure μ, and in particular $\psi=\varphi_\mu$ is genuinely the transform of a probability measure. A sequence in a metric space all of whose subsequential limits coincide with a single point converges to that point, so $\mu_n\Rightarrow\mu$. The multivariate case is identical: run (3.13) coordinatewise in each t_j to get tightness in $\mathbb{R}^d$, then repeat the Prokhorov–uniqueness endgame verbatim. See [62, Thm. 3.3.6], [84, Thm. 5.3] for the same result. $\square$

With inversion and continuity both proved, the two analytic facts the Fourier route demanded are in hand: inversion makes the transform a faithful encoding of the law, and Lévy's theorem makes it a continuous one, with continuity-at-the-origin standing in for tightness. Everything is now staged for the payoff. The central limit theorem becomes, from here, a short computation, and it is worth previewing its three moves so the next subsection reads as the harvest it is. First, factor the characteristic function of a centered and $N^{-1/2}$-rescaled sum into a product of N identical factors, using the independence-into-multiplication property (3). Second, expand one such factor near the origin by the Taylor formula (3.10), where — the summands being centered with variance σ^2 — only the term $-\frac{1}{2}(\sigma^2/N)t^2$ survives to leading order. Third, pass the N-fold product to its exponential limit through the elementary fact that $(1+z_N/N)^N\to e^z$ whenever $z_N\to z$, proved inline where it is used. The limit is $e^{-\sigma^2t^2/2}$, which is continuous at $t=0$, so Lévy's continuity theorem in the form (b) just proved identifies it without further work as the characteristic function of the Gaussian $\mathcal{N}(0,\sigma^2)$ and upgrades the convergence of transforms into the weak convergence of laws. We turn to that computation now.

3.4.2 The central limit theorem

Everything is now staged, and this subsection is the harvest. The two analytic facts the Fourier route demanded are in hand: inversion (lemma 3.20) makes the characteristic function a faithful encoding of a law, and Lévy's continuity theorem (theorem 3.21) makes that encoding *continuous*, so that a convergent sequence of transforms whose limit is continuous at the origin names a unique limiting measure and forces weak convergence of the laws. With those two facts the central limit theorem ceases to be an analytic struggle and becomes a short computation, and the reason is exactly the diagonalization of convolution recorded in proposition 3.19(3): the law of a sum of independent variables is a convolution, hopeless to track directly, but its transform is a *product*, and a product of N nearly-equal factors is precisely the kind of object whose $N \to \infty$ behaviour an exponential limit controls. The proof below is nothing more than carrying out that idea — factor, expand, exponentiate, identify — and its payoff is the single quantitative fact on which the entire mean field justification rests: the fluctuation of an average of N independent agents is of size $N^{-1/2}$ and, after rescaling by $\sqrt{N}$, is *Gaussian*, regardless of the microscopic law the agents are drawn from.

Theorem 3.22 (Central limit theorem, i.i.d. case)**.** *Let $(X_i)_{i\geq 1}$ be i.i.d. real random variables with $\mathbb{E}[X_1] = m_\star$ and $\mathrm{Var}(X_1) = \sigma^2 \in (0,\infty)$. Then*

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} (X_i - m_\star) \;\Rightarrow\; \mathcal{N}(0,\sigma^2) \qquad (N \to \infty),$$

where $\mathcal{N}(0,\sigma^2)$ is the centred Gaussian law with variance σ^2, characteristic function $t \mapsto e^{-\sigma^2 t^2/2}$.

Before the proof we isolate its one genuinely analytic ingredient as a standalone lemma, because the proof of the theorem compresses it into a parenthetical and it deserves to be seen cleanly. It is the complex, moving-base generalization of the freshman identity $(1 + z/N)^N \to e^z$: we need it not for a fixed base $1 + z/N$ but for a base $1 + z_N/N$ whose numerator z_N itself drifts (here because the Taylor remainder in the transform depends on N), and whose value is complex (because a characteristic function is). Both generalizations are free, and the principal-branch logarithm is what makes them so.

Lemma 3.23 (Exponential limit of a perturbed power)**.** *Let $(z_N)_{N\geq 1}$ be a sequence of complex numbers with $z_N \to z \in \mathbb{C}$. Then*

$$\Bigl(1 + \frac{z_N}{N}\Bigr)^N \;\longrightarrow\; e^z \qquad (N \to \infty).$$

Proof. A convergent sequence is bounded, so $M := \sup_N |z_N| < \infty$, and hence $|z_N/N| \leq M/N \leq \frac{1}{2}$ for all $N \geq 2M$. From here on take N that large; then the base $1 + z_N/N$ lies within distance $\frac{1}{2}$ of 1, in particular in the open right half-plane, safely off the negative real axis where the principal logarithm log has its branch cut. On that region the principal branch is analytic and satisfies $(1 + w)^N = \exp\left(N \log(1 + w)\right)$ for the genuine N-th power, with the power series $\log(1 + w) =$

$\sum_{k\geq 1}(-1)^{k-1}w^k/k$. Subtracting its linear term and estimating the tail,

$$|\log(1+w)-w| = \Big|\sum_{k\geq 2}\frac{(-1)^{k-1}w^k}{k}\Big| \leq \sum_{k\geq 2}\frac{|w|^k}{k} \leq \frac{|w|^2}{2}\sum_{k\geq 0}|w|^k = \frac{|w|^2}{2(1-|w|)} \leq |w|^2 \qquad (|w|\leq \tfrac{1}{2}),$$

the last step because $1-|w|\geq \frac{1}{2}$. Apply this with $w = z_N/N$ and multiply by N:

$$N\log\Big(1+\frac{z_N}{N}\Big) = z_N + N\,r_N, \qquad |r_N| \leq \Big|\frac{z_N}{N}\Big|^2, \quad \text{so} \quad |N\,r_N| \leq \frac{|z_N|^2}{N} \leq \frac{M^2}{N} \xrightarrow[N\to\infty]{} 0.$$

Therefore $N\log(1+z_N/N) = z_N + Nr_N \to z+0 = z$, and since the exponential is continuous, $(1+z_N/N)^N = \exp\left(N\log(1+z_N/N)\right) \to e^z$. The principal branch was essential exactly once: it gave a single-valued logarithm with the quadratic remainder bound, so that the error Nr_N accumulated over N factors stays of size $O(1/N)$ rather than threatening to blow up. □

With the exponential limit dispatched, the central limit theorem is three short moves: center, factor, expand-and-exponentiate.

Proof of theorem 3.22. Centering. The theorem concerns the sum of the deviations $X_i - m_\star$, so set $Y_i = X_i - m_\star$ at the outset. Subtracting the mean makes the Y_i i.i.d. with $\mathbb{E}[Y_i] = 0$ and $\mathbb{E}[Y_i^2] = \mathrm{Var}(X_1) = \sigma^2$, and this centering is not cosmetic: it is what removes the first-order term from the Taylor expansion below, leaving the variance as the leading datum. Write $\varphi(t) = \varphi_{Y_1}(t)$ for the common characteristic function of the centered variables and $S_N = N^{-1/2}\sum_{i\leq N} Y_i$ for the rescaled sum, so that the claim is $\varphi_{S_N}(t) \to e^{-\sigma^2t^2/2}$ followed by an appeal to continuity.

The product form. Here independence pays off. The summands $Y_i/\sqrt{N}$ are independent, so the transform of their sum factorizes by proposition 3.19(3); and because the Y_i are identically distributed, every factor is the *same* function φ evaluated at the common rescaled frequency $t/\sqrt{N}$. Concretely, $\varphi_{Y_i/\sqrt{N}}(t) = \mathbb{E}\Big[e^{i(t/\sqrt{N})Y_i}\Big] = \varphi(t/\sqrt{N})$ for each i, and multiplying the N identical factors,

$$\varphi_{S_N}(t) = \mathbb{E}\Big[\exp\Big(it\,\tfrac{1}{\sqrt{N}}\textstyle\sum_{i\leq N} Y_i\Big)\Big] = \prod_{i=1}^{N}\mathbb{E}\Big[e^{i(t/\sqrt{N})Y_i}\Big] = \Big(\varphi(t/\sqrt{N})\Big)^N.$$

This is the whole reason for working with transforms: the intractable N-fold convolution that is the law of S_N has become the N-th power of a single number, an object the exponential limit was built to handle.

The second-order expansion. Fix t and let $N\to\infty$, so the argument $t/\sqrt{N}\to 0$ slides into the origin, precisely where (3.10) controls φ. Since $Y_1 \in L^2$ with mean 0 and second moment σ^2, that expansion with its first-order term gone reads, at the point $t/\sqrt{N}$,

$$\varphi(t/\sqrt{N}) = 1 + i\,\frac{t}{\sqrt{N}}\underbrace{\mathbb{E}[Y_1]}_{=0} - \frac{1}{2}\Big(\frac{t}{\sqrt{N}}\Big)^2\underbrace{\mathbb{E}\big[Y_1^2\big]}_{=\sigma^2} + o\Big(\frac{t^2}{N}\Big) = 1 - \frac{\sigma^2t^2}{2N} + o\Big(\frac{1}{N}\Big),$$

where with t fixed the Peano remainder $o((t/\sqrt{N})^2)$ is $o(1/N)$. The centering has done its work: the $i(t/\sqrt{N})\mathbb{E}[Y_1]$ term, which would have been of order $N^{-1/2}$ and dominated everything, has vanished, and what governs the factor to leading order is the variance.

Exponentiation. Cast the factor in the shape the lemma wants by writing $\varphi(t/\sqrt{N}) = 1 + z_N/N$ with

$$z_N \;=\; N\Big(\varphi(t/\sqrt{N}) - 1\Big) \;=\; -\tfrac{1}{2}\sigma^2 t^2 + o(1) \;\xrightarrow[N\to\infty]{}\; z := -\tfrac{1}{2}\sigma^2 t^2.$$

Then $\varphi_{S_N}(t) = (1 + z_N/N)^N$, and lemma 3.23 applies verbatim, giving

$$\varphi_{S_N}(t) \;\longrightarrow\; e^z = e^{-\sigma^2 t^2/2} \qquad \text{for every } t \in \mathbb{R}.$$

Identification of the limit. The limit function $t \mapsto e^{-\sigma^2 t^2/2}$ is continuous at $t = 0$ — indeed everywhere — so it meets the one hypothesis Lévy's continuity theorem in the form theorem 3.21(b) imposes. That theorem therefore certifies, with no further computation, two things at once: that the limit *is* the characteristic function of an honest probability measure, and that this measure is the weak limit of the laws of S_N. It remains only to name it. That $e^{-\sigma^2 t^2/2}$ is the characteristic function of $\mathcal{N}(0, \sigma^2)$ is the completion of a single Gaussian integral — a prerequisite computation: with the density $(2\pi\sigma^2)^{-1/2} e^{-x^2/2\sigma^2}$ one completes the square in $\int e^{itx - x^2/2\sigma^2}\,\mathrm{d}x$ and reads off $e^{-\sigma^2 t^2/2}$ — and uniqueness (proposition 3.19(4)) leaves no other candidate carrying that transform. Hence $S_N \Rightarrow \mathcal{N}(0, \sigma^2)$, which is the theorem. □

It is worth pausing on *why* the variance, and nothing finer, survives, because that is the message the central limit theorem actually carries and the one the mean field theory exploits. Suppose the Y_i had a finite third moment $\mathbb{E}[Y_1^3] = \kappa$. Carrying the Taylor expansion (proposition 3.19(5)) one order further, the single factor would read

$$\varphi(t/\sqrt{N}) = 1 - \frac{\sigma^2 t^2}{2N} \underbrace{- \frac{i\kappa\, t^3}{6\, N^{3/2}}}_{\text{third moment}} + o\Big(\frac{1}{N}\Big),$$

and the third-moment term enters at order $N^{-3/2}$. When the N factors are multiplied — equivalently, when one passes to $z_N = N(\varphi(t/\sqrt{N}) - 1)$ — that term contributes $N \cdot O(N^{-3/2}) = O(N^{-1/2})$, which vanishes in the limit; and the same accounting kills every moment beyond the second, since the p-th moment sits at order $N^{-p/2}$ and survives multiplication by N only when $p \le 2$. The $\sqrt{N}$ rescaling is calibrated exactly so that the second moment lands at the critical order N^{-1}, neither blowing up (as the uncentered mean would) nor washing out (as every higher moment does). This is the content of the word *universality*: the limit law depends on the microscopic distribution of the Y_i *only* through its variance σ^2, and two ensembles with wildly different microscopic laws but the same variance produce the *same* Gaussian fluctuation. The Gaussian is not one distribution among many that happens to arise; it is the unique fixed shape that the rescaling selects, the only thing that can survive the addition of many small independent contributions.

This bare i.i.d. statement is the seed, but it is not yet phrased in the language the mean field theory speaks. The variance computation of section 3.1 already told us that the empirical average of N independent agents concentrates on its mean at rate $N^{-1/2}$; the theorem just proved upgrades that second-moment fact into a full distributional law, naming the Gaussian profile of the residual. The next subsection cashes this in. It takes a scalar observable of a mean field ensemble — a test function averaged over N independent agents drawn from the population law m — and reads theorem 3.22 as the precise statement that the observable equals its mean field value plus a fluctuation of size

$N^{-1/2}$ and Gaussian shape. That is the non-interacting skeleton of propagation of chaos: the clean picture in which the agents are genuinely independent, against which chapter 7 and chapter 13 will later measure the interacting refinement, where the coupling makes them only *asymptotically* independent and the fluctuation analysis becomes correspondingly delicate.

3.4.3 Fluctuations of a mean field observable

We can now state precisely what "the mean field is deterministic and fluctuations are $O(N^{-1/2})$" means, in the cleanest setting: an ensemble of N independent agents and a single scalar observable. This is the non-interacting skeleton of propagation of chaos; the interacting refinement, where the agents are coupled and only *asymptotically* independent, is the subject of chapter 7 and chapter 13.

The almost-sure side of the statement is a strong law of large numbers. The full Kolmogorov strong law holds under only $\mathbb{E}[|X_1|] < \infty$, but its proof (truncation, Kolmogorov's maximal inequality, Kronecker's lemma) is heavier than this book needs: every application here — to bounded test functions and to the Gaussian thread — has random variables with all moments finite. We therefore prove the version that the Borel–Cantelli lemma of this chapter delivers in a few lines, under a finite fourth moment, and use *it* as the in-book strong law.

Proposition 3.24 (Strong law under a fourth-moment bound)**.** *Let $(X_i)_{i\geq 1}$ be i.i.d. real random variables with $\mathbb{E}[X_1^4] < \infty$ and mean $m_\star = \mathbb{E}[X_1]$. Then $\overline{X}_N = \frac{1}{N}\sum_{i\leq N} X_i \to m_\star$ almost surely.*

Proof. Replacing X_i by $X_i - m_\star$ (which preserves independence and a finite fourth moment), we may assume $m_\star = 0$. Put $S_N = \sum_{i\leq N} X_i$. Expanding,

$$\mathbb{E}\left[S_N^4\right] = \sum_{i,j,k,l\leq N} \mathbb{E}[X_i X_j X_k X_l]\,.$$

By independence and $\mathbb{E}[X_i] = 0$, a term vanishes unless every index that appears does so at least twice; the surviving terms are the N terms $\mathbb{E}[X_i^4]$ and, for each ordered pair $i \neq j$, the term $\mathbb{E}[X_i^2]\,\mathbb{E}\left[X_j^2\right]$ counted with multiplicity 3 (the number of ways to split four ordered positions into two like-indexed pairs), i.e. $3N(N-1)$ such terms in all. Hence, using $\mathbb{E}[X_1^2]^2 \leq \mathbb{E}[X_1^4]$ (Cauchy–Schwarz),

$$\mathbb{E}\left[S_N^4\right] = N\,\mathbb{E}\left[X_1^4\right] + 3N(N-1)\mathbb{E}\left[X_1^2\right]^2 \leq C\,N^2, \qquad C := \mathbb{E}\left[X_1^4\right] + 3\mathbb{E}\left[X_1^4\right] = 4\mathbb{E}\left[X_1^4\right].$$

By Markov's inequality, for each $\varepsilon > 0$,

$$\mathbb{P}(\left|\overline{X}_N\right| > \varepsilon) = \mathbb{P}(S_N^4 > N^4\varepsilon^4) \leq \frac{\mathbb{E}[S_N^4]}{N^4\varepsilon^4} \leq \frac{C}{N^2\varepsilon^4},$$

which is summable in N. The first Borel–Cantelli lemma (theorem 3.27(i)) gives $\mathbb{P}(\left|\overline{X}_N\right| > \varepsilon \text{ i.o.}) = 0$; intersecting over $\varepsilon = 1/m$, $m \in \mathbb{N}$, yields $\overline{X}_N \to 0$ a.s.. This is the unbounded-variable extension of the bounded strong law lemma 1.37, by the identical fourth-moment method. $\square$

Corollary 3.25 (Law of large numbers and Gaussian fluctuations)**.** *Let $(X_i)_{i\geq 1}$ be i.i.d. real random variables with mean $m_\star = \mathbb{E}[X_1]$ and finite variance $\sigma^2 = \mathrm{Var}(X_1) \in (0,\infty)$, and set $\overline{X}_N = \frac{1}{N}\sum_{i=1}^N X_i$. Then*

$$\sqrt{N}\,(\overline{X}_N - m_\star) \;\Rightarrow\; \mathcal{N}(0,\sigma^2),$$

so $\overline{X}_N = m_\star + N^{-1/2}\,\xi_N$ with $\xi_N \Rightarrow \mathcal{N}(0,\sigma^2)$: the empirical average sits at its deterministic limit $m_\star$ up to a random correction of typical size $\sigma\,N^{-1/2}$ with a universal Gaussian profile. If in addition $\mathbb{E}[X_1^4] < \infty$ then the convergence to $m_\star$ is almost sure, $\overline{X}_N \xrightarrow{a.s.} m_\star$.

Proof. The fluctuation statement is theorem 3.22 after dividing by $\sqrt{N}$: $\sqrt{N}(\overline{X}_N - m_\star) = N^{-1/2}\sum_{i\leq N}(X_i - m_\star)$. Under the extra fourth-moment hypothesis, the almost-sure convergence is proposition 3.24. ☐

Physics note

This corollary is the rigorous content of *self-averaging* and of the suppression of fluctuations in the thermodynamic limit. An intensive observable of N independent constituents — here the per-agent average $\overline{X}_N$ — converges to its ensemble value $m_\star$, and the relative fluctuation $\sqrt{\mathrm{Var}(\overline{X}_N)}/\,|m_\star| = \sigma/(|m_\star|\,\sqrt{N})$ vanishes as $N^{-1/2}$, with the central limit theorem upgrading the variance estimate to a full Gaussian law for the rescaled deviation. In a mean field game the "constituents" are agents and the observable is the population's empirical distribution; the entire justification of replacing the N-player game by its $N = \infty$ limit rests on this $N^{-1/2}$ smallness, and the deviation of the finite game's equilibrium from the mean field equilibrium inherits the same scale. The one genuine subtlety, deferred to chapter 7, is that the agents are *not* exactly independent at finite N — the common field couples them — so the clean i.i.d. statement above must be replaced by propagation of chaos, which proves asymptotic independence and recovers the same $N^{-1/2}$ rate.

It is the empirical *measure*, not just the empirical mean, that the mean field theory tracks. The following worked computation upgrades corollary 3.25 from a single observable to the full random measure, and connects to the Gaussian initial law of the carried LQG thread.

Worked computation 3.26 (Empirical measure of an i.i.d. sample and its fluctuations)**.** Let $X_1,\dots,X_N$ be i.i.d. draws from a probability measure $\mu \in \mathcal{P}(\mathbb{R}^d)$, and form the random empirical measure

$$\mu^N = \frac{1}{N}\sum_{i=1}^N \delta_{X_i} \in \mathcal{P}(\mathbb{R}^d).$$

We claim two facts.

(1) Almost-sure weak convergence. As $N \to \infty$, $\mu^N \Rightarrow \mu$ almost surely. Indeed, fix $f \in C_b(\mathbb{R}^d)$. The random variables $f(X_i)$ are i.i.d. and *bounded* ($|f| \leq \|f\|_\infty$), hence have a finite fourth moment, so the in-book strong law proposition 3.24 gives $\mu^N(f) = \frac{1}{N}\sum_i f(X_i) \to \mu(f)$ a.s.. The null set on which this fails may depend on f; to obtain a *single* null set valid for all $f \in C_b$ simultaneously — which is what $\mu^N \Rightarrow \mu$ requires — we use the countable convergence-determining family for $\mathbb{R}^d$ proved in chapter 1: there is a fixed countable set $\{f_k\} \subseteq C_b(\mathbb{R}^d)$ such that $\nu_n(f_k) \to \nu(f_k)$ for all k

implies $\nu_n \Rightarrow \nu$ (lemma 1.34). Applying the strong law to each f_k and intersecting the countably many null sets gives a single full-measure event on which $\mu^N(f_k) \to \mu(f_k)$ for every k; on that event $\mu^N \Rightarrow \mu$ by lemma 1.34. This almost-sure weak law of large numbers for empirical measures is Varadarajan's theorem, and the metrizability of theorem 3.12 is what makes "a single null set works for all test functions" the right formulation.

(2) Gaussian fluctuations of a linear observable. Fix a single $f \in C_b(\mathbb{R}^d)$ with $\sigma_f^2 := \mathrm{Var}_\mu(f) = \mu(f^2) - \mu(f)^2 \in (0, \infty)$. Then, directly from theorem 3.22 applied to the i.i.d. variables $f(X_i)$,

$$\sqrt{N}\,\big(\mu^N(f) - \mu(f)\big) \;\Rightarrow\; \mathcal{N}(0, \sigma_f^2). \tag{3.14}$$

Thus each smooth "measurement" of the empirical measure fluctuates about its limit at scale $N^{-1/2}$ with a Gaussian profile whose variance is the μ-variance of the test function. (Letting f range over a family yields a Gaussian *field* indexed by test functions, the empirical-measure central limit theorem; we do not need that refinement here and defer it to chapter 13.)

Connection to Thread A. In the linear–quadratic–Gaussian thread the initial law m_0 of the state (the distribution $X_0 \sim m_0$ fixed in the thread specification) is taken Gaussian; we write it as $m_0 = \mathcal{N}(\bar{m}_0, v_0)$ on $\mathbb{R}$, with $\bar{m}_0$ its mean (consistent with the population-mean symbol $\bar{m}_t$ of the thread) and $v_0 > 0$ its *variance* (the same variance-parametrization v_t used for the evolving Gaussian in chapter 4 and chapter 14; we deliberately avoid a standard-deviation symbol that would clash with the noise volatility σ of the thread). Sampling $X_1, \ldots, X_N \sim m_0$ i.i.d. models an N-agent initialization. A Gaussian has all moments finite, so corollary 3.25 applies directly to the unbounded observable $f(x) = x$ (no truncation is needed): the empirical mean obeys

$$\overline{X}_N \xrightarrow{\text{a.s.}} \bar{m}_0, \qquad \sqrt{N}\,\big(\overline{X}_N - \bar{m}_0\big) \;\Rightarrow\; \mathcal{N}(0, v_0),$$

so the random initial mean that an actual N-agent system would feed into the forward dynamics differs from the idealized mean field value $\bar{m}_0$ by a Gaussian fluctuation of size $\sqrt{v_0}\,N^{-1/2}$. This is the precise sense in which the deterministic LQG mean field equations of chapter 14 are the $N = \infty$ limit of a fluctuating finite system, and it fixes the scale of the correction that chapter 13 controls.

3.5 Two concentration tools: Borel–Cantelli and Azuma–Hoeffding

The central limit theorem describes the *typical* size of a fluctuation; later chapters also need control of *rare* fluctuations and of *almost-sure* behaviour. Two elementary but indispensable tools provide this. The first, the Borel–Cantelli lemmas, converts summability of probabilities into almost-sure statements; we use it in chapter 19 to upgrade convergence in probability of sampled homomorphism densities to almost-sure convergence. The second, the Azuma–Hoeffding inequality, gives exponential tail bounds for functions that depend on many independent inputs with bounded sensitivity to each; it is the engine behind the concentration of motif counts in W-random graphs (chapter 19). Both are stated and proved here so that the development is self-contained.

3.5.1 The Borel–Cantelli lemmas

For a sequence of events $(A_n)_{n\geq 1}$ write $\{A_n \text{ i.o.}\} = \limsup_n A_n = \bigcap_{N\geq 1}\bigcup_{n\geq N} A_n$ for the event that infinitely many of the A_n occur.

Theorem 3.27 (Borel–Cantelli)**.** *Let $(A_n)_{n\geq 1}$ be events on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.*

(i) If $\sum_{n\geq 1} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$; equivalently, almost surely only finitely many A_n occur.

(ii) If the A_n are independent *and $\sum_{n\geq 1} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.*

Proof. (i) For every N, monotonicity and countable subadditivity give $\mathbb{P}(\bigcup_{n\geq N} A_n) \leq \sum_{n\geq N} \mathbb{P}(A_n)$, and the right-hand side is the tail of a convergent series, hence tends to 0 as $N \to \infty$. Since $\{A_n \text{ i.o.}\} \subseteq \bigcup_{n\geq N} A_n$ for every N, we get $\mathbb{P}(A_n \text{ i.o.}) \leq \inf_N \sum_{n\geq N} \mathbb{P}(A_n) = 0$.

(ii) It suffices to show $\mathbb{P}(\bigcup_{n\geq N} A_n) = 1$ for each N, i.e. $\mathbb{P}(\bigcap_{n\geq N} A_n^{\mathrm{c}}) = 0$. By independence and the elementary bound $1 - x \leq e^{-x}$, for any $M > N$,

$$\mathbb{P}\Big(\bigcap_{n=N}^{M} A_n^{\mathrm{c}}\Big) = \prod_{n=N}^{M} (1 - \mathbb{P}(A_n)) \leq \exp\Big(-\sum_{n=N}^{M} \mathbb{P}(A_n)\Big) \xrightarrow[M\to\infty]{} 0,$$

because the exponent diverges. Letting $M \to \infty$ gives $\mathbb{P}(\bigcap_{n\geq N} A_n^{\mathrm{c}}) = 0$, as required. □

3.5.2 The Azuma–Hoeffding inequality

Recall (it is developed in detail in section 3.6) that a sequence $(M_i)_{i\geq 0}$ adapted to a filtration $(\mathcal{F}_i)_{i\geq 0}$ is a *martingale* if $\mathbb{E}|M_i| < \infty$ and $\mathbb{E}[M_i \mid \mathcal{F}_{i-1}] = M_{i-1}$. The differences $D_i = M_i - M_{i-1}$ then satisfy $\mathbb{E}[D_i \mid \mathcal{F}_{i-1}] = 0$.

Theorem 3.28 (Azuma–Hoeffding)**.** *Let $(M_i)_{0\leq i\leq n}$ be a martingale with respect to $(\mathcal{F}_i)$ whose increments $D_i = M_i - M_{i-1}$ are bounded, $|D_i| \leq c_i$ almost surely for constants $c_i > 0$. Then for every $t > 0$,*

$$\mathbb{P}(M_n - M_0 \geq t) \;\leq\; \exp\Big(-\frac{t^2}{2\sum_{i=1}^{n} c_i^2}\Big),$$

and the same bound holds for $\mathbb{P}(M_n - M_0 \leq -t)$; consequently $\mathbb{P}(|M_n - M_0| \geq t) \leq 2\exp(-t^2/(2\sum_i c_i^2))$.

Proof. The key estimate is Hoeffding's lemma in conditional form: if D is a random variable with $\mathbb{E}[D \mid \mathcal{F}'] = 0$ and $|D| \leq c$ almost surely, then for every $\lambda \in \mathbb{R}$,

$$\mathbb{E}\Big[e^{\lambda D} \mid \mathcal{F}'\Big] \;\leq\; e^{\lambda^2 c^2/2} \qquad \text{a.s.} \tag{3.15}$$

Granting (3.15), fix $\lambda > 0$ and write $S_k = M_k - M_0 = \sum_{i\leq k} D_i$. Conditioning on $\mathcal{F}_{n-1}$ and using $S_n = S_{n-1} + D_n$ with S_{n-1} being $\mathcal{F}_{n-1}$-measurable,

$$\mathbb{E}\Big[e^{\lambda S_n}\Big] = \mathbb{E}\Big[e^{\lambda S_{n-1}}\,\mathbb{E}\Big[e^{\lambda D_n} \mid \mathcal{F}_{n-1}\Big]\Big] \leq e^{\lambda^2 c_n^2/2}\,\mathbb{E}\Big[e^{\lambda S_{n-1}}\Big].$$

Iterating down to $S_0 = 0$ yields $\mathbb{E}\Big[e^{\lambda S_n}\Big] \le \exp\big(\frac{\lambda^2}{2}\sum_{i=1}^n c_i^2\big)$. By Markov's inequality applied to $e^{\lambda S_n}$,

$$\mathbb{P}(S_n \ge t) \le e^{-\lambda t}\,\mathbb{E}\Big[e^{\lambda S_n}\Big] \le \exp\Big(-\lambda t + \tfrac{\lambda^2}{2}\textstyle\sum_i c_i^2\Big).$$

Optimizing over $\lambda > 0$ gives the minimizer $\lambda = t/\sum_i c_i^2$ and the stated bound. The lower-tail bound follows by applying the result to the martingale $-M_i$, and the two-sided bound by a union bound.

It remains to prove (3.15). By convexity of $x \mapsto e^{\lambda x}$ on $[-c, c]$, $e^{\lambda x} \le \frac{c-x}{2c}e^{-\lambda c} + \frac{c+x}{2c}e^{\lambda c}$; taking the conditional expectation and using $\mathbb{E}[D \mid \mathcal{F}'] = 0$ gives $\mathbb{E}\Big[e^{\lambda D} \mid \mathcal{F}'\Big] \le \frac{1}{2}(e^{\lambda c} + e^{-\lambda c}) = \cosh(\lambda c)$. The elementary inequality $\cosh(u) \le e^{u^2/2}$ (compare the power series term by term, $1/(2k)! \le 1/(2^k k!)$) with $u = \lambda c$ yields (3.15). □

Remark 3.29. The form used in applications is the *bounded-differences* (McDiarmid) inequality: if $Z = g(X_1, \dots, X_n)$ with independent X_i and g changes by at most c_i when its ith argument is altered, then the Doob martingale $M_i = \mathbb{E}[Z \mid X_1, \dots, X_i]$ has increments bounded by c_i, so theorem 3.28 gives $\mathbb{P}(|Z - \mathbb{E}Z| \ge t) \le 2\exp(-t^2/(2\sum_i c_i^2))$. This is exactly the device by which chapter 19 controls the deviation of a sampled homomorphism density from its mean.

3.6 Conditional expectation as orthogonal projection

The Azuma–Hoeffding inequality of the previous section was stated for martingales, and a martingale is built on conditional expectation — the one construction this chapter has so far invoked only on credit. We discharge that debt now, with the last of the four tools. Conditional expectation is the rigorous version of "average X given the information $\mathcal{G}$," it is the building block of filtrations, martingales, and the backward stochastic differential equations of chapter 9 and chapter 23, and — the point we emphasize, because it makes the abstract definition transparent to a reader who knows Hilbert spaces — in the L^2 case it is *nothing but the orthogonal projection of chapter 2.*

3.6.1 The L^2 construction

Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra. The space $L^2(\Omega, \mathcal{G}, \mathbb{P})$ of (equivalence classes of) square-integrable, $\mathcal{G}$-measurable random variables is a *closed* linear subspace of the Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$ equipped with the inner product $\langle X, Y\rangle = \mathbb{E}[XY]$: closedness holds because an L^2-limit of $\mathcal{G}$-measurable functions has a $\mathcal{G}$-measurable representative (an a.s. limit along a subsequence). Closed subspaces of a Hilbert space carry an orthogonal projection (the projection-onto-a-closed-convex-set theorem, theorem 2.5), and this is the definition.

Definition 3.30 (Conditional expectation, L^2 case)**.** For $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, the *conditional expectation* $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace $L^2(\Omega, \mathcal{G}, \mathbb{P})$: the unique $\mathcal{G}$-measurable $Y \in L^2$ minimizing $\mathbb{E}[(X - Y)^2]$, equivalently the unique $\mathcal{G}$-measurable $Y \in L^2$ satisfying the orthogonality relations

$$\mathbb{E}[(X - Y)\,Z] = 0 \qquad \text{for every } Z \in L^2(\Omega, \mathcal{G}, \mathbb{P}). \tag{3.16}$$

Taking $Z = \mathbf{1}_A$ for $A \in \mathcal{G}$, the defining relation (3.16) reads

$$\int_A \mathbb{E}[X \mid \mathcal{G}] \, \mathrm{d}\mathbb{P} = \int_A X \, \mathrm{d}\mathbb{P} \qquad \text{for all } A \in \mathcal{G}, \tag{3.17}$$

which is the form most textbooks take as the *definition* of conditional expectation. We have instead *derived* it from orthogonal projection, which explains both the existence (the projection theorem) and the variational meaning (best mean-square $\mathcal{G}$-measurable predictor of X). The geometry is the whole content: $\mathbb{E}[X \mid \mathcal{G}]$ is the foot of the perpendicular from X to the subspace of variables one is "allowed to know," and the residual $X - \mathbb{E}[X \mid \mathcal{G}]$ is orthogonal to everything knowable.

3.6.2 Extension to L^1 and the basic properties

The L^2 projection extends to all integrable X, because it is an L^1-contraction.

Theorem 3.31 (Existence, uniqueness, and properties of conditional expectation)**.** *Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra and $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$. There exists a $\mathbb{P}$-a.s. unique $\mathcal{G}$-measurable integrable random variable $\mathbb{E}[X \mid \mathcal{G}]$ satisfying* (3.17)*, agreeing with definition 3.30 when $X \in L^2$. It satisfies, almost surely:*

1. (Linearity) $\mathbb{E}[aX + bX' \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[X' \mid \mathcal{G}]$*;*
2. (Positivity and contraction) $X \geq 0 \Rightarrow \mathbb{E}[X \mid \mathcal{G}] \geq 0$*, and* $\mathbb{E}[\,|\mathbb{E}[X \mid \mathcal{G}]|\,] \leq \mathbb{E}[|X|]$*;*
3. (Tower property) *if* $\mathcal{H} \subseteq \mathcal{G}$ *then* $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$*; in particular* $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$*;*
4. (Taking out what is known) *if Z is $\mathcal{G}$-measurable and $ZX \in L^1$, $Z \in L^\infty$, then* $\mathbb{E}[ZX \mid \mathcal{G}] = Z\,\mathbb{E}[X \mid \mathcal{G}]$*;*
5. (Role of independence) *if* $\sigma(X) \perp\!\!\!\perp \mathcal{G}$ *then* $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$*;*
6. (Conditional Jensen) *for convex* $\phi : \mathbb{R} \to \mathbb{R}$ *with* $\phi(X) \in L^1$*,* $\phi\big(\mathbb{E}[X \mid \mathcal{G}]\big) \leq \mathbb{E}[\phi(X) \mid \mathcal{G}]$.

Proof. Existence/uniqueness. For $X \in L^2$ the variable $\mathbb{E}[X \mid \mathcal{G}]$ of definition 3.30 exists and satisfies (3.17). For $X \geq 0$ in L^1, approximate by $X_k = X \wedge k \in L^2$; the projections $\mathbb{E}[X_k \mid \mathcal{G}]$ are non-decreasing in k (positivity, proved next) and their limit Y is $\mathcal{G}$-measurable with $\int_A Y = \int_A X$ for $A \in \mathcal{G}$ by monotone convergence on both sides of (3.17); in particular $\mathbb{E}[Y] = \mathbb{E}[X] < \infty$, so $Y \in L^1$. For general $X \in L^1$ set $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X^+ \mid \mathcal{G}] - \mathbb{E}[X^- \mid \mathcal{G}]$. Uniqueness: if Y, Y' are $\mathcal{G}$-measurable and both satisfy (3.17), then $\int_A (Y - Y') \, \mathrm{d}\mathbb{P} = 0$ for all $A \in \mathcal{G}$; taking $A = \{Y - Y' > 0\} \in \mathcal{G}$ and $A = \{Y - Y' < 0\} \in \mathcal{G}$ forces $Y = Y'$ a.s..

Positivity is built into the construction: if $X \geq 0$ then on $A = \{\mathbb{E}[X \mid \mathcal{G}] < 0\} \in \mathcal{G}$ one has $\int_A \mathbb{E}[X \mid \mathcal{G}] = \int_A X \geq 0$, impossible unless $\mathbb{P}(A) = 0$. The contraction follows from $|\mathbb{E}[X \mid \mathcal{G}]| \leq \mathbb{E}[X^+ \mid \mathcal{G}] + \mathbb{E}[X^- \mid \mathcal{G}] = \mathbb{E}[|X| \mid \mathcal{G}]$ and taking expectations via (3.17) with $A = \Omega$.

Linearity is inherited from the linearity of the projection (on L^2) and of the integral characterization (on L^1). *Tower*: the right side is $\mathcal{H}$-measurable and, for $A \in \mathcal{H} \subseteq \mathcal{G}$, $\int_A \mathbb{E}[X \mid \mathcal{H}] = \int_A X = \int_A \mathbb{E}[X \mid \mathcal{G}]$ by applying (3.17) twice; uniqueness then identifies it with $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}]$. *Taking out* *what is known*: for $Z = \mathbf{1}_B$, $B \in \mathcal{G}$, and $A \in \mathcal{G}$, $\int_A Z\,\mathbb{E}[X \mid \mathcal{G}] = \int_{A\cap B} \mathbb{E}[X \mid \mathcal{G}] = \int_{A\cap B} X = \int_A ZX$, so $Z\mathbb{E}[X \mid \mathcal{G}]$ satisfies the defining property of $\mathbb{E}[ZX \mid \mathcal{G}]$; extend to simple, then bounded $\mathcal{G}$-measurable Z by linearity and dominated convergence. *Independence*: if $\sigma(X) \perp\!\!\!\perp \mathcal{G}$ then for $A \in \mathcal{G}$, $\int_A X = \mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[X]\ \mathbb{P}(A) = \int_A \mathbb{E}[X]$ by (3.3), so the constant $\mathbb{E}[X]$ satisfies the defining property. *Conditional Jensen*: write the convex ϕ as the supremum of a countable family of affine minorants $\phi(x) \geq a_k x + b_k$; apply linearity and positivity to each, $\mathbb{E}[\phi(X) \mid \mathcal{G}] \geq a_k\mathbb{E}[X \mid \mathcal{G}] + b_k$, and take the supremum over k. Full treatment: [132, Ch. 9], [84, Ch. 8]. □

Remark 3.32 (Why this is the prerequisite it is)**.** The orthogonality (3.16) is the abstract martingale property: a process $M_t = \mathbb{E}[X \mid \mathcal{F}_t]$ along an increasing family of σ-algebras (a filtration $\mathbb{F} = (\mathcal{F}_t)$, defined in chapter 5) has orthogonal increments and is the canonical example of a martingale. The tower property is the consistency that makes the conditional law of a Markov process well-defined, and "taking out what is known" is the rule by which the diffusion term of an Itô integral has conditional mean zero. In a backward stochastic differential equation the unknown is precisely a pair (Y_t, Z_t) realizing a prescribed terminal value as a conditional expectation evolving backward in time; the contraction property (2) is what makes the fixed-point map of chapter 9 well-posed. Nothing in the present chapter uses a filtration, but every property proved here is consumed there.

3.6.3 Regular conditional distributions and disintegration

Conditional expectation tells us the average of a single integrable variable given $\mathcal{G}$. For many purposes — in particular for the gluing of couplings used to build the Wasserstein metric in chapter 4 — one wants the entire *conditional law*: a genuine probability measure $\kappa(\omega, \cdot)$ on the state space that plays the role of "the distribution of X given the information $\mathcal{G}$," measurably in ω. On $\mathbb{R}^d$ such an object always exists, and it is built from conditional expectations of indicator functions by the same construction that turns a distribution function into a measure. We record it here because chapter 4 imports it by name.

Theorem 3.33 (Regular conditional distributions on $\mathbb{R}^d$)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra, and $X : \Omega \to \mathbb{R}^d$ a random variable. Then there is a map $\kappa : \Omega \times \mathcal{B}(\mathbb{R}^d) \to [0, 1]$, the* regular conditional distribution *of X given $\mathcal{G}$, such that*

(i) *for each ω, $B \mapsto \kappa(\omega, B)$ is a probability measure on $\mathbb{R}^d$;*

(ii) *for each Borel B, $\omega \mapsto \kappa(\omega, B)$ is $\mathcal{G}$-measurable and equals $\mathbb{P}(X \in B \mid \mathcal{G})$ a.s.;*

and consequently $\mathbb{E}[g(X) \mid \mathcal{G}] = \int_{\mathbb{R}^d} g\, \mathrm{d}\kappa(\cdot,)$ a.s. for every bounded Borel g. In particular, if $\pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ has second marginal ν, there is a Borel kernel $y \mapsto \pi_y \in \mathcal{P}(\mathbb{R}^d)$ with

$$\int_{\mathbb{R}^d\times\mathbb{R}^d} h(x, y)\ \mathrm{d}\pi(x, y) = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} h(x, y)\, \pi_y(\mathrm{d}x)\, \nu(\mathrm{d}y) \qquad (h \textit{ bounded Borel}), \tag{3.18}$$

the disintegration *of π over its second marginal.*

Proof. For each rational vector $q \in \mathbb{Q}^d$ choose a version $F(q, \cdot)$ of the conditional expectation $\mathbb{E}[\mathbf{1}\{X \leq q\} \mid \mathcal{G}]$ (coordinatewise order $X \leq q$). Using the almost-sure linearity, positivity, and conditional monotone convergence of theorem 3.31 on the countable index set $\mathbb{Q}^d$, there is a $\mathbb{P}$-null set off which, simultaneously for all rationals: $F(q,\omega) \le F(q',\omega)$ whenever $q \le q'$; $F(q,\omega) \to 0$ as some coordinate $q_j \to -\infty$ and $F(q,\omega) \to 1$ as all $q_j \to +\infty$ (apply conditional monotone convergence to $\mathbf{1}\{X \le q\} \downarrow 0$ and $\uparrow 1$); and $F(q,\omega) = \lim_{q' \downarrow q,\, q' \in \mathbb{Q}^d} F(q',\omega)$ (right-continuity, again by conditional dominated convergence). Off this null set define, for $x \in \mathbb{R}^d$, $\widehat{F}(x,\omega) = \inf\{F(q,\omega) : q \in \mathbb{Q}^d,\ q \ge x\}$; then $\widehat{F}(\cdot,\omega)$ is a bona fide d-dimensional distribution function (nondecreasing, right-continuous, with the correct limits at $\pm\infty$), and by the distribution-function–to–measure correspondence (chapter 1, Lebesgue–Stieltjes) it is the cumulative function of a unique probability measure $\kappa(\omega,\cdot)$ on $\mathbb{R}^d$. On the null set set $\kappa(\omega,\cdot)$ equal to any fixed probability measure. This gives (i). For (ii), $\omega \mapsto \kappa(\omega,(-\infty,q]) = F(q,\omega)$ is $\mathcal{G}$-measurable and a version of $\mathbb{P}(X \le q \mid \mathcal{G})$ for each rational q; the class of Borel B with $\omega \mapsto \kappa(\omega,B)$ $\mathcal{G}$-measurable and equal to $\mathbb{P}(X \in B \mid \mathcal{G})$ a.s. is a λ-system containing the π-system of rational lower-left orthants, hence is all of $\mathcal{B}(\mathbb{R}^d)$ by the π–λ theorem (theorem 1.9). The identity $\mathbb{E}[g(X) \mid \mathcal{G}] = \int g \, \mathrm{d}\kappa$ follows for indicators, then for simple, then for bounded Borel g by linearity and dominated convergence.

For the disintegration statement, realize π as the law of a pair (X,Y) of $\mathbb{R}^d$-valued variables on some probability space (e.g. the identity maps on $(\mathbb{R}^d \times \mathbb{R}^d, \pi)$) and take $\mathcal{G} = \sigma(Y)$. By the Doob–Dynkin lemma a $\sigma(Y)$-measurable function is a Borel function of Y, so the kernel $\kappa(\cdot,B) = \mathbb{P}(X \in B \mid Y)$ has the form $\pi_Y(B)$ for a Borel map $y \mapsto \pi_y \in \mathcal{P}(\mathbb{R}^d)$. Then for bounded Borel h, $\mathbb{E}[h(X,Y)] = \mathbb{E}[\mathbb{E}[h(\cdot,Y)(X) \mid Y]] = \mathbb{E}[\int h(x,Y)\pi_Y(\mathrm{d}x)]$, which is exactly (3.18) since $\mathcal{L}(Y) = \nu$. □

Rigor ledger

Proved.

- Expectation as integration against the law, the change-of-variables formula (3.1), and the characterizations of independence (3.3)–(3.4) (section 3.1).
- The hierarchy of modes of convergence (lemma 3.6), including the uniform-continuity argument for convergence in probability $\Rightarrow$ in distribution and the Borel–Cantelli subsequence.
- The **Portmanteau theorem** (theorem 3.9) in full, all five equivalences, via the metric Urysohn approximation lemma 3.8 and a layer-cake/continuity-set argument; the **continuous mapping theorem** (corollary 3.11).
- **Metrization of weak convergence** by the bounded-Lipschitz metric, with $\mathcal{P}(S)$ Polish when S is (theorem 3.12); **Ulam** tightness of a single measure (lemma 3.15); and the **Skorokhod representation theorem** (theorem 3.13).
- **Prokhorov's theorem** (theorem 3.16) on a Polish space, *both* directions in full: the limit measure is built as an inner-regular content on the tightness compacts and extended by Carathéodory.
- Characteristic-function properties (proposition 3.19) including the **Lévy inversion/uniqueness formula** (lemma 3.20); **Lévy's continuity theorem** (theorem 3.21) and

the **i.i.d. central limit theorem** (theorem 3.22) in full via the Fourier route; the fourth-moment **strong law** (proposition 3.24); the $N^{-1/2}$ Gaussian fluctuation statement (corollary 3.25) and its measure-valued form (worked computation 3.26).

- **Conditional expectation** as orthogonal projection (definition 3.30), its L^1 extension and full property list (theorem 3.31), and **regular conditional distributions / disintegration** on $\mathbb{R}^d$ (theorem 3.33).

Assumed (imported).

- Lebesgue integration, monotone/dominated convergence, Fubini–Tonelli, Markov's inequality, the π–λ theorem, the Carathéodory extension theorem, the distribution-function–to–measure correspondence, and the countable convergence-determining family on $\mathbb{R}^d$ (lemma 1.34), all from chapter 1; the bounded strong law lemma 1.37 (whose fourth-moment method we reuse in proposition 3.24).
- Hilbert-space geometry and the projection-onto-a-closed-convex-set theorem (theorem 2.5) from chapter 2. *(The Riesz–Markov representation $C(S)^* = \mathcal{M}(S)$ is **not** imported: it appears only as classical context in the notation watch of section 3.2.2 and is used nowhere as a tool.)*

Deferred.

- Explicit alternative metrics for weak convergence (Lévy–Prokhorov) and the refinement of the bounded-Lipschitz metric by $\mathcal{W}_2$: chapter 4; [61, Ch. 11].
- The empirical-measure central limit theorem (Gaussian field over test functions) and propagation of chaos: chapter 7, chapter 13.
- Filtrations, martingales, and the use of conditional expectation in BSDEs: chapter 5, chapter 9, chapter 23.

Open. Nothing at this foundational level: the theorems here are classical and complete. The genuinely open questions enter where these tools are applied — quantitative propagation-of-chaos rates beyond $N^{-1/2}$ for singular or non-exchangeable interactions, and tightness on the field-of-measures spaces of the graphon theory (chapter 24).

Bibliographic notes

The treatment of probability spaces, expectation, independence, and the characteristic-function proof of the central limit theorem follows the standard graduate accounts; [62] and [84] are the closest references, and [132] is the most readable source for conditional expectation as a projection and for the martingale viewpoint of remark 3.32. The theory of weak convergence of probability measures on metric spaces — the Portmanteau theorem, tightness, and Prokhorov's theorem — is developed canonically in [23]; our theorem 3.9 and theorem 3.16 are his Theorems 2.1 and 5.1–5.2, and the

Urysohn approximation lemma 3.8 is the standard device behind the closed-set inequality. [61] gives the metric theory of $\mathcal{P}(S)$, including the bounded-Lipschitz and Lévy–Prokhorov metrics quoted in section 3.2.2, and is the reference for the separability reductions used in worked computation 3.26 (Varadarajan's theorem). The Lévy continuity theorem and the tightness-from-continuity-at-the-origin argument are in [62, Ch. 3] and [84, Ch. 5]. The reader oriented toward the sequel should note that everything here is consumed almost immediately: weak convergence and tightness power the existence theory of mean field games in [36] and [40], the $N^{-1/2}$ fluctuation scale is the quantitative heart of propagation of chaos in [40, Vol. I, Ch. 1], and conditional expectation underlies the FBSDE formulation there and its graphon extension in [39]. The optimal-transport refinement of the weak topology, in which the mean field genuinely lives, is taken up next in chapter 4 following [131].

Chapter 4

Wasserstein Geometry of Probability Measures

The previous chapter (chapter 3) taught us to regard a probability measure as a *point* and to speak of one sequence of measures converging to another. Weak convergence, tightness, and Prokhorov's theorem gave us a topology on the set of probability measures, and that topology is already enough to power many compactness arguments. But a topology is not a geometry. It tells us when two measures are *close*, not *how far apart* they are, and it offers no notion of a straight line between them. Both omissions will bite, and it is worth seeing concretely what the missing structure will look like before we build it. "How far apart" will acquire a sharp and useful meaning: the distance between two measures will be the *cheapest cost of physically transporting* one pile of probability mass onto the other — an optimal-transport cost. And "a straight line" between μ and ν will turn out *not* to be the obvious pointwise average $(1-t)\mu + t\nu$ — which merely fades one measure out while fading the other in — but a *displacement geodesic* that slides each grain of mass steadily along a straight path from its position in μ to its destination in ν. Neither object can be seen from the weak topology alone, and both will be indispensable below.

For mean field games we will need the geometry, not just the topology. The equilibrium of a mean field game is a fixed point of a map that sends one population law to another, and to locate such a fixed point three quantitative operations are unavoidable: we must *measure distances* between laws, *estimate how much the map stretches* those distances, and sometimes *interpolate continuously* from one law to another along a controlled path. Each of the three is the seed of a concrete payoff this chapter is building toward, and naming the payoffs now — before proving anything, since the game has not even been defined — lets the reader carry a sense of purpose into the construction. Measuring distances yields $\mathcal{W}_2$-*compactness*, the hypothesis under which the Schauder and Kakutani fixed-point theorems of chapter 10 produce an equilibrium at all. Estimating stretch yields $\mathcal{W}_2$-*contraction*, the mechanism behind the uniqueness arguments of chapter 11. And interpolating yields *displacement convexity*, the form of convexity along the geodesics above that underlies the Lasry–Lions monotonicity of chapter 11. This chapter builds the single metric in which all three are expressed: the quadratic Wasserstein distance $\mathcal{W}_2$ on the space $\mathcal{P}_2(\mathbb{R}^d)$ of probability measures with finite second moment.

We assume from chapter 1 the language of measures, push-forwards, and integration; from chapter 2 the rudiments of Hilbert spaces and the Riesz representation theorem (theorem 2.6); and from chapter 3 weak convergence (definition 3.7), tightness (section 3.3), and Prokhorov's theorem (theorem 3.16), which we invoke without re-proof. The one notion of differential calculus we will need beyond chapter 2 — Fréchet differentiability of a functional on a Hilbert space — is recalled in place where it is used, in section 4.4, so that the chapter is self-contained on this point.

Looking forward, the metric and the two notions of differentiation built here are the working surface of nearly everything that follows: the best-response map of chapter 8 and chapter 10 is studied as a map on $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$; the Lasry–Lions monotonicity of chapter 11 is a statement about the flat derivative of the coupling; the master equation of chapter 12 is a partial differential equation whose unknown is differentiated in the Lions sense developed in section 4.4; and the Gaussian distance computed in section 4.5 is the geometric skeleton of the linear–quadratic–Gaussian (LQG) thread, reused in chapter 6 and chapter 14. The route there has five stages. We begin in section 4.1 with the transport problem itself — Monge's deterministic dream of a transport map, its failure, Kantorovich's relaxation to couplings, and the definition of $\mathcal{W}_2$ that falls out of it. Section 4.2 then earns the word "metric": it verifies that $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is a complete separable metric space, identifies $\mathcal{W}_2$-convergence with weak convergence *plus* convergence of second moments, and exhibits the displacement geodesics promised above. With the geometry in hand, section 4.3 pauses to explain — still informally, because the game has not yet been defined — why a mean field game is naturally posed in exactly this space. Section 4.4 then develops the two derivatives with respect to a measure argument, the flat (linear functional) derivative and the Lions derivative, slowly and with the density-functional analogy as the on-ramp. Finally section 4.5 carries out the worked computation that the LQG thread leans on: the closed form for $\mathcal{W}_2$ between two Gaussians.

4.1 Monge, Kantorovich, and the transport distance

This section builds the distance from the ground up, in three movements that mirror the historical and logical development of optimal transport. First we pose the problem in Monge's original terms — find the cheapest *map* that rearranges one mass distribution into another — and watch it fail: a map cannot split mass, so the problem can be infeasible outright and is in any case non-convex, with neither existence nor uniqueness guaranteed. Then we repair the defect with Kantorovich's relaxation, which replaces the rigid map by a flexible *coupling* and turns the question into a well-posed linear program; its optimal value, taken with squared-distance cost, *is* the quadratic Wasserstein distance $\mathcal{W}_2$. Finally we extract the two structural facts that the rest of the book will draw on: Kantorovich duality, which re-expresses $\mathcal{W}_2^2$ as a supremum over pairs of potentials, and Brenier's theorem, which says that when the source measure has a density the optimal coupling is after all carried by a map — and not just any map, but the gradient of a convex function. The first two subsections deliver the candidate distance; the third delivers the structure that makes it computable and analyzable.

One choice pervades all of this and deserves to be flagged before it is silently made: the price of moving a grain of mass from x to y will be the *squared* Euclidean distance $|x - y|^2$, not the distance itself nor any other power. Three reasons single out this cost, and each is repaid later.

First, it produces a distance that genuinely extends the Euclidean metric of $\mathbb{R}^d$: two Dirac masses δ_a and δ_b end up exactly $|a-b|$ apart (theorem 4.7), so $\mathbb{R}^d$ embeds isometrically into the space of measures and the new geometry contains the old one. Second, it gives the problem a Hilbertian, convex-analytic structure — the optimal map is the gradient of a convex potential (theorem 4.6), the natural interpolation is a geodesic in an L^2 sense, and the whole theory inherits the comfort of quadratic forms — whereas a general cost $|x-y|^p$ has no such linear-algebraic backbone. Third, it couples cleanly to the quadratic running costs of the linear–quadratic control problems of chapter 6, so that the geometry of the population and the cost paid by the individual agent speak one and the same quadratic language. With that choice fixed, we turn to the problem it defines.

4.1.1 The transport problem

With the squared cost fixed, we can make precise the phrase that has been guiding us: the *cheapest way to turn the pile μ into the landscape ν*. The right way to feel the problem before formalizing it is the picture optimal transport was born from. Imagine a pile of sand on $\mathbb{R}^d$ whose mass is distributed according to a probability measure μ: where μ puts a lot of weight there is a lot of sand, where it puts little there is a thin scattering, and the total amount of sand is one unit. We are asked to redeposit every grain so that the relocated pile matches a second prescribed shape, a target landscape described by another probability measure ν — perhaps a hole we must exactly fill, or an embankment we must exactly raise. We are free to move each grain wherever we like, but movement is not free: relocating a grain from position x to position y costs $c(x,y)$, and the bill for the whole operation is the sum (the integral) of these grain-by-grain costs. The transport problem asks for the cheapest such relocation.

Throughout this book the price of moving a grain from x to y is the squared Euclidean distance $c(x,y) = |x-y|^2$. This is the choice flagged and justified in the umbrella to this section: it is what makes the resulting distance a genuine extension of the Euclidean metric of $\mathbb{R}^d$, what gives the problem its Hilbertian convex-analytic backbone, and what makes it speak the same quadratic language as the running costs of the linear–quadratic control problems of chapter 6 (the three reasons are laid out in full there, and we do not repeat them). What matters for the present subsection is only that the cost is fixed, finite for finite displacements, and increasing in the distance moved, so that "cheapest" genuinely rewards moving each grain as little as possible.

Gaspard Monge posed this problem in 1781 in the most concrete terms imaginable: name, for each starting position, the single place its grain is to go. Formally, a *transport map* is a measurable function $T\colon \mathbb{R}^d \to \mathbb{R}^d$, and the prescription "the grain now at x is carried to $T(x)$" is a complete, deterministic relocation rule. The rule is not allowed to be arbitrary: when every grain has been moved according to T, the resulting heap must be exactly the target ν, not merely something resembling it. Spelled out, the mass that T deposits into any region B is precisely the mass that started in the preimage $T^{-1}(B)$ — all the grains whose destination lies in B — and this deposited mass must equal $\nu(B)$ for every region B. That is exactly the push-forward condition introduced in chapter 1: ν is the push-forward of μ under T, written $T_{\#}\mu = \nu$ and defined by $(T_{\#}\mu)(B) = \mu(T^{-1}(B))$ for every Borel set B. Read it as a bookkeeping identity — every grain is reassigned by T, and the reassigned pile is ν on the nose — and the constraint becomes intuitive rather than symbolic. Among all maps meeting this constraint, Monge sought the one of least total cost. The *Monge problem* is therefore

$$\inf\left\{\int_{\mathbb{R}^d} |x - T(x)|^2 \ \mu(\mathrm{d}x) \ : \ T_{\#}\mu = \nu\right\}, \tag{4.1}$$

where $|x - T(x)|^2$ is the cost incurred by the grain starting at x and the integral against μ totals it over the whole pile.

This formulation is irresistibly intuitive — and, it turns out, treacherous. The trouble is that a single-valued function is too rigid an object to describe how mass can actually need to flow, and the next box makes the failure unmistakable on the smallest possible example before we go looking for a cure.

Heuristic (non-rigorous): Monge's treachery: a map cannot split mass

The fatal defect is that a transport *map* sends each source point to one and only one destination, whereas the target may demand that the mass sitting at a single source point be divided and sent to several places. Take the smallest example that exhibits this. Let the source be a single unit atom, $\mu = \delta_{x_0}$ — all the sand piled at one point x_0 — and let the target split that unit of mass evenly between two distinct locations, $\nu = \frac{1}{2}\delta_{y_1} + \frac{1}{2}\delta_{y_2}$ with $y_1 \neq y_2$. A map T must assign to the source point x_0 exactly one value $T(x_0)$, so the entire unit of mass arrives at that single point: $T_{\#}\delta_{x_0} = \delta_{T(x_0)}$, a lone atom of mass one. But ν is not a lone atom; it carries mass $\frac{1}{2}$ at y_1 and mass $\frac{1}{2}$ at y_2. No choice of $T(x_0)$ can reproduce it — sending the grain to y_1 leaves y_2 empty, sending it to y_2 leaves y_1 empty, and a function cannot return both y_1 and y_2 *at once*. There is therefore *no* admissible T at all: the constraint set in (4.1) is *empty*, and the infimum is taken over the empty set (by convention $+\infty$). The Monge problem here is not merely hard to solve; it has nothing to be solved over. The obstruction is generic: whenever μ carries an atom heavier than the heaviest atom of ν, the mass at that point must split, and no map can split it.

A second, subtler defect spoils even the cases where maps do exist. As an optimization problem the Monge problem is badly behaved on two counts. Its objective $T \mapsto \int |x - T(x)|^2 \,\mathrm{d}\mu$ is *nonlinear* in the unknown T — the displacement $T(x)$ enters squared, so the cost is a quadratic, not a linear, function of the map, and the leverage one enjoys with linear objectives is unavailable. And its feasible set $\{T : T_{\#}\mu = \nu\}$ is *nonconvex*: if T_0 and T_1 both push μ to ν, their average $\frac{1}{2}(T_0 + T_1)$ generally does *not*, because pushing forward does not commute with averaging maps — the midpoint map sends each x to the midpoint of two destinations, which lands mass in places neither T_0 nor T_1 visited and need not reconstitute ν. (In the split-mass example the two pathologies fuse: the feasible set is not merely nonconvex, it is empty.) With a nonlinear objective minimized over a nonconvex — and possibly empty — constraint set, the standard existence and uniqueness guarantees of optimization simply do not apply.

Both defects trace to the same root: we insisted that mass move *deterministically*, one source point to one target point. What the split-mass example shows we need is the freedom to let the grain at x_0 go *half to y_1 and half to y_2* — to spread a source point's mass probabilistically over several destinations. Kantorovich's relaxation, taken up next, grants exactly this freedom by replacing the rigid map with a flexible joint law on source–target pairs.

The example has done its work: it has turned a vague unease about Monge's formulation into a concrete, felt need. A description of transport that forbids a source point from sending its mass to more than one destination is simply too narrow to model the flows we must allow. The remedy is to stop tracking "where does the grain at x go?" and instead track "how much mass travels from near x to near y?" — a quantity that can perfectly well be positive for two different destinations y sharing the same source x. Formalizing that quantity, as a joint law whose marginals are μ and ν, is the business of the Kantorovich relaxation, to which we now turn.

4.1.2 The Kantorovich relaxation

The split-mass example left us with a precise diagnosis. The Monge problem failed not because transport is impossible — physically there is no obstacle to shovelling the unit of sand at x_0 into two half-piles and carrying one to y_1 and the other to y_2 — but because the *language* of maps cannot express a split: a single-valued function T is forced to send the grain at x_0 to exactly one place. The cure suggested by the example was to stop insisting that mass move deterministically and instead allow a source point to spread its mass probabilistically over several destinations. Kantorovich's 1942 idea [85] is exactly this, and it is disarmingly simple: describe a transport scheme not by a map that names one destination per source, but by a *bookkeeping table* that records, for every pair of locations (x, y), how much mass travels from near x to near y. Such a table is itself a probability measure — a joint law on source–target pairs — and the two questions "how much mass leaves from each x?" and "how much mass arrives at each y?" are answered by its two marginals, which we require to be μ and ν respectively. This is the notion of a coupling.

Definition 4.1 (Couplings)**.** For $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$, the set of couplings is

$$\Pi(\mu, \nu) \;=\; \Big\{\pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d) \;:\; \pi(A \times \mathbb{R}^d) = \mu(A),\; \pi(\mathbb{R}^d \times B) = \nu(B) \text{ for all Borel } A, B\Big\}.$$

Every symbol here carries an operational meaning, and it is worth reading the definition slowly as the randomized transport scheme it encodes. The object $\pi(\mathrm{d}x, \mathrm{d}y)$ is the infinitesimal amount of mass dispatched from a neighborhood of the source point x to a neighborhood of the target point y; summing (integrating) it over all (x, y) gives one unit, because π is a probability measure and all the sand is accounted for. The first marginal constraint, $\pi(A \times \mathbb{R}^d) = \mu(A)$, fixes the total mass leaving the region A: it collects every grain whose source lies in A, regardless of where that grain ends up (that is what the unrestricted second slot $\mathbb{R}^d$ means), and demands the total equal $\mu(A)$. In words: *all of μ leaves, exactly as μ prescribes.* The second constraint, $\pi(\mathbb{R}^d \times B) = \nu(B)$, collects every grain whose destination lies in B, regardless of its source, and demands the total equal $\nu(B)$: *all of ν arrives, exactly as ν prescribes.* A coupling is thus a complete dispatch ledger that respects both the supply profile μ and the demand profile ν, while leaving entirely free how supply is routed to demand — and crucially, nothing forbids the mass sitting at a single source x from being shared among many targets y. The split that no map could perform is, for a coupling, simply a row of the ledger with more than one nonzero entry.

Two facts about this construction make precise the sense in which it both *contains* Monge and *repairs* him. First, $\Pi(\mu,\nu)$ is *never empty* — the relaxation can never be infeasible the way the Monge problem was. The witness is the product, or independent, coupling $\mu\otimes\nu$, which routes mass from x to y in proportion to $\mu(\mathrm{d}x)\,\nu(\mathrm{d}y)$, as though source and destination were drawn independently. Its marginals are μ and ν by the very definition of a product measure — integrating out the second coordinate of $\mu\otimes\nu$ returns μ, integrating out the first returns ν — so $\mu\otimes\nu\in\Pi(\mu,\nu)$ for *every* pair μ,ν. There is always at least one admissible plan, however expensive; in the split-mass example, where the Monge constraint set was literally empty, the independent coupling $\delta_{x_0}\otimes(\frac{1}{2}\delta_{y_1}+\frac{1}{2}\delta_{y_2})$ does the obvious thing, sending half the atom at x_0 to y_1 and half to y_2. Second, whenever a transport map *does* exist, Kantorovich sees it: a map T with $T_{\#}\mu=\nu$ corresponds to the special coupling $\pi_T=(\mathrm{Id},T)_{\#}\mu$, the push-forward of μ under $x\mapsto(x,T(x))$. This π_T is concentrated on the graph $\left\{(x,T(x)) : x\in\mathbb{R}^d\right\}$ — all its mass sits on the curve traced by T, so it assigns to each source x the *single* destination $T(x)$, with no splitting — and one checks directly that its cost $\int|x-y|^2\ \pi_T(\mathrm{d}x,\mathrm{d}y)=\int|x-T(x)|^2\ \mu(\mathrm{d}x)$ reproduces the Monge cost exactly. The graph-supported couplings are therefore a copy of the Monge maps sitting inside the larger set $\Pi(\mu,\nu)$, and the Kantorovich problem minimizes over a strictly wider class: it admits, in addition, all the plans that are *not* supported on a graph — the genuinely split ones. This is the precise meaning of the slogan that Kantorovich *relaxes* Monge. Minimizing over a larger feasible set can only lower the infimum, so $\mathcal{W}_2(\mu,\nu)^2$ defined below is automatically no greater than the Monge value, and one of the payoffs of theorem 4.6 will be that, under a mild condition, the two values coincide.

With couplings in hand we can finally write down the distance the chapter is built around.

Definition 4.2 (Kantorovich problem and the quadratic Wasserstein distance)**.** For $\mu,\nu\in\mathcal{P}(\mathbb{R}^d)$ define the optimal transport cost

$$\mathcal{W}_2(\mu,\nu)^2 \;=\; \inf_{\pi\in\Pi(\mu,\nu)}\int_{\mathbb{R}^d\times\mathbb{R}^d}|x-y|^2\ \pi(\mathrm{d}x,\mathrm{d}y). \tag{4.2}$$

Restricted to the set $\mathcal{P}_2(\mathbb{R}^d)=\left\{\mu\in\mathcal{P}(\mathbb{R}^d) : \int|x|^2\,\mathrm{d}\mu<\infty\right\}$ of measures with finite second moment, $\mathcal{W}_2$ is called the *quadratic Wasserstein distance.*

The infimum runs over the dispatch ledgers, and the integrand $|x-y|^2$ bills each infinitesimal shipment by the squared distance it travels; $\mathcal{W}_2(\mu,\nu)^2$ is the cheapest total bill. Two things must be checked before this deserves the name "distance," and they explain the restriction to $\mathcal{P}_2(\mathbb{R}^d)$ that the definition quietly imposes: the value must be *finite*, and the infimum must be *attained.* We deal with finiteness now and attainment next.

Finiteness is where the second-moment hypothesis pays for itself, and the argument is short enough to follow term by term. We need only exhibit *one* coupling of finite cost, since the infimum is then no larger; the most convenient candidate, the independent coupling $\mu\otimes\nu$, is the natural one to test. Its cost is $\int_{\mathbb{R}^d\times\mathbb{R}^d}|x-y|^2\ \mu(\mathrm{d}x)\,\nu(\mathrm{d}y)$, and the elementary inequality $|x-y|^2\le 2\,|x|^2+2\,|y|^2$ — itself just the expansion $|x-y|^2=|x|^2-2\langle x,y\rangle+|y|^2$ followed by $-2\langle x,y\rangle\le|x|^2+|y|^2$ — bounds the integrand by a sum in which the two variables have separated. Integrating the separated bound against the product measure, the $|x|^2$ term integrates the μ marginal (the ν in y contributing its total mass 1) and the $|y|^2$ term integrates the ν marginal, giving

$$\mathcal{W}_2(\mu,\nu)^2 \;\le\; \int_{\mathbb{R}^d\times\mathbb{R}^d} |x-y|^2\,\mu(\mathrm{d}x)\,\nu(\mathrm{d}y) \;\le\; 2\int_{\mathbb{R}^d} |x|^2\,\mu(\mathrm{d}x) + 2\int_{\mathbb{R}^d} |y|^2\,\nu(\mathrm{d}y).$$

Both integrals on the right are finite precisely because $\mu,\nu \in \mathcal{P}_2(\mathbb{R}^d)$, so $\mathcal{W}_2(\mu,\nu) < \infty$. This is the whole reason the second-moment condition is the right membership test: it is the exact amount of tail control that guarantees a finite transport bill between any two members of the space.

Drop that control and the value can genuinely blow up. On the larger space $\mathcal{P}(\mathbb{R}^d)$ the infimum in (4.2) can equal $+\infty$, and the picture is concrete: imagine μ and ν each spreading a sliver of mass arbitrarily far out along the axis — heavy-tailed laws whose second moment diverges, like a Cauchy distribution. No matter how cleverly one routes the dispatch, the far-flung grains of μ must be carried somewhere, and the squared-distance bill for relocating mass that lives infinitely far from the origin is itself infinite; *every* coupling costs $+\infty$, so the infimum is $+\infty$ and $\mathcal{W}_2$ is not even a real number. A quantity that can be $+\infty$ cannot be a metric. This is exactly why the metric space of interest is $\mathcal{P}_2(\mathbb{R}^d)$ and not $\mathcal{P}(\mathbb{R}^d)$: the finite-second-moment restriction is not a technical fussiness but the precise boundary between where the transport distance is a usable number and where it degenerates.

The second thing to verify is that the infimum is actually achieved — that there is a best ledger, not merely a sequence of ever-cheaper ones approaching the infimum from above. Unlike the Monge problem, whose feasible set could be empty or whose minimizing sequence of maps could fail to converge to a map, the Kantorovich problem always has a solution, and the reason is that couplings live in a space where the compactness tools of chapter 3 apply.

Proposition 4.3 (Existence of an optimal coupling)**.** *For all $\mu,\nu \in \mathcal{P}(\mathbb{R}^d)$ the infimum in (4.2) is attained by some $\pi^\star \in \Pi(\mu,\nu)$.*

This is the first place in the book where the weak-convergence machinery of chapter 3 does real work, so rather than cite the result we narrate the argument and let each invoked theorem announce why it is needed. The strategy is the *direct method* of the calculus of variations, whose skeleton is always the same three ribs: take a minimizing sequence; extract a convergent subsequence using compactness; pass the cost to the limit using lower semicontinuity, and conclude that the limit is a minimizer. For this to run, the feasible set must be compact (so a limit exists), it must be closed (so the limit is still feasible), and the cost must not jump up in the limit (so the limiting plan is at least as cheap). We supply the three ingredients in turn.

Proof sketch. Compactness of the feasible set. The first rib needs a convergent subsequence of any sequence $\pi_n \in \Pi(\mu,\nu)$, and the tool for that is Prokhorov's theorem (theorem 3.16), which trades relative weak compactness for tightness. So we verify that $\Pi(\mu,\nu)$ is tight — and the elegant point is that a *single* compact set works for every coupling at once, because the marginals are fixed in advance. Given $\varepsilon > 0$, use the tightness of the two fixed measures μ and ν individually (a single Borel measure on a Polish space is automatically tight, section 3.3) to choose one compact $K \subset \mathbb{R}^d$ with $\mu(K^c) < \varepsilon/2$ and $\nu(K^c) < \varepsilon/2$. Then for *any* $\pi \in \Pi(\mu,\nu)$ the mass escaping the compact box

$K \times K$ is controlled by the two marginals: a pair (x, y) lies outside $K \times K$ only if $x \notin K$ or $y \notin K$, so a union bound gives

$$\pi\big((K \times K)^c\big) \;\le\; \pi(K^c \times \mathbb{R}^d) + \pi(\mathbb{R}^d \times K^c) \;=\; \mu(K^c) + \nu(K^c) \;<\; \varepsilon,$$

where the middle equality is just the marginal constraints. Since $K \times K$ is compact and the same K works uniformly in π, the family $\Pi(\mu, \nu)$ is tight, hence relatively weakly compact by Prokhorov.

Closedness of the feasible set. The second rib needs the weak limit of feasible couplings to be feasible. The marginal constraints are exactly continuity requirements: testing $\pi(A \times \mathbb{R}^d) = \mu(A)$ against a bounded continuous f on the first factor reads $\int f(x)\,\pi(\mathrm{d}x, \mathrm{d}y) = \int f\;\mathrm{d}\mu$, a condition of the form "π integrates this fixed bounded continuous function to this fixed value." Such conditions pass to weak limits by the very definition of weak convergence (definition 3.7): if $\pi_n \Rightarrow \pi$ and each π_n has first marginal μ, then $\int f\;\mathrm{d}\mu = \int f(x)\,\pi_n \to \int f(x)\,\pi$ for every bounded continuous f, so π has first marginal μ too, and likewise for ν. Hence $\Pi(\mu, \nu)$ is weakly closed, and the subsequential limit of any minimizing sequence is again a coupling.

Lower semicontinuity of the cost. The third rib needs the cost not to increase in the limit: along $\pi_n \Rightarrow \pi$ we want $\int |x - y|^2\;\pi \le \liminf_n \int |x - y|^2\;\pi_n$. The only obstacle is that $|x - y|^2$ is unbounded, so we cannot test against it directly — weak convergence only controls integrals of *bounded* continuous functions. The standard remedy makes the claim genuinely easy to see. Truncate the cost at height M by setting $g_M(x, y) = \min\big(|x - y|^2, M\big)$; each g_M is bounded and continuous, and the g_M increase pointwise to $|x - y|^2$ as $M \uparrow \infty$. For fixed M, weak convergence gives $\int g_M\,\pi_n \to \int g_M\,\pi$ outright. Since $|x - y|^2 \ge g_M$ pointwise, every term of the sequence obeys $\int |x - y|^2\;\pi_n \ge \int g_M\,\pi_n$, and taking lim inf on both sides yields $\liminf_n \int |x - y|^2\;\pi_n \ge \int g_M\,\pi$. This holds for every M; letting $M \uparrow \infty$ and applying the monotone convergence theorem to the right-hand side ($g_M \uparrow |x - y|^2$) gives $\int g_M\,\pi \uparrow \int |x - y|^2\;\pi$, whence $\liminf_n \int |x - y|^2\;\pi_n \ge \int |x - y|^2\;\pi$. (This truncate-and-pass-to-the-limit move is the content of the Portmanteau theorem's clause that a nonnegative lower semicontinuous integrand is weakly lower semicontinuous; we have just exhibited it explicitly rather than quoting it.)

Conclusion. Let $\pi_n \in \Pi(\mu, \nu)$ be a minimizing sequence, so $\int |x - y|^2\;\pi_n \to \mathcal{W}_2(\mu, \nu)^2$. By tightness and Prokhorov a subsequence converges weakly, $\pi_{n_k} \Rightarrow \pi^\star$; by closedness $\pi^\star \in \Pi(\mu, \nu)$; and by lower semicontinuity $\int |x - y|^2\;\pi^\star \le \liminf_k \int |x - y|^2\;\pi_{n_k} = \mathcal{W}_2(\mu, \nu)^2$. The reverse inequality holds because $\pi^\star$ is feasible, so the two are equal and $\pi^\star$ attains the infimum. Full detail is in [131, Thm. 4.1]. ☐

It is worth pausing on what made this work where Monge failed. The obstruction in the Monge problem was that the feasible *maps* formed a set too rigid to be compact — a limit of maps need not be a map. Couplings dissolve that rigidity: they form a tight, weakly closed, hence weakly compact subset of probability measures on the product space, precisely the setting the direct method demands. Relaxing from maps to plans bought us not only feasibility (the set is nonempty) but also the compactness that delivers a minimizer. From here on $\mathcal{W}_2(\mu, \nu)$ is a well-defined nonnegative real number, attained at an *optimal coupling* $\pi^\star$ whose existence we may now freely invoke.

Remark 4.4 (Why squared distance, and the family $\mathcal{W}_p$)**.** The squared cost is one choice in a family, and seeing the alternatives sharpens why we commit to it. For any exponent $p \ge 1$ one defines, on the space $\mathcal{P}_p$ of measures with finite p-th moment, the transport distance $\mathcal{W}_p(\mu,\nu)^p = \inf_\pi \int |x-y|^p \,\mathrm{d}\pi$; the existence proof above goes through verbatim, since it used only that the integrand is nonnegative, continuous, and coercive. The two cases that matter are $p=1$ and $p=2$, and they trade off cleanly. The exponent $p=1$ bills each shipment by the distance itself, $|x-y|$; this is the Monge–Kantorovich–Rubinstein distance, and its hallmark is a duality against 1-Lipschitz functions (the Kantorovich–Rubinstein formula), so $\mathcal{W}_1$ is the natural gauge whenever one only needs to control how much a Lipschitz observable can change between two populations. The exponent $p=2$ bills by the *squared* distance, and the squaring is exactly what injects Hilbert-space structure into the problem: the cost is a quadratic form in the displacement, optimal plans turn out to be carried by gradients of convex functions (Brenier, theorem 4.6 below), the natural interpolation is an L^2 geodesic, and the whole theory inherits the linear-algebraic comfort of inner products — none of which survives for a generic p. That Hilbertian backbone is why $p=2$ is the working choice of this book: it linearizes neatly on Gaussians (section 4.5) and it speaks the same quadratic language as the running costs of the control problems of chapter 6. The two exponents are not unrelated: Hölder's inequality applied to the optimal $\mathcal{W}_2$-coupling gives $\int |x-y| \;\pi^\star \le \left(\int |x-y|^2 \;\pi^\star\right)^{1/2}$, hence $\mathcal{W}_1(\mu,\nu) \le \mathcal{W}_2(\mu,\nu)$. Controlling $\mathcal{W}_2$ therefore automatically controls $\mathcal{W}_1$ but not conversely — $\mathcal{W}_2$-closeness is the *stronger* statement, which is the right way round for us, since every later contraction and continuity estimate is asked to deliver the more demanding bound.

We now have what the next two threads of the argument need. The distance $\mathcal{W}_2$ is a well-defined, finite, *attained* minimization problem over couplings. Two structural consequences follow, and the chapter pursues each in turn. First, the Kantorovich problem is a *linear program* — a linear functional of π minimized over the convex set $\Pi(\mu,\nu)$ cut out by linear marginal constraints — and every linear program has a dual; the next subsection extracts that dual and reads off from it the form of the optimal map. Second, the very existence of an optimal coupling, guaranteed by proposition 4.3, is the lever behind the metric axioms: the proof that $\mathcal{W}_2$ satisfies the triangle inequality (theorem 4.7) will *glue* two optimal couplings along a shared marginal, an operation that presupposes the optimal couplings exist to be glued. With the minimization now well posed and attained, we turn first to the duality.

4.1.3 Duality and the optimal map

The previous subsection handed us a $\mathcal{W}_2$ that is at last a fully legitimate object: by proposition 4.3 the infimum in (4.2) is finite and, crucially, *attained*, so $\mathcal{W}_2(\mu,\nu)$ is an honest nonnegative number realized by some optimal coupling $\pi^\star$. A well-posed minimization is an invitation to look at its dual, and here the invitation is irresistible, because the Kantorovich problem is not just any minimization: it is a *linear program*. The quantity being minimized, $\pi \mapsto \int |x-y|^2 \,\mathrm{d}\pi$, is *linear* in the unknown π; the feasible set $\Pi(\mu,\nu)$ is the intersection of the convex cone of nonnegative measures with the two *affine* marginal constraints $\pi(\cdot \times \mathbb{R}^d) = \mu$ and $\pi(\mathbb{R}^d \times \cdot) = \nu$. A linear objective over a convex set cut out by affine constraints is, by definition, a linear program, and every linear program has a companion — its dual — obtained by pricing the constraints. This is exactly the linearity that the rigid Monge problem of (4.1) did not have (its objective was quadratic in the map T), and it is the structural dividend of Kantorovich's relaxation that we now collect.

Two questions motivate the work. First, can we compute or bound $\mathcal{W}_2$ without ever exhibiting an optimal plan, working instead with cheaper, lower-dimensional objects? The dual answers yes: it re-expresses $\mathcal{W}_2(\mu,\nu)^2$ as a supremum over pairs of *functions* on $\mathbb{R}^d$ — the Kantorovich potentials — rather than measures on $\mathbb{R}^d \times \mathbb{R}^d$, and any admissible pair already furnishes a *lower* bound, the mirror image of the upper bounds that test couplings supplied in the last subsection. Second, and more important for us, what is the *form* of the optimal plan? The dual turns out to encode it: the potentials are the data from which Brenier's theorem will reconstruct an explicit optimal *map*. We care about the form, not merely the value, for two downstream reasons that the rest of the book will cash in. The potentials themselves are the analytic objects on which the Lasry–Lions monotonicity and uniqueness argument of chapter 11 operates — that argument tests a coupling against potentials of exactly this kind. And the optimal map is the raw material of geometry: it is what theorem 4.9 slides along to build displacement geodesics, and what section 4.5 guesses in closed form for Gaussians. We record both facts here, at the source.

Theorem 4.5 (Kantorovich duality)**.** *For $\mu,\nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2(\mu,\nu)^2 \;=\; \sup\left\{\int_{\mathbb{R}^d} \varphi\ \mathrm{d}\mu + \int_{\mathbb{R}^d} \psi\ \mathrm{d}\nu \;:\; \varphi(x)+\psi(y) \le |x-y|^2 \text{ for all } x,y\right\}, \qquad (4.3)$$

the supremum taken over pairs (φ,ψ) of continuous functions of at most quadratic growth. The optimum is attained by a pair of Kantorovich potentials*, which may be taken to be c-conjugate to one another.*

The mechanism is convex duality, and it repays being run by hand, because watching the dual constraint *emerge* from the primal is what makes the formula intuitive rather than magical. We use the standard Lagrangian recipe: relax each equality constraint by introducing a multiplier and adding to the objective a term that rewards or penalizes its violation. Here the marginal constraints are not two scalar equations but two *continua* of them — $\pi(A\times\mathbb{R}^d)=\mu(A)$ is one equation for every Borel set A, equivalently $\int f(x)\ \mathrm{d}\pi = \int f\ \mathrm{d}\mu$ for every test function f — so the multiplier dual to each marginal is itself a *function* on $\mathbb{R}^d$, not a number. Call them φ (dual to the μ-marginal) and ψ (dual to the ν-marginal): one continuous function per marginal. Adjoining the two constraints to the cost with these multipliers gives the Lagrangian

$$\mathcal{L}(\pi;\varphi,\psi) \;=\; \int |x-y|^2\,\mathrm{d}\pi + \Big(\int \varphi\ \mathrm{d}\mu - \int \varphi(x)\ \mathrm{d}\pi(x,y)\Big) + \Big(\int \psi\ \mathrm{d}\nu - \int \psi(y)\ \mathrm{d}\pi(x,y)\Big),$$

where each parenthesized term vanishes precisely when the corresponding marginal constraint holds, so that on the feasible set $\mathcal{L}$ agrees with the transport cost. Regrouping the $\mathrm{d}\pi$-integrals collects the multiplier contributions into a single integrand,

$$\mathcal{L}(\pi;\varphi,\psi) \;=\; \int \varphi\ \mathrm{d}\mu + \int \psi\ \mathrm{d}\nu + \int \Big(|x-y|^2 - \varphi(x) - \psi(y)\Big)\,\mathrm{d}\pi(x,y),$$

and the dual function is obtained by minimizing this over all admissible π, now with the marginal constraints *dropped* and only $\pi \ge 0$ retained. The minimization is transparent term by term. The first two integrals do not involve π and pass through untouched. In the last integral π is free to place mass wherever it likes among nonnegative measures: if the bracket $|x-y|^2 - \varphi(x) - \psi(y)$ dipped below zero at even a single pair (x_0,y_0), we could pile arbitrarily much mass there and drive the integral, hence $\mathcal{L}$, to $-\infty$. A finite dual value therefore *requires* the bracket to be nonnegative everywhere,

$$\varphi(x) + \psi(y) \ \le\ |x-y|^2 \qquad \text{for all } x, y,$$

which is exactly the dual feasibility constraint in (4.3). And once that constraint holds, the cheapest choice is to put *no* mass where the bracket is strictly positive, leaving the last integral equal to zero; the Lagrangian then collapses to precisely

$$\int \varphi \,\mathrm{d}\mu + \int \psi \,\mathrm{d}\nu,$$

the dual objective. So the dual is born by a single clean move: free minimization over nonnegative π converts the soft penalty term into the hard pointwise constraint $\varphi + \psi \le |\cdot - \cdot|^2$ and erases the π-dependence entirely.

One half of (4.3) is now free. *Weak duality* — that the dual supremum never exceeds the primal infimum — needs no theory at all: for any feasible (φ, ψ) and any coupling $\pi \in \Pi(\mu, \nu)$, integrate the pointwise inequality $\varphi(x) + \psi(y) \le |x-y|^2$ against π and use the marginals to read off the left-hand side,

$$\int \varphi \,\mathrm{d}\mu + \int \psi \,\mathrm{d}\nu \ =\ \int \big(\varphi(x) + \psi(y)\big)\,\mathrm{d}\pi \ \le\ \int |x-y|^2 \,\mathrm{d}\pi.$$

Taking the supremum over (φ, ψ) on the left and the infimum over π on the right gives (dual sup) $\le$ (primal inf) $= \mathcal{W}_2(\mu, \nu)^2$. This already delivers the practical dividend advertised above: *every* admissible potential pair is a certified lower bound on $\mathcal{W}_2^2$, dual to the way every admissible coupling was an upper bound. The substantive content of the theorem is the *reverse* inequality — *strong* duality, the assertion that the gap is zero and the two optima meet. This is not automatic for infinite-dimensional linear programs; it is the conclusion of the Fenchel–Rockafellar duality theorem applied to this convex pair, whose engine is the Hahn–Banach separation of the convex value-function epigraph from a supporting hyperplane. We do not leave it on a citation: a complete, self-contained Hahn–Banach proof is given in appendix D.1, and the classical treatment is [131, Ch. 5]. The attainment of an optimal pair, and the fact that the two potentials may be taken c-conjugate (each recoverable from the other through the cost), come out of the same argument.

We now extract the form of the optimal plan, and the route is to undo the quadratic cost by completing the square. Expand the cost

$$|x-y|^2 \ =\ |x|^2 - 2\langle x, y\rangle + |y|^2\,,$$

so that the dual constraint $\varphi(x) + \psi(y) \le |x-y|^2$ reads $\varphi(x) + \psi(y) \le |x|^2 - 2\langle x, y\rangle + |y|^2$. The pure-quadratic pieces $|x|^2$ and $|y|^2$ depend on only one variable each, so they can be absorbed into the potentials. Define

$$\phi(x) \ =\ \tfrac{1}{2}\big(\,|x|^2 - \varphi(x)\big), \qquad \tilde{\phi}(y) \ =\ \tfrac{1}{2}\big(\,|y|^2 - \psi(y)\big);$$

substituting $\varphi(x) = |x|^2 - 2\phi(x)$ and $\psi(y) = |y|^2 - 2\tilde{\phi}(y)$, the $|x|^2$ and $|y|^2$ terms cancel from both sides and the constraint becomes, after dividing by -2 (which flips the inequality),

$$\langle x, y\rangle \ \le\ \phi(x) + \tilde{\phi}(y) \qquad \text{for all } x, y.$$

This is the defining inequality of a pair of *convex conjugates*: the tightest $\tilde{\phi}$ compatible with it is the Legendre transform $\phi^*(y) = \sup_x (\langle x, y\rangle - \phi(x))$, and at the optimal c-conjugate pair one has $\tilde{\phi} = \phi^*$ with ϕ convex. The quadratic cost has thus turned the Kantorovich potentials into a convex potential and its conjugate — and a convex potential is something we can differentiate. That observation is the whole content of the following structural theorem.

Theorem 4.6 (Brenier's theorem). *Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ and suppose μ is absolutely continuous with respect to Lebesgue measure. Then:*

1. *there is a μ-almost everywhere unique optimal coupling for* (4.2)*, and it is induced by a map, i.e. $\pi^\star = (\mathrm{Id}, T)_\# \mu$;*
2. *this optimal map is the gradient of a convex function, $T = \nabla\phi$ with $\phi\colon \mathbb{R}^d \to \mathbb{R}$ convex, and it is the μ-a.e. unique map of gradient-of-convex form pushing μ to ν;*
3. *consequently the Monge infimum* (4.1) *equals the Kantorovich infimum* (4.2)*, and $\mathcal{W}_2(\mu, \nu)^2 = \int |x - \nabla\phi(x)|^2 \, \mathrm{d}\mu$.*

Proof sketch. Begin from the optimal Kantorovich potentials (φ, ψ) of theorem 4.5 and pass, as just above, to the convex potential $\phi(x) = \frac{1}{2}(|x|^2 - \varphi(x))$, so that dual feasibility becomes the Fenchel–Young inequality

$$\langle x, y\rangle \;\le\; \phi(x) + \phi^*(y) \qquad \text{for all } x, y, \tag{4.4}$$

with ϕ^* the Legendre transform. The key is that strong duality forces this inequality to be an *equality* exactly on the support of the optimal plan $\pi^\star$. Indeed, integrating (4.4) against $\pi^\star$ and using that (φ, ψ) and $\pi^\star$ are jointly optimal (no duality gap) makes the integrated inequality an equality; since the integrand $\phi(x) + \phi^*(y) - \langle x, y\rangle$ is everywhere nonnegative, an integral that vanishes forces it to vanish $\pi^\star$-almost everywhere. This is the complementary-slackness content of strong duality. Now recall the meaning of equality in Fenchel–Young: it is the *definition* of the subdifferential,

$$\langle x, y\rangle = \phi(x) + \phi^*(y) \quad\Longleftrightarrow\quad y \in \partial\phi(x),$$

where $\partial\phi(x)$ is the set of slopes of supporting hyperplanes to the graph of the convex function ϕ at x. So for $\pi^\star$-almost every pair (x, y) the second coordinate is a subgradient of ϕ at the first: the plan is carried by the (a priori multivalued) subdifferential map $x \mapsto \partial\phi(x)$.

It remains to collapse the multivalued $\partial\phi$ to a genuine single-valued map, and this is exactly where the hypothesis that μ be absolutely continuous earns its place. A finite convex function on $\mathbb{R}^d$ is locally Lipschitz, hence differentiable Lebesgue-almost everywhere by Rademacher's theorem; at every such point of differentiability the subdifferential shrinks to the single vector $\nabla\phi(x)$, and the exceptional set where ϕ has a genuine corner (and $\partial\phi(x)$ is a nontrivial set of slopes) is Lebesgue-null. Because μ is absolutely continuous, it assigns that Lebesgue-null exceptional set measure zero, so for μ-almost every x the subdifferential is the singleton $\{\nabla\phi(x)\}$ and the relation $y \in \partial\phi(x)$ becomes the equation $y = \nabla\phi(x)$. Hence $\pi^\star$ is concentrated on the graph of the single map $T = \nabla\phi$, i.e. $\pi^\star = (\mathrm{Id}, T)_\#\mu$, which is claim (i)–(ii); the cost identity in (iii) is then just the transport cost of this graph coupling, and the Monge–Kantorovich values coincide because a graph-supported plan *is* a map. Drop absolute continuity and this last step fails: if $\mu = \delta_{x_0}$ is a Dirac, the single source point may sit at a corner of ϕ where $\partial\phi(x_0)$ is a whole segment of admissible destinations, and the optimal plan genuinely splits — the very split-mass obstruction of eq. (4.1) reappears, which is why the density hypothesis is not a convenience but the precise dividing line, and why it is flagged Open in the ledger. Uniqueness is read off the same potential: any two optimal maps of gradient-of-convex form must both be subgradients of a Kantorovich potential pinned down (up to an additive constant) by the dual problem, so their gradients agree μ-a.e. and the maps coincide. The complete argument, including the regularity that makes the potentials of quadratic growth, is Brenier's [30]; see also [131, Thm. 2.12] and [117, Ch. 1]. □

Brenier's theorem is the rigorous vindication of Monge. Monge's deterministic dream — a single map that names each grain's destination — failed in general because a map cannot split mass; Kantorovich repaired the failure by relaxing to plans; and Brenier now closes the circle by saying that *when the source has a density the relaxation changes nothing*: the cheapest randomized plan is in fact deterministic, supported on the graph of an honest map, and that map is the gradient of a convex potential. A gradient of a convex function is the multidimensional analogue of a monotone increasing function on the line (in one dimension ϕ convex means $\phi' = T$ nondecreasing), so the optimal map is the correct higher-dimensional notion of a *monotone rearrangement* of μ into ν — the mass is shuffled in the order-preserving way, never folding back on itself.

This structural fact is the technique the rest of the chapter and the LQG thread inherit. Because optimality is guaranteed for *any* gradient-of-convex map pushing μ to ν, one can compute $\mathcal{W}_2$ by *guessing* such a map and verifying it transports correctly, with no separate optimality check — and the simplest guess is an affine map, which is the gradient of the quadratic $\phi(x) = \frac{1}{2}\langle x, Ax\rangle + \langle b, x\rangle$ when A is symmetric positive definite. This is exactly the move section 4.5 makes to pin down the Gaussian distance: it posits $T(x) = m_1 + A(x - m_0)$ precisely because Brenier certifies that a gradient-of-convex transport is optimal, reducing the whole computation to linear algebra on the covariance matrices. The convex potential ϕ itself, meanwhile, is the object the geometry is built on: theorem 4.9 interpolates μ toward ν by sliding each grain along the straight segment $x \mapsto (1-t)x + t\,\nabla\phi(x)$, turning the optimal map into the displacement geodesic that the next section promises. And the potentials (φ, ψ) are the same analytic objects on which the Lasry–Lions monotonicity argument of chapter 11 will operate. With the value of $\mathcal{W}_2$ certified finite and attained, and now its optimal plan exhibited as a gradient map, we have every structural fact the metric needs; the next section asks whether $\mathcal{W}_2$ truly deserves the name distance, and what its topology and its straight lines are.

4.2 The metric space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$

The previous section left us with $\mathcal{W}_2$ as a genuine number — finite on $\mathcal{P}_2(\mathbb{R}^d)$, always attained by some optimal coupling, and, thanks to Brenier and Kantorovich duality, in the regular case realized by an honest transport map $T = \nabla\varphi$ pushing μ onto ν. That is enough to compute $\mathcal{W}_2(\mu, \nu)$ for any single pair, and it is exactly the input we now build on. But a number assigned to pairs is not yet a geometry. To run the mean field arguments of the later chapters we need to know that $\mathcal{W}_2$ organizes *all* of $\mathcal{P}_2(\mathbb{R}^d)$ into a space we can navigate: that "close in $\mathcal{W}_2$" is a coherent notion of nearness, that the space is large enough to support limiting arguments yet small enough to be handled, and that between any two laws there is a well-defined notion of a straight line. This section promotes $\mathcal{W}_2$ from a pairwise cost to a bona fide metric geometry, and it does so in three movements, each answering one of those needs and each cashed out later in the book.

First, the *metric axioms* (theorem 4.7): symmetry, nonnegativity, vanishing exactly on the diagonal, and — the one substantive point — the triangle inequality, which will follow by gluing optimal couplings through a common middle marginal and reading the result as Minkowski's inequality for a displacement random variable. This is the legitimacy step; without it the word "distance," and every estimate of the form $\mathcal{W}_2(\text{output of A}) \leq L\,\mathcal{W}_2(\text{input of A})$ that a contraction argument will lean on, would be empty. Second, the *topology*: we identify exactly what convergence in $\mathcal{W}_2$ means — weak convergence *plus* convergence of second moments, no more and no less — and show that $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is *Polish*, that is, complete and separable. Completeness is what lets a Cauchy sequence of approximate population laws actually converge to a limit law inside the space, and separability gives the countable scaffolding behind compactness; together they are the standing hypotheses under which the existence theory of chapter 10 produces a mean field equilibrium as a genuine limit rather than a formal one. Third, the *geodesics*: we show that any two measures are joined by a constant-speed shortest path — the displacement interpolation $t \mapsto ((1-t)\,\mathrm{Id} + tT)_{\#}\mu$ traced along Brenier's optimal map — so that $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is not merely a metric space but a geodesic one. These straight lines are the curves along which functionals will turn out to be convex, and that *displacement convexity* is the engine behind the uniqueness and stability arguments of chapters 11 and 12.

We take the three in that order, since each rests on the one before: the axioms make $\mathcal{W}_2$ a distance, the distance has a topology and a completion, and only on a complete metric space is it meaningful to ask whether shortest paths exist. We begin with the axioms, taking as given the existence of optimal couplings established in proposition 4.3.

4.2.1 The metric axioms

The umbrella to this section listed three things $\mathcal{W}_2$ must deliver before it can carry the geometry of the later chapters, and the very first — logically prior to any talk of topology or geodesics — was simply that $\mathcal{W}_2$ be a *distance* at all. That is the question we settle here. We have a nonnegative number $\mathcal{W}_2(\mu,\nu)$ attached to every ordered pair, finite on $\mathcal{P}_2(\mathbb{R}^d)$ and, by proposition 4.3, always realized by an optimal coupling $\pi^\star$; what we do not yet know is that this number behaves like a distance — that it is symmetric, that it vanishes *only* when the two measures coincide, and that it obeys the triangle inequality. The optimal couplings whose existence we fought to secure in section 4.1 are precisely the tool that makes all three checks go through, and the third one — the triangle inequality — is where they do their real work, through an operation called *gluing* that splices two optimal plans along the population they share.

Theorem 4.7 ($\mathcal{W}_2$ is a metric)**.** *$\mathcal{W}_2$ is a metric on $\mathcal{P}_2(\mathbb{R}^d)$: it is symmetric, finite, nonnegative, vanishes exactly on the diagonal, and satisfies the triangle inequality.*

Proof. The first four properties are quickly disposed of, and only the fifth has any content. Finiteness on $\mathcal{P}_2(\mathbb{R}^d)$ was established by the second-moment bound after definition 4.2. Symmetry is built into the cost: the reflection $(x, y) \mapsto (y, x)$ maps $\Pi(\mu, \nu)$ bijectively onto $\Pi(\nu, \mu)$ — it just swaps which marginal is called "source" and which "target" — and it leaves the integrand $|x-y|^2 = |y-x|^2$ untouched, so the two infima are equal and $\mathcal{W}_2(\mu, \nu) = \mathcal{W}_2(\nu, \mu)$. Nonnegativity is immediate from $|x-y|^2 \geq 0$. For vanishing on the diagonal, if $\mu = \nu$ then the *diagonal* coupling $\pi = (\mathrm{Id}, \mathrm{Id})_{\#}\mu$ — the law of the pair (X, X) for $X \sim \mu$, which places all its mass on $\{x = y\}$ — is admissible (both its marginals are μ) and costs $\int |x-x|^2 \, \mu(\mathrm{d}x) = 0$, so $\mathcal{W}_2(\mu, \mu) = 0$.

The converse — that $\mathcal{W}_2(\mu, \nu) = 0$ forces $\mu = \nu$ — is the point at which the *attainment* of the infimum earns its keep, and it is worth spelling out the step the skeleton draft compressed. Suppose $\mathcal{W}_2(\mu, \nu) = 0$. Because the infimum is attained (proposition 4.3), there is an actual optimal coupling $\pi^\star \in \Pi(\mu, \nu)$ with $\int |x-y|^2 \, \pi^\star(\mathrm{d}x, \mathrm{d}y) = 0$. An integral of the nonnegative function $|x-y|^2$ that vanishes forces the integrand to vanish $\pi^\star$-almost everywhere, so $\pi^\star$ is concentrated on the diagonal $\{x = y\}$: with probability one a sample $(X, Y) \sim \pi^\star$ has $X = Y$. But now the two marginals of $\pi^\star$ are the laws of X and of Y, and these are *the same random variable* (they agree almost surely), so they have the same law. The first marginal is μ and the second is ν by admissibility, hence $\mu = \nu$. This is the meaning of the terse phrase "the marginals of a diagonal measure coincide": a measure living on $\{x = y\}$ cannot tell its two coordinates apart, so its two marginals are one and the same.

The triangle inequality is the substantive axiom, and the difficulty it poses is concrete: $\mathcal{W}_2(\mu_1, \mu_2)$ comes with an optimal plan coupling μ_1 to μ_2, and $\mathcal{W}_2(\mu_2, \mu_3)$ with a separate optimal plan coupling μ_2 to μ_3, but $\mathcal{W}_2(\mu_1, \mu_3)$ asks for a plan coupling μ_1 directly to μ_3, and the two plans we hold do not obviously combine into one. The device that combines them is the *gluing lemma*: it manufactures, out of the two given plans, a single law on the *triple* product $(\mathbb{R}^d)^3$ that contains both as marginals, after which the intermediate variable can simply be forgotten. Let $\mu_1, \mu_2, \mu_3 \in \mathcal{P}_2(\mathbb{R}^d)$, let $\pi_{12} \in \Pi(\mu_1, \mu_2)$ be an optimal coupling and $\pi_{23} \in \Pi(\mu_2, \mu_3)$ an optimal coupling.

The two plans meet on μ_2, the law of the middle population, and that shared marginal is the hinge on which they are joined. To exploit it we *disintegrate* each plan over its μ_2-coordinate. Disintegration (the regular-conditional-distribution theorem on Polish spaces, theorem 3.33) says that a joint law can be sliced into its conditional laws: there is a measurable family $K_{12}(x_2, \mathrm{d}x_1)$, a probability measure on the first factor for μ_2-almost every x_2, that records the *conditional law of the source* x_1 *given that the middle point is* x_2, and reassembling these slices against μ_2 reconstructs π_{12}:

$$\pi_{12}(\mathrm{d}x_1, \mathrm{d}x_2) = \mu_2(\mathrm{d}x_2)\, K_{12}(x_2, \mathrm{d}x_1).$$

Likewise $\pi_{23}(\mathrm{d}x_2, \mathrm{d}x_3) = \mu_2(\mathrm{d}x_2)\, K_{23}(x_2, \mathrm{d}x_3)$, with $K_{23}(x_2, \mathrm{d}x_3)$ the conditional law of the target x_3 given the middle point x_2. Both kernels are anchored to the *same* variable x_2 distributed as μ_2, and that is what lets us run them side by side. Glue the two conditionals over the common middle by declaring x_1 and x_3 *conditionally independent given* x_2 — draw $x_2 \sim \mu_2$, then draw the source from $K_{12}(x_2, \cdot)$ and the target from $K_{23}(x_2, \cdot)$ independently — which is the measure on the triple product

$$\pi_{123}(\mathrm{d}x_1, \mathrm{d}x_2, \mathrm{d}x_3) \;=\; \mu_2(\mathrm{d}x_2)\, K_{12}(x_2, \mathrm{d}x_1)\, K_{23}(x_2, \mathrm{d}x_3).$$

This π_{123} is exactly engineered to carry both original plans inside it, which we now verify marginal by marginal. Integrating out x_3 uses $\int K_{23}(x_2, \mathrm{d}x_3) = 1$ (each conditional is a probability measure) and returns $\mu_2(\mathrm{d}x_2)\, K_{12}(x_2, \mathrm{d}x_1) = \pi_{12}$, so the $(1, 2)$ marginal of π_{123} is π_{12}. Symmetrically, integrating out x_1 uses $\int K_{12}(x_2, \mathrm{d}x_1) = 1$ and returns π_{23}, so the $(2,3)$ marginal is π_{23}. Finally the $(1,3)$ marginal, obtained by integrating out the middle variable x_2, is some measure $\pi_{13} \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$; its first marginal is the first marginal of π_{12}, namely μ_1, and its second marginal is the second marginal of π_{23}, namely μ_3, so $\pi_{13} \in \Pi(\mu_1, \mu_3)$ is a *legitimate* (if not necessarily optimal) coupling of the two end populations. The gluing has produced, out of two plans that did not directly meet, a single plan from μ_1 to μ_3.

It remains only to read off the inequality, and here is where the squared cost shows its hand. Regard x_1, x_2, x_3 as $\mathbb{R}^d$-valued random variables on the probability space $\left((\mathbb{R}^d)^3, \pi_{123}\right)$. The chain of inequalities is

$$\mathcal{W}_2(\mu_1, \mu_3) \;\le\; \|x_1 - x_3\|_{L^2(\pi_{123})} \;\le\; \|x_1 - x_2\|_{L^2(\pi_{123})} + \|x_2 - x_3\|_{L^2(\pi_{123})} \;=\; \mathcal{W}_2(\mu_1, \mu_2) + \mathcal{W}_2(\mu_2, \mu_3),$$

and each link deserves a word, because each carries part of the argument. The first inequality is the definition of $\mathcal{W}_2$ as an *infimum*: π_{13} is *a* coupling of μ_1 and μ_3, not in general the optimal one, so the cheapest cost $\mathcal{W}_2(\mu_1, \mu_3)^2$ cannot exceed the cost $\int |x_1 - x_3|^2 \, \pi_{123} = \|x_1 - x_3\|_{L^2}^2$ of this particular plan — we pay an honest price for using a suboptimal coupling, and the price is exactly what we can bound. The middle inequality is *Minkowski's inequality* (the triangle inequality for the $L^2(\pi_{123})$ norm) applied to the algebraic identity $x_1 - x_3 = (x_1 - x_2) + (x_2 - x_3)$: the displacement from source to target is the sum of the displacement source-to-middle and middle-to-target, and the L^2 norm of a sum is at most the sum of the norms. This is the single step that makes the whole inequality work, and it works *only* because the cost is a squared norm, so that $\mathcal{W}_2$ is an honest L^2-norm of a displacement to which Minkowski applies; a non-Hilbertian cost would have no triangle inequality to invoke here. The closing equality returns from the triple product to the original two-marginal problems: because the $(1,2)$ marginal of π_{123} is π_{12}, we have $\|x_1 - x_2\|_{L^2(\pi_{123})}^2 = \int |x_1 - x_2|^2 \, \pi_{12}$, and this equals $\mathcal{W}_2(\mu_1, \mu_2)^2$ precisely because π_{12} was chosen *optimal*; the same for π_{23} and $\mathcal{W}_2(\mu_2, \mu_3)$. Optimality of the two given plans is therefore not decorative — it is what turns the two L^2 norms back into the two distances we wanted on the right-hand side. $\square$

The Minkowski step repays a second look, because it explains a feature of $\mathcal{W}_2$ that will recur throughout the book: its distinctly *Euclidean flavor*. We have just seen that the triangle inequality for $\mathcal{W}_2$ is not proved by some measure-theoretic trick special to probability measures — it is literally the triangle inequality for the L^2 norm of the displacement random variable, transported up to the level of measures through the optimal couplings. The geometry of $\mathcal{P}_2(\mathbb{R}^d)$ inherits its shape from the geometry of L^2, and through L^2 from the inner-product geometry of $\mathbb{R}^d$ itself. This is the deep reason $\mathcal{W}_2$ will keep admitting Pythagorean-looking identities and quadratic expansions in the chapters ahead — it was built, one optimal coupling at a time, out of the Euclidean norm. The two computations that follow make that inheritance completely explicit on the simplest measures of all.

The first anchors $\mathcal{W}_2$ to the Euclidean distance it came from. Between two Dirac masses δ_a and δ_b there is only *one* coupling to choose from: any $\pi \in \Pi(\delta_a, \delta_b)$ must have first marginal δ_a and second marginal δ_b, which forces all its mass onto the single point (a, b), so $\Pi(\delta_a, \delta_b) = \left\{\delta_{(a,b)}\right\}$ is a singleton and there is nothing to optimize. Its cost is the cost of that lone shipment,

$$\mathcal{W}_2(\delta_a, \delta_b)^2 \;=\; \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 \, \delta_{(a,b)}(\mathrm{d}x, \mathrm{d}y) \;=\; |a - b|^2, \qquad \text{i.e.} \quad \mathcal{W}_2(\delta_a, \delta_b) = |a - b|. \tag{4.5}$$

The map $a \mapsto \delta_a$ therefore embeds $\mathbb{R}^d$ *isometrically* into $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$: distances between point masses are exactly the Euclidean distances between the points they sit on. The Wasserstein metric is a genuine extension of the underlying geometry of $\mathbb{R}^d$, with a faithful copy of ordinary space sitting inside the space of measures — precisely the property the squared cost was chosen to secure, as promised in the umbrella to section 4.1.

The second computation anchors the *ambient set.* Fix the reference point 0 and ask how far a general $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ sits from the Dirac mass δ_0 there. Again the coupling is forced: any $\pi \in \Pi(\mu, \delta_0)$ has second marginal δ_0, so it is concentrated on $\{y = 0\}$ and can only be $\mu \otimes \delta_0$ — every grain of μ, wherever it sits, must be shipped to the one available destination 0. There is again nothing to optimize, and the cost is

$$\mathcal{W}_2(\mu, \delta_0)^2 \;=\; \int_{\mathbb{R}^d} |x - 0|^2 \; \mu(\mathrm{d}x) \;=\; \int_{\mathbb{R}^d} |x|^2 \; \mu(\mathrm{d}x). \tag{4.6}$$

The squared distance from μ to the origin Dirac is exactly the second moment of μ. This is the cleanest possible confirmation that $\mathcal{P}_2(\mathbb{R}^d)$ is the right ambient set: membership in $\mathcal{P}_2(\mathbb{R}^d)$ means finite second moment, and we have just learned that finite second moment means *finite $\mathcal{W}_2$-distance to a fixed reference point.* The defining membership condition of the space and finiteness of the metric to a basepoint are one and the same fact, so $\mathcal{P}_2(\mathbb{R}^d)$ is exactly the set of measures the geometry can reach — no smaller, no larger.

With theorem 4.7 the word "metric" is earned, and the identity (4.6) is more than a sanity check: it is the lever the next subsection turns. Having a legitimate distance, we can finally ask which sequences it declares convergent and whether the space is complete and separable. The Dirac identity $\mathcal{W}_2(\mu, \delta_0)^2 = \int |x|^2 \, \mathrm{d}\mu$ is precisely what lets us read second-moment information directly off $\mathcal{W}_2$: a $\mathcal{W}_2$-convergent sequence stays at bounded distance from δ_0, hence has uniformly controlled second moments, and this is the bridge by which $\mathcal{W}_2$-convergence will be shown to mean weak convergence *together with* convergence of second moments. We turn to that characterization, and to completeness and separability, next.

4.2.2 Completeness, separability, and the link to weak convergence

Having earned the name "metric," we can finally interrogate the geometry it generates: which sequences of laws does $\mathcal{W}_2$ declare convergent, and is the resulting space well enough behaved to host the limiting arguments the later chapters will run inside it? Two structural questions decide the matter. *Completeness* asks whether every sequence that ought to converge — every $\mathcal{W}_2$-Cauchy sequence — actually has a limit sitting inside $\mathcal{P}_2(\mathbb{R}^d)$, with no points missing from the space; without it no fixed-point iteration or approximation scheme could be trusted to land somewhere. *Separability* asks whether the space, uncountable though it is, can be approximated to arbitrary precision from a single countable list of laws. Together the two properties make $\mathcal{P}_2(\mathbb{R}^d)$ a *Polish* space, and that is the exact standing hypothesis the existence theory of chapter 10 is built on. But underneath both questions lies a more basic one we must settle first: what does it even *mean* for measures to converge in $\mathcal{W}_2$? The tool that unlocks this is already in our hands. The identity (4.6), $\mathcal{W}_2(\mu, \delta_0)^2 = \int |x|^2 \, \mathrm{d}\mu$, says that distance to the reference Dirac *is* the second moment, so any control on $\mathcal{W}_2$ immediately hands us control on second moments — and this, we will see, is precisely the surplus that separates $\mathcal{W}_2$-convergence from the weaker weak convergence of chapter 3.

Theorem 4.8 (Topology of $\mathcal{W}_2$)**.** *$(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is a complete separable metric space. Moreover, for $\mu_n, \mu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2(\mu_n, \mu) \to 0 \quad \iff \quad \begin{cases} \mu_n \Rightarrow \mu \text{ (weak convergence, definition 3.7), and} \\ \int_{\mathbb{R}^d} |x|^2 \, \mu_n(\mathrm{d}x) \longrightarrow \int_{\mathbb{R}^d} |x|^2 \, \mu(\mathrm{d}x). \end{cases} \tag{4.7}$$

Equivalently, $\mathcal{W}_2$-convergence is weak convergence together with uniform integrability of the second moments.

The two topological claims are corollaries of the convergence characterization (4.7), so that is where the work goes. The characterization says something sharp and easy to misremember: $\mathcal{W}_2$-convergence is weak convergence *with one rider*, that the second moments converge to the *right* limit, not merely stay bounded. We prove the two implications separately, because they draw on genuinely different machinery — the forward direction on the representation theorem of chapter 3, the reverse on the duality of section 4.1 — and then read completeness and separability off the same ideas.

Proof sketch. Weak plus moments imply $\mathcal{W}_2$ (the implication $\Leftarrow$). Suppose $\mu_n \Rightarrow \mu$ with $\int |x|^2 \, \mathrm{d}\mu_n \to \int |x|^2 \, \mathrm{d}\mu$. The difficulty is one of language: weak convergence is a statement about integrals of *test functions*, whereas $\mathcal{W}_2(\mu_n, \mu)^2$ is the cost of an optimal *coupling*, and there is no test function in sight that we are allowed to feed the unbounded integrand $|x-y|^2$. Skorokhod's representation theorem (theorem 3.13) bridges exactly this gap by manufacturing one concrete coupling that already does most of the work: it produces, on a single probability space, random variables X_n with $\mathcal{L}(X_n) = \mu_n$ and X with $\mathcal{L}(X) = \mu$, such that the abstract weak convergence $\mu_n \Rightarrow \mu$ is realized as honest pointwise convergence, $X_n \to X$ almost surely. The joint law of (X_n, X) is a coupling of μ_n and μ, hence admissible in the Kantorovich problem (4.2), so it bounds the optimum from above:

$$\mathcal{W}_2(\mu_n, \mu)^2 \;\le\; \mathbb{E}\big[\, |X_n - X|^2 \,\big].$$

Everything now reduces to showing the right-hand side tends to 0. We already know $|X_n - X|^2 \to 0$ almost surely. The temptation is to pass the limit straight inside the expectation, but almost-sure convergence *never* licenses that on its own: the quantity $|X_n - X|^2$ could be large on an event of shrinking probability yet carrying non-vanishing expectation — mass leaking off to infinity. What forbids the leak is precisely the hypothesis we have not yet spent, the convergence of the second moments. The clean tool is the *Vitali convergence theorem*: almost-sure convergence upgrades to L^1 convergence exactly when the sequence is *uniformly integrable*, so it suffices to show that $\left\{|X_n - X|^2\right\}_n$ is uniformly integrable. Dominate it by the elementary bound $|X_n - X|^2 \le 2\,|X_n|^2 + 2\,|X|^2$; it is therefore enough that $\left\{|X_n|^2\right\}_n$ and the single variable $|X|^2$ be uniformly integrable. The latter is integrable, hence trivially uniformly integrable. For the former we cash in the saved hypothesis: $|X_n|^2 \to |X|^2$ almost surely and, by assumption, $\mathbb{E}\,|X_n|^2 \to \mathbb{E}\,|X|^2 < \infty$. A sequence of nonnegative random variables that converges almost surely and whose expectations converge to the (finite) expectation of the limit is automatically uniformly integrable — this is the converse half of the Scheffé / dominated-convergence circle of ideas. The intuition is exactly the no-leak picture: Fatou's lemma already forces $\mathbb{E}\,|X|^2 \leq \liminf \mathbb{E}\,|X_n|^2$, and demanding *equality* of the limits leaves no room for any second-moment mass to escape to infinity, which is what uniform integrability asserts. So $\left\{|X_n|^2\right\}$ is uniformly integrable, the dominating family $2\,|X_n|^2 + 2\,|X|^2$ is uniformly integrable, $\left\{|X_n - X|^2\right\}$ inherits it, and Vitali delivers $\mathbb{E}\,|X_n - X|^2 \to 0$, hence $\mathcal{W}_2(\mu_n, \mu) \to 0$. The moral worth carrying away — because the whole topology of $\mathcal{W}_2$ rests on it — is that *convergence of second moments is exactly the uniform integrability that almost-sure convergence was missing*: it is the precise quantum of tail control needed to move a limit past the integral sign.

$\mathcal{W}_2$ implies weak plus moments (the implication $\Rightarrow$). Now suppose $\mathcal{W}_2(\mu_n, \mu) \to 0$; we must extract both riders. For weak convergence, descend to the first-order distance. Hölder's inequality gave $\mathcal{W}_1 \leq \mathcal{W}_2$ (remark 4.4), so $\mathcal{W}_1(\mu_n, \mu) \to 0$ as well, and the Kantorovich–Rubinstein duality — the $p = 1$ dual recorded in remark 4.4, which tests against 1-Lipschitz functions — reads

$$\mathcal{W}_1(\mu_n, \mu) \;=\; \sup_{f \text{ 1-Lipschitz}} \Big(\int f \,\mathrm{d}\mu_n - \int f \,\mathrm{d}\mu \Big).$$

Thus $\mathcal{W}_1 \to 0$ forces $\int f \,\mathrm{d}\mu_n \to \int f \,\mathrm{d}\mu$ uniformly over all 1-Lipschitz f, in particular for every *bounded* Lipschitz f; and bounded Lipschitz functions are convergence-determining for the weak topology (the Lipschitz clause of the Portmanteau theorem of chapter 3), so $\mu_n \Rightarrow \mu$. For the second moments, play the reverse triangle inequality off the basepoint δ_0:

$$\big|\, \mathcal{W}_2(\mu_n, \delta_0) - \mathcal{W}_2(\mu, \delta_0) \,\big| \;\leq\; \mathcal{W}_2(\mu_n, \mu) \;\to\; 0,$$

so $\mathcal{W}_2(\mu_n, \delta_0) \to \mathcal{W}_2(\mu, \delta_0)$; squaring and invoking (4.6) turns this into $\int |x|^2 \,\mathrm{d}\mu_n \to \int |x|^2 \,\mathrm{d}\mu$. Both riders are in hand. Notice how lightly this direction runs: distance to a single fixed reference point already carries all the second-moment information — the entire payoff of the identity we paused to prove in the previous subsection.

Completeness. Let (μ_n) be $\mathcal{W}_2$-Cauchy; we must produce a limit lying in $\mathcal{P}_2(\mathbb{R}^d)$. A Cauchy sequence is bounded, so $D := \sup_n \mathcal{W}_2(\mu_n, \delta_0) < \infty$, and (4.6) reads this off as a uniform second-moment bound, $\sup_n \int |x|^2 \,\mathrm{d}\mu_n \leq D^2$. A uniform second-moment bound is exactly what Markov's inequality converts into tightness: for every $R > 0$,

$$\mu_n\big(\{|x| > R\} \big) \;\leq\; \frac{1}{R^2} \int |x|^2 \,\mathrm{d}\mu_n \;\leq\; \frac{D^2}{R^2},$$

which is below any prescribed ε once R is large, *uniformly in* n, and the closed ball $\{|x| \leq R\}$ is compact in $\mathbb{R}^d$. So $\{\mu_n\}$ is tight, and Prokhorov's theorem (theorem 3.16) extracts a weakly convergent subsequence $\mu_{n_k} \Rightarrow \mu$. This candidate limit lies in $\mathcal{P}_2(\mathbb{R}^d)$, because the second moment is weakly lower semicontinuous — truncate $|x|^2$ at height M, pass the bounded continuous truncation through the weak limit, then let $M \uparrow \infty$ by monotone convergence, the very move that gave lower semicontinuity of the cost in proposition 4.3 — so $\int |x|^2 \,\mathrm{d}\mu \leq \liminf_k \int |x|^2 \,\mathrm{d}\mu_{n_k} \leq D^2 < \infty$. It remains to promote this subsequential *weak* limit to a full $\mathcal{W}_2$ limit, and here the Cauchy property does work that mere existence of a cluster point cannot. The functional $(\alpha, \beta) \mapsto \mathcal{W}_2(\alpha, \beta)$ is itself lower semicontinuous under weak convergence of its two arguments (again the lower-semicontinuity-of-the-cost mechanism of proposition 4.3, now applied to a glued sequence of couplings). Fix $\varepsilon > 0$ and take N from the Cauchy property so that $\mathcal{W}_2(\mu_p, \mu_q) < \varepsilon$ whenever $p, q \geq N$. Hold $m \geq N$ fixed and let $k \to \infty$ in $\mathcal{W}_2(\mu_m, \mu_{n_k})$: since the constant sequence μ_m converges weakly to μ_m and $\mu_{n_k} \Rightarrow \mu$, lower semicontinuity gives

$$\mathcal{W}_2(\mu_m, \mu) \;\leq\; \liminf_k \mathcal{W}_2(\mu_m, \mu_{n_k}) \;\leq\; \varepsilon,$$

the last bound because $n_k \geq N$ eventually. As $m \geq N$ was arbitrary, $\mathcal{W}_2(\mu_m, \mu) \leq \varepsilon$ for all $m \geq N$; that is, $\mathcal{W}_2(\mu_m, \mu) \to 0$. The Cauchy sequence converges, and its limit belongs to the space, so $\mathcal{P}_2(\mathbb{R}^d)$ is complete.

Separability. We must exhibit a countable $\mathcal{W}_2$-dense set. The natural candidate is the collection $\mathcal{D}$ of finitely supported measures $\sum_{i=1}^N q_i \, \delta_{x_i}$ whose atoms sit at rational points ($x_i \in \mathbb{Q}^d$) and whose weights are rational and sum to one ($q_i \in \mathbb{Q}_{>0}$, $\sum q_i = 1$). There are only countably many finite rational point sets and countably many rational weight vectors, so $\mathcal{D}$ is countable. Density is a two-step approximation that the metric makes transparent, and both steps are again paid for by the second-moment bound. Given $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ and $\varepsilon > 0$, first *amputate the tail*: because the total second moment is finite, $\int_{|x|>R} |x|^2 \, \mathrm{d}\mu \to 0$ as $R \uparrow \infty$, and transporting the outlying mass radially inward onto the ball B_R costs in $\mathcal{W}_2^2$ no more than that vanishing tail integral, so a compactly supported μ' lies within $\varepsilon/2$ of μ. Then *discretize*: partition B_R into finitely many cells of diameter below δ, place at a rational point inside each cell an atom of rational weight approximating the μ'-mass of that cell, and round the last weight so that the weights sum to one. The map carrying each cell's mass to its chosen rational representative moves no grain farther than δ, so the resulting rational measure sits within δ (plus a vanishing rounding error) of μ' in $\mathcal{W}_2$. Choosing δ and the rounding small enough puts it within $\varepsilon/2$ of μ', hence within ε of μ. Thus $\mathcal{D}$ is dense, and $\mathcal{P}_2(\mathbb{R}^d)$ is separable. Together with completeness this makes $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ a *Polish* space — the property theorem 4.8 advertises and chapter 10 requires. The fully detailed arguments, including the regularity behind the weak lower semicontinuity of $\mathcal{W}_2$ used above, are in [131, Thm. 6.9, Thm. 6.18]; see also [7, Prop. 7.1.5]. We have given the complete skeleton; only routine measure-theoretic bookkeeping has been deferred to those references. □

Notation watch: $\mathcal{W}_2$-convergence is strictly stronger than weak convergence

It is worth dwelling on the gap between the two convergences, because every later "continuity in $\mathcal{W}_2$" claim trades on it. Weak convergence and $\mathcal{W}_2$-convergence are *not* the same notion; $\mathcal{W}_2$ strictly refines it, and a single example exhibits the whole difference. Let

$$\mu_n \;=\; \Big(1 - \tfrac{1}{n}\Big)\delta_0 + \tfrac{1}{n}\,\delta_n,$$

a unit of mass nearly all parked at the origin, with a sliver of weight $1/n$ flung out to the far point n. Test it weakly against any bounded continuous f:

$$\int f \,\mathrm{d}\mu_n = \Big(1 - \tfrac{1}{n}\Big) f(0) + \tfrac{1}{n} f(n) \;\longrightarrow\; f(0),$$

because f is bounded and so the far term $\frac{1}{n} f(n)$ is at most $(\sup|f|)/n \to 0$. Hence $\mu_n \Rightarrow \delta_0$: to the weak topology the sequence simply settles onto the origin. But weigh it with $\mathcal{W}_2$. Its second moment is

$$\int |x|^2 \, \mathrm{d}\mu_n = \left(1 - \tfrac{1}{n}\right) \cdot 0 + \tfrac{1}{n} \cdot n^2 = n,$$

so by (4.6) we have $\mathcal{W}_2(\mu_n, \delta_0)^2 = n \to \infty$: not only does μ_n fail to $\mathcal{W}_2$-converge to δ_0, it runs off to *infinite* distance from it. The mechanism is exactly the one (4.7) predicts. The far atom's *mass* $1/n$ vanishes — and vanishing mass is all the weak topology, testing against bounded functions, can perceive — yet $\mathcal{W}_2$ bills by squared distance, so the product (mass) $\times$ (distance)^2 $= \frac{1}{n} \cdot n^2 = n$ grows without bound. $\mathcal{W}_2$ sees the mass escaping to infinity that weak convergence is structurally blind to. This is the geometric content of the second-moment rider in (4.7): it is precisely a *no-escape-of-mass-to-infinity* condition. The meta-lesson propagates through the entire book: whenever a later chapter asserts that a map or a functional is continuous "in $\mathcal{W}_2$," it is claiming *strictly more* than weak continuity, and the surplus is always the same thing — uniform control of the tails, the guarantee that no second-moment mass leaks away to infinity in the limit.

That $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is Polish, together with the sharper reading of what $\mathcal{W}_2$-convergence controls, are exactly the structural facts the next two strands of the book stand on. The "why a mean field game lives here" discussion of section 4.3 will need a usable description of the *compact* sets in this space, and the proof above hands it over ready-made: combining theorem 4.8 with Prokhorov, a subset of $\mathcal{P}_2(\mathbb{R}^d)$ is $\mathcal{W}_2$-precompact precisely when it is *tight and* has *uniformly integrable second moments* — tightness keeping mass from spreading out and sliding off the support, uniform integrability keeping second-moment mass from escaping to infinity along the lines the example just dramatized. The contrast with the completeness argument is instructive and not accidental: tightness alone needed only *bounded* second moments via Markov, but genuine $\mathcal{W}_2$-precompactness, where the limit retains *all* of its second-moment mass, demands the strictly stronger uniform integrability. That two-part criterion is the hypothesis under which the Schauder and Kakutani fixed-point theorems of chapter 10 will produce a mean field equilibrium, so we have — without yet having defined the game — already isolated the compactness it will require. Completeness is the other handoff. Geodesics, the straight lines the chapter still owes, can only be built in a space where Cauchy sequences converge, since a displacement geodesic is constructed as a limit of ever-finer interpolations and must have somewhere to land. The final movement of this section now takes the optimal Brenier map of theorem 4.6 and slides mass along it to draw those straight lines (theorem 4.9), turning the complete metric space we have just secured into a genuine geodesic geometry.

4.2.3 Geodesics and displacement interpolation

The previous subsection secured the last structural fact a geometry of straight lines needs: $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is complete, so Cauchy sequences have somewhere to land and a curve built as a limit of ever-finer interpolations cannot fall out of the space. We now spend that completeness. A metric space worthy of the name does more than measure distances; it tells us how to travel between two points along the shortest path, and the goal of this subsection is to exhibit those shortest paths for $\mathcal{W}_2$. The raw material is already in hand: the Brenier optimal map $T = \nabla\phi$ of theorem 4.6, which names, for each grain of μ_0, the single destination it should occupy in μ_1. Sliding each grain steadily toward its destination will draw the straight line — but only after we have understood why the *obvious* candidate for a straight line is the wrong one.

Fix the target to aim at. In any metric space, a *constant-speed geodesic* from μ_0 to μ_1 is a curve $[0,1] \ni t \mapsto \mu_t$ with endpoints μ_0 and μ_1 that satisfies

$$\mathcal{W}_2(\mu_s,\mu_t) \;=\; |t-s|\;\mathcal{W}_2(\mu_0,\mu_1) \qquad \text{for all } s,t \in [0,1].$$

The condition says exactly what "straight line traversed at constant speed" should mean: the distance covered between two times is proportional to the elapsed time, with the proportionality constant equal to the total length $\mathcal{W}_2(\mu_0,\mu_1)$. No detours (otherwise the distance between the endpoints would be strictly less than the length of the curve), and no loitering or sprinting (otherwise the speed would vary). A space in which *every* pair of points is joined by such a curve is called a geodesic space, and we are about to prove $(\mathcal{P}_2(\mathbb{R}^d),\mathcal{W}_2)$ is one.

The naive guess is to average the two measures pointwise, $t \mapsto (1-t)\mu_0 + t\mu_1$, and it is instructive precisely because it is *wrong*. The defect is conceptual, not technical: pointwise averaging *mixes* the two distributions rather than *transporting* one into the other. At an intermediate time t the measure $(1-t)\mu_0 + t\mu_1$ is a superposition that keeps a fraction $1-t$ of the mass sitting exactly where μ_0 left it and fades in a fraction t of the mass exactly where μ_1 wants it; no grain ever moves. The cleanest place to feel the failure is two Dirac masses. Take $\mu_0 = \delta_a$ and $\mu_1 = \delta_b$ with $a \neq b$, so that $\mathcal{W}_2(\delta_a,\delta_b) = |a-b|$ (the optimal plan has no choice but to carry the single atom from a to b). The linear interpolation is $\mu_t^{\mathrm{lin}} = (1-t)\delta_a + t\delta_b$, a measure with an atom of mass $1-t$ parked at a and an atom of mass t parked at b — a barbell whose two weights merely shift, never the locations. Its distance from the start is governed by the transport that must still carry the mass t stranded at b all the way back across the gap, giving $\mathcal{W}_2(\mu_0,\mu_t^{\mathrm{lin}})^2 = t\;|a-b|^2$, hence $\mathcal{W}_2(\mu_0,\mu_t^{\mathrm{lin}}) = \sqrt{t}\;|a-b|$. Compare this with the constant-speed requirement $\mathcal{W}_2(\mu_0,\mu_t) = t\;|a-b|$: the linear curve covers $\sqrt{t}$ of the distance in time t, far too much too early, and is emphatically not a geodesic. The transport picture is the correct one: a genuine straight line should pick up the atom at a and *glide* it to b, so that at time t it sits at the intermediate point $(1-t)a + tb$ as a single atom $\delta_{(1-t)a+tb}$ — and that curve does satisfy $\mathcal{W}_2(\mu_0,\mu_t) = t\;|a-b|$. Mixing splits the mass and keeps it pinned at the two ends; transporting moves the mass through the middle. McCann's construction is nothing but the systematic version of "glide each atom to its destination," with the Brenier map supplying the destinations.

Theorem 4.9 (McCann's displacement interpolation)**.** *Let $\mu_0,\mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$ with μ_0 absolutely continuous, and let $T = \nabla\phi$ be the Brenier optimal map from μ_0 to μ_1 (theorem 4.6). For $t \in [0,1]$ set $T_t = (1-t)\,\mathrm{Id} + t\,T$ and $\mu_t = (T_t)_{\#}\mu_0$. Then $t \mapsto \mu_t$ is a constant-speed geodesic joining μ_0 to μ_1, and every $\mathcal{W}_2$-geodesic between two measures arises this way. In particular $(\mathcal{P}_2(\mathbb{R}^d),\mathcal{W}_2)$ is a geodesic space.*

Read the construction one grain at a time before reading the proof. The grain that starts at x is destined, by Brenier, for $T(x)$. The interpolating map $T_t = (1-t)\,\mathrm{Id} + tT$ places that grain, at time t, at $T_t(x) = (1-t)x + tT(x) = x + t\,(T(x) - x)$, which is exactly the point a fraction t of the way along the straight segment from x to $T(x)$. So T_t slides *every* grain simultaneously along its own straight segment, each at the constant velocity $T(x) - x$ and hence at the constant speed $|T(x) - x|$, the length of the segment it must cover in unit time. The interpolant $\mu_t = (T_t)_\# \mu_0$ is the snapshot of the whole population mid-glide. At $t = 0$, $T_0 = \mathrm{Id}$ and μ_0 is recovered; at $t = 1$, $T_1 = T$ and $\mu_1 = T_\# \mu_0$ is recovered, so the endpoints are right by construction. The content of the theorem is that this in-between curve is genuinely straight in $\mathcal{W}_2$.

Proof sketch. The whole verification rests on a single structural fact: the maps T_t are themselves optimal maps, so their transport cost *is* a squared Wasserstein distance and we may compute lengths by reading off displacements. To see it, fix $0 \le s \le t \le 1$ and ask for the map that carries the snapshot μ_s to the later snapshot μ_t. A grain occupying the position $T_s(x) = x + s\,(T(x) - x)$ at time s must occupy $T_t(x) = x + t\,(T(x) - x)$ at time t, so the map taking the former to the latter advances it by the fixed fraction $(t - s)/(1 - s)$ of its *remaining* segment: writing $y = T_s(x)$, the rearrangement sends $y \mapsto y + \frac{t-s}{1-s}\,(T(x) - T_s(x))$, the same direction $T(x) - x$ scaled down. This rearrangement is again a gradient of a convex function. Indeed each T_t is the gradient of $\phi_t(x) = (1-t)\frac{1}{2}\,|x|^2 + t\,\phi(x)$, a convex combination of the convex quadratic $\frac{1}{2}\,|x|^2$ (whose gradient is Id) and the Brenier potential ϕ (whose gradient is T); a convex combination of convex functions is convex, so ϕ_t is convex and $T_t = \nabla\phi_t$ is a gradient-of-convex map — this is the precise sense in which "the interpolant of gradients of convex functions remains a gradient of a convex function." The map from μ_s to μ_t is, up to the change of variables $y = T_s(x)$, the gradient of the corresponding interpolated potential, hence gradient-of-convex, hence *optimal* by the optimality half of Brenier's theorem (theorem 4.6: any gradient-of-convex map pushing one measure to another is the optimal one). Optimality is what lets us equate the transport cost of this map with $\mathcal{W}_2(\mu_s, \mu_t)^2$ rather than merely bounding it from above. Now compute that cost by pushing the displacement back to μ_0: the grain x moves from $T_s(x)$ to $T_t(x)$, a displacement of $T_t(x) - T_s(x) = (t - s)\,(T(x) - x)$, so

$$\mathcal{W}_2(\mu_s, \mu_t)^2 \;=\; \int_{\mathbb{R}^d} |T_t(x) - T_s(x)|^2\;\mu_0(\mathrm{d}x) \;=\; (t-s)^2 \int_{\mathbb{R}^d} |x - T(x)|^2\;\mu_0(\mathrm{d}x) \;=\; (t-s)^2\,\mathcal{W}_2(\mu_0, \mu_1)^2,$$

where the last equality is the Brenier cost identity for the endpoints (theorem 4.6(iii)). Taking square roots gives $\mathcal{W}_2(\mu_s, \mu_t) = (t - s)\,\mathcal{W}_2(\mu_0, \mu_1)$, which is exactly the constant-speed geodesic property. The converse — that every $\mathcal{W}_2$-geodesic is of this displacement form, so there are no exotic shortest paths we have missed — is the substance of McCann's theorem [101]; we give the argument in full in-book in appendix D.3, and the classical treatments are [131, Ch. 7] and [7, Ch. 7]. □

It is worth saying plainly what distinguishes the right curve from the wrong one, since the distinction is the whole pedagogical point. Linear interpolation $(1 - t)\mu_0 + t\mu_1$ averages the two measures *in the space of measures*; McCann's interpolation $(T_t)_\# \mu_0$ averages the two configurations *in the underlying space* $\mathbb{R}^d$, by moving each grain along a straight line there. The first is a fade between two photographs; the second is the motion the photographs are snapshots of. Only the latter respects the quadratic transport cost, because only the latter never pays to move mass it could have left in place.

This displacement geometry is not a curiosity collected for completeness; it is the form in which convexity enters every later stability argument. A functional $\mathcal{F}$ on $\mathcal{P}_2(\mathbb{R}^d)$ is called *displacement convex* when $t \mapsto \mathcal{F}(\mu_t)$ is convex along the geodesics μ_t just constructed — convex along the *transport* interpolation, not along the naive linear one, and the two notions genuinely differ. This is

McCann's notion, and it is the right one for us precisely because the maps in a mean field game move populations by transport. Displacement convexity of the coupling is what powers the Lasry–Lions monotonicity and displacement-monotonicity arguments that deliver uniqueness of the equilibrium in chapter 11, and it is the natural convexity in which the master equation of chapter 12 is well posed. We will meet the phenomenon in fully explicit form in section 4.5: the displacement interpolation between two Gaussians is again a Gaussian, its mean and covariance sliding affinely from one endpoint to the other, whereas the linear interpolation between two well-separated Gaussians is a two-bump mixture that is not even unimodal — the cleanest possible witness that mixing and transporting are different operations.

There is one more description of $\mathcal{W}_2$, dynamic rather than static, that completes the picture and earns the word "geodesic" a second, physical reading. So far we have built the shortest path by specifying *where each grain goes* (the Brenier map) and interpolating. The dynamic formulation instead specifies *how the whole density flows in time* and asks for the flow of least effort. Recasting $\mathcal{W}_2$ as such a least-action principle is valuable to us for a specific downstream reason: the constraint that bookkeeps a flowing density is the continuity equation $\partial_t\mu + \operatorname{div}(\mu v) = 0$, and that same equation is the one chapter 5 will derive — as the Fokker–Planck equation — for the population density of a mean field game. Seeing it appear here, as the constraint in a transport problem, is the first sighting of the conservation law the rest of the book runs on.

Theorem 4.10 (Benamou–Brenier dynamic formula)**.** *For $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2(\mu_0, \mu_1)^2 \;=\; \inf\left\{\int_0^1\!\!\int_{\mathbb{R}^d} |v_t(x)|^2 \; \mu_t(\mathrm{d}x)\; \mathrm{d}t\right\}, \tag{4.8}$$

the infimum over all curves $(\mu_t)_{t\in[0,1]}$ and velocity fields (v_t) solving the continuity equation $\partial_t\mu_t + \operatorname{div}(\mu_t v_t) = 0$ in the distributional sense, with endpoints μ_0, μ_1. The minimizing curve is the displacement geodesic of theorem 4.9; when μ_0 is absolutely continuous the interpolating map $T_t = (1-t)\operatorname{Id} + tT$ is invertible for every $t \in [0,1)$ (it is the gradient of a strictly convex function for $t < 1$), so the velocity field $v_t = (T - \operatorname{Id}) \circ T_t^{-1}$ is defined μ_t-a.e. along the geodesic; only the endpoint $t = 1$ may fail to be invertible, and it carries no mass of the time integral.

Two pieces of the statement deserve unpacking, because each is a small lesson in its own right. The first is the continuity-equation constraint $\partial_t\mu_t + \operatorname{div}(\mu_t v_t) = 0$. It is not an extra hypothesis imposed for convenience; it is exactly the statement that mass is conserved while it flows. If μ_t is the density of a population moving with velocity field v_t, then $\mu_t v_t$ is the flux — the rate at which mass crosses a surface — and the divergence theorem turns "no mass is created or destroyed, only transported" into the local balance $\partial_t\mu_t = -\operatorname{div}(\mu_t v_t)$. So the admissible curves in (4.8) are precisely the ones along which probability is shuffled around without leaking, and the formula reads: *$\mathcal{W}_2^2$ is the least total squared-speed an honest mass-conserving flow must spend to get from μ_0 to μ_1.* The second is the invertibility clause, which is the dynamic counterpart of "each grain glides along its own segment." To read off the velocity at a point z occupied at time t, we must know which grain is there, i.e. we must invert T_t to find its source $x = T_t^{-1}(z)$ and then assign it the velocity $T(x) - x = (T - \operatorname{Id})(x)$ it has carried since $t = 0$; this is the formula $v_t = (T - \operatorname{Id}) \circ T_t^{-1}$. The inversion is legitimate for every $t < 1$ because $T_t = \nabla\phi_t$ with $\phi_t = (1-t)\frac{1}{2}|\cdot|^2 + t\phi$ *strictly* convex (the $(1-t)\frac{1}{2}|\cdot|^2$ summand contributes a strictly positive curvature as long as $1 - t > 0$), and the gradient of a strictly convex function is injective. Only at the terminal instant $t = 1$ does the strict-convexity cushion vanish and $T_1 = T$ possibly fold mass together; but a single time-slice contributes nothing to the time integral $\int_0^1(\cdots)\,\mathrm{d}t$, so the velocity field is well defined exactly where it needs to be. The subtlety is real — the flow description can break at the very last instant — but it is harmless, and noting why is more honest than waving it away.

We do not reproduce the proof of (4.8) here; it is the Benamou–Brenier theorem [18], and we give a self-contained proof in-book in appendix D.4 (with the classical treatments in [131, Ch. 8] and [117, Ch. 6]). We record it because it recasts $\mathcal{W}_2$ as a least-action principle and because the continuity equation $\partial_t\mu + \operatorname{div}(\mu v) = 0$ is exactly the conservation law that governs the population density in a mean field game.

Physics note

The Benamou–Brenier formula is the only place in this chapter where a physical reading genuinely earns its keep, so we develop it once, carefully, and then put it down. Read μ_t as the time-dependent density of a fluid and v_t as its Eulerian velocity field — the velocity measured at a fixed point in space, not following a particular parcel. Then $\int |v_t|^2\,\mathrm{d}\mu_t$ is the integral of (speed^2) against mass, which is *twice* the kinetic energy $\frac{1}{2}\int |v_t|^2\,\mathrm{d}\mu_t$ of the fluid at the instant t, and the double integral $\int_0^1 \int |v_t|^2\,\mathrm{d}\mu_t\,\mathrm{d}t$ is the time-integrated kinetic energy — the *action* of a pressureless fluid flow (no internal energy, no pressure term, only inertia) that carries μ_0 to μ_1. Equation (4.8) then says that $\mathcal{W}_2^2$ is the *minimal action* over all such flows, the least-effort way to rearrange the fluid; and the continuity equation $\partial_t\mu + \operatorname{div}(\mu v) = 0$ is, as above, mass conservation. The least-action flow has a vivid shape: every parcel travels in a straight line at constant velocity — free transport, no acceleration, no forces — which is the fluid-dynamical face of the word "geodesic" and matches grain-by-grain the McCann picture of theorem 4.9. This is, moreover, the *same* continuity equation that chapter 5 obtains as the Fokker–Planck equation for a population of diffusing agents; the only difference is the provenance of the velocity field — there it is generated by the agents' drift and diffusion, here it is chosen to minimize action. That shared form is why the analogy is worth one paragraph. But the analogy ends at exactly this boundary, and it is important to flag it: $\mathcal{W}_2$ is a *geometry*, a way of measuring how far apart two laws are, not a *dynamics* prescribing how a law must evolve. The fluid is a heuristic for reading the formula, and nothing in the mathematics below — no metric axiom, no compactness statement, no later existence or uniqueness argument — depends on the fluid picture being taken literally.

With displacement geodesics constructed and the dynamic formula in hand, the geometry promised in the chapter's umbrella is complete: $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is a complete, separable, geodesic metric space whose convergence is weak convergence plus second moments and whose straight lines transport mass rather than mix it. We hand three things forward. To section 4.3 we hand the finished arena — with this section's geodesics joining the Polish structure and the compactness criterion of the previous one — so that the standing promise of the chapter preamble, that a mean field game is naturally posed in exactly this space, can finally be made good and the need for the metric (not merely the topology) named precisely. To chapter 5 we hand the continuity equation $\partial_t\mu + \operatorname{div}(\mu v) = 0$, which reappears there as Fokker–Planck. And to chapter 11 and chapter 12 we hand displacement interpolation itself, the curve along which the displacement convexity behind Lasry–Lions monotonicity and the master equation is measured. The concrete payoff of all of this — displacement interpolation between Gaussians staying Gaussian while linear interpolation loses even unimodality — waits for us in section 4.5.

4.3 Why mean field games live in $\mathcal{P}_2(\mathbb{R}^d)$

The previous section finished assembling the geometry the chapter set out to build: by theorem 4.8 and theorem 4.9, $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is a complete, separable, geodesic metric space — a finished arena. The chapter preamble promised that a mean field game would turn out to be naturally posed in exactly this arena, and that the metric — not merely the underlying topology — would be the structure every later existence and uniqueness argument leans on. We can now make that promise good. The game itself is constructed only in chapter 8 and chapter 9, so everything here remains deliberately informal: we are not yet defining an equilibrium, only explaining, in three movements, why the space we have built is the one the definition will need. The first movement says what the mean field *is* — a single point of $\mathcal{P}_2(\mathbb{R}^d)$. The second says what the strategic problem *does* — it turns that point into another point, through a map of the space to itself. The third says why the *metric*, and not just the topology, is what makes that map analyzable, and it untangles the two Lipschitz hypotheses that the rest of the book will lean on but that are perpetually confused with one another.

The mean field is a point. Begin with the finite system the game will be an idealization of. Imagine N agents who are *exchangeable* — interchangeable in their dynamics and costs, distinguished only by their states $X_1, \dots, X_N \in \mathbb{R}^d$ — and suppose, as is the defining structural assumption of a mean field model, that no single agent feels the others individually: each agent's cost and dynamics depend on the rest of the population only through the *statistics* of where everyone is, never through the names attached to the positions. The object that records exactly those statistics, and discards exactly the names, is the *empirical measure*

$$\mu^N = \tfrac{1}{N} \sum_{i=1}^{N} \delta_{X_i}$$

introduced in chapter 1 and chapter 3: it places a lump of mass $1/N$ at each occupied state, so that asking "what fraction of agents sit in the region A?" is answered by $\mu^N(A)$, and the labelling of agents has been forgotten. Two features of this object are worth saying out loud, because together they are the whole point of the movement. First, the empirical measure is *one element of* $\mathcal{P}_2(\mathbb{R}^d)$: it is a bona fide probability measure, and it has finite second moment $\int |x|^2 \ \mu^N(\mathrm{d}x) = \frac{1}{N} \sum_i |X_i|^2$ as soon as the finitely many states are finite, so it is a genuine point of the metric space built above, not a list of N numbers. A configuration of a hundred agents and a configuration of a million agents are, in this view, both just points of the same space, comparable by the same $\mathcal{W}_2$ ruler. Second, as $N \to \infty$ this point stops fluctuating: under the propagation-of-chaos mechanism developed in chapter 7, when the agents become asymptotically independent and identically distributed the random empirical measure μ^N converges to a *deterministic* limit $m \in \mathcal{P}_2(\mathbb{R}^d)$, the *mean field* — the smooth population law that the cloud of N atoms is sampling.

What makes $\mathcal{W}_2$ — rather than the bare weak topology of chapter 3 — the right gauge for this convergence is that $\mathcal{W}_2$ measures precisely the two things one needs controlled at once. When the X_i are i.i.d. draws from m with finite second moment, the law of large numbers in the space of measures gives $\mathcal{W}_2(\mu^N, m) \to 0$ almost surely, and by theorem 4.8 this single statement unpacks into the conjunction of *two* convergences: the empirical distribution converges weakly to m (the right fraction of mass accumulates in every region), *and* its second moment $\frac{1}{N}\sum_i |X_i|^2$ converges to $\int |x|^2\,\mathrm{d}m$ (no mass quietly leaks out to infinity). The second clause is not a technicality one could drop. It is exactly the failure mode the weak topology cannot see: a sequence can converge weakly while a vanishing fraction of its mass runs off to infinity carrying a diverging amount of "energy" $|x|^2$ — think of $(1-\frac{1}{N})m + \frac{1}{N}\delta_{Ne_1}$, weakly indistinguishable from m in the limit yet with a second moment that blows up like N. A mean field game prices agents by quadratic-type costs and couples them through moments of the population, so a notion of "the population is close to m" that tolerates this leakage is useless: the cost would not converge. $\mathcal{W}_2$-closeness forbids the leakage by construction, which is why it, and not weak convergence alone, is the convergence in which propagation of chaos is correctly stated. This is the precise content of the Appendix-A slogan that *the mean field is a point in $\mathcal{P}_2(\mathbb{R}^d)$, not a list of agents*: the limiting object of the N-agent system is a single point of the very metric space we have just finished building, and the metric we put on that space is the one under which the finite systems actually converge to it.

Best response is a map $\mathcal{P}_2(\mathbb{R}^d) \to \mathcal{P}_2(\mathbb{R}^d)$. With the population identified as a point of $\mathcal{P}_2(\mathbb{R}^d)$, the strategic content of the game becomes a statement about a *map* of that space to itself, and it is worth tracing the loop slowly because the rest of Part II is the study of this map. Freeze a candidate population law $m \in \mathcal{P}_2(\mathbb{R}^d)$ — in the time-dependent problem, a whole flow $t \mapsto m_t$ of such laws — and hand it to a single *representative agent* as a fixed feature of her environment: she treats m as given, as though the rest of the world were already settled into that law and would not react to what she does (legitimate in the limit, because one agent among infinitely many has no individual effect on the aggregate). Against this frozen background she solves an honest *optimal control problem* of the kind built in chapter 6: she chooses her control to minimize a cost whose coefficients — her drift b, her volatility σ, her running cost L — depend on m. That optimization returns an optimal feedback, propagating her state forward under it produces a stochastic process, and that process has, at each time, a *law*. Collecting those laws is the output measure (or flow of measures). The assignment of this output to the input m is the *best-response map*

$$\mathsf{S}\colon \mathcal{P}_2(\mathbb{R}^d) \longrightarrow \mathcal{P}_2(\mathbb{R}^d), \qquad m \longmapsto \mathsf{S}(m) = \mathcal{L}\big(\text{optimally controlled state under } m\big).$$

The loop closes on itself: we began with a population law, let a representative agent optimize against it, and read off the law her optimal behaviour generates — another population law. A *mean field equilibrium* is a population at which the loop is stationary, a fixed point $m^\star = \mathsf{S}(m^\star)$: a law that reproduces itself when every agent best-responds to it, so that the environment each agent optimized against is exactly the environment their collective optimal behaviour creates. (In the LQG thread of section 4.5, chapter 6 and chapter 8 this loop becomes wholly explicit and stays inside one family: freeze a Gaussian m with mean $\bar{m}$, the agent's optimal feedback is linear, the controlled state stays Gaussian, and $\mathsf{S}(m)$ is again Gaussian — so the fixed-point equation collapses to a coupled system of ordinary differential equations for the mean and the covariance.)

There is a genuine and useful analogy here to the self-consistent field of mean-field statistical mechanics, where one posits an average field, lets each particle respond to it, and demands that the averaged response reproduce the field one started from — and it is the same fixed-point shape, field $\mapsto$ response $\mapsto$ field. But the analogy must be drawn with its crucial difference flagged, the difference Part II returns to repeatedly: in statistical mechanics the field is fixed by a *thermal average*, each particle settling into the Boltzmann weight of the ambient field, whereas here the field is fixed by an *optimization*, each agent settling into the strategically best response to it. The self-consistency is the same; the mechanism that closes it is averaging in one case and *rational best response* in the other, and that replacement of an average by an argmin is what makes mean field game equilibria capable of the multiplicity and the symmetry-breaking phenomena studied later. We do not press the analogy further now — the game is not yet defined — but the structural lesson is the one this section exists to deliver: *equilibrium is a fixed point of a self-map of* $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$.

Why the metric, and not just the topology. A fixed point of a self-map is exactly the kind of object the metric, and not merely the topology, is needed to locate. To prove that S has a fixed point at all, and that in good cases it has only one, we need quantitative control of how S moves points around, and "quantitative" means metric: we must measure $\mathcal{W}_2$-distances between populations and estimate how S stretches them. There are two classical routes to a fixed point, and the metric enters each in a different but equally essential way.

The first route is *existence by topological fixed-point theorem.* The Schauder and Kakutani theorems invoked in chapter 10 guarantee a fixed point of a continuous self-map provided it is run on a *compact, convex* domain — intuitively, the map must shuffle points around inside a set small enough that it cannot escape to the boundary, so that continuity forces a point to land on itself. Convexity of subsets of $\mathcal{P}_2(\mathbb{R}^d)$ is cheap (linear mixtures of measures are measures). Compactness is the substance, and here the metric does the work: the relevant compactness is $\mathcal{W}_2$-compactness, which theorem 4.8 characterized concretely as *tightness together with uniform integrability of the second moments.* That second clause is again the no-leakage condition, now imposed uniformly over a whole family of candidate populations, and it is exactly what a weak-topology compactness statement would omit and what an existence proof cannot do without. So the very hypothesis that powers the existence theorems of Part II is a metric statement about $\mathcal{P}_2(\mathbb{R}^d)$ that the bare topology cannot phrase.

The second route is *uniqueness*, and it splits into the two mechanisms the chapter preamble named. One is the Lasry–Lions monotonicity of chapter 11 — the standing assumption (A-Mon) on the couplings — which forces any two equilibria to coincide by a convexity argument run along the displacement geodesics of theorem 4.9; that this argument is even *phrased* along geodesics is already a debt to the metric geometry of this chapter. The other, available in the more favourable regime, is a genuine *contraction estimate*

$$\mathcal{W}_2(\mathsf{S}(m), \mathsf{S}(m')) \;\le\; L\, \mathcal{W}_2(m, m'), \qquad L < 1,$$

which makes S a contraction of the complete metric space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ and hands its unique fixed point straight to the Banach theorem. A contraction estimate is unthinkable without a metric on both sides of the inequality: it is, letter for letter, a statement that S shrinks $\mathcal{W}_2$-distances, and there is no weaker topological surrogate for it. This is the sharpest form of the chapter's thesis that we need the geometry and not just the topology.

Both the existence and the contraction routes ask the same thing of the data — that S depend continuously, and quantitatively, on its argument m — and meeting that demand requires two *distinct* kinds of regularity that are perpetually conflated. It is worth separating them cleanly here, at the seam where the metric is introduced, since the confusion they cause is the single most common seam error in the subject. The first is the standing assumption (A-Lip) of appendix A.6: the drift b, the volatility σ, and the running cost L are Lipschitz *in the state* variable, uniformly in time and in the measure argument. This is a statement about how the data of a *single* agent's control problem vary as her *own* state moves, with the population held fixed; it is what makes the control problem of chapter 6 well-posed in the first place (unique solution of the state equation, a Lipschitz value function), and it says nothing whatever about the measure. The second — the one the metric of this chapter was built to express — is a Lipschitz dependence *in the measure argument itself*, measured in $\mathcal{W}_2$:

under (A-LipM) (measure-Lipschitz dependence), there is a constant C with

$$\big|b(x,m)-b(x,m')\big| \le C\,\mathcal{W}_2(m,m'), \qquad \text{and likewise for } \sigma \text{ and } L,$$

uniformly in the state and time variables.

(A-LipM) is a genuinely separate hypothesis from (A-Lip), and the cleanest way to feel that neither implies the other is to watch each hold while the other is vacuous. Take a drift $b(x,m)=\sin x$ that ignores the population entirely: it is Lipschitz in x, so (A-Lip) holds, while its dependence on m is constant, so (A-LipM) holds trivially with $C=0$ — the state-Lipschitz bound certifies nothing about populations because there is no population dependence to certify. Now take instead a drift $b(x,m)=\bar{m}:=\int y\,m(\mathrm{d}y)$ that depends *only* on the population mean: it is constant in x, so (A-Lip) is again trivial, yet its measure dependence is the whole story, and it satisfies (A-LipM) precisely because the mean is a $\mathcal{W}_2$-Lipschitz functional, $|\bar{m}-\bar{m}'|\le \mathcal{W}_2(m,m')$. The two assumptions constrain variation in *orthogonal* arguments — one across an agent's own states with the crowd frozen, the other across crowds with the state frozen — and a bound in one direction is silent about the other. One can even break (A-LipM) while keeping (A-Lip) by letting the data depend on m through a non-$\mathcal{W}_2$-Lipschitz statistic; then the control problem is well-posed for each fixed population but its solution need not vary continuously as the population is perturbed in $\mathcal{W}_2$, and S can fail to be continuous — exactly the failure that wrecks both routes to a fixed point.

It is therefore (A-LipM), and not (A-Lip), that makes the representative agent's control problem depend continuously on m, and hence (A-LipM) that delivers the continuity of S in $\mathcal{W}_2$ on which Schauder and Kakutani depend and the contraction constant L which Banach turns into uniqueness; it is invoked explicitly in the arguments of chapters 8, 10 and 11. The metric is, in the end, not a luxury layered on top of the topology but the structure those arguments are written in: every continuity and every contraction statement about the best-response map of a mean field game is an inequality between $\mathcal{W}_2$-distances, and the entire fixed-point apparatus of Part II would have nothing to grip without it.

This is the resolution of the chapter's standing promise. But it also exposes the next gap. The data of the game — the drift, the cost, the coupling — are not points of $\mathcal{P}_2(\mathbb{R}^d)$ but *functionals* of such a point, depending on the *whole* measure m. The metric tells us when two populations are close and how a map between them stretches distances; it does not yet tell us how to take a derivative in the measure argument. Supplying that missing calculus — in fact the two complementary derivatives the subject uses — is the business of the next section.

4.4 Differentiating functionals of a measure

The previous section ended by exposing a gap the metric alone cannot close, and naming it is the cleanest way to open this one. The data of a mean field game — the running cost, the value, the coupling — are not points of $\mathcal{P}_2(\mathbb{R}^d)$ but *functionals* of such a point: a scalar cost $F(m)$, a value u that depends on the whole population law, a coupling $G[m]$. The geometry built in the previous sections tells us when two such populations are close and how far a map between them can stretch their distance, and the named assumption (A-LipM) pins down the crude first-order variation of these functionals in $\mathcal{W}_2$. But to linearize the best-response map, to run the chain rule that converts a fixed-point equation into a differential equation, and above all to derive the master equation of chapter 12, we need genuine *calculus*: a way to differentiate a number that depends on a measure with respect to that measure. Supplying that calculus is the second pillar this chapter erects, standing beside the metric, and every existence, uniqueness, and master-equation argument later in the book silently rests on both.

There are two ways to perturb a probability measure at first order, and they give two genuinely different derivatives. One can *add* an infinitesimal sliver of mass at a point x — and, to keep the total mass equal to one, remove an equal sliver elsewhere; the first-order response of F to this is the *flat*, or linear functional, derivative $\frac{\delta F}{\delta m}(m, x)$, a scalar field over $\mathbb{R}^d$. Or one can *move* the mass that is already present, nudging each grain a little along a vector field; the first-order response to this is the *Lions* derivative $\partial_\mu F(m)(x)$, a vector field over $\mathbb{R}^d$ telling each grain which way it is worth being pushed. The distinction is not bookkeeping pedantry; it is forced on us by the dynamics. A stochastic flow X_t transports the population law $m_t = \mathcal{L}(X_t)$ by *moving* mass along trajectories, never by creating or destroying it, so the Itô formula for a functional $F(m_t)$ of such a flow — the chain rule from which the master equation is assembled — is necessarily written through the move-mass derivative, the Lions derivative; the add-mass derivative answers a different, static question and is the wrong tool there. Yet the two derivatives are not independent. As we will prove, the Lions derivative is exactly the spatial *gradient* of the flat derivative, $\partial_\mu F(m)(x) = \nabla_x \frac{\delta F}{\delta m}(m, x)$, so the scalar "energy" field and the vector "force" field are two readings of a single object. That gradient relation is the bridge between the two subsections below, and it is what lets a physicist import the entire apparatus from a place he already knows it: the first derivative is the density-functional first variation $\delta E/\delta \rho$ he has computed many times, and the Fréchet differentiability the second one rests on is recalled in place, in definition 4.14, so the development here stands on its own. We take the flat derivative first, on the density-functional on-ramp.

4.4.1 The flat derivative, via the density-functional analogy

We promised in the umbrella to this section that a physicist already owns the first of the two derivatives, and we now cash that promise. The opening move is the simpler of the two perturbations: hold the population in place and *add* an infinitesimal sliver of mass at a point, asking how the functional responds. This is precisely the question a condensed-matter reader answers every time he varies an energy functional with respect to a density, and the flat derivative will turn out to be the optimal-transport name for the object he calls $\delta E/\delta\rho$. So rather than define the flat derivative from nothing, we begin on the density-functional on-ramp, transcribe the familiar first variation, and then make explicit the single bookkeeping subtlety — mass conservation — that distinguishes a probability measure from a general density.

Heuristic (non-rigorous): The on-ramp: density-functional derivatives

A condensed-matter reader has differentiated functionals of a density many times. In density-functional theory one writes an energy $E[\rho]$ as a functional of an electron density $\rho(x)$ and asks how it responds when the density is nudged by a small profile η in a fixed direction, with strength ϵ. The first-order response defines the functional derivative $\delta E/\delta\rho(x)$ through

$$E[\rho + \epsilon\,\eta] - E[\rho] \;=\; \epsilon \int \frac{\delta E}{\delta \rho}(x)\,\eta(x)\;\mathrm{d}x + o(\epsilon).$$

Read this as the statement that, to first order in ϵ, the change in the energy is a *linear* functional of the perturbation η, and the kernel of that linear functional — the weight $\delta E/\delta\rho(x)$ attached to a sliver of mass deposited at x — is the derivative. It is therefore itself a *function of position*, a field over configuration space rather than a single number: it reports, location by location, the marginal energetic price of adding mass there. Physically it is the field thermodynamically conjugate to the density — a local chemical potential — and that is the right mental image to carry: the flat derivative will be the conjugate field to the population law, the landscape of marginal costs that the population feels.

The flat derivative of a functional on $\mathcal{P}_2(\mathbb{R}^d)$ is exactly this object, with one bookkeeping subtlety forced by the constraint that we perturb a *probability* measure, not an arbitrary density. To stay inside $\mathcal{P}_2(\mathbb{R}^d)$ the perturbed object $m + \epsilon\eta$ must again be a probability measure, so the sliver η may not change the total mass: it must integrate to zero, $\int \eta\;\mathrm{d}x = 0$. The honest way to produce such an η is to take it to be the difference $\eta = m' - m$ of two probability measures, which automatically has total mass $1 - 1 = 0$ — adding a little mass somewhere is paid for by removing an equal amount elsewhere. The consequence is that the derivative $\delta F/\delta m(x)$ is only ever *tested against zero-mean perturbations*, and a function tested only against zero-mean weights cannot detect a constant added to itself. The flat derivative is therefore defined only up to an additive constant. Everything past this one subtlety is the DFT computation the reader already knows.

Definition 4.11 (Flat / linear functional derivative)**.** A functional $F\colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ has a *flat derivative* (linear functional derivative) if there is a continuous map $\frac{\delta F}{\delta m}(\cdot,\cdot)\colon \mathcal{P}_2(\mathbb{R}^d) \times \mathbb{R}^d \to \mathbb{R}$, of at most quadratic growth in x locally uniformly in m, such that for all $m, m' \in \mathcal{P}_2(\mathbb{R}^d)$,

$$F(m') - F(m) \;=\; \int_0^1\!\!\int_{\mathbb{R}^d} \frac{\delta F}{\delta m}(m_s, x)\,(m' - m)(\mathrm{d}x)\;\mathrm{d}s, \qquad m_s := (1-s)\,m + s\,m'. \tag{4.9}$$

Equivalently, in differential form, for every $m' \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\,F(m + \epsilon(m' - m))\Big|_{\epsilon=0} \;=\; \int_{\mathbb{R}^d} \frac{\delta F}{\delta m}(m, x)\,(m' - m)(\mathrm{d}x). \tag{4.10}$$

We fix the additive constant by the normalization $\int_{\mathbb{R}^d} \frac{\delta F}{\delta m}(m, x)\, m(\mathrm{d}x) = 0$.

The definition offers the derivative in two equivalent guises, and it is worth being explicit about why each is present and how they imply one another, because the book uses them for different jobs. Equation (4.10) is the definition in its simplest form: it is the direct transcription of the DFT response above, with the perturbation direction $m' - m$ playing the exact role of η and ϵ the role of the strength. It is an *infinitesimal* statement — it pins down only the slope of F at m in the single straight direction pointing toward m' — and that is all one needs to read the derivative off a worked functional, which is why every computation below starts from it.

Equation (4.9) is the same derivative integrated up. Consider the straight segment of measures $m_s = (1-s)m + s\,m'$, $s \in [0,1]$, which interpolates linearly from m at $s = 0$ to m' at $s = 1$ (this is the flat interpolation, not the displacement geodesic of theorem 4.9; here it is only a bookkeeping path). Along it the scalar function $g(s) := F(m_s)$ has, by (4.10) applied at the base point m_s in the direction $m' - m = \frac{\mathrm{d}}{\mathrm{d}s} m_s$, the derivative $g'(s) = \int_{\mathbb{R}^d} \frac{\delta F}{\delta m}(m_s, x)\,(m' - m)(\mathrm{d}x)$; the ordinary fundamental theorem of calculus $F(m') - F(m) = g(1) - g(0) = \int_0^1 g'(s)\,\mathrm{d}s$ is then exactly (4.9). The reason to keep this integrated form on hand is that it is an *exact* identity for a *finite* difference $F(m') - F(m)$, not merely an $o(\epsilon)$ approximation: it expresses the full increment of F between two measures as a single integral of the derivative along the connecting segment. That is precisely the lever needed to turn a bound on the derivative into a bound on F itself — if $\frac{\delta F}{\delta m}(m_s, x)$ is, say, Lipschitz in x uniformly in s, then (4.9) converts that pointwise control directly into a $\mathcal{W}_2$-modulus of continuity for F, the kind of estimate the monotonicity and stability arguments of chapter 11 are built from. The differential form gives the derivative; the integral form is how the derivative controls finite changes.

The normalization removes the additive-constant ambiguity by demanding that the derivative have zero average against the base measure, $\int_{\mathbb{R}^d} \frac{\delta F}{\delta m}(m, x)\, m(\mathrm{d}x) = 0$ — in DFT terms, a choice of reference level for the chemical potential. It is a gauge fixing, harmless and reversible: it neither adds nor removes information, it merely selects one representative from the family of admissible derivatives so that formulas can be compared.

Notation watch

The additive-constant gauge deserves to be made fully concrete, because it is a frequent source of apparent contradictions between sources. Because $(m' - m)$ is a signed measure of total mass zero, adding any constant c to $\frac{\delta F}{\delta m}(m, \cdot)$ leaves (4.10) unchanged: $\int c\,(m' - m)(\mathrm{d}x) = c \cdot 0 = 0$, the constant being annihilated by the zero total mass of the perturbation. The flat derivative is therefore intrinsically defined only up to an additive constant. This is the exact analogue of the freedom a physicist already lives with in a scalar potential: a potential energy, or an electrostatic potential, is physical only through its *differences*, and one is free to slide its zero to any convenient reference; what is measurable — the force, the work done between two configurations — is untouched by that shift. Here the role of "force" is played by the integral against a zero-mean perturbation, which sees only differences of $\frac{\delta F}{\delta m}(m,\cdot)$ across x, never its absolute level. The normalization $\int \frac{\delta F}{\delta m}(m,x)\,m(\mathrm{d}x) = 0$ is one convenient choice of reference — it sets the m-average of the derivative to zero, the analogue of grounding the potential at the population's own mean level — but it is one gauge among many, chosen only so that distinct formulas can be lined up.

The practical warning is for cross-source comparison. Different texts fix the reference differently — some normalize against m as we do, some against Lebesgue measure, some leave the constant free — so two correct expressions for the same $\frac{\delta F}{\delta m}(m,\cdot)$ can differ by a function of m alone (constant in x, though it may depend on the base measure). When a formula here disagrees with one elsewhere by such an x-independent term, that is a gauge discrepancy, not an error: subtract its m-average and the two agree.

Example 4.12 (The three functionals we actually meet)**.**

1. *Linear functionals.* If $F(m) = \int_{\mathbb{R}^d} \varphi \,\mathrm{d}m$ for a fixed φ of quadratic growth, then (4.10) gives $\frac{\delta F}{\delta m}(m,x) = \varphi(x)$ (before normalization; $\varphi(x) - \int \varphi \,\mathrm{d}m$ after). The flat derivative of a linear functional is its integrand, just as in DFT the functional derivative of $\int V\rho$ is V.

2. *Smooth functions of moments.* If $F(m) = \xi\big(\langle\varphi_1, m\rangle, \dots, \langle\varphi_k, m\rangle\big)$ with $\langle\varphi_i, m\rangle = \int \varphi_i \,\mathrm{d}m$ and $\xi \in C^1(\mathbb{R}^k)$, the computation is the ordinary multivariable chain rule, with the directional form (4.10) doing the differentiation. Each scalar moment is itself linear in m, so along $m + \epsilon(m' - m)$ it moves at constant rate, $\frac{\mathrm{d}}{\mathrm{d}\epsilon}\langle\varphi_i, m + \epsilon(m' - m)\rangle = \langle\varphi_i, m' - m\rangle = \int \varphi_i\,(m' - m)(\mathrm{d}x)$, by case (1). Feeding these k inner derivatives into ξ through the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon} F\big(m + \epsilon(m' - m)\big)\Big|_{\epsilon=0} = \sum_{i=1}^{k} \partial_i \xi\big(\langle\varphi_1, m\rangle, \dots, \langle\varphi_k, m\rangle\big) \int_{\mathbb{R}^d} \varphi_i(x)\,(m' - m)(\mathrm{d}x),$$

   and matching this against $\int \frac{\delta F}{\delta m}(m,x)\,(m' - m)(\mathrm{d}x)$ reads off

$$\frac{\delta F}{\delta m}(m,x) = \sum_{i=1}^{k} \partial_i \xi\big(\langle\varphi_1, m\rangle, \dots, \langle\varphi_k, m\rangle\big)\,\varphi_i(x).$$

   Note that the partial derivatives $\partial_i \xi$ are evaluated at the *current* moments of m, so unlike case (1) the flat derivative now depends on m — nonlinearity in F has entered through ξ even though each φ_i is probed linearly. This is the workhorse class: most explicit couplings in the book are smooth functions of a few moments of m, the running cost of the LQG thread being a function of the mean alone (remark 4.18).

3. *Quadratic interaction functionals.* If $F(m) = \frac{1}{2}\iint_{\mathbb{R}^d \times \mathbb{R}^d} W(x,y)\,m(\mathrm{d}x)\,m(\mathrm{d}y)$ with W symmetric and bounded, write $\eta = m' - m$ and expand $F(m + \epsilon\eta)$ in powers of ϵ. Because the

integrand is bilinear in its two measure slots, substituting $m+\epsilon\eta$ into each slot and multiplying out produces exactly four terms, graded by their power of ϵ:

$$F(m+\epsilon\eta) = \tfrac{1}{2}\iint W\,(m+\epsilon\eta)(\mathrm{d}x)\,(m+\epsilon\eta)(\mathrm{d}y) = \underbrace{\tfrac{1}{2}\iint W\,m\,m}_{F(m)} + \epsilon\,\tfrac{1}{2}\iint W\,[\eta\,m + m\,\eta] + \epsilon^2\,\tfrac{1}{2}\iint W\,\eta\,\eta,$$

where the arguments (x, y) and the differentials are suppressed for legibility. The quadratic structure makes the expansion *terminate* at order ϵ^2 with no remainder — there is no $o(\epsilon^2)$ — which is special to interaction energies and is why they are the cleanest nontrivial example. The symmetry $W(x, y) = W(y, x)$ makes the two cross terms in the bracket equal — relabelling $(x, y) \leftrightarrow (y, x)$ turns one into the other — so they combine into $2\iint W\,\eta(\mathrm{d}x)\,m(\mathrm{d}y)$ and the prefactor $\frac{1}{2}$ cancels. Reading off the coefficient of ϵ as $\int \frac{\delta F}{\delta m}(m, x)\,\eta(\mathrm{d}x)$ identifies

$$\frac{\delta F}{\delta m}(m, x) = \int_{\mathbb{R}^d} W(x, y)\,m(\mathrm{d}y)$$

(before normalization; the ϵ^2 term is the genuinely nonlinear remainder and does not enter the first variation). The flat derivative of a pairwise-interaction energy is the *mean field generated by m through the kernel W* — the single-particle potential felt at x when every other grain of the population contributes its share $W(x, y)$, averaged against where the population actually sits. This is the self-consistent field of mean field theory, recovered here as a flat derivative: a scalar energy landscape over $\mathbb{R}^d$, one value per location x.

This is not a coincidence of notation, and it is worth flagging the connection precisely, with its one caveat. The map $m \mapsto \int W(x, y)\,m(\mathrm{d}y)$ is the action of the integral operator $\mathbb{W}$ of appendix A.5 on the measure m, so the flat derivative of a pairwise-interaction energy *is* the graphon coupling operator applied to the population — the static prototype of the nonlocal coupling that drives every graphon mean field game of Part IV. The caveat is the type-variable warning of the notation conventions (appendix A.5): here $x \in \mathbb{R}^d$ is a *spatial state* and $W(x, y)$ is an interaction kernel on physical space, whereas in Part IV the first argument of the kernel is the *type* $x \in [0, 1]$ — an exchangeable label, not a position — and $\mathbb{W}$ there is the Hilbert–Schmidt operator $(\mathbb{W}f)(x) = \int_0^1 W(x, y)f(y)\,\mathrm{d}y$ on $L^2[0, 1]$. The algebra of this example is identical in the two settings; only the meaning of the variable the kernel is indexed by changes. Keeping that distinction in view now is what lets the reader meet the graphon coupling in Part IV as the same object computed here rather than a new one. (The notation collides deliberately: the same letter x that is a point of $\mathbb{R}^d$ throughout this chapter is, in Part IV, reused for the type $x \in [0, 1]$; nothing in the algebra changes, but it is the reader's job to remember which meaning is in force.)

Remark 4.13 (Why the flat derivative is the right object for monotonicity)**.** The flat derivative is precisely the object in which the uniqueness theory of chapter 11 is phrased, which is worth flagging now so the reader meets it there as an old friend. The Lasry–Lions monotonicity condition that forces a mean field game to have a *unique* equilibrium asks that the coupling functional F satisfy

$$\int_{\mathbb{R}^d} \Big(\frac{\delta F}{\delta m}(m, x) - \frac{\delta F}{\delta m}(m', x)\Big)\,(m - m')(\mathrm{d}x) \;\geq\; 0 \qquad \text{for all } m, m' \in \mathcal{P}_2(\mathbb{R}^d),$$

a statement entirely about the flat derivative, tested against the mass-conserving difference $m - m'$. The gauge freedom is harmless here for the very reason it was harmless in definition 4.11: an additive constant in $\frac{\delta F}{\delta m}(m,\cdot)$ integrates to zero against the zero-mean difference $m - m'$, so the inequality does not depend on the choice of normalization. Example 3 above is the prototype — a pairwise interaction through a symmetric kernel W is monotone exactly when W is of positive type — and this static positivity is the seed that chapter 11 grows into a full uniqueness proof. It is the add-mass derivative, not the move-mass one, that monotonicity wants, because the condition compares two whole populations against each other rather than following one as it flows.

We have, then, the first of our two derivatives in hand, and the interaction example is the right note to close on, because it shows exactly what the flat derivative delivers and what it withholds. It delivers a *field*: a scalar energy landscape $\frac{\delta F}{\delta m}(m,\cdot)$ over $\mathbb{R}^d$, the self-consistent potential the population generates, one number per location. That is the natural answer to the static question "what does it cost to add a grain here?" But the dynamics of a mean field game never add grains; a stochastic flow X_t transports the law $m_t = \mathcal{L}(X_t)$ by *moving* the mass already present, and a grain that moves feels not a potential but a *force* — the spatial gradient of that potential. The flat derivative therefore points beyond itself: to differentiate F along a flow we will need to take the gradient $\nabla_x \frac{\delta F}{\delta m}(m,x)$, converting the energy field of the interaction example into the force field that actually pushes the population. That gradient is the Lions derivative, and constructing it directly — by the sharper perturbation that moves mass rather than adding it — is the business of the next subsection.

4.4.2 The Lions derivative, via lifting to random variables

The previous subsection closed by pointing past itself: the flat derivative $\frac{\delta F}{\delta m}(m,\cdot)$ delivers a scalar *energy field* over $\mathbb{R}^d$, answering the static question "how does F change if I *add* a little mass at x?" — whereas the dynamics of a mean field game never add mass, a stochastic flow X_t transporting the law $m_t = \mathcal{L}(X_t)$ by *moving* the grains already present. So the question we must now answer is the sharper one: how does F change if I *move* the mass that is already there, nudging each grain a little? Its answer is the *Lions derivative* $\partial_\mu F(m)(x)$, a vector field over $\mathbb{R}^d$ assigning to each location the direction it is worth pushing the grain sitting there. The end of this subsection will prove that this move-mass derivative is exactly the spatial gradient of the add-mass derivative, so that the energy field and the force field are two readings of one object; but first we must construct it, and that takes a genuinely new device.

The distinction is not pedantry — it is forced on us by the very calculation the master equation is assembled from. The chain rule for a function $F(m_t)$ of the *law* $m_t = \mathcal{L}(X_t)$ of an Itô process — the Itô formula along a flow of measures — is necessarily stated through the Lions derivative, because what a stochastic flow changes is *where* the mass sits, not *how much* of it there is. That law-level chain rule is built on the ordinary Itô formula of theorem 5.9; its measure-flow form, together with the second-order Lions calculus it requires, is the engine of the master equation of chapter 12 and is developed in full in appendix D.6. We are here laying its first-order foundation.

The obstacle to differentiating F directly is that its domain $\mathcal{P}_2(\mathbb{R}^d)$ is not a linear space: one cannot add two probability measures and stay a probability measure, so there is no ambient vector space in which to form difference quotients in the usual way. The flat derivative dodged this by perturbing along the one direction that *does* stay admissible, the mass-conserving segment $m+\epsilon(m'-m)$. Lions's device dodges it more radically and more powerfully: it *lifts* the functional off the curved space of measures and onto a flat Hilbert space, where the entire differential calculus of chapter 2 is already waiting. Fix once and for all an atomless probability space $(\Omega, \mathcal{F}, \mathbb{P})$ rich enough to carry, for every $m \in \mathcal{P}_2(\mathbb{R}^d)$, at least one random variable with law m; the canonical choice is $\Omega = [0,1]$ with Lebesgue measure, which is atomless and supports a variable of any prescribed law (compose the desired quantile function with the identity coordinate). The *lift* of $F\colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ is the functional

$$\widetilde{F}\colon L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R}^d) \to \mathbb{R}, \qquad \widetilde{F}(X) = F(\mathcal{L}(X)),$$

which evaluates F on the law of whatever random variable it is fed. Its domain $H = L^2(\Omega; \mathbb{R}^d)$ is exactly the Hilbert space of square-integrable random variables built in chapter 2 — a genuine vector space with inner product $\langle X, Y\rangle_H = \mathbb{E}[X \cdot Y]$ — and on it $\widetilde{F}$ is just an ordinary functional, of the kind we know how to differentiate. We have traded a single point of the awkward nonlinear space $\mathcal{P}_2(\mathbb{R}^d)$ for a whole *fibre* of representatives in H: every X with $\mathcal{L}(X) = m$ is a lift of the same m, so $\widetilde{F}$ is constant along each fibre. That redundancy looks wasteful, but it is precisely the slack that makes the construction work, and the structure theorem below is the statement that the answer does not depend on which representative we picked.

We differentiate $\widetilde{F}$ in the Fréchet sense, recalled here in place so the section stands on its own. A functional $G\colon H \to \mathbb{R}$ on a Hilbert space is *Fréchet differentiable* at $X \in H$ if its increment is, to leading order, a *linear* function of the displacement: there is a bounded linear map $DG(X)\colon H \to \mathbb{R}$ with

$$G(X+Y) - G(X) - DG(X)[Y] \;=\; o\big(\|Y\|_H\big) \qquad \text{as } \|Y\|_H \to 0.$$

This is word for word the definition of the total derivative from multivariable calculus, with the finite-dimensional gradient promoted to a bounded linear functional on H and the little-o measured in the L^2 norm; the boundedness of $DG(X)$ is the infinite-dimensional stand-in for "the gradient is a finite vector." By the Riesz representation theorem (theorem 2.6), every bounded linear functional on a Hilbert space is inner product against a unique vector, so the differential $DG(X)$ is realized by a unique *gradient vector*, which we also write $DG(X) \in H$, through $DG(X)[Y] = \langle DG(X), Y\rangle_H = \mathbb{E}\big[DG(X)\cdot Y\big]$. The gradient is thus itself a square-integrable random variable. Lions's observation is that when $G = \widetilde{F}$ is the lift of a functional of the *law*, this gradient random variable is not arbitrary: it is a deterministic function of the underlying variable X alone, and one that depends on X only through its law.

Definition 4.14 (Lions / L-derivative)**.** $F\colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ is *L-differentiable* (Lions differentiable) at m if its lift $\widetilde{F}$ is Fréchet differentiable at some (hence any) X with $\mathcal{L}(X) = m$. In that case the gradient $D\widetilde{F}(X) \in L^2(\Omega; \mathbb{R}^d)$, identified with an element of L^2 via the Riesz isomorphism of chapter 2, takes the form

$$D\widetilde{F}(X) \;=\; \partial_\mu F(m)(X) \qquad \text{a.s.,} \tag{4.11}$$

for a function $\partial_\mu F(m)\colon \mathbb{R}^d \to \mathbb{R}^d$ in $L^2(m; \mathbb{R}^d)$, called the *Lions derivative* of F at m. It depends on X only through its law m.

That the gradient depends on X only through its law — the content packed into (4.11) — is a genuine theorem, not a definition: it follows because $\widetilde{F}$ is invariant under measure-preserving rearrangements of Ω (two random variables with the same law have the same $\widetilde{F}$-value), and a rearrangement-invariant smooth functional has a rearrangement-equivariant gradient. The full statement and proof are due to Lions's Collège de France lectures, reported in [40, Ch. 5]; we state and prove it in full in-book as theorem D.5.

The two derivatives are not independent. The flat derivative is scalar-valued and lives on $\mathbb{R}^d$; the Lions derivative is $\mathbb{R}^d$-valued. The precise link, anticipated in appendix A.3, is that the Lions derivative is the *spatial gradient* of the flat derivative.

Proposition 4.15 (Flat and Lions derivatives are related by a gradient)**.** *Suppose F has a flat derivative $\frac{\delta F}{\delta m}(m,\cdot)$ that is continuously differentiable in x with at most linear growth of its gradient. Then F is L-differentiable and*

$$\partial_\mu F(m)(x) \;=\; \nabla_x \frac{\delta F}{\delta m}(m,x). \tag{4.12}$$

Proof sketch. Let X have law m, and let $Y \in L^2(\Omega;\mathbb{R}^d)$ be a perturbation direction; write $m_\epsilon = \mathcal{L}(X+\epsilon Y)$ for the curve of laws it traces as ϵ varies. The plan is to compute $\frac{\mathrm{d}}{\mathrm{d}\epsilon}\widetilde{F}(X+\epsilon Y)|_{\epsilon=0} = \frac{\mathrm{d}}{\mathrm{d}\epsilon}F(m_\epsilon)|_{\epsilon=0}$ in two ways and match them. Differentiating F along the curve m_ϵ through its flat derivative (4.10) pairs $\frac{\delta F}{\delta m}(m,\cdot)$ with the *velocity* of the curve — the rate at which mass is being redistributed — and that velocity is read off one test function at a time. For any smooth g,

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\int_{\mathbb{R}^d} g \;\mathrm{d}m_\epsilon\Big|_{\epsilon=0} = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\mathbb{E}\big[g(X+\epsilon Y)\big]\Big|_{\epsilon=0} = \mathbb{E}\big[\nabla g(X)\cdot Y\big],$$

where the first equality is the definition of $m_\epsilon = \mathcal{L}(X+\epsilon Y)$ and the second is the ordinary chain rule inside the expectation. So, to first order, the curve m_ϵ moves by transporting mass with a velocity tied to Y, and a gradient lands on whatever test function it is paired with. Taking $g = \frac{\delta F}{\delta m}(m,\cdot)$ — legitimate because it is C^1 in x by hypothesis — and feeding this velocity into the flat-derivative chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}F(m_\epsilon)\Big|_{\epsilon=0} = \mathbb{E}\Big[\nabla_x \frac{\delta F}{\delta m}(m,X)\cdot Y\Big].$$

The gradient sits on $\frac{\delta F}{\delta m}(m,\cdot)$ precisely because the curve's velocity carried it off the test function and onto the derivative — this is the "integration by parts" of the informal statement — and it is exactly here that the at-most-linear-growth hypothesis on $\nabla_x\frac{\delta F}{\delta m}(m,x)$ is spent: it keeps the expectation finite for every $Y \in L^2(\Omega;\mathbb{R}^d)$, so that the right-hand side is a genuine bounded linear functional of Y. The left side is, by the definition of the Fréchet gradient and the Riesz identification of theorem 2.6, $\langle D\widetilde{F}(X), Y\rangle_{L^2} = \mathbb{E}[D\widetilde{F}(X)\cdot Y]$. Since the two expressions agree for *every* direction Y, their integrands agree almost surely,

$$D\widetilde{F}(X) = \nabla_x \frac{\delta F}{\delta m}(m,X) \quad \text{a.s.},$$

which is the structure (4.11) read with $\partial_\mu F(m)(x) = \nabla_x\frac{\delta F}{\delta m}(m,x)$ — the claimed relation (4.12). Note in passing that the gradient automatically depends on X only through its law m, as definition 4.14 demands, since $\nabla_x\frac{\delta F}{\delta m}(m,\cdot)$ is a fixed function on $\mathbb{R}^d$ evaluated at X. The full justification of the curve-differentiation step, including the regularity that makes it rigorous, is [40, Sec. 5.4]. ☐

Example 4.16 (Lions derivatives of the three functionals)**.** Applying (4.12) to example 4.12:

$$F(m) = \int \varphi \, \mathrm{d}m \;\Rightarrow\; \partial_\mu F(m)(x) = \nabla\varphi(x); \qquad F(m) = \tfrac{1}{2}\iint W \, \mathrm{d}m \, \mathrm{d}m \;\Rightarrow\; \partial_\mu F(m)(x) = \int \nabla_x W(x,y)\, m(\mathrm{d}y).$$

The Lions derivative of the linear functional is the gradient of its integrand; the Lions derivative of the interaction energy is the gradient of the self-consistent field — the *force* that the population exerts on a particle at x. This is the form in which the coupling enters the Hamiltonian of the mean field game, and it is the bridge from the flat derivative (an energy) to the Lions derivative (a force) that the reader should keep.

We stop the development at first order. The full second-order calculus on $\mathcal{P}_2(\mathbb{R}^d)$ — the Itô formula along flows of measures, which requires a *second* Lions derivative $\partial^2_\mu F$ measuring how the force field itself responds when the population moves — is the engine of the master equation of chapter 12, and the complete construction of both derivatives, with their interrelations and regularity theory, is collected in appendix D.6, which rests in turn on the ordinary Itô formula theorem 5.9 of the next chapter. What this section carries forward is exactly three things: the flat derivative definition 4.11, the Lions derivative definition 4.14, and the gradient relation (4.12) that ties them together.

These three are not abstractions awaiting a distant payoff; the very next section spends them on the model the book carries throughout. In the Gaussian computation that follows, the closing remark remark 4.18 returns to the first-moment coupling of the LQG thread and writes down both derivatives explicitly — the flat derivative an affine field $x \mapsto q\, s^2 \bar{m} - q\, s\, x$, the Lions derivative a constant force $-q\, s\, \mathbf{1}$ — so that the energy-to-force passage of example 4.16 is instantiated concretely on the running example. With the metric of the earlier sections and the differential calculus of this one both in hand, $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is now a space we can both measure and differentiate over, and the remaining section turns to the one computation the whole LQG thread leans on.

4.5 Worked computation: $\mathcal{W}_2$ between two Gaussians

Worked computation 4.17 (The Bures–Wasserstein distance; Thread A geometric backbone)**.** The LQG thread (Thread A, appendix A.7) has the defining feature that the population law is always Gaussian: the linear state SDE of eq. (5.31) with Gaussian initial data propagates a normal law $m_t = \mathcal{N}(\bar{m}_t, v_t)$ with mean $\bar{m}_t$ and variance v_t — the Gaussian flow derived explicitly in worked computation 5.17, with moments (5.28) — and the control of chapter 6 keeps it Gaussian. To measure how far apart two population states are — which is what every $\mathcal{W}_2$-contraction estimate in chapter 14 requires — we need $\mathcal{W}_2$ between two Gaussians in closed form. We derive it now; it is the geometric skeleton on which the explicit LQG fixed point is later built.

Notation watch

Throughout this section we denote the *mean vector* of a Gaussian by $\bar{m}$ (matching the population-mean notation $\bar{m}_t$ of worked computation 5.17 and Thread A) and reserve m for the measure itself, $\mu_i = \mathcal{N}(\bar{m}_i, \Sigma_i)$; do not read $\bar{m}_i$ as a measure. In the scalar specialization below, a Gaussian *standard deviation* appears as $\sqrt{v_i}$ where v_i is the variance. We deliberately avoid the symbol σ for a standard deviation here: in chapter 5 and Thread A, $\sigma = \sigma$ is the reserved noise/volatility coefficient of the dynamics, and both meanings will be in scope together downstream in chapter 14.

Setup. Let $\mu_0 = \mathcal{N}(\bar{m}_0, \Sigma_0)$ and $\mu_1 = \mathcal{N}(\bar{m}_1, \Sigma_1)$ be two Gaussians on $\mathbb{R}^d$ with mean vectors $\bar{m}_0, \bar{m}_1 \in \mathbb{R}^d$ and symmetric positive-definite covariances Σ_0, Σ_1. We claim

$$\mathcal{W}_2(\mu_0, \mu_1)^2 \;=\; |\bar{m}_0 - \bar{m}_1|^2 \;+\; \operatorname{tr}\Big(\Sigma_0 + \Sigma_1 - 2\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Big), \tag{4.13}$$

where $\Sigma^{1/2}$ denotes the unique symmetric positive-definite square root. The mean contributes a flat Euclidean term; the covariances contribute the squared *Bures distance* $d_B(\Sigma_0, \Sigma_1)^2 = \operatorname{tr}\big(\Sigma_0 + \Sigma_1 - 2(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2})^{1/2}\big)$.

The optimal map is affine. The strategy is to exhibit a transport map of the gradient-of-convex form demanded by Brenier's theorem (theorem 4.6); by that theorem it is automatically optimal, so its transport cost *is* $\mathcal{W}_2^2$. Try an affine map

$$T(x) = \bar{m}_1 + A\,(x - \bar{m}_0), \qquad A \text{ symmetric positive definite},$$

which is the gradient of the convex quadratic $\phi(x) = \langle \bar{m}_1, x\rangle + \frac{1}{2}\langle x - \bar{m}_0, A(x - \bar{m}_0)\rangle$ (convex since $A \succ 0$). We must choose A so that T pushes μ_0 to μ_1. An affine image of a Gaussian is Gaussian with transformed parameters: $T_{\#}\mu_0 = \mathcal{N}(\bar{m}_1,\, A\Sigma_0 A)$ (the mean is $T(\bar{m}_0) = \bar{m}_1$ and the covariance is $A\Sigma_0 A^\top = A\Sigma_0 A$ since A is symmetric). The push-forward equals μ_1 iff

$$A\,\Sigma_0\,A = \Sigma_1. \tag{4.14}$$

Set

$$A \;=\; \Sigma_0^{-1/2}\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Sigma_0^{-1/2}.$$

This A is symmetric and positive definite (it is a congruence of a positive-definite square root). It satisfies (4.14): writing $M = \Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}$,

$$A\Sigma_0 A = \Sigma_0^{-1/2}M^{1/2}\Sigma_0^{-1/2}\,\Sigma_0\,\Sigma_0^{-1/2}M^{1/2}\Sigma_0^{-1/2} = \Sigma_0^{-1/2}M^{1/2}M^{1/2}\Sigma_0^{-1/2} = \Sigma_0^{-1/2}M\Sigma_0^{-1/2} = \Sigma_1,$$

using $\Sigma_0^{-1/2}\Sigma_0\Sigma_0^{-1/2} = I$ and $M^{1/2}M^{1/2} = M$. Thus $T = \nabla\phi$ is a gradient of a convex function pushing μ_0 to μ_1, and by theorem 4.6 it is the optimal transport map.

Computing the cost. Let $X \sim \mu_0$, so $\mathbb{E}[X] = \bar{m}_0$ and $\operatorname{Cov}(X) = \Sigma_0$. The displacement is

$$X - T(X) = (X - \bar{m}_0) - A(X - \bar{m}_0) + (\bar{m}_0 - \bar{m}_1) = (I - A)(X - \bar{m}_0) - (\bar{m}_1 - \bar{m}_0).$$

Since $\mathbb{E}[(I - A)(X - \bar{m}_0)] = 0$, the cross term vanishes in expectation and

$$\mathcal{W}_2(\mu_0, \mu_1)^2 = \mathbb{E}[|X - T(X)|^2] = |\bar{m}_1 - \bar{m}_0|^2 + \mathbb{E}\big[(X - \bar{m}_0)^\top (I - A)^2 (X - \bar{m}_0)\big] = |\bar{m}_1 - \bar{m}_0|^2 + \operatorname{tr}\big((I - A)^2\Sigma_0\big).$$

Expand $\operatorname{tr}\left((I-A)^2\Sigma_0\right) = \operatorname{tr}\left((I-2A+A^2)\Sigma_0\right) = \operatorname{tr}(\Sigma_0) - 2\operatorname{tr}(A\Sigma_0) + \operatorname{tr}(A\Sigma_0 A)$, where we used symmetry of A and Σ_0 to write $\operatorname{tr}(A^2\Sigma_0) = \operatorname{tr}(A\Sigma_0 A)$. By (4.14) the last trace is $\operatorname{tr}(\Sigma_1)$. For the middle trace, the cyclic property gives

$$\operatorname{tr}(A\Sigma_0) = \operatorname{tr}\Big(\Sigma_0^{-1/2} M^{1/2} \Sigma_0^{-1/2} \Sigma_0\Big) = \operatorname{tr}\Big(\Sigma_0^{-1/2} M^{1/2} \Sigma_0^{1/2}\Big) = \operatorname{tr}\Big(M^{1/2}\Big) = \operatorname{tr}\Big(\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Big).$$

Collecting the three pieces,

$$\operatorname{tr}\big((I-A)^2\Sigma_0\big) = \operatorname{tr}(\Sigma_0) + \operatorname{tr}(\Sigma_1) - 2\operatorname{tr}\Big(\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Big),$$

which together with the mean term is exactly (4.13). □

The scalar case — the Thread A backbone. In one dimension the covariances are variances, $\Sigma_i = v_i$, and the matrix square roots are ordinary square roots: $\big(\sqrt{v_0}\, v_1 \sqrt{v_0}\big)^{1/2} = \sqrt{v_0}\sqrt{v_1}$. Formula (4.13) collapses to the strikingly simple

$$\mathcal{W}_2\big(\mathcal{N}(\bar m_0, v_0), \mathcal{N}(\bar m_1, v_1)\big)^2 \;=\; (\bar m_0 - \bar m_1)^2 + \big(\sqrt{v_0} - \sqrt{v_1}\big)^2. \tag{4.15}$$

That is: the $\mathcal{W}_2$ distance between two scalar Gaussians is the ordinary Euclidean distance between the points $(\bar m_0, \sqrt{v_0})$ and $(\bar m_1, \sqrt{v_1})$ in the mean–standard-deviation half-plane $\mathbb{R} \times (0, \infty)$. The space of scalar Gaussians, with $\mathcal{W}_2$, is flat. This is the precise sense in which the LQG thread is geometrically trivial and hence fully solvable: as the population law $m_t = \mathcal{N}(\bar m_t, v_t)$ evolves, its motion in $(\mathcal{P}_2(\mathbb{R}), \mathcal{W}_2)$ is just the motion of the point $(\bar m_t, \sqrt{v_t})$ in a Euclidean half-plane, and a $\mathcal{W}_2$-contraction of the best-response map becomes an ordinary contraction in two real coordinates. Chapter 14 exploits exactly this when it reduces the LQG fixed point to a scalar Riccati-plus-mean system.

Geodesics stay Gaussian. The displacement interpolation of theorem 4.9 between μ_0 and μ_1 uses $T_t = (1-t)I + tA$, an affine map, so $\mu_t = (T_t)_\#\mu_0$ is again Gaussian: $\mu_t = \mathcal{N}\big((1-t)\bar m_0 + t\bar m_1,\ ((1-t)I + tA)\Sigma_0((1-t)I + tA)\big)$. The Gaussian family is geodesically closed in $\mathcal{W}_2$. By contrast the linear interpolation $(1-t)\mu_0 + t\mu_1$ is a *mixture* of two Gaussians — generically bimodal, not Gaussian at all — which visibly illustrates the warning of section 4.2 that the $\mathcal{W}_2$-geodesic is the displacement curve, not the straight line in the vector space of measures. The mean travels linearly, $(1-t)\bar m_0 + t\bar m_1$, while the standard deviation in the scalar case travels linearly too, $\sqrt{v_t} = (1-t)\sqrt{v_0} + t\sqrt{v_1}$, consistent with the flat geometry of (4.15).

Remark 4.18 (Connection to the static interaction functional)**.** Combining this section with example 4.12(3) closes a small loop that recurs in the LQG and graphon threads. When the running cost penalizes the distance of an agent from the population mean — as the term $\frac{q}{2}(X_t - s\,\bar m_t)^2$ does in Thread A (appendix A.7) — the cost felt by the representative agent is a smooth function of the first moment $\bar m = \int y\, m(\mathrm{d}y)$ of the population. Its flat derivative is the linear field $y \mapsto -q\, s\,(x - s\bar m)\, y$ on the population variable y (a special case of example 4.12(2), with the single moment $\varphi_1(y) = y$ and own state x held fixed), and its Lions derivative — the population gradient of that field, proposition 4.15 — is the constant force $-q\, s\,(x - s\bar m) = q\, s^2 \bar m - q\, s\, x$, independent of y. The whole strategic coupling of the LQG game is thus a first-moment functional, and the Gaussian distance (4.15) measures exactly the two coordinates — mean and spread — that this coupling can move.

Rigor ledger

Proved (in full or by self-contained computation).

- $\mathcal{W}_2$ is a metric on $\mathcal{P}_2(\mathbb{R}^d)$ (theorem 4.7), with the triangle inequality via the gluing lemma; the isometric embedding $a \mapsto \delta_a$ and the identity $\mathcal{W}_2(\mu, \delta_0)^2 = \int |x|^2 \, \mathrm{d}\mu$.
- The closed form for $\mathcal{W}_2$ between two Gaussians, (4.13), with a complete derivation via the affine Brenier map; the scalar specialization (4.15) and the Gaussian-closed geodesics.
- The flat/Lions derivatives of the three model functionals (examples 4.12 and 4.16) and the gradient relation $\partial_\mu F = \nabla \, \delta F/\delta m$ (proposition 4.15, sketch).

Assumed (imported, used without re-proof).

- Weak convergence (definition 3.7), tightness (section 3.3), Prokhorov's theorem (theorem 3.16), Skorokhod's representation theorem (theorem 3.13), and the disintegration / regular-conditional-kernel theorem (theorem 3.33), all from chapter 3; the Riesz representation theorem (theorem 2.6) from chapter 2 (Fréchet differentiability is recalled locally in section 4.4); push-forward and integration from chapter 1.
- The named assumption (A-LipM) (measure-Lipschitz dependence in $\mathcal{W}_2$, appendix A.6), used in section 4.3 to assert continuity of the best-response map; it is distinct from the state-Lipschitz (A-Lip), and its full force is spent in chapters 8, 10 and 11.

Deferred (stated, proof pointed elsewhere).

- Existence of optimal couplings (proposition 4.3) and Kantorovich duality (theorem 4.5): mechanism named (direct method; convex duality), full in-book proof of duality in appendix D.1 (and [131, Ch. 4–5]).
- Brenier's theorem (theorem 4.6) and McCann displacement interpolation (theorem 4.9): sketched; full in-book proofs in appendices D.2 and D.3 (originals [30, 101]).
- The Polish property and the $\mathcal{W}_2$/weak-plus-second-moment equivalence (theorem 4.8): sketched via Skorokhod and Prokhorov; full proof [131, Thm. 6.9, 6.18].
- The Benamou–Brenier dynamic formula (theorem 4.10): stated, with full in-book proof in appendix D.4 (original [18]).
- The rearrangement-invariance structure theorem behind the Lions derivative (definition 4.14): proved in-book as theorem D.5. The second-order measure calculus (the Itô formula along a flow of measures) it feeds into is developed in appendix D.6 and applied in chapter 12 [40, Ch. 5].

Open / honestly flagged.

- Brenier's theorem as stated requires the source measure to be absolutely continuous; the optimal plan need not be a map when μ_0 has atoms (e.g. transporting a Dirac), and the displacement geodesic construction inherits this caveat. The book's mean field populations are typically nondegenerate ((A-Nondeg) ensures densities), so the map form applies, but this is an assumption, not a free lunch.

- Everything here is the $p = 2$, dense, finite-dimensional ($\mathbb{R}^d$) theory. The infinite-dimensional and L^p-graphon settings of Parts III–IV need the type-indexed analogue $x \mapsto m^x$, whose metric structure is built later and is *not* a special case of the present $\mathcal{P}_2(\mathbb{R}^d)$.

Bibliographic notes

The definitive modern reference for optimal transport is Villani's *Optimal Transport: Old and New* [131]; our theorems 4.5, 4.6, 4.8 and 4.9 follow its Chapters 4–7, and the second-moment characterization (4.7) is its Theorem 6.9. Santambrogio's *Optimal Transport for Applied Mathematicians* [117] is the most accessible entry point and gives the cleanest treatment of the dynamic (Benamou–Brenier) picture. Ambrosio, Gigli, and Savaré [7] is the authority on the metric and differential structure of $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, including geodesics and the absolutely continuous curves underlying theorem 4.10. The relaxation from maps to plans is Kantorovich's [85]; the structure theorem that the optimal plan is a gradient of a convex map is Brenier's [30], with displacement interpolation and displacement convexity due to McCann [101]. The dynamic fluid-mechanical formula is Benamou and Brenier [18].

The two derivatives of section 4.4 are developed in full in Carmona and Delarue [40, Ch. 5], which is also the source for the lift construction and the relation $\partial_\mu F = \nabla\, \delta F/\delta m$; the master-equation payoff is their Volume II [41] and our chapter 12. The density-functional analogy that opens that section is meant only as a mnemonic; the flat derivative is the same first-variation object a physicist computes for an energy functional, and no statistical mechanics is imported beyond the notation.

The Gaussian distance (4.13) is classical, going back to Olkin–Pukelsheim and to Givens and Shortt [78], who computed the quadratic Wasserstein distance between Gaussians; the covariance term is the Bures distance familiar from quantum information, and the pair $(\mathcal{P}_2(\mathbb{R}^d) \cap \{\text{Gaussians}\}, \mathcal{W}_2)$ is the Bures–Wasserstein manifold. Its appearance here is purely as the explicit backbone of the LQG thread, continued in chapter 6 and chapter 14.

Chapter 5

Stochastic Analysis: From Langevin to Itô and Fokker–Planck

A physicist meets the Langevin equation

$$\dot{x}(t) \;=\; -\,V'\big(x(t)\big) \;+\; \xi(t), \qquad \big\langle \xi(t)\,\xi(t')\big\rangle = 2D\,\delta(t-t'),$$

long before meeting the words "filtration" or "martingale," and treats it with complete confidence: one writes a Fokker–Planck equation for the density, finds its steady state, computes correlation functions, and moves on. The trouble is that, taken literally, the equation is nonsense. The driving force ξ is white noise — a "function" with infinite variance at each instant and no values, only a distribution — and the trajectory $x(t)$ it produces is continuous but nowhere differentiable, so $\dot{x}$ does not exist. The manipulations nonetheless give correct answers because they are shadows of a rigorous calculus. This chapter builds that calculus. The reward is not pedantry: when in Part II the drift will itself be a control chosen by an optimizing agent, and the noise coefficient will be allowed to depend on the state, the naive Langevin rules acquire genuinely ambiguous terms (the Itô–Stratonovich question), and only the rigorous theory tells us which reading the mathematics of mean field games selects.

One technical fact does all the work, and it is worth naming at the outset so the reader carries a single organizing idea rather than a list of seven topics. A Brownian increment over a window of length $\mathrm{d}t$ has standard deviation $\sqrt{\mathrm{d}t}$, so its *square* is of size $\mathrm{d}t$ — and, summed over a window, these squares stop fluctuating and accumulate *deterministically.* The whole chapter is this one fact, "$(\mathrm{d}W)^2 = \mathrm{d}t$," carried wherever it leads. It is why the ordinary Riemann–Stieltjes integral fails and a new (Itô) integral is needed; it is the surviving second-order term in Itô's formula, the term that *is* diffusion; it is what turns the formally nonsensical Langevin equation into a well-posed stochastic differential equation; and it is the Laplacian inside the generator that drives the Fokker–Planck equation forward (carrying densities) and the Kolmogorov backward equation in reverse (carrying observables). Seven topics, one engine; keep it in view.

We assume the probabilistic foundations of chapter 3: a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, random variables and their laws $\mathcal{L}(\cdot)$, expectation $\mathbb{E}$ as integration, conditional expectation as orthogonal projection, the modes of convergence, and the notion of a filtration $\mathbb{F} = (\mathcal{F}_t)_{t\geq 0}$ encoding the information available by time t. From chapter 4 we borrow only the space $\mathcal{P}_2(\mathbb{R}^d)$ of probability measures with finite second moment and the language of densities as points in it; the Fokker–Planck equation we derive here is the law of motion of such a point, and it is the forward half of the mean field game system whose backward half — the Hamilton–Jacobi–Bellman equation — is built in chapter 6. Indeed the Kolmogorov backward equation of section 5.5 is the linear ancestor of HJB, and we flag the kinship where it appears.

The plan is linear, and each waypoint is forced by the one before it. Section 5.1 constructs Brownian motion and isolates the single fact that drives everything: its quadratic variation is finite and deterministic, so that "$(\mathrm{d}W)^2 = \mathrm{d}t$." *Because* that fact gives Brownian paths infinite ordinary variation, the Riemann–Stieltjes integral cannot make sense of $\int H\, \mathrm{d}W$, and section 5.2 must build a new one — the Itô integral, whose left-endpoint convention is also exactly what separates Itô from Stratonovich. *Because* the same fact leaves a second-order term that refuses to vanish, the change-of-variables rule acquires a correction, and section 5.3 proves Itô's formula, whose extra term *is* diffusion. *Armed* with that calculus, section 5.4 solves stochastic differential equations under the standing Lipschitz and growth assumptions — the Langevin equation, at last well-posed. Passing from one trajectory to the evolving law of the whole ensemble, section 5.5 extracts the generator — in which the second-order term reappears as a Laplacian — and derives the Fokker–Planck equation twice, as the adjoint of that generator and as a continuity equation for a probability current, alongside its backward twin. Section 5.6 is the Feynman–Kac bridge that ties path expectations to those linear PDEs, and section 5.7 runs every one of these tools on the Ornstein–Uhlenbeck process — its explicit Gaussian law, its Fokker–Planck steady state, and the spectral decomposition of its generator — the one fully explicit model that is the linear backbone of the LQG thread (Thread A) carried through the book.

Notation watch

Throughout Parts I–II, with no graphon in scope, Brownian motion is written W_t (rendered W_t), following standard probability usage, and the generator is $\mathcal{A}$ with adjoint $\mathcal{A}^*$ (appendix A). From Part III onward, where the graphon kernel $W(x, y)$ occupies the letter W, Brownian motion is renamed B_t. Every Brownian motion in this chapter is W.

5.1 Brownian motion

The preamble left a promissory note: the Langevin force is to be made rigorous, and the whole calculus of this chapter is to be built on the noise it idealizes. But one cannot integrate against a noise one has not yet constructed. Before there can be an Itô integral $\int H\, \mathrm{d}W$, an Itô formula, or a Fokker–Planck equation, there must be a W — a single, canonical random object that earns the name "white-noise bath." Building it, and reading off the one structural fact ($\sqrt{\mathrm{d}t}$ increments) that everything later will exploit, is the business of this section.

What should that object be? A physicist's idealized thermal bath has exactly three statistical properties and one regularity, and the art is to demand these and *nothing more*. The kicks it delivers carry no memory of the past; they have no preferred direction and the same statistics at every instant; and they are so numerous and so independent on any time scale that, summed over a window, they pile up into a Gaussian. Finally, the displacement they integrate to is continuous — a real particle does not teleport. Promote these four to axioms and one arrives at the following definition. We will see immediately afterward that they pin the object down completely, *except* for the single nontrivial question of whether the continuity in the last axiom can be had at all.

Definition 5.1 (Brownian motion)**.** A real-valued stochastic process $W = (W_t)_{t\geq 0}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ is a (standard, one-dimensional) *Brownian motion* if

1. $W_0 = 0$ almost surely;
2. *independent increments*: for $0 \leq t_0 < t_1 < \cdots < t_n$ the increments $W_{t_1} - W_{t_0}, \ldots, W_{t_n} - W_{t_{n-1}}$ are independent;
3. *stationary Gaussian increments*: for $0 \leq s < t$, $W_t - W_s \sim N(0,\, t-s)$;
4. the sample path $t \mapsto W_t(\omega)$ is continuous for $\mathbb{P}$-almost every ω.

A d-dimensional Brownian motion is a vector $W = (W^1, \ldots, W^d)$ of independent one-dimensional Brownian motions; then $W_t - W_s \sim N(0, (t-s)I_d)$. We always take W adapted to a filtration $\mathbb{F}$ for which increments $W_t - W_s$ are independent of $\mathcal{F}_s$; the *Brownian filtration* generated by W (augmented by null sets) is the canonical choice.

Take the four axioms one at a time, because each is a physical statement before it is a mathematical one. Axiom (1), $W_0 = 0$, is not an assumption but a *choice of origin*: W_t records the displacement accumulated since the start, and a particle launched from x is simply $x + W_t$. Axiom (2), independent increments, is *memorylessness*: the kick delivered over $[s,t]$ is statistically unrelated to the entire history of the path up to time s. This is the strongest form of the Markov property, and it is the precise content of the physicist's slogan that white noise is "δ-correlated in time" — disjoint time windows share nothing. Axiom (3) carries the most weight and splits into three readings. *Stationary*: the law of an increment depends only on the elapsed time $t-s$, never on the absolute time s, so the bath delivers identical statistics in every equal window — a *fixed energy per unit time. Gaussian and centered*: an increment is the accumulation of vastly many independent infinitesimal kicks, so the central limit theorem makes it Gaussian, and the absence of any preferred direction makes its mean zero. *Variance $t-s$*: the variance grows *linearly* in the elapsed time. This last clause is the seed of the entire chapter. It says the typical magnitude of the displacement over a window of length $\tau = t-s$ is $\sqrt{\mathrm{Var}} = \sqrt{\tau}$, not τ — the $\sqrt{\mathrm{d}t}$ scaling foregrounded in the chapter preamble. Everything downstream is this one clause carried forward: it is what makes the path non-differentiable (section 5.1.1), what makes the quadratic variation deterministic (section 5.1.2), and ultimately what forces "$(\mathrm{d}W)^2 = \mathrm{d}t$." Axiom (4), continuity, is *the one regularity we keep*: we ask that sample paths be continuous and demand nothing smoother. The next subsection shows that this is also the most one *may* ask — for an object obeying axiom (3), differentiability is already self-contradictory.

Why do axioms (1)–(3) determine the process, while axiom (4) is of an entirely different character? Fix instants $0 \leq t_1 < \cdots < t_n$. The vector $(W_{t_1}, \ldots, W_{t_n})$ is a fixed invertible linear image of the increment vector $(W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}})$ (with $t_0 = 0$), whose entries are, by axioms (2)–(3), *independent* centered Gaussians. A linear image of a Gaussian vector is Gaussian, so the joint law is centered Gaussian, and a centered Gaussian law is determined by its covariance alone. That covariance is one line. For $s \le t$ write $W_t = W_s + (W_t - W_s)$; the increment $W_t - W_s$ is independent of $W_s = W_s - W_0$ (a disjoint, earlier increment, with mean zero), so

$$\mathrm{Cov}(W_s, W_t) = \mathrm{Var}(W_s) + \mathrm{Cov}\left(W_s,\, W_t - W_s\right) = \mathrm{Var}(W_s) + 0 = s,$$

and by symmetry in the two arguments,

$$\mathrm{Cov}(W_s, W_t) = s \wedge t. \tag{5.1}$$

Thus axioms (1)–(3) fix *every* finite-dimensional distribution: the vector $(W_{t_1}, \dots, W_{t_n})$ is the centered Gaussian with covariance matrix $\left(t_i \wedge t_j\right)_{i,j}$, and the entire law on finitely many coordinates is pinned. Axiom (4) is invisible at this level. A finite-dimensional distribution constrains the path at finitely many instants; continuity is a statement about the path at *all* t at once — an uncountable conjunction that no finite-dimensional law can even express. This gap is exactly why existence is a genuine theorem and not a definition.

The content of existence is therefore concentrated entirely in axiom (4). We hold a consistent family of finite-dimensional Gaussian laws — consistent precisely because the covariance $s \wedge t$ is symmetric and positive semidefinite, so the laws prescribed on overlapping finite sets of times agree on their common coordinates. Kolmogorov's extension theorem turns any such consistent family into a single honest probability measure on the path space $\mathbb{R}^{[0,\infty)}$ that realizes all the prescribed marginals simultaneously. But the natural σ-algebra on that path space is generated by the cylinder sets — events that constrain the path at countably many times — and "$t \mapsto W_t$ is continuous" is not such an event. The raw process Kolmogorov returns can be, and on $\mathbb{R}^{[0,\infty)}$ generically is, pathologically irregular; continuity does not come for free and must be installed by hand, by passing to a *modification* (a process equal to the original at each fixed t almost surely, but with better paths). Installing it is the theorem.

Theorem 5.2 (Existence)**.** *There exists a probability space carrying a process satisfying definition 5.1.*

Proof sketch. Two standard routes; we name the mechanism of each, filling in enough of the bookkeeping to see where continuity is paid for, with the two beyond-prerequisite theorems involved proved in full in appendix C.

(i) Kolmogorov extension plus a continuity theorem. The first step is the extension just described: the covariance $s \wedge t$ is symmetric and positive semidefinite, the Gaussian finite-dimensional laws are consistent, and Kolmogorov's extension theorem (theorem C.8) realizes them on $\mathbb{R}^{[0,\infty)}$. To repair the paths one invokes the *Kolmogorov–Chentsov continuity theorem* (theorem C.9): if a process obeys a moment bound $\mathbb{E}\,|W_t - W_s|^p \le C\,|t-s|^{1+\gamma}$ for some $p, \gamma, C > 0$ — this is hypothesis (C.6) — then it admits a modification whose paths are locally Hölder continuous of *every* order $\alpha < \gamma/p$. Brownian increments supply such bounds in abundance, because all their moments are explicit: from $W_t - W_s \sim N(0, t-s)$ and the Gaussian moment formula,

$$\mathbb{E}\,|W_t - W_s|^{2k} = (2k-1)!!\,|t-s|^k, \qquad k \in \mathbb{N}.$$

Matching this against the hypothesis means reading off the exponents: $p = 2k$ on the left, and $1+\gamma = k$ on the right, i.e. $\gamma = k-1$. The Hölder exponent the theorem then delivers is any

$$\alpha < \frac{\gamma}{p} = \frac{k-1}{2k} = \frac{1}{2} - \frac{1}{2k}.$$

Two things are worth reading off this little computation. First, the *fourth* moment already suffices for continuity: at $k=2$ it gives $\mathbb{E}\,|W_t - W_s|^4 = 3\,|t-s|^2$, hence $1+\gamma = 2 > 1$, and the theorem applies. A single polynomial moment bound with exponent strictly above 1 is all that continuity ever costs; the fourth moment only certifies Hölder regularity up to $\alpha < \frac{1}{4}$, but continuity itself is already secured at that stage. Second, the *higher* moments are precisely what sharpen the exponent: as $k \to \infty$ the bound $\frac{1}{2} - \frac{1}{2k}$ climbs toward $\frac{1}{2}$, so every $\alpha < \frac{1}{2}$ is attainable, the sharp regularity recorded in remark C.10. The mechanism inside Kolmogorov–Chentsov is a Borel–Cantelli argument on a dyadic grid: the moment bound makes the probability that the increment across a dyadic interval of length 2^{-n} exceeds $2^{-n\alpha}$ summable in n, so by Borel–Cantelli only finitely many such over-shoots occur almost surely, and controlling the dyadic oscillations at every scale controls the modulus of continuity.

(ii) Lévy–Ciesielski explicit series. On $[0,1]$, let (h_n) be the Haar basis of $L^2[0,1]$ and (ϕ_n) the Schauder functions $\phi_n(t) = \int_0^t h_n$ — the "tent" primitives of the Haar functions. If (ξ_n) are i.i.d. $N(0,1)$ then the candidate is

$$W_t = \sum_n \xi_n\, \phi_n(t).$$

That it has the right law is a one-line Parseval computation. The increments are Gaussian, being limits of linear combinations of the ξ_n, and using $\phi_n(t) = \int_0^t h_n = \langle \mathbf{1}_{[0,t]}, h_n\rangle$ together with the completeness of the Haar basis,

$$\mathrm{Cov}(W_s, W_t) = \sum_n \phi_n(s)\,\phi_n(t) = \sum_n \langle \mathbf{1}_{[0,s]}, h_n\rangle\langle \mathbf{1}_{[0,t]}, h_n\rangle = \langle \mathbf{1}_{[0,s]}, \mathbf{1}_{[0,t]}\rangle = s \wedge t,$$

exactly the covariance (5.1); with mean zero and Gaussian increments, all of axioms (1)–(3) follow. This is the construction worth carrying as a mental picture: it exhibits Brownian motion as a random series whose coefficients are independent Gaussians of *equal variance in every mode*. That flat spectrum — no Haar frequency privileged over another — is the precise sense in which white noise is "equal weight in every mode," and W is simply its running integral, the ϕ_n being the antiderivatives of the h_n. Continuity, the property that did not come free in route (i), is here *built into* the mode of convergence: the series converges *uniformly* in t almost surely, and a uniform limit of continuous functions is continuous. Uniform convergence is the one delicate point, and it is won by a Borel–Cantelli estimate. Group the terms by Haar level: level n contributes 2^n Schauder tents with *disjoint* supports and common height of order $2^{-n/2}$. The level's contribution to the supremum $\sup_t |\cdot|$ is therefore at most $\big(\max_j |\xi_{n,j}|\big) \cdot 2^{-n/2}$, where the maximum runs over 2^n independent standard Gaussians. Such a maximum grows only like $\sqrt{n}$, and $\mathbb{P}\big(\max_j |\xi_{n,j}| > c\sqrt{n}\big)$ is summable in n once c is large; so by Borel–Cantelli the level-n contribution is eventually below $c\sqrt{n}\,2^{-n/2}$, and $\sum_n \sqrt{n}\,2^{-n/2} < \infty$. The geometric decay $2^{-n/2}$ coming from the shrinking tent heights overwhelms the slow $\sqrt{n}$ growth of the Gaussian maxima, and that summability is exactly the uniform convergence. Finally, concatenating independent copies on the intervals $[k, k+1]$ extends W from $[0,1]$ to all of $[0,\infty)$. $\square$

Physics note

Axiom (3) is the fluctuation–dissipation content in disguise. White noise $\xi(t)$ "is" the distributional time derivative $\mathrm{d}W_t/\,\mathrm{d}t$; its formal covariance $\langle \xi(t)\xi(t')\rangle = \delta(t-t')$ is nothing but the statement that $\mathrm{Var}(W_t - W_s) = t - s$ — the same linear-variance clause, read in frequency-free language. A bath at a temperature set by a diffusion constant D enters the dynamics later *only* through the volatility σ multiplying $\mathrm{d}W$; the noise process itself is universal. Every model in this book — LQG, graphon-LQG, the general nondegenerate diffusion — drives its state with the *same* canonical W, whose law (Wiener measure) carries no parameters at all. All physical scales (temperature, mobility, the diffusion constant, the friction) sit in the coefficients b, σ of the SDE, never in W, and the fluctuation–dissipation dictionary $\sigma^2/2 = D = k_\mathrm{B}\tau_\mathrm{temp}$ (made explicit for the harmonic case in section 5.7) is the statement that fixes those coefficients from the bath. Keeping W canonical and parameter-free, and loading every physical constant into b, σ, is precisely the bookkeeping the rigorous theory enforces — and it is what lets one universal object serve every model at once.

Two features of this construction must be carried forward into the next subsections. The covariance (5.1) and the explicit Gaussian moments $\mathbb{E}(W_t - W_s)^2 = t - s$, $\mathbb{E}(W_t - W_s)^4 = 3(t-s)^2$ are not byproducts of the existence proof; they are the working machinery of what follows. The quadratic-variation computation of section 5.1.2 runs entirely on these two moments — the very ones that bought us continuity here — so the same Gaussian arithmetic that built W will also extract the fact "$(\mathrm{d}W)^2 = \mathrm{d}t$" that drives the rest of the chapter. And continuity itself is a cliff edge. Axiom (4) is the *most* regularity a Brownian path possesses: we secured Hölder exponents up to but never including $\frac{1}{2}$, and the next subsection shows that this ceiling is real. Pushing past it to differentiability is impossible, because the $\sqrt{\mathrm{d}t}$ scaling of axiom (3) makes the difference quotient $(W_{t+h} - W_t)/h \sim 1/\sqrt{h}$ diverge. We turn now to pinning down exactly how rough the paths are, from above and from below.

5.1.1 Non-differentiability

The previous section left us at a cliff edge: continuity, axiom (4), is the *most* regularity a Brownian path possesses, and the existence proof already hinted why, securing Hölder exponents that climbed toward $\frac{1}{2}$ but never reached it. We now stand at that ceiling and pin down exactly how rough the paths are, and we do it from both sides at once. There is a positive statement — the paths are Hölder continuous of *every* order strictly below $\frac{1}{2}$, so they are not as wild as they could be — and a negative statement — they are Hölder of *no* order at or above $\frac{1}{2}$, and in the sharpest reading they are differentiable at no point whatsoever. The two bounds pincer the true regularity into the half-open window $[\,0, \frac{1}{2})$, and the gap they leave closes onto the single number $\frac{1}{2}$ that will organize the rest of the chapter. The reader should treat everything in this subsection as scaffolding for one takeaway: increments scale as $\sqrt{\mathrm{d}t}$, not $\mathrm{d}t$.

Theorem 5.3 (Path roughness)**.** *With probability one:*

1. *for every $\alpha < \frac{1}{2}$ the path $t \mapsto W_t$ is locally α-Hölder continuous;*

2. *the path is* nowhere *differentiable; more strongly, for every t the upper and lower derivates are $+\infty$ and $-\infty$.*

The two parts are the two halves of one picture, and it is worth saying what each costs and what each buys before sketching the proof. Part (1) is the *positive* half: it is a regularity *guarantee*, the best one available, and it is the half we will actually use. It is precisely the Hölder conclusion already extracted from the existence construction, where higher and higher Brownian moments fed into Kolmogorov–Chentsov pushed the attainable exponent up the ladder $\frac{1}{2} - \frac{1}{2k}$ toward, but never to, $\frac{1}{2}$. Part (2) is the *negative* half: it is an *obstruction*, certifying that the ceiling in part (1) is genuine and cannot be raised — not to differentiability, not even to Hölder-$\frac{1}{2}$. Where part (1) flows from the *smallness* of typical increments (they are no larger than $|t-s|^\alpha$ for $\alpha < \frac{1}{2}$), part (2) flows from their refusal to be *too small*: a $\sqrt{h}$-sized fluctuation cannot shrink to the $O(h)$ a derivative would demand.

Proof sketch. Part (1) is the Hölder conclusion of the Kolmogorov–Chentsov theorem (theorem C.9 and remark C.10) applied exactly as in route (i) of theorem 5.2: the Gaussian moments $\mathbb{E}\,|W_t - W_s|^{2k} = (2k-1)!!\,|t-s|^k$ verify the Chentsov hypothesis with $p = 2k$ and $1+\gamma = k$, delivering local Hölder continuity of every order $\alpha < \frac{1}{2} - \frac{1}{2k}$, and letting $k \to \infty$ gives every $\alpha < \frac{1}{2}$. This is the only part of the theorem the rest of the chapter relies on, and it is fully earned by the existence proof.

Part (2) is the Paley–Wiener–Zygmund theorem; we sketch only the mechanism, which is a contradiction between two scalings. Suppose, toward a contradiction, that the path were differentiable at some instant t_0 — or even merely had a bounded derivate there, so that $|W_{t_0+h} - W_{t_0}| \le L\,|h|$ for all small h and some finite L. A differentiable function has increments of order h: linear in the time step. But a Brownian increment is $W_{t_0+h} - W_{t_0} \sim N(0,h)$, whose typical size is $\sqrt{h}$. For small h the two scales are wildly mismatched, $\sqrt{h} \gg h$ — a fluctuation of size $\sqrt{h}$ is, relative to the $L\,|h|$ a derivative would permit, larger by the diverging factor $1/(L\sqrt{h}) \to \infty$. So the differentiability bound forces the increment to be *atypically small*, by an ever-growing margin as $h \downarrow 0$, and one must show that no point t_0 in $[0,1]$ can sustain such a run of improbably small increments. This is the dyadic-covering Borel–Cantelli step: cover $[0,1]$ by the 2^n dyadic intervals of length 2^{-n}; a derivate bounded by L at any t_0 would force *several consecutive* dyadic increments straddling t_0 to be of order 2^{-n} rather than their typical $2^{-n/2}$; the probability that any one dyadic interval shows so small an increment is itself small, the probability that a whole short block of them does so simultaneously is far smaller, and these probabilities sum to a convergent series in n. Borel–Cantelli then says that almost surely only finitely many such coincidences ever occur, so the bound can hold at no t_0 at all, and the event "differentiable somewhere" has probability zero. The careful covering count is carried out in [86, Thm. 2.9.18]. □

A word on *why we prove part (2) at all*, said plainly because it governs how much weight the reader should place on it: nowhere-differentiability is included *only* to complete the picture of path regularity, and it is *never* used anywhere downstream in this book. Not one later argument — not the Itô integral, not Itô's formula, not the Fokker–Planck equation — invokes the fact that paths fail to be differentiable. What the machinery downstream actually consumes is the *positive* regularity of part (1) together with the quadratic variation of the next subsection. We record part (2) because a reader who has just been told "$\dot{W}$ does not exist" deserves to see that claim honestly certified rather than asserted; but having certified it, we set it aside. The load the theorem really carries is one number and one scaling, not the dramatic nowhere-differentiability headline.

That scaling is the morally indispensable heuristic, and it is the single thing to carry out of this subsection: a Brownian increment over a window of length $\mathrm{d}t$ has size $\sqrt{\mathrm{d}t}$, not $\mathrm{d}t$. Both halves of the theorem are this one fact wearing different clothes. Part (1) holds because $\sqrt{h} = h^{1/2}$ is, for $\alpha < \frac{1}{2}$, smaller than h^α once h is small, so increments are controlled at every Hölder exponent below $\frac{1}{2}$; part (2) holds because $\sqrt{h}$ is, on the other hand, far *larger* than the h a derivative would require, so no difference quotient $(W_{t+h} - W_t)/h \sim 1/\sqrt{h}$ can settle to a finite limit — it diverges. The same $\sqrt{\mathrm{d}t}$ thus simultaneously caps the regularity from above and forbids differentiability from below.

This is precisely why $\mathrm{d}W$ cannot be manipulated by the ordinary rules of calculus, and it is the first warning that the integral $\int H \, \mathrm{d}W$ we will need cannot be a Riemann–Stieltjes integral: that classical construction integrates against the increments of a function of *finite* first variation, and a path whose increments are of size $\sqrt{\mathrm{d}t}$ accumulates infinite first variation on every interval (a point the next subsection makes precise). But the $\sqrt{\mathrm{d}t}$ scaling is, so far, only a statement about roughness — about what *fails*. Its real power is constructive, and emerges the moment we stop looking at a single increment and look instead at the *sum of their squares*. Squaring converts the inconvenient $\sqrt{\mathrm{d}t}$ into a clean $\mathrm{d}t$, and summing many such $\mathrm{d}t$'s, as the next subsection shows, turns a fluctuating random quantity into a deterministic one. We hand this heuristic to the quadratic-variation computation, which asks exactly that question: what does increments-of-size-$\sqrt{\mathrm{d}t}$ imply for the running sum of squared increments — the very quantity ordinary calculus throws away as higher order?

5.1.2 Quadratic variation

The previous subsection handed us a single heuristic, scrubbed of its dramatic packaging: a Brownian increment over a window of length $\mathrm{d}t$ has size $\sqrt{\mathrm{d}t}$, not $\mathrm{d}t$. So far this has been a statement about *roughness* — about what fails, why the difference quotient diverges and the path is nowhere smooth. Now we put the same fact to constructive use, and the trick is to stop watching a single increment and watch instead the running sum of their *squares*. Squaring is exactly the operation that converts the awkward $\sqrt{\mathrm{d}t}$ into a clean $\mathrm{d}t$: if ΔW is of order $\sqrt{\mathrm{d}t}$, then $(\Delta W)^2$ is of order $\mathrm{d}t$, the same order as the time step itself. Ordinary calculus throws this term away — for a differentiable path the squared increments are $O(\mathrm{d}t^2)$, genuinely higher order, and their sum vanishes in the continuum limit. For Brownian motion they are $O(\mathrm{d}t)$, of the *same* order as the spacing, so their sum is a sum of k_n terms each of size t/k_n, and it has every right to converge to something finite rather than to zero. The question this subsection answers is: to *what*? The answer is the clean, deterministic identity that is the technical engine of the entire chapter. Ordinary differentiable functions have zero quadratic variation; Brownian motion has quadratic variation equal to the elapsed time, and — crucially — it does so not on average but with the randomness washed out, the limit being a deterministic constant rather than a random variable.

Theorem 5.4 (Quadratic variation)**.** *Fix $t > 0$ and a sequence of partitions $\pi_n : 0 = t_0^n < t_1^n <$*

$\cdots < t^n_{k_n} = t$ with mesh $\|\pi_n\| = \max_i(t^n_{i+1} - t^n_i) \to 0$. Then

$$V_n \; := \; \sum_{i=0}^{k_n-1} \big(W_{t^n_{i+1}} - W_{t^n_i}\big)^2 \; \xrightarrow[n\to\infty]{L^2(\mathbb{P})} \; t. \tag{5.2}$$

If in addition $\sum_n \|\pi_n\| < \infty$ (e.g. dyadic refinements) the convergence is almost sure. One writes the limit as the quadratic variation $\langle W\rangle_t = t$.

The statement carries two layers, and it is worth separating them. The first is that the random quantity V_n has the right *average*: its mean is exactly t for every partition, fine or coarse. That much is just the linear-variance axiom restated. The second, and the substantive one, is that the *fluctuations of V_n around that mean die*: the variance of V_n collapses to zero as the mesh refines, so in the limit V_n is no longer random at all. This is the precise sense in which "the squares accumulate deterministically." The proof is exactly this two-step story — compute the mean, then kill the variance — and every ingredient is one of the two Gaussian moments that the existence construction already put on the table.

Proof. Write $\Delta_i W = W_{t^n_{i+1}} - W_{t^n_i}$ for the ith increment and $\Delta_i t = t^n_{i+1} - t^n_i$ for the length of the ith subinterval, so that $V_n = \sum_i (\Delta_i W)^2$ and $\sum_i \Delta_i t = t$.

The mean is t, exactly. Each increment is $\Delta_i W \sim N(0, \Delta_i t)$ by axiom (3), so $\mathbb{E}(\Delta_i W)^2 = \mathrm{Var}(\Delta_i W) = \Delta_i t$. Linearity of expectation then gives

$$\mathbb{E}V_n = \sum_{i=0}^{k_n-1} \mathbb{E}(\Delta_i W)^2 = \sum_i \Delta_i t = t,$$

a telescoping of the subinterval lengths. No independence and no fine structure of the partition is used here: $\mathbb{E}V_n = t$ holds term by term, for every n. The whole content of the theorem is therefore that V_n does not merely average to t but *concentrates* on t, and concentration is controlled by the variance.

The variance vanishes. Here independence does the work. Because the increments $\Delta_i W$ sit over disjoint time windows, axiom (2) makes them mutually independent, and hence so are their squares $(\Delta_i W)^2$; the variance of a sum of independent terms is the sum of the variances, with no cross-covariances surviving:

$$\mathrm{Var}\, V_n = \sum_i \mathrm{Var}\left((\Delta_i W)^2\right) = \sum_i \Big(\mathbb{E}(\Delta_i W)^4 - \big(\mathbb{E}(\Delta_i W)^2\big)^2\Big) = \sum_i \Big(\mathbb{E}(\Delta_i W)^4 - (\Delta_i t)^2\Big).$$

The fourth moment is the second Gaussian fact we already recorded: for $Z \sim N(0, \tau)$ the Gaussian moment formula gives $\mathbb{E}Z^4 = 3\tau^2$, the "3" being the number of pairings in Wick's theorem (or simply $(2\cdot 2 - 1)!! = 3$). With $\tau = \Delta_i t$ this turns each summand into $3(\Delta_i t)^2 - (\Delta_i t)^2 = 2(\Delta_i t)^2$, so

$$\mathrm{Var}\, V_n = \sum_i \left(3(\Delta_i t)^2 - (\Delta_i t)^2\right) = 2\sum_i (\Delta_i t)^2.$$

This sum is quadratic in the subinterval lengths, and that is precisely what makes it small: one factor of $\Delta_i t$ can be bounded by the mesh and pulled out, leaving the other factors to telescope.

Concretely, $\Delta_i t \le \|\pi_n\|$ for every i, so

$$\operatorname{Var} V_n = 2\sum_i (\Delta_i t)^2 \le 2\ \|\pi_n\| \sum_i \Delta_i t = 2\ \|\pi_n\|\ t \xrightarrow[n\to\infty]{} 0, \tag{5.3}$$

since $\sum_i \Delta_i t = t$ is fixed while the mesh $\|\pi_n\| \to 0$ by hypothesis. This is the decisive estimate: it is the refinement of the partition — many windows, each short — that drives the fluctuations to zero, and the mesh factor $\|\pi_n\|$ is exactly the lever. Combining the two steps, $\mathbb{E}\,(V_n - t)^2 = \operatorname{Var} V_n + (\mathbb{E}V_n - t)^2 = \operatorname{Var} V_n \to 0$, which is the asserted convergence (5.2) in $L^2(\mathbb{P})$.

Upgrade from L^2 to almost sure. The L^2 statement says $V_n \to t$ *on average* over ω; it does not by itself say that, for a fixed realization of the path, the numbers $V_n(\omega)$ settle down. To get that we trade the L^2 bound for a pathwise one, and the bridge is Chebyshev's inequality. For any $\varepsilon > 0$,

$$\mathbb{P}(\,|V_n - t| > \varepsilon)\ \le\ \frac{\mathbb{E}\,(V_n - t)^2}{\varepsilon^2}\ =\ \frac{\operatorname{Var} V_n}{\varepsilon^2}\ \le\ \frac{2t}{\varepsilon^2}\ \|\pi_n\|\,.$$

Under the extra hypothesis $\sum_n \|\pi_n\| < \infty$ — met, for instance, by dyadic refinements where $\|\pi_n\| = t\,2^{-n}$ and the geometric series converges — the right-hand side is summable in n, so $\sum_n \mathbb{P}(|V_n - t| > \varepsilon) < \infty$. The first Borel–Cantelli lemma then says that, with probability one, the event $\{|V_n - t| > \varepsilon\}$ occurs for only finitely many n; equivalently $\limsup_n |V_n - t| \le \varepsilon$ almost surely. Letting ε run through a sequence tending to 0 (and intersecting the countably many full-probability events) gives $V_n(\omega) \to t$ for almost every ω. The summability of the meshes is exactly what this argument consumes: without it the per-n probabilities need not add up, and one recovers only convergence along a subsequence rather than the full almost-sure statement. □

Before drawing the two corollaries, it is worth pausing on the contrast that makes the result surprising at all. Run the identical computation on a continuously differentiable path f. There $\Delta_i f = f'(\xi_i)\,\Delta_i t$ by the mean value theorem, so $(\Delta_i f)^2 = f'(\xi_i)^2(\Delta_i t)^2 \le \big(\sup|f'|\big)^2\ \|\pi_n\|\ \Delta_i t$, and summing gives a quadratic variation bounded by $\big(\sup|f'|\big)^2\ \|\pi_n\|\ t \to 0$: a smooth path has *zero* quadratic variation. The squared increments of a smooth function are genuinely $O(\mathrm{d}t^2)$ and wash out, exactly as ordinary calculus assumes. Brownian motion sits at the boundary where this assumption breaks: its increments are larger, of order $\sqrt{\mathrm{d}t}$, just enough to make the squared increments $O(\mathrm{d}t)$ and their sum survive. The quadratic variation t is the clean numerical signature of that borderline roughness.

Two corollaries now fix the intuition, and they are the two things this subsection hands forward. The first is a *negative* result. A Brownian path has *infinite* first variation on every interval — that is, $\sup_\pi \sum_i |\Delta_i W| = \infty$, the total length traced out by the path is unbounded. The reason is a one-line squeeze against the quadratic variation we just computed. On any partition,

$$V_n = \sum_i (\Delta_i W)^2\ \le\ \Big(\max_i |\Delta_i W|\Big) \sum_i |\Delta_i W|\,.$$

As the mesh refines, the path's continuity forces $\max_i |\Delta_i W| \to 0$, while the left side $V_n \to t > 0$. If the first-variation sum $\sum_i |\Delta_i W|$ stayed bounded, the right side would be (something $\to 0$) times (something bounded), hence $\to 0$, contradicting $V_n \to t$. So $\sum_i |\Delta_i W|$ must diverge: the first variation is infinite. This is precisely the obstruction the previous subsection promised, now made quantitative. The classical Riemann–Stieltjes integral $\int f \; \mathrm{d}g$ is built by summing f against the increments $\mathrm{d}g$ and is guaranteed to converge *only* when the integrator g has finite first variation, so that the increments can be controlled in absolute value. Brownian motion fails that single hypothesis by an infinite margin. There is therefore *no* Riemann–Stieltjes reading of $\int H \; \mathrm{d}W$, and the integral we will need in the next section must be built on a different principle entirely. The roughness that looked like a defect is the reason a genuinely new construction is forced.

The second corollary is the *positive* tool, and it is the one every later computation in the chapter will run on. Operationally, the identity $\langle W \rangle_t = t$ is the rigorous content of the symbolic multiplication table

$$(\mathrm{d}W_t)^2 = \mathrm{d}t, \qquad \mathrm{d}W_t \; \mathrm{d}t = 0, \qquad (\mathrm{d}t)^2 = 0,$$

which we will use repeatedly and which is justified precisely by theorem 5.4. Each entry is the limiting behavior of one kind of product of increments, read off the theorem. The first, $(\mathrm{d}W)^2 = \mathrm{d}t$, is theorem 5.4 itself in differential shorthand: summed over a window, $(\Delta W)^2$ behaves like the deterministic Δt, so under the integral sign one may *replace* $(\mathrm{d}W)^2$ by $\mathrm{d}t$. The third, $(\mathrm{d}t)^2 = 0$, is the smooth-path computation above: squared time increments are genuinely higher order and contribute nothing, just as in ordinary calculus. The middle entry, $\mathrm{d}W \; \mathrm{d}t = 0$, is the cross term, and it vanishes because it is intermediate in size: the product of a $\sqrt{\mathrm{d}t}$ object with a $\mathrm{d}t$ object is of order $\mathrm{d}t^{3/2}$, higher than first order, so summed over a window $\sum_i |\Delta_i W| \; \Delta_i t \le \|\pi_n\| \sum_i |\Delta_i W|$ behaves like $\|\pi_n\|^{1/2} \to 0$ once the typical $\sum_i |\Delta_i W|$ of order $\|\pi_n\|^{-1/2}$ is accounted for, and it too drops. The single moral, and the one to carry into the next sections, is that among all the second-order products only the *square* of the Brownian increment survives to first order, and it survives stripped of its randomness, as a deterministic $\mathrm{d}t$. That is the whole operational meaning of the quadratic-variation theorem, and it is what the Itô isometry, Itô's formula, and the generator will each be, in turn, an application of.

We close, then, by handing the next section two complementary things. The negative result — infinite first variation, hence no Riemann–Stieltjes integral — tells us that $\int_0^T H_t \; \mathrm{d}W_t$ cannot be defined by the classical recipe and forces a genuinely new construction. The positive result — the multiplication table $(\mathrm{d}W)^2 = \mathrm{d}t$, $\mathrm{d}W \; \mathrm{d}t = 0$, $(\mathrm{d}t)^2 = 0$ — is the deterministic arithmetic that new construction will be built to respect and that every computation downstream will exploit. The obstruction even points to where the freedom lies: if the Riemann–Stieltjes limit does not exist because the integrator is too rough, then the value of an approximating sum $\sum_i H_{\tau_i}(W_{t_{i+1}} - W_{t_i})$ can depend on *where* in the subinterval $[t_i, t_{i+1}]$ we sample the integrand H. Ordinary calculus is indifferent to that choice; here it is a real fork in the road. The next section turns that obstruction into a design decision — which evaluation point τ_i to fix — and shows that the answer the mathematics of mean field games selects is the left endpoint.

5.2 The Itô integral

The previous section closed by handing us an obstruction dressed as an opportunity. The negative corollary of the quadratic-variation theorem (theorem 5.4) was that a Brownian path has *infinite* first variation on every interval, and the classical Riemann–Stieltjes integral $\int f \; \mathrm{d}g$ converges in general only when the integrator g has *finite* first variation, so that the increments it is summed against can be controlled in absolute value. Brownian motion fails that single hypothesis by an infinite margin, so the object we actually need — $\int_0^T H_t \, \mathrm{d}W_t$, the running gain of an integrand H against the increments of W — has no Riemann–Stieltjes reading. We need this integral and cannot do without it: in the Langevin equation the noise enters precisely as $\sigma \, \mathrm{d}W$, and to allow a volatility that depends on the state, or later on a control, we must be able to integrate a genuinely *random*, possibly history-dependent integrand H_t against $\mathrm{d}W$. The classical machine refuses. But the *way* it refuses tells us how to build its replacement, and reading that off is the business of this short hinge section before the two subsections that do the construction.

Look at what actually goes wrong. To integrate H against $\mathrm{d}W$ one forms the approximating sums

$$\sum_i H_{\tau_i}(W_{t_{i+1}} - W_{t_i}), \qquad \tau_i \in [t_i, t_{i+1}],$$

where τ_i is a sample point chosen somewhere inside the ith subinterval, and asks whether they converge as the mesh refines. For an integrator of *finite* variation the answer is yes *and* the limit does not depend on where the τ_i are placed: moving each τ_i within its subinterval changes the sum by at most $\sum_i \left(\sup_{[t_i,t_{i+1}]} H - \inf_{[t_i,t_{i+1}]} H\right) |\Delta_i g|$, which is squeezed to zero by the continuity of H against the *bounded* total variation of g. That placement-independence is exactly the gift of finite variation, and it is why ordinary calculus never even raises the question of *where* in $[t_i, t_{i+1}]$ to evaluate the integrand: the question has no consequences. The moment the integrator is Brownian, the bounded-variation factor that did the squeezing is gone, and the question acquires consequences.

It is worth seeing, in one line, that the ambiguity is not a fastidious worry at the edge of some pathological example but a real and generically large effect — and that its size is once again the engine of the whole chapter. Take the simplest nontrivial integrand, $H_t = W_t$ itself, and compare the two extreme placements, the left endpoint $\tau_i = t_i$ and the right endpoint $\tau_i = t_{i+1}$. Their difference is

$$\sum_i W_{t_{i+1}}(W_{t_{i+1}} - W_{t_i}) - \sum_i W_{t_i}(W_{t_{i+1}} - W_{t_i}) = \sum_i \left(W_{t_{i+1}} - W_{t_i}\right)^2 \xrightarrow[\|\pi\|\to 0]{} T,$$

the quadratic variation of theorem 5.4. The left- and right-endpoint conventions do not agree in the limit; they disagree by the full deterministic amount T, and the gap between them is exactly the surviving second-order quantity $(\mathrm{d}W)^2 = \mathrm{d}t$ that the last subsection isolated. For a smooth integrator this same difference would be a sum of squared increments of order $\sum_i (\Delta_i g)^2 \to 0$ and the two conventions would coincide; for W it does not vanish. The evaluation point is therefore a genuine fork in the road, and *which* fork we take is not forced by anything analytic — it is a modelling decision we get to make, and different decisions give different, internally consistent calculi.

We make it by asking what the integrand *means*. When H_t is a strategy — a bet sized at time t, or, in Part II, a control chosen by an agent — it must be fixed using only information available *up to* time t; one cannot stake a wager on an increment one has not yet seen. The placement that honours this is the *left* endpoint, $\tau_i = t_i$: the integrand is locked in *before* the increment $W_{t_{i+1}} - W_{t_i}$ it multiplies is revealed. This is the Itô convention, and the whole construction to come is the systematic exploitation of that one choice. It carries two enormous payoffs and exacts a single, well-defined price; because the rest of the section and section 5.3 will repeatedly call back to this trade, we state it crisply here.

The *first payoff* is the martingale property. Because the bet H_{t_i} is decided before the fair, mean-zero increment it is multiplied against, each term contributes zero expected gain conditional on the past, and the running integral $t \mapsto \int_0^t H \, \mathrm{d}W$ inherits no systematic drift: it is a martingale, a fair game. This is the structural prize the mathematics of mean field games will demand — it is what lets stochastic integrals interlock with conditional expectations, optional stopping, and the verification arguments of chapter 6, and it is special to the left endpoint, as the $W \, \mathrm{d}W$ computation above already hints (the right-endpoint sum would carry a deterministic upward bias of size T). The *second payoff* is causality: the integral is assembled out of *non-anticipating* bets, the only admissible integrands once H is a control, so the Itô integral is automatically the right object for optimization under uncertainty rather than an analytic convenience retrofitted to it. The *one cost* is that the ordinary chain rule fails. The very fact that made the endpoint matter — that $(\mathrm{d}W)^2$ accumulates to a deterministic $\mathrm{d}t$ instead of vanishing — means that $\mathrm{d}f(W_t)$ is *not* $f'(W_t) \, \mathrm{d}W_t$; a second-order term survives, and the change-of-variables formula must be corrected to account for it. We have, in effect, bought the martingale property at the price of a modified calculus. Paying that price in full — deriving the corrected chain rule, Itô's formula, whose extra term *is* diffusion — is the work of section 5.3; here we only flag that the cost is real and exactly localized.

Two subsections discharge this programme. Section 5.2.1 performs the construction: it begins with *simple* integrands whose coefficients are decided before each increment — the literal "bet locked in at t_i" objects, made precise by demanding that the coefficient on $[t_i, t_{i+1}]$ be measurable with respect to the information available at t_i — defines the integral for them by the obvious finite sum, and then extends it to all square-integrable non-anticipating integrands by an isometry that the left-endpoint choice is exactly what makes available. Section 5.2.2 then returns to the fork in the road and follows the *other* branch: evaluating at the midpoint rather than the left endpoint yields the *Stratonovich* integral, a different but equally consistent calculus that trades the martingale property away in exchange for restoring the ordinary chain rule — the mirror image of the same trade — and we will compute the explicit correction term that converts one into the other.

5.2.1 Construction and the Itô isometry

The hinge section ended by committing to a single modelling choice: the integrand is to be locked in *before* the increment it multiplies, because that is what a non-anticipating bet — or, in Part II, a control chosen on the basis of information available so far — actually is. The construction now executes that commitment to the letter. The objects on which the integral is completely unambiguous are exactly those for which the integrand is piecewise constant in time and each constant value is fixed before its window opens; these are the literal "bet decided at t_i, then held fixed across $[t_i, t_{i+1}]$" processes, and everything else will be reached from them by a limit.

Let $\mathbb{F}$ be the Brownian filtration. Call a process H *simple* if there is a partition $0 = t_0 < \cdots < t_n = T$ and $\mathcal{F}_{t_i}$-measurable, square-integrable random variables η_i with $H_t = \sum_{i=0}^{n-1} \eta_i \, \mathbf{1}_{(t_i, t_{i+1}]}(t)$. For such H define, with no ambiguity,

$$I(H) \;:=\; \int_0^T H_t \, \mathrm{d}W_t \;=\; \sum_{i=0}^{n-1} \eta_i \left(W_{t_{i+1}} - W_{t_i} \right).$$

There is nothing to choose here and nothing to take a limit of: the integral of a simple process is a finite sum, the coefficient η_i being held constant across the window $(t_i, t_{i+1}]$ exactly as the left-endpoint convention prescribes. The single hypothesis that earns its keep is the $\mathcal{F}_{t_i}$-measurability of η_i. Spelled out, it says η_i is a function of the path *up to time* t_i alone and is therefore *independent* of the future increment $W_{t_{i+1}} - W_{t_i}$, which lives over $(t_i, t_{i+1}]$ and, by the independent-increments axiom (2) of definition 5.1, knows nothing of $\mathcal{F}_{t_i}$. This is the engine of the entire construction, and it works in two ways at once. First, it makes each term *mean zero*: a known coefficient times a fresh, fair, mean-zero kick has expectation zero. Second — and this is the deeper half — it makes distinct terms *uncorrelated*, because in any product of two terms the later increment is still a fresh fair kick relative to everything multiplying it. Mean zero plus uncorrelated is precisely the structure that turns the second moment of the sum into a sum of second moments, with no cross-terms surviving, and that collapse is the Itô isometry. We isolate it in a moment.

To house the limit we will take, let $\mathcal{M}^2[0,T]$ denote the space of progressively measurable processes H with $\mathbb{E}\int_0^T H_t^2 \, \mathrm{d}t < \infty$, a Hilbert space under that norm. The phrase *progressively measurable* deserves to be unpacked rather than waved through, since it is the minimal causal regularity the theory needs. Adaptedness, imported from chapter 3, asks that each *random variable* H_t be $\mathcal{F}_t$-measurable — a constraint at each fixed instant. Progressive measurability strengthens this to a statement about the integrand as a function of (s,ω) *jointly*: for every t, the restriction of $(s,\omega) \mapsto H_s(\omega)$ to $[0,t] \times \Omega$ is measurable for the product σ-algebra $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$. The point of the upgrade is purely to make the *running* integral $t \mapsto \int_0^t H \, \mathrm{d}W$ a bona fide process — measurable jointly in time and chance, so that one may later integrate it again, stop it, or take its expectation without measurability anxieties. It is exactly the path-space refinement of adaptedness, no more, and the reader should not read into it any analytic strength beyond causality. The reassurance worth carrying is that for the integrands we ever actually use — continuous adapted processes, or left-continuous adapted ones like the simple integrands above — progressive measurability is *automatic*: an adapted process with left- or right-continuous paths is always progressively measurable, so the hypothesis never has to be checked by hand in practice. It is a technical hygiene condition, not a modelling restriction.

Lemma 5.5 (Itô isometry on simple processes)**.** *For simple H, $\mathbb{E}\, I(H) = 0$ and $\mathbb{E}\, I(H)^2 = \mathbb{E}\int_0^T H_t^2 \, \mathrm{d}t$.*

Proof. Write $\Delta_i W = W_{t_{i+1}} - W_{t_i}$ for the increments. The mean is the easy half. Each term $\eta_i \, \Delta_i W$ has expectation zero, because η_i is $\mathcal{F}_{t_i}$-measurable while $\Delta_i W$ is independent of $\mathcal{F}_{t_i}$ with mean zero; conditioning on $\mathcal{F}_{t_i}$ and pulling out the known factor, $\mathbb{E}[\eta_i \, \Delta_i W] = \mathbb{E}[\eta_i \, \mathbb{E}[\Delta_i W \mid \mathcal{F}_{t_i}]] = \mathbb{E}[\eta_i \cdot 0] = 0$, and summing gives $\mathbb{E}\, I(H) = 0$.

For the second moment, expand the square of the sum into diagonal and off-diagonal parts,

$$\mathbb{E}\, I(H)^2 = \sum_i \mathbb{E}[\eta_i^2 (\Delta_i W)^2] + 2 \sum_{i<j} \mathbb{E}[\eta_i \, \Delta_i W \; \eta_j \, \Delta_j W].$$

The off-diagonal terms vanish, and this is where the causality hypothesis does its work. Fix $i < j$, so that $t_{i+1} \le t_j$ and the three factors η_i, $\Delta_i W$ and η_j are all $\mathcal{F}_{t_j}$-measurable, while the remaining factor $\Delta_j W$ is independent of $\mathcal{F}_{t_j}$ with mean zero. Conditioning on $\mathcal{F}_{t_j}$ and using the tower property, the three known factors come out of the conditional expectation:

$$\begin{aligned}\mathbb{E}[\eta_i\,\Delta_i W\;\eta_j\,\Delta_j W] &= \mathbb{E}\Big[\,\mathbb{E}[\eta_i\,\Delta_i W\;\eta_j\,\Delta_j W\,|\,\mathcal{F}_{t_j}]\Big] \\ &= \mathbb{E}\Big[\,\eta_i\,\Delta_i W\;\eta_j\;\mathbb{E}[\Delta_j W\mid\mathcal{F}_{t_j}]\Big] = \mathbb{E}\big[\,\eta_i\,\Delta_i W\;\eta_j\cdot 0\,\big] = 0.\end{aligned}$$

Every cross term is killed by the single fact that the later increment is fresh relative to the past it is paired against — the uncorrelatedness promised above, now made explicit. Only the diagonal survives, and each diagonal term is evaluated the same way, by conditioning on $\mathcal{F}_{t_i}$: since η_i^2 is $\mathcal{F}_{t_i}$-measurable and $\mathbb{E}[(\Delta_i W)^2\mid\mathcal{F}_{t_i}]=\mathrm{Var}(\Delta_i W)=t_{i+1}-t_i$ by independence and axiom (3),

$$\mathbb{E}[\eta_i^2(\Delta_i W)^2]=\mathbb{E}[\eta_i^2\,\mathbb{E}[(\Delta_i W)^2\mid\mathcal{F}_{t_i}]]=\mathbb{E}[\eta_i^2]\,(t_{i+1}-t_i)=\mathbb{E}\int_{t_i}^{t_{i+1}}\eta_i^2\;\mathrm{d}t,$$

the last step because η_i is constant across $(t_i,t_{i+1}]$. Summing over i and recognizing $\sum_i\eta_i^2\,\mathbf{1}_{(t_i,t_{i+1}]}=H_t^2$ gives $\mathbb{E}\,I(H)^2=\mathbb{E}\int_0^T H_t^2\;\mathrm{d}t$, the claim. □

Read structurally, the lemma says the map $I: H\mapsto I(H)$ is a *linear isometry* from the simple processes, normed by $\mathcal{M}^2$, into $L^2(\mathbb{P})$: it carries the $\mathcal{M}^2$-length $\big(\mathbb{E}\int_0^T H^2\;\mathrm{d}t\big)^{1/2}$ of an integrand exactly onto the L^2-length of its integral, $\big(\mathbb{E}\,I(H)^2\big)^{1/2}$, with no distortion. This is precisely the lever that lets us reach *arbitrary* square-integrable non-anticipating integrands without ever choosing an evaluation point again. The argument is the standard isometric-extension principle of Hilbert-space theory, and it is worth stating in the abstract once so the reader sees there is no sleight of hand. A linear isometry defined on a *dense* subspace of a metric space, taking values in a *complete* space, extends uniquely to a linear isometry on the whole space: given any $H\in\mathcal{M}^2[0,T]$, choose simple $H^{(n)}\to H$; the images $I(H^{(n)})$ form a Cauchy sequence in $L^2(\mathbb{P})$ because $\big\|I(H^{(n)})-I(H^{(m)})\big\|_{L^2}=\big\|H^{(n)}-H^{(m)}\big\|_{\mathcal{M}^2}\to 0$ by the isometry applied to the difference; $L^2(\mathbb{P})$ is complete, so the limit exists; and it is independent of the approximating sequence, again by the isometry, since two approximants of the same H differ in $\mathcal{M}^2$-norm by something tending to zero. Two hypotheses make this run, and each is genuinely used: that the simple processes are *dense* in $\mathcal{M}^2[0,T]$ — so that every integrand is the limit of bets we already know how to evaluate (lemma C.6, proved by the standard three-step truncate–mollify–discretize approximation: truncate to bound the integrand, mollify in time to make it continuous, then sample on a fine grid to make it simple) — and that the target $L^2(\mathbb{P})$ is *complete*, so that the Cauchy images actually converge to something. The extension is the Itô integral, and because the construction is by isometry, the isometry identity itself is inherited automatically by the limit.

Definition 5.6 (Itô integral)**.** For $H\in\mathcal{M}^2[0,T]$ the *Itô integral* $\int_0^T H_t\;\mathrm{d}W_t$ is the $L^2(\mathbb{P})$ limit of $I(H^{(n)})$ for any sequence of simple processes $H^{(n)}\to H$ in $\mathcal{M}^2[0,T]$. The indefinite integral $M_t:=\int_0^t H_s\;\mathrm{d}W_s$ is defined for $t\in[0,T]$ and has a continuous modification.

The continuity claim in the definition is not free, and it is worth seeing why it holds, because it is what makes M a *path*, the thing Itô's formula will differentiate. The integrals of simple integrands are manifestly continuous in t — they are finite sums of continuous Brownian increments — and they are martingales. The danger is that the L^2 limit defining $\int_0^t H\;\mathrm{d}W$ holds for each fixed t

separately, and an uncountable family of pointwise-in-t limits need not assemble into a continuous path; we saw exactly this gap for Brownian motion itself in section 5.1, where continuity had to be installed by a modification rather than read off the finite-dimensional laws. The repair here is Doob's maximal inequality (theorem C.2), which for a continuous martingale N controls the *whole path* by the endpoint, $\mathbb{E}\big[\sup_{t\le T}|N_t|^2\big] \le 4\,\mathbb{E}\,|N_T|^2$. Applied to the differences $M^{(n)} - M^{(m)}$ of the approximating martingales, it upgrades the $\mathcal{M}^2$-Cauchy property into *uniform-in-t* L^2 convergence of the paths $t \mapsto M_t^{(n)}$; a sequence converging uniformly almost surely along a subsequence has a continuous limit, and that limit is the desired continuous modification of M. Doob's inequality is thus the device that propagates continuity from the simple integrands to their L^2 limit — the maximal inequality converting endpoint control into path control.

With the integral built and its indefinite version a genuine continuous process, we can collect the three structural facts the rest of the chapter will live on. The first is the isometry, now in polarized (bilinear) form; the second is the martingale property, the prize the left-endpoint convention was chosen to secure; the third is the quadratic variation, which records that the multiplication rule $(\mathrm{d}W)^2 = \mathrm{d}t$ survives the weighting by H as $\mathrm{d}\langle M\rangle = H^2\,\mathrm{d}t$.

Theorem 5.7 (Itô isometry and martingale property)**.** *For $H, K \in \mathcal{M}^2[0,T]$:*

1. $\mathbb{E}\Big[\int_0^T H\,\mathrm{d}W\Big] = 0$ *and* $\mathbb{E}\Big[\int_0^T H\,\mathrm{d}W \int_0^T K\,\mathrm{d}W\Big] = \mathbb{E}\int_0^T H_t K_t\,\mathrm{d}t$ *(polarized isometry);*
2. *the process $M_t = \int_0^t H_s\,\mathrm{d}W_s$ is a continuous $\mathbb{F}$-martingale: $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ for $s \le t$;*
3. *its quadratic variation is $\langle M\rangle_t = \int_0^t H_s^2\,\mathrm{d}s$, the natural generalization of theorem 5.4.*

Proof sketch. Each statement holds termwise for simple integrands and then passes to the L^2 limit; the only content is *why* each survives that limit, and in every case it is because the relevant operation is L^2-continuous.

(1) The polarized isometry. For simple H the squared-norm identity is exactly lemma 5.5; replacing H by $H+K$, $H-K$, and subtracting (the polarization identity $4\langle a,b\rangle = \|a+b\|^2 - \|a-b\|^2$) turns it into the bilinear form stated. To extend to general $H, K \in \mathcal{M}^2[0,T]$, take simple $H^{(n)} \to H$ and $K^{(n)} \to K$ in $\mathcal{M}^2$. Then $I(H^{(n)}) \to \int_0^T H\,\mathrm{d}W$ and $I(K^{(n)}) \to \int_0^T K\,\mathrm{d}W$ in $L^2(\mathbb{P})$ by construction, and *both sides* of the bilinear identity are continuous: the left side is the L^2 inner product, which is jointly continuous in its two arguments, and the right side is $\mathbb{E}\int_0^T H^{(n)}K^{(n)}$, which converges to $\mathbb{E}\int_0^T HK$ because $\mathcal{M}^2$-convergence is convergence in $L^2([0,T]\times\Omega)$ and the inner product there is again continuous. Equating the two limits gives the identity for H, K.

(2) The martingale property. For simple H and $s \le t$ the increment $M_t - M_s = \int_s^t H\,\mathrm{d}W$ is a sum of terms $\eta_i\,\Delta_i W$ each of which has zero conditional expectation given $\mathcal{F}_s$ — precisely the calculation of lemma 5.5, the coefficient being decided before its increment — so $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$. To keep this under the limit, fix $s \le t$ and approximate by simple $H^{(n)} \to H$, so $M_t^{(n)} \to M_t$ and $M_s^{(n)} \to M_s$ in $L^2(\mathbb{P})$. The conditional expectation $\mathbb{E}[\,\cdot \mid \mathcal{F}_s]$ is the orthogonal projection of $L^2(\mathbb{P})$ onto $L^2(\Omega, \mathcal{F}_s, \mathbb{P})$, hence an L^2-*contraction*: it cannot increase L^2-distance, so $M_t^{(n)} \to M_t$ forces $\mathbb{E}[M_t^{(n)} \mid \mathcal{F}_s] \to \mathbb{E}[M_t \mid \mathcal{F}_s]$ in L^2. The left side equals $M_s^{(n)} \to M_s$; equating the two limits gives $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ almost surely. Continuity of M was established in definition 5.6.

(3) The quadratic variation. For simple H, on a window $(t_i, t_{i+1}]$ where $H \equiv \eta_i$ the integral increment is $\eta_i \, \Delta_i W$; refining that window into a subgrid and summing squared increments, the $\mathcal{F}_{t_i}$-measurable constant η_i^2 factors out and theorem 5.4 applied increment-by-increment turns the squared Brownian increments into the deterministic length, so the window contributes $\eta_i^2(t_{i+1} - t_i) = \int_{t_i}^{t_{i+1}} H_s^2 \, \mathrm{d}s$; summing windows gives $\langle M \rangle_t = \int_0^t H_s^2 \, \mathrm{d}s$. This is exactly $\mathrm{d}\langle M \rangle = H^2 \, \mathrm{d}t$: the rule $(\mathrm{d}W)^2 = \mathrm{d}t$ weighted by the integrand. Passing to the L^2 limit, the full statement — together with the localization that lets one drop the standing $\mathbb{E}\int H^2 \, \mathrm{d}t < \infty$ to merely $\int_0^T H^2 \, \mathrm{d}t < \infty$ almost surely, at the price of M becoming a *local* martingale defined through stopping times — is carried out in [86, Ch. 3]. □

It is worth reading the martingale property of part (2) slowly, because it is the structural prize the entire left-endpoint construction was built to secure, and it will be invoked by name many times downstream. In words: the running gain from a non-anticipating bet placed against a fair noise has, conditionally on everything known so far, zero expected increment. The bet H_s is sized using only the past, the noise $\mathrm{d}W_s$ it rides on is fair and fresh, and so no systematic drift can accumulate — the integral is a fair game in the exact technical sense $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$. This single property is what lets stochastic integrals interlock with conditional expectations, and it underwrites at least three later developments the reader should keep in view. It is the foundation of *optimal stopping*, where the martingale/supermartingale dichotomy decides when to act. It is the engine of the *verification theorems* of chapter 6, where one shows a candidate value function is optimal precisely by arranging that the stochastic-integral term in its Itô expansion is a martingale and hence drops under expectation, leaving an inequality that pins down the optimum. And it is the defining structural requirement of the *backward stochastic differential equations* of chapter 9, whose solutions are built to be martingales adapted forward in time. Each of these is, at bottom, an exploitation of the fact that the Itô integral carries no drift.

Two of the three prizes are exactly what the next steps consume, and it is worth naming the hand-off explicitly. Section 5.3, which derives Itô's formula, runs entirely on the quadratic variation of part (3): the corrected chain rule for $f(X_t)$ acquires its extra second-order term as $\frac{1}{2}f'' \, \mathrm{d}\langle X \rangle$, and for an Itô process $\mathrm{d}X = b \, \mathrm{d}t + \sigma \, \mathrm{d}W$ that bracket is precisely $\mathrm{d}\langle X \rangle = \sigma^2 \, \mathrm{d}t$ — the weighted $(\mathrm{d}W)^2 = \mathrm{d}t$ we just established, with $H = \sigma$. The martingale property of part (2), meanwhile, is the pivot of the comparison in section 5.2.2: the choice that bought it — the left endpoint — is the very choice we are about to revisit, and the contrast there is exactly martingale-but-corrected-chain-rule (Itô) against ordinary-chain-rule-but-not-a-martingale (Stratonovich). We hand both facts forward and turn to the fork in the road we deliberately left open.

5.2.2 Itô versus Stratonovich

The left endpoint was a choice. That is the loose thread we deliberately left at the fork of section 5.2: the sample point τ_i could be placed anywhere inside $[t_i, t_{i+1}]$, the placement genuinely mattered — the left- and right-endpoint conventions disagreed in the limit by the full quadratic variation T — and we picked the left endpoint because it is the only *non-anticipating* placement, the one that locks the bet in before the increment it rides on is revealed. That single choice is what purchased the martingale property of theorem 5.7(2), the structural prize the whole construction was built around. But a choice that could have gone otherwise demands an accounting of what the other branches buy and what they cost. The right endpoint is no real competitor — it peeks at the future and is unusable as a strategy — so the honest alternative is the *symmetric* one: evaluate at the *midpoint* $\tau_i = \frac{1}{2}(t_i + t_{i+1})$, straddling the increment it multiplies rather than sitting to one side of it. Sampling there yields a genuinely different object, the *Stratonovich* integral, written $\int_0^T H_t \circ \mathrm{d}W_t$ with a small circle to distinguish it from the Itô integral $\int_0^T H_t \, \mathrm{d}W_t$.

The trade is the exact mirror of the one we made for Itô, and it is worth naming before any computation. The midpoint sees half of each increment before committing H, so it is no longer non-anticipating, and the cancellation that made each Itô term a fair, mean-zero bet against a fresh kick is broken: the Stratonovich integral is *not* a martingale. What it gains in exchange is the one thing Itô had to give up. Because the symmetric evaluation effectively splits the surviving second-order term evenly between the two sides of the increment, it folds exactly the Itô correction back into the integral, and the Stratonovich calculus obeys the *ordinary* chain rule of elementary calculus — $\mathrm{d}f(X) = f'(X) \circ \mathrm{d}X$, no extra term. Martingale-but-corrected-chain-rule versus ordinary-chain-rule-but-no-martingale: the two calculi are the two faces of the single fact $(\mathrm{d}W)^2 = \mathrm{d}t$, and one cannot have both faces at once. Our task in this subsection is to make the bridge between them quantitative — to compute, for a smooth volatility σ of the path, the precise correction term that converts a Stratonovich integral into an Itô one — and then to decide which calculus a physicist's Langevin equation actually denotes.

The correction is carried by a single new object, the cross-variation, and we define it before stating the conversion because it is the natural bilinear companion of the quadratic variation that already runs the chapter. Quadratic variation measured how a process accumulates its *own* squared increments; cross-variation measures how *two* processes accumulate the products of their increments, and like the quadratic variation it is forced to a deterministic limit by the $(\mathrm{d}W)^2 = \mathrm{d}t$ engine. For two Itô processes $\mathrm{d}X = b^X \, \mathrm{d}t + \sigma^X \, \mathrm{d}W$ and $\mathrm{d}Y = b^Y \, \mathrm{d}t + \sigma^Y \, \mathrm{d}W$ driven by the same Brownian motion, the *cross-variation* (or covariation) $\langle X, Y \rangle$ is defined by polarization from the quadratic variation,

$$\langle X, Y \rangle \;:=\; \tfrac{1}{2}\big(\langle X+Y \rangle - \langle X \rangle - \langle Y \rangle\big) \;=\; \lim_{\|\pi\| \to 0} \sum_i (\Delta_i X)(\Delta_i Y) \quad (\text{limit in } \mathbb{P}), \tag{5.4}$$

the in-probability limit of sums of products of increments along refining partitions. The two expressions agree for the same reason the variance of a sum decomposes: the elementary polarization identity $ab = \frac{1}{2}\big((a+b)^2 - a^2 - b^2\big)$, applied increment by increment with $a = \Delta_i X$ and $b = \Delta_i Y$, turns $\sum_i (\Delta_i X)(\Delta_i Y)$ into $\frac{1}{2}$ times $\sum_i (\Delta_i X + \Delta_i Y)^2 - \sum_i (\Delta_i X)^2 - \sum_i (\Delta_i Y)^2$, and each of those three sums converges to a quadratic variation — of $X + Y$, of X, of Y respectively — so the combination converges to $\frac{1}{2}(\langle X+Y \rangle - \langle X \rangle - \langle Y \rangle)$. Defining cross-variation this way is what makes $(X, Y) \mapsto \langle X, Y \rangle$ *bilinear and symmetric*, the bracket inheriting from the inner-product structure of the polarization formula exactly the rules a product should obey. Its value is read straight off the multiplication table. Since $\mathrm{d}\langle X \rangle = (\sigma^X)^2 \, \mathrm{d}t$ by theorem 5.7(3), applying that to $X + Y$ (whose volatility is $\sigma^X + \sigma^Y$) and expanding the square, $\mathrm{d}\langle X + Y \rangle = (\sigma^X + \sigma^Y)^2 \, \mathrm{d}t = \big((\sigma^X)^2 + 2\sigma^X \sigma^Y + (\sigma^Y)^2\big) \, \mathrm{d}t$, the pure squares cancel against $\mathrm{d}\langle X \rangle$ and $\mathrm{d}\langle Y \rangle$ in (5.4), leaving the cross term: the bilinear multiplication-table rule

$$\mathrm{d}\langle X, Y \rangle_t = \sigma_t^X \, \sigma_t^Y \, \mathrm{d}t.$$

This is simply "$\mathrm{d}X\ \mathrm{d}Y = \sigma^X\sigma^Y\ (\mathrm{d}W)^2 = \sigma^X\sigma^Y\ \mathrm{d}t$," the drift parts contributing nothing because $\mathrm{d}t\ \mathrm{d}W$ and $(\mathrm{d}t)^2$ both vanish in the table. The one instance we actually need is the specialization to $Y = W$ itself, which has unit volatility $\sigma^Y \equiv 1$ and no drift:

$$\mathrm{d}\langle X, W\rangle_t = \sigma_t^X\ \mathrm{d}t, \tag{5.5}$$

the rate at which a process co-varies with its own driving noise being exactly its volatility. This is the only cross-variation that appears below.

Proposition 5.8 (Itô–Stratonovich conversion)**.** *Let X be an Itô process (section 5.3) and $\sigma \in C^1(\mathbb{R})$. Then*

$$\int_0^T \sigma(X_t) \circ \mathrm{d}W_t \;=\; \int_0^T \sigma(X_t)\ \mathrm{d}W_t \;+\; \tfrac{1}{2}\int_0^T \sigma'(X_t)\ \mathrm{d}\langle X, W\rangle_t. \tag{5.6}$$

In particular, for the Stratonovich SDE $\mathrm{d}X_t = b(X_t)\ \mathrm{d}t + \sigma(X_t) \circ \mathrm{d}W_t$, the equivalent Itô SDE carries an extra drift,

$$\mathrm{d}X_t = \Big(b(X_t) + \tfrac{1}{2}\,\sigma(X_t)\,\sigma'(X_t)\Big)\,\mathrm{d}t + \sigma(X_t)\ \mathrm{d}W_t. \tag{5.7}$$

Proof sketch. The whole content is the difference between two Riemann sums for the *same* integrand that differ only in where the integrand is sampled. The Stratonovich integral is the limit of the *midpoint* sums $\sum_i \sigma(X_{\tau_i})\,\Delta_i W$ with $\tau_i = \frac{1}{2}(t_i + t_{i+1})$ and $\Delta_i W = W_{t_{i+1}} - W_{t_i}$, while the Itô integral is the limit of the left-endpoint sums $\sum_i \sigma(X_{t_i})\,\Delta_i W$. Their difference is therefore

$$D_n \;=\; \sum_i \left[\sigma(X_{\tau_i}) - \sigma(X_{t_i})\right] \Delta_i W,$$

and the entire conversion is the evaluation of $\lim_n D_n$. Expand the bracket to first order in the displacement $X_{\tau_i} - X_{t_i}$ of the path over the *first half* of the subinterval. By Taylor, $\sigma(X_{\tau_i}) - \sigma(X_{t_i}) = \sigma'(X_{t_i})\,(X_{\tau_i} - X_{t_i}) + O\big((X_{\tau_i} - X_{t_i})^2\big)$, and along an Itô process the half-step displacement is dominated by its noise part, $X_{\tau_i} - X_{t_i} = \sigma(X_{t_i})\,(W_{\tau_i} - W_{t_i}) + O(\Delta_i t)$ — the drift contributes only $b(X_{t_i})\,(\tau_i - t_i) = O(\Delta_i t)$, an order smaller than the $O(\sqrt{\Delta_i t})$ noise term and negligible in the sum. Substituting,

$$\sigma(X_{\tau_i}) - \sigma(X_{t_i}) \;=\; \sigma'(X_{t_i})\,\sigma(X_{t_i})\,(W_{\tau_i} - W_{t_i}) \;+\; \text{(lower order)},$$

so, writing $g_i := \sigma'(X_{t_i})\,\sigma(X_{t_i})$ for the (history-known) $\mathcal{F}_{t_i}$-measurable coefficient and splitting the full increment as $\Delta_i W = (W_{\tau_i} - W_{t_i}) + (W_{t_{i+1}} - W_{\tau_i})$ into its two half-window pieces,

$$D_n \;=\; \underbrace{\sum_i g_i\,(W_{\tau_i} - W_{t_i})(W_{t_{i+1}} - W_{\tau_i})}_{S_n^{\text{cross}}} \;+\; \underbrace{\sum_i g_i\,(W_{\tau_i} - W_{t_i})^2}_{S_n^{\text{sq}}} \;+\; \text{(lower order)}.$$

Two sums, with opposite fates, and it is exactly here that the proof-sketch's $\approx$ signs must be made to earn their keep.

The cross-term sum S_n^{cross} vanishes in L^2. In each summand the two Brownian factors $W_{\tau_i} - W_{t_i}$ and $W_{t_{i+1}} - W_{\tau_i}$ live over the *disjoint* half-windows $[t_i, \tau_i]$ and $[\tau_i, t_{i+1}]$, so by the independent-increments axiom they are independent of each other and of the $\mathcal{F}_{t_i}$-measurable coefficient g_i, and each has mean zero. We compute the second moment $\mathbb{E}\,(S_n^{\text{cross}})^2 = \sum_{i,j} \mathbb{E}[\,g_i g_j\,(W_{\tau_i} - W_{t_i})(W_{t_{i+1}} - W_{\tau_i})(W_{\tau_j} - W_{t_j})(W_{t_{j+1}} - W_{\tau_j})\,]$ and show every term is either zero or small. For $i \neq j$, say $i < j$, the latest factor $W_{t_{j+1}} - W_{\tau_j}$ is independent, with mean zero, of all five remaining factors (they are all measurable with respect to the past up to τ_j); conditioning on that past pulls it out as $\mathbb{E}[W_{t_{j+1}} - W_{\tau_j} \mid \mathcal{F}_{\tau_j}] = 0$, so every off-diagonal term vanishes, exactly as the cross terms did in the Itô isometry. Only the diagonal survives, and on it the two half-increments are independent with $\mathbb{E}(W_{\tau_i} - W_{t_i})^2 = \mathbb{E}(W_{t_{i+1}} - W_{\tau_i})^2 = \frac{1}{2}\Delta_i t$, so

$$\mathbb{E}\,(S_n^{\text{cross}})^2 = \sum_i \mathbb{E}[g_i^2]\,\tfrac{1}{2}\Delta_i t \cdot \tfrac{1}{2}\Delta_i t = \tfrac{1}{4}\sum_i \mathbb{E}[g_i^2]\,(\Delta_i t)^2 \;\le\; \tfrac{1}{4}\,\big(\sup_i \mathbb{E}[g_i^2]\big)\,\|\pi_n\| \sum_i \Delta_i t \;\xrightarrow[n\to\infty]{}\; 0,$$

the same mesh-times-quadratic estimate (5.3) that killed the variance of the quadratic variation: one factor $\Delta_i t$ is bounded by the mesh $\|\pi_n\|$ and pulled out, the other telescopes to T, and $\|\pi_n\| \to 0$. So $S_n^{\text{cross}} \to 0$ in L^2. The cross-term carries no limit: it is built from products of *independent* increments, never from a square, and only squares survive.

The squared-increment sum S_n^{sq} gives the $\frac{1}{2}$. Here the factor *is* a square, $(W_{\tau_i} - W_{t_i})^2$, and squares accumulate deterministically. The increment runs over the half-window $[t_i, \tau_i]$ of length $\tau_i - t_i = \frac{1}{2}\Delta_i t$, so by the quadratic-variation law theorem 5.4 — applied to this refining family of *half*-windows, against the continuous adapted weight $g_i = \sigma'(X_{t_i})\sigma(X_{t_i})$ — the squared half-increments may be replaced by the deterministic half-lengths:

$$S_n^{\text{sq}} = \sum_i g_i\,(W_{\tau_i} - W_{t_i})^2 \;\xrightarrow[n\to\infty]{}\; \int_0^T \sigma'(X_t)\,\sigma(X_t)\;\mathrm{d}\Big(\tfrac{1}{2}t\Big) = \tfrac{1}{2}\int_0^T \sigma'(X_t)\,\sigma(X_t)\;\mathrm{d}t.$$

The $\frac{1}{2}$ is not a convention or a fudge: it is precisely the fraction of the full step $\Delta_i t$ that the *half*-increment's square accumulates. Had we sampled at a general point $\tau_i = t_i + \theta\,\Delta_i t$ with $\theta \in [0,1]$, the surviving sum would converge to $\theta \int_0^T \sigma'\sigma\;\mathrm{d}t$; the left endpoint ($\theta = 0$) recovers the Itô integral with no correction, the right endpoint ($\theta = 1$) gives the full $\int \sigma'\sigma$, and the midpoint ($\theta = \frac{1}{2}$) splits the difference, which is the symmetric choice that defines Stratonovich. The whole family of calculi is parametrized by where in the increment one stands.

Collecting the two limits, $D_n \to \frac{1}{2}\int_0^T \sigma'(X_t)\,\sigma(X_t)\;\mathrm{d}t$, and using the cross-variation identity (5.5) with $\sigma^X = \sigma(X)$ to write $\sigma(X_t)\;\mathrm{d}t = \mathrm{d}\langle X, W\rangle_t$, this is $\frac{1}{2}\int_0^T \sigma'(X_t)\;\mathrm{d}\langle X, W\rangle_t$, which is precisely the asserted conversion (5.6). Equation (5.7) is then the same identity applied to the volatility term of the SDE: the Stratonovich diffusion $\sigma(X) \circ \mathrm{d}W$ equals the Itô diffusion $\sigma(X)\;\mathrm{d}W$ plus the surviving drift $\frac{1}{2}\sigma'(X)\;\mathrm{d}\langle X, W\rangle = \frac{1}{2}\sigma(X)\sigma'(X)\;\mathrm{d}t$, so a model written in Stratonovich form acquires, when rewritten in Itô form, exactly the extra drift $\frac{1}{2}\sigma\sigma'$. □

The correction $\frac{1}{2}\sigma\sigma'$ deserves a name and a careful reading, because it is the crux of the whole comparison: it is the *noise-induced drift*, a systematic motion that the multiplicative coupling of the state to the noise manufactures out of pure fluctuation. Its structure tells the entire story. It is built from σ', the rate at which the noise amplitude changes with the state, multiplied by σ itself, the amplitude. Where the noise is stronger on one side than the other, the random kicks are systematically larger in one direction, and over many kicks that asymmetry rectifies into a net drift — a real, observable bias, not a bookkeeping artifact. The decisive consequence, and the one this whole book will lean on, is the conditional under which it *disappears*: the moment σ is a *constant*,

$\sigma' \equiv 0$ and the noise-induced drift is identically zero. *Additive* noise — the state coupled to the bath through a fixed amplitude, never modulated by the state — feels no correction at all, and the Itô and Stratonovich integrals literally coincide. Only *multiplicative* noise, where σ varies with the state, makes the two calculi part company. Hold onto this: it is exactly why the linear–quadratic threads of this book are spared the entire question.

With the conversion in hand we can state precisely what the two calculi are to each other, and it is not a rivalry. Each is internally consistent; each integrates the same integrands against the same Brownian motion; they simply make opposite choices at the one fork the roughness of W opened up, and they pay complementary prices. The trade is forced, and it is the same $(\mathrm{d}W)^2 = \mathrm{d}t$ surviving term seen from two sides: *the Itô integral is a martingale but obeys a corrected chain rule; the Stratonovich integral obeys the ordinary chain rule but is not a martingale.* One cannot have both. The martingale property came from locking the bet in before its increment — the left endpoint — and that same non-anticipation is exactly what forces the surviving second-order term to land entirely on the chain rule as a correction, because the integrand never gets to "see" the half of the increment that would symmetrize it away. Stratonovich's midpoint absorbs that half-increment back into the integral, which is why its chain rule is the ordinary one — but in seeing half of the increment it forfeits the conditional fairness that made Itô a martingale. There is no third convention escaping the dilemma; the two are the two stable resolutions of a single tension.

Notation watch

Which calculus does a physicist's Langevin equation mean? When the noise amplitude depends on the state — multiplicative noise — the bare equation $\dot{x} = b(x) + \sigma(x)\,\xi$ is *ambiguous*, and the ambiguity is exactly the $\frac{1}{2}\sigma\sigma'$ term in eq. (5.7). The resolution is physical, not conventional: if ξ is idealized from a real bath with a short but nonzero correlation time, then—this we invoke as physical motivation, not as a step in the book's rigorous development—a classical result of Wong and Zakai [133] asserts that the zero-correlation-time limit of the (ordinary, pathwise) solutions converges to the *Stratonovich* solution. The heuristic is immediate: a smooth (colored-noise) approximation of W has finite variation, so it obeys the *ordinary* chain rule, and the calculus that obeys the ordinary chain rule is precisely the Stratonovich one. So a Langevin equation obtained as a white-noise idealization of colored noise is Stratonovich; a model written down directly as a discrete-time non-anticipating recursion is Itô. The mathematics of stochastic control and of mean field games is built on the Itô integral, because optimization needs the martingale property; we therefore use Itô throughout, and translate any Stratonovich physical model via eq. (5.7). **For the LQG and graphon-LQG threads the noise is additive (σ constant), so the two calculi coincide and the question is moot** — one reason the linear–quadratic world is so clean.

The convention is now settled, and we carry it forward unambiguously: from here on every stochastic integral, SDE, and generator in this book is read in the *Itô* sense, chosen for the martingale property that optimization and the verification arguments of chapter 6 cannot do without. A physical model handed to us in Stratonovich form is not a competitor but a translation problem: we convert it once, through the drift correction (5.7), replacing $\sigma \circ \mathrm{d}W$ by $\sigma\ \mathrm{d}W$ and adding $\frac{1}{2}\sigma\sigma'\ \mathrm{d}t$ to the drift, and thereafter work entirely in Itô. For the additive-noise threads that run through the book this translation is the identity, and the distinction never resurfaces.

Settling the convention also discharges the promissory note left open since the construction of the integral. The price of the left endpoint, we said, was that the ordinary chain rule fails: $\mathrm{d}\sigma(W_t)$ is not $\sigma'(W_t)\ \mathrm{d}W_t$, and a second-order term survives. The conversion just computed is the first concrete sighting of that surviving term — the very $\frac{1}{2}$ that distinguishes the two calculi is the same $\frac{1}{2}$ that the corrected chain rule will carry — but here it appeared only as the gap between two integrals of a fixed integrand. What we have *not* yet done is repair the chain rule itself: write down, for a general smooth f and a general Itô process, the exact differential of $f(X_t)$ with its correction term in place. That is the task of the next section. It works at the level of the Itô-process notation $\mathrm{d}X = b\ \mathrm{d}t + \sigma\ \mathrm{d}W$ that the cross-variation rule (5.5) already presumed and that theorem 5.7 made legitimate, and it turns the morality of "$(\mathrm{d}W)^2 = \mathrm{d}t$" into a clean second-order calculus. The extra term it produces is no longer a nuisance to be converted away: it *is* diffusion, the bridge from a single trajectory to the Fokker–Planck flow of the whole ensemble. We turn to Itô's formula.

5.3 Itô's formula

The Itô-versus-Stratonovich subsection settled a convention and, in doing so, signed a promissory note. We chose the left endpoint, bought the martingale property with it, and paid for it in a single coin we named but did not yet mint: the ordinary chain rule fails, and a corrected one must take its place. The conversion computation of section 5.2.2 already showed the *shape* of the correction — a term proportional to the surviving second-order quantity $(\mathrm{d}W)^2 = \mathrm{d}t$, the very fact the symmetric Stratonovich evaluation folds silently back into the integral while the left-endpoint Itô integral leaves it standing in the open. What remains is to make that correction exact and universal: to write down, once and for all, how *any* smooth function of *any* Itô process evolves. That statement is Itô's formula, and it is the analytic heart of the chapter. Here the one fact carried since the quadratic-variation theorem — that squared Brownian increments accumulate to a deterministic $\mathrm{d}t$ rather than vanishing — finally surfaces as a concrete differential operator, the $\frac{1}{2}\sigma^2\partial_{xx}$ term that ordinary calculus would have discarded as higher order. We will see at the end of the section that this term *is* diffusion; everything downstream is its echo.

We state the rule for the natural class of processes the Itô integral was built to produce. Define an *Itô process* to be one of the form

$$X_t = X_0 + \int_0^t b_s\ \mathrm{d}s + \int_0^t \sigma_s\ \mathrm{d}W_s, \qquad \text{abbreviated} \quad \mathrm{d}X_t = b_t\ \mathrm{d}t + \sigma_t\ \mathrm{d}W_t, \tag{5.8}$$

with b progressively measurable and integrable and $\sigma \in \mathcal{M}^2$ — a drift piece accumulated as an ordinary (Lebesgue) integral and a noise piece accumulated as the Itô integral of section 5.2.1. This is exactly the right level of generality for two reasons. It is broad enough to contain every state process this book will meet: the solution of an SDE in section 5.4 is of this form, and so is the controlled state of Part II. And it is, as the theorem will show, *closed* under the operation we are about to define — a smooth function of an Itô process is again an Itô process — which is precisely what makes Itô calculus a calculus rather than a collection of one-off tricks: one may apply the change-of-variables rule and then apply it again to the result, exactly as one chains the ordinary chain rule.

Before stating the rule we read off the one structural fact an Itô process carries with it, its quadratic variation. The abbreviation $\mathrm{d}X_t = b_t\ \mathrm{d}t + \sigma_t\ \mathrm{d}W_t$ splits each increment into a smooth part of order $\mathrm{d}t$ and a rough part of order $\sqrt{\mathrm{d}t}$, and squaring sorts them by the multiplication table of section 5.1.2. The drift contributes nothing: the integral $\int_0^t b_s\ \mathrm{d}s$ has finite first variation, so by the smooth-path computation of theorem 5.4 its own quadratic variation is zero, and its cross-variation with the noise piece vanishes as well — this is the entry $\mathrm{d}W\ \mathrm{d}t = 0$, the cross term that is intermediate in size and drops. Only the square of the noise piece survives, and by $(\mathrm{d}W)^2 = \mathrm{d}t$ it survives stripped of its randomness. Thus

$$\mathrm{d}\langle X\rangle_t = \sigma_t^2\ \mathrm{d}t, \tag{5.9}$$

inherited from theorem 5.7(3): *the drift, having finite first variation, contributes nothing, and the entire quadratic variation of an Itô process is carried by its volatility.* This single line is what the formula below will contract against the second derivative of F; it is worth keeping (5.9) in view as the one piece of data that distinguishes the Itô chain rule from the ordinary one.

With the process and its quadratic variation in hand, ask the question the rule answers. We rarely care about the state X_t in isolation; we care about smooth observables of it — an energy $\frac{1}{2}X_t^2$, a payoff $F(t, X_t)$, a test function integrated against the law. How does such an observable evolve? In ordinary calculus the answer is the chain rule, $\mathrm{d}F = \partial_t F\ \mathrm{d}t + \partial_x F\ \mathrm{d}X$, and nothing more. For an Itô process that answer is wrong, and wrong by exactly one term.

Theorem 5.9 (Itô's formula)**.** *Let X be the Itô process* (5.8) *and let $F \in C^{1,2}([0,T]\times\mathbb{R})$. Then $F(t, X_t)$ is again an Itô process and*

$$\mathrm{d}F(t,X_t) = \Big(\partial_t F + b_t\,\partial_x F + \tfrac{1}{2}\,\sigma_t^2\,\partial_{xx}F\Big)(t,X_t)\ \mathrm{d}t\ +\ \sigma_t\,\partial_x F(t,X_t)\ \mathrm{d}W_t. \tag{5.10}$$

In d dimensions, with $\mathrm{d}X_t = b_t\ \mathrm{d}t + \sigma_t\ \mathrm{d}W_t$, $b_t \in \mathbb{R}^d$, $\sigma_t \in \mathbb{R}^{d\times d}$, and $F \in C^{1,2}([0,T]\times\mathbb{R}^d)$,

$$\mathrm{d}F(t,X_t) = \Big(\partial_t F + b_t\cdot\nabla F + \tfrac{1}{2}\ \mathrm{tr}\big(\sigma_t\sigma_t^\top\,\mathrm{D}^2\,F\big)\Big)\,\mathrm{d}t + \big(\nabla F\big)^\top\sigma_t\ \mathrm{d}W_t, \tag{5.11}$$

all derivatives evaluated at (t, X_t).

Proof sketch. The whole argument is one Taylor expansion read through the multiplication table; we give its skeleton and intuition in full and defer only the bookkeeping of the remainder to appendix C.5, where the localization is done carefully. Fix a partition $0 = t_0 < \cdots < t_k = t$ of $[0,t]$, write $\Delta_i t$, $\Delta_i X$, $\Delta_i W$ for the increments across $[t_i, t_{i+1}]$, and telescope: $F(t,X_t) - F(0,X_0) = \sum_i \big[F(t_{i+1}, X_{t_{i+1}}) - F(t_i, X_{t_i})\big]$. Because $F \in C^{1,2}$, expand each summand to *second* order — first order in time, second order in space, since the space variable is the rough one —

$$F(t_{i+1}, X_{t_{i+1}}) - F(t_i, X_{t_i}) = \partial_t F\,\Delta_i t + \partial_x F\,\Delta_i X + \tfrac{1}{2}\,\partial_{xx}F\,(\Delta_i X)^2 + R_i,$$

with all derivatives at (t_i, X_{t_i}) and R_i the Taylor remainder. The proof is the fate of the three explicit groups and the vanishing of $\sum_i R_i$, taken in turn.

The first two groups give ordinary calculus. The time sum $\sum_i \partial_t F\, \Delta_i t$ is a Riemann sum and converges to $\int_0^t \partial_t F(s, X_s)\, \mathrm{d}s$. The first-order space sum $\sum_i \partial_x F\, \Delta_i X$ converges to $\int_0^t \partial_x F\, \mathrm{d}X_s$ by the very definition of the Itô integral: it is a left-endpoint sum of a non-anticipating integrand against the increments $\Delta_i X$, and splitting $\mathrm{d}X = b\, \mathrm{d}t + \sigma\, \mathrm{d}W$ resolves it into a b-Riemann piece $\int_0^t \partial_x F\, b_s\, \mathrm{d}s$ and an Itô piece $\int_0^t \partial_x F\, \sigma_s\, \mathrm{d}W_s$. So far this is exactly the ordinary chain rule $\mathrm{d}F = \partial_t F\, \mathrm{d}t + \partial_x F\, \mathrm{d}X$; had the path been smooth, the story would end here.

The third group is the entire content of the formula. It does not vanish. Substitute $\Delta_i X = b_{t_i}\Delta_i t + \sigma_{t_i}\Delta_i W + (\text{higher order})$ and square. By the multiplication table the $\Delta_i t$ pieces are genuinely higher order — the $(\Delta_i t)^2$ and $\Delta_i t\, \Delta_i W$ products sum to zero — so the only surviving piece is $(\Delta_i X)^2 \approx \sigma_{t_i}^2 (\Delta_i W)^2$. Now the quadratic-variation theorem does its work. Were $(\Delta_i W)^2$ of order $(\Delta_i t)^2$, as it is for a differentiable path, the sum $\sum_i \frac{1}{2}\partial_{xx} F\, \sigma_{t_i}^2 (\Delta_i W)^2$ would be $O(\|\pi\|)$ and vanish, recovering ordinary calculus. But $(\Delta_i W)^2 \sim \Delta_i t$ by theorem 5.4, and what the theorem proved is that these squared increments may be *replaced* by the deterministic $\Delta_i t$ inside a sum, with the randomness washing out. Hence

$$\sum_i \tfrac{1}{2}\, \partial_{xx} F\, \sigma_{t_i}^2\, (\Delta_i W)^2 \;\longrightarrow\; \tfrac{1}{2} \int_0^t \partial_{xx} F(s, X_s)\, \sigma_s^2\, \mathrm{d}s,$$

not to zero. This is the surviving second-order term, and it is the whole difference between the two calculi: in ordinary calculus the second-order term is $o(\Delta t)$ and drops; here $(\Delta W)^2 \sim \Delta t$, so it persists with weight $\frac{1}{2}\sigma^2$. Adding the three limits gives (5.10).

The remainder, the only delicate point. It remains to show $\sum_i R_i \to 0$, and this is where the heuristic "$(\mathrm{d}W)^2 = \mathrm{d}t$" must be upgraded to a genuine convergence rather than a slogan. Two issues are folded together and we separate them. First, *localization*: F and its derivatives are continuous but not *a priori* bounded, and b, σ may be unbounded, so one first stops the process at the exit time of a large ball and at a level where the coefficients are bounded, proves the identity there, and removes the localization by letting the ball grow — the routine but unilluminating part deferred to appendix C.5. Granting that reduction, X takes values in a compact set K, all relevant derivatives live on the compact $[0, t] \times K$, and the substantive estimate is clean. The dangerous remainder is the second-order one: the exact Taylor formula writes the third group with $\partial_{xx} F$ evaluated not at (t_i, X_{t_i}) but at some intermediate point θ_i on the segment between the two endpoints, so the error made by freezing it at the left endpoint is

$$\sum_i \tfrac{1}{2} \big[\partial_{xx} F(\theta_i) - \partial_{xx} F(t_i, X_{t_i})\big] (\Delta_i X)^2.$$

Here is the decisive point. On the compact $[0, t] \times K$ the map $\partial_{xx} F$ is *uniformly* continuous, and as the mesh refines, path-continuity forces each θ_i to lie within a vanishing distance of (t_i, X_{t_i}); so given $\varepsilon > 0$, once the mesh is fine enough the bracket is uniformly below ε for every i. The remainder sum is then bounded in absolute value by $\frac{1}{2}\varepsilon \sum_i (\Delta_i X)^2$, and the surviving factor $\sum_i (\Delta_i X)^2$ is precisely the quadratic-variation sum, which converges in L^1 to $\langle X \rangle_t = \int_0^t \sigma_s^2\, \mathrm{d}s$ and is in particular bounded in L^1 uniformly in the partition. Thus the whole remainder is $\leq \frac{1}{2}\varepsilon \cdot (\text{a fixed } L^1 \text{ bound})$, and since ε was arbitrary it tends to zero in L^1. This is the exact step that converts the heuristic $(\mathrm{d}W)^2 = \mathrm{d}t$ into honest convergence: uniform continuity of $\partial_{xx} F$ on compacta supplies the small factor ε, and the quadratic-variation estimate (5.3) supplies the bounded factor $\sum_i (\Delta_i X)^2$, and the product is forced to zero. The lower-order pieces of R_i — those carrying a $\Delta_i t$ — vanish faster, by the same $\mathrm{d}W\ \mathrm{d}t = 0$, $(\mathrm{d}t)^2 = 0$ bookkeeping. The complete localized argument is appendix C.5; [106, Thm. 4.1.6] is the textbook reference.

The d-dimensional case is identical, with one bookkeeping change at the second-order term. The scalar $\frac{1}{2}\partial_{xx}F\,(\Delta_i X)^2$ becomes the quadratic form $\frac{1}{2}\sum_{k,l}\partial_{x_k x_l}F\,(\Delta_i X)_k(\Delta_i X)_l$, i.e. the Hessian $\mathrm{D}^2 F$ contracted against the outer product $\Delta_i X\,\Delta_i X^\top$. Reading that outer product through the multiplication table, and using that the components $W^1, \dots, W^d$ are *independent* Brownian motions, the cross-variations obey $\mathrm{d}W^i\ \mathrm{d}W^j = \delta_{ij}\ \mathrm{d}t$ — the off-diagonal $i \neq j$ entries vanish because independent Brownian motions have zero covariation (the polarization (5.4) of two independent, hence uncorrelated, increments), while the diagonal entries are each $\mathrm{d}t$. Hence $\mathrm{d}X\ \mathrm{d}X^\top = \sigma\sigma^\top\ \mathrm{d}t$, the matrix $\sigma\sigma^\top$ playing the role of the scalar σ^2, and the contracted second-order term is $\frac{1}{2}\ \mathrm{tr}\big(\sigma\sigma^\top\,\mathrm{D}^2 F\big)\ \mathrm{d}t$, which is (5.11). □

Heuristic (non-rigorous)

In practice one never re-runs the partition argument. The working mnemonic is purely mechanical: write the formal second-order Taylor expansion and then simplify $(\mathrm{d}X)^2$ by the multiplication table $(\mathrm{d}W)^2 = \mathrm{d}t$, $\mathrm{d}W\ \mathrm{d}t = 0$, $(\mathrm{d}t)^2 = 0$,

$$\mathrm{d}F = \partial_t F\ \mathrm{d}t + \partial_x F\ \mathrm{d}X + \tfrac{1}{2}\partial_{xx}F\,(\mathrm{d}X)^2, \qquad (\mathrm{d}X)^2 = (b\ \mathrm{d}t + \sigma\ \mathrm{d}W)^2 = \sigma^2\ \mathrm{d}t.$$

Keeping the expansion to first order in $\mathrm{d}t$ and to *second* order in $\mathrm{d}W$, and then collapsing $(\mathrm{d}W)^2 \to \mathrm{d}t$, reproduces (5.10) in one line. What licenses each of these symbolic moves is not formal bravado but the theorems just proved: the collapse $(\mathrm{d}W)^2 \to \mathrm{d}t$ is the quadratic-variation theorem theorem 5.4, the vanishing of $\mathrm{d}W\ \mathrm{d}t$ and $(\mathrm{d}t)^2$ is the finite first variation of the smooth pieces, and the legitimacy of stopping the Taylor expansion at second order is the L^1 remainder control above. The mnemonic is exact precisely because each entry of the table is a theorem. Every appearance of the Itô second-order term in this book — the Laplacian in Fokker–Planck, the $\frac{1}{2}\sigma^2\partial_{xx}$ in the generator, the Hessian penalty in stochastic HJB — is this one surviving term, and no other source of it exists in the whole development.

It is worth isolating, on the smallest possible example, exactly what the second-order term *means* — and what physics one would get wrong without it. Take the simplest nonlinear observable, $F(x) = x^2$, of the simplest Itô process, $\mathrm{d}X = \sigma\ \mathrm{d}W$ with σ constant (pure noise, no drift): this is just a Brownian particle with volatility σ and no restoring force. Here $\partial_x F = 2x$ and $\partial_{xx}F = 2$, so theorem 5.9 reads

$$\mathrm{d}(X_t^2) = \partial_x F\,\sigma\ \mathrm{d}W_t + \tfrac{1}{2}\partial_{xx}F\,\sigma^2\ \mathrm{d}t = 2X_t\,\sigma\ \mathrm{d}W_t + \sigma^2\ \mathrm{d}t.$$

Now take expectations. The first term is an Itô integral, hence a martingale, hence of zero mean by theorem 5.7(2); only the deterministic correction survives, and integrating $\mathrm{d}\,\mathbb{E}X_t^2 = \sigma^2\ \mathrm{d}t$ gives

$$\mathbb{E}X_t^2 = \mathbb{E}X_0^2 + \sigma^2\,t.$$

The second moment — the variance, if X_0 is centered — grows *linearly* in time at rate σ^2. A probability packet that started as a spike spreads, its width $\sqrt{\mathbb{E}X_t^2}$ swelling like $\sqrt{t}$, and that linear-in-time growth of the variance is the defining signature of diffusion. Itô's correction term is its entire source. Set the correction to zero — apply the naive chain rule $\mathrm{d}(X^2) = 2X\ \mathrm{d}X = 2X\,\sigma\ \mathrm{d}W$ — and one would conclude $\mathbb{E}X_t^2 = \mathbb{E}X_0^2$, a *conserved* variance: a diffusing particle that never spreads, which is simply the wrong physics. The discrepancy between the two answers is exactly the missing $\sigma^2 t$, the integral of the surviving second-order term. And this rate σ^2 is no accident of the example: it is the very constant that will reappear in section 5.5 as the diffusion coefficient multiplying the Laplacian in the Fokker–Planck equation, where the same term, now acting on a density rather than a single trajectory, drives the packet's spread. The Fokker–Planck diffusion constant is nothing but Itô's second-order term carried through.

This closes the loop the chapter opened. We can now hand Itô's formula forward as the single universal lever the remaining sections will all pull. Section 5.4 needs it immediately and in two roles at once: together with the Itô integral just constructed it is what gives the differential $\mathrm{d}X = b(t,X)\ \mathrm{d}t + \sigma(t,X)\ \mathrm{d}W$ — the rigorous form of the Langevin nonsense $\dot{x} = -V' + \xi$ the preamble warned about — its honest meaning as the integral equation (5.8) with state-dependent coefficients, and it is the engine of the moment estimates (apply the formula to $F(x) = |x|^2$, take expectations, and the second-order term produces exactly the Grönwall inequality that yields existence, uniqueness, and a priori bounds). Beyond that, the same formula generates the rest of the chapter: applying it to $F(X_t)$ and taking expectations isolates the generator $\mathcal{A} = b\,\partial_x + \frac{1}{2}\sigma^2\partial_{xx}$ (Dynkin's formula), in which the second-order term is now a differential operator; the generator's appearance in section 5.5 drives the Fokker–Planck equation forward and the Kolmogorov backward equation in reverse; and in section 5.6 the formula applied to a suitably weighted functional produces the martingale whose constant expectation is the Feynman–Kac bridge between path expectations and linear PDEs. In every one of these the extra term is the same $\frac{1}{2}\sigma^2\partial_{xx}$ we just minted: to repeat the flag, *every Itô second-order term in this book is this one surviving term.* The next section spends it first, on giving the Langevin equation a rigorous solution.

5.4 Stochastic differential equations

The previous section handed us Itô's formula as the universal lever of the chapter, and named the very first thing it would lift: it gives the differential $\mathrm{d}X = b(t,X)\ \mathrm{d}t + \sigma(t,X)\ \mathrm{d}W$ — the rigorous form of the Langevin nonsense $\dot{x} = -V' + \xi$ the preamble warned about — its honest meaning as an integral equation, and it is the engine of the moment estimates that make that equation solvable. We now spend the lever on exactly that. The whole construction of this chapter has been building toward a single payoff: a precise sense in which the physicist's Langevin equation has a solution, unique, defined for all time, and adapted to the noise that drives it. We are now in a position to deliver it. The Itô integral of section 5.2 supplies the meaning of the noise term $\int \sigma\ \mathrm{d}W$; Itô's formula of section 5.3 supplies the calculus that controls it; and with both in hand the equation that looked like nonsense in the chapter preamble becomes a well-posed object with a genuine existence-and-uniqueness theorem, proved by a method — Picard iteration in a space of processes — that is the template for every existence theorem the book builds downstream.

A stochastic differential equation prescribes the infinitesimal drift and volatility as functions of the current state:

$$\mathrm{d}X_t = b(t, X_t)\ \mathrm{d}t + \sigma(t, X_t)\ \mathrm{d}W_t, \qquad X_0 = \zeta. \tag{5.12}$$

The differential notation is, as always in this chapter, shorthand: the only meaning (5.12) carries is the integral equation

$$X_t = \zeta + \int_0^t b(s, X_s)\ \mathrm{d}s + \int_0^t \sigma(s, X_s)\ \mathrm{d}W_s, \qquad 0 \le t \le T, \tag{5.13}$$

in which the first integral is an ordinary (pathwise Lebesgue) integral and the second is the Itô integral built in section 5.2. Writing the equation this way is already the whole conceptual advance over the physicist's $\dot{x} = -V' + \xi$: there is no $\dot{X}$ and no ξ anywhere in (5.13), only increments of W accumulated against a non-anticipating integrand, which is the one form of noise the previous sections gave a rigorous meaning. The equation is an implicit one — the unknown process X appears on both sides, inside the coefficients of the very integrals that produce it — and that is precisely why a solution must be *constructed* rather than written down, and why the construction is a fixed-point problem.

Before constructing anything we must be clear about what counts as a solution, because there are two genuinely different notions and the difference is not pedantic — it decides which problems the theory can even pose. A *strong solution* is a continuous, $\mathbb{F}$-adapted process X satisfying (5.13) for all $t \in [0, T]$ almost surely, on a *given* probability space carrying a *given* Brownian motion W. The data are fixed in advance — the space $(\Omega, \mathcal{F}, \mathbb{P})$, the filtration $\mathbb{F}$, and the driving noise W — and the solution X must be built as a functional of that prescribed noise: at each time t, the value X_t is determined by the path of W up to time t and nothing else. A *weak solution*, by contrast, asks only that there *exist* some probability space carrying *some* pair (X, W) such that W is a Brownian motion and X solves (5.13); here the noise is an output of the construction, free to be chosen along with X, and only the *joint law* of the pair is pinned down. Every strong solution is a weak solution, but not conversely: there are equations (Tanaka's $\mathrm{d}X = \mathrm{sgn}(X)\ \mathrm{d}W$ is the standard witness) that have a weak solution with the right law and yet no solution adapted to any prescribed W.

We will always want strong solutions, and the reason is the one that organizes all of Part II. In a control problem the drift is not given to us; it is *chosen* by an optimizing agent as a feedback law $b(t, x) = b_0(t, x) + \alpha(t, x)$, a function of the current state that the agent computes from the information available so far. For such a law to be implementable the agent must be reacting to a noise that is given to her — revealed increment by increment as time runs — not co-designing the noise to suit her strategy. A weak solution, which is permitted to build a convenient W around the desired X, would let the "controller" clairvoyantly arrange the randomness; that is not a control problem at all. The strong formulation, with its fixed space and fixed W and its demand that X_t be a non-anticipating functional of the noise, is exactly the structure in which a feedback control — and the verification theorem of chapter 6 that certifies it optimal — makes sense. This is the same causality requirement that selected the Itô integral over Stratonovich in section 5.2, now reappearing one level up: bets are sized before the increment they ride on, and solutions are built from noise that is handed to us, not manufactured to taste. Existence and uniqueness of the strong solution hold under exactly the standing assumptions of the book, invoked here by their tags (A-Lip) and (A-Growth).

Theorem 5.10 (Well-posedness of SDEs)**.** *Let $b : [0,T] \times \mathbb{R}^d \to \mathbb{R}^d$ and $\sigma : [0,T] \times \mathbb{R}^d \to \mathbb{R}^{d\times d}$ be measurable and satisfy, uniformly in $t \in [0,T]$:*

- (A-Lip) *global Lipschitz in the state: there is K with $|b(t,x) - b(t,x')| + \|\sigma(t,x) - \sigma(t,x')\| \le K\,|x - x'|$;*
- (A-Growth) *linear growth: $|b(t,x)| + \|\sigma(t,x)\| \le K\,(1 + |x|)$.*

Let $\zeta \in L^2(\Omega, \mathcal{F}_0, \mathbb{P})$ be independent of W. Then (5.12) has a strong solution X, it is pathwise unique (any two solutions agree a.s. for all t), and it satisfies the moment bound

$$\mathbb{E}\Big[\sup_{0\le t\le T} |X_t|^2\Big] \le C\big(1 + \mathbb{E}\,|\zeta|^2\big)e^{CT} \tag{5.14}$$

for a constant $C = C(K,T,d)$.

Proof. We give the argument in full rather than as a sketch, because its skeleton — recast the equation as a fixed point, iterate, and force the iterates to be Cauchy by a single contraction estimate closed with Grönwall — is the exact template reused for the McKean–Vlasov equation of chapter 7 and the forward–backward systems of chapter 9. Every later existence theorem in the book is this proof with more moving parts; it repays seeing once, slowly.

The arena: a Banach space of processes. A fixed-point argument needs a complete metric space on which the map whose fixed point we seek is a contraction. The natural arena here is the space $\mathcal{S}^2$ of continuous, $\mathbb{F}$-adapted processes X on $[0,T]$ for which the norm

$$\|X\|^2_{\mathcal{S}^2} \;:=\; \mathbb{E}\Big[\sup_{0\le t\le T} |X_t|^2\Big]$$

is finite. This is not a casual choice. The supremum *inside* the expectation is what will let us control the entire path at once, and it is precisely the quantity Doob's maximal inequality is built to bound; the square and the expectation make $\mathcal{S}^2$ a Hilbert-like L^2 space of processes, complete because $L^2(\Omega; C[0,T])$ is complete. Completeness is the property we are buying: a Cauchy sequence of iterates will have a limit *in this same space*, automatically a continuous adapted process, with no separate regularity argument needed. Define the map Φ on $\mathcal{S}^2$ by

$$(\Phi X)_t \;=\; \zeta + \int_0^t b(s, X_s)\;\mathrm{d}s + \int_0^t \sigma(s, X_s)\;\mathrm{d}W_s;$$

a solution of (5.13) is exactly a fixed point $\Phi X = X$. That Φ maps $\mathcal{S}^2$ into itself — that ΦX is again square-integrable with a finite sup-norm — is the role of the linear-growth bound (A-Growth): it keeps the coefficients from inflating a bounded input into an unbounded output, and it is the same estimate that, applied to X itself, will produce the moment bound (5.14) at the end.

The iteration. Start from the constant process $X^0_t \equiv \zeta$ and define

$$X^{n+1}_t \;=\; (\Phi X^n)_t \;=\; \zeta + \int_0^t b(s, X^n_s)\;\mathrm{d}s + \int_0^t \sigma(s, X^n_s)\;\mathrm{d}W_s. \tag{5.15}$$

Each X^{n+1} is continuous (both integrals are continuous in their upper limit), adapted (the integrands depend only on the past), and in $\mathcal{S}^2$ (by the growth bound and induction). The plan is to show the consecutive differences $X^{n+1} - X^n$ shrink so fast that the series $\sum_n (X^{n+1} - X^n)$ converges in $\mathcal{S}^2$; its partial sums are the iterates, so the iterates converge.

The contraction estimate, ingredient by ingredient. Subtract two consecutive instances of (5.15). The initial datum ζ cancels, leaving a drift difference and a diffusion difference:

$$X_t^{n+1} - X_t^n = \underbrace{\int_0^t \left(b(s, X_s^n) - b(s, X_s^{n-1})\right) \mathrm{d}s}_{\text{drift part}} + \underbrace{\int_0^t \left(\sigma(s, X_s^n) - \sigma(s, X_s^{n-1})\right) \mathrm{d}W_s}_{\text{diffusion part}}.$$

We want to bound $\mathbb{E} \sup_{s \le t} \left|X_s^{n+1} - X_s^n\right|^2$, and four elementary moves do it, each pulling its weight.

(i) Split the two parts. The square of a sum is controlled by twice the sum of squares, $|a+b|^2 \le 2|a|^2 + 2|b|^2$ — the cheapest way to decouple the drift and diffusion contributions so each can be estimated by the tool suited to it:

$$\sup_{s \le t} \left|X_s^{n+1} - X_s^n\right|^2 \le 2 \sup_{s \le t} \left| \int_0^s (b(r, X_r^n) - b(r, X_r^{n-1})) \,\mathrm{d}r \right|^2 + 2 \sup_{s \le t} \left| \int_0^s (\sigma(r, X_r^n) - \sigma(r, X_r^{n-1})) \,\mathrm{d}W_r \right|^2.$$

(ii) Cauchy–Schwarz on the drift. The drift integral is an ordinary integral against $\mathrm{d}r$, and Cauchy–Schwarz (in the form $|\int_0^s g \,\mathrm{d}r|^2 \le s \int_0^s g^2 \,\mathrm{d}r \le T \int_0^t g^2 \,\mathrm{d}r$) trades the integral of a difference for the integral of its square, at the cost of a factor T:

$$\sup_{s \le t} \left| \int_0^s (b(r, X_r^n) - b(r, X_r^{n-1})) \,\mathrm{d}r \right|^2 \le T \int_0^t \left| b(r, X_r^n) - b(r, X_r^{n-1}) \right|^2 \mathrm{d}r.$$

(iii) Doob on the diffusion. The diffusion integral is an Itô integral, and the supremum over s of its square is exactly what Doob's L^2 maximal inequality (theorem C.2) controls. Combined with the Itô isometry (theorem 5.7(1)) it gives the maximal form

$$\mathbb{E} \sup_{s \le t} \left| \int_0^s H_r \,\mathrm{d}W_r \right|^2 \;\le\; 4\, \mathbb{E} \int_0^t H_r^2 \,\mathrm{d}r,$$

the bound (C.4) — the factor 4 is Doob's constant $(p/(p-1))^2$ at $p = 2$. Applied to $H_r = \sigma(r, X_r^n) - \sigma(r, X_r^{n-1})$, this again converts a martingale into the $\mathrm{d}r$-integral of a squared difference, which is the form we want. This step is the whole reason we worked in $\mathcal{S}^2$ with the supremum inside the expectation: only the maximal inequality lets us put the $\sup_s$ *inside* and still close.

(iv) (A-Lip) closes the loop. Both surviving integrands are now squared differences of a coefficient evaluated at X_r^n versus X_r^{n-1}, and this is exactly what the Lipschitz hypothesis was made to bound: (A-Lip) gives $|b(r,x) - b(r,x')|^2 \le K^2 |x - x'|^2$ and likewise for σ, so each integrand is at most $K^2 |X_r^n - X_r^{n-1}|^2$. Without (A-Lip) the right-hand side would still involve the coefficients, and the inequality could not be iterated; with it, the estimate becomes a self-referential bound on the differences alone. Taking expectations and collecting the constants from (i)–(iv) into a single $C_K = C_K(K, T, d)$,

$$\mathbb{E} \sup_{s \le t} \left|X_s^{n+1} - X_s^n\right|^2 \;\le\; C_K \int_0^t \mathbb{E} \sup_{r \le s} \left|X_r^n - X_r^{n-1}\right|^2 \mathrm{d}s, \tag{5.16}$$

where the inner $\sup_{r\le s}$ is inserted by monotonicity (the integrand at time s is at most its running supremum). This is the contraction estimate: the size of the $(n+1)$th difference at time t is controlled by the time-integral of the nth.

Summing the series. Write $g_n(t) = \mathbb{E}\sup_{s\le t}\left|X_s^n - X_s^{n-1}\right|^2$, so that (5.16) reads $g_{n+1}(t) \le C_K \int_0^t g_n(s)\,\mathrm{d}s$. Starting from a finite base bound $g_1(t) \le C_0$ (the first difference $X^1 - X^0 = \Phi\zeta - \zeta$ is in $\mathcal{S}^2$ by the growth bound), feeding the inequality into itself n times integrates the constant against itself n times:

$$g_{n+1}(t) \le C_0\,\frac{(C_K t)^n}{n!}, \qquad \text{hence} \qquad \left\|X^{n+1} - X^n\right\|_{\mathcal{S}^2}^2 = g_{n+1}(T) \le C_0\,\frac{(C_K T)^n}{n!}.$$

This is the payoff of carrying the time-integral rather than a bare constant: the repeated integration manufactures the factorial $n!$ in the denominator, and the resulting series $\sum_n \left(C_0 (C_K T)^n/n!\right)^{1/2}$ converges (it is dominated by $\sum_n (C_K T)^{n/2}/\sqrt{n!}$, summable since the factorial outruns any geometric). A geometric bound alone would have given convergence only for T small; the factorial gives it for *every* T, which is why no smallness or restart is needed. By the triangle inequality the partial sums $X^N = \zeta + \sum_{n=0}^{N-1}(X^{n+1} - X^n)$ form a Cauchy sequence in the complete space $\mathcal{S}^2$, so they converge to a limit $X \in \mathcal{S}^2$. Passing to the limit in (5.15) — legitimate because the drift and diffusion integral maps are continuous on $\mathcal{S}^2$, the diffusion map again by the Itô isometry — shows $\Phi X = X$: the limit solves (5.13).

Uniqueness, by Grönwall. The same contraction estimate, now applied to two genuine solutions X, Y rather than consecutive iterates, gives for $\phi(t) := \mathbb{E}\sup_{s\le t}|X_s - Y_s|^2$ the inequality $\phi(t) \le C_K \int_0^t \phi(s)\,\mathrm{d}s$. This is the integral inequality Grönwall's lemma is built to kill: with zero additive part, lemma C.1 (the case $A = 0$) forces $\phi \equiv 0$, so $X_t = Y_t$ for all t almost surely, and by continuity the two paths coincide identically — pathwise uniqueness. Note that this half of the proof uses only (A-Lip), never the growth bound: Lipschitz continuity alone guarantees that solutions cannot separate.

The moment bound, again by Grönwall. Finally, run the very same four ingredients on X against the zero process instead of on a difference of iterates. Now it is (A-Growth) that supplies the pointwise bound $|b(s, X_s)|^2 + \|\sigma(s, X_s)\|^2 \le 2K^2(1 + |X_s|^2)$, and steps (i)–(iii) yield

$$\mathbb{E}\sup_{s\le t}|X_s|^2 \;\le\; C\big(1 + \mathbb{E}\,|\zeta|^2\,\big) + C\int_0^t \mathbb{E}\sup_{r\le s}|X_r|^2\,\mathrm{d}s.$$

This is Grönwall with a nonzero additive part $A = C(1+\mathbb{E}\,|\zeta|^2)$; lemma C.1 then gives $\mathbb{E}\sup_{s\le t}|X_s|^2 \le C(1 + \mathbb{E}\,|\zeta|^2)\,e^{Ct}$, which at $t = T$ is exactly the asserted moment bound (5.14). The exponential is the signature of Grönwall: linear-growth coefficients can amplify the second moment, but only at a bounded exponential rate, never fast enough to escape to infinity in finite time. That finiteness is the analytic content of "no explosion," and the next remark reads it as physics. (For attribution, the classical reference for this theorem is [106, Thm. 5.2.1].) □

Remark 5.11 (The rigorous Langevin equation)**.** We can now redeem the promissory note of the chapter preamble. Equation (5.12) *is* the Langevin equation, no longer a formal expression to be handled with care but a theorem with a solution. The physicist's $\dot{x} = -V'(x) + \sqrt{2D}\,\xi$ is the special case $b(x) = -V'(x)$, $\sigma \equiv \sqrt{2D}$, read in integral (Itô) form

$$\mathrm{d}X_t = -V'(X_t)\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}W_t,$$

and theorem 5.10 is the precise statement that, whenever $-V'$ is Lipschitz with linear growth, this equation has a unique strong solution on all of $[0,T]$. The white noise ξ that had no values and the derivative $\dot{x}$ that did not exist have both been replaced by the one object the chapter built — the Itô integral against W — and the nonsense is gone.

Two points deserve to be drawn out, one a reassurance and one the genuine content of the hypotheses. First the reassurance: because the noise here is *additive* (σ a constant $\sqrt{2D}$, independent of the state), the Itô–Stratonovich question that section 5.2.2 warned about simply does not arise. The correction term that distinguishes the two calculi is proportional to $\sigma\,\partial_x\sigma$, which vanishes identically when σ is constant; the Itô and Stratonovich readings of an additive-noise equation coincide term for term. This is exactly why the physicist's manipulations of the overdamped Langevin equation were never wrong despite never naming a convention, and it is why the entire LQG thread (Thread A), whose noise is additive throughout, is immune to the subtlety: one may compute as the physicist always did and the rigorous theory ratifies it verbatim. The ambiguity returns only when the volatility is allowed to depend on the state, which is precisely where Part II will have to be careful.

Now the content of the hypotheses, which is best seen by asking what breaks without them. The Lipschitz hypothesis (A-Lip) is what buys *uniqueness* — it is the only thing the uniqueness half of the proof used — and a force $-V'$ that is merely locally Lipschitz still gives a unique solution, but possibly only up to a random *explosion time* $\tau < \infty$ at which $|X_t| \to \infty$. The growth bound (A-Growth) is what rules the explosion out, and the cleanest way to see why is to delete the noise entirely and watch the deterministic flow. Take the one-dimensional outward drift $b(x) = x^2$ — equivalently the anti-confining potential $V = -x^3/3$, a hill that steepens without bound as the particle runs downhill. The noiseless equation $\dot{x} = x^2$ separates, $\int x^{-2}\,\mathrm{d}x = \int \mathrm{d}t$, giving $x(t) = x_0/(1 - x_0 t)$: starting from any $x_0 > 0$ the solution reaches $+\infty$ at the finite time $t^* = 1/x_0$ and ceases to exist. No noise is needed for this catastrophe; a drift that grows faster than linearly *outward* drives the state to infinity in finite time on its own, and adding noise only jostles the timing. (A-Growth)'s linear bound $|b(x)| \le K(1 + |x|)$ forbids exactly this runaway: the worst the drift can do is amplify $|X|$ exponentially, which is the e^{CT} of the moment bound (5.14), and an exponentially growing quantity is finite at every finite time. The growth bound is the analytic guarantee of *no explosion*.

But — and this is the point most easily mistaken — (A-Growth) is *sufficient*, not *necessary*. It forbids *all* superlinear drifts, inward and outward alike, whereas only the outward ones cause trouble. A confining superlinear potential such as $V(x) = x^4$ has inward drift $b = -V' = -4x^3$, which violates the linear-growth bound by a wide margin and yet produces a perfectly global, non-exploding, pathwise-unique diffusion. The reason is the sign: a strong *restoring* force pulls a wandering particle ever more firmly back toward the origin, the steeper the potential the firmer the leash, so far from escaping to infinity the process is confined and in fact relaxes to the Boltzmann invariant density $\propto e^{-2V/\sigma^2}$ — the stationary state the physicist expects, and the one section 5.7 computes explicitly for the harmonic case. The honest statement is therefore that (A-Growth) is a convenient uniform sufficient condition that captures the well-posed regime by a single clean inequality, at the cost of excluding some confining models that are perfectly well-behaved for the opposite reason; the book adopts it because a uniform hypothesis keeps every downstream estimate clean, and notes here that the genuinely dangerous case is outward superlinear growth alone.

One further hypothesis, not needed for existence but decisive for everything the population sections will do, concerns the *nondegeneracy* of the noise. If the diffusion is uniformly nondegenerate — assumption (A-Nondeg), $\sigma\sigma^\top \succeq \lambda I$ with $\lambda > 0$, meaning the noise pushes in every direction with at least a floor of intensity — then the qualitative behaviour of the law of X_t changes completely. The well-posedness theorem already gives us the process X and hence, at each time, its law $m_t = \mathcal{L}(X_t)$ as a point in $\mathcal{P}_2(\mathbb{R}^d)$. Nondegeneracy upgrades that law from an abstract measure to a smooth object: for every $t > 0$ the measure m_t has a strictly positive density that is smooth in both the state and the time, the transition density of theorem C.12. The mechanism is parabolic regularization — the same smoothing that the heat equation performs on rough initial data, here driven by the Laplacian that the nondegenerate noise contributes to the generator — and it is what allows the Fokker–Planck equation of the next section to be read classically, as an honest PDE for a smooth density, rather than only in a weak or distributional sense. This is no idle refinement: (A-Nondeg) is the source of the regularity on which the existence theory of the coupled mean field game systems in chapters 8 and 10 rests, and we invoke it there by name and flag carefully the degenerate situations where it must be dropped.

This completes the passage from the formal Langevin equation to a rigorous diffusion. We now hold three objects the next section needs and will run with: the well-posed strong solution X itself, a single trajectory driven by the canonical noise; its law $m_t = \mathcal{L}(X_t)$, the evolving point in $\mathcal{P}_2(\mathbb{R}^d)$ that the Fokker–Planck equation will track; and, under (A-Nondeg), the smooth positive transition density that lets that evolution be written as a classical PDE. Section 5.5 takes exactly these and asks the population-level question the chapter has been building toward — not how a single particle moves, but how the law of the whole ensemble moves — and the bridge it uses is the tool we spent this section deploying: Itô's formula, with its martingale part averaged out, turning the trajectory-level dynamics of X into a deterministic evolution equation for m_t.

5.5 The generator, Fokker–Planck, and its backward twin

The previous section completed the passage from the formal Langevin equation to a rigorous diffusion and handed us three objects to run with: the well-posed strong solution X, a single trajectory driven by the canonical noise; its law $m_t = \mathcal{L}(X_t)$, the evolving point in $\mathcal{P}_2(\mathbb{R}^d)$; and, under (A-Nondeg), the smooth positive transition density that lets that evolution be read as a classical PDE. We now spend them. The whole chapter so far has tracked *one* trajectory; we change the question. We no longer ask how a single particle moves but how the *law of the entire ensemble* moves — not X_t but m_t. This is the population-level question the chapter has been building toward, and it is the question mean field games are made of: an agent is one trajectory, but the mean field that couples the agents is the law m_t, and the equation that propagates that law is the forward half of the MFG system. The bridge from the one to the many is the tool we spent the last section deploying — Itô's formula, with its martingale part averaged out — and the operator that bridge isolates is the *generator*. This section extracts the generator, proves the one identity (Dynkin's) that turns trajectory dynamics into an evolution law for m_t, and then forks: the adjoint of the generator drives the density forward (Fokker–Planck), the generator itself drives observables backward (Kolmogorov). Those two forks are the two subsections that follow.

Both a value and a density now enter the same chapter, so we fix the orientation convention once, here, on first joint use, as appendix A prescribes and as we will obey for the rest of the book: *value functions are solved backward, carrying terminal data at $t = T$ down to the present; densities evolve forward, carrying initial data at $t = 0$ up to the future.* The density $m_t = \mathcal{L}(X_t)$ of this section is the forward object par excellence, launched from m_0 and pushed ahead in time; the value $u(t, x)$ of section 5.5.3 is the backward object, pinned by a terminal payoff and pulled back. The two flow in opposite directions and meet in the middle: that opposition is the geometry of the MFG system, and it is no accident that the forward equation will be driven by $\mathcal{A}^*$ and the backward one by $\mathcal{A}$ — the generator and its adjoint are the two faces of a single integration-by-parts identity, and naming that identity now is precisely what makes the next two subsections read as two readings of one fact rather than two unrelated derivations.

So fix, for this section, the time-homogeneous SDE $\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}W_t$ under theorem 5.10, and let $m_t = \mathcal{L}(X_t) \in \mathcal{P}(\mathbb{R}^d)$ be the law of the state at time t, with $m_0 = \mathcal{L}(\zeta)$. The generator is the operator that answers the population question infinitesimally. Before the formula, the idea: probe the ensemble with a smooth test observable ϕ and watch its average $\mathbb{E}\,\phi(X_t)$. At $t = 0$ this is $\int \phi\,\mathrm{d}m_0$; an instant later it is $\int \phi\,\mathrm{d}m_h$; the *rate* at which it changes is the infinitesimal motion of the law, read through the observable ϕ. Write $P_h\phi(x) = \mathbb{E}\big[\phi(X_h) \mid X_0 = x\big]$ for the average of ϕ over the cloud of places the particle has reached after time h, started from x. As $h \downarrow 0$ the cloud collapses back to the point x, so $P_h\phi \to \phi$; the generator is the *first-order* content of that collapse, the limit

$$(\mathcal{A}\phi)(x) \;=\; \lim_{h\downarrow 0} \frac{P_h\phi(x) - \phi(x)}{h} \;=\; \frac{\mathrm{d}}{\mathrm{d}h}\Big|_{h=0} \mathbb{E}\big[\phi(X_h) \mid X_0 = x\big],$$

the instantaneous rate of change of the expected observable per unit time, started from x. This limit is exactly the drift term that survived in Itô's formula once the martingale part was averaged away — the second-order term the whole chapter has been chasing, now wearing the clothes of a differential operator. To see which operator, run the mnemonic of section 5.3: expand $\phi(X_h)$ to first order in $\mathrm{d}t$ and second order in $\mathrm{d}W$, and collapse $(\mathrm{d}W)^2 \to \mathrm{d}t$ by the multiplication table. The first-order drift contributes $b\cdot\nabla\phi\,\mathrm{d}t$; the surviving second-order term contributes $\frac{1}{2}\sum_{ij} a_{ij}\partial_{ij}\phi\,\mathrm{d}t$ with $a = \sigma\sigma^\top$; the martingale increment averages to zero under the conditional expectation. What is left, divided by h and sent to $h \downarrow 0$, is precisely the following operator. We elevate it to a definition.

Definition 5.12 (Generator)**.** The *infinitesimal generator* of the diffusion X is the second-order operator, acting on $\phi \in C_c^2(\mathbb{R}^d)$,

$$(\mathcal{A}\phi)(x) \;=\; b(x)\cdot\nabla\phi(x) + \tfrac{1}{2}\,\mathrm{tr}\big(\sigma(x)\sigma(x)^\top\,\mathrm{D}^2\,\phi(x)\big) \;=\; \sum_i b_i\,\partial_i\phi + \tfrac{1}{2}\sum_{i,j} a_{ij}\,\partial_{ij}\phi, \tag{5.17}$$

where $a(x) = \sigma(x)\sigma(x)^\top$ is the *diffusion matrix.* This is the convention fixed in appendix A.

Two features of the symbol and the formula deserve a word before we use them. The symbol itself is a deliberate choice: we write the generator $\mathcal{A}$ (rendering $\mathcal{A}$), not the more common L, because

L would collide with the law operator $\mathcal{L} = \mathcal{L}$ that is everywhere in this book; appendix A fixes $\mathcal{A}$ for the generator and $\mathcal{A}^*$ for its adjoint, and we hold to it. As for the formula, read its two pieces against the SDE that produced them. The drift b enters *linearly*, through the first-order operator $b\cdot\nabla$, the transport term: it tilts the average of ϕ in the direction the deterministic flow is carrying the particle. The volatility σ enters *quadratically*, and only through the combination $a = \sigma\sigma^\top$, the diffusion matrix — never through σ alone. This is the signature of the Itô second-order term: the noise is felt by expectations only through the variance it injects, and variance is the *square* of the volatility, exactly the $(\mathrm{d}W)^2 = \mathrm{d}t$ that has driven the chapter. That a particle and its mirror image $\sigma \mapsto -\sigma$ generate the identical diffusion is the differential-operator shadow of the fact that W and $-W$ have the same law. The half in front of the second-order term is the same $\frac{1}{2}$ that sits in Itô's formula, inherited verbatim. In the scalar case the operator is just $\mathcal{A}\phi = b\,\phi' + \frac{1}{2}\sigma^2\phi''$, and the smallest instance is the most telling: with $b = 0$ and σ constant it is $\frac{1}{2}\sigma^2\partial_{xx}$, half the Laplacian scaled by the diffusion constant — the heat-equation operator. The generator of pure Brownian motion *is* the heat semigroup's generator, and the appearance of the Laplacian here is the promised reincarnation of $(\mathrm{d}W)^2 = \mathrm{d}t$ as a differential operator.

It is worth instantiating (5.17) on the very observable that opened the discussion of the second-order term. Take the scalar pure-noise process $\mathrm{d}X = \sigma\ \mathrm{d}W$ with σ constant and the observable $\phi(x) = x^2$. Then $\phi' = 2x$, $\phi'' = 2$, and the generator returns $\mathcal{A}\phi = \frac{1}{2}\sigma^2 \cdot 2 = \sigma^2$, a constant. Read through the limit definition, this says $\frac{\mathrm{d}}{\mathrm{d}h}\big|_0 \mathbb{E}[X_h^2 \mid X_0 = x] = \sigma^2$: the expected square grows at rate σ^2 from every starting point, which is exactly the $\mathbb{E}X_t^2 = \mathbb{E}X_0^2 + \sigma^2 t$ we computed by hand in section 5.3. The generator has simply packaged that computation as an operator, and the diffusion rate σ^2 that there signalled the spreading of the packet is here the value of $\mathcal{A}$ on x^2. Same fact, same constant, one level more abstract.

With the generator named, the identity that turns it into an evolution law for m_t is one line of Itô plus one expectation. The generator was *defined* as the time derivative of the expected observable started from a fixed point; Dynkin's formula is the global, integrated version of that statement along the actual flow. Apply the multidimensional Itô formula (5.11) to $\phi(X_t)$ — with no explicit t-dependence, so $\partial_t\phi = 0$ — and the differential of $\phi(X_t)$ splits into exactly the drift and martingale pieces the generator was built to separate.

Proposition 5.13 (Dynkin's formula)**.** *For $\phi \in C_c^2(\mathbb{R}^d)$, $\mathrm{d}\,\phi(X_t) = (\mathcal{A}\phi)(X_t)\ \mathrm{d}t + (\nabla\phi)^\top\sigma\ \mathrm{d}W_t$, and consequently*

$$\mathbb{E}\,\phi(X_t) = \mathbb{E}\,\phi(X_0) + \mathbb{E}\int_0^t (\mathcal{A}\phi)(X_s)\ \mathrm{d}s, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\,\mathbb{E}\,\phi(X_t) = \mathbb{E}\,(\mathcal{A}\phi)(X_t) = \int_{\mathbb{R}^d} (\mathcal{A}\phi)\ \mathrm{d}m_t. \tag{5.18}$$

Proof. The differential statement is Itô's formula (5.11) read off the generator. Applied to $\phi(X_t)$, that formula produces a first-order drift term, the full contraction of the Hessian against $\mathrm{d}X\ \mathrm{d}X^\top = a\ \mathrm{d}t$, and the gradient paired against the noise:

$$\mathrm{d}\,\phi(X_t) = \nabla\phi(X_t)^\top\,\mathrm{d}X_t + \tfrac{1}{2}\ \mathrm{tr}\big(\sigma\sigma^\top\,\mathrm{D}^2\,\phi(X_t)\big)\,\mathrm{d}t = \underbrace{\Big(b\cdot\nabla\phi + \tfrac{1}{2}\,\mathrm{tr}(a\ \mathrm{D}^2\,\phi)\Big)}_{=(\mathcal{A}\phi)(X_t)}\,\mathrm{d}t + (\nabla\phi)^\top\sigma\ \mathrm{d}W_t,$$

where in the last step the $b\ \mathrm{d}t$ part of $\mathrm{d}X$ combined with the Hessian term to assemble exactly the generator (5.17), and the $\sigma\ \mathrm{d}W$ part of $\mathrm{d}X$ left the stated martingale increment. Integrating from 0 to t,

$$\phi(X_t) = \phi(X_0) + \int_0^t (\mathcal{A}\phi)(X_s)\ \mathrm{d}s + \int_0^t (\nabla\phi)^\top \sigma\ \mathrm{d}W_s.$$

Now take expectations, and this is the step that does the work. The final term is an Itô integral of the integrand $\nabla\phi(X_s)^\top\sigma(X_s)$; this integrand is bounded — both $\nabla\phi$ (compactly supported, $\phi \in C_c^2$) and σ on the support of ϕ are bounded — hence square-integrable on $[0,t]$, so by the Itô isometry (theorem 5.7) the integral is a genuine martingale started at 0, and its expectation is therefore 0. The martingale part is precisely what the average annihilates; this is the analytic content of "averaging out the noise," and it is the same fact that made the second moment computation of section 5.3 collapse to its drift. What survives is

$$\mathbb{E}\,\phi(X_t) = \mathbb{E}\,\phi(X_0) + \mathbb{E}\int_0^t (\mathcal{A}\phi)(X_s)\ \mathrm{d}s,$$

the first equality in (5.18); the boundedness of $\mathcal{A}\phi$ (again $\phi \in C_c^2$ with bounded coefficients on its support) lets Fubini exchange the expectation and the time integral. Differentiating in t by the fundamental theorem of calculus gives $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\,\phi(X_t) = \mathbb{E}\,(\mathcal{A}\phi)(X_t)$, and rewriting the expectation as integration against the law $m_t = \mathcal{L}(X_t)$ — which is all $\mathbb{E}\,(\mathcal{A}\phi)(X_t) = \int_{\mathbb{R}^d}(\mathcal{A}\phi)\ \mathrm{d}m_t$ says — finishes the proposition. □

The compact support of ϕ is not decoration: it is what lets us drop the Itô integral (boundedness, hence the martingale's zero mean), invoke Fubini, and, in the next subsection, integrate by parts with no boundary terms. A test class with this much room to spare is exactly the right register in which to state the identity; one extends to wider classes of ϕ afterward by approximation, never by weakening the test class in the derivation itself.

Read the last equality of (5.18) as the equation of motion of the law, written weakly. It says: for *every* test observable ϕ, the rate of change of its average against m_t equals the average of $\mathcal{A}\phi$ against m_t,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} \phi\ \mathrm{d}m_t = \int_{\mathbb{R}^d} (\mathcal{A}\phi)\ \mathrm{d}m_t.$$

This single identity is the hinge of the entire section, and the two subsections that follow are nothing but its two readings. The generator $\mathcal{A}$ acts on the test function ϕ; but the unknown we want an equation *for* is the measure m_t, not the test function. To get there we must move $\mathcal{A}$ off ϕ and onto m_t — that is, integrate by parts — and the operator that results, the one for which $\int(\mathcal{A}\phi)\ \mathrm{d}m = \int \phi\,(\mathcal{A}^*m)$ holds for all ϕ, is by definition the formal adjoint $\mathcal{A}^*$. The very same identity, left as it stands with $\mathcal{A}$ acting on ϕ, is instead the seed of the backward equation, where ϕ is promoted to an evolving observable and $\mathcal{A}$ stays put. So the fork is precise: section 5.5.1 takes (5.18) and integrates by parts, transferring all derivatives from ϕ onto the density to produce $\mathcal{A}^*m_t$ and the forward Fokker–Planck equation $\partial_t m = \mathcal{A}^*m$; section 5.5.3 keeps the generator where it is and lets it drive observables backward through $\mathcal{A}u$. Two faces of one integration by parts, the adjoint relation $\int(\mathcal{A}\phi)\,m = \int \phi\,(\mathcal{A}^*m)$ being the mirror between them. We turn first to the forward face.

5.5.1 Fokker–Planck as the adjoint of the generator

The previous subsection ended at the weak equation of motion of the law, the last equality of Dynkin's identity (5.18),

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d}\phi\ \mathrm{d}m_t = \int_{\mathbb{R}^d}(\mathcal{A}\phi)\ \mathrm{d}m_t, \qquad \text{for every } \phi \in C_c^2(\mathbb{R}^d),$$

and posed the structural complaint that makes this subsection necessary: the generator $\mathcal{A}$ is sitting on the *test function* ϕ, whereas the object we actually want a closed evolution law for is the *measure* m_t. As long as the derivatives in $\mathcal{A}\phi$ fall on ϕ, the identity is a statement about averages of observables, not an equation for the density. The remedy is the oldest move in the analysis of weak formulations: integrate by parts and transfer every derivative off ϕ and onto m_t. What comes back is a differential operator acting on the density — the adjoint $\mathcal{A}^*$ — and with it the forward Fokker–Planck equation. Assume henceforth that m_t has a density, written $m_t(x)$ by abuse of notation, so that $\mathrm{d}m_t = m_t(x)\ \mathrm{d}x$ and the pairings become honest Lebesgue integrals against $\mathrm{d}x$; the existence of that density for $t > 0$ is part of the theorem below, and at the level of distributions the computation goes through verbatim with m_t read as a measure.

Carry out the integration by parts term by term, because the two pieces of the generator (5.17) integrate by parts a *different number of times* and it is exactly the bookkeeping of how many derivatives land where that defines $\mathcal{A}^*$. Write $\mathcal{A}\phi = \sum_i b_i\,\partial_i\phi + \frac{1}{2}\sum_{ij} a_{ij}\,\partial_{ij}\phi$ and pair against m_t. Throughout, every boundary term produced by integration by parts vanishes: $\phi \in C_c^2(\mathbb{R}^d)$ has compact support, so ϕ and all its derivatives are identically zero outside a bounded set, and the boundary integrals at infinity have nothing to evaluate. This is the single place the compact support of the test class is spent, and it is why proposition 5.13 was careful to keep ϕ in C_c^2 rather than a wider class.

The drift term, integrated by parts once. The first-order piece carries one derivative on ϕ, so one integration by parts moves it onto m_t. For each i,

$$\int_{\mathbb{R}^d} b_i\,(\partial_i\phi)\,m_t\ \mathrm{d}x = -\int_{\mathbb{R}^d}\phi\,\partial_i\big(b_i\,m_t\big)\ \mathrm{d}x,$$

the sign flipping with the single integration by parts and the boundary term gone by compact support. Summing on i assembles a divergence,

$$\sum_i\int_{\mathbb{R}^d} b_i\,(\partial_i\phi)\,m_t\ \mathrm{d}x = -\int_{\mathbb{R}^d}\phi\sum_i\partial_i\big(b_i\,m_t\big)\ \mathrm{d}x = -\int_{\mathbb{R}^d}\phi\ \mathrm{div}\big(b\,m_t\big)\ \mathrm{d}x.$$

The transport operator $b\cdot\nabla$ has thus become, on the density side, the *negative divergence* $-\operatorname{div}(b\,\cdot)$ — one derivative, one sign change.

The diffusion term, integrated by parts twice. The second-order piece carries two derivatives on ϕ, so it takes two integrations by parts to clear them, and each one contributes a sign. Fix i, j and move the two derivatives across one at a time. The first, in the variable x_i, sends $\partial_i(\partial_j\phi)$ onto $a_{ij}m_t$ with a sign:

$$\int_{\mathbb{R}^d} a_{ij}\,(\partial_i\partial_j\phi)\,m_t\ \mathrm{d}x = -\int_{\mathbb{R}^d}(\partial_j\phi)\,\partial_i\big(a_{ij}\,m_t\big)\ \mathrm{d}x.$$

The second, in the variable x_j, moves the remaining ∂_j off ϕ, flipping the sign back to plus:

$$-\int_{\mathbb{R}^d} (\partial_j \phi)\, \partial_i(a_{ij}\, m_t)\ \mathrm{d}x = +\int_{\mathbb{R}^d} \phi\, \partial_j \partial_i(a_{ij}\, m_t)\ \mathrm{d}x = \int_{\mathbb{R}^d} \phi\, \partial_{ij}(a_{ij}\, m_t)\ \mathrm{d}x.$$

Two integrations by parts, two sign flips, net plus — so the second-order operator returns to the density side with its sign *unchanged*, in contrast to the drift. Restoring the factor $\frac{1}{2}$ and the sum over i, j,

$$\tfrac{1}{2} \sum_{ij} \int_{\mathbb{R}^d} a_{ij}\, (\partial_{ij}\phi)\, m_t\ \mathrm{d}x = \tfrac{1}{2} \sum_{ij} \int_{\mathbb{R}^d} \phi\, \partial_{ij}(a_{ij}\, m_t)\ \mathrm{d}x.$$

Assembling the adjoint. Adding the two computations collects every derivative onto m_t and leaves ϕ bare under the integral:

$$\int_{\mathbb{R}^d} (\mathcal{A}\phi)\, m_t\ \mathrm{d}x = \int_{\mathbb{R}^d} \phi \Big(-\mathrm{div}(b\, m_t) + \tfrac{1}{2} \sum_{ij} \partial_{ij}(a_{ij}\, m_t) \Big)\, \mathrm{d}x \ :=\ \int_{\mathbb{R}^d} \phi\, (\mathcal{A}^* m_t)\ \mathrm{d}x. \tag{5.19}$$

The bracketed operator is, by this very identity, the *formal* $L^2(\mathrm{d}x)$*-adjoint* $\mathcal{A}^*$ of the generator: writing $\langle f, g \rangle = \int_{\mathbb{R}^d} fg\ \mathrm{d}x$ for the (unweighted) $L^2(\mathrm{d}x)$ pairing, (5.19) is exactly $\langle \mathcal{A}\phi, m_t \rangle = \langle \phi, \mathcal{A}^* m_t \rangle$, which is the defining relation of an adjoint. The qualifier "formal" is doing honest work and should be read literally: we have defined $\mathcal{A}^*$ only as a differential operator, by where the derivatives land when one integrates by parts against the flat Lebesgue measure $\mathrm{d}x$, and we have done so on the convenient pairing of a smooth compactly supported ϕ with a smooth enough m_t. We have *not* specified the domains, the function spaces, or the self-adjointness questions that would make $\mathcal{A}^*$ a genuine Hilbert-space adjoint of a densely defined operator; those analytic questions are real but separate, and the differential identity (5.19) is all the forward equation needs. Concretely, in one space dimension with $a = \sigma^2$ the recipe reads off at a glance: the generator $\mathcal{A}\phi = b\,\phi' + \frac{1}{2}\sigma^2\phi''$ has adjoint

$$\mathcal{A}^* m = -(b\, m)' + \tfrac{1}{2}(\sigma^2 m)'',$$

one prime with a minus from the transport term, two primes with a plus from the diffusion term — the scalar shadow of the general $\mathcal{A}^*$ in (5.19).

Now the logical step that converts a pairing identity into a partial differential equation, and it is worth stating slowly because the whole derivation hinges on the quantifier. Insert (5.19) into the Dynkin equation of motion: for every $\phi \in C_c^2(\mathbb{R}^d)$,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^d} \phi\, m_t\ \mathrm{d}x = \int_{\mathbb{R}^d} (\mathcal{A}\phi)\, m_t\ \mathrm{d}x = \int_{\mathbb{R}^d} \phi\, (\mathcal{A}^* m_t)\ \mathrm{d}x,$$

and exchanging the time derivative with the spatial integral on the left (justified, as in proposition 5.13, by the boundedness afforded by compact support) turns this into $\int \phi\, \partial_t m_t\ \mathrm{d}x = \int \phi\, (\mathcal{A}^* m_t)\ \mathrm{d}x$, that is

$$\int_{\mathbb{R}^d} \phi\, (\partial_t m_t - \mathcal{A}^* m_t)\ \mathrm{d}x = 0 \qquad \text{for } \textit{every } \phi \in C_c^2(\mathbb{R}^d).$$

A function (or distribution) that integrates to zero against *every* compactly supported test function must itself be zero — this is the fundamental lemma of the calculus of variations, the same du Bois-Reymond argument used throughout weak formulations. Hence the bracket vanishes and the density satisfies $\partial_t m_t = \mathcal{A}^* m_t$ pointwise (in the distributional sense, for now). The clause "holds for every test ϕ" is precisely what upgrades a family of scalar identities, one per observable, into a single equation for the unknown m_t: it is because the relation is forced for *all* ϕ that ϕ can be cancelled and an equation in m_t alone survives. This is the abstract derivation of Fokker–Planck promised in the section opening, obtained with no probabilistic input beyond Dynkin's formula and no analysis beyond integration by parts.

Theorem 5.14 (Fokker–Planck / Kolmogorov forward equation)**.** *Under theorem 5.10 and (A-Nondeg), the laws $m_t = \mathcal{L}(X_t)$ have densities for $t > 0$ and solve, in the classical sense for $t > 0$ and in the distributional sense for $t \geq 0$, the* Fokker–Planck equation

$$\partial_t m \;=\; \mathcal{A}^* m \;=\; -\operatorname{div}(b\, m) + \tfrac{1}{2}\sum_{i,j} \partial_{ij}(a_{ij}\, m), \qquad m\big|_{t=0} = m_0, \tag{5.20}$$

where $\mathcal{A}^*$ is the formal $L^2(\mathrm{d}x)$-adjoint of the generator $\mathcal{A}$ and $a = \sigma\sigma^\top$.

Proof sketch. The two regularity claims in the statement are won by two completely different mechanisms, and separating them is the whole point of the sketch. The *distributional* form is already in hand and costs nothing beyond what precedes the theorem: (5.19) is an algebraic consequence of integration by parts and holds for all $t \geq 0$ and all $\phi \in C_c^2(\mathbb{R}^d)$ with no regularity of m_t assumed, since every derivative in it acts on m_t only *inside* the integral, against a smooth ϕ — that is exactly what "distributional solution" means. So $\partial_t m = \mathcal{A}^* m$ holds weakly directly from Dynkin's formula (5.18), with no appeal to (A-Nondeg) at all. This is worth emphasising because it locates precisely what the nondegeneracy hypothesis is and is not for: it is *not* needed to write the equation, only to know that its weak solution is a genuine smooth function.

That upgrade — from distribution to classical, smooth, strictly positive density — is where (A-Nondeg) earns its place, and the reason is structural. The assumption $a = \sigma\sigma^\top \succeq \lambda I$ with $\lambda > 0$ is exactly the statement that the second-order operator $\mathcal{A}^*$ is *uniformly elliptic*: its principal symbol $\frac{1}{2}\sum_{ij} a_{ij}\xi_i\xi_j \;\geq\; \frac{\lambda}{2}\,|\xi|^2$ is bounded below by a positive multiple of $|\xi|^2$ in every direction, so the diffusion genuinely spreads mass in all d directions and no direction is starved of noise. Appending the time derivative ∂_t to a uniformly elliptic spatial operator makes (5.20) a *uniformly parabolic* equation, and uniform parabolicity is precisely the hypothesis under which the linear theory delivers a smooth, strictly positive fundamental solution. That fundamental solution is nothing other than the transition density of X — the kernel $p(s, x; t, y)$ giving the probability of moving from x at time s to a neighbourhood of y at time t — and $m_t(y) = \int p(0, x; t, y)\, m_0(\mathrm{d}x)$ inherits its smoothness and positivity for every $t > 0$. We invoke this as the parabolic-regularity theorem theorem C.12(1),(3): part (1) furnishes the smooth positive fundamental solution of the uniformly parabolic equation, and part (3) identifies it as the transition density solving (5.20) classically for $t > 0$. The role of the cited theorem in the argument is thus sharply circumscribed — it supplies smoothness and nothing else — so we defer its proof to appendix C without losing the thread here.

The reason the hypothesis cannot simply be dropped is instructive, and it is the same fact viewed from the failure side. If a is allowed to degenerate — to lose rank in some direction at some point — ellipticity fails, the equation is merely *hypo*elliptic at best, and the smoothing the theorem provides can be lost: the law may concentrate on a lower dimensional set and fail to have a density at all (a deterministic flow, $\sigma \equiv 0$, keeps a point mass a point mass forever), or have one that is not smooth. Smoothness of m_t is not a generic gift of the dynamics; it is bought specifically by the noise reaching every direction, which is what uniform ellipticity guarantees. The degenerate case is not hopeless — Hörmander's bracket conditions can restore smoothness when the noise plus the drift together span all directions — but it lies outside (A-Nondeg) and we flag it only where it is needed later. (For attribution and the physics-facing treatment of the Fokker–Planck equation see [106, Ch. 8] and [115, 109].) $\square$

It is worth recording what (5.20) is, in the arc of the book, now that it stands derived. The operator $\mathcal{A}^*$ inherited the surviving second-order term of Itô's formula — the diffusion piece $\frac{1}{2}\sum_{ij}\partial_{ij}(a_{ij}\,\cdot)$ that is $(\mathrm{d}W)^2 = \mathrm{d}t$ promoted first to a Laplacian inside the generator (5.17) and now to its adjoint — so the Fokker–Planck equation is the chapter's organizing fact carried one final step, from a property of a single path to the law of motion of the entire ensemble's density. And that density evolves *forward*: (5.20) is posed with initial data m_0 at $t = 0$ and propagated to $t > 0$, in keeping with the global convention (appendix A) that densities flow forward while value functions flow backward. This is the forward half of the mean field game system; its backward twin, driven by $\mathcal{A}$ rather than $\mathcal{A}^*$, is built in section 5.5.3, and the Hamilton–Jacobi–Bellman equation of chapter 6 is its nonlinear descendant.

We have, however, paid for the abstract derivation in transparency. The adjoint form (5.20), with its second derivatives $\partial_{ij}(a_{ij}m)$ landing on the product $a_{ij}m$, is correct but opaque: one cannot read off from it the conservation of probability that physically must hold, nor see the flux of mass that is doing the transporting. That information is not lost — it is merely folded into the two-derivative diffusion term. The next subsection unfolds it, by distributing exactly one of those two derivatives back out to expose the probability current J, and rewriting the same $\mathcal{A}^*m$ in the manifestly conservation-law form $\partial_t m + \operatorname{div} J = 0$. The two subsections are deliberately the same equation read two ways: the adjoint reading we have just completed, and the continuity reading to come.

5.5.2 Fokker–Planck as a continuity equation

The previous subsection left us holding the equation $\partial_t m = \mathcal{A}^*m$ in the adjoint form (5.20), and noted in closing that this form, while correct, hides the two facts a physicist most wants to see: that probability is conserved, and that it is conserved because mass is *transported* by a current. The information is not absent — it is folded into the two-derivative diffusion term $\frac{1}{2}\sum_{ij}\partial_{ij}(a_{ij}m)$, where both derivatives sit outside the product $a_{ij}m$. The remedy is a single, almost trivial algebraic move: distribute exactly *one* of those two derivatives back out, leaving the other in place. This is the same equation read a second way, and the reading is the one the forward equation of every mean field game will wear.

Carry out the move explicitly. The second-order operator is a double divergence, and we peel off the outer one. Writing the ith component of the inner vector field as $\big(\operatorname{div}(a\,m)\big)_i = \sum_j \partial_j(a_{ij}m)$, the diffusion term is

$$\frac{1}{2}\sum_{i,j}\partial_{ij}(a_{ij}m) = \frac{1}{2}\sum_i \partial_i\Big(\sum_j \partial_j(a_{ij}m)\Big) = \sum_i \partial_i\Big(\tfrac{1}{2}\big(\operatorname{div}(a\,m)\big)_i\Big) = \operatorname{div}\Big(\tfrac{1}{2}\operatorname{div}(a\,m)\Big).$$

Every step here is just bookkeeping on the index i: the operator $\sum_{ij} \partial_{ij}$ is $\sum_i \partial_i\big(\sum_j \partial_j\big)$, so factoring out the outer ∂_i exhibits the whole term as the divergence of the vector field whose ith entry is $\frac{1}{2}\sum_j \partial_j(a_{ij}m)$. The drift term in (5.20) is already a divergence, $-\operatorname{div}(b\,m)$, so collecting the two and pulling the common div to the front turns the entire right-hand side $\mathcal{A}^* m$ into a single divergence, $-\operatorname{div} J$. Reading (5.20) as $\partial_t m = -\operatorname{div} J$ is precisely the continuity equation

$$\partial_t m + \operatorname{div} J = 0, \qquad J(x,t) \;=\; b(x)\,m(x,t) \;-\; \tfrac{1}{2}\,\operatorname{div}\big(a(x)\,m(x,t)\big), \tag{5.21}$$

where the vector field J is the *probability current*.

The current has exactly the two pieces the two terms of $\mathcal{A}^*$ contributed, and each has a name and a physical reading. The first, $b\,m$, is the *drift current* (or convective current): it is the density m carried bodily along the deterministic velocity field b, the same flux $\rho\mathbf{v}$ that appears when a fluid of density ρ is swept along at velocity $\mathbf{v}$. Where the drift points, mass is advected. The second, $-\frac{1}{2}\operatorname{div}(a\,m)$, is the *diffusion current*, and it is a Fick-type flux: in the constant-coefficient case $a = \sigma^2 I$ it reduces to $-\frac{\sigma^2}{2}\nabla m$, a flux proportional to *minus* the gradient of the density, i.e. mass flows from where there is more of it to where there is less. This is the gradient term of Itô's formula $(\mathrm{d}W)^2 = \mathrm{d}t$ wearing its final costume: the surviving second-order piece, which became a Laplacian in the generator and a double divergence in the adjoint, is here a downhill flux of probability. The drift current can push mass uphill against the density gradient; the diffusion current always relaxes it downhill; and the competition between the two is what selects the stationary profile we extract below.

Now the conservation reading, which is the entire payoff of having written the right-hand side as $-\operatorname{div} J$. Locally, $\partial_t m + \operatorname{div} J = 0$ says that the only way the density at a point can change is by a net imbalance of current flowing in and out of an infinitesimal neighbourhood of it: probability is neither created nor destroyed, only relocated. To see the content rather than assert it, integrate the local law over any fixed region $\Omega \subseteq \mathbb{R}^d$ with nice boundary and apply the divergence theorem to the flux term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega m(x,t)\;\mathrm{d}x = -\int_\Omega \operatorname{div} J\;\mathrm{d}x = -\int_{\partial\Omega} J\cdot\mathbf{n}\;\mathrm{d}S.$$

The rate of change of the mass inside Ω equals minus the total flux of J through the boundary $\partial\Omega$ — mass leaves Ω only by crossing its surface, at the rate the current carries it across. Nothing wells up in the interior. Taking $\Omega = \mathbb{R}^d$, the boundary recedes to infinity, and because X_t has finite second moment the density and its current decay there fast enough that the surface integral vanishes; hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} m(x,t)\;\mathrm{d}x = 0, \qquad \text{so} \qquad \int_{\mathbb{R}^d} m(x,t)\;\mathrm{d}x = \int_{\mathbb{R}^d} m_0(x)\;\mathrm{d}x = 1 \quad \text{for all } t. \tag{5.22}$$

Total probability is preserved *exactly*, for all time, as a law of motion for a genuine law must guarantee. This is not a regularity bonus or an approximate statement; it is built into the divergence structure of $\mathcal{A}^*$ and would survive even if the solution were only distributional. The continuity form is what makes it visible in one line, where the adjoint form (5.20) concealed it inside two derivatives.

Physics note

Equation (5.21) is *the* reason the Fokker–Planck equation feels familiar to a physicist: it is the continuity equation $\partial_t \rho + \operatorname{div} \mathbf{j} = 0$ for a probability "fluid" of density m, with a current J that splits into a convective part $b\,m$ and a diffusive (Fick's-law) part $-\frac{1}{2}\operatorname{div}(a\,m)$, exactly as the particle-number current of a drifting, diffusing gas does. Reading the equation this way lets us classify its *steady states* cleanly, and the classification turns on a distinction that is easy to state and easy to get wrong.

A stationary density m_∞ is one with $\partial_t m_\infty = 0$, which by (5.21) means

$$\operatorname{div} J_\infty = 0 :$$

the steady current is *divergence-free*. This is weaker than the current vanishing. There is a stronger condition,

$$J_\infty \equiv 0 \quad \text{everywhere},$$

called *zero current* or *detailed balance*: not only is there no net accumulation anywhere, there is no flow at all. Detailed balance is the hallmark of a *reversible* steady state — run the dynamics backward in time and the statistics are unchanged — and it is strictly more than stationarity. The gap between the two is the space of divergence-free currents that do not vanish: a stationary state can support steady *circulating* currents, loops of probability flux that close on themselves so that $\operatorname{div} J_\infty = 0$ while $J_\infty \neq 0$, like a fluid stirred into a permanent eddy that transports nothing on net but never comes to rest.

When are the two conditions the same? Precisely when the drift is a gradient. Write $a = \sigma^2 I$ for clarity (the scalar diffusion case). The zero-current equation reads $b\,m_\infty = \frac{\sigma^2}{2}\nabla m_\infty$, i.e. $b = \frac{\sigma^2}{2}\nabla \log m_\infty$ wherever $m_\infty > 0$ — so b would have to be a gradient, the gradient of $\frac{\sigma^2}{2}\log m_\infty$. If the drift is *not* a gradient, no choice of m_∞ can make J_∞ vanish identically, and the stationary state, if one exists, must carry circulating current. Conversely, if $b = -\nabla V$ is a gradient field one can solve $J_\infty = 0$ outright and detailed balance holds. Two regimes bracket this:

- In *one dimension* a drift is *always* a gradient — any scalar $b(x)$ is the derivative of its own antiderivative $\int^x b$ — so stationarity and zero current coincide automatically, and every 1D steady state is reversible. There is no room for a circulating loop on the line.

- In *higher dimensions* a non-gradient drift (one with nonzero curl, $\partial_i b_j \neq \partial_j b_i$) generically forces $J_\infty \neq 0$: the steady state is a non-equilibrium one, sustained by persistent probability circulation. This is the mathematical home of nonequilibrium steady states, and it is invisible in 1D.

The gradient case is the one with a closed-form steady density, and it is worth integrating the zero-current equation in full because the same calculation reappears in section 5.7. In one dimension with constant $a = \sigma^2$, zero current is the *first-order* ODE

$$J_\infty(x) = b(x)\,m_\infty(x) - \frac{\sigma^2}{2}\,m_\infty'(x) = 0,$$

a separable equation. Dividing by m_∞ (positive) gives $\dfrac{m_\infty'}{m_\infty} = \dfrac{2}{\sigma^2}\,b$, that is $\dfrac{\mathrm{d}}{\mathrm{d}x}\log m_\infty = \dfrac{2}{\sigma^2}b$, and integrating from a reference point yields the *Boltzmann–Gibbs form*

$$m_\infty(x) \;\propto\; \exp\Big(\frac{2}{\sigma^2}\int^x b(y)\,\mathrm{d}y\Big),$$

the proportionality constant fixed afterward by $\int m_\infty = 1$. Specialize to a gradient drift $b = -V'$, where V is a confining potential. Then $\int^x b = -V(x)$ up to a constant absorbed into normalization, and the steady density collapses to the Gibbs weight

$$m_\infty(x) \;\propto\; \exp\Big(-\frac{2V(x)}{\sigma^2}\Big).$$

This is the equilibrium of the overdamped Langevin dynamics $\mathrm{d}X_t = -V'(X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, and it is exactly the Boltzmann distribution $e^{-V/k_\mathrm{B}\tau}$ of statistical mechanics once one reads off the dictionary. Comparing exponents, $\dfrac{2V}{\sigma^2} = \dfrac{V}{k_\mathrm{B}\tau_\mathrm{temp}}$, which fixes

$$\frac{\sigma^2}{2} \;=\; D \;=\; k_\mathrm{B}\,\tau_\mathrm{temp}\;:$$

half the squared volatility is the diffusion constant D, and that diffusion constant is the temperature in energy units — the fluctuation–dissipation reading already flagged for the canonical noise in section 5.1. The noise strength σ and the potential V between them set the equilibrium; the bath is universal and only the coefficients carry the physics. We run this computation on the explicit harmonic potential $V(x) = \frac{1}{2}\theta x^2$ in section 5.7, where $J_\infty = 0$ returns the Gaussian invariant law of the Ornstein–Uhlenbeck process.

Finally, the current J is not a parochial feature of this chapter; it has the last word in Part II. The forward equation of the mean field game system (chapter 8) is *exactly* (5.21), with one change: the drift b is no longer given but is replaced by the agents' *optimal feedback*, the control each agent selects by solving the backward problem. The population is then transported by the current generated by its own best response, and the fixed point that closes the MFG loop is the demand that this self-generated current carry the density to the very distribution the agents assumed when optimizing.

This continuity reading hands two things forward. To the Ornstein–Uhlenbeck computation of section 5.7 it hands the steady-state machinery just assembled: the worked model extracts its invariant density not by solving the full time-dependent equation but by setting the current to zero, $J_\infty = 0$, and integrating the resulting first-order ODE exactly as above. And to the next subsection it hands the structural reading itself. We have now pushed densities *forward* in two equivalent forms — as the adjoint $\partial_t m = \mathcal{A}^* m$ and as the conservation law $\partial_t m + \operatorname{div} J = 0$ — and established that this is the forward half of the mean field game system. The matching backward half remains: instead of pushing densities forward under $\mathcal{A}^*$, one pushes *observables* backward under $\mathcal{A}$ itself. That is the Kolmogorov backward equation, and it is where we turn now.

5.5.3 The Kolmogorov backward equation

The forward equation propagates densities from initial data; its twin propagates *observables* from terminal data, backward in time, and is governed by $\mathcal{A}$ itself rather than its adjoint. Fix a terminal payoff ψ and define $u(t,x) = \mathbb{E}[\psi(X_T) \mid X_t = x]$, the expected terminal observable given the present state. By the Markov property of the diffusion (proposition C.11, which makes the conditional expectation a deterministic function of the present state alone) and Dynkin's formula, u solves a backward parabolic equation.

Theorem 5.15 (Kolmogorov backward equation)**.** *Let $\psi \in C_b^2(\mathbb{R}^d)$ and, under theorem 5.10, set $u(t,x) = \mathbb{E}[\psi(X_T) \mid X_t = x]$ for the time-homogeneous diffusion. Then $u \in C^{1,2}$ and*

$$\partial_t u + \mathcal{A}u = 0 \quad \text{on } [0,T) \times \mathbb{R}^d, \qquad u(T,\cdot) = \psi. \tag{5.23}$$

Proof sketch. By the Markov property (proposition C.11), the tower property, and time-homogeneity, $u(t,X_t) = \mathbb{E}[\psi(X_T) \mid \mathcal{F}_t]$ is a martingale. Apply Itô's formula (5.11) to $u(t,X_t)$; the differential splits into a drift part $(\partial_t u + \mathcal{A}u)(t,X_t)\,\mathrm{d}t$ and a martingale part. The drift part is itself a continuous finite-variation process, so $u(t,X_t)$ minus its martingale part is a continuous finite-variation martingale, which must be constant (lemma C.5); hence the drift vanishes, $\partial_t u + \mathcal{A}u = 0$, with the terminal value read off at $t = T$. Smoothness of u for $t < T$ is parabolic regularity under (A-Nondeg), theorem C.12(2) (attribution: [106, Ch. 8]). □

Notation watch

Two equations, two directions, one structural lesson. The forward equation (5.20) carries densities m from $t = 0$ *forward* and is driven by $\mathcal{A}^*$; the backward equation (5.23) carries values u from $t = T$ *backward* and is driven by $\mathcal{A}$. This is precisely the time-orientation convention of appendix A: **densities flow forward, values flow backward.** The Hamilton–Jacobi–Bellman equation of chapter 6 is the *nonlinear* descendant of (5.23) — replace the passive observable ψ by an optimized cost-to-go and the linear $\mathcal{A}$ by a Hamiltonian $\sup_\alpha$ over controls, and (5.23) becomes HJB. Reading the linear backward equation now is how we pre-load that intuition: there is nothing mysterious about "integrating backward from a terminal condition" — it is just conditioning on the present and looking at the future payoff. We deliberately write this passive linear object with the same symbol u that definition 6.3 reserves for the optimized value function: the backward solution here is the linear precursor that optimization will promote into the value (the passive observable ψ becoming an optimized cost-to-go, the linear $\mathcal{A}$ becoming a $\sup_\alpha$ Hamiltonian), and we announce the overload rather than spring it. The mean field game system is then exactly the pairing of the forward (5.20) with a backward HJB, coupled through m.

5.6 The Feynman–Kac formula

The backward equation of section 5.5.3 already gave one stochastic representation of a PDE solution: its value $u(t,x) = \mathbb{E}[\psi(X_T) \mid X_t = x]$ solved $\partial_t u + \mathcal{A}u = 0$ by averaging a terminal payoff along the diffusion. Feynman–Kac is that same bridge widened. It generalizes the backward equation by allowing a running cost and a discounting/killing rate, and is the precise correspondence between expectations of functionals of a diffusion and solutions of linear parabolic PDEs. It is the linear, exactly-solvable prototype of every "stochastic representation of a PDE solution" in the book.

Theorem 5.16 (Feynman–Kac)**.** *Let X solve $\mathrm{d}X_s = b(X_s)\,\mathrm{d}s + \sigma(X_s)\,\mathrm{d}W_s$ under theorem 5.10, with generator $\mathcal{A}$. Let $c \in C(\mathbb{R}^d)$ be bounded below, $f \in C(\mathbb{R}^d)$ and $\psi \in C(\mathbb{R}^d)$ of at most polynomial growth. Suppose $u \in C^{1,2}([0,T]\times\mathbb{R}^d)$, of at most polynomial growth in x uniformly on $[0,T]$, solves the terminal-value problem*

$$\partial_t u + \mathcal{A}u - c(x)\,u + f(x) = 0 \quad \text{on } [0,T)\times\mathbb{R}^d, \qquad u(T,\cdot) = \psi. \tag{5.24}$$

Then u has the stochastic representation

$$u(t,x) = \mathbb{E}\left[\psi(X_T)\,e^{-\int_t^T c(X_r)\,\mathrm{d}r} + \int_t^T f(X_s)\,e^{-\int_t^s c(X_r)\,\mathrm{d}r}\,\mathrm{d}s \;\Big|\; X_t = x\right]. \tag{5.25}$$

In particular such a u is unique. (When $c \equiv 0$, $f \equiv 0$ this is exactly the backward equation theorem 5.15.)

Proof. Fix (t,x) and abbreviate the discount factor $D_s = e^{-\int_t^s c(X_r)\,\mathrm{d}r}$, so $\mathrm{d}D_s = -c(X_s)D_s\,\mathrm{d}s$. Consider the process $Y_s = D_s\,u(s,X_s) + \int_t^s f(X_r)D_r\,\mathrm{d}r$ on $s \in [t,T]$. By the product rule and Itô's formula (5.11),

$$\mathrm{d}Y_s = D_s\Big(\partial_t u + \mathcal{A}u - c\,u + f\Big)(s,X_s)\,\mathrm{d}s + D_s(\nabla u)^\top \sigma\,\mathrm{d}W_s.$$

The drift bracket vanishes by the PDE (5.24), so Y is a local martingale; the growth hypotheses on u, f, c together with the moment bound (5.14) make it a true martingale on $[t,T]$. Therefore $\mathbb{E}[Y_T \mid X_t = x] = Y_t = u(t,x)$. Writing out Y_T with $u(T,X_T) = \psi(X_T)$ and $D_t = 1$ gives exactly (5.25). ∎

The formula reads PDE-solving as path-averaging: to evaluate u at (t,x), launch the diffusion from x, accumulate the source f and the terminal payoff ψ along each path weighted by the survival factor $e^{-\int c}$, and average. A physicist recognizes the structure of a (Euclidean, imaginary-time) path integral: (5.25) is $\langle x \mid e^{-(T-t)\hat{H}} \mid \cdot\rangle$ with $\hat{H} = -\mathcal{A} + c$ the "Hamiltonian," $\mathcal{A}$ the kinetic/diffusive part and c the potential, and the expectation over Brownian paths the Wiener-measure sum over histories. The two directions of solvability — run the PDE backward, or run the paths forward and average — are the PDE and probabilistic faces of one object, a duality that recurs as the analytic (chapter 8) versus probabilistic (chapter 9) formulations of the mean field game system.

5.7 Worked computation: the Ornstein–Uhlenbeck process

We close by carrying out every step of the chapter on the one model where all of it is explicit, and which is the linear backbone of the LQG thread (Thread A). Consider the scalar linear SDE

$$\mathrm{d}X_t = -\theta\,X_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \theta > 0,\ \sigma > 0,\ X_0 \sim m_0, \tag{5.26}$$

the *Ornstein–Uhlenbeck* (OU) process: a particle in a harmonic well $V(x) = \frac{1}{2}\theta x^2$ under additive thermal noise, equivalently the overdamped Langevin dynamics of a spring. In the parameter names of Thread A (eq. (5.31) below) this is the linear state equation $\mathrm{d}X_t = (aX_t + b\alpha_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ with state-feedback parameter $a = -\theta < 0$ and the control switched off, $\alpha \equiv 0$ (here a is the scalar Thread-A state-feedback coefficient, not the diffusion matrix $a = \sigma\sigma^\top$ of definition 5.12; in this scalar additive-noise model the latter is simply the constant σ^2); it is exactly the closed-loop process that the optimal LQG feedback will produce in chapters 6 and 14, which is why we precompute its law, its steady state, and the spectrum of its generator here.

Worked computation 5.17 (Ornstein–Uhlenbeck: law, steady state, spectrum)**.**

(a) Explicit solution and Gaussian law. The coefficients are affine, so (A-Lip) and (A-Growth) hold globally and theorem 5.10 gives a unique solution. To solve it, apply Itô's formula (5.10) to $F(t,x) = e^{\theta t}x$, an integrating factor: $\partial_t F = \theta e^{\theta t}x$, $\partial_x F = e^{\theta t}$, $\partial_{xx}F = 0$, so

$$\mathrm{d}(e^{\theta t}X_t) = \theta e^{\theta t}X_t\,\mathrm{d}t + e^{\theta t}\,\mathrm{d}X_t = e^{\theta t}\sigma\,\mathrm{d}W_t$$

(the drift cancels exactly; there is no Itô correction since $\partial_{xx}F = 0$). Integrating,

$$X_t = e^{-\theta t}X_0 + \sigma\int_0^t e^{-\theta(t-s)}\,\mathrm{d}W_s. \tag{5.27}$$

The Itô integral of a deterministic integrand is a centered Gaussian (corollary C.15: it is the L^2-limit of linear combinations of independent Gaussian increments, and an L^2-limit of Gaussians is Gaussian), so conditionally on X_0 the law of X_t is Gaussian with

$$\mathbb{E}[X_t \mid X_0] = e^{-\theta t}X_0, \qquad \mathrm{Var}[X_t \mid X_0] = \sigma^2\int_0^t e^{-2\theta(t-s)}\,\mathrm{d}s = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t}),$$

where the variance used the Itô isometry theorem 5.7(1) to turn the stochastic integral's second moment into the deterministic time integral. If m_0 has mean μ_0 and variance v_0, then $m_t = \mathcal{L}(X_t)$ is Gaussian with

$$\mu_t = e^{-\theta t}\mu_0, \qquad v_t = e^{-2\theta t}v_0 + \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t}). \tag{5.28}$$

Both the mean and the deviation of the variance from its limit relax exponentially, the mean at rate θ and the variance at rate 2θ.

(b) Fokker–Planck and its steady state. The generator and its adjoint are, from (5.17),

$$\mathcal{A}\phi = -\theta x\,\phi' + \tfrac{\sigma^2}{2}\,\phi'', \qquad \mathcal{A}^* m = \partial_x(\theta x\,m) + \tfrac{\sigma^2}{2}\,\partial_{xx}m,$$

so the Fokker–Planck equation (5.20) is $\partial_t m = \partial_x(\theta x\,m) + \frac{\sigma^2}{2}\partial_{xx}m$. One checks directly that the Gaussian density with moments (5.28) solves it; we extract the steady state from the zero-current condition of (5.21). The current is $J = -\theta x\,m - \frac{\sigma^2}{2}\partial_x m$, and $J \equiv 0$ gives the separable ODE $\partial_x m/m = -2\theta x/\sigma^2$, hence

$$m_\infty(x) = \sqrt{\frac{\theta}{\pi\sigma^2}}\,\exp\Big(-\frac{\theta x^2}{\sigma^2}\Big) \;=\; N\Big(0,\,\frac{\sigma^2}{2\theta}\Big), \tag{5.29}$$

the centered Gaussian of variance $v_\infty = \sigma^2/2\theta$. This is the $t \to \infty$ limit of (5.28) for any initial law, and it is the fluctuation–dissipation balance: noise strength σ^2 pumping variance in, restoring force θ draining it out, equilibrium variance their ratio (up to the factor 2). With the dictionary $\sigma^2/2 = D$ this is $v_\infty = D/\theta$, the equipartition result $\frac{1}{2}\theta\langle x^2\rangle = \frac{1}{2}D$ for a harmonic mode.

(c) Spectral decomposition of the generator. On the weighted space $L^2(m_\infty)$ the generator $\mathcal{A}$ is symmetric and negative—it is the $L^2(m_\infty)$-symmetric Dirichlet form of the gradient drift, with $\langle \mathcal{A}\phi, \psi\rangle_{m_\infty} = -\frac{\sigma^2}{2}\langle \phi', \psi'\rangle_{m_\infty}$, the boundary terms killed by the Gaussian decay—and its spectrum is purely the point spectrum $\{-n\theta\}_{n\ge 0}$. (The unbounded $\mathcal{A}$ is *not* compact, so this is not the compact-operator spectral theorem of chapter 2; the rigorous diagonalization, including completeness of the eigenbasis, is appendix C.11, especially lemma C.16.) Its eigenfunctions are the Hermite polynomials H_n (H_n here being the Hermite polynomials, not the control Hamiltonian H of later chapters) orthogonal with respect to the Gaussian weight $m_\infty = N(0, v_\infty)$, and

$$\mathcal{A}\, H_n = -n\,\theta\, H_n, \qquad n = 0, 1, 2, \ldots, \tag{5.30}$$

so the eigenvalues are $\lambda_n = -n\theta$. Indeed $H_0 \equiv 1$ ($\lambda_0 = 0$, the stationary mode), $H_1(x) = x$ with $\mathcal{A}x = -\theta x$ ($\lambda_1 = -\theta$), and one verifies (5.30) inductively from the Hermite recurrence. The $\{H_n\}$ form an *orthogonal* basis of $L^2(m_\infty)$ (complete by lemma C.16; orthonormal only after dividing each by $\|H_n\|_{L^2(m_\infty)}$—with the normalization $H_1(x) = x$ one has $\|H_1\|_{L^2(m_\infty)} = \sqrt{v_\infty} \neq 1$), so any observable expands as $\phi = \sum_n \hat{\phi}_n H_n$ with $\hat{\phi}_n = \langle \phi, H_n\rangle_{m_\infty} / \|H_n\|^2_{L^2(m_\infty)}$ and evolves as

$$\mathbb{E}[\phi(X_t) \mid X_0 = x] = (e^{t\mathcal{A}}\phi)(x) = \sum_{n\ge 0} \hat{\phi}_n\, e^{-n\theta t}\, H_n(x),$$

a sum of exponentially relaxing modes. The *spectral gap* is $\lambda_0 - \lambda_1 = \theta$: every centered observable decays to its equilibrium average at least as fast as $e^{-\theta t}$, matching the relaxation rate of the mean in (5.28). The forward operator $\mathcal{A}^*$ on $L^2(\mathrm{d}x)$ has the same eigenvalues with eigenfunctions $m_\infty \cdot H_n$; the slowest non-trivial forward mode, $m_\infty H_1 \propto x\, m_\infty$, is the relaxation of the mean, again at rate θ.

Remark 5.18 (Hand-off to the LQG thread)**.** The Thread-A representative agent obeys

$$\mathrm{d}X_t = \big(aX_t + b\,\alpha_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad X_0 \sim m_0, \tag{5.31}$$

with the parameter names fixed in the thread specification (a state feedback, b control gain, σ noise). The OU computation above is the $\alpha \equiv 0$, $a < 0$ instance; in chapter 6 the optimal feedback turns out to be *affine* in the state, $\alpha^*(t, x) = -b\big(P(t)x + r(t)\big)$, where $P(t)$ is the value's curvature solving a scalar Riccati equation (6.24) and $r(t)$ is an offset driven by the crowd coupling (eq. (6.27)). In the *uncoupled* instance ($\bar{m} \equiv 0$, so $r \equiv 0$) this is the homogeneous-linear feedback $\alpha^*_t = -b\,P(t)X_t$ and the closed-loop dynamics $\mathrm{d}X_t = (a - b^2P(t))X_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ are again of OU type (time-inhomogeneous); a nonzero coupling adds the inhomogeneous drift $-b^2r(t)$ but keeps the drift *affine*, which is all that matters here. Either way the Gaussian-flow formulas (5.28)–(5.29) are exactly what makes the LQG mean field game closeable in chapter 14: an affine drift maps Gaussians to Gaussians, so a Gaussian initial law stays Gaussian and the population m_t is pinned down by its mean $\bar{m}_t$ (the μ_t of (5.28), read now as the population mean) and its variance alone. The additive noise also discharges the Itô–Stratonovich question (section 5.2.2) for the whole thread. Note that from Part III the driving noise is renamed B_t; here, in scope of no graphon, it is W_t.

Rigor ledger

Proved (in full or by complete sketch with pointer).

- Existence of Brownian motion (theorem 5.2); path roughness (theorem 5.3); quadratic variation $\langle W \rangle_t = t$ in L^2 and a.s. (theorem 5.4, full proof).
- The Itô isometry on simple processes (lemma 5.5, full proof) and the construction, isometry and martingale property of the Itô integral (theorem 5.7).
- Itô's formula (theorem 5.9); the Itô–Stratonovich conversion (5.6)–(5.7).
- SDE well-posedness under (A-Lip),(A-Growth) with the moment bound, by Picard iteration and Grönwall (theorem 5.10, mechanism in full).
- Dynkin's formula (proposition 5.13); the Fokker–Planck equation in both the adjoint form (theorem 5.14) and the continuity-equation form (5.21); the Kolmogorov backward equation (theorem 5.15); the Feynman–Kac formula (theorem 5.16, full martingale proof).
- The Ornstein–Uhlenbeck law, Fokker–Planck steady state, and generator spectrum (worked computation 5.17, fully explicit).

Assumed (named standing assumptions, invoked here).

- (A-Lip) and (A-Growth): used for SDE well-posedness and the moment bound; (A-Lip) for additive-noise Langevin is the no-explosion condition (remark 5.11).
- (A-Nondeg): used to pass from distributional to classical Fokker–Planck and backward equations (existence and smoothness of transition densities); flagged wherever invoked. The OU model satisfies it with $\sigma^2 > 0$.

Deferred (with pointer; each proved in-book in appendix C).

- The foundational facts the chapter body states but does not stop to prove are all proved in the stochastic-calculus appendix appendix C: Grönwall's inequality (lemma C.1); Doob's L^2 maximal inequality and the $p = 2$ Burkholder–Davis–Gundy bound (theorems C.2 and C.3) with localization to local martingales (remark C.4); the constancy of a continuous finite-variation martingale (lemma C.5); density of simple processes (lemma C.6); the Kolmogorov extension and Kolmogorov–Chentsov continuity theorems (theorems C.8 and C.9); the Markov property of SDE solutions (proposition C.11); the Wiener-integral Gaussianity (corollary C.15); and completeness of the Hermite basis with the OU spectral resolution (lemma C.16 and appendix C.11). Itô's formula itself is proved in full in appendix C.5.
- Parabolic regularity for (5.20) and (5.23) under (A-Nondeg)—existence and smoothness of the transition density—is theorem C.12 (attribution: [106, Ch. 8], [109, 115]).

- Paley–Wiener–Zygmund nowhere-differentiability (theorem 5.3(2)), which is not used downstream, is sketched in the body with the standard dyadic Borel–Cantelli argument cited for attribution to [86, 114].
- Girsanov's theorem and the martingale representation theorem (not used in this chapter; needed for the probabilistic/BSDE formulation of chapter 9): supplied where first needed, with appendix C and [86] as the home.

Open / beyond scope here.

- Degenerate ($\sigma\sigma^\top$ rank-deficient) diffusions: hypoellipticity (Hörmander conditions) replaces (A-Nondeg); treated only where flagged in later chapters.
- Well-posedness under merely measurable or only locally Lipschitz drift (weak solutions, Krylov–Veretennikov regularization): outside the standing assumptions and not used in the book's main line.
- The *nonlinear* Fokker–Planck equation, where the drift depends on the law m_t itself (McKean–Vlasov), which makes (5.20) nonlinear; built in chapter 7 and coupled to a backward HJB to form the MFG system in chapter 8.

Bibliographic notes

The roadmap text for this chapter is [106], Chapters 3–5, from which the SDE well-posedness theorem (theorem 5.10), Itô's formula, and the generator/Fokker–Planck material are adapted; the level of rigor and the proof of Itô's formula follow [106, Ch. 4–5, 8]. The definitive reference for the construction of Brownian motion, the Itô integral as an L^2-isometry, the martingale theory behind theorem 5.7, and the localization and Burkholder–Davis–Gundy estimates used in the well-posedness proof is [86]; [114] is the complementary continuous-martingale account, and is the cleanest source for the Lévy–Ciesielski construction and the fine path properties of theorem 5.3. The Itô–Stratonovich correction and its physical origin via the zero-correlation-time limit are due to [133]. For the Fokker–Planck equation read as a continuity equation, for the probability-current and detailed-balance viewpoint, and for the spectral theory of the Ornstein–Uhlenbeck generator (Hermite eigenfunctions, spectral gap, exponential relaxation) the physics-facing references are [115] and [109]; the latter develops the OU spectral decomposition of worked computation 5.17 in detail and connects it to the Langevin equation in the form used here. The backward equation and the Feynman–Kac formula are the linear ancestors of the Hamilton–Jacobi–Bellman theory of chapter 6 and of the analytic mean field game system of chapter 8; the forward Fokker–Planck equation, with the optimal feedback in place of a fixed drift, is its forward half. The nonlinear (McKean–Vlasov) Fokker–Planck equation and propagation of chaos, which turn this chapter's linear forward equation into the engine of the mean field limit, are taken up in chapter 7 following [40].

Chapter 6

Optimal Control: From Least Action to Hamilton–Jacobi–Bellman

The previous two chapters built the two halves of the language we now fuse. From chapter 4 we take the metric space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ of probability measures with finite second moment — the eventual home of the "mean field" — and the flat and Lions derivatives on it. From chapter 5 we take Itô calculus, the generator $\mathcal{A}$ of a diffusion, its formal adjoint $\mathcal{A}^*$ driving the Fokker–Planck equation, and the Feynman–Kac bridge between expectations of diffusion functionals and linear parabolic equations. This chapter adds the missing ingredient of a *game*: an agent who does not merely diffuse but *chooses*. We study how a single agent optimizes, postponing all interaction. Here m enters only as a frozen *cost-coupling parameter*, an inert measure argument of the running and terminal costs. There are two distinct ways the population will later cease to be inert, and they belong to different chapters: chapter 7 switches on the *descriptive* law-dependence of the drift (a particle's coefficients depend on the law of the ensemble, but no one optimizes), while chapter 8 switches on the *strategic* cost-coupling proper, promoting the frozen m here into the unknown of a Nash fixed point. This chapter sets up exactly the object chapter 8 will close.

The reader trained in physics arrives with a powerful and, for this subject, partly misleading reflex: the principle of least action. Classical trajectories extremize an action functional, and one solves for them by the Euler–Lagrange equations, a *boundary-value problem* with data clamped at both ends of the time interval. Optimal control begins in exactly this language — section 6.1 is a calculus-of-variations recap in the reader's own dialect — but then executes a pivot that must be internalized slowly. Dynamic programming computes not a single optimal path but a *field*, the value function $u(t, x)$, the optimal cost-to-go from every state at every time; and it computes this field by integrating *backward* from a terminal condition. The Euler–Lagrange/least-action reflex must be retrained into the Hamilton–Jacobi–Bellman (HJB) reflex. Because this retraining is, in the author's experience, the single largest conceptual hurdle of Part I, section 6.3 gives the backward direction of time a full conceptual treatment rather than a one-line aside.

It is worth making the contrast vivid before any machinery starts, because everything in this chapter is organized around it. Picture first the physicist's object: a single optimal trajectory, one curve γ^* leaving a fixed start x_0, pinned into existence by extremizing an action and traced out forward in time like a self-contained worldline. Now picture the dynamic programmer's object: not a curve at all, but an entire landscape $u(t, x)$ that assigns to every state x and every instant t the least cost still payable from there — a field defined over the whole state space at once, and built not by launching forward from a start but by discharging a terminal obligation *backward* from $t = T$. The first object answers "what is the single best path from here?"; the second answers "what is the best achievable cost from *anywhere*, and hence what should I do next from wherever I happen to find myself?" These are not two solutions to one problem; they are answers to two different questions, and the chapter's central demand on the reader is to stop reaching for the first picture and learn to inhabit the second — because only the second survives the introduction of noise, and, in chapter 8, of a reacting population that an open-loop plan could never anticipate.

Here is the route. Section 6.1 recalls the calculus of variations and isolates the distinction between *open-loop* and *feedback* controls. Section 6.2 states and sketches the dynamic programming principle and defines the value function. Section 6.3 derives the HJB equation, dwells on the terminal condition and the backward integration, and proves a verification theorem. Section 6.4 develops the Pontryagin maximum principle, identifies the costate with the gradient of the value, and previews its stochastic incarnation as a backward stochastic differential equation (BSDE), the object that will return in chapter 9. Section 6.5 installs the diffusion term, writes the second-order HJB equation in the control Hamiltonian $H(x, p, m)$, and reads off the optimal feedback $\alpha^* = -\partial_p H$. Section 6.6 introduces viscosity solutions, the correct weak notion when the value function develops kinks. Section 6.7 solves the linear–quadratic–Gaussian (LQG) single-agent problem to its scalar Riccati equation, launching Thread A — the explicit backbone carried through the whole book.

Throughout, we hold to the global time convention recorded in appendix A, and it pays to see why the convention is load-bearing rather than cosmetic. The value function $u = u(t, x)$ carries its data at the *terminal* time $t = T$ — where, with no time left and no decision to make, the optimal cost is forced to be the terminal cost itself — and is computed *backward* toward the present. In this chapter no population density evolves yet, but the moment one does (from chapter 7 onward) it carries *initial* data at $t = 0$ and runs *forward*, because a density is pushed from a known present into an as-yet-unknown future. These two opposed directions are not a clash to be reconciled away: when chapter 8 couples them, the forward density and the backward value are welded at opposite ends of $[0, T]$, and that very opposition is what makes a mean field game a boundary-value problem in time rather than an initial-value one. Mistaking which end carries which datum does not merely flip a sign — it silently replaces a well-posed problem with an ill-posed one — which is why we fix the convention now and never relax it. We are in Part I, no graphon is in scope, and Brownian motion is accordingly written W following appendix A; the symbol W is reserved for the graphon kernel that enters only from Part III, and keeping W for Brownian motion here forestalls a collision that would otherwise have to be untangled later.

6.1 Calculus of variations and the two ways to specify a control

We begin where the physicist already lives, and we begin there on purpose. The abstract apparatus assembled in the preamble — the Wasserstein geometry of chapter 4, the Itô calculus and generators of chapter 5 — stays inert until it is tied to an object the reader already trusts completely, and that object is the action functional. Before an agent can play a game it must first be able to optimize, and the physicist's native instrument for optimization is the principle of least action. So we lay that instrument out in full, not because any of it is unfamiliar, but precisely so that we can watch, over the rest of the chapter, exactly which part of it has to be torn out and replaced. The reflex we are about to exhibit — find a single extremal curve by solving a boundary-value problem with data clamped at the two ends of time — is correct and powerful; it is also the very thing the dynamic-programming pivot will have to retrain.

Fix a horizon $T \in (0, \infty)$, a terminal cost $g \colon \mathbb{R}^d \to \mathbb{R}$, and a running cost — a *Lagrangian* — $L \colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$, written $L(x, v)$ with v the velocity slot. For an absolutely continuous curve $\gamma \colon [0, T] \to \mathbb{R}^d$ with $\gamma(0) = x_0$ fixed, define the action

$$\mathcal{J}[\gamma] \;=\; \int_0^T L(\gamma(t), \dot\gamma(t))\, \mathrm{d}t \;+\; g(\gamma(T)). \tag{6.1}$$

The classical problem is to minimize $\mathcal{J}$ over such curves.

To do so we make the one move that locates any stationary point: perturb the candidate and demand that the first-order response vanish. Replace a candidate γ^* by a nearby competitor $\gamma^* + \epsilon\eta$, where η is a fixed perturbation *shape* and ϵ a scalar that dials its amplitude, and require the action to be stationary in ϵ at $\epsilon = 0$. The only delicate point is which perturbations η count as admissible, and it is not a technicality but a modelling decision with visible consequences. The left endpoint is *clamped*: the agent genuinely starts at x_0, so every competitor must start there too, which forces $\eta(0) = 0$. The right endpoint is *free*: we did not pin down where the curve must end — instead we attached a price $g(\gamma(T))$ to wherever it does end — so $\eta(T)$ is unconstrained, and the optimizer is permitted to choose its own terminal position. That asymmetry between the two ends is the entire origin of the boundary term we are about to meet, and it is the first appearance of a pattern — a datum imposed at one end of time, a cost discharged at the other — that will organize this whole chapter. The following is the standard first-variation computation; we record it to fix notation and signs.

Theorem 6.1 (Euler–Lagrange equation and transversality)**.** *Let $L \in C^2(\mathbb{R}^d \times \mathbb{R}^d)$ and $g \in C^1(\mathbb{R}^d)$, and let γ^* be a minimizer of* (6.1) *over C^1 curves with $\gamma(0) = x_0$ and free right endpoint. Then on $[0, T]$,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\, \partial_v L(\gamma^*(t), \dot\gamma^*(t)) \;=\; \partial_x L(\gamma^*(t), \dot\gamma^*(t)), \tag{6.2}$$

together with the transversality condition $\partial_v L(\gamma^*(T), \dot\gamma^*(T)) + \nabla g(\gamma^*(T)) = 0$.

Proof. Let $\eta \in C^1([0, T]; \mathbb{R}^d)$ with $\eta(0) = 0$ enact the clamped-left, free-right choice just discussed. Since γ^* is a minimizer, the scalar function $\epsilon \mapsto \mathcal{J}[\gamma^* + \epsilon\eta]$ attains a minimum at $\epsilon = 0$, so its derivative there vanishes. We compute that derivative by differentiating under the integral sign — legitimate because $L \in C^2$, so the integrand and its ϵ-derivative are jointly continuous on the compact set $[0, T] \times \{|\epsilon| \le 1\}$ and dominated. The chain rule applied to $L(\gamma^* + \epsilon\eta, \dot\gamma^* + \epsilon\dot\eta)$ gives, at $\epsilon = 0$,

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\, L(\gamma^* + \epsilon\eta, \dot\gamma^* + \epsilon\dot\eta)\Big|_0 = \partial_x L(\gamma^*, \dot\gamma^*) \cdot \eta + \partial_v L(\gamma^*, \dot\gamma^*) \cdot \dot\eta,$$

the two terms recording how the action responds to displacing the position and to displacing the velocity. The terminal term contributes $\nabla g(\gamma^*(T)) \cdot \eta(T)$, again by the chain rule applied to $g(\gamma^*(T) + \epsilon\eta(T))$. Now the position perturbation enters through η but the velocity perturbation through $\dot{\eta}$; to compare them on equal footing we integrate the velocity term by parts, trading the derivative off η and onto $\partial_v L$ at the cost of a boundary contribution:

$$\int_0^T \partial_v L \cdot \dot{\eta} \, \mathrm{d}t = \Big[\partial_v L \cdot \eta\Big]_0^T - \int_0^T \left(\tfrac{\mathrm{d}}{\mathrm{d}t}\partial_v L\right) \cdot \eta \, \mathrm{d}t.$$

The boundary bracket vanishes at $t = 0$ because $\eta(0) = 0$ — this is exactly where the clamped left endpoint is spent — and at $t = T$ it leaves $\partial_v L\big|_{t=T} \cdot \eta(T)$. Collecting the surviving pieces,

$$0 = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\mathcal{J}[\gamma^* + \epsilon\eta]\Big|_0 = \int_0^T \Big(\partial_x L - \frac{\mathrm{d}}{\mathrm{d}t}\partial_v L\Big) \cdot \eta \, \mathrm{d}t + \Big(\partial_v L\big|_{t=T} + \nabla g(\gamma^*(T))\Big) \cdot \eta(T).$$

This identity must hold for *every* admissible η, and we exploit that freedom in two stages. First take η compactly supported in the open interval $(0, T)$, so that $\eta(T) = 0$ and only the integral survives; since the bracketed coefficient is continuous and integrates to zero against every such η, the fundamental lemma of the calculus of variations forces it to vanish pointwise, which is (6.2). (The fundamental lemma is the elementary fact that a continuous function orthogonal in L^2 to all smooth compactly supported test functions is identically zero — were it nonzero at a point, a bump test function concentrated there would integrate against it to something nonzero.) With the integral term annihilated for *all* η, the leftover boundary term must now vanish on its own; choosing an η with $\eta(T) \neq 0$ shows its coefficient is zero, which is the transversality condition. The two conditions are genuinely independent because the two families of perturbations — those that vanish at T and those that do not — probe the action independently. □

Three features of (6.2) now matter for everything that follows, and each is a crack through which the dynamic-programming pivot will eventually be driven.

The first feature is a matter of counting, and it is the structural fingerprint the physicist already recognizes from least action. Once the time derivative on the left is expanded by the chain rule, $\frac{\mathrm{d}}{\mathrm{d}t}\partial_v L$ contains $\ddot{\gamma}$, so (6.2) is a *second-order* ordinary differential equation for the $\mathbb{R}^d$-valued curve γ. The general solution of a second-order ODE in $\mathbb{R}^d$ carries exactly $2d$ free integration constants — morally, a free choice of position and a free choice of velocity at any one instant. The decisive question is how those $2d$ constants get pinned down. An *initial*-value problem would fix all $2d$ of them at $t = 0$, by prescribing both $\gamma(0)$ and $\dot{\gamma}(0)$, and then integrate forward uniquely; this is the Newtonian habit. But our problem does no such thing. It supplies the d components of $\gamma(0) = x_0$ at the *left* end and the d scalar components of the transversality condition $\partial_v L + \nabla g = 0$ at the *right* end. The $2d$ constants are split, d at each end of the interval, and a problem whose data are split across the two ends of time is by definition a *two-point boundary-value problem.* It is worth naming the discomfort this causes, because it is the discomfort the rest of the chapter exists to resolve: a boundary-value problem cannot be solved by integrating forward from the present, because the present does not carry enough information — half the data live in the future. Dynamic programming will dissolve exactly this difficulty, by trading the single curve for a field that already encodes the future obligation everywhere.

The second feature is a change of variables that converts the second-order equation into a first-order system, and it is the reader's first meeting with an object that will recur, under three different names, throughout the book. Define the *conjugate momentum* by the Legendre transform

$$p \;:=\; \partial_v L(x, v),$$

which is precisely the slot that appeared in the transversality condition. When L is strictly convex in v this relation is invertible, $v = v(x, p)$, and one sets the physical Hamiltonian $H_{\text{phys}}(x, p) = p\cdot v(x, p) - L(x, v(x, p))$. In these variables the lone second-order equation (6.2) splits into Hamilton's two first-order canonical equations,

$$\dot{x} = \partial_p H_{\text{phys}}(x, p), \qquad \dot{p} = -\partial_x H_{\text{phys}}(x, p),$$

as one checks directly: the first is just the inversion $v = v(x, p)$ rewritten through $\partial_p H_{\text{phys}} = v$, and the second reproduces (6.2) once one uses $\partial_x H_{\text{phys}} = -\partial_x L$ and $p = \partial_v L$. To make this concrete, take the textbook mechanical Lagrangian $L(x, v) = \frac{1}{2}|v|^2 - V(x)$. Then $p = \partial_v L = v$, the Legendre transform gives $H_{\text{phys}} = \frac{1}{2}|p|^2 + V(x)$ — the familiar total energy — and the canonical equations read $\dot{x} = p$, $\dot{p} = -\nabla V(x)$, which is Newton's second law $\ddot{x} = -\nabla V$ written as a first-order system. The momentum p is not, however, a passing convenience of classical mechanics. In section 6.4 the same object reappears as the *costate* of the Pontryagin maximum principle, the adjoint variable conjugate to the state; and proposition 6.12 will prove that along an optimal trajectory it coincides with the gradient ∇u of the value function we are about to construct. Three names — conjugate momentum here, costate there, value-gradient in the dynamic-programming picture — for one quantity. The reader who keeps that identification in view will watch the apparent multiplicity of methods in this chapter collapse into a single story.

The third feature is the limitation that motivates the entire remainder of the chapter, and it is the one the physicist is least inclined to notice, because it is a feature, not a bug, of least action. The Euler–Lagrange machinery, run to completion, delivers *one curve*: the optimal trajectory γ^* emanating from the one specific start x_0. It answers the question "given that I begin at x_0, what is the single best path?" and it answers nothing else. In particular it offers no guidance whatever to an agent that finds itself, at some intermediate time, at a state it had not planned to occupy — because of a disturbance, a modelling error, or, decisively for us, the random kick of a noise term. To such an agent the precomputed optimal curve from x_0 is simply irrelevant; what it needs is a *rule*, a prescription that returns a good action from *any* state at *any* time. The calculus of variations contains no such object. Manufacturing one — a field of optimal responses defined over the whole state space, in place of a lone optimal worldline — is exactly the task dynamic programming was invented for, and recognizing that this is what we want is the conceptual hinge of the chapter.

Before we can build that field, though, we must widen the problem itself. The calculus of variations lets the agent choose its curve directly, but an agent in an engineered or strategic setting does not actuate its position; it actuates a *control* — a throttle, an applied force, a trading rate — and the state responds through dynamics. The next step is therefore to reorganize the very same minimization around a dynamical constraint, so that the object chosen is a control rather than a trajectory. Only once the problem is posed that way does the distinction between committing to a plan in advance and reacting to the realized state — between, as we will call them, open-loop and feedback controls — acquire any teeth. We turn to that reformulation now.

6.1.1 From curves to controlled dynamics

The reformulation promised at the end of the previous section is a small change of emphasis with large consequences. In the calculus of variations the agent reached into the space of curves and pulled out the one it wanted; the curve was the primitive object of choice, and the dynamics — the relation $\dot{\gamma} = v$ between position and velocity — was a triviality so transparent it never had to be named. Optimal control keeps the same minimization but moves the dynamics to center stage. An agent in an engineered or strategic setting does not author its trajectory by fiat; it has its hands on a small number of knobs — a throttle, an applied force, a trading rate — and the state of the world responds to those settings through laws the agent does not get to rewrite. What the agent actually selects, then, is not a worldline but a schedule of knob settings, a *control*; the trajectory is whatever the control causes. Reorganizing the problem this way is what will eventually let us replace "find the one best curve from x_0" with "find the best response from every state," because once the object of choice is a rule for setting the knobs, that rule can be asked to specify a setting at any state at all, not merely along one favored path.

Concretely, the agent chooses a *control* α_t taking values in an action set $A \subseteq \mathbb{R}^k$, and the state responds through a controlled ordinary differential equation

$$\dot{X}_t \ = \ b(X_t, \alpha_t), \qquad X_0 = x_0, \tag{6.3}$$

with $b\colon \mathbb{R}^d \times A \to \mathbb{R}^d$ the drift. Read (6.3) as a machine: feed it the control history α and the start x_0, and it manufactures the state trajectory X uniquely (under the Lipschitz hypotheses inherited from chapter 5, which we will name (A-Lip), (A-Growth) once they are needed). The objective the agent minimizes is a functional of the control, with the state appearing only as the response (6.3) forces on it,

$$J(\alpha) \ = \ \int_0^T L(X_t, \alpha_t)\ \mathrm{d}t + g(X_T). \tag{6.4}$$

The running cost $L(X_t, \alpha_t)$ now prices both where the agent is and how hard it is pushing the knobs — fuel burned, effort spent, distance from a target — and $g(X_T)$ prices where it ends up. The notation $J(\alpha)$, with the control as the explicit argument and the trajectory suppressed, is the whole point: the trajectory is no longer free to vary independently, so it is bookkeeping, not choice.

That (6.3)–(6.4) genuinely *contains* the calculus of variations, rather than merely resembling it, is worth seeing explicitly, because it licenses everything earlier in the chapter as a special case of everything later. Take the drift to be the identity in the velocity slot, $b(x, v) = v$, and let the action set be all of $\mathbb{R}^d$, $A = \mathbb{R}^d$. Then (6.3) reads $\dot{X}_t = \alpha_t$: the control *is* the velocity, the agent steers its position directly with no intervening dynamics, and substituting $\alpha_t = \dot{X}_t$ into (6.4) reproduces the action (6.1) verbatim, with X playing the role of γ. The Euler–Lagrange problem is thus the corner of the control problem in which the knob is the velocity itself and there are no limits on how fast one may go.

Leaving that corner is exactly what makes the theory apply to systems the calculus of variations cannot describe, and the two ways of leaving it are independent. First, the drift need not equal the control: $b(x, a)$ can be any map, so the knob may act on the state only indirectly, through several integrations. The textbook witness is the double integrator, a unit mass on a line whose state is the pair $X = (\text{position}, \text{velocity})$ and whose control is the applied force a, with $b((q,v),a) = (v,a)$. Here the agent sets the force, but the force moves the *velocity*, which in turn moves the *position*; the control reaches the quantity one cares about only at second hand, and there is no way to recast the problem as a direct minimization over position-curves with an unconstrained velocity. Second, the action set may be bounded: $A = [-1,1]^k$ or a ball encodes a throttle that saturates, a force a motor cannot exceed, a trading rate a market will not absorb. A bounded A has no counterpart at all in (6.1), where the velocity ranged over all of $\mathbb{R}^d$; yet it is precisely the boundedness of the knobs that makes the optimal policy interesting (it may push a knob to its limit and hold it there, a bang-bang regime invisible to unconstrained least action), and it is what will later let a strategic agent have a genuinely constrained set of responses to the population it faces. These two generalizations — indirect actuation and bounded actuation — are what carry the theory out of pure mechanics and into the engineered and strategic problems this book is ultimately about.

With the object of choice now a control rather than a curve, a distinction opens up that was invisible before, and it is the conceptual fulcrum of the chapter: *what is the control allowed to depend on?* There are two honest answers, and they are not interchangeable.

Definition 6.2 (Open-loop and feedback controls)**.** An *open-loop* control is a measurable map $t \mapsto \alpha_t \in A$ of time alone, chosen at the outset. A *feedback* (closed-loop) control is a measurable map $(t,x) \mapsto \phi(t,x) \in A$; it is realized along trajectories by $\alpha_t = \phi(t, X_t)$, where X solves (6.3) with this substitution.

The two definitions encode two postures toward the future. An open-loop control is a *plan*: the agent writes down, at time zero, the entire schedule $t \mapsto \alpha_t$ of what it will do at every future instant, seals it, and executes it no matter what — a flight plan filed before takeoff, a sequence of moves committed to in advance. It is a curve in the action set, indexed by time alone, and it carries no provision for surprise because it never consults the state. A feedback control is a *rule*: the agent commits not to a schedule of actions but to a policy $\phi(t,x)$ that says, "if I find myself at state x at time t, I do $\phi(t,x)$." The action is then synthesized on the fly along the realized trajectory, $\alpha_t = \phi(t, X_t)$, by reading the current state and looking it up in the rule. The defining difference is informational: the open-loop agent acts blind to where it is, the feedback agent acts on it.

In the deterministic problem (6.3)–(6.4) this difference, real as it is in principle, costs nothing: the two notions produce the *same* optimal cost. The reason is that determinism leaves nothing to react to. Once the start x_0 and a feedback rule ϕ are fixed, the closed-loop trajectory X is completely determined — (6.3) with $\alpha_t = \phi(t, X_t)$ has a unique solution — so the realized control history $t \mapsto \phi(t, X_t)$ is itself a fixed function of time, i.e. an open-loop control that produces the identical trajectory and hence the identical cost. Conversely the optimal open-loop control, replayed along its own optimal trajectory, can be repackaged as a feedback rule that happens to be evaluated only on that one path. In a world with no noise the agent gains nothing by watching the state, because the state never tells it anything it did not already know at time zero: the future is a foregone conclusion, and a sealed plan forfeits no information because there is no information to forfeit. This is why the calculus of variations, which produces only the open-loop optimal schedule from the single start x_0, suffered no visible penalty for it.

Switch on noise and the equivalence collapses — and it collapses for a reason worth stating now, because it is the entire motivation for the stochastic theory of section 6.5. Replace (6.3) by a stochastic differential equation $\mathrm{d}X_t = b(X_t, \alpha_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}W_t$, so that the trajectory is now genuinely random and the agent's actual position at time t is not knowable in advance. An open-loop control, fixed at time zero, must commit to its entire schedule before a single Brownian increment is realized; it is betting on an average future and cannot adjust when the noise pushes the state somewhere the plan did not anticipate. A feedback control, by contrast, reads the realized X_t at each instant — noise included — and responds to where the agent *actually* is rather than where it was projected to be. This is the value of information made precise: feedback conditions on strictly more than open-loop (the entire realized history of the state, not just the clock), and since the open-loop schedules are exactly the feedback rules that ignore their state argument, the feedback optimum can only improve on the open-loop optimum, generically strictly. The gap is not a technical artifact; it is the difference between a plan and a policy in the presence of surprise, and a positive gap is the rule, not the exception, the moment $\sigma \not\equiv 0$. Section 6.7 will exhibit this concretely: in the linear–quadratic–Gaussian problem the optimal feedback is a linear rule in the current state, with a time-dependent gain set by a Riccati equation, that tracks the noisy state continuously — and no schedule fixed in advance can match its expected cost once the noise has anywhere to push.

This is the decisive observation for the rest of the book, and it tilts the whole development toward one of the two routes through optimal control. Dynamic programming is intrinsically a *feedback* theory: as we will see, it does not search for an optimal schedule from one start but constructs an object from which the optimal feedback rule $\phi(t, x)$ can be read off at *every* state x and *every* time t at once. That is precisely the rule, valid from everywhere, that the previous section identified as the thing genuinely wanted and that the calculus of variations could not supply — and it is precisely the field that the next section sets out to build. The Euler–Lagrange and Pontryagin routes, by contrast, characterize a single optimal trajectory at a time and are open-loop at heart; they remain indispensable (the Pontryagin costate returns in section 6.4 and again in chapter 9), but they are not the natural backbone for what is coming. For in a mean field game the "state" an agent must react to is not just its own noisy position but a whole population that is itself evolving randomly in time; reacting to a moving, uncertain environment is exactly what feedback does and open-loop cannot. An agent facing a fluctuating crowd needs a rule that prescribes a response from every configuration it might encounter, which is once more a field over states rather than a curve through them.

To produce such a field we cannot keep treating the single problem launched from the lone start x_0 in isolation; we must consider, all at once, the problem launched from every present time and every present state, and ask how their answers fit together. Embedding the single problem in that family — and discovering that the resulting family of optimal costs obeys one self-referential equation — is the program of the next section, to which we now turn.

6.2 Dynamic programming and the value function

The previous section left us with a sharp demand and no instrument to meet it. We argued that what a noisy or strategic agent genuinely needs is not the single optimal worldline launched from one start x_0, but a *rule* valid from every state at every time — a field of best responses, $\phi(t, x)$, that tells the agent what to do from wherever it actually finds itself. The calculus of variations and the Pontryagin route both produce the wrong shape of object: one trajectory, answering one question. To manufacture a field we must stop privileging the single start x_0 and instead solve, all at once, the entire *family* of control problems obtained by letting the agent begin at an arbitrary present time t and an arbitrary present state x. The optimal costs of that family, collected into one function of (t, x), will turn out to obey a single self-referential equation; and it is that equation, not any one trajectory, that secretly encodes the feedback rule at every state simultaneously. This section builds the family, names its optimal-cost function the value function, and extracts the identity. The identity is the engine of the chapter: the next section will crank it to an infinitesimal time step and watch a differential equation — the HJB equation — fall out.

So fix the data (b, L, g, T, A) once and for all, exactly as in (6.3)–(6.4), and embed the original problem in the promised family. The original problem started the clock at $t = 0$ from the state x_0; we now let it start at any $(t, x) \in [0, T] \times \mathbb{R}^d$. For an admissible control α on the remaining interval $[t, T]$, let $X^{t,x,\alpha}$ denote the solution of the controlled ODE $\dot{X}_s = b(X_s, \alpha_s)$ on $[t, T]$ launched from $X_t = x$ — the superscript records all three pieces of data the trajectory depends on, the start time, the start state, and the control. Pricing this trajectory over what remains of the horizon gives the *cost-to-go*,

$$J(t, x; \alpha) \;=\; \int_t^T L(X_s^{t,x,\alpha}, \alpha_s)\, \mathrm{d}s \;+\; g(X_T^{t,x,\alpha}).$$

This is the literal generalization of the cost $J(\alpha)$ of (6.4): that earlier object is recovered as the corner $J(0, x_0; \alpha)$, the integral now running from the present t rather than from 0 and the trajectory now hung from the present state x. Nothing new has been assumed; we have merely refused to commit to a single starting configuration, carrying the start (t, x) as a free pair of arguments. The whole subject turns on taking that refusal seriously.

Definition 6.3 (Value function)**.** The *value function* is the optimal cost-to-go,

$$u(t, x) \;=\; \inf_{\alpha} J(t, x; \alpha), \qquad (t, x) \in [0, T] \times \mathbb{R}^d,$$

the infimum taken over admissible controls on $[t, T]$. By construction $u(T, x) = g(x)$.

Two features of this definition deserve to be dwelt on before we use it, because they are exactly the features that separate dynamic programming from least action. The first is that u is a *field*, not a number. The calculus of variations, run from x_0, returned a single scalar — the minimal action of one curve. The value function returns a scalar *at every point* of the space-time slab $[0, T] \times \mathbb{R}^d$; it is the optimal cost-to-go evaluated at each conceivable present configuration, the entire landscape we contrasted with the lone worldline in the chapter's opening. Asking for u is asking for vastly more than asking for one optimal trajectory — and, paradoxically, asking for more is what makes the problem tractable, because the surplus information is precisely what lets the answers at neighboring points constrain one another. The second feature is the clause $u(T, x) = g(x)$, which is not an extra hypothesis but a forced consequence of the definition. At the terminal time the remaining interval $[T, T]$ has zero length: the integral in $J(T, x; \alpha)$ is empty, no control can be exercised, and the cost-to-go collapses to $g(x)$ for every admissible α. The infimum of a constant is that constant, so $u(T, \cdot) = g$ with nothing to optimize. This is the one slice of the field that is known outright, with no computation; the entire backward machinery of the next section exists to propagate this terminal datum to earlier times. We will return to its load-bearing role when we explain why HJB integrates backward.

It is worth fixing the picture with the smallest possible instance. Take $b \equiv 0$, so the state never moves and $X_s^{t,x,\alpha} \equiv x$, and take $L(x, a) = \frac{1}{2}|a|^2$ with $A = \mathbb{R}^d$. Then the cost-to-go is $J(t, x; \alpha) = \int_t^T \frac{1}{2}|\alpha_s|^2 \, \mathrm{d}s + g(x)$, the cheapest choice is plainly $\alpha \equiv 0$, and $u(t, x) = g(x)$ for *all* t: with motion forbidden and effort costly, the agent does nothing and simply pays the terminal price it is stuck with. Trivial as it is, this example already exhibits u as a function of the full pair (t, x) carrying the terminal data g, and it is the degenerate seed of every Riccati computation to come. The interesting content appears the instant the drift lets the agent *change* x at a price, so that paying running cost now can buy a cheaper terminal position later — the tension the value function is built to resolve.

That resolution is governed by one identity. Heuristically it says that optimizing over the whole interval $[t, T]$ can be done in two stages: optimize the first short leg $[t, t + h]$ and the remaining tail $[t + h, T]$ jointly, and the tail's contribution, *once re-optimized from wherever the first leg lands*, is nothing but the value function again, evaluated at the new time $t + h$ and the new state X_{t+h}. The decision over the whole collapses onto a decision over the first leg plus a lookup of u at its endpoint. This is the dynamic programming principle, and the surprise — the thing that makes it an equation rather than a tautology — is that the very same function u stands on both sides.

Theorem 6.4 (Dynamic programming principle, DPP)**.** *Under* (A-Lip) *and* (A-Growth) *on b and L, for every (t, x) and every $h \in [0, T - t]$,*

$$u(t, x) \;=\; \inf_{\alpha} \left\{ \int_t^{t+h} L(X_s^{t,x,\alpha}, \alpha_s) \, \mathrm{d}s \;+\; u\big(t + h,\, X_{t+h}^{t,x,\alpha}\big) \right\}. \tag{6.5}$$

Proof. An identity is two inequalities, and here the two carry genuinely different mechanisms — one is nearly free, the other needs a measurable-selection argument — so it is worth seeing exactly where the labor falls. Throughout, X solves the controlled ODE (6.3), so once the control on $[t, t + h]$ is fixed the endpoint $x' := X_{t+h}^{t,x,\alpha}$ is a *deterministic* reachable state, not a random variable; the measurability subtlety below is therefore the measurability of an ε-optimal tail control as a function of the reachable state x', and has nothing to do with randomness. (The genuinely random-endpoint version, where X_{t+h} is a random variable and one must select the tail measurably along sample paths, is the stochastic extension of section 6.5.)

The inequality $\geq$ (no selection needed). This direction says the left side cannot exceed the right, and it follows simply because any single competitor's cost overestimates an infimum. Let α be *any* admissible control on the whole of $[t, T]$, with state X, and write $x' = X_{t+h}$ for where its first leg lands. Because the cost-to-go is an integral, it splits at the intermediate time $t + h$ with no remainder:

$$J(t, x; \alpha) = \int_t^{t+h} L(X_s, \alpha_s) \, \mathrm{d}s + J(t + h, x'; \alpha|_{[t+h,T]}) \;\geq\; \int_t^{t+h} L(X_s, \alpha_s) \, \mathrm{d}s + u(t + h, x').$$

The single inequality is the entire crux of this direction, and it is almost free: the tail $\alpha|_{[t+h,T]}$ is *one* admissible competitor for the control problem started at $(t + h, x')$, so its cost-to-go is at least the infimum $u(t+h,x')$ over all such tails. We did not need to construct anything — restricting a control we already hold to its second half is automatic. The right-hand side is now bounded below by its own infimum over the first leg, and since α was arbitrary we may take $\inf_\alpha$ on the left, which yields "$\geq$". The reason no selection is needed is that we are throwing competitors *away* (passing to an infimum), and discarding never requires a measurable choice.

The inequality $\leq$ (selection needed). The opposite direction is where the work lives, and one can see in advance why. We must show the left side is no larger than the right, i.e. that for an essentially arbitrary first leg we can *assemble* a full control on $[t,T]$ whose total cost comes within ε of running that first leg and then paying the optimal tail-value $u(t+h,x')$. Assembling, unlike discarding, forces us to actually produce a tail control — and not just one tail, but one for *every* possible landing point x', glued together into a single control that is legitimate as an input to the dynamics. The gluing is the obstruction: the ε-optimal tail depends on x', and unless that dependence is measurable the concatenated control is not an admissible process and the construction is empty. Everything therefore hinges on choosing the tails measurably in the landing point, which is exactly what the next lemma supplies. We prove it in-book rather than invoke the general selection theorem, because under our Lipschitz standing assumptions an elementary partition argument suffices.

Lemma 6.5 (Measurable ε-optimal selection)**.** *Under* (A-Lip)*,* (A-Growth) *there is, for each $\tau \in [t,T]$ and each $\varepsilon > 0$, a Borel-measurable map $x' \mapsto \alpha^{x'}$ from states to admissible controls on $[\tau,T]$ with $J(\tau,x';\alpha^{x'}) \leq u(\tau,x') + \varepsilon$ for every x'.*

Proof. The single difficulty is measurability: for each *fixed* state x' an ε-optimal tail certainly exists, simply because $u(\tau,x')$ is an infimum and an infimum is approached to within any ε by some competitor. What is not automatic is that these competitors, one per state, can be chosen to depend Borel-measurably on x'; a careless pointwise choice could assemble a wildly non-measurable map. The way around this is to exploit continuity: if the optimal cost varies *continuously* in the starting state, then a control good at one base point is still nearly good at all nearby states, so we may reuse it on a whole neighborhood and choose only countably many controls in total.

The continuity we need comes from the dynamics being well-conditioned in the initial state, *uniformly over the control.* Fix any control α and two starting states x', y', and run (6.3) from each under the *same* α. Subtracting the two integral equations and using the (A-Lip) Lipschitz bound on b in the state, with constant $\mathrm{Lip}(b)$ *independent of the control argument*,

$$\left|X_s^{\tau,x',\alpha} - X_s^{\tau,y',\alpha}\right| \leq |x'-y'| + \mathrm{Lip}(b)\int_\tau^s \left|X_r^{\tau,x',\alpha} - X_r^{\tau,y',\alpha}\right| \,\mathrm{d}r,$$

and Grönwall's inequality turns this into $\left|X_s^{\tau,x',\alpha} - X_s^{\tau,y',\alpha}\right| \leq e^{\mathrm{Lip}(b)(s-\tau)}\,|x'-y'| \leq e^{\mathrm{Lip}(b)T}\,|x'-y'|$. The decisive point is that the amplification constant $e^{\mathrm{Lip}(b)T}$ depends only on the data, *not* on which control was used: the two flows started from nearby states stay close at a rate the control cannot worsen. Feeding this into the cost-to-go, where L and g are Lipschitz by (A-Lip), makes $x' \mapsto J(\tau,x';\alpha)$ Lipschitz with a constant that is again the same for every control — the family $\{J(\tau,\cdot;\alpha)\}_\alpha$ is *equi-Lipschitz.* Taking the infimum over α preserves a common Lipschitz bound (an infimum of functions sharing one modulus of continuity inherits that modulus), so $x' \mapsto u(\tau,x')$ is itself Lipschitz, hence continuous. Write ω for the common modulus of continuity of the whole equi-Lipschitz family — it bounds the oscillation of *every* $J(\tau,\cdot\,;\alpha)$ and of $u(\tau,\cdot)$ alike — and note once more that its uniformity over α is exactly what the Grönwall constant's control-independence bought us.

Now the partition. Cover $\mathbb{R}^d$ by a countable Borel partition $\{B_k\}$ into sets of diameter small enough that $\omega(\operatorname{diam} B_k) \le \varepsilon/2$; since one common modulus ω governs every cost-to-go in the family and the value, this single diameter threshold simultaneously caps the oscillation of all of them below $\varepsilon/2$ across each B_k — this is the content of the otherwise terse phrase "the common modulus of continuity gives oscillation below $\varepsilon/2$," and the word "common" is the whole point. In each cell pick a base point $x_k \in B_k$ and, using that $u(\tau, x_k)$ is an infimum, a control $\alpha^{(k)}$ that is $(\varepsilon/2)$-optimal there, $J(\tau, x_k; \alpha^{(k)}) \le u(\tau, x_k) + \varepsilon/2$. Define the selection by reusing each base-point control across its whole cell: $\alpha^{x'} = \alpha^{(k)}$ for $x' \in B_k$. This map is Borel because it is constant on each set of a countable Borel partition. To check it does its job, take $x' \in B_k$ and chain three estimates — transport the cost from x' to x_k, use $(\varepsilon/2)$-optimality at x_k, then transport the value back from x_k to x':

$$J(\tau, x'; \alpha^{(k)}) \le J(\tau, x_k; \alpha^{(k)}) + \tfrac{\varepsilon}{2} \le u(\tau, x_k) + \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} \le u(\tau, x') + \varepsilon,$$

the first and third steps each spending one ω-oscillation bound (below $\varepsilon/2$) on the cell B_k — the first on $J(\tau,\cdot\,;\alpha^{(k)})$, the third on $u(\tau,\cdot)$ — and the middle step spending the base-point optimality. The total slack is $\varepsilon/2 + \varepsilon/2 = \varepsilon$ only because the same ω bounds both oscillations; an oscillation budget that varied with the control would not close. This is the claim. The argument is the elementary in-book substitute for the general Kuratowski–Ryll-Nardzewski measurable-selection theorem, which would deliver a measurable ε-optimal selection with no continuity at all; the continuity afforded by (A-Lip) is precisely what lets us replace that abstract machinery with a countable partition and a triangle inequality. □

With the selection in hand the second inequality is immediate. Apply lemma 6.5 at $\tau = t + h$. Given any control on $[t, t+h]$ producing endpoint $x' = X_{t+h}$, concatenate it with the selected tail $\alpha^{x'}$ on $[t+h, T]$ — the concatenation is admissible precisely because $x' \mapsto \alpha^{x'}$ is measurable, so the glued process is a legitimate control. Its total cost is $\int_t^{t+h} L \,\mathrm{d}s + J(t+h, x'; \alpha^{x'}) \le \int_t^{t+h} L \,\mathrm{d}s + u(t+h, x') + \varepsilon$. Taking the infimum over the first leg and letting $\varepsilon \downarrow 0$ gives "$\le$", completing the proof. The classical references [136, Ch. III] and [68, Ch. I–II] carry out the general (random-endpoint, stochastic) version with the full measurable-selection apparatus. □

Remark 6.6 (The DPP is motivation; the rigorous spine is verification)**.** We have just spent real effort proving the DPP, so it is important to be candid about how much weight it is asked to bear downstream: *none.* No later result in this book — not the HJB equation we are about to derive, not the verification theorem, not the coupled system of chapter 8, nothing in the chapters beyond — logically rests on theorem 6.4. The DPP earns its place here for one purpose only: it is the heuristic *source* of the HJB equation. In the next section we expand (6.5) for small h and read off a partial differential equation, and that expansion is a genuine derivation, the kind that tells you what equation to write down and why it has the form it does. But a derivation is not a proof of correctness, and we will not pretend it is. The moment we have the candidate PDE, we change our footing entirely: we *forget* how we guessed it and prove, from scratch and by Itô's formula alone, that any sufficiently smooth solution of that PDE is the value function and that its feedback is optimal. That is the verification theorem theorem 6.10, and it is the load-bearing spine of the chapter. The practical upshot for the reader is liberating: every concern one might have about the DPP — whether the infimum is attained, whether the selection can be made along random paths in the stochastic case, whether u is regular enough for the expansion to be valid — can be set aside, because none of it is on the critical path. A reader who prefers to treat (6.5) as a purely formal identity, a mnemonic for where HJB comes from, loses no rigor whatsoever. We prove the DPP honestly because honest motivation is worth having; we route the rigor through verification because that is the route that needs no selection lemma, no a priori regularity of u, and no leap of faith about the $h \downarrow 0$ limit.

Heuristic (non-rigorous)

Read (6.5) as a bookkeeping tautology with teeth. The optimal cost from (t, x) equals the cost incurred over an arbitrarily short first leg $[t, t + h]$ plus the optimal cost from wherever that leg lands. Phrased as a slogan: optimality decomposes across time, because the tail of an optimal plan is itself optimal — if it were not, one could improve the tail and thereby improve the whole, contradicting optimality of the plan. That much really is almost a definition. The *content*, the part that is not tautological, is that the function appearing on the right under $u(t + h, \cdot)$ is the *same* function u that appears alone on the left. The identity relates u to itself across a time step, and a relation of an unknown to itself across an infinitesimal step is precisely what a differential equation is. Letting $h \downarrow 0$ will convert this fixed-point identity into the HJB equation. A physicist has met this move before: a self-consistency relation — the magnetization that must equal the response it induces, the field that must reproduce the field it generates — becomes, in the continuum or small-step limit, a differential (or integral) equation for the self-consistent quantity. The DPP is the self-consistency relation of optimal control; HJB is its continuum limit.

We now have the family, the field u, and the self-referential identity that binds the field's values at neighboring times. Only one operation remains to extract a usable equation from (6.5): take the first leg infinitesimally short and expand. That $h \downarrow 0$ limit — the move the heuristic just promised, converting a self-consistency relation into a differential equation — is the entire business of the next section, which opens by performing exactly this expansion and turning the DPP into the Hamilton–Jacobi–Bellman equation.

6.3 The Hamilton–Jacobi–Bellman equation and backward time

We arrive at the chapter's climax holding exactly one thing: the dynamic programming identity (6.5), which binds the value u to itself across a time step of length h. The previous section's closing heuristic already told us what to do with it: a relation of an unknown to itself across an infinitesimal step *is* a differential equation in disguise, the self-consistency relation of optimal control awaiting its continuum limit. We now take $h \downarrow 0$ and watch that limit, the Hamilton–Jacobi–Bellman (HJB) equation, condense out. This is the conversion the heuristic box anticipated, and it is the move that finally replaces the physicist's least-action reflex — extremize an action, solve a two-point boundary-value problem for one optimal curve — with the dynamic-programmer's: compute a whole field $u(t,x)$ by integrating a single PDE backward from terminal data.

The work falls into three stages, and it is worth seeing in advance that they are three movements of one argument rather than three separate results. The first stage, the *formal derivation*, simply performs the limit: granting ourselves provisionally that u is smooth, we Taylor-expand (6.5) in h, divide, let $h \downarrow 0$, and read off the PDE together with the object that organizes everything downstream, the Bellman Hamiltonian (6.6). This stage is honest but incomplete in a way we will name as we go: it is a *derivation*, the kind of calculation that tells you what equation to write and why it has the shape it does, but it assumes the very smoothness of u that is not given to us in advance, and it trusts an $h \downarrow 0$ limit we have not controlled. The second stage steps back from the algebra to confront the single feature of the resulting equation that most violates physical expectation: (6.7) carries its data at the *final* time and is solved *backward*. This is the central retraining of the chapter, and it earns a subsection of its own — *why the time runs backward* — because mistaking which end of the interval carries the datum does not flip a sign, it silently exchanges a well-posed problem for an ill-posed one. The third stage, the *verification theorem*, repays the debt the first stage incurred. Having let the DPP *suggest* the PDE, we discard the derivation entirely and turn it around into a guess-and-check: we forget where the equation came from, posit a smooth solution, and prove from Itô's formula alone that any such solution is the value function and that its feedback is optimal. This is the move flagged already in remark 6.6 — the DPP is motivation, the rigorous spine is verification — and it is the engine that every explicit computation in this book, the LQG worked example of section 6.7 included, will actually run.

So the architecture is: a formal derivation that assumes smoothness to find the equation; a conceptual treatment that explains the equation's strangest structural feature; and a verification theorem that turns the whole derivation around, requiring smoothness now only of a *candidate* we are free to guess, never of the value function we did not yet know. The verification stage is what makes the provisional smoothness of the first stage harmless: whatever the derivation assumed in order to write the equation down, the verification theorem re-establishes from scratch, by a route that needs no selection lemma, no a priori regularity of u, and no leap of faith about the limit. We begin, then, with the derivation, granting ourselves exactly the provisional smoothness the verification theorem will later let us take back.

6.3.1 Formal derivation

We now execute, in full, the $h \downarrow 0$ limit just described, expanding the dynamic programming identity (6.5) to read off the equation its self-reference becomes. To run the expansion at all we must be able to differentiate u, so we grant ourselves, *provisionally*, that $u \in C^1([0,T] \times \mathbb{R}^d)$. This is the one debt the derivation incurs, and it is worth marking it precisely as a debt: nothing so far guarantees that the infimum of a family of costs is even once differentiable in (t,x), and in section 6.6 we will meet explicit problems where it is not. The whole point of the verification theorem of theorem 6.10 is to retire this debt — to re-derive everything below demanding smoothness only of a *candidate* we are free to construct, never of the unknown u itself. For now we simply assume it and proceed.

Fix a base point (t,x). The infimum in (6.5) ranges over all admissible controls on $[t,t+h]$, but for the purpose of extracting the leading behaviour as $h \downarrow 0$ it suffices to probe it with the simplest possible competitors: hold a single *constant* action $\alpha_s \equiv a \in A$ across the short interval $[t, t+h]$. The justification is that over a vanishing interval only the action at the *instant* t can register at first order — any admissible control is, to leading order in h, indistinguishable from the constant equal to its value at t — so infimizing over constants reproduces the same leading term as infimizing over all controls. With a frozen, evaluate the two pieces of the right side of (6.5) in turn.

First the state increment. The controlled dynamics (6.3) integrate to $X_{t+h} = x + \int_t^{t+h} b(X_s, a)\, \mathrm{d}s$. Over the short interval the trajectory has barely left x, so $X_s \to x$ as $s \downarrow t$ and, by continuity of b, the integrand is $b(x,a) + o(1)$; the integral is therefore

$$X_{t+h} = x + h\, b(x,a) + o(h).$$

Next the value at the displaced endpoint. Taylor-expanding the C^1 function u about (t,x) in *both* arguments — the time has advanced by h, the state by $X_{t+h} - x = h\, b(x,a) + o(h)$ — and using the chain rule,

$$u(t+h, X_{t+h}) = u(t,x) + h\, \partial_t u(t,x) + (X_{t+h} - x)\cdot \nabla u(t,x) + o(h) = u(t,x) + h\big(\partial_t u(t,x) + b(x,a)\cdot \nabla u(t,x)\big) + o(h),$$

where the second equality substitutes the increment just computed and folds the $o(h)$-sized state error, dotted into the bounded gradient, into the remainder. Finally the running cost over the leg is $\int_t^{t+h} L(X_s, a)\, \mathrm{d}s = h\, L(x,a) + o(h)$ by the same "integrand is nearly constant" estimate that produced the state increment. Assembling the two pieces, the bracketed quantity inside the infimum of (6.5) is

$$\int_t^{t+h} L(X_s, a)\, \mathrm{d}s + u(t+h, X_{t+h}) = u(t,x) + h\, \partial_t u(t,x) + h\big(b(x,a)\cdot \nabla u(t,x) + L(x,a)\big) + o(h).$$

Now read off the structure. The term $u(t,x)$ on the right is exactly the left side of (6.5), so it cancels; the term $h\, \partial_t u(t,x)$ carries no dependence on the action a, so it slips outside the infimum untouched; and what is left under the infimum is the single a-dependent group $b(x,a)\cdot \nabla u(t,x) + L(x,a)$. The identity (6.5) thus collapses to

$$0 = h\, \partial_t u(t,x) + \inf_{a\in A} \Big\{ h\, b(x,a)\cdot \nabla u(t,x) + h\, L(x,a) \Big\} + o(h).$$

Divide through by h and let $h \downarrow 0$; the $o(h)/h$ remainder vanishes and we obtain the pointwise relation

$$0 = \partial_t u(t,x) + \inf_{a\in A} \Big\{ b(x,a)\cdot \nabla u(t,x) + L(x,a) \Big\}.$$

This is already the HJB equation; it remains only to name the minimization it contains. It is convenient, and consistent with the conventions of appendix A, to absorb that infimum into a Hamiltonian — precisely the partial Legendre transform that, in the calculus-of-variations corner of section 6.1, turned the running cost into the physical Hamiltonian. We define the *(Bellman) Hamiltonian*

$$H(x,p) \;:=\; \sup_{a\in A} \Big\{ -b(x,a)\cdot p - L(x,a) \Big\} \;=\; -\inf_{a\in A} \{ b(x,a)\cdot p + L(x,a) \}. \tag{6.6}$$

Then $\inf_a\{b \cdot \nabla u + L\} = -H(x, \nabla u)$, and the displayed identity is the *Hamilton–Jacobi–Bellman equation*

$$-\partial_t u(t,x) + H\big(x, \nabla u(t,x)\big) = 0, \qquad u(T,x) = g(x). \tag{6.7}$$

We must now say a word about what does the analytic work in this construction, because it is unusually easy — and a common error — to credit it to the wrong hypothesis. The temptation is to declare "H is convex in p" and treat that convexity as the operative assumption. It is not, and seeing exactly why is the single most important hypothesis-clarification in this chapter.

Start from the fact that convexity in p is *free*. For each fixed action a the map $p \mapsto -b(x,a) \cdot p - L(x,a)$ is *affine* — a constant plus a linear term in p. The Hamiltonian (6.6) is the pointwise supremum of this family over $a \in A$, and a pointwise supremum of affine functions is convex *always*, with no hypothesis on the data whatsoever: the supremum of a collection of straight lines is the upper envelope of those lines, and an upper envelope of lines is by inspection a convex curve. So the convexity of H in p is a structural identity, true for every b and L, attained by no assumption. A property that holds unconditionally cannot be the content of an assumption; if we want a hypothesis to buy us something, we must look for what convexity *fails* to deliver.

What it fails to deliver is the thing we actually need: a single, well-defined optimal action, varying smoothly with (x,p), so that the feedback (6.8) is a genuine function and the Hamiltonian is differentiable. A merely convex function can have *corners*. At a corner a convex function is continuous but not differentiable; in place of a gradient it carries a whole *set* of supporting slopes, its subdifferential $\partial_p H$, and through the envelope correspondence each such slope answers to a distinct maximizing action. A set-valued $\partial_p H$ is, read through (6.8), exactly a *non-unique feedback*: at such a p the agent is told to play any one of several actions, with no principled way to choose. The smallest honest illustration takes a finite action set. Let $A = \{-1, +1\}$, $d = 1$, $b(x,a) = a$, and $L \equiv 0$, so that $H(x,p) = \sup_{a \in \{-1,1\}}\{-a\,p\} = |p|$. This H is convex — it is the upper envelope of the two lines $-p$ and $+p$ — and it is perfectly differentiable away from the origin, where the maximizing action is the unique $a^* = -\operatorname{sgn}(p)$. But at $p = 0$ the two lines cross: H has a kink, $\partial_p H(x,0) = [-1,1]$ is a full interval, and *both* actions $a = \pm 1$ attain the supremum. The feedback is genuinely ambiguous there. Convexity did nothing to prevent this; the corner is compatible with it.

The mechanism that rules corners out lives one level down, not in H as a function of p but in the *action* map being optimized inside it,

$$a \;\longmapsto\; \mathfrak{H}(x,a,p) \;=\; b(x,a) \cdot p + L(x,a),$$

whose infimum over a defines $H(x,p) = -\inf_a \mathfrak{H}(x,a,p)$. Two properties of this map, in the variable a, are what we genuinely require, and they are logically independent. First, *coercivity*: $\mathfrak{H}(x,a,p) \to +\infty$ as $|a| \to \infty$, which we inherit from the superlinear growth recorded in (A-Growth). Coercivity guarantees the infimum is actually *attained*, and attained on a bounded set — a minimizing sequence cannot escape to infinity, so it has a convergent subsequence whose limit, by continuity, is a minimizer. (In the finite-set example coercivity is vacuous and there is nothing to attain in the interior; that degeneracy is precisely what left room for the tie at $p = 0$.) Second, *strict* convexity of $a \mapsto \mathfrak{H}(x,a,p)$: this forces the minimizer to be *unique*, for if two distinct points both minimized, their midpoint would, by strict convexity, lie strictly below the common minimum value — an impossibility. Coercivity supplies existence, strict convexity supplies uniqueness; together they make $a^*(x,p) = \arg\min_a \mathfrak{H}(x,a,p)$ a single-valued map, and (as the envelope lemma below makes precise, through a positive definite Hessian) a C^1 one. A unique, smoothly varying minimizer is in turn exactly what makes H differentiable in p, with the single-valued gradient $\partial_p H(x,p) = -b(x, a^*(x,p))$ of lemma 6.8 replacing the set-valued subdifferential a corner would have forced.

This is the classical duality between strict convexity on one side of a Legendre transform and differentiability on the other, and it pinpoints where the genuine assumption sits. *Strict convexity plus coercivity in the control a* is the operative hypothesis; the convexity of H in p is its dual shadow, not its source. We nonetheless give that shadow a name. Throughout the book the hypothesis is consumed downstream in the form "H is convex in p" — it is in that form that the verification theorem of theorem 6.10 compares a candidate against the true value, and in that form that the monotonicity and comparison arguments behind uniqueness in the coupled system of chapter 8 invoke it — so we tag the convexity of H in p as (A-Conv) and quote it by that tag where it is used. The reader should simply carry the translation: the *tag* (A-Conv) is the convexity-in-p in which the hypothesis is *spent*, while its *operative content*, the thing that has to be assumed of the data (b, L) to make any of it true, is strict convexity and coercivity in the action a. With that settled we may record the unique minimizer as the candidate optimal feedback.

Definition 6.7 (Optimal feedback)**.** Under (A-Growth) and strict convexity in the control of $a \mapsto b(x,a) \cdot p + L(x,a)$ (the operative content of (A-Conv); see the discussion above), for each (x,p) let $a^*(x,p) = \arg\min_{a \in A}\{b(x,a) \cdot p + L(x,a)\}$ be the unique minimizer in (6.6). The candidate *optimal feedback* is

$$\alpha^*(t,x) \;=\; a^*\big(x, \nabla u(t,x)\big). \tag{6.8}$$

The differentiability of H used throughout — and the formula $\partial_p H = -b(\cdot, a^*)$ on which the costate identity of section 6.4 and the feedback $\alpha^* = -\partial_p H$ of section 6.5 both rest — is not an appeal to an external "envelope theorem" but a one-line consequence of the first-order condition at the minimizer. We record it in-book.

Lemma 6.8 (Envelope identity for the Bellman Hamiltonian)**.** *Let $b(x,\cdot), L(x,\cdot) \in C^2(A)$ with $A \subseteq \mathbb{R}^k$ open, and assume that for (x,p) in an open set the map $\mathfrak{H}(x,\cdot,p) := b(x,\cdot)\cdot p + L(x,\cdot)$ has a unique interior minimizer $a^*(x,p)$ at which the Hessian $\partial_{aa}\mathfrak{H}(x,a^*,p)$ is positive definite (both guaranteed by the strict convexity and coercivity of definition 6.7). Then a^* is C^1 in (x,p), the Bellman Hamiltonian $H(x,p) = -\mathfrak{H}(x, a^*(x,p), p)$ is C^1, and*

$$\partial_p H(x,p) = -b\big(x, a^*(x,p)\big), \qquad \partial_x H(x,p) = -\partial_x \mathfrak{H}\big(x, a^*(x,p), p\big). \tag{6.9}$$

Proof. The argument has two moves: first establish that the minimizer a^* depends smoothly on the parameters, then differentiate the optimal value and watch a term die.

Smooth dependence of the minimizer. Because $a^*(x,p)$ is an *interior* minimizer of the smooth map $a \mapsto \mathfrak{H}(x,a,p)$, it is a stationary point: the first-order condition

$$\partial_a \mathfrak{H}\big(x, a^*(x,p), p\big) = 0$$

holds identically in (x,p) (an interior extremum of a differentiable function has vanishing gradient in the optimized variable; were $\partial_a \mathfrak{H} \neq 0$ one could decrease $\mathfrak{H}$ by an infinitesimal step in the direction $-\partial_a \mathfrak{H}$, contradicting minimality). Read this as an implicit equation $F(x,p,a) := \partial_a \mathfrak{H}(x,a,p) = 0$ defining a in terms of (x,p). Its Jacobian in the unknown a is $\partial_a F = \partial_{aa}\mathfrak{H}(x,a^*,p)$, the Hessian, which is positive definite by hypothesis and hence invertible. The implicit function theorem therefore solves $F = 0$ locally for a C^1 map $(x,p) \mapsto a^*(x,p)$. This is where the positive definite Hessian earns its keep: it is the nondegeneracy that upgrades "the minimizer exists and is unique" to "the minimizer moves smoothly as we tilt the data," which is what the chain rule below will require.

Differentiating the optimal value: the envelope cancellation. Now differentiate $H(x,p) = -\mathfrak{H}(x,a^*(x,p),p)$. The subtlety is that (x,p) enters in *two* ways — explicitly, through the first and third slots of $\mathfrak{H}$, and implicitly, through the minimizer $a^*(x,p)$ in the second slot — so the chain rule produces an explicit term and an implicit term. Take p first:

$$\partial_p H = -\partial_p \mathfrak{H}(x,a^*,p) - \underbrace{\partial_a \mathfrak{H}(x,a^*,p)}_{=0}\,\partial_p a^* = -\partial_p \mathfrak{H}(x,a^*,p) = -b(x,a^*),$$

where the implicit term — the one carrying the derivative of the minimizer — is annihilated outright by the first-order condition $\partial_a \mathfrak{H}(x,a^*,p) = 0$, and the surviving explicit term is evaluated using $\partial_p \mathfrak{H}(x,a,p) = \partial_p\big(b(x,a)\cdot p + L(x,a)\big) = b(x,a)$. This cancellation of the implicit term *is* the envelope theorem: at an optimum the value is insensitive, to first order, to the motion of the optimizer, because the optimizer sits where the gradient in a already vanishes. The same computation in x runs verbatim. Differentiating $H(x,p) = -\mathfrak{H}(x,a^*(x,p),p)$ in x,

$$\partial_x H = -\partial_x \mathfrak{H}(x,a^*,p) - \underbrace{\partial_a \mathfrak{H}(x,a^*,p)}_{=0}\,\partial_x a^* = -\partial_x \mathfrak{H}(x,a^*,p) = -\big(\partial_x b(x,a^*)\,p + \partial_x L(x,a^*)\big),$$

the implicit term $\partial_a \mathfrak{H} \cdot \partial_x a^*$ vanishing for the identical reason — the first-order condition kills it whichever parameter we vary — and the explicit term read off from $\partial_x \mathfrak{H}(x,a,p) = \partial_x b(x,a)\,p + \partial_x L(x,a)$. The essential phenomenon is that $\partial_x H$ picks up only the *explicit* dependence of $\mathfrak{H}$ on x, never the dependence routed through the moving minimizer; that is the precise sense in which the value's x-derivative "does not see" how a^* shifts. Finally, both right-hand sides in (6.9) are continuous in (x,p), since a^* is continuous (indeed C^1) and b, L are C^2; a function with continuous first partials is C^1, so $H \in C^1$, as claimed. $\square$

Notation watch

Two Hamiltonians, two sign habits, and one genuine reader trap. This book runs *three* objects all called "Hamiltonian," and confusing their signs is the most reliable way to derive a wrong equation, so we lay them side by side.

The *Bellman* Hamiltonian (6.6), $H(x,p) = -\inf_a\{b(x,a)\cdot p + L(x,a)\}$, is built with its outer minus sign for exactly one reason: it makes the HJB equation read $-\partial_t u + H(x,\nabla u) = 0$, the form used uniformly in this book and in the mean field game literature ([93]; [36]), and the form into which the coupled system of chapter 8 will slot the forward Fokker–Planck equation. Many control texts instead carry the bare *Pontryagin* Hamiltonian $\mathfrak{H}(x,a,p) = b(x,a)\cdot p + L(x,a)$ of section 6.4, retaining the action a as an explicit argument, and write

the HJB equation as $\partial_t u + \inf_a \mathfrak{H} = 0$. The two descriptions are the *same* equation: they are related by $H(x,p) = -\inf_a \mathfrak{H}(x,a,p)$, the single outer sign converting $\inf_a \mathfrak{H}$ into $-H$ and, simultaneously, $+\partial_t u$ into $-\partial_t u$. That one sign is load-bearing. Misplace it and convexity in p silently becomes concavity, the infimum over actions becomes a supremum, and the envelope formula $\partial_p H = -b(\cdot, a^*)$ flips its sign — after which the feedback $\alpha^* = -\partial_p H$ of section 6.5 points the agent in exactly the wrong direction.
The classical-mechanics reader carries a third sign habit that must be quarantined. The physical Hamiltonian of section 6.1, $H_{\text{phys}}(x,p) = \sup_v \{p \cdot v - L\}$, is a Legendre transform with a $+p \cdot v$ inside it; the Bellman Hamiltonian (6.6) carries $-b \cdot p$ instead. The extra minus sign is not a convention one is free to set: it is the price of optimizing a *cost* that is to be *minimized* rather than an action that is to be *extremized*. In the calculus-of-variations corner $b(x,a) = a$ and the actions are velocities, so the two transforms differ *only* by that sign, and one can watch the discrepancy enter exactly where "least cost" replaces "stationary action."
The safe discipline is to trust no remembered sign at all: write down (6.7) and (6.6) as fixed in this book, and check every later formula against them rather than against a Hamiltonian half-remembered from mechanics.

These two formulas — the differentiability of H and the envelope identity $\partial_p H = -b(\cdot, a^*)$ — are the export of this subsection. They are what the costate identity of section 6.4 will lean on when it identifies the Pontryagin adjoint with ∇u, and what section 6.5 will lean on when it reads the optimal feedback straight off the Hamiltonian as $\alpha^* = -\partial_p H$. One piece of (6.7), however, we have written down but not yet earned: the terminal clause $u(T, \cdot) = g$, with the equation to be integrated *backward* from it. We took that data for granted in the derivation, where it dropped out of definition 6.3 for free; but it sits in flat contradiction with every forward-in-time evolution the physicist trusts, and a contradiction of that magnitude deserves to be explained rather than merely stated. The next subsection does exactly that — it confronts the backward direction of time head-on, and contrasts it with the time-symmetric, two-point boundary-value structure of the Euler–Lagrange picture of section 6.1 that the chapter opened from.

6.3.2 Why the time runs backward

We closed the formal derivation owing one clause. Equation (6.7) was assembled from the dynamic programming identity by a perfectly forward-looking calculation — expand a short first leg, divide by h, take a limit — and yet the equation it produced carries its data at the *final* time, $u(T, \cdot) = g$, and must be integrated *backward* toward $t = 0$. That terminal clause dropped out of definition 6.3 for free and we wrote it down without comment; but it sits in flat contradiction with every forward-in-time evolution the physicist trusts, and the preamble of this chapter promised that we would not dispatch the contradiction in a sentence. This is the promised reckoning. In the author's experience the backward arrow of HJB is the single largest conceptual hurdle of Part I, and the right response to a genuine conceptual hurdle is not to step lightly over it but to stand on it until it stops feeling like one. So we belabor it deliberately, and we begin by sharpening exactly what it is in tension with.

The tension is with the picture the chapter opened from. In the Euler–Lagrange account of section 6.1, time carries no preferred direction at all. The optimal curve γ^* solves the second-order equation (6.2), and one pins it down by clamping data at *both* ends of the interval — the start $\gamma(0) = x_0$ at the left, the transversality condition $\partial_v L + \nabla g = 0$ at the right — which is precisely the two-point boundary-value structure we dwelt on there. Half the data live in the present and half in the future, and there is no operational sense in which one "integrates the action forward": the equation knows nothing of a flow of time, only of a curve simultaneously anchored at its two ends. A physicist is entirely at home with this. It is the shape of every boundary-value problem in classical field theory, and the time-symmetry is not a defect but a reflection of the time-reversal invariance of the underlying variational principle. Dynamic programming now breaks that symmetry, and breaks it on purpose, and the first thing to absorb is that the break is a feature of the *question* being asked, not an artifact of the method.

So ask the question dynamic programming actually answers. The value $u(t, x)$ is, by definition 6.3, the least cost the agent can still incur *from here on*: "standing now at state x, with exactly $T - t$ of time remaining, what is the cheapest total cost I can still arrange to pay over what is left of the horizon?" Read it slowly, because the phrase "what is left" is the whole of the matter. The value at (t, x) is a fact about the *future* of t — about the interval $[t, T]$ and nothing earlier — and it is utterly indifferent to how the agent arrived at x or to what happened before t. The past is a sunk cost; only the remaining obligation is priced. A quantity defined as "the best I can still do over the time that remains" is one whose natural reference point is not the start of the interval but its *end*, because the amount of time that remains is measured from the end.

That reorientation toward the end of the interval is what selects the terminal time as the place to begin. Among all instants t, there is exactly one at which the question answers itself with no optimization whatsoever, and it is $t = T$. At the terminal time there is no time left, hence no decision to make, hence no infimum to compute: the agent is simply handed whatever terminal cost its position carries, and $u(T, x) = g(x)$ is *forced*, not chosen. This is the one slice of the entire space-time field u that is known outright, before any equation is solved — the boundary datum of definition 6.3 re-encountered as the natural starting surface of the computation. Every other slice $u(t, \cdot)$ with $t < T$ genuinely requires the optimization to be carried out; only the terminal slice is given. If one is going to build the field at all, one has no choice but to build it outward from the only place it is already pinned down.

The cleanest way to feel why "outward from T" means "backward in t" is to discretize the clock and watch the field fill in. Replace $[0, T]$ by the finitely many instants $T, T - h, T - 2h, \ldots, 0$ and imagine, at each instant, a layer of states. The terminal layer $u(T, \cdot) = g$ is known. To find the value one step earlier, at $T - h$, the agent standing at a state x contemplates each available action, pays the running cost it incurs over the step, lands at a new state, and there reads off the *already-known* value $u(T, \cdot)$; minimizing over actions gives $u(T - h, x)$. The layer at $T - h$ is thus computed purely from the layer at T. Repeat: the layer at $T - 2h$ is computed from the layer at $T - h$, and so on, each layer drawing only on the one immediately to its future, until the recursion reaches $t = 0$. This is exactly backward induction on a decision tree — the elementary algorithm one runs on any finite-horizon problem, filling the table from the leaves toward the root — and it is the discrete shadow of the dynamic programming identity (6.5) we have been wielding all along. The HJB equation (6.7) is the $h \downarrow 0$ limit of precisely this recursion, and it inherits the recursion's direction: information flows from later layers to earlier ones, never the reverse.

The differential equation wears that direction openly once one looks at what it relates. Rearranged, (6.7) reads $\partial_t u(t, \cdot) = H(x, \nabla u(t, \cdot))$: it expresses the rate of change of the value as the present recedes purely in terms of u and its spatial gradient ∇u *at the same instant* t. The equation is local in time — it ties the field at t to its own time-derivative there — so it propagates a known slice to an adjacent one, and the only slice we know to seed it with is the terminal one. Knowing the whole field $u(t+h, \cdot)$ at a slightly later time determines, through this local-in-time relation, the field $u(t, \cdot)$ a step earlier. The terminal condition is the sole surface on which the data are unambiguous; it is therefore the surface one must launch from; and from a terminal surface the only direction one can march is toward smaller t. The backward arrow is not imposed on HJB from outside — it is the equation's honest report of the fact that a cost-to-go is anchored at the end of the horizon.

Set this beside the forward problems the physicist knows in his bones, because the contrast is the lesson. A Newtonian trajectory is launched from a present position and velocity and pushed into an as-yet-unwritten future. The heat equation takes a present temperature profile and diffuses it forward. The Fokker–Planck flow of chapter 5 takes a present probability density and transports it forward along the dynamics. Every one of these is an *initial*-value problem, and they share a single epistemic posture: the system *knows its present state* and computes its future from it. HJB inverts that posture exactly. It does not know its present value — that is the unknown it is solving for — but it knows its *future obligation*, the terminal cost g it will be made to pay, and it pulls that obligation backward into the present. A forward evolution propagates a known present into an unknown future; HJB propagates a known future into an unknown present. Once stated that baldly the backward arrow stops being mysterious: a quantity that prices the future cannot help but be computed from the future.

It is worth defusing, before it can take hold, a fear that the word "backward" tends to provoke in anyone who remembers the backward heat equation. That equation — the heat equation run in reverse — is the textbook ill-posed problem: it amplifies high-frequency noise without bound and has no solution for generic data. One might worry that HJB, integrated backward, suffers the same pathology. It does not, and seeing why is clarifying. "Backward" here is a statement about *physical* time only. Introduce the reversed clock $\tau := T - t$, which runs from 0 at the terminal time to T at the present; then $\partial_t = -\partial_\tau$, the terminal datum at $t = T$ becomes an *initial* datum at $\tau = 0$, and HJB becomes an ordinary *forward* evolution in τ launched from g. The terminal-value problem in t and the initial-value problem in τ are the same problem in two dressings, and the second is manifestly well-posed: one integrates forward in τ from known data, exactly as for any initial-value problem. (When the diffusion of section 6.5 is switched on, this is the content of the sign on the second-order term: HJB carries the Laplacian with precisely the sign that makes the τ-evolution a forward, well-posed parabolic equation, the genuine *adjoint partner* of the forward Fokker–Planck, not the catastrophic backward heat equation.) The backward arrow is bookkeeping relative to the agent's clock, not a symptom of ill-posedness; HJB is as well-posed as the heat equation, merely oriented the other way in physical time.

This pairing of one backward equation against one forward equation is not an idiosyncrasy of the single-agent problem; it is the structural skeleton of the entire subject, and naming it now is the main forward-pointing payoff of this subsection. When chapter 8 finally releases the population m

from the frozen, inert role it plays throughout this chapter and lets it become the unknown of a Nash fixed point, two evolutions will run simultaneously on the same interval $[0, T]$ and in opposite temporal directions. The population density m will satisfy a *forward* Fokker–Planck equation carrying its data at $t = 0$ — because a density, like any of the physicist's initial-value flows, knows its present and is pushed into the future. The value u will satisfy the *backward* HJB equation carrying its data at $t = T$ — because a cost-to-go knows its future obligation and is pulled into the present. The two are welded at opposite ends of the interval: the density's datum is clamped at the left end, the value's datum at the right, and each equation's coefficients depend on the other's unknown. A system specified by data split across the two ends of time is a boundary-value problem in time, not an initial-value one — the very two-endpoint structure we met in the Euler–Lagrange problem of section 6.1, now lifted from a single curve to a coupled pair of evolution equations. That opposed-arrow welding has no initial-value analogue, it cannot be integrated straight through in either direction, and it is exactly what makes a mean field game the delicate object it is. The reader should fix the picture now and carry it as the recurring structural motif of the whole book: *forward density, backward value, welded at the ends* is the signature by which one recognizes an MFG on sight.

Physics note

The classical limit makes the kinship visible, and it is worth tracing because it locates exactly what Bellman added to a structure the physicist already owns. The Hamilton–Jacobi equation of analytical mechanics, $\partial_t S + H_{\text{phys}}(x, \nabla S) = 0$ for Hamilton's principal function S, is (6.7) stripped of the optimization: there is no $\inf_a$ because the "control" — the velocity — has already been eliminated once and for all by the Legendre transform that defined H_{phys} in section 6.1, and the sign on the Hamiltonian differs because S is an action to be *extremized* rather than a cost to be *minimized*. Bellman's contribution is to put the optimization back, but to put it back *inside* the Hamiltonian: H in (6.6) is itself a Legendre-type transform of the running cost over the admissible controls, so that the single elimination of the velocity in mechanics is replaced by a genuine minimization over a possibly bounded, possibly indirectly-acting control set. Strip that minimization away — take the calculus-of-variations corner $b(x, a) = a$, $A = \mathbb{R}^d$, where the control is the velocity and the infimum is attained by the same stationarity that the Legendre transform performs — and HJB collapses back onto Hamilton–Jacobi. The principal function S is the un-optimized ancestor; the value u is an optimized principal function. The kinship runs further. The eikonal equation $|\nabla S| = n(x)$ of geometric optics is the stationary, first-order specialization of Hamilton–Jacobi, its characteristics are light rays, and its solutions are notorious for developing gradient kinks where rays cross — the *caustics* of optics, the bright cusped envelopes one sees at the bottom of a teacup. At such a crossing the principal function is continuous but its gradient is multivalued, precisely the corner phenomenon we saw (6.6) permit when the maximizing action ties. This is the same kink that will force us, in section 6.6, to abandon classical solutions of HJB for the weak notion of viscosity solutions: the caustic of geometric optics and the shock of the value function are one phenomenon, met first by the physicist in optics and about to be met again, with the optimization reinstated, in control.

That last observation is the hinge to the two subsections that follow. The kinks the physicsnote just flagged — the places where rays cross and ∇u ceases to exist — are exactly the failure of smoothness that section 6.6 will be built to handle, and we hand the caustic phenomenon forward to it intact. More immediately, the whole of this subsection, and indeed of the formal derivation that preceded it, has quietly leaned on u being differentiable enough to write $\partial_t u$ and ∇u and to expand the dynamic programming identity to first order. The physicsnote has now warned us, in the reader's own language of caustics, that this smoothness is not automatic — the value can crease — and the formal derivation already confessed that it had merely *assumed* $u \in C^1$ and trusted an uncontrolled $h \downarrow 0$ limit. Two debts, then, are outstanding: the derivation assumed regularity it did not establish, and the limit it took was never justified. The next subsection retires both at a stroke by inverting the logic. Rather than derive HJB from the value function and hope the value is smooth, it will *guess* a smooth solution of the equation and prove directly — by Itô's formula alone, with no appeal to the dynamic programming principle and no a priori regularity of u — that any such solution must be the value function and that its feedback is optimal. That clean, fully rigorous guess-and-check is the verification theorem.

6.3.3 The verification theorem

The previous subsection closed by tallying two debts that the formal derivation had quietly run up. First, every step of that derivation — writing $\partial_t u$ and ∇u, Taylor-expanding the dynamic programming identity to first order — presumed that u was differentiable, and the physicsnote's caustics have just warned us that the value can crease, so that smoothness is precisely what is *not* given to us in advance. Second, even granting smoothness, the passage $h \downarrow 0$ that produced the PDE was never controlled; we divided by h and waved the remainder away. A calculation carrying two unpaid debts is a fine *derivation* — it tells us what equation to write and why it has the shape it does — but it is not a proof that the equation governs the value function. The honest move now is not to patch the derivation but to abandon it.

The device is a logical inversion that the reader has met before in mathematics under the name guess-and-check, and it is worth pausing on why it is the clean route rather than a shortcut. The derivation runs *forward*, from the value function to the PDE: assume u exists and is smooth, manipulate it, and arrive at HJB. Inverting it means running *backward*, from the PDE to the value function: *forget* where the equation came from, hand ourselves any sufficiently smooth function w that happens to solve it, and prove from scratch that this w must coincide with u and that the feedback read off from w is optimal. The payoff of the inversion is total. We never again need to know that u is smooth — we need to know that *some* solution w is smooth, and that is a hypothesis we can simply impose and then verify in each concrete problem by exhibiting the w. We never take a limit, so there is no $h \downarrow 0$ to justify. And — this is the decisive economy, the one flagged already in remark 6.6 — we never invoke the dynamic programming principle at all: no infimum need be shown to be attained, no ε-optimal control need be selected measurably along random paths. The entire argument rests on a single tool, Itô's formula, applied once. That is why we were free to treat the whole of section 6.2 as motivation: the rigorous spine of the chapter is the theorem below, and it stands on Itô alone.

We state and prove the theorem in full stochastic generality from the outset, even though no noise has yet been switched on, and this costs us nothing — a point worth making explicit because it looks like premature abstraction and is not. The deterministic control problem of sections 6.1 to 6.3 is exactly the special case $\sigma \equiv 0$ of a controlled diffusion: kill the noise coefficient and the SDE (6.10) below collapses to the controlled ODE (6.3), the second-order term $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 w)$ in the HJB equation (6.11) vanishes, and the equation reduces to the first-order HJB (6.7). Crucially, the proof does not branch on whether σ is zero. Itô's formula carries the second-order term along automatically, and that term is the *only* place the noise enters the argument; setting $\sigma \equiv 0$ simply deletes it, turning every expectation into an evaluation along the deterministic flow and the stochastic integral into the zero process. Writing the stochastic statement therefore subsumes the deterministic one at no extra length, and it has the further virtue of being exactly the form section 6.5 will reach for when the diffusion is made permanent. We pay the generality once, here.

So fix the data and consider the controlled diffusion, with W an $\mathbb{R}^d$-valued Brownian motion on a filtered probability space satisfying the usual conditions (chapter 5),

$$\mathrm{d}X_s = b(X_s, \alpha_s)\,\mathrm{d}s + \sigma(X_s)\,\mathrm{d}W_s, \qquad X_t = x, \tag{6.10}$$

and cost $J(t,x;\alpha) = \mathbb{E}\Big[\int_t^T L(X_s,\alpha_s)\,\mathrm{d}s + g(X_T)\Big]$, the expectation having replaced the deterministic evaluation because the trajectory is now random. Before we can speak of optimizing over controls we must say precisely which controls are in the running, and each clause of the definition that follows is forced by a distinct demand of the problem rather than chosen for convenience.

Definition 6.9 (Admissible controls)**.** An *admissible* control on $[t,T]$ is a progressively measurable process $\alpha\colon [t,T]\times\Omega \to A$ for which (6.10) has a unique strong solution and $\mathbb{E}\Big[\int_t^T |L(X_s,\alpha_s)|\,\mathrm{d}s + |g(X_T)|\Big] < \infty$. The class is denoted $\mathcal{U}_t$.

Read the three requirements as three separate guarantees, each indispensable. That the control take values in A is the modelling constraint — the agent may only set its knobs within the physically available action set. That it be *progressively measurable* is the informational and analytic backbone: it says the control at time s may depend on the noise observed up to time s but not on the future, which is what makes a feedback law $\alpha_s = \phi(s, X_s)$ legitimate and, just as importantly, what makes the Itô integral $\int \sigma\,\mathrm{d}W$ in (6.10) *defined at all*, since the Itô construction requires an adapted integrand. That (6.10) have a *unique strong solution* is what entitles us to speak of "the" trajectory the control induces — without it the object X, and hence the cost, would be ambiguous; under the standing Lipschitz and growth hypotheses (A-Lip), (A-Growth) this is exactly the well-posedness guaranteed by theorem 5.10. The final integrability clause is what makes the cost $J(t,x;\alpha)$ a genuine finite number rather than a meaningless $\infty - \infty$, so that the infimum defining u is taken over a set of real-valued competitors. Only with all three in force is the optimization problem we are about to solve even posed.

Theorem 6.10 (Verification)**.** *Assume* (A-Lip)*,* (A-Growth)*,* (A-Conv)*. Let $w \in C^{1,2}([0,T]\times\mathbb{R}^d)$ have at most polynomial growth in x, uniformly in t, and solve the stochastic HJB equation*

$$-\partial_t w - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 w\big) + H(x,\nabla w) = 0 \quad \text{on } [0,T)\times\mathbb{R}^d, \qquad w(T,\cdot) = g, \tag{6.11}$$

with H as in (6.6). Then:

(i) *$w(t,x) \le J(t,x;\alpha)$ for every (t,x) and every $\alpha \in \mathcal{U}_t$; hence $w \le u$.*

(ii) *If the feedback (6.8) built from w, $\alpha^*_s = a^*(X^*_s, \nabla w(s, X^*_s))$, is admissible and the closed-loop equation (6.10) has a unique strong solution X^*, then equality holds: $w(t,x) = u(t,x) = J(t,x;\alpha^*)$, and α^* is optimal.*

Proof. The whole proof is a single application of Itô's formula followed by a sign inequality, a substitution, and a passage to expectations; we narrate it slowly because each move is elementary and the accumulation is the point. Fix (t,x) and an arbitrary $\alpha \in \mathcal{U}_t$ with induced state X. Because $w \in C^{1,2}$ we may apply Itô's formula (theorem 5.9) to the process $s \mapsto w(s, X_s)$ on $[t,T]$; with the drift and diffusion of (6.10) inserted into the Itô correction, it reads

$$w(T, X_T) = w(t,x) + \int_t^T \Big(\partial_s w + b(X_s, \alpha_s) \cdot \nabla w + \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 w)\Big)(s, X_s)\, \mathrm{d}s + \int_t^T \nabla w \cdot \sigma \, \mathrm{d}W_s.$$

This decomposes the change in w along the trajectory into a $\mathrm{d}s$ (bounded-variation) part, carrying the time derivative, the first-order transport $b \cdot \nabla w$, and the second-order Itô term $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 w)$, plus a stochastic-integral part $\int \nabla w \cdot \sigma \, \mathrm{d}W$ that is, a priori, only a *local* martingale. We will dispose of the stochastic part last; first we extract a one-sided bound on the $\mathrm{d}s$-integrand.

The bound comes for free from the very definition of the Hamiltonian. Recall from (6.6) that $H(x,p) = -\inf_{a \in A}\{b(x,a)\cdot p + L(x,a)\}$; the infimum being a lower bound for each competitor, every action value $a \in A$ — in particular the value $a = \alpha_s$ realized by our control at time s — satisfies

$$b(X_s, a) \cdot \nabla w(s, X_s) + L(X_s, a) \;\ge\; -H\big(X_s, \nabla w(s, X_s)\big),$$

with equality precisely when a is the minimizer $a^*(X_s, \nabla w(s, X_s))$ of (6.6). This is the entire role of the Hamiltonian inequality: it is nothing more than "the value of a function at a point is at least its infimum," but it is exactly the lever that converts the control-dependent integrand into a control-independent one. Now spend the PDE. The HJB equation (6.11) says that on $[t,T)$, $\partial_s w + \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 w) = H(\cdot, \nabla w)$ identically; substitute this into the $\mathrm{d}s$-integrand to eliminate the time- and second-order derivatives in favour of H, and then apply the Hamiltonian inequality:

$$\partial_s w + b \cdot \nabla w + \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 w) = H(\cdot, \nabla w) + b \cdot \nabla w \;\ge\; -L(X_s, \alpha_s),$$

where the inequality rearranges $b \cdot \nabla w + L \ge -H$, and is an equality iff $\alpha_s = a^*$. The $\mathrm{d}s$-integrand of the Itô expansion is thus bounded below by $-L(X_s, \alpha_s)$, pointwise in s and ω — which is the crux: we have traded the unknown w-derivatives for the known running cost, at the price of one inequality that we know how to make tight.

It remains to take expectations, and here the one genuinely technical step appears: we must know that the stochastic integral $M_\cdot := \int_t^\cdot \nabla w(s, X_s) \cdot \sigma(X_s)\, \mathrm{d}W_s$ has zero expectation at $s = T$. Itô's formula delivers M only as a *local* martingale, and a local martingale need not have constant (or even finite) expectation; what we need is that M is a *true* martingale, for which it suffices that its integrand be square-integrable in the sense $\mathbb{E}\Big[\int_t^T |\nabla w(s, X_s)\, \sigma(X_s)|^2 \, \mathrm{d}s\Big] < \infty$ — this is the L^2 bound that, via the Itô isometry, makes M a genuine L^2-bounded martingale, whence $\mathbb{E}[M_T] = \mathbb{E}[M_t] = 0$. That this bound holds is a short chain of growth estimates, and it is worth assembling explicitly because it is the one place the hypotheses of the theorem and the inheritance from chapter 5 are all spent at once. The hypothesis that w has at most polynomial growth in x uniformly in t passes to its gradient: $|\nabla w(s, y)| \leq C(1 + |y|^q)$ for some exponent q and constant C. The diffusion coefficient has at most linear growth, $|\sigma(y)| \leq C(1 + |y|)$, which is part of the Lipschitz hypothesis (A-Lip) (a globally Lipschitz function grows at most linearly). Multiplying and squaring, the integrand obeys

$$|\nabla w(s, X_s)\, \sigma(X_s)|^2 \;\leq\; C\big(1 + |X_s|\big)^2 \big(1 + |X_s|^q\big)^2 \;\leq\; C'\big(1 + |X_s|^{2q+2}\big),$$

a fixed polynomial in $|X_s|$. Finally the state itself has finite moments of every order, uniformly on the compact interval: $\sup_{s\in[t,T]} \mathbb{E}\big[|X_s|^{2q+2}\big] < \infty$, which is exactly the moment estimate that accompanies strong well-posedness of the SDE under (A-Lip), (A-Growth) in theorem 5.10. Taking expectations of the displayed bound and integrating over the finite interval $[t, T]$ therefore yields $\mathbb{E}\Big[\int_t^T |\nabla w\, \sigma|^2 \, \mathrm{d}s\Big] < \infty$, closing the chain: polynomial growth of ∇w, linear growth of σ, and finite moments of X combine to put the integrand in L^2, which upgrades M from local to true martingale and kills its expectation. (When $\sigma \equiv 0$ the integrand is identically zero and this entire paragraph is vacuous; the martingale subtlety is purely the noise's contribution, consistent with our remark that the diffusion is the only place randomness enters.)

With the stochastic term's expectation gone, take $\mathbb{E}[\cdot]$ across the Itô identity, use the terminal condition $w(T, \cdot) = g$, and discard the now-favourable sign of the ds-integrand:

$$\mathbb{E}[g(X_T)] = w(t,x) + \mathbb{E}\left[\int_t^T \big(\partial_s w + b\cdot\nabla w + \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 w)\big)\,\mathrm{d}s\right] \;\geq\; w(t,x) - \mathbb{E}\left[\int_t^T L(X_s,\alpha_s)\;\mathrm{d}s\right].$$

Rearranging, $w(t, x) \leq \mathbb{E}\Big[\int_t^T L \,\mathrm{d}s + g(X_T)\Big] = J(t, x; \alpha)$. Since $\alpha \in \mathcal{U}_t$ was arbitrary, taking the infimum over admissible controls gives $w(t, x) \leq u(t, x)$, which is (i): the smooth subsolution-and-solution w underbounds the cost of *every* strategy at once, hence underbounds the value.

For (ii) we run the identical computation along the closed loop and watch each inequality tighten to an equality. Let α^* be the feedback (6.8) built from w, $\alpha^*_s = a^*(X^*_s, \nabla w(s, X^*_s))$, applied along its own state X^* (admissible, and generating a unique strong solution, by hypothesis). At every time s the realized action α^*_s is by construction the minimizer in (6.6), so the Hamiltonian inequality holds with *equality*: $b \cdot \nabla w + L = -H(\cdot, \nabla w)$ along X^*. Consequently the ds-integrand equals exactly $-L(X^*_s, \alpha^*_s)$, the same L^2 bound makes the stochastic term a martingale of zero expectation, and the chain above runs with "$\geq$" replaced throughout by "$=$," giving $w(t, x) = J(t, x; \alpha^*)$. Now assemble: (i) gives $w \leq u$; the definition of u as an infimum over admissible controls gives $u(t, x) \leq J(t, x; \alpha^*) = w(t, x)$ because α^* is one such control; the two inequalities pinch, so $w(t, x) = u(t, x) = J(t, x; \alpha^*)$ and α^* attains the value, i.e. is optimal. Nothing beyond Itô's formula and the moment bound entered at any step — in particular the dynamic programming principle was never invoked, vindicating the claim of remark 6.6 that verification, not the DPP, is the chapter's load-bearing spine. □

The shape of this argument, not just its conclusion, is the lasting deliverable, and it is worth abstracting into a recipe because it is the play every explicit computation in this book will run. The theorem turns the solution of an optimal control problem into three mechanical steps. First, *propose an ansatz*: write down a candidate w of a form suggested by the problem's symmetry or structure, with free functions or constants to be determined. Second, *check the PDE*: insert the ansatz into the HJB equation (6.11) and into the terminal condition $w(T,\cdot) = g$, and let the requirement that it solve them pin down the free data — typically reducing the PDE to a system of ordinary differential equations the ansatz must satisfy. Third, *read off the feedback*: the minimizer $a^*(x, \nabla w)$ in the Hamiltonian, evaluated at the verified w, *is* the optimal feedback (6.8), delivered for free as a byproduct of having solved the PDE. No separate optimization over controls is ever performed; the verification theorem certifies that solving the PDE *is* solving the control problem. Section 6.7 runs precisely this play on the linear–quadratic–Gaussian atom: the ansatz is a quadratic $w(t,x) = \frac{1}{2}P(t)x^2 + \dots$, checking the PDE collapses it to the scalar Riccati equation for $P(t)$, and the optimal feedback falls out as a linear rule in the state. Every later explicit solution in the book is a variation on these three steps.

The one price the theorem exacts is the hypothesis it cannot do without: $w \in C^{1,2}$. The proof applied Itô's formula to $w(s, X_s)$, and Itô's formula needs one time derivative and two space derivatives to exist — without them the very first line of the argument is meaningless. For the second-order problems of section 6.5, where a nondegenerate diffusion smooths the value function, this regularity is typically available and the theorem applies as stated. But for first-order problems — the deterministic case $\sigma \equiv 0$, which is where the physicist's least-action instinct lives — it generically fails: the value function creases along the caustics the physicsnote exhibited, ∇u ceases to exist on a nonempty set, and there is simply no classical w to feed the theorem. The verification machinery, for all its rigour, is then silent. Repairing this — recovering a notion of solution, and a verification principle, that survives the kinks — is exactly the task of section 6.6, where viscosity solutions replace the pointwise derivatives by a test-function inequality that asks nothing of u beyond continuity. Before turning the diffusion on or weakening the solution concept, however, we pause to develop the complementary, trajectory-wise picture this field-based theory has so far suppressed: where verification produces the entire value field $u(t,x)$ at once, the Pontryagin maximum principle of the next section isolates a single optimal path and characterizes it through a forward–backward system, reuniting us with the costate $p = \nabla u$ glimpsed back in section 6.1.

6.4 The Pontryagin maximum principle and the costate

The verification theorem delivered the field method in its finished form: solve one partial differential equation over the whole space-time slab and read the optimal feedback off its gradient at every state at once. That global reach is the method's strength — and, measured against the trajectory the agent actually traverses, also its extravagance. An agent launched from a fixed x_0 rides a single optimal curve X^*; the value field $u(t,x)$ attaches a number to the uncountably many states the agent never visits, and computing all of them merely to characterize the one path it does follow is, from the trajectory's point of view, a detour. Dynamic programming is in this precise sense irreducibly global. The *Pontryagin maximum principle* (PMP) is its local, trajectory-wise counterpart: it bypasses the field entirely and pins down the single optimal trajectory directly, through a coupled system of ordinary differential equations that live only *along that one path.* Where verification answers "what is the best cost from anywhere?" and extracts the agent's path as an afterthought, the PMP asks only "what equations does the agent's own optimal path satisfy?" and never leaves it.

There are three reasons to develop this complementary picture even though HJB already settles the problem in principle, and each will pay off later. The first is genealogical: the PMP is the direct descendant of the Euler–Lagrange and Hamilton canonical equations the physicist already owns from section 6.1. The conjugate momentum $p = \partial_v L$ introduced there, which we flagged would return under a second name, is about to return as the PMP *costate* — the adjoint variable conjugate to the state — so the trajectory method is the reader's least-action reflex reorganized for control, not a foreign import. The second is structural and computational: the PMP replaces a PDE over all of state space by a system of ODEs along a single curve, an object far cheaper to solve and, decisively, the only one of the two pictures that survives intact into the *stochastic* setting of chapter 9. There the costate ODE becomes a backward stochastic differential equation and the forward–backward pair becomes the FBSDE that is the computational backbone of the entire probabilistic approach to mean field games. The third is the conceptual payload, and it is what this section exists to deliver: the costate p turns out to coincide, along the optimal path, with the gradient ∇u of the Bellman value. That identity is the bridge welding the local and global methods into one story, and it is the object on which chapter 9's BSDE will rest. We treat the deterministic problem (6.3)–(6.4) here and preview the stochastic case at the section's end.

The instrument is the Hamiltonian we have already met. Recall from the notationwatch of section 6.3 the *Pontryagin Hamiltonian*

$$\mathfrak{H}(x, a, p) \;=\; b(x, a) \cdot p + L(x, a),$$

which retains the action a as an explicit argument rather than optimizing it away, so that the Bellman Hamiltonian of (6.6) is its optimized negative, $H(x, p) = -\inf_a \mathfrak{H}(x, a, p)$. The single sign that relates the two will recur at every step below, so it is worth holding fixed now: $\mathfrak{H}$ is the thing the optimal action *minimizes*, and H is minus that minimum. With this object in hand the maximum principle reads as follows.

Theorem 6.11 (Pontryagin maximum principle, deterministic)**.** *Assume $b, L, g \in C^1$ with b, L satisfying* (A-Lip)*,* (A-Growth)*, and let α^* be an optimal control with state X^*. Then there is an absolutely continuous* costate $p\colon [0,T] \to \mathbb{R}^d$ *solving the backward* adjoint equation

$$\dot p_t = -\partial_x \mathfrak{H}(X_t^*, \alpha_t^*, p_t) = -\big(\partial_x b(X_t^*, \alpha_t^*)\big)^{\top} p_t - \partial_x L(X_t^*, \alpha_t^*), \qquad p_T = \nabla g(X_T^*), \tag{6.12}$$

and the optimal control minimizes the Pontryagin Hamiltonian pointwise,

$$\alpha_t^* \in \arg\min_{a \in A} \mathfrak{H}\big(X_t^*, a, p_t\big), \qquad a.e.\ t \in [0, T]. \tag{6.13}$$

Read the theorem as a description of a single self-consistent loop in time, and notice that its data are split across the two ends of the interval exactly as the Euler–Lagrange boundary-value problem of section 6.1 was. The state X^* runs *forward* from the known start $X_0^* = x_0$, integrating the controlled dynamics (6.3). The costate p runs *backward* from the terminal value $p_T = \nabla g(X_T^*)$, integrating the adjoint equation (6.12). And the two are not solved in isolation: at each instant the pointwise minimization (6.13) reads off the action α_t^* from the current pair (X_t^*, p_t), and that action feeds back into both the forward drift and the backward adjoint. The system is therefore genuinely *forward–backward and coupled*: one cannot run the state forward without knowing the control, the control is set by the costate, and the costate is fixed by a condition at the far end of time that depends on where the state lands. This circular dependence — a datum clamped at $t = 0$, another clamped at $t = T$, and a minimization knitting them together instant by instant — is precisely the two-point boundary-value structure the chapter's opening warned could not be integrated by marching forward from the present. It is the same structure that will reappear, welded to a forward population density, as the mean field game system of chapter 8; meeting it here on a single trajectory is the cleanest possible rehearsal.

One honesty is owed immediately. We state the maximum principle but do not prove that the adjoint equation and the minimum condition are *necessary* — that every optimal control must satisfy them. The clean route to necessity is the needle, or spike, variation: one perturbs the optimal control on a vanishingly short time interval by an arbitrary admissible value, propagates the resulting first-order perturbation of the state forward through the linearized dynamics, and demands that the induced first-order change in the cost be nonnegative; the costate is exactly the adjoint (transpose-linearized) variable that converts that forward sensitivity into the backward equation (6.12), and the arbitrariness of the needle's value forces the pointwise minimization (6.13). The argument is carried out completely in [136, Ch. 2] and [97], and we send the reader there rather than reproduce it. This is a deliberate, honest deferral, not a gap we are papering over, and the reason is that *nothing later in this book depends on the necessity of the deterministic PMP*. What the development actually consumes is two things, and neither needs the spike-variation argument. The first is the in-book costate identity $p_t = \nabla u(t, X_t^*)$ of proposition 6.12, which we prove below directly from the HJB equation — it *constructs* a costate with all the asserted properties out of the value function, rather than asserting one must exist. The second is the stochastic FBSDE machinery of chapter 9 (and again chapter 23), which is built there on its own self-contained foundation and does not import theorem 6.11. We therefore spend our effort only on the short argument that exposes what the costate *means*, because that meaning, and not the necessity proof, is the conceptual payload and the thread that ties the PMP back to HJB and forward to the BSDE.

Proposition 6.12 (Costate is the gradient of the value)**.** *Suppose the value function u of definition 6.3 is $C^{1,2}$ and the optimal feedback* (6.8) *generates the optimal trajectory X^*. Then $p_t := \nabla u(t, X_t^*)$ solves the adjoint equation* (6.12)*, and* (6.13) *holds.*

Proof. The whole proof is a single application of the chain rule, but it turns on two bookkeeping facts and one cancellation, so we spell each out. The strategy is to *define* $p_t := \nabla u(t, X_t^*)$ — the spatial gradient of the value, read along the optimal trajectory — and then simply differentiate this definition in t, using HJB to evaluate the time derivative of u that appears. Everything the theorem asserts about the costate will fall out of that one differentiation.

We will use the two envelope identities (6.9) of lemma 6.8. They apply here because, along the optimal path, the action that minimizes the Pontryagin Hamiltonian is the optimal control itself, $a^*(X_t^*, p_t) = \alpha_t^*$ — this is exactly what the feedback (6.8) encodes — so the two identities, evaluated at this minimizer, read

$$\partial_p H(x, p) = -b\big(x, a^*(x, p)\big), \qquad \partial_x H(x, p) = -\partial_x \mathfrak{H}\big(x, a^*(x, p), p\big).$$

The first says the slope of the optimized Hamiltonian in the momentum slot is minus the optimal drift; the second says its slope in the state slot ignores the dependence routed through the moving minimizer (the content of the envelope theorem) and equals minus the bare state-derivative of $\mathfrak{H}$. Both will be spent below, the first now and the second at the very end.

Step 1: rewrite the optimal velocity through the Hamiltonian. The optimal state obeys its own dynamics, $\dot{X}_t^* = b(X_t^*, \alpha_t^*)$. Substituting $\alpha_t^* = a^*(X_t^*, p_t)$ and using the first envelope identity converts this into a statement purely about H:

$$\dot{X}_t^* = b\big(X_t^*, a^*(X_t^*, p_t)\big) = -\partial_p H\big(X_t^*, \nabla u(t, X_t^*)\big). \tag{6.14}$$

This is the first place the canonical pairing surfaces: the optimal velocity is read off the p-gradient of the Hamiltonian, evaluated at $p = \nabla u$.

Step 2: differentiate the costate along the trajectory. Now differentiate the definition $p_t = \nabla u(t, X_t^*)$ in t. Here ∇u is evaluated at the moving point (t, X_t^*), so the total t-derivative collects an explicit time-derivative and a chain-rule contribution from the motion of X_t^*:

$$\dot{p}_t = \partial_t(\nabla u)(t, X_t^*) + \mathrm{D}^2\, u(t, X_t^*)\, \dot{X}_t^*.$$

The Hessian $\mathrm{D}^2\, u = \partial_x(\nabla u)$ is the spatial Jacobian of the vector field ∇u, and it multiplies the velocity $\dot{X}_t^*$ exactly as the chain rule for a moving evaluation point demands. Both terms are well defined precisely because $u \in C^{1,2}$ — this is the hypothesis at work: $C^{1,2}$ guarantees $\mathrm{D}^2\, u$ exists and is continuous, and it guarantees the mixed partials commute, $\partial_t(\nabla u) = \nabla(\partial_t u)$, which we use to rewrite the first term:

$$\dot{p}_t = \nabla(\partial_t u)(t, X_t^*) + \mathrm{D}^2\, u(t, X_t^*)\, \dot{X}_t^*. \tag{6.15}$$

Without $C^{1,2}$ neither $\mathrm{D}^2\, u$ nor the interchange of derivatives is licensed, and the whole computation collapses — which is exactly why the deterministic, first-order case where u creases will instead need the costate to be reconstructed by the weak machinery flagged for section 6.6, and the noisy, smoothing case to be handled by the BSDE below.

Step 3: evaluate $\nabla(\partial_t u)$ from HJB. The HJB equation (6.7), rearranged into the evolution form $\partial_t u(t, x) = H\big(x, \nabla u(t, x)\big)$, gives the time-derivative of u as a function of x. Take its spatial gradient. The right-hand side depends on x in *two* ways — directly through the first slot of H, and indirectly through the second slot $p = \nabla u(t, x)$ — so the chain rule produces two terms:

$$\nabla(\partial_t u)(t, x) = \partial_x H(x, \nabla u) + \mathrm{D}^2\, u(t, x)\, \partial_p H(x, \nabla u),$$

where the second term arises because $\partial_x(\nabla u) = \mathrm{D}^2\, u$ is the Jacobian of the inner argument ∇u, multiplying the p-slope $\partial_p H$ of the outer function.

Step 4: the cancellation. Insert this expression for $\nabla(\partial_t u)$ into (6.15), evaluating everything at $x = X_t^*$:

$$\dot{p}_t = \underbrace{\partial_x H(X_t^*, \nabla u)}_{\text{from Step 3}} + \mathrm{D}^2\, u(t, X_t^*)\, \partial_p H(X_t^*, \nabla u) + \mathrm{D}^2\, u(t, X_t^*)\, \dot{X}_t^*.$$

Now Step 1 is spent: $\dot{X}_t^* = -\partial_p H(X_t^*, \nabla u)$ by (6.14), so the third term is exactly the negative of the second. The two Hessian terms cancel identically, and not by accident — the Hessian appears once through the curvature of the value field along the moving trajectory (Step 2) and once through the indirect x-dependence of HJB (Step 3), and the canonical relation $\dot{X}^* = -\partial_p H$ is precisely the statement that these two appearances are equal and opposite. What remains is

$$\dot{p}_t = \partial_x H\big(X_t^*, \nabla u(t, X_t^*)\big) = -\partial_x \mathfrak{H}\big(X_t^*, \alpha_t^*, p_t\big),$$

the last equality being the second envelope identity. This is exactly the adjoint equation (6.12).

Step 5: the boundary and minimum conditions. The terminal value is read straight off the definition together with the terminal datum of HJB: since $u(T, \cdot) = g$, taking gradients gives $\nabla u(T, \cdot) = \nabla g$, hence $p_T = \nabla u(T, X_T^*) = \nabla g(X_T^*)$ — the costate inherits its endpoint condition from the value's, at the same far end of time. Finally, the minimum condition (6.13) holds because $\alpha_t^* = a^*(X_t^*, p_t)$ is by construction the minimizer of $a \mapsto \mathfrak{H}(X_t^*, a, p_t)$, which is the definition of a^*. This completes the verification that the constructed $p_t = \nabla u(t, X_t^*)$ is a costate in the sense of theorem 6.11. □

The identification $p_t = \nabla u(t, X_t^*)$ is the conceptual payload of the section, and it deserves to be read slowly, because it collapses what looked like two independent theories into one. The PMP costate — the adjoint variable that the spike-variation argument would have produced as a Lagrange multiplier enforcing the dynamics — is nothing other than the gradient of the Bellman value, sampled along the optimal path. This is the precise content of the slogan "momentum is conjugate to value." Trace its three incarnations: in section 6.1 the conjugate momentum was $p = \partial_v L$, the slot dual to velocity in the Legendre transform; here it is the PMP costate, the variable dual to the state in the adjoint equation; and now we learn that both equal ∇u, the sensitivity of the optimal cost-to-go to a displacement of the current state. One quantity, three faces — Legendre momentum, Pontryagin costate, value-gradient — and the identity above is the hinge that makes the local trajectory method and the global field method two readings of a single object rather than rival computations. Economically it is the marginal cost of state: p_t measures how much the remaining optimal cost changes if the agent is nudged off the optimal path at time t, which is exactly why it is the natural "price" of the state and why the running minimization (6.13) weighs each action by $b \cdot p$, the rate at which that action moves the state against its price.

The identity also explains, with no further computation, the one structural feature of the adjoint equation that would otherwise look arbitrary: *why the costate runs backward.* A first-order ODE needs its data at one end, and (6.12) takes them at $t = T$, not $t = 0$. The reason is now transparent — the costate is ∇u, and u itself is the backward object of the chapter, born at the terminal time from $u(T, \cdot) = g$ and propagated toward the present. Differentiating that terminal datum in space gives $p_T = \nabla g(X_T^*)$, so the costate is pinned at exactly the end where the value is, and inherits its backward direction wholesale. The forward state and the backward costate are not two unrelated equations that happen to be glued by (6.13); they are the trajectory-wise shadow of the chapter's central asymmetry — initial data pushed forward for the thing that flows, terminal data discharged backward for the thing that prices — and that very pairing is what chapter 8 will promote into the coupled forward–backward system of a mean field game.

Remark 6.13 (The envelope identity and $\alpha^* = -\partial_p H$)**.** The first envelope identity $\partial_p H(x, p) = -b\big(x, a^*(x, p)\big)$ used in the proof is worth isolating and handling with care, because it is the general form of the compact formula recorded in appendix A as $\alpha^* = -\partial_p H$ — and that compact formula is true only under a normalization that is easy to forget, generating a recurring sign-and-scaling error if it is applied blindly. The point of this remark is to make the distinction load-bearing rather than incidental.

The clean statement is that the envelope identity always delivers the optimal *drift*, never directly the optimal *control*. What it says is

$$b(x, a^*(x,p)) = -\partial_p H(x,p);$$

the left side is the velocity the optimal action induces, and only after one inverts the map $a \mapsto b(x,a)$ does a statement about the control itself emerge. Two regimes follow.

In the *canonical normalization*, where the control simply *is* the drift, $b(x,a) = a$ with $A = \mathbb{R}^d$, the inversion is the identity and the drift formula collapses onto the control: $a^*(x,p) = -\partial_p H(x,p)$ literally, and the optimal feedback is $\alpha^*(t,x) = -\partial_p H\big(x, \nabla u(t,x)\big)$. This is the form appendix A records and the form we will use whenever stating the general MFG theory in chapter 8 onward, precisely because in that canonical normalization the bookkeeping is clean and the control is the gradient of the Hamiltonian with a sign.

In any genuinely *control-affine* case the inversion is nontrivial and the collapse fails. Take the scalar LQG drift of section 6.7, $b(x,\alpha) = ax + b\,\alpha$, in which an uncontrolled state drift ax is present and a gain b multiplies the control, with running cost $L(x,\alpha) = \frac{1}{2}\alpha^2$ (a state-cost term plays no role in this extraction and we drop it). Then $\mathfrak{H}(x,\alpha,p) = (ax + b\alpha)p + \frac{1}{2}\alpha^2$, and minimizing over α gives the first-order condition $bp + \alpha^* = 0$, so $\alpha^* = -b\,p$. The induced optimal drift is $ax + b\alpha^* = ax - b^2 p$. Let us check this against the envelope identity by computing H directly:

$$H(x,p) = -\mathfrak{H}(x,\alpha^*,p) = -\big[(ax - b^2 p)\,p + \tfrac{1}{2}b^2p^2\big] = -axp + \tfrac{1}{2}b^2p^2, \qquad \text{so} \quad -\partial_p H(x,p) = ax - b^2 p.$$

Indeed $-\partial_p H = ax - b^2 p$ equals the optimal *drift* $ax + b\alpha^*$, exactly as the identity promises. But the optimal *control* is $\alpha^* = -bp$, and these two objects are *not* equal: solving $ax + b\alpha^* = -\partial_p H$ for the control gives $\alpha^* = \big(-\partial_p H - ax\big)/b$, the uncontrolled drift ax subtracted off and the gain b divided out. The naive reading $\alpha^* = -\partial_p H$ would here assert $-bp = ax - b^2 p$, which is false for general x, p.

The micro-example pins down exactly when the compact formula is safe. Equality $\alpha^* = -\partial_p H$ requires $-bp = ax - b^2 p$ identically in (x,p), which forces *both* $a = 0$ (no uncontrolled drift) *and* $b^2 = b$, i.e. $b = 1$ (unit gain) — that is, the full canonical normalization $b(x,\alpha) = \alpha$. It is *not* enough to have $b = 1$ alone: at $b = 1$ with $a \neq 0$ one has $-\partial_p H = ax - p$ while $\alpha^* = -p$, still separated by the surviving term ax. This is the trap the remark exists to defuse: "$b = 1$" is not the canonical normalization, "no uncontrolled drift *and* $b = 1$" is. We therefore adopt the canonical drift $b = \alpha$ whenever we state the general theory and want $\alpha^* = -\partial_p H$ to hold verbatim, and we revert to the explicit maximizer $\alpha^* = \big(-\partial_p H - ax\big)/b$ the moment a gain or an uncontrolled drift is in play — as it will be throughout the LQG computation of section 6.7, where a and b are both generically nonzero.

The two identities just exhibited are exactly what the rest of the chapter consumes. The envelope relation $b(x,a^*) = -\partial_p H$ is the formula section 6.5 will read off as the optimal feedback once the diffusion is switched on permanently, and the costate identification $p = \nabla u$ is the seed of the FBSDE of chapter 9. But both were derived under the deterministic, noiseless dynamics, where the costate obeyed a plain backward ODE with a terminal value $p_T = \nabla g(X_T^*)$ that the agent, knowing X^* exactly, could compute in advance. The instant noise enters, that last clause breaks: X_T^* becomes random and unknown until the terminal time, so p_T depends on the future, yet the costate must remain adapted to the information available now. Reconciling a future-determined terminal value with a present-measurable process is the question the next subsection takes up, and its answer — a martingale correction that turns the backward ODE into a backward *stochastic* differential equation — is where the costate story crosses into the stochastic setting that chapter 9 will inhabit.

6.4.1 Stochastic costate: a BSDE preview

The previous subsection left the costate story poised at a fracture, and this subsection is about how the fracture is repaired rather than papered over. In the deterministic problem the adjoint equation (6.12) was an ordinary backward ODE, and integrating it caused no anxiety: the agent knew its optimal trajectory X^* exactly, so the terminal value $p_T = \nabla g(X_T^*)$ was a fixed vector available in advance, and one simply ran the ODE from $t = T$ down to the present. The moment the dynamics become the controlled diffusion (6.10), every clause of that sentence fails at once. The trajectory X^* is now a stochastic process adapted to the filtration $\mathbb{F} = (\mathcal{F}_t)$ generated by the driving Brownian motion W; its terminal position X_T^* is not a number but an $\mathcal{F}_T$-measurable random variable, depending on the *entire* future of the noise path; and therefore so is the terminal datum $p_T = \nabla g(X_T^*)$. Yet whatever the costate becomes, it cannot be allowed to lose the one property that made it usable in the first place — it must stay *adapted*, measurable with respect to the information $\mathcal{F}_t$ actually in hand at time t. The costate is, after all, the price the agent reads off to set its feedback control through $\alpha^* = a^*(X^*, p)$, and a control is by definition a process that may consult only the past, never the future. An adapted costate with a terminal value that lives in the future is the exact tension we must resolve.

To see why it really is a tension and not a triviality, imagine ignoring it: keep the backward ODE form pathwise, $\dot{p}_t = -\partial_x \mathfrak{H}(X_t^*, \alpha_t^*, p_t)$, and integrate it sample path by sample path starting from $p_T = \nabla g(X_T^*)$. The resulting p_t would be a deterministic functional of the terminal value and of the path traversed back from T to t; in particular it would encode X_T^*, and so it would "know" at time $t < T$ where the noise is going to have pushed the state by the terminal time. Such a process looks into the future: it is $\mathcal{F}_T$-measurable, not $\mathcal{F}_t$-measurable, and is therefore inadmissible as a costate. Pathwise backward integration, the natural deterministic reflex, is exactly what adaptedness forbids.

The repair is to stop insisting that the equation hold pathwise and instead ask only that an *adapted* process solve it in integrated form — at the price of admitting a second unknown. The cleanest way to see why a second unknown is forced, and what it must be, is the degenerate model case in which the drift is switched off: seek an adapted process p with $\mathrm{d}p_t = Z_t \, \mathrm{d}W_t$ and prescribed terminal value $p_T = \xi$, where ξ is any $\mathcal{F}_T$-measurable random variable. A process with zero drift and an Itô-integral increment is a martingale, so $p_t = \mathbb{E}[\xi \mid \mathcal{F}_t]$ is forced — and this conditional expectation is, by construction, adapted: it is the best estimate of the future datum ξ given only the information available now, with the genuine future dependence averaged out. Adaptedness is thus recovered automatically, but at a cost: the martingale $t \mapsto \mathbb{E}[\xi \mid \mathcal{F}_t]$ is not constant, it has increments, and naming those increments requires a process Z with $\mathbb{E}[\xi \mid \mathcal{F}_t] = \mathbb{E}[\xi] + \int_0^t Z_s \, \mathrm{d}W_s$. That such a Z exists — that *every* square-integrable $\mathcal{F}_T$-measurable random variable is the terminal value of a stochastic integral against W — is precisely the content of the *martingale representation theorem*, the one beyond-prerequisite tool this preview leans on and which chapter 9 develops in full before using it. The structural lesson survives the reinstatement of the drift: a backward equation that fixes an $\mathcal{F}_T$-measurable terminal datum and demands an adapted solution is over-determined for a single unknown — one cannot in general both hit a future target and refuse to look ahead — and the martingale integrand Z is exactly the extra degree of freedom that reconciles the two. Contrast the forward SDE (6.10), where one fixes the *initial* value, lets the increments be driven by the noise, and accepts whatever terminal value emerges: there the lone unknown is the path. Reverse the arrow of time and the bookkeeping changes shape — the terminal value is now the datum, adaptedness is the constraint, and the pair (p, Z), not p alone, is the unknown.

Restoring the drift gives the object this subsection exists to name, a *backward stochastic differential equation* (BSDE). One feature of it has no deterministic shadow and must be put in by hand: when the volatility σ is genuinely state-dependent, the backward drift is governed not by the bare deterministic Pontryagin Hamiltonian $\mathfrak{H}$ but by its *stochastic completion*

$$H^{\mathrm{s}}(x, a, p, Z) \;=\; b(x, a) \cdot p + \operatorname{tr}\big(\sigma(x)^\top Z\big) + L(x, a),$$

which couples the volatility to the new integrand through the extra term $\operatorname{tr}(\sigma^\top Z)$. With that completion the adjoint reads

$$\mathrm{d}p_t = -\partial_x \mathfrak{H}(X_t^*, \alpha_t^*, p_t)\;\mathrm{d}t - \big(\partial_x \sigma(X_t^*)\big)^\top Z_t\;\mathrm{d}t + Z_t\;\mathrm{d}W_t, \qquad p_T = \nabla g(X_T^*), \tag{6.16}$$

whose $\mathrm{d}t$ part is exactly $-\partial_x H^{\mathrm{s}}(X_t^*, \alpha_t^*, p_t, Z_t)$ — the i-th component of the volatility correction being $\operatorname{tr}\big((\partial_{x_i}\sigma)^\top Z_t\big)$. A *solution* is now an adapted *pair* (p, Z): the costate p carries the deterministic adjoint drift $-\partial_x \mathfrak{H}$ *corrected* by the state-dependence of the volatility, while the new integrand Z supplies, instant by instant, the martingale correction $Z_t \,\mathrm{d}W_t$ that keeps p adapted as it is steered toward the random terminal value. Strip the noise ($\sigma \equiv 0$, hence $Z \equiv 0$, the volatility correction gone, and $\mathrm{d}W$ absent) and (6.16) collapses back to the ordinary adjoint ODE (6.12) of the deterministic theory — the BSDE is its honest stochastic completion, not a different animal.

What is the correction Z, concretely? The costate identity of proposition 6.12 answers the question without any new optimization. In the present nondegenerate-diffusion setting the noise smooths the value function, so the regularity hypothesis $u \in C^{1,2}$ that the identity needs is exactly what the diffusion delivers, and we may take $p_t = \nabla u(t, X_t^*)$ as before. Now read ∇u as a (vector-valued) function of (t, x) and apply Itô's formula (theorem 5.9) to the process $t \mapsto \nabla u(t, X_t^*)$ along the optimal diffusion (6.10). The differential splits, as every Itô expansion does, into a finite-variation ("$\mathrm{d}t$") part and a martingale ("$\mathrm{d}W$") part,

$$\mathrm{d}\big[\nabla u(t, X_t^*)\big] = (\text{finite-variation drift})\;\mathrm{d}t \;+\; \underbrace{\mathrm{D}^2\, u(t, X_t^*)\, \sigma(X_t^*)}_{\text{martingale integrand}}\,\mathrm{d}W_t,$$

all coefficients evaluated at (t, X_t^*). The martingale integrand is the only part we need to pin down explicitly, and it follows from the scalar Itô formula applied component by component: for the i-th component $\partial_i u$ of the field, the diffusion part of $\mathrm{d}\big[\partial_i u(t, X_t^*)\big]$ is $\nabla(\partial_i u) \cdot \sigma \, \mathrm{d}W_t$, and the row vector $\nabla(\partial_i u)$ is exactly the i-th row of the Hessian $\mathrm{D}^2 u$. Stacking the d components, the martingale part is governed by the Jacobian of the field ∇u — which is exactly $\mathrm{D}^2 u$ — contracted against the volatility σ through which the Brownian increment enters the state. The finite-variation drift, by contrast, involves third derivatives of u and we do not expand it: its content is fixed for us by HJB, as we note below. Matching the martingale part against the $Z_t \, \mathrm{d}W_t$ term of (6.16) identifies the second unknown outright,

$$Z_t = \mathrm{D}^2 u(t, X_t^*) \, \sigma(X_t^*), \tag{6.17}$$

the Hessian of the value times the volatility; the finite-variation part of the same Itô expansion reproduces the full BSDE drift $-\partial_x \mathfrak{H} - (\partial_x \sigma)^\top Z$ once HJB is used to evaluate $\partial_t \nabla u$. Its first term is exactly the cancellation already performed in proposition 6.12; its second is new — it is precisely the second-order Itô term $\frac{1}{2} \operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u)$ differentiated through the state-dependence of σ, which equals $(\partial_x \sigma)^\top Z$ and is invisible to the deterministic proposition, where $\sigma \equiv 0$ killed it — so the two pieces of the Itô expansion deliver the two pieces of the BSDE in one stroke. A word of care about (6.17): in dimension $d > 1$ the matrix $Z_t = \mathrm{D}^2 u \, \sigma$ is *not* symmetric, so the ordering of the factors is load-bearing — Z_t is $\mathrm{D}^2 u \, \sigma$ and not $\sigma^\top \mathrm{D}^2 u = (\mathrm{D}^2 u \, \sigma)^\top$, which is a genuinely different matrix even though $\mathrm{D}^2 u$ itself is symmetric. The distinction evaporates only in the scalar Thread A, where $\mathrm{D}^2 u = u_{xx}$ and $\sigma = \sigma$ are numbers, their product $Z_t = \sigma \, u_{xx}$ is a number, and a number equals its own transpose; one should not let the scalar case lull one into dropping the transpose-bookkeeping when the explicit single-agent atom is later widened to vector states.

The identification (6.17) is what entitles us to call (6.16) the *probabilistic face* of the second-order HJB equation (6.11). Recall the linear Feynman–Kac bridge of chapter 5: the solution of a *linear* parabolic PDE is represented as an expectation of a functional of a diffusion. The BSDE is the nonlinear, backward generalization of that same bridge. Its solution $(p, Z) = (\nabla u, \mathrm{D}^2 u \, \sigma)$ encodes the gradient and curvature of the solution u of the *nonlinear* HJB equation (6.11), with the Hamiltonian nonlinearity $\partial_x H^{\mathrm{s}}$ sitting in the BSDE driver where the linear potential sat in Feynman–Kac. Where the PDE (6.11) describes the value as a deterministic field obeying a second-order equation, the BSDE (6.16) describes the very same information probabilistically, as an adapted pair riding the optimal trajectory — two faces, analytic and probabilistic, of one object. This is the dual incarnation promised back in section 6.1 when the conjugate momentum first appeared: the costate that began as a Legendre slot, became a Pontryagin adjoint, and was identified with ∇u now acquires its final form as the leading component of a backward stochastic system.

We deliberately do no more than name the object here, and it is worth being explicit about exactly what has been deferred so the preview stays honest. We have not proved that the BSDE (6.16) has a solution: existence of the adapted pair (p, Z) rests on the martingale representation theorem invoked above, which is itself developed — stated and proved — only in chapter 9. Nor have we coupled the backward costate to the forward state: in (6.16) the trajectory X^* was held fixed and the equation solved for p afterward, but the optimal X^* is itself generated by the feedback $\alpha^* = a^*(X^*, p)$ built from that same p, so state and costate are in truth a single *forward–backward stochastic differential equation* (FBSDE) — forward in X^*, backward in (p, Z), knitted by the minimizer. That FBSDE, and the further McKean–Vlasov coupling that arises when the terminal and running costs are allowed to depend on the law of the population, is the entire subject of chapter 9, the probabilistic counterpart to the analytic mean field game system chapter 8 builds from

HJB. For the main line of this chapter the message to carry forward is simply this: the second-order HJB equation we are about to make permanent in the next section is not the only way to encode the optimizing agent. It has a dual probabilistic incarnation as a backward stochastic equation for the costate, and the two will be welded into the same fabric once the population is released from its frozen state.

6.5 Stochastic control, the control Hamiltonian, and optimal feedback

The previous subsection closed by naming the BSDE as the *probabilistic face* of a second-order HJB equation, and promised that the analytic original would be installed permanently here. We redeem that promise now. Throughout sections 6.1 to 6.4 the diffusion was a guest: it appeared in the verification theorem theorem 6.10 only so that setting $\sigma \equiv 0$ would recover the deterministic theory, and it was switched off again the moment we wanted a Pontryagin trajectory. From here on it is the host. We fix a genuinely nondegenerate volatility and never turn it off, for three reasons the chapter has been accumulating. Noise is the generic situation, not a perturbation of a noiseless ideal; feedback strictly beats open-loop precisely when the realized state can be kicked off its plan, as definition 6.2 argued; and — decisively for the rigour of everything below — a nondegenerate diffusion is exactly what *smooths* the value function, supplying the $C^{1,2}$ regularity that the verification theorem theorem 6.10 demanded as a hypothesis but could not itself produce. The deterministic, first-order theory had to leave that hypothesis dangling, to be rescued later by viscosity solutions; with the diffusion made permanent we can, for the first time, *prove* it.

There is a second, equally deliberate change of furniture, and it is the one that aims the whole section at chapter 8. We now let the running and terminal costs carry a *population law* $m \in \mathcal{P}_2(\mathbb{R}^d)$ as an extra argument: $L(x, a, m)$ and $g(x, m)$. The reader should resist reading anything dynamic into this. At this stage m is an *inert parameter* — a fixed probability measure, chosen once and held frozen, exactly as a coupling constant is held fixed while one solves a single-particle problem. Nothing in this section determines m; no equilibrium condition is imposed on it; the agent optimizes against it as against any other fixed feature of its cost. The only purpose of carrying the slot now is bookkeeping with foresight. When chapter 8 finally releases m from its frozen state — promoting it from a parameter into the unknown of a Nash fixed point, tied back to the agents through their own optimal motion — it will couple to a value function and a feedback that already have the m-slot machined into them. We pay that small notational cost here so that the welding in chapter 8 is a matter of closing a loop, not re-deriving the object. The agent's state accordingly solves the controlled SDE (6.10) with drift $b(x, a)$ and volatility $\sigma(x)$ — a control-independent diffusion, the standard mean field game setting, in which the agent steers the drift but cannot dial the noise.

The instrument is the same Bellman Hamiltonian (6.6) we built in section 6.3, now wearing the measure as a passive third slot — the *control Hamiltonian*

$$H(x, p, m) \;=\; \sup_{a \in A} \Big\{ -b(x, a) \cdot p - L(x, a, m) \Big\}, \tag{6.18}$$

the partial Legendre transform of the running cost in the action, convex in p under (A-Conv) for exactly the reason rehearsed at (6.6): a supremum of affine functions of p is convex with no hypothesis at all, while the genuine content of (A-Conv) — strict convexity and coercivity in the action a — is what keeps the maximizer single-valued. The measure m rides along untouched by the optimization. For each frozen m, (6.18) is *literally* (6.6) with the cost $L(\cdot,\cdot,m)$, so every envelope identity of lemma 6.8 holds verbatim at fixed m — in particular $\partial_p H(x,p,m) = -b\big(x, a^*(x,p,m)\big)$ with a^* the unique maximizing action. The m-slot is inert in precisely the technical sense that it never enters the supremum; it labels the cost, it is not a variable of the control problem.

Switching the noise on changes exactly one line of the dynamic programming derivation, and it is worth seeing which. Introduce the *controlled generator*

$$\mathcal{A}^a\phi(x) \;=\; b(x,a)\cdot\nabla\phi(x) + \tfrac{1}{2}\,\mathrm{tr}\big(\sigma(x)\sigma(x)^\top\,\mathrm{D}^2\,\phi(x)\big),$$

which is nothing but the diffusion generator $\mathcal{A}$ of definition 5.12 with its drift slot pinned to the frozen action value a: one generator for each setting of the knob, the family $\{\mathcal{A}^a\}_{a\in A}$ indexing the infinitesimal motions the agent has available. Now re-run the short-leg expansion of section 6.3 along the controlled diffusion (6.10) in place of the controlled ODE. The only alteration is in how the value at the displaced endpoint accumulates. In the deterministic case the Taylor step gave $u(t+h, X_{t+h}) - u(t,x) \approx h(\partial_t u + b\cdot\nabla u)$, the first-order displacement $X_{t+h} - x \approx h\,b$ dotted into ∇u. With Brownian increments present that displacement also has a *quadratic* part that survives at order h: $\mathbb{E}\Big[(X_{t+h}-x)(X_{t+h}-x)^\top\Big] \approx h\,\sigma\sigma^\top$, the Itô correction of theorem 5.9, contributing $\frac{1}{2}h\ \mathrm{tr}(\sigma\sigma^\top\,\mathrm{D}^2\,u)$ to the expected increment at the same order. Compactly, the expected leg contributes $h\big(\partial_t u + \mathcal{A}^a u\big) + o(h)$, with the entire action-dependence collected into $\mathcal{A}^a u + L(x,a,m)$. Dividing by h, letting $h\downarrow 0$, and infimizing over the constant action exactly as before yields the *second-order Hamilton–Jacobi–Bellman equation*

$$-\partial_t u - \tfrac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top\,\mathrm{D}^2\,u) + H(x,\nabla u, m) = 0, \qquad u(T,x) = g(x,m). \tag{6.19}$$

This is the verification theorem's stochastic HJB (6.11) with the measure argument restored, and it is the same equation read two ways. In *generator form* it is

$$-\partial_t u - \inf_{a\in A}\big\{\mathcal{A}^a u + L(\cdot,a,m)\big\} = 0,$$

which displays the dynamic-programming content directly — the optimal infinitesimal cost rate is the best, over actions, of "the generator acting on the value plus the running cost" — and collapses to (6.19) the instant the infimum is carried out, since the second-order term carries no a and factors out: $\inf_a\{\mathcal{A}^a u + L\} = \frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top\,\mathrm{D}^2\,u) + \inf_a\{b\cdot\nabla u + L\} = \frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top\,\mathrm{D}^2\,u) - H(x,\nabla u,m)$, the last step being the definition (6.18) of H as minus that infimum. The two forms are one statement; we keep the generator form in view because it is the form chapter 8 will set opposite a Fokker–Planck operator $\mathcal{A}^*$, the formal adjoint of the generator of the optimally controlled diffusion.

The structural novelty of (6.19) is the second-order term, and under (A-Nondeg) it is exactly what rescues the regularity the first-order theory could not supply. Nondegeneracy is the statement that $\sigma(x)\sigma(x)^\top \succeq \lambda\,\mathrm{Id}$ uniformly in x for some $\lambda > 0$ — the noise genuinely pushes in every direction — which is to say the operator $\frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top\,\mathrm{D}^2\,\cdot)$ is *uniformly elliptic*. Read backward in time from its terminal datum, (6.19) is then a uniformly parabolic (backward) quasilinear PDE, and parabolic equations *regularize*: an instantaneous diffusion smears any incipient kink in u across space, so that even nonsmooth terminal data $g(\cdot, m)$ are propagated, away from $t = T$, into a value function as smooth as the coefficients permit. This is the analytic content of the slogan that noise smooths the value, and it is the hinge on which the entire section turns: without it u could crease — as it does in the deterministic case — and the verification theorem would have no $C^{1,2}$ object to certify. The precise statement is the following classical existence/regularity result, whose in-book home is the parabolic theory of theorem C.12.

Theorem 6.14 (Classical solvability of the nondegenerate HJB equation)**.** *Assume* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv) *and, in addition, that* $\sigma\sigma^\top$ *and the data* $(x, m) \mapsto H(x, p, m)$*,* $g(\cdot, m)$ *are locally Hölder continuous in* x *(and* H *is* C^1 *in* p*, automatic from lemma 6.8). Then the HJB equation* (6.19) *has a unique solution* $u \in C^{1,2}([0, T) \times \mathbb{R}^d) \cap C([0, T] \times \mathbb{R}^d)$ *with polynomial growth in* x*, uniformly in* t*.*

Two clauses of the hypothesis deserve to be unpacked, because each marks exactly where the proof would fail without it, and together they pin down the precise — and limited — role (A-Nondeg) plays in this book.

Why the Hölder hypotheses cannot be dropped. Uniform parabolicity is necessary but not sufficient for a *classical* solution. Assume only (A-Nondeg) together with merely Lipschitz coefficients — the standing hypotheses of the chapter — and the strongest regularity available is the Krylov–Safonov theory: solutions exist in the parabolic Sobolev class $W_p^{2,1}$ for every $p < \infty$ and are Hölder continuous, but their second spatial derivatives and their time derivative live in L^p, not in C^0. A $W_p^{2,1}$ function need not be twice *continuously* differentiable; its Hessian is defined only almost everywhere and may genuinely fail to be continuous, so it is not a $C^{1,2}$ solution. This is not a cosmetic shortfall. The verification theorem's very first line applies Itô's formula to $u(s, X_s)$, and Itô's formula is *false* for functions whose second derivatives exist only in the L^p sense — it requires a continuous Hessian to control the quadratic variation term pointwise along the path. A merely $W_p^{2,1}$ value cannot be fed to the theorem at all. Closing the gap between $W_p^{2,1}$ and $C^{1,2}$ requires more than ellipticity: it requires the coefficients themselves to carry a modulus of continuity, which is exactly what the local Hölder hypothesis on $\sigma\sigma^\top$, on $H(\cdot, p, m)$, and on $g(\cdot, m)$ supplies. That is why theorem 6.14 cannot be stated under the Lipschitz standing assumptions alone, and why the extra Hölder clause is not decorative.

The standard route, and why it closes. With Hölder data the construction is the textbook freeze-and-iterate argument, and it is worth tracing its logic, because the same scheme recurs wherever a quasilinear parabolic equation is solved. The obstruction in (6.19) is that the principal nonlinearity enters through $H(x, \nabla u, m)$, i.e. through the unknown's own gradient. *Freeze* that gradient: given a fixed continuous vector field $q(t, x)$, replace $H(x, \nabla u, m)$ by $H(x, q, m)$ to obtain a *linear* uniformly parabolic equation for a new unknown w,

$$-\partial_t w - \tfrac{1}{2}\operatorname{tr}\bigl(\sigma\sigma^\top \mathrm{D}^2 w\bigr) + H(x, q, m) = 0, \qquad w(T, \cdot) = g(\cdot, m),$$

in which $H(x, q, m)$ is now a *known* Hölder-continuous source term, not a nonlinearity. To this linear problem the parametrix/Schauder theory of theorem C.12 applies directly: it produces a unique solution $w \in C^{1,2}$ with Gaussian bounds and — crucially — an interior Schauder estimate bounding $\|w\|_{C^{1,2}}$, hence in particular $\|\nabla w\|$, by the Hölder norm of the data. Define the map $\mathcal{T} \colon q \mapsto \nabla w$

that sends a frozen gradient to the gradient of the resulting linear solution. A *fixed point* of $\mathcal{T}$ — a field q with $q = \nabla w$ — is precisely a classical solution of the quasilinear equation (6.19), because then w solves the equation with its own gradient reinserted. That $\mathcal{T}$ has a fixed point is the Schauder fixed-point theorem (equivalently, a method-of-continuity argument): the Schauder estimate shows $\mathcal{T}$ maps a suitable closed bounded convex set of Hölder fields into itself, and that it is *compact* — it gains Hölder regularity, so its image is precompact in the weaker norm — whereupon a continuous compact self-map of such a set has a fixed point by theorem C.12's functional-analytic companion. For the iteration to be well-defined the source $H(x, q, m)$ must depend on q continuously, and for the standard estimates C^1-ly — which is exactly what (A-Conv) guarantees through lemma 6.8, where strict convexity in the action made H a C^1 function of its momentum slot. Drop (A-Conv) and H could carry corners in p, $\mathcal{T}$ would lose the regularity the fixed-point argument consumes, and the scheme would stall. The full construction is Ladyzhenskaya–Solonnikov–Uraltseva [92] and [68, Ch. VI]; we import the quasilinear statement and prove its linear engine, theorem C.12, in-book.

The payoff is precisely the regularity the verification theorem was missing. Granted theorem 6.14, the solution u of (6.19) is $C^{1,2}$, so theorem 6.10 applies on the nose: this u *is* the value function, and the feedback built from it is genuinely optimal — in particular the LQG computation of section 6.7, whose quadratic ansatz is manifestly $C^{1,2}$, inherits this rigour wholesale. This is the precise and limited role (A-Nondeg) plays for us: nondegenerate noise smooths the value function into the classical class, and thereby makes the entire verification programme rigorous *without any recourse to viscosity solutions*. The weak machinery of section 6.6 is, for nondegenerate problems, a luxury one can do without — and it is exactly when (A-Nondeg) fails, or the noise vanishes outright, that the luxury becomes a necessity. Flagging that contrast is how the present section sets up the next.

Remark 6.15 (Optimal feedback)**.** With u now genuinely $C^{1,2}$, the feedback formula of section 6.4 is no longer a formal manipulation but a licensed prescription, and it is the object chapter 8 will consume. In the canonical normalization $b(x, a) = a$ — the normalization remark 6.13 singled out as the only one in which the envelope identity collapses the optimal *drift* onto the optimal *control* — the optimal feedback is read straight off the Hamiltonian's momentum gradient,

$$\alpha^*(t, x) \;=\; -\,\partial_p H\big(x, \nabla u(t, x), m\big),$$

the formula recorded in appendix A. (Outside that normalization one keeps the explicit maximizer $a^*\big(x, \nabla u(t, x), m\big)$ of (6.18) and inverts the drift, exactly as remark 6.13 insisted; the LQG computation of section 6.7, with a gain and an uncontrolled drift both present, is precisely such a case, and reading $\alpha^* = -\partial_p H$ off it blindly would be the sign-and-scaling error that remark exists to forestall.)

This single formula is the seam along which chapter 8 will weld the two halves of a mean field game, and it is worth stating the closed loop in advance so that the present section's frozen-m picture is recognized as one frame of it. The value function solves the *backward* HJB equation (6.19) against a frozen m; its gradient yields the feedback $\alpha^*(t, x) = -\partial_p H(x, \nabla u, m)$; that feedback *is* the drift of the representative agent, so substituting it into the *forward* Fokker–Planck equation of chapter 5 transports an initial population density along the optimally controlled flow. The transported density is a new measure flow $(m_t)_{t\in[0,T]}$ — and the loop *closes* the instant one demands that this transported population coincide with the very m the agent optimized against, feeding back into the costs through the m-slot of (6.18). Freezing m, as we have done throughout this chapter, is exactly the act of cutting that loop open at one point so the single-agent problem can be solved in isolation; chapter 8 reconnects it, and the fixed-point condition that the transported law equal the m in the costs becomes the Nash equilibrium. Everything in this section has been built so that the reconnection requires no re-derivation, only the release of the frozen parameter.

One technical point underwrites the whole construction and is easy to misattribute, so we restate it precisely. What makes $\partial_p H$ single-valued — hence α^* a genuine function rather than a set of competing actions — is the strict convexity and coercivity of the action map $a \mapsto b(x,a)\cdot p + L(x,a,m)$, the operative content of (A-Conv) isolated in lemma 6.8. It is *not* the convexity of H in p that does this work. As section 6.3 showed at length, H is convex in p unconditionally, being a supremum of affine functions, and a merely convex function can still carry corners at which the feedback is ambiguous — the $H = |p|$ example there had a full interval of optimal actions at $p = 0$ despite perfect convexity. Convexity of H in p is the dual *shadow* of strict convexity in a, a consequence and not the cause; the substantive hypothesis lives one level down, in the action map, exactly where section 6.3 located it.

Remark 6.16 (Discounting and infinite horizon)**.** A last variant of the equation deserves to be put on the table now — not for its own sake, but because it will be the cleanest possible arena for the weak-solution theory of the next section. Many problems, especially the stationary mean field games of later chapters, run over an infinite horizon and weight future costs by a *discount rate* $\beta = \beta \geq 0$, replacing the running cost L by $e^{-\beta s}L$ in the objective. Two equivalent bookkeepings turn this into a PDE. One may keep the finite-horizon equation and carry the exponential explicitly; or, more cleanly, one factors the discount out of the value by writing $u(s,x) = e^{-\beta s}\tilde{u}(s,x)$, whereupon differentiating the weight $e^{-\beta s}$ produces a zeroth-order term $\beta\,\tilde{u}$ alongside the usual transport and diffusion terms. In the genuinely stationary regime the horizon is infinite and the coefficients are time-homogeneous, so the agent's situation at every instant is identical: the value cannot depend on the absolute clock, $\partial_t u = 0$, and there is no terminal time at which to impose data. The stationary, infinite-horizon value $u(x) = \inf_\alpha \mathbb{E}\Big[\int_0^\infty e^{-\beta s} L(X_s, \alpha_s, m)\,\mathrm{d}s\Big]$ accordingly solves the *time-independent* HJB equation

$$\beta\, u - \tfrac{1}{2}\operatorname{tr}\!\big(\sigma\sigma^\top \mathrm{D}^2 u\big) + H(x, \nabla u, m) = 0, \tag{6.20}$$

with no time derivative and no terminal condition — the discount term $\beta\, u$ standing in for the vanished $-\partial_t u$ and, when $\beta > 0$, supplying the strict monotonicity in the u-slot that a comparison principle will need in place of a horizon. (Per appendix A, β is always a discount rate here, never an inverse temperature; its resemblance to a mass term in a screened equation is a serviceable mnemonic but carries no thermodynamic content.)

Equation (6.20) is the object we hand forward. Stripped of the time variable and the terminal layer, it isolates the spatial structure of HJB — the competition between the elliptic diffusion term and the convex Hamiltonian — in its barest form, which is exactly why section 6.6 adopts it as the test bed on which to introduce viscosity solutions. And the hand-off is pointed, not incidental. The present section earned a *classical* $C^{1,2}$ solution only because (A-Nondeg) held and the diffusion smoothed the value. The next section confronts the complementary case — degenerate noise, or none at all — where that smoothing is unavailable, u is in general only Lipschitz and creases along the caustics the chapter's physicsnote foreshadowed, and the verification theorem's $C^{1,2}$ hypothesis simply cannot be met. There the classical solvability proved here is a luxury one has lost, and a weaker notion of solution, asking nothing of u beyond continuity, must take its place. That is the notion section 6.6 builds, with (6.20) as its proving ground.

6.6 Viscosity solutions: the correct weak notion

The verification theorem demands a $C^{1,2}$ solution. For first-order Hamilton–Jacobi equations — the degenerate case $\sigma \equiv 0$, and the eikonal equation of the physics note — the value function is in general only Lipschitz: it develops gradient discontinuities at points where two competing optimal trajectories meet, the control analogue of shock formation. At such kinks no classical solution exists, yet the value function is perfectly well-defined and is the object we want. We need a notion of solution that (i) makes sense for merely continuous functions, (ii) selects the value function uniquely among the many a.e.-solutions, and (iii) is stable under limits so that approximations converge to it. *Viscosity solutions*, introduced by Crandall and Lions, are that notion. We give the definition for the stationary equation (6.20) — cleaner because it has no time boundary — and indicate the parabolic modification.

Heuristic (non-rigorous)

The idea is to move all derivatives onto smooth test functions, as with distributions, but respecting the *one-sidedness* that the equation's ellipticity prescribes. At a point where u has an upward kink (a local max of $u - \varphi$ for smooth φ touching from above), the would-be second derivative is $-\infty$ in some direction; the equation, which contains $-\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 \cdot)$ and is therefore *decreasing* in the Hessian, can only be asked to hold as an inequality of the favourable sign there. Demanding the two complementary one-sided inequalities at all such touchings pins the function down exactly.

Write the equation as $F(x, u, \nabla u, \mathrm{D}^2 u) = 0$ with, from (6.20),

$$F(x, r, p, M) = \beta\, r - \tfrac{1}{2} \operatorname{tr}\big(\sigma(x)\sigma(x)^\top M\big) + H(x, p, m).$$

This F is *degenerate elliptic*: $F(x, r, p, M) \le F(x, r, p, M')$ whenever $M \succeq M'$ (the map $M \mapsto -\operatorname{tr}(\sigma\sigma^\top M)$ is nonincreasing in the PSD order). This monotonicity is the structural fact that makes the definition consistent.

Definition 6.17 (Viscosity solution)**.** Let F be degenerate elliptic and $u\colon \mathbb{R}^d \to \mathbb{R}$ continuous.

- u is a *viscosity subsolution* if for every $\varphi \in C^2$ and every local maximum point x_0 of $u - \varphi$,

$$F\big(x_0, u(x_0), \nabla\varphi(x_0), \mathrm{D}^2 \varphi(x_0)\big) \le 0.$$

- u is a *viscosity supersolution* if for every $\varphi \in C^2$ and every local minimum point x_0 of $u - \varphi$,

$$F\big(x_0, u(x_0), \nabla\varphi(x_0), \mathrm{D}^2 \varphi(x_0)\big) \ge 0.$$

- u is a *viscosity solution* if it is both.

For the time-dependent equation (6.19) one uses the same definition with (t, x) in place of x, test functions $\varphi \in C^{1,2}$, the operator $-\partial_t\varphi - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2\varphi) + H(x, \nabla\varphi, m)$ in place of F, and the terminal condition $u(T, \cdot) = g$ imposed classically.

Two sanity checks anchor the definition. First, *consistency*: if $u \in C^2$ is a classical solution, it is a viscosity solution, and conversely a C^2 viscosity solution solves the equation pointwise (at a touching point of a C^2 function, $\nabla\varphi = \nabla u$ and $\mathrm{D}^2\varphi \succeq \mathrm{D}^2 u$ at a max, $\preceq$ at a min, and degenerate ellipticity converts the two inequalities into the pointwise equation). Second, the inequalities point the "right" way precisely because of the sign in F; reversing the convention on F swaps sub- and supersolutions, a classic source of error.

The reason the definition is the *correct* one is that it restores uniqueness, which a.e.-solutions spectacularly lack. The mechanism is a comparison principle.

Theorem 6.18 (Comparison and uniqueness)**.** *Assume* (A-Lip)*,* (A-Growth) *on the data and* $\beta > 0$ *(or a finite horizon with prescribed terminal data). Let* $\overline{u}$ *be a bounded upper-semicontinuous viscosity subsolution and* $\underline{u}$ *a bounded lower-semicontinuous viscosity supersolution of* $F = 0$*. Then* $\overline{u} \le \underline{u}$ *on* $\mathbb{R}^d$*. Consequently the equation has at most one bounded continuous viscosity solution, and it coincides with the value function.*

The proof is the technical heart of the theory — it runs through the doubling-of-variables device and the Crandall–Ishii lemma on second-order superjets — and rather than relegate it to a citation we give it an in-book home: appendix E develops the comparison principle and the Crandall–Ishii lemma in full, including the precise structural conditions on F that make it work (uniform continuity in x at fixed jet, and the discount or horizon that provides the strict monotonicity in r). The classical references are the "User's Guide" [52], with [12] for the first-order/deterministic theory and [68, Ch. IV–V] for the second-order stochastic case.

Remark 6.19 (Where viscosity solutions are and are not needed)**.** Under (A-Nondeg) the second-order HJB equation (6.19) is uniformly parabolic and has a classical $C^{1,2}$ solution, so for the nondegenerate mean field games that occupy most of this book viscosity theory is a luxury, not a necessity — the verification theorem suffices. Its indispensability appears in three places we will meet: degenerate or first-order models where (A-Nondeg) fails; the rigorous justification that the $C^{1,2}$-classical solution *is* the value when one cannot assume smoothness a priori; and, most importantly later, as the template for the notion of solution of the master equation on $\mathcal{P}_2(\mathbb{R}^d)$ in chapter 12, where no classical calculus is available and the viscosity philosophy — test functions touching from one side — is the only game in town.

6.7 Worked computation: the LQG single agent and its Riccati equation

The viscosity detour behind us, we return to the nondegenerate, classically solvable backbone — where, as remark 6.19 stressed, the verification theorem alone suffices — and solve the first layer of Thread A, the linear–quadratic–Gaussian problem fixed in the project's thread specification, by running the verification programme of theorem 6.10 on an explicit ansatz. This is the atom from which the whole of Part II is built: in chapters 8 and 14 the frozen mean $\bar{m}$ below becomes the unknown of a fixed-point equation, but the single-agent solver — HJB reduces to a scalar Riccati ODE — is exactly the computation performed here.

Worked computation 6.20 (LQG single agent, Thread A layer 1)**.** The representative agent has scalar state $X_t \in \mathbb{R}$ driven by

$$\mathrm{d}X_t = \big(aX_t + b\,\alpha_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad X_0 \sim m_0, \tag{6.21}$$

with real constants a (state feedback), b (control gain), $\sigma > 0$ (noise), and W a scalar Brownian motion. The cost over $[0,T]$ is

$$J(\alpha) = \mathbb{E}\left[\int_0^T \Big(\tfrac{1}{2}\alpha_t^2 + \tfrac{q}{2}\big(X_t - s\,\bar{m}_t\big)^2\Big)\,\mathrm{d}t + \tfrac{q_T}{2}\big(X_T - s_T\,\bar{m}_T\big)^2\right], \tag{6.22}$$

with weights $q, q_T \geq 0$ and coupling constants s, s_T. In this single-agent layer the population mean $\bar{m}_t$ is frozen as a given continuous function of time (in chapter 14 it is determined self-consistently). The control set is $A = \mathbb{R}$.

Hamiltonian and HJB. Here the symbol a is the Thread A *state-feedback constant* of (6.21); we therefore write the action/maximization variable as $\alpha = \alpha$ (the control), reserving a for the constant. The drift is $ax + b\alpha$ and the running cost is $L(x,\alpha) = \frac{1}{2}\alpha^2 + \frac{q}{2}(x - s\bar{m}_t)^2$, so the control Hamiltonian (6.18) is computed by completing the square in α:

$$H_t(x,p) = \sup_{\alpha\in\mathbb{R}}\Big\{ -(ax+b\alpha)p - \tfrac{1}{2}\alpha^2 - \tfrac{q}{2}(x-s\bar{m}_t)^2\Big\} = -axp + \tfrac{1}{2}b^2p^2 - \tfrac{q}{2}\big(x - s\bar{m}_t\big)^2,$$

the supremum attained at $\alpha^* = -bp$. (Note the gain b: the optimal control is $-bp$, not $-p$; this is the general control-affine case of remark 6.13.) The HJB equation (6.19) reads

$$-\partial_t u - \tfrac{\sigma^2}{2}\partial_{xx}u - ax\,\partial_x u + \tfrac{b^2}{2}\big(\partial_x u\big)^2 - \tfrac{q}{2}\big(x - s\bar{m}_t\big)^2 = 0, \qquad u(T,x) = \tfrac{q_T}{2}\big(x - s_T\bar{m}_T\big)^2. \tag{6.23}$$

Quadratic ansatz and the Riccati system. The terminal data are quadratic in x and the equation preserves quadratic profiles, so we posit

$$u(t,x) = \tfrac{1}{2}P(t)\,x^2 + R(t)\,x + S(t),$$

with $\partial_x u = Px + R$, $\partial_{xx}u = P$, and $\partial_t u = \frac{1}{2}\dot{P}x^2 + \dot{R}x + \dot{S}$. Substituting into (6.23) and matching powers of x gives three ordinary differential equations. The x^2 coefficient yields the scalar *Riccati equation*

$$\dot{P}(t) = b^2P(t)^2 - 2a\,P(t) - q, \qquad P(T) = q_T, \tag{6.24}$$

a closed, autonomous, backward ODE for the curvature of the value function. The x^1 coefficient gives a linear equation for R driven by the (frozen) population mean,

$$\dot{R}(t) = (b^2 P(t) - a)R(t) + q\,s\,\bar{m}_t, \qquad R(T) = -q_T\,s_T\,\bar{m}_T, \tag{6.25}$$

and the x^0 coefficient gives the offset, with the noise entering only here,

$$\dot{S}(t) = \tfrac{b^2}{2}R(t)^2 - \tfrac{\sigma^2}{2}P(t) - \tfrac{q}{2}s^2\bar{m}_t^2, \qquad S(T) = \tfrac{q_T}{2}s_T^2\bar{m}_T^2. \tag{6.26}$$

Optimal feedback and certainty equivalence. By remark 6.13 the optimal feedback is $\alpha^* = -b\,\partial_x u$, i.e. the *linear* feedback law

$$\alpha^*(t,x) = -b\big(P(t)\,x + R(t)\big). \tag{6.27}$$

The closed-loop state (6.21) is then a linear SDE with Gaussian solution, the fact that will make the Fokker–Planck step of chapters 8 and 14 explicit. Three structural observations close the computation. First, the feedback gain $bP(t)$ is governed entirely by the Riccati equation (6.24), which does *not* see the noise σ or the coupling $\bar{m}$: this is *certainty equivalence*, the optimal gain is the same as in the deterministic ($\sigma = 0$) problem. Second, the noise enters only the constant offset S through the $-\frac{\sigma^2}{2}P$ term in (6.26); it raises the cost-to-go but leaves the policy unchanged. Third, the coupling to the crowd enters only the linear part R, an inhomogeneous forcing by $\bar{m}_t$ in (6.25) — the seed that, once $\bar{m}$ is no longer frozen but is itself the mean of the closed-loop law, will couple back through (6.25) and close the mean field game.

Solvability of the Riccati equation. Equation (6.24) is a scalar Riccati ODE with constant coefficients; it is solved in closed form by the standard linearization $P = -\dot{\Psi}/(b^2\Psi)$, which turns it into the linear second-order equation $\ddot{\Psi} + 2a\dot{\Psi} - b^2 q\,\Psi = 0$ with constant coefficients, hence elementary exponentials. Because $q, q_T \geq 0$, a comparison argument shows $P(t) \geq 0$ on $[0,T]$ and that no finite-time blow-up occurs when integrating backward from $P(T) = q_T$: the quadratic value function stays convex, consistent with (A-Conv) and with the well-posedness guaranteed by theorem 6.10. The explicit exponential solution, its long-horizon limit $P(t) \to \bar{P}$ as $T - t \to \infty$ (the stable root of $b^2\bar{P}^2 - 2a\bar{P} - q = 0$, i.e. the algebraic Riccati equation), and the matrix generalization are taken up when the thread is completed in chapter 14.

This is the promised atom. Everything explicit in Part II's LQG analysis — Gaussian population flows, closed Riccati–mean systems, the uniqueness/multiplicity phase diagram — is assembled from (6.24)–(6.27) by switching on the coupling that chapters 7 and 8 introduce.

Rigor ledger

Proved.

- Euler–Lagrange equation and transversality for the action functional (theorem 6.1).
- The verification theorem (theorem 6.10): a $C^{1,2}$ solution of the (stochastic) HJB equation with terminal data is the value function and its feedback is optimal — proved in full via Itô's formula.

- The dynamic programming principle (theorem 6.4), including the measurable ε-optimal selection (lemma 6.5) that the deterministic case needs — proved in-book.
- The envelope identity $\partial_p H = -b(\cdot, a^*)$, $\partial_x H = -\partial_x \mathfrak{H}(\cdot, a^*, \cdot)$ for the Bellman Hamiltonian (lemma 6.8), the single-valued feedback and the costate identity both resting on it.
- Identification of the Pontryagin costate with the gradient of the value, $p_t = \nabla u(t, X_t^*)$, and derivation of the adjoint equation from HJB (proposition 6.12).
- Reduction of the LQG single-agent problem to the scalar Riccati system (6.24)–(6.26) with linear optimal feedback (6.27) (worked computation 6.20).

Assumed (referenced by tag).

- (A-Lip), (A-Growth): well-posedness and moment bounds of the controlled state equation (theorem 5.10), used to make the Itô integral a martingale in theorem 6.10.
- (A-Conv): strict convexity/coercivity in the control (equivalently convexity of H in p), hence the single-valued feedback of lemma 6.8 and applicability of verification.
- (A-Nondeg) plus local Hölder continuity of the data: uniform parabolicity giving the classical $C^{1,2}$ value function of theorem 6.14, so that verification applies without viscosity theory.

Deferred (with in-book home).

- Classical solvability of the nondegenerate HJB equation (theorem 6.14): the linear parabolic input is proved in theorem C.12; the quasilinear closure (Schauder fixed point) follows [92], [68, Ch. VI].
- Necessity in the Pontryagin maximum principle theorem 6.11 (spike-variation argument): not used elsewhere in the book — only the in-book costate identity proposition 6.12 and the FBSDE development of chapter 9 are relied upon — so its proof is left to [136, Ch. 2], [97] as further reading.
- The comparison principle for viscosity solutions theorem 6.18 (doubling of variables, Crandall–Ishii lemma): proved in appendix E; classical references [52], [12], [68].
- The adjoint BSDE (6.16), the martingale representation theorem behind its integrand Z, and the FBSDE it generates: developed in chapter 9.

Open / beyond this chapter.

- Existence and uniqueness once $\bar{m}$ (here frozen) is determined by an equilibrium fixed point — the mean field game proper (chapters 8, 10, 11 and 14).
- Optimal control under control-dependent or degenerate diffusion, and the regularity of the value function there, where (A-Nondeg) fails and viscosity theory is essential.

Part I retrospective: the six tools

This chapter closes Part I, and with it the construction of the analytic vocabulary the rest of the book speaks. It is worth pausing to name the six foundational tools we have built and, for each, the single load-bearing result it contributes and the downstream mean-field-game (MFG) or graphon-mean-field-game (GMFG) theorem it powers. Part II begins, in chapter 7, by switching on the first coupling; everything it does is assembled from these six.

- **Chapter 1 — measure and the thermodynamic limit.** The passage from empirical measures of N agents to a deterministic limiting law is what makes "the population" a single mathematical object $m \in \mathcal{P}_2(\mathbb{R}^d)$. This underwrites the very statement of an MFG and reappears as the law-of-large-numbers leg of the propagation-of-chaos and finite-N convergence theorems (chapter 10, chapter 25).
- **Chapter 2 — functional analysis and operators.** Banach–Alaoglu, compact self-adjoint spectral theory, and the fixed-point theorems (Schauder/Kakutani) are the abstract existence engines. They power the existence of MFG equilibria by fixed point (chapter 10) and, through the spectral theory of the graphon operator $\mathbb{W}$, the entire reduction machinery of Part IV (chapter 20, chapter 24).
- **Chapter 3 — probability and weak convergence.** Prokhorov tightness and the topology of weak convergence give the compactness in which equilibria are extracted as limits; this is the analytic heart of the existence proof (chapter 10) and of the stability of equilibria under approximation.
- **Chapter 4 — Wasserstein geometry and derivatives on $\mathcal{P}_2(\mathbb{R}^d)$.** The metric $\mathcal{W}_2$ and the flat/Lions derivatives turn "a function of the population" into a differentiable object. These are exactly the calculus in which the master equation on measure space is written (chapter 12), and the monotonicity (A-Mon) that delivers uniqueness (chapter 11) is a Wasserstein-level statement.
- **Chapter 5 — Itô calculus and Fokker–Planck.** Itô's formula (theorem 5.9), the generator $\mathcal{A}$ and its adjoint $\mathcal{A}^*$, and the forward Fokker–Planck equation supply the *forward* leg of the MFG system: the equation the optimally-controlled population obeys (chapter 8), and the SDE well-posedness (theorem 5.10) on which every probabilistic argument rests.
- **Chapter 6 — optimal control and HJB (this chapter).** The verification theorem (theorem 6.10) and the HJB equation supply the *backward* leg: the equation the representative agent's value function obeys and the optimal feedback $\alpha^* = -\partial_p H$ that closes the loop. Welded to the forward leg of chapter 5 at opposite ends of $[0, T]$, this is the forward–backward MFG system of chapter 8, its probabilistic FBSDE face in chapter 9, and the single-agent atom of the LQG thread carried to its fixed point in chapter 14.

The recurring structural signature — a backward HJB welded to a forward Fokker–Planck, closed by a self-consistency (fixed-point) condition on m — is visible already in this single-agent chapter, where m sits frozen. Part II's task is to release it.

Bibliographic notes

The calculus-of-variations recap of section 6.1 and the deterministic control theory of sections 6.1 and 6.4 follow [97], the concise text the book's plan designates for this chapter; the Pontryagin maximum principle originates with [112]. Dynamic programming and the value function (section 6.2) are due to [17]; the modern treatment of the dynamic programming principle, the verification theorem, and stochastic control we follow is [68] and [136, especially Ch. 2–4], the latter also our reference for the stochastic maximum principle and the adjoint BSDE previewed in section 6.4. The Itô-calculus inputs to the verification proof are from chapter 5 (see [106]). Viscosity solutions (section 6.6) were introduced by Crandall and Lions; the indispensable reference is the "User's Guide" [52], with [12] for the first-order/deterministic theory and [68] for the second-order stochastic theory. The sign and structural conventions for the Hamiltonian, and the forward–backward coupling that this single-agent chapter sets up, are aligned with the mean field game literature we enter next: [93], [36], and [40]. The LQG worked computation continues the linear–quadratic line of [82] and is carried to its mean field fixed point in chapter 14.

Part II

Classical Mean Field Games

Chapter 7

McKean–Vlasov Dynamics and Propagation of Chaos

Picture a single particle adrift in a large swarm. Its own random noise jostles it from instant to instant, but it also feels the others—not any one of them in particular, only their aggregate, the cloud of positions in which it is immersed. Ask the physicist's question: what does that one particle feel as the swarm grows? If each neighbor pushed with a force of fixed size, the total exerted by the other $N-1$ would diverge as $N \to \infty$ and no limit could exist; if the coupling were too weak, the neighbors' influence would fade to nothing and the particle would simply diffuse alone. Exactly one scaling threads between these extremes: let every pair interact at strength $O(1/N)$, so that the force on our particle is an *average* of N contributions rather than a sum. An average of N comparable terms is $O(1)$—neither divergent nor negligible—and, by the law of large numbers, an average of many nearly independent contributions ceases to fluctuate and settles onto its mean. That single observation is the entire physical content of the mean-field limit: the right coupling converts the felt environment into a deterministic, self-generated field, and the swarm's empirical density—not the labels of its members—becomes the only thing to which each particle responds. This chapter makes that reflex precise and quantitative, and then asks what, if anything, it leaves out.

This chapter is the hinge between the toolbox of Part I and the mean field games of Part II. It studies a population of N stochastic particles that interact only through their *collective statistics*, and it makes rigorous the physicist's reflex that, as $N \to \infty$, the fluctuations of those statistics are suppressed and each particle comes to feel a smooth, deterministic, self-generated field. The limiting single-particle law is governed by a *McKean–Vlasov* stochastic differential equation, in which the law of the solution feeds back into its own coefficients, and equivalently by a *nonlinear Fokker–Planck* equation. The precise sense in which finite-N marginals factorize is *propagation of chaos*, which we prove by Sznitman's coupling with a quantitative $O(1/\sqrt{N})$ rate. These are not three separate topics but three faces of a single object—the law of one representative particle in the limit—and the sections are ordered so that meeting one face makes the next inevitable: the finite-N system forces the empirical measure into the foreground (section 7.1); freezing that measure to a deterministic flow yields the McKean–Vlasov equation (section 7.2); its marginals obey the nonlinear Fokker–Planck equation (section 7.4); and propagation of chaos (section 7.3) is the theorem that the finite-N reality and its idealized limit genuinely coincide. The reader should feel the four as one argument unfolding, not as a list of four results.

The descriptive theory of this chapter draws principally on the analytic tools of Part I. From chapter 4 we use the metric space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ of probability measures with finite second moment, in which the population state lives and in which best responses will later be shown continuous. From chapter 5 we use the theory of Itô stochastic differential equations (existence and uniqueness under Lipschitz and linear-growth coefficients, Itô's formula, the Feynman–Kac/backward-Kolmogorov representation of linear parabolic equations, and the Fokker–Planck/Kolmogorov forward equation as the adjoint evolution of densities), together with the Ornstein–Uhlenbeck worked example. From chapter 3 we use weak convergence of probability measures, tightness, and Prokhorov's theorem; from chapter 2 the Banach contraction principle (theorem 2.36); and from chapter 1 the empirical measure $\mu^N = \frac{1}{N}\sum_i \delta_{X_i}$ and its law of large numbers. One further import is strategic rather than analytic and is confined to the descriptive-versus-strategic pivot of section 7.5: from chapter 6 we borrow the value function, the Hamilton–Jacobi–Bellman equation, and the optimal feedback $\alpha^* = -\partial_p H$ of a single-agent control problem, used there only to contrast the prescribed drift of McKean–Vlasov dynamics with the optimized drift of a game. Throughout, Brownian motion is written W (rendering W); no graphon is present in this chapter, so there is no clash with the graphon kernel, which enters only from chapter 15 onward.

The chapter closes on its single most important conceptual point, the one that motivates all of Part II: McKean–Vlasov dynamics are *descriptive*. Each particle blindly obeys a prescribed law of motion that happens to depend on the population law, and the self-consistency $m_t = \mathcal{L}(X_t)$ is a statistical closure, not a decision. A mean field game replaces the prescribed drift by the *optimal feedback of a control problem*: the population law becomes the output of a Nash equilibrium rather than of an averaging postulate. We isolate exactly what that strategic layer adds, and point to where the next chapters build it (chapter 8, chapter 9, and, for the genuinely harder "propagation of chaos for equilibria," chapter 13).

Notation watch

Two notational collisions are worth naming at the outset, because each arises from a symbol that two well-entrenched literatures both insist on. The first is W: stochastic analysis universally writes Brownian motion W_t (for Wiener), while graphon theory universally writes the kernel $W(x, y)$ (for the limit object of dense graphs). Both are too standard to renounce, so the book keeps each in the part where it reigns and switches letters at the seam. In this chapter $W = W_t$ (and W_t^i for the ith particle) denotes Brownian motion, following the standard usage of the stochastic chapters. The letter W is *not* a graphon here; the graphon kernel and its operator $\mathbb{W}$ do not appear until Part III, where Brownian motion switches to B_t to free the letter W. See the notation watch in appendix A. The second collision is b, again forced by two standard usages: control theory writes the control gain b in the linear dynamics $\mathrm{d}X = (aX + b\alpha)\,\mathrm{d}t + \cdots$, while the diffusion literature writes the drift field $b(\cdot, \cdot)$. The scalar b in the carried linear–quadratic thread (worked computation 7.18) is the *control gain* fixed in the thread specification, whereas elsewhere $b(\cdot, \cdot)$ denotes the drift *field* (rendering the same letter b). To keep the two from colliding, in the linear–quadratic worked computation we never write the drift through the macro $b(\cdot,\cdot)$: there the drift is displayed explicitly as a formula in x and the population mean, precisely so that b appears only as the control gain.

7.1 Weakly interacting particle systems

The picture with which the chapter opened—one particle immersed in a swarm, feeling only the aggregate of its neighbors—must now become an equation, and the slogan "every pair interacts at strength $O(1/N)$" must become a precise Itô system. That is the business of this section. We fix the finite-N object whose limit the rest of the chapter studies, install the empirical-measure language in which the whole development is conducted, and prove the two structural facts on which everything downstream rests: the system is well-posed, and it is *exchangeable*, so that no particle's identity can influence the dynamics—only the collective configuration can. By the end the reader will hold a unique, label-blind N-body process, and the obvious next move—freeze the collective configuration to a deterministic flow and watch a single particle—will set up the McKean–Vlasov equation of section 7.2.

Consider N indistinguishable particles, the ith with state $X_t^{i,N} \in \mathbb{R}^d$, each driven by its own independent noise but coupled to all the others through the population's empirical distribution. (We keep the superscript N on $X_t^{i,N}$ because the law of the ith particle genuinely depends on how many companions it has; the same physical particle sits in a different bath when N changes.) The defining feature of a *weakly interacting* (or *mean-field*) system is the scaling of the coupling: every pair of particles interacts with strength $O(1/N)$, so that the *total* force exerted on any one particle by the other $N-1$ remains $O(1)$ as $N \to \infty$. This is precisely the scaling under which a single particle feels a well-defined average field rather than a sum that diverges or vanishes, exactly as the opening argument demanded. Before writing the dynamics we name the object through which the coupling acts.

Definition 7.1 (Empirical measure)**.** For states $x_1, \dots, x_N \in \mathbb{R}^d$ the *empirical measure* is the random probability measure

$$\mu^N \; := \; \frac{1}{N} \sum_{j=1}^{N} \delta_{x_j} \; \in \; \mathcal{P}(\mathbb{R}^d).$$

For a particle configuration $X_t^{1,N}, \dots, X_t^{N,N}$ we write $\mu_t^N = \frac{1}{N} \sum_{j=1}^N \delta_{X_t^{j,N}}$. If the x_j have a common finite second moment then $\mu^N \in \mathcal{P}_2(\mathbb{R}^d)$ and serves as a point in the Wasserstein space of chapter 4.

This is the same empirical measure introduced in chapter 1, now in its dynamical role. It is the rigorous incarnation of the physicist's "density profile" or "occupation distribution": integrating a test function against it, $\int \phi \, \mathrm{d}\mu^N = \frac{1}{N} \sum_j \phi(x_j)$, returns the population average of ϕ, which is exactly what a measurement of the swarm reports. Two features make it the right state variable. First, it *discards labels*: μ^N records how the population is distributed across state space but forgets which particle sits where, so any quantity expressed through μ^N is automatically symmetric under relabeling—a point that will return as exchangeability. Second, when the x_j are i.i.d. samples from a law m, the empirical measure is the object whose convergence $\mu^N \to m$ *is* the law of large numbers (chapter 1); this is the precise sense in which a fluctuating microscopic configuration sharpens, as N grows, onto a deterministic macroscopic density. The mean field limit of this chapter is that same law of large numbers, made to operate not on a fixed sample but on the interacting dynamics themselves. For now μ_t^N is the channel through which each particle perceives the rest, and we can write the dynamics.

Definition 7.2 (Weakly interacting particle system)**.** Let $b : \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}^d$ and a constant matrix $\sigma \in \mathbb{R}^{d\times d}$ be given, and let $W^1, \dots, W^N$ be independent $\mathbb{R}^d$-valued Brownian motions on a filtered probability space, independent of the i.i.d. initial data $X_0^{1,N}, \dots, X_0^{N,N} \sim m_0 \in \mathcal{P}_2(\mathbb{R}^d)$. The *weakly interacting particle system* is the coupled Itô system, for $i = 1, \dots, N$,

$$\mathrm{d}X_t^{i,N} \;=\; b\big(X_t^{i,N},\; \mu_t^N\big)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t^i, \qquad \mu_t^N = \frac{1}{N}\sum_{j=1}^N \delta_{X_t^{j,N}}. \tag{7.1}$$

The interaction is called *pairwise* when the drift has the form

$$b(x,\mu) = \int_{\mathbb{R}^d} b_0(x,y)\,\mu(\mathrm{d}y), \qquad \text{so that} \qquad b\big(X_t^{i,N}, \mu_t^N\big) = \frac{1}{N}\sum_{j=1}^N b_0\big(X_t^{i,N}, X_t^{j,N}\big), \tag{7.2}$$

for a fixed kernel $b_0 : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$.

Read (7.1) slowly: each particle has its *own* noise W^i—the W^i are independent, so absent the drift the particles would never meet—and the *only* place the others enter is the second argument μ_t^N of the common drift field b. The coupling is therefore entirely mediated by the empirical measure; this is what “mean-field interaction” means precisely. The pairwise special case (7.2) is the form in which the $O(1/N)$ scaling is laid bare, and it is worth spelling out why. Substituting $\mu_t^N = \frac{1}{N}\sum_j \delta_{X_t^{j,N}}$ into $b(x,\mu) = \int b_0(x,y)\,\mu(\mathrm{d}y)$ turns the integral into the displayed sum: the drift on particle i is

$$b\big(X_t^{i,N}, \mu_t^N\big) = \frac{1}{N}\sum_{j=1}^N b_0\big(X_t^{i,N}, X_t^{j,N}\big) = \underbrace{\frac{1}{N} b_0\big(X_t^{i,N}, X_t^{j,N}\big)}_{\text{one neighbor: } O(1/N)} \times N \text{ terms.}$$

Each individual neighbor j contributes a force $\frac{1}{N} b_0(X^i, X^j)$ of size $O(1/N)$—this is the “$O(1/N)$ pairwise strength” of the slogan—while the *average* of the N such contributions is $O(1)$, because an average of N bounded terms does not grow with N. Had we instead summed undiscounted forces $b_0(X^i, X^j)$ (strength $O(1)$ per pair), the total would be $O(N)$ and would blow up in the limit; had we discounted by $1/N^2$, each particle’s influence and the total alike would vanish. The $1/N$ normalization is the unique choice that keeps the felt field both finite and nontrivial—the discrete shadow of the preamble’s scaling argument. We will use the pairwise form repeatedly, both because it is the cleanest carrier of the scaling and because it is the setting in which the propagation-of-chaos rate of section 7.3 comes out dimension-free.

A word on the constant diffusion. Taking σ a fixed matrix, rather than a field $\sigma(x,\mu)$, is a deliberate simplification made for transparency, not a limitation of the theory, and it is worth being explicit about why we can afford it here. The entire difficulty of the mean-field limit lives in the *drift*: it is the drift that couples the particles, and it is the drift's dependence on μ^N that must be controlled as the empirical measure fluctuates. A measure-dependent volatility adds a second coupled coefficient but no new conceptual obstacle—the Itô isometry converts a diffusion mismatch into exactly the same kind of $\mathcal{W}_2$-Lipschitz estimate the drift already demands. We therefore carry the constant σ through the main arguments to keep the algebra legible, and flag at each stage (see remark 7.10) precisely what a state- and measure-dependent $\sigma(x, \mu)$, obeying the same Lipschitz and nondegeneracy conditions, would change; the only genuinely non-routine point is that the coupling constructions of section 7.3 must then be made on the *driving* Brownian motion rather than pathwise, which remains legitimate. Nothing below relies on σ being constant beyond this convenience.

We now fix the hypotheses under which the system is studied. They are the standing assumptions of the thread specification (appendix A), specialized to what the McKean–Vlasov theory needs—which turns out to be Lipschitz dependence of the drift on *both* of its arguments, the state and the measure, simultaneously.

Assumption 7.3 (Standing hypotheses for this chapter)**.** There is a constant $L > 0$ such that for all $x, x' \in \mathbb{R}^d$ and $\mu, \mu' \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\left|b(x,\mu) - b(x',\mu')\right| \;\le\; L\big(\left|x - x'\right| + \mathcal{W}_2(\mu,\mu')\big), \tag{A-Lip}$$

$$\left|b(x,\mu)\right| \;\le\; L\Big(1 + |x| + \Big(\textstyle\int |y|^2\, \mu(\mathrm{d}y)\Big)^{1/2}\Big), \tag{A-Growth}$$

and the diffusion is uniformly nondegenerate,

$$\sigma\sigma^\top \succeq \lambda I, \qquad \lambda > 0. \tag{A-Nondeg}$$

Each of the three conditions earns its place for a definite reason, and it repays the reader to see the "why" before any of them is used. Take (A-Lip) first. The thread specification states (A-Lip) as "Lipschitz in the state, uniformly in the measure"; here it is strengthened to *joint* Lipschitz continuity in (x, μ) with respect to the combined distance $|\cdot| + \mathcal{W}_2$, and holding the measure fixed recovers the thread's version verbatim. The strengthening is not decorative. Every well-posedness and propagation-of-chaos argument in this chapter is, at bottom, the same contraction: one estimates how much the drift changes when *both* a particle's position and the surrounding measure are perturbed, and one needs that change to be controlled by the perturbations of *both* inputs at once. Were b Lipschitz in x but only, say, continuous in μ, the difference $b(x, \mu) - b(x', \mu')$ could not be bounded by $L\mathcal{W}_2(\mu, \mu')$ in the measure slot, the Grönwall step in theorem 7.9 would have nothing to feed on, and the fixed-point map closing the McKean–Vlasov equation would lose its contraction. Joint Lipschitzness in $|\cdot| + \mathcal{W}_2$ is exactly the hypothesis that makes a perturbation of the law as controllable as a perturbation of the position—the indispensable ingredient of every contraction below.

The growth bound (A-Growth) is the (A-Growth) of the thread specification: the drift grows at most linearly in the state and in the population's root-second-moment. Its job is custodial rather than structural—it prevents the solution from exploding in finite time and guarantees the second moments $\mathbb{E}\Big[|X_t|^2\Big]$ stay finite, which is what lets Itô's formula and the moment estimates run. It buys existence on the whole interval $[0, T]$, not merely up to a blow-up time.

Nondegeneracy (A-Nondeg) is the (A-Nondeg) of the thread specification, and it stands apart from the other two: it is held *in reserve.* We draw attention now to the fact that it plays *no role* in the existence, uniqueness, or chaos results of sections 7.2 to 7.3—those rest on (A-Lip)–(A-Growth) alone. Its sole function is to supply parabolic regularization: a uniformly elliptic diffusion smooths the law, so that under (A-Nondeg) the limiting marginal m_t acquires a strictly positive C^∞ density solving the nonlinear Fokker–Planck equation classically (section 7.4). We keep it visible in the standing list because it is part of the thread's standing hypotheses, but we will be scrupulous about when it is and is not invoked, precisely so the reader can see which conclusions survive its removal (all the probabilistic ones do).

It is reassuring—and important for the worked examples—that the abstract hypothesis (A-Lip) is automatic for the pairwise drifts (7.2) whenever the kernel is Lipschitz, and that this can be checked by hand without any heavy machinery. The point deserves emphasis because the natural tool, Kantorovich–Rubinstein duality, is *not* needed: a direct coupling estimate does the whole job and keeps the chapter self-contained.

Lemma 7.4 (Pairwise Lipschitz kernels satisfy (A-Lip))**.** *If $b_0 : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$ is Lipschitz, with $\ell_x := \mathrm{Lip}(b_0(x,\cdot))$ its Lipschitz constant in the second variable, then for every fixed x the map $\mu \mapsto \int b_0(x,\cdot)\,\mathrm{d}\mu$ is ℓ_x-Lipschitz in $\mathcal{W}_2$:*

$$\Big| \int b_0(x,y)\,\mu(\mathrm{d}y) - \int b_0(x,y')\,\mu'(\mathrm{d}y')\Big| \;\le\; \ell_x\,\mathcal{W}_2(\mu,\mu') \qquad (\mu,\mu' \in \mathcal{P}_2(\mathbb{R}^d)).$$

Consequently a pairwise drift (7.2) with b_0 Lipschitz in both *arguments satisfies (A-Lip), and if in addition b_0 has linear growth it satisfies (A-Growth).*

Proof. Let π be *any* coupling of μ and μ', that is, any probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with first marginal μ and second marginal μ'. Because the two integrals are the two marginals of $\int b_0(x,\cdot)\,\mathrm{d}\pi$ read in the respective slots,

$$\int b_0(x,y)\,\mu(\mathrm{d}y) - \int b_0(x,y')\,\mu'(\mathrm{d}y') = \int_{\mathbb{R}^d\times\mathbb{R}^d} \big[b_0(x,y) - b_0(x,y')\big]\,\pi(\mathrm{d}y,\mathrm{d}y').$$

Taking norms inside the integral and using the Lipschitz bound $|b_0(x,y) - b_0(x,y')| \le \ell_x\,|y-y'|$,

$$\Big| \int b_0(x,y)\,\mu(\mathrm{d}y) - \int b_0(x,y')\,\mu'(\mathrm{d}y')\Big| \le \ell_x \int |y-y'|\;\pi(\mathrm{d}y,\mathrm{d}y') \le \ell_x \Big(\int |y-y'|^2\;\pi(\mathrm{d}y,\mathrm{d}y')\Big)^{1/2},$$

the last step by the Cauchy–Schwarz (Jensen) inequality $\mathbb{E}[|Z|] \le \mathbb{E}\big[|Z|^2\big]^{1/2}$ applied under π. Now take the infimum over all couplings π: the right-hand square root is, by the very definition of the Wasserstein metric (chapter 4), minimized at the optimal coupling to $\mathcal{W}_2(\mu,\mu')$. This proves the displayed Lipschitz bound. For the consequence: combining it with the Lipschitz bound of b_0 in its first variable gives $|b(x,\mu) - b(x',\mu')| \le \mathrm{Lip}_1(b_0)\,|x-x'| + \ell\,\mathcal{W}_2(\mu,\mu')$ with $\ell := \sup_x \ell_x$, which is (A-Lip) with $L = \max(\mathrm{Lip}_1(b_0),\ell)$; the growth claim is the same estimate applied to $b_0(x,y)$ against $b_0(0,0)$. ☐

No appeal to Kantorovich–Rubinstein duality was made; an admissible coupling and Cauchy–Schwarz sufficed, and the argument is entirely internal to the Wasserstein machinery of chapter 4.

The same idea—bound a difference of measure-integrals by choosing a convenient coupling—is about to do heavier lifting, and in the one special situation we need most it gives a strikingly clean bound that we isolate as a standalone fact, because it will be used four times over in this chapter and the next.

Lemma 7.5 (Index-matching bound for empirical measures)**.** *Let $x_1, \dots, x_N$ and $x'_1, \dots, x'_N$ be two configurations in $\mathbb{R}^d$, with empirical measures $\mu = \frac{1}{N}\sum_j \delta_{x_j}$ and $\mu' = \frac{1}{N}\sum_j \delta_{x'_j}$. Then*

$$\mathcal{W}_2(\mu, \mu')^2 \;\le\; \frac{1}{N}\sum_{j=1}^{N} \left|x_j - x'_j\right|^2. \tag{7.3}$$

Proof. Match the jth atom of μ to the jth atom of μ'. Formally, set

$$\pi := \frac{1}{N}\sum_{j=1}^{N} \delta_{(x_j, x'_j)} \;\in\; \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d).$$

Its first marginal is $\frac{1}{N}\sum_j \delta_{x_j} = \mu$ and its second marginal is $\frac{1}{N}\sum_j \delta_{x'_j} = \mu'$, so π is an admissible coupling of μ and μ'. By the definition of $\mathcal{W}_2$ as the infimum of $\big(\int |y - y'|^2 \, \mathrm{d}\pi\big)^{1/2}$ over couplings, *any* admissible coupling furnishes an upper bound, and this one gives

$$\mathcal{W}_2(\mu, \mu')^2 \;\le\; \int_{\mathbb{R}^d \times \mathbb{R}^d} |y - y'|^2 \, \pi(\mathrm{d}y, \mathrm{d}y') = \frac{1}{N}\sum_{j=1}^{N} \left|x_j - x'_j\right|^2,$$

which is (7.3). □

The inequality (7.3) says something one should keep in mind: matching atoms by their *index* is rarely the optimal transport plan—a cleverer pairing could move less mass—so $\mathcal{W}_2$ between two empirical measures is generally *strictly* smaller than the index-matched root-mean-square displacement. But for our purposes the inequality is exactly the right tool, because in every estimate below the two configurations will be two evolutions of the *same* labeled particles (the true system and a coupled copy), and we will already control $\frac{1}{N}\sum_j \left|x_j - x'_j\right|^2$ particle by particle. The lemma converts that particle-wise control into control of the Wasserstein distance between the two empirical measures, with no constant lost. It recurs in the proof of proposition 7.6 immediately below, in the contraction estimate of theorem 7.9, and twice inside the Sznitman coupling of theorem 7.15; establishing it once, carefully, is what makes those later appearances one-liners.

We can now dispatch the foundational facts about the N-body system. Two are needed: that (7.1) has a unique solution at all, and that this solution is exchangeable—the rigorous form of "the particles are indistinguishable." The second is the one that earns the empirical-measure language its keep: because the coefficients see the configuration only through the symmetric object μ^N, no particle's label can break into the dynamics, and consequently any statistic we ever compute is a symmetric function of the configuration. This is what licenses passing, in the limit, from N coupled particles to a single representative one.

Proposition 7.6 (Well-posedness of the N-particle system)**.** *Under assumption 7.3, the system (7.1) has a unique strong solution on $[0,T]$, and the solution is exchangeable: for every permutation π of $\{1,\dots,N\}$ the law of $(X^{\pi(1),N},\dots,X^{\pi(N),N})$ equals that of $(X^{1,N},\dots,X^{N,N})$.*

Proof. Well-posedness. The point is to recognize (7.1) as a single $\mathbb{R}^{Nd}$-valued Itô equation with globally Lipschitz, linear-growth coefficients, so that the standard theory of chapter 5 applies verbatim. Stack the states into $\mathbf{X}=(X^{1,N},\dots,X^{N,N})\in\mathbb{R}^{Nd}$ and write the stacked drift block by block,

$$\mathbf{b}(\mathbf{X}):=\big(b(X^i,\mu^N[\mathbf{X}])\big)_{i=1}^N\in\mathbb{R}^{Nd},\qquad \mu^N[\mathbf{X}]:=\frac{1}{N}\sum_{j=1}^N\delta_{X^j},$$

and let the stacked noise be $\mathbf{B}=(W^1,\dots,W^N)$ with the block-diagonal diffusion $\operatorname{diag}(\sigma,\dots,\sigma)$. Then (7.1) reads $\mathrm{d}\mathbf{X}_t=\mathbf{b}(\mathbf{X}_t)\,\mathrm{d}t+\operatorname{diag}(\sigma)\,\mathrm{d}\mathbf{B}_t$, an ordinary Itô SDE on $\mathbb{R}^{Nd}$. It remains to check $\mathbf{b}$ is globally Lipschitz and of linear growth.

Fix two stacked configurations $\mathbf{X},\mathbf{X}'$, with empirical measures μ^N,μ'^N. For the ith block, the joint Lipschitz bound (A-Lip) gives

$$\left|\mathbf{b}(\mathbf{X})_i-\mathbf{b}(\mathbf{X}')_i\right|=\left|b(X^i,\mu^N)-b(X'^i,\mu'^N)\right|\le L\big(\left|X^i-X'^i\right|+\mathcal{W}_2(\mu^N,\mu'^N)\big).$$

The two empirical measures are built from the same number N of atoms, indexed the same way, so lemma 7.5 applies and bounds the measure term *independently of i*:

$$\mathcal{W}_2(\mu^N,\mu'^N)^2\le\frac{1}{N}\sum_{j=1}^N\left|X^j-X'^j\right|^2=\frac{1}{N}\,\left|\mathbf{X}-\mathbf{X}'\right|^2.$$

Substituting and using $(a+b)^2\le 2a^2+2b^2$,

$$\left|\mathbf{b}(\mathbf{X})_i-\mathbf{b}(\mathbf{X}')_i\right|^2\le 2L^2\Big(\left|X^i-X'^i\right|^2+\tfrac{1}{N}\left|\mathbf{X}-\mathbf{X}'\right|^2\Big).$$

Now sum over the N blocks $i=1,\dots,N$. The first term sums to $\sum_i\left|X^i-X'^i\right|^2=\left|\mathbf{X}-\mathbf{X}'\right|^2$, and the second—being *independent of i*—is added N times, contributing $N\cdot\frac{1}{N}\left|\mathbf{X}-\mathbf{X}'\right|^2=\left|\mathbf{X}-\mathbf{X}'\right|^2$. This cancellation of the factor N against the $\frac{1}{N}$ normalization of the empirical measure is the crux: it is exactly why summing the per-block bound over all i leaves a constant that does not grow with N. Hence

$$\left|\mathbf{b}(\mathbf{X})-\mathbf{b}(\mathbf{X}')\right|^2=\sum_{i=1}^N\left|\mathbf{b}(\mathbf{X})_i-\mathbf{b}(\mathbf{X}')_i\right|^2\le 2L^2\big(\left|\mathbf{X}-\mathbf{X}'\right|^2+\left|\mathbf{X}-\mathbf{X}'\right|^2\big)=4L^2\left|\mathbf{X}-\mathbf{X}'\right|^2,$$

so $\mathbf{b}$ is globally Lipschitz on $\mathbb{R}^{Nd}$ with constant $2L$, *uniformly in N*. The identical computation with (A-Growth) in place of (A-Lip) (comparing $\mathbf{b}(\mathbf{X})$ to 0) gives the linear-growth bound $|\mathbf{b}(\mathbf{X})|\le C(1+|\mathbf{X}|)$. The block-diagonal diffusion is constant, hence trivially Lipschitz and bounded. The strong existence-and-uniqueness theorem for Itô SDEs with Lipschitz, linear-growth coefficients (chapter 5) therefore furnishes a unique strong solution $\mathbf{X}$ on $[0,T]$, adapted to the filtration generated by the W^i and the initial data, with $\sup_{t\le T}\mathbb{E}\Big[\left|\mathbf{X}_t\right|^2\Big]<\infty$.

Exchangeability. Fix a permutation σ of $\{1,\dots,N\}$ and consider the relabeled process $Y^i:=X^{\sigma(i),N}$. We claim $\mathbf{Y}=(Y^1,\dots,Y^N)$ solves the *same* system (7.1), but driven by the relabeled inputs $\widetilde{W}^i := W^{\sigma(i)}$ and $Y_0^i = X_0^{\sigma(i),N}$. Indeed, the empirical measure is symmetric in its atoms, $\frac{1}{N}\sum_j \delta_{Y^j} = \frac{1}{N}\sum_j \delta_{X^{\sigma(j)}} = \frac{1}{N}\sum_k \delta_{X^k} = \mu^N$, so permuting the labels leaves μ_t^N unchanged; applying the ith equation of (7.1) at index $\sigma(i)$ gives

$$\mathrm{d}Y_t^i = \mathrm{d}X_t^{\sigma(i),N} = b\big(X_t^{\sigma(i),N}, \mu_t^N\big)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t^{\sigma(i)} = b\big(Y_t^i, \mu_t^N\big)\ \mathrm{d}t + \sigma\ \mathrm{d}\widetilde{W}_t^i,$$

which is precisely the ith equation of the system with the tilde'd inputs. Now use that the inputs are exchangeable in distribution: the initial data $X_0^{i,N}$ are i.i.d. and the noises W^i are i.i.d., so the relabeled inputs $(X_0^{\sigma(i),N}, W^{\sigma(i)})_i$ have the *same joint law* as the original inputs $(X_0^{i,N}, W^i)_i$. Since the solution of the SDE is a measurable functional of its inputs and the system the two solve is identical, the strong uniqueness just established forces $\mathbf{Y}$ to be that same functional of the relabeled inputs; equality of the input laws then gives $\mathcal{L}(\mathbf{Y}) = \mathcal{L}(\mathbf{X})$, i.e. $\mathcal{L}(X^{\sigma(1),N}, \dots, X^{\sigma(N),N}) = \mathcal{L}(X^{1,N}, \dots, X^{N,N})$. This is exchangeability. The mechanism is worth naming once: exchangeability is *forced*, not assumed, and it is forced precisely because the coefficients depend on the configuration only through the label-blind object μ^N while the i.i.d. inputs are themselves exchangeable. □

We now have what the next section needs: a unique, exchangeable N-body process (7.1), governed by a drift that is jointly Lipschitz in state and measure (A-Lip), and the structural principle that *only the empirical measure can enter the dynamics*—no particle's identity ever does. That principle is the hinge. Exchangeability says all N particles are statistically interchangeable, so to understand the swarm it suffices to understand one representative particle; and the only thing that particle reacts to is μ_t^N. The natural move, then, is to ask what happens to a single particle when the empirical measure it feels stops fluctuating—when, by the law of large numbers anticipated in the preamble, μ_t^N settles onto a deterministic flow of measures $t \mapsto m_t$. Freezing μ_t^N to such an m_t decouples the chosen particle from the rest and leaves it obeying a single SDE whose drift depends on m_t; imposing that m_t be the law the particle thereby generates closes the loop. That is the McKean–Vlasov equation, and constructing it is the work of section 7.2.

7.2 The McKean–Vlasov stochastic differential equation

The structural principle that closed section 7.1—that the only thing any particle ever reacts to is the empirical measure μ_t^N, never the identity of its neighbors—is now ready to do its work. A single representative particle's dynamics are pinned down once we know the flow of measures it feels, and exchangeability guarantees that one such representative already describes the whole swarm. So we make precisely the move the previous section set up. We follow one particle, say $i = 1$, and we imagine that the law of large numbers anticipated in the chapter preamble has already done its job: that the fluctuating random measure μ_t^N has settled, as $N \to \infty$, onto a *deterministic* flow of probability measures $t \mapsto m_t$. This is not yet a theorem—that the genuine μ_t^N really does concentrate on such a flow is the content of propagation of chaos in section 7.3—but it is the right heuristic to follow, because it tells us exactly what equation the limiting particle must satisfy, and that equation will turn out to be well-posed in its own right.

Carry the heuristic through. With μ_t^N replaced by a frozen deterministic flow m_t, the first particle's equation in (7.1) loses its coupling to the others entirely: the noises W^j of the other particles no longer enter, and what remains is a *single* Itô SDE,

$$\mathrm{d}X_t^1 = b(X_t^1,\, m_t)\;\mathrm{d}t + \sigma\;\mathrm{d}W_t^1,$$

in which the population appears only through the external field m_t. But m_t cannot be chosen freely. We posited it as the limit of $\mu_t^N = \frac{1}{N}\sum_j \delta_{X_t^{j,N}}$, and in the limit, by exchangeability, every particle has the same law as the first; so the deterministic flow the particle feels must coincide with the law the particle itself generates: $m_t = \mathcal{L}(X_t^1)$. This self-consistency condition is the entire substance of the mean-field limit. Dropping the now-redundant index 1, we arrive at an equation whose drift coefficient depends on the law of its own solution.

Definition 7.7 (McKean–Vlasov SDE)**.** A *McKean–Vlasov* (or *nonlinear*, in McKean's sense) SDE is

$$\mathrm{d}X_t \;=\; b\big(X_t,\; \mathcal{L}(X_t)\big)\;\mathrm{d}t \;+\; \sigma\;\mathrm{d}W_t, \qquad X_0 \sim m_0, \tag{7.4}$$

where $\mathcal{L}(X_t) \in \mathcal{P}_2(\mathbb{R}^d)$ is the law of the solution at time t. A *strong solution* is a process X adapted to the filtration generated by W and X_0, with $\sup_{t\le T}\mathbb{E}\Big[|X_t|^2\Big] < \infty$, satisfying (7.4) together with the self-consistency $m_t = \mathcal{L}(X_t)$ for all t.

It is worth dwelling on the word "nonlinear," because it names something genuinely new and is easy to misread. It does *not* refer to nonlinearity in the state variable x: the drift $b(\cdot, m_t)$ may perfectly well be linear in x, as it is in the carried linear–quadratic thread, and (7.4) is still called nonlinear. The nonlinearity is in the *law*. To see the structure plainly, contrast it with the ordinary SDEs of chapter 5. There the coefficients were prescribed functions of (t, x) fixed in advance: one specified $b(t, x)$ and $\sigma(t, x)$ as data, and the equation $\mathrm{d}X_t = b(t, X_t)\;\mathrm{d}t + \sigma(t, X_t)\;\mathrm{d}W_t$ then *determined* the law $\mathcal{L}(X_t)$ as an output, with no feedback. In (7.4) the law is no longer a passive output: it is plugged back into the very coefficient that drives the equation. The unknown is therefore not just a path $t \mapsto X_t$ but a path *together with* the flow of laws it induces, and the two must be consistent. Said operationally, (7.4) is a *fixed-point equation in path space*: one seeks a process X such that the map $X \mapsto \mathcal{L}(X)$, fed into the drift, reproduces a process with that very law. It is this self-referential map $X \mapsto \mathcal{L}(X)$—an operation on the *distribution* of the unknown, not on its values—that makes the equation nonlinear in McKean's probabilistic sense, and that places it outside the scope of chapter 5. The remainder of this section shows that, despite the self-reference, the equation has one and only one solution, on *any* finite horizon.

Remark 7.8 (The decoupled case as a sanity check)**.** The cleanest way to feel what the law-dependence adds is to switch it off. Suppose the drift does not in fact depend on its measure argument, $b(x, \mu) = b_0(x)$. Then (7.4) degenerates to the ordinary SDE $\mathrm{d}X_t = b_0(X_t)\;\mathrm{d}t + \sigma\;\mathrm{d}W_t$ of chapter 5, the self-consistency condition $m_t = \mathcal{L}(X_t)$ becomes a tautology that imposes nothing, and there is no fixed point to solve—the law is read off at the end and never fed back. Everything genuinely new in this chapter lives in the *coupling* of the drift to m_t, which is exactly the ingredient the fixed-point argument below must control. This is the same lesson the $O(1/N)$ scaling taught at the level of the particle system, now seen at the level of the limit equation.

The strategy for solving (7.4) is dictated by the fixed-point reading we just gave it. If a McKean–Vlasov solution is a flow of measures that reproduces itself, then we should build the map "flow of measures $\mapsto$ the flow it reproduces" explicitly and hunt for its fixed point. That map is the conceptual engine of the entire well-posedness theory, and it is constructed by the one device the heuristic already used: *freeze the measure*. Given any candidate flow μ_t, freezing it turns the daunting law-dependent (7.4) into a tame ordinary SDE—law-dependence gone, coefficients now honest functions of (t, x) to which the full machinery of chapter 5 applies—which we solve, and whose marginal flow we read back off. A flow that comes back to itself under this procedure is a McKean–Vlasov solution; a flow that does not is not. The theorem below makes this precise and, crucially, shows that the procedure contracts.

Theorem 7.9 (Well-posedness of the McKean–Vlasov SDE)**.** *Under assumption 7.3 and* $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$*, the McKean–Vlasov SDE* (7.4) *has a unique strong solution on* $[0, T]$*. Its flow of marginals* $t \mapsto m_t = \mathcal{L}(X_t)$ *lies in* $C([0, T]; \mathcal{P}_2(\mathbb{R}^d))$*, and* $\sup_{t\le T} \mathbb{E}\Big[|X_t|^2\Big] \le C\big(1 + \mathbb{E}\Big[|X_0|^2\Big]\big)e^{CT}$.

Proof. We run a Picard fixed point on the space $\mathcal{C} := C([0, T]; \mathcal{P}_2(\mathbb{R}^d))$ of continuous flows of second-moment measures, metrized by $\mathbf{d}_t(\mu, \nu) := \sup_{s\le t} \mathcal{W}_2(\mu_s, \nu_s)$. The proof has two steps: build the freeze-solve-read-off map Φ and verify it lands inside $\mathcal{C}$ (Step 1); then show that a *power* of Φ contracts, no matter how large T is (Step 2).

Step 1 (the decoupled map Φ). Fix a flow $\mu = (\mu_s)_{s\le T} \in \mathcal{C}$ and freeze the measure argument of the drift at it. This produces the *ordinary* (non-McKean) SDE

$$\mathrm{d}X^\mu_t = b\big(X^\mu_t, \mu_t\big)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t, \qquad X^\mu_0 = X_0 \sim m_0, \tag{7.5}$$

in which μ_t is now a *given* time-dependent input, not an unknown. The coefficient $(t, x) \mapsto b(x, \mu_t)$ inherits from (A-Lip)–(A-Growth) both a Lipschitz bound in x (with constant L, uniformly in t, since freezing the second slot only removes a term from (A-Lip)) and a linear-growth bound in x (the measure-dependent piece $\big(\int |y|^2\, \mu_t(\mathrm{d}y)\big)^{1/2}$ is finite and bounded in t because $\mu \in \mathcal{C}$ has uniformly bounded second moments, so it contributes only a bounded additive constant). The classical existence and uniqueness theory for Itô SDEs with Lipschitz, linear-growth coefficients (chapter 5) therefore furnishes a unique strong solution X^μ on $[0, T]$ with $\sup_{t\le T} \mathbb{E}\Big[|X^\mu_t|^2\Big] < \infty$. Define the map by reading off its marginal flow,

$$\Phi(\mu)_t := \mathcal{L}(X^\mu_t) \in \mathcal{P}_2(\mathbb{R}^d).$$

We must check that $\Phi(\mu)$ is itself a continuous flow, i.e. $\Phi(\mu) \in \mathcal{C}$; the skeleton draft delegated this to chapter 5, but it is short enough to give in full, and seeing it makes the L^2-continuity concrete. Fix $0 \le s < t \le T$. From the integral form of (7.5),

$$X^\mu_t - X^\mu_s = \int_s^t b(X^\mu_r, \mu_r)\ \mathrm{d}r + \sigma\,(W_t - W_s),$$

so by $(a + b)^2 \le 2a^2 + 2b^2$ and the Cauchy–Schwarz inequality applied to the time integral,

$$\mathbb{E}\Big[|X^\mu_t - X^\mu_s|^2\Big] \le 2(t - s)\int_s^t \mathbb{E}\Big[|b(X^\mu_r, \mu_r)|^2\Big]\,\mathrm{d}r + 2\,\mathbb{E}\Big[|\sigma(W_t - W_s)|^2\Big].$$

The drift integrand is bounded uniformly in $r \le T$ by the growth bound (A-Growth) together with $\sup_{r\le T}\mathbb{E}\Big[|X^\mu_r|^2\Big] < \infty$ and $\sup_{r\le T}\int |y|^2\,\mu_r(\mathrm{d}y) < \infty$, say by a constant K; and the Brownian increment contributes $\mathbb{E}\Big[|\sigma(W_t - W_s)|^2\Big] = \operatorname{tr}(\sigma\sigma^\top)(t-s)$. Hence

$$\mathbb{E}\Big[|X^\mu_t - X^\mu_s|^2\Big] \le 2K(t-s)^2 + 2\operatorname{tr}(\sigma\sigma^\top)(t-s) \xrightarrow[t\to s]{} 0.$$

Finally, the *same-path* pair (X^μ_s, X^μ_t) is an admissible coupling of $\Phi(\mu)_s$ and $\Phi(\mu)_t$, so by the very definition of the Wasserstein distance $\mathcal{W}_2(\Phi(\mu)_s,\Phi(\mu)_t)^2 \le \mathbb{E}\Big[|X^\mu_t - X^\mu_s|^2\Big] \to 0$. Thus $t\mapsto \Phi(\mu)_t$ is $\mathcal{W}_2$-continuous, $\Phi(\mu)\in\mathcal{C}$, and we have a genuine self-map $\Phi:\mathcal{C}\to\mathcal{C}$.

The point of the construction is its design feature: a flow μ is a *fixed point* $\Phi(\mu)=\mu$ if and only if the frozen SDE (7.5) is fed the very law it produces, $\mu_t = \mathcal{L}(X^\mu_t)$, which is exactly the self-consistency (7.4) demands; and in that case X^μ solves the McKean–Vlasov equation. Solving (7.4) is therefore *identical* to finding a fixed point of Φ, and nothing has been lost in the translation.

Step 2 (a power of Φ contracts, for any T). We estimate how much Φ separates two inputs. Take two flows $\mu,\nu\in\mathcal{C}$ and—this is the decisive device—drive the two frozen SDEs (7.5) by the *same* Brownian motion W and start them from the *same* initial datum X_0. Then the stochastic integral $\sigma\,\mathrm{d}W_t$ and the initial condition are *identical* in the two equations and cancel exactly in the difference $X^\mu_t - X^\nu_t$; this is what makes the estimate pathwise and removes the Itô integral from view altogether:

$$X^\mu_t - X^\nu_t = \int_0^t \big[b(X^\mu_s,\mu_s) - b(X^\nu_s,\nu_s)\big]\,\mathrm{d}s.$$

Apply the joint Lipschitz bound (A-Lip) inside the integral, then Cauchy–Schwarz in time (which puts a factor t out front), then $(a+b)^2 \le 2a^2+2b^2$ to split the position discrepancy from the measure discrepancy:

$$|X^\mu_t - X^\nu_t|^2 \le t\int_0^t L^2\big(|X^\mu_s - X^\nu_s| + \mathcal{W}_2(\mu_s,\nu_s)\big)^2\,\mathrm{d}s \le 2TL^2\int_0^t\big(|X^\mu_s - X^\nu_s|^2 + \mathcal{W}_2(\mu_s,\nu_s)^2\big)\,\mathrm{d}s.$$

Now take expectations and write $g(t) := \mathbb{E}\Big[|X^\mu_t - X^\nu_t|^2\Big]$. With $B := 2TL^2$ this reads

$$g(t) \;\le\; B\int_0^t g(s)\,\mathrm{d}s \;+\; B\int_0^t \mathcal{W}_2(\mu_s,\nu_s)^2\,\mathrm{d}s.$$

The second integral is a known forcing term, controlled by the separation of the inputs: writing $h(s) := \sup_{r\le s}\mathcal{W}_2(\mu_r,\nu_r)^2 = \mathbf{d}_s(\mu,\nu)^2$, which is nondecreasing, the forcing is at most $a(t) := B\int_0^t h(s)\,\mathrm{d}s$, itself nondecreasing. So $g(t)\le a(t) + B\int_0^t g(s)\,\mathrm{d}s$ with a nondecreasing, and the integral form of Grönwall's inequality (lemma C.1) gives $g(t)\le a(t)\,e^{Bt}$. Spelling out the bookkeeping with $t\le T$ and $B = 2TL^2$,

$$g(t) \;\le\; e^{Bt}\,B\int_0^t h(s)\,\mathrm{d}s \;\le\; \underbrace{2TL^2\,e^{2T^2L^2}}_{=:C}\int_0^t h(s)\,\mathrm{d}s,$$

the exponent being $Bt \le BT = 2T^2L^2$; this is exactly where the displayed constant $C = 2TL^2e^{2T^2L^2}$ comes from. Finally, the common-noise pair (X^μ_s, X^ν_s) is an admissible coupling of $\Phi(\mu)_s$ and $\Phi(\nu)_s$, so $\mathcal{W}_2(\Phi(\mu)_s, \Phi(\nu)_s)^2 \le g(s)$; taking the supremum over $s \le t$ (the right side is already nondecreasing in t) yields the basic one-step estimate

$$\mathbf{d}_t\big(\Phi(\mu), \Phi(\nu)\big)^2 \;=\; \sup_{s\le t} \mathcal{W}_2\big(\Phi(\mu)_s, \Phi(\nu)_s\big)^2 \;\le\; C\int_0^t \mathbf{d}_s(\mu,\nu)^2 \,\mathrm{d}s, \qquad C = 2TL^2 e^{2T^2L^2}. \tag{7.6}$$

A single application of (7.6) is *not* a contraction unless $CT < 1$, that is, unless T is small—the familiar obstruction that a naive Picard iteration overcomes only on a short horizon. The escape is to iterate (7.6) and exploit the time integral, which manufactures a shrinking factorial. Apply it once to $\Phi\mu, \Phi\nu$ and substitute (7.6) again for the integrand:

$$\mathbf{d}_t(\Phi^2\mu, \Phi^2\nu)^2 \le C\int_0^t \mathbf{d}_s(\Phi\mu,\Phi\nu)^2 \,\mathrm{d}s \le C\int_0^t \Big(C\int_0^s \mathbf{d}_r(\mu,\nu)^2 \,\mathrm{d}r\Big)\,\mathrm{d}s \le C^2\,\mathbf{d}_T(\mu,\nu)^2 \int_0^t s\,\mathrm{d}s = C^2\,\mathbf{d}_T(\mu,\nu)^2\,\frac{t^2}{2},$$

where we bounded $\mathbf{d}_r(\mu,\nu)^2 \le \mathbf{d}_T(\mu,\nu)^2$ and did the inner integral. The same step repeated n times produces the nested time integral $\int_0^t\int_0^{s_1}\cdots\int_0^{s_{n-1}} \mathrm{d}s_n\cdots\mathrm{d}s_1 = t^n/n!$, hence by induction

$$\mathbf{d}_t(\Phi^n\mu, \Phi^n\nu)^2 \;\le\; \frac{(Ct)^n}{n!}\,\mathbf{d}_T(\mu,\nu)^2, \qquad \text{and at } t = T, \qquad \mathbf{d}_T(\Phi^n\mu, \Phi^n\nu)^2 \;\le\; \frac{(CT)^n}{n!}\,\mathbf{d}_T(\mu,\nu)^2.$$

The constant C and the horizon T are fixed, but $(CT)^n/n! \to 0$ as $n \to \infty$ *whatever* the value of CT, because the factorial outruns any geometric growth. Choose n large enough that $(CT)^n/n! < 1$; then Φ^n is a strict contraction on the complete metric space $(\mathcal{C}, \mathbf{d}_T)$. By the version of the Banach fixed point theorem for maps a power of which contracts (theorem 2.36), Φ has a unique fixed point $m \in \mathcal{C}$. By Step 1 this fixed point is precisely the marginal flow of the unique strong solution of (7.4), and conversely any solution's flow is such a fixed point, so the solution exists and is unique.

Moment bound. It remains to bound the solution's second moment. Apply Itô's formula to $|X_t|^2$ along (7.4): the martingale part has zero expectation, leaving

$$\mathbb{E}\Big[|X_t|^2\Big] = \mathbb{E}\Big[|X_0|^2\Big] + \mathbb{E}\Big[\int_0^t \big(2X_s\cdot b(X_s,m_s) + \mathrm{tr}(\sigma\sigma^\top)\big)\,\mathrm{d}s\Big].$$

The growth bound (A-Growth) with $m_s = \mathcal{L}(X_s)$, so that $\int |y|^2\, m_s(\mathrm{d}y) = \mathbb{E}\Big[|X_s|^2\Big]$, gives $2X_s\cdot b(X_s, m_s) \le C'(1 + |X_s|^2 + \mathbb{E}\Big[|X_s|^2\Big])$ after Young's inequality; taking expectations and writing $\psi(t) = \mathbb{E}\Big[|X_t|^2\Big]$ yields $\psi(t) \le \mathbb{E}\Big[|X_0|^2\Big] + C''\int_0^t (1+\psi(s))\,\mathrm{d}s$, and Grönwall (lemma C.1) closes it as $\sup_{t\le T}\mathbb{E}\Big[|X_t|^2\Big] \le C(1+\mathbb{E}\Big[|X_0|^2\Big])e^{CT}$. □

Remark 7.10 (Where the hypotheses bite)**.** It repays attention to track exactly which of the standing hypotheses (A-Lip)–(A-Nondeg) the proof consumed, because the answer cleanly separates the probabilistic core of the theory from a regularity refinement that comes later. Three points stand out.

First, *nondegeneracy* (A-Nondeg) *was never used.* The diffusion entered the argument only as the term that *cancels*: the common-noise, common-initial-data coupling of Step 2 was engineered precisely so that $\sigma\,\mathrm{d}W$ drops out of the difference $X^\mu - X^\nu$, leaving the drift estimate, which calls on (A-Lip)–(A-Growth) alone. Existence and uniqueness of the McKean–Vlasov solution—and, in section 7.3, propagation of chaos—therefore hold for *degenerate* σ as well, even $\sigma = 0$. Nondegeneracy is held in reserve for a single purpose: in section 7.4 it makes the generator uniformly elliptic, so that the marginal flow m_t acquires a smooth strictly positive density solving the nonlinear Fokker–Planck equation classically. The probabilistic edifice of this section does not see it.

Second, the *global-in-T* character of the result is not an accident of small data or a short horizon—it is structural, and the mechanism is the nilpotent-kernel iteration $(CT)^n/n! \to 0$. A direct contraction would force $CT < 1$ and confine us to short times, after which one would have to restart and patch intervals together. The factorial removes that restriction outright: *some* power Φ^n contracts no matter how large C and T are, and a map a power of which contracts already has a unique fixed point. This is why theorem 7.9 asserts well-posedness on an arbitrary finite horizon with no smallness assumption whatsoever, and it is the same Volterra-type mechanism that will let the coupled estimates of section 7.3 run all the way to time T.

Third, the constant volatility was, as promised in section 7.1, a convenience and not a crutch. For a measure-dependent volatility $\sigma(x, \mu)$ the argument repeats almost verbatim, with one bookkeeping change and one structural one. The bookkeeping change: the difference $X_t^\mu - X_t^\nu$ now retains a stochastic-integral term $\int_0^t \left[\sigma(X_s^\mu, \mu_s) - \sigma(X_s^\nu, \nu_s)\right] \mathrm{d}W_s$, and taking expectations of its square is handled by the Itô isometry, which converts it into $\int_0^t \mathbb{E}\Big[|\sigma(X_s^\mu, \mu_s) - \sigma(X_s^\nu, \nu_s)|^2\Big]\,\mathrm{d}s$; under a joint Lipschitz bound on σ this is in turn bounded by another const $\cdot \int_0^t \big(\mathbb{E}\Big[|X_s^\mu - X_s^\nu|^2\Big] + \mathcal{W}_2(\mu_s, \nu_s)^2\big)\,\mathrm{d}s$, a term of *exactly the same shape* as the drift contribution—in particular it adds the extra $\int_0^t \mathcal{W}_2(\mu_s, \nu_s)^2\,\mathrm{d}s$ flagged in section 7.1—which the same Grönwall, and the same factorial iteration, absorb with no new idea, merely a larger constant C. The structural change is the one to watch: when the volatility differs between the two equations one can no longer cancel the noise by sharing the path W pointwise, because the two integrands multiply the *same* increment $\mathrm{d}W_s$ by different matrices and the difference no longer vanishes. The coupling that makes the Itô-isometry estimate legitimate must therefore be moved one level down, onto the *driving* Brownian motion: one couples the two solutions by feeding them the same W and measures the residual through the isometry, rather than by demanding the diffusion terms agree pathwise. That is still a valid coupling and still produces an admissible coupling of the two marginal laws, so every estimate above goes through; it is simply the honest place where "share the noise" must be reinterpreted once the noise coefficient itself depends on the unknown law.

The deliverable of this section is now in hand, and it is worth stating plainly what we carry forward, because the next two sections lean on different faces of it. We have a unique strong solution X of the McKean–Vlasov equation (7.4) on the whole interval $[0, T]$, and—above all—its *deterministic* marginal flow $t \mapsto m_t = \mathcal{L}(X_t)$, the fixed point of Φ living in $C\big([0, T]; \mathcal{P}_2(\mathbb{R}^d)\big)$. This single flow is the hinge of the rest of the chapter. Section 7.3 runs the genuine N-body system (7.1) against N independent copies of *this* McKean–Vlasov process, all sharing this same m_t, and asks the converse-facing question the heuristic left open: in what precise sense, and how fast, do the real interacting particles become independent draws from m_t? And section 7.4 takes the very same flow m_t and characterizes it no longer probabilistically but analytically, as the solution of a nonlinear Fokker–Planck PDE. The law of the one representative particle, constructed here by a fixed point in Wasserstein path space, is thus the common object that the chapter's remaining faces—chaos and Fokker–Planck—will each describe in their own language.

7.3 Propagation of chaos

The limit flow $t \mapsto m_t$ that section 7.2 delivered was constructed *forward*: we posited that the empirical measure had already collapsed onto a deterministic flow, followed one particle, and closed the self-consistency loop. But positing the collapse is not proving it. We must now turn around and face the converse-facing question the heuristic quietly deferred: take the genuine, finite, fully coupled N-body system (7.1), the one whose empirical measure really does fluctuate; in what exact sense, and how fast, do its particles become independent copies of *this* limit law m_t? The forward heuristic told us what equation the limit must satisfy; the present section makes the limit itself a theorem, and it must do so by comparing the two systems that the forward argument kept apart—the interacting reality and its idealized, decoupled surrogate.

Before we can prove anything we must say precisely what is to be proved. "The particles become independent draws from m_t" is a slogan, not a statement: at every finite N the particles are emphatically *not* independent—they are coupled through μ_t^N, and particle 1's position carries information about particle 2's through their shared history. Independence can only be an asymptotic property, and the standard way to make it precise is to fix a finite group of particles and ask whether *that* group becomes independent as the population around it grows. This is the notion of *chaoticity*.

Definition 7.11 (m-chaoticity)**.** A sequence of exchangeable $\mathbb{R}^d$-valued random vectors $\left(X^{1,N}, \dots, X^{N,N}\right)_{N\geq 1}$ is *m-chaotic*, for a fixed $m \in \mathcal{P}(\mathbb{R}^d)$, if for every fixed $k \in \mathbb{N}$ the law of the first k coordinates converges weakly to the product measure,

$$\mathcal{L}(X^{1,N}, \dots, X^{k,N}) \xrightarrow[N\to\infty]{} m^{\otimes k}.$$

Read the definition with care, because the order of quantifiers is the entire content. We fix the group size k *first* and let $N \to \infty$ *after*, so k stays bounded while the surrounding population diverges; the appearance of the *product* measure $m^{\otimes k} = m \otimes \cdots \otimes m$ encodes two assertions at once. That the limiting law factorizes across the k coordinates is asymptotic *independence*: any fixed handful of particles decorrelates as the crowd they are embedded in grows. That each factor is the *same* m is asymptotic *identical distribution*: every particle is, in the limit, a copy of the one representative law. Together: any finite group of particles becomes asymptotically i.i.d. with common law m. By exchangeability (proposition 7.6) there is no loss in always taking the *first* k coordinates—any other k have the same joint law—so the single index k controls the whole tower of finite marginals. The case $k = 2$ already carries the essential physics: it says the pair correlation $\mathcal{L}(X^{1,N}, X^{2,N})$ factorizes into $m \otimes m$, i.e. the connected correlation between any two tagged particles vanishes in the limit. Chaoticity is exactly the statement that all such finite-order correlations switch off, leaving a product state.

There is a second, equivalent way to express the same idea, and it is the one a physicist reaches for first, because it is a law of large numbers. Instead of tracking k tagged particles, watch the empirical measure $\mu^N = \frac{1}{N}\sum_j \delta_{X^{j,N}}$—the "density profile" of the swarm—and ask whether this *random* measure concentrates on the *deterministic* m. The next lemma records that these two formulations, factorization of finite marginals and concentration of the empirical measure, are one and the same. We pause to prove it not because the main theorem will lean on it—it will not, as we explain at the close—but because the equivalence is exactly the bridge between the probabilist's language of chaoticity and the physicist's language of a sharpening density, and seeing why they coincide is genuinely clarifying.

Lemma 7.12 (Chaoticity equals the empirical law of large numbers)**.** *For an exchangeable array $(X^{1,N},\dots,X^{N,N})_{N\geq 1}$ and $m\in\mathcal{P}(\mathbb{R}^d)$, the following are equivalent: (i) the array is m-chaotic; (ii) the random empirical measure $\mu^N=\frac{1}{N}\sum_{j=1}^N\delta_{X^{j,N}}$ converges to the deterministic limit m in probability (in the weak topology of $\mathcal{P}(\mathbb{R}^d)$).*

Proof. Weak convergence is tested by bounded continuous functions, and the topology of $\mathcal{P}(\mathbb{R}^d)$ is metrized by countably many such test functions (chapter 3); it therefore suffices to work with a fixed $\phi\in C_b(\mathbb{R}^d)$ and to show that $\langle\mu^N,\phi\rangle\to\langle m,\phi\rangle$ in probability is equivalent to chaoticity, where $\langle\mu,\phi\rangle:=\int\phi\,\mathrm{d}\mu$. Convergence of the random scalar $\langle\mu^N,\phi\rangle=\frac{1}{N}\sum_j\phi(X^{j,N})$—a sample average of ϕ over the population—to the deterministic number $\langle m,\phi\rangle$ is the law of large numbers in its recognizable form, and this is what we relate to factorization of marginals.

(i)⇒(ii). The strategy is the one the physicist expects of a law of large numbers: show the random average has the right mean and a variance that collapses. Both the mean and the variance of $\langle\mu^N,\phi\rangle$ are determined by the one- and two-particle marginals, and chaoticity controls exactly those. For the mean, exchangeability makes every summand identically distributed, so $\mathbb{E}\Big[\langle\mu^N,\phi\rangle\Big]=\frac{1}{N}\sum_j\mathbb{E}\Big[\phi(X^{j,N})\Big]=\mathbb{E}\Big[\phi(X^{1,N})\Big]$, which tends to $\langle m,\phi\rangle$ by the $k=1$ marginal convergence $\mathcal{L}(X^{1,N})\to m$. For the second moment, expand the square of the sum into a double sum and separate the diagonal from the off-diagonal indices:

$$\mathbb{E}\Big[\langle\mu^N,\phi\rangle^2\Big]=\frac{1}{N^2}\sum_{j,\ell}\mathbb{E}\Big[\phi(X^{j,N})\phi(X^{\ell,N})\Big]=\frac{1}{N}\mathbb{E}\Big[\phi(X^{1,N})^2\Big]+\frac{N-1}{N}\mathbb{E}\Big[\phi(X^{1,N})\phi(X^{2,N})\Big].$$

The bookkeeping behind this split is the heart of the computation, so let us spell it out. The double sum has N^2 terms. Of these, the N *diagonal* terms $j=\ell$ each equal $\mathbb{E}\Big[\phi(X^{j,N})^2\Big]=\mathbb{E}\Big[\phi(X^{1,N})^2\Big]$ by exchangeability; divided by N^2 they contribute $N\cdot\frac{1}{N^2}\mathbb{E}\Big[\phi(X^{1,N})^2\Big]=\frac{1}{N}\mathbb{E}\Big[\phi(X^{1,N})^2\Big]$. The remaining $N(N-1)$ *off-diagonal* ordered pairs $j\neq\ell$ each equal $\mathbb{E}\Big[\phi(X^{1,N})\phi(X^{2,N})\Big]$, again by exchangeability (any two distinct indices have the same pair law as $1,2$); divided by N^2 they contribute $\frac{N(N-1)}{N^2}\mathbb{E}\Big[\phi(X^{1,N})\phi(X^{2,N})\Big]=\frac{N-1}{N}\mathbb{E}\Big[\phi(X^{1,N})\phi(X^{2,N})\Big]$. That is the displayed identity. Now pass to the limit term by term. The diagonal piece is $O(1/N)$ because ϕ is bounded, hence $\mathbb{E}\Big[\phi(X^{1,N})^2\Big]\leq\|\phi\|_\infty^2$; it vanishes—this is precisely the $1/N$ suppression of self-interaction, the same factor that made one neighbor's force negligible in section 7.1. The off-diagonal piece has prefactor $\frac{N-1}{N}\to 1$ and, by the $k=2$ marginal convergence $\mathcal{L}(X^{1,N},X^{2,N})\to m\otimes m$ tested against the bounded continuous function $(x,y)\mapsto\phi(x)\phi(y)$, its expectation tends to $\iint\phi(x)\phi(y)\,m(\mathrm{d}x)m(\mathrm{d}y)=\langle m,\phi\rangle^2$. Hence $\mathbb{E}\Big[\langle\mu^N,\phi\rangle^2\Big]\to\langle m,\phi\rangle^2$, while $\mathbb{E}\Big[\langle\mu^N,\phi\rangle\Big]\to\langle m,\phi\rangle$; subtracting the square of the second limit from the first gives $\operatorname{Var}\langle\mu^N,\phi\rangle=\mathbb{E}\Big[\langle\mu^N,\phi\rangle^2\Big]-\mathbb{E}\Big[\langle\mu^N,\phi\rangle\Big]^2\to\langle m,\phi\rangle^2-\langle m,\phi\rangle^2=0$. A random variable whose mean converges to $\langle m,\phi\rangle$ and whose variance vanishes converges to $\langle m,\phi\rangle$ in L^2, hence in probability by Chebyshev. As ϕ was an arbitrary bounded continuous test function, this is (ii). The mechanism is exactly the decorrelation built into chaoticity: the only thing that could keep $\operatorname{Var}\langle\mu^N,\phi\rangle$ from vanishing is a persistent pair correlation, and the $k=2$ factorization is precisely the statement that there is none.

(ii)⇒(i). We must run this in reverse: from concentration of the empirical measure recover factorization of every fixed k-marginal. Fix k and bounded continuous $\phi_1, \ldots, \phi_k \in C_b(\mathbb{R}^d)$; it suffices to show $\mathbb{E}\Big[\prod_{r=1}^k \phi_r(X^{r,N})\Big] \to \prod_{r=1}^k \langle m, \phi_r\rangle$, since products $\prod_r \phi_r$ generate the test functions for weak convergence of the k-marginal to $m^{\otimes k}$. The link to the empirical measure is the product $\prod_{r=1}^k \langle \mu^N, \phi_r\rangle$, which expands into an average over *all* ordered k-tuples of indices,

$$\prod_{r=1}^k \langle \mu^N, \phi_r\rangle = \frac{1}{N^k} \sum_{i_1,\ldots,i_k=1}^N \prod_{r=1}^k \phi_r(X^{i_r,N}),$$

the tagged-particle expectation $\mathbb{E}\Big[\prod_r \phi_r(X^{r,N})\Big]$ being the contribution of one *distinct*, increasingly ordered tuple. Now count: of the N^k ordered k-tuples $(i_1, \ldots, i_k)$, the number with all entries *distinct* is the falling factorial $N(N-1)\cdots(N-k+1) = N^k(1 - O(1/N))$, so for fixed k the distinct tuples make up a fraction $1 - O(1/N)$ of all tuples, and the tuples with at least one repeated index are only an $O(1/N)$ fraction. By exchangeability every distinct tuple contributes the *same* value $\mathbb{E}\Big[\prod_r \phi_r(X^{r,N})\Big]$, while every tuple—distinct or not—is bounded in absolute value by $\prod_r \|\phi_r\|_\infty$. Splitting the average over tuples into the distinct ones and the rest therefore gives

$$\mathbb{E}\left[\prod_{r=1}^k \langle \mu^N, \phi_r\rangle\right] = \frac{N(N-1)\cdots(N-k+1)}{N^k}\, \mathbb{E}\left[\prod_{r=1}^k \phi_r(X^{r,N})\right] \; + \; O(1/N),$$

where the $O(1/N)$ collects the $O(N^{k-1})$ repeated-index tuples, each $O(1/N^k) \cdot \prod_r \|\phi_r\|_\infty$ after normalization. Since the prefactor is $1 - O(1/N)$ and the tagged expectation is bounded, we may absorb its deviation from 1 into the error and solve for the quantity we want:

$$\mathbb{E}\left[\prod_{r=1}^k \phi_r(X^{r,N})\right] = \mathbb{E}\left[\prod_{r=1}^k \langle \mu^N, \phi_r\rangle\right] + O(1/N).$$

It remains to pass to the limit on the right. By (ii) each scalar $\langle \mu^N, \phi_r\rangle \to \langle m, \phi_r\rangle$ in probability, and a finite product of random variables converging in probability converges in probability to the product of the limits; hence $\prod_{r=1}^k \langle \mu^N, \phi_r\rangle \to \prod_{r=1}^k \langle m, \phi_r\rangle$ in probability. This product is, moreover, uniformly bounded by $\prod_r \|\phi_r\|_\infty$, a single deterministic constant, so the *bounded* convergence theorem upgrades convergence in probability of the integrand to convergence of its expectation: $\mathbb{E}\Big[\prod_r \langle \mu^N, \phi_r\rangle\Big] \to \prod_r \langle m, \phi_r\rangle$. Combined with the $O(1/N)$ display, $\mathbb{E}\Big[\prod_r \phi_r(X^{r,N})\Big] \to \prod_r \langle m, \phi_r\rangle$. As k and the ϕ_r were arbitrary, the k-marginal converges to $m^{\otimes k}$, which is (i). □

A remark on the standing of this lemma, so the reader weights it correctly. It is recorded *for orientation*: it certifies that the probabilist's chaoticity and the physicist's law-of-large-numbers concentration are interchangeable, and it shows that the entire tower of finite marginals is governed by the single random object μ^N. But the main result of this chapter will *not* pass through it. The theorem we are about to prove establishes something strictly stronger and more quantitative than either (i) or (ii): rather than weak convergence of marginals or convergence in probability of the empirical measure, it controls the $\mathcal{W}_2$ distance between the genuine k-marginal and the product law $m_t^{\otimes k}$, with an explicit rate. Weak convergence and $\mathcal{W}_2$ convergence of marginals will both fall out of that single estimate as immediate corollaries, so we have no need to route the argument through lemma 7.12; the lemma stands as a conceptual map of the terrain, not as a load-bearing beam.

How, then, will the stronger statement be reached? Not by manipulating laws and marginals abstractly, but by a strikingly concrete device. The target we must hit is $\mathcal{L}(X_t^{1,N},\dots,X_t^{k,N}) \to m_t^{\otimes k}$, ideally in $\mathcal{W}_2$ and with a rate. The idea—Sznitman's—is to build, on the *same* probability space as the real particles, an explicit family of N *independent* McKean–Vlasov copies, each an exact draw from the limit law m_t, and to make each copy shadow its real counterpart as closely as possible by feeding them the *same* Brownian motion and the *same* initial datum. If we can then show that the real particle and its idealized twin stay pathwise close—that their mean-square distance is $O(1/N)$—then the k-marginal of the real system is forced to within $O(1/\sqrt{N})$ of the k-marginal of the i.i.d. copies, which is exactly $m_t^{\otimes k}$. Chaoticity, in this approach, is not proved by checking factorization directly; it is read off from a single pathwise coupling estimate, and that estimate carries its own rate for free. Constructing the coupling and running the estimate is the work of the next subsection.

7.3.1 Sznitman's coupling

We left the previous section having reduced everything to one concrete task. The target is $\mathcal{L}(X_t^{1,N},\dots,X_t^{k,N}) \to m_t^{\otimes k}$, and we resolved to reach it not by checking factorization of marginals abstractly but by the device Sznitman invented for exactly this purpose: build, on the very same probability space as the real particles, a companion system of N *independent* McKean–Vlasov processes—one twin shadowing each real particle—and feed each twin the *same* Brownian motion and the *same* initial datum as the particle it shadows. The twins are exact draws from the limit law m_t; if we can show that a real particle and its twin never drift more than $O(N^{-1/2})$ apart in L^2, then the real k-marginal is squeezed to within $O(N^{-1/2})$ of the k-marginal of the twins, which is $m_t^{\otimes k}$ by independence. The whole of chaoticity, with its rate, then falls out of a single pathwise distance estimate. This subsection constructs that companion system, isolates the one variance estimate that makes the bound dimension-free, and runs the comparison to its conclusion.

Why must we use independent McKean–Vlasov copies, and not, say, N copies of the *interacting* system? Because the goal is to approximate the product law $m_t^{\otimes k}$, and the cleanest object whose marginals are literally $m_t^{\otimes k}$ is a family of genuinely independent processes each distributed as m_t. The single move that makes such a family *useful*—rather than a separate population living on its own probability space—is to refuse it fresh randomness: each copy is driven by the Brownian motion and started from the initial datum of the real particle it is paired with. That sharing is what converts a distributional comparison (hard) into a pathwise comparison (a Grönwall estimate), because when two equations are driven by the same noise their Itô integrals are identical and cancel in the difference, leaving an ordinary integral inequality. We met this exact trick already, twice: in the contraction of theorem 7.9, where common noise removed the stochastic integral from the one-step estimate, and it is about to do the same work here.

Definition 7.13 (The coupled limit system)**.** On the same probability space carrying (7.1), define for $i = 1,\dots,N$ the independent copies

$$\mathrm{d}\bar{X}_t^i = b(\bar{X}_t^i,\ m_t)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t^i, \qquad \bar{X}_0^i = X_0^{i,N}, \quad m_t = \mathcal{L}(\bar{X}_t^i), \tag{7.7}$$

where m_t is the (common, deterministic) flow of the McKean–Vlasov solution from theorem 7.9, the *same* Brownian motion W^i drives the ith copy and the ith particle, and the *same* initial datum is shared. By construction the $\bar{X}^i$ are i.i.d. McKean–Vlasov processes, each with marginal law m_t at time t. We write $\bar{\mu}_t^N := \frac{1}{N}\sum_{j=1}^N \delta_{\bar{X}_t^j}$ for their empirical measure.

It is worth pausing on why the $\bar{X}^i$ are *exactly* independent and *exactly* m_t-distributed, since this is what makes them a legitimate yardstick. Look at (7.7): the flow m_t is frozen—it is the deterministic McKean–Vlasov marginal flow, the *same* for every i, and it does *not* depend on the other copies. So each $\bar{X}^i$ solves an *ordinary* (non-McKean) SDE with a given time-dependent coefficient $(s,x) \mapsto b(x, m_s)$, driven only by its own W^i and its own initial datum $X_0^{i,N}$. The N pairs $(X_0^{i,N}, W^i)$ are independent across i (the initial data are i.i.d. and the Brownian motions are independent, by definition 7.2), and each $\bar{X}^i$ is a fixed measurable functional of its own pair alone; independent inputs through the same functional yield independent outputs, so the $\bar{X}^i$ are i.i.d. Moreover each $\bar{X}^i$ is, individually, a strong solution of the McKean–Vlasov SDE (7.4): its driving law m_s coincides with its own marginal $\mathcal{L}(\bar{X}_s^i)$ precisely because m is the self-consistent fixed point of theorem 7.9 and $\bar{X}_0^i \sim m_0$. Hence $\mathcal{L}(\bar{X}_t^i) = m_t$ for every i and t, and the joint law of any k of them is exactly $m_t^{\otimes k}$. The twins are, in short, an honest i.i.d. realization of the limit, embedded in the same world as the interacting particles so that we may subtract one from the other path by path.

The whole argument turns on a single probabilistic estimate, the *fluctuation variance*, which quantifies how far the empirical measure $\bar{\mu}_t^N$ of N i.i.d. samples sits from its mean m_t *when read through the drift*. It is the place where the law of large numbers enters quantitatively, and—this is the point that propagates into every constant below—for pairwise bounded interactions the estimate is *dimension-free*: the bound depends on the sup-norm of the interaction kernel but not at all on the dimension d of the state space. The reason, which the proof will lay bare, is that the drift reads the empirical measure only through a bounded scalar pairing, so the relevant quantity is an average of bounded $\mathbb{R}^d$-valued summands and its variance is governed by the elementary i.i.d. variance bound, with no geometry of $\mathbb{R}^d$ intervening.

Lemma 7.14 (Fluctuation variance)**.** *Let $b_0 : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$ be bounded measurable with $\|b_0\|_\infty := \operatorname{ess\,sup}|b_0| < \infty$, let $Z_1, \dots, Z_N$ be i.i.d. with law $\rho \in \mathcal{P}(\mathbb{R}^d)$, and set $\varphi(x) := \int_{\mathbb{R}^d} b_0(x,z)\,\rho(\mathrm{d}z)$. Then*

$$\mathbb{E}\Big[\Big|\tfrac{1}{N}\textstyle\sum_{j=1}^N b_0(Z_i, Z_j) - \varphi(Z_i)\Big|^2\Big] \;\le\; \frac{4\,\|b_0\|_\infty^2}{N} \qquad \textit{for each fixed } i.$$

Before the proof, read the statement physically. The quantity $\frac{1}{N}\sum_j b_0(Z_i, Z_j)$ is the empirical average force felt by sample Z_i from the whole population of i.i.d. samples, while $\varphi(Z_i) = \int b_0(Z_i, z)\,\rho(\mathrm{d}z)$ is the *mean-field* force it would feel against the true law ρ. The lemma says the gap between the two—the fluctuation of the realized field about its ideal—has mean square at most $4\|b_0\|_\infty^2/N$. This is a law of large numbers with an explicit $O(1/N)$ variance and a clean constant, and it is the engine of the rate: the $N^{-1/2}$ that appears in the final chaos bound is the square root of this N^{-1}.

Proof. Fix i. The one subtlety is that the index $j = i$ appears in the sum, so the summand $b_0(Z_i, Z_i)$ is *not* independent of the others given Z_i; we therefore peel it off and treat the genuinely off-diagonal part separately. Split

$$\tfrac{1}{N}\sum_{j=1}^{N} b_0(Z_i,Z_j)-\varphi(Z_i)=\underbrace{\tfrac{1}{N}\sum_{j\neq i}\left[b_0(Z_i,Z_j)-\varphi(Z_i)\right]}_{=:S}+\underbrace{\tfrac{1}{N}\left[b_0(Z_i,Z_i)-\varphi(Z_i)\right]}_{=:D}.$$

We now compute the conditional second moment given Z_i, because conditioning on Z_i is exactly what turns the off-diagonal summands into centered, independent contributions. Conditionally on Z_i, each Z_j with $j\neq i$ is still distributed as ρ and is independent of Z_i, so $\mathbb{E}[b_0(Z_i,Z_j)\mid Z_i]=\int b_0(Z_i,z)\,\rho(\mathrm{d}z)=\varphi(Z_i)$. Thus each summand $b_0(Z_i,Z_j)-\varphi(Z_i)$ of S has conditional mean zero—it is *centered* given Z_i—and the summands are conditionally i.i.d. across $j\neq i$. Variances of sums of conditionally independent centered terms add, so

$$\mathbb{E}\Big[|S|^2\mid Z_i\Big]=\frac{1}{N^2}\sum_{j\neq i}\mathbb{E}\Big[|b_0(Z_i,Z_j)-\varphi(Z_i)|^2\mid Z_i\Big]\le\frac{N-1}{N^2}\,\|b_0\|_\infty^2,$$

using for the bound that a conditional variance never exceeds the conditional second moment, $\mathbb{E}\Big[|b_0(Z_i,Z_j)-\varphi(Z_i)|^2\mid Z_i\Big]\le\mathbb{E}\Big[|b_0(Z_i,Z_j)|^2\mid Z_i\Big]\le\|b_0\|_\infty^2$ (centering can only decrease the second moment). The diagonal term D is a function of Z_i alone, hence Z_i-measurable, and $|b_0(Z_i,Z_i)-\varphi(Z_i)|\le 2\,\|b_0\|_\infty$ gives $|D|\le 2\,\|b_0\|_\infty/N$, so $|D|^2\le 4\,\|b_0\|_\infty^2/N^2$.

Here is the step that earns the sharp constant. Given Z_i, the term S is centered (mean zero) while D is deterministic (a constant), so their conditional cross term $2\,\mathbb{E}[\langle S,D\rangle\mid Z_i]=2\,\langle\mathbb{E}[S\mid Z_i]\,,D\rangle=0$ *vanishes*. The conditional second moment of the sum is therefore the bare sum of the two conditional second moments, with no cross contribution:

$$\mathbb{E}\Big[|S+D|^2\mid Z_i\Big]=\mathbb{E}\Big[|S|^2\mid Z_i\Big]+|D|^2\le\frac{N-1}{N^2}\,\|b_0\|_\infty^2+\frac{4}{N^2}\,\|b_0\|_\infty^2=\frac{N+3}{N^2}\,\|b_0\|_\infty^2\le\frac{4\,\|b_0\|_\infty^2}{N}\quad(N\ge 1),$$

the last inequality because $N+3\le 4N$ for $N\ge 1$. Taking expectations over Z_i removes the conditioning and gives the claim. The parenthetical contrast is worth keeping in view: had we bounded crudely by $|S+D|^2\le 2\,|S|^2+2\,|D|^2$, the doubling would have produced a constant near $2+8/N$ rather than 4; it is precisely the *exact* cancellation of the conditional cross term—legitimate because conditioning on Z_i makes S centered and D frozen—that lets us add the two pieces without loss and land on the clean 4. □

The variance scales as $1/N$ *independently of the dimension* d: this is the rigorous version of the central-limit statement "fluctuations of an average of N independent contributions are $O(N^{-1/2})$." Notice exactly where d failed to appear: the only bound used on b_0 was its scalar sup-norm $\|b_0\|_\infty$, and the only probabilistic fact used was that variances of independent centered terms add. Neither sees the dimension. This is the structural reason the coupling delivers a dimension-free rate, and it is the first of the two facts the next subsection will scrutinize: it holds because the drift reads the population through a *bounded pairwise* kernel, and we will see there that relaxing boundedness—passing to the full Wasserstein geometry of the empirical measure—reintroduces a dimension-dependent statistical rate. For now the lemma is exactly the quantitative heart of the mean-field limit, and we are ready to spend it.

Theorem 7.15 (Propagation of chaos, Sznitman)**.** *Assume the pairwise form* (7.2) *with b_0 bounded and Lipschitz (so* (A-Lip)*–*(A-Growth) *hold), $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$, and let $X^{i,N}$ and the coupled copies $\bar{X}^i$ be as in* (7.1) *and* (7.7)*. Then there is a constant $C = C(T, L, \|b_0\|_\infty)$, independent of N and of d, such that*

$$\max_{1\le i\le N} \mathbb{E}\left[\sup_{0\le t\le T} |X_t^{i,N} - \bar{X}_t^i|^2 \right] \le \frac{C}{N}. \tag{7.8}$$

Consequently, for each fixed k and $t \in [0,T]$,

$$\mathcal{W}_2\Big(\mathcal{L}(X_t^{1,N}, \dots, X_t^{k,N}),\, m_t^{\otimes k}\Big) \le \sqrt{\tfrac{Ck}{N}},$$

so the particle system is m_t-chaotic, and each single particle satisfies $\mathcal{W}_2(\mathcal{L}(X_t^{i,N}), m_t) \le \sqrt{C/N} = O(N^{-1/2})$.

Proof. Set $\Delta_t^i := X_t^{i,N} - \bar{X}_t^i$, the gap between the ith real particle and its twin. Because the two systems (7.1) and (7.7) are driven by the *same* Brownian motion W^i and started from the *same* datum $X_0^{i,N}$, the diffusion term $\sigma\, \mathrm{d}W_t^i$ and the initial condition are identical and cancel in the difference; only the drift integral survives:

$$\Delta_t^i = \int_0^t \Big[b(X_s^{i,N}, \mu_s^N) - b(\bar{X}_s^i, m_s) \Big]\, \mathrm{d}s.$$

This is the payoff of sharing the noise: the comparison is now a deterministic-looking integral inequality, and the stochastic integral has disappeared from view. The drift difference confronts two changes at once—the state moved from $\bar{X}_s^i$ to $X_s^{i,N}$, *and* the measure moved from m_s to μ_s^N—so we separate them by inserting two intermediate terms, changing one argument at a time, and along the way replacing the true limit measure m_s first by the twins' empirical measure $\bar{\mu}_s^N$ and then by the real empirical measure μ_s^N:

$$\begin{aligned} b(X_s^{i,N}, \mu_s^N) - b(\bar{X}_s^i, m_s) = &\underbrace{b(X_s^{i,N}, \mu_s^N) - b(\bar{X}_s^i, \mu_s^N)}_{(A)} \\ &+ \underbrace{b(\bar{X}_s^i, \mu_s^N) - b(\bar{X}_s^i, \bar{\mu}_s^N)}_{(B)} + \underbrace{b(\bar{X}_s^i, \bar{\mu}_s^N) - b(\bar{X}_s^i, m_s)}_{(C)}. \end{aligned}$$

Each of the three terms isolates one mechanism, and each is controlled by exactly one of the tools assembled earlier.

Term (A) (state mismatch). Here only the first argument changes, the measure being held at μ_s^N, so the state-Lipschitz part of (A-Lip)—equivalently, the Lipschitzness of b_0 in its first argument under the pairwise form (7.2)—bounds it directly by the gap it is meant to measure,

$$|(A)| \le L\, \left| X_s^{i,N} - \bar{X}_s^i \right| = L\, \left| \Delta_s^i \right|.$$

This is the self-coupling term: it feeds the particle's own discrepancy back into the estimate and will be absorbed by Grönwall.

Term (B) (empirical-measure mismatch, relabel-matched). Here the state is frozen at $\bar{X}_s^i$ and only the measure changes, from the real empirical measure μ_s^N to the twins' empirical measure $\bar{\mu}_s^N$. Both are empirical measures of N atoms indexed the same way—the real atoms $X_s^{j,N}$ versus the twin atoms $\bar{X}_s^j$—so the index-matching bound lemma 7.5 applies, $\mathcal{W}_2(\mu_s^N, \bar{\mu}_s^N)^2 \le \frac{1}{N}\sum_j \left|X_s^{j,N} - \bar{X}_s^j\right|^2 = \frac{1}{N}\sum_j \left|\Delta_s^j\right|^2$. Bounding the drift difference through the pairwise structure gives, in its rawest form, the relabel-matched sum

$$|(B)| \le \frac{L}{N}\sum_{j=1}^N \left|X_s^{j,N} - \bar{X}_s^j\right| = \frac{L}{N}\sum_j \left|\Delta_s^j\right|,$$

the inequality being the pairwise structure used directly: $|(B)| = \left|\frac{1}{N}\sum_j [b_0(\bar{X}_s^i, X_s^{j,N}) - b_0(\bar{X}_s^i, \bar{X}_s^j)]\right| \le \frac{L}{N}\sum_j \left|\Delta_s^j\right|$ by Lipschitzness of b_0 in its second argument. The same control follows from the measure-Lipschitz part of (A-Lip) and the index-matching bound *after squaring*, $|(B)|^2 \le L^2\, \mathcal{W}_2(\mu_s^N, \bar{\mu}_s^N)^2 \le \frac{L^2}{N}\sum_j \left|\Delta_s^j\right|^2$, which is the form we will use in a moment. This term couples particle i to *all* the others through the averaged discrepancy; the crucial point, which exchangeability will exploit, is that it is the *same* average for every i.

Term (C) (the fluctuation). Here both the state $\bar{X}_s^i$ and the kernel are evaluated on the *twins*, and the only change is from their empirical measure $\bar{\mu}_s^N$ to its mean m_s. Written out under (7.2),

$$(C) = b(\bar{X}_s^i, \bar{\mu}_s^N) - b(\bar{X}_s^i, m_s) = \frac{1}{N}\sum_{j=1}^N b_0(\bar{X}_s^i, \bar{X}_s^j) - \int b_0(\bar{X}_s^i, z)\, m_s(\mathrm{d}z),$$

which is *exactly* the object of lemma 7.14 with $Z_j = \bar{X}_s^j$ (i.i.d. with law $\rho = m_s$, by definition 7.13), $\varphi = b(\cdot, m_s)$, and the tagged index i. Abbreviating $G_s^i := (C)$, the fluctuation lemma delivers, with no dependence on d,

$$\mathbb{E}\left[\left|G_s^i\right|^2\right] \le \frac{4\,\|b_0\|_\infty^2}{N}.$$

This is the only term that does not vanish as the gaps Δ^j shrink: it is the irreducible statistical noise of sampling m_s by N points, and it is what ultimately sets the $O(1/N)$ floor of the whole estimate.

Closing the loop by Grönwall. Collect the three bounds. From the integral form, for any $r \le t$, the path increment $\Delta_r^i = \int_0^r [(A) + (B) + (C)]\, \mathrm{d}s$ satisfies, by the Cauchy–Schwarz inequality in time $\left(\int_0^r f\, \mathrm{d}s\right)^2 \le r\int_0^r f^2\, \mathrm{d}s \le T\int_0^t f^2\, \mathrm{d}s$ (using $r \le t \le T$ and extending the nonnegative integrand to $[0, t]$) followed by $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ on the integrand, the uniform bound

$$\sup_{r\le t}\left|\Delta_r^i\right|^2 \le 3T\int_0^t \left[\, |(A)|^2 + |(B)|^2 + |(C)|^2 \,\right] \mathrm{d}s \le 3T\int_0^t \left[L^2\left|\Delta_s^i\right|^2 + \tfrac{L^2}{N}\sum_j \left|\Delta_s^j\right|^2 + \left|G_s^i\right|^2\right] \mathrm{d}s.$$

Two of the three substitutions are immediate; the middle one used the Jensen (power-mean) inequality $\left(\frac{1}{N}\sum_j |\Delta_s^j|\right)^2 \le \frac{1}{N}\sum_j |\Delta_s^j|^2$, which is just convexity of $t \mapsto t^2$ applied to the average, converting the squared *average of magnitudes* from (B) into the *average of squared* magnitudes we can track. Now take expectations. Here exchangeability does the symmetrizing: because the system (7.1) and its coupled copies are exchangeable (proposition 7.6), the gaps Δ^j are identically distributed, so $\mathbb{E}\left[|\Delta_s^i|^2\right]$ is the same for every i and equals the average,

$$\mathbb{E}\left[\left|\Delta_s^i\right|^2\right] = \mathbb{E}\left[\tfrac{1}{N}\sum_j |\Delta_s^j|^2\right] =: \varepsilon_s.$$

This is the decisive simplification: the two different-looking terms $|(A)|^2$ and $|(B)|^2$, one carrying the particle's own gap and one the population average, collapse to the *same* scalar ε_s in expectation. Substituting and using $\mathbb{E}\Big[\big|G_s^i\big|^2\Big] \le 4\,\|b_0\|_\infty^2/N$,

$$\mathbb{E}\left[\sup_{r\le t}\left|\Delta_r^i\right|^2\right] \le 3T\int_0^t\Big(\underbrace{L^2\varepsilon_s}_{(A)}+\underbrace{L^2\varepsilon_s}_{(B)}+\underbrace{\tfrac{4\|b_0\|_\infty^2}{N}}_{(C)}\Big)\,\mathrm{d}s = 3T\int_0^t\Big(2L^2\varepsilon_s+\tfrac{4\|b_0\|_\infty^2}{N}\Big)\,\mathrm{d}s,$$

where the (B) contribution is $\mathbb{E}\Big[\frac{L^2}{N}\sum_j|\Delta_s^j|^2\Big] = L^2\mathbb{E}\Big[\frac{1}{N}\sum_j|\Delta_s^j|^2\Big] = L^2\varepsilon_s$, the same value as (A), so the two add to $2L^2\varepsilon_s$. Now invoke the elementary inequality $\varepsilon_t \le \mathbb{E}\Big[\sup_{r\le t}|\Delta_r^i|^2\Big]$ (the time-t value never exceeds the running supremum) to feed the left side back into the right, obtaining a closed Grönwall inequality for $u(t) := \mathbb{E}\Big[\sup_{r\le t}|\Delta_r^i|^2\Big]$:

$$u(t) \le 6TL^2\int_0^t u(s)\,\mathrm{d}s + \frac{12\,T\,\|b_0\|_\infty^2}{N}\,t.$$

Here the forcing term came from $3T\cdot t\cdot 4\,\|b_0\|_\infty^2/N$, and since $t\le T$ it is bounded by the nondecreasing constant $a := 12\,T^2\,\|b_0\|_\infty^2/N$; the linear coefficient is $B := 6TL^2$. The integral form of Grönwall's lemma (lemma C.1) with nondecreasing forcing gives $u(t)\le a\,e^{Bt}$, and at $t=T$ the exponent is $BT = 6TL^2\cdot T = 6T^2L^2$, so

$$\mathbb{E}\left[\sup_{t\le T}\left|\Delta_t^i\right|^2\right] \;\le\; \frac{12\,T^2\,\|b_0\|_\infty^2}{N}\,e^{6T^2L^2} \;=:\; \frac{C}{N},$$

which is (7.8), with the constant $C = 12\,T^2\,\|b_0\|_\infty^2\,e^{6T^2L^2}$ displayed in full. It depends on T, on the Lipschitz constant L, and on the sup-norm of the kernel—but not on N and, by the dimension-freeness of lemma 7.14, not on d.

Marginal consequences. Fix k and t. The paired array $((X_t^{1,N},\dots,X_t^{k,N}),(\bar X_t^1,\dots,\bar X_t^k))$, viewed as a single random element of $(\mathbb{R}^d)^k\times(\mathbb{R}^d)^k = \mathbb{R}^{dk}\times\mathbb{R}^{dk}$, is an *admissible coupling* of $\mathcal{L}(X_t^{1,N},\dots,X_t^{k,N})$ and $\mathcal{L}(\bar X_t^1,\dots,\bar X_t^k)$, since its two marginals are exactly those two laws. By the defining property of $\mathcal{W}_2$ on $\mathbb{R}^{dk}$—any coupling furnishes an upper bound—and the fact that squared Euclidean distance on $\mathbb{R}^{dk}$ is the sum of the per-particle squared distances,

$$\mathcal{W}_2\Big(\mathcal{L}(X_t^{1:k,N}),\,\mathcal{L}(\bar X_t^{1:k})\Big)^2 \le \sum_{i=1}^k\mathbb{E}\Big[\Big|X_t^{i,N}-\bar X_t^i\Big|^2\Big] = \sum_{i=1}^k\mathbb{E}\Big[\Big|\Delta_t^i\Big|^2\Big] \le \frac{Ck}{N},$$

each summand bounded by C/N from (7.8). But the twins are i.i.d. with law m_t, so $\mathcal{L}(\bar X_t^{1:k}) = m_t^{\otimes k}$ exactly, and the displayed inequality is the asserted $\mathcal{W}_2$ bound. Letting $N\to\infty$ with k fixed sends the right side to zero, so the k-marginal converges to $m_t^{\otimes k}$ in $\mathcal{W}_2$, hence weakly; this is precisely m_t-chaoticity (definition 7.11), now with the explicit rate $\sqrt{Ck/N}$. The case $k=1$ is the single-particle statement $\mathcal{W}_2(\mathcal{L}(X_t^{i,N}),m_t)\le\sqrt{C/N} = O(N^{-1/2})$. □

The estimate (7.8) is the precise, quantitative form of "fluctuations vanish in the thermodynamic limit": two pictures of the same particle—one embedded in the genuine N-body system, one evolving in the idealized mean field m_t—stay within L^2-distance $O(N^{-1/2})$ of each other, uniformly on $[0,T]$, and the constant does not deteriorate with the dimension of the state. Two features of what was just proved deserve to be flagged now, because the next subsection will hold the estimate up to honest scrutiny against exactly them. First, (7.8) bounds the distance between each *particle* and its *copy*—a trajectory-versus-twin bound—and not, at least not yet, the distance $\mathcal{W}_2(\mu_t^N, m_t)$ between the empirical measure and the limit; the two are related but not identical, and conflating them would overstate what we have. Second, the clean dimension-free constant $C = 12\,T^2\,\|b_0\|_\infty^2\,e^{6T^2L^2}$ rode entirely on the *bounded pairwise* structure, which is what let lemma 7.14 avoid the geometry of $\mathbb{R}^d$; and that same constant grows—quadratically in T inside the polynomial, exponentially in T^2 inside the exponential—so the bound is a statement about *bounded horizons*, not all time. Both points are taken up squarely in section 7.3.2. Before that, it is worth seeing the same result through the older eyes of kinetic theory, where it answers a question Boltzmann could only assume away.

Heuristic (non-rigorous): Molecular chaos and the unclosed hierarchy

The probabilist's coupling has a kinetic-theory shadow worth seeing, and reading it makes plain what propagation of chaos accomplishes that a physicist had long wanted. Start from the exact microscopic description. The joint density $f_N(t, x_1, \ldots, x_N)$ of the full N-body configuration of (7.1) obeys a *linear* Fokker–Planck equation in Nd dimensions—linear because, viewed as a single diffusion on $\mathbb{R}^{Nd}$ (the stacked process of proposition 7.6), the system is an ordinary Itô SDE whose forward Kolmogorov equation is the linear continuity-plus-diffusion equation of chapter 5. The catch is its dimension: Nd is astronomical, and one wants instead the evolution of the low-order marginals. So integrate out particles $2, \ldots, N$ to obtain an equation for the one-particle marginal $f_N^{(1)}(t, x_1) = \int f_N \,\mathrm{d}x_2 \cdots \mathrm{d}x_N$. Here the interaction strikes back: the drift on particle 1 is $\frac{1}{N}\sum_j b_0(x_1, x_j)$, and integrating its $j \neq 1$ terms against f_N produces a contribution involving the *two*-particle marginal $f_N^{(2)}(t, x_1, x_2)$—the equation for $f_N^{(1)}$ is not closed, because the force on one particle depends on where a second particle is. Seeking to close it by writing the equation for $f_N^{(2)}$ only defers the problem: that equation calls on $f_N^{(3)}$, which calls on $f_N^{(4)}$, and so on up an infinite tower. This is the BBGKY-type hierarchy: each marginal's evolution is enslaved to the next higher one, and no finite truncation is exact.

The mean-field *Stosszahlansatz*—Boltzmann's "molecular chaos" assumption, *Stosszahlansatz* meaning literally "collision-number hypothesis"—cuts the tower by *fiat*: it postulates that, in the limit, distinct particles are statistically independent, so the two-particle marginal factorizes, $f_N^{(2)}(t, x, y) \approx f^{(1)}(t, x)\, f^{(1)}(t, y)$. Substituting this factorization into the first equation of the hierarchy replaces the unknown $f_N^{(2)}$ by a product of the unknown $f^{(1)}$ with itself, and the result is a single *self-contained nonlinear* equation for $f^{(1)}$ alone—precisely the nonlinear Fokker–Planck equation that section 7.4 will write down and study, in which the drift depends on the very density being evolved. Closure is bought at the price of nonlinearity: the price of forgetting the higher marginals is that the first marginal must now feed back on itself.

The status of that factorization is the whole point. Boltzmann *assumed* it—it was a physical hypothesis about decorrelation, plausible but unproven, and the source of a century of debate about its validity and its compatibility with microscopic reversibility. What theorem 7.15 supplies is a *proof* of a quantitative, propagating version of exactly this hypothesis for the weakly interacting diffusions at hand: the k-marginals do factorize in the limit ($\mathcal{L}(X_t^{1:k,N}) \to m_t^{\otimes k}$), the factorization is not merely asymptotic but comes with the explicit rate $\sqrt{Ck/N}$, and—this is what "propagating" means—it holds *uniformly in time* across the whole horizon $[0,T]$, not just at the initial instant where it was posited. Chaos put in at $t=0$ (the i.i.d. initial data $X_0^{i,N} \sim m_0$) is shown to persist for all $t \le T$ despite the interaction's constant tendency to build up correlations. The one honest caveat is the horizon: the constant C grows with T (section 7.3.2), so what is proved is molecular chaos on bounded time intervals, with no claim, from this argument, about $t \to \infty$. This is the sense in which propagation of chaos is the rigorous incarnation of Boltzmann's *Stosszahlansatz*: the closure the physicist assumed to derive a kinetic equation is, here, a theorem with a rate.

7.3.2 Honest accounting of the rate and its dimension dependence

The proof of theorem 7.15 closed on two notes of caution, deliberately left hanging, and an honest theory must now make good on them. The estimate (7.8) is a beautiful thing—an $O(1/\sqrt{N})$ bound with a constant that does not so much as glance at the dimension d—and precisely because it is so clean it invites overstatement. There are exactly two ways a careless reader could claim more than was proved, and this subsection forecloses both. The first concerns *what* is being measured: the theorem bounds the distance between a particle and its mean-field copy, which is *not* the same as the distance between the empirical measure and its limit, and the difference between the two is exactly where the curse of dimension hides. The second concerns *for how long*: the constant C carries a factor $e^{6T^2L^2}$, so the bound is strong on any fixed horizon but degrades catastrophically as $T \to \infty$, and nothing in the argument survives uniformly in time. We take the two in turn, and in doing so we draw the cleanest possible line between what this book proves from the inside and what it merely quotes from the outside for orientation—a line that matters, because the whole point of the self-containedness discipline is that no conclusion here should secretly lean on an unproved external fact.

First caveat: trajectory rate versus empirical-measure rate. Theorem 7.15 controls $\mathbb{E}\Big[\sup_t \Big|X_t^{i,N} - \bar{X}_t^i\Big|^2\Big]$, the mean-square gap between the ith genuine particle and its mean-field twin $\bar{X}^i$. It is tempting to read off from this a rate for the quantity one most often wants in practice—the speed at which the *empirical measure* $\mu_t^N = \frac{1}{N}\sum_j \delta_{X_t^{j,N}}$ of the real system approaches the limit law m_t in $\mathcal{W}_2$—but the two are genuinely different questions, and the theorem settles only the first. To see exactly how they relate, insert the empirical measure $\bar{\mu}_t^N := \frac{1}{N}\sum_j \delta_{\bar{X}_t^j}$ of the *twins* as an intermediary and apply the triangle inequality for $\mathcal{W}_2$:

$$\mathcal{W}_2(\mu_t^N, m_t) \;\le\; \underbrace{\mathcal{W}_2(\mu_t^N, \bar{\mu}_t^N)}_{\text{(coupling)}} + \underbrace{\mathcal{W}_2(\bar{\mu}_t^N, m_t)}_{\text{(statistics)}}. \tag{7.9}$$

The split is the whole story, because the two terms are answered by two utterly different mechanisms. The first compares the empirical measure of the real particles to the empirical measure of their twins—two clouds of N atoms indexed the same way—so the index-matching bound lemma 7.5 applies atom by atom and the coupling controls it directly:

$$\mathbb{E}\Big[\mathcal{W}_2(\mu_t^N, \bar{\mu}_t^N)^2\Big] \;\le\; \mathbb{E}\Big[\tfrac{1}{N}\textstyle\sum_{j=1}^N \Big|X_t^{j,N} - \bar{X}_t^j\Big|^2\Big] = \mathbb{E}\Big[\tfrac{1}{N}\textstyle\sum_j \Big|\Delta_t^j\Big|^2\Big] \;\le\; \dfrac{C}{N},$$

the last step by exchangeability ($\mathbb{E}\Big[\Big|\Delta_t^j\Big|^2\Big]$ is the same for every j and bounded by C/N from (7.8)). This term is therefore $O(1/N)$ in mean square, *dimension-free*, inherited wholesale from the trajectory bound. The second term carries no particle dynamics at all: $\bar{\mu}_t^N$ is the empirical measure of N *i.i.d.* samples from m_t (the twins are i.i.d. with law m_t by construction), so $\mathcal{W}_2(\bar{\mu}_t^N, m_t)$ is precisely the textbook quantity "how fast does the empirical measure of an i.i.d. sample converge to its parent law in Wasserstein distance?" That is a pure question of statistics, with nothing to do with the interaction, the coupling, or the SDE—and it is the term where the dimension reappears with a vengeance.

Remark 7.16 (The statistical rate and the curse of dimension)**.** The contrast packed into the two terms of (7.9) is the central conceptual lesson of this subsection, and it deserves to be stated as such rather than buried: the *coupling* term is dimension-free, while the *statistical* term is dimension-cursed, and understanding why is understanding what the Sznitman bound really bought us. We begin with the external fact that quantifies the curse. We quote it purely for orientation; *no result of this chapter depends on it*, a point we return to and underline below. For N i.i.d. samples from a measure ρ on $\mathbb{R}^d$ possessing a finite moment of order $q > 4$, the sharp rate of Fournier and Guillin [69] is

$$\mathbb{E}\Big[\mathcal{W}_2(\bar{\mu}^N, \rho)^2\Big] \;\lesssim\; \begin{cases} N^{-1/2}, & d < 4, \\ N^{-1/2}\log N, & d = 4, \\ N^{-2/d}, & d > 4. \end{cases} \tag{7.10}$$

Two features of (7.10) matter for us. The first is the moment caveat, which we state honestly rather than suppress: the displayed clean cases require the parent law to have a finite moment of order $q > 4$; for a thinner tail with only $2 < q \le 4$ finite moments an extra additive term $N^{-(q-2)/q}$, slower than $N^{-1/2}$, appears and degrades the $d < 4$ line. The second is the dimensional phase transition: below $d = 4$ the empirical measure converges at the central-limit rate $N^{-1/2}$, but above it the rate deteriorates to $N^{-2/d}$, which for, say, $d = 10$ is only $N^{-1/5}$—one would need $N \approx 10^{10}$ samples to match what $N \approx 100$ achieves in the plane. This is the celebrated *curse of dimension* for empirical measures: filling out a d-dimensional law to Wasserstein accuracy ε needs on the order of ε^{-d} samples, because $\mathcal{W}_2$ asks the samples to reproduce the parent's *geometry*, and covering a d-dimensional shape with little balls is exponentially expensive in d.

Here is the resolution of the apparent paradox—how can (7.8) be dimension-free when (7.10) is not? The answer is that the two bounds measure the drift's dependence on the population through entirely different lenses. In the Sznitman proof the population entered the estimate *only* through term (C), the fluctuation, and there the drift read the empirical measure not through its full Wasserstein geometry but through the scalar averages $\frac{1}{N}\sum_j b_0(\bar{X}_s^i, \bar{X}_s^j) = \int b_0(\bar{X}_s^i, z)\,\bar{\mu}_s^N(\mathrm{d}z)$. A scalar average of a *bounded* kernel is a one-dimensional random variable no matter how large d is, and its fluctuation about its mean is $O(N^{-1/2})$ by the elementary variance count of lemma 7.14—a count that, as we stressed there, used only the sup-norm $\|b_0\|_\infty$ and the additivity of variances of independent terms, neither of which can see the dimension. The drift, in other words, never asked the empirical measure to be close to m_s in $\mathcal{W}_2$; it asked only that a handful of bounded scalar moments of $\bar{\mu}_s^N$ be close to those of m_s, and *that* weaker demand is met at the dimension-free central-limit rate. The full Wasserstein distance $\mathcal{W}_2(\bar{\mu}_s^N, m_s)$ is a far more stringent thing to control—it demands the whole geometry, not a few bounded test functions—and it is that stringency, not anything about the particle system, that the Fournier–Guillin rate prices in. The lesson is precise and worth carrying forward: *a dimension-free propagation-of-chaos rate is bought by the bounded, averaged way the drift reads the population, and is lost exactly when one insists on the full empirical-measure geometry.*

This also marks the boundary of the dimension-free regime, which honesty requires us to draw. The escape hatch was the bounded pairwise structure (7.2): it is what let term (C) be a bounded scalar average. For a *general* drift $b(x, \mu)$ that is merely Lipschitz in the measure slot—no pairwise kernel, no boundedness—the intermediate-term decomposition of the Sznitman proof still goes through unchanged up to term (C), but $(C) = b(\bar{X}_s^i, \bar{\mu}_s^N) - b(\bar{X}_s^i, m_s)$ can no longer be routed through lemma 7.14; the only tool left is the measure-Lipschitz bound itself,

$$|(C)| \;\le\; L\,\mathcal{W}_2(\bar{\mu}_s^N, m_s),$$

and the Grönwall step then closes on $\max_i \mathbb{E}\Big[\sup_t |\Delta_t^i|^2\Big] \le C \int_0^T \mathbb{E}\Big[\mathcal{W}_2(\bar{\mu}_s^N, m_s)^2\Big]\,\mathrm{d}s$. At this point the forcing is exactly the i.i.d. empirical-measure error of (7.9), and the Fournier–Guillin rate (7.10) becomes the best available input: the propagation-of-chaos rate then inherits the curse, degrading to $N^{-2/d}$ in high dimension. So the dimension-freeness is a property of bounded pairwise interactions, not of propagation of chaos as such.

We close the remark by foregrounding the honesty point, because it is the reason (7.10) appears here at all. *Not one in-book result rests on the Fournier–Guillin estimate.* It is quoted only to explain what would happen outside the bounded-pairwise setting and to put a name to the curse; the actual rate proved in this chapter, the clean $O(1/\sqrt{N})$ of theorem 7.15, rests entirely on the in-book lemma 7.14, whose proof we gave in full and which needs nothing beyond elementary variance bookkeeping. The self-containedness of the chapter is therefore intact: a reader who struck (7.10) and this whole remark from the text would lose a piece of orientation but not a single theorem. We flag the matter explicitly so that the $O(1/\sqrt{N})$ rate is never silently transported to settings (high dimension, general Lipschitz drift, empirical-measure convergence) where it has not been earned.

Second caveat: the horizon constant and the absence of long-time uniformity. The second place the bound could be overstated is in time. The constant in (7.8) was displayed in full as $C = 12\,T^2\,\|b_0\|_\infty^2\,e^{6T^2L^2}$, and the exponential factor $e^{6T^2L^2}$ is not a blemish to be polished away—it is the honest signature of the Grönwall argument that produced it. Recall where it came from: the self-coupling term (A) fed each particle's own gap back into the estimate, and the relabel-matched term (B) fed in the population-averaged gap, so the closed inequality for $u(t) = \mathbb{E}\Big[\sup_{r\le t} |\Delta_r^i|^2\Big]$ carried a linear coefficient $B = 6TL^2$ and Grönwall paid for it with $e^{BT} = e^{6T^2L^2}$. The bound is thus excellent for any *fixed* horizon—on $[0, T]$ it is genuinely $O(1/\sqrt{N})$, dimension-free, with a constant one can write down—but it says nothing whatsoever uniform in T: as $T \to \infty$ the prefactor explodes, and for N fixed the guarantee is vacuous once T is large enough that $e^{6T^2L^2}/N \gtrsim 1$. Reading the estimate as a statement valid for all time would be precisely the overclaim this subsection exists to prevent.

Whether propagation of chaos in fact holds *uniformly* in time is a real and harder question, and the Grönwall route is structurally incapable of answering it, because a worst-case Lipschitz bound only ever lets correlations grow. Obtaining a horizon-free rate requires a genuinely different input: some mechanism that makes the dynamics *contract* rather than merely fail to expand too fast—a convex or strongly monotone drift, a confining potential, a dissipativity condition—so that the correlations built up by the interaction are continually damped instead of accumulating. That is a separate theory with its own hypotheses, and we do not develop it here; we will meet the relevant structural conditions, and the long-time behaviour they buy, when uniform-in-time estimates become indispensable for the equilibrium analysis of chapter 13. For the descriptive theory of the present chapter, bounded horizons are exactly the right setting, and within them the rate is honest and sharp.

With both caveats squarely on the table, the descriptive analytic story of the limit is now complete and honestly told. We have a unique limiting law—the McKean–Vlasov solution of section 7.2—and we know, at the level of *trajectories*, the precise sense and rate in which the finite-N reality converges to it: each particle tracks its mean-field copy to L^2-accuracy $O(N^{-1/2})$ on any fixed horizon, dimension-free, with everything that claim does and does not assert pinned down. The flow of marginals $t \mapsto m_t$ has thereby earned two independent identities. It is the law of the representative particle, constructed by the fixed point of theorem 7.9; and it is the concentration limit of the fluctuating empirical measure, certified by the propagation-of-chaos estimates just audited. There remains one face of this single limiting object still to draw—its evolution as a measure in its own right—and it is where we turn next: section 7.4 takes the very same deterministic flow m_t and asks for its evolution equation, recovering the nonlinear Fokker–Planck equation as the forward (adjoint) law of the McKean–Vlasov diffusion and, under the nondegeneracy (A-Nondeg) held in reserve until now, as a genuine PDE for a smooth density.

7.4 The nonlinear Fokker–Planck equation

By now the deterministic flow of measures $t \mapsto m_t$ has been met from two directions, and they agree. Section 7.2 produced it as the marginal law $m_t = \mathcal{L}(X_t)$ of the unique McKean–Vlasov solution, a continuous curve in $\big(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2\big)$ obtained as the fixed point of the freeze-solve-read-off map Φ; sections 7.3 to 7.3.2 then certified that this same m_t is the genuine $N \to \infty$ limit of the fluctuating empirical measure μ_t^N, with a quantitative $O(1/\sqrt{N})$ rate. What we have so far *not* done is describe how m_t moves in its own right—as a point in measure space tracing out a curve, obeying its own equation, with no particle and no representative path in sight. This is the third and final face of the single limiting object the chapter promised: a partial differential equation for the marginal flow. Supplying it is the business of this section, and it completes the triangle SDE $\leftrightarrow$ law $\leftrightarrow$ PDE whose three vertices are three descriptions of one thing.

The viewpoint that produces such an equation is the *forward*, or adjoint, one, and it is worth recalling why a diffusion law has two complementary evolution equations rather than one. A test function carried by the process, $\phi(X_t)$, evolves under the generator $\mathcal{A}$; this is the *backward* picture, in which one propagates observables. Dually, the distribution of the process—the density that the observables are integrated against—evolves under the formal adjoint $\mathcal{A}^*$; this is the *forward* picture, in which one propagates the state of the population. We follow throughout the global convention fixed in appendix A: *densities evolve forward*, from initial data m_0 specified at $t = 0$, driven by $\mathcal{A}^*$; value functions (which will arrive only with the game of Part II) evolve backward from terminal data at $t = T$, driven by $\mathcal{A}$. The marginal flow m_t is a density in exactly this sense, so it is an object the forward equation governs. The only feature that distinguishes the present situation from the ordinary Fokker–Planck equation of chapter 5 is the one the whole chapter has been organized around: the coefficients of the generator depend on the very law they transport.

To say this precisely we must first name the generator at a *frozen* measure argument, because that is the only honest linear object available—the full generator's measure-dependence is what makes the equation nonlinear, and one cannot adjoin a nonlinear operator term by term until the measure is held fixed. For a fixed $m \in \mathcal{P}_2(\mathbb{R}^d)$ the McKean–Vlasov diffusion $\mathrm{d}X_t = b(X_t, m)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t$ is an ordinary Itô diffusion with constant volatility, and its generator is the second-order elliptic operator

$$(\mathcal{A}_m\phi)(x) = b(x, m) \cdot \nabla\phi(x) + \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top\, \mathrm{D}^2\, \phi(x)\big),$$

the first term a transport (drift) term, the second a diffusion term with the constant matrix $\sigma\sigma^\top$. Its formal adjoint $\mathcal{A}_m^*$, obtained by integrating $\int(\mathcal{A}_m\phi)\,\rho\ \mathrm{d}x = \int \phi\,(\mathcal{A}_m^*\rho)\ \mathrm{d}x$ by parts twice against a smooth density ρ, is

$$(\mathcal{A}_m^*\rho)(x) = -\,\mathrm{div}\big(b(x, m)\,\rho(x)\big) + \tfrac{1}{2}\sum_{i,j}(\sigma\sigma^\top)_{ij}\,\partial^2_{ij}\rho(x),$$

the transport term acquiring a divergence (the sign flip of one integration by parts) and the diffusion term returning to itself (two integrations by parts restoring the sign). The nonlinear Fokker–Planck equation is what results when one evaluates this frozen adjoint at the very measure being transported, $m = m_t$. We can now state the theorem, which packages three claims: that m_t solves this equation weakly, that it is the *unique* such weak solution, and that under nondegeneracy the solution is in fact a classical, smooth, strictly positive density.

Theorem 7.17 (Nonlinear Fokker–Planck equation)**.** *Under assumption 7.3 and $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$, the marginal flow $m_t = \mathcal{L}(X_t)$ of the unique solution of (7.4) is the unique weak solution in $C([0, T]; \mathcal{P}_2(\mathbb{R}^d))$ of the* nonlinear Fokker–Planck *(McKean–Vlasov) equation*

$$\partial_t m_t = \mathcal{A}_{m_t}^* m_t = -\,\mathrm{div}\big(b(\cdot, m_t)\, m_t\big) + \tfrac{1}{2}\sum_{i,j}(\sigma\sigma^\top)_{ij}\,\partial^2_{ij} m_t, \qquad m_t\big|_{t=0} = m_0, \tag{7.11}$$

in the sense that for every $\phi \in C_c^\infty(\mathbb{R}^d)$ and $t \in [0, T]$,

$$\int_{\mathbb{R}^d}\phi\ \mathrm{d}m_t = \int_{\mathbb{R}^d}\phi\ \mathrm{d}m_0 + \int_0^t\!\int_{\mathbb{R}^d}\Big(b(x, m_s)\cdot\nabla\phi(x) + \tfrac{1}{2}\,\mathrm{tr}\left(\sigma\sigma^\top\,\mathrm{D}^2\,\phi(x)\right)\Big)\, m_s(\mathrm{d}x)\ \mathrm{d}s. \tag{7.12}$$

Under (A-Nondeg), for every $t > 0$ the measure m_t has a strictly positive density in $C^\infty(\mathbb{R}^d)$ that solves (7.11) classically.

Proof. From SDE to PDE. The existence half is a direct application of Itô's formula and costs almost nothing, which is the whole point of the forward viewpoint: the SDE already *is* the law's evolution, one need only read it through a test function. Take any $\phi \in C_c^\infty(\mathbb{R}^d)$ and apply Itô's formula (theorem 5.9) to the process $\phi(X_t)$ along the McKean–Vlasov solution. Because the diffusion has the generator $\mathcal{A}_{m_s}$ at time s (its measure slot being the genuine marginal $m_s = \mathcal{L}(X_s)$), this gives

$$\phi(X_t) = \phi(X_0) + \int_0^t \mathcal{A}_{m_s}\phi(X_s)\, \mathrm{d}s + \int_0^t \nabla\phi(X_s)^\top \sigma\, \mathrm{d}W_s,$$

the Itô correction $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2\phi)$ already absorbed into $\mathcal{A}_{m_s}\phi$. The last term is a stochastic integral, and the crux is that it is a *true* martingale, not merely a local one—for only then does its expectation vanish. This is where the two standing controls earn their keep: its quadratic variation is $\int_0^t \left|\sigma^\top\nabla\phi(X_s)\right|^2 \mathrm{d}s$, whose expectation is bounded by $\|\sigma\|^2\, \|\nabla\phi\|_\infty^2\, t < \infty$ because $\nabla\phi$ is bounded (here $\phi \in C_c^\infty$ suffices) and, more generally, finite second moments $\sup_{t\le T}\mathbb{E}\Big[|X_t|^2\Big] < \infty$ (theorem 7.9) keep every such integrand integrable. A stochastic integral with square-integrable integrand is a genuine martingale of mean zero, so taking expectations annihilates the last term. Using $\mathbb{E}[\phi(X_s)] = \int \phi\, \mathrm{d}m_s$ on the remaining terms and $m_s = \mathcal{L}(X_s)$ inside the generator yields exactly the weak identity (7.12). The strong form (7.11) is nothing more than its distributional restatement: (7.12) says $\frac{\mathrm{d}}{\mathrm{d}t}\int \phi\, \mathrm{d}m_t = \int \mathcal{A}_{m_t}\phi\, \mathrm{d}m_t = \int \phi\,(\mathcal{A}_{m_t}^* m_t)$ for all test ϕ, which is the definition of $\partial_t m = \mathcal{A}_m^* m$ in the sense of distributions. Existence of a weak solution is thereby settled, and with no use whatsoever of (A-Nondeg): the forward equation holds for the law of *any* Itô diffusion, degenerate or not.

Uniqueness, by duality against the backward equation. Uniqueness is the substantial half, and it is where the chapter's most technical sketch lives. The danger is real: a nonlinear forward PDE can in principle have several weak solutions sharing the same datum, and nothing in the existence argument rules that out. Our strategy converts the question back into one we have already answered. Let $\tilde{m} \in C([0,T];\mathcal{P}_2(\mathbb{R}^d))$ be *any* weak solution of (7.12) with datum m_0. We will show that $\tilde{m}$ is forced to be a fixed point of the very map Φ of theorem 7.9—the freeze-solve-read-off map whose *unique* fixed point is the marginal flow m. Since Φ has exactly one fixed point, $\tilde{m} = m$ follows, and uniqueness for the PDE is inherited from the contraction already proved for the SDE. The device that achieves this is a self-contained *Holmgren duality*: against the forward equation we test not a fixed function but a cleverly chosen *time-dependent* one—the solution of the corresponding backward equation—engineered so that the two evolutions annihilate each other. No external "superposition principle" (the abstract theorem that every weak solution of a Fokker–Planck equation is the law of some diffusion) is invoked; we build the bridge by hand, and (A-Nondeg) is precisely what makes the bridge load-bearing, by guaranteeing the backward solution is smooth enough to test against.

Begin by *freezing the coefficient along the candidate.* Set $\beta_s(x) := b(x,\tilde{m}_s)$; this turns the nonlinear drift into a prescribed, honestly (s,x)-dependent field, because the measure slot has been filled in advance by the *given* curve $\tilde{m}$ we are testing. The freezing inherits good properties from the standing hypotheses: since $s \mapsto \tilde{m}_s$ is $\mathcal{W}_2$-continuous (it lies in $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$) and b is jointly Lipschitz (A-Lip), the field β is continuous in (s,x), globally L-Lipschitz in x uniformly in s, and of linear growth uniformly in s (the growth bound (A-Growth) contributes the moment $\left(\int |y|^2\, \tilde{m}_s(\mathrm{d}y)\right)^{1/2}$, finite and bounded because $\sup_s \int |y|^2\, \tilde{m}_s(\mathrm{d}y) < \infty$ on the compact interval).

These are exactly the regularity properties a linear SDE needs to be well-posed and a linear parabolic equation needs to be solvable.

Now build the test object. Fix a terminal time $\tau \in (0, T]$ and a target observable $\psi \in C_c^\infty(\mathbb{R}^d)$, and let $X^{s,x}$ denote the solution of the *frozen linear* SDE $\mathrm{d}X_r = \beta_r(X_r)\,\mathrm{d}r + \sigma\,\mathrm{d}W_r$ on $[s, \tau]$ started from $X_s = x$. This is an ordinary SDE—no law-dependence remains—so the theory of chapter 5 applies directly. Define the candidate test function as the expected terminal payoff,

$$v(s,x) := \mathbb{E}\big[\psi(X_\tau^{s,x})\big]\,,$$

the value, started at (s, x), of observing ψ at the fixed terminal time τ. This is the Feynman–Kac functional of the frozen problem, and the reason for defining it this way is that it automatically solves the backward equation, which is the equation we need to cancel against the forward one. Indeed, by the backward Kolmogorov / Feynman–Kac representation (theorems 5.15 and 5.16), v solves the *backward* Kolmogorov equation of the frozen diffusion,

$$\partial_s v + \mathcal{A}_{\tilde{m}_s} v = \partial_s v + \beta_s \cdot \nabla v + \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 v\big) = 0 \quad \text{on } [0,\tau), \qquad v(\tau,\cdot) = \psi, \tag{7.13}$$

with the generator at time s being precisely the frozen generator $\mathcal{A}_{\tilde{m}_s}$ that appears in the weak form (7.12). This matching of operators—the backward equation for v carries the *same* $\mathcal{A}_{\tilde{m}_s}$ that the forward weak form pairs against $\tilde{m}_s$—is the entire mechanism of the duality, and it is why v was built from this particular frozen drift and no other.

For the cancellation to be legitimate, v must be smooth enough to serve as a test function in (7.12), and this is the single place where nondegeneracy is indispensable. Under (A-Nondeg) the operator $\mathcal{A}_{\tilde{m}_s}$ is *uniformly parabolic*—its principal part $\frac{1}{2}\sigma\sigma^\top \succeq \frac{\lambda}{2} I$ is strictly positive definite—so the backward equation (7.13) regularizes its terminal datum. Concretely, the interior parabolic regularity theorem theorem C.12(2), applied locally after truncating β outside a large ball (a truncation that changes the equation only far from where we look and so does not affect interior smoothness), upgrades the merely bounded datum ψ to a classical solution $v \in C^{1,2}([0,\tau] \times \mathbb{R}^d)$. Three quantitative bounds come with it, and we will use all three. First, $|v| \le \|\psi\|_\infty$, since v is an average of values of ψ. Second, ∇v and $\mathrm{D}^2 v$ are bounded and continuous, which one can see probabilistically: $v(s,x) = \mathbb{E}[\psi(X_\tau^{s,x})]$ and the flow $x \mapsto X_\tau^{s,x}$ has an L^2-bounded derivative (the Lipschitz dependence of an SDE solution on its initial datum, chapter 5), so differentiating under the expectation against $\psi \in C_c^\infty$ and $\mathrm{D}^2\psi \in C_c^\infty$ produces bounded derivatives of v. Had the diffusion been allowed to degenerate, the regularization would fail, v might be only Lipschitz, $\mathrm{D}^2 v$ would not exist, and there would be no C^2 object to test against the weak form—this is the precise sense in which (A-Nondeg) "makes the duality work." In sum, for each s the function $v(s,\cdot)$ is a bounded C^2 test function with bounded derivatives, exactly the class (7.12) admits once we account for the explicit s-dependence and the non-compact support, which is the next step.

Now test the weak formulation against this time-dependent function and watch the two evolutions cancel. We want to differentiate the scalar map $s \mapsto \int_{\mathbb{R}^d} v(s,\cdot)\,\mathrm{d}\tilde{m}_s$, which couples two motions: the explicit time-dependence of v and the time-dependence of the measure $\tilde{m}_s$. The weak identity (7.12) as stated applies to a *time-independent* test function of *compact support*, whereas $v(s,\cdot)$ has both explicit s-dependence and full (non-compact) support. Two routine approximations remove exactly these two gaps, and because the section's blueprint asks that "routine" be earned, we carry them out.

(i) The compact-support cutoff. Choose cutoffs $\chi_R \in C_c^\infty(\mathbb{R}^d)$ with $\chi_R \equiv 1$ on $\{|x| \le R\}$, $0 \le \chi_R \le 1$, supported in $\{|x| \le 2R\}$, and with $|\nabla \chi_R| \le c/R$ and $\left|\mathrm{D}^2 \chi_R\right| \le c/R^2$ (rescaled copies of one fixed bump). For each fixed s the product $v(s,\cdot)\chi_R \in C_c^\infty(\mathbb{R}^d)$ is an admissible test function, so (7.12) applies to it verbatim. We must check that letting $R \to \infty$ recovers the same identity with $v(s,\cdot)$ itself, i.e. that the cutoff error vanishes. The error is concentrated on the annulus $\{R \le |x| \le 2R\}$, where χ_R is not identically 1, and it enters only through the generator term: writing out $\mathcal{A}_{\tilde m_s}(v\chi_R) - \chi_R \mathcal{A}_{\tilde m_s} v$, the extra contributions are the terms in which at least one derivative falls on χ_R, namely

$$b(x, \tilde m_s) \cdot v\, \nabla \chi_R + \mathrm{tr}(\sigma\sigma^\top \nabla v \otimes \nabla \chi_R) + \tfrac{1}{2}\, v\, \mathrm{tr}(\sigma\sigma^\top \mathrm{D}^2 \chi_R),$$

all supported on $\{R \le |x| \le 2R\}$. On that annulus $|\nabla \chi_R| \le c/R$ and $\left|\mathrm{D}^2 \chi_R\right| \le c/R^2$, while $|v| \le \|\psi\|_\infty$ and $|\nabla v|$ are bounded (from the regularity step) and $|b(x, \tilde m_s)| \le L(1 + |x| + M)$ grows at most linearly by (A-Growth), with $M := \sup_s (\int |y|^2\, \tilde m_s)^{1/2}$ finite. Hence the whole extra integrand is bounded pointwise by $C\, \frac{1}{R}(1 + |x|)$ on the annulus (the dominant term being the drift one, of size $\frac{c}{R} \cdot |x|$; the Hessian term is even smaller, $O(1/R^2)$), and integrating against $\tilde m_s$,

$$\left| \int_{\mathbb{R}^d} [\mathcal{A}_{\tilde m_s}(v\chi_R) - \chi_R \mathcal{A}_{\tilde m_s} v]\, \mathrm{d}\tilde m_s \right| \;\le\; \frac{c}{R} \int_{\{|x| \ge R\}} (1 + |x|)\, \tilde m_s(\mathrm{d}x) \xrightarrow[R\to\infty]{} 0,$$

the limit being zero because $\sup_s \int (1 + |x|^2)\, \tilde m_s(\mathrm{d}x) < \infty$ forces the tail integral $\int_{|x| \ge R} (1 + |x|)\, \mathrm{d}\tilde m_s$ to vanish as $R \to \infty$ (uniformly in s, by dominated convergence against the finite uniform second-moment bound). This is exactly the point at which (A-Growth) pays for itself: it both bounds the drift on the annulus and, through the finite second moments it guarantees, kills the tail. Together with the elementary facts $\int v\chi_R\, \mathrm{d}\tilde m_s \to \int v\, \mathrm{d}\tilde m_s$ and $\int \chi_R \mathcal{A}_{\tilde m_s} v\, \mathrm{d}\tilde m_s \to \int \mathcal{A}_{\tilde m_s} v\, \mathrm{d}\tilde m_s$ (dominated convergence, since the integrands are bounded), the cutoff limit shows that (7.12) holds with the full, non-compactly-supported $v(s,\cdot)$ in place of a compactly supported ϕ.

(ii) The explicit time-dependence. It remains to differentiate $F(s) := \int_{\mathbb{R}^d} v(s,\cdot)\, \mathrm{d}\tilde m_s$ in s. Two effects contribute, and the product rule separates them: the change of v at frozen measure, $\partial_s v$, and the change of the measure at frozen v, governed by the weak form. Rigorously, fix s and write the increment $F(s+h) - F(s)$ as

$$\underbrace{\int [v(s+h,\cdot) - v(s,\cdot)]\, \mathrm{d}\tilde m_{s+h}}_{\to \int \partial_s v\, \mathrm{d}\tilde m_s} + \underbrace{\left(\int v(s,\cdot)\, \mathrm{d}\tilde m_{s+h} - \int v(s,\cdot)\, \mathrm{d}\tilde m_s \right)}_{\to \int \mathcal{A}_{\tilde m_s} v\, \mathrm{d}\tilde m_s \text{ by (7.12)}}.$$

The first bracket, divided by h, converges to $\int \partial_s v\, \mathrm{d}\tilde m_s$ because $\partial_s v$ is continuous and bounded on the relevant compacts (it equals $-\mathcal{A}_{\tilde m_s} v$, bounded by the regularity step) and $\tilde m_{s+h} \to \tilde m_s$ weakly; the second bracket, divided by h, converges to $\int \mathcal{A}_{\tilde m_s} v\, \mathrm{d}\tilde m_s$ by the extended weak identity just established, applied to the fixed test function $v(s,\cdot)$. Both limits are legitimate precisely because v and its first two spatial derivatives, and $\partial_s v$, are bounded and continuous—the dividend of nondegeneracy. Thus $s \mapsto F(s)$ is absolutely continuous with

$$\frac{\mathrm{d}}{\mathrm{d}s} \int_{\mathbb{R}^d} v(s,\cdot)\, \mathrm{d}\tilde m_s = \int_{\mathbb{R}^d} (\partial_s v + \mathcal{A}_{\tilde m_s} v)\, \mathrm{d}\tilde m_s = 0,$$

the integrand vanishing *identically* by the backward equation (7.13). This is the cancellation the construction was designed to produce: v was chosen so that its own backward evolution exactly annuls the forward evolution of $\tilde{m}$.

A constant function—that is what F now is—may be evaluated at the two endpoints, and the two endpoints are exactly the two laws we want to compare. At $s = \tau$, the terminal datum $v(\tau, \cdot) = \psi$ gives $F(\tau) = \int_{\mathbb{R}^d} \psi \, \mathrm{d}\tilde{m}_\tau$. At $s = 0$, we read off the duality identity in full:

$$\int_{\mathbb{R}^d} \psi \, \mathrm{d}\tilde{m}_\tau \;=\; F(\tau) = F(0) = \int_{\mathbb{R}^d} v(0, \cdot) \, \mathrm{d}m_0 \;=\; \int_{\mathbb{R}^d} \mathbb{E}\Big[\psi(X_\tau^{0,x})\Big] \, m_0(\mathrm{d}x) \;=\; \mathbb{E}\Big[\psi(\widehat{X}_\tau)\Big] \;=\; \int_{\mathbb{R}^d} \psi \, \mathrm{d}\,\Phi(\tilde{m})_\tau.$$

Each equality earns its place: the first is $F(\tau) = F(0)$ (F constant); the second is the definition of $F(0)$ with datum m_0; the third unfolds $v(0, x) = \mathbb{E}\big[\psi(X_\tau^{0,x})\big]$; the fourth recognizes that averaging the started-at-x frozen process over $x \sim m_0$ is the same as running the single frozen-drift process $\widehat{X}$ from a random start $X_0 \sim m_0$ (the tower property); and the last is the *definition* of Φ—running the frozen SDE with drift $\beta_s = b(\cdot, \tilde{m}_s)$ from m_0 and reading off its time-τ law is exactly $\Phi(\tilde{m})_\tau$. Since $\psi \in C_c^\infty$ was arbitrary, two measures that integrate every such ψ to the same number are equal, so $\tilde{m}_\tau = \Phi(\tilde{m})_\tau$ for every $\tau \in [0, T]$. That is to say, $\tilde{m}$ is a fixed point of Φ. But Φ has the *unique* fixed point m (theorem 7.9), so $\tilde{m} = m$, and weak solutions are unique. (Note where the two hypotheses split the labor: (A-Nondeg) delivered the smooth v that made the test legitimate, while (A-Growth) guaranteed no blow-up and the finite moments that drove the cutoff limit. Remove the former and the duality has nothing to test against; remove the latter and the tail error need not vanish.)

Regularity. The last claim of the theorem is a bonus that costs nothing once uniqueness has pinned m down, and it is the one place where nondegeneracy buys not just a proof technique but an improved *conclusion*. With $m_s = \mathcal{L}(X_s)$ now established as *the* determined flow, the coefficient $b(\cdot, m_s)$ is no longer an unknown feeding back on itself: it is a fixed, known field. So (7.11) read forward is now an honest *linear* Fokker–Planck equation, with coefficient $b(\cdot, m_s)$—Lipschitz in x and continuous (hence locally Hölder) in (s, x) on bounded sets—and uniformly elliptic principal part $\frac{1}{2}\sigma\sigma^\top \succeq \frac{\lambda}{2} I$ supplied once more by (A-Nondeg). The forward parabolic regularity theorem theorem C.12(3), applied locally to accommodate the linear growth of the drift, then yields that for every $t > 0$ the measure m_t possesses a density in $C^\infty(\mathbb{R}^d)$ solving (7.11) in the classical (pointwise) sense, and that this density is *strictly positive* everywhere, by the parabolic strong maximum principle recorded in the same theorem. The strict positivity has a clean physical reading: a nondegenerate diffusion instantaneously spreads mass to every point of space, so no region can stay empty for any positive time—the hallmark of an irreducible, fully smoothing dynamics. Without (A-Nondeg) none of this survives: a degenerate or vanishing diffusion can leave m_t singular (think of a deterministic flow concentrating mass on a lower-dimensional set), and the equation would hold only in the weak, measure-valued sense established above. □

It is worth pausing on what the theorem says structurally, because it isolates the single source of difficulty and points to the worked computation that exploits its collapse. Equation (7.11) is *nonlinear* for one reason and one reason only: the drift $b(\cdot, m_t)$—and, in the general state- and measure-dependent volatility case flagged in remark 7.10, the diffusion as well—depends on the very solution m_t it transports. Strip that dependence away and the equation is the ordinary linear Fokker–Planck equation of chapter 5. The dependence is, in general, on the *entire* measure m_t, which is what makes the equation a genuinely infinite-dimensional object. But in the cases that admit closed solutions the coupling factors through only *finitely many* functionals of m_t—most often a few moments—and then the nonlinearity collapses dramatically: the law-feedback becomes an ordinary feedback through those finitely many scalars, which themselves close a small system of ODEs. The carried linear–quadratic thread is the paradigm. There the drift couples to m_t only through its first moment $\bar{m}_t = \int x\, m_t(\mathrm{d}x)$, the Gaussian form of the law is preserved, and the infinite-dimensional PDE (7.11) collapses onto a two-scalar (mean and variance) flow solved in closed form. We make this completely explicit in the descriptive baseline of worked computation 7.18 in the next section; for now it is enough to register that "finitely many moments" is exactly the structural escape hatch from infinite dimensions.

Physics note: The self-consistent field

Read (7.11) as a physicist reads any density equation: it is a *continuity equation* for probability mass, $\partial_t m_t + \operatorname{div} J_t = 0$, with current $J_t = b(\cdot, m_t)\, m_t - \frac{1}{2}\sigma\sigma^\top \nabla m_t$ assembled from a convective flux $b(\cdot, m_t)\, m_t$ and a diffusive (Fick) flux $-\frac{1}{2}\sigma\sigma^\top \nabla m_t$. What makes it more than a passive transport equation is that the drift generating the convective current is itself *built from the population it transports*. This is exactly the logical shape of a *self-consistent* (mean) field. The sanctioned analogy is the Curie–Weiss ferromagnet: there each spin feels an effective field proportional to the average magnetization of all the spins, while that average magnetization is in turn fixed by the distribution of spins the field induces—a loop closed by the self-consistency equation $m = \tanh(\beta J m)$ that the reader has surely met. Here the same loop appears, with the average magnetization replaced by the population law m_t and the self-consistency equation replaced by the closure $m_t = \mathcal{L}(X_t)$: the field a particle feels is generated by the law of the ensemble, which is generated by the field. The picture is genuinely illuminating, and it is the only mean-field analogy this chapter sanctions. But it carries one sharp limitation that must be stated now, because the entire next section turns on it. In the Curie–Weiss magnet, and here, the self-consistency is an *averaging postulate*: the response law (the spin's alignment tendency; our drift b) is *given*, fixed by the physics, and the fixed point merely makes the felt field consistent with the induced distribution. In a mean field game, the analogous closure is the output of an *optimization*: each agent's "response law" is not given but chosen to be a best reply, and the population law is selected by a Nash equilibrium rather than by an average. The Curie–Weiss picture is therefore exact for the descriptive theory of this chapter and must be *retired the moment strategy enters*—which is precisely the hinge the next section pries on.

With the third face of the limiting object now in hand, the descriptive theory is complete: a single particle's law $\mathcal{L}(X_t)$, a single representative SDE (7.4), and a single nonlinear PDE (7.11) are three exact descriptions of one object, tied together by the self-consistency closure $m_t = \mathcal{L}(X_t)$, and that closure is—as the physicsnote has just stressed—an *averaging postulate*: the drift is prescribed, and the law feeds back into it mechanically, with nothing optimized and nothing chosen. This is the whole of what the descriptive theory asserts, and also the whole of what it does *not*. The next section takes the physicsnote's parting warning as its organizing question: holding this entire self-consistent edifice fixed, what single ingredient must change to turn it into a *game*? Drawing that dichotomy sharply, against the precisely characterized baseline this chapter has built, is the pivot to Part II.

7.5 The pivot: descriptive dynamics versus strategic equilibrium

The physicsnote that closed section 7.4 left us with a warning, and this section is built to act on it. We have, by now, a complete and self-contained theory of a population evolving under a *prescribed* mean-field law of motion: one representative law $\mathcal{L}(X_t)$, one representative SDE (7.4), one nonlinear Fokker–Planck equation (7.11), all three locked together by the closure $m_t = \mathcal{L}(X_t)$ and made quantitative by the Sznitman coupling of section 7.3. This is the descriptive baseline, and it is finished. But the physicsnote insisted that its central closure is an *averaging postulate*—the drift is fixed by the physics and the fixed point merely makes the felt field consistent with the induced law, with nothing optimized and nothing chosen. The whole of Part II is the single modification that retires that postulate, and the purpose of this section is to isolate exactly what the modification is. We hold the entire self-consistent edifice of this chapter fixed and ask precisely one question: what minimal ingredient must change to turn it into a *game*? The answer is a clean dichotomy—a prescribed, mechanically-obeyed drift with a statistical closure, versus an optimized drift with a Nash closure—and drawing it sharply, against the baseline we have so carefully built, is the reason this chapter exists. The reader who finishes this section will meet the mean field game of the next chapters already knowing the one and only thing it adds.

What McKean–Vlasov is. Look again at the McKean–Vlasov equation (7.4) and ask what kind of object the drift $b(x, m)$ is. It is *given*—exogenous data, fixed before the problem begins, exactly as the response law of a Curie–Weiss spin is fixed by the physics. Each particle obeys it the way a planet obeys gravity: mechanically, with no latitude. No particle chooses anything; there is no cost functional, no objective, no notion of one trajectory being "better" than another and therefore preferred. The sole equation that closes the system is the statistical consistency $m_t = \mathcal{L}(X_t)$, and this consistency carries no optimality content whatsoever—it is the bookkeeping identity that the law a particle feels must equal the law the ensemble of such particles produces, nothing more. We will call a closure of this kind *descriptive* (or, borrowing the language of the modeller, *cooperative*): a field determined by an averaging rule, and a fixed point that is a self-consistency condition with no agent behind it weighing alternatives. Said in a phrase one should carry into Part II: McKean–Vlasov dynamics describe how a crowd *that already knows how to move* spreads out; they are silent on *why* it moves that way, because the "why" was supplied as data. It is exactly that silent "why" that a game will fill in.

What a mean field game adds. The modification is a single substitution, and it is best stated as a slogan before it is written as equations: a mean field game replaces the prescribed drift by the *optimal feedback of a control problem solved against the population.* Where McKean–Vlasov handed the drift to the particle, a game makes the particle *derive* its drift by solving an optimization—and then demands that the population this optimized behaviour generates be the very population it was optimized against. To see how the two halves of that sentence become mathematics, fix a flow

$m = (m_t)$ and endow the representative agent not with a law of motion but with a *controlled* law of motion and a cost to minimize,

$$\mathrm{d}X_t = b(X_t, \alpha_t, m_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad J(\alpha; m) = \mathbb{E}\left[\int_0^T L(X_t, \alpha_t, m_t)\,\mathrm{d}t + g(X_T, m_T)\right],$$

where α_t is now the agent's *control*—a quantity it is free to choose at each instant—L is a running cost and g a terminal cost. Two features of this setup are worth dwelling on, because they are exactly the seams along which the new structure will appear. First, the agent treats the population flow m as *frozen*: it optimizes against a fixed environment, as though the rest of the crowd were a backdrop it cannot influence. This is not a sleight of hand but the defining approximation of a mean field model, and it is legitimate precisely because a single agent among a continuum carries zero mass and cannot move the aggregate—the same vanishing-of-one-among-many that drove the $O(1/N)$ scaling at the start of the chapter, now read on the strategic side. Second, with m frozen the agent faces an honest single-agent stochastic control problem, exactly the object built in chapter 6; and that theory solves it *backward* in time. One introduces the value function $u(t,x)$, the minimal cost-to-go from state x at time t, which satisfies a Hamilton–Jacobi–Bellman equation carrying a *terminal* datum $u(T,\cdot) = g$; solving that equation backward produces, under the convexity hypothesis (A-Conv), the optimal feedback

$$\alpha^*(t,x) = -\partial_p H(x, \nabla u(t,x), m_t)$$

in the canonical normalization $b(x,a) = a$ in which the control *is* the drift. (For a control-affine drift, such as the carried LQG thread with gain $b \neq 1$, one uses instead the maximizer a^* of the Hamiltonian, and $-\partial_p H$ is then the optimal *drift* rather than the control itself; see remark 6.13. The distinction is notational, not conceptual, and we keep it visible because the LQG worked computation below turns on a gain $b \neq 1$.) To keep the comparison with chapter 8 clean we have let the population enter both the dynamics and the costs; chapter 8 develops the standard case in which m enters through the costs alone, the measure-dependent drift being a routine extension. Now substitute the optimizer back into the dynamics. The agent's drift becomes

$$b^*(x, m_t) := b(x, \alpha^*(t,x), m_t),$$

and this is once again a McKean–Vlasov drift—a function of state and population law—but of a completely different provenance: it is *endogenous*, the output of an optimization rather than an assumption handed down with the model. Up to here we have only solved one agent's problem against a fixed m. The game proper is the closing of the loop: the demand that the population this optimal feedback actually generates coincide with the population m the agents took as given when they optimized,

$$\underbrace{u \text{ solves backward HJB driven by } m,}_{\text{optimization}} \qquad \underbrace{m_t = \mathcal{L}(X_t^*), \quad \partial_t m_t = \mathcal{A}_m^* m_t \text{ with drift } b^*}_{\text{consistency}}. \tag{7.14}$$

The two structural novelties, isolated. Set (7.14) beside the McKean–Vlasov fixed point of theorem 7.9—the demand that the frozen SDE, fed the law it produces, reproduces that law—and the difference resolves into exactly two structural additions. They are independent of each other, and it is worth naming them separately because the chapters of Part II are organized around them.

The first is a *new unknown carrying a new direction of time.* The value function u has no McKean–Vlasov counterpart at all: the descriptive theory never solves an optimization, so it never produces a cost-to-go. And u does not merely add an unknown; it adds an unknown that runs the *wrong way.* It is pinned down not by an initial condition but by the *terminal* datum $u(T, \cdot) = g$—the cost an agent pays for where it ends up—and is therefore solved *backward*, from $t = T$ toward $t = 0$. Meanwhile the population density m still obeys the *forward* Fokker–Planck equation (7.11), propagated from its initial law m_0 at $t = 0$. The two are coupled—u feeds the optimal drift b^* into the equation for m, and m feeds the environment into the HJB equation for u—so the mean field game is not a forward evolution at all but a genuinely *forward–backward* boundary-value problem, with its data prescribed at *opposite ends* of the interval $[0, T]$ and no initial-value formulation. This is a categorical change from everything in this chapter: McKean–Vlasov is a pure forward problem, well-posed by marching from $t = 0$, whereas a game cannot be marched in either direction alone because neither end carries complete data. This forward–backward coupling is the structural signature of the entire subject (appendix A); it is developed analytically, as the coupled HJB/Fokker–Planck system, in chapter 8, and probabilistically, as a McKean–Vlasov forward–backward SDE, in chapter 9.

The second addition is that the closure is now *equilibrium, not consistency.* In the descriptive theory the fixed point asserts a bookkeeping identity: the law generated equals the law fed in. In (7.14) the fixed point asserts something with moral content—it is a *Nash equilibrium*, characterized by the property that *no agent can lower its own cost by a unilateral deviation.* The two coincide as equations only because the agents happen to be playing best responses; conceptually they could not be further apart. A consistency condition is symmetric and cooperative, an averaging postulate with a single natural solution; an equilibrium condition selects the drift b^* as the *argmin* of a functional, and selection by optimization is exactly the operation that can fail to be single-valued. Hence a mean field game equilibrium can be *non-unique*, can *bifurcate* as parameters cross thresholds—the strategic analogue of symmetry breaking—and its uniqueness, when it holds, is bought by structural *monotonicity* conditions (A-Mon) that simply have no place in the descriptive theory, where the closure was unique for free. Uniqueness, its failure, and the monotonicity that restores it are the business of chapter 11.

And one novelty deferred. There is a third change the reader should be warned about now, even though we cannot address it here, because it is easy to assume this chapter has already done more than it has. Propagation of chaos as we proved it is propagation of chaos for a *fixed* dynamics. The particles in (7.1) follow a feedback that was prescribed in advance, and theorem 7.15 shows that their empirical measure concentrates on the McKean–Vlasov law at rate $O(1/\sqrt{N})$. The corresponding statement in a finite N-player *game* is genuinely harder, and the reason is precisely the forward–backward coupling just isolated: in a game each agent best-responds to the *actual* empirical measure of its $N - 1$ opponents, so the optimal feedback is itself a functional of that fluctuating measure—the control and the measure it reacts to fluctuate together, and the convenient separation of "prescribed drift" from "fluctuating population" that made the Sznitman coupling work is gone. Proving that the N-player Nash equilibria nonetheless converge to the mean field game equilibrium is therefore a strictly deeper theorem, "propagation of chaos for equilibria," and the same analysis delivers the ε-Nash property—the statement that the limiting mean field strategy, played in the finite game, is approximately optimal with an error vanishing as $N \to \infty$, which is what licenses using the tractable limit as a surrogate for the intractable finite game. That is the deepest result of Part II and is proved in chapter 13. The Sznitman coupling of this chapter is its *descriptive prototype*: the right tool, applied in the one setting—fixed feedback—where the population can be frozen first and reasoned about afterward.

Worked computation 7.18 (The descriptive baseline of the LQG thread)**.** The cleanest way to feel the dichotomy just drawn is to watch it operate on a model where every object is computable in closed form. We instantiate the whole discussion on Thread A, the linear–quadratic–Gaussian model of the thread specification, but deliberately in its *pre-strategic* form: we put the feedback in by hand rather than deriving it from an optimization, so that the present computation is exactly the descriptive half of the forward–backward loop, and the next chapters can be read as "adding the optimization that selects the feedback constant we here choose arbitrarily." Holding the strategic half in abeyance is the whole point—it lets us see precisely which features of the LQG mean field game are already present at the descriptive level (the Gaussian flow, the moment closure, the $O(1/N)$ chaos) and which are genuinely added by the game (the time-varying optimal gain, the fixed-point closure of the mean).

The scalar state of the representative agent obeys the linear dynamics of the thread, with parameters a, b, σ, s fixed as in the specification,

$$\mathrm{d}X_t = \big(aX_t + b\,\alpha_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad X_0 \sim m_0,$$

where a is the intrinsic state drift, b the control gain, σ the volatility, and $s \in [0,1]$ the strength with which an agent is drawn toward the population. In place of an optimized control we *prescribe* the mean-reverting linear feedback

$$\alpha_t = -\theta(X_t - s\,\bar{m}_t), \qquad \bar{m}_t := \int x\, m_t(\mathrm{d}x) = \mathbb{E}[X_t]\,,$$

with $\bar{m}_t$ the population mean and $\theta \ge 0$ a chosen gain: the agent is pushed back toward the moving target $s\,\bar{m}_t$ at rate θ, a caricature of "follow the crowd" that is linear in the state and couples to the population through its mean alone. Substituting this feedback into the dynamics and collecting the terms linear in X_t gives the linear McKean–Vlasov SDE

$$\mathrm{d}X_t = \Big[(a - b\theta)X_t + b\theta s\,\bar{m}_t\Big]\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \bar{m}_t = \mathbb{E}[X_t]\,, \tag{7.15}$$

whose drift, $(a-b\theta)x+b\theta s\int y\,m(\mathrm{d}y)$, couples to the law only through its first moment $\int y\,m(\mathrm{d}y)$—the cleanest possible mean-field interaction, the very "coupling through finitely many moments" that section 7.4 flagged as the structural escape hatch from infinite dimensions. (As promised in the notation watch at the head of the chapter, we display this drift explicitly as a formula in x and the mean, never through the macro $b(\cdot,\cdot)$, precisely so that the symbol b denotes only the control gain and cannot be misread as the drift field.)

The law stays Gaussian and the nonlinear Fokker–Planck closes on two scalars. Because the drift is affine in x and the noise additive and Gaussian, one expects a Gaussian law to stay Gaussian; the work is to show that its two parameters close a self-contained system, and to see the mean-field nonlinearity collapse onto a single scalar. Start with the mean. Taking expectations in (7.15), and using that $\mathbb{E}[\sigma\,\mathrm{d}W_t] = 0$ and $\mathbb{E}[X_t] = \bar{m}_t$, the law-feedback term $b\theta s\,\bar{m}_t$ is already deterministic and passes through the expectation unchanged, so

$$\dot{\bar{m}}_t = (a - b\theta)\bar{m}_t + b\theta s\,\bar{m}_t = \big(a - b\theta(1-s)\big)\bar{m}_t, \qquad \Longrightarrow \qquad \bar{m}_t = \bar{m}_0\,e^{(a-b\theta(1-s))t}.$$

This is the entire nonlinearity of the problem, reduced to one linear scalar ODE: the self-consistent mean evolves at the modified rate $a - b\theta(1-s)$, in which the mean reversion $-b\theta$ is partially undone by the factor s measuring how much the target moves with the crowd. Now subtract this deterministic mean to center the process. The fluctuation $Y_t := X_t - \bar{m}_t$ obeys $\mathrm{d}Y_t = \mathrm{d}X_t - \dot{\bar{m}}_t\,\mathrm{d}t = (a - b\theta)Y_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, where one should notice what has happened: the mean-field term $b\theta s\,\bar{m}_t$, being a function of the mean alone, was exactly absorbed by the centering and has *cancelled*. What remains is an ordinary Ornstein–Uhlenbeck equation (chapter 5) with no population coupling at all. Its variance $v_t := \mathrm{Var}(X_t) = \mathbb{E}\big[Y_t^2\big]$ therefore solves the same linear ODE an uncoupled OU process would—by Itô's formula on Y_t^2, $\mathrm{d}(Y_t^2) = \big(2(a-b\theta)Y_t^2 + \sigma^2\big)\,\mathrm{d}t + 2\sigma Y_t\,\mathrm{d}W_t$, and taking expectations kills the martingale term—

$$\dot{v}_t = 2(a - b\theta)v_t + \sigma^2, \qquad v_0 = \mathrm{Var}(X_0).$$

Hence if $m_0 = \mathcal{N}(\bar{m}_0, v_0)$ then $m_t = \mathcal{N}(\bar{m}_t, v_t)$ for every t, with $(\bar{m}_t, v_t)$ given by the two ODEs above, and the infinite-dimensional nonlinear Fokker–Planck equation (7.11) is solved *in closed form* by this two-parameter Gaussian flow. The point to register is the division of labor: the "nonlinearity" is carried entirely by the scalar mean $\bar{m}_t$ through the convective term, while the variance equation never sees the coupling at all. An infinite-dimensional self-consistency has collapsed onto a single self-referential scalar—exactly the analytic transparency the LQG thread exists to provide, and a concrete instance of the moment-closure mechanism described abstractly at the end of section 7.4.

Propagation of chaos, with an exact constant. The descriptive baseline is also where we can watch the chaos estimate run with every constant known. The N-particle version of the model is (7.1) with drift $(a - b\theta)x + b\theta s\,\frac{1}{N}\sum_j X^{j,N}$, a pairwise interaction whose kernel is $(x, y) \mapsto (a - b\theta)x + b\theta s\,y$ in the sense of (7.2) (again written out so the gain b stands alone). This kernel is *linear in y*, hence *unbounded*, so the bounded-interaction hypothesis under which theorem 7.15 was proved literally fails: the linear LQG model sits outside the hypotheses of our general theorem. This is not a defect to apologize for but an instructive case, because the Gaussian structure supplies the moment control the boundedness hypothesis was only ever a proxy for, and it supplies it *exactly*.

Recall the structure of the Sznitman coupling: one runs each particle $X^{i,N}$ against an independent copy $\bar{X}^i$ of the McKean–Vlasov limit driven by the *same* Brownian motion W^i, and the only obstruction to their agreement is a fluctuation term measuring how far the empirical mean of the coupled copies sits from the true limiting mean. Here that fluctuation term is, exactly,

$$G_s^i = b\theta s\Big(\tfrac{1}{N}\textstyle\sum_{j=1}^N \bar{X}_s^j - \bar{m}_s\Big), \qquad \mathbb{E}\Big[\big|G_s^i\big|^2\Big] = (b\theta s)^2\,\tfrac{v_s}{N},$$

and the second identity is where Gaussianity does its work: the $\bar{X}_s^j$ are i.i.d. $\mathcal{N}(\bar{m}_s, v_s)$, so their sample mean $\frac{1}{N}\sum_j \bar{X}_s^j$ has mean $\bar{m}_s$ and variance v_s/N *exactly*—not up to a constant, but on the nose—whence $\mathbb{E}\big[|G_s^i|^2\big] = (b\theta s)^2 v_s/N$. The unbounded kernel costs us nothing because we never needed a uniform bound on it; we needed control of one centered sample-mean fluctuation, and the Gaussian law hands that over precisely.

The linearity buys a second, sharper, dividend: it lets us bypass the general quadratic-in-T Grönwall estimate of theorem 7.15 entirely. Because the noise is additive, it cancels in the difference $\Delta_t^i := X_t^{i,N} - \bar{X}_t^i$ between the true particle and its coupled limit, leaving a *pathwise linear* ODE,

$$\dot{\Delta}_t^i = (a - b\theta)\Delta_t^i + b\theta s\,\tfrac{1}{N}\textstyle\sum_{j=1}^N \Delta_t^j + G_t^i,$$

in which the term $\frac{1}{N}\sum_j \Delta_t^j$ is the mean coupling and G_t^i the now-quantified forcing. A direct one-dimensional Grönwall applied to this linear system—rather than the Lipschitz-and-square-the-Wasserstein machinery used in general—turns the exact forcing bound into the dimension-free rate with a genuine e^{CT} (not e^{CT^2}) time dependence and an explicit prefactor,

$$\max_{1\le i\le N} \mathbb{E}\Big[\sup_{t\le T}\Big|X_t^{i,N} - \bar{X}_t^i\Big|^2\Big] \;\le\; \frac{(b\theta s)^2\big(\sup_{t\le T} v_t\big)\,T}{N}\, e^{C(a,b,\theta)\,T}.$$

This is the carried-thread incarnation of the $O(1/N)$ bound of theorem 7.15, valid here even though the kernel is unbounded, and with the chapter's promised dimension-freeness made fully explicit (the model is scalar, $d = 1$, and no dimensional constant appears). The improvement from e^{CT^2} to e^{CT} in the exponent is genuinely a feature of the linear drift—a constant-coefficient linear ODE has an exponential, not a Gaussian, comparison function—and not of the general coupling bound, whose exponent must carry the $e^{6T^2L^2}$ of theorem 7.15.

Where the game layer enters. It remains to say exactly which line of the computation above was the arbitrary one, so that the reader can see the seam at which Part II cuts. Everything we did took the gain θ—and indeed the entire mean-reverting form of the feedback—as *given*. That is the descriptive postulate, and it is the one thing a game removes. The optimization that *selects* the feedback was, in fact, already carried out for the single agent in chapter 6: minimizing the representative agent's quadratic cost $J(\alpha) = \mathbb{E}\Big[\int_0^T\big(\frac{1}{2}\alpha_t^2 + \frac{q}{2}(X_t - s\bar{m}_t)^2\big)\,\mathrm{d}t + \cdots\Big]$ over a *frozen* mean trajectory $\bar{m}_t$ yields, by the Riccati theory developed there, not a constant-gain law but the affine optimal feedback

$$\alpha^*(t,x) = -b\big(P(t)\,x + r(t)\big),$$

in which $P(t)$ is the curvature of the value function—the solution of the scalar Riccati ODE—and the offset $r(t)$ solves a separate forced linear ODE that carries the coupling to the frozen mean $\bar{m}_t$ (eqs. (6.24), (6.25) and (6.27)). Compare this optimal law with the prescribed one $\alpha = -\theta(x - s\bar{m}_t)$ and the gap is exactly located: the optimal curvature gain $bP(t)$ is *time-varying*, sweeping toward the terminal data as $t \to T$, whereas our θ was a single constant; and the optimal offset $-br(t)$ is determined by its own ODE rather than being frozen at $b\theta s\,\bar{m}_t$. The constant-θ feedback is thus a deliberately simplified stand-in for the genuinely optimal time-varying law, and the two coincide only in the stationary, long-horizon limit where the Riccati curvature settles to its constant value $P(t) \to \bar{P}$ and the time-dependence washes out. Put the two missing halves together and the reading is complete. The backward, optimization half—the value function, the HJB equation, the feedback $-b(P(t)x + r(t))$—was already supplied for a single agent in chapter 6. The present worked computation is the forward, consistency half: with the feedback frozen, it is exactly the forward leg of (7.14), the Fokker–Planck propagation of the law generated by a fixed drift. What the next chapters (chapters 8 and 14) add is the one ingredient neither half contains on its own—the fixed-point closure $\bar{m}_t = \mathbb{E}[X_t^*]$ that demands the mean trajectory the agents optimize against be self-consistent with the one their optimal feedback generates, welding the backward optimization to the forward consistency into the single forward–backward equilibrium this section has been pointing toward all along. With that hand-off the chapter has done its work: it has delivered a complete descriptive theory and, in the same breath, the precisely drawn baseline against which the strategic layer of Part II is defined.

Rigor ledger

Proved.

- Well-posedness of the weakly interacting N-particle system (7.1) and its exchangeability (proposition 7.6).
- Existence and uniqueness of the McKean–Vlasov SDE (7.4) by a Picard contraction in $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$, global in T (theorem 7.9).
- The fluctuation-variance estimate (lemma 7.14) and, from it, propagation of chaos by Sznitman's coupling with the dimension-free rate $\max_i \mathbb{E}\left[\sup_t \left|X_t^{i,N} - \bar{X}_t^i\right|^2\right] \leq C/N$, hence $O(1/\sqrt{N})$ Wasserstein convergence of finite marginals (theorem 7.15).
- The nonlinear Fokker–Planck equation (7.11) as the forward law of the McKean–Vlasov solution, with uniqueness and (under (A-Nondeg)) smoothness (theorem 7.17), and its closed Gaussian solution for the LQG thread (worked computation 7.18).

Assumed.

- The standing hypotheses assumption 7.3: joint Lipschitz (A-Lip), linear growth (A-Growth), and nondegeneracy (A-Nondeg); plus boundedness of b_0 for the clean $O(1/\sqrt{N})$ rate.
- The Itô-SDE well-posedness and Itô formula of chapter 5, the Wasserstein geometry of chapter 4, and the weak-convergence/tightness apparatus of chapter 3.

Deferred.

- The classical linear parabolic regularity under (A-Nondeg)—the smooth, strictly positive transition density and the $C^{1,2}$ solvability of the backward and forward equations—on which both the uniqueness duality and the smoothness in theorem 7.17 rest: stated and proved (parametrix construction) in theorem C.12. Uniqueness itself is now established in-book by Holmgren duality, with *no* appeal to the Ambrosio–Figalli–Trevisan superposition principle (which would, however, give the analogous statement in the degenerate case; see the bibliographic notes).
- The sharp statistical (Fournier–Guillin) rate of empirical-measure convergence in $\mathcal{W}_2$ (remark 7.16) is quoted for orientation only; *no* in-book result depends on it.
- The strategic layer—HJB/optimal feedback (chapter 6), the forward–backward mean field game system (chapter 8, chapter 9), and propagation of chaos for Nash equilibria with the ε-Nash property (chapter 13).

Open.

- Long-time ($T \to \infty$) uniform-in-time propagation of chaos beyond the e^{CT^2} horizon dependence: requires structural (convexity/contractivity) hypotheses and is genuinely delicate; only special cases are known.
- Quantitative propagation of chaos for *singular* or merely measure-Lipschitz interactions at the optimal dimension-free rate, where the bounded-pairwise mechanism of lemma 7.14 is unavailable.

Bibliographic notes

The McKean–Vlasov equation and the nonlinear-diffusion viewpoint originate with [103], who associated a class of nonlinear parabolic equations with interacting diffusions; the kinetic-theory roots (molecular chaos, the BBGKY hierarchy) trace to Boltzmann and were placed on a probabilistic footing by Kac. The definitive modern treatment of propagation of chaos by coupling, including theorem 7.15 in the bounded-Lipschitz pairwise case, is Sznitman's Saint-Flour course [129]; our well-posedness proof (theorem 7.9) and coupling estimate follow that source and its presentation in [40, Vol. I, Ch. 1], which we recommend as the chapter's companion. The sharp dimension-dependent rate for the convergence of empirical measures in Wasserstein distance (remark 7.16) is due to [69], quoted in remark 7.16 for orientation only. Our uniqueness argument for the nonlinear Fokker–Planck equation is the elementary Holmgren duality against the smooth solution of the backward Kolmogorov equation, self-contained once the parabolic regularity of theorem C.12 is in hand; it uses only nondegeneracy (A-Nondeg). For *degenerate* diffusions, where this duality is unavailable, uniqueness instead rests on the Ambrosio–Figalli–Trevisan superposition principle, treated with full proofs in [40, Vol. I, Ch. 5]; see also [131] for the optimal-transport background of chapter 4. The descriptive-versus-strategic pivot of section 7.5, and the closed-loop McKean–Vlasov structure that the optimization induces, are the founding insight of the mean field game literature [93, 82]; the systematic development we point forward to is [36] (PDE) and [40] (probabilistic), with the convergence problem for N-player Nash equilibria—the strategic analogue of this chapter—in [37].

Chapter 8

The Mean Field Game System: Analytic Formulation

This chapter assembles, for the first time in the book, the full strategic mean field game. Up to now the two halves have been built separately. In chapter 6 we solved the *single* optimally controlled diffusion: dynamic programming produced a value function $u(t,x)$, computed backward from a terminal cost, satisfying a Hamilton–Jacobi–Bellman (HJB) equation whose Hamiltonian $H(x,p,m)$ packages the agent's instantaneous optimization. In chapter 5 and chapter 7 we learned how a *population* of diffusions transports its own law: the density $m(t,x)$ runs forward from an initial profile m_0 and obeys a Fokker–Planck (Kolmogorov forward) equation $\partial_t m = \mathcal{A}^* m$, where $\mathcal{A}^*$ is the formal adjoint of the generator $\mathcal{A}$. In chapter 7 these two ideas were glued in a *descriptive* way: in McKean–Vlasov dynamics the particles follow a *prescribed* dependence of their law, and propagation of chaos guarantees that the empirical measure converges to a deterministic flow.

The mean field game adds the one ingredient McKean–Vlasov lacks, and the addition is a genuine conceptual jump rather than a technical extension — it is the hinge on which all of Part II turns. In a McKean–Vlasov system the dependence of each particle on its own law is *prescribed* from outside: the drift is a given function of (X_t, m_t), and propagation of chaos then *computes* the flow m_t for us, as the deterministic forward evolution that the particles passively realize. The law is an output, but an output of a dynamics we have already fixed, and it can be obtained by integrating forward from a single initial slice. In a mean field game the agents do not follow the law, they *optimize against it*: each treats $t \mapsto m_t$ as the description of a crowd it cannot itself move, computes its own best response to it, and the crowd is in turn the aggregate of exactly those best responses. The law is therefore the *fixed point* of the best-response map, not the endpoint of a prescribed flow — and a fixed point cannot be marched forward from an initial slice, because each agent's optimal response at any instant already depends on the entire future of the crowd. The mean field is no longer postulated; it is the output of a Nash fixed point. A physicist will recognize the closure: this is the self-consistent field of mean field statistical mechanics — the field that moves each particle being the field the particles themselves generate — with one decisive substitution, that the relation closing field to configuration is no longer a thermal average but an *optimization*, each agent's best response taking the place of the Boltzmann weight. The analytic expression of this fixed point is a pair of coupled partial differential equations — a backward HJB driven by the density, and a forward Fokker–Planck driven by the feedback extracted from the value — with data specified at *opposite ends* of the time interval. This forward–backward, two-point boundary structure is the structural signature of the subject and has, as appendix A stresses, no initial-value analogue.

The chapter then follows the logic of that fixed point one consequence at a time. section 8.1 earns the system rather than postulating it: it derives the coupled pair as the $N \to \infty$ limit of an honest N-player Nash equilibrium and then tightens the heuristic into a precise candidate. Once we hold a candidate we must say exactly what it is and why it deserves the name equilibrium, so section 8.2 states the coupled system, gives the two readings — the HJB as a reverse-time value diffusion, the Fokker–Planck as conservation of mass — that organize the rest of Part II, and proves the verification statement that makes solving the two PDEs the very same act as playing a best response. That verification exposes the one feature that makes the object genuinely hard, namely that its data sit at opposite ends of time; section 8.3 therefore dwells on the forward–backward boundary structure and explains why no march from a single time slice can reach the solution. The only way past that obstruction is to solve for the whole flow at once as a fixed point, which is precisely why section 8.4 recasts the system as a fixed point of a best-response map S — the formulation that the existence theory of chapter 10 and the uniqueness theory of chapter 11 actually consume. With the abstract object in hand we ask what governs its behavior, and the answer is the coupling: section 8.5 classifies how the population enters — separable versus non-separable Hamiltonians, local versus nonlocal interactions, and congestion — because each class will demand different hypotheses downstream. For the best-behaved of these classes the strategic problem turns out to hide a single scalar objective: section 8.6 exhibits the variational (potential-game) structure, showing that for monotone separable couplings the whole system is the Euler–Lagrange system of one convex functional, the mean field analogue of an effective action and the conceptual bridge to the Ginzburg–Landau picture of Part IV. Finally section 8.7 advances Thread A to its second layer: the linear–quadratic–Gaussian (LQG) MFG, solved analytically through HJB $\to$ Riccati, Fokker–Planck $\to$ Gaussian flow, and a closed forward–backward fixed-point system for the population mean. The probabilistic counterpart of everything here — the same equilibrium written as a McKean–Vlasov forward–backward stochastic differential equation — is the subject of chapter 9.

Notation watch

Throughout, $u = u(t, x)$ is the value function: it carries *terminal* data $u(T, \cdot) = g$ and is computed *backward* in time. The density $m = m(t, x)$ carries *initial* data $m(0, \cdot) = m_0$ and is transported *forward*. The generator $\mathcal{A}$ acts on test functions (it governs u); its adjoint $\mathcal{A}^*$ acts on densities (it governs m). We fix once the sign of the Hamiltonian: H is a *supremum* (a Legendre transform), so that the *optimal drift* of the controlled diffusion is $b(x, \alpha^*) = -\partial_p H$ (the envelope identity of remark 6.13). In the canonical normalization $b(x, \alpha) = \alpha$, $A = \mathbb{R}^d$ — in which the control *is* the drift — this reads simply $\alpha^* = -\partial_p H$, the form recorded in appendix A; when a control gain is present (as in the LQG thread, whose drift is $ax + b\alpha$) the control is recovered from the maximizer and $-\partial_p H$ remains the optimal *drift* transported by Fokker–Planck. With these conventions the diffusion enters the HJB with a $+\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u)$ that has been moved to the left-hand side as $-\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u)$; the same operator enters

Fokker–Planck unchanged because $\mathcal{A}^*$ has the *same* second-order part as $\mathcal{A}$. Verify every sign below against conservation of mass $\frac{\mathrm{d}}{\mathrm{d}t}\int m \,\mathrm{d}x = 0$, never against a remembered Hamiltonian sign.

8.1 From N-player Nash to the mean field game

The preamble made a promise: the mean field is not a datum we hand the agents but an object they collectively produce, the law of the population being the fixed point of everyone's best response to it. That promise is at bottom a claim about a limit, and the honest way to earn it is to begin where the game is wholly unambiguous — a finite roster of players, each solving a genuine optimization against the others — and watch what survives as the roster grows. This section carries out that program in two deliberate steps. First we write the finite strategic game and its equilibrium conditions down exactly, with no approximation: N players, each minimizing its own cost, coupled only through the crowd they jointly form, with a Nash equilibrium characterized by a coupled system of N Hamilton–Jacobi–Bellman equations. Then we send $N \to \infty$ heuristically and read off the limiting object, postponing to chapter 13 the proof that the limit really is what we read off. The payoff of the second step is the *form* of the equations that section 8.2 will crystallize as definition 8.1; the discipline of the first is what guarantees that this form is *forced* on us rather than guessed.

What lets the limit close at all is a single structural feature of the finite game, and it is worth naming in advance because both subsections will lean on it: *anonymity*. Each player cares about the others only through their empirical distribution, never through their individual identities, and the players are statistically interchangeable — *exchangeable*. Because the coupling is anonymous and the players exchangeable, the influence of any one rival on any other is only $O(1/N)$, so no single opponent matters in the limit and the law of large numbers can convert the high-dimensional random environment a player faces into a single *deterministic* flow of measures. Anonymity is thus not a modelling convenience but the precise hypothesis under which the N-body coupling decouples: the first subsection uses it to justify writing each cost through the empirical measure, and the second uses it to replace that fluctuating empirical measure by its deterministic mean field limit. We turn first to the finite game, assembling it as an N-fold copy of the single controlled diffusion of chapter 6 coupled through the empirical measure of definition 7.1.

8.1.1 The finite game and its Nash system

We begin, as the section promised, by assembling the finite game out of pieces the reader already owns, adding nothing new except the one channel through which the pieces are allowed to touch. The atom of the construction is the single optimally controlled diffusion of chapter 6: one agent, a state in $\mathbb{R}^d$, a drift it steers through its control, an independent Brownian forcing, and a cost it minimizes. The finite game is nothing more than N synchronous copies of that atom, run on the same clock $[0, T]$, made to interact through the cost in the one way anonymity permits. Building it this way — as an N-fold assembly of an object we have already solved, rather than as a fresh high-dimensional problem — is what will make the eventual collapse to a single representative agent legible: every term we are about to write down is a term from chapter 6 with an index pasted on.

Fix, then, N symmetric players. Player i has state $X_t^i \in \mathbb{R}^d$ governed by the controlled diffusion

$$\mathrm{d}X_t^i \;=\; b\big(X_t^i, \alpha_t^i\big)\ \mathrm{d}t \;+\; \sigma\ \mathrm{d}W_t^i, \qquad X_0^i \sim m_0 \text{ i.i.d.}, \tag{8.1}$$

which is exactly the single state equation of chapter 6 carrying a player label: one independent noise W^i per player, the same drift law b and the same volatility σ for everyone (this is what *symmetric* means), and a common initial law m_0 from which the N starting states are drawn independently. The control $\alpha_t^i \in A$ is chosen in *feedback* form $\alpha_t^i = \alpha^i(t, X_t^1, \ldots, X_t^N)$, a function of the current time and the *entire* configuration of all N states. That the feedback is allowed to read every coordinate, not just the player's own X_t^i, is not a luxury but a necessity: because player i's cost will depend on where the others are, and the others diffuse randomly, an agent who optimizes must in principle condition its instantaneous action on the whole observed state of the game. Were the controls instead chosen in isolation, each looking only at its own state and its own cost, eq. (8.1) would describe N unrelated copies of the chapter 6 problem and there would be nothing to solve jointly. What welds the copies into a game is the structure of the cost.

The coupling between players is *anonymous* in the precise sense flagged above: player i cares about the others only through their *leave-one-out* empirical measure

$$\mu_t^{N,i} \;:=\; \frac{1}{N-1}\sum_{j\neq i}\delta_{X_t^j} \;\in\; \mathcal{P}(\mathbb{R}^d),$$

which is the empirical measure of definition 7.1 — a normalized sum of point masses, one per agent — formed here from the *other* $N-1$ players, with player i's own atom $\delta_{X_t^i}$ deliberately omitted. Two features of this object carry the whole modelling content, and both have a clean physical reading. First, the cost sees the others only through the *distribution* of their states, never through their labels: relabelling the opponents, or swapping which body sits at which point, leaves $\mu_t^{N,i}$ and hence the cost unchanged. The agent perceives a *crowd*, a cloud of mass spread over $\mathbb{R}^d$, not a roster of named rivals — this is precisely the anonymity (exchangeability) that the law of large numbers will later convert into a single deterministic field. Second, the sum runs over $j \neq i$: the agent reacts to the field generated by *everyone else* and not to a field that includes its own contribution. Physically an agent responds to the crowd around it, not to itself; mathematically, leaving the ith atom out is what prevents the self-interaction that would otherwise make "best response to the population" a circular instruction even before the limit. As for the choice of prefactor, $\frac{1}{N-1}$ rather than $\frac{1}{N}$ is the only normalization that makes $\mu_t^{N,i}$ a genuine probability measure once the ith atom has been removed ($N-1$ unit masses divided by $N-1$ integrate to one); had we kept the $\frac{1}{N}$ weight of definition 7.1 the leave-one-out sum would carry total mass $\frac{N-1}{N}$, and the two conventions therefore differ by a relative factor $\frac{N-1}{N} = 1 - \frac{1}{N}$, i.e. by $O(1/N)$. Since every conclusion we draw is a statement about the $N \to \infty$ limit, this $O(1/N)$ discrepancy washes out and the choice is purely one of bookkeeping convenience; we take the normalization that keeps each agent's perceived crowd a bona fide probability law.

With the perceived crowd in hand, player i minimizes the chapter 6-type cost

$$J^i(\alpha^1, \ldots, \alpha^N) \;=\; \mathbb{E}\left[\int_0^T L\big(X_t^i, \alpha_t^i, \mu_t^{N,i}\big)\ \mathrm{d}t \;+\; g\big(X_T^i, \mu_T^{N,i}\big)\right], \tag{8.2}$$

identical in shape to the single-agent cost functional of chapter 6 — a running Lagrangian L integrated to the horizon plus a terminal cost g — except that both L and g now take the perceived crowd $\mu_t^{N,i}$ as a third argument. This is the only place the other players enter J^i, and they enter it anonymously. The notation $J^i(\alpha^1, \dots, \alpha^N)$ makes the strategic structure explicit: player i's realized cost is a functional not only of its own control but of the *whole profile*, because the trajectories X^j that build $\mu_t^{N,i}$ are themselves driven by the other players' controls α^j. Player i can move only α^i; the rest of the profile is its environment.

This is what turns N optimizations into a *game*, and the equilibrium notion is the one that makes each agent's choice consistent with the others'. A *Nash equilibrium* is a profile $(\alpha^{1,*}, \dots, \alpha^{N,*})$ from which no single player can gain by unilaterally deviating: for each i and every admissible alternative control α^i,

$$J^i(\alpha^{1,*}, \dots, \alpha^{i,*}, \dots, \alpha^{N,*}) \;\le\; J^i(\alpha^{1,*}, \dots, \alpha^{i}, \dots, \alpha^{N,*}).$$

Read the inequality literally: hold every other player at its equilibrium strategy $\alpha^{j,*}$, $j \neq i$, let player i alone replace $\alpha^{i,*}$ by any admissible α^i, and the cost cannot go down. Each agent, given what everyone else is doing, is *already* playing a cost-minimizing response; there is no profitable unilateral deviation, hence no regret. Crucially the comparison is one player at a time, with the rest frozen — it is not a statement about jointly optimizing the profile, which is exactly why a Nash equilibrium is in general not the minimizer of any single objective.

It is the "rest frozen" clause that lets us solve for the equilibrium with the tools of chapter 6. Fix i and freeze the others' equilibrium strategies $\alpha^{j,*}$, $j \neq i$. From player i's vantage the states X^j, $j \neq i$, are now *driven processes* with known feedback laws, and player i faces an ordinary stochastic control problem: choose α^i to minimize eq. (8.2) subject to the dynamics eq. (8.1). The only difference from the lone agent of chapter 6 is that the relevant state is no longer X_t^i alone but the full vector $(X_t^1, \dots, X_t^N)$, since the cost depends on the others' positions and those positions evolve. The dynamic programming principle of chapter 6 therefore produces a value function $v^{N,i}(t, x_1, \dots, x_N)$ of *all* N states, satisfying an HJB equation in Nd spatial variables: player i's own coordinate x_i enters through the optimized Hamiltonian, while every other coordinate x_j enters through the *transport* generated by that player's frozen optimal drift. Writing the HJB equation for each i and recording that all N are coupled — each $v^{N,i}$ depends on all states, and the drift carrying coordinate x_j is the maximizer of player j's own problem — yields the *Nash system*: for $i = 1, \dots, N$,

$$-\partial_t v^{N,i} - \sum_{j=1}^{N} \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2_{x_j} v^{N,i}\big) - \sum_{j\neq i} b\big(x_j, \alpha^{j,*}\big)\cdot\nabla_{x_j} v^{N,i} + H\big(x_i, \nabla_{x_i} v^{N,i}, \mu^{N,i}\big) \;=\; 0, \tag{8.3}$$

with terminal data $v^{N,i}(T, \cdot) = g(x_i, \mu^{N,i})$. The three blocks of eq. (8.3) are worth reading one at a time. The Hessian sum $\sum_j \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2_{x_j} v^{N,i})$ is the second-order generator of all N independent noises acting on $v^{N,i}$ — one copy of chapter 6's diffusion term per coordinate, present because $v^{N,i}$ must average over every player's Brownian fluctuation. The last term $H(x_i, \nabla_{x_i} v^{N,i}, \mu^{N,i})$ is player i's *own* optimization, collapsed into the Hamiltonian

$$H(x,p,m) \;:=\; \sup_{\alpha\in A}\Big\{ -\, b(x,\alpha)\cdot p - L(x,\alpha,m)\Big\}, \tag{8.4}$$

the Legendre-type transform introduced in chapter 6: maximizing $-b(x,\alpha)\cdot p - L(x,\alpha,m)$ over the admissible controls $\alpha \in A$ is precisely the pointwise best-response computation of dynamic programming, packaging "pick the control that best trades drift against running cost" into a single function of state x, costate $p = \nabla_{x_i} v^{N,i}$, and perceived crowd $m = \mu^{N,i}$. Only the ith coordinate is optimized here, because player i controls only α^i. The middle block is the heart of the matter: the cross terms

$$\sum_{j \neq i} b(x_j, \alpha^{j,*}) \cdot \nabla_{x_j} v^{N,i}$$

transport $v^{N,i}$ along the closed-loop drift of *every other* player. Here $\alpha^{j,*}$ is not a free variable but player j's *own* optimal control — the maximizer of player j's Hamiltonian, i.e. a function of $\nabla_{x_j} v^{N,j}$ — so the equation for $v^{N,i}$ is laced through with the gradients of the *other* players' value functions. This is the exact mechanism by which the N HJB equations cease to be independent: each one's drift terms are built from the others' solutions, and the maximization defining $\alpha^{j,*}$ makes that dependence nonlinear. The cross terms, with $\alpha^{j,*}$ itself a maximizer, are precisely what make eq. (8.3) a genuinely coupled system of N nonlinear PDEs in Nd variables rather than N separate equations.

And that is where the finite game becomes analytically hopeless. A single $v^{N,i}$ is a function on $\mathbb{R}^{Nd}$, a space whose dimension grows *linearly in the number of players*: for N agents in $d = 1$ one must solve N coupled nonlinear PDEs each posed in N spatial dimensions, and for any population worth calling a crowd — thousands, millions — the state space $\mathbb{R}^{Nd}$ defeats every grid, every basis, every direct numerical scheme by sheer dimensional explosion, quite apart from the nonlinear coupling that forbids solving the equations one at a time. There is no realistic N for which one can write the solution down or compute it; one can only set the system up and pray. The escape, and the whole content of the mean field idea, is to stop fighting the dimension and instead let it work *for* us: send $N \to \infty$ and bet that the anonymous, $O(1/N)$ coupling — no single rival mattering — collapses the fluctuating Nd-dimensional environment into one self-consistent field that a single representative agent optimizes against. We turn to that limit next, fixing attention on a single representative player and watching the crowd it faces freeze into a deterministic flow of measures.

8.1.2 The heuristic limit

We left the finite game in an analytic dead end: a system of N coupled nonlinear HJB equations on $\mathbb{R}^{Nd}$, each value function $v^{N,i}$ laced through the gradients of every other player's value, the whole thing growing more intractable with each agent we add. The mean field idea is to turn that growth into an asset. The very feature that made the finite system hopeless — that every player is coupled to a huge crowd of others — is, after the limit, what *simplifies* it, because anonymity makes no single member of that crowd matter and the law of large numbers can then freeze the crowd into a deterministic background. The argument that does this is a chain of approximations rather than a theorem, and the discipline of the present subsection is to keep that distinction honest: we run the chain to read off the *form* of the limiting object, marking each link as heuristic, and we defer to later chapters the three places where a link must be earned. So fix attention, as the section promised, on a single representative player — call it player 1, with no loss of generality since the players are interchangeable — and watch the Nd-dimensional random environment it faces collapse, link by link, into a single deterministic flow of measures that it optimizes against. The collapse runs in five moves — exchangeability, the law of large numbers, the reduction to a single-agent control problem, the envelope read-off of the feedback, and the loop-closing consistency demand — and we take them in turn inside the heuristic box, then immediately audit what the box quietly assumed.

Heuristic (non-rigorous): The mean field collapse

Step 1: exchangeability makes the rivals conditionally i.i.d. Fix attention on player 1 and suppose the equilibrium is *symmetric*, so that all N players use the same feedback law — a natural ansatz, since the players are statistically identical (same dynamics, same cost, same initial law) and there is no label by which the equilibrium could prefer one to another. Now ask what the other players look like from player 1's seat. Each rival X^j obeys the same controlled diffusion eq. (8.1), driven by its own independent noise W^j and reacting to the same anonymous crowd; and crucially, because the coupling is anonymous and the influence of any single agent on any other is only $O(1/N)$ (section 8.1), the rivals are tied to one another only through the faint, shared field they all read, not through any direct pairwise bond. To leading order in $1/N$ that residual dependence is negligible, and conditionally on the common environment the states $X_t^2, \ldots, X_t^N$ are *i.i.d.* samples from one common one-particle law m_t. This is the exchangeable structure of chapter 7 read at the level of the game: identical agents weakly coupled to a common field behave, in the bulk, like independent copies of a single representative diffusion.

Step 2: the law of large numbers freezes the crowd. Player 1 does not see the rivals individually; it sees only their empirical measure $\mu_t^{N,1} = \frac{1}{N-1} \sum_{j \neq 1} \delta_{X_t^j}$, the perceived crowd of section 8.1. But an empirical measure of $N-1$ samples that are i.i.d. to leading order is precisely the object the law of large numbers governs: as $N \to \infty$ its random fluctuations are $O(N^{-1/2})$ and wash out, and $\mu_t^{N,1}$ concentrates on its mean, the common law m_t. Made rigorous, this concentration is exactly the *propagation of chaos* of chapter 7: the random empirical flow converges to the deterministic flow $m_t = \mathcal{L}(X_t^1)$. The decisive consequence is qualitative, not just quantitative. *Before* the limit, player 1 optimizes inside a fluctuating, high-dimensional, stochastic environment — the joint motion of $N-1$ other diffusers. *After* the limit, that environment has frozen into a single *fixed, deterministic* curve in the space of measures, $t \mapsto m_t$. The randomness of the crowd has been integrated away; what remains is a known, non-random schedule of population states.

Step 3: the representative agent solves an ordinary control problem. Once $t \mapsto m_t$ is a fixed, known function of time, player 1 no longer faces a game at all — it faces the *single-agent* stochastic control problem of chapter 6, with the population appearing as nothing more than a prescribed time-dependent parameter inside its cost. Substituting m_t for the perceived crowd in eq. (8.2), player 1 minimizes $\mathbb{E}\Big[\int_0^T L(X_t, \alpha_t, m_t)\, \mathrm{d}t + g(X_T, m_T)\Big]$ over its own control, subject to its own diffusion — a problem in the *one* state X_t, not in the joint configuration of all N players. The Nd-dimensional value function $v^{N,1}(t, x_1, \ldots, x_N)$ of the finite game has collapsed to a value $u(t,x)$ of player 1's own state alone, and dynamic programming (chapter 6) gives it as the solution of a *single* HJB equation in d variables,

$$-\partial_t u - \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) + H(x, \nabla u, m_t) = 0, \qquad u(T, \cdot) = g(\cdot, m_T),$$

with the same Hamiltonian eq. (8.4) as before, now carrying the frozen flow m_t as a parameter in its third slot and in the terminal datum. This is the entire dimensional payoff of the limit:

N coupled equations on $\mathbb{R}^{Nd}$ have become *one* equation on $\mathbb{R}^d$.

Step 4: the envelope identity reads off the optimal drift. The control problem of Step 3 is solved by the pointwise maximizer in the Hamiltonian eq. (8.4), and what the forward dynamics will need is not the control itself but the *closed-loop drift* that control induces. The envelope theorem (remark 6.13) supplies it without re-solving the maximization: differentiating $H(x,p,m) = \sup_\alpha\{-b(x,\alpha)\cdot p - L\}$ in p, only the explicit p-dependence survives at the maximizer, so the optimal feedback $\alpha^*(t,x)$ induces the closed-loop drift

$$b(x,\alpha^*(t,x)) = -\partial_p H(x, \nabla u(t,x), m_t).$$

A caveat on normalization, repeated here because it is the single most common sign trap of the subject: it is the optimal *drift* that equals $-\partial_p H$, not in general the control. The two coincide — $\alpha^* = -\partial_p H$ — only in the canonical normalization $b(x,\alpha)=\alpha$, $A=\mathbb{R}^d$, in which the control *is* the drift; when a control gain is present (the LQG drift $aX + b\alpha$ of the carried thread is the standing example) the control and the transported drift differ by that gain, and it is the drift, not the control, that Fokker–Planck moves.

Step 5: closing the loop — the consistency condition. So far m_t has been an *assumed* input: we froze the crowd, then computed player 1's best response to it. But the crowd is not external. If *every* agent is identical and each plays the same optimal feedback α^* extracted in Step 4, then the common law they collectively generate is not free to be anything — it must be the law of the diffusion actually driven by that feedback. That is, m must solve the Fokker–Planck (Kolmogorov forward) equation for the closed-loop drift $-\partial_p H$,

$$\partial_t m - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 m\big) - \operatorname{div}\Big(m\,\partial_p H(x,\nabla u, m_t)\Big) = 0, \qquad m(0,\cdot) = m_0,$$

the forward transport of the initial population m_0 along the very feedback that the HJB equation produced. The equilibrium is the demand that these two appearances of m be *the same*: the flow assumed when computing the value must equal the flow produced when the resulting feedback is played out,

$$m \;=\; \mathcal{L}\big(\text{closed-loop diffusion driven by } \alpha^*[m]\big).$$

This is the *consistency* or *Nash certainty equivalence* condition. It is the analytic incarnation of "no agent regrets its choice given what the crowd does, and the crowd is exactly the aggregate of those choices." The HJB equation of Step 3 and the Fokker–Planck equation of Step 5, locked together by this self-consistency, are the mean field game system.

The collapse is seductive, and the prose inside the box moved quickly past three places where a genuine theorem is needed rather than a plausibility argument; honesty about what we have and have not established is the whole point of marking the derivation heuristic. The first gap is the convergence $\mu_t^{N,1} \to m_t$ itself — but *at the equilibrium.* Step 2 invoked propagation of chaos, yet the propagation of chaos of chapter 7 was proved for a *prescribed* McKean–Vlasov dynamics, where the feedback is fixed from outside. Here the feedback is the strategic best response, which itself depends on the very limit law we are trying to establish; the law of large numbers must therefore be made to hold along a Nash profile, with the optimization and the concentration closing on each other simultaneously. That is a strictly deeper statement — the central convergence theorem of Part II — and it is proved only in chapter 13. The second gap is the decoupling of the value functions: Step 3 simply asserted that the Nd-variable $v^{N,1}(t, x_1, \ldots, x_N)$ converges to a function $u(t, x_1)$ of player 1's own coordinate alone, the other coordinates' dependence dissolving into the deterministic parameter m_t. This too requires proof, and it too belongs to the convergence analysis. The third gap is the most basic: Step 5 wrote down a fixed-point demand, but a demand is not a solution. Nothing so far guarantees that any flow m satisfies the consistency condition — that the best-response map has a fixed point at all. Existence is the charge of chapter 10.

It is worth being completely explicit about the resulting division of labor, because it is the pedagogical contract not just of this subsection but of the whole chapter. section 8.1.2 is to be read as the *derivation of an ansatz*, and nothing more: it tells us the *form* the limiting object must take — a backward HJB driven by the population coupled to a forward Fokker–Planck driven by the feedback, pinned by data at opposite ends of $[0, T]$ — but it does *not*, on its own, establish that this object is the genuine $N \to \infty$ limit of the finite Nash equilibria. We deliberately split the burden in two. The claim that the coupled pair is *internally consistent* — that solving the two PDEs really does encode a representative agent best-responding to a population that reproduces itself — is something we can and will verify *directly*, by the verification theorem of section 8.2, with no appeal to any limit and with the finite game nowhere in sight. The separate claim that this internally consistent object is the correct *limit* of the N-player game — that the three gaps above can be closed — is postponed to the convergence theory of chapter 13. The reader should therefore treat everything that follows in this chapter as the study of a self-standing analytic system, justified on its own terms as the equilibrium condition of a representative agent, and should not regard the heuristic's limit-talk as cashed in until chapter 13 does so. What the heuristic has bought us is precise and limited: the *shape* of the equations, forced on us by an argument we trust enough to build on, even though two of its links are still promissory notes.

With that shape in hand, the next section takes over the task of making it exact. It will fix the standing data, state the coupled pair as a precise definition rather than a heuristic readout, read the two equations in the two complementary ways that organize the rest of Part II — the HJB as a reverse-time value diffusion, the Fokker–Planck as conservation of mass — and then discharge the first half of the contract just drawn, proving the verification statement: that this ansatz is internally consistent, a Nash-consistent best response, without yet invoking the limit.

8.2 The coupled system and its two readings

The previous section handed us a shape and a promissory note. The mean field collapse of section 8.1.2 produced, link by heuristic link, a coupled pair — a backward HJB driven by the population, a forward Fokker–Planck driven by the feedback the value generates — but it produced that pair as an *ansatz*, the form an equilibrium must take, not yet a defined object we may reason about on its own terms. The closing discussion there split the work of honoring that ansatz into two clauses: that the coupled pair is *internally consistent* — that solving the two PDEs really does encode a representative agent best-responding to a population that reproduces itself — is provable here and now, directly, with the finite game nowhere in sight; that this consistent object is the genuine $N \to \infty$ *limit* of the Nash equilibria is deferred to the convergence theory of chapter 13. This section discharges the first clause. Our itinerary is four steps and we name them in advance so the reader can hold the whole section in view: first we fix the standing data, pinning down exactly which ingredients are given and under which hypotheses, so that "solve the system" has an unambiguous meaning; then we state the coupled pair as a precise definition rather than a heuristic readout; then we read that one pair of equations in two complementary ways, because — as we are about to argue — the same two equations teach two different lessons depending on which end of time one stands at; and finally we verify the coupling, proving the consistency clause by importing the representative-agent verification theorem of chapter 6.

The two readings deserve a word of motivation before we earn the right to give them, since at first sight reading the same system twice looks like indulgence. It is not. The HJB equation eq. (8.5a) and the Fokker–Planck equation eq. (8.5b) are not two facts about one population; they are two *directions* through the same population, and each direction is natural for exactly one of the fields. Read the HJB equation as a value flow and it runs *against* the clock: data prescribed at the deadline $t = T$, information about the optimal cost-to-go diffusing backward out of the terminal cost as the horizon recedes — a reaction–diffusion equation in reversed time, smoothing the kinks a first-order Hamilton–Jacobi equation would otherwise sharpen. Read the Fokker–Planck equation as a mass flow and it runs *with* the clock: data prescribed at the start $t = 0$, an honest continuity equation transporting a conserved unit of probability forward along the optimal drift while diffusion spreads it. Neither reading is dispensable, because the analytic difficulty of the subject lives precisely in the clash between them — the value wants to be solved from the future, the density from the past — and one cannot feel the force of that clash without first seeing each equation in the time-direction that makes it elementary. Stating both readings is also what exposes how *little* couples the two equations: the HJB needs the population only through m_t, the Fokker–Planck needs the value only through ∇u (via the optimal drift), and nothing else passes between them. That economy — a single measure flowing one way, a single gradient flowing the other — is the whole interface of the mean field game, and it is what makes the fixed-point reformulation of section 8.4 tractable.

Before any of this we must say what is given. The next subsection therefore pauses on the standing data, picking up the general pivot of chapter 7 — where the controlled drift was allowed to depend on the population law along with the state and control — and specializing it to the setting in which the whole rest of the chapter works: a *measure-independent* drift $b(x, \alpha)$, so that the population enters the agent's problem *only through the cost*. This is the standard analytic arena, it already contains the carried LQG thread, and it is the cleanest stage on which to watch the two readings and the verification play out; we record there exactly why the specialization costs us nothing structural and where the more general measure-dependent drift would re-enter.

8.2.1 Standing data and assumptions

The section opened by promising to say what is given before we say what is to be solved, and this is where we pay that debt. A definition that began "a pair (u, m) solving the following two equations" would be hollow until the ingredients of those equations — the state space, the cost, the drift, the noise, and the hypotheses that make the operations in them legitimate — were pinned down. So we pause here to fix the entire stage. Nothing in this subsection is a theorem; its job is to make the next subsection's definition stateable without a single parenthetical caveat, so that when the coupled system appears it appears clean, every object already named and every hypothesis already in force. The reader who wants the equations first may skim to definition 8.1 and refer back; but the price of doing so is that the "why" of each assumption — which is exactly what the narration doctrine of this book asks us to surface — lives here.

We fix the data of the representative-agent problem. The state space is $\mathbb{R}^d$. A *flow of measures* is a map $m\colon [0,T] \to \mathcal{P}_2(\mathbb{R}^d)$, $t \mapsto m_t$, continuous for the metric $\mathcal{W}_2$ (chapter 4); it is the deterministic background the representative agent optimizes against, the frozen crowd that section 8.1.2 manufactured from the law of large numbers. The running cost $L\colon \mathbb{R}^d \times A \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$, the drift $b\colon \mathbb{R}^d \times A \to \mathbb{R}^d$, the (constant, for simplicity of exposition) volatility $\sigma \in \mathbb{R}^{d\times d}$, and the terminal cost $g\colon \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ are given. The Hamiltonian $H(x, p, m)$ is defined by eq. (8.4).

One modelling choice has already been made silently in the way these data types are written, and it is worth dragging into the light because the rest of the chapter rides on it. In chapter 7 we worked with the *general* pivot, allowing the controlled drift to read the population law directly, $b = b(X_t, \alpha_t, m_t)$: the crowd could push each agent mechanically, not merely tax it. Here we specialize to a *measure-independent* drift $b(x, \alpha)$, with the population stripped out of the state equation entirely. The consequence is a clean division of labor between the two ways the crowd can act on an agent. In the general model the population enters both the dynamics and the cost; in the present one it enters *only through the cost*, $L(x, \alpha, m)$ and $g(x, m)$, while the law of motion an agent obeys depends on no one but itself. This is not a loss of generality worth worrying about, for three reasons we record now so they need not be relitigated later. First, it is the standard analytic arena for the subject: with the drift inert in m, the Hamiltonian's p-dependence is a fixed Legendre transform and only its m-slot moves with the crowd, which is what makes the two readings of the next subsections — and the variational structure of section 8.6 — come out as cleanly as they do. Second, it already contains the object we have been carrying: the LQG thread's drift $aX + b\alpha$ (worked computation 8.12) is measure-independent, the population there acting purely through a quadratic cost that penalizes deviation from the mean. Third, the genuinely more general case — the *extended* MFG, where the drift does see m — is a routine generalization rather than a new phenomenon: it thickens the coupling between the two equations (the Fokker–Planck drift then depends on m a second time, through b itself) but changes none of the structural features — the forward–backward split, the fixed-point reformulation, the verification logic — that this chapter is built to expose. We carry it out where a result actually needs it and otherwise keep the lighter notation.

With the data fixed we impose the standing hypotheses of the book (appendix A), invoked by their global tags so that the same names mean the same things across all of Part II. The list reads like bookkeeping, but each entry is doing a specific job, and the surest way to understand an assumption is to ask what collapses if it is dropped. We therefore state each tag together with the failure it forestalls.

(A-Lip) b, σ, L, g are Lipschitz in the state, uniformly in time, control, and the measure argument (the latter measured in $\mathcal{W}_2$). Lipschitz control of the coefficients is what keeps the maps in this chapter from being merely formal: it is what makes the closed-loop stochastic differential equation of eq. (8.5b) have a unique strong solution rather than branch or explode, and — in the *joint* (x, m)-Lipschitz form of assumption 7.3, which is how we read the tag wherever the fixed-point and verification arguments below require it — it is what makes the best-response map of section 8.4 continuous in the crowd, so that a small change in the population m produces only a small change in the agent's optimal response. Without a modulus of continuity in m there is no hope of a fixed point, since the very map whose fixed point we seek could jump. The tag (*A-Lip*) thus carries the same meaning throughout Part II.

(A-Growth) b, σ have at most linear growth in the state and L is coercive in the control — it grows fast enough in α (superlinearly, in the canonical normalization where the control *is* the drift) that steering harder is eventually never worth its price. This coercivity is exactly what guarantees that the supremum defining the Hamiltonian in eq. (8.4) is *attained at a finite control and is itself finite*: were L to grow only sublinearly in α, the agent could buy unboundedly favorable drift $-b(x, \alpha) \cdot p$ for asymptotically free, the supremum would run off to $+\infty$, and $H(x, p, m)$ would not even be a well-defined function, let alone one we could differentiate to read off a feedback. The linear growth of b, σ plays the partner role on the forward side: it keeps the closed-loop diffusion from escaping to infinity in finite time, so that mass is conserved and the flow m_t stays a genuine probability measure (the tightness invoked in eq. (8.9)).

(A-Nondeg) $\sigma\sigma^\top \succeq \lambda I$ for some $\lambda > 0$. This is the assumption that buys us *classical* solutions on both ends, and it is the one a physicist should read most physically: a strictly positive amount of noise in every direction means the second-order operator $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2\, \cdot)$ is uniformly elliptic, so both equations of eq. (8.5) are uniformly parabolic and their solutions are smoothed by diffusion rather than left to develop the singularities a first-order equation would. Concretely, on the backward side the $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2\, u)$ term regularizes the Hamilton–Jacobi nonlinearity into a classical $C^{1,2}$ value function (theorem 6.14); on the forward side the same operator turns the transport of m_0 into a smooth density solving eq. (8.5b) (theorem 5.14). Drop nondegeneracy and both statements weaken at once: the value function survives only in the viscosity sense developed in chapter 6, and the density may concentrate onto lower-dimensional sets where it has no Lebesgue representative. We assume $\lambda > 0$ throughout this chapter precisely so that "the density $m(t, \cdot)$" in definition 8.1 is an honest function we may insert into the cost.

(A-Conv) $L(x, \cdot, m)$ is *strictly* convex in the control. This is the most easily misread of the four, so we separate its two halves carefully. Convexity of the Hamiltonian $H(x, \cdot, m)$ in the costate p is *free*: by its definition eq. (8.4), $H(x, p, m)$ is a pointwise supremum over α of the maps $p \mapsto -b(x, \alpha) \cdot p - L(x, \alpha, m)$, each of which is *affine* in p; and a supremum of affine functions is convex no matter what, with no hypothesis on L at all. So if one wanted only "H convex in p" there would be nothing to assume. The content of the tag lies entirely on the other side: it is the *strict* convexity of L in α that we are buying, and what it buys is a *unique* maximizer in eq. (8.4). Uniqueness is exactly what the rest of the machinery needs — a unique maximizer makes the optimal drift $-\partial_p H$, and hence the feedback α^*, a single-valued function of (t, x), so that eq. (8.5b) transports along a well-defined velocity field and the verification theorem of chapter 6 applies. A merely convex L would leave H convex but possibly non-differentiable in p, the maximizing control set-valued at the kinks, and "the" optimal drift undefined precisely where it matters.

The distinction in (*A-Conv*) between what is free and what is assumed is subtle enough to deserve a concrete instance. Work in the canonical normalization $b(x,\alpha)=\alpha$, where $H(x,p,m) = \sup_\alpha\{-\alpha\cdot p - L(x,\alpha,m)\}$ is literally the Legendre transform of L in its control slot, and take $d=1$ with the bounded control set $A=[-1,1]$. Choose the running cost $L(\alpha)=|\alpha|$, which is convex but *not strictly* convex (it is flat-sided: affine on $[-1,0]$ and on $[0,1]$). The Hamiltonian is then

$$H(p) = \sup_{\alpha\in[-1,1]} \big\{ -\alpha\, p - |\alpha| \big\} = (|p|-1)_+,$$

a perfectly good convex function — convexity came for free, exactly as advertised, even though L was not strictly convex. But look at the maximizer. For $|p|<1$ the unique optimal control is $\alpha^*=0$; for $|p|>1$ it is the saturated value $\alpha^* = -\operatorname{sgn}(p)$. At the borderline costate $p=1$, however, the objective $-\alpha\, p - |\alpha| = -\alpha - (-\alpha) = 0$ is *identically zero for every* $\alpha\in[-1,0]$: the entire segment maximizes, and the optimal control is the whole interval $[-1,0]$ rather than a point. This is precisely the kink of H at $|p|=1$, where H fails to be differentiable and $-\partial_p H$ — the drift Fokker–Planck is supposed to transport along — has no single value. Strict convexity of L is exactly the hypothesis that erases such flat segments: it bends the cost strictly upward in α, so the supremum is attained at one control, H becomes C^1 in p, and the feedback is single-valued everywhere. Replace $|\alpha|$ by $\frac{1}{2}\alpha^2$ and the pathology vanishes: $H(p)=\frac{1}{2}p^2$ is smooth and the maximizer $\alpha^*=-p$ is unique for every p — the familiar LQG arithmetic of the carried thread. The micro-example is thus the whole point of (*A-Conv*) in miniature: convexity of H is automatic; single-valued feedback is not, and it is the strict convexity of L that secures it.

One assumption is conspicuously absent from this list, and its absence is deliberate rather than an oversight. Monotonicity **(A-Mon)** — the Lasry–Lions condition on how the cost responds to the crowd — is *not* needed to *write* the mean field game system, nor to *solve* it for a single equilibrium. Everything in this section and the next, the definition, the two readings, the verification theorem, the fixed-point reformulation, and the existence of a solution, runs on **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)** alone. What **(A-Mon)** governs is a strictly later and different question: whether the equilibrium so produced is *the only one*, and whether the strategic system secretly minimizes a single convex energy. The first is the uniqueness theory of chapter 11; the second is the variational (potential-game) structure of section 8.6, where monotonicity is exactly the convexity that turns the forward–backward pair into the Euler–Lagrange system of one functional. Keeping monotonicity in reserve is not pedantry: it sharpens the reader's map of the subject by separating the hypotheses that make the equilibrium *exist and make sense* from the single extra hypothesis that makes it *unique and variational*. We therefore omit it here and flag its entry explicitly each time it is switched on downstream.

With the stage fully set — state space, data, Hamiltonian, and the four standing hypotheses all fixed and motivated, and monotonicity consciously held back — we have earned the right to state the system itself. The next subsection does nothing but record, as a precise definition, the coupled pair that section 8.1.2 produced heuristically; because every hypothesis it relies on is now in force by name, that definition can be written cleanly, with no caveat re-litigated, as the object of study for the remainder of the chapter.

8.2.2 The mean field game system

The stage is now fully set. The previous subsection fixed the data (b, σ, L, g), defined the Hamiltonian H by eq. (8.4), and switched on the four standing hypotheses **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, each motivated by the failure it forestalls. Nothing is left undefined and no caveat remains to be relitigated. We have therefore earned the right to do the one thing this subsection exists to do: *record*, as a clean mathematical object, the coupled pair that the mean field collapse of section 8.1.2 produced heuristically. The heuristic already did the creative work — it told us the limiting equilibrium must be a backward HJB driven by the population locked to a forward Fokker–Planck driven by the feedback the value generates, with data at opposite ends of $[0, T]$. What remains is not to discover that form but to pin it down precisely enough to reason about on its own terms, with the finite game and the limit nowhere in sight. That is the content of the following definition; everything after it in this chapter is a study of the object the definition names.

Definition 8.1 (MFG PDE system)**.** A pair (u, m), with $u\colon [0, T] \times \mathbb{R}^d \to \mathbb{R}$ sufficiently smooth and $m\colon [0, T] \to \mathcal{P}_2(\mathbb{R}^d)$ a flow of measures with densities $m(t, \cdot)$, is a *(classical) solution of the mean field game system* with data (m_0, g) if

$$-\partial_t u - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 u\big) + H(x, \nabla u, m_t) = 0, \qquad u(T, x) = g(x, m_T), \tag{8.5a}$$

$$\partial_t m - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 m\big) - \operatorname{div}\Big(m\, \partial_p H(x, \nabla u, m_t)\Big) = 0, \qquad m(0, x) = m_0(x), \tag{8.5b}$$

hold on $(0, T) \times \mathbb{R}^d$, where D^2 and div act in the state variable x.

Two equations, two fields, two time directions: the definition is best absorbed not as a black box of symbols but as a sentence one can read aloud. Read as a whole, eq. (8.5) couples a *value* u that is computed backward from the deadline to a *density* m that is transported forward from the start, and the only places the two fields touch are the few we are about to isolate. Before turning each equation over we say what it is in words; after the display we say what its single point of contact with the other field is, because it is exactly the smallness of that contact — the "strikingly economical" coupling that will become a refrain — that the rest of Part II exploits.

Take eq. (8.5a) first. It is the backward Hamilton–Jacobi–Bellman equation of the representative agent, the very equation dynamic programming produced in chapter 6, now carrying a frozen population as a parameter. Its terminal datum $u(T, \cdot) = g(\cdot, m_T)$ sits at the end of the interval and information about the optimal cost-to-go propagates from there backward into $t < T$; the nonlinearity lives entirely in the Hamiltonian $H(x, \nabla u, m_t)$, which eq. (8.4) defined as the agent's pointwise best-response computation. The decisive observation is where the crowd appears in this equation, and where it does *not*. The population m_t enters eq. (8.5a) in exactly two slots — the third argument of H and the measure argument of the terminal cost g — and *nowhere else*. It does not touch the agent's dynamics (the drift b is measure-independent, by the standing specialization of the previous subsection), it does not touch the diffusion operator $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u)$, and it does not touch the costate ∇u directly. From the value's point of view the entire population is a single time-dependent parameter dropped into its running and terminal costs. Hold $t \mapsto m_t$ fixed and eq. (8.5a) is an ordinary single-agent HJB equation of the kind chapter 6 solved; the whole strategic content is smuggled in through that one parametric slot.

Now eq. (8.5b). It is the forward Fokker–Planck (Kolmogorov forward) equation for the density of the closed-loop process: the law of the diffusion the representative agent actually realizes once it plays its optimal feedback. Its datum $m(0,\cdot) = m_0$ sits at the *start* of the interval and the density is pushed forward from there, the mirror image of the HJB's backward sweep. And just as the HJB touched the population in only one place, the Fokker–Planck touches the value in only one place: through the drift that carries the mass, the *optimal drift*

$$b\big(x, \alpha^*(t,x)\big) \;=\; -\,\partial_p H(x, \nabla u(t,x), m_t), \qquad \alpha^*(t,x) \;=\; \operatorname*{arg\,max}_{\alpha\in A} \Big\{ -b(x,\alpha)\cdot\nabla u(t,x) - L(x,\alpha,m_t) \Big\}, \tag{8.6}$$

which depends on the value only through its gradient ∇u. Nothing else of u reaches the forward equation: not its level, not its time derivative, only the spatial gradient, and that only as it is funnelled through $\partial_p H$.

The equality in eq. (8.6) is the *envelope identity* of remark 6.13, and because it is the hinge on which the coupling turns it is worth re-deriving in place rather than merely cited. The Hamiltonian eq. (8.4) is a supremum over controls,

$$H(x,p,m) = \sup_{\alpha\in A} \Big\{ -b(x,\alpha)\cdot p - L(x,\alpha,m) \Big\} = -b(x,\alpha^*)\cdot p - L(x,\alpha^*,m),$$

where $\alpha^* = \alpha^*(x,p,m)$ is the maximizer, single-valued and depending smoothly on (x,p,m) because **(A-Conv)** makes the bracket strictly concave in α with a unique stationary point. Differentiate the right-hand side in p. Naively there are two channels: the explicit p multiplying $-b(x,\alpha^*)$, and the implicit dependence of α^* on p. The second channel vanishes — this is the whole content of the envelope theorem. By the chain rule its contribution is

$$\partial_\alpha \Big[-b(x,\alpha)\cdot p - L(x,\alpha,m) \Big]\Big|_{\alpha=\alpha^*} \cdot\; \partial_p \alpha^*,$$

and the first factor is precisely the first-order optimality condition that *defines* α^*, hence is zero at an interior maximizer. (At a boundary maximizer the same conclusion holds by the standard envelope argument; under **(A-Conv)** together with the coercivity in **(A-Growth)** the maximizer is interior.) Only the explicit channel survives, and it gives at once

$$\partial_p H(x,p,m) \;=\; -\,b\big(x, \alpha^*(x,p,m)\big). \tag{8.7}$$

Evaluating eq. (8.7) at the costate $p = \nabla u(t,x)$ and the frozen crowd $m = m_t$ is exactly eq. (8.6): the gradient of the Hamiltonian *is* the (negative of the) closed-loop drift, recovered without ever re-solving the maximization. This is what makes the forward equation cheap to form from the backward one: one differentiates H, a function already in hand, instead of redoing the agent's optimization at every point.

A warning is built into eq. (8.6), and it is the single most common sign trap of the subject, so we state it sharply and then check it by hand. What m transports is the optimal *drift* $b(x,\alpha^*)$, and that drift equals $-\partial_p H$ — it is *not*, in general, the optimal control α^* itself. The two are distinct objects: α^* is the action the agent chooses, while $b(x, \alpha^*)$ is the velocity that action induces on the state, and only the latter appears in a transport equation, because Fokker–Planck moves mass along velocities, not along control values. The two coincide *exactly* in the canonical normalization $b(x, \alpha) = \alpha,\ A = \mathbb{R}^d$ — the convention of the general theory and of appendix A — in which the control *is* the drift and so $\alpha^* = -\partial_p H$ reads directly off eq. (8.7). The moment a control gain is present they part ways by exactly that gain.

The carried LQG thread is the cleanest place to watch them part, and the computation is short enough to run end to end. There (worked computation 8.12) the drift is $b(x, \alpha) = ax + b\,\alpha$ with control gain b, and the running cost is $L_0(\alpha) = \frac{1}{2}\alpha^2$, so with $p = \partial_x u$ the bracket in eq. (8.6) is

$$-b(x, \alpha)\, p - L_0(\alpha) \;=\; -(ax + b\,\alpha)\, p - \tfrac{1}{2}\alpha^2.$$

Maximizing in α kills the derivative $-b\,p - \alpha = 0$, so the optimal control is

$$\alpha^* = -b\,p = -b\,\partial_x u.$$

The transported drift is a *different* quantity: substitute this α^* back into b,

$$b(x, \alpha^*) = ax + b\,\alpha^* = ax + b\,(-b\,p) = ax - b^2\,\partial_x u \;=\; -\partial_p H,$$

which one may double-check directly from the closed form $H(x, p) = -ax\,p + \frac{1}{2}b^2p^2 + (m\text{-dependent cost})$, whence $\partial_p H = -ax + b^2 p$ and $-\partial_p H = ax - b^2\partial_x u$, the same thing. So the control the agent picks, $\alpha^* = -b\,\partial_x u$, and the drift the crowd transports along, $ax - b^2\partial_x u$, differ both by the gain b (the control is the velocity's controllable part divided through by b) and by the uncontrolled mean-reversion ax that is present in the velocity but absent from the control. Confusing the two would put the wrong velocity into eq. (8.5b) and violate conservation of mass; the envelope identity is precisely the bookkeeping that keeps drift and control straight.

We can now state the structural fact that the two readings of the next two subsections, and the fixed-point reformulation of section 8.4, all lean on. The coupling between eq. (8.5a) and eq. (8.5b) is *strikingly economical*: the backward equation needs the population only through the measure flow m_t, the forward equation needs the value only through the gradient ∇u (funnelled through $\partial_p H$), and *nothing else passes between them.* The entire interface of the mean field game is one measure flowing one way and one gradient flowing the other. This economy is not a cosmetic observation. It is what will let us, in section 8.4, collapse the whole coupled system onto a single best-response map on the space of measure flows — feeding m into the HJB, reading ∇u out of it, pushing the induced drift through Fokker–Planck, and asking for a fixed point — because so little information crosses the seam that the seam can be parametrized by the population flow alone. With the system stated and its interface laid bare, we turn to reading the two equations one at a time, each in the time direction that makes it elementary. We begin with the backward HJB equation, whose parabolic character is easiest to see by running the clock in reverse.

8.2.3 Reading one: HJB as a reverse-time value diffusion

We take the backward equation first, as promised, and the task of this first reading is to dissolve an unease the equation provokes the moment one looks at it as written. In eq. (8.5a) the time derivative carries a minus sign and the diffusion term carries a minus sign too: the agent's value obeys $-\partial_t u - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) + H = 0$. A reader schooled on the heat equation $\partial_t \theta = \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 \theta)$ should flinch here, because moving both minus signs to the other side displays a Laplacian entering with the *wrong* sign relative to a forward-running time derivative — the signature of the *backward* heat equation, the canonical ill-posed problem in which fine features are amplified rather than smoothed and generic data cannot be evolved at all. Were that the correct reading, the HJB equation would be hopeless, and with it the whole left half of the mean field game. It is not the correct reading, and the cleanest way to see why is to stop fighting the clock and run it backward: the equation is not a heat equation evolved into its past but a perfectly ordinary diffusion evolved into its future, provided one agrees to call "the future" the direction in which the deadline recedes.

Make the substitution explicit. The value is anchored by data at the terminal time $t = T$ and computed toward earlier times, so the natural clock for it measures *time remaining until the deadline*. Set

$$\tau \;=\; T - t, \qquad \tilde{u}(\tau, x) \;=\; u(T - \tau, x),$$

so that $\tau = 0$ is the deadline $t = T$ and $\tau = T$ is the present $t = 0$; as the physical clock t runs down from T to 0, the reversed clock τ runs *up* from 0 to T. The only term that feels the reversal is the time derivative, by the chain rule $\partial_\tau \tilde{u} = -\partial_t u$; the spatial operators D^2 and ∇ act in x and are untouched, and the frozen population is simply read at the corresponding instant, $m_t = m_{T-\tau}$. Substituting $-\partial_t u = \partial_\tau \tilde{u}$ into eq. (8.5a) and moving the diffusion and Hamiltonian to the right turns the backward HJB equation into

$$\partial_\tau \tilde{u} = \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 \tilde{u}) - H(x, \nabla\tilde{u}, m_{T-\tau}), \qquad \tilde{u}(0, x) = g(x, m_T).$$

The single sign flip on the time derivative has done everything. The diffusion term now sits on the right with the *same* sign as the time derivative on the left — exactly the configuration of the ordinary heat equation — and the data, which in the original variables sat awkwardly at the end of the interval, now sit at $\tau = 0$, the natural starting line for a forward problem. In the reversed time τ the equation is therefore a genuine *forward parabolic* (reaction–diffusion) equation: the second-order part is the stabilizing diffusion that smears features out, and **(A-Nondeg)**, which makes $\sigma\sigma^\top \succeq \lambda I$, is precisely the uniform ellipticity that guarantees it smears in every direction at a strictly positive rate. The unease was an artifact of insisting on the physical clock. Read against the deadline, the value diffuses the way temperature does.

This gives the first reading its slogan. The terminal cost $g(\cdot, m_T)$ is the initial datum at $\tau = 0$ — the heat distribution the agent is handed at the deadline — and as τ grows, i.e. as the deadline recedes into the future and the agent has more time to act, the value *diffuses out from the terminal cost*. The cost-to-go at a present state is, in this picture, the terminal penalty seen through a lengthening lens of diffusive smoothing and the deterministic transport we turn to next: the further the deadline, the more the sharp profile of g has spread and been reshaped. The reaction–diffusion analogy is exact down to the two mechanisms a reaction–diffusion equation always combines. The $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 \tilde{u})$ term is the *diffusion*: it is the agent's own Brownian noise, integrated into the value because the cost-to-go must average over every future fluctuation of the state, and it acts to spread and smooth. The $-H(x, \nabla\tilde{u}, m_{T-\tau})$ term is the *reaction*: it is nonlinear in the gradient $\nabla\tilde{u}$ and it is the Hamilton–Jacobi transport of cost-to-go, the local bookkeeping by which the agent's optimal trade-off of drift against running cost — packaged once and for all into the Hamiltonian eq. (8.4) — pushes information about future penalties along the characteristics that the optimal feedback carves out. Diffusion blurs; the Hamiltonian transports and reshapes; together they propagate the terminal cost backward into a smooth field of optimal costs-to-go.

It is worth dwelling on why the reaction term, left to itself, would be the source of trouble, because that is where the role of **(A-Nondeg)** becomes physical rather than bookkeeping. Strip the diffusion away — set $\sigma = 0$ — and what remains, $\partial_\tau \tilde{u} = -H(x, \nabla \tilde{u}, \cdot)$, is a pure first-order Hamilton–Jacobi equation. Such equations are the deterministic limit of optimal control, and they are notorious for one behavior: even from perfectly smooth initial data their characteristics cross in finite time, and at the crossing the gradient $\nabla \tilde{u}$ becomes multivalued and the solution develops *kinks* — corners where u is continuous but ∇u jumps, the value analogue of a shock forming in a compressible flow. A classical (twice-differentiable) solution simply ceases to exist past the first crossing, and one is forced into the weaker notion of viscosity solutions to continue at all. Now restore the diffusion. The second-order term acts on any incipient kink exactly as heat conduction acts on a temperature corner: a discontinuity in the gradient is a near-singularity in the Hessian, and $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 \tilde{u})$ responds to it with an overwhelming smoothing flux that rounds the corner off before it can sharpen into a genuine singularity. Under **(A-Nondeg)** this happens uniformly and in every direction, and the upshot is that the strictly parabolic HJB equation admits a classical $C^{1,2}$ solution where its first-order shadow would only admit a kinked one — precisely the regularity theorem 6.14 delivers and that definition 8.1 quietly assumed when it spoke of u as "sufficiently smooth" and of ∇u as a genuine velocity to feed the forward equation.

This is the mechanism worth naming, because it recurs throughout the subject: *vanishing viscosity made non-vanishing.* In the classical theory of first-order Hamilton–Jacobi equations one *adds* an artificial viscosity $\varepsilon\Delta$, solves the regularized parabolic problem, and sends $\varepsilon \to 0$; the limit, taken along this vanishing-viscosity sequence, is what *defines* the viscosity solution and selects the physically correct kinked profile. The HJB equation of a genuinely noisy control problem skips the limit entirely. Its viscosity is not an artifice introduced to regularize and then removed; it is the agent's real diffusion $\frac{1}{2}\sigma\sigma^\top$, present in the model from the start and *never* sent to zero. The regularization that the classical theory must engineer by hand, **(A-Nondeg)** hands us for free, with $\lambda > 0$ the guaranteed floor on how much smoothing is always available. That is the precise sense in which the noise of the control problem is "vanishing viscosity made non-vanishing," and it is why, in the nondegenerate setting of this chapter, we may work with honest classical solutions and never mention viscosity solutions again. Should one ever drop **(A-Nondeg)** — allow the diffusion to degenerate in some direction, or pass to a deterministic ($\sigma = 0$) game — the smoothing floor collapses, kinks reappear, and the correct notion of solution reverts to the viscosity solutions developed in chapter 6; everything downstream then carries over in that weaker sense.

The HJB equation is, in short, a value equation that flows *against* the clock: its data live at the deadline and its information propagates from the future into the present, a forward diffusion only once one measures time by how much of it remains. That orientation is not a quirk to be normalized away — it is half of the structural tension this chapter is building toward. For the companion field runs the other way. The density m knows nothing of deadlines; it is handed its initial profile m_0 at $t = 0$ and pushed forward by the very feedback that the value, flowing backward, has just produced.

The next subsection reads eq. (8.5b) in *its* natural direction, recasting it not as a value diffusion but as a continuity equation that conserves mass while transporting it forward along the optimal drift. Holding the two readings side by side will make the asymmetry impossible to miss: one equation is solved from the future, the other from the past, and they are chained to each other at both ends of $[0,T]$. That collision of opposed time directions — the value pulled backward from T, the mass pushed forward from 0, each feeding the other — is the forward–backward tension that section 8.3 will name as the structural signature of the entire subject.

8.2.4 Reading two: Fokker–Planck as conservation of mass

The first reading left the value flowing against the clock, its data pinned at the deadline. Its companion runs the other way. The density m is handed an initial profile m_0 at $t=0$ and pushed forward, and the right way to read eq. (8.5b) is therefore not as a backward value diffusion but in the time-direction natural to a population: as a statement about how a fixed unit of probability is carried around $\mathbb{R}^d$ as time advances. A population density is not an arbitrary field. It must satisfy two non-negotiable constraints to deserve the name at all — it must stay nonnegative (there is no such thing as a negative amount of crowd at a point) and its total mass must stay equal to one (the agents do not multiply or vanish; the same unit of probability is merely rearranged). The task of this second reading is to make both constraints *manifest* rather than hope for them, and the device that does so is the oldest bookkeeping in physics: write the evolution as a conservation law, a continuity equation in which the only thing that ever happens to the density is that it flows. Once eq. (8.5b) is in that form the two constraints fall out almost for free, the first from the divergence theorem and the second from the meaning of the object being transported.

Recast eq. (8.5b), then, as a continuity equation. A continuity equation says that the rate of change of a density at a point is minus the divergence of a *current* — the local flux of stuff through that point — so we must first exhibit the current that eq. (8.5b) is implicitly moving mass along. Define the *probability current*

$$\mathbf{J}(t,x) \;:=\; \underbrace{m(t,x)\,b\big(x,\alpha^*(t,x)\big)}_{\text{advective}} \;-\; \underbrace{\tfrac{1}{2}\,\sigma\sigma^\top\nabla m(t,x)}_{\text{diffusive}}, \tag{8.8}$$

where we used the envelope identity eq. (8.6) to write the optimal drift as $b(x,\alpha^*) = -\partial_p H(x,\nabla u, m_t)$. The two pieces are the two physically distinct ways mass moves, and it is worth naming each. The first, $m\,b(x,\alpha^*)$, is the *advective* (convective) current: a parcel of probability at x is carried bodily along the optimal velocity field $b(x,\alpha^*)$, exactly as a dye is swept along by a fluid, the local flux being density times velocity. The second, $-\frac{1}{2}\sigma\sigma^\top\nabla m$, is the *diffusive* (Fick) current: noise drives mass down its own gradient, from where the crowd is dense toward where it is sparse, at a rate set by the diffusion matrix $\frac{1}{2}\sigma\sigma^\top$. The minus sign is Fick's law made literal — the flux opposes the gradient — and it is this term, present precisely because the agents diffuse, that will later supply the smoothing the existence theory leans on.

Notation watch

The current is set in *boldface* $\mathbf{J}$ deliberately, to keep it apart from the plain letter J, which throughout the book denotes a *cost functional*: the agent's objective $J(\alpha)$ of eq. (8.22) on the LQG thread, and the per-player cost $J^i(\alpha^1, \ldots, \alpha^N)$ of eq. (8.2) carried from chapter 7. The two objects could not be more different in kind — one is a vector field on $\mathbb{R}^d$, the other a real number assigned to a control — but they share a glyph, and the boldface is the only thing that disambiguates them. We never write the current without the boldface, and never the cost functional with it.

With the current named, the claim is that eq. (8.5b) *is* the continuity equation for it,

$$\partial_t m + \operatorname{div} \mathbf{J} = 0, \tag{8.9}$$

and this is a one-line verification worth doing in the open, because it is exactly where a sign error would hide. Take the divergence of eq. (8.8) term by term. The diffusive piece gives, since $\sigma\sigma^\top$ is a constant matrix,

$$\operatorname{div}\Big(\tfrac{1}{2}\sigma\sigma^\top \nabla m\Big) = \sum_{i,j} \tfrac{1}{2}(\sigma\sigma^\top)_{ij}\, \partial_{x_i}\partial_{x_j} m = \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top\, \mathrm{D}^2\, m\big),$$

the very second-order operator appearing in eq. (8.5b). The advective piece gives $\operatorname{div}(m\, b(x, \alpha^*))$, and using $b(x, \alpha^*) = -\partial_p H$ this is exactly the transport term of eq. (8.5b) with its sign reversed: $\operatorname{div}(m\, b(x, \alpha^*)) = -\operatorname{div}(m\, \partial_p H)$. Assembling,

$$\partial_t m + \operatorname{div} \mathbf{J} = \partial_t m - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top\, \mathrm{D}^2\, m\big) - \operatorname{div}\big(m\, \partial_p H(x, \nabla u, m_t)\big),$$

which is precisely the left-hand side of eq. (8.5b). So setting $\partial_t m + \operatorname{div} \mathbf{J} = 0$ *is* the Fokker–Planck equation; the continuity form is not an analogy but a literal rewriting. (This is the place to honor the chapter's standing dictum: every sign in eq. (8.8) should be checked against this identity, never against a remembered formula, because it is the conservation law that the signs must serve.)

Now reap the first constraint. Conservation of total mass is the statement that a continuity equation makes automatic, and the argument is the divergence theorem applied to all of $\mathbb{R}^d$. Integrate eq. (8.9) over the whole space and exchange the time derivative with the spatial integral:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} m(t,x)\, \mathrm{d}x = \int_{\mathbb{R}^d} \partial_t m\, \mathrm{d}x = -\int_{\mathbb{R}^d} \operatorname{div}\mathbf{J}\, \mathrm{d}x = -\lim_{R\to\infty}\int_{\partial B_R} \mathbf{J}\cdot\mathbf{n}\, \mathrm{d}S,$$

the last step being the divergence theorem on the ball B_R of radius R, with $\mathbf{n}$ the outward normal. Everything hinges on that boundary flux vanishing as $R \to \infty$, and this is exactly where a standing hypothesis does load-bearing work rather than decoration. The flux $\int_{\partial B_R} \mathbf{J}\cdot\mathbf{n}\, \mathrm{d}S$ measures the rate at which probability crosses the sphere of radius R outward; it tends to zero only if essentially no mass — and no current — reaches arbitrarily large radii. That is precisely the content of *tightness*: the family $\{m(t,\cdot)\}_{t\in[0,T]}$ keeps all but a vanishing fraction of its mass inside a fixed compact set, with both density and gradient decaying at infinity fast enough to kill the surface integral. Tightness is in turn what **(A-Growth)** buys: the at-most-linear growth of b and σ prevents the closed-loop diffusion from carrying mass off to infinity in finite time, yielding the uniform moment bounds that, by the tightness criterion of chapter 3, confine the flow. With the boundary term gone we are left with

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} m(t,x)\ \mathrm{d}x = 0,$$

so the total mass is a constant of the motion; since m_0 is a probability density with $\int m_0 = 1$, every later slice $m(t,\cdot)$ integrates to one as well. Mass does not leak: the unit of probability handed in at $t = 0$ is conserved exactly, only rearranged. This is the first half of "m genuinely transports a probability density."

The second half is nonnegativity, and here the situation is unusually comfortable: there are *two* independent routes to it, and neither requires us to invent any new maximum-principle machinery, because each leans on a fact we have already established for this exact operator. They are worth giving separately, since they illuminate the conclusion from opposite sides — one says nonnegativity is a fact about *probability*, the other that it is a fact about *parabolic PDEs* — and they agree only because theorem 5.14 identifies the two solutions they speak about.

The first route is probabilistic, and it gets nonnegativity for free from the *meaning* of the solution. By the Kolmogorov-forward representation of theorem 5.14, the unique solution of eq. (8.5b) with initial probability density m_0 is nothing other than the family of time-marginal laws of the closed-loop diffusion

$$\mathrm{d}X_t = b\big(X_t, \alpha^*(t,X_t)\big)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t, \qquad X_0 \sim m_0,$$

that is, $m_t = \mathcal{L}(X_t)$ — this stochastic differential equation having a unique strong solution because **(A-Lip)** and **(A-Growth)** make its drift Lipschitz and non-exploding. But a law is, by the very definition of a probability measure, nonnegative: $m_t(E) = \mathbb{P}(X_t \in E) \geq 0$ for every Borel set E, so its density is nonnegative wherever it exists. Nonnegativity here is not something one proves about the PDE at all; it is inherited from the fact that the PDE is solved by a *law*. No comparison argument, no maximum principle, nothing beyond reading off what theorem 5.14 says the solution is.

The second route forgets the probabilistic representation entirely and treats eq. (8.5b) as the linear PDE it manifestly is. Once the value function has been solved and the feedback $\alpha^*(t,x)$ frozen, the velocity field $b(\cdot,\alpha^*(t,\cdot))$ is a *known* coefficient — and a Lipschitz one, because **(A-Nondeg)** makes the value classical (theorem 6.14) and hence its gradient, through which the feedback is read, Lipschitz in space. The equation

$$\partial_t m = \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top\,\mathrm{D}^2\, m\big) - \mathrm{div}\big(m\, b(\cdot,\alpha^*)\big)$$

is then a *linear, uniformly parabolic* equation in the single unknown m, uniform parabolicity being exactly **(A-Nondeg)**'s $\sigma\sigma^\top \succeq \lambda I$. Such operators obey a weak comparison principle: two solutions ordered at the initial time stay ordered for all later times. Apply it with the constant function $m \equiv 0$, which is itself a solution, as the lower barrier. At $t = 0$ we have $m_0 \geq 0$; comparison propagates this ordering forward, giving $m(t,\cdot) \geq 0$ for every t. This needs "no separate maximum-principle apparatus" in the precise sense that the comparison principle for this very operator is already part of the linear parabolic theory packaged in theorem 5.14 — we import it, we do not re-erect it — and uniform parabolicity, the single hypothesis it requires, is one we have assumed from the outset.

Putting the two halves together, eq. (8.5b) does exactly what a population density must: it conserves probability (mass stays one, by the divergence theorem and **(A-Growth)**) and keeps it nonnegative (by either route above), all while transporting the crowd along the optimal feedback and diffusing it at rate $\frac{1}{2}\sigma\sigma^\top$. The forward half of the mean field game is, in this reading, a bona fide conservation law for a genuine probability density — not a formal adjoint written down for symmetry with the HJB, but an honest transport of mass.

Physics note

The Fokker–Planck equation eq. (8.9) is a continuity equation for a conserved density, its current $\mathbf{J}$ split into an advective (drift) part and a Fick (diffusive) part — the same anatomy as the continuity equation of any transported scalar. What is unusual, and is the whole reason a physicist should pause here, is the *origin of the advective velocity.* In ordinary transport theory the drift is an externally prescribed field; here it is generated *self-consistently*: $b(x, \alpha^*) = -\partial_p H(x, \nabla u, m_t)$ is read off from the value u, which was itself computed knowing the very flow m it now transports. The field that moves the crowd is the field the crowd itself generates. This is precisely the self-consistent-field closure of mean field statistical mechanics — the Curie–Weiss magnet feeling the mean magnetization it collectively produces — but with one decisive substitution. In the thermal case the relation that closes field to configuration is an *averaging* postulate (the mean field is a Boltzmann-weighted expectation); here it is an *optimization.* The crowd's drift is the gradient of everyone's best strategy, and everyone's best strategy is computed against the crowd's drift. Replacing the Boltzmann average by a best response is the entire conceptual distance between equilibrium statistical mechanics and a mean field *game.*

The two readings now stand side by side, and the picture they jointly produce is the one the next section must come to terms with: a value flowing backward from the deadline and, locked to it, a conserved probability density flowing forward from the start, the density driven by a drift that the value manufactures and the value shaped by a population that the density *is.* We have called this drift "self-consistent" and the loop "closed," but those are so far slogans — the physicsnote earns them only at the level of a suggestive closure relation. What we have genuinely established is narrower and firmer: that the forward equation, fed a feedback from *any* value function, transports a probability density honestly. Whether the loop actually closes — whether a pair (u, m) solving the two equations together really encodes a representative agent best-responding to a crowd that reproduces itself, rather than two equations that merely happen to share symbols — is a separate claim that a pair of PDEs does not certify on its own. The next subsection takes up exactly that worry, making "self-consistent" precise as a genuine fixed-point property via the verification theorem; and the forward–backward collision we have just laid bare — one equation solved from the future, the other from the past — is what section 8.3 will name as the structural signature of the whole subject.

8.2.5 Why the coupling is legitimate: verification

Reading Two left us with a picture and a debt. The picture is the self-consistent field of the chapter's preamble made nearly tangible: a conserved unit of probability transported forward along a drift that the value function manufactures, while the value in turn is shaped by the very population the density carries. We have been calling that drift "self-consistent" and the loop "closed" since section 8.1.2, but as the close of the previous subsection warned, those are still slogans. A pair of partial differential equations does not become an equilibrium merely because we wrote the two fields with names that share symbols; nothing forces the m that the agent optimizes *against* in eq. (8.5a) to be the same object as the m that the optimal feedback *produces* through eq. (8.5b). We have only *posited* that coincidence, in definition 8.1, by writing the same letter twice. The worry is therefore precise and it must be discharged before the system earns the name we have given it: *is solving the coupled pair really the same act as a representative agent best-responding to a crowd that reproduces itself?* The instrument that settles the question is the verification theorem of stochastic control — imported wholesale from chapter 6 and applied with the population flow held fixed, so that no appeal to any limit and no trace of the finite game enters. The strategy is exactly the one the heuristic anticipated: freeze the crowd, certify that the agent's response to it is genuinely optimal, and then check that the law that response generates is the crowd we froze.

Proposition 8.2 (Representative-agent optimality)**.** *Assume **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**. Let $m = (m_t)_{t\in[0,T]}$ be a fixed flow of measures and let u be the classical solution of the backward HJB equation eq. (8.5a) with this m. Then $u(t,x)$ is the value function of the representative agent's control problem against the environment m, namely*

$$u(t,x) = \inf_{\alpha} \ \mathbb{E}\left[\int_t^T L(X_s, \alpha_s, m_s)\ \mathrm{d}s + g(X_T, m_T)\ \Big|\ X_t = x\right],$$

the infimum over admissible feedback controls of the diffusion $\mathrm{d}X_s = b(X_s,\alpha_s)\ \mathrm{d}s + \sigma\ \mathrm{d}W_s$; and the feedback $\alpha^*(t,x)$ of eq. (8.6), whose closed-loop drift is $b(x,\alpha^*(t,x)) = -\partial_p H(x,\nabla u(t,x),m_t)$, is optimal. Consequently the law of the optimally controlled diffusion solves the Fokker–Planck equation eq. (8.5b).

The hypotheses are exactly the four standing ones of the chapter, and it is worth keeping in view — since the proof leans on each in a different place — which clause does which job: nondegeneracy makes the HJB classically solvable so that Itô's formula may be applied to u at all; convexity makes the maximizer in the Hamiltonian unique so that "the" optimal control is a function rather than a set; Lipschitz and growth control keep the closed-loop dynamics from exploding and keep the cost integrals finite. The argument below is a sketch by design — the two engines it runs on, the verification theorem and the forward-equation theorem, are proved in full in chapter 6 and chapter 5 — but it is an honest sketch: we write out the Itô-to-optimality chain explicitly, because it is short and it is exactly where the HJB equation does its work, and we are careful about the one genuinely delicate step, the well-posedness of the closed-loop equation, rather than waving it through.

Proof sketch. Step 0: the value function exists and is regular. Existence of the classical solution u is not at issue here, but it is the platform the rest stands on, so we name it precisely. Hold the flow m fixed. Then the backward HJB equation eq. (8.5a) is a single semilinear parabolic equation in which the population appears nowhere as an unknown: it enters only through the two slots $H(x, \nabla u, m_t)$ and $g(x, m_T)$, i.e. as a *prescribed* time-dependent coefficient and a prescribed terminal datum. Because $t \mapsto m_t$ is $\mathcal{W}_2$-continuous and H, g are Lipschitz in m for $\mathcal{W}_2$ by **(A-Lip)**, that coefficient is merely a Lipschitz-in-time inhomogeneity — it adds no new nonlinearity and no spatial roughness. Under **(A-Nondeg)** the second-order operator is uniformly elliptic, so the equation is uniformly parabolic, and **(A-Conv)** and **(A-Growth)** make H smooth in p with controlled growth; the classical-solvability theorem theorem 6.14 then delivers a unique $C^{1,2}$ solution u of polynomial growth. The proof there is written for a frozen measure m, but it uses of m only that it enters as a coefficient with the stated regularity, so it applies verbatim with $m \mapsto m_t$ a given continuous flow. The point to carry forward is that u is genuinely $C^{1,2}$ — continuously differentiable in time and twice in space — which is exactly the regularity Itô's formula will demand.

Step 1: Itô along an arbitrary control gives $u \le J$. Fix the start (t, x) and let $\alpha = (\alpha_s)_{s\in[t,T]}$ be *any* admissible feedback control, driving the diffusion $\mathrm{d}X_s = b(X_s, \alpha_s)\,\mathrm{d}s + \sigma\,\mathrm{d}W_s$ from $X_t = x$. Write $J(\alpha)$ for the cost in the Proposition. Apply Itô's formula (theorem 5.9) to the $C^{1,2}$ function $s \mapsto u(s, X_s)$:

$$\mathrm{d}\,u(s, X_s) = \Big[\partial_s u + b(X_s, \alpha_s)\cdot\nabla u + \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 u\big)\Big]\,\mathrm{d}s + \nabla u\cdot\sigma\,\mathrm{d}W_s.$$

The stochastic integral is a martingale — ∇u has polynomial growth and the diffusion has finite moments under **(A-Growth)** — so taking expectations and integrating from t to T kills it:

$$\mathbb{E}[u(T, X_T)] - u(t, x) = \mathbb{E}\left[\int_t^T \Big[\partial_s u + b(X_s, \alpha_s)\cdot\nabla u + \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 u\big)\Big]\,\mathrm{d}s\right].$$

Now invoke the HJB equation eq. (8.5a) in the rearranged form $\partial_s u + \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) = H(X_s, \nabla u, m_s)$, and the defining inequality of the Hamiltonian eq. (8.4), which says that for *every* control value α_s,

$$H(X_s, \nabla u, m_s) \;\ge\; -b(X_s, \alpha_s)\cdot\nabla u - L(X_s, \alpha_s, m_s),$$

because H is the supremum of exactly that bracket over admissible controls. Substituting, the integrand obeys $\partial_s u + b\cdot\nabla u + \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) = H + b\cdot\nabla u \ge -L(X_s, \alpha_s, m_s)$, whence

$$\mathbb{E}[u(T, X_T)] - u(t, x) \;\ge\; -\mathbb{E}\left[\int_t^T L(X_s, \alpha_s, m_s)\,\mathrm{d}s\right].$$

Using the terminal datum $u(T, \cdot) = g(\cdot, m_T)$ and rearranging gives the clean inequality

$$u(t, x) \;\le\; \mathbb{E}\left[\int_t^T L(X_s, \alpha_s, m_s)\,\mathrm{d}s + g(X_T, m_T)\right] = J(\alpha).$$

The value function thus underbounds the cost of every admissible control — which is half of what "value function" means.

*Step 2: equality at the pointwise maximizer, and where **(A-Conv)** enters.* The inequality of Step 1 came from a single inequality, the one between H and the bracket, and it is an equality precisely when the running control achieves the supremum in eq. (8.4) at each instant. That is, choose the feedback $\alpha_s = \alpha^*(s, X_s)$ with α^* the maximizer of eq. (8.6); then the bracket equals H identically along the path, every inequality above becomes an equality, and $u(t,x) = J(\alpha^*)$. Combined with Step 1 this is exactly $u(t,x) = \inf_\alpha J(\alpha) = J(\alpha^*)$: u is the value function and α^* is optimal. Here is where strict convexity earns its keep. For the recipe "play the maximizer" to define an honest feedback we need the maximizer to be a single point, not a flat set; **(A-Conv)** — strict convexity of $L(x,\cdot,m)$ in the control — makes the bracket strictly concave in α, so the argmax in eq. (8.6) is unique, the optimal drift is the single value $-\partial_p H$ by the envelope identity, and $\alpha^*(s,\cdot)$ is a genuine function. Without it the maximizing control could be set-valued at the kinks of H, exactly the pathology the micro-example of section 8.2 exhibited, and "the" optimal feedback would be undefined precisely where the forward equation needs a velocity field.

Step 3: the closed-loop law solves Fokker–Planck. It remains to identify the density the optimal agent produces. Substitute the optimal feedback into the dynamics: the representative agent, playing α^*, realizes the *closed-loop* diffusion

$$\mathrm{d}X_s = b\big(X_s, \alpha^*(s, X_s)\big)\,\mathrm{d}s + \sigma\,\mathrm{d}W_s, \qquad X_0 \sim m_0.$$

By the Kolmogorov-forward (Fokker–Planck) theorem of chapter 5 (theorem 5.14), the law of this diffusion has, under **(A-Nondeg)**, a density $\widetilde{m}(s,\cdot)$ solving $\partial_s \widetilde{m} = \mathcal{A}^*_{\alpha^*}\widetilde{m}$, where $\mathcal{A}^*_{\alpha^*}$ is the formal adjoint of the generator of the closed-loop process — the same second-order part $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2\,\cdot)$ as on the backward side, plus the transport $-\operatorname{div}(\,\cdot\, b(x,\alpha^*))$. Now make the one substitution that fuses the two equations: by eq. (8.6) the closed-loop drift *is* the optimal drift, $b(x,\alpha^*) = -\partial_p H(x, \nabla u, m_s)$, so the forward equation for $\widetilde{m}$ reads $\partial_s \widetilde{m} - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2\,\widetilde{m}) - \operatorname{div}(\widetilde{m}\,\partial_p H(x,\nabla u, m_s)) = 0$ — which is letter for letter the Fokker–Planck equation eq. (8.5b), with the same initial datum m_0. The law the optimal agent generates therefore solves the very equation the density was asked to solve.

Step 4: the delicate point — the closed-loop equation is well-posed. Steps 1–3 quietly assumed that the closed-loop diffusion of Step 3 exists, is unique, and has a single well-defined law to speak of. That is not free, because the drift $x \mapsto b(x, \alpha^*(s,x))$ is itself built by an optimization and could in principle be too rough for the SDE to have a unique strong solution. This is the one step worth pausing on. The optimal drift is the composition $-\partial_p H(x, \nabla u(s,x), m_s)$ of three ingredients: the map $p \mapsto -\partial_p H(x,p,m)$, which is locally Lipschitz because **(A-Conv)** makes H continuously differentiable in p with a Lipschitz gradient; the spatial gradient $\nabla u(s,\cdot)$, which is Lipschitz in x because u is $C^{1,2}$ by Step 0; and the frozen slice m_s, which is a constant in x. A composition of Lipschitz maps is Lipschitz, so $x \mapsto b(x,\alpha^*(s,x))$ is Lipschitz in x, uniformly on $[0,T]$. The standard strong-existence-and-uniqueness theorem for SDEs with Lipschitz drift and (here constant, hence Lipschitz) diffusion then applies: the closed-loop equation has a unique strong solution, its law is well-defined, and Step 3's identification is legitimate. It is precisely the marriage of **(A-Conv)** (smoothness of H in p) with the $C^{1,2}$ regularity of u from **(A-Nondeg)** that supplies this Lipschitz feedback; drop either and the closed-loop drift could lose the regularity that makes "the law of the optimal agent" a single object at all. The full arguments behind the two imported theorems are in chapter 6 (verification) and chapter 5 (forward equation); what we have done here is assemble them against the frozen flow and check the seam where they meet. □

The proposition is exactly the rigorous content of the heuristic's "close the loop," and it is worth saying plainly what has and has not been shown. We did *not* prove that the system has a solution —

m was held fixed throughout, an input we never had to produce. What we proved is the *internal consistency* promised in section 8.2: *if* a pair (u, m) solves eq. (8.5), then the two appearances of m are not an accident of notation but a genuine equilibrium identity — the flow the agent optimizes against is reproduced exactly by the flow its optimal response generates. The double use of the letter m in definition 8.1 is thereby vindicated: a solution of the coupled pair is a flow m such that, when the representative agent best-responds to it, the law it itself generates is m again. The crowd one optimizes against and the crowd one's optimization produces are the same crowd. That is a fixed point, the subject of section 8.4 — and the obstruction standing between this self-map and a march in time, namely that its two equations carry their data at opposite ends of $[0, T]$, is exactly the forward–backward structure that the next section now isolates as the structural signature of the whole subject.

8.3 The forward–backward boundary-value problem

The verification proposition of the last section left us with a sentence to live with: a solution of eq. (8.5) is a flow m such that, when the representative agent best-responds to it, the law its optimization generates is m again — the crowd one optimizes against and the crowd one's optimization produces are the same crowd. That is a self-consistency, and we already saw, in the two readings of section 8.2, that the two equations realizing it run in *opposite* temporal directions. The HJB equation eq. (8.5a) is a value flow read *backward*: its datum sits at the deadline $t = T$, and the optimal cost-to-go diffuses out of the terminal cost g as the horizon recedes toward the present. The Fokker–Planck equation eq. (8.5b) is a mass flow read *forward*: its datum sits at the start $t = 0$, and the initial population m_0 is transported along the optimal drift as the clock advances. The self-consistent flow that verification certified is therefore stitched together from two halves whose data are pinned at *opposite ends* of $[0, T]$.

We pause on this structure — and the whole of the present section does nothing but pause on it — because it is the single feature that sets a mean field game apart from every evolution problem the reader has solved by marching a known state forward, and because it is the seed from which essentially all of the analytic difficulty in Parts II and IV grows. Existence (chapter 10), uniqueness (chapter 11), the master equation that linearizes the fixed point (Part IV), and the convergence of the N-player Nash equilibria (chapter 13) are each hard for the *same* reason, and that reason is the two-point structure we now isolate. It is worth getting it into the bones before any of those chapters lean on it, so we will resist the temptation to state it as a one-line slogan and instead watch, concretely, exactly where an attempt to solve the system the only way one knows how — by integration from a single time slice — breaks down.

So imagine you sit at $t = 0$ holding the one thing the problem actually hands you at that instant: the initial population m_0. You would like to do what every initial-value solver does, namely produce the state an infinitesimal step later and repeat. The Fokker–Planck equation eq. (8.5b) is even willing to oblige in *form* — it is a forward parabolic equation, exactly the kind one integrates from $t = 0$ — but to take the step it demands the velocity field along which to transport the mass, the optimal drift $-\partial_p H(x, \nabla u, m_0)$. That drift is built from $\nabla u(0, \cdot)$, the gradient of the value function *now*. So you turn to the HJB equation to supply $u(0, \cdot)$ — and there the march dies. The HJB equation eq. (8.5a) does not determine u at the present from anything in the present; it determines $u(0,\cdot)$ only as the result of integrating *backward* from the terminal condition $u(T,\cdot) = g(\cdot, m_T)$, and that backward integration reads the population density m_t at *every* intermediate time $t \in [0,T]$ as it goes, because the Hamiltonian's third slot is fed m_t all along the way. To know $u(0,\cdot)$ you must already know the entire future flow $\{m_t\}_{t\in[0,T]}$ — which is the very object you were trying to construct, step by step, starting from the one slice m_0 you possess. You have come full circle without advancing the clock by a single instant: the first forward step needs a value that needs the whole future of the mass that you have not yet built. The symmetric attempt fares no better. Sit instead at $t = T$ holding g, the one datum the value equation offers, and try to integrate the HJB equation backward; at the first step it asks for m_T, the terminal population, which is the *output* of running the forward equation eq. (8.5b) from $t = 0$ all the way to T along a drift that depends on the very u you are trying to compute. Each sweep, in either direction, stalls at its first step on a global-in-time demand for the other field. This is the obstruction in its rawest form, and the following remark records it.

Remark 8.3 (No initial-value reduction)**.** One cannot march eq. (8.5) forward (or backward) in time from a single time slice. To step the density forward via eq. (8.5b) one needs the feedback, hence ∇u at the *current* time; but u at the current time is determined by the HJB equation *integrated backward from* $t = T$, which requires knowing m_t over the *entire* future interval $[t,T]$. Symmetrically, to integrate the HJB equation backward one must already know the future density, which is produced by the forward equation, which in turn needs the value. Neither equation can be advanced one step without the other's *global-in-time* solution already in hand. There is no Cauchy datum at any single time from which both fields evolve: the problem is genuinely a two-point boundary-value problem *in function space*, in which one seeks the pair (u,m) on the whole space–time cylinder $[0,T]\times\mathbb{R}^d$ at once, subject to one constraint (m_0) at the past end and a different constraint (g) at the future end. The forward–backward coupling is not a Cauchy problem with awkward data; it has no initial-value reduction at all.

It is worth being precise about which familiar tool this defeats, because naming the failure points to the cure. The workhorse for ordinary evolution equations is the Picard iteration that underlies the Cauchy–Lipschitz theorem: given the state at one time, one writes the solution as a fixed point of the integral operator "initial datum plus the time-integral of the vector field," and a Lipschitz bound makes that operator a contraction on a short time interval, so the solution exists, is unique, and is obtained by literally iterating short forward steps. Every ingredient of that argument presumes a *Cauchy datum*: a single slice from which the integral operator builds the future. The mean field game system has no such slice to feed the iteration — as the remark shows, no time carries data for *both* fields — so the Picard/Cauchy–Lipschitz machinery does not merely converge slowly or under stronger hypotheses; it does not even get started, because the object it iterates, "the solution as a function of one initial condition," does not exist here. The local-in-time contraction that tames ODEs and parabolic Cauchy problems is simply the wrong shape for a problem whose constraints live at two separated times.

The way past the obstruction is therefore not to repair the march but to abandon it, and to solve for the whole flow at once. The standard device — set up here only far enough to motivate it, made precise in the next section and deployed in chapter 10 — decouples the two ends by turning the self-consistency demand into a search for a fixed point of a map on the space of *density flows* alone. The idea is to break the circle the remark exposed by cutting it at the population: *guess* an entire flow $m = \{m_t\}_{t\in[0,T]}$, treat it as the frozen background of section 8.2, and let each equation run in the one direction that is now unobstructed. With m given over all of $[0,T]$, the HJB equation eq. (8.5a) *can* be integrated backward from g — its third slot is now a known schedule, so the obstruction of the remark is gone — yielding a value u and, by the envelope identity, a feedback drift $-\partial_p H(x, \nabla u, m_t)$. With that feedback in hand the Fokker–Planck equation eq. (8.5b) *can* be integrated forward from m_0, transporting the initial population along the now-known drift to produce a *new* flow, which we will call $\mathsf{S}[m]$. The guess and the output need not agree; a solution of eq. (8.5) is exactly a flow on which they do — a flow left invariant by this guess-solve-read-transport cycle, $m = \mathsf{S}[m]$. One has converted a two-point boundary-value problem with no initial-value reduction into a *fixed-point problem for a single self-map* of the space of flows, at the price of having to find the right flow rather than march to it. That existence of a fixed point is then secured not by contraction but by compactness: the smoothing of **(A-Nondeg)** and the tightness furnished by the mass and moment bounds (chapter 3) make S a continuous self-map of a compact convex set of flows, and a topological fixed-point theorem of the Schauder/Kakutani family (chapter 2) then forces a fixed point to exist. The essential point for the present section is conceptual, and we state it flatly so it is not mistaken: the fixed-point reformulation does *not* remove the forward–backward structure, it *encodes* it. The whole difficulty of the two-point problem has been packed into the single map S, whose evaluation *contains* one backward HJB sweep and one forward Fokker–Planck sweep glued at their data; the fixed points of S are, by construction, precisely the solutions of the two-point problem with its data at both ends. We trade a problem we cannot integrate for a problem we cannot contract, and earn existence from compactness instead of from iteration.

That trade also reveals the true mathematical genus of the object, and it is the one the closing of this section wants to leave the reader holding. A problem whose data sit at two separated ends of an interval, with no evolution from a single slice, is not a Cauchy problem at all; it is a *boundary-value problem*, and boundary-value problems are the native habitat of the calculus of variations, where the stationarity condition of a functional over a time interval generically produces equations constrained at both endpoints rather than initialized at one. The forward–backward mean field game system is the strategic instance of exactly this pattern: data at $t = 0$ and $t = T$, a coupled pair stationary in between. The next two sections cash this observation in along the two routes it opens. section 8.4 takes the fixed-point road just sketched and makes it precise — naming the space of flows and the uniform-Wasserstein metric on it, and defining the best-response map S whose fixed points we have just argued are the equilibria — so that the existence and uniqueness theories of chapter 10 and chapter 11 have a rigorous object to act on. section 8.6 takes the variational road and shows that the analogy with the calculus of variations is, for the special but central class of potential (monotone, separable) couplings, no analogy at all: there the forward–backward pair is *literally* the Euler–Lagrange system of a single convex functional on flows, and solving the mean field game is minimizing one effective action. In the general, non-potential case no such global functional exists and the system remains an irreducibly strategic Nash fixed point — the "action" rendered stationary is each agent's own cost against the crowd, not any quantity the population jointly extremizes — which is precisely why the fixed-point formulation, not a minimization, is the load-bearing one for the rest of Part II.

8.4 The Nash fixed-point reformulation

The previous section ended by abandoning the march and pointing to a different kind of solution. We saw that the forward–backward system eq. (8.5) carries no Cauchy datum at any single time, so the only honest way to reach a solution is to stop integrating from one slice and instead solve for the *entire* flow at once, as the invariant of a single self-map: guess a whole population history m, run the HJB backward and the Fokker–Planck forward against it, and demand that what comes out be what went in. That sketch was deliberately informal — it named no space, no metric, and no map with a domain — and the present section pays the debt. It is also the section in which the verification proposition of section 8.2 finally cashes out structurally. proposition 8.2 taught us to read a solution of the mean field game as a flow that the representative agent *reproduces*: optimize against m, push the initial population forward along the resulting feedback, and recover m itself. That sentence already describes a fixed point in everything but name — "the input flow equals the output flow." What remained was to give the map whose input and output those are a precise home, so that "fixed point" becomes a theorem about an operator on a metric space rather than a turn of phrase. That is exactly what we do now, and the payoff is that the whole of Part II — existence, uniqueness, the master equation, the convergence of the finite game — will from here on speak about *this one operator* rather than about a pair of coupled PDEs.

We begin with the space the flows live in, because the map has nowhere to act until it is named. A *flow of measures*, recall from section 8.2, is a continuous curve $t \mapsto m_t$ in the Wasserstein space $\mathcal{P}_2(\mathbb{R}^d)$; the set of all such curves is

$$\mathfrak{C} \;:=\; C\big([0,T];\mathcal{P}_2(\mathbb{R}^d)\big),$$

the candidate population histories among which an equilibrium is to be found. We metrize it by the *uniform-in-time Wasserstein distance*

$$d_{\mathfrak{C}}(m,m') \;:=\; \sup_{t\in[0,T]} \mathcal{W}_2(m_t,m'_t), \tag{8.10}$$

the sup over the horizon of the slice-by-slice Wasserstein distance built in chapter 4. Two readings of eq. (8.10) are worth keeping in mind, because both will be used downstream. As a *metric*, the uniform distance makes $\mathfrak{C}$ a complete metric space: a Cauchy sequence of flows is uniformly Cauchy in $\mathcal{W}_2$, hence converges at each time to a limit measure, and the uniform control forces the limit curve to be continuous, so it lies back in $\mathfrak{C}$ — completeness being exactly what a contraction argument for uniqueness (chapter 11) will want underfoot. As a *topology*, it is strong enough that the coefficients of our equations, Lipschitz in m by **(A-Lip)**, vary continuously along it, yet weak enough that the bounded sets of flows our estimates produce are precompact — which is exactly what a compactness argument for existence (chapter 10) will want. The same metric thus serves both of the two routes to a fixed point, and choosing it now is what lets the later chapters simply invoke "the space of flows" without re-litigating its structure.

Notation watch

We write the space of flows with the *fraktur* $\mathfrak{C}$, never with the script $\mathcal{C}$. The script letter $\mathcal{C}$ is reserved for the Thread B per-type cost functional of chapter 20, an unrelated object living on the type space $[0,1]$; using distinct alphabets here forestalls a collision at that seam, where flows of measures and per-type costs appear on the same page.

With the stage named we can build the map. The construction is nothing more than one turn of the guess–solve–read–transport cycle of section 8.3, and we lay it out as three steps, citing the well-posedness of each so that the output is a bona fide element of $\mathfrak{C}$ and the map is genuinely a self-map. Define the *best-response map*

$$\mathsf{S}\colon \mathfrak{C} \to \mathfrak{C}$$

as follows. Given a flow $m \in \mathfrak{C}$:

1. *Solve the backward HJB equation against this m.* Freeze the flow in the third slot of the Hamiltonian and in the terminal datum and solve eq. (8.5a) backward from $u(T,\cdot) = g(\cdot, m_T)$, obtaining the representative value $u^{[m]}$. This is exactly the single-agent problem of chapter 6 with a known time-dependent parameter, so under the standing assumptions a unique $C^{1,2}$ classical solution of polynomial growth exists by theorem 6.14 — **(A-Nondeg)** supplying the parabolic smoothing that makes it classical, **(A-Conv)** and **(A-Growth)** making the Hamiltonian well defined and differentiable — and it is the agent's genuine value function by proposition 8.2.

2. *Read off the optimal feedback by the envelope identity.* From $u^{[m]}$ form the optimal control $\alpha^*(t,x) = \arg\max_\alpha\{-b(x,\alpha)\cdot\nabla u^{[m]}(t,x) - L(x,\alpha,m_t)\}$, single-valued by **(A-Conv)**, whose induced closed-loop drift is, by the envelope identity eq. (8.6), $b(x,\alpha^*(t,x)) = -\partial_p H(x, \nabla u^{[m]}(t,x), m_t)$. No optimization is re-solved here: the drift is obtained by differentiating the Hamiltonian already computed in step 1.

3. *Transport the initial population forward along that feedback.* Solve the forward Fokker–Planck equation
$$\partial_t \tilde m - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2\,\tilde m\big) - \operatorname{div}\big(\tilde m\, b(x,\alpha^*)\big) = 0, \qquad \tilde m_0 = m_0,$$
which, with the drift of step 2 now a *known* velocity field, is a *linear* parabolic Cauchy problem; under **(A-Nondeg)** and the Lipschitz, linear-growth control of the drift it has a unique solution that is a continuous flow of probability measures with finite second moment (theorem 5.14), so $\tilde m \in \mathfrak{C}$. Set $\mathsf{S}[m] := \tilde m$.

Each step is well posed precisely because the input m has frozen the one slot that made the coupled system intractable: in step 1 the population is a known schedule, so the HJB obstruction of remark 8.3 is gone and the backward sweep runs; in step 3 the feedback is a known field, so the Fokker–Planck is linear and the forward sweep runs. The map S thus packages exactly one backward sweep and one forward sweep, glued at the seam where step 2 converts value into drift. It is a genuine self-map of $\mathfrak{C}$ — flows in, flows out — and the equilibria are exactly the flows it leaves alone.

Proposition 8.4 (Equivalence of the two formulations). *Under **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, a flow $m \in \mathfrak{C}$ is a fixed point of S if and only if, with $u = u^{[m]}$, the pair (u, m) solves the mean field game system eq. (8.5). The map S encodes one full forward–backward sweep; its fixed points are exactly the mean field equilibria.*

The proposition is the hinge of the whole reformulation: it certifies that nothing is lost in trading the PDE system for the operator, that the two problems have *the same* solution set. The proof is short — it is little more than reading the construction of S in both directions — but each direction says something we want, and naming what each says is the point.

Proof. ($\Rightarrow$, a solution is a fixed point.) Suppose (u, m) solves eq. (8.5). We must show $\mathsf{S}[m] = m$, i.e. that feeding m through the three steps returns m unchanged. Because u already solves the HJB equation eq. (8.5a) driven by this very m, and that equation has a unique solution of the admissible class (theorem 6.14), step 1 recovers precisely $u = u^{[m]}$; step 2 then forms the feedback eq. (8.6) from this u. Now compare the forward problem solved in step 3 with the Fokker–Planck equation eq. (8.5b) that the hypothesis says m already satisfies: they are the *identical* linear Cauchy problem — same initial datum m_0, same second-order operator, same drift $b(x, \alpha^*)$. A linear parabolic Cauchy problem with these coefficients has, under **(A-Nondeg)** and the Lipschitz drift, a *unique* solution (theorem 5.14, chapter 5); since both m and $\tilde{m} = \mathsf{S}[m]$ solve it, they coincide, and $\mathsf{S}[m] = m$. This direction is what makes the reformulation *lossless* in the safe sense: every equilibrium we already believe in is captured as a fixed point, so searching for fixed points can never miss a solution. The uniqueness of the linear forward equation is the load-bearing fact — it is what forbids S from sending an equilibrium flow anywhere other than back to itself.

($\Leftarrow$, a fixed point is a solution.) Suppose conversely $\mathsf{S}[m] = m$. Set $u = u^{[m]}$, the value produced by step 1; by definition of step 1 it solves the backward HJB equation eq. (8.5a) with this m in the coupling slot, which is the first equation of the system. By definition of step 3, the output $\tilde{m} = \mathsf{S}[m]$ solves the forward Fokker–Planck equation eq. (8.5b) with the feedback eq. (8.6) read off u; but the fixed-point hypothesis says $\tilde{m} = m$, so it is m itself that solves eq. (8.5b) with this feedback — the second equation of the system. The pair (u, m) therefore satisfies both equations of eq. (8.5) simultaneously, with the value's coupling slot and the density's transport drift built from the *same* m and the *same* u. This direction is what makes the reformulation *faithful*: it produces no spurious solutions, every fixed point being a genuine equilibrium. The two directions together say the solution sets coincide exactly, which is precisely the licence to study the operator in place of the PDEs. □

The reduction the proposition certifies is, conceptually, the central move of the analytic theory, and it is worth saying plainly why it is the *right* move rather than a mere repackaging. The difficulty of eq. (8.5), isolated in section 8.3, was global-in-time: the value at any instant depends on the whole future of the density, and the density at any instant on the whole past of the feedback, so no local-in-time iteration can even begin. What S does is absorb that entire global coupling *into the definition of one evaluation.* A single application of S already performs a complete backward sweep and a complete forward sweep — it digests the global-in-time entanglement once and for all — so that the only thing left to ask of the flow is a *pointwise* (in $\mathfrak{C}$) condition: that it be a fixed point.

We have not made the coupling disappear; we have repackaged a problem with *no* solvable temporal direction as a problem about a single operator that has no temporal direction at all, only an input and an output. That repackaging is the right move because fixed points of self-maps of nice metric spaces are objects we have powerful, general tools for, whereas forward–backward PDE systems are not. As section 8.3 warned, the trade is from a problem we cannot *integrate* to one we cannot in general *contract*; existence comes not from iterating S to convergence but from the topological fact that **(A-Nondeg)** makes S a *continuous* and in fact *compactifying* self-map — its parabolic smoothing maps bounded sets of flows into precompact ones — so that a Schauder/Kakutani theorem can force a fixed point without any contraction at all. The single hypothesis **(A-Nondeg)** is thus doing double duty: it makes *each* application of S well posed (classical value in step 1, smooth density in step 3) and it makes the *family* of outputs compact, which is what existence will consume.

This is the formulation the rest of Part II actually runs on, and it is worth recording, chapter by chapter, which property of S each downstream result consumes, because that mapping is the reason this section exists. Existence (chapter 10) is a *fixed-point theorem* for S: it uses the compactness just described to produce, with no uniqueness claim, at least one m with $\mathsf{S}[m] = m$. Uniqueness (chapter 11) is a *contraction or monotonicity* statement about S: under the extra Lasry–Lions monotonicity **(A-Mon)** deliberately held in reserve in section 8.2, the map either contracts in the metric eq. (8.10) on short horizons or obeys a global monotonicity estimate, and in either case its fixed point is unique. The master equation (chapter 12) studies how the fixed point *moves* when the data move: it differentiates the dependence of the value $u^{[m]}$ — and hence of S — on the initial measure of the flow, turning the fixed-point relation into a single PDE on the space of measures. And the convergence theory (chapter 13) closes the last promissory note of section 8.1.2 by showing that the finite Nash system eq. (8.3) converges, as $N \to \infty$, to a fixed point of this same S — so that the operator built here is literally the limit object the N-player equilibria approach. In every one of these the coupled PDE system has receded from view and the self-map S stands in its place; proposition 8.4 is the warrant that doing so loses nothing.

We do not prove existence here — that is chapter 10's charge — but proposition 8.4 has already bought the decisive thing: the analytic problem has been reduced from "solve a forward–backward PDE system with data at opposite ends" to "find a fixed point of a self-map of the complete metric space $\mathfrak{C}$," with the regularizing assumption **(A-Nondeg)** making each application of S a well-posed and compactifying operation. That is the object the rest of Part II consumes.

Two roads lead out of this reformulation, and the next sections take them in turn. The only population-dependence left in S sits in a single place — the slot where m enters the Hamiltonian's third argument in step 1's backward HJB, together with the terminal cost — and so the qualitative behavior of the whole operator is governed entirely by *how* the crowd enters there. That observation is the seed of the couplings taxonomy of section 8.5, which classifies the ways m can sit in H (separable versus non-separable, local versus nonlocal, congestion) precisely because each way reshapes step 1, and hence S, differently. The second road is the variational one: for the best-behaved couplings the fixed points of S turn out to be not merely equilibria but *minimizers* of a single convex energy on $\mathfrak{C}$, so that the search for a fixed point becomes a search for the bottom of an effective action — the subject of section 8.6. We hand the operator S and the fixed-point viewpoint to both.

Notation watch

The map S is the best-response operator Φ of appendix A; it acts on *flows* of measures, not on single measures. In stationary (ergodic) MFG and in the static graphon games of chapter 20 the same letter S denotes a self-map of a single measure (or strategy profile); the dynamic best response here is its time-resolved version. We keep the notation S throughout and let the domain disambiguate.

8.5 The structure of the couplings

The fixed-point reformulation of section 8.4 left us with a single self-map S and one sharp observation about it: of all the data feeding the best-response operator, the population m enters in exactly one place. It sits in the third slot of the Hamiltonian inside step 1's backward HJB, together with the terminal cost, and nowhere else — the volatility, the control set, the agent's own p-dependence are all blind to the crowd. The whole strategic content of the mean field game is therefore funneled through a single channel: *how m* occupies that slot. Two equilibria built from the same agent dynamics but a different placement of the crowd in H are, as analytic objects, different animals; two built from the same placement are siblings, whatever the rest of the data. It follows that before we ask any harder question about eq. (8.5) we should classify the channel itself — and that is the business of this section.

The classification is not taxonomy for its own sake. Each distinction it draws is the answer to an *operational* question that the rest of Part II will force us to ask, and the reader should read the three subsections below as the branches of a decision tree to be rerun on every new model, not as a glossary of names to memorize. The first question is structural: does the strategic system secretly minimize a single scalar energy, so that the search for an equilibrium becomes the search for the bottom of an effective action? That hinges on whether the crowd and the control *decouple* inside H — whether the Hamiltonian splits into a p-part and an m-part — and it is the axis of *separable versus non-separable* couplings. The second question is about uniqueness and well-posedness: through what spatial averaging does an agent at x feel the crowd — pointwise through the local density value $m(x)$, or smeared through an integral against the whole measure? That is the axis of *local versus nonlocal* couplings, and it controls both how much regularity the coupling supplies for free and which monotonicity arguments are even available. The third question is dynamical and the most physical: does the crowd merely *tax* an agent's position, or does it change the *price of moving* — making a dense region expensive to traverse? That is *congestion*, the canonical source of non-separability, and it is where the cleanest structural simplifications break down. Variational structure, uniqueness, numerics: the three axes are the three operational questions in the order a practitioner meets them.

These are also exactly the questions the downstream chapters answer, so the taxonomy is the scaffold on which their hypotheses hang. Separability is the precondition the variational section needs and is therefore the first axis we draw, handed straight to section 8.6, which will show that a separable coupling that is in addition *monotone* turns the forward–backward pair into the Euler–Lagrange system of one convex functional — separability supplying the splitting, monotonicity the convexity. The local/nonlocal distinction and the congestion distinction are what the existence and uniqueness theory of chapters 10 and 11 sort their hypotheses by: nonlocal smoothing couplings are the friendly regime where fixed points are easiest to produce and monotonicity is easiest to verify, local and congestive couplings the demanding regime where extra care — and stronger assumptions — is the price of admission. The same two distinctions return, lifted onto a network, when chapter 22 replaces the anonymous crowd by a graphon-kernel coupling $W(x, y)$ and the classification must be redone with the kernel in hand. We take the three axes in turn. The first and most consequential is separability, because it alone decides whether the entire apparatus of section 8.6 is available; we isolate the question of whether H splits in the next subsection.

8.5.1 Separable versus non-separable Hamiltonians

We begin the classification, as the section promised, with the first and most consequential axis, the one that alone decides whether the entire variational apparatus of section 8.6 is even available. The question is the narrowest one can ask about how the crowd sits inside the Hamiltonian, and it is purely structural: in the third slot of $H(x, p, m)$ the crowd m and the costate p are, in principle, entangled — the population could change not only the level of cost an agent pays for sitting at x but the very *rate* at which H responds to its momentum, $\partial_p H$, which is the closed-loop drift the forward equation transports along. Everything in this subsection turns on whether that entanglement is present or absent. So before any finer distinction — before we ask through what spatial averaging the crowd is felt, or whether it taxes position or penalizes motion — we isolate the cleanest possible separation: does H split into a piece that owns all the p-dependence and a piece that owns all the m-dependence, with no term mixing the two? When it does, the agent's optimization and its interaction with the crowd decouple inside H, and a long chain of simplifications follows; when it does not, that chain breaks at its first link. We make the split precise and then trace the chain.

Definition 8.5 (Separable coupling)**.** The Hamiltonian is *separable* if it splits as

$$H(x, p, m) \;=\; H_0(x, p) \;-\; F(x, m), \tag{8.11}$$

with H_0 carrying all the p-dependence and $F\colon \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ the *cost coupling*. Otherwise H is *non-separable*.

Read the two summands for what they each mean to the agent. The term $H_0(x, p)$ is the agent's *own* optimization, untouched by the crowd: it is exactly the measure-free Legendre transform $H_0(x, p) = \sup_\alpha\{-b(x, \alpha) \cdot p - L_0(x, \alpha)\}$ of a running cost $L_0(x, \alpha)$ that depends on state and control alone, so the entire p-dependence of H — the part that knows about steering, momentum, and best response — lives here and nowhere else. The term $-F(x, m)$ is the *cost coupling*: an additive levy the agent at x pays for the configuration of the crowd it finds itself in, a levy it has no control authority over. The agent cannot change $F(x, m)$ by steering, because F contains no α and no p; moving its momentum around changes H_0 but leaves the coupling term exactly where it was. Separability is precisely the statement that the population enters as a cost the agent *suffers* but cannot *negotiate* through its control. The minus sign is bookkeeping fixed by our Legendre convention (eq. (8.4) defines H as a supremum, so an additive running cost enters H with a flipped sign); we will see it convert F into an honest, correctly-signed source on the right-hand side of the HJB equation in a moment.

At the level of the data this splitting is nothing but an additive decomposition of the running cost itself,

$$L(x, \alpha, m) \;=\; L_0(x, \alpha) \;+\; F(x, m), \tag{8.12}$$

a control-dependent part L_0 that the agent pays for the effort of steering, and a population-interaction part F that it pays for the company it keeps. Feeding eq. (8.12) through the Legendre transform eq. (8.4) makes the splitting of H automatic: since $F(x, m)$ carries no α, it comes straight out of the supremum,

$$H(x, p, m) = \sup_{\alpha} \{ -b(x, \alpha) \cdot p - L_0(x, \alpha) - F(x, m) \} = \underbrace{\sup_{\alpha} \{ -b(x, \alpha) \cdot p - L_0(x, \alpha) \}}_{H_0(x,p)} - F(x, m),$$

which is exactly eq. (8.11). So "separable Hamiltonian" and "additively separable running cost" are one and the same hypothesis viewed on the two sides of the Legendre transform; one may impose it on L and read it off H, or the reverse.

With the cost coupling pulled out of the supremum, the HJB equation eq. (8.5a) of definition 8.1 acquires a particularly transparent form. Substituting eq. (8.11) and moving the now-p-free coupling term across the equality,

$$-\partial_t u - \tfrac{1}{2} \operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) + H_0(x, \nabla u) \;=\; F(x, m_t), \qquad u(T, x) = G(x, m_T), \tag{8.13}$$

where G is the terminal coupling (the separable terminal cost $g(x, m) = G(x, m)$). The structure on display is the cleanest possible: the left-hand side is a *closed*, measure-free Hamilton–Jacobi–Bellman operator in u alone — the operator of a lone agent with no crowd in sight — and the population has been quarantined entirely onto the right-hand side, where the cost coupling $F(x, m_t)$ acts as a prescribed, time-dependent *source* and $G(x, m_T)$ as the terminal datum. The crowd no longer reaches into the nonlinearity; it only forces the equation from outside, the way a known inhomogeneity forces a linear PDE.

This relocation of the crowd to the source is the hinge of the whole subsection, so we state its consequence as sharply as it deserves. The optimal feedback is read off the Hamiltonian by the envelope identity (remark 6.13, recorded in the notation watch at the head of this chapter) as the closed-loop drift $b(x, \alpha^*) = -\partial_p H(x, \nabla u, m_t)$. In the separable case the m-dependence of H sits entirely in the additive term $-F(x, m)$, which carries no p; differentiating in p annihilates it,

$$-\partial_p H(x, p, m) = -\partial_p\big(H_0(x, p) - F(x, m)\big) = -\partial_p H_0(x, p),$$

with the right-hand side a function of (x, p) *with no m left in it.* Hence the optimal drift, and (in the canonical normalization $b(x, \alpha) = \alpha$) the optimal feedback itself,

$$\alpha^*(t, x) \;=\; -\partial_p H_0\big(x, \nabla u(t, x)\big), \tag{8.14}$$

no longer depends on m at all. The population still shapes the value function u — it forces eq. (8.13) through the source F, so ∇u certainly feels the crowd — but it does so only *indirectly*, through the single channel ∇u, and the law mapping ∇u to a drift is fixed once and for all by the agent's own cost L_0. There is no *second*, direct dependence of the drift on m.

It is worth saying in plain words why this single fact is "the decisive simplification" rather than a cosmetic one, because the payoff is collected only two sections later and the reader should carry the reason there. Look at what the forward Fokker–Planck equation eq. (8.5b) now transports along. Its drift is precisely the closed-loop drift $-\partial_p H$, which by eq. (8.14) is a function of $(x, \nabla u)$ and of *no other appearance of* m. Write the momentum field $w = m\,\alpha^* = -m\,\partial_p H_0(x, \nabla u)$ that the density carries; because the velocity $-\partial_p H_0(x, \nabla u)$ does not see m, the field w is *linear* in m at fixed u. This is exactly the property the variational reformulation of section 8.6 will need and exploit: there the forward equation is not solved but imposed as the continuity *constraint* eq. (8.19), $\partial_t m - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 m) + \operatorname{div} w = 0$, and that constraint is an *affine* (hence convex) condition on the pair (m, w) *only because* w depends linearly on m. The separable feedback's m-independence is what makes "transport the population along the optimal drift" a linear constraint rather than a nonlinear one, and a linear constraint over a convex objective is what lets the entire strategic system collapse into the minimization of one convex energy. Separability is, in this precise sense, the structural precondition that the whole potential-game picture of section 8.6 silently assumes; without it the constraint set is curved and the convex-optimization machinery has nothing to grip.

A one-line instance on the carried thread makes the m-independence concrete. In the LQG arithmetic of Thread A (worked computation 8.12), in canonical normalization the agent's own cost is $L_0(x, \alpha) = \frac{1}{2}|\alpha|^2$, so $H_0(x, p) = \frac{1}{2}|p|^2$ and $-\partial_p H_0 = -p$; the population enters only through the mean-tracking cost $F(x, m) = \frac{q}{2}(x - s\,\bar{m})^2$ with $\bar{m} = \int \xi\, m(\mathrm{d}\xi)$, which carries no control. The optimal feedback is therefore $\alpha^* = -\nabla u$, manifestly free of m: the crowd's mean $\bar{m}$ moves the source term of the HJB equation and hence bends the value function, but the rule "steer opposite to the value gradient" is the same whatever the crowd does. This is exactly why the thread will close in section 8.7 as two backward Riccati equations (the value side, forced by the crowd) plus one forward quadrature for the mean (the density side, transported along an m-free drift), and not as a single tangled nonlinearity.

The contrast with the non-separable case is now sharp, and it is a genuine structural collapse rather than a quantitative complication. If $H(x, p, m)$ does not split, then $\partial_p H$ retains its m-argument, and the feedback

$$\alpha^*(t, x) \;=\; -\partial_p H\big(x, \nabla u(t, x), m_t\big) \tag{8.15}$$

depends on the density *directly* — not only through ∇u but through a second, irreducible appearance of m in the drift law itself. Now follow that m into the forward equation. The Fokker–Planck drift in eq. (8.5b) is this same $-\partial_p H(x, \nabla u, m_t)$, so the transport equation for m carries m in its own velocity field: the unknown appears both as the transported density and inside the law that transports it. The forward equation is no longer a *linear* transport driven by an externally supplied drift but a genuinely *nonlinear* (quasilinear) equation in m, of McKean–Vlasov type, and the momentum field $w = m\,\alpha^* = -m\,\partial_p H(x, \nabla u, m)$ is no longer linear in m. The clean separation of labor that made the separable HJB a forced linear equation and the separable Fokker–Planck a linear constraint is gone: the two equations are coupled through *two* channels in each direction rather than one, and the affine constraint set of the variational problem becomes curved. Non-separable Hamiltonians arise exactly when the crowd changes the *price of moving* rather than merely the cost of being somewhere — when a dense region is expensive to traverse, so the very gain on the control depends on the local density — and the canonical such model is congestion, the subject of the last subsection below. For all such couplings the variational structure of section 8.6 generally fails to exist, and both the uniqueness theory of chapter 11 and the numerical schemes that lean on the potential formulation must be rebuilt on weaker footing.

This first axis thus cleaves the subject in two, and the cleavage organizes everything that follows. On one side sits the separable class, where the feedback is m-independent, the forward equation is a linear constraint, and the door to a convex energy stands open: we hand this class first to the next subsection, which subdivides it by asking *how* the now-quarantined coupling $F(x, m)$ reads the crowd — pointwise through the local density value or smeared through an integral — and then, in its fully developed form, to the variational section that the linearity of the constraint was built to serve. On the other side sits the non-separable class, carrying the warning we have just issued: when the crowd reaches into the drift, the structure that makes the problem tractable dissolves. We pass that warning directly to the congestion subsection, where the leading non-separable model is studied as precisely the place where these simplifications break down.

8.5.2 Local versus nonlocal couplings

Within the separable class, the coupling $F(x, m)$ is *nonlocal* when it depends on m through an integral against a kernel, and *local* when it depends only on the pointwise value of the density at x.

Example 8.6 (Nonlocal coupling)**.** A typical nonlocal cost coupling is a convolution,

$$F(x,m) \;=\; (\phi * m)(x) \;=\; \int_{\mathbb{R}^d} \phi(x-y)\, m(\mathrm{d}y),$$

with an interaction potential ϕ. This F is bounded and Lipschitz in $\mathcal{W}_2$ whenever ϕ is bounded and Lipschitz, so it sits comfortably inside **(A-Lip)**. It is the form most amenable to the existence theory because it is continuous in the *weak* topology, which tightness controls. The mean coupling of Thread A, $F(x, m) = \frac{q}{2}(x - s\,\bar{m})^2$ with $\bar{m} = \int \xi\, m(\mathrm{d}\xi)$, is nonlocal of this kind: it sees m only through its mean.

Example 8.7 (Local coupling)**.** A local coupling has the form $F(x, m) = f(x, m(x))$, depending on the *density value* $m(x)$ at the point x. The archetype is a power law $f(x, \rho) = \rho^\gamma$, modelling an aversion to crowded states that grows with local density. Local couplings are analytically harder: $m \mapsto m(x)$ is not continuous for $\mathcal{W}_2$ (a measure need not even have a density), so existence requires the parabolic smoothing of **(A-Nondeg)** to first guarantee that m_t has a density, and the coupling is then read against that density. They are central to congestion-free crowd-motion models and to the variational theory, where $f(x, \cdot)$ being increasing is exactly the local form of the Lasry–Lions monotonicity **(A-Mon)**.

The distinction matters operationally: nonlocal couplings are weakly continuous and play well with compactness arguments; local couplings demand that one first establish density bounds. We will see both. The graphon coupling of Part IV — the aggregate $(\mathbb{W}\, m)(x) = \int_0^1 W(x, y)\, \langle \cdot, m^y \rangle \;\mathrm{d}y$

of chapter 22 — is a structured nonlocal coupling *in the type variable*, and it is precisely the spatial (type) nonlocality through W that the whole graphon programme is about.

8.5.3 Congestion

The last and most physical of the three axes returns us to the non-separable class that the first subsection set aside with a warning. Congestion is its canonical instance: the crowd raises not the cost of *being* somewhere but the price of *moving* through a dense region, and it is exactly here that the clean separable structure of the preceding subsections breaks down.

Definition 8.8 (Congestion coupling)**.** A *congestion* model is a non-separable Hamiltonian in which the population modulates the cost of control, the canonical form being

$$H(x,p,m) \;=\; \frac{1}{c(x,m)}\,H_0(x,p) \;-\; F(x,m), \qquad c(x,m)>0, \tag{8.16}$$

where $c(x,m)$ is large where m is concentrated, so that moving through a dense region is expensive.

Congestion encodes the realistic feature that, in a crowd, motion itself is costly — one moves slowly through a dense region. Because m multiplies the momentum-dependent part, the optimal feedback $\alpha^* = -\partial_p H = -c(x,m_t)^{-1}\partial_p H_0(x,\nabla u)$ depends on the density, and eq. (8.5b) acquires a drift that is nonlinear in m through c. Congestion problems lie outside the separable/variational framework of section 8.6 in general; their well-posedness theory (existence under structural conditions on c, and the special "congestion-monotone" cases where uniqueness survives) is genuinely harder and is treated in chapters 10 and 11. We flag this as a place where the clean dual structure of the chapter does not extend, rather than papering over the gap.

8.6 Variational and potential-game structure

The taxonomy of section 8.5 closed on a contrast: separability is the structural precondition for a convex energy, and congestion is the model where that precondition dissolves. We now collect the reward separability was held in reserve to buy. For a large and important subclass of couplings — separable, with monotone interaction — the mean field game system eq. (8.5) is not just a coupled pair of equations but the *optimality system of a single minimization.* Such MFGs are called *potential* (or *variational*) games: the strategic forward–backward problem collapses to a convex optimal-control problem on the space of density flows. This is the analytic backbone of the numerical methods of chapter 28 and the conceptual bridge to the Ginzburg–Landau picture that organizes Part IV.

8.6.1 The potential and the energy functional

We work in the separable case eq. (8.11) with dynamics affine in the control through a *constant, invertible* gain: $b(x,\alpha) = B\,\alpha$ with $B \in \mathbb{R}^{d\times d}$ invertible and $A = \mathbb{R}^d$. Relabelling the control

$\alpha \mapsto B\,\alpha$ (a linear change of variables, with L_0 transported accordingly) then normalizes the gain to the identity, so without loss of generality we take $b(x, \alpha) = \alpha$: the agent directly controls its velocity. This is the genuine scope of the variational reduction. A drift with a state-dependent gain $B(x)$, or one not affine in the control, does *not* reduce this way and lies outside the potential framework of this section; we treat it only where it is needed and flag it as such. Take the Lagrangian $L_0(x, \alpha)$ convex in α with Legendre dual $H_0(x, p) = \sup_\alpha\{-\alpha \cdot p - L_0(x, \alpha)\}$. The crucial hypothesis is that the couplings are themselves *gradients*.

Definition 8.9 (Potential coupling)**.** The couplings F, G are of *potential type* if there exist functionals $\mathcal{F}, \mathcal{G} \colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ whose flat (linear functional) derivatives, in the sense of chapter 4 and appendix A, recover them:

$$F(x, m) = \frac{\delta \mathcal{F}}{\delta m}(m)(x), \qquad G(x, m) = \frac{\delta \mathcal{G}}{\delta m}(m)(x). \tag{8.17}$$

The role of the flat derivative is exactly the one a condensed-matter reader expects of $\delta\mathcal{F}/\delta\rho(x)$ in density-functional theory: it is the "one-body potential" generated by the interaction functional $\mathcal{F}$ at the configuration m. With couplings of potential type, define on flows (m, w) — where $w(t, x) \in \mathbb{R}^d$ is the momentum field (mass times velocity) satisfying the continuity constraint — the *energy functional*

$$\mathcal{J}(m, w) \;=\; \int_0^T\!\!\int_{\mathbb{R}^d} m(t, x)\, L_0\Big(x, \tfrac{w(t,x)}{m(t,x)}\Big)\,\mathrm{d}x\;\mathrm{d}t \;+\; \int_0^T \mathcal{F}(m_t)\;\mathrm{d}t \;+\; \mathcal{G}(m_T), \tag{8.18}$$

to be minimized over pairs (m, w) obeying the (linear) continuity constraint

$$\partial_t m - \tfrac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top\,\mathrm{D}^2\, m) + \mathrm{div}\, w = 0, \qquad m(0, \cdot) = m_0. \tag{8.19}$$

The first term of eq. (8.18) is the total control cost rewritten in terms of the momentum $w = m\,\alpha$; the integrand $(m, w) \mapsto m\,L_0(x, w/m)$ is the *perspective function* of L_0, jointly convex and lower semicontinuous in (m, w) (extended by 0 when $m = w = 0$ and $+\infty$ when $m = 0 \neq w$; we prove these two properties in proposition B.2). The second and third terms are the accumulated and terminal interaction energies.

The reader who has met the phrase "minimize kinetic energy subject to a continuity equation" will recognize eq. (8.18)–eq. (8.19) as a *dynamic optimal-transport* problem of Benamou–Brenier type (appendix D.4), enriched by the interaction terms $\mathcal{F}, \mathcal{G}$. With the quadratic choice $L_0(x, \alpha) = \frac{1}{2}|\alpha|^2$ and no interaction, the kinetic term is exactly the Benamou–Brenier action whose minimum is $\mathcal{W}_2^2$. The one structural difference is the second-order term $-\frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top\,\mathrm{D}^2\, m)$ inside the constraint eq. (8.19): it is the diffusion, and it is precisely what turns the *deterministic* transport problem into its *viscous* (entropic) version, the dynamic-transport face of the Fokker–Planck equation. The MFG energy is thus a viscous Benamou–Brenier problem with interaction, and this is the precise sense in which $\mathcal{J}$ is an "effective action" for the population.

Lemma 8.10 (Convexity of the energy)**.** *Assume $L_0(x, \cdot)$ is convex and that $\mathcal{F}, \mathcal{G}$ are convex on $\mathcal{P}_2(\mathbb{R}^d)$. Then $\mathcal{J}$ is convex on the (convex) set of (m, w) satisfying eq. (8.19). Convexity of $\mathcal{F}$ is equivalent to the Lasry–Lions monotonicity* ***(A-Mon)*** *of the coupling $F = \frac{\delta\mathcal{F}}{\delta m}(m)(x)$:*

$$\int_{\mathbb{R}^d} (F(x, m_1) - F(x, m_2))\;\mathrm{d}(m_1 - m_2)(x) \;\geq\; 0 \quad \textit{for all } m_1, m_2, \tag{8.20}$$

and likewise for $\mathcal{G}, G$.

Proof sketch. Joint convexity (and lower semicontinuity) of $(m,w) \mapsto m\,L_0(x, w/m)$ is the perspective construction, proved in proposition B.2; integrating in (t,x) preserves convexity, and the constraint set eq. (8.19) is affine, hence convex. For the interaction terms, a functional $\mathcal{F}$ with flat derivative F is convex along the segment $m_\theta = (1-\theta)m_2 + \theta m_1$ iff $\theta \mapsto \frac{\mathrm{d}}{\mathrm{d}\theta}\mathcal{F}(m_\theta) = \int F(x, m_\theta)\,\mathrm{d}(m_1 - m_2)(x)$ is nondecreasing; differentiating once more and the fundamental theorem of calculus turn this into the monotonicity inequality eq. (8.20). The full identification of **(A-Mon)** with convexity is carried out in chapter 11; we use it here only to state the hypothesis in the form the variational structure needs. $\square$

8.6.2 The Euler–Lagrange system is the MFG system

Theorem 8.11 (Potential MFG / variational characterization)**.** *Assume **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, control-affine dynamics, potential couplings (definition 8.9), and $\mathcal{F}, \mathcal{G}$ convex (equivalently, **(A-Mon)**). If (m^*, w^*) minimizes $\mathcal{J}$ of eq. (8.18) subject to eq. (8.19), then there is a function u (the Lagrange multiplier of the constraint eq. (8.19)) such that, writing $m := m^*$,*

1. *the optimal momentum is $w^* = -\,m\,\partial_p H_0(x, \nabla u)$, i.e. the velocity is the optimal feedback $\alpha^* = -\partial_p H_0(x, \nabla u)$;*
2. *u solves the backward HJB equation eq. (8.13) with couplings $F = \frac{\delta \mathcal{F}}{\delta m}(m)(x)$, $G = \frac{\delta \mathcal{G}}{\delta m}(m)(x)$;*
3. *m solves the forward Fokker–Planck equation eq. (8.5b).*

That is, (u, m) solves the mean field game system eq. (8.5). Conversely, under the stated convexity, any classical solution of eq. (8.5) arises this way. The equilibrium density *m^* is unique whenever the interaction is* strictly *monotone — i.e. $\mathcal{F}$ (and $\mathcal{G}$) is strictly convex, so that eq. (8.20) is strict for $m_1 \neq m_2$ — and u is then determined on $\{m^* > 0\}$. We do not claim strict convexity of the energy $\mathcal{J}$ itself: its kinetic integrand $(m,w) \mapsto m\,L_0(x, w/m)$ is positively* 1*-homogeneous and hence never strictly convex, so uniqueness must come from the coupling, not from the kinetic term. The precise statement and proof of uniqueness under strict monotonicity are chapter 11's.*

Proof sketch. Introduce a multiplier $u(t,x)$ for the constraint eq. (8.19) and form the (variational) Lagrangian functional — we write it $\mathfrak{L}$, never $\mathcal{L}$, since the glyph $\mathcal{L} = \mathcal{L}$ is reserved in this book for the law of a random variable (appendix A) and is used in that sense in this very chapter —

$$\mathfrak{L}(m, w, u) = \mathcal{J}(m, w) - \int_0^T \int_{\mathbb{R}^d} u\Big(\partial_t m - \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 m) + \operatorname{div} w\Big)\,\mathrm{d}x\,\mathrm{d}t.$$

Integrating by parts moves the derivatives onto u; the boundary terms at $t = T$ produce the terminal condition $u(T, \cdot) = G(\cdot, m_T)$, those at $t = 0$ are killed by the fixed initial datum. Stationarity in w gives, pointwise, $\partial_w[mL_0(x, w/m)] = -\nabla u$: the energy contributes $\partial_\alpha L_0(x, w/m)$ (the inner $1/m$ cancels the outer m), while the constraint term $-\iint u \operatorname{div} w$ contributes $+\nabla u$ after integrating $\operatorname{div} w$ by parts, and the two must cancel. Hence $\partial_\alpha L_0(x, \alpha^*) = -\nabla u$ with $\alpha^* = w/m$; by Legendre duality (the maximizer of $H_0(x,p) = \sup_\alpha\{-\alpha \cdot p - L_0(x,\alpha)\}$ satisfies $\partial_\alpha L_0(x,\alpha^*) = -p$) this is exactly $\alpha^* = -\partial_p H_0(x,\nabla u)$, which is claim (1) and shows the velocity is the optimal feedback. Stationarity in m gives, after using the flat-derivative identity $\frac{\delta\mathcal{F}}{\delta m}(m)(x) = F(x,m)$, the HJB equation eq. (8.13), claim (2): the ∂_t, D^2 terms come from the constraint, the $L_0 + \alpha^* \cdot \nabla u = -H_0(x,\nabla u)$ combination is the Legendre identity (evaluating $H_0 = -\alpha^* \cdot \nabla u - L_0$ at the maximizer), and F is the source. Stationarity in u returns the constraint eq. (8.19), which with $w^* = m\,\alpha^*$ is exactly the Fokker–Planck equation eq. (8.5b), claim (3). Convexity (lemma 8.10) makes the stationary point a global minimizer; strict monotonicity of the coupling then pins down the equilibrium density uniquely (chapter 11), the kinetic term being only convex, never strictly so. The rigorous version (existence of a minimizer by the direct method, regularity of the multiplier, and the duality with the *dual* HJB problem) is carried out in chapter 10 and, for the variational numerics, chapter 28. The original references are Lasry–Lions and Cardaliaguet's notes. □

Physics note

theorem 8.11 says that a potential MFG is governed by a single functional $\mathcal{J}$ whose stationary points are the equilibria — exactly the status of an effective action or Ginzburg–Landau free energy, with m the order-parameter field and u the conjugate (multiplier) field. Convexity of $\mathcal{J}$, equivalently Lasry–Lions monotonicity eq. (8.20), is the analogue of thermodynamic stability: a convex free energy has a unique minimizer, so the equilibrium is unique. When the interaction is *attractive* enough to break convexity — the population is rewarded for concentrating — $\mathcal{J}$ acquires multiple stationary points and uniqueness fails, the mean field counterpart of a symmetry-breaking transition. chapter 11 and the LQG phase diagram of chapter 14 make this precise. The reader is cautioned that this variational picture is a *feature of potential couplings*, not of MFG in general: a non-separable Hamiltonian (congestion, definition 8.8) has no such single governing functional, and the equilibrium is then irreducibly a Nash fixed point rather than a minimizer.

8.7 Worked computation: the LQG mean field game, analytic fixed point

With the equilibrium's abstract anatomy now complete — the verification theorem, the Nash fixed point, the coupling taxonomy, and the variational backbone of the potential case — we close the chapter by making all of it explicit on the carried thread. We now advance Thread A to its second layer (appendix A). Layer 1, the single-agent LQG problem with the population mean held fixed, was solved in chapter 6 and produced a scalar Riccati equation for the curvature of the value. Here we *close the loop*: we impose the mean field consistency of proposition 8.4 and exhibit the resulting forward–backward system in fully explicit, closed form. The computation is the analytic skeleton on which the entire LQG capstone chapter 14 is built, and it instantiates every abstract feature of this chapter — backward HJB, forward Fokker–Planck, two-point boundary data, separable nonlocal coupling — on a problem one can solve with pen and paper.

Worked computation 8.12 (LQG MFG fixed point)**.** Recall Thread A (appendix A): the representative agent in $\mathbb{R}$ has scalar state

$$\mathrm{d}X_t = \big(a\,X_t + b\,\alpha_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad X_0 \sim m_0, \tag{8.21}$$

and minimizes, against the population-mean flow $\bar{m}_t := \int \xi\, m_t(\mathrm{d}\xi)$,

$$J(\alpha) = \mathbb{E}\Bigg[\int_0^T \Big(\tfrac{1}{2}\alpha_t^2 + \tfrac{q}{2}\big(X_t - s\,\bar{m}_t\big)^2\Big)\,\mathrm{d}t + \tfrac{q_T}{2}\big(X_T - s_T\,\bar{m}_T\big)^2\Bigg], \tag{8.22}$$

with $q, q_T \geq 0$ and parameters a, b, σ, s, s_T, T as fixed in the thread specification. This is a separable, nonlocal coupling (definition 8.5 and example 8.6): the population enters only through its mean $\bar{m}_t$.

Step 1: HJB → Riccati. The Hamiltonian (the dynamics eq. (8.21) have drift $ax + b\,\alpha$, with b the control gain, and running cost $L_0 = \frac{1}{2}\alpha^2$) gives optimal feedback $\alpha^* = -b\,\partial_x u$ and the HJB equation eq. (8.5a)

$$\partial_t u + a\,x\,\partial_x u + \tfrac{\sigma^2}{2}\,\partial_{xx}u - \tfrac{b^2}{2}\,(\partial_x u)^2 + \tfrac{q}{2}\big(x - s\,\bar{m}_t\big)^2 = 0, \qquad u(T,x) = \tfrac{q_T}{2}\big(x - s_T\,\bar{m}_T\big)^2. \tag{8.23}$$

Because the data are quadratic in x and the equation preserves quadratics, we take the ansatz $u(t,x) = \frac{1}{2}P(t)\,x^2 + r(t)\,x + \chi(t)$. Substituting and matching powers of x yields three ordinary differential equations. The x^2 coefficient gives the scalar *control Riccati equation*

$$\dot{P} = b^2P^2 - 2aP - q, \qquad P(T) = q_T, \tag{8.24}$$

which is decoupled from the population: it is the same curvature equation as in the single-agent problem of chapter 6, and under $q, q_T \geq 0$ it has a unique, nonnegative, bounded solution on $[0,T]$ (no finite-time blow-up backward from q_T, because it is the value of a well-posed convex problem). The x^1 coefficient gives the *offset equation*

$$\dot{r} = (b^2P - a)r + q\,s\,\bar{m}_t, \qquad r(T) = -q_T\,s_T\,\bar{m}_T, \tag{8.25}$$

which carries *all* the population dependence; the x^0 coefficient determines $\chi(t)$ by a quadrature and is irrelevant to the equilibrium mean.

Step 2: Fokker–Planck → Gaussian flow. The optimal feedback is linear, $\alpha^*(t,x) = -b\,\partial_x u = -b\big(P(t)x + r(t)\big)$, so the closed-loop state is the linear (Ornstein–Uhlenbeck-type) diffusion

$$\mathrm{d}X_t = \Big[\big(a - b^2P(t)\big)X_t - b^2r(t)\Big]\,\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{8.26}$$

A linear SDE driven by additive Gaussian noise keeps a Gaussian law whenever m_0 is Gaussian (chapter 5); thus $m_t = \mathcal{N}(\bar{m}_t, V_t)$ is fully described by its mean and variance, which solve the closed linear moment equations obtained from eq. (8.26),

$$\dot{\bar{m}}_t = \big(a - b^2P(t)\big)\,\bar{m}_t - b^2r(t), \qquad \bar{m}_0 = \textstyle\int \xi\, m_0(\mathrm{d}\xi), \tag{8.27}$$

$$\dot{V}_t = 2\big(a - b^2P(t)\big)\,V_t + \sigma^2, \qquad V_0 = \mathrm{Var}(m_0). \tag{8.28}$$

The variance equation eq. (8.28) is a *linear* (Lyapunov) ODE that depends only on P and not on the coupling, so the spread of the population is slaved to the Riccati solution and decouples entirely from the fixed point; the Fokker–Planck equation eq. (8.5b) for m_t reduces, on the Gaussian manifold, to exactly eq. (8.27)–eq. (8.28).

Step 3: the closed fixed-point system. The mean field consistency of proposition 8.4 is the demand that the $\bar{m}_t$ appearing as a parameter in the HJB-derived offset eq. (8.25) be the *same* $\bar{m}_t$ produced by the Fokker–Planck mean eq. (8.27). Imposing it couples eq. (8.25) and eq. (8.27) into the linear forward–backward two-point boundary-value problem

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\bar{m}_t\\ r_t\end{pmatrix} = \begin{pmatrix}k(t) & -b^2\\ q\,s & -k(t)\end{pmatrix}\begin{pmatrix}\bar{m}_t\\ r_t\end{pmatrix}, \qquad k(t) := a - b^2 P(t), \tag{8.29}$$

with the boundary data split across the two ends,

$$\bar{m}_0 = \textstyle\int \xi\, m_0(\mathrm{d}\xi) \ \ \text{(initial)}, \qquad r_T = -q_T\, s_T\, \bar{m}_T \ \ \text{(terminal)}. \tag{8.30}$$

This is the analytic signature of the mean field game in its barest form: a *forward* unknown $\bar{m}$ pinned at $t = 0$, a *backward* unknown r pinned at $t = T$, coupled at every instant. It is exactly the scalar shadow of the forward–backward structure of section 8.3.

Step 4: solving the fixed point in closed form. The two-point problem eq. (8.29)–eq. (8.30) is solved by a *decoupling field*: seek a scalar function $\pi(t)$ with $r_t = \pi(t)\,\bar{m}_t$ for all t. Differentiating and matching against eq. (8.29) gives a second, lower scalar Riccati equation

$$\dot{\pi} = b^2\pi^2 - 2k(t)\,\pi + q\,s = b^2\pi^2 - 2\big(a - b^2 P(t)\big)\pi + q\,s, \qquad \pi(T) = -q_T\, s_T, \tag{8.31}$$

whose terminal datum is read off from eq. (8.30). Given π, the equilibrium mean is the explicit forward quadrature

$$\bar{m}_t = \bar{m}_0\,\exp\Big(\int_0^t \big(k(\tau) - b^2\pi(\tau)\big)\,\mathrm{d}\tau\Big), \tag{8.32}$$

and then $r_t = \pi(t)\bar{m}_t$, V_t from eq. (8.28), and the full Gaussian flow $m_t = \mathcal{N}(\bar{m}_t, V_t)$ are all explicit. The triple $(P, \pi, \bar{m})$ — two backward Riccati equations and one forward quadrature — *is* the LQG mean field equilibrium in closed form. This is the promised "HJB → Riccati, FP → Gaussian flow, closed fixed-point system."

Existence, uniqueness, and the seam to the phase diagram. The decoupling eq. (8.31) succeeds, and the equilibrium is unique, precisely as long as π does not blow up in finite time on $[0, T]$. Since eq. (8.29) is linear, the two-point problem has a unique solution unless the homogeneous problem has a nontrivial one — a conjugate point — which is exactly a finite-time blow-up of π. For a *repulsive/aligning* coupling in the stabilizing regime ($q\,s \geq 0$ with s not too large) the quadratic term keeps π bounded and the fixed point is unique; for sufficiently strong *attractive* coupling (s large of the destabilizing sign, large horizon T) π can escape to $-\infty$ before $t = 0$, signalling loss of uniqueness — the multiplicity (symmetry-breaking) regime. We do not chart the boundary here: the full phase diagram in (s, q, T), its identification with the failure of **(A-Mon)**/convexity of theorem 8.11, and the matrix-valued generalization are the business of chapter 11 (uniqueness) and chapter 14 (the LQG capstone). The numerical reproduction of eq. (8.31)–eq. (8.32) by a forward–backward solver is carried out, as a validation of both the analytics and the scheme, in chapter 28.

Remark 8.13 (What the worked computation isolates)**.** Three structural facts of this chapter are visible in worked computation 8.12 with no analytic clutter. First, the *decoupling* of the value's curvature (P, the Riccati equation) from the population: P solves the *same* equation as in the single-agent problem, because curvature is a property of the cost geometry, not of the crowd. Second, the population enters only through the *offset* r and the *mean* $\bar{m}$, which form the genuinely game-theoretic forward–backward pair eq. (8.29); this is the LQG incarnation of "HJB needs m, FP needs ∇u, nothing else passes." Third, the variance flow eq. (8.28) is entirely slaved, confirming that in the Gaussian world the only nontrivial fixed point is in the first moment — the order parameter $\bar{m}$. These features survive, suitably generalized, into the graphon LQG of chapter 22, where the scalar mean $\bar{m}_t$ becomes a type-profile $\bar{m}_t(x)$ and the coupling $q\,s$ becomes the graphon operator $q\,s\,\mathbb{W}$.

Rigor ledger

Proved (or proved modulo cited inputs).

- The MFG PDE system eq. (8.5): backward HJB driven by m, forward Fokker–Planck driven by the feedback $-\partial_p H$; its two readings, value-diffusion (section 8.2) and mass-conservation eq. (8.9).
- Representative-agent optimality proposition 8.2 and the equivalence of the PDE system with the Nash fixed point of the best-response map S (proposition 8.4), under **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**.
- The coupling taxonomy: separable/non-separable, local/nonlocal, congestion (section 8.5).
- The variational characterization of potential MFG (theorem 8.11): under potential couplings and convexity (equivalently **(A-Mon)**), the system is the Euler–Lagrange system of the convex energy eq. (8.18).
- Thread A, layer 2: the LQG equilibrium in closed form (worked computation 8.12), the triple $(P, \pi, \bar{m})$ with explicit quadrature eq. (8.32).

Assumed.

- The standing assumptions **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)** for the verification and well-posedness of each half-sweep; **(A-Mon)** only for the convexity in theorem 8.11.
- The single-step well-posedness of the HJB equation (chapter 6) and of the Fokker–Planck equation (chapter 5); propagation of chaos as the justification of section 8.1.2 (chapter 7).

Deferred.

- Existence of a fixed point of S, i.e. of a solution to eq. (8.5): chapter 10 (Schauder/Kakutani, with compactness from chapters 2 and 3).

- Uniqueness and the precise statement of **(A-Mon)**; non-uniqueness/multiplicity: chapter 11; LQG phase diagram in (s, q, T): chapter 14.
- Rigorous direct-method proof of theorem 8.11 and the dual HJB problem; variational numerics: chapter 28.
- The probabilistic (FBSDE) rendering of the same equilibrium: chapter 9; the master-equation linearization: chapter 12; the $N \to \infty$ convergence of eq. (8.3) to the system: chapter 13.
- Degenerate diffusion (dropping **(A-Nondeg)**) and the viscosity-solution formulation: chapter 6 and Appendix E.

Open (at the research frontier).

- Well-posedness of *non-separable* (congestion) MFG beyond structural special cases; sharp conditions on $c(x, m)$ in eq. (8.16).
- Uniqueness outside the monotone regime: the non-convex landscape of theorem 8.11 and the classification of its multiple equilibria.
- Regularity for *local* couplings with strong (super-linear) growth at the boundary of the integrability allowed by **(A-Nondeg)**.

Bibliographic notes

The mean field game system in the form eq. (8.5) originates with Lasry and Lions [93], who introduced the coupled HJB–Fokker–Planck pair, the monotonicity condition for uniqueness, and the variational interpretation of the potential case; and, independently and in the engineering literature, with Huang, Caines, and Malhamé [82] under the name *Nash certainty equivalence*, which is precisely the consistency closure of section 8.1.2. The PDE viewpoint adopted in this chapter — verification against a fixed flow, best-response fixed point, the forward–backward structure, separable/local/nonlocal couplings, and the variational formulation — follows Cardaliaguet's lecture notes [36], the standard analytic reference for Part II; our theorem 8.11 is his treatment of variational MFG, itself going back to [93]. The probabilistic counterpart, to which chapter 9 is devoted, is developed at length in Carmona and Delarue [40], whose Chapters 3–4 give the FBSDE formulation and the two-pictures correspondence we preview here. Congestion models and their well-posedness subtleties, flagged in definition 8.8, trace to the later Lasry–Lions program and are surveyed in [40]; the variational/primal–dual structure underlies the numerical methods of Achdou and Capuzzo-Dolcetta [1], taken up in chapter 28. The forward–backward boundary structure and the convergence of the N-player Nash system eq. (8.3) to the limit eq. (8.5) are the subject of the master-equation monograph of Cardaliaguet, Delarue, Lasry, and Lions [37], the reference for chapters 12 and 13. The LQG worked computation worked computation 8.12 is the mean field specialization of the classical linear–quadratic theory; its mean field form is in [82, 93] and is carried to its phase diagram in chapter 14.

Chapter 9

The Mean Field Game System: Probabilistic Formulation

In chapter 8 we wrote the mean field game equilibrium as a coupled pair of deterministic partial differential equations: a backward Hamilton–Jacobi–Bellman equation for the representative agent's value $u(t,x)$, driven by the population density $m(t,x)$, and a forward Fokker–Planck equation for m, driven by the optimal feedback extracted from u. That is the *analytic* picture of an equilibrium, and it is complete: the pair (u,m) pins the equilibrium down with nothing left over. It is fair to ask why one would want a second description at all.

The reason is that the analytic picture, complete as it is, conceals exactly the objects the rest of the book most needs to handle. It speaks of *fields* on state space, $u(t,x)$ and $m(t,x)$, and says nothing directly about the *trajectories* — the actual paths along which agents move. Yet four of this development's eventual destinations are statements about paths, not fields. The first is the particle limit: a mean field equilibrium is the $N \to \infty$ limit of N-player Nash play, and that convergence is a statement about coupling each player's trajectory to a single limiting one (chapter 13), a statement a pair of densities cannot even phrase. The second is degenerate — or vanishing — diffusion, where the HJB equation may possess no classical solution and the Fokker–Planck no density, while the controlled process and its law remain perfectly well defined. The third is simulation: forward–backward Monte Carlo schemes (chapter 28) integrate paths, not grids, and reach dimensions where finite differences cannot follow. The fourth, furthest off, is the graphon extension of chapter 23, where the state becomes a continuum-indexed family of *processes* coupled through an operator. All four are invisible to two PDEs on $[0,T] \times \mathbb{R}^d$ and native to a description that follows the paths themselves. This chapter builds that description: the very same equilibrium encoded not by two PDEs but by a single coupled *forward–backward stochastic differential equation* (FBSDE) of McKean–Vlasov type, living on path space. The reward, redeemed in section 9.3, is a precise dictionary translating each picture into the other; previewing the motivation now is what makes that dictionary land as a promise kept rather than a surprise sprung.

The two legs of this single equation can be named before they are built. The forward leg is the optimally controlled state process X_t, the trajectory the representative agent actually follows. The backward leg is its *adjoint*, or costate: the stochastic incarnation of the Pontryagin momentum previewed in chapter 6, the variable conjugate to the state that prices, instant by instant, how sensitively the agent's remaining cost responds to being displaced now. In the deterministic control of chapter 6 this costate was a single curve integrated backward from the terminal time. In the presence of noise it cannot stay that simple. A process pinned by data at T and required to remain *adapted* — to depend only on the past of the Brownian motion, never on its future — cannot in general be a deterministic function of the current state. Restoring adaptedness forces a *second* unknown into the backward equation: a process Z_t recording the instantaneous correlation between the costate's fluctuations and the driving noise. The stochastic costate is therefore the *pair* (Y, Z), not Y alone, and the forced appearance of Z — a consequence of the martingale representation theorem (theorem C.17) — is the first genuinely stochastic subtlety of the chapter. We flag it now so that when Z surfaces unbidden in the adjoint equation of section 9.1 it reads as a debt foreseen rather than a rabbit pulled from a hat.

The tools are imported wholesale from earlier chapters. The Itô calculus and stochastic differential equations come from chapter 5; the two backward-in-time results that the adjoint formulation rests on — the martingale representation theorem for the Brownian filtration (theorem C.17) and the Pardoux–Peng existence and uniqueness theorem for backward stochastic differential equations under a Lipschitz driver (theorem C.18) — are proved in full in the stochastic-calculus reference appendix C; the Pontryagin maximum principle and the identification of the costate with the gradient of the value function come from chapter 6; the McKean–Vlasov self-consistency condition — the law of the process feeding back into its own dynamics — comes from chapter 7. The reader should keep chapter 8 open throughout, since the entire payoff of this chapter is the precise dictionary, assembled in section 9.3, between the FBSDE objects here and the PDE objects there. It is worth being equally precise about what this chapter does *not* do. It *sets up* the fixed-point structure of the equilibrium but does not prove that the fixed point exists: the existence and uniqueness questions that structure raises are stated here, with their three governing regimes named, but settled only in chapters 10 and 11. And it *opens* a road it does not yet travel — the measure-dependent decoupling field introduced in section 9.4 is the gateway to the master equation of chapter 12. Knowing in advance what is owed, and to which chapter, lets the reader read the deferrals here as signposts rather than gaps.

We work throughout on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{0\le t\le T}, \mathbb{P})$ supporting a k-dimensional Brownian motion $W = (W_t)_{0\le t\le T}$, with $\mathbb{F}$ the (completed, right-continuous) filtration it generates, augmented by the law of the initial condition. As fixed by appendix A and used throughout Parts I–II, Brownian motion carries the letter W (rendered W); no graphon is present in this chapter, so there is no clash. Relative to chapter 8, which works with time-homogeneous coefficients and a constant diffusion matrix, this chapter admits three mild generalizations, each a deliberate choice made for a reason rather than for generality's sake. We allow *time-dependent* coefficients, because the dictionary we are after must cover genuinely time-inhomogeneous models — the LQG thread among them, whose Riccati flow is time-varying — and nothing in the probabilistic argument resists the extra argument. We allow a state- and measure-dependent volatility $\sigma(t, x, m) \in \mathbb{R}^{d\times k}$, so that the noise an agent feels may depend on where it stands and on the crowd around it. But we keep that volatility *uncontrolled*: the control α does not enter σ. This last restriction is the consequential one, and it earns its keep — an uncontrolled diffusion is exactly what keeps the adjoint system first order, a single backward equation for the pair (Y, Z) rather than the additional matrix-valued second-order adjoint that a control-dependent volatility would force. The only change to the Hamiltonian's role is the extra time argument, $H(t,x,p,m)$ in place of $H(x,p,m)$. We retain the global time convention: the value function and its stochastic avatar, the costate, are anchored by *terminal* data at $t=T$ and run *backward*; the state process and the population law are anchored by *initial* data at $t=0$ and run *forward*. The coupling of data at opposite ends of $[0,T]$ is the structural signature of the problem and survives translation into every one of its equivalent forms.

9.1 The representative agent and the stochastic maximum principle

The preamble made two promises this section now redeems: that the costate of the probabilistic picture is the stochastic Pontryagin momentum of chapter 6, and that the equilibrium will be encoded by following individual paths. Both are easiest to secure if we first, deliberately, decline to solve the equilibrium at all. A mean field game asks two things at once — that each agent optimize against the population, and that the population be the aggregate of those very agents — and the two demands are entangled, because the object an agent optimizes against is precisely what its optimization is meant to produce. The standard way to break such a circle is to cut it in two. We *freeze* the population — treat it as a fixed, exogenous environment the agent cannot influence — and solve the resulting pure control problem to completion; only afterward, in section 9.2, do we *close the loop* and demand that the frozen environment coincide with the law the optimization generates. Freezing first is not a stopgap to be apologized for. The frozen problem is a clean, classical stochastic control problem, and its complete solution — an optimal feedback together with an adjoint pair — is exactly the raw material the closure will weld into the McKean–Vlasov FBSDE. Solve the optimization in isolation, and the fixed point of the next section becomes a matter of substitution.

This section carries out the freeze-and-solve step in three movements, mirrored in its three subsections. We first lay down the control problem an agent faces against a fixed population flow: the admissible controls, the cost it pays, the state it steers, and — decisively — the standing assumption that the volatility is uncontrolled, already flagged in the preamble as what keeps the adjoint first order. We then introduce the generalized Pontryagin Hamiltonian, whose pointwise minimizer is the candidate optimal feedback $\hat{\alpha}(t,x,m,y,z)$, and the backward adjoint equation that produces the costate pair (Y,Z) — the equation in which the second unknown Z, foreshadowed above, is forced into being. Finally we prove that the resulting stochastic maximum principle is both necessary and, under convexity, sufficient, so that the search for an optimum collapses into the solution of a coupled system of stochastic equations. The feedback map $\hat{\alpha}$ and the adjoint pair (Y,Z) assembled here are exactly what section 9.2 takes in hand, replacing the frozen flow by the self-consistent law $\mathrm{Law}(X_t)$ to close the loop. To begin we must say precisely what "the frozen population" is; the next subsection makes it concrete as a deterministic curve of probability measures that the agent reads as its environment.

9.1.1 The control problem against a frozen population

We begin exactly where the preceding overview left off: with the resolution to freeze the population before solving anything, and the promise to make the word "frozen" concrete. This subsection carries out that first step — the freeze — probabilistically, and its entire content is to fix, with no ambiguity left over, every object the maximum principle of the next subsection will act on: the environment the agent inhabits, the controls it may choose, the state those controls steer, and the cost it pays. Nothing here is optimized yet; we are laying the table. section 9.2 will later delete the word "frozen" and impose self-consistency, but that step is a substitution, and it can only be a clean substitution if the frozen problem is first pinned down completely.

The frozen population is the least mysterious object in the chapter. It is a deterministic flow of probability measures

$$\mu = (\mu_t)_{0\le t\le T}, \qquad \mu_t \in \mathcal{P}_2(\mathbb{R}^d),$$

which the agent regards as fixed scenery: a movie of how the crowd is distributed across state space, one frame μ_t per instant. Two words in that line carry weight. *Deterministic*, because the representative agent faces no uncertainty about the crowd — the only randomness it contends with is its own Brownian noise; the population curve is a known datum, read off in advance. And valued in $\mathcal{P}_2(\mathbb{R}^d)$, the space of Borel probability measures on $\mathbb{R}^d$ with finite second moment $\int |x|^2\, \mu_t(\mathrm{d}x) < \infty$, because the costs below will measure the agent's state against population statistics — in the LQG thread, literally against the population mean $\int x\, \mu_t(\mathrm{d}x)$ — and those statistics exist precisely when the measures have a finite second moment. Demanding $\mathcal{P}_2(\mathbb{R}^d)$ from the outset is therefore not fastidiousness; it is the minimal regularity that makes the cost functional even well defined, and it is the same space on which the Wasserstein metric $\mathcal{W}_2$ of chapter 4 lives, the metric in which the fixed-point maps of chapter 10 will later be shown continuous.

Against this scenery the agent controls an $\mathbb{R}^d$-valued state process through an $\mathbb{F}$-progressively measurable control $\alpha = (\alpha_t)_{0\le t\le T}$ taking values in a closed convex action set $A \subseteq \mathbb{R}^\ell$. Progressive measurability is the precise rendering of the physical demand that the agent act on information it actually possesses: α_t may depend on the trajectory observed up to time t but not on the future of the noise. Given μ and α, the state solves the Itô SDE

$$\mathrm{d}X_t \;=\; b(t, X_t, \mu_t, \alpha_t)\; \mathrm{d}t \;+\; \sigma(t, X_t, \mu_t)\; \mathrm{d}W_t, \qquad X_0 \sim m_0 \in \mathcal{P}_2(\mathbb{R}^d), \tag{9.1}$$

with drift $b : [0, T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \times A \to \mathbb{R}^d$ and volatility $\sigma : [0, T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}^{d\times k}$. Read the equation as a balance of two influences on the agent's motion: a steerable drift b, through which the chosen control bends the trajectory and through which the frozen crowd μ_t exerts its pull, and an unsteerable diffusion $\sigma\; \mathrm{d}W_t$, the irreducible noise. The volatility is *uncontrolled*: α does not enter σ. This is the standing assumption of the present chapter, flagged already in the preamble, and is what keeps the adjoint system first order — a single backward equation for the pair (Y, Z) rather than the additional matrix-valued second-order adjoint that a control-dependent volatility would force. The controlled-diffusion case is set aside and revisited only where it matters (remark 9.7).

That eq. (9.1) actually defines a process at all is a debt owed to chapter 5. Under **(A-Lip)** and **(A-Growth)** — Lipschitz and linear-growth coefficients, uniformly in (t, μ) — the SDE well-posedness theorem proved there delivers a unique strong solution adapted to $\mathbb{F}$, for every admissible control, and with it the moment bound

$$\mathbb{E}\Big[\sup_{0\le t\le T}|X_t|^2\Big] < \infty.$$

This bound is not a technical afterthought; it is the load-bearing estimate of everything downstream. It is what guarantees the cost eq. (9.2) is finite rather than $+\infty$, so that comparing two controls is meaningful at all. It is what places the optimal state in the space $\mathcal{S}^2$ of square-integrable, sup-norm-controlled processes (chapter 5), the space in which the backward adjoint of definition 9.3 will be required to live, and the space in which the Itô product-rule manipulations of the maximum-principle proofs have integrable, genuinely-martingale stochastic integrals — without it those integrals need not have zero expectation and the entire duality argument collapses. And it is the seed of the tightness estimates by which chapter 10 extracts limits of best responses. A uniform second moment, in short, is the currency in which the whole probabilistic apparatus is denominated.

With the environment and the state fixed, we can say precisely which controls the agent is allowed to use and what each one costs.

Definition 9.1 (Admissible controls)**.** The set $\mathbb{A}$ of *admissible controls* (the blackboard $\mathbb{A} = \mathbb{A}$, to be kept distinct from the italic action set $A = A$ in which controls take values, and from the calligraphic generator $\mathcal{A} = \mathcal{A}$) consists of all $\mathbb{F}$-progressively measurable processes $\alpha : [0,T]\times\Omega \to A$ with $\mathbb{E}\int_0^T |\alpha_t|^2 \,\mathrm{d}t < \infty$. For $\alpha \in \mathbb{A}$ and the frozen flow μ, the agent's cost is

$$J^\mu(\alpha) \;=\; \mathbb{E}\left[\int_0^T L(t, X_t, \mu_t, \alpha_t)\,\mathrm{d}t \;+\; g(X_T, \mu_T)\right], \tag{9.2}$$

with running cost L and terminal cost g, where X solves eq. (9.1). A control $\hat\alpha \in \mathbb{A}$ is *optimal against* μ if $J^\mu(\hat\alpha) \le J^\mu(\alpha)$ for all $\alpha \in \mathbb{A}$.

The parenthetical in the definition deserves to be read slowly, because three different typefaces of the letter "A" are in play at once and confusing them is the single easiest way to misread the rest of the chapter. The blackboard $\mathbb{A} = \mathbb{A}$ is the *class of admissible control processes* — an infinite-dimensional set of strategies, the search space of the optimization. The italic $A = A$ is the *action set*, a fixed subset of $\mathbb{R}^\ell$ in which a control *value* α_t lives at each instant; it is finite-dimensional and pointwise. And the calligraphic $\mathcal{A} = \mathcal{A}$, which surfaces once we write the generator of X, is a *differential operator*, not a set at all. A process belongs to $\mathbb{A}$, a value to A, and a function to the domain of $\mathcal{A}$; the glyphs are deliberately distinct so the three never collide on the page. The square integrability $\mathbb{E}\int_0^T |\alpha_t|^2 \,\mathrm{d}t < \infty$ is the entrance fee for membership in $\mathbb{A}$: it is exactly what is needed for the controlled drift $b(t, X_t, \mu_t, \alpha_t)$ to be integrable in (t, ω) under the linear growth of **(A-Growth)**, and hence for the state SDE and the cost integral to make sense together.

It remains to say under which hypotheses on the coefficients this problem is not merely posed but solvable cleanly, and to say it concretely rather than by tag, since these are the hypotheses the maximum principle leans on at every step. We impose throughout this section **(A-Lip)**, **(A-Growth)**, and the coercivity/convexity package **(A-Conv)**, and it is worth spelling out what each one buys and what would break in its absence. **(A-Lip)** — the coefficients b, σ, L, g Lipschitz in the state, uniformly in (t, μ) — is what chapter 5 converts into existence and uniqueness of the state process: without a Lipschitz drift the SDE can split into several solutions or explode in finite time, and there would be no single X to attach a cost to. The same Lipschitz bound is what later makes the optimal feedback $\hat{\alpha}$ Lipschitz and the best-response map a contraction in the small-horizon regime, so this hypothesis is felt twice over. **(A-Growth)** — at most linear growth in the state, together with the coercivity of the cost — does two complementary jobs: linear growth forbids the trajectory from escaping to infinity and secures the moment bound $\mathbb{E}[\sup_t |X_t|^2] < \infty$ just discussed, while coercivity of L in the control (the running cost grows without bound as $|\alpha| \to \infty$) forbids the optimizer from buying arbitrarily large control for free, which is what makes a minimizer exist rather than the infimum being approached only in a limit of ever-wilder controls. Strip coercivity away and the cost can be driven down without attaining its bound; strip linear growth away and even the cost of a fixed control can be infinite. **(A-Conv)** is the structural heart of the section and is best stated concretely: we ask that b and σ be affine in (x, α), that $L(t, \cdot, \mu, \cdot)$ be convex in (x, α) with the strong convexity in α that supplies the coercivity above, and that $g(\cdot, \mu)$ be convex and C^1 in x.

The reason this particular package is the right one repays a moment's reflection, because it is exactly what upgrades the maximum principle from a *necessary* condition into a *sufficient* one. The stochastic Pontryagin principle, derived next, will produce a candidate $\hat{\alpha}$ satisfying a first-order stationarity condition; in general such a stationary point is only a *necessary* feature of an optimum, no more than calculus guarantees a vanishing derivative at a minimum. What promotes stationarity to genuine optimality is convexity, and here it must hold jointly in the state and the control, not in the control alone. The affineness of the dynamics is precisely what makes this joint convexity propagate correctly: when b and σ are affine in (x, α), the cost J^μ inherits convexity from the convexity of L and g, because a convex cost composed with an affine state map is again convex — a curved dynamics could bend a convex cost into a non-convex functional and destroy the implication. The strong (uniformly positive-curvature) convexity in α then does the sharper work: it makes the pointwise Hamiltonian minimization of the next subsection have a *unique* minimizer that depends Lipschitz-continuously on its arguments, which is what lets the feedback map $\hat{\alpha}(t, x, m, y, z)$ be substituted back into the dynamics without losing well-posedness. This is the precise probabilistic counterpart of the convexity that makes the HJB Hamiltonian $H(t, x, p, m)$ of chapter 8 convex in p, hence well defined as a Legendre transform: the two statements are the same hypothesis seen in two coordinate systems, and the reconciliation is made explicit in remark 9.11. Drop joint convexity and the maximum principle survives only as a necessary condition, leaving a possible gap between stationary points and minima that a separate verification argument would have to close.

A glance at the LQG thread shows the package is not restrictive in practice but is satisfied on the nose by the canonical example. There the scalar state obeys $\mathrm{d}X_t = (aX_t + b\,\alpha_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ — drift affine in (x, α), volatility constant and a fortiori uncontrolled — and the cost integrand is $\frac{1}{2}\alpha_t^2 + \frac{q}{2}(X_t - s\,\bar{\mu}_t)^2$ with terminal $\frac{q_T}{2}(X_T - s_T\bar{\mu}_T)^2$, which is strongly convex in α (curvature 1, uniformly positive) and convex in x for $q, q_T \geq 0$. Here $\bar{\mu}_t = \int x\,\mu_t(\mathrm{d}x)$ is the frozen population mean, the only statistic of μ the agent sees. Every clause of **(A-Conv)** holds verbatim, the Hamiltonian minimization of the next subsection will be an unconstrained quadratic with an explicit linear minimizer, and the whole machinery collapses to the Riccati system this thread tracks chapter by chapter. The general theory we are assembling is, in this instance, guaranteed to reproduce that closed form.

Everything the maximum principle needs is now on the table: the environment μ, the admissible class $\mathbb{A}$, the controlled state eq. (9.1), and the cost eq. (9.2) whose minimizer we seek. The optimization itself is the single requirement

$$J^\mu(\hat{\alpha}) \;\le\; J^\mu(\alpha) \qquad \text{for all } \alpha \in \mathbb{A},$$

and as it stands it is a statement about an infinite-dimensional search over strategies — intractable head-on. The next subsection performs the decisive change of viewpoint that the maximum principle prescribes: it reorganizes this global comparison of costs into a *pointwise* condition, instant by instant, by introducing a single function — the Pontryagin Hamiltonian — of the state, the costate, and the control, against which the optimal action can be selected one moment at a time. The costate pair (Y, Z) that prices the state in that function is the object we turn to now.

9.1.2 The Pontryagin Hamiltonian and the adjoint equation

The previous subsection left the optimization in its starkest possible form: find $\hat{\alpha} \in \mathbb{A}$ with $J^\mu(\hat{\alpha}) \le J^\mu(\alpha)$ for *every* competing admissible control. As a literal instruction this is hopeless — it asks us to weigh one strategy against an entire infinite-dimensional space of rivals, each of them a whole stochastic process. The maximum principle's decisive move is to refuse this global comparison and replace it with a *local* one. Rather than compare complete strategies, it prices the consequences of a single instant's decision and asks only that, moment by moment, the agent take the locally cheapest action. The bookkeeping that makes this trade honest is carried by one scalar function of the state, a *costate*, and the control: the generalized Pontryagin Hamiltonian. This subsection builds that function, reads off from it the candidate optimal feedback, and then constructs the backward equation that supplies the costate. The proof that local optimality really is global optimality is deferred to section 9.1; here we are assembling, with care, the objects those theorems will act on.

The word "costate" deserves a pause before any formula, because it is the conceptual hinge of the chapter. Recall from chapter 6 that in deterministic optimal control the adjoint variable — the costate — is the *momentum conjugate to the state*: at each instant it is the marginal cost-to-go, the sensitivity of the remaining optimal cost to an infinitesimal displacement of the current position. It is exactly what lets a purely local calculation feel the entire future, because it has already compressed the future into a single price per unit of displacement. The Hamiltonian is then nothing but the instantaneous rate at which total cost accrues, written so that this price is explicit: the running cost paid right now, plus the future cost the agent's motion is about to incur, the latter obtained by pairing the agent's velocity against the costate. Minimizing this rate over the control at each instant is the local surrogate for minimizing the global cost. In the deterministic theory the costate was a single curve $y(t)$ integrated backward from the terminal time. The genuinely new phenomenon in the stochastic problem, foreshadowed in the chapter preamble, is that the costate can no longer be a single curve: it becomes a *pair* (Y, Z) solving a backward equation, and the second component Z is the price of keeping the costate honest about not being allowed to see the future. We build the Hamiltonian first and earn the pair (Y, Z) immediately afterward.

Definition 9.2 (Generalized Hamiltonian)**.** The *generalized (Pontryagin) Hamiltonian* of the control problem eq. (9.1)–eq. (9.2) is the function

$$\mathcal{H} : [0,T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \times \mathbb{R}^d \times \mathbb{R}^{d\times k} \times A \longrightarrow \mathbb{R},$$

$$\mathcal{H}(t,x,m,y,z,\alpha) \;=\; \langle b(t,x,m,\alpha), y\rangle \;+\; \mathrm{tr}\big(\sigma(t,x,m)^\top z\big) \;+\; L(t,x,m,\alpha). \tag{9.3}$$

Under **(A-Conv)** the map $\alpha \mapsto \mathcal{H}(t,x,m,y,z,\alpha)$ is strictly convex and coercive, so it has a unique minimizer

$$\hat{\alpha}(t,x,m,y,z) \;=\; \operatorname*{arg\,min}_{\alpha\in A}\; \mathcal{H}(t,x,m,y,z,\alpha), \tag{9.4}$$

which is a Lipschitz function of (x,m,y,z) (a consequence of strong convexity and the implicit function theorem; see chapter 6). The associated *reduced Hamiltonian* is $\widehat{\mathcal{H}}(t,x,m,y,z) = \mathcal{H}\big(t,x,m,y,z,\hat{\alpha}(t,x,m,y,z)\big)$.

The formula eq. (9.3) is best read not as a definition to memorize but as three priced ledgers added together, each measuring one channel through which the present instant charges the future. Take them in turn. The first term, $\langle b(t,x,m,\alpha), y\rangle$, is the *cost of motion*. The drift b is the controllable velocity of the state — the rate at which x is about to move — and y is the costate, the marginal cost of displacement; their inner product is therefore the rate at which the agent's chosen motion accrues future cost. This is the single term through which the control reaches into the future: by tilting b the agent buys displacement, and y is the price tag. It is the exact stochastic echo of the $\langle b,p\rangle$ term in the deterministic Pontryagin Hamiltonian of chapter 6. The third term, $L(t,x,m,\alpha)$, is the *cost of the present*: the running cost actually disbursed over the current instant, the only toll the agent pays in cash rather than in future consequences. The middle term, $\mathrm{tr}(\sigma^\top z) = \langle \sigma, z\rangle_F$, is the subtle one and is the term with no deterministic ancestor. It pairs the volatility σ against a second costate z, the component that prices not the displacement of the state but the *fluctuation* of the costate against the driving noise. It is the Hamiltonian's record of how randomness couples to the agent's accounting of the future; concretely it is the avatar, in costate language, of the second-order term $\frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top D^2 u)$ that the Itô correction contributes to the HJB equation, a correspondence we make exact in section 9.3. Crucially, in the present chapter the volatility is uncontrolled — σ does not depend on α — so this middle term is a *constant* as far as the minimization over α is concerned. The agent's choice sees only the first and third terms; the noise channel contributes to the dynamics of the costate but not to the selection of the action. This is precisely the simplification the standing uncontrolled-diffusion assumption was introduced to buy, and it is why the adjoint system stays first order (remark 9.7).

Because the minimization eq. (9.4) only ever sees the α-dependent part $\langle b(t,x,m,\alpha), y\rangle + L(t,x,m,\alpha)$, the regularity of the feedback $\hat{\alpha}$ is governed entirely by the curvature of this reduced object in α. Here is where the strong convexity packaged into **(A-Conv)** does its sharper work, and it repays spelling out the mechanism rather than leaving it as a parenthetical citation. Strong convexity means the Hessian $\partial^2_{\alpha\alpha}\mathcal{H}$ is bounded below by a fixed $\alpha > 0$ times the identity, uniformly in all arguments; in particular it is everywhere invertible. Suppose first that the minimizer is interior to A, so that it is pinned by the stationarity condition

$$\partial_\alpha \mathcal{H}\big(t,x,m,y,z,\hat{\alpha}(t,x,m,y,z)\big) = 0.$$

This is an implicit equation for $\hat{\alpha}$ in terms of the parameters (x, m, y, z), and its α-Jacobian is exactly the invertible Hessian $\partial^2_{\alpha\alpha}\mathcal{H} \succeq \alpha I$. The implicit function theorem therefore solves it for $\hat{\alpha}$ as a function of the parameters, and differentiating the stationarity relation gives, schematically, $\partial_{(x,m,y,z)}\hat{\alpha} = -\left(\partial^2_{\alpha\alpha}\mathcal{H}\right)^{-1}\partial_{(x,m,y,z)}\partial_\alpha\mathcal{H}$. The inverse Hessian is bounded by $1/\alpha$ and the mixed derivatives are bounded by the Lipschitz constants of the coefficients (**(A-Lip)**), so $\hat{\alpha}$ is Lipschitz in (x, m, y, z) with a constant that scales like (coefficient Lipschitz constant)$/\alpha$. When the minimizer sits on the boundary of the convex set A, the same conclusion holds with the stationarity equality replaced by the variational inequality and the gradient step by the projection onto A, which is itself 1-Lipschitz; strong convexity again furnishes the uniform constant. The upshot is the single fact the rest of the chapter leans on without further comment: the feedback $\hat{\alpha}(t, x, m, y, z)$ is a genuine Lipschitz function of its arguments, so substituting it back into the dynamics produces coefficients that still satisfy **(A-Lip)** and the forward–backward system stays well posed. Take away the uniform lower bound $\alpha > 0$ and the minimizer can jump discontinuously as the parameters cross a flat spot of the Hamiltonian, and the substitution that defines the FBSDE would no longer return a Lipschitz problem.

It is worth seeing the abstraction collapse onto something explicit, so let the LQG thread instantiate every clause. There $b = aX + b\,\alpha$, the volatility σ is a constant scalar σ, and $L = \frac{1}{2}\alpha^2 + \frac{q}{2}(x - s\bar{\mu})^2$, so the Hamiltonian eq. (9.3) reads

$$\mathcal{H}(t, x, \mu, y, z, \alpha) = (ax + b\,\alpha)\,y + \sigma z + \tfrac{1}{2}\alpha^2 + \tfrac{q}{2}(x - s\bar{\mu})^2.$$

The middle term σz carries no α, as advertised; the minimization sees only $b\,\alpha\,y + \frac{1}{2}\alpha^2$, a scalar parabola of curvature 1 (the strong-convexity constant $\alpha = 1$ here), and the stationarity condition $by + \alpha = 0$ gives the closed-form feedback

$$\hat{\alpha}(t, x, \mu, y, z) = -b\,y.$$

The feedback is *linear* in the costate y — manifestly Lipschitz with constant $|b| = |b|/\alpha$, in exact agreement with the bound just derived — and it depends on neither x nor z, a degeneracy special to the quadratic case. Reading it physically: the optimal control is to push against the momentum, with gain set by the coupling b between control and state. This is the seed of the Riccati closed form the thread carries, and it is reassuring to see the general machinery reproduce it without effort.

Notation watch

The symbol $H = H$ is reserved, in chapter 8 and appendix A, for the *HJB* Hamiltonian $H(x, p, m)$ (here carrying the extra time argument, $H(t, x, p, m)$), a function of the state, the costate variable p, and the measure, convex in p, with optimal feedback $\alpha^* = -\partial_p H$. The object in eq. (9.3) is a *different* function — the Pontryagin Hamiltonian, a function of state, costate *and control* — carried by the shared macro $\mathcal{H}$ (a calligraphic $\mathcal{H}$, distinct from the fraktur $\mathfrak{H}$ used for the *deterministic* Pontryagin Hamiltonian of chapter 6). The two are reconciled in the canonical case in remark 9.11; in brief, the reduced Hamiltonian $\widehat{\mathcal{H}}$ equals $-H$ once the z-term is suppressed and y is read as p. Do not confuse $\mathcal{H}$ (this chapter, contains α) with H (chapter 8, control already optimized out). A separate glyph clash to keep in mind: the calligraphic $\mathcal{H}$ of $\mathcal{H}$ is unrelated to the process space $\mathcal{H}^2 = \mathcal{H}^2$

(square-integrable adapted processes, chapter 5) that appears wherever the BSDE solution (Y, Z) is housed; the superscript 2 always distinguishes them.

The notation watch is not a clerical aside but a map of the chapter's central translation, and it is worth restating its content in words. Three Hamiltonians are circulating, and they differ precisely by how much optimization has already been performed. The deterministic Pontryagin Hamiltonian $\mathfrak{H}$ of chapter 6 still carries the control and lives without noise. The generalized Hamiltonian $\mathcal{H}$ of eq. (9.3) is its stochastic upgrade: it still carries the control α as a free argument — that is the whole point, since we are about to *minimize* over it — and it has acquired the extra noise channel $\mathrm{tr}(\sigma^\top z)$. The HJB Hamiltonian H of chapter 8 sits at the far end of the same pipeline, with the control already optimized out: it is a function of state, costate p, and measure only. The reduced Hamiltonian $\widehat{\mathcal{H}}$ of definition 9.2 is the bridge between them — it is $\mathcal{H}$ after the minimization, hence a function of (t, x, m, y, z) with no control left, and the notation watch records the punchline that $\widehat{\mathcal{H}}$ coincides with $-H$ once the noise term is dropped and the probabilistic costate y is identified with the analytic costate p. Holding these three apart now, while they are merely notation, is what lets the dictionary of section 9.3 read as the discovery of a single object in three guises rather than a collision of symbols. The final caution in the box — that the calligraphic $\mathcal{H} = \mathcal{H}$ has nothing to do with the process space $\mathcal{H}^2 = \mathcal{H}^2$ where the costate will be required to live — is exactly the kind of glyph clash that, unflagged, derails a first reading; the superscript carries the whole distinction.

With the Hamiltonian in hand we have the candidate feedback $\hat{\alpha}(t, x, m, y, z)$, but it is useless until we can supply its arguments. The state x and measure m are given along the trajectory; the costate pair (y, z) is not. We now need an equation whose solution *is* the costate — a process (Y_t, Z_t) that, fed into the feedback at each instant, yields the optimal action. The costate must track the marginal cost-to-go along the path, so its terminal value is forced: at the final time there is no future left, and the only sensitivity of remaining cost to the state is the gradient of the terminal cost $\partial_x g$. Its rate of change, run backward from there, must be the rate at which the Hamiltonian varies with the state — the stochastic counterpart of Hamilton's canonical equation $\dot{y} = -\partial_x \mathfrak{H}$ from chapter 6, the equation of motion for momentum. This is the content of the next definition; the only stochastic novelty, the appearance of Z, we explain immediately after.

Definition 9.3 (Adjoint BSDE)**.** Let $\alpha \in \mathbb{A}$ with state X solving eq. (9.1). The *adjoint process* (costate) associated with α is the pair of $\mathbb{F}$-progressively measurable processes (Y, Z), Y valued in $\mathbb{R}^d$ and Z in $\mathbb{R}^{d\times k}$, solving the backward stochastic differential equation

$$\mathrm{d}Y_t \;=\; -\,\partial_x \mathcal{H}(t, X_t, \mu_t, Y_t, Z_t, \alpha_t)\;\mathrm{d}t \;+\; Z_t\;\mathrm{d}W_t, \qquad Y_T \;=\; \partial_x g(X_T, \mu_T), \tag{9.5}$$

where $\partial_x \mathcal{H}$ denotes the gradient in the state argument x alone, holding (m, y, z, α) fixed. Under **(A-Lip)** the driver is Lipschitz in (y, z) and the terminal datum is square-integrable, so by the Pardoux–Peng existence/uniqueness theorem for backward stochastic differential equations (theorem C.18, proved in full in the stochastic-calculus reference) there is a unique solution (Y, Z) in $\mathcal{S}^2 \times \mathcal{H}^2$.

Two features of eq. (9.5) deserve to be drawn out, because together they are what make the adjoint a *backward stochastic* differential equation rather than an ordinary one read in reverse, and the distinction is the first place the stochastic theory genuinely departs from the deterministic skeleton of chapter 6.

The first feature is the *direction of solution*. The equation is anchored at the terminal time: its data is the gradient $\partial_x g(X_T, \mu_T)$ of the terminal cost, and it is solved from $t = T$ back toward $t = 0$. This is not a stylistic choice but a structural necessity, and it is the same necessity that orients the HJB equation it shadows. The costate measures the sensitivity of *remaining* cost to the current state; "remaining" means the cost still to be paid between now and T, so the natural place to know it exactly is at T, where nothing remains and the sensitivity is simply the gradient of the terminal penalty. Information about cost-to-go propagates from the future into the present, which on the page is a flow backward in time. The value function of chapter 8 runs backward for precisely this reason, and the costate, being its gradient (as section 9.3 will prove), inherits the orientation. Forward data would over-determine the problem: one cannot freely prescribe both where the costate starts and the terminal cost it must match.

The second feature is the genuinely stochastic one, and it is the subtlety the chapter preamble promised would surface here. The component Z is *not* prescribed anywhere; it is part of the unknown, solved for alongside Y. To see why it must be there, try to do without it. If we simply integrated the driver backward from the terminal datum, as one would a time-reversed ODE, setting $Y_t = \partial_x g(X_T, \mu_T) + \int_t^T \partial_x \mathcal{H} \, \mathrm{d}s$, then Y_t would depend on the entire trajectory of X — and hence of the Brownian motion — all the way out to the terminal time T. Such a Y_t *anticipates the future*: it is not measurable with respect to $\mathcal{F}_t$, the information available at time t, and so it is not an admissible adapted process. But the costate, as the price the agent attaches to its position *now*, cannot depend on noise that has not yet occurred; adaptedness is non-negotiable. The resolution is to add a corrective martingale term $Z_t \, \mathrm{d}W_t$ whose job is exactly to absorb the part of the terminal datum that looks into the future, redistributing it instant by instant so that what remains, Y_t, depends only on the past. That such a Z exists, and is unique, is the content of the martingale representation theorem (theorem C.17): every square-integrable terminal random variable can be written as its conditional-expectation process plus a single stochastic integral against W, and the integrand of that integral is Z. So Z is not an extra modelling choice we make; it is forced into existence by the collision between terminal pinning and adaptedness, and it is the price of insisting that the costate not cheat by peeking ahead. This is the source of every subtlety that separates a BSDE from a time-reversed ODE, and it is why the stochastic costate is irreducibly a *pair* (Y, Z). The well-posedness of the pair — existence and uniqueness in $\mathcal{S}^2 \times \mathcal{H}^2$ under the Lipschitz driver — is delivered by the Pardoux–Peng theorem (theorem C.18) invoked in the definition; it is the backward analogue of the forward SDE well-posedness imported in section 9.1, and it is what guarantees the adjoint is a bona fide object to compute with rather than a formal expression.

Having built the Hamiltonian, the feedback, and the adjoint, we can now state in one deliberately non-rigorous breath why these are the right objects — why the costate is a momentum and why it runs backward. The argument below is the heuristic engine behind the theorems of section 9.1; reading it first turns those proofs from formal manipulations into a story whose ending is already understood.

Heuristic (non-rigorous)

Why is the costate a momentum, and why does it run backward? Freeze the population and imagine perturbing the optimal control on a vanishingly short interval $[t, t+\epsilon]$ by a "needle" (spike) variation: on that tiny window the agent deviates to some arbitrary action a, and everywhere else it goes back to playing optimally. The first-order change in total cost splits cleanly into two pieces. The first, (i), is the *immediate* change in running cost accumulated over the spike, of order $\big(L(t, X_t, \mu_t, a) - L(t, X_t, \mu_t, \hat{\alpha}_t)\big)\epsilon$ — the price paid on the spot for acting differently. The second, (ii), is subtler and is where the costate is born. By acting differently over $[t, t+\epsilon]$ the agent nudges the state, leaving it at $X_{t+\epsilon}$ displaced by roughly $\big(b(t, X_t, \mu_t, a) - b(t, X_t, \mu_t, \hat{\alpha}_t)\big)\epsilon$ from where it would have been. From $t+\epsilon$ onward the agent resumes optimal play, but it does so from a *shifted* starting point, and that shift propagates through all remaining cost. The total downstream penalty of a unit displacement at time t is, by definition, the sensitivity of the optimal cost-to-go to the current state — the gradient of the value function. Call it Y_t. This is the sense in which the costate is a momentum: it is the exchange rate converting a displacement of the state *now* into units of cost paid *later*, exactly the conjugate-to-position role momentum plays in mechanics. Piece (ii) is then $\langle Y_t, b(t, X_t, \mu_t, a) - b(t, X_t, \mu_t, \hat{\alpha}_t)\rangle\epsilon$, and adding it to piece (i) shows the total first-order cost change is governed by the difference of the generalized Hamiltonian $\mathcal{H}$ evaluated at a versus at $\hat{\alpha}_t$. Optimality forbids this difference from being negative for any a, which is exactly the statement that $\hat{\alpha}_t$ *minimizes* $\mathcal{H}$ pointwise — the minimum principle. Now trace how the sensitivity Y_t itself evolves. Because it measures cost still to come, its natural anchor is the terminal time, where the only cost left is the terminal penalty and the sensitivity is simply $\partial_x g$. Moving the anchor inward, the sensitivity changes at the rate the Hamiltonian changes with the state, $-\partial_x \mathcal{H}$, giving a transport equation that runs *backward* from T — the adjoint equation. The future is what the sensitivity sees, so the future is the direction from which its data must come. Finally, why the second component Z_t? In the stochastic world the sensitivity Y_t is itself a random process, buffeted by the same Brownian motion that drives the state, and adaptedness forbids it from anticipating that noise. The piece of Y's fluctuation that is instantaneously correlated with $\mathrm{d}W_t$ cannot be carried by the drift; it must appear as a separate martingale term, and its integrand is Z_t. So Z is the costate's noise sensitivity, forced into the ledger by the same demand for non-anticipation that orients the equation backward. The next two theorems convert each of these heuristic strokes — the pointwise minimization, the backward adjoint, the forced Z — into honest stochastic analysis.

Everything the maximum principle requires is now assembled. The generalized Hamiltonian eq. (9.3) converts the choice of action into a pointwise minimization; its minimizer eq. (9.4) is the candidate optimal feedback $\hat{\alpha}(t, x, m, y, z)$, Lipschitz by strong convexity; and the adjoint BSDE eq. (9.5) produces the costate pair (Y, Z) that the feedback consumes. The heuristic just given asserts that these fit together in exactly one way: along an optimal trajectory the control minimizes $\mathcal{H}$ instant by instant, with the costate priced by the adjoint run backward from $\partial_x g$. That assertion is, so far, only motivated. What the next subsection owes is its proof, in two directions and with the measure argument held frozen at μ throughout. The *necessary* direction will show that any optimal $\hat{\alpha}$ must satisfy the pointwise minimum condition

$$\mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \hat{\alpha}_t) = \min_{\alpha \in A} \mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \alpha),$$

turning the needle-variation heuristic into a rigorous duality computation through the adjoint. The *sufficient* direction will show that, under the convexity of **(A-Conv)**, this same condition is enough to *guarantee* optimality — so that solving the coupled state–adjoint system is not merely a property of the optimum but a complete characterization of it. With necessity and sufficiency both in hand, the search for an optimal control collapses entirely into the solution of a system of stochastic equations, which is exactly the raw material section 9.2 will close into the McKean–Vlasov FBSDE.

9.1.3 Necessity and sufficiency

The previous subsection ended with a promise and a heuristic. The promise was that the three objects we had assembled — the generalized Hamiltonian eq. (9.3), the feedback $\hat{\alpha}(t, x, m, y, z)$ it selects eq. (9.4), and the adjoint pair (Y, Z) the backward equation eq. (9.5) produces — fit together in exactly one way: along an optimal trajectory the control minimizes $\mathcal{H}$ instant by instant against a costate priced by the adjoint. The heuristic, run through a needle variation, made that fit *plausible*. This subsection makes it *true*, and it does so in both directions, with the population argument held frozen at μ from start to finish — we are still inside the frozen problem of section 9.1, optimizing a single agent against fixed scenery, and the word "frozen" will not be deleted until section 9.2.

The two directions answer two different questions and lean on two different hypotheses, and it is worth separating them before either proof begins. *Necessity* asks: if a control is optimal, must it satisfy the minimum principle? This is a first-order question — it differentiates the cost at the optimum and reads off that the derivative cannot point downhill — so it needs the coefficients to be differentiable, but it needs no convexity. *Sufficiency* asks the converse: if a control satisfies the minimum principle, is it optimal? A vanishing first derivative is, in general, only a stationarity certificate, not a proof of minimality; what upgrades it is convexity, exactly as a stationary point of a convex function on the line is automatically its global minimum. So sufficiency is where **(A-Conv)** finally earns the central billing it was given in section 9.1.

Both proofs, for all that they answer opposite questions, are powered by a single mechanical device, and recognizing it once makes both arguments legible. That device is the Itô product rule applied to the pairing $\langle \hat{Y}_t, \cdot \rangle$ of the adjoint against a state-side perturbation — the variational process V in the necessity proof, the genuine displacement $\delta X = X - \hat{X}$ in the sufficiency proof. In each case the adjoint was *built* eq. (9.5) precisely so that, when paired against the state side through Itô, the troublesome $\partial_x \mathcal{H}$ drift it carries cancels the state-sensitivity terms the pairing generates, leaving behind only the control-direction derivative $\partial_\alpha \mathcal{H}$ — the one quantity the agent actually governs. This cancellation is the whole point of the adjoint, and seeing it execute, term by term, is the content of what follows. We carry out the necessity proof first.

Theorem 9.4 (Stochastic maximum principle, necessary condition)**.** *Assume **(A-Lip), (A-Growth), (A-Conv)**, and that the coefficients b, σ, L, g are continuously differentiable in (x, α) with* *derivatives of at most linear growth. Let* $\hat{\alpha} \in \mathbb{A}$ *be optimal against the frozen flow* μ*, with optimal state* $\hat{X}$ *solving eq.* (9.1) *and adjoint* $(\hat{Y}, \hat{Z})$ *solving eq.* (9.5) *along* $(\hat{X}, \hat{\alpha})$*. Then,* $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$*-almost everywhere,*

$$\mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \hat{\alpha}_t) \;=\; \min_{\alpha \in A} \mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \alpha), \tag{9.6}$$

equivalently $\hat{\alpha}_t = \hat{\alpha}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t)$ *with the minimizer of eq.* (9.4)*.*

Proof. This is the stochastic Pontryagin principle, the deterministic skeleton of which was given in theorem 6.11 and proposition 6.12; we carry out the stochastic argument in full, with the measure argument frozen at μ throughout. The proof has four movements — a variation, a differentiation, a duality, and a selection — and we take them in that order.

Before the first, a word on *which* variation, because the choice is dictated by the geometry of the action set and it is what makes the present argument shorter than the textbook one. To probe whether $\hat{\alpha}$ is optimal we must perturb it and watch the cost respond. There are two classical ways to perturb a control. A *needle* (or spike) variation replaces $\hat{\alpha}$ by an arbitrary value $a \in A$ on a vanishingly short time window $[t, t+\epsilon]$ and leaves it untouched elsewhere; this is the device one is *forced* to use when A is non-convex, because then the straight-line blend of two admissible controls may leave the action set, and a large deviation on a tiny set is the only admissible perturbation available. A *convex* (Gâteaux) variation instead blends globally, $\alpha^\theta = \hat{\alpha} + \theta(\alpha - \hat{\alpha})$, nudging the control a little at *every* instant toward a rival α. The blend is admissible only if the action set is convex — otherwise $\hat{\alpha}_t + \theta(\alpha_t - \hat{\alpha}_t)$ can fall outside A — and here it is, by hypothesis. Convexity of A is therefore exactly the licence to use the gentler, globally smooth variation, which differentiates cleanly in θ and never requires the delicate $\sqrt{\epsilon}$ bookkeeping of the spike. (The needle variation yields the *same* minimum principle and does not need convexity of A; we simply do not need it, and say so to flag that the convexity of A is a convenience here, not a load-bearing hypothesis of the conclusion.)

Variation and variational equation. Fix any rival $\alpha \in \mathbb{A}$ and, for $\theta \in (0,1]$, set $\alpha^\theta = \hat{\alpha} + \theta(\alpha - \hat{\alpha})$, which lies in $\mathbb{A}$ by convexity of A; write the perturbation direction $\delta\alpha_t = \alpha_t - \hat{\alpha}_t$ and let X^θ be the state driven by α^θ. As $\theta \downarrow 0$ the perturbed state X^θ collapses back onto the optimal state $\hat{X}$, and the question is how fast and in what direction. The answer is the first-order sensitivity of the state to the control perturbation, the *variational process*

$$V_t \;=\; \lim_{\theta \downarrow 0} \frac{X_t^\theta - \hat{X}_t}{\theta},$$

the stochastic analogue of the Jacobi field that propagates an initial nudge through a flow. Differentiating the state SDE eq. (9.1) in θ at $\theta = 0$ — formally, by the chain rule, with the drift and volatility expanded to first order about $(\hat{X}_t, \hat{\alpha}_t)$ — gives the linear SDE this limit must obey,

$$\mathrm{d}V_t = \left[\partial_x b\, V_t + \partial_\alpha b\, \delta\alpha_t\right] \mathrm{d}t + (\mathrm{D}_x\sigma)[V_t]\; \mathrm{d}W_t, \qquad V_0 = 0, \tag{9.7}$$

with initial value $V_0 = 0$ because every X^θ starts from the same $X_0 \sim m_0$, so the perturbation has not yet had time to act. The drift of V has two sources, visible term by term: $\partial_x b\, V_t$ is how the already-accumulated displacement V_t feeds forward through the state-dependence of the drift, and $\partial_\alpha b\, \delta\alpha_t$ is the fresh push injected at time t by the control perturbation itself. The diffusion term $(\mathrm{D}_x\sigma)[V_t]$ is the directional derivative of the volatility in the direction of the displacement, the matrix $\sum_\ell \partial_{x_\ell}\sigma\,(V_t)_\ell \in \mathbb{R}^{d\times k}$; the control perturbation does *not* appear here, because the volatility is uncontrolled — the first place in this proof where the standing assumption pays off, and the reason $\delta\alpha$ enters V only through the drift. All coefficients $\partial_x b, \partial_\alpha b, \mathrm{D}_x\sigma$ are evaluated along the optimal trajectory $(t, \hat{X}_t, \mu_t, \hat{\alpha}_t)$, so they are bounded processes by **(A-Lip)**. That the rescaled differences genuinely converge to V in $\mathcal{S}^2$, and not merely formally, is where the affineness inside **(A-Conv)** is used: when b, σ are affine in (x, α) the first-order expansion is *exact*, with no remainder to control, so the limit is immediate; for merely smooth coefficients one would estimate a second-order remainder, which the affine case spares us. Equation eq. (9.7) then has bounded coefficients and a zero-control diffusion of linear growth, hence a unique solution $V \in \mathcal{S}^2$ with $\mathbb{E}\sup_{t\le T}|V_t|^2 < \infty$ by the linear SDE estimate of chapter 5.

Differentiation of the cost. With the state sensitivity V in hand we can differentiate the cost $J^\mu(\alpha^\theta)$ at $\theta = 0$. Recall from eq. (9.2) that J^μ is the expectation of a running integral of L plus a terminal g; the perturbation θ enters $J^\mu(\alpha^\theta)$ through two channels, directly via the explicit control argument $\alpha_t^\theta = \hat{\alpha}_t + \theta\,\delta\alpha_t$ and indirectly via the perturbed state X_t^θ, whose θ-sensitivity is exactly V. The chain rule therefore produces one term per channel. Since the integrands L, g are C^1 with derivatives of at most linear growth, and $X^\theta \to \hat{X}$ in $\mathcal{S}^2$ with difference quotients dominated by the $\mathcal{S}^2$-bounded V, dominated convergence legitimizes differentiating under the expectation, and we obtain

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\bigg|_{\theta=0^+} J^\mu(\alpha^\theta) = \mathbb{E}\left[\int_0^T \Big(\langle \partial_x L, V_t\rangle + \langle \partial_\alpha L, \delta\alpha_t\rangle\Big)\,\mathrm{d}t + \langle \partial_x g(\hat{X}_T, \mu_T), V_T\rangle\right], \tag{9.8}$$

all cost derivatives evaluated along $(t, \hat{X}_t, \mu_t, \hat{\alpha}_t)$. Read the three terms: $\langle \partial_x L, V_t\rangle$ is the running cost's response to the *displaced state*, $\langle \partial_\alpha L, \delta\alpha_t\rangle$ its response to the *altered control*, and $\langle \partial_x g, V_T\rangle$ the terminal cost's response to the displaced final state. Two of these three — the two carrying the state sensitivity V — are exactly what the adjoint is engineered to absorb. Optimality of $\hat{\alpha}$ forces $J^\mu(\alpha^\theta) \ge J^\mu(\hat{\alpha})$ for every $\theta \in (0, 1]$; dividing by $\theta > 0$ and letting $\theta \downarrow 0$, the one-sided derivative on the right of eq. (9.8) is therefore ≥ 0. This nonnegativity is the entire content of optimality at first order; everything that follows is bookkeeping to extract from it a *pointwise* statement.

Duality through the adjoint. The cost derivative eq. (9.8) still carries the state sensitivity V in two of its three terms, and V is an awkward object — it depends on the perturbation $\delta\alpha$ through the whole history of eq. (9.7). The adjoint exists to launder it away. The instrument is the Itô product rule applied to the scalar process $\langle \hat{Y}_t, V_t\rangle$, the running pairing of the costate against the state sensitivity. Writing the product rule out in full, with $\mathrm{d}\hat{Y}$ from eq. (9.5) and $\mathrm{d}V$ from eq. (9.7),

$$\mathrm{d}\langle \hat{Y}_t, V_t\rangle = \langle \mathrm{d}\hat{Y}_t, V_t\rangle + \langle \hat{Y}_t, \mathrm{d}V_t\rangle + \mathrm{d}\langle \hat{Y}, V,\rangle\, t,$$

three pieces: the costate's increment paired with V, the costate paired with V's increment, and the quadratic covariation of the two. Substitute each. The first piece contributes $-\langle \partial_x \mathcal{H}_t, V_t\rangle\ \mathrm{d}t$ plus a $\mathrm{d}W$ term from $Z_t\ \mathrm{d}W_t$; the second contributes $\langle \hat{Y}_t, \partial_x b\, V_t + \partial_\alpha b\,\delta\alpha_t\rangle\ \mathrm{d}t$ plus a $\mathrm{d}W$ term from $(\mathrm{D}_x\sigma)[V_t]$. The covariation is the term with no deterministic analogue: both $\hat{Y}$ and V are driven by the *same* Brownian motion, $\hat{Y}$ through $Z_t\ \mathrm{d}W_t$ and V through $(\mathrm{D}_x\sigma)[V_t]\ \mathrm{d}W_t$, so their brackets multiply and sum over the k noise components, giving the Frobenius pairing

$$\mathrm{d}\langle \hat{Y}, V,\rangle\, t = \mathrm{tr}\big((\mathrm{D}_x\sigma)[V_t]\,\hat{Z}_t^\top\big)\ \mathrm{d}t = \langle \hat{Z}_t, (\mathrm{D}_x\sigma)[V_t]\rangle_F\ \mathrm{d}t.$$

Now integrate from 0 to T and take expectations. The two $\mathrm{d}W$ contributions are genuine martingale integrals — their integrands $\langle Z_t, V_t\rangle$ and $\langle \hat{Y}_t, (\mathrm{D}_x\sigma)[V_t]\rangle$ are integrable precisely because $V \in \mathcal{S}^2$, $\hat{Y} \in \mathcal{S}^2$ and $\hat{Z} \in \mathcal{H}^2$, which is the moment bound of section 9.1 doing its load-bearing work — so they have zero expectation; and $V_0 = 0$ kills the lower endpoint. What survives is

$$\mathbb{E}\Big[\langle \hat{Y}_T, V_T\rangle\Big] = \mathbb{E}\left[\int_0^T \Big(- \langle \partial_x \mathcal{H}_t, V_t\rangle + \langle \hat{Y}_t, \partial_x b\, V_t + \partial_\alpha b\, \delta\alpha_t\rangle + \langle \hat{Z}_t, (\mathrm{D}_x\sigma)[V_t]\rangle_F \Big)\,\mathrm{d}t\right],$$

where $\partial_x\mathcal{H}_t = \partial_x\mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \hat{\alpha}_t)$. The cancellation is now visible. Differentiate the definition eq. (9.3) of $\mathcal{H}$ in the state x, holding (y, z, α) fixed:

$$\partial_x\mathcal{H} = \underbrace{(\partial_x b)^\top \hat{Y}}_{\partial_x\langle b, \hat{Y}\rangle} + \underbrace{\langle \hat{Z}, (\mathrm{D}_x\sigma)[\,\cdot\,]\rangle_F}_{\partial_x\,\mathrm{tr}(\sigma^\top \hat{Z})} + \partial_x L,$$

the three pieces being the x-gradients of the three terms of the Hamiltonian in turn. Pair this against V_t: the first piece is $\langle \hat{Y}_t, \partial_x b\, V_t\rangle$ (moving the transpose across the inner product), and it cancels the matching term from the second product piece; the middle piece is $\langle \hat{Z}_t, (\mathrm{D}_x\sigma)[V_t]\rangle_F$, and it cancels the Frobenius covariation term exactly. The only survivor from $-\langle \partial_x\mathcal{H}_t, V_t\rangle$ is $-\langle \partial_x L, V_t\rangle$, and the only survivor from the second product piece is $\langle \hat{Y}_t, \partial_\alpha b\, \delta\alpha_t\rangle = \langle (\partial_\alpha b)^\top \hat{Y}_t, \delta\alpha_t\rangle$. Finally use the terminal condition $\hat{Y}_T = \partial_x g(\hat{X}_T, \mu_T)$ on the left, and the identity reads

$$\mathbb{E}[\langle \partial_x g, V_T\rangle] = \mathbb{E}\left[\int_0^T \Big(- \langle \partial_x L, V_t\rangle + \langle (\partial_\alpha b)^\top \hat{Y}_t, \delta\alpha_t\rangle \Big)\,\mathrm{d}t\right]. \tag{9.9}$$

This is the payoff of having built the adjoint exactly as we did: the troublesome terminal term $\langle \partial_x g, V_T\rangle$ on the left, and the troublesome running state-sensitivity $\langle \partial_x L, V_t\rangle$ on the right, are now expressed through the adjoint, with the control perturbation $\delta\alpha$ entering only through the single clean term $\langle (\partial_\alpha b)^\top \hat{Y}_t, \delta\alpha_t\rangle$.

Collapse. Now feed the duality eq. (9.9) back into the cost derivative eq. (9.8). The terminal term $\langle \partial_x g, V_T\rangle$ in eq. (9.8) is, by eq. (9.9), equal to the right-hand integral, so substituting gives

$$0 \le \frac{\mathrm{d}}{\mathrm{d}\theta}\Big|_{0+} J^\mu(\alpha^\theta) = \mathbb{E}\left[\int_0^T \Big(\underbrace{\langle \partial_x L, V_t\rangle}_{\text{from } eq.\ (9.8)} + \langle \partial_\alpha L, \delta\alpha_t\rangle \underbrace{- \langle \partial_x L, V_t\rangle}_{\text{from } eq.\ (9.9)} + \langle (\partial_\alpha b)^\top \hat{Y}_t, \delta\alpha_t\rangle \Big)\,\mathrm{d}t\right].$$

The whole purpose of the adjoint is now consummated: the two $\langle \partial_x L, V_t\rangle$ terms — the running cost's response to the state displacement, entering once from the cost derivative and once, with opposite sign, from the duality — cancel identically, and with them goes every appearance of the state sensitivity V. What is left contains only the control direction $\delta\alpha_t = \alpha_t - \hat{\alpha}_t$:

$$0 \le \frac{\mathrm{d}}{\mathrm{d}\theta}\Big|_{0+} J^\mu(\alpha^\theta) = \mathbb{E}\left[\int_0^T \langle \partial_\alpha L + (\partial_\alpha b)^\top \hat{Y}_t, \delta\alpha_t\rangle\,\mathrm{d}t\right] = \mathbb{E}\left[\int_0^T \langle \partial_\alpha \mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \hat{\alpha}_t), \alpha_t - \hat{\alpha}_t\rangle\,\mathrm{d}t\right].$$

The last equality is again just the definition eq. (9.3) read in the control argument: differentiating $\mathcal{H} = \langle b, y\rangle + \mathrm{tr}(\sigma^\top z) + L$ in α gives $\partial_\alpha\mathcal{H} = (\partial_\alpha b)^\top y + \partial_\alpha L$, with *no* contribution from the middle term $\mathrm{tr}(\sigma^\top z)$ because σ does not depend on the control. This is the second dividend of the uncontrolled-diffusion assumption: the very quantity the agent is steering, $\partial_\alpha\mathcal{H}$, sees the noise channel not at all. We have arrived at the clean variational inequality $\mathbb{E}\int_0^T \langle \partial_\alpha \mathcal{H}_t, \alpha_t - \hat{\alpha}_t \rangle \, \mathrm{d}t \geq 0$, valid for every admissible α — a single scalar inequality that still, however, averages over space and time. The last movement turns it pointwise.

Pointwise minimization. The inequality $\mathbb{E}\int_0^T \langle \partial_\alpha \mathcal{H}_t, \alpha_t - \hat{\alpha}_t \rangle \, \mathrm{d}t \geq 0$ holds for *every* admissible $\alpha \in \mathbb{A}$, and we now squeeze a pointwise statement out of this freedom in the choice of α. The argument is by contradiction. Suppose the desired pointwise inequality failed — suppose that on some set $G \subseteq [0,T] \times \Omega$ of positive $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$ measure there existed, at each $(t, \omega) \in G$, an action $a = a(t, \omega) \in A$ with

$$\langle \partial_\alpha \mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \hat{\alpha}_t), a - \hat{\alpha}_t \rangle < 0.$$

Because $\partial_\alpha \mathcal{H}$ is continuous in its arguments and $(\hat{X}, \hat{Y}, \hat{Z})$ are progressively measurable, a measurable-selection theorem furnishes such an $a(t, \omega)$ as a progressively measurable function on G — this is the technical role of measurability, to guarantee the pointwise improving direction can be assembled into a bona fide *process*. Define the rival control $\alpha_t = a(t, \omega)$ on G and $\alpha_t = \hat{\alpha}_t$ off G; it is admissible (it takes values in A and inherits square-integrability from $\hat{\alpha}$ and the boundedness of the selection). For this particular rival the integrand $\langle \partial_\alpha \mathcal{H}_t, \alpha_t - \hat{\alpha}_t \rangle$ is strictly negative on G and zero off it, so the expectation $\mathbb{E}\int_0^T \langle \partial_\alpha \mathcal{H}_t, \alpha_t - \hat{\alpha}_t \rangle \, \mathrm{d}t < 0$, contradicting the variational inequality we just proved. Hence no such G exists, and

$$\langle \partial_\alpha \mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \hat{\alpha}_t), a - \hat{\alpha}_t \rangle \geq 0 \qquad \text{for all } a \in A, \quad dt \otimes \mathrm{d}\mathbb{P}\text{-a.e.} \tag{9.10}$$

It remains only to recognize eq. (9.10) for what it is. It says that at $\hat{\alpha}_t$ no admissible direction $a - \hat{\alpha}_t$ is a descent direction for $\alpha \mapsto \mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \alpha)$. For a convex function on a convex set — and under **(A-Conv)** this map is convex on A — "no admissible descent direction" is not merely necessary for a minimum but *equivalent* to it: a convex function lies above its supporting hyperplane at $\hat{\alpha}_t$, so a nonnegative directional derivative toward every other point forces $\mathcal{H}(\cdots, a) \geq \mathcal{H}(\cdots, \hat{\alpha}_t)$ for all a. That is exactly the minimum principle eq. (9.6), and equivalently $\hat{\alpha}_t = \hat{\alpha}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t)$ with the unique minimizer of eq. (9.4). The optimal control is thus pinned, pathwise and instant by instant, by the costate the adjoint supplies. □

So much for necessity. We have shown that *if* a control is optimal it must solve the minimum principle — but a skeptic is entitled to ask whether this is worth anything operationally. Necessity alone licenses no algorithm: it tells us what an optimum looks like, not that anything looking like that is an optimum. There could in principle be controls satisfying eq. (9.6) that are saddle points or worse, and a separate *verification* argument — typically a comparison against a candidate value function solving HJB — would be needed to certify any of them. The next theorem removes that need entirely. Under the convexity already assumed in **(A-Conv)**, the minimum principle is not only necessary but *sufficient*: any control solving it *is* optimal, full stop. This is the precise sense in which the probabilistic formulation earns its keep. It converts the optimization — an infinite-dimensional search over strategies — into the solution of a system of coupled stochastic equations, the forward state and the backward adjoint, with the minimum principle as the algebraic tie between them, and no auxiliary value function to be guessed or verified. The engine of the proof is the same Itô pairing as before, run this time on the genuine displacement $\delta X = X - \hat{X}$ rather than the infinitesimal V, and the role played at first order by differentiation is played here, globally, by convexity's defining inequality — that a convex function lies above each of its tangents.

Theorem 9.5 (Stochastic maximum principle, sufficient condition)**.** *Assume* ***(A-Lip)***, ***(A-Growth)*** *and the following convexity (a precise form of* ***(A-Conv)****): for each (t,μ) the map $(x,\alpha)\mapsto \mathcal{H}(t,x,\mu_t,y,z,\alpha)$ is convex for every (y,z), and $x\mapsto g(x,\mu_T)$ is convex. Let $\hat\alpha\in\mathbb{A}$, with state $\hat X$ and adjoint $(\hat Y,\hat Z)$, satisfy the minimum condition eq. (9.6) $\mathrm{d}t\otimes\mathrm{d}\mathbb{P}$-a.e. Then $\hat\alpha$ is optimal against μ: $J^\mu(\hat\alpha)\le J^\mu(\alpha)$ for all $\alpha\in\mathbb{A}$.*

Proof. The strategy is to bound the cost gap $J^\mu(\alpha)-J^\mu(\hat\alpha)$ from below by zero for an *arbitrary* competitor α, using two convexity inequalities glued together by an Itô pairing. Fix $\alpha\in\mathbb{A}$ with state X, and write the genuine displacement $\delta X_t = X_t-\hat X_t$ — not infinitesimal now, but the full difference between the rival trajectory and the candidate. Suppress the frozen argument μ_t for readability and abbreviate the Hamiltonian along the two trajectories, $\mathcal{H}_t[\alpha]=\mathcal{H}(t,X_t,\mu_t,\hat Y_t,\hat Z_t,\alpha_t)$ (rival state and control, but *candidate* costate) and $\hat{\mathcal{H}}_t=\mathcal{H}(t,\hat X_t,\mu_t,\hat Y_t,\hat Z_t,\hat\alpha_t)$ (everything at the candidate). From the definition eq. (9.2) of the cost,

$$J^\mu(\alpha)-J^\mu(\hat\alpha)=\mathbb{E}\left[\int_0^T\big(L(t,X_t,\alpha_t)-L(t,\hat X_t,\hat\alpha_t)\big)\,\mathrm{d}t+g(X_T)-g(\hat X_T)\right].$$

We bound the terminal and running pieces separately, and the adjoint is what stitches them.

The terminal piece, by convexity of g. A convex differentiable function lies above its tangent plane: $g(X_T)\ge g(\hat X_T)+\langle\partial_x g(\hat X_T),X_T-\hat X_T\rangle$. Since $\hat Y_T=\partial_x g(\hat X_T)$ is exactly the adjoint's terminal datum, this reads

$$g(X_T)-g(\hat X_T)\ \ge\ \langle\hat Y_T,\delta X_T\rangle. \tag{9.11}$$

Convexity of the terminal cost has thus traded an unknown terminal difference for a linear functional of δX_T paired against the costate — precisely the form the Itô pairing can process.

The pairing, by Itô. Apply the product rule to $\langle\hat Y_t,\delta X_t\rangle$, with $\mathrm{d}(\delta X_t)=\mathrm{d}X_t-\mathrm{d}\hat X_t$ supplied by eq. (9.1) at (X,α) and at $(\hat X,\hat\alpha)$, and $\mathrm{d}\hat Y_t$ by eq. (9.5). As in the necessity proof the expansion has three pieces. The drift of $\hat Y$ contributes $-\langle\partial_x\hat{\mathcal{H}}_t,\delta X_t\rangle\,\mathrm{d}t$; the drift of δX contributes $\langle\hat Y_t,b(t,X_t,\alpha_t)-b(t,\hat X_t,\hat\alpha_t)\rangle\,\mathrm{d}t$; and the covariation of the two martingale parts — $\hat Z_t\,\mathrm{d}W_t$ against $(\sigma(t,X_t)-\sigma(t,\hat X_t))\,\mathrm{d}W_t$ — contributes the Frobenius term $\mathrm{tr}[(\sigma(t,X_t)-\sigma(t,\hat X_t))^\top\hat Z_t]\,\mathrm{d}t$. The two genuine stochastic integrals are martingales by the $\mathcal{S}^2\times\mathcal{H}^2$ integrability and so drop out under expectation; $\delta X_0=0$ kills the lower endpoint. Hence

$$\mathbb{E}\Big[\langle\hat Y_T,\delta X_T\rangle\Big]=\mathbb{E}\left[\int_0^T\Big(\langle\hat Y_t,b(t,X_t,\alpha_t)-b(t,\hat X_t,\hat\alpha_t)\rangle+\mathrm{tr}\big[(\sigma(t,X_t)-\sigma(t,\hat X_t))^\top\hat Z_t\big]-\langle\partial_x\hat{\mathcal{H}}_t,\delta X_t\rangle\Big)\,\mathrm{d}t\right]. \tag{9.12}$$

The first two terms inside the integral are not arbitrary: they are exactly the drift and volatility channels of the Hamiltonian *difference*. Writing out $\mathcal{H}_t[\alpha]=\langle b(t,X_t,\alpha_t),\hat Y_t\rangle+\mathrm{tr}(\sigma(t,X_t)^\top\hat Z_t)+L(t,X_t,\alpha_t)$ and subtracting the same with hats, the inner-product and trace terms reproduce precisely the first two integrands of eq. (9.12), and the leftover is the running-cost difference:

$$\langle\hat Y_t,b(t,X_t,\alpha_t)-b(t,\hat X_t,\hat\alpha_t)\rangle+\mathrm{tr}\big[(\sigma(t,X_t)-\sigma(t,\hat X_t))^\top\hat Z_t\big]=\mathcal{H}_t[\alpha]-\hat{\mathcal{H}}_t-\big(L(t,X_t,\alpha_t)-L(t,\hat X_t,\hat\alpha_t)\big).$$

(The volatility carries no control, so $\sigma(t,X_t)$ versus $\sigma(t,\hat X_t)$ differ only through the state; this is why the trace difference is a pure δX effect.)

Assembling the bound. Now combine. Substitute eq. (9.12) for $\mathbb{E}\langle \hat{Y}_T, \delta X_T\rangle$ into the terminal inequality eq. (9.11), then insert the result into the cost gap. The running-cost difference $L(t, X_t, \alpha_t) - L(t, \hat{X}_t, \hat{\alpha}_t)$ appears once in the cost gap with a plus sign and once inside eq. (9.12) with a minus sign, so the two cancel exactly — the mirror image of the $\partial_x L$ cancellation in the necessity proof — and what remains is

$$J^\mu(\alpha) - J^\mu(\hat{\alpha}) \;\geq\; \mathbb{E}\left[\int_0^T \Big(\mathcal{H}_t[\alpha] - \hat{\mathcal{H}}_t - \langle \partial_x \hat{\mathcal{H}}_t, \delta X_t\rangle\Big)\, \mathrm{d}t\right].$$

The cost gap is now controlled entirely by the Hamiltonian, with no trace of the running cost or the terminal cost left over: the adjoint has folded both into $\mathcal{H}$.

Closing with the second convexity inequality and the minimum principle. It remains to show the integrand is nonnegative. Here the *joint* convexity of $(x, \alpha) \mapsto \mathcal{H}(t, x, \mu_t, \hat{Y}_t, \hat{Z}_t, \alpha)$ enters — and note it must be joint, in the state and control together, for what follows; convexity in the control alone would not suffice, since the bound carries a genuine state displacement δX_t as well as a control displacement, and it is exactly the affineness of b, σ inside **(A-Conv)** that makes $\mathcal{H}$ jointly convex in (x, α) and so licenses the joint inequality. The tangent-plane inequality for this convex function, expanded about the candidate point $(\hat{X}_t, \hat{\alpha}_t)$ and evaluated at (X_t, α_t), gives

$$\mathcal{H}_t[\alpha] - \hat{\mathcal{H}}_t \;\geq\; \langle \partial_x \hat{\mathcal{H}}_t, \delta X_t\rangle + \langle \partial_\alpha \hat{\mathcal{H}}_t, \alpha_t - \hat{\alpha}_t\rangle.$$

The first term on the right is exactly the $-\langle \partial_x \hat{\mathcal{H}}_t, \delta X_t\rangle$ sitting (with opposite sign) in the bound above, so it cancels, and the integrand is bounded below by $\langle \partial_\alpha \hat{\mathcal{H}}_t, \alpha_t - \hat{\alpha}_t\rangle$. But this last quantity is nonnegative for *every* competitor, and that is precisely the hypothesis we are handed: the minimum condition eq. (9.6) says $\hat{\alpha}_t$ minimizes $\alpha \mapsto \mathcal{H}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t, \alpha)$ over the convex set A, so its directional derivative toward any admissible α_t is ≥ 0, i.e. $\langle \partial_\alpha \hat{\mathcal{H}}_t, \alpha_t - \hat{\alpha}_t\rangle \geq 0$. The integrand is therefore ≥ 0 pointwise, its expectation is ≥ 0, and $J^\mu(\alpha) \geq J^\mu(\hat{\alpha})$. Since α was arbitrary, $\hat{\alpha}$ is optimal. □

The two theorems together close the circle the section opened. theorem 9.4 shows that optimality *implies* the coupled system — the forward state eq. (9.1), the backward adjoint eq. (9.5), and the feedback tie $\hat{\alpha}_t = \hat{\alpha}(t, \hat{X}_t, \mu_t, \hat{Y}_t, \hat{Z}_t)$ — and theorem 9.5 shows the system *implies* optimality. Solving the system and optimizing are, under **(A-Conv)**, the same act. What we have manufactured, then, is a *solution operator*: a map that takes a frozen population flow μ and returns the unique optimal response of the representative agent, packaged as the triple $(\hat{X}, \hat{Y}, \hat{Z})$ together with the feedback it induces. This operator is the entire deliverable of section 9.1. It is also, exactly, the raw material the next section welds shut: section 9.2 will demand that the law $\mathrm{Law}(\hat{X}_t)$ of the optimal state *coincide* with the very flow μ_t the agent optimized against, turning the solution operator into a fixed-point problem and the frozen parameter into a self-consistent unknown. Before we strike the freeze, two remarks fix the precise seam at which that closure happens — what changes, and what must not.

Remark 9.6 (MFG versus mean field control: a decisive seam)**.** There is a single line in the adjoint equation eq. (9.5) that silently encodes the difference between the two great cousins of this theory, and it deserves to be made loud, because the FBSDE the next section builds inherits the choice. The adjoint differentiates $\mathcal{H}$ in the state x *only*, with the measure μ_t held frozen — there is no ∂_μ term.

That is exactly right for the mean field *game*. In a game the representative agent is a *price-taker*: it is one negligible atom in a continuum, its own deviation moves the population law not at all, and so when it computes how a displacement of its state changes its future cost it is entitled to treat μ as inert scenery. The population's response is not denied — it is simply deferred, imposed afterward and from the outside as the self-consistency condition $\mu_t = \mathrm{Law}(X_t)$ of section 9.2, never felt by the optimizer during its optimization.

Mean field *control* — the cooperative problem, in which a single central planner chooses one feedback rule and applies it to the *whole* population, optimizing the population's average cost — is a different beast precisely here. The planner is not a price-taker: when it perturbs the common control it moves every agent at once, hence moves the law μ_t itself, hence changes the cost that the law-dependence of L, b, σ feeds back to everyone. Differentiating the cost now picks up that channel, and the adjoint equation acquires an extra drift term,

$$\widetilde{\mathbb{E}}[\partial_\mu \mathcal{H}(t, \widetilde{X}_t, m_t, \widetilde{Y}_t, \widetilde{Z}_t, \widetilde{\alpha}_t)(X_t)],$$

where $\partial_\mu = \partial_\mu$ is the Lions derivative of chapter 4 — the derivative of a function of a *measure*, which is itself a function of a probe point in state space — and $\widetilde{\mathbb{E}}$ is expectation over an *independent copy* $(\widetilde{X}, \widetilde{Y}, \widetilde{Z}, \widetilde{\alpha})$ of the solution, drawn from the same law m_t but independent of the trajectory being written.

The argument bookkeeping in that term is the subtle part, and it is worth slowing down to read it, because the tilde and the evaluation point play opposite roles and confusing them inverts the meaning. The object $\partial_\mu \mathcal{H}(t, \cdot, m_t, \cdot)(\cdot)$ is a function of *two* kinds of state argument: the ordinary arguments inside $\mathcal{H}$, and the Lions "probe" argument that comes after the parenthesis. The inner arguments $(\widetilde{X}_t, \widetilde{Y}_t, \widetilde{Z}_t, \widetilde{\alpha}_t)$ all wear tildes: they belong to the independent copy, the *representative other agent* whose cost is being differentiated, and $\widetilde{\mathbb{E}}$ averages them out over the population — this is the planner summing the marginal harm its perturbation does to a typical member of the crowd. The Lions evaluation point, by contrast, is X_t *without* a tilde: it is the state of the very equation being written, the agent whose adjoint this is. In words: "average, over a random other agent $\widetilde{X}$, the sensitivity of that agent's Hamiltonian to the population mass located at my own current position X_t." Swap the tilde onto X_t and you would instead be asking how a random agent's cost responds to mass at a random other location — a different, and wrong, quantity. The asymmetry — integrate the copy, evaluate at the self — is the signature of the Lions derivative and the whole content of the extra term.

The LQG thread makes the gap concrete and quantitative. Take the running cost $\frac{q}{2}(x - s\bar{\mu}_t)^2$ with $\bar{\mu}_t = \int \xi \, \mu_t(\mathrm{d}\xi)$ the population mean. In the game, the adjoint sees only ∂_x of this, namely $q(x - s\bar{\mu}_t)$, with $\bar{\mu}_t$ frozen. In the control problem the planner additionally registers that nudging the common control shifts $\bar{\mu}_t$, and the Lions derivative of $\frac{q}{2}(x - s\bar{\mu})^2$ in the mean direction contributes an extra term proportional to qs, evaluated by averaging the other agents' costates. The upshot is that the effective mean-reversion coefficient in the control adjoint is shifted relative to the game adjoint — in the symmetric quadratic case the game coupling s is effectively replaced by something like $s(2 - s)$ or, in the standard normalization, the coupling is *doubled* — and the two equilibria genuinely differ. This is the probabilistic root of the "price of anarchy" between competitive and cooperative mean field solutions: same Hamiltonian, different adjoint, different fixed point. For the development that follows the moral is a single sentence: the system eq. (9.13) we are about to close is the *game* FBSDE, built on the ∂_μ-free adjoint, and every well-posedness and master-equation statement downstream is a statement about the game; the control problem is its near twin, separated from it by exactly the term displayed above.

Remark 9.7 (Controlled diffusion: why the first-order adjoint suffices only when σ is uncontrolled)**.** The standing assumption that the volatility σ does not depend on the control was invoked three times in the proofs above — it deleted $\delta\alpha$ from the diffusion of the variational process eq. (9.7), it killed the control-dependence of $\partial_\alpha \mathcal{H}$, and it made the trace term in eq. (9.12) a pure state effect. It is worth being explicit about what would go wrong without it, because the breakdown is structural, not a mere loss of convenience. Suppose $\sigma = \sigma(t, x, m, \alpha)$. Then perturbing the control perturbs the *noise* the agent feels, and a perturbation of the noise is a far more violent thing than a perturbation of the drift: in the Itô calculus the diffusion contributes at order $\sqrt{\mathrm{d}t}$, so its square contributes at order $\mathrm{d}t$, the same order as the drift. Concretely, a needle variation of size $a - \hat{\alpha}$ on a window of length ϵ now changes the state at second order through the quadratic variation, by an amount that does *not* vanish faster than the first-order terms. The clean first-order expansion that produced the variational process and the single duality eq. (9.9) is no longer the whole story; a second-order term survives, and the minimum principle eq. (9.6) acquires an extra piece quadratic in $\sigma(t, x, m, a) - \sigma(t, x, m, \hat{\alpha})$.

That quadratic piece must be *priced*, and a price is a costate. The new term measures the cost of injected randomness, and the cost of randomness is set by the *curvature* of the cost-to-go — exactly the second derivative D^2u of the value function, since a convex cost-to-go penalizes spread. So a *second* adjoint appears, a matrix-valued pair (P, Q) solving a second-order BSDE whose terminal datum is the Hessian $\partial^2_{xx} g$ and whose role is to carry D^2u along the trajectory, just as the first adjoint (Y, Z) carries the gradient ∇u. The Hamiltonian to be minimized then gains a Frobenius term $\frac{1}{2}\operatorname{tr}[(\sigma(t, x, m, \alpha) - \hat{\sigma})^\top P\,(\sigma(t, x, m, \alpha) - \hat{\sigma})]$ coupling the control to this curvature, and the pointwise minimization eq. (9.4) couples to P. This is the general Peng stochastic maximum principle, genuinely heavier machinery: two coupled adjoint systems, one first order and one second order, rather than the single (Y, Z) we have worked with. We exclude it deliberately. The uncontrolled-diffusion assumption is what keeps the entire chapter's adjoint apparatus first order — one BSDE, one costate pair — and it is no real loss of generality for our purposes: the LQG thread has constant volatility, and the overwhelming majority of mean field models of interest, where noise is an exogenous environmental input rather than a decision variable, satisfy it on the nose. The reader who needs the controlled-volatility theory will find the second-order maximum principle and its matrix BSDE developed in full in [40] (Vol. I, Ch. 4); for this book it is a road deliberately not taken, flagged here so that the first-order adjoint of eq. (9.13) reads as a considered restriction rather than an oversight.

9.2 The McKean–Vlasov FBSDE and the fixed point in path space

The previous section discharged, in full, the half of the mean field problem that can be solved in isolation. With the population held frozen at an arbitrary flow μ, the stochastic maximum principle

— necessity and sufficiency both established — collapsed the agent's infinite-dimensional optimization into the solution of a coupled state–adjoint system, and it did more than characterize a candidate: under **(A-Conv)** it certified that the minimizer of the pointwise Hamiltonian *is* the optimum. What we hold in hand, therefore, is not a heuristic but a verified solution operator. To any frozen scenery μ it assigns the optimally controlled state, its law, and the costate that prices it; feeding a flow in at one end, it returns the best response to that flow at the other. Everything in the present section is the consequence of a single act performed on this operator: we delete the word "frozen."

Deleting it is the whole of the mean field game. A frozen μ was always a fiction — a placeholder standing in for the crowd while we pretended the agent could not influence it — and the equilibrium demand is precisely that the fiction become true of itself. The agent optimizes against μ and, in doing so, generates a state process X with its own law $\mathrm{Law}(X_t)$; self-consistency is the requirement that this generated law be the very scenery optimized against, $\mu_t = \mathrm{Law}(X_t)$ at every instant. This is the McKean–Vlasov closure of chapter 7, but imposed now on a process of a new kind. In chapter 7 self-consistency was asked of a *prescribed* dynamics: the drift was given, and the fixed point only had to make the law feeding that drift agree with the law the drift produced. Here the dynamics being made self-consistent is itself the output of an optimization — the drift carries the optimal feedback $\hat{\alpha}$, which depends on the costate, which depends through its terminal gradient on the very law we are closing. The closure therefore wraps a fixed point in the measure around a problem that is already a fixed point in time, and it is worth being precise, before any formulas, about how many distinct knots this ties. We are closing the *game* FBSDE — the system built on the ∂_μ-free adjoint isolated in the mean-field-game-versus-control remark just above — not its cooperative twin; every well-posedness and master-equation statement that follows is a statement about the game.

The object the closure produces, assembled in definition 9.8 below, stacks three difficulties in one equation, and naming them in advance is what keeps the rest of the section from reading as a list of unrelated technical hurdles. The first is *forward–backward coupling*: the state runs forward from its initial law m_0 while the adjoint runs backward from its terminal gradient, and the two are chained through the feedback, so that data pinned at opposite ends of $[0, T]$ must be reconciled along a single trajectory — the two-point boundary tension that is the governing signature of the whole chapter. The second is *distributional coupling*: the coefficients see not just the trajectory but its law $\mathrm{Law}(X_t)$, so the unknown is not one path but a path together with the measure it induces, and that measure feeds back into the very equations that generate it. The third is the *implicit feedback*: the drift and the adjoint driver both contain $\hat{\alpha}$, which is defined only as the argmin of a minimization, so even pointwise the system is specified implicitly rather than in closed form. Each difficulty would be serious alone; what makes the mean field game FBSDE its own subject is that the three are nested — the distributional coupling sits outside the forward–backward coupling, which sits outside the implicit feedback — so that resolving one does not unwrap the others. The strong convexity of **(A-Conv)** already tamed the innermost knot in section 9.1, rendering $\hat{\alpha}$ a genuine Lipschitz function of its arguments; the outer two are the business of the existence theory, and the honest accounting of when they can be undone is the well-posedness trichotomy this section closes on.

The section develops these in order. Definition 9.8 substitutes the optimal feedback and imposes $\mu_t = \mathrm{Law}(X_t)$ to write the McKean–Vlasov FBSDE itself, the central equation eq. (9.13) of the chapter; a notation watch there records a seam we must not let pass silently, that the probabilistic law $m_t = \mathrm{Law}(X_t)$ appearing in those coefficients is the *same* object as the population density $m(t,\cdot)$ of chapter 8, the two coinciding wherever the diffusion is nondegenerate and the probabilistic law being the more primitive when it is not. We then give the equilibrium its two complementary fixed-point readings — one a fixed point of the best-response map on flows of measures, the geometry closest to the PDE picture of chapter 8 and to the Schauder/Kakutani arguments of chapter 10; the other a fixed point directly on path space, the natural home of the contraction and continuation methods and of the propagation of chaos of chapter 13. Finally we state, honestly and without reproving them, the three regimes under which the fixed point is known to exist: small horizon (or equivalently small coupling) by contraction, arbitrary horizon under Lasry–Lions monotonicity by continuation, and arbitrary horizon without monotonicity by compactness at the cost of possible multiplicity. This chapter sets the fixed-point structure up; chapters 10 and 11 settle whether it has a solution. The equilibrium object the closure delivers is exactly what section 9.3 will read in its second coordinate system, identifying the costate with the value gradient and proving the probabilistic and analytic pictures to be one equilibrium twice described; and the two-point boundary tension named above, which no translation removes, is precisely what the decoupling field of section 9.4 is built to resolve. We begin by performing the substitution and imposing the consistency that turn the verified solution operator into a closed fixed point.

9.2.1 Closing the loop

We rejoin the development exactly where the maximum principle left it. With the population flow μ frozen at an arbitrary curve of measures, the agent has solved its problem in full: the optimally controlled state obeys eq. (9.1), the costate is the adjoint pair (Y,Z) from eq. (9.5), and the two are tied together by the feedback $\hat{\alpha}_t = \hat{\alpha}(t, X_t, \mu_t, Y_t, Z_t)$, the unique pointwise minimizer of the generalized Hamiltonian. This is a complete, verified solution operator: hand it a frozen scenery μ and it returns the best response to that scenery. Turning it into a mean field game is the work of two operations performed in a fixed order, and pulling them apart is what makes the closure a clean substitution rather than a leap.

The first operation is *substitution*, and it is pure bookkeeping. In the frozen problem the action α_t was an independent unknown, ranging over the admissible class $\mathbb{A}$ while we searched for its optimal value. Once the maximum principle has identified that value as $\hat{\alpha}_t = \hat{\alpha}(t, X_t, \mu_t, Y_t, Z_t)$, we may simply write the feedback in wherever a control appears — in the state drift $b(t, X_t, \mu_t, \alpha_t)$ and in the adjoint driver $\partial_x \mathcal{H}(t, X_t, \mu_t, Y_t, Z_t, \alpha_t)$. The control ceases to be a free variable; the only unknowns surviving the substitution are the triple (X,Y,Z) and the flow they evolve against. This step changes nothing about the problem's content. It merely records, in the equations themselves, the optimality already proved.

The second operation is the entire physical content of the mean field game, and it is the one the frozen problem was built to suspend. The flow μ was always a fiction: a placeholder for the crowd, held fixed so that a single agent could optimize as though it could not move the crowd at all. Equilibrium is the demand that the fiction come true of itself — that the scenery the agent optimized against be reproduced, exactly, by the agents optimizing. The optimally controlled state

X has a law $\mathrm{Law}(X_t)$ at each instant, and self-consistency is the requirement

$$\mu_t \;=\; \mathrm{Law}(X_t), \qquad 0 \le t \le T,$$

imposed at every time simultaneously. This is the McKean–Vlasov closure of chapter 7, but the process it now closes is of a new kind. In chapter 7 the dynamics being made self-consistent was *prescribed*: the drift was given, and the only fixed point was to make the law feeding the drift agree with the law the drift produces. Here the dynamics is itself the output of an optimization. Its drift carries $\hat{\alpha}$, which depends on the costate Y, whose terminal value $\partial_x g(X_T, m_T)$ depends in turn on the very law m_T we are closing. Substitute, then impose consistency, and the verified solution operator becomes a closed system that determines its own input.

Definition 9.8 (The MFG FBSDE)**.** Substituting the optimal feedback eq. (9.4) and imposing the consistency $\mu_t = \mathrm{Law}(X_t)$, a *mean field game equilibrium in probabilistic form* is a tuple (X, Y, Z) of $\mathbb{F}$-progressively measurable processes, together with the flow $m_t = \mathrm{Law}(X_t)$, solving the McKean–Vlasov forward–backward system

$$\begin{cases} \mathrm{d}X_t = b\big(t, X_t, m_t, \hat{\alpha}_t\big)\ \mathrm{d}t + \sigma\big(t, X_t, m_t\big)\ \mathrm{d}W_t, & X_0 \sim m_0, \\ \mathrm{d}Y_t = -\,\partial_x \mathcal{H}\big(t, X_t, m_t, Y_t, Z_t, \hat{\alpha}_t\big)\ \mathrm{d}t + Z_t\ \mathrm{d}W_t, & Y_T = \partial_x g\big(X_T, m_T\big), \\ \ \hat{\alpha}_t = \hat{\alpha}\big(t, X_t, m_t, Y_t, Z_t\big), & m_t = \mathrm{Law}(X_t). \end{cases} \tag{9.13}$$

The qualifier "McKean–Vlasov" refers to the dependence of the coefficients on the *law* $m_t = \mathrm{Law}(X_t)$ of the solution itself; this is what makes eq. (9.13) a distributed fixed point rather than a system for a single trajectory.

The system eq. (9.13) is the probabilistic mean field game, the central object of the chapter. Before turning it over to the existence theory, it is worth dwelling on the single adjective that distinguishes it from every forward–backward system we have written so far — "McKean–Vlasov" — because it is precisely this feature that lifts the equation out of the realm of one trajectory and makes it a problem about a whole ensemble at once.

In an ordinary (forward–backward) stochastic differential equation, the coefficients at time t depend on the solution only through its current *value* $X_t(\omega)$ along the single sample path ω. Such an equation can be solved one realization at a time: fix the Brownian path, and the coefficients become known functions of the unknown trajectory, with no reference to any other realization. The frozen-flow system of section 9.1 was of exactly this kind, which is why the maximum principle could solve it pathwise against a μ supplied from outside. The closure destroys that independence. After we impose $m_t = \mathrm{Law}(X_t)$, the coefficients depend not on the value $X_t(\omega)$ but on the *law* of X_t — a single deterministic measure obtained by averaging over *all* realizations ω at once. No trajectory can now be solved in isolation, because the equation that governs it refers to the distribution of the entire ensemble it belongs to. The unknown is therefore not a path but a path-together-with-its-own-law, and the solution must satisfy the self-referential demand that the law it induces be the law its coefficients already presumed. This is what is meant by calling eq. (9.13) a *distributed* fixed point: the fixed-point condition is not on a number or a function but on a probability measure spread across the whole population, and it is reflexive — the measure must reproduce itself. The LQG thread makes the abstraction tangible. There the only statistic of m_t the coefficients see is the population mean $\bar{m}_t = \int x \, m_t(\mathrm{d}x) = \mathbb{E}[X_t]$, so the state drift of a single agent reads $aX_t - b^2 Y_t$ with no overt mean, but the running and terminal costs pull the agent toward $s\bar{m}_t$ and $s_T \bar{m}_T$, and through the adjoint this places $\mathbb{E}[X_t]$ — an average over the ensemble — inside the equation each member of the ensemble obeys. Every path is coupled to the mean of all paths; solving for one requires knowing the law of all.

With the closure understood, we can name the obstructions it erects. The system stacks three distinct difficulties, and they are not independent hurdles to be cleared in any order but *nested* shells, each enclosing the next, so that resolving one does not by itself unwrap the others. Reading them off the three lines of eq. (9.13) is the cleanest way to see both what each one is and how it sits relative to the rest.

The innermost is the *implicit feedback*, visible in the third line of eq. (9.13). The control $\hat{\alpha}_t = \hat{\alpha}(t, X_t, m_t, Y_t, Z_t)$ that drives the state and enters the adjoint driver is not given by a closed formula; it is defined only as the argmin eq. (9.4) of the Hamiltonian minimization. Even pointwise, then — even with the trajectory, the law, and the costate all in hand at a single instant — the system specifies its own coefficients only implicitly, through an optimization that must be solved before the dynamics can be evaluated. This is the knot that **(A-Conv)** already untied in section 9.1: strong convexity made the minimizer unique and Lipschitz in (x, m, y, z), so that substituting it back returns coefficients still satisfying **(A-Lip)**. The innermost shell is, by the time we reach this section, already open. What remains is to undo the two that enclose it.

The middle shell is the *forward–backward coupling*, the structural signature the whole chapter is organized around. The first line of eq. (9.13) runs the state forward from its initial law m_0; the second runs the adjoint backward from its terminal gradient $\partial_x g(X_T, m_T)$; and the feedback chains them, since the forward drift needs Y_t to know which way to push while the backward terminal datum needs X_T to know where to start. Data are thus pinned at *opposite ends* of $[0, T]$ and must be reconciled along a single trajectory — a two-point boundary-value problem in time, not an initial-value one that can be integrated in a single sweep. This is the tension flagged in the chapter preamble as surviving every translation of the equilibrium, and it is exactly what the decoupling field of section 9.4 will be built to resolve, by expressing the backward unknown as a function of the forward state and collapsing the two-point structure into a single forward equation.

The outermost shell is the *distributional coupling* introduced by the closure itself. Through $m_t = \mathrm{Law}(X_t)$ the law appears as an argument of the drift, the diffusion, the Hamiltonian, and the terminal gradient, so the forward–backward system of the inner two shells is not solved once but must be solved *consistently with the measure it generates*. Even if one could, for a given candidate flow, resolve the implicit feedback and the forward–backward coupling and produce a solution (X, Y, Z), that solution would induce a law $\mathrm{Law}(X_t)$ which need not equal the flow one started from; equilibrium demands they agree. This is the shell that sits strictly outside the other two — it wraps an entire forward–backward solve inside a further fixed point on the space of measure flows — and it is the one neither convexity nor a decoupling field removes. Settling it is the business of the existence theory, and the honest accounting of when it can be undone is the well-posedness trichotomy this section closes on.

Notation watch

The law $m_t = \mathrm{Law}(X_t)$ in eq. (9.13) is the *same* object as the population density $m(t,\cdot) = m(t,\cdot)$ of chapter 8: where the PDE chapter writes a density on $\mathbb{R}^d$, here we write the law of a random variable. Under **(A-Nondeg)** this law has a density and the two coincide; without nondegeneracy the probabilistic object is the more primitive one, since $\mathrm{Law}(X_t)$ is defined whether or not a density exists. We use m_t for the law and $m(t,\cdot)$ for its density interchangeably, flagging only where the distinction bites.

This is not a pedantic distinction but one that decides which of the two pictures is the more fundamental, and the answer matters for the whole programme of the chapter. When the diffusion is uniformly nondegenerate — **(A-Nondeg)**, $\sigma\sigma^\top \succeq \lambda I$ for some $\lambda > 0$ — the noise smooths the law instantly: for every $t > 0$ the measure $\mathrm{Law}(X_t)$ is absolutely continuous with respect to Lebesgue measure and possesses a density, so that $m_t(\mathrm{d}x) = m(t,x)\,\mathrm{d}x$ and the probabilistic law of the present chapter coincides with the Fokker–Planck density $m(t,\cdot)$ of chapter 8 as the same object written in two notations. In that regime nothing is lost by sliding between "law" and "density." But the coincidence is a privilege of nondegeneracy, not a structural truth. Strip the smoothing away — let the diffusion degenerate, or vanish outright — and $\mathrm{Law}(X_t)$ can be singular: in the noiseless limit it is a Dirac mass transported along a deterministic flow, a measure with no density at all, on which the Fokker–Planck equation has no classical meaning. The probabilistic object survives this collapse untouched, because $\mathrm{Law}(X_t)$ is defined for any random variable whether or not a density exists; it is the more primitive of the two. This is one of the four motivations the chapter preamble offered for building the probabilistic picture in the first place — the case where the analytic picture has no density to speak of while the controlled process and its law remain perfectly well defined — and the notationwatch is where that promised payoff first shows its hand. Throughout this chapter the diffusion is nondegenerate and we treat m_t and $m(t,\cdot)$ as one; the reader should simply remember which one is load-bearing should the floor of nondegeneracy ever be removed.

The system eq. (9.13) is now assembled in full: a single McKean–Vlasov forward–backward equation, its feedback substituted in and its population law closed back on itself, carrying in one object the three nested couplings just anatomized. What it does not yet carry is an answer to the question the existence theory must ask of it. We have written the equilibrium as a fixed point, but "fixed point" is so far only a word; to feed the equation to the contraction, continuation, and compactness arguments that establish solvability, we must say in precisely *what sense* it is a fixed point — on what space, of what map — and there turns out to be more than one honest answer, each suited to a different downstream proof. The next subsection takes eq. (9.13) in hand and reads it as a fixed-point problem in two complementary ways: as a fixed point of the best-response map on flows of measures, the geometry nearest the PDE picture of chapter 8, and as a fixed point directly on path space, the natural home of the contraction and propagation-of-chaos methods to come.

9.2.2 The fixed point in path space

The previous subsection finished assembling eq. (9.13) and handed it on with a single unanswered question. We have called the equilibrium a fixed point, but that word has so far done no work: a fixed point is a point left invariant by a map, and we have named neither the map nor the space it acts on. Until we do, the system is an equation with a self-referential clause $m_t = \mathrm{Law}(X_t)$ and no apparatus to test it against. The business of this subsection is to convert that self-reference into an honest fixed-point problem — to exhibit a map whose invariant points are exactly the equilibria of eq. (9.13) — so that the existence theory of the next subsection has something concrete to act on. The reason we must work a little here, rather than write down "the" fixed-point map and move on, is that the distributional coupling can be cut in more than one place, and the cut one chooses decides which downstream proof becomes natural and which becomes awkward. There turn out to be two honest cuts, giving two fixed-point formulations of the same equilibrium, and the practical content of this subsection is not only to state both but to say precisely *which later chapter uses which, and why.* Both are used in the existence theory of chapter 10 and the propagation-of-chaos analysis of chapter 13; neither is more correct than the other, but each is adapted to a different machine.

The flow-of-measures fixed point. The first cut severs the loop at the marginal flow, and it reproduces, in probabilistic clothing, the best-response operator the analytic chapter already built. Recall from section 8.4 and proposition 8.4 that the mean field game equilibrium was characterized there as a fixed point of a map S that took a candidate population flow, solved the representative agent's optimization against it, and returned the population the optimizers generate. That map was realized in chapter 8 by the HJB/Fokker–Planck sweep — solve HJB against the frozen flow for the value, extract the feedback, push the density forward by Fokker–Planck. We now realize *the very same operator* through the FBSDE instead of the PDE pair, and the equivalence of the two realizations is one instance of the dictionary section 9.3 will prove in full. The construction is exactly the freeze-and-solve of section 9.1 read as a map. Given an input flow

$$\mu = (\mu_t)_{0 \le t \le T} \in C\big([0,T]; \mathcal{P}_2(\mathbb{R}^d)\big),$$

a continuous curve of second-moment measures, one freezes it as exogenous scenery and solves the *decoupled* forward–backward system obtained from eq. (9.13) by replacing every McKean–Vlasov occurrence of $m_t = \mathrm{Law}(X_t)$ with the frozen μ_t. The point of the substitution is that it severs precisely the outermost shell anatomized in the previous subsection: with μ frozen, the law no longer feeds back on itself, and what remains is the *standard* (not McKean–Vlasov) FBSDE of the single-agent problem of section 9.1, whose forward and backward legs are now coupled only to each other and not to their own distribution. That decoupled system is solvable by the single-agent theory already in hand; let X^μ denote its optimal state, and output the flow of its time-marginals,

$$\mathsf{S}(\mu) \;=\; \big(\mathrm{Law}(X_t^\mu)\big)_{0 \le t \le T}. \tag{9.14}$$

The operator S is the best response made literal: hand it a guess for the crowd, and it returns the crowd that the optimal responders to that guess actually form. A flow m is a mean field game equilibrium precisely when the guess is self-reproducing, that is when it is a fixed point,

$$\mathsf{S}(m) = m,$$

for then the population the agents form against m is m itself — which is the closure condition $m_t = \mathrm{Law}(X_t)$ of eq. (9.13) stated as the invariance of a map.

This is the formulation that sits nearest chapter 8, and that is exactly why it is the one the Schauder/Kakutani existence arguments of chapter 10 are built on. A topological fixed-point theorem of that kind asks for two things and nothing more: a convex compact set carried into itself, and a continuous map carrying it. The flow-of-measures picture supplies both in the most transparent way, because its unknown is a low-dimensional, geometrically tame object — a curve in the Wasserstein space $\mathcal{P}_2(\mathbb{R}^d)$ — on which compactness and continuity have concrete, checkable meanings, rather than a measure on the infinite-dimensional path space whose compactness is harder to certify. This is the WHY behind preferring it downstream, and it splits into the two analytic inputs that chapter 10 owns. The first is *continuity of* S *in the Wasserstein metric*: if two frozen flows μ and μ' are close in $\sup_t \mathcal{W}_2(\mu_t, \mu'_t)$, the decoupled FBSDEs they drive have nearby coefficients, so by the stability of SDEs and BSDEs in their data (chapter 5) their optimal states are close in $\mathcal{S}^2$, hence their marginal laws are close in $\mathcal{W}_2$ — this is what makes S continuous, and it is the estimate chapter 10 makes quantitative. The second is the *compactness* that lets a fixed-point theorem bite at all: the candidate flows must be confined to a set that is compact in the topology of uniform Wasserstein convergence, which by Prokhorov's theorem means the marginals $\mathsf{S}(\mu)_t$ must be *tight* — their mass cannot leak to infinity — uniformly in t, together with an equicontinuity in t that prevents the curve from jittering. Tightness is bought from the uniform second-moment bound $\mathbb{E}[\sup_t |X_t|^2] < \infty$ established in section 9.1; equicontinuity from the Hölder-in-time estimate the SDE supplies. Both inputs, finally, lean on the regularizing effect of **(A-Nondeg)**: it is the nondegenerate noise that smooths the marginals enough to keep the best-response map continuous rather than merely measurable across the kinks where the feedback's optimization changes regime, and that prevents the output laws from concentrating into singular limits that would escape the compact set. We only flag these here and name their owner; chapter 10 develops each into a theorem.

The path-space fixed point. The second cut severs the loop one level higher, at the law of the *whole trajectory* rather than at its time-marginals, and although it encodes the same equilibrium it is the formulation the harder existence machinery prefers. Read eq. (9.13) now as a single McKean–Vlasov FBSDE and seek a fixed point directly on path space. One fixes a candidate law of the entire forward path,

$$\mathbf{m} = \mathrm{Law}(X) \in \mathcal{P}_2(C([0,T];\mathbb{R}^d)),$$

a single probability measure on the space of continuous trajectories — not a curve of marginals but one object living over the whole horizon at once — solves the standard FBSDE that results when this $\mathbf{m}$ is frozen into the coefficients, reads off the law of the forward component of the solution, and demands that it agree with $\mathbf{m}$. The two cuts are related by marginalization: projecting a path-space law $\mathbf{m}$ onto its value at time t recovers the flow μ_t, so a path-space fixed point projects to a flow-of-measures fixed point, while the converse requires the extra information of how the marginals are tied together along trajectories. Why climb to the heavier object? Because two of the chapter's eventual destinations are statements about trajectories that the marginal flow cannot even phrase, and on path space they become natural. The contraction and continuation methods of the next subsection are estimates on the solution *process* — they bound $\mathcal{S}^2$-distances between whole paths produced by competing input laws, and the cleanest such bound lives in the metric $\mathcal{W}_2$ on $\mathcal{P}_2(C([0,T];\mathbb{R}^d))$, which compares trajectories end to end rather than slice by slice; the forward–backward stability that drives the contraction is genuinely a path-space estimate and only descends to marginals as a corollary. And the propagation of chaos of chapter 13 is, by its very statement, a path-space convergence: the empirical measure of the N players' *trajectories* converges to $\mathbf{m}$, and each player's path couples to an independent copy of the McKean–Vlasov solution path. The limit object that N-particle Nash play converges to is precisely this path-space law, so the formulation in which the limit theorem is stated, and proved, must be the one whose unknown is $\mathbf{m}$. The marginal flow is the right stage for compactness and the PDE comparison; the path-space law is the right stage for contraction, continuation, and the particle limit — and pinning this division of labor down is exactly what lets the next subsection name, for each of its three regimes, the object the regime acts on without re-deriving it.

With both cuts on the table we can record, once and for all, what a solution *is*, in the function spaces the rest of the chapter will demand of it. The definition fixes the regularity of all three unknowns of eq. (9.13) simultaneously, so that every estimate downstream has a normed home to live in.

Definition 9.9 (Solution of the MFG FBSDE)**.** A *solution* of eq. (9.13) on $[0, T]$ is a triple (X, Y, Z) with $X \in \mathcal{S}^2(\mathbb{R}^d)$, $Y \in \mathcal{S}^2(\mathbb{R}^d)$, $Z \in \mathcal{H}^2(\mathbb{R}^{d\times k})$, adapted to $\mathbb{F}$, such that the integral forms of the forward and backward equations hold $\mathbb{P}$-a.s. with $\hat{\alpha}$ given by eq. (9.4) and $m_t = \mathrm{Law}(X_t)$ for all $t \in [0, T]$. The spaces $\mathcal{S}^2$ (square-integrable, with sup-norm) and $\mathcal{H}^2$ (square-integrable in $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$) are those of chapter 5.

The choice of spaces in definition 9.9 is not bookkeeping; each is forced by what its unknown must do. The forward state and the backward Y are asked to live in $\mathcal{S}^2$ — processes with $\mathbb{E}[\sup_t |\cdot|^2] < \infty$, finite second moment in the *sup* norm — because both are compared pathwise: the state's sup-moment is what we just saw delivers tightness of the marginals $\mathrm{Law}(X_t)$ for the flow-of-measures fixed point, and Y's membership in $\mathcal{S}^2$ is the backward analogue established for the adjoint in section 9.1, the regularity in which the dictionary $Y_t = \nabla u(t, X_t)$ of section 9.3 will be read. The martingale integrand Z asks only for the weaker $\mathcal{H}^2$ — square-integrability in $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$, with no sup — because Z is never evaluated at a single instant but only ever integrated against $\mathrm{d}W$, and the Itô isometry weighs it precisely in that $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$ norm; demanding a sup-bound on Z would be both unnecessary and, in general, false. That the closure $m_t = \mathrm{Law}(X_t)$ is imposed *for every* $t \in [0, T]$, and not merely almost everywhere, is what ties the triple back to a genuine flow $m \in C([0, T]; \mathcal{P}_2(\mathbb{R}^d))$ — continuity in t coming for free from $X \in \mathcal{S}^2$ — so that the solution of definition 9.9 and the fixed point $\mathsf{S}(m) = m$ of eq. (9.14) are two names for one equilibrium, the path-space triple on one side and its projected marginal flow on the other.

These are the objects the existence theory must now act on: the best-response map S of eq. (9.14) on the flow space $C([0, T]; \mathcal{P}_2(\mathbb{R}^d))$, its path-space refinement on $\mathcal{P}_2(C([0, T]; \mathbb{R}^d))$, and the solution triple $(X, Y, Z) \in \mathcal{S}^2 \times \mathcal{S}^2 \times \mathcal{H}^2$ of definition 9.9 they are required to produce. We have written the equilibrium as a fixed point in two precise senses and matched each to its downstream proof, but we have not asked the one question that decides whether any of this machinery turns over: does a fixed point exist at all? The closure that made the system self-consistent is also what makes its solvability genuinely in doubt — the decoupled FBSDE can already blow up before the McKean–Vlasov layer is even imposed — so the answer cannot be free. The next subsection asks it honestly and finds not one answer but three, each buying existence through a different mechanism acting on one of the two fixed-point objects just defined, and each at a different cost.

9.2.3 Well-posedness: three regimes

The previous subsection handed us the equilibrium written as a fixed point in two precise senses — the best-response map S on the flow space $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$ of eq. (9.14), and its path-space refinement on $\mathcal{P}_2(C([0,T];\mathbb{R}^d))$ — together with the solution triple $(X,Y,Z) \in \mathcal{S}^2 \times \mathcal{S}^2 \times \mathcal{H}^2$ of definition 9.9 they must produce. Writing the equilibrium as a fixed point is a genuine achievement, but it is not yet a proof that anything sits at the fixed point. The honest question, postponed until now precisely so that we could pose it against concrete objects, is whether eq. (9.13) has a solution at all. It does not come for free, and it is worth being exact about why, because the reason is the same forward–backward tension that has run through the whole chapter.

The obstruction is not the McKean–Vlasov closure — it is already present once μ is frozen. Strip the self-consistency away and look at the *decoupled* FBSDE, an ordinary (non-McKean–Vlasov) forward–backward system with μ held fixed. One might expect this to be harmless: a forward SDE with Lipschitz coefficients has a unique global solution by chapter 5, and a backward SDE with a Lipschitz driver has a unique global solution by Pardoux–Peng (theorem C.18). Each leg, in isolation, is globally well posed on any horizon. What can fail is their *coupling*. The forward drift depends on the costate Y through the feedback $\hat{\alpha}$, and the costate is pinned at the terminal time; the backward equation depends in turn on the forward state. The cleanest way to see the damage this two-way dependence can do is to anticipate the decoupling field $U(t,x)$ of section 9.4, the function for which $Y_t = U(t,X_t)$: substituting this ansatz into eq. (9.13) and applying Itô turns the coupled system into a single quasilinear parabolic PDE for U, a Riccati-type equation. And Riccati equations blow up. The scalar caricature $\dot{u} = u^2$, with smooth bounded data, escapes to $+\infty$ in finite time; the genuine decoupling-field equation inherits exactly this nonlinearity from the feedback's dependence on Y, so that for a long enough horizon T, or a strong enough coupling between the legs, U — and with it ∇u — can cease to exist while each leg taken alone remains perfectly tame. Finite-time blow-up of the decoupling field is the precise mechanism by which a globally Lipschitz forward–backward system can have *no* global solution, and it is the obstruction every regime below must defeat. The McKean–Vlasov layer, far from healing this, can only compound it: the closure feeds the law of the forward state back into the very coefficients whose blow-up we are trying to avert.

Against this obstruction we have three honest answers, not one. Each secures a solution through a different mechanism, acting on a different one of the fixed-point objects just recalled, and each pays a different price — in the size of the horizon, in a structural hypothesis on the couplings, or in the loss of uniqueness. We state all three together and then read off, regime by regime, what mechanism is at work and what it costs.

Theorem 9.10 (Well-posedness of the MFG FBSDE)**.** *Grant **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, and **(A-Conv)**, so that the feedback $\hat{\alpha}$ of eq.* (9.4) *is well defined and Lipschitz. Then the McKean–Vlasov FBSDE eq.* (9.13) *admits a solution (X,Y,Z) in the sense of definition 9.9 in each of the following regimes:*

(i) (Small horizon / small coupling.) *There is $T_0 > 0$, depending only on the Lipschitz constants,* *such that for $T \leq T_0$ the solution exists and is unique. The mechanism is a contraction of the best-response map* S *on* $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$*.*

(ii) (Monotonicity.) *If in addition the Lasry–Lions monotonicity **(A-Mon)** holds for the cost couplings (chapter 11), then a solution exists and is unique for* every *horizon T, by the continuation-in-parameter method along the monotone structure.*

(iii) (Compactness, no monotonicity.) *Without monotonicity but with bounded, continuous couplings, a solution exists for every T (uniqueness may fail) via a Schauder fixed point for* S*; this is the existence theorem of chapter 10.*

Proof sketch and pointers. The three regimes are proved in full in chapters 10 and 11; the purpose here is not to reprove them but to expose the mechanism each one runs on, and to mark precisely where the self-contained argument of this chapter ends and the imported argument of the neighbouring chapters begins. Read this sketch as a map of three roads, with the destinations owned elsewhere.

Regime (i): contraction. This is the cheapest mechanism and the one that lives closest to objects already in hand. The best-response map eq. (9.14) takes a candidate flow μ, solves the *decoupled* control problem against it — which is well posed by the single-agent theory of section 9.1 and the Pardoux–Peng theorem, exactly the "each leg alone is fine" fact from the discussion above — and returns the law of the resulting optimal state. A solution of the MFG is a fixed point $\mathsf{S}(\mu) = \mu$. The claim is that on a short horizon S is a strict contraction in a suitably weighted Wasserstein-sup metric on $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$, so that Banach's theorem supplies a unique fixed point. The mechanism is propagation control: perturb the input flow by a small Wasserstein amount δ, and ask how far the output flow moves. The perturbation reaches the output only through the measure-arguments of the coefficients, is transported along the state and adjoint by their standard stability estimates in the coefficients (chapter 5) amplified by the Lipschitz feedback $\hat{\alpha}$, and accumulates over the interval $[0,T]$. Each of these steps contributes a bounded factor, and the accumulation contributes the length of the interval, so the output moves by at most $C\,T\,\delta$ for an explicit constant C built from the Lipschitz data. When $C\,T < 1$ the map contracts; setting $T_0 := 1/C$ gives the threshold of the theorem, and Banach delivers existence *and* uniqueness in one stroke — the only regime that gets both from a single argument. This is also the precise sense in which "small coupling" and "small horizon" are interchangeable. The constant C *absorbs the Lipschitz constants of the couplings* $\partial_m b, \partial_m L, \partial_m g$ — the derivatives of the coefficients in their measure argument, which are exactly the channels through which a perturbation of the input flow becomes a perturbation of the output — so the contraction constant is the product (coupling Lipschitz constant)$\times T$ up to fixed structural factors. Shrinking the coupling shrinks C and restores $C\,T < 1$ at a *fixed*, possibly large, horizon T, exactly as shrinking T restores it at fixed coupling: the same product is being driven below 1, and it is indifferent to which factor one shrinks. The full contraction estimate, with the weight and the constant made explicit, is carried out in chapter 10; primary source [40] (Vol. I, Ch. 4, the small-time theorem).

Regime (ii): continuation under monotonicity. The contraction of (i) is local in the horizon by its very nature — the factor T cannot be argued away — so reaching arbitrary T demands a different idea. One embeds eq. (9.13) in a one-parameter family $(\mathcal{F}_\lambda)_{\lambda\in[0,1]}$ that interpolates between a trivially solvable system at $\lambda = 0$ (the couplings switched off, so the legs decouple and each is globally well posed) and the target system at $\lambda = 1$. Let $S \subseteq [0, 1]$ be the set of parameters for which $\mathcal{F}_\lambda$ is solvable. The continuation method shows S is both open and closed in $[0, 1]$ and nonempty (it contains 0); since $[0, 1]$ is connected, $S = [0, 1]$ and in particular $1 \in S$. Openness is the easy half: near a solvable parameter the small-horizon contraction of (i), applied to the *increment* in λ rather than to the whole problem, lets one step a short distance in λ by the same Banach argument, so S contains a neighbourhood of each of its points. Closedness is the half that needs structure, and it is exactly where **(A-Mon)** enters. To pass to a limit of solvable parameters one needs the solutions not to run off to infinity as λ approaches the boundary of S — precisely the finite-time blow-up the opening warned of, now threatening in the parameter direction rather than the time direction. The Lasry–Lions monotonicity supplies the *a priori bound* that forecloses it: pairing the difference of two solutions against the difference of their adjoints and integrating by parts — the same computation that drives the Lasry–Lions uniqueness proof of chapter 11 — produces a sign-definite quantity, and monotonicity forces that sign to bound the solution uniformly in λ. With the a priori bound in hand the blow-up cannot occur, S is closed, and the system is solvable, and in fact uniquely solvable, for *every* horizon. The price is structural: **(A-Mon)** is a genuine hypothesis on the couplings — crowd-aversion rather than crowd-seeking — and models that violate it (imitation, congestion with the wrong sign) fall outside this regime and into the next. This is the Carmona–Delarue continuation method [40] (Vol. I, Ch. 4); the monotonicity hypothesis and the pairing identity that powers the a priori bound are stated and proved in full in chapter 11.

Regime (iii): compactness without monotonicity. When neither the horizon is small nor the couplings monotone, contraction and continuation both fail, and one falls back on the weakest tool that still produces a fixed point: Schauder. One restricts S to a convex compact set $\mathcal{K}$ of flows and shows it maps $\mathcal{K}$ continuously into itself, whereupon the Schauder–Kakutani theorem of chapter 2 furnishes a fixed point inside $\mathcal{K}$. The compactness is not free; it is bought from the diffusion. Under **(A-Nondeg)** the noise is uniformly nondegenerate, which regularizes the marginal laws $\mathrm{Law}(X_t)$ and, together with the tightness estimates of chapter 3, makes the relevant set of flows precompact in the Wasserstein-sup topology — this is the same parabolic-smoothing role the nondegeneracy plays for the decoupling field in theorem 9.14. Continuity of S follows from the stability of the single-agent problem in its data. What Schauder gives, however, is strictly less than what Banach gave in (i): it produces *a* fixed point but no contraction, and therefore *no* claim of uniqueness. This is exactly where uniqueness is lost. Two distinct flows can both be fixed points of S, and without monotonicity to enforce the sign that would collapse them, nothing rules this out — the multiplicity is real, and it is the symmetry-breaking phenomenon previewed in chapter 11 and made explicit for the LQG thread in chapter 14. The argument and the precise function-space setting are the content of chapter 10; we state the result here only to place the FBSDE on the same well-posedness footing as the PDE system, and defer the construction to avoid duplicating that chapter. □

It is worth stating cleanly what this subsection has and has not done, because the division of labor is itself part of the architecture. We have deliberately *not* reproved existence: chapter 10 owns the small-horizon and Schauder constructions, chapter 11 owns the monotone continuation, and our job here was to set up the FBSDE — its fixed-point objects, its hypotheses, its blow-up obstruction — so that the proofs there have something precise to act on. What *has* been established intrinsically, and owes nothing to the later chapters, is the equivalence of the three readings of the equilibrium assembled across this chapter (the single-agent stochastic maximum principle of section 9.1, the flow-of-measures fixed point, and the path-space fixed point of section 9.2), together with the exact hypotheses under which each well-posedness mechanism turns over: small horizon or small coupling for contraction, **(A-Mon)** for monotone continuation, and **(A-Nondeg)** with bounded continuous couplings for compactness. Uniqueness, where it holds, is revisited under **(A-Mon)** in chapter 11; multiplicity, where monotonicity fails, is the symmetry-breaking phenomenon previewed there and made explicit for the LQG thread in chapter 14.

The payoff of having paid this price is that the FBSDE now stands on exactly the same well-posedness footing as the PDE pair of chapter 8: three regimes — short horizon, monotone coupling, nondegenerate compactness — guarantee a solution in each picture, and they are not merely analogous regimes but, as the next section will show, the *same* regimes read in two coordinate systems. That is the claim we are now in a position to make good. With both descriptions known to possess solutions under the same conditions, the question is no longer whether each exists but whether the two are one, and that is precisely section 9.3, where the dictionary $Y_t = \nabla u(t, X_t)$ identifies the probabilistic equilibrium with the analytic one. The blow-up obstruction we met at the head of this subsection — the forward–backward coupling that can destroy a global solution — is not forgotten in that translation either: it is exactly what the *decoupling field* of section 9.4 is built to resolve, and the small-horizon and monotone mechanisms named here reappear there as the two regimes in which that field exists, the device standing behind both existence arguments and, once the population is promoted from frozen parameter to genuine argument, at the doorstep of the master equation of chapter 12.

9.3 Two pictures of one equilibrium

The chapter opened by setting the probabilistic picture deliberately against the analytic one of chapter 8 and promising, as the reward for the detour through path space, a precise dictionary between them. We are now in a position to redeem that promise. The previous subsection did the last piece of preparatory work: it placed the FBSDE eq. (9.13) on exactly the same well-posedness footing as the PDE pair of chapter 8, with the very same three regimes — short horizon, monotone coupling, nondegenerate compactness — delivering a solution in each picture. That coincidence is suggestive, but on its own it proves nothing about identity. Two genuinely different objects can happen to exist under the same hypotheses; that both the field system and the process system are solvable precisely when the horizon is short, or the coupling monotone, or the diffusion nondegenerate, does not yet show that a solution of one *is* a solution of the other. The decisive question, the one this section settles, is sharper than coexistence: are the two descriptions the same equilibrium written twice, related by an explicit and invertible change of coordinates? The answer is yes, and the change of coordinates is the dictionary $Y_t = \nabla u(t, X_t)$ this section now assembles.

It is worth stating the thesis of the section in its strongest form, because the two subsections that follow exist precisely to substantiate it. The naive expectation is hierarchical: one of the two descriptions is the “real” one and the other a derived convenience, a shadow cast by the fundamental object onto a more tractable surface. The claim here is the opposite, and it is more symmetric.

Neither the fields (u, m) on state space nor the processes (X, Y, Z) on path space is the underlying reality of which the other is an image. They are two coordinate systems on the one set of equilibria, related by an invertible dictionary, and choosing between them is choosing a chart, not choosing what is true. What separates them is not fundamentality but *transparency*: each chart renders one structural feature of the equilibrium manifest while leaving another implicit. The value gradient is the cleanest illustration, and it is no accident that it is also the hinge of the whole dictionary. In the PDE picture ∇u is a *derived* quantity — one must first solve the Hamilton–Jacobi–Bellman equation for the field u and only then differentiate it — whereas in the FBSDE picture the same object appears as a *primitive* unknown, the backward process Y solved for directly alongside the state. Conversely, the smoothness of u, which the parabolic theory of chapter 8 reads off almost for free, is buried in the path picture, recoverable only through the decoupling field. The same equilibrium content sits in plain sight in one chart and in the shadows in the other, and which is which flips from object to object. That is the precise sense in which neither picture is more fundamental and each makes a different aspect transparent.

The section makes this thesis good in two movements, one per subsection. The first, *The dictionary*, exhibits the literal translation. Its entire weight rests on a single identity — the costate is the gradient of the value function evaluated along the optimal trajectory, $Y_t = \nabla u(t, X_t)$, with its curvature companion $Z_t = \mathrm{D}^2\, u\, \sigma$ — from which the reconciliation of the two Hamiltonians, the agreement of the two optimal feedbacks, and the literal equality of the two population laws all follow. This identity is the dictionary the chapter preamble promised, and proving it is what turns the structural analogy of the previous paragraph into a theorem. The second movement, *What each picture sees better*, substantiates the other half of the thesis: it lays out the division of labor between the charts — which structural questions the field picture answers with the least friction, which the path picture answers with the least friction — and in doing so it motivates the FBSDE machinery that the later chapters lean on, from the propagation of chaos of chapter 13 to the Monte Carlo solvers of chapter 28.

The hinge identity does more than close the present chapter; it is the seed of everything that comes after it. Once $Y_t = \nabla u(t, X_t)$ is established with the population flow frozen, section 9.4 promotes it into the *decoupling field* $U(t, x) = \nabla u(t, x)$, the function that collapses the two-point boundary structure of the FBSDE — forward initial law at $t = 0$, backward terminal gradient at $t = T$, the single tension that has shadowed every form of the equilibrium in this chapter — into one Markovian forward equation. Promote the population in turn from frozen parameter to genuine argument, and that same decoupling field becomes the master field standing at the doorstep of chapter 12. The boxed equation of the next subsection is thus not only the chapter's payoff but the first stone of the road to the master equation. To lay it we must first put the analytic picture back on the table — the HJB and Fokker–Planck equations of chapter 8, rewritten in the sign convention native to the adjoint of this chapter — and then ask, field by field, where each of its objects goes under the translation. That is the task the next subsection takes up.

9.3.1 The dictionary

The previous section put the thesis on the table and named the task that will prove it: take the analytic picture of chapter 8, set it beside the path picture assembled in section 9.2, and ask — field by field, process by process — where each object of the one goes under translation into the other. This subsection performs that translation, and the discovery it reports is that the entire correspondence is governed by a *single* identity, the costate read as a value gradient, from which every other matching follows as a corollary. We begin by laying the two pictures side by side in one notation, so that the question "what maps to what" can even be posed sharply; only then do we apply Itô's formula to the candidate translation and watch it solve the adjoint equation of its own accord.

To pose the question sharply we must first be in a regime where both pictures are literally available as classical objects, because the dictionary we are about to write evaluates *derivatives* of the value function along trajectories, and those derivatives must exist to be evaluated. We therefore assume for this subsection the regularity under which the analytic picture is classical: the HJB value function $u(t,x) = u(t,x)$ of chapter 8 is $C^{1,2}$ on $[0,T) \times \mathbb{R}^d$ — once continuously differentiable in time and twice in space — and $\sigma = \sigma(t,x)$ is uncontrolled, the standing restriction of the whole chapter. Each half of this hypothesis earns its place. The $C^{1,2}$ smoothness is exactly what makes $\nabla u(t,x)$ and $\mathrm{D}^2 u(t,x)$ well-defined functions, so that the candidate processes $Y_t = \nabla u(t,X_t)$ and $Z_t = \mathrm{D}^2 u(t,X_t)\,\sigma(t,X_t)$ are objects one can form at all; the uncontrolled volatility is what keeps the adjoint first order, so that the costate is the pair (Y,Z) rather than the larger second-order adjoint a control-dependent σ would force (remark 9.7). Under the nondegeneracy hypothesis **(A-Nondeg)** the $C^{1,2}$ regularity is not an extra assumption but a consequence: the interior parabolic regularity of theorem C.12 extracts it from the equation itself, the noise smoothing the value into a classical solution. In the degenerate case no such bootstrap is available and the smoothness must be posited outright, which is what we do here, in full awareness that section 9.4 will later have to recover it by other means.

With the regularity fixed, we put the analytic equilibrium back on the table. Recall from chapter 8 that the equilibrium value solves the backward HJB equation. We write it here in the sign convention *opposite* to eq. (8.5a) — one obtains the present form by multiplying that equation through by -1, so the two are line-for-line identical and nothing mathematical has changed — because this is the convention in which the HJB drift will line up term for term with the adjoint driver of eq. (9.5), sparing us a sign chase in the middle of the Itô computation. In this convention,

$$\partial_t u + \tfrac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top \mathrm{D}^2 u) - H(t,x,\nabla u, m_t) = 0, \qquad u(T,x) = g(x,m_T), \tag{9.15}$$

with the HJB Hamiltonian $H = H$ of appendix A (here carrying the extra time argument, as flagged in the chapter preamble). The population density, in turn, evolves by the forward Fokker–Planck equation $\partial_t m_t = \mathcal{A}_t^* m_t$, with m_0 given, where the generator $\mathcal{A}_t$ carries the optimal feedback drift $b(t,x,m_t,\alpha^*(t,x))$ and $\alpha^* = -\partial_p H(t,x,\nabla u, m_t)$.

These two equations carry the same two-end tension that has shadowed every form of the equilibrium in this chapter: the value u is pinned at the *terminal* time $t = T$ by $g(x,m_T)$ and runs backward, while the density m is pinned at the *initial* time $t = 0$ by m_0 and runs forward, the two coupled through the feedback α^* that u generates and the law m that u reads. The FBSDE eq. (9.13) carries the identical tension — backward Y pinned at T, forward X pinned at 0 — which is the first sign that the translation we are about to make is not a coincidence but a change of coordinates that preserves structure. The question this subsection answers is now concrete enough to state as a table: the analytic chart holds two *fields* on state space, the value $u(t,x)$ and the density $m(t,x)$, while the path chart holds three *processes* along the trajectory, the state X_t, the costate Y_t, and its noise sensitivity Z_t. The density m_t is the same object in both — it is literally $\mathrm{Law}(X_t)$ either way, as the closure of section 9.2 already arranged — so the only genuine unknowns in the dictionary are how the value field u and its derivatives produce the backward processes Y and Z. We claim, and now prove, that the answer is the most economical one imaginable: Y is the gradient of u read along X, and Z is its Hessian, contracted with the volatility. Everything else — the agreement of the two Hamiltonians, the agreement of the two feedbacks — is bookkeeping once this is secured.

Before we can match the HJB equation eq. (9.15) to the adjoint eq. (9.5) we must settle one piece of translation that is logically prior to the Itô computation, because the two equations are written in terms of two *different* Hamiltonians. The HJB equation carries $H(t,x,p,m)$, the analytic Hamiltonian of chapter 8 with the control already optimized out and a free costate slot p; the adjoint carries $\mathcal{H}(t,x,m,y,z,\alpha)$, the Pontryagin Hamiltonian of definition 9.2 with the control still present. The notation watch following definition 9.2 already warned that these are distinct functions related by a Legendre transform; we now make that relation quantitative in the canonical case, since it is exactly the substitution the proof will invoke when it differentiates the HJB drift.

Remark 9.11 (Reconciling the two Hamiltonians)**.** In the canonical case $b(t,x,m,\alpha)=\alpha$ (the control is the drift) and σ constant, the reduced Pontryagin Hamiltonian of definition 9.2 is $\widehat{\mathcal{H}}(t,x,m,y)=\min_\alpha\{\langle\alpha,y\rangle+L(t,x,m,\alpha)\}=-H(t,x,y,m)$, where $H(t,x,p,m)=\sup_\alpha\{-\langle\alpha,p\rangle-L(t,x,m,\alpha)\}$ is exactly the convex-conjugate HJB Hamiltonian of chapter 8. The optimal feedbacks agree: $\hat\alpha(t,x,m,y)=-\partial_p H(t,x,y,m)=\alpha^*$ once $y=p$. The sign flip is the Legendre-duality bookkeeping and nothing more; both conventions encode the same optimization.

The single line worth dwelling on is the sign flip $\widehat{\mathcal{H}}=-H$, because it is not an accident of one convention but the visible fingerprint of Legendre duality, and seeing why it is there dissolves any worry that the two pictures might disagree. The Pontryagin Hamiltonian is built by *minimizing* the running cost plus the priced motion, $\widehat{\mathcal{H}}(t,x,m,y)=\min_\alpha\{\langle\alpha,y\rangle+L\}$; the HJB Hamiltonian is, by the convention of chapter 8, the *maximum* of the negated quantity, $H(t,x,p,m)=\sup_\alpha\{-\langle\alpha,p\rangle-L\}$. A minimum of f is minus the maximum of $-f$, so the two are negatives of each other the instant one sets $p=y$ — the two costate variables, the probabilistic y and the analytic p, are the same momentum wearing different letters. The flip is therefore forced by the choice to define the analytic Hamiltonian as a *convex* function of p (a supremum of affine functions is convex), the choice that made the Legendre transform of chapter 8 well posed in the first place. That the optimal feedbacks nonetheless agree, $\hat\alpha=-\partial_p H=\alpha^*$, is the reassurance that the sign flip is bookkeeping and not substance: differentiating a Legendre pair returns the same argmin from either side. We will use this remark in the proof in exactly the form $H=-\widehat{\mathcal{H}}$, to turn a gradient of the HJB Hamiltonian into a gradient of the Pontryagin driver.

We can now state the dictionary as a proposition and prove it. The statement has two halves that face in opposite directions, and it is worth flagging the asymmetry before reading it. The forward half — from the analytic solution to the probabilistic one — is elementary: it is a single application of Itô's formula, and we prove it here in full. The converse half — from the probabilistic solution back to the analytic one — is genuinely harder, because it must reconstruct a *field* on all of state space out of processes that only ever visit the trajectory, and that reconstruction is exactly the work of the decoupling field of section 9.4; we state it here and point to where it is discharged. The honest content provable on the spot is the forward half, and it is where the boxed identity is born.

Proposition 9.12 (FBSDE–PDE dictionary)**.** *Under the regularity above, let $m = (m_t)$ be an equilibrium flow and u the corresponding HJB solution eq.* (9.15)*. Let X solve the forward equation of eq.* (9.13) *with this m, and set*

$$Y_t \;=\; \nabla u(t, X_t), \qquad Z_t \;=\; \mathrm{D}^2\, u(t, X_t)\, \sigma(t, X_t). \tag{9.16}$$

Then (X, Y, Z) solves the FBSDE eq. (9.13) *with the same $m_t = \mathrm{Law}(X_t)$. Conversely, if (X, Y, Z, m) solves eq.* (9.13) *and the FBSDE admits a sufficiently regular decoupling field (section 9.4), then that field's x-gradient reconstructs ∇u and u solves eq.* (9.15)*. The two formulations are equivalent representations of the same equilibrium.*

Proof. The whole forward half is one application of Itô's formula, organized so that the HJB equation supplies exactly the cancellation the adjoint needs. We carry it out in four steps: differentiate the candidate Y_t by Itô to read off both its martingale part and its drift; differentiate the HJB equation in x to get a second expression for that drift; match the two; and check the terminal datum.

Step 1: Itô on $Y_t = \nabla u(t, X_t)$, and the birth of Z. Treat Y componentwise: its i-th component is the scalar $Y_t^i = \partial_{x_i} u(t, X_t)$, to which the scalar Itô formula of chapter 5 applies along the forward dynamics $\mathrm{d}X_t = b\ \mathrm{d}t + \sigma\ \mathrm{d}W_t$ of eq. (9.13),

$$\mathrm{d}Y_t^i = (\partial_t + \mathcal{A}_t)(\partial_{x_i} u)(t, X_t)\ \mathrm{d}t + (\nabla\, \partial_{x_i} u)(t, X_t)^\top \sigma(t, X_t)\ \mathrm{d}W_t,$$

where $\mathcal{A}_t f \;=\; \langle b, \nabla f\rangle + \frac{1}{2}\, \mathrm{tr}(\sigma\sigma^\top\, \mathrm{D}^2 f)$ is the generator of X from chapter 5. Stacking the d components, the drift is the vector $(\partial_t \nabla u + \mathcal{A}_t \nabla u)(t, X_t)$ — the generator acting entrywise on the gradient — and the martingale integrand is the $d \times k$ matrix whose i-th row is $(\nabla\, \partial_{x_i} u)^\top \sigma$, namely the Jacobian of ∇u (which is the Hessian $\mathrm{D}^2\, u$) times the volatility:

$$\mathrm{d}Y_t = (\partial_t \nabla u + \mathcal{A}_t \nabla u)(t, X_t)\ \mathrm{d}t + (\,\mathrm{D}^2\, u\, \sigma)(t, X_t)\ \mathrm{d}W_t.$$

Comparing the martingale term against the $Z_t\ \mathrm{d}W_t$ of the adjoint eq. (9.5) forces $Z_t = \mathrm{D}^2\, u(t, X_t)\, \sigma(t, X_t)$ — the second line of eq. (9.16). This is not an extra postulate: once $Y = \nabla u$, the martingale representation theorem (theorem C.17) leaves the integrand Z with no freedom, and Itô computes it for us as the curvature $\mathrm{D}^2\, u$ read along the path and contracted against the noise. The shape is worth pausing on, because it is the single most common transcription error in this calculation. Since $\mathrm{D}^2\, u \in \mathbb{R}^{d\times d}$ and $\sigma \in \mathbb{R}^{d\times k}$, the product $\mathrm{D}^2\, u\, \sigma$ lies in $\mathbb{R}^{d\times k}$, which is exactly the shape required for $Z_t\ \mathrm{d}W_t$ to be an $\mathbb{R}^d$ vector when $\mathrm{d}W_t \in \mathbb{R}^k$ — a $(d \times k)$ matrix against a k-vector gives a d-vector, matching $\mathrm{d}Y_t \in \mathbb{R}^d$. The transposed object $\sigma^\top\, \mathrm{D}^2\, u$ is $k \times d$ and would make the dimensions inconsistent; it is the scalar-state identity written down without minding the order of the factors, and it coincides with the correct expression only in the degenerate case $d = k = 1$, where every matrix is a scalar and order does not matter. We flag the trap precisely because the $d = k = 1$ LQG thread, the example one instinctively checks against, is exactly the case that cannot detect it.

Step 2: differentiate the HJB equation in x. We now need a second formula for the drift $\partial_t \nabla u + \mathcal{A}_t \nabla u$, and the HJB equation eq. (9.15) supplies it once we differentiate it in the state. That this is permitted is the one analytic debt of the proof: differentiating eq. (9.15) a further time in x requires $u \in C^{1,3}$ in the relevant sense, one derivative beyond the $C^{1,2}$ already assumed. Under **(A-Nondeg)** this is free — the interior parabolic regularity of theorem C.12 bootstraps the HJB solution to as many spatial derivatives as the coefficients allow, the nondegenerate noise smoothing u at every interior point — while in the degenerate case it is part of the standing classical-regularity hypothesis of this subsection. Granting it, apply ∇ to eq. (9.15). The time and second-order terms differentiate directly; the Hamiltonian term $H(t, x, \nabla u, m_t)$ must be differentiated through *both* of its x-dependences — the explicit one and the one routed through the slot ∇u — by the chain rule:

$$\nabla_x \big[H(t, x, \nabla u, m_t)\big] = \underbrace{(\partial_x H)(t, x, \nabla u, m_t)}_{\text{explicit } x} + \underbrace{\mathrm{D}^2 u\, (\partial_p H)(t, x, \nabla u, m_t)}_{\text{through } \nabla u},$$

the second term being the Hessian (the Jacobian of ∇u) contracted with the costate-gradient $\partial_p H$. Differentiating the whole equation therefore gives

$$\partial_t \nabla u + \tfrac{1}{2} \nabla\, \operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) - \partial_x H - \mathrm{D}^2 u\, \partial_p H = 0.$$

Step 3: match the two drifts, and watch the cancellation. Write the Step-1 drift as $\partial_t \nabla u + \mathrm{D}^2 u\, b + \frac{1}{2} \operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2(\nabla u))$, where the middle term is the first-order part of $\mathcal{A}_t \nabla u$ — the generator's drift acting entrywise on the gradient, which stacks the components $\langle b, \nabla \partial_{x_i} u\rangle$ into the vector $\mathrm{D}^2 u\, b$ — and the last is its second-order part acting entrywise. Substitute the Step-2 expression for $\partial_t \nabla u$. Along the optimal path the forward drift is the optimal feedback $b = \alpha^*(t, X_t) = -\partial_p H$ (the canonical feedback of remark 9.11). Hence $\mathrm{D}^2 u\, b = -\mathrm{D}^2 u\, \partial_p H$, which *exactly cancels* the $+\mathrm{D}^2 u\, \partial_p H$ carried in from Step 2. This cancellation is the whole mechanism: the second-order sensitivity of the value, $\mathrm{D}^2 u\, \partial_p H$, is precisely what the agent's optimal motion is steering against, and the two annihilate. What survives is

$$\text{drift of } \mathrm{d}Y_t = \partial_x H + \Big[\tfrac{1}{2} \operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2(\nabla u)\big) - \tfrac{1}{2} \nabla\, \operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 u\big)\Big].$$

The bracket is the difference between differentiating the second-order term *after* the generator has acted entrywise and *before*; the two disagree only where ∇ falls on the x-dependent volatility σ, and that residue is exactly minus the x-derivative of the Frobenius pairing $\operatorname{tr}(\sigma^\top Z)$ with $Z = \mathrm{D}^2 u\, \sigma$ held fixed — the noise channel of the Pontryagin Hamiltonian. (When σ is constant in x, as in the LQG thread, the bracket vanishes identically and this subtlety disappears.) Finally, by remark 9.11 the analytic Hamiltonian is the negated reduced Pontryagin one, $H = -\widehat{\mathcal{H}}$, and the envelope theorem gives $\partial_x H = -\partial_x L$ evaluated at the optimizer; assembling the surviving terms,

$$\text{drift of } \mathrm{d}Y_t = -\partial_x L - \partial_x \operatorname{tr}(\sigma^\top Z) = -\partial_x \mathcal{H}(t, X_t, m_t, Y_t, Z_t, \hat{\alpha}_t), \qquad \hat{\alpha}_t = \alpha^*(t, X_t),$$

since in the canonical case $b = \alpha$ is x-independent so $\partial_x \langle b, y\rangle = 0$ and $\partial_x \mathcal{H} = \partial_x L + \partial_x \operatorname{tr}(\sigma^\top z)$. This is precisely the driver of the adjoint eq. (9.5).

Step 4: the terminal datum. At $t = T$ the dictionary gives $Y_T = \nabla u(T, X_T) = \nabla_x g(X_T, m_T)$, using the terminal condition $u(T, x) = g(x, m_T)$ of eq. (9.15); this is exactly the terminal datum $Y_T = \partial_x g(X_T, m_T)$ of eq. (9.5). With the driver matched in Step 3 and the terminal value matched here, the pair (Y, Z) of eq. (9.16) solves the same backward equation as the adjoint, and by the uniqueness half of Pardoux–Peng (theorem C.18) it *is* that adjoint. Together with the forward leg, which X solves by hypothesis with $m_t = \mathrm{Law}(X_t)$, the triple (X, Y, Z) solves the FBSDE eq. (9.13). This proves the forward direction.

The converse. The reverse implication — that any sufficiently regular FBSDE solution arises from an HJB solution this way — cannot be a single Itô stroke, because it must manufacture a field u on *all* of $\mathbb{R}^d$ out of a process Y that lives only on the random trajectory. The bridge is the decoupling field of section 9.4: a deterministic function $U(t, x)$ representing the backward process as $Y_t = U(t, X_t)$, whose existence and regularity are the content of theorem 9.14 and whose in-book proof is the small-horizon contraction of chapter 10 together with its monotone extension in chapter 11. Given such a U, its x-antiderivative reconstructs u with $\nabla u = U$, and running the four steps above backward shows that this u solves eq. (9.15); uniqueness of the BSDE solution (theorem C.18) then makes the two constructions genuinely inverse to each other, so the correspondence is a bijection of equilibria and not merely a one-way map. The general form of the dictionary — controlled diffusion, common noise, the second-order adjoint these force — is carried out in [40] (Vol. I, Ch. 3–4); our standing uncontrolled-diffusion assumption is exactly what spares us that apparatus here. □

The proof produced two identities, but only the first is worth committing to memory, because the second is its derivative and the rest of the dictionary is its corollary. The entry to keep is

$\boxed{Y_t = \nabla u(t, X_t)}$ the costate is the gradient of the value function, evaluated along the optimal trajectory.

Read it as the resolution of the very puzzle the chapter opened with. The value gradient ∇u is, in the analytic picture, the marginal cost-to-go — how much the agent's remaining cost rises if its position is nudged — and the Pontryagin costate Y was *defined* in chapter 6 to be exactly that marginal cost-to-go, the momentum conjugate to the state. The box says these two readings of "marginal cost of displacement" are literally the same number, one computed as a field and then sampled at X_t, the other carried as a process. The momentum the agent prices its position with *is* the slope of the value landscape it stands on. This is why Y appears in the FBSDE as a primitive unknown while in the PDE it is a derived quantity: the path picture solves for the slope directly and never forms the field u at all.

The second component $Z_t = \mathrm{D}^2 u(t, X_t)\, \sigma(t, X_t)$ is then forced, and it has an equally clean reading: it is the *curvature* of the value function, the Hessian $\mathrm{D}^2 u$, sampled along the path and projected onto the noise directions by σ. Where Y measures the slope of the cost landscape, Z measures how that slope bends as the noise jostles the state — the rate at which the marginal cost-to-go itself fluctuates. The LQG thread makes this concrete and is the cleanest possible check. There $u(t, x) = \frac{1}{2} P_t x^2 + \dots$ is quadratic in the state, so its Hessian $\mathrm{D}^2 u = P_t$ is the scalar Riccati coefficient — *deterministic*, independent of x — and the dictionary gives $Z_t = P_t\, \sigma$, a deterministic process equal to the volatility times the Riccati gain. That Z is deterministic in the LQG case is not a coincidence but the signature of quadratic value: a parabola has constant curvature, so its second derivative carries no randomness, and the noise sensitivity of the costate collapses to a time-function. We verify this Riccati-for-Riccati in the worked computation worked computation 9.16, where the FBSDE's backward coefficient is matched against the HJB's Riccati flow and the two are seen to be the same equation.

With the box and its curvature companion in hand, the remaining matchings are immediate. The optimal feedback in the two pictures, $\hat{\alpha}(t, X_t, m_t, Y_t, Z_t)$ from the Pontryagin minimization and $\alpha^*(t, X_t) = -\partial_p H(t, X_t, \nabla u, m_t)$ from the HJB Legendre transform, coincide the moment $Y = \nabla u$ is substituted, by the feedback agreement of remark 9.11. The population law is not merely matched but *identical* in both pictures — it is $\mathrm{Law}(X_t)$ either way, the same measure under one name — as the notation watch of definition 9.2 already recorded. So the entire equilibrium content lines up: the PDE picture stores it as two *fields* on state space, (u, m); the FBSDE picture stores it as two coupled *processes* on path space, (X, Y) with the forced companion Z; and the single bridge between the two storehouses is "evaluate the field along the random trajectory." The two-end tension — terminal data on the backward object, initial data on the forward one — is preserved verbatim by the translation, which is the structural reason the dictionary could be an exact change of coordinates rather than an approximation.

Physics note

The relationship between the two formulations is, structurally, the relationship between the Schrödinger and Heisenberg pictures of a dynamical theory. In the analytic (PDE) picture the unknowns are *fields* on configuration space — the value $u(t, x)$ and the density $m(t, x)$ — evolving by deterministic equations, like a wavefunction obeying a PDE while the configuration label x stays put. In the probabilistic (FBSDE) picture the state x itself becomes a *process* X_t that moves, and the dynamical content is carried by operator-valued (here, process-valued) objects X_t, Y_t, Z_t evaluated along that motion; the field u has been "pushed onto the trajectory" via $Y_t = \nabla u(t, X_t)$, just as a Heisenberg operator is a Schrödinger operator conjugated by the flow. The two pictures compute identical observables — here, the equilibrium cost and the law m_t — and the choice between them is a choice of convenience: fields when one wants regularity and PDE technology, processes when one wants probabilistic representations, Monte Carlo, or the N-player coupling of chapter 13. As always, the analogy is organizational only; it proves nothing, and the equivalence is the rigorous content of proposition 9.12.

The boxed identity $Y_t = \nabla u(t, X_t)$ is the chapter's payoff, but it is also a handoff in two directions, and naming both now is what lets the rest of the development read as consequences rather than fresh starts. Forward in this chapter, it is the seed of section 9.4: there we hold t and x fixed rather than riding the trajectory, turning the pathwise identity into the deterministic *decoupling field* $U(t, x) = \nabla u(t, x)$ of eq. (9.18), the function that collapses the FBSDE's two-point boundary structure into a single Markovian forward equation and, once the population is promoted from frozen parameter to genuine argument, becomes the master field at the doorstep of chapter 12. Sideways into the threads, it is what the worked computation worked computation 9.16 will check explicitly, matching the FBSDE's backward coefficient to the HJB's Riccati flow and confirming the dictionary Riccati-for-Riccati in the one case where every object is closed-form. But before either of those, the equivalence we have just proved raises a practical question it does not answer: if the two pictures are genuinely the same equilibrium, which one should a working analyst reach for, and why? Having shown the charts interchangeable, we now ask what each makes easy — the subject of the next subsection.

9.3.2 What each picture sees better

The dictionary just proved settles a question of principle: the field picture (u, m) of chapter 8 and the path picture (X, Y, Z) of this chapter are not two theories of the equilibrium but two coordinate systems on one. Yet an equivalence of content is not an equivalence of convenience, and the moment the two charts are known to describe the same object a practical question presses forward that the equivalence theorem itself does not answer: faced with a concrete task — proving a regularity estimate, establishing the particle limit, writing a solver — which chart should the working analyst open, and why? The honest answer is that it depends entirely on the task, because the very feature that makes the translation exact also guarantees that each object of the equilibrium sits in plain view in one chart and in the shadows in the other. This subsection makes the division of labor explicit, not as a matter of taste but as a structural fact about which questions become *easy* in which chart, so that the chapters ahead can reach for the right one deliberately rather than by habit. The single tension that has organized the whole chapter — forward data pinned at $t = 0$, backward data pinned at $t = T$ — is what the choice ultimately turns on: each picture makes one end of that tension transparent and pays for it by obscuring the other.

Begin with the field picture, because its strengths are the ones the reader has already met. The unknowns there are honest functions on $[0, T] \times \mathbb{R}^d$, and functions on a fixed domain are exactly what the machinery of partial differential equations was built to control. This is not a vague slogan; it is the reason three specific and otherwise hard-won kinds of information are nearly free in the analytic chart. The first is *regularity.* Parabolic Schauder and energy estimates — the statements that a solution of the HJB or Fokker–Planck equation is as smooth as its data and its coefficients permit, with quantitative Hölder or Sobolev bounds — are theorems about fields, proved by differentiating the equation and testing it against itself on the state domain. They have no direct pathwise meaning: a single trajectory X_t is nowhere differentiable in t, so "the solution is $C^{2,\alpha}$ in x" is a sentence the path picture cannot even pronounce without first reconstructing the field $u(t, x) = u(t, \cdot)$ off the trajectory. When one needs to know that the value function is twice differentiable — which is, not incidentally, exactly the hypothesis under which the dictionary $Y_t = \nabla u(t, X_t)$ of section 9.3 was established — one proves it on the field and then transports the conclusion to the path, never the reverse. The second is the *maximum principle*: the assertion that a sub/supersolution of a parabolic equation attains its extremes on the parabolic boundary, from which comparison, uniqueness, and a priori sup-norm bounds all flow. It is a pointwise statement about a field touching a barrier from one side, geometrically vivid on the function $u(t, x)$ and with no clean counterpart in a description that has dissolved the field into a bundle of paths. The third is the *variational and potential structure* that appears when the Hamiltonian is separable: in that case the MFG system is the optimality condition of a single convex functional on the space of flows, and the whole equilibrium can be read as a minimizer. That reading lives entirely on the fields — it is a statement about a functional of (u, m) — and it is the seam through which numerical optimization and Γ-convergence arguments enter.

These three are not an arbitrary catalogue; they converge on the one place the book most needs the field picture, which is *monotonicity*. The Lasry–Lions monotonicity condition that drives the uniqueness theory of chapter 11 is the statement that the coupling $\int (f(t,x,m) - f(t,x,m'))\,(m - m')(\mathrm{d}x) \geq 0$ has a definite sign — a pairing of two *measures* tested against the difference of two *fields*. Its entire force is extracted by integrating the difference of two copies of the HJB/Fokker–Planck system against each other over state space and watching the cross terms cancel by the divergence theorem, leaving a signed quantity. Every ingredient of that computation — the integration by parts, the cancellation, the resulting energy identity — is a manipulation of fields on $\mathbb{R}^d$, and attempting it on trajectories would mean reconstructing precisely the densities and gradients the field picture hands over for free. This is why chapter 11 is written almost entirely in the analytic chart: monotonicity is a field phenomenon, and the path picture would only get in its way. The same loyalty governs computation. Finite-difference and semi-Lagrangian schemes (chapter 28) discretize the HJB and Fokker–Planck equations directly, placing the unknowns (u,m) on a grid over state space and stepping the coupled system in time; they inherit, grid-cell by grid-cell, exactly the monotonicity and comparison structure just described, which is what makes them provably convergent. For low-dimensional state spaces these grid methods are the gold standard, and they are native to the fields. The price of all this, and it is the price the rest of the subsection is about, is that the field chart makes the *forward* end of the equilibrium — the actual law of the actual paths — into a derived quantity, recovered only by solving a second PDE, and says nothing at all about individual trajectories.

Turn now to the path picture, whose strengths are the exact complement: it makes the *trajectories* primitive and pays by relegating the fields to derived objects. The reason this matters is that several of the book's most important destinations are, at bottom, statements about paths that the field chart cannot phrase. The cleanest is the *particle limit* of chapter 13. A mean field equilibrium is the $N \to \infty$ limit of N-player Nash play, and the precise form of that convergence — propagation of chaos — is the assertion that each player's optimal trajectory $X_t^{i,N}$ can be coupled, on the *same* probability space, to an independent copy of the limiting McKean–Vlasov process X_t built in section 9.2, so that $\mathbb{E}\sup_t |X_t^{i,N} - X_t|^2 \to 0$. This is a statement about two processes being close *pathwise*; it lives or dies on a coupling of trajectories, and a pair of densities on $\mathbb{R}^d$ cannot so much as state it, because two laws being close says nothing about whether the underlying paths can be realized near one another. The FBSDE, carrying the trajectory as its forward unknown, is the only chart in which the coupling estimate can even be written, and the moment bound $\mathbb{E}[\sup_t |X_t|^2] < \infty$ that has underwritten every estimate of this chapter is precisely the currency in which that limit is controlled. The path picture is not a convenience here; it is a precondition.

The second destination is *degenerate or vanishing diffusion.* The dictionary $Y_t = \nabla u(t, X_t)$ was proved under enough regularity for u to be twice differentiable, but that regularity is a luxury the field chart cannot always afford. When the volatility σ degenerates — loses rank in some directions, or shrinks toward zero in a vanishing-viscosity limit — the HJB equation loses its parabolicity and may possess no classical solution, and the Fokker–Planck equation may propagate a singular measure with no density at all. At that point the field picture has no objects to manipulate: $u(t,x)$ exists only as a viscosity solution, m_t only as a measure, and the smooth manipulations of the analytic chart stall. The path picture, by contrast, never required a density or a smooth field. The controlled process X_t and its law $\mathrm{Law}(X_t)$ remain perfectly well defined as soon as the coefficients are Lipschitz, entirely independent of whether σ is nondegenerate, and the FBSDE continues to encode the equilibrium when the PDE system has ceased to make classical sense. This robustness is structural, not incidental: the probabilistic objects are built from the trajectory outward and never pass through the field, so they survive exactly the degeneracies that destroy the field.

The third destination is *simulation in high dimension*, and it is the computational mirror image of the field chart's grid methods. The finite-difference schemes praised above discretize state space, and their cost grows like the number of grid points, which is exponential in the state dimension d — the curse of dimensionality, fatal beyond a handful of dimensions. Forward–backward Monte Carlo solvers (chapter 28) escape it by discretizing *paths* instead of grids: they simulate the forward state X_t by an Euler scheme and solve the backward (Y, Z) by regression or a deep-network parametrization along the simulated trajectories, with a cost that scales with the number of sample paths rather than the volume of a grid. These schemes are statements about the FBSDE in its own native terms — they integrate the very forward–backward system this chapter constructed — and they reach dimensions where no finite-difference grid can follow. Here the two-end tension of the equilibrium reappears as the central numerical difficulty: the forward simulation needs an initial condition the backward equation has not yet supplied, and it is exactly the decoupling field of the next section that the better schemes approximate to break the deadlock.

The fourth and furthest destination is the *graphon generalization* of chapter 23, and it is the one that most decisively favors the path picture. There the single representative agent is replaced by a continuum of types $x \in [0, 1]$, each carrying its own state process X_t^x, and the mean field coupling is replaced by an operator coupling through the graphon: agent x feels not the global law but the kernel-weighted aggregate $(\mathbb{W}\, m)(x) = \int_0^1 W(x, x')\, m^{x'}\, \mathrm{d}x'$ of its neighbors' laws. In the field chart this would demand a *family* of HJB/Fokker–Planck pairs indexed by a continuum, coupled through $\mathbb{W}$ — an object that is awkward even to write down and worse to estimate. In the path chart the generalization is almost transparent: one simply lets the state become the type-indexed family of processes $(X_t^x)_{x\in[0,1]}$ and replaces the self-consistent law $\mathrm{Law}(X_t)$ feeding back into the dynamics by its graphon average $\mathbb{W}\ \mathrm{Law}(X_t^{\cdot})$, leaving the entire forward–backward architecture of this chapter structurally intact. The McKean–Vlasov closure becomes a graphon closure; the adjoint, the Hamiltonian, and the fixed-point reading all survive the substitution with only the coupling relabelled. That the probabilistic formulation absorbs the graphon with so little friction, while the analytic one strains, is the clearest single piece of evidence that the path picture is the right vehicle for Part III, and it is the reason this chapter was placed where it is in the arc of the book.

The balance sheet is therefore clean and worth stating in one breath. The field picture owns regularity, comparison, monotonicity, and low-dimensional grids — everything that flows from having honest functions on a fixed state space, and everything chapter 11 and the finite-difference half of chapter 28 will lean on. The path picture owns the particle limit, robustness under degeneracy, high-dimensional Monte Carlo, and the graphon extension — everything that flows from carrying the trajectory itself, and everything chapter 13, chapter 23, and the simulation half of chapter 28 will lean on. Neither is more fundamental; the dictionary guarantees that anything provable in one is true in the other, but not that it is equally *visible*. The rest of Part II, and the graphon chapters beyond it, are built on the probabilistic machinery assembled here precisely because their questions are path questions, and it is worth the reader's carrying forward the habit of asking, before opening either chart, which end of the equilibrium the task lives on.

That question — which end — returns us to the structural tension the chapter opened with and has never resolved: the forward state X is pinned by its initial law at $t = 0$, the backward adjoint (Y, Z) by its terminal gradient at $t = T$, and every translation between the charts has carried this two-point coupling along untouched rather than dissolved it. The field picture hides the tension by never forming the trajectory; the path picture wears it on its sleeve, as the boundary-value structure of the FBSDE itself. Confronting it head-on is the business of the next section. The device that resolves it lives on the FBSDE side — it is the path picture's own answer to its own difficulty — and it is nothing other than the boxed identity $Y_t = \nabla u(t, X_t)$ read in a new way: hold t and x fixed instead of riding the trajectory, and the pathwise relation becomes a deterministic function expressing the backward unknown as a function of the forward one. That function is the *decoupling field*, and it collapses the two-point boundary problem into a single Markovian forward equation. We take it up now.

9.4 Decoupling fields and the road to the master equation

That function — the device the previous section named to resolve the FBSDE's two-point tension, the forward unknown X pinned at $t = 0$ against the backward unknown (Y, Z) pinned at $t = T$, and the object that turns out to be the seed of the master equation of chapter 12 — is the *decoupling field*: a deterministic map expressing the backward unknown through the forward one. We define it, exhibit the collapse it induces, and identify it with the value gradient.

Definition 9.13 (Decoupling field)**.** Consider the FBSDE eq. (9.13) with the population flow $m = (m_t)$ held fixed at an equilibrium (so the system is now a standard, non-McKean–Vlasov FBSDE). A measurable function $U : [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$ is a *decoupling field* for this FBSDE if the unique solution satisfies

$$Y_t \;=\; U(t, X_t) \qquad \text{for all } t \in [0, T],\ \mathbb{P}\text{-a.s.} \tag{9.17}$$

When U is differentiable in x, the second adjoint component is $Z_t = \partial_x U(t, X_t)\, \sigma(t, X_t)$, with $\partial_x U \in \mathbb{R}^{d\times d}$ the Jacobian of U (so $Z_t \in \mathbb{R}^{d\times k}$, consistent with definition 9.3).

The decoupling field, when it exists, collapses the two-point boundary structure into a single forward equation: substituting eq. (9.17) into the forward dynamics of eq. (9.13) makes X a self-contained (Markov) diffusion with drift $b(t, x, m_t, \hat{\alpha}(t, x, m_t, U(t, x), \partial_x U\, \sigma))$, and the backward equation is then solved *automatically* by $Y_t = U(t, X_t)$. Comparing with the dictionary eq. (9.16), the decoupling field of the MFG FBSDE is exactly the gradient of the HJB value function,

$$U(t, x) \;=\; \nabla u(t, x). \tag{9.18}$$

This is the precise sense in which "solving the FBSDE" and "solving the HJB equation" are the same act: the decoupling field is the value gradient, and the Markovian forward equation it produces is the optimally controlled state SDE whose law solves Fokker–Planck.

Theorem 9.14 (Existence of the decoupling field)**.** *Grant **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, with the equilibrium flow m fixed. Then there is a horizon $T_0 > 0$ such that for $T \le T_0$ the decoupled FBSDE eq.* (9.13) *has a unique Lipschitz decoupling field U, given by eq.* (9.18)*; under **(A-Mon)** the field exists and is Lipschitz for every T. The map $(t, x) \mapsto U(t, x)$ is the classical or viscosity gradient of the HJB solution, and eq.* (9.17) *holds.*

Proof sketch. For small T this is the four-step scheme / contraction argument of Ma–Protter–Yong and Delarue: one posits the relation $Y_t = U(t, X_t)$, derives by Itô a quasilinear parabolic PDE for U (a system, with the HJB equation eq. (9.15) as the potential whose gradient is U), and solves it by contraction in a space of Lipschitz fields, the horizon bound controlling the contraction constant. The nondegeneracy **(A-Nondeg)** supplies the parabolic smoothing that makes U Lipschitz; under **(A-Mon)** an a priori gradient bound replaces the small-horizon restriction and the field extends to all T, exactly as in theorem 9.10(ii). The identification $U = \nabla u$ is proposition 9.12. The in-book home of the full argument is the existence theory of the neighbouring chapters: the small-horizon contraction is carried out in chapter 10, the monotone a priori gradient bound that extends the field to every horizon in chapter 11, and the measure-dependent (master-field) version in chapter 12; this theorem records the result and the mechanism, and the construction itself is owned there. The Ma–Protter–Yong four-step scheme and the Delarue contraction are the original sources [40] (Vol. I, Ch. 4), with the master-equation construction in [37]; the global-in-T degenerate case is delicate and we do not claim it here. □

9.4.1 From decoupling field to master field

So far the decoupling field $U(t, x)$ was built with the equilibrium flow m *frozen.* The deeper object — and the subject of chapter 12 — is obtained by letting the field depend on the population state itself. Suppose that for every initial population $m \in \mathcal{P}_2(\mathbb{R}^d)$ (started at the current time t) the MFG has a unique equilibrium, and define the *master field*

$$\mathcal{V}(t, x, m) \;=\; u^{(t,m)}(t, x), \tag{9.19}$$

the equilibrium value at state x when the population, currently distributed as m, evolves forward in its own equilibrium from time t. Its x-gradient $\mathcal{U}(t, x, m) = \nabla_x \mathcal{V}(t, x, m)$ is the *measure-dependent decoupling field*: along the equilibrium it satisfies

$$Y_t \;=\; \mathcal{U}(t, X_t, m_t) \;=\; \nabla_x \mathcal{V}(t, X_t, m_t), \tag{9.20}$$

which upgrades the frozen-flow relation eq. (9.17) by making the dependence on the population explicit. The function $\mathcal{V}$ is no longer a function on $[0, T] \times \mathbb{R}^d$ but on $[0, T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d)$; differentiating the consistency relation eq. (9.20) along the equilibrium dynamics, using the Lions derivative ∂_μ of chapter 4 for the measure argument, produces a single equation for $\mathcal{V}$ — the *master equation* — that subsumes the entire forward–backward system into one Hamilton–Jacobi equation on the space of measures. That derivation, its well-posedness in the monotone regime, and its role as the generating object from which both the FBSDE and the PDE system descend by the method of characteristics, are the content of chapter 12. We have arrived exactly at its doorstep: the master field is the decoupling field with the population promoted from a frozen parameter to a genuine argument.

Remark 9.15. The promotion in eq. (9.19)–eq. (9.20) is the structural reason the master equation restores the Markov property at the level of the population. The frozen-flow decoupling field $U(t,x)$ knows only the current agent state; the master field $\mathcal{U}(t,x,m)$ knows the current agent state *and* the current population, which is the full state of the mean field system. This is the minimal enrichment that turns the forward–backward, non-Markov-in-m MFG into a forward, Markov-in-(x,m) flow — and it is indispensable once a common noise shocks the population (chapter 26), since then m_t is itself random and only the master field tracks it.

9.5 Worked computation: the LQG thread as an FBSDE

We now carry Thread A (the linear–quadratic–Gaussian mean field game, fixed in the project thread specification) through the probabilistic formulation, and check that it reproduces the analytic closed form of chapters 8 and 14. This is the FBSDE incarnation of the same equilibrium; every Riccati coefficient that appears matches the one obtained from the HJB equation, as the dictionary eq. (9.16) demands.

Worked computation 9.16 (Thread A: the scalar LQG MFG in FBSDE form)**.** Recall the scalar Thread A model: state, for the representative agent,

$$\mathrm{d}X_t = \big(aX_t + b\,\alpha_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad X_0 \sim m_0,$$

and cost over $[0,T]$, against the population mean $\bar{m}_t = \int \xi\, m_t(\mathrm{d}\xi) = \mathbb{E}[X_t]$,

$$J(\alpha) = \mathbb{E}\left[\int_0^T \Big(\tfrac{1}{2}\alpha_t^2 + \tfrac{q}{2}\big(X_t - s\,\bar{m}_t\big)^2\Big)\,\mathrm{d}t + \tfrac{q_T}{2}\big(X_T - s_T\,\bar{m}_T\big)^2\right],$$

with $a, b, \sigma \in \mathbb{R}$, $q, q_T \geq 0$, and coupling strengths s, s_T. The diffusion is constant (hence **(A-Nondeg)** for $\sigma \neq 0$) and the data are convex, so the sufficient maximum principle theorem 9.5 applies.

Hamiltonian and feedback. With costate y the generalized Hamiltonian eq. (9.3) is

$$\mathcal{H}(t,x,y,z,\alpha) = (ax + b\alpha)\,y + \sigma\,z + \tfrac{1}{2}\alpha^2 + \tfrac{q}{2}(x - s\bar{m}_t)^2,$$

minimized over α at $\partial_\alpha \mathcal{H} = by + \alpha = 0$, i.e.

$$\hat{\alpha}(t,x,y) = -b\,y. \tag{9.21}$$

The optimal feedback is *linear in the costate* — the probabilistic signature of a quadratic value function.

The MFG FBSDE. Substituting eq. (9.21) and the state derivative $\partial_x \mathcal{H} = ay + q(x - s\bar{m}_t)$ into eq. (9.13) gives the scalar McKean–Vlasov FBSDE

$$\begin{cases} \mathrm{d}X_t = \big(aX_t - b^2 Y_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, & X_0 \sim m_0, \\ \mathrm{d}Y_t = -\big(aY_t + q(X_t - s\bar{m}_t)\big)\,\mathrm{d}t + Z_t\,\mathrm{d}W_t, & Y_T = q_T\big(X_T - s_T\bar{m}_T\big), \\ \bar{m}_t = \mathbb{E}[X_t]. \end{cases} \tag{9.22}$$

Affine decoupling ansatz. Because the system is linear and the coupling enters only through the deterministic mean $\bar{m}_t$, we posit the affine decoupling field

$$Y_t = P_t\,X_t + r_t, \tag{9.23}$$

with deterministic P_t, r_t. Applying Itô to eq. (9.23) along the forward dynamics and matching with the backward dynamics of eq. (9.22) term by term: the martingale parts give $Z_t = P_t\sigma$, the coefficient of X_t gives the scalar Riccati equation

$$\dot{P}_t = b^2 P_t^2 - 2aP_t - q, \qquad P_T = q_T, \tag{9.24}$$

and the constant term gives the linear offset equation

$$\dot{r}_t = (b^2 P_t - a)r_t + q\,s\,\bar{m}_t, \qquad r_T = -q_T\,s_T\,\bar{m}_T. \tag{9.25}$$

Equation eq. (9.24) is *identical* to the Riccati equation for the curvature of the HJB value function in chapters 6 and 14: indeed, by the dictionary eq. (9.16), $Y_t = \partial_x u(t, X_t)$ with $u(t,x) = \frac{1}{2}P_t x^2 + r_t x + (\text{const})$, so P_t is exactly the value's second derivative and $Z_t = \sigma\,\partial_{xx}u = \sigma\,P_t$ matches the second line of eq. (9.16). The probabilistic and analytic pictures share one Riccati equation.

Closing the fixed point. Taking expectations of the forward equation in eq. (9.22) and using eq. (9.23) with $\mathbb{E}[Y_t] = P_t\bar{m}_t + r_t$ yields the mean evolution

$$\dot{\bar{m}}_t = (a - b^2 P_t)\bar{m}_t - b^2 r_t, \qquad \bar{m}_0 = \int \xi\, m_0(\mathrm{d}\xi). \tag{9.26}$$

Equations eq. (9.25) and eq. (9.26) form a *linear forward–backward two-point boundary value problem* in $(\bar{m}_t, r_t)$ — the deterministic shadow of the FBSDE, with $\bar{m}$ pinned at $t = 0$ and r at $t = T$. This is exactly the deterministic forward–backward system whose existence and uniqueness the LQG companion of chapter 10 analyzes; the curvature Riccati P_t, the offset r_t, and the mean $\bar{m}_t$ here are the same objects, in the same canonical symbols, that the explicit LQG fixed point of chapters 10 and 14 produces from the HJB/Fokker–Planck side. It is itself solved by a second decoupling: since the system is homogeneous and the terminal datum $r_T = -q_T s_T \bar{m}_T$ is proportional to $\bar{m}_T$, the ansatz $r_t = \pi_t\bar{m}_t$ reduces it to a second scalar Riccati,

$$\dot{\pi}_t = b^2\pi_t^2 - 2(a - b^2 P_t)\pi_t + q\,s, \qquad \pi_T = -q_T\,s_T, \tag{9.27}$$

after which the mean closes into a single linear ODE

$$\dot{\bar{m}}_t = (a - b^2 P_t - b^2\pi_t)\bar{m}_t, \qquad \bar{m}_0 \text{ given}, \tag{9.28}$$

solved in closed form by an exponential of the time integral of its coefficient. The full equilibrium is thus $(P_t, \pi_t, \bar{m}_t)$ from eq. (9.24), eq. (9.27), eq. (9.28), with $r_t = \pi_t \bar{m}_t$ and the feedback $\hat{\alpha}_t = -b(P_t X_t + \pi_t \bar{m}_t)$.

Consistency and the Gaussian law. With this linear feedback the forward equation is an Ornstein–Uhlenbeck process with Gaussian initial law, so X_t is Gaussian for all t and $m_t = \mathrm{Law}(X_t)$ is the Gaussian $\mathcal{N}(\bar{m}_t, v_t)$ whose variance v_t solves the linear (Lyapunov) ODE $\dot{v}_t = 2(a - b^2 P_t)v_t + \sigma^2$, $v_0 = \mathrm{Var}(m_0)$ — decoupled from the mean, as it must be, since the cost couples only to $\bar{m}_t$. This reproduces, from the probabilistic side, exactly the closed-form LQG equilibrium that chapter 14 obtains analytically; the carried thread is consistent across the two pictures, Riccati for Riccati. The sign regime in which eq. (9.27) admits a global solution (no finite-time blow-up of π_t) is the uniqueness regime; its boundary, where multiplicity sets in, is the phase diagram analyzed in chapters 11 and 14.

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Proved (in this chapter).

- The stochastic maximum principle for the representative agent against a frozen population: a full proof of necessity by the convex-variation/adjoint-duality argument (theorem 9.4) and a full proof of sufficiency under convexity (theorem 9.5), with the adjoint BSDE eq. (9.5) and the generalized Hamiltonian eq. (9.3).
- The FBSDE–PDE dictionary proposition 9.12: $Y_t = \nabla u(t, X_t)$, $Z_t = \mathrm{D}^2\, u(t, X_t)\, \sigma(t, X_t)$ (valued in $\mathbb{R}^{d\times k}$), with the reconciliation of the Pontryagin and HJB Hamiltonians (remark 9.11).
- The identification of the frozen-flow decoupling field with the value gradient, $U = \nabla u$ (eq. (9.18)), and the promotion to the measure-dependent master field (eq. (9.19)–eq. (9.20)).
- The complete LQG thread in FBSDE form (worked computation 9.16), with its two Riccati equations and closed-form mean, shown to share the Riccati curvature P_t with the analytic solution of chapters 8 and 14.

Assumed (named standing hypotheses).

- **(A-Lip)**, **(A-Growth)** for SDE/BSDE well-posedness and Lipschitz feedback; **(A-Conv)** for the existence and Lipschitz regularity of the Hamiltonian minimizer and for sufficiency of the maximum principle; **(A-Nondeg)** for parabolic smoothing of the decoupling field; **(A-Mon)** for global-in-T well-posedness and uniqueness.
- Uncontrolled diffusion $\sigma = \sigma(t, x, m)$ throughout (controlled diffusion deferred, remark 9.7).

Deferred (with pointers).

- Existence/uniqueness of the MFG FBSDE: stated as theorem 9.10 in three regimes; the Schauder/compactness proof is chapter 10, the monotone continuation proof and the pairing identity are chapter 11; primary source [40].
- Existence and Lipschitz regularity of the decoupling field (theorem 9.14): the small-horizon contraction is owned by chapter 10, the monotone a priori gradient bound extending it to every horizon by chapter 11, and the measure-dependent (master-field) version by chapter 12; the global-in-T degenerate case is not claimed here. Primary sources [40, 37].
- The master equation derived from the master field eq. (9.20): chapter 12.
- BSDE theory — the Pardoux–Peng existence/uniqueness theorem (theorem C.18) and the martingale representation theorem (theorem C.17) — is proved in full in the stochastic-calculus reference appendix C.

Open / honestly flagged.

- Controlled-diffusion MFG requires the second-order adjoint and is not treated here; general well-posedness there remains partial.
- Uniqueness outside the monotone or small-horizon regimes is genuinely open in general; theorem 9.10(iii) gives existence only, and the LQG thread (worked computation 9.16) shows multiplicity is a real possibility once eq. (9.27) loses its global solution.

Bibliographic notes

The probabilistic approach to mean field games via forward–backward stochastic differential equations is due to Carmona and Delarue, and the canonical reference for every result of this chapter is their two-volume monograph [40, 41]; Volume I, Chapters 1–4, contains the stochastic maximum principle for mean field control and games, the McKean–Vlasov FBSDE, the small-time and monotone well-posedness theorems, and the decoupling-field analysis, all in greater generality (controlled diffusions, common noise) than we have presented. The stochastic Pontryagin principle and the adjoint BSDE rest on the backward SDE theory of Pardoux and Peng, proved in full in appendix C; the decoupling-field (four-step / continuation) method for FBSDEs originates with Ma, Protter and Yong and with Delarue, and is exposited in [40]. The equivalence of the probabilistic and analytic formulations, and the identification of the costate with the gradient of the value function, is the FBSDE counterpart of the Lasry–Lions PDE system [93] treated analytically in chapter 8 following [36]; the Nash-certainty-equivalence origin of the same fixed point is [82]. The master field and the passage from the decoupling field to the master equation, sketched at the end of this chapter, are developed rigorously in [37] and in chapter 12. The LQG worked computation follows the explicit linear–quadratic theory that chapter 14 treats as the capstone of Part II; its probabilistic form, with the two Riccati equations, is standard and appears in [40] (Vol. I, Ch. 3) and the references therein. The graphon generalization of the FBSDE formulation, in which the representative agent becomes a continuum-indexed family coupled through the operator $\mathbb{W}$, is the subject of chapter 23.

Chapter 10

Existence of Mean Field Game Equilibria

The two preceding chapters built the mean field game (MFG) system in its two complementary pictures: the analytic, partial-differential picture of a backward Hamilton–Jacobi–Bellman (HJB) equation coupled to a forward Fokker–Planck (FP) equation (chapter 8), and the probabilistic, path-space picture of a forward–backward stochastic differential equation of McKean–Vlasov type (chapter 9). Each of those chapters wrote down, with care, what an equilibrium *is*: a population profile together with the optimal response to it that reproduces that very profile. What neither chapter did — and neither could, with the tools it carried — is take the next step and assert that such a profile can in fact be found. That step is the entire business of the present chapter, and it is worth pausing to see that it is a genuinely different kind of statement. To write the equations an object must satisfy is to *describe* the object; to assert that those equations have a solution is to claim the object *exists*, and the second claim does not follow from the first. One can write perfectly sensible equations that nothing satisfies — a polynomial with no real root, an ordinary differential equation whose solution escapes to infinity before the horizon is reached, a self-consistency relation with no self-consistent value. The coupled HJB–FP system of chapter 8 and the McKean–Vlasov FBSDE of chapter 9 are descriptions of exactly this kind: exact, but silent on whether anything answers to them. Supplying the missing existential quantifier — *there exists* a flow of measures that is its own best response — is the one task of this chapter.

We supply it the way one always must when no formula for the solution is on offer: not by constructing the equilibrium, but by recasting the equilibrium condition as a fixed point of a single map — the *best-response map* — and proving abstractly that the map has a fixed point. The instruments are the ones Part I assembled for exactly this purpose: the topological fixed-point theorems of chapter 2 (Schauder for single-valued maps, Kakutani–Fan–Glicksberg for set-valued ones), the compactness machinery of weak convergence and tightness from chapter 3, and the Wasserstein geometry of chapter 4, which turns the space of population profiles into a metric space on which those theorems can act. A topological fixed-point theorem is a deliberately modest instrument: it never tells us *which* profile is the equilibrium, never produces a formula, and says nothing about uniqueness. It certifies only that the set of equilibria is nonempty. That certificate is precisely, and only, what we are after; uniqueness is a separate question, deferred in full to chapter 11.

The architecture of the argument is one every physicist already owns under another name: it is the self-consistent field. Posit a field — here a candidate population profile m. Solve the one-body problem in that field: compute how a single representative agent, facing the frozen population m, optimally responds. Now let the entire ensemble respond in the identical way, since the agents are interchangeable, and read off the new profile that this collective response generates. Equilibrium is the demand that the profile read off at the end equal the profile posited at the start. This is the Curie–Weiss loop — guess a magnetisation, compute the molecular field it produces, solve the single-spin problem in that field, read off the average magnetisation, and require it to match the guess — transplanted from a static thermal average to a dynamic optimisation over an entire trajectory. The spine of the whole chapter is this loop, made into a map $m \mapsto \mathsf{S}(m)$ whose fixed points are the equilibria.

Everything subtle is concentrated in the single word *equals*. Asking two numbers to be equal is trivial; asking two elements of an infinite-dimensional space of measure-valued trajectories to be equal, when the map relating them is nonlinear and — this is the crux — in general *not* a contraction, is not. Were S a contraction, the self-consistent-field iteration $m^{(0)}$, $m^{(1)} = \mathsf{S}(m^{(0)}), \ldots$ would converge to a unique field and there would be nothing to prove beyond Banach's theorem; but generic MFGs admit no such contraction, the iteration may wander, and the field may have several self-consistent values. When contraction fails, the only remaining lever is *compactness*: one shows that the iteration cannot escape some compact set of profiles, so that the orbit must accumulate somewhere, and one arranges that an accumulation point is forced to be fixed. Compactness in a space of measures is never free — it must be manufactured — and the single structural hypothesis that manufactures it is the nondegeneracy of the diffusion, (A-Nondeg). *This is the hero of the chapter, and the reader should watch for it.* Here is its first entrance: the parabolic smoothing that a nondegenerate diffusion provides is exactly what converts a merely bounded, merely measurable optimal response into a *precompact* family of responses — one that Arzelà–Ascoli and Prokhorov can trap in a compact set (section 10.2).

That same hypothesis is then asked to do a second, independent job, and the reader should watch for it a second time. Trapping the orbit in a compact set delivers an accumulation point; but to conclude that the accumulation point is a *fixed* point one needs the map S to be *continuous*, so that the limit may be passed through it. Continuity of the best-response map is again a parabolic-regularity statement — it rests on the continuous dependence of the HJB solution on its frozen data, which is false without ellipticity — and so (A-Nondeg) returns, in section 10.3, to supply it. The hypothesis thus earns its billing twice over: once to make the arena compact, once to make the map continuous on it. Strip it away and both halves of the Schauder argument collapse at a single stroke; this is also why the LQG thread, whose diffusion is nondegenerate but whose *data* are unbounded, will need separate handling rather than falling out as a corollary.

With the strategy fixed, the chapter proceeds in four movements, each buying a piece of the argument at a stated price. Section 10.1 (this section) fixes the state space of candidate equilibria — flows of probability measures — and defines the best-response map precisely; its cost is a deliberate choice of topology, the order-one Wasserstein distance in place of the order-two distance natural in chapter 9, and its purchase is a sharply posed fixed-point question together with the single/set-valued dichotomy that organises everything after. Section 10.2 produces the convex compact set on which the map will be shown to act, from the moment and equicontinuity estimates that a bounded optimal feedback buys, with tightness (chapter 3) as the engine; its cost is an a priori bound on the feedback — exactly what fails for unbounded LQG data — and its purchase is the arena Schauder requires. Section 10.3 proves the map continuous, the one place where parabolic regularity (A-Nondeg) is genuinely indispensable, and assembles the Schauder existence theorem; its cost is the restriction to bounded, regularising couplings, and its purchase is existence *with no monotonicity assumed* and hence — honestly — no uniqueness claimed. Section 10.4 treats the harder case in which the optimal feedback is not unique: we relax controls to randomised ones, pass to a set-valued best response, invoke the Kakutani–Fan–Glicksberg theorem, and recover a genuine feedback by measurable selection; its cost is passage to weak equilibria carried on laws rather than flows, and its purchase is existence under dramatically weaker hypotheses. Throughout, uniqueness is *not* our concern — it is the subject of chapter 11 and rests on the Lasry–Lions monotonicity condition (A-Mon); here a game may well have many equilibria, and our theorems guarantee only that their set is nonempty. Section 10.5 closes the chapter by returning to the linear–quadratic–Gaussian (LQG) thread (Thread A): it shows that the thread *violates* the abstract theorem's hypotheses — its data are unbounded, so the compactness construction does not apply — yet collapses to a finite-dimensional linear two-point problem with a sharper existence-and-uniqueness statement, confirming that the explicit Gaussian flow built earlier is indeed *an* equilibrium and, at short horizon, the unique one. This is a deliberate counterpoint: the abstract theorem is not the last word, and the place it fails to reach is the very place where the multiplicity and symmetry-breaking story of chapter 11 and chapter 14 will begin.

Throughout this chapter no graphon is present, so we follow the Part I–II convention of appendix A and write Brownian motion as W_t.

10.1 Equilibrium as a fixed point

Turning the self-consistency loop into a fixed-point question takes four steps, and the four subsections below are precisely those steps, in the order any fixed-point argument needs them: the objects, the space they inhabit, the map, and the structure of the map. We first recall — not redefine — the data of the game and the coupled backward-HJB / forward-FP system that the analytic picture of chapter 8 hands us, since that system is the substrate on which the map is built and the place where the two pictures of chapters 8 and 9 are seen to describe one and the same equilibrium. We then choose the space in which candidate equilibria are sought — a space of measure-valued trajectories — and explain why its topology is the one metrised by the order-one Wasserstein distance rather than the order-two distance natural to the L^2 estimates of chapter 9. With the data and the arena fixed we define the best-response map S itself, as the two-stage operation *solve the frozen-flow HJB equation, then run the closed-loop dynamics it dictates*, and record that a fixed point of S is exactly an equilibrium. Finally we observe that S is a genuine single-valued map or only a set-valued correspondence according to whether the optimal feedback is unique — a dichotomy that cleaves the rest of the chapter into the Schauder route (sections 10.2 and 10.3) and the Kakutani route (section 10.4). What this section hands forward is therefore a precisely defined map S on the ambient space $\mathcal{M}$, together with the demand — discharged in the next section — for a nonempty convex compact set on which to run it, and, until section 10.4, the standing restriction to the single-valued, strictly convex regime.

10.1.1 The data of the game and the MFG system

The chapter preamble committed to a claim, and this subsection's task is to make it concrete enough to build on. The claim is that the analytic backward-HJB / forward-FP system of chapter 8 and the probabilistic McKean–Vlasov FBSDE of chapter 9 are not two equilibria but two descriptions of one: a single self-consistency condition, viewed in two coordinate systems. To turn that condition into a fixed point we need one concrete substrate on which to erect the best-response map, and of the two pictures the analytic one is the more convenient launching pad — it presents the equilibrium as a pair of coupled equations whose coupling we can literally *freeze*, solve, and re-read. So we begin by recalling, deliberately rather than redefining, the objects that chapters 8 and 9 already own; the reader who has those chapters in hand should find nothing new here except the framing.

The representative agent has state $X_t \in \mathbb{R}^d$ on the horizon $[0, T]$, with controlled dynamics

$$\mathrm{d}X_t \;=\; -\,\partial_p H\big(X_t, \nabla u(t, X_t)\big)\;\mathrm{d}t \;+\; \sigma(X_t)\;\mathrm{d}W_t, \qquad X_0 \sim m_0, \tag{10.1}$$

where $\sigma : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ is the volatility and $H = H(x, p)$ is the Hamiltonian of the control problem. It is worth pausing on why the drift is the particular expression $-\partial_p H$ rather than some free control still to be chosen: the drift in eq. (10.1) is *already optimised*. In the canonical MFG the agent steers its own drift directly, $\mathrm{d}X_t = \alpha_t\;\mathrm{d}t + \sigma(X_t)\;\mathrm{d}W_t$, paying a running cost $L(x, \alpha)$ for the effort, and the Hamiltonian is the Legendre transform of that cost in the control,

$$H(x, p) \;=\; \sup_{\alpha}\big\{-\alpha\cdot p - L(x, \alpha)\big\}.$$

Differentiating under the envelope theorem — only the explicit p-dependence survives, because the α-derivative vanishes at the maximiser — gives $\partial_p H(x, p) = -\alpha^*(x, p)$, so the maximising control is $\alpha^*(x, p) = -\partial_p H(x, p)$. The verification theorem of chapter 6, valid under convexity (A-Conv), certifies that substituting this feedback (evaluated along the value function's gradient $p = \nabla u(t, X_t)$) into the dynamics is genuinely optimal. That substitution is exactly what eq. (10.1) records: the open-loop control has been replaced by the closed-loop optimal-feedback drift $-\partial_p H(\cdot, \nabla u)$, the optimal-feedback map of appendix A. The state equation thus already encodes the agent's best response to whatever value function u it is handed.

What makes this a *game* rather than a lone control problem is that the value function u is not given from outside; it is shaped by the population. The population is described by a *flow of measures* $t \mapsto m_t$ — the time-indexed distribution of agents over state space — and the representative agent's costs are coupled to it in two places. The *running coupling* $F[m]$ records how the instantaneous cost an agent pays at time t depends on where everyone else is then; the *terminal coupling* $G[m]$ records how the cost charged at the horizon depends on the final configuration m_T. We write both with the measure inside square brackets and the spatial argument outside, $F[m](x)$ and $G[m](x)$, a deliberate piece of notation: the brackets signal a *functional*, generally nonlocal, dependence on the entire measure m, while x is the ordinary pointwise evaluation of the resulting function of state. This is the same bracket-outside convention under which the Lasry–Lions pairing is written in the thread specification and in chapter 11, and F, G are precisely the objects whose monotonicity is named (A-Mon) and exploited there for uniqueness — a property we neither assume nor use in the present chapter.

Assembling the optimised dynamics and the population coupling, the MFG system recalled from chapter 8 is the coupled pair

$$-\partial_t u - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 u\big) + H(x, \nabla u) = F[m_t](x), \qquad u(T, \cdot) = G[m_T](\cdot), \tag{10.2}$$

$$\partial_t m - \tfrac{1}{2}\sum_{i,j} \partial^2_{ij}\big((\sigma\sigma^\top)_{ij}\, m\big) - \operatorname{div}\big(m\, \partial_p H(x, \nabla u)\big) = 0, \qquad m_0 \text{ given.} \tag{10.3}$$

The two equations are mirror images that move in opposite directions in time. As stressed in appendix A and chapter 8, the HJB equation eq. (10.2) is solved *backward*: its data live at the terminal time $t = T$, where $u(T, \cdot) = G[m_T]$, and the value propagates information about the future cost landscape back toward the present, telling each agent how to steer now in anticipation of where the crowd will be later. The Fokker–Planck (continuity) equation eq. (10.3) is solved *forward*: its datum is the initial distribution m_0 at $t = 0$, and it transports the population's density ahead in time under the very drift $-\partial_p H(x, \nabla u)$ that the value function dictates. The data therefore sit at *opposite ends* of the interval $[0, T]$, and the equations are bolted together — m feeds the HJB source and terminal data, u feeds the FP drift — across that whole span. It is this two-point, backward-forward coupling, and not either parabolic equation taken alone, that is the defining structural feature of the game: each equation in isolation is a textbook problem, but the demand that they hold *simultaneously*, with the forward density generating the backward cost that in turn generates the forward density, is precisely the self-consistency that the chapter throughline identifies with a fixed point. The feedback closing the loop is

$$\alpha^*(t, x) \;=\; -\,\partial_p H\big(x, \nabla u(t, x)\big), \tag{10.4}$$

the optimal-feedback map $\alpha^* = -\partial_p H$ of appendix A: a single well-defined function — the unique maximiser — when (A-Conv) holds with *strict* convexity, and a possibly set-valued object when convexity is merely weak. That dichotomy, latent here in the $\arg\max$ behind eq. (10.4), is the fork that organises the whole chapter, sending the strictly convex case down the Schauder route and the degenerate case down the Kakutani route of section 10.4.

Notation watch: Where the measure sits, and the structural restriction it carries

The master convention of appendix A and the systems of chapters 8 and 9 carry the measure *inside* the Hamiltonian, $H = H(x, p, m)$ with a single terminal cost $g(x, m)$. Throughout this chapter we use the equivalent *separable* decomposition that pulls the measure-dependence out into the additive couplings,

$$H(x, p, m) \;=\; H(x, p) \;+\; F[m](x), \qquad g(x, m) \;=\; G[m](x),$$

so that the frozen-flow Hamiltonian $H(x, p)$ is genuinely measure-free and the entire coupling is transparent in F, G. This is not a cosmetic rewriting, and it is worth spelling out so that a reader crossing chapter 9→chapter 10 is not ambushed. In chapter 9 the central object was a Hamiltonian of *three* arguments, $H(x,p,m)$, the population entering on the same footing as state and momentum; here, abruptly, the Hamiltonian appearing in eqs. (10.1) to (10.4) has only *two*, $H(x,p)$. Taken at face value the central object would seem to have silently shed an argument — to have changed arity between two chapters that are supposed to describe the same game — and a careful reader would be right to be alarmed. The reconciliation is mechanical but must be stated. Separability means the measure enters *additively* and through the *state slot only*: the three-argument Hamiltonian of chapters 8 and 9 is recovered as $H_{chapter\ 8}(x,p,m) = H(x,p)+F[m](x)$, with $G[m] := g(\cdot,m)$, the second "missing" argument having been relocated into the explicit source term $F[m]$ of the HJB equation rather than deleted. Because the added piece $F[m](x)$ does not depend on p, the momentum derivative is untouched, $\partial_p H_{chapter\ 8}(x,p,m) = \partial_p H(x,p)$: the optimal-feedback drift eq. (10.4) is exactly the measure-free one, and the population's only influence on an agent's motion is through the additive cost landscape $F[m]$, never through the agent's responsiveness to its own momentum. This is the standard *separable* (non-congestion) setting of analytic MFG and is exactly the class isolated in definition 8.5. What it *excludes* is precisely the case where the crowd modulates the momentum dependence itself — an m-dependent control cost $L(x,\alpha,m)$ that makes manoeuvring more expensive in a dense region, i.e. true congestion — for there $H(x,p,m)$ is genuinely non-additive in m, the drift $\partial_p H(x,p,m)$ feels the population directly, and the clean freeze-and-solve structure below breaks down. Such couplings demand the displacement-monotonicity refinements developed in chapter 11, and we defer them there; all existence statements in the present chapter are for the separable class unless stated otherwise.

The separable structure is not merely a bookkeeping convenience: it is what makes the best-response map definable in the first place. Read eq. (10.2) once more with the population trajectory m held *fixed* inside the couplings. Then the right-hand source $(t,x) \mapsto F[m_t](x)$ and the terminal data $x \mapsto G[m_T](x)$ are simply *given* functions, carrying no live dependence on the unknown u, and the HJB equation *decouples* from the Fokker–Planck equation: it becomes a single backward parabolic problem with prescribed data, of exactly the kind chapters 6 and 8 know how to solve. This frozen-flow viewpoint — freeze m inside F and G, watch the HJB equation detach — is the hinge on which the entire construction turns, and the next two subsections build directly on it: solving the frozen problem and then transporting the resulting law forward is what the best-response map *does*. But before we can speak of freezing a flow and varying it, we must say precisely what kind of object a candidate flow m is, and in what space it lives — with what notion of distance two candidate populations are deemed close. That is the question the next subsection takes up.

10.1.2 The space of candidate equilibria

The previous subsection left us with a recipe we are not yet equipped to execute: freeze the population trajectory m inside the couplings F and G, watch the HJB equation detach into a decoupled backward parabolic problem, solve it, and transport the resulting law forward. To make that recipe into a *map* — something we can iterate, bound, and hand to a fixed-point theorem — we must first pin down what kind of object is being frozen and varied. What, precisely, is a "candidate population trajectory"? In what set does it range? And, the question that turns out to govern everything downstream, with what notion of nearness are two such candidates judged close? This last choice is not a technicality to be settled in passing. The instruments we intend to deploy — Schauder, Kakutani–Fan–Glicksberg — speak entirely in the vocabulary of a topology: compactness, continuity, and convexity are all properties of a set or a map *relative to* a notion of convergence, and a topology chosen too strong will leave our candidate set non-compact, while one chosen too weak will leave the map discontinuous. The present subsection fixes the arena and its metric, and — because the natural-looking choice is the wrong one here — explains at length why we depart from the distance that served chapter 9 so well.

A candidate equilibrium is, before anything else, a population trajectory. We make this the primary unknown.

Definition 10.1 (Flows of measures)**.** For $p \geq 1$, let $\mathcal{P}_p(\mathbb{R}^d)$ carry the Wasserstein distance $\mathcal{W}_p$ of chapter 4. A *flow of measures* is a continuous map $m = (m_t)_{t\in[0,T]} \in C\big([0,T];\mathcal{P}_p(\mathbb{R}^d)\big)$, the latter equipped with the uniform metric

$$\mathsf{d}_p(m, m') \;=\; \sup_{t\in[0,T]} \mathcal{W}_p(m_t, m'_t).$$

We write $\mathcal{M} := C\big([0,T];\mathcal{P}_1(\mathbb{R}^d)\big)$ for the ambient space of the existence argument and reserve the second moment as an a priori bound rather than as part of the topology.

Three features of definition 10.1 deserve to be read slowly, because each is a decision rather than a reflex. First, the unknown is a *trajectory of measures*, not a single measure: an equilibrium of a dynamic game is a whole history $t \mapsto m_t$ of how the crowd is distributed at each instant, and the self-consistency we are chasing must hold at every time simultaneously. Second, we demand the trajectory be *continuous* in time — no instantaneous jumps in the population — and we measure two trajectories by the *uniform* (sup-over-time) distance d_p, the supremum of the instantaneous Wasserstein gaps. The uniform metric is the right one because it makes $C([0,T];\mathcal{P}_p)$ *complete* whenever $\mathcal{P}_p$ is: a d_p-Cauchy sequence of flows is, instant by instant, a $\mathcal{W}_p$-Cauchy sequence of measures with a uniform-in-time modulus, so the pointwise limits exist and inherit continuity, and the limit is again a flow. Completeness is not optional decoration; it is what lets us pass to limits inside the space rather than falling out of it, and every compactness statement below is ultimately a statement about which subsets of this complete space are precompact. Third — and this is the decision the rest of the subsection defends — the ambient space is built on $\mathcal{P}_1$, with $\mathcal{W}_1$ supplying the topology, while the second moment is held back as a separate, quantitative bound that never enters the metric.

That third decision is worth stating as a slogan before we justify it: $\mathcal{W}_1$ *is the topology; the second moment is a leash.* The topology is the weak notion of nearness in which we will demand compactness and continuity; the leash is a strong quantitative estimate — a uniform ceiling on $\int |x|^2\, m_t(\mathrm{d}x)$ — that we carry along to prevent mass from escaping to infinity, but which we deliberately refuse to fold into the definition of "close." The reason for the split is structural and will recur at every step. Compactness in a space of measures is manufactured from a moment bound through tightness and Prokhorov's theorem (chapter 3); a uniform second moment is exactly the lever that buys it. But a moment bound buys *tightness*, which is relative compactness for *weak* convergence, and weak convergence is a weak topology. If we were to insist on a strong metric — one demanding that the leash itself converge — we would be asking for precisely the kind of convergence that tightness does not deliver, and the compactness we worked to obtain would evaporate at the last step. Keeping the moment out of the metric is what lets the leash do its one job (trapping mass) without contaminating the topology in which limits are taken.

The choice of $\mathcal{W}_1$ for that topology is not merely the path of least resistance; it pays off twice, and both payoffs are cashed in the sections that follow. The first payoff is that, on exactly the sets we will work over, $\mathcal{W}_1$-convergence and weak convergence are *the same thing.* On any family of measures with uniformly integrable first moments — in particular any family carrying a uniform bound on a moment of order strictly greater than one, and our second-moment leash is such a bound — $\mathcal{W}_1$ metrises weak convergence (theorem D.11). The practical consequence is decisive: when tightness and Prokhorov hand us a weakly convergent subsequence, that subsequence is *automatically* $\mathcal{W}_1$-convergent, with no further work. Compactness in the metric we actually care about thus comes free the instant we have a moment bound, which is precisely why the moment bound is the only nontrivial input to the compact-set construction of section 10.2. The second payoff is dual in character. By the Kantorovich–Rubinstein representation (theorem D.10), $\mathcal{W}_1(\mu,\nu)$ equals the supremum of $\int f \, \mathrm{d}(\mu-\nu)$ over all 1-Lipschitz test functions f. This is exactly the language the stability estimates of section 10.3 speak. Those estimates control $\mathsf{S}(m)$ by coupling two controlled processes and bounding $\mathbb{E}|X_t^m - X_t^{m'}|$ — a first-moment, 1-Lipschitz-tested quantity — so the metric and the estimate are matched at the source: $\mathcal{W}_1$ is, by duality, the smallest distance the coupling estimate can possibly produce, and no order is wasted converting between them.

Remark 10.2 (Why not $\mathcal{W}_2$, the distance of chapter 9?)**.** A reader fresh from chapter 9 has every right to expect the order-two distance here too. There the FBSDE well-posedness of theorem 9.10 ran a *small-time contraction* of the best-response map in the $\mathcal{W}_2$-sup metric on $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$, and that was the natural choice: the probabilistic estimates were L^2 estimates — Itô isometries, second-moment Grönwall bounds — and $\mathcal{W}_2$, being the L^2 transport cost, is the metric in which an L^2 contraction reads off cleanly. When the engine is contraction, one wants the *strongest* metric in which the map still contracts, because a contraction in a stronger metric is a stronger statement and the completeness needed for Banach's theorem comes for free. The compactness route inverts this preference exactly. Here there is no contraction to exploit; the engine is Prokhorov, and what we need is for our bounded candidate set to be *precompact.* Precompactness is easiest in the *weakest* reasonable metric, and so we want the weaker distance, not the stronger one.

The reason the strong distance actively fails is worth stating in full, since it is the crux of the whole topological choice. On a set of measures bounded only in the second moment, $\mathcal{W}_2$-convergence is strictly stronger than weak convergence: $\mathcal{W}_2(\mu_n,\mu)\to 0$ holds if and only if $\mu_n\to\mu$ weakly *and* the second moments converge, $\int |x|^2 \, \mathrm{d}\mu_n \to \int |x|^2 \, \mathrm{d}\mu$, the latter amounting to the demand that no second-moment mass leak out along the tails — and a uniform *ceiling* on the second moment does nothing to force the second moments to *converge*, since mass can slosh out to large radii while staying under the ceiling. Consequently the weakly convergent subsequences that Prokhorov manufactures from our moment bound need not be $\mathcal{W}_2$-convergent at all: the compactness we can honestly produce is $\mathcal{W}_1$-compactness, never $\mathcal{W}_2$-compactness, on a merely second-moment-bounded set. (To metrise weak convergence by $\mathcal{W}_2$ one would need a uniform bound on a moment of order strictly greater than *two*, a leash we have no a priori means to tighten.) There is a second, independent reason to avoid $\mathcal{W}_2$ even setting compactness aside: $\mathcal{W}_2$-continuity of the best-response map is more than we can cheaply guarantee, since it would require controlling the convergence of second moments of the output law under perturbations of the frozen flow, whereas the 1-Lipschitz coupling estimate delivers $\mathcal{W}_1$-continuity directly and for free. Both halves of the Schauder argument — compactness of the arena and continuity of the map — therefore point the same way, toward the order-one distance.

With definition 10.1 we have fixed the arena. The ambient space is $\mathcal{M} = C([0,T];\mathcal{P}_1(\mathbb{R}^d))$, complete under the uniform order-one metric d_1; the second moment rides along as an a priori leash that will, in section 10.2, be spent on tightness and on equicontinuity in time to carve a convex compact subset out of $\mathcal{M}$. This is exactly the concrete domain and codomain the next subsection needs: it takes the freeze-and-solve recipe of the frozen-flow viewpoint and reads it as an honest self-map of $\mathcal{M}$ — freeze a flow $m \in \mathcal{M}$, solve the decoupled HJB equation it sources, close the loop by transporting m_0 forward under the optimal feedback, and land on a new flow $\mathsf{S}(m) \in \mathcal{M}$ — in the very metric d_1 in which its continuity (section 10.3) and the invariant set's compactness (section 10.2) will be proved.

10.1.3 The best-response map

The previous subsection fixed the arena: the ambient space $\mathcal{M} = C([0,T];\mathcal{P}_1(\mathbb{R}^d))$, complete under the uniform order-one metric d_1, with the second moment carried along as an a priori leash. We now have a domain and a codomain, and a precise notion of when two candidate populations are close. What we do *not* yet have is the object whose fixed points we are after: the map that turns one candidate flow into another. Building that map is the business of this subsection, and it amounts to nothing more than executing, as an honest self-map of $\mathcal{M}$, the freeze-and-solve recipe that the frozen-flow viewpoint of section 10.1 has been promising. The recipe has three movements — freeze, solve, close the loop — and we walk through each in turn, recording at every step why the operation is licensed by results already in hand, so that the map we end with is not a formal stipulation but a genuinely constructed object.

Freeze. Fix a candidate flow $m \in \mathcal{M}$ and substitute it into the coupling terms of eq. (10.2) wherever the population appears. The running source collapses to the *given* function $(t,x) \mapsto F[m_t](x)$ — a prescribed inhomogeneity, no longer carrying any live dependence on the unknown value u — and the terminal data collapses to the given function $x \mapsto G[m_T](x)$. This is exactly the decoupling that separability bought us in section 10.1: because the measure entered F and G additively and through the state slot only, freezing m removes the population from the equation entirely except as known data, and the two-point backward–forward coupling that made the system a *game* is, for the moment, broken.

Solve. What remains is a single equation. The HJB equation eq. (10.2), with its couplings frozen, is now a *decoupled*, semilinear, backward parabolic problem with prescribed source and terminal data — precisely the class chapters 6 and 8 are equipped to handle. Here (A-Nondeg) makes its first working appearance in the construction: nondegeneracy of $\sigma\sigma^\top$ renders the equation *uniformly parabolic*, which is what guarantees a classical $C^{1,2}$ solution rather than a mere viscosity solution, and a classical solution is what we need for the gradient ∇u in the feedback to exist pointwise. The control-theoretic existence and verification theory of chapter 6, valid under the convexity (A-Conv), then returns two things: a value function u^m solving the frozen HJB equation, and — through the optimal-feedback formula eq. (10.4) read off from u^m — an optimal feedback $\alpha^{*,m}(t,x) = -\partial_p H(x, \nabla u^m(t,x))$. The verification theorem is what certifies that this feedback is not merely a formal critical point but genuinely the optimal response of a representative agent who believes the population will follow m.

Close the loop. The agent now *acts* on this optimal feedback, and so does every other agent, since they are interchangeable. Feeding $\alpha^{*,m}$ into the closed-loop dynamics eq. (10.1) gives a well-posed stochastic differential equation with drift $-\partial_p H(x, \nabla u^m)$ — bounded and (under the standing Lipschitz hypotheses) Lipschitz in x, hence admitting a unique strong solution X^m started from $X_0^m \sim m_0$. Equivalently, and this is the analytic face of the same operation, one solves the Fokker–Planck equation eq. (10.3) with that *given* drift and initial datum m_0; with the drift now prescribed, eq. (10.3) is a *linear* continuity equation, no longer coupled to anything. Either way the output is a new flow of measures, the law of the optimally controlled state at each instant. Call it $\mathsf{S}(m)$. The loop has closed: we posited m, the population responded optimally to it, and $\mathsf{S}(m)$ is the profile that response generates.

That this output is again a legitimate element of $\mathcal{M}$ — a continuous flow in $\mathcal{P}_1$ — is not automatic, and it is precisely what the compactness section will verify: the moment bound of lemma 10.6 keeps $\mathsf{S}(m)_t$ in $\mathcal{P}_1$ at every instant, and the time equicontinuity of lemma 10.7 makes $t \mapsto \mathsf{S}(m)_t$ continuous in $\mathcal{W}_1$. For now we take those for granted and record the map.

We have, in effect, re-derived on the analytic side the best-response map already introduced in the path-space picture of eq. (9.14). We re-present it here for the ambient space $\mathcal{M}$ of definition 10.1.

Definition 10.3 (Best-response map and equilibrium)**.** The *best-response map* (or *Nash map*) is the assignment

$$\mathsf{S} : \mathcal{M} \longrightarrow \mathcal{M}, \qquad \mathsf{S}(m) \; = \; \Big(\mathrm{Law}\,(X_t^m) \Big)_{t \in [0,T]},$$

where X^m solves the closed-loop SDE eq. (10.1) with $u = u^m$ the solution of the decoupled HJB equation eq. (10.2) for the frozen flow m, and $X_0^m \sim m_0$. A *mean field game equilibrium* is a flow $m^\star \in \mathcal{M}$ with

$$\mathsf{S}(m^\star) \; = \; m^\star.$$

This is the same object as eq. (9.14); only the topology in which we shall run the fixed point differs. Theorem 9.10(i) ran the *small-time* contraction in the $\mathcal{W}_2$-sup metric on $C([0,T]; \mathcal{P}_2(\mathbb{R}^d))$, the natural home for the L^2 probabilistic estimates there; here, where the engine is compactness rather than contraction, we deliberately work in the weaker $\mathcal{W}_1$-sup metric of definition 10.1 (one always has $\mathcal{W}_1 \le \mathcal{W}_2$, and on sets carrying a uniform bound on a moment of order strictly greater than one — in particular the uniform second-moment bounds in force here — $\mathcal{W}_1$ metrises weak convergence, chapter 4; note that $\mathcal{W}_2$ itself does *not* metrise weak convergence on merely second-moment-bounded sets, which is precisely why the compactness argument needs the weaker distance), and this is what the Kantorovich–Rubinstein duality below will exploit.

A fixed point of S is exactly a solution of the coupled system eq. (10.2)–eq. (10.3): the flow used to compute the value equals the flow that the resulting optimal behaviour generates, which is the self-consistency condition. The whole of the existence question is therefore:

does the map S of definition 10.3 have a fixed point?

Heuristic (non-rigorous)

The map S *is* the self-consistent-field (SCF) iteration of mean field theory, transplanted from a static average to a dynamic optimisation. Run it as an actual loop. Start from any guessed population trajectory $m^{(0)} \in \mathcal{M}$ — the constant flow $m_t^{(0)} \equiv m_0$ will do. Solve the one-body problem in that frozen field, i.e. the decoupled HJB equation eq. (10.2) with couplings $F[m_t^{(0)}]$, $G[m_T^{(0)}]$, to obtain the value $u^{(0)} = u^{m^{(0)}}$ and the feedback it dictates; let the whole interchangeable ensemble respond by that feedback and read off the law it generates,

$$m^{(1)} = \mathsf{S}(m^{(0)}), \qquad m^{(2)} = \mathsf{S}(m^{(1)}), \qquad \dots, \qquad m^{(k+1)} = \mathsf{S}(m^{(k)}).$$

This is the SCF orbit. Two regimes are possible, and the whole strategic choice of the chapter is which one we are in. *(i) Contraction.* If S is a contraction in d_1, that is $\mathsf{d}_1(\mathsf{S}(m), \mathsf{S}(m')) \le \kappa\, \mathsf{d}_1(m, m')$ with some $\kappa < 1$, then the orbit is Cauchy — $\mathsf{d}_1(m^{(k+1)}, m^{(k)}) \le \kappa^k \mathsf{d}_1(m^{(1)}, m^{(0)})$ — so by completeness of $\mathcal{M}$ it *converges*, and its limit is the *unique* fixed point: existence and uniqueness at one stroke, by Banach rather than Schauder, with the iteration itself a convergent algorithm for the equilibrium. This is the favourable regime — monotone or weakly coupled data, or a short horizon T that starves the feedback–density feedback loop of time to amplify — and we will watch it operate concretely for the LQG thread at short horizon in section 10.5, where S collapses to a genuine contraction on a finite-dimensional parameter and the fixed point can be solved for in closed form. *(ii) No contraction.* Generic MFGs supply no such $\kappa < 1$. Then the orbit $(m^{(k)})$ carries *no* convergence guarantee: it may oscillate, cycle, or drift, and the field may possess several distinct self-consistent values among which the iteration never settles. All one can still salvage is the weaker fact that the orbit cannot *escape* — the moment and equicontinuity bounds of section 10.2 confine every $m^{(k)}$ to one compact set $\mathcal{K}$ — so the sequence must at least *accumulate* somewhere in $\mathcal{K}$, even if it never converges as a whole. Compactness buys an accumulation point; continuity of S (section 10.3) then forces that accumulation point to be *fixed*. This is the Schauder/Kakutani route, and the gap between "converges" and "merely stays in a compact set" is exactly the gap between Banach's theorem and the topological fixed-point theorems that the rest of the chapter is built to cross.

Physics note

The SCF loop $m \mapsto \mathsf{S}(m)$ is the exact dynamic analogue of the Curie–Weiss self-consistency loop, but the one-body problem sitting inside each iteration is of a fundamentally different character, and it is worth being precise about the difference. In Curie–Weiss mean field theory the self-consistent magnetisation $m = \tanh(\beta_{\mathrm{T}} J m)$ is fixed by a *static* averaging postulate: posit a molecular field Jm, solve the single-spin problem in it — a one-line thermal average, $\langle \sigma \rangle = \tanh(\beta_{\mathrm{T}} J m)$ — and demand consistency. The one-body solve is an *instantaneous* computation; there is no time in it, and the only equation is the scalar transcendental fixed point.

Here the one-body solve is not instantaneous. To compute a single representative agent's optimal response to a frozen flow m one must solve the HJB equation eq. (10.2) on the entire interval $[0, T]$ — a *boundary-value problem in time*, not a static minimisation. The value $u^m(t, x)$ obeys a backward parabolic equation whose only data sit at the terminal time, $u^m(T, \cdot) = G[m_T]$, and the solution propagates that terminal information backward to determine how the agent should steer *now* in anticipation of the cost it will face *later*: the optimal action at time t is fixed by events at all times $s > t$. The density, meanwhile, is transported *forward* by eq. (10.3) from its datum m_0 at $t = 0$. So even at the level of a single SCF iteration — before any self-consistency is imposed — the computation already threads information in both directions of time: backward through the value, forward through the law, with the data pinned at *opposite ends* of $[0, T]$. The static thermal average has been replaced by a two-point problem in time, and the fixed-point demand $\mathsf{S}(m) = m$ then couples this backward object to this forward one. This is why no equilibrium thermodynamics furnishes the relevant intuition: a thermal average is memoryless and instantaneous, whereas an optimal control is anticipatory and spread over the horizon. The SCF *outer* loop is faithfully Curie–Weiss; the *inner* solve is a boundary-value problem with no thermal-equilibrium counterpart. Multiplicity of self-consistent fields, when it occurs, is nonetheless the genuine dynamic counterpart of symmetry breaking, and we return to it in chapter 11.

The existence question is now sharply posed, and we have a constructed map to ask it of. But before we can deploy any fixed-point theorem against S, one structural feature of the construction must be confronted, because it determines *which* theorem we may use. The whole of S pivots on the feedback eq. (10.4), $\alpha^{*,m}(t, x) = -\partial_p H(x, \nabla u^m(t, x))$, and we quietly assumed in the *solve* step that this expression names a single well-defined function — that $\partial_p H(x, \cdot)$ returns one value, not a set. When it does, S is a genuine single-valued map and Schauder's theorem is the right instrument; the compact invariant set of section 10.2 and the continuity estimate of section 10.3 then complete the argument. When it does not — when the $\arg\max$ hidden inside $\partial_p H$ can return several optimal actions — S ceases to be a map at all and becomes a set-valued correspondence, and a different, set-valued fixed-point theorem must take over. The next subsection examines the feedback formula precisely where this fork lies, asking exactly when its $\arg\max$ is a singleton and when it is not, and so cleaves the remainder of the chapter into the Schauder route and the Kakutani route.

10.1.4 Two regimes: single-valued versus set-valued response

The previous subsection ended by isolating the one feature of the construction that decides *which* fixed-point theorem we are entitled to deploy. Everything in S funnels through the feedback eq. (10.4), $\alpha^{*,m}(t, x) = -\partial_p H(x, \nabla u^m(t, x))$, and that expression is innocent only if it names *one* number for each (t, x). But $-\partial_p H(x, p)$ is not a primitive datum of the game; it is shorthand for an optimisation.

Recall from the derivation of eq. (10.1) that the Hamiltonian is the Legendre–Fenchel transform of the running cost in the control, $H(x,p) = \sup_{a\in A}\{-a\cdot p - L(x,a)\}$, so that the maximising action is the *argmax*

$$\alpha^*(x,p) \;\in\; \arg\max_{a\in A} \{ -\; a\cdot p - L(x,a)\}, \qquad p = \nabla u^m(t,x), \tag{10.5}$$

and the clean differential identity $\alpha^* = -\partial_p H$ is licensed *only* when this arg max is a singleton and $H(x,\cdot)$ is consequently differentiable. The question this subsection settles is exactly: when is the set in eq. (10.5) a single point, and when can it contain several? The answer is the fork in the road for the entire chapter, because a single point makes S a bona fide *map* and hands the argument to Schauder, whereas a genuine set makes S a *correspondence* and forces the heavier set-valued machinery.

When the argmax is a singleton. The decisive sufficient condition is *strict* convexity of the running cost $L(x,\cdot)$ — equivalently, strict convexity of the Hamiltonian $H(x,\cdot)$ in the momentum, the strong form of (A-Conv). The picture is the elementary one behind the Legendre transform: maximising the concave function $a \mapsto -a\cdot p - L(x,a)$ over the action set is locating the tangency where a hyperplane of slope $-p$ first touches the strictly concave graph, and a *strictly* concave graph admits exactly one such tangency. Strict convexity therefore makes $H(x,\cdot)$ not merely differentiable but C^1 with a single-valued gradient, so eq. (10.5) collapses to the lone maximiser $\alpha^*(x,p) = -\partial_p H(x,p)$, a genuine function of (x,p). The carried LQG thread is the cleanest instance: there $L(x,a) = \frac{1}{2}|a|^2$ (up to the linear state coupling), the Hamiltonian is the quadratic $H(x,p) = \frac{1}{2}|p|^2 + \cdots$, strictly convex in p with the explicit unique maximiser $\alpha^* = -p$, and no ambiguity can arise anywhere. Whenever the cost is uniformly convex the feedback $\alpha^{*,m}$ is thus a single well-defined function, X^m is the unique strong solution of one closed-loop SDE, and $\mathsf{S}(m)$ is a single flow — the regime in which S is honestly a self-map of $\mathcal{M}$.

Three ways single-valuedness can fail. Drop strict convexity and three distinct mechanisms can each return more than one optimal action on a set of (t,x) of positive measure; it is worth seeing each as a separate picture, because they are repaired by different devices in section 10.4, and the reader who can name the failure can anticipate the cure.

(i) *Merely convex H (a flat maximiser set).* If $L(x,\cdot)$ is convex but not *strictly* so, its graph carries an affine stretch; dually, $H(x,\cdot)$ has a corner, or equivalently $-\partial_p H$ is replaced by a set-valued subdifferential. The tangent hyperplane of slope $-p$ can then kiss the graph along a whole flat segment, and eq. (10.5) is that entire segment — a convex set of equally-optimal actions rather than a point. The agent is genuinely indifferent among a range of drifts. *Picture: a Hamiltonian with a kink, its argmax an interval.* Here the maximiser set is already convex and nonempty, so nothing needs convexifying; what is needed is only to *pick* one optimiser per (t,x) in a measurable way, and that is the job of the measurable-selection step.

(ii) *Non-convex action set (the optimiser sits between two actions).* If A itself is not convex — two isolated allowed drifts, say, with nothing in between — the set of *attainable* averaged dynamics need not be convex, and the maximisation can prefer a value strictly between two pure actions that the agent is not permitted to play. This is precisely the obstruction by which a finite game can lack a pure-strategy Nash equilibrium: the best response "wants to mix." *Picture: a double-well payoff with the optimum stranded on the forbidden ridge between the wells.* Selection alone cannot help, because no single admissible action is optimal; the repair must first *convexify* the problem by allowing the agent to randomise — the relaxed (Young-measure) controls of definition 10.17 — restoring an optimiser in the convex hull before any selection is attempted.

(iii) *Degenerate diffusion (no classical gradient to feed the formula).* If (A-Nondeg) fails and $\sigma\sigma^\top$ is allowed to degenerate, the HJB equation eq. (10.2) loses its uniform parabolicity and u^m is in general only a *viscosity* solution, possessing no classical gradient ∇u^m at every point. The feedback eq. (10.4) then has no pointwise meaning on the (possibly large) set where u^m is merely Lipschitz with a kink, and $\arg\max$ in eq. (10.5) is read against a momentum p that is itself set-valued. *Picture: a value landscape with creases, the gradient — and hence the prescribed drift — undefined along each crease.* This is the failure that cannot be wished away by any convexity of H; it is repaired by abandoning the deterministic-flow formulation altogether and seeking a *weak* equilibrium carried on laws, where the control enters only through the martingale problem of definition 10.18 and need never be read off a gradient.

In all three cases the conclusion is the same in form: "the" optimal feedback is no longer a function, the assignment $m \mapsto \mathsf{S}(m)$ returns a *set* of admissible response flows rather than a single one, and S has degenerated from a map into an upper-hemicontinuous correspondence. Schauder's theorem, which speaks only of single-valued maps, then has nothing to act on, and a fixed point must instead be extracted by a set-valued theorem — the Kakutani–Fan–Glicksberg theorem of chapter 2 — after which an honest, executable feedback is recovered by measurable selection. That is the entire programme of section 10.4, and the two devices it deploys map onto the three failures exactly as flagged above: relaxation cures the non-convexity of (ii) and the gradient-lessness of (iii) by enlarging the strategy space, while selection cures the residual indeterminacy of (i) — and the leftover ambiguity in (ii)–(iii) — by choosing one optimiser measurably.

This dichotomy is the organising decision of the chapter, and we make it explicit so the later turn to set-valued analysis arrives as a planned manoeuvre rather than a surprise. The two routes diverge here and do not meet again until the closing synthesis: under *strict* convexity (A-Conv) together with nondegeneracy (A-Nondeg) — the regime in which all three failure modes are simultaneously excluded — the feedback is a single function, S is a continuous self-map, and the Schauder route applies; dropping either hypothesis admits one or more of the failures and routes the argument through Kakutani. *For the next two sections we therefore adopt the single-valued regime as a standing assumption*: we assume throughout sections 10.2 and 10.3 that (A-Nondeg) and the strict form of (A-Conv) hold, so that $\alpha^{*,m}$ is a genuine function and S a genuine map, and we build the Schauder argument for that map without further comment on set-valuedness. The set-valued machinery is deferred in its entirety to section 10.4, which drops both crutches and shows that an equilibrium survives all three failures above. With the regime now fixed, the immediate task is the one posed at the close of section 10.1: Schauder needs a nonempty convex compact set carried into itself, and producing that invariant set — in the single-valued regime just adopted — is the business of section 10.2, to which we now turn.

10.2 Compactness: the invariant set for the best-response map

The previous section closed by adopting the single-valued, strictly convex regime as a standing assumption and naming the debt it left outstanding. Schauder's theorem (chapter 2) certifies a fixed point only for a *continuous* self-map of a *nonempty convex compact* set, and so far we possess neither half of that hypothesis: S acts on the whole ambient space $\mathcal{M} = C([0,T];\mathcal{P}_1(\mathbb{R}^d))$, which is complete but hopelessly too large to be compact. In $\mathcal{M}$ mass can run off to infinity and a flow can wiggle in time arbitrarily fast, so there is nothing for Schauder to grip. The two missing pieces split into two cleanly separated tasks, and the chapter splits with them: the present section manufactures the set $\mathcal{K}$ — nonempty, convex, compact, and carried into itself by S — and section 10.3 supplies the continuity of S on it. This answers, in the affirmative, exactly the question left hanging at the end of section 10.1: yes, S does possess an invariant compact set, and here is how to build one. We work throughout in the single-valued regime just fixed, so that S is a genuine map and the containment $\mathsf{S}(\mathcal{K}) \subseteq \mathcal{K}$ is an honest statement about where points go, not where sets go.

The strategy for building $\mathcal{K}$ is the one the throughline has been signalling all along. No contraction is available, so compactness cannot be inherited from the dynamics — it must be *manufactured*, and the raw material is a pair of *a priori bounds* on the output flows: a ceiling on how far their mass can stray from the origin (a moment bound), and a ceiling on how fast they can move in time (an equicontinuity bound). A flow that is at once spatially confined and temporally slow simply cannot escape a compact set; that is the entire mechanism, and the remainder of the section is the labour of producing the two bounds and reading off the compactness they force. Both bounds, remarkably, descend from a single quantitative fact about the closed-loop dynamics — that the optimal drift is uniformly bounded — so we isolate that fact first and give it a name. We do so deliberately, because it is the one ingredient of the whole construction that can fail, and *does* fail for the LQG thread, forcing the separate treatment of section 10.5.

Assumption 10.4 (Bounded optimal feedback)**.** There is a constant $M < \infty$ such that, for every flow $m \in \mathcal{M}$, the optimal feedback drift is bounded:

$$\left\|\partial_p H(\cdot, \nabla u^m(t,\cdot))\right\|_\infty \;\leq\; M, \qquad t \in [0,T].$$

Assumption 10.4 reads, on its face, like an extra demand bolted onto the game — a fresh restriction on the data, asserted precisely where it happens to be convenient. If it genuinely were that, it would be a weakness in the argument, a fact assumed at the point of use rather than earned. It is not. Assumption 10.4 is a *consequence* of the primitive hypotheses already in force, and following the deduction is worth the few lines it takes, for two reasons: it shows the boundedness is bought rather than postulated, and it is the first place in the construction where the chapter's hero, nondegeneracy (A-Nondeg), does load-bearing work. The drift to be bounded is $-\partial_p H(x, \nabla u^m)$ from eq. (10.4), a composition of two pieces: the fixed map $\partial_p H$, which the standing growth hypotheses keep under control on any bounded set of momenta, and the gradient ∇u^m of the frozen-flow value, which is the only piece that carries any dependence on the population trajectory m. Bounding the drift uniformly in m therefore reduces cleanly to bounding ∇u^m uniformly in m — a Lipschitz bound on the value function that must hold no matter which population trajectory was frozen to source it. That uniform Lipschitz bound is the content of the next lemma; we state it, read assumption 10.4 off it, and record the mechanism of its proof while deferring the full parabolic estimate to appendix E.

Lemma 10.5 (Uniform gradient bound for the value)**.** *Assume (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), that the couplings F, G are bounded with $F[\mu](\cdot), G[\mu](\cdot)$ uniformly Lipschitz in x uniformly in μ, and that $p \mapsto \partial_p H(x,p)$ is continuous with at most linear growth. Then there is a constant $L_u < \infty$, depending only on the data and on T but* not *on the frozen flow m, such that the HJB solution u^m of eq. (10.2) satisfies $\|\nabla u^m(t,\cdot)\|_\infty \le L_u$ for all $t \in [0,T]$. Consequently assumption 10.4 holds with $M = \sup_{|p|\le L_u,\, x} |\partial_p H(x,p)|$.*

Proof. The estimate is the uniform-in-m Lipschitz bound for the value function, proved in full as theorem E.17 of appendix E; here we give the mechanism, because seeing where the constant L_u comes from is exactly seeing where the boundedness of the couplings and nondegeneracy (A-Nondeg) enter. The cleanest route is through the stochastic-control representation of the value (theorem 6.10),

$$u^m(t,x) \;=\; \inf_\alpha \; \mathbb{E}_{t,x}\Big[\int_t^T \big(L(X_r,\alpha_r) + F[m_r](X_r)\big)\, \mathrm{d}r \;+\; G[m_T](X_T) \Big],$$

the infimum over admissible controls of the running-plus-terminal cost of the controlled process started at $X_t = x$. To bound the Lipschitz constant of $x \mapsto u^m(t,x)$ we compare the values at two starting points x and x' by a *synchronous coupling*: run both trajectories with the *same* control and driven by the *same* Brownian path, so that they differ *only* through their initial condition. The separation $\Delta_r := X_r - X'_r$ then carries no independent noise increment — the $\sigma\, \mathrm{d}W$ terms enter only through the state-Lipschitz discrepancy of σ — and (A-Lip), the uniform state-Lipschitz property of the drift and diffusion with constant Λ, gives by Grönwall's inequality (lemma C.1) the pathwise bound

$$|\Delta_r| \;\le\; e^{\Lambda(r-t)}\, |x - x'|, \qquad r \in [t,T].$$

This is where the exponential is born: two synchronously-coupled trajectories spread apart at most exponentially in elapsed time, at the rate Λ that measures how strongly the dynamics can amplify a state discrepancy, so a gap present at time t is magnified by the factor $e^{\Lambda(r-t)}$ by time r. Feeding this into the value representation, and using that the *same* control makes the $L(X_r,\alpha_r)$ contributions cancel between the two trajectories so that only the couplings remain,

$$|u^m(t,x) - u^m(t,x')| \;\le\; \mathbb{E}_{t,x}\Big[\int_t^T \|\nabla_x F[m_r]\|_\infty\, |\Delta_r|\, \mathrm{d}r + \|\nabla G[m_T]\|_\infty\, |\Delta_T| \Big] \;\le\; \Big(\int_t^T e^{\Lambda(r-t)} \|\nabla_x F[m_r]\|_\infty\, \mathrm{d}r + e^{\Lambda(T-t}$$

The terminal datum sits furthest in the future, at distance $T - t$, and therefore suffers the *full* amplification $e^{\Lambda(T-t)}$; the running source at an intermediate time r suffers only $e^{\Lambda(r-t)} \le e^{\Lambda(T-t)}$, and replacing it by the worst-case terminal factor pulls the exponential out of the integral. Dividing by $|x - x'|$ and letting $x' \to x$ identifies the left side with $\|\nabla u^m(t,\cdot)\|_\infty$ and yields exactly the displayed bound

$$\|\nabla u^m(t,\cdot)\|_\infty \;\le\; e^{\Lambda(T-t)} \Big(\|\nabla G[m_T]\|_\infty + \int_t^T \|\nabla_x F[m_r]\|_\infty\, \mathrm{d}r \Big) \;\le\; L_u.$$

Two features make this the lemma we want, and each corresponds to one hypothesis. First, the bound is *uniform in m*: the frozen flow enters only through $\nabla_x F[m_r]$ and $\nabla G[m_T]$, and these are controlled by the Lipschitz constants of F and G in the state *uniformly in the measure argument* — exactly the boundedness/Lipschitz hypothesis on the couplings — so that $L_u = e^{\Lambda T}(\mathrm{Lip}_x G + T\,\mathrm{Lip}_x F)$ depends on the data and on T but on no property whatsoever of the particular trajectory m that was frozen. This uniformity is the whole point: it is what will make the moment and equicontinuity bounds below hold with one set of constants across the entire candidate set. Second, the representation itself is licensed by (A-Nondeg): nondegeneracy of $\sigma\sigma^\top$ renders the frozen HJB equation uniformly parabolic, hence its solution u^m a genuine classical $C^{1,2}$ solution (theorem C.12), so that ∇u^m exists pointwise and the feedback eq. (10.4) — and with it the synchronous-coupling control used above — actually denotes something. Strip away (A-Nondeg) and u^m degenerates to a mere viscosity solution with a kinked, possibly set-valued gradient, and the very first line of the argument loses its meaning. The PDE-side counterpart of the same estimate is the classical Bernstein gradient bound, obtained by differentiating the equation and applying the maximum principle to $|\nabla u^m|^2$; both routes are carried out in theorem E.17, and either delivers the same constant L_u. Finally, setting $M = \sup_{|p|\le L_u,\,x} |\partial_p H(x,p)|$ — finite because $\partial_p H(x,\cdot)$ has at most linear growth and is evaluated only on the bounded momentum range $|p| \le L_u$ — yields assumption 10.4. Cf. [36, 93, Ch. 3] for exactly this use inside an MFG existence proof. □

Notation watch

It is essential to be honest about the scope of assumption 10.4 and lemma 10.5, because the price they exact is exactly the price the carried thread cannot pay. The deduction above needed two things: couplings F, G that are *bounded* and Lipschitz in the state (uniformly in the measure), and a $\partial_p H$ that stays bounded on the range of ∇u. The LQG thread of appendix A (Thread A) violates *both*. Its running coupling $F[m](x) = \frac{q}{2}(x - s\bar{m})^2$ is *quadratic* in x, hence neither bounded nor globally Lipschitz; and its optimal feedback works out to the affine law $-bPx - b\,r$, which is *linear*, hence unbounded, in x. Neither the uniform gradient bound L_u nor the drift ceiling M survives: a linear feedback transports mass at a rate that grows with $|x|$, with no a priori ceiling. The abstract existence theorem assembled in this section therefore simply does *not* reach Thread A. This is not an oversight to be patched but a structural fact to be respected: LQG existence is recovered separately, by the explicit linear-Gaussian construction of section 10.5, where the very unboundedness that defeats the compactness argument is turned to advantage through the closed-form Riccati/mean reduction. The reader should carry this flag forward; it is the seam along which the abstract and the explicit halves of the chapter are stitched.

With assumption 10.4 now established as a consequence of the data, the uniform drift ceiling M is in hand, and the two a priori bounds that will define $\mathcal{K}$ follow from it by short and entirely classical stochastic-calculus estimates. The first controls *space* — it caps the second moment of every output flow, keeping mass from escaping to infinity — and the second controls *time* — it caps how fast an output flow can move, making each a uniformly continuous curve in $\mathcal{W}_1$. We prove them in turn; neither uses anything beyond the drift bound $|b^m| \le M$, the diffusion bound $\|\sigma\|_\infty \le \Sigma$, and the Itô isometry.

Lemma 10.6 (Uniform moment bound). *Assume (A-Nondeg) with $\|\sigma\|_\infty \le \Sigma$ and assumption 10.4.*

Let $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Then for every $m \in \mathcal{M}$ the output flow $\mathsf{S}(m)$ satisfies

$$\sup_{t\in[0,T]} \int_{\mathbb{R}^d} |x|^2\, \mathsf{S}(m)_t(\mathrm{d}x) \;\le\; R^2 := C\Big(1 + \textstyle\int |x|^2 m_0(\mathrm{d}x) + M^2T^2 + \Sigma^2 T\Big),$$

with C a universal constant.

Proof. Let X^m be the closed-loop process of definition 10.3, with drift $b^m(t,x) = -\partial_p H(x, \nabla u^m(t,x))$, so that $|b^m| \le M$ pointwise by assumption 10.4. Written in integral form the process is

$$X_t^m \;=\; X_0 \;+\; \underbrace{\int_0^t b^m(s, X_s^m)\,\mathrm{d}s}_{\text{drift}} \;+\; \underbrace{\int_0^t \sigma(X_s^m)\,\mathrm{d}W_s}_{\text{martingale}},$$

a sum of three terms, and we bound the size of each. The elementary inequality $|a+b+c|^2 \le 3(|a|^2+|b|^2+|c|^2)$ — a one-line consequence of Cauchy–Schwarz, or of the convexity of $r \mapsto r^2$ applied to the average $\frac{1}{3}(a+b+c)$ — separates the three contributions:

$$|X_t^m|^2 \;\le\; 3|X_0|^2 \;+\; 3\Big|\int_0^t b^m(s, X_s^m)\,\mathrm{d}s\Big|^2 \;+\; 3\Big|\int_0^t \sigma(X_s^m)\,\mathrm{d}W_s\Big|^2.$$

The drift integral is controlled deterministically: since $|b^m| \le M$, Jensen's inequality (equivalently Cauchy–Schwarz in time) gives $\big|\int_0^t b^m\,\mathrm{d}s\big|^2 \le t\int_0^t |b^m|^2\,\mathrm{d}s \le t\cdot tM^2 = M^2t^2$, with no expectation needed. The martingale integral is controlled in mean square by the *Itô isometry*, which converts the square of a stochastic integral into the time-integral of the squared integrand,

$$\mathbb{E}\Big|\int_0^t \sigma(X_s^m)\,\mathrm{d}W_s\Big|^2 \;=\; \mathbb{E}\int_0^t \operatorname{tr}\big(\sigma\sigma^\top(X_s^m)\big)\,\mathrm{d}s \;\le\; \Sigma^2 t,$$

the last step using the diffusion bound $\|\sigma\|_\infty \le \Sigma$ so that $\operatorname{tr}(\sigma\sigma^\top) \le \Sigma^2$ pointwise. Taking expectations in the separated inequality and inserting these two estimates,

$$\mathbb{E}\Big[|X_t^m|^2\Big] \;\le\; 3\,\mathbb{E}\Big[|X_0|^2\Big] + 3\,M^2t^2 + 3\,\Sigma^2 t.$$

Taking the supremum over $t \in [0,T]$ and recalling $\operatorname{Law}(X_0) = m_0$ gives the claim with $C = 3$ (any larger constant absorbing the 1+ also works). Notice how each of the three data quantities surfaces in R^2 for a transparent reason: the initial spread $\int |x|^2 m_0$ through X_0, the transport budget M^2T^2 through a drift that can push for time T at speed M, and the diffusive spread $\Sigma^2 T$ through the noise accumulated over $[0,T]$. □

The moment bound is the spatial half of the construction: it confines the output flows to a $\mathcal{W}_1$-bounded region of $\mathcal{P}_1$ at every instant. The temporal half is next — a bound on how far an output flow can travel in $\mathcal{W}_1$ over a short time interval. The same drift ceiling M and diffusion ceiling Σ that controlled the moment now control the increment, and the only new idea is to read $\mathcal{W}_1$ as a coupling distance so that the displacement of the *law* is bounded by the expected displacement of the *process*.

Lemma 10.7 (Equicontinuity in time)**.** *Under the hypotheses of lemma 10.6, every output flow $\mathsf{S}(m)$ satisfies*

$$\mathcal{W}_1\big(\mathsf{S}(m)_s, \mathsf{S}(m)_t\big) \;\le\; \omega(|t-s|) := M\,|t-s| + \Sigma\sqrt{|t-s|}, \qquad s,t \in [0,T].$$

Proof. The first step turns a question about measures into a question about the process. The pair (X^m_s, X^m_t) is, by construction, a coupling of the two marginals $\mathrm{Law}(X^m_s) = \mathsf{S}(m)_s$ and $\mathrm{Law}(X^m_t) = \mathsf{S}(m)_t$, and since $\mathcal{W}_1$ is the *infimum* of $\mathbb{E}|U - V|$ over all couplings (U, V) of its two arguments (chapter 4), this particular coupling furnishes an upper bound:

$$\mathcal{W}_1(\mathsf{S}(m)_s, \mathsf{S}(m)_t) \;\le\; \mathbb{E}[|X^m_t - X^m_s|]\,.$$

It remains to bound the expected increment of the process. For $s < t$, subtracting the integral forms at the two times leaves $X^m_t - X^m_s = \int_s^t b^m(r, X^m_r)\,\mathrm{d}r + \int_s^t \sigma(X^m_r)\,\mathrm{d}W_r$, so by the triangle inequality

$$\mathbb{E}[|X^m_t - X^m_s|] \;\le\; \mathbb{E}[*]\int_s^t |b^m(r, X^m_r)|\,\mathrm{d}r + \mathbb{E}[*]\left|\int_s^t \sigma(X^m_r)\,\mathrm{d}W_r\right|.$$

The drift term is immediate from $|b^m| \le M$: it is at most $M(t-s)$. The martingale term needs the one genuine estimate. We cannot apply the Itô isometry directly, because that governs the *second* moment of a stochastic integral, not the first; so we first raise to the square root by Jensen's inequality (the concavity of $r \mapsto \sqrt{r}$, equivalently Cauchy–Schwarz on $\mathbb{E}|\cdot| = \mathbb{E}|\cdot|\cdot 1$) and *then* apply the isometry:

$$\mathbb{E}[*]\left|\int_s^t \sigma(X^m_r)\,\mathrm{d}W_r\right| \;\le\; \left(\mathbb{E}\left|\int_s^t \sigma(X^m_r)\,\mathrm{d}W_r\right|^2\right)^{1/2} \;=\; \left(\mathbb{E}\int_s^t \mathrm{tr}(\sigma\sigma^\top(X^m_r))\,\mathrm{d}r\right)^{1/2} \;\le\; \Sigma\sqrt{t-s}.$$

(The Burkholder–Davis–Gundy inequality of appendix A would give the same $\sqrt{t-s}$ scaling more robustly, controlling $\mathbb{E}\sup$ of the increment rather than the increment at the endpoint; the elementary Jensen-plus-isometry route suffices here.) Combining the two terms gives $\mathbb{E}[|X^m_t - X^m_s|] \le M(t-s) + \Sigma\sqrt{t-s}$, which is $\omega(|t-s|)$. □

The shape of the modulus repays a second look. The two terms scale differently in the time gap: the drift contributes a Lipschitz piece $M|t-s|$, but the diffusion contributes a *half-Hölder* piece $\Sigma\sqrt{|t-s|}$, and for small $|t-s|$ the square root dominates. This $\sqrt{\cdot}$ is the unmistakable fingerprint of Brownian motion — mass spreads like $\sqrt{\text{time}}$, not like time — and it is the reason the output flows are $\frac{1}{2}$-Hölder rather than Lipschitz in time. Decisively, the modulus ω is *independent of* m: every output flow obeys the same modulus, set once and for all by the two constants M and Σ. That uniformity is precisely what an equicontinuity hypothesis demands, and it is the second of the two a priori bounds we set out to collect. With space controlled by lemma 10.6 and time controlled by lemma 10.7, both uniformly in m, we can now assemble the compact set.

Lemma 10.8 (A convex compact invariant set)**.** *With R from lemma 10.6 and ω from lemma 10.7, define*

$$\mathcal{K} \;=\; \Big\{ m \in \mathcal{M} :\; m_0 \textit{ fixed},\; \sup_t \int |x|^2 m_t(\mathrm{d}x) \le R^2,\; \mathcal{W}_1(m_s, m_t) \le \omega(|t-s|)\ \forall s,t \Big\}.$$

Then $\mathcal{K}$ is nonempty, convex, and compact in $(\mathcal{M}, \mathsf{d}_1)$, and $\mathsf{S}(\mathcal{K}) \subseteq \mathcal{K}$.

Proof. There are four things to check — nonempty, self-map, convex, compact — and the first three are quick; the weight is in compactness, which is where the moment bound is finally spent.

Nonempty. The constant flow $m_t \equiv m_0$ lies in $\mathcal{K}$: it has the prescribed initial value, its second moment $\int |x|^2 m_0(\mathrm{d}x)$ is $\le R^2$ by the definition of R, and its $\mathcal{W}_1$ increments vanish, so the equicontinuity bound $0 \le \omega(|t-s|)$ holds trivially. So $\mathcal{K} \neq \varnothing$, and Schauder will not be applied to the empty set — a precaution that is not idle, since an invariant set carved out by inequalities can perfectly well turn out to be empty.

Self-map. This is exactly what the two preceding lemmas were for. Lemma 10.6 says every output flow $\mathsf{S}(m)$ obeys the second-moment ceiling R^2, and lemma 10.7 says it obeys the modulus ω; both hold for every $m \in \mathcal{M}$, in particular for every $m \in \mathcal{K}$. Together with $\mathsf{S}(m)_0 = \mathrm{Law}(X_0) = m_0$ this is precisely the statement that $\mathsf{S}(m) \in \mathcal{K}$. Hence $\mathsf{S}(\mathcal{K}) \subseteq \mathcal{K}$. (Note the asymmetry that makes the construction work: the two lemmas bound the *output* regardless of the input, so the image of the *whole* ambient space already lands in $\mathcal{K}$; the restriction to $\mathcal{K}$ is needed only to have a compact convex domain for Schauder, not to make the containment hold.)

Convex. Take two members m, m' and a weight $\theta \in [0,1]$, and form the pointwise mixture $m_t^\theta = \theta m_t + (1-\theta) m'_t$. Each defining constraint survives the mixture, for an elementary reason in each case. The initial value is right because $m_0^\theta = \theta m_0 + (1-\theta) m_0 = m_0$. The second-moment ceiling survives because $\mu \mapsto \int |x|^2 \,\mathrm{d}\mu$ is *linear* in μ, so the second moment of the mixture is the mixture of the second moments, each $\le R^2$, hence $\le R^2$. The equicontinuity ceiling survives because $\mathcal{W}_1$ is *jointly convex*,

$$\mathcal{W}_1(\theta\mu_1 + (1-\theta)\nu_1,\ \theta\mu_2 + (1-\theta)\nu_2) \ \le\ \theta\, \mathcal{W}_1(\mu_1, \mu_2) + (1-\theta)\, \mathcal{W}_1(\nu_1, \nu_2),$$

which one sees by *superposing* the two optimal transport plans (a convex combination of couplings is again a coupling of the corresponding mixed marginals, and its cost is the mixture of the two costs). Applying this with $\mu_i = m$-values and $\nu_i = m'$-values at times s, t bounds $\mathcal{W}_1(m_s^\theta, m_t^\theta)$ by $\theta\,\omega + (1-\theta)\,\omega = \omega$. So $m^\theta \in \mathcal{K}$, and $\mathcal{K}$ is convex — the property Schauder needs in order to have a target for the would-be fixed point.

Compact. This is the heart of the matter. Compactness in $(\mathcal{M}, \mathsf{d}_1)$ is compactness of a family of *curves* $t \mapsto m_t$ from the compact interval $[0,T]$ into the metric space $(\mathcal{P}_1, \mathcal{W}_1)$, and the right tool is the generalized Arzelà–Ascoli theorem (theorem B.24): a family of maps from a compact interval into a complete metric space is relatively compact in the uniform metric provided it is (i) uniformly equicontinuous and (ii) pointwise relatively compact, the target space being (iii) complete. We verify the three inputs.

(i) Uniform equicontinuity is built into $\mathcal{K}$ by fiat: every member obeys the single modulus ω of lemma 10.7, which is what "uniform" means here.

(ii) Pointwise relative compactness is where the second moment is cashed, and it is worth doing in two stages because the moment buys two distinct things. Fix a time t and consider the slice $\mathcal{K}_t := \{m_t : m \in \mathcal{K}\} \subseteq \mathcal{P}_1(\mathbb{R}^d)$; we must show it is relatively compact *in* $\mathcal{W}_1$. *Stage one (weak relative compactness).* The uniform bound $\int |x|^2\, m_t(\mathrm{d}x) \le R^2$ forces *tightness* by Markov's inequality: the mass that any member places beyond radius ρ is

$$m_t(\{|x| > \rho\}) \ \le\ \frac{1}{\rho^2} \int_{|x|>\rho} |x|^2\, m_t(\mathrm{d}x) \ \le\ \frac{R^2}{\rho^2} \ \xrightarrow[\rho\to\infty]{}\ 0,$$

uniformly over $\mathcal{K}_t$, so the slice is uniformly tight, and Prokhorov's theorem (chapter 3) converts uniform tightness into relative compactness *for weak convergence*. *Stage two (the $\mathcal{W}_1$ upgrade).* Weak relative compactness is not yet what we need, because on $\mathcal{P}_1(\mathbb{R}^d)$ the metric $\mathcal{W}_1$ is *strictly stronger* than weak convergence in general: a weakly convergent sequence can fail to converge in $\mathcal{W}_1$ if first-moment mass leaks out along the tails. The exact characterization (theorem D.12) is that $\mathcal{W}_1(\mu_n, \mu) \to 0$ holds iff $\mu_n \to \mu$ weakly *and* $\int |x| \, \mathrm{d}\mu_n \to \int |x| \, \mathrm{d}\mu$, i.e. weak convergence plus convergence of the first moments; and the second condition follows automatically from the first precisely when the first moments are *uniformly integrable* (Vitali's theorem, the integrable upgrade of dominated convergence). Uniform integrability is exactly the extra mileage the second moment buys for free: for any member,

$$\int_{|x|>\rho} |x| \, m_t(\mathrm{d}x) \;\le\; \frac{1}{\rho} \int_{|x|>\rho} |x|^2 \, m_t(\mathrm{d}x) \;\le\; \frac{R^2}{\rho} \;\xrightarrow[\rho\to\infty]{}\; 0,$$

uniformly over $\mathcal{K}_t$ — the tails of the first moment are uniformly small because a second moment is uniformly bounded. Hence any weakly convergent sequence extracted in stage one is, by Vitali, also first-moment convergent, and therefore $\mathcal{W}_1$-convergent. Weak relative compactness has been upgraded to $\mathcal{W}_1$ relative compactness, and this upgrade is the entire reason the construction is carried out in $\mathcal{P}_1$ with $\mathcal{W}_1$ rather than in $\mathcal{P}_2(\mathbb{R}^d)$ with $\mathcal{W}_2$: a second-moment *ceiling* controls the first-moment tails but not the second-moment tails, so it upgrades weak compactness to $\mathcal{W}_1$ but *not* to $\mathcal{W}_2$.

(iii) Completeness of the target: $(\mathcal{P}_1, \mathcal{W}_1)$ is a complete metric space (theorem D.11), as recorded when we fixed the arena in definition 10.1.

By theorem B.24, inputs (i)–(iii) make $\mathcal{K}$ *relatively* compact in $(\mathcal{M}, \mathsf{d}_1)$: every sequence in $\mathcal{K}$ has a subsequence converging in d_1 to *some* flow in $\mathcal{M}$. To finish we must promote relative compactness to compactness, i.e. show $\mathcal{K}$ is *closed*, so that the limit flow lies back in $\mathcal{K}$ rather than escaping it. Let $m^{(n)} \in \mathcal{K}$ with $\mathsf{d}_1(m^{(n)}, m) \to 0$; we check each defining constraint passes to the limit m. The initial value is fixed because evaluation at $t = 0$ is d_1-continuous, so $m_0 = \lim m_0^{(n)} = m_0$. The equicontinuity bound passes by the triangle inequality: $\mathcal{W}_1(m_s, m_t) \le \mathcal{W}_1(m_s, m_s^{(n)}) + \mathcal{W}_1(m_s^{(n)}, m_t^{(n)}) + \mathcal{W}_1(m_t^{(n)}, m_t) \;\le\; 2\,\mathsf{d}_1(m^{(n)}, m) + \omega(|t-s|) \to \omega(|t-s|)$. The delicate constraint is the second-moment ceiling, and here is the one place the choice to use an *upper bound* rather than an equality pays off decisively. Convergence in $\mathcal{W}_1$ does *not* imply convergence of second moments — mass can slide outward in the limit — so we cannot expect $\int |x|^2 m_t^{(n)} \to \int |x|^2 m_t$. What we *do* have is one-sided: the second moment is *lower-semicontinuous* under $\mathcal{W}_1$ (indeed under weak) convergence, by Fatou's lemma applied to the nonnegative lower-semicontinuous integrand $|x|^2$ (theorem D.11), whence

$$\int |x|^2 \, m_t(\mathrm{d}x) \;\le\; \liminf_n \int |x|^2 \, m_t^{(n)}(\mathrm{d}x) \;\le\; R^2.$$

Lower semicontinuity points in exactly the helpful direction: in the limit the second moment can only *drop*, never jump above the ceiling, so a ceiling constraint is automatically preserved. (Had we instead pinned the second moment to a fixed *value*, the limit could have fallen short and left $\mathcal{K}$; the inequality is what makes $\mathcal{K}$ closed.) All three constraints survive, so $m \in \mathcal{K}$, $\mathcal{K}$ is closed, and a closed relatively compact subset of the complete space $(\mathcal{M}, \mathsf{d}_1)$ is compact. □

Remark 10.9 (Where tightness does the work)**.** It is worth standing back to see which single ingredient did the real work, because once it is isolated the whole compactness edifice can be held in one thought. Strip away the bookkeeping and the only non-elementary input above is Prokhorov's theorem — the equivalence "uniform tightness $\Leftrightarrow$ relative weak compactness" of chapter 3 — applied to the time-slices, with everything else (Markov, Vitali, Fatou, Arzelà–Ascoli) elementary glue. The lesson to carry forward is a slogan: *compactness in the space of population profiles is not a gift of the dynamics; it is bought, and the currency is a second moment.* The two a priori bounds spend that currency on two complementary confinements. The drift ceiling M controls how far mass can be transported per unit time, which is equicontinuity — confinement in *time*. The initial second moment together with the diffusion ceiling Σ control how far mass can be from the origin at all, which is tightness — confinement in *space*. Confine a family of curves in both space and time, uniformly, and Arzelà–Ascoli traps it in a compact set. Remove the moment bound and the purchase fails at the source: this is exactly what happens for the LQG thread, where the linear feedback drives the support outward at a rate proportional to its own extent, with no a priori ceiling, so that no fixed R confines the iterates and tightness is unavailable. The compactness construction of this section therefore genuinely does not apply there, and section 10.5 must — and does — exploit the explicit linear structure in its place, recovering existence by closed-form algebra rather than by trapping.

We have now manufactured the compact arena that Schauder's first demand requires. The set

$$\mathcal{K} \;=\; \Big\{\, m\in\mathcal{M}:\; m_0 \text{ fixed},\; \sup_t \int |x|^2 m_t(\mathrm{d}x)\le R^2,\; \mathcal{W}_1(m_s,m_t)\le\omega(|t-s|)\ \forall s,t \,\Big\}$$

of lemma 10.8 is a nonempty, convex, compact subset of $(\mathcal{M},\mathsf{d}_1)$ that S maps into itself: a concrete arena on which the fixed-point theorem can act. Everything Schauder asks of the *domain* is in hand. What remains is everything it asks of the *map*: that S, restricted to $\mathcal{K}$, be *continuous* in the metric d_1. That single missing ingredient — no formality, but the second and most demanding place the chapter's hero (A-Nondeg) does its work — is the business of the next section. We take $\mathcal{K}$ as fixed from here on and turn to proving S continuous on it.

10.3 Continuity of the best response and the Schauder theorem

Section 10.2 discharged the first of Schauder's two demands: it produced a nonempty convex compact set $\mathcal{K}\subset(\mathcal{M},\mathsf{d}_1)$ that S carries into itself, the arena on which a fixed point can be sought. But Schauder's theorem asks for *two* things, and a compact convex invariant domain is only the first; the second is that the map acting on that domain be *continuous*. The two demands are not interchangeable, and continuity is no afterthought: a discontinuous self-map of a compact convex set can wander forever without ever sitting still — continuity is precisely what forbids the orbit from accumulating at a point that the map then hurls elsewhere, and it is what lets one pass a limit *through* the map to certify that an accumulation point is fixed. Without it the compactness we laboured for in section 10.2 buys nothing at all. Supplying that one missing ingredient, and then assembling it with the compact set into the existence theorem, is the entire business of this section. It is the analytic heart of the chapter, and the second place the diffusion's nondegeneracy (A-Nondeg) is genuinely indispensable.

Why (A-Nondeg) must return *here* is the single sentence the reader should carry through everything below: *continuity of* S *factors through the stability of the frozen HJB solve, and that stability is a parabolic-regularity statement that is false without ellipticity.* Unpacked: to know that two nearby population profiles m and m' produce nearby responses $\mathsf{S}(m)$ and $\mathsf{S}(m')$, we must know that the value functions u^m and $u^{m'}$ they source are close — and, more delicately, that their *gradients* are close, since it is ∇u^m, not u^m itself, that drives the closed-loop dynamics. That the gradient of the HJB solution depends continuously on the equation's frozen data is exactly a parabolic estimate: it is the interior Schauder bound that uniform parabolicity supplies and that a degenerate diffusion destroys. Strip the ellipticity and u^m collapses to a mere viscosity solution whose gradient can jump across creases under arbitrarily small perturbations of the data; the response then need not depend continuously on the profile at all, and the fixed-point argument loses its second leg. So the hero of the chapter earns its billing a second time over: once, in section 10.2, to make the arena compact; here, to make the map continuous on it. Strip (A-Nondeg) away and *both* legs of Schauder fail at a single stroke — which is also the structural reason the LQG thread, nondegenerate but unbounded in its data, will not fall out as a corollary of the theorem assembled here.

The section proceeds in two movements, matching its two subsections. The first proves continuity by *factoring* the best-response map into the three elementary operations it is literally built from and showing each continuous in turn (the stability lemmas), then packaging the three back into continuity of S itself (proposition 10.13). The second feeds that proposition, together with the compact invariant set already in hand, into Schauder's theorem and reads off existence (theorem 10.14) — pausing to state honestly what the theorem costs, and to record that the probabilistic FBSDE route of chapter 9 delivers the identical conclusion through the two-pictures dictionary.

10.3.1 Stability of the decoupled problems

Continuity of S would be a formidable thing to attack head-on — the map sends a measure-valued trajectory to a measure-valued trajectory through a nonlinear, nonlocal, infinite-dimensional pipeline — were it not for one structural fact that makes the attack routine: S is not a monolith but a *composition* of three far simpler operations, and a composition of continuous maps is continuous. The strategy is therefore the oldest one in analysis, divide and conquer: write S as the chain of arrows it is literally assembled from in definition 10.3, prove each link continuous in isolation, and concatenate. The chain is

$$m \xmapsto{\text{(HJB)}} u^m \xmapsto{\textit{eq. } (10.4)} b^m = -\partial_p H(\cdot, \nabla u^m) \xmapsto{\text{(FP / SDE)}} \mathsf{S}(m) :$$

solve the frozen-flow HJB equation to obtain the value u^m; *differentiate and substitute* into the optimal-feedback formula eq. (10.4) to obtain the closed-loop drift b^m; *transport* the initial law m_0 forward under that drift (equivalently, solve the now-linear Fokker–Planck equation) to obtain the output flow. The three arrows are of three quite different characters — a parabolic PDE solve, a pointwise algebraic substitution, and a stochastic transport — and we treat their continuity separately, in that order. The first is the hard one and the only place ellipticity is consumed; the middle is an exercise in the continuity of an explicit function; the third is a synchronous-coupling argument. We establish each on $\mathcal{K}$, where the uniform a priori bounds of section 10.2 are in force.

It is worth flagging at the outset that each of the three links, examined closely, leans on exactly one of the three structural luxuries that the previous section's hand-off named as *crutches* — the conveniences the single-valued Schauder argument is permitted, and which section 10.4 must then do without. The first arrow leans on (A-Nondeg) to make the HJB solve regular and stable; the second leans on *strict* convexity of $H(x, \cdot)$ to keep the feedback a single-valued continuous function of the gradient; the third leans on the *Lipschitz* character of the resulting drift to run a pathwise coupling. As we walk the chain we will pause at each link to mark precisely which crutch it uses and what fails if it is removed, so that the reader reaches section 10.4 already knowing the three weaknesses that section is built to repair.

The first arrow: stability of the HJB solve. The whole difficulty is concentrated here. We must show that perturbing the frozen population m slightly perturbs the value u^m slightly — and, crucially, perturbs its *gradient* slightly, since it is ∇u^m, not u^m, that closes the loop. This is a *stability* statement for a backward parabolic equation under perturbation of its data, and it is exactly here that uniform parabolicity does its work: it is what upgrades a bare C^1-in-x ceiling on the value into a compactness statement in $C^{1,2}$ strong enough to pass to the limit in the equation *and* in its gradient. Absent (A-Nondeg) the lemma below has no estimate to invoke.

Lemma 10.10 (Stability of HJB in the frozen flow)**.** *Under the hypotheses of lemma 10.5, suppose the couplings are continuous for weak convergence: $\mu_n \to \mu$ weakly with $\sup_n \int |x|^2 \,\mathrm{d}\mu_n < \infty$ implies $F[\mu_n](\cdot) \to F[\mu](\cdot)$ and $G[\mu_n](\cdot) \to G[\mu](\cdot)$ locally uniformly. If $m^{(n)} \to m$ in $(\mathcal{K}, \mathsf{d}_1)$, then $u^{m^{(n)}} \to u^m$ and $\nabla u^{m^{(n)}} \to \nabla u^m$ locally uniformly on $[0,T] \times \mathbb{R}^d$.*

Proof. The argument has three movements: manufacture a uniform regularity bound (here ellipticity is spent), feed in the convergence of the data (here d_1-convergence is spent), and close by compactness and uniqueness.

Step 1: a uniform interior Hölder bound on the family. By lemma 10.5 every member of the family obeys the *same* gradient ceiling, $\|\nabla u^{m^{(n)}}(t,\cdot)\|_\infty \le L_u$, with L_u independent of n, of t, and of the frozen flow — this uniformity is exactly what lets one estimate the whole sequence with a single constant. On any parabolic cylinder $Q = (t_0, t_1) \times B_\rho$ compactly contained in $[0,T) \times \mathbb{R}^d$, the frozen HJB equation eq. (10.2) is, for each n, the linear uniformly parabolic equation

$$-\partial_t u^{(n)} - \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top \mathrm{D}^2\, u^{(n)}\big) \;=\; g^{(n)}(t,x), \qquad g^{(n)}(t,x) \;:=\; F[m_t^{(n)}](x) - H\big(x, \nabla u^{(n)}(t,x)\big),$$

where we have moved the semilinear term to the right and written $u^{(n)} = u^{m^{(n)}}$. The gradient ceiling is what makes this right-hand side *bounded uniformly in n*: the coupling F is bounded by hypothesis, and $H(x,\cdot)$ is evaluated only on the ball $|p| \le L_u$, where it too is bounded. (A-Nondeg) supplies a uniform ellipticity sandwich $\lambda I \preceq \sigma\sigma^\top \preceq \Lambda I$ with $\lambda > 0$ independent of x. A bounded right-hand side, uniform parabolicity, and bounded terminal/lateral data give first a uniform interior Hölder bound on $u^{(n)}$ and $\nabla u^{(n)}$; with $\nabla u^{(n)}$ now Hölder, $g^{(n)}$ is itself Hölder (uniformly in n, since H is locally Lipschitz on the fixed ball $|p| \le L_u$), and a single application of the interior parabolic Schauder estimate theorem E.19 promotes the family to a uniform bound in the Hölder space $C^{1+\theta/2,\,2+\theta}_{\mathrm{loc}}(Q)$ — a bound on the value, its time derivative, and its spatial derivatives up to second order, uniform in n. This is the one step that consumes ellipticity: strip away the uniform lower bound $\lambda > 0$ and there is no interior Schauder estimate, hence nothing to keep the second derivatives bounded as the data are perturbed.

Step 2: the frozen data converge locally uniformly. Step 1 is blind to the limit; to identify it we must use the convergence of the inputs, and this is where d_1-convergence of the flows is cashed. By hypothesis $m^{(n)} \to m$ in $(\mathcal{K}, \mathsf{d}_1)$, that is $\sup_t \mathcal{W}_1(m_t^{(n)}, m_t) \to 0$; in particular, at each fixed time t, $m_t^{(n)} \to m_t$ in $\mathcal{W}_1$ and hence weakly, while the uniform second-moment ceiling R^2 in force on $\mathcal{K}$ (lemma 10.6) keeps the masses tight and their first moments uniformly integrable. The assumed coupling continuity for weak convergence then converts this weak convergence of the populations into locally uniform convergence of the *data* they source: $F[m_t^{(n)}](\cdot) \to F[m_t](\cdot)$ and $G[m_T^{(n)}](\cdot) \to G[m_T](\cdot)$, locally uniformly in x. So both the right-hand source and the terminal datum of the frozen HJB equation converge, locally uniformly, to those of the limiting frozen equation.

Step 3: extract a limit, identify it, and conclude for the whole sequence. By Step 1 the family is bounded in $C^{1+\theta/2,\,2+\theta}_{\mathrm{loc}}$, so by the Arzelà–Ascoli theorem (theorem B.24, in its real-valued form) it is precompact in $C^{1,2}_{\mathrm{loc}}$: some subsequence $u^{(n_k)}$ converges in $C^{1,2}_{\mathrm{loc}}$ to a limit $\bar{u}$, together with its first spatial derivative and the derivatives appearing in the equation. Passing to the limit in eq. (10.2) along this subsequence — the second-order terms converge because the convergence is in $C^{1,2}_{\mathrm{loc}}$, and the data converge by Step 2 — shows that $\bar{u}$ solves the HJB equation with the limiting frozen flow m. But that decoupled equation has *at most one* solution: by the classical comparison principle for the uniformly parabolic semilinear HJB (a fortiori the viscosity comparison principle theorem E.14), $\bar{u} = u^m$. Every subsequential limit is therefore the *same* function u^m, and the whole sequence converges to it; the elementary reason is spelled out in the paragraph below. In particular $\nabla u^{(n)} \to \nabla u^m$ locally uniformly, which is the gradient convergence the next arrow needs. The interior Schauder estimate and the comparison principle are proved in appendix E; in the viscosity formulation, should one ever drop to that level, the identical conclusion follows from the stability of viscosity solutions under local uniform convergence of the data — if $u^{(n_k)}$ converges locally uniformly and the data converge locally uniformly, the limit is again a viscosity solution, and comparison again pins it to u^m (appendix E; [36, §2–3]). □

The shape of that argument is worth naming, because it recurs throughout the chapter: it is a *compactness–uniqueness* argument, and it is the standard way to prove a continuous dependence one cannot estimate directly. One never bounds $u^{m^{(n)}} - u^m$ by hand. Instead the uniform parabolic estimate makes the sequence $\{u^{m^{(n)}}\}$ precompact in $C^{1,2}_{\mathrm{loc}}$; every limit point is pinned to the *same* flow u^m by uniqueness for the decoupled equation; and the elementary topological fact that "a sequence all of whose subsequential limits coincide, in a space where the sequence is precompact, converges to that common value" forces the whole sequence to converge.

That last fact is the one piece of the argument that is purely topological, and since it is used silently again and again it is worth spelling out once, because it is the precise mechanism by which subsequential information becomes whole-sequence information. Suppose, for contradiction, that $u^{(n)} \not\to u^m$ in $C^{1,2}_{\mathrm{loc}}$. Then convergence fails on some compact cylinder Q: there is an $\varepsilon > 0$ and an infinite set of indices $n_1 < n_2 < \cdots$ along which $\|u^{(n_k)} - u^m\|_{C^{1,2}(Q)} \geq \varepsilon$ for every k. But the *sub*sequence $\{u^{(n_k)}\}$ satisfies the hypotheses of Step 1 verbatim — the uniform Hölder bound knows nothing of which indices we kept — so it is itself precompact in $C^{1,2}_{\mathrm{loc}}$ and admits a further subsequence converging to some limit; by Steps 2–3 that limit can only be u^m. Along that further subsequence, then, $\|u^{(n_{k_j})} - u^m\|_{C^{1,2}(Q)} \to 0$, contradicting the standing lower bound $\geq \varepsilon$. No such bad cylinder can exist, so the whole sequence converges. The device works for exactly two reasons, and both are supplied by (A-Nondeg): *(i)* precompactness, so that *every* subsequence has a convergent sub-subsequence (Step 1), and *(ii)* uniqueness of the limit, so that all those sub-subsequential limits coincide (Step 3). Either one alone is useless; together they upgrade "some subsequence converges to u^m" into "the whole sequence converges to u^m," which is the genuinely continuous-dependence statement the lemma claims.

The single load-bearing hypothesis at both junctures is, again, uniform parabolicity: it produces the $C^{1,2}$ bound through the interior Schauder estimate, and it underwrites the comparison principle that delivers uniqueness. Delete (A-Nondeg) and both supports collapse at once — there is no uniform $C^{1,2}$ bound from which to extract a limit, and no classical comparison to identify it — which is the precise sense in which this arrow, and with it the continuity of S, is simply false in the degenerate case. This is the second appearance of the chapter's hero promised in the section opening, and it is the load-bearing one.

The middle arrow: continuity of the feedback substitution. This link is almost free, and it is the one place strict convexity — rather than nondegeneracy — is spent. The arrow $\nabla u^m \mapsto b^m = -\partial_p H(\cdot, \nabla u^m)$ is nothing but pointwise composition with the fixed function $p \mapsto -\partial_p H(x,p)$. Under the strict form of (A-Conv) that function is single-valued and continuous — it is the unique Legendre maximiser of eq. (10.5), and the argmax of a strictly concave family depends continuously on its slope. The elementary reason behind that last clause is the maximum theorem specialised to a unique maximiser: the maximiser correspondence of a continuous, jointly concave objective is always upper hemicontinuous, and an upper hemicontinuous correspondence whose values are *singletons* is exactly a continuous function. Strict concavity is what collapses the values to singletons; without it the correspondence could jump, and continuity would fail. So the locally uniform convergence $\nabla u^{m^{(n)}} \to \nabla u^m$ delivered by the first arrow passes straight through to locally uniform convergence $b^{(n)} \to b$ of the drifts: composing a locally uniformly convergent sequence of gradients with the fixed continuous map $-\partial_p H$ preserves local uniform convergence, since on each compact set $-\partial_p H$ is uniformly continuous. Were H only weakly convex this very arrow would already be set-valued, and we would be travelling the Kakutani road of section 10.4 rather than the Schauder one — single-valuedness here is precisely one of the crutches the present argument leans on and the next section knocks away. With the drifts now known to converge, only the third arrow remains: that convergent drifts transport the common initial law m_0 to convergent output flows.

Lemma 10.11 (Stability of the Fokker–Planck / closed-loop SDE)**.** *Let $b^{(n)} \to b$ locally uniformly with $\sup_n \|b^{(n)}\|_\infty \leq M$, let each $b^{(n)}$ and the limit b be Lipschitz in x uniformly in t (as the interior Schauder estimate theorem E.19 provides for the smoothing-coupling class, where $b^{(n)} = -\partial_p H(\cdot, \nabla u^{m^{(n)}})$ with $\nabla u^{m^{(n)}}$ uniformly C^1 in x), and let σ satisfy (A-Nondeg) and (A-Lip). Let $X^{(n)}$, X solve the SDEs with drifts $b^{(n)}$, b, diffusion σ, and common initial law m_0. Then* $\mathrm{Law}(X^{(n)}) \to \mathrm{Law}(X)$ *in $C([0,T];\mathcal{P}_1)$, i.e.* $\mathsf{d}_1\big(\mathrm{Law}(X^{(n)}_\cdot), \mathrm{Law}(X_\cdot)\big) \to 0$.

Proof. We use the *direct* (synchronous) *coupling*: drive $X^{(n)}$ and X by the *same* Brownian motion W, started from the *same* draw $X_0 \sim m_0$, so that the two processes differ only through their drifts. Both SDEs are well-posed: their drifts are bounded (by M) and Lipschitz in x, and σ is Lipschitz and nondegenerate, so theorem 5.10 gives unique strong solutions — boundedness of the drift removes any growth obstruction. Writing $D_t^{(n)} = X_t^{(n)} - X_t$ and subtracting the two integral equations,

$$D_t^{(n)} = \underbrace{\int_0^t \Big(b^{(n)}(s, X_s^{(n)}) - b(s, X_s)\Big)\,\mathrm{d}s}_{\text{drift gap}} \;+\; \underbrace{\int_0^t \Big(\sigma(X_s^{(n)}) - \sigma(X_s)\Big)\,\mathrm{d}W_s}_{\text{martingale}}. \tag{10.6}$$

The very form of the martingale term is the dividend of synchronous coupling, and it is worth dwelling on. Because both processes carry the *common* diffusion σ against the *common* Brownian motion, subtracting them leaves the single difference $\sigma(X_s^{(n)}) - \sigma(X_s)$, whose integrand is *small in the gap* — of order $\Lambda_\sigma |D_s^{(n)}|$ by the Lipschitz property of σ — and which therefore vanishes as the processes come together. Had we instead driven $X^{(n)}$ and X by *independent* Brownian motions, subtraction would have left two stochastic integrals against different noises that do *not* combine; their quadratic variation would be of order $\sigma\sigma^\top$, an $O(1)$ floor that no amount of drift closeness could erase, and the estimate below would be impossible. The synchronous coupling is precisely the device that makes the noise cancel down to a gap-controlled remainder; this is also exactly the device that *fails* in the relaxed setting of remark 10.12, where there is no common pathwise driver to couple against.

Now estimate. The drift gap is split by the standard *add-and-subtract* trick: insert and remove $b^{(n)}(s, X_s)$,

$$b^{(n)}(s, X_s^{(n)}) - b(s, X_s) = \underbrace{\big[b^{(n)}(s, X_s^{(n)}) - b^{(n)}(s, X_s)\big]}_{\text{self-improving: Lipschitz in the gap}} + \underbrace{\big[b^{(n)}(s, X_s) - b(s, X_s)\big]}_{\text{data: } b^{(n)}\to b \text{ along the fixed process } X},$$

which separates the part that can be folded back into $|D^{(n)}|$ from the part that measures how far the input drifts are from their limit. The first bracket is bounded by $\Lambda_b |D_s^{(n)}|$ using $\mathrm{Lip}_x b^{(n)} \le \Lambda_b$ (uniform in n); the second is the genuine forcing, evaluated along the *fixed* limit process X, and carries no $D^{(n)}$. Square the identity eq. (10.6), take the running supremum, and use $|a+b+c|^2 \le 3(|a|^2+|b|^2+|c|^2)$ to treat the three contributions separately. On the self-improving drift term, Cauchy–Schwarz in time gives $\big(\int_0^t \Lambda_b |D_s^{(n)}|\,\mathrm{d}s\big)^2 \le \Lambda_b^2\, t \int_0^t |D_s^{(n)}|^2\,\mathrm{d}s \le \Lambda_b^2\, T \int_0^t |D_s^{(n)}|^2\,\mathrm{d}s$; on the data term, the same Cauchy–Schwarz gives a factor T in front of $\int_0^T |b^{(n)}(s, X_s) - b(s, X_s)|^2\,\mathrm{d}s$; on the martingale term, the Burkholder–Davis–Gundy inequality (theorem C.3) replaces the supremum of the stochastic integral by its quadratic variation, $\mathbb{E}\sup_{r\le t} |\int_0^r (\sigma(X^{(n)}) - \sigma(X))\,\mathrm{d}W|^2 \le c_{\mathrm{BDG}}\, \mathbb{E}\int_0^t |\sigma(X_s^{(n)}) - \sigma(X_s)|^2\,\mathrm{d}s \le c_{\mathrm{BDG}}\,\Lambda_\sigma^2\, \mathbb{E}\int_0^t |D_s^{(n)}|^2\,\mathrm{d}s$ by the Lipschitz bound on σ. Collecting the three,

$$\mathbb{E}\sup_{r\le t} |D_r^{(n)}|^2 \;\le\; C\,\mathbb{E}\int_0^t |D_s^{(n)}|^2\,\mathrm{d}s \;+\; C\,\mathbb{E}\int_0^T \big|b^{(n)}(s, X_s) - b(s, X_s)\big|^2\,\mathrm{d}s,$$

where the single constant $C = C(\Lambda_b, \Lambda_\sigma, T) = 3\max\{\Lambda_b^2\, T + c_{\mathrm{BDG}}\Lambda_\sigma^2,\ T\}$ collects, transparently, the three sources: the factor 3 from the triple square, the term $\Lambda_b^2 T$ from the drift-Lipschitz constant and the Cauchy–Schwarz time factor, the term $c_{\mathrm{BDG}}\Lambda_\sigma^2$ from the universal BDG constant times the diffusion-Lipschitz constant, and the lone T from the data term. Crucially C depends only on the structural constants $\Lambda_b, \Lambda_\sigma$ and the horizon T — never on n — so the same constant governs every member of the sequence. Grönwall's inequality (lemma C.1) applied to $\phi(t) = \mathbb{E}\sup_{r\le t}|D_r^{(n)}|^2$, whose self-referential term is $C\int_0^t \phi$, then gives

$$\mathbb{E}\sup_{r\le T}|D_r^{(n)}|^2 \;\le\; C\,e^{CT}\,\varepsilon_n, \qquad \varepsilon_n \;:=\; \mathbb{E}\int_0^T \left|b^{(n)}(s,X_s) - b(s,X_s)\right|^2 \,\mathrm{d}s.$$

It remains to see that the forcing $\varepsilon_n \to 0$, and this is the one place the hypothesis $b^{(n)} \to b$ *locally uniformly* is used. Fix ω. The path $s \mapsto X_s(\omega)$ is continuous on $[0,T]$, so its image is a *compact* subset of $[0,T]\times\mathbb{R}^d$; local uniform convergence of $b^{(n)}$ to b is uniform on that compact image, whence $\sup_{s\le T}|b^{(n)}(s,X_s(\omega)) - b(s,X_s(\omega))| \to 0$, and in particular the integrand tends to 0 for every s and $\mathcal{P}()$-a.e. ω. The integrand is dominated, uniformly in n, by $(2M)^2$ — both drifts are bounded by M — which is an integrable majorant on the finite-measure space $[0,T]\times\Omega$. The dominated convergence theorem therefore yields $\varepsilon_n \to 0$. Finally, since the synchronous coupling realises both marginals on one probability space, it is an admissible coupling of $\mathrm{Law}(X_t^{(n)})$ and $\mathrm{Law}(X_t)$, so by the very definition of $\mathcal{W}_1$ together with Jensen,

$$\mathcal{W}_1\big(\mathrm{Law}(X_t^{(n)}), \mathrm{Law}(X_t)\big) \;\le\; \mathbb{E}\,|D_t^{(n)}| \;\le\; \Big(\mathbb{E}\sup_{r\le T}|D_r^{(n)}|^2\Big)^{1/2} \;\le\; \big(C\,e^{CT}\,\varepsilon_n\big)^{1/2}$$

uniformly in t. Taking the supremum over $t\in[0,T]$ shows $\mathsf{d}_1\big(\mathrm{Law}(X_\cdot^{(n)}), \mathrm{Law}(X_\cdot)\big) \le (C\,e^{CT}\varepsilon_n)^{1/2} \to 0$, which is the claimed d_1-convergence. □

The coupling argument just run leaned, decisively, on the drift being *Lipschitz* in x: that Lipschitz constant Λ_b is what made the add-and-subtract trick produce the self-improving term $C\,\mathbb{E}\int_0^t |D_s^{(n)}|^2\,\mathrm{d}s$ that Grönwall absorbs into exponential growth. Without it the first bracket of the drift gap could not be folded back into $|D^{(n)}|$, there would be no differential inequality to close, and the whole pathwise estimate would evaporate. This is the third and last of the three crutches the single-valued argument rests on — nondegeneracy for the HJB solve, strict convexity for the feedback, and now Lipschitz drift for the transport — and, like the other two, it is a luxury the rough case of section 10.4 cannot afford. There the optimal feedback is read off a value function that need not be differentiable, so the drift is at best bounded and measurable, never Lipschitz, and the synchronous coupling has nothing to contract. The device that replaces it is different in kind — it abandons the pathwise coupling altogether and argues stability at the level of *laws* — and we record it now, in its own right, because naming it here is what makes the structure of section 10.4 inevitable rather than ad hoc: that section is not a fresh start but precisely the substitution of this one tool for the one the present lemma used.

Remark 10.12 (The martingale-problem route, for the relaxed setting)**.** Suppose the limiting drift is only *bounded* and locally-uniformly approximated — not Lipschitz, so that eq. (10.6) no longer closes. The repair keeps the same goal, $\mathrm{Law}(X^{(n)}) \to \mathrm{Law}(X)$, but proves it without ever forming the pathwise difference $D^{(n)}$, through *martingale-problem stability*, in three moves that exactly mirror the compactness–uniqueness skeleton of lemma 10.10. *First* (compactness), the laws $\{\mathrm{Law}(X^{(n)})\}$ on path space $C([0,T];\mathbb{R}^d)$ are tight: the uniform drift and diffusion bounds give the Kolmogorov–Chentsov moment estimate theorem C.9 on increments, which is the path-space analogue of the equicontinuity that bought tightness in section 10.2, so Prokhorov supplies weakly convergent subsequences. *Second* (identification), every weak limit solves the martingale problem for the limiting generator — the defining martingales of the approximants pass to the limit because the test functions are smooth and the coefficients converge — so each subsequential limit is a solution of the limiting dynamics. *Third* (uniqueness), that martingale problem is *well-posed* (unique in law) under (A-Nondeg) by theorem C.21: nondegeneracy, the very hypothesis that bought regularity on the analytic side, here buys uniqueness in law on the probabilistic side. With all subsequential limits forced to coincide, the same "every subsequence has a further subsequence to the same limit" device used above promotes weak convergence of subsequences to convergence of the full sequence, and the time-marginals then converge in $\mathcal{W}_1$ via the uniform moment bound of lemma 10.6. This is not an optional alternative proof of the present lemma; it is the *only* route available once the feedback ceases to be Lipschitz, and it is the engine that section 10.4 will run — in its controlled, set-valued form, where the drift is replaced by a relaxed control and the martingale problem becomes a controlled martingale problem — which is exactly why we develop the controlled version in appendix C.13. The reader should therefore read this remark not as a digression but as the advertisement for section 10.4: it isolates the single mechanism that must change, and shows that everything else — compactness, identify the limit, uniqueness upgrades subsequences to the whole sequence — is structurally identical to the argument just completed.

All three arrows are now established: the HJB solve is stable (lemma 10.10), the feedback substitution is continuous under strict convexity, and the closed-loop transport is stable (lemma 10.11). Concatenating them costs nothing further — the work was in the links. The proposition below simply chases a convergent sequence of profiles through the chain, arrow by arrow, and reads off that the outputs converge in d_1; it is the bookkeeping that joins the three stability statements into the one continuity statement Schauder demands.

Proposition 10.13 (Continuity of the best-response map)**.** *Under the standing hypotheses (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), the boundedness and continuity of the couplings F, G assumed in lemmas 10.5 and 10.10, and strict convexity of $H(x,\cdot)$ (so that the feedback eq.* (10.4) *is single-valued and continuous in ∇u), the map $\mathsf{S} : \mathcal{K} \to \mathcal{K}$ is continuous.*

Proof. Continuity is sequential continuity on the metric space $(\mathcal{K}, \mathsf{d}_1)$, so let $m^{(n)} \to m$ in $\mathcal{K}$ and chase the convergence along the three arrows of the chain.

First arrow. By lemma 10.10 — whose hypotheses are exactly items (A-Lip)–(A-Conv) together with the boundedness and weak-continuity of F, G assumed here — $u^{m^{(n)}} \to u^m$ and, what we actually use, $\nabla u^{m^{(n)}} \to \nabla u^m$ locally uniformly on $[0,T] \times \mathbb{R}^d$.

Middle arrow. Strict convexity of $H(x,\cdot)$ makes $p \mapsto \partial_p H(x,p)$ a single-valued continuous function (the unique maximiser of eq. (10.5)), so composing the locally uniformly convergent gradients with it preserves local uniform convergence: the drifts $b^{(n)} = -\partial_p H(\cdot, \nabla u^{m^{(n)}})$ converge locally uniformly to $b = -\partial_p H(\cdot, \nabla u^m)$. Two uniform bounds ride along, and the third arrow needs both: first, $\|b^{(n)}\|_\infty \le M$ for every n by assumption 10.4 (the gradients sit inside the ball $|p| \le L_u$ on which $\partial_p H$ is bounded by M); second, the interior Schauder estimate theorem E.19 bounds $\{u^{m^{(n)}}\}$ in

$C^{1,2}$ uniformly in n, so each $\nabla u^{m^{(n)}}$, and hence each $b^{(n)}$, is Lipschitz in x with one constant Λ_b uniform in n and t.

Third arrow. The drifts $b^{(n)}, b$ are thus bounded by M, Lipschitz in x with the common constant Λ_b, locally uniformly convergent, and paired with the nondegenerate Lipschitz diffusion σ and the common initial law m_0 — precisely the hypotheses of lemma 10.11. That lemma delivers $\mathsf{S}(m^{(n)}) = \mathrm{Law}(X^{(n)}_\cdot) \to \mathrm{Law}(X_\cdot) = \mathsf{S}(m)$ in d_1. Since $\mathsf{S}(m^{(n)}) \to \mathsf{S}(m)$ for every d_1-convergent input sequence, S is continuous on $\mathcal{K}$. $\square$

This is the deliverable of the section. The best-response map S, restricted to the compact convex set $\mathcal{K}$ that section 10.2 carved out, is a continuous self-map in the order-one metric d_1 — proved by factoring it into the three arrows of definition 10.3 and showing each stable, with (A-Nondeg) consumed once at the HJB solve and once more in the comparison principle, strict convexity consumed once at the feedback, and the Lipschitz drift consumed once at the transport. The compactness of the arena was the first of Schauder's two demands and was settled in section 10.2; the continuity just proved is the second. With both now in hand, nothing stands between us and a fixed point but the act of quoting Schauder's theorem — which is exactly what the next subsection does.

10.3.2 The existence theorem

The previous subsection handed us the last missing ingredient. Factoring S into its three constituent arrows and proving each link stable, proposition 10.13 established that S, restricted to the compact convex set $\mathcal{K}$, is *continuous* in the order-one metric d_1. That was the second of Schauder's two demands; the first — a nonempty convex compact invariant arena $\mathcal{K}$ with $\mathsf{S}(\mathcal{K}) \subseteq \mathcal{K}$ — was carved out of $\mathcal{M}$ back in section 10.2. The two demands now hold *simultaneously, on the very same set*, and each was paid for at one of the two places (A-Nondeg) was spent: the moment-and-equicontinuity construction that made $\mathcal{K}$ compact, and the parabolic stability of the HJB solve that made S continuous. The fixed-point theorem of chapter 2 exists precisely to convert "compact convex domain" plus "continuous self-map" into "has a fixed point," so the mathematical work is behind us; what remains is to quote the theorem correctly and then read off what its fixed point *means.*

"Quote it correctly" hides one genuine wrinkle, and it is the only new labour in this subsection. Schauder's theorem as stated and proved in chapter 2 lives on a *Banach space*: its proof runs by finite-dimensional projection followed by Brouwer's theorem, and both of those steps need the linear and the normed structure of a vector space. But $\mathcal{M} = C([0,T]; \mathcal{P}_1(\mathbb{R}^d))$ is a space of *measure-valued curves*, and a probability measure is not, on its face, an element of any vector space — one cannot add two probability measures and stay a probability measure. Before Schauder can act we must therefore exhibit $\mathcal{M}$ as a compact convex subset *of an honest Banach space*, in a way that does not disturb the d_1-topology in which compactness and continuity were proved. That realisation, through the Kantorovich–Rubinstein normed space of signed measures, is the one technical move in the proof below; once it is in place the theorem drops out. We state the existence theorem, prove it by this route, and then spend the rest of the subsection being honest about what it delivers — in particular that it asserts existence with *no* uniqueness, and that through the two-pictures dictionary it is verbatim the existence statement promised on the probabilistic side in chapter 9.

Theorem 10.14 (Existence of MFG equilibria, regularising couplings)**.** *Assume:*

(i) *(A-Lip), (A-Growth), (A-Nondeg), (A-Conv), with $H(x,\cdot)$ strictly convex and C^1, and $\partial_p H$ of at most linear growth in p;*

(ii) *the couplings F, G are bounded, uniformly Lipschitz in x uniformly in the measure argument, and continuous for weak convergence of second-moment-bounded measures (as in lemma 10.10);*

(iii) $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$.

Then the best-response map S has a fixed point $m^\star \in \mathcal{K}$. Equivalently, the MFG system eq. (10.2)–eq. (10.3) admits a solution $(u^\star, m^\star)$ with $m^\star \in C([0,T]; \mathcal{P}_2(\mathbb{R}^d))$, $u^\star$ Lipschitz in x uniformly in t, and $m^\star$ uniformly $\frac{1}{2}$-Hölder in time for $\mathcal{W}_1$.

Proof. The two hypotheses Schauder consumes are already proved. By lemma 10.8, $\mathcal{K}$ is a nonempty convex compact subset of the metric space $(\mathcal{M}, \mathsf{d}_1)$, and $\mathsf{S}(\mathcal{K}) \subseteq \mathcal{K}$; by proposition 10.13, $\mathsf{S}\big|_{\mathcal{K}}$ is continuous in d_1. Everything below is the single act of placing these facts in a form Schauder accepts.

Realising $\mathcal{K}$ inside a Banach space. As flagged above, $\mathcal{P}_1(\mathbb{R}^d)$ is not a vector space, so we must first locate it inside one. Enlarge it to the real vector space of finite signed Borel measures of finite first moment; this is a genuine linear space — signed measures add and scale freely — and $\mathcal{P}_1(\mathbb{R}^d)$ sits inside it as a *convex* subset, since a convex combination $\theta\mu + (1-\theta)\nu$ of probability measures is again a probability measure, and a difference $\mu - \nu$ of two of them is a signed measure of total mass zero. Norm this vector space by the dual-bounded-Lipschitz (Kantorovich–Rubinstein) norm

$$\|\nu\|_{\mathrm{KR}} = \sup\Big\{ \textstyle\int \phi \, \mathrm{d}\nu : \ \mathrm{Lip}(\phi) \le 1, \ \|\phi\|_\infty \le 1 \Big\},$$

and write $(\mathcal{M}_{\mathrm{KR}}, \|\cdot\|_{\mathrm{KR}})$ for the resulting normed (Banach, after completion) space (theorem D.10). The point of this particular norm is that it reproduces the transport geometry we have been working in. By Kantorovich–Rubinstein duality, $\mathcal{W}_1(\mu,\nu) = \sup\{\int \phi \, \mathrm{d}(\mu-\nu) : \mathrm{Lip}(\phi) \le 1\}$, the same supremum but without the sup-norm cap on the test function. Because the both-caps test class is the smaller of the two, one has the one-sided bound $\|\mu - \nu\|_{\mathrm{KR}} \le \mathcal{W}_1(\mu,\nu)$ for all μ, ν, so $\|\cdot\|_{\mathrm{KR}}$ is in general *not* equal to $\mathcal{W}_1$ but the (weaker) dual-bounded-Lipschitz distance, which metrises weak convergence. On the second-moment-bounded, tight set $\mathcal{K}$, however, $\mathcal{W}_1$ *also* metrises weak convergence (theorem D.11), so the additional cap is immaterial for the *topology* it induces: the two distances are equivalent there, and $\mathcal{P}_1(\mathbb{R}^d)$ embeds into $\mathcal{M}_{\mathrm{KR}}$ as a homeomorphism onto its image, with $\mathcal{W}_1$-convergence on a second-moment-bounded set coinciding with $\|\cdot\|_{\mathrm{KR}}$-convergence. Lifting from measures to curves, $\mathcal{M} = C([0,T]; \mathcal{P}_1)$ embeds in the Banach space $C([0,T]; \mathcal{M}_{\mathrm{KR}})$ under the uniform-in-time norm, whose induced topology agrees on $\mathcal{K}$ with that of the order-one metric d_1 of definition 10.1. Under this embedding the set $\mathcal{K}$ keeps both properties verbatim: it is *convex* because convexity is an affine notion preserved by the linear embedding (a convex combination of flows is computed time-slice by time-slice and stays in $\mathcal{K}$ by lemma 10.8), and it is *compact* because

d_1-compactness is a topological notion and the embedding is a homeomorphism onto its image. The gain from doing the bookkeeping this way is concrete: because $\mathcal{K}$ now lives in a *normed* space, we may invoke the clean compact-convex Schauder theorem theorem 2.38 *as stated*, without reaching for its locally convex (Tychonoff–Schauder) generalisation — the version one is forced into when the only available ambient structure is the weak topology on measures, which is locally convex but not normable. Here the Kantorovich–Rubinstein norm hands us normability for free, so the lighter theorem suffices.

Applying Schauder. The ambient space $C([0,T];\mathcal{M}_{\mathrm{KR}})$ is Banach, $\mathcal{K}$ is a compact convex subset of it, and $\mathsf{S}:\mathcal{K}\to\mathcal{K}$ is a continuous self-map. These are exactly the hypotheses of theorem 2.38, which therefore applies directly and produces a point $m^\star\in\mathcal{K}$ with $\mathsf{S}(m^\star)=m^\star$.

Unwinding the fixed point into a solution of the MFG system. A fixed point of S is more than an abstract certificate; spelled out through definition 10.3 it *is* a solution of the coupled HJB–FP system, and it is worth tracing the implication slot by slot. Recall that $\mathsf{S}(m)$ was built by freezing the population inside the couplings, solving the resulting decoupled HJB equation for a value u^m, reading off the optimal feedback, and transporting m_0 forward under it. Set $u^\star := u^{m^\star}$. By construction $u^\star$ solves the HJB equation eq. (10.2) whose source and terminal data are $F[m_t^\star]$ and $G[m_T^\star]$ — but because $m^\star$ is the *fixed* flow, those frozen couplings are no longer a provisional guess: they are the genuine equilibrium couplings, so $u^\star$ solves the full, self-referential HJB equation of the MFG system, not merely a decoupled surrogate. The optimal feedback $\alpha^{*,\star}=-\partial_p H(\cdot,\nabla u^\star)$ it generates drives the closed-loop SDE eq. (10.1), whose time-marginals are by definition $\mathsf{S}(m^\star)$. The fixed-point equation $\mathsf{S}(m^\star)=m^\star$ says precisely that those time-marginals *are* $m^\star$ itself, so $m^\star$ is the law of the optimally controlled state and hence solves the Fokker–Planck equation eq. (10.3) with the drift generated by $u^\star$ and initial datum m_0. The two equations, deliberately decoupled throughout the construction by the freezing device, are thus *re-coupled exactly at the fixed point*: the flow used to compute the value equals the flow the value's optimal response generates. That re-coupling is the self-consistency condition the chapter set out to solve, and $(u^\star,m^\star)$ is an MFG equilibrium in the sense of definition 10.3.

Regularity of the solution. The asserted regularity is inherited from the two constructions the fixed point was extracted from. That $m^\star\in C([0,T];\mathcal{P}_2(\mathbb{R}^d))$ with a uniform $\frac{1}{2}$-Hölder modulus in time for $\mathcal{W}_1$ is the membership in $\mathcal{K}$, whose defining moment and equicontinuity bounds are lemmas 10.6 and 10.7; that $u^\star$ is Lipschitz in x uniformly in t is the gradient ceiling $\|\nabla u^{m^\star}\|_\infty\le L_u$ of lemma 10.5, which held uniformly over *every* frozen flow and so in particular at $m^\star$. ☐

Remark 10.15 (What the theorem buys, and what it costs)**.** Every fixed-point existence theorem is a bargain, and it pays to read the price tag. The cost of theorem 10.14 is concentrated in hypothesis (ii): the couplings F,G must be *bounded and smoothing*. This is no idle technical convenience — it is exactly the property that made the gradient ceiling lemma 10.5 uniform over all frozen flows m, and that uniform ceiling was the seed of everything downstream (the moment and equicontinuity bounds defining $\mathcal{K}$, and the uniform $C^{1,2}$ estimate underwriting continuity). A coupling whose value or gradient could blow up as the population concentrates would break that ceiling at the source and the whole construction with it. The class that qualifies is the *nonlocal/regularising* one, and it is worth keeping three concrete admissible templates in view. *(a) Convolution couplings,* $F[m](x)=(\phi*m)(x)=\int\phi(x-y)\,m(\mathrm{d}y)$ with a bounded, Lipschitz kernel ϕ: the convolution smooths the measure into a function whose sup norm and Lipschitz constant are controlled by ϕ alone, uniformly in m, no matter how singular m becomes. *(b) Scalar-moment couplings,* $F[m](x) = f(x, \int \eta \, \mathrm{d}m)$ with η bounded and f bounded and Lipschitz: here the population enters only through a finite-dimensional bounded statistic $\int \eta \, \mathrm{d}m$, so F inherits f's boundedness and regularity directly. *(c) Mollified local couplings,* $F[m](x) = f(x, (\rho_\varepsilon * m)(x))$ with f bounded Lipschitz and ρ_ε a fixed mollifier — a regularised stand-in for the genuinely local cost $f(x, m(x))$ that depends on the pointwise density, which is itself *not* admissible because $m \mapsto m(x)$ is neither bounded nor weakly continuous. What the theorem *buys* in exchange is substantial: a classical-in-x, Lipschitz value function $u^\star$ and a time-continuous equilibrium flow $m^\star$, obtained with *no monotonicity hypothesis whatsoever* on the couplings. The silence on monotonicity is the honest cost on the other side of the ledger: with no (A-Mon) in force the theorem claims existence and *only* existence, and the equilibrium it produces may be one of several. This is the standard division of labour in the analytic MFG literature [93, 36]: nondegeneracy plus bounded smoothing couplings $\Rightarrow$ existence by Schauder, here; monotonicity (A-Mon) $\Rightarrow$ uniqueness, separately, in chapter 11. The two relaxations the theorem still owes both trace to a hypothesis it could not drop — (ii)'s boundedness and (i)'s strict convexity — and each is bought back elsewhere: *unbounded data* costs the compactness construction and is recovered by linear structure for the LQG thread in section 10.5, while *non-unique feedback* costs single-valuedness and is recovered by set-valued fixed-point theory in section 10.4.

Remark 10.16 (The probabilistic route gives the same conclusion)**.** The fixed point can equally be sought on flows directly in the FBSDE picture of chapter 9: freeze m, solve the McKean–Vlasov-decoupled forward–backward system for (X^m, Y^m, Z^m), and set $\mathsf{S}(m) = \mathrm{Law}(X^m_\cdot)$. Schauder applies verbatim once the same compactness (tightness of $\{\mathrm{Law}(X^m_\cdot)\}$) and continuity (stability of FBSDE solutions in the frozen flow) are in hand; this is the route of [40, Ch. 3–4, Vol. I]. We do not need to re-run the fixed point in the probabilistic picture, however: the FBSDE–PDE dictionary proposition 9.12 identifies the law of the closed-loop state computed from (X^m, Y^m, Z^m) with the law computed from the HJB value u^m, so the equilibrium flow $m^\star$ produced above *is* a solution of the McKean–Vlasov FBSDE of chapter 9, and conversely. This is exactly what theorem 9.10(iii) promises when it calls the Schauder fixed point "the existence theorem of chapter 10": the PDE and probabilistic existence proofs are two coordinatisations of the one Schauder argument, transported across the dictionary, mirroring the "two pictures" theme of chapter 9.

This completes the single-valued existence story, and before moving on it is worth tallying exactly what it leaned on — because the two sections that follow are each defined precisely by one support this argument could not do without. The Schauder argument just assembled rested on three crutches, each entering at an identifiable point above. *Single-valued feedback*: the optimal-feedback formula eq. (10.4) had to name one drift, not a set, for S to be a genuine map rather than a correspondence and for Schauder even to have an object to act on. *Strict convexity* of $H(x, \cdot)$: this is what made the middle arrow of the factorization continuous, rendering the Legendre maximiser a single continuous function of momentum. *Lipschitz drift*: this is what powered the synchronous-coupling proof of lemma 10.11, supplying the self-improving Grönwall estimate that delivered transport stability. Above these three sat a fourth hypothesis, hidden in plain sight in (ii) and flagged in remark 10.15: the *boundedness* of the data, without which the gradient ceiling — and hence the entire compactness arena $\mathcal{K}$ — never gets off the ground.

Two of the chapter's promised relaxations are therefore now precisely owed, and we name both so that each later section arrives pre-motivated rather than as a fresh start. The first is *unbounded data*. The LQG thread carried throughout the book has linear-quadratic costs whose data grow without bound in the state, so it violates hypothesis (ii) outright and falls outside theorem 10.14 entirely — the compactness construction simply does not apply to it. Section 10.5 shows that this is no obstruction in disguise: the very linearity that breaks boundedness collapses the fixed-point problem to a finite-dimensional two-point linear system with a sharper existence-and-uniqueness statement, recovering existence by structure where the abstract theorem could not reach. The second is *non-unique feedback*. When the first three crutches fail together — a merely convex Hamiltonian with a flat maximiser set, a non-convex action set whose optimiser wants to mix, or a degenerate diffusion whose value is only a viscosity solution with no pointwise gradient — the formula eq. (10.4) returns a *set* of optimal drifts and S ceases to be a map at all. Section 10.4 confronts exactly this case and shows that an equilibrium survives nonetheless, at the price of relaxing controls, replacing the single-valued S by a set-valued best response settled by the Kakutani–Fan–Glicksberg theorem in place of Schauder, and recovering an honest, executable feedback after the fact by measurable selection. The next section takes up the second of these; the LQG counterpoint waits for section 10.5.

10.4 Relaxed equilibria and measurable selection

The previous section closed by tallying the three crutches the Schauder argument leaned on and naming the two relaxations it still owed; this section discharges the second of them. The route of sections 10.2 and 10.3 delivered an equilibrium only in the favourable regime where the best-response map S is a genuine single-valued, continuous self-map of $\mathcal{M}$ — the regime in which a nondegenerate diffusion (A-Nondeg) and a *strictly* convex Hamiltonian (the strong form of (A-Conv)) cooperate to make the optimal-feedback formula eq. (10.4) name exactly one drift for each (t, x). That is the smooth side of the single-valued/set-valued dichotomy posed in section 10.1. We now cross to the rough side. We drop the two supports that the existence theorem flagged as removable — *single-valuedness* of the feedback and the *strict convexity* of H that guaranteed it — and ask what survives when S is no longer a map at all but a *correspondence*, returning a whole set of optimal drifts for each (t, x) in place of a single one.

It helps first to see that the three distinct ways single-valuedness failed, catalogued in the "two regimes" analysis of section 10.1, are at bottom one obstruction. The three mechanisms there were a merely convex Hamiltonian whose argmax in eq. (10.5) is a flat segment rather than a point; a non-convex action set A on which the optimiser is stranded strictly between two admissible drifts; and a degenerate diffusion that, with (A-Nondeg) abandoned, leaves u a viscosity solution with no classical gradient to feed eq. (10.4). Different as the three pictures look, they share a single geometric root: the set of *attainable* instantaneous dynamics — the drifts an agent can actually produce, weighed against their cost — need not be convex, and over a non-convex attainable set the cost-optimal velocity can be one that no single admissible action realises. When that happens no deterministic feedback is optimal, and the Schauder construction, which presupposed exactly one closed-loop SDE to transport the population forward, has nothing to transport. The maximiser set in eq. (10.4) fails to be a singleton not by accident but because the underlying optimisation has genuinely no pure solution.

This is not a new pathology peculiar to mean field games; it is the oldest difficulty in game theory, and it has a classical cure, which we carry as the motivation for everything below. A finite game can fail to possess a *pure*-strategy Nash equilibrium for exactly this reason — matching pennies is the textbook witness: against any deterministic choice the opponent has a strict deterministic counter, so the pure best-response correspondence jumps from one extreme to the other and closes on nothing, and no pure profile is left fixed. Von Neumann's resolution, the founding move of the subject, was to *enlarge the strategy space*: permit each player a *mixed* strategy, a probability distribution over actions rather than one action. Mixing convexifies. Recomputed over distributions of actions, the attainable payoffs fill out the convex hull of the pure ones; the best-response correspondence acquires nonempty convex closed values and an upper-semicontinuous graph; and a fixed point reappears — which is precisely the structure Nash fed to Kakutani's theorem to prove that every finite game has an equilibrium in mixed strategies. The relaxation we are about to perform is this very move, transplanted to a continuous-time control problem. The mean field analogue of a mixed strategy is a *relaxed control*: at each instant the agent plays not a single action but a probability measure over actions, and the state is driven by the resulting *averaged* drift. Averaging *is* convexification; it returns to the attainable dynamics exactly the convexity that the non-convex modes destroyed, and with it an optimiser and the upper-semicontinuity that a fixed-point theorem demands. The slogan to carry through the section is that *relaxed controls are to a mean field game what von Neumann's mixed strategies are to a finite one.*

With that motivation fixed, the section proceeds in three movements, and it is worth saying in advance which failure mode each repairs, since the two devices are not interchangeable. The first subsection performs the von Neumann move itself: it defines relaxed (Young-measure) controls and the associated notion of a *weak*, or relaxed, equilibrium, carried — in the path-space style of chapter 9 — as a single law on trajectory-and-control space rather than as a deterministic flow paired with a feedback. The convexity that randomisation buys is, crucially, visible only one level up: it is manifest on the *joint* law of state and control, where mixing two candidate equilibria is literally averaging two measures, and *not* on the state-marginal flow alone, on which the law of the controlled path is not an affine image of the control. That is why the fixed point must migrate off the flow m and onto the joint law Q — the business of the second subsection, which erects the set-valued best response on joint laws and settles its fixed point with the Kakutani–Fan–Glicksberg theorem (theorem 2.40) in place of Schauder, through its three checks: nonempty convex values, closed graph, and a convex compact domain. Relaxation is what cures the genuinely non-convex modes, where it manufactures an optimiser that did not exist before. It does *not*, by itself, return the deterministic feedback an agent must ultimately execute, nor does it resolve the residual indifference of the flat-maximiser mode, where the optimiser set was already convex and nonempty and the only difficulty was *choosing* one element of it. Both of those are the task of the third subsection: a *measurable selection* plucks a single optimiser from the maximiser set at each (t, x), coherently enough across (t, x) to define a genuine feedback, by the classical selection theorems (theorem B.29). Selection without relaxation disposes of a flat maximiser set; relaxation without selection convexifies but does not yet execute; only together do they close the dichotomy that has organised the whole chapter.

At the end of these three movements the abstract existence theory is complete along both of the branches it was ever going to need for *bounded* data: Schauder for the smooth, single-valued regime of theorem 10.14, and Kakutani for the rough, set-valued regime of theorem 10.19, with an honest feedback recovered in either case. The one kind of game that escapes both branches is one whose data are *unbounded*, for then the compactness arena itself never gets built — and that is exactly the carried LQG thread, which section 10.5 will show slips out of the entire compactness apparatus only to be caught, far more tightly, by its own linear structure. We begin with the cure itself.

10.4.1 Relaxed controls and weak equilibria

Before the formalism, the picture — recalled in one line from the section opening: over a non-convex attainable set the optimiser "wants to sit between two actions," and von Neumann's cure is to let the agent *randomise*, playing a probability distribution over actions rather than a single action. The averaged dynamics then range over the *convex hull* of the pure drifts, the value function becomes upper-semicontinuous, and an optimiser exists. A randomisation rule, time-indexed, is exactly a measure on $[0,T] \times A$ projecting to Lebesgue measure in time — a *relaxed control.* With this motivation we state the definition.

Definition 10.17 (Relaxed control and relaxed feedback)**.** A *relaxed (open-loop) control* on the action set $A \subseteq \mathbb{R}^k$ is a measure ρ on $[0,T] \times A$ whose time-marginal is Lebesgue measure, i.e. $\rho(\mathrm{d}t, \mathrm{d}a) = \rho_t(\mathrm{d}a)\ \mathrm{d}t$ with each $\rho_t \in \mathcal{P}(A)$. A *relaxed feedback* is a measurable map $(t,x) \mapsto \rho_{t,x} \in \mathcal{P}(A)$. The relaxed state dynamics replace the drift $b(x,a)$ by its average $\bar{b}(t,x) = \int_A b(x,a)\, \rho_{t,x}(\mathrm{d}a)$ and the running cost likewise.

One subtlety deserves emphasis, lest relaxation seem to change the problem: under (A-Nondeg) and convexity (A-Conv) the relaxed and strict problems have the *same* value and the same optimal feedback (no genuine randomisation is needed — a Dirac relaxed control $\rho_{t,x} = \delta_{\alpha^*(t,x)}$ is optimal), so relaxation changes nothing in the regime of theorem 10.14 and only buys generality where that theorem does not apply — precisely the merely-convex or degenerate case at hand.

Definition 10.18 (Weak / relaxed MFG equilibrium)**.** A *weak MFG equilibrium* is a probability measure P on the canonical path-and-control space $\Omega = C([0,T];\mathbb{R}^d) \times \{\text{relaxed controls}\}$ such that, writing $m_t = P \circ (X_t)^{-1}$ for its state time-marginals:

(a) *Consistency:* under P the canonical process solves the controlled martingale problem associated with eq. (10.1) driven by the relaxed control, with $X_0 \sim m_0$;

(b) *Optimality:* P-almost every realisation of the control is optimal for the cost functional in which the population is frozen to the flow $(m_t)_t$ that P itself generates.

Definition 10.18 is the path-space formulation of equilibrium owned conceptually by chapter 9: instead of a deterministic flow m and a feedback, one carries a single law P on trajectories, and the self-consistency is that the population flow read off P is the very flow against which P is optimal. It subsumes definition 10.3: a strong equilibrium $(u^\star, m^\star)$ induces a weak equilibrium

$P^\star = \mathrm{Law}(X^\star_\cdot, \delta_{\alpha^*})$. The advantage is that weak equilibria exist under dramatically weaker hypotheses, because the fixed point is now sought for a set-valued map on a space of *laws*, where Kakutani's machinery is available.

10.4.2 The set-valued best response and Kakutani–Fan–Glicksberg

The correct ambient object for the set-valued fixed point is *not* the flow m but the *joint law* of state and relaxed control, because it is on joint laws — not on their state-marginals — that convexity is manifest (one literally mixes the measures). Let $\mathcal{Q}$ be the set of probability measures Q on the path-and-control space $\Omega = C([0,T];\mathbb{R}^d) \times \mathcal{R}$, where $\mathcal{R}$ is the (compact, metrisable) space of relaxed controls of definition 10.17, such that: Q has time-marginal flow $m^Q_t = Q \circ X_t^{-1}$ lying in the convex compact set $\mathcal{P}_{\mathrm{e}}$ of flows built as in lemma 10.8 (or, on a compact state space such as the torus $\mathbb{T}^d$, automatically), $Q \circ X_0^{-1} = m_0$, and under Q the canonical process solves the controlled martingale problem of definition 10.18(a). The set $\mathcal{Q}$ is convex and compact for the topology of weak convergence (the constraints — moment bounds, fixed initial law, the martingale-problem identity — are all convex and closed). Define the *set-valued best response* on $\mathcal{Q}$,

$$R(Q) \;=\; \Big\{ P \in \mathcal{Q} \;:\; P \text{ is relaxed-optimal against the frozen flow } m^Q \Big\} \subseteq \mathcal{Q}, \tag{10.7}$$

i.e. $P \in R(Q)$ iff the control marginal of P is, P-a.s., optimal for the cost in which the population is frozen to m^Q (and P solves the same martingale problem). A fixed point $Q^\star \in R(Q^\star)$ is a Q that is optimal against its own flow $m^{Q^\star}$ — exactly a weak MFG equilibrium (definition 10.18), whose equilibrium flow is the state-marginal $m^\star = m^{Q^\star}$.

The fixed-point theorem for correspondences is the Kakutani–Fan–Glicksberg theorem, which the book proves in theorem 2.40: *a correspondence $\Psi : \mathcal{C} \rightrightarrows \mathcal{C}$ on a nonempty convex compact subset $\mathcal{C}$ of a locally convex Hausdorff space, with nonempty convex closed values and closed graph, has a fixed point.* (The weak topology on $\mathcal{Q}$ is not normable, so the genuinely locally convex version proved there — not merely the Banach Schauder of theorem 2.38 — is what is needed here.)

Theorem 10.19 (Existence of weak MFG equilibria)**.** *Assume (A-Lip), (A-Growth) and (A-Nondeg), that the action set A is compact (or that coercivity (A-Growth) confines optimal relaxed controls to a compact set), that the drift, volatility and running/terminal costs are continuous in (x, a, m) with the cost convex in the relaxed sense, and that the couplings are continuous for weak convergence. Then the correspondence R of eq.* (10.7) *satisfies the hypotheses of theorem 2.40, and hence the game admits a weak MFG equilibrium in the sense of definition 10.18.*

Proof. Three claims must be checked for R on the convex compact set $\mathcal{Q}$; each is a named in-book result, applied to the controlled martingale problem. *(1) Nonempty values.* For each frozen Q (hence frozen flow m^Q) the relaxed control problem attains its optimum: by theorem C.22 the relaxed admissible set is compact in the Young-measure topology and the relaxed cost is lower semicontinuous, so a minimiser exists, and lifting it to a joint law gives a nonempty $R(Q) \subseteq \mathcal{Q}$. *(2) Convex values.* If $P_0, P_1 \in R(Q)$ and $\theta \in [0,1]$, the mixture $P_\theta = \theta P_0 + (1-\theta)P_1$ is again a probability measure on Ω; it solves the same (linear-in-P) martingale-problem identity, lies in $\mathcal{Q}$, and has cost

$\theta\,\mathrm{cost}(P_0) + (1-\theta)\,\mathrm{cost}(P_1)$, which equals the common optimal value, so $P_\theta \in R(Q)$. Convexity holds *because we mix the joint laws*: it is precisely to make this step legitimate that the fixed point is run on $\mathcal{Q}$ rather than on the state-marginal flows, on which $\mathrm{Law}(X_\cdot)$ is not an affine image of the control. *(3) Closed graph.* If $Q^{(n)} \to Q$ and $P^{(n)} \in R(Q^{(n)})$ with $P^{(n)} \to P$, then by stability of the controlled martingale problem (theorem C.23; the limit solves the martingale problem for the limiting frozen flow, using (A-Nondeg) for well-posedness and the assumed continuity of coefficients and couplings) the limit P is admissible, and by Berge's maximum theorem (theorem B.27; upper semicontinuity of the value and closedness of the arg min correspondence under the continuous data) P is optimal against m^Q, so $P \in R(Q)$. Compactness of $\mathcal{Q}$ was noted above (or follows from compactness of the state space). Theorem 2.40 then provides $Q^\star \in R(Q^\star)$, a weak equilibrium with flow $m^\star = m^{Q^\star}$. The controlled-martingale-problem framework and its Young-measure inputs are developed in appendix C.13; this is the route of Lacker [88] and [40, Ch. 3, Vol. I]. □

10.4.3 From a weak equilibrium back to a feedback: measurable selection

Theorem 10.19 produces a law $P^\star$, possibly carrying a genuinely randomised control. For applications one wants a *feedback* — a function $(t,x) \mapsto \alpha^*(t,x)$ that an agent can actually execute. When is the relaxed equilibrium realisable as a (Markov, deterministic) feedback? The obstruction is exactly the non-uniqueness of the maximiser in eq. (10.4): the optimiser correspondence

$$(t,x) \;\longmapsto\; A^\star(t,x) \;=\; \operatorname*{arg\,max}_{a\in A} \big\{ -\langle b(x,a), p^\star(t,x)\rangle - L(x,a,m_t^\star)\big\}, \qquad p^\star = \nabla u^\star,$$

may be a set, not a point. The maximand here is exactly the integrand whose supremum *defines* the Hamiltonian: with $H(x,p,m) = \sup_{a\in A}\{-\langle b(x,a),p\rangle - L(x,a,m)\}$ the Legendre–Fenchel transform introduced in section 10.1, $A^\star(t,x)$ is the set of maximisers attaining $H(x,p^\star,m_t^\star)$, and when $H(x,\cdot)$ is strictly convex this set collapses to the singleton $\{-\partial_p H\}$, recovering the single-valued feedback α^* of eq. (10.4). In the rough case one must choose one element of $A^\star(t,x)$ for each (t,x), *measurably*. This is a selection problem, and it is solved by the classical selection theorems.

Lemma 10.20 (Measurable selection of the optimal feedback)**.** *Suppose A is a Polish space, the map $(t,x) \mapsto A^\star(t,x)$ has nonempty closed values, and it is measurable in the sense that $\{(t,x) : A^\star(t,x) \cap U \neq \emptyset\}$ is measurable for every open $U \subseteq A$ (a consequence of continuity of $(t,x,a) \mapsto$ the maximand and measurability of $p^\star = \nabla u^\star$). Then $A^\star$ admits a measurable selection: there is a measurable $\alpha^* : [0,T]\times\mathbb{R}^d \to A$ with $\alpha^*(t,x) \in A^\star(t,x)$ for all (t,x).*

Proof. This is an instance of the Kuratowski–Ryll-Nardzewski selection theorem (theorem B.29): a measurable correspondence with nonempty closed values into a Polish space admits a measurable selection. Its hypotheses are met here because the maximand is jointly continuous in a and Carathéodory-measurable in (t,x) through $p^\star(t,x) = \nabla u^\star(t,x)$, so $A^\star$ is a closed-valued measurable multifunction. (When $A \subseteq \mathbb{R}^k$ and the maximand is continuous, one does not even need the general theorem: the explicit *lexicographic* selection "take the optimiser of smallest first coordinate, then smallest second, . . . " is itself measurable — the elementary Filippov selection of control theory — and we give that self-contained argument in theorem B.29's proof as the Euclidean special case.) □

Remark 10.21 (The role of nondegeneracy, restated)**.** Under (A-Nondeg) *and* strict convexity (A-Conv), $A^\star(t,x)$ is a singleton for a.e. (t,x): the value function is once differentiable in x (parabolic regularity) and the strictly convex Hamiltonian has a unique maximiser, so the selection problem is vacuous and the feedback eq. (10.4) is already a function. Selection theory is needed exactly when one of these two is dropped — degenerate diffusion (only viscosity u, gradient may not exist classically) or non-strictly-convex H (flat maximiser set). The chapter's two halves are thus the two sides of a single dichotomy: *nondegenerate + strictly convex* gives a continuous single-valued S and Schauder (theorem 10.14); *neither* gives an upper-hemicontinuous set-valued S, Kakutani, and a measurably selected feedback (theorem 10.19 and lemma 10.20).

10.5 Thread A: the LQG equilibrium is the equilibrium

The abstract existence theory is now complete on both of the branches it was built for — Schauder for the smooth, single-valued regime (theorem 10.14) and Kakutani for the rough, set-valued one (theorem 10.19) — but, as the previous section warned, both branches presuppose data bounded enough to build the compactness arena, and exactly one carried example slips out of both. We close the chapter by honouring the carried-thread duty for it: we apply the existence viewpoint to the LQG mean field game (Thread A) and confirm that the explicit Gaussian flow — constructed at the level of Riccati equations in chapter 6 (single agent) and assembled into a fixed point in chapter 14 — is genuinely *an* MFG equilibrium, and that at short horizon it is the *unique* one. As flagged in the notation watch of section 10.2, the LQG data are unbounded, so theorem 10.14 does not apply directly; the linear structure supplies its own, sharper, existence proof.

Worked computation 10.22 (The LQG best response collapses to a scalar fixed point)**.** Recall the scalar Thread A model: the representative agent has

$$\mathrm{d}X_t = (aX_t + b\,\alpha_t)\;\mathrm{d}t + \sigma\;\mathrm{d}W_t, \qquad X_0 \sim m_0,$$

and cost

$$J(\alpha) = \mathbb{E}[*] \int_0^T \left(\tfrac{1}{2}\alpha_t^2 + \tfrac{q}{2}\,(X_t - s\,\bar m_t)^2\right) \mathrm{d}t + \tfrac{q_T}{2}\,(X_T - s_T \bar m_T)^2,$$

with $\bar m_t = \int x\, m_t(\mathrm{d}x)$ the population mean and parameters $a, b, \sigma, q, q_T \geq 0, s, s_T, T$ as fixed in the thread specification. The coupling enters *only* through the scalar function $t \mapsto \bar m_t$; this is the structural windfall of the linear model.

Step 1 — the single-agent best response (frozen $\bar m$). For a frozen continuous mean $\bar m : [0,T] \to \mathbb{R}$, the verification theorem of chapter 6 gives a value function quadratic in the state,

$$u(t,x) = \tfrac{1}{2} P(t)\,x^2 + r(t)\,x + \chi(t),$$

whose coefficients solve, backward from $t = T$,

$$-\dot P = 2aP - b^2 P^2 + q, \qquad\qquad P(T) = q_T, \tag{10.8}$$

$$-\dot r = (a - b^2 P)\,r - q\,s\,\bar m_t, \qquad\qquad r(T) = -q_T\, s_T\, \bar m_T, \tag{10.9}$$

and χ a quadrature that does not affect the feedback. Here $P(t)$ is the Riccati curvature and $r(t)$ the linear (decoupling) offset of the value, in the canonical thread-A symbols fixed in the thread specification; this is the deterministic shadow of the forward–backward stochastic system eq. (9.22) of worked computation 9.16, with the costate curvature there named P and the costate offset named r identically. The Riccati equation eq. (10.8) is decoupled from the population (it is the single-agent curvature of chapter 6); under $q, q_T \geq 0$ it has a unique bounded nonnegative solution P on $[0, T]$. The optimal feedback is linear,

$$\alpha^*(t, x) = -b\,\partial_x u(t, x) = -b\big(P(t)x + r(t)\big),$$

so the closed-loop dynamics are the linear SDE

$$\mathrm{d}X_t = \big[(a - b^2P(t))X_t - b^2 r(t)\big]\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Step 2 — the output flow is Gaussian, and its mean closes. Because the closed loop is linear with additive Gaussian noise and m_0 is (taken) Gaussian, $m_t = \mathsf{S}(m)_t$ is Gaussian for all t: $m_t = \mathcal{N}(\hat{m}(t), V(t))$. The mean and variance solve

$$\begin{aligned} \dot{\hat{m}}(t) &= (a - b^2P(t))\,\hat{m}(t) - b^2 r(t), & \hat{m}(0) &= \bar{m}_0 := \textstyle\int x\, m_0(\mathrm{d}x), \\ \dot{V}(t) &= 2(a - b^2P(t))\,V(t) + \sigma^2, & V(0) &= \mathrm{Var}(m_0). \end{aligned} \tag{10.10}$$

The variance equation is *autonomous* — it does not see the coupling — so V is determined once and for all by P; the entire population interaction lives in the mean.

Step 3 — the fixed point is a scalar linear two-point problem. Equilibrium demands $\mathsf{S}(m) = m$, i.e. $\hat{m} \equiv \bar{m}$. Imposing this in eq. (10.10) and coupling it to eq. (10.9) gives the closed linear forward–backward system for the pair $(\bar{m}, r)$,

$$\begin{cases} \dot{\bar{m}}(t) = (a - b^2P(t))\,\bar{m}(t) - b^2 r(t), & \bar{m}(0) = \bar{m}_0, \\ -\dot{r}(t) = (a - b^2P(t))\,r(t) - q\,s\,\bar{m}(t), & r(T) = -q_T\,s_T\,\bar{m}(T), \end{cases} \tag{10.11}$$

a two-point boundary-value problem with data split between $t = 0$ and $t = T$ — the finite-dimensional shadow of the forward–backward MFG structure, identical term for term to the deterministic reduction eq. (9.22) of the FBSDE in worked computation 9.16. Any solution $(\bar{m}^\star, r^\star)$ of eq. (10.11) yields the equilibrium flow $m^\star_t = \mathcal{N}(\bar{m}^\star(t), V(t))$; this is precisely the explicit closed form assembled in chapter 14.

It remains to show that eq. (10.11) has a solution — existence of the LQG equilibrium — and to confirm that the closed-form flow is *the* equilibrium in the natural class.

Proposition 10.23 (Existence and small-horizon uniqueness for LQG). *Fix Gaussian $m_0 \in \mathcal{P}_2(\mathbb{R})$ and parameters with $q, q_T \geq 0$. Let P be the unique nonnegative bounded solution of eq. (10.8) on $[0, T]$.*

(i) *The map $\bar{m} \mapsto \hat{m} =: \Psi(\bar{m})$ defined by solving eq. (10.9) then eq. (10.10) is affine and continuous on $C([0, T]; \mathbb{R})$; it can be written $\Psi(\bar{m}) = c + \mathsf{L}\bar{m}$ with c the response to the initial datum $\bar{m}_0$ and L a bounded linear operator whose norm satisfies*

$$\|\mathsf{L}\| \;\leq\; \Gamma\, e^{2\gamma T}\big(|s_T| q_T\, T + |s| q\, T^2\big) \;\leq\; \Gamma\, e^{2\gamma T}\big(|s| q + |s_T| q_T\big) \max(T, T^2)$$

for constants Γ, γ *depending only on* a, b *and* $\sup_{[0,T]} P$ *(in particular independent of* T*, since* $\sup_{[0,T]} P$ *stays bounded as* T *grows). In particular* $\|\mathsf{L}\| \to 0$ *as* $T \to 0$.

(ii) *If* T *is small enough that* $\|\mathsf{L}\| < 1$*, then* Ψ *is a contraction, the fixed-point equation* $\bar{m} = \Psi(\bar{m})$ *— equivalently the boundary-value problem eq.* (10.11) *— has a* unique *solution* $\bar{m}^\star = (I - \mathsf{L})^{-1} c$*, and hence the LQG game has the unique equilibrium flow* $m_t^\star = \mathcal{N}(\bar{m}^\star(t), V(t))$.

(iii) *For arbitrary* T*, eq.* (10.11) *is a linear two-point boundary-value problem; it has a unique solution whenever the operator* $I - \mathsf{L}$ *is invertible, i.e. unless the associated homogeneous problem (*$\bar{m}_0 = 0$*,* $\bar{m} \equiv \mathsf{L}\bar{m}$*) has a nontrivial solution — a non-generic "conjugate-point" degeneracy. The set of horizons* T *at which* $I - \mathsf{L}$ *is singular is discrete (eq.* (10.12) *below), so for all but at most a discrete set of* T *the equilibrium exists and is unique. At an exceptional* T *the linear problem is* singular *and behaves dichotomously: writing* $\Psi(\bar{m}) = c + \mathsf{L}\bar{m}$*, the equilibrium equation* $(I - \mathsf{L})\bar{m} = c$ *has* no *solution when* $c \notin \mathrm{Range}(I - \mathsf{L})$ *(non-existence of an LQG equilibrium at that horizon) and a whole affine family of solutions when* $c \in \mathrm{Range}(I - \mathsf{L})$ *(genuine multiplicity). Only the latter is the multiplicity / symmetry-breaking case taken up for the explicit threshold in proposition 11.8 of chapter 11 and, in time-dependent closed form, in chapter 14; non-uniqueness or non-existence is in either case a measure-zero, discrete-in-*T *phenomenon, not an open region.*

Proof. (i) Solve eq. (10.9) by the backward variation-of-constants formula. Writing $\Theta(t) = \exp\big(\int_0^t (a - b^2 P(\tau))\,\mathrm{d}\tau\big)$ for the closed-loop fundamental solution, the linear ODE eq. (10.9) for the offset r integrates to

$$r(t) = \frac{\Theta(t)}{\Theta(T)}\big(-q_T s_T\,\bar{m}(T)\big) + \Theta(t)\int_t^T \frac{q\,s\,\bar{m}(\tau)}{\Theta(\tau)}\,\mathrm{d}\tau,$$

which is linear in $\bar{m}$ with no constant term; hence $r = \mathsf{L}_r\bar{m}$ for a bounded linear L_r. The two summands are bounded separately: the boundary term by $e^{\gamma T} q_T |s_T|$ and the integral term by $e^{\gamma T} q|s|\,T$ (the exponential ratios $\Theta(t)/\Theta(\tau)$ are bounded by $e^{\gamma T}$ with $\gamma = \sup_{[0,T]} |a - b^2 P|$, and the τ-integral over $[t, T]$ contributes the extra factor T), so $\|\mathsf{L}_r\| \le e^{\gamma T}(q_T|s_T| + q|s|\,T)$. Then $\hat{m}$ solves eq. (10.10), a forward linear ODE with source $-b^2 r = -b^2\mathsf{L}_r\bar{m}$ and initial datum $\bar{m}_0$; variation-of-constants again gives $\hat{m} = c + \mathsf{L}\bar{m}$ with $c(t) = \Theta(t)\bar{m}_0$ (independent of the input $\bar{m}$) and $\mathsf{L} = -b^2\,\mathcal{I} \circ \mathsf{L}_r$, where the bounded forward solution operator $\mathcal{I}$ contributes a further factor $b^2 e^{\gamma T} T$. Composing,

$$\|\mathsf{L}\| \;\le\; b^2 e^{2\gamma T}\big(q_T|s_T|\,T + q|s|\,T^2\big),$$

which is the bound of part (i) with $\Gamma = b^2$. The running-coupling term carries *two* powers of T (one from each of the backward and forward integrations) and the terminal term only one; both vanish as $T \to 0$. *(ii)* For T small the right side is < 1; the contraction mapping theorem (theorem 2.36) gives the unique fixed point $\bar{m}^\star = (I - \mathsf{L})^{-1} c = \sum_{k\ge 0} \mathsf{L}^k c$. *(iii)* The system eq. (10.11) is linear with split boundary data; by the standard theory of linear two-point boundary-value problems it is solvable, and uniquely, iff $I - \mathsf{L}$ is invertible. Concretely, let $\Phi(t) \in \mathbb{R}^{2\times 2}$ be the fundamental matrix of the homogeneous first-order system obtained by stacking $(\bar{m}, r)$ in eq. (10.11) (forget the inhomogeneous data $\bar{m}_0$ and the terminal source). The split boundary conditions $\bar{m}(0) = 0$ and

$r(T) + q_T s_T \bar{m}(T) = 0$ have a nontrivial common solution iff

$$\Delta(T) \;:=\; \det \begin{pmatrix} e_{\bar{m}}^{\top}\,\Phi(0) \\ (e_r + q_T s_T e_{\bar{m}})^{\top}\,\Phi(T) \end{pmatrix} \;=\; 0, \tag{10.12}$$

where $e_{\bar{m}}, e_r$ are the coordinate vectors. The entries of Φ are solutions of a linear ODE with real-analytic (indeed smooth) coefficients in t, hence $T \mapsto \Delta(T)$ is real-analytic and not identically zero (for small T it is nonzero by part (ii)), so its zeros are isolated. Away from them $I - \mathsf{L}$ is invertible and the equilibrium exists and is unique; at a zero, the dichotomy of the proposition statement holds by the Fredholm alternative for the finite-rank perturbation $I - \mathsf{L}$ of the identity. $\square$

Heuristic (non-rigorous): Conjugate points: the geometric content of eq. (10.12)

The two-point problem eq. (10.11) asks the forward mean-flow (started at $\bar{m}_0$, $t = 0$) and the backward adjoint-flow (started from the terminal coupling, $t = T$) to match in the middle. For most horizons they match in exactly one way. The vanishing of $\Delta(T)$ is precisely the event that the forward and backward linear flows become *compatible along a one-parameter family* — a zero mode of the linearised boundary-value operator. This is the same phenomenon as a *conjugate point* in the classical calculus of variations, where the second-variation operator on $[0, T]$ acquires a kernel and the minimiser ceases to be unique. Here it is the finite-dimensional precursor of the genuine symmetry-breaking threshold of the full MFG, developed for the explicit phase boundary in proposition 11.8 (chapter 11) and chapter 14. Note that $\Delta(T) = 0$ is a codimension-one (discrete-in-T) event: multiplicity is a knife-edge, never an interval.

Remark 10.24 (Reconciling with the abstract theory)**.** Proposition 10.23 is the honest LQG existence statement, and it differs in spirit from theorem 10.14 in exactly the way the notation watch promised. The LQG coupling is unbounded, so the uniform moment/gradient bounds underpinning lemma 10.8 are unavailable — yet the *linear–Gaussian* structure collapses the infinite-dimensional fixed point on $\mathcal{M}$ to a *finite* (indeed scalar-flow) linear problem eq. (10.11), where existence is read off the theory of linear boundary-value problems and short-horizon uniqueness off a contraction. Two lessons survive to the general theory. First, the relevant "mean field" here is not the whole measure flow but its mean $\bar{m}_t$: the best response factors through a finite-dimensional sufficient statistic, the Gaussian structure being preserved by S. Second, uniqueness is *not* automatic — it holds for small T (or, as chapter 11 will show, whenever the coupling sign makes the problem Lasry–Lions monotone, e.g. $s \le 0$, "aversion to the crowd"), and can fail at large T with attractive coupling, exactly the regime in which the homogeneous problem of part (iii) acquires a nontrivial solution. The explicit uniqueness/multiplicity threshold for the stationary-tuned reduction — the condition $b^2 qs = A_\infty^2$ and the critical horizons of eq. (11.14) — is derived in proposition 11.8 of chapter 11; the full time-dependent phase diagram in (s, q, T) for general $q_T \neq P_\infty$ is the capstone of chapter 14.

Rigor ledger

Proved.

- Equilibria of the MFG system eq. (10.2)–eq. (10.3) are exactly the fixed points of the best-response map S (definition 10.3).
- Under (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) with strictly convex H and *bounded, smoothing* couplings, S maps a convex compact set of flows $\mathcal{K}$ continuously into itself, so a fixed point exists by Schauder (theorem 10.14). Compactness of $\mathcal{K}$ rests on a uniform second moment (tightness, Prokhorov, chapter 3) plus uniform $\frac{1}{2}$-Hölder time-regularity (lemmas 10.6 to 10.8); continuity rests on parabolic stability of HJB (A-Nondeg) and SDE/martingale-problem stability (lemmas 10.10 and 10.11 and proposition 10.13).
- When the feedback is non-unique (degenerate diffusion or non-strict convexity), relaxed/weak equilibria (definition 10.18) exist via the Kakutani–Fan–Glicksberg theorem (theorem 2.40) applied to a set-valued best response on *joint* path-and-control laws (theorem 10.19), whose nonempty/convex/closed-graph inputs are the in-book relaxed-control existence and stability results theorems C.22 and C.23 and Berge's theorem theorem B.27; a measurable feedback is recovered by the Kuratowski–Ryll-Nardzewski selection (lemma 10.20, proved at theorem B.29).
- The LQG thread: the explicit Gaussian flow is an equilibrium; the fixed point is the scalar linear two-point problem eq. (10.11); it has a unique solution at short horizon (contraction) and for all but a discrete set of T (worked computation 10.22 and proposition 10.23).

Assumed.

- The named standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) as stated in appendix A; bounded, uniformly-Lipschitz, weakly-continuous couplings for the Schauder theorem; compact action set (or coercive confinement) for the relaxed theorem.
- Assumption 10.4 (bounded optimal feedback), derived from (A-Nondeg) plus bounded smoothing couplings in lemma 10.5 but *not* valid for the unbounded LQG data.

Deferred (each to a named in-book proof; external citations are for attribution only).

- The uniform parabolic gradient (Bernstein) estimate underlying lemma 10.5, the interior Schauder estimate and the classical comparison principle underlying lemma 10.10: proved in appendix E (theorems E.14, E.17 and E.19), on the classical solvability/transition-density theory of theorem C.12; cf. [36, 93].
- The generalized Arzelà–Ascoli theorem (theorem B.24) and the $\mathcal{W}_1$ Wasserstein facts — Kantorovich–Rubinstein duality, metrisation of weak convergence, completeness, and tightness/uniform-integrability compactness — (theorems D.10 to D.12) used in lemma 10.8.

- For the smoothing-coupling (Schauder) route the closed-loop SDE stability (lemma 10.11) is proved in-book by direct coupling (theorems 5.10 and C.3 and lemma C.1); the martingale-problem well-posedness and relaxed-control compactness/stability needed for the weak route are theorems C.21 to C.23 of appendix C.13, with Berge's theorem theorem B.27 and the Kuratowski–Ryll-Nardzewski selection theorem B.29; cf. [88, 40].
- *Uniqueness* of equilibria in general: to chapter 11 (Lasry–Lions monotonicity (A-Mon)); the explicit LQG uniqueness/multiplicity threshold to proposition 11.8, the full phase diagram to chapter 14.
- Convergence of N-player Nash equilibria to these MFG equilibria: to chapter 13.

Open.

- Existence with *degenerate* diffusion and *unbounded, non-smoothing* couplings simultaneously, without passing to relaxed/weak equilibria, is delicate and only partially understood; the strong-equilibrium (genuine feedback) theory there is incomplete.
- Sharp characterisation of the exceptional horizons T in proposition 10.23(iii) across the full parameter range, and the precise boundary between the existence-only and existence-and-uniqueness regimes, is the subject of chapters 11 and 14; for general (non-LQG) attractive couplings it is open.

Bibliographic notes

The fixed-point strategy for MFG existence originates with Lasry and Lions [93], who established existence for the analytic system under nondegenerate diffusion and regularising couplings; the Schauder argument as organised here — best response on flows of measures, compactness from parabolic gradient bounds and tightness, continuity from HJB stability — follows Cardaliaguet's lecture notes [36], which remain the cleanest analytic reference and to which the deferred parabolic estimates of lemmas 10.5 and 10.10 should be traced. The probabilistic counterpart (remark 10.16), in which the same Schauder fixed point is run on path-space laws via FBSDE stability, is developed in Carmona–Delarue, Volume I [40, Ch. 3–4], which also contains the Kakutani-based existence of equilibria with non-unique controls. The weak/relaxed equilibrium formulation of section 10.4, the controlled-martingale-problem set-up, and the Kakutani–Fan–Glicksberg existence theorem in that setting are due to Lacker [88]; we prove the relaxed-control compactness and stability and the controlled-martingale-problem well-posedness in appendix C.13. The fixed-point, maximum, and measurable-selection theorems themselves (theorems 2.40, B.27 and B.29) are classical and may be found in Aliprantis–Border [6, Ch. 17–18] and Granas–Dugundji [79]. The Wasserstein compactness inputs (tightness $\Rightarrow$ $\mathcal{W}_1$-relative compactness) are theorem D.12, proved from chapters 3 and 4; cf. [131]. The LQG worked computation continues the Nash-certainty-equivalence line of Huang, Caines and Malhamé [82]; its boundary-value/Riccati form eq. (10.11) and the multiplicity threshold are taken up again in chapters 11 and 14.

Chapter 11

Uniqueness and the Lasry–Lions Monotonicity Condition

The previous chapter (chapter 10) earned us *at least one* mean field game equilibrium: under boundedness, Lipschitz, and nondegeneracy hypotheses a fixed point of the best-response map S exists for regularising couplings (theorem 10.14), and for the carried linear–quadratic thread the equilibrium is constructed explicitly (proposition 10.23). Existence, however, is only half of what a useful theory owes us, and on its own it is close to epistemically empty. A mean field game is supposed to be a predictive object: feed it the data of a population – the dynamics, the costs, the initial distribution – and it returns the self-consistent future, the trajectory along which every agent's anticipation of the crowd is borne out by the crowd that those anticipations produce. If that return value is a *set* rather than a point, the model has told us only that the future will be one of several mutually incompatible things, and it has given us no rule for saying which. Two equilibria are two contradictory forecasts from one and the same dataset; absent a further selection principle, a model with several equilibria predicts nothing at all. A theory that aspires to forecast must therefore deliver not merely *an* equilibrium but *the* equilibrium. This chapter asks when it can, and isolates the single structural hypothesis under which uniqueness holds for *arbitrary* time horizon: the *Lasry–Lions monotonicity condition*, the precise form of the standing assumption tagged **(A-Mon)** in appendix A.6. That one hypothesis has a dual face we will meet from both sides. Read physically, it is the statement that the interaction is *repulsive* – agents are crowd-averse, each paying to sit where others already sit – and such an interaction convexifies the equilibrium energy landscape, leaving a single basin and nothing to choose between. Read analytically, it is the exact sign that an old and beautiful duality computation needs in order to turn the forward–backward coupling into a sum of manifestly nonnegative terms; the proof it buys is three lines of integration by parts and is one of the most illuminating arguments in the subject.

We import the mean field game system in its analytic form from chapter 8 – the backward Hamilton–Jacobi–Bellman (HJB) equation for the value $u = u(t, x)$, solved *backward* from terminal data at $t = T$, coupled to the forward Fokker–Planck (FP) equation for the population density $m = m(t, x)$, solved *forward* from initial data at $t = 0$. As fixed once and for all in appendix A.1, value runs backward and mass runs forward: the optimizer reasons from the terminal payoff back to now, while the crowd is carried by its history from the initial distribution onward. This opposition of arrows is not a notational accident but the structural signature of the problem, and it is precisely the reason uniqueness is delicate – there is no single time direction along which to march the coupled pair and watch errors decay, as the rest of this chapter makes explicit. From chapter 6 we import the control Hamiltonian $H(x,p)$, convex in the adjoint variable p; that convexity (assumption **(A-Conv)**) is what makes the optimal feedback single-valued and, as we will see, contributes one of the favorable signs in the uniqueness identity for free. From chapter 4 we import the flat (linear functional) derivative $\frac{\delta \cdot}{\delta m}(m,x)$ and the geometry of $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, the language in which "the coupling is the derivative of an energy" and "monotone means convex" are made precise. And we carry forward the Linear–Quadratic–Gaussian (LQG) thread (Thread A), whose parameters $a, b, \sigma, q, q_T, s, s_T, T$ are fixed in the thread specification and whose mean-coupling strength s will turn out to be the exact knob that tunes the system across the uniqueness boundary.

The chapter has a single arc, and it is worth seeing the whole of it before the first symbol. We begin (section 11.1) by establishing the *problem*: why uniqueness is not automatic, why the forward–backward structure defeats the contraction arguments that settle ordinary initial-value problems, and – through an attractive-coupling thought experiment – why a *sign* condition is the natural cure. That section ends by demanding exactly the object the next one supplies: section 11.2 pins the vague "repulsive enough" down to the checkable, horizon-free inequality eq. (11.3), the Lasry–Lions monotonicity condition, and interprets it as a crowd-averse coupling that convexifies the equilibrium landscape. Section 11.3 then spends that hypothesis: it proves the uniqueness theorem by the Lasry–Lions duality computation, given in full because it is the prototype every later uniqueness theorem in the book imitates. Having seen *why* monotonicity works, we ask what happens without it: section 11.4 reads the loss of uniqueness as a genuine symmetry-breaking / phase-coexistence phenomenon, sudden rather than gradual, with a Curie–Weiss self-consistency lurking underneath. Section 11.5 makes all of this computable on Thread A: we reduce the LQG mean field game to a deterministic two-point boundary-value problem for the population mean and read off its exact uniqueness/multiplicity phase diagram in (s, q, T), including the soft band where monotonicity fails yet uniqueness survives. Finally section 11.6 surveys the modern refinements – displacement monotonicity and weak monotonicity – that reach beyond the Lasry–Lions regime, and section 11.7 assembles the full menu of uniqueness hypotheses and locates the boundary of the uniqueness region. The destination is not local: the monotone regime is exactly the regime in which the master equation of chapter 12 is well posed, and the graphon uniqueness theory of chapter 24 is built on the operator generalization of the very computation given here. The short duality identity of this chapter is therefore load-bearing for much of the book that follows.

11.1 Why uniqueness is not automatic

The preamble flagged the value-backward/mass-forward structure as the reason uniqueness is delicate. It is time to turn that asserted signature into an argument, because seeing exactly *how* the opposed arrows defeat the usual uniqueness machinery is what tells us what a cure must look like.

Begin with the machinery that does work, so the contrast is sharp. In a standard *initial-value* problem – a forward stochastic differential equation, a forward parabolic PDE – Lipschitz data buy uniqueness for free by a Grönwall estimate. The mechanism is worth recalling in one line, because everything below is about why it is unavailable here. Suppose two solutions y_1, y_2 of $\dot{y} = b(t, y)$ share the same initial datum, with b Lipschitz in y with constant C. Then their difference $\delta := y_1 - y_2$ satisfies $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}|\delta|^2 = \delta \cdot (b(t, y_1) - b(t, y_2)) \le C|\delta|^2$, and since $|\delta(0)|^2 = 0$, Grönwall's inequality forces $|\delta(t)|^2 \le |\delta(0)|^2 e^{2Ct} = 0$ for all t. The argument has exactly one moving part: a single time direction, with the data nailed down at one end, along which any discrepancy is propagated and shown to stay zero. Uniqueness is automatic precisely because the problem has a direction to march.

The mean field game system has no such direction, and this is the whole difficulty. It is a *forward–backward* system: the value u is pinned at the *final* time T, the density m is pinned at the *initial* time 0, and the two are coupled through the entire interval $[0, T]$. Neither field can be marched alone. If we try to integrate the pair forward from $t = 0$ we know $m(0) = m_0$ but not $u(0)$, because u is determined by data at the far end; if we try to integrate backward from $t = T$ we know $u(T)$ but not $m(T)$, because m is determined by data at the near end. The honest way to phrase the problem is as a *shooting* problem – guess the missing datum at one end, integrate to the other, and adjust the guess until the boundary condition there is met – and the map from guess to mismatch can perfectly well vanish at more than one guess. That is multiplicity. The Grönwall trick collapses because the difference $\bar{u} = u_1 - u_2$ of two value functions starts at zero at the *terminal* time while the difference $\bar{m} = m_1 - m_2$ of two densities starts at zero at the *initial* time; there is no single instant at which *both* discrepancies are known to vanish, so no Grönwall loop closes. Concretely, the feedback is genuinely two-sided: agents best-respond to the population they *anticipate* over the whole horizon, and the population is the aggregate of exactly those best responses, so a long horizon gives the loop time to support several mutually consistent configurations – several anticipations, each self-fulfilling. (The one regime that survives is a short horizon, where the round trip is too brief for the loop to wind up; we return to it below and make the threshold $C\,T < 1$ explicit.) The technical reading of this chapter is: find the one structural condition under which, even though no Grönwall direction exists, a *different* bookkeeping device – a time-integrated pairing of the two discrepancies against each other – still forces them to vanish.

To set up that bookkeeping we first fix notation, recalling from chapter 8 the mean field game system in the analytic form we will use throughout. Write $\nu := \frac{1}{2}\sigma^2 > 0$ for the diffusion coefficient; the strict positivity here is exactly **(A-Nondeg)**, and although the uniqueness argument will use almost nothing about ν, nondegeneracy reappears at the very last step to guarantee that the linear equations with frozen coefficients are themselves uniquely solvable, so we keep it in view from the start. Let the coupling enter through two functionals

$$F\colon \mathcal{P}_2(\mathbb{R}^d) \to C(\mathbb{R}^d) \cap L^1(\mathbb{R}^d), \qquad G\colon \mathcal{P}_2(\mathbb{R}^d) \to C(\mathbb{R}^d) \cap L^1(\mathbb{R}^d),$$

the running coupling and the terminal coupling. Each takes a population measure m and returns a function of the state x: the number $F(x, m)$ is the extra running cost an agent sitting at x pays because the rest of the population is distributed as m, and $G(x, m)$ is the analogous terminal cost. This is how the crowd is felt in the single agent's problem – not as a force, but as a state-dependent surcharge that the population's own shape dictates. Two structural facts about the codomain are doing quiet work and are recorded here once. Continuity, $F(\cdot, m) \in C(\mathbb{R}^d)$, is what lets the cost be evaluated pointwise along trajectories. Integrability, $F(\cdot, m) \in L^1(\mathbb{R}^d)$, is what we will need in section 11.2: the monotonicity condition eq. (11.3) is built from the pairing

$\int_{\mathbb{R}^d} (F(x,m) - F(x,m'))\,(m-m')(\mathrm{d}x)$, an integral of a cost *difference* against the finite signed measure $m - m'$, and it is the L^1 membership of $F(\cdot, m)$ that makes this integral absolutely convergent rather than a formal symbol. The same membership is what makes the time-integrated pairing in the uniqueness proof of section 11.3 a genuine number; we will not flag it again, but it is the reason every integral in the chapter is allowed to be written down. A classical solution of the mean field game system is a pair $(u, m) = (u, m)$ solving

$$-\partial_t u - \nu\,\Delta u + H(x, \nabla u) = F(x, m_t), \qquad u(T, \cdot) = G(x, m_T), \tag{11.1}$$

$$\partial_t m - \nu\,\Delta m - \operatorname{div}(m\,\partial_p H(x, \nabla u)) = 0, \qquad m(0, \cdot) = m_0, \tag{11.2}$$

on $[0, T] \times \mathbb{R}^d$. It pays to read the two equations slowly, because every sign in them is about to matter. The first, eq. (11.1), is the HJB equation of the representative agent who treats the population path $m_\cdot$ as given exogenous data: $u(t, x)$ is the optimal cost-to-go from state x at time t, the term $H(x, \nabla u)$ is the minimized Hamiltonian encoding the agent's best instantaneous trade-off, $F(x, m_t)$ is the crowd surcharge entering as a source, and the terminal datum $u(T, \cdot) = G(\cdot, m_T)$ records the final payoff. The signs $-\partial_t u$ and $-\nu\Delta u$ are the fingerprints of a *backward* equation: data are given at $t = T$ and the equation is integrated toward smaller t, so that the heat operator regularizes in the correct (backward) time direction. The second, eq. (11.2), is the Fokker–Planck (forward Kolmogorov) equation for the density of the closed-loop state, integrated *forward* from $m(0, \cdot) = m_0$; here $+\partial_t m$ and the same $-\nu\Delta m$ make the diffusion regularize forward in time. The two operators differ only by the orientation of time – this is the value-backward, mass-forward opposition of arrows in its sharpest form.

The link that closes the loop is the drift. The agent's optimal feedback control is $\alpha^* = -\partial_p H(x, \nabla u)$ – the minus sign is the convention of appendix A.4, where the Hamiltonian is the Legendre transform of the running cost and its p-gradient returns the optimizing velocity with a sign set so that the agent descends the cost. An agent following this feedback moves with velocity α^*, so the *population* of such agents is transported by exactly the vector field $\alpha^* = -\partial_p H(x, \nabla u)$. That is why the drift appearing in eq. (11.2) is not some independent datum but is forced to equal $\partial_p H(x, \nabla u)$: writing the transport term as $-\operatorname{div}(m\,\partial_p H(x, \nabla u))$ is the same as writing $+\operatorname{div}(m\,\alpha^*)$ on the other side, i.e. eq. (11.2) is the continuity equation $\partial_t m + \operatorname{div}(m\,\alpha^*) = \nu\Delta m$ with the agents' own optimal velocity. The forward equation is slaved to the backward one through this drift, and the backward equation is slaved to the forward one through the coupling F; that mutual slaving, with the two boundary data at opposite ends, *is* the forward–backward coupling. We collect eqs. (11.1) and (11.2) and refer to them jointly as the *MFG system*.

Notation watch: Constant scalar diffusion, and why the argument does not need it

For transparency we have written the second-order terms with a *constant scalar* diffusion, $\nu\Delta u$ and $\nu\Delta m$ with $\nu = \frac{1}{2}\sigma^2$. It is worth being explicit that this is a convenience of display and not a hypothesis the theorem leans on, because a reader could reasonably worry that constancy is smuggling something in. Chapter 10 carries the existence theory for a genuinely state-dependent, matrix volatility $\sigma(x)$, with generator $\mathcal{A}u = \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top\,\mathrm{D}^2\,u)$ and Fokker–Planck operator $\mathcal{A}^* m = \frac{1}{2}\sum_{ij}\partial^2_{ij}((\sigma\sigma^\top)_{ij} m)$. The two operators are formal adjoints by construction: $\mathcal{A}^*$ is built so that $\int(\mathcal{A}\varphi)\,\psi = \int\varphi\,(\mathcal{A}^*\psi)$ for test functions φ, ψ, which is just integration by parts twice, moving both derivatives off φ and onto ψ. Now look ahead to where the diffusion enters the uniqueness proof. It enters in exactly one place: the diffusion contributions to $\frac{\mathrm{d}}{\mathrm{d}t}\int \bar{u}\,\bar{m}$ must cancel, i.e. we need

$$\int(-\mathcal{A}\bar{u})\,\bar{m} + \int\bar{u}\,(\mathcal{A}^*\bar{m}) = -\int(\mathcal{A}\bar{u})\,\bar{m} + \int(\mathcal{A}\bar{u})\,\bar{m} = 0,$$

where the first equality is the adjoint relation applied to $\varphi = \bar{u}$, $\psi = \bar{m}$, and the sign flip comes from the backward orientation of the HJB diffusion against the forward orientation of the FP diffusion. (For the scalar Laplacian this is the Green's identity $\int(-\nu\Delta\bar{u})\bar{m} + \int\bar{u}(\nu\Delta\bar{m}) = 0$ used in section 11.3.) Nothing in this cancellation looks at *whether* σ is scalar, constant, or even continuous in x – it uses only that the operator driving the value and the operator driving the mass are formal adjoints of each other, paired against sufficiently decaying fields. That self-adjoint pairing is the entire footprint of the diffusion on the argument; constancy and scalarity are never invoked. We keep the scalar ν only to declutter the displays, and every statement extends verbatim to the matrix-volatility existence class of chapter 10 – under **(A-Nondeg)** for the final solvability step and sufficient decay for the integrations by parts – by replacing $\nu\Delta$ with $\mathcal{A}$ and $\mathcal{A}^*$ throughout.

With the system fixed, we can say concretely how multiplicity is born, and the cleanest way is to watch the feedback loop run in the one regime where it misbehaves: an *attractive* coupling, where an agent prefers to sit where the crowd already is.

Heuristic (non-rigorous)

The whole difficulty sits in the two boundary conditions pointing in opposite directions, and an attractive coupling lets us see it turn into multiplicity in a small number of steps. Suppose the couplings are attractive: $F(x, m)$ is *lower* where m has more mass, so an agent's cost is reduced by being near the crowd. Trace the loop one revolution. *(1)* Posit an anticipation: every agent believes the population will concentrate near a point x_0. *(2)* Best response to it: since sitting near x_0 is now cheap, the optimal feedback $\alpha^* = -\partial_p H(x, \nabla u)$ steers each agent toward x_0. *(3)* Aggregate the responses: the Fokker–Planck equation eq. (11.2) transports the realized population toward x_0, so it does concentrate there. *(4)* Compare with the anticipation: the realized concentration near x_0 matches – indeed reinforces – the one we posited, so the anticipation is self-consistent and we have an equilibrium. The crucial point is step (3) feeding step (1): concentration begets cheapness begets concentration. This is a positive feedback, and a positive feedback has no reason to have a unique fixed point. Run the identical argument with a *different* anchor $x_1 \neq x_0$ and it closes just as well: believe in x_1, drift to x_1, realize x_1. Two anticipations, two self-fulfilling populations, two equilibria – the best-response map S has more than one fixed point. This is precisely the Curie–Weiss self-reinforcement of a mean-field ferromagnet, where the belief "the magnetization is m" produces, through each spin aligning with the mean field it generates, a realized magnetization that can sit at more than one self-consistent value once the coupling is strong enough; section 11.4 makes that analogy exact. What forbids the runaway is the opposite sign: a *repulsive* coupling, for which $F(x,m)$ is *higher* where m concentrates, so that step (3) feeds back into step (1) with a minus – every attempt to concentrate raises the local cost and pushes mass back out, damping the loop instead of amplifying it. Monotonicity, defined in the next section, is the rigorous form of "the coupling is repulsive enough that this damping wins at every horizon."

The phrase "the damping wins" can be made quantitative, and it is worth previewing how, because it tells us which quantity the eventual proof will track. One cannot hope to control $\bar{u} = u_1 - u_2$ and $\bar{m} = m_1 - m_2$ separately – each is pinned at the wrong end for a Grönwall march, as we saw. The object that *is* controllable is their pairing $\int_{\mathbb{R}^d} \bar{u}\,\bar{m}\,\mathrm{d}x$, the natural duality between a difference of value functions (a function of the state) and a difference of populations (a signed measure on the state). This scalar knows about both discrepancies at once and, crucially, vanishes automatically at $t = 0$ (where $\bar{m} = 0$) so that its behavior is governed by the terminal end alone. The content of the uniqueness theorem in section 11.3 is that, under a repulsive sign condition on the coupling, the time evolution of $\int \bar{u}\,\bar{m}$ has a definite sign that traps it between zero and zero, forcing $\bar{u}$ and $\bar{m}$ to coincide – a contraction not in any fixed norm but in this particular bilinear pairing, and one that holds *uniformly in* T because the favorable signs accumulate monotonically over $[0,T]$ rather than competing against a growing exponential. The reader can hold onto $\int \bar{u}\,\bar{m}$ as the single quantity the whole chapter is built to control; the monotonicity condition is engineered to be exactly what gives it the right sign.

Before formalizing that condition, two clarifications guard against false hope. First, uniqueness can hold without monotonicity if the horizon is short: on $[0,T]$ with T small the forward–backward coupling is a genuine contraction *regardless* of the coupling's sign, because the round trip described above is too brief for the feedback loop to wind up. This is not new to this chapter – it is precisely the small-horizon regime of the McKean–Vlasov FBSDE proved in theorem 9.10(i). There the best-response map S is shown to be Lipschitz with constant $C\,T$, where C aggregates the Lipschitz constants of the data, so that S is a genuine contraction the moment $C\,T < 1$, and the Banach fixed-point theorem then forces a unique equilibrium; this is the honest reappearance, in the present language, of the Grönwall mechanism we said was unavailable – it is available, but only on a horizon short enough that the contraction factor stays below one, and the admissible window $T < 1/C$ shrinks as the coupling (and hence C) strengthens. We revisit this estimate in section 11.7, and Thread A will display the window explicitly as the band $T < T_c$. Monotonicity is exactly what removes the horizon restriction and upgrades uniqueness to *all* T.

Second, monotonicity is *sufficient but not necessary*: there are couplings that violate it outright yet still yield a unique equilibrium at every horizon. This is not a minor caveat but a quantitative fact we will be able to measure. The LQG thread of section 11.5 exhibits a whole *soft band* of parameters – attractive couplings that are anti-monotone by the sign computation eq. (11.10), yet too weak relative to the individual cost curvature to support multiplicity – in which uniqueness holds for every T despite the failure of **(A-Mon)**. The gap between the monotonicity threshold and the true multiplicity threshold is the precise width of this band, and it is the precise measure of how conservative the Lasry–Lions condition is. None of this diminishes the condition's value: its worth is that it is *checkable* (a single inequality, verifiable directly on the data), *horizon-free* (it asks nothing about T), and exactly the structure under which the deeper objects of the theory – the potential/variational formulation of chapter 8 and the master equation of chapter 12 – become well behaved.

We have thus generated a precise demand. The heuristic told us we need a coupling that is "repulsive enough"; the preview of $\int \bar{u}\,\bar{m}$ told us this must mean "repulsive enough that a certain pairing of cost differences against population differences has a definite sign"; and the clarifications told us the condition must be horizon-free and checkable, strong enough to defeat the feedback loop at every T but understood to be merely sufficient. What is missing is the inequality itself: a clean, checkable, T-independent sign condition on F and G that says "pushing mass into a region raises the cost there." Supplying that inequality – pinning the word repulsive to the pairing $\int (F(x,m) - F(x,m'))(m-m')(\mathrm{d}x)$ whose absolute convergence we have just secured through the L^1 membership of the couplings – is the business of the next section, and it is the definition of Lasry–Lions monotonicity to which we now turn.

11.2 The Lasry–Lions monotonicity condition

The previous section closed by manufacturing a precise want: a coupling "repulsive enough" that the pairing of cost differences against population differences carries a definite sign, written as an inequality that is *checkable* (verifiable on the data alone) and *horizon-free* (silent about T) – strong enough to defeat the feedback loop at every T yet understood from the outset to be only sufficient. It also secured the lone piece of regularity that lets such a pairing be a genuine number rather than a formal symbol: the L^1 membership of the couplings, which makes $\int (F(x,m)-F(x,m'))(m-m')(\mathrm{d}x)$ an absolutely convergent integral. This section supplies the inequality those demands were converging on, names it, and – because the sign in it is the single thing a physicist is most likely to read backward – spends the rest of its length pinning down what the inequality *means*, along two independent axes: an operator-theoretic reading (a cost paired against the shift that produced it) and a geometric reading (convexity of an energy along straight-line interpolations of populations). Each axis is realized by a concrete model class, and the geometric one is the bridge to the variational picture of chapter 8 that the proof in the next section will lean on.

Definition 11.1 (Lasry–Lions monotonicity)**.** A coupling functional $F\colon \mathcal{P}_2(\mathbb{R}^d) \to C(\mathbb{R}^d) \cap L^1$ is *(Lasry–Lions) monotone* if for all $m, m' \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} (F(x,m) - F(x,m'))\,(m-m')(\mathrm{d}x) \;\geq\; 0. \tag{11.3}$$

It is *strictly* monotone if equality in eq. (11.3) holds only when $m = m'$. The standing assumption **(A-Mon)** is that the running coupling F and the terminal coupling G are both monotone in this sense.

This is the rigorous form of "repulsive enough," and it is worth dwelling on why it has exactly the shape it does. Read the integrand factor by factor. The signed measure $\Delta := m - m'$ records *how the population was shifted*: it is positive where the new configuration m has piled up extra mass relative to the old m', and negative where mass was vacated. It has zero total mass, $\int \mathrm{d}\Delta = 0$, because both m and m' are probability measures – a shift moves mass around without creating or destroying it. The function $F(x, m) - F(x, m')$ records *the change in cost* that this very shift induced at each location x. The pairing in eq. (11.3) is then the natural duality between a cost (a function of the state) and a population shift (a finite signed measure on the state), and the inequality demands that the change in cost, *weighted by the shift that produced it*, be nonnegative on average. The crowd-averse content is now exact: where the shift deposited extra mass ($\mathrm{d}\Delta > 0$), the induced cost should have gone *up* ($F(x, m) - F(x, m') > 0$), so that the product is positive; where mass was vacated, the cost should have gone down, and the product is again positive. A coupling for which crowding a region makes that region more expensive will have the two factors aligned in sign at every x, and the integral cannot be negative. This is the sign the feedback-loop heuristic asked for, now written as a single scalar inequality on the data.

There is a second, equivalent reading that explains why the analytic machinery of the next section works at all. Regard $m \mapsto F(\cdot, m)$ as a map sending a point of $\mathcal{P}_2(\mathbb{R}^d)$ to an element of the dual space of (signed measures tested against) functions of the state. Inequality eq. (11.3) is precisely the statement that this map is a *monotone operator* in the sense of nonlinear functional analysis: pairing the increment of the operator, $F(\cdot, m) - F(\cdot, m')$, against the increment of its argument, $m - m'$, yields a nonnegative number. Monotone operators are exactly the class for which the "energy decreases along the flow" bookkeeping of the previous section – a contraction not in a norm but in a bilinear pairing – has teeth, and this is the structural reason the same three-line computation will close in section 11.3 even though no Grönwall direction exists. The reader does not need the general theory of monotone operators; what matters is that eq. (11.3) is its defining inequality specialized to our F, and that this is the abstract slot the uniqueness proof plugs into.

Notation watch

The single most common slip a physicist makes here is a sign, and it is worth stating the convention four ways so that none of them surprises later. Monotonicity is a statement about *costs*, with the sign fixed by eq. (11.3), and throughout this chapter F and G are costs: the agent *minimizes*, so larger F is worse. With that fixed:

- A *monotone* coupling is one for which crowding a region *raises* the cost there. It is the *stabilizing*, *crowd-averse*, *repulsive* case – congestion, aversion to company – and it is the regime in which uniqueness holds at every horizon. Monotone = repulsive = stabilizing.

- An *anti-monotone* (or non-monotone) coupling is one for which crowding a region *lowers* the cost there. It is the *attractive*, *aggregating*, *imitative* case – the agent is rewarded for company – and it is precisely the regime where uniqueness can fail.

The trap is this: a physicist trained to think of an *attractive potential* as the "nice," energy-minimizing, ground-state-having case, or to think in terms of a *payoff to be maximized* rather than a cost to be minimized, will instinctively call attraction stabilizing. In the MFG cost convention it is the reverse. Attraction is *anti*-monotone and is where the landscape becomes nonconvex and equilibria proliferate; repulsion is monotone and is where the landscape stays convex and the equilibrium is unique. If at any later point a payoff $-F$ or a free energy $-\mathcal{F}$ is written, the inequality eq. (11.3) flips, and the maximization convention would call the *concave* case the good one; we never switch conventions inside this chapter, but the reader translating to a Hamiltonian-energy language elsewhere should carry the minus sign explicitly.

Two model classes turn the abstract inequality into something one can recognize on sight. The first is the most elementary and exhibits the pointwise mechanism in its purest form; the second exposes the geometric content – that monotonicity is convexity of an energy – and is the bridge to the variational reading.

Proposition 11.2 (Local couplings)**.** *Let F be local, $F(x,m) = f(x, m(x))$, defined for m with a density and with $f(x,\cdot)$ nondecreasing in its second argument for every x. Then F is monotone, and strictly monotone if each $f(x,\cdot)$ is strictly increasing.*

Proof. Because the coupling is local, the cost at x sees the population only through its density $m(x)$ at that same point, and the entire integrand factorizes location by location. Fix two densities m, m' and look at a single point x. The two factors of the integrand are $f(x, m(x)) - f(x, m'(x))$ and $m(x) - m'(x)$. If $m(x) \geq m'(x)$ then, since $f(x,\cdot)$ is nondecreasing, $f(x, m(x)) \geq f(x, m'(x))$, so both factors are ≥ 0 and their product is ≥ 0; if instead $m(x) \leq m'(x)$ then $f(x, m(x)) \leq f(x, m'(x))$, both factors are ≤ 0, and their product is again ≥ 0. Either way the integrand $\big(f(x, m(x)) - f(x, m'(x))\big)\big(m(x) - m'(x)\big)$ is nonnegative *pointwise* – a product of two factors that a monotone $f(x,\cdot)$ forces to share a sign. Integrating a pointwise-nonnegative function over x keeps the sign, so $\int \big(f(x,m) - f(x,m')\big)(m - m')\,\mathrm{d}x \geq 0$, which is eq. (11.3). If each $f(x,\cdot)$ is *strictly* increasing the product is strictly positive wherever $m(x) \neq m'(x)$ and zero only where $m(x) = m'(x)$; the integral therefore vanishes only when $m = m'$ a.e., i.e. $m = m'$, which is strict monotonicity. ☐

This is the congestion or crowd-aversion model, and a concrete instance fixes it in mind. Take $f(x,\rho) = \rho$, so that $F(x,m) = m(x)$: the running cost an agent pays for sitting at x is exactly the local density of the crowd there, a flat per-capita congestion charge. Then $f(x,\cdot)$ is strictly increasing, the proposition gives strict monotonicity, and the pairing is the manifestly nonnegative $\int \big(m(x) - m'(x)\big)^2\,\mathrm{d}x = \|m - m'\|_{L^2}^2$ – the squared L^2 distance between the two populations, which is zero only when they coincide. The physical reading is immediate: doubling the crowd at a point doubles the toll there, so any attempt to over-concentrate is self-penalizing, and there is no way for two distinct populations to both be self-consistent. The opposite sign, $f(x,\cdot)$ *decreasing* – for instance $f(x,\rho) = -\rho$, a per-capita reward for company – is aggregation or imitation; the same pairing comes out as $-\|m - m'\|_{L^2}^2 \leq 0$, the integrand is nonpositive pointwise, and the coupling is anti-monotone. The single scalar $\int (F(x,m) - F(x,m'))(m - m')$ has flipped sign with the monotonicity of f, exactly as the notationwatch warned.

The second model class exposes the geometric content of monotonicity: it is exactly convexity of an energy. This is the reading that makes "unique equilibrium" intelligible as "unique minimizer of a convex functional," and it is the one the next section ultimately formalizes.

Proposition 11.3 (Potential couplings and convex energy)**.** *Suppose F is the flat derivative of a functional $\mathcal{F}\colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$,*

$$F(x, m) = \frac{\delta \mathcal{F}}{\delta m}(m, x),$$

in the sense of appendix A.3. Then F is monotone if and only if $\mathcal{F}$ is convex along linear interpolations of measures, i.e. $\epsilon \mapsto \mathcal{F}\big((1-\epsilon)m' + \epsilon m\big)$ is convex on $[0, 1]$ for all m, m'.

Proof. The idea is to reduce a statement about a functional on the infinite-dimensional space $\mathcal{P}_2(\mathbb{R}^d)$ to a statement about an ordinary scalar function of one real variable, by restricting $\mathcal{F}$ to a straight segment of measures and reading convexity off the sign of the second derivative. Write

$$m_\epsilon := (1-\epsilon)m' + \epsilon m = m' + \epsilon(m - m'), \qquad \varphi(\epsilon) := \mathcal{F}(m_\epsilon),$$

so that $\epsilon \mapsto m_\epsilon$ is the affine path from m' (at $\epsilon = 0$) to m (at $\epsilon = 1$). Its velocity is the constant signed measure $m - m'$, a finite measure of zero total mass: as ϵ advances, the population is interpolated linearly, neither gaining nor losing total mass. The function φ is a real function on $[0, 1]$, and the proposition is the assertion that φ is convex for every choice of endpoints m, m' exactly when F is monotone.

First derivative. By the defining property of the flat derivative (appendix A.3), which is precisely the chain rule for an affine perturbation of the argument, differentiating $\mathcal{F}$ along the constant-velocity path produces the flat derivative integrated against that velocity:

$$\varphi'(\epsilon) = \int \frac{\delta \mathcal{F}}{\delta m}(m_\epsilon, x)\,(m - m')(\mathrm{d}x) = \int F(x, m_\epsilon)\,(m - m')(\mathrm{d}x). \tag{11.4}$$

Already eq. (11.4) contains the monotonicity pairing in disguise: it is the cost $F(\cdot, m_\epsilon)$ paired against the population shift $m - m'$, evaluated at the intermediate configuration m_ϵ.

Second derivative. To differentiate eq. (11.4) a second time we need the rate of change of the integrand $x \mapsto F(x, m_\epsilon)$ as ϵ advances. The argument m_ϵ again moves with constant velocity $m - m'$, so this is once more a flat derivative – but now of the map $m \mapsto F(x, m)$, which is itself the first flat derivative of $\mathcal{F}$. Differentiating a flat derivative again is what *defines* the second flat derivative, and we make the iteration explicit. Hold the first slot x fixed and regard $m \mapsto F(x, m) = \frac{\delta \mathcal{F}}{\delta m}(m, x)$ as a scalar functional of m; its flat derivative, taken in m at a probe point y, is by definition the kernel $\dfrac{\delta^2 \mathcal{F}}{\delta m^2}(m)(x, y)$, characterized by the requirement that for every affine direction $m - m'$,

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon} F(x, m_\epsilon) = \int \frac{\delta^2 \mathcal{F}}{\delta m^2}(m_\epsilon)(x, y)\,(m - m')(\mathrm{d}y). \tag{11.5}$$

In words: eq. (11.5) is just eq. (11.4) applied one level down, to the functional $F(x, \cdot)$ in place of $\mathcal{F}$, with the probe variable renamed y to keep it distinct from the fixed first slot x. The kernel is *symmetric*, $\dfrac{\delta^2 \mathcal{F}}{\delta m^2}(m)(x, y) = \dfrac{\delta^2 \mathcal{F}}{\delta m^2}(m)(y, x)$, and the symmetry is Clairaut's theorem – the equality of mixed second partials – transported to this setting. To see it, perturb $\mathcal{F}$ in two independent directions at once, $m \mapsto m + \epsilon\,\delta_x + \eta\,\delta_y$ (formally; rigorously one uses smooth bump approximations of the Dirac masses), and form $\Psi(\epsilon, \eta) := \mathcal{F}(m + \epsilon\,\delta_x + \eta\,\delta_y)$. Differentiating in ϵ first then in η recovers the second flat derivative with the slots in one order, $\partial_\eta \partial_\epsilon \Psi(0, 0) = \frac{\delta^2 \mathcal{F}}{\delta m^2}(m)(x, y)$; doing it in the other order gives $\partial_\epsilon\partial_\eta\Psi(0,0) = \frac{\delta^2\mathcal{F}}{\delta m^2}(m)(y,x)$. Since Ψ is a smooth function of the two real variables (ϵ,η), Clairaut's theorem equates the two mixed partials, and the two slot orders therefore agree. This symmetric kernel is part of the second-order differential calculus on $\mathcal{P}_2(\mathbb{R}^d)$ assembled in chapter 4 and appendix D; here we need only the elementary iterate eq. (11.5) and the right to differentiate eq. (11.4) under the integral sign. That last right is genuine and must be licensed, not assumed: differentiation under the integral is the statement that the difference quotients of $F(x,m_\epsilon)$ in ϵ converge to eq. (11.5) *and* are dominated, uniformly in ϵ, by a fixed integrable function so that the limit may be taken inside the integral. Both hold because the second flat derivative $\delta^2\mathcal{F}/\delta m^2$ is assumed bounded and continuous on the line segment $\{m_\epsilon : \epsilon\in[0,1]\}$, which is a compact subset of $\mathcal{P}_2(\mathbb{R}^d)$: boundedness furnishes the integrable dominating function (a constant times the total variation of $m-m'$, which is finite), continuity furnishes the pointwise convergence of the difference quotients, and dominated convergence then justifies passing $\mathrm{d}/\mathrm{d}\epsilon$ through the x-integral in eq. (11.4). Carrying this out and inserting eq. (11.5),

$$\varphi''(\epsilon) = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\int F(x,m_\epsilon)\,(m-m')(\mathrm{d}x) = \iint \frac{\delta^2\mathcal{F}}{\delta m^2}(m_\epsilon)(x,y)\,(m-m')(\mathrm{d}x)\,(m-m')(\mathrm{d}y).$$

This is the symmetric bilinear form $\delta^2\mathcal{F}/\delta m^2$ evaluated on the *same* signed measure $m-m'$ in both arguments – a quadratic form in the population shift.

Both directions. Convexity of φ for every pair m,m' is, for a twice-differentiable function of one real variable, exactly $\varphi''\geq 0$ on $[0,1]$; by the display this is the nonnegativity of the quadratic form $\delta^2\mathcal{F}/\delta m^2$ on all signed measures of zero total mass (every such measure arises as $m-m'$ for suitable populations, up to scaling). Monotonicity eq. (11.3) is the difference-quotient form of the very same statement, and the fundamental theorem of calculus interpolates between them. Indeed, with $\Delta := m-m'$, the monotonicity pairing is the increment of φ' across the segment:

$$\int \big(F(x,m)-F(x,m')\big)\,\mathrm{d}\Delta = \varphi'(1)-\varphi'(0) = \int_0^1 \varphi''(\epsilon)\,\mathrm{d}\epsilon,$$

where the first equality is eq. (11.4) read at the two endpoints $\epsilon=1$ (giving $\int F(\cdot,m)\,\mathrm{d}\Delta$) and $\epsilon=0$ (giving $\int F(\cdot,m')\,\mathrm{d}\Delta$), and the second is the fundamental theorem applied to φ'. If $\varphi''\geq 0$ the right-hand integral is ≥ 0, so the pairing is ≥ 0 and F is monotone – convexity implies monotonicity. Conversely, suppose F is monotone. Apply eq. (11.3) not to the full segment but to a short sub-segment from m_ϵ to $m_{\epsilon+h}$, whose population shift is $h(m-m')$; the pairing is then $h^2(\varphi''(\epsilon)+o(1))\geq 0$, and dividing by h^2 and letting $h\to 0$ gives $\varphi''(\epsilon)\geq 0$ at almost every ϵ. Hence monotonicity of F for all pairs forces $\varphi''\geq 0$, i.e. convexity of $\mathcal{F}$ along every segment. The two properties are equivalent. □

Proposition 11.3 is the bridge to the variational picture of chapter 8, and it is worth spelling out the bridge, because the closing intuition of this chapter – that monotone coupling means a convex equilibrium landscape with a single basin – rests on it. In the separable, control-affine setting of section 8.6, where the running cost splits into a control part and a population part and the couplings are of potential type ($F=\frac{\delta\mathcal{F}}{\delta m}(m,x)$, $G=\frac{\delta\mathcal{G}}{\delta m}(m,x)$ for energies $\mathcal{F},\mathcal{G}$, as in definition 8.9), the entire forward–backward MFG system eqs. (11.1) and (11.2) is not merely a coupled pair of equations but the *Euler–Lagrange optimality system of one scalar functional.* That functional is the energy $\mathcal{J}(m,w)$ of eq. (8.18): a kinetic (control) cost accumulated along a density flow, plus the running and terminal interaction energies $\int_0^T \mathcal{F}(m_t)\,\mathrm{d}t + \mathcal{G}(m_T)$, minimized over density flows (m, w) that obey the continuity constraint. The precise content of "Euler–Lagrange" is that stationarity of $\mathcal{J}$ – setting its first variation to zero, with the value function u appearing as the Lagrange multiplier enforcing the continuity constraint – reproduces the HJB equation eq. (11.1) for u and the Fokker–Planck equation eq. (11.2) for m exactly (theorem 8.11). A solution of the MFG system is therefore a critical point of a single energy, and the strategic forward–backward problem collapses to one minimization.

Now feed proposition 11.3 into this. The interaction energies $\mathcal{F}, \mathcal{G}$ are convex along linear interpolations of measures precisely when their flat derivatives F, G are monotone; the kinetic term of $\mathcal{J}$ is convex on its own (it is the perspective function of a convex Lagrangian, jointly convex in the density–momentum pair, lemma 8.10), and the continuity constraint cuts out a convex set of flows. So under **(A-Mon)** the energy $\mathcal{J}$ is a convex functional on a convex domain, and a convex functional on a convex domain has its minimizers forming a convex set – and a *single* minimizing density once the convexity is strict, which strict monotonicity of the coupling supplies (theorem 8.11). Uniqueness of the equilibrium is then nothing more exotic than uniqueness of the minimizer of a convex energy: a single basin, with nothing to choose between because there is nowhere else for the landscape to dip. This is the geometric reading promised at the section's outset, and it is the picture to carry into the proof. The general uniqueness theorem of section 11.3 does *not* require a potential structure – it works for any monotone couplings, separable Hamiltonian or not – but it is exactly this convex-landscape intuition that the duality computation there will make rigorous without ever naming the energy: *monotone coupling = convex equilibrium landscape = unique equilibrium.*

We hand the next section two things it will spend. First, the monotonicity inequality eq. (11.3) as a usable tool with its sign nailed down: paired against a population difference, a difference of monotone costs is ≥ 0, and in the time-integrated identity of the proof this is exactly the term that enters with a favorable (uniqueness-forcing) contribution. Second, the convex-energy reading just developed, which tells us *in advance* what the proof must find – that two candidate equilibria cannot both sit at the bottom of a single convex bowl – so that the duality computation of section 11.3 reads not as a trick but as the concrete realization of "a convex landscape has one minimum."

11.3 The uniqueness theorem

The previous section handed us two things to spend, and this section spends them. The first was the monotonicity inequality eq. (11.3) with its sign nailed down: a difference of monotone costs, paired against the population difference that produced it, is ≥ 0. The second was the picture that inequality encodes – *monotone coupling = convex equilibrium landscape* – which told us in advance what any honest proof must discover, that two candidate equilibria cannot both sit at the bottom of a single convex bowl. We now make that picture rigorous. The remarkable feature of the argument is that it never names the energy whose convexity proposition 11.3 exhibited; it does not even require the couplings to be of potential type. It works directly on the forward–backward system eqs. (11.1) and (11.2), extracting convexity wherever it is hidden – in the Hamiltonian through **(A-Conv)**, in the couplings through **(A-Mon)** – and assembling those hidden convexities into a single identity that pins the two solutions together. The argument is short, exact, and worth presenting in full because it is the prototype for every uniqueness theorem in the book: the graphon version of chapter 24 is this same computation with the scalar pairing replaced by an operator pairing, and the master-equation monotonicity of chapter 12 is its differential shadow. Its engine is a single time-integrated pairing – the quantity $\int \bar{u}\,\bar{m}$ flagged in section 11.1 as the one scalar that knows about both discrepancies at once – whose time derivative, after one Green's identity and one integration by parts, turns out to be a sum of manifestly signed terms.

Before the theorem we isolate the one elementary fact about convex functions that the proof consumes. Convexity **(A-Conv)** of the Hamiltonian was imported in the preamble as a hypothesis on $H(x,\cdot)$; the proof needs it in a specific, quantitative form – not as "the graph curves upward" but as a nonnegative *gap* between the Hamiltonian and its own linearization. That gap is the Bregman divergence, and the lemma below is nothing more than the statement that convexity makes it nonnegative. It is the analytic operationalization of the p-convexity imported in **(A-Conv)**: every later appearance of convexity in the proof enters through this one quantity.

Lemma 11.4 (Bregman nonnegativity)**.** *Under **(A-Conv)**, for every x the Bregman divergence of $H(x,\cdot)$,*

$$D_H(x;p,q) := H(x,p) - H(x,q) - \partial_p H(x,q)\cdot(p-q),$$

satisfies $D_H(x;p,q) \geq 0$ for all $p,q \in \mathbb{R}^d$. If $H(x,\cdot)$ is strictly convex, then $D_H(x;p,q) = 0$ if and only if $p = q$.

Proof. The three terms of $D_H(x;p,q)$ are, in order, the value of the function at p, the value of its tangent (first-order Taylor) approximation built at q, and – assembled with the indicated signs – the gap between them. To see this, fix x and abbreviate $h(p) := H(x,p)$. The tangent plane to the graph of h at the base point q is the affine function

$$\ell_q(p) := h(q) + \partial_p h(q)\cdot(p-q),$$

the unique affine function agreeing with h in value and gradient at q. With this notation $D_H(x;p,q) = h(p) - \ell_q(p)$ is exactly the vertical gap at the point p between the graph of h and the tangent plane drawn at q. Convexity of h is, by the first-order characterization of convexity, precisely the statement that the graph never dips below any of its tangent planes: $h(p) \geq \ell_q(p)$ for every pair p,q. Hence the gap is nonnegative, $D_H(x;p,q) \geq 0$, which is the claim. If h is *strictly* convex the graph meets each tangent plane only at the single point of tangency, so the gap is strictly positive whenever $p \neq q$ and vanishes if and only if $p = q$. (For twice-differentiable h the same fact reads off Taylor's theorem with integral remainder,

$$D_H(x;p,q) = \int_0^1 (1-\theta)\,(p-q)^\top\,\partial_{pp}^2 h(q+\theta(p-q))\,(p-q)\;\mathrm{d}\theta,$$

a manifestly nonnegative integral of the convex – hence positive-semidefinite – Hessian against the squared increment, strictly positive once that Hessian is positive definite.) One feature will matter in the proof: the divergence is in general *not* symmetric in its two arguments, since it measures the gap from a designated base point q. The uniqueness computation will use both orderings, $D_H(x;\nabla u_1,\nabla u_2)$ and $D_H(x;\nabla u_2,\nabla u_1)$, one weighted by each of the two densities. □

Theorem 11.5 (Lasry–Lions uniqueness)**.** *Assume* ***(A-Lip)****,* ***(A-Nondeg)****, and* ***(A-Conv)*** *with $H(x,\cdot)$ strictly convex, and assume* ***(A-Mon)****: the couplings F and G are monotone in the sense of definition 11.1. Then the MFG system eqs.* (11.1) *and* (11.2) *has at most one classical solution (u,m) for which the formal integrations by parts below are justified — concretely, u of at most quadratic growth with ∇u of at most linear growth in x, and m with finite second moment and enough spatial decay (e.g. Gaussian-type tails) that the products $\bar u\,\bar m$, $\nabla\bar u\cdot m_i\partial_p H_i$ and the boundary terms at spatial infinity vanish. This regularity is the one delivered by chapter 10: for bounded regularising couplings the Lipschitz-in-x value of theorem 10.14 (with the gradient bound of lemma 10.5), and for the Gaussian LQG thread of worked computation 11.7 the explicit quadratic value and Gaussian density of proposition 10.23, for which the polynomial growth of ∇u is dominated by the Gaussian decay of m.*

The list of growth and decay conditions in the statement looks fussy, but each clause pays for one specific manipulation in the proof, and it is worth reading the hypothesis as a checklist of exactly the integrations by parts to come. There are three. First, the pairing $\int \bar u\,\bar m$ must itself be a finite number that can be differentiated in time: this asks that the product $\bar u\,\bar m$ be integrable and that the boundary term at spatial infinity in $\frac{\mathrm{d}}{\mathrm{d}t}\int \bar u\,\bar m$ vanish, which holds once $\bar u$ grows no faster than polynomially while $\bar m$ – a difference of two densities of finite second moment with fast tails – decays fast enough to overwhelm that polynomial. Second, the Green's identity that cancels the diffusion integrates by parts *twice*, transferring two derivatives from $\bar u$ across to $\bar m$ and back; each transfer drops a surface term of the form $\oint_{|x|=R}(\nabla\bar u\,\bar m - \bar u\,\nabla\bar m)$, and these vanish as $R\to\infty$ only because $\nabla\bar u$ is at most linearly growing while m_i and ∇m_i carry Gaussian-type decay. Third, the single integration by parts of the divergence (transport) term moves one derivative from $\bar u$ onto the flux $m_i\,\partial_p H_i$, dropping the surface term $\oint_{|x|=R}\bar u\,m_i\,\partial_p H_i$, which again dies because the at-most-quadratic $\bar u$ and the at-most-linearly-growing $\partial_p H_i$ are crushed by the decay of m_i. None of these clauses is idle: a value function growing like $e^{|x|}$, or a density with merely algebraic tails, could leave a nonzero flux at infinity and break the very cancellations the proof is built on. The reason we may *assert* the conditions rather than hope for them is that chapter 10 already delivered them on exactly the solution classes we care about. For bounded regularising couplings, theorem 10.14 produces a value u globally Lipschitz in x – so ∇u is bounded, a fortiori of linear growth – with the explicit gradient bound of lemma 10.5, while the Fokker–Planck density it transports has finite second moment with tail decay inherited from the nondegenerate diffusion. For the LQG thread of worked computation 11.7 the situation is cleaner still and entirely explicit: the value is the exact quadratic $u(t,x)=\frac{1}{2}P(t)x^2+r(t)x+c(t)$, so ∇u is exactly affine in x, and the density m_t is exactly Gaussian, so its Gaussian decay dominates that affine growth at every order and every boundary term is not merely small but identically convergent. The hypotheses of the theorem are thus not an extra burden placed on the reader; they are a property the previous chapter's solutions *have*, recorded here so that the integrations by parts below are licensed rather than formal.

Proof. Suppose, against the conclusion, that the system admits *two* classical solutions (u_1,m_1) and (u_2,m_2) – both with the regularity just discussed – and form their differences

$$\bar u := u_1 - u_2, \qquad \bar m := m_1 - m_2.$$

Everything hinges on the single scalar $t \mapsto \int_{\mathbb{R}^d} \bar{u}(t,x)\,\bar{m}(t,x)\;\mathrm{d}x$, the pairing of the two discrepancies announced in section 11.1. The plan is to compute its time derivative honestly from the differential equations (Steps 1–3), then integrate that derivative over $[0,T]$ and evaluate the two endpoint contributions from the boundary data (Step 4). The endpoint sum will turn out to be nonnegative, while the integrated derivative is a sum of terms each of which the hypotheses force to be nonpositive; the only way a nonnegative number can equal a sum of nonpositive numbers is for every one of them to vanish, and that collapse is what forces the two solutions to coincide (Step 5).

Step 1: the equations for the differences. Because both pairs solve the *same* system, the difference of two copies of an equation kills every term common to them and leaves only the nonlinear discrepancies. Subtract the two copies of the HJB equation eq. (11.1): the linear operator $-\partial_t - \nu\Delta$ and the terminal rule subtract cleanly, while the Hamiltonian and the coupling, being nonlinear, leave *differences* rather than zero,

$$\partial_t \bar{u} = -\nu\Delta\bar{u} + \big(H(x,\nabla u_1) - H(x,\nabla u_2)\big) - \big(F(x,m_1) - F(x,m_2)\big),$$

where $\partial_t \bar{u}$ has been moved to the left in forward-time form. Likewise subtract the two copies of the Fokker–Planck equation eq. (11.2); here the diffusion is linear and subtracts, while the transport term leaves the difference of the two probability fluxes,

$$\partial_t \bar{m} = \nu\Delta\bar{m} + \operatorname{div}\big(m_1\,\partial_p H(x,\nabla u_1) - m_2\,\partial_p H(x,\nabla u_2)\big).$$

We will not solve either equation; the point is only to pair them against each other.

Step 2: differentiate the pairing and watch the diffusion cancel. Differentiate the pairing under the integral sign – licensed by the decay hypotheses, which make the integrand and its t-derivative dominated – using the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \bar{u}\,\bar{m} = \int \partial_t\bar{u}\;\bar{m} + \int \bar{u}\;\partial_t\bar{m},$$

and substitute the two difference equations of Step 1. This produces four kinds of term: the two diffusion terms, the Hamiltonian-difference term, the transport (divergence) term, and the coupling-difference term. Take them in turn. The two diffusion contributions are $\int(-\nu\Delta\bar{u})\bar{m}$ from $\partial_t\bar{u}$ and $\int \bar{u}(\nu\Delta\bar{m})$ from $\partial_t\bar{m}$, and they cancel *against each other* – this is the one place the opposed time-orientations of the two equations pay a dividend. Integrating by parts twice (once to move a gradient off $\bar{u}$ onto $\bar{m}$, a second time to move it back, with the surface terms at infinity vanishing by the assumed decay) is Green's second identity,

$$\int(-\nu\Delta\bar{u})\,\bar{m} + \int \bar{u}\,(\nu\Delta\bar{m}) = -\nu\int \Delta\bar{u}\,\bar{m} + \nu\int \bar{u}\,\Delta\bar{m} = 0,$$

since $\int \Delta\bar{u}\,\bar{m} = \int \bar{u}\,\Delta\bar{m}$ for sufficiently decaying fields. The diffusion has removed itself from the computation entirely, at no sign cost – the "diffusion cancels for free" phenomenon we return to after the proof. The transport term $\int \bar{u}\,\operatorname{div}(\cdot)$ is handled by a *single* integration by parts, moving the divergence off the flux and onto $\bar{u}$ as a gradient (again the surface term at infinity vanishes by decay),

$$\int \bar{u}\;\operatorname{div}(m_1\partial_p H_1 - m_2\partial_p H_2) = -\int \nabla\bar{u}\cdot(m_1\partial_p H_1 - m_2\partial_p H_2),$$

where from here on we abbreviate $H_i := H(x, \nabla u_i)$ and $\partial_p H_i := \partial_p H(x, \nabla u_i)$. Collecting the surviving terms – the Hamiltonian difference paired with $\bar{m}$, the transported flux paired with $\nabla \bar{u}$, and the coupling difference paired with $\bar{m}$ – gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \int \bar{u}\,\bar{m} = \underbrace{\int (H_1 - H_2)\,\bar{m} - \int \nabla \bar{u} \cdot (m_1 \partial_p H_1 - m_2 \partial_p H_2)}_{=:\ \mathcal{B}(t)} - \int (F(x, m_1) - F(x, m_2))\,\bar{m}. \tag{11.6}$$

Step 3: the Hamiltonian term is a sum of Bregman divergences. The bracket $\mathcal{B}(t)$ in eq. (11.6) is where convexity **(A-Conv)** earns its keep. Write out its two pieces with $\bar{m} = m_1 - m_2$ in the first and $\nabla \bar{u} = \nabla u_1 - \nabla u_2$ in the second, and sort the resulting terms by which density, m_1 or m_2, they multiply. The Hamiltonian-difference term contributes $(H_1 - H_2)$ against m_1 and against $-m_2$; the flux term contributes $-(\nabla u_1 - \nabla u_2) \cdot \partial_p H_1$ against m_1 (from the $m_1 \partial_p H_1$ piece) and $+(\nabla u_1 - \nabla u_2) \cdot \partial_p H_2$ against m_2 (from the $-m_2 \partial_p H_2$ piece). Grouping by density,

$$\mathcal{B}(t) = \int m_1 \Big[(H_1 - H_2) - (\nabla u_1 - \nabla u_2) \cdot \partial_p H_1\Big] - \int m_2 \Big[(H_1 - H_2) - (\nabla u_1 - \nabla u_2) \cdot \partial_p H_2\Big].$$

Now read each bracket against the definition $D_H(x; p, q) = H(x, p) - H(x, q) - \partial_p H(x, q) \cdot (p - q)$ of lemma 11.4, and the two orderings the lemma warned about appear. For the first bracket take base point $q = \nabla u_1$ and evaluation point $p = \nabla u_2$: then $H(x, p) - H(x, q) = H_2 - H_1$ and $\partial_p H(x, q) \cdot (p - q) = \partial_p H_1 \cdot (\nabla u_2 - \nabla u_1)$, so $D_H(x; \nabla u_2, \nabla u_1) = (H_2 - H_1) - \partial_p H_1 \cdot (\nabla u_2 - \nabla u_1)$, which is exactly the *negative* of the first bracket; the first bracket is therefore $-D_H(x; \nabla u_2, \nabla u_1)$. For the second bracket take $q = \nabla u_2$, $p = \nabla u_1$: then $D_H(x; \nabla u_1, \nabla u_2) = (H_1 - H_2) - \partial_p H_2 \cdot (\nabla u_1 - \nabla u_2)$, which is exactly the second bracket, so the second bracket is $+D_H(x; \nabla u_1, \nabla u_2)$. This is precisely why the Bregman divergence had to be kept asymmetric: the two halves of $\mathcal{B}$ use the two opposite base points. Substituting both,

$$\mathcal{B}(t) = -\int m_1\, D_H(x; \nabla u_2, \nabla u_1)\ \mathrm{d}x\ -\ \int m_2\, D_H(x; \nabla u_1, \nabla u_2)\ \mathrm{d}x\ \le\ 0, \tag{11.7}$$

The sign is now forced, and it is worth stating slowly. Each density is nonnegative, $m_1, m_2 \ge 0$ (they are population densities), and each Bregman divergence is nonnegative, $D_H \ge 0$, by lemma 11.4 – which is just **(A-Conv)** in operational form. A sum of two nonnegative integrands enters $\mathcal{B}(t)$ with two minus signs, so $\mathcal{B}(t) \le 0$ for every t, *independently of anything about the couplings.* This is the favorable sign that the agents' own strictly convex optimization supplies for free.

Step 4: integrate in time and account for the boundary data. We have $\frac{\mathrm{d}}{\mathrm{d}t} \int \bar{u}\,\bar{m}$; recover the pairing itself by integrating eq. (11.6) over the whole horizon $[0, T]$, so the left-hand side telescopes to the difference of the pairing at the two endpoints, $\left[\int \bar{u}\,\bar{m}\right]_{t=T} - \left[\int \bar{u}\,\bar{m}\right]_{t=0}$. This is where the two boundary data, pinned at opposite ends – the structural obstruction of section 11.1 – finally enter, and they enter with the right signs. At the initial time the two populations share the prescribed datum, $m_1(0) = m_2(0) = m_0$, so $\bar{m}(0) = 0$ and the entire pairing vanishes at $t = 0$ regardless of $\bar{u}(0)$: the near-end boundary term costs us nothing, and the value's unknown initial behaviour never has to be controlled. At the terminal time the two value functions share the terminal *rule* $u_i(T, \cdot) = G(\cdot, m_i(T))$ but not its value, because the terminal populations $m_i(T)$ may differ; thus

$\bar{u}(T) = G(x, m_1(T)) - G(x, m_2(T))$, and the far-end boundary term is the pairing of this terminal cost difference against the terminal population difference,

$$\Big[\int \bar{u}\,\bar{m}\Big]_{t=T} = \int \left(G(x, m_1(T)) - G(x, m_2(T))\right)\left(m_1(T) - m_2(T)\right)\,\mathrm{d}x \;\geq\; 0.$$

That this is ≥ 0 is exactly monotonicity of the terminal coupling G (**(A-Mon)**): it is the inequality eq. (11.3) applied to G at the two terminal populations $m_1(T), m_2(T)$. So the left-hand side of the time-integrated identity equals the nonnegative terminal term minus zero. Therefore

$$0 \;\leq\; \int \left(G(x, m_1(T)) - G(x, m_2(T))\right)(m_1 - m_2)(T) = \int_0^T \mathcal{B}(t)\,\mathrm{d}t - \int_0^T\!\!\int \left(F(x, m_1) - F(x, m_2)\right)(m_1 - m_2)\,\mathrm{d}x\,\mathrm{d}t. \tag{11.8}$$

Read eq. (11.8) as the balance it is. The left-hand side is ≥ 0 by the terminal monotonicity just used. The right-hand side is a sum of two contributions, and *both* are ≤ 0: the first, $\int_0^T \mathcal{B}(t)\,\mathrm{d}t$, is nonpositive because $\mathcal{B}(t) \leq 0$ pointwise in t by eq. (11.7) (the Bregman/convexity sign); the second, $-\int_0^T \int (F(x, m_1) - F(x, m_2))(m_1 - m_2)$, is nonpositive because the integral it negates is ≥ 0 by monotonicity of the running coupling F (eq. (11.3), now carrying an overall minus). We have thus trapped a nonnegative quantity between zero and a sum of two nonpositive quantities. The squeeze is decisive: a number that is simultaneously ≥ 0 and equal to a sum of terms each ≤ 0 can only be zero, and each nonpositive term must *itself* be zero (a sum of nonpositive numbers vanishes only if every summand does). Hence all three pieces vanish separately:

$$\int_0^T \mathcal{B}(t)\,\mathrm{d}t = 0, \qquad \int_0^T\!\!\int \left(F(x, m_1) - F(x, m_2)\right)(m_1 - m_2)\,\mathrm{d}x\,\mathrm{d}t = 0, \qquad \int \left(G(x, m_{1,T}) - G(x, m_{2,T})\right)(m_1 - m_2)(T) \tag{11.9}$$

Step 5: from the vanishing identity back to $m_1 = m_2$ and $u_1 = u_2$. The three equalities of eq. (11.9) are scalar statements – three integrals are zero – and the work of this step is to upgrade them to the pointwise conclusion that the two solutions coincide. Begin with the Bregman one. Since $\int_0^T \mathcal{B}(t)\,\mathrm{d}t = 0$ while $-\mathcal{B}(t)$ is, by eq. (11.7), the x-integral of the two manifestly nonnegative densities $m_1\,D_H(x; \nabla u_2, \nabla u_1)$ and $m_2\,D_H(x; \nabla u_1, \nabla u_2)$, the time integral of a nonnegative integrand can vanish only if that integrand vanishes almost everywhere. Hence

$$m_1(t,x)\,D_H(x; \nabla u_2, \nabla u_1) = 0 \quad \text{and} \quad m_2(t,x)\,D_H(x; \nabla u_1, \nabla u_2) = 0 \qquad \text{for a.e. } (t,x).$$

Now invoke *strict* convexity, the hypothesis not yet spent. By the strict clause of lemma 11.4, $D_H(x; p, q) = 0$ forces $p = q$; so at any point where $m_1 > 0$ the first equality forces $\nabla u_2 = \nabla u_1$, and at any point where $m_2 > 0$ the second forces the same equality $\nabla u_1 = \nabla u_2$. The two value gradients therefore agree *wherever either population carries mass* – precisely on the set that transport can see. Pushed through the optimal-feedback map this says the two closed-loop drifts coincide there, $\partial_p H(x, \nabla u_1) = \partial_p H(x, \nabla u_2)$ on $\{m_1 > 0\} \cup \{m_2 > 0\}$. (Off that set the drift is irrelevant, since the Fokker–Planck flux $m_i\,\partial_p H_i$ is multiplied by a density that is zero there.)

This is exactly the input needed to collapse the two forward equations into one. Each m_i solves eq. (11.2) with its own drift $\partial_p H_i$; but on the union of supports – the only place either flux is nonzero – those drifts are equal, and both equations carry the same diffusion ν and the same initial datum m_0. So m_1 and m_2 are two solutions of *one and the same* linear Fokker–Planck equation: a forward Kolmogorov equation whose drift and diffusion are now *frozen*, i.e. state-dependent but no longer solution-dependent. The logical chain that closes the argument is the imported well-posedness of that linear equation, and it is worth naming link by link, because this is the only place the existence-theory of the earlier chapters is consumed. The frozen Fokker–Planck equation is the forward (Kolmogorov) equation of the single, decoupled SDE

$$\mathrm{d}X_t = -\,\partial_p H\big(X_t, \nabla u(t, X_t)\big)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t, \qquad X_0 \sim m_0,$$

with the now-common drift $-\partial_p H$ and diffusion σ. Under **(A-Lip)** that drift is Lipschitz and under **(A-Nondeg)** the diffusion is nondegenerate, so this SDE is well posed and possesses a *unique solution in law* by theorem 5.10. The time-marginals $t \mapsto \mathrm{Law}(X_t)$ of that law are a solution of the Fokker–Planck equation, and by theorem 5.14 the Fokker–Planck equation has a *unique* measure-valued solution issuing from m_0 – so any density that solves it must be those marginals. Both m_1 and m_2 solve it from the same m_0; each therefore equals the common marginal law, and $m_1 \equiv m_2$. This is the precise sense in which "the Fokker–Planck marginal inherits the SDE's uniqueness": well-posedness in law upstream (theorem 5.10) is transported to uniqueness of the density downstream (theorem 5.14), with nondegeneracy guaranteeing a genuine density to compare in the first place.

With the densities now equal, the couplings they generate are equal too: $F(x, m_1) = F(x, m_2)$ for every t, and $G(x, m_1(T)) = G(x, m_2(T))$. Feed this back into the backward equation. Both u_1 and u_2 now solve the *same* HJB equation eq. (11.1) – identical source $F(\cdot, m)$, identical Hamiltonian, identical terminal datum $G(\cdot, m(T))$ – so their difference $\bar{u}$ solves a homogeneous backward parabolic equation with zero terminal datum. The comparison principle for that equation (theorem 6.18), applied once to $\bar{u}$ and once to $-\bar{u}$, forces $\bar{u} \le 0$ and $\bar{u} \ge 0$, hence $\bar{u} \equiv 0$ and $u_1 \equiv u_2$. (This is the one place the HJB well-posedness theory of chapter 6 is invoked – not to construct the value, which we already hold, but to certify that the value is *determined* by its data, so that equal data give equal values.) The two solutions we started from agree in both components; the system has at most one classical solution. $\square$

The argument is three lines of integration by parts, but it rewards a second, slower look, because it is unusually transparent about which hypothesis does which job – and that transparency is exactly what the rest of the chapter exploits. Four ingredients went in, and they played strictly separated roles.

First, the diffusion cancelled for free. The two second-order terms met as a Green's-identity pair and annihilated each other before any sign question could arise; nondegeneracy **(A-Nondeg)** was never used to control them, and the entire computation is silent about the size, shape, or even constancy of ν until the final step. Nondegeneracy re-enters only in Step 5, and only to certify that the *linear*, frozen-coefficient Fokker–Planck and HJB equations are uniquely solvable – a well-posedness role, not a sign role. For the purposes of uniqueness the diffusion is a spectator.

Second, convexity of the Hamiltonian was *always* favorable and *always* present. The Bregman term eq. (11.7) is the analytic price of the agents' optimization being genuinely strict, and **(A-Conv)** guarantees it carries the uniqueness-forcing sign $\mathcal{B} \le 0$ *no matter what the couplings are*. It is a structural surplus the single-agent problem contributes to the population problem unconditionally; it never fights us.

Third – and this is the crux – the *only* coupling-dependent signs in the whole computation live in the boundary identity eq. (11.8), and they are supplied by monotonicity alone. The running term $-\int_0^T\int(F_1-F_2)(m_1-m_2)$ and the terminal term $\int(G_1-G_2)(m_1-m_2)$ are the two places where the data about *how the crowd is felt* enters, and **(A-Mon)** is precisely the hypothesis that fixes their signs the favorable way. Everything else in the proof would run verbatim for an attractive coupling; these two terms are the entire footprint of the interaction's sign on uniqueness.

Fourth, and consequently, removing monotonicity does not break a step of the proof – it removes a sign. Reread eq. (11.8) with F and G *anti*-monotone: the running and terminal pairings now point the wrong way, the right-hand side is no longer a sum of nonpositive terms, and the squeeze that trapped everything at zero simply unclamps. A negative coupling term can be balanced against the still-favorable Bregman term, the net can stay finite, and two genuinely distinct solutions (u,m) can satisfy the identity at once. This is not a weakness of the argument to be patched; the proof has correctly located the exact membrane across which uniqueness must fail. The identity eq. (11.8) is a sign ledger, and monotonicity is the single entry that keeps it balanced.

That last observation is the hinge into the next section. We have learned not merely that monotonicity *suffices* for uniqueness, but *where* – in one identity – its work is concentrated: the boundary identity eq. (11.8) is the one place the coupling's sign is read, and it is exactly that identity which loses its definite sign the moment the coupling turns anti-monotone. The natural question is then no longer analytic but structural: if the sign can fail, *what kind of object is the multiple equilibrium that the failure permits?* Answering that is the business of section 11.4, which reads the sign failure physically, as a symmetry-breaking threshold rather than a mere loss of a constant.

Remark 11.6 (Why the clean statement is the undiscounted finite-horizon one)**.** It is worth seeing precisely where discounting disturbs this bookkeeping, because the answer explains why theorem 11.5 is stated without a discount and why that is the version chapter 12 consumes. A discounted running cost replaces the HJB equation eq. (11.1) by $-\partial_t u+\beta\,u-\nu\Delta u+H(x,\nabla u)=F(x,m_t)$, i.e. it adds the linear zeroth-order term $\beta\,u$ with the discount rate $\beta=\beta>0$ of appendix A.1. Run the proof again. The difference equation for $\bar u$ now carries an extra $+\beta\,\bar u$, so differentiating the pairing produces one new contribution,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int\bar u\,\bar m\ \supset\ \beta\int\bar u\,\bar m,$$

and after the time integration of Step 4 the identity eq. (11.8) acquires the extra term $\beta\int_0^T\int\bar u\,\bar m\ \mathrm{d}x\ \mathrm{d}t$. Here is the difficulty in one sentence: this new term is the time integral of the very pairing $\int\bar u\,\bar m$ whose sign the proof was built to *deduce*, not a quantity known to be signed along the way. The other three terms of eq. (11.8) were each pinned – the boundary term nonnegative by **(A-Mon)**, the running and Bregman terms nonpositive by **(A-Mon)** and **(A-Conv)** – but $\bar u\,\bar m$ inherits no sign from either hypothesis, because the pairing can be positive at some times and negative at others. So the squeeze no longer closes: a genuinely indefinite term has entered the ledger, and the clean "nonnegative = sum of nonpositives ⇒ all vanish" collapse is lost. One can still recover uniqueness under extra smallness of β or of the horizon, but the finite-horizon, undiscounted statement is the one that needs *no* such qualification, and that unconditional cleanliness is exactly why we made it the theorem.

Two honest escapes deserve mention. For the discounted *ergodic* (stationary) problem the difficulty evaporates rather than worsens: there is no time integration at all, the forward–backward pair collapses to a single coupled elliptic system, and the monotonicity computation runs on that stationary pair directly and is in fact cleaner – the offending $\int \bar{u}\,\bar{m}$ term has nowhere to accumulate (see [93, 36]). And in the explicitly solvable LQG thread of section 11.5 the indefinite term is computable in closed form, so discounting can be carried there by hand. But for the general finite-horizon theory we keep to the undiscounted statement, which is precisely the regime in which the master equation of chapter 12 is shown well posed: the monotonicity that buys uniqueness here is the same hypothesis under which that equation's characteristics do not cross, and chapter 12 imports theorem 11.5 in exactly this undiscounted finite-horizon form.

11.4 Failure of uniqueness as symmetry breaking

The proof ended on a structural observation rather than an analytic one. We learned not just that monotonicity suffices for uniqueness but *where*, in the whole forward–backward apparatus, its entire work is concentrated: in the single boundary entry of the sign ledger eq. (11.8). When the coupling is anti-monotone – attractive, imitative, aggregating – that one entry flips, the right-hand side stops being a sum of nonpositive terms, the squeeze that trapped $\int \bar{u}\,\bar{m}$ between zero and zero unclamps, and two genuinely distinct solutions can satisfy the identity at once. The proof has thus located, to the nearest sign, the membrane across which uniqueness must fail. But locating the membrane is not the same as understanding what lies on the far side, and that is the question this section takes up. The analytic statement "the boundary term changes sign" needs to be turned into a structural one: *what kind of object is the multiple equilibrium that the sign failure permits?* Is it a pathology – two solutions that happen, accidentally, to coincide in the pairing – or is it something with a shape, a phenomenon a physicist would recognize on sight?

It is the latter, and naming it correctly is what explains the most striking feature of the transition. A multiple equilibrium is a set of mutually incompatible self-consistent populations, each one an anticipation that the crowd it generates exactly fulfils; the heuristic of section 11.1 already glimpsed two of them, the belief in a concentration at x_0 and the belief in a concentration at $x_1 \neq x_0$, each closing the loop "concentration begets cheapness begets concentration" on its own anchor. What that heuristic did not yet say is that these coexisting solutions do not switch on gradually as the attraction is dialed up. They appear all at once, at a sharp value of the coupling, and below that value there is genuinely only one. Loss of uniqueness is a *threshold* phenomenon, not a continuous softening, and the reason is the same reason a ferromagnet magnetizes at a critical temperature and not before: the multiple equilibrium is *spontaneous symmetry breaking*. This is the chapter's one sanctioned physical picture, and it earns its place precisely because it predicts the suddenness that the analytic sign-flip, read literally, would leave unexplained.

To set up the picture, reduce the feedback loop to the one scalar it turns on. Take the attraction to act through the population mean – each agent is rewarded for sitting near $\bar{m}_t = \int \xi\, m_t(\mathrm{d}\xi)$, the LQG coupling of section 11.5. Then the entire self-consistent problem projects onto a single condition on that mean: given an anticipated mean $\bar{m}$, every agent best-responds to a crowd anchored there, the aggregate of those responses has its own mean, and equilibrium is the demand that the produced mean equal the anticipated one. This is a scalar fixed-point equation $\bar{m} = \mathcal{R}(\bar{m})$, and the number of equilibria is just the number of its fixed points. Whether that number is one or several is decided by a single derivative, and that is exactly the Curie–Weiss situation.

Physics note: Multiplicity as broken symmetry

Consider the attractive mean-coupling just described: each agent prefers to be near the population mean. The self-consistent condition on the mean is a fixed-point equation of the schematic form $\bar{m} = \mathcal{R}(\bar{m})$, where $\mathcal{R}$ is the population mean produced when every agent best-responds to a crowd anchored at $\bar{m}$. This is structurally the Curie–Weiss self-consistency $m = \tanh(\beta_{\mathrm{T}}(h + Jm))$ for the magnetization of a mean-field ferromagnet, the response curve $\mathcal{R}$ playing the role of $m \mapsto \tanh(\beta_{\mathrm{T}}(h + Jm))$ and the attraction strength playing the role of the coupling $\beta_{\mathrm{T}} J$. Take the symmetric case ($h = 0$, no external bias), so that $\bar{m} = 0$ – the unmagnetized, "everyone sits at the origin" configuration – always solves $\mathcal{R}(0) = 0$. The fate of that symmetric solution is governed by the slope $\mathcal{R}'(0)$. *Below threshold*, $\mathcal{R}'(0) < 1$: the response curve crosses the diagonal only once, $\mathcal{R}$ is a contraction near the symmetric point, and $\bar{m} = 0$ is the unique fixed point – weak attraction cannot overcome each agent's private incentives and the only consistent crowd is the symmetric one. *Above threshold*, $\mathcal{R}'(0) > 1$: the response curve now leaves the diagonal steeper than 45° at the origin, so the symmetric solution is no longer a contraction but a repeller, and the curve must recross the diagonal on either side. A *pair* of new fixed points $\pm\bar{m}_*$ appears, born together out of the destabilized symmetric one – the analog of spontaneous magnetization, where below the Curie temperature a nonzero $\pm M$ appears in place of the demagnetized state. The agents have collectively committed to clustering left or clustering right; the equilibrium has *broken the symmetry* of the symmetric anticipation that the model itself respects. The transition is sharp – it is the crossing $\mathcal{R}'(0) = 1$, an instability of a single mode, not a gradual blurring – which is exactly why uniqueness fails suddenly.

Crucially, the static fixed-point equation $\bar{m} = \mathcal{R}(\bar{m})$ does *not*, by itself, say which branch is realized: $+\bar{m}_*$ and $-\bar{m}_*$ enter on a perfectly equal footing, just as tanh gives no reason to prefer up-magnetization to down. Selecting one requires extra input beyond the equilibrium condition – a *selection principle* – and the candidates are exactly the ones that will recur in this book: dynamic stability (keep only equilibria that are attracting under a learning or best-response dynamics), a vanishing-viscosity or entropic refinement (add a small regularizer, solve, and take the regularizer to zero, the limit picking a distinguished branch), or the boundary-layer analysis of the master equation in chapter 12 (which resolves the choice on a fast time scale the finite-dimensional system hides). Each is the game-theoretic counterpart of how a physical ferromagnet picks a magnetization sign: by its history, or by an infinitesimal external field that tilts the otherwise balanced double well. The monotone (repulsive) regime is the one with no such ambiguity because it has *positive susceptibility everywhere*: there crowding a region always raises its cost, every perturbation is restored rather than amplified, $\mathcal{R}'$ never reaches 1, and there is simply no symmetric solution to destabilize and no symmetry to break. That is the same statement, in magnetic language, as the convex-energy reading of section 11.2: by proposition 11.3 a monotone coupling makes the equilibrium energy convex along interpolations of populations, so its landscape is a single convex basin with one minimizer. "Positive susceptibility," "convex landscape," and "single basin" are three names for one fact – a stable, restoring response – and the double well with its two minima is the picture of what monotonicity forbids.

It bears repeating, because the style of this book insists on it, that the magnetic picture is here to *motivate* and nothing more: not one step of the uniqueness theorem leaned on it, and the threshold it predicts will in the next section be derived from the equations alone. The analogy's job is to make the right expectation – a sharp transition, a born-in-pairs branch structure, an unselected sign – feel inevitable before we compute it. With that expectation in hand, strip the analogy away and ask what survives as mathematics.

The mathematical residue is sharper, and more favorable, than the general picture might suggest, and it is the content of the next section. In the explicitly solvable LQG model the response map $\mathcal{R}$ is not a transcendental curve like tanh but *linear*: the self-consistency condition on the mean reduces to a linear two-point boundary-value problem in time – a forward equation for the mean trajectory slaved to a backward equation for the optimal feedback, closed by data at opposite ends. Linearity is a windfall, because a linear self-consistency problem has either exactly one solution or a whole subspace of them, with no middle ground: multiplicity appears *precisely* when that boundary-value problem becomes singular, i.e. when its homogeneous version acquires a nonzero solution. The condition for singularity is therefore an exact equation on the parameters, not an inequality to be estimated. And it is more selective than mere anti-monotonicity. A singular boundary-value problem requires the coupled forward–backward dynamics to be *oscillatory* rather than exponential – the homogeneous solution must be able to start and end at zero without being identically zero, which a monotone, sign-definite system can never do – and oscillation sets in only when the attraction is not merely present but *strong relative to the curvature of the individual cost*, strong enough to overpower each agent's private pull toward its own optimum. This is the quantitative content of the soft band promised in section 11.1: between the failure of monotonicity (the sign computation eq. (11.10) turning the wrong way) and the true onset of multiplicity (the boundary-value problem actually going singular) lies a measurable gap of attractive-but-weak couplings where the equilibrium is still unique at every horizon. The threshold separating the two is sharp and computable.

That sets the mandate for section 11.5 exactly. The symmetry-breaking story has promised a threshold that is at once sharp – a clean equality on the parameters (s, q, T), not a conservative bound – and computable in closed form. Delivering it is now a matter of doing the linear algebra the analogy stood in for: reduce the LQG mean field game to its linear two-point boundary-value problem for the population mean, and read off, from the determinant that governs that problem's solvability, the exact locus where it goes singular. The next section carries out precisely that reduction and produces the uniqueness/multiplicity phase diagram in (s, q, T), soft band and all.

11.5 Worked computation: the LQG uniqueness phase diagram

The previous section ended by issuing a mandate and standing aside. Its symmetry-breaking picture predicted – on the strength of an analogy, not yet a computation – that the loss of uniqueness in an attractive mean field game is a *threshold* phenomenon: sharp rather than gradual, with equilibria born in pairs out of a destabilized symmetric solution, the way a ferromagnet magnetizes at its Curie point. And it promised that the carried LQG thread, being exactly solvable, would deliver that threshold as a clean equation on the parameters rather than an estimate. This section makes good on that promise. We advance Thread A by carrying the LQG mean field game to the point where its uniqueness/multiplicity boundary can be read off in closed form, and we discover that the boundary is sharper and more structured than the abstract monotonicity criterion would lead one to expect: not one threshold but two, separated by a measurable gap, and beyond the second a discrete ladder of resonant horizons rather than a solid region of failure.

This is the "sign/uniqueness" layer of the thread promised in the thread specification; the full closed-form solution – value, density, and Riccati system together – is assembled in the capstone chapter 14, and here we extract only what bears on uniqueness. The payoff is a concrete witness, computable to the last constant, for the chapter's central qualitative claim: that Lasry–Lions monotonicity is *sufficient but not necessary.* We will see the monotone region, the region where monotonicity fails yet uniqueness survives, and the region where it genuinely fails, all three drawn on one diagram in the parameters (s, q, T), with σ – the noise – nowhere to be found.

Worked computation 11.7 (LQG mean field game: reduction to a mean boundary-value problem)**.** Recall the scalar LQG model of Thread A. The representative agent has state

$$\mathrm{d}X_t = (aX_t + b\,\alpha_t)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t, \qquad X_0 \sim m_0,$$

and minimizes, against a population mean path $\bar m_t := \int \xi\, m_t(\mathrm{d}\xi)$,

$$J(\alpha) = \mathbb{E}\left[\int_0^T \left(\tfrac{1}{2}\alpha_t^2 + \tfrac{q}{2}\,(X_t - s\,\bar m_t)^2\right) \mathrm{d}t + \tfrac{q_T}{2}\,(X_T - s_T\,\bar m_T)^2\right],$$

with $q, q_T \geq 0$, and $s, s_T \in \mathbb{R}$ the mean-coupling strengths. The running coupling, read as a functional of the population, is $F(x,m) = \frac{q}{2}(x - s\,\bar m)^2$ with $\bar m = \int \xi\, m(\mathrm{d}\xi)$.

Monotonicity of the LQG coupling. Before touching the dynamics it pays to locate the model on the map drawn in section 11.2: is this coupling monotone, and for which parameters? The answer will tell us in advance which slice of parameter space the general theorem theorem 11.5 already covers, so that the work of this section is confined to the complement. The test is definition 11.1: pair the cost difference against the population difference and read off the sign. Take two populations m, m' with means $\bar m, \bar m'$, and abbreviate the *shift in the mean* by $\Delta := \bar m - \bar m' = \int \xi\,(m - m')(\mathrm{d}\xi)$. The coupling $F(x,m) = \frac{q}{2}(x - s\bar m)^2$ depends on the population only through its mean, so its difference is a quadratic in x whose coefficients see m, m' only through $\bar m, \bar m'$. Expanding the two squares and cancelling the common $\frac{q}{2}x^2$,

$$F(x,m) - F(x,m') = \tfrac{q}{2}\Big[(x - s\bar m)^2 - (x - s\bar m')^2\Big] = \tfrac{q}{2}\Big(-2s\,x\,\Delta + s^2(\bar m - \bar m')(\bar m + \bar m')\Big),$$

where the cross term $-2sx(\bar m - \bar m') = -2sx\Delta$ is linear in x and the remaining term $s^2(\bar m^2 - \bar m'^2)$ is a constant in x. Now integrate against the signed measure $m - m'$. Two elementary facts about that measure do all the work: it has zero total mass, $\int(m - m') = 0$, because both are probability measures, which kills the constant term outright; and its first moment is exactly the mean shift,

$\int x\,(m-m')(\mathrm{d}x) = \Delta$, which converts the linear term into $-2s\Delta \cdot \Delta$. Hence only the cross term survives,

$$\int \big(F(x,m) - F(x,m')\big)\,(m-m')(\mathrm{d}x) = \tfrac{q}{2}\big(-2s\,\Delta^2\big) = -\,q\,s\,(\bar m - \bar m')^2. \tag{11.10}$$

The entire monotonicity content of the LQG model is compressed into this one line, and its sign is transparent: the right-hand side is a fixed negative multiple, $-qs$, of the perfect square $(\bar m - \bar m')^2$. The pairing is therefore ≥ 0 for *all* pairs of populations precisely when $qs \leq 0$, i.e., with $q > 0$ fixed, precisely when $s \leq 0$. This is the crowd-averse regime: with $s < 0$ the agent's target $s\bar m$ sits on the *opposite* side of the origin from the crowd, so the cost pushes each agent away from where the others are, and the coupling is monotone (repulsive) in the exact sense of definition 11.1. The imitative regime $s > 0$, where the agent is drawn *toward* $s\bar m$ and hence toward the crowd, makes eq. (11.10) negative and is anti-monotone – the attractive, multiplicity-prone case the symmetry-breaking section warned about. (The terminal coupling $G(x,m) = \frac{q_T}{2}(x - s_T\bar m)^2$ obeys the identical computation with s_T in place of s.) The verdict from the general theory is immediate and one-directional: by theorem 11.5, $s \leq 0$ together with $s_T \leq 0$ already guarantees a unique equilibrium at *every* horizon T, with no further computation. What that theorem says nothing about is the half-line $s > 0$, and it is precisely there – in the anti-monotone imitative regime – that the interesting question lives: exactly how far into $s > 0$ does uniqueness survive before the feedback loop finally winds up? Answering that is what the rest of the reduction is for.

The HJB and the reduction. We now run the representative agent's optimization with the mean path $\bar m_\cdot$ treated as given exogenous data, exactly as the MFG system eq. (11.1) prescribes, and then close the loop by demanding that the realized mean equal the anticipated one. Substituting the LQG Hamiltonian – minimizing $\frac{1}{2}\alpha^2 + (ax + b\alpha)\partial_x u$ over α gives $\alpha^* = -b\,\partial_x u$ and minimized value $ax\,\partial_x u - \frac{b^2}{2}(\partial_x u)^2$ – the HJB eq. (11.1) for this model reads

$$-\partial_t u = \tfrac{q}{2}(x - s\bar m_t)^2 + a\,x\,\partial_x u - \tfrac{b^2}{2}(\partial_x u)^2 + \tfrac{\sigma^2}{2}\partial_{xx} u, \qquad u(T,x) = \tfrac{q_T}{2}(x - s_T\bar m_T)^2,$$

with optimal feedback $\alpha^* = -b\,\partial_x u$. Every term on the right is at most quadratic in x (the source $\frac{q}{2}(x - s\bar m_t)^2$ is quadratic, the drift term $ax\,\partial_x u$ multiplies x by a derivative, the quadratic term $(\partial_x u)^2$ squares a derivative), and the terminal datum is quadratic; so the value cannot generate higher powers, and the quadratic structure is preserved by the ansatz

$$u(t,x) = \tfrac{1}{2}P(t)\,x^2 + r(t)\,x + c(t).$$

Here, in the carried-thread notation fixed for the LQG solution quantities, $P(t)$ is the value's Riccati curvature – the quadratic coefficient, the same object written P in chapter 10 for the single-agent problem and in chapter 12 for the master equation – and $r(t)$ is its linear coefficient, the only coefficient through which the population mean will turn out to act. With $\partial_x u = Px + r$ and $\partial_{xx} u = P$, substitute and collect by powers of x. The coefficient of x^2 gives, after dividing the matched relation $-\frac{1}{2}\dot P = -\frac{b^2}{2}P^2 + aP + \frac{q}{2}$ by $-\frac{1}{2}$, the scalar Riccati equation

$$\dot P = b^2 P^2 - 2a\,P - q, \qquad P(T) = q_T, \tag{11.11}$$

whose terminal datum $P(T) = q_T$ is read off the x^2 coefficient of $u(T,\cdot)$. The decisive feature of eq. (11.11) is that its right-hand side contains *no* $\bar m$: the curvature equation is *autonomous and* *decoupled* from the mean, a Riccati ODE the agent could solve before knowing anything about the crowd. Matching the coefficient of x^1 – which collects $-\dot{r}$ on the left and $-qs\bar{m}_t + ar - b^2Pr$ on the right – gives the *linear* equation for the linear coefficient,

$$\dot{r} = q\, s\, \bar{m}_t - \big(a - b^2 P(t)\big)\, r, \qquad r(T) = -q_T s_T\, \bar{m}_T; \tag{11.12}$$

the source $qs\bar{m}_t$ is where the crowd finally enters, and the terminal datum $r(T) = -q_T s_T \bar{m}_T$ is the linear coefficient of $\frac{q_T}{2}(x - s_T\bar{m}_T)^2$. The constant term gives an equation for $c(t)$ that feeds back nowhere – c shifts the value by an x-independent amount and changes neither the feedback $\alpha^* = -b(Px + r)$ nor the dynamics – so we discard it. To close the loop we compute the realized mean. Inserting the feedback into the state equation, the closed-loop dynamics are $\mathrm{d}X_t = \big((a - b^2P)X_t - b^2 r\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, an Ornstein–Uhlenbeck-type linear SDE; taking expectations annihilates the martingale $\sigma\,\mathrm{d}W_t$ and leaves the forward linear ODE for the mean,

$$\dot{\bar{m}}_t = \big(a - b^2 P(t)\big)\, \bar{m}_t - b^2\, r(t), \qquad \bar{m}_0 = \int x\, m_0(\mathrm{d}x). \tag{11.13}$$

Read eqs. (11.12) and (11.13) together and the structure of the whole problem is laid bare. The curvature P is determined once and for all by eq. (11.11), independent of everything else; with P in hand, the remaining unknowns are the pair $(r, \bar{m})$, governed by two coupled *linear* ODEs – a backward one for r (terminal datum) slaved to $\bar{m}$, a forward one for $\bar{m}$ (initial datum) slaved to r. The most consequential observation is what is *absent*: the noise level σ enters only the variance of X_t – it appears nowhere in eqs. (11.11) to (11.13), which see only the *expectation* of the state. The diffusion stirs each agent's position but leaves the mean trajectory, and hence the self-consistency condition, untouched. We have therefore reduced the question of how many equilibria the stochastic mean field game admits to a question about a *deterministic* two-variable linear system: **the uniqueness question is entirely governed by the deterministic forward–backward pair $(r, \bar{m})$.** This is the precise vindication, in the solvable thread, of the general remark from section 11.1 that the uniqueness proof barely used the diffusion; here it does not use it at all.

Uniqueness as a two-point boundary-value problem. The pair $(r, \bar{m})$ is exactly the kind of object the chapter has been circling: two linear ODEs with their data pinned at *opposite* ends – $\bar{m}$ at $t = 0$, r at $t = T$ – and coupled through the interior, the same value-backward/mass-forward opposition that defeated the Grönwall march in section 11.1, now stripped down to two scalars. This is a linear two-point boundary-value problem (BVP), and writing it in matrix form makes its solvability a question of linear algebra. Stacking $(\bar{m}, r)$ and writing the closed-loop rate inline as $a - b^2P(t)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\bar{m}\\ r\end{pmatrix} = M(t)\begin{pmatrix}\bar{m}\\ r\end{pmatrix}, \qquad M(t) = \begin{pmatrix} a - b^2P(t) & -b^2 \\ qs & -(a - b^2P(t)) \end{pmatrix}, \qquad \bar{m}_0 \text{ given}, \quad r(T) + q_T s_T\, \bar{m}(T) = 0,$$

where the off-diagonal entries are read directly off eq. (11.13) (the $-b^2 r$ source for $\bar{m}$) and eq. (11.12) (the $qs\,\bar{m}$ source for r), and the two diagonal entries are equal and opposite because r runs with rate $-(a - b^2P)$ where $\bar{m}$ runs with $+(a - b^2P)$. Note in passing that $M(t)$ is *traceless*, $\operatorname{tr} M(t) = 0$; this is not an accident but the forward–backward antisymmetry of the system, and it will be the single algebraic fact that drives the threshold analysis below. The boundary conditions are an initial datum on $\bar{m}$ and a terminal linear relation tying $r(T)$ to $\bar{m}(T)$. Now comes the structural point that makes the LQG case so clean. The map from the initial datum $\bar{m}_0$ to the solution is *affine*: solve the homogeneous flow, then add the particular response to $\bar{m}_0$. Consequently the equilibrium is unique if and only if the associated *homogeneous* BVP – same equations and same terminal relation but with $\bar{m}_0 = 0$ – has only the trivial solution $(r, \bar{m}) \equiv 0$. An affine system whose homogeneous part has only the zero solution is invertible and returns exactly one answer for each datum; an affine system whose homogeneous part has a nonzero solution returns either no answer or an entire one-parameter family. This is the linear-algebra dichotomy with no middle ground that section 11.4 promised: *multiplicity is exactly singularity of the homogeneous BVP*. It admits a nonzero $(r, \bar{m})$ that starts at $\bar{m}_0 = 0$ yet is not identically zero – a self-consistent oscillation of the mean that needs no external push to sustain it – and at such a parameter value the inhomogeneous problem degenerates, the LQG signature of equilibrium multiplicity. The phase boundary we are after is therefore the locus in parameter space where this homogeneous problem first goes singular.

To turn this into an explicit phase boundary we must contend with the fact that $M(t)$ has time-dependent entries through $P(t)$, which would force us to track the full Riccati flow. The autonomy of the Riccati equation eq. (11.11) lets us sidestep this cleanly. Its right-hand side is a fixed quadratic in P, so it has *equilibria* – the roots of the algebraic Riccati equation $b^2P^2 - 2aP - q = 0$ – and the curvature relaxes onto one of them. The relevant root is the *stabilizing* one,

$$P_\infty = \frac{a + \sqrt{a^2 + b^2 q}}{b^2}, \qquad A_\infty := a - b^2 P_\infty = -\sqrt{a^2 + b^2 q} < 0,$$

chosen by the $+$ sign of the discriminant so that the resulting closed-loop rate $A_\infty = a - b^2 P_\infty$ is *negative*: this is the root that makes the controlled state mean-revert rather than blow up, and the strict negativity $A_\infty < 0$ (which holds for every $q > 0$, since then $\sqrt{a^2 + b^2 q} > |a|$) is the single sign that the whole threshold proof will lean on. The number A_∞ is the constant closed-loop rate at the stationary curvature. If we now tune the terminal weight to $q_T = P_\infty$, the terminal datum of eq. (11.11) already sits at the equilibrium, so the solution is *exactly* constant, $P(t) \equiv P_\infty$ for all t, and therefore $a - b^2 P(t) \equiv A_\infty$: the matrix $M(t)$ loses its time dependence and the BVP acquires constant coefficients, which is what makes a closed-form threshold possible. This tuning costs us no generality of substance: for a generic terminal weight $q_T \neq P_\infty$ the curvature $P(t)$ is a nonconstant Riccati trajectory that relaxes toward P_∞ as one integrates away from $t = T$, the matrix $M(t)$ is correspondingly time-dependent, and the same analysis goes through with the propagator of the variable-coefficient system in place of e^{Mt} – qualitatively identical, threshold included, but no longer expressible in elementary functions, and so deferred to the numerical treatment of chapter 14. Tuning $q_T = P_\infty$ buys the closed form and nothing else. We also set $s_T = 0$, removing the terminal coupling so that the terminal relation collapses to the clean homogeneous condition $r(T) = 0$; this isolates the running coupling s as the sole knob and is again a convenience, not a restriction, since a nonzero s_T only shifts the terminal line $r(T) = -q_T s_T \bar{m}(T)$ without changing the oscillatory/non-oscillatory dichotomy that decides multiplicity.

Proposition 11.8 (LQG uniqueness/multiplicity threshold)**.** *In the LQG mean field game of worked computation 11.7 with $q > 0$, terminal weight $q_T = P_\infty$, and $s_T = 0$, define*

$$\Omega^2 := b^2 q\, s - A_\infty^2 = b^2 q\,(s - 1) - a^2.$$

1. *If* $\Omega^2 \le 0$ *– in particular whenever* $s \le 1+\dfrac{a^2}{b^2 q}$*, and a fortiori for every* $s \le 0$ *– the homogeneous BVP has only the trivial solution and the equilibrium is* unique for all horizons T*.*

2. *If* $\Omega^2 > 0$ *– equivalently* $s > 1 + \dfrac{a^2}{b^2 q}$ *– the homogeneous BVP has a nontrivial solution* exactly *at the discrete sequence of horizons* T_n *(*$n = 1, 2, \dots$*) solving*

$$\tan(\Omega T) = \frac{\Omega}{A_\infty}, \tag{11.14}$$

and only there. At each T_n *the equilibrium is non-unique (or fails to exist); for every other* $T > 0$*, including every* T *strictly between consecutive resonances, the equilibrium is* unique*. The smallest resonance* $T_c := T_1$ *marks the first loss of horizon-free uniqueness; for* $a = 0$ *it is* $T_c = \left(\pi - \arctan\sqrt{s-1}\,\right)/\left(b\sqrt{q(s-1)}\right)$*.*

Proof. The whole proof is the solution of one linear constant-coefficient system, read off through its propagator, with the sign of Ω^2 deciding among four qualitatively different behaviors. We begin by finding the propagator. The matrix M is traceless, $\operatorname{tr} M = 0$, so its two eigenvalues are $\pm\lambda$ with $\lambda^2 = -\det M$; and

$$\det M = -(a - b^2 P_\infty)^2 - (-b^2)(qs) = -A_\infty^2 + b^2 qs,$$

so that $\lambda^2 = -\det M = A_\infty^2 - b^2 qs = -\Omega^2$, with $\Omega^2 := b^2 qs - A_\infty^2$ as in the statement (using $A_\infty^2 = a^2 + b^2 q$ gives the stated form $\Omega^2 = b^2 q(s-1) - a^2$). Because M is traceless it satisfies its own characteristic equation $M^2 = -\det M\, I = -\Omega^2 I$ by the Cayley–Hamilton theorem – the 2×2 statement that $M^2 - (\operatorname{tr} M)M + (\det M)I = 0$. This single relation is what makes the matrix exponential elementary: powers of M collapse onto $\{I, M\}$. Summing the exponential series in even and odd powers,

$$e^{Mt} = \sum_{k\ge 0} \frac{(Mt)^k}{k!} = \Big(\sum_{j\ge 0} \frac{(-\Omega^2)^j t^{2j}}{(2j)!}\Big) I + \Big(\sum_{j\ge 0} \frac{(-\Omega^2)^j t^{2j+1}}{(2j+1)!}\Big) M,$$

where we used $M^{2j} = (-\Omega^2)^j I$ and $M^{2j+1} = (-\Omega^2)^j M$. When $\Omega^2 > 0$ the two bracketed series are exactly $\cos(\Omega t)$ and $\Omega^{-1}\sin(\Omega t)$, so

$$e^{Mt} = \cos(\Omega t)\, I + \Omega^{-1} \sin(\Omega t)\, M;$$

when $\Omega^2 < 0$ the alternating signs become non-alternating and the same series are $\cosh(\omega t)$ and $\omega^{-1}\sinh(\omega t)$ with $\omega^2 := -\Omega^2 > 0$; and the borderline $\Omega^2 = 0$ is the limit $e^{Mt} = I + Mt$. The sign of Ω^2 thus controls whether the deterministic mean–feedback loop oscillates or relaxes, and that is the entire physics of the threshold.

Now impose the boundary conditions on the *homogeneous* BVP. We must produce a nonzero solution with $\bar m_0 = 0$, so start at $(\bar m_0, r_0) = (0, r_0)$ with $r_0 \ne 0$ and propagate to $t = T$ by e^{MT}. The first column of e^{MT} multiplies the (vanishing) initial mean, so the $qs\,\bar m_0$ pathway contributes nothing and only the r_0-column survives. Reading the second (r) component of $e^{MT}(0, r_0)^\top$ off the propagator – the I term contributes $\cos(\Omega T)\, r_0$ and the M term contributes $\Omega^{-1}\sin(\Omega T)$ times the lower-right entry $-A_\infty$ of M acting on r_0 – gives, in the oscillatory case,

$$r(T) = r_0\Big[\cos(\Omega T) - \tfrac{A_\infty}{\Omega}\sin(\Omega T)\Big].$$

With $s_T = 0$ the terminal relation is simply $r(T) = 0$, and since $r_0 \neq 0$ this is solvable exactly when the bracket vanishes, $\cos(\Omega T) = \frac{A_\infty}{\Omega}\sin(\Omega T)$, i.e. $\tan(\Omega T) = \Omega/A_\infty$, which is eq. (11.14). A nontrivial homogeneous solution exists if and only if the horizon T solves that transcendental equation; everything below is the case analysis of when it has roots and how they are distributed.

Hyperbolic case $\Omega^2 < 0$. The loop relaxes rather than oscillates. With $\omega = \sqrt{-\Omega^2} > 0$ real, the same propagator computation with $\cosh, \sinh$ in place of $\cos, \sin$ yields

$$r(T) = r_0\Big[\cosh(\omega T) - \tfrac{A_\infty}{\omega}\sinh(\omega T)\Big].$$

Now the two negatives conspire in our favor: $A_\infty < 0$ makes $-\frac{A_\infty}{\omega} > 0$, while $\cosh(\omega T) \geq 1 > 0$ and $\sinh(\omega T) > 0$ for $T > 0$, so the bracket is a sum of two strictly positive terms and is > 0 for *every* $T > 0$. (Equivalently, $\tanh(\omega T) \in (0, 1)$ never reaches the negative value Ω/A_∞ that eq. (11.14) would demand.) The bracket never vanishes, the only homogeneous solution is the trivial one, and the equilibrium is unique at every horizon. This is part (i): the condition $\Omega^2 < 0$ is $b^2 q(s-1) < a^2$, i.e. $s < 1 + a^2/(b^2 q)$, and in particular every $s \leq 0$ gives $\Omega^2 = b^2 q(s-1) - a^2 < -a^2 \leq 0$ with room to spare, recovering the monotone region from the other direction.

Marginal case $\Omega^2 = 0$. Exactly on the borderline $b^2 qs = A_\infty^2$ the two eigenvalues $\pm\lambda$ have collided at the origin, $\det M = 0$, and M is *nilpotent*: Cayley–Hamilton gives $M^2 = -\Omega^2 I = 0$ because M is traceless and now singular. The exponential series truncates after two terms, $e^{Mt} = I + Mt$, and the second component is the *linear* (non-oscillatory) function

$$r(T) = r_0\,(1 - A_\infty T).$$

Since $A_\infty < 0$, the factor $1 - A_\infty T = 1 + |A_\infty| T > 0$ for every $T > 0$; the bracket again never vanishes, the homogeneous BVP is trivial, and uniqueness holds on the borderline too. With the hyperbolic and marginal cases together we have established uniqueness for all of $\Omega^2 \leq 0$, completing part (i).

Oscillatory case $\Omega^2 > 0$. Now the eigenvalues have re-emerged on the imaginary axis as $\pm i\Omega$, the loop genuinely oscillates, and the bracket can vanish. Write

$$h(T) := \cos(\Omega T) - \tfrac{A_\infty}{\Omega}\sin(\Omega T),$$

the function whose zeros are the resonant horizons. It is a single sinusoid: combining the cosine and sine into one phase-shifted cosine, $h(T) = \mathcal{A}\cos(\Omega T + \varphi)$ with amplitude

$$\mathcal{A} = \sqrt{1 + \tfrac{A_\infty^2}{\Omega^2}} \;>\; 1,$$

the inequality being strict because $A_\infty \neq 0$ contributes a strictly positive second term under the root. The amplitude exceeding 1 is the decisive quantitative fact, and it is worth seeing why it forces zeros to exist: $h(0) = \cos 0 - \frac{A_\infty}{\Omega}\sin 0 = 1$, so the sinusoid *starts* at the value 1, which lies strictly inside its range $[-\mathcal{A}, \mathcal{A}]$ since $\mathcal{A} > 1$; a sinusoid that starts strictly between its extremes must subsequently reach 0, and being periodic it returns to 0 once per half-period thereafter. Concretely the zeros of $\mathcal{A}\cos(\Omega T + \varphi)$ occur where $\Omega T + \varphi = \frac{\pi}{2} + k\pi$, i.e. at $T = (\frac{\pi}{2} - \varphi + k\pi)/\Omega$, a *discrete arithmetic progression* $T_1 < T_2 < \cdots$ with common spacing π/Ω (one zero per half-period, as a non-degenerate sinusoid has). These T_n are precisely the positive solutions of $\tan(\Omega T) = \Omega/A_\infty$, eq. (11.14). The homogeneous BVP is singular exactly at the T_n – where $h(T_n) = 0$ – and at no other horizon: for any $T \notin \{T_n\}$, and in particular for every T in an open interval (T_n, T_{n+1}) strictly between consecutive resonances, $h(T) \neq 0$, the relevant propagator entry is nonzero, the affine BVP is invertible, and the equilibrium is unique. Hence even in the strongly imitative regime multiplicity does not fill an interval of horizons; it is confined to a *countable, evenly spaced, measure-zero* set of resonant horizons, with uniqueness restored on every gap between them. This is the LQG realization of the general principle that multiple equilibria are non-generic, and it dovetails exactly with proposition 10.23(iii), where these same T_n appear from the existence side as the *conjugate points* of the linear two-point problem – the horizons at which the forward–backward shooting map drops rank. Existence (Ch. 10) sees them as the horizons where the explicit equilibrium formula develops a $0/0$; uniqueness (here) sees them as the horizons where the homogeneous problem acquires a nonzero solution; they are one and the same discrete ladder.

It remains to evaluate the smallest resonance $T_c = T_1$ in closed form in the cleanest subcase $a = 0$ (no intrinsic drift), where the algebra is fully transparent. Then $A_\infty = -\sqrt{b^2 q} = -b\sqrt{q}$ and $\Omega = \sqrt{b^2 q(s-1)} = b\sqrt{q(s-1)}$, so the gain b and the weight $\sqrt{q}$ cancel in the ratio, $\Omega/A_\infty = -\sqrt{s-1}$, and eq. (11.14) becomes $\tan(\Omega T) = -\sqrt{s-1}$. Since the right-hand side is negative, the smallest positive root has ΩT in the second quadrant: $\Omega T_c = \pi - \arctan\sqrt{s-1}$. Dividing by $\Omega = b\sqrt{q(s-1)}$ gives the closed form $T_c = \big(\pi - \arctan\sqrt{s-1}\big)/\big(b\sqrt{q(s-1)}\big)$ recorded in the statement. As a sanity check on the geometry: at the onset $s \to 1^+$ we have $\Omega \to 0^+$, the second-quadrant angle $\pi - \arctan\sqrt{s-1} \to \pi$, and $T_c \to \infty$ – the first resonance recedes to infinite horizon exactly as the oscillatory regime is entered, so the transition into multiplicity is continuous in the horizon even though it is sharp in s. $\square$

Proposition 11.8 has done more than locate a single threshold; it has partitioned the parameter half-plane into three qualitatively distinct regions, and it is worth stating them as a clean trichotomy because each carries a different lesson for the chapter's thesis. As s increases from $-\infty$ at fixed q, a, b, the coupling passes from monotone, through an anti-monotone-but-still-unique band, into the genuinely multiplicity-prone regime; the schematic in fig. 11.1 draws the result in the (s, T) plane.

- **Monotone region, $s \le 0$.** Crowd aversion: the agent is pushed away from the mean. The coupling satisfies **(A-Mon)** by eq. (11.10) (which is ≥ 0 exactly when $s \le 0$), and uniqueness holds for all (q, T). Here the two halves of the chapter meet: the general duality theorem theorem 11.5 and the explicit BVP computation proposition 11.8(i) both certify uniqueness, by completely different routes – one an integration-by-parts identity, the other a non-oscillatory propagator – and they agree, as they must, since they are reading the same fact off the same model.

- **Soft anti-monotone region, $0 < s \le 1 + a^2/(b^2 q)$.** Imitation, but weak relative to the individual cost curvature. Now the agent is drawn *toward* the mean, so the coupling is anti-monotone and eq. (11.10) is strictly negative: the Lasry–Lions condition **(A-Mon)** *fails*, and theorem 11.5 has nothing to say. Yet the equilibrium remains unique at every horizon T, because $\Omega^2 = b^2 q(s-1) - a^2 \le 0$ keeps the homogeneous BVP non-oscillatory (the hyperbolic/marginal cases of the proof), so the loop relaxes instead of resonating. This band is the concrete, computable witness the chapter has been promising that *monotonicity is sufficient but not necessary*: an explicit open set of attractive couplings, of width $s_\star = 1 + a^2/(b^2 q)$ in s, on which monotonicity is violated and uniqueness nonetheless holds for all T. It also quantifies *how badly* the abstract criterion overshoots, which the third structural fact below makes precise.

- **Resonant horizons,** $s > 1 + a^2/(b^2 q)$**.** Strong imitation. The agent now overshoots the crowd hard enough that the closed loop turns oscillatory (Ω real), and the homogeneous BVP becomes singular – but only at the *discrete* horizons $T_n(s, q)$ of eq. (11.14), spaced π/Ω apart. At each T_n the equilibrium is non-unique (or fails to exist); at every other horizon, including each open interval (T_n, T_{n+1}) between resonances, it remains unique. The careful reading is therefore that what is lost beyond $s_\star = 1 + a^2/(b^2 q)$ is not uniqueness on a range of horizons but *unconditional*, horizon-free uniqueness: a discrete ladder of resonant horizons $T_n(s)$ switches on, the first rung being $T_c = T_1$, and T_c decreases as s increases – stronger attraction brings the first resonance sooner, the first rung sliding down toward $t = 0$ as the imitation stiffens. These are exactly the conjugate points of proposition 10.23(iii) seen from the uniqueness side.

Three structural facts deserve emphasis, because each is a transferable lesson rather than an LQG accident. First, the boundary is governed entirely by the *deterministic* pair $(r, \bar{m})$: the noise level σ dropped out of eqs. (11.12) and (11.13) and never reappeared, so the phase diagram lives in (s, q, T) alone, exactly as the thread specification anticipated, with σ an inert spectator that broadens each agent's marginal without moving the mean or shifting a single threshold. Uniqueness of a *stochastic* game is decided by a *deterministic* two-point problem. Second, the transition is genuinely a threshold and not a gradual erosion, and the proof exhibits the mechanism: horizon-free uniqueness persists intact as s grows, untouched, right up to $b^2 q s = A_\infty^2$, where the real eigenvalue pair $\pm\omega$ of M *coalesces* at the origin (the marginal nilpotent case) and then re-emerges on the imaginary axis as $\pm i\Omega$, converting relaxation into oscillation. It is this eigenvalue collision – a spectral event at a precise parameter value – that makes the loss of uniqueness abrupt, and it is the exact mathematical content of the "spontaneous symmetry breaking" language of section 11.4: the symmetric solution loses stability not by softening but by a pair of modes crossing into oscillation. Third, the monotonicity threshold ($s = 0$) and the true multiplicity threshold ($s_\star = 1 + a^2/(b^2 q)$) are *different* numbers, separated by a band of width at least 1 in s – exactly 1 when $a = 0$, wider when the intrinsic drift a is present. That gap is not a defect of our analysis but the precise, model-by-model measure of how *conservative* the Lasry–Lions condition is: it forbids the entire half-line $s > 0$, whereas the truth forbids only $s > s_\star$, so the condition discards a whole band of well-behaved attractive couplings to buy the convenience of a single checkable inequality. A sharper, less conservative structural hypothesis would shave into that band – which is precisely the question the next section takes up.

Remark 11.9 (Interpreting the critical attraction $s = 1$)**.** The threshold $s_\star = 1 + a^2/(b^2 q)$ has a clean reading, cleanest in the driftless case $a = 0$ where it sits exactly at $s = 1$. Recall that the agent's running cost is $\frac{q}{2}(x - s\bar{m})^2$, so the agent is penalized for deviating from the moving target $s\,\bar{m}$. The value $s = 1$ is the precise point at which that target *tracks the population mean* *one-for-one*: each agent wants to sit exactly where the average agent sits. Below it, $s < 1$, the target is a fractional copy of the mean – the agent chases a damped image of the crowd, $s\bar{m}$ moves less than $\bar{m}$, and any collective drift is partially resisted, so the feedback is self-correcting and a unique consistent mean survives at every horizon. At $s > 1$ the agent *overshoots*: the target $s\bar{m}$ moves *further* than the crowd it imitates, so a small collective lean to the right makes everyone want to be even further right than the crowd already is, and the response amplifies the displacement instead of damping it. That is the self-amplifying feedback the symmetry-breaking section described, and it becomes fatal to horizon-free uniqueness once the horizon is long enough for the overshoot to compound – which is exactly why the first resonance T_c is finite for $s > 1$ and recedes to $+\infty$ as $s \downarrow 1$. The correction term $a^2/(b^2 q)$ raises the bar above 1 by exactly the amount the agent's private incentives stiffen it against the crowd: a large individual penalty q for straying from one's own target, or a strong intrinsic mean-reversion encoded in a, both anchor the agent to its private optimum and demand a more aggressive overshoot before the collective mode can run away. The ratio $a^2/(b^2 q)$ is the dimensionless measure of private stiffness relative to control authority b, and it is precisely the width of the soft band.

The figure makes the trichotomy legible at a glance, and it is worth reading deliberately because the geometry of the third region is easy to misdraw. Figure 11.1 plots the (s, T) plane at fixed q, a, b. To the left of $s = 0$ is the monotone half-plane, unique for every T; the vertical strip $0 < s \leq s_\star$ is the soft band, still unique for every T despite violating monotonicity, so the entire region $s \leq s_\star$ is a solid block of uniqueness with no internal structure. The essential point is what happens for $s > s_\star$: the failure of uniqueness there is *not* a shaded region but a family of *curves*. Each resonant horizon $T_n(s)$ traces one curve $T = T_n(s)$ sweeping upward as $s \downarrow s_\star$ (where every T_n escapes to $+\infty$) and descending as s grows; the curves do not fill area, and the open gaps between consecutive curves $T_n(s) < T < T_{n+1}(s)$ are themselves regions of uniqueness. Uniqueness is thus the generic (open, full-measure) condition everywhere on the diagram, and multiplicity lives only on the measure-zero curves – the lowest of which, $T_c(s) = T_1(s)$, is the boundary of horizon-free uniqueness and the one a modeller working at a fixed long horizon meets first as the imitation strength is dialed up.

We have, in this one solvable model, the entire phase structure the chapter set out to expose: the monotone region where the general theorem applies, the soft band that proves the theorem's hypothesis is merely sufficient, and the resonant ladder that shows how – and how suddenly – uniqueness finally fails. But the soft band leaves a pointed question hanging. The Lasry–Lions condition discarded the whole half-line $s > 0$ because it tests convexity of the equilibrium energy along *linear* interpolations of populations, the flat mixing $m_\epsilon = (1-\epsilon)m' + \epsilon m$ of proposition 11.3; and the LQG computation has just shown that this flat notion of "straight line" is conservative – it condemns couplings that are in fact perfectly well behaved. That should provoke the suspicion that linear interpolation is not the only, or even the most natural, notion of a straight path between two populations. On $\mathcal{P}_2(\mathbb{R}^d)$ there is a second one: the Wasserstein geodesic, which slides mass along optimal transport routes rather than fading one density into another. Convexity along *that* family of straight lines is a different, incomparable condition – displacement monotonicity – and it buys uniqueness in regimes the flat condition cannot reach, just as the flat condition reaches regimes it cannot. The LQG soft band is the concrete evidence that there is room, and demand, for such alternative structural hypotheses. The next section takes up exactly this: what other, genuinely

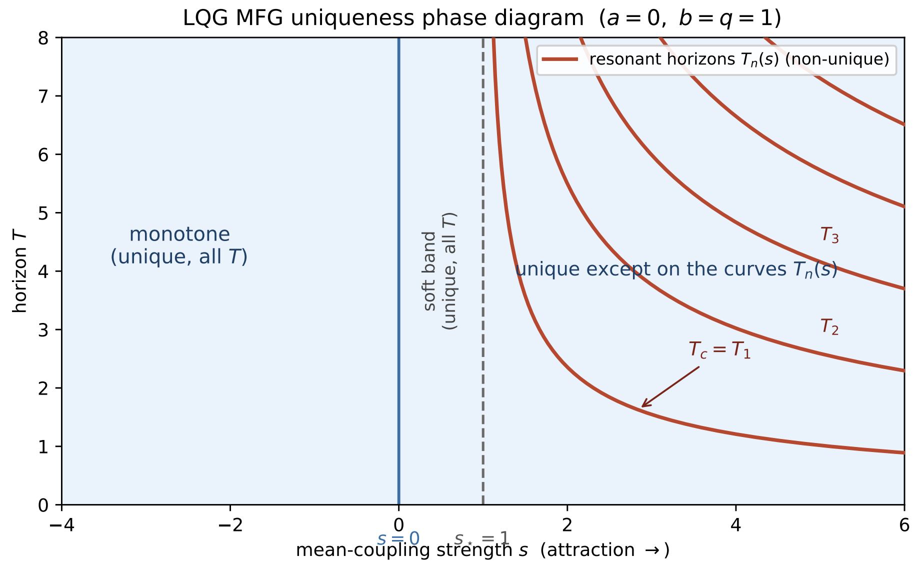

Figure 11.1: Schematic uniqueness phase diagram of the LQG mean field game in the (s, T) plane at fixed q, a, b (proposition 11.8). For $s \leq 0$ the coupling is Lasry–Lions monotone and the equilibrium is unique for all T. A soft band $0 < s \leq 1 + a^2/(b^2 q)$ remains unique for all T despite violating monotonicity. Beyond $s_\star = 1 + a^2/(b^2 q)$ the equilibrium is unique except on a *discrete* family of resonant horizon curves $T_n(s)$ (the solutions of eq. (11.14), drawn as curves, not a filled region), where uniqueness fails; the lowest curve $T_c(s) = T_1(s)$ marks the first loss of horizon-free uniqueness and decreases in s. Between consecutive curves the equilibrium is again unique, in agreement with proposition 10.23(iii).

different, convexities on the space of populations purchase uniqueness, and how flat (linear-mixing) monotonicity sits beside displacement (mass-transport) monotonicity as one of several incomparable sufficient conditions.

11.6 Displacement monotonicity and modern refinements

The LQG soft band left us with a sharpened complaint rather than a loose end. The Lasry–Lions condition condemned the entire imitation half-line $s > 0$, yet the explicit computation showed that a whole sub-band of those couplings produces a unique equilibrium at every horizon. The condition is, demonstrably, conservative – and the diagnosis offered at the close of the previous section located the conservatism precisely: **(A-Mon)** tests convexity of the equilibrium energy only along *linear* interpolations of populations, the flat mixing $m_\epsilon = (1 - \epsilon)m' + \epsilon m$ of proposition 11.3. That is one notion of a straight line between two populations, but it is not the only one, and on the space of probability measures it is arguably not even the most natural one. Linear mixing fades one density into another without moving a single particle: to interpolate "half-way" from m' to m it simply averages the two densities pointwise, leaving mass exactly where it already sat in either configuration and merely re-weighting it. The geometry of $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ – the geometry the whole transport picture of chapter 4 was built to respect – prefers a different straight line, one in which mass is physically *carried* from where m' puts it to where m wants it, each parcel travelling the shortest route. This section takes up the structural conditions that ride on that second straight line, contrasts them honestly with the flat condition the chapter's theorem actually consumes, and – because the full theory is a recent and technical development – is explicit about where the boundary of what this book proves lies.

Recall from chapter 4 that $\mathcal{W}_2$ is not an arbitrary metric on $\mathcal{P}_2(\mathbb{R}^d)$ but a *geodesic* distance: between any two measures there is a constant-speed shortest path, and that path is got by optimal transport, not by mixing. Concretely, if T is the optimal map pushing m' forward to m (the Monge map, when it exists), the constant-speed geodesic is the *McCann displacement interpolation*

$$\mu_\theta \;=\; \big((1-\theta)\,\mathrm{Id} + \theta\, T\big)_\sharp\, m', \qquad \theta \in [0,1],$$

constructed in full in appendix D (appendix D.3): each particle starting at x under m' slides along the straight segment from x to its destination $T(x)$, and μ_θ is the snapshot of the whole cloud at fraction θ of the journey. This is a genuinely different curve in $\mathcal{P}_2(\mathbb{R}^d)$ from the linear segment m_ϵ. The cleanest way to see the difference is two equal point masses: with $m' = \delta_a$ and $m = \delta_b$, linear mixing gives $m_\epsilon = (1-\epsilon)\delta_a + \epsilon\delta_b$, a pair of spikes whose *weights* shift from a to b while the spikes themselves never move; displacement interpolation gives $\mu_\theta = \delta_{(1-\theta)a+\theta b}$, a single spike that *travels* from a to b. Mixing reweights in place; transport moves. Convexity of an energy along these two families of straight lines is therefore convexity in two different senses, and there is no reason to expect them to agree.

McCann's insight was that many energies that are emphatically *not* convex along linear mixing are convex along displacement – the internal (entropy-type) energies, the potential energies $\int V\,\mathrm{d}m$ with V convex, and the interaction energies $\iint K\,\mathrm{d}m\,\mathrm{d}m$ with K convex all become convex once one interpolates by transport rather than by mixing. This is the convexity that the geometry of $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ sees, and it is the natural input to any argument that itself lives on that geometry – in particular to the master equation, whose unknown is a function *on* $\mathcal{P}_2(\mathbb{R}^d)$ and whose well-posedness is controlled by how energies bend along the metric's own geodesics. We can now state the condition for orientation.

Definition 11.10 (Displacement convexity and displacement monotonicity)**.** A functional $\mathcal{F}\colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ is *displacement convex* if for every constant-speed Wasserstein geodesic $(\mu_\theta)_{\theta\in[0,1]}$ in $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ — the McCann displacement interpolation constructed in appendix D (appendix D.3) — the map $\theta \mapsto \mathcal{F}(\mu_\theta)$ is convex. A coupling $F = \delta\mathcal{F}/\delta m$ is then called *displacement monotone*; equivalently (for F regular enough that the L-derivative $\partial_\mu F$ of appendix A.3 exists), displacement monotonicity is the positivity of the second variation of $\mathcal{F}$ along geodesics, a condition on $\partial_\mu F$ and its spatial gradient rather than on F paired with $m - m'$.

The last clause is the one a physicist should pause on, because it is exactly where the two monotonicities part company, and it is worth saying *why* the displacement condition falls on the spatial gradient of the L-derivative rather than on the bare cost difference $F(\cdot, m) - F(\cdot, m')$ that eq. (11.3) pairs against $m - m'$. The reason is the one the point-mass picture already made visible: along a geodesic, mass moves. Differentiate $\theta \mapsto \mathcal{F}(\mu_\theta)$ once and one gets, by the chain rule on $\mathcal{P}_2(\mathbb{R}^d)$ (appendix A.3), the flat derivative $F(\cdot, \mu_\theta)$ paired against the velocity of the curve; but the velocity of a displacement geodesic is not a re-weighting, it is a transport field $\nabla\phi$ (the gradient of a Kantorovich potential), so the pairing is against the *motion* of mass. Differentiating a second time then asks how the cost itself changes as each particle is carried along its segment, and a transported particle samples the cost at shifting *locations*: what enters is therefore the rate of change of F in the state variable, i.e. the spatial gradient ∂_x of the L-derivative $\partial_\mu F$, contracted twice against the transport field. The schematic shape of the second variation is

$$\frac{\mathrm{d}^2}{\mathrm{d}\theta^2}\,\mathcal{F}(\mu_\theta) \;=\; \int \Big\langle \partial_x\partial_\mu F(x,\mu_\theta)\,\nabla\phi(x),\,\nabla\phi(x)\Big\rangle\,\mu_\theta(\mathrm{d}x) \;+\; (\text{lower-order transport terms}),$$

and displacement monotonicity is the demand that this quadratic form in the displacement field $\nabla\phi$ be nonnegative. Contrast the flat second variation $\varphi''(\epsilon) = \iint \frac{\delta^2\mathcal{F}}{\delta m^2}(m_\epsilon)(x,y)\,(m-m')(\mathrm{d}x)(m-m')(\mathrm{d}y)$ of proposition 11.3, which never differentiates F in space at all, because under linear mixing nothing moves in space – only the weights change, so only the bare values of the cost (through the kernel $\delta^2\mathcal{F}/\delta m^2$) are sampled. *Mixing sees cost values; transport sees cost gradients.* That single sentence is the whole reason the two conditions are different objects rather than two encodings of one object, and it is why one is a condition on F paired with $m-m'$ while the other is a condition on $\partial_x\partial_\mu F$.

Notation watch

Lasry–Lions ("flat") monotonicity and displacement monotonicity are *not* nested: neither implies the other. A coupling can be flat-monotone but not displacement-monotone and vice versa. The two conditions coincide for the trivial (decoupled) case and overlap for specific structured couplings, but in general they cut out different – both non-empty – regions of model space. It is a genuine error to treat "monotone" as unambiguous; one must say *which* monotonicity.

The non-nesting is not an abstract caution; one model class exhibits it on a single line of algebra, and it is worth doing because it makes the two conditions concretely incomparable rather than merely "not known to be comparable." Take a pure interaction energy $\mathcal{F}(m) = \frac{1}{2}\iint K(x-y)\,m(\mathrm{d}x)\,m(\mathrm{d}y)$ with K even, whose flat derivative is the convolution $F(x,m) = (K*m)(x)$. Its flat monotonicity pairing collapses to a single quadratic form in the population shift $\Delta = m - m'$,

$$\int \big(F(x,m) - F(x,m')\big)\,\Delta(\mathrm{d}x) = \iint K(x-y)\,\Delta(\mathrm{d}x)\,\Delta(\mathrm{d}y),$$

which is nonnegative for *every* zero-mass Δ precisely when K is of positive type – by Bochner's theorem, when its Fourier transform $\widehat{K} \geq 0$. Displacement convexity of the very same $\mathcal{F}$, on the other hand, is McCann's criterion that the interaction kernel K be *convex* as a function on $\mathbb{R}^d$. These two requirements – positive-definite versus convex – are logically independent, and one can fail while the other holds. A smooth bump such as a centered Gaussian $K(z) = e^{-|z|^2/2}$ has $\widehat{K} \geq 0$, so it is flat-monotone, yet it is concave in a neighborhood of the origin and hence *not* convex, so it is not displacement-monotone. Conversely the quadratic kernel $K(z) = |z|^2$ is convex, hence displacement-monotone, but its flat pairing evaluates (using $\int \Delta = 0$, so the $|x|^2$ and $|y|^2$ pieces drop) to $\iint |x-y|^2 \Delta(\mathrm{d}x)\Delta(\mathrm{d}y) = -2\big|\int x\,\Delta(\mathrm{d}x)\big|^2 \leq 0$, which is strictly negative as soon as m, m' differ in mean – it is flat *anti*-monotone. So here are two couplings, each satisfying exactly one of the two conditions and violating the other. The conditions are genuinely incomparable, and "F

is monotone" is, without a qualifier, an ambiguous sentence. (Note this is the cleanest separating example, not a claim that the two notions never meet: an entropy or a strictly increasing local congestion cost can be both at once. The point is only that membership in one class is no evidence of membership in the other.)

With the geometry and the separation in hand, the operational dichotomy is clean, and it is the one piece of this section the rest of the chapter actually uses. Lasry–Lions monotonicity is what theorem 11.5 consumes: it is the sign that makes the boundary pairing $\int \bar{u}\,\bar{m}$ accumulate favorably and delivers uniqueness of the MFG *system* eqs. (11.1) and (11.2) itself, working directly on the coupled PDEs in physical state space. Displacement monotonicity is the analogue one level up, on the space of measures: it is the sign that makes the relevant second-order differential operator on $\mathcal{P}_2(\mathbb{R}^d)$ – the one whose vanishing is the *master equation* of chapter 12 – monotone, and thereby delivers well-posedness of that equation, and through it uniqueness, in regimes the flat condition cannot certify. The headline such regime is the *non-separable* Hamiltonian, where the running cost couples the control and the measure together (the agent's optimal velocity itself depends on the crowd, not merely its cost), so that the clean splitting into a control part and a population part that underwrites the flat-monotone variational picture of chapter 8 is simply unavailable; certain congestion models and the master equation under common noise sit in the same camp. There the duality identity of section 11.3 has no monotone term to extract, and one must instead ask the energy to bend the right way along the metric's own geodesics – which is displacement convexity.

Here the chapter reaches the boundary of what it proves, and the style of the book obliges us to mark that boundary rather than paper over it. The master equation is built and shown well posed *under Lasry–Lions monotonicity* in chapter 12, in-book and in full; that is the route this text certifies. The *displacement*-monotone theory – the proof that the condition just stated yields a monotone master operator and hence well-posedness for non-separable couplings – is a more recent and considerably more technical development, and we do not reproduce it. The precise functional inequality is a positivity condition on $\partial_x \partial_\mu F$ and its symmetrization, paired against arbitrary square-integrable displacement fields (the $\nabla\phi$ of the second-variation formula above); the argument that it propagates to monotonicity of the master operator, and from there to a unique classical solution, is due to Gangbo–Mészáros and collaborators and to Mou–Zhang [75, 76, 104], and it rests on the displacement-convexity calculus on $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ assembled in appendix D. We state the condition for orientation and attribute the theory to those sources deliberately: this is a survey-and-attribute passage, not a bare "see [reference]" standing in for an argument the chapter pretends to make. The honest reason it is out of scope is concrete rather than rhetorical – the proof needs the full second-order differential calculus on Wasserstein space (Lions derivatives, the chain rule along geodesics, and the master operator's coercivity estimates) at a level of technical machinery this chapter has not built and chapter 12 builds only for the separable, flat-monotone case. A reader who wants the theorem in force should read it as the displacement analogue of theorem 11.5, with the same architecture (a monotone pairing forces two solutions to coincide) transplanted from the system in state space to the master equation on measure space.

A fourth route rounds out the picture, and unlike displacement monotonicity it is not a different structure but a quantitative softening of the one we already proved. *Lasry–Lions monotonicity up to a small constant* – equivalently *weak monotonicity* – asks only that the coupling be monotone *after* subtracting a controlled non-monotone perturbation: eq. (11.3) is allowed to fail, but by no more than a fixed deficit. Uniqueness then survives provided that deficit is dominated, either by the diffusion's regularizing strength (a nondegenerate ν smooths the feedback loop and can absorb a small wrong-sign coupling) or by a short enough horizon (too little time for the residual instability to wind up). This is exactly the quantitative bridge between the chapter's two extremes. At one end sits the unconditional, horizon-free uniqueness of theorem 11.5, which demands the full sign condition and asks nothing of T; at the other sits the short-horizon contraction of theorem 9.10(i), which demands no sign condition but caps T. Weak monotonicity interpolates: the more nearly monotone the coupling, the longer the horizon that can be tolerated, recovering the unconditional theorem as the deficit shrinks to zero and the short-horizon theorem as it grows. This is the weak-monotonicity programme of Ahuja and others [4], and it is precisely the soft-band phenomenon the LQG thread exhibited, now named as a general mechanism: the band of attractive-but-too-weak couplings that the explicit computation found uniquely solvable at every horizon is exactly the regime where the non-monotone defect is small enough for the diffusion to swallow it.

We have thus put on the table two further routes to uniqueness beyond the short-horizon contraction and the Lasry–Lions condition the chapter has spent: displacement monotonicity, a genuinely different and incomparable structural hypothesis that reaches the non-separable models the flat condition cannot, and weak monotonicity, a quantitative relaxation that trades a small non-monotone defect against diffusion or horizon. Four conditions now stand assembled – short-horizon, Lasry–Lions, displacement, weak – each buying uniqueness at a different price in structure and a different reach in T. The next section lays them side by side and reads off the boundary of the uniqueness region they jointly map out, ordered from the cheapest and most local to the structural and horizon-free.

11.7 The boundary of the uniqueness regime

The previous section finished assembling the cast: four routes to a unique equilibrium — short-horizon contraction, Lasry–Lions monotonicity, displacement monotonicity, and weak monotonicity — each buying the same conclusion at a different price. It is time to make good on the promise to lay them side by side and read off, from the comparison, where the uniqueness region ends and what lies past its edge. The right way to see the four conditions is not as a list of unrelated sufficient hypotheses but as points along a single axis of trade: at one end, conditions that ask almost nothing of the *structure* of the coupling but demand a short *horizon*; at the other, conditions that ask a definite structural sign but then give uniqueness for *every* horizon. Each route pays for what it spares. Reading them in that order — cheapest and most local first, structural and horizon-free last — is reading down the chapter's own argument, from the elementary contraction of section 11.1 to the duality identity of section 11.3 and the refinements of section 11.6.

The first route asks for no structure at all and pays in horizon. **Short horizon (any coupling).** If T is small, the forward–backward best-response map S is a contraction on $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$: its Lipschitz constant is $C\,T$, where C aggregates the Lipschitz constants of the data (the couplings F, G and the Hamiltonian's velocity field), so the moment $C\,T < 1$ the Banach fixed-point theorem forces a unique fixed point. This is the small-horizon well-posedness already proved in theorem 9.10(i), and it needs *no sign condition* whatever — attractive or repulsive, the round trip from initial mass to terminal value and back is simply too brief for the feedback loop to wind up. It is the honest reappearance of the Grönwall mechanism that section 11.1 declared unavailable: available after all, but only on a horizon clamped below $1/C$. That clamp is exactly the defect. The admissible window $T < 1/C$ *shrinks as the coupling strengthens*, because a stronger interaction raises C, and it collapses to nothing in the strongly attractive limit where $C \to \infty$. The LQG thread displays the window concretely: the band $T < T_c$ certified by proposition 11.8 holds for every imitation strength s, but the upper edge $T_c = (\pi - \arctan\sqrt{s-1})/(b\sqrt{q(s-1)})$ descends as s climbs past its critical value, so the short-horizon guarantee buys less and less time precisely where one would most want it. Short-horizon uniqueness is therefore the universal but local route: good for any model, good only briefly.

The second route reverses the trade. **Lasry–Lions monotonicity (any horizon).** Under **(A-Mon)** — the single inequality eq. (11.3), that crowding a region raises its cost — uniqueness holds for *all* T, by theorem 11.5. There is no horizon restriction to shrink, because the favorable signs in the duality identity accumulate monotonically over $[0, T]$ rather than racing a growing exponential; lengthening the horizon only adds more nonnegative terms to a quantity already trapped between zero and zero. This is the chapter's workhorse, and its value is threefold and worth restating in one breath: it is *checkable* (a single pairing, verifiable on the data without solving anything), *horizon-free* (it never mentions T), and exactly the structural hypothesis under which the deeper machinery of the book runs. The variational picture of chapter 8 is well behaved precisely here — monotone coupling is convexity of the equilibrium energy, so the Euler–Lagrange system has one critical point because a convex bowl has one bottom — and the master equation of chapter 12 is well posed precisely here, for the same reason read one level up. The price is honest and was paid in full in section 11.5: monotonicity is sufficient, not necessary, and it is conservative by exactly the width of the LQG soft band.

The third route is not a relaxation of the second but a genuine alternative, reaching models the second cannot certify. **Displacement monotonicity (any horizon).** As section 11.6 established, this is an *incomparable* structural hypothesis — convexity of the energy along Wasserstein geodesics rather than along linear mixtures, a condition on the spatial gradient $\partial_x \partial_\mu F$ rather than on F paired with $m - m'$ — neither implying nor implied by **(A-Mon)** (definition 11.10). It too gives uniqueness at every horizon, and its reach is the *non-separable* Hamiltonians where the agent's optimal velocity itself depends on the crowd, so that the clean control/population splitting underwriting the flat-monotone variational picture is unavailable and the duality identity of section 11.3 has no monotone term to extract. Uniqueness there comes instead through a displacement-monotone master-equation argument, attributed in section 11.6 to [75, 104] and *not reproduced in-book*: the proof needs the full second-order differential calculus on $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ at a level of machinery this chapter has not built. It enters the menu as the structural route to non-separable models, with the explicit caveat that the book certifies the flat-monotone case and surveys, rather than proves, this one.

The fourth route is the quantitative bridge between the first two, and it is what the LQG computation was secretly an instance of. **Weak / approximate monotonicity.** Here eq. (11.3) is allowed to fail, but by no more than a controlled deficit; uniqueness survives provided that deficit is dominated — by the diffusion's regularizing strength (a nondegenerate ν smooths the feedback loop and absorbs a small wrong-sign coupling) or by a short enough horizon (too little time for the residual instability to compound) [4]. This is a continuous interpolation along the very axis the menu is ordered by: as the non-monotone defect shrinks to zero it recovers the unconditional horizon-free theorem of route (2), and as the defect grows it degrades to the horizon-capped contraction of route (1). The soft band of section 11.5 — the attractive-but-too-weak couplings $0 < s \leq s_\star$ that the explicit computation found uniquely solvable at every horizon — is exactly this mechanism made visible: a small anti-monotone defect, swallowed by the private cost curvature q and the diffusion, leaving uniqueness intact where **(A-Mon)** had already given up.

This is the whole interior of the uniqueness region, and the four routes between them map it out. What the menu also makes visible is its *boundary* — and here the LQG phase diagram is the indispensable witness, because it shows the boundary is not approached asymptotically but reached at a sharp, computable threshold. Past $s = s_\star$, at the resonant horizon $T_c(s) = T_1(s)$ of proposition 11.8, uniqueness fails abruptly: the loss is an eigenvalue collision, a spectral event at a precise parameter value, not a gradual erosion. Outside the menu, then, lies genuine multiplicity, and the natural next question is the one this chapter cannot fully answer. *At* and *beyond* the boundary, where several mutually incompatible equilibria coexist, which one does the world play? This is the subject of *selection theory*, and the candidate principles are several: one may keep only the dynamically stable equilibria (those robust to small perturbations of belief), or take the vanishing-viscosity limit (the configuration selected as the noise that smooths the landscape is sent to zero), or single out the equilibrium consistent with the classical solution of the master equation where one exists. Each is reasonable, none is universal, and they need not agree. It must be stated plainly, in keeping with the book's refusal to paper over its edges: *a complete and general selection principle for non-monotone mean field games is not known.* It is one of the genuine open problems of the subject, revisited in chapter 31; the honest summary is that uniqueness, where this chapter secures it, is a structural gift, and where the structure fails the theory presently offers a forecast only up to a finite menu of self-consistent futures it cannot yet rank.

There is, however, one direction in which the boundary becomes *sharper* rather than more mysterious, and it is the direction the rest of the book travels: replacing the global mean coupling by a network. This chapter's entire phase structure was organized around a single scalar knob, the imitation strength s, with the critical attraction sitting at $s = 1$ in the driftless case — the point, recorded in remark 11.9, at which each agent's target $s\bar{m}$ tracks the population mean one-for-one, so that a small collective lean reproduces itself instead of damping out. The graphon generalization of chapter 24 keeps exactly this physics but indexes it by the coupling *kernel*. There the agent is drawn not toward the global mean $\bar{m}$ but toward the graphon-aggregated neighbor mean $(\mathbb{W}\,\bar{m})(x) = \int_0^1 W(x,y)\,\bar{m}(y)\,\mathrm{d}y$, weighting the crowd by network proximity, and the self-consistent collective lean is no longer a single number but a *mode* on the type space $[0,1]$. The runaway that destroys uniqueness happens when the strategic gain s amplifies some such mode faster than the private cost curvature can restore it; and which mode runs away first is settled by the operator $\mathbb{W}$. The fastest-growing mode is the leading eigenvector of $\mathbb{W}$ — the network-centrality (Perron) profile that weights each type by how strongly the kernel funnels the crowd toward it — and the threshold is governed by the corresponding eigenvalue, hence by the operator norm $\|\mathbb{W}\|$ (the spectral radius for the self-adjoint kernels of **(A-Reg-W)**). Heuristically, the scalar resonance $s = 1$ becomes the operator competition $s\,\|\mathbb{W}\| \asymp 1$: uniqueness is the regime where strategic monotonicity wins against the kernel's amplification, exactly the operator-norm smallness recorded as **(A-Coupling)**, and the critical coupling at which it fails is a symmetry-breaking threshold tied to the spectral radius of $\mathbb{W}$ — the same physics as this chapter's $s = 1$ line, now reading off network centrality rather than a single mean. The constant graphon $W \equiv c$ collapses the operator back to multiplication by c and recovers Thread A's scalar threshold with s rescaled by c, which is the consistency check that the two pictures are one. We state this only as the teaser it is: chapter 24 carries the actual theorem, the precise constant in **(A-Coupling)**, and the proof, which is this chapter's duality computation with the scalar pairing $\int \bar{u}\,\bar{m}$ promoted to a pairing in $L^2[0,1]$ against the operator $\mathbb{W}$.

That promotion is the chapter's exit, and it is worth naming what travels. The single duality identity proved in section 11.3 — one Green's identity, one integration by parts, a sum of manifestly signed terms — is not a self-contained curiosity but the seed of the book's later well-posedness theory. It hands forward to chapter 12 the fact that the master equation is well posed *exactly* in the monotone regime mapped here, with the chapter's convex-landscape reading becoming the monotonicity of the master operator; and it hands forward to chapter 24 the operator generalization just previewed, where the same computation, run in L^2 against $\mathbb{W}$, yields graphon uniqueness up to the spectral threshold. The throughline that opened the chapter — existence gives *an* equilibrium, a predictive theory needs *the* equilibrium — thus closes not at this chapter's edge but at the entrance to the two that build on it: the geometry of where uniqueness lives, why it can fail suddenly, and how to compute the boundary, is the geometry those later chapters inherit.

Rigor ledger

Proved.

- theorem 11.5: under **(A-Lip)**, **(A-Nondeg)**, **(A-Conv)** with strict convexity, and **(A-Mon)**, the MFG system eqs. (11.1) and (11.2) has at most one classical solution, via the Lasry–Lions duality identity eq. (11.8).
- propositions 11.2 and 11.3: local crowd-averse couplings are monotone; potential couplings are monotone iff their energy is flat-convex.
- worked computation 11.7 and proposition 11.8: the LQG mean field game reduces to a deterministic two-point BVP for $(r, \bar{m})$; its coupling is monotone iff $s \le 0$ (eq. (11.10)); the exact loss of horizon-free uniqueness occurs at $b^2 qs = A_\infty^2$, beyond which non-uniqueness sits only on the discrete resonant horizons eq. (11.14), giving the three-regime phase diagram in (s, q, T), independent of σ.

Assumed.

- Existence of a classical solution with the decay/regularity making the integrations by parts valid – imported from theorem 10.14 (bounded regularising couplings, with the value's x-Lipschitz bound lemma 10.5) and proposition 10.23 (the Gaussian LQG thread); the small-horizon contraction is theorem 9.10(i).
- Uniqueness in law for the linear SDE / Fokker–Planck equation with fixed coefficients (theorems 5.10 and 5.14) and the comparison/uniqueness principle for the backward HJB equation (theorem 6.18), used to close Step 5 of theorem 11.5.

- The second-order flat-derivative calculus on $\mathcal{P}_2(\mathbb{R}^d)$ underlying proposition 11.3 (chapter 4 and appendix D); the displacement-interpolation geometry of definition 11.10 (appendix D.3).

Deferred.

- Displacement monotonicity: precise functional inequality and its use for master-equation well-posedness with non-separable Hamiltonians – surveyed in section 11.6 and attributed to [75, 76, 104]; the displacement-convexity calculus it rests on is in appendix D. Not reproduced in-book.
- Discounted/ergodic monotonicity argument (remark 11.6) → [93].
- The graphon uniqueness theorem and the $\|\mathbb{W}\|$ threshold → chapter 24.
- The full time-dependent ($q_T \neq P_\infty$) LQG threshold, solved numerically → chapters 14 and 28.

Open.

- A general selection principle among multiple equilibria in the non-monotone regime (chapter 31).
- Sharp characterization of the full uniqueness region for non-separable Hamiltonians beyond the displacement-monotone class.

Bibliographic notes

The monotonicity condition and the duality proof of uniqueness originate with Lasry and Lions [93]; the lecture-note treatment we have followed most closely, including the finite-horizon computation of theorem 11.5 and the integration-by-parts structure, is Cardaliaguet [36]. The probabilistic counterpart and the role of monotonicity in the well-posedness of mean field FBSDEs and the master equation are developed in Carmona–Delarue [40, 41], and the master-equation well-posedness under monotonicity is the central theme of Cardaliaguet–Delarue–Lasry–Lions [37], to which chapter 12 is the introduction.

Displacement convexity is McCann's notion from optimal transport [131]; its deployment as a monotonicity condition for mean field games – displacement monotonicity – and the resulting global well-posedness of master equations for non-separable Hamiltonians are due to Gangbo and Mészáros and collaborators [75, 76] and, in the non-separable setting, to Mou and Zhang [104]. The weak/approximate monotonicity condition under common noise is Ahuja [4].

The LQG phase analysis of section 11.5 is the uniqueness layer of the linear–quadratic thread completed in chapter 14; the reduction of LQG mean field games to coupled Riccati/mean systems is classical, going back to the Nash-certainty-equivalence framework of Huang, Caines, and Malhamé [82]. The reading of equilibrium multiplicity as spontaneous symmetry breaking, and of the monotone regime as the convex, positive-susceptibility regime, is folklore in the mathematical-physics literature on mean field games; we have kept it to a single physics note, and the graphon refinement – multiplicity at a spectral/centrality threshold – is taken up in chapters 20 and 24.

Chapter 12

The Master Equation

Up to this point the mean field game has presented itself as a *system*, and in two equivalent guises. The first is analytic: a backward Hamilton–Jacobi–Bellman equation for the representative agent's value $u = u(t, x)$, coupled to a forward Fokker–Planck equation for the population density $m = m(t, x)$, with data fixed at *opposite* ends of the interval $[0, T]$ — the value pinned by a terminal condition at $t = T$, the density by an initial condition at $t = 0$ (chapter 8). The second is probabilistic: a McKean–Vlasov forward–backward stochastic differential equation in path space, in which a forward state process and a backward adjoint are locked together by the self-consistency demand that the law of the state coincide with the measure the cost reacts to (chapter 9). We solved this system for existence (chapter 10) and, under Lasry–Lions monotonicity (A-Mon), for uniqueness (chapter 11: the precise monotonicity inequality is definition 11.1 and the uniqueness theorem is theorem 11.5, both imported by label below). Both guises share one defining feature. The unknown is a *pair* of objects glued along a forward–backward seam, with split boundary data, and to know the answer at any interior time one must solve the two halves *together*, propagating in opposite directions until they are made consistent. The population enters the agent's problem only as a curve that has already been computed.

This chapter performs a genuine change of viewpoint. It compresses the entire apparatus into a *single* scalar equation for one unknown function

$$U : [0, T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \longrightarrow \mathbb{R},$$

the *master equation.* Its value $U(t, x, m)$ is the equilibrium cost-to-go of an agent who finds herself at state x at time t when the *entire population distribution* is m — a third, free argument that the bare value function $u(t, x)$ never carried. What is *gained* is precisely that argument: one object that answers the cost-to-go question for *every* conceivable population, not merely the one realized along a particular equilibrium, and that is therefore Markovian in the true state of the game and differentiable in the population. The forward–backward seam dissolves, because the population is no longer something solved for in advance but an explicit input slot. What is *paid* is the arena. The new unknown is no longer a function of a finite-dimensional state but a function on the infinite-dimensional Wasserstein space $\mathcal{P}_2(\mathbb{R}^d)$, and to even write its equation we must learn to differentiate in the measure argument. The tools for that are the flat and Lions derivatives built in chapter 4 and recorded in appendix A.3; they are the literal price of admission, and we assemble them in section 12.2.

This is not mere economy of bookkeeping; it buys three things the system cannot, and the chapter is organized to deliver each as a promise kept. *First*, a Markovian, single-equation picture. Because U carries the population as an argument, it is a value function in the genuine game state (x, m), and its equation turns out to be a Hamilton–Jacobi equation posed on the space of measures whose method of characteristics is *exactly* the MFG system of chapter 8 — the two coupled PDEs are recovered as the ODEs one integrates along the characteristic flows of the single equation (sections 12.4 and 12.5). *Second*, the engine of finite-player convergence. The very differentiability that the system lacks — the derivative of the value in the population direction — is what supplies the smooth "mean field expansion" of the N-player Nash value, and through it the convergence of the N-player game to the mean field limit with quantitative, optimal-order rates (chapter 13). *Third*, and decisively, the only formulation that survives *common noise* — a shock felt simultaneously by every agent. Under common noise the population measure stops being a deterministic curve and becomes a genuine stochastic process; the forward–backward system, which presumed a predetermined population trajectory, loses its meaning, whereas the master equation merely acquires second-order derivatives in m and keeps its form (section 12.7). The first section makes the case for the first of these by exhibiting, sharply, what goes wrong without U; the rest of the chapter delivers the other two.

Two standing conventions govern everything below, and it is worth foregrounding them because the master equation lives at the meeting point of both directions of time. Value functions are solved *backward*, carrying terminal data at $t = T$; population densities evolve *forward*, carrying initial data at $t = 0$ (appendix A.1). The master function U is a value, so it inherits a *terminal* condition at $t = T$, namely the terminal coupling G; but its measure argument is a density, so along any equilibrium that argument is transported forward from $t = 0$. This double bookkeeping — backward in the value, forward in the population it is evaluated against — is the entire content of the chapter in miniature, and the reader should keep it in view.

We also fix, once, the dynamics. We work in the canonical setting of chapter 8, in which the controlled drift is the control itself and the diffusion is an additive constant,

$$\mathrm{d}X_t \;=\; \alpha_t \,\mathrm{d}t \;+\; \sigma \,\mathrm{d}W_t, \qquad X_0 \sim m_0, \tag{12.1}$$

with σ constant and nondegenerate (A-Nondeg). This is chosen deliberately as the clean vehicle: with the drift equal to the control and σ a constant, every term in the master equation has an unobstructed interpretation, and the new measure-derivative content is not hidden behind state-dependent coefficients. Nothing structural is lost by it — the general Lipschitz drift of the standing assumptions, a state- and measure-dependent diffusion, and a non-separable Hamiltonian all produce the *same* equation with the coefficients evaluated in the obvious places — and that general case is carried in full in the worked remark (remark 12.4), which is also the form chapter 13 imports. Treat the canonical setting as the lit stage and the worked remark as the proof that the lighting changes nothing.

The chapter is a map in three movements. We first *derive* the master equation without common noise and prove its exact equivalence with the system (section 12.2–section 12.6): the differentiation machinery, the derivation, the reading of the result as a Hamilton–Jacobi equation on $\mathcal{P}_2(\mathbb{R}^d)$ whose characteristics are the MFG system, and the existence/uniqueness question that its first-order-in-m structure forces. We then *turn on common noise* and record the modifications it forces (section 12.7), deferring the full graphon-and-common-noise theory to chapter 26. Finally we make everything *explicit* on the carried LQG thread (section 12.8): we compute its decoupling field, exhibit the master equation as a finite system of Riccati equations, and verify that it reproduces the fixed point of chapter 8 exactly. We begin, in the next section, with the question that justifies the whole undertaking: why one equation, and what precisely is wrong with the value function we already have.

12.1 Why a single equation? The non-Markovian obstruction

Recall the equilibrium value function of the classical MFG. Fix an initial population $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Solving the MFG system produces a forward flow of measures $(m_t)_{t\in[0,T]}$ and a value $u(t,x)$ satisfying

$$-\partial_t u - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 u\big) + H(x,\nabla u) = F(x,m_t), \qquad u(T,x) = G(x,m_T), \tag{12.2a}$$
$$\partial_t m - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 m\big) - \operatorname{div}\big(m\,\partial_p H(x,\nabla u)\big) = 0, \qquad m_0 \text{ given}, \tag{12.2b}$$

the system of chapter 8. Here $H(x,p)$ is the Hamiltonian, convex in p under (A-Conv), and the optimal feedback is $\alpha^*(x,p) = -\partial_p H(x,p)$ (appendix A.4); F and G are the running and terminal couplings to the population, monotone under (A-Mon).

The function $u(t,x)$ answers the question "what is my optimal cost-to-go from (t,x)?" — but read the small print. It answers it *only for the one population flow generated by the particular initial condition* m_0. Built into the very numbers $u(t,x)$ is the assumption that the crowd around the agent is, and always will be, the equilibrium slice m_t obtained by solving eq. (12.2b) forward from that fixed m_0. Change m_0 and the whole function u changes; u is indexed, silently, by an initial population it never displays.

This is a real defect, not a cosmetic one, and it is worth making concrete. Imagine the equilibrium crowd at time t is the configuration m_t, and ask the agent a counterfactual: "suppose that, when you arrive at x at time t, you instead find the population already massed elsewhere, in some $m \neq m_t$ — a denser cluster to your left, say, raising the congestion cost of moving there. What is your optimal cost-to-go now?" The bare value function $u(t,x)$ has *no answer*. It was computed against m_t and only m_t; it has no value attached to the contingency $m \neq m_t$, because that population never appears anywhere in its construction. The agent who genuinely faced such a crowd would re-optimize — her continuation cost, her optimal feedback, everything downstream depends on the population she actually sees — but u cannot express the dependence, having frozen it. In dynamic-programming language, $u(t,x)$ is therefore *not a value function in a Markov state.* The true state of the game is the pair (x,m) — one's own position *and* the population that prices one's choices — and a value function deserving the name must be a function of both. The bare u knows only the realized slice $m = m_t$; it is a value function on a one-dimensional curve drawn through an infinite-dimensional state space, not a value function on that state space. Restoring the missing argument is the entire task of the chapter, and the object that restores it is $U(t,x,m)$.

Heuristic (non-rigorous)

The right way to feel the gap is the difference between a *trajectory* and a *flow*. Take an ordinary differential equation $\dot{y} = v(y)$. Solving it from one fixed initial point y_0 produces a single curve $t \mapsto y(t)$ — one trajectory, valid for that one start. The *flow map* $(t, y_0) \mapsto y(t; y_0)$ is a different and larger object: it returns the solution started from *every* initial point at once. It is the flow map, not any single trajectory, that is the genuinely Markovian object, because it can answer the question "where will I be in time s?" from *wherever* I happen to find myself now — it composes with itself, $y(t+s; y_0) = y(t;\, y(s; y_0))$, which is exactly the semigroup property a Markov state must have. A single trajectory cannot compose with itself; it only knows the one orbit it traces.

The two formulations of the mean field game stand in exactly this relation. The MFG system eq. (12.2), solved from a fixed m_0, is the single *trajectory*: the one equilibrium flow through that one initial population, with $u(t,x)$ its value read off along the way. The master equation is the *flow map*. What we want is a function $U(t,x,m)$ that, *whatever* population m is presented at time t — realized or counterfactual — returns

$$U(t,x,m) \;=\; \text{the equilibrium cost-to-go of an agent at } x \text{ when the crowd is } m.$$

Here m is a free input slot, ranging over all of $\mathcal{P}_2(\mathbb{R}^d)$, precisely the slot the bare u lacked. The connection between the two is then a single, sharp identity: along the actual equilibrium flow (m_t) generated by the given m_0, the master function evaluated at the running population must reproduce the bare value,

$$u(t,x) \;=\; U(t,x,m_t), \qquad t \in [0,T]. \tag{12.3}$$

This is the flow-map-restricted-to-a-trajectory statement made literal: u is the slice of U taken along the one characteristic curve $m_\cdot$, just as a single ODE solution is the flow map evaluated along one orbit. The bare value function is recovered from the master function by evaluation; the master function is the strictly richer object that also values every off-equilibrium crowd. And here is the lever that turns this picture into an equation: because eq. (12.3) holds for *all* t, *differentiating this identity in* t must be consistent with the HJB equation that u already satisfies, and forcing that consistency — once we know how U responds to the motion of its measure argument — will produce a closed equation for U itself. That differentiation is the derivation of the master equation, carried out in section 12.3.

So far one might object that the flow map is a luxury: if only m_0 is ever realized, why value crowds that never occur? The objection collapses the moment one introduces *common noise*, and it is worth seeing why, because common noise turns the flow map from a convenience into a logical necessity. In the games of chapter 8 each agent carries her own independent Brownian motion, the idiosyncratic shocks average out across the continuum, and the population m_t evolves *deterministically* — it is a single curve one can compute in advance and then hold fixed while each agent optimizes against it. That is exactly what makes the bare $u(t,x)$ tenable: there is only ever one slice to value.

Common noise destroys that. Suppose, on top of her private shocks, every agent is buffeted by one *shared* Brownian motion W^0 — a market-wide shock, a common weather, an aggregate sentiment. This shock does not average out: it pushes the entire population the same way at the same instant, so the aggregate m_t is itself jostled randomly. The population ceases to be a deterministic curve and becomes a *random measure*, a stochastic process $t \mapsto m_t$ adapted to the common-noise filtration, taking a genuinely different value on each realization of W^0. Now "the realized slice" is not a single object that can be solved for ahead of time; *which* population the agent will face at time t is not known until the shared shock has played out, and the same agent at the same (t, x) may confront a whole distribution of possible crowds. A value function that hard-codes one trajectory m_t literally cannot be written down, because there is no one trajectory. What *can* be written down is a function $U(t, x, m)$ that has already valued *every* contingency m in advance; one then simply evaluates it at the population the shared shock actually produces. The per-crowd object is no longer an upgrade — it is the only thing that survives. We make this precise, and see common noise add the promised second-order m-derivatives to the master equation, in section 12.7.

Physics note

The master equation is the mean field game's *effective action* viewed as a generating functional. In a many-body field theory one integrates out microscopic degrees of freedom to obtain a functional $\Gamma[m]$ of the order-parameter configuration m, whose functional derivatives reproduce the physical correlation and response functions. Here $U(t, x, m)$ plays the role of Γ: its x-gradient $\nabla_x U$ is the equilibrium "momentum" (the decoupling field of section 12.5), and its measure-derivative $\frac{\delta U}{\delta m}(t, x, m, y)$ (with (t, x) frozen) is the linear response of one agent's value to an infinitesimal reweighting of the crowd. The analogy is structural and motivational only; unlike a thermal effective action, U encodes a *strategic* (Nash) fixed point with no variational principle behind it in the non-potential case, and nothing below relies on the picture.

We now have the program in hand: the object we want is the flow map $U(t, x, m)$, valuing every crowd; its tie to the equilibrium we already understand is the restriction identity eq. (12.3), $u(t, x) = U(t, x, m_t)$; and the route to a closed equation for U is, as flagged twice above, to *differentiate this identity in* t and demand consistency with the HJB equation. The one ingredient still missing is the meaning of that differentiation, for the right-hand side $U(t, x, m_t)$ depends on t not only explicitly, in its first slot, but *implicitly*, through the population m_t that the Fokker–Planck equation eq. (12.2b) transports in time. Differentiating it therefore demands a chain rule on the space of measures — a rule for how a smooth functional of the population changes as the population is carried along a Fokker–Planck flow. Building that chain rule, from the flat and Lions derivatives, is exactly the business of the next section.

12.2 Differentiation along a flow of measures

The previous section left us with one concrete obligation. The restriction identity $u(t, x) = U(t, x, m_t)$ ties the bare value to the master function along the equilibrium flow, and the plan announced there — twice — is to extract a closed equation for U by *differentiating this identity in* t and demanding consistency with the HJB equation that u already obeys. The left-hand side is a function of t through (t, x) alone, and we know how to differentiate it. The right-hand side is where the work is: t enters $U(t, x, m_t)$ *twice*, once explicitly in the first slot and once *implicitly* through the population m_t, which is being carried along in time by the Fokker–Planck equation. So before we can differentiate the identity we must answer a single self-contained question: how does a smooth functional of the population change as the population is transported by a Fokker–Planck flow? That question is a chain rule on the space of measures, and assembling it is the entire business of this section. Everything downstream in the chapter is this one lemma applied — with a particular choice of functional and a particular drift.

A chain rule needs a notion of derivative, and on $\mathcal{P}_2(\mathbb{R}^d)$ there are two of them, because a functional of a measure can be perturbed in two genuinely different ways. We recall both from chapter 4 and appendix A.3, but it pays to recover the intuition first, since the two derivatives play visibly different roles in the formula we are after.

The obstacle is that $\mathcal{P}_2(\mathbb{R}^d)$ is not a vector space: one cannot add two probability measures and stay normalized, so the naive difference quotient that would define a directional derivative has no home. The repair is to notice that while m itself is not a vector, the *difference* $m' - m$ of two probability measures *is* a perfectly good signed measure of total mass zero — an admissible *direction* of perturbation that preserves total mass. Differentiating Ψ along the straight segment $\epsilon \mapsto m + \epsilon(m' - m)$, which remains in $\mathcal{P}_2(\mathbb{R}^d)$ for small $\epsilon \geq 0$, is then a legitimate directional derivative, and representing that derivative as an integral against a single function $\frac{\delta\Psi}{\delta m}(m, \cdot)$ is the content of the *flat* (linear functional) derivative. Its one-line meaning: $\frac{\delta\Psi}{\delta m}(m, y)$ is the marginal cost, as registered by Ψ, of placing an extra infinitesimal speck of mass at the location y when the current configuration is m. It is a function of a *location* y, and it is defined only up to an additive constant, because shifting it by a constant integrates to zero against any mass-zero direction $m' - m$ and so cannot be detected by the perturbation.

The second derivative answers a different physical question. The flat derivative reports what happens when we *add* mass at y; but in a transport flow mass is never created, only *moved*. To see how Ψ responds when an existing speck of mass at y is nudged in a spatial direction, we differentiate the marginal-cost landscape $y \mapsto \frac{\delta\Psi}{\delta m}(m, y)$ in y. Its gradient is the *Lions derivative* $\partial_\mu \Psi(m)(y) = \nabla_y \frac{\delta\Psi}{\delta m}(m, y)$, a *vector* attached to each location y — the local "force," the direction in which displacing the mass sitting at y most increases Ψ. This is exactly the object built to pair against a velocity: if the mass at y moves with velocity $\mathbf{b}(y)$, the rate of change of Ψ contributed by that motion is the inner product $\partial_\mu \Psi(m)(y) \cdot \mathbf{b}(y)$, accumulated over the population. The flat derivative integrates against *added* mass; the Lions derivative contracts against *transport* velocities. That division of labor is the whole reason both objects appear in the chain rule below.

Definition 12.1 (Flat and Lions derivatives; recalled)**.** Let $\Psi : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$. Its *flat (linear functional) derivative* is the function $\frac{\delta\Psi}{\delta m}(m, \cdot) : \mathbb{R}^d \to \mathbb{R}$, defined up to an additive constant by

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\, \Psi(m + \epsilon(m' - m))\Big|_{\epsilon=0} \;=\; \int_{\mathbb{R}^d} \frac{\delta\Psi}{\delta m}(m, y)\,(m' - m)(\mathrm{d}y) \qquad \text{for all } m' \in \mathcal{P}_2(\mathbb{R}^d).$$

When $y \mapsto \frac{\delta\Psi}{\delta m}(m, y)$ is differentiable, the *Lions (L-)derivative* is its gradient,

$$\partial_\mu \Psi(m)(y) \;=\; \nabla_y \frac{\delta\Psi}{\delta m}(m, y) \;\in\; \mathbb{R}^d,$$

as fixed in appendix A.3. We say Ψ is *fully* $\mathcal{C}^2$ if these exist, are jointly continuous in (m, y), and $y \mapsto \partial_\mu \Psi(m)(y)$ is itself $\mathcal{C}^1$ with Jacobian $\partial_y \partial_\mu \Psi(m)(y) \in \mathbb{R}^{d\times d}$. We call this *fully* $\mathcal{C}^2$ because it involves *two spatial* differentiations of a *single* flat derivative; it is a weaker requirement than the genuine second *measure* derivative $\partial^2_\mu \Psi$ (the Lions derivative of $y \mapsto \partial_\mu \Psi(m)(y)$ in the measure argument, an element of $\mathbb{R}^{d\times d}$ indexed by a *second* integration variable), which does not enter the master equation until common noise (section 12.7). Keep the two apart: the notation watch after theorem 12.3 catalogues all three "second derivatives" that appear.

The "fully $\mathcal{C}^2$" notion deserves a careful word, because it is the precise regularity the no-common-noise master equation will demand, and the distinction it draws is load-bearing for the rest of the chapter. It asks for *two spatial* differentiations of a *single* flat derivative: first ∇_y to pass from $\frac{\delta \Psi}{\delta m}(m, \cdot)$ to the Lions field $\partial_\mu \Psi(m)(\cdot)$, then a second ∇_y to form the Jacobian $\partial_y \partial_\mu \Psi(m)(y)$, an ordinary $d \times d$ matrix at each location y. Both differentiations act on the *spatial* variable y; the measure argument m is differentiated only *once*, when the flat derivative is taken. Contrast this with the genuine second *measure* derivative $\partial^2_\mu \Psi$, obtained by differentiating the Lions field $y \mapsto \partial_\mu \Psi(m)(y)$ a second time *in the measure direction*. That operation introduces a *second* integration variable y' — it asks how the force at y responds to reweighting the crowd near y' — and produces an object $\partial^2_\mu \Psi(m)(y, y')$ indexed by two locations, a true Hessian on $\mathcal{P}_2(\mathbb{R}^d)$. "Fully $\mathcal{C}^2$" is strictly weaker: a Jacobian in one variable, not a Hessian in two. The reason the weaker notion suffices here, and the reason the distinction matters at all, is the same: in the chain rule below the only second-order content comes from the *idiosyncratic* diffusion $\sigma \, \mathrm{d}W_t$ that each agent carries privately, which spreads mass locally and therefore couples to the single-variable Jacobian $\partial_y \partial_\mu \Psi$. There is, in the no-common-noise world, no mechanism that correlates the random motion of mass at y with the random motion of mass at a distant y', so the two-point object $\partial^2_\mu \Psi$ never appears. Common noise supplies exactly that mechanism — one shared shock moves all of the mass together, correlating y and y' — and it is precisely the term $\frac{1}{2} \iint \partial^2_\mu U(t, x, m)(y, y') \, \mathrm{d}m(y) \, \mathrm{d}m(y')$ that it adds to the equation (section 12.7). The gap between "two spatial derivatives of one flat derivative" and "a genuine second measure derivative" is thus the exact gap between the equation without common noise and the equation with it.

We apply these derivatives to a functional of the form $\Psi(m) = U(t, x, m)$ with (t, x) frozen; the resulting objects carry (t, x) as inert parameters and a fresh integration variable $y \in \mathbb{R}^d$, written $\frac{\delta U}{\delta m}(t, x, m, y)$ and $\partial_\mu U(t, x, m)(y)$. With the two derivatives in hand we can state the chain rule. The next lemma is the Itô–Dynkin formula on $\mathcal{P}_2(\mathbb{R}^d)$ specialized to a *deterministic* Fokker–Planck flow — no randomness in m itself, only the transport-plus-diffusion of its mass — and it is the single engine on which the whole chapter runs.

Lemma 12.2 (Chain rule along a Fokker–Planck flow)**.** *Let $\Psi : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ be fully $\mathcal{C}^2$ in the sense of definition 12.1, with derivatives bounded and Lipschitz on bounded sets. Let $(m_t)_{t\in[0,T]}$ be a weak solution of the Fokker–Planck equation*

$$\partial_t m_t \;=\; \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top \, \mathrm{D}^2 \, m_t\big) \;-\; \mathrm{div}(m_t \, \mathbf{b}_t),$$

driven by a bounded measurable drift field $\mathbf{b}_t : \mathbb{R}^d \to \mathbb{R}^d$ with $\int_0^T \int |\mathbf{b}_t|^2 \, \mathrm{d}m_t \, \mathrm{d}t < \infty$. Then $t \mapsto \Psi(m_t)$

is absolutely continuous and

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\Psi(m_t) \;=\; \int_{\mathbb{R}^d} \partial_\mu \Psi(m_t)(y)\cdot \mathbf{b}_t(y)\, m_t(\mathrm{d}y) \;+\; \tfrac{1}{2}\int_{\mathbb{R}^d} \mathrm{tr}\Big(\sigma\sigma^\top\, \partial_y \partial_\mu \Psi(m_t)(y)\Big)\, m_t(\mathrm{d}y). \tag{12.4}$$

Proof sketch. The argument is three moves — differentiate, substitute, integrate by parts twice — and we spell out the algebra so that both terms of eq. (12.4) can be seen to emerge from their respective pieces of the Fokker–Planck operator. The full justification of every analytic step (absolute continuity, the differentiation under the integral, the boundary terms) is the Itô formula for flows of measures, theorem D.7 (appendix D.6), whose first-order half is stated already in chapter 4; we are specializing that theorem to the case of a deterministic flow and recording the resulting expectation against m_t.

Move 1: differentiate through the flat derivative. Because Ψ is fully $\mathcal{C}^2$ and $t \mapsto m_t$ is absolutely continuous, $t \mapsto \Psi(m_t)$ is absolutely continuous, and its time derivative is the first variation of Ψ contracted against the rate of change of its argument. The increment $m_{t+h} - m_t$ is, to first order in h, the mass-zero signed measure $h\,\partial_t m_t$, so the directional-derivative identity of definition 12.1 applies with the perturbation direction $\partial_t m_t$ in place of $m' - m$, giving

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\Psi(m_t) \;=\; \int_{\mathbb{R}^d} \frac{\delta\Psi}{\delta m}(m_t, y)\, \partial_t m_t(\mathrm{d}y).$$

This is the entire role of the flat derivative: it converts "how Ψ changes" into a *linear* functional of "how the measure changes," and the Fokker–Planck equation tells us exactly what the latter is.

Move 2: substitute the Fokker–Planck right-hand side. Replacing $\partial_t m_t$ by the two terms of the Fokker–Planck operator splits the integral cleanly into a transport piece and a diffusion piece,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\Psi(m_t) \;=\; \underbrace{\int \frac{\delta\Psi}{\delta m}(m_t, y)\Big[-\mathrm{div}(m_t\, \mathbf{b}_t)\Big](\mathrm{d}y)}_{\text{transport}} \;+\; \underbrace{\int \frac{\delta\Psi}{\delta m}(m_t, y)\, \tfrac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top\, \mathrm{D}^2_y\, m_t)(\mathrm{d}y)}_{\text{diffusion}},$$

each of which we now integrate by parts to move the derivatives off the (irregular) measure m_t and onto the (smooth) flat derivative.

Move 3a: the transport piece (one integration by parts). The integrand is a function times a divergence. Writing the integral against Lebesgue density and using the divergence theorem — the boundary term at infinity vanishing because $\frac{\delta\Psi}{\delta m}(m_t,\cdot)$ and its gradient are bounded while $m_t\,\mathbf{b}_t$ has the finite-energy decay guaranteed by $\int |\mathbf{b}_t|^2\, \mathrm{d}m_t < \infty$ — one derivative passes from the divergence onto the test function with a single sign change:

$$\int \frac{\delta\Psi}{\delta m}(m_t, y)\Big[-\mathrm{div}(m_t\, \mathbf{b}_t)\Big](\mathrm{d}y) \;=\; \int \nabla_y \frac{\delta\Psi}{\delta m}(m_t, y)\cdot \mathbf{b}_t(y)\, m_t(\mathrm{d}y).$$

The gradient that appears is, by definition, the Lions derivative, $\nabla_y \frac{\delta\Psi}{\delta m}(m_t,\cdot) = \partial_\mu \Psi(m_t)$, so the transport piece is exactly $\int \partial_\mu \Psi(m_t)(y)\cdot \mathbf{b}_t(y)\, m_t(\mathrm{d}y)$ — the velocity field $\mathbf{b}_t$ contracted against the Lions force, summed over the population, precisely the pairing the Lions derivative was constructed for.

Move 3b: the diffusion piece (two integrations by parts). Here $\mathrm{D}^2_y\, m_t$ carries two spatial derivatives on the measure; integrating by parts *twice* (again with vanishing boundary terms, now using the boundedness of the second spatial derivatives of $\frac{\delta\Psi}{\delta m}(m_t,\cdot))$ transfers both onto the flat derivative with no net sign change, and the constant matrix $\sigma\sigma^\top$ rides along under the trace:

$$\int \frac{\delta\Psi}{\delta m}(m_t,y)\,\tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top\,\mathrm{D}_y^2\,m_t\big)(\mathrm{d}y) \;=\; \tfrac{1}{2}\int \mathrm{tr}\Big(\sigma\sigma^\top\,\mathrm{D}_y^2\,\frac{\delta\Psi}{\delta m}(m_t,y)\Big)\,m_t(\mathrm{d}y).$$

The Hessian now appearing is the spatial Hessian of the flat derivative, which is precisely the Jacobian of the Lions field, $\mathrm{D}_y^2\,\frac{\delta\Psi}{\delta m}(m_t,\cdot) = \partial_y\partial_\mu\Psi(m_t)$ — the single-variable second derivative whose existence "fully $\mathcal{C}^2$" was designed to guarantee. (This is the only place the second spatial derivative is used, and it is why no genuine $\partial_\mu^2\Psi$ is needed: the constant $\sigma\sigma^\top$ spreads mass locally, at one point y at a time.) Adding the two pieces gives eq. (12.4).

The boundedness and Lipschitz hypotheses are what make every integral finite and legitimize both the differentiation under the integral sign in Move 1 and the discarding of boundary terms in Move 3; the case of unbounded drifts is recovered by truncating $\mathbf{b}_t$ and passing to the limit using the finite-energy bound, carried out in full in theorem D.7, of which eq. (12.4) is the deterministic-time, expectation-against-m_t specialization. The result is classical; see [41], Ch. 5, for attribution and the general Itô calculus on measure space. □

The two terms of eq. (12.4) now have transparent meaning, and it is worth pausing on each because both will reappear, character for character, in the master equation. The first, $\int \partial_\mu\Psi(m_t)(y)\cdot \mathbf{b}_t(y)\,m_t(\mathrm{d}y)$, is the *advective transport* of Ψ by the drift $\mathbf{b}_t$: the crowd flows along $\mathbf{b}_t$, and Ψ changes at the rate at which the Lions force pushes in the direction of that flow. It is first order — one gradient on Ψ, one velocity — exactly the structure of a transport (continuity-equation) term. The second, $\frac{1}{2}\int \mathrm{tr}(\sigma\sigma^\top\,\partial_y\partial_\mu\Psi(m_t)(y))\,m_t(\mathrm{d}y)$, is the *spreading* of mass by the idiosyncratic diffusion: it is the measure-space avatar of the Laplacian $\frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top\,\mathrm{D}^2)$, with the finite-dimensional Hessian replaced by the Lions-field Jacobian $\partial_y\partial_\mu\Psi$. It is second order, and as anticipated it needs only the single-variable Jacobian, never the two-point $\partial_\mu^2\Psi$ — the signature of the no-common-noise setting.

This is precisely the engine the derivation needs, and we can already read off how it will be fired. In the next section we will apply lemma 12.2 not to an arbitrary functional Ψ and an arbitrary drift $\mathbf{b}_t$, but to the specific functional $\Psi = U(t_0,x,\cdot)$ — the master function with its time and state slots frozen — transported along the *equilibrium* flow, whose Fokker–Planck drift is the optimal feedback $\mathbf{b}_t = -\partial_p H(\,\cdot\,,\nabla_x U)$ read off from eq. (12.2b). With that identification, the transport term becomes the crowd drifting under the common optimal control and the diffusion term becomes the crowd spreading under the idiosyncratic noise — the two new lines of the master equation. Differentiating the flow-consistency identity in t by exactly this lemma, and forcing the result to agree with the HJB equation, is the derivation we now carry out.

12.3 Derivation of the master equation

The previous section handed us the single engine the whole chapter runs on: the chain rule lemma 12.2, which says how a smooth functional of the population changes as that population is transported along a Fokker–Planck flow, with the equilibrium feedback $\mathbf{b}_t = -\partial_p H$ already identified as the drift to be plugged in. We now execute, to the letter, the plan announced twice in section 12.1: write down the master function as a flow map, record the flow-consistency identity $u(t,x) = U(t,x,m_t)$ that ties it to the bare value, *differentiate that identity in* t using exactly lemma 12.2, and force the result to agree with the HJB equation u already satisfies. What falls out is a single closed equation for U. This section is the technical heart of the chapter, and it is in large part bookkeeping — but bookkeeping in which every sign and every evaluation point must be tracked, because that is exactly where the structure of the master equation lives.

Begin with the construction of U. The heuristic asked for a function that values *every* crowd, not merely the one realized along a particular equilibrium; we manufacture such a function by solving the game from every conceivable starting configuration and recording the answer. Fix an initial datum $(t_0, m) \in [0,T] \times \mathcal{P}_2(\mathbb{R}^d)$ and pose the MFG system eq. (12.2) on the sub-interval $[t_0, T]$ with the population pinned to m at the initial time, $m_{t_0} = m$. Solving it produces a forward flow of measures and a backward value, and the master function is defined to be that value read off at the very instant its own problem begins.

For eq. (12.5) below to define a genuine *function* of (t_0, x, m) — one number, not a set of competing candidates — the sub-interval problem must have exactly one solution at each datum, and this is precisely where the two pillars of Part II become load-bearing. It is worth naming the distinct role of each. *Existence* of a solution to the sub-interval problem holds under the standing assumptions, by the fixed-point argument of chapter 10; without it U would be undefined at some data. *Uniqueness* — and this is the subtle one — must hold for *every* horizon $[t_0, T]$ and *every* initial measure m, not just for the full interval $[0,T]$ and the one physical m_0; it is supplied by Lasry–Lions monotonicity (A-Mon) through theorem 11.5. Without uniqueness the construction would be multivalued: two distinct equilibria issuing from the same (t_0, m) would assign two distinct "values" to the same crowd, and there would be no well-defined number $U(t_0, x, m)$ at all. We therefore assume (A-Mon) throughout this section, not as a technical convenience but as the exact hypothesis that renders the flow map single-valued. (That uniqueness is needed at every horizon, and not only at the full one, is the reason chapter 11 took the trouble to prove it for arbitrary terminal time: the master function reaches into every sub-interval at once.) Denote by $(m_s^{t_0,m})_{s \in [t_0,T]}$ the resulting forward flow and by $u^{t_0,m}(s,x)$ the resulting value, and define the candidate master function by evaluating the value at the initial time of its own problem:

$$U(t_0, x, m) \;:=\; u^{t_0,m}(t_0, x), \qquad (t_0, x, m) \in [0,T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d). \tag{12.5}$$

Taking $t_0 = T$ leaves no interval over which to evolve: the value at the terminal time is simply the terminal coupling, so $U(T, x, m) = G(x, m)$, and this will furnish the terminal condition of the equation we are about to derive.

The one property of U that powers the entire derivation is a *flow-consistency* (semigroup) identity, and it is worth seeing *why* it holds before we put it to work. Solve the game once from (t_0, m), run the population forward to some later time $t \geq t_0$, and freeze the picture there: the crowd is now $m_t^{t_0,m}$ and the agent's continuation value from (t,x) is $u^{t_0,m}(t,x)$. But the tail of an equilibrium on $[t_0, T]$ is itself an equilibrium on the shorter interval $[t, T]$ — nothing in the agent's remaining optimization, nor in the population's remaining evolution, remembers *how* the configuration $m_t^{t_0,m}$ was reached, only that it is the population now present. By the very uniqueness we just invoked, that tail *is* the unique solution of the game restarted from $(t, m_t^{t_0,m})$, and so its value at the restart instant is, by definition eq. (12.5), exactly $U(t, x, m_t^{t_0,m})$. Reading the value off at time t along the original flow must therefore agree with solving afresh from the configuration reached at t:

$$u^{t_0,m}(t,x) \;=\; U\big(t,x,m_t^{t_0,m}\big) \qquad \text{for all } t \in [t_0,T]. \tag{12.6}$$

This is the precise, fully decorated form of the heuristic identity "$u(t,x) = U(t,x,m_t)$" of eq. (12.3): the bare value is the master function evaluated along its own equilibrium flow, i.e. along the single characteristic curve $m_\cdot^{t_0,m}$ that the initial datum traces through the infinite-dimensional state space. Notice that every dependence on the initial datum (t_0, m) visible on the left-hand side is carried, on the right, entirely inside the measure slot $m_t^{t_0,m}$ — which is the whole payoff of having promoted the population to a free argument. Granting that U is fully $\mathcal{C}^2$ in m (definition 12.1) and $\mathcal{C}^{1,2}$ in (t,x) — a regularity we state here as a hypothesis and return to in section 12.6, where we discuss what verifying it would require — we now differentiate eq. (12.6) in t and substitute the result into the HJB equation. Both sides depend on t: the left through (t,x) alone, the right through t *twice* (explicitly in the first slot, implicitly through the transported measure), and reconciling the two is the whole of the derivation.

Theorem 12.3 (The master equation)**.** *Assume the standing assumptions* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv)*,* (A-Mon)*, and suppose the function U defined by eq.* (12.5) *is fully $\mathcal{C}^2$ in the measure argument (definition 12.1) and $\mathcal{C}^{1,2}$ in (t,x), with derivatives bounded and locally Lipschitz. Then U solves, on $[0,T) \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d)$, the* master equation

$$\begin{aligned} -\,\partial_t U(t,x,m) - \tfrac12 \operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}_x^2 U\big) \;+\; H\big(x, \nabla_x U\big)& \\ +\int_{\mathbb{R}^d} \partial_\mu U(t,x,m)(y)\cdot \partial_p H\big(y, \nabla_x U(t,y,m)\big)\, m(\mathrm{d}y)& \\ -\,\tfrac12 \int_{\mathbb{R}^d} \operatorname{tr}\Big(\sigma\sigma^\top\, \partial_y\partial_\mu U(t,x,m)(y)\Big)\, m(\mathrm{d}y) \;=\; F(x,m)&, \end{aligned} \tag{12.7}$$

with terminal condition

$$U(T,x,m) \;=\; G(x,m). \tag{12.8}$$

Conversely, if U is a classical solution of eq. (12.7)*–eq.* (12.8) *with the stated regularity, then for each (t_0, m) the pair*

$$u(t,x) := U\big(t,x,m_t\big), \qquad m_t \textit{ the solution of eq. (12.2b) with } \nabla u(t,x) = \nabla_x U(t,x,m_t),\; m_{t_0} = m,$$

solves the MFG system eq. (12.2) *on $[t_0, T]$.*

Proof. From the system to the equation. Fix (t_0, m) and write $m_t = m_t^{t_0,m}$ for the equilibrium flow it generates. By the Fokker–Planck half of the system eq. (12.2b), this flow is transported by the optimal feedback drift

$$\mathbf{b}_t(y) \;=\; -\partial_p H\big(y, \nabla u^{t_0,m}(t,y)\big),$$

the common velocity field under which the crowd moves once each agent plays her optimal control $\alpha^* = -\partial_p H$. This is exactly the kind of drift the chain rule lemma 12.2 was built to handle, and the equilibrium flow is exactly the kind of Fokker–Planck flow it takes as input; the derivation is the single act of pointing that lemma at the right functional.

Differentiate the flow-consistency identity eq. (12.6) in t and evaluate at $t = t_0$. The left-hand side $u^{t_0,m}(t,x)$ depends on t through its first slot only, so its derivative is the ordinary time-partial $\partial_t u^{t_0,m}(t_0,x)$. The right-hand side $U(t,x,m_t)$ depends on t in two ways at once, and the total derivative is the sum of the two contributions. The *explicit* dependence, through the first slot of U, contributes $\partial_t U(t_0,x,m)$. The *implicit* dependence, through the transported measure m_t, is precisely a derivative of the form $\frac{\mathrm{d}}{\mathrm{d}t}\Psi(m_t)$ with the frozen-(t_0,x) functional $\Psi(\cdot) = U(t_0,x,\cdot)$, and lemma 12.2 evaluates it with the drift $\mathbf{b}_{t_0}$. Adding the two,

$$\partial_t u^{t_0,m}(t_0,x) = \partial_t U(t_0,x,m) + \int \partial_\mu U(t_0,x,m)(y)\cdot \mathbf{b}_{t_0}(y)\, m(\mathrm{d}y) + \frac{1}{2}\int \mathrm{tr}\big(\sigma\sigma^\top \partial_y \partial_\mu U(t_0,x,m)(y)\big)\, m(\mathrm{d}y),$$

the first integral being the chain rule's transport term and the second its diffusion term, each evaluated at the initial measure $m_{t_0} = m$.

Three substitutions now collapse this into the master equation. First, evaluating the flow-consistency identity itself at $t = t_0$ gives $u^{t_0,m}(t_0,x) = U(t_0,x,m)$, and because this holds as an identity in x we may differentiate it in x to read off the spatial derivatives at the initial instant,

$$\nabla u^{t_0,m}(t_0,x) = \nabla_x U(t_0,x,m), \qquad \mathrm{D}^2\, u^{t_0,m}(t_0,x) = \mathrm{D}_x^2\, U(t_0,x,m).$$

(These hold only *at* $t = t_0$, where the measure slot is frozen at m; the differentiation in x does not touch the population argument, which is why no measure-derivative of U appears here.) Second, the feedback drift is rewritten in terms of U using the first of these, $\mathbf{b}_{t_0}(y) = -\partial_p H\big(y, \nabla u^{t_0,m}(t_0,y)\big) = -\partial_p H\big(y, \nabla_x U(t_0,y,m)\big)$ — note that the momentum slot is filled by $\nabla_x U$ evaluated at the *integration point* y, not at the agent's own x, since it is the crowd at y that moves with velocity $\mathbf{b}_{t_0}(y)$. Third, insert all of this into the HJB equation eq. (12.2a) written at $t = t_0$,

$$-\partial_t u^{t_0,m}(t_0,x) - \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top\, \mathrm{D}^2\, u^{t_0,m}(t_0,x)\big) + H\big(x, \nabla u^{t_0,m}(t_0,x)\big) = F(x,m).$$

Replace $-\partial_t u^{t_0,m}(t_0,x)$ by minus the three-term expression above (so that $-\partial_t U$ and the two integrals enter with their signs flipped), replace $\mathrm{D}^2\, u$ by $\mathrm{D}_x^2\, U$ and ∇u by $\nabla_x U$, and substitute the feedback drift. The result is exactly the master equation eq. (12.7) at the point (t_0,x,m). Since (t_0,m) was an arbitrary datum, the equation holds at every point of $[0,T) \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d)$; the terminal condition eq. (12.8) is the construction eq. (12.5) read at $t_0 = T$.

It is worth dwelling on the sign of the transport term, since this is the one place the bookkeeping could mislead. In the chain rule the implicit term entered the *right* side as $+\int \partial_\mu U \cdot \mathbf{b}_{t_0}\, m(\mathrm{d}y)$, i.e. as part of $\partial_t u$. Moving $\partial_t u$ to the left of the HJB flips its sign, turning it into $-\int \partial_\mu U \cdot \mathbf{b}_{t_0}\, m(\mathrm{d}y)$; and substituting $\mathbf{b}_{t_0} = -\partial_p H$ flips it once more, producing the $+\int \partial_\mu U \cdot \partial_p H\, m(\mathrm{d}y)$ that appears on the second line of eq. (12.7). The two sign reversals — one from transposing across the HJB equation, one from the explicit minus in the optimal feedback — compose to a plus. In short: the $+\,\partial_\mu U \cdot \partial_p H$ of the master equation *is* the $-\,\partial_\mu U \cdot \mathbf{b}$ of the chain rule eq. (12.4), carried to the left-hand side. The diffusion term carries a single sign flip (it has no feedback factor to flip a second time), which is why it appears as $-\frac{1}{2}\int \mathrm{tr}(\sigma\sigma^\top \partial_y \partial_\mu U)\, m(\mathrm{d}y)$ on the third line.

From the equation to the system. For the converse we run the same computation backward, and the only genuinely new point to establish is that the object we differentiate exists — that the flow m_t is well defined. Let U be a classical solution of eq. (12.7)–eq. (12.8) with the stated regularity, fix a datum (t_0, m), and consider the Fokker–Planck equation eq. (12.2b) closed by the feedback $-\partial_p H(x, \nabla_x U(t, x, m_t))$ with $m_{t_0} = m$. This is no longer an externally given flow but a *McKean–Vlasov* flow: the drift at (t, x) depends on the current law m_t through the third slot of U, so the equation is self-referential and its well-posedness needs an argument. It is supplied by the regularity of U: because U is fully $\mathcal{C}^2$ in m and $\mathcal{C}^{1,2}$ in (t, x) with derivatives bounded and locally Lipschitz, the map $(x, m) \mapsto -\partial_p H(x, \nabla_x U(t, x, m))$ is Lipschitz in the state x and in the measure m (for $\mathcal{W}_2$) uniformly in t — here the convexity (A-Conv) makes $\partial_p H$ well defined and (A-Lip) controls its growth. A Lipschitz McKean–Vlasov drift is exactly the hypothesis of theorem 7.9, which delivers a unique flow (m_t); the underlying pathwise SDE is well posed by theorem 5.10, and nondegeneracy (A-Nondeg) upgrades m_t to a classical solution of the parabolic Fokker–Planck equation via the Schauder estimates of chapter 8, so the flow has the smoothness the chain rule demands. With m_t in hand, set $u(t, x) := U(t, x, m_t)$. Differentiating in x along the flow gives $\nabla u = \nabla_x U$ and $\mathrm{D}^2 u = \mathrm{D}^2_x U$, so the feedback that defined the flow is precisely $-\partial_p H(x, \nabla u)$, which is to say (u, m) satisfies the Fokker–Planck equation eq. (12.2b) by construction. It remains to recover the HJB equation. Differentiate $u(t, x) = U(t, x, m_t)$ in t by the chain rule lemma 12.2, exactly as in the forward direction but now at a generic time t rather than only at the initial instant, and subtract the master equation eq. (12.7): the explicit $\partial_t U$, the transport integral, and the diffusion integral cancel term for term against their counterparts in the master equation, leaving precisely

$$-\partial_t u - \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) + H(x, \nabla u) = F(x, m_t),$$

the HJB equation eq. (12.2a). The terminal condition $u(T, \cdot) = U(T, \cdot, m_T) = G(\cdot, m_T)$ is the terminal coupling eq. (12.8). Thus (u, m) solves the MFG system eq. (12.2) on $[t_0, T]$, and the equivalence is complete in both directions. $\square$

This converse direction deserves a forward glance, because it is more than a tidy symmetry. What it shows is that a single classical solution U of the master equation *manufactures* an equilibrium of the game from *any* starting datum (t_0, m), by closing the Fokker–Planck equation with the feedback $-\partial_p H(x, \nabla_x U)$ and propagating. The resulting flow (m_t) is the characteristic curve through (t_0, m), and $u(t, x) = U(t, x, m_t)$ is the value read along it. That is the seed of the method of characteristics developed in section 12.5: the master equation is solved by these flows, and along each of them it degenerates into the coupled HJB/Fokker–Planck system. For now we have what the chapter's throughline most needed — the equation itself, and its exact equivalence with the system in both directions.

With the equation derived and proved equivalent, it pays to read its anatomy line by line, because the three lines of eq. (12.7) have three different physical jobs, and keeping them straight is what makes the equation legible. The first line — $-\partial_t U - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2_x U) + H(x, \nabla_x U)$ — is an *ordinary* second-order HJB operator in the finite-dimensional state x, with the measure m held frozen as an inert parameter: if one pretended the population could not move, this is the entire equation, the classical Bellman operator of a single agent facing a fixed crowd. The second line is the first correction for the fact that the crowd *does* move: it is a *transport* term, $\int \partial_\mu U(t, x, m)(y) \cdot \partial_p H(y, \nabla_x U(t, y, m))\, m(\mathrm{d}y)$, recording how the agent's value drifts as the whole population flows along the common optimal feedback $-\partial_p H$. The third line is the second correction: a *diffusion* term, $-\frac{1}{2}\int \operatorname{tr}(\sigma\sigma^\top \partial_y \partial_\mu U)\, m(\mathrm{d}y)$, recording how the value changes as each agent's idiosyncratic noise $\sigma\, \mathrm{d}W_t$ spreads the crowd locally.

First line: the agent optimizes against a frozen population. Second line: the population drifts. Third line: the population spreads. The second and third lines, both absent from the bare HJB eq. (12.2a) that knew only the realized slice, are exactly the price and the payoff of carrying m as a live argument.

The third line, the so-called *extra diffusion*, is the term most worth understanding, because it is precisely what separates the master equation from a naive "frozen-population HJB" in which one would simply hold m fixed and solve the first line alone. When does it actually contribute? The term is built from $\partial_y \partial_\mu U$, the spatial Jacobian of the Lions field, so it vanishes exactly when $\partial_\mu U(t, x, m)(y)$ is *constant in the integration variable* y. There is a clean dichotomy here that is easy to misstate, so we state it carefully. If the coupling sees the population only through finitely many moments, the term *collapses to a finite-dimensional expression* — a sum over those moments — but it does *not* vanish; it vanishes identically only in the strictly narrower case of *mean-field* (first-moment) coupling, where U depends on m only through its mean $\int y\, m(\mathrm{d}y)$, the flat derivative $\delta U/\delta m$ is then affine in y, its gradient $\partial_\mu U$ is *constant* in y, and so $\partial_y \partial_\mu U \equiv 0$. The contrast between "one moment" and "two moments" makes the point sharp. Take a second-moment coupling, $U = \phi(M_2(m))$ with $M_2(m) = \int y^2\, m(\mathrm{d}y)$. Its flat derivative is computed by perturbing m: since M_2 is linear in m, $\delta M_2/\delta m = y^2$, and the chain rule for the flat derivative gives $\delta U/\delta m = \phi'(M_2)\, y^2 = \partial_{M_2} U\, y^2$. Differentiating once in y gives the Lions field $\partial_\mu U = 2\, \partial_{M_2} U\, y$, which is *linear* in y, not constant; differentiating a second time gives the Jacobian $\partial_y \partial_\mu U = 2\, \partial_{M_2} U$, a nonzero constant. The extra-diffusion term is therefore

$$-\tfrac{1}{2} \int \operatorname{tr}\!\big(\sigma\sigma^\top\, 2\, \partial_{M_2} U\big)\, m(\mathrm{d}y) \;=\; -\, \operatorname{tr}(\sigma\sigma^\top)\, \partial_{M_2} U,$$

generically nonzero: a second-moment coupling feels the crowd's spreading. The first-moment coupling does not, because the Lions field there has no y-dependence to differentiate. So it is the mean-coupling structure — not the weaker property of depending on finitely many moments — that kills the term. We will see this concretely on the LQG thread in section 12.8, where the coupling is through the mean, $\partial_\mu U$ is constant in y, $\partial_y \partial_\mu U \equiv 0$, and the extra-diffusion line drops out, leaving the Riccati system that reproduces the chapter 8 fixed point.

Notation watch

The master equation is first order in the measure variable m but *second order* in the state x, and it carries yet another second-order object in y inside the crowd integral. Three "second derivatives" thus circulate through this chapter, they look alike on the page, and conflating them is the single most common error in reading the equation, so we catalogue them once and for all.

1. $\mathrm{D}_x^2 U(t, x, m)$ — the ordinary $d \times d$ Hessian in the agent's *own* state x, with t and m frozen. It sits on the first line of eq. (12.7) and is the agent's own idiosyncratic diffusion acting on her own value. It is a finite-dimensional second derivative; no measure differentiation is involved.

2. $\partial_y \partial_\mu U(t, x, m)(y)$ — the $d \times d$ Jacobian, in the integration variable y, of the Lions field $y \mapsto \partial_\mu U(t, x, m)(y)$. It is *two spatial* derivatives of a *single* flat derivative (first ∇_y to make the Lions field, then ∇_y again), and it sits on the third line, contracted against

$\sigma\sigma^\top$ and integrated against m. It is the crowd's spatial spreading. It is the "fully $\mathcal{C}^2$" content of definition 12.1 and, despite its name, differentiates the measure only *once*.

3. $\partial_\mu^2 U(t,x,m)(y,y')$ — the genuine second *measure* derivative, the Lions derivative of the Lions field taken *in the measure direction*, indexed by *two* distinct locations y, y'. This is a true Hessian on $\mathcal{P}_2(\mathbb{R}^d)$. It does *not* appear in eq. (12.7) at all; it enters only under common noise (section 12.7), where the shared shock correlates the motion of mass at y with the motion of mass at y'.

Only the third is a second derivative in $\mathcal{P}_2(\mathbb{R}^d)$ in the strong sense. The first acts on the agent, the second on the spatial spreading of the crowd, the third on two-point measure correlations; keep them apart and the equation reads cleanly.

Remark 12.4 (General coefficients and non-separable couplings: the form imported by chapter 13)**.** We displayed eq. (12.7) in the canonical *separated* setting of eq. (12.1) — control-as-drift, constant σ, additive couplings F, G, and a Hamiltonian $H(x,p)$ carrying no measure — because it is the cleanest vehicle for the structure: every term has an unobstructed reading and the new measure-derivative content is not hidden behind state-dependent coefficients. But it is essential, both for honesty and because chapter 13 needs the general form, to see that the canonical setting was a choice of lighting and not a structural assumption. Look back at the derivation: nowhere did it use separability of the Hamiltonian or constancy of σ. The only ingredients were the chain rule lemma 12.2 — which holds for *any* Lipschitz feedback drift and *any* (possibly x- and m-dependent) diffusion — and the substitution of the HJB equation, which likewise never assumed the coupling was additive. The construction therefore extends verbatim. For a general controlled drift $b(x,m,\alpha)$, a measure- and state-dependent diffusion $\sigma(x,m)$, and a *non-separable* Hamiltonian $H(x,p,m)$ — written in the canonical argument order of appendix A.4, convex in p under (A-Conv), with the running coupling now absorbed into H so that no separate F survives on the right — the master equation reads

$$\begin{aligned} &-\partial_t U - \tfrac{1}{2}\operatorname{tr}\big(\sigma(x,m)\sigma(x,m)^\top \mathrm{D}_x^2 U\big) \;+\; H(x,\nabla_x U, m) \\ &\qquad + \int_{\mathbb{R}^d} \partial_\mu U(t,x,m)(y)\cdot \partial_p H\big(y, \nabla_x U(t,y,m), m\big)\, m(\mathrm{d}y) \\ &\qquad - \tfrac{1}{2}\int_{\mathbb{R}^d} \operatorname{tr}\Big(\sigma(y,m)\sigma(y,m)^\top\, \partial_y\partial_\mu U(t,x,m)(y)\Big)\, m(\mathrm{d}y) \;=\; 0, \end{aligned} \tag{12.9}$$

with $U(T,x,m) = G(x,m)$. The correspondence with the separated equation is term by term, and worth spelling out because each modification lands in exactly the slot the derivation predicts. The first-line Hamiltonian gains a measure argument, $H(x,p) \rightsquigarrow H(x,p,m)$, since the running cost may now depend on the crowd directly rather than only through an additive F; the first-line diffusion becomes x- and m-dependent, $\sigma \rightsquigarrow \sigma(x,m)$, evaluated at the agent's own state. In the transport integral on the second line, the momentum derivative $\partial_p H$ likewise acquires the measure slot and is evaluated, as before, at the *integration point* y — it is $\partial_p H(y, \nabla_x U(t,y,m), m)$, the crowd at y moving with its own optimal velocity. In the diffusion integral on the third line the crucial point is the *evaluation point of the diffusion*: it is $\sigma(y,m)$, the diffusion of the mass sitting at y, not $\sigma(x,m)$, because the chain rule spreads each speck of crowd by its own local noise. This is exactly how lemma 12.2 produced the term, with $\sigma\sigma^\top$ riding under the trace at the integration variable, and it is the one place where a careless reader might wrongly write the agent's own $\sigma(x,m)$. Finally the right-hand side is now 0, because the coupling has been folded into H. The separated equation eq. (12.7) is recovered as the special case $H(x,p,m) = H(x,p)$, $\sigma(x,m) = \sigma$, with the coupling pulled back out as $+F(x,m)$ on the right — the two forms coincide precisely when the coupling is additively separable from the Hamiltonian. This general-coefficient form eq. (12.9) is the one chapter 13 evaluates, term by term, at the configuration $(t, x_i, \mu^{N,-i})$ of player i against the empirical measure of the others, in order to compare the N-player Nash system with the master equation and extract the convergence rate; chapter 13 imports it by theorem 12.8 and the present remark, so we have kept its coefficients fully general here precisely so that the later chapter need not re-derive them.

We have what the chapter most needed: the master equation eq. (12.7), derived by differentiating the flow-consistency identity, and proved exactly equivalent to the MFG system in both directions. It is time to stop computing and step back to read the *shape* of what we produced. Notice that the transport term $\int \partial_\mu U \cdot \partial_p H \, m(\mathrm{d}y)$ is the pairing of a measure-gradient $\partial_\mu U$ against a velocity field $-\partial_p H$ — precisely the structure the chain rule taught us to recognize, and precisely the structure of a Hamilton–Jacobi equation in which the "gradient" is a Wasserstein gradient and the "spatial" variable is the population itself. That reading, which turns the master equation into a Hamilton–Jacobi equation on $\mathcal{P}_2(\mathbb{R}^d)$ and prepares the method of characteristics, is the business of the next section; and the converse direction we just proved — a classical U manufacturing an equilibrium from each datum — is the seed of the characteristics theorem two sections later.

12.4 The master equation as a Hamilton–Jacobi equation on $\mathcal{P}_2(\mathbb{R}^d)$

The previous section closed by noticing the shape of the transport term — a measure-gradient $\partial_\mu U$ paired against the velocity field $-\partial_p H$. We now take that observation seriously, because it is not an accident of bookkeeping but the signature of what the master equation *is*. Read line by line, eq. (12.7) looks like an HJB operator in x decorated with two crowd integrals; read as a whole, with the agent's own coordinate x demoted to a passive label and the population m promoted to the variable that matters, it is a Hamilton–Jacobi equation in its own right — one whose spatial variable is the distribution and whose gradient is the Lions derivative. Making that reading explicit is the work of this section, and it pays off twice: it explains the *form* of the equation, and, more importantly, it predicts in advance the exact difficulties the well-posedness theory will have to overcome.

To see the shape, strip eq. (12.7) of its x-dependence for a moment and regard U as a function of (t,m) alone, gathering every term that involves the population — the frozen-agent HJB operator, the transport integral, and the diffusion integral — into a single effective *measure-Hamiltonian* $\mathcal{H}$. The equation then collapses to

$$-\partial_t U + \mathcal{H}(m, \partial_\mu U, \partial_y \partial_\mu U) = 0, \qquad U(T,\cdot) = G, \tag{12.10}$$

a backward, terminal-value Hamilton–Jacobi(–Bellman) equation in which the "state" is the measure m, the data are pinned at the terminal time $t = T$ by the coupling G (the value inherits its backward orientation from being a value, exactly the double bookkeeping flagged in the chapter preamble), and the "gradient" fed into $\mathcal{H}$ is the Lions derivative $\partial_\mu U$, with the spatial Jacobian $\partial_y\partial_\mu U$ supplying the second-order content. The arguments of $\mathcal{H}$ are therefore the population and its first and (one) second derivative, in exact analogy with a finite-dimensional Hamilton–Jacobi equation $-\partial_t U + \mathcal{H}(x, \nabla U, \mathrm{D}^2 U) = 0$, with m in the role of x and $\partial_\mu U$ in the role of ∇U.

This is not a metaphor, and it is worth pinning down why, because the precise sense in which $\partial_\mu U$ is a *gradient* is what licenses the whole analogy. Recall from chapter 4 the dynamic, Benamou–Brenier picture of the Wasserstein metric: the squared distance $\mathcal{W}_2(m_0, m_1)^2$ is the minimal kinetic energy $\int_0^1 \int |v_t|^2 \, \mathrm{d}m_t \, \mathrm{d}t$ of a velocity field v_t that transports m_0 to m_1 along the continuity equation $\partial_t m_t + \mathrm{div}(m_t v_t) = 0$. That construction equips $\mathcal{P}_2(\mathbb{R}^d)$ with a Riemannian-like structure in which the tangent space at m is a space of velocity fields, the cotangent objects are gradients of functions, and the pairing between a cotangent gradient $\partial_\mu \Psi(m)$ and a tangent velocity v is exactly

$$\langle \partial_\mu \Psi(m), \, v \rangle_m \; = \; \int_{\mathbb{R}^d} \partial_\mu \Psi(m)(y) \cdot v(y) \, m(\mathrm{d}y).$$

The transport term of the master equation is *precisely* this pairing, with the velocity field $v = -\partial_p H(\cdot, \nabla_x U)$ — the common optimal feedback under which the crowd flows. So when we call $\partial_\mu U$ the Wasserstein gradient of U we are not speaking loosely: it is the genuine Riemannian gradient of the dynamic metric of chapter 4, and the transport term is the genuine inner product of that gradient with the controlled velocity. The master equation is thereby revealed as the Hamilton–Jacobi equation of an optimal-control problem whose *state* is the entire distribution, whose controlled dynamics are the Fokker–Planck equation, and whose running and terminal costs are F and G — a control problem played on $\mathcal{P}_2(\mathbb{R}^d)$ rather than on $\mathbb{R}^d$.

With the shape identified, the same picture immediately predicts where the analysis will be hard, and this is the more valuable half of the reading. The difficulty is that the base space is $\mathcal{P}_2(\mathbb{R}^d)$, an infinite-dimensional, non-flat metric space that lacks almost every comfort the finite-dimensional theory relies on. There is no *local compactness*: a bounded sequence of measures need not have a $\mathcal{W}_2$-convergent subsequence, so the compactness arguments that underwrite existence of viscosity solutions in $\mathbb{R}^d$ have no direct analogue. There is no *canonical reference measure* — no Lebesgue measure on $\mathcal{P}_2(\mathbb{R}^d)$ — against which one could integrate, average, or define a heat kernel. And the space is genuinely *curved* in the Wasserstein sense, so even the linear structure that makes difference quotients meaningful is purchased only through the tangent-space construction recalled above. Each of these is an obstacle; but the decisive one, the hinge on which the entire well-posedness discussion turns, is the absence of a smoothing operator.

It is worth dwelling on this, because it is the single structural fact that controls everything downstream and is reused, in inverted form, under common noise. In a finite-dimensional HJB equation the second-order term $\frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top \mathrm{D}^2 U)$ is not cosmetic: it is what makes the equation *parabolic*. It is the term that smooths irregular data, that enforces a comparison principle, and that lets characteristics of the first-order part cross harmlessly — where they would otherwise collide and form a shock, the diffusion blurs the would-be singularity into a smooth profile, and a classical solution survives globally in time. Ask now for the analogous smoothing in the measure direction of eq. (12.7), the term that would diffuse U as a function of m, and there is none. The only second-order object in the equation is $\mathrm{D}_x^2 U$, the Hessian in the agent's *own* finite-dimensional coordinate x; it sits on the first line, it smooths U in x, and it is blind to the measure slot. The would-be "Laplacian on $\mathcal{P}_2(\mathbb{R}^d)$" — a non-negative second-order operator in m that would parabolically regularize U in the population direction — is simply absent. (It would require precisely the two-point measure Hessian $\partial_\mu^2 U(t,x,m)(y,y')$ catalogued in the notation watch after theorem 12.3, integrated against $m \otimes m$, and that object does not appear in the no-common-noise equation at all.) Heuristically: the idiosyncratic noise $\sigma\, \mathrm{d}W_t$ that each agent carries privately spreads each speck of mass *locally*, at one point y at a time, which is why it produced only the single-point Jacobian $\partial_y \partial_\mu U$ in the chain rule; it never correlates the random motion of distant specks, and so it cannot diffuse the *shape* of m as a whole. Local spreading of mass is not diffusion of the distribution.

The structural punchline follows at once, and it is the organizing fact of the next two sections. Because the genuine measure-dependence of eq. (12.7) enters only through the first derivative $\partial_\mu U$ and the single spatial Jacobian $\partial_y \partial_\mu U$ — never through a true second *measure* derivative — the master equation is, *in the m-direction*, a *first-order*, transport-type Hamilton–Jacobi equation. And a first-order Hamilton–Jacobi equation comes with a well-known liability: it is solved by characteristics, and characteristics generically *cross.* When two characteristic curves carrying different values of U collide, the solution becomes multivalued and the classical solution breaks down — the measure-space analogue of a shock. With no diffusion in m to smear such a collision away, nothing in the equation itself prevents it. One therefore expects a global classical solution *only* under a structural condition on the data strong enough to keep the characteristics from ever crossing in the first place. That condition is monotonicity — the very Lasry–Lions inequality (A-Mon) that bought uniqueness for the system in chapter 11 — now reappearing in its master-equation role as the hypothesis that organizes the characteristic flow so that distinct populations propagate to distinct values without collision. The cost of being first-order in m is exactly this: there is no free smoothing, so the regularity of U must be *purchased from structure* rather than supplied by the equation.

This also locates, in advance, the one place the missing Laplacian will be restored. When common noise is switched on in section 12.7, the shared shock moves all of the mass together and so correlates the motion of the speck at y with that of the speck at a distant y'; that correlation is precisely the two-point object $\partial_\mu^2 U(t,x,m)(y,y')$, and it enters the master equation through a new term $\frac{1}{2}\iint \partial_\mu^2 U(t,x,m)(y,y')\, \mathrm{d}m(y)\, \mathrm{d}m(y')$ (the shared-shock diffusion of section 12.7 riding inside it). That term *is* the Laplacian on $\mathcal{P}_2(\mathbb{R}^d)$ the no-common-noise equation lacked: a non-negative second-order operator in the measure direction that finally makes the master equation parabolic in m and restores, in the infinite-dimensional setting, the smoothing the finite-dimensional theory took for granted. The reader should carry forward the slogan that the present section establishes: *no common noise, first order in m, regularity bought from monotonicity; common noise, second order in m, regularity supplied by the equation.* The two regimes sit on opposite sides of exactly this divide.

Remark 12.5 (Potential games and a genuine variational principle)**.** There is one structurally favored case in which the control-problem reading of eq. (12.10) becomes literal rather than analogical, and it is worth isolating because it is the cleanest place the entire master-equation picture closes the loop with the chapter's opening physics note. Suppose the couplings derive from a *potential*: there are functionals $\mathcal{F}, \mathcal{G}$ on $\mathcal{P}_2(\mathbb{R}^d)$ with $F(x,m) = \frac{\delta \mathcal{F}}{\delta m}(m,x)$ and $G(x,m) = \frac{\delta \mathcal{G}}{\delta m}(m,x)$ (the variational/potential MFG of chapter 8). Then the equilibrium condition is no longer merely a Nash fixed point but the stationarity condition of a *single scalar functional* on the space of population trajectories, and the master equation is the Hamilton–Jacobi equation of that one control problem on $\mathcal{P}_2(\mathbb{R}^d)$. In this case U is itself a derivative, $U = \frac{\delta \mathcal{U}}{\delta m}(m, \cdot)$, of a value functional $\mathcal{U}$ on measures: the per-agent value is the first variation of an aggregate value, exactly as a one-particle quantity is a functional derivative of a free energy. This is the cleanest incarnation of the "effective action" picture sketched in the opening physics note of section 12.1: there U was offered as the analogue of an effective action $\Gamma[m]$ whose functional derivatives reproduce physical responses, with the caveat that the analogy carried no genuine variational principle in general. The potential case is precisely where that caveat lifts — there really is an action $\mathcal{U}$ whose stationary points are the equilibria, and U really is its derivative, so the field-theoretic picture is not just suggestive but exact. In the general, non-potential case no such scalar potential exists: the strategic (Nash) fixed point cannot be written as the critical point of any functional, and the master equation, *not* a variational problem, is the primary object. This is why the development that follows is built around the equation directly and does not lean on a minimization principle — it must work even when none is available.

We have read the equation's shape and, from that shape, anticipated its central difficulty: being first order in the measure, it must be solved along characteristics, and only monotonicity can keep those characteristics from crossing. Both halves of that sentence are now promissory notes the next two sections redeem. The phrase "solved along characteristics" is made literal first: the very next section exhibits the characteristic curves — they are the Fokker–Planck flows (m_t) already identified in the derivation — and shows that along them the single master equation degenerates into the coupled HJB/Fokker–Planck system, the unifying diagram that displays the three faces of the mean field game as one object. Only once those characteristics are in hand does the second phrase, "monotonicity prevents the characteristics from crossing," acquire a precise meaning; we therefore defer the well-posedness theory to section 12.6 and make the claim precise there, after exhibiting the characteristics themselves.

12.5 Method of characteristics: the unifying diagram

The previous section ended on a promissory note: read in the measure direction the master equation is first order, and a first-order Hamilton–Jacobi equation "is solved by characteristics." We now cash that promise. The base space $\mathcal{P}_2(\mathbb{R}^d)$ is identified, the candidate characteristic curves — the Fokker–Planck flows (m_t) — are already in hand from the derivation, and all that remains is to make the slogan literal: to exhibit the curves, write down the ODEs the PDE collapses to along them, and recognize those ODEs.

It pays to recall first what the method of characteristics *is*, in the finite-dimensional setting where it is elementary, because the measure-space version is the same idea transcribed onto a larger base space. Take a first-order Hamilton–Jacobi equation in $\mathbb{R}^d$, say $-\partial_t u + H(x, \nabla u) = 0$ with data at $t = T$. The method observes that one need not solve the PDE on all of space at once; instead one launches *characteristic curves* $t \mapsto x_t$ through the base space $\mathbb{R}^d$, chosen so that along each curve the partial differential equation degenerates into a closed system of *ordinary* differential equations for the position x_t, the value $u_t = u(t, x_t)$, and the momentum $p_t = \nabla u(t, x_t)$ — the Hamilton equations $\dot{x}_t = \partial_p H$, $\dot{p}_t = -\partial_x H$, with u_t recovered by integrating along the way. The PDE is then reconstructed by sweeping out the whole space with these curves: through each point passes one characteristic, and the value there is read off the ODE solution. The base space is where the curves live; the characteristic system is the bundle of ODEs they satisfy. That is the entire mechanism, and it is purely local to each curve.

The master equation is this same mechanism with the base space enlarged from $\mathbb{R}^d$ to the space of populations $\mathcal{P}_2(\mathbb{R}^d)$. A characteristic is now a curve $t \mapsto m_t$ *through the space of measures*, and the claim — which the derivation of section 12.3 has, in effect, already proved — is that the curves along which the master equation collapses to ODEs are exactly the *Fokker–Planck flows* (m_t) generated by the equilibrium feedback, and that the "ODEs" obtained along them are exactly the coupled HJB/Fokker–Planck system eq. (12.2) of chapter 8. (They are ODEs in the same generalized sense as before: a closed evolution along the characteristic, here a pair of PDEs in the agent's own coordinate x rather than scalar ODEs, because the base point m_t is itself infinite-dimensional.) Thus the single scalar equation for U, restricted to one characteristic, *is* the mean field game system; and this is the precise sense in which the three formulations of the mean field game — the master equation, the coupled HJB/Fokker–Planck system, and the McKean–Vlasov FBSDE of chapter 9 — are not three theories but one object viewed three ways. The theorem below states the equivalence in both of its remaining faces: the system (part 1, the converse of the derivation theorem) and the probabilistic FBSDE (part 2, where $\nabla_x U$ is revealed as the decoupling field).

Theorem 12.6 (Characteristics of the master equation)**.** *Let U be a classical solution of eq.* (12.7)*–eq.* (12.8) *with the regularity of theorem 12.3. Fix (t_0, m) and let $(m_t)_{t \geq t_0}$ be the Fokker–Planck flow generated by the feedback $\mathbf{b}_t(\cdot) = -\partial_p H(\cdot, \nabla_x U(t, \cdot, m_t))$, $m_{t_0} = m$. Then:*

1. *the value $u(t, x) := U(t, x, m_t)$ and the flow m_t solve the MFG system eq.* (12.2) *on $[t_0, T]$* (theorem 12.3)*;*

2. *equivalently, the McKean–Vlasov FBSDE of chapter 9 for the optimally controlled state X_t, its adjoint Y_t, and martingale integrand Z_t is solved by*

$$Y_t \;=\; \nabla_x U\big(t, X_t, m_t\big), \qquad Z_t \;=\; \mathrm{D}_x^2\, U\big(t, X_t, m_t\big)\, \sigma, \qquad m_t = \mathcal{L}(X_t), \tag{12.11}$$

so that $\nabla_x U$ is the decoupling field *of the FBSDE.*

Proof sketch. Part (1) is nothing new: it is exactly the converse direction of theorem 12.3, which showed that closing the Fokker–Planck equation eq. (12.2b) with the feedback $-\partial_p H(\cdot, \nabla_x U)$ and setting $u(t, x) = U(t, x, m_t)$ recovers the full MFG system. We restate it here only to display it as the *first* face of the characteristic picture — restricting U to the curve $m_\cdot$ produces the system — so that the second face, the probabilistic one, can be set beside it.

For part (2) we connect U to the McKean–Vlasov FBSDE through the dictionary already built in chapter 9, and the only work is a substitution. Recall the two ingredients from there. First, the stochastic maximum principle of chapter 9 represents the equilibrium of the game, for a *fixed* population flow (m_t), through a forward–backward pair: a forward state X_t driven by its optimal feedback control, and a backward adjoint (Y_t, Z_t) enforcing optimality, with Y_t the costate (the shadow price of position) and Z_t the integrand of its martingale part. Second — and this is the bridge to the PDE world — the nonlinear Feynman–Kac relation of chapter 5, applied to the HJB equation that u satisfies, identifies that probabilistic adjoint with spatial derivatives of the analytic value function. This is the FBSDE–PDE dictionary of proposition 9.12, namely eq. (9.16),

$$Y_t \;=\; \nabla u(t, X_t), \qquad Z_t \;=\; \sigma^\top \mathrm{D}^2 u(t, X_t),$$

which says: the costate is the gradient of the value at the random state, and the martingale integrand is its Hessian dressed by the diffusion. (Heuristically this is just Itô's formula applied to $t \mapsto \nabla u(t, X_t)$: the drift of the costate must match $-\partial_x H$ by the adjoint equation, and the diffusion coefficient of $\nabla u(t, X_t)$ is precisely $\mathrm{D}^2 u$ times σ, which is what Z_t collects.)

Now feed in the master function. Along the equilibrium flow generated by (t_0, m) the flow-consistency identity eq. (12.6) reads $u(t, x) = U(t, x, m_t)$, and differentiating it in x — which, as in the derivation, does not touch the frozen measure slot — gives $\nabla u(t, x) = \nabla_x U(t, x, m_t)$ and $\mathrm{D}^2 u(t, x) = \mathrm{D}^2_x U(t, x, m_t)$. Substituting these into the dictionary above, and using $m_t = \mathcal{L}(X_t)$ (the self-consistency that the state's law *is* the population it reacts to), turns the costate and integrand into evaluations of U at the running state and running population,

$$Y_t \;=\; \nabla_x U(t, X_t, m_t), \qquad Z_t \;=\; \sigma^\top \mathrm{D}^2_x U(t, X_t, m_t),$$

which is eq. (12.11) (modulo the transpose discussed below). That completes the identification; the substance is entirely in the two ch. 9 ingredients, and the master function contributes only the replacement of (t, x)-derivatives of u by the corresponding x-derivatives of U evaluated against the live population.

It remains to say what is *gained* by writing the adjoint this way, because this is the point of part (2). To call $\nabla_x U$ the *decoupling field* of the FBSDE is to make a structural claim, not just to record a formula. A forward–backward system is, a priori, genuinely coupled: the backward adjoint Y_t is defined by a terminal condition and an equation running from T down to t, so it appears to depend on the entire future of the path, while the forward state needs Y_t *now* to choose its drift. A *decoupling field* is a deterministic function $\phi(t, x)$ such that $Y_t = \phi(t, X_t)$ pathwise — the backward component is slaved to the present value of the forward component through a fixed function, with no dependence on the future beyond what X_t already carries. Its existence is exactly what breaks the forward–backward circularity: substituting $Y_t = \phi(t, X_t)$ into the forward drift closes the state equation into an ordinary (McKean–Vlasov) SDE one can integrate forward, after which the backward part is determined. Equation (12.11) exhibits such a function: $Y_t = \nabla_x U(t, X_t, m_t)$ is precisely of the form "deterministic function of the present state" (and present population), matching the definition definition 9.13. That this representation is not merely available but is what *well-poses* the system is the content of the two existence theorems of chapter 9: theorem 9.14, which produces the decoupling field and shows it is Lipschitz, and theorem 9.10, which uses that Lipschitz field to solve the FBSDE uniquely. The master equation thus does more than match the FBSDE — its x-gradient *is* the object whose existence makes the FBSDE solvable, which is why the master picture and the probabilistic picture stand or fall together. For attribution and the general theory of decoupling fields see [41], Ch. 4–5.

We record one notational reconciliation, because the two ch. 9 forms and the form natural here differ by a transpose. Equation (9.16) writes the martingale integrand as $Z_t = \sigma^\top \mathrm{D}_x^2 U(t, X_t, m_t)$, whereas eq. (12.11) writes $Z_t = \mathrm{D}_x^2 U(t, X_t, m_t)\, \sigma$. The difference is exactly the order of the constant matrix σ and the Hessian. In the canonical setting of this chapter σ is a constant symmetric matrix and $\mathrm{D}_x^2 U$ is symmetric (mixed partials of U in x commute), so $\sigma^\top \mathrm{D}_x^2 U = \sigma\, \mathrm{D}_x^2 U = \mathrm{D}_x^2 U\, \sigma$ and the two expressions literally coincide; the reconciliation is a triviality here and matters only when one later allows a non-symmetric or state-dependent σ. We keep eq. (12.11) in the form $\mathrm{D}_x^2 U\, \sigma$, since that is the order produced directly by Itô's formula applied to $Y_t = \nabla_x U(t, X_t, m_t)$: the diffusion of X_t is $\sigma\, \mathrm{d}W_t$, and differentiating the vector field $\nabla_x U$ along it contracts $\mathrm{D}_x^2 U$ against σ on the right. □

The whole correspondence is summarized by the following diagram, the central object of Part II. It has four corners — two analytic on the left, two probabilistic on the right; two "per-crowd" on top, two "single-equilibrium" on the bottom — and two arrows in each direction, and the entire content of this chapter is the statement that the diagram *commutes*: every path from the master equation in the top-left to the FBSDE in the bottom-right yields the same object.

$$
\boxed{
\begin{tikzcd}
\begin{array}{c}\textbf{Master equation}\\ U(t,x,m)\end{array} \arrow[r, "\text{restrict to } m_t"] \arrow[d, "\nabla_x"] & \begin{array}{c}\textbf{MFG system}\\ (u,m) = (U(\cdot,\cdot,m_\cdot),\, m_\cdot)\end{array} \arrow[d, "\text{Feynman–Kac}"] \\
\begin{array}{c}\textbf{Decoupling field}\\ \nabla_x U(t,x,m)\end{array} \arrow[r, "\text{evaluate along } m_t,\ X_t\sim m_t"] & \begin{array}{c}\textbf{McKean–Vlasov FBSDE}\\ (X_t, Y_t, Z_t)\end{array}
\end{tikzcd}
}
\tag{12.12}
$$

Read *horizontally*, the two rows are the two ways of restricting a per-crowd object to a single realized equilibrium. Along the *top* row, the arrow "restrict to m_t" is the flow-consistency identity eq. (12.6) itself: one takes the master function $U(t, x, m)$, which knows the value of *every* crowd, and evaluates its measure slot along the one characteristic flow $m_\cdot$ launched from a given m_0; the result is the MFG system's value $u(t, x) = U(t, x, m_t)$, paired with the flow $m_\cdot$ that produced it. The free argument m collapses to the realized trajectory, and the flow map becomes a single orbit — exactly the "trajectory inside the flow map" picture of section 12.1. Along the *bottom* row, the arrow "evaluate along m_t, $X_t \sim m_t$" is the same restriction performed on the decoupling field: one takes the deterministic field $\nabla_x U(t, x, m)$ and reads it along the random state X_t and its law $m_t = \mathcal{L}(X_t)$, producing the FBSDE adjoint $Y_t = \nabla_x U(t, X_t, m_t)$ of eq. (12.11). Each row says the same thing at its own level: the left object is the per-crowd master version, the right object is its restriction to one equilibrium.

Read *vertically*, the two columns are the two passages from the analytic world to the probabilistic one. The *left* arrow, labeled ∇_x, is the operation of differentiating the master function in the agent's own coordinate: it sends the value U to the field $\nabla_x U$, i.e. it turns the master *equation* into the *decoupling field* — the gradient of a value is a momentum, the costate the agent acts on. The *right* arrow, labeled Feynman–Kac, is the same passage carried out on the restricted objects: the nonlinear Feynman–Kac dictionary of chapter 9 (built on chapter 5) sends the analytic MFG value (u, m) to its probabilistic representation (X_t, Y_t, Z_t). Commutativity of the square is then precisely eq. (12.11): differentiating first and then restricting (go down, then right) gives $\nabla_x U$ evaluated along m_t, X_t; restricting first and then applying Feynman–Kac (go right, then down) gives $\nabla u(t, X_t) = \nabla_x U(t, X_t, m_t)$; the two agree. Master equation, system, and FBSDE are the four corners of one commuting diagram, and there is no "primary" formulation among them at the level of a single fixed m_0 — each is recovered from any other by one of these four arrows.

And yet there *is* a reason to prefer the top-left corner, and the diagram makes it visible even though the system in the right column already answers the equilibrium question for a single m_0. The reason is the one structural asymmetry the diagram hides: only the left-column objects, U and $\nabla_x U$, are functions of the population m, and therefore only they can be *differentiated* in m. The MFG value $u(t, x)$ and the FBSDE triple in the right column have already had the measure slot frozen to the realized m_t; the dependence on the crowd has been evaluated away, and with it any possibility of asking how the value would change were the crowd slightly different. That measure-derivative is not an idle refinement — it is the engine of the two payoffs the rest of Part II is built to collect. *First*, finite-player convergence with the optimal rate. The N-player Nash value differs from the mean field value because the empirical population of N agents is a perturbation, of size $O(1/\sqrt{N})$ in $\mathcal{W}_2$, of the limiting m; expanding the N-player value around the mean field one requires a *smooth* dependence of the value on the population to Taylor-expand in, and that smoothness is exactly the Lions differentiability of U. The first variation $\partial_\mu U$ supplies the leading correction and the second variation controls the remainder, yielding convergence of the N-player game to the mean field limit at the optimal rate $O(1/N)$ — the "mean field expansion" that the bare system, having no m-argument to expand in, simply cannot write down (chapter 13). *Second*, stability of the equilibrium under perturbations of the initial population: if m_0 is nudged, the entire equilibrium flow and value move, and the linear response of the value to such a nudge is, by definition, the measure-derivative $\partial_\mu U$. The master function thus carries its own sensitivity analysis, where the system carries none. In both cases the lever is the same single fact — U is differentiable in m, the system is not — and it is the reason the per-crowd object earns its more demanding arena.

Everything in this section, however, has been stated *conditionally*. The characteristic equivalence of theorem 12.6, the decoupling identity eq. (12.11), the commuting diagram, and the differentiability that powers the two payoffs all presume that a classical solution U with the regularity of theorem 12.3 actually exists. We have shown what such a U *does* — it manufactures the system and the FBSDE from any datum, and its x-gradient is the decoupling field — but not that it *exists*, nor that it is unique. That is the one debt the chapter still owes, and the structure read off in section 12.4 has already warned us it will not be cheap: being first order in the measure, the master equation is solved along the very characteristics we have now exhibited, and first-order characteristics generically cross. Where two characteristic flows carrying different values collide, the diagram ceases to commute single-valuedly: the same crowd is assigned two values, U becomes multivalued, and the classical solution — with it, the whole construction of this section — breaks down. That collision is exactly the multiplicity of MFG equilibria, and the structural condition that forbids it is the monotonicity (A-Mon) we have invoked all along. The next section confronts this head on: it asks whether a regular U exists and is unique, and shows that monotonicity is precisely what keeps the characteristics from crossing, purchasing the regularity that the absence of measure-diffusion denies the equation for free. The commuting diagram, meanwhile, is not retired once well-posedness is settled; it is the object chapter 13 exploits, where the differentiability of U along the top-left corner becomes the mean field expansion of the N-player game.

12.6 Well-posedness in the monotone regime

Section 12.5 left the chapter with a single unpaid debt, and named the warning that comes with it. Everything established there about the master function — the equivalence with the system and the FBSDE (theorem 12.3), the equilibrium decoupling field (eq. (12.11)), the commuting diagram, and the differentiability in m that chapter 13 will exploit — was proved *conditionally*, granting that a classical solution U with the regularity demanded in theorem 12.3 actually exists. We have shown, at length, what such a U *does*; we have not shown that it *exists*, nor that it is *unique*. This section pays that debt, and the warning is that it cannot be paid for free.

The warning is structural, and it is worth stating in its honest, unsoftened form because it is what the whole section is organized against. Being first order in the measure argument, the master equation eq. (12.7) carries no diffusion in m — no second-order $\partial_\mu^2 U$ term to smooth the dependence on the population — and a first-order Hamilton–Jacobi equation is solved *along* its characteristics, which are exactly the MFG flows exhibited in section 12.5. But the characteristics of a first-order Hamilton–Jacobi equation generically *cross* in finite time. This is the same elementary mechanism by which a smooth scalar conservation law develops a shock: two characteristic curves carrying incompatible data run into the same point, and the would-be classical solution becomes multivalued there. Translate that collision back through the unifying diagram and it acquires a precise game-theoretic name. Two characteristics arriving at a common configuration m at a common time t means two distinct equilibrium flows — issuing from *different* initial data — both passing through the *same* population m and assigning it *different* continuation values $U(t, \cdot, m)$. That is exactly the multiplicity of MFG equilibria, the same m_0 supporting more than one equilibrium, studied in chapters 11 and 14; in the LQG thread it surfaces as a Riccati blow-up (chapter 14). Crossing of characteristics and non-uniqueness of equilibria are therefore not analogues but the *same event* seen from the two sides of the diagram, and either one destroys the single-valued U that the construction eq. (12.5) requires: where two values collide, $U(t, \cdot, m)$ has no well-defined number to report.

The substantive analytic question is thus not merely whether U is smooth but whether the characteristics can be kept from ever crossing, for *every* horizon and *every* initial population the construction reaches into. The structural condition that forbids the crossing is Lasry–Lions monotonicity (A-Mon), and one can already see, before any estimate, why monotonicity is the right hypothesis: it is precisely the inequality under which chapter 11 proved equilibria unique, and uniqueness of equilibria *is* non-crossing of characteristics. We therefore proceed by two routes, and we are honest at the outset about which parts of each are reproduced here and which are imported. The first route, through monotonicity, holds for *all* horizons but rests at one step on an all-horizon regularity estimate we state and attribute precisely rather than prove (theorem 12.8, remark 12.9). The second route abandons monotonicity entirely and proves *everything* in book, at the cost of a short horizon (proposition 12.10); it is the load-bearing self-contained result of the section, and we give its contraction in full. We begin with the hypotheses both routes draw on.

Assumption 12.7 (Standing hypotheses for well-posedness)**.** In addition to (A-Lip), (A-Growth), (A-Nondeg), and (A-Mon), assume the strict form of (A-Conv) — the Hamiltonian H is *strictly* convex in p (uniformly in x), as required by the uniqueness theorem theorem 11.5 that we import in Step 1 below — and that H and the couplings F, G are sufficiently smooth in all arguments, with $F(\cdot, m), G(\cdot, m)$ and their derivatives bounded uniformly in m. The monotonicity (A-Mon) is the Lasry–Lions condition of definition 11.1: recall (eq. (11.3)) that F is monotone when, for all $m, m' \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} \left(F(x,m) - F(x,m')\right)(m - m')(\mathrm{d}x) \;\geq\; 0, \tag{12.13}$$

and likewise for G; we reproduce the inequality only for the reader's convenience, the object itself being definition 11.1. (A-Nondeg) provides the parabolic regularization in x.

Each clause of assumption 12.7 earns its place against a specific step of the proof, and it is worth pairing them now so that nothing in the assumption looks like decoration. The *strict* convexity of H in p is the hypothesis of the all-horizon uniqueness theorem theorem 11.5: the Lasry–Lions argument turns the monotonicity inequality eq. (12.13) into a *strict* sign only because strict convexity makes the Hamiltonian's contribution to the same pairing strictly definite, and without it two equilibria could differ on a set the monotonicity inequality cannot see. That is exactly Step 1 below, where uniqueness of the system at every horizon is what makes the construction eq. (12.5) single-valued; so strict (A-Conv) is the precise hypothesis that turns the candidate U from a multivalued recipe into a function. The *uniform-in-m* smoothness and boundedness of F, G and their derivatives is what lets the estimates of Steps 2 and 3 be uniform in the population argument — the master function must be controlled across *all* of $\mathcal{P}_2(\mathbb{R}^d)$ at once, not measure by measure, since its derivatives in m are the whole point. Finally, nondegeneracy (A-Nondeg) supplies the parabolic regularization in the spatial variable: it is what upgrades the merely Lipschitz objects produced by the fixed-point and linearization arguments to genuine classical x-derivatives $\nabla_x U$, $\mathrm{D}_x^2 U$, and it is also the smoothing that, in Step 2, absorbs the cross terms left over after the monotone coupling has been extracted with its favorable sign.

Theorem 12.8 (Well-posedness of the master equation, monotone regime)**.** *Under assumption 12.7 there exists a unique function $U : [0,T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ that is $\mathcal{C}^{1,2}$ in (t,x), fully $\mathcal{C}^2$ in m in the sense of definition 12.1, and that solves the master equation eq. (12.7) with terminal condition eq. (12.8) classically on $[0,T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d)$. Its derivatives are controlled uniformly: $\nabla_x U$ and $\partial_\mu U$ are bounded and Lipschitz uniformly in (t,x,m), and U moreover admits a bounded, Lipschitz* second-order *Lions derivative $\partial_\mu^2 U(t,x,m)(y,y')$ and a bounded, Lipschitz mixed derivative $\nabla_x \partial_\mu U(t,x,m)(y)$, both uniform in (t,x,m,y,y'). (These second-order objects do not enter the no-common-noise master equation eq. (12.7), but they exist as regularity of U and are exported for the finite-player expansion of chapter 13 and for the common-noise equation of section 12.7.) Moreover U is given by the construction eq. (12.5), and for every (t_0, m) the MFG system eq. (12.2) has a unique solution, whose value is the corresponding slice of U.*

Proof sketch and where the full argument lives. There are two routes; both pass through monotonicity. We describe the *characteristics* route, which is the most illuminating and the one the rest of the chapter has set up: build U from the system by eq. (12.5), show the construction is single-valued and smooth, and read back the equation. The argument is four steps — make the recipe a function, control its first m-derivative, control its second, then verify and close uniqueness — and we are explicit at the end about which step is reproduced here and which is imported.

Step 1 (global solvability of the system, so that U is a function at all). The construction eq. (12.5) reads $U(t_0, x, m)$ off the equilibrium of the game restarted from (t_0, m); for this to assign *one* number we need the restarted game to have exactly one equilibrium, at *every* restart datum (t_0, m) and over *every* horizon $[t_0, T]$. Existence holds under the standing assumptions by the fixed-point argument of chapter 10. Uniqueness is the delicate half, and it is where monotonicity does its work: by theorem 11.5, under (A-Mon) and the strict convexity of assumption 12.7 the MFG system eq. (12.2) has a unique classical solution on $[t_0, T]$, and crucially this holds for *all* horizons, not merely the physical one. The mechanism is exactly the no-crossing picture of the opening: the Lasry–Lions identity forces the a priori estimates on (u, m) to remain bounded *uniformly in the length of the interval* — monotonicity drains the energy that would otherwise let two equilibria separate as the horizon grows — so characteristics that start apart never collide and characteristics that arrive together must have coincided all along. (This is why chapter 11 took the trouble to prove uniqueness at arbitrary terminal time: the master function reaches into every sub-interval at once, and a uniqueness theorem for the single interval $[0, T]$ would not suffice.) With uniqueness in hand, eq. (12.5) defines U pointwise and single-valuedly, and strict (A-Conv) is seen to be load-bearing precisely here: it is the hypothesis of theorem 11.5 without which the recipe would be multivalued and there would be no function U to differentiate.

Step 2 (Lipschitz estimate in the measure). Next one shows U is Lipschitz in m for the $\mathcal{W}_2$ distance, uniformly in (t, x) — the bound that will make the first measure derivative $\partial_\mu U$ exist and stay bounded. Fix two nearby initial populations m, m', run the two equilibria (u, m_t) and (u', m'_t) they generate, and test how far apart their values can drift. The driver is again the monotonicity inequality eq. (12.13), deployed exactly as in the uniqueness proof of chapter 11: form the time derivative of the pairing $\int (u - u')\, \mathrm{d}(m_t - m'_t)$, substitute the two HJB equations for $\partial_t(u - u')$ and the two Fokker–Planck equations for $\partial_t(m_t - m'_t)$, and integrate by parts. The Hamiltonian terms collect, by strict convexity, into a nonnegative quadratic in $\nabla(u - u')$ weighted by $m_t + m'_t$; the coupling terms collect into $\int (F(\cdot, m_t) - F(\cdot, m'_t))\, \mathrm{d}(m_t - m'_t)$, which is ≥ 0 by eq. (12.13); and what remains — the cross terms generated by integrating the second-order operator by parts — is absorbed by the parabolic dissipation that (A-Nondeg) supplies, since $\sigma\sigma^\top$ is uniformly positive. The favorable signs on the two principal terms turn the pairing into a Grönwall inequality controlling $\sup_t \mathcal{W}_2(m_t, m'_t)$ and $\sup_t \|u - u'\|$ by $\mathcal{W}_2(m, m')$, uniformly in the horizon. Reading the bound at $t = t_0$ and recalling $U(t_0, x, m) = u(t_0, x)$ gives the $\mathcal{W}_2$-Lipschitz estimate on U, hence the boundedness of $\partial_\mu U$.

Step 3 (differentiability in the measure, to second order). Lipschitz continuity gives a derivative only almost everywhere; to obtain a genuine, continuous $\partial_\mu U$ one differentiates the construction itself. Perturb the initial population infinitesimally in an admissible mass-zero direction, $m \to m + \epsilon(m' - m)$, and differentiate the MFG system eq. (12.2) in ϵ at $\epsilon = 0$. Because the unperturbed (u, m_t) is fixed, the result is a *linear* forward–backward system — the *linearized MFG system* — for the pair of first variations $(\dot{u}, \dot{m}_t)$, with the linearized couplings $\frac{\delta F}{\delta m}(m_t, \cdot)$, $\frac{\delta G}{\delta m}(m_T, \cdot)$ as sources. Its unique solvability is once more a consequence of monotonicity, now in its linearized (Hessian) form: the same Lasry–

Lions pairing applied to the linear system shows the homogeneous problem has only the zero solution, so the inhomogeneous one is uniquely solvable. The solution map $(m' - m) \mapsto \dot{u}(t_0, \cdot)$ is linear and bounded, and that is by definition the flat derivative $\frac{\delta U}{\delta m}(t_0, \cdot, m, \cdot)$; its spatial gradient is $\partial_\mu U$, and the Step-2 estimate makes it uniformly bounded and continuous in (t, x, m, y). Differentiating the linearized system *once more* — a second perturbation, in an independent direction $m'' - m$ — produces a *second* linear system for the second variations, solvable by the same monotonicity, and its solution map is the second-order Lions derivative $\partial_\mu^2 U(t, x, m)(y, y')$ together with the mixed derivative $\nabla_x \partial_\mu U(t, x, m)(y)$. These inherit uniform boundedness and Lipschitz continuity from the same estimates. They do not appear in the no-common-noise master equation eq. (12.7), but they are genuine regularity of U, and they are exactly the objects exported to chapter 13 for the finite-player expansion and to section 12.7 for the common-noise equation. Throughout, the parabolic smoothing of (A-Nondeg) upgrades the variations from weak to classical x-derivatives, so $\nabla_x U$ and $\mathrm{D}_x^2 U$ are honest pointwise objects.

Step 4 (verification and uniqueness of the master solution). With U constructed and shown regular, theorem 12.3 applies verbatim: its forward direction shows this U solves the master equation eq. (12.7) with terminal data eq. (12.8). Uniqueness of the classical solution is then immediate from the characteristic representation theorem 12.6. Any classical solution of eq. (12.7) must, by the converse direction of theorem 12.3, agree along each characteristic with the value of the corresponding system equilibrium, which Step 1 showed is unique; and the characteristics issuing from the data $(t_0, \cdot)$, as (t_0, m) ranges over $[0, T] \times \mathcal{P}_2(\mathbb{R}^d)$, *fill* the domain — every point (t, x, m) lies on the characteristic restarted from its own (t, m). A classical solution pinned to the unique system value on a set that fills the space is unique. This closes the construction.

Steps 1 and 4 are fully in-book: existence and all-horizon uniqueness of the system are chapters 10 and 11, and the verification and uniqueness of the master solution from the characteristics are theorems 12.3 and 12.6, both proved here in full. What we do *not* reproduce is the quantitative core of Steps 2 and 3 — the precise function spaces and the linearized-system $\mathcal{W}_2$-Lipschitz and differentiability estimates that show the constructed U is genuinely twice differentiable in m with the stated uniform bounds. We have given the *mechanism* of those steps honestly — the monotone pairing, the favorable signs, the linearized systems — but the passage from mechanism to uniform all-horizon estimate is a monograph-length undertaking. It is carried out for the non-degenerate monotone case in [37] (their main theorem) and, by the probabilistic decoupling-field method, in [41], Ch. 5. □

Remark 12.9 (Status of theorem 12.8: an acknowledged imported result)**.** This is the one place in the chapter where a load-bearing argument is imported rather than reproduced, and the book's self-containedness doctrine requires us to say so plainly and to draw the line exactly. On the in-book side: the *equivalence* of the three formulations (theorem 12.3), the *characteristic representation* that powers the verification and uniqueness (theorem 12.6), the *mechanism* of monotonicity — the no-crossing picture and the Lasry–Lions pairing that gives Steps 1–4 their structure — and the entire *short-horizon* well-posedness (proposition 12.10 below) are proved here in full. On the imported side, and *only* on it: the all-horizon, uniform $\mathcal{C}^2$-in-m regularity estimates that turn the mechanism of Steps 2–3 into quantitative bounds. We do not prove those, and we do not pretend to; we cite them precisely, to [37] (their main theorem, the non-degenerate monotone case) and to [41], Ch. 5 (the probabilistic decoupling-field route). The reason is genuine rather than expedient: these estimates are themselves a research monograph's worth of parabolic and Wasserstein-calculus analysis, on the same footing as the handful of other foundational results the book imports, and reproducing them would bury rather than illuminate the master equation's structure, which is this chapter's subject. The honest reader's takeaway is the boundary itself: nothing about *what* the master equation is, or *why* it is equivalent to the system, or *how* monotonicity prevents the breakdown, is taken on faith — only the hard regularity number at the end of an all-horizon estimate is.

The import in remark 12.9 can be retired entirely, at one cost: a short horizon. We now give a complete, self-contained proof of well-posedness for T small that uses no monotonicity whatsoever — the second of the two regimes remark 12.11 will name. It is the same contraction that produced short-time existence for the MFG system in chapter 10 and short-time uniqueness in section 11.7, now packaged at the level of the master function, and it is the load-bearing self-contained result of the section: where Steps 2–3 above imported their estimate, here every constant is exhibited. The single mechanism is that on a short interval the best-response map cannot move its input far, because everything that drives it is an integral over the short window $[t_0, T]$ of bounded integrands.

Proposition 12.10 (Short-horizon well-posedness of the master equation)**.** *Assume* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*, the strict form of* (A-Conv)*, and that H, F, G are bounded with bounded, Lipschitz first and second derivatives in all arguments (*no monotonicity is assumed*). There is $T_\star > 0$, depending only on these bounds and the Lipschitz constants, such that for every horizon $T \le T_\star$ the master function U of eq.* (12.5) *is well defined, single-valued, $\mathcal{C}^{1,2}$ in (t,x), and fully $\mathcal{C}^2$ in m with $\nabla_x U$, $\partial_\mu U$, $\partial_\mu^2 U$, $\nabla_x \partial_\mu U$ bounded and Lipschitz uniformly in (t,x,m); it solves eq.* (12.7)*–eq.* (12.8) *classically and is the unique such solution.*

Proof. The proof is a contraction on the decoupling field, run on the Banach space the rest of the book already provides. Let $\mathcal{B}$ be the space of maps $\Phi : [0,T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}^d$ that are bounded, Lipschitz in x and (for $\mathcal{W}_2$) in m, with norm $\|\Phi\|_{\mathcal{B}} = \sup|\Phi| + \mathrm{Lip}_x(\Phi) + \mathrm{Lip}_m(\Phi)$; $\mathcal{B}$ is complete. Define the map S as follows. Given a candidate decoupling field $\Phi \in \mathcal{B}$ and an initial datum (t_0, m), let $(m_t)_{t \ge t_0}$ be the Fokker–Planck flow generated by the feedback $-\partial_p H(\cdot, \Phi(t, \cdot, m_t))$ with $m_{t_0} = m$ — well posed and Lipschitz-dependent on the data by theorem 7.9 and (A-Nondeg) — and let $u(t,x)$ solve the backward HJB eq. (12.2a) along this fixed flow; this is a *linear* (in the sense of fixed-coefficient) backward parabolic problem with terminal data $G(\cdot, m_T)$, uniquely and classically solvable with Schauder estimates under (A-Nondeg). Set $(\mathsf{S}\Phi)(t_0, x, m) := \nabla_x u(t_0, x)$.

A fixed point $\Phi = \mathsf{S}\Phi$ is, by construction, a decoupling field whose induced (u, m) solves the full MFG system from every (t_0, m): the flow m_t was built from the feedback $-\partial_p H(\cdot, \Phi)$ and u solves the HJB along it, and the fixed-point identity $\Phi = \nabla_x u$ is exactly the decoupling relation $\nabla u = \nabla_x U$ that ties the two halves of the system together. Equivalently, U defined by $\nabla_x U = \Phi$ and $U(t_0, x, m) = u(t_0, x)$ solves the master equation, by the computation of theorem 12.3 read backwards. So everything reduces to one assertion: S is a contraction on $\mathcal{B}$ for small T. We prove it by exhibiting the contraction constant, because that constant is the entire content of the short-horizon theory.

Decompose S into its two constituent maps, $\mathsf{S} = \mathsf{G} \circ \mathsf{F}$, where $\mathsf{F} : \Phi \mapsto (m_t)$ is the forward flow construction and $\mathsf{G} : (m_t) \mapsto \nabla_x u(t_0, \cdot)$ is the backward solve. The smallness for short T enters entirely through F, and it is worth seeing exactly why, since this is the mechanism the whole proposition turns on. Take two candidate fields Φ, Φ' and the two flows $(m_t), (m'_t)$ they generate from the *same* initial datum $m_{t_0} = m'_{t_0} = m$. At (t, x) the two drifts are $-\partial_p H(x, \Phi(t, x, m_t))$ and $-\partial_p H(x, \Phi'(t, x, m'_t))$, and their difference splits into a part charged to $\Phi - \Phi'$ and a part charged to $m_t - m'_t$,

$$\left|\partial_p H(x, \Phi(t, x, m_t)) - \partial_p H(x, \Phi'(t, x, m'_t))\right| \;\le\; L\,\|\Phi - \Phi'\|_{\mathcal{B}} \;+\; L'\,\mathcal{W}_2(m_t, m'_t),$$

the constants L, L' furnished by the bounded second derivatives of H together with the Lipschitz-in-m norm built into $\mathcal{B}$. Feeding this into the standard stability estimate for the Fokker–Planck/McKean–Vlasov flow (theorem 7.9, with the diffusion nondegenerate by (A-Nondeg)) and applying Grönwall over $[t_0, t]$ gives

$$\mathcal{W}_2(m_t, m'_t) \;\le\; \int_{t_0}^{t} \Big(L\,\|\Phi - \Phi'\|_{\mathcal{B}} + L'\,\mathcal{W}_2(m_s, m'_s)\Big)\,\mathrm{d}s \;\le\; C_1\,(t - t_0)\,\|\Phi - \Phi'\|_{\mathcal{B}}\,,$$

the crucial $(t - t_0)$ factor coming from precisely the fact that the two flows are *pinned to the same m at t_0*: they can separate only by accumulating the drift discrepancy over the elapsed window, so their separation is an integral over $[t_0, t]$ and vanishes as $t \downarrow t_0$. Thus F is Lipschitz with constant $C_1(T - t_0)$.

The backward solve G carries no such smallness on its own — as $T \downarrow t_0$ the value $u(t_0, \cdot)$ tends to the terminal datum $G(\cdot, m)$ and its gradient to ∇G, an $O(1)$ object — but it does not need any, because all of Φ's influence has already passed through F. What G must supply is only ordinary Lipschitz dependence on its *flow* input. Writing $w = u - u'$ for the difference of the two backward solutions along (m_t) and (m'_t), w solves a linear backward parabolic equation whose terminal datum is $G(\cdot, m_T) - G(\cdot, m'_T)$ and whose source is $F(\cdot, m_t) - F(\cdot, m'_t)$; both are bounded in sup-norm by $L'' \sup_t \mathcal{W}_2(m_t, m'_t)$ using the Lipschitz couplings, and the Schauder estimates of (A-Nondeg) bound $\nabla_x w(t_0, \cdot)$ by the same quantity. Hence G is Lipschitz with an $O(1)$ constant C_2 in $\sup_t \mathcal{W}_2(m_t, m'_t)$. (A part of the source, the running-data integral, in fact contributes its own $O(T - t_0)$ factor for the same window reason; we do not need it, the F-factor already sufficing.) Composing the two,

$$\|\mathsf{S}\Phi - \mathsf{S}\Phi'\|_{\mathcal{B}} \;\le\; C_2 \cdot C_1(T - t_0)\,\|\Phi - \Phi'\|_{\mathcal{B}} \;=:\; C\,(T - t_0)\,\|\Phi - \Phi'\|_{\mathcal{B}} \;\le\; C\,T\,\|\Phi - \Phi'\|_{\mathcal{B}}\,.$$

Therefore, for $T \le T_\star := 1/(2C)$, the map S is a $\frac{1}{2}$-contraction on the closed ball of $\mathcal{B}$ that its (horizon-uniform) bounds leave invariant. By the Banach fixed-point theorem (theorem 2.36) it has a unique fixed point $\Phi^\star = \nabla_x U$, and this delivers existence and uniqueness of the smooth master solution. Notice that $T_\star$ depends only on C, hence only on the sup-norms and Lipschitz constants of H, F, G and their first two derivatives — exactly the data named in the proposition, and nothing about monotonicity.

Differentiability in m to second order is obtained by running the *same* contraction on a larger Banach space that records the m-derivatives alongside Φ. Differentiating the fixed-point identity $\mathsf{S}\Phi = \Phi$ formally in the measure direction — legitimate because the coefficients H, F, G are $\mathcal{C}^2$ — yields linear fixed-point equations for $\partial_\mu \Phi$ and $\partial^2_\mu \Phi$ in which the linear operator is the derivative of S and therefore inherits the *same* contraction factor $C(T - t_0)$. For $T \le T_\star$ these linearized fixed-point problems are themselves contractions, so $\partial_\mu \Phi$ and $\partial_\mu^2 \Phi$ exist, are their unique solutions, and inherit the uniform bounds of $\Phi^\star$; the mixed $\nabla_x \partial_\mu U$ comes out of the same system. Parabolic regularity of the backward solve upgrades the x-dependence to $\mathcal{C}^{1,2}$. This is a complete, self-contained proof of well-posedness on a short horizon: no estimate has been imported, and every constant traces to the bounds in the hypotheses. The sole role of monotonicity in theorem 12.8 is to remove the restriction $T \leq T_\star$ — to make the no-crossing of characteristics hold for all horizons rather than only short ones. □

Remark 12.11 (Beyond monotonicity, and the short-horizon alternative)**.** Two honest caveats frame what we have, and what we have not, proved. First, monotonicity is *sufficient but not necessary*: there are perfectly well-behaved non-monotone problems whose master solution is smooth, but no general well-posedness theory covers them, and the boundary is sharp in the worst case — in the non-monotone regime classical solutions can fail to exist *exactly* when MFG equilibria become non-unique, which is the characteristics crossing of this section's opening realized in an example. Chapter 14 exhibits precisely this for the LQG thread, where the loss of a classical U shows up as a finite-time blow-up of the Riccati matrix that parametrizes it. Second, monotonicity is not the only escape: even without it one always has a *local*, short-horizon classical solution, by the contraction proposition 12.10, for which every constant was exhibited above. The two facts fit together cleanly — short horizon buys smoothness unconditionally, monotonicity buys it for all horizons — and they are two instances of a dichotomy that recurs across the whole book: "small coupling or short horizon," where a contraction closes the argument with explicit constants, versus "monotone for all horizons," where a structural sign condition does the work no contraction can. We will meet this same pair again for graphon games, where the role of the smallness parameter is played by the operator norm of the graphon: well-posedness holds either below the operator-norm threshold (A-Coupling) — the graphon analogue of small coupling or short horizon — or under a monotonicity condition on the kernel, and chapter 24 draws the threshold exactly.

This settles the well-posedness question for the theory built on Lasry–Lions monotonicity (A-Mon), and it is worth being precise about what "the theory built on (A-Mon)" quietly assumed. The monotonicity inequality eq. (12.13) pairs a *coupling* difference $F(\cdot, m) - F(\cdot, m')$ against a *measure* difference $m - m'$; it never touches the Hamiltonian, because in the setting of this section the population enters only through the additive couplings $F(x, m), G(x, m)$ while the Hamiltonian $H(x, p)$ carries no measure. That is the *separable* structure, and Lasry–Lions monotonicity is exactly tailored to it: the favorable sign in Steps 2–3 came from the coupling and the Hamiltonian sitting in separate, independently-signed terms of the pairing. The moment the Hamiltonian itself depends on the population — $H(x, p, m)$, with the crowd coupling directly to an agent's own velocity, as it must in any genuine congestion or trading model — those terms no longer separate, the Lasry–Lions pairing mixes Hamiltonian and coupling contributions, and the sign that protected the characteristics can be spoiled by their interaction. The monotonicity method breaks at exactly the place where the coupling and the Hamiltonian are forced to cooperate. Chapter 11 flagged this non-separable case and deferred the operative condition; that condition — *displacement* monotonicity, the structural hypothesis that restores a favorable sign when H and the coupling can no longer be signed apart — is the subject of the subsection that follows.

12.6.1 Displacement monotonicity and non-separable Hamiltonians

The closing paragraph of section 12.6 located the exact fault line, and this subsection works on it. Recall the breakdown just diagnosed: the moment the Hamiltonian itself depends on the crowd — $H(x, p, m)$, as in any honest congestion or trading model where the cost of moving rises with the local density one must push through — the Lasry–Lions pairing of Steps 2–3 no longer splits into a Hamiltonian term and a coupling term that can be signed *independently*; the two contributions merge, and the sign that protected the characteristics from crossing can be destroyed by their cross-interaction. Chapter 11 foresaw this, flagged the non-separable case as outside the reach of its uniqueness theorem, and deferred to the present chapter the precise condition that takes over together with the proof that it does for the master operator what Lasry–Lions did in the separable theory. The condition is *displacement monotonicity*, and we now supply both items: the definition, and the proof that it makes the master operator monotone.

The conceptual move behind the new condition is a change of *geometry* on the space of measures, and it pays to see it clearly before writing anything down, because the two monotonicity conditions are two ways of asking the *same* object to be increasing along two *different* families of paths through $\mathcal{P}_2(\mathbb{R}^d)$. Lasry–Lions monotonicity lives in the *flat* geometry: it interpolates between two populations along the straight segment $\epsilon \mapsto m + \epsilon(m' - m)$ of signed measures — the same linear interpolation that defined the flat derivative in definition 12.1 — and asks that the cost pair favorably with the mass-zero direction $m' - m$. Mass is reweighted in place; nothing moves. The displacement geometry interpolates instead along a *Wasserstein geodesic*: it lifts the two measures to random variables X, X' with $\mathcal{L}(X) = m$, $\mathcal{L}(X') = m'$ and slides along the L^2 segment $\epsilon \mapsto (1 - \epsilon)X + \epsilon X'$, whose law is the displacement (optimal transport) interpolation between m and m'. Now mass is genuinely *transported*: each particle at position $X(\omega)$ glides to $X'(\omega)$. The natural object to ask about along such a path is not how the cost pairs with a reweighting but how its *spatial gradient* — the force felt by a particle — pairs with the *displacement* $X - X'$ that carries the particle. That is exactly what eq. (12.14) below will demand: the lifted gradient field $X \mapsto \nabla_x\Theta(X, \mathcal{L}X)$ should be a monotone operator on $L^2(\Omega; \mathbb{R}^d)$, increasing along L^2 segments just as a monotone function on the line increases along intervals. The reader should hold onto the slogan: flat monotonicity signs a *reweighting* against a *cost*; displacement monotonicity signs a *transport* against a *force*.

Definition 12.12 (Displacement monotonicity)**.** A function $\Theta : \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$, of class $\mathcal{C}^1$ in x, is *displacement monotone* if for every pair of square-integrable $\mathbb{R}^d$-valued random variables X, X' on a common atomless probability space,

$$\mathbb{E}\big[\langle \nabla_x\Theta(X, \mathcal{L}(X)) - \nabla_x\Theta(X', \mathcal{L}(X')), X - X'\rangle\big] \;\geq\; 0. \tag{12.14}$$

Equivalently, the lifted map $X \mapsto \nabla_x\Theta(X, \mathcal{L}(X))$ is a monotone operator on $L^2(\Omega; \mathbb{R}^d)$. The standing data F, G are *displacement monotone* when eq. (12.14) holds for each of them.

Three features of the definition deserve to be drawn out, because each is exactly what makes eq. (12.14) the right inequality and not a cosmetic variant of the flat one. First, the two arguments of $\nabla_x\Theta$ move *together*: in $\nabla_x\Theta(X, \mathcal{L}X)$ the spatial slot is the random variable X and the measure slot is its *own* law, so as we slide X toward X' both slots vary, and the inequality controls the joint response of the gradient to this coupled motion. This is why the lift to L^2 is not optional bookkeeping but the content of the condition: only on $L^2(\Omega;\mathbb{R}^d)$ can "spatial position" and "population" be varied by the single object X. Second, the pairing is with $X - X'$, a vector in L^2, and the natural dual of a *force* $\nabla_x\Theta$ is precisely a *displacement*; the dimensions match a transport, not a reweighting. Third, atomlessness of the probability space is what guarantees that *every* pair of laws (m, m') is representable by some pair (X, X') — including, crucially, by the optimal-transport coupling that realizes $\mathcal{W}_2(m, m')$ — so quantifying over random variables genuinely quantifies over all of Wasserstein space and over all geodesics through it. A useful micro-check: in the degenerate case where $\Theta(x, m) = \theta(x)$ carries no measure dependence, eq. (12.14) reduces to $\mathbb{E}\langle\nabla\theta(X) - \nabla\theta(X'), X - X'\rangle \geq 0$, which is exactly the statement that $\nabla\theta$ is a monotone vector field, i.e. that θ is convex. Displacement monotonicity is thus the measure-coupled generalization of "the gradient is monotone, the function is convex," and the next remark makes that parallel precise.

Remark 12.13 (Displacement versus Lasry–Lions)**.** The two conditions are genuinely different, and neither implies the other; it is worth being exact about how they differ, because the difference is the whole reason a second condition is needed. Definition 11.1 pairs the increment of the *cost itself, across two measures*, $F(\cdot, m) - F(\cdot, m')$, against the signed measure $m - m'$ in the flat duality — a pairing of a function against a reweighting, integrated over space. Equation (12.14) pairs the increment of the *state-gradient of the cost*, $\nabla_x F(X, m) - \nabla_x F(X', m')$, against the displacement $X - X'$ of the underlying random variables — a pairing of a force against a transport, averaged over Ω. The objects paired are different (cost versus gradient-of-cost), the directions paired against are different ($m - m'$ versus $X - X'$), and the duality is different (flat $L^2(\mathbb{R}^d)$ versus lifted $L^2(\Omega)$). There is no algebraic identity converting one into the other.

The cleanest way to *see* that neither implies the other is the potential case, which also exhibits the exact parallel between the two conditions and their two convexities. Suppose the coupling derives from a potential, $F(x, m) = \frac{\delta\mathcal{F}}{\delta m}(m, x)$ for some functional $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ — the situation in every variational (potential) MFG. Then the two monotonicities of F are precisely the two *convexities* of $\mathcal{F}$:

- Lasry–Lions (flat) monotonicity of F $\iff$ *flat convexity* of $\mathcal{F}$, i.e. $\epsilon \mapsto \mathcal{F}\big(m + \epsilon(m' - m)\big)$ is convex along linear interpolations of measures;
- displacement monotonicity of F $\iff$ *displacement convexity* of $\mathcal{F}$, i.e. the lift $\epsilon \mapsto \mathcal{F}\big(\mathcal{L}((1-\epsilon)X + \epsilon X')\big)$ is convex along L^2 segments — McCann's notion of convexity along Wasserstein geodesics.

In each line the monotonicity of the gradient $\delta\mathcal{F}/\delta m$ is the convexity of $\mathcal{F}$ along the corresponding family of paths; the two lines differ only in *which* paths. That these are independent is now concrete. Take the interaction energy $\mathcal{F}(m) = \frac{1}{2}\iint_{\mathbb{R}^d\times\mathbb{R}^d} K(x-y)\, m(\mathrm{d}x)\, m(\mathrm{d}y)$ with an even kernel K, whose flat derivative is the convolution $F(x, m) = (K * m)(x)$. Flat convexity of $\mathcal{F}$ holds exactly when K is a *positive-definite* kernel ($\widehat{K} \geq 0$), which is the classical Lasry–Lions monotonicity of a convolution coupling; displacement convexity holds exactly when K is *convex* as a function on $\mathbb{R}^d$ (McCann's criterion). A convex kernel need not be positive-definite and a positive-definite kernel need not be convex — the Gaussian $K(z) = e^{-|z|^2}$ is positive-definite but not convex, while a convex but oscillation-free even kernel can fail positive-definiteness — so the two conditions select genuinely different classes of couplings, neither containing the other.

The decisive point, and the reason the displacement condition is the one this subsection needs, is *compatibility with a p-dependent, m-coupled Hamiltonian.* Lasry–Lions monotonicity was built to sign a term in which the coupling and the Hamiltonian appear *separately*; it has no purchase once the population enters through the momentum slot. Displacement monotonicity, by contrast, pairs against the displacement $X - X'$ — and, as the proof of proposition 12.14 will show, the difference of two FBSDE trajectories generates *exactly* pairings of this form, gradient-increment against state-increment, with the population slot riding along as $\mathcal{L}(X)$ versus $\mathcal{L}(X')$. It is the condition the non-separable computation naturally produces and can therefore consume, where the flat one is left with no term to attach to.

With the geometry fixed and the condition stated, the payoff is the same structural conclusion as in the separable theory — a *monotone decoupling field*, the no-crossing of characteristics that section 12.6 identified as the heart of well-posedness — but now reached through the displacement pairing rather than the flat one. We prove it in the canonical control-as-drift setting eq. (12.1) of this chapter, where the Hamiltonian is $H(x,p) = \frac{1}{2}|p|^2$ and the entire population coupling sits in F, G; in this setting every term is explicit and the role of each displacement-monotonicity hypothesis can be watched as it enters. The restriction to $\frac{1}{2}|p|^2$ is for transparency, not necessity: for a Hamiltonian carrying its own x- and m-dependence the same pairing is run, with H itself required to satisfy the joint displacement-monotone analogue (monotonicity of $(X,P) \mapsto (\nabla_x H, -\nabla_p H)$ in the lifted variables), and the all-horizon theory under that hypothesis is the subject of [76] and [104], which we cite for the full quantitative estimates.

Proposition 12.14 (Displacement monotonicity yields a monotone master operator)**.** *Work in the canonical setting eq.* (12.1) *with σ constant nondegenerate and $H(x,p) = \frac{1}{2}|p|^2$, and suppose F and G are displacement monotone (definition 12.12) with bounded Lipschitz x-gradients. Let (X_t, Y_t) and (X'_t, Y'_t) be two solutions of the MFG FBSDE of chapter 9 on $[0,T]$, driven by the* same *Brownian motion, with $m_t = \mathcal{L}(X_t)$ and $m'_t = \mathcal{L}(X'_t)$. Then*

$$\mathbb{E}\big[\langle Y_0 - Y'_0, X_0 - X'_0\rangle\big] \;\geq\; \int_0^T \mathbb{E}\,\left|Y_t - Y'_t\right|^2 \mathrm{d}t \;\geq\; 0. \tag{12.15}$$

In particular, whenever the two equilibria start from the same state ($X_0 = X'_0$ a.s.) they coincide, so the FBSDE has at most one solution and the master function eq. (12.5) *is single-valued; and writing $Y_t = \nabla_x U(t, X_t, m_t)$* (eq. (12.11))*, the decoupling field $\nabla_x U$ is a* monotone *operator. This is the structural input that well-poses the master equation for non-separable couplings, replacing Lasry–Lions monotonicity in theorem 12.8.*

Proof. The whole proof is a single application of Itô's product rule to the inner product of the two trajectory-differences, and the displacement-monotonicity hypotheses enter at exactly the two boundary-and-running terms that the product rule produces. We track every step, because the sign of each term is the entire content.

The two FBSDEs. In the canonical setting the optimal feedback is $\alpha^* = -\partial_p H(x, Y) = -Y$ (since $\partial_p H = p$ for $H = \frac{1}{2}|p|^2$), so the MFG FBSDE of chapter 9 reads

$$\mathrm{d}X_t = -Y_t\ \mathrm{d}t + \sigma\ \mathrm{d}W_t, \qquad \mathrm{d}Y_t = -\nabla_x F(X_t, m_t)\ \mathrm{d}t + Z_t\ \mathrm{d}W_t, \qquad Y_T = \nabla_x G(X_T, m_T),$$

with $m_t = \mathcal{L}(X_t)$. Here $\nabla_x H \equiv 0$ because $H = \frac{1}{2}|p|^2$ does not depend on x, so the backward driver, which in general is $\nabla_x H(X_t, Y_t) - \nabla_x F(X_t, m_t)$ by the adjoint (stochastic Pontryagin) computation of chapter 9, collapses to $-\nabla_x F(X_t, m_t)$. The second pair (X'_t, Y'_t) solves the same system with $m'_t = \mathcal{L}X'_t$. Write $\Delta X_t = X_t - X'_t$ and $\Delta Y_t = Y_t - Y'_t$ for the trajectory differences, and $\Delta Z_t = Z_t - Z'_t$.

Why ΔX has no martingale part. Subtract the two forward equations. The drift contributes $-(Y_t - Y'_t)\ \mathrm{d}t = -\Delta Y_t\ \mathrm{d}t$. The diffusion contributes $\sigma\ \mathrm{d}W_t - \sigma\ \mathrm{d}W_t$, which is *identically zero*: this is precisely where the two hypotheses "same Brownian motion" and "σ constant" are spent. Because σ is a deterministic, state-independent constant and the *same* W drives both forward equations, the two stochastic integrals are literally the same process and cancel. Were $\sigma = \sigma(x, m)$ state- or measure-dependent, the difference would leave a residual $(\sigma(X_t, m_t) - \sigma(X'_t, m'_t))\ \mathrm{d}W_t$ and ΔX would carry a martingale part, re-introducing a quadratic-variation term below; the canonical constant-σ setting is exactly what removes it. Hence

$$\mathrm{d}\Delta X_t = -\Delta Y_t\ \mathrm{d}t,$$

a process of finite variation in t (absolutely continuous paths), with no $\mathrm{d}W$ term.

Itô's product rule. Apply the product rule (theorem 5.9) to the scalar $\langle \Delta X_t, \Delta Y_t\rangle$:

$$\mathrm{d}\langle \Delta X_t, \Delta Y_t\rangle = \langle \mathrm{d}\Delta X_t, \Delta Y_t\rangle + \langle \Delta X_t, \mathrm{d}\Delta Y_t\rangle + \mathrm{d}\langle \Delta X, \Delta Y,_\rangle\, t.$$

The last term is the quadratic co-variation of the two difference processes. *It vanishes*: co-variation only sees the martingale (Brownian) parts of the two processes, and ΔX has none — it is of finite variation, by the previous paragraph — so $\mathrm{d}\langle \Delta X, \Delta Y,_\rangle\, t = 0$ regardless of the martingale part $\Delta Z_t\ \mathrm{d}W_t$ carried by ΔY. (This is the second dividend of the canceling diffusions: not only does ΔX acquire smooth paths, but the cross-variation between forward and backward differences drops out, leaving a clean ordinary-calculus product rule.) Substituting $\mathrm{d}\Delta X_t = -\Delta Y_t\ \mathrm{d}t$ and $\mathrm{d}\Delta Y_t = -(\nabla_x F(X_t, m_t) - \nabla_x F(X'_t, m'_t))\,\mathrm{d}t + \Delta Z_t\ \mathrm{d}W_t$,

$$\mathrm{d}\langle \Delta X_t, \Delta Y_t\rangle = \Big(-|\Delta Y_t|^2 - \langle \Delta X_t, \nabla_x F(X_t, m_t) - \nabla_x F(X'_t, m'_t)\rangle \Big)\,\mathrm{d}t + \langle \Delta X_t, \Delta Z_t\ \mathrm{d}W_t\rangle,$$

the first inner product giving $\langle -\Delta Y_t, \Delta Y_t\rangle = -|\Delta Y_t|^2$.

Expectation and integration. The term $\langle \Delta X_t, \Delta Z_t\ \mathrm{d}W_t\rangle$ is a genuine Itô (martingale) increment, so under the square-integrability carried by the FBSDE solutions its expectation is zero. Taking $\mathbb{E}$ and integrating from 0 to T,

$$\mathbb{E}\langle \Delta X_T, \Delta Y_T\rangle - \mathbb{E}\langle \Delta X_0, \Delta Y_0\rangle = -\int_0^T \mathbb{E}\ |\Delta Y_t|^2\ \mathrm{d}t - \int_0^T \mathbb{E}\langle \Delta X_t, \nabla_x F(X_t, m_t) - \nabla_x F(X'_t, m'_t)\rangle\ \mathrm{d}t.$$

At the terminal time the backward data give $\Delta Y_T = \nabla_x G(X_T, m_T) - \nabla_x G(X'_T, m'_T)$, so $\mathbb{E}\langle \Delta X_T, \Delta Y_T\rangle$ is exactly the displacement-monotonicity bracket for G evaluated on the pair (X_T, X'_T). Solving the displayed identity for the quantity at time 0,

$$\mathbb{E}\langle \Delta X_0, \Delta Y_0\rangle = \underbrace{\mathbb{E}\langle \Delta X_T, \nabla_x G(X_T, m_T) - \nabla_x G(X_T', m_T')\rangle}_{\geq 0} + \int_0^T \mathbb{E}\ |\Delta Y_t|^2\ \mathrm{d}t + \int_0^T \underbrace{\mathbb{E}\langle \Delta X_t, \nabla_x F(X_t, m_t) - \nabla_x F(X_t',}_{\geq 0}$$

The sign of each term. The terminal bracket is nonnegative because G is displacement monotone: it is precisely eq. (12.14) applied with $\Theta = G$ and the pair (X_T, X_T'), whose laws are $m_T = \mathcal{L}(X_T)$ and $m_T' = \mathcal{L}(X_T')$ — so both slots of $\nabla_x G$ are filled consistently, the spatial slot by the random variable and the measure slot by its own law, exactly as the definition requires. The running integrand is nonnegative for the same reason with $\Theta = F$ and the pair (X_t, X_t') at each t. The middle term $\int_0^T \mathbb{E}\,|\Delta Y_t|^2\ \mathrm{d}t$ is manifestly nonnegative. Dropping the two displacement-monotone brackets (each ≥ 0) from the right-hand side can only decrease it, which yields the asserted chain

$$\mathbb{E}\langle \Delta X_0, \Delta Y_0\rangle \ \geq\ \int_0^T \mathbb{E}\ |\Delta Y_t|^2\ \mathrm{d}t \ \geq\ 0,$$

that is, eq. (12.15). This is where the two hypotheses pay off: the *terminal* displacement monotonicity of G controls the boundary bracket, the *running* displacement monotonicity of F controls the integrand, and together they isolate the nonnegative middle term — exactly the role Lasry–Lions monotonicity played for the separable couplings, now discharged by the displacement condition.

Uniqueness. Suppose the two equilibria start from the same state, $X_0 = X_0'$ a.s., so $\Delta X_0 = 0$ and the left-hand side of eq. (12.15) is 0. The chain then forces $\int_0^T \mathbb{E}\,|\Delta Y_t|^2\ \mathrm{d}t = 0$, hence $\Delta Y_t = 0$ for a.e. t, a.s.; that is, $Y \equiv Y'$. Feeding this back into the forward difference, $\mathrm{d}\Delta X_t = -\Delta Y_t\ \mathrm{d}t = 0$ with $\Delta X_0 = 0$ gives $\Delta X_t \equiv 0$, i.e. $X \equiv X'$. The FBSDE issuing from a given initial law has at most one solution, so the construction eq. (12.5) assigns a single value and the master function is well defined — the same single-valuedness that monotonicity bought in the separable case, now secured without it.

Monotonicity of the master operator. Finally, evaluate the FBSDE through its decoupling field $Y_t = \nabla_x U(t, X_t, m_t)$ (eq. (12.11)) at $t = 0$, where $X_0 \sim m$ and $X_0' \sim m'$. Then $Y_0 = \nabla_x U(0, X_0, m)$ and $Y_0' = \nabla_x U(0, X_0', m')$, so the left-hand side of eq. (12.15) reads

$$\mathbb{E}\langle X_0 - X_0', \nabla_x U(0, X_0, m) - \nabla_x U(0, X_0', m')\rangle \ \geq\ 0.$$

This is exactly eq. (12.14) for the lifted gradient field $X \mapsto \nabla_x U(0, X, \mathcal{L}X)$: the decoupling field $\nabla_x U$ is a monotone operator on $L^2(\Omega; \mathbb{R}^d)$. The displacement monotonicity of the data has propagated, through the FBSDE, into displacement monotonicity of the master function's own gradient. □

Thus displacement monotonicity is the precise functional inequality eq. (12.14), and proposition 12.14 is the promised proof that it makes the master operator monotone — the two deliverables chapter 11 forwarded here. The monotone-operator conclusion is the exact non-separable substitute for the Lasry–Lions input of theorem 12.8: it is what forbids the characteristics from crossing when the Hamiltonian couples to the crowd, and hence what keeps the construction eq. (12.5) single-valued at every horizon. As in the separable case, the passage from this structural sign to a full all-horizon classical well-posedness theorem — the uniform $\mathcal{C}^2$-in-m regularity estimates — is a monograph-scale undertaking we do not reproduce; it is carried out, for displacement-monotone non-separable data, in [76] and [104], and we import it on *exactly* the acknowledged-exception footing of remark 12.9: the structure (the monotone pairing just proved) is in book, only the terminal regularity number is cited. With this the well-posedness arc closes. Both regimes are settled — the separable case by Lasry–Lions monotonicity, the non-separable case by displacement monotonicity, each with its short-horizon unconditional fallback (proposition 12.10) and each with its all-horizon estimate honestly flagged as imported — and in every case the master function exists, is unique, and carries the differentiability in m that the rest of the program needs.

Every result up to this point, however, has presumed one thing that the next section overturns: that the population m_t is a *deterministic* curve, a single trajectory the construction eq. (12.5) could solve for in advance. That is what made the bare value function tenable and the flow map an upgrade rather than a necessity. The next section turns on a shock felt by every agent at once — *common noise* — and watches the population become a genuine stochastic process in $\mathcal{P}_2(\mathbb{R}^d)$: this is the regime, promised since section 12.1, that the master equation was built for, the one in which it ceases to be a convenience and becomes the only formulation that survives. We reach it now with well-posedness in hand.

12.7 Common noise: why the master equation becomes indispensable

We have reached the section the whole construction was pointing at. The well-posedness arc is closed: in both the separable and the non-separable regimes the master function exists, is single-valued at every horizon, and carries the differentiability in m the program needs. But every one of those results — the construction eq. (12.5), the flow-consistency identity eq. (12.6), the entire equivalence with the system — rested on a silent hypothesis: that the population m_t is a *deterministic* curve, one trajectory the construction could solve for in advance and then hold fixed while each agent optimizes against it. That hypothesis is exactly what made the bare value function $u(t, x)$ tenable, and it is exactly what made the master function look, to a skeptic, like a luxury: if only one population flow is ever realized, why carry a value for every contingency? Section 12.1 answered that the answer would arrive under common noise, and at every "we return to this under common noise" pointer since, the debt has been deferred to here. We now pay it. We turn on a shock felt by every agent at once, watch the population become a genuine stochastic process in $\mathcal{P}_2(\mathbb{R}^d)$, and find that the deterministic forward–backward system loses its meaning while the master function — having already valued *every* contingent crowd in advance — is the only formulation left standing. The price the master equation extracts for this is the second-order measure derivative it now acquires: a genuine Laplacian on Wasserstein space.

The setup: a population that has become random

Suppose all agents are driven, in addition to their private noises, by a single common Brownian motion W^0:

$$dX_t \;=\; \alpha_t \, dt \;+\; \sigma \, dW_t \;+\; \sigma_0 \, dW_t^0. \tag{12.16}$$

The private W_t is the agent's own idiosyncratic shock, independent across the continuum; the new term $\sigma_0 \,\mathrm{d}W_t^0$ is the same Brownian increment for everyone, scaled by the common-noise intensity σ_0. The distinction is the whole story. Idiosyncratic shocks average out across the population — by a law-of-large-numbers effect each agent's private wiggle is washed away in the aggregate, which is precisely why, in chapter 8 and throughout the deterministic theory, the population obeyed an *ordinary* (non-random) Fokker–Planck equation. The common shock does not average out: it pushes the entire population the same way at the same instant, so the aggregate is itself jostled randomly.

Concretely, the relevant population is now the *conditional* law

$$m_t \;=\; \mathcal{L}(X_t \mid \mathcal{F}_t^0),$$

the distribution of the representative state given the history $\mathcal{F}^0 = (\mathcal{F}_t^0)$ of the common noise up to time t. Conditioning on $\mathcal{F}_t^0$ integrates out the private shocks (which, being independent of W^0, still average to a smooth density) but *retains* the common one; what survives is a measure that depends on the realized path of W^0, and so varies from one realization to another. The object $t \mapsto m_t$ is therefore not a curve in $\mathcal{P}_2(\mathbb{R}^d)$ but a *stochastic process taking values in* $\mathcal{P}_2(\mathbb{R}^d)$ — a random measure, adapted to $\mathcal{F}^0$, with a genuinely different trajectory on each draw of the shared shock.

This single fact dismantles the deterministic forward–backward system eq. (12.2). That system presupposed a predetermined population trajectory: the Fokker–Planck equation eq. (12.2b) was solved forward to produce *one* curve $m_\cdot$, against which the backward HJB eq. (12.2a) was then solved. Strip away the determinism and the construction is incoherent on both ends. There is no longer a single curve m_t to solve for — *which* population the agent will face at time t is not known until the shared shock has played out, so there is nothing to pin the forward equation to a unique solution. And the backward value can no longer be a deterministic function solved in one sweep from $t = T$: an agent's optimal continuation must *react* to the population she actually finds herself in, a population revealed only as the common noise unfolds, so her value process must be *adapted* to information that arrives going forward in time. A backward function pinned to a single foreknown trajectory m_t literally cannot be written down, because there is no one trajectory to pin it to. This is the obstruction section 12.1 promised in advance, now made sharp.

What *can* be written down is precisely $U(t, x, m)$. The master function was built to value every contingency $m \in \mathcal{P}_2(\mathbb{R}^d)$, with the population as a free input slot rather than a solved-for curve; so it does not care that the realized population has become random. One simply *composes* it with the random measure, forming $u_t(x) = U(t, x, m_t)$, and reads off the agent's value along whatever crowd the shared shock actually produces. The per-crowd object, optional in the deterministic world, is here the only thing that survives. The remaining task is to find the equation it must satisfy, and the surprise — the content of the rest of this section — is that turning m_t into a stochastic process forces *second-order* terms in the measure direction that the deterministic equation never carried.

The engine: Itô–Wentzell on $\mathcal{P}_2(\mathbb{R}^d)$, and the terms it produces

In the deterministic theory the equation for U came from differentiating the flow-consistency identity $u_t(x) = U(t, x, m_t)$ in time with the chain rule lemma 12.2. The same strategy works here, but the chain rule must be upgraded: m_t is now a *stochastic* flow, so we need the Itô formula for a functional of a random measure — an Itô–Wentzell formula on $\mathcal{P}_2(\mathbb{R}^d)$. (“Wentzell” because, exactly as in finite dimensions, differentiating $f(t, X_t)$ when *both* the function's time-slot and its random argument carry the same Brownian increment produces an extra cross-term beyond the ordinary Itô formula; here the common W^0 enters both U's state argument $x = X_t$ and its measure argument m_t.) Applying it to $u_t(x) = U(t, x, m_t)$ along the conditional flow, and demanding consistency with the agent's now-adapted HJB equation exactly as before, produces the old master equation eq. (12.7) plus several new families of terms, all governed by the common-noise coefficient σ_0. Three mechanisms, traced one at a time, account for every one of them; the reader should hold the three-second-derivatives notation watch after theorem 12.3 alongside, because each mechanism activates a different one of the objects catalogued there.

Mechanism one: the agent's own extra diffusion. The representative agent's state X_t now carries, on top of its private $\sigma \, \mathrm{d}W_t$, the common increment $\sigma_0 \, \mathrm{d}W_t^0$. To her own value $x \mapsto U(t, x, m)$ this is simply additional diffusion of her own coordinate, and Itô's formula in the state variable adds the corresponding second-order term in x. With m frozen, the quadratic variation of X_t gains $\sigma_0\sigma_0^\top \, \mathrm{d}t$, and the value picks up $-\frac{1}{2} \operatorname{tr}(\sigma_0\sigma_0^\top \, \mathrm{D}_x^2 U)$ — the same Hessian $\mathrm{D}_x^2 U$ that already sat on the first line of eq. (12.7), now fed the extra diffusion. This is the first object of the notation watch (the ordinary finite-dimensional Hessian in the agent's own state); nothing about the measure is involved yet.

Mechanism two: the population spreads with the full diffusion. This is the subtle one, and it is where the coefficient *upgrade* lives. Ask how the conditional law m_t itself evolves. Each agent diffuses with both noises, but conditioning on $\mathcal{F}_t^0$ does *not* remove the common one — W^0 is $\mathcal{F}_t^0$-measurable only as a *path*, not in its future increments, and the spreading it induces is felt by the conditional density just as the private noise is. The conditional Fokker–Planck equation for m_t therefore carries the *full* diffusion $\sigma\sigma^\top + \sigma_0\sigma_0^\top$, not merely the private $\sigma\sigma^\top$ of the deterministic theory: the common noise spreads the population exactly as it spreads the individual. Now recall that the deterministic master equation's extra-diffusion term, $-\frac{1}{2} \int \operatorname{tr}(\sigma\sigma^\top \, \partial_y \partial_\mu U) \, m(\mathrm{d}y)$, was produced by the chain rule lemma 12.2 from precisely this population-spreading piece of the Fokker–Planck operator (Move 3b of its proof). Feed that same machinery the upgraded diffusion and the coefficient is upgraded with it, $\sigma\sigma^\top \to \sigma\sigma^\top + \sigma_0\sigma_0^\top$, so the equation gains the *additional* term $-\frac{1}{2} \int \operatorname{tr}(\sigma_0\sigma_0^\top \, \partial_y \partial_\mu U) \, m(\mathrm{d}y)$. The object differentiated is again the single-variable Jacobian $\partial_y \partial_\mu U$ — the second entry of the notation watch, “fully $\mathcal{C}^2$” content, the crowd's *spatial* spreading — not yet a genuine measure Hessian. The upgrade is invisible in any term-counting of the equation; it shows up only in the coefficient, and the consistency check below is designed precisely to make it visible.

Mechanism three: the common mode correlates distinct agents. Here is the genuinely new structure, and the reason the deterministic theory could get away with single-variable derivatives. In the no-common-noise world the random motion of the mass at a point y was statistically independent of the random motion of the mass at a distant point y' — private shocks are independent across agents — so there was no mechanism correlating the two, and the two-point measure Hessian $\partial_\mu^2 U$ never appeared (this is exactly the gap discussed at definition 12.1). Common noise supplies the missing mechanism. One shared increment $\sigma_0 \, \mathrm{d}W_t^0$ moves *all* of the mass together: the speck at y and the speck at y' receive the identical shock, so their motions are perfectly correlated through W^0. The Itô–Wentzell formula records this correlation as a second derivative of U in the measure direction — the third object of the notation watch, the true Hessian on $\mathcal{P}_2(\mathbb{R}^d)$ indexed by two locations,

$\partial_\mu^2 U(t,x,m)(y,y')$ — contracted against $\sigma_0\sigma_0^\top$ and integrated against the crowd *twice*, once in y and once in y'. There is moreover a *cross* contribution from the same shared increment correlating the agent's own state $x = X_t$ with the crowd point y: both move under W^0, and the cross-variation produces a mixed derivative $\nabla_x \partial_\mu U(t,x,m)(y)$, a spatial gradient in the agent's coordinate of the Lions field in the crowd's coordinate. Each of these is a hallmark of the Itô–Wentzell (as opposed to plain Itô) calculus: they are exactly the terms that appear because the same Brownian motion sits in two slots of U at once.

Collecting the four contributions, the master equation acquires the additional terms

$$
\begin{aligned}
&-\tfrac{1}{2}\operatorname{tr}\big(\sigma_0\sigma_0^\top\, \mathrm{D}_x^2\, U\big) - \tfrac{1}{2}\int_{\mathbb{R}^d} \operatorname{tr}\big(\sigma_0\sigma_0^\top\, \partial_y\partial_\mu U(t,x,m)(y)\big)\, m(\mathrm{d}y)\\
&\quad -\int_{\mathbb{R}^d} \operatorname{tr}\big(\sigma_0\sigma_0^\top\, \nabla_x\partial_\mu U(t,x,m)(y)\big)\, m(\mathrm{d}y)\\
&\quad -\tfrac{1}{2}\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \operatorname{tr}\big(\sigma_0\sigma_0^\top\, \partial_\mu^2 U(t,x,m)(y,y')\big)\, m(\mathrm{d}y)\, m(\mathrm{d}y'),
\end{aligned}
\tag{12.17}
$$

added to the left-hand side of eq. (12.7), where $\partial_\mu^2 U$ is the second Lions derivative. Reading them off line by line against the three mechanisms: the lone $\mathrm{D}_x^2\, U$ term is mechanism one, the agent's own extra diffusion. The first integral, $-\frac{1}{2}\int \operatorname{tr}(\sigma_0\sigma_0^\top\, \partial_y\partial_\mu U)\, m(\mathrm{d}y)$, is mechanism two, the population self-spreading — the coefficient upgrade, with the same single-variable Jacobian that already appeared in eq. (12.7), now fed the common diffusion. The second integral, $-\int \operatorname{tr}(\sigma_0\sigma_0^\top\, \nabla_x\partial_\mu U)\, m(\mathrm{d}y)$, is the cross term, coupling the agent's own state x to the crowd point y through the shared shock. And the double integral, $-\frac{1}{2}\iint \operatorname{tr}(\sigma_0\sigma_0^\top\, \partial_\mu^2 U)\, m(\mathrm{d}y)\, m(\mathrm{d}y')$, is the genuine *Laplacian on* $\mathcal{P}_2(\mathbb{R}^d)$: the only term in the entire chapter built from the true second *measure* derivative, switched on precisely because the common mode correlates distinct agents. The first and last are the diagonal and off-diagonal pieces of one second-order operator in the measure — the agent–crowd and crowd–crowd correlations that the single shared Brownian motion induces.

A consistency check: the pure-translation functional

The coefficient upgrade of mechanism two is the easiest of the new effects to overlook — it hides inside an existing term rather than adding a visibly new one — so it is worth a self-contained check that isolates it and confirms its coefficient is exactly right. Take the simplest nontrivial regime: no control and no private noise, every agent driven only by the common shock,

$$
\mathrm{d}X_t \;=\; \sigma_0\, \mathrm{d}W_t^0,
$$

and let U value the crowd alone through a smooth pairing, $U(t,x,m) = \int_{\mathbb{R}^d} \phi(x,y)\, m(\mathrm{d}y)$ — a *pure-translation functional*, linear in m. We compute its drift two ways and demand they agree.

Directly. With dynamics $\mathrm{d}X_t = \sigma_0\, \mathrm{d}W_t^0$ every agent receives the *same* increment, so $X_t = X_0 + \sigma_0 W_t^0$ and the conditional law is a rigid translation of its initial value, $m_t = (\tau_{\sigma_0 W_t^0})_\#\, m_0$ with $\tau_z(y) = y + z$. Hence

$$
U(t,x,m_t) \;=\; \int_{\mathbb{R}^d} \phi(x,\, y + \sigma_0 W_t^0)\, m_0(\mathrm{d}y) \;=\; g(\sigma_0 W_t^0), \qquad g(z) := \int_{\mathbb{R}^d} \phi(x, y+z)\, m_0(\mathrm{d}y),
$$

a smooth function of the single $\mathbb{R}^d$-valued process $\sigma_0 W_t^0$. Ordinary finite-dimensional Itô gives the drift $\frac{1}{2}\operatorname{tr}(\sigma_0\sigma_0^\top \mathrm{D}_z^2 g)$, and since $\mathrm{D}_z^2 g(z) = \int \mathrm{D}_y^2 \phi(x, y+z)\, m_0(\mathrm{d}y) = \int \mathrm{D}_y^2 \phi(x,y)\, m_t(\mathrm{d}y)$ by the change of variables, the drift of $U(t,x,m_t)$ is

$$\frac{1}{2}\int_{\mathbb{R}^d} \operatorname{tr}\big(\sigma_0\sigma_0^\top \mathrm{D}_y^2 \phi(x,y)\big)\, m_t(\mathrm{d}y).$$

From the new terms. For the linear functional $U(t,x,m) = \int \phi(x,y)\, m(\mathrm{d}y)$ the derivatives are immediate: the flat derivative is $\frac{\delta U}{\delta m}(t,x,m,y) = \phi(x,y)$, the Lions field is $\partial_\mu U(t,x,m)(y) = \nabla_y \phi(x,y)$, and its Jacobian is $\partial_y \partial_\mu U(t,x,m)(y) = \mathrm{D}_y^2 \phi(x,y)$. Crucially, because U is *linear* in m its flat derivative $\phi(x,y)$ does not depend on m at all, so its further measure derivative vanishes identically: $\partial_\mu^2 U \equiv 0$. The Laplacian term is therefore *off*. The remaining new terms involving the agent's own coordinate — the $\mathrm{D}_x^2 U$ term and the cross term $\nabla_x \partial_\mu U$ — play no role here either, for two reasons worth keeping apart. First, this check holds the agent's state x fixed and follows only the motion of the measure argument m_t, so the own-state second-order terms, which track the agent's *own* diffusion, do not enter the drift we are comparing. Second — to make the isolation airtight rather than merely out of scope — we take ϕ depending on the crowd coordinate y alone, whereupon they vanish identically, $\mathrm{D}_x^2 U = 0$ and $\nabla_x \partial_\mu U = 0$. For a *general* $\phi(x,y)$ these own-state terms, $\int \mathrm{D}_x^2 \phi(x,y)\, m(\mathrm{d}y)$ and $\nabla_x \nabla_y \phi(x,y)$, are nonzero; specializing to $\phi = \phi(y)$ removes them by construction, leaving the self-spreading term as the sole survivor. The *only* surviving new term is the coefficient upgrade,

$$-\frac{1}{2}\int_{\mathbb{R}^d} \operatorname{tr}\big(\sigma_0\sigma_0^\top \partial_y \partial_\mu U(t,x,m)(y)\big)\, m(\mathrm{d}y) \;=\; -\frac{1}{2}\int_{\mathbb{R}^d} \operatorname{tr}\big(\sigma_0\sigma_0^\top \mathrm{D}_y^2 \phi(x,y)\big)\, m(\mathrm{d}y),$$

which is exactly minus the directly-computed drift — the sign being the one that, carried to the left-hand side against $-\partial_t U$, reproduces it. The two computations agree, and the agreement is delicate: had we *not* upgraded the coefficient (had we kept the bare $\sigma\sigma^\top$, which is zero here because there is no private noise), this term would be absent and the equation would predict zero drift, contradicting the direct calculation. The self-spreading term is precisely the one that would go missing without the upgrade of mechanism two, and the check pins its coefficient at $\sigma_0\sigma_0^\top$.

Blessing and difficulty

The new second-order structure cuts both ways, and it is worth naming the duality squarely. On the one hand it is a *blessing*. In section 12.4 we observed that the deterministic master equation, read as a Hamilton–Jacobi equation on $\mathcal{P}_2(\mathbb{R}^d)$, was first-order in the measure and hence offered no smoothing in that direction — the source of the well-posedness difficulties section 12.6 had to handle by monotonicity rather than by parabolic regularization. Common noise repairs exactly that. The genuine Laplacian $\partial_\mu^2 U$ it adds makes the master equation *parabolic on the measure space*, supplying the second-order regularization that was missing; nondegeneracy of σ_0 smooths U in m the way nondegeneracy of σ smooths an ordinary HJB equation in x. On the other hand it is a genuine *difficulty*. The second measure derivative is the analytically heaviest object in the theory — a function of two crowd locations (y, y') at every (t,x,m), whose estimates and even whose precise function spaces are delicate — and it is the reason the common-noise theory is a separate and harder undertaking than the deterministic one. The same term that regularizes the equation is the term that makes it hard to control.

We have therefore stated the *structure* of the common-noise master equation and interpreted every term, but we have not proved it, and we are honest about why. The Itô–Wentzell formula on $\mathcal{P}_2(\mathbb{R}^d)$ used to obtain eq. (12.17) is a beyond-prerequisite result: it is the stochastic counterpart of the Itô formula for flows of measures theorem D.7 on which the deterministic chain rule lemma 12.2 rested, generalized from a deterministic flow to one driven by the common Brownian motion. Its in-book proof, together with the well-posedness theory that the formula presupposes — the role of nondegeneracy of σ_0 in regularizing U, the precise function spaces for $\partial_\mu^2 U$, and the extension to the graphon setting — is carried out in chapter 26, where common noise is treated systematically. The present section is therefore explicitly *conditional* on chapter 26: we have used the master equation's per-contingency design to argue *why* it, and it alone, survives a random population, and we have read off the terms the Itô–Wentzell formula must produce, deferring the formula's proof to its proper home. The references [37] and [41], Ch. 4 and 7, carry the derivation out in full and are cited for attribution, never in place of the in-book argument.

Physics note

Common noise is the closest the mean field story comes to a genuine fluctuating order-parameter field: a single global mode W_t^0 shakes the whole population coherently, so the macroscopic state m_t itself becomes a stochastic process rather than a deterministic background. The second-measure-derivative term in eq. (12.17) is the diffusion of that global mode — the order-parameter "propagator" — and it is exactly what was missing from the bare mean-field (deterministic-m) description. As before, this is orientation only; the quantitative content is in chapter 26.

This completes the case the chapter set out to make. We have shown, in full generality, that the master function is not a convenience but the only formulation that survives a random population, and we have read off the second-order measure structure that common noise forces upon it. It remains to descend from this height to a single fully explicit model and watch the entire apparatus — the decoupling field, the transport and diffusion integrals, the flow-consistency that ties U to the bare value — collapse into a handful of scalar ordinary differential equations one can solve by hand. We do this on the carried LQG thread. There the coupling is through the population *mean*, the master function depends on m only through that single scalar, and the abstract derivatives of this chapter become finite-dimensional calculus: the genuine second measure derivative of common noise does not arise (we take no common noise), and — the point flagged in section 12.3 and revisited just above — the extra-diffusion integral $\int \operatorname{tr}(\sigma\sigma^\top \partial_y \partial_\mu U)\, m(\mathrm{d}y)$ *vanishes outright*, because mean coupling makes the Lions field constant in y and its Jacobian zero. What is left is a pair of Riccati ordinary differential equations, and we verify that they reproduce the chapter 8 fixed point exactly.

12.8 Worked computation: the LQG decoupling field and its master equation

The chapter has been built at altitude. We argued first *why* a population-feedback object is forced (the non-Markovian obstruction of section 12.1), assembled the single analytic engine that powers everything (the chain rule along a Fokker–Planck flow, lemma 12.2), used it to *derive* the master equation and prove its exact equivalence with the MFG system (theorem 12.3), reread that equation as a Hamilton–Jacobi equation on $\mathcal{P}_2(\mathbb{R}^d)$ whose characteristics *are* the system (sections 12.4 and 12.5), confronted the well-posedness its first-order-in-m structure forces (section 12.6), and finally shown that common noise makes U not a convenience but the only survivor (section 12.7). Every one of these statements was about an abstract function on an infinite-dimensional space. The promise repeatedly deferred — and the way to be certain the apparatus is real and not merely formal — is to descend to a model where one can write U down in closed form, watch each abstract term either compute itself out or visibly vanish, and check by hand that the answer is the one Part II already obtained by other means. That model is the carried LQG thread, and this section delivers the descent.

We advance Thread A to its third layer (appendix A.7). The first two layers are already in hand: the single-agent control problem was solved in chapter 6 (Layer 1), producing the scalar curvature Riccati equation for the value of one agent facing a frozen crowd; and the MFG fixed point was solved in chapter 8 (Layer 2), producing the coupled backward-curvature / forward-mean system whose self-consistency pins the equilibrium. Layer 3, the present task, is to write the *master* function and the decoupling field for the same model, reduce them to ordinary differential equations, and verify that restricting the master function to the equilibrium flow returns the Layer-2 system with not a sign out of place. The model is the scalar linear–quadratic–Gaussian MFG with the fixed parameter names of the thread specification (appendix A.7): the representative agent's state obeys

$$\mathrm{d}X_t = (aX_t + b\,\alpha_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad X_0 \sim m_0, \tag{12.18}$$

and she chooses α to minimize, with population mean $\bar{m}_t = \int \xi\, m_t(\mathrm{d}\xi)$,

$$J(\alpha) = \mathbb{E}\left[\int_0^T \left(\tfrac{1}{2}\alpha_t^2 + \tfrac{q}{2}\big(X_t - s\,\bar{m}_t\big)^2\right)\mathrm{d}t + \tfrac{q_T}{2}\big(X_T - s_T\,\bar{m}_T\big)^2\right]. \tag{12.19}$$

Each symbol carries the meaning fixed once and for all in the thread specification: a is the uncontrolled *state feedback* (the rate at which the state drifts on its own), b the *control gain* (how strongly the agent's action moves the state), σ the constant *noise* amplitude; $q \geq 0$ and $q_T \geq 0$ are the running and terminal *state-cost weights* that penalize deviation from a target; and s, s_T are the *mean-coupling* strengths that set where the target sits — each agent is drawn toward (for $s > 0$) or repelled from (for $s < 0$) the population mean $s\,\bar{m}_t$. The single feature of eq. (12.19) on which everything below turns is that the agent feels the crowd *only through its mean* $\bar{m}_t$, a single scalar functional of the measure m_t, and not through any finer feature of its shape. Every measure functional we shall meet therefore depends on m solely through $\bar{m}$. This is exactly the mean-coupling structure singled out in the discussion following theorem 12.3, and it is what makes the LQG master equation collapse from a PDE on $\mathcal{P}_2(\mathbb{R}^d)$ to a PDE in one extra scalar variable; it is the cleanest possible illustration of theorem 12.3, and — as we are about to see term by term — the case in which the extra-diffusion line of the master equation drops out entirely.

One point of bookkeeping must be settled before any HJB equation is written, because the canonical setting eq. (12.1) of this chapter took the drift to be the control itself, whereas the thread dynamics eq. (12.18) also carry an *uncontrolled* linear drift aX. Conflating the two is the easiest slip to make here, so we flag it.

Notation watch

The dynamics eq. (12.18) carry an uncontrolled linear drift aX, so the relation $\alpha^* = -\partial_p H$ of appendix A.4 (stated for the canonical control-as-drift normalization $a = 0, b = 1$ of eq. (12.1)) applies to the *control part* only. The agent's instantaneous Hamiltonian is obtained by minimizing the control-dependent part of the pre-Hamiltonian over α: with adjoint variable p, she minimizes $b\,\alpha\,p + \frac{1}{2}\alpha^2$, a strictly convex scalar quadratic whose minimizer is $\alpha^* = -b\,p$ and whose minimum value is $-\frac{b^2}{2}p^2$. The full HJB Hamiltonian then *bundles in* the uncontrolled drift axp on top of this,

$$H_{\mathrm{LQG}}(x, p) = -axp + \tfrac{b^2}{2}p^2,$$

so that $\partial_p H_{\mathrm{LQG}} = -ax + b^2 p$ — whose negative is *not* the optimal control $\alpha^* = -bp$ but the optimal control *plus the uncontrolled drift*, exactly the total velocity of the state. This is the correct object to feed into the Fokker–Planck transport, and it is the source of the a-terms that will accompany every b^2P below. We carry the general (a, b) to keep the thread intact; setting $a = 0, b = 1$ collapses H_{LQG} to the canonical $\frac{1}{2}p^2$ and recovers the normalization of eq. (12.1) throughout.

The HJB operator and the equilibrium ansatz

We begin on familiar ground, with the single-agent Hamilton–Jacobi–Bellman equation, because the master equation will be built on top of it and the ansatz is dictated by its structure. Fix the population mean as a known curve $\bar{m}_t$ for the moment (this is the Layer-1 posture: optimize one agent against a frozen crowd). The optimal feedback computed in the notation watch is $\alpha^* = -b\,\partial_x u$, and substituting it into the dynamic-programming equation for eq. (12.18)–eq. (12.19) — the uncontrolled drift axu_x entering linearly, the control contributing its minimized value $-\frac{b^2}{2}(\partial_x u)^2$, the noise its Laplacian $\frac{1}{2}\sigma^2 \partial_{xx} u$, and the running cost its quadratic — gives

$$\partial_t u + ax\,\partial_x u - \tfrac{b^2}{2}(\partial_x u)^2 + \tfrac{1}{2}\sigma^2\,\partial_{xx} u + \tfrac{q}{2}\big(x - s\,\bar{m}_t\big)^2 = 0, \qquad u(T, x) = \tfrac{q_T}{2}\big(x - s_T\,\bar{m}_T\big)^2. \quad (12.20)$$

The shape of the solution is legible from the equation itself. The terminal datum is a quadratic in x, and the only x-dependent forcing is the quadratic running cost; the operator preserves the class of quadratics-in-x (a quadratic has constant second derivative, so the nonlinear term $(\partial_x u)^2$ stays quadratic and the Laplacian stays constant). Quadratic data therefore propagate quadratically, and the value is quadratic in x for every t. The mean $\bar{m}_t$ enters eq. (12.20) only *linearly* (through the cross-term $-qs\,x\,\bar{m}_t$ and a $\bar{m}_t^2$ term that touches no power of x above the constant), so the dependence on the crowd is affine. Promoting the frozen $\bar{m}_t$ back to a free population argument m, we are led to posit a master function quadratic in x and affine in $\bar{m}$,

$$U(t,x,m) = \tfrac{1}{2}\,P(t)\,x^2 + \big(\pi(t)\,\bar{m} + \zeta(t)\big)\,x + c(t,\bar{m}), \qquad \bar{m} = \int \xi\, m(\mathrm{d}\xi), \tag{12.21}$$

with three time-dependent coefficients to be determined — the curvature $P(t)$, the population-sensitivity $\pi(t)$, and a constant-in-x offset $\zeta(t)$ — together with a purely x-independent term $c(t,\bar{m})$ that will turn out to be inert for the decoupling field.

Two notational conventions in eq. (12.21) deserve to be stated cleanly, since the whole computation rides on them. First, the *shadow notation*: because U here depends on the measure m only through the scalar $\bar{m} = \int \xi\, m(\mathrm{d}\xi)$, we write $U(t,x,\bar{m})$ for the value of $U(t,x,m)$ and let the same letter U denote both the genuine functional on $\mathcal{P}_2(\mathbb{R}^d)$ and its mean-reduced *shadow* on $\mathbb{R}$. (One could carry a separate symbol $\widehat{U}(t,x,\bar{m})$ for the shadow; we do not, since no confusion arises once the reduction is announced, and the single letter keeps the kinship between the two objects visible.) Under this convention $\partial_{\bar{m}} U$ is an *ordinary* partial derivative in a real variable. Second, and this is the payoff, the abstract *measure* derivatives of U collapse to that ordinary one. The reason is a one-line computation that is worth seeing, because it is the exact mechanism by which the mean-coupling structure trivializes the hard part of the master equation. The functional $m \mapsto \bar{m} = \int \xi\, m(\mathrm{d}\xi)$ is *linear* in m, so its flat derivative is read off directly from definition 12.1: perturbing along $m' - m$,

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon} \int \xi\, \big(m + \epsilon(m'-m)\big)(\mathrm{d}\xi)\Big|_{\epsilon=0} = \int \xi\, (m'-m)(\mathrm{d}\xi) = \int y\, (m'-m)(\mathrm{d}y),$$

so $\frac{\delta(\bar{m})}{\delta m}(m,y) = y$ identically. By the chain rule for the flat derivative, any functional of the form $m \mapsto U(t,x,\bar{m})$ has flat derivative $\frac{\delta U}{\delta m}(t,x,m,y) = \partial_{\bar{m}} U(t,x,\bar{m}) \cdot y$ — the ordinary mean-derivative times the bare coordinate y. Differentiating once in y to form the Lions field, the linear factor y becomes a *constant*: the Lions derivative is constant in y,

$$\partial_\mu U(t,x,m)(y) = \nabla_y \frac{\delta U}{\delta m}(t,x,m,y) = \partial_{\bar{m}} U(t,x,\bar{m}) = \pi(t)\,x + \partial_{\bar{m}} c(t,\bar{m}), \tag{12.22}$$

the last equality reading $\partial_{\bar{m}} U$ off the ansatz eq. (12.21). A field that is constant in y has vanishing Jacobian: $\partial_y \partial_\mu U \equiv 0$, and the extra-diffusion line of the master equation eq. (12.7) — $-\frac{1}{2}\int \mathrm{tr}(\sigma\sigma^\top \partial_y \partial_\mu U)\, m(\mathrm{d}y)$ — *vanishes identically.*

This is not a lucky cancellation special to the algebra; it is the concrete realization of the general claim made in deriving the master equation. Recall the discussion following theorem 12.3: the extra-diffusion term measures how the agent's value responds to the *spreading* of the crowd by idiosyncratic noise, and it is nonzero precisely when the Lions field bends in y — when the agent cares about more than the crowd's location of mass. There the contrast was drawn sharply against a *second-moment* coupling, for which $\frac{\delta U}{\delta m}(m,y) \propto y^2$, the Lions field $\partial_\mu U \propto y$ is linear and not constant, its Jacobian $\partial_y \partial_\mu U$ is a nonzero constant, and the extra-diffusion term survives: a crowd whose variance the agent feels does register the crowd's spreading. *Mean* coupling is exactly the borderline case in which the flat derivative is affine in y, the Lions field is flat, and the spreading goes unfelt. The LQG thread is thus the worked instance of the principle that it is mean coupling — not the weaker property of depending on finitely many moments — that annihilates the second-order measure term; we now watch the rest of the equation reduce.

Reduction of the master equation to Riccati ODEs

With the second-order measure term gone, the master equation eq. (12.7) reduces to a transport-augmented HJB in the two real variables $(x, \bar{m})$, and the only nonstandard term left is the transport integral. Here too mean coupling simplifies it to an ordinary calculus object. The transport line of eq. (12.7) pairs the Lions field against the equilibrium feedback drift, integrated over the crowd; but the equilibrium population is the closed loop of eq. (12.18) under $\alpha^* = -b\,\partial_x U = -b(Px + \pi\bar{m} + \zeta)$, and what the transport term tracks is precisely the motion of the mean. Taking expectations in eq. (12.18) kills the martingale $\sigma\,\mathrm{d}W_t$ and leaves the *closed-loop mean drift*

$$V(t,\bar{m}) := \frac{\mathrm{d}}{\mathrm{d}t}\bar{m} = a\,\bar{m} + b\,\mathbb{E}[\alpha_t^*] = (a - b^2P)\,\bar{m} - b^2(\pi\bar{m} + \zeta) = (a - b^2P - b^2\pi)\,\bar{m} - b^2\zeta, \quad (12.23)$$

where we used $\mathbb{E}[X_t] = \bar{m}$ and $\mathbb{E}[\alpha_t^*] = -b(P\bar{m} + \pi\bar{m} + \zeta)$ since α^* is affine in X_t. Now substitute the flow-consistency relation $u(t,x) = U(t,x,m_t)$ of eq. (12.6) into the HJB equation eq. (12.20). The only subtlety is the time derivative: u depends on t both explicitly and through the transported mean, so by the chain rule $\partial_t u = \partial_t U + \partial_{\bar{m}} U \cdot V$. This is nothing but the finite-dimensional shadow of lemma 12.2 — the transport term $\int \partial_\mu U \cdot \mathbf{b}\,\mathrm{d}m$ has collapsed, via eq. (12.22) and the affine feedback, to the single product $\partial_{\bar{m}} U \cdot V$, and the second-order measure term is absent because $\partial_y \partial_\mu U \equiv 0$. Carrying this out yields the *reduced master equation*

$$\partial_t U + \partial_{\bar{m}} U \cdot V(t,\bar{m}) + ax\,\partial_x U - \tfrac{b^2}{2}(\partial_x U)^2 + \tfrac{1}{2}\sigma^2\,\partial_{xx} U + \tfrac{q}{2}(x - s\,\bar{m})^2 = 0, \quad (12.24)$$

a genuine PDE in $(t, x, \bar{m})$ — the master equation of this section, shorn of all infinite-dimensional content.

It remains to insert the ansatz eq. (12.21) and match powers of x, which turns eq. (12.24) into a closed system of ODEs for (P, π, ζ). It is worth doing the bookkeeping in the open, since the structure of the Riccati system lives in exactly how the terms sort by degree. From the ansatz,

$$\partial_t U = \tfrac{1}{2}\dot{P}\,x^2 + (\dot{\pi}\,\bar{m} + \dot{\zeta})\,x + \partial_t c, \qquad \partial_x U = P\,x + \pi\bar{m} + \zeta, \qquad \partial_{xx} U = P, \qquad \partial_{\bar{m}} U = \pi\,x + \partial_{\bar{m}} c.$$

Substituting and expanding the two nonlinear pieces,

$$ax\,\partial_x U = aP\,x^2 + a(\pi\bar{m} + \zeta)\,x, \qquad -\tfrac{b^2}{2}(\partial_x U)^2 = -\tfrac{b^2}{2}P^2x^2 - b^2P(\pi\bar{m} + \zeta)\,x - \tfrac{b^2}{2}(\pi\bar{m} + \zeta)^2,$$

together with $\partial_{\bar{m}} U \cdot V = \pi V\,x + \partial_{\bar{m}} c \cdot V$ and $\frac{q}{2}(x - s\bar{m})^2 = \frac{q}{2}x^2 - qs\bar{m}\,x + \frac{q}{2}s^2\bar{m}^2$, we collect by power of x.

The x^2 coefficient gathers $\frac{1}{2}\dot{P}$, aP, $-\frac{b^2}{2}P^2$, and $\frac{1}{2}\sigma^2\partial_{xx}U$'s contribution to x^2 (which is zero, since $\partial_{xx}U = P$ is constant in x) and $\frac{q}{2}$:

$$\tfrac{1}{2}\dot{P} + aP - \tfrac{b^2}{2}P^2 + \tfrac{q}{2} = 0 \quad \Longleftrightarrow \quad \dot{P} = b^2P^2 - 2aP - q,$$

the single-agent curvature Riccati, decoupled from everything else. The x^1 coefficient gathers $\dot{\pi}\bar{m} + \dot{\zeta}$, the transport contribution πV, the drift term $a(\pi\bar{m} + \zeta)$, the nonlinear term $-b^2P(\pi\bar{m} + \zeta)$, and $-qs\bar{m}$; substituting $V = (a - b^2P - b^2\pi)\bar{m} - b^2\zeta$ and separating the part proportional to $\bar{m}$ from the part independent of it splits this single line into the π- and ζ-equations. The $\bar{m}$-proportional part reads

$$\dot{\pi} + \underbrace{(a - b^2P - b^2\pi)\pi}_{\text{from } \pi V} + a\pi - b^2P\pi - qs = 0 \iff \dot{\pi} = b^2\pi^2 - 2(a - b^2P)\pi + qs,$$

and the $\bar{m}$-independent part reads

$$\dot{\zeta} + \underbrace{(-b^2\pi)\,\zeta}_{\text{from } \pi V} + a\zeta - b^2P\zeta = 0 \iff \dot{\zeta} = -(a - b^2P - b^2\pi)\,\zeta.$$

Finally the x^0 coefficient collects $\partial_t c$, $\partial_{\bar{m}}c \cdot V$, $\frac{1}{2}\sigma^2 P$, $-\frac{b^2}{2}(\pi\bar{m} + \zeta)^2$, and $\frac{q}{2}s^2\bar{m}^2$ into a scalar PDE for $c(t, \bar{m})$; it is decoupled *forward* from (P, π, ζ) — it is driven by them but never feeds back, since c carries no power of x and so contributes nothing to the decoupling field $\nabla_x U$. We record it as solved and set it aside. Reading off the terminal data from $u(T,x) = \frac{q_T}{2}(x - s_T\bar{m}_T)^2 = \frac{q_T}{2}x^2 - q_T s_T \bar{m}\, x + \cdots$, the three coefficient equations are the decoupled backward Riccati system

$$\dot{P} = b^2P^2 - 2aP - q, \qquad P(T) = q_T, \tag{12.25a}$$
$$\dot{\pi} = b^2\pi^2 - 2(a - b^2P)\pi + q\,s, \qquad \pi(T) = -q_T s_T, \tag{12.25b}$$
$$\dot{\zeta} = -(a - b^2P - b^2\pi)\,\zeta, \qquad \zeta(T) = 0. \tag{12.25c}$$

The system is triangular and reads off cleanly. Equation eq. (12.25c) for ζ is linear and homogeneous with zero terminal data, so $\zeta \equiv 0$ identically: the decoupling field carries *no* constant term. This is not an accident of the parameters but a *centering symmetry.* The model eq. (12.18)–eq. (12.19) is equivariant under the simultaneous translation $X \mapsto X + c$, $\bar{m} \mapsto \bar{m} + c$ (it shifts the target $s\bar{m}$ and the state together), so a population centered at $\bar{m} = 0$ must produce an equilibrium whose linear-in-x value vanishes at $x = 0$; the only translation-invariant constant the offset could be is zero, and ζ inherits it. With ζ gone, only P and π survive, each solving a scalar backward Riccati equation. The curvature P is the *familiar single-agent value curvature* of chapter 6 — note that eq. (12.25a) makes no reference whatever to the coupling parameters s, s_T, so P is exactly the Layer-1 object, untouched by the crowd. The coefficient π is the genuinely *new* quantity that Layer 3 contributes: the *sensitivity of an agent's marginal value to the population mean*, the rate at which her optimal momentum tilts as the whole crowd shifts. Its equation eq. (12.25b) is forced by the coupling — the source term qs and the terminal datum $-q_T s_T$ are exactly where s, s_T enter — and it is driven by P through the closed-loop rate $a - b^2P$, but it does not feed back into P. Reading off the x-gradient of the ansatz eq. (12.21) with $\zeta \equiv 0$, the *LQG decoupling field* is the affine map

$$\nabla_x U(t, x, m) = P(t)\,x + \pi(t)\,\bar{m}, \tag{12.26}$$

the explicit, fully computed instance of the abstract object the chapter has been chasing: a function on $\mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d)$ that returns the equilibrium momentum at x for *every* crowd m, here linear in own state and in crowd mean. By the characteristics theorem theorem 12.6 this is exactly the relation $Y_t = P(t)X_t + \pi(t)\bar{m}_t$ between the FBSDE's adjoint Y_t and its forward components — the decoupling field of the LQG forward–backward system made fully explicit, with P and π its two scalar ingredients.

Consistency with the MFG fixed point of chapter 8

The construction is complete; the question that justifies the whole chapter is whether it is *correct* — whether the master function, restricted to the realized equilibrium flow, reproduces the Layer-2 fixed-point system that chapter 8 obtained by solving the coupled forward–backward problem directly. This is the LQG instance of the equivalence theorem 12.3, and on a model this explicit we can check it by hand, with every cancellation in view. The bridge is the restriction identity $u(t,x) = U(t,x,m_t)$: along the realized flow the master function's x-gradient must equal the Layer-2 value-gradient. From eq. (12.26), $\partial_x u(t,x) = P(t)\,x + \pi(t)\,\bar{m}_t$, so if we define the *linear offset along the equilibrium*

$$r(t) := \pi(t)\,\bar{m}_t,$$

then $\partial_x u(t,x) = P(t)\,x + r(t)$, which is precisely the form of the Layer-2 value-gradient. We use the canonical Thread-A solution symbols pinned in the specification (appendix A.7): $P(t)$ is the value-curvature Riccati unknown and $r(t)$ the linear offset — the very objects $P(t)$ and $r(t)$ of the uniqueness analysis of chapter 11 (the curvature Riccati eq. (11.11) and the offset eq. (11.12)). The single new piece of information the master equation supplies, beyond what Layer 2 already knew, is the *factorization* $r(t) = \pi(t)\,\bar{m}_t$: the equilibrium offset is the population-sensitivity coefficient π — a quantity defined off-equilibrium, for every crowd — evaluated at the realized mean $\bar{m}_t$. The master picture explains *why* r has the dynamics it does, by deriving them from π and the mean's motion.

To carry out the check we need the realized mean. From eq. (12.23) with $\zeta \equiv 0$, the equilibrium mean solves the closed linear forward ODE

$$\dot{\bar{m}}_t = (a - b^2 P - b^2\pi)\,\bar{m}_t, \qquad \bar{m}_0 \text{ given}, \tag{12.27}$$

exactly the forward half of the Layer-2 system. The offset r is a product of two time-dependent factors, so its dynamics follow from the product rule, and this is the climax of the verification — the one place where one must watch the nonlinear Riccati terms cancel. Differentiating $r = \pi\bar{m}$ and substituting the π-equation eq. (12.25b) for $\dot{\pi}$ and the mean equation eq. (12.27) for $\dot{\bar{m}}$,

$$\begin{aligned}
\dot{r} = \dot{\pi}\,\bar{m} + \pi\,\dot{\bar{m}} &= \big[\,b^2\pi^2 - 2(a - b^2P)\pi + qs\,\big]\bar{m} \;+\; \pi\,(a - b^2P - b^2\pi)\,\bar{m}\\
&= \big[\,b^2\pi^2 - 2(a - b^2P)\pi + qs + (a - b^2P)\pi - b^2\pi^2\,\big]\bar{m}\\
&= \big[\,-(a - b^2P)\pi + qs\,\big]\bar{m}\\
&= -(a - b^2P)\,(\pi\bar{m}) + q\,s\,\bar{m}_t \;=\; -(a - b^2P)\,r + q\,s\,\bar{m}_t.
\end{aligned}$$

The two nonlinear terms $+b^2\pi^2$ (from $\dot{\pi}$) and $-b^2\pi^2$ (from $\pi\dot{\bar{m}}$) cancel exactly, and the linear-in-π terms $-2(a - b^2P)\pi$ and $+(a - b^2P)\pi$ combine to $-(a - b^2P)\pi$; what was a *nonlinear* Riccati equation for π becomes, after multiplication by the mean, a *linear* backward equation for r. The terminal value transports the same way, $r(T) = \pi(T)\,\bar{m}_T = -q_T s_T\,\bar{m}_T$. This is *exactly* the Layer-2 forward–backward LQG system of chapter 8: the backward offset equation $\dot{r} = -(a - b^2P)\,r + qs\,\bar{m}$ with terminal data $-q_T s_T \bar{m}_T$, coupled to the forward mean equation eq. (12.27), with the curvature P solving its own eq. (12.25a). Not a coefficient, not a sign, not a boundary value is out of place. The cancellation of $b^2\pi^2$ is the whole content of the equivalence made arithmetic: the master equation's nonlinearity in the off-equilibrium sensitivity π is precisely what is needed for the on-equilibrium offset r to obey the linear forward–backward dynamics of the fixed point. We record the result.

Worked computation 12.15 (LQG master equation and its consistency)**.** For the scalar LQG MFG eq. (12.18)–eq. (12.19), the master function is the quadratic eq. (12.21) with $\zeta \equiv 0$ and (P, π) solving the decoupled backward Riccati pair eq. (12.25); the extra-diffusion term of the general master equation eq. (12.7) vanishes identically because $\partial_y \partial_\mu U \equiv 0$. The decoupling field is the affine map eq. (12.26). Restricting U to the equilibrium flow m_t via $r(t) = \pi(t)\bar{m}_t$ reproduces, with no discrepancy, the coupled Riccati-plus-mean fixed-point system of chapter 8 (Thread A, Layer 2), including its terminal data. The two pictures agree exactly.

There is one last dividend to collect, and it closes a loop the chapter opened twice: the abstract well-posedness question of section 12.6 acquires here a completely explicit, scalar face. The master equation exists globally exactly when its coefficients do, and on the LQG thread the only coefficient that can fail to exist is π.

Remark 12.16 (Where uniqueness enters)**.** The Riccati equation eq. (12.25b) for the population sensitivity π is the LQG incarnation of the well-posedness question of theorem 12.8. The curvature P, solving eq. (12.25a), never blows up backward from $P(T) = q_T \geq 0$ when $q \geq 0$: it is the benign single-agent object, bounded on every horizon. All the well-posedness drama lives in π. Tracing the chain of equivalences the chapter built: a global classical solution of the master equation exists iff its coefficient π exists on all of $[0, T]$ (the ansatz is the solution); by theorem 12.3 the master solution is equivalent to the MFG system; and by theorem 12.8 a global master solution is exactly uniqueness of the MFG equilibrium for every initial mean $\bar{m}_0$. Hence

$$\text{global existence of } \pi \iff \text{global master solution} \iff \text{equilibrium uniqueness.}$$

For parameters in the monotone/uniqueness regime — in particular for s, s_T small enough that the coupling respects Lasry–Lions monotonicity (A-Mon) — the source term qs and terminal datum $-q_T s_T$ are mild, and the backward Riccati solution π stays finite for all T. When s is strong enough to violate monotonicity the source can overwhelm the quadratic damping $b^2\pi^2$, π blows up to $-\infty$ in finite backward time, the ansatz eq. (12.21) ceases to define a function on the whole interval, the master solution fails to exist, and — by the same chain of equivalences read in reverse — equilibria multiply. This is the LQG face of remark 12.11: the blow-up of π in eq. (12.25b) occurs precisely at the (s, q, T) threshold where the forward–backward boundary-value problem of chapter 11 becomes singular. The closed-form threshold and the resulting phase diagram in (s, q, T) are already given in proposition 11.8 (fig. 11.1); they are completed for general terminal weight q_T in chapter 14. The lesson is that the entire well-posedness theory of this chapter, abstract as it looked, is encoded for the LQG thread in whether a single scalar Riccati trajectory reaches the terminal time without escaping to infinity.

This worked computation closes the argument of the chapter. We set out to show that the mean field game's backward–forward apparatus collapses into one scalar object $U(t, x, m)$, and that the collapse is a genuine upgrade rather than mere economy. The LQG thread has now made the claim concrete and checkable: the master equation is not only unifying but *computable*, reducing on this model to the two scalar Riccati equations eq. (12.25) whose restriction to the equilibrium flow reproduces the chapter 8 fixed point with no discrepancy at all, and exhibiting the affine decoupling field eq. (12.26) explicitly. Three threads run out of this section into the rest of the book. First, the second-order regularity of U that we computed here trivially — the curvature P and the smoothness in x and m — is the very input that chapter 13 requires to turn the master function into a quantitative N-player expansion: the differentiability that the bare value function lacked, made explicit on the thread, is what powers convergence with optimal rates. Second, the (s, q, T) blow-up of remark 12.16 hands the uniqueness-versus-multiplicity boundary to the phase-diagram analysis of chapter 11 and its completion for general q_T in chapter 14, where the LQG Riccati threshold becomes the sharp, computable instance of the abstract dichotomy. Third, and most ambitiously, the entire chapter — the derivation, the Hamilton–Jacobi reading, the common-noise second-order term we read off but did not prove, and this explicit LQG reduction — is lifted in chapter 26 to the graphon setting with common noise, where the population mean $\bar{m}$ becomes a type-indexed field aggregated through the kernel W and driven by a shared shock, and where the master equation regains the genuine second measure derivative that mean coupling here annihilated. The scalar Riccati pair of this section is the seed from which that far larger structure grows.

Rigor ledger

Proved.

- The chain rule for fully $\mathcal{C}^2$ functionals along a Fokker–Planck flow (lemma 12.2), with the two-term transport/diffusion structure; its full justification is the Itô formula on $\mathcal{P}_2(\mathbb{R}^d)$ proved in theorem D.7.
- The equivalence of the master equation with the MFG system: the master equation eq. (12.7)–eq. (12.8) holds for the constructed U, and conversely classical solutions generate solutions of the MFG system (theorem 12.3); the general-coefficient/non-separable form eq. (12.9) extends it (remark 12.4).
- The method of characteristics: Fokker–Planck flows are the characteristic curves, $\nabla_x U$ is the FBSDE decoupling field, and the master/system/FBSDE pictures coincide (theorem 12.6, diagram eq. (12.12)).
- *Short-horizon* well-posedness of the master equation, with full $\mathcal{C}^{1,2}/\mathcal{C}^2$ regularity, by a self-contained contraction argument assuming *no* monotonicity (proposition 12.10).
- Displacement monotonicity: the precise functional inequality definition 12.12 and the proof that it makes the master operator monotone (proposition 12.14), the deliverables forwarded from chapter 11 for non-separable Hamiltonians.
- The LQG master equation in closed form: a vanishing extra-diffusion term, two scalar backward Riccati equations eq. (12.25) in the canonical symbols P, r (appendix A.7), the affine decoupling field eq. (12.26), and exact consistency with the Layer-2 fixed point of chapter 8 (worked computation 12.15).

Assumed (imported).

- The MFG system and the optimal-feedback relation $\alpha^* = -\partial_p H$, the Hamiltonian $H(x, p, m)$ in the canonical argument order, convex (here strictly) in p under (A-Conv), and the variational/potential structure (chapter 8, appendix A.4).

- The McKean–Vlasov FBSDE, the maximum principle, and the FBSDE–PDE dictionary $Y_t = \nabla u(t, X_t)$, $Z_t = \mathrm{D}^2\, u(t, X_t)\sigma$ (proposition 9.12, with the decoupling-field existence theorem 9.14; chapter 5 for Itô).

- Existence of MFG equilibria (chapter 10); Lasry–Lions monotonicity (A-Mon) of definition 11.1 with its uniqueness consequence theorem 11.5; the McKean–Vlasov flow well-posedness theorem 7.9; standing assumptions (A-Lip),(A-Growth),(A-Nondeg),(A-Conv) of appendix A.6.

- The flat and Lions derivatives and the Itô formula on $\mathcal{P}_2(\mathbb{R}^d)$ (chapter 4, appendix A.3, theorem D.7).

- *Exported* for downstream use: the constructed U carries bounded, Lipschitz second-order Lions derivatives $\partial_\mu^2 U$, $\nabla_x \partial_\mu U$ (theorem 12.8), imported by the $O(1/N)$ expansion of chapter 13.

Deferred.

- The *all-horizon* classical well-posedness of eq. (12.7) in the monotone regime (theorem 12.8): the equivalence, the mechanism, and the short-horizon case are proved in-book; only the all-horizon linearized-system $\mathcal{W}_2$-estimates and second-order m-differentiability are imported, as the acknowledged exception of remark 12.9, from [37] and [41], Ch. 5.

- The all-horizon well-posedness under *displacement* (rather than Lasry–Lions) monotonicity for non-separable Hamiltonians: imported on the same footing from [76] and [104].

- The common-noise master equation eq. (12.17): structure only here, conditional on the Itô–Wentzell formula on $\mathcal{P}_2(\mathbb{R}^d)$, whose in-book proof and the full well-posedness theory are chapter 26 (with [37, 41] for attribution).

- The quantitative N-player convergence powered by U (rate $O(1/N)$): chapter 13.

Open.

- Well-posedness outside the monotone (and outside the short-horizon) regime: no general theory; classical solutions can fail exactly where equilibria multiply (remark 12.11, chapter 14).

- Master-equation theory under common noise on *sparse*/L^p graphon limits, flagged for chapters 26 and 27 and the research map of chapter 31.

Bibliographic notes

The master equation was introduced by Lions in his Collège de France lectures and recorded in [36]; the foundational text is [37], which proves classical well-posedness in the monotone, non-degenerate regime and uses it to settle the convergence problem from the N-player game (the source for theorem 12.8 and for the $O(1/N)$ rates forwarded to chapter 13). The probabilistic theory — the Itô calculus on $\mathcal{P}_2(\mathbb{R}^d)$ of lemma 12.2, the decoupling-field viewpoint of theorem 12.6, and the common-noise master equation of section 12.7 — is developed systematically in [41], especially Chapters 4–5, building on the calculus on Wasserstein space first organized for this purpose in [40]. The Lasry–Lions monotonicity condition eq. (12.13) and the multiplicity phenomenon when it fails originate in [93]; the geometry of $\mathcal{P}_2(\mathbb{R}^d)$ underlying the Hamilton–Jacobi reading of section 12.4 follows [131]. *Displacement monotonicity* (definition 12.12) as the structural condition for the master equation with non-separable Hamiltonians is due to Gangbo and Mészáros [75] (the displacement-convex potential case) and Gangbo, Mészáros, Mou and Zhang [76] and Mou and Zhang [104] (the general non-separable case); our proposition 12.14 is their monotone-decoupling-field computation in the canonical setting. The LQG master equation and decoupling field of section 12.8 are classical folklore for linear–quadratic mean field games; the reduction to Riccati systems and the consistency check are carried out in the LQG chapters of [40] and instantiated for this book's thread in chapter 14. The graphon lift of the entire chapter, together with common noise across a kernel, is the subject of chapter 26.

Chapter 13

From N-Player Games to the Mean Field Limit

The mean field game system of chapters 8 and 9, its solution theory (chapters 10 and 11), and the master equation of chapter 12 were all *postulated* as the right description of a large population of strategic agents. We argued for them by analogy, by the self-consistent-field reflex, and by the sheer cleanliness of the limit object — but we never certified that the limit is the limit *of anything.* A physicist should be uneasy here, and the unease is the right instinct. We have spent five chapters studying an effective field theory — a single representative agent moving in a self-generated mean field — without ever exhibiting the finite, microscopic system it is supposed to coarse-grain. Nothing so far rules out that the MFG system is an internally consistent fiction with no finite-N preimage, or that it is the limit of a population governed by different microscopic rules than the ones we care about. The analogue in statistical mechanics would be to write down a Landau free energy and never check that any honest Hamiltonian flows to it. This chapter supplies the missing certificate. It begins from a genuine, finite, non-cooperative game — N agents, each minimizing its own cost, each best-responding to the other $N-1$ — and asks in what precise sense, and at what rate, this microscopic game is approximated by the mean field game (MFG) as $N \to \infty$. Only once that is done is the postulate retired and the limit object earned.

Two facts make the question subtle, and both are the point of the chapter; we flag them now without proof so that the reader feels the stakes before the machinery arrives. The first is that the finite game is itself a perfectly sharp object, not a vague crowd. A Markovian Nash equilibrium of the N-player game is encoded *exactly* by a coupled system of N Hamilton–Jacobi–Bellman (HJB) equations on the Nd-dimensional configuration space $(\mathbb{R}^d)^N$ — the *Nash system.* This system is sharp but hopeless: it contains the full truth about the finite game, and it is computationally and analytically intractable for any N beyond a handful, because its unknowns live on a space whose dimension grows with the number of players and whose N equations are mutually coupled through the gradients of one another's solutions. The mean field is precisely the cure — it replaces N functions on $(\mathbb{R}^d)^N$ by one function on $\mathbb{R}^d$ driven by a deterministic field — and the chapter's first job (section 13.2) is to write the Nash system down and read off, term by term, exactly which features the limit dissolves and which it must keep.

The second fact is that the limit can be taken in two opposite directions, and they are *not* equally easy — their asymmetry is the single most important conceptual seam of the chapter, and it has a one-word cause: whether the interaction is *exogenous* or *endogenous.* In the *top-down* direction one starts from the MFG solution, hands its decentralized feedback to each of the N players, and asks how far short of equilibrium the resulting profile falls. Here the interaction is exogenous: the feedback is fixed in advance by the limit problem, so the N states evolve by a *prescribed* McKean–Vlasov dynamics, and the only thing to check is that the empirical measure of that dynamics concentrates — propagation of chaos for a fixed law of motion, exactly the phenomenon of chapter 7. This direction is cheap and yields the ε-Nash property. In the *bottom-up* direction one starts from the *actual* Nash equilibria of the N-player game and asks whether they converge to the MFG. Here the interaction is endogenous: the feedback each agent uses is itself the unknown of the Nash system, so the empirical measure one must show concentrates is generated by feedbacks that already depend on it. The chaos is no longer an input to a known dynamics but an *output* of the equilibrium — *propagation of chaos for Nash equilibria* — and there is no fixed system to which chapter 7 can be applied. Getting a rate for it is exactly what the regularity of the master equation buys; getting mere convergence, without a rate, costs only a compactness argument. Holding this exogenous-versus-endogenous distinction firmly is the key to not conflating the two halves of the chapter.

A word on standing conventions, so the reader carries the chapter's dependency skeleton before entering it. Throughout we work on the horizon $[0, T]$ and keep the convention of appendix A: value functions carry terminal data and are solved *backward* from $t = T$, while densities carry initial data and flow *forward* from $t = 0$ — the backward value/forward density pairing that the whole forward–backward structure of the theory rests on. No graphon is in scope anywhere in this chapter, so the letter W is free to denote Brownian motion, and we write W^i for the i-th agent's independent noise; the switch to B for Brownian motion, forced by the reappearance of the graphon kernel W, happens only from Part III onward. The chapter is a confluence of nearly everything proved so far, and it is worth naming the imports explicitly. From chapters 8 and 9 we take the MFG system and the representative McKean–Vlasov dynamics whose law is the mean field; from chapter 6 the single-agent stochastic control and verification machinery that the Nash system reduces to agent by agent; from chapter 7 the propagation of chaos that powers the top-down direction; from chapter 11 the Lasry–Lions monotonicity condition (A-Mon) and the uniqueness of the MFG solution it delivers; from chapter 12 the master field $U(t, x, m)$ and the strong regularity that makes it usable as an ansatz; and the linear–quadratic–Gaussian (LQG) thread is carried in from chapter 6 and forward into chapter 14. In exchange the chapter exports three results, and the reader may keep them as the three milestones to head for: the ε-Nash property of the mean field strategy (theorem 13.6), the qualitative convergence of equilibria (theorem 13.9), and the quantitative convergence with rate (theorem 13.13). chapter 14 instantiates all three explicitly on the LQG model, where everything is a Riccati ODE, and chapter 25 generalizes them to networks.

13.1 The N-player stochastic differential game

The chapter preamble promised to begin from a genuine, finite, non-cooperative game; this is where that game is written down once and frozen. Everything that follows — the Nash system, the ε-Nash property, the two convergence theorems — is a statement about the single object defined in this section, so it is worth fixing its data with care and saying out loud why each modelling choice is the one a mean field limit demands.

We fix the data of the finite game once and use it for the rest of the chapter. There are N agents indexed $i \in \{1, \dots, N\}$. Agent i has a private state $X_t^i \in \mathbb{R}^d$ driven by an admissible control α^i and by an independent d-dimensional Brownian motion W^i, with the W^i mutually independent and independent of the initial states X_0^i, which are i.i.d. with common law $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$. The interaction is *anonymous*: every agent feels the others only through the *empirical measure*

$$\mu_t^N \;:=\; \frac{1}{N}\sum_{j=1}^{N} \delta_{X_t^j} \;\in\; \mathcal{P}(\mathbb{R}^d), \tag{13.1}$$

the rigorous occupation profile introduced in chapter 1 and the carrier of the mean field throughout chapters 7 and 8.

The choice to let agents interact *only* through μ_t^N is not a simplifying convenience; it is the precise structural hypothesis without which no mean field limit can exist, and it deserves to be understood before we go on. Anonymity says that an agent's drift and cost depend on the others not through their identities or their labelled positions $(X^1, \dots, \widehat{X^i}, \dots, X^N)$ but only through the unordered cloud of where everyone is — the histogram, not the seating chart. Formally, the coefficients are invariant under permutations of the other agents' indices, so they can be written as functions of the empirical measure alone. Two things flow from this. First, the game is *exchangeable*: relabelling the agents leaves the entire system of dynamics and costs unchanged, which is exactly the symmetry that lets us speak of a single "representative" agent in the limit, since no agent is distinguished. Second, and more importantly, anonymity is what makes the limit *stable in* N: an empirical-measure functional such as $\int \phi \, \mathrm{d}\mu_t^N = \frac{1}{N}\sum_j \phi(X_t^j)$ is an intensive quantity — an average, not a sum — so it has a finite, N-independent limit as the population grows, and adding one more agent perturbs it by only $O(1/N)$. A non-anonymous coupling, say a drift that singled out agent 1, would either blow up or retain a memory of microscopic detail that no mean field could carry. This is the same intensive-versus-extensive distinction that underlies every thermodynamic limit, and it is why the empirical measure — introduced in chapter 1 as the rigorous meaning of "the state of the population" and used in chapters 7 and 8 as the argument the mean field converges to — is the only object through which the interaction may legitimately flow. The dynamics and costs are

$$\mathrm{d}X_t^i = b(X_t^i,\, \mu_t^N,\, \alpha_t^i)\; \mathrm{d}t \;+\; \sigma(X_t^i,\, \mu_t^N)\; \mathrm{d}W_t^i, \qquad X_0^i \sim m_0, \tag{13.2}$$

$$J^{N,i}(\alpha^1, \dots, \alpha^N) = \mathbb{E}\left[\int_0^T L(X_t^i,\, \mu_t^N,\, \alpha_t^i)\; \mathrm{d}t \;+\; g(X_T^i,\, \mu_T^N)\right]. \tag{13.3}$$

The coefficients b, σ, L, g are the *same* functions for every agent — the game is symmetric (exchangeable) — and they satisfy the standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) of appendix A, with the measure arguments measured in $\mathcal{W}_2$. We write $\mathcal{A}^\alpha$ for the controlled generator of a single state,

$$(\mathcal{A}^\alpha \phi)(x) = b(x, m, \alpha) \cdot \nabla\phi(x) + \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top(x, m)\; \mathrm{D}^2\, \phi(x)\big),$$

the operator that, by Itô's formula (theorem 5.9), gives the instantaneous expected rate of change of $\phi(X_t)$ along the controlled state; its first-order part carries the drift and its second-order part the diffusion. The diffusion coefficient enters this generator only through the symmetric positive-semidefinite matrix $\sigma\sigma^\top(x,m)$ — never through σ alone — because the second-order operator is $\frac{1}{2}\sum_{kl}(\sigma\sigma^\top)_{kl}\partial_{kl}$, so it is $\sigma\sigma^\top$, the local covariance of the noise, that is the analytically meaningful object. It would be standard shorthand to christen it $a(x,m) := \sigma\sigma^\top(x,m)$, exactly as one does in the general PDE literature, and we keep it spelled out instead, precisely as in chapter 12, for a deliberate notational reason that the reader should bank now: the symbol a is reserved across the whole book for the *state-feedback coefficient* of the LQG Thread A, the scalar in $\mathrm{d}X = (aX + b\alpha)\,\mathrm{d}t + \sigma\,\mathrm{d}W$. Were we to write a for the diffusion matrix here, the worked LQG computation later in this chapter would suffer a genuine collision — the same letter would mean $\sigma\sigma^\top$ in the general theory and the linear drift gain in the thread it specializes to. We therefore never abbreviate $\sigma\sigma^\top$, and the notationwatch preceding worked computation 13.8 restates the reservation at the seam where it bites.

Notation watch

Two empirical measures appear, differing by which agents are counted, and the gap between them is small but not always negligible — so it is worth getting the bookkeeping right once. The full empirical measure μ_t^N of eq. (13.1) includes agent i itself, all N atoms. When we isolate the environment *seen by* agent i — the population of competitors it actually responds to — we use the empirical measure of the *others*,

$$\mu_t^{N,-i} \;:=\; \frac{1}{N-1}\sum_{j\neq i}\delta_{X_t^j}.$$

Passing from one to the other deletes a single atom of mass $1/N$ and renormalizes the remaining $N-1$ atoms from weight $1/N$ to weight $1/(N-1)$. For the purpose of *rates* the two are interchangeable: their discrepancy is $o(r_N)$, negligible relative to the sampling rate r_N of lemma 13.5. But *how* negligible depends on the metric in which we measure it, and that dependence carries genuine dimensional intuition that recurs below.

In the weakest pairings — testing against a bounded function, or measuring in the flat/$\mathcal{W}_1$ distance — the discrepancy is a clean $O(1/N)$. Indeed, for a 1-Lipschitz (or bounded) test function ϕ,

$$\Big|\int\phi\;\mathrm{d}\mu_t^N - \int\phi\;\mathrm{d}\mu_t^{N,-i}\Big| = \Big|\tfrac{1}{N}\phi(X_t^i) - \tfrac{1}{N(N-1)}\sum_{j\neq i}\phi(X_t^j)\Big| = O(\tfrac{1}{N}),$$

because removing a single normalized atom and rescaling moves any average by at most a $1/N$ fraction. This $O(1/N)$ statement is exactly the one the value estimate eq. (13.22) uses, since the value comparison pairs the measures against the (Lipschitz) derivatives of the master field, i.e. against the flat/$\mathcal{W}_1$ structure. There the with-self/without-self distinction is beneath the resolution of the theorem.

In $\mathcal{W}_2$ the picture is subtler, and the subtlety is dimensional. $\mathcal{W}_2$ is an *optimal-transport* distance: it does not merely reweight mass, it must *move* it, and the cost of redistributing the deleted atom's $1/N$ of mass over the cloud depends on how spread out the cloud is, which is a function of dimension. The empirical measure of N i.i.d. points fills $\mathbb{R}^d$ at a typical inter-point spacing of order $N^{-1/d}$, so re-transporting an excess $1/N$ of mass can require displacements that, in dimension $d \geq 2$, exceed the naive $1/N$: the $\mathcal{W}_2$ discrepancy is then larger than the flat/$\mathcal{W}_1$ value $1/N$, though it stays strictly below the sampling error r_N. This is the same geometric effect that makes N samples resolve a measure only at the dimension-dependent Fournier–Guillin rate r_N of eq. (13.12) rather than at $1/N$: in high dimension, empty space between samples is the bottleneck. What rescues us is only that even this larger $\mathcal{W}_2$ discrepancy remains $o(r_N)$ in *every* dimension — the deletion of one atom is always asymptotically smaller than the sampling error of the whole cloud — which is all the estimates below require. We therefore keep the with-self/without-self distinction only where the $O(1/N)$ term is itself the quantity under study, as in the master-equation expansion of section 13.5, and elsewhere use whichever of μ_t^N, $\mu_t^{N,-i}$ reads more naturally.

We must now say precisely what an agent is allowed to know and do, and this is a genuinely consequential choice, not boilerplate. In the single-agent control of chapter 6 the information structure was immaterial: an optimally controlled isolated agent does equally well whether it commits to a deterministic schedule in advance or reacts to its own state in feedback, because the two coincide along the optimal path. In a game this is no longer so. What agent i may condition on — its own past noise, its own current state, the entire configuration of all players — determines what counts as a "deviation," hence which profiles are in equilibrium, and, as we will see in remark 13.10, it can even change the object the equilibria converge to. We therefore fix three nested information classes.

Definition 13.1 (Admissible strategies)**.** Let $\mathbb{F}^N = (\mathcal{F}_t^N)_{t\in[0,T]}$ be the filtration generated by all initial states and all Brownian motions.

1. An *open-loop* strategy for agent i is an $\mathbb{F}^N$-progressively measurable process $\alpha^i = (\alpha_t^i)$ valued in the action set $A \subseteq \mathbb{R}^k$ with $\mathbb{E}\int_0^T |\alpha_t^i|^2 \, \mathrm{d}t < \infty$.

2. A *Markovian* (closed-loop, feedback) strategy is given by a measurable map $\pi^i : [0,T]\times(\mathbb{R}^d)^N \to A$ used in feedback form $\alpha_t^i = \pi^i(t, \boldsymbol{X}_t)$, where $\boldsymbol{X}_t = (X_t^1, \dots, X_t^N)$; it is admissible if the closed-loop system eq. (13.2) is well-posed and the resulting control is square-integrable.

3. A *decentralized* (distributed) Markovian strategy is one in feedback form $\alpha_t^i = \pi(t, X_t^i)$ depending on agent i's own state alone.

We write $\boldsymbol{\alpha} = (\alpha^1, \dots, \alpha^N)$ for a profile and $\boldsymbol{\alpha}^{-i}$ for the profile of all agents but i.

The three classes are best read as three answers to the question "what does agent i watch?". An *open-loop* agent watches the raw randomness: it commits to a control process adapted to the filtration $\mathbb{F}^N$ generated by all the noises and initial states, so in principle it may react to the entire history of every Brownian path, but it does so through a fixed, pre-planned dependence on that randomness rather than on the realized positions. A *Markovian* or closed-loop agent watches the whole current configuration $\boldsymbol{X}_t$: it uses a feedback law $\pi^i(t, \boldsymbol{X}_t)$ that reads off every player's present state and responds instantaneously, the natural notion of a strategy that "sees the board." A *decentralized* agent watches only its own state X_t^i: its feedback $\pi(t, X_t^i)$ ignores the others entirely, the radically local information structure that a single representative agent in the mean field limit would in any case be left with, since in the limit the rest of the population has collapsed into the deterministic field m_t and there is nothing agent-specific left to watch. These are genuinely different powers, and the inclusions are strict: decentralized $\subset$ Markovian, while open- and closed-loop are not nested but encode different commitments. The decentralized class is the one the MFG hands us for free and the one we will test in the top-down direction; the Markovian class is the one in which the Nash system of section 13.2 is naturally posed; and the gap between open-loop and closed-loop equilibria is the source of a real subtlety in the *limit* — closed-loop equilibria can carry configuration-mediated correlations that survive as extra randomness in the limit, so that the two classes need not converge to the same object. We flag this now and return to it, with the resolution under monotonicity, in remark 13.10.

Having fixed *what an agent may know and do*, we can say what it means for a profile of such strategies to be self-consistent — for no agent, given the others' play, to wish to move. This is the equilibrium concept, and its ε-relaxation, that the rest of the chapter compares against the mean field.

Definition 13.2 (Nash and ε-Nash equilibria)**.** A profile $\boldsymbol{\alpha}^*$ is a *Nash equilibrium* of the N-player game eq. (13.2)–eq. (13.3) (within a given admissibility class) if no agent can strictly improve by a unilateral deviation:

$$J^{N,i}(\alpha^{*,i}, \boldsymbol{\alpha}^{*,-i}) \;\le\; J^{N,i}(\alpha^{i}, \boldsymbol{\alpha}^{*,-i}) \qquad \text{for all } i \text{ and all admissible } \alpha^i.$$

For $\varepsilon \ge 0$, the profile is an ε*-Nash equilibrium* if

$$J^{N,i}(\alpha^{*,i}, \boldsymbol{\alpha}^{*,-i}) \;\le\; J^{N,i}(\alpha^{i}, \boldsymbol{\alpha}^{*,-i}) \;+\; \varepsilon \qquad \text{for all } i \text{ and all admissible } \alpha^i. \tag{13.4}$$

That is, no unilateral deviation can lower an agent's cost by more than ε.

It is worth being clear about why we carry the slack ε at all, rather than chasing exact Nash equilibria of the finite game. The ε-relaxation is not a technical convenience to be removed later; it is the *only* honest target for the mean field strategy, and the reason is a fluctuation count. The decentralized feedback the MFG hands each agent, $\widehat{\alpha}(t, X_t^i)$, is the exact best response of a representative agent against the *deterministic* limiting field m_t. But in a finite game agent i does not face m_t; it faces the random empirical measure $\mu_t^{N,-i}$ of its $N-1$ competitors. By the central limit theorem, this empirical field deviates from its mean m_t by fluctuations of size $O(1/\sqrt{N})$: averaging $N-1$ independent contributions suppresses the variance like $1/N$, so the standard deviation of any smooth functional of $\mu_t^{N,-i}$ around its mean is of order $N^{-1/2}$, not zero. A strategy tuned to the mean field is, by construction, blind to these fluctuations — it does not even take $\mu_t^{N,-i}$ as an input. An agent that *did* watch the realized field could, in principle, exploit those $O(1/\sqrt{N})$ wiggles and shave a little off its cost, so the mean field strategy is generically *not* an exact best response, and the symmetric profile in which everyone plays it is generically *not* an exact Nash equilibrium, at any finite N. What saves the construction is that the cost is smooth in the field, so an $O(1/\sqrt{N})$ perturbation of the field can change the attainable cost by at most $O(1/\sqrt{N})$ — and, when the mean field control is a critical point, only by the square, $O(1/N)$. Either way the exploitable gain is a vanishing ε_N, never exactly zero. Demanding exact equilibrium would therefore be demanding the wrong thing; the ε-Nash inequality eq. (13.4) is the relaxation that exactly matches what the mean field strategy can deliver. Quantifying the smallest admissible ε — showing $\varepsilon_N \to 0$ and at what rate — is the content of section 13.4.

With the finite game fixed, the information classes pinned down, and the equilibrium concept (exact and relaxed) defined, we have said abstractly *what* a Markovian Nash equilibrium is but not *which equations* it must satisfy. Answering that question — writing down the system of coupled Hamilton–Jacobi–Bellman equations a Markovian equilibrium obeys — is the business of the next section.

13.2 The Nash system: N coupled HJB equations

The previous section closed by sharpening a question it could only pose: we have said abstractly *what* a Markovian Nash equilibrium is — a feedback profile from which no agent wishes to deviate — but not *which equations* its value functions must obey. For a single isolated optimizer that question had a clean answer in chapter 6: dynamic programming compressed the entire control problem into one Hamilton–Jacobi–Bellman (HJB) equation for the value function. We now have not one optimizer but N of them, each best-responding to the other $N-1$, and the natural guess is that equilibrium is encoded by N HJB equations, one per agent, knotted together by the fact that each agent's optimal play depends on what the others do. That guess is correct, and making it precise is the business of this section. The resulting object — the *Nash system* — is the finite-N analogue of the single HJB of chapter 6, and writing it down is the cleanest way to see two things at once: that the finite game is perfectly well-posed in principle, and that it is hopeless in practice. Both readings will matter, because the entire rest of the chapter is the project of escaping the second while keeping faith with the first.

Before any system can be written, we need the right Hamiltonian, and getting its conventions straight now will spare us sign confusion at every later step. Recall from chapters 6 and 8 that the role of the Hamiltonian is to pre-compute, once and for all, an agent's optimal instantaneous response to a given "momentum" — the local sensitivity $p = \nabla u$ of its value function to its own state. We use the convex (Lions) convention fixed in appendix A, with the canonical argument order $H(x, p, m)$ — state, then momentum, then measure — so that the standing convexity assumption (A-Conv) reads simply "H convex in p":

$$H(x,p,m) \;:=\; \sup_{\alpha\in A}\big\{-\,p\cdot b(x,m,\alpha)-L(x,m,\alpha)\big\}, \qquad \alpha^*(x,p,m) \;:=\; \operatorname*{arg\,max}_{\alpha\in A}\big\{-\,p\cdot b(x,m,\alpha)-L(x,m,\alpha)\big\}, \tag{13.5}$$

so that the closed-loop optimal drift of a single agent is $b(x, m, \alpha^*(x, p, m)) = -\partial_p H(x, p, m)$.

Three features of this definition deserve a sentence each, because each is a deliberate choice rather than an accident of typography. First, the *minus signs*. An agent *minimizes* cost, so the quantity it wants small is $p \cdot b + L$ — the drift term measures how the agent's motion changes its own cost-to-go through $p = \nabla u$, the running term the instantaneous price of acting. Taking the supremum of the *negated* bracket is exactly minimizing the bracket itself, and packaging the minimization as a sup is what lets us read off convexity for free: a supremum of functions affine in p is convex in p, so

(A-Conv) holds structurally rather than by fiat. Under (A-Conv) — strict convexity of L in α with b affine in α — the bracket is strictly concave in α, so the maximizer $\alpha^*(x,p,m)$ is *unique* and, by the implicit function theorem applied to its first-order condition, Lipschitz in p on bounded sets. That regularity is not cosmetic: it is exactly what makes the equilibrium feedbacks $\widehat{\alpha}^j$ below well-defined Lipschitz functions of the value gradients, hence what makes the closed-loop Nash system a genuine PDE rather than a differential inclusion. Second, the envelope identity $b(x,m,\alpha^*) = -\partial_p H$ is just the statement that one may differentiate the maximized bracket in p as if α^* were frozen — the derivative through α^* vanishes at the optimum by the first-order condition. It is the device by which the optimal *drift* re-enters the Fokker–Planck transport in the heuristic below, and we will use it without further comment.

Third, and most important for the bookkeeping ahead, this is the *measure-loaded* Hamiltonian: the population coupling enters through the measure slot m of b and L, is carried inside the bracket, and is therefore absorbed wholesale into $H(x,p,m)$. Keeping the measure slot explicit is not pedantry — the N-player running cost eq. (13.3) need not split cleanly into a control part and a population part, and where it does not, the coupling genuinely lives inside the optimization. It is worth seeing, however, exactly how this reconciles with the more familiar separated convention of chapter 12, where the coupling sat on the right-hand side of the HJB as a source. Suppose the data are *additively separable*: the running cost splits as $L(x,m,\alpha) = L_0(x,\alpha) + F(x,m)$, with the control-dependent part L_0 and the drift $b(x,\alpha)$ both free of the measure, while the entire population dependence is collected in the cost $F(x,m)$. Then the measure term is a constant as far as the maximization over α is concerned, and it pulls straight out of the sup:

$$H(x,p,m) = \sup_\alpha \{-p\cdot b(x,\alpha) - L_0(x,\alpha) - F(x,m)\} = \underbrace{\sup_\alpha \{ -p\cdot b(x,\alpha) - L_0(x,\alpha)\}}_{:=H_0(x,p)} - F(x,m) = H_0(x,p) - F(x,m)$$

Substituting this factorization into the single-agent HJB $-\partial_t u - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) + H(x,\nabla u, m) = 0$ moves F to the other side and reproduces verbatim the separated MFG–HJB of chapter 12 (eq. (12.2a)), $-\partial_t u - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u) + H_0(x,\nabla u) = F(x,m_t)$, with the population coupling reappearing as the source on the right and the terminal cost $g(x,m_T)$ playing the role of $G(x,m_T)$ there. The two conventions therefore agree *exactly* on the separable class; the measure-loaded form is strictly more general, and we carry it throughout precisely because the N-player coupling need not be separable. With the Hamiltonian fixed, we can state the system that a Markovian equilibrium must satisfy.

Proposition 13.3 (The Nash system)**.** *Suppose the N-player game admits a Markovian Nash equilibrium whose value functions*

$$v^{N,i}(t,\boldsymbol{x}) \;:=\; J^{N,i}(\alpha^{*,i},\boldsymbol{\alpha}^{*,-i}) \text{ started from } \boldsymbol{X}_t = \boldsymbol{x} \text{ at time } t, \qquad \boldsymbol{x} = (x_1,\dots,x_N) \in (\mathbb{R}^d)^N,$$

are of class $C^{1,2}$. Then, writing $\mu^N_{\boldsymbol{x}} = \frac{1}{N}\sum_k \delta_{x_k}$, the family $(v^{N,i})_{i=1}^N$ solves on $[0,T)\times(\mathbb{R}^d)^N$ the coupled system

$$-\partial_t v^{N,i} \;-\; \sum_{j=1}^N \tfrac{1}{2}\operatorname{tr}\Big(\sigma\sigma^\top(x_j,\mu^N_{\boldsymbol{x}})\, \mathrm{D}^2_{x_j} v^{N,i}\Big) \;-\; \sum_{j\neq i} b\Big(x_j,\mu^N_{\boldsymbol{x}},\, \widehat{\alpha}^j(t,\boldsymbol{x})\Big)\cdot\nabla_{x_j} v^{N,i} \;+\; H\Big(x_i,\nabla_{x_i} v^{N,i},\mu^N_{\boldsymbol{x}}\Big) \;=\; 0, \tag{13.6}$$

with terminal data $v^{N,i}(T,\boldsymbol{x}) = g(x_i, \mu^N_{\boldsymbol{x}})$*, where the equilibrium feedback of agent* j *is* $\widehat{\alpha}^j(t,\boldsymbol{x}) = \alpha^*(x_j, \nabla_{x_j} v^{N,j}(t,\boldsymbol{x}), \mu^N_{\boldsymbol{x}})$*. Conversely, a classical solution of eq.* (13.6) *with the indicated feedbacks verifies a Markovian Nash equilibrium.*

Proof sketch. The whole argument turns on a single device — "freeze the others" — and the subtle point is that the freezing must be *self-consistent* to deliver an equilibrium rather than merely a best response to an arbitrary environment. We unfold it in three movements.

Freeze, and a control problem appears. Fix one agent i and pretend, for the moment, that the other $N-1$ agents are committed to their equilibrium feedbacks $\widehat{\alpha}^j$, $j \neq i$. The definition of Nash equilibrium (definition 13.2) says precisely that, against this frozen environment, agent i's own equilibrium control is its *optimal* control — it has no profitable deviation. So $v^{N,i}$, the equilibrium cost-to-go of agent i, is the *value function of a single, ordinary stochastic control problem*, the one agent i solves when everyone else is glued to $\widehat{\alpha}^j$. The state of this control problem is the full configuration $\boldsymbol{X} = (X^1, \dots, X^N) \in (\mathbb{R}^d)^N$, but only the i-th block is steerable: agent i controls X^i through α, while each X^j with $j \neq i$ is an *uncontrolled* diffusion carrying the now fixed drift $b(x_j, \mu^N, \widehat{\alpha}^j)$ and the diffusion $\sigma(x_j, \mu^N)$. This is the conceptual move that makes everything tractable: an N-player game has been reduced, agent by agent, to N ordinary single-controller problems, each posed on the high-dimensional space $(\mathbb{R}^d)^N$ because each agent must track where everyone else is in order to know the cost it pays.

Dynamic programming gives an HJB on $(\mathbb{R}^d)^N$*.* To this single control problem we apply the dynamic programming principle of chapter 6 verbatim, only with the state space enlarged from $\mathbb{R}^d$ to $(\mathbb{R}^d)^N$. The generator of the configuration splits into the part agent i controls and the part it merely watches: the diffusion contributes a second-order term summed over *all* N blocks (every state, controlled or not, diffuses), the frozen drifts of the others contribute first-order transport terms summed over $j \neq i$, and agent i's own drift sits inside an optimization. Dynamic programming yields $\partial_t v^{N,i} + \sum_j \frac{1}{2} \operatorname{tr}(\sigma\sigma^\top(x_j, \mu^N)\, \mathrm{D}^2_{x_j} v^{N,i}) + \sum_{j\neq i} b(x_j, \mu^N, \widehat{\alpha}^j) \cdot \nabla_{x_j} v^{N,i} + \min_\alpha\{b(x_i, \mu^N, \alpha) \cdot \nabla_{x_i} v^{N,i} + L(x_i, \mu^N, \alpha)\} = 0$, where the inner $\min_\alpha$ is agent i optimizing its own instantaneous trade-off between steering $(b \cdot \nabla_{x_i} v^{N,i})$ and paying (L). By the very definition eq. (13.5) of the Hamiltonian as a *maximized negated* bracket, this inner minimum is $\min_\alpha\{\cdots\} = -H(x_i, \nabla_{x_i} v^{N,i}, \mu^N)$, and the minimizing control is read off as $\alpha^*(x_i, \nabla_{x_i} v^{N,i}, \mu^N)$. Substituting and moving the sign gives exactly the i-th equation of eq. (13.6).

Self-consistency upgrades best response to equilibrium. So far we have only used that agent i optimizes against *some* frozen feedbacks $\widehat{\alpha}^j$; the equation we obtained would hold against any fixed environment. What makes the family $(v^{N,i})_i$ an *equilibrium* rather than a collection of unrelated best responses is the closure condition that the minimizer we just computed for agent i, $\widehat{\alpha}^i = \alpha^*(x_i, \nabla_{x_i} v^{N,i}, \mu^N)$, must coincide with the very feedback the *other* agents froze for i when they solved their own problems. Each agent's optimal response, computed holding the others fixed, reproduces the feedback the others had assumed: the freezing is a fixed point, not an arbitrary choice. This is the finite-N incarnation of the self-consistency that runs through the whole subject — the same logic that in the limit will force m_t to be the law of the agent that optimally responds to m_t. It is also why the system is *simultaneous*: agent i's equation contains $\nabla_{x_j} v^{N,j}$ through $\widehat{\alpha}^j$, so no agent's value can be solved without all the others' at once.

The converse — that a classical solution of eq. (13.6) with the indicated feedbacks really does verify a Markovian Nash equilibrium — is the verification theorem of chapter 6 applied agent by agent: given a $C^{1,2}$ solution, the Itô expansion of $v^{N,i}$ along any admissible deviation of agent i shows the equilibrium feedback is optimal, with the Hamiltonian's defining inequality supplying the one-sided bound. Nondegeneracy (A-Nondeg) is what licenses all of this: it supplies the parabolic regularity (appendix C.9) that makes the candidate value $C^{1,2}$ in the first place and legitimizes the Itô expansion (Itô's formula, theorem 5.9). Full statements are in [40, Ch. 2] and [37, Ch. 1]. $\square$

Remark 13.4 (Why the Nash system is intractable)**.** The Nash system eq. (13.6) is, in the sense just proved, the *complete* answer to the finite game: it contains every Markovian equilibrium, exactly, for every N. And yet it is useless as something one could actually solve once N is more than a handful. It is worth diagnosing precisely *why*, because the diagnosis is the specification of the cure: each defect is repaired in a definite way by the mean field limit, and seeing the pairing now is the cleanest way to understand what the limit is *for*.

The first defect is *dimension*. Each value $v^{N,i}$ is a function on $[0,T] \times (\mathbb{R}^d)^N$, so a naive grid resolving it to fixed accuracy has a number of nodes exponential in Nd — the curse of dimensionality in its purest form, and it is fatal already at moderate N and d. The mean field limit attacks this head-on: it replaces the N functions on the Nd-dimensional configuration space by a *single* function $u(t,x)$ on the fixed, N-independent space $[0,T] \times \mathbb{R}^d$, together with one density flow m_t that carries everything the configuration used to carry. The dimension of the unknown stops growing with the population; this is the whole computational payoff of passing to the limit.

The second defect is *coupling*, and it is more insidious than dimension because it would survive even in low dimension. Through the equilibrium feedbacks $\widehat{\alpha}^j = \alpha^*(x_j, \nabla_{x_j} v^{N,j}, \mu^N)$, the equation for $v^{N,i}$ depends on the *gradients of the other value functions* $v^{N,j}$, $j \neq i$. The N equations are therefore genuinely simultaneous: one cannot solve for $v^{N,1}$, then $v^{N,2}$, and so on, because each presupposes all the others. This is the analytic fingerprint of the self-consistency in the proof above, and it is precisely what the limit dissolves: the mean field decouples the N-fold mutual dependence into a single forward–backward *pair*, the HJB for u driven by m_t and the Fokker–Planck equation for m_t driven by u, with the $N-1$ cross-gradient couplings collapsed into one scalar field. One representative agent against a field replaces N agents against each other.

The third feature is not a defect to be cured but a structure to be respected, and naming it guards against a natural misreading. One might hope the limit also sweeps away the awkward two-point boundary structure — value data prescribed at the terminal time T, density data at the initial time 0 — but it does not, and it should not. After the limit the terminal data on u and the initial data on m do not disappear; they become exactly the forward–backward MFG system of chapter 8. The forward–backward coupling is intrinsic to the problem — it expresses that rational agents anticipate the future while the crowd is pushed by the past — and any honest limit object must keep it. So the limit cures the two things that scale badly with N (the dimension and the N-fold simultaneous coupling) and preserves the one thing that is structural (the backward-value / forward-density pairing).

One simplification of the Nash system is available even before any limit, and it is worth isolating because it is exactly the lever the master equation will pull. Because the game is exchangeable —

the coefficients are invariant under relabelling the agents (section 13.1) — the N value functions are not genuinely distinct: they are all the *same* function of their arguments, evaluated at permuted configurations. Concretely one may seek $v^{N,i}(t,\boldsymbol{x}) = v^N(t, x_i, (x_j)_{j\neq i})$ with a single symmetric function v^N that singles out the "tagged" coordinate x_i and is symmetric in the remaining $N-1$, so that the entire family $(v^{N,i})_{i=1}^N$ is recovered by moving which coordinate is tagged. This collapses N unknowns to one, and it is the structural fact the master equation exploits: the master field $U(t,x,m)$ of chapter 12 is morally the $N\to\infty$ avatar of this one symmetric function, with the $N-1$ spectator coordinates absorbed into the measure argument m. But — and this is the point one must not overstate — exchangeability alone does *not* defeat the dimension. The single symmetric function v^N still lives on $(\mathbb{R}^d)^N$; it still depends on all N positions, just in a permutation-invariant way, and resolving a symmetric function of N arguments is still exponentially hard in N. What exchangeability buys is the *hope* that this symmetric function has a limit as $N\to\infty$ — a function of one tagged point and a measure — and turning that hope into a theorem with a rate is the work the master equation does. Exchangeability is the doorway; the master equation is what walks through it.

Heuristic (non-rigorous)

The mean field substitution is the self-consistent-field closure familiar from statistical mechanics — a single particle replaced by a particle in an averaged, self-generated field — transported here to a *strategic* setting where the field is generated by optimizers rather than by passive matter. Run it on eq. (13.6) and watch which terms survive. Suppose that as $N\to\infty$ the empirical measure μ_t^N ceases to fluctuate and converges to a deterministic flow $m_t\in\mathcal{P}(\mathbb{R}^d)$. Then every appearance of $\mu_{\boldsymbol{x}}^N$ in the Nash system, $\mu_{\boldsymbol{x}}^N \rightsquigarrow m_t$, becomes a frozen, exogenous datum: the agent no longer reacts to a random crowd but to a prescribed field. The diffusion term and the Hamiltonian term, which involve only agent i's own coordinate x_i, pass to the limit unproblematically. The delicate terms are the cross terms $\sum_{j\neq i} b(x_j, m_t,\cdot)\cdot\nabla_{x_j} v^{N,i}$, the ones encoding how agent i's value responds to the *motion of the others*, and the temptation is to wave them away as small. That temptation must be resisted, and resisting it correctly is the entire subtlety of the limit.

Here is the trap. One might argue that each cross term is negligible: agent j is a single atom of weight $1/N$ in the population, so surely its individual influence on agent i is $O(1/N)$ and may be dropped. The first half is right and the conclusion is wrong. Each summand is indeed $O(1/N)$ — but there are $N-1$ of them, and $(N-1)\cdot O(1/N) = O(1)$. The sum does *not* vanish by smallness; it is an order-one quantity, the collective influence of the whole crowd, and any argument that discards it "because each agent is small" is discarding the mean field itself. This is exactly the intensive-versus-extensive distinction of section 13.1: no single agent matters, but the average over all of them matters at leading order.

So why is there no analogue of these cross terms in the limiting single-agent HJB? The reason is *structural*, not a matter of magnitude. In the limit, the object that replaces $v^{N,i}$ is the representative value $u(t,x_i)$, and — this is the whole content of passing to a representative agent — it depends on agent i's *own* state x_i and the exogenous field m_t alone, with no residual dependence on the labelled positions x_j of the spectators. Therefore $\nabla_{x_j} u = 0$

identically for every $j \neq i$: the cross terms still have order-one total coefficient, but in the limit they act on a function that is *flat* in the spectator coordinates, so each $b \cdot \nabla_{x_j} u$ is zero not because the coefficient is small but because there is nothing for it to differentiate. The crowd's influence has not been neglected; it has been *relocated*. Having stripped the value of its dependence on individual spectators, we paid for that simplification by making the value depend on the *measure* m_t instead — and that is exactly where the order-one collective effect re-emerges.

It is worth saying precisely where, because this prefigures the quantitative theory of section 13.5.3 and is the seam at which this heuristic becomes a theorem. Promote the representative value to the master field $U(t, x, m)$ of chapter 12, which restores the measure as an explicit argument. Differentiating U in its measure slot is the Lions derivative $\partial_\mu U(t, x, m)(\cdot)$, and the chain rule for how U changes as the population is transported by the drift b produces exactly a term $\int b(y, m, \cdot) \cdot \partial_\mu U(t, x, m)(y)\, m(\mathrm{d}y)$. This integral *is* the continuum limit of the order-one average $\frac{1}{N} \sum_{j \neq i} b(x_j, m_t, \cdot) \cdot [\,\cdot\,]$ that the structural flatness made vanish from the x_j-slots: what looked like $N - 1$ small cross terms acting on a flat value reappears, intact and order-one, as a single transport integral acting through the measure derivative. The master equation of chapter 12 is precisely the equation that carries this term; section 13.5.3 will turn the informal chain rule here into the rigorous expansion that bounds its residual against the Nash system by $O(1/N)$. Nothing was lost in the limit — the crowd's order-one influence was merely rewritten from a sum over labelled neighbours into an integral against the population.

What remains, then, for the typical agent is a *single* HJB equation, the representative value moving in the self-consistent field,

$$-\partial_t u - \tfrac{1}{2} \operatorname{tr}\left(\sigma\sigma^\top(x, m_t)\, \mathrm{D}^2 u\right) + H(x, \nabla u, m_t) = 0, \qquad u(T, \cdot) = g(\cdot, m_T),$$

driven by the field m_t. The closure is the strategic version of self-consistency: the field m_t that the agent best-responds to must itself be the law of that optimally responding agent, which is the Fokker–Planck equation of chapter 8 and turns the bare postulate "$\mu_t^N \to m_t$" into a genuine fixed-point condition — the field generates the optimal behaviour that in turn regenerates the field. The one non-rigorous move in all of this is the very first: the suppression of fluctuations, the assertion that μ_t^N ceases to wobble and freezes to m_t. Making that move precise — with the right notion of equilibrium, controlling the suppressed fluctuations, and extracting a rate — is not a detail but the whole remaining content of the chapter.

We have arrived at the chapter's central tension, and it is worth naming sharply before we act on it. The Nash system is a *sharp* object: it captures the finite game exactly, with nothing approximated and nothing lost. It is also a *hopeless* one: its unknowns live on a space whose dimension grows with the population, and its equations are bound together by mutual self-consistency. The heuristic has just shown us the escape we want — one representative agent, one HJB, one transported field — but it bought that escape with an act of faith, the suppression of fluctuations, that no theorem yet underwrites. The remaining question is therefore not *what* we want the limit to be, but *how* the passage to it is to be justified. And here the structure of the problem forces a fork, because the limit can be approached from either end of the finite-to-infinite passage, and the two ends are of sharply unequal difficulty. One may start from the clean limit object and test it against the finite game; or one may start from the finite game's actual equilibria and ask whether they converge. Whether the interaction one must control is handed to us in advance or emerges from the very fixed point we are solving for makes these two routes profoundly different in cost. Organizing the escape into these two opposite directions — and seeing exactly why one is cheap and the other deep — is the task of the next section.

13.3 The two limits: top-down and bottom-up

The heuristic of the previous section left exactly one debt unpaid, and the whole remaining chapter is the schedule on which it is repaid. It delivered the object we want — one representative agent, one HJB, one transported field — but it delivered it by an act of faith: the single unjustified line was the assertion that the empirical measure μ_t^N stops fluctuating and freezes onto a deterministic flow m_t. Suppress the fluctuations and everything else in the heuristic follows mechanically; but suppressing them is precisely the step a theorem must earn and a heuristic merely assumes. What divides this subject's wishes from its theorems is, almost entirely, the discipline with which that one fluctuation-suppression step is discharged. So the live question is no longer *what* the limit should be — the Nash system told us the finite truth, and the heuristic told us the shape of its coarse-graining — but *how* the suppression of fluctuations is to be made rigorous against the sharp-but-hopeless finite object we are trying to escape.

The first thing to see is that this can be done in two opposite directions, according to which end of the finite-to-infinite passage one holds fixed, and that the two are not two proofs of one fact but two genuinely different tasks of sharply unequal difficulty. It pays to separate them cleanly before proving anything, because the literature sometimes runs them together, and running them together is exactly how one mistakes the cheap half of the chapter for the whole of it. The cause of the asymmetry has a one-word name we have already met: whether the interaction whose fluctuations we must suppress is *exogenous* — fixed in advance, handed to us by the limit problem — or *endogenous* — itself the unknown we are solving for. The full contrast is drawn in section 13.5.1; here we lay out the map.

Top-down (ε-Nash): testing a known field. The first direction holds the *infinite* end fixed. Start from a *solution* of the MFG system — a pair (u, m), with u solving the backward HJB driven by the field and m the forward law of the optimally controlled representative agent (chapters 8 and 9). This pair is assumed already in hand: the entire fixed-point problem of chapters 8, 9 and 11 has been solved *once*, on the fixed space $\mathbb{R}^d$, before any finite game is even considered. Out of it the MFG manufactures a single decentralized feedback,

$$\widehat{\alpha}(t,x) \;:=\; \alpha^*(x,\, \nabla u(t,x),\, m_t), \tag{13.7}$$

a map of an agent's *own* state and time alone — no spectator coordinates, no empirical measure, nothing agent-specific to watch but oneself. Now perform the only experiment that needs no further theory: hand this one feedback to each of the N players and let them play it simultaneously. *Question:* how far short of a Nash equilibrium of the finite game does the resulting symmetric profile fall?

What makes this direction cheap is that handing every agent the *same* fixed feedback closes the N-body loop into a dynamics that is already fully understood. With $\widehat{\alpha}$ frozen, the drift of each X^i becomes $b\big(X_t^i, \mu_t^N, \widehat{\alpha}(t, X_t^i)\big)$ — a prescribed function of the agent's own state and the empirical measure, carrying no other agent's control as an unknown. That is exactly a McKean–Vlasov interacting particle system of the kind chapter 7 governs: N exchangeable diffusions coupled only through their own empirical measure, with a *given* law of motion. The fluctuation-suppression step is then not an article of faith but a theorem we already own — propagation of chaos for a fixed dynamics — and it applies verbatim, with no new idea required. The empirical measure of this system concentrates on the flow m_t at the sampling rate, and the answer to the question (section 13.4) is "within $\varepsilon_N \to 0$."

This direction is constructive, cheap, and — for an engineer rather than a theorist — the practically decisive one. It does not ask the finite game to disclose its own equilibria; it *manufactures* an explicit, decentralized, near-optimal strategy and then certifies it. Compute the mean field solution once, on the fixed space $\mathbb{R}^d$, and deploy the very same feedback on every large finite population, with a guarantee that no agent can gain more than ε_N by deviating. The entire N-dependence has been quarantined into the single scalar ε_N; the strategy itself does not depend on N at all. This is the sense in which the mean field, computed once, is a usable approximate equilibrium for an entire family of finite games at a stroke.

Bottom-up (convergence of equilibria): an emergent field. The second direction holds the *finite* end fixed, and it is the foil to the first in every respect. Start now from the *actual* Nash equilibria of the N-player game — the solutions $v^{N,i}$ of the Nash system eq. (13.6), together with their equilibrium feedbacks $\widehat{\alpha}^j$ and the law of the equilibrium state process they generate. *Question:* do these converge, as $N \to \infty$, to the MFG solution — the values $v^{N,i}$ to u, the empirical measure of the equilibrium to m_t, the finite marginals to products of m_t? The answer (section 13.5) is again "yes," but it is a yes of an entirely different order, and the reason is that here the interaction is *endogenous*.

There is no fixed dynamics to feed chapter 7. In the top-down direction the feedback was an *input*, frozen before the particles moved; here the feedback each agent uses is *itself the unknown*, the gradient $\nabla_{x_j} v^{N,j}$ of a value function that is being solved for at the same time. The law whose fluctuations we must suppress is the law *of the equilibrium* — the very object we are trying to characterize — so we are asked to show that something concentrates before we are even allowed to name what it concentrates on. The self-reference is the crux, and it is worth stating in one breath: the empirical measure μ_t^N is generated by feedbacks that, at every instant, depend on μ_t^N. One therefore cannot first fix the dynamics and then average, as the McKean–Vlasov picture demands; the averaging and the dynamics are pinned down *together*, by the fixed point, and neither is available without the other. Chaos, in this direction, is not a hypothesis imposed on a known law of motion but an *emergent property of the equilibrium* — propagation of chaos for Nash equilibria — and there is simply no kernel to hand to chapter 7.

Because the chaos must be *extracted* from the fixed point rather than *assumed* of the dynamics, two genuinely different routes are open, and it is worth fixing the map now so that the later sections read as a plan rather than a pair of surprises. The first is a *compactness* route: assume as little as possible, show that the family of N-player equilibria is tight, extract a convergent subsequence, and identify every one of its limit points as a solution of the MFG system; under the Lasry–Lions monotonicity (A-Mon) of chapter 11 that solution is unique, so a single limit point forces the whole family to converge. This buys *qualitative* convergence — the limit exists and is the MFG — but no rate, because a compactness argument never sees how fast. The second is the *master-equation* route: take the regular solution $U(t, x, m)$ of chapter 12 as an ansatz for the $v^{N,i}$, substitute it into the Nash system, and bound the residual it leaves behind. The regularity of U is exactly what converts that residual into an $O(1/N)$ error, and so delivers a *rate*. The first route is robust and cheap in hypotheses but mute about speed; the second is expensive in hypotheses — it presupposes a smooth master field, which exists only in the monotone regime — but quantitative. section 13.5.1 draws the contrast in full; section 13.5.3 carries out the master-equation expansion.

Physics note

The two directions are the strategic counterparts of two familiar but quite distinct moves in many-body physics, and naming the physics sharpens why one is so much cheaper than the other. The top-down direction is the *test-particle* calculation. One freezes the self-consistent field at its mean-field value m_t, drops a single tagged particle into it, and asks only how that one particle fares. The single thing the tagged agent gets wrong is that the field it *actually* samples is not the smooth m_t but the granular empirical field $\mu_t^{N,-i}$ assembled from its $N-1$ neighbours — and that field, being an average of $N-1$ independent contributions, departs from its mean only by central-limit fluctuations of size $N^{-1/2}$. Here is the decisive point: because the mean field strategy is optimal against the *average* field, it sits at a critical point of the cost in the field variable, so a fluctuation of order $N^{-1/2}$ about that critical point moves the attainable cost only at *second* order, and the tagged agent's exploitable gain is the small ε_N of section 13.4. A probe in a bath, in other words, barely notices the bath's graininess, and the little it notices it cannot easily exploit.

The bottom-up direction asks the strictly harder thing: not whether a probe survives the bath, but whether the bath is *itself* mean-field. The claim is that the fully interacting N-body equilibrium — every agent simultaneously best-responding, all correlations switched on, no field frozen anywhere — is nonetheless described, as $N \to \infty$, by the same single-particle field m_t. That is a genuine self-averaging statement about the interacting system, of the kind that in equilibrium statistical mechanics one earns only with real work (cluster expansions, correlation-decay estimates), not a perturbation of a known background. The whole gap between the two directions is the gap between a one-body and a true many-body computation. The master equation is what bridges it: a sufficiently regular $U(t, x, m)$ controls the equilibrium's correlations tightly enough to upgrade the probe-in-a-bath statement into the bath-is-mean-field statement, and to do so with a rate. That bridge is the business of section 13.5.3; everything before it is the cheaper, test-particle half.

With the map drawn, we take the two directions in order of difficulty, easiest first. The next section makes the top-down promise precise and turns the slogan "within $\varepsilon_N \to 0$" into a theorem with an explicit rate. The bottom-up direction — convergence of the genuine equilibria, first by compactness and then by the master equation — waits for section 13.5, once the cheap direction has shown us what a rate even looks like when the dynamics are handed to us for free.

13.4 The ε-Nash property of the mean field strategy

The previous section drew the map and bound us to walking the cheap road first; this section is that road, taken from one end to the other. The plan it handed us was completely explicit, and we execute it in exactly its order: freeze the mean field feedback, hand the one map to all N players at once, recognize the closed loop as a McKean–Vlasov system of precisely the kind chapter 7 already governs, and so turn the slogan "within $\varepsilon_N \to 0$" into a theorem with a rate. The work will divide cleanly. A single probabilistic input — propagation of chaos for a fixed dynamics, lemma 13.5 — carries all the analysis; after it, the ε-Nash theorem is little more than a triangle inequality between three control problems.

Fix, then, an MFG solution (u, m) as in section 13.3 — with u the backward value and m the forward flow it is self-consistent with — and the decentralized feedback $\widehat{\alpha}$ it manufactures, eq. (13.7). The strategy we put on trial is the *mean field profile*: the symmetric profile in which every one of the N agents, watching nothing but its own state, plays the same own-state feedback,

$$\alpha_t^i \;=\; \widehat{\alpha}(t, X_t^i), \qquad i = 1, \dots, N. \tag{13.8}$$

Nobody consults the empirical measure, nobody consults a spectator's position; each agent simply runs the map that a lone representative agent would run against the deterministic field m_t. The question of the section is the one posed in section 13.3: by how much can a single agent improve on this profile by deviating, and does that margin vanish as $N \to \infty$?

Everything cheap about this direction flows from one structural observation, and it repays making slowly, because it is precisely the feature the bottom-up direction will *lack*. Install the common feedback in every agent at once: substitute $\alpha_t^i = \widehat{\alpha}(t, X_t^i)$ into the state dynamics eq. (13.2). The drift of agent i collapses to

$$b(X_t^i, \mu_t^N, \widehat{\alpha}(t, X_t^i)) \;=\; \widetilde{b}(X_t^i, \mu_t^N), \qquad \widetilde{b}(x, m) := b(x, m, \widehat{\alpha}(t, x)),$$

and its diffusion is $\sigma(X_t^i, \mu_t^N)$ as before. Read off the arguments on the right-hand side: each is either agent i's *own* state X_t^i or the empirical measure μ_t^N of the whole cloud — and nothing else. In particular *no other agent's control survives as an unknown.* This is the decisive contrast with the Nash system eq. (13.6), whose drift carried the equilibrium feedbacks $\widehat{\alpha}^j = \alpha^*(x_j, \nabla_{x_j} v^{N,j}, \mu^N)$ and so dragged in the gradients of every other value function: there the law of motion was knotted to the very equilibrium one was still solving for. Here, because $\widehat{\alpha}$ was fixed once and for all by the already-solved limit problem, the law of motion is *prescribed* in advance. The closed-loop state system eq. (13.2) under the mean field profile eq. (13.8) is therefore exactly a *McKean–Vlasov interacting particle system* of the type studied in chapter 7: N exchangeable diffusions with common coefficients $(\widetilde{b}, \sigma)$, coupled solely through their shared empirical measure, with a given — not endogenous — law of motion.

For such a system chapter 7 tells us exactly what to expect. As $N \to \infty$ the empirical measure ceases to fluctuate and freezes onto a deterministic flow, while a single tagged particle decouples from the rest and follows the *McKean–Vlasov SDE of the representative agent*,

$$\mathrm{d}\bar{X}_t = \widetilde{b}(\bar{X}_t, m_t)\ \mathrm{d}t + \sigma(\bar{X}_t, m_t)\ \mathrm{d}W_t, \qquad \mathcal{L}(\bar{X}_t) = m_t, \tag{13.9}$$

in which the empirical measure has been replaced by the agent's own time-marginal law. Now comes the point that makes the construction self-consistent rather than circular, and it is worth stating with care. The flow m_t that appears in eq. (13.9) is *not* a new unknown to be solved for; it is the *same* m we started from. By construction of the MFG solution, $\widehat{\alpha}$ is the optimal feedback of the representative agent against m, and the second (Fokker–Planck) equation of the MFG system of chapter 8 states precisely that m_t is the law of the state process driven by that feedback — which is eq. (13.9) verbatim. So the consistency condition $\mathcal{L}(\bar{X}_t) = m_t$ holds by the very fixed-point equation that defined the MFG solution: the limiting law of the particle system *is* the mean field flow, by construction (chapters 8 and 9), with nothing left to verify. All that now separates the finite profile from this clean limit is the suppression of the empirical measure's fluctuations — and that, for a fixed law of motion, is a theorem we already own. We isolate it as the single analytic input that the rest of the section will consume.

Lemma 13.5 (Propagation of chaos for the mean field profile)**.** *Assume* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv) *and that the feedback $\widehat{\alpha}$ is bounded and Lipschitz in x uniformly in t, so that $\widetilde{b}(\cdot,\cdot)$ is Lipschitz in (x, m) in the sense of* (A-Lip)*. Let $(X^i)_{i\le N}$ solve eq.* (13.2) *under the mean field profile eq.* (13.8) *and let $(\bar{X}^i)_{i\le N}$ be i.i.d. copies of eq.* (13.9) *driven by the same Brownian motions and initial states. Then*

$$\max_{i\le N}\ \mathbb{E}\Big[\sup_{t\le T}\big|X^i_t - \bar{X}^i_t\big|^2\Big]\ \le\ C\,r_N^2, \qquad \mathbb{E}\Big[\sup_{t\le T}\mathcal{W}_2\big(\mu^N_t,\, m_t\big)^2\Big]\ \le\ C\,r_N^2, \tag{13.10}$$

where C depends on T, the Lipschitz constants and $\int |x|^2\, m_0(\mathrm{d}x)$ but not on N. Here r_N denotes the $\mathcal{W}_2$ sampling rate *of N i.i.d. draws, fixed once and for all as the root-mean-square empirical-measure error,*

$$r_N^2\ :=\ \sup_{\rho}\ \mathbb{E}\Big[\mathcal{W}_2\big(\bar{\mu}^N, \rho\big)^2\Big]\,, \tag{13.11}$$

the supremum over the laws $\rho = m_t$, $t \le T$ (each with a finite moment of order $q > 4$; under (A-Growth) *the flow m_t inherits the moments of m_0, so this is a mild requirement on the initial law). It is crucial that r_N is defined through the* second *moment $\mathbb{E}\mathcal{W}_2^2$, the quantity that the synchronous coupling actually produces, and not through $\mathbb{E}\mathcal{W}_2$ (squaring the latter would run Jensen's inequality the wrong way). This is the* Fournier–Guillin *rate, proved in-book — via a dyadic multiresolution comparison and a binomial-variance bound — in theorem D.16 (appendix D; the role of the rate in propagation of chaos is also recorded in remark 7.16; attribution [69]):*

$$r_N^2\ \lesssim\ \begin{cases} N^{-1/2}, & d < 4,\\ N^{-1/2}\log N, & d = 4,\\ N^{-2/d}, & d > 4,\end{cases} \qquad \textit{so that } r_N \lesssim N^{-1/4} \textit{ for } d \le 3. \tag{13.12}$$

The same constant and rate govern every appearance of r_N below.

Proof sketch. This is the synchronous-coupling argument of chapter 7 (Sznitman's coupling) applied to the fixed coefficients $\widetilde{b}, \sigma$; we recall its shape to make visible *which* part of the rate is structural.

Subtract eq. (13.9) from the closed loop of eq. (13.2) for the same i and apply Itô's formula (theorem 5.9) to $|X_t^i - \bar{X}_t^i|^2$; the stochastic-integral part is a martingale (the diffusion coefficients are matched, so it carries the difference $\sigma(X^i, \mu^N) - \sigma(\bar{X}^i, m_t)$), and the drift part together with the Lipschitz bound on $\widetilde{b}, \sigma$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\Big[|X_t^i - \bar{X}_t^i|^2\Big] \le C\,\mathbb{E}\Big[|X_t^i - \bar{X}_t^i|^2\Big] + C\,\mathbb{E}\Big[\mathcal{W}_2(\mu_t^N, \bar{\mu}_t^N)^2\Big] + C\,\mathbb{E}\Big[\mathcal{W}_2(\bar{\mu}_t^N, m_t)^2\Big],$$

where $\bar{\mu}_t^N = \frac{1}{N}\sum_j \delta_{\bar{X}_t^j}$. The middle term is bounded by $\frac{1}{N}\sum_j \mathbb{E}|X_t^j - \bar{X}_t^j|^2$ (coupling the empirical measures atom by atom: with μ_t^N and $\bar{\mu}_t^N$ supported on the paired points X_t^j and $\bar{X}_t^j$, the diagonal coupling that matches atom j to atom j is one admissible transport plan, so $\mathcal{W}_2(\mu_t^N, \bar{\mu}_t^N)^2 \le \frac{1}{N}\sum_j |X_t^j - \bar{X}_t^j|^2$, and by exchangeability each summand has the same expectation as the i-th). The last term is the *sampling error* of N i.i.d. draws from m_t — here is where driving both systems by the *same* Brownian motions and initial states pays off, for it makes the $\bar{X}_t^j$ genuinely i.i.d. samples of m_t, so $\mathbb{E}\mathcal{W}_2(\bar{\mu}_t^N, m_t)^2$ is exactly r_N^2 of eq. (13.12) (theorem D.16). Grönwall's inequality (appendix C.1) closes the mean-square estimate. Passing from $\mathbb{E}|\cdot|^2$ to the *supremum-in-time* bound $\mathbb{E}\sup_{t\le T}|\cdot|^2$ asserted in eq. (13.10) is the one place a maximal inequality is needed: one applies the Burkholder–Davis–Gundy / Doob L^2 inequality (theorem C.2) to the martingale part before invoking Grönwall, exactly as in the SDE well-posedness estimate of theorem 5.10.

The shape of this estimate repays one more sentence, because it is the whole moral of the top-down direction. Two sources of error entered, and they are of genuinely different character. The middle (coupling) term is the discrepancy between the interacting particles X^j and their decoupled limits $\bar{X}^j$; it is exponentially controlled by Grönwall, and — this is the point — it would vanish *identically* if the particles were already independent, for then $X^j \equiv \bar{X}^j$ and the term is zero. It measures only the residual interaction, and the interaction is precisely what Grönwall dissolves. The last term is different in kind: r_N is the error one commits in replacing a measure m_t by the empirical measure of N *exact* i.i.d. samples of it, and it is present even for a non-interacting cloud, even when the coupling term is exactly zero. It is therefore irreducible — no improvement in the coupling can beat it — and it, not the interaction, is what sets the rate in eq. (13.10). The dimension thresholds of eq. (13.12) are the fingerprint of this irreducible term alone: N samples resolve a measure at the slow, dimension-dependent Fournier–Guillin rate because in high dimension the empty space between samples is the bottleneck, exactly the geometry discussed in the notationwatch of section 13.1. One last subtlety, easy to lose and costly to get wrong: the synchronous coupling natively produces second moments, $\mathbb{E}\mathcal{W}_2^2$, and that is exactly the quantity eq. (13.11) defines r_N^2 to be. Defining r_N instead through $\mathbb{E}\mathcal{W}_2$ and squaring would invoke Jensen in the wrong direction — $(\mathbb{E}\mathcal{W}_2)^2 \le \mathbb{E}\mathcal{W}_2^2$, so the squared first moment is the *weaker* bound and would not close the loop — which is why the lemma is careful to anchor r_N to the second moment the coupling actually delivers. Full details: [40, Ch. 1], [69]. ☐

Theorem 13.6 (ε-Nash property of the mean field strategy)**.** *Assume* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv)*, and that the MFG solution* (u, m) *has a feedback* $\widehat{\alpha}$ *that is bounded and Lipschitz in* x*, uniformly in* t*. Assume moreover that the running and terminal costs are Lipschitz in the measure argument in* $\mathcal{W}_2$*, uniformly in* (x, α) *on the relevant range* (part of (A-Lip))*. Then the mean field profile eq.* (13.8) *is an* ε_N*-Nash equilibrium of the* N*-player game in the open-loop class,* *with*

$$\varepsilon_N \;\le\; C\, r_N \;\xrightarrow[N\to\infty]{}\; 0, \tag{13.13}$$

r_N the sampling rate eq. (13.12) of lemma 13.5 and C independent of N. In particular $\varepsilon_N \le C\, N^{-1/4}$ when $d \le 3$.

A word on the deviation class, because information structure was flagged in definition 13.1 as genuinely consequential and we should be precise about which choice is at work here. The inequality eq. (13.13) quantifies over *open-loop* deviations: the tagged agent may use any square-integrable control adapted to the noise, but the $N-1$ others stay glued to $\widehat{\alpha}$. The reason the bound is so robust to this choice is exactly the mechanism isolated in lemma 13.5. A single deviator is one atom of vanishing weight $1/N$ in the empirical measure, so the field it experiences — generated by the $N-1$ others, who keep playing $\widehat{\alpha}$ — is unaffected, to leading order, by *how* that one agent plays; the deviator cannot, by deviating, materially move the environment it is responding to. The same conclusion therefore holds verbatim against *closed-loop* (feedback) deviations in the bounded-Lipschitz class, by the identical estimate applied to the closed-loop McKean–Vlasov system the deviator now induces [40, Ch. 6]. We state the open-loop version only because it is the weakest hypothesis under which the cost comparison of the proof is immediate, and because it is the one an engineer cares about: it certifies the mean field strategy against the broadest reasonable class of unilateral defections.

It is worth separating this *deviation-class* question sharply from a second, genuinely different question that the same words "open-loop" and "closed-loop" also name, lest the two be conflated. Here the class enters only through what a single deviator is permitted to watch while everyone else holds a *fixed* profile; and because the field is insensitive to a lone atom, the answer does not depend on the class. The *equilibrium-class* question is different in kind: it asks in which class the equilibrium *itself* is sought — whether the N players are all best-responding in open-loop or all in closed-loop — and there the two classes do not agree. Closed-loop Nash equilibria can sustain configuration-mediated correlations among the players that survive into the limit as genuine extra randomness, so that open- and closed-loop equilibria need not even converge to the same object. That divergence is a property of the *bottom-up* direction, where the feedback is endogenous and the field is no longer immune to how the population plays, and it is taken up under monotonicity in remark 13.10. The present theorem lives entirely on the deviation-class side, where the distinction is invisible, precisely because the profile under test is held fixed.

Proof. Fix a tagged agent i and a candidate deviation $\alpha^i = \nu$, an admissible open-loop control, while every other agent keeps $\widehat{\alpha}$. The proof is a triangle inequality between three control problems for agent i: (i) the finite game with the deviation; (ii) the same with $\widehat{\alpha}$; (iii) the limiting MFG control problem against the frozen field m.

Step 0: reduction to deviations of bounded cost. The first move is one of pure economy: we may discard, for free, every deviation that does not even threaten the inequality we are trying to prove. The ε-Nash inequality eq. (13.4) is a statement only about deviations that could *help* the deviator, so it suffices to establish the regret bound for controls ν satisfying

$$J^{N,i}(\nu, \widehat{\alpha}^{-i}) \;\le\; J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i});$$

any ν violating this is already worse than $\widehat{\alpha}$, so it satisfies eq. (13.4) trivially, for every $\varepsilon_N \geq 0$, and needs no argument. This restriction is not mere tidiness; it is exactly what makes the field estimate of Step 1 *uniform*, and without it the proof would have a real gap. Admissibility alone — the bare square-integrability $\mathbb{E}\int_0^T |\nu_t|^2\,\mathrm{d}t < \infty$ of definition 13.1 — supplies *no* uniform-in-ν moment bound on the state: a sequence of admissible controls can send $\mathbb{E}\sup_t |X_t^i|^2$ to infinity. Such a runaway deviator would place into μ_t^N a single atom sitting at an uncontrolled position X_t^i, of weight $1/N$ but of $\mathcal{W}_2$-displacement $\frac{1}{N}|X_t^i|$ that need not be small at all, and that one wandering atom could by itself destroy the closeness of the empirical field to m_t on which Step 1 depends. The cost-no-larger reduction closes exactly this loophole, and the following chain is what turns it into the moment bound Step 1 needs.

First, the no-deviation cost is itself bounded. Under the mean field profile the states have second moments bounded uniformly in N: the limiting representative agent eq. (13.9) inherits finite moments from m_0 through (A-Growth), and lemma 13.5 transfers them to the finite system, so $J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i}) \leq C$ with C independent of N. Every retained deviation therefore obeys $J^{N,i}(\nu, \widehat{\alpha}^{-i}) \leq C$ as well. Second, coercivity converts this cost bound into a control bound. By the coercivity built into (A-Growth) the running cost dominates a positive multiple of the control energy, $L(x, m, \alpha) \geq \kappa|\alpha|^2 - C(1 + |x|^2)$ for some $\kappa > 0$, so dropping the nonnegative terminal cost and rearranging eq. (13.3),

$$\kappa\,\mathbb{E}\int_0^T |\nu_t|^2\ \mathrm{d}t \ \leq\ J^{N,i}(\nu, \widehat{\alpha}^{-i}) + C\,\mathbb{E}\int_0^T \left(1 + |X_t^i|^2\right)\ \mathrm{d}t.$$

Third, the state moment is in turn controlled by the control energy, through Itô and Grönwall. Applying Itô's formula (theorem 5.9) to $|X_t^i|^2$ along the deviated dynamics eq. (13.2) and taking expectations annihilates the martingale part, leaving

$$\mathbb{E}|X_t^i|^2 \ =\ \mathbb{E}|X_0^i|^2 + \mathbb{E}\int_0^t \Big(2\,X_s^i\cdot b(X_s^i, \mu_s^N, \nu_s) + \mathrm{tr}\ \sigma\sigma^\top(X_s^i, \mu_s^N)\Big)\,\mathrm{d}s.$$

The at most linear growth of b and σ in (x, α) from (A-Growth), together with the elementary $2\,x\cdot b \leq |x|^2 + |b|^2$, bounds the integrand by $C\big(1 + |X_s^i|^2 + |\nu_s|^2\big)$, whence

$$\mathbb{E}|X_t^i|^2 \ \leq\ C\Big(1 + \mathbb{E}\int_0^T |\nu_s|^2\ \mathrm{d}s\Big) + C\int_0^t \mathbb{E}|X_s^i|^2\ \mathrm{d}s.$$

Now close the loop between the second and third displays. The third bounds the state moment by the control energy plus a self-term; Grönwall's inequality (appendix C.1) removes the self-term and gives $\sup_{t\leq T}\mathbb{E}|X_t^i|^2 \leq C\big(1 + \mathbb{E}\int_0^T |\nu_s|^2\,\mathrm{d}s\big)$. Feeding this back into the second display, the $\mathbb{E}\int(1 + |X^i|^2)$ term is now itself bounded by a constant plus the control energy, and (with T finite, after the standard absorption of a small multiple of $\mathbb{E}\int|\nu|^2$ from the right) one obtains first $\mathbb{E}\int_0^T |\nu_t|^2\,\mathrm{d}t \leq C$ and then $\mathbb{E}\sup_{t\leq T}|X_t^i|^2 \leq C$ — the supremum coming from a final application of the Burkholder–Davis–Gundy/Doob bound (theorem C.2) to the martingale part, exactly as in theorem 5.10. The constant C is uniform in N and across the retained deviations ν. This single uniform moment bound is all that Step 0 exports, and it is precisely what licenses the treatment of the deviating agent as one negligible atom in Step 1.

Step 1: the deviator's environment is close to the mean field. When agent i deviates, the other $N - 1$ agents still run the mean field profile, so by lemma 13.5 applied to those $N - 1$ agents (whose dynamics involve ν only through the contribution of the single atom X^i to μ^N, of $\mathcal{W}_2$-size $\frac{1}{N}|X^i|$, controlled by the uniform moment bound of Step 0) one has $\mathbb{E}\sup_t \mathcal{W}_2(\mu_t^N, m_t)^2 \le C\, r_N^2$, uniformly over the retained deviations ν. The uniformity is the crucial point: with the moment of X^i controlled, the bound on the field seen by i does not depend on how i plays, because i is a single atom of vanishing weight.

Step 2: cost continuity in the field. We now compare the deviator's *actual* cost in the finite game with the cost it would pay in the clean limiting problem. Let $J_i^{\mathrm{MFG}}(\nu)$ be the representative-agent cost of using the same control ν against the *frozen* field m — that is, eq. (13.3) with μ_t^N replaced everywhere by the deterministic m_t and the state driven by $b(\cdot, m_t, \nu)$. The two problems differ in one respect only: the deviator in the finite game feels the random empirical field μ_t^N, while its limiting counterpart feels the deterministic m_t; the control ν and the driving noise are the same in both. The claim is

$$\Big| J^{N,i}(\nu, \widehat{\alpha}^{-i}) - J_i^{\mathrm{MFG}}(\nu) \Big| \;\le\; C\, r_N \qquad \text{uniformly over the retained } \nu.$$

The mechanism is the $\mathcal{W}_2$-Lipschitz dependence of b, σ, L, g on their measure argument, propagated through the dynamics. Couple the two state processes by the same ν and the same Brownian motion; their difference is driven only by the discrepancy in the measure slot, so Itô on the squared difference and the $\mathcal{W}_2$-Lipschitz bounds give, after Grönwall (appendix C.1), a pathwise-in-mean control of order $\mathbb{E}\sup_t \mathcal{W}_2(\mu_t^N, m_t)^2$ on the squared state gap. Inserting this into the two costs — whose integrands again differ by a $\mathcal{W}_2$-Lipschitz amount in the measure and a Lipschitz amount in the state — and invoking the propagation-of-chaos bound eq. (13.10) of lemma 13.5 collapses the whole discrepancy to $C\big(\mathbb{E}\sup_t \mathcal{W}_2(\mu_t^N, m_t)^2\big)^{1/2} \le C\, r_N$. The square root is what turns the r_N^2 of eq. (13.10) into the r_N quoted here; it is also why the final rate of the theorem is r_N rather than its square. The same bound holds, by the identical argument, for the no-deviation choice $\nu = \widehat{\alpha}$.

Step 3: optimality in the limit. This is the one and only place where the strategic content of the theorem lives, and it is elementary. By the very construction of the MFG solution, $\widehat{\alpha}$ is the *optimal* control of the representative agent against the frozen field m (chapters 6 and 8) — that is what it means for (u, m) to solve the MFG. Hence $J_i^{\mathrm{MFG}}(\widehat{\alpha}) \le J_i^{\mathrm{MFG}}(\nu)$ for *every* admissible ν: against the limiting field, no control beats the mean field control, by definition. Everything that was hard was analytic (Steps 0–2); the game theory is this one inequality.

It remains only to chain the three estimates, and the logic is a clean triangle. Read off the deviator's regret by passing to the limiting problem, using optimality there, and passing back:

$$J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i}) \;\overset{\text{Step 2}}{\le}\; J_i^{\mathrm{MFG}}(\widehat{\alpha}) + C r_N \;\overset{\text{Step 3}}{\le}\; J_i^{\mathrm{MFG}}(\nu) + C r_N \;\overset{\text{Step 2}}{\le}\; J^{N,i}(\nu, \widehat{\alpha}^{-i}) + 2C r_N.$$

The first inequality replaces the finite cost of playing $\widehat{\alpha}$ by its limiting value (Step 2); the second uses that $\widehat{\alpha}$ is optimal in the limit (Step 3); the third returns the limiting cost of the deviation ν to its finite value (Step 2 again, now for ν). The two field-replacement errors each cost one $C\, r_N$ and add. The chain says exactly that playing $\widehat{\alpha}$ costs no more than the best deviation plus $2C\, r_N$ — which is eq. (13.4) with $\varepsilon_N = 2C\, r_N$, and eq. (13.13) follows. □

Remark 13.7 (What the proof did *not* use)**.** The argument never touched the Nash system eq. (13.6) and never assumed anything about the existence of finite-N equilibria. Its only probabilistic input was lemma 13.5, propagation of chaos for the *single, fixed* dynamics induced by the common feedback $\widehat{\alpha}$. In the language of section 13.3 this is why the top-down direction is the cheap one: the chaos is hypothesized of a known McKean–Vlasov system, exactly as in chapter 7, and the strategic content is confined to the elementary optimality statement of Step 3. The historical origin is the *Nash certainty equivalence* principle of Huang, Caines and Malhamé [82], where the ε-Nash property was first established for linear–quadratic populations; see also [40, Ch. 6]. That nothing here touched the Nash system is worth holding onto: it is the exact measure of how much the bottom-up direction of section 13.5 will have to supply on its own, where the field is endogenous and no fixed McKean–Vlasov dynamics can be handed to chapter 7.

The theorem leaves the rate at the generic sampling level r_N — as slow as $N^{-1/4}$ in low dimension by eq. (13.12) — because that is the price of replacing a measure by N samples for an *arbitrary* model satisfying the standing assumptions. It is natural to ask whether a model with more structure can do better, and the question is not idle: the bound $\varepsilon_N \le C\, r_N$ is sharp only insofar as the sampling error genuinely propagates into the cost, and in a sufficiently rigid model it may not. The cleanest place to see this is the LQG thread, where every object is a Gaussian and every estimate is a moment computation, and there the generic r_N collapses all the way to $1/N$. We turn to that computation now, both to advance Thread A by one layer and to watch, in a case where everything is explicit, exactly how the special structure beats the generic rate.

Worked computation: ε-Nash for Thread A on N agents

Notation watch

For the LQG thread the symbols a, b, σ revert to the Thread A *scalars* fixed in appendix A and the thread specification: a is the state-feedback coefficient, b the control gain, $\sigma = \sigma$ the noise level (so $\mathrm{d}X = (aX + b\alpha)\,\mathrm{d}t + \sigma\,\mathrm{d}W$). In particular a here is *not* the diffusion matrix; in the general sections of this chapter we never abbreviated $\sigma\sigma^\top$ by a, precisely to keep this slot free for the Thread A parameter. We import the solution symbols of chapter 14 verbatim — curvature $P(t)$, linear offset $r(t)$, constant $c(t)$ — as fixed canonically in the thread specification; no symbol is renamed across the seam.

Worked computation 13.8. Theorem 13.6 delivered, for an arbitrary model meeting the standing assumptions, the bound $\varepsilon_N \le C\, r_N$ — the mean field strategy is within the sampling rate of a Nash equilibrium. The present computation asks the sharpening question raised just above: what does the *explicit* LQG model do to that generic estimate? The answer is that it collapses the whole way to the optimal $1/N$, and the value of doing the calculation by hand is that one can watch *why* — the collapse happens in two clean stages, each of which has a name we will reuse. The first stage is that the imitation cost feels the population only through a *scalar* statistic, its mean, whose fluctuations obey the central limit theorem at the sharp rate $N^{-1/2}$ rather than the coarser measure rate r_N. The second, and decisive, stage is the envelope/first-order-optimality phenomenon: because the mean field control is a *critical point* of the agent's cost, a target fluctuation of size $N^{-1/2}$ costs only its *square*, N^{-1}, in value. Both stages are made exact — not merely asymptotic

— by the Gaussian-quadratic rigidity of the thread, which is exactly what makes Thread A the cleanest laboratory for the mechanism. This is the ε-Nash layer of Thread A; chapter 14 supplies the closed-form MFG solution (eqs. (14.7), (14.8) and (14.10)) whose single-agent Riccati/feedback pair we import here.

The finite game. Scalar states $X_t^i \in \mathbb{R}$, $i \leq N$,

$$\mathrm{d}X_t^i = \big(a\, X_t^i + b\, \alpha_t^i\big)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t^i, \qquad X_0^i \sim m_0 \text{ i.i.d.}, \tag{13.14}$$

with the imitation cost coupled to the empirical mean $\bar{X}_t^N := \frac{1}{N}\sum_j X_t^j$,

$$J^{N,i}(\boldsymbol{\alpha}) = \mathbb{E}\Bigg[\int_0^T \Big(\tfrac{1}{2}(\alpha_t^i)^2 + \tfrac{q}{2}\big(X_t^i - s\,\bar{X}_t^N\big)^2\Big)\,\mathrm{d}t + \tfrac{q_T}{2}\big(X_T^i - s_T\,\bar{X}_T^N\big)^2\Bigg], \tag{13.15}$$

the parameters $a, b, \sigma, q \geq 0, q_T \geq 0, s, s_T$ being those fixed for Thread A. One structural feature is worth flagging before it is used, because the whole sharpening turns on it: in this model the population enters only the *cost* eq. (13.15), through the imitation penalty $\frac{q}{2}(X^i - s\bar{X}^N)^2$, and never the *dynamics* eq. (13.14), whose drift $aX^i + b\alpha^i$ and diffusion σ are agent-private. We exploit this twice below: once to make the closed-loop states exactly independent (the empirical measure cannot survive in a dynamics it never entered), and once to confine the coupling to a single scalar statistic, the empirical mean $\bar{X}_t^N$, that the cost reads off.

The mean field strategy. The single-agent Riccati reduction of chapter 6 gives the LQG MFG value as the quadratic $u(t,x) = \frac{1}{2}P(t)x^2 + r(t)x + c(t)$, the curvature P solving the scalar Riccati equation $\dot{P} = b^2P^2 - 2aP - q$, $P(T) = q_T$ (eq. (14.7)), and the optimal feedback the affine map

$$\widehat{\alpha}(t,x) = -\,b\big(P(t)\,x + r(t)\big) \tag{13.16}$$

(eq. (14.10)), where the offset r solves the linear equation eq. (14.8) forced by the mean-field mean $\bar{m}_t = \int \xi\, m_t(\mathrm{d}\xi)$. The *explicit* closure of the fixed point — the coupled linear system tying r to $\bar{m}_t$ — is the content of chapter 14 and is genuinely borrowed forward (the dependency graph has Ch. 13 $\to$ Ch. 14). We isolate exactly what is used here and borrow no more: the single-agent Riccati/feedback pair eq. (13.16), already available from chapter 6, together with the abstract fact that $\bar{m}_t$ is a *deterministic* function of time. The explicit forms of r and $\bar{m}_t$ are never invoked; all we use is that eq. (13.16) is the optimal representative-agent control against the frozen target $s\bar{m}_t$.

Exact independence. Now hand the single feedback eq. (13.16) to all N agents and watch what becomes of the coupled system. The feedback is a function of an agent's *own* state and time alone, $\widehat{\alpha}(t, X_t^i) = -b(P(t)X_t^i + r(t))$, so substituting it into the dynamics eq. (13.14) — which never carried the population to begin with — yields, for every agent,

$$\mathrm{d}X_t^i = \big((a - b^2P(t))X_t^i - b^2r(t)\big)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t^i. \tag{13.17}$$

Read the right-hand side carefully: the drift is affine in X_t^i with a deterministic, i-independent time profile (recall that $\bar{m}_t$, hence $r(t)$, is a fixed function of time, the one fact we borrow from chapter 14), and the noise is the agent's private W^i. The empirical measure has *disappeared* from the dynamics. There is simply no channel left through which agent j's position can influence agent

i's motion: the only place the population ever appeared was the cost, and the closed-loop *dynamics* do not see the cost. Consequently the N states solve N decoupled linear SDEs with independent noises and i.i.d. initial data, and so $X_t^1, \dots, X_t^N$ are *exactly* i.i.d. Gaussian processes at every t — each Gaussian because a linear SDE driven by Gaussian noise from a Gaussian start stays Gaussian, each identically distributed because the coefficients are common, and mutually independent because nothing couples them. Their common law has mean $\mathbb{E}X_t^i = \bar{m}_t$ (the closed-loop mean solves the same linear ODE that defines $\bar{m}_t$, started from the same data) and a common variance $V(t)$ obeying the scalar Lyapunov ODE $\dot{V} = 2(a - b^2P)V + \sigma^2$, $V(0) = \mathrm{Var}(m_0)$, read off by Itô from eq. (13.17). This is propagation of chaos in its most extreme conceivable form: not asymptotic factorization with an r_N remainder, as in lemma 13.5, but *exact* independence and *exact* Gaussianity at every finite N, with no remainder at all.

It is essential to see that this exactness is a property of the *decentralized mean field profile*, not of the LQG game as such, and that it *fails* for the game's true Nash feedback — the contrast is precisely the seam the chapter is built around. The true finite-N equilibrium feedback solves the Nash system eq. (13.6); it is a function of the entire configuration $\boldsymbol{x} = (x_1, \dots, x_N)$, because an agent who is allowed to watch the board responds to where the crowd actually is. Substituting such a feedback into the dynamics would leave each agent's drift depending on *all* the states, and the empirical measure μ_t^N would survive in the closed-loop system as a genuine coupling. The N states would then be neither independent nor Gaussian; their chaos would be an *emergent* property to be proved, with an r_N-type remainder, exactly the harder task of section 13.5. What buys the exact independence here is that the MFG strategy has already thrown away the agent's right to watch the others: it conditions on X_t^i alone, so when it is plugged back into a dynamics that the population never entered, no coupling can reappear. The decentralization that made the strategy suboptimal (it ignores the exploitable $N^{-1/2}$ wiggles) is the very thing that makes the closed-loop states exactly i.i.d. This is the cleanest illustration in the book of why the top-down direction is cheap: the dynamics it must analyze is a *fixed* McKean–Vlasov system, here so rigid that the chaos is not merely asymptotic but literal.

With the states exactly i.i.d. Gaussian, the statistic the cost actually reads off is trivial to control. The imitation penalty couples agent i only to the empirical mean

$$\bar{X}_t^N = \frac{1}{N}\sum_j X_t^j, \qquad \mathbb{E}\bar{X}_t^N = \bar{m}_t, \qquad \mathrm{Var}\left(\bar{X}_t^N\right) = \frac{1}{N^2}\sum_j \mathrm{Var}(X_t^j) = \frac{V(t)}{N},$$

the variance computing exactly — not up to constants — because the N summands are independent with common variance $V(t)$, so their average has variance $V(t)/N$. Hence the field fluctuation seen by the cost, $e_t := \bar{X}_t^N - \bar{m}_t$, is a centred Gaussian with $\mathbb{E}e_t^2 = V(t)/N$ exactly. This is the heart of the first stage of the collapse, and it is worth stating its mechanism in one sentence: e_t is the fluctuation of a *scalar average* of N independent samples, which the central limit theorem pins at the sharp rate $N^{-1/2}$, whereas r_N of lemma 13.5 is the rate at which the *whole empirical measure* converges, and resolving an entire probability measure from N samples is strictly harder than resolving one of its moments. In $d = 1$ the gap is already stark — $r_N \sim N^{-1/4}$ against $N^{-1/2}$ — and it is dimensional in origin: the measure rate degrades with dimension (eq. (13.12)) because samples must fill space, while the mean of a scalar average does not care about dimension at all. The Gaussianity upgrades the CLT heuristic $O(N^{-1/2})$ to an exact identity, but the operative point is structural: the LQG cost is blind to every feature of the population except its mean, and that one scalar is controlled at the finer rate. It is this scalar fluctuation e_t, never the full empirical measure, that the quadratic cost will see in the regret calculation.

The deviator's regret. We now estimate the one thing the ε-Nash property demands: how much an agent could save by abandoning the mean field strategy for its true best response, given that everyone else holds to it. Freeze all agents $j \neq i$ at eq. (13.16); their states are then the exactly i.i.d. Gaussians of the previous step, oblivious to whatever agent i does (agent i's own deviation cannot infect them, since the population entered nobody's *dynamics*). Agent i, free to optimize, still pays the imitation cost eq. (13.15), and so it faces a tracking problem: keep X_t^i near the *random* target

$$\tau_t \;=\; s\,\bar{X}_t^N \;=\; s\,\bar{m}_t \;+\; s\,e_t^{-i} \;+\; \frac{s}{N}X_t^i, \qquad e_t^{-i} := \frac{1}{N}\sum_{j\neq i} X_t^j - \frac{N-1}{N}\bar{m}_t.$$

The decomposition isolates the three pieces of the target the agent sees. The first, $s\bar{m}_t$, is the deterministic field the mean field strategy was tuned to. The second, $s\,e_t^{-i}$, is the fluctuation of the $N-1$ competitors' mean about that field; since those $N-1$ states are i.i.d. with variance $V(t)$, one has $\mathbb{E}|e_t^{-i}|^2 = \frac{N-1}{N^2}V(t) = O(1/N)$, the now-familiar CLT scale. The third, $\frac{s}{N}X_t^i$, is agent i's own contribution to the empirical mean — the price of counting oneself in $\bar{X}^N$ — and it is smaller still, $\mathbb{E}|\frac{s}{N}X_t^i|^2 = O(1/N^2)$, using the uniform second-moment bound of Step 0 of theorem 13.6. The whole target discrepancy is therefore

$$\tau_t - s\,\bar{m}_t \;=\; s\,e_t^{-i} + \frac{s}{N}X_t^i, \qquad \mathbb{E}\left[\int_0^T \left|\tau_t - s\,\bar{m}_t\right|^2\,\mathrm{d}t\right] \;=\; O\!\left(\tfrac{1}{N}\right), \tag{13.18}$$

dominated by the $O(1/N)$ competitor fluctuation. So the deviator solves a scalar LQ tracking problem — the dynamics eq. (13.14) and cost eq. (13.15) with the random target τ_t in place of the deterministic $s\bar{m}_t$ — whose target differs from the one the mean field strategy was built for by an L^2-amount of size $N^{-1/2}$. The question is how that $N^{-1/2}$ target error translates into lost value, and this is where the quadratic structure pays off twice over.

The first payoff is an exact regret identity. Write $\alpha^{i,*}$ for the deviator's genuine optimal control and $\widehat{\alpha}_t^i = \widehat{\alpha}(t, X_t^i)$ for the mean field control it is tempted to abandon. Because the running control cost is the pure square $\frac{1}{2}\alpha^2$ and the drift is affine in the control, the integrand the agent minimizes, $\frac{1}{2}\nu^2 + b\,\nu\,\partial_x v$, is an exact parabola in the control value ν; its minimizer is $\alpha^{i,*} = -b\,\partial_x v$, and the excess of *any* other choice ν over that minimizer is the completed square $\frac{1}{2}(\nu - \alpha^{i,*})^2$, with the linear term cancelling identically. Integrating this pointwise identity along the controlled trajectory and taking expectations — the verification computation of chapter 6, in which the value-function terms telescope — gives the regret of any admissible ν as a clean quadratic,

$$J^{N,i}(\nu, \widehat{\alpha}^{-i}) - \inf_{\alpha} J^{N,i}(\alpha, \widehat{\alpha}^{-i}) \;=\; \tfrac{1}{2}\,\mathbb{E}\left[\int_0^T \left|\nu_t - \alpha_t^{i,*}\right|^2\,\mathrm{d}t\right]. \tag{13.19}$$

The crucial feature is the *absence of a first-order term.* There is no contribution linear in $\nu - \alpha^{i,*}$, because $\alpha^{i,*}$ is a critical point of the cost: at an optimum the first variation vanishes, so the leading response of the cost to a control perturbation is quadratic, not linear. This is the envelope/first-order-optimality phenomenon in its most transparent form, and it is exactly what will convert an

$N^{-1/2}$ control error into an N^{-1} value loss. Note that this argument needs nothing about *how* far $\widehat{\alpha}^{\,i}$ is from optimal; it needs only that the comparison is made against the true optimizer, so that the linear term is gone. We will meet the very same cancellation again, in a less elementary guise, when the master field's first-order optimality kills the $O(N^{-1/2})$ term in the value expansion of theorem 13.13.

The second payoff is that the control gap is itself $O(N^{-1/2})$, so that its square in eq. (13.19) is $O(1/N)$ — and this is where the LQ tracker's structure must be unpacked rather than asserted. The deviator's value function is again quadratic in its own state, $v(t,x) = \frac{1}{2}P(t)x^2 + r_t^* x + \cdots$, and the point to appreciate is that the *curvature* $P(t)$ is unchanged from the mean field problem: P solves the same Riccati equation $\dot{P} = b^2P^2 - 2aP - q$, $P(T) = q_T$, because the Riccati equation is fixed by the quadratic-in-state and control structure of eq. (13.14)–eq. (13.15), and shifting the *target* of a tracking cost moves only its linear-in-state part, never its curvature. All the deviation's information therefore lives in the linear offset r^*, and the optimal control reads

$$\alpha_t^{i,*} = -b\big(P(t)X_t^i + r_t^*\big), \qquad \text{against} \qquad \widehat{\alpha}_t^{\,i} = -b\big(P(t)X_t^i + r(t)\big),$$

so the entire control gap is carried by the offsets, $\widehat{\alpha}_t^{\,i} - \alpha_t^{i,*} = b\,(r_t^* - r(t))$. The two offsets solve the *same* linear ODE — the one driving the offset in any LQ tracker against this dynamics, with homogeneous part set by the closed-loop drift $a - b^2P(t)$ and forcing proportional to the cost weight q times the coupling s times the target — but with different forcings: r is forced by the deterministic field $s\bar{m}_t$, while r^* is forced by the deviator's *conditional* prediction of its target, $\mathbb{E}[\tau_\cdot \mid \mathcal{F}_t]$, since an optimally controlled agent steers toward the best forecast of where the crowd will be, not toward its realized position. (That r^* tracks the conditional mean, and not the raw target, is the one genuinely stochastic-control feature here: the offset is the solution of a backward equation, hence forward-looking.) Subtracting the two linear ODEs, the difference $r_t^* - r(t)$ solves the *homogeneous* linear equation forced solely by the target discrepancy $\mathbb{E}[\tau_\cdot \mid \mathcal{F}_t] - s\bar{m}_\cdot$. A scalar linear ODE on the bounded interval $[0,T]$ propagates its forcing with a bounded Green's function, so Grönwall's inequality (or, equally, the explicit variation-of-constants representation) gives the L^2 stability bound

$$\mathbb{E}\bigg[\int_0^T \big|r_t^* - r(t)\big|^2 \,\mathrm{d}t\bigg] \;\le\; C\,\mathbb{E}\bigg[\int_0^T \big|\tau_t - s\,\bar{m}_t\big|^2 \,\mathrm{d}t\bigg] \;=\; O\Big(\tfrac{1}{N}\Big),$$

where the last equality is eq. (13.18) (the conditional expectation only contracts the L^2 norm, by Jensen, so it cannot enlarge the discrepancy). The forcing-to-offset constant C is the square of the linear ODE's forcing gain, hence proportional to q^2s^2 — a factor qs for each of the two copies of the forcing in the squared norm — which is the promised non-trivial scaling, not a single power of q.

Assembling the two payoffs, substitute $\nu = \widehat{\alpha}^{\,i}$ into the regret identity eq. (13.19):

$$J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i}) - \inf_{\alpha} J^{N,i}(\alpha, \widehat{\alpha}^{-i}) = \tfrac{1}{2}\,\mathbb{E}\bigg[\int_0^T b^2\big|r_t^* - r(t)\big|^2 \,\mathrm{d}t\bigg] \le \tfrac{1}{2}b^2C\,\mathbb{E}\bigg[\int_0^T \big|\tau_t - s\bar{m}_t\big|^2 \mathrm{d}t\bigg] = O\Big(\tfrac{1}{N}\Big),$$

uniformly in i by exchangeability. This is the quadratic gain, and taking the supremum over agents gives the sharpened ε-Nash rate

$$\varepsilon_N^{\mathrm{LQG}} \;=\; \sup_i \Big[J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i}) - \inf_{\alpha} J^{N,i}(\alpha, \widehat{\alpha}^{-i})\Big] \;\le\; \frac{C}{N}, \qquad C = C' \sup_{t\le T} V(t), \tag{13.20}$$

the constant assembled exactly as the chain above tracked it: the regret carries one factor b^2 from the control gap $b(r^* - r)$, the offset-stability bound contributes $q^2 s^2$, and the horizon enters through Grönwall, so that $C' \asymp q^2 b^2 s^2$ up to a T-dependent factor, multiplying the variance scale $\sup_{t \le T} V(t)$ of the population. We do not display C' in closed form; its *shape* — quadratic in each of b, q, s, never linear — is the informative part, and it is the fingerprint of the two squarings: one from the CLT (e_t enters through its variance) and one from the envelope (the offset enters the regret through its square). *Interpretation.* Stand back from the algebra and read off what governed the

rate. The generic bound of theorem 13.6 promises only $\varepsilon_N = O(r_N) = O(N^{-1/4})$ in this $d = 1$ problem, because it routes *everything* through the full $\mathcal{W}_2$ distance of the empirical measure — it charges the agent the cost of resolving the entire population, which in one dimension converges no faster than $N^{-1/4}$. The LQG structure beat that by a full power and a quarter, $N^{-1/4} \to N^{-1}$, and it did so in two stages, each of which is a quantity squared. First, the imitation cost eq. (13.15) sees the population only through its *scalar mean* $\bar{X}_t^N$, whose fluctuation e_t is controlled at the finer central-limit rate $N^{-1/2}$; the cost never needs the whole measure, so it never pays the measure rate. Second — the decisive step — the envelope phenomenon made exact by the quadratic cost: because the mean field control is a *critical point* of the agent's cost against its target, the first-order response of the cost to a target perturbation vanishes, and an $N^{-1/2}$ fluctuation costs only its square, N^{-1}, in value. The two squarings compose: a field fluctuation of size $N^{-1/2}$, felt through a cost that responds quadratically, yields a regret of size $(N^{-1/2})^2 = N^{-1}$.

The moral is the one to carry out of the whole top-down arc. The ε-Nash rate is governed not by the *size* of the field fluctuations an agent endures but by *how its cost responds to them* — and at an optimum the response is quadratic, never linear. A strategy sitting at a critical point is first-order insensitive to the very fluctuations it ignores, so the price of ignoring them is second order. This is why a decentralized agent, blind to the $N^{-1/2}$ wiggles of the crowd, nonetheless loses only N^{-1} by its blindness. Nothing in the argument was special to Thread A except that the Gaussian-quadratic rigidity made both squarings *exact* rather than asymptotic; the mechanism itself is general, and we have isolated it here in the one model where it can be watched without remainder.

This envelope cancellation — a fluctuation costing its square because the comparison is anchored at a critical point — is not a curiosity of the LQG example but the single mechanism the chapter will lean on again at its quantitative climax. When theorem 13.13 establishes that the finite-N *values* converge to the master field at the rate $1/N$, faster than the measures whose laws they encode, the engine will be exactly this: the master field U is itself a first-order-optimal object, so the $O(N^{-1/2})$ fluctuation of the empirical measure enters the value comparison only through its square, leaving an $O(1/N)$ residual. The worked computation has given us, in the simplest possible setting, the intuition for why values converge faster than the configurations they summarize. With that intuition banked — and with remark 13.7 having recorded that this entire top-down arc never once touched the Nash system — we have exhausted the cheap direction. Everything so far started from a *known* field and tested it; nothing yet has confronted the actual equilibria of the finite game, where the field is not handed to us but is the unknown we are solving for. That is the harder, deeper half of the chapter, and it is where we turn next.

13.5 Convergence of Nash equilibria and propagation of chaos for Nash

The whole top-down arc just completed shared one feature that made it cheap, and naming it sharply is the right way to feel the change of altitude we are about to make. There the dynamics were *given*: we took the MFG's decentralized feedback $\widehat{\alpha}$, a map fixed once and for all by a problem solved on $\mathbb{R}^d$ before any finite game was in view, handed the same map to all N players, and asked only how far short of equilibrium the resulting profile fell. Because the feedback was external and frozen, the N states evolved by a *prescribed* McKean–Vlasov law, the empirical measure concentrated by the propagation of chaos of chapter 7 applied verbatim, and the envelope cancellation isolated in worked computation 13.8 did the rest — the agent's blindness to the crowd's $O(N^{-1/2})$ wiggles cost only their square. At no point did that argument so much as glance at the Nash system. We now drop exactly that crutch.

The bottom-up direction starts from the other end. We take the *actual* finite Nash equilibria — the value functions $v^{N,i}$ solving the coupled Nash system eq. (13.6), the equilibrium feedbacks $\widehat{\alpha}^j = \alpha^*(x_j, \nabla_{x_j} v^{N,j}, \mu^N)$ they generate, and the law of the state process $\boldsymbol{X}_t = (X_t^1, \dots, X_t^N)$ those feedbacks drive — and ask whether, as $N \to \infty$, this microscopic equilibrium really does relax onto the mean field object the heuristic of section 13.2 promised. Three convergences are at stake, and it is worth listing them so the reader knows what a theorem in this section must deliver: that the equilibrium *values* $v^{N,i}$ converge to the master field $u = U(\cdot, \cdot, m)$ of chapter 12; that the (random) *empirical measure* μ^N of the equilibrium states converges to the deterministic MFG flow m; and that the finite-dimensional *marginals* factorize, $\mathcal{L}(X_t^1, \dots, X_t^k) \to m_t^{\otimes k}$, the equilibrium becoming asymptotically chaotic. This last is the substantive content and it has its own name — *propagation of chaos for Nash equilibria*.

This is the deepest result of Part II, and the honesty the section owes the reader begins with admitting that it does not come in a single grade. There are two routes to it, of sharply different character, and we will be candid throughout about the divide between them. The first is *qualitative*: under little more than what makes the games well-posed and the MFG unique, a compactness argument extracts convergent subsequences and identifies their limits as MFG solutions — but it yields no rate, and it must be careful about *which* solution is selected. The second is *quantitative*: at the price of a strong hypothesis, the regularity of the master equation, one obtains genuine convergence *with a rate*, and indeed learns that the values converge faster ($1/N$) than the measures whose laws they encode (the sampling rate r_N). Sections 13.5.2 and 13.5.3 execute these two routes in turn; theorems 13.9 and 13.13 are the two milestones they reach. Quantitative control, we will see, is a monotone-regime luxury that the regularity of U buys, not a birthright of the convergence itself.

Before either route is opened, though, one objection must be dispatched, because without it a reader who has just watched chapter 7 power the entire top-down argument will reasonably ask why the same machine cannot simply be run again. It cannot, and the reason is the exogenous-versus-endogenous seam that has organized the whole chapter — here it finally becomes decisive rather than merely foreshadowed. The next subsection makes that gap fully explicit, so that when the convergence machinery does arrive we already understand why it has to be new.

13.5.1 Why this is not chapter 7 again

section 13.3 named the bottom-up problem's *endogeneity* as the one-word cause of the chapter's central asymmetry, and the opening of section 13.5 promised to discharge, before any machinery, the objection it provokes: a reader who has just watched chapter 7 carry the entire top-down argument is entitled to ask why the same machine cannot simply be run again. It is worth dwelling on the gap between the propagation of chaos of chapter 7 and what the bottom-up problem demands, because it is the single most important conceptual distinction in the chapter, and because seeing it clearly is what tells us that the convergence machinery to come must be genuinely new rather than a relabelling of what we already have. The cleanest way to feel the gap is to draw the two situations side by side.

The chapter 7 picture: chaos as the output of a prescribed evolution. In chapter 7 the dynamics are *given*. One writes a McKean–Vlasov particle system — N particles whose drift depends on the others only through the empirical measure μ_t^N, and does so through a *fixed*, externally specified interaction kernel $b(x, m)$ that is no part of the unknown –

$$\mathrm{d}X_t^i = b\big(X_t^i, \mu_t^N\big)\ \mathrm{d}t + \sigma\big(X_t^i, \mu_t^N\big)\ \mathrm{d}W_t^i,$$

— and proves that as $N \to \infty$ the empirical measure converges to the deterministic flow m_t solving the nonlinear Fokker–Planck equation $\partial_t m_t = \mathcal{A}_{m_t}^* m_t$ associated with that same kernel. The logical order here is the whole point. The evolution law is written down first, in full, with every coefficient known; the empirical measure is then *computed* along it; and chaos — the asymptotic factorization $\mathcal{L}(X_t^1, \dots, X_t^k) \to m_t^{\otimes k}$ — is a *theorem about that prescribed evolution*, established by the coupling-to-i.i.d. argument that bounds each particle against an independent copy driven by the limiting field. The law m_t is the *output* of a dynamics fixed in advance. Nothing in the proof ever has to *solve* for the interaction; it is handed in. Exactly this is what powered the top-down half of the present chapter. There the chapter 7 machine ran verbatim through lemma 13.5, with the fixed kernel being the MFG's own decentralized feedback $\widehat{\alpha}(t, x)$ — a map manufactured once, on $\mathbb{R}^d$, by a problem solved before any finite game was in view, and then frozen and handed identically to all N players. The feedback was external data; the dynamics it generated were a closed McKean–Vlasov system; the chaos was the output. At no point did that argument so much as consult the Nash system.

The bottom-up picture: a self-referential interaction. In the bottom-up problem the dynamics are *not given* — and this single difference is what severs the connection to chapter 7. The states still move by

$$\mathrm{d}X_t^i = b\big(X_t^i, \mu_t^N, \widehat{\alpha}^{\,i}(t, \boldsymbol{X}_t)\big)\ \mathrm{d}t + \sigma\big(X_t^i, \mu_t^N\big)\ \mathrm{d}W_t^i,$$

but the feedback driving them is no longer external. The equilibrium feedback of agent j,

$$\widehat{\alpha}^{\,j}(t, \boldsymbol{X}_t) = \alpha^*\big(X_t^j, \nabla_{x_j} v^{N,j}(t, \boldsymbol{X}_t), \mu_t^N\big),$$

depends through $\nabla_{x_j} v^{N,j}$ on the *entire configuration* $\boldsymbol{X}_t$, not on X_t^j alone: an agent permitted to watch the board responds to where *every* other agent is. And $v^{N,j}$ is not a coefficient one may look up — it is precisely *the unknown of the Nash system* eq. (13.6), the very object whose limit we are trying to certify. The empirical measure of the equilibrium is therefore a *self-referential* object: it is the empirical measure of states whose drifts are equilibrium feedbacks, and those feedbacks are functions — through the coupled value-gradients — of that same empirical measure. The interaction does not sit above the dynamics as a given; it *is* the equilibrium we are solving for. There is consequently no fixed McKean–Vlasov system to feed into chapter 7: the kernel that the chapter 7 proof requires as input is here itself an output, and one cannot prove convergence of a fixed-point by an argument that presumes the fixed point already resolved.

Why the correlations are genuinely on. The practical face of this endogeneity is that, at finite N, the equilibrium states $X_t^1, \dots, X_t^N$ are genuinely *correlated*. This is the point at which the contrast with the worked computation bites hardest, and the two cases are worth setting literally one line apart. Under the *decentralized* mean field profile of worked computation 13.8, the LQG closed-loop drift collapsed to $b = (a - b^2 P(t))X_t^i - b^2 r(t)$ (eq. (13.17)): the empirical measure dropped out of the dynamics entirely, the N states solved N *decoupled* linear SDEs with private noises, and they were therefore *exactly* i.i.d. Gaussian at every t and every finite N — propagation of chaos with no remainder at all. Replace that decentralized feedback by the game's *true* Nash feedback — the solution of eq. (13.6), a function of the whole configuration $\boldsymbol{x}$ — and the very term that vanished reappears: each agent's drift now depends on all the states, the empirical measure μ_t^N survives in the closed-loop system as a live coupling, and the N states are neither independent nor Gaussian. The exact independence of the worked computation was never a feature of the LQG game; it was a feature of the decentralized profile, which had already *thrown away* the agent's right to watch the others, so that when it was plugged back into a dynamics the population never entered, no coupling could reappear. The true Nash feedback keeps that right, and with it keeps the correlations. Exact independence is the special case; correlation is the generic truth of an endogenous equilibrium.

Establishing that these correlations nonetheless *vanish* as $N \to \infty$ — that the endogenous equilibrium self-averages, so that the self-referential empirical measure does after all concentrate on the deterministic MFG flow m_t and the marginals factorize — is the content of *propagation of chaos for Nash equilibria.* Here is the thesis to carry out of this subsection: *chaos is now an emergent property of a fixed point, not a hypothesis on a dynamics.* In chapter 7 chaos was deduced from a known evolution; here it must be *produced* from the equilibrium condition itself, with no prior dynamics to lean on.

Two routes accomplish this, and they pay for the missing dynamics in two different currencies. The first is *qualitative*: lacking a fixed evolution to track, one gives up on tracking and instead extracts convergence by *compactness* — showing the family of equilibria is precompact and that every limit point solves the MFG system, which under uniqueness pins the whole family. It asks almost nothing beyond well-posedness and buys no rate. The second is *quantitative*: one supplies the dynamics that the finite game refused to hand over, in the form of the *master equation* of chapter 12, using the master field U as an ansatz whose residual against the Nash system is $O(1/N)$ — turning the emergent chaos into a controlled expansion at the price of the regularity of U. We take these in turn, beginning with the qualitative route, which trades any rate for the most economical hypotheses.

13.5.2 Qualitative convergence by compactness (Lacker's route)

The previous subsection left us with a sharply posed task and an explicit warning that the obvious tool is unavailable: the equilibrium empirical measure is self-referential, so the chaos we want must be *produced* from the fixed-point condition itself rather than deduced from a dynamics fixed in advance. There is no McKean–Vlasov system to hand to chapter 7, because the very kernel that argument would consume as input is here an output of the equilibrium. We now discharge that task in the cheapest currency the problem admits — pure compactness — and we pay honestly for paying so little. The idea is to stop trying to *track* the equilibrium evolution (there is none to track) and instead to argue indirectly: show that the family of equilibria, viewed as random objects on a suitable space, cannot escape to infinity (tightness), so that some subsequence converges; show that whatever it converges to must satisfy the defining relations of a mean field equilibrium (identification); and then, when the limit problem has a unique solution, conclude that the whole family converges to it. This is a classic move — it is how one proves existence and convergence for many self-consistent systems where no contraction or explicit solution is available — and its great virtue is that it asks almost nothing of the data.

Concretely, this first route assumes essentially only what already makes the finite games well-posed and the limiting MFG unique. In exchange it surrenders two things, and naming both now is what keeps the bookkeeping with the next subsection honest. First, it produces *no rate*: a compactness argument exhibits a convergent subsequence but never a modulus, so we learn *that* the equilibria converge without learning *how fast*. Second — and this is the more interesting cost — it forces us to be careful about *which* MFG solution the equilibria converge to. The compactness machine certifies that every limit point solves the mean field equilibrium relations, but in the closed-loop information class the set of objects satisfying those relations can be strictly larger than the Markovian MFG solution of chapter 8, and the argument cannot by itself rule out the extra ones. Both deficits are exactly what the second route will buy back, at the price of strong regularity; here we accept them as the fair price of minimal hypotheses.

Theorem 13.9 (Limit points of N-player equilibria are MFG solutions)**.** *Assume* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv)*, with the coefficients continuous and the costs bounded below and coercive enough for relaxed controls to be compact. For each N let $\boldsymbol{\alpha}^{*,N}$ be an ε_N-Nash equilibrium of the N-player game with $\varepsilon_N \to 0$, and let $P^N = \mathcal{L}(\mu^N)$ be the law of the (random) empirical measure flow of the equilibrium states on path space. Then:*

1. *the family $(P^N)_N$ is tight, and every weak limit point is supported on the set of* weak MFG solutions *(mean field equilibria in the relaxed/martingale-problem sense, defined in definition G.5);*

2. *if in addition the Lasry–Lions monotonicity condition* (A-Mon) *of chapter 11 holds, so that the MFG solution (u, m) is unique, then $\mu^N \to m$ in distribution (the limit is deterministic), and the finite marginals propagate chaos: $\mathcal{L}(X_t^{1,N}, \dots, X_t^{k,N}) \to m_t^{\otimes k}$ weakly for each fixed k.*

Proof sketch; full argument in appendix G. The mechanism is a compactness–and–identification argument in the spirit of the martingale-problem method, due in this generality to Lacker [89, 90] (see also Fischer [67] for the open-loop case and [40, Ch. 6] for an exposition). The full argument, with every estimate carried out, lives in appendix G; importing it wholesale here would drown the idea, so our task in the main text is to narrate the four moving parts — *what* each does, *why* it is needed, and *where* it is discharged — without reproducing the appendix's proofs. The deferral is deliberate and we keep it honest: each step below names the appendix result that supplies what we only motivate.

Part one: lift, and relax the controls. The states $X^{i,N}$ are not the natural objects to take limits of, because an equilibrium is a statement about *states and the controls that drive them* jointly, and controls are the fragile part. One therefore lifts each equilibrium to a single probability measure on the space of state–control *paths*, $C([0,T];\mathbb{R}^d) \times$ (control trajectories), recording where an agent went and what it did along the way. The delicate point is the control coordinate: ordinary A-valued controls form a space with no compactness to offer — a sequence of controls can oscillate ever faster without converging to any control — so a naive limit of the lifted measures may simply fail to exist. The repair is to enlarge the control space to the *relaxed* (measure-valued) controls of definition G.1: instead of a point $\alpha_t \in A$ at each instant one allows a probability measure $q_t(\mathrm{d}\alpha)$ over actions, which is exactly the closure that makes fast oscillations converge to a genuine limiting object (a chattering control becomes, in the limit, an honest mixture). On this enlarged space the relevant set of trajectories is *compact* (lemma G.2), which is the whole reason for the detour — compactness is the engine of the argument, and the relaxation is the price of admission. Within this lift, the two defining properties of an equilibrium become two relations on the lifted measure: each agent's optimality is encoded as a *martingale (Hamiltonian) condition* expressing that no deviation improves its cost (definition G.5), and the anonymous interaction is encoded as a *consistency constraint* tying the law the agent responds to back to the empirical measure of the population. A *weak MFG solution* is by definition any limit object satisfying both.

Part two: tightness on path space. With the lift in place we must produce a convergent subsequence, and this is the first genuinely substantive estimate. The objects in play, $P^N = \mathcal{L}(\mu^N)$, are laws *on path space*: each is the distribution of the random measure-valued trajectory $t \mapsto \mu_t^N$. Tightness here is therefore strictly more than the Prokhorov-on-$\mathbb{R}^d$ statement one might reflexively reach for — controlling the spread of the time-t marginals is not enough, because a family of measures on $C([0,T];\mathbb{R}^d)$ can have perfectly tight marginals at every fixed t and still escape to infinity through wilder and wilder *oscillation in time*. The path-space criterion (Arzelà–Ascoli in probabilistic form) demands two ingredients: uniform control of the marginals *and* uniform control of the temporal modulus of continuity. The first comes for free from (A-Growth) and the cost coercivity, which bound the second moments of the states and of the (relaxed) controls uniformly in N. The second is where (A-Nondeg) earns its place: feeding the uniform nondegeneracy and the bounded control moments into a Kolmogorov–Chentsov modulus estimate (theorem C.9, in the tightness form proposition G.3) yields a uniform bound on the increments $\mathbb{E}|X_t^{i,N} - X_s^{i,N}|^p \le C|t-s|^{1+\alpha}$ that forbids the trajectories from wiggling too fast on average. Together these give tightness on $C([0,T];\mathbb{R}^d)$, and Prokhorov's theorem (theorem 3.16) extracts a weakly convergent subsequence $P^{N_k} \to P^\infty$. Note what nondegeneracy is doing: the same diffusion that supplied parabolic regularity to the Nash system in section 13.2 supplies, on the probabilistic side, the regularity of paths that makes them precompact. Without it (in the degenerate case) this step needs a substitute and the clean statement can fail.

Part three: pass to the limit in optimality. A convergent subsequence is not yet an MFG solution; we must show the limit P^∞ inherits both defining relations. The optimality relation is where the hypothesis $\varepsilon_N \to 0$ is spent. Each finite-N profile is only an ε_N-Nash equilibrium, so along the sequence each agent is optimal only up to a slack ε_N; the martingale optimality condition holds with an error that vanishes as $\varepsilon_N \to 0$. Taking the limit therefore turns *asymptotic* optimality into *exact* optimality of the limiting object (lemma G.6): in the limit no deviation improves on the limiting control, which is precisely the Hamiltonian martingale condition of definition G.5 with no slack. This is also where the relaxation pays off a second time — because the limiting control may be a genuine mixture, the optimality statement is naturally phrased for relaxed controls, and the martingale formulation is exactly what survives the weak limit (test functions and the generator pass through, the relaxed control integrates against the generator continuously).

Part four: pass to the limit in consistency, and identify. The second relation, consistency, is the fixed-point constraint that the law the representative agent feels equals the law its own optimal motion generates. Passing it to the limit along the convergent subsequence identifies P^∞ as supported on weak MFG solutions (theorem G.8): the limiting empirical-measure flow is the law of the limiting optimally controlled state, which is the definition of a mean field equilibrium. This establishes part (1) of the theorem — tightness plus identification — with no appeal whatsoever to the Nash system's regularity, indeed with no need for the Nash equilibria to be unique or even Markovian; relaxed controls absorb any non-uniqueness of best responses. That robustness is the qualitative route's signature strength.

The monotone upgrade. Part (2) is what turns "every limit point is *a* weak MFG solution" into genuine convergence and chaos. Under the Lasry–Lions monotonicity (A-Mon) of chapter 11 the weak MFG solution is *unique* and, being unique, *deterministic*: the random limit measure P^∞ can charge only one point of $\mathcal{P}(C([0,T];\mathbb{R}^d))$, namely the Dirac mass at the unique flow m (theorem G.7). Two consequences follow. First, since every subsequential limit is forced to be this same point, the usual subsequence argument promotes subsequential convergence to convergence of the *full* sequence: $\mu^N \to m$ in distribution. Second — and this is the step worth dwelling on — convergence in distribution *to a constant* is a much stronger statement than convergence to a random limit. A sequence of random measures converging in law to a deterministic m must concentrate: its fluctuations vanish, so the random empirical measure becomes asymptotically non-random. This is exactly what propagation of chaos asserts, and the standard equivalence (convergence of the empirical measure to a deterministic $m \iff$ factorization of the k-marginals) upgrades $\mu^N \to m$ into $\mathcal{L}(X_t^{1,N}, \dots, X_t^{k,N}) \to m_t^{\otimes k}$ for every fixed k. The endogenous correlations that section 13.5.1 insisted were genuinely present at finite N have, in the monotone regime, been shown to wash out in the limit — not by tracking them, but by squeezing them between compactness and uniqueness. What the argument never delivers is the *rate* at which they wash out: compactness exhibits a limit but no modulus, and uniqueness pins the limit but says nothing about the speed of approach. That missing modulus is precisely the debt the next route is contracted to pay. □

Remark 13.10 (Open-loop versus closed-loop: a genuine subtlety)**.** The informational class of definition 13.1 — which we flagged on first sight as "not boilerplate" — is precisely where it stops being innocuous, and the qualitative theorem above is the right place to confront it, because the breadth of the limit set it certifies depends on which class one started in. For *open-loop* equilibria the picture is the clean one: each agent commits to a control adapted to the raw noise, the lifted measures carry no configuration-mediated coupling, and every limit point is a genuine MFG solution — under monotonicity, the unique one [67, 89]. For *closed-loop* (Markovian) equilibria, where each agent reads the whole board $\boldsymbol{X}_t$ and responds to it, Lacker [90] discovered that the set of attainable limits is strictly *larger*. The mechanism is exactly the endogenous correlation of section 13.5.1: because every agent's feedback depends on the common configuration, the agents are coupled through a shared signal, and that coupling can transmit a common randomness that does *not* average away. The limit may then be a *weak* mean field equilibrium in which the field m_t is itself random — driven by a common noise the feedbacks manufactured among themselves — rather than the deterministic, Markovian MFG solution of chapter 8. The compactness argument is blind to the distinction: such weak equilibria satisfy every relation the identification step checks, so they sit inside the certified limit set with full right.

The inclusion fails in the other direction too, and this is the terser half worth a sentence of intuition. Not every MFG solution arises as a closed-loop limit, because a limiting field is only realizable if some sequence of genuine finite-N equilibria approaches it, and finite-N individual rationality is a real constraint: an MFG fixed point that is, say, dynamically unstable under the finite best-response correspondence may have no N-player equilibrium in any neighbourhood of it, so it survives as a solution of the limit equations while being unreachable from below. The limit set and the MFG-solution set therefore *overlap without nesting* in general — the closed-loop limit set can contain extra weak equilibria and can miss some Markovian ones. Under monotonicity (A-Mon) both pathologies vanish at once: uniqueness collapses the weak equilibria onto the single deterministic flow (the common noise has nothing to be random about when the fixed point is unique), and the unique solution is then trivially the only candidate, so every route — open-loop, closed-loop, deterministic — agrees. This is one more entry in the running ledger of why the monotone regime is the comfortable one. The point to carry forward is that this entire phenomenon is what the master-equation route of section 13.5.3 *excludes by fiat*: by presupposing a single sufficiently regular master field U it builds the Markovian solution into its ansatz from the start, so the weak, common-noise equilibria are never on the table to begin with. The clean Markovian limit is a hypothesis there, not a conclusion — which is exactly the regularity one is paying for.

This is the natural seam at which to change routes. The qualitative argument has delivered exactly what minimal hypotheses can buy and no more: under near-well-posedness it certifies that the N-player equilibria are precompact and that their limits solve the mean field equilibrium relations, and under monotonicity it pins the limit and upgrades convergence to propagation of chaos. But it has left two debts unpaid, and they are the same two we flagged at the outset. It gives *no rate* — compactness sees a limit but never a modulus, so we cannot say convergence happens at the sampling rate r_N, or at $1/N$, or at all quickly. And in the closed-loop class it does not by itself *select* the Markovian solution — the remark just showed the certified limit set can be strictly larger than the deterministic MFG. To buy back both — a quantitative rate and the selection of the Markovian solution — we must stop asking so little of the data. We must supply, from outside, the very dynamics the finite game refused to hand over: a sufficiently regular solution of the master equation of chapter 12, used as an ansatz whose residual against the Nash system we can control term by term. That is the strong hypothesis the next route pays, and what it buys in return is the subject of section 13.5.3.

13.5.3 Quantitative convergence via the master equation (the CDLL route)

The qualitative route handed us its conclusion together with its two deficits — *no rate*, and, in the closed-loop class, no *selection* of the Markovian solution (remark 13.10 showed the certified limit set can be strictly larger than the deterministic MFG) — and the second route is built to answer exactly those two. Both deficits trace to the same poverty of hypothesis: asking only for near-well-posedness, the qualitative route never supplied the finite game with the dynamics it refused to hand over. The remedy is to stop asking so little. We now *pay* a strong hypothesis — the existence of a sufficiently regular solution of the master equation of chapter 12 — and we will see it buy back precisely the two missing things: a quantitative rate, and the selection of the Markovian solution (the regular master field is a deterministic, Markovian object, so building the proof on it excludes the weak common-noise equilibria by fiat, as remark 13.10 already anticipated). This is the resolution of the *convergence problem* by Cardaliaguet, Delarue, Lasry and Lions [37]; see also [41, Ch. 6]. It delivers a rate for the values *and* a rate for propagation of chaos, and it is the one genuinely new mechanism of the chapter: every earlier result either reused chapter 7 verbatim (the top-down arc) or extracted a limit without a modulus (the qualitative route), whereas here we manufacture the rate from the regularity of U.

The strategy in one sentence, before any formalism: use the master field U as a candidate solution of the Nash system, measure how badly it fails to solve that system exactly, show the failure is $O(1/N)$, and then use a stability estimate to convert that $O(1/N)$ residual into an $O(1/N)$ error on the genuine Nash values. Everything below is the careful execution of these four moves, and every hypothesis we now load onto U is there to make one of them go through.

Recall from chapter 12 the master field $U : [0,T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$, solving the master equation eq. (12.7), whose value $U(t,x,m)$ is the optimal cost-to-go of a representative agent at state x when the population is in configuration m. We import its regularity by the specific theorem that carries it, theorem 12.8: under (A-Mon) and the smoothness of the data there, U is $\mathcal{C}^{1,2}$ in (t,x) and *fully* $\mathcal{C}^2$ in the measure variable in the sense of definition 12.1, with $\nabla_x U$ and $\partial_\mu U(t,x,m)(\cdot)$ bounded and Lipschitz uniformly in (t,x,m), and with the mixed derivative $\partial_x \partial_\mu U$ and the *second Lions derivative* $\partial^2_\mu U$ bounded. This regularity is the entire input below, and it is worth spending a moment on *why each piece is needed*, because the proof is little more than the orderly cashing-in of these derivatives, and a reader who knows in advance what each one pays for will see the bookkeeping as inevitable rather than arbitrary.

The smoothness in (t,x) is the cheapest demand: $\partial_t U$ and the spatial $\mathrm{D}^2_x U$ are simply the terms one needs to even write the Nash operator applied to the candidate, the analogue of asking that a function be twice differentiable before substituting it into a heat equation. The first measure derivative $\partial_\mu U$ is the load-bearing one. When we differentiate the candidate $U(t,x_i,\cdot)$ in a *spectator* coordinate x_j, the chain rule sends that derivative through the measure slot, and $\partial_\mu U$ is the object it produces; its *boundedness* is what keeps each of the $N-1$ cross-gradients $O(1/N)$, and its

Lipschitzness in the measure is what lets us, in the final trajectory step, replace one empirical measure by another (or by m_t) at controlled cost. The mixed derivative $\partial_x\partial_\mu U$ appears the instant we take a *second* spectator derivative: differentiating $\partial_\mu U(\cdot)(x_j)$ again in x_j hits its explicit spatial argument, and this term assembles — after summing the $N-1$ diagonal Hessian blocks — the master equation's extra second-order transport contribution $\int \operatorname{tr}(\sigma\sigma^\top\, \partial_x\partial_\mu U)\,\mathrm{d}m$; it must be bounded for that $O(1)$ contribution to be finite and to match term-by-term. Finally the second Lions derivative $\partial^2_\mu U$ is the subtle one, and the whole rate hinges on its boundedness. The same second spectator derivative also hits the measure slot of $\partial_\mu U(\cdot)(x_j)$, producing a $\partial^2_\mu U$ term weighted not by $1/(N-1)$ but by $1/(N-1)^2$. It is precisely *because* $\partial^2_\mu U$ is bounded that this doubly-suppressed term is genuinely $O(1/N)$ rather than a hidden $O(1)$ contaminant — and so it is $\partial^2_\mu U$, the very object chapter 12 worked hardest to construct, that certifies the residual is small. Strip away any one of these bounds and a term in the residual fails to be controlled; this is the sense in which the regularity of U *is* the rate.

Notation watch: Two points of contact with chapter 12

Two seams with chapter 12 deserve an explicit word, because the residual computation below leans on both. *(i) The second Lions derivative.* This is the seam most easily misread, so we draw it carefully. The phrase "fully $\mathcal{C}^2$ in m" of definition 12.1 is doing double duty. Part of it is the spatial regularity $\partial_x\partial_\mu U$ — one measure derivative, then a *spatial* derivative of the function it returns — and that part is not what we mean here. The other part is the genuine *second measure* derivative $\partial^2_\mu U(t,x,m)(y,y')$, obtained by differentiating $\partial_\mu U$ once more *in its measure slot*: it measures how the sensitivity of U to mass at y itself changes as one adds mass at y'. This is a strictly stronger and harder object; chapter 12 does not get it for free from the first derivative but constructs it by iterating the linearized MFG system one further time (Step 3 of the proof of theorem 12.8), the same step that the common noise theory needs. We lean on its *boundedness* in Steps 2–3 below, and the reason is exactly the weighting count flagged above: a second derivative in a single spectator coordinate x_j produces a $\partial^2_\mu U(\cdots)(x_j,x_j)$ term carrying the prefactor $1/(N-1)^2$, and only because $\partial^2_\mu U$ is bounded does that doubly-suppressed term sum — over the $N-1$ diagonal blocks — to something $O((N-1)\cdot(N-1)^{-2}) = O(1/N)$ rather than $O(1)$. Were $\partial^2_\mu U$ merely to *exist* without a uniform bound, this would be the one place the rate could silently break. We therefore import theorem 12.8 *including* that second-order bound, as recorded in its standing hypotheses, and not merely the first-order regularity that the phrase "$\mathcal{C}^2$ in m" might at a glance suggest. *(ii) General coefficients.* chapter 12 displays the master equation eq. (12.7) in its canonical form — control-as-drift, constant nondegenerate diffusion σ, additively separated coupling $F(x,m)$, Hamiltonian $H(x,p)$. The N-player data of eq. (13.2)–eq. (13.3) are more general: drift $b(x,m,\alpha)$, measure-dependent diffusion $\sigma(x,m)$, and a non-separated Hamiltonian $H(x,p,m)$. The master equation extends to this setting *verbatim* in structure: the Itô–Dynkin chain rule on $\mathcal{P}_2(\mathbb{R}^d)$ (lemma 12.2) is indifferent to where the measure dependence sits, so the only changes are that the transport term carries $\partial_p H(y,\nabla_x U,m)$, the extra-diffusion term carries $\sigma\sigma^\top(y,m)$, and the local Hamiltonian carries the third slot $H(x,\nabla_x U,m)$. No new term is created; on the separable subclass the equation collapses back to eq. (12.7) with F on the right. We use the general-coefficient version throughout and flag here that it is the harmless extension of the equation chapter 12 actually wrote.

Before the theorem we isolate the one differential-geometric identity its proof turns on: how an *ordinary* derivative in a particle's position acts on a functional of the empirical measure. The whole proof of theorem 13.13 is, at bottom, a translation problem. The master equation speaks the language of measure derivatives $\partial_\mu U$, $\partial_\mu^2 U$ — infinitesimal calculus on $\mathcal{P}_2(\mathbb{R}^d)$. The Nash system speaks the language of ordinary partial derivatives ∇_{x_j}, $\mathrm{D}^2_{x_j}$ — calculus on the finite configuration space $(\mathbb{R}^d)^N$. To check that U almost solves the Nash system we must dictionary one into the other, and the dictionary is the following lemma. It is this identity — not the generic chain rule — that converts the master equation into a statement about the Nash system, and the factors of $1/M$ it produces are the very factors that turn "$N-1$ small terms" into "one $O(1)$ integral plus an $O(1/N)$ remainder."

The mechanism behind it is worth seeing before the symbols arrive, because it is entirely concrete. The empirical measure $\nu^M = \frac{1}{M}\sum_k \delta_{y_k}$ assigns each particle a weight $1/M$. Nudge a single particle, $y_j \mapsto y_j + h$, and the measure changes by $\nu^M \mapsto \nu^M + \frac{1}{M}(\delta_{y_j+h} - \delta_{y_j})$: one atom of mass $1/M$ slides from y_j to $y_j + h$. So an ordinary derivative in y_j is a measure perturbation *of size* $1/M$, and that explains, before any computation, why every spatial derivative of F^M carries a factor $1/M$. The first derivative is clean: one factor $1/M$ times the measure derivative of F evaluated at y_j. The second derivative is where the structure that drives the rate first appears, because the single coordinate y_j now sits in *two* places at once — it is the point at which $\partial_\mu F$ is read off, *and* it is one of the atoms making up the measure ν^M at which $\partial_\mu F$ is evaluated. Differentiating again must hit both, and the two hits carry different powers of $1/M$: hitting the explicit evaluation point gives another $1/M$, while hitting the measure slot — a *second* one-atom perturbation — gives a further $1/M$ on top of the one already there, hence $1/M^2$. The Hessian therefore splits, structurally and unavoidably, into a $1/M$ piece and a $1/M^2$ piece. Keep that split in view; it is the entire reason values will converge at $1/N$.

Lemma 13.11 (Finite-dimensional restriction of a measure functional)**.** *Let $F : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ be fully $\mathcal{C}^2$ in the sense of definition 12.1. For M points $y_1, \dots, y_M \in \mathbb{R}^d$ set $\nu^M = \frac{1}{M}\sum_{k=1}^M \delta_{y_k}$ and $F^M(y_1, \dots, y_M) := F(\nu^M)$. Then F^M is twice continuously differentiable and, for each j,*

$$\nabla_{y_j} F^M = \frac{1}{M}\,\partial_\mu F(\nu^M)(y_j), \qquad \mathrm{D}^2_{y_j} F^M = \frac{1}{M}\,\partial_y \partial_\mu F(\nu^M)(y_j) + \frac{1}{M^2}\,\partial_\mu^2 F(\nu^M)(y_j, y_j). \tag{13.21}$$

Proof. Move a single atom: replacing y_j by $y_j + h$ changes ν^M by the signed measure $\frac{1}{M}(\delta_{y_j+h} - \delta_{y_j})$. By the definition of the flat derivative (definition 12.1; appendix A.3), to first order in h,

$$F^M(\dots, y_j + h, \dots) - F^M(\dots, y_j, \dots) = \frac{1}{M}\Big(\frac{\delta F}{\delta m}(\nu^M, y_j + h) - \frac{\delta F}{\delta m}(\nu^M, y_j)\Big) + o(|h|),$$

so $\nabla_{y_j} F^M = \frac{1}{M}\nabla_y \frac{\delta F}{\delta m}(\nu^M, y)\big|_{y=y_j} = \frac{1}{M}\partial_\mu F(\nu^M)(y_j)$, using $\partial_\mu F = \nabla_y\, \delta F/\delta m$ (definition 4.14). Differentiating this in y_j once more, the position y_j now enters in *two* places — the explicit argument of $\partial_\mu F(\nu^M)(\cdot)$ and, through ν^M, the measure slot. The first dependence yields $\frac{1}{M}\partial_y \partial_\mu F(\nu^M)(y_j)$; the second, by the same one-atom perturbation applied to the measure slot of $\partial_\mu F(\nu^M)(y_j)$ and the relation $\partial_\mu^2 F = \partial_\mu[\partial_\mu F]$, yields $\frac{1}{M^2}\partial_\mu^2 F(\nu^M)(y_j, y_j)$. This is eq. (13.21); full continuity of the derivatives is the "fully $\mathcal{C}^2$" hypothesis. ∎

The second ingredient is the analytic heart of the bottom-up rate, and it is the *one* place where monotonicity (A-Mon) does genuine work. Everything else in this subsection is differential calculus and bookkeeping; the smallness of the residual is geometry, not dynamics. But a small residual is worthless on its own — a nearly-solving candidate is only close to the true solution if the system does not *amplify* small discrepancies, and amplification is exactly what a forward–backward system, with its anticipatory coupling, threatens to do. The linearized Nash system is a coupled backward–forward pair; left to itself such a pair can have the residual feed the perturbation and the perturbation feed back into the residual, so that an $O(1/N)$ source drives an $O(1)$ response and the rate collapses. What forbids that feedback loop is monotonicity. The stability lemma says that the Nash system, linearized about its solution, is *dissipative*: a small source produces a proportionally small response, with a constant that does not deteriorate as N grows — the N-independence being indispensable, since an N-dependent constant $C_N \sim N$ would eat the whole $1/N$ and leave no rate at all. We state and prove it before using it, so that Step 4 of the convergence proof is an import rather than an assertion.

Lemma 13.12 (Stability of the linearized Nash system)**.** *Assume* (A-Lip)*,* (A-Nondeg)*,* (A-Conv)*,* (A-Mon)*. Let* $(\phi^{N,i})_{i\le N}$ *solve the linearization of the Nash system eq.* (13.6) *about its solution* $(v^{N,i})_{i\le N}$*, with zero terminal data* $\phi^{N,i}(T,\cdot)=0$ *and sources* $(R^{N,i})_{i\le N}$ *on the right-hand side. Then*

$$\max_{i\le N}\sup_{(t,\boldsymbol{x})}\big|\phi^{N,i}(t,\boldsymbol{x})\big| \;\le\; C\,\max_{i\le N}\sup_{(t,\boldsymbol{x})}\big|R^{N,i}(t,\boldsymbol{x})\big|,$$

with C depending on T, the Lipschitz and nondegeneracy constants, and the monotonicity modulus, but not *on N.*

Proof. Linearizing eq. (13.6) about $(v^{N,i})$ produces, for each i, a linear backward Kolmogorov equation for $\phi^{N,i}$ on $(\mathbb{R}^d)^N$: the second-order part is the same nondegenerate operator $\sum_j \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2_{x_j}\,\cdot)$, the first-order part transports along the (frozen) equilibrium drifts $b(x_j,\mu^N,\widehat{\alpha}^{\,j})$, and the coupling enters through the derivative of the equilibrium feedbacks $\widehat{\alpha}^{\,j}$ and of the Hamiltonian in the measure slot — precisely the terms that make the N equations simultaneous. The terminal data vanish and the source is $R^{N,i}$.

The estimate is the N-particle incarnation of the Lasry–Lions uniqueness computation of chapter 11, and it is worth seeing that it is the *same* computation, only with the single representative agent replaced by the configuration of N. Recall the shape of that argument: to prove MFG uniqueness one took two solutions, subtracted, paired the difference of the HJB equations against the difference of the Fokker–Planck equations, integrated by parts, and watched the transport and diffusion terms annihilate by duality while monotonicity gave the one surviving cross term a sign that forced the difference to vanish. Here we do the same to a *single* solution's linearization. Linearizing eq. (13.6) produces the backward perturbations $(\phi^{N,i})$ above; alongside them, linearize the forward equation for the equilibrium state density m^N_t in the same direction, obtaining a forward (Fokker–Planck) perturbation ρ^N_t. Now form the duality pairing $\frac{\mathrm{d}}{\mathrm{d}t}\langle\phi^{N,i},\rho^N\rangle$ and integrate by parts in $\boldsymbol{x}$.

The pairing is engineered precisely so that the leading operators cancel. The backward equation contributes $+\mathcal{A}\phi^{N,i}$ paired against ρ^N; the forward equation contributes $\phi^{N,i}$ paired against $\mathcal{A}^*\rho^N$, the formal adjoint of the *same* second-order/transport generator $\mathcal{A} = \sum_j [\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2_{x_j}) + b \cdot \nabla_{x_j}]$. By the very definition of the adjoint, $\langle \mathcal{A}\phi, \rho\rangle = \langle \phi, \mathcal{A}^*\rho\rangle$, so every transport term and every second-order term cancels between the two pairings — this is the HJB↔FP duality that the backward-value/forward-density structure was built to deliver, exactly the cancellation of chapter 11. What is *not* a transport or diffusion term, and therefore survives, is the coupling: the piece in which $\phi^{N,j}$ feeds the equation for $\phi^{N,i}$ through the measure derivative of the data, paired against the density perturbation it generates. This surviving term is precisely the linearized form of the Lasry–Lions coupling, and the monotonicity inequality eq. (11.3) of (A-Mon) says it carries a *favorable* sign: monotone coupling means an agent dislikes crowding toward where the perturbation pushes mass, so the cross term enters the energy balance with the sign that *drains* energy rather than feeding it. Concretely, the time-derivative of the paired energy is bounded by the source contribution *minus* a nonnegative monotone term — the monotone term can only help. Monotonicity thus closes off the one channel by which the backward–forward coupling could pump the perturbation: the linearized system is dissipative, not amplifying.

With the dangerous coupling sign-defeated, what remains is a one-directional estimate. The pairing reduces to a Grönwall inequality (appendix C.1) driven by the source $R^{N,i}$ alone, and nondegeneracy (A-Nondeg) supplies the parabolic smoothing (appendix C.9) that controls the first-order cross terms uniformly in N — uniformly because the smoothing constant depends on the ellipticity of $\sigma\sigma^\top$ and the Lipschitz data, none of which sees N. The sup bound follows with a constant independent of N. We are honest about what we have and have not done here: the *mechanism* — which sign survives the duality, why monotonicity makes it favorable, and that this is exactly what turns an $O(1/N)$ source into an $O(1/N)$ response with an N-free constant — is the content displayed above and stands complete in-book as the linearization of the chapter 11 computation. The remaining labour is purely the tracking of that constant through the parabolic estimates on $(\mathbb{R}^d)^N$, uniformly in the dimension N, and that bookkeeping is the technical core of [37], to which we defer it without apology, the way one defers the constant in a Sobolev inequality once its existence and shape are established. □

Theorem 13.13 (Convergence with rate)**.** *Assume* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv)*,* (A-Mon)*, and that the master equation of chapter 12 has a solution U with the regularity stated above. Let $(v^{N,i})_{i\le N}$ be the solution of the Nash system eq.* (13.6) *and let $(X^{i,N})_{i\le N}$ be the equilibrium state processes. Then there is a constant C, independent of N and i, such that:*

1. (Values, rate $1/N$.) *For all $(t, \boldsymbol{x})$,*

$$\left| v^{N,i}(t,\boldsymbol{x}) - U\big(t, x_i, \mu^{N,-i}_{\boldsymbol{x}}\big) \right| \le \frac{C}{N}, \qquad \mu^{N,-i}_{\boldsymbol{x}} = \frac{1}{N-1}\sum_{j\ne i} \delta_{x_j}. \tag{13.22}$$

2. (Trajectories / chaos with sampling rate.) *There exist i.i.d. copies $(\bar{X}^i)_{i\le N}$ of the MFG McKean–Vlasov solution eq.* (13.9)*, on the same probability space, with*

$$\max_{i\le N} \mathbb{E}\left[\sup_{t\le T} \big|X^{i,N}_t - \bar{X}^i_t\big|^2\right] \le C\, r_N^2, \qquad \mathbb{E}\left[\sup_{t\le T} \mathcal{W}_2\big(\mu^N_t, m_t\big)^2\right] \le C\, r_N^2, \tag{13.23}$$

with r_N the $\mathcal{W}_2$ sampling rate eq. (13.12) *of N i.i.d. draws from m_t (theorem D.16; attribution [69]). Consequently, coupling the $X^{i,N}$ to the i.i.d. copies $\bar{X}^i$ atom by atom, the k-marginal laws* *converge with the natural L^2 rate: since $\mathcal{W}_2(\mathcal{L}(X^{1,N},\dots,X^{k,N}),m^{\otimes k})^2 \le \sum_{i\le k}\mathbb{E}\sup_t|X^{i,N}_t - \bar X^i_t|^2 \le C\,k\,r_N^2$, one has $\mathcal{W}_2(\mathcal{L}(X^{1,N},\dots,X^{k,N}),m^{\otimes k}) \le C\sqrt{k}\,r_N$.*

Proof sketch. The proof is the one genuinely new mechanism of the chapter, so we name its parts. The idea, stated once more now that the tools are in hand, is to use the master field U as an *ansatz* for the Nash system, bound the residual by which it fails to solve that system exactly, and convert that residual into the value error by stability. Five steps: build the candidate (Step 1); differentiate its measure argument and watch the dictionary of lemma 13.11 turn $N-1$ small derivatives into one $O(1)$ integral plus an $O(1/N)$ remainder (Step 2); invoke the master equation so the $O(1)$ part annihilates term-by-term, leaving only the remainder as a source (Step 3); apply the stability lemma to convert the $O(1/N)$ source into the $O(1/N)$ value bound (Step 4); and pass from values to trajectories by Itô, Grönwall and Doob, with the sampling rate r_N the irreducible residue (Step 5). Steps 1–4 are the genuinely new content; Step 5 reuses the machinery of the top-down arc but feeds it an input the top-down arc had to *assume*.

Step 1: the projected ansatz. Define $w^{N,i}(t,\boldsymbol{x}) := U(t,x_i,\mu^{N,-i}_{\boldsymbol{x}})$, the master field read at agent i's own state and the empirical measure of the others. This is the natural finite-N projection of the limit object: U wants a representative state and a population law, and the configuration $\boldsymbol{x}$ supplies both — the tagged coordinate x_i for the state, the empirical measure $\mu^{N,-i}$ of the *others* for the law. (We use $\mu^{N,-i}$, excluding agent i itself, both because that is the population agent i actually faces and because it is the choice under which the dictionary lemma 13.11, with its $M=N-1$ atoms, applies cleanly to the spectator coordinates.) The aim is to show that this $w^{N,i}$ *almost* solves the Nash system eq. (13.6), missing it only by a source of size $O(1/N)$ — after which Step 4 will say that almost-solving means being within $O(1/N)$ of the true solution.

Step 2: differentiate the empirical-measure argument. This is the technical heart, and it is worth doing the bookkeeping in full, because every later step is a consequence of which power of $1/N$ each term carries. We must substitute $w^{N,i}$ into the Nash operator, and to do that we need all the configuration derivatives of $w^{N,i}$. The tagged coordinate x_i is easy: $w^{N,i}$ depends on it through the *state* slot of U, so $\nabla_{x_i}w^{N,i} = \nabla_x U(t,x_i,\mu^{N,-i})$ plus a measure-slot contribution of order $1/N$ that we absorb into the remainder (deleting/relabelling one atom shifts $\mu^{N,-i}$ by $O(1/N)$ and ∇_x does not change that). The interesting derivatives are in the *spectator* coordinates x_j, $j\ne i$, which enter $w^{N,i}$ *only* through the measure argument $\mu^{N,-i} = \frac{1}{N-1}\sum_{k\ne i}\delta_{x_k}$. This is exactly the situation of lemma 13.11: apply it to the functional $F = U(t,x_i,\cdot)$ with the $M=N-1$ points $(x_k)_{k\ne i}$, so that $\nu^M = \mu^{N,-i}$. For each $j\ne i$ the dictionary returns

$$\nabla_{x_j}w^{N,i} = \frac{1}{N-1}\,\partial_\mu U(t,x_i,\mu^{N,-i})(x_j), \qquad \mathrm{D}^2_{x_j}\,w^{N,i} = \frac{1}{N-1}\,\partial_x\partial_\mu U(t,x_i,\mu^{N,-i})(x_j) + \frac{1}{(N-1)^2}\,\partial^2_\mu U(\cdots)(x_j,x_j)$$

Now feed these into the two spectator sums of the Nash operator and watch the powers of $1/N$ do their work.

The transport sum reconstitutes an integral. Each cross-gradient is individually $O(1/N)$ — the factor $1/(N-1)$ in front, with $\partial_\mu U$ bounded. Naively one might therefore hope to drop them, and that hope is the same trap as in the heuristic of section 13.2: there are $N-1$ of them, and $(N-1)\cdot O(1/N) = O(1)$. They do not vanish; they *assemble*. Writing out the Nash transport sum with the gradient just computed,

$$\sum_{j\neq i} b(x_j, \mu^{N,-i}, \widehat{\alpha}^j) \cdot \nabla_{x_j} w^{N,i} = \sum_{j\neq i} b(x_j, \mu^{N,-i}, \widehat{\alpha}^j) \cdot \frac{1}{N-1}\, \partial_\mu U(t, x_i, \mu^{N,-i})(x_j),$$

and now the prefactor $1/(N-1)$ in front of a sum over the $N-1$ spectators is *precisely* the weight that turns the sum into an average, i.e. into an integral against $\mu^{N,-i}$:

$$\sum_{j\neq i} b(x_j, \cdot, \cdot) \cdot \nabla_{x_j} w^{N,i} = \int b(y, \mu^{N,-i}, \cdot) \cdot \partial_\mu U(t, x_i, \mu^{N,-i})(y)\, \mu^{N,-i}(\mathrm{d}y).$$

This is the term-by-term reconstitution the prose promised: the $O(1)$ collective influence of the crowd, which the heuristic relocated from the spectator slots into the measure derivative, has now literally re-emerged as the master equation's transport integral $\int b \cdot \partial_\mu U \, \mathrm{d}m$. Nothing was approximated in this identity — it is an exact rewriting — which is why the $O(1)$ part of the residual will cancel exactly, not just to leading order.

The Hessian sum splits into $O(1) + O(1/N)$. The same arithmetic, applied to the diagonal second-order blocks, is where the two powers of $1/N$ part company. Summing the Hessian formula over $j \neq i$ against the diffusion,

$$\sum_{j\neq i} \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top(x_j, \mu^{N,-i})\, \mathrm{D}^2_{x_j} w^{N,i}\big) = \underbrace{\sum_{j\neq i} \frac{1}{N-1}\, \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top(x_j, \cdot)\, \partial_x\partial_\mu U(\cdots)(x_j)\big)}_{\text{(I)}} + \underbrace{\sum_{j\neq i} \frac{1}{(N-1)^2}\, \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top(x_j, \cdot)\, \partial^2_\mu U(\cdot}_{\text{(II)}}$$

Term (I) carries a single $1/(N-1)$ over a sum of $N-1$ bounded summands, so by the same averaging it is $O(1)$ and assembles into the master equation's second transport-type contribution $\int \frac{1}{2}\operatorname{tr}\left(\sigma\sigma^\top(y, \cdot)\, \partial_x\partial_\mu U(t, x_i, \cdot)(y)\right) \mu^{N,-i}(\mathrm{d}y)$ — this is the term whose finiteness required the *boundedness* of $\partial_x\partial_\mu U$ flagged earlier. Term (II) carries $1/(N-1)^2$ over a sum of $N-1$ summands, so it is $(N-1)\cdot(N-1)^{-2}\cdot O(1) = O(1/N)$ — *provided* the summands are bounded, which is exactly the boundedness of $\partial^2_\mu U$ imported from theorem 12.8. Here is the payoff of the one-atom-perturbation count of lemma 13.11: the Hessian's $1/M$ piece ($\partial_x\partial_\mu U$) survives into the limiting equation as an $O(1)$ structural term, while its $1/M^2$ piece ($\partial^2_\mu U$) is doubly suppressed and falls into the $O(1/N)$ remainder. The split is not a happy accident; it is the precise reason values converge at $1/N$, and Step 3 now collects the two halves.

Step 3: invoke the master equation. Now assemble the Nash operator applied to $w^{N,i}$ and separate the $O(1)$ skeleton from the $O(1/N)$ debris. Collecting the $O(1)$ pieces from Step 2 — the t-derivative $\partial_t U$, the tagged-state diffusion $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top(x_i, \cdot)\,\mathrm{D}^2_x U)$, the reconstituted transport integral $\int b \cdot \partial_\mu U \, \mathrm{d}\mu^{N,-i}$, the reconstituted extra-diffusion integral $\int \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \partial_x\partial_\mu U)\, \mathrm{d}\mu^{N,-i}$ from term (I), and the local Hamiltonian $H(x_i, \nabla_x U, \mu^{N,-i})$ — the combination

$$-\partial_t w^{N,i} - \sum_j \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top\, \mathrm{D}^2_{x_j} w^{N,i}) - \sum_{j\neq i} b \cdot \nabla_{x_j} w^{N,i} + H(x_i, \nabla_{x_i} w^{N,i}, \mu^{N,-i})$$

is, *term by term*, exactly the left-hand side of the general-coefficient master equation of chapter 12 evaluated at the point $(t, x_i, \mu^{N,-i})$. This is the moment the whole construction was built for: the master equation was *defined* to be the equation U satisfies, with a ∂_t, a state-diffusion, a transport integral against $\partial_\mu U$, an extra-diffusion integral against $\partial_x\partial_\mu U$, and a local Hamiltonian — and Step 2's reconstitution produced one Nash term for each master term, in the same place, with the same coefficient. (This is precisely where we cash the general-coefficient extension of the notation-watch: the Nash data carry $b(x, m, \alpha)$, measure-dependent $\sigma(x, m)$ and a non-separated $H(x, p, m)$, and the master equation we match against is the verbatim-structure extension that carries those same three slots; on the separable subclass it would collapse to eq. (12.7) with F on the right, but we never need that collapse.) Because U solves that equation (theorem 12.8, in the extended form recorded in the notation-watch above), this entire $O(1)$ skeleton *vanishes*. What is left is exactly the debris Step 2 set aside: the $O(1/N)$ remainder from term (II) (the $\partial^2_\mu U$ contribution, $O(1/N)$ by its boundedness), the $O(1/N)$ measure-slot correction to $\nabla_{x_i} w^{N,i}$, and the $O(1/N)$ from replacing $\mu^{N,-i}$ by μ^N where the coefficients carry the full population. Each is bounded by C/N using the boundedness of $\partial^2_\mu U$, $\partial_x\partial_\mu U$ and the Lipschitzness of $\partial_\mu U$. Hence $w^{N,i}$ solves eq. (13.6) *up to* a source $R^{N,i}$ of size $\|R^{N,i}\|_\infty \leq C/N$: the master field almost solves the Nash system, and "almost" is quantitatively $O(1/N)$.

Step 4: stability of the Nash system. The error we care about is $\phi^{N,i} = v^{N,i} - w^{N,i}$, the gap between the true Nash value and the master-field candidate. Subtract the two equations: $v^{N,i}$ solves the Nash system exactly, $w^{N,i}$ solves it up to the source $R^{N,i}$, and both share the terminal data $g(x_i, \mu^{N,-i})$ at $t = T$ (the master field's terminal condition is exactly the Nash terminal cost), so $\phi^{N,i}$ solves the *linearized* Nash system with zero terminal data and source of size C/N. This is the precise hypothesis of lemma 13.12, which now does its single decisive job: dissipativity, the favorable monotone sign, converts the $O(1/N)$ source directly into an $O(1/N)$ response with an N-independent constant, yielding the value bound eq. (13.22). Without stability the small residual would be useless — Step 3 alone says only that $w^{N,i}$ nearly solves an equation, not that it is near the solution. It is monotonicity that licenses the inference from the first to the second, and this is the step at which (A-Mon) is indispensable and the reason remark 13.15 flags the entire rate as a monotone-regime result.

Step 5: from values to trajectories. The value bound is the hard-won input; turning it into a trajectory and chaos rate is now a controlled comparison of SDEs, the same machinery as the top-down lemma 13.5 but fed a fact that the top-down arc had to take on faith. The link is the feedback. The equilibrium agent i plays $\widehat{\alpha}^i = \alpha^*(X^{i,N}, \nabla_{x_i} v^{N,i}, \mu^N)$, built from the *true* Nash gradient $\nabla_{x_i} v^{N,i}$; the MFG copy plays $\widehat{\alpha}(X^{i,N}) = \alpha^*(X^{i,N}, \nabla u, m_t)$, built from the *limit* gradient ∇u. By Step 4, $v^{N,i}$ and $w^{N,i} = U(t, x_i, \mu^{N,-i})$ agree to $O(1/N)$; differentiating that agreement in x_i (parabolic gradient estimates upgrade the C^0 value bound to a gradient bound) and using $\nabla_x U(t, x, m_t) = \nabla u(t, x)$ at the limit measure, the two feedbacks differ by $O(1/N)$ plus the sampling discrepancy $\mathcal{W}_2(\mu^N_t, m_t)$ — here the Lipschitzness of α^* and of $\partial_\mu U$ in the measure is what bounds the effect of moving the measure argument. This is the crux that distinguishes the two routes, and it deserves to be said flatly: in lemma 13.5 the closeness of the played feedback to the MFG feedback was *built in* — the agents were *handed* the decentralized MFG control by hypothesis — whereas here the agents play their own selfish equilibrium feedbacks, and their closeness to the MFG feedback is *derived*, the output of Steps 1–4 and ultimately of the master equation's regularity. That derivation is precisely what "propagation of chaos for Nash equilibria" means with a rate attached.

With the feedbacks shown $O(1/N)$-close, compare the equilibrium states $X^{i,N}$ with i.i.d. MFG copies $\bar{X}^i$ (solutions of the representative dynamics eq. (13.9)) driven by the *same* Brownian motions. Subtracting the two SDEs and applying Itô (theorem 5.9), the difference of drifts has two sources: the $O(1/N)$ feedback gap just bounded, and the genuine population discrepancy $\mathcal{W}_2(\mu_t^N, m_t)$ entering through the Lipschitzness of b and $\partial_\mu U$ in the measure. A Grönwall argument (appendix C.1) closes the differential inequality, and Doob's maximal inequality (theorem C.2) upgrades it to the sup-in-time bound of eq. (13.23). The irreducible term is the sampling error $r_N = \mathbb{E}\,\mathcal{W}_2(\frac{1}{N}\sum \delta_{\bar{X}_t^i}, m_t)$ — *not* the $1/N$ of the values, because resolving the population by N samples is a genuinely harder task than evaluating a smooth functional of it, exactly the asymmetry remark 13.14 will dissect. The $1/N$ feedback gap is asymptotically negligible against r_N, so the trajectory rate is r_N, precisely as in the top-down lemma 13.5 — but now standing on a derived rather than an assumed feedback closeness. The complete argument, with the constants tracked through the parabolic estimates on $(\mathbb{R}^d)^N$, occupies [37] and [41, Ch. 6]. □

Remark 13.14 (Reading the two rates)**.** The asymmetry between $1/N$ for values eq. (13.22) and r_N (the sampling rate eq. (13.12), e.g. $N^{-1/4}$ for $d \leq 3$) for trajectories eq. (13.23) is not a defect of the proof; it is structural, it is the central quantitative lesson of the chapter, and it is worth internalizing through the mechanism that produced it. A value is a *scalar functional* of the population — one number, $U(t, x, m)$, read off the cloud. Expand it around the limit measure: the leading fluctuation of the empirical measure about its mean m_t is $O(N^{-1/2})$ by the central limit theorem, and it enters U *linearly* through the first measure derivative $\partial_\mu U$. But that linear term integrates a *centered* fluctuation — the deviations of the N sampled atoms from their common law average to zero, $\frac{1}{N}\sum_j[\delta_{X^j} - m]$ being mean-zero by construction — so the $O(N^{-1/2})$ contribution *cancels in expectation*, and the first surviving contribution is the *second*-order $\partial_\mu^2 U$ term, which is quadratic in the $O(N^{-1/2})$ fluctuation and hence $O(1/N)$.

This is precisely the envelope cancellation of the LQG worked computation, and the identification is exact, not analogical. There (eq. (13.20)) the agent's cost sat at a *critical point* in its target, so the first-order response to an $O(N^{-1/2})$ target fluctuation vanished and only the square, $O(1/N)$, survived — "a fluctuation costing its square because the comparison is anchored at a critical point." Here the master field U *is* that critical-point object, the value of an already-optimized agent, so its first-order sensitivity to a centered population fluctuation averages away by the same first-order-optimality, and again only the $\partial_\mu^2 U$ square survives at $O(1/N)$. The two squarings of the LQG calculation — one from the CLT, one from the envelope — are the same two we just ran abstractly: $N^{-1/2}$ from sampling, squared to $1/N$ by the second-derivative structure of a value at an optimum. The empirical measure itself enjoys no such cancellation. It is not a scalar at an optimum but the population to be *resolved*, and resolving a d-dimensional law by N samples costs the Fournier–Guillin rate r_N in every metric that sees space, with no envelope to rescue it — in high dimension the empty space between samples is the bottleneck, and averaging cannot fill it. Thus the moral: values converge faster than the objects whose laws they encode, because a value averages the population while a measure must reconstruct it. The same dichotomy, with the same envelope explanation, will reappear in the network setting of chapter 25, where it must be run a second time against the graph limit.

Remark 13.15 (What is conditional)**.** theorem 13.13 is only as strong as its hypothesis, and intellectual honesty demands we say exactly how conditional it is. The whole edifice rests on the existence of a *regular* solution U of the master equation — $\mathcal{C}^{1,2}$ in (t,x), fully $\mathcal{C}^2$ in m with bounded $\partial_\mu U$, $\partial_x\partial_\mu U$, $\partial_\mu^2 U$. chapter 12 delivers exactly this, but it delivers it *under* (A-Mon) and additional smoothness of the data; the monotonicity is not decoration but the very condition under which the linearized MFG system, iterated to build $\partial_\mu U$ and $\partial_\mu^2 U$, stays well-posed. Strip away monotonicity and two things break at once, both of which we have already seen. First, the master equation may have *no* smooth solution: outside the monotone regime characteristics of the underlying forward–backward system can cross, the value can develop shocks in the measure variable, and there is then no regular U to serve as ansatz — Step 1 has no candidate to build. Second, even if one foregoes the rate and retreats to the qualitative theorem 13.9, the closed-loop limit set can contain the weak, common-noise equilibria of remark 13.10, so the *selection* of the Markovian solution is also lost. Monotonicity, in other words, is doing double duty throughout this subsection: it is what makes the ansatz *exist* (via chapter 12) and what makes the residual-to-error conversion *work* (via the dissipativity of lemma 13.12). The honest summary is the slogan to carry forward: *rates are a monotone-regime luxury.* Outside that regime the quantitative theory is genuinely open, and even the qualitative selection of the limit is delicate. The sharp thresholds — how much non-monotonicity the smooth master field tolerates before it cracks — are part of the open landscape catalogued in chapter 31.

We have closed the loop the chapter set out to close. Chapters 8–12 postulated the MFG system and the master equation as the right description of a large strategic population; this chapter has earned them, by exhibiting the finite N-player game they coarse-grain and certifying the passage in both directions. We export three results, the three milestones promised in the preamble. The ε-Nash property (theorem 13.6): the decentralized MFG feedback is an ε_N-equilibrium of the finite game, with $\varepsilon_N \to 0$ at the sampling rate — the cheap top-down certificate that the limit object is at least *playable* at finite N. Qualitative convergence (theorem 13.9): the actual Nash equilibria are precompact and their limits solve the MFG relations, the minimal-hypothesis bottom-up statement, with the open/closed-loop caveat attached. And quantitative convergence (theorem 13.13): under monotonicity and a regular master field, the values converge at $1/N$ and the trajectories and chaos at the sampling rate r_N, with the closeness of equilibrium-to-MFG feedback no longer assumed but *derived* — propagation of chaos for Nash equilibria with a rate.

Two things travel onward from here as the chapter's lasting deposits, and both recur. The first is the dichotomy *values converge faster than the measures whose laws they encode* ($1/N$ versus r_N), powered by the envelope cancellation that remark 13.14 traced from the LQG worked computation through to the $\partial_\mu^2 U$ term: a value at an optimum is first-order blind to centered population fluctuations, a measure is not. The second is the caveat that all of this quantitative control is a *monotone-regime luxury*, bought with the regularity of U and forfeit where that regularity fails. chapter 14 now instantiates the entire arc on the LQG model, where U is an explicit Riccati quadratic and every rate above becomes a closed-form constant one can read off — the laboratory in which the abstract $1/N$-versus-r_N split is verified by hand. chapter 25 then runs the same mechanism a second time in the network setting, where a graph limit is superimposed on the mean field limit and the values-faster-than-measures dichotomy, together with its monotone-regime caveat, reappears against *two* limits at once. The machinery built here is exactly what those chapters inherit.

Rigor ledger

Proved (in full, in this chapter).

- The Nash system eq. (13.6): the PDE characterization of Markovian Nash equilibria as N coupled HJB equations on $(\mathbb{R}^d)^N$ (proposition 13.3), and the precise sense in which it is intractable (remark 13.4).
- The ε-Nash property of the mean field strategy with rate $\varepsilon_N \leq Cr_N$ (theorem 13.6), via propagation of chaos for the fixed feedback (lemma 13.5); and its LQG sharpening to $\varepsilon_N \leq C/N$ (worked computation 13.8, eq. (13.20)).
- The finite-dimensional restriction identity eq. (13.21) (lemma 13.11) and the linearized-Nash-system stability lemma (lemma 13.12), the two ingredients on which the master-equation residual estimate (Steps 1–4 of theorem 13.13) rests; assembled in-book, they reduce theorem 13.13 to imported regularity.

Assumed (imported by `cref` to an in-book proof, or hypothesized).

- Existence and feedback regularity of the MFG solution (u, m) and the representative McKean–Vlasov dynamics (chapters 8 and 9); Itô's formula (theorem 5.9), Grönwall (appendix C.1), and the Doob/Burkholder–Davis–Gundy maximal inequality (theorem C.2); the standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv).
- The Fournier–Guillin sampling rate eq. (13.12), proved in-book in theorem D.16 (appendix D) by a dyadic multiresolution comparison of $\mathcal{W}_2$ with the per-cube mass discrepancies and a binomial-variance estimate; remark 7.16 records the same rate's role in propagation of chaos. Attribution [69].
- For theorem 13.13: a regular solution U of the master equation with bounded $\partial_\mu^2 U$, imported by theorem 12.8 (chapter 12), together with monotonicity (A-Mon) from chapter 11 (eq. (11.3)).

Deferred (stated here, full in-book proof in a named appendix).

- The qualitative convergence theorem 13.9 (compactness/martingale method): full argument — relaxed-control compactness, path-space tightness, martingale-optimality identification, and uniqueness under (A-Mon) — in appendix G. Attribution: [89, 90, 67], [40, Ch. 6].
- The quantitative convergence theorem 13.13: the residual computation and stability are in-book (Steps 1–4, lemmas 13.11 and 13.12); the detailed constant bookkeeping is [37], [41, Ch. 6].
- The master-equation regularity it relies on: theorem 12.8 (chapter 12) and Appendix D.

Open.

- Rates of convergence outside the monotone regime, where the master equation need not be smooth; sharp ε-Nash and convergence thresholds (chapter 31).

- The full characterization of closed-loop limits and the weak MFG solutions they may select (remark 13.10); when, beyond monotonicity, closed-loop limits coincide with the Markovian MFG.
- Dimension-free trajectory rates: whether the r_N in eq. (13.23) can be improved below the sampling barrier for special structures other than LQG.

Bibliographic notes

The ε-Nash property and the mean field heuristic for large symmetric games originate with the Nash certainty equivalence principle of Huang, Caines and Malhamé [82] on the engineering side and, independently, with Lasry and Lions [93] on the analytic side; both established the top-down direction for special structures, and the linear–quadratic case of worked computation 13.8 is essentially theirs. The systematic probabilistic treatment of the N-player game, the Nash system, and the ε-Nash theorem in the generality of theorem 13.6 is in Carmona and Delarue [40, Vol. I, Ch. 6]; the propagation of chaos input lemma 13.5 traces to Sznitman's coupling (chapter 7), with the Wasserstein sampling rate eq. (13.12) proved in-book in theorem D.16 (Fournier and Guillin [69]).

The convergence of N-player Nash equilibria to the MFG — the bottom-up direction — has two strands. The compactness/martingale-problem strand, giving qualitative convergence under minimal assumptions and clarifying the open-loop versus closed-loop distinction, is due to Lacker [89, 90] and Fischer [67]; [90] in particular is the source of the subtlety in remark 13.10. We give the full compactness-and-identification argument in-book in appendix G. The quantitative strand, resolving the convergence problem with the $1/N$ value rate eq. (13.22) through the regularity of the master equation, is the monograph of Cardaliaguet, Delarue, Lasry and Lions [37], with an alternative exposition in Carmona and Delarue [41, Vol. II, Ch. 6]. The master equation itself is developed in chapter 12; the LQG instantiation of every result here is carried out in chapter 14, and the network generalization — where a second limit, graph convergence, must be taken jointly — is the subject of chapter 25.

Chapter 14

Capstone: The Linear–Quadratic–Gaussian Mean Field Game

This chapter is where the abstractions of Part II become a calculation one can carry out with a pencil. We take the single carried example of the book—Thread A, the linear–quadratic–Gaussian (LQG) mean field game fixed once and for all in the project's thread specification—and solve it *end to end*. For this one model every abstract object built in the preceding chapters turns out to be an elementary function of time. The backward Hamilton–Jacobi–Bellman (HJB) equation collapses to a scalar Riccati ordinary differential equation; the forward Fokker–Planck equation collapses to a pair of ordinary differential equations for the mean and variance of a Gaussian; and the mean field fixed point—the self-consistency between what each agent optimally does and what the population it best-responds to actually is—becomes a linear two-point boundary value problem whose solvability we read off from a single scalar $D(T)$. Nothing is hidden inside an existence theorem invoked from afar; one can check every claim by hand. And out of that one solvability scalar falls a genuine *phase diagram*: a curve in parameter space separating a regime of unique equilibrium from a regime where the equilibrium degenerates.

The single sharp result the chapter is built around is that *uniqueness of equilibrium outlives the monotonicity that is supposed to guarantee it.* To feel why this is a surprise, recall the expectation Part II sets up. The Lasry–Lions monotonicity condition (A-Mon) is the one general-purpose hypothesis that delivers uniqueness across the whole theory, so the natural guess is that uniqueness lives exactly where monotonicity does and fails the instant it lapses. Monotonicity, read physically, is repulsion: a congestion coupling that spreads agents apart, the kind of interaction one expects to stabilize a single equilibrium. Its opposite is imitation, an attractive coupling that lets the population clump—the ferromagnetic case in which one fully expects the agents to be able to organize themselves into several distinct self-consistent histories, a symmetry breaking with multiple equilibria, the moment the sign of the coupling flips. The result of the explicit calculation is that it does *not* flip at the sign change. Uniqueness survives the entire non-monotone (imitation) band, all the way up to a finite imitation threshold $s = 1$, and is lost only beyond it, at a conjugate horizon

$T_c(s)$ of the forward–backward system. Monotonicity, that is, is sufficient but strictly not necessary; the true uniqueness boundary is a spectral, conjugate-point event, set by a competition between the pull of imitation and the restoring stiffness of the convex control cost, not by the sign of the coupling. The gap between the monotone region and that true boundary is the whole point of the chapter.

We reach this by importing, and instantiating one by one, the machinery of the preceding chapters; the chapter is best read as an assembly of previously built atoms rather than a fresh derivation. The single-agent HJB and its verification theorem come from chapter 6; the coupled HJB–Fokker–Planck system is the analytic mean field game system of chapter 8; its probabilistic shadow, a forward–backward stochastic differential equation (FBSDE), is from chapter 9; existence is the fixed-point argument of chapter 10; uniqueness and the monotonicity condition are from chapter 11; the master equation is from chapter 12; and the finite-player justification is from chapter 13. Each will reappear below not as a citation but as an explicit object one can write down. The standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), and (A-Mon) are used by name throughout, and we check honestly which hold for LQG and which must be replaced by model-specific linear–quadratic arguments. That audit is itself informative: LQG is the cleanest example in the book precisely because it sits at the *boundary* of several of these generic hypotheses. Its diffusion is nondegenerate only because we keep $\sigma > 0$; its costs are coercive but, being quadratic, only locally Lipschitz and so outside the bounded-Lipschitz hypothesis under which the generic existence theorem is stated; and its coupling is monotone for one sign of s and not the other. Each generic hypothesis is met in the simplest possible way or fails in a single, diagnosable way—which is exactly what makes the example clean and the audit worth performing carefully.

We follow the global convention recalled in appendix A: the value function $u = u(t, x)$ carries terminal data at $t = T$ and is computed backward in time, while the density $m = m(t, x)$ carries initial data at $t = 0$ and evolves forward. Because no graphon is present, Brownian motion is written W, following the Part II convention. With those conventions fixed, the last two sections of the chapter look forward. We record the matrix (multidimensional) version of every formula, and we explain in what precise sense this chapter is the *template* for the graphon mean field game of Part IV: replacing the scalar population mean by the graphon-aggregated field $(\mathbb{W}\,\bar{m}_t)(x)$, indexed by the type $x \in [0, 1]$, turns every Riccati equation here into a type-indexed family, and the rank-one kernel collapses that family back to a single scalar order parameter. Those generalizations are carried out in chapter 20 and chapter 22; here we build the object they generalize.

14.1 The model, and which assumptions survive

We restate Thread A exactly as fixed in the thread specification, so that nothing is silently redefined and the reader can match every symbol below against the master list. The representative agent has a scalar state $X_t \in \mathbb{R}$ obeying the controlled linear stochastic differential equation

$$\mathrm{d}X_t \;=\; \big(a\,X_t + b\,\alpha_t\big)\;\mathrm{d}t \;+\; \sigma\;\mathrm{d}W_t, \qquad X_0 \sim m_0 \in \mathcal{P}_2(\mathbb{R}), \tag{14.1}$$

driven by a one-dimensional Brownian motion W, with constant coefficients $a \in \mathbb{R}$, $b \in \mathbb{R} \setminus \{0\}$, and $\sigma \geq 0$. Each coefficient carries a concrete meaning that recurs throughout the chapter. The *state feedback* a is the agent's intrinsic linear drift: $a > 0$ pushes the state away from the origin, a runaway tendency the control must fight, while $a < 0$ pulls it back, a mean reversion the control can ride. The *control gain* b measures how much steering authority one unit of control buys; we insist $b \neq 0$ so that the control actually reaches the state, for if b vanished the agent would be a passive spectator with nothing to optimize. It is the *square* b^2, the product of one factor of b converting control into drift and a second converting marginal value into optimal control, that will appear in every closed-loop formula below. The *noise amplitude* $\sigma \geq 0$ sets the size of the idiosyncratic Brownian shock each agent absorbs. The control α_t is feedback, $\alpha_t = \alpha(t, X_t)$, in the admissible class of chapter 6.

The agent is coupled to the population only through the population mean

$$\bar{m}_t \;:=\; \int_{\mathbb{R}} \xi \, m_t(\mathrm{d}\xi), \tag{14.2}$$

and through nothing else—not the variance, not any higher moment—which is precisely what will let the entire population layer reduce to the motion of a single scalar. The agent pays the cost

$$J(\alpha) \;=\; \mathbb{E}\bigg[\int_0^T \Big(\tfrac{1}{2}\,\alpha_t^2 + \tfrac{q}{2}\big(X_t - s\,\bar{m}_t\big)^2\Big)\,\mathrm{d}t \;+\; \tfrac{q_T}{2}\big(X_T - s_T\,\bar{m}_T\big)^2\bigg], \tag{14.3}$$

which is worth reading term by term. The control cost $\frac{1}{2}\alpha_t^2$ charges the agent for acting: steering is never free, and the strictly convex quadratic shape of this charge is exactly what will make the optimal control unique and linear. The running state cost $\frac{q}{2}(X_t - s\bar{m}_t)^2$ penalizes, with weight $q \geq 0$, the squared distance between the agent's state and the *moving target* $s\,\bar{m}_t$: the agent would like to sit not at the origin but at s times the crowd mean. The terminal cost imposes the same penalty at the horizon, with its own weight $q_T \geq 0$ and its own target $s_T\,\bar{m}_T$. The weights q, q_T thus set *how badly* the agent wants to track its target, and the *mean-coupling strengths* $s, s_T \in \mathbb{R}$ set *what* that target is—where, relative to the crowd, the agent is trying to be.

The sign of s is the whole story of this chapter, and it is worth dwelling on. When $s > 0$ the target $s\bar{m}_t$ sits on the same side of the origin as the crowd, so the agent is *drawn toward* the population: this is *imitation*, the wish to be where everyone else is—to hold the popular asset, to adopt the common opinion, to crowd into the busy venue. When $s < 0$ the target sits on the opposite side, so the agent is *repelled* from the crowd: this is *congestion* or aversion, the wish to dodge the jam, to differentiate, to flee the overbooked resource. The borderline case $s = 0$ severs the coupling entirely and returns the plain single-agent problem of chapter 6. Everything quantitative that follows—whether the equilibrium is unique, and if not at what horizon it ceases to be—turns on this sign and on its magnitude. There is no discount in the carried thread; we comment on the discounted variant in remark 14.14.

What should it mean to have "solved" this game? No agent controls the mean $\bar{m}_t$; each takes it as given and optimizes against it, and yet the mean is nothing but the aggregate of what all the agents do. An equilibrium is the value of $\bar{m}_\cdot$ at which these two readings of it coincide: the flow each agent best-responds to is the very flow their best responses produce. This is the mean field specialization of the Nash idea—no agent can improve unilaterally, and beliefs about the population match the population itself—and it is naturally a fixed-point statement in two parts.

Definition 14.1 (LQG mean field game equilibrium)**.** A *mean field equilibrium* for (14.1)–(14.3) is a deterministic flow $t \mapsto \bar{m}_t$ on $[0, T]$ together with a feedback control α^* such that (i) α^* minimizes $J(\alpha)$ when the flow $\bar{m}.$ in (14.3) is held fixed, and (ii) the mean of the law of the resulting optimally controlled state X^* reproduces the flow, $\mathbb{E}[X_t^*] = \bar{m}_t$ for all $t \in [0, T]$. We call the pair (u, m) solving the associated HJB–Fokker–Planck system (section 14.4) the equilibrium in PDE form.

The two clauses are exactly the two arrows of the self-consistency loop. Clause (i) is the *optimization* arrow: freeze the population flow $\bar{m}.$, so that the cost (14.3) becomes an ordinary single-agent control problem, and let the agent best-respond to it. Clause (ii) is the *aggregation* arrow: take the population of such optimizers, all started from m_0 and all using the resulting optimal feedback, and ask what mean they actually generate. A flow is an equilibrium when chasing these two arrows around the loop returns it unchanged. All of the game-theoretic content—the fact that this is a game and not a single optimization—lives in the demand that the loop close; the rest of the chapter is the mechanics of closing it. The pair (u, m) named in the definition are the backward value function and forward density of the HJB–Fokker–Planck system we assemble in section 14.4.

Before computing, we audit the standing assumptions against this model, assumption by assumption. This audit is the honest accounting the rigor doctrine demands, and here it earns its keep concretely: hypothesis by hypothesis it doubles as a map of which parts of the Part II theory we may quote directly for LQG and which we must replace with a hands-on, model-specific argument.

Begin with nondegeneracy. The diffusion here is the scalar $\sigma\sigma^\top = \sigma^2$, and **(A-Nondeg)**—uniform nondegeneracy $\sigma^2 \geq \lambda > 0$—holds exactly when $\sigma > 0$. We assume $\sigma > 0$ throughout the main development. Its job is to keep the Fokker–Planck equation genuinely parabolic, so that the population law carries a smooth Gaussian density that spreads in time rather than a collapsing deterministic flow; without it the forward equation degenerates to a pure transport equation and the density can concentrate on a moving point. The backward Riccati structure, by contrast, never sees σ, so the deterministic limit $\sigma = 0$ remains solvable—we flag it where the distinction bites—but it is degenerate as a diffusion and forfeits the parabolic regularization (A-Nondeg) buys.

Convexity is equally immediate. The control enters the dynamics (14.1) linearly and is charged the strictly convex cost $\frac{1}{2}\alpha^2$, so the minimization over α that defines the Hamiltonian is the minimization of a strictly convex quadratic and has a unique minimizer. This is **(A-Conv)**: the Hamiltonian is convex in the adjoint variable, the optimal feedback is unique and linear in the state, and the verification theorem of chapter 6 applies verbatim. Strip the convexity away—make the control charge concave, or let the control enter the dynamics nonlinearly—and the infimum over α need be neither attained nor unique, the feedback would be ill-defined, and the clean reduction of the next section could not even begin.

The subtle entry in the audit is the pair **(A-Lip)** and **(A-Growth)**, and it repays being slow about, because the chapter leans on its conclusion twice more. The *dynamics* are no trouble: the drift $(x, \alpha) \mapsto ax + b\alpha$ is affine, hence globally Lipschitz, so the coefficient side of (A-Lip) holds outright. The *costs* are where the difficulty hides. The generic existence theorem of chapter 10 is packaged under *bounded, globally Lipschitz* couplings—a hypothesis chosen so that the best-response map is a continuous self-map of a fixed compact set of measures, with uniform a priori bounds handed over for free, to which a Schauder fixed-point argument then applies. The LQG costs satisfy neither half of that hypothesis: the quadratic $\frac{q}{2}(x - s\bar{m})^2$ is unbounded, and its gradient grows linearly in x, so it is only *locally* Lipschitz, with a Lipschitz constant that runs off to infinity with the state. The generic theorem therefore does not apply to LQG—not because the model is pathological, but because its costs grow at infinity, which is the normal state of affairs for a quadratic regulator. What stands in for the missing boundedness is *coercivity*, and that is the content of **(A-Growth)**: the running cost is coercive in the control, tending to $+\infty$ as $|\alpha| \to \infty$, which is exactly what makes the Legendre transform defining the Hamiltonian finite rather than $-\infty$, and the state cost is coercive in x, supplying the a priori control on the state that boundedness would otherwise have provided. Coercivity keeps the problem well posed, but it is not the form in which the chapter 10 theorem is stated, and we deliberately decline to pause and reprove a more general existence theorem to absorb it. Instead we cash in the one luxury LQG affords: its quadratic structure collapses the whole fixed point to a linear two-point boundary value problem, and we read existence and uniqueness straight off the solvability of *that*—a single scalar $D(T)$ being nonzero. This is the precise substitution behind the posture, repeated in section 14.4 and recorded in the ledger, that for LQG we *solve* the system rather than *invoke* the generic existence theorem.

That leaves monotonicity, **(A-Mon)**, whose status is not a footnote but one of the destinations of the chapter. Lasry–Lions monotonicity is the single general-purpose hypothesis that buys uniqueness in Part II, so the natural question is just whether our coupling has it. The coupling is the m-dependent part of the cost, $F[m](x) = \frac{q}{2}(x - s\bar{m})^2$, and one expects on physical grounds that its sign should decide the matter: a *repulsive* coupling, which spreads agents apart, is the prototype of a monotone, stabilizing interaction, whereas an *attractive* one, which lets agents clump together, is the prototype of the non-monotone interaction under which several self-consistent populations might coexist. That expectation is exactly right, and pinning it down quantitatively is a one-line computation we defer to lemma 14.8: (A-Mon) holds if and only if $qs \leq 0$, hence—since $q, q_T \geq 0$—precisely for the congestion sign $s \leq 0$ (and $s_T \leq 0$), and it fails for the imitation sign $s > 0$. We flag it now and prove it later because the gap between where (A-Mon) lapses and where uniqueness actually fails is the headline of the chapter; the audit needs only to record that monotonicity is a sign condition on s, not a property we may assume across the board.

With the model fixed and the audit complete, we adopt for the rest of the chapter the posture the audit forced on us: for LQG we will *solve* the mean field game directly, layer by layer, rather than *invoke* generic theorems whose hypotheses it sits just outside. The assembly begins in the next section exactly where Part II began, with a single agent. We freeze the population flow $\bar{m}_\cdot$, so that clause (i) of definition 14.1 becomes an ordinary control problem, and read off the representative agent's value function from the single-agent machinery of chapter 6—carrying the coupling term $s\bar{m}_t$ along as an external forcing—while closing the loop, clause (ii), waits for section 14.4. Before we start, one notation hazard deserves a flag.

Notation watch

The control gain b in (14.1) shares the macro name with the generic drift symbol b of the earlier chapters; in this chapter b is *always* the scalar control gain and a the scalar state feedback. This matters precisely to a reader importing the chapter 6 single-agent formulas, where b may have stood for the (uncontrolled) drift coefficient: here the uncontrolled drift is

a, the control multiplier is b, and the combination that governs every closed-loop quantity below is b^2. The discount macro $\beta = \beta$ does not appear in the main thread ($\beta = 0$); when we restore it in remark 14.14 it is a discount rate, never an inverse temperature, per appendix A.

14.2 The representative agent's HJB equation and the Riccati reduction

The previous section fixed the model, completed the assumption audit, and handed us its standing posture; we now begin the layer-by-layer solve exactly where Part II began, with a single agent. Freezing $\bar{m}_\cdot$ turns clause (i) of definition 14.1—the optimization arrow—into an ordinary stochastic control problem, the atom assembled in chapter 6, in which the coupling to the crowd survives only as a prescribed, time-dependent shift $s\,\bar{m}_t$ of the running target. We carry that shift along untouched and read off the representative agent's value function directly from the single-agent machinery of chapter 6 rather than invoking a generic theorem whose hypotheses LQG sits just outside; closing the loop, clause (ii), waits for section 14.4. The value function

$$u(t,x) \;=\; \inf_{\alpha}\; \mathbb{E}\biggl[\int_t^T \Bigl(\tfrac{1}{2}\alpha_r^2 + \tfrac{q}{2}(X_r - s\bar{m}_r)^2\Bigr)\,\mathrm{d}r + \tfrac{q_T}{2}(X_T - s_T\bar{m}_T)^2 \,\Big|\, X_t = x\biggr]$$

satisfies the dynamic programming principle: the value at (t,x) equals the best the agent can do over an infinitesimal step $[t, t+\mathrm{d}t]$, paying the running cost incurred on the step and inheriting the value at its random endpoint. Expanding that principle with Itô's formula and dividing by $\mathrm{d}t$—the derivation carried out once and for all in chapter 6, which we cite rather than repeat—turns it into a backward partial differential equation, while the convexity hypothesis (A-Conv) guarantees both that the pointwise minimization inside it is solved by a genuine minimizer and that the feedback so obtained is in fact optimal (the verification half of the chapter 6 theorem). The outcome is the backward HJB equation

$$\partial_t u + ax\,\partial_x u + \tfrac{\sigma^2}{2}\,\partial_{xx}u - \tfrac{1}{2}b^2\bigl(\partial_x u\bigr)^2 + \tfrac{q}{2}\bigl(x - s\bar{m}_t\bigr)^2 = 0, \qquad u(T,x) = \tfrac{q_T}{2}\bigl(x - s_T\bar{m}_T\bigr)^2. \tag{14.4}$$

The one nonlinear term here, $-\frac{1}{2}b^2(\partial_x u)^2$, repays being seen assembled rather than quoted, because it is where the control is eliminated. The control enters the dynamic programming equation only through the Hamiltonian $\inf_{\alpha\in\mathbb{R}}\{\,b\,\alpha\,\partial_x u + \frac{1}{2}\alpha^2\,\}$: the first term is the contribution $b\,\alpha\,\partial_x u$ of the control to the drift acting on the marginal value $\partial_x u$, and the second is the running charge $\frac{1}{2}\alpha^2$ for exerting it. This is a strictly convex quadratic in the scalar α, so its minimum is located by setting the derivative in α to zero, $b\,\partial_x u + \alpha = 0$, i.e. $\alpha = -b\,\partial_x u$. Substituting that minimizer back into the brace,

$$b\,(-b\,\partial_x u)\,\partial_x u + \tfrac{1}{2}(-b\,\partial_x u)^2 \;=\; -b^2(\partial_x u)^2 + \tfrac{1}{2}b^2(\partial_x u)^2 \;=\; -\tfrac{1}{2}b^2(\partial_x u)^2,$$

which is exactly the nonlinear term in (14.4): the convex charge has been Legendre-traded for a concave term in the gradient. The minimizer is the optimal feedback

$$\alpha^*(t,x) \;=\; -\,b\,\partial_x u(t,x), \tag{14.5}$$

which reads off the optimal action at every state as a fixed multiple $-b$ of the marginal value: the agent steers down the value gradient, harder where value is more sensitive to the state, and the gain of that response is set by the control authority b.

Reading the control off the value function has reduced everything to solving the single scalar PDE (14.4) for u. We do not need its general solution; we need only notice that the equation cannot carry us out of a very small family of functions. Its terminal datum $\frac{q_T}{2}(x - s_T\bar{m}_T)^2$ is a quadratic polynomial in x, and every operation on the left of (14.4) maps quadratics to quadratics: ∂_t acts coefficient-wise, the drift operator $x\,\partial_x$ and the diffusion ∂_{xx} preserve (do not raise) the polynomial degree, the square $(\partial_x u)^2$ of a degree-one gradient is again quadratic, and the source $\frac{q}{2}(x - s\bar{m}_t)^2$ is quadratic outright. A quadratic terminal datum propagated backward by an equation that preserves quadratics stays quadratic for all earlier times. This—not a lucky guess—is the structural reason to seek u in the form

$$u(t,x) \;=\; \tfrac{1}{2}P(t)\,x^2 + r(t)\,x + c(t), \tag{14.6}$$

with the three time-dependent coefficients $P, r, c : [0,T] \to \mathbb{R}$ to be found. The ansatz reduces every derivative in (14.4) to these functions and their rates of change: $\partial_x u = P(t)x + r(t)$ is linear, $\partial_{xx}u = P(t)$ is constant in x, and $\partial_t u = \frac{1}{2}\dot{P}\,x^2 + \dot{r}\,x + \dot{c}$. Substituting and grouping the left-hand side by powers of x is mechanical but worth doing in the open. The nonlinear term contributes $-\frac{1}{2}b^2(Px+r)^2 = -\frac{1}{2}b^2P^2x^2 - b^2Pr\,x - \frac{1}{2}b^2r^2$, the drift term contributes $ax(Px+r) = aPx^2 + ar\,x$, the diffusion term contributes $\frac{\sigma^2}{2}P$ (a pure x^0 term), and the source contributes $\frac{q}{2}(x - s\bar{m}_t)^2 = \frac{q}{2}x^2 - qs\bar{m}_t\,x + \frac{q}{2}s^2\bar{m}_t^2$. Collecting by degree, (14.4) becomes

$$\Big(\tfrac{1}{2}\dot{P} + aP - \tfrac{1}{2}b^2P^2 + \tfrac{q}{2}\Big)x^2 + \Big(\dot{r} + ar - b^2Pr - qs\bar{m}_t\Big)x + \Big(\dot{c} + \tfrac{\sigma^2}{2}P - \tfrac{1}{2}b^2r^2 + \tfrac{q}{2}s^2\bar{m}_t^2\Big) = 0.$$

Because this must hold at every state x, and three powers of x are linearly independent functions, each parenthesis must vanish on its own. That is all "matching powers" means here: one PDE in (t,x) splits into three ordinary differential equations in t, one per power of x, with their terminal data obtained by expanding the quadratic terminal datum $\frac{q_T}{2}(x - s_T\bar{m}_T)^2 = \frac{q_T}{2}x^2 - q_Ts_T\bar{m}_T\,x + \frac{q_T}{2}s_T^2\bar{m}_T^2$ in the same way and reading off the coefficient of each power. The coefficient of x^2 yields the scalar *Riccati equation*

$$\dot{P} \;=\; b^2P^2 - 2aP - q, \qquad P(T) = q_T, \tag{14.7}$$

the coefficient of x^1 yields the *linear offset equation*

$$\dot{r} \;=\; (b^2P - a)\,r + q\,s\,\bar{m}_t, \qquad r(T) = -\,q_T\,s_T\,\bar{m}_T, \tag{14.8}$$

and the coefficient of x^0 yields the scalar constant

$$\dot{c} \;=\; \tfrac{1}{2}b^2r^2 - \tfrac{q}{2}s^2\,\bar{m}_t^2 - \tfrac{\sigma^2}{2}P, \qquad c(T) = \tfrac{q_T}{2}s_T^2\,\bar{m}_T^2. \tag{14.9}$$

Three observations organize everything downstream. First, the constant c is a genuine spectator, and it is worth saying once why we may discard it with a clear conscience. Its equation (14.9) is read by no other equation—P and r are determined entirely without reference to c—and the optimal feedback (14.5) depends on the value function only through its gradient $\partial_x u = Px + r$, in which c does not appear at all. The constant c merely raises or lowers the value function uniformly across all states x, bookkeeping the accumulated running cost paid along the way; and an additive level shift of u changes no agent's optimal action, just as adding a constant to an energy moves no equilibrium. We therefore record (14.9) for completeness and ignore it hereafter, secure that nothing we care about depends on it.

Second—and this is the structural fact the rest of the chapter turns on—the population has not touched the curvature. Equation (14.7) for P contains no $\bar{m}_t$ whatsoever; it is *verbatim* the single-agent Riccati equation of chapter 6, the very equation one writes for the isolated control problem with no crowd at all. The mean field has entered the value function in exactly two places, both linear in x: the inhomogeneous forcing $q\,s\,\bar{m}_t$ in the offset equation (14.8), and the terminal datum $r(T) = -q_T s_T \bar{m}_T$. This is the precise content of the slogan *the mean field shifts the linear-in-x part of the value function and leaves the curvature alone*: P is a population-blind constant of the model, computed once and for all, while every trace of the game's self-consistency is funneled into the single linear channel r. We will exploit that clean split repeatedly—most decisively in section 14.4, where it is exactly what makes the closed fixed point a *linear* two-point boundary value problem rather than a nonlinear one, and through that linearity what turns uniqueness into a question of linear algebra.

Third, with P thus decoupled, the only thing standing between us and the offset equation is the assurance that P exists, stays finite on all of $[0,T]$, and supplies a well-behaved coefficient $b^2P - a$ to drive r. A Riccati equation is the canonical object that can blow up in finite time, so this assurance is not automatic and must be earned. We earn it now.

Theorem 14.2 (Global solvability of the Riccati equation)**.** *Assume $q \geq 0$, $q_T \geq 0$, and $b \neq 0$. Then the terminal-value problem (14.7) has a unique solution $P \in C^1([0,T])$, defined on the whole interval, with $P(t) \geq 0$ for all $t \in [0,T]$. Consequently the gain $b^2P(t) - a$ in (14.8) is a bounded continuous coefficient, and r is the unique solution of a linear inhomogeneous ordinary differential equation once $\bar{m}_\cdot$ is fixed.*

Proof. Pass to reversed time $\tau = T - t$ and set $\Pi(\tau) = P(T-\tau)$, which converts the terminal-value problem into an initial-value problem run forward in τ: since $\mathrm{d}\Pi/\,\mathrm{d}\tau = -\dot{P}$,

$$\frac{\mathrm{d}\Pi}{\mathrm{d}\tau} = -(b^2\Pi^2 - 2a\Pi - q) = q + 2a\Pi - b^2\Pi^2, \qquad \Pi(0) = P(T) = q_T \geq 0.$$

Read the right-hand side as a function of Π alone, the fixed quadratic $h(\Pi) = -b^2\Pi^2 + 2a\Pi + q$. Because $b \neq 0$ its leading coefficient $-b^2$ is strictly negative, so h is a *downward-opening parabola*: it is positive strictly between its two real roots and negative outside them. Solving $b^2\Pi^2 - 2a\Pi - q = 0$ gives those roots, $\Pi_\pm = \big(a \pm \sqrt{a^2 + b^2 q}\big)/b^2$, and since $q \geq 0$ forces $a^2 + b^2 q \geq a^2$, i.e. $\sqrt{a^2 + b^2 q} \geq |a|$, they straddle the origin: $\Pi_- \leq 0 \leq \Pi_+$, one root at or below zero and one at or above it.

The two standard tools now do genuinely different jobs, and it clarifies the argument to keep them apart. *Local* existence and uniqueness are Picard–Lindelöf: h is a polynomial, hence locally Lipschitz, so through each initial value passes exactly one trajectory, defined on some maximal interval of τ. That is all Picard–Lindelöf delivers—on its own it does not forbid the maximal interval from being finite, which is precisely the finite-time blow-up a Riccati equation is notorious for. *Global* existence is the extra mileage, and it comes from a trapping (comparison) argument that consumes the cost signs. The two constant functions $\Pi \equiv \Pi_-$ and $\Pi \equiv \Pi_+$ are themselves solutions, because h vanishes there and so $\mathrm{d}\Pi/\,\mathrm{d}\tau = 0$; and by the uniqueness just secured no other trajectory may ever touch either of them, since a point of contact would be a point through which two distinct solutions pass. The constant solutions are impassable barriers.

It remains to follow the actual trajectory $\Pi(0) = q_T$, and there are two cases, both pinned by those barriers. If $q_T \in [0, \Pi_+]$, the trajectory starts inside $[0, \Pi_+] \subseteq [\Pi_-, \Pi_+]$, where $h \geq 0$. It cannot exit upward across Π_+ (a barrier), and the nonnegative field pushes it upward or holds it still, so Π is monotonically nondecreasing and bounded above by Π_+; a monotone bounded solution cannot escape to infinity in finite time, so it exists for all $\tau \geq 0$ and remains in $[0, \Pi_+]$. If instead $q_T > \Pi_+$, the trajectory starts *above* the upper root, where $h < 0$, so Π is strictly decreasing; it descends toward the barrier Π_+, which it cannot cross, hence is bounded below by Π_+ and again global, relaxing monotonically down to Π_+. In neither case can Π run off to $-\infty$: reaching any value below Π_+ from a start at or above it would require first crossing the barrier $\Pi \equiv \Pi_+$, which is forbidden. Therefore Π, and with it $P(t) = \Pi(T - t)$, is defined on the whole interval and nonnegative throughout—$P \in [0, \Pi_+]$ in the first case, $P \geq \Pi_+ \geq 0$ in the second.

The mechanism deserves a sentence of emphasis, because it recurs and because it is easy to attribute to the wrong hypothesis. What excludes blow-up is not any nondegeneracy or parabolic smoothing—there is no diffusion anywhere in this scalar ODE; σ does not even appear in it—but the *sign* of the cost data. The condition $q \geq 0$ is what places one root at or below zero and makes h nonnegative across a neighborhood of the nonnegative axis, and $q_T \geq 0$ is what starts the trajectory on the safe side of the barrier. These are the signs of a genuine cost, penalties rather than subsidies, and they are exactly what a Riccati equation needs to stay finite; reverse them and the downward parabola would propel the trajectory off to $-\infty$ in finite time. Finally, with P now a bounded continuous function on $[0, T]$, the gain $b^2 P(t) - a$ in (14.8) is a bounded continuous coefficient, so for any continuous $\bar{m}_{\cdot}$ the offset equation is a linear inhomogeneous ODE with bounded coefficients and continuous forcing $q\,s\,\bar{m}_t$, and variation of constants delivers its unique global solution. □

Remark 14.3 (Where nondegeneracy is, and is not, used)**.** It is worth being explicit about a division of labor that is easy to blur. The well-posedness just proved leaned entirely on the cost signs $q, q_T \geq 0$ and not at all on the noise: σ is absent from the Riccati equation (14.7), so theorem 14.2 holds word for word in the deterministic limit $\sigma = 0$, and the curvature P, the closed-loop gain $b^2 P - a$, and the whole backward layer are insensitive to whether the agents are noisy. Nondegeneracy (A-Nondeg) is a hypothesis the *forward* equation will need, not this one. In section 14.3 the closed-loop dynamics are pushed forward into a population law, and there $\sigma > 0$ is what keeps the Fokker–Planck equation genuinely parabolic, so that the law carries a smooth Gaussian density that spreads in time. At $\sigma = 0$ the same dynamics collapse to a deterministic transport that can concentrate the entire population onto a single moving point—a Dirac flow rather than a density. The contrast is sharp and worth holding onto: the backward Riccati layer is governed by the convex cost structure and is blind to noise, whereas the forward density layer is governed by the diffusion and is exactly where nondegeneracy earns its keep.

With the backward layer solved, we have in hand everything the population side will ask for. The optimal control is the explicit linear feedback

$$\alpha^*(t, x) = -\,b\,\partial_x u(t, x) = -\,b\big(P(t)\,x + r(t)\big),$$

in which P is the population-blind Riccati curvature, just shown to be globally defined and nonnegative, and r is the single linear channel through which the mean field acts. Inserting this feedback into the state dynamics (14.1) replaces the agent's open-loop drift by a closed-loop drift carrying the bounded, continuous gain $a - b^2P(t)$, the signature of a stable linear—Ornstein–Uhlenbeck-type—law whenever that gain is negative. Carrying out that substitution is precisely the task of the next section: it inserts the feedback (14.5) into the state equation, closes the loop on the population side, and turns the forward Fokker–Planck equation into a pair of scalar ordinary differential equations for the mean and variance of a Gaussian flow.

14.3 The Fokker–Planck equation and the Gaussian flow

The backward layer is finished, and it handed us two things to work with: the explicit linear feedback $\alpha^*(t,x) = -b(P(t)x + r(t))$, and the guarantee from theorem 14.2 that the curvature P is globally defined and nonnegative, so that the closed-loop gain $a - b^2P(t)$ is a bounded, continuous coefficient. That is precisely the input the population side has been waiting for. So far each agent has been an isolated optimizer best-responding to a *frozen* flow $\bar{m}_\cdot$; that is clause (i) of definition 14.1, the optimization arrow, and it is done. We now turn the second arrow, clause (ii)—aggregation—and ask what population such optimizers actually generate. The instrument for that is the forward equation of chapter 8: take the whole population, all started from m_0, let every agent steer with the optimal feedback just computed, and propagate the law of the state forward in time. Inserting the feedback into the dynamics is what *closes the loop* on the population side—it replaces the open-loop drift, in which the control was a free variable, by a self-determined closed-loop drift—and the remainder of this section extracts from that closed loop the two scalar ordinary differential equations to which the entire population layer collapses.

Insert the optimal feedback (14.5), which with the ansatz (14.6) reads

$$\alpha^*(t,x) \;=\; -\,b\big(P(t)\,x + r(t)\big), \tag{14.10}$$

into the state dynamics (14.1). The open-loop drift $aX_t + b\alpha_t$ becomes $aX_t + b\cdot\big(-b(P(t)X_t + r(t))\big) = (a - b^2P(t))X_t - b^2r(t)$, so the state now obeys the linear SDE

$$\mathrm{d}X_t \;=\; \big((a - b^2P(t))\,X_t - b^2r(t)\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \tag{14.11}$$

an Ornstein–Uhlenbeck process with the time-dependent *closed-loop gain* $a - b^2P(t)$ (compare the constant-coefficient case of chapter 5). The new drift is worth reading slowly, because every feature of the population flow is already legible in it. The agent's intrinsic drift coefficient a has been bent by the control into $a - b^2P(t)$; since $P(t) \ge 0$ by theorem 14.2, the feedback can only *subtract* from the drift, never add to it—the optimal control is a restoring force that pulls the state back toward where the running cost prefers it, the mean-reverting signature of a stable linear law whenever $a - b^2P(t) < 0$. The constant-in-x term $-b^2r(t)$ is the only place the crowd enters: it is the trace of the mean field, the single channel r through which the population displaces the equilibrium point of that restoring force away from the origin. And the noise $\sigma\,\mathrm{d}W_t$ is left exactly as it was, because the control acted only on the drift. Notice that this is the same closed-loop gain $a - b^2P(t)$ the HJB section already flagged on handing the feedback over; we now let it drive a law forward.

Two general facts about *linear* stochastic differential equations turn this closed loop into something completely explicit, and it is worth seeing why each holds before we lean on them. The first is that linearity preserves Gaussianity: if the initial state X_0 is Gaussian, then X_t is Gaussian at every later time. The reason is structural, not a coincidence of the model. A linear SDE can be solved by an integrating factor—we carry this out in the proof below—and the solution expresses X_t as a fixed deterministic shift, plus a fixed deterministic multiple of X_0, plus a stochastic integral of a *deterministic* integrand against the Brownian motion. Each of those three pieces is Gaussian: the shift trivially, the multiple of X_0 because rescaling and translating a Gaussian leaves it Gaussian, and the Wiener integral because it is an L^2-limit of Gaussian Riemann–Stieltjes sums and so inherits Gaussianity. They are assembled *affinely*, and an affine combination of jointly Gaussian quantities is again Gaussian. A nonlinear drift would in general distort a Gaussian into a law with skew and heavy tails, propagating information into every moment; linearity is exactly what forbids that and collapses the infinite-dimensional law onto its first two moments. The second fact is that the law m_t of X_t solves the forward Kolmogorov—or Fokker–Planck—equation associated with (14.11),

$$\partial_t m \;=\; \mathcal{A}^* m \;=\; -\,\partial_x\big[\big((a-b^2P(t))x - b^2 r(t)\big)\, m\big] + \tfrac{\sigma^2}{2}\,\partial_{xx} m, \qquad m(0,\cdot) = m_0, \tag{14.12}$$

the forward equation of chapter 8, with $\mathcal{A}^*$ the adjoint of the closed-loop generator in the sign convention of appendix A. Read physically, this is a continuity equation for probability: the first term on the right is the transport of mass by the closed-loop drift—the divergence of a probability current—and the second is the diffusive spreading the noise σ supplies. It is the conservation law for the total probability, the forward dual of the backward HJB we have just solved, and the pair of them (backward value, forward density) is the analytic mean field game system of chapter 8. For the linear–quadratic model, however, we never have to treat (14.12) as a genuine partial differential equation. The first fact—Gaussianity—guarantees that m_t is pinned down by just its mean and variance, so the PDE is equivalent to two scalar ordinary differential equations for those two numbers. Deriving them, and proving that the law really is Gaussian and not merely Gaussian in its first two moments, is the content of the next theorem.

Theorem 14.4 (Gaussian flow)**.** *Suppose $m_0 = \mathcal{N}(\bar m_0, V_0)$ is Gaussian with mean $\bar m_0$ and variance $V_0 \ge 0$, and $\sigma > 0$. Then for the closed-loop dynamics (14.11) the law m_t is Gaussian for all $t \in [0,T]$, $m_t = \mathcal{N}(\bar m_t, V_t)$, with mean and variance solving the decoupled ordinary differential equations*

$$\dot{\bar m}_t = (a - b^2 P(t))\,\bar m_t - b^2 r(t), \qquad\qquad \bar m_0 = \int \xi\, m_0(\mathrm{d}\xi), \tag{14.13}$$

$$\dot V_t = 2\,(a - b^2 P(t))\,V_t + \sigma^2, \qquad\qquad V_0 = \mathrm{Var}(m_0). \tag{14.14}$$

The variance equation (14.14) is closed (it does not see the mean coupling), is linear, and has the unique positive solution $V_t = \phi(t)^2\big(V_0 + \sigma^2\int_0^t \phi(\tau)^{-2}\,\mathrm{d}\tau\big)$ with the scalar fundamental solution $\phi(t) = \exp\!\int_0^t (a - b^2P(\tau))\,\mathrm{d}\tau$.

Proof. Write $g(t) := a - b^2P(t)$ for the closed-loop gain (a local abbreviation, used only inside this proof). We argue in two stages: first extract the moment ODEs, which need only Itô's formula and an expectation; then upgrade "first two moments" to "genuinely Gaussian."

The moment equations. Apply Itô's formula (theorem 5.9) to X_t under (14.11). For the identity map there is no second-derivative term, so $\mathrm{d}X_t = (gX_t - b^2 r)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t$; taking expectations kills the Itô integral (a martingale with zero mean) and, because the drift is affine, lets the expectation pass through: $\dot{\bar{m}}_t = \mathbb{E}[gX_t - b^2 r] = g\,\bar{m}_t - b^2 r$, which is (14.13). For the second moment apply Itô to $f(x) = x^2$, where now the $\frac{\sigma^2}{2} f'' = \sigma^2$ term is present:

$$\mathrm{d}(X_t^2) = \big(2X_t(gX_t - b^2 r) + \sigma^2\big)\ \mathrm{d}t + 2\sigma X_t\ \mathrm{d}W_t.$$

Taking expectations again removes the martingale and gives $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[X_t^2] = 2g\,\mathbb{E}[X_t^2] - 2b^2 r\,\bar{m}_t + \sigma^2$. The variance is $V_t = \mathbb{E}[X_t^2] - \bar{m}_t^2$, so $\dot{V}_t = \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[X_t^2] - 2\bar{m}_t\dot{\bar{m}}_t$; substituting $\dot{\bar{m}}_t = g\bar{m}_t - b^2 r$ makes the cross terms cancel, $-2b^2 r\,\bar{m}_t - 2\bar{m}_t(g\bar{m}_t - b^2 r) = -2g\,\bar{m}_t^2$, leaving $\dot{V}_t = 2g\,\mathbb{E}[X_t^2] - 2g\,\bar{m}_t^2 + \sigma^2 = 2g\,V_t + \sigma^2$, which is (14.14). The cancellation is the whole reason the variance equation comes out *closed*: the inhomogeneous forcing $-b^2 r$, which carries the mean field, enters $\mathbb{E}[X_t^2]$ and $\bar{m}_t^2$ with equal and opposite weight and so leaves no residue in the centered second moment. We return to this point after the proof.

Gaussianity. It remains to show that m_t is genuinely *Gaussian*, not merely that its first two moments evolve as stated—two different laws can share a mean and a variance. This is the time-inhomogeneous version of the Ornstein–Uhlenbeck computation carried out in worked computation 5.17 for constant coefficients and forward-promised for the LQG thread in remark 5.18; we now redeem that promise. Introduce the (deterministic, strictly positive) integrating factor $\phi(t) = \exp\!\int_0^t g(\tau)\ \mathrm{d}\tau$, which satisfies $\dot{\phi} = g\phi$ and $\phi(0) = 1$. Because ϕ^{-1} is deterministic and of bounded variation, Itô's product rule applied to $\phi(t)^{-1}X_t$ has no quadratic-covariation term, and using $\mathrm{d}(\phi^{-1}) = -g\phi^{-1}\,\mathrm{d}t$,

$$\mathrm{d}(\phi(t)^{-1}X_t) = \phi^{-1}\,\mathrm{d}X_t - g\phi^{-1}X_t\ \mathrm{d}t = \phi^{-1}\big[(gX_t - b^2 r)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t\big] - g\phi^{-1}X_t\ \mathrm{d}t = \phi^{-1}(-b^2 r\ \mathrm{d}t + \sigma\ \mathrm{d}W_t),$$

the gX_t terms cancelling by design. Integrating from 0 to t and multiplying back by $\phi(t)$ gives the explicit solution

$$X_t \;=\; \phi(t)\Big(X_0 \;-\; b^2\int_0^t \phi(\tau)^{-1} r(\tau)\ \mathrm{d}\tau \;+\; \sigma\int_0^t \phi(\tau)^{-1}\ \mathrm{d}W_\tau\Big).$$

The right-hand side is *affine* in the pair $(X_0,\, W_{[0,t]})$: a deterministic shift $-b^2\phi(t)\int_0^t \phi^{-1} r$, a fixed deterministic multiple $\phi(t)$ of the Gaussian initial datum X_0, and the Itô integral $\int_0^t \phi(\tau)^{-1}\,\mathrm{d}W_\tau$ of a *deterministic* integrand ϕ^{-1}. This last object is exactly the Wiener integral covered by the Itô isometry (theorem 5.7): being a deterministic-integrand stochastic integral it is a centered Gaussian random variable, with variance

$$\mathrm{Var}\Big(\int_0^t \phi(\tau)^{-1}\,\mathrm{d}W_\tau\Big) = \mathbb{E}\Big[\Big(\int_0^t \phi(\tau)^{-1}\,\mathrm{d}W_\tau\Big)^2\Big] = \int_0^t \phi(\tau)^{-2}\ \mathrm{d}\tau,$$

the isometry trading the square of the integral for the integral of the square of the integrand. Moreover this integral is *independent* of X_0: it is built entirely from the Brownian increments on $[0,t]$, and X_0 (an $\mathcal{F}_0$-measurable initial condition) is independent of the driving Brownian motion by the standing setup of (14.1). An affine image of *independent* Gaussians is Gaussian—the sum of independent Gaussians is Gaussian and the affine map only rescales and shifts—so $X_t \sim \mathcal{N}(\bar{m}_t, V_t)$. Reading off its mean and variance reproduces (14.13)–(14.14), and because the two independent contributions add their variances, $\operatorname{Var}(X_t) = \phi(t)^2(V_0 + \sigma^2 \int_0^t \phi(\tau)^{-2}\,\mathrm{d}\tau)$, which is the announced closed form for V_t. Nondegeneracy $\sigma > 0$ makes the integral $\int_0^t \phi^{-2} > 0$ strictly positive for $t > 0$, so $V_t > 0$ even when the population starts as a point mass, $V_0 = 0$: the diffusion immediately regularizes the Dirac initial datum into a genuine spreading Gaussian, which is the population-level meaning of (A-Nondeg) flagged in remark 14.3. At $\sigma = 0$ the variance stays $V_t = \phi(t)^2 V_0$ and a point mass would remain a point mass, transported but never spread. □

The closed form for the variance is worth reading for the meaning of the fundamental solution ϕ. The factor $\phi(t) = \exp\int_0^t (a - b^2 P(\tau))\,\mathrm{d}\tau$ is the deterministic flow of the closed loop: it is the factor by which an initial displacement is amplified ($\phi > 1$) or contracted ($\phi < 1$) by time t under the gain $a - b^2 P$, the solution of the noiseless homogeneous equation $\dot{y} = g(t)y$. The variance formula then says exactly what one would expect of an Ornstein–Uhlenbeck process with a moving stiffness: the initial spread V_0 is carried forward and rescaled by ϕ^2, while the noise injected in each past instant $\mathrm{d}\tau$ contributes $\sigma^2\,\mathrm{d}\tau$ of fresh variance that is itself rescaled by $\phi(t)^2/\phi(\tau)^2$ over the remaining time—the $\phi(\tau)^{-2}$ inside the integral and the overall $\phi(t)^2$ together accomplishing precisely that propagation. When the closed loop is stable, $g < 0$, the two effects balance into a finite stationary spread; this is the Gaussian flow that theorem 14.4 delivers and that the equilibrium law inherits.

We can now name the punchline. The entire population layer of Thread A has reduced to two scalar ordinary differential equations, (14.13) for the mean and (14.14) for the variance, and these two are not on equal footing. The variance V_t is a *spectator*: its equation (14.14) reads only the closed-loop gain $g = a - b^2 P$, which the backward layer already fixed once and for all, and the noise level σ; it never sees the offset r and hence never sees the mean coupling. Once P is known, V_t is determined—it is the explicit $\phi^2(V_0 + \sigma^2 \int \phi^{-2})$ above—and it then sits inertly, feeding back into nothing, because the only way the population reaches into the cost (14.3) is through the mean $\bar{m}_t$, and the variance is invisible there. So although the population law m_t is a full Gaussian, all of its *game-theoretic* content is concentrated in the single scalar $\bar{m}_t$. And the mean does not stand alone either: its equation (14.13) is driven by the offset $r(t)$, which is in turn the backward variable governed by (14.8) and driven by $\bar{m}_t$. The two of them, $\bar{m}_t$ and its conjugate offset $r(t)$, form a closed forward–backward pair, and everything that makes this a *game* rather than a control problem lives in their mutual dependence. That is the exact hinge the next section opens on.

Physics note

The mean $\bar{m}_t$ plays the role of a self-consistent (mean) field, exactly as in Curie–Weiss theory. In the ferromagnet each spin feels an effective field built from the average magnetization, adjusts to it, and the adjusted spins in turn *produce* that average—the field is whatever makes this loop consistent. Here the same loop runs with optimization in place of thermal averaging: each agent feels the crowd mean $\bar{m}_t$ through the tracking cost $\frac{q}{2}(X_t - s\bar{m}_t)^2$, best-responds to it via the feedback (14.10), and the population of best responders generates a mean through (14.13); equilibrium is the value of $\bar{m}_\cdot$ at which the field an agent responds to equals the field the responses produce. The mean-coupling strength s is the analogue of the coupling constant: $s > 0$ (imitation) is the ferromagnetic, aligning case in which one half-expects the population to be able to organize into more than one consistent history, exactly the symmetry-breaking, multiple-equilibria scenario, while $s < 0$ (congestion) is the antiferromagnetic, spreading case. The decisive observation of this section is the analogue of a familiar mean-field statement: the variance V_t decouples completely from the order parameter $\bar{m}_t$. In the magnet this is the statement that, at the mean-field (Gaussian) level of this quadratic theory, the *fluctuations* do not renormalize the order parameter—the width of the distribution rides along passively and never feeds back to shift the self-consistent field. It is precisely because the theory is quadratic and the coupling is through the mean alone that fluctuation corrections to the order parameter vanish at this level; in a non-Gaussian (non-quadratic) theory the higher moments would re-enter and the clean scalar self-consistency we are about to exploit would be lost. This decoupling is what licenses the next section to formalize the entire game as a self-consistency condition on $\bar{m}_t$ and $r(t)$ alone.

14.4 Closing the loop: the fixed-point system

The previous section left us with a sharp summary of where all the content now lives. The full Gaussian population law has been distilled to a single scalar, the mean $\bar{m}_t$, because that is the only feature of the crowd the cost (14.3) can feel; and that mean does not move on its own but is driven, through (14.13), by the offset $r(t)$. The offset, in turn, is the backward variable produced by the value function, driven through (14.8) by the very same mean. We are therefore holding two relations that point at each other. The HJB layer delivered r *as a function of* $\bar{m}_\cdot$: freeze the population flow and the offset equation (14.8) integrates backward to give r. The Fokker–Planck layer delivered $\bar{m}$ *as a function of* r: freeze the offset and the mean equation (14.13) integrates forward to give $\bar{m}$. Each was derived holding the other's driver fixed—that is exactly the frozen-flow posture of clauses (i) and (ii) of definition 14.1, taken one at a time. An equilibrium is the demand that both hold *at once*: the offset the agents compute against a mean must be the offset that, fed back through the population, regenerates that same mean. Closing the loop is nothing but making these two arrows mutually consistent, and the rest of this section does it and reads off the price of admission.

Consistency is immediate to write down because both relations are linear and local in time. Stack the two scalars into the vector $w(t) = (\bar{m}_t, r(t))^\top$. The mean equation (14.13), $\dot{\bar{m}}_t = (a - b^2P(t))\bar{m}_t - b^2 r(t)$, supplies the top row of a 2×2 system, and the offset equation (14.8), which we rewrite as $\dot{r} = (b^2P - a)r + qs\,\bar{m}_t = qs\,\bar{m}_t - (a - b^2P(t))r$, supplies the bottom row. There is no longer any frozen datum to carry: every appearance of $\bar{m}$ and of r now stands for the running unknowns themselves. Collecting (14.8), (14.13), and the boundary data, the equilibrium mean and offset solve the *linear two-point boundary value problem*

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\bar{m}_t \\ r(t)\end{pmatrix} = \underbrace{\begin{pmatrix} a - b^2P(t) & -b^2 \\ qs & -(a - b^2P(t))\end{pmatrix}}_{=:\,M(t)}\begin{pmatrix}\bar{m}_t \\ r(t)\end{pmatrix}, \qquad \bar{m}_0 = \int \xi\, m_0(\mathrm{d}\xi)\ \text{given}, \quad r(T) = -q_T s_T\,\bar{m}_T, \tag{14.15}$$

with the closed-loop gain $a - b^2P(t)$ from the already-solved Riccati equation (14.7). This is the entire mean field game system of chapter 8, reduced for LQG to four scalar ODEs (the Riccati equation (14.7) for P, the variance equation (14.14) for V, and the coupled pair (14.15) for $(\bar{m}, r)$).

The boundary data of (14.15) repay a careful look, because they are the visible face of a structure Part II kept insisting was the heart of the difficulty. The data are *split* across the two ends of the interval and across the two variables: the forward variable $\bar{m}$ is pinned at the *initial* time, $\bar{m}_0 = \int \xi\, m_0(\mathrm{d}\xi)$, because the population starts from the prescribed distribution m_0; the backward variable r is pinned at the *terminal* time, $r(T) = -q_T s_T \bar{m}_T$, because the offset inherits its terminal value from the terminal cost. Neither end carries both pieces of data, so one cannot simply integrate (14.15) forward from a known $w(0)$ as one would an ordinary initial-value problem: the second component of $w(0)$, the initial offset $r(0)$, is unknown, and it must be chosen so that the trajectory it launches lands on the prescribed terminal value $r(T)$ many time units later. This is precisely the forward–backward coupling that, in the general nonlinear theory of chapter 9, forced the heavy machinery of FBSDEs and decoupling fields: there is no initial-value analogue, no way to march the system forward without already knowing something fixed only at the far end. What the quadratic structure has bought us is that here, stripped of every nonlinearity, this forward–backward obstruction is *nothing more exotic than a linear two-point boundary value problem*—a shooting problem for two scalar ODEs. The whole conceptual weight of Part II survives, but its analytic difficulty has evaporated into a question we can settle with a determinant.

The reason it is genuinely a *linear* problem, and not merely a problem we have linearized, is the decoupling established in section 14.2: the Riccati equation (14.7) for the curvature P contains no $\bar{m}$, so P is computed once and for all before the loop is ever closed, and theorem 14.2 guarantees it is globally defined and nonnegative. The closed-loop gain $a - b^2 P(t)$ that appears in $M(t)$ is therefore a *known, bounded, continuous coefficient*, not an unknown to be solved for jointly with $(\bar{m}, r)$. Had the curvature instead depended on the mean—as it would the instant the cost stopped being quadratic—the matrix $M(t)$ would itself depend on the solution and (14.15) would be a genuinely nonlinear system, with no fundamental matrix and no clean solvability scalar. It is exactly the slogan of section 14.2, that the mean field shifts the linear-in-x part of the value and leaves the curvature alone, that demotes the coupling to the off-diagonal entries of a *fixed* linear operator. Consequently the question of whether the equilibrium exists and is unique is no longer a question of analysis but of linear algebra: does the linear map sending the unknown initial offset to the realized terminal mismatch have a trivial kernel? We settle it now.

Lemma 14.5 (Solvability of the linear BVP)**.** *Let $\Psi(t)$ be the fundamental matrix of* (14.15)*, i.e. $\dot{\Psi} = M(t)\Psi$, $\Psi(0) = I$, and write its entries $\Psi(t) = \big(\Psi_{ij}(t)\big)_{i,j=1,2}$. Define the scalar*

$$D(T) \ := \ \Psi_{22}(T) + q_T s_T\, \Psi_{12}(T). \tag{14.16}$$

Then (14.15) *has, for every initial mean $\bar{m}_0$,*

1. *a* unique *solution if $D(T) \neq 0$;*
2. *no solution or a one-parameter family of solutions if $D(T) = 0$ (the homogeneous problem with $\bar{m}_0 = 0$ then has a nontrivial solution).*

In case (1) the unknown initial offset is $r(0) = -\big(\Psi_{21}(T) + q_T s_T\, \Psi_{11}(T)\big)\bar{m}_0 / D(T)$, and the whole equilibrium is then determined.

Proof. The strategy is the shooting method made exact. We have one unknown at the initial time, the offset $\rho := r(0)$; once it is chosen, the entire trajectory is determined, because (14.15) is a linear initial-value problem in forward time as soon as $w(0)$ is fully specified. We then impose the single terminal constraint and watch it become one scalar equation for the one scalar unknown ρ.

Write $w(t) = (\bar{m}_t,\, r(t))^\top$. Because (14.15) is linear with $M(t)$ a known coefficient, its solution is transported from the initial data by the fundamental matrix, $w(t) = \Psi(t)\, w(0)$, where Ψ solves $\dot{\Psi} = M(t)\Psi$ with $\Psi(0) = I$ and exists on all of $[0, T]$ because M is bounded and continuous (here theorem 14.2 earns its keep, by keeping $a - b^2 P$ finite). The initial vector is $w(0) = (\bar{m}_0, \rho)^\top$ with $\bar{m}_0$ prescribed and ρ the unknown to be pinned. Reading off the two components of $w(T) = \Psi(T)w(0)$,

$$\bar{m}_T = \Psi_{11}(T)\,\bar{m}_0 + \Psi_{12}(T)\,\rho, \qquad r(T) = \Psi_{21}(T)\,\bar{m}_0 + \Psi_{22}(T)\,\rho,$$

both terminal values are explicit affine functions of the single unknown ρ. Now impose the terminal boundary condition $r(T) = -q_T s_T\,\bar{m}_T$, substituting both expressions:

$$\Psi_{21}(T)\,\bar{m}_0 + \Psi_{22}(T)\,\rho \;=\; -\,q_T s_T\big(\Psi_{11}(T)\,\bar{m}_0 + \Psi_{12}(T)\,\rho\big).$$

Collect the two terms carrying the unknown ρ on the left and the two terms carrying the datum $\bar{m}_0$ on the right:

$$\big(\Psi_{22}(T) + q_T s_T\,\Psi_{12}(T)\big)\,\rho \;=\; -\big(\Psi_{21}(T) + q_T s_T\,\Psi_{11}(T)\big)\,\bar{m}_0.$$

The coefficient multiplying ρ is exactly the scalar $D(T)$ defined in (14.16); the whole solvability question has been compressed into whether this one number is nonzero. This is the 1×1 linear equation $D(T)\,\rho = (\text{const}) \cdot \bar{m}_0$ to which the entire forward–backward system reduces, and its three cases are the textbook trichotomy for a scalar linear equation.

If $D(T) \neq 0$, the equation determines ρ uniquely,

$$\rho = r(0) = -\,\frac{\Psi_{21}(T) + q_T s_T\,\Psi_{11}(T)}{D(T)}\,\bar{m}_0,$$

which is the stated formula; feeding this $r(0)$ back into $w(t) = \Psi(t)(\bar{m}_0, \rho)^\top$ produces the unique trajectory $(\bar{m}_\cdot, r(\cdot))$, and with it the whole equilibrium, proving claim (1). If instead $D(T) = 0$, the left side no longer depends on ρ at all, and the equation reads $0 = -\big(\Psi_{21}(T) + q_T s_T \Psi_{11}(T)\big)\bar{m}_0$. This is a constraint on the *datum* $\bar{m}_0$, not on ρ. When the right side happens to vanish, every ρ satisfies the equation, so the BVP has a one-parameter family of solutions indexed by the free initial offset; when the right side does not vanish, no ρ works and the BVP has no solution. In particular the homogeneous problem, $\bar{m}_0 = 0$, always falls in the first sub-case: its terminal constraint is satisfied by every ρ, so any $\rho \neq 0$ generates a nontrivial solution $(\bar{m}_\cdot, r(\cdot))$ of the homogeneous BVP. The vanishing of $D(T)$ is thus precisely the existence of a nonzero homogeneous mode, which is claim (2). □

The lemma is best digested on the trivial coupling first, as a sanity check that the machinery reproduces what it must. Set $s = 0$ (and $s_T = 0$): the agents do not track the crowd at all, and the game degenerates to the isolated control problem of chapter 6. Then the lower-left entry qs of $M(t)$ vanishes, M is upper triangular, and the fundamental matrix inherits the triangular structure, with diagonal entry $\Psi_{22}(t) = \exp\big(-\int_0^t (a - b^2 P(\tau))\,\mathrm{d}\tau\big) = \phi(t)^{-1}$, a strictly positive number. Since

$q_T s_T = 0$ as well, $D(T) = \Psi_{22}(T) > 0$ for every horizon, so by the lemma the equilibrium is always unique; and the offset numerator $\Psi_{21}(T) + q_T s_T \Psi_{11}(T) = 0$ forces $r(0) = 0$, hence $r \equiv 0$ throughout. The mean then evolves freely as $\dot{\bar{m}}_t = (a - b^2 P(t))\bar{m}_t$ with no forcing, exactly the optimally-controlled mean of a single agent ignoring the population. The closed loop, in other words, recovers the uncoupled solution the instant the coupling is switched off—the kind of consistency check that confirms the off-diagonal entry qs is genuinely the entire game. Everything interesting in the chapter is the story of what that one entry does once it is nonzero, and the whole of it is metered by the single scalar $D(T)$.

Theorem 14.6 (Existence of the LQG equilibrium)**.** *Under $q, q_T \geq 0$, $b \neq 0$, $\sigma > 0$, and whenever $D(T) \neq 0$ in (14.16), the LQG mean field game (14.1)–(14.3) has a unique equilibrium in the sense of definition 14.1. The equilibrium value function is the quadratic (14.6) with P solving (14.7) and r the BVP solution; the equilibrium control is the linear feedback (14.10); and the equilibrium population law m_t is the affine pushforward of m_0 under the explicit solution map of (14.11), with mean $\bar{m}_t$ from (14.13). If in addition $m_0 = \mathcal{N}(\bar{m}_0, V_0)$ is Gaussian, m_t is the Gaussian flow $m_t = \mathcal{N}(\bar{m}_t, V_t)$ of theorem 14.4.*

The theorem is, at this point, an act of bookkeeping: every piece was manufactured in an earlier section, and the proof only checks that assembling them satisfies the two clauses of the equilibrium definition. What deserves emphasis is not the mechanics but the route. In the general theory of chapter 10, existence is a fixed-point theorem—one builds a continuous self-map of a compact set of measures and invokes Schauder—and that route was closed to LQG, as section 14.1 took pains to explain, because the quadratic cost is unbounded and only locally Lipschitz and so sits outside the bounded-Lipschitz hypothesis under which the generic theorem is stated. Here we never take that route. We *solve* the system, and existence is read off the one number $D(T)$ being nonzero. This is the substitution promised in the model section: in place of the coercivity-versus-boundedness gap that would have demanded a more general existence theorem, the quadratic structure hands us a linear BVP whose solvability we can simply compute.

Proof. Every object is already in hand; we collect them in the order the loop closes. theorem 14.2 gives the curvature P globally on $[0, T]$ and with it the bounded continuous closed-loop gain $a - b^2 P(\cdot)$, so the coefficient matrix $M(t)$ of (14.15) is well defined. lemma 14.5 with $D(T) \neq 0$ then gives a unique pair $(\bar{m}_\cdot, r(\cdot))$ solving the two-point boundary value problem. From r we recover the value function as the quadratic (14.6) (with c the irrelevant spectator of (14.9)) and the feedback as the linear law (14.10). It remains to verify that this candidate genuinely satisfies both clauses of definition 14.1. Clause (i), optimality against the frozen flow, is the verification theorem of chapter 6: with $\bar{m}_\cdot$ held fixed the feedback $\alpha^* = -b(P\bar{m} + r)$ was constructed precisely as the minimizer of the single-agent problem, and (A-Conv) guarantees it is the true optimum, not merely a critical point. Clause (ii), reproduction of the flow, is the construction of (14.13): pushing the population through the closed-loop dynamics (14.11) yields a state whose mean obeys exactly the mean equation that the top row of (14.15) solved, so the realized mean coincides with the $\bar{m}_\cdot$ the agents best-responded to. The two arrows therefore close on this pair, and it is an equilibrium. Because the mean of a linear SDE depends on the initial law only through its mean (theorem 14.4), the equilibrium $\bar{m}_\cdot$ is fixed regardless of the shape of m_0; the equilibrium law m_t is the affine pushforward of m_0 under the explicit solution map of (14.11), and when m_0 is Gaussian, theorem 14.4 identifies it with the Gaussian flow $\mathcal{N}(\bar{m}_t, V_t)$. Uniqueness is inherited verbatim from the uniqueness of ρ in lemma 14.5: a second equilibrium would supply a second admissible $r(0)$, which $D(T) \neq 0$ forbids. Existence and uniqueness are thus obtained directly from the solvability of the linear BVP, bypassing the Schauder argument of chapter 10 entirely. □

The fixed-point theorem we have just proved replaced an abstract dichotomy—contraction or Schauder—with the invertibility of a single linear map. It is worth making fully explicit what that replacement is, because it is the conceptual payoff of the whole reduction and the precise sense in which nonuniqueness, when it comes, is a *spectral* event rather than a failure of hypotheses.

Remark 14.7 (The Schauder argument collapses to a 1×1 invertibility)**.** It is instructive to track what the abstract existence machinery becomes here, line by line. The best-response map S of chapter 10 sends a candidate mean flow $\bar{m}.$ to the mean of the optimally controlled population that best-responds to it. For LQG this map is *affine* in its argument: given $\bar{m}.$, one solves the offset equation (14.8) for r (linear, with $\bar{m}.$ as inhomogeneous forcing) and then the mean equation (14.13) for the new mean (linear, with that r as forcing), and the composition of two affine maps is affine. Write it as $\mathsf{S}(\bar{m}.) = L\,\bar{m}. + \beta$, with L the linear part and β the $\bar{m}.$-independent piece carried by the terminal data. An equilibrium is a fixed point, $\bar{m}. = \mathsf{S}(\bar{m}.)$, i.e. $(I - L)\bar{m}. = \beta$, and the elementary theory of affine maps says such a fixed point exists and is unique for every β exactly when $I - L$ is invertible—when 1 is not an eigenvalue of the linear part L. That is the spectral condition the general theory leaves abstract; we have computed it.

The computation is the elimination already performed in lemma 14.5, now read as a statement about $I - L$. The map S acts on an infinite-dimensional space of time-dependent flows $\bar{m}.$, so $I - L$ is at first sight an operator on a function space. But the LQG dynamics collapse it. Every flow that can possibly arise as $\mathsf{S}(\bar{m}.)$ is the top component of a solution of the *linear* system (14.15), hence is determined by just two numbers, the initial data $(\bar{m}_0, r(0))$, through $w(t) = \Psi(t)w(0)$; and $\bar{m}_0$ is fixed by the problem. *Eliminating* r—that is, solving the offset equation in closed form via the fundamental matrix rather than carrying it as an independent function—leaves a single free scalar, the initial offset $\rho = r(0)$, and a single scalar equation it must satisfy, namely $D(T)\,\rho = (\text{const})\,\bar{m}_0$ from the lemma's proof. The operator $I - L$, restricted to the one-dimensional space of genuine freedom that survives the elimination, *is* multiplication by $D(T)$: it is the 1×1 linear map $\rho \mapsto D(T)\,\rho$. Invertibility of $I - L$ is therefore literally the condition $D(T) \neq 0$, and the contraction/Schauder dichotomy of the general theory has collapsed to the invertibility of a 1×1 matrix. When $D(T) = 0$, the kernel is nontrivial—there is a homogeneous mode, a direction in which the best response neither contracts nor expands but fixes, and the equilibrium ceases to be pinned. This is why nonuniqueness in mean field games is a genuine *spectral* phenomenon, an eigenvalue of the linearized best-response operator crossing 1, and not an artifact of weak or generic hypotheses: it is visible here, in the cleanest possible model, as a single computable scalar passing through zero.

The entire question of the LQG equilibrium has thus been distilled to a single binary, $D(T) \neq 0$ versus $D(T) = 0$, carrying with it the spectral reading just given: $D(T) = 0$ is exactly the moment the linearized best-response operator $I - L$ becomes singular, a homogeneous mode opens, and the equilibrium loses its uniqueness. What the present section cannot say is *where*, in the space of model parameters $(s, q, T, \dots)$, the protective inequality $D(T) \neq 0$ holds and where it first fails. That is now a concrete and answerable question—compute the fundamental matrix $\Psi(T)$, form the scalar $D(T)$, and locate its zeros—and it is the question the next section takes up. There we turn the binary "unique iff $D(T) \neq 0$" into the geometry of parameter space, showing first that the monotone region $s \leq 0$ lies safely inside $\{D(T) \neq 0\}$ for every horizon, and then, by computing $D(T)$ in closed form, exactly how far beyond monotonicity the uniqueness region actually extends before the first conjugate zero of $D(T)$ closes it off.

14.5 Uniqueness, monotonicity, and the phase diagram

The previous section distilled the entire LQG equilibrium to one binary, $D(T) \neq 0$ versus $D(T) = 0$, and read that binary spectrally: by remark 14.7, $D(T)$ is the determinant of the 1×1 linear map to which the best-response operator $I - L$ collapses, so $D(T) \neq 0$ is exactly the statement that $I - L$ is invertible, that no homogeneous mode is open, and that the equilibrium is uniquely pinned. That is a clean criterion, but it is a criterion evaluated at a *single* point of the model: fix the parameters $(s, q, T, \dots)$, compute $D(T)$, and you learn whether *that* game has a unique equilibrium. It says nothing yet about the shape of the answer as the parameters move. The job of this section is to promote the pointwise binary into a global picture—to ask not "is $D(T) \neq 0$ here?" but "*where*, in the space of model parameters, does the protective inequality $D(T) \neq 0$ hold, and along what surface does it first fail?" The answer to that question is a curve in parameter space separating a region of guaranteed uniqueness from a region where the equilibrium degenerates, and that curve is the phase diagram the chapter has been built to draw.

We attack the question from two complementary directions, and the tension between them is the substance of the chapter. The first attack is *structural and global.* It does not compute $D(T)$ at all; instead it invokes the one general-purpose hypothesis that the whole of Part II offers for uniqueness—the Lasry–Lions monotonicity condition (A-Mon)—and asks where our coupling satisfies it. Where it does, the abstract uniqueness theorem of chapter 11 applies wholesale and forces $D(T) \neq 0$ *for every horizon T at once*, with no calculation and no reference to the fine structure of the dynamics. This is the strength of the structural attack: it is horizon-uniform and robust, buying an entire half-line of safe parameters in a single stroke. But it has an equally sharp weakness. Monotonicity is a *sufficient* condition; it certifies a region inside which uniqueness is guaranteed, but it is silent about everything outside that region. When monotonicity lapses, the structural attack simply stops speaking—it does not assert that uniqueness fails, only that this particular guarantee no longer covers us. By itself it can never locate the true boundary, because it never looks past its own sufficient condition.

The second attack is *explicit and exact*, and it is the one only a fully solvable model affords. Rather than certify a safe region from above, it computes $D(T)$ in closed form and watches for the precise instant the scalar passes through zero. Where the structural attack hands back a conservative inclusion, the exact computation hands back the boundary itself: the first horizon at which $D(T)$ vanishes, as a function of the coupling, is the genuine edge of the uniqueness region, not a sufficient surrogate for it. The whole point of running both attacks is that their two answers *do not coincide*, and the gap between them is the headline of the chapter. One expects, on the physical reading of section 14.1, that monotonicity and uniqueness should change hands together at the sign flip $s = 0$—repulsion stabilizing, imitation destabilizing. The explicit calculation will show instead that the true uniqueness boundary sits *strictly beyond* the monotone region: uniqueness survives the entire non-monotone (imitation) band, all the way to a finite imitation threshold, and is lost only afterward, at a conjugate horizon $T_c(s)$ of the forward–backward system. Monotonicity is sufficient but not necessary, and measuring exactly how far the necessary boundary lies past the sufficient one is what the two-pronged strategy delivers. The interface between the two answers—the monotone surface on one side, the first conjugate zero of $D(T)$ on the other—is the phase boundary, and the band between them is the territory the chapter was written to expose.

The two subsections execute the two attacks in turn. Section 14.5.1 carries out the structural half: it recalls from chapter 11 the precise monotonicity inequality, instantiates it on the LQG running coupling $F[m](x) = \frac{q}{2}(x - s\bar{m})^2$ to find exactly which sign of s makes the coupling monotone (the one-line computation flagged as lemma 14.8 back in the assumption audit), and then shows that in the monotone regime $s \leq 0$ the Lasry–Lions argument forces $D(T) \neq 0$ for every horizon. Section 14.5.2 carries out the explicit half: in a clean normalization it computes $D(T)$ in closed form and locates the critical horizon where it first vanishes, drawing the phase diagram and confirming that its boundary falls in the non-monotone band $s > 0$ rather than at the sign change $s = 0$. We take the structural attack first, both because it is the lighter computation and because pinning down precisely *where* monotonicity holds is what makes the surprise of the second subsection legible: the reader needs the monotone region drawn before the uniqueness region can be seen to overspill it.

14.5.1 When LQG is monotone

We now run the first of the two attacks the parent section laid out—the structural, horizon-uniform one—and it is worth restating exactly what it promises before we cash it in. The whole of Part II offers a single general-purpose hypothesis for uniqueness, the Lasry–Lions monotonicity condition (A-Mon), and where it holds it forces uniqueness *with no calculation at all*: the abstract theorem of chapter 11 applies wholesale, and in our reduced language that means the protective scalar $D(T)$ cannot vanish at *any* horizon. The strategy of this subsection is therefore in two beats. First, find precisely which LQG couplings are monotone—this is a one-line computation that locates the monotone regime exactly at $s \leq 0$, the congestion sign. Second, prove that throughout that regime $D(T) \neq 0$ for every T, so that the repulsive band is certified unique at every horizon in a single stroke. The reason to be precise about *where* monotonicity holds is that this boundary is the foil for the whole chapter: only once the monotone region is drawn exactly can the next subsection's surprise—that the uniqueness region overspills it—be seen for what it is.

Recall from chapter 11 the form of the hypothesis. The Lasry–Lions monotonicity condition for a coupling $F[m](x)$ is the inequality

$$\int_{\mathbb{R}} (F[m](x) - F[m'](x))\,(m - m')(\mathrm{d}x) \geq 0 \qquad \text{for all } m, m' \in \mathcal{P}_2(\mathbb{R}), \tag{14.17}$$

and similarly for the terminal coupling. It is worth a sentence on what this asks. The signed measure

$m-m'$ is the displacement of one population from another; $F[m]-F[m']$ is the resulting change in the cost landscape each agent feels. The pairing in (14.17) measures the correlation between the displacement and the cost change it induces: monotonicity demands that moving mass from m' to m *raises* the cost where the mass went, so that the interaction penalizes agglomeration and pushes populations apart. That is the precise meaning of "repulsion stabilizes," and it is exactly the property that, in chapter 11, made two equilibria collapse onto one. For LQG the coupling is the m-dependent part of the cost (14.3), the running term $F[m](x)=\frac{q}{2}(x-s\bar{m})^2$ with $\bar{m}=\int\xi\,m(\mathrm{d}\xi)$, and because it depends on m only through the scalar mean $\bar{m}$ the pairing collapses to something one can evaluate in a single line.

Lemma 14.8 (Monotonicity of the LQG coupling)**.** *The LQG running coupling $F[m](x)=\frac{q}{2}(x-s\bar{m})^2$ satisfies the Lasry–Lions monotonicity condition (A-Mon) if and only if $qs\le 0$; explicitly,*

$$\int_{\mathbb{R}}\big(F[m]-F[m']\big)\,\mathrm{d}(m-m')\;=\;-\,q\,s\,(\bar{m}-\bar{m}')^2. \tag{14.18}$$

The same computation gives the terminal coupling monotone iff $q_T s_T\le 0$. In particular, since $q,q_T\ge 0$, monotonicity holds exactly for $s\le 0$ and $s_T\le 0$ (repulsive/congestion coupling) and fails for $s>0$ (attractive/imitation coupling).

Proof. Everything hinges on the algebraic shape of the integrand, so we expand it first and integrate second. Subtract the two cost landscapes, taking the common factor $\frac{q}{2}$ out front and expanding the difference of squares,

$$F[m](x)-F[m'](x)=\tfrac{q}{2}\big[(x-s\bar{m})^2-(x-s\bar{m}')^2\big]=\tfrac{q}{2}\big[-2xs(\bar{m}-\bar{m}')+s^2(\bar{m}^2-\bar{m}'^2)\big].$$

The point of writing it this way is that the x^2 terms have cancelled outright—both populations share the same curvature $\frac{q}{2}x^2$, since the mean coupling shifts the parabola but never reshapes it—leaving only a piece *linear* in x and a piece *constant* in x. That split is decisive once we integrate against the signed measure $(m-m')(\mathrm{d}x)$, because the two pieces meet that measure very differently.

The constant-in-x piece, $\frac{q}{2}s^2(\bar{m}^2-\bar{m}'^2)$, carries no x at all, so integrating it against $(m-m')$ pulls the constant out and leaves the bare total mass $\int_{\mathbb{R}}(m-m')(\mathrm{d}x)=1-1=0$: both m and m' are probability measures, so their difference is a signed measure of total mass zero, and any quantity that does not vary with x is annihilated by it. This is the structural reason the imitation *level* drops out and only the imitation *slope* survives. The linear-in-x piece, $-q\,s(\bar{m}-\bar{m}')\,x$, does see the measure: integrating x against $(m-m')$ returns the difference of means, $\int_{\mathbb{R}}x\,(m-m')(\mathrm{d}x)=\bar{m}-\bar{m}'$, by the very definition (14.2) of the mean. Assembling the two contributions, only the linear one remains,

$$\int_{\mathbb{R}}\big(F[m]-F[m']\big)\,\mathrm{d}(m-m')=\tfrac{q}{2}\cdot\big(-2s(\bar{m}-\bar{m}')\big)\cdot(\bar{m}-\bar{m}')=-\,q\,s\,(\bar{m}-\bar{m}')^2,$$

which is the announced defect (14.18). Now read off the sign. The factor $(\bar{m}-\bar{m}')^2$ is a square, hence ≥ 0 and strictly positive whenever the two populations have distinct means; the prefactor is $-qs$. The pairing is therefore ≥ 0 for *every* admissible pair m,m' if and only if $-qs\ge 0$, i.e. $qs\le 0$. Since the cost weight is a genuine penalty, $q\ge 0$, this is exactly the sign condition $s\le 0$. The identical computation with q_T,s_T in place of q,s gives the terminal defect $-q_Ts_T(\bar{m}_T-\bar{m}'_T)^2$ and the terminal condition $s_T\le 0$. ☐

The lemma deserves a moment's interpretation, because the clean form of (14.18) is exactly the quantitative version of the physical slogan that opened this subsection. The Lasry–Lions pairing is not merely nonnegative or negative for LQG; it is the explicit number $-qs(\bar{m}-\bar{m}')^2$, a *monotonicity defect* that measures by how much the coupling fails (or, when negative, satisfies) the condition. For the congestion sign $s<0$ the defect is strictly positive whenever two candidate populations disagree on their mean, and that strict positivity is precisely the wedge the uniqueness argument will drive between any two distinct equilibria. For the imitation sign $s>0$ the defect goes the wrong way: a population displaced toward higher mean *lowers* the cost where the mass went, rewarding agglomeration, which is the analytic fingerprint of the attractive interaction under which multiple self-consistent populations might coexist. At $s=0$ the defect vanishes identically, as it must, since the coupling is switched off and the game degenerates to the isolated control problem. What we have established is thus a sharp dichotomy in the single parameter s: the LQG coupling is Lasry–Lions monotone precisely on the half-line $s \le 0$, and we now show that this entire half-line is certified unique at every horizon.

Theorem 14.9 (Monotone uniqueness)**.** *If $s \le 0$ and $s_T \le 0$, then $D(T) \neq 0$ for every $T>0$, and the LQG equilibrium is unique for every horizon and every initial condition.*

Before the proof, a word on why we give one at all, when lemma 14.8 has just verified (A-Mon) and the abstract theorem of chapter 11 would hand us uniqueness directly. The reason is that the abstract theorem speaks in the wrong currency for this chapter. It concludes "the two equilibrium *measures* coincide," whereas the entire phase analysis is metered by the scalar $D(T)$, and what we need is the sharper statement that $D(T) \neq 0$—that the homogeneous linear BVP has no nontrivial mode—throughout the monotone region. The two are equivalent by lemma 14.5, but the equivalence runs through the very object we want to control, so it is cleaner, and more illuminating, to prove the $D(T) \neq 0$ statement head-on with an energy identity built directly on the reduced system (14.15). The argument is the LQG shadow of the general Lasry–Lions energy estimate—the one that pairs the two equilibria against $F[m_1]-F[m_2]$ and plays the monotone sign against the convex sign of the Hamiltonian (A-Conv)—but here it collapses to a single scalar function of time whose derivative we can write in one line.

Proof. We argue the contrapositive packaged through $D(T)$. By lemma 14.5, $D(T)=0$ holds if and only if the *homogeneous* two-point boundary value problem—system (14.15) with the data $\bar{m}_0=0$ and $r(T)=-q_T s_T \bar{m}_T$—admits a nontrivial solution $(\bar{m}_\cdot, r(\cdot))$. It is therefore enough to show that under $s \le 0$, $s_T \le 0$ the homogeneous problem has *only* the trivial solution; the conclusion $D(T) \neq 0$ then follows for every T, and uniqueness of the equilibrium follows from theorem 14.6. So suppose $(\bar{m}_\cdot, r(\cdot))$ solves the homogeneous BVP, and consider the scalar

$$\mathcal{E}(t) \;:=\; \bar{m}_t\, r(t), \tag{14.19}$$

the product of the two BVP components—an "energy" in the sense that its monotone behavior will trap the system at zero. Everything reduces to differentiating it.

The energy identity. Abbreviate the closed-loop gain by $g(t) := a - b^2 P(t)$, exactly the quantity theorem 14.2 guaranteed to be bounded and continuous. In this notation the homogeneous system

(14.15) is the pair

$$\dot{\bar m}_t = g(t)\,\bar m_t - b^2\,r(t), \qquad \dot r(t) = q\,s\,\bar m_t - g(t)\,r(t),$$

where we have used that the matrix $M(t)$ of (14.15) has lower-left entry qs and that its two diagonal entries are exactly g and its negative, $a - b^2 P$ on top and $-(a - b^2 P)$ on the bottom. That equal-and-opposite diagonal—the substitution $2(b^2 P - a) = -2g$ written into both rows—is the structural fact that makes the next line clean, so we keep it in view. Differentiating (14.19) by the product rule and inserting the two equations,

$$\dot{\mathcal{E}} = \dot{\bar m}_t\,r + \bar m_t\,\dot r = \big(g\,\bar m_t - b^2 r\big)\,r + \bar m_t\big(qs\,\bar m_t - g\,r\big) = \underbrace{g\,\bar m_t r - g\,\bar m_t r}_{=\,0} - b^2 r^2 + qs\,\bar m_t^2.$$

The two cross terms $+g\,\bar m_t r$ and $-g\,\bar m_t r$ cancel *precisely* because the diagonal entries of M are negatives of one another; this is where the antisymmetric diagonal earns its keep, and it is what leaves a derivative free of the gain g altogether:

$$\dot{\mathcal{E}}(t) \;=\; -\,b^2\,r(t)^2 \;+\; q\,s\,\bar m_t^2. \tag{14.20}$$

Now read the signs. The first term $-b^2 r^2$ is ≤ 0 unconditionally, since $b^2 > 0$. The second term $qs\,\bar m_t^2$ is the product of $q \ge 0$, of $s \le 0$ (the monotone hypothesis, here doing its one and only job), and of the square $\bar m_t^2 \ge 0$, hence also ≤ 0. Therefore $\dot{\mathcal{E}} \le 0$ on all of $[0, T]$: the energy is nonincreasing.

The endpoint pincer. A nonincreasing function is squeezed to zero if its value at the later time is at least its value at the earlier time. We check exactly that. At the initial time the homogeneous datum $\bar m_0 = 0$ gives

$$\mathcal{E}(0) = \bar m_0\,r(0) = 0,$$

whatever the (unknown) initial offset $r(0)$ is. At the terminal time the boundary condition $r(T) = -q_T s_T\,\bar m_T$ gives

$$\mathcal{E}(T) = \bar m_T\,r(T) = -q_T s_T\,\bar m_T^2.$$

Here the terminal hypothesis enters: $q_T \ge 0$ and $s_T \le 0$ make $-q_T s_T \ge 0$, and $\bar m_T^2 \ge 0$, so $\mathcal{E}(T) \ge 0$. We thus have a nonincreasing function with $\mathcal{E}(0) = 0$ at the left end and $\mathcal{E}(T) \ge 0$ at the right end. But a nonincreasing function cannot rise, so $\mathcal{E}(T) \le \mathcal{E}(0) = 0$; combined with $\mathcal{E}(T) \ge 0$ this forces $\mathcal{E}(T) = 0$, and then $\mathcal{E}$ is pinned between its two equal endpoint values and must be *constant*, $\mathcal{E} \equiv 0$, with $\dot{\mathcal{E}} \equiv 0$ throughout. This is the pincer: monotonicity bounds the energy from above by its initial value 0, while the terminal sign condition bounds it from below by 0, and the two bounds collide.

Collapse to the trivial solution. With $\dot{\mathcal{E}} \equiv 0$, the identity (14.20) reads $0 = -b^2 r^2 + qs\,\bar m^2$ at every time. Both terms on the right are ≤ 0 and they sum to zero, so each vanishes separately: $b^2 r(t)^2 \equiv 0$ and $qs\,\bar m_t^2 \equiv 0$. Since $b \ne 0$, the first gives $r \equiv 0$. Feeding $r \equiv 0$ back into the first BVP equation leaves $\dot{\bar m}_t = g(t)\,\bar m_t$, a homogeneous linear ODE for $\bar m$ with the initial value $\bar m_0 = 0$; its unique solution is $\bar m \equiv 0$. Hence the only homogeneous solution is the trivial one $(\bar m_\cdot, r(\cdot)) \equiv (0, 0)$. By lemma 14.5 this means $D(T) \ne 0$, and it holds for *every* $T > 0$ because nothing in the argument referred to the length of the horizon—the energy pincer closes at whatever time the interval ends. Uniqueness of the equilibrium for every horizon and every initial condition is then theorem 14.6. $\square$

Theorem 14.9 is the rigorous content of the slogan "repulsion stabilizes the equilibrium," and it is worth pausing on exactly how the slogan is realized in the proof. The stabilizing mechanism is not a vague pressure but the precise sign in (14.20): when the coupling is repulsive, $s \le 0$, the imitation term $qs\,\bar{m}^2$ joins the dissipative control term $-b^2r^2$ on the *same* side of zero, and the energy $\mathcal{E}$ can only fall. With the terminal data on the safe side as well, the only history the system can sustain is the flat one, and no second equilibrium can open. Congestion, that is, drains the homogeneous mode; there is no direction in which the best response can fail to pin the mean.

But now read what the theorem pointedly does *not* claim, because this is the narrative hinge of the entire chapter. It establishes that $s \le 0$ is *sufficient* for uniqueness at every horizon. It says nothing whatsoever about $s > 0$. The natural reflex—encouraged by the physical picture of section 14.1, where attraction lets the population clump and one expects symmetry breaking the instant the coupling flips sign—is to read the boundary of the monotone region, $s = 0$, as the boundary of uniqueness too, so that the slightest imitation would already split the equilibrium. The proof gives no warrant for that reading. Inspect (14.20) again: for small $s > 0$ the term $qs\,\bar{m}^2$ is positive, but it is competing against the strictly negative $-b^2r^2$, and which one wins is a *quantitative* contest between the pull of imitation and the restoring stiffness of the convex control cost—a contest the monotone argument, which only ever needed the easy case where both terms have the same sign, never even had to adjudicate. Monotonicity is sufficient precisely because it makes the contest trivial; it is silent past $s = 0$ precisely because there the contest becomes real. The honest conclusion is therefore the sharp one: *monotonicity is sufficient but not necessary*, and the true edge of uniqueness must be found by actually settling the competition, not by reading off the sign of s.

That is exactly the computation the next subsection performs. Armed with the certainty that the whole repulsive half-line $s \le 0$ is safe at every horizon—the foil now firmly drawn—we push directly into the non-monotone band $s > 0$, compute $D(T)$ in closed form rather than merely bound its sign, and watch where it *first* vanishes. The outcome, which the energy identity above could not have anticipated, is that the uniqueness region overspills the monotone region completely: it covers the entire imitation band up to a finite threshold $s = 1$, and fails only beyond it, at a finite conjugate horizon $T_c(s)$. We turn to that explicit calculation now.

14.5.2 An exactly solvable phase boundary

The previous subsection drew the foil and then, in its closing paragraphs, told us exactly what is left to do. The monotone half-line $s \le 0$ is certified unique at every horizon, but the energy identity (14.20), $\dot{\mathcal{E}} = -b^2r^2 + qs\,\bar{m}^2$, showed that the certificate works only because for $s \le 0$ both terms share a sign. Push into the imitation band $s > 0$ and the two terms oppose each other: the imitation reward $qs\,\bar{m}^2 > 0$ pulls the energy up while the control dissipation $-b^2r^2 < 0$ drags it down, and which prevails is a genuine quantitative contest between the pull of imitation and the restoring stiffness of the convex control cost. Monotonicity never had to adjudicate that contest; the present subsection settles it outright, by computing the solvability scalar $D(T)$ of lemma 14.5 in closed form and watching for the precise instant it passes through zero.

To get a closed form we need the coefficient matrix $M(t)$ of the boundary value problem (14.15) to be *constant* in time, and the only t-dependence in $M(t)$ enters through the closed-loop gain $a - b^2 P(t)$. So we choose the model parameters to make the Riccati curvature P constant. This is a convenience, not a cheat: it fine-tunes the *numbers* so that the phase boundary becomes an elementary function of (s, T), but it does not manufacture the qualitative structure. As we will see, everything in the phase diagram is controlled by the sign of $\det M = \omega^2(s-1)$, and that determinant changes sign at $s = 1$ for any choice of the remaining parameters; the normalization merely strips away the inessential constants so the conjugate-point collision can be read off by hand. We carry out the computation as the chapter's centerpiece.

Worked computation 14.10 (The LQG phase diagram)**.** Take $a = 0$, $b = 1$, running weight $q = \omega^2$ with $\omega > 0$, terminal weight $q_T = \omega$, and no terminal coupling, $s_T = 0$; the running coupling s and the horizon T are the free knobs, and $\sigma > 0$ is arbitrary. Each choice earns its keep. Setting $a = 0$ removes the intrinsic drift, so the only stiffness in the problem is the one the control supplies, and $b = 1$ sets the control authority to unity so that $b^2 P$ is just P. The pairing $q = \omega^2$, $q_T = \omega$ is the one fine-tuning that matters: it places the terminal datum of the Riccati equation (14.7) *exactly* on the equation's stable equilibrium, which freezes the curvature for all time. To see this, substitute the chosen weights into (14.7), which with $a = 0$, $b = 1$, $q = \omega^2$ reads

$$\dot{P} = P^2 - \omega^2, \qquad P(T) = q_T = \omega.$$

Its stationary points are the roots of $P^2 - \omega^2 = 0$, namely $P = \pm\omega$, and these are precisely the roots $\Pi_\pm = (a \pm \sqrt{a^2 + b^2 q})/b^2 = \pm\omega$ of theorem 14.2 with the present constants. The constant function $P(t) \equiv \omega$ satisfies the equation ($\dot{P} = 0$ and $\omega^2 - \omega^2 = 0$) and meets the terminal value $P(T) = \omega$; since the right-hand side $P^2 - \omega^2$ is a polynomial, hence locally Lipschitz, Picard–Lindelöf makes the solution through the terminal value unique, so $P(t) \equiv \omega$ is *the* solution—no other trajectory can share its terminal point. The choice $q_T = \omega$ is exactly what lands the terminal datum on the stable root $P_+ = \omega$ rather than off it; had we taken any other q_T, the curvature would relax toward ω as t runs backward from T but would not be constant, the closed-loop gain $a - b^2 P(t)$ would be genuinely time-dependent, and $D(T)$ would cease to be elementary—without, however, moving the sign change that organizes the diagram. With the curvature frozen, the closed-loop gain is the constant $a - b^2 P(t) \equiv 0 - \omega = -\omega$, and the boundary value problem (14.15) becomes the *constant-coefficient* system

$$\dot{\bar{m}}_t = -\omega\, \bar{m}_t - r(t), \qquad \dot{r}(t) = \omega^2 s\, \bar{m}_t + \omega\, r(t), \qquad \bar{m}_0 \text{ given}, \quad r(T) = 0, \tag{14.21}$$

where the terminal condition is $r(T) = -q_T s_T \bar{m}_T = 0$ because $s_T = 0$. Its coefficient matrix is $M = \left(\begin{smallmatrix} -\omega & -1 \\ \omega^2 s & \omega \end{smallmatrix}\right)$, obtained by reading the gain $a - b^2 P \equiv -\omega$ into the top-left slot, $-b^2 = -1$ into the top-right, the coupling $qs = \omega^2 s$ into the bottom-left, and the negated gain $+\omega$ into the bottom-right. Two invariants of M run the entire analysis. Its trace is $-\omega + \omega = 0$—the equal-and-opposite diagonal we have met before, the structural signature of the forward–backward pairing—and its determinant is

$$\det M = (-\omega)(\omega) - (-1)(\omega^2 s) = -\omega^2 + \omega^2 s = \omega^2(s-1).$$

Because the trace vanishes, the characteristic polynomial $\lambda^2 - (\operatorname{tr} M)\lambda + \det M = \lambda^2 + \det M$ has roots $\lambda = \pm\sqrt{-\det M} = \pm\,\omega\sqrt{1-s}$. These are *real* when $s < 1$ and *imaginary* when $s > 1$, and they collide at the origin precisely at $s = 1$, where $\det M = 0$. This single sign change of $\det M$ at

$s = 1$—a pair of real eigenvalues turning into a conjugate imaginary pair, hyperbolic flow turning into rotation—is the controlling event of the whole phase diagram, and the two regimes it separates are exactly the two cases we now develop.

Regime $s \le 1$ (includes all monotone $s \le 0$). Here $\det M = \omega^2(s-1) \le 0$, the eigenvalues are real, and the flow is hyperbolic. Write $\nu = \omega\sqrt{1-s} \ge 0$, so that $\nu^2 = \omega^2(1-s) = -\det M$. Because M is constant, the fundamental matrix is just the matrix exponential $\Psi(t) = e^{Mt}$, and the vanishing trace makes this exponential elementary. By the Cayley–Hamilton theorem M satisfies its own characteristic equation $M^2 + (\det M)I = 0$, i.e.

$$M^2 = -(\det M)\, I = \nu^2 I,$$

so every even power of M is a scalar multiple of the identity, $M^{2k} = \nu^{2k} I$, and every odd power is the same scalar times M, $M^{2k+1} = \nu^{2k} M$. Splitting the exponential series into its even and odd parts and resumming,

$$e^{Mt} = \sum_{k\ge 0} \frac{(Mt)^{2k}}{(2k)!} + \sum_{k\ge 0} \frac{(Mt)^{2k+1}}{(2k+1)!} = \Big(\sum_{k\ge 0} \frac{(\nu t)^{2k}}{(2k)!}\Big) I + \frac{1}{\nu}\Big(\sum_{k\ge 0} \frac{(\nu t)^{2k+1}}{(2k+1)!}\Big) M,$$

the two scalar series are exactly $\cosh(\nu t)$ and $\sinh(\nu t)$, so

$$\Psi(t) = \cosh(\nu t)\, I + \nu^{-1} \sinh(\nu t)\, M. \tag{14.22}$$

The factor $\nu^{-1}\sinh(\nu t)$ is well defined even at the collision $\nu = 0$, where $s = 1$ and M is nilpotent ($M^2 = 0$): there $\nu^{-1}\sinh(\nu t) \to t$ and (14.22) degenerates to the correct $\Psi(t) = I + tM$. This is the one closed form that covers the whole regime $s \le 1$ continuously, including its right endpoint. Now read off the solvability scalar. Since $s_T = 0$ kills the second term of (14.16), $D(T)$ is simply the lower-right entry $\Psi_{22}(T)$, and from (14.22) that entry is $\cosh(\nu T)$ (from the I part) plus $\nu^{-1}\sinh(\nu T)$ times $M_{22} = \omega$:

$$D(T) = \Psi_{22}(T) = \cosh(\nu T) + \tfrac{\omega}{\nu} \sinh(\nu T) > 0 \quad \text{for all } T > 0.$$

Every term here is strictly positive for $T > 0$—$\cosh \ge 1$, and $\sinh(\nu T)$, ω, ν are all positive—so $D(T)$ can never vanish: there is no horizon at which the BVP becomes singular. By lemma 14.5 the equilibrium is therefore *unique for every horizon*, throughout the whole band $s \le 1$. The decisive point is that this band reaches past the monotone region: monotonicity already failed for every $0 < s \le 1$ (lemma 14.8), yet uniqueness persists undisturbed across that entire imitation strip. This is the chapter's headline made literal—uniqueness outlives monotonicity—and the closed form $D(T) = \cosh(\nu T) + (\omega/\nu)\sinh(\nu T)$ is the explicit witness the energy argument of the previous subsection could never have produced, because the energy argument stopped speaking the moment s crossed zero.

Regime $s > 1$ (strong imitation). Now $\det M = \omega^2(s-1) > 0$, the eigenvalues $\pm i\mu$ with $\mu = \omega\sqrt{s-1} > 0$ are imaginary, and the hyperbolic flow becomes a rotation. The Cayley–Hamilton resummation runs exactly as before, but with the sign of M^2 reversed: now $M^2 = -(\det M)I = -\mu^2 I$, so $M^{2k} = (-1)^k \mu^{2k} I$ and $M^{2k+1} = (-1)^k \mu^{2k} M$, and the same two series resum to the *circular* functions in place of the hyperbolic ones,

$$\Psi(t) = e^{Mt} = \cos(\mu t)\, I + \mu^{-1} \sin(\mu t)\, M.$$

Reading off the lower-right entry just as in the previous regime—$\cos(\mu t)$ from I plus $\mu^{-1}\sin(\mu t)$ times $M_{22} = \omega$—and again using $s_T = 0$ to discard the second term of (14.16), the solvability scalar is now an oscillating function of the horizon,

$$D(T) = \Psi_{22}(T) = \cos(\mu T) + \tfrac{\omega}{\mu}\sin(\mu T). \tag{14.23}$$

Unlike its hyperbolic cousin, this $D(T)$ must vanish: it is a sinusoid of amplitude $\sqrt{1+\omega^2/\mu^2} > 0$, so it crosses zero periodically. The first crossing is the critical horizon. Setting (14.23) to zero and dividing by $\cos(\mu T)$ (legitimate at the first zero, where cos has not yet vanished) gives $1 + (\omega/\mu)\tan(\mu T) = 0$, i.e.

$$\tan(\mu T_c) = -\frac{\mu}{\omega}.$$

Because $\mu/\omega > 0$, the right-hand side is negative, and the smallest positive angle whose tangent is negative lies in the second quadrant: writing $\arctan(\mu/\omega) \in (0, \frac{\pi}{2})$ for the principal value and using $\tan(\pi - \theta) = -\tan\theta$, the first solution is $\mu T_c = \pi - \arctan(\mu/\omega)$. Dividing by μ,

$$T_c(s) \;=\; \frac{1}{\mu}\Big(\pi - \arctan\tfrac{\mu}{\omega}\Big), \qquad \mu = \omega\sqrt{s-1}. \tag{14.24}$$

For every horizon shorter than this, $T < T_c(s)$, the scalar $D(T)$ has not yet reached its first zero, so $D(T) \neq 0$ and the equilibrium is unique by lemma 14.5. At $T = T_c(s)$ the scalar vanishes, and lemma 14.5(2) takes over. What happens there is not one phenomenon but two, and which one the reader sees depends entirely on the prescribed initial mean $\bar{m}_0$. The two cases are the two sub-cases of the lemma's trichotomy, and they deserve to be separated cleanly because they are physically opposite.

The symmetry-breaking face ($\bar{m}_0 = 0$). Suppose the population starts centered, $\bar{m}_0 = 0$. The lemma's terminal constraint $D(T)\rho = -(\Psi_{21}(T) + q_T s_T \Psi_{11}(T))\bar{m}_0$ has a right-hand side proportional to $\bar{m}_0$, hence *zero*; at $T = T_c$ its left-hand side $D(T_c)\rho$ is also zero for every ρ, so the equation $0 = 0$ imposes no condition at all on the initial offset. Every value of $\rho = r(0)$ is admissible, and each nonzero choice launches, through $w(t) = \Psi(t)(0, \rho)^\top$, a distinct nontrivial trajectory $(\bar{m}_\cdot, r(\cdot))$ that solves the homogeneous BVP and reproduces $\bar{m}_0 = 0$. The symmetric equilibrium $(\bar{m}_\cdot, r) \equiv 0$ is still present, but it is now surrounded by a *one-parameter family* of asymmetric companions: the mean is no longer pinned: the population can self-consistently organize itself onto a continuum of distinct centered-but-drifting histories. This is a continuous symmetry breaking, the conjugate-point analogue of a pitchfork, and it is the first appearance in the chapter of genuine multiplicity.

The non-existence face (generic $\bar{m}_0 \neq 0$). Suppose instead the population starts off center, $\bar{m}_0 \neq 0$, as a generic datum will. Now the right-hand side of the same constraint need not vanish, and at $T = T_c$ the equation reads $0 = -\Psi_{21}(T_c)\,\bar{m}_0$ (the $q_T s_T$ term being absent since $s_T = 0$). This is no longer an equation for ρ—the unknown has dropped out entirely—but a *constraint on the datum* $\bar{m}_0$, and it is satisfied only if $\Psi_{21}(T_c) = 0$. We check that it is not. From the oscillatory fundamental matrix, the lower-left entry is $\Psi_{21}(t) = \mu^{-1}\sin(\mu t)\,M_{21} = \omega^2 s\,\mu^{-1}\sin(\mu t)$, and at $t = T_c$ the angle $\mu T_c = \pi - \arctan(\mu/\omega)$ lies in $(\frac{\pi}{2}, \pi)$, so $\sin(\mu T_c) > 0$; together with $\omega^2 s > 0$ this makes $\Psi_{21}(T_c) \neq 0$. The constraint $\Psi_{21}(T_c)\,\bar{m}_0 = 0$ therefore fails for every $\bar{m}_0 \neq 0$: no initial offset can steer the trajectory onto the required terminal value, and *no equilibrium exists* at $T = T_c$. The same conjugate horizon that opened a continuum of equilibria for the centered population annihilates the equilibrium altogether for an off-center one. These are the two faces of the single event $D(T_c) = 0$: at $\bar{m}_0 = 0$ the homogeneous mode is a free direction and multiplicity blooms; at $\bar{m}_0 \neq 0$ that same mode obstructs solvability and the problem is overdetermined.

Past the first conjugate horizon the sinusoid (14.23) continues to oscillate. For $T > T_c(s)$, away from the discrete set of further zeros—the successive solutions of $\tan(\mu T) = -\mu/\omega$, spaced by π/μ in the angle—$D(T)$ is again nonzero and the equilibrium again unique, though D has changed sign on crossing T_c, so the comparative statics (the sign of the offset $r(0)$ as a function of $\bar{m}_0$ in lemma 14.5) reverse across each conjugate horizon. The geometry of the boundary is governed by the two limits of (14.24). As $s \downarrow 1$, $\mu = \omega\sqrt{s-1} \to 0$, the arctangent vanishes, and $T_c \sim \pi/\mu \to \infty$: the boundary recedes to infinite horizon, smoothly matching the hyperbolic regime where $D(T) > 0$ at *every* finite horizon. As $s \to \infty$, $\mu \to \infty$, $\arctan(\mu/\omega) \to \frac{\pi}{2}$, and $T_c \sim \pi/(2\mu) \to 0$: arbitrarily strong imitation destabilizes arbitrarily fast, the unique-region shrinking to a vanishing sliver of short horizons.

It remains to account for the cost stiffness q, the third axis of the (s, q, T) phase diagram the thread specification asks for, which the normalization $q = \omega^2$ has folded into the single symbol $\omega = \sqrt{q}$. Far from being lost, that axis is carried by an exact scaling that we can read straight out of (14.24), and the cleanest way to see it is to track what depends on q and what does not. The angle inside the arctangent is

$$\frac{\mu}{\omega} = \frac{\omega\sqrt{s-1}}{\omega} = \sqrt{s-1},$$

which is *independent of* q: it is a pure function of the coupling s. The only q-dependence in T_c therefore sits in the prefactor $1/\mu = 1/(\omega\sqrt{s-1}) = 1/(\sqrt{q}\,\sqrt{s-1})$, so the critical horizon factorizes as

$$T_c(s, q) = \frac{1}{\sqrt{q}}\;\underbrace{\frac{\pi - \arctan\sqrt{s-1}}{\sqrt{s-1}}}_{=:\,C(s)}, \qquad \text{i.e.} \qquad \sqrt{q}\,T_c = C(s) \text{ depends only on } s. \tag{14.25}$$

This is the promised invariance, now demonstrated rather than asserted. The product $\sqrt{q}\,T_c$ is a function of s alone, so the boundary is unchanged under the one-parameter rescaling $(q, T) \mapsto (\kappa^2 q,\, \kappa^{-1}T)$: scaling the stiffness up by κ^2 and the horizon down by κ leaves $\sqrt{q}\,T$ fixed and hence maps the critical surface to itself. Concretely, stronger state-cost stiffness q *contracts* the unique-equilibrium region in T as $1/\sqrt{q}$, pushing the first conjugate point earlier—faithful to intuition, since a stiffer cost makes the offset r respond more violently to the mean and so reaches the destabilizing resonance sooner. The practical consequence is that the third axis is redundant: the single (s, T) slice drawn at $\omega = 1$ (that is, $q = 1$) in fig. 14.1 already captures the full three-parameter picture, provided one reads the horizon in units of $1/\sqrt{q}$. The curve (14.24)—equivalently $T = C(s)/\sqrt{q}$—is what fig. 14.1 plots.

The worked computation has handed us, in closed form, everything the chapter set out to find: a single elementary scalar $D(T)$, positive for all horizons when $s \leq 1$ and a sinusoid with a computable first zero $T_c(s)$ when $s > 1$. We now record the conclusion as a theorem, taking care to state it no more strongly than the closed forms warrant. The temptation is to write "unique if and only if $T < T_c(s)$," but the oscillatory $D(T)$ forbids that biconditional: D vanishes only on a *discrete* set of horizons and is nonzero on each open interval between them, so uniqueness returns past the first conjugate point. The precise statement keeps this in view.

Theorem 14.11 (Phase boundary)**.** *In the normalization of worked computation 14.10, the LQG mean field game has a unique equilibrium for* every *horizon $T > 0$ when $s \le 1$. When $s > 1$ it has a unique equilibrium for all $T < T_c(s)$, with $T_c(s)$ in (14.24) the* smallest *horizon at which uniqueness is lost—the first conjugate point; for $T > T_c(s)$ the equilibrium is again unique for every T* except *the discrete set of further conjugate horizons, the successive zeros of $D(T)$ in (14.23). The locus $\{T = T_c(s) : s > 1\}$ is the phase boundary: it is where uniqueness is* first *lost, at the first conjugate point of the forward–backward system, and is precisely where the linearized best-response operator $I - (\textit{linear part})$ of remark 14.7 first becomes singular.*

Proof. By lemma 14.5 the equilibrium is unique exactly when $D(T) \ne 0$, so the claim is a statement about the zero set of the closed forms for $D(T)$ in worked computation 14.10. For $s \le 1$, $D(T) > 0$ for all T, giving uniqueness at every horizon. For $s > 1$, $D(T) = \cos(\mu T) + \frac{\omega}{\mu}\sin(\mu T)$ in (14.23) is a nonzero sinusoid of T; its zeros are the simple, discrete solutions of $\tan(\mu T) = -\mu/\omega$, the smallest of which is $T_c(s)$ in (14.24). Hence $D(T) \ne 0$ for all $T < T_c(s)$ (uniqueness), $D(T_c) = 0$ (uniqueness lost), and for $T > T_c$ one has $D(T) \ne 0$ off the discrete set of further zeros, where theorem 14.6 again gives a unique equilibrium. Uniqueness therefore fails only on the measure-zero set $\{D(T) = 0\}$, consistent with remark 14.12; the biconditional "unique $\Leftrightarrow T < T_c$" would be false, since uniqueness holds on each open interval between consecutive conjugate horizons. ☐

Physics note

It is worth naming the physical shape of the conjugate-point event, because it is the same one that recurs whenever uniqueness fails in a mean field game. The loss of uniqueness at $T = T_c$ is a *continuous symmetry breaking* of the forward–backward problem. Below the threshold the only self-consistent history of a centered population is the symmetric one, $\bar{m}_\cdot \equiv 0$; at the conjugate horizon the homogeneous mode of lemma 14.5(2) opens, and a continuum of asymmetric histories appears around it (the symmetry-breaking face computed above). This is the exact dynamic analogue of the Curie–Weiss ferromagnet acquiring a continuum of magnetized states below its critical temperature: the symmetric state does not disappear, it merely ceases to be the only one, and the new ones branch off it continuously. The imitation coupling s plays the role of inverse temperature, and the conjugate horizon $T_c(s)$ traces the phase boundary in the (s, T) plane.

The one quantitative lesson the calculation forces—and the lesson that the naive physical guess of section 14.1 got wrong—is that the threshold sits at $s = 1$, not at the sign change $s = 0$. The instability requires a *finite* critical coupling, and the closed forms say exactly why. Imitation contributes the destabilizing term $qs\,\bar{m}^2$ to the energy (14.20), but it must overcome a restoring stiffness supplied by two sources at once: the convex control cost $\frac{1}{2}\alpha^2$, which charges the population for moving, and the Riccati curvature $P \equiv \omega$, which is the accumulated strength of that charge fed back through the value function. In the normalization, those two combine into the gap $\det M = \omega^2(s-1)$, and the gap stays on the stable side until

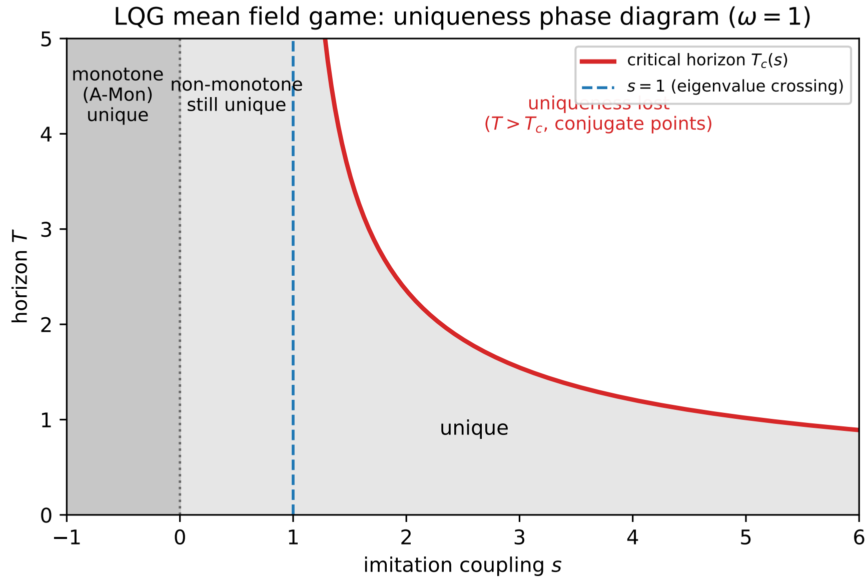

Figure 14.1: The LQG mean field game phase diagram in the normalization of worked computation 14.10 ($a = 0$, $b = 1$, $q = \omega^2$, $q_T = \omega$, $s_T = 0$, $\omega = 1$). For $s \le 1$ the equilibrium is unique at every horizon. For $s > 1$ (strong imitation) uniqueness holds below the critical horizon $T_c(s)$ of (14.24) and is first lost on the boundary; the boundary diverges as $s \downarrow 1$ and collapses to zero as $s \to \infty$. The monotone region $s \le 0$ of theorem 14.9 is shaded; note that uniqueness extends well beyond it, up to $s = 1$.

imitation is strong enough to close it—at $s = 1$. Repulsion, $s \le 0$, contributes a term of the *stabilizing* sign and so, like a screening interaction that can only add to the restoring force, can never reach the threshold from below: this is precisely the physical content of the monotone-uniqueness theorem theorem 14.9. The finite threshold is not an artifact of the normalization; it is the statement that a convex regulator carries an intrinsic stiffness, and imitation must pay a finite price to break it.

Remark 14.12 (Honest scope of "multiplicity")**.** A word of caution is owed about how much the phase boundary actually delivers, because it is easy to over-read fig. 14.1 as a map of a region densely filled with multiple equilibria. It is not. In the strictly linear–quadratic model the degeneracy is confined to the conjugate hypersurfaces $\{D(T) = 0\}$—the curve $T = T_c(s)$ and its successors—and these form a *measure-zero* set in the parameter space (s, q, T). Off them, at every parameter point not landing exactly on a conjugate surface, $D(T) \ne 0$ and theorem 14.6 returns a unique equilibrium. So what the phase boundary marks is not an open basin of many coexisting equilibria but the *onset* of the mechanism that produces them: it is the precursor of multiplicity, the precise locus where the linearized best-response operator first acquires a kernel, not a region where multiplicity is already realized.

The reason the linear model only ever sees the precursor, and not an open multiplicity region, is that its best-response map is exactly affine (remark 14.7), so the singular set of $I - L$ is the zero set of a single analytic scalar $D(T)$, which is generically a hypersurface. Two standard perturbations thicken that hypersurface into a genuine open region, and naming them shows what the linear calculation has isolated. First, reinstating a terminal coupling $s_T \neq 0$ restores the second term $q_T s_T \Psi_{12}(T)$ in (14.16), and for suitable signs this can hold $D(T)$ at zero across an *interval* of horizons rather than at isolated points. Second, and more fundamentally, any nonlinearity in the coupling—a cost that depends on the population *variance*, a non-quadratic congestion term, anything that makes the curvature P feel the mean—destroys the affineness of the best-response map, so that the conjugate point becomes a genuine fold or pitchfork bifurcation at which several solution branches are born and persist on an open neighborhood. The LQG calculation isolates the linear seed of exactly this bifurcation; the bifurcated picture, with its open multiplicity regions, belongs to chapter 11 and the references there. The honest summary is thus that the shaded "unique" region of fig. 14.1 is *exact*, the conjugate boundary is exact, and the boundary is to be read as the threshold of an instability rather than as the edge of a filled basin.

We have now drawn the phase diagram the chapter was built to produce, and the single object it turns on deserves to be named once more before we move on, because the next section is organized entirely around it. The boundary $\{T = T_c(s)\}$ is the *first conjugate point* of the forward–backward system (14.15): the smallest horizon at which the homogeneous two-point problem acquires a nontrivial mode and the solvability scalar $D(T)$ first vanishes. By the spectral reading of remark 14.7, that same event is the moment the linearized best-response operator $I - L$ first becomes singular—an eigenvalue of the best-response linearization crossing through the value that opens a kernel. One event, two names: a conjugate point of a boundary value problem, and a singularity of a fixed-point operator. The chapter's final move is to show that this is not a coincidence of two bookkeeping schemes but a single fact every layer of Part II records in its own language. The next section takes the equilibrium we have just pinned down and the critical event we have just located, and exhibits how the single-agent problem, the analytic HJB–Fokker–Planck system, the FBSDE, the fixed-point map, and—most strikingly —the master equation's decoupling field each see precisely this equilibrium and precisely this $D(T_c) = 0$, the decoupling field blowing up exactly where the best-response operator goes singular. We turn to that assembly now.

14.6 Every layer of Part II, on one model

The previous section ended by giving the phase boundary two names for one event: the first conjugate point of the forward–backward system (14.15), where the solvability scalar $D(T)$ first vanishes, and the singularity of the linearized best-response operator $I - L$ of remark 14.7, where one eigenvalue first crosses the value that opens a kernel. We claimed there that this coincidence is not an artifact of two bookkeeping schemes but a single fact that every layer of Part II records in its own dialect. Redeeming that claim is the business of this section, and it is also where the capstone keeps the promise the chapter preamble made: that each chapter of Part II would reappear here not as a citation but as an *explicit object* computed on the one model. We now make that assembly, walking the layers in the order the theory was built. Two threads run through the walk and should be watched at every stage. The first is that each layer sees *the same equilibrium*—the quadratic value

(14.6), the Gaussian flow $\mathcal{N}(\bar{m}_t, V_t)$, the offset r solving the boundary value problem—packaged in whatever language that chapter speaks. The second, and the sharper one, is that each layer also sees *the same critical event*: the conjugate point at $T_c(s)$ where $D(T)$ vanishes is, in the analytic picture a loss of solvability, in the probabilistic picture a singularity of the decoupling field, in the fixed-point picture the moment $I - L$ acquires a kernel, and—most strikingly, and proved in full below—in the master-equation picture the finite-time blow-up of a second Riccati coefficient Q. By the end of the section these will be visibly one event, and the chapter's assembly will be complete.

Single-agent control (chapter 6). The first layer is the one with no game in it at all, and it is the atom out of which every later layer is built. Freeze the population flow at a constant, $\bar{m}_t \equiv \bar{m}$; then the cost (14.3) no longer couples one agent to another, and the coupling term $s\bar{m}$ survives only as a fixed external target. What remains is precisely the single-agent linear–quadratic regulator of chapter 6: the HJB equation (14.4) with $\bar{m}_t$ a prescribed constant, whose value function is the quadratic (14.6), with curvature P from the Riccati equation (14.7) and offset r from the now-decoupled linear equation (14.8), and whose minimizer is the linear-state feedback (14.10), the classical LQ controller. Everything the rest of the chapter does to this atom is carried by the two channels the audit of section 14.2 already isolated: the curvature P is *population-blind*, computed once and for all from the cost data and globally defined by theorem 14.2, while the entire trace of the crowd is funneled into the single linear channel r. The critical event of the chapter cannot yet be seen at this layer, precisely because there is no loop to close—freezing $\bar{m}$ severs the very self-consistency whose breakdown is the conjugate point. The atom is healthy at every horizon; the disease lives in the assembly, which is the rest of Part II.

The analytic MFG system (chapter 8). Unfreeze the mean and the atoms must talk to one another; the instrument for that conversation is the coupled HJB–Fokker–Planck system of chapter 8, and for LQG it is sitting in front of us already. The pair $\{(14.4)$ backward, (14.12) forward$\}$ *is* that system, specialized to a quadratic value u and a Gaussian density m. What chapter 8 insisted was the structural heart of the difficulty is here made visible in the cleanest possible form: the two equations run in *opposite directions of time*. The HJB equation (14.4) carries terminal data at $t = T$ and is integrated backward, because value is computed by looking ahead to the cost still to be paid; the Fokker–Planck equation (14.12) carries initial data at $t = 0$ and is integrated forward, because the population is pushed ahead from where it started. Neither can be solved without the other, and the only place they touch is the scalar pair $(\bar{m}_t, r(t))$: the forward equation needs the backward variable r to know its closed-loop drift, and the backward equation needs the forward variable $\bar{m}$ to know its target. This is the same observation that organized section 14.3 and section 14.4, now read as the Part II layer it instantiates. And it already tells us where to look for the critical event: it cannot live in (14.4) alone, which is globally well posed by theorem 14.2, nor in (14.12) alone, which is a benign linear Fokker–Planck equation; it can only live in the *handshake* between them, the demand that the forward $\bar{m}$ and the backward r be mutually consistent. That handshake is the next layer.

The probabilistic FBSDE (chapter 9). The same handshake has a probabilistic shadow, and seeing the two pictures land on the identical boundary value problem is worth the few lines of algebra it takes. The stochastic maximum principle of chapter 9 attaches to the state (14.1) an *adjoint* (co-state) process Y_t tracking the marginal value of the state along the optimal trajectory; for the quadratic value (14.6) it is the gradient read along the path,

$$Y_t = \partial_x u(t, X_t) = P(t)\, X_t + r(t),$$

which is *affine in the state*. This is the LQG instance of the general decoupling field of chapter 9: the object that expresses the backward unknown as a deterministic function of the forward state, and so disentangles the forward–backward coupling. The pair (X, Y) solves the McKean–Vlasov forward–backward SDE

$$\mathrm{d}X_t = (aX_t - b^2 Y_t)\;\mathrm{d}t + \sigma\;\mathrm{d}W_t, \qquad \mathrm{d}Y_t = -\big(aY_t + q(X_t - s\bar{m}_t)\big)\;\mathrm{d}t + Z_t\;\mathrm{d}W_t, \quad Y_T = q_T(X_T - s_T\bar{m}_T),$$

the forward equation being (14.1) with the optimal control $\alpha^* = -bY_t$ inserted, and the backward equation the adjoint (Pontryagin) equation, coupled to the population through $\bar{m}_t$. Now take expectations and watch (14.15) fall out. Because both drifts are affine and the Itô integrals are zero-mean martingales, the expectation passes through; writing $\bar{m}_t = \mathbb{E}[X_t]$ and using the affine adjoint to get $\mathbb{E}[Y_t] = P(t)\bar{m}_t + r(t)$, the forward equation gives

$$\dot{\bar{m}}_t = a\,\bar{m}_t - b^2\,\mathbb{E}[Y_t] = a\,\bar{m}_t - b^2(P\bar{m}_t + r) = (a - b^2P)\,\bar{m}_t - b^2 r,$$

which is exactly the top row of (14.15), the mean ODE (14.13). For the bottom row, take expectations of the backward equation, $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[Y_t] = -a\,\mathbb{E}[Y_t] - q(\bar{m}_t - s\bar{m}_t)$, and substitute $\mathbb{E}[Y_t] = P\bar{m}_t + r$ on the left, differentiating by the product rule, $\dot{P}\bar{m}_t + P\dot{\bar{m}}_t + \dot{r}$. Replacing $\dot{P}$ by the Riccati right-hand side (14.7) and $\dot{\bar{m}}_t$ by the mean ODE just obtained, the curvature terms $\dot{P}\bar{m} + P\dot{\bar{m}} = (-aP - q)\bar{m} - b^2Pr$ collect, and after cancelling the common $-aP\bar{m} - q\bar{m}$ against the right-hand side the residue is precisely

$$\dot{r} = (b^2P - a)\,r + qs\,\bar{m}_t,$$

the offset equation (14.8), i.e. the bottom row of (14.15). So the probabilistic and analytic pictures do not merely resemble each other; taking expectations of the FBSDE *reproduces* (14.15) *on the nose*. The reason the two pictures agree is structural and worth stating once: the affine decoupling field $Y_t = PX_t + r$ is nothing but the gradient $\partial_x u$ of the quadratic value, so the probabilistic adjoint and the analytic offset are the same object viewed from two sides—one as a pathwise co-state, the other as a coefficient of the value function. The critical event reappears here too, now wearing its chapter 9 name: the conjugate point $T_c(s)$ is exactly the horizon at which this decoupling field ceases to exist globally, the FBSDE losing its unique solvability where $D(T)$ vanishes.

Existence (chapter 10). The general theory establishes existence by a fixed-point argument: chapter 10 builds the best-response map S, which sends a candidate mean flow to the mean the best-responding population actually produces, and finds an equilibrium as a fixed point $\bar{m}_\cdot = \mathsf{S}(\bar{m}_\cdot)$ via Schauder—a continuous self-map of a compact convex set of measures must fix a point. For LQG that whole apparatus collapses, and the collapse is instructive rather than merely convenient. As remark 14.7 showed, the LQG best-response map is exactly *affine*, $\mathsf{S}(\bar{m}_\cdot) = L\,\bar{m}_\cdot + \beta$, because it is the composition of the two linear solves (14.8) and (14.13); an affine fixed point $(I - L)\bar{m}_\cdot = \beta$ exists and is unique exactly when $I - L$ is invertible, and the elimination of lemma 14.5 showed that $I - L$, restricted to the one direction of genuine freedom that survives, is multiplication by the single scalar $D(T)$. So existence and uniqueness reduce to the one inequality $D(T) \neq 0$, recorded in theorem 14.6. The compactness and continuity inputs the Schauder argument needs are still present—they are what make S well defined—but they have become inert: the quadratic structure trivializes the search for a fixed point to the invertibility of a 1×1 linear map. And once again the critical event is visible in this layer's language: $D(T) = 0$ is precisely where $I - L$ stops being invertible, where the affine fixed point ceases to be uniquely determined. Existence and uniqueness do not fail because a hypothesis of the abstract theorem lapses; they fail because a computable eigenvalue crosses through zero.

Uniqueness and monotonicity (chapter 11). The general uniqueness theory of chapter 11 runs on the Lasry–Lions monotonicity condition, and the LQG model makes both the condition and its limits perfectly explicit. lemma 14.8 evaluated the monotonicity pairing in closed form and found it equal to the single number $-qs(\bar{m} - \bar{m}')^2$, a *monotonicity defect* whose sign locates the monotone regime exactly at the congestion half-line $s \leq 0$. On that half-line theorem 14.9 certifies uniqueness at every horizon, and it does so by the LQG shadow of the chapter 11 energy estimate: the scalar energy $\mathcal{E}(t) = \bar{m}_t r(t)$ obeys the one-line identity $\dot{\mathcal{E}} = -b^2 r^2 + qs\,\bar{m}^2$, both of whose terms share the safe sign when $s \leq 0$, trapping the only homogeneous mode at zero. This is the structural, horizon-uniform half of the phase story. Its limit is equally explicit, and it is the hinge of the chapter: the energy argument needs both terms of $\dot{\mathcal{E}}$ to share a sign, which they do only for $s \leq 0$; the instant s turns positive the imitation term $qs\,\bar{m}^2$ opposes the dissipation $-b^2 r^2$, the certificate falls silent, and only the explicit computation of $D(T)$ can say what happens. The phase boundary theorem 14.11 is exactly the locus where this monotone protection lapses and is not replaced—the first conjugate horizon $T_c(s)$, sitting strictly beyond the monotone region, in the band $0 < s \leq 1$ where (A-Mon) has already failed but uniqueness still holds, and on past $s = 1$. Monotonicity, in this layer's language, is the sufficient condition whose boundary the true uniqueness boundary overspills.

The master equation (chapter 12). We arrive at the deepest layer, and the one in which the critical event of the chapter shows its true face. The master equation of chapter 12 replaces the coupled HJB–Fokker–Planck *system* by a single scalar PDE $U(t, x, m)$ on $[0, T] \times \mathbb{R} \times \mathcal{P}_2(\mathbb{R})$, whose value $U(t, x, m)$ is the equilibrium cost of an agent currently at state x when the population it faces is the measure m. Its virtue is that it carries the *whole* equilibrium correspondence in one object: differentiating U in x gives the feedback for every population state at once, so U knows the answer before the forward–backward loop is ever run. For LQG that object collapses dramatically, for the same reason the cost did: U can depend on the population m only through the scalar mean $\mu = \bar{m}$, since that is the only feature of the crowd the cost (14.3) can feel. And, just as in section 14.2, the master equation preserves quadratics, so it admits the quadratic-in-(x, μ) ansatz

$$U(t, x, \mu) = \tfrac{1}{2}P(t)x^2 + Q(t)\,x\mu + \tfrac{1}{2}R(t)\,\mu^2 + (\text{terms independent of } x). \tag{14.26}$$

Three coefficients now carry the model, where the value function (14.6) had two. The coefficient of x^2 obeys the *same* Riccati equation (14.7) for P it always has—the curvature is population-blind in the master equation exactly as it was in the value function, a fact worth pausing on, since it means the diagonal Riccati layer is literally identical across all of Part II. The genuinely new object is the cross coefficient $Q(t)$ of $x\mu$, which measures how the agent's marginal value responds to a shift of the population mean. Substituting (14.26) into the master equation and matching the coefficient of $x\mu$—the master-equation operator feeds the $\frac{1}{2}Px^2$ term, the $Qx\mu$ term, and the mean-derivative of U into this slot—produces a *second* Riccati equation,

$$\dot{Q} = b^2Q^2 + 2(b^2P - a)Q + qs, \qquad Q(T) = -q_T s_T, \tag{14.27}$$

whose terminal datum is read from the terminal coupling in (14.3) exactly as $r(T)$ was. The remaining coefficient $R(t)$ of μ^2 solves a linear equation driven by P and Q, and it plays no role we have to track: like the spectator constant $c(t)$ of (14.9) in the value function, R (with the x-independent remainder) merely shifts U uniformly along μ, drops out of the feedback $-b\,\partial_x U$ because that feedback sees only the x-derivative, and so cannot touch the equilibrium or the phase boundary. It is the master-equation analogue of the spectator c, and we record it and move on.

The reason Q is the object to watch is the decoupling identity, and verifying it is the deepest cross-layer computation in the chapter, so we carry out every substitution rather than assert the result. The decoupling-field theorem of chapter 12 (theorem 12.6, equation (12.11)) says that the master field, evaluated along the *equilibrium* population flow m_t, reproduces the value function of the equilibrium: $u(t, x) = U(t, x, \bar{m}_t)$. Reading off the coefficient of x^1 on each side—$r(t)$ on the left from (14.6), $Q(t)\bar{m}_t$ on the right from (14.26)—this asserts the identity

$$r(t) \;=\; Q(t)\,\bar{m}_t. \tag{14.28}$$

We check that (14.28) is consistent with the equations r, Q, and $\bar{m}$ already satisfy. Abbreviate the closed-loop gain by $g := a - b^2P$, so that $b^2P - a = -g$ and in particular $2(b^2P - a) = -2g$; this single rewriting turns the Q-Riccati (14.27) into $\dot{Q} = b^2Q^2 - 2gQ + qs$ and is the only algebra the verification needs. Differentiate the candidate identity (14.28) by the product rule and substitute the two evolution laws—the Q-Riccati in this g-form, and the mean ODE (14.13) with r itself replaced by $Q\bar{m}$, so that $\dot{\bar{m}} = g\bar{m} - b^2r = g\bar{m} - b^2Q\bar{m}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}(Q\bar{m}) = \dot{Q}\,\bar{m} + Q\,\dot{\bar{m}} = (b^2Q^2 - 2gQ + qs)\bar{m} + Q(g\bar{m} - b^2Q\bar{m}).$$

Expand the second product, $Q(g\bar{m} - b^2Q\bar{m}) = gQ\,\bar{m} - b^2Q^2\bar{m}$, and collect by inspection. The two quadratic-in-Q pieces are equal and opposite, $+b^2Q^2\bar{m}$ from the Riccati and $-b^2Q^2\bar{m}$ from the mean term, and *cancel*; the linear-in-Q pieces combine as $-2gQ\bar{m} + gQ\bar{m} = -gQ\bar{m}$; and the source $qs\,\bar{m}$ rides along. What survives is

$$\frac{\mathrm{d}}{\mathrm{d}t}(Q\bar{m}) = (b^2Q^2 - 2gQ + qs + gQ - b^2Q^2)\bar{m} = (-gQ + qs)\bar{m}.$$

Now compute, independently, what the offset equation (14.8) demands of $\dot{r}$ once we impose $r = Q\bar{m}$. In g-notation that equation is $\dot{r} = (b^2P - a)r + qs\,\bar{m} = -g\,r + qs\,\bar{m}$, and substituting $r = Q\bar{m}$ on its right-hand side gives

$$\dot{r} = -g\,Q\bar{m} + qs\,\bar{m} = (-gQ + qs)\bar{m}.$$

The two expressions agree: the time-derivative of $Q\bar{m}$ forced by the Q-Riccati and the mean ODE is identical to the $\dot{r}$ forced by the offset equation, and at $t = T$ the data match as well, since

$r(T) = -q_T s_T \bar{m}_T = Q(T)\bar{m}_T$. By uniqueness for the scalar linear ODE, $r(t) = Q(t)\bar{m}_t$ holds throughout, exactly as the ch12 worked computation worked computation 12.15 records from the master-equation side. The crucial point of the cancellation is that the b^2Q^2 term—the only nonlinearity—is killed precisely by the $-b^2Q\bar{m}$ feedback in the mean ODE; the master equation's second Riccati and the equilibrium's offset are two readings of the very same linear channel r.

Now the identity (14.28) pays off, because it converts the entire phase question into a statement about a single Riccati equation. Since $r = Q\bar{m}$ along equilibrium, the offset stays finite exactly as long as Q does, and the no-blow-up⇔well-posedness correspondence proved for LQG in remark 12.16 sharpens this to: *the equilibrium is unique on* $[0,T]$ *exactly when* Q *stays finite on* $(0,T]$. Where Q blows up, the decoupling field of the master equation ceases to exist, and the equilibrium correspondence the master field was supposed to carry breaks down. It remains to locate that blow-up, and the normalization of worked computation 14.10 makes it elementary. There $a = 0$, $b = 1$, $q = \omega^2$, $P \equiv \omega$, and $s_T = 0$, so the terminal datum is $Q(T) = -q_T s_T = 0$ and the Q-Riccati (14.27) becomes, completing the square,

$$\dot{Q} = Q^2 + 2\omega Q + \omega^2 s = (Q+\omega)^2 + \omega^2(s-1).$$

For $s > 1$ the right-hand side is bounded below by the strictly positive constant $\omega^2(s-1) > 0$: the field has *no real stationary point*, because $(Q+\omega)^2 = -\omega^2(s-1) < 0$ is unsolvable, so nothing can arrest the trajectory. Set $u = Q + \omega$ and $\mu = \omega\sqrt{s-1}$, so that $\dot{u} = u^2 + \mu^2$ with $u(T) = \omega$; this is the elementary Riccati whose solution is $u(t) = \mu\tan(\mu(t-t_0))$, with the phase fixed by $u(T) = \omega$, i.e. $\mu(T-t_0) = \arctan(\omega/\mu)$. Running *backward* from T, the tangent drives $u \to -\infty$ when its argument reaches $-\frac{\pi}{2}$, so Q blows up to $-\infty$ a backward-time

$$T - t_* = \frac{1}{\mu}\Big(\tfrac{\pi}{2} + \arctan\tfrac{\omega}{\mu}\Big) = \frac{1}{\mu}\Big(\pi - \arctan\tfrac{\mu}{\omega}\Big)$$

before the horizon, where we used $\arctan(\omega/\mu) = \frac{\pi}{2} - \arctan(\mu/\omega)$. This is exactly $T_c(s)$ of (14.24): the second Riccati coefficient Q first diverges at precisely the horizon at which $D(T)$ first vanishes. The three statements are now visibly one. The conjugate point of the forward–backward boundary value problem (14.15), the singularity of the linearized best-response operator $I - L$ of remark 14.7, and the finite-time blow-up of the master-equation decoupling field $r = Q\bar{m}$ are the same event, seen in three languages—loss of solvability, loss of invertibility, and divergence of a Riccati coefficient—at the one horizon $T_c(s)$. This is the cross-layer identification the section was written to make.

Finite-player games and the ε-Nash property (chapter 13). The last layer closes the circle back to the finite population the mean field game was an idealization of. Take the genuine N-player game in which agent i pays the cost (14.3) with the population mean $\bar{m}_t$ replaced by the *empirical* mean $\bar{X}_t^N = \frac{1}{N}\sum_j X_t^j$ of the realized states. This game is itself exactly solvable, and watching how the solve goes is what makes the mean-field limit concrete rather than merely asserted. The lever is symmetry. Agent i's running cost $\frac{1}{2}\alpha_i^2 + \frac{q}{2}(X_i - s\bar{X}^N)^2$ is invariant under any permutation of the other $N-1$ labels, because they enter only through the symmetric average $\bar{X}^N$; this *exchangeability* forces the equilibrium value of agent i to depend on the full state vector $\mathbf{X} = (X_1, \dots, X_N)$ only through the pair $(X_i, \bar{X}^N)$—its own state and the crowd average, nothing finer. Since the coupled

HJB system of the N control problems again preserves quadratics, the value of agent i must take the form

$$V^i(t,\mathbf{X}) = \tfrac{1}{2}P^N(t)\,X_i^2 + L^N(t)\,X_i\,\bar{X}^N + \tfrac{1}{2}\,\Pi^N(t)\,(\bar{X}^N)^2 + (\text{terms independent of } X_i), \qquad (14.29)$$

with three time-dependent scalars (P^N, L^N, Π^N) common to all agents by symmetry. The structure of this quadratic is dictated, not assumed: its Hessian in $\mathbf{X}$ is $P^N(t)\,I$, the diagonal self-curvature, plus a rank-one matrix in the all-ones direction $\mathbf{1} = (1,\dots,1)^\top$, inevitable because $\bar{X}^N = \frac{1}{N}\mathbf{1}^\top\mathbf{X}$ couples the agents only through that one direction. This "$P^N I + \text{rank-one}$" Hessian is the finite-N fingerprint of mean-field coupling, and it is the object the matrix calculation of the next section will generalize. Differentiating (14.29) gives the symmetric feedback

$$\alpha_i = -b\,\partial_{X_i}V^i = -b\big(P^N(t)\,X_i + r^N(t)\,\bar{X}^N\big), \qquad r^N = L^N + \tfrac{1}{N}(L^N + \Pi^N).$$

The combination $r^N = L^N + \frac{1}{N}(L^N + \Pi^N)$ repays a careful look, because the entire mean-field limit is hiding in its $\frac{1}{N}$ term. When agent i differentiates $\bar{X}^N = \frac{1}{N}\sum_j X_j$ with respect to its own X_i, it does not get zero: it owns a $\frac{1}{N}$ share of the very average it is best-responding to, and that self-interaction feeds the $X_i\bar{X}^N$ and $(\bar{X}^N)^2$ terms of (14.29) back into its own gradient with weight $\frac{1}{N}$, producing exactly the $\frac{1}{N}(L^N + \Pi^N)$ correction. Substituting this feedback into the N HJB equations and matching the X_i^2, $X_i\bar{X}^N$, and $(\bar{X}^N)^2$ coefficients closes a finite, explicitly solvable system of scalar Riccati ODEs for (P^N, L^N, Π^N), in which every coupling term carries a bookkept power of $\frac{1}{N}$ tracking how much of the population an individual agent constitutes. The mean-field equations are now the $N \to \infty$ limit read off by dropping those $O(1/N)$ self-interaction terms. Drop them and the diagonal curvature decouples: P^N solves the *same* population-blind Riccati equation (14.7) that P does—the curvature is identical across this layer too, up to the $O(1/N)$ correction—so $P^N \to P$, while the offset combination r^N converges to the mean-field offset r of (14.8), each at rate $O(1/N)$. In the limit the all-ones block decouples from the diagonal and the symmetric feedback collapses onto the mean-field controller (14.10): a single agent best-responding to the deterministic flow $\bar{m}_t$ rather than to the fluctuating empirical mean.

The converse direction is the one that justifies the whole enterprise, and the LQG structure makes its rate sharp. Deploy the *mean-field* feedback (14.10)—computed once, ignorant of N—in the actual N-player game; the claim of chapter 13 is that this is an ε-Nash equilibrium, no agent able to gain more than ε by deviating. The generic theorem theorem 13.6 promises only $\varepsilon = O(r_N) = O(N^{-1/2})$ here, that being the fluctuation scale at which the empirical mean of N samples differs from the true mean $\bar{m}_t$ it is standing in for. But for the quadratic cost this generic rate is pessimistic, and the reason is an *envelope* (critical-point) identity worth spelling out, since it is easy to cite and hard to see. A deviating agent's potential gain is the gap between its cost under the mean-field feedback and its cost under the best reply to the realized empirical configuration. Both costs are evaluated at the *same* population perturbation, of size $\delta = O(N^{-1/2})$, the discrepancy between empirical and true mean. Now the mean-field feedback is the exact minimizer of the agent's cost against the true mean, so the first variation of that cost in the control vanishes there—that is the first-order optimality condition. When the population is perturbed by δ, re-optimizing the agent's own control would move it by $O(\delta)$, but because the cost is stationary in the control at the mean-field optimum, that $O(\delta)$ adjustment changes the value only at *second* order, $O(\delta^2)$. The first-order regret—the term linear in δ—is killed by the envelope identity, leaving the deviating agent's advantage at

$O(\delta^2) = O((N^{-1/2})^2) = O(1/N)$. The LQG worked computation worked computation 13.8 carries this cancellation out term by term and pins the constant, upgrading the gap to

$$\varepsilon = \varepsilon_N^{\mathrm{LQG}} \leq \frac{C}{N} \;=\; O(1/N) \tag{14.30}$$

by eq. (13.20)—one full order smaller than the generic bound. This is the quantitative justification, on the carried model, for the substitution at the heart of the whole subject: solving the single mean-field control problem in place of the intractable coupled system of N HJB equations costs an error of only $O(1/N)$, not the naive $O(N^{-1/2})$, precisely because the mean-field feedback sits at a critical point of each agent's cost. See theorem 13.6 and worked computation 13.8 and [82, 40].

The assembly the chapter preamble announced is now complete. Every layer of Part II has reappeared on the one model as an explicit object—the single-agent value and its Riccati curvature, the backward HJB paired with the forward Fokker–Planck, the FBSDE with its affine adjoint, the affine best-response map and its solvability scalar, the monotonicity defect and energy identity, the master equation with its second Riccati Q, and the N-player Riccati system with its $O(1/N)$ corrections—and each has been seen to view the same equilibrium and the same conjugate-point event at $T_c(s)$. What we have *not* yet examined is what, in all of this, ever truly used the state being a single scalar. The answer, as the next section observes, is almost nothing: every reduction above is an identity in linear algebra that survives verbatim when the scalar state becomes a vector and the scalars $(a, b, P, Q, r, \bar{m})$ become matrices and vectors—the one exception being the global solvability of the Riccati equation, whose comparison argument used the ordering of the real line and must be repaired matricially. The final section makes exactly that passage: it rewrites every formula of this chapter in the matrix LQG, repairs the Riccati theory in the multidimensional setting, and then performs the one substitution—replacing the population mean by the graphon-aggregated field $(\mathbb{W}\,\bar{m}_t)(x)$ indexed by type $x \in [0,1]$—that turns this entire single-model picture into the template Part IV generalizes. We turn to that lift now.

14.7 The matrix LQG, and the template for graphons

Matrix LQG. Nothing above used scalarity in an essential way, though the global theory of the Riccati equation does require a genuinely matricial argument, which we supply. For a state $X_t \in \mathbb{R}^d$ with dynamics $\mathrm{d}X_t = (AX_t + B\alpha_t)\,\mathrm{d}t + \Sigma\,\mathrm{d}W_t$ ($A \in \mathbb{R}^{d\times d}$, $B \in \mathbb{R}^{d\times k}$, $\Sigma \in \mathbb{R}^{d\times d'}$, W a d'-dimensional Brownian motion) and cost $\mathbb{E}\int_0^T (\frac{1}{2}\alpha_t^\top R\,\alpha_t + \frac{1}{2}(X_t - S\bar{m}_t)^\top Q(X_t - S\bar{m}_t))\,\mathrm{d}t + \dots$ with cost-weight matrices $R \succ 0$ and $Q \succeq 0$ (these uppercase Q, R are the constant multidimensional cost weights of this section, *not* the time-dependent scalar master-equation coefficients $Q(t), R(t)$ of (14.26)–(14.27)), the value is $u(t,x) = \frac{1}{2}x^\top P(t)x + r(t)^\top x + c(t)$ with the *matrix Riccati equation*

$$\dot{P} = PBR^{-1}B^\top P - A^\top P - PA - Q, \qquad P(T) = Q_T \succeq 0, \tag{14.31}$$

the offset $\dot{r} = (PBR^{-1}B^\top - A^\top)r + QS\bar{m}_t$ with terminal $r(T) = -Q_T S_T \bar{m}_T$, the optimal feedback $\alpha^* = -R^{-1}B^\top(Px + r)$, and the Gaussian flow with mean $\dot{\bar{m}} = (A - BR^{-1}B^\top P)\bar{m} - BR^{-1}B^\top r$ and covariance matrix $\mathbf{V}(t) \in \mathbb{S}^d$ solving the matrix Lyapunov equation

$$\dot{\mathbf{V}} = (A - BR^{-1}B^\top P)\mathbf{V} + \mathbf{V}(A - BR^{-1}B^\top P)^\top + \Sigma\Sigma^\top, \qquad \mathbf{V}(0) = \mathrm{Var}(m_0).$$

The BVP (14.15) becomes a $2d$-dimensional linear two-point problem and $D(T)$ a determinant; the monotonicity defect of lemma 14.8 becomes $-(\bar m - \bar m')^\top QS(\bar m - \bar m')$, monotone precisely when $QS + S^\top Q \preceq 0$, i.e. never for $S = I$, $Q \succ 0$, matching the scalar sign $qs \le 0$ once s is read as a coupling matrix. The one place where the scalar trapping argument of theorem 14.2 does *not* carry over verbatim is global solvability: the field of (14.31) is quadratic on the cone $\mathbb{S}^d$ of symmetric matrices and there is no scalar parabola to trap. The correct generalization is a matrix comparison argument, which we now give in full so that the multidimensional thread is as self-contained as the scalar one. (This discharges the forward reference of chapter 6, which proves the scalar case and defers the matrix Riccati global theory to this chapter.)

Theorem 14.13 (Global solvability of the matrix Riccati equation)**.** *Let $A \in \mathbb{R}^{d\times d}$, $B \in \mathbb{R}^{d\times k}$, $R \succ 0$, and $Q, Q_T \in \mathbb{S}^d$ with $Q \succeq 0$, $Q_T \succeq 0$. Then (14.31) has a unique solution $P \in C^1([0,T];\mathbb{S}^d)$ defined on the whole interval, and $0 \preceq P(t) \preceq \bar P(t)$ for all $t \in [0,T]$, where*

$$\bar P(t) = \Phi_A(T,t)^\top Q_T\,\Phi_A(T,t) + \int_t^T \Phi_A(\tau,t)^\top Q\,\Phi_A(\tau,t)\,\mathrm{d}\tau \tag{14.32}$$

is the cost-to-go of the zero-control problem, with $\Phi_A(\tau,t)$ the state-transition matrix of $\dot y = Ay$ (so $\partial_\tau \Phi_A(\tau,t) = A\,\Phi_A(\tau,t)$, $\Phi_A(t,t) = I$). In particular the closed-loop gain $A - BR^{-1}B^\top P(t)$ is a bounded continuous coefficient.

Proof. The right-hand side of (14.31) is a quadratic polynomial in P, hence locally Lipschitz on $\mathbb{S}^d$, so Picard–Lindelöf gives a unique maximal C^1 solution backward from $P(T) = Q_T$; it stays symmetric because the field maps $\mathbb{S}^d$ to $\mathbb{S}^d$ and the terminal datum is symmetric. Two comparisons bound it on its interval of existence.

Lower bound $P \succeq 0$. Set $K(t) = R^{-1}B^\top P(t)$ and let $\Phi_K(\tau,t)$ be the transition matrix of the closed loop $\dot y = (A - BK)y$. Differentiating

$$\widetilde P(t) \;:=\; \Phi_K(T,t)^\top Q_T\,\Phi_K(T,t) + \int_t^T \Phi_K(\tau,t)^\top\big(Q + K(\tau)^\top R\,K(\tau)\big)\Phi_K(\tau,t)\,\mathrm{d}\tau$$

shows it solves $-\dot{\widetilde P} = (A-BK)^\top \widetilde P + \widetilde P(A - BK) + Q + K^\top RK$, $\widetilde P(T) = Q_T$; expanding and substituting $K = R^{-1}B^\top P$ reduces this to (14.31), so by uniqueness $\widetilde P \equiv P$. As an integral of positive semidefinite matrices ($Q \succeq 0$, $K^\top RK \succeq 0$, $Q_T \succeq 0$), $P(t) = \widetilde P(t) \succeq 0$.

Upper bound $P \preceq \bar P$. The matrix $\bar P$ of (14.32) solves the Lyapunov equation $-\dot{\bar P} = A^\top \bar P + \bar P A + Q$, $\bar P(T) = Q_T$ (differentiate (14.32)). Hence the gap $\Delta = \bar P - P$ satisfies $-\dot\Delta = A^\top\Delta + \Delta A + PBR^{-1}B^\top P$, $\Delta(T) = 0$, whose integrating-factor representation $\Delta(t) = \int_t^T \Phi_A(\tau,t)^\top\big(PBR^{-1}B^\top P\big)(\tau)\,\Phi_A(\tau,t)\,\mathrm{d}\tau$ is again an integral of PSD matrices, so $\Delta(t) \succeq 0$, i.e. $P(t) \preceq \bar P(t)$.

Thus $0 \preceq P(t) \preceq \bar P(t)$ wherever P exists, and $\bar P$ is continuous, hence bounded, on $[0,T]$. A maximal solution of an ODE that stays in a bounded set cannot blow up in finite time, so the maximal interval is all of $[0,T]$, and P is global. Boundedness of P makes $A - BR^{-1}B^\top P$ a bounded continuous coefficient, so the offset and mean equations are linear with bounded coefficients exactly as in the scalar case. ∎

With theorem 14.13 in hand, every layer of this chapter generalizes to the matrix setting with scalars replaced by matrices, sign conditions by definiteness, and the scalar trapping of theorem 14.2 by the comparison (14.32).

Template for graphons (Thread B). The graphon mean field game of Part IV is obtained from this chapter by one substitution: the scalar population mean $\bar{m}_t$ is replaced by the *graphon-aggregated neighbor mean*

$$z_t(x) = (\mathbb{W}\,\bar{m}_t)(x) = \int_0^1 W(x,y)\,\bar{m}_t(y)\,\mathrm{d}y,$$

where $\mathbb{W}$ is the graphon integral operator constructed in chapter 15 (with the underlying operator theory in chapter 2), and now there is a continuum of representative agents indexed by type $x \in [0,1]$, each with its own state X_t^x and its own value function u^x (this is Thread B, whose dynamic development occupies chapter 20 and chapter 22 under (A-Reg-W)). The cost couples a type-x agent to $z_t(x)$ in place of $\bar{m}_t$. The HJB ansatz becomes type-indexed, $u^x(t,x) = \frac{1}{2}P(t)x^2 + r^x(t)x + c^x(t)$, and the curvature Riccati (14.7) is *the same for every type* (it does not see the coupling), while the offset equation (14.8) becomes a family coupled across types through $\mathbb{W}$: $\dot{r}^x = (b^2P - a)r^x + qs\,(\mathbb{W}\bar{m}_t)(x)$. The two reductions promised in the thread specification are now transparent. A constant kernel $W \equiv c$ makes $(\mathbb{W}\bar{m}_t)(x) = c\int_0^1 \bar{m}_t(y)\,\mathrm{d}y$ independent of x and recovers Thread A with rescaled coupling $s \mapsto cs$. A rank-one kernel $W(x,y) = f(x)f(y)$ gives $(\mathbb{W}\bar{m}_t)(x) = f(x)\cdot\varrho_t$ with the *scalar order parameter* $\varrho_t = \int_0^1 f(y)\bar{m}_t(y)\,\mathrm{d}y$ (the canonical symbol $\varrho = \varrho$ of the thread specification, inherited unchanged by chapters 15, 20 and 22), collapsing the entire type-indexed family back to a one-parameter self-consistency equation for ϱ_t—the cleanest graphon worked example, and the direct descendant of the present chapter's scalar BVP. The phase boundary of theorem 14.11 then reappears with s rescaled by the leading eigenvalue of $\mathbb{W}$, which is the precise sense in which the operator norm $\|\mathbb{W}\|$ and the smallness condition (A-Coupling) control uniqueness in Part IV; the threshold $s = 1$ here becomes a threshold on $s\,\|\mathbb{W}\|$ there. This is the template the capstone exists to provide.

Remark 14.14 (Restoring a discount rate)**.** If a discount $\beta = \beta > 0$ is included, with running cost $e^{-\beta t}(\frac{1}{2}\alpha^2 + F)$ or equivalently an infinite-horizon stationary problem, the Riccati equation (14.7) acquires a term $+\beta P$ and, in the stationary limit $T \to \infty$, becomes the algebraic Riccati equation $b^2P^2 - (2a - \beta)P - q = 0$ with the stabilizing root selected. The mean–offset system (14.15) becomes autonomous and the phase analysis of section 14.5 is read off the eigenvalues of the constant matrix M directly; the discount shifts the threshold but does not change the qualitative picture. We keep $\beta = 0$ in the carried thread for definiteness, as fixed in the thread specification.

Rigor ledger

Proved.

- Global solvability and nonnegativity of the scalar Riccati equation under $q, q_T \geq 0$ (theorem 14.2), and of the matrix Riccati equation under $Q, Q_T \succeq 0$, $R \succ 0$ by a symmetric-matrix comparison argument (theorem 14.13, discharging the forward reference of chapter 6); the closed-loop population is an exactly Gaussian flow with decoupled mean and variance ODEs (theorem 14.4).

- Reduction of the LQG equilibrium to the linear two-point BVP (14.15); existence and uniqueness iff the scalar solvability condition $D(T) \neq 0$ holds (lemma 14.5, theorem 14.6).
- The LQG Lasry–Lions monotonicity defect equals $-qs(\bar{m} - \bar{m}')^2$, so (A-Mon) holds iff $s \leq 0$, $s_T \leq 0$ (lemma 14.8); in that regime the equilibrium is unique for every horizon (theorem 14.9).
- A closed-form phase boundary: in the normalization of worked computation 14.10, uniqueness for all T when $s \leq 1$; for $s > 1$ uniqueness for all $T < T_c(s) = \mu^{-1}(\pi - \arctan(\mu/\omega))$, $\mu = \omega\sqrt{s-1}$, with T_c the first conjugate horizon where uniqueness is lost and the discrete successive zeros of D thereafter (theorem 14.11); equivalently, the master-equation coefficient Q of (14.27) first blows up at T_c.

Assumed.

- (A-Nondeg) via $\sigma > 0$ for the Gaussian-density statements; (A-Conv) automatically from the convex quadratic control cost; (A-Growth) coercivity in place of the bounded-Lipschitz cost hypothesis of the generic existence theorem. The linear-quadratic verification theorem of chapter 6 is used without reproof.

Deferred.

- The generic existence (chapter 10), uniqueness/monotonicity (chapter 11), and master equation (chapter 12); here instantiated, not reproved. The N-to-mean-field convergence is imported from chapter 13: the generic ε-Nash bound is $O(N^{-1/2})$ (theorem 13.6), but for LQG the envelope cancellation sharpens the equilibrium gap to $O(1/N)$ (worked computation 13.8, eq. (13.20)), the rate quoted in (14.30).
- The bifurcated multiplicity picture beyond the linear conjugate surfaces (terminal coupling and nonlinear couplings), flagged in remark 14.12 and treated in chapter 11.

Open.

- Sharp characterization of the open multiplicity regions for non-quadratic congestion couplings (no closed form analogous to (14.24) is known); the general non-monotone uniqueness threshold remains a research question (chapter 11, and forward to the graphon case chapter 24).

Bibliographic notes

The linear–quadratic–Gaussian mean field game is the model on which the subject was first made fully explicit. The Nash certainty equivalence formulation and the LQG equilibrium with its decoupled Riccati system are due to Huang, Caines, and Malhamé [82]; the same circle of ideas appears, from the analytic side, in Lasry and Lions [93] and in Cardaliaguet's lecture notes [36]. The probabilistic (FBSDE/decoupling-field) treatment of LQG, including the affine decoupling field used in section 14.6, is developed systematically in Carmona and Delarue [40]; the master-equation reduction (14.26)–(14.27) and the identification of conjugate points with blow-up of the decoupling field follow the master equation theory of [37, 41]. The Lasry–Lions monotonicity computation of lemma 14.8 and the resulting uniqueness theorem are the LQG specialization of [93] as presented in [36]; the reading of multiplicity as a symmetry-breaking/conjugate-point phenomenon is standard in the calculus of variations and is the deterministic shadow of the bifurcation analysis surveyed in [40]. The finite-player convergence and ε-Nash statements specialize [40, 37]. The graphon template of section 14.7 anticipates [34, 108]; the rank-one collapse to a scalar order parameter is the mean-field-game counterpart of the Curie–Weiss closure and is carried out in chapter 20 and chapter 22. Numerical confirmation of the closed-form Riccati answer by a forward–backward solver is the subject of chapter 28, following [1].

Part III

Graphons and Graph Limits

Chapter 15

From Adjacency Matrices to Graphons

This chapter opens Part III and changes the subject. Parts I and II built mean field games on the premise of *homogeneity*: every agent is exchangeable, every agent feels the same population through the same anonymous average, and the "mean field" is a single point $m \in \mathcal{P}_2(\mathbb{R}^d)$ (appendix A.2). Real interacting populations are not like that. An agent in a social, economic, or epidemiological network feels a *weighted* aggregate of *specific* neighbors, and which neighbors matter is fixed by a graph. The program of Part III is to take the $N \to \infty$ limit not of an anonymous crowd but of a structured network, and to find the object that survives that limit. That object is the *graphon*, and this chapter constructs it from the most elementary starting point—an adjacency matrix viewed as a black-and-white image.

We assume very little from earlier chapters here, and what we assume is foundational. From chapter 1 we take Lebesgue measure on $[0, 1]$ and on $[0, 1]^2$, the Lebesgue spaces L^p and their completeness (definition 1.29 and theorem 1.31), and the Fubini–Tonelli theorem (theorem 1.27), which is what licenses the double integrals $\int\int W(x, y)\,(\cdots)\,\mathrm{d}x\,\mathrm{d}y$ that pervade the subject. From chapter 2 we take the geometry of the Hilbert space $L^2[0, 1]$ and—crucially—the fact, already proved there, that the integral operator of a bounded symmetric kernel is a self-adjoint Hilbert–Schmidt operator with a clean spectral theory (definition 2.23, proposition 2.24, theorem 2.26, and worked computation 2.28). That the operator side of the story is already in place is deliberate: it lets this chapter concentrate on the *combinatorial* and *probabilistic* content—how a finite network becomes a kernel—and import the functional analysis when we need it.

The chapter is organized as a single passage from the discrete to the continuum, each stop forced by the one before. We begin in section 15.1 with the pixel picture: an $N \times N$ adjacency matrix read literally as a black-and-white image on the unit square—a $\{0, 1\}$-valued step function whose grid we refine as the network grows. That picture turns out to carry two blemishes, an artefactual grid and an ambiguous labeling, and section 15.2 distils what survives once both are discarded into the definition of a graphon, fixing along the way the type-label reading of the variable $x \in [0, 1]$ with a strong notation watch and restating the Brownian-motion-versus-graphon letter clash that governs all of Part III. To make the refining-grid passage precise as far as is honest at this stage, section 15.3 isolates the one analytic operation the limit rests on—block-averaging, or *stepping*—and shows it is an orthogonal projection, while pointing to where the genuine convergence theorems live (chapters 16 and 17). With the object and its coarse-graining in hand, section 15.4 is a gallery of the graphons the rest of the book leans on—the Erdős–Rényi constant kernel, the stochastic block models, and the rank-one "Curie–Weiss" kernel $W(x, y) = f(x)f(y)$, which carries Thread B and which we develop in a worked computation. Section 15.5 then asks whether node- and edge-weighted generalizations enlarge the theory and finds that, in the dense bounded regime, they do not—they are absorbed by relabeling. Finally section 15.6 returns to the integral operator $\mathbb{W}$, now read as the coupling that will appear in *every* graphon mean field game equation, and establishes the elementary bounds—degree function, Schur estimate, positivity—we use repeatedly in Part IV.

We do not, in this chapter, develop the topology of graphon space or prove that graph sequences converge; that is the business of chapter 16 (homomorphism densities) and chapter 17 (the cut metric and the Lovász–Szegedy compactness theorem). What we do is build the object and learn to recognize it.

Notation watch

The type variable x is not space and not momentum. From here to the end of the book the variable $x \in [0, 1]$ is the *type* or *label* of an agent: an exchangeable index of its kind or its position *in the network*, and nothing more. It is not a spatial coordinate, not a lattice site, and not a momentum. The kernel $W(x, y)$ will often look like a translation-noninvariant hopping amplitude between "sites" x and y, and a reader with a condensed-matter reflex will be tempted to read $|x - y|$ as a distance. Resist this: the interval $[0, 1]$ carries no metric of its own that W respects, and indeed chapter 17 will show that relabeling the types by any measure-preserving map leaves the network unchanged. The number $|x - y|$ has no intrinsic meaning. When a formula invites the spatial reading, we write "type" explicitly. This restates the standing convention of appendix A.5.

Notation watch

Brownian motion and the graphon both want the letter W. In Parts I–II, where no graphon appears, Brownian motion was W_t (rendered W_t), following the standard stochastic-analysis usage of chapter 5. From this chapter onward the letter W is *reserved for the graphon* $W = W(x, y)$ and its operator $\mathbb{W} = \mathbb{W}$. Brownian motion, when it reappears in the dynamic theory of Part IV, is written B_t. Thus in any equation that contains a graphon, W is the kernel and B is the noise; there is no ambiguity because the two never compete for the symbol. This restates the convention of appendix A.5 and is announced once more at the start of Part IV. No Brownian motion appears in the present chapter, which is entirely static.

15.1 The pixel picture

The preamble promised to build the graphon from the most elementary starting point—an adjacency matrix viewed as a black-and-white image—while keeping the variable $x \in [0, 1]$ in its role as a *type* or *label* rather than a point of space. We now make good on both promises with a single move from the discrete to the continuum: take a finite labeled graph, lay its adjacency matrix flat on the unit square as a literal picture, and watch what the picture does as the graph grows. The starting point is therefore a finite network and its adjacency matrix. Before anything else we fix terminology, since the word "graph" is used loosely across fields and the entire construction depends on which object we mean.

Definition 15.1 (Finite graph and adjacency matrix)**.** A *finite simple graph* G is a pair (V, E) with vertex set $V = \{1, 2, \ldots, n\}$ and edge set E a set of unordered pairs of distinct vertices. Its *adjacency matrix* is the symmetric $n \times n$ matrix $A = A(G)$ with entries

$$A_{ij} = \begin{cases} 1 & \text{if } \{i, j\} \in E, \\ 0 & \text{otherwise,} \end{cases} \qquad A_{ii} = 0.$$

The number $n = |V|$ is the *order* of G. A *weighted graph* replaces $\{0, 1\}$ by allowed entries in $[0, 1]$ (an edge *probability* or *intensity*) or in $\mathbb{R}$ (a signed *edge weight*); the matrix remains symmetric with zero diagonal unless loops are explicitly admitted.

The adjacency matrix is a complete description of G—but only *once a labeling of the vertices is chosen.* The integers $1, \ldots, n$ are bookkeeping; nothing intrinsic distinguishes vertex 3 from vertex 7, and a different clerk recording the same network would in general write down a different matrix. Concretely, a relabeling is a permutation $\pi \in S_n$ of $\{1, \ldots, n\}$, and it produces the conjugated matrix $A^{\pi}_{ij} = A_{\pi(i)\pi(j)}$, which describes the *same* graph in the new coordinates. Take the path $1-2-3$ with edges $\{1, 2\}, \{2, 3\}$: its adjacency matrix has the central vertex 2 joined to both ends. Relabel by swapping the names of 1 and 2 and the matrix changes—the entry A_{12} that was 1 now sits in a different place—yet the object on the page is still a path on three vertices. What does *not* change is anything genuinely about the network: the degree sequence $(1, 2, 1)$, the number of edges, whether the graph is connected. A single matrix entry such as A_{12} is meaningful only relative to a labeling; a graph *invariant* is, by definition, a function of the whole conjugacy class $\{A^{\pi} : \pi \in S_n\}$ rather than of A itself.

This relabeling freedom is not a nuisance to be suppressed; it is the central structural fact of the whole theory. The reason is that the limit object we are chasing inherits it intact. Just as a finite network is an adjacency matrix *up to vertex permutation*, the graphon will turn out to be a kernel *up to a measure-preserving relabeling of* $[0, 1]$—the continuum analogue of the symmetric group S_n, with permutations of n blocks replaced by measure-preserving maps of the interval (the precise statement is remark 15.5, and the full quotient theory, built on the cut metric, is the business of chapter 17). Because of this every quantity we construct in the chapters to come—edge density, spectrum, homomorphism density, and ultimately the graphon mean field equilibrium itself—will be engineered to be invariant under relabeling, for otherwise it could not be a property of "the network" at all. For now we simply record the demand: any statement we make must survive $A \mapsto A^{\pi}$.

To take a continuum limit we must put the discrete index $i \in \{1, \ldots, n\}$ onto a continuum. The device is to slice the unit interval into n equal pieces and read the matrix as a picture.

Definition 15.2 (Pixel graphon of a graph)**.** Let G be a (possibly weighted) graph of order n with symmetric matrix $A \in [0, 1]^{n \times n}$. Partition $[0, 1]$ into the n half-open intervals

$$I_i^{(n)} = \left(\tfrac{i-1}{n}, \tfrac{i}{n}\right], \qquad i = 1, \ldots, n$$

(with $I_1^{(n)}$ taken to include 0). The *pixel graphon* (or *empirical graphon*) of G is the function $W^G\colon [0,1]^2 \to [0,1]$,

$$W^G(x,y) = A_{\iota_n(x)\,\iota_n(y)}, \qquad \iota_n(x) = \max(\lceil nx \rceil, 1) \in \{1,\dots,n\},$$

that is, $W^G(x,y) = A_{ij}$ whenever $x \in I_i^{(n)}$ and $y \in I_j^{(n)}$.

The map ι_n is nothing more than "which subinterval does x fall into," read from the inside out: nx rescales $[0,1]$ to $[0,n]$, the ceiling $\lceil nx \rceil$ rounds up to the next integer, and the outer $\max(\,\cdot\,,1)$ exists only to repair the single point $x=0$, sending it to interval 1 instead of to the empty index 0. Thus $x \in I_i^{(n)} = (\frac{i-1}{n}, \frac{i}{n}]$ yields $\iota_n(x) = i$, and the kernel W^G merely looks up the matrix entry A_{ij} of the two blocks that x and y land in. In particular ι_n is constant on each interval, so W^G is constant on each cell—a fact we use at once.

The name is exact: if one draws the unit square, divides it into an $n \times n$ grid of congruent cells $I_i^{(n)} \times I_j^{(n)}$, and shades cell (i,j) black when $A_{ij} = 1$ and white when $A_{ij} = 0$, the resulting image *is* the graph W^G of the function W^G. The adjacency matrix has become a black-and-white picture on the unit square, and a weighted graph becomes a greyscale picture valued in $[0,1]$. The diagonal cells, where $A_{ii} = 0$, form a vanishingly thin white band that will play no measure-theoretic role.

Proposition 15.3 (A pixel graphon is a symmetric measurable kernel)**.** *For any graph G of order n, the pixel graphon W^G is a symmetric, measurable function* $[0,1]^2 \to [0,1]$*; if G is unweighted then W^G is* $\{0,1\}$*-valued. Its integral over the square is the average matrix entry,*

$$\int_{[0,1]^2} W^G(x,y)\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{n^2}\sum_{i,j=1}^{n} A_{ij},$$

which for an unweighted graph equals the edge density $2\,|E|\,/n^2$.

Proof. Consider measurability first. By construction W^G takes the constant value A_{ij} on the rectangle $I_i^{(n)} \times I_j^{(n)}$, and these n^2 rectangles—each a product of two half-open intervals, hence a measurable subset of $[0,1]^2$—partition the square. For any value c, the preimage $\left\{W^G = c\right\}$ is the union of those finitely many cells on which $A_{ij} = c$, a finite union of measurable sets and therefore measurable. A function that takes only finitely many values, each on a measurable set, is by definition a (measurable) simple function, so W^G is measurable. Symmetry is inherited entrywise: for any (x,y),

$$W^G(x,y) = A_{\iota_n(x)\,\iota_n(y)} = A_{\iota_n(y)\,\iota_n(x)} = W^G(y,x),$$

the middle equality being the matrix symmetry $A_{ij} = A_{ji}$. The range of W^G is exactly the set of matrix entries, which is contained in $[0,1]$ for a weighted graph and equals $\{0,1\}$ in the unweighted case.

For the integral, the integral of a simple function is the sum of its values weighted by the measures of the sets on which they are attained (eq. (1.4)). Each cell $I_i^{(n)} \times I_j^{(n)}$ is a product of two intervals of length $1/n$ and so has area $1/n^2$; summing over the n^2 cells gives

$$\int_{[0,1]^2} W^G(x,y)\,\mathrm{d}x\,\mathrm{d}y = \sum_{i,j=1}^{n} A_{ij}\,|I_i^{(n)} \times I_j^{(n)}| = \sum_{i,j=1}^{n} A_{ij}\cdot\frac{1}{n^2} = \frac{1}{n^2}\sum_{i,j=1}^{n} A_{ij}.$$

For an unweighted graph this double sum counts each edge twice: an edge $\{i, j\} \in E$ has $i \neq j$ and contributes the two symmetric entries $A_{ij} = A_{ji} = 1$, while the diagonal contributes nothing because $A_{ii} = 0$. Hence $\sum_{i,j} A_{ij} = 2\,|E|$ and the integral is $2\,|E|\,/n^2$. □

The number $2\,|E|\,/n^2$, the *edge density*, deserves a name now, because it is the first and simplest of the graph invariants that survive the limit. It is manifestly relabeling-invariant—$\sum_{ij} A^{\pi}_{ij} = \sum_{ij} A_{ij}$, since permuting the indices only reorders the terms of the sum—so, unlike a bare entry A_{12}, it is a genuine property of the network and passes the test demanded above. Three readings of this one scalar foreshadow three strands of the book. Combinatorially it is the edge count $|E|$ normalized by n^2. Probabilistically it is the chance that a uniformly random ordered pair of vertices (sampled with replacement) happens to be joined by an edge; in the language of chapter 16 this is the homomorphism density $t(K_2, G)$ of a single edge K_2, the network's most basic "moment," and the entire convergence theory of chapter 16 is built by promoting this one count to the family of counts of all small subgraphs. Analytically—anticipating section 15.6—it is the quadratic form $\langle \mathbb{W}^G \mathbf{1}, \mathbf{1} \rangle = \int_{[0,1]^2} W^G$, the response of the constant unit field $\mathbf{1}$ to the coupling operator. That a single number can be computed as an edge count, as a homomorphism density, and as a quadratic form of $\mathbb{W}$ is the first small instance of the chapter's main lesson: the network reaches everything downstream only through the operator $\mathbb{W}$.

Refining the grid. A single graph gives a single picture at resolution n. The limit object appears when we let the network *grow*. Consider a sequence $(G_n)_{n \geq 1}$ of graphs with $|V(G_n)| \to \infty$ and form the pixel graphons W^{G_n}. Each is a step function on a finer and finer grid. The qualitative expectation—made rigorous only in chapter 17—is that, after the relabeling ambiguity is quotiented out, the sequence of pictures settles down to a limiting greyscale image W on the unit square. That limiting image is the graphon. The following heuristic states the picture; the honest mathematics behind it is deferred, and we are explicit about the deferral.

Heuristic (non-rigorous): The refining-grid limit

Imagine a growing sequence of networks whose macroscopic wiring stabilizes: the fraction of edges between "the first third" and "the last quarter" of the vertices (in some consistent labeling) converges, and likewise for every pair of macroscopic blocks. Then the pixel pictures W^{G_n}, viewed at any fixed coarse resolution, stop changing. Sending the coarse resolution to infinity after the network size, one expects a limiting kernel $W\colon [0,1]^2 \to [0,1]$ whose block averages reproduce all these limiting frequencies. The two obstructions to turning this into a theorem are (i) the labeling: "the first third of the vertices" is not invariant, so the limit can only exist *up to a measure-preserving relabeling of* $[0, 1]$; and (ii) the topology: "the pictures stop changing" must be measured in a norm coarse enough to ignore microscopic reshuffling but fine enough to be a metric; the obvious candidate, the L^2 distance between the pixel pictures, turns out to be too fine—it never forgets the microscopic $0/1$ texture, a concrete failure we exhibit in remark 15.8. Both obstructions are resolved by the cut metric $\delta_\square$ of chapter 17, and the existence of the limit for *every* bounded sequence is the Lovász–Szegedy compactness theorem (chapter 17). Until then we treat "$W^{G_n} \to W$" as a picture, not a proof.

The remainder of this chapter does not depend on section 15.1: we will define the graphon as an object in its own right and study the ones we need by direct computation, leaving the convergence theory to the chapters built for it. But the heuristic has done its job of pointing at the target. We now hold a completely concrete object—the pixel graphon W^G, a symmetric measurable step function $[0,1]^2 \to [0,1]$ whose integral is the edge density—and we have located its only two defects, both of them artefacts of how we drew the picture rather than features of the network: the $n \times n$ *grid*, an accident of the chosen resolution, and the *labeling*, an accident of the chosen vertex names. The next section performs exactly the distillation these two defects demand. It keeps the three properties that are real—symmetry, measurability, and the value range $[0,1]$—discards the grid, and remembers that the labeling is to be quotiented away, arriving at the definition of a graphon and recording that this combinatorial construction agrees on the nose with the operator-side definition of definition 2.23.

15.2 The graphon

The previous section left us holding a completely concrete object together with a precise instruction for what to do with it. The object is the pixel graphon W^G of proposition 15.3: a symmetric, measurable step function $[0,1]^2 \to [0,1]$, the adjacency matrix laid flat as a black-and-white picture. The instruction was the distillation that closed section 15.1: of the properties this picture carries, keep the three that are genuine features of the network—*symmetry*, *measurability*, and the *value range* $[0,1]$—and discard the two that are artefacts of how we happened to draw it, the $n \times n$ grid and the chosen labeling. The grid we throw away now, simply by no longer demanding that the kernel be piecewise constant; the labeling we cannot throw away by fiat, so we carry it as a quotient to be imposed, and the closing remark of this section makes that quotient precise. What is left when the grid is gone is the definition of a graphon.

It pays to be explicit about why exactly those three properties survive and nothing else does. Symmetry survived the proof of proposition 15.3 as the entrywise identity $A_{ij} = A_{ji}$, the algebraic fingerprint of an undirected edge. Measurability survived because a function constant on finitely many rectangles is a simple function, and simple functions are the building blocks of the integral—without it the double integrals $\iint W$ that compute every quantity we care about would be meaningless. The range $[0,1]$ survived because each value is a matrix entry, an edge indicator or probability. The grid did *not* survive: it was the resolution n of one particular picture, and the limit we are chasing is precisely what is left after the resolution is sent to infinity. So we keep the three and drop the grid by allowing *any* symmetric measurable kernel, not only the piecewise-constant ones. This is the residue of proposition 15.3, and it is also—no coincidence, since the operator side was designed to receive it—word for word the kernel already introduced in definition 2.23. We restate it here in this chapter's combinatorial spirit, as the central object of Part III.

Definition 15.4 (Graphon)**.** A *graphon* is a symmetric measurable function

$$W\colon [0,1]^2 \longrightarrow [0,1], \qquad W(x,y) = W(y,x) \ \text{ for a.e. } (x,y).$$

The value $W(x, y)$ is interpreted as the *probability (or intensity) of a link between an agent of type x and an agent of type y.* We write $\mathcal{W}_0$ for the set of all such graphons. This is exactly the symmetric, bounded, measurable kernel of assumption (A-Reg-W) (appendix A.6), and it agrees with definition 2.23.

The definition is short, and that brevity hides three reading conventions that one must internalize before the object behaves the way the rest of the book assumes. None of the three is a new hypothesis; each is a clarification of how to interpret the data already present in definition 15.4. We take them in turn.

The reference measure is Lebesgue, the "uniform distribution of types." Nothing in the symbol $W : [0, 1]^2 \to [0, 1]$ names a measure, yet the moment we integrate—and we only ever use W by integrating—a measure on the type space $[0, 1]$ is in play, and the convention is that it is Lebesgue measure. Reading $x \in [0, 1]$ as the type of an agent, this says the types are spread *uniformly*: a fraction $|B|$ of the population has type in any measurable set $B \subseteq [0, 1]$. The interpretation of the kernel follows by sampling. Draw an agent's type x uniformly at random; then the slice $y \mapsto W(x, y)$ is precisely the *connection density* of that agent across the population—the probability, per unit of type-mass, that it links to a partner of type y—and its total mass $\int_0^1 W(x, y)\, \mathrm{d}y$ is the expected fraction of the population the agent connects to, the continuum degree we will name in section 15.6. One should resist hearing "uniform" as a modeling restriction: it is a *normalization*, not a loss of generality. Real data come with a non-uniform distribution of types—some kinds of agent are common, others rare—and section 15.5 shows that such node weights are absorbed into the kernel itself, by re-coordinatizing the type space so that the weighting becomes uniform again; chapter 17 identifies the freedom in that re-coordinatization with the relabeling freedom of remark 15.5 below. Uniformity is the gauge we fix, not a feature of the world.

Equality of graphons is almost-everywhere equality. The symmetry condition in definition 15.4 already reads "for a.e. (x, y)," and that qualifier is not a fussy technicality but the correct notion of identity for the whole theory. The reason is uniform across the book: W never enters a single formula except through an integral—through the operator $\mathbb{W}$ (definition 2.23), through the homomorphism densities of chapter 16, through the cut norm of chapter 17—and an integral cannot distinguish two kernels that differ only on a Lebesgue-null set. Two graphons that agree almost everywhere therefore produce identical operators, identical densities, identical equilibria; they are, for every purpose here, the *same* graphon. Accordingly we speak of "the" graphon while quietly working with its equivalence class in $L^2([0, 1]^2)$ or $L^\infty([0, 1]^2)$. The most useful instance of this licence is the diagonal: the set $\{x = y\}$ is null in $[0, 1]^2$, so the value of W along it—whether we record self-links $W(x, x)$ or set them to zero in imitation of the zero diagonal of an adjacency matrix—changes no integral and hence changes nothing. We will exploit this in section 15.5 to dispose of self-loops in one line; for now the point is only that the diagonal never matters.

Symmetry is the continuum imprint of undirected edges, and it is what makes $\mathbb{W}$ self-adjoint. The condition $W(x, y) = W(y, x)$ is the limit of the matrix identity $A_{ij} = A_{ji}$, which encoded that an edge between i and j is the same edge as the one between j and i—there is no orientation. In the continuum this single equation buys a great deal of structure. It is exactly the condition under which the integral operator $\mathbb{W}$ of definition 2.23 is *self-adjoint* on $L^2[0, 1]$ (proposition 2.24): for any test fields g, h, $\langle \mathbb{W}g, h\rangle = \int\int W(x,y)g(y)h(x)\, \mathrm{d}y\, \mathrm{d}x$, and swapping the names of the dummy variables together with the symmetry of W turns this into $\langle g, \mathbb{W}h\rangle$. Self-adjointness is not decoration: it is what guarantees, via the spectral theorem already proved in theorem 2.26, that $\mathbb{W}$ has a *real* spectrum and an orthonormal basis of eigenfunctions—the "modes" of the network that organize all of chapter 18 and through which the equilibria of Part IV are diagonalized. Drop symmetry and the edges acquire an orientation: one has a *directed* network, a non-self-adjoint operator with a possibly complex spectrum, and none of the dense undirected machinery of this book applies. That entire theory sits outside our scope precisely because we keep this one equation.

The two notation watches at the head of the chapter already govern the variable x and the letter W, and we do not repeat them. What remains is the single structural point that the rest of Part III is built to formalize, and it is the most important conceptual statement of the chapter: the labeling we deferred when discarding the grid must now be quotiented away, and doing so is what turns "a kernel" into "a network."

Remark 15.5 (The network is the graphon *up to relabeling*)**.** A finite graph and its vertex relabelings $A \mapsto A^\pi$ all describe one and the same network; their pixel graphons W^G are *distinct* functions on the square, related by permuting the blocks $I_i^{(n)}$ that the matrix indices were laid into. Passing to the continuum, ask what plays the role of the relabeling $\pi \in S_n$. A permutation of n equal blocks shuffles the interval $[0,1]$ while preserving the length of every block, hence preserving Lebesgue measure; its continuum analogue is therefore a *measure-preserving bijection* $\varphi\colon [0,1] \to [0,1]$, the smallest closure of the finite permutations that still respects the uniform type distribution fixed above. (One should picture the chain $\pi \in S_n \rightsquigarrow$ "rename which type sits at which point of $[0,1]$, keeping the amount of each type fixed"; the limiting object is exactly a measure-preserving map.) Such a φ acts on a graphon by relabeling both arguments,

$$W^\varphi(x,y) = W(\varphi(x), \varphi(y)),$$

the precise continuum image of $A^\pi_{ij} = A_{\pi(i)\pi(j)}$. Two graphons related by such a φ—or, more generally, lying at *cut distance zero*, which is the fully general statement and the one chapter 17 proves to be the right notion—encode the same network and must therefore yield the same edge density, the same spectrum, the same homomorphism densities, and the same graphon mean field equilibrium. The honest object of the theory is consequently not a single representative $W \in \mathcal{W}_0$ but its entire relabeling class; the cut metric of chapter 17 is built precisely to be a genuine distance on these classes.

This imposes a working discipline that we adopt from here on. We will always compute with some chosen representative W, because one must write *a* function down to integrate it, but every quantity we are entitled to call a property of "the network" must come out the same for W and for W^φ—it must be φ-invariant. The edge density passes this test (relabeling only reshuffles the mass under the integral); the spectrum passes it ($\mathbb{W}$ and $\mathbb{W}^\varphi$ are unitarily equivalent); a bare pointwise value does *not*. The cautionary micro-example to keep in mind is the value at a single point: "W at $x = \frac{1}{2}$" is not a property of the network at all, because a measure-preserving φ can carry $\frac{1}{2}$ to any other point while leaving the network untouched, and in any case $\left\{x = \frac{1}{2}\right\}$ is null and may be edited freely by the previous remark. The rule, stated once and used throughout, is: *treat any feature of a chosen representative that is not φ-invariant as meaningless.* It is the same demand we recorded for finite graphs in section 15.1—"every statement must survive $A \mapsto A^{\pi}$"—now lifted to the continuum.

This relabeling demand is not only a discipline on what we may compute; it also shapes what the next section must build. We have a clean abstract graphon—a symmetric measurable kernel read up to measure-preserving relabeling—but we have not yet honored the promise of section 15.1, that pixel pictures W^{G_n} of a growing network actually converge to such a kernel. That convergence cannot be proved here, and the reason is now sharp: because the network is the graphon only up to φ, "the pictures stop changing" can only be measured after the labeling has been quotiented away, which forces a coarse-resolution comparison rather than a pointwise one. What we *can* do honestly at this stage is isolate and analyze the single operation that such a comparison rests on—viewing a kernel at a coarse resolution by averaging it over blocks, an operation that must, to be admissible, respect the block structure rather than the individual labels. That operation is the stepping operator, and we turn to it now.

15.3 From pixel pictures to graphons: the stepping operator

The previous section closed by converting an open-ended convergence question into a single sharp demand. Because the network is the graphon only *up to* a measure-preserving relabeling φ (remark 15.5), the statement "the pixel pictures W^{G_n} stop changing" cannot be read pointwise: a pointwise comparison sees the labels, and the labels are exactly the data we are not entitled to see. It must be read at a *coarse resolution*—compare two pictures only after each has been averaged over a common family of patches, so that reshuffling labels *within* a patch is invisible while the gross between-patch frequencies are still compared. We cannot yet prove that the patch-averaged pictures converge; that is the cut metric of chapter 17 and the compactness theorem behind it. What we *can* do, fully and honestly, is isolate the one operation that "average over patches" names, define it, and understand it completely. That operation is *stepping*, and the pleasant surprise is that it is nothing more exotic than orthogonal projection in the Hilbert space $L^2([0,1]^2)$ whose geometry chapter 2 already built.

Stepping earns this section for three reasons beyond the convergence question it clarifies. It is the rigorous form of "view the picture at a coarse resolution"; it is the exact bridge between general graphons and the *step graphons*—the stochastic block models—we are about to catalogue in section 15.4, since stepping a graphon *produces* a step graphon; and it is the analytic heart of the weak regularity lemma of chapter 19, the theorem that a *bounded* number of patches already suffices. We develop it now as a self-contained piece of L^2 geometry and harvest the consequences immediately.

To average "over patches" we must first say what a patch is, and then say what a function looks like once it can no longer resolve anything finer than a patch. A patch is a block of types: we cut $[0,1]$ into finitely many measurable pieces $P_1,\ldots,P_k$, and a *rectangle* $P_i \times P_j$ of the square is the set of type-pairs (x,y) with x in block i and y in block j. A kernel that is constant on each such rectangle is one that *cannot tell two types apart once they share a block*—it resolves the type space only down to the partition $\mathcal{P}$. These are the step graphons, and the set of all of them (symmetric or not) is a linear subspace $\mathcal{S}_{\mathcal{P}}$ of $L^2([0,1]^2)$: a sum of two block-constant kernels is block-constant, and so is a scalar multiple. Averaging a kernel over each block—replacing its values on $P_i \times P_j$ by their mean—is then the natural way to push it into $\mathcal{S}_{\mathcal{P}}$, and we will see in a moment that it is the *best* such push, the nearest point of $\mathcal{S}_{\mathcal{P}}$ in L^2. That is the whole content of stepping; we now make it precise.

Definition 15.6 (Step graphons and the stepping operator)**.** Let $\mathcal{P} = \{P_1, \dots, P_k\}$ be a finite partition of $[0,1]$ into measurable sets of positive measure $|P_i| > 0$ (the calligraphic $\mathcal{P}$ here denotes a partition, not a probability-measure set $\mathcal{P}(\cdot)$). A graphon is a *step graphon adapted to* $\mathcal{P}$ if it is (a.e.) constant on every block $P_i \times P_j$; write $\mathcal{S}_{\mathcal{P}} \subset L^2([0,1]^2)$ for the (closed) subspace of all functions, symmetric or not, that are constant on every $P_i \times P_j$. The *stepping operator* $\Pi_{\mathcal{P}} \colon L^2([0,1]^2) \to \mathcal{S}_{\mathcal{P}}$ replaces a kernel by its block averages,

$$(\Pi_{\mathcal{P}} W)(x,y) = \frac{1}{|P_i|\ |P_j|} \int_{P_i \times P_j} W(s,t)\ \mathrm{d}s\ \mathrm{d}t \qquad \text{for } (x,y) \in P_i \times P_j.$$

The definition asserts that $\Pi_{\mathcal{P}}$ lands in $\mathcal{S}_{\mathcal{P}}$ and calls it the "stepping operator," but three claims are doing the real work and each must be earned. First, that $\Pi_{\mathcal{P}}$ is not merely *a* map into the step graphons but the canonical one—the orthogonal projection, the genuinely nearest step graphon—which is what makes the block average the right notion of "viewing at coarse resolution" rather than one arbitrary choice among many. Second, that stepping keeps a graphon a graphon: feed it a symmetric $[0,1]$-valued kernel and it returns one, so the operation stays inside the world of networks. Third, that it conserves the masses we care about, the block frequencies and hence the edge density, so that coarse-graining loses fine texture but not gross bookkeeping—exactly the property a "view at coarse resolution" must have to be useful. The next proposition establishes all three at once, and its proof is short because the first claim does almost all the work: once $\Pi_{\mathcal{P}}$ is identified as an orthogonal projection, the rest is reading off standard projection properties.

Proposition 15.7 (Stepping is orthogonal projection)**.** *Let $\mathcal{P}$ be a finite partition of $[0,1]$ into blocks of positive measure. Then:*

(i) $\Pi_{\mathcal{P}}$ *is the orthogonal projection of $L^2([0,1]^2)$ onto $\mathcal{S}_{\mathcal{P}}$; in particular $\Pi_{\mathcal{P}}^2 = \Pi_{\mathcal{P}}$, $\Pi_{\mathcal{P}}$ is self-adjoint, and $\|\Pi_{\mathcal{P}} W\|_{L^2} \le \|W\|_{L^2}$.*

(ii) *If W is a graphon (symmetric, $[0,1]$-valued) then so is $\Pi_{\mathcal{P}} W$, and it is a step graphon adapted to $\mathcal{P}$.*

(iii) *Stepping preserves all block masses, in particular the edge density: for every i, j,*

$$\int_{P_i \times P_j} \Pi_{\mathcal{P}} W = \int_{P_i \times P_j} W, \qquad \textit{hence} \qquad \int_{[0,1]^2} \Pi_{\mathcal{P}} W = \int_{[0,1]^2} W.$$

Proof. Everything follows from recognizing $\Pi_{\mathcal{P}}$ as the orthogonal projection onto $\mathcal{S}_{\mathcal{P}}$; the one piece of labour is to identify the abstract projection with the concrete block-average formula, and for that we write down an explicit orthogonal basis of $\mathcal{S}_{\mathcal{P}}$.

An orthogonal basis of $\mathcal{S}_\mathcal{P}$. For each ordered pair (i,j) with $1 \le i,j \le k$ let $g_{ij} = \mathbf{1}_{P_i \times P_j}$ be the indicator of the block. A function lies in $\mathcal{S}_\mathcal{P}$ exactly when it is constant on every block, i.e. when it can be written $\sum_{i,j} c_{ij} g_{ij}$ for some scalars c_{ij}; so the g_{ij} span $\mathcal{S}_\mathcal{P}$, and as there are only k^2 of them, $\dim \mathcal{S}_\mathcal{P} = k^2 < \infty$. A finite-dimensional subspace of a normed space is automatically closed, which is why $\mathcal{S}_\mathcal{P}$ deserves the label "closed" attached to it in definition 15.6—we do not have to prove closedness by hand. The basis is orthogonal: if $(i,j) \neq (i',j')$ the rectangles $P_i \times P_j$ and $P_{i'} \times P_{j'}$ are disjoint (because the P_i partition $[0,1]$), so the product $g_{ij}\, g_{i'j'}$ vanishes identically and $\langle g_{ij}, g_{i'j'} \rangle = 0$; while

$$\|g_{ij}\|_{L^2}^2 = \int_{[0,1]^2} \mathbf{1}_{P_i \times P_j} = |P_i|\;|P_j| > 0,$$

the strict positivity being precisely where the hypothesis $|P_i| > 0$ is used—a null block would contribute a zero basis vector and an undefined $0/0$ in the average below.

(i) The projection is the block average. Since $\mathcal{S}_\mathcal{P}$ is closed, the orthogonal projection $P_{\mathcal{S}_\mathcal{P}}$ of $L^2([0,1]^2)$ onto it exists and is unique (theorem 2.5). Because $\{g_{ij}\}$ is an orthogonal basis of the finite-dimensional space $\mathcal{S}_\mathcal{P}$, that projection is computed coordinate by coordinate—this is the Fourier/Parseval expansion (theorem 2.8) applied *inside* $\mathcal{S}_\mathcal{P}$, not in all of $L^2([0,1]^2)$:

$$P_{\mathcal{S}_\mathcal{P}} W = \sum_{i,j} \frac{\langle W, g_{ij} \rangle}{\|g_{ij}\|_{L^2}^2}\, g_{ij}.$$

Now read off the two factors. The numerator is $\langle W, g_{ij} \rangle = \int_{[0,1]^2} W\, \mathbf{1}_{P_i \times P_j} = \int_{P_i \times P_j} W$, the mass of W on the block; dividing by the denominator $\|g_{ij}\|_{L^2}^2 = |P_i|\,|P_j|$ converts that mass into the *average* value of W over the block. So the coefficient of g_{ij} is exactly $\frac{1}{|P_i||P_j|} \int_{P_i \times P_j} W$, and since g_{ij} is the indicator that is 1 precisely when $(x,y) \in P_i \times P_j$,

$$(P_{\mathcal{S}_\mathcal{P}} W)(x,y) = \frac{1}{|P_i|\,|P_j|} \int_{P_i \times P_j} W \qquad \text{for } (x,y) \in P_i \times P_j,$$

which is the block-average formula of definition 15.6. Hence $\Pi_\mathcal{P} = P_{\mathcal{S}_\mathcal{P}}$. The three properties listed in (i) are now the standard attributes of an orthogonal projection (theorem 2.5), read in this language: idempotence $\Pi_\mathcal{P}^2 = \Pi_\mathcal{P}$ (projecting an already block-constant kernel returns it unchanged, since its block average is itself), self-adjointness $\langle \Pi_\mathcal{P} W, H \rangle = \langle W, \Pi_\mathcal{P} H \rangle$, and the contraction $\|\Pi_\mathcal{P} W\|_{L^2} \le \|W\|_{L^2}$, which is the Pythagorean statement that a projection discards the component orthogonal to $\mathcal{S}_\mathcal{P}$ and keeps only part of the squared norm.

(ii) Stepping keeps a graphon a graphon. Three things must be checked: the output is constant on blocks, lies in $[0,1]$, and is symmetric. Block-constancy is immediate, since it is exactly what membership in $\mathcal{S}_\mathcal{P}$ means and we have just shown $\Pi_\mathcal{P} W \in \mathcal{S}_\mathcal{P}$. The range $[0,1]$ is preserved because each output value is an *average* of input values, and an average cannot escape the interval its inputs lie in: if $0 \le W \le 1$ almost everywhere then, by monotonicity of the integral,

$$0 \le \frac{1}{|P_i|\,|P_j|} \int_{P_i \times P_j} W \le \frac{1}{|P_i|\,|P_j|} \int_{P_i \times P_j} 1 = 1.$$

Symmetry needs the partition to be used symmetrically—the same blocks $P_1, \dots, P_k$ index both arguments—together with Fubini. Fix $(x,y) \in P_i \times P_j$, so that the reflected point (y,x) lies in $P_j \times P_i$. The value of $\Pi_\mathcal{P} W$ at (x,y) is the average of $W(s,t)$ over $(s,t) \in P_i \times P_j$; its value at (y,x)

is the average of $W(s,t)$ over $(s,t) \in P_j \times P_i$. In the second average substitute $W(s,t) = W(t,s)$ (symmetry of the input) and rename the two dummy variables $s \leftrightarrow t$; this is legitimate because Fubini–Tonelli (theorem 1.27) lets the double integral be evaluated in either order, and the swap carries the rectangle $P_j \times P_i$ onto $P_i \times P_j$. The second average therefore equals the first, so $(\Pi_{\mathcal{P}} W)(x,y) = (\Pi_{\mathcal{P}} W)(y,x)$ and the stepped kernel is symmetric. (Were the input *not* symmetric, or were rows and columns partitioned differently, this argument would fail—which is exactly why both hypotheses are named.)

(iii) Stepping preserves every block mass. Fix a block $P_i \times P_j$. On it $\Pi_{\mathcal{P}} W$ is the constant $c_{ij} = \frac{1}{|P_i||P_j|} \int_{P_i \times P_j} W$, so integrating that constant over the block—area $|P_i|\,|P_j|$—returns the original mass,

$$\int_{P_i \times P_j} \Pi_{\mathcal{P}} W = c_{ij}\ |P_i|\,|P_j| = \int_{P_i \times P_j} W,$$

the area cancelling the normalization exactly. Summing over the k^2 blocks, which tile $[0,1]^2$, gives $\int_{[0,1]^2} \Pi_{\mathcal{P}} W = \int_{[0,1]^2} W$, so in particular the edge density (proposition 15.3) is unchanged. There is a slicker, projection-flavoured way to see both statements at once, worth recording because it generalizes. Each block indicator $g_{ij} = \mathbf{1}_{P_i \times P_j}$ is itself block-constant, $g_{ij} \in \mathcal{S}_{\mathcal{P}}$, hence fixed by the projection: $\Pi_{\mathcal{P}} g_{ij} = g_{ij}$. Self-adjointness then gives

$$\int_{P_i \times P_j} \Pi_{\mathcal{P}} W = \langle \Pi_{\mathcal{P}} W, g_{ij} \rangle = \langle W, \Pi_{\mathcal{P}} g_{ij} \rangle = \langle W, g_{ij} \rangle = \int_{P_i \times P_j} W,$$

the per-block conservation; taking $\mathbf{1} = \sum_{i,j} g_{ij} \in \mathcal{S}_{\mathcal{P}}$ in place of a single g_{ij} (the constant kernel is block-constant too, so $\Pi_{\mathcal{P}} \mathbf{1} = \mathbf{1}$) gives the total-mass conservation $\langle \Pi_{\mathcal{P}} W, \mathbf{1} \rangle = \langle W, \mathbf{1} \rangle$ in one line. $\square$

Two consequences orient the rest of Part III, and both come straight from reading the projection geometrically.

Coarse-graining can only lose L^2-energy. Part (i) gives the contraction $\|\Pi_{\mathcal{P}} W\|_{L^2} \le \|W\|_{L^2}$, and the projection picture makes the case of equality sharp. Split W into its block-constant part and the orthogonal remainder,

$$W = \Pi_{\mathcal{P}} W + (W - \Pi_{\mathcal{P}} W), \qquad \|W\|_{L^2}^2 = \|\Pi_{\mathcal{P}} W\|_{L^2}^2 + \|W - \Pi_{\mathcal{P}} W\|_{L^2}^2\,,$$

the second equation being the Pythagorean theorem, valid because the two summands are orthogonal by construction of the projection. The contraction is now visible, and so is its equality case: $\|\Pi_{\mathcal{P}} W\|_{L^2} = \|W\|_{L^2}$ holds *iff* the remainder term has zero norm, i.e. iff $W = \Pi_{\mathcal{P}} W$ already lies in $\mathcal{S}_{\mathcal{P}}$—iff W was already a step graphon adapted to $\mathcal{P}$. (The one-line reason: a vector's norm equals the norm of its projection precisely when the vector lies in the subspace, for the discarded orthogonal component is then zero.) The energy that is lost, $\|W - \Pi_{\mathcal{P}} W\|_{L^2}^2$, is exactly the total within-block variance of W—the fine texture the resolution $\mathcal{P}$ cannot resolve. This is the precise sense in which the stochastic block models of the next section are the *finite-resolution* graphons: each is a fixed point of some $\Pi_{\mathcal{P}}$, the full kernel is recovered only in the limit of ever-finer partitions, and the approximation is always monotone—stepping moves *toward* the data and never overshoots it. It is also why the approximation runs in one direction only, general graphons *by* step graphons and not the reverse: the step graphons are the finite-dimensional subspaces $\mathcal{S}_{\mathcal{P}}$ we project *onto*, and there is no operation carrying a step graphon "up" to a generic one.

Two micro-examples fix the scale. The coarsest partition $\mathcal{P} = \{[0,1]\}$, a single block, collapses every graphon to the *constant* kernel equal to its global average $\int_{[0,1]^2} W$—the edge density, the most lossy resolution there is, and exactly the Erdős–Rényi graphon of example 15.9. A two-block partition $\mathcal{P} = \{P_1, P_2\}$ sends W to the 2×2 array of its four block averages, the smallest nontrivial stochastic block model. Refining $\mathcal{P}$ interpolates between this caricature and the kernel itself, each refinement recovering more energy.

Pixel graphons are steps to the equipartition. The pixel graphon W^G of a graph of order n is *itself* a step graphon, adapted to the equipartition $\mathcal{P}_n = \left\{I_1^{(n)}, \ldots, I_n^{(n)}\right\}$ into the n equal cells of definition 15.2—indeed that is how it was built, constant on each cell $I_i^{(n)} \times I_j^{(n)}$. So W^G is a fixed point of $\Pi_{\mathcal{P}_n}$ (stepping it to its own grid changes nothing), and stepping an arbitrary graphon W to $\mathcal{P}_n$—averaging it over the $n \times n$ grid of congruent cells—is exactly "rendering W at resolution n," the finite-resolution snapshot that section 15.1 pictured the refining grid producing. The honest convergence statement deferred there is thus, in part, a statement about how $\Pi_{\mathcal{P}_n} W$ behaves as $n \to \infty$. The content of the weak regularity lemma (chapter 19) will sharpen this dramatically: a *bounded* number of blocks—independent of how complicated W is—already approximates any graphon to a prescribed accuracy in the cut norm. That is the rigorous form of the physicist's expectation that a network, however large, has only a finite "effective complexity": a few well-chosen patches capture all of its macroscopic structure.

The same projection geometry that makes stepping so clean also exposes, by a single explicit computation, why the natural first guess for a convergence norm cannot work—and thereby motivates the genuinely coarse norm of chapter 17.

Remark 15.8 (Why a coarser norm is unavoidable)**.** One might hope to run section 15.1 directly in L^2: declare $W^{G_n} \to W$ when $\left\|W^{G_n} - W\right\|_{L^2} \to 0$. This fails, and instructively so. Sample a graph G_n with n vertices by placing each edge independently with probability $\frac{1}{2}$ (an Erdős–Rényi draw, example 15.9). The pixel graphons W^{G_n} are $\{0,1\}$-valued and "should" converge to the constant graphon $W \equiv \frac{1}{2}$—the block frequencies do settle to $\frac{1}{2}$ over every fixed patch. Yet the L^2 distance never moves. Here is the computation in full: because W^{G_n} takes only the values 0 and 1, at every point of the square $\left(W^{G_n} - \frac{1}{2}\right)^2 = \left(\pm \frac{1}{2}\right)^2 = \frac{1}{4}$ *identically*, so

$$\left\|W^{G_n} - \tfrac{1}{2}\right\|_{L^2}^2 = \int_{[0,1]^2} \left(W^{G_n} - \tfrac{1}{2}\right)^2 = \frac{1}{4}$$

for every n, no matter how large the network and no matter how perfectly its block frequencies have stabilized. The L^2 norm is comparing the two pictures pixel by pixel, and at the pixel level a fair-coin graph and the flat grey kernel disagree maximally everywhere—one is black or white, the other is uniformly grey. It sees the microscopic 0/1 texture forever; it is too fine to be the topology of graph limits. The cure is to integrate *before* measuring discrepancy. The cut norm of chapter 17 pairs $W^{G_n} - \frac{1}{2}$ against indicator patches $\mathbf{1}_{S \times T}$ and only then takes the absolute value; over any fixed patch the $\pm\frac{1}{2}$ fluctuations average to nearly zero by the law of large numbers, so the cut distance *does* tend to 0. It measures agreement of patch averages, not of pixels, and is by design blind to the texture that L^2 can never forget. This is the concrete failure promised in section 15.1, and it is why the metric of graph limits is the cut metric and not L^2.

With stepping understood we have exactly the vocabulary the next section needs. We have learned that the step graphons are the finite-resolution objects—fixed points of the projections $\Pi_{\mathcal{P}}$, dense among all graphons in the coarse cut sense, and the universal target of approximation—which is precisely the role the *stochastic block models* of the gallery will play; that a pixel picture is the step of a kernel to an equipartition, so the discrete and the continuum sit on one ladder of resolutions; and that energy is lost monotonically under coarse-graining, the framing that will explain why the constant, step, and rank-one kernels are the canonical exactly-solvable classes—they are the kernels whose operator $\mathbb{W}$ has the lowest, most rigid spectral structure, and on which the coarse-graining picture is exact rather than approximate. The L^2-too-fine failure we just exhibited is the seed of the cut metric of chapter 17. We now turn to the gallery itself, where each kernel is met as a formula, drawn as a pixel picture, and tied forward to the mean field game it parametrizes.

15.4 A gallery of graphons

The stepping section left us with a sharp way to read the graphons we are about to meet. It identified the step graphons as the *finite-resolution* objects—fixed points of the projections $\Pi_{\mathcal{P}}$, the targets of all coarse-graining—and it promised, at its close, that constant, step, and rank-one kernels are the canonical exactly-solvable classes because their operator $\mathbb{W}$ carries the lowest and most rigid spectral structure. This section makes good on that promise by exhibiting the three classes one at a time. Each is met the same way, and the reader should watch for the recurring template: a graphon is first written down as a *kernel* (a formula for $W(x, y)$), then *seen* through its pixel picture (the black-and-white or greyscale image of definition 15.2, which makes its structure visible at a glance), and finally *tied forward* to a specific mean field game in Part IV that it parametrizes and renders explicitly solvable. Kernel, picture, game: that is the arc each subsection traces.

The organizing claim—and the thesis that makes this a single section rather than three unrelated vignettes—is that constant, step, and rank-one are *not coincidentally* the three classes on which the integral operator $\mathbb{W}$ is exactly diagonalizable (worked computation 2.28). The spectral theorem of theorem 2.26 guarantees every graphon a real spectrum and an orthonormal eigenbasis, but for a generic kernel that basis is infinite and inexplicit; one knows it exists without being able to write it down. The three gallery members are exactly the cases where one *can* write it down, because each has an operator of finite rank with eigenfunctions read straight off the formula: the constant kernel has rank one with the constant mode, a k-block step graphon has rank at most k with piecewise-constant modes, and the rank-one kernel $W(x, y) = f(x)f(y)$ has rank one with the single mode f. Exact diagonalizability is not a curiosity of these kernels; it is precisely *what makes the corresponding game solvable.* The graphon enters every equation of Part IV only through $\mathbb{W}$ (the point section 15.6 will press), so a $\mathbb{W}$ whose action is captured by a handful of modes collapses the type-continuum of coupled equilibrium conditions onto a handful of scalar unknowns, and a problem in finitely many scalars is one we can actually close.

The number of those scalars is the spine that runs through all three subsections, and it is fixed by a single integer: the rank of $\mathbb{W}$ equals the number of *order parameters* needed to close the game. A rank-one operator yields one scalar order parameter, a rank-k step operator yields k of them, and the structureless constant kernel—rank one again—yields the single scalar that the homogeneous theory of Parts I–II already lived on. We state this here only as the expectation that organizes the gallery; the `physicsnote` accompanying the stochastic block models in section 15.4.2 makes it precise through the statistical-mechanics dictionary, and chapter 18 turns it into a theorem. The cleanest worked instance is the rank-one collapse $z(x) = \varrho\, f(x)$ of section 15.4.3, where the entire type-indexed coupling field reduces to a single number ϱ; that collapse is the seed of Thread B, and it is the reason the rank-one graphon is the example we return to whenever an explicit graphon mean field game is wanted.

We open with the structureless extreme. The constant graphon—uniform grey across the whole square, the simplest formula and the simplest picture there is—is the first and plainest instance of the kernel-picture-game template, and at the same time it is the bridge back to where Part III began: it is the single degenerate point at which the homogeneous, anonymous-crowd theory of Parts I–II reappears inside the graphon theory. We meet it now.

15.4.1 Erdős–Rényi: the constant graphon

The gallery preamble promised to open with the structureless extreme, and it is the natural place to start for the same reason that one calibrates an instrument against a blank: structure is only legible against a background that has none. The constant graphon *is* that background. It is uniform grey across the entire square—the simplest formula $W \equiv p$ and the simplest picture, a single flat level—and it is the first instance of the kernel–picture–game template the gallery traces. It is also, by the spine of the section, the most rigid member of the catalogue on the operator side: its $\mathbb{W}$ has the lowest possible nontrivial rank, namely one, and a single mode handles its entire action. We will see that this rank-one structure is exactly what collapses the type-continuum of the game down to one scalar, and that the scalar in question is the very global mean on which the homogeneous theory of Parts I–II already lived. The constant graphon is thus simultaneously the plainest example we can write and the precise bridge back to where Part III began.

Example 15.9 (Constant graphon)**.** Fix $p \in [0,1]$ and let $W \equiv p$ on $[0,1]^2$. This is the *constant* (or *Erdős–Rényi*) graphon. Its pixel-picture origin is the dense random graph $G(n,p)$, in which each of the $\binom{n}{2}$ possible edges is present independently with probability p: as $n \to \infty$, the pixel graphons $W^{G(n,p)}$ concentrate around the flat greyscale image at level p (the precise sampling statement, with concentration rates, is chapter 19). Every type connects to every other type with the same probability p; there is no structure, no notion of centrality, and no distinguished type. The graphon is invariant under *every* measure-preserving relabeling, the extreme case of remark 15.5.

Each clause of the example earns a word, because the constant kernel is the degenerate limit of several distinct things the rest of the chapter measures, and degeneracy is most instructive when one sees exactly what has collapsed. The *origin* is the dense Erdős–Rényi graph $G(n,p)$: drawing each of the $\binom{n}{2}$ edges as an independent fair-or-biased coin with heads-probability p, the resulting pixel graphon $W^{G(n,p)}$ is a random $\{0,1\}$-valued checkerboard, black with frequency p and white with frequency $1-p$. No individual picture is flat—it is maximally textured, every cell pure black or pure white—yet as n grows the *block frequencies* settle: over any fixed patch the proportion of black cells tends to p by the law of large numbers, so the network, read at any coarse resolution, looks like uniform grey at level p. This is the same phenomenon we already met head-on in remark 15.8, where the $p = \frac{1}{2}$ case showed that the convergence $W^{G(n,p)} \to p$ is real but invisible to the L^2 norm and visible only after one integrates against patches; the honest concentration statement, with rates, is the sampling theorem of chapter 19. The *absence of structure* is then exact rather than informal: because the single value p is attached to every pair of types alike, the slice $y \mapsto W(x,y) \equiv p$ is the same constant function for every x, so no type connects more, or to anyone in particular, more than any other. There is no high-degree hub and no low-degree fringe—no centrality—and no type that the kernel can tell apart from the rest. That last point is the *invariance under every relabeling*, and it is worth seeing as the genuine extreme case of remark 15.5: a generic graphon is fixed only by those measure-preserving φ that happen to respect its structure, and remark 15.5 warned us to discard any feature not so fixed; the constant kernel is the unique graphon fixed by *all* of them, since $W^{\varphi}(x,y) = W(\varphi(x), \varphi(y)) = p = W(x,y)$ for any measure-preserving φ whatsoever. Having no structure to break, it is the one kernel for which the relabeling quotient does nothing at all—which is precisely what "structureless" means made precise.

This structurelessness is not a dead end; it is the bridge back to the homogeneous theory of Parts I–II, and the bridge is built entirely out of the operator $\mathbb{W}$. We saw in the gallery preamble that the graphon reaches every equation of Part IV only through $\mathbb{W}$, and that the rank of $\mathbb{W}$ counts the order parameters the game needs; so to locate the constant graphon inside the homogeneous theory we have only to compute its operator and count. That computation is already on record as worked computation 2.28(a), and it is one line:

$$(\mathbb{W}f)(x) = \int_0^1 W(x,y)\, f(y)\, \mathrm{d}y = \int_0^1 p\, f(y)\, \mathrm{d}y = p \int_0^1 f(y)\, \mathrm{d}y = p\, \langle f, \mathbf{1} \rangle,$$

the constant p pulling outside the integral and the remaining $\int_0^1 f$ being exactly the inner product of f with the constant unit field $\mathbf{1}$ (which has $\|\mathbf{1}\|_{L^2} = 1$ on $[0,1]$, so it is already a unit eigenvector). Read this output. Whatever field f goes in, what comes out is a *constant* function—the value $p\,\langle f, \mathbf{1} \rangle$, the same number at every type x—so the range of $\mathbb{W}$ is the one-dimensional line of constants and $\mathbb{W}$ has *rank one*, the minimum a nonzero coupling can have. Its spectral picture is correspondingly the most rigid possible: $\mathbb{W}\mathbf{1} = p\,\langle \mathbf{1}, \mathbf{1} \rangle\, \mathbf{1} = p\,\mathbf{1}$, so $\mathbf{1}$ is an eigenfunction with eigenvalue p; and any field g orthogonal to the constants, $\langle g, \mathbf{1} \rangle = 0$, is annihilated, $\mathbb{W}g = 0$. The whole spectrum is therefore $\{p, 0, 0, \ldots\}$: a single nonzero eigenvalue p carried by the constant mode, every other mode killed. This is worked computation 2.28(a) read for its meaning rather than its value, and it is the rank-one collapse of the section's spine in its very simplest form—one nonzero mode, hence (looking ahead to chapter 18) one order parameter.

The reduction to the homogeneous game is now immediate, and it is worth spelling out so that the claim "the homogeneous theory sits inside as one point" is earned rather than asserted. Suppose the population is described by a type-indexed mean field $x \mapsto \bar{m}(x)$ and each agent is coupled to its neighbors through $\mathbb{W}$, feeling the aggregate $z(x) = (\mathbb{W}\bar{m})(x)$—this is exactly the graphon-aggregated neighbor mean that Thread B builds its games on, where in general $z(x)$ depends on the agent's own type x because different types see different neighborhoods. For the constant graphon the computation above makes that dependence vanish:

$$z(x) = (\mathbb{W}\bar{m})(x) = p \int_0^1 \bar{m}(y)\, \mathrm{d}y = p\, \langle \bar{m}, \mathbf{1} \rangle,$$

a single number, *the same for every type* x. Every agent, regardless of its own label, feels one and the same scalar aggregate $p \int_0^1 \bar{m}$. That is precisely the *anonymous global-mean coupling* of the classical mean field game of Thread A (appendix A.7): a crowd in which no agent sees a distinguished neighborhood, only the population average, weighted here by the connection probability p. The type variable x has become a passenger—it indexes the agents but no longer enters the coupling—so the type-continuum of equilibrium conditions, one per x, collapses to the single self-consistent condition on the scalar $\langle \bar{m}, \mathbf{1} \rangle$ that the homogeneous theory already solved. The constant graphon therefore does not merely *resemble* the homogeneous MFG of chapter 14; it *is* that game, recovered as the degenerate, structureless special case in which $\mathbb{W}$ has rank one and the only role left for the kernel is the constant p, which rescales the coupling strength (replacing the homogeneous coupling weight s of Thread A by $p\, s$). We will carry out this reduction explicitly inside the dynamic theory, checking that the graphon HJB–Fokker–Planck system over the kernel degenerates to the homogeneous Riccati system of Thread A, in chapter 22.

With the structureless point understood, the natural next move is to add the least structure that is still structure. The constant graphon is the partition $\mathcal{P} = \{[0, 1]\}$ of definition 15.6 with a single block—one community, every pair joined at the same density p—and the obvious generalization is to cut $[0, 1]$ into k blocks and allow each pair of blocks its own density. That turns the single number p into a $k \times k$ array of densities and the single global mean into one aggregate per community, while keeping everything else—the exact diagonalizability, the finite count of order parameters—intact. The constant graphon is the $k = 1$ corner of that picture, and the next subsection occupies the rest of it.

15.4.2 Stochastic block models: step graphons

The constant graphon left us at the $k = 1$ corner of definition 15.6: a single block, one community, every pair of types joined at the same density p. The previous subsection closed by naming the obvious next move—cut $[0, 1]$ into k blocks and let each pair of blocks carry its own density—and this subsection occupies the rest of that picture. The object we get is the coarsest structure that is still structure: finitely many *communities*, each ordered pair of them connected at its own density. We have already built this object once, abstractly, in definition 15.6: a graphon constant on every rectangle $P_r \times P_s$ of a finite partition is exactly a step graphon. What is new here is not the kernel but the *reading*. A step graphon, read as a recipe for generating a random network—place an edge between an agent of community r and an agent of community s independently with probability equal to the block value—is what network science calls the *stochastic block model*, the canonical model of a network with communities. The stepping projection of proposition 15.7 then turns out to do more than relate these two objects: it makes the SBM the *universal finite-resolution approximation* to every graphon, which is the structural reason the model recurs throughout the book and supplies the bridge to the finite multi-population games of chapter 30. We first fix the data, then read off the picture, then earn each of those claims in turn.

Example 15.10 (Stochastic block model)**.** Fix an integer $k \geq 1$, a partition $[0,1] = \bigsqcup_{r=1}^{k} C_r$ into measurable *communities* of sizes $|C_r| = \rho_r > 0$ with $\sum_r \rho_r = 1$, and a symmetric matrix $\mathsf{B} = (\mathsf{B}_{rs}) \in [0,1]^{k\times k}$ of inter-community connection probabilities (the sans-serif B keeps the block matrix distinct from the Brownian motion B of Part IV). The associated *stochastic block model graphon* is the step graphon

$$W^{\mathrm{SBM}}(x,y) = \mathsf{B}_{rs} \qquad \text{for } x \in C_r,\ y \in C_s.$$

It is adapted to the partition $\{C_1, \dots, C_k\}$ in the sense of definition 15.6.

The data come in three pieces and each has a plain meaning. The integer k is the number of communities; the masses $\rho_r = |C_r|$ record what fraction of the population belongs to each, so that $\sum_r \rho_r = 1$ simply says every agent belongs somewhere; and the block matrix B records the density of links between communities, with B_{rs} the probability that a type drawn from C_r and a type drawn from C_s are joined. The symmetry $\mathsf{B}_{rs} = \mathsf{B}_{sr}$ is forced—it is exactly the graphon symmetry $W(x,y) = W(y,x)$ restricted to blocks, the continuum imprint of undirected edges discussed at definition 15.4—and the entries lie in $[0,1]$ because they are probabilities. One should note the deliberate notational care: the block matrix is written B in sans serif precisely so that it never collides with the Brownian motion B_t that the dynamic theory of Part IV will run on top of these kernels, a clash the chapter's opening notation watch flagged and this is the place it would otherwise bite.

The pixel picture makes the structure legible at a glance, and it is the cleanest instance of the kernel–picture template the gallery follows. Laying W^{SBM} flat on the unit square as in definition 15.2 produces a $k \times k$ array of solid grey rectangles: the rectangle $C_r \times C_s$ has dimensions $\rho_r \times \rho_s$ and is shaded at the single uniform level B_{rs}. Nothing varies inside a rectangle—that is what "step" means—so the whole image is a checkerboard of flat grey tiles, no texture within any tile. The two kinds of tile carry the two kinds of network structure. The *diagonal* blocks B_{rr} are within-community densities: a large B_{rr} means agents of community r link densely among themselves, the signature of an *assortative*, community-structured network where like associates with like. The *off-diagonal* blocks B_{rs} with $r \neq s$ are cross-community densities, the rate at which the communities reach across to one another; making the off-diagonals small relative to the diagonals sharpens the community structure, while making them large (or larger than the diagonals, the *disassortative* case) dissolves or inverts it. The constant graphon of example 15.9 is the degenerate picture with a single tile filling the whole square.

Every step graphon is an SBM, and stepping any graphon produces one. The example calls W^{SBM} a step graphon, but the relationship runs both ways and is worth making fully explicit, because it is what promotes the SBM from one model among many to the canonical one. Both directions are read straight off proposition 15.7.

For the forward direction, suppose we are handed an arbitrary step graphon—any symmetric $[0,1]$-valued kernel that is constant on every rectangle of some finite partition $\mathcal{P} = \{P_1, \dots, P_k\}$ of $[0,1]$. Then it *is* an SBM graphon, with no work beyond renaming: take the communities to be the blocks themselves, $C_r = P_r$, with masses $\rho_r = |P_r| > 0$; and let B_{rs} be the constant value the kernel takes on $P_r \times P_s$. Symmetry of the kernel hands us $\mathsf{B}_{rs} = \mathsf{B}_{sr}$, and the values lie in $[0,1]$ because the kernel does, so (k, ρ, B) is legitimate SBM data and the kernel is exactly its W^{SBM}. There is no daylight between "step graphon" and "SBM graphon"—they are the same objects under two names, one analytic and one generative.

For the converse direction, take *any* graphon W, structured however badly, and any finite partition $\mathcal{P}$. Stepping it produces an SBM. Concretely, $\Pi_{\mathcal{P}} W$ is by proposition 15.7(ii) again a symmetric $[0,1]$-valued kernel constant on every block—a step graphon—so by the forward direction it is the SBM graphon of some data, and definition 15.6 tells us exactly what that data is: the block matrix entries are the block averages

$$\mathsf{B}_{rs} = \frac{1}{|P_r|\ |P_s|} \int_{P_r \times P_s} W,$$

the mean density of W across the pair of communities. Thus stepping is the operation that *coarse-grains any network into a stochastic block model*, reading each cross-community density as the average it should be. The forward direction says the SBMs are precisely the fixed points of the stepping projections; the converse says every graphon casts an SBM shadow at every resolution.

Why the SBM is universal, not merely one model. These two directions would be a tidy dictionary and nothing more if the shadow were a poor likeness of the original. The content that makes the SBM *universal* is that the likeness can be made as faithful as one wishes with a *bounded* number of communities. This is the weak regularity lemma of chapter 19: for every tolerance $\varepsilon > 0$ there is a partition into at most $k(\varepsilon)$ blocks—a bound depending on ε alone, not on how intricate W is—whose stepped kernel $\Pi_{\mathcal{P}} W$ approximates W to within ε in the cut norm of chapter 17. In the language just established: *every* graphon is cut-close to a stochastic block model with a controlled number of communities, so the SBMs are *dense* in graphon space under the cut metric. This is the structural reason the SBM is not one network model competing with others but the universal finite-resolution approximation, the continuum echo of the physicist's expectation—recorded already at proposition 15.7—that any network, however large, has only a finite effective complexity that a handful of well-chosen communities already captures. It is also why the model is worth studying in its own right: results proved for SBMs, if they are stable under cut-distance limits, propagate to all graphons.

Two reductions: the constant kernel and the finite type space. With the correspondence in hand, two specializations anchor the SBM for everything downstream. The first is the floor we came in on. With $k = 1$ there is a single community $C_1 = [0,1]$ of mass $\rho_1 = 1$ and a 1×1 block matrix $\mathsf{B} = (p)$, and the SBM graphon is the constant kernel $W \equiv p$ of example 15.9. The SBM thus contains the Erdős–Rényi graphon as its $k = 1$ corner, exactly as promised, and inherits its rank-one collapse as the simplest case of what follows.

The second specialization is the one that carries the SBM into Part IV, and it comes from reading the communities not as blocks of a continuum but as a *finite type space* in their own right. When a graphon is constant on communities, an agent's type matters only through *which community it belongs to*: two types in the same C_r are coupled to the rest of the network identically, so the continuum of types $x \in [0,1]$ degenerates to the finite set $\{1,\dots,k\}$ of community labels. Concretely, watch what the operator $\mathbb{W}^{\mathrm{SBM}}$ does to a field f. For a type $x \in C_r$,

$$(\mathbb{W}^{\mathrm{SBM}} f)(x) = \int_0^1 W^{\mathrm{SBM}}(x,y)\, f(y)\; \mathrm{d}y = \sum_{s=1}^{k} \mathsf{B}_{rs} \int_{C_s} f(y)\; \mathrm{d}y = \sum_{s=1}^{k} \mathsf{B}_{rs}\, \rho_s\, \bar{f}_s, \qquad \bar{f}_s := \frac{1}{\rho_s} \int_{C_s} f,$$

where the integral splits over the communities because W^{SBM} is the constant B_{rs} on C_s for $x \in C_r$, and $\bar{f}_s$ is the average of f over community s. Two things are visible at once. The output value $\sum_s \mathsf{B}_{rs}\rho_s \bar{f}_s$ depends on x *only through its community label* r, so $\mathbb{W}^{\mathrm{SBM}} f$ is itself constant on communities; and it depends on the input f *only through the* k *averages* $(\bar{f}_1,\dots,\bar{f}_k)$. The operator therefore factors through the k-dimensional space of functions constant on the communities—it forgets everything about f except its k community means—so it has *rank at most* k. This is the SBM instance of the section's spine: the rank of $\mathbb{W}$ is bounded by the number of communities.

The consequence for the game is the bridge to chapter 30. The SBM graphon is precisely the data of a finite *multi-population* mean field game: k populations of relative sizes ρ_r, the r-th confined to community C_r, coupled to one another with strengths B_{rs}. Because $\mathbb{W}^{\mathrm{SBM}}$ depends on a population profile $\bar{m}$ only through its k community aggregates, the type-indexed coupling field an agent feels collapses, for an agent of community r, to the single number

$$z_r = (\mathbb{W}^{\mathrm{SBM}} \bar{m})\big|_{C_r} = \sum_{s=1}^{k} \mathsf{B}_{rs}\, \rho_s\, \bar{m}_s,$$

a B-weighted combination of the k community aggregates $\bar{m}_s$. The continuum of equilibrium conditions, one per type x, therefore reduces to k conditions, one per community, and the graphon mean field game on an SBM *collapses to a coupled system of* k *classical mean field games*—one per community—interacting only through the $k \times k$ matrix B. The k scalars $\bar{m}_1,\dots,\bar{m}_k$ (more precisely the k aggregates that close the system) are the order parameters of the problem. This is the bridge between the continuum graphon theory and the finite multi-population MFG of chapter 30; we set it up here and exploit it in the dynamic reduction of chapter 22.

That last count—k communities, rank at most k, k order parameters—is no accident, and it is worth naming the statistical-mechanics picture that explains why the three numbers always agree, because that picture is the hidden thesis of the whole gallery.

Physics note

The three exactly solvable kernels of this section have precise statistical-mechanics counterparts, and the analogy is worth one box because it correctly predicts the order parameter in each case—and because the pattern it reveals is the chapter's central claim. Start with the constant graphon. It is the all-to-all (Curie–Weiss) coupling $J_{ij} \equiv J/N$: every spin is coupled equally to every other, each feels the same mean field $J\,m$ with the single scalar magnetization $m = N^{-1}\sum_i s_i$, and that one number closes the self-consistency equation $m = \tanh(\beta J m)$. One scalar suffices because the coupling has *rank one*: J_{ij} is the constant matrix, whose only nonzero mode is the uniform vector. Next the block graphon. It is a *multi-species* Curie–Weiss model: the spins are split into k sublattices, the coupling B_{rs} depends only on which pair of species two spins belong to, and the theory closes on a *vector* order parameter $(m_1, \ldots, m_k)$ of one magnetization per species, coupled through B exactly as the k community aggregates $\bar{m}_r$ above are coupled through it. Here the coupling has rank at most k—it sees a spin only through its species—and one needs exactly k order parameters. Finally the rank-one graphon $W(x, y) = f(x)f(y)$ of the next subsection is the single-pattern Hopfield (Mattis) model, with separable coupling $J_{ij} = \xi_i\xi_j$ built from one stored pattern ξ; its order parameter is the *single* scalar overlap $m = N^{-1}\sum_i \xi_i s_i$ with that pattern, and the coupling, being an outer product, again has rank one. The pattern is now unmistakable, and it is the load-bearing point: in each case *the rank of the coupling operator equals the number of order parameters needed to close the mean field equations*—rank one gives one scalar, rank k gives k, because the order parameters are precisely the projections of the population onto the nonzero modes of the coupling, and a field that the operator cannot resolve cannot enter the closed equations. This is exactly the prediction the spectral theory of chapter 18 will turn into a theorem, and it is why the rank-one graphon is the cleanest graphon mean field game there is. One honest caveat keeps the analogy in its place: it predicts the *number and identity* of the order parameters, nothing more. The equilibrium values below are fixed by a *Nash* condition—each agent best-responds to the aggregate it feels—not by a thermal average minimizing a free energy, and the two need not coincide. The dictionary tells us how many scalars to track; it does not tell us the equations they solve.

The spine the physicsnote names—rank of $\mathbb{W}$ equals the number of order parameters—now points the way down. The SBM realizes it at general rank k, with a vector of k aggregates closing the game. The sharpest instance is the floor of that scale: a coupling operator of rank *one*, where the entire type-indexed coupling field collapses not to k scalars but to a *single* one. That is the rank-one graphon, the cleanest exactly-solvable kernel and the one that carries Thread B, and we meet it next.

15.4.3 The rank-one (Curie–Weiss) graphon and a scalar order parameter

The stochastic block model realized the chapter's spine—rank of $\mathbb{W}$ equals the number of order parameters that close the game—at general rank k: a k-community kernel is resolved by the operator into at most k modes, and the game closes on a vector of k aggregates. Push that spine to its floor. The smallest nontrivial rank is one, and a coupling operator of rank one carries the cleanest possible payload: not k scalars but a *single* one. This is the rank-one graphon, the simplest non-constant kernel there is, built from a single type profile f; it is the graphon on which the type-continuum game first collapses to one scalar—the order parameter ϱ that carries Thread B—and so it is both the simplest structured network and the worked example we return to whenever an explicit graphon mean field game is wanted. The previous subsection handed it to us as the $k = 1$ end of the rank scale; we now show that this one end is special enough to deserve its own development, because at rank one the entire apparatus of coupled type-indexed control problems folds into a single fixed-point equation in a single real unknown.

Example 15.11 (Rank-one graphon)**.** Let $f\colon [0, 1] \to [0, 1]$ be measurable (a *type profile*), and set

$$W(x, y) = f(x)\, f(y).$$

This is symmetric, measurable, and $[0,1]$-valued (since f is), hence a graphon: the *rank-one* or *Curie–Weiss* graphon. The value $f(x)$ is the propensity of type x to form links; an edge between x and y forms with probability proportional to the product of their propensities. In the sampling picture (chapter 19) this is the expected-degree / Chung–Lu model, and it is the graphon analogue of a configuration model in which each vertex carries a single "activity" $f(x)$.

The reading to internalize is that a single number per type controls everything. Each type carries one scalar, its propensity $f(x) \in [0,1]$ to engage at all, and the kernel says that two types connect in proportion to the product of their propensities and in no other way: there is no residual "chemistry" between specific pairs, no term in W that survives factoring out $f(x)$ and $f(y)$. A high-f type is a hub—it links readily to everyone, and most readily to other hubs—while a type with $f(x) \approx 0$ is nearly isolated, linking weakly to all. To make the multiplicative law concrete, two types with propensities 0.9 and 0.2 connect with probability 0.18, the same probability with which a single type of propensity 0.6 links to one of propensity 0.3; only the product is observable. The degree of a type follows the same one-number logic. The expected fraction of the population that a type-x agent connects to is, by definition 15.4's sampling reading,

$$d_W(x) = \int_0^1 W(x,y)\ \mathrm{d}y = f(x)\int_0^1 f(y)\ \mathrm{d}y = \bar{f}\, f(x), \qquad \bar{f} := \int_0^1 f,$$

so the degree function is the profile itself rescaled by its own mean $\bar{f}$: the *expected-degree* or *Chung–Lu* reading, in which a vertex's connectivity is prescribed by a single activity parameter and the network is the maximum-entropy graph with those expected degrees. Sampling an actual finite graph from this kernel—placing each edge independently with probability $f(x_i)f(x_j)$ over i.i.d. types—is deferred to chapter 19; here the kernel is studied as an object in its own right, and what we extract from it is purely analytic.

This is the graphon that carries Thread B (the graphon LQG game of appendix A.7) and the one on which the entire Part IV machine becomes a one-parameter problem. The mechanism is the rank-one collapse already recorded operator-side in worked computation 2.28(b); we now make it explicit as the coupling that a type-x agent will feel, which is the form in which it enters every graphon mean field game equation. The worked computation that follows is the chapter's Thread-B installment, and it is worth doing in full because every later rank-one calculation in the book—static in chapter 20, dynamic in chapter 22, well-posed in chapter 24—begins from exactly this collapse.

Worked computation 15.12 (Rank-one collapse to a scalar order parameter — Thread B foreshadow)**.** Carry the standing graphon LQG model of appendix A.7 (Thread B). There a type-x agent does not feel the global mean but the *graphon-aggregated neighbor mean*

$$z(x) = (\mathbb{W}\bar{m})(x) = \int_0^1 W(x,y)\,\bar{m}(y)\ \mathrm{d}y, \qquad \bar{m}(y) = \int_{\mathbb{R}} \xi\, m^y(\mathrm{d}\xi),$$

where $m^y \in \mathcal{P}(\mathbb{R})$ is the state distribution of type-y agents and $\bar{m}(y)$ its mean (we suppress the time argument, which plays no role in the algebra). This $z(x)$ is the sole channel through which the network reaches the type-x agent: the agent solves a private control problem in which the population enters only as the scalar field $z(x)$ it is drawn toward. Without further structure $z(x)$ is a genuine function of x—one coupling value per type, a continuum of them, all entangled because

$\bar{m}$ is itself the equilibrium output of every type's problem. Specialize now to the rank-one kernel $W(x, y) = f(x)f(y)$ and watch the continuum collapse. Substituting and *pulling the x-dependent factor out of the y-integral*—legitimate because $f(x)$ does not depend on the integration variable y—gives

$$z(x) = \int_0^1 f(x)\, f(y)\, \bar{m}(y)\ \mathrm{d}y = f(x) \underbrace{\int_0^1 f(y)\, \bar{m}(y)\ \mathrm{d}y}_{=:\ \varrho} = \varrho\, f(x), \qquad \varrho = \langle f, \bar{m}\rangle_{L^2[0,1]}.$$

The integral that remains, $\int_0^1 f(y)\bar{m}(y)\ \mathrm{d}y$, contains no x at all: it is one fixed real number, the L^2 inner product $\varrho = \langle f, \bar{m}\rangle$ of the profile against the population mean field. The entire type-indexed coupling field $x \mapsto z(x)$ has therefore collapsed onto a *single scalar* ϱ multiplying the fixed, known profile f. Every agent, regardless of type, feels the *same* order parameter ϱ, only rescaled by the deterministic gain $f(x)$ it was born with: a type with $f(x) = 0$ is decoupled—it feels nothing and contributes nothing to ϱ—while a type with large $f(x)$ is strongly exposed and, symmetrically, weighs heavily in ϱ. The profile f thus plays a double role, as the gain through which a type *feels* the aggregate and the weight with which it *enters* the aggregate, and these are the same f precisely because the kernel is the symmetric outer product $f \otimes f$. The one number ϱ is the only population statistic anyone needs; the full field $\bar{m}(\cdot)$, an element of an infinite-dimensional space, matters only through its single projection onto f. We write the scalar with the book's order-parameter symbol ϱ (appendix A.7); the dynamic chapters chapters 20 and 22 carry the same ϱ, never an η or θ.

This is the graphon analogue of the Curie–Weiss closure of Thread A, and it is what makes the rank-one graphon the cleanest worked graphon mean field game. The self-consistency that chapter 22 will close is now visible in one move. Each agent's problem depends on its type only through the scalar coupling $\varrho f(x)$ together with its own type data—its initial law m_0^x and any type-dependent coefficients—so once the single number ϱ is declared, type x's problem is a *decoupled* standard control problem with a known target $\varrho f(x)$. Solving it returns that type's optimal behavior and hence its equilibrium state mean, which therefore depends on the type and on the one scalar it was told:

$$\bar{m}(x) = \Psi(x, \varrho),$$

where Ψ is the best-response map of the type-x control problem (in the LQG case of chapter 22 it is explicit, linear in ϱ, with coefficients built from the Riccati data). Feeding this back into the *definition* of ϱ closes the loop: ϱ was $\langle f, \bar{m}\rangle = \int_0^1 f(x)\bar{m}(x)\ \mathrm{d}x$, and substituting $\bar{m}(x) = \Psi(x, \varrho)$ turns that definition into the *one-dimensional* fixed-point equation

$$\varrho = \int_0^1 f(x)\, \Psi(x, \varrho)\ \mathrm{d}x \ =:\ \Gamma(\varrho).$$

A continuum of coupled control problems, one per type, has been reduced to a single scalar equation $\varrho = \Gamma(\varrho)$ for one real unknown—exactly the reduction promised for the rank-one kernel in appendix A.7. This is the entire payoff of rank one: the equilibrium is a root of a real function Γ of a real variable, to be located by the elementary fixed-point tools of one dimension rather than by an infinite-dimensional fixed point over the space of type-indexed fields.

It is worth confirming that this machinery degenerates correctly to the homogeneous theory. Take the constant graphon, which is the rank-one kernel with the *flat* profile $f \equiv \sqrt{p}$ (so that

$W(x,y) = \sqrt{p}\,\sqrt{p} = p$ recovers the Erdős–Rényi value of example 15.9). Then the order parameter is

$$\varrho = \langle f, \bar{m} \rangle = \int_0^1 \sqrt{p}\,\bar{m}(y)\,\mathrm{d}y = \sqrt{p}\,\langle \mathbf{1}, \bar{m} \rangle = \sqrt{p}\,\langle \bar{m} \rangle, \qquad \langle \bar{m} \rangle := \int_0^1 \bar{m}(y)\,\mathrm{d}y,$$

where $\langle \bar{m} \rangle$ is the single *global* population mean, averaged over all types. The coupling each agent feels then collapses one step further, losing even its type dependence:

$$z(x) = \varrho\, f(x) = \big(\sqrt{p}\,\langle \bar{m} \rangle\big)\sqrt{p} = p\,\langle \bar{m} \rangle,$$

the *same* number for every type. This is precisely the anonymous global-mean coupling of Thread A—each agent is drawn toward the one population average $\langle \bar{m} \rangle$—now with the interaction strength rescaled by the edge density $p = \|\mathbb{W}\|$, matching the constant-graphon aggregate computed in chapter 14. The graphon machinery has reproduced the homogeneous mean field game exactly, as it must, with p as the only trace the network leaves. The rank-one collapse is thus the genuine generalization of Thread A's mean field closure: a flat f gives one type-independent scalar, a non-flat f gives one scalar weighted by the propensity profile, and in both cases a single real order parameter closes the game.

We carry out the dynamic version of this computation, with the explicit LQG best response Ψ and its Riccati ingredients, in chapter 22; the existence and uniqueness of the scalar fixed point $\varrho = \Gamma(\varrho)$ is controlled by the operator norm $\|\mathbb{W}\| = \|f\|_{L^2}^2$—the single nonzero eigenvalue of the rank-one operator, computed in worked computation 2.28(b)—through the smallness assumption (A-Coupling), and is treated in chapter 24. Here we have only run the algebra of the collapse, but it is the algebra on which the whole rank-one thread rests.

The rank-one example also exhibits, in the simplest possible setting, the link between *centrality* and *exposure* that drives the static and dynamic graphon games of Part IV, and the collapse just performed explains why the link is so tight here. The integral operator $\mathbb{W}$ of the rank-one kernel has a single nonzero eigenvalue $\|f\|_{L^2}^2$ with eigenfunction $e_1 = f/\,\|f\|_{L^2}$ (worked computation 2.28(b))—the normalized profile f itself, and the only mode the network possesses. Centrality, meaning how prominently a type sits in the network's dominant mode, is read off from e_1 and so is governed by $f(x)$. Exposure, meaning how strongly a type feels the aggregate $z(x) = \varrho f(x)$, is governed by the *same* $f(x)$, as the worked computation showed directly. The two notions therefore coincide pointwise: $f(x)$ answers both "how central is type x" and "how strongly does type x feel the crowd," because with a single mode there is nothing to distinguish them—the one eigenfunction does both jobs. This coincidence is special to rank one. For a general graphon the spectrum has many modes, centrality and exposure are then governed by the full eigenbasis rather than a single function, and they separate; that spectral story is the subject of chapter 18, and the precise map from centrality to equilibrium effort is made into a theorem in chapter 20.

The whole gallery just assembled—the constant kernel, the block models, and this rank-one collapse—was built under two normalizations we have so far taken for granted: the type distribution is uniform on $[0,1]$ (every type profile f, every block, was weighted by Lebesgue measure), and the kernel is valued in $[0,1]$ (every propensity, every block probability, was an honest edge probability). Real data violate both—some kinds of agent are common and others rare, and edge weights can exceed one or be unbounded—so the natural next question is whether loosening these two normalizations enlarges the theory or merely re-coordinatizes it. That is exactly what the next section takes up: node weights, which relax the uniform type distribution, and edge weights, which relax the $[0,1]$ range.

15.5 Node- and edge-weighted generalizations

The gallery's closing paragraph put the two standing normalizations of definition 15.4 on the table and asked whether dropping them enlarges the theory. Both deserve the scrutiny. The uniform type distribution hid that real populations are not uniform in type—some kinds of agent are common, others vanishingly rare—and the $[0,1]$ range hid that real interactions need not be confined to it: a tie can be antagonistic, hence negative, or a transaction arbitrarily large, hence unbounded. The honest question is whether dropping these two normalizations produces a genuinely larger theory or merely a re-coordinatization of the one we have. The answer sorts the two relaxations cleanly, and each points at a named later development. Loosening uniformity—attaching *node weights*—adds nothing new in the dense theory: the weighting is absorbed back into the kernel, and what looks like a multi-population model is the same graphon read on a re-coordinatized type space. Loosening the range—admitting *edge weights*—splits in two: a benign bounded regime in which the whole machinery survives word for word, and a genuinely harder unbounded regime that no bounded kernel can model and that forces the sparse theory of chapter 27. We take the two relaxations in turn, and close by disposing of self-loops in a single line.

Node weights: a non-uniform type distribution. Begin with uniformity. A *node-weighted* graph attaches to each vertex i a weight $\rho_i > 0$, normalized so that $\sum_i \rho_i = 1$, recording what fraction of the population is of that kind—a model in which not all vertices count equally, as when a single "vertex" in a coarse description already stands for a whole sub-population. The pixel construction of definition 15.2 adapts immediately, and the only change is that the tiles are no longer congruent: instead of cutting $[0,1]$ into n equal cells we give vertex i the interval J_i of *length* ρ_i, laying the J_i end to end to tile $[0,1]$ (so $J_i = (\rho_1 + \cdots + \rho_{i-1},\, \rho_1 + \cdots + \rho_i]$), and we set $W(x,y) = A_{ij}$ for $x \in J_i$, $y \in J_j$. A heavily weighted vertex now occupies a correspondingly wide stripe of the square, so that integrating against Lebesgue measure on $[0,1]$ automatically weights each vertex by its ρ_i—the unequal tiling has encoded the node weights into the *geometry* of the picture. Passing to the limit, the surviving object is no longer a bare kernel but a *pair* (W,ν): a symmetric kernel W together with a probability measure ν on the type space $[0,1]$, the *type distribution*, recording how the population is spread across types. The constant from the unweighted theory, where ν was Lebesgue, has become a free piece of data.

The central claim is that this apparent enlargement is illusory in the dense theory: one can always *push the node weights into the kernel* and return to the uniform convention, recovering a plain graphon on $([0,1],\mathrm{Leb})$ that carries the same network content. The mechanism by which this is done depends on whether ν has atoms, and the two cases are genuinely different in character—one is a smooth change of variables, the other a piecewise tiling—so we narrate them separately.

The atomless case: re-coordinatize by the quantile transform. Suppose first that ν has no atoms, so no single type carries positive mass and the population is smeared continuously across $[0,1]$. Its cumulative distribution function

$$F_\nu(x) = \nu([0,x])$$

is then continuous and nondecreasing, running from 0 to 1 as x sweeps $[0,1]$. The point of F_ν is that it *spreads ν out to be uniform*: a sub-interval on which ν piles up a lot of mass is stretched by F_ν into a long sub-interval of the image, while a region ν barely charges is compressed, and the net effect is to even the mass out perfectly. This is the *probability-integral transform* of undergraduate probability—if a random type x is drawn from ν, then $F_\nu(x)$ is uniformly distributed on $[0,1]$—and stated as an identity it says that F_ν is a measure-preserving map from $([0,1],\nu)$ to $([0,1],\mathrm{Leb})$,

$$\nu(F_\nu^{-1}(B)) = \mathrm{Leb}(B) \qquad \text{for every Borel } B \subseteq [0,1], \tag{15.1}$$

i.e. the pushforward $(F_\nu)_\# \,\nu$ is Lebesgue measure. (The inverse here is the quantile function, the left-continuous inverse of F_ν; on the ν-null set where F_ν is flat the choice of inverse is immaterial precisely because that set carries no ν-mass.) Now transport the kernel along this map: define

$$\widetilde{W}(u,v) = W\big(F_\nu^{-1}(u),\, F_\nu^{-1}(v)\big), \qquad (u,v) \in [0,1]^2.$$

By the change-of-variables identity that (15.1) encodes, every double integral of W against $\nu \otimes \nu$ equals the corresponding integral of $\widetilde{W}$ against Leb$\otimes$Leb, so $\widetilde{W}$ is an ordinary graphon on $([0,1],\mathrm{Leb})$ producing exactly the same operator, the same edge density, and the same homomorphism densities as the weighted pair (W,ν). The node weights have vanished into the relabeling of the type axis. One scope note is owed here, in the self-contained spirit of the book: this is the *elementary CDF transform on the interval* and nothing more. We are using only that a continuous nondecreasing F_ν on $[0,1]$ has a quantile inverse satisfying (15.1); we are *not* invoking the general theorem that any two atomless standard Borel probability spaces are measurably isomorphic, which is a heavier statement we neither need nor prove.

The atomic case: tile by sub-intervals. The quantile transform stalls the instant ν has an atom, and it is worth seeing exactly why, because the obstruction is real and not a defect of the argument. If a single type x_i carries positive mass ρ_i, then any measure-preserving map to $([0,1],\mathrm{Leb})$ would have to send that one point to a set of Lebesgue measure $\rho_i > 0$; but a *bijection* sends a point to a point, and a point has Lebesgue measure zero. No measure-preserving bijection can therefore exist. This is precisely the case that matters in practice, since a genuinely finite type space—k populations, k communities—is purely atomic. The resolution is not a smooth change of variables but the interval-tiling we already built: spread the atom of mass ρ_i over the sub-interval J_i of length ρ_i and read the kernel as the step graphon that is constant on each block $J_i \times J_j$. A point of positive ν-mass becomes an interval of equal Lebesgue measure, and the map $x \mapsto J_i$ is many-to-one (not a bijection) exactly where it must be. The canonical instance is the stochastic block model of example 15.10 read as a k-population model: ν puts mass ρ_r on community C_r, the tiling assigns C_r an interval of length ρ_r, and the result is the step graphon of definition 15.6 with unequal blocks. The unequal block sizes *are* the node weights, now stored in Lebesgue geometry.

Either route—the quantile transform when ν is atomless, the interval tiling when it is atomic, and a routine combination on the mixed case—lands the weighted pair (W,ν) back on $([0,1],\mathrm{Leb})$ with its entire network content intact. The conclusion is the one promised at the head of the section: *node weights add no genuinely new object in the dense theory.* They are absorbed by re-coordinatizing the type axis, which is the very relabeling freedom of remark 15.5—the continuum analogue of permuting vertices, here used to permute the *amounts* of each type rather than their names. Accordingly we keep, as we have throughout, the convention that unless stated otherwise types are distributed uniformly on $[0, 1]$; a non-uniform ν is never an obstacle, only a reminder to absorb it first.

Edge weights: kernels beyond $[0, 1]$**.** Loosening the second normalization is not so benign, and the theory genuinely forks. An *edge-weighted* graph replaces the entries $A_{ij} \in \{0, 1\}$ by real numbers $w_{ij} \in \mathbb{R}$, modeling interactions that an edge probability cannot express: a negative weight for an antagonistic or competitive tie, a large positive weight for a heavy traffic volume or a large transaction. Its pixel picture is a symmetric measurable kernel now valued in $\mathbb{R}$ rather than $[0, 1]$, and whether this stays inside the theory of this book turns entirely on whether that kernel is *bounded.*

If it is—if $W : [0, 1]^2 \to \mathbb{R}$ is symmetric with $\|W\|_{L^\infty} \le M$ for some finite M—then nothing of substance changes. Every argument of this chapter, every line of the operator theory of chapter 18, and the stepping operator of proposition 15.7 go through verbatim with the range $[0, 1]$ replaced by $[-M, M]$. The reason is that none of those arguments used the *positivity* of the kernel or the specific upper bound 1; they used only symmetry and a finite sup-norm. Concretely, $\mathbb{W}$ is still self-adjoint Hilbert–Schmidt (proposition 2.24 invoked boundedness and symmetry alone, never $W \ge 0$), its spectrum is still real, and the stepping projection still returns a step kernel because a block average of values in $[-M, M]$ again lies in $[-M, M]$—the only property of the interval the averaging used was convexity, which $[-M, M]$ shares with $[0, 1]$. So the bounded symmetric kernel is the natural home of the whole dense theory, $[0, 1]$ being merely the special case of a propensity. What the signed case *adds* is not a difficulty but a genuine modeling distinction that the $[0, 1]$ convention had hidden. The Lasry–Lions monotonicity condition on a coupling (appendix A.6, (A-Mon)) asks for a definite sign, and with signed kernels that sign is a real choice: a positive coupling makes the interaction *attractive* (agents are drawn toward the aggregate they feel), a negative one makes it *repulsive* (they are pushed away, as in a congestion or competition term). Whether a graphon game has a unique equilibrium or several can hinge on exactly this sign, and that is where the distinction is cashed out, in chapters 20 and 24.

The fork's other branch is the unbounded one, and here the theory of this chapter genuinely runs out. Some networks cannot be modeled by any bounded kernel at all: sparse, heavy-tailed networks, in which a few hub types carry unboundedly many connections, have a degree distribution with no finite ceiling, and a bounded W produces only bounded degrees (we will define the degree function in section 15.6 and see it is $\int_0^1 W(x, \cdot)$, manifestly at most M). Capturing such networks forces a kernel with $\|W\|_{L^\infty} = \infty$, controlled instead only in some $L^p([0, 1]^2)$ with $p < 2$. The threshold genuinely sits at $p = 2$: on the finite measure space $[0, 1]^2$ the inclusions run the other way, $L^p \subseteq L^2$ for every $p \ge 2$, so a kernel that is still square-integrable remains Hilbert–Schmidt and hence a bounded operator on L^2—it is only when the kernel drops below square-integrability, into L^p with $p < 2$ (or is merely measurable), that the picture can break. The price of falling below that threshold is that the clean Hilbert–Schmidt picture degrades: such a kernel need no longer induce a bounded operator on $L^2[0,1]$, so the self-adjoint compact operator on which all of chapter 18 rests is no longer guaranteed, and the function-analytic repairs—operator-norm versus L^p-norm control of $\mathbb{W}$, and the revised existence and uniqueness theory that follows—are deferred, honestly, to chapter 27. We do not attempt them here. The order of development is deliberate and we state it plainly: the dense bounded theory comes first because it is the regime in which complete theorems exist, and the sparse unbounded theory is built later as a separate enterprise rather than smuggled in as a special case.

Loops and the diagonal. One last piece of data a real network might carry is self-loops, $A_{ii} \neq 0$, an agent's interaction with itself. In the dense graphon theory these are simply invisible, and the reason is the almost-everywhere principle already established in section 15.2: a self-loop changes the kernel only on the diagonal $\{x = y\}$, which is a Lebesgue-null subset of $[0,1]^2$, and a null set changes no integral. Hence loops alter *no* value of $\mathbb{W}$, *no* edge density, and *no* homomorphism density—every quantity we ever compute is an integral, and the integral cannot see the diagonal. We may therefore record self-links or zero them out at will, exactly as we did when disposing of the diagonal in definition 15.4. Loops return to relevance only in the sparse theory of chapter 27, where the reference measure is no longer Lebesgue and the diagonal can carry positive mass; in the dense bounded world of this chapter they cost us nothing and we ignore them.

With both relaxations accounted for, the graphon is back in its normalized form, and we can say precisely what we hand the next section. Node weights have been re-coordinatized away, so the type distribution is uniform on $[0,1]$; edge weights have been sorted, and we keep the branch on which the theory is complete, the *bounded symmetric* kernels, with the unbounded L^p regime quarantined in chapter 27. What remains is exactly the object the operator theory wants: a symmetric kernel, bounded, read on $[0,1]$ with Lebesgue measure—and, by proposition 2.24, precisely the clean self-adjoint Hilbert–Schmidt operator $\mathbb{W}$ that the final section now anatomizes as the single channel through which network structure enters every graphon mean field game.

15.6 The graphon operator: the coupling in every GMFG equation

The previous section handed us the graphon in its fully normalized form: a symmetric kernel, bounded, read on $[0,1]$ against Lebesgue measure—node weights re-coordinatized away into uniformity, edge weights restricted to the bounded branch on which the theory is complete. By proposition 2.24 this is exactly the data that makes the integral operator $\mathbb{W}$ a clean self-adjoint Hilbert–Schmidt map on $L^2[0,1]$. The whole chapter has been a march toward this operator: a network became an adjacency matrix, the matrix became a pixel picture, the picture's limit became a kernel, and now the kernel becomes the one operator the rest of the book actually uses. This closing section says *why* that operator, and nothing else about the network, is what the games of Part IV will see.

The claim is sharp and worth stating before any formalism, because it organizes everything that follows. In every graphon mean field game to come—static (chapter 20), dynamic (chapters 22 and 23), at equilibrium (chapter 24)—a type-x agent never feels the network "directly." It feels a single scalar, the value at its own type of a population aggregate, and that aggregate is always built the same way: take whatever statistic $h\colon [0,1] \to \mathbb{R}$ the agents respond to—a mean state, a density of chosen actions, a congestion level, indexed by the type of the agent producing it—and pass it through the kernel,

$$z(x) = (\mathbb{W}\, h)(x) = \int_0^1 W(x, y)\, h(y)\ \mathrm{d}y.$$

The integral is a weighted average of the statistic over the whole population, the weight $W(x, y)$ being precisely the propensity of a type-x agent to be linked to a type-y one. Where a classical mean field game would hand the agent the flat population average $\int h$—every other agent counted equally—the graphon game hands it the *network-weighted* average $(\mathbb{W}h)(x)$, and the difference between the two is the entire content of "the agents interact through a network" rather than through an anonymous crowd. The constant graphon $W \equiv p$ collapses $(\mathbb{W}h)(x) = p \int h$ back to a multiple of the flat average, which is exactly why the classical theory is the constant-graphon special case.

The decisive structural fact is the one the displayed formula already shows: the graphon enters *only* through the linear map $h \mapsto \mathbb{W}h$. No equation of Part IV ever evaluates W at a point, or asks for any feature of the kernel that the operator does not already carry. Consequently, "understand the network's effect on the game" and "understand $\mathbb{W}$ as a linear operator on type-fields" are the same task, and we can dispatch the first by doing the second once and for all. We do exactly that here: assemble the elementary operator facts—norm, degree, positivity, monotonicity—that the later chapters invoke by name, leaving the finer spectral anatomy, which deserves its own development, to chapter 18.

We recall the operator from definition 2.23 rather than redefine it.

Definition 15.13 (Graphon operator, recalled)**.** The *graphon operator* $\mathbb{W}\colon L^2[0,1] \to L^2[0,1]$ of a graphon W is $(\mathbb{W}f)(x) = \int_0^1 W(x, y) f(y)\ \mathrm{d}y$ (definition 2.23). Under (A-Reg-W) it is self-adjoint Hilbert–Schmidt on $L^2[0,1]$ with $\|\mathbb{W}\| \le \|\mathbb{W}\|_{\mathrm{HS}} = \|W\|_{L^2([0,1]^2)} \le 1$ (proposition 2.24), and its spectrum is real, discrete away from 0, and treated in full in chapter 18.

The one network statistic we attach to $\mathbb{W}$ now, because it controls the coupling bounds of Part IV directly, is the degree function.

Definition 15.14 (Degree function)**.** The *degree function* of a graphon W is

$$d_W(x) = \int_0^1 W(x, y)\ \mathrm{d}y = (\mathbb{W}\mathbf{1})(x),$$

the expected fraction of the population to which a type-x agent links. For a graphon, $d_W(x) \in [0, 1]$.

Three readings make this object concrete. First, its meaning: $d_W(x)$ is the result of applying the coupling to the constant unit field, $\mathbb{W}\mathbf{1}$, which is the special case of the aggregate $z = \mathbb{W}h$ in which every agent contributes the same unit statistic $h \equiv 1$. It is therefore the total network "weight" a type-x agent feels—the expected fraction of the population it links to—and nothing else. Second, its range: because $0 \le W \le 1$, the average $\int_0^1 W(x, y)\ \mathrm{d}y$ of a quantity bounded by 1 over a set of total measure 1 lands in $[0, 1]$, so d_W is itself a $[0, 1]$-valued function on the type space—a degree, sensibly, is a fraction. Third, its provenance: it is the exact continuum limit of the normalized vertex degrees $\frac{1}{n}\sum_j A_{ij}$ of the underlying graphs. For a pixel graphon W^G this is visible by direct computation—on the block $I_i^{(n)}$ the kernel equals A_{ij} for $y \in I_j^{(n)}$, so

$$d_{W^G}(x) = \int_0^1 A_{i\,\iota_n(y)}\,\mathrm{d}y = \sum_{j=1}^n A_{ij}\left|I_j^{(n)}\right| = \frac{1}{n}\sum_{j=1}^n A_{ij} \qquad (x \in I_i^{(n)}),$$

so d_{W^G} is constant on each block $I_i^{(n)}$ and equals there exactly the degree fraction of vertex i. The degree function is the macroscopic shadow of the degree sequence.

Averaging the degree over the population closes a loop we opened in section 15.1. By Fubini–Tonelli,

$$\int_0^1 d_W(x)\,\mathrm{d}x = \int_0^1\int_0^1 W(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_{[0,1]^2} W = \langle \mathbb{W}\mathbf{1}, \mathbf{1}\rangle,$$

which says three things are one and the same number: the mean degree, the edge density of proposition 15.3, and the quadratic form of $\mathbb{W}$ on the constant field. The edge density that we first met as an edge count $2\,|E|\,/n^2$, then as a homomorphism density, now appears a third time in purely operator language as $\langle \mathbb{W}\mathbf{1}, \mathbf{1}\rangle$—the same scalar read off three faces of the same object, exactly the chapter's recurring lesson that the network reaches everything downstream through $\mathbb{W}$ alone.

The degree function is not only a summary statistic; it gives the sharpest elementary bound on the operator norm, by a Schur-test argument that we include in full because it is short, self-contained, and used directly in the contraction estimates of chapter 24.

Lemma 15.15 (Degree (Schur) bound on the operator norm)**.** *For any graphon W (more generally any symmetric kernel with $W \geq 0$),*

$$\|\mathbb{W}\| \;\leq\; \operatorname*{ess\,sup}_{x\in[0,1]} d_W(x) \;=\; \|d_W\|_{L^\infty} \;\leq\; \|W\|_{L^\infty} \leq 1.$$

Moreover $\mathbb{W}\mathbf{1} = d_W$, so if $d_W \equiv c$ is constant then $\|\mathbb{W}\| = c$ and the bound is attained.

Proof. Write $D = \|d_W\|_{L^\infty}$, so that $d_W(x) \leq D$ for a.e. x. We must show $\|\mathbb{W}f\|_{L^2} \leq D\,\|f\|_{L^2}$ for every $f \in L^2[0,1]$, since the operator norm is by definition the smallest such constant.

The one idea in the proof is to split the kernel symmetrically before applying Cauchy–Schwarz, so that one factor of W supplies the degree and the other stays married to f. Concretely, inside the integrand of $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y$ factor $W = W^{1/2}\cdot W^{1/2}$ (legitimate because $W \geq 0$, the one place positivity of the kernel is used) and regard the two halves as the two functions of y entering the Cauchy–Schwarz inequality $\left(\int gh\right)^2 \leq \left(\int g^2\right)\left(\int h^2\right)$, with $g = W(x,\cdot)^{1/2}$ and $h = W(x,\cdot)^{1/2}f$. Then $g^2 = W$ and $h^2 = Wf^2$, so for a.e. x,

$$\big((\mathbb{W}f)(x)\big)^2 = \left(\int_0^1 W(x,y)^{1/2}\cdot W(x,y)^{1/2}f(y)\,\mathrm{d}y\right)^2 \leq \underbrace{\left(\int_0^1 W(x,y)\,\mathrm{d}y\right)}_{=d_W(x)}\left(\int_0^1 W(x,y)f(y)^2\,\mathrm{d}y\right),$$

the first factor being exactly the degree $d_W(x)$. This is the pointwise inequality

$$(\mathbb{W}f)(x)^2 \leq d_W(x)\int_0^1 W(x,y)f(y)^2\,\mathrm{d}y.$$

Now integrate over $x \in [0,1]$. Bounding the degree factor by its supremum $d_W(x) \le D$ and then exchanging the order of integration—Fubini–Tonelli (theorem 1.27) applies because the integrand $W f^2 \ge 0$ is nonnegative, so Tonelli licenses the swap unconditionally—gives

$$\|\mathbb{W}f\|_{L^2}^2 = \int_0^1 (\mathbb{W}f)(x)^2 \,\mathrm{d}x \le D \int_0^1\int_0^1 W(x,y)f(y)^2 \,\mathrm{d}y\,\mathrm{d}x = D\int_0^1 f(y)^2\Big(\int_0^1 W(x,y)\,\mathrm{d}x\Big)\,\mathrm{d}y.$$

The inner integral is where symmetry earns its keep: $\int_0^1 W(x,y)\,\mathrm{d}x = \int_0^1 W(y,x)\,\mathrm{d}x = d_W(y)$, because integrating the kernel over its *first* slot would not be the degree, but symmetry $W(x,y) = W(y,x)$ turns it into an integral over the second slot, which is. Hence

$$\|\mathbb{W}f\|_{L^2}^2 \le D\int_0^1 f(y)^2\, d_W(y)\,\mathrm{d}y \le D\cdot D\int_0^1 f(y)^2\,\mathrm{d}y = D^2\,\|f\|_{L^2}^2\,,$$

using $d_W \le D$ a second time. Taking square roots gives $\|\mathbb{W}f\|_{L^2} \le D\,\|f\|_{L^2}$, hence $\|\mathbb{W}\| \le D = \|d_W\|_{L^\infty}$. The remaining inequalities are pointwise: $d_W(x) = \int_0^1 W(x,y)\,\mathrm{d}y \le \|W\|_{L^\infty}\int_0^1 \mathrm{d}y = \|W\|_{L^\infty} \le 1$, so $\|d_W\|_{L^\infty} \le \|W\|_{L^\infty} \le 1$.

For the attainment, recall $\mathbb{W}\mathbf{1} = d_W$ (definition 15.14). If the degree is constant, $d_W \equiv c$, then $\mathbb{W}\mathbf{1} = c\mathbf{1}$, so the constant field is an eigenvector with eigenvalue c; since $\|\mathbf{1}\|_{L^2} = 1$ this forces $\|\mathbb{W}\| \ge \|\mathbb{W}\mathbf{1}\|_{L^2} = c$, which together with the upper bound $\|\mathbb{W}\| \le \|d_W\|_{L^\infty} = c$ gives $\|\mathbb{W}\| = c$. The bound is exact precisely when the network is degree-regular. □

Two worked checks calibrate the lemma, one where it is exact and one where it is not. For the constant (Erdős–Rényi) graphon $W \equiv p$ the degree is constant, $d_W \equiv p$, so the attainment clause applies verbatim and the lemma returns $\|\mathbb{W}\| = p$ on the nose—recovering the operator norm computed spectrally in worked computation 2.28(a). This is the degree-regular case, and the Schur bound is tight exactly because every row of the kernel has the same mass.

The rank-one (Curie–Weiss) graphon $W(x,y) = f(x)f(y)$ of section 15.4.3 shows the other side. Its degree is $d_W(x) = f(x)\int_0^1 f$, no longer constant unless f is, so the Schur bound reads

$$\|\mathbb{W}\| \le \|d_W\|_{L^\infty} = \|f\|_{L^\infty}\int_0^1 f,$$

whereas the exact operator norm, found in worked computation 2.28(b) by noting that f itself is the lone eigenfunction, is $\|\mathbb{W}\| = \|f\|_{L^2}^2 = \int_0^1 f^2$. Cauchy–Schwarz guarantees $\int f^2 \le \|f\|_\infty \int f$, so the bound is always correct; it is tight only when f is constant. A single number makes the gap concrete. Take $f(x) = x$, so $W(x,y) = xy$ is a legitimate graphon (≤ 1 on the square). Then $\int_0^1 f = \frac{1}{2}$, $\|f\|_\infty = 1$, and $\int_0^1 f^2 = \frac{1}{3}$, so the Schur bound gives $\|\mathbb{W}\| \le \frac{1}{2}$ while the truth is $\|\mathbb{W}\| = \frac{1}{3}$: a correct estimate that overshoots by fifty percent. The reason is structural and worth naming, because it recurs whenever one uses a row-sum bound. The Schur estimate controls the operator by the *largest* row mass $\sup_x d_W(x)$, as if the worst-connected type set the norm; but the true norm is a global, energy-weighted quantity that the high-degree types dominate only in proportion to how much of the eigenvector's mass sits on them. Off the degree-regular case the heaviest row overstates the operator, and the lemma cheerfully reports a ceiling rather than the value.

Even so, a ceiling is exactly what Part IV usually needs, and that is the lemma's real payoff. The existence and uniqueness theorems of the graphon games invoke an abstract smallness condition on the coupling, (A-Coupling): $\|\mathbb{W}\| < \kappa$ for a threshold κ set by the cost data. As stated this is a condition on a spectral radius, awkward to verify for a kernel given by a formula or estimated from data. The Schur bound converts it into something one can read off the network's degrees: if *every* type has degree fraction below κ, that is $\|d_W\|_{L^\infty} < \kappa$, then $\|\mathbb{W}\| < \kappa$ automatically and (A-Coupling) holds. A verifiable degree condition replaces an abstract operator-norm condition, and the over-conservatism we just exhibited is the harmless price of that convenience—a sufficient condition can afford to be generous.

Two further structural properties of $\mathbb{W}$ are used throughout Part IV. They are not deep—both follow in a line from the fact that $W \geq 0$ and that integration against a nonnegative kernel is an averaging—but they are exactly the properties that the existence theory leans on, so it is worth isolating them and seeing precisely why each holds. The first says the coupling cannot turn a nonnegative statistic negative, nor reverse an inequality between two statistics; the second says it cannot amplify a uniformly bounded statistic. Both are statements that $\mathbb{W}$ *respects order*, which is what monotone fixed-point arguments require of it.

Proposition 15.16 (Positivity and monotonicity of the coupling)**.** *Let W be a graphon and $\mathbb{W}$ its operator.*

(i) (Positivity preserving.) *If $f \geq 0$ a.e. then $\mathbb{W}f \geq 0$ a.e.; consequently $f \leq g$ a.e. implies $\mathbb{W}f \leq \mathbb{W}g$ a.e.*

(ii) (L^∞ bound.) *$\mathbb{W}$ maps $L^\infty[0,1]$ into itself with $\|\mathbb{W}f\|_{L^\infty} \leq \|d_W\|_{L^\infty} \|f\|_{L^\infty} \leq \|f\|_{L^\infty}$; in particular $\mathbb{W}$ preserves the order interval $[0,1]$ of fields when $d_W \leq 1$.*

Proof. (i) Suppose $f \geq 0$ a.e. Since the kernel is nonnegative, the integrand $W(x,y)f(y)$ is a product of two a.e.-nonnegative quantities, hence ≥ 0 for a.e. y; an integral of a nonnegative function is nonnegative, so $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y \geq 0$ for a.e. x, which is $\mathbb{W}f \geq 0$. Monotonicity is this positivity together with linearity. If $f \leq g$ a.e., then $g - f \geq 0$ a.e., so by what we just proved $\mathbb{W}(g-f) \geq 0$; but $\mathbb{W}$ is linear, so $\mathbb{W}(g-f) = \mathbb{W}g - \mathbb{W}f$, and therefore $\mathbb{W}g - \mathbb{W}f \geq 0$, i.e. $\mathbb{W}f \leq \mathbb{W}g$ a.e. The order on type-fields is preserved.

(ii) Let $f \in L^\infty[0,1]$. For a.e. x, pull the absolute value inside the integral and use $W \geq 0$ to keep it as a weight:

$$|(\mathbb{W}f)(x)| = \Big|\int_0^1 W(x,y)f(y)\,\mathrm{d}y\Big| \leq \int_0^1 W(x,y)\,|f(y)|\,\mathrm{d}y \leq \|f\|_{L^\infty} \int_0^1 W(x,y)\,\mathrm{d}y = \|f\|_{L^\infty}\, d_W(x),$$

the second inequality replacing $|f(y)|$ by its a.e. upper bound $\|f\|_{L^\infty}$ under the nonnegative weight $W(x,\cdot)$, and the final equality being the definition of the degree. Taking the essential supremum over x gives $\|\mathbb{W}f\|_{L^\infty} \leq \|f\|_{L^\infty}\|d_W\|_{L^\infty}$, and since $\|d_W\|_{L^\infty} \leq 1$ for a graphon, $\mathbb{W}$ is a contraction on $L^\infty[0,1]$. In particular, if $0 \leq f \leq 1$ a.e. then by (i) and (ii) $0 \leq \mathbb{W}f \leq \|d_W\|_{L^\infty} \leq 1$, so $\mathbb{W}$ maps the order interval of $[0,1]$-valued fields into itself. □

Each of these properties has a named destination. Positivity guarantees that the graphon aggregate of a nonnegative population statistic—a density, a congestion level—is again nonnegative, the elementary consistency without which the forward Fokker–Planck side of the dynamic system of chapter 22 would be ill-posed. Monotonicity is the order-theoretic input to the monotone fixed-point methods that construct equilibria there and in chapter 24: those arguments iterate the best-response map and need $\mathbb{W}$ to preserve the order between successive iterates, which is exactly what proposition 15.16(i) provides. The degree/Schur bound (lemma 15.15) is what turns the abstract coupling threshold (A-Coupling) into the checkable degree condition that the uniqueness proof of chapter 24 actually verifies. And the self-adjoint Hilbert–Schmidt structure recalled in definition 15.13—real spectrum, orthonormal eigenmodes, discreteness away from zero—is the entry point for chapter 18, where the network is read through the eigenvalues and eigenfunctions of $\mathbb{W}$: spectral gap as mixing rate, top eigenvector as centrality, and finite-rank truncation as a principled low-dimensional reduction of the game.

That last point is the chapter's hidden spine, and it is worth stating as the operator's parting lesson. Because the game sees the network only through $z = \mathbb{W}h$, the effective dimensionality of a graphon mean field game is the rank of the coupling, not the size of the network: the number of order parameters needed to close the game equals the number of modes of $\mathbb{W}$ that the dynamics excite. The rank-one (Curie–Weiss) graphon $W = f(x)f(y)$ is the cleanest instance and the seed of Thread B—there $(\mathbb{W}h)(x) = f(x)\int_0^1 f(y)h(y)\,\mathrm{d}y$ collapses to $z(x) = \varrho\, f(x)$ with the *single* scalar order parameter $\varrho = \langle f, h\rangle$, so an entire continuum of type-indexed agents closes on one number. The spectral reading of chapter 18 is the general form of this collapse: a coupling of rank r closes on r order parameters.

With this the operator side of the graphon is fully assembled, and with it the chapter. We have built one object three ways—a network as a matrix, a matrix as a picture, a picture's limit as a kernel—and shown that all three feed a single linear map $\mathbb{W}$ on type-fields: self-adjoint, Hilbert–Schmidt, positivity-preserving, monotone, of norm at most one, with degree-controlled operator norm. This is the entire interface between "the network" and "the game." Every equation of Part IV is, structurally, a classical mean field game with the anonymous global mean $\int h$ replaced by the network-weighted aggregate $\mathbb{W}h$, and the chapters that follow are, in this light, the systematic study of that one substitution: chapter 18 diagonalizes $\mathbb{W}$, chapters 20, 22 and 23 install it inside the static and dynamic equilibrium systems, and chapter 24 closes those systems under the coupling-smallness threshold this section made verifiable. The graphon has stopped being a picture and become an operator; from here on we compute with it.

Rigor ledger

Proved (in this chapter).

- Every pixel graphon W^G is a symmetric measurable graphon, $\{0,1\}$-valued for an unweighted graph, with $\int_{[0,1]^2} W^G = 2\,|E|\,/n^2$ the edge density (proposition 15.3).
- The stepping operator $\Pi_{\mathcal{P}}$ is the orthogonal projection of $L^2([0,1]^2)$ onto step functions adapted to $\mathcal{P}$; it preserves symmetry, the range $[0,1]$, and all block masses (proposition 15.7). Pixel graphons are exactly the steps to the equipartition.
- Rank-one collapse: for $W(x,y) = f(x)f(y)$ the graphon aggregate $z(x) = (\mathbb{W}\bar{m})(x) = \varrho f(x)$ reduces to a single scalar order parameter $\varrho = \langle f, \bar{m}\rangle$, turning the type-continuum

game into a one-dimensional fixed-point equation (worked computation 15.12); the constant graphon $f \equiv \sqrt{p}$ recovers the classical global-mean coupling $z \equiv p\,\langle \bar{m} \rangle$.

- Degree (Schur) bound $\|\mathbb{W}\| \le \|d_W\|_{L^\infty} \le 1$, attained for constant degree (lemma 15.15); positivity and monotonicity of $\mathbb{W}$, and its L^∞ bound (proposition 15.16).

Assumed (imported, proved elsewhere in the book).

- Lebesgue measure, L^p completeness, and Fubini–Tonelli on $[0,1]$ and $[0,1]^2$ (definition 1.29 and theorems 1.27 and 1.31).
- Hilbert-space geometry: orthogonal projection, Parseval expansion (theorems 2.5 and 2.8).
- That $\mathbb{W}$ is self-adjoint Hilbert–Schmidt with $\|\mathbb{W}\| \le \|W\|_{L^2} \le 1$, and its spectral theorem (definition 2.23, proposition 2.24, and theorem 2.26); the constant and rank-one spectra (worked computation 2.28).
- The standing graphon LQG model and its coupling $z(x) = (\mathbb{W}\bar{m})(x)$ (appendix A.7); standing assumptions (A-Reg-W), (A-Coupling), (A-Mon) (appendix A.6).

Deferred (to a named later chapter).

- The precise meaning and proof of "$W^{G_n} \to W$": homomorphism-density (left) convergence (chapter 16); the cut metric $\delta_\square$, the relabeling quotient, and Lovász–Szegedy compactness (chapter 17).
- The spectral reading of networks (centrality, gap) and low-rank reductions (chapter 18).
- Sampling $G(n, W)$ and concentration; the weak regularity lemma as bounded-complexity step-graphon approximation (chapter 19).
- The dynamic rank-one fixed point with explicit LQG best response, and existence/uniqueness via $\|\mathbb{W}\|$ (chapters 22 and 24); the block-graphon = multi-population reduction (chapter 30).
- Unbounded L^p/sparse kernels and the repair of the Hilbert–Schmidt theory (chapter 27).

Open (genuinely, beyond this book's settled theory).

- Identifiability: recovering a graphon (its relabeling class) from a single observed finite network is only partially understood, and quantitative rates degrade with sparsity; flagged again in chapters 19 and 31.
- A fully satisfactory limit theory for sparse, heavy-tailed networks (graphexes, L^p graphons) remains an active research frontier (chapters 27 and 31).

Bibliographic notes

The pixel picture and the definition of a graphon as a symmetric measurable kernel on $[0,1]^2$ originate in the dense graph-limit theory of Lovász and Szegedy [100] and Borgs, Chayes, Lovász, Sós and Vesztergombi [29]; the authoritative reference for everything in this chapter and the next four is Lovász's monograph [99], with the survey [98] a gentler first pass that this chapter follows in spirit. The reading of the type variable as an *exchangeable* label is made precise by the Aldous–Hoover representation of exchangeable arrays, surveyed for this audience by Diaconis and Janson [60], who also explain why two graphons related by a measure-preserving map encode the same random graph—the analytic root of remark 15.5, taken up properly with the cut metric in chapter 17.

The stepping operator of definition 15.6 is the conditional-expectation / block-averaging operator that underlies the weak and strong regularity lemmas; its role as the engine of approximation by stochastic block models is developed in [99, Ch. 9, Ch. 14], and we return to it in chapter 19. The stochastic block model as a step graphon, and node/edge weights as weighted graph limits, are treated in [99, Ch. 7–8] and [29]; the expected-degree (Chung–Lu) reading of the rank-one graphon is standard in the random-graph literature.

The operator viewpoint—$\mathbb{W}$ as the self-adjoint Hilbert–Schmidt coupling, the degree function, the Schur bound—is the bridge to the mean field game theory of Part IV and is emphasized in the graphon-game line of work: Parise and Ozdaglar's static graphon games [108], the stochastic static graphon games of Carmona, Cooney, Graves and Laurière [39], and the dynamic graphon mean field games of Caines and Huang [34, 33], where the rank-one and block reductions of worked computation 15.12 and example 15.10 are exactly the tractable models. The constant-graphon reduction to the classical homogeneous theory connects back to the foundational mean field game papers of Lasry and Lions [93] and Huang, Caines and Malhamé [82] and to the LQG capstone of chapter 14.

Chapter 16

Homomorphism Densities and the Topology of Convergence

In chapter 15 a graphon was introduced (definition 15.4) as the limiting object of a sequence of dense graphs: a symmetric measurable kernel $W\colon [0,1]^2 \to [0,1]$, with the type variable $x \in [0,1]$ an exchangeable label rather than a spatial coordinate (the standing convention, restated in appendix A.5). What that chapter did *not* settle is the sense in which a sequence of finite graphs (G_n) "converges" to a graphon. The pixel picture — redraw the adjacency matrix as a black-and-white image on $[0,1]^2$ and refine the grid — is suggestive but treacherous, because it depends on an arbitrary choice of vertex order. Take the three-vertex path a–b–c, a centre b joined to two leaves. List its vertices with the centre in the middle, in the order (a,b,c), and the two edges sit just off the main diagonal, a tidy band hugging it; list them with the centre last, in the order (a,c,b), and those same two edges migrate into the far row and column, an L pushed against the border of the image. One graph, two orderings, two visibly different pictures — yet nothing about the graph has changed, only the names of its vertices. The pathology runs the other way too: a single graphon admits a whole continuum of equally valid pixel pictures, obtained by permuting — more generally, measure-preservingly rearranging — the type interval. The image is an artifact of labelling, not a property of the network. We need a notion of convergence that is blind to relabelling and that captures only the *statistical* content of a network.

This chapter supplies that notion. The central observable is the *homomorphism density* $t(F,G)$, the frequency with which a fixed small pattern F — an edge, a triangle, a path, a star — appears, in the appropriate counting sense, inside a host graph G. These densities are the *moments* of a graph: a countable family of numbers in $[0,1]$ that summarize its structure exactly as the moments $\int t^k \,\mu(\mathrm{d}t)$ summarize a probability measure on the line. This analogy is the organizing idea of the whole chapter, so it is worth stating sharply once. A probability measure on a bounded interval is pinned down completely by the single sequence of its moments — the determinate case of the classical moment problem — and in precisely the same way a graph, or its limiting graphon, is pinned down by the single family of its homomorphism densities, up to the one feature these numbers cannot resolve: the labelling of its vertices. The mean of a measure is its crudest moment; the edge density is its graph analogue. The higher monomials t^k are the higher moments; the larger motifs F are the larger graph moments. Nothing beyond the full family is needed, and that is what makes a *moment theory of graphs* possible. We develop three things in turn. First (section 16.1) we define homomorphism densities for finite graphs and for graphons, prove the elementary counting estimates that relate the several natural normalizations, and show that the graphon densities extend the finite ones with no seam. Second (section 16.2) we define *left-convergence* of a graph sequence as convergence of *all* motif densities, exhibit the topology this induces as a product ("moment") topology, and record both the easy Tychonoff compactness of the profile space and the deep theorem of Lovász and Szegedy — proved in chapter 17 and appendix F — that every left-convergent sequence has a graphon limit. The bridge between motif densities and the metric of chapter 17 is the *counting lemma*, which we state and prove here. Third (section 16.3) we treat the *moment problem for graphs*: exactly which sequences of numbers arise as the homomorphism densities of some graphon. The answer — normalization, multiplicativity, and reflection positivity — is the graph-theoretic twin of the classical theorem that a sequence of reals is a moment sequence iff its Hankel matrices are positive semidefinite.

Throughout, W denotes a graphon satisfying **(A-Reg-W)**: symmetric, measurable, and bounded, here with values in $[0, 1]$. The operator viewpoint $\mathbb{W}$ (definition 2.23) returns in chapter 18; the sampling of finite graphs from W, and the concentration of the densities defined below, are the subject of chapter 19. We use no result from the mean field game chapters; this chapter is pure graph-limit theory, and it is the vocabulary in which the convergence statements of Part IV (chapter 25) are eventually phrased.

16.1 Homomorphisms and homomorphism densities

The preamble promised a single observable — the homomorphism density — and called it "the moments of a graph." This section cashes that promise out concretely, building the observable from the ground up. A *homomorphism* is the formal version of the most naive thing one can do with two graphs: take a small fixed pattern F — an edge, a triangle, a path, a star — and try to lay it down inside a large host G, sending the vertices of F to vertices of G so that every edge of the pattern falls on an actual edge of the host. Counting how many such placements exist, and normalizing so that the count becomes a frequency, *is* the observable. There is nothing more to it than counting how often the small pattern fits.

One choice in that sentence deserves a word of warning before we make it, because it looks like a technicality and is in fact the source of all the good algebra to come: we count *maps*, not literal subgraph copies. A homomorphism is free to send two vertices of F to the same vertex of G, and free to send a non-edge of F onto an edge of G; it is forbidden only from sending an edge of F onto a non-edge. This is more permissive than the geometric notion of "a copy of F sitting inside G," and the extra permissiveness is exactly what we want. Counting all maps gives a quantity that is a clean sum over an unconstrained product set, so it factorizes over disjoint unions of patterns and behaves algebraically like a moment or a partition function; and it is the form that lifts without friction to a graphon, where "a uniformly chosen vertex" becomes "a uniformly chosen type" and "is an edge" becomes "carries the weight $W(x, y)$." The stricter, geometrically faithful counts — injective and induced — differ from the map-count only by the rare placements that collapse two vertices onto one, a correction of relative size $O(1/n)$ that we will quantify exactly and then never worry about again.

The section therefore proceeds in three movements, finite graphs first and graphons second with no seam between them. We *count*: fix two finite graphs and tally the maps from one into the other, in the three increasingly strict senses just sketched. We *normalize*: divide each raw count by the number of maps of its kind so that what remains is a frequency in $[0, 1]$, and we reconcile the three normalizations by an elementary counting estimate showing they agree in the large-host limit. Finally we *lift*: transport the probabilistic reading of the density verbatim to a graphon, where the count becomes an integral over the type cube, and we prove that this graphon density restricts back to the finite one exactly — the pixel graphon of a finite graph has precisely the densities of that graph. By the end the reader should be convinced that $F \mapsto t(F, \cdot\,)$ is one and the same well-defined functional, valued in $[0, 1]$ and multiplicative over disjoint unions, whether its second argument is a finite graph or a graphon. That functional, ranged over all patterns F, is the *homomorphism profile*, and it is the object section 16.2 will turn into a notion of convergence. We begin with the rawest layer of all: counting the maps between two fixed finite graphs.

16.1.1 Counting maps between finite graphs

The observable promised above is, at bottom, a count, and a count needs two ingredients: two graphs and a notion of what it means to lay one down inside the other. We fix both now, in the most elementary form there is, with both graphs finite — a small fixed pattern F and a large host G — and we isolate the single most basic question that everything later in the chapter is built upon: in how many ways can F be placed inside G? The subtlety, and the reason this short subsection earns its own heading, is that "placed inside" is not one notion but three, of increasing strictness, and the differences among them are exactly what the rest of the section will have to reconcile. The three integers defined here — hom, inj, ind — are the raw material; the densities, the moment profile, and the entire convergence theory are refinements of them.

Let F and G be finite simple graphs, with vertex sets $V(F), V(G)$ and edge sets $E(F), E(G)$; write $k = |V(F)|$ and $n = |V(G)|$. A simple graph has no loops and no multiple edges, and edges are unordered pairs $\{i, j\}$, which we abbreviate ij. We write $|E(F)|$ for the number of edges of F; chapter 17 writes the same quantity $e(F)$. The two notations name one and the same integer, and we keep both on purpose. The set-theoretic form $|E(F)|$ is the natural one here, where we are counting maps and partitioning them by their fibres and the edge *set* is an object we point at directly; the lighter $e(F)$ reads better in chapter 17, where the number of edges surfaces only as a Lipschitz constant multiplying a cut norm. This is a deliberate cross-chapter convention rather than an inconsistency to be smoothed over, and wherever the seam could cause confusion we say which form is meant.

Definition 16.1 (Homomorphism)**.** A *homomorphism* from F to G is a map $\varphi\colon V(F) \to V(G)$ that preserves adjacency: $ij \in E(F)$ implies $\varphi(i)\varphi(j) \in E(G)$. Write $\hom(F, G)$ for the number of homomorphisms. An *injective homomorphism* is one that is also injective on vertices; write $\mathrm{inj}(F, G)$ for their number. An *embedding* is an injective homomorphism φ that additionally reflects non-adjacency, $ij \notin E(F) \Rightarrow \varphi(i)\varphi(j) \notin E(G)$; write $\mathrm{ind}(F, G)$ for their number.

The three definitions are a chain of progressively tighter demands, and it pays to say in words what each successive clause forbids and what that costs. A bare homomorphism is allowed to do two things that offend the geometric eye. It may *collapse*: send several vertices of F to one vertex of G, so that the placed pattern is degenerate — a triangle laid down with two of its corners on the same point. The consequence is that hom overcounts honest copies by including these pinched, lower-dimensional placements; harmless in the limit, since they are vanishingly rare, but the precise source of the $O(1/n)$ corrections tracked in lemma 16.6. A homomorphism may also *create an edge*: send a non-adjacent pair of F onto an adjacent pair of G, since the adjacency-preservation clause constrains only the edges of F and is silent about its non-edges. The consequence is that hom cannot distinguish F from any graph obtained by deleting some of its edges; it registers only a lower bound on the structure actually present. The one liberty even a bare homomorphism is denied is the converse of the first: it may never send an edge of F onto a non-edge of G, for that is exactly the adjacency it is required to preserve, and this single prohibition is what makes the count nontrivial. Demanding injectivity withdraws the first licence (no collapsing), so inj counts only non-degenerate placements of all k vertices on distinct hosts. Demanding in addition that non-adjacency be reflected withdraws the second (no created edges), so an embedding lands F on a set of k vertices of G whose *induced* subgraph is exactly F — neither more edges nor fewer — which is why ind is the count of *induced* copies and the one faithful to the geometric picture of "F sits inside G." The chain $\mathrm{ind}(F, G) \le \mathrm{inj}(F, G) \le \mathrm{hom}(F, G)$ records the price of each added demand.

A single tiny host makes all three visible at once.

Example 16.2 (Three kinds of placement in one picture)**.** Let $F = P_3$ be the path 1–2–3 (edges 12 and 23, with 13 its single non-edge), and let the host G be the *paw*: a triangle on $\{a, b, c\}$ together with one extra vertex d joined only to a. Three placements of P_3 exhibit the three notions exactly.

- *A collapsing homomorphism.* Send $1 \mapsto a$, $2 \mapsto b$, $3 \mapsto a$. The edges go $12 \mapsto ab$ and $23 \mapsto ba$, both present in G, so adjacency is preserved and this is a homomorphism; but 1 and 3 land on the same vertex a, so it is not injective. It is counted by $\mathrm{hom}(P_3, G)$ and not by $\mathrm{inj}(P_3, G)$.

- *An injection that creates an edge.* Send $1 \mapsto b$, $2 \mapsto a$, $3 \mapsto c$. The three images are distinct and the edges go $12 \mapsto ba$, $23 \mapsto ac$, both present, so this is an injective homomorphism. But the non-edge 13 of P_3 maps to bc, which *is* an edge of the triangle: the map manufactures an adjacency the pattern did not have. It is counted by $\mathrm{inj}(P_3, G)$ and not by $\mathrm{ind}(P_3, G)$.

- *An embedding.* Send $1 \mapsto d$, $2 \mapsto a$, $3 \mapsto b$. The edges go $12 \mapsto da$ and $23 \mapsto ab$, both present; the images are distinct; and the non-edge 13 maps to db, which is genuinely a non-edge of the paw. Non-adjacency is reflected, so the induced subgraph of G on $\{d, a, b\}$ is exactly the path d–a–b: a faithful induced copy of P_3. It is counted by $\mathrm{ind}(P_3, G)$.

The same example shows the strictness is not free. The path P_3 has injective homomorphisms into the bare triangle K_3 — for instance $1 \mapsto a$, $2 \mapsto b$, $3 \mapsto c$ — yet *no* embedding into K_3 at all, because K_3 has no non-adjacent pair on which to realize the non-edge 13. Thus $\mathrm{ind}(P_3, K_3) = 0$ while $\mathrm{inj}(P_3, K_3) > 0$: the gap between the counts is a real loss of copies, not an artefact of bookkeeping.

It is worth saying now why we will nonetheless make the most permissive count, hom, the protagonist, even though ind is the geometrically faithful one. The reason was flagged in the section preamble and is borne out by example 16.2: hom counts maps over an *unconstrained* domain — every one of the n^k functions $V(F) \to V(G)$ is a candidate, with no injectivity side-condition coupling the choices of where the vertices go — and that absence of constraint is exactly what will let the count factorize over disjoint unions and lift, with no friction, to an integral against a graphon. The strict counts inj and ind purchase geometric fidelity at the cost of this clean product structure, and we will recover them from hom by an inclusion–exclusion costing only $O(1/n)$. For the moment the single point to carry forward is that the three integers are genuinely distinct objects.

Remark 16.3 (Why "left"?)**.** The asymmetry of the symbol $\hom(F, G)$ is not cosmetic, and the name we are about to attach to the whole programme reads it off literally. We hold the pattern F on the *left* fixed and small — an edge, a triangle, a star — and let the host G on the *right* grow without bound; the numbers we watch are $\hom(F, \cdot)$ as F ranges over all small patterns. Convergence governed by these numbers is therefore called *left*-convergence, and the densities $t(F, \cdot)$ are, in this reading, local statistics of the host: each fixed small F probes how often its motif appears, and growing G is what lets those frequencies settle down. This is the regime of *moments*, and it is the one this chapter pursues.

The terminology earns its qualifier only by contrast with the genuinely different theory got by swapping the roles. Fix the graph on the *right* — call it the target H — and let the graph on the *left* grow: the count $\hom(G, H)$ is then the number of adjacency-preserving maps $V(G) \to V(H)$, which is the number of *H-colourings* of G. Taking $H = K_q$, the complete graph on q vertices, counts the proper q-colourings of G (no edge monochromatic); other targets count independent sets, and weighting the vertices and edges of H turns $\hom(G, H)$ into the partition function of a statistical-mechanical spin system living on G. Letting G grow on the left while H stays fixed is *right*-convergence, and it organizes a parallel limit theory keyed to these partition functions rather than to motif frequencies. The two theories are not one limit seen twice; they answer different questions — "how often does this small motif occur?" versus "how many colourings does this large graph admit?" — and the dense graph-limit machinery of this book lives entirely on the left. We name our notion "left-convergence" so that the road not taken stays clearly marked; the right theory is signposted in the bibliographic notes and not pursued further.

These three integers are the bedrock, but they are not yet *comparable*. Each of $\hom(F, G)$, $\mathrm{inj}(F, G)$, $\mathrm{ind}(F, G)$ grows with the size n of the host — there are simply more ways to place a pattern in a larger graph — so a raw count carries the host's size baked into it and says nothing on its own about how *densely* the motif occurs. A four-vertex host and a million-vertex host cannot be set on a common axis until that size dependence is divided out. Doing so is the business of the next subsection: it turns each of the three counts into a frequency in $[0, 1]$ by dividing by the number of candidate maps of its kind, and then reconciles the three normalizations against one another.

16.1.2 The three densities and their normalizations

The previous subsection ended on an obstruction: the three integers $\hom(F, G)$, $\mathrm{inj}(F, G)$, $\mathrm{ind}(F, G)$ all swell as the host grows, so a raw count from a four-vertex host and a raw count from a million-vertex host live on incommensurable scales and cannot be compared as they stand. The cure is the one the moment analogy already anticipates. A moment $\int t^k \, \mu(\mathrm{d}t)$ is an *average*, not a sum — it has the size of μ divided out — and that is exactly why moments of different measures sit on a common axis. We do the same here: divide each count by the number of candidate maps of its own kind, turning a count into the *fraction* of candidates that succeed, a pure number in $[0, 1]$ that no longer carries the host's size baked in.

What "number of candidate maps of its own kind" means is dictated by which random object is natural for each count, and the two counts split here. For hom the domain of candidates is *all* maps $V(F) \to V(G)$: there is no injectivity constraint, so the k vertices of F are placed completely independently, each one sent to a host vertex chosen uniformly from the n available, and the number of such maps is $n \cdot n \cdots n = n^k$. Dividing by n^k is therefore not an arbitrary rescaling but the unique one that reads $\hom(F, G)$ as a probability over the natural product experiment "drop each of the k pattern-vertices onto an independent uniform host-vertex." For inj and ind the candidates are instead the *injections* $V(F) \hookrightarrow V(G)$: here the placements of the k vertices are coupled by the requirement that no two collide, and a uniformly random injection — place the first vertex among n choices, the second among the $n - 1$ that remain, and so on — is the right random object. The number of injections is the falling factorial $(n)_k = n(n-1)\cdots(n-k+1)$, so that is the matching denominator. Each normalization is the size of its own candidate set; nothing else would turn the count into an honest frequency.

Definition 16.4 (Homomorphism densities)**.** For finite simple graphs F (with $|V(F)| = k \geq 1$) and G (with $|V(G)| = n \geq 1$) define the *homomorphism density*, *injective* density, and *induced* density

$$t(F, G) \;=\; \frac{\hom(F, G)}{n^k}, \qquad t_{\mathrm{inj}}(F, G) \;=\; \frac{\mathrm{inj}(F, G)}{(n)_k}, \qquad t_{\mathrm{ind}}(F, G) \;=\; \frac{\mathrm{ind}(F, G)}{(n)_k}.$$

Each lies in $[0, 1]$. Equivalently, $t(F, G)$ is the probability that a map $\varphi\colon V(F) \to V(G)$ chosen uniformly at random (each of the k vertices sent to an independent uniform vertex of G) is a homomorphism; t_{inj} and t_{ind} are the corresponding probabilities for a uniformly random *injection.*

This probabilistic reading is the single sentence to carry through the rest of the chapter, so it is worth saying slowly. To evaluate $t(F, G)$, imagine the pattern F printed on a transparency and dropped at random onto the host G: independently for each vertex i of F, pick a uniform host vertex $\varphi(i)$, with no attempt to keep the picks distinct. The transparency "lands on actual edges" when every edge of F falls on an edge of G — exactly the homomorphism condition — and $t(F, G)$ is the chance that it does. It is the frequency with which a randomly thrown copy of the motif fits. The injective density is the same experiment played with a uniformly random *injection* in place of a free map, so the thrown copy is forbidden to pinch two of its vertices onto one host vertex; the induced density asks the stricter question still, that the landing site induce *exactly* F. Throughout the book the bare symbol $t(F, G)$ — the macro `\homden` — denotes the hom-normalized density of this definition, with degenerate maps included; the injective and induced variants never lose their subscripts. Keeping that discipline matters because the three agree in the limit but differ in every finite exact identity.

It pays to run the experiment once, on the smallest pattern, and watch the normalizations reconcile. Take $F = K_2$, the single edge on vertices $\{1, 2\}$. A map φ is a homomorphism precisely when $\varphi(1)\varphi(2)$ is an edge of G, and the number of ordered pairs landing on an edge is $\hom(K_2, G) = 2\,|E(G)|$ — each undirected edge uv is hit by the two ordered pairs (u, v) and (v, u). Dividing by the n^2 candidate maps,

$$t(K_2, G) \;=\; \frac{2\,|E(G)|}{n^2},$$

the *hom-normalized edge density.* Combinatorics, on the other hand, usually reports the edge density as the fraction of *unordered distinct* pairs that are edges, $|E(G)|\,/\binom{n}{2} = 2\,|E(G)|\,/(n(n-1))$, which is exactly $t_{\mathrm{inj}}(K_2, G)$: an injective map of K_2 is an ordered pair of distinct vertices, of which there are $(n)_2 = n(n-1)$, and $2\,|E(G)|$ of them are edges. The two edge densities therefore differ only by the substitution of n^2 for $n(n-1)$ in the denominator, and that difference is precisely the n *diagonal* pairs (u, u). These diagonal maps are the collapsing homomorphisms of K_2; since G is loopless none of them is ever an edge, so the n^2 denominator counts n candidates that can never succeed while the $n(n-1)$ denominator omits them. Quantitatively

$$t(K_2, G) \;=\; \frac{2\,|E(G)|}{n^2} \;=\; \frac{n-1}{n}\cdot\frac{2\,|E(G)|}{n(n-1)} \;=\; \left(1-\tfrac{1}{n}\right) t_{\mathrm{inj}}(K_2, G),$$

so the hom- and combinatorial edge densities agree up to a factor $1 - 1/n$, an $O(1/n)$ discrepancy that comes entirely from the diagonal and vanishes as the host grows. This is the K_2 instance of the general reconciliation between $t(F, \cdot)$ and $t_{\mathrm{inj}}(F, \cdot)$ made precise in lemma 16.6; for $F = K_2$ the lemma's constant is $\binom{2}{2}/n = 1/n$, and indeed $|t(K_2, G) - t_{\mathrm{inj}}(K_2, G)| = t_{\mathrm{inj}}(K_2, G)/n \le 1/n$.

For $F = K_3$ the triangle, $t(K_3, G)$ is the triangle density; for F a path or a star, one recovers the path and star densities. These few examples already show why the densities deserve the name *motif frequencies*, and the next example assembles a whole catalogue of them, organized so that a second reading of the same numbers — as traces of powers of the adjacency matrix — comes into view.

Example 16.5 (A catalogue of small motifs)**.** Let $A = A(G)$ be the adjacency matrix of G and $\mathbf{1}$ the all-ones vector. Because a homomorphism is just a vertex map under which every pattern edge lands on a host edge, the homomorphism count is a sum, over all placements, of a product of adjacency entries — one factor $A_{\varphi(i)\varphi(j)}$ per edge ij of F — and for the small patterns this sum collapses into linear algebra. Counting homomorphisms is counting walks:

$$\hom(K_2, G) = \mathbf{1}^\top A\,\mathbf{1}, \qquad \hom(P_3, G) = \mathbf{1}^\top A^2\,\mathbf{1}, \qquad \hom(K_3, G) = \operatorname{tr} A^3,$$

where P_3 is the path on three vertices (two edges). Each identity is the bookkeeping made explicit. For the edge, $\hom(K_2, G) = \sum_{a,b} A_{ab} = \mathbf{1}^\top A\,\mathbf{1}$, which is the $2\,|E(G)|$ already met above. For the path 1–2–3, summing the product $A_{ab}A_{bc}$ over the images $a = \varphi(1)$, $b = \varphi(2)$, $c = \varphi(3)$ and doing the b sum last,

$$\hom(P_3, G) = \sum_{a,b,c} A_{ab}A_{bc} = \sum_{a,c}\Big(\sum_{b} A_{ab}A_{bc}\Big) = \sum_{a,c}(A^2)_{ac} = \mathbf{1}^\top A^2\,\mathbf{1},$$

so the path density tallies walks of length two; the inner sum is the number of common neighbours of a and c, and grouping by the centre b instead gives the equivalent count $\sum_b \deg(b)^2$. The triangle is the case where the spectral reading first becomes unmistakable. Summing $A_{ab}A_{bc}A_{ca}$ over $a = \varphi(1)$, $b = \varphi(2)$, $c = \varphi(3)$,

$$\hom(K_3, G) = \sum_{a,b,c} A_{ab}A_{bc}A_{ca} = \sum_a \Big(\sum_{b,c} A_{ab}A_{bc}A_{ca}\Big) = \sum_a (A^3)_{aa} = \operatorname{tr} A^3,$$

because $(A^3)_{aa} = \sum_{b,c} A_{ab}A_{bc}A_{ca}$ counts the closed walks $a \to b \to c \to a$ of length three, and the trace sums them over their basepoint. In a simple, loopless graph a closed 3-walk can only visit three *distinct* vertices that are pairwise adjacent — a triangle — traversed from any of its 3 vertices in either of 2 directions, so $\operatorname{tr} A^3$ equals 6 times the number of triangles. (This is also why, for the triangle alone, no correction is needed: a collapsing map of K_3 would force a pattern edge onto a diagonal entry $A_{aa} = 0$, so every homomorphism of K_3 is automatically injective and $\hom(K_3, G) = \operatorname{inj}(K_3, G)$ exactly.) More generally hom of a path is a sum of entries of a power of A, and hom of a cycle C_ℓ is the closed-ℓ-walk count

$$\hom(C_\ell, G) = \operatorname{tr} A^\ell = \sum_i \lambda_i^\ell,$$

the ℓ-th power sum of the eigenvalues $\lambda_1, \ldots, \lambda_n$ of A — that is, the ℓ-th moment of the adjacency spectrum. The densities are these quantities divided by $n^{|V(F)|}$. This is the first sign of a theme chapter 18 will develop in full: the cycle densities $t(C_\ell, G)$ are, up to the normalization $n^{-\ell}$, exactly the spectral power sums $\sum_i \lambda_i^\ell$, so the homomorphism profile and the spectrum of the graph operator are not rival summaries of G but the *same* information read through two different windows — the combinatorial window of motif counts and the analytic window of eigenvalues. We will exploit that identification when the spectral picture takes over in chapter 18.

Two loose threads are left deliberately hanging for the next subsection to take up. The first is quantitative: the K_2 computation above showed $t(K_2, \cdot)$ and $t_{\text{inj}}(K_2, \cdot)$ differing only through the diagonal — the collapsing maps — and the same is true of every pattern F, the two densities parting ways solely on the homomorphisms that send two vertices of F to one vertex of G. The next subsection counts those degenerate maps and shows there are only $O(n^{k-1})$ of them out of n^k, a vanishing fraction, so the three normalizations define one and the same limit. The second thread is the spectral one just opened: that densities and eigenvalues are one dataset in two coordinate systems, a remark chapter 18 will turn into a working tool.

16.1.3 Comparing the normalizations: the first counting lemma

The previous subsection left us holding two normalizations of the same motif count — the hom density $t(F, G)$, which divides by all n^k maps, and the injective density $t_{\text{inj}}(F, G)$, which divides by the $(n)_k$ injections — together with the observation, borne out explicitly on the edge K_2, that the two part ways on exactly one population of maps: the *degenerate* ones, the homomorphisms that send two distinct vertices of F to a single vertex of G. The hom count includes them, the injective count does not, and the entire discrepancy between the two densities is the weight of this degenerate population. So the question of whether the normalizations matter in the limit is the wholly concrete question of how many degenerate maps there are. We answer it now, and the answer is reassuring: there are very few of them.

The heuristic is a counting of dimensions, and it is worth saying in words before any symbols, because the whole lemma is just this sentence made exact. A map $\varphi\colon V(F) \to V(G)$ is a free choice of k images, one for each of the k vertices of F, drawn from the n vertices of G; there are n^k such choices, and one should think of the exponent k as a "dimension." To be *degenerate* a map must make at least one coincidence — two prescribed vertices forced to share an image — and a single such coincidence ties two of the k free choices together into one, dropping the dimension from k to $k-1$. There are only $\binom{k}{2}$ pairs of vertices that could be the ones to collide, so the degenerate maps live on $\binom{k}{2}$ many $(k-1)$-dimensional slices inside a k-dimensional box: a set of relative size $O(n^{k-1})/n^k = O(1/n)$, a vanishing fraction as the host grows. The injectivity constraint that distinguishes t_{inj} from $t(F,\cdot)$ therefore costs nothing in the limit; it only removes a measure-zero skin of collapsed placements. The next lemma turns "$O(1/n)$" into the explicit constant $\binom{k}{2}/n$, with a sharp two-sided bound. It is the prototype of every "counting lemma" in the subject — a statement that two ways of counting motifs differ by a controllable error — and unlike the deeper cut-norm counting lemma of section 16.2 it is completely elementary, nothing but bookkeeping on finite sets.

Lemma 16.6 (Hom versus injective density)**.** *For all finite simple graphs F (with $|V(F)| = k$) and G (with $|V(G)| = n \geq 1$),*

$$0 \;\leq\; t(F,G) - \frac{(n)_k}{n^k}\, t_{\mathrm{inj}}(F,G) \;\leq\; \binom{k}{2}\frac{1}{n}, \qquad \textit{hence} \qquad \bigl|\, t(F,G) - t_{\mathrm{inj}}(F,G)\,\bigr| \;\leq\; \binom{k}{2}\frac{1}{n}.$$

Proof. The argument has three movements: split the hom count into its injective and degenerate parts (this gives the left inequality and the first half of the right one), bound the degenerate part by a union bound over pairs, and finally combine the two error terms watching their signs — the last step is where the constant stays at $\binom{k}{2}/n$ rather than doubling.

Partition by fibres. Sort the n^k maps $\varphi\colon V(F) \to V(G)$ according to how they collapse, that is, by the partition π of the domain $V(F)$ whose blocks are the nonempty fibres $\varphi^{-1}(v)$, $v \in V(G)$. Two vertices of F land in the same block of π exactly when φ sends them to the same vertex of G. The *injective* maps are precisely those whose partition is the finest one, $\hat{0}$, all blocks singletons — no two vertices share an image; every other partition records at least one coincidence and marks a degenerate map. This sorts the maps into two camps, injective and degenerate, with no overlap and nothing left out. Now restrict to homomorphisms. Among the injective maps, the ones that are homomorphisms are by definition counted by $\mathrm{inj}(F,G)$, and the total number of injective maps (homomorphism or not) is the falling factorial $(n)_k = n(n-1)\cdots(n-k+1)$, so by the very definition $t_{\mathrm{inj}}(F,G) = \mathrm{inj}(F,G)/(n)_k$ the injective homomorphisms number $(n)_k\, t_{\mathrm{inj}}(F,G)$. The remaining homomorphisms are exactly the degenerate ones, a nonnegative count. Hence

$$\mathrm{hom}(F,G) \;=\; \underbrace{(n)_k\, t_{\mathrm{inj}}(F,G)}_{\text{injective homs}} \;+\; \underbrace{\#\{\text{degenerate homomorphisms}\}}_{\geq 0} \;\geq\; (n)_k\, t_{\mathrm{inj}}(F,G).$$

Dividing through by n^k and recalling $t(F,G) = \mathrm{hom}(F,G)/n^k$ gives the left inequality $t(F,G) -$

$\frac{(n)_k}{n^k} t_{\rm inj}(F,G) \geq 0$ at once: the hom density is the injective contribution *plus* the nonnegative degenerate surplus.

Union bound over the $\binom{k}{2}$ pairs. It remains to bound that surplus from above. A degenerate *homomorphism* is in particular a degenerate *map*, so it suffices to bound the number of degenerate maps, ignoring the adjacency condition entirely (dropping a constraint can only enlarge the count, which is all we want for an upper bound). A degenerate map identifies at least one pair of distinct vertices of F; fix one offending pair $\{a,b\} \subseteq V(F)$ and count the maps that send a and b to a common image. Such a map is determined by: the common image of a and b, which is one vertex of G (n choices); and the images of the other $k-2$ vertices, free among n each (n^{k-2} choices). That is $n \cdot n^{k-2} = n^{k-1}$ maps identifying the fixed pair $\{a,b\}$ — the "lost" factor of n is precisely the one degree of freedom the coincidence removes. Every degenerate map identifies at least one pair, so it is caught by at least one term of the union over the $\binom{k}{2}$ pairs $\{a,b\}$; the union bound (which only overcounts, since a map collapsing several pairs is counted once per collapsed pair) gives

$$\#\{\text{degenerate maps}\} \;\leq\; \sum_{\{a,b\}\subseteq V(F)} n^{k-1} \;=\; \binom{k}{2} n^{k-1}.$$

Dividing by n^k,

$$t(F,G) - \frac{(n)_k}{n^k}\, t_{\rm inj}(F,G) \;=\; \frac{\#\{\text{degenerate homomorphisms}\}}{n^k} \;\leq\; \frac{\#\{\text{degenerate maps}\}}{n^k} \;\leq\; \binom{k}{2}\frac{1}{n},$$

which is the right inequality of the first display. The same union bound, applied without the homomorphism restriction, also controls the normalization ratio itself: the maps that are *not* injective number $n^k - (n)_k$, and the count just made bounds them by $\binom{k}{2} n^{k-1}$, so

$$0 \;\leq\; 1 - \frac{(n)_k}{n^k} \;=\; \frac{n^k - (n)_k}{n^k} \;\leq\; \binom{k}{2}\frac{1}{n}. \tag{16.1}$$

(One can see (16.1) directly by expanding the product $(n)_k/n^k = \prod_{j=0}^{k-1}(1-j/n)$: it is a product of factors in $[0,1]$, hence at most 1, and at least $1 - \sum_{j=0}^{k-1} j/n = 1 - \binom{k}{2}/n$ by the elementary inequality $\prod(1-x_j) \geq 1 - \sum x_j$ for $x_j \in [0,1]$. The two routes agree because $\binom{k}{2} = \sum_{j=0}^{k-1} j$.)

Sign cancellation: keeping the constant at $\binom{k}{2}/n$. We now have two separate $O(1/n)$ errors in hand — one from replacing the inj-count by its degenerate-inclusive hom-count, one from replacing the denominator $(n)_k$ by n^k — and the lazy move is to pass to the two-sided bound by the triangle inequality, adding the two corrections to get $2\binom{k}{2}/n$. That is correct but not sharp, and the slack is worth removing because it exposes a structural point: the two corrections push in *opposite* directions and partly annihilate. To see it, write the gap we actually care about as the sum of exactly these two pieces, by adding and subtracting $\frac{(n)_k}{n^k} t_{\rm inj}(F,G)$:

$$t(F,G) - t_{\rm inj}(F,G) = \underbrace{\Big(t(F,G) - \tfrac{(n)_k}{n^k} t_{\rm inj}(F,G)\Big)}_{=:A} + \underbrace{t_{\rm inj}(F,G)\Big(\tfrac{(n)_k}{n^k} - 1\Big)}_{=:B}.$$

These are the two corrections, isolated. The first, A, is the degenerate surplus: by the left inequality and the right inequality already proved it satisfies $A \in [0, \binom{k}{2}/n]$ — it is nonnegative (the hom count includes extra maps) and no larger than $\binom{k}{2}/n$. The second, B, is the denominator correction: since $t_{\mathrm{inj}}(F,G) \in [0,1]$ and, by (16.1), the factor $\frac{(n)_k}{n^k} - 1$ lies in $[-\binom{k}{2}/n,\, 0]$, the product B is *non-positive*, $B \le 0$, with magnitude $|B| = t_{\mathrm{inj}}(F,G)\,(1 - \frac{(n)_k}{n^k}) \le \binom{k}{2}/n$. So A lives in $[0, \binom{k}{2}/n]$ and B lives in $[-\binom{k}{2}/n, 0]$: one is a nonnegative number at most $\binom{k}{2}/n$, the other a nonpositive number at least $-\binom{k}{2}/n$. Their sum cannot escape the interval spanned by those two ranges,

$$t(F,G) - t_{\mathrm{inj}}(F,G) \;=\; A + B \;\in\; \Big[-\frac{\binom{k}{2}}{n},\; \frac{\binom{k}{2}}{n}\Big],$$

because the most positive the sum can be is A at its ceiling with B at 0 (value $\binom{k}{2}/n$), and the most negative is B at its floor with A at 0 (value $-\binom{k}{2}/n$). This is the stated two-sided bound $|t(F,G) - t_{\mathrm{inj}}(F,G)| \le \binom{k}{2}/n$, and the sign bookkeeping — $A \ge 0$ paired with $B \le 0$ — is exactly what saved the factor of two the naive addition would have spent. $\square$

It is worth pausing on what the constant says. The bound $\binom{k}{2}/n$ depends on the pattern F *only* through its number of vertices k, not through its edges: a k-vertex tree and a k-vertex clique incur the same coincidence correction, because the correction counts collisions of *vertices*, of which there are $\binom{k}{2}$ candidate pairs whatever the edge set. And the dependence on the host is the single clean factor $1/n$. So for each fixed pattern the gap between the two densities is $O(1/n)$ uniformly in everything else, which is precisely the strength one needs to conclude that the choice of normalization is invisible to the limit.

Remark 16.7 (All three densities define the same convergence)**.** Fix a pattern F with k vertices and let the hosts grow, $n = |V(G_n)| \to \infty$. Then $\binom{k}{2}/n \to 0$, so lemma 16.6 pins $t(F, G_n)$ and $t_{\mathrm{inj}}(F, G_n)$ together: their difference is squeezed to zero, and therefore one of the two sequences converges if and only if the other does, to the same limit. The hom-normalization and the injective normalization are thus interchangeable as far as left-convergence is concerned — they certify exactly the same convergent sequences and assign exactly the same limits.

The third family, the induced densities t_{ind}, is tied to the injective ones not by an estimate but by an exact, host-independent *linear change of basis*, so it converges in lockstep as well. The relationship is inclusion–exclusion on the non-edges. Recall the distinction: an injective homomorphism of F must carry every *edge* of F onto an edge of G but says nothing about non-edges, whereas an induced embedding must in addition carry every *non-edge* of F onto a non-edge of G. So $\mathrm{inj}(F,G)$ counts the injective maps realizing *at least* the edges of F — they may accidentally realize some non-edges of F as edges of G too — while $\mathrm{ind}(F,G)$ counts those realizing *exactly* the edges of F and no more. Sorting the injective maps by which extra non-edges they happen to fill in expresses each inj as a sum of ind's, and inverting that triangular relation (the inclusion–exclusion sign pattern) expresses each ind as an alternating sum of inj's over the graphs obtained from F by promoting some of its non-edges to edges. Both directions are finite integer-coefficient combinations with coefficients depending only on F, never on n.

The smallest instance makes the change of basis concrete. Take F on two labelled vertices $\{1,2\}$. There are two such graphs: the single edge K_2 and the edgeless pair $\overline{K_2}$ (two isolated vertices). An injective map of either into a host G is just an ordered pair of distinct vertices (u,v). For K_2, an injective homomorphism requires uv to be an edge, while an induced embedding requires the induced subgraph on $\{u, v\}$ to be exactly one edge — which is the same condition, $uv \in E(G)$ — so here $\mathrm{inj}(K_2, G) = \mathrm{ind}(K_2, G)$, both equal to the number of ordered adjacent pairs $2\,|E(G)|$. For $\overline{K_2}$, by contrast, an injective homomorphism imposes *no* edge condition (there are no edges to preserve), so $\mathrm{inj}(\overline{K_2}, G) = (n)_2$ counts *all* ordered distinct pairs, both adjacent and non-adjacent; whereas an induced embedding of $\overline{K_2}$ demands that the one non-edge 12 map to a non-edge, i.e. $uv \notin E(G)$, so $\mathrm{ind}(\overline{K_2}, G)$ counts only the ordered *non*-adjacent pairs. The two are related by the partition of all distinct pairs into adjacent and non-adjacent:

$$\mathrm{inj}(\overline{K_2}, G) \;=\; (n)_2 \;=\; \underbrace{2\,|E(G)|}_{\mathrm{ind}(K_2,G)} \;+\; \underbrace{\mathrm{ind}(\overline{K_2}, G)}_{\text{non-adjacent pairs}}\;, \qquad \text{i.e.} \qquad \mathrm{ind}(\overline{K_2}, G) \;=\; \mathrm{inj}(\overline{K_2}, G) - \mathrm{inj}(K_2, G).$$

That last identity is exactly inclusion–exclusion: the induced count of the *non*-edge is the injective count of the non-edge (all pairs) minus the injective count of the graph with that non-edge promoted to an edge (the adjacent pairs). Dividing by $(n)_2$ turns it into the density identity $t_{\mathrm{ind}}(\overline{K_2}, G) = 1 - t_{\mathrm{inj}}(K_2, G)$, the two-vertex shadow of the general alternating sum. (Reassuringly $t_{\mathrm{inj}}(K_2, G)$ is the ordinary edge density, so $t_{\mathrm{ind}}(\overline{K_2}, G)$ is the non-edge density, and the two sum to one, as they must.)

Because every such conversion is a fixed finite linear combination, a sequence along which all $t_{\mathrm{inj}}(F, \cdot)$ converge is one along which all $t_{\mathrm{ind}}(F, \cdot)$ converge, and conversely; combined with the first paragraph, all three families — $t(F, \cdot)$, t_{inj}, t_{ind} — certify the same convergent sequences with the same limits. We may therefore work with whichever normalization is most convenient and switch between them at will. Of the three, $t(F, \cdot)$ is the one we promote to canonical status, and for a reason the counting estimate alone does not reveal: it is the only normalization that extends cleanly to the continuum. Dividing by $(n)_k$ presupposes a finite vertex set to count injections in, and there is no falling factorial of a graphon; but dividing by n^k corresponds to drawing each image *independently and uniformly*, an experiment that transfers verbatim to a graphon with "uniform vertex" replaced by "uniform point of the type interval." That is the construction the next subsection carries out, and it is why the hom-normalization, and not its stricter cousins, is the one we lift.

16.1.4 Homomorphism densities of a graphon

The previous subsection closed with a verdict: of the three normalizations only the hom-normalization survives the passage to the continuum, because dividing by n^k encodes the experiment "drop each pattern-vertex onto an independent uniform host-vertex," and that experiment has a verbatim continuum analogue while the falling-factorial normalization of the injective and induced densities does not. We now cash that verdict in. The whole content of this subsection is that one sentence of the probabilistic reading carries over word for word, with only two substitutions in the dictionary, and that once it does, every algebraic property the finite density enjoyed reappears on the graphon side for the same reason it held before.

Recall the reading. To evaluate $t(F, G)$ we threw the pattern F at random onto the host: independently for each vertex i of F we chose a uniform host-vertex $\varphi(i)$, and $t(F, G)$ was the chance that every edge of F landed on an edge of G. A graphon has no vertices to land on — its "vertices" form the continuum $[0, 1]$ of types — and it has no sharp notion of "is an edge": between two types x, y it records only the weight $W(x, y) \in [0, 1]$, the probability that an edge is present in a graph sampled from it. Both gaps close with a single substitution each. For "uniform vertex of G" read "uniform point ξ_i of the type interval $[0, 1]$"; the n equally likely host vertices become the continuum of equally likely types, and a uniform draw on $\{1, \ldots, n\}$ becomes the uniform (Lebesgue) law on $[0, 1]$. For "the edge $\varphi(i)\varphi(j)$ is present" — an event, worth 1 when it occurs and 0 when it does not — read "carry the weight $W(\xi_i, \xi_j)$," the $[0, 1]$-valued soft indicator that replaces the hard 0/1 adjacency. The hard question "did every edge of F land on an edge?" — a product of 0/1 indicators, itself 0 or 1 — thereby softens into the product $\prod_{ij \in E(F)} W(\xi_i, \xi_j)$ of weights in $[0, 1]$, and the probability that the throw succeeds becomes the *expectation* of that product. Making the two substitutions in the definition of $t(F, G)$ and reading the resulting average as an integral against Lebesgue measure on the type cube is the entire definition that follows.

Definition 16.8 (Homomorphism density into a graphon)**.** Let F be a finite simple graph with $V(F) = \{1, \ldots, k\}$ and let W satisfy **(A-Reg-W)**. The *homomorphism density of F in W* is

$$t(F, W) \;=\; \int_{[0,1]^k} \prod_{ij \in E(F)} W(x_i, x_j) \;\mathrm{d}x_1 \cdots \mathrm{d}x_k \;=\; \mathbb{E}\left[\prod_{ij \in E(F)} W(\xi_i, \xi_j)\right],$$

where $\xi_1, \ldots, \xi_k$ are i.i.d. uniform on $[0, 1]$. For the one-vertex graph K_1 and the edgeless graph $\overline{K_r}$ on r vertices the empty product is 1, so $t(K_1, W) = t(\overline{K_r}, W) = 1$.

Three small points need checking before the symbol on the left is entitled to exist, and at a physicist's level of rigor each is a one-line verification rather than a hurdle. First, the integrand is a genuine measurable function on $[0, 1]^k$: each factor $W(x_i, x_j)$ is the composition of the measurable kernel W with the coordinate projection $(x_1, \ldots, x_k) \mapsto (x_i, x_j)$, hence measurable, and a finite product of measurable functions is measurable, so the whole integrand is. Second, the integral makes sense and may be computed in any order: the integrand is nonnegative (a product of factors in $[0, 1]$) and the domain $[0, 1]^k$ carries the product of k copies of Lebesgue probability measure, so Tonelli's theorem (theorem 1.27) applies — it is precisely the theorem licensing the interchange of a multiple integral with iterated single integrals for a nonnegative measurable integrand, with no integrability hypothesis needed because nonnegativity already forbids the $\infty - \infty$ ambiguity. This is what lets us write the k-fold integral as the expectation over k i.i.d. uniform draws, and it is the only analytic fact the definition rests on. Third, the value lies in $[0, 1]$: the integrand is bounded between 0 (some factor could vanish) and 1 (every factor is at most 1) pointwise, and the integral of a function valued in $[0, 1]$ against a probability measure is again in $[0, 1]$. So $t(F, W)$ is a well-defined number in $[0, 1]$, exactly as its finite predecessor was.

The expectation form on the right is the one that earns the word "moment." $t(F, W)$ is the expected value of a product of kernel evaluations, one factor per edge of F, at independent uniform insertion points — precisely the structure of a joint moment of the random field $(x, y) \mapsto W(x, y)$. The simplest case fixes the picture: for the single edge $F = K_2$, $t(K_2, W) = \int_{[0,1]^2} W(x, y)\, \mathrm{d}x\, \mathrm{d}y$ is the mean edge weight, the continuum edge density and the crudest moment of all, exactly as the mean is the crudest moment of a measure on the line. The one degenerate case worth dwelling on is the *empty* product, flagged in the definition: a graph with no edges contributes no factors, and the empty product is 1 by convention, so the integrand is the constant function 1 and its integral over the probability space $[0,1]^k$ is 1. Hence $t(K_1, W) = t(\overline{K_r}, W) = 1$ for the one-vertex graph and for the edgeless graph on any number r of vertices. This is not a boundary convention chosen for tidiness; it is forced, and it matters. A pattern with no edges imposes no constraint on the throw, so the throw succeeds with certainty, and the density must be 1 — the same value the finite count $\hom(\overline{K_r}, G)/n^r = n^r/n^r = 1$ returns, since every one of the n^r maps of an edgeless pattern is vacuously a homomorphism. We will need $t(\overline{K_r}) = 1$ as the normalization anchor of the moment problem in section 16.3, where it is the graph analogue of the zeroth moment $\int \mu(\mathrm{d}t) = 1$ of a probability measure.

We have now defined the same symbol $t(F, \cdot)$ twice over — once with a finite graph in the second slot, once with a graphon — and an honest reader should worry that the two definitions might disagree where they overlap. They do not, and the cleanest way to see it is to feed the graphon definition the graphon that *is* a finite graph and check that the integral collapses back to the finite count. There is a canonical such graphon. Recall the *pixel graphon* (empirical graphon) W^G of a graph G on vertex set $\{1, \ldots, n\}$ (definition 15.2 and proposition 15.3): partition $[0,1]$ into n consecutive intervals $I_1, \ldots, I_n$ of equal length $1/n$ and set $W^G(x, y) = A_{ij}$ for $x \in I_i,\ y \in I_j$, where $A = A(G)$ is the adjacency matrix. (It is the step graphon adapted to the equipartition, in the sense of definition 15.6; we keep the name "pixel graphon" of chapter 15 and reserve "step graphon" for the general partition.) It is the literal black-and-white image of the adjacency matrix painted on the unit square, each entry A_{ij} smeared uniformly over the little block $I_i \times I_j$. If the two definitions are to deserve the same name, then evaluating the graphon density on this image must return the finite density of G exactly, with no boundary term, no leftover $1/n$, no seam between the discrete and continuous worlds. That exact agreement is the next proposition, and the reason it holds without any correction is worth watching slowly, because it is the technical heart of the claim that "graphons extend graphs."

Proposition 16.9 (Consistency of the two definitions)**.** *For every finite simple graph F and every finite simple graph G,*

$$t(F, W^G) \;=\; t(F, G).$$

Proof. Write $k = |V(F)|$, $n = |V(G)|$. The argument is a single change of variable: each uniform type splits into a discrete part the pixel graphon sees and a continuous part it cannot see, and once the invisible part is integrated out the integral becomes the finite sum.

One uniform type = one uniform vertex times one ignored position. Fix a single index i and look at the uniform draw $\xi_i \in [0,1]$. The interval $I_v = [(v-1)/n,\, v/n)$ containing ξ_i is determined by which of the n equal blocks ξ_i falls into; write $\varphi(i) \in \{1, \ldots, n\}$ for that block index. Because the blocks all have the same length $1/n$, the index $\varphi(i)$ is *uniform* on $\{1, \ldots, n\}$: $\mathbb{P}(\xi_i \in I_v) = 1/n$ for each v. And conditionally on $\varphi(i) = v$, the leftover $n\xi_i - (v-1)$ — the position of ξ_i *within* its block, rescaled to $[0,1]$ — is uniform on $[0,1]$ and independent of v. So a single uniform type decomposes, with no loss and no remainder, into a uniform vertex $\varphi(i)$ *times* an independent uniform position-in-interval. The pixel graphon was built to be constant on each block $I_i \times I_j$, so it depends on (ξ_i, ξ_j) *only* through the block indices $(\varphi(i), \varphi(j))$ and is completely blind to the positions-within-intervals. This is the "no seam" mechanism in one line: the continuous degrees of freedom the graphon definition integrates over are exactly the degrees of freedom the pixel graphon was designed to ignore, so integrating them out costs nothing.

Collapsing the integral. Do the k draws jointly. The vector $(\xi_1, \dots, \xi_k)$ i.i.d. uniform on $[0,1]$ is, by the previous paragraph applied coordinatewise, the same as an i.i.d. uniform vertex map $\varphi\colon V(F) \to V(G)$ together with k independent ignored positions. On the event $\{\xi_i \in I_{\varphi(i)}\ \forall i\}$ — an event of probability n^{-k}, since the k block indices are independent and each is one particular value out of n — the integrand takes the value $\prod_{ij\in E(F)} W^G(\xi_i, \xi_j) = \prod_{ij\in E(F)} A_{\varphi(i)\varphi(j)}$, which no longer depends on the positions and is 1 exactly when every edge ij of F has $A_{\varphi(i)\varphi(j)} = 1$ — that is, exactly when φ is a homomorphism $F \to G$ — and 0 otherwise. Summing the value n^{-k} over the n^k equally likely block-index patterns φ and weighting each by its 0/1 integrand,

$$t(F, W^G) = \mathbb{E}\Big[\prod_{ij\in E(F)} W^G(\xi_i, \xi_j)\Big] = \frac{1}{n^k} \sum_{\varphi\colon V(F)\to V(G)} \prod_{ij\in E(F)} A_{\varphi(i)\varphi(j)} = \frac{\hom(F, G)}{n^k} = t(F, G). \quad \square$$

The factor $1/n^k$ is the probability weight of each block pattern; the sum counts the homomorphisms; their product is the finite density, with not a single error term left over.

So nothing is lost in passing to the continuum: the same functional $F \mapsto t(F, \cdot)$ reads a finite graph and a graphon, and on the pixel graphon of a finite graph it returns precisely that graph's densities. The finite theory is not approximated by the graphon theory; it is a special case of it, sitting exactly inside with no rounding. With consistency secured we may treat $t(F, \cdot)$ as one object and ask after its structure. Two identities will be used over and over — multiplicativity over disjoint unions, and the monotone bounds — and both follow from the integral form by the same one-move argument that powered the consistency proof: factor what is independent, dominate what is bounded.

Proposition 16.10 (Multiplicativity and monotone bounds)**.** *Let W satisfy* ***(A-Reg-W)*** *with values in $[0,1]$, and let F, F_1, F_2 be finite simple graphs.*

(i) (Multiplicativity over disjoint unions.) *If $F = F_1 \sqcup F_2$ is the disjoint union, then $t(F, W) = t(F_1, W)\, t(F_2, W)$. In particular, if F has connected components $F^{(1)}, \dots, F^{(c)}$ then $t(F, W) = \prod_{a=1}^{c} t(F^{(a)}, W)$.*

(ii) (Edge-monotone decay.) *Adding an edge to F cannot increase its density: if $F' \supseteq F$ on the same vertex set, then $t(F', W) \le t(F, W)$; and the trivial bounds $0 \le t(F, W) \le 1$ always hold.*

Proof. (i) Multiplicativity. Write $F = F_1 \sqcup F_2$ with vertex sets $V_1 = V(F_1)$ and $V_2 = V(F_2)$ disjoint, $|V_1| = k_1$, $|V_2| = k_2$, so $|V(F)| = k_1 + k_2$. Every edge of F lies wholly inside F_1 or wholly inside F_2 — a disjoint union creates no edge across the two parts — so the edge product splits cleanly along the partition of vertices,

$$\prod_{ij\in E(F)} W(\xi_i, \xi_j) = \Big(\prod_{ij\in E(F_1)} W(\xi_i, \xi_j)\Big)\Big(\prod_{ij\in E(F_2)} W(\xi_i, \xi_j)\Big).$$

The first factor is a function of the variables $(\xi_i)_{i\in V_1}$ alone, the second of $(\xi_i)_{i\in V_2}$ alone; the two variable blocks are disjoint, hence the corresponding i.i.d. draws are independent. The expectation of a product of functions of independent variables is the product of the expectations — which at the level of integrals is exactly Tonelli (theorem 1.27) splitting the (k_1+k_2)-fold integral over $[0,1]^{V_1} \times [0,1]^{V_2}$ into the product of a k_1-fold and a k_2-fold integral. Therefore $t(F,W) = t(F_1,W)\,t(F_2,W)$. A one-line instance: two disjoint edges form $F = K_2 \sqcup K_2$, and $t(K_2 \sqcup K_2, W) = t(K_2,W)^2$ — the chance that two independently thrown edges both land on weight is the square of the chance for one, precisely because the two throws share no vertex and so cannot interfere. Iterating the split over the connected components $F^{(1)},\dots,F^{(c)}$ of F gives $t(F,W) = \prod_{a=1}^{c} t(F^{(a)},W)$, since the components are pairwise vertex-disjoint by definition. This is the property that makes the homomorphism profile behave like a sequence of *moments* of a product structure, and it is the one section 16.3 will lean on hardest: it reduces the entire profile to its values on connected patterns, the indecomposable "letters" from which every density is a product.

(ii) Edge-monotone decay. Let $F' \supseteq F$ have the same vertex set, with F' obtained from F by adding one edge ab (the general case follows by adding edges one at a time). The integrand of $t(F',W)$ is that of $t(F,W)$ multiplied by the single extra factor $W(x_a,x_b)$, which takes values in $[0,1]$. Multiplying a nonnegative integrand by a factor that nowhere exceeds 1 can only shrink it pointwise, so the integrand of $t(F',W)$ is everywhere $\le$ that of $t(F,W)$; integrating preserves the inequality, giving $t(F',W) \le t(F,W)$. A one-line instance: the triangle K_3 is the path P_3 with the closing edge 13 added, so $t(K_3,W) \le t(P_3,W)$ — demanding the third edge also land on weight can only make the throw harder. The same pointwise estimate, applied to the bare integrand itself, gives the universal bounds: the integrand is a product of factors in $[0,1]$, hence pointwise in $[0,1]$, hence so is its integral against a probability measure, which is the assertion $0 \le t(F,W) \le 1$. (The upper bound is the empty-edge extreme of monotonicity — deleting all edges leaves the constant integrand 1 — and the lower bound is nonnegativity of a nonnegative integrand.) $\square$

Notation watch

Three different "densities" — $t(F,\cdot)$, t_{inj}, t_{ind} — and one symbol t. In this book $t(F,G)$ *always* means the homomorphism density (non-injective maps included), matching the macro `\homden` fixed in appendix A.5. The injective and induced variants always carry the subscripts $t_{\mathrm{inj}}, t_{\mathrm{ind}}$. The distinction is immaterial for convergence (remark 16.7) but *not* for exact identities: $t(K_3,G)$ and $t_{\mathrm{inj}}(K_3,G)$ already differ, though *not* because of any degenerate surplus. Collapsing any pair of the triangle would force a pattern edge onto a diagonal entry $A_{aa}=0$, so on a loopless host K_3 admits *no* surviving degenerate homomorphism and $\hom(K_3,G) = \mathrm{inj}(K_3,G) = 6\cdot\#\{\text{triangles}\}$ exactly; the two densities differ only in their denominators, n^3 versus the falling factorial $(n)_3$. (Genuine degenerate homomorphisms appear only for patterns with a non-adjacent pair, on which a collapse need not land an edge on the diagonal.) When an exact small-graph count matters — as in the moment problem of section 16.3, where multiplicativity is used — it is the hom-normalization that is algebraically clean, because $t(\cdot,W)$ is multiplicative while $t_{\mathrm{inj}}(\cdot,W)$ is not. The reason is exactly the independence exploited in proposition 16.10 (i): the free maps that hom ranges over factor across a disjoint union because the two blocks of vertices are chosen independently, with nothing forbidding a vertex of F_1 from colliding with a vertex of F_2. The injective normalization breaks precisely this independence — a uniform injection of $F_1 \sqcup F_2$ must keep *all* images distinct, so the placement of F_2 is constrained by where F_1 already went, and the joint count does not factor into the two separate injective counts. Multiplicativity is therefore not a cosmetic convenience but the structural reason the hom-normalization is the one promoted to canonical status and carried into the moment problem; its failure for t_{inj} is the same coupling the falling-factorial denominator records.

Physics note

A physicist will recognize $t(F, W) = \mathbb{E}\prod_{ij\in E(F)} W(\xi_i, \xi_j)$ as a connected or disconnected *correlation function* of the field W evaluated at random insertion points, with the graph F playing the role of a Feynman diagram: vertices are integration points, edges are propagators $W(x, y)$, and the density is the diagram's value. Multiplicativity over disjoint unions (proposition 16.10 (i)) is then the factorization of vacuum diagrams into connected pieces, and the statement that "a graph is determined by all its motif densities" is the statement that a translation-invariant field theory is determined by all its correlation functions. The analogy is exact at the level of bookkeeping and stops there: there is no action, no temperature, and the "field" W is a fixed deterministic kernel, not a fluctuating degree of freedom. We use the picture only to organize the diagrams, never to compute.

We have arrived at a single, uniform object. The functional $F \mapsto t(F, \cdot)$ is now defined by the *same* formula — an average of an edge-product over independent uniform arguments — on finite graphs and on graphons alike, agrees with the elementary count on pixel graphons with no seam (proposition 16.9), is multiplicative over disjoint unions and monotone under adding edges (proposition 16.10), and takes values in $[0, 1]$ with the edgeless patterns normalized to 1. Every finite graph and every graphon thus carries a whole *vector* of these numbers, one coordinate per pattern F. Once an object is summarized by a vector of numbers, the question "what should it mean for a sequence of such objects to converge?" has an almost forced answer — converge coordinate by coordinate, motif by motif — and that is exactly the candidate the next section adopts and examines. We hand section 16.2 this functional, together with the one structural property it will most want kept in view: multiplicativity, which says the profile is determined by its values on connected patterns and which resurfaces as a defining constraint when section 16.3 asks which vectors of numbers are graphon profiles at all.

16.2 Left-convergence and the moment topology

Section 16.1 handed us a single, well-tempered observable. The map $F \mapsto t(F, \cdot)$ is now one and the same functional — valued in $[0, 1]$, multiplicative over disjoint unions, blind to relabelling — whether its second argument is a finite graph or a graphon. We have, in other words, manufactured for every network a whole *vector* of numbers: one motif frequency for each pattern F, the graph's moments. The question this chapter was built to answer can now be asked in the only form that does the observable justice. We declared at the outset that the right notion of convergence must be blind to vertex labels and must capture only statistical content; we have just produced a labelling-blind, purely statistical summary of every graph; so the candidate definition of convergence writes itself. Two graphs are close when their moments are close, and a sequence converges when *every* one of its motif frequencies settles down. Convergence of graphs becomes coordinatewise convergence of their moment vectors. That single move — promoting the per-pattern densities into one point of an infinite-dimensional cube and reading convergence off coordinate by coordinate — is the central act of the chapter, and the present section carries it out and then takes the measure of what it costs and what it buys.

The section proceeds in four beats, and it is worth laying them out in advance so the reader sees the arc rather than four disconnected results. *First* we make the definition itself: assemble the densities $t(F, W)$ over all F into a single point of $[0, 1]^{\mathcal{F}_0}$, the homomorphism profile, and define *left-convergence* as coordinatewise convergence of these profiles — pausing only to see why the test family must be *all* of $\mathcal{F}_0$ and not some thrifty subset. *Second* we identify the topology this definition carries: coordinatewise convergence on a countable product is the product ("moment") topology, and because the index set $\mathcal{F}_0$ is countable it is metrizable by an explicit weighted-sum metric, which makes left-convergence an ordinary metric convergence rather than something exotic. *Third* we confront, honestly, the gap between what this soft topology gives for free and what we actually want. Compactness of the profile cube is cheap — a countable diagonal extraction shows every graph sequence has a left-convergent subsequence — but it delivers only an abstract limit *point* of moment vectors, and says nothing about whether that point is the profile of a genuine graphon. Closing that gap is the Lovász–Szegedy realization theorem, which we state precisely and locate, deferring its proof to chapter 17 and appendix F. *Fourth* we prove the one piece of the deferred machinery that belongs here and is entirely elementary: the *counting lemma*, which bounds the change in any motif density by the cut-norm distance between graphons, and so reveals the moment map as Lipschitz — the bridge to the metric theory of chapter 17.

Holding those four beats together is a single strategic asymmetry, and stating it now will keep the reader oriented through everything that follows. The map from a graphon to its moment profile is *easy*: the counting lemma shows it is not merely continuous but Lipschitz, and its proof is three lines of telescoping and a convexity argument. The map the other way — from a profile back to a graphon that realizes it — is *deep*: it is the existence half of the Lovász–Szegedy theorem, and the honest accounting of which results it rests on (the weak regularity lemma, the inverse counting lemma) places its proof squarely in later chapters. The whole moment theory of graphs lives in the tension between these two directions, the forward map cheap and the inverse map the entire substance of the subject. This section proves the cheap direction in full and signposts the deep one with care; the genuinely hard inverse question — exactly which profiles are realizable, and whether the realizing graphon is unique — is what section 16.3 will take up as the moment problem. We begin with the definition, assembling the per-graph densities into a single profile and declaring convergence to be coordinatewise convergence of profiles.

16.2.1 Definition and the choice of test family

The section preamble left us with a vector of numbers attached to every network — one motif frequency $t(F, W)$ for each pattern F — and the resolve to call two networks close when these vectors are close. To turn that resolve into a definition we first need a single home for the vector. The patterns F are indexed by $\mathcal{F}_0$, the set of all finite simple graphs taken up to isomorphism; each entry is a density living in $[0,1]$; so the whole vector is a point of the product cube $[0,1]^{\mathcal{F}_0}$, one coordinate axis per motif. Assembling the per-pattern densities of section 16.1 into one such point is the move that converts a graph, or a graphon, into a *single* object in a fixed space where all networks can be compared on a common footing, and it is the first thing to write down.

Let $\mathcal{F}_0$ denote the (countable) set of all finite simple graphs up to isomorphism. Each graphon W, and each finite graph G, gives a point

$$t(\,\cdot\,,W) \;=\; \big(t(F,W)\big)_{F\in\mathcal{F}_0} \;\in\; [0,1]^{\mathcal{F}_0},$$

its *homomorphism profile*. Two remarks make this an honest definition rather than a piece of notation. That $\mathcal{F}_0$ is *countable* is what keeps the cube tractable: there are finitely many graphs on each fixed number of vertices, and a countable union of finite sets is countable, so the profile is a sequence of numbers, not an uncountable tuple — the fact the next subsection leans on to metrize everything. That we take graphs *up to isomorphism* is forced by the whole programme: the density $t(F,W)$ depends only on the isomorphism class of F (relabelling the pattern permutes the integration variables of definition 16.8 and leaves the integral unchanged), so isomorphic patterns would contribute identical coordinates and listing them separately would only pad the cube with redundant axes. With the home fixed, convergence of graphs is declared to be convergence of these profiles, read off one coordinate at a time.

Definition 16.11 (Left-convergence)**.** A sequence (G_n) of finite simple graphs with $|V(G_n)|\to\infty$ is *left-convergent* if the limit $\lim_n t(F,G_n)$ exists in $[0,1]$ for every $F\in\mathcal{F}_0$. The sequence *converges to the graphon W* if

$$\lim_{n\to\infty} t(F,G_n) \;=\; t(F,W) \qquad \text{for every } F\in\mathcal{F}_0.$$

By remark 16.7 the same notion results if $t(F,\cdot)$ is replaced throughout by $t_{\mathrm{inj}}(F,\cdot)$ or $t_{\mathrm{ind}}(F,\cdot)$. More generally, a sequence of *graphons* (W_n) is *left-convergent* if $\lim_n t(F,W_n)$ exists for every $F\in\mathcal{F}_0$, and converges to W if $t(F,W_n)\to t(F,W)$ for every F; a sequence of finite graphs is the special case $W_n = W^{G_n}$ via proposition 16.9, and we freely identify G_n with its pixel graphon W^{G_n} when applying graphon-level statements to finite networks.

Three features of this definition repay a slower look: the growth hypothesis on the hosts, the robustness to the choice of normalization, and the seamless passage between finite graphs and graphons. The fourth and most important — why the test family must be *all* of $\mathcal{F}_0$ — we take up last and at length, since it is the heart of the matter.

The hypothesis $|V(G_n)|\to\infty$ is not a throwaway regularity condition; it is what makes left-convergence a statement about *limits of graphs* rather than about graphs themselves, and it is exactly tuned to the counting estimate of section 16.1. Two things break without it. The cheap one is that it rules out the degenerate sequence that never grows: the constant sequence $G_n\equiv G$ trivially has all densities convergent, but it converges to the finite graph G, which is not a graphon limit in any interesting sense — a single dense graph is not the kind of object the theory is built to produce. Demanding $|V(G_n)|\to\infty$ excludes this and every sequence with a bounded subsequence of orders, so that a left-convergent sequence is genuinely a sequence of ever-larger networks settling toward a continuum object. The deeper reason is quantitative and is the one the growth hypothesis was reverse-engineered from. Lemma 16.6 showed that for a k-vertex pattern the hom-, injective-, and induced-normalizations of the density disagree only by the diagonal-coincidence correction $\binom{k}{2}/n$, where $n = |V(G_n)|$; this correction is precisely the weight of the collapsing maps that pinch two vertices of F onto one vertex of G_n. It tends to zero *if and only if* $n \to \infty$. So $|V(G_n)| \to \infty$ is exactly the condition under which the three normalizations become asymptotically equal — the condition that makes the next clause of the definition true.

That next clause is the robustness recorded in the definition: by remark 16.7 one may read $t(F, \cdot)$, or $t_{\mathrm{inj}}(F, \cdot)$, or $t_{\mathrm{ind}}(F, \cdot)$, and obtain the same left-convergent sequences with the same limits. This is more than a convenience. It says that left-convergence is a property of the *geometry* of the sequence, not of the bookkeeping convention we happen to favour: whether we count all maps, only injective maps, or only induced copies, the verdict on which sequences converge and to what is identical, because (by the growth hypothesis just discussed) the three counts are squeezed together as $n \to \infty$. We are therefore free to compute with whichever normalization is most convenient for a given pattern — $t(,)$ for its clean product structure, t_{ind} when a genuine induced-copy count is wanted — and the notion of convergence does not notice the switch.

The third feature is the one that lets the whole apparatus address finite networks and continuum limits in one breath. Proposition 16.9 established that the density of F into a finite graph G and the density of F into its pixel graphon W^G are literally the same number, $t(F, G) = t(F, W^G)$. Consequently a finite graph *is* a graphon for every purpose internal to this theory: its profile is the profile of a step kernel. This is why the definition can phrase left-convergence for a sequence of graphons and then recover the finite-graph case merely by setting $W_n = W^{G_n}$, and why we will, without further comment, write graphon-level statements (the counting lemma, the cut-norm estimates of chapter 17) and apply them to finite networks by passing to their pixel graphons. We keep the *graphon*-sequence version of the definition in full view, not just the finite-graph one, because the later sections need it: the realization theorem produces a limiting graphon, the moment problem ranges over all graphons, and the worked computation at the end of the chapter compares sequences of graphons directly.

We come now to the decision that gives the definition its content, the one the section preamble flagged as essential: the profile records a density for *every* pattern $F \in \mathcal{F}_0$, and left-convergence demands that *all* of them converge. Why not economize — test only against the edge K_2, or against edges and triangles, and call a sequence convergent when those few coordinates settle? The answer is the moment analogy taken seriously, and it is worth making concrete rather than asserting. A single moment of a probability measure — say the mean $\int t\, \mu(\mathrm{d}t)$ — pins down one number about μ and leaves everything else free; two measures with the same mean can be wildly different, and to separate them one needs the higher moments $\int t^k\, \mu(\mathrm{d}t)$. The edge density $t(K_2, \cdot)$ is exactly the mean: it is the crudest, lowest moment of the graph, and a host of structurally unalike networks share it.

The sharpest illustration is the one this chapter's threads return to. Compare two graphons of the same edge density $\frac{1}{2}$. The first is the constant graphon $W \equiv \frac{1}{2}$, the homogeneous "quasirandom" kernel in which every pair of types is joined with the same weight $\frac{1}{2}$ and there is no community structure at all; a large graph sampled from it looks like an Erdős–Rényi graph $G(n, \frac{1}{2})$, edges scattered without correlation. The second is the single-block kernel $W_{\mathrm{cl}} = \mathbf{1}[\,x, y \le c\,]$ with $c = 1/\sqrt{2}$: a clique of relative size c sitting in an otherwise empty network, the extreme of clustered structure. Their edge densities coincide,

$$t(K_2, W) = \tfrac{1}{2}, \qquad t(K_2, W_{\mathrm{cl}}) = \int_{[0,1]^2} \mathbf{1}[\,x, y \le c\,]\ \mathrm{d}x\ \mathrm{d}y = c^2 = \tfrac{1}{2},$$

so the edge test — the mean — cannot tell them apart: it sees two networks of identical overall link frequency and reports a single number that is blind to whether those links are spread evenly or piled into one dense lump. The very next moment exposes the difference. The triangle density of the homogeneous kernel is $t(K_3, W) = (\frac{1}{2})^3 = \frac{1}{8} = 0.125$, the product of three independent edge weights, while for the clustered kernel $t(K_3, W_{\mathrm{cl}}) = c^3 = 2^{-3/2} \approx 0.354$, nearly three times larger, because in a clique every triple of present-edges closes into a triangle whereas in the homogeneous graph triangles must form by coincidence. What the edge density alone cannot see is precisely this: *the correlation between edges* — whether the presence of xy and yy' as edges makes xx' more likely to be an edge too. That correlation is a higher-order statistic, and reading it off requires the higher motif K_3, exactly as detecting the spread of a measure beyond its mean requires the second moment. Stopping at edges, or at edges and triangles, or at any finite slate of motifs, always leaves some pair of graphons that the chosen slate cannot separate but a slightly larger motif can; only the *entire* family $\mathcal{F}_0$ resolves every structural distinction the densities are capable of detecting. The full profile is the graph analog of testing a measure against *all* monomials t^k, not against the mean alone — and just as the full sequence of monomials determines a compactly supported measure (the determinate moment problem of section 16.3), the full family of motif densities is what stands a chance of determining the graphon. That "stands a chance" is the open question the chapter is built to settle; for now it fixes the test family once and for all as the whole of $\mathcal{F}_0$.

This leaves us holding a concrete object — the profile space $[0,1]^{\mathcal{F}_0}$ — and a concrete notion of convergence inside it — coordinatewise convergence of profiles, one motif at a time. Neither has yet been given any topological standing: we have a definition of convergent sequences but no metric, no guarantee that this convergence is tame enough to support the compactness and continuity arguments the rest of the section will run on. Supplying that standing is the next task. Because $\mathcal{F}_0$ is countable, coordinatewise convergence on the cube $[0,1]^{\mathcal{F}_0}$ is nothing more exotic than convergence in a countable product topology, and a countable product of intervals is metrizable; the next subsection makes this precise, exhibits an explicit metric, and harvests the cheap compactness that the product structure hands over for free.

16.2.2 The induced topology is a product topology, and it is metrizable

The previous subsection left us with exactly the raw materials of a topology and nothing yet assembled from them: a fixed home, the cube $[0,1]^{\mathcal{F}_0}$, and a rule for declaring sequences convergent, namely coordinatewise convergence of profiles — $t(F, G_n) \to t(F, W)$ separately for each motif F. "Convergent in every coordinate" is not yet a metric statement; it is a statement about a *product* of topologies, one copy of $[0,1]$ for each axis F, with the product (also called the *moment*) topology being by definition the coarsest one in which all the coordinate projections are continuous.

Convergence in that topology is precisely coordinatewise convergence, so the notion of definition 16.11 is, verbatim, convergence in the product topology on $[0,1]^{\mathcal{F}_0}$. The single fact that turns this abstract identification into something usable is the one the preceding subsection was at pains to record: $\mathcal{F}_0$ is *countable*. A countable product of metric spaces is metrizable, so the product topology on our cube is not some exotic object on which only nets converge — it is induced by an honest distance function, which we can write down by hand. Doing so, and harvesting the compactness that comes with it, is the whole of this subsection.

The recipe for the metric is the standard one for any countable product, and it is worth seeing why it must take the shape it does. We want a single number measuring the discrepancy between two whole profiles, built from the per-coordinate discrepancies $|t(F_j,W)-t(F_j,W')|$. Each of those is already a number in $[0,1]$, so a sum of them would be the obvious candidate — but the sum over the infinitely many motifs F_j need not converge. The fix is to damp the later coordinates with summable weights, and the cleanest summable weights are the powers 2^{-j}, whose total is 1. Damping by 2^{-j} does no harm to the topology — every coordinate still gets a strictly positive weight, so no axis is ignored — while it guarantees the sum converges and stays bounded by 1. With this in mind the definition writes itself. Because $\mathcal{F}_0$ is countable we may enumerate it $F_1, F_2, F_3, \ldots$ (say, by increasing number of vertices and then by number of edges, so that the enumeration is concrete and the early, structurally important motifs $K_2, P_3, K_3, \ldots$ carry the heaviest weights), and form the weighted sum.

Definition 16.12 (A left-convergence metric)**.** Fix an enumeration $\mathcal{F}_0=\{F_1,F_2,\ldots\}$. For graphons W, W' (or for finite graphs via their step graphons) set

$$d_{\mathrm{left}}(W,W') \;=\; \sum_{j\geq 1} 2^{-j}\,\big|\,t(F_j,W)-t(F_j,W')\,\big|.$$

The choice of weights and of enumeration is a matter of convenience, not of substance: any other strictly positive summable weights (w_j), or any other listing of $\mathcal{F}_0$, would give a different distance function but the *same* topology, because what the metric has to reproduce — and all it has to reproduce — is coordinatewise convergence, which sees only that each weight is positive, never its precise value. The next proposition makes that claim exact and, in the same breath, names the one defect that keeps d_{left} from being a genuine metric.

Proposition 16.13 (Metrization of the moment topology)**.** *d_{left} is a pseudometric on graphons taking values in $[0,1]$, and a sequence (G_n) converges to W in the sense of definition 16.11 if and only if $d_{\mathrm{left}}(G_n,W)\to 0$. It is the restriction to homomorphism profiles of the product topology on $[0,1]^{\mathcal{F}_0}$, which is compact by Tychonoff's theorem and metrizable because $\mathcal{F}_0$ is countable.*

Proof. That d_{left} has the formal properties of a distance is read off termwise. Each summand $\rho_j(W,W') := |t(F_j,W)-t(F_j,W')|$ is the absolute difference of two coordinates, hence symmetric, non-negative, and subject to the one-dimensional triangle inequality $\rho_j(W,W'')\leq \rho_j(W,W')+\rho_j(W',W'')$; multiplying by the positive weight 2^{-j} and summing preserves all three, so d_{left} is symmetric, non-negative, and satisfies the triangle inequality. Each ρ_j lies in $[0,1]$ because both densities do, so the series is dominated termwise by $\sum_{j\geq 1} 2^{-j}=1$ and $d_{\mathrm{left}}\leq 1$; in particular the sum always converges and the function is well defined. The one axiom that *fails* — that $d_{\text{left}}(W, W') = 0$ should force $W = W'$ — is exactly what makes it a *pseudo*metric, and we return to it below.

The heart of the statement is that $d_{\text{left}} \to 0$ is equivalent to coordinatewise convergence, and this is the general head-and-tail argument for a weighted sum of bounded coordinate pseudometrics, which we give in full because it recurs whenever a countable product is metrized.

Convergence of all coordinates forces $d_{\text{left}} \to 0$. Suppose $\rho_j(G_n, W) \to 0$ as $n \to \infty$ for every fixed j — that is, every motif density converges. Fix $\varepsilon > 0$. Split the sum at an index J into a head and a tail,

$$d_{\text{left}}(G_n, W) \;=\; \underbrace{\sum_{j=1}^{J} 2^{-j} \rho_j(G_n, W)}_{\text{head}} \;+\; \underbrace{\sum_{j>J} 2^{-j} \rho_j(G_n, W)}_{\text{tail}}.$$

The tail is controlled with no reference to n at all: each $\rho_j \le 1$, so $\sum_{j>J} 2^{-j} \rho_j \le \sum_{j>J} 2^{-j} = 2^{-J}$, and choosing J large enough that $2^{-J} < \varepsilon/2$ makes the tail smaller than $\varepsilon/2$ for *every* n. With J now fixed, the head is a sum of *finitely many* terms, each of which tends to 0 by hypothesis (a finite sum of null sequences is null); so there is an N beyond which the head is $< \varepsilon/2$. For $n \ge N$ both pieces are below $\varepsilon/2$ and $d_{\text{left}}(G_n, W) < \varepsilon$. This is the crux: the damping weights make the infinite tail uniformly negligible, after which only finitely many coordinates have to be driven down, which the hypothesis does one at a time.

$d_{\text{left}} \to 0$ *forces convergence of every coordinate.* This direction is even shorter, and is where the positivity of each weight is used. Every summand is non-negative, so any single one is bounded above by the whole sum:

$$2^{-j} \rho_j(G_n, W) \;\le\; \sum_{i \ge 1} 2^{-i} \rho_i(G_n, W) \;=\; d_{\text{left}}(G_n, W).$$

Hence for each fixed j, if $d_{\text{left}}(G_n, W) \to 0$ then $\rho_j(G_n, W) \le 2^j \, d_{\text{left}}(G_n, W) \to 0$ — the factor 2^j is a constant once j is fixed — so that coordinate converges. Since j was arbitrary, all coordinates converge.

The two directions together say that $d_{\text{left}}(G_n, W) \to 0$ holds if and only if $t(F, G_n) \to t(F, W)$ for every $F \in \mathcal{F}_0$, which is verbatim the definition of left-convergence (definition 16.11). So d_{left} metrizes exactly the convergence we declared, and that convergence is convergence in the product topology on the cube $[0,1]^{\mathcal{F}_0}$: the moment topology is metrizable, with d_{left} an explicit metric for it. The cube itself is compact in this topology — a product of the compact intervals $[0, 1]$ — a fact one may file under Tychonoff's theorem, and metrizable because $\mathcal{F}_0$ is countable. (We will see in remark 16.14 that the only part of this compactness ever used is the elementary sequential part, for which the full Tychonoff theorem is overkill.) □

It remains to confront the failed axiom, the one that demotes d_{left} from metric to *pseudo*metric: two graphons can be at zero distance without being equal. By construction $d_{\text{left}}(W, W') = 0$ means every coordinate pseudometric vanishes, i.e. $t(F, W) = t(F, W')$ for *all* $F \in \mathcal{F}_0$: the two graphons have identical homomorphism profiles. That this can happen for genuinely different kernels $W \ne W'$ is not a defect of our metric but a feature of the objects: homomorphism densities are, by design, blind to anything the moment family cannot resolve, and that includes the labelling of types. Two graphons with the same profile are called *weakly isomorphic*; chapter 17 shows this is exactly the relation $\delta_\square(W, W') = 0$ (proposition 17.7), so that $d_{\text{left}} = 0$ and $\delta_\square = 0$ cut out the same equivalence, and quotienting graphon space by it — the construction chapter 17 carries out — promotes d_{left} to a true metric on the quotient. The pseudometric is therefore precisely as fine as the densities allow and no finer; it is the correct object, and the loss of separation is the price, paid honestly, for relabelling-blindness.

It is worth being exact about what "relabelling-blind" covers, because the obvious guess undersells it. One certainly expects the profile to be unchanged when types are relabelled by a single measure-preserving *bijection* $\sigma\colon [0,1] \to [0,1]$ — replacing W by $W^\sigma(x,y) = W(\sigma x, \sigma y)$ merely renames the integration variables in definition 16.8 and leaves every integral fixed. But weak isomorphism is strictly coarser than that: it identifies graphons related by measure-preserving maps that need not be invertible. A clean witness is the *doubling* map $\sigma(x) = 2x \bmod 1$, which pushes Lebesgue measure forward to Lebesgue measure but is two-to-one, not a bijection. For any graphon W the pulled-back kernel $W'(x,y) = W(\sigma x, \sigma y)$ has, by the change-of-variables formula coordinate by coordinate,

$$t(F, W') = \int_{[0,1]^k} \prod_{ij \in E(F)} W(\sigma x_i, \sigma x_j)\, \mathrm{d}x = \int_{[0,1]^k} \prod_{ij \in E(F)} W(u_i, u_j)\, \mathrm{d}u = t(F, W)$$

for every F — the substitution $u_i = \sigma x_i$ sends each factor of Lebesgue measure to itself — so W and W' are weakly isomorphic, yet no single bijection of $[0,1]$ relates them in general, since σ collapses two type-intervals onto one. This is the concrete shadow of a fact chapter 17 treats with care: the infimum defining the cut distance $\delta_\square$ need not be attained by a bijection, so "$\delta_\square = 0$" is genuinely the equivalence generated by such non-invertible couplings, of which "differ by a relabelling bijection" is only the cleanest special case. We flag the distinction here and lean on it there; for the present purpose all that matters is that d_{left} separates graphons exactly up to weak isomorphism, no more and no less.

With the topology pinned down and metrized, we collect the dividend the product structure pays out for free: a cheap compactness, strong enough to manufacture limit points and weak enough to leave the real work undone.

Remark 16.14 (Compactness for free, realization is the hard part)**.** There is an immediate but *weak* compactness statement: every sequence of finite graphs (G_n) has a subsequence (G_{n_ℓ}) whose profile $\big(t(F, G_{n_\ell})\big)_F$ converges coordinatewise to *some* point $\tau \in [0,1]^{\mathcal{F}_0}$ — i.e. every graph sequence has a left-convergent subsequence. The mechanism is the classical *diagonal extraction*, and it deserves to be seen in the open because it, and not any heavy theorem, is what is actually at work. Enumerate $\mathcal{F}_0 = \{F_1, F_2, \dots\}$ as before. The first coordinate $t(F_1, G_n)$ is a sequence in the compact interval $[0,1]$, so by Bolzano–Weierstrass some subsequence $(G_n^{(1)})$ makes it converge. Inside that subsequence the second coordinate $t(F_2, G_n^{(1)})$ again lives in $[0,1]$, so a further subsequence $(G_n^{(2)})$ makes it converge too — and the first coordinate, being a subsequence of a convergent sequence, still converges. Continuing, we obtain nested subsequences $(G_n^{(1)}) \supseteq (G_n^{(2)}) \supseteq \cdots$, the m-th of which converges in the first m coordinates. The diagonal sequence $G_{n_\ell} := G_\ell^{(\ell)}$ is, from its m-th term on, a subsequence of $(G_n^{(m)})$, hence converges in coordinate m for *every* m; that is, its whole profile converges coordinatewise to a limit $\tau \in [0,1]^{\mathcal{F}_0}$. The entire argument uses only that each coordinate ranges over the compact interval $[0,1]$, invoked *countably* many times — nothing more.

It is precisely this point that we want to keep honest, because it is easy to dress the same conclusion in grander clothing. One may, if one likes, attach the name "Tychonoff" to the compactness of the full cube $[0,1]^{\mathcal{F}_0}$, and the proposition above did so. But the unrestricted Tychonoff theorem — an *arbitrary* product of compacts is compact — is a statement about possibly uncountable products that delivers only *topological* compactness (every open cover has a finite subcover), from which sequential compactness does not follow for free in a non-metrizable space, and whose proof leans on the axiom of choice. None of that machinery is needed here. Our index set $\mathcal{F}_0$ is countable, the cube is metrizable, topological and sequential compactness coincide, and the sequential statement is supplied outright by the diagonal argument over a countable family of compact intervals. The downgrade is deliberate: "Tychonoff" is at most a name for a fact we prove by hand, and the operative content is the countable diagonal extraction, never the uncountable product theorem.

What this compactness gives, then, is a limit *point*; what it conspicuously does *not* give is a limit *object*. The extracted τ is an element of the abstract cube $[0,1]^{\mathcal{F}_0}$ — a coherent table of numbers, one per motif, arising as a coordinatewise limit of honest graph profiles. Nothing in the extraction certifies that τ is itself the profile $t(\cdot, W)$ of any graphon: a priori it is merely a point in the cube, an abstract limit of moment vectors, and the cube is vastly larger than the set of profiles sitting inside it. The gap between "τ exists as a limit of profiles" and "τ *is* a profile" is the entire remaining difficulty, and it is a real one — the analogue, in the moment picture, of asking not merely whether a sequence of moment vectors has a limit but whether that limiting sequence of numbers is the moment sequence of an actual measure. Cheap compactness answers the first; only a realization theorem can answer the second.

That realization is the one thing the product topology cannot hand us, and supplying it is the next order of business. The subsequential compactness just proved guarantees an abstract limit profile $\tau \in [0,1]^{\mathcal{F}_0}$ for every graph sequence, but leaves open whether $\tau = t(\cdot, W)$ for some graphon W — whether the limit table of motif frequencies is genuinely realized by a continuum network. The Lovász–Szegedy realization theorem asserts that it always is: every left-convergent sequence, and a fortiori every such limit point τ, is the homomorphism profile of an actual graphon. We state it precisely next, locate its proof in the cut-metric machinery of chapter 17 and appendix F, and isolate the one piece of its supporting argument — the counting lemma — that belongs in this chapter.

16.2.3 The Lovász–Szegedy realization theorem

The previous subsection left us holding an abstract limit and nothing to anchor it to. Diagonal extraction (remark 16.14) manufactures, for every graph sequence, a coordinatewise limit point $\tau \in [0,1]^{\mathcal{F}_0}$ — a coherent table of numbers, one frequency per motif, got as the limit of honest graph profiles — but it certifies nothing about whether that table is the profile of any *object*. The cube $[0,1]^{\mathcal{F}_0}$ is enormous; the profiles $t(\cdot, W)$ of actual graphons form a thin, intricately constrained subset of it, and a priori a limit of points in that subset could perfectly well escape it, landing on a number-table that no graphon produces. Closing exactly this gap — promoting the abstract limit vector τ to a statement that $\tau = t(\cdot, W)$ for a genuine continuum network W — is the content of the theorem this subsection is built around. It is the assertion that the thin constrained subset of realizable profiles is already *closed*: a limit of graphon profiles is again a graphon profile.

Theorem 16.15 (Every left-convergent sequence has a graphon limit; Lovász–Szegedy)**.** *For every left-convergent sequence (G_n) of finite simple graphs with $|V(G_n)| \to \infty$ there exists a graphon W satisfying **(A-Reg-W)** (symmetric, measurable, $[0,1]$-valued) such that*

$$\lim_{n\to\infty} t(F, G_n) \;=\; t(F, W) \qquad \textit{for every } F \in \mathcal{F}_0.$$

The graphon W is unique up to weak isomorphism, i.e. up to $\delta_\square = 0$ (proposition 17.7).

This is the founding theorem of dense graph limit theory: it is what makes the whole moment programme more than a bookkeeping device. Everything in this chapter up to now has been, in a strict sense, formal — we defined a profile, declared a topology, metrized it, and extracted abstract limits — and none of it required that the limits be anything but points in a cube. The theorem is what installs the *object* on the other side of the limit, and that is precisely where the depth lies. It is worth being emphatic about what is and is not hard here, because the two are easy to conflate. That the limit *exists as a number-vector* is the cheap part, already in hand: it is nothing but Bolzano–Weierstrass applied countably many times, the diagonal argument of remark 16.14, and it would hold for any countable family of $[0,1]$-valued statistics whatever, with no graph theory in it at all. What is deep is the *realization* — that this particular limit vector is not some exotic point of the cube but is hit by an actual symmetric measurable kernel on $[0,1]^2$. The analogy that organizes the chapter says it sharply: producing a candidate moment sequence as a limit of genuine moment sequences is elementary; proving that the candidate is *the moment sequence of an actual measure* is the whole moment problem. The realization theorem is the graph-limit instance of that second, substantial step, and it is the reason a left-convergent graph sequence has a bona fide limiting network rather than merely a limiting list of frequencies.

We do not prove theorem 16.15 in this chapter. Per the self-containedness discipline, "we do not prove it here" must come with a precise pointer to where in this book it *is* proved, and it does: its natural proof runs through the cut metric $\delta_\square$ and the compactness of the resulting quotient space, machinery that is developed in chapter 17 and carried to a complete proof in appendix F. What belongs here is an honest map of that dependency, so the reader knows exactly which load-bearing facts are imported and from where. Chapter 17 supplies two ingredients. The first is an *equivalence*: left-convergence of a graph sequence is the same thing as convergence in the cut metric $\delta_\square$, up to relabelling — the moment topology of this chapter and the metric topology of chapter 17 are, on graphon space, one and the same. The second is a *compactness*: the quotient graphon space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ — the space of graphons modulo weak isomorphism, equipped with $\delta_\square$ — is compact, in particular complete, so that every Cauchy sequence in it converges to a point of it.

Granting those two imports, the realization theorem assembles itself in a short chain, and it is worth walking the links so the logical shape is unmistakable. Start with a left-convergent (G_n). By the *equivalence*, its profiles converging coordinatewise means the G_n form a $\delta_\square$-Cauchy sequence in the quotient space: motif frequencies settling down is the same news as the graphs bunching together in cut metric. By the *compactness* (hence completeness) of $(\widetilde{\mathcal{W}}_0, \delta_\square)$, that Cauchy sequence has a genuine limit there — a graphon W, defined up to weak isomorphism, with $\delta_\square(G_n, W) \to 0$. Applying the *equivalence* once more, now in the reverse reading, cut-metric convergence to W forces $t(F, G_n) \to t(F, W)$ for every motif F. So the abstract limit τ is realized: $\tau = t(\cdot, W)$, and the uniqueness up to weak isomorphism is exactly the statement that $\delta_\square$ identifies graphons precisely when their profiles agree (proposition 17.7). The realization theorem is, in this reading, the round trip *left-convergent* $\Rightarrow$ $\delta_\square$*-Cauchy* $\Rightarrow$ $\delta_\square$*-convergent to* W $\Rightarrow$ *convergent in density*, with the equivalence used at both ends and the compactness supplying the limit in the middle.

The equivalence on which this turns is itself two implications of very unequal difficulty, and keeping them separate is the chapter's honest accounting of what it proves versus what it imports. One direction is *easy*: closeness in $\delta_\square$ forces closeness of all motif densities — if two graphons are near in cut metric then every homomorphism frequency is near. This is the counting lemma, and it is completely elementary, a few lines of telescoping and a cut-norm bound. It is the one piece of the entire realization argument that lives in this chapter, and we prove it in the very next subsection. The other direction is *hard*: closeness of *all* motif densities forces closeness in $\delta_\square$ — if every frequency agrees then the graphons are near in cut metric. This is the *inverse counting lemma*, the genuine content behind the moment-uniqueness theorem theorem 17.11, and it is not elementary at all: it rests on the weak regularity lemma (theorem 17.16), the structural result that any graphon is approximated in cut norm by a bounded-complexity step function, which is the single deepest import of the whole development. The asymmetry is the usual one in a moment theory. That cut-close profiles are density-close is the forward, "moments are continuous" direction; that density-close profiles are cut-close is the inverse, "the moments determine the object" direction, and inverting a moment map is always where the work is. Both halves, and the compactness, are proved in full in chapter 17 and appendix F; this chapter contributes the forward half and imports the rest with the pointers just given.

Notation watch

The label "Lovász–Szegedy theorem" is attached in this book to two distinct but linked statements: the *realization* theorem theorem 16.15 (every left-convergent graph sequence has a graphon limit) and the *compactness* theorem of chapter 17 (the quotient graphon space is compact in $\delta_\square$). They are equivalent given the metric equivalence of chapter 17; we prove the compactness version in appendix F and derive the realization version from it (the chain above). When we write "by Lovász–Szegedy" we mean whichever of the two is being invoked, and we always say which.

With the founding theorem stated and its logical debts itemized, exactly one of those debts is ours to discharge here: the easy half of the equivalence, that cut-closeness implies density-closeness. That is the counting lemma, and proving it is the business of the next subsection — the single piece of the realization theorem's supporting argument that belongs in this chapter, and the one we turn to now.

16.2.4 The counting lemma

We turn now to the single debt the realization theorem leaves for this chapter to settle in full. The preceding subsection split the equivalence between left-convergence and cut-metric convergence into two implications of wildly unequal difficulty and reserved exactly one of them for proof here: the *easy* direction, that closeness in cut norm forces closeness of every motif frequency. That implication is the *counting lemma*, and it has a sharp quantitative shape. Homomorphism densities are not merely continuous in the cut norm; they are *Lipschitz* in it, with an explicit and almost embarrassingly small Lipschitz constant — the number of edges of the pattern, and nothing else. Two graphons that are within ε of one another in cut norm have all of their edge densities within ε, all of their triangle densities within 3ε, all of their F-densities within $|E(F)|\ \varepsilon$. This is the whole of the forward half of the equivalence, and it is genuinely elementary: a telescoping sum and one application of the defining property of the cut norm.

It is worth saying at the outset why so plain a lemma earns the title "workhorse." Read it in the language of the map Θ that the chapter has been building toward, the assignment $W \mapsto \big(t(F,W)\big)_{F\in\mathcal{F}_0}$ that sends a graphon to its moment profile. The counting lemma is precisely the statement that *each coordinate* of Θ is Lipschitz, and therefore that Θ is a continuous map from cut-metric graphon space into the profile cube $[0,1]^{\mathcal{F}_0}$: the moments move continuously when the network moves. That is the property that lets the densities serve as a genuine *coordinate system* on graphon space — coordinates that respond continuously to the object they describe, so that controlling the object controls every coordinate at once. Without it, "convergence of all motif frequencies" and "convergence of the underlying networks" would be unrelated notions, and the moment programme would have no traction on the geometry of chapter 17. With it, one half of their identification is immediate, and only the deep half remains to be imported.

We first recall the cut norm; its detailed study, and the metric $\delta_\square$ obtained by also optimizing over relabellings, are the business of chapter 17 (definitions 17.1 and 17.6).

Definition 16.16 (Cut norm; recalled from chapter 17)**.** For a bounded measurable symmetric kernel $H\colon [0,1]^2 \to \mathbb{R}$ the *cut norm* is

$$\|H\|_\square \;=\; \sup_{S,T\subseteq[0,1]} \left| \int_{S\times T} H(x,y)\ \mathrm{d}x\ \mathrm{d}y \right|,$$

the supremum over measurable S,T. We use one elementary fact, proved in chapter 17: the supremum is unchanged if the indicators $\mathbf{1}_S, \mathbf{1}_T$ are replaced by arbitrary measurable functions $g,h\colon [0,1]\to[0,1]$,

$$\sup_{g,h\colon [0,1]\to[0,1]} \left| \int_{[0,1]^2} H(x,y)\, g(x)\, h(y)\ \mathrm{d}x\ \mathrm{d}y \right| \;=\; \|H\|_\square\,, \tag{$\star$}$$

because the bilinear functional $(g,h)\mapsto \int H\,g\,h$ is, for h fixed, linear in g and hence maximized over the convex set $\{g : [0,1]\to[0,1]\}$ at an extreme point $g=\mathbf{1}_S$, and symmetrically in h.

The identity ($\star$) is the only fact about the cut norm the proof will use, so it pays to see clearly why it holds; the argument is a clean instance of the principle that a linear functional on a convex set is maximized at an extreme point. One inequality is free: indicators $\mathbf{1}_S, \mathbf{1}_T$ are in particular $[0,1]$-valued functions, so the supremum over arbitrary $g, h\colon [0,1] \to [0,1]$ is taken over a *larger* family than the supremum over indicators and can only be at least as big — the left side of ($\star$) is $\geq \|H\|_\square$. The substance is the reverse inequality, that enlarging the test family buys nothing. Fix $h\colon [0,1] \to [0,1]$ and consider the functional $g \mapsto \Phi_h(g) = \int_{[0,1]^2} H(x,y)\, g(x)\, h(y)\; \mathrm{d}x\; \mathrm{d}y$. It is *linear* in g, and its domain $K = \{g\colon [0,1] \to [0,1]\}$ is a convex set: a convex combination of two functions valued in $[0,1]$ is again valued in $[0,1]$. A linear functional on a convex set attains its maximum at an extreme point of that set, and the extreme points of K are exactly the $\{0,1\}$-valued functions — the indicators $\mathbf{1}_S$ of measurable sets S — because any g taking a value strictly between 0 and 1 on a positive-measure set can be perturbed up and down on that set, writing g as the midpoint of two distinct admissible functions and so failing to be extreme. Hence $\sup_{g \in K} \Phi_h(g) = \sup_S \Phi_h(\mathbf{1}_S)$, and the same value is attained when we additionally minimize, so the supremum of $|\Phi_h|$ over $g \in K$ equals its supremum over indicators $g = \mathbf{1}_S$. Now repeat the argument in the second slot: with $g = \mathbf{1}_S$ frozen, $h \mapsto \int H\, \mathbf{1}_S\, h$ is again linear in h over the convex set K, maximized at $h = \mathbf{1}_T$. Optimizing the two variables in turn drives both to indicators without ever exceeding $\|H\|_\square$, which is exactly ($\star$). The reason we want the relaxed form, rather than the cleaner-looking indicator form, is entirely practical and will surface in the proof below: the factors that the counting lemma peels off a motif integrand are naturally $[0,1]$-valued functions — products of kernel evaluations — not sharp indicators, and ($\star$) lets us feed them straight into the cut norm with no rounding step.

With the cut norm and its one usable property in hand, here is the lemma. The constant is the edge count $|E(F)|$ — the same integer written $e(F)$ in chapter 17 — and the proof will show that this is no coincidence: we shall trade W for W' one edge at a time, paying a cut norm for each, so the price is exactly one $\|W - W'\|_\square$ per edge.

Lemma 16.17 (Counting lemma)**.** *Let W, W' be graphons with values in $[0,1]$ and let F be a finite simple graph with $|E(F)|$ edges. Then*

$$\left|\, t(F,W) - t(F,W')\, \right| \;\leq\; |E(F)| \cdot \|W - W'\|_\square \,.$$

Proof. Enumerate the edges of F as $e_1, \ldots, e_M$ with $M = |E(F)|$. We pass from W to W' one edge at a time, defining hybrid integrands in which the first r edges carry the kernel W' and the remaining $M - r$ carry W:

$$I_r \;=\; \int_{[0,1]^k} \Big(\prod_{s \leq r} W'(x_{e_s})\Big)\Big(\prod_{s > r} W(x_{e_s})\Big)\; \mathrm{d}x_1 \cdots \mathrm{d}x_k, \qquad k = |V(F)|\,,$$

where x_e abbreviates the pair of coordinates indexed by the endpoints of edge e. Then $I_0 = t(F,W)$ and $I_M = t(F,W')$, and by telescoping

$$t(F,W) - t(F,W') \;=\; \sum_{r=1}^{M} (I_{r-1} - I_r).$$

Fix r and write $e_r = \{a,b\}$. The difference $I_{r-1} - I_r$ replaces the single factor $W(x_a, x_b)$ by $W'(x_a, x_b)$; thus

$$I_{r-1} - I_r \;=\; \int_{[0,1]^k} \; (W - W')(x_a, x_b) \; \prod_{e \neq e_r} [\;\cdot\;](x_e)\; \mathrm{d}x_1 \cdots \mathrm{d}x_k,$$

where each remaining factor is W or W' according to the hybrid and lies in $[0,1]$. The decisive move — and it is the whole proof — is to bound the inner (x_a, x_b)-integral *before* integrating any of the other coordinates, holding them frozen. The order matters, and it is worth being concrete about exactly what goes wrong if one yields to the temptation to integrate the others out first. Take $F = K_3$ the triangle, with vertices a, b, c and $e_r = \{a, b\}$ the edge being swapped, so the two surviving factors sit on the edges $\{a, c\}$ and $\{b, c\}$. If we integrate the lone remaining coordinate x_c out first, those two factors combine into

$$\int_{[0,1]} W(x_a, x_c)\, W(x_c, x_b)\; \mathrm{d}x_c \;=\; (\mathbb{W}^2)(x_a, x_b),$$

the kernel of the squared integral operator — a function of the *pair* (x_a, x_b) that does *not* split as a product $g(x_a)\, h(x_b)$. The common neighbour c has glued the two coordinates together: physically, a and b are no longer independently coupled to their surroundings but share a neighbour, and integrating that neighbour out first bakes the sharing into an inseparable function of (x_a, x_b). Against such a function the extreme-point identity ($\star$), which can only see integrands of the product form $g(x_a)h(x_b)$, has no purchase, and the bound $\|W - W'\|_\square$ is unavailable. The remedy is to do nothing clever: leave x_c (and every other surviving coordinate) *unintegrated*, fixed at a value, and bound the two-dimensional (x_a, x_b)-integral first. With x_c held fixed, $W(x_a, x_c)$ is a function of x_a alone and $W(x_c, x_b)$ a function of x_b alone — the product structure is restored precisely because we declined to integrate the coordinate that couples them. So fix the remaining coordinates at any value $z = (x_u)_{u \neq a,b} \in [0,1]^{k-2}$. With z held fixed, every factor other than the removed one depends on *at most one* of x_a, x_b: a factor $W^{(\prime)}(x_a, x_u)$ on an edge incident to a is, for fixed x_u, a function of x_a alone; a factor on an edge incident to b is a function of x_b alone; a factor on an edge incident to neither (both endpoints in z) is a constant. (No surviving factor touches both x_a and x_b, since e_r is the *unique* edge on the pair $\{a, b\}$ in the simple graph F.) Collecting these,

$$\prod_{e \neq e_r} [\;\cdot\;](x_e) \;=\; g_z(x_a)\, h_z(x_b), \qquad g_z, h_z \colon [0,1] \to [0,1],$$

where g_z gathers the (constant) factors among z together with those on edges incident to a, and h_z those on edges incident to b; both are products of $[0,1]$-valued factors and hence valued in $[0,1]$. By the extreme-point identity ($\star$), applied for each fixed z,

$$\left| \int_{[0,1]^2} (W - W')(x_a, x_b)\, g_z(x_a)\, h_z(x_b)\; \mathrm{d}x_a\; \mathrm{d}x_b \right| \;\leq\; \|W - W'\|_\square\,,$$

a bound *uniform* in z. Integrating this uniform bound over $z \in [0,1]^{k-2}$, a probability space (Tonelli applies, the integrand being bounded and measurable), gives

$$|I_{r-1} - I_r| \;\leq\; \int_{[0,1]^{k-2}} \|W - W'\|_\square \; \mathrm{d}z \;=\; \|W - W'\|_\square\,.$$

Summing the M telescoped terms gives $|I_0 - I_M| \leq M\, \|W - W'\|_\square = |E(F)|\, \|W - W'\|_\square$, which is the claim. $\square$

It is worth registering what the Lipschitz constant $|E(F)|$ does and does not depend on. Unlike the first counting lemma of lemma 16.6, whose constant $\binom{k}{2}/n$ was governed by the number of *vertices* of F (because there it was vertex-collisions being counted), the constant here is the number of *edges.* The reason is structural and visible in the telescope: we paid one cut norm for each edge whose kernel we swapped, and a motif with M edges requires M swaps. A k-vertex tree (with $k-1$ edges) and a k-vertex clique (with $\binom{k}{2}$ edges) therefore incur very different counting-lemma errors even though they have the same number of vertices — the clique, being edge-heavy, is the more sensitive probe of cut-norm discrepancy. Note also what is absent: no dependence on a host size n (these are graphons, with no n), and no constant larger than 1 in front — the estimate is as tight as a telescope of M unit-Lipschitz steps can be. This cleanness is exactly what makes Θ Lipschitz coordinate by coordinate, and it is the property the next corollary harvests.

Corollary 16.18 (Cut-convergence implies left-convergence)**.** *If a sequence of graphons W_n satisfies $\|W_n - W\|_\square \to 0$, then $t(F, W_n) \to t(F, W)$ for every finite simple graph F; that is, cut-norm convergence implies left-convergence. The same holds with $\|\cdot\|_\square$ replaced by the relabelling-invariant cut metric $\delta_\square$, since each $t(F, \cdot)$ is invariant under measure-preserving relabelling.*

Proof. The first assertion is immediate from lemma 16.17: for each fixed F the right-hand side $|E(F)|\,\|W_n - W\|_\square \to 0$, the constant $|E(F)|$ being a fixed integer, so the left-hand side $|t(F, W_n) - t(F, W)|$ is squeezed to 0 as well. Since F was arbitrary, every motif density converges, which is by definition left-convergence.

For the second assertion we must check that each density $t(F, \cdot)$ is blind to relabelling, so that the estimate survives when $\|\cdot\|_\square$ is replaced by its relabelling-optimized refinement $\delta_\square$. A relabelling of a graphon is a measure-preserving map $\sigma\colon [0,1] \to [0,1]$ acting by $W^\sigma(x, y) = W(\sigma x, \sigma y)$ — the continuum analogue of permuting the vertices of a finite graph, which is exactly the relabelling ambiguity the whole chapter set out to quotient away. Its effect on a density is none at all: in the defining integral

$$t(F, W^\sigma) = \int_{[0,1]^k} \prod_{ij \in E(F)} W(\sigma x_i, \sigma x_j)\; \mathrm{d}x_1 \cdots \mathrm{d}x_k,$$

substitute $x_i' = \sigma x_i$ in each of the k coordinates. Because σ is measure-preserving, each single-coordinate change of variables leaves Lebesgue measure invariant, and the integral becomes $\int \prod W(x_i', x_j')\; \mathrm{d}x_1' \cdots \mathrm{d}x_k' = t(F, W)$ verbatim. So $t(F, W^\sigma) = t(F, W)$ for every relabelling σ: the densities are precisely the relabelling-invariant content the chapter promised they would be. Consequently the counting-lemma bound holds with W, W' each replaced by any of their relabelled copies, and taking the infimum of the right-hand side over all relabellings turns $\|W - W'\|_\square$ into the cut *metric* $\delta_\square(W, W')$ (definition 17.6) while the left-hand side, being already relabelling-invariant, is untouched. Thus $\delta_\square(W_n, W) \to 0$ likewise forces left-convergence. $\square$

Corollary 16.18 is the "easy direction" promised after theorem 16.15, and with it the chapter has discharged in full its one share of the realization theorem's supporting argument. The map $\Theta\colon W \mapsto \big(t(F, W)\big)_{F \in \mathcal{F}_0}$ from graphons to moment profiles is now known to be continuous — Lipschitz in each coordinate — and this is exactly the fact section 16.3 will pick up: once Θ is continuous and (modulo relabelling, by the import of chapter 17) injective, the only remaining structural question about it is the description of its *range*, which is the moment problem for graphs.

Before turning there it is worth dwelling on the asymmetry the corollary exposes, because it is the central quantitative fact of the whole dense theory and the reason the two convergence notions behave so differently as *metrics* even though they generate the same *topology*. The forward direction we have just proved is clean and uniform: a Lipschitz bound with constant $|E(F)|$, valid for every pair of graphons, no logarithms, no qualifications. The converse — that left-convergence implies cut-convergence — is true as a statement about topologies but *false* at the level of any comparably clean inequality. There is no bound of the form $\|W - W'\|_\square \leq C \sum_F w_F \, |t(F,W) - t(F,W')|$ with fixed summable weights w_F holding uniformly over all graphons; one cannot recover the cut distance from a fixed weighted combination of density gaps. The correct converse is necessarily quantitative-and-truncated, the *inverse counting lemma* that underlies the moment-uniqueness theorem theorem 17.11 (proved in appendix F): matching all of the *finitely many* densities $t(F,\cdot)$ with $|V(F)| \leq k$ forces the cut distance down only to $O(1/\sqrt{\log k})$. The contrast could hardly be sharper — to make two graphons cut-close to within ε through their moments alone one must control all motifs up to roughly e^{1/ε^2} vertices, an astronomically slow rate, whereas the forward bound delivers ε-closeness of any single density for free from ε-closeness in cut norm. That a moment map is Lipschitz forward but only logarithmically invertible is the graph instance of a phenomenon familiar throughout moment theory: reading off a moment is a bounded, stable operation, but reconstructing the object from its moments is delicate and ill-conditioned. It is exactly this gap that lets left-convergence and cut-convergence coincide as topologies (chapter 17) — the same convergent sequences, the same limits — while remaining wildly inequivalent as metrics, and it is why one direction of the equivalence is a four-line telescope and the other needs the full weight of the regularity lemma.

Heuristic (non-rigorous)

Why should "all motif densities" be enough to pin down a graphon, when even a single density like the edge density is so lossy? The picture is the same as for the classical moment problem, and holding the two side by side makes the chapter's organizing analogy do real work rather than serve as decoration. One moment $\int t\,\mu(\mathrm{d}t)$ fixes only the mean of a measure — two very different distributions can share it — but the whole sequence $\int t^k\,\mu(\mathrm{d}t)$, $k \geq 0$, fixes μ entirely whenever the moment problem is determinate, which on a bounded interval such as $[0,1]$ it always is. The same lossiness and the same eventual rigidity recur for graphs. The edge density $t(K_2, W) = \int W$ is the graph mean, and like a mean it is hopelessly coarse: it cannot tell a graphon with its edges clustered into a dense community from one with the same number of edges spread uniformly. But the full family $\big(t(F,W)\big)_{F\in\mathcal{F}_0}$ — which contains, for *every* k, all the k-vertex patterns, the graph analogue of all the monomials t^k — is rich enough to resolve the graphon down to the one ambiguity it *cannot* and *should* not see: how the type interval $[0,1]$ is labelled. That residual ambiguity is the whole reason the chapter went looking for relabelling-blind observables in the first place, and it is the precise graph counterpart of a measure being determined "up to nothing" once its moments are fixed.

The two halves of the moment programme line up just as cleanly across the analogy. The counting lemma of this subsection is the *forward* half — the moment map $\Theta\colon W \mapsto (t(F,W))_F$ is continuous, exactly as $\mu \mapsto (\int t^k \,\mathrm{d}\mu)_k$ is — and it was the cheap half, a telescope. The regularity lemma of chapter 17 supplies the *inverse* half — the moments determine the object, exactly as a determinate moment sequence determines μ — and it is the expensive half, as inverting a moment map always is. This is the same forward-easy/inverse-hard asymmetry the previous paragraphs measured as a Lipschitz bound against an $O(1/\sqrt{\log k})$ rate. Finally, the chapter's title is meant literally. "The topology of convergence" is not a figure of speech: left-convergence *is* the initial topology generated by the countable family of moment functionals $t(F, \cdot)$ — the coarsest topology on graphon space making every one of those functionals continuous — and the counting lemma is the verification that the cut metric is at least fine enough to belong to it. A graph, read through this chapter, is its sequence of moments; convergence of graphs is convergence of those moments; and that is the moment theory of graphs the chapter set out to build.

16.3 The moment problem for graphs

The previous section ended by assembling, out of all the motif densities at once, a single object: the map that sends a graphon to its entire homomorphism profile. Writing $\mathcal{F}_0$ for the countable set of (isomorphism classes of) finite simple graphs — the index set of all admissible patterns, the graph analogue of the exponents $k = 0, 1, 2, \dots$ that label the monomials t^k — this map is

$$\Theta\colon\ W \ \longmapsto\ \big(t(F, W)\big)_{F\in\mathcal{F}_0} \ \in\ [0,1]^{\mathcal{F}_0}.$$

It takes a graphon to the sequence of all its moments at once, one coordinate $t(F, W)$ per pattern F, and lands in the product cube $[0,1]^{\mathcal{F}_0}$ — the *moment profile space*, topologized in section 16.2 as the product ("moment") topology in which left-convergence is exactly coordinatewise convergence. The whole chapter has been, at bottom, the study of this one map. Two of its three structural features are now in hand, and they are precisely the two that a moment map should have. First, Θ is *continuous*: this is corollary 16.18, the forward half of the counting lemma, which made each coordinate $t(F, \cdot)$ Lipschitz in the cut metric with constant $|E(F)|$, so that cut-close graphons have profiles close in every coordinate. Reading off a moment is a stable operation, just as $\mu \mapsto \int t^k \, \mathrm{d}\mu$ is continuous in the weak topology. Second, Θ is *injective modulo relabelling*: this is the content chapter 17 will supply through the inverse counting lemma and the regularity method — two graphons with identical profiles differ only by a measure-preserving relabelling of the type interval, the one ambiguity the densities are built to ignore and *should* ignore. The moment profile determines the graphon up to nothing else. So Θ is, in the relabelling-quotiented sense, a continuous injection.

Once a map is known to be a continuous injection, exactly one structural question about it remains open, and it is the natural one: what is its *range*? Which points of the target are actually hit? This is the *moment problem for graphs*. Spelled out: which sequences $\tau = (\tau_F)_{F\in\mathcal{F}_0}$ of numbers in $[0, 1]$ are genuinely of the form $\tau = \Theta(W)$ for some graphon W — and which are impostors, number sequences that look like profiles but are realized by no graphon at all? Not every sequence can be a profile: the coordinates are heavily constrained by the fact that they all come from integrating products of *one and the same* kernel, and the moment problem is the project of writing those constraints down exactly, turning the implicit definition "τ lies in the image of Θ" into an explicit, checkable list of conditions on the numbers τ_F themselves.

This is the precise graph counterpart of the oldest question in classical moment theory, and the analogy is not a loose slogan but a structural correspondence that will dictate the exact form of the answer, so it is worth laying the classical side out in full. On the line one asks: given a sequence of reals $(s_k)_{k\geq 0}$, when is there a probability measure μ with $s_k = \int t^k \, \mu(\mathrm{d}t)$ for every k? The classical theorem (Hamburger on $\mathbb{R}$, Stieltjes on $[0,\infty)$, Hausdorff on the bounded interval $[0,1]$, which is our case) answers it with two conditions. The first is a *normalization*: $s_0 = \int 1 \, \mathrm{d}\mu = 1$, since μ is a probability measure. The second is a *positive-semidefiniteness*: the infinite Hankel matrix $H = (s_{i+j})_{i,j\geq 0}$ must be positive semidefinite. The reason is a one-line Gram computation that we will replay almost verbatim on the graph side. For any real coefficients $c_0, \ldots, c_N$,

$$\sum_{i,j} c_i c_j \, s_{i+j} = \sum_{i,j} c_i c_j \int t^{i+j} \, \mu(\mathrm{d}t) = \int \Big(\sum_i c_i t^i \Big)^2 \mu(\mathrm{d}t) \ \geq \ 0,$$

so the Hankel matrix is positive semidefinite precisely *because* it is the Gram matrix of the monomials $1, t, t^2, \ldots$ in the inner product $\langle f, g \rangle = \int fg \, \mathrm{d}\mu$ of $L^2(\mu)$. The crucial structural input that lets this work is that the monomials multiply, $t^i \cdot t^j = t^{i+j}$, so that the product of two basis functions is again a basis function and the inner product of t^i with t^j is itself a moment, s_{i+j}. On a bounded interval these two conditions (suitably stated) are also *sufficient*, and the moment problem is *determinate*: the measure is unique. That determinacy is what makes "a measure is its sequence of moments" a faithful slogan on $[0,1]$, and it is the exact analogue of the injectivity of Θ we just imported from chapter 17.

The graph problem has the very same shape, with one genuine new feature, and naming that feature in advance organizes the entire section. The answer will again be a list of conditions on the candidate sequence τ, and it will have *three* parts. Two of them are the direct translations of the classical pair. A *normalization*, $\tau_{K_1} = 1$, is the total-mass condition: the single vertex K_1 is the empty product, the graph analogue of the monomial $t^0 = 1$, and its density is forced to be 1. A *positive-semidefiniteness* will again be a Gram-matrix constraint — the requirement that certain symmetric matrices built from τ be positive semidefinite, by exactly the same one-line argument as above, with monomials replaced by partially-integrated graph densities and the inner product still that of an L^2 space. This is the condition we will call *reflection positivity*, and constructing the right matrices for it is the one piece of new machinery the section must build. The third condition is the new feature, and it is new only because the algebra of graph motifs is richer than the algebra of monomials. The monomials form a *single* multiplicative chain, $t^i \cdot t^j = t^{i+j}$, indexed by one integer; the finite graphs form the free commutative monoid on the *connected* graphs under disjoint union, a far larger algebra. The profile must respect that algebra: a disjoint union of patterns must have the product of their densities, $\tau_{F_1 \sqcup F_2} = \tau_{F_1}\tau_{F_2}$. In the scalar case this multiplicativity is invisible because it is built into the indexing — there is only one way to multiply powers of t — but for graphs it is a separate, explicit *algebra-homomorphism* condition that a number sequence can perfectly well violate. (The classical scalar Hausdorff problem is recovered as the *rank-one shadow* of the graph problem, the slice on which the connection matrices collapse to ordinary Hankel matrices; we point this out where it surfaces in the characterization.) So the three-part answer is: normalization, multiplicativity, and reflection positivity — the graph-theoretic twin of "$s_0 = 1$ and the Hankel matrix is positive semidefinite," with multiplicativity the extra clause the larger algebra demands.

The section is organized around establishing exactly that these three conditions are *necessary and sufficient.* We take them in the order the difficulty dictates. First (section 16.3 below, the subsection on necessary conditions) we verify necessity, which is the cheap direction — and two of the three are already proved, so the real work is to build the reflection-positivity machinery, the partially labelled graphs and their gluing that play the role of the monomials and their multiplication, and to run the Gram computation in the graph setting. Then we state the full characterization, whose *sufficiency* half is one of the two genuinely deep theorems of the chapter — the Lovász–Szegedy / Freedman–Lovász–Schrijver realization theorem — precisely located and its in-book proof pointed to appendix F. Finally we close the chapter with a worked computation that makes the whole apparatus concrete, evaluating the profiles of the three graphons that anchor the book's running threads. We begin with the necessary conditions, where two of the three are already in our hands.

16.3.1 Necessary conditions

The section preamble promised a three-part answer — normalization, multiplicativity, reflection positivity — and said we would meet the conditions in the order their difficulty dictates: necessity before sufficiency, and within necessity the two cheap conditions before the one that takes work. Two of the three are, in fact, already theorems of this chapter; we have only to read them off in the language of the candidate sequence $\tau = (\tau_F)_{F\in\mathcal{F}_0}$ and note that a genuine profile cannot escape them. *Normalization* is the requirement $\tau_{K_1} = 1$ (and, more generally, $\tau_{\overline{K_r}} = 1$ for every edgeless graph on r vertices). It holds for any profile because the one-vertex pattern K_1 contributes the empty edge-product, so definition 16.8 forces $t(K_1, W) = 1$ for *every* graphon: this is the total-mass condition, the exact graph twin of the classical $s_0 = \int 1 \, \mathrm{d}\mu = 1$. *Multiplicativity* is the requirement $\tau_{F_1\sqcup F_2} = \tau_{F_1}\tau_{F_2}$ for disjoint unions, and it holds for any profile by proposition 16.10 (i), the factorization of the density of a disjoint union into the product of the densities of its parts. Both are settled, and nothing more need be said about them as standalone facts. The entire labour of this subsection is the third condition, the positivity, which has no ready-made predecessor and must be constructed from scratch.

Why it must be *built* rather than merely cited is worth a sentence, for that sentence dictates exactly what to build. Recall the one-line proof of the classical positivity condition, displayed in the section opening: for the Hankel matrix $H = (s_{i+j})$ of a measure μ, $\sum_{i,j} c_i c_j \, s_{i+j} = \int \big(\sum_i c_i t^i\big)^2 \, \mathrm{d}\mu \geq 0$. That argument turned on exactly one structural fact — that the monomials *multiply*, $t^i \cdot t^j = t^{i+j}$, so the $L^2(\mu)$ inner product $\int t^i t^j \, \mathrm{d}\mu$ of two basis functions is *itself* a moment, namely the entry s_{i+j}. The matrix was positive semidefinite for the single reason that it was a Gram matrix. To replay this on the graph side we therefore need two ingredients that our vocabulary does not yet contain: objects that play the part of the monomials t^i — functions in some L^2 space, whose pairwise products we can integrate — and an operation on those objects that plays the part of multiplication, sending the pair (t^i, t^j) to the single object t^{i+j} whose density is again a coordinate of τ. *Partially labelled graphs* supply the first and their *gluing* supplies the second. We now define both, verify that gluing really does correspond to pointwise multiplication of functions — the graph version of $t^i \cdot t^j = t^{i+j}$ — and then the Gram argument transcribes almost verbatim.

The guiding image is this. A monomial t^i, before we integrate it against μ, is a *function* of the free variable t; only the act of integration collapses it to the number s_i. Its graph analogue should likewise be a homomorphism density with some of its variables left *free* rather than integrated out — a function, not yet a number. The way to leave k of a pattern's type-variables free is to single out k of its vertices and refuse to integrate over their types; the labels $1, \dots, k$ are simply names for those free variables.

Definition 16.19 (k-labelled graphs and gluing)**.** A *k-labelled graph* is a finite simple graph in which k vertices are singled out and assigned the distinct labels $1, \dots, k$ (one vertex per label); the remaining vertices are unlabelled. We call it *label-edge-free* if no edge joins two labelled vertices. Given two k-labelled graphs F_1, F_2, their *gluing* (or product) $F_1 F_2$ is formed by taking the disjoint union and then identifying the label-i vertex of F_1 with the label-i vertex of F_2, for each $i = 1, \dots, k$. When both F_1 and F_2 are label-edge-free this identification creates no multiple edge — the only edges that could coincide would join two labelled vertices, and there are none — so $F_1 F_2$ is again a simple, label-edge-free k-labelled graph. This is the only case we shall use; for $k = 0$ the gluing is the disjoint union. The *unlabelling* $[\![F]\!]$ of a k-labelled graph forgets the labels, yielding an ordinary graph in $\mathcal{F}_0$.

It pays to run the gluing once on a concrete pair, both because the operation is easier seen than described and because the example we choose will reappear as the smallest instance of the positivity to come. Take $k = 2$ and let F_1 be the 2*-labelled cherry*: labelled vertices 1 and 2, one unlabelled vertex u joined to both, and no edge between 1 and 2. It is label-edge-free, since its only edges are $1u$ and $2u$. Let F_2 be a second, independent copy of the same cherry, with its own unlabelled apex w joined to 1 and 2. To glue, lay the two side by side and weld together their like-labelled vertices: the two label-1 vertices become a single vertex 1, the two label-2 vertices become a single vertex 2, while u and w stay distinct (they carry no labels and so are never identified). The result $F_1 F_2$ is the graph on $\{1, 2, u, w\}$ with edges $1u$, $2u$, $1w$, $2w$ — precisely the four-cycle $1-u-2-w-1$, that is, $[\![F_1 F_2]\!] = C_4$. Two cherries glued at their tips make a quadrilateral. Notice that the weld created no double edge: the only place a coincidence could have occurred is between the labelled vertices 1 and 2, and neither cherry has an edge there.

That last observation is the whole reason for the label-edge-free clause, and it deserves to be made sharp, because without it the clean algebra below silently breaks. Gluing identifies like-labelled vertices; if F_1 carried the labelled edge 12 and F_2 carried it too, the weld would lay one atop the other and a *simple* graph, having no notion of a doubled edge, would record the union as a single edge 12. The edge-count bookkeeping would then be off by one: an edge present in both factors would survive only once in the glue. Restricting to label-edge-free graphs forbids exactly this collision — there are no labelled edges to double — so every edge of F_1 and every edge of F_2 survives *intact and unmerged* in $F_1 F_2$, and the edge set of the glue is the honest disjoint union of the two edge sets. This is what makes the algebraic identity below an *exact* equality rather than one that quietly drops a repeated factor; we return to what that dropped factor would have been once we can name it.

With the geometric operation in place we can finally write down the "monomial as a function" that motivated it. For a graphon W and a k-labelled graph F, define the *partially integrated density* by holding the labelled vertices at fixed types $x_1, \dots, x_k$ and integrating out only the unlabelled ones:

$$t_{x_1,\dots,x_k}(F,W) \;=\; \int_{[0,1]^{V(F)\setminus\{1,\dots,k\}}} \prod_{ij\in E(F)} W(x_i,x_j) \prod_{u \text{ unlabelled}} \mathrm{d}x_u.$$

The very same Tonelli/measurability checks that validated definition 16.8 apply here to the unlabelled variables alone, so this is a genuine measurable function of $(x_1,\dots,x_k)\in[0,1]^k$ with values in $[0,1]$ — the analogue of the function $t\mapsto t^i$, not yet the number. Its single most important property is that integrating away the remaining (labelled) variables collapses it back to an ordinary density,

$$\int_{[0,1]^k} t_{x_1,\dots,x_k}(F,W)\;\mathrm{d}x_1\cdots\mathrm{d}x_k \;=\; t([\![F]\!],W),$$

exactly as $\int t^i\;\mathrm{d}\mu = s_i$. For the running example, the 2-labelled cherry F_1 has the single unlabelled apex u to integrate out, so $t_{x_1,x_2}(F_1,W)=\int_{[0,1]} W(x_1,z)W(x_2,z)\;\mathrm{d}z$, the codegree kernel measuring how strongly types x_1 and x_2 share a common neighbour.

Now the key algebraic fact, the graph counterpart of $t^i\cdot t^j = t^{i+j}$: *for label-edge-free* F_1,F_2, gluing corresponds to *pointwise multiplication* of these functions,

$$t_{x_1,\dots,x_k}(F_1\,F_2,W) \;=\; t_{x_1,\dots,x_k}(F_1,W)\cdot t_{x_1,\dots,x_k}(F_2,W). \tag{16.2}$$

The proof is bookkeeping plus one application of Tonelli, and every clause of the label-edge-free hypothesis is used. Because F_1 and F_2 are label-edge-free, their gluing merges no edges, so the edge set of $F_1\,F_2$ is the disjoint union $E(F_1)\sqcup E(F_2)$ and the integrand of the left-hand side factors cleanly,

$$\prod_{ij\in E(F_1F_2)} W(x_i,x_j) = \Big(\prod_{ij\in E(F_1)} W(x_i,x_j)\Big)\Big(\prod_{ij\in E(F_2)} W(x_i,x_j)\Big),$$

with no factor appearing in both brackets. The labelled types $x_1,\dots,x_k$ are held fixed and shared by the two brackets, but the *unlabelled* vertices of F_1 and those of F_2 are disjoint sets of integration variables — this is the second place the gluing keeps the two factors apart — so the first bracket is a function of $(x_\bullet;\text{unlabelled of } F_1)$ only and the second of $(x_\bullet;\text{unlabelled of } F_2)$ only. With the shared labels frozen, the integral over the joint unlabelled variables is an integral of a product of functions of disjoint variable blocks, and Tonelli (theorem 1.27) splits it into the product of the two separate integrals — which are exactly $t_{x_\bullet}(F_1,W)$ and $t_{x_\bullet}(F_2,W)$. That is (16.2). On the running example it reads $\int_{[0,1]^2}\big(\int_[0,1]W(x_1,z)W(x_2,z)\;\mathrm{d}z\big)^2\,\mathrm{d}x_1\,\mathrm{d}x_2 = t(C_4,W)$: the square of the codegree kernel, integrated over the two free types, is the four-cycle density — the glue C_4 read off as the product of two cherries.

We can now name the factor the label-edge-free restriction protected us from. Were a labelled edge ij present in *both* factors, the simple glue $F_1\,F_2$ would carry the weight $W(x_i,x_j)$ once, whereas the product on the right of (16.2) would carry it twice, as $W(x_i,x_j)^2$ — and the identity would fail by exactly that squared factor. Keeping such repeated weights honestly is possible, but it requires enlarging "graph" to a formal object that remembers edge multiplicities: the algebra of *quantum graphs*, where a doubled labelled edge genuinely means W^2. That algebra is developed in appendix F; for the necessity argument at hand we do not need it, because the label-edge-free restriction removes every labelled edge and so makes (16.2) an exact equality on the nose. This is the single piece of bookkeeping the whole positivity argument rests on.

With multiplication in hand we can assemble the graph Hankel matrix. In the classical problem the Hankel matrix had entry s_{i+j} in row i, column j — the moment of the *product* $t^i \cdot t^j$. The graph version indexes rows and columns by labelled graphs and puts in each slot the density of the *glued* graph.

Definition 16.20 (Connection matrix and reflection positivity)**.** Let $\tau = (\tau_F)_{F\in\mathcal{F}_0}$ and extend τ to k-labelled graphs by $\tau(F) := \tau_{[\![F]\!]}$. For each $k \geq 0$ the *k-th connection matrix* $M_k(\tau)$ is the (infinite, symmetric) matrix whose rows and columns are indexed by the *label-edge-free* k-labelled graphs — so that the gluing $F_1\,F_2$ of any two index graphs is again simple and $\tau(F_1\,F_2) = \tau_{[\![F_1\,F_2]\!]}$ is unambiguous — and whose entry is

$$\big(M_k(\tau)\big)_{F_1,F_2} \;=\; \tau(F_1\,F_2).$$

We say τ is *reflection positive* if every finite principal submatrix of every $M_k(\tau)$ is positive semidefinite. (Restricting the index set to label-edge-free graphs is the standard indexing for the moment problem: it is exactly what makes the Gram identity of proposition 16.21 hold on the nose, and it loses no necessary positivity constraint.)

Three features of this matrix are worth pinning down before we prove anything, because each is the graph echo of a familiar feature of the scalar Hankel matrix. First, it is symmetric: gluing is a symmetric operation, $F_1\,F_2 = F_2\,F_1$ as labelled graphs, so $(M_k)_{F_1,F_2} = (M_k)_{F_2,F_1}$, just as $s_{i+j} = s_{j+i}$. Second, its entries depend on the index graphs *only through their glued unlabelling*, the way a Hankel entry depends on (i,j) only through the sum $i+j$ — many different pairs of labelled graphs can glue to the same unlabelled graph and hence share an entry, the graph analogue of the constant anti-diagonals that give Hankel matrices their name. Third, the matrix is infinite (there are infinitely many label-edge-free k-labelled graphs), so positive semidefiniteness is imposed in the only sense available for infinite matrices: *every* finite principal submatrix — every choice of finitely many index graphs — must be positive semidefinite. That is the meaning of "reflection positive." For $k=0$ there are no labels to hold fixed, gluing is plain disjoint union, and $M_0(\tau)$ has entry $\tau_{F_1\sqcup F_2}$; multiplicativity already forces this to factor as $\tau_{F_1}\tau_{F_2}$, so M_0 is the rank-one matrix of an outer product and its positivity is automatic — the genuine content of reflection positivity lives at $k \geq 1$, where the labels couple the factors and the matrix can have higher rank. We will see this rank-one degeneracy resurface in the next subsection as precisely the slice on which the graph problem collapses to the classical scalar one.

Proposition 16.21 (Necessity of reflection positivity)**.** *If $\tau = \Theta(W)$ for a graphon W, then τ is reflection positive.*

Proof. The argument is the classical Gram computation with monomials replaced by partially integrated densities and the line measure μ replaced by Lebesgue measure on the type cube; the gluing identity (16.2) is what makes every step go through. Fix the level k and any finite list of *label-edge-free* k-labelled graphs $F^{(1)},\dots,F^{(N)}$ — this names an arbitrary finite principal submatrix of $M_k(\tau)$ — and let $c = (c_1,\dots,c_N) \in \mathbb{R}^N$ be arbitrary coefficients. To each index graph attach the function it represents,

$$g_a(x_1,\dots,x_k) \;=\; t_{x_1,\dots,x_k}(F^{(a)},W), \qquad a = 1,\dots,N,$$

which lies in $L^2([0,1]^k)$: it is bounded by 1, hence square-integrable against the probability measure on the cube. These g_a are the graph monomials. Now read off the matrix entry. Because each $F^{(a)}$ is label-edge-free, so is every glue $F^{(a)}F^{(b)}$ (gluing two label-edge-free graphs cannot create a labelled edge), so (16.2) applies to the pair and the product $g_a\, g_b$ is itself a partially integrated density — that of the glued graph. Integrating out the labels then turns it into the unlabelled density, which is by definition the matrix entry:

$$(M_k(\tau))_{F^{(a)},F^{(b)}} = \tau(F^{(a)}F^{(b)}) = t([\![F^{(a)}F^{(b)}]\!], W) = \int_{[0,1]^k} t_{\mathbf{x}_\bullet}(F^{(a)}F^{(b)}, W)\, \mathrm{d}x = \int_{[0,1]^k} g_a\, g_b = \langle g_a, g_b\rangle_{L^2([0,1]^k)}.$$

Here the first equality is the definition of τ on labelled graphs, the second is $\tau = \Theta(W)$ unpacked, the third recovers an ordinary density by integrating out the labels of the glue, the fourth is the gluing identity (16.2), and the last is the definition of the L^2 inner product. So the chosen principal submatrix is exactly the Gram matrix of the vectors $g_1, \ldots, g_N$ in $L^2([0,1]^k)$ — the same statement as "the Hankel matrix is the Gram matrix of the monomials," transcribed. A Gram matrix is positive semidefinite for one reason, that quadratic forms in it are squared norms:

$$c^\top M_k(\tau)\, c = \sum_{a,b=1}^{N} c_a c_b \langle g_a, g_b\rangle = \Big\langle \sum_a c_a g_a,\ \sum_b c_b g_b \Big\rangle = \Big\| \sum_a c_a g_a \Big\|^2_{L^2([0,1]^k)} \ \geq\ 0,$$

using bilinearity of the inner product to pull the sums inside. Since k, the list $F^{(1)}, \ldots, F^{(N)}$, and c were arbitrary, every finite principal submatrix of every $M_k(\tau)$ is positive semidefinite, which is reflection positivity. $\square$

The smallest case of this proof is the running example, and it is worth seeing the abstraction land on it. At $k = 2$ take the single index graph $F^{(1)}$ = the 2-labelled cherry; the 1×1 submatrix is the diagonal entry $\tau(F^{(1)}F^{(1)}) = t(C_4, W)$, and the Gram identity reads $t(C_4, W) = \|g_1\|^2_{L^2} \geq 0$ with $g_1(x_1, x_2) = \int_[0,1] W(x_1, z)W(x_2, z)\, \mathrm{d}z$. So the nonnegativity of the four-cycle density is, read this way, nothing but the statement that the codegree kernel has a nonnegative squared L^2 norm — the positivity of a single diagonal entry of M_2. The full reflection positivity is this observation made simultaneously for every finite collection of labelled graphs at every level.

This completes the three necessary conditions, and it is worth standing back to name what the last one is. Proposition 16.21 is the graph-theoretic shadow of the elementary half of the Hamburger/Hausdorff moment theorem: the moments of a measure form a positive-semidefinite Hankel matrix because that matrix is the Gram matrix of the monomials $1, t, t^2, \ldots$ in $L^2(\mu)$. Here the connection matrix $M_k(\tau)$ *is* the graph Hankel matrix; the gluing of labelled graphs *is* the multiplication of monomials, $t^i \cdot t^j = t^{i+j}$; and the partially integrated densities g_a *are* the monomials, now functions on the type cube rather than on the line. The classical statement is the rank-one, single-variable face of this one, the slice on which the connection matrices degenerate to ordinary Hankel matrices — a correspondence we locate precisely in the characterization that follows. We have shown that a genuine profile $\tau = \Theta(W)$ *must* be normalized, multiplicative, and reflection positive. The striking and far deeper fact, which the next subsection states and which closes the moment problem, is that these three necessary conditions are also *sufficient*: any number sequence passing all three is realized by some graphon. The Gram/Hankel identification just made is the hook on which that converse hangs, for it is what lets the whole characterization be read as the graph moment theorem.

16.3.2 The characterization

The content of the moment problem is that these three necessary conditions are also *sufficient*.

Theorem 16.22 (Moment characterization of graphon profiles; Lovász–Szegedy, Freedman–Lovász–Schrijver)**.** *Let $\tau = (\tau_F)_{F \in \mathcal{F}_0}$ be a family of real numbers. Then $\tau = \Theta(W)$ for some graphon W satisfying* ***(A-Reg-W)*** (*symmetric, measurable,* $[0,1]$*-valued*) *if and only if*

1. (Normalization) $\tau_{K_1} = 1$*;*
2. (Multiplicativity) $\tau_{F_1 \sqcup F_2} = \tau_{F_1}\,\tau_{F_2}$ *for all $F_1, F_2 \in \mathcal{F}_0$;*
3. (Reflection positivity) *every connection matrix $M_k(\tau)$, $k \geq 0$, is positive semidefinite.*

Moreover the graphon is then unique up to weak isomorphism ($\delta_\square = 0$, proposition 17.7), and automatically $0 \leq \tau_F \leq 1$ *for all F.*

Proof sketch, with the gap located. Necessity is proposition 16.10 (i), $t(K_1, W) = 1$, and proposition 16.21; the bound $\tau_F \in [0,1]$ is proposition 16.10 (ii).

For sufficiency one builds the graphon out of the positivity data, exactly as the classical moment problem builds a measure out of positive-semidefinite Hankel matrices via a Gram–Schmidt / spectral construction. The mechanism is the following. Reflection positivity of $M_k(\tau)$ endows the (real) vector space spanned by label-edge-free k-labelled graphs, modulo the kernel of the bilinear form $\langle F_1, F_2\rangle := \tau(F_1F_2)$, with a genuine inner product; gluing makes this space a commutative algebra and the form is a state on it. The Gelfand–Naimark–Segal construction turns the pair (algebra, state) into multiplication operators on a Hilbert space; analyzing the cases $k = 1$ and $k = 2$ produces a probability space (Ω, ν) carrying the "random type" and a symmetric measurable kernel w on $\Omega \times \Omega$ with values in $[0,1]$ representing the edge variable, such that $\tau_F = \int \prod_{ij \in E(F)} w$. A measure-theoretic isomorphism of (Ω, ν) with the standard atomless space $([0,1], \mathrm{Leb})$ — any standard atomless probability space is so isomorphic — then transports w to a graphon W on $[0,1]^2$ with $\Theta(W) = \tau$. Uniqueness up to weak isomorphism is the same statement as the injectivity of Θ modulo $\delta_\square = 0$, namely the cut-metric uniqueness theorem theorem 17.11 (proved in appendix F).

Honest status of this direction. Unlike the realization theorem theorem 16.15 — whose proof, via cut-metric compactness, is carried out in full in appendix F — the analytic heart of the present sufficiency is *not* reproduced in this book. The steps we have only sketched are exactly the hard ones: that the GNS algebra is realized by a single symmetric two-variable kernel rather than by a higher-order object, that the kernel can be taken $[0,1]$-valued and measurable, and the requisite measure-isomorphism of the constructed space with $([0,1], \mathrm{Leb})$. These are the content of the Lovász–Szegedy / Freedman–Lovász–Schrijver theorem (reflection positivity and "rank connectivity"), which we *state and use as a cited external result*: [100, 70] and [99, Ch. 5, Ch. 11]. We flag this explicitly so the reader is not misled: the "only if" half (necessity) and the rank-one shadow (the Hausdorff moment problem, appendix F.8) are proved in this book; the general "if" half is the one place in this chapter where we rely on an attribution rather than an in-book proof, and it is used nowhere downstream. $\square$

Remark 16.23 (The classical moment problem as the k-independent shadow)**.** Specialize to *rank-one* data. Suppose all densities depend on F only through its degree sequence; worked computation 16.24 below shows this is exactly the case $W(x,y) = f(x)f(y)$, for which $\tau_F = \prod_{i\in V(F)} c_{\deg_F(i)}$ with $c_d = \int_0^1 f^d$. In this rank-one sector the moment problem collapses to a classical one: the realizable profiles are exactly those for which $(c_d)_{d\geq 0}$ is a *Hausdorff moment sequence*, $c_d = \int_0^1 r^d\,\rho(\mathrm{d}r)$ for a probability measure ρ on $[0,1]$ (here $\rho = f_{\#}\mathrm{Leb}$, the law of $f(\xi)$ for ξ uniform). Two halves of this are fully in-book. The elementary *necessity* direction — a realizable (c_d) has positive-semidefinite Hankel matrices $(c_{i+j})_{i,j}$ — is the rank-one shadow of the Gram argument of proposition 16.21, since $\sum_{i,j} a_i a_j c_{i+j} = \int_0^1 \big(\sum_i a_i r^i\big)^2 \rho(\mathrm{d}r) \geq 0$. The *sufficiency* direction — a Hausdorff sequence is realized by some f, hence some rank-one graphon — together with the classical characterization by complete monotonicity, is the Hausdorff moment theorem, which we prove from scratch in appendix F.8. Thus, unlike the general graph moment problem, its rank-one sector is established here without any external appeal: the graph moment problem genuinely contains the classical one as its rank-one shadow, and that is the cleanest sense in which homomorphism densities "are moments."

16.3.3 A worked computation: motif densities of structured graphons

We close by computing $t(F,W)$ in closed form for the three graphons that anchor the rest of the book, and reading off what their homomorphism profiles look like. The rank-one case is the kernel of Thread B (the graphon-LQG congestion model, appendix A.7), and the constant case is its Thread A reduction; computing their motif densities here is the graph-limit-theoretic side of those threads.

Worked computation 16.24 (Constant, rank-one, and block graphons)**. (i) Constant graphon (Erdős–Rényi limit).** Let $W \equiv p \in [0,1]$ be the constant graphon of example 15.9. Then every edge contributes the constant factor p and the integral is trivial:

$$t(F,p) \;=\; p^{|E(F)|}.$$

The profile depends on F only through its edge count. This is the limit of the dense Erdős–Rényi graphs $G(n,p)$ (sampling and concentration in chapter 19), and via proposition 16.9 it is the homomorphism profile of any quasirandom sequence of density p. In the language of appendix A.7 this is the constant-kernel reduction $W \equiv c$ that returns Thread B to the homogeneous Thread A: no type heterogeneity survives, and the only surviving "moment" is the single number p.

(ii) Rank-one (Curie–Weiss) graphon. Let $W(x,y) = f(x)f(y)$ with $f\colon [0,1] \to [0,1]$ measurable (so $W \in [0,1]$ and **(A-Reg-W)** holds): the rank-one graphon of example 15.11, the kernel of Thread B. Group the product over edges by vertices: each vertex i appears in exactly $\deg_F(i)$ edges, contributing a factor $f(x_i)$ each time, so

$$\prod_{ij\in E(F)} W(x_i,x_j) = \prod_{ij\in E(F)} f(x_i)f(x_j) = \prod_{i\in V(F)} f(x_i)^{\deg_F(i)}.$$

The variables separate, and with the *moment sequence* $c_d := \int_0^1 f(t)^d\,\mathrm{d}t$ ($c_0 = 1$, $c_1 = \int f$, and $0 \leq c_{d+1} \leq c_d \leq 1$),

$$t(F,W) \;=\; \prod_{i\in V(F)} c_{\deg_F(i)}. \tag{16.3}$$

Thus the entire homomorphism profile of a rank-one graphon is the multiplicative function of the degree sequence built from the single moment sequence (c_d). Reading off motifs: the edge density is $t(K_2, W) = c_1^2$ $(= (\int f)^2)$; the triangle density is $t(K_3, W) = c_2^3$; the d-star S_d (one centre of degree d, d leaves of degree 1) has $t(S_d, W) = c_d\, c_1^d$; a path P_ℓ with two endpoints of degree 1 and interior vertices of degree 2 has $t(P_\ell, W) = c_1^2 c_2^{\ell-2}$. The profile is *not* a pure power of the edge density — unlike the constant case (i) — and that departure is the detectable signature of type heterogeneity. The cleanest instance is the cherry P_3 (two edges, centre of degree 2): the quasirandom baseline at the same edge density $p = t(K_2, W) = c_1^2$ would give $p^2 = c_1^4$, whereas

$$t(P_3, W) \;=\; c_2\, c_1^2 \;\geq\; c_1^2 \cdot c_1^2 \;=\; t(K_2, W)^2,$$

because $c_2 = \int f^2 \geq \big(\int f\big)^2 = c_1^2$ by Cauchy–Schwarz, with equality iff f is constant a.e. Thus rank-one heterogeneity produces a *cherry/path excess* controlled by the variance $c_2 - c_1^2 = \mathrm{Var}(f)$: degree heterogeneity inflates the count of two-paths above the quasirandom value, and the size of the excess is exactly the variance of the type-profile f, a first quantitative signature of heterogeneity that the moment data exposes.

By remark 16.23 the realizability of (16.3) as a graphon profile is exactly the Hausdorff moment condition on (c_d), with $\rho = f_\#\mathrm{Leb}$; a left-convergent sequence of graphs converging to a rank-one graphon is therefore a sequence whose empirical degree-moment sequences converge to a Hausdorff moment sequence. This is the moment problem of section 16.3 made completely explicit on Thread B's kernel.

(iii) Block graphon (stochastic block model limit). Recall the stochastic block model graphon of example 15.10: communities $C_1, \ldots, C_q \subseteq [0,1]$ of sizes $\rho_r = |C_r|$ with $\sum_r \rho_r = 1$, and a symmetric matrix $\mathsf{B} \in [0,1]^{q\times q}$ of inter-community connection probabilities, with $W(x,y) = \mathsf{B}_{rs}$ for $x \in C_r, y \in C_s$. (We write q for the number of communities, reserving $k = |V(F)|$ for the size of the pattern as everywhere else in this chapter; the community sizes ρ_r and blocks C_r and the block matrix B are exactly the data (k, ρ, B) exported by example 15.10, with its k renamed q here.) Then the integral becomes a finite sum over block-assignments $\psi\colon V(F) \to \{1, \ldots, q\}$:

$$t(F, W) = \sum_{\psi\colon V(F)\to[q]} \Big(\prod_{i\in V(F)} \rho_{\psi(i)} \Big) \prod_{ij\in E(F)} \mathsf{B}_{\psi(i)\psi(j)},$$

a polynomial of degree $|E(F)|$ in the entries of B weighted by the block sizes. Taking q blocks recovers the finite multi-population structure (the bridge to multi-class models in chapter 30); taking $q = 1$ recovers case (i). Note that (16.3) is the special case in which $\mathsf{B}_{rs} = \phi_r\phi_s$ is itself rank one, with $c_d = \sum_r \rho_r \phi_r^d$.

Rigor ledger

Proved.

- Definition of hom, inj, ind and the densities $t(F, G)$, t_{inj}, t_{ind} (definitions 16.1 and 16.4); the elementary counting lemma $|t(F, G) - t_{\mathrm{inj}}(F, G)| \leq \binom{k}{2}/n$ (lemma 16.6), hence all three normalizations define the same convergence (remark 16.7).

- Homomorphism density into a graphon as a moment $\mathbb{E}\prod W(\xi_i,\xi_j)$ (definition 16.8); consistency $t(F,W_G)=t(F,G)$ (proposition 16.9); multiplicativity over disjoint unions and the $[0,1]$ bound (proposition 16.10).
- Left-convergence as coordinatewise convergence of all motif densities (definition 16.11); the moment topology is metrizable and sits inside the compact cube $[0,1]^{\mathcal{F}_0}$, giving subsequential left-convergence of *every* graph sequence by Tychonoff (proposition 16.13 and remark 16.14).
- The counting lemma $|t(F,W)-t(F,W')|\le|E(F)|\;\|W-W'\|_\square$ (lemma 16.17) and its corollary that cut-convergence implies left-convergence (corollary 16.18).
- Necessity of normalization, multiplicativity, and reflection positivity for a graphon profile (proposition 16.21); closed-form profiles of constant, rank-one, and block graphons (worked computation 16.24).

Assumed. Graphons obey **(A-Reg-W)** (symmetric, measurable, $[0,1]$-valued). Tonelli/-Fubini (theorem 1.27). The pixel (empirical) graphon W^G of a finite graph, imported from definition 15.2 and proposition 15.3, together with the constant, rank-one, and stochastic-block-model graphons of examples 15.9 to 15.11. The cut norm $\|\cdot\|_\square$ and its extreme-point identity ($\star$) are recalled from chapter 17. **Deferred.**

- The Lovász–Szegedy realization theorem (theorem 16.15): proof via cut-metric compactness in chapter 17 and appendix F.
- Sufficiency in the moment characterization (theorem 16.22): the Freedman–Lovász–Schrijver / GNS construction of a graphon from reflection-positive data — and with it the measure-isomorphism of the constructed atomless space with $([0,1],\mathrm{Leb})$ — is *not* reproduced in this book and is cited as an external result ([100, 70, 99]); it is used nowhere downstream. Its rank-one shadow, the Hausdorff moment theorem, *is* proved in-book (theorem F.14).
- The weak regularity lemma (stated and proved as theorem 17.16) and the inverse counting lemma it powers (the converse of corollary 16.18, proved via theorem 17.11 in appendix F); the sampling/concentration consequences are chapter 19.

Open. Effective/quantitative inverse counting — the sharp dependence of cut-distance on the number of densities matched — remains far from tight; the best bounds are $O(1/\sqrt{\log k})$ and are not known to be improvable in general. Outside the dense regime the entire moment philosophy fails: for sparse sequences ($|E(G_n)|=o(|V(G_n)|^2)$) all $t(F,G_n)\to 0$ for F with an edge, so motif densities carry no information and a different notion of convergence is required (chapters 19 and 27).

Bibliographic notes

Homomorphism densities, left-convergence, and the realization theorem are due to Lovász and Szegedy [100]; the systematic metric theory — subgraph frequencies, the counting lemma, and the equivalence with the cut metric — was developed by Borgs, Chayes, Lovász, Sós, and Vesztergombi [29]. The textbook account we follow is Lovász's monograph [99, Ch. 7] for densities and left-convergence, with the moment characterization and its reflection-positivity proof in [99, Ch. 5, Ch. 11]; the reflection-positivity / rank-connectivity circle of ideas originates with Freedman, Lovász, and Schrijver [70]. The survey [98] is the gentlest entry point and the source of the "moments of a graph" slogan. The counting lemma in the form of lemma 16.17 and the extreme-point identity ($\star$) are standard; see [99, §4.4, §10.2]. The classical moment-problem analogy invoked in remark 16.23 (Hausdorff's theorem, positive-semidefinite Hankel matrices) uses only the bounded-interval (Hausdorff) case, which we prove from scratch in theorem F.14 via the complete-monotonicity criterion rather than relying on an external moment-problem reference. The forward references are load-bearing: the cut metric, the inverse counting lemma, and the proof of compactness are the subject of chapter 17 (with the full Lovász–Szegedy proof in appendix F), the spectral/operator reading of W is chapter 18, and sampling W-random graphs $G(n, W)$ with concentration of the densities defined here is chapter 19. The whole apparatus is used in Part IV — most directly in chapter 25, where convergence of a finite network's equilibria to the graphon mean field game equilibrium is phrased through left-/cut-convergence of the underlying graphs.

Chapter 17

The Cut Metric and Compactness of Graphon Space

The previous two chapters built the objects we now learn to measure. Chapter 15 introduced the graphon as the limit of an adjacency matrix: a symmetric measurable kernel $W\colon [0,1]^2 \to [0,1]$, with the type variable $x \in [0,1]$ an exchangeable label rather than a spatial coordinate, and with the integral operator $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y$ that will couple agents across types in every graphon mean field game. Chapter 16 then attached to each graphon a sequence of *observables*: the homomorphism densities $t(F, W)$, the continuum analogues of subgraph frequencies, and it declared a sequence of graphons *left-convergent* when all of these observables converge. That gave graphons a notion of convergence — a topology. But a topology assembled out of the convergence of countably many numbers is a curiously thin thing to hold in the hand. It tells us *whether* a sequence eventually settles down, yet it offers no *single number* measuring how far apart two graphons stand: no way to declare that this pair is closer than that one, no ball of radius ϵ to draw around a candidate limit, no triangle inequality to chain estimates through. With only a topology we cannot form a Cauchy sequence, cannot speak of completeness, cannot run the extract-a-convergent-subsequence arguments on which the whole of analysis turns. Chapter 16, in short, handed us a topology but no *metric space* — and it is on a metric space, not a bare topology, that the existence and convergence theorems of Part III will have to stand. Supplying that missing distance, and extracting from it the one structural fact that makes the whole graphon mean field theory go through, is the business of this chapter.

The distance is the *cut metric* $\delta_\square$, built from the *cut norm* $\|\cdot\|_\square$: at last a single number $\delta_\square(W, W')$ that says how far apart two graphons are, in which left-convergence becomes ordinary convergence and in which the set of all graphons becomes a space we can do analysis on. Its defining feature, and the reason it is none of the distances one would first reach for, is coarseness: it deliberately ignores the fine-grained, microscopic placement of edges and registers only the macroscopic, averaged structure — precisely the structure that survives relabeling of the types and survives sampling a finite network from the graphon. We will see that the cut metric is the right distance in three reinforcing senses, and the reader should treat the first three sections of this chapter as a guided ascent through them toward a fourth. First, it metrizes left-convergence: a sequence converges in $\delta_\square$

(after the unavoidable quotient by relabelings) if and only if all its homomorphism densities converge (theorem 17.12), so the topology of chapter 16 and the geometry of this chapter are literally one and the same. Second, the quotient of the set of graphons by the relation "$\delta_\square$-distance zero" is a genuine metric space, the true *space of graphons*, whose points are the intrinsic networks themselves rather than their arbitrary labelings. Third, and decisively, that metric space is *compact* (theorem 17.13, Lovász–Szegedy).

It is this third fact that is the real destination, and the reader should keep it in view as the keystone the whole chapter is shaped to support. Compactness is what a later existence proof reaches for when it must extract a convergent subsequence of approximate equilibria, or pass an equilibrium to a limit network; it is the input from Part III that makes the graphon mean field game existence theorem of chapter 24 and the finite-network convergence theorem of chapter 25 possible, and it is the reason a single limit object can stand in for an entire convergent sequence of growing networks. The first two senses are not ends in themselves: the metrization and the quotient are the scaffolding without which the word "compact" has nothing to attach to, for one cannot ask whether a space is compact before knowing it is a metric space and knowing what its points are. Sections one through three raise that scaffolding; section four pays it off.

To carry the program out we lean throughout on what earlier chapters established, and it is worth seeing each import as a tool earmarked for a definite job rather than a line in a manifest. From chapter 15 we take the graphon and its integral operator $\mathbb{W}$. From chapter 16 we take the homomorphism density $t(F, \cdot)$, the notion of left-convergence, and the *counting lemma* bounding density differences by the cut distance — the very bridge that will deliver the first sense, metrization. From chapter 1 come the everyday tools for integrating a kernel over patches of types: measurable sets, L^1 and L^2, product measure and Fubini. From chapter 2 come the Hilbert space $L^2([0,1]^2)$ and orthogonal projection, together with the Banach–Alaoglu theorem that will supply completeness. And from chapter 3 comes conditional expectation as the L^2 projection onto a sub-σ-algebra, which is the engine of the regularity argument behind compactness. Looking forward, chapter 18 reads the same graphon as a self-adjoint Hilbert–Schmidt operator and compares cut convergence with spectral convergence; chapter 19 returns to the weak regularity lemma proved here and develops its sampling consequences; and chapters 24 and 25 consume the compactness theorem directly — which is the ultimate reason we spend a whole chapter earning it.

The plan is as follows. Section 17.1 defines the cut norm, gives its three equivalent forms, and places it among the norms the reader already owns. Section 17.2 introduces measure-preserving relabelings, defines the cut metric, and constructs the metric space of graphons as a quotient. Section 17.3 proves the equivalence of cut convergence and left-convergence. Section 17.4 states and proves (the total-boundedness half in full, the completeness half by an honest sketch with the remainder deferred to appendix F) the compactness theorem, and explains why it is the structural keystone of Part III. A worked computation on the rank-one and block graphons of the carried threads runs alongside.

17.1 The cut norm

Before we can quotient by relabelings and arrive at a metric space of graphons, we need the raw material the cut metric is assembled from: the right notion of *size* for a kernel. A distance, after all, is the size of a difference, so the question "how far apart are two graphons?" rests on the prior question "how big is a kernel?", and the coarseness we are about to demand of the distance must already be present in this notion of size. The three subsections that follow answer the prior question. We first confront the ordinary norms the reader already owns and see exactly why each is too fine for the job; then we define the cut norm and record its three interchangeable forms; and finally we place it among the familiar norms by a short chain of comparison inequalities. Only size is at issue here — the quotient by relabelings that turns this norm into the cut metric is deferred to section 17.2.

17.1.1 Why not an ordinary norm

We have just agreed that the cut metric will be the size of a difference, so before anything else we must settle what "size of a kernel" should mean. A graphon is a function on $[0,1]^2$, and for measuring functions the reader already owns an entire toolbox: the L^p norms $\|W\|_p = (\int_{[0,1]^2} |W|^p)^{1/p}$ for $1 \leq p < \infty$, and the sup norm $\|W\|_\infty$. The honest question is why any of these will not do. If one of them already metrized left-convergence there would be nothing to invent, and we could lift it to a metric on graphons by the relabeling quotient of the next section and be done. The thesis of this subsection is that none of them can: every ordinary norm is *too fine*, reacting to features of the kernel that no macroscopic observable — no homomorphism density, no sampled subgraph count, no application of the coupling operator $\mathbb{W}$ — can detect. A distance built from such a norm would call two networks far apart that every experiment would call the same. We make this concrete with the example that drives the entire chapter, and then read off from its failure precisely what a usable notion of size must do instead.

Two networks that agree on everything but the wiring. Fix a density $p \in (0,1)$, say $p = \frac{1}{2}$ for definiteness, and build two large graphs G_N and G'_N on the same N vertices in the same way: for each of the $\binom{N}{2}$ pairs of nodes flip an independent fair coin and join the pair iff the coin shows heads, doing this once for G_N and a second, *independent* time for G'_N. The two graphs are statistically identical. Both have edge density $\to \frac{1}{2}$; both have triangle density $\to (\frac{1}{2})^3 = \frac{1}{8}$; more generally, for any fixed simple graph F, both have F-density $\to (\frac{1}{2})^{e(F)}$, because each of the $e(F)$ required edges is independently present with probability $\frac{1}{2}$. They share the same (lack of) community structure — a single homogeneous block — and the same degree profile, each vertex having degree $\frac{N}{2}(1+o(1))$. By every coarse, observable measure G_N and G'_N are the same network, and both are converging, in the sense of chapter 16, to the same limit: the constant graphon $W \equiv \frac{1}{2}$, the featureless "half-density" kernel that is their common smooth average. There is, macroscopically, only one object here.

Now view the two graphs as kernels. Order the vertices $1, \ldots, N$, cut $[0,1]$ into N equal subintervals $I_1, \ldots, I_N$, and let W_N be the step kernel equal to the (i,j) entry of G_N's adjacency matrix on the tile $I_i \times I_j$, and likewise W'_N for G'_N (this is the standard identification of a finite graph with its step graphon, chapter 16). On each tile both kernels are constant, taking a value in $\{0,1\}$. They *disagree* on tile $I_i \times I_j$ exactly when one coin came up heads and the other tails, an event of probability $2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$, and on a disagreeing tile one kernel is 0 while the other is 1, so their difference there has absolute value 1. Consequently a positive fraction — here about half — of the total area of $[0,1]^2$ carries $|W_N - W_N'| = 1$. This is everything the ordinary norms need to register a chasm:

$$\|W_N - W_N'\|_\infty = 1, \qquad \|W_N - W_N'\|_1 = 2p(1-p) + o(1) = \tfrac{1}{2} + o(1), \tag{17.1}$$

and the same lower bound $\|W_N - W_N'\|_p \geq \|W_N - W_N'\|_1 \geq \frac{1}{2} + o(1)$ holds for every $p \geq 1$. The sup norm is pinned at its maximum possible value, and not one of the L^p norms decays: in each of them the two kernels stay a fixed positive distance apart for all N, and the same is true of either kernel against the smooth limit $W \equiv \frac{1}{2}$ (there $\left|W_N - \frac{1}{2}\right| = \frac{1}{2}$ everywhere, so $\left\|W_N - \frac{1}{2}\right\|_p = \frac{1}{2}$ for all N and all p). The ordinary norms see three completely different functions where every observable sees one.

Why nothing observable notices the gap. The mismatch is not a defect of the example but a structural fact about what the observables are. A homomorphism density is an *integral* of products of W over many type variables; the coupling that drives every graphon mean field game enters only through the operator $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y$, again an integral against W; and sampling a finite network reads off block frequencies, integrals once more. *Every* quantity the theory will ever ask of a graphon is obtained by integrating it against bounded test functions, never by evaluating it at a point. And integration is exactly the operation that annihilates the coin-flip noise. To see this quantitatively, take any two fixed sizeable patches of types $S, T \subseteq [0,1]$ and watch the total interaction $\int_S \int_T W_N$ across them. It is the sum of the roughly $N^2\lambda(S)\lambda(T)$ independent $\{0,1\}$ entries lying in the corresponding block, each weighted by the tile area $1/N^2$; by the law of large numbers this sum is $p\,\lambda(S)\lambda(T)$ up to a fluctuation whose standard deviation is of order $\sqrt{N^2}/N^2 = 1/N$. The two noisy kernels and their smooth average therefore satisfy

$$\int_S \int_T W_N \;=\; \int_S \int_T W_N' + O(\tfrac{1}{N}) \;=\; \int_S \int_T \left(\tfrac{1}{2}\right) + O(\tfrac{1}{N}) \;=\; \tfrac{1}{2}\,\lambda(S)\lambda(T) + O(\tfrac{1}{N}),$$

uniformly over the patches. Averaging over any patch of positive measure kills the fluctuation: the ± 1 disagreements that the sup norm magnified are uncorrelated, so over a region containing order N^2 of them they cancel down to order $1/N$. The microscopic wiring is loud pointwise and silent on every patch. *This* is the sense in which the noisy network is close to its smooth average — close in exactly the currency the observables are denominated in — even though every L^p norm insists they are far.

Reading off the cure. The example does more than condemn the ordinary norms; it tells us what to put in their place. The fatal move was to measure $W_N - W_N'$ pointwise, because pointwise the coin-flip noise is full strength. The repair is to measure it the way the observables do: by its integrals over regions, never by its values at points. If we deem two kernels close precisely when their total interaction $\int_S \int_T$ matches across *every* pair of sizeable patches S, T, then the computation above guarantees that our new notion of size will report the noisy network as close to its smooth average, while still refusing to identify genuinely different mesoscopic structures: two block models with different block densities differ on the patches aligned to their blocks, and no choice of integration region can hide that. The block-level information that homomorphism densities, sampling, and the operator $\mathbb{W}$ all depend on is exactly the patch-integral information we propose to keep; the pointwise fluctuations that no observable depends on are exactly what patch integration throws away. We crystallize the design goal in a box.

Heuristic (non-rigorous)

The cure is to measure a kernel not by its pointwise size but by the size of its *integrals over large regions*. Two kernels should be deemed close when, for *every* pair of sizeable patches $S, T \subseteq [0,1]$ of types, the total interaction $\int_S \int_T$ between those patches is nearly the same. This throws away pointwise fluctuations — they average out over any patch of positive measure — while retaining exactly the block-level information that homomorphism densities, sampling, and the mean field coupling $\mathbb{W}$ all depend on. A norm built this way will judge the noisy network above to be *close* to its smooth average, which is what we want.

The design goal is now stated in one line: *measure the size of a kernel by the largest total interaction it accumulates over any single pair of sizeable patches of types.* The next subsection turns this sentence, almost verbatim, into a definition — taking the supremum of $|\int_S \int_T W|$ over all measurable patches S, T — and then records the two equivalent forms of the resulting quantity that make it usable.

17.1.2 Definition and equivalent forms

The previous subsection ended with a design goal so concrete it almost reads as a formula: *measure the size of a kernel by the largest total interaction it accumulates over any single pair of sizeable patches of types.* We promised to turn that sentence, nearly verbatim, into a definition, and that is the whole content of the present subsection. The phrase "total interaction over the patches S and T" is the double integral $\int_S \int_T U$; "largest" is a supremum over all admissible patches; and the absolute value is there because we care how *far* from balanced the interaction is, in either direction. Reading those three clauses off the design goal produces the cut norm with no further invention. Once it is written down we owe the reader two things before moving on: an explanation of the combinatorial name it carries, and — the real work of the subsection — two alternative descriptions of the very same number, one a smoothed-out test-function form that is far easier to compute with, the other an operator norm that opens the door to the spectral viewpoint of chapter 18.

Throughout, a *kernel* is a bounded measurable function $U\colon [0,1]^2 \to \mathbb{R}$, not necessarily symmetric and not necessarily $[0,1]$-valued; a *graphon* (chapter 15) is a symmetric kernel with values in $[0,1]$. We allow the wider class of signed, possibly asymmetric kernels deliberately: the cut metric will be built from $\|U - W\|_\square$, a *difference* of two graphons, and such a difference is signed (it takes values in $[-1,1]$) even when the two graphons themselves are not. We write λ for Lebesgue measure on $[0,1]$ and integrate against it throughout.

Definition 17.1 (Cut norm)**.** For a kernel $U\colon [0,1]^2 \to \mathbb{R}$ the *cut norm* is

$$\|U\|_\square \;=\; \sup_{S,T} \left| \int_S \int_T U(x,y)\, \mathrm{d}x\, \mathrm{d}y \right|, \tag{17.2}$$

the supremum taken over all pairs of measurable sets $S, T \subseteq [0,1]$.

The name is combinatorial in origin, and it is worth pausing to picture the graph it comes from. Suppose U is the *centered* adjacency kernel of a finite graph G: on the tile $I_i \times I_j$ it equals $a_{ij} - p$, where $a_{ij} \in \{0,1\}$ records whether the edge ij is present and p is the ambient edge density we have subtracted off. Pick two groups of vertices, encoded by the type sets S and T. Then $\int_S \int_T U$ is (up to the uniform tile area) the number of edges running between group S and group T *minus* the number one would expect at density p — the *edge imbalance*, or discrepancy, of the bipartition-like "cut" that singles out S on one side and T on the other. A perfectly typical, structureless graph has a tiny imbalance across every cut; a graph with hidden community structure has some pair S, T across which far more (or far fewer) edges run than density alone predicts. The cut norm hunts over all cuts for the most anomalous one and reports the size of that anomaly. This is exactly the mesoscopic information the previous subsection argued we should keep: it is blind to which individual pair of vertices is wired, but acute to how densely whole blocks of vertices interact. With the object and its name in hand, we record the two reformulations that make it usable.

Lemma 17.2 (Three forms of the cut norm)**.** *For every kernel U,*

$$\|U\|_\square \;=\; \sup_{u,v\colon [0,1]\to[0,1]} \left| \int_{[0,1]^2} u(x)\, U(x,y)\, v(y)\, \mathrm{d}x\, \mathrm{d}y \right|, \tag{17.3}$$

the supremum over all measurable $[0,1]$-valued functions, and moreover

$$\|U\|_\square \;\le\; \|U\|_{\infty\to1} \;:=\; \sup_{\|u\|_\infty \le 1,\ \|v\|_\infty \le 1} \left| \int_{[0,1]^2} u\, U\, v \right| \;\le\; 4\, \|U\|_\square. \tag{17.4}$$

Before the proof, a word on what each line claims and why one would want it. The original eq. (17.2) ranges over *sets* S, T — combinatorial objects, sharp-edged indicators. Form eq. (17.3) relaxes the two indicators $\mathbf{1}_S, \mathbf{1}_T$ to arbitrary $[0,1]$-valued "soft" test functions u, v, which one may picture as fuzzy or fractional memberships in the two groups. This looks like a strictly larger supremum — we are optimizing over a vastly bigger set of competitors — and the lemma's small surprise is that it is not larger at all: softening the membership buys nothing, the optimum already sits at a crisp set. That equality is what makes eq. (17.3) the *analytic workhorse*: a supremum over a convex set of smooth functions is something calculus can attack (one can differentiate in u and v, integrate by parts, apply Cauchy–Schwarz), whereas a supremum over raw sets is combinatorially rigid. Form eq. (17.4) then drops the sign constraint entirely, letting u, v range over the unit ball of L^∞ (values in $[-1,1]$); the resulting quantity $\|U\|_{\infty\to1}$ is, as we explain afterward, exactly the operator norm of $\mathbb{W}_U$ viewed as a map $L^\infty([0,1]) \to L^1([0,1])$, and the lemma says it agrees with the cut norm up to a factor of 4. We prove the two displays in turn.

Proof. The test-function form eq. (17.3)*.* One inequality is free: indicators are themselves $[0,1]$-valued, so taking $u = \mathbf{1}_S$ and $v = \mathbf{1}_T$ in eq. (17.3) reproduces $\int_S \int_T U$, and ranging over all S, T

shows the right side of eq. (17.3) is *at least* $\|U\|_\square$. The substance is the reverse: that allowing fuzzy test functions never beats the best crisp pair. We prove this by an elementary *pointwise* maximization — optimizing the integrand separately at each type — and we draw attention to the fact that it needs nothing beyond Fubini's theorem. One could instead invoke the general principle that a linear functional on a convex set attains its maximum at an extreme point, together with the identification of the extreme points of the $[0,1]$-valued unit ball as the indicators; but that route imports extreme-point and weak-$*$ compactness machinery we have no wish to assume here. The pointwise argument is deliberately self-contained, and it is also more transparent, because it exhibits the optimal u and v explicitly. Here it is.

Fix any $v\colon [0,1] \to [0,1]$ and contract the kernel against it in the second variable, setting

$$g(x) \;:=\; \int_[0,1]U(x,y)\,v(y)\;\mathrm{d}y.$$

Because U is bounded, g lies in $L^1([0,1])$ (this is the one place Fubini is used: it lets us write $\int_{[0,1]^2} u\,U\,v \;=\; \int_[0,1]u(x)\,g(x)\;\mathrm{d}x$, turning the double integral into a single integral against the legitimately defined function g). Now ask how large $\int u\,g$ can be made as u ranges over $[0,1]$-valued functions. The integrand $u(x)\,g(x)$ is maximized *at each point x independently*: where $g(x) > 0$ we want $u(x)$ as large as allowed, namely $u(x) = 1$; where $g(x) < 0$ we want $u(x) = 0$; where $g(x) = 0$ the choice is immaterial. Concretely,

$$\int_[0,1]u(x)\,g(x)\;\mathrm{d}x \;\le\; \int_[0,1]u(x)\,g(x)^+\;\mathrm{d}x \;\le\; \int_[0,1]g(x)^+\;\mathrm{d}x \;=\; \int_{\{g>0\}} g\;\mathrm{d}x,$$

where the first inequality drops the (nonpositive) contribution of g^- and the second uses $u \le 1$; both become equalities at the admissible choice $u = \mathbf{1}_{\{g>0\}}$. Hence $\sup_{0\le u\le 1} \int u\,g = \int_{\{g>0\}} g = \int_S g$ with $S = \{g > 0\}$, and the supremum is genuinely attained, at the indicator $\mathbf{1}_S$. Since the defining supremum in eq. (17.3) carries an absolute value, we may also need the most *negative* value of $\int u\,g$; applying the same reasoning to $-g$ shows its optimum is the indicator $\mathbf{1}_{\{g<0\}}$. Either way, for every fixed v the best u is an indicator. Consequently the joint supremum over u, v loses nothing by restricting u to indicators:

$$\sup_{u,v\in[0,1]} \left| \int_{[0,1]^2} u\,U\,v \right| \;=\; \sup_{S,\,v\in[0,1]} \left| \int_S \Big(\int_[0,1]U(x,y)\,v(y)\;\mathrm{d}y \Big)\,\mathrm{d}x \right|.$$

Now repeat the identical move in the *other* variable — this is the step the skeleton compressed, so we spell it out. Fix the set S just produced and contract the kernel against its indicator in the *first* variable, setting

$$h(y) \;:=\; \int_S U(x,y)\;\mathrm{d}x \;\in\; L^1([0,1]),$$

again integrable because U is bounded. The inner expression above is $\int_S \int_[0,1]U\,v = \int_[0,1]h(y)\,v(y)\;\mathrm{d}y$, a single integral against h, and we maximize it over $v \in [0,1]$ by the very same pointwise argument: the optimum is $v = \mathbf{1}_{\{h>0\}}$ for the most positive value and $v = \mathbf{1}_{\{h<0\}}$ for the most negative, so in either case $v = \mathbf{1}_T$ for a set T $(= \{h > 0\}$ or $\{h < 0\})$. Substituting back, the joint supremum is attained at a *pair* of indicators $\mathbf{1}_S, \mathbf{1}_T$, and its value is $|\int_S \int_T U|$ for those sets, which is by definition no larger than $\|U\|_\square$. We have shown each side of eq. (17.3) dominates the other, so they are equal.

Note that the only analytic fact used was that contracting a bounded kernel against a bounded function yields an L^1 function (Fubini); no compactness, no extreme points, no appeal to attainment of suprema over infinite-dimensional balls.

The operator form eq. (17.4). The left inequality $\|U\|_\square \le \|U\|_{\infty\to 1}$ is again free: indicators have $\|\cdot\|_\infty \le 1$, so they are admissible competitors in the larger supremum defining $\|U\|_{\infty\to 1}$, which therefore dominates the supremum over indicators, i.e. $\|U\|_\square$. The right inequality is where the factor 4 is born, and it comes from splitting each signed test function into its two nonnegative halves. Given u, v with $\|u\|_\infty \le 1$ and $\|v\|_\infty \le 1$, write $u = u_+ - u_-$ and $v = v_+ - v_-$ for the positive and negative parts $u_\pm = \max(\pm u, 0)$, and likewise for v. The point of this decomposition is that each of u_+, u_-, v_+, v_- is now a *genuinely* $[0,1]$*-valued* function (the negative part of a function bounded by 1 in absolute value is itself bounded between 0 and 1), so each is an admissible test function for eq. (17.3). Expanding the bilinear integrand,

$$\int_{[0,1]^2} u\,U\,v = \int u_+\,U\,v_+ \;-\; \int u_+\,U\,v_- \;-\; \int u_-\,U\,v_+ \;+\; \int u_-\,U\,v_-,$$

a sum of four terms, each of the form $\int([0,1]\text{-fn})\,U\,([0,1]\text{-fn})$ and hence bounded in absolute value by $\|U\|_\square$ thanks to the form eq. (17.3) just proved. By the triangle inequality the whole sum is at most $4\,\|U\|_\square$ in absolute value,

$$\left|\int_{[0,1]^2} u\,U\,v\right| \;\le\; 4\,\|U\|_\square\,,$$

and taking the supremum over all admissible u, v gives $\|U\|_{\infty\to 1} \le 4\,\|U\|_\square$. The four summands are exactly the four sign combinations $(\pm,\pm)$, which is precisely where the constant 4 comes from; it is harmless because it is a fixed universal factor that washes out of every convergence statement. $\square$

It is worth being explicit about why we carry *both* forms forward rather than collapsing to one. Form eq. (17.3) is the one we actually compute with in the rest of the chapter: relabeling invariance, subadditivity, the worked block-graphon computation, and the energy-increment step of the regularity lemma are all cleanest when phrased as an optimization over $[0,1]$-valued test functions, because those test functions can be manipulated with ordinary analysis. Form eq. (17.4) is the one we hand to the future: it says that, up to the universal factor 4, the cut norm of U *is* the operator norm of $\mathbb{W}_U$ as a map from $L^\infty([0,1])$ to $L^1([0,1])$,

$$\tfrac{1}{4}\,\|U\|_{\infty\to 1} \;\le\; \|U\|_\square \;\le\; \|U\|_{\infty\to 1}\,,$$

which recasts a question about set discrepancies as a question about an operator. That reinterpretation is the bridge to chapter 18, where the same graphon is read as a self-adjoint Hilbert–Schmidt operator and cut convergence is compared with spectral convergence; the very next subsection already begins to use it, controlling $\|U\|_\square$ by the $L^2 \to L^2$ operator norm in order to place the cut norm among the familiar norms.

Notation watch

We will use three names that are easy to conflate, and the distinctions are load-bearing for the rest of the chapter. The *cut norm* $\|\cdot\|_\square$ of definition 17.1 is a genuine norm on kernels —

it is what this subsection has constructed and characterized. The *cut distance* between two kernels means simply the cut norm of their difference, $\|U - U'\|_\square$; it is a distance between *labeled* kernels and still sees relabelings. The *cut metric* $\delta_\square$, defined in section 17.2, is obtained from the cut distance by *minimizing over relabelings*; it is only a *pseudometric* on graphons (distinct relabelings of one graphon are at distance zero) and becomes a true metric only after passing to the quotient space of proposition 17.7. Whenever the text says "norm," "distance," or "metric," it means precisely one of these three, and the reader should not silently substitute one for another.

17.1.3 Where the cut norm sits

We now have two interchangeable working descriptions of the cut norm — the test-function form eq. (17.3) that we will compute with, and the operator-norm reading eq. (17.4) that we will hand to chapter 18. What we do not yet have is a sense of *scale*: when we said in the chapter's opening that the cut norm is "deliberately coarse," and when the noisy-network example showed an L^1 chasm collapsing to a vanishing patch-integral, we were asserting a quantitative relationship between $\|\cdot\|_\square$ and the norms the reader already owns without ever pinning it down. This subsection pins it down. The cut norm does not float free of the familiar L^p scale; it sits at the *bottom* of it, dominated by every one of them, and it is in addition controlled from above by the spectral ($L^2 \to L^2$ operator) norm. Locating it in this lattice does two jobs at once. It converts the heuristic "coarser than any pointwise norm" into a chain of clean inequalities we can quote later — the regularity argument of section 17.4 will need exactly the comparison $\|\cdot\|_\square \le \|\cdot\|_2$ to run its energy bookkeeping — and, almost as a by-product, it lets us discharge the one obligation still outstanding from the definition: that $\|\cdot\|_\square$ really is a *norm*, non-degeneracy included. The comparison and the norm axioms travel together, so we state them in a single lemma.

Lemma 17.3 (Comparison inequalities)**.** *For every kernel U,*

$$\|U\|_\square \;\le\; \|U\|_1 \;\le\; \|U\|_2 \;\le\; \|U\|_\infty\,, \qquad \textit{and} \qquad \|U\|_\square \;\le\; \|\mathbb{W}_U\|_{2\to 2}\,, \tag{17.5}$$

where $\|\mathbb{W}_U\|_{2\to2}$ is the operator norm of $\mathbb{W}_U$ on $L^2([0,1])$. In particular $\|\cdot\|_\square$ is a norm: it is positively homogeneous and subadditive by eq. (17.3), and $\|U\|_\square = 0$ forces $\int_S \int_T U = 0$ for all S, T, hence $U = 0$ almost everywhere.

The lemma asserts four comparisons and three norm axioms, and it is worth saying in advance which of these is routine and which carries weight. The chain $\|U\|_\square \le \|U\|_1 \le \|U\|_2 \le \|U\|_\infty$ is bookkeeping — each link is a one-line application of a standard integral inequality on the probability space $([0,1]^2, \lambda \otimes \lambda)$ — but it is the bookkeeping that makes the word "coarsest" precise. The spectral bound $\|U\|_\square \le \|\mathbb{W}_U\|_{2\to2}$ is the same kind of one-liner with a Hilbert-space flavour, and its equality structure is something we will want to revisit. The genuinely substantive clause is the last: that $\|U\|_\square = 0$ forces $U = 0$ almost everywhere, the *non-degeneracy* that separates a true norm from a mere seminorm. It is substantive because vanishing of *every* patch integral $\int_S \int_T U$ is, on its face, far weaker than vanishing of U pointwise, and bridging that gap is exactly the measure-theoretic content of the claim. We take the four comparisons first and then spend the bulk of the proof on non-degeneracy.

Proof. The first inequality. For any measurable S, T the double integral is dominated by the integral of the absolute value over all of $[0,1]^2$, simply because enlarging the region of integration and dropping the cancellation between positive and negative parts can only increase the modulus:

$$\Big|\int_S\int_T U(x,y)\ \mathrm{d}x\ \mathrm{d}y\Big| \ \le\ \int_S\int_T |U| \ \le\ \int_{[0,1]^2} |U| \ =\ \|U\|_1\,.$$

This holds for every S, T, so the supremum over S, T — which is $\|U\|_\square$ by eq. (17.2) — is also at most $\|U\|_1$. This first link is where the coarseness is born: the cut norm is a constrained, region-by-region accounting that can never exceed the unconstrained total mass $\|U\|_1$, and it falls strictly below it precisely when U changes sign at fine scales, so that the cancellations the L^1 norm forbids are exactly the ones the cut norm exploits.

The L^p chain. The remaining links $\|U\|_1 \le \|U\|_2 \le \|U\|_\infty$ are the monotonicity of L^p norms in p on a *probability* space, and it is worth recalling why the total mass being 1 is what makes them point this way. For $\|U\|_1 \le \|U\|_2$, apply the Cauchy–Schwarz inequality to the pair $|U|$ and the constant function 1 on $([0,1]^2, \lambda\otimes\lambda)$:

$$\|U\|_1 = \int_{[0,1]^2} |U|\cdot 1 \ \le\ \Big(\int_{[0,1]^2} |U|^2\Big)^{1/2}\Big(\int_{[0,1]^2} 1^2\Big)^{1/2} \ =\ \|U\|_2\cdot\lambda([0,1]^2)^{1/2} = \|U\|_2\,,$$

the last equality because $\lambda\otimes\lambda$ assigns total mass 1. Equivalently this is Jensen's inequality for the convex function $t\mapsto t^2$ applied to the averaging $\int |U|\ \mathrm{d}(\lambda\otimes\lambda)$. For $\|U\|_2 \le \|U\|_\infty$ there is nothing to optimize: $|U| \le \|U\|_\infty$ pointwise, so $\int |U|^2 \le \|U\|_\infty^2 \int 1 = \|U\|_\infty^2$, and taking square roots gives the bound. On a space of total mass larger than 1 these inequalities would acquire mass-dependent constants and could even reverse; it is the normalization $\lambda([0,1]) = 1$, inherited from reading the type variable as an exchangeable label rather than a length, that makes the familiar ordering hold cleanly.

The spectral bound. Return to the set form eq. (17.2) and read the inner double integral as a Hilbert-space pairing. By Fubini,

$$\int_S\int_T U(x,y)\ \mathrm{d}x\ \mathrm{d}y = \int_S\Big(\int_T U(x,y)\ \mathrm{d}y\Big)\,\mathrm{d}x = \int_[0,1]\mathbf{1}_S(x)\,(\mathbb{W}_U\mathbf{1}_T)(x)\ \mathrm{d}x = \langle \mathbf{1}_S, \mathbb{W}_U\mathbf{1}_T\rangle,$$

since $(\mathbb{W}_U\mathbf{1}_T)(x) = \int_[0,1]U(x,y)\mathbf{1}_T(y)\,\mathrm{d}y = \int_T U(x,y)\,\mathrm{d}y$ is exactly the kernel operator applied to the indicator of T. Now Cauchy–Schwarz on $L^2([0,1])$, followed by the definition of the operator norm, gives

$$|\langle \mathbf{1}_S, \mathbb{W}_U\mathbf{1}_T\rangle| \ \le\ \|\mathbf{1}_S\|_2\ \|\mathbb{W}_U\mathbf{1}_T\|_2 \ \le\ \|\mathbf{1}_S\|_2\ \|\mathbb{W}_U\|_{2\to 2}\ \|\mathbf{1}_T\|_2\,.$$

The two indicator norms are harmless: $\|\mathbf{1}_S\|_2 = (\int_[0,1]\mathbf{1}_S)^{1/2} = \lambda(S)^{1/2} \le 1$ and likewise $\|\mathbf{1}_T\|_2 \le 1$, so the right-hand side is at most $\|\mathbb{W}_U\|_{2\to 2}$. This bound is uniform in S, T, so the supremum $\|U\|_\square$ is bounded by $\|\mathbb{W}_U\|_{2\to 2}$ as claimed. We flag the two places slack can enter, because the worked computation later in the chapter will close exactly this gap on a rank-one kernel: slack appears when the optimal S, T have measure strictly below 1 (so $\|\mathbf{1}_S\|_2 < 1$), and when the indicators $\mathbf{1}_S, \mathbf{1}_T$ fail to align with the top singular functions of $\mathbb{W}_U$. For a rank-one kernel $W(x,y) = f(x)f(y)$ the operator $\mathbb{W}_U$ has a single nontrivial singular direction, and we will see the cut norm track $\|\mathbb{W}_U\|_{2\to 2}$ closely precisely because the best cut aligns with that one direction; the inequality here is the loose version of an estimate that becomes nearly sharp there.

Positive homogeneity and subadditivity. These two norm axioms are inherited verbatim from the test-function form eq. (17.3), because that form exhibits $\|U\|_\square$ as a supremum of linear functionals of U. For a scalar c, every integrand scales as $\int u\,(cU)\,v = c\int u\,U\,v$, so $\|cU\|_\square = \sup_{u,v}|c|\,|\int u\,U\,v| = |c|\,\|U\|_\square$. For subadditivity, fix any admissible u, v and use the triangle inequality for the absolute value pointwise inside the supremum,

$$\Big|\int_{[0,1]^2} u\,(U{+}U')\,v\Big| = \Big|\int_{[0,1]^2} u\,U\,v{+}\int_{[0,1]^2} u\,U'\,v\Big| \;\le\; \Big|\int_{[0,1]^2} u\,U\,v\Big|{+}\Big|\int_{[0,1]^2} u\,U'\,v\Big| \;\le\; \|U\|_\square{+}\|U'\|_\square,$$

and taking the supremum over u, v on the left gives $\|U+U'\|_\square \le \|U\|_\square + \|U'\|_\square$. The mechanism is the standard one — a supremum of subadditive quantities is subadditive — and it works only because the *same* pair u, v is used for both kernels before the supremum is taken; this is the property the next section will lean on directly.

Non-degeneracy. It remains to show that $\|U\|_\square = 0$ forces $U = 0$ almost everywhere. The hypothesis says $\int_S\int_T U = 0$ for *every* pair of measurable sets S, T, and the task is to upgrade this family of scalar equations into pointwise vanishing of U. We do it in two stages, peeling off one variable at a time. Fix a measurable set T and contract U against its indicator in the second variable, defining

$$g_T(x) \;:=\; \int_T U(x,y)\;\mathrm{d}y,$$

which lies in $L^1([0,1])$ because U is bounded. The hypothesis, read for this fixed T and all S, says exactly that $\int_S g_T = 0$ for every measurable S. A function whose integral over every measurable set vanishes must vanish almost everywhere — take $S = \{g_T > 0\}$ to get $\int g_T^+ = 0$, hence $g_T^+ = 0$ a.e., and $S = \{g_T < 0\}$ to get $g_T^- = 0$ a.e. — so $g_T = 0$ a.e. on $[0,1]$. Thus for *each* fixed T there is a conull set of types x on which $\int_T U(x,y)\;\mathrm{d}y = 0$.

Here is the delicate point the skeleton compressed, and it is genuinely the crux. The conull set just produced *depends on* T, and there are uncountably many measurable sets T; an uncountable union of null sets need not be null, so we cannot simply say "off a null set, $\int_T U(x,\cdot) = 0$ for all T" and be done. The repair is to work with only *countably* many T, chosen densely enough to determine all the rest. Let $\mathcal{A}$ be a countable algebra of measurable subsets of $[0,1]$ that generates the Borel σ-algebra — for instance the algebra of finite unions of dyadic subintervals $[\,k2^{-n},(k+1)2^{-n})$. For each $T \in \mathcal{A}$ the previous paragraph furnishes a null set $Z_T \subseteq [0,1]$ off which $g_T(x) = 0$. Because $\mathcal{A}$ is countable, the union

$$Z \;:=\; \bigcup_{T\in\mathcal{A}} Z_T$$

is a *countable* union of null sets and hence itself null; this is the single line where countability does all the work, and it is exactly what an uncountable index set would deny us. On the conull complement $[0,1]\setminus Z$ we have achieved simultaneity: for *every* $x \notin Z$ and *every* $T \in \mathcal{A}$ at once, $\int_T U(x,y)\;\mathrm{d}y = 0$.

Now fix any single such $x \notin Z$ and consider the set function $T \mapsto \mu_x(T) := \int_T U(x,y)\;\mathrm{d}y$. Since $U(x,\cdot) \in L^1([0,1])$, this μ_x is a finite signed measure, absolutely continuous with respect to λ, with density $U(x,\cdot)$. We have just shown μ_x vanishes on the generating algebra $\mathcal{A}$. A finite signed measure that vanishes on an algebra generating the σ-algebra vanishes on the entire σ-algebra — this is the uniqueness half of the Carathéodory/Dynkin extension theorem, the statement that a measure is determined by its values on a generating π-system. Hence $\mu_x(T) = 0$ for *all* Borel T, not merely those in $\mathcal{A}$, and a signed measure that vanishes on every set has zero density: $U(x,y) = 0$ for λ-a.e. y. This holds for every $x \notin Z$, i.e. for a.e. x, and so by Fubini $U = 0$ for $(\lambda \otimes \lambda)$-a.e. (x,y), which is what non-degeneracy demands. Together with homogeneity and subadditivity this completes the verification that $\|\cdot\|_\square$ is a genuine norm. $\square$

A short concrete check makes the non-degeneracy argument feel less abstract. Suppose $U(x,y) = \operatorname{sgn}(\sin 2\pi Nx)\operatorname{sgn}(\sin 2\pi Ny)$, a checkerboard of ± 1 tiles of side $1/2N$. Its L^1 norm is 1 for every N, so it is "large" pointwise; yet against any dyadic block $S \times T$ coarser than the tiles the $+1$ and -1 cells cancel in pairs and $\int_S \int_T U \to 0$. The non-degeneracy proof is the assurance that this near-cancellation can never become *exact* for all S, T unless U is genuinely zero: as long as some patch integral survives, $\|U\|_\square > 0$. The cut norm is coarse but not blind.

The first chain of eq. (17.5) is, emphatically, a one-way street, and that is the entire reason this chapter exists. None of its links can be reversed: the noisy-versus-smooth construction of the heuristic furnishes the witness, with $\|W_N - W'_N\|_\square \to 0$ while $\|W_N - W'_N\|_1$ stays pinned near $\frac{1}{2}$ for all N, so the cut norm can be arbitrarily small where the L^1 norm is bounded below. A finer norm would refuse to let those two sequences approach each other; the cut norm lets them, and in doing so it carves out room for objects — limits of microscopically noisy but macroscopically smooth networks — that the L^p topologies simply cannot hold. That extra room is not a defect to be apologized for; it is the space the cut norm makes, and the space on which the rest of Part III will do its work.

With $\|\cdot\|_\square$ now established as a bona fide norm — coarse, well-placed in the lattice of familiar norms, and in particular *subadditive* — we have precisely the raw material the next section requires. Subadditivity is the property that will become the triangle inequality of the cut metric once we minimize over relabelings, and having a fixed, fully-understood norm in hand is what lets us treat the quotient by relabelings cleanly rather than as a tangle of two complications at once. What remains is to confront the redundancy the norm still carries: it is a distance between *labeled* kernels, and the labeling of types is an artifact we have no right to keep. Removing it is the task of section 17.2.

17.2 Relabelings and the cut metric

The previous section closed with a fixed, fully-understood object in hand: $\|\cdot\|_\square$ is a genuine norm on kernels, coarse enough to ignore the microscopic wiring noise that no observable sees, and — by lemma 17.5, which we are about to state and prove — already blind to relabelings of the type coordinate. That coarseness is exactly half of what we need. The cut norm tells us how big a kernel is in a way that respects the observables; what it does not yet give us is a *distance between networks*. For $\|U - W\|_\square$, the cut norm of a difference, is still a distance between *labeled* kernels, and the labeling of types is precisely the kind of artifact this chapter exists to discard. The type variable $x \in [0,1]$ is, from chapter 15 onward, an exchangeable index — a name we hand to an agent's network position, with no intrinsic meaning of its own. Two graphons that differ only by a measure-preserving rearrangement of those names encode one and the same network, yet $\|U - W\|_\square$ will in general report them as far apart, because the difference $U - W$ compares the kernels name-by-name rather than network-to-network. This residual sensitivity to an arbitrary labeling is the last redundancy standing between us and a true distance, and removing it is the entire business of this section.

The program is short to state and occupies the rest of the section in three steps. First, we make "relabeling" precise: it is a measure-preserving bijection φ of $[0, 1]$, acting on kernels by the pull-back $U^\varphi(x, y) = U(\varphi(x), \varphi(y))$, and we check that the cut norm and the homomorphism densities — everything observable — are invariant under it, so that relabeling really is a symmetry the theory cannot detect. Second, we quotient that symmetry away in the only manner a norm permits: the *cut metric* $\delta_\square(U, W) = \inf_\varphi \|U - W^\varphi\|_\square$ takes the smallest cut distance achievable by relabeling one graphon to align it as closely as possible with the other, and passing to the quotient by the relation "$\delta_\square$-distance zero" converts this infimum from a mere pseudometric on labeled graphons into a genuine metric on the intrinsic networks themselves — the *space of graphons* on which all the analysis of this chapter, and of Part III, will take place. Third, we read the construction back as a statement about information: the cut metric remembers a graphon's block-averaged, mesoscopic structure and forgets everything finer, an interpretation that previews the regularity lemma of section 17.4 and explains in advance why the resulting space turns out to be small enough to be compact. We take the three steps in turn, beginning with why the labels deserve to be thrown away at all.

17.2.1 The type labels are arbitrary

Before we may promote the cut norm to a distance between networks we must interrogate it for the symmetries it is obliged to respect. A distance is only as meaningful as the quantities it declines to separate, and the cut norm inherits, from the very meaning of the type variable, one symmetry it has no right to violate: a relabeling of the types must leave it untouched. Establishing that — and that the homomorphism densities of chapter 16 share the same blindness — is the formal license we need before we may quotient relabelings away. This short subsection earns that license.

The conceptual heart is a fact carried forward from chapter 15, and it is worth restating in full because everything in this section turns on it. The type variable $x \in [0, 1]$ is an *exchangeable label*, not a spatial coordinate. When the graphon arose as the limit of an adjacency matrix, the points of $[0, 1]$ inherited the role of the vertex indices $1, \ldots, N$, and those indices were never anything but names: relabeling the vertices of a graph — calling vertex 3 "vertex 17" and vice versa — produces a different adjacency matrix that nonetheless encodes the very same graph. Nothing about the network changes; only the bookkeeping does. In the continuum the analogue of "permute the vertex indices" is a *measure-preserving rearrangement* of the coordinate x, and the analogue of "same graph, different matrix" is "same network, different kernel." Two graphons related by such a rearrangement are not merely close: they are the identical object wearing different name tags, and a theory that could tell them apart would be reading meaning into a labeling that carries none. The distance we are building must therefore be blind to relabelings; and it can be made blind only because the raw material it is built from — the cut norm and the densities — is *already* blind, which is the content of the lemma below. We first make "relabeling" precise.

Definition 17.4 (Relabeling and pull-back)**.** A *measure-preserving bijection* of $[0,1]$ is a bijection $\varphi\colon [0,1] \to [0,1]$ such that both φ and φ^{-1} are measurable and measure-preserving: $\lambda(\varphi^{-1}(A)) = \lambda(A)$ for all measurable A. For a kernel U and such a φ the *pull-back* is

$$U^{\varphi}(x,y) \;:=\; U\big(\varphi(x),\varphi(y)\big).$$

Each clause of this definition pulls its weight, and it pays to see why before using it. That φ is a *bijection* encodes that a relabeling is a pure renaming: every type receives exactly one new name and every name is used exactly once, so the operation is reversible and loses no type. That φ is *measure-preserving* encodes that the renaming does not distort how much of the population sits where: under the exchangeable reading, $\lambda(A)$ is the fraction of agents whose type lies in A, and a legitimate relabeling must keep that fraction fixed. A bijection that, say, compressed the entire left half $[0,\frac{1}{2})$ into the tiny interval $[0,\frac{1}{100})$ would be a renaming on paper but would silently reweight the population, inflating the influence of some types and deflating others; the measure-preserving condition forbids exactly this and is what guarantees the integrals below are untouched. The *pull-back* U^{φ} is then nothing but "rename the types by φ and read off the kernel in the new names": its value at the renamed pair (x,y) is the old kernel's value at the original pair $(\varphi(x),\varphi(y))$. As a one-line illustration, take φ to be the half-swap $\varphi(x) = x + \frac{1}{2} \pmod 1$, which exchanges $[0,\frac{1}{2})$ with $[\frac{1}{2},1)$ and is plainly a measure-preserving bijection. If W is a two-block graphon with one density on the diagonal blocks and another off-diagonal, then W^{φ} is the same two-block graphon with the two blocks interchanged — the networks are manifestly the same, only the block sitting on $[0,\frac{1}{2})$ has traded places with the one on $[\frac{1}{2},1)$. Pull-back is the formal expression of precisely this relabeling, and the claim we now prove is that no observable can see it.

Lemma 17.5 (Invariance under relabeling)**.** *For every kernel U and every measure-preserving bijection φ, $\|U^{\varphi}\|_{\square} = \|U\|_{\square}$, and for every finite simple graph F, $t(F,W^{\varphi}) = t(F,W)$.*

Proof. The cut norm. Start from the set form eq. (17.2) and evaluate the inner double integral for the pulled-back kernel. For any measurable S,T,

$$\int_S\int_T U^{\varphi}(x,y)\;\mathrm{d}x\;\mathrm{d}y = \int_S\int_T U\big(\varphi(x),\varphi(y)\big)\;\mathrm{d}x\;\mathrm{d}y,$$

and we change variables $x' = \varphi(x)$, $y' = \varphi(y)$. Because φ is a measure-preserving bijection, it preserves λ in each variable separately and hence the product measure $\lambda\otimes\lambda$ on $[0,1]^2$; the change of variables therefore introduces no Jacobian factor, $\mathrm{d}x'\;\mathrm{d}y' = \mathrm{d}x\;\mathrm{d}y$. As x ranges over S the new variable $x' = \varphi(x)$ ranges over the image $\varphi(S)$, and likewise y' ranges over $\varphi(T)$, so

$$\int_S\int_T U^{\varphi}(x,y)\;\mathrm{d}x\;\mathrm{d}y = \int_{\varphi(S)}\int_{\varphi(T)} U(x',y')\;\mathrm{d}x'\;\mathrm{d}y'. \tag{17.6}$$

Thus the discrepancy that U^{φ} accumulates over the cut (S,T) is *identical* to the discrepancy that U itself accumulates over the relabeled cut $(\varphi(S),\varphi(T))$: relabeling the kernel merely moves the test sets, it does not alter any of the numbers being maximized over.

It remains to see that moving the test sets does not shrink or enlarge the pool of numbers the supremum ranges over — this is the step the reader should not let slip past, because the whole invariance rests on it. The map $(S,T) \mapsto (\varphi(S), \varphi(T))$ is a bijection from pairs of measurable sets onto pairs of measurable sets. It lands in measurable pairs because φ^{-1} is measurable, so $\varphi(S) = (\varphi^{-1})^{-1}(S)$ is measurable; and it is onto because any prescribed pair (S', T') is hit by taking $S = \varphi^{-1}(S')$, $T = \varphi^{-1}(T')$, which are themselves measurable for the same reason. Consequently, as (S,T) runs over *all* measurable pairs, $(\varphi(S), \varphi(T))$ runs over exactly the same collection — all measurable pairs — and the two suprema are taken over an identical set of real numbers. Combining this with eq. (17.6),

$$\|U^\varphi\|_\square = \sup_{S,T} \Big| \int_S \int_T U^\varphi \Big| = \sup_{S,T} \Big| \int_{\varphi(S)} \int_{\varphi(T)} U \Big| = \sup_{S',T'} \Big| \int_{S'} \int_{T'} U \Big| = \|U\|_\square \,,$$

where the third equality is precisely the change of index $(S', T') = (\varphi(S), \varphi(T))$ just justified. This is the relabeling invariance of the cut norm.

The homomorphism densities. The same change of variables, now applied in every type coordinate at once, settles the densities. Recall from chapter 16 that

$$t(F, W) = \int_{[0,1]^{V(F)}} \prod_{ij \in E(F)} W(x_i, x_j) \prod_{k \in V(F)} \mathrm{d}x_k,$$

an integral over one copy of $[0,1]$ for each vertex of F. Replacing W by W^φ turns each factor into $W^\varphi(x_i, x_j) = W(\varphi(x_i), \varphi(x_j))$, so the whole integrand is the original integrand evaluated at the renamed coordinates $x'_k = \varphi(x_k)$. Substituting $x'_k = \varphi(x_k)$ simultaneously for every $k \in V(F)$ preserves the product measure $\prod_k \mathrm{d}x_k$ on $[0,1]^{V(F)}$, one measure-preserving factor per coordinate, and carries the domain $[0,1]^{V(F)}$ onto itself because φ is a bijection of $[0,1]$. Hence

$$t(F, W^\varphi) = \int_{[0,1]^{V(F)}} \prod_{ij \in E(F)} W(x'_i, x'_j) \prod_k \mathrm{d}x'_k = t(F, W),$$

identical to the original. Every observable the theory recognizes — the cut norm and all the subgraph densities — is therefore deaf to relabeling. $\square$

This lemma is small, but it is the workhorse of everything that follows, and we will lean on it so often in the next subsection that it is worth flagging its uses in advance. It is what makes the cut metric $\delta_\square(U, W) = \inf_\varphi \|U - W^\varphi\|_\square$ of the next subsection *symmetric*: swapping the roles of U and W amounts to pulling the difference kernel back by φ^{-1}, and lemma 17.5 says that pull-back costs nothing, so the two infima coincide. It is what powers the *triangle inequality*: two near-optimal relabelings are chained through the cocycle identity $W^{\psi\circ\varphi} = (W^\psi)^\varphi$, and invariance is what lets us align the middle graphon by φ without paying for it. And far downstream, in the total-boundedness half of the compactness theorem (section 17.4), it is what permits us to *relabel each part of a partition onto a subinterval* of $[0,1]$ when we tidy a step-function approximant into standard form: because relabeling changes no observable and leaves the cut distance to the original graphon at zero, such normalizations are free. Invariance, in short, is the permission slip the rest of the chapter spends repeatedly. With it secured, we may at last quotient the relabelings away and build the genuine distance between networks — the task of the next subsection, which uses this lemma at every turn to define a distance blind to labeling: the best alignment of U to W in cut norm.

17.2.2 The cut metric and the space of graphons

The previous subsection ended by handing us a permission slip: relabeling changes no observable, because both the cut norm and every homomorphism density are invariant under a measure-preserving bijection of the types (lemma 17.5). We spend that slip now. The cut norm of a difference, $\|U - W\|_\square$, is a distance between *labeled* kernels; it still sees the arbitrary name tags on the types and will report two relabelings of one network as far apart. We want instead a distance blind to labeling: a single number measuring how far apart the two *networks* stand, with the bookkeeping of names quotiented away. The construction is forced by what we have just proved. Since every relabeling W^φ of W encodes the identical network, the only honest way to ask "how far is U from W?" is to ask how close U can be brought to W after we are allowed to relabel W as favorably as we like — to slide W's types around by the best alignment and then measure the residual gap in cut norm. "As favorably as we like" is an optimization, and the optimal value is an infimum over all relabelings. That single recipe — best alignment, then cut norm of what remains — is the cut metric.

Definition 17.6 (Cut metric)**.** Let $\mathcal{W}_0$ denote the set of all graphons (symmetric measurable $W\colon [0,1]^2 \to [0,1]$). For $U, W \in \mathcal{W}_0$ the *cut metric* is

$$\delta_\square(U, W) \;=\; \inf_{\varphi} \; \|U - W^\varphi\|_\square\,, \tag{17.7}$$

the infimum over all measure-preserving bijections φ of $[0,1]$.

The cut metric measures how close U can be brought to W in cut norm by the best relabeling of W's types. One should read $\delta_\square(U, W)$ as the score of an alignment game: the player is handed the freedom to renumber W's types by any measure-preserving bijection, tries to make the renumbered W^φ overlap U as exactly as the cut norm can detect, and the cut metric records the smallest cut-norm mismatch any renumbering can achieve. If U and W are the same network in different clothing, some φ undresses the disguise and drives the mismatch to zero; if they are genuinely different mesoscopic structures, no φ can hide the difference and the infimum stays positive. That this prescription deserves the word "metric" is the content of the next proposition, and the proof has exactly one subtle moment, which we flag in advance so the reader knows where to look. The infimum in eq. (17.7) ranges over an infinite-dimensional, noncompact set of bijections and *need not be attained*: there may be no single best relabeling, only a sequence of ever-better ones. A careless argument would want to grab the optimal φ and manipulate it, and would stall when no such φ exists. The remedy is that we never need the optimum itself. Everything — symmetry and the triangle inequality both — goes through using only two facts already in hand: that measure-preserving bijections *compose* to give another measure-preserving bijection, and that the cut norm is *invariant* under any one of them. Working with relabelings that are merely near-optimal, and composing them, sidesteps attainment entirely.

Proposition 17.7 ($\delta_\square$ is a pseudometric; the metric space of graphons)**.** *The cut metric $\delta_\square$ is a pseudometric on $\mathcal{W}_0$: it is symmetric, nonnegative, satisfies $\delta_\square(W, W) = 0$, and obeys the triangle inequality $\delta_\square(U, W) \le \delta_\square(U, V) + \delta_\square(V, W)$. The relation*

$$U \sim W \quad\Longleftrightarrow\quad \delta_\square(U, W) = 0 \tag{17.8}$$

is an equivalence relation, called weak isomorphism*, and on the quotient set* $\widetilde{\mathcal{W}}_0 := \mathcal{W}_0/\sim$ *the induced function is a genuine metric. We call* $(\widetilde{\mathcal{W}}_0, \delta_\square)$ *the* metric space of graphons*.*

Proof. Nonnegativity and reflexivity. These are immediate and we dispose of them first. Each competitor $\|U - W^\varphi\|_\square$ is a norm and hence ≥ 0, so their infimum $\delta_\square(U, W) \geq 0$. Taking the single competitor $\varphi = \mathrm{id}$ in eq. (17.7) gives $\delta_\square(W, W) \leq \|W - W\|_\square = 0$, and combined with nonnegativity this forces $\delta_\square(W, W) = 0$. So far nothing delicate has happened; the two substantive axioms are symmetry and the triangle inequality, and both are where the no-attainment subtlety must be navigated.

Symmetry. The definition eq. (17.7) is visibly asymmetric in U and W: it relabels the second argument and leaves the first alone. We must check the choice does not matter. The mechanism is the substitution $\varphi \mapsto \varphi^{-1}$, which is legitimate because the inverse of a measure-preserving bijection is again one (built into definition 17.4), so φ and φ^{-1} range over the same index set and the infimum is unchanged if we reparametrize by the inverse. The content is then a single identity: for every φ,

$$\|U - W^\varphi\|_\square \;=\; \left\|(U - W^\varphi)^{\varphi^{-1}}\right\|_\square \;=\; \left\|U^{\varphi^{-1}} - W\right\|_\square. \tag{17.9}$$

The first equality is relabeling invariance (lemma 17.5) applied to the single kernel $U - W^\varphi$ pulled back by φ^{-1}: pulling a kernel back costs nothing in cut norm. The second equality unwinds the pull-back through the two cocycle facts $(U)^{\varphi^{-1}} = U^{\varphi^{-1}}$ and $(W^\varphi)^{\varphi^{-1}} = W^{\varphi\circ\varphi^{-1}} = W^{\mathrm{id}} = W$ (the cocycle identity eq. (17.10), derived just below, with the relabeling φ and its undoing φ^{-1} cancelling exactly on the second term). Now take the infimum of eq. (17.9) over φ: the left side gives $\delta_\square(U, W)$, while the right side, as φ^{-1} sweeps over all measure-preserving bijections, gives $\inf_\psi \left\|U^\psi - W\right\|_\square$. Renaming the mute variable and noting $\left\|U^\psi - W\right\|_\square = \left\|(U^\psi - W)^{\psi^{-1}}\right\|_\square = \left\|U - W^{\psi^{-1}}\right\|_\square$ by the same invariance, this last infimum is exactly $\delta_\square(W, U)$. Hence $\delta_\square(U, W) = \delta_\square(W, U)$. Note that we never produced an optimal relabeling; we only matched the two index sets and used that pull-back is free, so the absence of an attained minimizer never entered.

Triangle inequality. Here is the point flagged above, and it is the only place the no-attainment subtlety bites: we cannot select optimal φ, ψ, so we work with *near*-optimal ones and lean on composition. Fix a tolerance $\eta > 0$. By the definition of the infimum, there exist measure-preserving bijections φ, ψ that come within η of the two pairwise cut metrics,

$$\|U - V^\varphi\|_\square < \delta_\square(U, V) + \eta, \qquad \left\|V - W^\psi\right\|_\square < \delta_\square(V, W) + \eta,$$

the first relabeling V to align with U, the second relabeling W to align with V. The idea is to chain these two alignments into a single alignment of W to U, routed through V. The chaining is the composition $\psi \circ \varphi$, which is again a measure-preserving bijection of $[0, 1]$ (the composite of two measure-preserving bijections is measure-preserving and bijective), so it is a *legitimate competitor* in the infimum defining $\delta_\square(U, W)$ — not necessarily the optimal one, but admissible, and that is all an infimum requires. To use it we need to know how a composite relabeling acts on a kernel, which is the cocycle identity. Reading definition 17.4 twice,

$$W^{\psi\circ\varphi}(x, y) = W\big((\psi\circ\varphi)(x), (\psi\circ\varphi)(y)\big) = W\big(\psi(\varphi(x)), \psi(\varphi(y))\big) = W^\psi\big(\varphi(x), \varphi(y)\big) = \big(W^\psi\big)^\varphi(x, y), \tag{17.10}$$

where the third equality is the definition of W^ψ evaluated at the inner pair $(\varphi(x), \varphi(y))$ and the fourth is the definition of pulling W^ψ back by φ. So $W^{\psi\circ\varphi} = (W^\psi)^\varphi$: applying ψ then φ to the kernel is the same as relabeling by the composite, with the order of the two reversed in the way pull-backs always reverse. Feeding the competitor $\psi \circ \varphi$ into the infimum and splitting the difference through V^φ with the *cut-norm* triangle inequality (lemma 17.3, the subadditivity proved in the previous section),

$$\delta_\square(U, W) \ \le\ \left\|U - W^{\psi\circ\varphi}\right\|_\square \ =\ \left\|U - (W^\psi)^\varphi\right\|_\square \ \le\ \|U - V^\varphi\|_\square + \left\|V^\varphi - (W^\psi)^\varphi\right\|_\square,$$

where we inserted and subtracted the same intermediate kernel V^φ, the relabeled copy of V. The first summand is already controlled by our choice of φ. The second summand is where invariance earns its keep: both kernels inside it are pulled back by the *same* φ, so their difference is $(V - W^\psi)^\varphi$, and pulling that single kernel back by φ leaves its cut norm unchanged,

$$\left\|V^\varphi - (W^\psi)^\varphi\right\|_\square = \left\|(V - W^\psi)^\varphi\right\|_\square = \left\|V - W^\psi\right\|_\square < \delta_\square(V, W) + \eta$$

by lemma 17.5 and our choice of ψ. In words: the detour through φ that we needed in order to compare U against the composite relabeling costs nothing on the V-to-W leg, because both terms of that leg ride along under the same φ and the cut norm cannot tell. Assembling the two summands,

$$\delta_\square(U, W) \ <\ \big(\delta_\square(U, V) + \eta\big) + \big(\delta_\square(V, W) + \eta\big) \ =\ \delta_\square(U, V) + \delta_\square(V, W) + 2\eta.$$

The left side does not depend on η, so letting $\eta \downarrow 0$ removes the slack and yields the triangle inequality $\delta_\square(U, W) \le \delta_\square(U, V) + \delta_\square(V, W)$. The $\eta \to 0$ limit is exactly the device that lets near-optimal competitors stand in for the unattainable optimum: at every positive η we used only honest, existing bijections, and the limit is taken on a clean numerical inequality after the relabelings have done their work.

From pseudometric to metric on the quotient. With symmetry, reflexivity, nonnegativity, and the triangle inequality established, $\delta_\square$ is a *pseudometric* on $\mathcal{W}_0$ — all the metric axioms except possibly the separation axiom, which can fail because distinct graphons (different labelings of one network) can sit at distance zero. The standard cure is to collapse the zero-distance ties, and we now check it works exactly as the abstract theory of pseudometrics promises. First, the zero-set relation eq. (17.8) is an equivalence relation. Reflexivity is $\delta_\square(W, W) = 0$; symmetry of the relation is symmetry of $\delta_\square$; and transitivity is the triangle inequality, for if $\delta_\square(U, V) = 0$ and $\delta_\square(V, W) = 0$ then $0 \le \delta_\square(U, W) \le 0 + 0 = 0$, forcing $U \sim W$. So $\sim$ partitions $\mathcal{W}_0$ into weak-isomorphism classes, and $\widetilde{\mathcal{W}}_0 = \mathcal{W}_0/\sim$ is the set of those classes.

Two things must be checked to call $\delta_\square$ a metric on $\widetilde{\mathcal{W}}_0$: that it is *well-defined* on classes (independent of which representatives we pick), and that it *separates* distinct classes. Well-definedness is the one terse step the reader should see spelled out, because it is the place the triangle inequality does double duty. Suppose $U \sim U'$ and $W \sim W'$, i.e. $\delta_\square(U, U') = 0$ and $\delta_\square(W, W') = 0$. We claim the two distances $\delta_\square(U, W)$ and $\delta_\square(U', W')$ agree. Applying the triangle inequality twice, hopping first from U to U' and then from W' to W,

$$\delta_\square(U, W) \ \le\ \delta_\square(U, U') + \delta_\square(U', W') + \delta_\square(W', W) \ =\ 0 + \delta_\square(U', W') + 0,$$

and the same chain run in the opposite direction gives $\delta_\square(U',W') \le \delta_\square(U,W)$. Subtracting, both differences are squeezed,

$$\left|\delta_\square(U,W) - \delta_\square(U',W')\right| \;\le\; \delta_\square(U,U') + \delta_\square(W,W') \;=\; 0,$$

so $\delta_\square(U,W) = \delta_\square(U',W')$: the distance depends only on the two classes, not on the chosen representatives, and therefore descends to a function on $\widetilde{\mathcal{W}}_0 \times \widetilde{\mathcal{W}}_0$. Separation is now automatic and is precisely the gain bought by quotienting: if two classes are at distance zero, then any representatives U, W satisfy $\delta_\square(U,W) = 0$, which by the very definition eq. (17.8) of $\sim$ means $U \sim W$, i.e. the two representatives lie in the *same* class. So distance zero on the quotient forces equality of classes — exactly the separation axiom that failed downstairs. The descended $\delta_\square$ inherits symmetry, the triangle inequality, and nonnegativity verbatim from upstairs (they are statements about the numerical values, which are unchanged), and it now also separates, so it is a genuine metric. Thus $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is a metric space. □

It is worth dwelling for a moment on what made the whole proof go through, because the same move will return when we prove completeness. At no point did we exhibit an optimal relabeling; the infimum in eq. (17.7) may have no minimizer, and we never asked it to. Symmetry needed only that pull-back is free and that φ and φ^{-1} index the same set; the triangle inequality needed only that two near-optimal relabelings *compose* into an admissible third one, with the cost of the composite split by the cut-norm triangle inequality and the detour absorbed by invariance. This near-optimal-bijection-plus-composition technique is precisely what the completeness half of the compactness theorem (proposition 17.18) will reuse to assemble a limiting alignment out of a Cauchy sequence of approximate ones. The reader who has followed the $\eta \to 0$ bookkeeping here will find that argument familiar.

There is a tempting alternative route that we deliberately decline, and honesty requires saying so explicitly rather than leaving the reader to wonder whether we missed it.

Remark 17.8 (The coupling reformulation)**.** The proof above establishes everything the chapter needs about the metric structure with bare hands — only invariance and composition, no compactness, no attained optimizer. It is worth knowing that $\delta_\square$ admits two equivalent reformulations, due to Borgs–Chayes–Lovász–Sós–Vesztergombi: enlarging the relabelings from bijections to *measure-preserving maps* (not necessarily invertible), and further to *couplings* π of λ with itself,

$$\delta_\square(U,W) = \inf_{\varphi,\psi} \left\| U^\varphi - W^\psi \right\|_\square = \inf_{\text{couplings } \pi} \sup_{S,T} \left| \int_{S\times T} (U - W)\, \mathrm{d}\pi^{\otimes 2} \right|.$$

The coupling form repairs the very defect we just learned to live with: its infimum *is* attained, because the set of couplings of λ with itself is convex and weak-$*$ compact, so a best coupling always exists where a best bijection may not. One could therefore rebuild the pseudometric proof on the coupling form and never confront the no-attainment subtlety at all. We do not, for two reasons. First, the coupling form costs us the weak-$*$ compactness machinery up front, whereas the bare-hands proof above asks for nothing beyond the previous section. Second, the bijection form is what the rest of the chapter actually manipulates — the total-boundedness argument relabels parts onto subintervals, and the completeness sketch chases near-optimal bijections — so it is the form worth becoming fluent in. The coupling reformulations are what one uses to identify weak isomorphism with measure-preserving isomorphism of the underlying kernels; we do not need them below — the completeness proof of proposition 17.18 uses only near-optimal bijections plus weak-$*$ compactness of the L^∞ ball — and refer the reader to [99, Ch. 8] for the equivalence.

Notation watch

The space of graphons is written $\mathcal{W}_0$ (set) and $\widetilde{\mathcal{W}}_0$ (quotient metric space), with subscript 0. Do not confuse $\mathcal{W}_0 = \mathcal{W}_0$ with the Wasserstein distance $\mathcal{W}_2 = \mathcal{W}_2$ of chapter 4: the letter $\mathcal{W}$ is overloaded in the literature, and we disambiguate purely by the subscript and by context. Points of $\widetilde{\mathcal{W}}_0$ are weak-isomorphism classes of graphons; we nonetheless write a class by any of its representatives W and only invoke the quotient when the distinction matters (as it does for the metric and for compactness).

We have now delivered the central object of the chapter: a genuine metric space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ whose points are intrinsic networks — graphons stripped of their arbitrary labelings — and whose distance is the best alignment in cut norm. This is the arena in which the rest of Part III operates. It is on $(\widetilde{\mathcal{W}}_0, \delta_\square)$, and only after we know it is a metric space with well-defined points, that the word *compact* will finally have something to attach to in section 17.4; the metrization and the quotient built here are exactly the scaffolding the chapter's opening promised, raised so that the keystone can be set. Two threads from this construction run forward explicitly. The equivalence relation of weak isomorphism, $\delta_\square(U, W) = 0$, is not just a technical device for separating the quotient: section 17.3 will show it has a completely observable characterization — two graphons are weakly isomorphic exactly when all their homomorphism densities agree — so the points of $\widetilde{\mathcal{W}}_0$ are precisely what the moment data of chapter 16 can tell apart, no finer and no coarser. And the near-optimal-bijection technique we used to dodge the missing minimizer will resurface intact in the completeness sketch of proposition 17.18, where a limiting alignment is built as a limit of approximate ones.

Before we can profitably study convergence or compactness on this space, though, we owe the reader a structural reading of the distance we have just built. We have a metric, but what does *closeness* in it actually mean about two networks? The cut-norm motivation of section 17.1 already hinted at the answer — block-level interaction is kept, pointwise wiring noise is discarded — and the relabeling quotient has now added a second layer of forgetting on top of it. The next subsection steps back from the construction to make this precise: it asks, of the metric space we now hold, exactly what the cut metric remembers about a graphon and exactly what it throws away.

17.2.3 Coarse-graining: what the cut metric remembers and forgets

We now hold a genuine metric space $(\widetilde{\mathcal{W}}_0, \delta_\square)$, and the previous subsection left us with a pointed question: when this metric calls two networks close, what is it actually asserting about them? The construction was driven by two acts of forgetting, stacked one on the other. The cut norm of section 17.1 was engineered to discard pointwise wiring noise while retaining block-level interaction — that was the whole lesson of the noisy-versus-smooth example. Then the relabeling quotient of section 17.2 added a second layer, forgetting which name each type happens to wear. The purpose of this short subsection is to fuse those two readings into a single structural picture of the distance, a picture that is worth installing in the reader's mind *before* the regularity lemma of section 17.4 makes it a theorem, because once one sees what closeness in $\delta_\square$ means it is no surprise at all that every graphon admits a low-complexity caricature. The slogan is that the cut metric is a *coarse-graining*: it remembers a graphon's mesoscopic, block-averaged structure and forgets everything finer.

Make the slogan concrete by spelling out the operation it names. Take a graphon W and a finite partition $\mathcal{P} = \{P_1, \ldots, P_k\}$ of the type interval $[0,1]$ into measurable parts — think of each part as a *community* of types, a cluster of agents we have decided to treat as interchangeable. Over each pair of parts $P_i \times P_j$ replace W by its average value there, the single number $\frac{1}{\lambda(P_i)\lambda(P_j)} \int_{P_i} \int_{P_j} W$ recording how densely community i interacts with community j on the whole. The result is a step graphon $W_{\mathcal{P}}$, a block-constant caricature carrying only k^2 numbers no matter how intricate W was to begin with. This block average is exactly the conditional expectation $\mathbb{E}[W \mid \mathcal{P} \otimes \mathcal{P}]$ that section 17.4 will build the regularity lemma on; here we need only its meaning, not yet its machinery. The coarse-graining claim is that, for a sufficiently fine choice of $\mathcal{P}$, this caricature is cut-close to the original: $\|W - W_{\mathcal{P}}\|_\square$ is small. Within each community-pair the fine-grained fluctuations of W around its block average are precisely the sign-changing, patch-cancelling structure the cut norm was designed not to see, so replacing W by the locally constant $W_{\mathcal{P}}$ costs almost nothing in cut norm even though it can cost a great deal in any L^p norm. That a single bounded number of blocks always suffices, uniformly over all graphons, is the regularity lemma; the present picture is its intuition installed in advance.

Run this reading back through the metric and one obtains a clean characterization of closeness. Two graphons are $\delta_\square$-close exactly when, after the best relabeling has been chosen to align them, their block averages agree to high accuracy across *every* sufficiently coarse partition simultaneously. The two quantifiers each carry one of the two acts of forgetting. The "best relabeling" is the cut metric's infimum over measure-preserving bijections from section 17.2: it discharges the arbitrariness of the labels, so that we compare the networks community-to-community rather than name-to-name. The "every sufficiently coarse partition" is the cut norm's supremum over patches from section 17.1, now read in block form: it demands that the two networks not merely agree on one favoured decomposition into communities but on all of them at once, so that no choice of how to cluster the types reveals a discrepancy. What this joint condition can detect and what it cannot is then sharply delimited. Microscopic wiring — which individual pair of types happens to be joined, the coin-flip randomness of the heuristic example — is invisible, because it is averaged away inside the blocks and contributes nothing to any block average. Mesoscopic structure — how densely whole communities of types interact, the block densities themselves — is exactly what survives, because it *is* the block averages, and two networks with genuinely different community interactions will differ on the partition aligned to those communities and so cannot be made $\delta_\square$-close by any relabeling. The cut metric forgets the wiring and remembers the community structure, and the phrase "every sufficiently coarse partition" is the precise form of that promise.

Physics note

The cut metric formalizes a move the condensed-matter reader has made many times: replacing a microscopic configuration by a block-averaged (coarse-grained) one and asking only that the coarse-grained descriptions agree. Two microscopically distinct spin configurations that flow to the same block-spin field under a Kadanoff decimation are, for the purpose of long-wavelength observables, the same configuration; the equivalence classes are the orbits of the coarse-graining map, just as the block-density caricature $W_{\mathcal{P}}$ stands in for W here. Weak isomorphism — $\delta_\square$-distance zero — is the graphon analogue of that identification: two kernels are declared the same precisely when no amount of block-averaging, under any relabeling of types, can tell them apart, exactly as two configurations are physically identical when no coarse-grained observable separates them. One caveat keeps the analogy honest, and it is not a quibble. The renormalization group *iterates* a fixed-ratio rescaling, generating a flow on coupling space along which one can ask which directions are relevant and which irrelevant; the cut metric does no such thing. It performs its averaging *once and at all scales simultaneously*, quantifying over every partition in a single supremum rather than coarse-graining step by step. There is consequently no flow, no fixed point, and no relevant/irrelevant dichotomy in the cut metric — the shared content is the equivalence-by-coarse-graining alone, and the analogy carries no mathematical weight beyond that intuition.

With this picture in place the next section's machinery should feel inevitable rather than imposed. We have asserted that a bounded number of blocks always suffices to approximate a graphon in cut norm; turning that assertion into a theorem is the content of the weak regularity lemma, and the picture above already names its three moving parts. The block averages are the step kernels $W_{\mathcal{P}}$; the measure of how much structure a partition has captured will be an *energy*, the squared L^2 norm of the block average; and the mechanism that drives the argument is *refinement* — whenever a partition's caricature is still cut-far from the graphon, one splits its parts to capture more energy, and because energy is bounded the splitting must stop. The phrase "every sufficiently coarse partition" that we used to characterize closeness is precisely the foothold that energy-increment mechanism exploits: a surviving cut-norm discrepancy is exactly a partition on which the block averages have not yet agreed, and that is the signal to refine. The worked computation of section 17.4 makes all of this fully explicit on the carried thread graphons, exhibiting the block approximation of a rank-one and a two-block kernel and watching its cut distance to the original shrink as the partition is refined. We turn first, however, to a prior question that the metric now lets us settle cleanly: whether convergence in this cut metric is the very same notion as the left-convergence of chapter 16, closing the loop the chapter's opening promised.

17.3 Cut convergence is left-convergence

The previous two sections did the constructive work this one cashes in. We now own a genuine metric space $(\widetilde{\mathcal{W}}_0, \delta_\square)$: its points are intrinsic networks — graphons read modulo the arbitrary labeling of types — and the cut metric assigns each pair of them a single number that, by design, registers block-averaged interaction and forgets microscopic wiring. The chapter's opening promised that this distance would be "right" in three reinforcing senses, and named the first of them as the keystone the next two sections raise scaffolding for: the cut metric should *metrize left-convergence*, so that the geometry built here and the topology imported from chapter 16 are not two competing notions of nearness but literally one. This section delivers that first sense in full. It is the moment the loop the preamble drew finally closes — where the homomorphism densities of chapter 16, which told us only *whether* a sequence settles down, are shown to agree exactly with the cut metric, which tells us *how far* two networks stand apart.

Before we can even ask whether the two notions coincide, we have to make them speak about the same objects, and here there is a vocabulary mismatch to reconcile. In chapter 16 left-convergence was defined for sequences of *finite simple graphs* (G_n) with $|V(G_n)| \to \infty$ (definition 16.11): the sequence converges when, for every fixed pattern F, the frequency $t(F, G_n)$ of F inside G_n settles to a limit. The cut metric, by contrast, lives on graphons. To compare them we extend left-convergence to graphons in the only natural way: a sequence (W_n) in $\mathcal{W}_0$ is *left-convergent* when the density $t(F, W_n)$ converges for every finite simple graph F. This is not a new idea grafted on but the same definition read on the continuum — the densities are the same observables, now evaluated on kernels rather than adjacency matrices — and chapter 16 already took the first step of this extension when it declared a graphon sequence convergent by exactly this clause.

The bridge that fuses the two readings into one is the identification of a finite graph with its step graphon, and because every application to growing networks routes through it, it is worth restating with care rather than waving past. Given a finite simple graph G on n vertices, cut $[0, 1]$ into n equal subintervals $I_1, \ldots, I_n$, one per vertex, and let the *step graphon* W^G be the symmetric kernel that takes the value 1 on the tile $I_i \times I_j$ when ij is an edge of G and 0 otherwise — the adjacency matrix of G "pixelated" into a function on $[0, 1]^2$. Proposition 16.9 established the one fact we need about this construction: it preserves *every* density, $t(F, W^G) = t(F, G)$ for all F, because integrating a product of kernel values over the uniform types reproduces, block by block, the finite average that counts copies of F in G. The combinatorial frequency and the analytic integral are the same number. This is what lets us treat a finite graph and its step graphon as interchangeable: any statement about the densities of one is verbatim a statement about the densities of the other, and we make that identification silently from here on, writing G where we mean W^G whenever a network is fed to a graphon-level construction.

The payoff of the identification is precisely the form chapter 25 will consume, so it is worth spelling out concretely. A sequence of growing finite networks $G_1, G_2, \ldots$ — the adjacency structures of larger and larger interacting populations — becomes, through $G_n \mapsto W^{G_n}$, a sequence of graphons sitting inside $\mathcal{W}_0$, with no loss of any observable information: $t(F, W^{G_n}) = t(F, G_n)$ for every F and every n. Consequently the finite networks left-converge in the sense of definition 16.11 if and only if the associated step graphons left-converge in the graphon sense just defined, and — once the theorem of this section is in hand — if and only if those step graphons converge in the cut metric to some limiting graphon W. That last clause is the one Part III leans on: it says a single graphon can stand in, in the precise quantitative sense of cut distance, for an entire convergent sequence of finite networks of unbounded size. The reader should keep this finite-to-continuum translation in view, because every later invocation of cut convergence for "a growing network" is shorthand for cut convergence of its step graphons.

With both notions now phrased for graphons, the remaining definition is the obvious one. A sequence (W_n) in $\mathcal{W}_0$ *converges in the cut metric* to W when $\delta_\square(W_n, W) \to 0$: the relabeled cut distance, minimized over alignments of the type coordinate, shrinks to zero. The central theorem of dense graph-limit theory — and the first of this chapter's three senses — is that this metric convergence and left-convergence are the very same notion (theorem 17.12). The proof splits cleanly along the difficulty of its two implications, and the rest of the section follows that split. One direction is easy: *cut convergence forces the densities to converge*, and it falls straight out of the counting lemma of chapter 16, which bounds each density difference by the cut distance. The converse is the deep half: *convergence of all densities forces cut convergence*. It cannot come from the counting lemma, which controls densities by cut distance and never the reverse; it needs two genuinely new ingredients, the *compactness* of $(\widetilde{\mathcal{W}}_0, \delta_\square)$ to manufacture a convergent subsequence out of thin air, and a *moment-uniqueness* principle to pin that subsequence's limit to the intended target. The reliance on compactness here is not incidental — it is exactly why the next section must prove it, and why this section forward-references theorem 17.13 as an input rather than supplying it now. We take the tractable half first.

17.3.1 The easy direction: from cut distance to densities

We take up the split announced at the close of the section preamble and dispatch the tractable half first, both because it is genuinely a corollary and because the very inequality that makes it easy will reveal, by what it does *not* say, why the converse must be hard. The claim is the forward implication of the equivalence: *cut convergence forces the densities to converge*. What makes it forward and not converse is the direction of a single inequality we already own in chapter 16. The cut metric was engineered to register exactly the block-averaged interaction that homomorphism densities read off; it should therefore be no surprise that controlling the cut distance controls every density. The instrument that turns "should be no surprise" into a one-line proof is the counting lemma, and the entire content of this subsection is to import that lemma honestly, read off what its Lipschitz constant means, and then watch the corollary fall out.

The counting lemma is the bridge of chapter 16, and we restate it here so the argument that consumes it is self-contained on the page, while being explicit that the proof lives there and is not reproduced. In its cut-norm form it says that homomorphism densities are Lipschitz functions of the kernel in the cut norm, with a Lipschitz constant that is not some abstract universal but the concrete combinatorial size of the test pattern: its number of edges.

Lemma 17.9 (Counting lemma; recall lemma 16.17)**.** *For every finite simple graph F and all graphons U, W,*

$$|t(F,U) - t(F,W)| \;\le\; e(F)\;\|U - W\|_\square\,, \tag{17.11}$$

where $e(F) = |E(F)|$ is the number of edges of F (the notation of chapter 16, restated here in the more compact $e(F)$). Consequently, by lemma 17.5 and the infimum over relabelings, $|t(F,U) - t(F,W)| \le e(F)\,\delta_\square(U,W)$.

The constant $e(F)$ is worth dwelling on, because it is exactly what one should expect and its expectedness is reassuring. The density $t(F,U)$ is a multilinear integral in which the kernel U appears once for each edge of F: replacing U by W everywhere is a substitution in $e(F)$ slots at once, and if a single slot's worth of substitution costs at most $\|U - W\|_\square$, then $e(F)$ slots cost at most $e(F)$ times that. For the simplest patterns this reads off instantly. When F is a single edge, $e(F) = 1$ and the density is the edge density $t(F, U) = \int_{[0,1]^2} U$; the lemma then says edge densities differ by at most $\|U - W\|_\square$, which is barely more than the definition of the cut norm specialized to $S = T = [0, 1]$. When F is a triangle, $e(F) = 3$ and triangle densities are 3-Lipschitz; for the complete graph K_4 the constant climbs to 6. The constant grows with the pattern's complexity — richer observables are more sensitive — but it never depends on U or W, which is precisely the uniformity that lets us push a single cut-distance bound through all patterns at once.

We import lemma 17.9 as a black box, and self-containedness obliges us to point at the in-book proof rather than wave at the literature: it is exactly the cut-norm counting lemma established as lemma 16.17, restated here only under a local number for ease of reference. The argument there is a *telescoping edge-by-edge swap*, and it is worth recalling its shape in one sentence so the reader trusts the black box for the right reason. One writes the difference $t(F, U) - t(F, W)$ as a sum of $e(F)$ intermediate differences, in each of which a single copy of U is replaced by W while the other edge-factors are held fixed; each such single-slot difference is an integral that is linear in the one swapped kernel $U - W$, with the remaining factors acting as a $[0, 1]$-valued test weight, so it is bounded by one application of the cut norm (in the test-function form eq. (17.3)); summing the $e(F)$ telescoping steps and applying the triangle inequality yields the factor $e(F)$. That is the whole mechanism, and chapter 16 carries it out in full; we do not reprove it here.

The second sentence of the lemma is the passage from the cut *norm* to the cut *metric*, and although it is short it is the step that makes the whole import usable on $\widetilde{\mathcal{W}}_0$, so we spell it out rather than let it pass as "consequently." The bound eq. (17.11) is stated for labeled kernels and their literal difference $U - W$; the cut metric, by contrast, is the infimum $\delta_\square(U, W) = \inf_\varphi \|U - W^\varphi\|_\square$ over all relabelings φ (definition 17.6). To bridge them we use that the densities themselves are relabeling-invariant: by lemma 17.5, $t(F, W^\varphi) = t(F, W)$ for every measure-preserving φ, because a relabeling renames the types without changing any integral. Hence for *every* φ we may replace W by its relabeling on the right of eq. (17.11) without touching the left:

$$|t(F, U) - t(F, W)| = |t(F, U) - t(F, W^\varphi)| \;\le\; e(F)\; \|U - W^\varphi\|_\square .$$

The left side does not depend on φ, so it is bounded by the infimum of the right side over all φ, which is exactly $e(F)\, \delta_\square(U, W)$. This is why the lemma's constant survives the quotient intact: the densities cannot tell relabelings apart, so they are controlled by the best alignment, not merely the literal one. With that in hand the corollary is immediate.

Proposition 17.10 (Cut convergence implies left-convergence)**.** *If $\delta_\square(W_n, W) \to 0$ then (W_n) is left-convergent, with $t(F, W_n) \to t(F, W)$ for every finite simple graph F.*

Proof. By lemma 17.9, $|t(F, W_n) - t(F, W)| \le e(F)\, \delta_\square(W_n, W) \to 0$. □

One concrete instance closes the loop with the example that opened the chapter. For the noisy networks W_N of section 17.1, built by coin-flips at density $\frac{1}{2}$, the cut distance to the smooth limit $W \equiv \frac{1}{2}$ tends to zero even as every L^p distance stays pinned; proposition 17.10 now certifies what we asserted there by hand, that every density $t(F, W_N)$ converges — here to $(\frac{1}{2})^{e(F)}$ — with the approach governed quantitatively by $e(F)\,\delta_\square(W_N, W)$. The macroscopic observables follow the cut metric, exactly as designed.

This establishes the implication (ii)⇒(i) of the equivalence — cut convergence implies left-convergence — and it is more than a tidy first half. The subsequence argument of the hard direction will invoke this very proposition *twice*: once to verify that the convergent subsequence compactness hands it actually carries the right densities, and again to confirm that the limit it lands on is the one the densities demanded. It is the workhorse the deep direction stands on, so it is well that it costs only the counting lemma.

But the proposition's proof also exposes, in the plainest possible way, why the deep direction cannot be had by the same means. Everything here flowed from the single inequality eq. (17.11), and that inequality is resolutely one-directional: it bounds density differences *by* cut distance, never cut distance by density differences. It says small cut distance forces matching densities, but it is silent on whether matching densities force small cut distance — and there is no companion inequality in the other direction to invoke, because none can be elementary in the way this one is. To go from infinitely many scalar coincidences (all the densities agree in the limit) back to a single geometric fact (the cut distances shrink) is to reconstruct a point of the metric space from its complete list of moments, and a list of numbers does not, on its own, manufacture a limit in a space. What is missing is precisely a guarantee that the space is not too large to be pinned down this way — a guarantee that any sequence has somewhere to converge, so that the matching moments have a target to select. That guarantee is *compactness*, and the next subsection supplies it as the engine of the converse: compactness manufactures a convergent subsequence out of the bounded sequence, and moment-uniqueness pins that subsequence's limit to W. The one-directionality we have just met is exactly the gap that compactness is built to close.

17.3.2 The hard direction and the equivalence theorem

The easy direction closed by naming, with some precision, the exact thing it could not do: the counting lemma bounds density differences *by* cut distance and is silent in the other direction, so nothing it offers turns the infinitely many scalar coincidences of matching densities back into a single geometric statement about distance. The converse we now face — that convergence of all homomorphism densities forces convergence in the cut metric — is therefore the deep half, and it must be bought with machinery the counting lemma cannot supply. The shape of the purchase was already visible at the end of the last subsection: what is missing is a guarantee that the space is not too large to be pinned down by its moments, so that a sequence whose densities settle has somewhere to go and only one place it can go. Two imported facts deliver exactly this, and it is worth naming the job each one does before we use it, because the proof will then read as nothing more than snapping the two together.

The first is *compactness* of the metric space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ (theorem 17.13, proved in the next section). Its role is existential: out of a bare sequence of graphons, with no candidate limit in sight, it manufactures a subsequence that actually converges in the cut metric to *some* graphon. This is the ingredient that supplies a target where the moments alone supply none — compactness produces a point of the space for the matching densities to select. The second is a *moment-uniqueness* principle:

the homomorphism densities determine a graphon completely, up to weak isomorphism, so that two graphons with identical densities are the same point of $\widetilde{\mathcal{W}}_0$. Its role is to *pin*: once compactness has handed us a convergent subsequence, moment-uniqueness forbids its limit from being anything other than the intended W. Compactness manufactures a limit; moment-uniqueness identifies it. Neither alone suffices — compactness without uniqueness would leave the limit ambiguous, uniqueness without compactness would leave no limit to identify — and the entire deep direction is the observation that, run together, they close the gap the counting lemma left open. We state the uniqueness principle first, since it is the ingredient not yet on the table, and then assemble the equivalence.

Theorem 17.11 (Moments determine the graphon)**.** *Two graphons U, W satisfy $t(F,U) = t(F,W)$ for every finite simple graph F if and only if $\delta_\square(U,W) = 0$, i.e. U and W are weakly isomorphic.*

Read the theorem as the assertion that the homomorphism densities are a *complete* set of invariants for a graphon: they separate the points of $\widetilde{\mathcal{W}}_0$ exactly, no two distinct networks sharing the same density table. The two directions are wildly unequal in depth, and it is worth being honest about which is which. The "if" direction — weak isomorphism implies equal densities — is no new work at all: it is proposition 17.10 applied to the *constant* sequence $W_n \equiv U$. For if $\delta_\square(U,W) = 0$ then trivially $\delta_\square(U,W) \to 0$, and the easy direction immediately yields $t(F,U) = t(F,W)$ for every F. The densities cannot tell weakly isomorphic graphons apart because they are relabeling-invariant, a fact we have owned since lemma 17.5. All the content sits in the "only if": that matching *all* subgraph densities forces the cut distance to be exactly zero. This is the graphon analogue of a classical *moment problem* — reconstructing a distribution from its complete list of moments — and the name is precise rather than decorative, as the mechanism makes plain.

That mechanism, which appendix F carries out in full and we sketch here only so the reader trusts the deferral for the right reason, runs through sampling — but through the *Bernoulli* sample, not a weighted one, and the distinction is the whole subtlety. Draw k types $x_1, \ldots, x_k$ independently and uniformly from $[0,1]$ and form the random *simple* graph $G(k,U)$ on k vertices that includes each edge $\{i,j\}$ independently with probability $U(x_i, x_j)$. The point is that the law of this $\{0,1\}$-valued graph is pinned down by the simple-graph densities. For a fixed target graph H on $[k]$, conditioning on the types gives $\mathbb{P}(G(k,U){=}H) = \mathbb{E} \prod_{\{i,j\} \in E(H)} U \prod_{\{i,j\} \notin E(H)} (1-U)$, and expanding the non-edge factors leaves each edge weight to the *first power only* — the edge indicator Z is $\{0,1\}$-valued, so $Z^2 = Z$ and no higher power can arise. Every term is thus a *multilinear* expected product $\mathbb{E} \prod_{ij \in E(F)} U(x_i, x_j) = t(F,U)$ over some simple graph F, so $\mathbb{P}(G(k,U){=}H)$ is a fixed finite linear combination of the densities $t(F,U)$ (the inclusion–exclusion passage from homomorphism to induced-subgraph counts). Equal densities for all F therefore force $G(k,U)$ and $G(k,W)$ to be *equal in law* for every k. It is essential that we run this through the Bernoulli sample: a simple graph F contributes only the multilinear moment $\mathbb{E} \prod_{ij \in E(F)} U(x_i, x_j)$, each edge to the first power, so the densities do *not* sweep out the genuinely higher moments such as $\mathbb{E}\, U(x_1, x_2)^2$ — those are *multigraph* (parallel-edge) densities, outside the simple-graph table — and hence do *not* determine the law of the weighted graph. The $\{0,1\}$ identity $Z^2 = Z$ is exactly what removes the higher moments and lets the densities fix the Bernoulli law outright. The bridge from this distributional equality back to the cut metric is the *second (Bernoulli) sampling lemma* (theorem F.9): a Bernoulli k-sample of a graphon is, with high probability, within cut distance $O(1/\sqrt{\log k})$ of the graphon it was drawn from, so $\mathbb{E}[\delta_\square(\cdot)] \to 0$ as $k \to \infty$. Pinching U and W between their common Bernoulli sample law gives $\delta_\square(U, W) \le \mathbb{E}[\delta_\square(U, U_{[k]})] + \mathbb{E}[\delta_\square(W_{[k]}, W)]$, where the middle term has been swapped to W using the equal sample laws; both terms are $O(1/\sqrt{\log k}) \to 0$, forcing $\delta_\square(U, W) = 0$. This is the theorem of Borgs–Chayes–Lovász; we defer its two quantitative inputs — the $\{0, 1\}$ determinacy of the Bernoulli sample law and the second sampling lemma — to appendix F (theorem F.11), flag it as one of the chapter's deep imports, and record it in the closing ledger as a deferral, with attribution [28, 99].

Theorem 17.12 (Cut convergence $\Leftrightarrow$ left-convergence)**.** *For graphons $W_n, W \in \mathcal{W}_0$ the following are equivalent:*

(i) $t(F, W_n) \to t(F, W)$ for every finite simple graph F (left-convergence);

(ii) $\delta_\square(W_n, W) \to 0$ (cut convergence).

Moreover every left-convergent sequence of graphons has a limit graphon: if all $t(F, W_n)$ converge, there exists $W \in \mathcal{W}_0$ with $\delta_\square(W_n, W) \to 0$.

Proof. The direction (ii) $\Rightarrow$ (i) is already in hand: it is precisely proposition 17.10, the easy direction. All the work is in (i) $\Rightarrow$ (ii).

Before the argument proper we isolate the topological principle it runs on, since it may be unfamiliar and it is doing the real organizing. *The subsequence principle.* In any metric space, a sequence (x_n) converges to a point x if and only if every subsequence of (x_n) has a *further* subsequence that converges to x. One direction is trivial: if the whole sequence converges to x, so does every sub-subsequence. The useful direction is the converse, and its contrapositive makes it transparent. If (x_n) did *not* converge to x, then some $\epsilon > 0$ is escaped infinitely often — there is a subsequence with $d(x_{n_k}, x) \ge \epsilon$ for all k — and that subsequence can have no further subsequence converging to x, since all of its terms sit a fixed distance ϵ away. So if every subsequence *does* admit a further subsequence converging to x, no such escaping subsequence can exist, and therefore $(x_n) \to x$. The principle is valuable in exactly the situation we are in: one cannot control the whole sequence directly, but one can always *extract* convergence from any piece of it — which is precisely the leverage compactness provides.

Now the argument. Assume $t(F, W_n) \to t(F, W)$ for every finite simple graph F; we must show $\delta_\square(W_n, W) \to 0$. By the subsequence principle it suffices to show that an *arbitrary* subsequence of (W_n) admits a further subsequence cut-converging to W. So fix any subsequence. Because $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is compact (theorem 17.13, proved in the next section and — we stress — proved with no appeal whatever to the present theorem), this subsequence, like every sequence in a compact metric space, has a convergent sub-subsequence: there are a graphon U and an extraction (W_{n_j}) with $\delta_\square(W_{n_j}, U) \to 0$. This U is the target compactness manufactures, and at this stage we have no idea what it is. To identify it, we play its densities off against the hypothesis from two sides. On the one hand, the easy direction (ii) $\Rightarrow$ (i) applied to the convergent sub-subsequence gives $t(F, W_{n_j}) \to t(F, U)$ for every F — the first of the two promised uses of proposition 17.10. On the other hand, (W_{n_j}) is a subsequence of the original sequence, along which the hypothesis already forces $t(F, W_{n_j}) \to t(F, W)$. A sequence of real numbers has at most one limit, so the two coincide: $t(F, U) = t(F, W)$ for every F. Now moment-uniqueness (theorem 17.11) fires: matching all densities forces $\delta_\square(U, W) = 0$, that is, U *is* W as a point of $\widetilde{\mathcal{W}}_0$. Substituting this back, $\delta_\square(W_{n_j}, W) = \delta_\square(W_{n_j}, U) \to 0$: the extracted sub-subsequence cut-converges to W itself. The initial subsequence was arbitrary, so every subsequence has a further subsequence cut-converging to W, and the subsequence principle delivers $\delta_\square(W_n, W) \to 0$. This is (i) $\Rightarrow$ (ii).

It remains to prove the "moreover": that a left-convergent sequence has a limit graphon even when *no* candidate W is named in advance. This is the same argument run one notch earlier, and it is worth seeing that the candidate limit was never essential — compactness supplies it. Suppose only that $t(F, W_n) \to \ell_F$ for each F, the limits ℓ_F being some prescribed real numbers with no graphon attached to them. By compactness (theorem 17.13) extract a single subsequence (W_{n_j}) cut-converging to some graphon $W \in \mathcal{W}_0$. By the easy direction once more — the second promised use — its densities are the limits along that subsequence, $t(F, W) = \lim_j t(F, W_{n_j}) = \ell_F$ for every F. Thus the manufactured W *realizes* the prescribed density table ℓ_F: the abstract list of numbers has been incarnated as an actual point of the space. With this concrete W now in hand as a target, the equivalence just proved applies verbatim — since $t(F, W_n) \to \ell_F = t(F, W)$ for all F, part (i) $\Rightarrow$ (ii) gives $\delta_\square(W_n, W) \to 0$, and the *whole* sequence, not merely the extracted subsequence, converges to W in the cut metric. We emphasize what was *not* used. At no point did we need the density numbers ℓ_F to form a Cauchy or otherwise well-behaved family beyond their mere convergence, and at no point did we need a limit handed to us in advance. Moment-uniqueness alone does the pinning, converting "the densities settle" into "the graphons converge." □

Two remarks fix the logical dependencies, and both matter for how the rest of the chapter is read.

The first concerns circularity, and it is essential enough to state flatly. The proof of the equivalence *uses* the compactness theorem of the next section — twice, once in each half of the argument — but the compactness theorem does *not* use the equivalence. When we come to prove theorem 17.13 in section 17.4, the argument will run entirely inside the cut metric: total boundedness through the weak regularity lemma's energy increments, and completeness through a weak-$*$ extraction, neither of which so much as mentions homomorphism densities or the present theorem. There is thus no hidden loop in which compactness is invoked to prove the equivalence while secretly leaning on it. The dependency is strictly one-way, compactness $\Rightarrow$ equivalence, and the reader may treat the forward reference to theorem 17.13 as an honest IOU rather than a circular plea.

The second tallies the genuinely deep inputs, and they are exactly three. The counting lemma (chapter 16) supplied the easy direction; moment-uniqueness (theorem 17.11, whose hard half is the graphon moment problem deferred to appendix F) supplied the pinning; and compactness (theorem 17.13) supplied the limit to be pinned. Everything else in this section — the subsequence principle, the uniqueness of limits of real sequences, the playing-off of two density computations against each other — is soft point-set topology that the reader could reconstruct unaided. The depth of the equivalence is borrowed entirely from those three results, and two of the three are already paid for: the counting lemma was proved in chapter 16, and moment-uniqueness is charged honestly to appendix F, where this chapter's ledger will record it as a deferral.

That leaves precisely one of the three still unpaid, and it is the load-bearing one. The entire converse — and with it the first of the chapter's three senses, that cut convergence and left-convergence are one and the same notion — rests on the compactness of $(\widetilde{\mathcal{W}}_0, \delta_\square)$, which so far we have only *invoked*. We have used it to manufacture convergent subsequences out of thin air and never once justified that it holds. A forward reference is a promissory note, and this note now falls due: the next section must earn compactness outright. It is also, and not by coincidence, the chapter's announced destination — the structural keystone of Part III, the fact that lets a single limit graphon stand in for an entire growing sequence of networks. We turn to paying for it now.

17.4 Compactness of graphon space

We arrive at the structural keystone of Part III. The previous section left compactness of $(\widetilde{\mathcal{W}}_0, \delta_\square)$ as the one promissory note still unpaid — it was invoked twice to manufacture convergent subsequences and never once justified — and that note now falls due. But compactness is far more than a lemma owed to an earlier proof. It is the destination the chapter's preamble promised from the first page: the single structural fact that the rest of the book actually consumes. When a later existence proof must extract a convergent subsequence of approximate equilibria, or pass an equilibrium to a limit network, it is reaching for exactly this property; it is what licenses a single limit graphon to stand in for an entire convergent sequence of growing networks, and so it is the input from Part III that makes the existence theorem of chapter 24 and the finite-network convergence theorem of chapter 25 go through. Paying the debt and reaching the destination are, it turns out, the same act. We perform it now.

Theorem 17.13 (Lovász–Szegedy compactness)**.** *The metric space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ of graphons modulo weak isomorphism is compact. Equivalently, every sequence of graphons has a subsequence that converges in the cut metric to a graphon.*

The two formulations are interchangeable because $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is a metric space, where compactness, sequential compactness, and "every sequence has a convergent subsequence" all coincide; we proved in section 17.2 that we are working in a genuine metric space, so we may use whichever face of compactness is convenient and will mostly use the sequential one. The reader should pause on how strong the claim is. The space $\mathcal{W}_0$ of all graphons is enormous — it contains every measurable symmetric kernel $[0,1]^2 \to [0,1]$, an infinite-dimensional set with no finiteness anywhere in sight — and yet the theorem asserts that after we quotient by relabelings and measure distance with $\delta_\square$, this vast set becomes as tame as a closed bounded interval. No analogue holds for any of the L^p metrics of section 17.1: the noisy-network sequence W_N of the chapter's opening has *no* L^1-convergent subsequence (any two of its members stay a fixed positive L^1 distance apart, so it is uniformly discrete), and a uniformly discrete infinite set is the furthest thing from compact. The coarsening from $\|\cdot\|_1$ to $\|\cdot\|_\square$ is precisely what collapses that discreteness: in the cut metric the same sequence converges to the constant graphon $\frac{1}{2}$, and the theorem says this rescue happens for *every* sequence, with no exception. Compactness is thus the deepest dividend of the coarseness we engineered into the cut norm; the first three sections were the investment, and this is the return.

How does one prove a space compact when its points are infinite-dimensional objects and no coordinates are available? Through the standard structural characterization of compactness in metric spaces, which trades the topological condition for two concrete metric conditions that can be checked by hand.

Proposition 17.14 (Metric compactness dichotomy)**.** *A metric space is compact if and only if it is* complete *and* totally bounded*.*

This is a standard fact of point-set topology — it is part of the prerequisite toolkit, and we use it rather than reprove it — but it is worth recalling what its two halves demand, because the entire remaining labor of the chapter is the verification of one of them in full and an honest sketch of the other. *Total boundedness* is a finiteness-of-resolution condition: for every tolerance $\epsilon > 0$ the whole space can be covered by *finitely many* balls of radius ϵ. It says the space has no infinite supply of mutually ϵ-separated points — that at any fixed resolution there are only finitely many genuinely distinct graphons. This is the half that fails so spectacularly for L^1 and must succeed for $\delta_\square$, and it is the half that carries all the real mathematics. *Completeness* is the no-holes condition: every Cauchy sequence actually converges to a point of the space, so that the limits total boundedness lets us approach are genuinely present and are themselves graphons rather than escaping to some larger ambient set. Total boundedness manufactures candidate limits; completeness guarantees they live in the space. Both are needed, and proposition 17.14 says they are exactly enough.

Here we make the rigor commitment of the section explicit, because the self-containedness doctrine of this book requires us to say plainly which arguments we carry out in full and which we sketch, and never to dress a sketch as a proof. *The total-boundedness half we prove completely.* Its engine is the *weak regularity lemma* — the statement that every graphon is approximated in cut norm, to any prescribed accuracy, by a step kernel with a bounded number of blocks depending only on the accuracy and not on the graphon — and that lemma, with its beautiful energy-increment argument, is short and illuminating enough that withholding it would be a disservice; we give it whole, and total boundedness then follows by counting the finitely many step kernels at a fixed resolution. *The completeness half we sketch honestly.* We will exhibit the correct mechanism — a weak-$*$ compactness extraction in $L^\infty([0,1]^2)$, powered by the Banach–Alaoglu theorem imported from chapter 2, producing a candidate limit kernel — and explain why it yields a Cauchy sequence's limit; but the quantitative heart, that the weak-$*$ limit is approached in the *cut* metric and not merely weak-$*$, rests on a sampling/martingale estimate whose full execution we defer to appendix F. The chapter's ledger will record that deferral by name, alongside the moment-uniqueness deferral the previous section already charged to appendix F. The reader should hold the asymmetry firmly in mind: total boundedness, the half that does the work and supplies the regularity lemma later chapters reuse, is self-contained here; completeness, the more routine functional-analytic half, is the one leaning on a deferred estimate.

The work ahead therefore organizes itself into a short sequence of subsections, each preparing the next. We begin by building the machinery the regularity lemma runs on: the *block average* of a kernel relative to a finite partition — which we will recognize as a conditional expectation onto the partition's product σ-algebra, cashing in the coarse-graining picture of section 17.2 — and the *energy* of a partition, the monotone scalar quantity whose bounded ascent will drive the whole argument.

With that machinery in place we state and prove the *weak regularity lemma* by the energy-increment mechanism: if a step kernel is still far from U in cut norm, the cut witness exposes a refinement that strictly increases the energy by a fixed amount, and since the energy is bounded the refinement can only happen boundedly often. From the lemma we read off *total boundedness* by a finite count of step kernels at a fixed scale. We then dispatch *completeness* by the weak-$*$ sketch and assemble the two halves, through proposition 17.14, into the compactness theorem. A subsection on *why this is the keystone* draws the line from the bare topological fact to its consumption in Part III, and a closing *worked computation* on the rank-one and block graphons of the carried thread makes the regularity approximation concrete by exhibiting, on an explicit kernel, the finite step kernel that approximates it in cut norm. We take the first step now, making block averaging precise.

17.4.1 Block averages and the energy of a partition

Notation watch

In this section and the next, $\mathcal{P}$ and $\mathcal{Q}$ denote finite *partitions* of the type interval $[0,1]$, not probability measures: the calligraphic $\mathcal{P}$ that elsewhere in the book (appendix A) names a space of probability measures is locally repurposed for a partition, and no probability-measure $\mathcal{P}$ appears in this chapter to collide with it.

Fix a measurable partition $\mathcal{P} = \{P_1, \dots, P_k\}$ of $[0,1]$ into parts of positive measure. The *step kernel* (block average) of a kernel U relative to $\mathcal{P}$ is the conditional expectation of U given the product σ-algebra $\mathcal{P} \otimes \mathcal{P}$ generated by the rectangles $P_i \times P_j$:

$$U_{\mathcal{P}}(x,y) \;=\; \frac{1}{\lambda(P_i)\lambda(P_j)} \int_{P_i}\int_{P_j} U(x',y')\,\mathrm{d}x'\,\mathrm{d}y'\,, \qquad (x,y) \in P_i \times P_j. \tag{17.12}$$

By chapter 3, $U_{\mathcal{P}} = \mathbb{E}[U \mid \mathcal{P}\otimes\mathcal{P}]$ is the orthogonal projection of U in $L^2([0,1]^2)$ onto the finite-dimensional subspace of functions constant on each block-pair $P_i \times P_j$. A *step graphon with k steps* is any graphon of this block-constant form, determined by the part sizes $a_i = \lambda(P_i)$ and a symmetric matrix $(\theta_{ij}) \in [0,1]^{k\times k}$ of block values.

The driving quantity is the *energy* (or index) of a partition,

$$\mathcal{E}_U(\mathcal{P}) \;:=\; \|U_{\mathcal{P}}\|_2^2 \;=\; \sum_{i,j} a_i a_j\,\theta_{ij}^2, \tag{17.13}$$

the squared L^2 norm of the block average. Two facts make it the right monotone:

- *Boundedness.* Since $U_{\mathcal{P}}$ is an L^2 projection of U, $\mathcal{E}_U(\mathcal{P}) = \|U_{\mathcal{P}}\|_2^2 \le \|U\|_2^2 \le \|U\|_\infty^2 \le 1$ for a graphon $U \in [0,1]$.
- *Monotonicity under refinement.* If $\mathcal{Q}$ refines $\mathcal{P}$ then the block-constant subspace for $\mathcal{Q}$ contains that for $\mathcal{P}$, so projecting onto the larger subspace can only lengthen the projection: by the Pythagorean theorem,
$$\|U_{\mathcal{Q}}\|_2^2 \;=\; \|U_{\mathcal{P}}\|_2^2 + \|U_{\mathcal{Q}} - U_{\mathcal{P}}\|_2^2\,, \tag{17.14}$$
because $U_{\mathcal{Q}} - U_{\mathcal{P}}$ is orthogonal to the $\mathcal{P}$-subspace (the tower property of conditional expectation).

17.4.2 The weak regularity lemma

The next lemma says: *whenever a block average still differs appreciably from the kernel in cut norm, one can refine the partition and capture a definite quantum of energy.* Because the energy is bounded, this cannot go on forever, and the process terminates at a partition whose block average is cut-close to the kernel. This is the Frieze–Kannan *weak regularity lemma* [72], the workhorse of the whole theory.

Lemma 17.15 (Energy increment)**.** *Let U be a kernel and $\mathcal{P}$ a partition of $[0,1]$ into k parts. If $\|U - U_{\mathcal{P}}\|_\square > \epsilon$, then there is a refinement $\mathcal{Q}$ of $\mathcal{P}$ into at most $4k$ parts with*

$$\mathcal{E}_U(\mathcal{Q}) \;\geq\; \mathcal{E}_U(\mathcal{P}) + \epsilon^2.$$

Proof. Write $R := U - U_{\mathcal{P}}$. By eq. (17.2) and the hypothesis there exist measurable $S, T \subseteq [0,1]$ with $|\int_S \int_T R| > \epsilon$. Let $\mathcal{Q}$ be the common refinement of $\mathcal{P}$ with the two-set partition $\{S, S^c\}$ and the two-set partition $\{T, T^c\}$; each original part is cut into at most four pieces, so $\mathcal{Q}$ has at most $4k$ parts, and $\mathbf{1}_S$ and $\mathbf{1}_T$ are constant on the parts of $\mathcal{Q}$, hence $\mathbf{1}_{S\times T}$ is $\mathcal{Q}\otimes\mathcal{Q}$-measurable.

Because $U_{\mathcal{Q}} = \mathbb{E}[U \mid \mathcal{Q}\otimes\mathcal{Q}]$ and $\mathbf{1}_{S\times T}$ is $\mathcal{Q}\otimes\mathcal{Q}$-measurable, the defining property of conditional expectation gives $\int_{S\times T}(U - U_{\mathcal{Q}}) = 0$, that is $\int_S\int_T U = \int_S \int_T U_{\mathcal{Q}}$. The same identity at the coarser level reads $\int_S\int_T U_{\mathcal{P}}$ *need not* equal $\int_S\int_T U$, and indeed

$$\int_S\int_T (U_{\mathcal{Q}} - U_{\mathcal{P}}) = \int_S\int_T (U - U_{\mathcal{P}}) = \int_S\int_T R, \qquad \text{so} \qquad |\langle U_{\mathcal{Q}} - U_{\mathcal{P}}, \mathbf{1}_{S\times T}\rangle| > \epsilon.$$

By Cauchy–Schwarz in $L^2([0,1]^2)$ and $\|\mathbf{1}_{S\times T}\|_2 = (\lambda(S)\lambda(T))^{1/2} \leq 1$,

$$\epsilon < |\langle U_{\mathcal{Q}} - U_{\mathcal{P}}, \mathbf{1}_{S\times T}\rangle| \leq \|U_{\mathcal{Q}} - U_{\mathcal{P}}\|_2 \; \|\mathbf{1}_{S\times T}\|_2 \leq \|U_{\mathcal{Q}} - U_{\mathcal{P}}\|_2\,,$$

so $\|U_{\mathcal{Q}} - U_{\mathcal{P}}\|_2^2 > \epsilon^2$. Substituting into the Pythagorean identity eq. (17.14) gives $\mathcal{E}_U(\mathcal{Q}) = \mathcal{E}_U(\mathcal{P}) + \|U_{\mathcal{Q}} - U_{\mathcal{P}}\|_2^2 > \mathcal{E}_U(\mathcal{P}) + \epsilon^2$. ☐

Theorem 17.16 (Weak regularity lemma)**.** *For every $\epsilon > 0$ and every graphon U there is a partition $\mathcal{P}$ of $[0,1]$ into at most $4^{\lceil 1/\epsilon^2\rceil}$ parts with*

$$\|U - U_{\mathcal{P}}\|_\square \;\leq\; \epsilon.$$

Consequently every graphon lies within cut distance ϵ of a step graphon with a number of steps bounded by a function of ϵ alone, independent of U.

Proof. Start from the trivial partition $\mathcal{P}_0 = \{[0,1]\}$ $(k=1)$, with energy $\mathcal{E}_U(\mathcal{P}_0) = \big(\int_{[0,1]^2} U\big)^2 \geq 0$. While the current partition $\mathcal{P}_r$ satisfies $\|U - U_{\mathcal{P}_r}\|_\square > \epsilon$, apply lemma 17.15 to obtain $\mathcal{P}_{r+1}$ with $\mathcal{E}_U(\mathcal{P}_{r+1}) \geq \mathcal{E}_U(\mathcal{P}_r) + \epsilon^2$ and at most four times as many parts. Since the energy is nonnegative, increases by at least ϵ^2 at each step, and never exceeds 1 (a graphon has $\|U\|_\infty \leq 1$), the loop runs for at most $r_* \leq \lceil 1/\epsilon^2\rceil$ steps before the stopping condition $\|U - U_{\mathcal{P}_r}\|_\square \leq \epsilon$ is met. The terminal partition has at most $4^{r_*} \leq 4^{\lceil 1/\epsilon^2\rceil}$ parts, and $U_{\mathcal{P}_r}$ is the desired step graphon. ☐

Theorem 17.16 says that any network, however intricate, admits a *bounded-complexity effective description* accurate at the level of all coarse observables: a partition of types into a number of communities that depends only on the target accuracy ϵ, not on the size or detail of the network, within which the kernel may be replaced by its block average. The number of blocks $4^{1/\epsilon^2}$ is enormous, which is the honest price of demanding accuracy in the strong cut norm uniformly; the lemma is a statement about *existence* of a coarse model, not about its parsimony.

17.4.3 Total boundedness

Total boundedness now follows by discretizing the finitely many parameters of a step graphon. Recall that $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is *totally bounded* if for every $\epsilon > 0$ it is covered by finitely many balls of radius ϵ.

Theorem 17.17 (Total boundedness)**.** *$(\widetilde{\mathcal{W}}_0, \delta_\square)$ is totally bounded.*

Proof. Fix $\epsilon > 0$ and set $k(\epsilon) = 4^{\lceil 1/\epsilon^2 \rceil}$. By theorem 17.16, every graphon U is within cut distance ϵ of a step graphon $U_{\mathcal{P}}$ with $k \le k(\epsilon)$ parts. A step graphon with k parts is determined up to relabeling (lemma 17.5: relabel each part onto a subinterval) by its vector of part sizes $a = (a_1, \dots, a_k)$, a probability vector, and its symmetric value matrix $\theta = (\theta_{ij}) \in [0,1]^{k \times k}$. We round both to finite grids.

Rounding the values. Replace each θ_{ij} by the nearest multiple of ϵ in $[0,1]$; call the result θ'. The corresponding step graphons differ by at most ϵ in L^∞, hence by at most ϵ in cut norm (by eq. (17.5), $\|\cdot\|_\square \le \|\cdot\|_1 \le \|\cdot\|_\infty$). There are at most $(\lceil 1/\epsilon \rceil + 1)^{k(k+1)/2}$ symmetric matrices θ' on this grid.

Rounding the sizes. Replace the size vector a by a vector a' whose entries are multiples of ϵ/k summing to 1 (round down and assign the remainder to one part); then $\sum_i |a_i - a_i'| \le \epsilon$. Changing the part sizes from a to a' while keeping the value matrix moves mass across blocks and changes the step graphon by at most $2\sum_i |a_i - a_i'| \le 2\epsilon$ in L^1, hence by at most 2ϵ in cut norm. There are finitely many such grid size-vectors for each k.

Combining, every graphon U is within cut distance $\epsilon + \epsilon + 2\epsilon = 4\epsilon$ of one of *finitely many* step graphons — those with $k \le k(\epsilon)$ parts, grid value matrix θ', and grid size vector a' — the total count being finite because each of k, θ', a' ranges over a finite set. This finite family is a 4ϵ-net for $(\widetilde{\mathcal{W}}_0, \delta_\square)$. As $\epsilon > 0$ was arbitrary, the space is totally bounded. $\square$

17.4.4 Completeness, and the conclusion

It remains to show that $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is complete: every $\delta_\square$-Cauchy sequence converges to a graphon. This is the analytic heart of the theorem; we give the mechanism and defer the fully quantified construction to appendix F.

Proposition 17.18 (Completeness; sketch, full proof in appendix F)**.** *$(\widetilde{\mathcal{W}}_0, \delta_\square)$ is complete.*

Proof sketch. Let (W_n) be $\delta_\square$-Cauchy. Passing to a subsequence we may assume $\delta_\square(W_n, W_{n+1}) < 2^{-n}$; since a Cauchy sequence with a convergent subsequence converges, it suffices to produce a limit of this subsequence. Using only that $\delta_\square$ is an infimum over measure-preserving bijections, choose for each n a *near-optimal* bijection and replace W_{n+1} by the corresponding representative, aligning the representatives so that $\|V_n - V_{n+1}\|_\square < 2^{-n}$ holds for the aligned kernels V_n (no attainment is needed — a bijection within 2^{-n} of the infimum exists by definition of the infimum). The aligned sequence (V_n) is thus Cauchy in the cut *norm*.

The limit is now extracted by functional analysis rather than by a martingale. The graphons form a subset of the closed unit ball of $L^\infty([0,1]^2)$, which is weak-$*$ compact by the Banach–Alaoglu theorem (imported from chapter 2); as $L^\infty([0,1]^2)$ is the dual of the separable space $L^1([0,1]^2)$, the ball is weak-$*$ *sequentially* compact, so a subsequence of (V_n) converges weak-$*$ to some $W \in L^\infty$. The graphon constraints $0 \le W \le 1$ and $W(x,y) = W(y,x)$ a.e. cut out a convex, weak-$*$ closed set (each is an intersection of half-spaces $\langle \cdot\,, h\rangle \le c$ with $h \in L^1$), so the limit W is again a graphon. Finally the cut-norm Cauchy estimates pass to the weak-$*$ limit: for fixed S, T the test function $\mathbf{1}_{S\times T}$ lies in L^1, so $|\langle V_n - W, \mathbf{1}_{S\times T}\rangle| = \lim_m |\langle V_n - V_m, \mathbf{1}_{S\times T}\rangle| \le \sup_{m\ge n} \|V_n - V_m\|_\square$, and taking the supremum over S, T gives $\|V_n - W\|_\square \to 0$, hence $\delta_\square(W_n, W) \to 0$. The fully quantified version — in particular the bookkeeping of the simultaneous alignments — is carried out in appendix F (theorem F.5), by exactly this weak-$*$ route. $\square$

Proof of theorem 17.13. By theorem 17.17 the space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is totally bounded, and by proposition 17.18 it is complete. A metric space that is both totally bounded and complete is compact (equivalently, sequentially compact: from total boundedness every sequence has a Cauchy subsequence, and completeness supplies its limit). Hence $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is compact. $\square$

17.4.5 Why this is the keystone

Compactness of $\widetilde{\mathcal{W}}_0$ is exactly the property a fixed-point or limiting argument needs when the unknown is itself a network. Three downstream uses, made precise later, show its weight.

- *Existence of equilibria with a varying network.* In chapter 24 a graphon mean field equilibrium is sought as a fixed point of a best-response map that depends continuously (in $\delta_\square$, under (A-Reg-W)) on the underlying graphon. Compactness of $\widetilde{\mathcal{W}}_0$ supplies the convergent subsequences of approximate equilibria that a Schauder/Kakutani argument (chapter 2, chapter 10) converts into an exact one; without it the space of admissible networks would be too large to close the argument.

- *Finite networks converge to a limit.* The final statement of theorem 17.12 guarantees that a left-convergent sequence of growing finite networks — the sampled graphs $G(N, W)$ of chapter 19, each identified with its step graphon $W_{G(N,W)}$ via proposition 16.9 so that the equivalence theorem applies verbatim — has an honest graphon limit. Compactness is what forbids a convergent stream of observables from corresponding to "no limiting object," so a single graphon can legitimately stand in for the whole sequence.

- *Stability under network perturbation.* Because homomorphism densities, and (under (A-Reg-W)) the equilibrium maps of Part IV, are continuous in $\delta_\square$, two networks that are $\delta_\square$-close yield close equilibria. Chapter 25 turns this into the statement that the graphon mean field equilibrium is an ϵ-Nash equilibrium on any large finite network drawn from a nearby graphon — the practical justification for using the limit theory to design interventions on real graphs.

Physics note

A compact metric space of order parameters is a comfortable place to do physics: every sequence of candidate states has a convergent subsequence, so variational and limiting arguments never leak their minimizers or accumulation points out of the space. The space of graphons is the order-parameter manifold for network structure, and theorem 17.13 says it is compact in the natural macroscopic metric. The reader should resist over-reading this as a thermodynamic statement: there is no energy functional or partition function on $\widetilde{\mathcal{W}}_0$ here, only a metric geometry. The payoff is structural, not dynamical — it is the guarantee that the existence machinery of Part IV has somewhere to land.

17.4.6 Worked computation: cut norm and approximation of the thread graphons

Worked computation 17.19 (Cut norm of the rank-one graphon and its block approximation)**.** We compute the cut norm in two cases tied to the carried threads, then watch a graphon being coarse-grained as in theorem 17.16. Recall (chapter 15) that the *constant* graphon $W \equiv c$ recovers the homogeneous (Thread A) coupling, while the *rank-one* graphon $W(x,y) = f(x)f(y)$, with $f\colon [0,1] \to [0,1]$ measurable, is the "Curie–Weiss" kernel of Thread B whose aggregate $\mathbb{W}\bar{m}$ collapses to a scalar order parameter times f.

(a) The constant graphon. For $W \equiv c$, $\int_S \int_T c = c\,\lambda(S)\lambda(T)$, maximized over $\lambda(S), \lambda(T) \le 1$ at $S = T = [0,1]$, so $\|c\|_\square = |c|$. The cut norm of a constant kernel is its absolute value, matching the L^1 norm; here the comparison eq. (17.5) is an equality because there is no fluctuation to discard.

(b) The rank-one graphon. With $W(x,y) = f(x)f(y)$, the function form eq. (17.3) factorizes:

$$\int_{[0,1]^2} u(x)\, f(x)f(y)\, v(y)\ \mathrm{d}x\,\mathrm{d}y = \Big(\int_[0,1] uf\Big)\Big(\int_[0,1] fv\Big).$$

Each factor is maximized over $u, v \in [0,1]$ by taking $u = v = \mathbf{1}_{\{f>0\}} = 1$ (as $f \ge 0$), giving $\int_[0,1] f$ in each. Hence

$$\|W\|_\square = \Big(\int_[0,1] f\Big)^2 = \bar{f}^{\,2}, \qquad \bar{f} := \int_[0,1] f.$$

The cut norm of a rank-one graphon is the square of the mean of its profile. Compare the spectral bound of lemma 17.3: the operator $\mathbb{W}$ here is the rank-one projection-like map $g \mapsto f\,\langle f, g\rangle$, with operator norm $\|\mathbb{W}\|_{2\to2} = \|f\|_2^2 = \int_[0,1] f^2$, and indeed $\bar{f}^{\,2} = (\int f)^2 \le \int f^2 = \|f\|_2^2$ by Cauchy–Schwarz, confirming $\|W\|_\square \le \|\mathbb{W}\|_{2\to2}$ with equality iff f is constant. This is exactly the regime in which Thread B reduces to Thread A: a constant profile f makes the rank-one and constant graphons coincide and saturates the comparison.

(c) Block approximation and total boundedness in action. Take the explicit profile $f(x) = x$, so $W(x, y) = xy$, a smooth rank-one graphon, and coarse-grain it on the uniform partition $\mathcal{P}_k$ of $[0, 1]$ into k equal intervals $P_i = [\frac{i-1}{k}, \frac{i}{k}]$. The block average eq. (17.12) is the product of the interval means, $W_{\mathcal{P}_k} = \theta_{ij}$ with $\theta_{ij} = (\frac{2i-1}{2k})(\frac{2j-1}{2k})$, a step graphon with k steps — a rank-one block model. The error kernel $R = W - W_{\mathcal{P}_k}$ factorizes within each block as $xy - \bar{x}_i\bar{y}_j$ where $\bar{x}_i = \frac{2i-1}{2k}$; on $P_i \times P_j$, $|R| \le |x - \bar{x}_i|\ y + \bar{x}_i\,|y - \bar{y}_j| \le \frac{1}{2k} + \frac{1}{2k} = \frac{1}{k}$, so $\|R\|_\infty \le 1/k$ and a fortiori $\|R\|_\square \le \|R\|_1 \le 1/k \to 0$. Thus the rank-one thread graphon is approximated in cut norm, at rate $1/k$, by rank-one block models with k steps — a concrete instance of the finite-net construction behind theorem 17.17, and the bridge by which the continuum Thread B kernel is replaced, with controlled cut-norm error, by a finite block graphon (the multi-population reduction revisited in chapter 22).

(d) Relabeling invisibility. Finally, let $\varphi(x) = 1 - x$, a measure-preserving involution, and set $W' = W^\varphi$, i.e. $W'(x, y) = (1 - x)(1 - y)$. As kernels $W \ne W'$ and $\|W - W'\|_1 > 0$, yet by construction $\delta_\square(W, W') = 0$: they are weakly isomorphic, the same point of $\widetilde{\mathcal{W}}_0$. Reversing the labeling of types — reading centrality from the right end of $[0, 1]$ instead of the left — changes nothing a homomorphism density or the mean field coupling can detect, exactly as lemma 17.5 predicts. This is the smallest nontrivial example of the quotient at work.

Rigor ledger

Proved (in full or by self-contained computation).

- The three equivalent forms of the cut norm (lemma 17.2), the comparison inequalities $\|\cdot\|_\square \le \|\cdot\|_1 \le \|\cdot\|_2 \le \|\cdot\|_\infty$ and $\|\cdot\|_\square \le \left\|\mathbb{W}_{(\cdot)}\right\|_{2\to2}$ (lemma 17.3), and that $\|\cdot\|_\square$ is a norm.
- Relabeling invariance of the cut norm and of homomorphism densities (lemma 17.5); that $\delta_\square$ is a pseudometric — symmetry, the triangle inequality (by composition of near-optimal bijections plus relabeling invariance), transitivity of weak isomorphism, and the induced genuine metric on the quotient $\widetilde{\mathcal{W}}_0$ — *in full* (proposition 17.7).
- The easy direction of the equivalence: cut convergence implies left-convergence (proposition 17.10), via the imported counting lemma.
- The energy-increment lemma (lemma 17.15) and the weak regularity lemma (theorem 17.16) *in full*, by the energy-increment argument.
- Total boundedness of $(\widetilde{\mathcal{W}}_0, \delta_\square)$ *in full* (theorem 17.17), by weak regularity plus a finite discretization of the step-graphon parameters.
- The compactness theorem (theorem 17.13) conditional on completeness, via "totally bounded + complete ⇒ compact."
- The worked computation (worked computation 17.19): cut norms of the constant and rank-one thread graphons, the $1/k$ block-approximation rate, and an explicit weak-isomorphism pair.

Assumed (imported, used without re-proof).

- The graphon and its operator $\mathbb{W}$ (chapter 15); the homomorphism density $t(F,\cdot)$ and left-convergence (chapter 16); the counting lemma eq. (17.11) (lemma 17.9, proved in chapter 16).
- Conditional expectation as L^2 orthogonal projection and the tower property (chapter 3); L^p spaces, Fubini, and the Pythagorean identity in $L^2([0,1]^2)$ (chapters 1 and 2); the metric-space fact "totally bounded + complete = compact" (elementary point-set topology).
- Standing assumption (A-Reg-W) (symmetric, measurable, bounded W; $\mathbb{W}$ self-adjoint Hilbert–Schmidt) wherever the operator-norm comparison or the Part IV continuity statements are invoked.

Deferred (stated, proof pointed elsewhere).

- The equivalent coupling/overlay reformulation of $\delta_\square$ and the attainment of its infimum (remark 17.8): *not used* in this chapter — the metric structure is proved directly above without it — and recorded only as the route that identifies weak isomorphism with measure-preserving isomorphism; see [99, Ch. 8].
- The moment-uniqueness theorem (theorem 17.11, "only if"): mechanism named (sampling lemma plus moment-determinacy of bounded variables); full in-book proof in appendix F (theorem F.11), with attribution [28, 99].
- Completeness of $(\widetilde{\mathcal{W}}_0, \delta_\square)$ (proposition 17.18): the mechanism — align representatives by near-optimal bijections, then extract a weak-$*$ limit graphon from the closed L^∞ ball by Banach–Alaoglu (chapter 2) — is given in the sketch above; the fully quantified construction is theorem F.5 in appendix F.

Open / honestly flagged.

- The whole development is the *dense*, bounded ($[0,1]$-valued) theory. For the sparse and L^p-graphon regimes of chapter 27 the cut norm must be renormalized and compactness in the present form *fails*; what survives is discussed there. The compactness theorem is a feature of dense graph limits specifically.
- The number of steps $4^{1/\epsilon^2}$ in theorem 17.16 is essentially best possible for the strong cut norm and is astronomically large; it makes the lemma a tool for existence and qualitative structure, not a practical compression. Quantitative regularity with usable bounds requires extra structure (e.g. bounded operator rank), taken up in chapters 18 and 19.

Bibliographic notes

The cut norm and cut metric, and the theorem that left-convergence of dense graph sequences coincides with convergence in the cut metric, are due to Borgs, Chayes, Lovász, Sós, and Vesztergombi [29]; our theorem 17.12 and proposition 17.7 follow their development, and the equivalent overlay/coupling formulations of $\delta_\square$ recorded in remark 17.8 are theirs; our in-chapter proof that $\delta_\square$ is a metric instead uses only composition of near-optimal bijections, so it does not invoke those formulations. The compactness of graphon space, theorem 17.13, is the Lovász–Szegedy theorem [100]; the survey [98] is the gentlest first reading, and the monograph [99] (especially Chapters 8, 9, and 11) is the comprehensive reference and the source for the full proofs we deferred to appendix F. The weak regularity lemma theorem 17.16, and the energy-increment argument lemma 17.15 that proves it, are due to Frieze and Kannan [72]; the formulation via conditional expectation and the L^2 projection energy is the one in [99, Ch. 9], and it is this argument, rather than Szemerédi's original regularity lemma, that we use because it is self-contained and yields the bounded-step approximation directly. The moment-uniqueness theorem theorem 17.11 — that homomorphism densities determine a graphon up to weak isomorphism — is proved by Borgs, Chayes, and Lovász [28]. The operator-norm reading of the cut norm via the $L^\infty \to L^1$ norm eq. (17.4) is standard and connects to the spectral theory of chapter 18. The use of graphon compactness as the structural input to existence and finite-network convergence in graphon mean field games — our chapters 24 and 25 — traces to the Caines–Huang program [33, 34] and the static graphon-game theory of Parise and Ozdaglar [108].

Chapter 18

Graphons as Operators: Spectral Theory

The previous three chapters built the graphon as a *combinatorial* and *metric* object: a symmetric kernel W on $[0,1]^2$ (chapter 15), the moments of which are the homomorphism densities $t(F, W)$ (chapter 16), and the geometry of which is the cut metric $\delta_\square$ that makes graphon space compact (chapter 17). This chapter develops the *third* face of the same object, the one that will do the analytic work in Part IV: the graphon as a *self-adjoint Hilbert–Schmidt operator*

$$(\mathbb{W}f)(x) = \int_0^1 W(x, y)f(y)\ \mathrm{d}y$$

on $L^2[0,1]$. The phrase "third face" is meant literally. Nothing on the page changes between these chapters — it is one and the same symmetric kernel $W(x, y)$ throughout — and what changes is only the question we put to it. Chapter 16 asked how many copies of a small graph the kernel contains; chapter 17 asked how far it sits from another kernel in the cut metric; now we ask what happens when we let it *act*, repeatedly, by integration against test functions. The spectrum is the answer to that last question. The reader should picture this chapter not as building a new object but as slotting a single new lens onto an already-familiar one: the kernel is fixed, and we are merely learning to read it spectrally.

That this is no new construction at all is worth stressing, because it sets the tone of the chapter as a collection of promised payoffs rather than a fresh development. The operator $\mathbb{W}$ was defined in definition 2.23, shown to be self-adjoint Hilbert–Schmidt in proposition 2.24, and diagonalised by the spectral theorem theorem 2.26; its spectrum was even computed in closed form for the constant and rank-one kernels in worked computation 2.28. Chapter 2 staged all of this deliberately and promised the payoff here. We now collect it.

Why does Part IV need this third face specifically, and not merely the combinatorial and metric ones already in hand? Because in a graphon mean field game the coupling between types *is* the operator $\mathbb{W}$. A player of type x feels the rest of the population not individually but through the kernel-weighted average $\int_0^1 W(x, y)\, m(y)\ \mathrm{d}y = (\mathbb{W}m)(x)$ of the others' states, so the map from a population profile m to the environment a type actually experiences is exactly $\mathbb{W}$ acting on

$L^2[0,1]$. Solving for equilibrium then amounts to inverting, iterating, or linearising this one operator, and every such operation is diagonal in the spectral basis. Understanding equilibria therefore reduces, again and again in the chapters to come, to understanding the spectrum of $\mathbb{W}$: which mode dominates the coupling, how fast the subdominant modes decay under repetition, and how the whole spectrum moves when the kernel is perturbed or sampled. A single identity makes the entire chapter cohere and deserves to be named at the outset — the power sums of the spectrum are the cycle homomorphism densities of chapter 16,

$$\sum_n \lambda_n^k \;=\; t(C_k, W).$$

This is the hinge of the chapter. It is the one statement that lets the combinatorial accounting of chapter 16 and the metric continuity of chapter 17 reach *into* the spectrum: cycle densities are objects those chapters already control completely, and the identity converts that control directly into control of the eigenvalues. Every convergence statement below, and several of the closed forms in later chapters, run through it.

The chapter has four movements. First (section 18.1) we assemble the spectral decomposition of $\mathbb{W}$ into a usable form: the eigenvalue sequence, the Mercer-type expansion of the kernel, and the identity that ties the spectrum back to chapter 16 — the k-th spectral moment is the homomorphism density of the k-cycle. Second (section 18.2) we read the leading eigenfunction as *eigenvector centrality* via an infinite-dimensional Perron–Frobenius theorem. Third (section 18.3) we read the *spectral gap* as the rate of mixing and the *near-degeneracies* of the top of the spectrum as community structure, working the two-block stochastic block model out by hand. Fourth (section 18.4) we settle the relationship between the metric of chapter 17 and the spectrum: cut convergence implies convergence of every individual eigenvalue, but is strictly weaker than operator-norm convergence, and only the stronger notion controls eigen*functions*. The final section (section 18.5) cashes the spectral theory into the carried Thread B: finite-rank graphons collapse the infinite-dimensional graphon mean field game to a finite system of scalar order parameters, and every graphon admits a controlled low-rank reduction. Throughout, W obeys the standing regularity assumption (A-Reg-W) of appendix A.6, so $\mathbb{W}$ is self-adjoint Hilbert–Schmidt; additional continuity of W is invoked only where named — chiefly in Mercer's theorem (theorem 18.2), and nowhere in the load-bearing moment identity.

Notation watch

This is the first chapter of Part IV in which a graphon and a stochastic noise will come to share the page, so we restate here the book's resolution of a standing letter clash (fixed in appendix A.5). Both Brownian motion and the graphon traditionally want the symbol W. The convention from Part III onward is that, *whenever a graphon W is in scope* — as it is everywhere from here to the end of the book — Brownian motion is written B (so B_t, with increments $\mathrm{d}B_t$), while W is reserved for the kernel W and $\mathbb{W}$ for its integral operator. No Brownian motion actually appears in the present chapter, which is purely spectral; we record the convention now so that the dynamic chapters built on this spectral theory inherit it without a second announcement.

18.1 The spectral decomposition of the graphon operator

We now record, in the notation we keep for the rest of Parts III and IV, what the spectral theorem of chapter 2 hands us. It is fair to ask why a decomposition already proved deserves a section of its own rather than a one-line reminder, and the honest answer is bookkeeping: everything downstream in this chapter — centrality, the gap, spectral convergence, low rank — is a *reading* of the very same eigenvalues and eigenfunctions, and those readings pull on the spectrum in incompatible ways unless the names are fixed once, carefully, in advance. Two of those readings sort the eigenvalues by size and two sort them by sign; one wants to evaluate the kernel pointwise and another must scrupulously avoid doing so. If we do not settle these distinctions now they will resurface as confusion exactly where the argument is most delicate. So this foundational section does nothing but lay down a usable working form for the spectral data — and it is genuinely foundational: nothing here is read for its own sake, only to be deployed later.

The section has three beats, and their order is not arbitrary. First (definition 18.1 and its surrounding discussion) we organise the output of the spectral theorem — the spectral expansion that the whole section orbits — and fix the *two orderings* of the eigenvalues, by modulus and by sign, that the rest of Part IV will switch between. Second we confront the natural question raised the moment one writes the kernel as a sum of eigenfunction products: does that sum converge *pointwise*, so that one may read off a value $W(x, y)$ or, more pointedly, a diagonal value $W(x, x)$? The classical answer is Mercer's theorem, and the reason it comes second is that its hypothesis — positive semidefiniteness — generically *fails* for graphons, and recognising this failure is what forces the third beat to be proved the way it is. Third and most important, we establish the moment identity $\sum_n \lambda_n^k = t(C_k, W)$, the hinge advertised in the chapter opening, by a route that pairs Hilbert–Schmidt iterates against each other and so never touches the diagonal — precisely because Mercer is unavailable. Mercer must therefore precede the moment identity: it is the obstacle the moment-identity proof is built to sidestep.

What this section exports is small to state and used everywhere. The named pair (λ_n, e_n) and the two orderings travel forward verbatim: section 18.2 reads the leading eigenfunction e_1 as centrality, and section 18.3 reads the pair λ_1, λ_2 as a gap. The moment identity and the uniform ℓ^2 tightness it delivers — only finitely many eigenvalues can be large, and the bound is the same for *every* graphon — are the entire engine of the spectral convergence theorem of section 18.4. With that map in hand we begin where every reading must, with the eigenvalues, the eigenfunctions, and how we agree to list them.

18.1.1 Eigenvalues, eigenfunctions, and orderings

The spectral theorem of chapter 2 has already done the hard analytic work; what is left for us here is the seemingly clerical but genuinely load-bearing task of writing its output in a form we can live with for the rest of Parts III and IV. The output is a single expansion, and everything later in the chapter is a way of reading it, so we begin by fixing it on the page and then deciding, once and carefully, how to list the numbers it contains. By proposition 2.24 the operator $\mathbb{W}$ is compact and self-adjoint on $L^2[0, 1]$, so the spectral theorem theorem 2.26 applies verbatim: there is an orthonormal system of eigenfunctions $\{e_n\} \subseteq L^2[0,1]$ and real eigenvalues $\{\lambda_n\}$, repeated according to multiplicity, with

$$\mathbb{W}f \;=\; \sum_n \lambda_n \langle f, e_n\rangle\, e_n, \qquad W(x,y) \;=\; \sum_n \lambda_n\, e_n(x)\, e_n(y) \;\text{ in } L^2([0,1]^2), \tag{18.1}$$

where $\langle f, g\rangle = \int_0^1 fg$ is the real L^2 inner product. This is the object the orderings will organise, so it is worth being slow about exactly what its two halves say and why they are the same statement.

The first equality is the operator form: it asserts that $\mathbb{W}$ acts on any test function f by resolving f into its components along the eigendirections e_n, scaling the n-th component by λ_n, and reassembling. This is the literal content of "diagonalisation" — in the basis $\{e_n\}$ the operator is multiplication by the sequence (λ_n). The second equality is the kernel form, and it is not a new fact but a translation of the first through the kernel–operator dictionary of theorem 2.22, which says that a Hilbert–Schmidt operator and its $L^2([0,1]^2)$ kernel determine each other. Apply that dictionary term by term. The n-th summand of the operator form is the rank-one operator $P_n\colon f \mapsto \langle f, e_n\rangle e_n$, the orthogonal projection onto the line $\mathbb{R}e_n$. Writing its action out,

$$(P_n f)(x) = \langle f, e_n\rangle\, e_n(x) = \Big(\int_0^1 f(y)e_n(y)\;\mathrm{d}y\Big)e_n(x) = \int_0^1 e_n(x)\, e_n(y)\, f(y)\;\mathrm{d}y,$$

we read off that P_n is integration against the kernel $e_n(x)e_n(y)$ — the outer product of e_n with itself. Summing with the weights λ_n gives precisely the kernel form of eq. (18.1). The only thing left to justify is that the infinite sum on the right actually converges as a function of two variables, and here the Hilbert–Schmidt structure pays for itself. The tensor products $\{e_n \otimes e_n\}$, where $(e_n \otimes e_n)(x,y) = e_n(x)e_n(y)$, are orthonormal in $L^2([0,1]^2)$ (they are part of the product orthonormal basis $\{e_m \otimes e_n\}$), so the partial sums of the kernel series have squared increments $\|\sum_{M<n\le N} \lambda_n\, e_n \otimes e_n\|^2_{L^2([0,1]^2)} = \sum_{M<n\le N} \lambda_n^2$. This is a Cauchy tail precisely when $\sum_n \lambda_n^2 < \infty$ — which holds, and with a value we already know: $\sum_n \lambda_n^2 = \|W\|^2_{L^2([0,1]^2)} = \|W\|^2_{\mathrm{HS}} < \infty$ by proposition 2.24. So the kernel series converges in $L^2([0,1]^2)$, and its limit is W because both sides represent the same operator. The convergence is in $L^2([0,1]^2)$ and *only* there for a general graphon; whether one may evaluate the series at a fixed point (x,y), in particular on the diagonal $x = y$, is a strictly harder question we postpone to theorem 18.2.

The same finiteness $\sum_n \lambda_n^2 < \infty$ also explains the two structural facts about the eigenvalues that we will lean on constantly: they *accumulate only at* 0, and each nonzero eigenvalue has *finite multiplicity*. Both are immediate from square-summability — for any threshold $\varepsilon > 0$, at most ε^{-2} of the λ_n can exceed ε in modulus, or their squares would already overflow the bound — which is just the concrete face of the compactness of $\mathbb{W}$ established in proposition 2.24. A compact self-adjoint operator has exactly this spectral shape: a discrete sequence of real eigenvalues piling up at 0, each repeated finitely often. We will return to this ℓ^2 tightness as the engine of spectral convergence; for now it is what guarantees that the lists we are about to write down are honest lists, with a genuine first entry, second entry, and so on.

That brings us to the listing itself, which is the real business of this subsection. There is no single canonical way to enumerate the λ_n, and the two natural ways disagree — not pedantically but substantively, because they answer different questions. We therefore fix both now, with names, and resolve once and for all which one each later section is using.

Definition 18.1 (Spectrum and its orderings)**.** Let W satisfy (A-Reg-W) and let $\mathbb{W}$ be its operator. The *spectrum* of W is the multiset of nonzero eigenvalues of $\mathbb{W}$, denoted $\mathrm{spec}(W)$. We list it two ways.

- *Modulus order* (as in theorem 2.26): $|\lambda_1| \geq |\lambda_2| \geq \cdots > 0$, with $\lambda_n \to 0$.
- *Signed order*: the positive eigenvalues listed decreasingly, $\lambda_1^+ \geq \lambda_2^+ \geq \cdots > 0$, and the negative eigenvalues listed increasingly, $\lambda_1^- \leq \lambda_2^- \leq \cdots < 0$ (either list may be finite or empty).

We write $\rho(\mathbb{W}) = \|\mathbb{W}\| = |\lambda_1| = \max(\lambda_1^+, -\lambda_1^-)$ for the spectral radius.

Why carry two orderings rather than pick one and convert? Because the conversion destroys exactly the information each ordering exists to preserve, and both pieces of information are indispensable downstream. The modulus order is what the spectral theorem natively hands us, and it is the right ledger whenever the quantity we are estimating depends only on *sizes*: the operator norm is $\|\mathbb{W}\| = |\lambda_1|$, the rate at which repeated averaging by $\mathbb{W}$ kills subdominant modes is governed by the ratio $|\lambda_2| / |\lambda_1|$, and the ℓ^2 tail $\sum_{n>N} \lambda_n^2$ that measures truncation error cares nothing for sign. In all of these the sign of an eigenvalue is invisible, and forcing signs into the bookkeeping would only clutter it. The signed order, by contrast, is what the Courant–Fischer min–max principle corollary 2.27 produces — it builds the positive eigenvalues by maximising the Rayleigh quotient $\langle \mathbb{W}g, g\rangle$ over successively smaller subspaces and the negative ones by minimising it — and it is the right ledger whenever the *sign* carries meaning. It does so exactly twice in this chapter, and both times decisively: the leading *positive* eigenvalue λ_1^+ is the one whose eigenfunction is nonnegative and reads as centrality (section 18.2), and the sign pattern at the top of the spectrum is what distinguishes a clustered community from a bipartite one (section 18.3).

The two orderings really can disagree, so it is worth seeing once how. Imagine a (signed, rank-three) kernel whose nonzero eigenvalues are $+0.6$, -0.4, and $+0.2$. In modulus order the list is $\lambda_1 = +0.6$, $\lambda_2 = -0.4$, $\lambda_3 = +0.2$, so "the second eigenvalue λ_2" is the *negative* number -0.4. In signed order the positive list is $\lambda_1^+ = +0.6$, $\lambda_2^+ = +0.2$ and the negative list is $\lambda_1^- = -0.4$, so "the second positive mode λ_2^+" is $+0.2$. The phrase "the second eigenvalue" is therefore ambiguous until the ordering is named — it denotes -0.4 under one convention and $+0.2$ under the other — and this is exactly the ambiguity that, left unresolved, would corrupt the gap estimates of section 18.3, where "the gap below the top" is a statement about $|\lambda_2|$ but "the principal gap between the two clustering modes" is a statement about λ_2^+. From here on the rule is simple: when a statement is about size we write $|\lambda_n|$ and mean the modulus order; when it is about a positive (or negative) mode we write λ_n^+ (or λ_n^-) and mean the signed order.

Notation watch

The two orderings genuinely differ and both are standard; the modulus order $|\lambda_1| \geq |\lambda_2| \geq \cdots$ is the output of theorem 2.26 and the signed orders $\lambda_1^+ \geq \lambda_2^+ \geq \cdots$, $\lambda_1^- \leq \lambda_2^- \leq \cdots$ are the output of Courant–Fischer corollary 2.27. What makes the sign worth tracking is a genuine piece of network meaning, which we spell out now because section 18.3 will depend on it. An eigenfunction e_n with a *positive* eigenvalue is reproduced with its own sign under averaging,

$\mathbb{W}e_n = \lambda_n e_n$ with $\lambda_n > 0$: the region where $e_n > 0$ is linked, through W, predominantly to itself, so the mode describes a *clump* of types that reinforce one another — "like attracts like," an assortative mode. An eigenfunction with a *negative* eigenvalue is reproduced with its sign *flipped*, $\mathbb{W}e_n = \lambda_n e_n$ with $\lambda_n < 0$: the region where $e_n > 0$ is linked predominantly to the region where $e_n < 0$, so the mode describes two camps that connect to each other rather than internally — "opposites attract," an alternating, bipartite mode. Positive eigenvalues are thus the spectral signature of community (assortative) structure and negative eigenvalues the signature of bipartite (disassortative) structure; this is the reading the two-block model of example 18.11 will make completely explicit, where the single nonzero off-constant eigenvalue is positive when within-block ties dominate and negative when cross-block ties do. When a statement is about size we use $|\lambda_n|$; when it is about the leading positive mode we write λ_1^+. For a *nonnegative* kernel, $W \geq 0$ — that is, for an actual graphon rather than a signed difference of two — the two top entries coincide, $\lambda_1^+ = |\lambda_1| = \rho(\mathbb{W})$, by the Perron–Frobenius theorem of section 18.2: a nonnegative kernel cannot hide a negative eigenvalue larger in modulus than its largest positive one, so its spectral radius is always carried by an assortative mode. (Disagreement at the very top therefore requires a signed kernel, of the kind that arises as a difference $W_n - W$ in the convergence arguments of section 18.4; for a graphon proper the orderings can still part company deeper down, as the micro-example above shows, but never at the leading entry.)

With the spectral data named and both orderings fixed, the spectral radius notation $\rho(\mathbb{W}) = |\lambda_1|$ and the signed/modulus vocabulary now travel forward unchanged: section 18.2 will read e_1 at $\lambda_1^+ = \rho(\mathbb{W})$ as centrality, and section 18.3 will read the gap below $|\lambda_1|$ as mixing speed. Before either, though, the kernel form of eq. (18.1) leaves one question dangling that we flagged and deferred: the expansion converges in $L^2([0,1]^2)$, but can it be evaluated at a single point — in particular on the diagonal, where it would compute $W(x,x)$ and, integrated, the trace? That is the next thing to settle, and the answer is more delicate than the L^2 convergence we have just secured.

18.1.2 Mercer's theorem: when the expansion is pointwise

We closed the previous subsection with a question we deliberately left dangling. The kernel form of eq. (18.1) expresses W as the sum $\sum_n \lambda_n \, e_n(x)e_n(y)$, but we proved that sum converges only in $L^2([0,1]^2)$ — as a statement about the function of *two* variables, integrated against test functions, with no claim about its value at any single point. The natural next question is whether we may say more: can the series be evaluated *pointwise*, at a fixed (x,y), and in particular on the *diagonal* $x = y$, where it would read off the number

$$W(x,x) \;\stackrel{?}{=}\; \sum_n \lambda_n \, e_n(x)^2.$$

This is not idle. The diagonal is exactly where the trace lives. If the diagonal identity held and we could integrate it term by term, we would obtain

$$\int_0^1 W(x,x)\;\mathrm{d}x \;=\; \sum_n \lambda_n \int_0^1 e_n(x)^2\;\mathrm{d}x \;=\; \sum_n \lambda_n,$$

the sum of the eigenvalues — the first spectral moment. The very next subsection is about spectral moments $\sum_n \lambda_n^k$, which we will identify with the cycle densities of chapter 16 and on which the entire convergence theory of section 18.4 depends. So the question "may we evaluate the eigenfunction expansion on the diagonal?" is the question "is the naive diagonal route to the spectral moments legitimate?" The answer turns out to be a sharp *no* for general graphons, and understanding precisely why is what dictates the diagonal-free strategy of the next proof. This subsection draws that line.

The L^2 convergence we already have is, on reflection, the most one can expect from square-summability of the λ_n alone, because L^2 convergence is exactly an ℓ^2 statement about coefficients and says nothing about the size of the summands at any individual point. To get *pointwise* control one needs the summands not to conspire — not to add up with alternating signs that cancel in the mean but blow up at a point — and the cleanest structural hypothesis that rules out such conspiracy is that every eigenvalue have the *same* sign. That is what positive semidefiniteness gives, and it is the hypothesis of Mercer's theorem. It is worth pausing on what the hypothesis means before we use it, because its name invites a confusion that this whole subsection exists to dispel.

An operator $\mathbb{W}$ is *positive semidefinite* when its quadratic form is never negative,

$$\langle \mathbb{W}f, f\rangle = \iint_{[0,1]^2} W(x,y)\, f(x)f(y)\, \mathrm{d}x\, \mathrm{d}y \;\geq\; 0 \qquad \text{for all } f \in L^2[0,1].$$

Diagonalising through eq. (18.1), $\langle \mathbb{W}f, f\rangle = \sum_n \lambda_n \langle f, e_n\rangle^2$, so positive semidefiniteness is exactly the requirement that *all eigenvalues be nonnegative*: choosing $f = e_m$ gives $\langle \mathbb{W}e_m, e_m\rangle = \lambda_m \geq 0$, and conversely if every $\lambda_n \geq 0$ the sum of nonnegative terms is nonnegative. This is a statement about the operator — about how W acts as an averaging kernel — and it is emphatically *not* the statement that W is nonnegative as a function. A graphon takes values in $[0,1]$, so $W(x,y) \geq 0$ pointwise; but pointwise nonnegativity of the kernel and nonnegativity of the quadratic form are different conditions, and a $[0,1]$-valued graphon routinely fails the latter. The reason is that f in the quadratic form ranges over *signed* test functions: a kernel that links a region where $f > 0$ to a region where $f < 0$ contributes a *negative* product $f(x)f(y)$ weighted by a positive $W(x,y)$, and if such cross-links dominate, the whole form goes negative. Concretely, take $f = \mathbf{1}_{I_1} - \mathbf{1}_{I_2}$ in the two-block kernel of example 18.11: a direct computation gives $\langle \mathbb{W}f, f\rangle = (p-q)/2$, which is strictly negative whenever cross-block ties are denser than within-block ties, $q > p$ — a perfectly ordinary, perfectly nonnegative graphon with a negative eigenvalue. We will return to this example below; for now it is the warning that the hypothesis of the theorem we are about to state is a genuine restriction, not an automatic feature of graphons.

With the hypothesis understood as a real constraint, we state the classical theorem. It is the statement that, precisely *when* the operator is positive semidefinite, the gap between L^2 and pointwise convergence closes completely.

Theorem 18.2 (Mercer)**.** *Suppose $W\colon [0,1]^2 \to \mathbb{R}$ is continuous, symmetric, and* positive semidefinite*, meaning $\langle \mathbb{W}f, f\rangle \geq 0$ for all $f \in L^2[0,1]$ (equivalently all eigenvalues are ≥ 0). Then the eigenfunctions e_n corresponding to nonzero eigenvalues may be taken continuous, and the expansion*

$$W(x,y) = \sum_n \lambda_n\, e_n(x)e_n(y)$$

converges absolutely and uniformly on $[0,1]^2$*. In particular the series converges on the diagonal,* $W(x,x) = \sum_n \lambda_n e_n(x)^2$*, and* $\int_0^1 W(x,x)\,\mathrm{d}x = \sum_n \lambda_n = \mathrm{tr}(\mathbb{W})$*.*

Proof sketch. The argument has three moves, and it is worth seeing how each one uses the hypotheses — continuity for the first, positive semidefiniteness for the second, square-summability for the third.

Move 1: the eigenfunctions are automatically continuous. We never assumed the e_n continuous; we get it for free from the eigenvalue equation. Rearranging $\mathbb{W}e_n = \lambda_n e_n$ for $\lambda_n \neq 0$,

$$e_n(x) = \frac{1}{\lambda_n}(\mathbb{W}e_n)(x) = \frac{1}{\lambda_n}\int_0^1 W(x,y)\,e_n(y)\,\mathrm{d}y.$$

The right-hand side is continuous in x: if $x_k \to x$ then $W(x_k,y) \to W(x,y)$ for every y by continuity of the kernel, and the integrands are dominated by the integrable bound $\|W\|_\infty\,|e_n(y)|$, so dominated convergence carries the limit inside the integral. Thus $e_n = \lambda_n^{-1}\mathbb{W}e_n$ is continuous, and we may speak of its pointwise values $e_n(x)^2$ without ambiguity. (This is the only place the continuity hypothesis on W is used; it manufactures continuous representatives of objects that the L^2 theory only knows up to a.e. equality.)

Move 2: positivity of the remainder forces monotone diagonal partial sums, and Dini finishes. Fix N and consider the truncation remainder

$$R_N(x,y) = W(x,y) - \sum_{n\le N}\lambda_n e_n(x)e_n(y).$$

As an operator, R_N is $\mathbb{W}$ with its top N spectral projections subtracted, so its eigenvalues are $\{\lambda_n : n > N\}$ together with zeros — and *all of these are* ≥ 0 *by positive semidefiniteness.* Hence R_N is itself a positive semidefinite operator, and the quadratic form of a positive operator is nonnegative on its diagonal in the following sense: testing against an approximate delta at x (or, rigorously, using that the continuous diagonal of a positive continuous kernel is nonnegative) gives $R_N(x,x) \ge 0$, i.e.

$$\sum_{n\le N}\lambda_n e_n(x)^2 \;\le\; W(x,x) \;\le\; \|W\|_\infty \qquad \text{for every } x\in[0,1] \text{ and every } N.$$

Now read what this says about the diagonal series $\sum_n \lambda_n e_n(x)^2$. Every term $\lambda_n e_n(x)^2$ is ≥ 0 (here again $\lambda_n \ge 0$ is doing the work), so for each fixed x the partial sums $S_N(x) = \sum_{n\le N}\lambda_n e_n(x)^2$ *increase with* N; and they are bounded above, uniformly in x, by $W(x,x)$. A bounded monotone sequence converges, so $S_N(x) \uparrow S(x)$ pointwise, with $S(x) \le W(x,x)$. This is exactly the configuration Dini's theorem governs: a sequence of continuous functions S_N (continuous by Move 1, each a finite sum of squares of continuous functions) increasing *monotonically* to a limit S that is itself continuous, on the *compact* interval $[0,1]$. Dini upgrades such monotone pointwise convergence to *uniform* convergence — the obstruction to uniformity is escaping mass that pools at a moving point, and monotonicity plus compactness forbids it. (That the limit S is continuous, which Dini requires, follows because $S = W(\cdot,\cdot)$ on the diagonal once we identify the limit in Move 3; one runs the standard argument by first establishing uniform convergence to *some* continuous function via the Cauchy criterion below and then naming it.) So $\sum_n \lambda_n e_n(x)^2$ converges uniformly in x.

Move 3: Cauchy–Schwarz transfers diagonal control to the full kernel, off the diagonal too. Uniform convergence on the diagonal is upgraded to absolute and uniform convergence of the two-variable series by a termwise Cauchy–Schwarz inequality on the tail. For $M < n \le N$,

$$\Big(\sum_{M<n\le N} \lambda_n e_n(x)e_n(y)\Big)^2 \;\le\; \Big(\sum_{M<n\le N} \lambda_n e_n(x)^2\Big)\Big(\sum_{M<n\le N} \lambda_n e_n(y)^2\Big),$$

which is legitimate precisely because the weights $\lambda_n \ge 0$ make $(u,v) \mapsto \sum \lambda_n u_n v_n$ a genuine inner product on coefficient tails. Both factors on the right are tails of the uniformly convergent diagonal series of Move 2, hence both are $\le \sup_x \sum_{n>M} \lambda_n e_n(x)^2 \to 0$ as $M \to \infty$, uniformly in x, y. So the kernel series satisfies the uniform Cauchy criterion and converges absolutely and uniformly on all of $[0,1]^2$. Its uniform limit is a continuous function that agrees with W in $L^2([0,1]^2)$ by eq. (18.1), hence agrees with the continuous W a.e., hence *everywhere* by continuity of both sides. Setting $x = y$ now licenses the diagonal identity and, integrating the uniform sum termwise, $\int_0^1 W(x,x)\,\mathrm{d}x = \sum_n \lambda_n = \mathrm{tr}(\mathbb{W})$. The classical reference is [113, Ch. VI]; see also [50]. ☐

Remark 18.3 (Graphons are not positive semidefinite in general)**.** This remark carries the real payload of the subsection, so we develop it carefully along three steps: the conceptual distinction that traps the unwary, an example where the diagonal identity survives anyway, and the genuine failure mode it does not.

The distinction. A graphon takes values in $[0,1]$, so $W \ge 0$ *pointwise* as a function on $[0,1]^2$. This is completely unrelated to positive *semidefiniteness* of the operator $\mathbb{W}$, which — as we saw in setting up Mercer — is the condition $\langle \mathbb{W}f, f\rangle \ge 0$ for all *signed* f, equivalently that every eigenvalue $\lambda_n \ge 0$. Pointwise nonnegativity of the kernel constrains its values; operator positivity constrains its spectrum; neither implies the other. The mismatch is structural, not exotic: most graphons one writes down have eigenvalues of both signs, and we already named why in section 18.1 — a negative eigenvalue is the spectral signature of a disassortative, bipartite-like mode, and such modes are ubiquitous. So Mercer's hypothesis is the exception, not the rule, among graphons.

An example where the identity holds anyway. Take the two-block model of example 18.11 with $p < q$. Its nonzero eigenvalues are $\lambda_1 = (p+q)/2 > 0$ and $\lambda_2 = (p-q)/2 < 0$, so it has a strictly negative eigenvalue, is not operator positive semidefinite, and is moreover discontinuous across the block boundary — it violates Mercer's hypotheses twice over, and the theorem simply does not apply to it. And yet the diagonal identity *still holds* for this kernel, which is the instructive point: Mercer is sufficient, not necessary. The reason is that this kernel is *finite rank*, so its eigenfunction series is a *finite* sum, which converges everywhere trivially, with no need of Dini or positivity. One checks the diagonal value by hand. The normalised eigenfunctions are $e_1 = \mathbf{1}$ (value 1 everywhere) and $e_2 = \mathbf{1}_{I_1} - \mathbf{1}_{I_2}$ (values ± 1), so $e_1(x)^2 = 1$ and $e_2(x)^2 = 1$ for a.e. x, and the finite diagonal sum is

$$\sum_n \lambda_n e_n(x)^2 = \lambda_1 \cdot 1 + \lambda_2 \cdot 1 = \frac{p+q}{2} + \frac{p-q}{2} = \frac{(p+q)+(p-q)}{2} = p = W(x,x),$$

the within-block value, and this holds regardless of the sign of $p - q$. So the negative eigenvalue does no harm here: finiteness of the sum, not positivity of its terms, is what makes a finite-rank diagonal expansion legitimate.

The genuine failure mode. What positivity actually protects against is the interaction of *infinitely many* sign-varying terms. For the diagonal identity to genuinely fail one needs an *infinite-rank*, continuous, non-positive-semidefinite kernel. There the entire mechanism of Mercer's proof collapses at Move 2: with eigenvalues of both signs the diagonal partial sums $S_N(x) = \sum_{n\le N} \lambda_n e_n(x)^2$ are *no longer monotone in* N — adding a term with $\lambda_n < 0$ decreases the sum — so they are not pinned between 0 and $W(x,x)$, Dini's hypothesis is unavailable, and there is no longer any reason for the series to converge to $W(x,x)$ at a given point. (It still converges in L^2, by eq. (18.1); what is lost is only the pointwise/diagonal value.) The positive and negative parts of the series can each diverge on the diagonal while their L^2 difference stays finite, and the would-be trace $\sum_n \lambda_n$ need not even converge absolutely. This is the real obstruction, and it is generic.

The consequence for what follows — read this as the instruction for the next proof. Because we cannot trust the diagonal evaluation $W(x,x) = \sum_n \lambda_n e_n(x)^2$ for a general graphon, and because the spectral moments we are about to compute would naively be read off exactly that diagonal, the next subsection must reach the trace-type identity by a route that *never touches the diagonal at all.* The device is to write each spectral power sum as a Hilbert–Schmidt inner product of two operator iterates $\mathbb{W}^a$ and $\mathbb{W}^b$ and evaluate that pairing as an honest $L^2([0,1]^2)$ integral of two kernels — an off-diagonal object to which the L^2 theory of eq. (18.1) applies unconditionally, with no appeal to Mercer, to positive semidefiniteness, or to pointwise convergence. That is precisely how proposition 18.4 is proved, and the present remark is the reason it is proved that way.

18.1.3 Spectral moments are cycle densities

The previous subsection ended with a precise instruction, and we now follow it. Mercer's theorem told us that the eigenfunction expansion may be summed pointwise — and in particular read off on the diagonal to give the trace — only when the operator is positive semidefinite, a hypothesis that *generically fails* for graphons. The naive route to the spectral moments, evaluate $\sum_n \lambda_n e_n(x)^2$ on the diagonal and integrate, is therefore closed to us. Remark 18.3 closed by naming the detour that *is* open: write each spectral power sum as a Hilbert–Schmidt inner product of two operator iterates and evaluate that pairing as an honest $L^2([0,1]^2)$ integral of two kernels — an off-diagonal object to which the L^2 convergence of eq. (18.1) applies with no Mercer, no positive semidefiniteness, no pointwise evaluation anywhere. This subsection carries out exactly that program, and the payoff is the hinge advertised in the chapter opening: the power sums of the spectrum *are* the cycle homomorphism densities of chapter 16. It is the single bridge across which the combinatorial machinery of chapter 16 and the metric continuity of chapter 17 reach into the spectrum, so it is worth proving slowly and completely.

Proposition 18.4 (Spectral moments)**.** *Let W satisfy* (A-Reg-W)*, with spectrum $\{\lambda_n\}$. For every integer $k \ge 2$ the spectral power sum converges absolutely and*

$$\sum_n \lambda_n^k \;=\; \int_{[0,1]^k} W(x_1,x_2)W(x_2,x_3)\cdots W(x_k,x_1)\;\mathrm{d}x_1\cdots \mathrm{d}x_k \;=\; t(C_k,W), \tag{18.2}$$

the homomorphism density of the k-cycle C_k (chapter 16). For $k=2$ this reads $\sum_n \lambda_n^2 = \|W\|^2_{L^2([0,1]^2)} = t(C_2,W)$.

Proof. The proof has a clean four-part architecture, and it is worth seeing the shape before the algebra: (i) the iterated operators $\mathbb{W}^j$ are again Hilbert–Schmidt, which keeps us inside the apparatus chapter 2 already built and away from trace-class theory the book has not developed; (ii) the kernel of $\mathbb{W}^j$ is the j-fold convolution of W; (iii) we split the exponent $k = a + b$ into two pieces, each at least 1 — this is the one place the hypothesis $k \geq 2$ is spent; and (iv) we compute a single Hilbert–Schmidt inner product $\langle \mathbb{W}^a, \mathbb{W}^b \rangle_{\mathrm{HS}}$ *two ways*, once spectrally and once as a kernel integral, and equate the answers. The spectral evaluation produces $\sum_n \lambda_n^k$; the integral evaluation produces the closed cycle. Throughout, the only convergence we ever invoke is the unconditional $L^2([0,1]^2)$ convergence of eq. (18.1); no eigenfunction is ever evaluated at a point, so Mercer plays no role.

Step (i): the iterates are Hilbert–Schmidt. Fix an integer $j \geq 1$. We claim the operator $\mathbb{W}^j$ is again Hilbert–Schmidt. Here $\mathbb{W}$ itself is Hilbert–Schmidt by proposition 2.24, and the remaining factor $\mathbb{W}^{j-1}$ is at least bounded (a power of a bounded operator). The Hilbert–Schmidt operators form a *two-sided ideal* inside the bounded operators: the product of a Hilbert–Schmidt operator with any bounded operator, on either side, is again Hilbert–Schmidt. So $\mathbb{W}^j = \mathbb{W} \cdot \mathbb{W}^{j-1}$ is Hilbert–Schmidt. This is exactly the property that lets us stay self-contained. To read $\sum_n \lambda_n^k$ as a trace of $\mathbb{W}^k$ directly would require the trace-class theory of chapter 2 — nuclear operators, the trace functional, and its continuity — which the book has deliberately not built. The ideal property gives us all we need with strictly less: every iterate lives in the *Hilbert*–Schmidt class, where the inner product $\langle \cdot, \cdot \rangle_{\mathrm{HS}}$ and its kernel-integral formula definition 2.21 are available, and that is the whole arena of the proof.

Step (ii): the kernel of the iterate is the convolution. By the spectral theorem theorem 2.26, $\mathbb{W}^j$ shares the eigenfunctions e_n of $\mathbb{W}$ and raises each eigenvalue to the j-th power, $\mathbb{W}^j e_n = \lambda_n^j e_n$. At the level of kernels, composing integral operators convolves their kernels: the kernel–operator dictionary theorem 2.22 says that if $\mathbb{W}$ has kernel W then $\mathbb{W}^j$ has the j-fold convolution kernel

$$W^{(j)}(x, y) = \int_{[0,1]^{j-1}} W(x, x_2) W(x_2, x_3) \cdots W(x_j, y) \, \mathrm{d}x_2 \cdots \mathrm{d}x_j,$$

with the convention $W^{(1)} = W$. The picture to carry is combinatorial: $W^{(j)}(x, y)$ is the total W-weight of all length-j walks from x to y, the continuum analogue of the matrix-power entry $(A^j)_{uv}$ counting length-j walks in a finite graph. The internal vertices $x_2, \ldots, x_j$ are summed (integrated) over; only the two endpoints x, y remain free.

Step (iii): split the exponent, and see where $k \geq 2$ is used. Write

$$k = a + b, \qquad a = \lfloor k/2 \rfloor \geq 1, \qquad b = \lceil k/2 \rceil \geq 1,$$

so both exponents are genuine positive integers and $a + b = k$. This is the precise point at which the hypothesis $k \geq 2$ is spent and cannot be relaxed: only for $k \geq 2$ can we split k into two summands each ≥ 1. The exponents must both be ≥ 1 for two independent reasons that recur below. First, $\mathbb{W}^a$ and $\mathbb{W}^b$ are then both genuine iterates (Step (i)), so they annihilate $\ker \mathbb{W}$ and the spectral sum below ranges only over the nonzero eigenvalues. Second, the integral evaluation needs each side to be an honest walk of length ≥ 1, so that gluing them produces a cycle with $k \geq 2$ edges. The forbidden case $k = 1$ would demand $a = 1, b = 0$, i.e. pairing $\mathbb{W}$ against the *identity* — which is not Hilbert–Schmidt on the infinite-dimensional $L^2[0,1]$ — and would land us right back on the diagonal trace $\sum_n \lambda_n$ that remark 18.3 showed need not even converge. The split is exactly the device that keeps us off the diagonal.

Step (iv), first evaluation: spectral. We compute the Hilbert–Schmidt inner product $\langle \mathbb{W}^a, \mathbb{W}^b \rangle_{\text{HS}}$ against a convenient orthonormal basis of $L^2[0,1]$: the eigenfunctions $\{e_n\}$ for the nonzero eigenvalues, completed by an orthonormal basis of $\ker \mathbb{W}$. On the kernel basis vectors both $\mathbb{W}^a$ and $\mathbb{W}^b$ vanish (since $a, b \geq 1$), so those terms contribute nothing, and only the e_n survive:

$$\langle \mathbb{W}^a, \mathbb{W}^b \rangle_{\text{HS}} = \sum_n \langle \mathbb{W}^a e_n, \mathbb{W}^b e_n \rangle = \sum_n \langle \lambda_n^a e_n, \lambda_n^b e_n \rangle = \sum_n \lambda_n^a \lambda_n^b \langle e_n, e_n \rangle = \sum_n \lambda_n^{a+b} = \sum_n \lambda_n^k,$$

where the middle steps use $\mathbb{W}^a e_n = \lambda_n^a e_n$ and $\mathbb{W}^b e_n = \lambda_n^b e_n$ from Step (ii) and the orthonormality $\langle e_n, e_n \rangle = 1$. This is the first of the two faces of the same number: the k-th power sum of the spectrum.

Why the sum converges absolutely. The series $\sum_n \lambda_n^k$ is unconditionally summable, and the bound is worth recording because it is also what makes the moment *finite and uniformly controlled* across all graphons. Every eigenvalue obeys $|\lambda_n| \leq \|\mathbb{W}\| \leq \|W\|_\infty$, the second inequality because the operator norm of an averaging kernel against a unit-mass type space is bounded by the supremum of the kernel (a Schur-test bound: $\|\mathbb{W}\| \leq \sup_x \int_0^1 |W(x,y)| \, \mathrm{d}y \leq \|W\|_\infty$). Peeling off two of the k factors and bounding the remaining $k-2$ each by $\|W\|_\infty$,

$$\sum_n |\lambda_n|^k = \sum_n |\lambda_n|^{k-2} \; \lambda_n^2 \;\leq\; \|W\|_\infty^{k-2} \sum_n \lambda_n^2 \;=\; \|W\|_\infty^{k-2} \; \|W\|_{\text{HS}}^2 \;<\; \infty,$$

finite for every $k \geq 2$ since $\sum_n \lambda_n^2 = \|W\|_{\text{HS}}^2 < \infty$ by proposition 2.24. The two retained factors are exactly the λ_n^2 that the $k = 2$ identity controls; the leftover $k-2$ factors are precisely why the simple-cycle range will be $k \geq 3$ rather than $k \geq 2$ when we hand off below. For a graphon proper, $\|W\|_\infty \leq 1$, so in fact $\sum_n |\lambda_n|^k \leq \sum_n \lambda_n^2 \leq 1$ for every $k \geq 2$ — the moments are bounded by 1, uniformly over all graphons at once.

Step (iv), second evaluation: kernel integral, and why it closes a cycle. The same inner product has a second face. By the integral form of the Hilbert–Schmidt inner product (definition 2.21), pairing two Hilbert–Schmidt operators equals pairing their kernels in $L^2([0,1]^2)$:

$$\langle \mathbb{W}^a, \mathbb{W}^b \rangle_{\text{HS}} = \int_{[0,1]^2} W^{(a)}(x,y) \, W^{(b)}(x,y) \; \mathrm{d}x \; \mathrm{d}y.$$

Now look at what this double integral is, geometrically. By Step (ii), $W^{(a)}(x,y)$ is the total weight of all length-a walks from x to y, and $W^{(b)}(x,y)$ the total weight of all length-b walks between the same two endpoints. The product, integrated over x and y, glues a length-a walk and a length-b walk *at both shared endpoints*. Here symmetry does the essential work: because $W(x,y) = W(y,x)$, the second walk reads equally well backwards, $W^{(b)}(x,y) = W^{(b)}(y,x)$, a length-$b$ walk from y *back to* x. So one factor carries us out from x to y in a steps, the other carries us home from y to x in b steps, and the two legs together form a single *closed* walk of $a + b = k$ edges that returns to its start. Writing out both convolutions and relabelling the integration variables $x = x_1$, $y = x_{a+1}$ together with the internal vertices in order, the glued object is precisely the closed k-cycle integral

$$\int_{[0,1]^k} W(x_1,x_2) W(x_2,x_3) \cdots W(x_k,x_1) \; \mathrm{d}x_1 \cdots \mathrm{d}x_k,$$

each consecutive pair (x_i, x_{i+1}) contributing one edge of W and the final pair (x_k, x_1) closing the loop. This is exactly $t(C_k, W)$, the homomorphism density of the k-cycle, by the definition of cycle density in chapter 16. The split exponent is what makes the geometry honest: gluing a walk to a walk along a shared endpoint *is* a cycle only when each walk has at least one edge, which is again the content of $a, b \geq 1$.

Conclusion. The two evaluations compute one and the same number $\langle \mathbb{W}^a, \mathbb{W}^b \rangle_{\mathrm{HS}}$, so equating them gives eq. (18.2), $\sum_n \lambda_n^k = t(C_k, W)$. The boundary case $k = 2$ uses $a = b = 1$, where both iterates are $\mathbb{W}$ itself and the identity degenerates to $\langle \mathbb{W}, \mathbb{W} \rangle_{\mathrm{HS}} = \sum_n \lambda_n^2 = \|W\|_{\mathrm{HS}}^2$ of proposition 2.24 on one side and $\int_{[0,1]^2} W(x,y)^2$, the density $t(C_2, W)$ of the doubled edge, on the other. We emphasise the cash value of the previous subsection's warning: *at no point did we evaluate an eigenfunction expansion on the diagonal*, and so at no point did we invoke Mercer's theorem or positive semidefiniteness. Every object handled was an off-diagonal $L^2([0,1]^2)$ pairing of kernels, where the unconditional convergence of eq. (18.1) is all that is needed. This is precisely the diagonal-free route remark 18.3 promised, and it is why the moment identity holds for *every* graphon, positive semidefinite or not. □

We extract from the $k = 2$ case the one quantitative fact the rest of the chapter runs on, and state it sharply enough to quote. Setting $k = 2$ in eq. (18.2) collapses three a priori distinct quantities into a single number: the sum of squared eigenvalues, the squared Hilbert–Schmidt (equivalently L^2) norm of the kernel, and the homomorphism density of the doubled edge C_2 all coincide,

$$\sum_n \lambda_n^2 = \|W\|_{\mathrm{HS}}^2 = t(C_2, W) \leq 1,$$

the final bound because $0 \leq W \leq 1$ forces $W^2 \leq W \leq 1$ pointwise, hence $\int_{[0,1]^2} W^2 \leq 1$. The spectrum of any graphon therefore lives in the unit ℓ^2 ball, and this immediately limits how many eigenvalues can be large.

Lemma 18.5 (Uniform ℓ^2 tightness of graphon spectra)**.** *For every graphon W and every threshold $\varepsilon > 0$, the number of eigenvalues λ_n with $|\lambda_n| > \varepsilon$ is at most ε^{-2}. The bound is the same for* every *graphon, with no dependence on W beyond the standing assumption* (A-Reg-W)*.*

The proof is one line: if more than ε^{-2} eigenvalues exceeded ε in modulus, their squares alone would sum to more than $\varepsilon^{-2} \cdot \varepsilon^2 = 1 \geq \sum_n \lambda_n^2$, a contradiction. The point worth underlining is the *uniformity*: the count is governed entirely by the universal bound $\sum_n \lambda_n^2 \leq 1$, so a single tightness estimate holds across the whole space of graphons at once. This is what makes the spectra of a cut-convergent sequence *simultaneously* tight, and it is the engine that powers the spectral convergence theorem of section 18.4: there one extracts a limiting eigenvalue list by a diagonal argument over indices that succeeds precisely because, uniformly in the sequence, only finitely many eigenvalues can sit above any given level (compare the "uniform tightness" step of lemma 18.13, which is this very count).

The moment identity hands section 18.4 a second thing, more subtle than the tightness bound but just as load-bearing, and it is a restriction rather than a gift. The convergence theorem will not be allowed to use the $k = 2$ moment. Cut convergence delivers convergence of $t(F, \cdot)$ only for *finite simple* graphs F (the counting lemma of chapter 16 is stated for simple graphs), and the doubled edge C_2 is not simple — it is the single edge traversed twice. Concretely, $t(C_2, \cdot) = \|\cdot\|_{\mathrm{HS}}^2$ is *not* cut-continuous: the right inequality $\|U\|_{2\to 2} \le \|U\|_{\mathrm{HS}}$ of the norm hierarchy eq. (18.5) is strict precisely because the Hilbert–Schmidt norm can escape cut control, so a sequence can converge in cut metric while its $k = 2$ moments do not converge. The simple cycles begin at $k = 3$ (the triangle), and it is exactly the power sums $\sum_n \lambda_n^k = t(C_k, W)$ for $k \ge 3$ that cut convergence will deliver. The role of the $k = 2$ identity in the convergence proof is thus inverted from what one might expect: it is never a convergence *hypothesis* but only the standing, universal *bound* $\sum_n \lambda_n^2 \le 1$ of lemma 18.5 — supplying tightness, never a limit. With the power sums $k \ge 3$ converging and the $k = 2$ bound clamping the whole spectrum into the ℓ^2 ball, section 18.4 will be able to pass from convergence of cycle densities to convergence of individual eigenvalues, eigenvalue by eigenvalue. That is the next movement.

18.2 The leading eigenfunction is eigenvector centrality

The first movement put the entire spectrum on the table and brought it under control: the eigenvalues $\{\lambda_n\}$ are the cycle homomorphism densities $\sum_n \lambda_n^k = t(C_k, W)$ of proposition 18.4, they are listed in the two orderings of definition 18.1, and the $k = 2$ identity clamps the whole sequence into the unit ℓ^2 ball (lemma 18.5). With the eigen*values* thus accounted for, the natural next question turns to the eigen*functions* — and not to all of them at once, but to the single one that matters most. This movement narrows the focus from the whole spectrum to its top entry λ_1^+ and the mode e_1 that sits there.

Why fasten on that one mode? Because in a graphon mean field game the coupling *is* the operator $\mathbb{W}$ (the chapter opening), and a population profile pushed repeatedly through $\mathbb{W}$ aligns itself, mode by mode, with the eigenfunction of largest eigenvalue: the subdominant modes decay relative to it at the rate $|\lambda_2| / |\lambda_1|$ we will quantify in section 18.3, so what survives in the limit of repeated averaging is a scalar multiple of e_1 alone. The leading eigenfunction is therefore the shape into which the network's coupling collapses everything else — the dominant channel through which any one type feels the entire population. An object that load-bearing deserves a name and a meaning, and network science supplies both.

The name is *eigenvector centrality*, and its informal definition is the oldest circular idea in the subject: a node is central to the degree that it is linked to central nodes. As it stands this is no definition at all, since it explains centrality in terms of itself; but the circularity is exactly the kind an eigenvalue equation dissolves. Write $c(x)$ for the centrality of type x and impose the self-referential demand literally — that $c(x)$ be proportional to the W-weighted average of its neighbours' centralities,

$$\int_0^1 W(x,y)\, c(y)\, \mathrm{d}y \;=\; \lambda\, c(x) \qquad \text{for some constant } \lambda,$$

which is precisely the eigenvalue equation $\mathbb{W}c = \lambda c$. The self-reference becomes a request for an eigenfunction. The only remaining question is *which* eigenfunction, of the many $\mathbb{W}$ possesses, carries the meaning — and two requirements of the interpretation, not of the algebra, single one out. A centrality must be *nonnegative*, since no node is "negatively central"; and on a connected network it must be *unique up to overall scale*, so that the ranking of types it induces is well defined rather than an artefact of which eigenfunction one happened to pick. The content of this section is that exactly one eigenfunction meets both requirements — the one belonging to the top eigenvalue — and that under the right hypothesis it is automatically strictly positive, so that every type receives a genuinely positive score.

Establishing this also discharges a promise left open in the first movement. The notationwatch of section 18.1 asserted that for a *nonnegative* kernel the two orderings of definition 18.1 agree at the very top, $\lambda_1^+ = |\lambda_1| = \rho(\mathbb{W})$ — a nonnegative kernel cannot conceal a negative eigenvalue larger in modulus than its largest positive one — and deferred the proof to here. It is part (1) of the theorem below. The leading positive mode and the spectral radius turn out to be the same number for a structural reason worth previewing: nonnegativity forces the extreme of the Rayleigh quotient to be attained at a nonnegative test function, and there the quadratic form is itself nonnegative, so the assortative top mode always wins the competition for the largest modulus. The vehicle for all of this is the Perron–Frobenius theorem, whose finite-graph version the reader already knows; the work is to lift it to the integral operator $\mathbb{W}$, and that lift needs exactly one continuum hypothesis — the analogue of a finite graph being connected. Identifying that hypothesis correctly is the first order of business.

18.2.1 Perron–Frobenius for the graphon operator

For a finite graph with nonnegative adjacency matrix A, the Perron–Frobenius theorem says that the largest eigenvalue is real and positive, that it is attained by a nonnegative eigenvector, and — if the graph is connected — that this eigenvector is unique up to scale and strictly positive. The entries of that eigenvector are precisely the *eigenvector centralities* of the nodes: the circular definition of the previous paragraphs, made into the components of a genuine eigenvector, with the proportionality constant λ identified as the leading eigenvalue.

It pays to watch the circular definition dissolve into an eigenequation once, in the smallest nontrivial finite case, because the graphon argument below is the very same move performed in $L^2[0,1]$ rather than in $\mathbb{R}^n$. Take the path on three vertices $1-2-3$, with adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The informal demand "each c_i is proportional, by a single shared constant, to the sum of its neighbours' centralities" reads $\lambda c_1 = c_2$, $\lambda c_2 = c_1 + c_3$, $\lambda c_3 = c_2$ — three coupled proportionalities with one common constant of proportionality, which is exactly the eigenequation $Ac = \lambda c$. Solving it, the three eigenvalues are $\lambda = \sqrt{2}$, 0, $-\sqrt{2}$. The largest, $\lambda_1 = \sqrt{2} > 0$, carries the eigenvector $c = (1, \sqrt{2}, 1)$: every entry strictly positive, with the middle vertex scored most central because it lies on every short path through the graph. Neither competitor is nonnegative — $\lambda = 0$ gives $(1, 0, -1)$ and $\lambda = -\sqrt{2}$ gives $(1, -\sqrt{2}, 1)$, each forced to change sign. So all three features the theorem promises are already visible in this 3×3 picture: a real positive top eigenvalue, an everywhere-positive eigenvector sitting at it, and a mandatory sign change in every other eigenvector. The work of this section is to show that these features survive *verbatim* when the finite matrix A is replaced by the integral operator $\mathbb{W}$ on the continuum of types.

We now lift the statement, verbatim in spirit, to the graphon operator $\mathbb{W}$. The only piece that does not transcribe word for word is the word "connected": a continuum of types has no edges to count, so connectedness must be re-expressed in a form that survives the passage from a finite vertex set to $[0, 1]$.

In a finite graph connectedness fails in exactly one way: the vertices split into two nonempty groups with no edges running between them, and after relabelling the adjacency matrix becomes block-diagonal. The continuum analogue replaces "two nonempty groups" by "two sets of positive measure" and "no edges between them" by "no kernel weight between them." That is the content of the next definition.

Definition 18.6 (Connected graphon)**.** A graphon W is *connected* (operator-theoretically: $\mathbb{W}$ is *irreducible*) if there is no measurable $S \subseteq [0, 1]$ with $0 < |S| < 1$ such that $W = 0$ almost everywhere on $S \times S^c$. Equivalently, $[0, 1]$ cannot be split into two parts of positive measure with no W-weight between them.

It is worth seeing concretely what connectedness buys, by watching the conclusion fail without it. Suppose W is *dis*connected: $[0, 1] = S \sqcup S^c$ with both pieces of positive measure and $W = 0$ a.e. on $S \times S^c$. Then $\mathbb{W}$ does not mix the two pieces at all — it splits as a direct sum $\mathbb{W} = \mathbb{W}_S \oplus \mathbb{W}_{S^c}$ of two independent operators, one acting on $L^2(S)$ and one on $L^2(S^c)$, exactly as a block-diagonal adjacency matrix decouples into its blocks. Each block carries its own leading eigenvalue and its own nonnegative leading eigenfunction, the latter supported entirely on its own piece. If the two block leading eigenvalues happen to coincide — as they do, for instance, when S and S^c host identical sub-graphons — then the top eigenspace of $\mathbb{W}$ is two-dimensional, and it contains a whole cone of nonnegative eigenfunctions: every nonnegative combination $\alpha\, e_S + \beta\, e_{S^c}$ of the two block modes is again a nonnegative leading eigenfunction. Centrality is then genuinely ambiguous. There is no principled way to weigh a node in the first component against a node in the second, precisely as one cannot rank vertices across the connected components of a disconnected graph: the relative scale of the two blocks is free. Connectedness is exactly the hypothesis that forbids this splitting and so restores the uniqueness the interpretation needs. With it stated we can give the theorem in full.

Theorem 18.7 (Perron–Frobenius / Krein–Rutman for graphons)**.** *Let W satisfy* (A-Reg-W) *with $W \geq 0$, and suppose W is not the zero graphon. Then:*

1. *the spectral radius $\rho(\mathbb{W}) = \|\mathbb{W}\|$ is itself an eigenvalue, the largest one, so $\lambda_1^+ = \rho(\mathbb{W}) = |\lambda_1| > 0$;*

2. *there is a corresponding eigenfunction $e_1 \geq 0$ almost everywhere;*

3. *if in addition W is connected, then λ_1^+ is simple (its eigenspace is one-dimensional) and $e_1 > 0$ almost everywhere; moreover e_1 is, up to sign and scaling, the only nonnegative eigenfunction of $\mathbb{W}$.*

Proof. (1)–(2): the top eigenvalue is positive and carries a nonnegative eigenfunction. By the Courant–Fischer principle corollary 2.27, the largest positive eigenvalue is the maximum of the

Rayleigh quotient over the unit sphere, $\lambda_1^+ = \max_{\|g\|=1}\langle \mathbb{W}g, g\rangle$, and compactness of $\mathbb{W}$ supplies an actual maximiser — some unit vector g attaining the supremum — exactly as in the construction of the first eigenpair inside the proof of theorem 2.26. That this maximum is genuinely *positive*, not merely nonnegative, uses that W is a nonzero nonnegative kernel: testing the quotient against the constant function $\mathbf{1}$ gives $\lambda_1^+ \geq \langle \mathbb{W}\mathbf{1}, \mathbf{1}\rangle = \iint_{[0,1]^2} W > 0$, the total edge weight, which is strictly positive because W is not the zero graphon.

The crucial step is that the maximiser can be taken nonnegative. Replacing g by its pointwise modulus $|g|$ does not change its norm, $\||g|\| = \|g\| = 1$, so $|g|$ is still admissible in the maximisation; and because $W \geq 0$, passing to moduli can only raise the quadratic form,

$$\langle \mathbb{W}\,|g|\,, |g|\rangle = \iint W(x,y)\,|g(x)|\;|g(y)|\;\mathrm{d}x\,\mathrm{d}y \;\geq\; \Big|\iint W(x,y)g(x)g(y)\;\mathrm{d}x\,\mathrm{d}y\Big| = |\langle \mathbb{W}g, g\rangle| = \lambda_1^+.$$

The inequality is the triangle inequality for the integral, valid term by term precisely because the weight $W(x,y)$ is nonnegative, so that $W\,|g(x)|\,|g(y)| \geq W\,g(x)g(y)$ pointwise. But λ_1^+ is the *maximum*, so $\langle \mathbb{W}\,|g|\,, |g|\rangle \leq \lambda_1^+$ as well; the two inequalities pinch, $\langle \mathbb{W}\,|g|\,, |g|\rangle = \lambda_1^+$, and $|g|$ is itself a maximiser. Set $e_1 := |g| \geq 0$.

A maximiser of the Rayleigh quotient on the unit sphere is automatically an eigenfunction for the top eigenvalue, and it is worth doing the variation explicitly, since it is the same stationarity computation that produces the leading eigenpair inside theorem 2.26. We are extremising $Q(h) = \langle \mathbb{W}h, h\rangle$ subject to $\|h\|^2 = 1$; introduce a Lagrange multiplier μ for the constraint and consider, at the maximiser e_1, an admissible perturbation $e_1 + \varepsilon\varphi$ with $\varphi \in L^2[0,1]$ arbitrary. Using self-adjointness of $\mathbb{W}$,

$$Q(e_1+\varepsilon\varphi) = \langle \mathbb{W}e_1, e_1\rangle + 2\varepsilon\,\langle \mathbb{W}e_1, \varphi\rangle + \varepsilon^2\langle \mathbb{W}\varphi,\varphi\rangle, \qquad \|e_1+\varepsilon\varphi\|^2 = 1 + 2\varepsilon\langle e_1,\varphi\rangle + \varepsilon^2\,\|\varphi\|^2\,,$$

so the first-order stationarity condition $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\big[Q - \mu\,\|\cdot\|^2\big]\big|_{\varepsilon=0} = 0$ reads $2\langle \mathbb{W}e_1, \varphi\rangle - 2\mu\langle e_1, \varphi\rangle = 0$, that is $\langle \mathbb{W}e_1 - \mu e_1, \varphi\rangle = 0$ for *every* φ. A vector orthogonal to all of $L^2[0,1]$ is zero, so $\mathbb{W}e_1 = \mu e_1$: the maximiser is an eigenfunction. Pairing this identity with e_1 and using $\|e_1\| = 1$ identifies the multiplier with the constrained maximum, $\mu = \langle \mathbb{W}e_1, e_1\rangle = \lambda_1^+$. Hence $\mathbb{W}e_1 = \lambda_1^+ e_1$ with $e_1 \geq 0$. Finally λ_1^+ equals the spectral radius. The operator norm of a self-adjoint operator is $\|\mathbb{W}\| = \sup_{\|g\|=1}|\langle \mathbb{W}g, g\rangle|$, and we have just exhibited the supremum being approached through nonnegative test functions, where $\langle \mathbb{W}g, g\rangle \geq 0$; so the extreme of the modulus is attained on the positive side, $\|\mathbb{W}\| = \lambda_1^+$. A nonnegative kernel therefore cannot hide a negative eigenvalue of larger modulus than its top positive one, and $\lambda_1^+ = |\lambda_1| = \rho(\mathbb{W})$ — the identity promised in the notationwatch of section 18.1 and used throughout the signed bookkeeping of definition 18.1.

(3): connectedness forces strict positivity, simplicity, and uniqueness. Assume now that W is connected. We prove the three claims in the order they depend on one another; strict positivity comes first because both simplicity and the orthogonality argument lean on it.

Strict positivity, $e_1 > 0$ a.e. Let $Z = \{e_1 = 0\}$ be the zero set of the leading eigenfunction and suppose, for contradiction, that $|Z| > 0$. For a.e. $x \in Z$ the eigenvalue equation reads

$$0 = \lambda_1^+\,e_1(x) = (\mathbb{W}e_1)(x) = \int_0^1 W(x,y)\,e_1(y)\;\mathrm{d}y.$$

The integrand is a product of two nonnegative functions, $W \geq 0$ and $e_1 \geq 0$, so a vanishing integral forces the integrand itself to vanish a.e.: for a.e. $x \in Z$ we have $W(x,y)\,e_1(y) = 0$ for a.e. y. Wherever $e_1(y) > 0$ — that is, for a.e. $y \in Z^c$ — this can only hold if $W(x,y) = 0$. We have shown $W = 0$ a.e. on $Z \times Z^c$. Were Z^c also of positive measure, this would split $[0,1]$ into the two positive-measure pieces Z and Z^c with no kernel weight between them — exactly the disconnection definition 18.6 forbids. Hence $|Z^c| = 0$, i.e. $|Z| = 1$; but that says $e_1 = 0$ a.e., contradicting $\|e_1\| = 1$. The only escape is $|Z| = 0$, so $e_1 > 0$ almost everywhere: on a connected network every type has strictly positive centrality.

Simplicity, $\dim\ker(\mathbb{W} - \lambda_1^+) = 1$. The strict positivity just established is the lever for everything that follows. Our goal is to show that every eigenfunction for λ_1^+ is a scalar multiple of e_1, so that the top eigenspace is a single line. Since $\mathbb{W}$ is real and self-adjoint, each eigenspace is spanned by real eigenfunctions — a complex one splits into real and imaginary parts that each separately solve $\mathbb{W}v = \lambda_1^+ v$ — so it suffices to rule out a *second* real eigenfunction independent of e_1. We argue by contradiction, and strict positivity is again the lever.

Suppose the top eigenspace had dimension at least two. Applying Gram–Schmidt inside it (and taking real parts) we could then produce a real eigenfunction ψ for λ_1^+ with $\|\psi\| = 1$ and $\psi \perp e_1$. The decisive observation is that $|\psi|$ is *also* an eigenfunction for λ_1^+. Because $W \geq 0$, the triangle inequality for the integral gives, pointwise a.e.,

$$\lambda_1^+\ |\psi(x)| = |(\mathbb{W}\psi)(x)| = \Big|\int_0^1 W(x,y)\,\psi(y)\ \mathrm{d}y\Big| \ \leq\ \int_0^1 W(x,y)\ |\psi(y)|\ \mathrm{d}y = (\mathbb{W}\,|\psi|)(x),$$

so $\mathbb{W}\,|\psi| \geq \lambda_1^+\,|\psi|$ a.e. Integrating this inequality against the nonnegative function $|\psi|$, and using $\||\psi|\| = \|\psi\| = 1$, gives $\langle \mathbb{W}\,|\psi|\,, |\psi|\rangle \geq \lambda_1^+$. But $\lambda_1^+ = \max_{\|g\|=1}\langle \mathbb{W}g, g\rangle$ is the constrained maximum of part (1)–(2), so the reverse inequality holds as well; the two pinch, $|\psi|$ is itself a unit maximiser, and the stationarity computation of part (1)–(2) upgrades any unit maximiser to an eigenfunction, $\mathbb{W}\,|\psi| = \lambda_1^+\,|\psi|$.

Now strict positivity closes the argument. The function $g := |\psi| - \psi$ is a difference of two eigenfunctions for λ_1^+, hence itself one, and it is nonnegative, $g \geq 0$ a.e. The strict-positivity dichotomy just proved therefore applies to it: either $g \equiv 0$ or $g > 0$ almost everywhere, with no middle ground. The orthogonality $\psi \perp e_1$ decides which branch holds. Since $e_1 > 0$ a.e. and $\psi \not\equiv 0$, the constraint $\int_0^1 \psi\, e_1 = 0$ forces ψ to be strictly positive on a set of positive measure (and strictly negative on another): were $\psi \geq 0$ a.e. the integral against the strictly positive weight e_1 could not vanish, and likewise for $\psi \leq 0$. On the set $\{\psi > 0\}$ we have $g = |\psi| - \psi = 0$, so g vanishes on a set of positive measure, and the dichotomy can then be satisfied only by $g \equiv 0$, i.e. $\psi = |\psi| \geq 0$ a.e. That contradicts ψ being strictly negative on a set of positive measure. Hence no real eigenfunction orthogonal to e_1 exists, the top eigenspace is one-dimensional, and λ_1^+ is simple. This is exactly where connectedness is indispensable: it is the hypothesis that powered the dichotomy, and without the dichotomy the orthogonal companion ψ could survive as a second eigenfunction independent of e_1 — precisely the two-dimensional top eigenspace the disconnected block-diagonal example exhibited above.

Uniqueness of the nonnegative eigenfunction. It remains to see that no *other* eigenfunction — one belonging to a different eigenvalue — can be nonnegative, so that e_1 is, up to sign and scale, the only nonnegative eigenfunction $\mathbb{W}$ possesses. Any e_n with $n \geq 2$ is orthogonal to e_1, $\int_0^1 e_n e_1 = 0$, by the orthonormality of the spectral basis. But $e_1 > 0$ a.e., so an integral $\int_0^1 e_n e_1 = 0$ against a strictly positive weight is impossible unless e_n takes both signs: e_n must change sign. Hence the leading mode is the unique nonnegative eigenfunction, exactly as the interpretation of centrality demanded. The general statement, of which this is the graphon instance, is the Krein–Rutman theorem on the positive eigenvector of a positivity-preserving compact operator [113, Ch. VI]; the graphon form is in [99, Ch. 7]. □

Definition 18.8 (Eigenvector centrality)**.** For a connected nonnegative graphon W, the *eigenvector centrality* is the unique nonnegative, L^2-normalised leading eigenfunction $c = e_1$, characterised by the self-consistency relation

$$c(x) \;=\; \frac{1}{\lambda_1^+} \int_0^1 W(x,y)\, c(y)\, \mathrm{d}y, \qquad c \geq 0, \quad \|c\|_{L^2} = 1. \tag{18.3}$$

Equation eq. (18.3) is the promised circular definition made rigorous: the centrality of type x is the W-weighted average of its neighbours' centralities, rescaled by λ_1^+. Read as a self-consistency condition it says that c is a fixed point of the normalised averaging operator $\mathbb{W}/\lambda_1^+$, and theorem 18.7 guarantees this fixed point both exists and — on a connected graphon — is the *only* nonnegative one, so the centrality ranking it induces is unambiguous. It is illuminating to see the fixed point solved in the one case where it is fully explicit, the rank-one kernel $W(x,y) = f(x)f(y)$ with $f \geq 0$ and $\|f\|_{L^2} > 0$. Here $(\mathbb{W}g)(x) = f(x)\langle f, g\rangle$, so any eigenfunction with nonzero eigenvalue must be proportional to f; normalising gives $c = f/\,\|f\|_{L^2}$ with leading eigenvalue $\lambda_1^+ = \|f\|_{L^2}^2$, and one checks eq. (18.3) directly: $\frac{1}{\lambda_1^+}\mathbb{W}c = \frac{1}{\|f\|^2} f\langle f, c\rangle = \frac{f}{\|f\|^2} \cdot \frac{\|f\|^2}{\|f\|} = \frac{f}{\|f\|} = c$. Centrality is just the profile f itself — the single coordinate that the whole rank-one network runs on.

That last computation is not a toy: it is the seed of the closed-form graphon mean field game carried as Thread B. In the static graphon games of chapter 20 a type's optimal action at equilibrium is, to leading order in the coupling strength, proportional to its eigenvector centrality $c(x)$ — the more central a type, the more population pressure it feels and the more effort it exerts in response. The reason a *closed-form* comparative static exists there is precisely eq. (18.3): because equilibrium effort solves the same linear fixed-point relation as centrality, the entire equilibrium profile is pinned by a single eigenfunction rather than by an infinite-dimensional system, and on a rank-one kernel that eigenfunction collapses to the scalar order parameter $\varrho = \langle f, \bar{m}\rangle$ of the thread, reducing the game to one self-consistent scalar equation. The existence and uniqueness supplied by theorem 18.7 are what make that reduction legitimate: without the Perron eigenvalue being simple and its eigenfunction strictly positive, "the" equilibrium effort profile would not be well defined. The centrality fixed point is thus handed forward to chapter 20 as the backbone of every static closed form to come.

Physics note

The eigenvalue problem $\mathbb{W}c = \lambda_1^+ c$ is the ground state of the single-particle "Hamiltonian" $-\mathbb{W}$ on the type line: the leading eigenfunction is the lowest-energy mode, and on a connected (irreducible) network it is nodeless — strictly positive — exactly as the ground-state wave-function of a Schrödinger operator with no internal boundary is nodeless. Centrality is thus the network's ground-state profile, the shape into which any nonnegative input relaxes under repeated coupling by $\mathbb{W}$. The higher eigenfunctions e_n, $n \geq 2$, are the excited modes; they must have nodes (sign changes), and section 18.3 reads those nodes as community boundaries. One caution keeps the analogy honest: the "type line" is the label interval $[0, 1]$ carrying *no* intrinsic geometry — $|x - y|$ has no meaning, the types being exchangeable labels rather than spatial positions (the standing convention of appendix A.5). The "Hamiltonian" $-\mathbb{W}$ is an integral, not a differential, operator, and "nodeless" here refers to sign, not to any spatial smoothness. The proof above uses no physics; the analogy is a mnemonic for which eigenfunction is which.

We have isolated the top of the spectrum and given it a meaning: the leading eigenvalue $\lambda_1^+ = \rho(\mathbb{W})$ is isolated and, on a connected graphon, simple, and its eigenfunction e_1 is the strictly positive eigenvector centrality. Two threads leave this section. The simplicity of λ_1^+ and the strict positivity of e_1 are precisely the non-degeneracy hypotheses that the Davis–Kahan perturbation bound of section 18.4 will require: a simple, isolated eigenvalue is exactly the situation in which the leading eigenfunction moves continuously — rather than jumping — when the kernel is perturbed, and that stability is what lets convergence of eigenvalues be upgraded to convergence of the centrality profile itself. The other thread is the gap. We have established that λ_1^+ sits strictly above the rest of the spectrum, but we have said nothing yet about *how far* above, and that distance — the gap between λ_1^+ and the next eigenvalue below it — is the quantity that controls how fast repeated averaging by $\mathbb{W}$ aligns a population with e_1, and, when it nearly closes, signals that the network is splitting into communities. Having read the top mode, we turn in section 18.3 to the gap beneath it.

18.3 The spectral gap: mixing and community structure

The previous section isolated the very top of the spectrum and gave it both a name and a meaning: on a connected graphon the leading eigenvalue $\lambda_1^+ = \rho(\mathbb{W})$ is simple and strictly separated from everything beneath it, and its eigenfunction e_1 is the strictly positive centrality profile into which the network relaxes under repeated averaging. But we left a question hanging at the close of that section. We argued that e_1 *governs* the network — that iterating $\mathbb{W}$ aligns any input with e_1 — without ever saying how quickly, or what happens when that alignment is slow. The physicsnote there sketched the shape of the answer: e_1 is the nodeless "ground state," and the higher eigenfunctions e_n, $n \geq 2$, are "excited modes" carrying nodes, with those nodes marking community boundaries. This section turns that picture into quantitative statements. The single quantity that controls everything is the *gap* between the leading eigenvalue and the next one below it, and the whole section is the elaboration of one dichotomy this gap encodes.

The dichotomy is this. Each round of averaging by $\mathbb{W}$ multiplies the n-th spectral component of a function by λ_n; relative to the leading component, which is multiplied by λ_1, the n-th component shrinks by the ratio λ_n/λ_1 at every round. If the gap is *large* — $|\lambda_2|$ sitting well below $|\lambda_1|$ — this ratio is small, every subdominant mode is crushed within a few rounds, and the network collapses spectrally onto the single mode e_1: it behaves as *one* community, and mixing is fast. If the gap is *small* — a second eigenvalue $|\lambda_2|$ nearly as large as $|\lambda_1|$ — the mode e_2 decays barely faster than e_1, survives many rounds of averaging, and constitutes a second persistent collective degree of freedom alongside centrality. A whole cluster of eigenvalues at the top is a whole cluster of long-lived modes, and that multiplicity of persistent modes *is* community structure: each near-top eigenfunction is the indicator of one community, its node the seam between two camps. Large gap means fast mixing onto a single dominant community; small gap means slow mixing and several coexisting communities. Everything in this section is that sentence made precise.

We develop it in two movements. First we make "suppresses geometrically" exact: power iteration — applying $\mathbb{W}$ over and over — is the elementary mechanism behind the relaxation onto e_1, and the gap ratio $|\lambda_2| / |\lambda_1|$ turns out to be exactly its convergence rate, so that the gap simply *is* the mixing rate. Second, in example 18.11, we make "community structure" concrete on the one graphon where every eigenvalue and eigenfunction can be written down by hand — the two-block stochastic block model — and watch the abstract statements become explicit formulas: the gap below the top is literally the cross-block tie density, the second eigenfunction is literally the ± 1 community indicator, and its sign is literally the community-detection rule. The closing paragraph then lifts that hand-computed instance to the general lesson the next two sections consume.

18.3.1 Power iteration and the meaning of the gap

The section lead made a promise and hid a word inside it. It claimed that each round of averaging by $\mathbb{W}$ suppresses every non-leading mode *geometrically* relative to the leading one, and on the strength of that single adverb built the entire dichotomy between fast mixing and community structure. But "geometrically" was left to stand on intuition, and the target of the relaxation — the leading eigenline $\mathbb{R}e_1$, the strictly positive centrality profile that section 18.2 singled out as the shape any nonnegative input relaxes into — was named without a rate. We now supply both: we make "geometrically" an exact inequality and we identify the gap ratio as precisely its rate, so that the gap and the mixing rate become the same number.

The cleanest way to do this is to watch the elementary process that singles out the leading mode e_1 in the first place: *power iteration*, the repeated application of $\mathbb{W}$ to an arbitrary starting function. Take any $f \in L^2[0,1]$ and resolve it in the eigenbasis, $f = \sum_n \langle f, e_n \rangle e_n$. Applying $\mathbb{W}$ once scales the n-th coordinate by λ_n (this is just the operator form of eq. (18.1)); applying it k times scales that coordinate by λ_n^k, so $\mathbb{W}^k f = \sum_n \lambda_n^k \langle f, e_n \rangle e_n$. The leading coordinate behaves like λ_1^k and dominates all the others, because every subdominant coordinate carries the extra factor $(\lambda_n/\lambda_1)^k$, which decays geometrically in k provided $|\lambda_n| < |\lambda_1|$. Divide out the overall λ_1^k and what remains after k rounds is e_1 plus an error controlled entirely by the largest surviving ratio $|\lambda_2| / |\lambda_1|$ — the slowest-decaying of the subdominant modes sets the pace, and the rest decay at least as fast. The single number $|\lambda_2| / |\lambda_1|$, or equivalently its complement, is therefore the entire story of how fast a population aligns with the centrality profile e_1 inherited from the previous section. We name it before we quantify it.

Definition 18.9 (Spectral gap)**.** For a graphon with $|\lambda_1| > |\lambda_2|$, the (multiplicative) *spectral gap* is $\gamma = 1 - |\lambda_2| / |\lambda_1| \in (0, 1]$, and the absolute gap is $|\lambda_1| - |\lambda_2|$. For a connected nonnegative graphon, where $|\lambda_1| = \lambda_1^+$, we also speak of the *principal gap* $\lambda_1^+ - \lambda_2^+$ between the two largest positive eigenvalues.

Three forms of the gap appear because three different questions use it, and they reprise the two orderings of definition 18.1. Each form is the right currency for exactly one of those questions, and pairing the wrong form with a question is the standard way to get a bound that is either dimensionally nonsensical or sign-blind. The *multiplicative* gap $\gamma = 1 - |\lambda_2| \,/\, |\lambda_1|$ is the dimensionless quantity that governs convergence *rates*: it is built from the modulus order, since a relaxation rate cares only about sizes, and it is what enters the bound below as $(1-\gamma)^k = (|\lambda_2| \,/\, |\lambda_1|)^k$. A rate must be a pure ratio — it multiplies itself once per round and has to be scale-free for $(\,\cdot\,)^k$ to make sense — so the multiplicative form, and only it, is what the mixing proposition can use. The *absolute* gap $|\lambda_1| - |\lambda_2|$, by contrast, carries the units of an eigenvalue and is the right currency for a different question entirely: the stability of the leading eigen*space* under a perturbation of the kernel. When W is nudged to a nearby kernel W', its leading eigenfunction does not jump but *rotates*, and the angle through which it rotates is bounded by the size of the perturbation *divided by the absolute gap* — this is the Davis–Kahan $\sin\Theta$ estimate that section 18.4 invokes to upgrade operator-norm closeness of two graphons into L^2 closeness of their centrality profiles. A ratio would be the wrong currency there, because a perturbation displaces eigenvalues by an *absolute* amount, not a fractional one; what protects the leading eigenvector is the absolute distance in the real line separating λ_1 from its nearest neighbour, the literal room the perturbation must traverse before it can confuse the two modes. The *principal* gap $\lambda_1^+ - \lambda_2^+$ is built from the signed order and tracks instead the separation between the two leading *assortative* modes; it is the quantity that says how cleanly a community splits, and it is the one we read off the block model. The three are easy to keep straight by their roles: multiplicative for rates, absolute for the stability of eigenvectors, principal for the assortative splitting visible only in nonnegative kernels. The definition demands the strict separation $|\lambda_1| > |\lambda_2|$ as a hypothesis precisely because everything that follows divides by it: with $|\lambda_1| = |\lambda_2|$ the leading mode is degenerate, there is no single e_1 to relax onto, and both the rate and the centrality profile become ill-posed — the failure we will watch happen explicitly when the two-block model decouples. The definition records the gap; the next proposition cashes it as a convergence rate, turning the heuristic count of factors $(\lambda_2/\lambda_1)^k$ into a clean L^2 estimate.

Proposition 18.10 (Mixing by power iteration)**.** *Suppose $|\lambda_1| > |\lambda_2|$ and let $P_1 = \langle\cdot, e_1\rangle e_1$ be the orthogonal projection onto the leading eigenline. Then for every $f \in L^2[0,1]$ and every $k \geq 1$,*

$$\left\| \lambda_1^{-k}\, \mathbb{W}^k f - \langle f, e_1\rangle e_1 \right\|_{L^2} \;\leq\; \Big(\frac{|\lambda_2|}{|\lambda_1|}\Big)^k \;\|f\|_{L^2}\,. \tag{18.4}$$

If moreover $\langle f, e_1\rangle \neq 0$, then $\mathbb{W}^k f / \left\|\mathbb{W}^k f\right\| \to \pm e_1$ in L^2 at the geometric rate $(|\lambda_2| \,/\, |\lambda_1|)^k$.

Proof. By eq. (18.1), $\mathbb{W}^k f = \sum_n \lambda_n^k \langle f, e_n\rangle e_n$, so $\lambda_1^{-k}\mathbb{W}^k f - \langle f, e_1\rangle e_1 = \sum_{n\geq 2} (\lambda_n/\lambda_1)^k \langle f, e_n\rangle e_n$. By orthonormality and $|\lambda_n| \leq |\lambda_2|$ for $n \geq 2$,

$$\Big\| \sum_{n\geq 2} (\lambda_n/\lambda_1)^k \langle f, e_n\rangle e_n \Big\|^2 = \sum_{n\geq 2} \Big(\frac{\lambda_n}{\lambda_1}\Big)^{2k} \langle f, e_n\rangle^2 \leq \Big(\frac{|\lambda_2|}{|\lambda_1|}\Big)^{2k} \sum_{n\geq 2} \langle f, e_n\rangle^2 \leq \Big(\frac{|\lambda_2|}{|\lambda_1|}\Big)^{2k} \|f\|^2\,,$$

which is eq. (18.4). Read the three inequalities in order: the equality is Parseval applied to the orthonormal e_n; the first inequality replaces each surviving ratio $|\lambda_n/\lambda_1| \leq |\lambda_2/\lambda_1|$ by the largest one, which is exactly the gap doing its work; the second drops the missing $n = 1$ term to extend the sum back to all of $\|f\|^2 = \sum_n \langle f, e_n\rangle^2$. Taking square roots gives the bound.

The normalised statement follows by tracking the norm itself. Splitting off the leading term, $\mathbb{W}^k f = \lambda_1^k \langle f, e_1\rangle e_1 + \sum_{n\geq 2} \lambda_n^k \langle f, e_n\rangle e_n$, and applying Parseval again, $\left\|\mathbb{W}^k f\right\|^2 = \lambda_1^{2k}\langle f, e_1\rangle^2 + \sum_{n\geq 2}\lambda_n^{2k}\langle f, e_n\rangle^2$. When $\langle f, e_1\rangle \neq 0$ the first term dominates and the tail is smaller by a factor at most $(|\lambda_2| / |\lambda_1|)^{2k} \to 0$, so $\left\|\mathbb{W}^k f\right\| = |\lambda_1|^k |\langle f, e_1\rangle| (1 + o(1))$. Dividing $\mathbb{W}^k f$ by this norm cancels the common λ_1^k and leaves a unit multiple of e_1 up to the same vanishing tail, so the normalised iterate converges to $\pm e_1$ at the geometric rate $(|\lambda_2| / |\lambda_1|)^k$. The lone hypothesis $\langle f, e_1\rangle \neq 0$ is the requirement that the seed have *some* overlap with the leading mode: $\mathbb{W}$ preserves the orthogonal complement of e_1, so a seed orthogonal to e_1 stays orthogonal forever and can never be steered toward it — the one starting direction power iteration cannot recover from. $\square$

It helps to watch the bound act on a concrete number before reading off its meaning. Suppose the network has a healthy gap, say $|\lambda_2| / |\lambda_1| = \frac{1}{2}$ — the second mode decays twice as fast as the first under each averaging. Then eq. (18.4) says the normalised iterate sits within $2^{-k}\|f\|$ of the centrality line after k rounds: within $\frac{1}{4}$ at $k = 2$, within $\frac{1}{32}$ at $k = 5$, within a thousandth at $k = 10$. So a mere handful of rounds of kernel-weighted averaging — five passes of "replace each type's state by the W-weighted average of its neighbours' states" — already drives an arbitrary starting profile to within a few percent of the eigenvector centrality e_1, *whatever* the starting profile was, provided only that it was not exactly orthogonal to e_1. This is the operational content of a large gap, and it is why centrality is a robust, quickly-computed summary of a well-connected network rather than a fragile spectral abstraction: power iteration reaches it in single-digit passes. The same arithmetic read in reverse is the warning the rest of the section develops. Halve the gap to $|\lambda_2| / |\lambda_1| = 0.9$ and the error after five rounds is $0.9^5 \approx 0.59$ — the iterate has barely moved, the second mode is still very much alive, and "the" centrality profile is not yet a meaningful summary because a second collective mode is sharing the stage with it.

So a *large* gap means fast relaxation onto the single dominant mode e_1: the network is, spectrally, "one community," and any influence diffuses to the centrality profile in a few rounds of averaging. A *small* gap — two or more eigenvalues near the top, nearly degenerate — means slow relaxation and the persistence of more than one collective mode. That persistence is community structure, and the eigenfunctions of the near-degenerate top eigenvalues are the community indicators.

It is worth attaching a number to "fast" and "slow," because the dependence on the gap is sharp. To drive the error eq. (18.4) below a tolerance ε takes

$$k \;\approx\; \frac{\log(1/\varepsilon)}{\log(|\lambda_1| / |\lambda_2|)} \;\approx\; \frac{\log(1/\varepsilon)}{\gamma}$$

rounds of averaging, the second approximation holding for a small multiplicative gap γ, where $\log(|\lambda_1| / |\lambda_2|) = -\log(1-\gamma) \approx \gamma$. The mixing time thus scales as the *inverse* gap and diverges exactly as the gap closes: a network with two nearly degenerate top modes takes order γ^{-1} rounds to forget which of the two it started in. And when the gap does close, what survives alongside e_1 is not noise but the next eigenfunction e_2 — the first excited mode, which by the Perron–Frobenius discussion of section 18.2 *must* change sign, since it is orthogonal to the strictly positive e_1. Its node, the type at which e_2 crosses zero, is the boundary between the two persistent collective modes. This is the precise sense in which a near-degeneracy at the top of the spectrum is community structure: a second mode that averaging cannot extinguish, partitioned by its single node into two camps.

The reading is still abstract — it names e_2 and its node without showing either — and it is most convincing where it can be checked against explicit formulas. The next subsection supplies exactly that. In the two-block stochastic block model the whole spectrum is two numbers, so the gap $|\lambda_1| - |\lambda_2|$ can be *computed* rather than estimated, and the second eigenfunction e_2 can be *written down* — and it turns out to be the ± 1 indicator of the two blocks, its single node sitting exactly on the seam between them. The promise of this subsection, that small gap equals near-degenerate top eigenfunctions equals community structure, becomes there a line of arithmetic. We now make all of it completely explicit on the one example where the arithmetic closes.

18.3.2 The two-block model, by hand

The dichotomy of the previous subsection is sharpest where it can be checked entirely by hand, and there is exactly one graphon simple enough for that yet rich enough to exhibit the phenomenon: the stochastic block model with two equal blocks. It is the hydrogen atom of community structure — low enough in rank that the entire spectrum is two numbers, structured enough that those two numbers are precisely the centrality eigenvalue and the gap, and the two eigenfunctions are precisely the uniform mode and the community indicator. Every general statement made above reappears here as an explicit formula. We will also return to this model repeatedly: it is the worked instance behind the convergence pathologies of section 18.4 — where a tiny perturbation splits its degeneracy and swings the eigenfunctions — and behind the rank-two reduction of section 18.5. We compute its spectrum from scratch.

Example 18.11 (Spectrum of the two-block stochastic block model)**.** Partition $[0,1] = I_1 \cup I_2$ into two halves $I_1 = [0, \frac{1}{2})$, $I_2 = [\frac{1}{2}, 1]$, and set

$$W(x,y) = \begin{cases} p, & x, y \text{ in the same block,} \\ q, & x, y \text{ in different blocks,} \end{cases} \qquad p, q \in [0,1].$$

Everything about this operator is dictated by a single structural fact: the kernel $W(x,y)$ depends on the pair (x,y) only through *which blocks* the two types fall in — it is constant on each of the four block-pair rectangles $I_i \times I_j$. Consequently the value $(\mathbb{W}f)(x) = \int_0^1 W(x,y) f(y)\, \mathrm{d}y$ can depend on x only through x's block: for *any* input f, the output $\mathbb{W}f$ is constant on I_1 and constant on I_2. So the range of $\mathbb{W}$ lies in the two-dimensional space $V = \operatorname{span}\{\mathbf{1}_{I_1}, \mathbf{1}_{I_2}\}$ of block-constant functions, and $\mathbb{W}$ has rank at most two. The complementary fact identifies the kernel of the operator. Call a function f *block-mean-zero* if it integrates to zero over each block separately, $\int_{I_1} f = \int_{I_2} f = 0$; such an f is annihilated outright, because for $x \in I_1$

$$(\mathbb{W}f)(x) = p \int_{I_1} f + q \int_{I_2} f = p \cdot 0 + q \cdot 0 = 0,$$

and symmetrically for $x \in I_2$. The block-mean-zero functions are precisely the orthogonal complement $V^\perp$ inside $L^2[0,1]$, so $V^\perp \subseteq \ker \mathbb{W}$ and the entire action of $\mathbb{W}$ is carried by its restriction to the two-dimensional V — the operator is genuinely a 2×2 matrix in disguise. On V,

$$\mathbb{W}\mathbf{1}_{I_1} = \tfrac{p}{2}\mathbf{1}_{I_1} + \tfrac{q}{2}\mathbf{1}_{I_2}, \qquad \mathbb{W}\mathbf{1}_{I_2} = \tfrac{q}{2}\mathbf{1}_{I_1} + \tfrac{p}{2}\mathbf{1}_{I_2},$$

because, e.g., $(\mathbb{W}\mathbf{1}_{I_1})(x) = \int_{I_1} W(x,y)\,\mathrm{d}y$ equals $p \cdot \frac{1}{2}$ for $x \in I_1$ (the kernel is p throughout $I_1 \times I_1$ and $|I_1| = \frac{1}{2}$) and $q \cdot \frac{1}{2}$ for $x \in I_2$ (the kernel is q throughout $I_2 \times I_1$). In the basis $(\mathbf{1}_{I_1}, \mathbf{1}_{I_2})$ the operator is the matrix $\frac{1}{2}\left(\begin{smallmatrix} p & q \\ q & p \end{smallmatrix}\right)$, whose eigenpairs are

$$\lambda_1 = \frac{p+q}{2} \;\text{ with }\; e_1 \propto \mathbf{1}_{I_1} + \mathbf{1}_{I_2} = \mathbf{1}, \qquad \lambda_2 = \frac{p-q}{2} \;\text{ with }\; e_2 \propto \mathbf{1}_{I_1} - \mathbf{1}_{I_2}.$$

Normalising in $L^2[0,1]$ gives $e_1 = \mathbf{1}$ (the constant, with $\|\mathbf{1}\| = 1$) and $e_2 = \mathbf{1}_{I_1} - \mathbf{1}_{I_2}$, the genuine ± 1 community indicator. Note the contrast with the block indicators $\mathbf{1}_{I_j}$, which have norm $\left\|\mathbf{1}_{I_j}\right\| = \sqrt{1/2}$ and so are *not* normalised: the difference $\mathbf{1}_{I_1} - \mathbf{1}_{I_2}$ takes the values ± 1, so its square is 1 a.e. and $\|\mathbf{1}_{I_1} - \mathbf{1}_{I_2}\|_{L^2} = 1$ already — which is precisely why the L^2-normalised second eigenfunction *is* the ± 1 indicator, with no rescaling. The rest of the spectrum is zero, and the two surviving eigenpairs already carry every reading the abstract discussion promised. We take them one at a time.

Centrality is uniform. The leading eigenfunction is the constant $e_1 = \mathbf{1}$, so every type carries the same eigenvector centrality of section 18.2: in a symmetric two-block model no type occupies a structurally privileged position, exactly as the block-interchange symmetry of the model demands. This is the degenerate, featureless case of the centrality reading, and its interest lies in what happens when the symmetry is broken. Make the blocks unequal in size, or give them unequal internal densities $p_1 \neq p_2$, and the 2×2 reduction loses its symmetry; its Perron eigenvector e_1 ceases to be constant and *tilts* toward the denser, larger block, assigning that block the higher centrality. That tilt is not a curiosity but the comparative static the equilibrium analysis of chapter 20 runs on: there e_1 becomes the equilibrium effort profile, and "which block works harder" is read off directly from the direction in which the broken symmetry tilts the leading eigenfunction.

The second eigenfunction is the community indicator. Because $e_2 = \mathbf{1}_{I_1} - \mathbf{1}_{I_2}$, its sign $\operatorname{sgn} e_2(x)$ returns $+1$ on I_1 and -1 on I_2, recovering the block of x exactly; and its single node — the seam $x = \frac{1}{2}$, where it jumps from $+1$ to -1 — is precisely the community boundary that the previous subsection located abstractly as the zero crossing of the first excited mode. Spectral community detection — compute the second eigenfunction, threshold it at zero, and assign each type to a community by the sign — is nothing more than this observation, and it is the continuum shadow of the Fiedler-vector heuristic that partitions a finite graph by the sign pattern of the second eigenvector of its adjacency (or Laplacian) matrix. Here the heuristic is not heuristic at all but exact: $\operatorname{sgn} e_2$ recovers the planted partition with no error, because the model has nothing but the partition in it.

Assortative versus disassortative, and a non-PSD witness. Whether e_2 reports a clustered community or a bipartite split is read off the *sign* of its eigenvalue $\lambda_2 = (p-q)/2$, and this is where the signed ordering of definition 18.1 earns its keep. When $p > q$ — within-block ties denser than cross-block ties, an assortative community structure — one has $\lambda_2 = (p-q)/2 > 0$: the community mode is reproduced with its own sign under averaging, each block linked predominantly to itself, the "like attracts like" picture. The principal gap between the two clustering modes is then

$$\lambda_1 - \lambda_2 = \frac{p+q}{2} - \frac{p-q}{2} = q,$$

small precisely when q is small, i.e. when the two communities are weakly linked — the slow-mixing regime of proposition 18.10, in which the blocks behave as nearly independent subnetworks and repeated averaging takes order q^{-1} rounds to forget which block it started in. When $p < q$ — cross-block ties denser, a bipartite-like, disassortative structure — the same eigenvalue turns negative, $\lambda_2 = (p-q)/2 < 0$: the mode now flips sign under averaging, each block linked predominantly to the *other*, "opposites attract," and the negative eigenvalue is the spectral signature of bipartiteness. This is the concrete realisation of the warning issued in remark 18.3. The kernel is a perfectly ordinary $[0,1]$-valued graphon, nonnegative as a function on $[0,1]^2$, yet its operator is *not* positive semidefinite, for the explicit test function $e_2 = \mathbf{1}_{I_1} - \mathbf{1}_{I_2}$ witnesses a negative quadratic form,

$$\langle \mathbb{W} e_2, e_2 \rangle = \lambda_2 \, \|e_2\|_{L^2}^2 = \lambda_2 = \frac{p-q}{2} < 0,$$

exactly as promised there. Pointwise nonnegativity of the kernel and positivity of its quadratic form are different conditions, and the disassortative two-block model is the cleanest place to watch them part company. In this regime the modulus ordering and the signed ordering disagree about which entry is "second": $|\lambda_1| = (p+q)/2$ still leads, but the gap that governs mixing is now

$$|\lambda_1| - |\lambda_2| = \frac{p+q}{2} - \frac{q-p}{2} = p,$$

which closes as the within-block density p vanishes — the limit of a perfect bipartite graphon with every tie crossing the seam.

Finally, two sanity checks against the spectral-moment identity proposition 18.4, worked through in full because the arithmetic is exactly the kind of bookkeeping the identity is supposed to make automatic. The first is $k = 2$, the squared Hilbert–Schmidt norm. Spectrally,

$$\sum_n \lambda_n^2 = \lambda_1^2 + \lambda_2^2 = \Big(\frac{p+q}{2}\Big)^2 + \Big(\frac{p-q}{2}\Big)^2 = \frac{(p+q)^2 + (p-q)^2}{4} = \frac{2p^2 + 2q^2}{4} = \frac{p^2+q^2}{2},$$

where the cross terms $\pm 2pq$ cancel in the numerator. Computed instead straight from the kernel, $\|W\|_{L^2([0,1]^2)}^2 = \iint_{[0,1]^2} W^2$ sums p^2 over the two same-block squares $I_1 \times I_1$ and $I_2 \times I_2$ (total area $\frac{1}{2}$) and q^2 over the two cross-block rectangles (total area $\frac{1}{2}$), giving $p^2 \cdot \frac{1}{2} + q^2 \cdot \frac{1}{2} = \frac{1}{2}(p^2 + q^2)$ — the two evaluations agree. The second check is $k = 3$, the triangle density. Spectrally, the odd cross terms now *survive* while the rest cancel:

$$\sum_n \lambda_n^3 = \lambda_1^3 + \lambda_2^3 = \frac{(p+q)^3 + (p-q)^3}{8} = \frac{2p^3 + 6pq^2}{8} = \frac{p^3 + 3pq^2}{4},$$

using $(p+q)^3 + (p-q)^3 = 2(p^3 + 3pq^2)$. Computed combinatorially as $t(C_3, W)$, the closed three-cycle integral, draw three independent uniform types: the $2^3 = 8$ block patterns each have probability $\frac{1}{8}$, the two monochromatic patterns ($I_1 I_1 I_1$ and $I_2 I_2 I_2$) contribute three within-block edges and hence p^3 apiece, and each of the remaining six patterns has one lone vertex, contributing one within-block edge and two crossing edges, pq^2 apiece, so $t(C_3, W) = \frac{1}{8}(2p^3 + 6pq^2) = \frac{1}{4}(p^3 + 3pq^2)$ — again the two evaluations agree. The third spectral moment is, on the nose, the probability that three random types close a triangle.

The block model is the cleanest case, but its lesson is general, and stating that generalisation correctly is the bridge out of this section. Read the two-block spectrum structurally rather than numerically: there were exactly two nonzero eigenvalues, separated by a gap from an infinite sea of zeros, and their two eigenfunctions — the uniform mode and the ± 1 indicator — between them spanned the space of functions constant on blocks, that is, the space of all possible community-level profiles. The general principle is the same with r in place of 2. Suppose a graphon's spectrum shows r eigenvalues clustered near the top and then a gap down to a tail that is small in the ℓ^2 sense controlled by proposition 18.4. By the power-iteration argument of proposition 18.10, the r top modes are exactly the ones averaging cannot extinguish while the tail decays away, so the network supports precisely r persistent collective degrees of freedom — "r communities plus noise" — and the span of the top r eigenfunctions is the space of macroscopic order parameters, the continuum replacement for the r block indicators. The gap is what makes the integer r well defined: it is the threshold that cleanly separates the structural modes from the noise, and without it "how many communities" has no answer.

This is not a metaphor but the exact hypothesis the remainder of the chapter runs on, and it leaves this section along two distinct seams. The first runs to section 18.5: if the tail below the gap is negligible, discarding it replaces the full operator by its rank-r truncation onto the top eigenfunctions, and section 18.5 shows that this collapses the infinite-dimensional graphon mean field game coupling to a *finite* system of r scalar order parameters — one per surviving mode — with the truncation error controlled by precisely the ℓ^2 tail the gap bounds. The two-block model is the case $r = 2$ (indeed $r = 1$ once one notes the constant mode is its only assortative one), the very template the low-rank section generalises, and the rank-one limit of that template is the single self-consistent scalar equation of Thread B. The second seam runs to section 18.4: the entire mixing-and-community picture rests on the leading eigenvalue being *simple* and *strictly separated* from the rest — the hypothesis $|\lambda_1| > |\lambda_2|$ that definition 18.9 demands and that example 18.11 meets whenever both p and q are positive. That same simple-eigenvalue-with-a-gap hypothesis is exactly what the Davis–Kahan perturbation theorem of section 18.4 will require in order to conclude that the leading eigenfunction — the centrality profile, and one rung down the community indicator — moves *continuously* rather than jumping when the kernel is perturbed or sampled. The two-block model previews why the hypothesis is indispensable: let the blocks decouple, $q \to 0$, and the leading eigenvalue acquires a second copy, $\lambda_1 = \lambda_2 = p/2$, so "the" centrality profile is no longer a single function but an entire two-dimensional eigenspace, and any infinitesimal asymmetry can swing the leading eigenfunction discontinuously from one block to the other. That degenerate instability is what section 18.4 must confront head-on, and the gap is precisely the quantity whose positivity forbids it. The spectral gap, in short, is at once the rate of mixing, the count of communities, and the licence for spectral stability.

18.4 Operator convergence versus cut convergence

The two-block model left us with a warning. When the gap above the second eigenvalue shrinks — when two modes of the spectrum approach one another, as the blocks of example 18.11 do as $q \to 0$ — the eigenvalues themselves stay perfectly well behaved, but the eigen*functions* become fragile: an infinitesimal, perfectly symmetric perturbation can rotate a near-degenerate eigenplane through a finite angle and swing "the" centrality profile discontinuously from one block to the other. Eigenvalues are robust; eigenfunctions are delicate, and the spectral gap is what stands between the two. That contrast is the seam the previous section explicitly handed forward, and honouring it is the whole business of this fourth movement. We must now make precise *in what sense* the spectrum of $\mathbb{W}$ depends continuously on the kernel W — separating the part of the spectrum that survives the weakest reasonable notion of closeness from the part that needs a far stronger one.

To do that we must first decide what "two graphons are close" even means, and the chapter has quietly accumulated *two* answers that do not agree. One is the cut metric $\delta_\square$ of chapter 17, under which graphon space is compact and under which sampling a network produces a kernel close to its limit; this is the notion the rest of the book will actually be handed by its compactness and sampling theorems, so it is the one we are stuck with whether we like it or not. The other is the family of operator-theoretic norms inherited from $\mathbb{W}$ — the operator (or $2 \to 2$) norm and the Hilbert–Schmidt norm — which measure closeness the way a spectral theorist would, by how nearly the two operators act alike on every test function. These notions are genuinely different, and the distance between them is exactly the distance between "the eigenvalues agree" and "the eigenfunctions agree." Pinning down that distance is the subject of this section.

The mechanism that will let us cross from one notion to the other is already in hand: it is the moment identity $\sum_n \lambda_n^k = t(C_k, W)$ of proposition 18.4. That identity is the hinge of the whole chapter precisely because it expresses a *spectral* quantity — a power sum of the eigenvalues — as a *cut-continuous* one, a cycle homomorphism density that chapter 17 already proved moves continuously in $\delta_\square$. Cut control of the kernel therefore transfers, through the moment identity, into control of the power sums of the spectrum, and from the power sums one can read back the individual eigenvalues. This is how the weakest notion of closeness still manages to pin every eigenvalue, and it is why the proof of the positive theorem below runs entirely through the moments rather than through any direct comparison of operators.

The architecture of the section is a strict *hierarchy of four notions of convergence*, each one provably weaker than the one before it, with a single positive theorem threading through to the bottom. Spelled out as an implication chain, convergence in the strongest norm forces convergence in each weaker one in turn,

$$\|W_n - W\|_{\mathrm{HS}} \to 0 \implies \|W_n - W\|_{2\to 2} \to 0 \implies \delta_\square(W_n, W) \to 0 \implies \lambda_k(W_n) \to \lambda_k(W) \text{ for every } k,$$

and at each arrow the converse fails — there is a concrete in-book sequence that converges in the right-hand sense but not the left. The first two arrows are the norm inequality $\|\cdot\|_\square \le \|\cdot\|_{2\to 2} \le \|\cdot\|_{\mathrm{HS}}$ of eq. (18.5), read as a statement about convergence; the third is the genuine theorem of this section, that cut convergence alone — the weakest, most easily attained notion — already forces every individual eigenvalue to converge. Eigen*functions*, crucially, sit *below the bottom* of this chain: none of these notions short of operator-norm convergence controls them, which is exactly the fragility the two-block model warned of, now stated as the failure of an implication.

Why insist on the full hierarchy rather than just working in the strongest norm throughout? Because we do not get to choose. Sampling a finite network from W and the compactness theorem theorem 17.13 both deliver convergence only in $\delta_\square$, the *weakest* of the four; nothing in those results gives operator-norm closeness for free. So the practically urgent question is precisely how much of the spectrum the weakest notion controls — and the answer, that it controls every eigenvalue but no eigenfunction, is what the three subsections establish in turn. The first sets down the three norms and proves the inequality eq. (18.5) that orders them, exhibiting an explicit witness that each inequality is strict. The second proves the one positive theorem, that cut convergence forces eigenvalue convergence, via the moment identity and a purely sequential fact about power sums of ℓ^2 sequences. The third confronts the eigenfunctions directly: a counterexample shows cut convergence cannot move them, and the Davis–Kahan theorem theorem 18.16 shows that operator-norm convergence *plus* a spectral gap can.

That final split is the message the section hands downstream, and it is worth naming its two destinations now. To chapter 19: the consistency of network sampling will be proved there at the level of the cut metric, so by this section it certifies that a sampled network has nearly the right *eigenvalues* — and, by itself, says nothing about its centrality profile. To chapter 25: a separate matrix-concentration argument will upgrade that cut consistency to operator-norm convergence of the empirical operator, at which point theorem 18.16 transports the leading eigenfunction — the eigenvector centrality of section 18.2, the object that becomes equilibrium effort in Part IV — from graphon to sampled network, provided the gap of section 18.3 is open. The eigenvalue/eigenfunction divide established here is thus the exact specification of what each later chapter must supply: cut convergence for the spectrum, operator-norm convergence for the centrality vector.

18.4.1 Three norms and an inequality

The section lead promised a hierarchy of four notions of closeness, strictly ordered, with a single positive theorem threading to the bottom. The top three rungs of that ladder are *norms* on kernels, and before we can speak of one notion being weaker than another we must put all three on the same footing and prove the inequality that orders them. That is the skeleton everything in this section hangs on, so we build it first and with care: three ways of measuring the size of a kernel U, a single chain $\|U\|_\square \le \|U\|_{2\to2} \le \|U\|_{\mathrm{HS}}$ relating them, and — the part that makes the hierarchy genuine rather than an artifact of loose constants — two explicit in-book sequences proving that *neither* inequality can be reversed even after rescaling. Each strictness will turn out to carry a specific message forward, so we do not merely assert the gaps; we manufacture them and read off what they mean.

The three norms measure size in three increasingly demanding ways, and it pays to name the intuition behind each before writing the symbols. For a kernel U on $[0,1]^2$ — signed, symmetric, measurable, bounded, exactly the class of *differences* $W_n - W$ we will care about — the weakest, the cut norm of chapter 17,

$$\|U\|_\square = \sup_{S,T\subseteq[0,1]} \Big| \int_{S\times T} U \Big|,$$

asks only how much mass U can accumulate over a single rectangle $S\times T$; it is blind to any structure finer than "how much net weight sits between this block of types and that one," which is exactly why it is the natural currency of the combinatorial and sampling theory and why it is so easily made small. The middle norm is the L^2 operator (or $2\to2$) norm

$$\|U\|_{2\to2} = \|\mathbb{W}_U\| = \sup_{\|g\|_{L^2}\le1} \|\mathbb{W}_U g\|_{L^2}\,, \qquad (\mathbb{W}_U g)(x) = \int_0^1 U(x,y)g(y)\;\mathrm{d}y,$$

where $\mathbb{W}_U$ is the parametric form of the integral operator $\mathbb{W} = \mathbb{W}$ of chapter 17; it asks how much U, acting as an operator, can stretch the *worst* unit test function, and so it sees not just block averages but the whole spectral action of U — for symmetric U it equals $\max_n |\lambda_n|$, the top eigenvalue in modulus. The strongest, the Hilbert–Schmidt norm

$$\|U\|_{\mathrm{HS}} = \|U\|_{L^2([0,1]^2)} = \Big(\int_{[0,1]^2} U^2\Big)^{1/2} = \Big(\sum_n \lambda_n^2\Big)^{1/2},$$

asks for the full L^2 energy of the kernel, equivalently the entire ℓ^2 length of the spectrum rather than just its largest entry. So the three norms are, spectrally, "one mode," "the largest mode," and "all the modes at once" — a genuine ordering of how much of the operator each one is allowed to feel. The basic comparison records that ordering:

$$\|U\|_{\square} \;\le\; \|U\|_{2\to 2} \;\le\; \|U\|_{\mathrm{HS}}\,. \tag{18.5}$$

Both inequalities have short, structural proofs, and we give them rather than wave at them. The *right* inequality is the statement that the operator norm of a Hilbert–Schmidt operator never exceeds its Hilbert–Schmidt norm, $\|\mathbb{W}_U\| \le \|\mathbb{W}_U\|_{\mathrm{HS}}$, which is definition 2.21; spectrally it is the triviality that the largest entry of a sequence is at most its ℓ^2 length, $\max_n |\lambda_n| \le (\sum_n \lambda_n^2)^{1/2}$. The *left* inequality is the operator-norm comparison lemma 17.3 of the previous chapter, and it is worth spelling out the one identity on which it turns, because that identity is the exact bridge between the rectangle-testing of the cut norm and the function-testing of the operator norm. For any measurable $S, T \subseteq [0,1]$, testing $\mathbb{W}_U$ against the two indicator functions $\mathbf{1}_S, \mathbf{1}_T$ reproduces the rectangle integral:

$$\int_{S\times T} U = \int_S \Big(\int_T U(x,y)\;\mathrm{d}y\Big)\,\mathrm{d}x = \int_0^1 \mathbf{1}_S(x)\,(\mathbb{W}_U\mathbf{1}_T)(x)\;\mathrm{d}x = \langle \mathbb{W}_U\mathbf{1}_T, \mathbf{1}_S\rangle.$$

Taking absolute values and applying Cauchy–Schwarz, then the definition of the operator norm,

$$\Big|\int_{S\times T} U\Big| = |\langle \mathbb{W}_U\mathbf{1}_T, \mathbf{1}_S\rangle| \le \|\mathbb{W}_U\mathbf{1}_T\|_{L^2}\;\|\mathbf{1}_S\|_{L^2} \le \|\mathbb{W}_U\|\;\|\mathbf{1}_T\|_{L^2}\;\|\mathbf{1}_S\|_{L^2} \le \|\mathbb{W}_U\|\,,$$

since $\|\mathbf{1}_S\|_{L^2} = |S|^{1/2} \le 1$ and likewise for T. Taking the supremum over S, T gives $\|U\|_{\square} \le \|U\|_{2\to 2}$. So the left inequality is, at bottom, the observation that indicators of sets are particular unit-ish test functions, and the cut norm optimises over only this impoverished family while the operator norm optimises over all of L^2.

That both inequalities in eq. (18.5) are *strict* — not reversible even up to a universal constant — is the whole point of the hierarchy, for it is what makes the four rungs genuinely distinct rather than the same notion wearing three names. We prove strictness the honest way, by exhibiting for each inequality a concrete sequence of kernels that drives the ratio of the two sides to its extreme. Each witness has been chosen so that the *reason* it works is the same phenomenon we will meet again downstream.

The left gap: a mean-zero spike. We seek symmetric kernels U_n with $\|U_n\|_{\square} \to 0$ but $\|U_n\|_{2\to 2} \not\to 0$ — a sequence small in cut but not in operator norm, witnessing that the left inequality cannot be reversed. The mechanism we want is a single eigenfunction that carries all the operator energy while spreading almost no net mass over any rectangle, and the cleanest way to build it is to make the eigenfunction itself thin and tall. Fix a sequence $\varepsilon_n \downarrow 0$ and define the mean-zero "spike"

$$\phi_n = \varepsilon_n^{-1/2}(\mathbf{1}_{[0,\varepsilon_n)} - \varepsilon_n),$$

a function that is large and positive on the short interval $[0, \varepsilon_n)$ and slightly negative elsewhere, with the constant $-\varepsilon_n$ chosen exactly to kill its mean. Three one-line computations pin down its size. Its integral vanishes, $\int_0^1 \phi_n = \varepsilon_n^{-1/2}(\varepsilon_n - \varepsilon_n) = 0$, so $\langle \phi_n, \mathbf{1} \rangle = 0$. Its L^2 energy is essentially one,

$$\|\phi_n\|_{L^2}^2 = \varepsilon_n^{-1} \int_0^1 \big(\mathbf{1}_{[0,\varepsilon_n)} - \varepsilon_n\big)^2 = \varepsilon_n^{-1}(\varepsilon_n - 2\varepsilon_n \cdot \varepsilon_n + \varepsilon_n^2) = 1 - \varepsilon_n \longrightarrow 1,$$

so $\|\phi_n\|_{L^2} = 1 + o(1)$. And, crucially for the cut estimate, its L^1 norm is small: the positive part contributes $\varepsilon_n^{-1/2}(1-\varepsilon_n)$ over a set of measure ε_n and the negative part contributes $\varepsilon_n^{-1/2}\varepsilon_n$ over a set of measure $1-\varepsilon_n$, so

$$\|\phi_n\|_{L^1} = \varepsilon_n^{-1/2}\Big[\varepsilon_n(1-\varepsilon_n) + \varepsilon_n(1-\varepsilon_n)\Big] = 2\,\varepsilon_n^{1/2}(1-\varepsilon_n) = O(\sqrt{\varepsilon_n}).$$

Now set $U_n(x,y) = \phi_n(x)\phi_n(y)$, the rank-one symmetric kernel whose operator is $g \mapsto \langle \phi_n, g\rangle\, \phi_n$. This operator has the single nonzero eigenvalue $\|\phi_n\|_{L^2}^2$ with eigenfunction ϕ_n, so its operator norm is

$$\|U_n\|_{2\to 2} = \|\phi_n\|_{L^2}^2 = 1 - \varepsilon_n \longrightarrow 1,$$

which stays bounded away from 0. The cut norm goes the other way, and to see it cleanly we isolate the elementary set-optimisation that governs a mean-zero function.

Lemma 18.12 (Optimal cut set of a mean-zero function). *Let $\phi \in L^1([0,1])$ have $\int_0^1 \phi = 0$. Then $\sup_{S\subseteq[0,1]} \big|\int_S \phi\big| = \frac{1}{2}\,\|\phi\|_{L^1}$, attained at $S = \{\phi > 0\}$.*

Proof. Write $\phi = \phi^+ - \phi^-$ for its positive and negative parts, so $\|\phi\|_{L^1} = \int \phi^+ + \int \phi^-$. The mean-zero hypothesis says $\int \phi^+ = \int \phi^-$, hence each equals $\frac{1}{2}\,\|\phi\|_{L^1}$. For any measurable S, $\int_S \phi = \int_S \phi^+ - \int_S \phi^-$ is the difference of two nonnegative numbers each at most $\int \phi^+ = \frac{1}{2}\,\|\phi\|_{L^1}$, so $\big|\int_S \phi\big| \le \frac{1}{2}\,\|\phi\|_{L^1}$; choosing $S = \{\phi > 0\}$ makes the second term vanish and the first equal $\int \phi^+$, attaining the bound (the sign-flipped optimum $S = \{\phi < 0\}$ attains $-\frac{1}{2}\,\|\phi\|_{L^1}$). $\square$

For the rank-one kernel the cut supremum factorises across the two indicators, $\int_{S\times T} U_n = \big(\int_S \phi_n\big)\big(\int_T \phi_n\big)$, so the joint supremum is the product of two single-set suprema, each governed by lemma 18.12:

$$\|U_n\|_\square = \Big(\sup_S \Big|\int_S \phi_n\Big|\Big)^2 = \Big(\tfrac{1}{2}\,\|\phi_n\|_{L^1}\Big)^2 = \Big(\varepsilon_n^{1/2}(1-\varepsilon_n)\Big)^2 = O(\varepsilon_n) \longrightarrow 0.$$

Thus $\|U_n\|_\square \to 0$ while $\|U_n\|_{2\to 2} \to 1$: the left inequality of eq. (18.5) cannot be reversed, even up to a constant, even within the rank-one kernels. The picture to keep is physical. The operator energy is concentrated entirely in the eigenfunction ϕ_n, which grows tall and thin and slides into the corner $x = 0$; the cut norm cannot feel it because over any rectangle the tall positive spike is almost exactly cancelled by the broad negative shelf the mean-zero correction installs. This is precisely the "eigenfunction concentrates and escapes" obstruction that makes *eigenfunctions* discontinuous under cut convergence, the failure mode the Davis–Kahan theory of theorem 18.16 is built to repair — and the same phenomenon behind the cospectral witness of remark 18.15 below.

The right gap: the half-graph. For the right inequality we want a single *fixed* symmetric kernel whose operator norm is strictly below its Hilbert–Schmidt norm, witnessing that the largest eigenvalue can be strictly smaller than the ℓ^2 length of the whole spectrum — which happens precisely when the spectral energy is spread across many modes rather than hoarded by one. The half-graph kernel

$$W(x,y) = \mathbf{1}\{x+y \le 1\}$$

is the standard such example, an honest graphon (values in $\{0,1\}$) supported on the lower triangle. Its Hilbert–Schmidt norm is read straight off the geometry: $W^2 = W$ is again the indicator of the triangle $\{x+y \le 1\}$, whose area is $\frac{1}{2}$, so

$$\|W\|_{\mathrm{HS}} = \|W\|_{L^2} = \Big(\int_{[0,1]^2} W^2\Big)^{1/2} = \frac{1}{\sqrt{2}}.$$

The operator norm requires the spectrum, and here it can be computed in closed form by turning the eigenvalue problem into an ODE — a small but genuinely instructive calculation, so we run it in full. Because W is symmetric, $\mathbb{W}$ is self-adjoint and $\|\mathbb{W}\|_{2\to 2} = \max_n |\lambda_n|$, so it suffices to find the eigenvalues. The eigenequation $(\mathbb{W}f)(x) = \lambda f(x)$ is, with the triangular support evaluated,

$$\int_0^{1-x} f(y)\,\mathrm{d}y = \lambda f(x), \qquad 0 \le x \le 1. \tag{18.6}$$

We assume $\lambda \neq 0$ (the kernel has infinitely many nonzero eigenvalues, computed below, so this loses nothing). Differentiating eq. (18.6) in x and using the fundamental theorem of calculus on the variable *upper* limit $1-x$ — whose derivative contributes the chain-rule factor -1 — gives

$$-f(1-x) = \lambda f'(x). \tag{18.7}$$

This already encodes the two boundary conditions we will need. Setting $x = 1$ in eq. (18.6) collapses the integral to $\int_0^0 = 0$, so $0 = \lambda f(1)$ and hence

$$f(1) = 0;$$

setting $x = 0$ in eq. (18.7) gives $\lambda f'(0) = -f(1) = 0$, so

$$f'(0) = 0.$$

Differentiating eq. (18.7) once more (again a chain-rule factor -1 from the argument $1-x$) yields $\lambda f''(x) = f'(1-x)$. To close the system, evaluate eq. (18.7) at the reflected point $1-x$, which reads $\lambda f'(1-x) = -f(x)$, i.e. $f'(1-x) = -f(x)/\lambda$; substituting,

$$\lambda f''(x) = f'(1-x) = -\frac{f(x)}{\lambda}, \qquad \text{that is} \qquad \lambda^2 f'' = -f.$$

The reflection has turned the nonlocal eigenequation into a plain harmonic oscillator. Its general solution is $f(x) = A\cos(x/\lambda) + B\sin(x/\lambda)$, and the two boundary conditions select the spectrum.

From $f'(x) = \lambda^{-1}\big(-A\sin(x/\lambda) + B\cos(x/\lambda)\big)$ the condition $f'(0) = 0$ forces $B = 0$, leaving $f(x) = A\cos(x/\lambda)$; the condition $f(1) = 0$ then forces $A\cos(1/\lambda) = 0$, and with $A \neq 0$ this is the quantisation

$$\cos(1/\lambda) = 0 \quad\Longleftrightarrow\quad \frac{1}{\lambda} = \frac{(2k-1)\pi}{2},\ k \in \mathbb{N} \quad\Longleftrightarrow\quad \lambda = \frac{2}{(2k-1)\pi}.$$

The eigenvalues are $\frac{2}{\pi}, \frac{2}{3\pi}, \frac{2}{5\pi}, \dots$ (with sign alternations from the reflection symmetry that do not affect the moduli), so the largest in modulus, and hence the operator norm, is the $k = 1$ value:

$$\|W\|_{2\to2} = \frac{2}{\pi} \approx 0.6366 \ < \ \frac{1}{\sqrt{2}} \approx 0.7071 = \|W\|_{\mathrm{HS}}\,.$$

The gap is strict, and one can see *why* from the eigenvalue list: the Hilbert–Schmidt norm collects $\sum_k \lambda_k^2 = \frac{4}{\pi^2}\sum_k (2k-1)^{-2} = \frac{4}{\pi^2}\cdot\frac{\pi^2}{8} = \frac{1}{2}$ over the whole tail of modes, while the operator norm sees only the leading one; the right inequality is strict exactly to the extent that the spectral energy fails to be rank one.

We have, then, the ordered chain eq. (18.5) together with a witness at each end proving both inequalities genuinely strict. The two strictnesses are not curiosities; they are the load-bearing facts the rest of the section spends. The strict *left* inequality — cut strictly weaker than operator norm, witnessed by a spike whose eigenfunction escapes — is precisely why eigen*functions* can fail to converge under cut convergence, and it is the gap the eigenfunction subsection theorem 18.16 must close with the stronger operator norm; this is the operator norm earning its place as the "stronger notion" the downstream theory needs. The strict *right* inequality — Hilbert–Schmidt strictly above operator norm, witnessed by the half-graph whose energy is spread across all modes — is the reason the squared length $\sum_n \lambda_n^2 = \|W\|_{\mathrm{HS}}^2 = t(C_2, W)$ is *not* cut-continuous: it lives at the strong end the cut norm cannot reach, which is exactly why the eigenvalue theorem of the next subsection must route around the C_2 moment and use only the simple-cycle moments $k \geq 3$. With the ladder built and both gaps shown genuine, we now turn to the good news hiding at its weak end: that the cut norm, feeble as the spike just showed it to be against eigenfunctions, nonetheless pins every individual eigenvalue.

18.4.2 Cut convergence forces eigenvalue convergence

The positive theorem is that the spectrum is continuous in the cut metric, eigenvalue by eigenvalue. The mechanism is the moment identity proposition 18.4: cut convergence is equivalent to convergence of all homomorphism densities (theorem 17.12), in particular of the cycle densities $t(C_k, W) = \sum_n \lambda_n^k$ for *simple* cycles $k \geq 3$, i.e. of those power sums of the spectrum; and the power sums $\{p_k\}_{k\geq3}$ of an ℓ^2 sequence already determine, and continuously control, its individual entries. We first isolate that last, purely sequential, fact. (The $k = 2$ power sum $\sum_n \lambda_n^2 = \|W\|_{\mathrm{HS}}^2$ is *deliberately excluded*: the doubled edge C_2 is not a simple graph, the counting lemma of chapter 16 does not apply to it, and indeed $t(C_2, \cdot)$ is *not* cut-continuous — the right inequality of eq. (18.5) is strict precisely because the Hilbert–Schmidt norm escapes cut control. We will not need it.)

Lemma 18.13 (Moments determine an ℓ^2 spectrum)**.** *For each $n \in \mathbb{N}$ let $\big(\mu_i^{(n)}\big)_i$ be a real sequence with entries in $[-1, 1]$ and $\sum_i \big(\mu_i^{(n)}\big)^2 \leq 1$, and let $(\mu_i)_i$ be another such sequence. Suppose the* *power sums converge:* $\sum_i \big(\mu_i^{(n)}\big)^k \to \sum_i \mu_i^k$ *as* $n \to \infty$*, for every integer* $k \geq 3$*. List each sequence in signed order (positives decreasing, negatives increasing). Then* $\mu_i^{(n)} \to \mu_i$ *for every* i *in each list.*

The hypothesis is imposed only for $k \geq 3$. The exponent $k = 2$ is omitted on purpose: it is the one power sum that will *not* be available from cut convergence (see the discussion above), and, as the proof shows, it is not needed — the uniform ℓ^2 bound $\sum_i (\mu_i^{(n)})^2 \leq 1$ does all the work that $k = 2$ would, and it is a standing bound, not a convergence hypothesis.

Proof. Uniform tightness. Because $\sum_i (\mu_i^{(n)})^2 \leq 1$, for any $\varepsilon > 0$ the number of indices with $\left|\mu_i^{(n)}\right| > \varepsilon$ is at most ε^{-2}, *uniformly in* n. Hence along any subsequence we may, by a diagonal extraction over i, assume $\mu_i^{(n)} \to c_i$ for every i (each lies in the compact $[-1, 1]$), and the limit list (c_i) again satisfies $\sum_i c_i^2 \leq 1$ with the same tightness.

Power sums pass to the limit. Fix $k \geq 3$. Split $\sum_i (\mu_i^{(n)})^k = \sum_{\left|\mu_i^{(n)}\right| > \varepsilon} + \sum_{\left|\mu_i^{(n)}\right| \leq \varepsilon}$. The tail is bounded by $\varepsilon^{k-2} \sum_i (\mu_i^{(n)})^2 \leq \varepsilon^{k-2}$ (here $k - 2 \geq 1$), uniformly in n, hence $\leq \varepsilon^{k-2}$ in the limit as well; the head has at most ε^{-2} terms, each converging to c_i^k. Letting $n \to \infty$ and then $\varepsilon \to 0$ gives $\sum_i (\mu_i^{(n)})^k \to \sum_i c_i^k$. Combining with the hypothesis, $\sum_i c_i^k = \sum_i \mu_i^k$ for every $k \geq 3$.

Power sums determine the sequence. Two ℓ^2 sequences in $[-1, 1]$ with equal power sums $p_k = \sum_i \mu_i^k$ for all $k \geq 3$ have the same entries (with multiplicity). Indeed the largest modulus is recovered as $|\mu|_{\max} = \lim_{k\to\infty} p_{2k}^{1/(2k)}$, and its multiplicity m as $m = \lim_k p_{2k}/\,|\mu|_{\max}^{2k}$; the sign multiplicities split by $p_{2k+1}/\,|\mu|_{\max}^{2k+1} \to m_+ - m_-$, the surplus of positive over negative entries of maximal modulus. Removing these finitely many entries decreases every p_k ($k \geq 3$) by a known amount and leaves an ℓ^2 sequence of strictly smaller maximal modulus; induction on the (discrete, accumulating only at 0) set of moduli recovers the entire sequence. Applying this to (c_i) and (μ_i) gives $c_i = \mu_i$ for all i.

Conclusion. Every subsequence of $(\mu^{(n)})$ has a further subsequence along which $\mu_i^{(n)} \to \mu_i$ for every i; hence the full sequence converges entrywise. □

Theorem 18.14 (Cut convergence implies spectral convergence)**.** *Let* W_n, W *satisfy* (A-Reg-W) *and suppose* $\delta_\square(W_n, W) \to 0$ *(equivalently, after measure-preserving relabellings,* $\|W_n - W\|_\square \to 0$*). Then for every* i*,*

$$\lambda_i^+(W_n) \to \lambda_i^+(W) \qquad \text{and} \qquad \lambda_i^-(W_n) \to \lambda_i^-(W),$$

the positive and negative eigenvalues each converging in signed order. In particular every spectral moment, the spectral radius, and the spectral gap (when the limiting top eigenvalue is simple) are continuous under cut convergence.

Proof. Cut convergence is equivalent to left-convergence, i.e. to convergence of every homomorphism density $t(F, \cdot)$ over *finite simple graphs* F (theorem 17.12; the counting lemma of chapter 16 gives the quantitative bound $|t(F, W_n) - t(F, W)| \leq e(F)\ \|W_n - W\|_\square$ after optimal relabelling, for simple F). Specialise to the *simple* cycles $F = C_k$, $k \geq 3$: by proposition 18.4, $t(C_k, W_n) = \sum_i \lambda_i(W_n)^k$ and $t(C_k, W) = \sum_i \lambda_i(W)^k$, so the power sums of the spectra converge for every $k \geq 3$. We do *not* use $k = 2$: the doubled edge C_2 is not a simple graph, the counting lemma does not cover it, and $t(C_2, \cdot) = \|\cdot\|_{\mathrm{HS}}^2$ is genuinely cut-discontinuous (a quasirandom sequence converging in cut to the constant $\frac{1}{2}$ has $t(C_2, \to)\frac{1}{2} \neq \frac{1}{4}$). This is exactly why lemma 18.13 was stated with hypotheses only for $k \geq 3$. Each spectrum lies in $[-1, 1]$ with the *standing* bound $\sum_i \lambda_i^2 \leq 1$ of proposition 18.4 (no convergence of the $k = 2$ sum is needed — only this uniform bound). Lemma 18.13 now yields entrywise convergence of the signed-ordered eigenvalues. Continuity of the spectral radius is the case $i = 1$; continuity of the gap follows when the limiting $|\lambda_1| > |\lambda_2|$ so that $|\lambda_1(W_n)| - |\lambda_2(W_n)| \to |\lambda_1(W)| - |\lambda_2(W)|$. □

Remark 18.15 (The converse fails: the spectrum is strictly coarser than the cut class)**.** Theorem 18.14 does *not* reverse. Two graphons can have identical spectra yet be at positive cut distance, so eigenvalue convergence does not imply cut convergence. The reason is structural and worth stating: by proposition 18.4 the spectrum encodes exactly the *closed-walk* (cycle) homomorphism densities, and nothing else. Tree densities — $t(P_3, W)$, the density of the two-edge path, for instance — are invisible to the spectrum, because a tree's homomorphism density is not a function of the λ_i alone. A self-contained witness is at hand. Compare the constant graphon $W_a \equiv \frac{1}{2}$ with the half-clique $W_b = \mathbf{1}_{A\times A}$, $A = [0, \frac{1}{2})$. Both have a single nonzero eigenvalue equal to $\frac{1}{2}$ (for W_a on the eigenfunction $\mathbf{1}$; for W_b on $\mathbf{1}_A$, since $\mathbb{W}_b\mathbf{1}_A = |A|\ \mathbf{1}_A = \frac{1}{2}\mathbf{1}_A$), so $\mathrm{spec}(W_a) = \mathrm{spec}(W_b) = \left\{\frac{1}{2}\right\}$ and *every* cycle density agrees, $t(C_k, W_a) = t(C_k, W_b) = (\frac{1}{2})^k$. Yet their path densities differ: writing $d(x) = \int_0^1 W(x, y)\, \mathrm{d}y$ for the degree function, $t(P_3, W) = \int_0^1 d(x)^2\, \mathrm{d}x$, and here $d_a \equiv \frac{1}{2}$ gives $t(P_3, W_a) = \frac{1}{4}$, while $d_b = \frac{1}{2}\mathbf{1}_A$ gives $t(P_3, W_b) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$. Different path densities force $\delta_\square(W_a, W_b) > 0$ (by the counting lemma): identical spectra, different cut class. This is also a sharp foil to the moment-uniqueness theorem theorem 17.11, which determines a graphon up to weak isomorphism — but only from *all* homomorphism densities; the cycle moments alone (equivalently, the spectrum) are a strict subset and so cannot. (Schwenk's cospectral non-isomorphic graphs and their graphon limits give a whole family of such witnesses.) Thus the implications

$$\|\cdot\|_{\mathrm{HS}}\text{-conv.} \implies \|\cdot\|_{2\to2}\text{-conv.} \implies \delta_\square\text{-conv.} \implies \text{eigenvalue conv.}$$

are all strict, the last by the cospectral witness above and the first two by the strictness of eq. (18.5). See also [99, Ch. 11] and [29].

18.4.3 Eigenfunctions need the stronger norm

Cut convergence controls eigen*values* but not eigen*functions*. The reason is visible already in example 18.11: a small symmetric perturbation can split a degenerate or near-degenerate pair and rotate its eigenplane arbitrarily while barely moving the eigenvalues. To pin an eigenfunction one needs (i) the stronger operator-norm convergence and (ii) a spectral gap isolating the eigenvalue. This is the content of the Davis–Kahan theorem, which we state in the form used in chapter 25 (finite-network convergence), where a matrix-concentration argument supplies the operator-norm bound, to certify that a sampled or perturbed network has nearly the same centrality profile as its graphon.

Theorem 18.16 (Davis–Kahan, leading-eigenfunction stability)**.** *Let $W, \widetilde{W}$ satisfy* (A-Reg-W) *with simple leading eigenvalue for W, and let $e_1, \widetilde{e}_1$ be unit leading eigenfunctions of $\mathbb{W}, \widetilde{\mathbb{W}}$. Let*

$\delta = |\lambda_1(W)| - |\lambda_2(W)| > 0$ be the absolute gap. If $\left\|\widetilde{\mathbb{W}} - \mathbb{W}\right\|_{2\to 2} \le \delta/2$, then the angle $\theta \in [0, \frac{\pi}{2}]$ between the two eigenlines obeys

$$\sin\theta \;\le\; \frac{2\left\|\widetilde{\mathbb{W}} - \mathbb{W}\right\|_{2\to 2}}{\delta}, \qquad \textit{where} \qquad \min_{\pm} \|\widetilde{e}_1 \mp e_1\| \;=\; 2\sin(\theta/2). \tag{18.8}$$

In particular, with the sign chosen to align them, since $2\sin(\theta/2) \le \sqrt{2}\sin\theta$ on $[0, \frac{\pi}{2}]$, $\|\widetilde{e}_1 - e_1\| \le 2\sqrt{2}\left\|\widetilde{\mathbb{W}} - \mathbb{W}\right\|_{2\to 2}/\delta$.

Proof sketch. Write $E = \widetilde{\mathbb{W}} - \mathbb{W}$ and decompose $\widetilde{e}_1 = (\cos\theta)e_1 + (\sin\theta)v$ with $v \perp e_1$, $\|v\| = 1$. Project the perturbed eigenequation $\widetilde{\mathbb{W}}\widetilde{e}_1 = \widetilde{\lambda}_1\widetilde{e}_1$ onto $e_1^\perp$. Using $\mathbb{W}v = \sum_{n\ge 2}\lambda_n\langle v, e_n\rangle e_n$ and that $\mathbb{W}e_1 = \lambda_1 e_1$ has no component in $e_1^\perp$, the projection reads $(\widetilde{\lambda}_1 I - \mathbb{W})\big|_{e_1^\perp}(\sin\theta)\,v =$ (projection of $E\widetilde{e}_1$). On $e_1^\perp$ the operator $\widetilde{\lambda}_1 I - \mathbb{W}$ is invertible with norm of its inverse at most $1/\operatorname{dist}(\widetilde{\lambda}_1, \{\lambda_n\}_{n\ge 2})$. For the eigenvalue stability we use corollary 2.27 directly: for any unit x, $\left|\langle\widetilde{\mathbb{W}}x, x\rangle - \langle\mathbb{W}x, x\rangle\right| = |\langle Ex, x\rangle| \le \|E\|$, so every min–max eigenvalue moves by at most $\|E\|$; in particular $\left|\widetilde{\lambda}_1 - \lambda_1\right| \le \|E\| \le \delta/2$. Since the nearest λ_n, $n \ge 2$, is at distance $\ge \delta$ from λ_1, that distance is at least $\delta/2$; hence $\sin\theta \le \|E\|\,\|\widetilde{e}_1\|/(\delta/2) = 2\|E\|/\delta$, which is eq. (18.8). The companion identity $\min_\pm \|\widetilde{e}_1 \mp e_1\| = 2\sin(\theta/2)$ is elementary geometry: for unit vectors at line-angle θ, $\|\widetilde{e}_1 \mp e_1\|^2 = 2 \mp 2\cos\theta$, minimised over the sign at $2 - 2\cos\theta = 4\sin^2(\theta/2)$. (The quantity $\sqrt{2}\sin\theta$ that appears in some statements is instead $\left\|P_1 - \widetilde{P}_1\right\|_{\mathrm{HS}}$, the Frobenius norm of the difference of the two rank-one spectral projectors, not the eigenvector distance.) Full statements and the general (subspace) version are in [56]; the symmetric-operator form is standard. □

The split between theorem 18.14 and theorem 18.16 is the practically important message of this section. Sampling a finite W-random network (chapter 19) yields, at the level we prove there, only *cut-metric* consistency — the empirical graphon converges to W in $\delta_\square$ — and by eq. (18.5) that is *not* enough to move the eigenfunctions. Upgrading to operator-norm convergence of the empirical operator requires a separate matrix-concentration argument, carried out in chapter 25; there it is strong enough to transport both eigenvalues *and* the centrality eigenfunction through theorem 18.16, provided a gap exists. When one only knows that two networks are macroscopically equivalent — close in $\delta_\square$, as the compactness theorem theorem 17.13 delivers — one is guaranteed matching spectra but *not* matching centralities. A small gap is exactly the regime where centrality is ill-conditioned, which is the spectral reason community detection is hard when communities are weakly separated.

18.5 Low-rank reductions and tractable graphon mean field games

We now collect the dividend that the whole chapter was built to pay: when the graphon has finite or rapidly decaying spectrum, the graphon mean field game coupling — an operator on an infinite-dimensional field of measures — collapses to a finite system of scalar order parameters. This is the analytic reason Thread B is solvable, and it is the template for every closed-form graphon model in Part IV.

18.5.1 Finite-rank graphons collapse the coupling

Definition 18.17 (Finite-rank graphon)**.** A graphon W has *rank* r if its operator $\mathbb{W}$ has exactly r nonzero eigenvalues, counted with multiplicity; equivalently

$$W(x,y) = \sum_{k=1}^{r} \lambda_k \, \phi_k(x) \, \phi_k(y), \qquad \langle \phi_k, \phi_\ell \rangle = \delta_{k\ell}, \tag{18.9}$$

for the orthonormal eigenfunctions $\phi_k = e_k$ and nonzero eigenvalues λ_k.

Block graphons (example 18.11, and the multi-block models of chapter 22) are finite rank: a graphon constant on the blocks of an r-part partition has rank at most r. The constant graphon ($r = 1$) and the rank-one graphon $W = f \otimes f$ of worked computation 2.28 are the smallest cases.

The reduction is the following. In the graphon mean field game (Thread B, appendix A.7), a type-x agent is coupled not to a global mean but to the *graphon-aggregated* neighbour field

$$z_t(x) = (\mathbb{W}\bar{m}_t)(x) = \int_0^1 W(x,y) \, \bar{m}_t(y) \, \mathrm{d}y, \qquad \bar{m}_t(y) = \int_{\mathbb{R}} \xi \, m_t^y(\mathrm{d}\xi), \tag{18.10}$$

where $\bar{m}_t(y)$ is the mean of the type-y population m_t^y (recall the convention of appendix A.1: the population field m evolves forward from m_0). The aggregate z_t is the only channel through which the kernel enters the dynamics. For a rank-r kernel eq. (18.9),

$$z_t(x) = \sum_{k=1}^{r} \lambda_k \, \phi_k(x) \, \varrho_k(t), \qquad \varrho_k(t) := \langle \phi_k, \bar{m}_t \rangle = \int_0^1 \phi_k(y) \, \bar{m}_t(y) \, \mathrm{d}y. \tag{18.11}$$

Thus the entire infinite-dimensional aggregate field $x \mapsto z_t(x)$ is determined, at each time, by the r scalars $\varrho_1(t), \dots, \varrho_r(t)$ — the projections of the population mean onto the eigenfunctions. We record the consequence.

Proposition 18.18 (Order-parameter reduction)**.** *Suppose the graphon has rank r as in eq.* (18.9)*, and consider the graphon mean field game of Thread B in which type-x agents interact only through the aggregate eq.* (18.10)*. Then the coupling is carried entirely by the finite-dimensional order-parameter vector $\varrho(t) = (\varrho_1(t), \dots, \varrho_r(t))$ of eq.* (18.11)*: each agent's optimal control and each type's population evolution depend on the population field only through $z_t(x) = \sum_k \lambda_k \phi_k(x) \varrho_k(t)$, and the mean field fixed point reduces to a self-consistent system of r scalar equations for $\varrho(t)$, closed by $\varrho_k(t) = \langle \phi_k, \bar{m}_t \rangle$ with $\bar{m}_t$ the mean of the optimally controlled population. In particular a rank-one kernel gives a single scalar order parameter.*

Proof. By eq. (18.11) the aggregate z_t is a known linear combination of the fixed profiles ϕ_k with time-dependent coefficients $\lambda_k \varrho_k(t)$, so the only data the agents' problems require from the population is the vector $\varrho(t)$. Conversely $\varrho_k(t)$ is, by definition, a linear functional of the population means $\bar{m}_t$, which are produced by the agents best-responding to z_t, hence to $\varrho(t)$. The mean field consistency requirement — that the population generating ϱ is the one optimally responding to the z built from ϱ — is therefore a fixed-point equation on the finite-dimensional $\varrho \in \mathbb{R}^r$ (as a function of time), rather than on the infinite-dimensional field of measures. The number of scalar equations equals the rank r. □

This is the precise sense in which "low rank ⇒ tractable." The full dynamic solution of the reduced system — the coupled Riccati and order-parameter equations — is carried out in chapter 22; here we have isolated the structural collapse that makes it possible, and we now instantiate it on the cleanest case to advance Thread B by one layer.

Worked computation 18.19 (Rank-one graphon: a single scalar order parameter)**.** We continue Thread B (appendix A.7) with the rank-one kernel $W(x,y) = f(x)f(y)$, where $f\colon [0,1] \to [0,1]$ (so that $f \geq 0$ and $W = f \otimes f$ is a genuine $[0,1]$-valued graphon), whose spectrum was computed in worked computation 2.28: a single nonzero eigenvalue $\lambda_1 = \|f\|_{L^2}^2$ with normalised eigenfunction $\phi_1 = f/\|f\|_{L^2}$, and $\ker \mathbb{W} = \{f\}^\perp$. Recall the notation convention in force from Part III onward (appendix A.1): with a graphon W in scope, Brownian motion is written B, so the type-x state obeys $\mathrm{d}X_t^x = (aX_t^x + b\,\alpha_t^x)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x$.

The aggregate collapses to one scalar. By eq. (18.10),

$$z_t(x) = \int_0^1 f(x)f(y)\,\bar{m}_t(y)\,\mathrm{d}y = f(x)\,\varrho(t), \qquad \varrho(t) := \langle f, \bar{m}_t\rangle = \int_0^1 f(y)\,\bar{m}_t(y)\,\mathrm{d}y,$$

a single scalar order parameter $\varrho(t)$ multiplying the fixed profile f. (In the eigenfunction normalisation of eq. (18.11) this is $z_t = \lambda_1\phi_1\,\varrho_1$ with $\phi_1 = f/\|f\|_{L^2}$, $\lambda_1 = \|f\|_{L^2}^2$, and $\varrho_1 = \langle \phi_1, \bar{m}_t\rangle = \varrho/\|f\|_{L^2}$; then $\lambda_1\phi_1 = \|f\|_{L^2}\,f$ and $\|f\|_{L^2}\,\varrho_1 = \varrho$, so $z_t = \lambda_1\phi_1\varrho_1 = \|f\|_{L^2}\,f \cdot \varrho/\|f\|_{L^2} = f\varrho$: the two bookkeepings agree at the level of z_t.) The type-x running cost of Thread B becomes

$$\tfrac{1}{2}(\alpha_t^x)^2 + \tfrac{q}{2}\big(X_t^x - s\,f(x)\,\varrho(t)\big)^2,$$

so a type-x agent solves a *scalar* linear–quadratic problem with a type-dependent target $s\,f(x)\,\varrho(t)$ that is the *same* time signal $\varrho(t)$ scaled by the agent's own centrality weight $f(x)$.

The fixed point is one self-consistent equation. Freezing $\varrho(\cdot)$, each type x faces a tracking LQG problem whose optimal feedback is linear and whose controlled mean $\bar{m}_t^x$ follows a closed linear ODE driven by the target $sf(x)\varrho(t)$; by linearity $\bar{m}_t^x = f(x)\,\psi_t[\varrho]$ for a type-independent functional $\psi_t[\varrho]$ of the order-parameter path (the response of a unit-centrality agent), plus a homogeneous term carrying the initial mean. Closing the loop through the definition of ϱ,

$$\varrho(t) = \langle f, \bar{m}_t\rangle = \int_0^1 f(x)\,\bar{m}_t^x\,\mathrm{d}x = \Big(\int_0^1 f(x)^2\,\mathrm{d}x\Big)\,\psi_t[\varrho] + (\text{initial term}) = \lambda_1\,\psi_t[\varrho] + (\text{initial term}),$$

a single scalar self-consistent equation for the path $\varrho(\cdot)$, with the coupling strength entering only through the scalar $\lambda_1 = \int f^2 = \|f\|_{L^2}^2 = \|\mathbb{W}\|$. This is the cleanest graphon mean field game: an entire continuum of interacting types, solved by one scalar fixed point, exactly as the constant-graphon reduction of Thread B recovers the global-mean LQG of Thread A. The operator-norm smallness condition (A-Coupling), $\|\mathbb{W}\| = \lambda_1 < \kappa$, is here a smallness condition on the single scalar $\int f^2$, and it is the hypothesis under which the scalar fixed point is a contraction and hence unique; the threshold is analysed in chapter 24, and the explicit Riccati form of ψ_t is completed in chapter 22.

18.5.2 Every graphon has a controlled low-rank reduction

Most graphons are not finite rank. But the spectral decomposition supplies a canonical sequence of finite-rank approximations, and the error is exactly the spectral tail. This is the Eckart–

Young theorem in the Hilbert–Schmidt setting, and it tells us when a graphon mean field game is *approximately* low-dimensional.

Proposition 18.20 (Best low-rank approximation). *Let W satisfy* (A-Reg-W) *with modulus-ordered spectrum $|\lambda_1| \geq |\lambda_2| \geq \cdots$ and eigenfunctions e_n. The rank-r truncation*

$$W_r(x,y) := \sum_{n=1}^{r} \lambda_n \, e_n(x) \, e_n(y)$$

is the minimiser of $\|W-U\|_{\mathrm{HS}}$ over all symmetric kernels U of rank at most r, with optimal error

$$\|W-W_r\|_{\mathrm{HS}}^2 = \sum_{n>r} \lambda_n^2, \qquad \|W-W_r\|_{2\to 2} = |\lambda_{r+1}| \,. \tag{18.12}$$

In particular $\|W-W_r\|_{\mathrm{HS}} \to 0$ and $\|W-W_r\|_{2\to 2} \to 0$ as $r \to \infty$ for every graphon.

Proof. The truncation $W - W_r$ has operator $\sum_{n>r} \lambda_n \langle \cdot, e_n \rangle e_n$, whose Hilbert–Schmidt norm squared is $\sum_{n>r} \lambda_n^2$ and whose operator norm is the largest remaining modulus $|\lambda_{r+1}|$, by eq. (18.1). It remains to show that W_r is optimal in both norms. We give a self-contained argument resting only on corollary 2.27, using *singular values*: for a compact self-adjoint operator A write $\sigma_k(A)$ for the k-th largest eigenvalue of the positive operator A^2, square-rooted, so $\sigma_k(A) = |\lambda_k(A)|$ is the k-th largest modulus and $\|A\|_{\mathrm{HS}}^2 = \sum_k \sigma_k(A)^2$. Applying the min–max identity corollary 2.27 to the positive operator A^2 gives the dual ("min") characterisation

$$\sigma_k(A) \;=\; \min_{\dim M = k-1} \; \max_{\substack{x \perp M \\ \|x\|=1}} \; \|Ax\| \,. \tag{18.13}$$

A one-line Weyl inequality. For compact self-adjoint A, B and indices $i, j \geq 1$, $\sigma_{i+j-1}(A+B) \leq \sigma_i(A) + \sigma_j(B)$: pick M_A, M_B achieving the minima in eq. (18.13) for $\sigma_i(A), \sigma_j(B)$, of dimensions $i-1, j-1$; then $M = M_A + M_B$ has dimension $\leq i+j-2$, and for unit $x \perp M$, $\|(A+B)x\| \leq \|Ax\| + \|Bx\| \leq \sigma_i(A) + \sigma_j(B)$, so the min over such M in eq. (18.13) for index $i+j-1$ is bounded by $\sigma_i(A) + \sigma_j(B)$.

Lower bounds. Let U be any symmetric kernel of rank $\leq r$, so its operator has $\sigma_{r+1}(\mathbb{W}_U) = 0$. Writing $W = (W-U) + U$ and applying the Weyl inequality with $A = W-U$, $B = U$ (and $i = j$, $j' = r+1$):

$$\sigma_{j+r}(W) \;\leq\; \sigma_j(W-U) + \sigma_{r+1}(\mathbb{W}_U) \;=\; \sigma_j(W-U) \qquad (j \geq 1).$$

Taking $j = 1$ gives the operator-norm bound $\|W-U\|_{2\to 2} = \sigma_1(W-U) \geq \sigma_{r+1}(W) = |\lambda_{r+1}|$. Summing the squares over $j \geq 1$ gives the Hilbert–Schmidt bound

$$\|W-U\|_{\mathrm{HS}}^2 = \sum_{j\geq 1} \sigma_j(W-U)^2 \;\geq\; \sum_{j \geq 1} \sigma_{j+r}(W)^2 = \sum_{n>r} \lambda_n^2.$$

Both lower bounds are attained by $U = W_r$, which realises the values computed in the first paragraph, so W_r is the minimiser in each norm with the stated error. (No appeal to an external Hoffman–Wielandt inequality is needed.) Convergence to 0 is the convergence of the tail of $\sum_n \lambda_n^2 < \infty$ and of $|\lambda_n| \to 0$. $\square$

Two honest caveats place this result against its neighbours. First, the truncation converges in the *strong* (Hilbert–Schmidt and operator) norms, hence *a fortiori* in cut by eq. (18.5); by proposition 18.18 the corresponding rank-r graphon mean field game is a finite, r-order-parameter system, and by the stability theory of chapters 24 and 25 its equilibrium approximates that of W with an error governed by the tail eq. (18.12). The reduction is therefore quantitatively controlled *per graphon*. Second, the control is *not* uniform over all graphons: the tail $\sum_{n>r} \lambda_n^2$ can decay arbitrarily slowly, so there is no single rank r that approximates every graphon to a fixed accuracy. The uniform statement — that every graphon is within cut distance ε of a step graphon of bounded complexity, with the bound depending only on ε — is a different theorem, the weak regularity lemma, proved in full in theorem 17.16 (packaged as the total boundedness of graphon space, theorem 17.17); it lives in the cut norm rather than the spectral one, and its sampling consequences are developed in chapter 19. The spectral truncation here is sharp and computable; the regularity-lemma truncation there is universal but coarser. Both feed the numerical discretisation of chapter 28, where the coupling integral $\mathbb{W}$ is replaced by a finite quadrature whose accuracy is read off from exactly these two estimates.

Rigor ledger

Proved.

- proposition 18.4: for $k \geq 2$ the k-th spectral moment equals the k-cycle homomorphism density, $\sum_n \lambda_n^k = t(C_k, W)$; in particular $\sum_n \lambda_n^2 = \|W\|_{\mathrm{HS}}^2 \leq 1$, giving uniform ℓ^2 tightness of all graphon spectra.

- theorem 18.7: for a connected nonnegative graphon the spectral radius is a simple eigenvalue with a strictly positive eigenfunction — the eigenvector centrality definition 18.8, the unique nonnegative fixed point of $\mathbb{W}/\lambda_1^+$.

- proposition 18.10: the multiplicative spectral gap controls the geometric rate of relaxation onto the leading mode; worked out, with assortative/disassortative signs and community detection, on the two-block model example 18.11.

- theorem 18.14 via lemma 18.13: cut convergence forces convergence of every eigenvalue in signed order, through the cycle-density/moment identity.

- proposition 18.18 and worked computation 18.19: a rank-r kernel reduces the graphon mean field game coupling to r scalar order parameters; the rank-one kernel reduces Thread B to a single self-consistent scalar equation with coupling $\lambda_1 = \|\mathbb{W}\| = \int f^2$.

- proposition 18.20: the spectral truncation is the best low-rank approximation in Hilbert–Schmidt and operator norm, with error equal to the spectral tail.

Assumed.

- (A-Reg-W): W symmetric, measurable, bounded, so $\mathbb{W}$ is self-adjoint Hilbert–Schmidt — imported from proposition 2.24; continuity of W assumed only in Mercer's theorem theorem 18.2.

- The spectral theorem and Courant–Fischer principle for compact self-adjoint operators (theorem 2.26 and corollary 2.27), and the Hilbert–Schmidt $=$ L^2-kernel isometry (theorem 2.22).
- Equivalence of cut convergence with left-convergence and the counting lemma $|t(F, W_n) - t(F, W)| \le e(F)\delta_\square(W_n, W)$, imported from chapters 16 and 17.

Deferred.

- Mercer's uniform convergence in the non-positive-semidefinite case (it fails; remark 18.3); the diagonal trace identity is instead obtained through the Hilbert–Schmidt iterates in proposition 18.4, with no appeal to trace-class theory.
- The general Krein–Rutman theorem (of which theorem 18.7 is the graphon instance, proved in full in-chapter) → [113, Ch. VI].
- The subspace Davis–Kahan theorem theorem 18.16 in full → [56]; used to transport centrality under sampling in chapter 25.
- The dynamic solution of the reduced order-parameter system (worked computation 18.19) and its Riccati form → chapter 22; the uniqueness threshold in $\|\mathbb{W}\|$ → chapter 24; the uniform cut-norm (weak regularity) truncation → theorem 17.16.
- Centrality ⇒ equilibrium effort, the static comparative static built on eq. (18.3) → chapter 20.

Open.

- Sharp two-sided relations between the cut metric and the operator norm on natural subclasses of graphons (beyond the strict, non-quantitative hierarchy of eq. (18.5)); when is cut convergence *quantitatively* upgradable to operator convergence?
- Identifiability of the eigenfunctions (not just eigenvalues) of a graphon from finite samples in the small-gap regime, where theorem 18.16 is uninformative; the spectral hardness of community detection near the detectability threshold.

Bibliographic notes

The identification of a graphon with a self-adjoint Hilbert–Schmidt operator, and the use of its spectrum as a graph invariant, is developed in Lovász's monograph [99, Ch. 7 and Ch. 11]; the continuity of the spectrum under cut convergence (our theorem 18.14) and the moment/cycle-density mechanism behind it are there and in [29]. The functional-analytic backbone — the spectral theorem, Mercer's theorem, the Krein–Rutman theorem, and Hilbert–Schmidt theory — is standard and we follow [113, 50, 31]; the staging in this book is sections 2.4 and 2.5, of which this chapter is the announced payoff. The Perron–Frobenius reading of eigenvector centrality, and its role in network games, is the entry point of the static graphon-game theory of Parise and Ozdaglar [108], taken up in chapter 20; the spectral view of community structure in stochastic block models, including the second-eigenvector (Fiedler) heuristic worked out in example 18.11, is classical spectral graph theory transported to the continuum. The eigenvector-perturbation bound theorem 18.16 is the symmetric case of the Davis–Kahan $\sin\theta$ theorem [56]; its use to transport spectral structure from a graphon to its sampled finite networks underlies the finite-network convergence theory of chapter 25. The low-rank reductions of section 18.5 are the operator-theoretic form of the rank-one and block-graphon models that recur throughout Part IV [34, 10], and the carried Thread B reduction of worked computation 18.19 is the spectral preview of the explicit dynamic solution in chapter 22.

Chapter 19

Sampling, Regularity, and the Road to Sparsity

The previous four chapters built the dense theory of graph limits as a closed mathematical world, each chapter supplying one faculty the next would need. From an adjacency matrix we passed to a graphon $W\colon [0,1]^2 \to [0,1]$, the single analytic object in which an entire growing family of networks could be held at once (chapter 15); we learned to read such an object through its motif densities $t(F, W)$, the observable moments by which one graphon is told from another (chapter 16); we metrized the resulting space by the cut metric $\delta_\square$ and found it compact, so that limits of networks are guaranteed to exist and to live in the same world (chapter 17); and we recognized W as a self-adjoint Hilbert–Schmidt operator $\mathbb{W}$ (the operator structure itself being proposition 2.24) whose spectrum reads off centrality and community structure (chapter 18). Each of these faculties is internal to the continuum: it lets us compare, classify, and diagonalize graphons among themselves. What none of them supplies is a passage between that continuum and an actual finite network of N nodes — and without such a passage the theory, however complete on its own terms, floats free of the data it is meant to explain.

Building that passage — the *bridge in both directions* between a graphon and a finite network — is the business of this chapter, and crossing it answers three concrete questions. Going outward: given a graphon, how does one *draw* from it a finite graph of any prescribed size? Once a sample is in hand, how faithfully does it *remember* the graphon it came from — do the sample's measurable statistics return the parent's? Going inward: how coarsely can a graphon itself be *compressed*, replaced by a finite, bounded-complexity surrogate, without losing its macroscopic identity? These are the outward, the fidelity, and the inward questions, and the chapter is built to take them in turn.

It does so in two complete and classical halves and an honest negative third. First, in sections 19.1 and 19.2 we define the W-random graph $G(N, W)$ and prove that the motifs of the sample concentrate around the motifs of the generating graphon at the Monte-Carlo rate $N^{-1/2}$; this is the sampling half of the bridge — the outward passage together with its fidelity — and it is the engine that will drive the finite-network convergence theorem of chapter 25. Second, in section 19.3 we prove the *weak regularity lemma*: every graphon, however intricate, is within cut distance ε of a step graphon whose number of steps depends only on ε and not on the graphon. This is the rigorous form of coarse-graining — the inward passage — carrying an explicit complexity bound, and it pins down exactly which features of a network are macroscopically meaningful. The two halves meet in section 19.4, where the law of the sampled graphs, the full list of motif densities, and cut distance are shown to be one and the same equivalence, so that a graphon is recoverable precisely up to the relabeling the cut metric was built to quotient out. Third, and not as an aside, section 19.5 states the price of admission: this entire apparatus is a theory of *dense* graphs, and it sees *every* sparse sequence of real networks as the single uninformative zero graphon while forcing every degree to grow like $\Theta(N)$. This failure is not a technicality to be patched; it is structural, and it is load-bearing motivation — it is the reason the sparse frameworks (L^p graphons, graphex) of chapter 27 have to be built at all.

This chapter assumes the dense graphon machinery of chapters 15 to 18 and carries no LQG thread; its job is to set up chapter 25 (finite-network convergence) and chapter 27 (sparsity). Throughout, the type variable $x \in [0,1]$ is the exchangeable agent label of chapter 15, never a spatial coordinate. No Brownian motion appears, so no W/B clash arises here; W is always the graphon.

19.1 The W-random graph $G(N,W)$

A graphon is, among other things, a recipe for generating random graphs of any size, and it is this generative reading that first turns the closed continuum of chapters 15 to 18 into something one can confront with finite data. We build the outward half of the bridge here. The recipe is the natural "continuum stochastic block model," and it is worth seeing it as the obvious limit of a familiar finite construction. In an ordinary stochastic block model one fixes a handful of blocks and a matrix B of inter-block connection probabilities, assigns each vertex to a block, and connects two vertices with the probability B records for their two blocks. A graphon performs exactly this with the finite block index replaced by the continuous type $x \in [0,1]$ and the probability matrix replaced by the kernel value $W(x,y)$. To draw a graph one therefore does two things in turn: first hand each of the N prospective vertices a type drawn uniformly from $[0,1]$, placing it somewhere in the continuum of kinds; then, for every pair, toss a single biased coin whose chance of coming up an edge is precisely the kernel value the two types prescribe. The first stage fixes a random geometry of types; the second wires that geometry up. We make this precise.

Definition 19.1 (W-random graph)**.** Let $W\colon [0,1]^2 \to [0,1]$ be a graphon, symmetric and measurable as in (A-Reg-W). Fix $N \in \mathbb{N}$. Draw i.i.d. *latent types* $U_1, \dots, U_N \sim \mathrm{Unif}[0,1]$, and, conditionally on the types, draw independent edge indicators

$$\xi_{ij} \mid (U_1, \dots, U_N) \;\sim\; \mathrm{Bernoulli}\big(W(U_i, U_j)\big), \qquad 1 \le i < j \le N,$$

with $\xi_{ji} = \xi_{ij}$ and no self-loops. The resulting simple graph on vertex set $[N] = \{1, \dots, N\}$, with edge set $\{\, ij : \xi_{ij} = 1 \,\}$, is the *$W$-random graph $G(N,W)$*.

A companion construction records the finite sample as a point of graphon space: the *induced step graphon* $W_{[N]}\colon [0,1]^2 \to [0,1]$, obtained from $G(N,W)$ by partitioning $[0,1]$ into N equal intervals $J_1, \dots, J_N$ and setting $W_{[N]}(x,y) = \xi_{ij}$ for $x \in J_i$, $y \in J_j$ (and 0 on the diagonal blocks $J_i \times J_i$, there being no self-loops). This is the pixel picture of definition 15.2 applied to the sample.

The two-stage structure of the definition repays a slow reading, because the whole of the next section turns on it. The latent types $U_1, \ldots, U_N$ are drawn once and then held fixed: they are the raw material on which everything downstream is conditional. Given them, the edge indicators are *genuinely* independent — each pair flips its own coin and no two coins look at one another — so the only channel through which different edges can become correlated is the types they share. A vertex with an unusual type tilts every coin incident to it in the same direction at once; that single shared cause is the entire source of dependence in the model, and isolating it is exactly what the concentration proof will do.

The companion object, the induced step graphon $W_{[N]}$, records the finished sample not as a graph on abstract vertices but as a point of the very space its parent inhabits. One chops $[0, 1]$ into N equal slots, gives vertex i the slot J_i, and paints the block $J_i \times J_j$ with the colour $\xi_{ij} \in \{0, 1\}$ of the edge that was or was not drawn — a literal black-and-white pixel image of the adjacency matrix. The diagonal blocks $J_i \times J_i$ are painted 0, encoding the convention that the model draws no self-loops; since these blocks together occupy only the fraction $N \cdot (1/N)^2 = 1/N$ of the unit square, the choice on the diagonal is invisible in the limit and any convention there would do. This is the pixel picture of definition 15.2, now applied to a random sample rather than to a fixed graph.

Placing the sample inside graphon space is what makes comparison a single number. A finite graph and a continuum kernel are, on their face, objects of different types and cannot be subtracted; but $W_{[N]}$ and W are *both* graphons, so $\delta_\square(W_{[N]}, W)$ is a well-defined distance measuring how much the sample has forgotten of its parent. Recall from chapter 17 the cut norm and cut metric, which we will use repeatedly:

$$\|U\|_\square \;=\; \sup_{S,T \subseteq [0,1]} \Big| \int_{S \times T} U(x, y)\; \mathrm{d}x\; \mathrm{d}y \Big|, \qquad \delta_\square(W, W') \;=\; \inf_{\varphi}\; \|W - W'^{\varphi}\|_\square\,, \tag{19.1}$$

the supremum over measurable S, T and the infimum over measure-preserving bijections φ of $[0, 1]$ (definitions 17.1 and 17.6). The cut norm asks only that two kernels agree on every large-scale "patch integral" $\int_{S \times T}$; it is blind to disagreements that average away when integrated against indicator sets. That is precisely the kind of disagreement a finite sample is forced to exhibit — at the pixel scale $1/N$, the sharp coins $\xi_{ij} \in \{0, 1\}$ can never match the smooth values $W(U_i, U_j) \in [0, 1]$ pointwise. The cut norm is in this sense the *weakest* reasonable norm on graphons, and weakest is exactly what we want: it is the coarsest topology in which sampling could possibly be consistent, because any finer norm would refuse to overlook the unavoidable pixel-scale noise of a finite draw. The infimum over relabelings φ quotients out the one ambiguity the model genuinely cannot resolve — which type label happened to land in which slot — so that $\delta_\square = 0$ means "the same network object up to renaming of types."

It helps to see the two extreme cases the construction sits between, because they are the two models the reader already knows and they bracket everything in between.

Remark 19.2 (Special cases anchor the construction)**.** When $W \equiv p$ is constant, the kernel value $W(U_i, U_j) = p$ does not depend on the types at all, so the first stage of the recipe is inert and every edge is an independent p-coin: $G(N, W)$ is precisely the Erdős–Rényi graph $G(N, p)$ (example 15.9). Here there is no geometry to remember, only an edge density. At the other end, when W is the step graphon of a stochastic block model (example 15.10) — constant on the blocks $V_a \times V_b$ with value the inter-block probability B_{ab} — the uniform type U_i does nothing but decide, by which interval it falls into, the block of vertex i; conditionally on the blocks the edges are again independent coins, now with block-dependent biases. This is exactly the classical planted-partition model. Between these two poles — a structureless single density and a finite palette of block densities — the general W-random graph interpolates continuously: a graphon is the limit in which the finite block index of an SBM is refined into the continuum of types, so that ER and SBM are not separate theories but the constant and piecewise-constant points of one family.

Physics note

The sampling map $W \mapsto G(N, W)$ is the network analogue of drawing a finite configuration from a continuum field, and its randomness comes at two temperatures. The latent types U_i are *quenched* disorder: frozen, vertex-attached randomness that fixes each agent's "kind" once and for all and is not averaged over while the edges are drawn. The edge coins ξ_{ij} are the *annealed* fluctuations living on top of that frozen geometry — thermal noise conditional on the quench. The order of the two stages (quench the types, then flip the edges) is exactly the order in which the concentration argument of section 19.2 will dismantle the randomness: it conditions on the types, controls the conditionally independent edge fluctuations first, and only then averages over the quenched geometry. The two contributions even separate by magnitude — the quenched types will turn out to carry the dominant sensitivity that sets the $N^{-1/2}$ rate, the annealed coins a smaller correction — so the picture is a genuine preview of the proof split, not merely a suggestive label. We do not lean on the analogy beyond this; the mathematics stands on the explicit variance bookkeeping to come.

We now have everything the bridge's outward end requires: a sample $G(N, W)$ of any size, and a way — the induced step graphon and the cut metric — to lay that sample alongside its parent and read off a single number for how far they sit apart. The obvious next question is the fidelity question promised in the preamble: now that a sample exists and can be compared to its parent, *how faithfully does it remember the parent's statistics?* The next section answers it for the observable moments of a network, its motif densities, and the answer is carried by exactly the quenched/annealed decomposition just described — realized concretely as a coupling that drives every edge coin from an independent uniform seed, so that the types and the seeds become the jointly independent coordinates the concentration estimate splits along.

19.2 Concentration of motif densities

That fidelity question — now that the comparison can actually be made, *how faithfully does a draw remember the graphon it came from?* — is what this section takes up, and it answers it in the currency in which a network is really measured: its motif statistics. The answer is the cleanest one could hope for: the statistics of the sample equal those of the parent up to a Monte-Carlo error of order $N^{-1/2}$ that is exponentially unlikely to be exceeded. This is the *first sampling lemma*, the fidelity half of the outward bridge, and it is the engine that will later drive the convergence of finite-network equilibria to their graphon limit in chapter 25. The whole argument is a concrete realization of the quenched/annealed split previewed at the close of section 19.1: we will freeze the types, control the conditionally independent edge noise that lives on top of them, and only then average over the frozen geometry — and a coupling, introduced in the proof, makes this order of operations literal.

To state the claim precisely we need the observables it concerns. Recall (chapter 16) that for a fixed simple graph F on vertex set $[k]$ with $e(F)$ edges — read F as a small pattern one looks for in a network, an edge, a triangle, a path, a star — the homomorphism density of F in a graph G on n vertices is

$$t(F,G) \;=\; \frac{\hom(F,G)}{n^k} \;=\; \frac{1}{n^k}\,\bigl|\{\,\phi\colon [k]\to[n] : \phi \text{ maps every edge of } F \text{ to an edge of } G\,\}\bigr|,$$

and that for the graphon W the corresponding density is

$$t(F,W) \;=\; \int_{[0,1]^k} \prod_{ab\in E(F)} W(x_a,x_b)\; \prod_{i=1}^{k} \mathrm{d}x_i.$$

The two formulas are one and the same average read in two registers. In the finite graph one averages a product of edge *indicators* over all n^k ways of dropping the k vertices of F onto $[n]$; in the graphon one averages a product of edge *probabilities* over all ways of dropping them into the continuum $[0,1]^k$, the discrete count $\hom(F,G)/n^k$ passing into the integral exactly as a Riemann sum passes into its limit. It is worth dwelling on why these deserve the name *moments* of a network, because the analogy organizes everything that follows. A probability law on the line is pinned down by its moments $\mathbb{E}[X^j]$; a graphon is pinned down — this is precisely what section 19.4 will establish — by the countable family $\{t(F,W)\}_F$ as F ranges over all finite graphs. Each F probes one mode of the network's structure: K_2 returns the edge density, the triangle K_3 the clustering, the star S_k the spread of the degree distribution, longer paths the connectivity at range. To say that a sample reproduces its parent's motif densities is therefore the network analogue of the elementary fact that an empirical distribution reproduces the moments of the law it was drawn from. The theorem below makes that analogue quantitative: the moments of the sample equal the moments of the parent up to a Monte-Carlo error that is exponentially unlikely to be large.

Before proving anything, it pays to see in advance why the result is true and where the $N^{-1/2}$ rate comes from, so that the formal proof reads as the execution of a plan rather than a sequence of estimates pulled from the air.

Heuristic (non-rigorous)

Fix F on k vertices. By definition $t(F,G(N,W))$ is an average of N^k terms, one per map $\phi\colon [k]\to[N]$, each term a product $\prod_{ab\in E(F)}\xi_{\phi(a)\phi(b)}$ of edge indicators lying in $\{0,1\}$. Were the N^k terms independent, the law of large numbers would pin the average at its mean. They are *not* independent — two maps that share a vertex share the coins incident to that vertex — but the dependence is *local*: altering what happens at a single vertex disturbs only the $O(N^{k-1})$ maps that pass through it, a vanishing fraction $O(1/N)$ of all maps. The average therefore behaves like an average of N weakly dependent blocks, not of N^k independent terms. Two consequences follow heuristically and will be made exact. First, its mean is, up to an

$O(1/N)$ correction from the diagonal maps that are forbidden a loop, exactly $t(F, W)$, because the expectation of one injective term is the continuum integral. Second, the fluctuation of an average of N weakly dependent bounded blocks is of order $N^{-1/2}$. The entire technical task is to turn "the dependence is local" into a rigorous tail bound.

The tool that converts locality of dependence into a Gaussian tail is the *method of bounded differences*, the quantitative form of the slogan "a function of many independent inputs, no one of which matters much on its own, is sharply concentrated about its mean." The plan it dictates is exactly three moves: write $t(F, G(N, W))$ as a function of a list of *independent* coordinates; check that perturbing any single coordinate moves the function by only a little; and read off the tail. We first record the inequality that performs the third move. It is the bounded-differences (McDiarmid) form of the Azuma–Hoeffding inequality already proved in chapter 3 (theorem 3.28 and remark 3.29); for completeness we supply the one nontrivial step — the bound on the martingale increments — which remark 3.29 only asserts.

Lemma 19.3 (McDiarmid's bounded-differences inequality)**.** *Let $Z_1, \dots, Z_n$ be independent random variables, Z_i taking values in a set $\mathcal{Z}_i$, and let $g\colon \prod_i \mathcal{Z}_i \to \mathbb{R}$ satisfy the bounded-differences condition: for each i there is a constant c_i such that for all z, z' differing only in the i-th coordinate, $|g(z) - g(z')| \le c_i$. Then for every $t > 0$,*

$$\mathbb{P}\Big(|g(Z) - \mathbb{E}[g(Z)]| \ge t \Big) \;\le\; 2\exp\Big(-\frac{t^2}{2\sum_{i=1}^n c_i^2} \Big).$$

Proof. The idea is to reveal the independent inputs $Z_1, \dots, Z_n$ one at a time and watch how the best current guess of g updates as each is disclosed; the bounded-differences hypothesis will cap the size of each update, and Azuma will turn capped updates into a Gaussian tail. Make this precise as a martingale. Let $\mathcal{F}_i = \sigma(Z_1, \dots, Z_i)$ record the information contained in the first i inputs (with $\mathcal{F}_0$ trivial), and set $M_i = \mathbb{E}[g(Z) \mid \mathcal{F}_i]$. Then $(M_i)_{0 \le i \le n}$ is the *Doob martingale* of $g(Z)$: it begins at the unconditional mean $M_0 = \mathbb{E}[g(Z)]$, when nothing has been revealed, and ends at the realized value $M_n = g(Z)$, when everything has, and it telescopes,

$$g(Z) - \mathbb{E}[g(Z)] \;=\; \sum_{i=1}^n (M_i - M_{i-1}).$$

So it suffices to confine each increment $M_i - M_{i-1}$ to an interval of length at most c_i and to invoke Azuma on the sum.

We claim each increment does lie in such an interval. The obstruction to comparing M_i with M_{i-1} directly is that M_{i-1} has already integrated out Z_i while M_i still retains it; to bring the two under a common roof we re-express the smaller σ-field's average as a *resampling*. Let Z_i' be an independent copy of Z_i, drawn from the same law and independent of every Z_j, and write $Z^{(i)} = (Z_1, \dots, Z_{i-1}, Z_i', Z_{i+1}, \dots, Z_n)$ for the configuration in which the i-th coordinate has been replaced by the fresh copy. Because Z_i' is independent of $\mathcal{F}_i$ and has the same law as Z_i, conditioning on $\mathcal{F}_i$ and then averaging over Z_i' is identical to conditioning on $\mathcal{F}_{i-1}$: swapping Z_i for Z_i' erases exactly the one new fact contained in $\mathcal{F}_i \setminus \mathcal{F}_{i-1}$ and leaves all earlier inputs untouched. Hence

$$M_{i-1} \;=\; \mathbb{E}[g(Z) \mid \mathcal{F}_{i-1}] \;=\; \mathbb{E}\Big[g(Z^{(i)}) \mid \mathcal{F}_i \Big],$$

and subtracting this from $M_i = \mathbb{E}[g(Z) \mid \mathcal{F}_i]$ places the two configurations under a single conditional expectation,

$$M_i - M_{i-1} \;=\; \mathbb{E}\Big[\, g(Z) - g(Z^{(i)}) \mid \mathcal{F}_i \Big]\,.$$

Now Z and $Z^{(i)}$ differ only in their i-th coordinate, so the bounded-differences hypothesis governs the inner difference directly. For each fixed value of $\mathcal{F}_i$, define $a_i = \inf_{z_i'}\big(g(Z) - g(Z_1, \ldots, z_i', \ldots, Z_n)\big)$ and let b_i be the corresponding supremum, both taken over the resampled value z_i'; the hypothesis $|g(z) - g(z')| \le c_i$ gives $b_i - a_i \le c_i$. Since $M_i - M_{i-1}$ is a conditional average of that inner quantity over Z_i', it lies in $[a_i, b_i]$, an interval of length at most c_i. One refinement pins the interval in place: a martingale increment is conditionally centered, $\mathbb{E}[M_i - M_{i-1} \mid \mathcal{F}_{i-1}] = 0$, so the interval $[a_i, b_i]$ — which contains the conditional average — must straddle 0. An interval of length at most c_i that contains 0 forces $|M_i - M_{i-1}| \le b_i - a_i \le c_i$ almost surely. This is the one nontrivial input that remark 3.29 merely asserted. With the increments so bounded, the Azuma–Hoeffding inequality (theorem 3.28) applied to the martingale (M_i) yields the stated two-sided estimate $2\exp(-t^2/(2\sum_i c_i^2))$, which is exactly the in-book bounded-differences inequality remark 3.29. (The textbook reference [102, §2] carries out the same computation with the sharper constant $2\exp(-2t^2/\sum_i c_i^2)$, which requires the range form of Hoeffding's lemma; we use only the weaker in-book constant, which suffices for every conclusion below.) □

With the tail in hand, the whole content of the first sampling lemma is the bookkeeping of the two preceding moves: identifying independent coordinates for $g = t(F, G(N, W))$ and bounding the sensitivity of g to each. We carry that out now.

Theorem 19.4 (First sampling lemma: motif concentration)**.** *Let W satisfy* (A-Reg-W) *and let F be a fixed simple graph with $k = v(F)$ vertices and $e(F)$ edges. Sample $G(N, W)$ as in definition 19.1. Then:*

1. *(Unbiasedness up to $O(1/N)$.)*

$$\big|\,\mathbb{E}[t(F, G(N, W))] - t(F, W)\,\big| \;\le\; \frac{\binom{k}{2}}{N}.$$

2. *(Concentration.) For every $\varepsilon > 0$,*

$$\mathbb{P}\Big(\big|\, t(F, G(N, W)) - t(F, W)\,\big| \ge \varepsilon\Big) \;\le\; 2\exp\Big(-\,\frac{\big(\varepsilon - \binom{k}{2}/N\big)_+^2}{2\big(k^2/N + 4\,e(F)^2/N^2\big)}\Big),$$

where $(\cdot)_+$ denotes the positive part. In particular, for each fixed F and $\varepsilon > 0$ there is $N_0 = N_0(k, \varepsilon)$ such that for all $N \ge N_0$,

$$\mathbb{P}\Big(\big|\, t(F, G(N, W)) - t(F, W)\,\big| \ge \varepsilon\Big) \;\le\; 2\exp\Big(-\,\frac{N\varepsilon^2}{16k^2}\Big),$$

and consequently $t(F, G(N, W)) \to t(F, W)$ almost surely as $N \to \infty$ for every fixed F.

Proof. Write the homomorphism density of the sample as an explicit function of all the randomness in the model,

$$g \;=\; t(F, G(N, W)) \;=\; \frac{1}{N^k} \sum_{\phi\colon [k]\to[N]} \;\prod_{ab\in E(F)} \xi_{\phi(a)\phi(b)},$$

where, for a map ϕ that sends some edge of F to a loop (i.e. $\phi(a) = \phi(b)$ for an edge ab), the corresponding indicator $\xi_{\phi(a)\phi(b)}$ is read as 0, since the sample carries no loops. We must show this g concentrates, and the heuristic already told us why to expect it: no single source of randomness governs more than an $O(1/N)$ fraction of the N^k summands.

A coupling that recovers independence. The method of bounded differences (lemma 19.3) demands that g be written as a function of *independent* coordinates, and the two obvious candidates — the types U_i and the coins ξ_{ij} — fail this on their own: by construction ξ_{ij} is Bernoulli$(W(U_i, U_j))$ *given* the types, so the coin's very distribution is steered by U_i and U_j, and the two coordinate families are entangled. Conditioning the coins on the types would restore independence but would also freeze the type randomness, which is precisely the fluctuation we must measure. The standard remedy is to drive every coin from its own external source of randomness, so that the entanglement is pushed into a deterministic rule rather than into the joint law of the inputs. Let $(\eta_{ij})_{1\le i<j\le N}$ be i.i.d. Unif$[0,1]$ *edge seeds*, independent of one another and of all the latent types, and realize each coin by thresholding its seed against the kernel value,

$$\xi_{ij} \;=\; \mathbf{1}\big[\eta_{ij} \le W(U_i, U_j)\big], \qquad \xi_{ji} = \xi_{ij}.$$

Because $\mathbb{P}\big(\eta_{ij} \le W(U_i, U_j) \mid U\big) = W(U_i, U_j)$ and the seeds are independent, this reproduces the law of definition 19.1 exactly: given the types, the ξ_{ij} are again independent Bernoulli variables with the prescribed means. The gain is structural. The model is now a *deterministic* function of the list

$$\big(U_1, \dots, U_N,\ (\eta_{ij})_{i<j}\big),$$

and these $N + \binom{N}{2}$ coordinates *are* jointly independent — all the dependence between coins has been absorbed into the deterministic thresholding, where the bounded-differences hypothesis can examine it one coordinate at a time. This is the quenched/annealed split of the physics note made literal: the U_i are the quenched geometry, the η_{ij} the annealed noise on top of it, and they are now genuinely separate, independent inputs. We apply lemma 19.3 to g regarded as a function of this list.

Mean (unbiasedness up to $O(1/N)$). We first locate the center of g and isolate the source of the $O(1/N)$ bias in (1). Condition on the types and sort the N^k maps by whether they are injective. For an *injective* ϕ the images $\phi(1), \dots, \phi(k)$ are distinct vertices, so the edges of F land on distinct pairs and the coins involved are conditionally independent; hence

$$\mathbb{E}\left[\prod_{ab\in E(F)} \xi_{\phi(a)\phi(b)} \;\middle|\; U\right] \;=\; \prod_{ab\in E(F)} W\big(U_{\phi(a)}, U_{\phi(b)}\big).$$

Averaging over the i.i.d. uniform types, each distinct type $U_{\phi(c)}$ becomes an independent uniform integration variable, and the product collapses to the continuum density,

$$\mathbb{E}\left[\prod_{ab\in E(F)} W\big(U_{\phi(a)}, U_{\phi(b)}\big)\right] \;=\; \int_{[0,1]^k} \prod_{ab\in E(F)} W(x_a, x_b) \prod_{c=1}^{k} \mathrm{d}x_c \;=\; t(F, W),$$

the same value for every injective ϕ. It remains to count the maps that fail injectivity. A map is non-injective exactly when two of its k coordinates coincide; the fraction of maps with at least one coincidence is at most $\binom{k}{2}/N$ (a factor $1/N$ for each of the $\binom{k}{2}$ pairs forced to collide). Each of the N^k summands lies in $[0,1]$, so whatever the non-injective terms contribute, they perturb the average by at most their fraction. Therefore

$$\bigl|\,\mathbb{E}[g] - t(F,W)\,\bigr| \;\le\; \frac{\binom{k}{2}}{N},$$

which is statement (1): the sample density is unbiased up to a $O(1/N)$ diagonal correction, exactly the price of the no-loops convention.

Bounded differences. We now verify the hypothesis of lemma 19.3 for each of the two kinds of coordinate, asking in each case how many of the N^k summands a single coordinate can possibly touch.

(i) A latent type U_i. Changing U_i alters only those terms of the sum whose map ϕ has i in its image. The number of maps $\phi\colon [k] \to [N]$ that hit a prescribed value i is at most $k\,N^{k-1}$ (choose which of the k coordinates equals i, then the remaining $k-1$ coordinates freely). Each term lies in $[0,1]$, so it can change by at most 1. Thus

$$c_{U_i} \;\le\; \frac{k\,N^{k-1}}{N^k} \;=\; \frac{k}{N}.$$

(ii) An edge seed η_{ij}. Changing η_{ij} flips at most the single coin ξ_{ij} (it can only move ξ_{ij} across its threshold), leaving every other coin and every type unchanged. It therefore alters only those terms whose map ϕ sends some edge $ab \in E(F)$ onto the ordered pair (i,j) or (j,i). For a fixed edge ab and a fixed orientation there are at most N^{k-2} such maps (two coordinates are pinned, $k-2$ are free); summing over the $e(F)$ edges and the two orientations gives at most $2\,e(F)\,N^{k-2}$ maps. Hence

$$c_{\eta_{ij}} \;\le\; \frac{2\,e(F)\,N^{k-2}}{N^k} \;=\; \frac{2\,e(F)}{N^2}.$$

Assembling the tail. The denominator in lemma 19.3 is the sum of the squared sensitivities, and the two coordinate families contribute to it on very different scales. There are N types, each with sensitivity at most k/N, and at most $N^2/2$ edge seeds, each with sensitivity at most $2e(F)/N^2$, so

$$\sum_i c_i^2 \;\le\; N\cdot\Bigl(\frac{k}{N}\Bigr)^2 + \frac{N^2}{2}\cdot\Bigl(\frac{2e(F)}{N^2}\Bigr)^2 \;=\; \frac{k^2}{N} + \frac{2\,e(F)^2}{N^2} \;\le\; \frac{k^2}{N} + \frac{4\,e(F)^2}{N^2}.$$

The first term, of order $1/N$, dominates; the seed term is smaller by a further factor $1/N$. This is already the $N^{-1/2}$ rate in disguise: a denominator of order $1/N$ in $\exp(-t^2/2\sum c_i^2)$ means deviations of size $t \sim N^{-1/2}$ are the natural scale of fluctuation. Applying lemma 19.3 with $t = \varepsilon - \binom{k}{2}/N$ — the slack $\binom{k}{2}/N$ absorbing the mean shift of (1), so that the bound is about the true target $t(F,W)$ rather than about $\mathbb{E}[g]$ — yields exactly statement (2). To read off the clean asymptotic form, fix k and ε and take N large. Once $N \ge k^2$ the edge-seed term obeys $4e(F)^2/N^2 \le k^2/N$ (using the crude $e(F) \le \binom{k}{2} \le k^2/2$), so the whole denominator $2(k^2/N + 4e(F)^2/N^2)$ is at most $4k^2/N$; and once N is large enough that $\binom{k}{2}/N \le \varepsilon/2$, the numerator obeys $\bigl(\varepsilon - \binom{k}{2}/N\bigr)^2 \ge \varepsilon^2/4$. Substituting these two estimates turns the exponent $-\bigl(\varepsilon - \binom{k}{2}/N\bigr)^2/2(k^2/N + 4e(F)^2/N^2)$ into $-(\varepsilon^2/4)/(4k^2/N) = -N\varepsilon^2/16k^2$, which is the simplified bound $2\exp(-N\varepsilon^2/16k^2)$. Finally, this right-hand side is summable in N (a geometric-in-N sequence), so the first Borel–Cantelli lemma (theorem 3.27) tells us that the event $\{\,|t(F, G(N, W)) - t(F, W)| \geq \varepsilon\,\}$ occurs for only finitely many N almost surely; as this holds for every $\varepsilon > 0$, the deviation tends to 0, upgrading convergence in probability to almost-sure convergence. □

The proof exhibits, term by term, the two-temperature decomposition promised in the physics note. The latent types contributed the dominant sensitivity k^2/N — the quenched geometry, frozen before any edge is drawn — while the edge seeds contributed only the lower-order $e(F)^2/N^2$ — the annealed fluctuations riding on a fixed geometry. It is the quenched part, the uncertainty in *which* types were drawn, that sets the $N^{-1/2}$ Monte-Carlo rate; the annealed coins are a genuine but subdominant correction. The split was not a heuristic gloss laid over the proof after the fact but the very seam along which the coupling cut the randomness into independent pieces.

Remark 19.5 (From motifs to the cut metric: the second sampling lemma)**.** Theorem 19.4 controls each *fixed* motif F, with constants that degrade as F grows. That is enough to recover any prescribed finite list of network statistics, but it is not enough to claim that the sample resembles its parent in *every* respect at once: nothing in the first sampling lemma stops the many small discrepancies, one per motif, from conspiring. The deeper statement, which controls all motifs simultaneously and is the input chapter 25 actually consumes, is that the *sampled graphon itself* converges to its parent in the cut metric — the single distance, introduced in section 19.1, that already bundles all motif comparisons into one number. The *second sampling lemma* of Borgs–Chayes–Lovász–Sós–Vesztergombi makes this quantitative: there is an absolute constant C such that, for the induced step graphon $W_{[N]}$ of definition 19.1,

$$\mathbb{P}\Big(\delta_\square(W_{[N]}, W) \;>\; \frac{C}{\sqrt{\log N}}\Big) \;\leq\; \exp\Big(-\,\frac{N}{C\log N}\Big). \tag{19.2}$$

Thus $\delta_\square(W_{[N]}, W) \to 0$ almost surely: a graphon is recovered in cut distance, up to relabeling and a slowly vanishing error, from a single large sample.

The slow $1/\sqrt{\log N}$ rate is striking next to the clean $N^{-1/2}$ of a single motif, and it is genuine rather than an artifact of the proof. The reason lies in what the cut norm asks of the sample. Where a single motif compares the kernels against *one* fixed test pattern, the cut norm $\left\|W_{[N]} - W\right\|_\square$ takes a *supremum* over all pairs of sets (S, T), and the sample must match its parent against that whole exponentially large family of tests at once. The right way to see where the logarithm comes from is to anticipate the next section. To pin a graphon to within cut-norm accuracy η over all test pairs simultaneously, one needs — by the weak regularity lemma of section 19.3 (theorem 19.8) — a step approximation with on the order of $4^{1/\eta^2}$ blocks; only at that resolution does the supremum over (S, T) become controllable. A sample of N vertices can populate and resolve such a block structure only when N comfortably exceeds the number of blocks, $N \gtrsim 4^{1/\eta^2}$. Inverting this relation between the achievable accuracy and the sample size, $1/\eta^2 \lesssim \log N$, gives exactly $\eta \gtrsim 1/\sqrt{\log N}$. The logarithm is thus the footprint of the exponential complexity that uniform cut-norm control demands; it is the same $4^{1/\varepsilon^2}$ bound the next section will prove, read here as a sampling rate, and it cannot be removed.

The proof is given in full in theorem F.9; we do not reproduce it here, but we record its assembly because it is exactly the two-piece split the heuristic predicts. One routes the comparison through the *weighted* sample — the step graphon $W_{\mathbb{H}(N,W)}$ whose block values are the conditional edge means $W(U_i, U_j)$ rather than the realized coins ξ_{ij} — by the triangle inequality,

$$\delta_\square(W_{[N]}, W) \;\le\; \left\| W_{[N]} - W_{\mathbb{H}(N,W)} \right\|_\square \;+\; \delta_\square(W_{\mathbb{H}(N,W)}, W),$$

and the two pieces are precisely the annealed and the quenched contributions seen above, now measured in cut norm. The first term is the cut norm of an array of conditionally independent, mean-zero edge fluctuations $\xi_{ij} - W(U_i, U_j)$ — the annealed noise on a frozen geometry; bounding it is where the union bound over the 2^{2N} test pairs (S, T) meets theorem 3.28, giving $O(N^{-1/2})$ with probability $1 - e^{-\Omega(N)}$, the exponentially small failure probability visible on the right of (19.2). The second term compares the *smooth* weighted sample to its parent — the quenched fluctuation in which types were drawn — and is controlled by the weighted sampling lemma (theorem F.7), which is where the slow $1/\sqrt{\log N}$ rate is incurred, together with the bounded-differences concentration of lemma 19.3 in the N types. The two rates then combine, the slower one prevailing. The result is due to Borgs–Chayes–Lovász–Sós–Vesztergombi [29]; see also [99, Ch. 10]. Chapter 25 uses (19.2), not the fixed-motif theorem 19.4, as the input to finite-network convergence — it is the uniformity over all test pairs, not the control of any one motif, that the limit theorem there requires.

Worked computation 19.6 (Sampling a rank-one graphon)**.** Take the rank-one kernel $W(x, y) = f(x)f(y)$ with a measurable $f\colon [0,1] \to [0,1]$ — the "Curie–Weiss" graphon of example 15.11, whose single nonzero eigenvalue $\lambda_1 = \|f\|_{L^2}^2$, with eigenfunction $f/\|f\|_{L^2}$, is computed in worked computation 2.28 (see also the operator discussion of worked computation 18.19). We instantiate the sampling theory on it explicitly. This kernel is also the carrier of Thread B, the Graphon-LQG model: the scalar order-parameter reduction we see here — everything collapsing onto the single number λ_1 and the linear functional $\bar f$ — previews the one-parameter self-consistent equation that chapters 20 and 22 build on the same rank-one structure.

Edge density. The kernel factorizes across its two arguments, so the double integral splits into a product of identical single integrals. With $\bar f = \int_0^1 f(x)\,\mathrm{d}x$,

$$t(K_2, W) \;=\; \int_{[0,1]^2} f(x)f(y)\,\mathrm{d}x\,\mathrm{d}y \;=\; \Big(\int_0^1 f(x)\,\mathrm{d}x\Big)\Big(\int_0^1 f(y)\,\mathrm{d}y\Big) \;=\; \bar f^{\,2}.$$

The edge density depends on the profile f only through its mean $\bar f$ — a first sign of the scalar reduction to come. By the first sampling lemma, $G(N, W)$ has expected edge density $\bar f^{\,2} + O(1/N)$, and its empirical edge density concentrates at $\bar f^{\,2}$ with fluctuations $O(N^{-1/2})$.

Triangles. For $F = K_3$ ($k = 3$, $e(F) = 3$) the three kernel factors, one per edge of the triangle, multiply to a separable integrand: each type variable x_c appears in exactly the two edges meeting at vertex c, contributing a factor $f(x_c)$ apiece, hence $f(x_c)^2$ in total. Regrouping by variable,

$$t(K_3, W) = \int_{[0,1]^3} \underbrace{f(x_1)f(x_2)}_{\text{edge }12}\,\underbrace{f(x_2)f(x_3)}_{\text{edge }23}\,\underbrace{f(x_3)f(x_1)}_{\text{edge }31}\,\mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3 = \prod_{c=1}^{3}\int_0^1 f(x_c)^2\,\mathrm{d}x_c = \Big(\int_0^1 f(x)^2\,\mathrm{d}x\Big)^3 = \lambda_1^3,$$

the cube of the leading eigenvalue $\lambda_1 = \|f\|_{L^2}^2$ — the rank-one spectral structure of worked computation 18.19 surfacing directly inside a motif count, with no spectral calculation required. The same separation extends to any motif: for any F the integrand factorizes across the type variables, so $t(F, W) = \prod_{c \in V(F)} \int_0^1 f(x)^{\deg_F(c)}\, dx$, a monomial in the moments $\int_0^1 f^d$ of the profile, each vertex of degree d contributing one factor $\int_0^1 f^d$ — so that in particular each leaf contributes a $\bar{f}$ and each degree-two vertex a λ_1. The concentration bound specializes, with $k = 3$ giving $16k^2 = 144$, to $\mathbb{P}(|t(K_3, G(N, W)) - \lambda_1^3| \geq \varepsilon) \leq 2\exp(-N\varepsilon^2/144)$ for N large.

Degrees. The degree function is the row sum of the kernel, and the same factorization makes it explicit. Conditional on $U_i = x$, the degree of vertex i is a sum of $N-1$ independent coins, $\mathrm{Binomial}\big(N-1,\, d_W(x)\big)$, with

$$d_W(x) \;=\; \int_0^1 W(x,y)\, dy \;=\; f(x)\int_0^1 f(y)\, dy \;=\; f(x)\,\bar{f}.$$

So a vertex of type x connects to a fraction $f(x)\bar{f}$ of the rest: the profile value $f(x)$ scales the whole degree, and the population mean $\bar{f}$ sets the overall level. Conditional on $U_i = x$ the incident coins are i.i.d. bounded with this mean, so the strong law for bounded summands (proposition 3.24) gives the normalized degree $\deg(i)/(N-1) \to f(U_i)\,\bar{f}$ almost surely; and the empirical distribution of normalized degrees converges weakly to the law of $f(U)\bar{f}$, $U \sim \mathrm{Unif}[0,1]$, by the same empirical-measure argument detailed in proposition 19.14(3). We will return to this in section 19.5: it already shows that every degree is $\Theta(N)$, the hallmark of a *dense* model, no matter how the profile f is chosen.

What the computation foreshadows. Across all three observables the entire profile f entered only through the two scalars $\bar{f} = \int f$ and $\lambda_1 = \int f^2$: the edge density is $\bar{f}^2$, the triangle density λ_1^3, the degree law that of $f(U)\bar{f}$. This collapse of an infinite-dimensional kernel onto a couple of scalar summaries is the rank-one structure that Thread B, the Graphon-LQG model, will exploit: when $W = ff^\top$ couples agents, the network coupling each agent feels reduces to a single scalar order parameter, and the graphon fixed point reduces to one self-consistent equation in that scalar — the rank-one reductions of chapters 20 and 22. We flag the parallel because it is the same algebraic mechanism, not because the thread is carried here: this chapter proves nothing about games, and the order parameter above is purely a statistic of the sampled graph. The point is only that the dimensional collapse the thread relies on is already visible, for free, in how a rank-one graphon samples.

Having shown that a sample faithfully records every macroscopic statistic of its parent — each fixed motif by theorem 19.4, and the whole kernel in cut distance by (19.2) — we have completed the outward bridge and its fidelity. But the rank-one computation just performed quietly raised a sharper question. There the entire content of an infinite-dimensional kernel was carried by two numbers; in general, how much of a graphon's macroscopic content is genuinely irreducible? If finite samples faithfully encode a graphon, and if the relevant topology is the cut norm in which that encoding is consistent, then how *few* parameters does the cut-norm-visible structure actually require? This is the reverse of the move we have been making. Instead of pushing a graphon outward to a finite sample, we now compress the graphon inward, replacing it by a finite, bounded-complexity surrogate that is cut-norm-indistinguishable from it. The next section answers the question in the strongest possible form — a bound on the number of parameters that depends on the target accuracy alone, not on the graphon — and it works in exactly the cut norm (19.1) this section has established as the relevant topology.

19.3 The weak regularity lemma: coarse-graining made rigorous

This section answers that question by reversing the direction of travel. Sampling went from a graphon to a finite graph; coarse-graining stays inside the graphon world but compresses, replacing an arbitrary graphon by a *finite, bounded-complexity* surrogate — a step graphon with controllably few steps — that is indistinguishable from the original at the resolution of the cut norm. And it is the *same* cut norm (19.1) throughout: we are not changing the yardstick, only running it backwards. Where sampling asked how well a finite draw remembers its continuum parent, coarse-graining asks how cheaply the continuum parent can be *described* so that no cut-norm test can tell the description from the original. The answer — a complexity bound depending on the target accuracy and nothing else — is the content of the regularity lemma, and it is what pins down which features of a network survive as macroscopically meaningful.

A word on which regularity lemma. The phrase usually summons Szemerédi's celebrated theorem, which partitions a graph into a bounded number of parts almost all of whose *pairs* behave like random bipartite graphs — a pair-by-pair, "ε-regular" guarantee of remarkable strength. That strength is expensive: the number of parts it forces is a tower of exponentials in $1/\varepsilon$, and (as remark 19.10 explains) the tower is provably necessary. We do not need it. Every use the cut-metric theory makes of regularity — the compactness theorem of chapter 17, the finite-network convergence of chapter 25 — asks only that one *global* integral functional, the cut norm, be made small, never that individual pairs be pseudorandom. The *weak* (Frieze–Kannan) form delivers exactly that weaker conclusion, and in exchange it costs only a clean $4^{1/\varepsilon^2}$ parts and admits a short, transparent Hilbert-space proof: the whole argument is a single monotone-energy bookkeeping in $L^2([0,1]^2)$. Asking for precisely what the applications consume, and no more, is what buys the tame bound; this is the trade we make deliberately, and remark 19.10 returns to weigh it.

The object that does the compressing is the simplest one imaginable: given a partition of the type space into finitely many parts, declare the kernel constant on every block of the induced grid, taking on each block the average of W there. "Constant on each block" is the defining feature of a step graphon, the discretized SBM-like kernels of example 15.10, and "the average" is not an arbitrary choice but the geometrically forced one. The functions that are constant on the blocks form a finite-dimensional subspace of $L^2([0,1]^2)$; replacing W by its blockwise average is exactly orthogonally *projecting* it onto that subspace, and the orthogonal projection is the unique closest point of the subspace in the L^2 sense — the best block-constant approximation of W there is. That the blockwise average and the L^2-nearest block-constant function coincide is the same elementary fact that the mean minimizes mean-square error, here applied one block at a time. Reading the projection probabilistically, it is the conditional expectation of W given the coarsened σ-algebra that can only see which block a point lies in. Holding these three descriptions — blockwise average, orthogonal projection, conditional expectation — as one object is what makes the proof short, because each lends it a different tool: averaging makes $W_{\mathcal{P}}$ a graphon, projection gives Pythagoras, conditioning gives the contraction.

Definition 19.7 (Partition and stepping operator)**.** A *partition* $\mathcal{P} = \{V_1, \ldots, V_q\}$ of $[0,1]$ into finitely many measurable parts of positive measure induces the closed subspace

$$\mathcal{S}_{\mathcal{P}} \;=\; \operatorname{span}\Big\{ \mathbf{1}_{V_i \times V_j} \;:\; 1 \le i, j \le q \Big\} \;\subseteq\; L^2([0,1]^2)$$

of functions constant on each block $V_i \times V_j$. The *stepping operator* $W \mapsto W_{\mathcal{P}}$ is the L^2-orthogonal projection onto $\mathcal{S}_{\mathcal{P}}$; concretely, for $x \in V_i$ and $y \in V_j$,

$$W_{\mathcal{P}}(x,y) \;=\; \frac{1}{|V_i|\;|V_j|}\int_{V_i\times V_j} W(s,t)\,\mathrm{d}s\,\mathrm{d}t \;=\; \mathbb{E}[W \mid \mathcal{P}\otimes\mathcal{P}]\,(x,y),$$

the average of W over the block. The result $W_{\mathcal{P}}$ is itself a graphon (symmetric, $[0,1]$-valued), called the *step graphon* of W with respect to $\mathcal{P}$; it has at most q steps. Because $W_{\mathcal{P}}$ is a conditional expectation it preserves the integral, $\int W_{\mathcal{P}} = \int W$, and it is a contraction on every L^p, $\|W_{\mathcal{P}}\|_p \le \|W\|_p$.

With the stepping operator in hand we need a quantity to drive the construction, and the right one is already implicit in the projection picture. Projecting onto a larger subspace can only move the projected vector closer to W, hence can only *lengthen* it; so the squared L^2 norm $\|W_{\mathcal{P}}\|_{L^2}^2$ of the step graphon — the "energy" captured by the partition $\mathcal{P}$ — can only increase when $\mathcal{P}$ is refined. This is the monotone that will count the steps. Two facts make it usable. First, it is bounded above: since $0 \le W \le 1$ we have $W^2 \le W$, so the total energy $\int W^2 \le \int W \le 1$ caps every captured energy from above. A quantity that increases at each step and cannot exceed 1 can take only finitely many steps — provided each step is not too small. Second, then, every refinement we make will be *forced* to raise the energy by a fixed amount: whenever the current step graphon still differs from W by more than ε in cut norm, we will exhibit a single refinement that converts that cut-norm deficit into an energy gain of at least ε^2. A bounded quantity climbing in jumps of at least ε^2 exhausts itself after at most $1/\varepsilon^2$ jumps, and at that point no cut-norm deficit can remain. The whole proof is the execution of this one idea; the only thing to verify is the conversion of deficit into energy, and that is pure Hilbert-space geometry.

Physics note

The energy $\|W_{\mathcal{P}}\|_{L^2}^2 = \int W_{\mathcal{P}}^2$ is the variance of the kernel captured at the resolution of the partition $\mathcal{P}$. Refining $\mathcal{P}$ is a renormalization-group step that resolves finer block structure; the energy is a monotone "captured-variance" functional that increases at each refinement and is bounded above by the total $\int W^2 \le 1$. The proof below is precisely the statement that you can only refine a bounded number of times before no large patch carries unresolved correlation — a quantitative stopping rule for block-spin coarse-graining. Loosely, the energy plays the role of an entropy- or c-function-like monotone along the refinement flow; this is an analogy only, and the mathematical content is entirely in the monotone increase (19.4) below.

Theorem 19.8 (Weak regularity lemma for graphons)**.** *Let $W\colon [0,1]^2 \to [0,1]$ be a graphon. For every $\varepsilon > 0$ there is a partition $\mathcal{P}$ of $[0,1]$ into at most*

$$q \;\le\; 4^{\lceil 1/\varepsilon^2\rceil}$$

parts such that the step graphon $W_{\mathcal{P}}$ satisfies

$$\|W - W_{\mathcal{P}}\|_{\square} \;\le\; \varepsilon.$$

The number of parts depends only on ε, not on W.

Proof. We build the partition greedily, increasing the captured energy by at least ε^2 at each step. Set $\mathcal{P}_0 = \{[0,1]\}$, the trivial partition, so that $W_{\mathcal{P}_0} \equiv \int W$ is constant.

Suppose after t steps we have a partition $\mathcal{P}_t$ with $\|W - W_{\mathcal{P}_t}\|_\square > \varepsilon$. By the definition (19.1) of the cut norm there exist measurable sets $S, T \subseteq [0,1]$ with

$$\Big|\int_{S\times T}(W - W_{\mathcal{P}_t})\Big| \;=\; |\langle W - W_{\mathcal{P}_t},\, \mathbf{1}_{S\times T}\rangle| \;>\; \varepsilon, \tag{19.3}$$

where $\langle\cdot,\cdot\rangle$ is the $L^2([0,1]^2)$ inner product. Refine $\mathcal{P}_t$ by the two sets S and T: replace each part V of $\mathcal{P}_t$ by the (at most four) nonempty sets among $V\cap S\cap T,\ V\cap S\cap T^c,\ V\cap S^c\cap T,\ V\cap S^c\cap T^c$. Call the result $\mathcal{P}_{t+1}$; it has at most 4 times as many parts as $\mathcal{P}_t$, and by construction $\mathbf{1}_{S\times T} \in \mathcal{S}_{\mathcal{P}_{t+1}}$.

We claim the energy jumps by at least ε^2:

$$\|W_{\mathcal{P}_{t+1}}\|_{L^2}^2 - \|W_{\mathcal{P}_t}\|_{L^2}^2 \;=\; \|W_{\mathcal{P}_{t+1}} - W_{\mathcal{P}_t}\|_{L^2}^2 \;>\; \varepsilon^2. \tag{19.4}$$

The first equality is Pythagoras: $W_{\mathcal{P}_t}$ is the projection of $W_{\mathcal{P}_{t+1}}$ onto the smaller space $\mathcal{S}_{\mathcal{P}_t} \subseteq \mathcal{S}_{\mathcal{P}_{t+1}}$, so $W_{\mathcal{P}_{t+1}} - W_{\mathcal{P}_t} \perp W_{\mathcal{P}_t}$. For the inequality, write $h = \mathbf{1}_{S\times T} \in \mathcal{S}_{\mathcal{P}_{t+1}}$, so that $\|h\|_{L^2} \le 1$. Using $W - W_{\mathcal{P}_t} \perp \mathcal{S}_{\mathcal{P}_t}$ and then $W - W_{\mathcal{P}_{t+1}} \perp \mathcal{S}_{\mathcal{P}_{t+1}} \ni h$,

$$\langle W - W_{\mathcal{P}_t}, h\rangle = \langle W_{\mathcal{P}_{t+1}} - W_{\mathcal{P}_t}, h\rangle \;\le\; \big\|W_{\mathcal{P}_{t+1}} - W_{\mathcal{P}_t}\big\|_{L^2}\,\|h\|_{L^2} \;\le\; \big\|W_{\mathcal{P}_{t+1}} - W_{\mathcal{P}_t}\big\|_{L^2}.$$

Indeed, $\langle W - W_{\mathcal{P}_t}, h\rangle = \langle W - W_{\mathcal{P}_{t+1}}, h\rangle + \langle W_{\mathcal{P}_{t+1}} - W_{\mathcal{P}_t}, h\rangle$, and the first summand vanishes because $h \in \mathcal{S}_{\mathcal{P}_{t+1}}$ while $W - W_{\mathcal{P}_{t+1}}$ is orthogonal to that subspace. Combining with (19.3), $\big\|W_{\mathcal{P}_{t+1}} - W_{\mathcal{P}_t}\big\|_{L^2} > \varepsilon$, which is (19.4).

Now sum: after m refinement steps, $\|W_{\mathcal{P}_m}\|_{L^2}^2 \ge \|W_{\mathcal{P}_0}\|_{L^2}^2 + m\,\varepsilon^2 \ge m\,\varepsilon^2$. But $\|W_{\mathcal{P}_m}\|_{L^2}^2 \le \|W\|_{L^2}^2 \le 1$, because $0 \le W \le 1$ implies $W^2 \le W \le 1$ and the stepping operator is an L^2 contraction. Hence the process must stop after at most $m \le \lceil 1/\varepsilon^2\rceil$ steps — and it can only stop because the cut-norm deficit (19.3) is no longer available, i.e. because $\|W - W_{\mathcal{P}_m}\|_\square \le \varepsilon$. Each step multiplies the number of parts by at most 4, so the terminal partition has at most $4^{\lceil 1/\varepsilon^2\rceil}$ parts. This is the assertion. $\square$

Two features of this proof deserve emphasis because they recur in the applications. The first is the most surprising thing about the statement, and it is worth saying plainly why it holds. The bound $4^{\lceil 1/\varepsilon^2\rceil}$ on the number of parts mentions neither W nor any notion of how big or how intricate W is: a kernel with structure on arbitrarily fine scales is approximated to within ε by the *same* bounded number of blocks as a featureless constant. The reason is visible in the proof: the energy budget that limits the number of steps is $\int W^2 \le 1$, a single number bounded for *every* graphon by the range $[0,1]$ alone, and each step spends at least ε^2 of it. The graphon's complexity never enters the count because the cut norm only ever measures W against block-constant test functions, and there are only so many orthogonal directions of size $\ge \varepsilon$ to be found before the bounded energy is used up. The complexity of the surrogate is thus set by the target accuracy alone — which is exactly what makes "the macroscopic content of a graphon is finite-dimensional" a theorem rather than a hope, and what the estimation discussion of section 19.4 will turn into a parameter count independent of sample size.

The second feature is that the proof is constructive in a usable sense. The partition is not produced by an abstract compactness or averaging argument; it is grown one block-refinement at a time, and each refinement is driven by a concrete witness — the pair of sets (S, T) at which the current surrogate most visibly fails the cut-norm test. One looks for the patch $S \times T$ on which the residual $W - W_{\mathcal{P}_t}$ still integrates to more than ε, cuts along S and T, and repeats; the algorithm halts precisely when no such patch survives. This greedy, witness-by-witness growth is the same structure a statistician exploits when fitting a stochastic block model to data, splitting groups along the most informative cut until no large discrepancy remains, and we return to it in section 19.4 as the reason weak regularity, not its tower-bound cousin, is the estimation-theoretic object of interest.

Corollary 19.9 (Bounded-complexity approximation in cut metric)**.** *Every graphon lies within cut distance ε of the set of step graphons with at most $4^{\lceil 1/\varepsilon^2 \rceil}$ steps. Equivalently, step graphons of bounded complexity are $\delta_\square$-dense in graphon space.*

Proof. Theorem 19.8 gives $\|W - W_{\mathcal{P}}\|_\square \le \varepsilon$, and $\delta_\square \le \|\cdot\|_\square$ on the diagonal (the infimum over relabelings in (19.1) includes the identity). The displayed bound on the number of parts is that of the theorem. $\square$

This corollary is the structural reason the cut metric is the "right" topology for the dense theory, and it closes a loop opened back in chapter 17. The cut metric is exactly the topology in which the genuinely finite-dimensional objects — bounded-complexity step graphons — are dense, and density of finite-parameter approximants is what makes the space small enough to be compact. Concretely: for each ε the corollary says the finite family of step graphons with at most $4^{\lceil 1/\varepsilon^2 \rceil}$ steps is an ε-net, every graphon sitting within cut distance ε of one of them. Since the values of a step graphon with q blocks live in the compact cube $[0, 1]^{q \times q}$, each such net is itself covered by finitely many cut-balls of any smaller radius, so the ε-nets can be taken finite; a metric space with a finite ε-net for every ε is totally bounded, and (the space being complete, by chapter 17) compact. The weak regularity lemma is therefore the workhorse behind the compactness theorem theorem 17.13: it is what manufactures the finite ε-nets that the abstract argument there consumes. The inward direction of the bridge and the compactness of the limit world are, in this sense, the same fact stated twice.

Remark 19.10 (Weak versus strong regularity)**.** It is worth being precise about what we gave up to get the tame bound, because the two regularity lemmas differ in exactly the way their applications do. Szemerédi's original lemma [128] produces an *equitable* partition (all parts of essentially equal size) almost every pair of whose parts is "ε-regular": on the block $V_i \times V_j$ the kernel behaves like a constant not merely *on average* but on every reasonably large sub-rectangle $S \times T$ with $S \subseteq V_i$, $T \subseteq V_j$, the edge density across any such sub-pair deviating from the block average by at most ε. This is a *local*, pair-by-pair pseudorandomness guarantee, and it is precisely what one needs for arguments that count copies of a fixed small graph by multiplying densities across its edges — the embedding and counting lemmas at the heart of extremal and additive combinatorics, where one must locate structure *inside* individual blocks. That strength is not free: the number of parts it forces is a tower of exponentials, an iterated exponential of height $\sim \varepsilon^{-5}$, and Gowers showed this tower is genuinely necessary, not an artifact of the proof [99, Ch. 9]. The reason is structural:

pinning every pair to within ε requires resolving correlations on all scales simultaneously, and each refinement to fix one bad pair can spoil others, so the partition must be refined astronomically many times.

The weak form keeps only the *global* consequence. Cut-norm control (19.3) asks that $W - W_{\mathcal{P}}$ integrate to nearly zero over *every* $S \times T$ drawn from the whole of $[0, 1]$ at once — a single supremum, not a guarantee block by block — and this is strictly weaker: it permits individual pairs to be irregular as long as their irregularities cancel in the global integral. That weaker conclusion is, however, exactly and only what the cut-metric theory consumes. Every use we make of regularity — the total boundedness that gives the compactness theorem of chapter 17 (corollary 19.9), and the finite-network convergence of chapter 25 — is phrased against the cut norm, which is itself a single global supremum over (S, T); none of it ever inspects a single block for pseudorandomness, because the cut metric was built precisely to forget that level of detail (section 19.1). Asking for a global integral to be small, rather than every pair to be pseudorandom, is what lets the energy increment argument spend only ε^2 per step against a budget of 1, and so caps the part count at the tame $4^{1/\varepsilon^2}$ instead of a tower. The lesson is a recurring one: match the strength of the tool to what the application actually reads off, and the bounds repay you. Here the dense theory only ever reads the cut norm, so the weak lemma is not a convenient weakening but the exactly-right instrument.

Both halves of the bridge are now in place: section 19.2 carried a graphon outward to faithful finite samples, and the present section has compressed an arbitrary graphon inward to a finite, cut-indistinguishable surrogate of accuracy-only complexity. The two are not yet joined. Sampling tells us a graphon's statistics are recoverable from data; regularity tells us only finitely many cut-visible parameters carry those statistics. The next section fuses them into a single statement of what is and is not recoverable — equating the law of the sampled graphs, the full list of motif densities, and cut distance zero — and reads the regularity-driven finite-parameter picture forward into the estimation question: precisely the bounded-complexity, cut-visible structure isolated here is the structure that can be learned, and nothing finer.

19.4 Identifiability and the reach of the dense theory

This section fuses the two halves of the bridge into the single question they were both circling. Once one can pass faithfully from a graphon to its samples and back from a graphon to a bounded-complexity surrogate, the natural thing to ask is the *inverse* question that any modeller eventually confronts: if I am handed the data a graphon generates — a growing family of random graphs, or equivalently all their measurable statistics — how much of the graphon have I actually pinned down? Is the kernel W recoverable from its output, and if not, exactly what residual ambiguity survives? This is the *identifiability* question, and it is the precise hinge between the two directions: it is meaningful only because sampling sends a graphon outward to observable data, and it is answerable only because the cut metric — the same metric in which the second sampling lemma made that data consistent — supplies the right notion of "the same kernel." The answer below settles it completely. The law of the random graphs a graphon generates determines it exactly, but only up to the one ambiguity the cut metric was deliberately built to quotient out: the renaming of the unobservable type labels.

Before stating it, it pays to lay out the three things the theorem will identify, because they are precisely the three vantage points the chapter has been accumulating. The first is the *generative* view: a graphon *is* the law of the random graphs $G(N, W)$ it produces, the full probabilistic recipe of definition 19.1, read across all sample sizes N at once — this is what an experimenter who could resample the network indefinitely would have access to. The second is the *moment* view: a graphon is the countable list of its motif densities $\{t(F, W)\}_F$, the observable averages of chapter 16 that section 19.2 showed a single large sample already estimates. The third is the *metric* view: a graphon is a point of the quotient space $\widetilde{\mathcal{W}}_0$ under the cut distance of chapter 17. The theorem says these three are one and the same object. That they coincide is what makes "recovering a graphon from data" a well-posed task at all: it guarantees that the experimenter's resampling, the statistician's moment estimates, and the geometer's cut distance can never disagree about whether two kernels are the same.

Theorem 19.11 (Identifiability up to weak isomorphism)**.** *Let W_1, W_2 be graphons satisfying* (A-Reg-W)*. The following are equivalent.*

1. *For every N, the random graphs $G(N, W_1)$ and $G(N, W_2)$ have the same distribution (as random graphs on $[N]$ up to isomorphism).*
2. *$t(F, W_1) = t(F, W_2)$ for every finite simple graph F.*
3. *$\delta_\square(W_1, W_2) = 0$; equivalently W_1 and W_2 are* weakly isomorphic *in the sense of eq. (17.8), i.e. equal as points of the quotient metric space $\widetilde{\mathcal{W}}_0$.*

Proof. The proof is a chain of two equivalences, each of which we have an in-book tool for; the work is in narrating why each link holds, since assembled correctly they close the loop between the generative, the moment, and the metric views.

(1)⇔(2)*: the sampling law and the motifs determine each other.* The law of $G(N, W)$ on the vertex set $[N]$ is, by inclusion–exclusion, completely encoded in the family of *subgraph-containment* probabilities $\mathbb{P}(F \subseteq G(N, W))$ as F ranges over labeled graphs on subsets of $[N]$: knowing, for every pattern F, the chance that the sampled graph contains it on a prescribed set of vertices is the same as knowing the probability of every individual graph, since the latter are signed sums of the former. So it suffices to evaluate one containment probability. Fix a labeled copy of F on distinct vertices $\phi(1), \ldots, \phi(v(F))$; the incident edge coins are then driven by *distinct* types, hence conditionally independent, and the computation is the very one that produced the mean in theorem 19.4:

$$\mathbb{P}\big(F \subseteq G(N, W)\big) \;=\; \mathbb{E}\Bigg[\prod_{ab \in E(F)} \xi_{\phi(a)\phi(b)}\Bigg] \;=\; \mathbb{E}\Bigg[\prod_{ab \in E(F)} W\big(U_{\phi(a)}, U_{\phi(b)}\big)\Bigg] \;=\; \int_{[0,1]^{v(F)}} \prod_{ab \in E(F)} W(x_a, x_b) \prod_c \mathrm{d}x_c \;=$$

The labeled copy sits on distinct vertices, so the types $U_{\phi(c)}$ are i.i.d. uniform and the conditional product integrates to precisely the homomorphism density $t(F, W)$ (the integral runs over all of $[0, 1]^{v(F)}$, the coincidence diagonal being null and hence contributing nothing, so the distinct-vertex restriction is invisible in the continuum). Thus every subgraph-containment probability *is* a motif density, and the sampling law across all N is determined by — and determines — the family $\{t(F,W)\}_F$. Two graphons therefore generate the same random graphs at every size if and only if they share every motif density. The converse half is also visible directly and dynamically, which is worth recording because it is the operational content the statistician relies on: by the first sampling lemma theorem 19.4, $t(F,G(N,W)) \to t(F,W)$ almost surely, so each density is literally recovered as the large-N limit of a statistic of the sampling law — the moments are measurable functionals of the law, as they must be for the equivalence to hold.

(2)⇔(3)*: matching all motifs is the same as vanishing cut distance.* This is the counting/inverse-counting pair established in chapters 16 and 17, and the two directions sit at very different depths. The forward direction is soft and quantitative: the *counting lemma* lemma 16.17 bounds the discrepancy of any fixed motif by the cut distance, $|t(F,W_1) - t(F,W_2)| \le e(F)\,\delta_\square(W_1,W_2)$, so $\delta_\square = 0$ immediately forces every density to agree — the metric view refines the moment view for free. The converse is the deep direction, and it is where the machinery of chapter 17 earns its keep: that matching *all* densities forces $\delta_\square = 0$ is the moment-uniqueness theorem theorem 17.11, equivalently the cut-/left-convergence equivalence theorem 17.12. One cannot read this off any single motif; it is proved by combining the sampling lemma with the compactness of $\widetilde{\mathcal{W}}_0$ — if two kernels had positive cut distance yet identical densities, one extracts cut-convergent sample sequences whose limits must coincide in every density but lie at positive distance, contradicting that densities separate points of the compact quotient. Both halves are carried out in full in theorems F.11 and F.12; we import them here rather than reprove them.

Closing the loop. Finally $\delta_\square(W_1,W_2) = 0$ is, by the very definition (17.8), the relation of weak isomorphism on the quotient space $\widetilde{\mathcal{W}}_0$: the points are graphons modulo cut distance zero, so $\delta_\square = 0$ says W_1 and W_2 are literally equal there. Statement (3) is therefore simultaneously the metric assertion (distance zero) and the equivalence-relation assertion (same point of $\widetilde{\mathcal{W}}_0$), and the chain (1)⇔(2)⇔(3) is complete. □

The theorem says *that* a graphon is determined up to $\delta_\square = 0$; it is natural to ask *what* that residual equivalence looks like concretely, as an explicit relation between two kernels. Here we reach a deliberate honesty boundary, and the next remark marks it carefully: there is a clean structural answer, but only half of it is within the reach of the machinery this book has built, and only that half is ever needed downstream.

Remark 19.12 (The concrete content of weak isomorphism)**.** One would like a picture of $\delta_\square(W_1,W_2) = 0$ as a literal renaming relating the two kernels point by point, not merely the statement that an infimum over relabelings vanishes. The Borgs–Chayes–Lovász structure theorem supplies exactly such a picture: $\delta_\square(W_1,W_2) = 0$ holds if and only if there are measure-preserving maps $\varphi_1, \varphi_2 \colon [0,1] \to [0,1]$ with

$$W_1(\varphi_1(x),\varphi_1(y)) = W_2(\varphi_2(x),\varphi_2(y)) \qquad \text{for a.e. } (x,y). \tag{19.5}$$

That is, after pulling both kernels back through suitable type-relabelings they become the same function almost everywhere — weak isomorphism is genuine isomorphism up to a common refinement of the two labelings. Allowing *two* maps rather than one is essential and is the whole subtlety: a measure-preserving *bijection* of $[0,1]$ need not exist between two kernels that are nonetheless cut-equivalent (for instance, a graphon and a twofold "stretch" of it that splits each type into two identical copies are weakly isomorphic but related by no single invertible relabeling), and the pair (φ_1, φ_2) into a common refinement is what repairs this.

The easy direction is in hand from the book's own machinery, and it is worth running the reasoning because it shows where the soft part lies. Suppose maps as in (19.5) exist. Because φ_1, φ_2 are measure-preserving, pulling a kernel back through one of them leaves *every* homomorphism density unchanged — a relabeling only renames the integration variables in $\int \prod W(x_a, x_b)\, \mathrm{d}x$, and a measure-preserving change of variables does not move the value of the integral. So W_1 has the same densities as $W_1 \circ \varphi_1$ (meaning the kernel $(x, y) \mapsto W_1(\varphi_1(x), \varphi_1(y))$), which by (19.5) equals $W_2 \circ \varphi_2$ a.e., which in turn has the same densities as W_2. Matching all densities then forces $\delta_\square = 0$ by the moment-uniqueness theorem theorem 17.11. The whole easy direction thus rides on two in-book facts already proved: density-invariance under measure-preserving maps, and moment-uniqueness.

The hard direction is the converse — that vanishing cut distance *produces* the exact maps (19.5), an almost-everywhere identification, rather than only the approximate, up-to-ε alignments that the cut metric directly delivers. This is genuinely deeper than anything the cut-metric theory of chapter 17 reaches. The cut distance gives, for each $\varepsilon > 0$, a relabeling aligning the kernels to within ε; manufacturing from this sequence of approximate alignments a *single* pair of exact measure-preserving maps requires a compactness-and-limit argument carried out not on graphons but on their richer *soft-graphon* (coupling) representatives, where one tracks not just the kernels but the measure couplings that realize the alignments and extracts a limiting coupling. That apparatus is developed in [28] and [99, Ch. 13], and we do not reproduce it. This is the one structural fact about weak isomorphism we state and attribute but leave uncertified in this book. The reason we can leave it is that nothing downstream needs it: neither the present chapter, nor the finite-network convergence of chapter 25, nor the sparse theory of chapter 27, ever invokes the explicit maps (19.5). What every later argument consumes is only the *metric* statement $\delta_\square(W_1, W_2) = 0$ of theorem 19.11(3) — equality as points of $\widetilde{\mathcal{W}}_0$ — and that statement *is* proved in-book, through the counting lemma and moment-uniqueness of chapters 16 and 17. We flag the gap rather than paper over it precisely so that the boundary between what this book certifies and what it cites stays crisp.

That the recovery is only up to this relabeling is the right place to pause, because it is easy to misread as a weakness of the theory when it is in fact a faithful reflection of what a network *is*. The type labels $x \in [0,1]$ are a modelling fiction: an actual network presents only its wiring — who is connected to whom — and never an intrinsic coordinate stamped on each node. Two graphons that differ by a measure-preserving renaming of types encode the identical distribution over wirings, as the proof of theorem 19.11(1) makes literal, so no amount of data, however much one resamples, could ever tell them apart; demanding that the theory distinguish them would be demanding that it recover information the network does not carry. The cut metric was built in chapter 17 to quotient out exactly this fictitious coordinate, and identifiability up to $\delta_\square = 0$ says it quotients out neither too much nor too little: everything observable is retained, everything unobservable is discarded. This is the same exchangeability of the type variable that has been a standing principle since chapter 15 — the agents' labels are interchangeable because the model is symmetric under permuting them —

now seen from the estimation side. Exchangeability is not a technical convenience here; it is the precise content of "the label is unobservable," and identifiability is its statistical shadow.

Estimation, and why weak regularity is the statistician's friend. Theorem 19.11 is an asymptotic, population-level statement: it concerns the law of the random graphs, or equivalently an infinite list of densities, both idealizations of unlimited data. The working statistician instead holds a *single* network G on n nodes and wants to estimate the kernel W that produced it. The first thing to settle is whether this is even possible, and the second sampling lemma (19.2) settles it in principle: the empirical graphon $W_{[n]}$ built from the one observed sample is $\delta_\square$-consistent, converging to W (up to relabeling) as $n \to \infty$. So consistency is not in doubt. The danger is of the opposite kind — overfitting — and seeing why exposes the role weak regularity plays. As an estimator, the raw $W_{[n]}$ has one free entry per pair of vertices, on the order of n^2 parameters, fit from the $\binom{n}{2}$ observed edge bits: it carries as many parameters as data points and does no averaging at all, so it reproduces the sampling noise rather than smoothing it away. An estimator that memorizes its data this way has irreducible variance; the empirical graphon is consistent only in the weak cut topology, and never in any pointwise sense, exactly because those n^2 entries never settle down individually.

The weak regularity lemma supplies the remedy, and it is the same finite-parameter picture the previous section handed forward. By corollary 19.9, the population graphon is within cut distance ε of a step graphon with at most $4^{\lceil 1/\varepsilon^2 \rceil}$ blocks — that is, a stochastic block model with $q(\varepsilon) = O(4^{1/\varepsilon^2})$ classes, carrying only $O(q(\varepsilon)^2)$ effective parameters, a number that depends on the target accuracy ε alone and *not* on the sample size n. Fitting such a bounded-complexity surrogate instead of the raw pixel array converts estimation into a genuine bias–variance tradeoff. Choosing q classes incurs an *approximation* (bias) error, which weak regularity caps: it shrinks as q grows, at the rate $\varepsilon \sim (\log q)^{-1/2}$ read off by inverting the block count. It also incurs an *estimation* (variance) error from fitting $O(q^2)$ block probabilities out of $\binom{n}{2}$ edges, which *grows* with q, on the order of q^2/n^2 per block and aggregating to a term increasing in q. The two pull in opposite directions, so an optimal block count $q = q_n$ grows slowly with n — fast enough to drive the bias down, slowly enough to keep the variance controlled — and at that choice the SBM estimator attains the optimal tradeoff. This is the rigorous basis of consistent graphon estimation, and the minimax analysis makes the regularity-driven block count explicit, the optimal q_n and the resulting rate being governed by precisely the $4^{1/\varepsilon^2}$ complexity that weak regularity furnishes [87]. The conceptual payoff is that coarse-graining and statistical identifiability are revealed to be one phenomenon seen from two sides. Weak regularity, run *inward*, says a graphon's cut-visible content is captured by boundedly many parameters; the estimation problem, run *from data*, says those same boundedly many parameters are exactly what a finite sample can learn. Only the bounded-complexity, cut-norm-visible structure is recoverable, and weak regularity is precisely the theorem that isolates it.

This closes the positive account. We have now delimited, sharply, what the dense theory *can* do: it recovers any graphon up to weak isomorphism — the exact and only ambiguity the data leave open — and it estimates that recoverable, cut-visible structure consistently and efficiently from a single large sample, with weak regularity setting the number of parameters worth fitting. Every faculty the chapter built — sampling outward, coarse-graining inward, and now the identifiability that fuses them — has been turned to a constructive end. But the same machinery has a sharp edge, and intellectual honesty requires us to run it the other way. The cut metric and the sampling lemmas that make all of this work were defined for *dense* graphons, bounded kernels with $\Theta(N^2)$ edges; the rank-one computation of section 19.2 already let slip that every degree in such a model grows like $\Theta(N)$. Real networks are not like this. The next section takes the very cut-metric and sampling apparatus just celebrated and shows that, applied outside the dense regime, it does not merely lose accuracy but goes structurally blind — collapsing every sparse network onto a single uninformative point. What the dense theory can recover, it recovers perfectly; what it is built to see is, it turns out, precisely not the world most networks live in.

19.5 Why the dense theory fails for real networks

Intellectual honesty now requires us to run the same machinery the other way and see where it breaks. This is the limitation that gives the chapter its subtitle, and we will not apologize for it — we will prove it. The cut metric and the two sampling lemmas that powered everything above were built for one regime, and outside it they do not merely lose accuracy: they go blind. Two precise theorems pin down how. The first shows that the cut metric collapses every sparse network sequence onto a single point; the second shows that the W-random model cannot generate a sparse network in the first place. Together they say the blindness is structural, present in both the limit notion and the generative recipe, and therefore not a defect any refinement *within* the dense framework could repair.

Everything in this chapter is a theory of *dense* graphs — graphs with $\Theta(N^2)$ edges, a positive fraction of the $\binom{N}{2}$ possible ones present. This is not an incidental feature one might relax; it is wired into the definitions at the ground floor. A graphon is a kernel $W\colon [0,1]^2 \to [0,1]$ bounded by 1, and a bounded kernel integrates to a positive number, so its edge density $t(K_2, W) = \int_{[0,1]^2} W$ is a constant of order one; multiplying by the $\binom{N}{2}$ pairs of a sample forces $\Theta(N^2)$ edges. The cut metric inherits the same ceiling, because it compares two such bounded kernels through patch integrals that are themselves order-one quantities. The rank-one computation of section 19.2 already let this slip: there we found $\deg(i)/(N-1) \to f(U_i)\bar{f}$, so every degree in that model is $\Theta(N)$ no matter how the profile f is chosen, and promised to return to the point here. We now make good on the promise in general.

The trouble is that real networks are the opposite of dense. Social graphs, the web hyperlink graph, power grids, citation and collaboration networks — across the board their edge counts grow linearly, or only slightly faster than linearly, in the number of nodes, so a typical vertex has $\Theta(1)$ neighbours, not $\Theta(N)$, and their degree distributions are heavy-tailed, with a handful of hubs whose degree dwarfs the median by orders of magnitude. A graph on a million nodes in which each node knows a few hundred others has on the order of 10^8 edges, not the 10^{12} a dense model would demand; the dense and the real differ not by a constant but by a growing factor of N. The dense theory does not approximate such a network poorly. It collapses it — and every network like it, however different they are from one another — to one and the same trivial limit. We prove this first.

Proposition 19.13 (Sparse sequences converge to the zero graphon)**.** *Let $(G_n)_{n\geq 1}$ be a sequence of finite simple graphs with $v(G_n) \to \infty$ and* subquadratic *edge count, $e(G_n) = o(v(G_n)^2)$. Then for* *every finite simple graph F with at least one edge,*

$$t(F, G_n) \longrightarrow 0,$$

and consequently the associated pixel graphons converge to the zero graphon, $\delta_\square(W^{G_n}, \mathbf{0}) \to 0$, where W^{G_n} is the pixel graphon of G_n in the sense of definition 15.2.

Proof. The whole argument is a single domination bound: any motif density is squeezed below the edge density, and the edge density is exactly the quantity the sparsity hypothesis kills. Fix F with an edge $ab \in E(F)$ and write $n = v(G_n)$. Every factor in the homomorphism product is an edge indicator lying in $[0,1]$, so discarding all factors but the one indexed by the chosen edge ab can only raise the product:

$$t(F, G_n) = \frac{1}{n^{v(F)}} \sum_{\phi} \prod_{cd \in E(F)} \mathbf{1}[\phi(c)\phi(d) \in E(G_n)] \;\leq\; \frac{1}{n^{v(F)}} \sum_{\phi} \mathbf{1}[\phi(a)\phi(b) \in E(G_n)] \;=\; t(K_2, G_n).$$

The last equality is a counting identity worth seeing explicitly. In the right-hand sum only the two coordinates $\phi(a), \phi(b)$ are constrained — they must land on an edge — while the remaining $v(F) - 2$ coordinates are entirely free, each ranging over all n vertices and so contributing a factor n. Those $v(F) - 2$ free factors produce $n^{v(F)-2}$, which against the normalization $n^{v(F)}$ leaves $n^{-2} \sum_{\phi(a),\phi(b)} \mathbf{1}[\,\cdot \in E(G_n)]$, exactly the edge density $t(K_2, G_n)$. And that density is precisely twice the edge count over the squared vertex count, $t(K_2, G_n) = 2e(G_n)/n^2$, which tends to 0 by the subquadratic hypothesis $e(G_n) = o(n^2)$. Hence $t(F, G_n) \to 0$ for every F with an edge: the sample retains no motif at all in the limit.

For the cut distance to the zero graphon there is nothing more to prove, because the cut norm of a nonnegative kernel is dominated by its L^1 norm, and here the L^1 norm is again the edge density. The pixel graphon W^{G_n} is nonnegative, so the supremum $\left\|W^{G_n}\right\|_\square = \sup_{S,T} \int_{S\times T} W^{G_n}$ from (19.1) is largest when $S = T = [0,1]$ and no cancellation can help; thus

$$\left\|W^{G_n}\right\|_\square \;\leq\; \left\|W^{G_n}\right\|_{L^1} \;=\; \int_{[0,1]^2} W^{G_n} \;=\; t(K_2, G_n) \;=\; \frac{2e(G_n)}{n^2} \;\to\; 0.$$

Finally $\delta_\square(W^{G_n}, \mathbf{0}) = \left\|W^{G_n}\right\|_\square$, because the zero graphon is fixed by every measure-preserving relabeling φ — $\mathbf{0}^\varphi = \mathbf{0}$ — so the infimum over φ in (19.1) changes nothing and is attained at the identity. Hence $\delta_\square(W^{G_n}, \mathbf{0}) \to 0$. □

The proposition is devastating in its simplicity, and it is worth pausing to feel the full weight of what the one-line domination bound just did. A sparse sequence of networks — no matter how intricate its internal architecture, no matter whether it is organized into tight communities, threaded by high-degree hubs, locally clustered into triangles, or stacked into hierarchies of subgraphs — is, in the cut metric, *indistinguishable in the limit from the empty graph.* All of that structure lives in the motif densities, and every single one of them was crushed to zero by the same bound, because each was trapped beneath an edge density that itself vanishes. The dense graph-limit machinery, asked for the limit object of such a sequence, returns $\mathbf{0}$ and in doing so throws away everything that made the sequence interesting. Take two sparse sequences as different as a growing collection of disjoint triangles and a growing sparse expander: their community structure, their connectivity, their clustering could not be more unlike, yet both have edge density tending to zero, so both limit to the very same zero graphon and the theory cannot tell them apart. The information loss is not partial and it is not graceful; it is total, and it falls exactly on the features one builds a network model to capture. The cut metric, so powerful within the dense world, simply has no resolution outside it.

One might still hope the fault lies only in the limit notion — in measuring with the cut metric — and that the generative model itself remains flexible enough to produce sparse samples that we then judge by some better yardstick. It does not. The second proposition closes that escape, by showing that the W-random model of definition 19.1 cannot emit a sparse graph at all: density is forced by the construction, before any metric is chosen.

Proposition 19.14 (The W-random model is intrinsically dense)**.** *Let W satisfy* (A-Reg-W) *with* $\int_{[0,1]^2} W > 0$*, and sample $G(N, W)$. Write $d_W(x) = \int_0^1 W(x, y)\,\mathrm{d}y$ for the degree function. Then:*

1. *conditionally on $U_i = x$, the degree of vertex i is* $\mathrm{Binomial}(N-1,\, d_W(x))$*, so* $\deg(i)/(N-1) \to d_W(U_i)$ *almost surely;*

2. *the average degree satisfies* $\mathbb{E}[\deg(i)] = (N-1)\, t(K_2, W)$*, which grows linearly in N; equivalently the edge count is* $\Theta(N^2)$ *almost surely;*

3. *the empirical degree distribution, rescaled by $N - 1$, converges weakly to the law of $d_W(U)$ for $U \sim \mathrm{Unif}[0, 1]$ — a fixed distribution* supported on $[0, 1]$ *that does not change shape with N.*

Proof. The three statements are three readings of the same fact, that conditional on its own type a vertex collects independent coins of a fixed bias. We establish that conditional law first and then push it through, in turn, to one vertex (a strong law), to the population mean (linearity of expectation), and to the empirical distribution (a weak-convergence argument).

The conditional degree law and (1). Fix vertex i and condition on $U_i = x$. For each $j \neq i$ the incident coin ξ_{ij} was drawn, in the model of definition 19.1, as Bernoulli($W(U_i, U_j)$) given the two types; with $U_i = x$ held fixed but $U_j \sim \mathrm{Unif}[0, 1]$ still random, marginalising over U_j (and over the edge seed) gives the unconditional success probability

$$\mathbb{P}(\xi_{ij} = 1 \mid U_i = x) = \int_0^1 W(x, y)\,\mathrm{d}y = d_W(x),$$

the row integral of the kernel, which is the degree function evaluated at x. The coins ξ_{ij}, $j \neq i$, are moreover independent of one another given $U_i = x$, since each draws on a distinct partner type U_j and a distinct seed; so the degree $\deg(i) = \sum_{j \neq i} \xi_{ij}$ is a sum of $N - 1$ i.i.d. Bernoulli($d_W(x)$) variables, that is Binomial($N - 1$, $d_W(x)$). Because those coins are i.i.d. and bounded in $[0, 1]$, the in-book strong law for bounded summands (proposition 3.24) applies to their average, giving $\deg(i)/(N - 1) \to d_W(U_i)$ almost surely. This is (1), and it is the same computation already met for the rank-one kernel in worked computation 19.6, where $d_W(x) = f(x)\bar{f}$; the present proposition is that worked computation stripped of its rank-one specialisation.

Linear mean degree and (2). Take expectations of the conditional law. Since the conditional mean of a Binomial$(N-1, d_W(U_i))$ is $(N-1)d_W(U_i)$, the tower property gives $\mathbb{E}[\deg(i)] = (N-1)\,\mathbb{E}[d_W(U_i)] = (N-1)\int_0^1 d_W(x)\,\mathrm{d}x = (N-1)\,t(K_2, W)$, the last equality because integrating the row integral over x reassembles the full double integral $\int_{[0,1]^2} W = t(K_2, W)$. The hypothesis $\int_{[0,1]^2} W > 0$ makes this a strictly positive multiple of $N-1$, so the mean degree grows *linearly* in N. Summing $\deg(i)$ over the N vertices and halving counts each edge once, $e(G(N,W)) = \frac{1}{2}\sum_i \deg(i)$, whose expectation is therefore $\frac{1}{2}N(N-1)t(K_2,W) = \Theta(N^2)$; and the edge density is exactly the motif density $t(K_2, G(N,W))$, which by the first sampling lemma (theorem 19.4 with $F = K_2$) concentrates at $t(K_2, W) > 0$. This upgrades the expectation to an almost-sure statement, $e(G(N,W)) = \Theta(N^2)$, which is (2): the model cannot help being dense.

The limiting degree shape and (3). This is the most delicate of the three, so we spell it out. We claim the empirical degree measure

$$\mu_N := \frac{1}{N}\sum_{i=1}^{N}\delta_{\deg(i)/(N-1)} \qquad \text{converges weakly to} \qquad \nu := \mathrm{Law}\big(d_W(U)\big), \quad U \sim \mathrm{Unif}[0,1].$$

One must be clear about what is and is not being claimed. We assert only *weak* convergence of the random measure μ_N — convergence of $\int h\,\mathrm{d}\mu_N$ for bounded continuous test functions h — not the stronger uniform convergence of cumulative distribution functions that Glivenko–Cantelli would supply; the weaker statement is all the proposition needs and all that is true at this level of generality, and flagging the distinction keeps us from over-claiming. By the bounded-Lipschitz form of the Portmanteau theorem (theorem 3.9), it suffices to show $\int h\,\mathrm{d}\mu_N \to \int h\,\mathrm{d}\nu$ in probability for every bounded Lipschitz h, say with Lipschitz constant L. We obtain this by interposing the "ideal" empirical average built from the exact degree functions, $\frac{1}{N}\sum_i h(d_W(U_i))$, and controlling the two gaps separately — a clean quenched/annealed split once more.

The first gap, from the ideal average to its mean, is a law of large numbers in the types. The summands $h(d_W(U_i))$ are i.i.d. (the U_i are) and bounded (as h is), so proposition 3.24 gives

$$\frac{1}{N}\sum_{i=1}^{N} h(d_W(U_i)) \;\longrightarrow\; \mathbb{E}[h(d_W(U))] \;=\; \int h\,\mathrm{d}\nu \qquad \text{almost surely,}$$

which is the right-hand target. The second gap, from the true empirical average $\int h\,\mathrm{d}\mu_N = \frac{1}{N}\sum_i h(\deg(i)/(N-1))$ to the ideal one, measures how far each realized normalized degree sits from its conditional mean. Conditional on U_i, the normalized degree $\deg(i)/(N-1)$ is the mean of $N-1$ i.i.d. Bernoulli$(d_W(U_i))$ coins; its conditional mean is exactly $d_W(U_i)$, and its conditional variance is $\frac{1}{N-1}d_W(U_i)\big(1-d_W(U_i)\big) \le \frac{1}{4(N-1)}$, the bound $p(1-p) \le \frac{1}{4}$ holding for every $p \in [0,1]$ and so *uniformly in* i regardless of the type drawn. By Jensen (or Cauchy–Schwarz), the mean absolute deviation is at most the square root of the variance,

$$\mathbb{E}\big[\,|\deg(i)/(N-1) - d_W(U_i)|\,\big] \;\le\; \Big(\tfrac{1}{4(N-1)}\Big)^{1/2} \;=\; \tfrac{1}{2}(N-1)^{-1/2},$$

uniformly in i. The Lipschitz bound on h now transfers this control from the degrees to the test averages: term by term $|h(\deg(i)/(N-1)) - h(d_W(U_i))| \le L\,|\deg(i)/(N-1) - d_W(U_i)|$, so averaging and taking expectations,

$$\mathbb{E}\Big[\Big|\int h\,\mathrm{d}\mu_N - \tfrac{1}{N}\textstyle\sum_i h(d_W(U_i))\Big|\Big] \;\le\; \dfrac{L}{N}\displaystyle\sum_{i=1}^{N}\mathbb{E}\big[\,|\deg(i)/(N-1) - d_W(U_i)|\,\big] \;\le\; \frac{L}{2\sqrt{N-1}} \;\longrightarrow\; 0,$$

where the uniformity in i is exactly what let the N identical bounds collapse to a single $O(N^{-1/2})$ term rather than accumulate. Convergence in L^1 forces convergence in probability, so this second gap vanishes in probability. Adding the two estimates — the first gap to $\int h \, \mathrm{d}\nu$ almost surely, the second gap to 0 in probability — gives $\int h \, \mathrm{d}\mu_N \to \int h \, \mathrm{d}\nu$ in probability for every bounded Lipschitz h, which by theorem 3.9 is precisely $\mu_N \Rightarrow \nu$. The limit law ν is a fixed distribution on $[0,1]$, determined by the kernel through its degree function alone and *independent of* N; this is (3). The argument is the same empirical-measure convergence invoked, without proof, at the end of worked computation 19.6, now discharged in full. □

Statement (3) repays a careful reading, because it draws the exact line between what the dense model can imitate and what it can never produce, and the line is sharper than one might first guess. On the permissive side, the dense model *can* reproduce any *shape* of rescaled degree distribution whatsoever. Given any target law on $[0,1]$, set d_W equal to its quantile function: then $d_W(U)$ for $U \sim \mathrm{Unif}[0,1]$ has exactly that law, and by (3) the limiting normalized-degree distribution matches it. So if one only ever looks at degrees *after* dividing by $N-1$, the model is completely flexible — bimodal, skewed, concentrated, spread; any silhouette is attainable.

The flexibility is an illusion produced entirely by that normalization, and statement (3) is precise about where it ends. What the dense model cannot do is the one thing every real network insists upon: it cannot make the *unnormalized* degrees small. The limit law ν lives on $[0,1]$ and describes $\deg(i)/(N-1)$, so a vertex landing at the point $d_W(U_i) = d$ has actual degree $\approx (N-1)d = \Theta(N)$. Every degree is forced to be $\Theta(N)$; the maximum degree is $\Theta(N)$, never sublinear; and the genuinely heavy, N-independent power-law tail that real networks exhibit in their *raw* degrees is unreachable, because rescaling by $N-1$ is the very operation that flattens any such tail into the bounded interval $[0,1]$ before a limit is taken. The model can match a degree silhouette and never a degree scale. And the culprit is named and unremovable: it is the boundedness $W \le 1$, which caps the row integral $d_W(x)$ at 1 and so caps every connection probability at order one, guaranteeing $\Theta(N)$ neighbours. That ceiling is not a modelling choice we made carelessly; it is the definition of a graphon. To escape it one must change the definition.

We have now diagnosed the failure in three named pieces, and each will be matched, in the next section, by a specific cure. The cut metric *trivializes* on sparse sequences (proposition 19.13), collapsing them all to $\mathbf{0}$ — so the metric must be rescaled before it can resolve anything. The W-random model is *intrinsically dense* (proposition 19.14), forcing $\Theta(N^2)$ edges — so the sampling recipe must carry a vanishing target density. And the bounded kernel $W \le 1$ is the *structural cause* of both, ruling out heavy tails by fiat — so the kernel must be allowed to leave L^∞. Three failure modes, three defects to repair: a trivializing metric, a density-forcing recipe, a bounded kernel. The next section takes them in exactly this form and shows how the two sparse frameworks — L^p graphons and graphex — are built precisely to mend them, so that what follows reads as the prescription answering this diagnosis rather than as a fresh topic.

19.6 The road to sparsity: L^p graphons and graphex

Section 19.5 did more than register a disappointment; it took care to name the failure in three separate pieces — a trivializing metric (proposition 19.13), a density-forcing recipe (proposition 19.14), and beneath both the structural cause, the boundedness $W \leq 1$ — each phrased so that a cure could be aimed at it directly, and the discipline of this section is to honour that framing. We do not present sparsity as a new subject that happens to follow the dense one. We present it as the prescription that the preceding diagnosis already wrote out, taking the three named defects in the exact form the last section handed them over and matching each to the repair built to mend it. These are not three independent complaints. They are three faces of one ceiling, which is why a repair worth the name cannot patch them one at a time but must change the sampling scale, the comparison metric, and the function class of the kernel together. Two related programmes do exactly this, by two different amounts of surgery. Both are developed in full in chapter 27; we describe here only what they are and why each lands on a specific failure mode above, and we prove nothing — the point of previewing them is to make precise the claim that sparsity is a genuine frontier and not a corner of the dense theory one might reach by a careful estimate.

Route 1: sparse and L^p graphons. The first and more conservative route keeps everything about the dense picture it possibly can — the unit-interval type space $[0, 1]$, the latent uniform types, the reading of the kernel as a recipe for biased coins — and changes only the one dial that the second proposition identified as the seat of the trouble: the scale at which the coins are biased. The W-random model forces density because the connection probability $W(x, y)$ is of order one, so a vertex meets an order-one fraction of the rest; the obvious remedy is to turn that probability down, uniformly and by a factor that shrinks with the size of the graph. One therefore attaches a vanishing *target density* $\rho_N \to 0$ to the model and connects the types x, y with probability $\min\{1, \rho_N W(x, y)\}$ in place of $W(x, y)$. The factor ρ_N does the dimming; the clip $\min\{1, \cdot\}$ is there only because a probability may not exceed 1, and where W is bounded it never bites for N large, since $\rho_N W \leq \rho_N \to 0$ — it becomes active precisely at the heavy-tailed hubs, the very places, as we shall see, where the unbounded kernels below put their mass.

The accounting is immediate and shows the dial doing exactly what was asked. A vertex of type x now connects to each of the $N - 1$ others with probability about $\rho_N\, d_W(x)$, where $d_W(x) = \int_0^1 W(x, y)\, \mathrm{d}y$ is the degree function of proposition 19.14, so its expected degree is about $(N - 1)\rho_N\, d_W(x)$ and the average over types is about $(N - 1)\rho_N\, t(K_2, W)$. Choosing $\rho_N = c/N$ holds this product at a constant, making the average degree $\Theta(1)$ — a genuinely sparse graph, $\Theta(N)$ edges in place of $\Theta(N^2)$ — and the family $\rho_N = N^{-\gamma}$, $0 < \gamma < 1$, sweeps the intermediate densities between the sparse and the dense extremes. So the density-forcing recipe is mended at a stroke. But the same vanishing factor that fixes the density breaks two other things that the dense theory took for granted, and repairing those is what makes Route 1 a programme rather than a one-line substitution; it is the content of chapter 27.

The first casualty is the metric. With connection probabilities of size ρ_N, the sampled kernel $W_{[N]}$ has entries of size $\rho_N \to 0$, so it is itself a sparse object and falls under proposition 19.13: its unrescaled cut-distance limit is the zero graphon, and all the structure one tuned W to carry is crushed exactly as before. The metric has not been fixed by lowering the density; it has been handed the very sequences it cannot see. The repair is to undo the dimming before comparing — to measure the *rescaled* empirical kernel $\rho_N^{-1}W_{[N]}$, whose entries are back at order one, against the parent W, working with the *stretched* (or normalized) cut distance $\delta_\square(\rho_N^{-1}W_{[N]}, W)$ [25, 26]. This is more than a cosmetic division. The bounded-kernel cut theory of chapter 17 — compactness, the equivalence of cut distance with motif convergence, the whole calculus on which the present chapter rested — was built for kernels valued in $[0, 1]$, and the rescaled kernel $\rho_N^{-1}W_{[N]}$ is unbounded, with spikes wherever the dimmed coins clustered. The cut theory does not apply verbatim; it has to be re-proved in a setting where the objects compared are no longer uniformly bounded.

The second casualty is deeper and explains why the heavy tails of real networks demand a new function class rather than a mere rescaling. A degree distribution with a genuine, N-independent power-law tail in its raw degrees requires connection probabilities that are themselves wildly unequal across types — a few types overwhelmingly more connective than the rest — and a kernel bounded by 1 simply cannot encode such disparity, as proposition 19.14 proved by capping the row integral. To carry it, the limiting kernel must be allowed to grow without bound, $W \in L^p([0, 1]^2)$ for some $1 \le p < \infty$ in place of $W \in L^\infty$. These are the L^p *graphons* (definition 27.5), and leaving L^∞ costs the analytic backbone of chapter 18. There the kernel was a self-adjoint Hilbert–Schmidt operator $\mathbb{W}$, and it was square-integrability, $W \in L^2([0, 1]^2)$, that made $\mathbb{W}$ Hilbert–Schmidt and hence compact, with the discrete real spectrum the centrality and community readings depended on. An L^p graphon need not be in L^2 at all; its operator $\mathbb{W}$ is in general neither Hilbert–Schmidt nor even bounded on L^2, so the compact-operator spectral theory of chapter 18 and the operator-norm smallness condition (A-Coupling) — the contraction hypothesis on which uniqueness turned — lose their meaning and must be replaced by genuinely L^p estimates. This is exactly the point at which the GMFG existence theory of chapter 24 has to be rebuilt for sparse couplings, the equilibrium fixed point reconstructed on an operator that no longer obeys the Hilbert-space bounds Part IV will assume; it is the analytic heart of chapter 27 and the reason the sparse story cannot be told as a footnote to the dense one.

It helps to anchor the abstraction in the one model that makes all of it concrete, and it is the sparse twin of the example this chapter has already leaned on. The Chung–Lu, or expected-degree, model [48] takes a rank-one sparse kernel $W(x, y) \propto w(x)w(y)$ with a *heavy-tailed* weight function w, sparsified by the factor ρ_N, and produces genuine power-law degree sequences on graphs with $\Theta(N)$ edges. It is the sparse analogue of the rank-one worked computation worked computation 19.6: there the bounded profile f collapsed every observable onto the two scalars $\bar{f}$ and λ_1 and forced $\Theta(N)$ degrees by the very boundedness of f; here the unbounded weight w is precisely what frees the degree tail that the bounded f could never reach, the rank-one algebra surviving but the function class changing under it. We return to it in chapter 27.

Route 2: graphex and exchangeable random measures. The second route asks a more radical question. Route 1 kept the finite type space $[0, 1]$ and patched the kernel and the metric around it; but $[0, 1]$ with Lebesgue measure is a *probability* space, a space of total mass one, and one may reasonably suspect that the forced density is built into that finiteness as much as into the boundedness of the kernel — that a model whose vertices are drawn from a unit-mass space of types will struggle to be sparse no matter how the coins are dimmed. Route 2 takes the suspicion seriously and abandons the unit-interval vertex set altogether. In place of N types drawn from $[0,1]$, one places a Poisson point process of vertices on the half-line $\mathbb{R}_+$, equipped now with the infinite-mass Lebesgue measure: a countable cloud of latent vertices, each carrying a label, of which only finitely many are *active* below any finite truncation level, and connects them through a symmetric kernel $W: \mathbb{R}_+^2 \to [0,1]$ — a *graphex* (definition 27.12). The infinite reservoir of latent vertices is what lets the active graph stay sparse while still admitting hubs of unbounded relative degree: there is always more room on the half-line for a heavy tail than a unit interval can hold.

The structural objects this construction produces are the *exchangeable random measures* of Caron–Fox and Veitch–Roy [45, 130], and their role here is exactly parallel to the one the Aldous–Hoover representation played for dense graphons. Aldous–Hoover characterized the dense limit by saying which random arrays are invariant under relabeling of the vertices, and identified those with graphons; the Caron–Fox and Veitch–Roy theory does the corresponding job one rung down in density, characterizing the sparse limit objects as the random measures on $\mathbb{R}_+^2$ that are invariant in the matching sense and representing them through a graphex. It is the sparse analogue of the exchangeability backbone of the dense theory, not a loose metaphor for it. The accompanying notion of limit is supplied by sampling: one truncates the point process at level s, reads off the finite graph of vertices active below s, and lets $s \to \infty$, obtaining *sampling convergence* (also called *stretched convergence*) under which sparse graphs with power-law degrees possess genuine, nontrivial limits [26] — the convergence the dense cut distance could not provide. And the payoff is that the graphex framework delivers, generically and without any fine-tuning of the kernel, exactly the catalogue of features that proposition 19.14 proved a W-random graph could never exhibit: $\Theta(N)$ edges, a maximum degree sublinear in N relative to the natural scaling, and heavy-tailed degree distributions whose tails do not flatten under rescaling. The dense theorem said these were impossible; the graphex says they are typical, which is the sharpest possible statement that the two regimes are different worlds. Building the mean-field game theory over such limits — the sparse counterpart of the entire Part IV development of chapters 21 to 24 — is the business of chapter 27.

Remark 19.15 (What is theorem and what is frontier)**.** The dense statements of this chapter (theorems 19.4, 19.8 and 19.11) are complete and classical. The sparse story is in active development. The graphex representation theory and sampling-convergence framework are on firm footing [45, 130, 26]; what remains substantially open, and is flagged as such in chapters 27 and 31, is the mean-field game theory *over* sparse limits: existence, uniqueness, and finite-network approximation for GMFG when the coupling operator is an unbounded L^p kernel or a graphex. We will be careful in chapter 27 to separate the proved from the conjectural there; for the present chapter the honest summary is that the dense theory is closed and exact within its regime, and structurally blind outside it.

With that, the whole sparse programme passes out of this chapter and into the rest of the book. The analytic machinery both routes demand — the L^p operator theory, the stretched cut distance, the graphex representation, and the equilibrium theory built on top of them — is the charge of chapter 27, and the explicit map of what within it is settled and what is still conjecture is laid out in the research survey of chapter 31. What we keep for the immediate purpose is only the clean line the frontier remark just drew, the line between the proved-and-classical and the still-open, because it is exactly that line, widened from this one chapter to the four pillars of the whole part, that the retrospective now takes up. The dense theory we are about to summarize is the part of the story that is finished; the sparse theory we have just sketched is the part that is not, and Part III ends by saying, pillar by pillar, precisely which is which.

Part III in retrospect

The frontier remark just drew a single clean line — between the dense theory that is finished and the sparse theory that is not — and promised to widen it. That widening is the work of this closing unit. The line was drawn for one corner of one chapter, the cut metric's treatment of subquadratic graphs; but the same line runs the length of Part III, and seen at full length it is not a defect at the edge of an otherwise seamless theory but the precise shape of the theory's domain. So it is worth standing back from the individual theorems and looking at the arc whole, pillar by pillar, to see exactly what was built, exactly what it is good for, and exactly where it stops — not as three separate verdicts but as one. Part III set out to turn "a network" into an analytic object one could do calculus on; it succeeded completely, and the boundary we are about to mark is the boundary of that success, proved from the inside.

Four faculties were built, each the prerequisite for the next, and together they are the calculus. The first was *representation*. We replaced the adjacency matrix of a growing family of graphs — a combinatorial object whose very dimension changes with N — by a single measurable kernel $W\colon [0,1]^2 \to [0,1]$ (chapter 15), so that an entire sequence of networks of every size could be held at once in one fixed analytic object. The second was *observation*. A kernel carries a labelling that no network actually exposes, and the motif densities $t(F,W)$ (chapter 16) are exactly the functionals that survive relabelling: they are the network's moments, the only coordinates by which one graphon is honestly told from another, and chapter 16 showed they determine everything observable. The third was *metric and compactness*. Equipping the space of kernels with the cut distance $\delta_\square$ (chapter 17) turned "a sequence of networks converges" from a hope into a theorem: the cut metric is the weakest yardstick fine enough to see motif densities and coarse enough to forgive pixel-scale noise, and in it graphon space is *compact*, so a limit always exists and always lives in the same world as the objects approaching it — there is no escape to a larger universe. The fourth was *spectrum*. Recognising W as a self-adjoint Hilbert–Schmidt operator $\mathbb{W}$ (chapter 18) let us diagonalise it, and its eigenvalues and eigenfunctions read off centrality, community structure, and the modes through which any dynamics on the network will couple. Representation, moments, metric, operator: each faculty is internal to the continuum, a way of comparing, classifying, and decomposing graphons among themselves.

What this chapter added is the faculty the other four presuppose but none supplies: a *passage* between that continuum and an actual finite network of N nodes, built in both directions. Outward, sampling: a graphon is a generative recipe, $W \rightsquigarrow G(N,W)$, and its samples faithfully remember it — each fixed motif of the sample concentrates on the parent's at the Monte-Carlo rate $N^{-1/2}$ (theorem 19.4), and the whole sampled kernel converges to the parent in cut distance ((19.2)). Inward, coarse-graining: every graphon, however intricate, sits within cut distance ε of a step graphon whose number of steps depends on ε alone (theorem 19.8), so that replacing a continuum by a bounded-complexity surrogate is rigorous and the count of macroscopically meaningful parameters is finite and explicit. And the two directions close on each other in a single identity: the law of the sampled graphs, the full list of motif densities, and cut distance zero are one and the same equivalence (theorem 19.11), so a graphon is recoverable from data precisely up to the relabelling the cut metric was built to quotient out — no finer, because that ambiguity is real, and no coarser, because everything else is identifiable. The bridge does not merely connect the continuum to finite data; it certifies that the connection loses nothing the theory was ever entitled to keep.

Set the ledger of what density buys against what it forbids, and keep the two columns honest. On the credit side, *within its regime* — graphs with $\Theta(N^2)$ edges — the four pillars plus the bridge are a single coherent calculus, and a remarkably complete one: limits of network sequences exist (compactness), are unique up to the one unavoidable relabelling ambiguity (identifiability), are observable through a canonical countable list of moments (motif densities), are decomposable along a spectrum (the operator), and are estimable from a single large sample at a known rate (the sampling lemmas). One does not often get a theory in which existence, uniqueness, observability, and statistical recoverability are all settled and all compatible; the dense graphon is such a theory. On the debit side stands a limitation we did not apologise for but *proved*, and proved twice over. Every subquadratic graph sequence — which is to say every sparse real network — is mapped by the unrescaled cut metric to the single uninformative zero graphon, collapsing all of them onto one structureless point (proposition 19.13); and the generative recipe runs the same blindness in reverse, forcing every vertex of $G(N, W)$ to a degree of order $\Theta(N)$ no matter how the kernel is chosen (proposition 19.14). These are not two failures but one, seen from the two ends of the bridge: the dense theory cannot see sparsity going in and cannot produce sparsity coming out, because boundedness of the kernel is the same hypothesis on both sides. The boundary of Part III is therefore the boundary of density itself — not an edge where the theorems grow weak, but the exact locus where the standing hypothesis that made all of them true runs out. It is a clean boundary, and a clean boundary is worth far more than a blurred one: it tells the next part of the book precisely where to build.

And that is the hand-off. Two doors open from here, and they open in opposite directions. Through the first, Part IV takes the dense operator $\mathbb{W}$ developed in this part — now equipped with its spectrum, its sampling theory, and its identifiability — and builds mean-field *games* on top of it: the analytic world we have made closed and complete becomes the arena in which strategic agents, coupled through $W(x, y)$, solve for equilibrium, and every faculty assembled here is consumed there as infrastructure. Through the second, chapter 27 reopens the analytic foundations themselves, replacing the bounded kernel by the unbounded L^p graphons and graphex objects that can finally represent the sparse, heavy-tailed networks the dense theory is structurally blind to — rebuilding, in a wider category, the very pillars we have just inventoried. The two doors are not in tension: the dense graphon is the right object to have understood first because it is where the moves are cleanest, and one carries the moves through both doors. That is the closing thesis of Part III, and it is an honest one. The dense graphon is not the final object of this book; it is the cleanest one, and the cleanest object is exactly the right place to have learned the moves.

Rigor ledger

Proved.

- The W-random graph $G(N, W)$, its weighted and step-graphon companions (definition 19.1).
- Motif concentration at the Monte-Carlo rate: unbiasedness up to $O(1/N)$ and the bounded-differences tail bound, hence a.s. convergence $t(F, G(N, W)) \to t(F, W)$ (theorem 19.4), with McDiarmid's inequality (lemma 19.3) reduced to Azuma.
- The weak regularity lemma: every graphon is within cut norm ε of a step graphon with at most $4^{\lceil 1/\varepsilon^2 \rceil}$ steps, via the energy-increment argument (theorem 19.8 and corollary 19.9).
- Subquadratic graph sequences converge to the zero graphon (proposition 19.13); the W-random model is intrinsically dense with $\Theta(N)$ degrees (proposition 19.14).

Assumed (imported).

- The cut norm/metric and the metric structure on graphons modulo weak isomorphism (definitions 17.1 and 17.6 and theorem 17.12); the counting lemma (lemma 16.17); compactness of graphon space (theorem 17.13); rank-one spectral structure (worked computation 18.19, with the spectrum computed in worked computation 2.28).
- Azuma–Hoeffding and Borel–Cantelli from probability foundations (theorems 3.27 and 3.28).
- Standing assumption (A-Reg-W) on W.

Deferred.

- The second sampling lemma (19.2) (Bernoulli sampling in cut distance), stated here with its proof deferred to theorem F.9, where it is assembled in full from the weighted sampling lemma and the in-book bounded-differences inequality; it is supplied to chapter 25 as the input to finite-network convergence (attribution [29, 99]).
- The explicit measure-preserving-map characterisation of weak isomorphism (remark 19.12, (19.5)): stated and attributed to [28, 99] but *not* certified in-book. Nothing downstream uses it; only the in-book metric statement $\delta_\square = 0$ (theorem 19.11(3)) is consumed.
- All sparse machinery — stretched cut metric, L^p graphons, graphex, sampling convergence — to chapter 27 (section 19.6).

Open.

- Mean-field game theory over sparse limits: well-posedness and finite-network approximation when $\mathbb{W}$ is an unbounded L^p kernel or a graphex (remark 19.15; chapters 27 and 31).
- Sharp constants and optimal scaling in the second sampling lemma for adaptively chosen test functions; minimax graphon estimation in the sparse regime beyond [87].

Bibliographic notes

The W-random graph and the two sampling lemmas are due to Lovász and Szegedy [100] and Borgs, Chayes, Lovász, Sós and Vesztergombi [29]; the definitive textbook treatment, including the logarithmic rate (19.2) and the diagonal corrections we absorbed into the $O(1/N)$ mean bias, is Lovász [99, Ch. 10–11]. The method of bounded differences we used for theorem 19.4 is McDiarmid's [102]; our explicit two-temperature sensitivity computation (latent types versus edge coins) follows the standard recipe but is written out for transparency. The weak regularity lemma originates with Frieze and Kannan [72] in matrix form; the Hilbert-space energy-increment proof for graphons given in theorem 19.8 is the one in [99, Ch. 9] and [100], and should be contrasted with Szemerédi's original lemma [128] and its tower-type lower bounds (see the discussion in [99]). Identifiability up to weak isomorphism (theorem 19.11) is the moment-uniqueness theorem of Borgs, Chayes and Lovász [28]; statistically, the connection between weak regularity and consistent block-model estimation is developed by Klopp, Tsybakov and Verzelen [87].

The sparsity discussion previews chapter 27. The honest diagnosis that bounded graphons force density is folklore but is made sharp by the degree computation of proposition 19.14; the rank-one heavy-tailed sparsification is the Chung–Lu expected-degree model [48]. Stretched/normalized cut metrics for sparse graphs are due to Bollobás and Riordan [25] and Borgs, Chayes, Cohn and Holden [26]; the graphex framework and its exchangeable random-measure representation are due to Caron and Fox [45] and Veitch and Roy [130], with sampling convergence developed in [26]. The mean-field game theory built on these foundations, and the careful separation of proved from conjectural that it requires, is the business of chapter 27 and the research map of chapter 31.

Part IV

The Graphon Mean Field Game Synthesis

Chapter 20

Static Graphon Games

This chapter opens Part IV, the synthesis of graph limits with mean field games, and it does so deliberately in the simplest possible setting: a *static* game, with no time, no diffusion, and no Hamilton–Jacobi machinery. Every later chapter of this part carries a forward equation coupled to a backward one across a kernel; here there is only an algebraic fixed point. The reward for this austerity is that the genuinely new object of the synthesis—a continuum of strategically interacting agents whose interaction is filtered through the graphon operator $\mathbb{W}$—appears stripped of every complication except the one that matters. The reader who understands how the operator norm $\|\mathbb{W}\|$ governs existence, uniqueness, and the relation between network centrality and equilibrium effort in this chapter will recognize exactly the same quantities, doing exactly the same work, when they reappear dressed in PDEs in chapters 22 and 24.

If the reader carries away a single object from this chapter, it should be the operator norm $\|\mathbb{W}\|$. It is one nonnegative number, computable from the kernel alone, and the entire theory organizes itself around it as around an order parameter. Existence of an equilibrium will cost nothing but the compactness of $\mathbb{W}$, and so will hold no matter how large $\|\mathbb{W}\|$ is. Uniqueness, by contrast, will turn on a single dimensionless inequality, $L_{\mathrm{br}}\ \|\mathbb{W}\| < 1$, in which the behavioral responsiveness L_{br} of an individual best response is multiplied by the structural amplification $\|\mathbb{W}\| = \lambda_1$ of the network; below the threshold the equilibrium is unique and equals a network centrality, and as the product climbs to one the equilibrium condenses onto the top eigenvector while its susceptibility diverges. Watching that one product cross one is the whole drama—and it is the same product, behavioral gain times spectral radius, that will control well-posedness when time and diffusion are switched back on in chapters 22 and 24. That coincidence is not a slogan but a statement we will be able to make precise by the end of the chapter (section 20.7): the static threshold is the literal stationary shadow of the dynamic one. This is the real reason it is worth spending an entire chapter on a game with no dynamics at all—the order parameter is already fully present, and present in closed form.

Three imports furnish everything we need, and it is worth saying at the outset what each one is *for*. The first is the network itself. From chapters 15 and 18 we take the graphon $W : [0,1]^2 \to [0,1]$ and its integral operator $\mathbb{W}$, under the standing regularity **(A-Reg-W)** that makes $\mathbb{W}$ a self-adjoint Hilbert–Schmidt operator on $L^2[0,1]$ (definition 2.23 and proposition 2.24); the spectral theorem for such operators (theorem 2.26) is what hands us the eigenvalues λ_n, and in particular the top one $\lambda_1 = \|\mathbb{W}\|$ that the whole chapter watches. Compactness of $\mathbb{W}$ does double duty here: it is the smoothing that will deliver existence, and the discreteness of the spectrum it guarantees is what makes "the threshold $L_{\mathrm{br}} \|\mathbb{W}\| = 1$" a sharp, attained edge rather than a supremum that is approached but never reached. The second import is a small toolbox of fixed-point theorems, because every equilibrium in this chapter is the fixed point of a best-response map and nothing more. From chapter 2 we take the three engines—Banach (theorem 2.36) for the contraction that gives uniqueness below threshold, Schauder (theorem 2.38) for the compactness-only existence that needs no smallness at all, and Kakutani (theorem 2.40) for the set-valued best responses that appear once strict concavity is dropped—together with the min–max characterization of eigenvalues (corollary 2.27) that we use to pin $\|\mathbb{W}\|$ to the top of the spectrum. The third import is a principle rather than an object: from chapter 11 we take the Lasry–Lions monotonicity philosophy (definition 11.1) and reuse its logic in a static, finite-dimensional-feeling guise, where it becomes the convexity of a potential and supplies a second, structural route to uniqueness that survives even when the coupling is *not* small. Finally, the carried thread we advance is Thread B, the graphon LQG model of appendix A.7: this chapter introduces its *static* incarnation and solves it in closed form on the constant, rank-one, and block kernels, so that every abstract claim is checked against a kernel on which the spectrum, the centrality, and the susceptibility can all be written down by hand.

The plan is linear, and each step is the natural next question raised by the one before it. Section 20.1, the present section, builds the static graphon game by taking a finite network game to its continuum-of-types limit, and in doing so isolates the central object: the *graphon aggregate* $z(x) = (\mathbb{W}s)(x)$, the single payoff-relevant statistic that each type feels. Once the game has been reduced to an equation in the profile s, the first thing to ask is whether it has a solution at all; sections 20.2 and 20.3 cast the equilibrium condition as a fixed point of a best-response map and answer yes, by compactness alone, signaling already that existence is the cheap part of the story. The expensive part—and the heart of the chapter—is section 20.4: a fixed point can fail to be unique once interactions are strong, and there we locate the exact dividing line, the operator norm $\|\mathbb{W}\|$ entering as the coupling strength that must lie below a threshold (assumption **(A-Coupling)**). We prove uniqueness twice, by contraction and by monotonicity, precisely in order to see the same threshold from two sides, the spectral and the variational. A unique equilibrium is worth little until we can say what it *is*; section 20.5 identifies it as a recognizable network centrality, turning the abstract fixed point into a concrete ranking of players—equilibrium effort is Katz–Bonacich centrality, near criticality it aligns with eigenvector centrality, and targeted interventions are ranked by centrality. Section 20.6 then earns all of this on the carried Thread B model, computing the spectrum, the centrality, and the divergent susceptibility in closed form on the constant, rank-one, and block kernels. Finally section 20.7 closes the loop with the rest of Part IV by explaining in what precise sense this entire static theory is the stationary shadow of the dynamic graphon mean field game of chapters 21, 22 and 24—the payoff for having watched $\|\mathbb{W}\|$ the whole way.

Notation watch

Two symbol clashes are resolved once here, and both are genuine: each letter has an established, non-negotiable meaning on *both* sides, so we cannot simply rename one away without breaking continuity either with a neighboring chapter or with the standard network-games literature.

(i) In this chapter s denotes a *strategy profile*, the function $x \mapsto s(x)$ recording the action chosen by each type; this matches the coverage of static network games and the aggregate $z(x) = \int_0^1 W(x,y)\, s(y)\, \mathrm{d}y$. In the carried Thread B of appendix A.7 the same letter s is a fixed scalar *coupling strength*, the weight pulling each agent toward its neighbors. The clash would bite in exactly one place—the Thread B worked computation of section 20.6, where both meanings are in play at once—and we defuse it there by writing the strategic variable as the stationary per-type mean $\bar{m}(x)$, which both frees s for its thread meaning and agrees with the canonical Thread B symbol; everywhere else the two never appear in the same formula. *(ii)* No Brownian motion appears in this static chapter, so W is unambiguously the graphon and we may use it without comment. This is a reprieve, not a resolution: the W versus B convention of appendix A.5 becomes relevant the moment noise returns, from chapter 21 onward, where Brownian motion is written B precisely so that W can keep meaning the kernel.

20.1 From a network game to a continuum of types

Before we can study the protagonist of the chapter we have to construct it, and the construction is the whole business of this section: we manufacture the static graphon game—a game played by a continuum of types, each feeling the rest of the population only through the aggregate $z(x) = (\mathbb{W}s)(x)$—out of a finite, finger-on-the-blackboard network game that needs no new ideas to write down. We do this in three deliberate moves, and it is worth knowing in advance why each one is there. First (section 20.1.1) we set up the *finite network game*: N players sitting at the nodes of a graph, each best-responding to the average action of its neighbors. This is the concrete, combinatorial object whose equilibria we ultimately care about, and it is where the one structural feature that survives everything to follow—that a player's payoff sees the rest of the population *only* through a neighbor aggregate, never through the unstructured full profile—first shows itself. Second (section 20.1.2) we pass to the *continuum of types*, letting $N \to \infty$ along a convergent dense sequence and watching the adjacency matrix dissolve into a kernel and the neighbor average dissolve into the graphon operator $\mathbb{W}$; this is the move that trades a combinatorial object growing without bound for a single fixed analytic one, and it is what turns the large-N behavior into something we can compute rather than merely simulate. Third (section 20.1.3) we pin down the *linear–quadratic benchmark*, the one payoff for which the best response is affine and equilibrium collapses to a linear operator equation; we need it because it is the model on which every later spectral statement—the uniqueness threshold, centrality, the divergent susceptibility—can be proved in a single line, and because it is the static face of the carried Thread B.

These three moves all serve one question, which the finite game is about to pose sharply. The defining nuisance of a network game is heterogeneity: a player buried in a dense core feels a large neighbor aggregate, a player out on the periphery feels a small one, and so the equilibrium is not one number repeated across players—as in the anonymous mean field game of Part II—but a profile indexed by network position. The reason to go to the continuum is emphatically *not* to wash this heterogeneity out; it must survive the limit, or the limit is worthless. The point is to carry it through intact while replacing the unwieldy $N \times N$ adjacency matrix, which grows without bound, by an object that does not. So the question the next subsection sets up sharply—and the one after it answers—is precisely this: how does core-versus-periphery heterogeneity persist as $N \to \infty$, and what single analytic object inherits the role that the adjacency matrix A played in the finite game?

20.1.1 The finite network game

We begin where every network model honestly begins: with a finite, combinatorial object one could in principle draw on a blackboard. The continuum game that is the protagonist of this chapter is not postulated out of thin air; it is the large-N shadow of a perfectly ordinary game played by finitely many people sitting at the nodes of a graph. Building that finite game first, and watching exactly which of its features are essential and which are accidents of finiteness, is what will tell us what the continuum limit must preserve. The single feature we will track above all others is the one the previous section flagged: *heterogeneity*, the fact that different players occupy structurally different positions and therefore feel different things. The finite game is where that heterogeneity is born, and the entire graphon apparatus exists to carry it intact across the limit.

Fix a finite simple graph on N nodes with symmetric adjacency matrix $A = (A_{ij}) \in \{0,1\}^{N\times N}$. Each node i is a player who chooses an action $s_i \in S \subseteq \mathbb{R}$, where S is the player's action set (an effort level, an investment, an opinion, a contribution to a public good). The defining feature of a *network game* is that player i's payoff depends on its own action and on the *aggregate of its neighbors' actions*,

$$z_i \;=\; \frac{1}{N}\sum_{j=1}^{N} A_{ij}\, s_j, \tag{20.1}$$

rather than on the full profile $(s_1,\ldots,s_N)$ in an unstructured way. Player i's payoff is

$$\pi_i(s_i, z_i) \qquad \text{with each } \pi_i :\; S\times\mathbb{R}\to\mathbb{R} \text{ continuous}, \tag{20.2}$$

depending on the profile of other players *only* through the scalar aggregate z_i. A game of this form is called an *aggregative* (network) game.

Why the $1/N$. The normalization in eq. (20.1) is not cosmetic, and it is worth being explicit about why the factor is $1/N$ rather than, say, 1 or $1/\deg(i)$. We are interested in dense graphs—graphs in which a typical node has on the order of N neighbors, a positive fraction of the population—because those are the graphs that have nontrivial graphon limits (chapter 15). On such a graph the unnormalized neighbor sum $\sum_j A_{ij}s_j$ counts a number of terms that grows like N; if the actions s_j stay $O(1)$, that bare sum diverges as the network grows, and with it every payoff that depends on it—the limit would be a game in which a single agent feels an infinitely strong pull from the population, which is not a game at all. Dividing by N turns the sum into an average and keeps the aggregate $O(1)$ as $N \to \infty$, so that the strategic problem each player faces converges to a finite, sensible limit. Concretely, z_i is the action s_j averaged over the whole population with the neighbor indicators A_{ij}/N as weights; these weights sum to the neighbor *fraction* $\deg(i)/N \in [0,1]$, which stays bounded, whereas the degree $\deg(i)$ itself grows. This is precisely the *dense-graph normalization* that the graphon limit of chapter 15 requires: it is the same $1/N$ that makes the rescaled adjacency matrix converge to a kernel W taking values in $[0, 1]$ rather than blowing up, and we will see in section 20.1.2 that under it the discrete aggregate eq. (20.1) is *exactly* an integral against the pixel graphon, with no error term at all. Choosing the local normalization $1/\deg(i)$ instead would also bound the aggregate, but it would erase exactly the information we care about—degree heterogeneity—by forcing every node to feel a neighbor average on the same scale; the $1/N$ convention deliberately keeps a high-degree node feeling *more* than a low-degree one, which is the whole point of a network game.

Core versus periphery. To see what that retained heterogeneity means concretely, take the smallest example that already exhibits it. Suppose a "core" node c and a "periphery" node p, and suppose for the moment that everyone plays the same action $s_j \equiv a$, but that c is connected to a fraction $\deg(c)/N = 0.8$ of the population while p is connected to only $\deg(p)/N = 0.05$. Then the aggregates they feel are

$$z_c \;=\; \frac{1}{N}\sum_j A_{cj}\, a \;=\; \frac{\deg(c)}{N}\, a \;=\; 0.8\, a, \qquad z_p \;=\; \frac{\deg(p)}{N}\, a \;=\; 0.05\, a,$$

a sixteen-fold difference in the very quantity their payoffs respond to, produced entirely by position and not at all by any difference in their preferences π_c, π_p. Even if the two players had *identical* payoff functions, they would best-respond to different aggregates and so, generically, choose different actions. This is the structural fact that an anonymous mean field model cannot represent: there, every agent feels the same population average, so position is invisible. Here position is everything, and it is the surviving content of the network. The job of the continuum limit is to keep $z_c \neq z_p$ alive as $N \to \infty$ while disposing of the unwieldy bookkeeping of the matrix A that produced them.

Why aggregative. It is the word *aggregative*—the restriction that π_i sees the rest of the population only through the single scalar z_i, never through the full vector $(s_1, \ldots, s_N)$ in an unstructured way—that makes all of this possible, and it deserves to be recognized as a genuine modeling hypothesis rather than a convenience. A general game on N players would let π_i depend separately on each of the $N - 1$ other actions; there is then no scalar summary of "what the neighbors are doing" to pass to a limit, and the object that would have to converge as $N \to \infty$ is a function of a growing number of arguments, with no stable limiting form. Aggregativeness collapses that $(N - 1)$-dimensional dependence onto one real number per player, and crucially does so *linearly* in the neighbors' actions through the adjacency weights. That is exactly the structure a graphon can carry: the linear neighbor-averaging $s \mapsto z$ is what becomes the linear integral operator $\mathbb{W}$ in the limit, and the scalar z_i is what becomes the limiting field $z(x) = (\mathbb{W}s)(x)$. Were the interaction not aggregative, there would be no $\mathbb{W}$ to find. Aggregativeness is thus the precise structural restriction that opens the door to the entire graphon analysis to come; the linear–quadratic payoff of section 20.1.3 will be the cleanest instance of it.

Equilibrium. With the game specified, equilibrium is the usual Nash notion, but it pays to write it out per node, because its form already encodes the heterogeneity we have been tracking. A Nash equilibrium is a profile $(s_1^*, \dots, s_N^*)$ in which no player can improve by unilaterally changing its own action, the rest held fixed:

$$s_i^* \in \arg\max_{a \in S} \pi_i\Big(a, \tfrac{1}{N} \textstyle\sum_j A_{ij}\, s_j^*\Big), \qquad i = 1, \dots, N. \tag{20.3}$$

Read this slowly, one node at a time. Player i takes the aggregate $z_i^* = \frac{1}{N} \sum_j A_{ij} s_j^*$ generated by everyone else's equilibrium actions as a fixed number, and chooses its own action a to maximize its payoff against that number; the equilibrium demand is that every player's chosen action be simultaneously consistent with the aggregate it helps create. Two things about eq. (20.3) are worth fixing in mind before we take any limit. First, it is a genuine *fixed-point* condition—the profile s^* appears on both sides, inside the aggregate on the right and as the maximizer on the left—and this fixed-point character is exactly what survives to the continuum and gets solved in sections 20.2 to 20.4. Second, and this is the point of the whole section, the equilibrium is *not* a single number repeated across players. In the anonymous mean field game of Part II every agent faces the same population field and an equilibrium is one action (or one distribution) common to all; here the index i runs over *network positions*, each carrying its own aggregate z_i^*, and so s^* is an N-vector that genuinely varies from node to node—a *profile indexed by position*, in which the core node c and the periphery node p of our micro-example will sit at different entries with different values. The unknown of a network game is a labeled profile, not a scalar; remembering this is the difference between the present theory and ordinary mean field theory.

The difficulty, and the reason the next section exists, is now sharply visible. Everything that makes the equilibrium eq. (20.3) interesting—the differing aggregates, the core/periphery split, the position-indexed profile—is carried by the matrix A, and A is a combinatorial object that grows without bound: it has N^2 entries, no limiting form as a matrix, and offers no analytic handle through which to compute anything about large networks. We do not want to study a sequence of ever-larger matrices; we want a single fixed object that inherits everything A was doing and that we can do calculus with—differentiate, invert, diagonalize. The previous section promised that such an object exists. What we need, then, is a device that replaces the heterogeneous combinatorial matrix A by an analytic kernel acting on a limiting profile, while keeping the neighbor-averaging structure—and with it the heterogeneity—exactly intact. Producing that device, and showing that the discrete aggregate eq. (20.1) passes cleanly into the graphon operator $\mathbb{W}$, is the business of the next subsection.

20.1.2 Passing to the continuum of types

The previous subsection ended with a precise want: an analytic object that inherits everything the adjacency matrix A was doing in the neighbor average eq. (20.1)—it must preserve the position-dependent heterogeneity, it must act linearly on the profile, and it must be a *single fixed* object rather than a sequence of ever-larger matrices. We have in fact already built exactly such an object, in a different context and for a different purpose, and the whole business of this subsection is to recognize that it is the right tool here. The device is the pixel picture of chapter 15, which taught us to read a finite adjacency matrix not as a combinatorial array but as a function on the unit square; once a profile is read the same way, the discrete neighbor average is *literally* an integral against that function, and the limit $N \to \infty$ replaces the function by a fixed kernel and the integral by the graphon operator $\mathbb{W}$. No new mechanism is introduced; we are only changing how we look at a sum we already have.

Let us make the dictionary explicit before compressing it into the box, because each line of it is an *identification*—a choice of how to encode discrete data as a function—and not a theorem requiring proof. There are exactly three encodings and one consequence. First, the network: the pixel graphon (definition 15.2) turns the matrix A into the step function on $[0,1]^2$ that is constant on each $\frac{1}{N} \times \frac{1}{N}$ tile and equal there to the corresponding matrix entry A_{ij}. Second, the players: we slice the type interval $[0,1]$ into N equal bands $I_i = (\frac{i-1}{N}, \frac{i}{N}]$ and declare that player i *is* the band of types I_i, so that "node" and "interval of types" become two names for the same thing. Third, the profile: the N numbers (s_i) become the step function s_N taking the constant value s_i throughout the band I_i. None of these three is an approximation; each is an exact re-encoding of finite data as a piecewise-constant function. The consequence is what makes the re-encoding worth the trouble: when the neighbor sum is rewritten band by band, every factor $1/N$ in eq. (20.1) is exactly the width $|I_j| = 1/N$ of a band of types, so the average turns into an honest integral with *no* leftover normalization—which is the concrete vindication of the dense $1/N$ convention chosen in section 20.1.1. The box records this calculation.

Heuristic (non-rigorous)

Recall from chapter 15 the pixel picture (the pixel graphon of definition 15.2): the adjacency matrix A of an N-node graph is the step function $W^A(x,y) = A_{\lceil Nx \rceil, \lceil Ny \rceil}$ on $[0,1]^2$, and a convergent dense sequence $A^{(N)}$ has a limit graphon $W : [0,1]^2 \to [0,1]$. Identify player i with the type interval $I_i = (\frac{i-1}{N}, \frac{i}{N}]$ and let the discrete strategy profile (s_i) become the step function $s_N(x) = s_{\lceil Nx \rceil}$. For $x \in I_i$ the integrand $y \mapsto W^A(x,y)\, s_N(y)$ is constant on each band $y \in I_j$, equal there to $A_{ij}s_j$, and that band has width $1/N$; summing the bands shows the finite aggregate eq. (20.1) is, exactly and with no error term,

$$z_i \;=\; \frac{1}{N}\sum_{j=1}^{N} A_{ij}s_j \;=\; \sum_{j=1}^{N} |I_j|\; A_{ij}s_j \;=\; \int_0^1 W^A(x,y)\, s_N(y)\; \mathrm{d}y \;=\; (\mathbb{W}^A s_N)(x) \quad \text{for } x \in I_i.$$

As $N \to \infty$ with $A^{(N)} \to W$ and $s_N \to s$, this is the integral $\int_0^1 W(x,y)s(y)\; \mathrm{d}y = (\mathbb{W}s)(x)$. The discrete neighbor average converges to the action of the graphon operator on the limiting strategy profile. The limit object is a game played by a *continuum* of players, one for each type $x \in [0,1]$, in which the payoff-relevant statistic felt by type x is the graphon aggregate $(\mathbb{W}s)(x)$.

It is worth separating cleanly what this box settles from what it merely asserts, because the two have very different status. What it *settles*, exactly and at every finite N, is the chain of equalities ending in $(\mathbb{W}^A s_N)(x)$: the neighbor average of a finite graph game *is* the graphon operator of the pixel graphon applied to the step-function profile, with no limit taken and no approximation made. That is pure bookkeeping. What it *asserts but does not prove* is the final sentence's passage $N \to \infty$: that the pixel operators $\mathbb{W}^{A^{(N)}}$ converge to a limiting $\mathbb{W}$ and the step profiles s_N to a limiting s in a topology strong enough to carry the integral through. That convergence is a genuine theorem with genuine hypotheses—it is where the cut metric of chapter 15 and the sampling theory of chapter 19 do their work—and we are deliberately *not* proving it here. The full statement, that graphon Nash equilibria of the finite games converge to graphon Nash equilibria of the limit, is the convergence theorem of chapter 25.

So rather than derive the continuum game, we take it as the primitive and study it on its own terms. This is the same methodological choice that organized Part II, and it is worth recalling why it was the right one there. The mean field game was posited and analyzed directly—its existence, uniqueness, and structure all established as statements about the limit problem itself—and only later, in chapter 13, did we circle back to justify it as the $N \to \infty$ shadow of the genuine N-player game. The justification is real and it matters, but it is a separate and harder question, and welding it onto the definition would bury exactly the clean algebraic structure we came here to expose. We therefore declare the continuum game by fiat, state its equilibrium notion, and spend the chapter solving it; the reader who wants the bridge back to finite networks will find it built in chapter 25. With that division of labor settled, here are the two definitions the rest of the chapter rests on.

Definition 20.1 (Static graphon game)**.** A *static graphon game* is a tuple $(W,\, S,\, (\pi_x)_{x\in[0,1]})$ where:

- $W : [0,1]^2 \to [0,1]$ is a graphon satisfying **(A-Reg-W)**, with integral operator $\mathbb{W}$ on $L^2[0,1]$ as in definition 2.23;
- $S \subseteq \mathbb{R}$ is the common action set, assumed nonempty, closed, and convex (an interval, possibly all of $\mathbb{R}$);
- for each type $x \in [0,1]$, $\pi_x : S \times \mathbb{R} \to \mathbb{R}$ is the payoff of a type-x player as a function of its own action $a \in S$ and the scalar aggregate $z \in \mathbb{R}$ it feels; the map $(x, a, z) \mapsto \pi_x(a, z)$ is jointly measurable and continuous in (a, z).

A *strategy profile* is a (measurable) function $s \in L^2[0,1]$ with $s(x) \in S$ for almost every x; we write $\mathcal{S} = \{s \in L^2[0,1] :\ s(x) \in S \text{ a.e.}\}$ for the set of admissible profiles. The *graphon aggregate* felt under the profile s is

$$z(x) \;=\; (\mathbb{W}s)(x) \;=\; \int_0^1 W(x,y)\, s(y)\; \mathrm{d}y. \tag{20.4}$$

The aggregate eq. (20.4) is the chapter's protagonist, and it repays a careful look. It is a *field*: not a single population number but one real value $z(x)$ for *each* type, obtained by averaging the whole population's actions against the row $W(x,\cdot)$ of the kernel. The row is the type's personal weighting of who counts as a neighbor, so a type whose row puts large weight on high-action types feels a large aggregate, and two types with different rows feel different aggregates even under the same profile. This is the continuum incarnation of the core-versus-periphery split of section 20.1.1: the heterogeneity that the matrix A carried has not been washed out by the limit, it has been absorbed into the type-dependence of the row, exactly as intended.

The map $s \mapsto z = \mathbb{W}s$ is linear—this is the residue of aggregativeness—and it has one further property that will quietly run the existence theory: $\mathbb{W}$ is compact (proposition 2.24), hence *smoothing*. The word deserves to be unpacked, because it is the engine of section 20.3. Diagonalize $\mathbb{W}$ in its eigenbasis (chapter 18); a profile $s = \sum_n c_n\phi_n$ is mapped to $\mathbb{W}s = \sum_n \lambda_n c_n\phi_n$, and since the eigenvalues $\lambda_n \to 0$, the high-index components—the rapidly oscillating, large-n modes in which neighboring types *disagree*—are multiplied by vanishingly small factors and suppressed in z. Averaging against a kernel row cannot manufacture fine structure that the row itself does not resolve, so disagreement at fine scales is damped: $\mathbb{W}$ takes a possibly wild profile s and returns a tamer field z. Concretely, this means that as s ranges over a merely bounded set of profiles, the images $\mathbb{W}s$ do not range over a merely bounded set but over a *precompact* one—bounded sets are compressed toward a finite-dimensional core by the decay of the λ_n. That precompactness is precisely the hypothesis a fixed-point theorem needs to locate an equilibrium without any smallness assumption on the coupling: in section 20.3 we will feed the precompact image of $\mathbb{W}$ into Schauder's theorem and get existence for free, no matter how large $\|\mathbb{W}\|$ is. The compactness of $\mathbb{W}$ is thus not a technical footnote but the structural reason equilibria exist; we flag it here so that when it is cashed in two sections later the move looks inevitable rather than clever.

Definition 20.2 (Graphon Nash equilibrium)**.** A strategy profile $s^* \in \mathcal{S}$ is a *graphon Nash equilibrium* (GNE) of the static graphon game if for almost every type $x \in [0,1]$,

$$s^*(x) \;\in\; \operatorname*{arg\,max}_{a\in S}\; \pi_x\big(a,\, (\mathbb{W}s^*)(x)\big). \tag{20.5}$$

That is, each type's action is a best response to the aggregate generated by the whole profile, and the profile is consistent with itself.

Condition eq. (20.5) looks like a pointwise optimization, but it is a fixed point in disguise, and seeing it as one is what unlocks the rest of the chapter. Read the loop it describes: a profile s^* generates, through $\mathbb{W}$, an aggregate field $z^* = \mathbb{W}s^*$; each type then best-responds to the value of that field at its own location; and the equilibrium demand is that the profile of those best responses be the profile s^* we started from. The unknown appears on both sides—inside the aggregate on the right, as the maximizer on the left—so s^* is required to reproduce itself after one pass through the loop "aggregate, then best-respond." This is exactly the self-consistency that defines a fixed point, and section 20.2 will make it literal by packaging the loop as a single map S whose fixed points are the GNE; from that point on, existence, uniqueness, and the spectral threshold are all questions about one operator.

The closure is the same one the reader met in the mean field game of chapter 8—each agent optimizes against a population field, and the field is itself the aggregate of the optimizers' choices—so a GNE is the static, graphon-filtered cousin of an MFG equilibrium. Two differences are worth naming, because they are precisely the features this chapter adds and subtracts. It *subtracts* time: there is no forward density evolving and no backward value function, only a single algebraic consistency condition, which is what makes everything provable in closed form. And it *adds* network structure through the kernel: the field a type feels is the *graphon-filtered* aggregate $(\mathbb{W}s)(x)$, one number per type, rather than the single population mean common to everyone in an anonymous model. That this is a genuine generalization, and not just heavier notation, is visible in one limit. Set $W \equiv 1$, the constant kernel in which every type is everyone's neighbor with equal weight; then the row $W(x,\cdot) \equiv 1$ does not depend on x, the aggregate collapses to the type-independent scalar mean $(\mathbb{W}s)(x) = \int_0^1 s(y)\,\mathrm{d}y$, and eq. (20.5) becomes an anonymous aggregative game in which every player faces the identical population average—the mean field model of Part II, recovered exactly. The graphon is the device that breaks this anonymity: any non-constant kernel makes the row, and hence the aggregate, depend on the type, so that position once again matters. Everything genuinely new in the chapter lives in that departure from $W \equiv 1$.

These two definitions are deliberately payoff-agnostic: π_x is any jointly measurable, concave-in-action payoff, and the GNE condition eq. (20.5) is stated for all of them at once. That generality is what the existence and uniqueness theorems will need, but it is also more than we can compute with. The next subsection therefore specializes the whole apparatus to the one payoff—linear marginal return, quadratic own cost—for which the best response is affine and the fixed-point condition eq. (20.5) collapses to a single linear operator equation in s^*. That linear–quadratic benchmark is where the abstract field $(\mathbb{W}s)(x)$ becomes something we can diagonalize, and it is the static face of the carried graphon LQG thread; it is what we turn to now.

20.1.3 The linear–quadratic benchmark

The definitions of the previous subsection were deliberately payoff-agnostic: π_x was allowed to be any jointly measurable, strictly concave-in-action payoff, and the equilibrium condition eq. (20.5) was written for all of them at once. That breadth is exactly what the existence and uniqueness theorems of sections 20.3 and 20.4 will need, and we will keep it. But breadth is the enemy of explicit computation, and the previous subsection closed by promising to cash some of it in. The point of the chapter is not merely to assert that the operator norm $\|\mathbb{W}\|$ governs the equilibrium; it is to make the reader *watch* it do so, in closed form. For that we need a payoff on which the best response is not just well defined but writable down, so that the self-consistency loop "aggregate, then best-respond" of definition 20.2 collapses from an abstract fixed point into an equation we can diagonalize. There is one canonical choice with this property, and most of the chapter is carried by it: the *linear–quadratic* (LQ) network game, in which the cost of one's own action is quadratic and the coupling to the neighbor aggregate is linear. Its virtue is that the best response comes out *affine*, and an affine best response turns the equilibrium condition into a single linear operator equation—which is precisely a spectral question about $\mathbb{W}$, and hence, $\mathbb{W}$ being the graphon operator, a question about network structure.

This is also the moment the carried thread re-enters, in its simplest possible dress. The LQ network game is the *static cousin* of the linear–quadratic–Gaussian (LQG) graphon thread that runs through the book (appendix A.7): strip that dynamic model of its time horizon, its state dynamics, and its Gaussian noise, keeping only the quadratic own-cost and the linear coupling through the kernel, and what is left is exactly the payoff below. It is at the same time the graphon lift of the canonical finite static network game of Parise and Ozdaglar [108], whose linear–quadratic model on a finite graph is the N-player object that section 20.1.2 has just taught us to read in the continuum. Everything we establish on this benchmark therefore sits between two relatives we can name: a finite-network ancestor on one side and, on the other, a dynamic descendant in chapters 22 and 24 of which—by the time we reach section 20.7—this static problem will turn out to be the literal stationary shadow.

Example 20.3 (Linear–quadratic graphon game)**.** Take the payoff

$$\pi_x(a,z) \;=\; -\tfrac{1}{2}\,a^2 \;+\; a\,(\alpha(x) + \mathsf{b}\,z), \tag{20.6}$$

with a measurable, bounded *standalone marginal return* $\alpha \in L^\infty[0,1]$ and a scalar *interaction coefficient* $\mathsf{b} \in \mathbb{R}$. Differentiating in the own action gives the marginal payoff and the curvature

$$\partial_a \pi_x(a,z) \;=\; -a + \alpha(x) + \mathsf{b}\,z, \qquad \partial_{aa}\pi_x(a,z) \;\equiv\; -1 \;<\; 0,$$

so $\pi_x(\cdot, z)$ is strictly concave and possesses a unique unconstrained maximizer over $a \in \mathbb{R}$, found by setting the marginal to zero:

$$a \;=\; \alpha(x) + \mathsf{b}\,z. \tag{20.7}$$

With $S = \mathbb{R}$ the best response is therefore the affine map $z \mapsto \alpha(x) + \mathsf{b}z$, and a GNE solves the *linear graphon equation*

$$s^*(x) \;=\; \alpha(x) \;+\; \mathsf{b}\,(\mathbb{W}s^*)(x), \qquad \text{i.e.} \qquad (I - \mathsf{b}\mathbb{W})\,s^* = \alpha. \tag{20.8}$$

With S a bounded interval $[\underline{s}, \bar{s}]$ the best response is the same expression clipped to S; we return to the clipped case in section 20.3.

It is worth reading the payoff eq. (20.6) one term at a time, because every structural feature the chapter exploits is already legible in it. The first term $-\frac{1}{2}a^2$ is a strictly convex *cost of own effort*: doing more is increasingly expensive, and its curvature $\partial_{aa}\pi_x \equiv -1$ is normalized to -1 once and for all, which simply fixes the scale on which we measure actions (any other constant curvature could be absorbed by rescaling a, α, and b). This strict concavity is not cosmetic: it is exactly the hypothesis that makes the maximizer eq. (20.7) *unique*, so that the best response is a genuine single-valued function rather than a set—the property that section 20.2 will isolate as the thing the whole fixed-point apparatus rests on. The second term, $a\,\alpha(x)$, is the private incentive to act, and the third, the cross term $\mathsf{b}\,az$, is the entire strategic content of the game: it is the only place where a player's payoff feels its neighbors at all, and it does so by tying the value of one's own action to the aggregate z those neighbors generate.

The sign of b decides the character of the interaction, and the two cases are worth naming with their behavioral readings because they recur throughout the chapter. When $\mathsf{b} > 0$, the cross term rewards aligning one's action with the neighbor aggregate—the marginal return $\partial_a\pi_x = -a+\alpha(x)+\mathsf{b}z$ rises with z, so a larger neighbor aggregate makes a player want to do *more*. Actions are then *strategic complements*: effort begets effort. This is the regime of R&D spillovers, of studying harder when one's classmates study harder, of adopting a technology once one's contacts have—the cases in which the network *amplifies* behavior, and, as we will see, the only regime in which the coupling can drive the equilibrium toward a spectral threshold. When $\mathsf{b} < 0$, the marginal return falls with z, so a player wants to do *less* when its neighbors do more. Actions are then *strategic substitutes*: this is the regime of local public goods and of free-riding, where one neighbor's contribution relieves the others of the need to contribute. The two signs will behave very differently at the uniqueness threshold of section 20.4, and the distinction is already encoded, with no further structure, in the single number b.

The passage from the pointwise maximizer eq. (20.7) to the operator equation eq. (20.8) is a single substitution, but it is the substitution on which the entire chapter turns, so we expand it. In equilibrium, each type x does not respond to an arbitrary aggregate but to the one it actually feels under the prevailing profile, which by eq. (20.4) is $z = (\mathbb{W}s^*)(x)$. Substituting this value of z into the maximizer eq. (20.7) yields, for almost every x,

$$s^*(x) \;=\; \alpha(x) \;+\; \mathsf{b}\,(\mathbb{W}s^*)(x).$$

Now collect the unknown on one side: subtracting $\mathsf{b}\,(\mathbb{W}s^*)(x)$ from both sides gives $s^*(x) - \mathsf{b}\,(\mathbb{W}s^*)(x) = \alpha(x)$. The left-hand side is precisely the value at x of the function $(I - \mathsf{b}\mathbb{W})s^*$, where I is the identity on $L^2[0,1]$ and $\mathsf{b}\mathbb{W}$ is the graphon operator scaled by b; since the identity holds for almost every x, it is an equality of functions, $(I - \mathsf{b}\mathbb{W})s^* = \alpha$, which is exactly eq. (20.8). The meaning of this rearrangement is the conceptual payoff of the LQ specialization: the equilibrium is no longer the output of a fixed-point *search*, with the unknown buried inside a maximization on the right, but the solution of a *linear inversion*,

$$s^* \;=\; (I - \mathsf{b}\mathbb{W})^{-1}\alpha \qquad \text{whenever the operator } (I - \mathsf{b}\mathbb{W}) \text{ is invertible.}$$

The operator $I - \mathsf{b}\mathbb{W}$ encodes the strategic feedback loop in one object: the identity is a player's own response to its private incentive, and $-\mathsf{b}\mathbb{W}$ is the correction induced by the fact that the neighbors are responding too, and to each other, all at once. Inverting it resolves that infinite regress of mutual best responses in closed form.

This is also the place to say what *standalone marginal return* means and why α earns the name, since it is the data the inversion acts on. Set $z = 0$ in eq. (20.7)—imagine a type momentarily cut off from the network, or equivalently with all its neighbors silent. The maximizer is then $a = \alpha(x)$, so $\alpha(x)$ is the action a type-x player would choose *in isolation*, driven purely by its own incentive with no strategic feedback. Equivalently, $\alpha(x) = \partial_a \pi_x(0,0)$ is the marginal return to the very first unit of action when a player stands alone: it is the intrinsic, network-free pull toward acting, the part of the incentive that owes nothing to who one's neighbors are. The full equilibrium effort eq. (20.8) is then read as this standalone return propagated and amplified through the network: $s^* = (I - \mathsf{b}\mathbb{W})^{-1}\alpha$ takes the bare per-type incentive α and dresses it with every order of neighbor feedback,

$$s^* \;=\; \alpha \;+\; \mathsf{b}\,\mathbb{W}\alpha \;+\; \mathsf{b}^2\,\mathbb{W}^2\alpha \;+\; \cdots$$

whenever the series converges—a Neumann expansion whose convergence is, once again, a question about $\mathbb{W}$, and which we will recognize in section 20.5 as Katz–Bonacich centrality.

The single equation eq. (20.8) thus already contains the whole story, and the remainder of the chapter is little more than its careful reading. Whether $(I - \mathsf{b}\mathbb{W})$ is invertible at all decides whether an equilibrium exists and is unique; the size of its inverse—how strongly $\|(I - \mathsf{b}\mathbb{W})^{-1}\|$ blows up—measures how violently the equilibrium responds to the incentives α, the susceptibility that will diverge at the threshold; and the way its inverse *acts* on α, mode by mode in the eigenbasis of $\mathbb{W}$, is what identifies the equilibrium profile with a recognizable network centrality. All three are spectral questions about $\mathsf{b}\mathbb{W}$, and since $\mathbb{W}$ is the graphon operator, all three are statements about *network structure*: the eigenvalue $\lambda_1 = \|\mathbb{W}\|$ that the chapter watches is exactly the quantity that controls when $1 - \mathsf{b}\lambda_1$ first hits zero and the inverse ceases to exist.

What this section bought through the affine best response, however, it bought on the strength of the LQ form alone, and we should not let the cleanest results of the theory rest on that piece of luck. Existence and uniqueness ought to follow from the *structure* of best responding to a graphon-filtered aggregate—from concavity and from the strength of the coupling—not from the accident that the maximizer happened to be linear in z. The next section therefore steps back from example 20.3 to a general concave payoff and makes the fixed-point picture precise once and for all: it packages the loop “aggregate, then best-respond” into a single map whose fixed points are the GNE, and it isolates the two properties of the scalar best response—that it is a well-defined function of the aggregate, and that it is Lipschitz with a definite constant—on which every existence and uniqueness argument actually turns. The affine LQ map of this section will reappear there as the special case whose Lipschitz constant is exactly $|\mathsf{b}|$, so that the dimensionless product $L_{\mathrm{br}}\ \|\mathbb{W}\|$ that governs the general theory specializes here to $|\mathsf{b}|\ \|\mathbb{W}\|$—the same $\mathsf{b}\mathbb{W}$ whose spectrum we have just learned to watch.

20.2 Best response and the equilibrium operator

The linear–quadratic benchmark closed by warning us not to let the cleanest results of the theory rest on a piece of luck. There the best response came out *affine*—the maximizer eq. (20.7) was literally $\alpha(x) + \mathsf{b}z$—and that affineness was exactly what collapsed the equilibrium condition into the single linear operator equation eq. (20.8), which we could then invert and diagonalize. But affineness is special to the quadratic own-cost; a general concave payoff has a best response that bends, and there is no operator equation to invert. If existence and uniqueness are to follow from the *structure* of the problem—from concavity and from the strength of the coupling—rather than from the LQ accident, we must do two things. First, package the self-consistency loop “aggregate, then best-respond” of definition 20.2 into a single map on profiles, so that a GNE is literally a fixed point of one operator. Second, identify the smallest set of properties of that operator on which the fixed-point theorems actually turn, and verify those properties from hypotheses on the payoff that are mild enough to cover every game we care about. That is the entire business of this section.

The payoff of doing this carefully is a clean separation of labor that runs through the rest of the chapter. We will find that the loop factors into two stages—a *linear* aggregation through $\mathbb{W}$, which we already understand completely, followed by a *pointwise, nonlinear* best response—and that everything the later theorems ask of the loop reduces to exactly two facts about the second stage: that it is Lipschitz in the aggregate, with a constant we can name, and that it is measurable in the type. The first fact will power the contraction that gives uniqueness; the second will guarantee that the loop is a genuine self-map of the profile space, so that there is something for a fixed-point theorem to act on. We begin by naming the loop.

Definition 20.4 (Best-response map)**.** For a static graphon game with strictly concave payoffs, the *type-x best response* to an aggregate value $z \in \mathbb{R}$ is the unique maximizer

$$\mathrm{br}_x(z) \;=\; \arg\max_{a\in S}\ \pi_x(a,z) \;\in\; S. \tag{20.9}$$

The *best-response map* $\mathsf{S} : \mathcal{S} \to \mathcal{S}$ acts on profiles by composing the aggregate with the pointwise best response,

$$\mathsf{S}(s)(x) \;=\; \mathrm{br}_x\big((\mathbb{W}s)(x)\big). \tag{20.10}$$

A profile s^* is a GNE if and only if it is a fixed point, $\mathsf{S}(s^*) = s^*$.

The decomposition implicit in eq. (20.10) is worth making explicit, because it is the organizing fact of the whole chapter. Read the formula right to left. A profile s is first sent to its aggregate field $\mathbb{W}s$—the linear, smoothing, completely understood step built in section 20.1—and only then is each type's scalar best response br_x applied to the value of that field at x. The second step is a *pointwise* operation: it acts on the field $x \mapsto (\mathbb{W}s)(x)$ one type at a time, with no coupling between types beyond whatever $\mathbb{W}$ has already supplied. An operation of this form—apply a x-dependent scalar function to the value of a function at each point—is a *Nemytskii* (or superposition) operator. Writing $N(v)(x) = \mathrm{br}_x(v(x))$ for it, the best-response map is the composition

$$\mathsf{S} \;=\; N \circ \mathbb{W}, \tag{20.11}$$

network amplification followed by individual response. This factorization is precisely why the network and the behavior can be analyzed separately, and it dictates the rest of the section: every property of S that the existence and uniqueness theorems require will be assembled from a property of the linear factor $\mathbb{W}$—which we already own, it is compact with norm $\|\mathbb{W}\| = \lambda_1$—and a property of the nonlinear factor N, which is in turn inherited pointwise from the scalar best responses br_x. The remaining burden of the section is therefore to establish two properties of the scalar map $z \mapsto \mathrm{br}_x(z)$: that it is Lipschitz in z, uniformly in x, with a constant we can write down; and that it is measurable enough in x that N carries admissible profiles to admissible profiles. Both hold automatically for the LQ payoff, where br_x is affine-then-clipped and nothing is in doubt; we now isolate the hypotheses on a *general* payoff that secure them.

Assumption 20.5 (Smooth strongly concave payoffs)**.** There are constants $0 < \theta \le \Theta < \infty$ and $\ell \ge 0$ such that for every type x the payoff $\pi_x(\cdot, z)$ is twice continuously differentiable on the interior of S with

$$-\Theta \;\le\; \partial_{aa}\pi_x(a,z) \;\le\; -\theta, \qquad \big|\partial_{az}\pi_x(a,z)\big| \;\le\; \ell,$$

uniformly in (a,z) and in x.

Each of the three constants controls a different feature of the best response, and it pays to say in words what each one buys and what would break without it, because the proof of lemma 20.6 below uses all three in distinct roles.

The lower bound $\theta > 0$ on own-action concavity is the load-bearing hypothesis. It says the payoff is *strongly* concave in the own action—curving down at a rate at least θ, uniformly across types and aggregates—and it does two things at once. First, strong concavity makes the maximizer in eq. (20.9) unique, so that br_x is a genuine single-valued function rather than a set-valued correspondence; this is what lets us speak of *the* best response at all. Second, and quantitatively, θ will sit in the *denominator* of the Lipschitz constant $L_{\mathrm{br}} = \ell/\theta$: it measures how firmly the own cost *anchors* the optimal action against being dragged around by the neighbors. A sharply curved payoff (large θ) pins the optimum down, so a change in the aggregate barely moves it; a nearly flat payoff (small

θ) lets the optimum slide freely. If θ were allowed to degenerate to zero—a merely concave but not *strongly* concave family—the maximizer could jump discontinuously as the aggregate varies, and the Lipschitz constant would blow up, destroying at one stroke the single-valuedness and the contraction on which the whole theory rests. This is also why the bound must be uniform in x: a single type with vanishing curvature would already wreck the uniform Lipschitz estimate that lemma 20.6 delivers.

The cross-sensitivity bound $\ell \geq 0$ on $|\partial_{az}\pi_x|$ measures the opposite effect—how strongly the marginal value of one's own action *responds* to the neighbors' aggregate z. It is the individual-level strength of the strategic coupling: $\partial_{az}\pi_x$ is exactly the rate at which raising the neighbor aggregate shifts the location of one's own optimum, so ℓ will sit in the *numerator* of L_{br}. If $\ell = 0$ the own optimum does not depend on the neighbors at all and the best response is constant in z—responsiveness zero, no game; the larger ℓ, the more violently the optimum tracks the aggregate. Were ℓ allowed to be unbounded, an arbitrarily small change in the aggregate could swing the optimal action arbitrarily far, the best response would fail to be Lipschitz in z, and there would be no finite constant to feed into the contraction—indeed the continuity of S itself would be in jeopardy.

The upper bound $\Theta < \infty$ on the curvature plays a quieter, supporting role, and it is important to be clear that it does *not* enter the rate constant L_{br} at all. Its job is regularity. Bounding $|\partial_{aa}\pi_x| \leq \Theta$ guarantees that π_x is genuinely C^2 with a *bounded* second derivative, so that the marginal payoff $a \mapsto \partial_a\pi_x(a,z)$ is itself globally Lipschitz in a (with constant Θ), not merely continuous. That finiteness is what licenses the clean mean-value step in the proof below: the intermediate curvature $\partial_{aa}\pi_x(\xi, z_1)$ that the mean value theorem produces is a *finite* number trapped in $[-\Theta, -\theta]$, never a runaway slope. Looking ahead, the same upper bound is what keeps the data of the Riccati and Hamiltonian systems of the dynamic chapters (chapters 22 and 24) uniformly regular when this payoff is differentiated through time. Without it the best response would still be Lipschitz with constant ℓ/θ—Θ buys no responsiveness—but the marginal could steepen without limit, and the smoothness we will silently rely on later would be gone.

For the linear–quadratic payoff eq. (20.6) all three constants are immediate: $\partial_{aa}\pi \equiv -1$ pins both curvature bounds to $\theta = \Theta = 1$, and $\partial_{az}\pi \equiv \mathsf{b}$ gives $\ell = |\mathsf{b}|$. The general rate $L_{\mathrm{br}} = \ell/\theta$ then specializes to exactly $|\mathsf{b}|$, recovering the interaction coefficient we already learned to watch in section 20.1.3—a first sanity check that the abstract constant is the right one. We now prove that it is, in fact, the Lipschitz constant for any payoff satisfying assumption 20.5.

Lemma 20.6 (Best response is Lipschitz with the right constant)**.** *Under assumption 20.5, each scalar best response* $\mathrm{br}_x : \mathbb{R} \to S$ *is well defined, single valued, and Lipschitz with constant*

$$L_{\mathrm{br}} := \frac{\ell}{\theta}, \tag{20.12}$$

uniformly in x*. For the LQ payoff* $L_{\mathrm{br}} = |\mathsf{b}|$.

Proof. Single-valuedness is the easy half. Because $\partial_{aa}\pi_x \leq -\theta < 0$, the map $a \mapsto \pi_x(a,z)$ is strongly concave on the closed convex set S, and a strongly concave function has at most one maximizer on a convex set; a maximizer exists because strong concavity forces $\pi_x(\cdot, z) \to -\infty$ at the ends of S

when S is unbounded, and a concave function attains its supremum on a closed interval otherwise. So $\mathrm{br}_x(z)$ is well defined. The content of the lemma is the quantitative bound, and we extract it by comparing the first-order optimality conditions *across two aggregates*.

Fix $z_1 < z_2$ and abbreviate $a_k = \mathrm{br}_x(z_k)$ for $k = 1, 2$. Each a_k maximizes $\pi_x(\cdot, z_k)$ over S, and the first-order condition for a maximizer on a closed convex set is the *variational inequality*: moving from a_k toward any other admissible point a cannot raise the payoff to first order,

$$\partial_a \pi_x(a_k, z_k)\,(a - a_k) \;\le\; 0 \qquad \text{for all } a \in S, \quad k = 1, 2. \tag{20.13}$$

(If a_k is interior to S the marginal vanishes and eq. (20.13) holds with equality; if a_k sits at an endpoint of S the marginal points outward and the inequality is strict—this is exactly the mechanism by which the estimate survives clipping to a bounded action set.) The trick is to test each instance against the *other* optimizer. Put $a = a_2$ in the $k = 1$ instance and $a = a_1$ in the $k = 2$ instance:

$$\partial_a \pi_x(a_1, z_1)\,(a_2 - a_1) \;\le\; 0, \qquad \partial_a \pi_x(a_2, z_2)\,(a_1 - a_2) \;\le\; 0.$$

The second carries $(a_1 - a_2) = -(a_2 - a_1)$, so adding the two and factoring out the common difference yields a single inequality,

$$\big(\partial_a \pi_x(a_1, z_1) - \partial_a \pi_x(a_2, z_2)\big)(a_2 - a_1) \;\le\; 0. \tag{20.14}$$

This is the discrete monotonicity of the optimality map; the whole task now is to read the curvature and the cross-sensitivity out of the marginal difference in eq. (20.14), and for that we travel from (a_1, z_1) to (a_2, z_2) *one coordinate at a time*. Add and subtract the mixed value $\partial_a \pi_x(a_2, z_1)$:

$$\partial_a \pi_x(a_1, z_1) - \partial_a \pi_x(a_2, z_2) = \underbrace{\big[\partial_a \pi_x(a_1, z_1) - \partial_a \pi_x(a_2, z_1)\big]}_{a \text{ varies, } z = z_1 \text{ fixed}} + \underbrace{\big[\partial_a \pi_x(a_2, z_1) - \partial_a \pi_x(a_2, z_2)\big]}_{z \text{ varies, } a = a_2 \text{ fixed}}.$$

Each bracket is the increment of a C^1 function in a single variable, so the mean value theorem applies to each on its own. In the first bracket only a moves, and the governing derivative is the curvature: there is an intermediate ξ between a_1 and a_2 with

$$\partial_a \pi_x(a_1, z_1) - \partial_a \pi_x(a_2, z_1) = \partial_{aa} \pi_x(\xi, z_1)\,(a_1 - a_2).$$

In the second bracket only z moves, and the governing derivative is the cross-partial: there is an intermediate ζ between z_1 and z_2 with

$$\partial_a \pi_x(a_2, z_1) - \partial_a \pi_x(a_2, z_2) = \partial_{az} \pi_x(a_2, \zeta)\,(z_1 - z_2).$$

(It is precisely the upper bound Θ of assumption 20.5 that makes these intermediate derivatives finite, and the same assumption that bounds them.) Substituting both into eq. (20.14),

$$\partial_{aa} \pi_x(\xi, z_1)\,(a_1 - a_2)(a_2 - a_1) + \partial_{az} \pi_x(a_2, \zeta)\,(z_1 - z_2)(a_2 - a_1) \;\le\; 0.$$

Now read the two terms. In the first, $(a_1 - a_2)(a_2 - a_1) = -(a_2 - a_1)^2$ and $\partial_{aa} \pi_x(\xi, z_1) \le -\theta$, so that first term is at least $\theta\,(a_2 - a_1)^2$; moving it to the right-hand side,

$$\theta\,(a_2 - a_1)^2 \;\le\; \partial_{az} \pi_x(a_2, \zeta)\,(z_2 - z_1)(a_2 - a_1) \;\le\; \ell\,|z_2 - z_1|\,|a_2 - a_1|,$$

the last step using $|\partial_{az}\pi_x| \le \ell$ and replacing the signed product by its absolute value. If $a_2 = a_1$ there is nothing to prove; otherwise divide both sides by $|a_2 - a_1| > 0$ to obtain

$$|a_2 - a_1| \;\le\; \frac{\ell}{\theta}\,|z_2 - z_1|,$$

which is exactly eq. (20.12). The constant earns its name in this final division: the cross-sensitivity ℓ in the numerator is the force tugging the optimum to follow the aggregate, while the curvature θ in the denominator is the restoring stiffness resisting that tug, so $L_{\mathrm{br}} = \ell/\theta$ is the ratio "pull over stiffness" that any constrained optimum must obey. It is the right constant because it is the only combination of the two local payoff data with the units of a sensitivity $\mathrm{d}a/\,\mathrm{d}z$, and the LQ value $L_{\mathrm{br}} = |\mathsf{b}|$ is its simplest instance. $\square$

We have the Lipschitz half of the two facts the section promised. The other half is measurability, and although it can sound like a technicality it is genuinely load-bearing. The best-response map was *defined* as a self-map of the admissible profiles, $\mathsf{S} : \mathcal{S} \to \mathcal{S}$ in definition 20.4; for that definition even to make sense the output $\mathsf{S}(s) = \mathrm{br}.\big((\mathbb{W}s)(\cdot)\big)$ must itself be a *measurable* function of x—otherwise it is not an element of $L^2[0,1]$ at all, and there is no profile space for any fixed-point theorem to act on. The danger is real precisely because we have left the LQ world: for a general payoff $\mathrm{br}_x(z)$ is defined *implicitly*, as the location of a maximum, with no formula exhibiting its dependence on x, so we cannot simply inspect it and pronounce it measurable. What we *can* do is exploit the one structural feature assumption 20.5 hands us—that the marginal payoff is strictly decreasing in a—to trade the implicit "argmax in x" for the explicit "the marginal is still positive," a condition manifestly measurable in x through the joint measurability of π. Carrying out that trade is the content of the next lemma.

Lemma 20.7 (The best response is a Carathéodory map)**.** *Under the joint measurability of* $(x, a, z) \mapsto \pi_x(a, z)$ *(definition 20.1) and assumption 20.5, the scalar best response* $(x, z) \mapsto \mathrm{br}_x(z)$ *is a* Carathéodory function*: measurable in x for each fixed z, and continuous in z for each fixed x. Consequently, for every measurable $v : [0,1] \to \mathbb{R}$ the superposition $x \mapsto \mathrm{br}_x(v(x))$ is measurable, and the Nemytskii operator $N : v \mapsto \big(x \mapsto \mathrm{br}_x(v(x))\big)$ sends measurable functions to measurable functions.*

Proof. There are three claims to discharge: continuity of $z \mapsto \mathrm{br}_x(z)$, measurability of $x \mapsto \mathrm{br}_x(z)$, and measurability of the superposition. The first is already in hand, and more: continuity in z is immediate from the Lipschitz bound eq. (20.12).

For measurability in x we show that every superlevel set of the maximizer is measurable. Fix the aggregate z and a threshold r in the interior of S. Under assumption 20.5 the marginal payoff $a \mapsto \partial_a \pi_x(a, z)$ is *strictly decreasing*, its derivative being $\partial_{aa}\pi_x \le -\theta < 0$, so the unique maximizer $\mathrm{br}_x(z)$ is located by a single sign change of the marginal: the marginal is positive to the left of the optimum and negative to its right. The optimum therefore lies strictly above r exactly when the marginal has not yet changed sign by $a = r$, i.e. is still positive there:

$$\{x :\ \mathrm{br}_x(z) > r\} \;=\; \{x :\ \partial_a \pi_x(r, z) > 0\} \tag{20.15}$$

(with the obvious modifications $\{x : \mathrm{br}_x(z) > r\} = [0,1]$ when $r < \inf S$, the threshold falling below every admissible action, and $= \emptyset$ when $r \geq \sup S$). The virtue of eq. (20.15) is that it has eliminated the implicit argmax on the left in favor of a condition on the right that involves only the *explicit* datum $\partial_a \pi_x(r,z)$, evaluated at the fixed point $a = r$. And that condition is measurable in x: for each fixed $h > 0$ the difference quotient $h^{-1}\big(\pi_x(r+h,z) - \pi_x(r,z)\big)$ is a measurable function of x by the joint measurability of $(x,a,z) \mapsto \pi_x$ assumed in definition 20.1, and $\partial_a \pi_x(r,z)$ is the pointwise limit of these as $h \downarrow 0$, hence measurable; its positivity set is therefore measurable. Since this holds for every threshold r, the function $x \mapsto \mathrm{br}_x(z)$ has measurable superlevel sets and is measurable. Combined with the continuity in z, this is exactly the Carathéodory property.

The Carathéodory property is what makes superposition safe, and we now cash it in. Let $v : [0,1] \to \mathbb{R}$ be measurable; we must show $x \mapsto \mathrm{br}_x(v(x))$ is measurable. Approximate v by simple functions $v_n = \sum_k c_{n,k}\, \mathbf{1}_{E_{n,k}}$ with $v_n \to v$ pointwise, where the $E_{n,k}$ are the disjoint measurable level sets on which v_n is constant (the standard simple-function construction). On each level set $E_{n,k}$ the argument fed to br_x is the *constant* $c_{n,k}$, so there the composed map is simply $x \mapsto \mathrm{br}_x(c_{n,k})$, which is measurable by the previous paragraph; the whole map $x \mapsto \mathrm{br}_x\big(v_n(x)\big)$ is the countable patching of these measurable pieces over the disjoint measurable sets $E_{n,k}$, hence measurable. Finally, by the continuity of $z \mapsto \mathrm{br}_x(z)$ established at the outset, $\mathrm{br}_x\big(v_n(x)\big) \to \mathrm{br}_x\big(v(x)\big)$ pointwise as $n \to \infty$, and a pointwise limit of measurable functions is measurable. Thus the superposition is measurable and N sends measurable functions to measurable functions, as claimed. $\square$

Remark 20.8 (The LQ case needs no abstract selection)**.** For the linear–quadratic benchmark example 20.3 the issue does not even arise: the best response is the explicit affine-then-clip map $\mathrm{br}_x(z) = \Pi_S\big(\alpha(x) + \mathsf{b}z\big)$, where Π_S is the metric projection onto the interval S; this is manifestly Carathéodory because α is measurable and Π_S is 1-Lipschitz. Lemma 20.7 is needed only to license the superposition operator for a *general* payoff, where the maximizer is not given in closed form; it replaces an appeal to an external measurable-maximum theorem with the elementary monotone-marginal argument available under assumption 20.5.

Step back and read what the two lemmas have jointly produced, because it is the hinge on which the next two sections turn. Lemma 20.6 says that the strategic responsiveness of an individual, measured by $L_{\mathrm{br}} = \ell/\theta$, is a fixed property of the *payoff* alone: it knows nothing about the graph, depending only on how sharply the own cost curves and how strongly the neighbor aggregate tugs at the optimum. The network enters the best-response map $\mathsf{S} = N \circ \mathbb{W}$ at exactly one place, the linear factor $\mathbb{W}$, and its entire strength is summarized by the single number $\|\mathbb{W}\| = \lambda_1$. Since S is the composition of a map of Lipschitz constant $\|\mathbb{W}\|$—the bounded linear operator $\mathbb{W}$—with a map of Lipschitz constant L_{br}—the Nemytskii operator N, whose pointwise ingredient is Lipschitz by lemma 20.6 and which inherits the same constant on L^2, since

$$\|N(v) - N(v')\|_{L^2}^2 = \int_0^1 \big|\mathrm{br}_x(v(x)) - \mathrm{br}_x(v'(x))\big|^2\, \mathrm{d}x \;\leq\; L_{\mathrm{br}}^2\, \|v - v'\|_{L^2}^2$$

by integrating the scalar bound—the composite is Lipschitz with constant the *product*,

$$\mathrm{Lip}(\mathsf{S}) \;\leq\; L_{\mathrm{br}}\, \|\mathbb{W}\|\,. \tag{20.16}$$

This single dimensionless number—behavioral gain times spectral radius—is the coupling strength of the entire theory, and the two sections that follow are precisely the two things one can do with it. When it is below one the composite contracts, and the fixed point is forced to be unique; that is section 20.4, where the product is handed over as the order parameter whose crossing of unity is the chapter's central event.

Existence, by contrast, will not even need the product to be small, and it is worth recording now exactly which facts of this section it consumes, so that the argument of section 20.3 reads as inevitable rather than clever. It needs two, both established above. First, that N is Lipschitz on $L^2[0,1]$ with constant L_{br} (the displayed integration of lemma 20.6) and is a genuine self-map of the admissible profiles (the measurability of lemma 20.7). Second, that $\mathbb{W}$ is compact, which is the standing import **(A-Reg-W)**. These combine through one elementary fact that the existence proof will use without comment, so we make it explicit here: *a Lipschitz map sends precompact sets to precompact sets.* The reason is that precompactness in a metric space is total boundedness—a set is precompact exactly when, for every $\varepsilon > 0$, it admits a finite cover by balls of radius ε—and a map N with $\|N(v) - N(v')\| \le L_{\mathrm{br}} \|v - v'\|$ turns a cover of a set A by ε-balls into a cover of the image $N(A)$ by $L_{\mathrm{br}}\varepsilon$-balls, simply by taking the images of the original centers: a point within ε of a center v_0 lands within $L_{\mathrm{br}}\varepsilon$ of $N(v_0)$. A finite ε-cover of A thus yields a finite $L_{\mathrm{br}}\varepsilon$-cover of $N(A)$, so total boundedness, and with it precompactness, is preserved. Apply this with $A = \mathbb{W}(K)$ for a bounded admissible set K: compactness of $\mathbb{W}$ sends the bounded K to a *precompact* set $\mathbb{W}(K)$, and the Lipschitz N then carries that precompact set to a precompact image $N(\mathbb{W}(K)) = \mathsf{S}(K)$. A continuous self-map with precompact image is exactly what Schauder's theorem demands—with no smallness of $L_{\mathrm{br}} \|\mathbb{W}\|$ asked anywhere—and that compactness-only existence is what we turn to next.

20.3 Existence of equilibrium

The previous section did all the work; this section spends it. Section 20.2 packaged the self-consistency loop of a graphon Nash equilibrium into a single map $\mathsf{S} = N \circ \mathbb{W}$ on the profile space $L^2[0,1]$, whose fixed points are exactly the equilibria (definition 20.4), and it handed us two facts about that map and nothing else. The first is that the nonlinear factor N, the Nemytskii operator built from the pointwise best responses, is Lipschitz on $L^2[0,1]$ with constant L_{br} (lemma 20.6) and a genuine self-map of the admissible profiles (lemma 20.7). The second is the standing import **(A-Reg-W)**: the linear factor $\mathbb{W}$ is compact. We closed that section by isolating the one elementary consequence on which everything here will turn—a Lipschitz map sends precompact sets to precompact sets—and by tracing the precompactness chain $\mathbb{W}$ compact $\Rightarrow$ $\mathbb{W}(K)$ precompact $\Rightarrow$ $N(\mathbb{W}(K))$ precompact all the way to the conclusion that S has precompact image. That is precisely the hypothesis a topological fixed-point theorem needs. So the existence proof is now a matter of assembling pieces we already own, and the assembly carries a message worth stating before we begin: *nowhere in it will the coupling be asked to be small.* Schauder's theorem cares that the image be precompact, not that the map contract, and precompactness was bought from the compactness of $\mathbb{W}$ alone—a structural property of the kernel that holds however large $\|\mathbb{W}\|$ is. The whole threshold drama of $L_{\mathrm{br}} \|\mathbb{W}\|$ crossing one, which the next section will stage, is therefore absent from existence by design. We prove existence twice, in two cases that differ only in which fixed-point engine they invoke, precisely so that the reader can see that the first case asks for no smallness whatever, and the second asks for it only to keep the action set's unboundedness from letting the equilibrium escape to infinity—not to secure the fixed point itself.

Theorem 20.9 (Existence of a graphon Nash equilibrium)**.** *Consider a static graphon game (definition 20.1) under **(A-Reg-W)** and assumption 20.5, and suppose either*

(i) the action set $S = [\underline{s}, \bar{s}]$ is a bounded interval; or

(ii) $S = \mathbb{R}$ and the standalone optimum is bounded, $\sup_x |\mathrm{br}_x(0)| < \infty$, together with the subcriticality condition $L_{\mathrm{br}}\ \|\mathbb{W}\| < 1$.

Then the game possesses at least one graphon Nash equilibrium $s^* \in \mathcal{S}$.

Proof. Case (i): bounded actions, by Schauder. The plan is the standard one for a continuous self-map with precompact image: exhibit a nonempty, convex, *compact* set carried into itself, and quote Schauder. The only subtlety—the one place the proof does more than recite definitions—is that the natural candidate set is precompact but not closed, so we must take a closed convex hull and argue it stays compact; that is where Mazur's theorem enters. We take the four ingredients in turn.

The candidate set is the order interval

$$K = \left\{ s \in L^2[0,1] : \ \underline{s} \leq s(x) \leq \bar{s} \text{ a.e.} \right\},$$

the admissible profiles for the bounded action set $S = [\underline{s}, \bar{s}]$. It is worth checking each of the four properties Schauder will eventually need, since each one uses a different feature of K. It is *nonempty*: the constant profile $s \equiv \underline{s}$ lies in it. It is *convex*: if $\underline{s} \leq s, s' \leq \bar{s}$ pointwise then $\underline{s} \leq \lambda s + (1-\lambda)s' \leq \bar{s}$ pointwise for every $\lambda \in [0,1]$, because a pointwise two-sided bound is preserved by averaging. It is *bounded*: every $s \in K$ satisfies $\|s\|_{L^2}^2 = \int_0^1 s(x)^2 \, \mathrm{d}x \leq \max(\underline{s}^2, \bar{s}^2)$, so K sits inside a ball of radius $\max(|\underline{s}|, |\bar{s}|)$. And it is *closed* in $L^2[0,1]$: if $s_n \in K$ and $s_n \to s$ in L^2, then a subsequence converges almost everywhere, and the pointwise bounds $\underline{s} \leq s_n \leq \bar{s}$ pass to the almost-everywhere limit, so $s \in K$. Thus K is a nonempty, closed, convex, bounded subset of the Hilbert space $L^2[0,1]$—but boundedness and closedness are not compactness in infinite dimensions, which is exactly why a fixed-point theorem cannot be applied to K directly and why the smoothing of $\mathbb{W}$ has work to do.

Self-map. We check $\mathsf{S}(K) \subseteq K$, so that there is an invariant set to begin with. This is immediate from the construction of S: by definition 20.4 the pointwise best response br_x is the maximizer of $\pi_x(\cdot, z)$ over $a \in S = [\underline{s}, \bar{s}]$, so it *takes values in* S by definition of an argmax over S. Hence for any s the profile $\mathsf{S}(s)(x) = \mathrm{br}_x\big((\mathbb{W}s)(x)\big)$ satisfies $\underline{s} \leq \mathsf{S}(s)(x) \leq \bar{s}$ for almost every x, i.e. $\mathsf{S}(s) \in K$. Note this used the boundedness of S and nothing else—no concavity, no smallness; clipping the response to the action interval is what keeps the loop inside K.

Precompactness of the image. This is the heart of the matter, and it is the precompactness chain assembled at the close of section 20.2, run once concretely. The operator $\mathbb{W}$ is compact—it is Hilbert–Schmidt by proposition 2.24 under **(A-Reg-W)**, and a compact operator by definition carries the bounded set K to a *precompact* set $\mathbb{W}(K) \subseteq L^2[0,1]$, one whose closure is compact. This is the link in the chain that consumes the bounded set K and produces something with finite-dimensional character: the eigenvalue decay $\lambda_n \to 0$ of section 20.1 compresses the bounded K toward a compact core. The Nemytskii operator $N : v \mapsto (x \mapsto \mathrm{br}_x(v(x)))$ is Lipschitz from $L^2[0,1]$ to $L^2[0,1]$ with constant L_{br} by lemma 20.6, since integrating the pointwise scalar Lipschitz bound gives

$$\|N(v) - N(v')\|_{L^2}^2 = \int_0^1 \left|\mathrm{br}_x(v(x)) - \mathrm{br}_x(v'(x))\right|^2 \mathrm{d}x \le L_{\mathrm{br}}^2 \int_0^1 \left|v(x) - v'(x)\right|^2 \mathrm{d}x = L_{\mathrm{br}}^2 \left\|v - v'\right\|_{L^2}^2 .$$

And a Lipschitz map sends precompact sets to precompact sets—the total-boundedness argument recorded at the end of section 20.2, that a finite ε-cover of the domain set maps to a finite $L_{\mathrm{br}}\varepsilon$-cover of the image. Applying this with the precompact set $\mathbb{W}(K)$ in the role of the domain set, the image

$$\mathsf{S}(K) = N(\mathbb{W}(K))$$

is precompact in $L^2[0,1]$. The structure to retain is that compactness was supplied entirely by the *linear* factor $\mathbb{W}$; the nonlinear factor N contributes no smoothing of its own, it merely *preserves* the precompactness that $\mathbb{W}$ created, because Lipschitz maps cannot inflate a totally bounded set into an unbounded one.

Continuity. The map $\mathsf{S} = N \circ \mathbb{W}$ is continuous on $L^2[0,1]$ as a composition of two continuous maps: $\mathbb{W}$ is a bounded linear operator, hence continuous, and N is Lipschitz by the display above, hence continuous. There is nothing more to verify; the factorization $\mathsf{S} = N \circ \mathbb{W}$ does the work.

The Mazur step and the conclusion. We now have a continuous self-map S of the nonempty closed convex set K, with *precompact* image—but Schauder's theorem (theorem 2.38) is a statement about a continuous self-map of a *compact* convex set, and K itself is not compact. The remedy is to shrink K to a compact convex set that S still maps into itself. Take

$$C := \overline{\mathrm{conv}}\, \mathsf{S}(K),$$

the closed convex hull of the image. The point that makes this work—and the reason the hull does not reintroduce the noncompactness we were trying to escape—is *Mazur's theorem*: in a Banach space (here the Hilbert space $L^2[0,1]$) the closed convex hull of a compact set is again compact. We apply it to the compact set $\overline{\mathsf{S}(K)}$ (the closure of the precompact image just shown to be precompact), whose closed convex hull is C; Mazur's theorem makes C compact, and it is convex and closed by construction. Two containments now close the argument. First, $C \subseteq K$: the image $\mathsf{S}(K)$ lies in K by the self-map property, and K is itself closed and convex, so K contains the *smallest* closed convex set containing $\mathsf{S}(K)$, which is exactly C. Second, $\mathsf{S}(C) \subseteq C$: since $C \subseteq K$ we have $\mathsf{S}(C) \subseteq \mathsf{S}(K) \subseteq \overline{\mathrm{conv}}\, \mathsf{S}(K) = C$. Thus S is a continuous map sending the nonempty, convex, *compact* set C into itself, and Schauder's fixed-point theorem (theorem 2.38) supplies a point $s^* \in C \subseteq K$ with $s^* = \mathsf{S}(s^*)$. By definition 20.4 that fixed point is a graphon Nash equilibrium. No hypothesis on the size of $L_{\mathrm{br}} \|\mathbb{W}\|$ entered anywhere.

Case (ii): unbounded actions, by an a priori bound. When $S = \mathbb{R}$ the order interval K above is all of $L^2[0,1]$, which is unbounded, so the candidate set is no longer bounded and the precompactness argument has nothing finite to chew on. The standard fix is to manufacture a bounded invariant set by hand—a ball that S cannot escape—and run the case-(i) machinery inside it. This is exactly where the subcriticality hypothesis $L_{\mathrm{br}}\,\|\mathbb{W}\| < 1$ earns its place: it is what prevents the strategic feedback from pushing the equilibrium off to infinity, and it is needed *only* for this confinement, not for the fixed point itself.

We produce the bound by estimating how far one application of S can move a profile away from the origin. Fix any profile $s \in L^2[0,1]$. At each type x, the Lipschitz bound of lemma 20.6 applied with the two aggregate values $(\mathbb{W}s)(x)$ and 0 gives

$$\big|\mathrm{br}_x\big((\mathbb{W}s)(x)\big) - \mathrm{br}_x(0)\big| \;\le\; L_{\mathrm{br}}\,\big|(\mathbb{W}s)(x)\big|.$$

Hence, pointwise, $\big|\mathsf{S}(s)(x)\big| \le \big|\mathrm{br}_x(0)\big| + L_{\mathrm{br}}\big|(\mathbb{W}s)(x)\big|$, and taking L^2 norms and using the triangle inequality together with the elementary fact that $\|\,|f|\,\|_{L^2} = \|f\|_{L^2}$,

$$\|\mathsf{S}(s)\|_{L^2} \;\le\; \|\mathrm{br}_{\cdot}(0)\|_{L^2} + L_{\mathrm{br}}\,\|\mathbb{W}s\|_{L^2} \;\le\; \|\mathrm{br}_{\cdot}(0)\|_{L^2} + L_{\mathrm{br}}\,\|\mathbb{W}\|\,\|s\|_{L^2}\,,$$

where the last step is the operator-norm inequality $\|\mathbb{W}s\|_{L^2} \le \|\mathbb{W}\|\,\|s\|_{L^2}$. Abbreviate the two constants of this estimate as

$$c_0 \;:=\; \|\mathrm{br}_{\cdot}(0)\|_{L^2} < \infty \qquad \text{and} \qquad \rho \;:=\; L_{\mathrm{br}}\,\|\mathbb{W}\| < 1,$$

where c_0 is finite precisely because of the standing hypothesis $\sup_x|\mathrm{br}_x(0)| < \infty$ on the standalone optimum (it bounds the integrand $|\mathrm{br}_x(0)|^2$ by a constant), and $\rho < 1$ is the subcriticality assumption. The estimate reads $\|\mathsf{S}(s)\|_{L^2} \le c_0 + \rho\,\|s\|_{L^2}$. Now seek a radius R for which the ball $B_R = \{s : \|s\|_{L^2} \le R\}$ is invariant: if $\|s\|_{L^2} \le R$ then $\|\mathsf{S}(s)\|_{L^2} \le c_0 + \rho R$, and this is $\le R$ exactly when $c_0 \le (1-\rho)R$, i.e. when

$$R \;\ge\; \frac{c_0}{1-\rho}.$$

Here is the one place $\rho < 1$ is indispensable: only because $1-\rho > 0$ is this threshold radius finite and positive; were $\rho \ge 1$ the affine bound $c_0 + \rho R$ would meet or exceed R for every R, and no ball would be invariant—the feedback could amplify a profile without limit on each pass. Take $R = c_0/(1-\rho)$. The ball B_R is nonempty, closed, convex, and bounded, S maps it into itself by the estimate just made, and on this bounded set the precompactness, continuity, and Mazur arguments of case (i) apply verbatim—$\mathbb{W}(B_R)$ is precompact because B_R is bounded and $\mathbb{W}$ compact, N preserves precompactness, and $\overline{\mathrm{conv}}\,\mathsf{S}(B_R)$ is a compact convex invariant subset of B_R. Schauder's theorem (theorem 2.38) again furnishes a fixed point $s^* = \mathsf{S}(s^*)$, a graphon Nash equilibrium. $\square$

Remark 20.10 (Two machines, one hypothesis in case (ii))**.** It is worth being honest that case (ii) was, in a sense, overkill, and saying precisely why illuminates the division of labor between the two fixed-point engines. Under the hypotheses of case (ii) the subcriticality $L_{\mathrm{br}}\,\|\mathbb{W}\| < 1$ is exactly the condition that makes S a *contraction*: by eq. (20.16) its Lipschitz constant is at most $L_{\mathrm{br}}\,\|\mathbb{W}\| = \rho < 1$. A contraction of a complete metric space needs no compactness at all—the Banach fixed-point theorem (theorem 2.36) already delivers a fixed point by iteration, and delivers it *unique*, which is more than Schauder ever gives. So for the unbounded case the honest minimal argument is Banach, not Schauder, and indeed that is exactly the route theorem 20.14 below will take to prove existence *and* uniqueness in one stroke. Why then run Schauder here? Purely for a uniform statement: writing the existence theorem so that both cases are concluded by the same theorem lets us assert "equilibria exist" under either hypothesis without bifurcating the proof at the level of the statement. The point of the pairing is to make visible that the two cases lean on the two engines for opposite reasons. The *genuinely* Schauder-only situation—the one that no contraction argument can reach—is the bounded-action case (i), where $S = [\underline{s}, \bar{s}]$ is compact but *no* smallness of the coupling is assumed, so S may fail to contract (indeed $L_{\mathrm{br}} \|\mathbb{W}\|$ may be enormous) and yet an equilibrium still exists because the bounded action set keeps the loop confined and compactness of $\mathbb{W}$ supplies the precompact image. That is the content worth carrying forward: *existence with arbitrarily strong coupling* is a real phenomenon, and it is bought by compactness, not by smallness.

Remark 20.11 (Set-valued best responses)**.** The whole argument leaned on strict (indeed strong) concavity at exactly one point: it made the maximizer in eq. (20.9) *unique*, so that br_x was a single-valued function and S a genuine map to which Schauder applies. Drop strong concavity—keep payoffs merely concave in the own action—and the maximizing set $\mathrm{br}_x(z) = \arg\max_{a \in S} \pi_x(a, z)$ can contain more than one point; it becomes a *correspondence*, convex- and compact-valued (the argmax of a concave function over an interval is a closed subinterval) and upper hemicontinuous in z (small perturbations of z cannot make the maximizing set jump outward). Existence survives this loss intact in case (i), at the cost of upgrading the fixed-point engine from Schauder to its set-valued generalization. One applies the Kakutani–Fan–Glicksberg theorem (theorem 2.40) to the set-valued map $s \mapsto N(\mathbb{W}s)$ on K: its values are convex because each $\mathrm{br}_x(z)$ is a convex set and convexity is preserved under the superposition and the linear aggregation, and it is upper hemicontinuous with precompact range because $\mathbb{W}$ is compact—the same precompactness chain as before, now feeding a set-valued rather than single-valued fixed-point theorem. A fixed point is then a profile s^* with $s^*(x) \in \mathrm{br}_x\big((\mathbb{W}s^*)(x)\big)$ a.e., a graphon Nash equilibrium in which ties in the best response are allowed to be broken type by type. This is the static, network-filtered analogue of the relaxed (mixed) equilibria of chapter 10: there, convexity was restored by randomizing over pure controls and Kakutani was applied in the space of measures; here, with no time and no dynamics, the convexification is the elementary one of allowing the best response to be set-valued, and the same theorem closes the loop.

The verdict of this section is a single clean separation, and it is the claim we hand to uniqueness. Existence of a graphon Nash equilibrium is *cheap and structural*: it costs only the compactness of $\mathbb{W}$—a regularity property of the kernel that holds however large the operator norm is—and the concavity of payoffs, and in the bounded-action case it asks for no constraint whatever on the strength of the interaction. The dimensionless product $L_{\mathrm{br}} \|\mathbb{W}\|$, the order parameter handed over by section 20.2, made not a single appearance in case (i) and entered case (ii) only to fence a wandering equilibrium back into a finite ball, never to secure its existence. The threshold phenomenon—the crossing of $L_{\mathrm{br}} \|\mathbb{W}\|$ through one, with everything it brings—therefore lives *entirely* in the question of *uniqueness*, which is what we turn to next: with existence settled for free, the operator norm reappears not as a cost of having an equilibrium but as the exact dividing line between having exactly one and having many.

20.4 Uniqueness and the operator-norm threshold

The verdict section 20.3 handed us was deliberately one-sided, and that one-sidedness is precisely what sets up the present section. An equilibrium always exists, bought by compactness of $\mathbb{W}$ and concavity of payoffs alone, with the dimensionless order parameter $L_{\mathrm{br}} \|\mathbb{W}\|$ never once deciding whether the self-consistency loop has a fixed point. That cleanly quarantines the threshold phenomenon. If the crossing of $L_{\mathrm{br}} \|\mathbb{W}\|$ through one is going to matter anywhere in this chapter—and the throughline insists it is the order parameter of the whole theory—it can only matter here, in the question of how *many* fixed points there are. Existence is settled and free; uniqueness is the lone remaining place where the operator norm can earn its keep.

And uniqueness is genuinely in doubt. Nothing in the equilibrium condition of definition 20.2 forces the loop "aggregate, then best-respond" to have only one solution. When the coupling is strong the loop can fold back on itself: a profile of high effort generates, through $\mathbb{W}$, a large aggregate, which—if best responses are sufficiently responsive to that aggregate—calls forth still higher effort, so that a self-reinforcing high-effort profile and a self-reinforcing low-effort profile may both be internally consistent at once. This is not a pathology peculiar to networks. It is exactly the mechanism by which a strongly coupled mean field game acquires multiple equilibria—the failure of uniqueness analyzed in section 11.4—now filtered through the kernel rather than through an anonymous population mean. The forward reference made there, that strong coupling breeds multiplicity, comes due in this section, and the graphon is what makes it quantitative: "strong" will turn out to mean precisely $L_{\mathrm{br}} \|\mathbb{W}\| \geq 1$, the order parameter at or past its critical value.

Uniqueness must therefore be *earned*, and there are two essentially different ways to earn it, which is why two theorems are coming rather than one. They differ along the very axis that organizes the chapter: the first route controls the *size* of the coupling, the second its *sign structure*. Stating both, and seeing that they locate the same boundary from opposite sides, is the point of proving uniqueness twice.

The first route is *contraction*, and it is a smallness condition (assumption **(A-Coupling)**). Recall from section 20.2 that the best-response map factors as $\mathsf{S} = N \circ \mathbb{W}$: a Lipschitz pointwise response with constant L_{br}, applied after the linear aggregation $\mathbb{W}$, whose own Lipschitz constant on L^2 is exactly the operator norm $\|\mathbb{W}\|$. The composite then contracts distances by at most the product $L_{\mathrm{br}} \|\mathbb{W}\|$, and once that product drops below one the Banach fixed-point theorem returns exactly one equilibrium, reached by geometric iteration. The reason this is the route the throughline cares about is that the contraction factor *is* the order parameter itself—no cruder estimate stands between the hypothesis and the conclusion. Two features make it sharp rather than merely sufficient. It is the spectral radius $\|\mathbb{W}\| = \lambda_1$, and not some larger norm of the kernel, that enters, because the worst case of the linear step is realized along the top eigenfunction and nowhere worse; and the boundary $L_{\mathrm{br}} \|\mathbb{W}\| = 1$ it identifies is genuinely *attained*, not merely approached, because the compactness of $\mathbb{W}$ makes λ_1 an honest maximum rather than an unreached supremum—the discreteness of the spectrum flagged in the chapter preamble doing exactly the work promised for it. At that exact value the underlying linear operator $I - \mathsf{b}\mathbb{W}$ of the LQ benchmark eq. (20.8) loses invertibility along the top eigenfunction, so the edge the contraction argument produces is the true edge of well-posedness and not an artifact of the proof technique. Contraction is thus the route that puts the operator norm on open display as the coupling strength whose climb to one is the chapter's central drama; we develop it in section 20.4.2.

The second route is *monotonicity*, and it trades smallness for structure. Rather than ask the coupling to be weak, it asks the operator encoding the equilibrium condition to be *monotone*—in the linear–quadratic case, that the quadratic potential whose stationary points are the equilibria be convex—and from convexity alone, by the uniqueness of a minimizer, it reads off a single equilibrium. The decisive difference from contraction is that a convexity certificate carries no smallness inside it: it can certify uniqueness at *any* coupling strength, arbitrarily far above the contraction threshold, provided the sign structure of the interaction cooperates. This is the exact static shadow of the Lasry–Lions monotonicity condition of definition 11.1, which in the dynamic theory secures a unique MFG equilibrium with no smallness assumption whatever; here, with time and noise stripped away and the potential merely quadratic, that condition collapses to the elementary positive-definiteness of a Hessian operator, but the logic is identical—monotone coupling defeats multiplicity unconditionally in its favorable sign. Monotonicity is therefore the structural, variational face of the same threshold the contraction route saw spectrally; we develop it in section 20.4.3.

That the two routes view one boundary from two sides is exactly what makes the comparison instructive, and it reproduces, in static and computable form, the attractive-versus-repulsive asymmetry of section 11.4. For the strategic-complements sign—the network amplifying behavior—the two thresholds will turn out to coincide on the top eigenvalue λ_1, so the contraction and convexity boundaries agree and the uniqueness edge is genuinely sharp. For the strategic-substitutes sign the monotone route reaches strictly further than contraction, governed by the weaker spectral edge rather than by λ_1, mirroring exactly the way the "crowd-averse" coupling of section 11.4 stays tame over a wide range while the "crowd-seeking" coupling is the one pinned to a sharp critical value. The section thus sees the same threshold twice—spectrally and variationally—and that double vision is the whole reason to carry both arguments.

Both routes, however, are stated in terms of a single number we have so far only written down rather than examined: the operator norm $\|\mathbb{W}\|$, which the contraction route makes the contraction factor and the monotone route makes the binding eigenvalue. Before either theorem can be stated—let alone shown to be sharp—we must therefore pin down exactly what that number *is* for the kernel at hand, a nonnegative symmetric graphon: whether it sits at the top of the spectrum, whether that top is attained, and what its eigenfunction looks like. That identification is the first order of business, and it is where we turn now.

20.4.1 The operator norm of a graphon

Both routes to uniqueness pivot on one number, and so far we have only written its symbol down. The contraction argument of section 20.4.2 will make $\|\mathbb{W}\|$ the very factor by which the equilibrium loop shrinks distances; the monotone argument of section 20.4.3 will make it the eigenvalue that first turns the equilibrium potential non-convex. Neither threshold can even be stated as a sharp, attained edge—let alone shown to be sharp—until we know what $\|\mathbb{W}\|$ *is* for the object actually in front of us, which is not a general bounded operator but a graphon: a kernel that is *symmetric* (so $\mathbb{W}$ is self-adjoint) and *nonnegative* (because W takes values in $[0, 1]$, an edge density). Those two properties are exactly what make the answer clean, and the answer is the seed of everything spectral in the chapter. The operator norm is the *top eigenvalue* λ_1; it is strictly positive; and its eigenfunction may be taken nonnegative—the profile we will christen *eigenvector centrality* and watch become the order parameter's direction in section 20.5 and the condensate at criticality in section 20.5.2. We state this as a lemma, prove it from the spectral theorem with the nonnegativity of the kernel doing the one decisive piece of work, and then watch it come out by hand on the two kernels the chapter computes with.

Lemma 20.12 (Operator norm equals the top eigenvalue)**.** *Under **(A-Reg-W)**, $\mathbb{W}$ is self-adjoint and compact on $L^2[0,1]$, so by the spectral theorem (theorem 2.26) it has a real, discrete spectrum of eigenvalues (λ_n) (listed with multiplicity), accumulating only at 0, together with an orthonormal eigenbasis (ϕ_n) of $(\ker \mathbb{W})^\perp$. Because the nonzero eigenvalues form a sequence converging to 0, both spectral edges are attained: we order the nonnegative and nonpositive eigenvalues separately,*

$$\lambda_1 \;\geq\; \lambda_2 \;\geq\; \cdots \;\geq\; 0 \;\geq\; \cdots \;\geq\; \lambda_{\min}, \qquad \lambda_1 := \sup_n \lambda_n, \quad \lambda_{\min} := \inf_n \lambda_n,$$

both extrema being achieved by compactness (a most-positive and a most-negative eigenvalue exist whenever $\mathbb{W}$ has eigenvalues of that sign; if $\mathbb{W} \geq 0$ we set $\lambda_{\min} := 0$). Because the kernel is nonnegative, the largest eigenvalue is positive and equals the operator norm and the spectral radius:

$$\|\mathbb{W}\| \;=\; \lambda_1 \;=\; \max_n |\lambda_n| \;>\; 0, \tag{20.17}$$

and the corresponding eigenfunction ϕ_1 may be taken nonnegative.

Proof. Self-adjointness and compactness of $\mathbb{W}$ are proposition 2.24: the kernel is symmetric, so $\mathbb{W} = \mathbb{W}^*$, and it is Hilbert–Schmidt, hence compact. The spectral theorem theorem 2.26 then supplies everything in the first two displays of the statement—a real spectrum, eigenvalues listed with multiplicity accumulating only at 0, and an orthonormal eigenbasis (ϕ_n) of $(\ker \mathbb{W})^\perp$—and it is precisely the accumulation *only* at 0 that lets us speak of a largest and a smallest eigenvalue: away from any neighborhood of 0 there are finitely many eigenvalues, so the supremum λ_1 and infimum $\lambda_{\min}$ are each achieved by an honest eigenvalue rather than merely approached. This attainment is not a technicality we can take or leave; it is what will make the uniqueness boundary $L_{\mathrm{br}}\,\|\mathbb{W}\| = 1$ a sharp edge in section 20.4.2 rather than an unreached supremum.

We need two facts: that the operator norm equals the maximum modulus among eigenvalues, and that this maximum modulus is delivered by a *positive* eigenvalue. The first is the quadratic-form (Courant–Fischer) characterization corollary 2.27, which it is worth restating in the only form we use. Expanding any unit vector in the eigenbasis as $f = \sum_n c_n \phi_n$ with $\sum_n |c_n|^2 = 1$, self-adjointness gives the Rayleigh quotient as a *weighted average of the eigenvalues*,

$$\langle f, \mathbb{W} f \rangle \;=\; \sum_n \lambda_n\, |c_n|^2, \tag{20.18}$$

the weights $|c_n|^2$ summing to one. A weighted average of the λ_n is largest when all the weight sits on the largest eigenvalue and smallest when it sits on the smallest, so the extremes of eq. (20.18) over unit f are the top and bottom of the spectrum, each attained at the corresponding eigenfunction:

$$\lambda_1 = \sup_{\|f\|=1} \langle f, \mathbb{W} f \rangle, \qquad \lambda_{\min} = \inf_{\|f\|=1} \langle f, \mathbb{W} f \rangle.$$

Because $\langle f, \mathbb{W}f\rangle \in [\lambda_{\min}, \lambda_1]$ for every unit f, the operator norm, which for a self-adjoint operator is $\|\mathbb{W}\| = \sup_{\|f\|=1} |\langle f, \mathbb{W}f\rangle|$, equals $\max(\lambda_1,\ |\lambda_{\min}|) = \max_n |\lambda_n|$. So far this used only self-adjointness and compactness; nothing yet decides whether the most negative or the most positive eigenvalue is larger in modulus. That is the one place the *nonnegativity* of the kernel enters.

Fix any $f \in L^2[0,1]$ and compare the quadratic form of f with that of its modulus $|f|$. Pointwise $f(x)f(y) \le |f(x)|\,|f(y)|$, and $W \ge 0$, so multiplying by the kernel preserves the inequality and integrating gives

$$\langle f, \mathbb{W}f\rangle = \iint W(x,y)\, f(x)f(y)\; \mathrm{d}x\,\mathrm{d}y \;\le\; \iint W(x,y)\,|f(x)|\,|f(y)|\; \mathrm{d}x\,\mathrm{d}y = \langle |f|, \mathbb{W}|f|\rangle. \quad (20.19)$$

The same pointwise bound applied to $-f(x)f(y) \le |f(x)|\,|f(y)|$ gives $-\langle f, \mathbb{W}f\rangle \le \langle |f|, \mathbb{W}|f|\rangle$ as well, so in fact $|\langle f, \mathbb{W}f\rangle| \le \langle |f|, \mathbb{W}|f|\rangle$: passing to $|f|$ can only raise the quadratic form, because it removes every cancellation between oppositely-signed regions of f, and a nonnegative kernel rewards alignment. Now take f to be a unit eigenfunction for the most negative eigenvalue, so $\langle f, \mathbb{W}f\rangle = \lambda_{\min}$ and $\||f|\| = \|f\| = 1$. The chain

$$\lambda_1 \;\ge\; \langle |f|, \mathbb{W}|f|\rangle \;\ge\; |\langle f, \mathbb{W}f\rangle| \;=\; |\lambda_{\min}|$$

reads off, left to right: λ_1 is the supremum of the Rayleigh quotient over unit vectors and $|f|$ is one such vector; then eq. (20.19) in its two-sided form; then the eigenvalue equation. Hence $\lambda_1 \ge |\lambda_{\min}|$, and therefore $\|\mathbb{W}\| = \max_n |\lambda_n| = \lambda_1$, a nonnegative number: the maximum modulus is carried by a positive (or zero) eigenvalue, never by a strictly more-negative one.

It remains to rule out the degenerate case $\lambda_1 = 0$, which would force *every* eigenvalue to vanish and hence, by the spectral theorem, $\mathbb{W} = 0$. But $\mathbb{W} \ne 0$ for any genuine graphon. The cleanest certificate is the Hilbert–Schmidt norm, which for a self-adjoint operator is the ℓ^2 norm of the spectrum,

$$\sum_n \lambda_n^2 \;=\; \|\mathbb{W}\|_{\mathrm{HS}}^2 \;=\; \iint W(x,y)^2\; \mathrm{d}x\,\mathrm{d}y \;=\; \|W\|_{L^2([0,1]^2)}^2 \;>\; 0,$$

strictly positive whenever W is not almost everywhere zero, i.e. whenever the network has any edges at all. If the λ_n^2 sum to a positive number they cannot all be zero, so $\lambda_1 = \max_n |\lambda_n| > 0$. (When the diagonal is integrable one may argue even more directly: $\operatorname{tr}\mathbb{W} = \int_0^1 W(x,x)\,\mathrm{d}x = \sum_n \lambda_n \ge 0$, and if this is strictly positive then a positive eigenvalue exists outright; but the Hilbert–Schmidt certificate needs nothing about the diagonal and is the one we lean on.)

Finally, the eigenfunction can be taken nonnegative. Let f be a unit eigenfunction for λ_1, so $\langle f, \mathbb{W}f\rangle = \lambda_1 = \sup_{\|g\|=1}\langle g, \mathbb{W}g\rangle$. By eq. (20.19), $\langle |f|, \mathbb{W}|f|\rangle \ge \langle f, \mathbb{W}f\rangle = \lambda_1$, and $|f|$ is again a unit vector; but λ_1 is the supremum, so equality holds and $|f|$ also maximizes the Rayleigh quotient. A maximizer of the Rayleigh quotient is an eigenfunction for the top eigenvalue, so $|f|$ is itself a λ_1-eigenfunction. Replacing f by $\phi_1 := |f| \ge 0$ therefore loses nothing, and the top eigenfunction may be taken nonnegative. □

It is worth dwelling on what the pair (λ_1, ϕ_1) *means* as network data, because both halves are about to do heavy lifting and the lemma has just earned us the right to read them structurally. The eigenvalue $\lambda_1 = \|\mathbb{W}\|$ is the *spectral radius* of the network operator: the single largest factor by which neighbor-averaging can amplify a profile, realized along ϕ_1 and nowhere stronger. This is exactly why it, and not some cruder norm of the kernel, is the right measure of "how strongly the network couples"—it is the worst case of the linear step $s \mapsto \mathbb{W}s$, which is precisely the quantity a contraction estimate must control.

The eigenfunction $\phi_1 \geq 0$ is the *eigenvector centrality* profile of the graphon, the continuum analogue of the Perron eigenvector of an adjacency matrix. The defining relation $\mathbb{W}\phi_1 = \lambda_1\phi_1$ says it is the profile reproduced *up to scale* by one round of neighbor averaging: if the population plays ϕ_1, then the aggregate each type feels, $(\mathbb{W}\phi_1)(x) = \lambda_1\phi_1(x)$, is the very same profile rescaled by λ_1. This is a self-consistency condition on importance—a type is central exactly insofar as it is connected to other central types—and it is the same fixed-point logic that defines eigenvector centrality on finite networks, now read off a kernel. That ϕ_1 comes out *nonnegative* is what makes this reading legitimate: a centrality ranking must assign each type a magnitude of the same sign, and the lemma guarantees the top mode can be chosen so, with no type's centrality fighting another's. The full spectral dictionary for network structure is the business of chapter 18; here we extract only the two facts the rest of the chapter runs on. First, $\|\mathbb{W}\| = \lambda_1$, so the dimensionless coupling $L_{\mathrm{br}}\,\|\mathbb{W}\|$ that section 20.2 handed forward is literally $L_{\mathrm{br}}\lambda_1$, a spectral quantity. Second, $\phi_1 \geq 0$ is a bona fide centrality profile—the "soft mode" of the network—which section 20.5.2 will show the equilibrium condenses onto as the coupling is driven to the threshold. The lemma is thus the hinge between the abstract order parameter and the concrete network ranking it secretly is.

Abstract as the lemma is, the threshold quantity $\|\mathbb{W}\| = \lambda_1$ is something we can write down in one line on the two kernels the chapter carries, and doing so now—before the threshold is used—turns the inequality $L_{\mathrm{br}}\,\|\mathbb{W}\| < 1$ into a concrete arithmetical condition the reader can already evaluate. Both computations exploit the same accident the lemma's degenerate case warned of: these kernels have rank one, so $\mathbb{W}$ has a single nonzero eigenvalue, which by the lemma must be the positive number $\lambda_1 = \|\mathbb{W}\|$.

Example 20.13 (The norm in the constant and rank-one cases)**.** *(a) Constant graphon.* Let $W \equiv c$ with $c \in (0,1]$, the kernel in which every type is everyone's neighbor with equal weight. Then for any $g \in L^2[0,1]$,

$$(\mathbb{W}g)(x) = c\int_0^1 g(y)\,\mathrm{d}y = c\,\langle g, \mathbf{1}\rangle\,\mathbf{1}(x),$$

where $\mathbf{1}$ is the constant function 1 (note $\|\mathbf{1}\| = 1$ since $\int_0^1 1 = 1$). The image is *always* a multiple of $\mathbf{1}$, so $\mathbb{W}$ has rank one. Testing $\mathbf{1}$ itself, $\mathbb{W}\mathbf{1} = c\,\langle \mathbf{1}, \mathbf{1}\rangle\,\mathbf{1} = c\,\mathbf{1}$, so $\mathbf{1}$ is an eigenfunction with eigenvalue c; every zero-mean function g has $\langle g, \mathbf{1}\rangle = 0$ and hence lies in $\ker\mathbb{W}$. The single nonzero eigenvalue is therefore $\lambda_1 = c$ with normalized nonnegative eigenfunction $\phi_1 = \mathbf{1}$, and $\|\mathbb{W}\| = c$ (worked computation 2.28(a)). The lemma is borne out: the maximum modulus sits on a positive eigenvalue, $\phi_1 \geq 0$, and $\lambda_1 = c = \|W\|_{L^2([0,1]^2)}$ as the Hilbert–Schmidt check predicts for a rank-one operator. The subcriticality condition becomes simply $L_{\mathrm{br}}\,c < 1$: the constant kernel couples through the population mean alone, and its strength is the edge density c.

(b) Rank-one graphon. Let $W(x,y) = f(x)f(y)$ with $f \in L^2[0,1]$, $f \geq 0$ (and bounded into $[0,1]$ so it is a graphon). Then

$$(\mathbb{W}g)(x) = f(x)\int_0^1 f(y)g(y)\,\mathrm{d}y = \langle f, g\rangle\,f(x),$$

again rank one, the image always a multiple of f. Testing f, $\mathbb{W}f = \langle f, f\rangle f = \|f\|_{L^2}^2 f$, so f is an eigenfunction with eigenvalue $\|f\|_{L^2}^2 = \int_0^1 f^2$, while everything orthogonal to f is in the kernel. The single nonzero eigenvalue is $\lambda_1 = \int_0^1 f^2$ with normalized eigenfunction $\phi_1 = f/\|f\|_{L^2} \geq 0$, so $\|\mathbb{W}\| = \|f\|_{L^2}^2$ (worked computation 2.28(b)). Here the centrality profile $\phi_1 \propto f$ is the type's own connectivity weight $f(x)$, exactly as one would want: in a rank-one network a type is central in proportion to how strongly it attaches to the single shared factor. The subcriticality condition reads $L_{\mathrm{br}} \|f\|_{L^2}^2 < 1$.

With $\|\mathbb{W}\| = \lambda_1$ now pinned down—a positive, attained top eigenvalue, computable in closed form on the kernels we care about—the dimensionless coupling constant that section 20.2 built but could not yet name becomes a genuinely *spectral* quantity. The contraction factor $L_{\mathrm{br}} \|\mathbb{W}\|$ of the best-response map is $L_{\mathrm{br}}\lambda_1$, and the hypothesis $L_{\mathrm{br}} \|\mathbb{W}\| < 1$ is the demand that the behavioral gain stay below the reciprocal $\kappa = 1/L_{\mathrm{br}}$ of the spectral radius—the network operating *subcritically*. That this boundary is attained, by the compactness the lemma invoked, is what will make the contraction edge the true edge of well-posedness rather than an artifact. The next subsection takes exactly this identity and turns the abstract contraction estimate of section 20.2 into the sharp spectral subcriticality threshold.

20.4.2 Uniqueness by contraction

The previous subsection finished the one piece of arithmetic the threshold was waiting on: it identified the operator norm of a graphon as its top eigenvalue, $\|\mathbb{W}\| = \lambda_1$ (lemma 20.12), an honest, attained maximum of the spectrum carried by a nonnegative eigenfunction ϕ_1. That identity is what turns the abstract contraction constant of section 20.2 into something we can read as network data. Recall the shape of the argument we are about to complete. The best-response map factors as $\mathsf{S} = N \circ \mathbb{W}$ (eq. (20.11)): a linear aggregation through the kernel, followed by a pointwise best response that is Lipschitz in the aggregate with constant $L_{\mathrm{br}} = \ell/\theta$ (lemma 20.6). A composition of Lipschitz maps contracts by at most the product of their constants, so S shrinks L^2 distances by at most $L_{\mathrm{br}} \cdot \|\mathbb{W}\|$. Until this subsection the second factor was a bare symbol; now it is λ_1, and the whole contraction estimate becomes a single *spectral* inequality, $L_{\mathrm{br}}\lambda_1 < 1$. Below that line the loop "aggregate, then best-respond" is a strict contraction and Banach's theorem hands us exactly one equilibrium, reached by geometric iteration; this is the chapter's central quantitative result, and we now state and prove it, then spend the rest of the subsection reading off what the threshold means.

Theorem 20.14 (Subcritical uniqueness)**.** *Consider a static graphon game under* ***(A-Reg-W)*** *and assumption 20.5, with the standalone optimum in L^2,* $\mathrm{br.}(0) \in L^2[0,1]$ *(automatic when S is bounded, and equal to* $\Pi_S\alpha \in L^\infty[0,1]$ *for the LQ game), and impose the coupling condition* ***(A-Coupling)*** *in the sharp form*

$$L_{\mathrm{br}} \|\mathbb{W}\| \;<\; 1, \qquad \textit{equivalently} \qquad \|\mathbb{W}\| \;<\; \kappa := \frac{1}{L_{\mathrm{br}}} = \frac{\theta}{\ell}. \tag{20.20}$$

Then S *maps* $\mathcal{S}$ *into* $\mathcal{S}$ *and is a contraction on* $L^2[0,1]$*, and the game has a* unique *graphon Nash equilibrium s^*. Moreover the fixed-point iteration* $s^{(k+1)} = \mathsf{S}(s^{(k)})$ *converges to s^* geometrically from any* $s^{(0)} \in \mathcal{S}$*, with* $\left\|s^{(k)} - s^*\right\|_{L^2} \leq (L_{\mathrm{br}} \|\mathbb{W}\|)^k \left\|s^{(0)} - s^*\right\|_{L^2}$.

Proof. That S maps $\mathcal{S}$ into itself is immediate from the construction of the factorization: for any admissible profile s the aggregate $\mathbb{W}s$ is a well-defined element of $L^2[0,1]$, and each br_x takes its value in the action set S, so $\mathsf{S}(s)(x) = \mathrm{br}_x((\mathbb{W}s)(x)) \in S$ for almost every x and $\mathsf{S}(s) \in \mathcal{S}$; the hypothesis $\mathrm{br}_\cdot(0) \in L^2$ together with the Lipschitz bound below keeps $\mathsf{S}(s)$ square-integrable. The content is the contraction estimate. Take two admissible profiles $s, s' \in \mathcal{S}$ and bound the L^2 distance between their images one type at a time, then in aggregate. Pointwise in x, the two images differ only through the aggregates fed to the *same* scalar map br_x, and that map is L_{br}-Lipschitz uniformly in x by lemma 20.6, so

$$\left|\mathrm{br}_x((\mathbb{W}s)(x)) - \mathrm{br}_x((\mathbb{W}s')(x))\right| \;\le\; L_{\mathrm{br}} \left|(\mathbb{W}s)(x) - (\mathbb{W}s')(x)\right| \;=\; L_{\mathrm{br}} \left|(\mathbb{W}(s-s'))(x)\right|,$$

the last equality by linearity of $\mathbb{W}$. Squaring this pointwise inequality and integrating over x carries it to the norm,

$$\left\|\mathsf{S}(s) - \mathsf{S}(s')\right\|_{L^2}^2 = \int_0^1 \left|\mathrm{br}_x((\mathbb{W}s)(x)) - \mathrm{br}_x((\mathbb{W}s')(x))\right|^2 \mathrm{d}x \;\le\; L_{\mathrm{br}}^2 \int_0^1 \left|(\mathbb{W}(s-s'))(x)\right|^2 \mathrm{d}x = L_{\mathrm{br}}^2 \left\|\mathbb{W}(s-s')\right\|_{L^2}^2 .$$

This is the pointwise, behavioral half of the bound; the structural half is the definition of the operator norm, $\|\mathbb{W}v\|_{L^2} \le \|\mathbb{W}\| \, \|v\|_{L^2}$ applied to $v = s - s'$, which by lemma 20.12 is $\|\mathbb{W}v\|_{L^2} \le \lambda_1 \|v\|_{L^2}$. Substituting,

$$\left\|\mathsf{S}(s) - \mathsf{S}(s')\right\|_{L^2}^2 \;\le\; L_{\mathrm{br}}^2 \, \|\mathbb{W}\|^2 \, \left\|s - s'\right\|_{L^2}^2 , \qquad \text{i.e.} \qquad \left\|\mathsf{S}(s) - \mathsf{S}(s')\right\|_{L^2} \;\le\; L_{\mathrm{br}} \, \|\mathbb{W}\| \, \left\|s - s'\right\|_{L^2} .$$

Thus S is Lipschitz on L^2 with constant $L_{\mathrm{br}} \|\mathbb{W}\|$, and the hypothesis eq. (20.20) makes that constant strictly less than one: S is a contraction. Its domain $\mathcal{S} = \{s \in L^2 : \; s(x) \in S \text{ a.e.}\}$ is a closed subset of the complete space $L^2[0,1]$ (closedness because S is closed and L^2-limits have a.e.-convergent subsequences), hence itself a complete metric space. The Banach fixed-point theorem (theorem 2.36) therefore furnishes a unique fixed point s^*, which by definition 20.4 is the unique GNE. The geometric rate is the standard by-product: iterating the contraction inequality on consecutive iterates $s^{(k)} = \mathsf{S}(s^{(k-1)})$ and using $\mathsf{S}(s^*) = s^*$ gives $\left\|s^{(k)} - s^*\right\|_{L^2} = \left\|\mathsf{S}(s^{(k-1)}) - \mathsf{S}(s^*)\right\|_{L^2} \;\le\; L_{\mathrm{br}} \|\mathbb{W}\| \left\|s^{(k-1)} - s^*\right\|_{L^2}$, and unwinding the recursion k times yields $\left\|s^{(k)} - s^*\right\|_{L^2} \le (L_{\mathrm{br}} \|\mathbb{W}\|)^k \left\|s^{(0)} - s^*\right\|_{L^2}$ from any starting profile $s^{(0)} \in \mathcal{S}$. □

This is the chapter's central quantitative result and the prototype for the whole of Part IV, and three features of the single inequality eq. (20.20) deserve to be drawn out, because each will recur, dressed in PDEs, in chapters 22 and 24.

First, the threshold is a loop gain—a competition between behavior and structure. The two factors in $L_{\mathrm{br}} \|\mathbb{W}\|$ are independent and measure entirely different things. The behavioral responsiveness $L_{\mathrm{br}} = \ell/\theta$ is a property of the *payoff* alone: it is the ratio of how strongly a player's marginal incentive reacts to its neighbors' aggregate (ℓ) to how firmly its own cost anchors its action (θ), and it owes nothing to who those neighbors are. The structural amplification $\|\mathbb{W}\| = \lambda_1$ is a property of the *network* alone: the largest factor by which one round of neighbor-averaging can magnify a profile, owing nothing to the payoff. Their product is the gain of the feedback loop that section 20.4 described: a perturbation of the profile is amplified λ_1-fold by the structural step $s \mapsto \mathbb{W}s$ and then L_{br}-fold by the behavioral step $z \mapsto \mathrm{br}_x(z)$ before returning as a perturbation of the profile, and the loop is stable—returns less than it was handed, so iteration damps out to a single fixed point—exactly when the round-trip gain $L_{\mathrm{br}}\lambda_1$ is below one. This is why responsiveness and connectivity are interchangeable in setting the threshold: a very stiff payoff (small L_{br}) tolerates a strongly coupled network (large λ_1), and a sparsely coupled network tolerates a twitchy payoff, but the *product* is what must stay subcritical. The same loop gain, behavioral responsiveness times spectral radius, is what will govern the well-posedness of the dynamic game once time and diffusion are restored.

Second, it is the spectral radius that enters, not a cruder norm. The estimate did not settle for any convenient bound on the size of the kernel—not the supremum $\|W\|_\infty$, not the Hilbert–Schmidt norm $\|W\|_{L^2([0,1]^2)} = \sqrt{\sum_n \lambda_n^2}$, each of which would also control the contraction but with room to spare. It is the *leading eigenvalue* λ_1, the smallest constant for which $\|\mathbb{W}v\| \leq \lambda_1 \|v\|$ holds for every v, that appears, and it does so because lemma 20.12 identified it as the exact operator norm: the worst case of the linear step is realized along the top eigenfunction ϕ_1 and nowhere worse. The practical consequence is that a network may carry a great many strong individual connections—a large $\|W\|_\infty$, many high-weight rows—and still operate subcritically, provided those connections do not conspire into a single large eigenvalue. Only coherent, mutually reinforcing structure, the kind that piles up into a large λ_1, threatens uniqueness; incoherent strength, spread across modes with eigenvalues that do not add, does not. This sharpness is exactly what cruder norms would miss, and it is why the chapter has insisted on watching the spectral radius and not the kernel's gross size.

Third, the bound is sharp in a structural sense—but in a sense worth stating precisely so as not to overclaim. It is *not* the claim that uniqueness fails the instant $L_{\mathrm{br}} \|\mathbb{W}\| \geq 1$; the very next subsection will exhibit couplings far above this line that still have a unique equilibrium, secured by sign structure rather than smallness. What is sharp is that no contraction-type argument—no argument that asks only the *magnitude* of the coupling to be small—can reach past the threshold, because at the threshold the linear inversion underlying the LQ benchmark genuinely breaks down for the worst-case sign and data. To see this concretely, specialize to the LQ game example 20.3, where $L_{\mathrm{br}} = |\mathsf{b}|$ and the GNE solves $(I - \mathsf{b}\mathbb{W})s = \alpha$, and take the strategic-complements sign $\mathsf{b} > 0$, so that the binding eigenvalue is the positive top one and $L_{\mathrm{br}} \|\mathbb{W}\| = \mathsf{b}\lambda_1$. At exactly $\mathsf{b}\lambda_1 = 1$ the operator $I - \mathsf{b}\mathbb{W}$ is singular: applying it to the top eigenfunction gives

$$(I - \mathsf{b}\mathbb{W})\phi_1 = \phi_1 - \mathsf{b}\,\mathbb{W}\phi_1 = \phi_1 - \mathsf{b}\lambda_1\phi_1 = (1 - \mathsf{b}\lambda_1)\phi_1 = 0,$$

so ϕ_1 lies in the kernel and the inverse $(I - \mathsf{b}\mathbb{W})^{-1}$ on which $s^* = (I - \mathsf{b}\mathbb{W})^{-1}\alpha$ relied no longer exists. Because $\mathsf{b}\mathbb{W}$ is compact, the Fredholm alternative applies to $I - \mathsf{b}\mathbb{W}$: the equation $(I - \mathsf{b}\mathbb{W})s = \alpha$ is solvable in L^2 if and only if α is orthogonal to the kernel span $\{\phi_1\}$, i.e. iff $\langle \alpha, \phi_1 \rangle = 0$. This is what splits the failure into its two faces.

The *generic* face is non-existence. For a typical standalone return α the overlap $\langle \alpha, \phi_1 \rangle$ is nonzero, the solvability condition fails, and the equation has *no* L^2 solution at all—there is no unconstrained equilibrium. One can watch this coming from below the threshold by tracking the ϕ_1-component of the equilibrium. Expanding $\alpha = \sum_n \langle \alpha, \phi_n \rangle \phi_n$ and inverting mode by mode (the diagonalization the LQ form permits), the equilibrium is

$$s^* = (I - \mathsf{b}\mathbb{W})^{-1}\alpha = \sum_n \frac{\langle \alpha, \phi_n \rangle}{1 - \mathsf{b}\lambda_n}\, \phi_n,$$

and the $n = 1$ term carries the denominator $1 - \mathsf{b}\lambda_1$, which vanishes as $\mathsf{b}\lambda_1 \uparrow 1$. The component of equilibrium effort along the centrality direction, $\langle \alpha, \phi_1 \rangle/(1 - \mathsf{b}\lambda_1)$, therefore diverges, and with it the whole profile: $\|s^*\| \to \infty$ as criticality is approached. This blow-up is the equilibrium condensing onto ϕ_1, the phenomenon theorem 20.19 makes precise; here it is the analytic fingerprint of the loop gain reaching one. The *non-generic* face is non-uniqueness. For the fine-tuned data $\alpha \perp \phi_1$ the solvability condition is met and a particular solution s exists, but now $\phi_1 \in \ker(I - \mathsf{b}\mathbb{W})$ means the entire line $s + t\phi_1$, $t \in \mathbb{R}$, solves the same equation—a one-parameter family of equilibria, genuine multiplicity along the soft mode. Either way the unconstrained LQ problem is ill posed exactly at the contraction edge, which is what we mean by calling the threshold structurally sharp: the operator-theoretic well-posedness it certifies is the true edge, not an artifact of the proof technique.

Two caveats keep this honest, and both point forward. The bounded-action (clipped) game retains an equilibrium regardless of where the coupling sits, because that existence was bought by compactness and Schauder's theorem in theorem 20.9, with no smallness assumed; what the threshold governs is uniqueness, not existence. And the sharpness just argued is sharpness *for the complements sign*. For strategic substitutes $\mathsf{b} < 0$ the product $L_{\mathrm{br}} \|\mathbb{W}\| = |\mathsf{b}|\, \lambda_1$ may well exceed one while $I - \mathsf{b}\mathbb{W}$ stays comfortably invertible—its spectrum is $\{1 - \mathsf{b}\lambda_n\}$, and with $\mathsf{b} < 0$ and λ_n of either sign the nearest approach to the singular value 0 is governed by the most *negative* eigenvalue $\lambda_{\min}$, not by λ_1, so the genuine well-posedness edge is the weaker $|\mathsf{b}|\, |\lambda_{\min}| < 1$. In that regime the contraction bound is merely sufficient and leaves a gap, and closing it requires the second, structural route to uniqueness. We return to both faces of the criticality failure in sections 20.5 and 20.6, where they are computed in closed form.

Physics note

The product $L_{\mathrm{br}} \|\mathbb{W}\|$ plays the role of a coupling constant approaching a critical value, and the threshold $L_{\mathrm{br}} \|\mathbb{W}\| = 1$ is a stability boundary of exactly the Curie–Weiss type, complete with a divergent susceptibility that we can compute in one line. Take the LQ game on the constant kernel $W \equiv c$, where every type feels the same population mean $\bar{s} = \int_0^1 s$ and the aggregate is the type-independent number $z^* = c\,\bar{s}$ (example 20.13(a)). The equilibrium condition $s^*(x) = \alpha(x) + \mathsf{b}z^*$ integrated over x gives a self-consistency equation for the mean alone,

$$\bar{s} = \bar{\alpha} + \mathsf{b}\, z^* = \bar{\alpha} + \mathsf{b}c\,\bar{s} \qquad \Longrightarrow \qquad \bar{s} = \frac{\bar{\alpha}}{1 - \mathsf{b}c}, \qquad z^* = \frac{c\,\bar{\alpha}}{1 - \mathsf{b}c},$$

with $\bar{\alpha} = \int_0^1 \alpha$. This is the mean-field self-consistency relation in its barest form: the population's average effort is its standalone average dressed by the factor $(1 - \mathsf{b}c)^{-1}$, the resummed feedback of everyone responding to everyone. Now probe it. Shift the standalone return by a uniform amount, $\alpha \mapsto \alpha + \delta\mathbf{1}$, so $\bar{\alpha} \mapsto \bar{\alpha} + \delta$; the response of the collective is

$$\chi \;:=\; \frac{\partial \bar{s}}{\partial \delta} \;=\; \frac{1}{1 - \mathsf{b}c} \;=\; \frac{1}{1 - \mathsf{b}\lambda_1},$$

the *susceptibility* of the population to a small common perturbation, and it *diverges* as $\mathsf{b}c = \mathsf{b}\lambda_1 \uparrow 1$. (For a general kernel the same computation in the eigenbasis gives mode-by-mode susceptibilities $(1 - \mathsf{b}\lambda_n)^{-1}$, of which the $n = 1$ term is the first and most violent to blow up—the divergent ϕ_1-component of s^* already met above.) This is the static, network-resolved version of the divergent susceptibility at a mean-field critical point: the collective response to a small common nudge blows up as the coupling reaches the inverse $1/\lambda_1$ of the top eigenvalue, while individual stiffness keeps every agent's own response finite. The eigenfunction ϕ_1 is the *soft mode*—the direction of vanishing restoring force, along which the response first diverges and onto which the equilibrium condenses—and it is the direction of incipient symmetry breaking: just past criticality the unconstrained problem can no longer pin the profile's ϕ_1-component, and which way it tips is decided by the data rather than the loop. We will see the very same eigenvalue threshold govern uniqueness of the *dynamic* graphon mean field game in chapter 24, with the spectral radius again playing the order parameter; that the static and dynamic thresholds coincide is the precise content, made good in section 20.7, of the claim that the static case is the dynamic theory's stationary shadow.

The contraction route has thus delivered uniqueness, but it has done so by paying a definite price: the coupling had to be *small*, $L_{\mathrm{br}}\,\|\mathbb{W}\| < 1$, and we have just seen that beyond that line—at least for the strategic-complements sign—genuine ill-posedness can set in. Two debts are left open by this, and the next two sections settle them in turn. The first is the gap the second caveat exposed: for strategic substitutes the contraction bound was merely sufficient, the true invertibility edge being the weaker $|\mathsf{b}|\,|\lambda_{\min}| < 1$, and more broadly one may ask whether uniqueness can ever survive *strong* coupling, far above the contraction threshold, when it is the network's sign *structure* rather than the coupling's smallness that does the work. That is precisely the question the next subsection takes up, trading the contraction's smallness price for a structural monotonicity condition that costs nothing in coupling strength when the signs cooperate. The second debt is owed to section 20.5: everything there—the reading of the equilibrium as Katz–Bonacich centrality, the convergent Neumann series, the comparative statics—presumes that $I - \mathsf{b}\mathbb{W}$ is invertible, and it is exactly the subcritical regime certified here, $\mathsf{b}\lambda_1 < 1$, that supplies that invertibility. The contraction theorem is thus the guarantee on which the centrality calculus quietly rests. We turn first to the structural route.

20.4.3 Uniqueness by monotonicity

The contraction theorem left a sharp but deliberately limited verdict: the equilibrium is unique below the line $L_{\mathrm{br}}\,\|\mathbb{W}\| < 1$, and that verdict was bought by insisting the coupling be *small*. Its closing caveats already hinted that smallness is not the only currency that can purchase uniqueness. For strategic substitutes the genuine invertibility edge was the weaker $|\mathsf{b}|\,|\lambda_{\min}| < 1$, leaving the contraction bound a strict overpayment; and more broadly one suspects that whenever the network's interaction *signs* cooperate the equilibrium should stay unique no matter how strong the coupling grows. The contraction subsection handed exactly this question forward—can uniqueness survive *strong* coupling through structure rather than smallness?—and the present subsection answers it, in the affirmative. The price is not waived but exchanged: in place of a bound on the coupling's *magnitude* we impose a condition on its *sign structure*, namely that the operator $I - \mathsf{b}\mathbb{W}$ encoding the equilibrium be positive definite. When that holds, uniqueness costs nothing at all in coupling strength—the equilibrium is unique however large b is.

This is the exact static mirror of the Lasry–Lions monotonicity condition of definition 11.1. There a monotone coupling defeated multiplicity of the dynamic mean field game regardless of the time horizon; here a monotone—positive-definite—equilibrium operator defeats multiplicity regardless of the coupling strength. As there, we phrase the condition as monotonicity of the operator that encodes the equilibrium, and as there the working mechanism is *convexity*: the equilibrium will turn out to be the minimizer of a single convex functional, and a strictly convex functional has at most one minimizer. The whole subsection is the unpacking of that one sentence.

The LQ game is a potential game. The structural fact that opens the variational route is that the LQ equilibrium condition is not merely *an* equation but a *gradient* equation: there is a single scalar functional on profiles whose stationarity reproduces the first-order condition of every type *at once*. A game with this property is called a *potential game*, and the functional is its *potential*. The payoff of recognizing one is immediate and large: the search for a fixed point of mutually-best-responding players—a problem with the unknown on both sides—collapses into the minimization of one ordinary function, to which the entire convex-analysis toolkit then applies, uniqueness of the minimizer included.

To exhibit the potential, recall the LQ equilibrium condition in operator form, $(I - \mathsf{b}\mathbb{W})s = \alpha$ (eq. (20.8)), and consider the quadratic functional

$$\mathcal{V}(s) \;=\; \tfrac{1}{2}\langle s, (I - \mathsf{b}\mathbb{W})s\rangle - \langle \alpha, s\rangle \;=\; \tfrac{1}{2}\,\|s\|_{L^2}^2 - \tfrac{\mathsf{b}}{2}\langle s, \mathbb{W}s\rangle - \langle \alpha, s\rangle. \tag{20.21}$$

The single ingredient that makes this work is that $\mathbb{W}$ is *self-adjoint*—this is the symmetry of the kernel, $W(x,y) = W(y,x)$, built into **(A-Reg-W)**—so that $I - \mathsf{b}\mathbb{W}$ is self-adjoint too, and a quadratic form built from a self-adjoint operator is its own gradient generator. To see this concretely, differentiate $\mathcal{V}$ along an arbitrary direction $v \in L^2[0,1]$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\mathcal{V}(s+tv) \;=\; \tfrac{1}{2}\big(\langle v, (I-\mathsf{b}\mathbb{W})s\rangle + \langle s, (I-\mathsf{b}\mathbb{W})v\rangle\big) - \langle \alpha, v\rangle \;=\; \langle (I-\mathsf{b}\mathbb{W})s - \alpha,\, v\rangle,$$

where the last step folds the two cross terms into one using $\langle s, (I-\mathsf{b}\mathbb{W})v\rangle = \langle (I-\mathsf{b}\mathbb{W})s, v\rangle$, which is *exactly* self-adjointness. Since v is arbitrary, the gradient is $\nabla\mathcal{V}(s) = (I-\mathsf{b}\mathbb{W})s - \alpha$, and its vanishing $\nabla\mathcal{V}(s) = 0$ is precisely the GNE equation $(I-\mathsf{b}\mathbb{W})s = \alpha$ of eq. (20.8). Thus the GNE are exactly the *critical points* of $\mathcal{V}$. It is worth pausing on the role of symmetry: had the kernel been asymmetric, $\mathbb{W}$ would not be self-adjoint, the two cross terms above would not combine, and $(I-\mathsf{b}\mathbb{W})s - \alpha$ would fail to be the gradient of *any* functional—there would be no potential and no variational route at all. The symmetry that **(A-Reg-W)** imposes is not a convenience here; it is the precise hypothesis that makes the game a potential game.

The second variation is simpler still and constant in s:

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\mathcal{V}(s+tv) \;=\; \langle v, (I-\mathsf{b}\mathbb{W})v\rangle,$$

so the Hessian of $\mathcal{V}$ is the fixed self-adjoint operator $H = I - \mathsf{b}\mathbb{W}$, independent of where it is evaluated. A quadratic functional with constant Hessian H is convex exactly when $H \geq 0$ and *strictly* (indeed uniformly) convex exactly when H is uniformly positive, $H \geq \eta I$ with $\eta > 0$; and a uniformly convex functional on a Hilbert space, being coercive, has a *unique* minimizer. The whole question of uniqueness therefore reduces to one spectral question—when is $H = I - \mathsf{b}\mathbb{W}$ positive definite?—and that question has a one-line answer in terms of the eigenvalues of $\mathbb{W}$. The Hessian of $\mathcal{V}$ is $I - \mathsf{b}\mathbb{W}$, which is positive definite if and only if $\mathsf{b}\lambda_n < 1$ for every eigenvalue λ_n.

Theorem 20.15 (Monotone uniqueness for the LQ game)**.** *For the LQ graphon game of example 20.3 with $S = \mathbb{R}$, the following are equivalent:*

(i) the functional $\mathcal{V}$ of eq. (20.21) is strictly convex on $L^2[0,1]$;

(ii) the operator $I - \mathsf{b}\mathbb{W}$ is positive definite, i.e. $\langle s, (I - \mathsf{b}\mathbb{W})s\rangle \geq \eta \|s\|^2$ for some $\eta > 0$;

(iii) $\mathsf{b}\,\lambda_n < 1$ for every eigenvalue λ_n of $\mathbb{W}$.

Under any of these, the GNE is unique and is the unique minimizer of $\mathcal{V}$. In particular, writing the condition (iii) as $\mathsf{b}\lambda_n < 1$ for the eigenvalue achieving $\sup_n \mathsf{b}\lambda_n$: for strategic complements ($\mathsf{b} > 0$) the binding eigenvalue is $\lambda_1 = \|\mathbb{W}\|$ and the condition is the spectral bound $\mathsf{b}\,\lambda_1 = \mathsf{b}\,\|\mathbb{W}\| < 1$; for strategic substitutes ($\mathsf{b} < 0$) the binding eigenvalue is the most negative one $\lambda_{\min}$, and the condition reads

$$\mathsf{b}\,\lambda_{\min} \;=\; |\mathsf{b}|\,|\lambda_{\min}| \;<\; 1.$$

This is not *automatic in general: a nonnegative symmetric graphon under **(A-Reg-W)** can have $\lambda_{\min} < 0$ (e.g. the complete-bipartite kernel $W(x,y) = \mathbf{1}_{x\leq 1/2}\mathbf{1}_{y>1/2} + \mathbf{1}_{x>1/2}\mathbf{1}_{y\leq 1/2}$ has $\|\mathbb{W}\| = \frac{1}{2}$ and $\lambda_{\min} = -\frac{1}{2}$), and then a sufficiently strong substitute coupling $|\mathsf{b}| \geq 1/|\lambda_{\min}|$ makes $I - \mathsf{b}\mathbb{W}$ indefinite, $\mathcal{V}$ non-convex, and the critical point a saddle rather than a minimizer. Substitutes are* unconditionally *unique only for positive semidefinite graphons ($\lambda_{\min} = 0$), where $\mathsf{b}\lambda_{\min} = 0 < 1$ for every $\mathsf{b} \leq 0$; the chapter's worked constant and rank-one kernels are PSD, so the claim is true there.*

Proof. The functional $\mathcal{V}$ is quadratic with the constant self-adjoint Hessian $H = I - \mathsf{b}\mathbb{W}$ computed above. We take the three equivalences in turn.

(i)⇔(ii). For a quadratic functional with constant Hessian H, strict convexity means $\langle s, Hs\rangle > 0$ for every $s \neq 0$, while (ii) is the uniform statement $\langle s, Hs\rangle \geq \eta\,\|s\|^2$ with $\eta > 0$. In a general infinite-dimensional Hilbert space these differ—a Hessian whose spectrum approached 0 would be strictly positive yet not uniformly so—but here they coincide, and the reason is exactly the compactness of $\mathbb{W}$ that the chapter keeps cashing in. By the spectral theorem for the compact self-adjoint operator $\mathbb{W}$ (theorem 2.26) its eigenvalues satisfy $\lambda_n \to 0$, so the spectral values $1 - \mathsf{b}\lambda_n$ of H converge to 1; their *only* possible accumulation point is $1 > 0$. Consequently, if each $1 - \mathsf{b}\lambda_n > 0$ then their infimum cannot be 0 (any subsequence tending to 0 would contradict convergence to 1), so the infimum is a strictly positive *minimum*, attained at an extreme eigenvalue. Strict positivity of the spectrum thus upgrades automatically to uniform positivity, giving (i)⇔(ii).

(ii)⇔(iii). The spectrum of $H = I - \mathsf{b}\mathbb{W}$ is obtained from that of $\mathbb{W}$ by the affine map $\mu \mapsto 1 - \mathsf{b}\mu$. Since $\mathbb{W}$ is compact self-adjoint, its spectrum is $\{\lambda_n\} \cup \{0\}$, the value 0 coming from the (generally infinite-dimensional) kernel of $\mathbb{W}$—the directions orthogonal to every eigenfunction, on which $\mathbb{W}$, and hence $\mathsf{b}\mathbb{W}$, acts as zero. Applying the affine map, the spectrum of H is $\{1 - \mathsf{b}\lambda_n\} \cup \{1\}$, where the isolated value 1 is the image of 0 and records that H acts as the identity on $\ker\mathbb{W}$. Because $1 > 0$ always, those kernel directions never threaten convexity; positivity of H is decided entirely

on the eigen-directions of $\mathbb{W}$. Hence $H \geq \eta I$ for some $\eta > 0$ iff $\inf_n(1 - \mathsf{b}\lambda_n) > 0$, which is iff $\sup_n \mathsf{b}\lambda_n < 1$, i.e. $\mathsf{b}\lambda_n < 1$ for every n—statement (iii).

Uniqueness. Under (i)–(iii) the functional $\mathcal{V}$ is uniformly convex, hence coercive, on the Hilbert space $L^2[0,1]$, and a coercive strictly convex functional has a unique minimizer. That minimizer is characterized by the vanishing of its gradient $\nabla\mathcal{V} = Hs - \alpha = 0$, which is exactly the GNE equation eq. (20.8); since the critical point is unique and is the global minimizer, the GNE is unique.

The binding eigenvalue. It remains to read condition (iii), $\sup_n \mathsf{b}\lambda_n < 1$, through the sign of b, using the spectral ordering $\lambda_1 = \|\mathbb{W}\| \geq \lambda_n \geq \lambda_{\min} \geq -\|\mathbb{W}\|$. When $\mathsf{b} > 0$ the map $\lambda \mapsto \mathsf{b}\lambda$ is increasing, so $\sup_n \mathsf{b}\lambda_n = \mathsf{b}\lambda_1$ is achieved at the *top* eigenvalue, and (iii) reads $\mathsf{b}\lambda_1 = \mathsf{b}\|\mathbb{W}\| < 1$. When $\mathsf{b} < 0$ the map is decreasing, so $\sup_n \mathsf{b}\lambda_n = \mathsf{b}\lambda_{\min}$ is achieved at the most *negative* eigenvalue, and (iii) reads $\mathsf{b}\lambda_{\min} = |\mathsf{b}|\,|\lambda_{\min}| < 1$. In the latter case the constraint is vacuous—and so holds for *every* $\mathsf{b} \leq 0$—exactly when $\lambda_{\min} = 0$, that is when $\mathbb{W}$ is positive semidefinite. □

The complete-bipartite assertion inside the theorem deserves the small computation that makes it concrete, because it is the cleanest demonstration that a nonnegative symmetric graphon really can carry a strictly negative eigenvalue—and so that the substitute case really is conditional. Write $g = \mathbf{1}_{x \leq 1/2}$ and $h = \mathbf{1}_{x > 1/2}$ for the indicators of the two halves of $[0,1]$; then the complete-bipartite kernel is the rank-two function $W(x,y) = g(x)h(y) + h(x)g(y)$, which is 1 when x and y lie on opposite sides of $\frac{1}{2}$ and 0 otherwise. Its operator acts by

$$(\mathbb{W}f)(x) \;=\; g(x)\langle h, f\rangle + h(x)\langle g, f\rangle,$$

so its range lies in $\operatorname{span}\{g,h\}$ and the entire spectrum, apart from the zero eigenvalue on $\operatorname{span}\{g,h\}^{\perp}$, is read off the 2×2 action on that span. Using the orthogonality $\langle g, h\rangle = 0$ and the norms $\langle g, g\rangle = \langle h, h\rangle = \frac{1}{2}$ (each half has length $\frac{1}{2}$), one finds $\mathbb{W}g = \frac{1}{2}\,h$ and $\mathbb{W}h = \frac{1}{2}\,g$, i.e. the matrix $\left(\begin{smallmatrix} 0 & 1/2 \\ 1/2 & 0 \end{smallmatrix}\right)$ in the basis $\{g,h\}$, whose eigenvalues are $\pm\frac{1}{2}$. The eigenfunction for $+\frac{1}{2}$ is the symmetric combination $g + h = \mathbf{1}$, the constant function: every type feels the same neighbor mass, and indeed $\mathbb{W}\mathbf{1} = \frac{1}{2}\mathbf{1}$, so $\lambda_1 = \|\mathbb{W}\| = \frac{1}{2}$. The eigenfunction for $-\frac{1}{2}$ is the antisymmetric combination $g - h$, which is $+1$ on the left half and -1 on the right: the two halves do *opposite* things, and $\mathbb{W}(g-h) = -\frac{1}{2}(g-h)$, so $\lambda_{\min} = -\frac{1}{2}$. This negative edge is the analytic trace of the bipartite structure—a node's neighbors all sit on the *other* side, so the natural mode is one in which the sides oppose one another. Now switch on a strong substitute coupling $\mathsf{b} < 0$ with $|\mathsf{b}| \geq 1/|\lambda_{\min}| = 2$. Along the antisymmetric mode $v = g - h$ the Hessian quadratic form is

$$\langle v, (I - \mathsf{b}\mathbb{W})v\rangle \;=\; (1 - \mathsf{b}\lambda_{\min})\,\|v\|^2 \;=\; \left(1 - |\mathsf{b}|\,\tfrac{1}{2}\right)\|v\|^2 \;\leq\; 0,$$

so $\mathcal{V}$ has a direction of nonpositive curvature: it is no longer convex, its critical point is a saddle rather than a minimizer, and uniqueness fails. Behaviorally this is exactly what a strong local public good on a bipartite network should do—when one side contributes heavily the other free-rides, and the roles can swap, so alongside the symmetric configuration there appear asymmetric ones in which one half works and the other coasts. The lesson the computation enforces is the hedged claim in the theorem: substitutes are *unconditionally* unique only when $\lambda_{\min} = 0$, i.e. for positive semidefinite graphons; the constant and rank-one kernels the chapter carries are PSD (a constant kernel is $c\,\mathbf{1}\otimes\mathbf{1}$, rank one and nonnegative definite; $W = ff$ is $f \otimes f \geq 0$), so on those the substitute claim is safe, but it is not a free general fact.

It is now worth stepping back to compare the two uniqueness routes side by side, because the comparison is not symmetric in the sign of the coupling, and that asymmetry is the conceptual heart of the subsection. The contraction route asked only the *magnitude* $L_{\mathrm{br}}\,\|\mathbb{W}\|$ to be small; the monotone route asks the *signed* quantity $\sup_n \mathsf{b}\lambda_n$ to be less than one. For complements these two demands turn out to be the very same line, while for substitutes the monotone demand is genuinely weaker—but only conditionally so. The following remark makes this precise and reads it as the static face of an asymmetry the reader already met in the dynamic theory.

Remark 20.16 (Two thresholds, one for complements)**.** For strategic complements the contraction threshold of theorem 20.14 ($\mathsf{b}\,\|\mathbb{W}\| < 1$, taking $L_{\mathrm{br}} = |\mathsf{b}| = \mathsf{b}$) and the monotone/convexity threshold of theorem 20.15(iii) ($\mathsf{b}\lambda_1 < 1$) *coincide*, because $\lambda_1 = \|\mathbb{W}\|$ by lemma 20.12. The two routes, spectral and variational, see the *same* edge from two sides: contraction sees the loop gain reaching one, convexity sees the soft mode ϕ_1 losing its restoring curvature, and they are the same event. The LQ complements game therefore has a sharp uniqueness boundary at $\mathsf{b}\,\|\mathbb{W}\| = 1$, with a unique GNE strictly below and the genuine multiplicity issues of section 20.4.2 at and above it; for the crowd-seeking sign there is exactly one threshold and nothing is gained by trading smallness for structure. For substitutes the picture differs. The monotone route is strictly stronger than contraction—it certifies uniqueness whenever $|\mathsf{b}|\,|\lambda_{\min}| < 1$, a strictly weaker requirement than the contraction bound $|\mathsf{b}|\,\|\mathbb{W}\| < 1$ precisely when $|\lambda_{\min}| < \lambda_1$—but it is *not* unconditional. It becomes the unconditional "all $\mathsf{b} < 0$" statement only for positive semidefinite graphons ($\lambda_{\min} = 0$); when the spectrum has a negative edge $\lambda_{\min} < 0$, a strong enough substitute coupling $|\mathsf{b}| \geq 1/|\lambda_{\min}|$ renders $I - \mathsf{b}\mathbb{W}$ indefinite, $\mathcal{V}$ non-convex, and *neither* route certifies uniqueness—exactly the complete-bipartite breakdown computed above. Modulo this caveat, the qualitative picture is the same asymmetry between attractive and repulsive coupling seen in the Lasry–Lions analysis of section 11.4: the "crowd-averse" (substitute) sign is tame over a wide range governed by the *weaker* spectral edge $|\lambda_{\min}|$ and is unconditionally tame when that edge sits at zero, while the "crowd-seeking" (complement) sign is constrained by the *top* eigenvalue λ_1 and has the single, sharp critical coupling at which the equilibrium begins to condense onto ϕ_1. The order parameter $\|\mathbb{W}\| = \lambda_1$ governs the dangerous, attractive side; the harmless, repulsive side is held off by the lesser edge $|\lambda_{\min}|$, which for the kernels we compute on is simply 0.

The monotone route also reaches beyond the linear–quadratic payoff, and it is worth recording how, both to place the present theorem in the larger architecture of the book and because the dynamic descendant in Part IV is built on exactly this idea. Nothing in the variational argument used the quadratic form of the payoff beyond two features: that the equilibrium condition is the gradient of *some* functional—the game is a potential game—and that this functional is convex. For a general aggregative game these become, respectively, a symmetry condition on the cross-partials of the payoff (so that a potential exists) and a monotonicity condition pitting the concavity of each agent's own-action response against the spectral amplification $\|\mathbb{W}\|$ of the coupling (so that the potential is convex); when both hold, uniqueness follows by the identical "strictly-convex-functionals-have-one-minimizer" argument, with the LQ Hessian $I - \mathsf{b}\mathbb{W}$ replaced by the second variation of the nonlinear potential. We do not develop the fully nonlinear monotone theory here—the LQ case already exhibits every structural feature we need—but it is precisely the static specialization of the Lasry–Lions monotonicity of definition 11.1, and its dynamic, measure-valued form, where the inner product $\langle\,\cdot\,,(I-\mathsf{b}\mathbb{W})\,\cdot\,\rangle$ is replaced by a monotone coupling functional on densities, is the engine of the existence-and-uniqueness theory of chapter 24. We record this correspondence in the chapter ledger.

It remains to discharge the debt the contraction subsection flagged toward the centrality harvest. Whichever route certifies it, the operative conclusion for what follows is the same: in the positive-definite regime—subcritical complements $0 \le \mathsf{b}\lambda_1 < 1$, or substitutes $\mathsf{b} < 0$ with $|\mathsf{b}||\lambda_{\min}| < 1$, which on the chapter's PSD kernels means *any* $\mathsf{b} < 0$—the operator $I - \mathsf{b}\mathbb{W}$ has spectrum bounded away from zero and is therefore boundedly invertible, with a unique GNE $s^* = (I - \mathsf{b}\mathbb{W})^{-1}\alpha$. And wherever the coupling is additionally subcritical in magnitude, $|\mathsf{b}|\,\|\mathbb{W}\| < 1$, that inverse is delivered by the convergent Neumann series $\sum_{k\ge 0} \mathsf{b}^k\mathbb{W}^k$—the resummed feedback of every order of neighbor response. This invertibility, and that series, are exactly the standing hypotheses on which the next section rests: section 20.5 takes the unique equilibrium $(I - \mathsf{b}\mathbb{W})^{-1}\alpha$ as given and asks not *whether* it exists but *what it is*, identifying it—through the very Neumann expansion just named—with a recognizable network centrality. Having certified the inverse from two sides, we are free to harvest it.

20.5 Centrality determines strategy

The previous two subsections did the hard work of certifying an inverse; this one spends it. Sections 20.4.2 and 20.4.3 closed by handing us a single operative fact, proved twice over—once by the contraction and once by the convexity of the potential—and it is the only thing the present section needs as input: in the regime they delimited (made precise in the standing hypotheses just below), the static LQ game of example 20.3 has a unique graphon Nash equilibrium given by the resolvent, $s^* = (I - \mathsf{b}\mathbb{W})^{-1}\alpha$. We no longer ask *whether* that equilibrium exists. We ask what it *is*—and the answer, the entire content of this section, is that it is a recognizable *network centrality*, a quantity a network scientist would have computed from the kernel W alone, with no game in sight.

Let us fix the standing regime precisely, because every statement below lives inside it. The payoff is the linear–quadratic one of eq. (20.6), the action set is the whole line $S = \mathbb{R}$ (so no clipping intervenes and the equilibrium is genuinely the linear solution eq. (20.8), not a constrained variant), and the coupling sits in the positive-definite window certified in section 20.4: either subcritical strategic complements, $0 \le \mathsf{b}\,\|\mathbb{W}\| < 1$, or strategic substitutes of any strength on the positive-semidefinite kernels of this chapter, $\mathsf{b} < 0$. By theorems 20.14 and 20.15 this is exactly the regime in which $I - \mathsf{b}\mathbb{W}$ is invertible, and wherever the coupling is moreover subcritical in magnitude, $|\mathsf{b}|\ \|\mathbb{W}\| < 1$, that inverse is delivered termwise by the convergent Neumann series $\sum_{k\ge 0} \mathsf{b}^k\mathbb{W}^k$. That series—the resummed feedback of every order of mutual best response, first written down at the end of section 20.1.3—is the machine the whole section runs on, and it is what makes the identification with a centrality not merely suggestive but literal.

Here is why that identification is worth a section rather than a sentence. Reading the equilibrium as a centrality vector is the move that converts three otherwise distinct strategic questions into one and the same spectral computation on $\mathbb{W}$. *Ranking*: who, in equilibrium, exerts the most effort? Once s^* is a centrality, this is answered by ordering players by a structural score read off the kernel, not by solving the game player by player. *Spillover prediction*: if one player's standalone incentive rises, whose equilibrium effort moves, and by how much? Since s^* depends on α through the linear operator $(I - \mathsf{b}\mathbb{W})^{-1}$, the entire propagation of a local shock through the network is encoded in that one resolvent, walk by walk. *Intervention design*: which player should a planner subsidize to raise aggregate effort the most? This becomes an optimization against the same operator, ranking targets by a centrality of their own. The single algebraic object $(I - \mathsf{b}\mathbb{W})^{-1}$ thus answers all three at once, and the order parameter $\|\mathbb{W}\| = \lambda_1$ that we have watched the whole chapter is precisely what controls how violently it responds—gently far below threshold, explosively as $\mathsf{b}\lambda_1 \to 1$.

The section delivers this in three movements, and it is worth seeing them in advance as one arc rather than three separate results. First (section 20.5.1) we *name* the centrality: the Neumann series identifies s^* exactly with the Katz–Bonacich centrality of the graphon, the b-discounted count of all weighted walks emanating from each type, so that equilibrium effort *is* a walk-counting score. Second (section 20.5.2) we push the coupling toward its critical value and watch that score *degenerate*: as $\mathsf{b}\lambda_1 \to 1$ the $k = 1$ eigen-direction ϕ_1 dominates the series, every type's effort aligns with the single top mode, and Katz–Bonacich centrality collapses onto eigenvector centrality while its overall magnitude—the susceptibility—diverges. Third (section 20.5.3) we put the centrality to work: differentiating s^* in the data turns comparative statics and optimal targeting into spectral computations, so the planner's best intervention is itself ranked by a centrality. Centrality, then criticality, then targeting: a recognizable score, its breakdown at the spectral edge, and its use as a design tool—all three are one reading of the resolvent $(I - \mathsf{b}\mathbb{W})^{-1}$.

We begin with the naming. The next subsection picks up the Neumann series just invoked and uses it to pin down exactly which centrality the equilibrium equals.

20.5.1 Equilibrium effort is Katz–Bonacich centrality

Definition 20.17 (Katz–Bonacich centrality)**.** For b with $|\mathsf{b}|\ \|\mathbb{W}\| < 1$ and a weight $\alpha \in L^2[0, 1]$, the *Katz–Bonacich centrality profile* with discount b and weights α is

$$\mathbf{c}_{\mathsf{b}}[\alpha] \;:=\; (I - \mathsf{b}\mathbb{W})^{-1}\alpha \;=\; \sum_{k=0}^{\infty} \mathsf{b}^k\, \mathbb{W}^k\, \alpha \;=\; \alpha + \mathsf{b}\mathbb{W}\alpha + \mathsf{b}^2\mathbb{W}^2\alpha + \cdots . \tag{20.22}$$

When $\alpha = \mathbf{1}$ we write $\mathbf{c}_{\mathsf{b}} := \mathbf{c}_{\mathsf{b}}[\mathbf{1}]$ for the *unweighted* centrality, which counts, for each type, the b-discounted total weight of all walks in the graphon emanating from that type.

The two expressions in eq. (20.22) agree, and both define a bounded operator, precisely because $\|\mathsf{b}\mathbb{W}\| = |\mathsf{b}|\ \|\mathbb{W}\| < 1$. We spell this out, since the Neumann series is the engine of the whole section. The bounded operators on $L^2[0, 1]$ form a Banach space under the operator norm, which is submultiplicative, $\left\|(\mathsf{b}\mathbb{W})^k\right\| \le \|\mathsf{b}\mathbb{W}\|^k$; hence the partial sums $S_M = \sum_{k=0}^{M} \mathsf{b}^k\mathbb{W}^k$ are Cauchy ($\|S_M - S_{M'}\| \le \sum_{k>M'} \|\mathsf{b}\mathbb{W}\|^k \to 0$) and converge in operator norm to some bounded S. Multiplying the telescoping identity $(I - \mathsf{b}\mathbb{W})S_M = I - (\mathsf{b}\mathbb{W})^{M+1}$ and its mirror by passing $M \to \infty$ (the remainder $(\mathsf{b}\mathbb{W})^{M+1} \to 0$ in norm) gives $(I - \mathsf{b}\mathbb{W})S = S(I - \mathsf{b}\mathbb{W}) = I$, so S is the two-sided bounded inverse $(I - \mathsf{b}\mathbb{W})^{-1}$. Equivalently, and consistently with the spectral computations to come: by the spectral theorem (theorem 2.26) $I - \mathsf{b}\mathbb{W}$ acts as the scalar $1 - \mathsf{b}\lambda_n$ on each eigenfunction ϕ_n and as 1 on

$\ker \mathbb{W}$, and since $|\mathsf{b}\lambda_n| \leq |\mathsf{b}|\,\|\mathbb{W}\| < 1$ each factor $1 - \mathsf{b}\lambda_n$ is bounded away from 0; its bounded inverse therefore acts as $1/(1-\mathsf{b}\lambda_n)$ on ϕ_n and as 1 on $\ker \mathbb{W}$, which is exactly the diagonal action of the series $\sum_k \mathsf{b}^k \mathbb{W}^k$ term by term. The term $\mathbb{W}^k\alpha$ is the aggregate, over length-k weighted walks, of the standalone returns, and b^k discounts long walks. This is exactly the discrete Katz–Bonacich centrality lifted to the graphon.

Theorem 20.18 (Centrality $\Rightarrow$ strategy). *In the subcritical LQ graphon game, the unique GNE is the Katz–Bonacich centrality profile*

$$s^* \;=\; (I - \mathsf{b}\mathbb{W})^{-1}\alpha \;=\; \mathbf{c}_{\mathsf{b}}[\alpha]. \tag{20.23}$$

If the standalone return is uniform, $\alpha \equiv \alpha_0$ constant, then $s^* = \alpha_0\,\mathbf{c}_{\mathsf{b}}$ is proportional to the unweighted centrality profile, and the equilibrium effort of a type is an increasing function of its centrality: types with more, and more central, neighbors exert more effort.

Proof. Equation (20.8) reads $(I - \mathsf{b}\mathbb{W})s^* = \alpha$, and under subcriticality $I - \mathsf{b}\mathbb{W}$ is boundedly invertible with inverse the Neumann series eq. (20.22), giving eq. (20.23). For $\alpha \equiv \alpha_0$, linearity gives $s^* = \alpha_0(I - \mathsf{b}\mathbb{W})^{-1}\mathbf{1} = \alpha_0\mathbf{c}_{\mathsf{b}}$. Monotonicity in centrality is the statement that the operator $(I - \mathsf{b}\mathbb{W})^{-1} = \sum_k \mathsf{b}^k\mathbb{W}^k$ has a nonnegative kernel when $\mathsf{b} \geq 0$ (each $\mathbb{W}^k$ has the nonnegative iterated kernel $W^{(k)} \geq 0$), so a type with pointwise larger row sums in every $W^{(k)}$ has pointwise larger s^*. $\square$

Notation watch

The discount b in the centrality eq. (20.22) is the *strategic interaction coefficient* of the game, not the discount *rate* $\beta = \beta$ reserved for intertemporal control problems (appendix A.4). In static graphon games no time and no β appear; the letter b is the game's own complementarity parameter. We use the distinct symbol b throughout precisely to avoid this collision.

20.5.2 Eigenvector centrality at criticality

As the coupling approaches its critical value, the centrality profile loses its dependence on the fine structure of α and aligns with a single network mode: the eigenvector centrality ϕ_1.

Theorem 20.19 (Emergence of eigenvector centrality). *Assume **(A-Reg-W)** with a* simple *top eigenvalue λ_1 (spectral gap $\lambda_1 > \lambda_2$), and let $\phi_1 \geq 0$ be its unit eigenfunction. Fix $\alpha \in L^2[0,1]$ with $\langle \alpha, \phi_1\rangle \neq 0$. Then for strategic complements $\mathsf{b} \uparrow 1/\lambda_1 = 1/\|\mathbb{W}\|$ the normalized equilibrium profile aligns with eigenvector centrality:*

$$\frac{s^*}{\|s^*\|_{L^2}} \xrightarrow[\mathsf{b}\uparrow 1/\lambda_1]{} \operatorname{sgn}(\langle \alpha, \phi_1\rangle)\,\phi_1 \qquad \textit{in } L^2[0,1], \tag{20.24}$$

and the equilibrium effort diverges, $\|s^*\|_{L^2} \to \infty$, with leading behavior

$$s^* \;=\; \frac{\langle \alpha, \phi_1\rangle}{1 - \mathsf{b}\lambda_1}\,\phi_1 \;+\; O(1) \qquad (\mathsf{b} \uparrow 1/\lambda_1). \tag{20.25}$$

Proof. Expand α and s^* in the eigenbasis (ϕ_n): writing $\alpha = \sum_n a_n\phi_n$ with $a_n = \langle\alpha, \phi_n\rangle$, the diagonal action of $(I - \mathsf{b}\mathbb{W})^{-1}$ gives

$$s^* = (I - \mathsf{b}\mathbb{W})^{-1}\alpha = \sum_n \frac{a_n}{1 - \mathsf{b}\lambda_n}\,\phi_n.$$

As $\mathsf{b} \uparrow 1/\lambda_1$ the $n = 1$ coefficient $a_1/(1 - \mathsf{b}\lambda_1)$ blows up, while every $n \geq 2$ coefficient stays bounded because $1 - \mathsf{b}\lambda_n \to 1 - \lambda_n/\lambda_1 > 0$ by the spectral gap. Splitting off the top mode,

$$s^* = \frac{a_1}{1 - \mathsf{b}\lambda_1}\phi_1 + r, \qquad \|r\|^2 = \sum_{n\geq 2} \frac{a_n^2}{(1 - \mathsf{b}\lambda_n)^2} \leq \frac{\|\alpha\|^2}{(1 - \lambda_2/\lambda_1)^2} = O(1),$$

which is eq. (20.25). Dividing by $\|s^*\|$ and letting $\mathsf{b} \uparrow 1/\lambda_1$, the bounded remainder is negligible against the diverging top term, yielding eq. (20.24) with the sign of $a_1 = \langle\alpha, \phi_1\rangle$. □

Physics note

Theorem 20.19 is the static graphon game's order-parameter statement. Below criticality the equilibrium reflects the whole spectrum of the network, weighted by how the standalone incentives α project onto each mode. As the coupling is tuned to the critical value $1/\lambda_1$, a single collective mode—the top eigenfunction, the eigenvector centrality ϕ_1—takes over, and the equilibrium profile becomes, up to normalization, *independent of the microscopic incentives* α, retaining only the scalar overlap $\langle\alpha, \phi_1\rangle$. This is condensation onto the soft mode: near the critical coupling the population organizes itself along the network's principal axis, exactly as a system near a continuous transition is dominated by its longest-wavelength fluctuation. The divergence $\|s^*\| \sim (1 - \mathsf{b}\lambda_1)^{-1}$ is the susceptibility divergence of the previous physics note, now resolved into the direction ϕ_1 along which it occurs.

20.5.3 Interventions and comparative statics

The closed form eq. (20.23) turns equilibrium analysis into linear algebra and makes precise how a planner who can perturb the standalone incentives should target the network. Suppose a planner adds an incentive perturbation $\delta\alpha \in L^2[0, 1]$ to the standalone returns, changing $\alpha \mapsto \alpha + \delta\alpha$. By linearity the equilibrium shifts by

$$\delta s^* \;=\; (I - \mathsf{b}\mathbb{W})^{-1}\,\delta\alpha \;=\; \mathbf{c}_{\mathsf{b}}[\delta\alpha], \tag{20.26}$$

the Katz–Bonacich centrality of the perturbation itself. The induced change in *aggregate equilibrium effort* $\Sigma(s) := \langle\mathbf{1}, s\rangle = \int_0^1 s$ is

$$\delta\Sigma = \langle\mathbf{1}, (I - \mathsf{b}\mathbb{W})^{-1}\delta\alpha\rangle = \langle(I - \mathsf{b}\mathbb{W})^{-1}\mathbf{1}, \delta\alpha\rangle = \langle\mathbf{c}_{\mathsf{b}}, \delta\alpha\rangle, \tag{20.27}$$

using self-adjointness of $(I - \mathsf{b}\mathbb{W})^{-1}$. Equation (20.27) is the key-player principle in graphon form: to maximize the aggregate response per unit of (suitably normalized) incentive, the planner should concentrate $\delta\alpha$ on the types of highest unweighted centrality $\mathbf{c}_{\mathsf{b}}$. We summarize.

Proposition 20.20 (Comparative statics and optimal targeting)**.** *In the subcritical LQ graphon game:*

(i) (Monotone response.) *For complements* $\mathsf{b} \geq 0$*, the equilibrium response operator* $(I - \mathsf{b}\mathbb{W})^{-1}$ *has a nonnegative kernel; hence* $\delta\alpha \geq 0$ *implies* $\delta s^* \geq 0$ *pointwise: raising any type's standalone return raises every type's equilibrium effort.*

(ii) (Centrality ranks the marginal value of targeting.) *The marginal effect on aggregate effort of an incentive placed on type* x_0 *(formally* $\delta\alpha = \varepsilon\, \delta_{x_0}$*, or a narrow bump around* x_0*) is, to first order,* $\varepsilon\, \mathbf{c}_{\mathsf{b}}(x_0)$*. The optimal type to target, under a unit budget* $\|\delta\alpha\|_{L^2} \leq 1$*, is the direction* $\delta\alpha = \mathbf{c}_{\mathsf{b}} / \|\mathbf{c}_{\mathsf{b}}\|$*, achieving* $\delta\Sigma = \|\mathbf{c}_{\mathsf{b}}\|_{L^2}$*.*

(iii) (Amplification grows with coupling.) *Each entry of* $\mathbf{c}_{\mathsf{b}}$ *is nondecreasing in* $\mathsf{b} \in [0, 1/\lambda_1)$*, and* $\|\mathbf{c}_{\mathsf{b}}\| \to \infty$ *as* $\mathsf{b} \uparrow 1/\lambda_1$*: the same intervention has unboundedly larger leverage as the network approaches criticality, with the leverage concentrating on the eigenvector-centrality direction* ϕ_1 *by theorem 20.19.*

Proof. (i) The Neumann kernel $\sum_k \mathsf{b}^k W^{(k)}$ is a sum of nonnegative iterated kernels for $\mathsf{b} \geq 0$; nonnegative kernels map nonnegative functions to nonnegative functions. (ii) is eq. (20.27) with $\delta\alpha$ a unit bump, optimized by Cauchy–Schwarz, whose equality case is $\delta\alpha \propto \mathbf{c}_{\mathsf{b}}$. (iii) Each entry of $\mathbf{c}_{\mathsf{b}} = \sum_k \mathsf{b}^k \mathbb{W}^k \mathbf{1}$ is a power series in b with nonnegative coefficients, hence nondecreasing; and projecting onto ϕ_1, $\langle \mathbf{c}_{\mathsf{b}}, \phi_1 \rangle = \langle \mathbf{1}, \phi_1 \rangle / (1 - \mathsf{b}\lambda_1) \to \infty$ forces $\|\mathbf{c}_{\mathsf{b}}\| \to \infty$, with the alignment from theorem 20.19. □

These statements are the static payoff of the whole construction: once the equilibrium is identified with a centrality vector, ranking players, predicting the sign and size of spillovers, and choosing where to intervene all become spectral computations on $\mathbb{W}$. The dynamic chapters inherit this vocabulary; the "key player" of proposition 20.20 becomes the optimally targeted type in the dynamic interventions of chapter 30.

20.6 Worked computation: the static Thread B game

We now introduce the static incarnation of Thread B (the graphon LQG model of appendix A.7) and solve it on the three diagnostic kernels. This is the chapter's carried-thread duty: it instantiates the abstract theory on the exact model that the dynamic chapters will carry forward, and it exhibits the aggregate $z(x) = (\mathbb{W}\bar{m})(x)$ in its simplest form.

20.6.1 The static reduction of Thread B

Worked computation 20.21 (Static Thread B: graphon-aggregate best response)**.** Recall the dynamic Thread B model of appendix A.7: a type-x agent controls a scalar state X_t^x and is penalized for deviating from $s\, z_t(x)$, where $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ is the graphon-aggregated neighbor mean and s is the fixed coupling strength. In the *static reduction* we freeze time: each type chooses a single stationary position $\bar{m}(x) \in \mathbb{R}$ (its equilibrium mean state), pays a quadratic cost of effort to hold a position away from a neutral target—which we normalize to a type-dependent ideal point $\eta(x)$—and a quadratic congestion/imitation cost for deviating from the aggregate it feels. Concretely, type x chooses $\bar{m}(x)$ to minimize the static cost

$$\mathcal{C}_x(\bar{m}(x); z(x)) \;=\; \tfrac{1}{2}\big(\bar{m}(x) - \eta(x)\big)^2 \;+\; \tfrac{q}{2}\big(\bar{m}(x) - s\,z(x)\big)^2, \qquad z(x) = (\mathbb{W}\bar{m})(x), \tag{20.28}$$

with $q \geq 0$ the relative weight of the congestion term and s the Thread B coupling strength. The first-order condition $\partial_{\bar{m}(x)}\mathcal{C}_x = 0$ reads $(\bar{m}(x) - \eta(x)) + q(\bar{m}(x) - s\,z(x)) = 0$, i.e.

$$\bar{m}(x) \;=\; \frac{1}{1+q}\,\eta(x) \;+\; \frac{qs}{1+q}\,(\mathbb{W}\bar{m})(x). \tag{20.29}$$

This is exactly the linear graphon equation eq. (20.8) with the dictionary

$$\text{strategy } s_{\text{gen}} \leftrightarrow \bar{m}, \qquad \alpha \leftrightarrow \frac{\eta}{1+q}, \qquad \mathsf{b} \leftrightarrow \frac{qs}{1+q}, \qquad L_{\text{br}} = |\mathsf{b}| = \frac{q|s|}{1+q}. \tag{20.30}$$

The static graphon game is therefore an LQ game in the sense of example 20.3, and all of theorems 20.14, 20.15, 20.18 and 20.19 and proposition 20.20 apply verbatim. The static Thread B equilibrium exists, and is unique, whenever the equilibrium operator $I - \mathsf{b}\mathbb{W}$ is invertible; by theorem 20.15 the clean sufficient condition is the spectral one

$$\mathsf{b}\,\lambda_1 \;=\; \frac{qs}{1+q}\,\lambda_1 \;<\; 1 \quad (s>0,\text{ complements}), \qquad |\mathsf{b}|\,|\lambda_{\min}| \;=\; \frac{q|s|}{1+q}\,|\lambda_{\min}| \;<\; 1 \quad (s<0,\text{ substitutes}), \tag{20.31}$$

the substitute branch being automatic for all $s \leq 0$ only when $\mathbb{W}$ is positive semidefinite ($\lambda_{\min} = 0$). When unique it is given by the centrality formula $\bar{m}^* = (I - \mathsf{b}\mathbb{W})^{-1}\frac{\eta}{1+q} = \frac{1}{1+q}\mathsf{c}_{\mathsf{b}}[\eta]$.

(a) Constant kernel $W \equiv c$ (Curie–Weiss / Thread A shadow). Here $\mathbb{W}\bar{m} = c\int_0^1 \bar{m} = c\,\overline{\bar{m}}$, the population grand mean, so the aggregate is the same scalar for every type and eq. (20.29) reduces to a single scalar self-consistency. With $\eta \equiv \eta_0$ constant the equilibrium is uniform, $\bar{m}^*(x) \equiv \mu^*$, where $\mu^* = \frac{\eta_0}{1+q} + \mathsf{b}c\,\mu^*$, giving

$$\mu^* = \frac{\eta_0/(1+q)}{1 - \mathsf{b}c} = \frac{\eta_0}{1+q-qsc}, \qquad \text{well defined iff } \mathsf{b}c = \frac{qsc}{1+q} < 1.$$

This is precisely the static, anonymous LQG closure—the Thread A mean-field fixed point with the global coupling rescaled by c—recovering the constant-graphon reduction promised in appendix A.7. The static uniqueness boundary here is $\mathsf{b}c = \frac{qsc}{1+q} = 1$, i.e. $s_{\text{crit}} = (1+q)/(qc)$. This is the static shadow of the Thread A critical attraction line $s = 1$ of proposition 11.8 (the $a = 0$ multiplicity threshold of remark 11.9) in a *structural*, not a numerical, sense: both mark the point at which the effective coupling reaches the inverse of the top eigenvalue, so that the self-consistent mean ceases to be uniquely determined. The two are not related by a direct substitution of parameters—Thread A's $s = 1$ is independent of q, whereas the static threshold $s_{\text{crit}} = (1+q)/(qc)$ depends on q and equals 1 (at $c = 1$) only in the strong-congestion limit $q \to \infty$, where $\mathsf{b} = \frac{qs}{1+q} \to s$. As section 20.7 explains, the static b is a Riccati composite of the dynamic data, not a literal (q, c) reading of s; the correspondence is the alignment of two "coupling hits the inverse top eigenvalue" thresholds, not a parameter identity.

(b) Rank-one kernel $W(x,y) = f(x)f(y)$ (the clean GMFG reduction). By worked computation 2.28(b), $\mathbb{W}g = \langle g, f\rangle f$ and $\|\mathbb{W}\| = \lambda_1 = \int_0^1 f^2$. The aggregate collapses to a single scalar *order parameter* times the fixed profile f: $z(x) = (\mathbb{W}\bar{m})(x) = \varrho\, f(x)$ with $\varrho := \langle \bar{m}, f\rangle$. Inserting into eq. (20.29),

$$\bar{m}(x) = \frac{\eta(x)}{1+q} + \mathsf{b}\,\varrho\, f(x).$$

Projecting onto f (taking $\langle \cdot, f\rangle$ of both sides) closes the system on the single scalar ϱ:

$$\varrho = \langle \bar{m}, f\rangle = \frac{\langle \eta, f\rangle}{1+q} + \mathsf{b}\,\varrho\,\|f\|_{L^2}^2 = \frac{\langle \eta, f\rangle}{1+q} + \mathsf{b}\lambda_1\,\varrho \quad\Longrightarrow\quad \varrho^* = \frac{\langle \eta, f\rangle}{(1+q)(1-\mathsf{b}\lambda_1)},$$

well defined exactly under eq. (20.31). The full equilibrium is then the explicit profile

$$\bar{m}^*(x) = \frac{\eta(x)}{1+q} + \frac{\mathsf{b}\,\langle \eta, f\rangle}{(1+q)(1-\mathsf{b}\lambda_1)}\, f(x), \tag{20.32}$$

a baseline term proportional to the type's own ideal point plus a network term proportional to the eigenvector centrality f with a coefficient that diverges as $\mathsf{b}\lambda_1 \uparrow 1$—the rank-one instance of theorem 20.19, here exact rather than asymptotic because there is only one nonzero mode. The whole continuum game has collapsed to a single scalar equation for ϱ^*: this is the cleanest worked graphon equilibrium, and it is the static skeleton of the dynamic rank-one reduction taken up in chapter 22.

(c) Block kernel (finite multi-population reduction). Let W be piecewise constant on a partition $[0,1] = \bigsqcup_{k=1}^K I_k$ with $|I_k| = p_k$ and $W(x,y) = \mathsf{B}_{k\ell}$ for $x \in I_k, y \in I_\ell$, the symmetric block (stochastic-block-model) graphon. Because W is block-constant, the aggregate felt by a type depends only on its block: for $x \in I_k$, $(\mathbb{W}\bar{m})(x) = \sum_\ell \mathsf{B}_{k\ell}\int_{I_\ell}\bar{m} = \sum_\ell \mathsf{B}_{k\ell}\, p_\ell\, \mathfrak{m}_\ell$ with $\mathfrak{m}_\ell := \frac{1}{p_\ell}\int_{I_\ell}\bar{m}$ the block-averaged mean. Hence the *aggregate* term in the first-order condition eq. (20.29) is block-constant, but the equilibrium profile itself is *not* piecewise constant unless η is: taking the block average of eq. (20.29) closes a $K \times K$ linear system on the block means,

$$\mathfrak{m}_k = \frac{\bar{\eta}_k}{1+q} + \mathsf{b}\sum_{\ell=1}^K \mathsf{B}_{k\ell}\, p_\ell\, \mathfrak{m}_\ell, \qquad \bar{\eta}_k = \frac{1}{p_k}\int_{I_k}\eta, \tag{20.33}$$

i.e. $(I - \mathsf{b}\,\mathsf{B}\,\mathrm{P})\,\mathbf{m} = \frac{1}{1+q}\bar{\boldsymbol{\eta}}$ with $\mathrm{P} = \mathrm{diag}(p_1, \ldots, p_K)$, and the full equilibrium is then recovered pointwise as

$$\bar{m}^*(x) = \frac{\eta(x)}{1+q} + \mathsf{b}\sum_{\ell=1}^K \mathsf{B}_{k\ell}\, p_\ell\, \mathfrak{m}_\ell \qquad (x \in I_k), \tag{20.34}$$

which carries the within-block variation of $\eta(x)$ on top of the block-constant network term; it collapses to the piecewise-constant profile $\bar{m}^* \equiv \mathfrak{m}_k$ on I_k exactly when η is itself block-constant. The block means $\mathfrak{m}_k$ thus solve the equilibrium of a finite K-population network game, with $\|\mathbb{W}\|$ now the top eigenvalue of the $K \times K$ matrix $\mathsf{B}\,\mathrm{P}$ (similar to the symmetric $\mathrm{P}^{1/2}\mathsf{B}\,\mathrm{P}^{1/2}$, hence real). Uniqueness of the block system holds for complements iff $\mathsf{b}\,\lambda_1(\mathsf{BP}) < 1$ and for substitutes iff $|\mathsf{b}|\,|\lambda_{\min}(\mathsf{BP})| < 1$; an SBM block matrix can have negative eigenvalues, so even within this worked example strong substitutes can break uniqueness, in accordance with theorem 20.15. The graphon game thus contains finite multi-population games as its block special case—the bridge to chapter 30—and the operator-norm threshold reduces to the spectral radius of a finite matrix.

The three reductions confirm the dictionary entries promised in appendix A.7: constant W gives back the anonymous (Thread A) closure, rank-one W collapses to a one-parameter self-consistent equation, and block W recovers finite multi-population games. In every case the existence/uniqueness boundary is $\mathsf{b}\,\|\mathbb{W}\| = 1$, the spectral threshold of theorem 20.14.

20.7 The static game as the stationary shadow

It remains to say in what precise sense this entire static theory is the "stationary shadow" of the dynamic graphon mean field game that occupies the rest of Part IV. The phrase is not merely an analogy; it is a limit, and identifying it now orients the reader for chapters 21, 22 and 24.

Consider the dynamic Thread B model with a discount rate $\beta > 0$ and an infinite horizon, or equivalently a long horizon $T \to \infty$ with ergodic averaging. The type-x agent solves a stationary stochastic control problem whose data depend on the population only through the time-frozen aggregate $z(x) = (\mathbb{W}\bar{m})(x)$, where $\bar{m}(y)$ is now the *stationary* mean of the type-y population. Two facts, both established in the classical theory and lifted to graphons in chapters 22 and 24, make the reduction exact in the LQG case:

1. For a fixed aggregate field $z(\cdot)$, the stationary LQG control problem of each type has an optimal feedback that is affine in the state, and the induced stationary population mean $\bar{m}(x)$ is an affine function of the felt aggregate $z(x)$ (the static reduction of the Riccati/mean system of section 14.4). This is the map $z(x) \mapsto \bar{m}(x)$ whose graph is the best response eq. (20.29).

2. Closing the loop, the stationary mean field must reproduce the aggregate that generated it: $z = \mathbb{W}\bar{m}$. This self-consistency is exactly the GNE condition eq. (20.5), with the per-type stationary mean $\bar{m}$ in the role of the strategy profile.

Thus the stationary graphon mean field equilibrium *is* a graphon Nash equilibrium of a static graphon game whose payoff is the type's stationary cost-to-hold-a-mean. The coupling coefficient b of the static game is the composite of the dynamic congestion weight, the discount rate, and the Riccati gain; the operator $\mathbb{W}$ is untouched. In particular the critical coupling $\mathsf{b}\,\|\mathbb{W}\| = 1$ of theorem 20.14 is the static face of the uniqueness threshold for the dynamic graphon mean field game, the threshold that reappears—now competing with strategic monotonicity—as assumption **(A-Coupling)** in chapter 24. This is why the static analysis is worth doing first: every structural feature of the dynamic theory (existence by compactness of $\mathbb{W}$, the spectral uniqueness threshold, centrality as the equilibrium order parameter, the susceptibility divergence at criticality) is already visible, and provable in a paragraph, before a single PDE is written down.

Remark 20.22 (What the static theory cannot see)**.** The reduction is honest about its losses. Freezing time discards the transient forward–backward coupling that is the structural signature of mean field games (appendix A.1): the static game has no terminal condition, no backward value propagation, and no Fokker–Planck transport. It can therefore say nothing about how an equilibrium is approached, about out-of-equilibrium dynamics, or about phenomena—common noise, time-inconsistency, finite-horizon boundary layers—that are invisible to the stationary fixed point. The static game is the t-independent skeleton; chapters 21 to 24 and 26 put the flesh back on.

Rigor ledger

Proved.

- theorem 20.9: under **(A-Reg-W)** and strong concavity (assumption 20.5), a static graphon game has a GNE—by Schauder (theorem 2.38) using compactness of $\mathbb{W}$, for bounded actions with no coupling restriction, and for unbounded actions under subcriticality. Set-valued version by Kakutani (remark 20.11).
- lemma 20.6: the best response is Lipschitz in the aggregate with constant $L_{\text{br}} = \ell/\theta$, $= |\mathsf{b}|$ for the LQ payoff.
- lemma 20.12: for a nonnegative symmetric kernel, $\|\mathbb{W}\| = \lambda_1 > 0$ with nonnegative top eigenfunction (eigenvector centrality).
- theorem 20.14: under **(A-Coupling)** in the form $L_{\text{br}} \|\mathbb{W}\| < 1$, the GNE is unique and the best-response iteration converges geometrically.
- theorem 20.15: for the LQ game, uniqueness $\Leftrightarrow$ positive definiteness of $I - \mathsf{b}\mathbb{W} \Leftrightarrow \mathsf{b}\lambda_n < 1$ for all n; complements share the threshold $\mathsf{b}\lambda_1 = \mathsf{b}\|\mathbb{W}\| < 1$ with contraction, while substitutes ($\mathsf{b} < 0$) are unique iff $|\mathsf{b}|\,|\lambda_{\min}| < 1$—unconditionally so only for positive semidefinite graphons ($\lambda_{\min} = 0$).
- theorems 20.18 and 20.19: the subcritical LQ equilibrium is Katz–Bonacich centrality $(I - \mathsf{b}\mathbb{W})^{-1}\alpha$; as $\mathsf{b} \uparrow 1/\lambda_1$ it aligns (after normalization) with eigenvector centrality ϕ_1 and diverges.
- proposition 20.20: monotone comparative statics, the centrality-ranked key-player principle, and divergence of intervention leverage at criticality.
- worked computation 20.21: the static Thread B game is LQ; solved in closed form on the constant, rank-one, and block kernels, with the spectral threshold $\mathsf{b}\|\mathbb{W}\| = 1$ in each case.

Assumed.

- **(A-Reg-W)**: W symmetric, measurable, bounded; $\mathbb{W}$ self-adjoint Hilbert–Schmidt (imported from chapters 2, 15 and 18; proposition 2.24).
- Concavity/strong-concavity of payoffs (assumption 20.5); for the LQ benchmark this is automatic.
- The spectral theorem and min–max for compact self-adjoint operators (theorem 2.26 and corollary 2.27) and the three fixed-point theorems (theorems 2.36, 2.38 and 2.40), all proved in chapter 2.

Deferred.

- The full graphon spectral theory and the rigorous standing of eigenvector centrality and the spectral gap → chapter 18.
- Justification of the static game as a limit of finite network games on sampled graphs $G(N, W)$ → chapters 19 and 25.
- The nonlinear monotone-operator uniqueness theory beyond the LQ potential case, and its measure-valued dynamic form → chapter 24 (static specialization of definition 11.1).
- The precise dynamic-to-static reduction (section 20.7): stationary graphon LQG control and the identification of b with the discounted Riccati gain → chapters 22 and 24.

Open.

- Equilibrium selection among multiple GNE in the supercritical complements regime $\mathsf{b}\,\|\mathbb{W}\| \geq 1$ (the static face of the selection problem of chapter 31).
- Identifiability and estimation of W from observed equilibrium efforts—using eq. (20.23) as an inverse problem—and its stability under cut-metric perturbation of the network (ties to chapters 17 and 25).

Bibliographic notes

The static graphon game, the graphon aggregate $\mathbb{W}s$, and the program of analyzing network games and interventions through the limiting kernel are due to Parise and Ozdaglar [108], whose linear–quadratic benchmark, contraction-based uniqueness under an operator-norm condition, and centrality-based intervention analysis are the backbone of sections 20.2, 20.4 and 20.5; our theorem 20.14 and the key-player principle proposition 20.20 follow their development, recast in the Hilbert-space language of chapter 2. The Katz–Bonacich centrality and its role as the equilibrium of a linear–quadratic network game with strategic complements originate in the network-games literature that [108] lifts to the graphon limit; the emergence of eigenvector centrality at criticality (theorem 20.19) is the spectral restatement we have given. The stochastic and measure-valued generalization of the static graphon game—where each type chooses a distribution rather than a scalar action—is the subject of Carmona, Cooney, Graves, and Laurière [39], "Stochastic graphon games: the static case," which connects this chapter to the probabilistic GMFG of chapter 23. The operator-theoretic facts we import—self-adjoint Hilbert–Schmidt structure, the spectral theorem, and the min–max characterization of the top eigenvalue—are standard and are proved in chapter 2 following [113, 50, 31]; the fixed-point theorems are from [79]. The reading of the critical coupling $\mathsf{b}\,\|\mathbb{W}\| = 1$ as a symmetry-breaking threshold continues the Lasry–Lions multiplicity discussion of chapter 11 [93, 36], and is taken up in its dynamic graphon form, with the competition between monotonicity and $\|\mathbb{W}\|$, in chapter 24 following the graphon mean field game program of Caines and Huang [33, 34]. For the graph-limit prerequisites—the pixel picture, block and rank-one graphons, and the operator $\mathbb{W}$—see chapters 15 and 18 and Lovász [99].

Chapter 21

Setting Up the Dynamic Graphon Mean Field Game

The previous chapter, chapter 20, studied *static* graphon games: a continuum of players indexed by a type $x \in [0,1]$, each choosing a one-shot action, coupled to a graphon aggregate $z(x) = \int_0^1 W(x,y)\, s(y)\, \mathrm{d}y$ of the others' actions, with equilibrium controlled by the operator norm $\|\mathbb{W}\|$ (lemma 20.12). That chapter is the *stationary shadow* (section 20.7) of the object we now build. A shadow is flat: nothing in it moves, each player commits to a single number, and the game is settled in one shot. The present chapter restores the missing dimension — time. We promote every player to a genuinely *dynamic* agent who carries a private state that diffuses on $[0,T]$ and is steered, moment by moment, by an optimally chosen control; the static action becomes a feedback policy, and the one-shot fixed point becomes a fixed point over an entire trajectory of population laws.

The purpose of the present chapter is narrow, and the narrowness is deliberate. We do *not* solve the resulting equilibrium here; we *set it up*. The distinction is worth dwelling on, because it is the organizing decision of the chapter. *Saying what an equilibrium is* means writing down, unambiguously, the agents, the population object, the space that object lives in, the topology on that space, the optimization each agent performs, and the consistency condition that closes the loop — so that the sentence "$(\boldsymbol{m}_t)_t$ is an equilibrium" acquires a definite truth value. *Proving that one exists* (and is unique, and can be computed) is a different and much harder task that presupposes the first is already done. We separate them on purpose, and the payoff of the separation is concrete: the next three chapters each consume *this* setup, untouched, and supply one engine apiece. chapter 22 turns the fixed point into a coupled backward–forward partial differential system and analyses it; chapter 23 recasts the same fixed point probabilistically as a system of graphon forward–backward stochastic differential equations (FBSDEs); chapter 24 proves existence and uniqueness. If the setup here were even slightly ambiguous — if the state space, the metric, or the coupling were left informal — those three chapters would each have to re-fix it, and they would risk fixing it differently. Investing the precision once, here, is what lets them agree (definitions 21.5, 21.8 and 21.16); the analytic engines then merely specialize the general agent class of definitions 21.1 and 21.6 to the control-affine, aggregate-in-cost subclass they solve in closed form.

The chapter has one destination and one test of honesty. The destination is the equilibrium definition (definition 21.16): the type-by-type Nash fixed point of the best-response map. The test is that the definition degenerate correctly, which we verify three times at the end — constant W must give back the classical mean field game of Part II, rank-one W must collapse the infinite field of measures to a single scalar self-consistency, and block W must reproduce finite multi-population mean field games. Everything between this opening and those checks exists to make the destination precise and the checks meaningful. The arc between them is a sequence of structural choices, each forced by the single move from one representative agent to a representative agent of every type, and each made honestly: the continuum of independent noises is carried not by a literal ensemble of trajectories but at the level of the deterministic field of laws (remark 21.2); that field is housed in a complete separable L^2-Wasserstein space (definition 21.8 and proposition 21.9); the graphon aggregate each agent feels is a $\|\mathbb{W}\|$-Lipschitz operation on that space, which is exactly where the coupling threshold $|s|\ \|\mathbb{W}\|$ is born (lemma 21.11 and remark 21.12); and equilibrium is the type-by-type Nash fixed point of the best-response map (definition 21.16).

To get there we import three streams, and it is worth saying in one breath why each is needed and where it will surface. The first is the Itô calculus of chapter 5: it furnishes the controlled stochastic differential equation that *is* the agent (theorem 5.10 guarantees the agent's state is a well-defined process), the generator $\mathcal{A}$ (definition 5.12) that will appear in the agent's HJB equation, and the Fokker–Planck equation (theorem 5.14) that pushes the agent's law forward in time and so generates the very population field the next agent responds to — without this stream there is no agent and no population. The second stream is classical mean field game theory (chapters 7, 8 and 14): the McKean–Vlasov mechanism (definition 7.7) is the exact template we are generalizing — one representative agent, one self-consistent measure — and its coupled HJB–Fokker–Planck system (eq. (8.5)) together with the fully solved linear–quadratic–Gaussian (LQG) fixed point (theorem 14.6) is the x-independent special case our setup must reproduce, which is exactly what the constant-graphon reduction will check; without this stream we would not know what "correct" means. The third stream is the graphon machinery of chapters 15, 17 and 18: the kernel W and its integral operator $\mathbb{W}$, the self-adjoint Hilbert–Schmidt structure granted by (A-Reg-W), and the rank-one spectral reduction. This is the stream that does the new work — it is what turns "one population" into "a field of populations coupled through a network" — and it surfaces the moment we define the aggregate an agent feels, returning in force when the operator norm $\|\mathbb{W}\|$ sets the coupling threshold. Beneath all three runs the Wasserstein geometry of $\mathcal{P}_2(\mathbb{R}^d)$ (chapter 4): it is the metric in which every population object is measured, and the reason the field space we build will be complete and separable.

Two global conventions are in force throughout, and because both a value function and a density appear in the equilibrium definition we restate the first one *and* motivate it. *Value functions $u = u$ carry terminal data at $t = T$ and are computed backward; population densities $m = m$ carry initial data at $t = 0$ and evolve forward.* The asymmetry is not bookkeeping; it reflects the arrow of causation in an optimization. An agent's value at time t is the best it can still achieve over the *remaining* horizon $[t, T]$, so it is determined by the future: the terminal cost is known and the value propagates *back* from $t = T$. A population's law at time t is the accumulated effect of where it started and how it has been driven over $[0, t]$, so it is determined by the past: the initial distribution is known and the density propagates *forward* from $t = 0$. Backward value, forward density; the two flows meet only at the fixed point, where the feedback computed backward generates exactly the law that flows forward. The second convention concerns a clash of letters that becomes acute precisely here, and we flag it before writing a single stochastic differential equation.

Notation watch

This is the first chapter of Part IV in which a graphon W and a Brownian motion appear in the same equation, so the global resolution announced in appendix A takes effect. *Whenever a graphon is in scope, the kernel is $W = W(x,y)$ and Brownian motion is written B_t.* We never write W_t for noise in this part. The driving noises of the agents are B^x, indexed by type; the kernel is W; the integral operator is $\mathbb{W}$. A reader arriving from Part II, where Brownian motion was W_t, should mentally rename it B_t from here on.

21.1 A continuum of controlled agents indexed by type

The preamble promised that this chapter would say precisely what the agents are. We begin there, with the single elementary unit out of which the whole game is assembled: one representative agent, of one type, and the equation its private state obeys. Everything later in the chapter — the field of measures, its topology, the equilibrium fixed point — is built on top of this unit, so it pays to install it carefully and to keep two conventions from the preamble live while we do.

The first convention is notational and was just flagged in chapter 21: the moment a graphon is in scope, the driving noise of an agent is written B_t, never W_t. The symbol W is reserved for the kernel. A reader who learned the single-agent theory of Part II with Brownian motion called W_t should perform the rename now, because the two W's are about to appear in the same line and only one of them can keep the letter.

The second convention is conceptual and far more consequential, so we restate it with the emphasis of chapter 15. The type label $x \in [0,1]$ is an *exchangeable label* — an index recording which "kind" of agent we are looking at, or equivalently where in the network it sits. It is *not* a spatial coordinate, and it is *not* a momentum. The agent's private state $X_t^x \in \mathbb{R}^d$ is what lives in space (or in whatever $\mathbb{R}^d$ models); the type x merely names the agent. In particular the numerical closeness of two labels — $|x-y|$ small — carries *no* intrinsic meaning whatsoever: relabelling types by a measure-preserving bijection of $[0,1]$ leaves the game unchanged, and all of the network's geometry, who interacts with whom and how strongly, is carried by the kernel $W(x,y)$ and by nothing else. If this still feels unnatural, the cleanest mental picture is the finite one: N agents, agent i assigned a type $\xi_i \in [0,1]$, the type doing no work except indexing the row and column of the interaction matrix $W(\xi_i, \xi_j)$.

With both conventions live we can state the one conceptual move that separates this chapter from Part II. The classical mean field game has *one* representative agent: every player is statistically identical, anonymous, and interacts with the rest only through a single population measure $m_t \in \mathcal{P}_2(\mathbb{R}^d)$. The graphon generalization replaces "one representative agent" by "a representative agent of *every* type." That is the entire leap. Each type now carries its own copy of the single-agent control problem, with its own initial distribution and its own driving noise, and the copies are tied together only through the kernel. Remarkably, almost nothing else changes at the level of a single agent's equation; all of the novelty hides in *what aggregate of the others* the agent responds to. We write the agent's dynamics now, motivate every coefficient, and then spend the rest of the chapter unpacking that aggregate.

It helps to read the controlled Itô equation below one term at a time before meeting it on the page. The state X_t^x is the private variable the type-x agent controls — a position, an inventory, an opinion, whatever the model is about. It changes for two reasons. The first is *drift*: a deterministic push $b(x, X_t^x, \alpha_t^x, \Theta_t(x))$ that the agent partly authors, because the control α_t^x enters it. This is the precise sense in which the agent "steers": it cannot touch the noise, but it can bias the drift, and steering *is* the choice of how to bias it. The drift also depends on the type x (different kinds of agent have different uncontrolled dynamics) and on the aggregate $\Theta_t(x)$ (the agent is pushed by what its neighbors are doing). The second reason the state changes is *noise*: an irreducible random buffeting $\sigma(x, X_t^x)\, \mathrm{d}B_t^x$ that the agent cannot influence at all, with its own type-x Brownian motion B^x and a diffusion coefficient σ that may depend on type and state but *not* on the control — the agent steers the mean, not the volatility. Finally the agent must start somewhere: its initial state is drawn from a type-dependent law m_0^x, because there is no reason different kinds of agent should begin life identically distributed, and the starting point is one of the few honest channels through which heterogeneity enters already at $t = 0$. With those four ingredients named, the equation reads as follows.

Definition 21.1 (The type-x representative agent)**.** Fix a horizon $T \in (0, \infty)$. For each type $x \in [0, 1]$ the representative agent of type x has a private state $X_t^x \in \mathbb{R}^d$ evolving on $[0, T]$ according to the controlled Itô equation

$$\mathrm{d}X_t^x \;=\; b\big(x,\, X_t^x,\, \alpha_t^x,\, \Theta_t(x)\big)\, \mathrm{d}t \;+\; \sigma\big(x,\, X_t^x\big)\, \mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x, \tag{21.1}$$

where $\alpha_t^x \in A \subseteq \mathbb{R}^k$ is the control exerted by the type-x agent, B^x is a d'-dimensional Brownian motion, $m_0^x \in \mathcal{P}_2(\mathbb{R}^d)$ is the type-x initial distribution, and $\Theta_t(x)$ is the *graphon-aggregated mean field* felt by type x, defined in section 21.2. The coefficients b, σ are assumed Borel measurable in x and to satisfy, uniformly in x, the standing assumptions (A-Lip), (A-Growth), (A-Nondeg) of assumption 7.3: Lipschitz and linear growth in (X, α) uniformly in (t, x, Θ), and $\sigma\sigma^\top \succeq \lambda I$ with $\lambda > 0$.

The three standing assumptions imported from assumption 7.3 are not decoration; each buys a specific guarantee, and it is worth recalling what breaks without it. (A-Lip) — Lipschitz dependence of b, σ on (X, α) uniformly in (t, x, Θ) — is what makes the SDE (21.1) have a unique strong solution (theorem 5.10); drop it and two agents started from the same point could split into genuinely different futures, so "the law of the state" would not even be well-defined. (A-Growth) — at most linear growth in the state — prevents the solution from running off to infinity in finite time and keeps its second moment finite, which is exactly what will let every law live in $\mathcal{P}_2(\mathbb{R}^d)$. (A-Nondeg) — $\sigma\sigma^\top \succeq \lambda I$ with $\lambda > 0$ — guarantees the noise genuinely pushes in every direction; it is the parabolicity that will later smooth the agent's value function and density, and it is the one assumption we will occasionally have to flag when it fails. That all three hold *uniformly in x* is what will let us aggregate over the continuum of types without the constants degenerating type by type.

So the whole novelty relative to Part II is concentrated in the single argument $\Theta_t(x)$: the agent's drift, and shortly its cost, depend not on one global population measure but on a *type-local* aggregate of the entire population, gathered through the kernel W. We have deliberately written $\Theta_t(x)$ as an opaque symbol, because it cannot be defined until we have said what the population *is* as an object — a field of measures, one per type — and that is the business of section 21.2. Before we get there, though, we owe the reader an honest account of a subtlety the equation (21.1) quietly introduced the instant we attached a *separate* Brownian motion B^x to each of a continuum of types. It is tempting to treat this as a harmless notational convenience. It is not, and pretending otherwise would poison everything downstream.

Remark 21.2 (On a continuum of independent Brownian motions)**.** Equation (21.1) posits, for each $x \in [0,1]$, an independent Brownian motion B^x. The naive reading is that there sits, on one probability space, a single jointly measurable random field $(x,\omega) \mapsto B^x_t(\omega)$ whose columns are independent and identically distributed across x. *No such object exists in the naive sense*, and the reason is a clean clash with Fubini's theorem that is worth seeing in full, because it dictates the entire way we will pose the game.

Consider the simplest caricature: a family $\{\zeta^x\}_{x\in[0,1]}$ of real random variables that are independent, identically distributed, and nondegenerate (say mean zero and variance $v > 0$), and suppose for contradiction that $(x,\omega) \mapsto \zeta^x(\omega)$ is jointly measurable on $[0,1] \times \Omega$ (Lebesgue $\times\,\mathbb{P}$). Joint measurability is precisely the hypothesis that Fubini–Tonelli needs. Apply it to the cross-type average $S(\omega) := \int_0^1 \zeta^x(\omega)\,\mathrm{d}x$. On one hand, $\mathbb{E}[S] = \int_0^1 \mathbb{E}[\zeta^x]\,\mathrm{d}x = 0$. On the other hand, computing the second moment by Fubini applied to the product,

$$\mathbb{E}\Big[S^2\Big] = \int_0^1\int_0^1 \mathbb{E}[\zeta^x\zeta^y]\;\mathrm{d}x\;\mathrm{d}y = \int_0^1\int_0^1 v\,\mathbf{1}\{x=y\}\;\mathrm{d}x\;\mathrm{d}y = 0,$$

because independence forces $\mathbb{E}[\zeta^x\zeta^y] = 0$ off the diagonal $\{x=y\}$, while the diagonal is a Lebesgue-null subset of the square and contributes nothing. So $\mathbb{E}\big[S^2\big] = 0$, i.e. $S = 0$ almost surely: the cross-type average of a continuum of independent, mean-zero, *nondegenerate* variables would have to be deterministically zero. That much is in fact the "exact law of large numbers" one might *want*. The trouble is that the very same joint measurability, fed through Fubini in the other order, forces almost every *sample path* $x \mapsto \zeta^x(\omega)$ to be a Lebesgue-measurable function of x — and an independent, identically nondegenerate continuum has, by a standard zero–one argument, almost no measurable sample paths at all (the cross-section oscillates like an uncountable sequence of fresh coin-flips and pins down no measurable structure in x). The two consequences of joint measurability are incompatible, so the hypothesis must be denied: there is no jointly measurable, i.i.d., nondegenerate family indexed by $[0,1]$ with Lebesgue measure, and hence the "exact LLN" $x \mapsto X^x_t$ cannot be realized on an ordinary product probability space. What an exact law of large numbers across types would *require* is exactly a space on which the cross-section $x \mapsto X^x_t(\omega)$ is, for almost every fixed ω, a measurable function whose empirical distribution equals the deterministic average $\int_0^1 \mathcal{L}(X^x_t)\,\mathrm{d}x$ of the marginals — an interchange of "average over x" with "expectation over ω" that an ordinary product space cannot supply. There are two honest resolutions, and we use the first throughout.

1. *Distributional (the one we adopt).* We never ask for a joint realization in the first place. For each fixed x, (21.1) is a single, perfectly ordinary, well-posed SDE on its own probability space (theorem 5.10); there is no continuum of coupled trajectories to make sense of. The population

is then described not by realized paths but by the *deterministic* object $x \mapsto m_t^x = \mathcal{L}(X_t^x)$ — the law of the type-x state, one probability measure per type. We will call this map the *field of laws*, and the next section (section 21.2) makes it the central object of the whole chapter: the population *is* this field, not any sample of trajectories. Measurability then becomes a statement about a deterministic map $[0,1] \to \mathcal{P}_2(\mathbb{R}^d)$, not about a random field, and as such it holds and is proved in lemma 21.13. This is precisely the level at which the classical mean field game is already posed — through $\mathcal{L}(X_t)$ rather than realized paths (definition 7.7) — so we are not adding fragility, only being explicit that the continuum lives in the deterministic field and never in a literal ensemble of paths.

2. *Fubini extension (when realizations are genuinely wanted).* If one truly wants a jointly measurable continuum of independent processes — for instance to state and use an exact law of large numbers across types as a *theorem* rather than a motivation — the honest device is a *Fubini extension* of the product space in the sense of Sun [125]: an enlargement of $[0,1] \times \Omega$ on which Fubini's theorem is postulated to hold and on which an essentially pairwise independent continuum *does* exist and *does* satisfy the exact LLN, precisely by sidestepping the obstruction above. We flag this route for completeness and *do not* rely on it. The finite-network theory of chapter 25 reaches the same conclusions by an unimpeachable route, keeping N genuine agents on a sampled graph $G(N, W)$ and passing rigorously to the limit, so nothing in this book ever needs a literal continuum of paths.

The upshot, which the reader should carry forward: no theorem anywhere below depends on a jointly measurable continuum of trajectories. Every statement is phrased through the deterministic field of laws, and that is what keeps the dynamic graphon game on solid measure-theoretic ground.

The abstract agent (21.1) is general by design, since chapters 22 and 23 will need room for nonlinear drifts and state-dependent volatilities. But the thread we carry through the book for concreteness — Thread B, the graphon-LQG model fixed in the thread specification and given its static incarnation in chapter 20 — is the simplest honest instance: scalar state, scalar control, affine dynamics, additive noise. It is worth writing out and checking, line by line, that it really is a special case of definition 21.1, both to anchor the abstractions and to rehearse the assumption checks we will lean on later.

Example 21.3 (Thread B: the graphon-LQG agent)**.** With the fixed parameter names of the thread specification — state feedback a, control gain b, noise level $\sigma = \sigma$ — the type-x agent obeys

$$\mathrm{d}X_t^x \;=\; \big(a\, X_t^x + b\, \alpha_t^x\big)\;\mathrm{d}t \;+\; \sigma\; \mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x, \tag{21.2}$$

a scalar state ($d = 1$) with scalar control. This is definition 21.1 with the affine drift $b(x, X, \alpha, \Theta) = aX + b\alpha$ and the constant diffusion $\sigma \equiv \sigma$. Two features deserve note. First, the drift here happens *not* to depend on the aggregate Θ: in this model the neighbors influence an agent only through its *cost*, not its dynamics, and the coupling will enter in example 21.7. The general definition 21.1 permits Θ in the drift, but Thread B does not exercise that freedom. Second, the volatility σ is constant — independent of type, state, and control — which is the cleanest setting in which nondegeneracy is transparent.

We check the three standing assumptions against this affine model, since the closed-form results of section 21.5 depend on their holding. (A-Nondeg) asks $\sigma\sigma^\top \succeq \lambda I$ for some $\lambda > 0$; in dimension one this reads $\sigma^2 \geq \lambda > 0$, which holds *iff* $\sigma \neq 0$ (one may then simply take $\lambda = \sigma^2$). The degenerate case $\sigma = 0$ turns the SDE into a controlled ODE with no parabolic smoothing and is exactly the situation our nondegeneracy hypothesis excludes. (A-Lip) asks Lipschitz dependence of b, σ on (X, α) uniformly in the remaining arguments: here $\partial_X b = a$ and $\partial_\alpha b = b$ are constants and $\sigma \equiv \sigma$ is constant, so the coefficients are globally Lipschitz with constant $\max\{|a|, |b|\}$ — the assumption holds trivially because the model is affine. (A-Growth) asks at most linear growth in the state: $|b(x, X, \alpha, \Theta)| \leq |a|\ |X| + |b|\ |\alpha|$ and $|\sigma| = |\sigma|$ are visibly affine in (X, α), so linear growth holds with no work. An affine model with additive noise is the generic example for which all three assumptions hold the instant one writes it down; the only genuine demand is the harmless $\sigma \neq 0$. This is the dynamic backbone we advance in section 21.5.

We have now said precisely what a single agent is: a well-posed controlled diffusion, one per type, with type-dependent starting law and its own noise, obeying the three standing assumptions uniformly in x. Exactly one symbol in that description remains undefined — the aggregate $\Theta_t(x)$ that the drift and the cost depend on. We deferred it on purpose, because pinning it down forces us first to say what the *population* is, and the population is no longer a single point of $\mathcal{P}_2(\mathbb{R}^d)$ as it was in Part II. It is the field of laws $x \mapsto m_t^x$ that remark 21.2 just placed at the centre of the construction. The next section builds that field-of-measures object, equips it with the W-weighting that turns it into the type-local aggregate $\Theta_t(x)$ each agent actually feels, and so closes the one gap left open here. With Θ in hand we will be able to write the agent's cost, its best response, and finally the equilibrium fixed point.

21.2 The graphon-coupled mean field: a field of measures

The previous section closed by leaving exactly one symbol undefined, and it left it undefined on purpose: the aggregate $\Theta_t(x)$ that the agent's drift (21.1) — and, in a moment, its cost — depends on. We promised the reader that $\Theta_t(x)$ is "the W-weighted average the type-x agent actually feels." But a weighted average *of what*, taken *over what object*? Answering that is the entire business of this section, and it forces us first to say what the population *is*, because the average is taken over the population and we have not yet pinned the population down as a mathematical object. This is where the dynamic graphon game parts company with the theory of Part II.

In Part II the population was a single point $m_t \in \mathcal{P}_2(\mathbb{R}^d)$: one probability measure summarising the entire anonymous crowd, and the representative agent responded to a scalar or measure functional of that one measure. There was nothing else to respond to. Every agent was statistically identical, so a single law described them all, and "the population at time t" was literally one element of $\mathcal{P}_2(\mathbb{R}^d)$. The moment we gave *every type* its own agent, its own initial law m_0^x, and its own noise B^x (definition 21.1), that single point dissolved. There is no longer one population law; there is one law *per type*, $m_t^x = \mathcal{L}(X_t^x)$, and the type-$x$ agent must be told which combination of these it feels. So the population has been lifted from a *point* of $\mathcal{P}_2(\mathbb{R}^d)$ to a *function of type into measures* — an entire x-indexed family of probability laws — and the aggregate $\Theta_t(x)$ will be a kernel-weighted average over that family. We give the family a name first, then build the average on top of it.

Definition 21.4 (Population field and first-moment field)**.** A *population field* is a map

$$\boldsymbol{m} \;=\; (\, m^x \,)_{x\in[0,1]}\,, \qquad m^x \in \mathcal{P}_2(\mathbb{R}^d),$$

assigning to each type x a probability measure with finite second moment. In the dynamic game the field is time-indexed, $\boldsymbol{m}_t = (m_t^x)_{x\in[0,1]}$ with $m_t^x = \mathcal{L}(X_t^x)$. Its *first-moment field* is the $\mathbb{R}^d$-valued function

$$\bar{m}_t(y) \;:=\; \int_{\mathbb{R}^d} \xi \, m_t^y(\mathrm{d}\xi) \;=\; \mathbb{E}[X_t^y]\,, \qquad y \in [0,1].$$

Two features of definition 21.4 deserve a word before we build on it. First, the field $\boldsymbol{m}_t$ carries strictly more information than any single average of it: it remembers the entire shape of each type's law, not just its mean. We single out the first-moment field $\bar{m}_t$ because Thread B couples only through means, but the construction never collapses the field to its first moment until a particular cost *asks* it to. Second, the field is a *deterministic* object — a fixed map from $[0,1]$ into $\mathcal{P}_2(\mathbb{R}^d)$ — exactly as remark 21.2 insisted: it is the field of laws, not any random ensemble of trajectories, and that is what keeps the continuum on solid measure-theoretic ground. The whole population, at each instant, *is* this map.

Now the agent. The type-x agent does not see the whole field $\boldsymbol{m}_t$, and it would be the wrong model if it did: an agent embedded in a network feels its neighbours, not the global census. What it feels is a W-weighted average of the others' laws, the weight $W(x,y)$ recording how strongly type x is exposed to type y. This single weighting is the only place the network geometry enters the model, and it is worth writing the resulting aggregate in two equivalent ways. One keeps the full measure-valued information and is the right object when an agent's cost is a general functional of its neighbours' *distributions*; the other retains only a scalar (or vector) observable and is the right object — and the computationally tractable one — when the cost depends on the neighbours only through, say, their mean. We will need both pictures repeatedly, so we set them down side by side.

Definition 21.5 (Graphon mean-field aggregate)**.** Let $\boldsymbol{m} = (m^x)_{x\in[0,1]}$ be a population field with $x \mapsto m^x$ Borel measurable into $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$. The *aggregate measure* felt by type x is the (in general sub-probability) finite measure on $\mathbb{R}^d$

$$(\mathbb{W} \circledast \boldsymbol{m})(x) \;:=\; \int_0^1 W(x,y)\, m^y \; \mathrm{d}y \;\in\; \mathcal{M}_+(\mathbb{R}^d), \tag{21.3}$$

defined by its action on bounded Borel test functions h, $\langle h, (\mathbb{W} \circledast \boldsymbol{m})(x)\rangle = \int_0^1 W(x,y)\langle h, m^y\rangle \; \mathrm{d}y$, where $\langle h, \nu\rangle = \int h \; \mathrm{d}\nu$. Its total mass is $(\mathbb{W}\mathbf{1})(x) = \int_0^1 W(x,y) \; \mathrm{d}y \le 1$, so $(\mathbb{W}\circledast\boldsymbol{m})(x)$ is a probability measure only when $\int_0^1 W(x,y)\,\mathrm{d}y = 1$. The pairing also extends to Borel observables of at most linear growth — in particular the *unbounded* identity $h(\xi) = \xi$ — because each $m^y \in \mathcal{P}_2(\mathbb{R}^d)$ has a finite first moment and W is bounded; this is the case that drives Thread B. When the coupling enters through a single such observable h, the relevant object is the *scalar (or vector) aggregate field*

$$\Theta_t(x) \;=\; (\mathbb{W}\,[\,\langle h, m_t^{\cdot}\rangle\,])(x) \;=\; \int_0^1 W(x,y)\,\langle h, m_t^y\rangle \; \mathrm{d}y. \tag{21.4}$$

For Thread B the observable is the identity $h(\xi) = \xi$, so $\langle h, m_t^y \rangle = \bar{m}_t(y)$ and the aggregate is exactly the graphon-smoothed first-moment field

$$z_t(x) \;:=\; (\mathbb{W}\,\bar{m}_t)(x) \;=\; \int_0^1 W(x,y)\,\bar{m}_t(y)\;\mathrm{d}y, \tag{21.5}$$

matching the aggregate of the thread specification and of the static game (worked computation 20.21).

The definition carries a deliberate two-level notation, $\circledast$ versus plain $\mathbb{W}$, and the distinction is worth making explicit because the two operators *look* the same on the page but act on different kinds of object. The symbol $\mathbb{W}\circledast$ in (21.3) is the integral operator $\mathbb{W}$ of definition 2.23 (whose spectral theory is developed in chapter 18) acting *measure-valuedly*: it integrates the measure-valued map $y \mapsto m^y$ against the kernel $W(x,\cdot)$, producing for each x a measure on $\mathbb{R}^d$. There is no average of real numbers here — the "values" being averaged are whole probability laws, and the integral $\int_0^1 W(x,y)\,m^y\;\mathrm{d}y$ is a measure, defined the only way a measure can be defined, by its action on test functions. When instead $\mathbb{W}$ is fed a genuine *scalar* field such as $\bar{m}_t$, the circled symbol is dropped and it is literally the ordinary operator of definition 2.23, $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y$, an honest weighted average of numbers. The two are consistent: pairing the measure-valued aggregate against an observable h and applying the scalar operator to the field $\langle h, m^{\cdot}\rangle$ give the same number, which is exactly the identity $\langle h, (\mathbb{W}\circledast \boldsymbol{m})(x)\rangle = (\mathbb{W}[\langle h, m^{\cdot}\rangle])(x)$ read off from (21.3)–(21.4). One should keep both pictures live: the aggregate is a *measure* when the cost is a general functional of the neighbours' distributions, and it *collapses* to $\mathbb{W}$ acting on a moment field the moment the cost depends on the neighbours only through a single observable — which is the LQG case, (21.5), and the one we actually compute with.

The one feature of (21.3) that has no analogue in Part II, and that a reader can easily misread, is that the aggregate measure is in general a *sub*-probability: its total mass is $(\mathbb{W}\mathbf{1})(x) = \int_0^1 W(x,y)\;\mathrm{d}y$, the row sum of the kernel, which the standing assumption $0 \le W \le 1$ caps at 1 but does not force to equal 1. This is not a defect of the definition; it is the correct bookkeeping of how *connected* type x is. A type with $\int_0^1 W(x,y)\,\mathrm{d}y < 1$ is weakly coupled to the rest of the network — it has few or faint neighbours — and it should feel a correspondingly *lighter* aggregate, a measure of mass less than one. Only when type x is fully connected, $\int_0^1 W(x,y)\,\mathrm{d}y = 1$, does $(\mathbb{W}\circledast\boldsymbol{m})(x)$ come out a bona fide probability measure. The lesson is that one must *not* renormalise the aggregate to total mass one by hand: doing so would erase exactly the heterogeneity in connectivity that the graphon is there to encode, and it would break the linearity of $\mathbb{W}$ on which every estimate below rests. The scalar form (21.4) inherits the same feature painlessly — it is simply a kernel-weighted (not kernel-*averaged*) sum of the observable, so a poorly connected type contributes a smaller number — which is one more reason Thread B prefers to work with it.

With the aggregate $\Theta_t(x)$ now a defined object — the scalar field (21.4), or in the LQG case $z_t(x)$ of (21.5) — the symbol that definition 21.1 carried as a placeholder has acquired its meaning, and we can finish specifying the game by writing down what each agent is trying to minimise. As in Part II, the type-x agent minimises an expected cost over its admissible controls, and it does so taking the population field flow $(\boldsymbol{m}_t)_{t\in[0,T]}$ — hence the aggregate flow $t \mapsto \Theta_t(x)$ — as *given*. This is the infinitesimal-agent convention inherited unchanged from mean field game theory: a single agent is one of a continuum and cannot, by itself, move the aggregate, so during its own optimisation the field is a fixed exogenous input. The self-consistency between the field an agent responds to and the field its optimal behaviour generates is imposed only later, at the equilibrium fixed point (definition 21.16); here the field is frozen and the agent simply solves a standard control problem against it.

Definition 21.6 (Cost functional)**.** Given a measurable field flow $\boldsymbol{m} = (\boldsymbol{m}_t)_{t\in[0,T]}$, an admissible control α^x (definition 21.14) for type x incurs the cost

$$J^x(\alpha^x;\boldsymbol{m}) \;=\; \mathbb{E}\left[\int_0^T L\big(x, X_t^x, \alpha_t^x, \Theta_t(x)\big)\;\mathrm{d}t \;+\; g\big(x, X_T^x, \Theta_T(x)\big)\right], \tag{21.6}$$

with running cost L and terminal cost g Borel in x and satisfying the convexity/coercivity needed for the Hamiltonian

$$H(x,X,p,\Theta) \;=\; \sup_{\alpha\in A}\big\{-\;p\cdot b(x,X,\alpha,\Theta) - L(x,X,\alpha,\Theta)\big\}$$

to be well-defined and convex in p under (A-Conv). The optimal feedback is the maximizer

$$\alpha^*(x,X,p,\Theta) \;=\; \operatorname*{arg\,max}_{\alpha\in A}\big\{-\;p\cdot b(x,X,\alpha,\Theta) - L(x,X,\alpha,\Theta)\big\}. \tag{21.7}$$

We caution that α^* is *not* $-\partial_p H$ when the drift is genuinely control-dependent: by the envelope theorem $\partial_p H = -b(x,X,\alpha^*,\Theta)$, so $-\partial_p H$ returns the optimal *drift*, not the control. The two coincide only in the reduced convention $b=\alpha$. For the control-affine drift $b = b_0(x,X,\Theta)+B\alpha$ with strongly convex running cost — the case of every worked example below — (21.7) is the explicit first-order condition $B^\top p + \partial_\alpha L = 0$, which for quadratic $L = \frac{1}{2}|\alpha|^2 + \cdots$ gives $\alpha^* = -B^\top p$ (scalar LQG: $\alpha^* = -bp$).

The cautionary clause inside definition 21.6 is compressed enough to be easy to misread, so we spell out the one-line derivation that justifies it; it is load-bearing because every HJB equation in chapters 22 and 23 is written with the Hamiltonian, and confusing α^* with $-\partial_p H$ would silently corrupt the optimal control whenever the drift is not the control itself. Write the bracket being maximised as $\mathcal{H}(x,X,p,\alpha,\Theta) := -\,p\cdot b(x,X,\alpha,\Theta) - L(x,X,\alpha,\Theta)$, so that $H(x,X,p,\Theta) = \mathcal{H}(x,X,p,\alpha^*(x,X,p,\Theta),\Theta)$ with α^* the maximiser. Differentiating in p and using the chain rule,

$$\partial_p H \;=\; \partial_p\mathcal{H}\big|_{\alpha=\alpha^*} \;+\; \big(\partial_\alpha\mathcal{H}\big|_{\alpha=\alpha^*}\big)\,\partial_p\alpha^* \;=\; -\,b(x,X,\alpha^*,\Theta) \;+\; 0,$$

where the second term vanishes by the first-order optimality condition $\partial_\alpha\mathcal{H}|_{\alpha=\alpha^*} = 0$ — this cancellation *is* the envelope theorem. Hence $\partial_p H = -b(x,X,\alpha^*,\Theta)$, so $-\partial_p H$ recovers the optimal *drift* $b(x,X,\alpha^*,\Theta)$, which equals the optimal *control* α^* only in the reduced convention $b=\alpha$ where drift and control are literally the same variable. In the control-affine setting $b = b_0(x,X,\Theta)+B\alpha$ the optimal drift is $b_0 + B\alpha^* = b_0 - BB^\top p$, manifestly not $\alpha^* = -B^\top p$ unless $B=I$ and $b_0 = 0$. We retain the explicit maximiser (21.7) as the definition of α^* precisely so that no such confusion can arise downstream.

Example 21.7 (Thread B: the graphon-LQG cost)**.** With $z_t(x)$ from (21.5) and parameters $q, q_T \geq 0$ (state-cost weights) and s, s_T (mean-coupling strengths), the type-x agent's cost is

$$J^x(\alpha^x;\boldsymbol{m}) = \mathbb{E}\left[\int_0^T \Big(\tfrac{1}{2}(\alpha_t^x)^2 + \tfrac{q}{2}\big(X_t^x - s\,z_t(x)\big)^2\Big)\,\mathrm{d}t + \tfrac{q_T}{2}\big(X_T^x - s_T\,z_T(x)\big)^2\right], \tag{21.8}$$

exactly the thread-specification cost, and the place where the kernel coupling finally reaches the agent: the neighbours enter only here, through the term $\frac{q}{2}(X_t^x - s\, z_t(x))^2$, which penalises the squared distance between the agent's own position and the graphon-smoothed average position $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ of its neighbours. For $s > 0$ this draws each agent *toward* the crowd (imitation/congestion-seeking); for $s < 0$ it pushes it *away* (anti-coordination); and s multiplies the operator norm $\|\mathbb{W}\|$ to form the coupling strength $|s|\ \|\mathbb{W}\|$ that remark 21.12 will identify as the control parameter of the whole theory. Note that in this thread the drift $b = aX + b\alpha$ is *free* of Θ, so the aggregate sits entirely in the cost — consistent with example 21.3, where we already observed that Thread B couples through cost and not through dynamics.

Let us compute the Hamiltonian explicitly, both to check convexity and to see the scalar-LQG HJB fall out. With $b = aX + b\alpha$ and $L = \frac{1}{2}\alpha^2 + \frac{q}{2}(X - s\Theta)^2$, the bracket in (21.7) is

$$-\,p\,(aX + b\alpha) - \tfrac{1}{2}\alpha^2 - \tfrac{q}{2}(X - s\Theta)^2 \;=\; \underbrace{-\,b\,p\,\alpha - \tfrac{1}{2}\alpha^2}_{\text{depends on } \alpha}\ \underbrace{-\,aXp - \tfrac{q}{2}(X - s\Theta)^2}_{\text{independent of } \alpha}.$$

The α-dependent part is a downward parabola in α (the coefficient of α^2 is $-\frac{1}{2} < 0$, which is the strong concavity that makes the supremum attained and unique); its maximiser solves the first-order condition $-bp - \alpha = 0$, giving $\alpha^* = -bp$, in agreement with the affine-quadratic rule $\alpha^* = -B^\top p$ of definition 21.6 with $B = b$. Substituting $\alpha = \alpha^* = -bp$ back, the α-dependent part becomes $-bp(-bp) - \frac{1}{2}(-bp)^2 = b^2p^2 - \frac{1}{2}b^2p^2 = \frac{1}{2}b^2p^2$, so

$$H(x, X, p, \Theta) \;=\; \tfrac{1}{2}b^2p^2 - aXp - \tfrac{q}{2}(X - s\Theta)^2.$$

This is convex in p — indeed $\partial_p^2 H = b^2 \geq 0$, strictly positive whenever the control is genuinely effective ($b \neq 0$) — so (A-Conv) holds and the verification theorem of chapter 6 applies. To obtain the agent's equation we feed this Hamiltonian into the best-response HJB of definition 21.15, which for a value function $u = u^x(t, X)$ reads $\partial_t u + \frac{\sigma^2}{2}\partial_{XX}u - H(x, X, \partial_X u, \Theta) = 0$ with terminal data from g. Writing $p = \partial_X u$ and $\Theta = z_t(x)$ and expanding $-H = -\frac{1}{2}b^2p^2 + aXp + \frac{q}{2}(X - sz)^2$ term by term,

$$\partial_t u + \tfrac{\sigma^2}{2}u_{XX} \;-\; \tfrac{1}{2}b^2u_X^2 + aXu_X + \tfrac{q}{2}\big(X - s\,z_t(x)\big)^2 \;=\; 0,$$

the standard scalar-LQG HJB equation. The only trace of the graphon is the single scalar $z_t(x)$ frozen inside the running cost; for each fixed type the agent therefore solves an ordinary one-dimensional LQG control problem, and the entire network coupling is carried by how $z_t(x)$ depends, through $\mathbb{W}$, on the field the other types generate. This is the precise sense in which the graphon game is, type by type, a classical LQG problem tied to its neighbours by one number — and it is the structure chapter 22 exploits to write the coupled HJB–FP system in closed form.

The definition (21.3) was written down by fiat, and the reader is entitled to ask where the integral against W *comes from* — why this particular average is the one a networked agent should feel. The honest answer is a finite-population story that the continuum object approximates, and it is worth telling because it is the picture chapter 25 later makes rigorous.

Heuristic (non-rigorous)

Where does the continuum coupling (21.3) come from? Take N agents sampled from the graphon: assign each i a type ξ_i i.i.d. uniform on $[0,1]$ and draw the edge $i \sim j$ independently with probability $W(\xi_i, \xi_j)$, the random-graph model $G(N, W)$ of chapter 19, recorded in the symmetric adjacency matrix $A = (A_{ij})$ with $A_{ij} \in \{0,1\}$. Suppose agent i feels the empirical average of an observable h over the agents it is actually connected to, normalised by N. Conditionally on the types, the edges incident to i are independent coin-flips with $\mathbb{E}[A_{ij} \mid \xi] = W(\xi_i, \xi_j)$, so the realised adjacency may be replaced by its conditional mean,

$$\frac{1}{N}\sum_{j=1}^{N} A_{ij}\, h(X_t^j) \;\approx\; \frac{1}{N}\sum_{j} W(\xi_i, \xi_j)\, h(X_t^j),$$

the first averaging — over the randomness of the graph at fixed types — concentrating the $\{0,1\}$ weights onto the smooth weights $W(\xi_i, \xi_j)$ as $N \to \infty$. Now a second averaging acts on the right-hand side. Conditionally on the types, the empirical measure $\frac{1}{N}\sum_j W(\xi_i, \xi_j)\delta_{X_t^j}$ is a sum of independent terms whose weights are fixed by the types and whose atoms X_t^j are drawn from the type-ξ_j law; grouping the j-sum by type, the law of large numbers across the agents replaces this empirical object by its expectation,

$$\frac{1}{N}\sum_{j} W(\xi_i, \xi_j)\, \delta_{X_t^j} \;\longrightarrow\; \int_0^1 W(\xi_i, y)\, m_t^y \,\mathrm{d}y \;=\; (\mathbb{W} \circledast \boldsymbol{m}_t)(\xi_i),$$

since the types ξ_j are i.i.d. uniform on $[0,1]$ (their empirical distribution tends to Lebesgue measure dy) while, given $\xi_j = y$, the state X_t^j has law m_t^y (so its contribution to the average of h is $\langle h, m_t^y \rangle$). Pairing against h recovers exactly $\Theta_t(\xi_i)$ of (21.4). The step that replaces the empirical neighbour measure by its expectation is the only inference here that is not a finite computation, and it is justified — *heuristically* — by a conditional law of large numbers: at fixed types the summands are conditionally independent with the right conditional means, so their normalised sum concentrates. The double averaging — once over the random graph, once over the agents of each type — is precisely what the rigorous propagation-of-chaos theorem of chapter 25 establishes, keeping N genuine agents throughout and never invoking the forbidden continuum of trajectories of remark 21.2. Here the argument only *motivates* the definition (21.3); nothing in the present chapter leans on it, and all rigor is deferred to chapter 25.

We have now assembled the full data of the dynamic graphon game: a representative agent of every type (definition 21.1), the field of measures that *is* the population (definition 21.4), the kernel-weighted aggregate $\Theta_t(x)$ each agent feels in both its measure-valued and scalar forms (definition 21.5), and the cost (definition 21.6) that closes the optimisation. Every one of these objects was built on top of the field $\boldsymbol{m}_t = (m_t^x)_{x \in [0,1]}$, which we have so far carried as a bare family of measures — with no home, and no way to measure two fields apart. The next section pays that debt, fixing the space the field inhabits and the metric that controls every limit one might take of it.

21.3 The state space of population fields and its topology

The previous section left us holding a promissory note. It assembled the entire data of the dynamic graphon game — the per-type agent (21.1), the field of measures $\boldsymbol{m}_t = (m_t^x)_{x\in[0,1]}$ that *is* the population (definition 21.4), the kernel-weighted aggregate $\Theta_t(x)$ (definition 21.5), and the cost (21.6) — but it treated the field as a bare family of probability measures, with no home and no notion of distance between two such families. That is a debt we must now pay, and it is the single most important structural decision of the chapter. The reason is not pedantry. Every claim the next three chapters wish to make is a claim about *limits* of fields: that the best response varies continuously with the field it responds to, that a sequence of approximate equilibria converges to a genuine one, that the best-response map has a fixed point at all. None of these sentences even acquires a truth value until we have said what space the field inhabits and which metric measures the gap between two fields. Fixing that space and that metric is what cashes the note.

The stakes are concrete because the choice is inherited, not re-made downstream. When chapter 24 runs a Schauder or Kakutani argument it will need the field space to be a complete, separable, convex subset of a Banach (or at least Polish) space; when chapter 22 states that the equilibrium pair $x \mapsto (u^x, m^x)$ is continuous in type it will need a *second*, finer space in which type-continuity is even expressible. Both of those chapters take the space we are about to define and use it verbatim, so a vague choice here would propagate as ambiguity — or, worse, as silent disagreement — through everything that follows. We therefore make the decision once, in full, and justify every clause of it. The section has three deliverables, and it is worth naming them in advance so the reader can see the arc: first the definition of the field space $(\boldsymbol{\mathcal{P}}_2, d_2)$ together with the proof that it is the right kind of space (complete and separable), and the finer continuous-in-type alternative alongside it; second — the technical heart — the proof that the aggregate operation $\boldsymbol{m} \mapsto \Theta$ is Lipschitz with constant exactly $\|\mathbb{W}\|$, which is where the coupling threshold of the whole theory is born; and third the measurability of the field of laws, the last loose end from remark 21.2. With those three in hand, every ingredient a fixed-point definition needs will be on the table, and the next section can simply assemble them.

What kind of object is a field, and how should we measure two fields apart? A field assigns to each type x a probability measure $m^x \in \mathcal{P}_2(\mathbb{R}^d)$, so it is a map from the type interval $[0,1]$ into the metric space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ — a curve, indexed not by time but by type, living in the space of laws. We already have a canonical way to measure two laws apart: the 2-Wasserstein distance $\mathcal{W}_2$ of chapter 4, which is the right metric here precisely because the agents carry second moments and the aggregate the cost depends on is a *moment* of the neighbours' laws. So at each fixed type the distance between two fields is $\mathcal{W}_2(m^x, (m')^x)$. The only remaining question is how to combine that type-by-type discrepancy into a single number measuring the two fields apart *as fields*, and here there is a genuine choice with consequences: do we take the *supremum* of the discrepancy over types, or its L^2 *average*? The supremum is the more demanding, stronger metric; the L^2 average is weaker but, as we will see, buys separability and the compactness one needs to run a fixed-point argument. We make the L^2 average the default and keep the supremum as a named alternative. The definition records the default.

Definition 21.8 (The space of L^2 population fields)**.** Fix a reference point $\delta_0 \in \mathcal{P}_2(\mathbb{R}^d)$ (the Dirac at the origin) and let $\boldsymbol{\delta}_0$ denote the constant field $x \mapsto \delta_0$. The *space of L^2 population fields* is

$$\boldsymbol{\mathcal{P}}_2 \ := \ \Big\{ \boldsymbol{m} = (m^x)_{x\in[0,1]} \ : \ x \mapsto m^x \text{ Borel measurable into } (\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2), \ \int_0^1 \mathcal{W}_2(m^x, \delta_0)^2 \,\mathrm{d}x < \infty \Big\}, \tag{21.9}$$

fields identified when they agree for Lebesgue-a.e. x. We metrize $\boldsymbol{\mathcal{P}}_2$ by the L^2-Wasserstein distance over types,

$$d_2(\boldsymbol{m}, \boldsymbol{m}') \ := \ \Big(\int_0^1 \mathcal{W}_2(m^x, (m')^x)^2 \,\mathrm{d}x \Big)^{1/2}. \tag{21.10}$$

Three clauses of this definition were chosen deliberately — the integrability requirement, the L^2 (rather than supremum) aggregation, and the a.e.-in-type identification — and each pays for a specific structural feature we will need. We take them in turn.

The first clause is the integrability requirement $\int_0^1 \mathcal{W}_2(m^x, \delta_0)^2 \,\mathrm{d}x < \infty$. Unwinding the definition of $\mathcal{W}_2$ against the Dirac at the origin, $\mathcal{W}_2(m^x, \delta_0)^2 = \int_{\mathbb{R}^d} |\xi|^2 \ m^x(\mathrm{d}\xi)$ is simply the second moment of the type-x law, so the condition reads $\int_0^1 \int_{\mathbb{R}^d} |\xi|^2 \ m^x(\mathrm{d}\xi) \ \mathrm{d}x < \infty$: a *uniform-in-type second moment*, finite when averaged across the population. This is not an idle integrability hypothesis; it is precisely the passport that admits the first-moment field $\bar{m}$ into the Hilbert space $L^2([0,1];\mathbb{R}^d)$ on which the operator $\mathbb{W}$ is defined and bounded (chapter 18). The chain is short and worth seeing now, since it is the crux of the Lipschitz lemma below. By Jensen's inequality the mean is controlled by the second moment pointwise in type, $|\bar{m}(x)|^2 = |\int \xi \, m^x(\mathrm{d}\xi)|^2 \le \int |\xi|^2 \ m^x(\mathrm{d}\xi) = \mathcal{W}_2(m^x, \delta_0)^2$; integrating over x turns the field's finite-second-moment condition into $\|\bar{m}\|^2_{L^2([0,1])} \le d_2(\boldsymbol{m}, \boldsymbol{\delta}_0)^2 < \infty$. Without the uniform second moment this last integral could diverge, $\bar{m}$ would fall outside $L^2([0,1])$, and the very expression $\mathbb{W}\bar{m}$ — the aggregate the agent feels — would be meaningless. The clause exists exactly so that the domain of $\mathbb{W}$ contains every field in $\boldsymbol{\mathcal{P}}_2$; we record the bound as lemma 21.11(i).

The second clause is the choice to aggregate the type-by-type discrepancy in L^2 rather than in the supremum. This is the choice that makes $\boldsymbol{\mathcal{P}}_2$ *separable*, and separability is not a luxury — it is what the compactness machinery of chapter 24 runs on. To see why the supremum would fail, recall that the uniform metric d_∞ of definition 21.10 below makes the field space behave like $C([0,1]; \mathcal{P}_2(\mathbb{R}^d))$ only on the continuous fields; on the full space of measurable fields the supremum metric is non-separable for the same reason $L^\infty([0,1])$ is non-separable — there are uncountably many fields pairwise a fixed distance apart (switch a field's value on a small interval and the supremum distance jumps by a constant no matter how small the interval). The L^2 average, by contrast, sees only the *measure* of the set on which two fields differ, so a small interval contributes a small amount, and the resulting space inherits separability from the separability of its two ingredients, $L^2([0,1])$ and $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ (proved as proposition 21.9). Separability in turn gives the sequential compactness that Schauder and Kakutani arguments need: a bounded set of fields has convergent subsequences in a metrizable-weak sense, and there is no room for that in a non-separable space. This is also exactly the dynamic analogue of the L^2 setting in which the static graphon game was posed (chapter 20), and it matches the function-space conventions of the graphon mean-field literature, Bayraktar–Wu–Zhang [16] and Aurell–Carmona–Laurière [10].

The third clause identifies two fields that agree for Lebesgue-almost-every type. This is harmless — indeed forced — because nothing in the game can distinguish two fields that differ only on a type-null set. The agent's dynamics (21.1) read off the field only through the aggregate (21.3), and that aggregate is an integral against Lebesgue measure, $\Theta(x) = \int_0^1 W(x,y)\langle h, m^y\rangle \,\mathrm{d}y$, which is blind to any change of the integrand on a null set of y. So two fields agreeing a.e. produce literally the same aggregate for every type, the same best responses, the same equilibrium condition. Identifying them is not throwing away information; it is refusing to record a distinction the model cannot feel — and it is what makes d_2 a genuine metric (so that $d_2(\boldsymbol{m}, \boldsymbol{m}') = 0$ forces $\boldsymbol{m} = \boldsymbol{m}'$ in $\boldsymbol{\mathcal{P}}_2$) rather than a mere pseudometric. This is the exact analogue of identifying functions in $L^2([0,1])$ that agree a.e., and for the same reason.

We claimed two properties of this space, completeness and separability, and leaned on them when justifying the L^2 choice. They are exactly the properties a fixed-point argument consumes — completeness so that Cauchy sequences of approximate solutions actually have limits to converge to, separability so that compactness arguments have countable scaffolding — so we owe a proof rather than an assertion. The proof is the Riesz–Fischer construction familiar from L^2 of real-valued functions, transported to a setting where the values are probability measures rather than numbers. That transport is the only subtlety: $\mathcal{P}_2(\mathbb{R}^d)$ is not a vector space, so we cannot add measures or invoke the usual normed-space completeness; we must run the argument purely metrically, using only that $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is itself complete. We do exactly that.

Proposition 21.9 (Completeness and separability)**.** $(\boldsymbol{\mathcal{P}}_2, d_2)$ *is a complete separable metric space.*

Proof. d_2 is a metric. Symmetry and the vanishing $d_2(\boldsymbol{m}, \boldsymbol{m}) = 0$ are immediate from the corresponding properties of $\mathcal{W}_2$; that $d_2(\boldsymbol{m}, \boldsymbol{m}') = 0$ forces equality in $\boldsymbol{\mathcal{P}}_2$ uses the a.e.-identification, since $\int_0^1 \mathcal{W}_2(m^x, (m')^x)^2 \,\mathrm{d}x = 0$ means $\mathcal{W}_2(m^x, (m')^x) = 0$ for a.e. x, i.e. the two fields agree a.e. The triangle inequality is the one axiom that needs the L^2 structure. Pointwise in x the $\mathcal{W}_2$ triangle inequality (theorem 4.7) gives $\mathcal{W}_2(m^x, (m'')^x) \le \mathcal{W}_2(m^x, (m')^x) + \mathcal{W}_2((m')^x, (m'')^x)$; taking the $L^2([0,1])$ norm of both sides and applying Minkowski's inequality in $L^2([0,1])$ to the right-hand sum yields $d_2(\boldsymbol{m}, \boldsymbol{m}'') \le d_2(\boldsymbol{m}, \boldsymbol{m}') + d_2(\boldsymbol{m}', \boldsymbol{m}'')$. So d_2 is a bona fide metric.

Completeness. The structural fact we exploit is that $\boldsymbol{\mathcal{P}}_2$ is the L^2-Bochner-type space $L^2([0,1]; (\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2))$ of square-integrable measurable maps into the Polish space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ (Polish by theorem 4.8). Completeness of such a space is the metric-target Riesz–Fischer argument, and we run all four of its steps. Let $\boldsymbol{m}^{(n)}$ be a d_2-Cauchy sequence. *(Step 1: extract a fast subsequence.)* Because the sequence is Cauchy, we may choose indices $n_1 < n_2 < \cdots$ with $d_2(\boldsymbol{m}^{(n_{k+1})}, \boldsymbol{m}^{(n_k)}) \le 2^{-k}$, so that $\sum_k d_2(\boldsymbol{m}^{(n_{k+1})}, \boldsymbol{m}^{(n_k)}) \le \sum_k 2^{-k} < \infty$; it suffices to show this subsequence converges, since a Cauchy sequence with a convergent subsequence converges to the same limit. Write the per-type increment $g_k(x) := \mathcal{W}_2(m^{(n_{k+1}),x}, m^{(n_k),x}) \ge 0$, so that $d_2(\boldsymbol{m}^{(n_{k+1})}, \boldsymbol{m}^{(n_k)}) = \|g_k\|_{L^2([0,1])}$. *(Step 2: Tonelli turns summable-in-L^2 into summable-a.e.)* The function $G := \sum_k g_k$ is a nonnegative measurable function on $[0,1]$, and by Minkowski's inequality together with the monotone convergence theorem, $\|G\|_{L^2([0,1])} \le \sum_k \|g_k\|_{L^2([0,1])} \le \sum_k 2^{-k} < \infty$. (Equivalently one applies Tonelli to the nonnegative double sum to interchange $\int_0^1$ and $\sum_k$.) A function in $L^2([0,1])$ is finite a.e., so $G(x) = \sum_k g_k(x) < \infty$ for Lebesgue-a.e. x; fix such a full-measure set $[0,1]_0$ of types. *(Step 3: pointwise Polish limit.)* For each $x \in [0,1]_0$ the tail sums $\sum_{k \ge K} g_k(x) \to 0$ as $K \to \infty$, so the sequence $(m^{(n_k),x})_k$ in $\mathcal{P}_2(\mathbb{R}^d)$ is $\mathcal{W}_2$-Cauchy: for $j > k$, $\mathcal{W}_2(m^{(n_j),x}, m^{(n_k),x}) \le \sum_{\ell \ge k} g_\ell(x) \to 0$. Since $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is complete, this Cauchy sequence has a limit, which we name m^x; on the null complement of $[0,1]_0$ set $m^x := \delta_0$ (the choice is immaterial by a.e.-identification). The limit map

$x \mapsto m^x$ is Borel measurable, being an a.e. pointwise limit of the Borel-measurable maps $x \mapsto m^{(n_k),x}$ (a pointwise limit of measurable maps into a metric space is measurable). *(Step 4: dominated convergence in L^2.)* It remains to upgrade the pointwise convergence to d_2 convergence. For a.e. x and every k, $\mathcal{W}_2(m^x, m^{(n_k),x}) = \lim_j \mathcal{W}_2(m^{(n_j),x}, m^{(n_k),x}) \le \sum_{\ell \ge k} g_\ell(x) \le G(x)$, a bound that is independent of k and square-integrable since $G \in L^2([0,1])$. The left-hand side tends to 0 pointwise as $k \to \infty$, and it is dominated by the fixed L^2 function G, so the dominated convergence theorem gives $d_2(\boldsymbol{m}, \boldsymbol{m}^{(n_k)})^2 = \int_0^1 \mathcal{W}_2(m^x, m^{(n_k),x})^2 \,\mathrm{d}x \to 0$. Thus $\boldsymbol{m}^{(n_k)} \to \boldsymbol{m}$ in $\boldsymbol{\mathcal{P}}_2$, and with it the whole Cauchy sequence. The space is complete.

Separability. A countable dense set is built from the two ingredient countable dense sets. Let $\mathcal{D} \subset \mathcal{P}_2(\mathbb{R}^d)$ be a countable $\mathcal{W}_2$-dense subset (it exists because $\mathcal{P}_2(\mathbb{R}^d)$ is Polish) and consider the fields that are x-step functions taking finitely many values in $\mathcal{D}$ with rational breakpoints, i.e. $\boldsymbol{m} = \sum_{i=1}^N \mu_i \mathbf{1}_{I_i}$ with $\mu_i \in \mathcal{D}$ and $\{I_i\}$ a partition of $[0,1]$ into intervals with rational endpoints. This family is countable. It is also d_2-dense: given any $\boldsymbol{m} \in \boldsymbol{\mathcal{P}}_2$ and $\varepsilon > 0$, the measurable map $x \mapsto m^x$ can be approximated in $L^2([0,1]; \mathcal{P}_2(\mathbb{R}^d))$ first by a simple (finitely-valued measurable) map within $\varepsilon/2$ — the standard simple-function approximation, valid because the target is separable and the second moment is integrable — and then each of its finitely many values replaced by a $\mathcal{D}$-element within $\mathcal{W}_2$-distance $\varepsilon/2$, and each of its level sets replaced by a rational-endpoint interval approximation, incurring a further L^2 error below $\varepsilon/2$; the triangle inequality gives total error below ε. Hence the countable family is dense and $\boldsymbol{\mathcal{P}}_2$ is separable. The identical metric-target Riesz–Fischer construction, carried out for curves into $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, appears in [7, Ch. 1], to which we refer for further reading and not for any step omitted here. □

The L^2 space $(\boldsymbol{\mathcal{P}}_2, d_2)$ is the right home for existence and compactness, but it is deliberately coarse: it sees a field only up to type-null sets and therefore cannot, on its own, express a statement like "the equilibrium law varies *continuously* with the type." Yet such statements are exactly what chapter 22 wants when the kernel itself is continuous in type — the regularity half of (A-Reg-W) — because then there is no reason for neighbouring types, wired together smoothly by W, to carry wildly different laws, and one expects, and wants to prove, that $x \mapsto (u^x, m^x)$ is continuous. To even pose that conclusion we need a space whose points *are* continuous fields and whose metric controls the supremum over types. So we record a second, finer space alongside the default. It is not a replacement — it is a strictly smaller space carrying a strictly stronger metric, used only where continuity-in-type is the output one is after.

Definition 21.10 (Type-continuous fields)**.** Let $\mathcal{C}_2 := C\big([0,1]; (\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)\big)$ be the set of $\mathcal{W}_2$-continuous maps $x \mapsto m^x$, metrized by the uniform distance

$$d_\infty(\boldsymbol{m}, \boldsymbol{m}') \;:=\; \sup_{x \in [0,1]} \mathcal{W}_2\big(m^x, (m')^x\big). \tag{21.11}$$

$(\mathcal{C}_2, d_\infty)$ is a complete metric space, it embeds continuously into $(\boldsymbol{\mathcal{P}}_2, d_2)$ via $d_2 \le d_\infty$, and it is the space in which we phrase continuity-in-type conclusions.

Each clause of definition 21.10 is quickly checked, and the checks expose exactly how the two spaces relate. Completeness of $(\mathcal{C}_2, d_\infty)$ is the classical fact that a uniform (i.e. d_∞-Cauchy) limit of continuous maps into a complete target is again continuous: if $\boldsymbol{m}^{(n)} \to \boldsymbol{m}$ uniformly and each

$\boldsymbol{m}^{(n)}$ is $\mathcal{W}_2$-continuous in x, then the limit inherits continuity by the standard three-ε argument, and the target completeness of $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ supplies the pointwise limits to which the uniform limit is assembled. The embedding is the elementary bound $d_2(\boldsymbol{m}, \boldsymbol{m}')^2 = \int_0^1 \mathcal{W}_2(m^x, (m')^x)^2 \,\mathrm{d}x \le \sup_x \mathcal{W}_2(m^x, (m')^x)^2 = d_\infty(\boldsymbol{m}, \boldsymbol{m}')^2$, an integral over a probability interval being dominated by the supremum of the integrand: convergence in d_∞ forces convergence in d_2, so the inclusion $\mathcal{C}_2 \hookrightarrow \boldsymbol{\mathcal{P}}_2$ is 1-Lipschitz. What the bound does *not* give — and this is the whole point of keeping two spaces — is any reverse control: d_2-convergence is far weaker than d_∞-convergence, and the two spaces are genuinely non-interchangeable. The next box states the convention and warns against conflating them.

Notation watch

The default working space of Part IV is $(\boldsymbol{\mathcal{P}}_2, d_2)$ of definition 21.8: existence and compactness arguments (chapter 24) live here. The uniform space $(\mathcal{C}_2, d_\infty)$ of definition 21.10 is used *only* when (A-Reg-W) supplies continuity of W in type and one wants continuity of $x \mapsto (u^x, m^x)$ as an output. The two are not interchangeable, and the reason is concrete: $\mathcal{C}_2$ is *not* d_2-closed in $\boldsymbol{\mathcal{P}}_2$. To exhibit this, take in dimension one the continuous fields $\boldsymbol{m}^{(n)}$ given by $m^x_{(n)} = \delta_{f_n(x)}$ with f_n a continuous ramp that climbs from 0 to 1 across the interval $[\frac{1}{2} - \frac{1}{n}, \frac{1}{2} + \frac{1}{n}]$ and is flat outside it. Each $\boldsymbol{m}^{(n)} \in \mathcal{C}_2$, since $x \mapsto \delta_{f_n(x)}$ is $\mathcal{W}_2$-continuous. But in d_2 they converge to the field $\boldsymbol{m}^{(\infty)} = \delta_{\mathbf{1}\{x>1/2\}}$, a Dirac that jumps from δ_0 to δ_1 across $x = \frac{1}{2}$: indeed $d_2(\boldsymbol{m}^{(n)}, \boldsymbol{m}^{(\infty)})^2 = \int_0^1 |f_n(x) - \mathbf{1}\{x > 1/2\}|^2 \,\mathrm{d}x \le 2/n \to 0$, because the integrand is supported on the shrinking ramp interval and bounded by 1 there. The d_2-limit $\boldsymbol{m}^{(\infty)}$ is plainly *discontinuous* in type, hence not in $\mathcal{C}_2$. So an L^2 limit of continuous fields need not be continuous, and a statement proved in one space is not automatically valid in the other: a compactness argument run in $(\boldsymbol{\mathcal{P}}_2, d_2)$ may produce a limit field that lies outside $\mathcal{C}_2$ unless continuity is separately propagated. For this reason each theorem in chapters 22 and 24 names which space it uses, and we make the default explicit here so that downstream chapters do not silently disagree.

We arrive at the technical heart of the section, and indeed of the chapter's structural argument. Everything so far has prepared a home for the field; we now show that the one operation the game performs on the field — forming the aggregate $\boldsymbol{m} \mapsto \Theta$ that each agent feels — is *Lipschitz* on that home, with a Lipschitz constant we can name exactly. This is the fact every later existence and uniqueness argument leans on, because a fixed-point map can only be controlled if each of its constituent steps is controlled, and the aggregate is the step through which the entire network coupling acts. The constant that emerges is the operator norm $\|\mathbb{W}\|$ of chapter 18, and it is no accident that it is precisely this quantity: $\|\mathbb{W}\|$ measures how much the kernel can amplify a perturbation of the field, and a perturbation of the field is exactly what propagates through the equilibrium loop. The moment we read off the constant $\|\mathbb{W}\|$ we will have located, in the dynamic theory, the birthplace of the coupling threshold (A-Coupling) that governs every uniqueness statement downstream.

The proof rests on one elementary inequality, and it is worth flagging in advance that we need only the *easy* half of the Kantorovich–Rubinstein picture — the half that holds for any coupling and invokes no duality theorem. We are spared the hard half (that the $\mathcal{W}_1$ distance equals the supremum over 1-Lipschitz test functions) because we do not need an identity; we need only an upper bound on the change in a single observable, and that bound falls straight out of the definition of a coupling. Keeping the argument elementary here is deliberate: it means the Lipschitz property of the aggregate rests on nothing deeper than the triangle inequality and Cauchy–Schwarz, and so survives into every generalization downstream.

Lemma 21.11 (Measurability and Lipschitz continuity of the aggregate)**.** *Assume* (A-Reg-W)*: W is symmetric, measurable, bounded, and $\mathbb{W}$ is self-adjoint Hilbert–Schmidt on $L^2([0,1])$. Let $\boldsymbol{m}, \boldsymbol{m}' \in \boldsymbol{\mathcal{P}}_2$.*

(i) *The first-moment field $\bar m \in L^2([0,1];\mathbb{R}^d)$, with $\|\bar m\|_{L^2} \le d_2(\boldsymbol{m}, \boldsymbol{\delta}_0)$, and the map $x \mapsto (\mathbb{W}\,\bar m)(x)$ is well-defined and lies in $L^2([0,1];\mathbb{R}^d)$.*

(ii) *For the first-moment coupling* (21.5)*,*

$$\|\,\mathbb{W}\bar m - \mathbb{W}\bar m'\,\|_{L^2([0,1])} \;\le\; \|\mathbb{W}\|\;\|\bar m - \bar m'\|_{L^2([0,1])} \;\le\; \|\mathbb{W}\|\;d_2(\boldsymbol{m}, \boldsymbol{m}'), \tag{21.12}$$

where $\|\mathbb{W}\|$ is the operator norm on $L^2([0,1])$. In particular the aggregate map $\boldsymbol{m} \mapsto \mathbb{W}\bar m$ is $\|\mathbb{W}\|$-Lipschitz from $(\boldsymbol{\mathcal{P}}_2, d_2)$ to $L^2([0,1];\mathbb{R}^d)$.

(iii) *More generally, for a coupling through a bounded 1-Lipschitz observable h the field $x \mapsto \langle h, m^x\rangle$ obeys the same estimate with $\bar m$ replaced by $\langle h, m^{\cdot}\rangle$, hence $\boldsymbol{m} \mapsto \Theta = \mathbb{W}[\langle h, m^{\cdot}\rangle]$ is $\|\mathbb{W}\|$-Lipschitz in d_2.*

Proof. (i) We first place the first-moment field in $L^2([0,1];\mathbb{R}^d)$, which is the half of the statement that the integrability clause of definition 21.8 was designed to deliver. Fix a type x. By Jensen's inequality applied to the convex function $\xi \mapsto |\xi|^2$, the mean is dominated by the root second moment, $|\bar m(x)| = |\int \xi\, m^x(\mathrm{d}\xi)| \le (\int |\xi|^2\; m^x(\mathrm{d}\xi))^{1/2}$, and the right-hand side is exactly $\mathcal{W}_2(m^x, \delta_0)$, since transporting any measure to the Dirac at the origin costs its root second moment. Squaring this pointwise bound and integrating over types, $\|\bar m\|^2_{L^2([0,1])} = \int_0^1 |\bar m(x)|^2\,\mathrm{d}x \le \int_0^1 \mathcal{W}_2(m^x, \delta_0)^2\,\mathrm{d}x = d_2(\boldsymbol{m}, \boldsymbol{\delta}_0)^2$, which is finite precisely because $\boldsymbol{m} \in \boldsymbol{\mathcal{P}}_2$. So $\bar m \in L^2([0,1];\mathbb{R}^d)$ with $\|\bar m\|_{L^2} \le d_2(\boldsymbol{m}, \boldsymbol{\delta}_0)$, the bound claimed in (i). Measurability of $x \mapsto \bar m(x)$ follows because $x \mapsto m^x$ is Borel measurable into $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ by hypothesis and the first-moment functional $\mu \mapsto \int \xi\,\mu(\mathrm{d}\xi)$ is $\mathcal{W}_2$-continuous on $\mathcal{P}_2(\mathbb{R}^d)$ (it is even $\mathcal{W}_2$-Lipschitz, by the very coupling bound (21.13) proved in (ii) with h the identity), so the composition is measurable. Finally, $\mathbb{W}$ is a bounded operator on $L^2([0,1])$ — under (A-Reg-W) it is Hilbert–Schmidt, hence bounded with $\|\mathbb{W}\| < \infty$ — and a bounded operator maps L^2 into L^2 coordinatewise, so $\mathbb{W}\bar m \in L^2([0,1];\mathbb{R}^d)$ is well-defined. This proves (i): the aggregate $\mathbb{W}\bar m$ is a genuine element of the Hilbert space in which the network coupling lives.

(ii) The estimate (21.12) is built from one pointwise-in-type inequality, which we isolate. For two probability measures $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ and any 1-Lipschitz $h : \mathbb{R}^d \to \mathbb{R}$,

$$|\langle h, \mu\rangle - \langle h, \nu\rangle| \;\le\; \mathcal{W}_1(\mu, \nu) \;\le\; \mathcal{W}_2(\mu, \nu). \tag{21.13}$$

Here is the entire argument, and it uses only the *easy* half of the Kantorovich–Rubinstein picture — no duality theorem. Let $\pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ be *any* coupling of (μ, ν), that is, any probability measure on the product with first marginal μ and second marginal ν. Then $\langle h, \mu\rangle = \int\!\!\int h(\xi)\,\pi(\mathrm{d}\xi, \mathrm{d}\xi')$ and

$\langle h, \nu\rangle = \iint h(\xi')\,\pi(\mathrm{d}\xi, \mathrm{d}\xi')$ — this is exactly what "marginal" means — so the two single integrals over μ and ν become a single integral over the coupling of the *difference*:

$$|\langle h, \mu\rangle - \langle h, \nu\rangle| = \Big|\iint (h(\xi) - h(\xi'))\,\pi(\mathrm{d}\xi, \mathrm{d}\xi')\Big| \le \iint |h(\xi) - h(\xi')|\;\pi(\mathrm{d}\xi, \mathrm{d}\xi') \le \iint |\xi - \xi'|\;\pi(\mathrm{d}\xi, \mathrm{d}\xi'),$$

the first inequality by Jensen (the modulus of an integral is at most the integral of the modulus) and the second by 1-Lipschitzness of h, $|h(\xi) - h(\xi')| \le |\xi - \xi'|$. The left-hand side does not depend on the coupling π, so we may take the infimum of the right-hand side over all couplings, which by definition is the 1-Wasserstein distance $\mathcal{W}_1(\mu, \nu) = \inf_\pi \int |\xi - \xi'|\,\mathrm{d}\pi$; this gives the first inequality in (21.13). The second, $\mathcal{W}_1 \le \mathcal{W}_2$, is one application of the Cauchy–Schwarz (Jensen) inequality to the $\mathcal{W}_2$-optimal coupling π^*: $\int |\xi - \xi'|\,\mathrm{d}\pi^* \le (\int |\xi - \xi'|^2\,\mathrm{d}\pi^*)^{1/2} = \mathcal{W}_2(\mu, \nu)$ (theorem 4.8). Nothing here used more than the existence of couplings and two elementary inequalities, which is exactly why the conclusion is so robust.

Now we pass from one type to the whole field. Take $h(\xi) = \xi$, applied componentwise — each coordinate of the identity is 1-Lipschitz — and set $\mu = m^x$, $\nu = (m')^x$, so that $\langle h, \mu\rangle = \bar{m}(x)$ and $\langle h, \nu\rangle = \bar{m}'(x)$. The pointwise inequality (21.13) becomes $|\bar{m}(x) - \bar{m}'(x)| \le \mathcal{W}_2(m^x, (m')^x)$; squaring and integrating over x,

$$\|\bar{m} - \bar{m}'\|^2_{L^2([0,1])} = \int_0^1 |\bar{m}(x) - \bar{m}'(x)|^2\,\mathrm{d}x \le \int_0^1 \mathcal{W}_2(m^x, (m')^x)^2\,\mathrm{d}x = d_2(\boldsymbol{m}, \boldsymbol{m}')^2.$$

So the map $\boldsymbol{m} \mapsto \bar{m}$ from $(\boldsymbol{\mathcal{P}}_2, d_2)$ to $L^2([0,1]; \mathbb{R}^d)$ is itself 1-Lipschitz. The final step is the only place the operator enters: $\mathbb{W}$ is bounded with operator norm $\|\mathbb{W}\|$, so by the definition of the operator norm, $\|\mathbb{W}\bar{m} - \mathbb{W}\bar{m}'\|_{L^2} = \|\mathbb{W}(\bar{m} - \bar{m}')\|_{L^2} \le \|\mathbb{W}\|\;\|\bar{m} - \bar{m}'\|_{L^2} \le \|\mathbb{W}\|\;d_2(\boldsymbol{m}, \boldsymbol{m}')$, which is exactly (21.12). The two factors compose with no loss: the field-to-mean map contributes its constant 1 and the operator contributes $\|\mathbb{W}\|$, so the aggregate $\boldsymbol{m} \mapsto \mathbb{W}\bar{m}$ is $\|\mathbb{W}\|$-Lipschitz from $(\boldsymbol{\mathcal{P}}_2, d_2)$ to $L^2([0,1]; \mathbb{R}^d)$, proving (ii).

(iii) The general first-order coupling is identical, because the only property of the identity used in (ii) was its 1-Lipschitzness. Replacing the identity by a bounded 1-Lipschitz observable h, the field $x \mapsto \langle h, m^x\rangle$ satisfies the same pointwise bound $|\langle h, m^x\rangle - \langle h, (m')^x\rangle| \le \mathcal{W}_2(m^x, (m')^x)$ from (21.13), hence the same L^2 contraction and the same $\|\mathbb{W}\|$ factor, so $\boldsymbol{m} \mapsto \Theta = \mathbb{W}[\langle h, m^{\cdot}\rangle]$ is $\|\mathbb{W}\|$-Lipschitz in d_2. The boundedness of h is what keeps $\langle h, m^{\cdot}\rangle$ a bounded (hence L^2) field without any moment assumption, while 1-Lipschitzness is what couples its variation to the $\mathcal{W}_2$ geometry; the identity of Thread B is the borderline case where h is unbounded but its growth is controlled by the finite second moment, which is why part (i) had to secure $\bar{m} \in L^2$ separately. □

Remark 21.12 (Where the coupling threshold comes from)**.** Lemma 21.11 is the dynamic origin of the operator-norm condition (A-Coupling), and it is worth doing the bookkeeping explicitly, because the product $|s|\;\|\mathbb{W}\|$ that will control every uniqueness theorem downstream is assembled from exactly two factors, contributed by two distinct parts of the model. Trace one loop of the best-response map and watch the constants accumulate. The loop has three legs. The aggregate leg $\boldsymbol{m} \mapsto \Theta$ is the one we just bounded: by the lemma it is Lipschitz with constant $\|\mathbb{W}\|$ — *this is the first factor*, and it comes purely from the network kernel, with no reference to the cost. The cost leg is where the coupling strength s enters: the agent feels the aggregate only through the term $\frac{q}{2}(X - s\,z_t(x))^2$ of (21.8), so a unit change in the aggregate $\Theta = z_t(x)$ shifts the optimization data by a factor $|s|$ — *this is the second factor*, and it comes purely from the cost, with no reference to the kernel. The remaining leg — solving the per-type control problem and pushing the law forward by Fokker–Planck — contributes a third constant L_{ctrl} that depends only on the *single-type* data (a, b, q, q_T, σ in Thread B) and not at all on the coupling. Composing the three legs, the best-response map is Lipschitz on $C([0,T]; \boldsymbol{\mathcal{P}}_2)$ with constant at most $L_{\mathrm{ctrl}} \cdot |s| \cdot \|\mathbb{W}\|$, and a contraction — hence a unique fixed point by Banach — whenever $L_{\mathrm{ctrl}}\ |s|\ \|\mathbb{W}\| < 1$, i.e. whenever the coupling product $|s|\ \|\mathbb{W}\|$ lies below the threshold $1/L_{\mathrm{ctrl}}$ fixed by the single-type control problem alone. The clean separation — one factor of $\|\mathbb{W}\|$ from the kernel, one factor of $|s|$ from the cost, and a coupling-independent budget set by the isolated agent — is the dynamic counterpart of the static subcriticality threshold of theorem 20.14 (whose linear–quadratic spectral form is eq. (20.31)), and it is the precise sense in which $|s|\ \|\mathbb{W}\|$ is the single control parameter of the whole theory. We will see the product collapse to a single scalar in the rank-one reduction worked computation 21.18, where $\|\mathbb{W}\|$ becomes the lone nonzero eigenvalue and the infinite field closes into one self-consistent equation with effective coupling $s\ \|\mathbb{W}\|$.

One loose end remains, and it is the one remark 21.2 explicitly deferred to this section. There we resolved the continuum-of-noises difficulty by declaring that the population is the *deterministic* field of laws $x \mapsto m_t^x = \mathcal{L}(X_t^x)$, never a random ensemble of trajectories, and we promised that the only measurability this demands is measurability of a deterministic map $[0,1] \to \mathcal{P}_2(\mathbb{R}^d)$ — a statement that holds and can be proved without any jointly measurable random field. We now make good on that promise. The point worth appreciating is how *cheaply* the measurability comes: we do not need a measurable-selection theorem, which is the usual heavy hammer for "does a measurable choice exist over a parameter," because the field of laws is not a *selection* but a *composition* — the type-dependent data is measurable by hypothesis, and the map from data to law is *continuous*, by a single Grönwall estimate. A continuous function of a measurable function is measurable, and that is the whole argument. The work is in establishing the continuity, and the device that establishes it — coupling two types through the *same* Brownian motion — is worth seeing in detail, because the same trick recurs throughout chapters 23 and 24.

Lemma 21.13 (Measurability of the field of laws)**.** *Suppose $x \mapsto m_0^x$ is Borel measurable into $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ with $\boldsymbol{m}_0 \in \boldsymbol{\mathcal{P}}_2$, the coefficients b, σ are Borel in x and obey* (A-Lip)*,* (A-Growth) *uniformly in x, the aggregate flow $t \mapsto \Theta_t(\cdot)$ is given and measurable in x, and a measurable-in-x admissible feedback is fixed. Then for each t the map $x \mapsto m_t^x = \mathcal{L}(X_t^x)$ is Borel measurable into $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, and $\boldsymbol{m}_t \in \boldsymbol{\mathcal{P}}_2$ with $\sup_{t \le T} d_2(\boldsymbol{m}_t, \boldsymbol{\delta}_0) < \infty$.*

Proof. Fix the measurable-in-x feedback ϕ^x, assumed jointly measurable in (x, t, X) and Lipschitz in X uniformly in (x, t) (the regularity the verification theorem of chapter 6 produces; for the LQG thread the feedback is linear in X). The closed-loop coefficients $\widehat{b}(x, t, X) := b(x, X, \phi^x(t, X), \Theta_t(x))$ and $\sigma(x, X)$ are then, under (A-Lip), (A-Growth), Lipschitz and of linear growth in X uniformly in (x, t). For each fixed x the closed-loop SDE (21.1) has a unique strong solution by theorem 5.10, with the second-moment bound

$$\mathbb{E}\Big[\sup_{t \le T} |X_t^x|^2\Big] \le C\big(1 + \mathbb{E}\Big[|X_0^x|^2\Big] + \sup_t |\Theta_t(x)|^2\big), \tag{21.14}$$

the constant C depending only on T and the Lipschitz/growth constants, hence uniform in x.

Stability in the data. We show that the law $\mathcal{L}(X_t^x)$ depends continuously on the type-dependent data $(m_0^x, \Theta.(x), \widehat{b}(x,\cdot,\cdot))$, by a single Grönwall estimate, and the mechanism is a coupling trick that deserves to be stated plainly. To compare the laws of two types x and x' in the $\mathcal{W}_2$ metric we must exhibit *some* coupling of those two laws and bound the expected squared distance under it; any coupling gives an upper bound on $\mathcal{W}_2$, and we are free to engineer a convenient one. The convenient choice is to run *both* closed-loop SDEs on one probability space, driven by the *same* Brownian motion B and with initial conditions X_0, X_0' realizing a $\mathcal{W}_2$-optimal coupling of the two initial laws $(m_0^x, m_0^{x'})$. Driving both by the identical noise is the whole device: it makes the stochastic integrals in the two equations *cancel* when subtracted, so that the difference process $D_t := X_t - X_t'$ is driven only by the discrepancy in the drifts and in the diffusion coefficients — the irreducible randomness, which we could not have controlled, has been arranged to subtract away, leaving a difference that obeys an almost deterministic estimate. (If the two had been driven by independent noises, D_t would carry the *sum* of two fresh Brownian motions and no Grönwall bound could close.)

Subtract the two closed-loop equations and write the difference as the sum of an initial term, a drift term, and a martingale term. Apply $|a+b+c|^2 \le 3(|a|^2+|b|^2+|c|^2)$, the Lipschitz bound on $\widehat{b}$ and σ in X (so that $\left|\widehat{b}(x,r,X_r) - \widehat{b}(x,r,X_r')\right| \le L\,|D_r|$ and likewise for σ), and Doob's L^2 maximal inequality to control the supremum of the martingale part by its terminal second moment (exactly as in the well-posedness proof of theorem 5.10). The one term that does not close into $|D|$ is the genuine *type* discrepancy: where the second equation should carry $\widehat{b}(x,r,X_r')$ it carries $\widehat{b}(x',r,X_r')$ instead, so we insert and subtract $\widehat{b}(x,r,X_r')$ by the triangle inequality, splitting that term into a Lipschitz-in-X piece (folded into the $|D|$ terms) and a residual measuring how far the two *coefficients* are at the same state. Collecting,

$$\mathbb{E}\left[\sup_{s\le t}|D_s|^2\right] \;\le\; C\Big(\mathbb{E}\Big[|X_0 - X_0'|^2\Big] + \int_0^t \mathbb{E}\left[\sup_{r\le s}|D_r|^2\right]\mathrm{d}s + \delta(x,x')\Big),$$

where the residual $\delta(x,x') := \int_0^T \big(\sup_X \frac{\left|\widehat{b}(x,r,X) - \widehat{b}(x',r,X)\right|^2}{1+|X|^2}\big)\,\mathrm{d}r$ collects the entire coefficient-and-aggregate discrepancy between the two types: it is large when the closed-loop drifts of types x and x' differ, which happens when their aggregates $\Theta.(x), \Theta.(x')$ or their raw coefficients differ, and it vanishes as the two data coincide. (The $1+|X|^2$ normalization is what keeps the supremum finite under the linear growth of $\widehat{b}$.) The first two terms on the right are the standard self-referential structure that Grönwall absorbs; the third is the genuine input. Grönwall's inequality (lemma C.1) applied to $t \mapsto \mathbb{E}\Big[\sup_{s\le t}|D_s|^2\Big]$ closes the estimate into

$$\mathcal{W}_2(m_t^x, m_t^{x'})^2 \;\le\; \mathbb{E}\left[\sup_{s\le t}|D_s|^2\right] \;\le\; C\,e^{CT}\Big(\mathcal{W}_2(m_0^x, m_0^{x'})^2 + \delta(x,x')\Big), \tag{21.15}$$

where the leftmost inequality holds because (X_t, X_t') is *a* coupling — not necessarily the optimal one — of $(m_t^x, m_t^{x'})$, so the actual $\mathcal{W}_2$ distance, an infimum over couplings, can only be smaller than the cost of this particular coupling; and the rightmost uses that the initial coupling *was* chosen optimal, $\mathbb{E}\Big[|X_0 - X_0'|^2\Big] = \mathcal{W}_2(m_0^x, m_0^{x'})^2$. Inequality (21.15) is the precise statement that the map from type-data to type-law is (locally Lipschitz, hence) continuous.

Measurability. Now the composition argument promised above runs in one line. By hypothesis the data map $x \mapsto \big(m_0^x, \Theta_\cdot(x), \widehat{b}(x,\cdot,\cdot)\big)$ is Borel measurable into the relevant data space (the initial law is measurable by assumption, the aggregate flow is measurable in x by hypothesis, and the closed-loop drift inherits measurability from b, the feedback, and the aggregate). Inequality (21.15) says that the map sending these data to the law $\mathcal{L}(X_t^x)$ is continuous: when two types' data are close — in initial law and in coefficient discrepancy δ — their laws are $\mathcal{W}_2$-close. The field of laws $x \mapsto m_t^x = \mathcal{L}(X_t^x)$ is therefore the composition of a Borel-measurable map (data of x) with a continuous map (data $\mapsto$ law), and a continuous function of a Borel-measurable function is Borel measurable. So $x \mapsto m_t^x$ is Borel measurable into $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ for each t. We emphasize what was *not* needed: *no measurable-selection theorem*, because we never had to select anything — the law is determined outright by the data, and its measurability is inherited from theirs. (We cite [6, Ch. III] only for the separate matter of selecting a measurable admissible feedback in the first place, not for the present composition step.) Finally, the field lands in $\boldsymbol{\mathcal{P}}_2$ with a uniform-in-time bound: integrating the second-moment estimate (21.14) over types, $\int_0^1 \mathcal{W}_2(m_t^x, \delta_0)^2\,\mathrm{d}x \le \int_0^1 \mathbb{E}\Big[|X_t^x|^2\Big]\,\mathrm{d}x \le C\big(1+\int_0^1 \mathbb{E}\Big[|X_0^x|^2\Big]\,\mathrm{d}x + \sup_t \|\Theta_t\|_{L^2}^2\big) < \infty$ for all $t \le T$, the right-hand side finite because $\boldsymbol{m}_0 \in \boldsymbol{\mathcal{P}}_2$ and the aggregate flow is square-integrable in type. Hence $\boldsymbol{m}_t \in \boldsymbol{\mathcal{P}}_2$ with $\sup_{t\le T} d_2(\boldsymbol{m}_t, \boldsymbol{\delta}_0) < \infty$, completing the proof. □

With this lemma the last structural debt of the chapter is paid. We now possess everything a fixed-point definition requires and nothing more is owed: a state space $(\boldsymbol{\mathcal{P}}_2, d_2)$ that is complete and separable (proposition 21.9), so that limits of fields exist and compactness has room to operate; its finer continuous-in-type companion $(\mathcal{C}_2, d_\infty)$ for the regularity statements of chapter 22; the aggregate $\boldsymbol{m} \mapsto \Theta$ shown to be a $\|\mathbb{W}\|$-Lipschitz operation on that space (lemma 21.11), which both makes the coupling a bona-fide continuous map and locates the threshold $|s|\ \|\mathbb{W}\|$ that will govern uniqueness (remark 21.12); and the guarantee that the field of laws an agent's optimal behaviour generates is itself a legitimate, measurable element of $\boldsymbol{\mathcal{P}}_2$ (lemma 21.13), so the best-response map maps the space into itself. These are precisely the ingredients a fixed-point question consumes, and the next section assembles them into the equilibrium concept — the type-by-type Nash fixed point toward which every construction in this chapter has been building.

21.4 The equilibrium concept

Every ingredient is now in place, and this section spends them. The previous section closed by paying "the last structural debt of the chapter": we have a state space $(\boldsymbol{\mathcal{P}}_2, d_2)$ that is complete and separable (proposition 21.9), a graphon aggregate $\boldsymbol{m} \mapsto \Theta$ that is a $\|\mathbb{W}\|$-Lipschitz operation on it (lemma 21.11), and the guarantee that the field of laws produced by an agent's optimal behaviour is itself a legitimate, measurable element of that space (lemma 21.13). Those three facts are not curiosities collected along the way; they are precisely the three things one must own before the question "does the best-response map have a fixed point?" is even meaningful — a good domain, a continuous coupling, and a map that lands back in its domain. This section is where they are assembled into a definition. We build it in three deliberate steps, each answering one question the definition cannot leave open: *which* controls an agent is allowed to use (the admissible class), *what* an agent does with a population it is shown (the best-response map), and *when* a population is consistent with its own response (the fixed point). The destination is the third, but it is empty without the first two.

The architecture we are about to write down is not new in shape — it is the fixed-point structure of Part II (definition 8.1), in which a single representative agent optimizes against a measure flow and equilibrium demands that the optimizer regenerate that very flow. What is new is the dimension of the unknown. In Part II the unknown was one curve $t \mapsto m_t$ in $\mathcal{P}_2(\mathbb{R}^d)$; here it is an entire curve $t \mapsto \boldsymbol{m}_t$ in the field space $\boldsymbol{\mathcal{P}}_2$, one measure per type at each instant. The leap from "one measure flow" to "field-of-measures flow" is the whole content of the generalization, and the reader should keep it in view as the single thing that distinguishes every definition below from its Part II ancestor.

One standing convention must be live throughout, because both a value function and a density appear in the same definition and they flow in opposite directions in time. *In the best-response problem the value function u^x is solved backward from terminal data at $t = T$; the induced law $\widetilde{m^x}$ flows forward from initial data at $t = 0$.* The two are computed by integrating two different equations in two different time directions, and they meet — are forced to be mutually consistent — only at the fixed point. Holding that asymmetry in mind makes the definitions below read as what they are: a backward computation and a forward computation, locked together by a self-consistency condition.

We start with the admissible class, because a control problem is not posed until one has said which controls are even on the table. The choice is not cosmetic: too large a class and the infimum in the best-response problem need not be attained by any usable object; too small a class and one risks excluding the genuine optimizer. The class we adopt is the standard one of stochastic control, and it is dictated by two requirements that the agent's information and its budget impose. First, a control must be *non-anticipating*: the type-x agent decides α_t^x knowing only what has happened up to time t — its own initial draw and the history of its own noise B^x — and never the future of the Brownian path. This is the content of progressive measurability with respect to the filtration generated by B^x and X_0^x; it is what makes the controlled SDE (21.1) a well-defined object in the first place, since the stochastic integral against B^x only makes sense for integrands that do not peek ahead. Second, a control must have *finite energy*, $\mathbb{E}\Big[\int_0^T |\alpha_t^x|^2 \,\mathrm{d}t\Big] < \infty$, which is exactly the integrability under which the quadratic running cost of (21.6) is finite and the state's second moment stays controlled — without it the cost can be $+\infty$ and "minimizing" it is vacuous.

Definition 21.14 (Admissible controls)**.** Fix a frozen, measurable field flow $(\boldsymbol{m}_t)_{t\in[0,T]}$, $\boldsymbol{m}_t \in \boldsymbol{\mathcal{P}}_2$, and the associated aggregate $\Theta_t(x)$. For each x, the set $\mathcal{U}^x$ of *admissible controls* consists of A-valued processes α^x progressively measurable with respect to the filtration generated by B^x and X_0^x, with $\mathbb{E}\Big[\int_0^T |\alpha_t^x|^2 \,\mathrm{d}t\Big] < \infty$. Markov (feedback) controls $\alpha_t^x = \phi^x(t, X_t^x)$ with ϕ^x measurable and of linear growth in the state form the subclass on which the verification theorem of chapter 6 produces optimizers under (A-Conv), (A-Nondeg).

The definition names a large class — all square-integrable progressively measurable processes — and then singles out a small subclass, the Markov or *feedback* controls $\alpha_t^x = \phi^x(t, X_t^x)$ that depend on the past only through the current state. It is worth being clear about why both appear. The large class is the honest arena of competition: when we say an agent does the best it can, "best" must mean best over *all* admissible strategies, including history-dependent ones, or the optimality claim is weaker than it sounds and the equilibrium would not deserve the name Nash. The small class is where the answer actually lives. The verification theorem of chapter 6 says that under convexity of the Hamiltonian (A-Conv) and nondegeneracy of the noise (A-Nondeg), the optimizer over the *large* class is in fact realized by a *feedback* of the small class — the one read off from the gradient of the value function — so that nothing is lost by the restriction. Nondegeneracy is what gives the value function enough regularity (a classical or suitably weak second derivative) for the verification argument to apply Itô's formula to it; convexity is what makes the pointwise maximizer of the Hamiltonian a genuine minimizer of the cost rather than a critical point of the wrong sign. This is the precise division of labour we will lean on at once: we pose the best-response problem over the large class so that "best response" means what it should, and we solve it in the small class because that is where the HJB equation delivers the optimizer in closed form.

With the admissible class fixed we can write down what a single agent does when it is shown a population. The construction is the heart of the section, and it is best read as the composition of three classical operations that we will name now and then unpack one at a time after the statement. The crucial conceptual point, inherited unchanged from mean field game theory and worth stating before any formula, is that during its own optimization the infinitesimal agent treats the population field — and hence the aggregate it feels — as *frozen, exogenous data.* A single agent is one of a continuum; it cannot, by acting differently, move the aggregate $\Theta_t(x)$, because that aggregate is a kernel-weighted average over a continuum of types and altering one type's behaviour on a set of measure zero changes nothing. So from the agent's vantage point the flow $t \mapsto \Theta_t(x)$ is a fixed input function, and the optimization it performs is an entirely *standard* stochastic control problem with that input baked into the running and terminal costs. The graphon has, for the duration of the agent's calculation, vanished into a known time-dependent coefficient. All of the network's self-referential difficulty is postponed to the fixed-point condition that closes the loop; here there is no loop, only an ordinary control problem solved against a given target.

Definition 21.15 (Best-response map)**.** Given a field flow $(\boldsymbol{m}_t)_t$ and the induced aggregate flow $(\Theta_t)_t$, the type-x agent solves

$$u^x(t,X) \;=\; \inf_{\alpha^x \in \mathcal{U}^x} \mathbb{E}\left[\int_t^T L(x, X_r^x, \alpha_r^x, \Theta_r(x))\,\mathrm{d}r + g(x, X_T^x, \Theta_T(x)) \;\middle|\; X_t^x = X\right],$$

a standard stochastic control problem (the aggregate is exogenous to the infinitesimal agent) whose value u^x solves the backward HJB equation $\partial_t u^x + \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u^x) - H(x, X, \nabla u^x, \Theta_t(x)) = 0$, $u^x(T,\cdot) = g(x,\cdot,\Theta_T(x))$, with optimal feedback the maximizer $\alpha^{x,*} = \alpha^*(x, X, \nabla u^x, \Theta_t(x))$ of (21.7) under (A-Conv) (for control-affine drift, $\alpha^{x,*} = -B^\top \nabla u^x$; in the scalar LQG thread, $-b\,\partial_X u^x$). Let $\widetilde{m^x}_t := \mathcal{L}(X_t^{x,*})$ be the law of the optimally controlled state, which evolves forward by the Fokker–Planck equation (theorem 5.14) $\partial_t \widetilde{m^x} = \mathcal{A}^{x,*}\widetilde{m^x}$, $\widetilde{m^x}_0 = m_0^x$, where $\mathcal{A}^{x,*}$ is the adjoint of the generator of the optimally controlled type-x diffusion. The *best-response map* is

$$\mathsf{S}:\; (\boldsymbol{m}_t)_t \;\longmapsto\; (\widetilde{\boldsymbol{m}}_t)_t, \qquad \widetilde{\boldsymbol{m}}_t = (\widetilde{m^x}_t)_{x\in[0,1]},$$

acting on the path space $C([0,T];\boldsymbol{\mathcal{P}}_2)$ (or $C([0,T];\mathcal{C}_2)$ under (A-Reg-W) continuity).

The definition is compressed by design, and it packs three logically distinct steps into one block.

Pulling them apart is what makes the map intelligible, so we take them in turn, and the order is exactly the order in which one computes a best response.

Step 1: solve the backward HJB equation (the value). For each fixed type x the agent's optimal cost-to-go from state X at time t is the value function $u^x(t, X)$, and dynamic programming turns the control problem into the second-order parabolic equation

$$\partial_t u^x + \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u^x) - H(x, X, \nabla u^x, \Theta_t(x)) = 0, \qquad u^x(T, \cdot) = g(x, \cdot, \Theta_T(x)),$$

solved *backward* from the terminal cost. This is where the standing convention earns its keep: the equation carries data at $t = T$ and is integrated toward $t = 0$, because the value at time t is the best the agent can still do over the *remaining* horizon $[t, T]$, and the only thing it knows for certain about that horizon is the terminal cost. The frozen aggregate enters only as the time-dependent coefficient $\Theta_t(x)$ sitting inside the Hamiltonian and the terminal data; with it frozen, this is precisely the HJB equation whose solvability and nondegenerate-parabolic regularity were established for a single agent, and nothing about the graphon is visible except through that one coefficient. (A-Nondeg) is what makes the $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u^x)$ term uniformly elliptic, which is exactly the smoothing that gives u^x the second derivative the next step needs.

Step 2: read off the optimal feedback (the verification theorem). Having the value, the verification theorem of chapter 6 certifies that the pointwise maximizer $\alpha^{x,*} = \alpha^*(x, X, \nabla u^x, \Theta_t(x))$ of (21.7) is not merely a candidate but the genuine optimizer over the full admissible class of definition 21.14. The logic is the one rehearsed there: substitute the candidate feedback into the cost, apply Itô's formula to $u^x(t, X_t^x)$ along the controlled state, and use the HJB equation to show the result is a true lower bound attained with equality exactly when the agent plays $\alpha^{x,*}$. Convexity (A-Conv) guarantees the maximizer is unique and depends measurably on (t, X), so $\alpha^{x,*}$ is a legitimate feedback in the Markov subclass — which is why that subclass was singled out in definition 21.14. In the control-affine case $b = b_0 + B\alpha$ the first-order condition collapses to the explicit $\alpha^{x,*} = -B^\top \nabla u^x$, and in the scalar LQG thread to $\alpha^{x,*} = -b\,\partial_X u^x$, the optimal control being literally proportional to the marginal value of state. We stress, as in definition 21.6, that this is the maximizer of the Hamiltonian and *not* $-\partial_p H$ when the drift is genuinely control-dependent; the two agree only when drift equals control.

Step 3: push the optimally controlled law forward (the Fokker–Planck adjoint). Closing the feedback loop turns the controlled SDE into an ordinary (uncontrolled) diffusion — the type-x agent now follows $\mathrm{d}X_t^{x,*} = b(x, X_t^{x,*}, \alpha_t^{x,*}, \Theta_t(x))\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x$ with the feedback substituted in — and the marginal law of that closed-loop diffusion evolves by the Fokker–Planck equation $\partial_t \widetilde{m^x} = \mathcal{A}^{x,*}\widetilde{m^x}$ of theorem 5.14, started from the given initial law m_0^x and integrated *forward*. Here $\mathcal{A}^{x,*}$ is the formal adjoint of the generator of the *optimally controlled* diffusion — it depends on u^x through the feedback, which is the only way the backward computation of Step 1 feeds into the forward computation of Step 3. This forward integration carries data at $t = 0$ and propagates toward $t = T$, the mirror image of Step 1, because a population's law at time t records where it started and how it has been driven over the *past* $[0, t]$. The output $\widetilde{m^x}_t = \mathcal{L}(X_t^{x,*})$ is the law the optimally responding type-x agents actually realize.

Assembling these three steps across all types gives the best-response map S: feed it a field flow, it solves one backward HJB per type, reads off one feedback per type, runs one forward Fokker–Planck per type, and returns the field of resulting laws. The backward/forward pairing is the structural signature of the whole construction and deserves to be stated plainly: S is the composition of a *backward* sweep that computes optimal behaviour from the future-facing value and a *forward* sweep that computes the law that behaviour generates from the past-facing initial data. The two sweeps are run in opposite time directions and are coupled only through the feedback. That S maps the path space $C([0,T];\boldsymbol{\mathcal{P}}_2)$ into itself — that each output law is square-integrable in type and the field depends continuously on time — is exactly the content of lemma 21.13, which is why that lemma was the last debt the previous section discharged before this one could be written.

The best-response map answered "what does an agent do with a population it is shown?" but it took the shown population as exogenous. An equilibrium is the population that is not arbitrary — the one an agent would, in fact, generate by responding optimally to it. This is the only place the loop closes, and closing it is a single demand: the field you optimize *against* must equal the field your optimization *produces*. That demand is a fixed-point equation for S, and it is the chapter's destination.

Definition 21.16 (Dynamic graphon mean field game equilibrium)**.** A field flow $(\boldsymbol{m}_t)_{t\in[0,T]} \in C([0,T];\boldsymbol{\mathcal{P}}_2)$ with $\boldsymbol{m}_0 = \boldsymbol{m}_0^{\text{given}}$ is a *dynamic graphon mean field game equilibrium* (a *GMFG solution*) if it is a fixed point of the best-response map,

$$\mathsf{S}\big[(\boldsymbol{m}_t)_t\big] \;=\; (\boldsymbol{m}_t)_t, \tag{21.16}$$

that is: for Lebesgue-a.e. type x, when every agent of type x uses the optimal feedback computed against the aggregate $\Theta_t(x) = (\mathbb{W} \circledast \boldsymbol{m}_t)(x)$ of the field $\boldsymbol{m}$, the law of its optimally controlled state coincides with m_t^x for all $t \in [0,T]$. Equivalently, the pair $(x \mapsto u^x,\, x \mapsto m^x)$ solves the coupled backward–forward system of chapter 22, with the only inter-type coupling entering through $\mathbb{W}$.

The content of (21.16) is the same Nash consistency as in classical mean field games: the population an agent optimizes against is the population that the optimizing agents generate. Nobody has an incentive to deviate, because each agent is already playing its best response to the realized population, and — being infinitesimal — its own deviation does not move that population. What is genuinely new is the *shape* of the consistency. In Part II it was a single equation, one measure flow equal to the law it induces. Here it is a continuum of consistency conditions, one per type, holding for Lebesgue-a.e. x simultaneously, and they are not independent: each type's optimization sees, through the aggregate $\Theta_t(x) = (\mathbb{W} \circledast \boldsymbol{m}_t)(x)$, a W-weighted average of *all* the others' distributions. So the only channel through which one type's equilibrium behaviour reaches another is the operator $\mathbb{W}$. Strip the kernel away — make W diagonal-like or constant — and the types decouple or fuse; that is exactly the lever the reduction checks of the next section will pull. The phrase to retain is: *Nash consistency, imposed type by type, with all inter-type coupling carried by* $\mathbb{W}$ *and nothing else.*

It is equally important to be explicit about what this definition deliberately does *not* do, because the chapter's discipline is to set up cleanly and defer honestly. We do not prove that a fixed point of (21.16) exists, nor that it is unique. That is the business of chapter 24, and its hypotheses are precisely the standing assumptions we have been accumulating — (A-Lip), (A-Nondeg), (A-Conv), (A-Reg-W) — together with a control on the coupling strength, either a smallness condition or a monotonicity condition ((A-Coupling), or a graphon Lasry–Lions condition, definition 11.1). The smallness threshold is no mystery: it is the $|s|\ \|\mathbb{W}\|$ already located by the Lipschitz constant of the aggregate (remark 21.12), which is exactly why that constant was worth computing two sections ago. Nor do we write the equilibrium as a system of equations here in any usable analytic form: the *partial differential* characterization — the coupled backward HJB and forward Fokker–Planck system obtained by letting the two sweeps of S run against a common field — is the subject of chapter 22, and the *probabilistic* characterization, as a system of graphon forward–backward stochastic differential equations, is the subject of chapter 23. The definition above is the fixed object all three of those chapters operate on; its job is to be unambiguous, not yet to be solved.

There is one physical picture that genuinely accelerates understanding of the fixed point, and it is worth one box. A reader with a statistical-mechanics background will already have felt that (21.16) looks like a self-consistency equation, and indeed it is one — of exactly the Curie–Weiss self-consistent-field kind, but with the spatial uniformity of Curie–Weiss broken by the kernel. We record the analogy with the two honesty cautions that keep it from overreaching.

Physics note

The fixed point (21.16) is a self-consistent-field condition with an internal flavor index. The symbols in this box are local to the analogy: m_{mag}, J_{CW} below are the Curie–Weiss magnetization and spin coupling, unrelated to the density macro m and the cost functional J^x of (21.6). In Curie–Weiss mean field theory each spin feels an effective field $h_{\mathrm{eff}} = J_{\mathrm{CW}}\, m_{\mathrm{mag}}$ set self-consistently by the magnetization $m_{\mathrm{mag}} = \tanh(\beta_{\mathrm{T}} h_{\mathrm{eff}})$ it produces; the field is uniform because every spin couples identically to every other. The graphon game keeps the self-consistency but *breaks the uniformity*: type x feels its own effective field $\Theta_t(x) = \int W(x,y)\langle h, m_t^y\rangle\, \mathrm{d}y$, weighted by the kernel W that records how strongly x is wired to y. The order parameter is now a *field* $x \mapsto \langle h, m^x\rangle$ rather than a scalar, and $\mathbb{W}$ plays the role of the (inhomogeneous, off-diagonal) coupling matrix J^{CW}_{xy}. Two cautions keep the analogy honest: the field here is fixed not by a thermal average but by a Nash optimization run backward in time, and there is no inverse-temperature parameter at all — this is a finite-horizon, undiscounted game (a discount would reweight time, not act as a temperature). The analogy is a picture, not a proof; every statement above stands without it.

The equilibrium is now defined. Every object the sentence "$(\boldsymbol{m}_t)_t$ is a dynamic graphon mean field game equilibrium" refers to has been pinned down: the agents, the field of measures, the space it lives in, the topology on that space, the optimization each type performs, and the type-by-type consistency that closes the loop. The destination promised in the chapter preamble has been reached. But a definition, however carefully assembled, is only as good as its faithfulness to the theories it claims to generalize, and we have so far only *asserted* that this one is the right one. The reader is entitled to demand proof of fitness, and the honest test is degeneration: a correct generalization must reproduce every special case it subsumes. The next section subjects definition 21.16 to exactly that test, along three slices of the kernel. Setting W constant should erase the type label entirely and hand back the classical mean field game of Part II; collapsing W to rank one should crush the infinite field of measures down to a single scalar self-consistency with effective coupling $s\ \|\mathbb{W}\|$, previewing the Curie–Weiss closure of chapter 22; and taking W piecewise constant in blocks should recover the finite multi-population mean field games. If the definition survives all three, it has earned the trust that the three downstream engines will place in it.

21.5 Reduction checks

With definition 21.16 in hand, the question facing us is no longer "what is a dynamic graphon equilibrium?" — that has a definite answer now — but the sharper and more honest "is it the right *it*?" A definition can be internally flawless and still generalize the wrong way; the only test that catches such a failure is degeneration. A faithful generalization must contain, as exact special cases, the theories it claims to subsume, and it must contain them *visibly* — not after a change of variables one has to be told to make, but by simply switching off the feature that distinguishes the new theory from the old. For us that feature is the kernel W. The whole novelty of chapter 21 is concentrated in the one object W that ties the continuum of types together; so the natural way to probe the construction is to dial W down to the degenerate shapes for which the answer is already known, and check that the machinery returns it.

We take three such slices, in increasing distance from the classical picture, and they are not chosen for variety: they are exactly the three reductions the thread specification fixes for Thread B (constant, rank-one, block), so that verifying them here is the same as discharging the obligations that specification records. The first slice is the constant kernel $W \equiv c$. This is the must-have backward-compatibility check: erasing all type structure should erase the type label itself and hand back the single-population mean field game of Part II, with the coupling merely rescaled by c. If the dynamic graphon game failed *this*, nothing else about it would matter, so it comes first and is the most basic. The second slice is the rank-one kernel $W(x, y) = f(x)f(y)$. Here the type structure survives — the game is genuinely heterogeneous — yet the infinite field of measures collapses onto a single scalar order parameter, giving the cleanest non-trivial worked example in the book and advancing Thread B to the dynamic Curie–Weiss closure that chapter 22 will solve in full; this is less a sanity check than a first genuine solution, which is why it earns a `workedcomp` rather than a remark. The third slice is the block kernel, piecewise constant on a partition of $[0, 1]$. It points the other way — not back to Part II but forward, to the finite multi-population mean field games — and so serves as a bridge: it shows the continuum theory sitting one specialization above the multi-type theory of chapter 30, the field of measures degenerating to a finite vector of population laws. Two backward-looking sanity checks and one forward-looking bridge; together they fence the construction in on both sides. We begin with the most elementary of the three, the constant kernel.

21.5.1 Constant graphon recovers the classical mean field game

The reduction section just promised that the most basic obligation on the construction is backward compatibility: switch off the kernel entirely — make every type interact with every other type at one uniform strength — and the elaborate field-of-measures machinery ought to fold back down to the single-population mean field game of Part II that we already understand. This is the "does the new theory contain what we already know?" check, and it is the right one to discharge first, because if it failed nothing else about the dynamic graphon game would be worth trusting. The constant kernel is exactly the switch: it is the unique graphon that records *no* network structure at all, every pair of types coupled identically, so it is the formal image of "the network does not matter, only the crowd does." We now turn the crank and watch the type label disappear.

The mechanism is visible already in the aggregate. Recall that the only place the network ever touches an agent is through the kernel-weighted average $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ of (21.5) — the graphon-smoothed first-moment field. Smoothing against a *constant* kernel is not smoothing at all; it is plain global averaging followed by a rescaling, and the type label, having nothing left to discriminate, drops out. The following remark makes this precise and reads off, one consequence at a time, that every distinguishing feature of the graphon game collapses onto its Part II value: the per-type problem becomes one problem, the operator norm becomes the bare number c, the threshold becomes the classical threshold, and the LQG cost becomes the Thread A cost with coupling s rescaled to sc.

Remark 21.17 (Constant W)**.** Let $W(x,y) \equiv c$ for a constant $c \in (0,1]$. The aggregate (21.5) becomes

$$z_t(x) \;=\; \int_0^1 c\,\bar{m}_t(y)\,\mathrm{d}y \;=\; c\,\bar{m}_t^{\mathrm{glob}}, \qquad \bar{m}_t^{\mathrm{glob}} := \int_0^1 \bar{m}_t(y)\,\mathrm{d}y, \tag{21.17}$$

which is *independent of* x: the integral against the constant c pulls c out and leaves the type-x-free global average $\bar{m}_t^{\mathrm{glob}} = \int_0^1 \bar{m}_t(y)\,\mathrm{d}y$, so every agent — regardless of its label — feels one and the same number, the global mean position scaled by c. The kernel argument x that $z_t(x)$ carried in general has simply been integrated away, because a constant kernel cannot tell one row from another: $W(x,\cdot) \equiv c$ for all x, so the weight every type places on the population is identical and the weighted average it computes is identical.

Two of the model's three sources of type-dependence are thereby gone. The coefficients b, σ and the cost weights q, q_T, s, s_T of Thread B are already type-independent (they were fixed constants from the start), and we have just seen the aggregate $z_t(x) \equiv c\,\bar{m}_t^{\mathrm{glob}}$ is type-independent too. The only remaining channel through which x could enter is the initial law m_0^x. Suppose for this check it is switched off as well, $m_0^x \equiv m_0$. Then the type-x control problem — HJB plus Fokker–Planck against the frozen aggregate — has *identical data for every* x: same dynamics, same cost, same initial condition, same scalar z_t inside the running cost. Identical optimization problems have a common solution, so the constant field $m_t^x \equiv m_t$ is consistent — it is a genuine equilibrium of the graphon game — and for it the global average reduces further, $\bar{m}_t^{\mathrm{glob}} = \int_0^1 \bar{m}_t(y)\,\mathrm{d}y = \bar{m}_t$, to the single population mean. The field of measures has collapsed to one measure m_t evolving in time, which is precisely the Part II population object.

A word of honesty is owed here, and it is exactly the subtlety a careful reader should be suspicious of. We have shown that the symmetric, type-independent field is *a* solution; we have *not* shown it is the *only* one. The argument established that identical data *admit* an identical-across-types solution, not that every solution must be type-independent. The distinction is not pedantic. Below the coupling threshold the best-response map is a contraction (remark 21.12), its fixed point is unique, and uniqueness forces the only fixed point to be the symmetric one we exhibited — there the caveat is vacuous. Above the threshold uniqueness can fail, and when it does the symmetric solution may coexist with type-*inhomogeneous* equilibria in which agents with the very same data nevertheless settle into different laws. This is genuine symmetry breaking: the data are invariant under relabelling types, yet a solution need not share that invariance, exactly as the self-consistent-field picture of section 21.4 and the phase diagram of chapter 14 warn. So the correct statement is the careful one — a symmetric equilibrium *exists* and recovers Part II; whether it is the only equilibrium is the same multiplicity question Part II already faces, neither created nor cured by passing to the constant graphon.

It remains to confirm the two quantitative fingerprints of Part II: the coupling threshold and the LQG cost. Both come from understanding $\mathbb{W}$ for the constant kernel, which is the simplest operator in the book. Acting on a field $f \in L^2([0,1])$,

$$(\mathbb{W}f)(x) \;=\; \int_0^1 c\, f(y)\, \mathrm{d}y \;=\; c\Big(\int_0^1 f\Big)\,\mathbf{1}(x),$$

so $\mathbb{W}$ sends *every* f to a multiple of the constant function $\mathbf{1}$: it is a rank-one operator with range $\operatorname{span}\{\mathbf{1}\}$. Its spectrum is read off in one line. On the constant function itself, $\mathbb{W}\mathbf{1} = c\big(\int_0^1 1\big)\mathbf{1} = c\,\mathbf{1}$, so $\mathbf{1}$ is an eigenfunction with eigenvalue c. On any mean-zero field, $\int_0^1 f = 0$ forces $\mathbb{W}f = 0$, so the entire orthogonal complement $\{f : \int f = 0\}$ is the kernel, eigenvalue 0. The spectrum is therefore $\{c, 0\}$ with the single nonzero eigenvalue c, and since $\mathbb{W}$ is self-adjoint its operator norm equals its largest eigenvalue in modulus, $\|\mathbb{W}\| = c > 0$. Feeding this into the threshold of remark 21.12, the control parameter $|s|\,\|\mathbb{W}\|$ becomes $|s|\,c$, which is exactly the rescaled coupling $|sc|$ of the classical theory: the constant graphon contributes nothing but the scalar c, multiplying the cost coupling s to the effective strength sc, in agreement with the Ch. 14 threshold.

The same rescaling closes the loop on the cost. Substituting the collapsed aggregate $z_t(x) = c\,\bar{m}_t$ into the Thread B cost (21.8), the running coupling term becomes $\frac{q}{2}\big(X_t^x - s\,c\,\bar{m}_t\big)^2$ and the terminal one $\frac{q_T}{2}\big(X_T^x - s_T\,c\,\bar{m}_T\big)^2$, so

$$J(\alpha; m) = \mathbb{E}\left[\int_0^T \Big(\tfrac{1}{2}\alpha_t^2 + \tfrac{q}{2}\big(X_t - (sc)\,\bar{m}_t\big)^2\Big)\,\mathrm{d}t + \tfrac{q_T}{2}\big(X_T - (s_T c)\,\bar{m}_T\big)^2\right], \tag{21.18}$$

which is verbatim the classical scalar LQG mean field game cost of Thread A (worked computation 8.12, section 14.1) with the mean-coupling strength s replaced throughout by sc (and s_T by $s_T c$). The type superscript has vanished because nothing in (21.18) depends on it any longer; the agent optimizes against the single population mean $\bar{m}_t$ exactly as in Part II. The dynamic graphon game of this chapter therefore contains the entire Part II theory, intact, as its constant-graphon slice, with the network contributing only the harmless scalar c that renormalizes the coupling.

The constant kernel was the trivial-network limit: with the geometry switched off, the field of measures degenerated to a single measure and the whole apparatus reproduced Part II with a rescaled coupling. The lesson is that type-dependence and the field structure are inseparable companions of a *nontrivial* kernel — erase the kernel's structure and both collapse together. The natural next question is whether they must collapse *together*: is there a kernel poor enough to be solved in closed form, yet rich enough that type-dependence genuinely survives while the field still closes onto a finite object? There is, and it is the rank-one kernel. It is the cleanest nontrivial graphon in the book: the type label does real work — different types feel genuinely different aggregates — yet the infinite field of measures once again collapses, this time not to a single measure but to a single *scalar* order parameter through which all types communicate. That is the Curie–Weiss closure we take up next, and it is where the spectral picture we just sketched for the constant kernel earns its keep, with $\|\mathbb{W}\|$ becoming the lone nonzero eigenvalue of a genuinely structured operator.

21.5.2 Rank-one graphon: a Curie–Weiss closure

The constant kernel left us with a question rather than a verdict. There, type-dependence and the field structure collapsed together only because the kernel had no structure to collapse; and we asked whether the two must always fall together, or whether some kernel could be poor enough to solve in closed form yet rich enough to keep the type label genuinely at work. The rank-one kernel is the affirmative answer, and it is the flagship computation of the chapter. Here different types feel genuinely different aggregates — the label discriminates, as it could not under a constant kernel — and yet the infinite field of measures collapses once more. This time it collapses not onto a single *measure* but onto a single *scalar*, an order parameter through which every type speaks to every other. That collapse is the dynamic analogue of the Curie–Weiss closure of section 21.4: the inhomogeneous self-consistent field of the general game, which there was an entire function of type, here folds back to one number, because the coupling operator $\mathbb{W}$ has rank one. We carry Thread B through it in full, and give it the most expansive treatment in the chapter, because it is the cleanest worked example of a graphon mean field game in the book.

The kernel is $W(x,y) = f(x)f(y)$ with $f \in L^2([0,1])$, $f \geq 0$: a single "shape function" f whose value $f(x)$ records how strongly type x both broadcasts to and listens to the rest of the network. Before computing anything we must separate two normalizations of f that are easy to conflate and that control entirely different things; keeping them straight is what keeps the rest of the section honest, and it is a distinction the static chapter could afford to gloss but the dynamic one cannot.

The first is a *validity* constraint, and it lives in L^∞. For W to be a graphon at all it must take values in $[0,1]$, and since $W(x,y) = f(x)f(y)$ its largest value is $\|f\|_\infty^2$, attained where f peaks. Graphon validity is therefore exactly $\|f\|_\infty^2 \leq 1$, a bound on the *supremum* of f. The L^2 norm will *not* police this: a tall narrow spike — f large on a tiny set and small elsewhere — can have $\|f\|_{L^2}$ as small as one likes while $\|f\|_\infty$, and hence the product $f(x)f(y)$ at the peak, blows past 1. The pointwise bound that makes W a legitimate kernel is an L^∞ statement, not an L^2 one.

The second is the *coupling* constraint, and it lives in L^2. By Mercer's theorem (theorem 18.2) the operator $\mathbb{W}$ of a rank-one kernel carries a single nonzero eigenvalue, and one reads it off directly: acting on any field $g \in L^2([0,1])$,

$$(\mathbb{W}g)(x) = \int_0^1 f(x)f(y)\,g(y)\,\mathrm{d}y = f(x)\,\langle f, g\rangle_{L^2},$$

so $\mathbb{W}$ sends every g to a multiple of f — its range is the single line $\operatorname{span}\{f\}$ — and on f itself $\mathbb{W}f = f\,\langle f, f\rangle_{L^2} = \|f\|_{L^2}^2\, f$. Thus f is the top eigenfunction with eigenvalue $\|f\|_{L^2}^2$, every direction orthogonal to f is annihilated, and because $\mathbb{W}$ is self-adjoint its operator norm is that lone eigenvalue, $\|\mathbb{W}\| = \|f\|_{L^2}^2$. So the operator norm — the quantity that, multiplied by the cost coupling s, will set the threshold of the whole theory — is the *square of the L^2 norm* of f, an integral quantity insensitive to spikes. The two norms answer two different questions: $\|f\|_\infty$ asks whether W is a legal kernel; $\|f\|_{L^2}$ asks how strongly that legal kernel couples. Conflating them — using $\|f\|_{L^2} \le 1$ as a stand-in for validity, or $\|f\|_\infty$ as a stand-in for coupling strength — would mis-state both, and the two need not even point the same way (the spike is valid yet weakly coupling; a broad gentle f may couple strongly yet sit comfortably inside $[0,1]$). One last bookkeeping point separates f from its normalization: the *eigenvector centrality* of lemma 20.12 and definition 18.8 — the network's profile of who-is-central — is the *unit* eigenfunction $\phi_1 = f/\|f\|_{L^2}$. We deliberately keep the *unnormalized* f throughout, because the amplitude that ϕ_1 discards in normalizing is precisely the coupling strength $\|f\|_{L^2}^2 = \|\mathbb{W}\|$, and that amplitude is exactly what the threshold will turn on. We now advance Thread B by showing that the entire field collapses to a single scalar order parameter — the dynamic analogue of the Curie–Weiss closure.

Worked computation 21.18 (Thread B: rank-one collapse to a scalar order parameter)**.** Take the graphon-LQG agent (21.2) with cost (21.8) and $W(x,y) = f(x)f(y)$. The aggregate (21.5) factorizes:

$$z_t(x) \;=\; \int_0^1 f(x)f(y)\,\bar{m}_t(y)\,\mathrm{d}y \;=\; f(x)\,\varrho_t, \qquad \varrho_t \;:=\; \int_0^1 f(y)\,\bar{m}_t(y)\,\mathrm{d}y \;=\; \langle f, \bar{m}_t\rangle_{L^2}, \tag{21.19}$$

where ϱ_t is the *scalar order parameter*: the projection $\langle f, \bar{m}_t\rangle_{L^2}$ of the first-moment field onto the shape function f, one real number that summarizes the entire field $\bar{m}_t(\cdot)$ as far as the coupling is concerned. It is the exact dynamic analogue of the static rank-one order parameter ϱ of worked computation 20.21 — the same projection onto f, now carrying a time index because the means evolve. The factorization $z_t(x) = f(x)\,\varrho_t$ is the whole point of rank one: the drive that type x feels is its own broadcast strength $f(x)$ times one common scalar ϱ_t shared by every type. No type needs to know the full field; it needs only one number and its own $f(x)$. This is precisely the Curie–Weiss collapse anticipated in section 21.4: the inhomogeneous effective field of the general game reduces to a single self-consistent scalar the instant the coupling operator has rank one, modulated per type only by the fixed profile f.

With ϱ_t frozen — the infinitesimal agent takes the field as given, definition 21.6 — substituting $z_t(x) = f(x)\varrho_t$ into the running coupling turns it into $\frac{q}{2}(X_t^x - s\,f(x)\,\varrho_t)^2$, so type x is simply tracking the known, exogenous, time-dependent target $s\,f(x)\,\varrho_t$. That is an ordinary scalar LQG problem with an affine target — the very single-population problem solved in closed form in chapter 14 — the only twist being that the target is scaled by the constant $f(x)$. Its value function is quadratic in the state, $u^x(t,X) = \frac{1}{2}P(t)X^2 + r_t(x)\,X + \cdots$, with $P(t)$ the scalar Riccati curvature (eq. (14.7), the same P as in eq. (8.24)) and $r_t(x)$ the linear offset (eq. (14.8)) that carries the tracking target. Crucially $P(t)$ is the *same* for every type: it depends only on a, b, q, q_T, not on where the target sits, because the quadratic curvature of an LQG value function is insensitive to the affine forcing. All the type-dependence is pushed into the offset $r_t(x)$. Feeding the costate $p = \partial_X u^x = P(t)X + r_t(x)$ into the affine-quadratic feedback rule $\alpha^* = -bp$ of definition 21.6, the optimal feedback is

$$\alpha_t^{x,*} \;=\; -b\,\big(P(t)\,X_t^x + r_t(x)\big), \tag{21.20}$$

a state-linear feedback whose slope $-bP(t)$ is shared across types and whose intercept $-b\,r_t(x)$ carries the type label. Inserting it into the agent's dynamics (21.2) and taking expectations — the noise averages out, $\mathbb{E}[\mathrm{d}B_t^x] = 0$ — the closed-loop mean $\bar{m}_t(x) = \mathbb{E}[X_t^x]$ obeys the thread-A mean evolution eq. (14.13), while the offset solves its own *backward* equation eq. (14.8):

$$\dot{\bar{m}}_t(x) = (a - b^2 P(t))\,\bar{m}_t(x) \;-\; b^2\, r_t(x), \qquad \bar{m}_0(x) = \bar{m}_0^x, \tag{21.21}$$

$$\dot{r}_t(x) = (b^2 P(t) - a)\, r_t(x) \;+\; q\, s\, z_t(x), \qquad r_T(x) = -\, q_T\, s_T\, z_T(x). \tag{21.22}$$

Two points deserve emphasis, because a naive reading miscloses the system, and both are exactly the places where the dynamic graphon game is subtler than its static shadow. The first concerns *what forces the mean*. One is tempted to read (21.21) as "the mean relaxes toward the target $s\, z_t(x)$," but that is wrong, and seeing why is worth the line of algebra. Take the agent's SDE (21.2), substitute the optimal feedback (21.20), and average:

$$\dot{\bar{m}}_t(x) = \mathbb{E}[a\, X_t^x + b\, \alpha_t^{x,*}] = a\, \bar{m}_t(x) - b^2\, \mathbb{E}[P(t)\, X_t^x + r_t(x)] = (a - b^2 P(t))\, \bar{m}_t(x) - b^2\, r_t(x),$$

since $P(t)$ and $r_t(x)$ are deterministic and $\mathbb{E}[X_t^x] = \bar{m}_t(x)$. The forcing is $-b^2 r_t(x)$, the *offset* of the value function — *not* the target $s\, z_t(x)$ itself. The target never appears in the forward equation; it enters only *indirectly*, by driving the backward offset equation (21.22), whose forcing $q\, s\, z_t(x)$ is where the coupling actually lives. So (21.21)–(21.22) is a genuine *forward–backward* pair — a two-point boundary-value problem, the mean carrying initial data $\bar{m}_0(x)$ at $t = 0$ and the offset terminal data $r_T(x)$ at $t = T$ — and not a local forward ODE one could integrate by marching from $t = 0$. Mistaking $-b^2 r_t$ for a multiple of $s\, z_t$ would turn the two-point problem into a one-point one and silently destroy the very coupling between backward optimization and forward flow that *is* the mean field game.

The second point is the closure itself, and it is what makes rank one tractable where a general kernel is not. The only way the type label enters the drive is through the factor $f(x)$ in $z_t(x) = f(x)\varrho_t$. Look at the offset equation (21.22): its forcing $q\, s\, z_t(x) = q\, s\, f(x)\varrho_t$ and its terminal datum $-q_T s_T z_T(x) = -q_T s_T f(x)\varrho_T$ are *both* proportional to $f(x)$, while the coefficient $b^2 P(t) - a$ multiplying $r_t(x)$ on the left is type-independent. A linear ODE whose forcing and terminal data are a fixed function $f(x)$ times scalar paths has a solution that is $f(x)$ times a scalar path: substituting the ansatz $r_t(x) = f(x)\widetilde{r}_t$ into (21.22) cancels the common factor $f(x)$ from every term, leaving one scalar equation for $\widetilde{r}_t$. Thus the offset inherits exactly the factorization of the drive,

$$r_t(x) \;=\; f(x)\, \widetilde{r}_t, \qquad \dot{\widetilde{r}}_t = (b^2 P(t) - a)\, \widetilde{r}_t + q\, s\, \varrho_t, \quad \widetilde{r}_T = -\, q_T\, s_T\, \varrho_T, \tag{21.23}$$

with $\widetilde{r}_t$ the single scalar offset path. The mean forcing in (21.21) is now $-b^2 r_t(x) = -b^2 f(x)\widetilde{r}_t$ — the same function $f(x)$ times one scalar path — the only genuinely type-dependent residue being the initial datum $\bar{m}_0^x$. Everything is poised to close: the field game is driven entirely by the two scalars ϱ_t and $\widetilde{r}_t$, so we need only their two equations. The equation for $\widetilde{r}_t$ we already have, in (21.23). The equation for ϱ_t comes from *projecting* the mean ODE onto f — applying $\langle f, \cdot \rangle_{L^2}$ to both sides of (21.21), which is legitimate because the coefficients $a - b^2 P(t)$ and b^2 are scalars independent of x and pass through the integral over types. The left side becomes $\langle f, \dot{\bar{m}}_t \rangle = \frac{\mathrm{d}}{\mathrm{d}t}\langle f, \bar{m}_t \rangle = \dot{\varrho}_t$, by the definition (21.19) of the order parameter; the first term on the right becomes $(a - b^2 P(t))\langle f, \bar{m}_t \rangle = (a - b^2 P(t))\varrho_t$; and the forcing term becomes

$$-b^2 \langle f, r_t \rangle_{L^2} = -b^2 \langle f, f\, \widetilde{r}_t \rangle_{L^2} = -b^2\, \widetilde{r}_t\, \langle f, f \rangle_{L^2} = -b^2\, \|f\|_{L^2}^2\, \widetilde{r}_t = -b^2\, \|\mathbb{W}\|\, \widetilde{r}_t,$$

where the lone nonzero eigenvalue $\langle f, f \rangle_{L^2} = \|f\|_{L^2}^2 = \|\mathbb{W}\|$ of the rank-one operator enters *precisely here*: the projection step is where the operator norm is born into the scalar system, the one place the network's coupling amplitude survives the collapse. Assembling the three pieces and keeping the offset equation (21.23) beside it, the entire infinite-dimensional field game has closed onto the *scalar pair* $(\varrho_t, \widetilde{r}_t)$:

$$\begin{cases} \dot{\varrho}_t = (a - b^2 P(t))\, \varrho_t - b^2\, \|\mathbb{W}\|\, \widetilde{r}_t, & \varrho_0 = \langle f, \bar{m}_0 \rangle_{L^2}, \\ \dot{\widetilde{r}}_t = (b^2 P(t) - a)\, \widetilde{r}_t + q\, s\, \varrho_t, & \widetilde{r}_T = -\, q_T\, s_T\, \varrho_T. \end{cases} \tag{21.24}$$

This is a 2×2 linear forward–backward boundary-value problem in scalars, and it is the heart of the rank-one reduction: an infinite field of measures, one per type in a continuum, has collapsed to a single self-consistency in the two numbers $(\varrho_t, \widetilde{r}_t)$. The diagonal of the system is the type-independent LQG data $\pm(a - b^2 P(t))$; all the network and cost coupling sits in the two off-diagonal entries, $b^2\, \|\mathbb{W}\|$ feeding the offset back into the mean and qs feeding the mean forward into the offset. The decisive observation is that *these two coefficients are not separately visible to the system* — only their product is. To see it, rescale the unknowns by any constant $\lambda \neq 0$, setting $\varrho'_t = \lambda \varrho_t$ and $\widetilde{r}'_t = \lambda^{-1} \widetilde{r}_t$. Substituting into (21.24), the ϱ-equation becomes $\dot{\varrho}'_t = (a - b^2 P(t))\varrho'_t - (b^2\, \|\mathbb{W}\|\, \lambda^2)\, \widetilde{r}'_t$ and the offset equation becomes $\dot{\widetilde{r}}'_t = (b^2 P(t) - a)\widetilde{r}'_t + (qs\, \lambda^{-2})\varrho'_t$: the rescaling carries the system to one of identical form but with the two off-diagonals multiplied by λ^2 and λ^{-2} — reciprocally. A solution of one system is λ times a solution of the other, so the two are equivalent, and any property that decides solvability or uniqueness cannot depend on $b^2\, \|\mathbb{W}\|$ and qs individually: it can depend only on the one combination the rescaling leaves fixed, the product $(b^2\, \|\mathbb{W}\|)(qs) = b^2 q\, (s\, \|\mathbb{W}\|)$. Stripping the intrinsic LQG weights b^2 and q, the kernel and the cost coupling reach the equilibrium *only* through the single *effective coupling*

$$s_{\text{eff}} \;=\; s\, \|\mathbb{W}\| \;=\; s\, \|f\|^2_{L^2}\,. \tag{21.25}$$

This is the punchline of the rank-one model, and its consequences point straight at the next two chapters. Because the entire equilibrium is governed by the scalar pair (21.24), and that pair feels the kernel and the cost coupling only through $s_{\text{eff}} = s\, \|\mathbb{W}\|$, both solvability and uniqueness of the GMFG equilibrium in the rank-one case reduce to a single scalar condition on $|s_{\text{eff}}|$: the non-degeneracy of the linear two-point boundary-value problem, which fails only once $|s_{\text{eff}}|$ grows large enough to make the forward–backward coupling resonate over the horizon $[0, T]$. This is exactly the dynamic shadow of the static rank-one threshold $|s|\, \|\mathbb{W}\|$ of eq. (20.31). The static game saw that combination as the point where the one-shot fixed point lost uniqueness; the dynamic game sees the *same* combination $s\, \|\mathbb{W}\|$ as the point where the two-point problem degenerates. That two problems as different as a one-shot game and a finite-horizon control problem turn on the *identical* grouping $s\, \|\mathbb{W}\|$ is the first concrete evidence for the general threshold $|s|\, \|\mathbb{W}\|$ that chapter 24 will prove governs the full field game, and it is the threshold intuition this section hands forward. We deliberately stop at the closure: we have *exhibited* the scalar pair (21.24) and isolated its single control parameter, but we have not *solved* it. Producing the Riccati curvature $P(\cdot)$ in closed form and inverting the two-point pair into an explicit solution operator is the work of chapter 22, which inherits (21.24) verbatim as its rank-one test case. Setting up the closure, not solving it, is this chapter's job.

Remark 21.19 (Consistency of the two reductions)**.** The two reductions we have now performed — constant and rank-one — are not independent checks; the rank-one computation strictly *contains* the constant one, and verifying the nesting is a small but reassuring consistency test. A constant kernel $W \equiv c$ is the rank-one kernel with the *constant* shape function $f \equiv \sqrt{c}$, since then $f(x)f(y) = c$. Watch every object of worked computation 21.18 specialize. The order parameter becomes

$$\varrho_t = \langle f, \bar{m}_t \rangle_{L^2} = \sqrt{c} \int_0^1 \bar{m}_t(y) \, \mathrm{d}y = \sqrt{c}\, \bar{m}_t^{\mathrm{glob}},$$

the global mean scaled by $\sqrt{c}$; the drive becomes $z_t(x) = f(x)\varrho_t = \sqrt{c} \cdot \sqrt{c}\, \bar{m}_t^{\mathrm{glob}} = c\, \bar{m}_t^{\mathrm{glob}}$, type-independent, exactly the collapsed aggregate (21.17) of the constant case. The two normalizations specialize consistently too: $\|f\|_\infty^2 = c \le 1$ reproduces the graphon-validity bound $c \le 1$, while $\|f\|_{L^2}^2 = \int_0^1 c \, \mathrm{d}y = c = \|\mathbb{W}\|$ reproduces the lone eigenvalue of the constant kernel computed in remark 21.17. The effective coupling (21.25) becomes $s_{\mathrm{eff}} = s\,\|f\|_{L^2}^2 = sc$, precisely the rescaled coupling of remark 21.17, and the scalar pair (21.24) becomes the single-population forward–backward system of Part II. The rank-one worked computation thus strictly contains the constant-graphon reduction as its degenerate-shape case, and both agree with Part II — the chapter's reductions nest correctly.

Rank one gave us the cleanest *one*-dimensional collapse: the whole field communicates through a single scalar, because the coupling operator $\mathbb{W}$ has exactly one nonzero eigenvalue. But "one" is special only as the smallest non-trivial rank. If the kernel instead has finite rank K — if there are K distinct shapes of interaction rather than one — the same projection argument that produced (21.24) should close the field game not onto a single scalar but onto a finite, K-dimensional system. The most transparent finite-rank kernel is the *block* graphon, piecewise constant on a partition of $[0, 1]$ into K groups, and it is the third and last of our reduction checks. Its character is different from the two before it. Where the constant kernel recovered Part II behind us and the rank-one kernel previewed the closed form of chapter 22 ahead, the block kernel looks forward: it reproduces the finite *multi-population* mean field games — the very setting from which the field-of-measures picture of this chapter was abstracted in the first place. We turn to it now, moving from a one-dimensional reduction to a K-dimensional one.

21.5.3 Block graphon recovers multi-population mean field games

Rank one gave us the smallest nontrivial collapse: one shape function, one nonzero eigenvalue, and an entire field of measures speaking through a single scalar. The projection argument that produced it, though, used nothing special about the number one. It used only that the kernel's range was finite-dimensional, so that the type label could enter the drive through finitely many fixed profiles and a linear ODE forced by such profiles would close on the corresponding finitely many scalars. Raise the rank from one to K — replace the single interaction shape by K distinct ones — and the same machinery should close the field game not onto a scalar but onto a K*-dimensional* system. The cleanest finite-rank kernel that realizes this is the *block* graphon: piecewise constant on a partition of $[0, 1]$ into K groups, the discrete caricature in which the continuum of types is bundled into K homogeneous populations and all the network records is the $K \times K$ table of inter-group coupling strengths. It is the third and final reduction, and we make it the last because its lesson points in the opposite direction from the first two. The constant kernel looked *backward*, recovering the Part II theory we already trusted; the rank-one kernel looked *sideways*, previewing the closed form that chapter 22 will solve. The block kernel looks *forward*, and it closes a circle: it reproduces the finite *multi-population* mean field games of chapter 30 — the very setting from which the field-of-measures picture of this chapter was abstracted in the first place. Multi-population MFG is not a rival framework that the graphon theory must be reconciled with; it *is* the graphon theory, read on its piecewise-constant slice. Verifying that is the content of the following remark, and it is the natural place to end the chapter's exposition, because it shows the new object degenerating not to something older and poorer but to a parallel theory of independent standing, recovered exactly.

Remark 21.20 (Block W and the multi-population reduction)**.** Partition $[0,1]$ into K consecutive intervals $I_1,\dots,I_K$ of lengths $\pi_1,\dots,\pi_K$ (so $\sum_k \pi_k = 1$; the π_k are the population *mass fractions*), and let W be constant, $W(x,y) = W_{k\ell}$, on each cell $I_k \times I_\ell$ — the stochastic block model graphon of chapter 15, whose data is the single symmetric matrix $(W_{k\ell})$ of how strongly group k couples to group ℓ. The type label has been coarsened: within a block it does no work, and across blocks it does exactly the work of naming which of the K populations an agent belongs to.

Suppose, in parallel with the constant-kernel check of remark 21.17, that the remaining type-dependent data are block-constant too: the initial laws $m_0^x \equiv m_0^{(k)}$ and any type-dependent coefficients are common to all x in a given block I_k. Then every agent in block I_k solves an identical control problem — identical dynamics, identical cost, identical initial law, and (as we are about to see) identical aggregate — so the game *admits* a block-constant equilibrium field,

$$m_t^x \;\equiv\; m_t^{(k)} \qquad \text{for } x \in I_k, \quad k = 1,\dots,K, \tag{21.26}$$

one law per population rather than one per type. The same honesty owed in the constant case is owed here, and for the same reason: (21.26) exhibits *a* solution, not necessarily the *only* one. The argument shows that block-constant data *admit* a block-constant fixed point, not that every fixed point must respect the block structure. Below the coupling threshold the best-response map is a contraction (remark 21.12), its fixed point is unique, and uniqueness forces it to be the block-constant one just exhibited; above the threshold the block-constant equilibrium may coexist with finer, block-*breaking* equilibria in which agents of one and the same population settle into different laws — the identical symmetry-breaking caveat that remark 21.17 flagged for the constant kernel, neither sharpened nor relieved by the passage to K blocks. With that caveat recorded, restrict attention to the block-constant slice and watch the field game become finite.

The reduction is driven, exactly as before, entirely through the aggregate. On a block-constant field the first moments are block-constant, $\bar m_t(y) \equiv \bar m_t^{(\ell)}$ for $y \in I_\ell$, and the graphon-smoothed aggregate (21.5) felt by a type $x \in I_k$ collapses to a finite sum by splitting the integral over types into its blocks:

$$z_t(x) \;=\; \int_0^1 W(x,y)\,\bar m_t(y)\,\mathrm{d}y \;=\; \sum_{\ell=1}^K \int_{I_\ell} W_{k\ell}\,\bar m_t^{(\ell)}\,\mathrm{d}y \;=\; \sum_{\ell=1}^K \pi_\ell\, W_{k\ell}\,\bar m_t^{(\ell)}, \qquad x \in I_k, \tag{21.27}$$

which depends on x only through its block index k, confirming the self-consistency of the ansatz (21.26): a block-constant field produces a block-constant drive, which feeds back into block-constant optimal behaviour. The factor π_ℓ in each term is the same sub-probability bookkeeping that (21.3) insisted on — a more populous neighbour group ℓ (larger mass π_ℓ) contributes proportionally more to the aggregate felt by group k — and it is exactly the weight that prevents one from reading $W_{k\ell}$ alone as the effective inter-group coupling. The coupling that actually reaches group k from group ℓ is the *mass-weighted* entry $\pi_\ell W_{k\ell}$.

On this slice every distinguishing object of the field game now carries a finite index. The infinite family $(m_t^x)_{x\in[0,1]}$ has become the K-tuple $(m_t^{(1)},\dots,m_t^{(K)})$; the aggregate $z_t(x)$ has become the K numbers $z_t^{(k)} = \sum_\ell \pi_\ell W_{k\ell}\bar m_t^{(\ell)}$; and the single per-type control problem against a frozen scalar drive has become K control problems, the kth tracking its own drive $s\,z_t^{(k)}$, coupled to the others only through the means $\bar m_t^{(\ell)}$ that enter that drive. This is precisely a *finite system of K coupled mean field games* — one representative agent per population, each responding to a weighted blend of all K population means — with *inter-population coupling matrix*

$$M \;=\; (\pi_\ell\, W_{k\ell})_{k,\ell=1}^{K}. \tag{21.28}$$

This is the multi-population MFG developed in chapter 30, recovered here not as an analogy but as the literal piecewise-constant slice of the graphon theory; its explicit backward–forward PDE form — K coupled Hamilton–Jacobi–Bellman equations driven by K Fokker–Planck equations, laced together through M — is written out in chapter 22, which inherits (21.27) as its block test case just as it inherits the scalar pair (21.24) as its rank-one one.

One subtlety in (21.28) deserves a clarifying word, because the matrix that governs the dynamics and the matrix that governs the *threshold* are not the same matrix, and conflating them would mislocate the coupling constant. The coupling matrix $M = (\pi_\ell W_{k\ell})$ that appears in the dynamics is, in general, *not symmetric*: its (k,ℓ) entry weights by the mass π_ℓ of the *source* group, so $M_{k\ell} \neq M_{\ell k}$ whenever $\pi_k \neq \pi_\ell$, even though the underlying kernel $W_{k\ell} = W_{\ell k}$ is symmetric. The threshold of remark 21.12, however, is set by the *operator norm* $\|\mathbb{W}\|$, and $\mathbb{W}$ is self-adjoint on $L^2([0,1])$, so the relevant finite object is $\mathbb{W}$'s representation in an *orthonormal* basis of the block-constant subspace, where it is automatically symmetric. The block-constant fields form a K-dimensional $\mathbb{W}$-invariant subspace of $L^2([0,1])$, spanned by the indicators $\mathbf{1}_{I_k}$; but those indicators are not orthonormal — in the L^2 inner product $\langle \mathbf{1}_{I_k}, \mathbf{1}_{I_\ell}\rangle_{L^2} = \pi_k\,\delta_{k\ell}$, so each has length $\sqrt{\pi_k}$. Normalizing to the orthonormal basis $e_k := \mathbf{1}_{I_k}/\sqrt{\pi_k}$ and reading off the matrix of $\mathbb{W}$ there,

$$\langle e_k, \mathbb{W} e_\ell\rangle_{L^2} = \frac{1}{\sqrt{\pi_k\pi_\ell}}\,\langle \mathbf{1}_{I_k}, \mathbb{W}\mathbf{1}_{I_\ell}\rangle_{L^2} = \frac{1}{\sqrt{\pi_k\pi_\ell}}\int_{I_k}\Big(\int_{I_\ell} W_{k\ell}\,\mathrm{d}y\Big)\,\mathrm{d}x = \frac{\pi_k\pi_\ell\,W_{k\ell}}{\sqrt{\pi_k\pi_\ell}} = \sqrt{\pi_k\pi_\ell}\,W_{k\ell},$$

so the operator $\mathbb{W}$ restricted to block-constant fields is the *symmetric* $K\times K$ matrix

$$S \;=\; (\sqrt{\pi_k\pi_\ell}\,W_{k\ell})_{k,\ell=1}^{K} \;=\; D^{1/2} M\, D^{-1/2}, \qquad D := \operatorname{diag}(\pi_1,\dots,\pi_K), \tag{21.29}$$

the displayed similarity $S = D^{1/2}MD^{-1/2}$ being a one-line check on the entries. Because S and M are *similar*, they share the same eigenvalues: the spectrum that decides solvability and multiplicity of the K-population system is the same whether one reads it off the asymmetric dynamical matrix M or its symmetrized cousin S. But only S is symmetric, and only for a symmetric matrix does the spectral *norm* (largest singular value) coincide with the spectral *radius* (largest eigenvalue modulus). The threshold is an operator-*norm* statement, $\|\mathbb{W}\|$ restricted to this subspace, so it is $\|S\|$ — the top eigenvalue of the symmetric matrix S — that is the relevant coupling constant, and using $\|M\|$ instead (its largest singular value, which can strictly exceed $\|S\|$ when the masses are unequal) would overstate the coupling. Thus the spectrally correct finite avatar of the kernel is the mass-symmetrized matrix (21.29), and $\|S\| = \|\mathbb{W}\|\,|_{\text{block}}$ multiplied by $|s|$ is the block instance of the universal control parameter $|s|\ \|\mathbb{W}\|$ of remark 21.12 — in full consistency with the threshold the chapter has tracked throughout.

With the block reduction in hand the chapter's single argument is complete. We set out to build one object — the dynamic graphon mean field game — precisely enough that the sentence "$(\boldsymbol{m}_t)_t$ is an equilibrium" has a definite truth value, and then to earn trust in that object by watching it degenerate correctly. It has now passed all three tests. The constant kernel folded the field of measures back to the single population law of Part II, with the network contributing only a scalar rescaling of the coupling (remark 21.17); the rank-one kernel kept the type label genuinely at work yet collapsed the infinite field onto a single self-consistent scalar with effective coupling $s\,\|\mathbb{W}\|$ (worked computation 21.18); and the block kernel has just reproduced the finite multi-population mean field game as the piecewise-constant slice, with mass-weighted coupling matrix (21.28) and spectrally relevant norm (21.29). In all three the same control parameter $|s|\ \|\mathbb{W}\|$ surfaced and the same "a symmetric equilibrium exists, uniqueness is the delicate question above threshold" caveat recurred — the consistent fingerprint of a single well-posed construction seen from three angles, not three separate constructions.

That is where the exposition ends and the engines take over. The setup assembled here — the representative agent of every type (definition 21.1), the field of measures that is the population (definition 21.4), the kernel-weighted aggregate (definition 21.5), the complete separable L^2-Wasserstein field space, the $\|\mathbb{W}\|$-Lipschitz aggregation that births the threshold (remark 21.12), and the type-by-type Nash fixed point that is the equilibrium (definition 21.16) — is now fixed and unambiguous, and the next three chapters consume it untouched. Chapter 22 turns the fixed point into the coupled Hamilton–Jacobi–Bellman–Fokker–Planck system and supplies, for Thread B, the Riccati curvature and the explicit solution operator that closes the rank-one and block reductions left open here; chapter 23 recasts the same fixed point probabilistically as a system of graphon forward–backward stochastic differential equations; and chapter 24 proves existence and uniqueness, turning the threshold $|s|\ \|\mathbb{W}\| < 1/L_{\text{ctrl}}$ from a heuristic into a theorem. What remains of this chapter is the bookkeeping that makes the hand-off honest: the ledger below records, line by line, what has been *proved* here, what has been *assumed*, what has been *deferred* to those three chapters and to the finite-N theory of chapter 25, and what genuinely remains *open*.

Rigor ledger

Proved.

- Well-posedness of the type-x controlled SDE (21.1) for each fixed type under (A-Lip), (A-Growth), (A-Nondeg), and measurability of the field of laws $x \mapsto m_t^x$ with uniform second moment (lemma 21.13).
- $(\boldsymbol{\mathcal{P}}_2, d_2)$ is a complete separable metric space and $(\mathcal{C}_2, d_\infty)$ a complete metric space embedding into it (proposition 21.9, definition 21.10).
- The graphon aggregate is well-defined and $\|\mathbb{W}\|$-Lipschitz on $(\boldsymbol{\mathcal{P}}_2, d_2)$ via the elementary

coupling bound $|\langle h, \mu\rangle - \langle h, \nu\rangle| \le \mathcal{W}_1 \le \mathcal{W}_2$ for 1-Lipschitz h (lemma 21.11), locating the coupling threshold at $|s| \, \|\mathbb{W}\|$.

- The constant-graphon reduction to classical MFG (remark 21.17); the rank-one collapse to a scalar self-consistency with effective coupling $s \, \|\mathbb{W}\|$ (worked computation 21.18); the block reduction to multi-population MFG (remark 21.20).

Assumed.

- The standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) on the agent dynamics and cost, and (A-Reg-W) on the kernel (used in lemma 21.11 and the spectral statements).
- Existence of optimizers for the per-type control problem via the verification theorem of chapter 6 under (A-Conv), (A-Nondeg).

Deferred.

- Existence and uniqueness of a fixed point of S (definition 21.16): chapter 24, under (A-Coupling) or a graphon Lasry–Lions condition (definition 11.1).
- The explicit coupled backward HJB / forward Fokker–Planck system across the kernel, the Riccati solution $P(\cdot)$, and the closed-form solution operator of the scalar forward–backward pair (21.24) used in worked computation 21.18: chapter 22.
- The probabilistic (graphon-FBSDE) characterization: chapter 23. The finite-network derivation underlying section 21.2: chapter 25.

Open.

- Whether the d_2 (existence-friendly) and d_∞ (regularity-friendly) topologies of section 21.3 can be reconciled into a single space carrying both compactness and continuity-in-type without extra kernel regularity, in the most general non-LQG setting.
- Rigorous status of an exact law of large numbers across a continuum of types at the level of realized trajectories (remark 21.2), beyond the Fubini-extension device of [125], in the controlled graphon setting.

Bibliographic notes

The dynamic graphon mean field game was introduced by Caines and Huang [33, 34], whose GMFG equations are the object this chapter sets up and chapter 22 solves; their formulation works with a type-indexed family of HJB–Fokker–Planck systems coupled through the graphon operator, precisely the structure of definition 21.16. The field-of-measures state space and its L^2-Wasserstein metric, our definition 21.8, follow the analytic setting used by Bayraktar, Wu and Zhang for graphon mean field systems [16] and by Aurell, Carmona and Laurière in the linear-quadratic graphon game [10], the latter being the closest published analogue of the Thread B model of example 21.3; the static counterpart, with the same aggregate $z(x)$, is the stochastic graphon game of Carmona, Cooney, Graves and Laurière [39] and the graphon games of Parise and Ozdaglar [108] treated in chapter 20. The subtlety of a continuum of independent noises (remark 21.2) is the exact-law-of-large-numbers problem resolved by Sun's Fubini extensions [125]; the route we actually take — posing the game through the deterministic field of laws — is the graphon lift of the McKean–Vlasov formulation of [103, 129] developed in chapter 7. The Wasserstein estimate underlying lemma 21.11 (the elementary coupling bound $|\langle h, \mu\rangle - \langle h, \nu\rangle| \le \mathcal{W}_1 \le \mathcal{W}_2$ for 1-Lipschitz h, the easy half of the Kantorovich–Rubinstein picture) uses only the definition of $\mathcal{W}_1$ and the ordering $\mathcal{W}_1 \le \mathcal{W}_2$ of chapter 4, following [131]; the completeness of $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ and the metric-target L^2 theory are in [7]. The reductions of section 21.5 trace back to the classical LQG mean field game of Huang, Caines and Malhamé [82] (constant coupling) and to the rank-one/Curie–Weiss picture flagged throughout the graphon literature; the multi-population limit of remark 21.20 is surveyed in the modern overview [11] and developed in chapter 30.

Chapter 22

The Coupled HJB–Fokker–Planck System Across a Kernel

In chapter 21 the dynamic graphon mean field game was *set up*: a continuum of representative agents indexed by a type $x \in [0,1]$, each carrying a private state and a controlled diffusion, coupled not to a single global population law but to a *field of measures* $x \mapsto m_t^x$ through the graphon operator $\mathbb{W}$. That was half the problem — it fixes who plays, what each agent controls, and what each pays, but not what they do. This chapter solves the other half: the actual system of equations the equilibrium must satisfy. It helps to say in advance what an *equation-level* answer even looks like, because it is neither a number nor a single curve. An equilibrium of a differential game is a pair of partial differential equations pinned at opposite ends of the clock — a backward equation for an agent's value, carrying *terminal* data at $t = T$, coupled to a forward equation for that agent's law, carrying *initial* data at $t = 0$ — and here there is one such pair for *every type at once*. The object we are after is therefore a continuum-indexed family of such pairs, parametrized by $x \in [0,1]$. This is the graphon lift of chapter 8, where the strategic fixed point of a homogeneous population condensed into exactly one backward Hamilton–Jacobi–Bellman (HJB) equation driven by the density and one forward Fokker–Planck (FP) equation driven by the optimal feedback, with data at the two ends of $[0,T]$. Here the identical dual structure holds per type, and the *only* new ingredient is that the types talk to one another — but solely through the kernel W. That last clause is the organizing promise of the whole chapter, and it is worth stating as a slogan: the coupling between types factors through a single linear object, the integral operator $\mathbb{W}$, and through nothing else. Every section below either exploits this fact or guards against over-reading it. The regularity section pushes continuity-in-type around the best-response loop precisely because $\mathbb{W}$ is linear and bounded; the special-structures section collapses the continuum of couplings to finitely many numbers precisely because $\mathbb{W}$ has a usable spectrum; the worked computation reads its leading eigenvalue as the single rescaling of the interaction strength; and the couplings section warns against the seductive but false reading of $\mathbb{W}$ as a local gradient coupling between neighbouring types. The result we are heading for is thus a continuum-indexed family of HJB–FP pairs, locked together by the integral operator $\mathbb{W}$ and by nothing else.

What we borrow from earlier chapters, we borrow for a specific later use, and it is worth naming the use rather than the citation. From chapter 8 we take the per-type *engine*: the analytic mean field game system (definition 8.1), the verification theorem (proposition 8.2) that certifies an HJB–FP pair as a genuine optimization rather than a formal pair of PDEs, the best-response reformulation (proposition 8.4) that turns the pair into a fixed point, and the coupling taxonomy of section 8.5 that classifies how the population may enter a cost. Once the field is frozen, each type *is* a Chapter 8 problem, so this is the whole single-type apparatus and we re-use it verbatim. From the existence theory of chapter 10 we take the two *stability* lemmas (lemmas 10.10 and 10.11), which bound how far the value and the density move when the coefficients and the environment move; these are exactly what powers section 22.3, where the perturbation in question is a change of *type* and the payoff is that the solution depends measurably, and continuously, on it. From chapter 18 we take the *spectral* picture of the graphon as a self-adjoint Hilbert–Schmidt operator; its eigenvalues and eigenfunctions are precisely what section 22.4 uses to collapse the infinite type-coupling to finitely many order parameters, and the rank-one corner of that collapse is what feeds the worked computation. The Lasry–Lions monotonicity condition (definition 11.1) we carry only to mark where uniqueness would call for it. And from the LQG capstone of chapter 14 we take the *template* the chapter's worked computation fills in: its closing section 14.7 already wrote the multi-population Riccati system, and our Thread B is that same calculation with the scalar population mean $\bar{m}_t$ replaced by the graphon-aggregated neighbor mean $(\mathbb{W}\bar{m}_t)(x)$, watching each Riccati equation become a type-indexed family. Two things this chapter deliberately does *not* do, stated here as the reader's map: we do not prove existence or uniqueness of the coupled fixed point — that is the charge of chapter 24, which consumes the system written below — and we do not give the probabilistic, forward–backward stochastic differential equation rendering of the same equilibrium, which is chapter 23 and will refer back to the system eq. (22.6) defined here.

The plan is as follows. section 22.1 states the graphon MFG system, derives it from the per-type optimization against a frozen field, gives the "coupling only through $\mathbb{W}$" reading, and casts it as a fixed point of a best-response map on fields of measures. section 22.2 compares the genuinely nonlocal (kernel) coupling that a graphon provides with the local (gradient) coupling of a spatial field theory, and develops the Ginzburg–Landau analogy that organizes the rest of Part IV — with an explicit warning about its one false note. section 22.3 establishes the regularity and measurability of the map $x \mapsto (u^x, m^x)$, the technical backbone that makes "a continuum of HJB–FP pairs" a well-defined object. section 22.4 treats the solvable structures: separable (finite-rank) kernels collapse the type-coupling to finitely many order parameters, and block graphons recover finite multi-population MFG, the explicit bridge to chapter 30. section 22.5 is the worked computation that advances Thread B: the rank-one graphon LQG, solved in closed form to a single scalar two-point boundary value problem with an effective coupling rescaled by the leading eigenvalue of $\mathbb{W}$.

Notation watch

Three conventions, all announced in appendix A and restated here because Part IV is the first place all of them collide in one equation — and each collision is a genuine hazard, not a pedantry. The system eq. (22.6) runs two clocks in opposite directions in the same display (so a sign or a boundary datum read off the wrong end is silently wrong), writes the kernel and the noise with letters that were historically the same (so W used loosely is ambiguous), and carries a type label beside a state variable in every function (so a derivative taken in the wrong argument is meaningless). A reader who lets any of the three slide will misread the equations; taken in turn: *(i) The two directions of time.* The value field $u^x = u^x(t,\xi)$ carries *terminal* data at $t = T$ and is computed *backward*; the density field $m^x = m^x(t,\xi)$ carries *initial* data at $t = 0$ and is transported *forward*. *(ii) Brownian motion versus graphon.* The kernel owns the letter $W = W$; Brownian motion is therefore written B_t throughout Part IV. *(iii) Type versus state.* The variable $x \in [0,1]$ is the exchangeable *type* label and carries no metric of its own (appendix A); the agent's *private state* lives in $\mathbb{R}^d$ and is written ξ (the running spatial variable) or X_t^x (the process). Thus $u^x(t,\xi)$ has the type as a superscript parameter and the state ξ as its PDE argument, and ∇ and D^2 act in ξ unless decorated otherwise. We never differentiate in x (there is, in general, no derivative in type to take).

22.1 The graphon mean field game system

This section delivers the chapter's central object — the graphon MFG system itself — in three deliberate steps, and it is worth seeing the arc before the formalism arrives. First we recall the standing data of chapter 21: the space of fields of measures, the two metrics it carries, and above all the graphon-aggregated neighbor distribution $(\mathbb{W}\boldsymbol{m}_t)(x)$, read in the only sense this chapter needs — the weak one. Nothing can be written down until this aggregate is pinned, because it is the one channel through which a type feels the rest of the population, and an equation is only as well-posed as the object that sits inside its nonlinearity. Second, with the data in hand, we run the mean-field heuristic one type at a time: freeze the field, let the type-x agent optimize against the frozen aggregate as if it were an ordinary prescribed environment, then close the loop by demanding that the law this agent generates reproduce the field it was optimizing against. Performed simultaneously for every type, this yields the coupled backward-HJB / forward-FP system that is the chapter's deliverable. Third, we recast that system in the form the rest of the book actually consumes: a solution is a fixed point of a best-response map on flows of fields, the graphon analogue of the homogeneous map of chapter 8, whose single genuinely new feature is that one application of it already requires the entire field over $[0,1]$ and therefore does not factor over types. We begin with the data.

22.1.1 Standing data

Before any equation can be written we must fix the arena it lives in, and the section opener promised exactly this: first recall the objects chapter 21 built — the field of measures, the two metrics it carries, and above all the graphon-aggregated neighbor distribution — because the equilibrium system is a statement *about* these objects and is no better defined than they are. We import them rather than redefine them, citing chapter 21 at each step.

The state of the whole population at one instant is not a single measure but a measure for every type. A *field of measures* is a map $\boldsymbol{m}\colon [0,1] \to \mathcal{P}_2(\mathbb{R}^d)$, $x \mapsto m^x$, measurable in x and with a uniformly bounded second moment, $\sup_x \int_{\mathbb{R}^d} |\xi|^2\, m^x(\mathrm{d}\xi) < \infty$ — this is the field space $\boldsymbol{\mathcal{P}}_2$ of definition 21.8. Both qualifiers earn their place. The measurability in x is the price of admission: without it the type-integral that defines the aggregate below would be a nonsense, since one cannot integrate a non-measurable family. The uniform second-moment bound is what keeps that integral landing back inside $\mathcal{P}_2(\mathbb{R}^d)$ rather than letting mass run off to infinity along some sequence of types. A *flow of fields* is the time-dependent version, a map $t \mapsto \boldsymbol{m}_t = (m_t^x)_{x\in[0,1]}$; this flow is the unknown the equilibrium system will solve for.

Two distinct metrics on fields will be used, and the reader should feel *why* there are two before either is written down, because the distinction is not pedantry — it is the reason this chapter and chapter 24 are phrased on different spaces. A field is a function of the type x valued in the metric space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, so to compare two fields we must somehow aggregate the per-type Wasserstein gaps $\mathcal{W}_2(m^x, (\boldsymbol{m}')^x)$ over x, and there are two natural ways to do that. The first averages them in L^2,

$$d_2(\boldsymbol{m}, \boldsymbol{m}') = \Big(\int_0^1 \mathcal{W}_2(m^x, (\boldsymbol{m}')^x)^2 \, \mathrm{d}x \Big)^{1/2},$$

the L^2-over-types Wasserstein metric on $\boldsymbol{\mathcal{P}}_2$ (the default of definition 21.8); the second takes their supremum,

$$d_\infty(\boldsymbol{m}, \boldsymbol{m}') = \sup_x \mathcal{W}_2(m^x, (\boldsymbol{m}')^x),$$

the uniform-in-type metric on the type-continuous subspace $\mathcal{C}_2$ of definition 21.10. They serve different masters. The L^2 metric d_2 is *separable* and tolerant: it permits the per-type measures to be merely measurable in x, and it counts two fields as close as soon as they are close *on average* over types, so a single badly-behaved type sitting on a Lebesgue-null set costs nothing. That tolerance is precisely what makes d_2 *compactness-friendly*, and compactness is the one thing chapter 24 cannot do without, since its entire job is to extract a convergent subsequence of near-equilibria; that is why chapter 24 lives on $(\boldsymbol{\mathcal{P}}_2, d_2)$. The uniform metric d_∞ is stricter: d_∞-closeness demands the two fields be close at *every* type simultaneously, with no bad type allowed to hide. That strictness is a liability for compactness but an asset for *stability*, which is this chapter's currency. The per-type estimates we inherit from chapter 10 are bounds that hold for each type separately and with a constant uniform across types — sup-in-type estimates — so they assemble with no loss into a clean d_∞ statement; phrasing them in d_2 would only throw the per-type sharpness away in an average. This is why every stability and regularity result below is phrased on $(\mathcal{C}_2, d_\infty)$ while chapter 24 switches back to $(\boldsymbol{\mathcal{P}}_2, d_2)$. The two are emphatically *not* interchangeable. One inequality always holds, $d_2 \le d_\infty$ (an average never exceeds a supremum), so d_∞-convergence implies d_2-convergence but never the reverse; and the gap has teeth, because $\mathcal{C}_2$ is *not* d_2-closed — a d_2-Cauchy sequence of type-continuous fields can converge to a field that is only measurable, its type-continuity dissolved in the averaging. The mental model to keep is therefore: d_∞ is the working metric of the present chapter, and d_2 is the metric chapter 24 drops to precisely when it needs compactness, paying for it by leaving $\mathcal{C}_2$. section 21.3 records this trade-off, and we name the metric at each result. In particular, calling a flow of fields *continuous* will always mean continuity for d_∞, i.e. $\sup_x \mathcal{W}_2(m_t^x, m_s^x) \to 0$ as $s \to t$ — the per-type laws move continuously in time, uniformly over types.

With the arena fixed we can describe the players. The type-x representative agent has private state $X_t^x \in \mathbb{R}^d$ obeying the controlled diffusion

$$\mathrm{d}X_t^x \;=\; b(x, X_t^x, \alpha_t^x)\ \mathrm{d}t \;+\; \sigma(x)\ \mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x, \tag{22.1}$$

with type-dependent drift $b\colon [0,1]\times\mathbb{R}^d\times A\to\mathbb{R}^d$, volatility $\sigma\colon[0,1]\to\mathbb{R}^{d\times d}$, independent Brownian motions B^x, and feedback control $\alpha^x_t=\alpha^x(t,X^x_t)$.

The single structural novelty relative to chapter 8 — the one ingredient that makes this a *graphon* mean field game and not a parametrized family of ordinary ones — is how the population enters the agent's cost. A type-x agent does not feel the global population. It feels the *graphon-aggregated neighbor distribution* (definition 21.5), the kernel-weighted average of the laws of all other types,

$$(\mathbb{W}\boldsymbol{m}_t)(x) \;:=\; \int_0^1 W(x,y)\, m_t^y \;\mathrm{d}y \;\in\; \mathcal{M}_+(\mathbb{R}^d). \tag{22.2}$$

This is the object every later section either exploits or guards against over-reading, so it is worth dwelling on what it is and how to read it honestly.

First, what kind of measure it is. The weights $W(x,y)$ lie in $[0,1]$ and integrate over y to the *degree* of type x, $(\mathbb{W}\mathbf{1})(x)=\int_0^1 W(x,y)\,\mathrm{d}y\le 1$. Since each m_t^y is a probability measure, the aggregate is a convex-like combination of probability measures with total weight at most one: its total mass is exactly that degree, so in general $(\mathbb{W}\boldsymbol{m}_t)(x)$ is a *sub-probability* measure, an element of $\mathcal{M}_+(\mathbb{R}^d)$ rather than $\mathcal{P}_2(\mathbb{R}^d)$. The missing mass is not a defect to be normalized away; it is information. A weakly-connected type — one for which $W(x,\cdot)$ is small — genuinely feels a fainter, lighter neighbor distribution than a hub, and the cost eq. (22.4) is built to register that difference. Only when every row of the kernel integrates to one (a doubly-stochastic, in particular the complete, graphon) does the aggregate stay a probability measure for every type.

Second, and this is the point of method, we read eq. (22.2) *weakly*, and the weak reading is the only sense this chapter ever uses. Taken at face value, the integral $\int_0^1 W(x,y)\,m_t^y\;\mathrm{d}y$ is an integral of a *measure-valued* integrand over y — to give it meaning directly one would have to develop vector-valued (Bochner) integration in the space of measures, fuss over which Banach space the measures sit in, and verify Bochner-measurability and integrability of $y\mapsto m_t^y$. None of that is necessary, because the integral is only ever tested against functions. We therefore *define* the aggregate by how it pairs with test functions: for a.e. x, $(\mathbb{W}\boldsymbol{m}_t)(x)$ is the unique finite Borel measure on $\mathbb{R}^d$ characterized by

$$\langle\varphi,(\mathbb{W}\boldsymbol{m}_t)(x)\rangle=\int_0^1 W(x,y)\,\langle\varphi,m_t^y\rangle\;\mathrm{d}y\qquad\text{for every bounded continuous }\varphi\colon\mathbb{R}^d\to\mathbb{R}. \tag{22.3}$$

The gain is that the right-hand side is now an ordinary *scalar* Lebesgue integral: the integrand $y\mapsto W(x,y)\langle\varphi,m_t^y\rangle$ is a bounded measurable real-valued function of y, nothing more, because $\langle\varphi,m_t^y\rangle=\int_{\mathbb{R}^d}\varphi\;\mathrm{d}m_t^y$ is a bounded scalar for each y and is measurable in y by the field's measurability. The whole vector-valued machinery collapses to scalar integration done one test function at a time, and that a single Borel measure is pinned down by its pairings against bounded continuous φ is just the fact that such φ are convergence-determining on $\mathbb{R}^d$ (chapter 3). This is exactly the measure-valued action $\mathbb{W}\circledast$ of definition 21.5; we drop the $\circledast$ decoration and write $\mathbb{W}$ for both the scalar action on $L^2[0,1]$ and this measure-valued action, the argument disambiguating which is meant.

That the prescription eq. (22.3) genuinely does define a finite measure — with mass ≤ 1 and uniformly bounded second moment — and that this measure depends *measurably* on x (indeed *continuously* when the kernel is), is not assumed here but *deferred*: it is exactly the content of section 22.3, where it is proved as lemma 22.8. We flag this as a forward promissory note rather than a standing hypothesis, because the measurability is what later closes the best-response loop $\mathsf{S}\colon \mathcal{C} \to \mathcal{C}$ of definition 22.5, and it deserves a proof rather than a stipulation.

With the aggregate in hand we can state how the population enters payoffs. The running cost L and terminal cost g of the type-x agent depend on the population *only* through the aggregate eq. (22.2):

$$J^x(\alpha^x) = \mathbb{E}\left[\int_0^T L\Big(x, X_t^x, \alpha_t^x, (\mathbb{W}\boldsymbol{m}_t)(x)\Big)\,\mathrm{d}t + g\Big(X_T^x, (\mathbb{W}\boldsymbol{m}_T)(x)\Big)\right]. \tag{22.4}$$

The single appearance of $(\mathbb{W}\boldsymbol{m}_t)(x)$ inside L and of $(\mathbb{W}\boldsymbol{m}_T)(x)$ inside g is the precise content of the slogan that "coupling acts across types only through $\mathbb{W}$": strip the aggregate out and each type would be an isolated control problem with no knowledge of any other; put it back and the only wire between types is the linear operator $\mathbb{W}$. Everything downstream turns on this single point of contact.

The optimization the type-x agent performs against its frozen environment is encoded, exactly as in the homogeneous theory, by a Hamiltonian. We define the per-type Hamiltonian, with the type x now carried as a parameter, by the same Legendre-type formula as eq. (8.4):

$$H(x,\xi,p,\nu) \;:=\; \sup_{a\in A}\Big\{ -b(x,\xi,a)\cdot p - L(x,\xi,a,\nu)\Big\}, \qquad \nu \in \mathcal{M}_+(\mathbb{R}^d), \tag{22.5}$$

so that its four slots are the type x, the state ξ, the costate p (the value gradient, in the equations below), and the aggregate measure ν, into which $(\mathbb{W}\boldsymbol{m}_t)(x)$ will be substituted. Note that the population variable lives in the cost slot only: the supremand depends on ν through L and not through b, which is the structural feature remark 22.1 isolates and the next subsection exploits when it freezes the field.

The hypotheses under which all of this is legitimate are the standing assumptions of appendix A, invoked here by tag but now demanded *uniformly in the type* — uniformity being the whole content of "a continuum of well-behaved problems" rather than "a problem for each type with no control across them." Each earns its keep, and it is worth saying what breaks without it.

(A-Lip) b, σ, L, g are Lipschitz in the state ξ, uniformly in (t, x, a) and in the measure argument (the latter for $\mathcal{W}_2$ on $\mathcal{M}_+$); and they are Lipschitz in the type x, uniformly in the remaining arguments. The state-Lipschitz half is the standard well-posedness input for the SDE eq. (22.1) and the per-type PDEs; the measure-Lipschitz half is what lets a small change in the aggregate produce only a small change in the cost, the engine of every stability estimate. The *type*-Lipschitz half is the genuinely new clause, and it is what powers section 22.3: without continuity of the data in x the solution field $x \mapsto (u^x, m^x)$ could jump arbitrarily, and "a continuum of HJB–FP pairs" would be a continuum in name only.

(A-Growth) linear growth in ξ and coercivity in a, uniformly in x, so the supremum in eq. (22.5) is attained and finite. Drop coercivity and the agent could push the control to infinity at vanishing marginal cost, leaving the Hamiltonian $+\infty$ and the maximizer undefined; drop linear growth and the second moment could blow up in finite time, ejecting m_t^x from $\mathcal{P}_2(\mathbb{R}^d)$ and breaking the moment bounds that keep the aggregate inside $\mathcal{M}_+(\mathbb{R}^d)$.

(A-Nondeg) $\sigma(x)\sigma(x)^\top \succeq \lambda I$ for some $\lambda > 0$ independent of x. This is the parabolic-regularization clause: a uniformly elliptic second-order term smooths both the backward HJB and the forward FP, giving classical solutions and, crucially, an honest *density* $m^x(t,\cdot)$ for the FP equation rather than a bare measure. Without it the FP solution can stay singular, the local couplings of section 22.2 (which read off a pointwise density) become ill-posed, and the stability constants degrade as $\lambda \downarrow 0$. The uniformity in x is what makes those constants assemble into the d_∞ estimates.

(A-Conv) $H(x,\xi,\cdot,\nu)$ convex in the costate, so the feedback $-\partial_p H$ is *single-valued.* This is what turns the optimal control into a genuine function of (t,ξ) rather than a set: with a unique maximizer the closed-loop drift of the FP equation is well-defined, the best-response map is a map and not a correspondence, and — the payoff for section 22.3 — the type-to-solution assignment needs no measurable-selection theorem, since there is nothing to select.

(A-Reg-W) $W\colon [0,1]^2 \to [0,1]$ symmetric, measurable, bounded; continuous in type where stated; $\mathbb{W}$ self-adjoint Hilbert–Schmidt on $L^2[0,1]$. Boundedness gives the sub-probability mass bound above; measurability is what makes eq. (22.3) a legitimate scalar integral; symmetry and Hilbert–Schmidtness are what hand $\mathbb{W}$ a real spectrum and an orthonormal eigenbasis, the spectral data that section 22.4 collapses to order parameters and section 22.5 reads off as a single eigenvalue. The continuity-in-type clause is invoked only where stated, for the continuous (as opposed to merely measurable) half of section 22.3.

Two further assumptions from the standing list are deliberately held back. The Lasry–Lions monotonicity **(A-Mon)** and the operator-norm smallness **(A-Coupling)** are levers for *uniqueness* of the fixed point, not for writing the system down, and this chapter neither proves uniqueness nor needs them; we defer both to chapter 24 and flag each use there.

Remark 22.1 (The subclass solved here)**.** It is worth being explicit that eqs. (22.1) and (22.4) are not the most general agent chapter 21 allows, and about why the restriction is exactly the one that lets the chapters 8 and 10 machinery transfer with no new analysis. The general representative agent of definitions 21.1 and 21.6 carries a richer signature: there the drift may itself depend on the aggregate, $b(x, X, \alpha, (\mathbb{W}\boldsymbol{m}_t)(x))$, and the diffusion may depend on the state, $\sigma(x, X)$. This chapter — and the existence theory of chapter 24 that consumes it — solves the subclass in which two simplifications hold. (i) The drift does *not* depend on the aggregate, so the population enters the agent's problem only through the cost, and hence only as the source term and terminal datum of the backward HJB; the population never touches the dynamics directly. (ii) The diffusion is state-independent, $\sigma = \sigma(x)$, so the second-order operator in eqs. (22.6a) and (22.6b) has coefficients that are constant in the state ξ (depending on the type but not on where the agent is).

These two are precisely what make the per-type theory of chapters 8 and 10 apply *verbatim.* Restriction (i) is the load-bearing one. Because the aggregate sits in the cost alone, freezing the field renders the HJB an ordinary backward equation with a prescribed time-dependent source $F(x,\xi,(\mathbb{W}\boldsymbol{m}_t)(x))$ and a prescribed terminal datum — a standard single-agent control problem in a fixed environment, to which the verification theorem applies unchanged. The forward FP equation it generates is then *linear* in m^x: its transport drift $-\partial_p H(x,\xi,\nabla u^x)$ is built from the value alone and does not see m^x, so the only nonlinearity in the whole coupled system is the benign one routed through the cost. Restriction (ii) keeps the elliptic part of both equations a constant-coefficient operator in ξ, which is what **(A-Nondeg)** needs to deliver classical solutions and densities uniformly in x.

Were restriction (i) dropped — the aggregate-dependent-drift, or *congestion*, case — the picture changes qualitatively. The optimal closed-loop drift would then read $-\partial_p H(x, \xi, \nabla u^x, (\mathbb{W}\boldsymbol{m}_t)(x))$, carrying the aggregate inside it, so the population would enter the forward transport *directly* and not merely through the value. The FP equation eq. (22.6b) would lose its linearity in m^x and become a genuinely nonlinear transport coupled across types, in which the very drift that moves m^x depends on the graphon average of all the m^y. That is a strictly harder problem — the clean per-type decoupling under a frozen field is gone — and we do not solve it here; it is logged at the research frontier in the closing ledger.

We now have everything the next subsection needs: the field space and its two metrics, the controlled state eq. (22.1), the weakly-read aggregate eq. (22.3) that is the sole channel of coupling, the cost eq. (22.4) that depends on the population only through it, and the per-type Hamiltonian eq. (22.5) into which the aggregate is substituted. With the data fixed, the move that produces the equilibrium system is the mean-field heuristic applied one type at a time — freeze the field, let the type-x agent optimize its Hamiltonian against the frozen aggregate as against any prescribed environment, then demand the resulting law reproduce the field. We carry out that freezing next.

22.1.2 The system

The previous subsection left us holding exactly what we need — the controlled state eq. (22.1), the weakly-read aggregate eq. (22.3), the cost eq. (22.4) that sees the population only through that aggregate, and the per-type Hamiltonian eq. (22.5) — and it flagged the move that turns this data into an equilibrium system: the mean-field heuristic, run one type at a time. We now carry out the freezing it promised, and it pays to develop it slowly, because the whole claim of the chapter is that this single heuristic survives the graphon lift untouched.

The homogeneous theory of chapter 8 reached its system through three beats: *freeze* the population, *solve* the representative agent's control problem against the frozen population, then *close* the loop by demanding that the law the agent generates coincide with the population it was handed (section 8.1.2). The only thing that could obstruct lifting these three beats to the graphon setting is the coupling. In the homogeneous problem the agent feels one global law m_t; here it feels the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$, an object assembled from the *entire* field $\boldsymbol{m}_t$ over all types. So the question that decides whether the lift goes through is sharp: once we freeze the field, does each individual type still face an *ordinary* control problem, or does the cross-type pooling leave behind some residual entanglement that no single-agent optimization can swallow?

It goes through cleanly, and seeing precisely why is the content of the freezing step. Fix a candidate flow of fields $\boldsymbol{m} = (\boldsymbol{m}_t)_{t\in[0,T]}$ — a full trial guess at the time-dependent population state, one measure m_t^x for every type and every instant. To *freeze* the field is to treat this entire flow as given data, inert for the duration of the agent's optimization, not as an unknown to be solved for. Now isolate a single type x. The only place the population touches the type-x agent's problem is the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ appearing in the running and terminal costs of eq. (22.4). But with the whole field $\boldsymbol{m}$ frozen, the curve

$$\nu_t^x \;:=\; (\mathbb{W}\boldsymbol{m}_t)(x), \qquad t \in [0,T],$$

is — for this one fixed x — just a prescribed flow of measures: a known, time-dependent environment, computed once and for all by the scalar integral eq. (22.3), with no remaining dependence on anything the type-x agent will go on to do. The agent cannot budge it. Its own law is a single input of Lebesgue-null weight to the type-integral defining every aggregate, so an individual deviation changes ν^y for *no* type y, its own included. This is the mean-field decoupling, and in the graphon setting it operates twice over: the agent is negligible within its type, and the type is negligible within the continuum.

With the environment ν^x frozen to a prescribed curve, the type-x agent's problem eq. (22.4) is an ordinary stochastic control problem — minimize an expected integral cost over feedback controls, subject to the state dynamics eq. (22.1) and a known time-dependent cost coefficient ν_t^x. Nothing graphon-specific survives inside it. The type x has become a *spectator parameter*: it selects which drift $b(x,\cdot,\cdot)$, which volatility $\sigma(x)$, which running cost $L(x,\cdot,\cdot,\cdot)$, and which environment ν^x the agent confronts, but inside the optimization it is an inert tag carried along for the ride, never differentiated, never coupled to anything — exactly the standing-data warning that we never take a derivative in x. This is verbatim the situation of chapter 8, and its resolution is verbatim Chapter 8's: the value u^x solves a backward HJB equation driven by the frozen environment, and the optimally controlled state has a law that flows forward by a Fokker–Planck equation driven by the optimal feedback. (That the freezing is a legitimate idealization of a large but finite network, rather than a fiction, is the content of chapter 21 and, quantitatively, of chapter 25.)

That disposes of the agent's half. The loop is closed by the self-consistency demand: the law the type-x agent generates must be the very field m^x we froze — and this is imposed *simultaneously for every type*. It is the simultaneity that re-welds the continuum of decoupled single-type problems into one coupled system. They are coupled because the environment $\nu^x = (\mathbb{W}\boldsymbol{m}_t)(x)$ that type x optimized against is built out of the laws m^y of all the other types, each of which is in turn being pinned by its own agent's optimization against its own aggregate. Imposing every closure at once is what locks the family together. We now write down exactly what that simultaneous closure says.

Definition 22.2 (Graphon MFG system)**.** A pair of fields $(\boldsymbol{u},\boldsymbol{m}) = \big((u^x)_{x\in[0,1]},(m^x)_{x\in[0,1]}\big)$, with each $u^x\colon [0,T]\times\mathbb{R}^d \to \mathbb{R}$ sufficiently smooth and each $m^x\colon [0,T]\to\mathcal{P}_2(\mathbb{R}^d)$ a flow with density $m^x(t,\cdot)$, is a *(classical) solution of the graphon mean field game system* with data $(\boldsymbol{m}_0, g)$ if, for almost every type $x\in[0,1]$, the backward HJB equation

$$-\partial_t u^x - \tfrac12\,\mathrm{tr}\big(\sigma(x)\sigma(x)^\top \mathrm{D}^2\, u^x\big) + H\Big(x,\xi,\nabla u^x,(\mathbb{W}\boldsymbol{m}_t)(x)\Big) = 0, \tag{22.6a}$$

$$\partial_t m^x - \tfrac12\,\mathrm{tr}\big(\sigma(x)\sigma(x)^\top \mathrm{D}^2\, m^x\big) - \mathrm{div}\Big(m^x\,\partial_p H\big(x,\xi,\nabla u^x,(\mathbb{W}\boldsymbol{m}_t)(x)\big)\Big) = 0, \tag{22.6b}$$

hold on $(0,T)\times\mathbb{R}^d$, with split boundary data

$$u^x(T,\xi) = g\Big(\xi,(\mathbb{W}\boldsymbol{m}_T)(x)\Big), \qquad m^x(0,\cdot) = m_0^x, \tag{22.7}$$

where D^2 and div act in the state variable ξ.

Read the system one type at a time and it is the Chapter 8 pair, with one substitution. Equation eq. (22.6a) is the type-x agent's backward HJB: its only contact with the rest of the population is the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ inside H and in the terminal datum. Equation eq. (22.6b) is the forward FP equation for the type-x density, transported by the type-x optimal *closed-loop drift*

$$b^{x,*}(t,\xi) \;:=\; -\,\partial_p H\Big(x,\xi,\nabla u^x(t,\xi),(\mathbb{W}\boldsymbol{m}_t)(x)\Big) \;=\; b(x,\xi,\alpha^{x,*}(t,\xi)), \tag{22.8}$$

where the optimal *control* is the Hamiltonian maximizer

$$\alpha^{x,*}(t,\xi) \;:=\; \operatorname*{arg\,max}_{a\in A}\Big\{-\,b(x,\xi,a)\cdot\nabla u^x(t,\xi) - L\big(x,\xi,a,(\mathbb{W}\boldsymbol{m}_t)(x)\big)\Big\}, \tag{22.9}$$

single-valued under **(A-Conv)**.

Before reading off the coupling, settle the boundary data, because the split in eq. (22.7) is the structural signature the notationwatch insisted on and is silently easy to get backwards. The two equations of the pair are pinned at *opposite* ends of the clock and run toward each other. The value carries *terminal* data $u^x(T,\xi) = g(\xi,(\mathbb{W}\boldsymbol{m}_T)(x))$ at the far end $t = T$ and is integrated *backward*: an agent values a state by what the future holds, so the value is anchored at the horizon and propagated back. The density carries *initial* data $m^x(0,\cdot) = m_0^x$ at the near end $t = 0$ and is transported *forward*: a population is anchored at where it starts and pushed ahead by the flow. This is hazard (i) of the notationwatch made concrete — two clocks in one display — and the cross-coupling is what forbids solving either equation alone: the backward HJB needs the field $\boldsymbol{m}_t$ (through the aggregate) at every t to fix its source and terminal datum, while the forward FP needs ∇u^x at every t to fix its drift, so neither end can be reached without the other. Reading the terminal datum off $t = 0$ or the initial datum off $t = T$ does not merely shuffle labels; it inverts the entire forward–backward structure and produces a different, wrong system.

Now the two objects in eqs. (22.8) and (22.9), which must not be conflated — and the warning that this is "not cosmetic" deserves the full reason. The optimal control $\alpha^{x,*}$ of eq. (22.9) is the *action* the agent chooses: an element of the control set A, the Hamiltonian maximizer. The closed-loop drift $b^{x,*}$ of eq. (22.8) is the *velocity* that action induces on the state, $b(x,\xi,\alpha^{x,*})$. The Fokker–Planck equation eq. (22.6b) transports the density with the drift, *not* the control, and this is forced rather than conventional: the density m^x is the law of the state X^x, the state moves at velocity $b(x,\xi,\alpha^{x,*})$, so the transport term of any continuity equation for that law is $\operatorname{div}(m^x\, b^{x,*})$. Putting the control there instead would be a category error — the control need not even be a velocity (it can be a force, a rate, a signal valued in an abstract A), so $\operatorname{div}(m^x\, \alpha^{x,*})$ may not even typecheck.

What licenses writing that drift as $-\partial_p H$, rather than as $b(x,\xi,\alpha^{x,*})$, is the envelope identity, and it is worth the one line. By definition eq. (22.5), $H(x,\xi,p,\nu) = \sup_a\{-b(x,\xi,a)\cdot p - L(x,\xi,a,\nu)\}$, with the supremum attained at $a = \alpha^{x,*}$ (single-valued by **(A-Conv)**). Differentiate the value of this maximization in the costate p. At the maximizer the first-order condition holds, so the term that would come from differentiating $\alpha^{x,*}(p)$ through the chain rule is multiplied by a vanishing gradient and drops out; only the *explicit* p-dependence of the supremand survives. Hence

$$\partial_p H(x,\xi,p,\nu) \;=\; -\,b(x,\xi,\alpha^{x,*}), \qquad \text{so} \qquad -\,\partial_p H \;=\; b(x,\xi,\alpha^{x,*}) \;=\; b^{x,*}.$$

The induced drift and minus the costate-gradient of the Hamiltonian are therefore *literally the same vector*, which is exactly what eq. (22.8) records in writing $b^{x,*} = -\partial_p H = b(x, \xi, \alpha^{x,*})$ in one breath, and why eq. (22.6b) carries $\partial_p H$. The control $\alpha^{x,*}$ is recovered separately by the argmax eq. (22.9); in general it does not appear in the transport at all.

The control and the drift coincide in exactly one circumstance: when the control enters the dynamics as a *unit-gain velocity*, $b(x, \xi, a) = a$, so that choosing the action *is* choosing the velocity. Then $b^{x,*} = \alpha^{x,*}$ and the distinction collapses. This is the special structure of much of the textbook LQG literature, which is precisely why the gap is invisible in the simplest models and must be flagged the instant one leaves them. Thread B (worked computation 22.15) is the cleanest place to watch it open up to nonzero width: its state eq. (22.1) has scalar drift $b(x, \xi, \alpha) = a\xi + b\,\alpha$, with a genuine control gain $b \neq 1$ and a state-feedback term $a\xi$ that the action does not set, so the optimal control is the linear feedback $\alpha^{x,*} = -b\,(P\xi + r^x)$ — set by the Riccati curvature P and the decoupling offset r^x — while the closed-loop drift is $b^{x,*} = a\xi + b\,\alpha^{x,*} = a\xi - b^2(P\xi + r^x)$. The two agree only in the degenerate normalization $a = 0$, $b = 1$, where the dynamics reduce to $\mathrm{d}X = \alpha\ \mathrm{d}t + \sigma\ \mathrm{d}B$ and the control is the velocity; for any other (a, b) the FP equation of Thread B transports with $a\xi - b^2(P\xi + r^x)$, and an agent who substituted the control would propagate the wrong Gaussian.

Step back and the architecture is exactly Chapter 8's, drawn once per type. Within a single type there are precisely two arrows, and *both are old*. The HJB needs the population: the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ feeds the Hamiltonian and the terminal datum, so the forward field drives the backward value — arrow one. The FP needs the value: the closed-loop drift $-\partial_p H(x, \xi, \nabla u^x)$ is built from ∇u^x, so the backward value drives the forward density — arrow two. These two arrows are the entire Chapter 8 mechanism, and nothing else passes between u^x and m^x within type x. The single new ingredient hides inside arrow one. In chapter 8 the population the HJB consumed was the agent's own law m_t; here it is *not* m_t^x but the graphon average $(\mathbb{W}\boldsymbol{m}_t)(x)$, which mixes in the laws of *all other* types. That one substitution — $m_t^x \rightsquigarrow (\mathbb{W}\boldsymbol{m}_t)(x)$ — is the *one new wire*, and it is the wire that couples the continuum of otherwise independent Chapter 8 pairs to one another. Crucially it is a *linear* wire: the map $\boldsymbol{m} \mapsto \mathbb{W}\boldsymbol{m}$ is the bounded linear operator $\mathbb{W}$, so the coupling across types — however vast the continuum of them — is carried entirely by one self-adjoint Hilbert–Schmidt operator and through nothing else.

Remark 22.3 (Where the homogeneous system sits inside this one)**.** The sanity check that frames the whole programme is to ask what definition 22.2 becomes for the most featureless kernel of all. Setting $W \equiv 1$ (the complete graphon) makes $(\mathbb{W}\boldsymbol{m}_t)(x) = \int_0^1 m_t^y\ \mathrm{d}y$ independent of x: every agent feels the *same* averaged population, the ordinary mean of all the type-laws. If in addition the data are type-independent — one common b, σ, L, g — then every per-type HJB–FP pair sees the same environment and starts from the same datum, so the entire field collapses to a single pair, recovering the homogeneous Chapter 8 system eq. (8.5) exactly. This is the anchor: the graphon system is a strict generalization that contains the old theory at one trivial value of the kernel, and nothing in the lift can have damaged the $W \equiv 1$ case. The remark also pins down what is genuinely new. As an operator on $L^2[0, 1]$, the constant kernel acts by $\mathbb{W}f = (\int_0^1 f)\,\mathbf{1} = \langle \mathbf{1}, f\rangle\,\mathbf{1}$ — it is the *rank-one orthogonal projection onto the constants* (the leading eigenfunction $\mathbf{1}$, eigenvalue 1, everything else annihilated). The entire graphon programme is the study of what $\mathbb{W}$ does once it is allowed to be anything other than that one projection: once it has more than one eigenvalue, a non-constant leading eigenfunction, a nontrivial spectrum. Every later section is a way of measuring that departure — section 22.4 reads off the finitely many surviving eigenvalues when $\mathbb{W}$ has finite rank, section 22.5 reads off the single leading one. The reduction is made quantitative in chapter 21 and reappears, with $W \equiv c$ rescaling the coupling by the scalar c, in the worked computation of section 22.5.

Two questions are now squarely open, and they are exactly the charge of the next subsection. The first is *legitimacy.* We built definition 22.2 by a heuristic — freeze, optimize, close — and a heuristic is not a theorem. Does a pair $(\boldsymbol{u}, \boldsymbol{m})$ solving these PDEs genuinely encode a type-by-type Nash-consistent equilibrium, one in which no agent of any type can lower its cost by deviating, or is it merely a formal family of coupled equations that happens to resemble one? Answering this is the work of the per-type verification theorem, which certifies each frozen-field problem as an honest optimization rather than a bare PDE. The second is to turn the reading we have been pressing — that the coupling across types is carried entirely by the linear operator $\mathbb{W}$ — from a slogan into a working object. The next subsection does both at once: it verifies the system type by type and recasts it as a fixed point of a field best-response map, where "coupling only through $\mathbb{W}$" becomes the concrete, load-bearing statement that a single application of the map already requires the whole field over $[0, 1]$, because $\mathbb{W}$ does, and therefore does not factor over types.

22.1.3 Per-type verification and the field best-response map

The previous subsection handed this one two debts, and they are paid together here. The first is *legitimacy*: definition 22.2 was assembled by a heuristic — freeze, optimize, close — and a heuristic is a recipe, not a certificate, so we owe a theorem that a pair $(\boldsymbol{u}, \boldsymbol{m})$ solving those PDEs genuinely encodes a type-by-type Nash-consistent equilibrium rather than a formal family of coupled equations that merely resembles one. The second is to turn the reading we have been pressing — that the coupling across types is carried entirely by the linear operator $\mathbb{W}$ and by nothing else — from a slogan into a working object. Both debts are discharged by the same observation, and it is the organizing claim of this subsection: once the field is frozen, the architecture of each per-type problem is exactly that of chapter 8, drawn once for every x, with a single new wire — the substitution of the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ for the agent's own law — threaded through it. Legitimacy then follows by importing chapter 8's verification theorem verbatim, once per type. And the slogan becomes precise the moment we package the per-type solves into one self-map and notice that this same new wire is exactly what stops the map from factoring over types: a single evaluation of it already needs the whole field, because $\mathbb{W}$ does.

Proposition 22.4 (Per-type optimality)**.** *Assume **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, **(A-Reg-W)**. Fix a flow of fields $\boldsymbol{m} = (\boldsymbol{m}_t)_{t\in[0,T]}$ and let, for each x, u^x be the classical solution of the backward HJB eq. (22.6a)–eq. (22.7) with environment $t \mapsto (\mathbb{W}\boldsymbol{m}_t)(x)$. Then $u^x(t, \xi)$ is the value function of the type-x agent's control problem eq. (22.4) against that fixed environment, the feedback control $\alpha^{x,*}$ of eq. (22.9) is optimal, and the law of the optimally controlled state eq. (22.1) — transported by the induced drift $b^{x,*}$ of eq. (22.8) — solves the forward FP equation* *eq.* (22.6b)*.*

Proof sketch. Fix the flow $\boldsymbol{m}$ and the type x. The argument is the verification theorem of chapter 8 run against a frozen environment, and it has three moves: turn the frozen field into a legitimate prescribed environment, use Itô and the HJB to sandwich the cost, and use the Kolmogorov-forward theorem to identify the closed-loop law.

The frozen aggregate is a prescribed continuous flow of measures. Set $\nu_t^x := (\mathbb{W}\boldsymbol{m}_t)(x)$. Because the whole flow $\boldsymbol{m}$ is held fixed, this curve is computed once and for all by the scalar pairing eq. (22.3) and carries no residual dependence on anything the agent will do. It lands in $\mathcal{M}_+(\mathbb{R}^d)$ — a sub-probability measure of mass $(\mathbb{W}\mathbf{1})(x) \leq 1$ with uniformly bounded second moment, by the mass and moment bounds of section 22.1.1 — and it is continuous in t. Indeed, for $s \to t$ and any bounded Lipschitz φ, the weak prescription eq. (22.3) gives

$$\langle \varphi, \nu_t^x - \nu_s^x \rangle = \int_0^1 W(x,y)\, \langle \varphi, m_t^y - m_s^y \rangle \, \mathrm{d}y,$$

whose right side is bounded, up to the Lipschitz constant of φ and the degree $(\mathbb{W}\mathbf{1})(x) \leq 1$, by $\sup_y \mathcal{W}_2(m_t^y, m_s^y)$ and hence vanishes by the d_∞-continuity of $\boldsymbol{m}$ in t together with the boundedness of $\mathbb{W}$. So ν^x is exactly the kind of object chapter 8 optimizes against: a known, time-dependent, continuous environment in $\mathcal{M}_+(\mathbb{R}^d)$, inert for the duration of the optimization. (The quantitative continuity, and its uniformity in x, is lemma 22.8 of section 22.3; here we need only that for one fixed x the flow is prescribed and continuous.)

Itô plus the HJB sandwich the cost. With ν^x frozen, the type-x problem eq. (22.4) is an ordinary stochastic control problem with data $\big(b(x,\cdot,\cdot), \sigma(x), L(x,\cdot,\cdot,\nu^x), g(\cdot,\nu_T^x)\big)$, to which proposition 8.2 applies verbatim. Take any admissible feedback α^x with state X^x solving eq. (22.1) from an arbitrary start $X_t^x = \xi$, and apply Itô's formula (theorem 5.9) to $s \mapsto u^x(s, X_s^x)$ on $[t,T]$. The finite-variation part of the resulting semimartingale carries the drift $\partial_s u^x + \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u^x) + b(x, X_s^x, \alpha_s^x) \cdot \nabla u^x$, while the stochastic integral against $\sigma(x)\,\mathrm{d}B^x$ is a true martingale of zero expectation under the growth and nondegeneracy bounds. Substitute the HJB eq. (22.6a), which rearranges to $\partial_s u^x + \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u^x) = H(x,\cdot,\nabla u^x, \nu^x)$, and recall that by the very definition eq. (22.5) of the Hamiltonian as a supremum, $H(x,\xi,p,\nu) \geq -b(x,\xi,a)\cdot p - L(x,\xi,a,\nu)$ for *every* action a, with equality precisely at the maximizer. Taking expectations and adding the terminal datum $u^x(T,\cdot) = g(\cdot,\nu_T^x)$ collapses the telescoping integral to the inequality $u^x(t,\xi) \leq J_t^x(\alpha^x)$, the value bounding the cost-to-go from (t,ξ) from below for every admissible control. The inequality is saturated exactly when the supremum is attained pointwise along the trajectory, i.e. when the agent plays the Hamiltonian maximizer $\alpha^{x,*}$ of eq. (22.9). Here **(A-Conv)** earns its keep: convexity of $H(x,\xi,\cdot,\nu)$ in the costate makes that maximizer *single-valued*, so $\alpha^{x,*}(s,\xi)$ is a genuine feedback and not a multivalued selection, and the chain $u^x(t,\xi) = J_t^x(\alpha^{x,*}) \leq J_t^x(\alpha^x)$ identifies u^x as the value function and $\alpha^{x,*}$ as optimal.

Kolmogorov-forward identifies the closed-loop law. Now feed the optimal feedback back into the dynamics. The optimally controlled state $X^{x,*}$ is the diffusion with drift $b(x,\xi,\alpha^{x,*}(s,\xi)) = b^{x,*}(s,\xi)$ — the induced closed-loop drift eq. (22.8), equal to $-\partial_p H$ by the envelope identity — and state-independent diffusion $\sigma(x)$. Its time-marginal law $\mathcal{L}(X_t^{x,*})$ therefore solves the Kolmogorov-forward (Fokker–Planck) equation (theorem 5.14) with that drift and that diffusion, which is exactly eq. (22.6b); under **(A-Nondeg)** the law carries a density $m^x(t,\cdot)$ and the equation holds classically. Started from the datum m_0^x, this closed-loop law is the forward field the system demands. This disposes of the optimality for each fixed x.

One point is held in reserve, and it is the one the proof statement flags. The estimates the three moves rest on — Schauder and parabolic bounds on u^x, hence on $\alpha^{x,*}$, imported from chapters 8 and 10 — come out with constants that depend on the data only through the bounds in **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)** and through the bound on ν^x, every one of which **(A-Reg-W)** and the standing assumptions make *uniform in* x. For the optimality statement at a single fixed x this uniformity is irrelevant — proposition 8.2 already delivers the conclusion type by type, with no need to know how the constant moves as x varies. Its role is later and structural: uniformity in x is precisely what upgrades these pointwise-in-type facts into *field* statements. A bound that holds for each x separately, but with a constant that could secretly blow up as x ranges over $[0,1]$, says nothing about the assembled field $x \mapsto (u^x, m^x)$ as a single object; a bound *uniform* in x assembles with no loss into a statement about that field in the sup-in-type metric d_∞ of section 22.1.1. Performing that upgrade is the entire business of section 22.3, and we flag here only that the per-type verification already carries the uniformity that section will cash in. □

proposition 22.4 certifies each frozen-field problem as an honest optimization, but it does so one type at a time and against a field that was merely *posited*. The equilibrium demand of chapter 21 is stronger, and it is what welds the types back together: the law each type generates as its optimal closed loop must be the very field that was frozen, and this must hold for every type at once. Read as a condition on the flow $\boldsymbol{m}$, this is self-consistency of the cleanest possible kind — run every type's optimization against the graphon average of $\boldsymbol{m}$, collect the resulting closed-loop laws into a new flow, and demand that the new flow be $\boldsymbol{m}$ again. A flow that reproduces itself under this operation is a *fixed point*, and a fixed point is exactly what an equilibrium is. To state this precisely we must name the space the operation acts on, and check it is the kind of space a fixed-point search can live on.

Let $\mathcal{C} := C([0,T];(\mathcal{C}_2,d_\infty))$ be the space of flows of fields, continuous in t for the uniform-in-type Wasserstein metric d_∞ of section 22.1.1 (definition 21.10); we also write $L^\infty_{\mathcal{W}_2}([0,1];\mathcal{P}_2(\mathbb{R}^d))$ for the one-instant field space $(\mathcal{C}_2,d_\infty)$. That $\mathcal{C}$ is a *complete* metric space — the one property a fixed-point search cannot do without, since completeness is what gives a Cauchy sequence of near-equilibria a limit to converge to — assembles from three imports, and it is worth chaining them in a single readable line rather than leaving them stacked as citations. Start at the bottom rung, one type at one instant: the value space $(\mathcal{P}_2(\mathbb{R}^d),\mathcal{W}_2)$ is complete, indeed Polish, by the completeness of the quadratic Wasserstein distance (theorem 4.8). Climb to fields at one instant: $(\mathcal{C}_2,d_\infty)$ is the type-continuous fields under the sup-in-type metric, and a uniform limit of continuous maps into a complete space is again a continuous map into that space, so $(\mathcal{C}_2,d_\infty)$ inherits completeness directly from $\mathcal{W}_2$ (definition 21.10). Climb to flows in time: for *any* complete metric space E, the path space $C([0,T];E)$ of continuous curves under the uniform-in-time metric is complete, by the identical uniform-limit argument now run in the time variable. Taking $E=(\mathcal{C}_2,d_\infty)$ chains the three rungs and delivers completeness of $\mathcal{C}$. This is the graphon analogue of the homogeneous flow space $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$ of chapter 10 (definition 10.1); the single change is that the fibre over one instant has grown from the measure space $\mathcal{P}_2(\mathbb{R}^d)$ to the field space $(\mathcal{C}_2,d_\infty)$, with completeness surviving the growth intact.

Definition 22.5 (Graphon best-response map)**.** The *graphon best-response map* $\mathsf{S}\colon \mathcal{C} \to \mathcal{C}$ sends a flow of fields $\boldsymbol{m}$ to $\mathsf{S}[\boldsymbol{m}] = \widetilde{\boldsymbol{m}}$ defined typewise by: (i) for each x, solve the backward HJB eq. (22.6a) with environment $(\mathbb{W}\boldsymbol{m}_t)(x)$ to get u^x; (ii) form the optimal control eq. (22.9) and the induced closed-loop drift $b^{x,*}$ of eq. (22.8); (iii) solve the forward FP equation eq. (22.6b) transported by that drift, from the datum m_0^x, to get $\widetilde{m^x}$.

Proposition 22.6 (Equivalence with the fixed point)**.** *Under **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, **(A-Reg-W)**, a flow of fields $\boldsymbol{m} \in \mathcal{C}$ is a fixed point of S if and only if, with $u^x = u^{x,[\boldsymbol{m}]}$, the pair $(\boldsymbol{u}, \boldsymbol{m})$ solves the graphon MFG system eq. (22.6)–eq. (22.7).*

Proof. The argument is proposition 8.4 applied typewise, and the typewise reduction is what makes it short. Fix x. By construction (definition 22.5), step (i) produces u^x as the solution of the backward HJB eq. (22.6a) with environment $(\mathbb{W}\boldsymbol{m}_t)(x)$, so this half of the system holds for $\mathsf{S}[\boldsymbol{m}]$ whether or not $\boldsymbol{m}$ is a fixed point — the backward equation is reproduced automatically. The content is therefore in the forward half. Step (iii) solves, for each x, the *linear* parabolic Cauchy problem eq. (22.6b) — linear because, under the model class of remark 22.1, the transport drift $b^{x,*}$ is built from u^x alone and does not see the density — with Lipschitz coefficients and a uniformly elliptic second-order part under **(A-Nondeg)** (chapter 5). Such a problem has a *unique* solution from the datum m_0^x. Uniqueness is the hinge of the equivalence: it means the produced field $\widetilde{m^x}$ is the one and only solution of eq. (22.6b) with the feedback built from u^x, so the typewise identity $\widetilde{m^x} = m^x$ holds if and only if the frozen field m^x *is* that solution — i.e. if and only if m^x solves the forward equation with its own induced drift. Demanding the identity at every x is thus demanding exactly eq. (22.6b) for the whole field, which together with the HJB from step (i) is the full system eq. (22.6)–eq. (22.7); the converse reads the same chain backward. One obligation is incurred along the way and discharged elsewhere: for $\mathsf{S}[\boldsymbol{m}]$ to be an element of $\mathcal{C}$ at all — a genuine field, depending measurably (indeed continuously) on x — the per-type solves of steps (i)–(iii) must assemble into a measurable family over $[0, 1]$. That the assembled field $x \mapsto (u^x, \widetilde{m^x})$ is measurable in x is precisely proposition 22.9 below, proved by the stability estimates of section 22.3; we invoke it here and prove it there. □

A fixed point of S in this sense is precisely an equilibrium of the dynamic graphon mean field game in the sense of definition 21.16; the two are the same object, here cast as a self-map. This is the formulation chapter 24 consumes, and the reduction it accomplishes is worth stating bluntly: the analytic problem has gone from "solve a continuum-indexed forward–backward PDE system" to "find a fixed point of a single self-map S on the complete metric space $\mathcal{C}$ of flows of fields." The forward–backward PDEs have not vanished — they live inside one *evaluation* of S — but the coupling has been packaged into the act of evaluating the map, which is where the entire difficulty now sits.

It is worth being exact about what that difficulty is, because it is the seam to chapter 24. Relative to the homogeneous reformulation proposition 8.4, nothing in the forward–backward structure is new: that structure is inherited type by type, two arrows per type, both of them Chapter 8's. What is new is that S *does not factor over types.* One could imagine, in the most favourable world, that evaluating S at type x required only the type-x slice m^x of the input — that the self-map were a product $\prod_x \mathsf{S}^x$ of independent per-type maps, each acting on its own fibre. Were that so, the continuum would be mere bookkeeping and each type a separate Chapter 8 problem solved in parallel, with no genuine interaction. It is *not* so, and the obstruction is the one new wire. Step (i) of definition 22.5 solves the type-x HJB against the environment $(\mathbb{W}\boldsymbol{m}_t)(x)$, and by eq. (22.2) that aggregate is an integral over *all* types y of m_t^y, weighted by $W(x, y)$. A single evaluation of S at one type already reads the entire field over $[0, 1]$. The types are welded together at the level of one application of the map, not merely at the fixed point.

This is exactly why the spectral data of $\mathbb{W}$ — and nothing finer about the kernel — governs the behaviour of S. The cross-type entanglement enters only through the linear map $\boldsymbol{m} \mapsto \mathbb{W}\boldsymbol{m}$, so the sensitivity of S to a perturbation of its input field is controlled by how much $\mathbb{W}$ can amplify that perturbation, and that amplification is measured by the spectrum and by nothing else. The operator norm $\|\mathbb{W}\|$ bounds the worst-case gain over all directions in type-space and therefore sets the budget for a *contraction* estimate: the per-type best response contracts the environment by a factor read off the Lipschitz data, the cross-type amplification $\|\mathbb{W}\|$ multiplies it, and whether S contracts at all turns on the product falling below one. The leading eigenvalue, and the eigenfunction along which it is attained, identify the single direction in type-space that is amplified most — the soft mode along which a near-equilibrium is least stable, and the first place a unique equilibrium can be lost. These are the two numbers the later sections read off: section 22.4 collapses the whole coupling to the finitely many eigenvalues that survive when $\mathbb{W}$ has finite rank, and section 22.5 reads off the leading one alone as the rescaling of the effective coupling strength. They are also why chapter 24 cannot secure existence and uniqueness from monotonicity alone: the Lasry–Lions condition **(A-Mon)** disciplines the *sign* of the per-type coupling but is blind to the *size* of the cross-type amplification, so chapter 24 must pair it with the operator-norm smallness **(A-Coupling)** — $\|\mathbb{W}\|$ small enough that the contraction survives the lift. Monotonicity and operator-norm smallness are the two halves of one lock, one key per axis of the coupling, and it is the non-factoring of S that forces both.

We record this as the central seam to chapter 24 and turn now to the structure of the coupling itself. The reformulation has isolated cleanly *what* is genuinely new — the coupling of the types, carried entirely by the integral operator $\mathbb{W}$ — but it has not yet said what that coupling *means*: whether it behaves like the short-range, neighbour-to-neighbour coupling a field theorist's reflexes expect, or like something else entirely. Pinning down that meaning, and guarding against its most natural misreading, is the work of the next section.

22.2 Couplings: nonlocal kernel versus local gradient

The previous section pressed a single reading until it became a slogan: within each type the system definition 22.2 is the Chapter 8 pair drawn once, and the *only* genuinely new content is that the types are coupled to one another — coupled through one object, the integral operator $\mathbb{W}$, and through nothing else. That reading is now established. What it is not yet is *understood*. Knowing that $\mathbb{W}$ is the sole wire between types tells us where the new physics lives; it does not tell us what kind of coupling that wire carries, nor how it relates to the couplings a physicist already has reflexes for. This section is the conceptual interlude that supplies those reflexes — and, just as importantly, disables the wrong ones — before the technical machinery of the following sections puts $\mathbb{W}$ to work.

There is one obstacle to clear thinking here, and naming it organizes everything that follows: the word "local" is overloaded. In a field theory "local" means a coupling that sees only the immediate neighbourhood of a point and is built from the field and its gradients there; in the mean field game literature inherited from chapter 8 "local" means a cost that reads the population density at the agent's own state rather than a smoothed average of it. These are different statements about different variables, and in the graphon setting both variables are present at once — the agent's private state $\xi \in \mathbb{R}^d$ and its type $x \in [0,1]$ — so a single undecorated "local" is ambiguous in a way it never was when only one of the two was in play. The entire payoff of getting $\mathbb{W}$ right hinges on refusing to let these two senses blur, because the operator $\mathbb{W}$ is emphatically nonlocal in one of them and the inherited Chapter 8 dichotomy lives entirely on the other.

The section makes its argument in three moves, and they are meant to be read as one. We first *disentangle the two localities*: we separate locality-in-state, which is old and entirely a matter of section 8.5 applied per type, from locality-in-type, which is the new axis and is where $\mathbb{W}$ acts, fixing once and for all which of the two carries the graphon's novelty. With the axes split we then *test the physical analogy* the splitting invites: the resemblance of the type variable to a spatial coordinate and of $\mathbb{W}$ to a Ginzburg–Landau gradient energy is strong enough to be worth taking seriously and — on exactly one point — wrong enough that taking it uncritically would mislead, so we develop it precisely far enough to read off where it holds and mark the single place it fails. Finally, with the picture calibrated, we *specialize to the tractable class*: the couplings whose dependence on the aggregate is separable, which is the structural hypothesis that makes the rest of the chapter computable and is the form every worked structure downstream will take.

The hand-off is therefore twofold. The two-axes distinction drawn in the first subsection is the coordinate system the Ginzburg–Landau discussion lives in — that analogy is a statement about the *type* axis specifically, and it cannot be stated, let alone judged, until state-locality and type-locality have been pried apart. And the separable-coupling class isolated in the last subsection is the standing structural assumption that section 22.4 and the worked computation of section 22.5 will consume: it is here that we earn the right to write the couplings the rest of the chapter solves. We begin by splitting the two meanings of "local" across the two variables.

22.2.1 Two axes of locality

The section opener left us with a warning rather than a result: the word "local" is overloaded, and until we split it across the two variables in play we cannot even state, let alone judge, the Ginzburg–Landau analogy the next subsection tests. We make that split now. The reason the overload is harmless in the homogeneous theory and dangerous here is simply arithmetic of variables. In chapter 8 there was exactly one variable a coupling could be local or nonlocal *in* — the agent's private state $\xi \in \mathbb{R}^d$ — so "local" had a single unambiguous referent. In the graphon setting there are two variables present in every function we write, the state $\xi \in \mathbb{R}^d$ and the type $x \in [0,1]$, and "local" can mean a statement about either. A coupling can be local in the state and nonlocal in the type, or the reverse, or both, or neither; these are four genuinely different things, and collapsing them into one undecorated word is the fastest route to confusing the old physics with the new. So we treat the two axes separately, name on each axis the local and the nonlocal options, and — this is the payoff — show that the entire novelty of the graphon lives on one axis and the entire inheritance from chapter 8 on the other.

Locality in the state $\xi \in \mathbb{R}^d$ — the inherited axis. This axis is old. It is the axis of section 8.5, and once the field is frozen it is inherited by each type verbatim, with the single substitution we have been tracking all chapter: the agent's own law m of the homogeneous theory is replaced by the type-x aggregate

$$\nu^x \;:=\; (\mathbb{W}\boldsymbol{m}_t)(x),$$

the kernel-weighted neighbour distribution of eq. (22.2). Holding the type x fixed, the question "how does the cost see ν^x?" is exactly the Chapter 8 question with ν^x in the role of m, and it has the same two answers.

A coupling is *nonlocal in the state* when it reads the aggregate only through a smoothed, spatially-averaged functional of it. The canonical instance is a convolution,

$$F(x,\xi,\nu^x) \;=\; (\phi * \nu^x)(\xi) \;=\; \int_{\mathbb{R}^d} \phi(\xi-\xi')\,\nu^x(\mathrm{d}\xi'),$$

with ϕ a bounded continuous interaction kernel on $\mathbb{R}^d$ (example 8.6): the agent at state ξ feels a ϕ-weighted average of where the neighbour population sits, never the population's value at the single point ξ. The defining analytic virtue is that such an F is *weakly continuous* in ν^x — if $\nu_n^x \rightharpoonup \nu^x$ weakly then $(\phi * \nu_n^x)(\xi) = \langle \phi(\xi-\cdot), \nu_n^x\rangle \to \langle \phi(\xi-\cdot), \nu^x\rangle$ for each ξ, because $\phi(\xi-\cdot)$ is a bounded continuous test function and weak convergence is precisely convergence of such pairings. Weak continuity is exactly what a compactness argument wants: a sequence of approximate fields that converges only weakly (all one ever gets cheaply, by Prokhorov, from a second-moment bound) still produces convergent costs, so the fixed-point machinery of chapter 24 can pass to the limit with no extra regularity. Nonlocal-in-state couplings are, in this precise sense, the compactness-friendly ones.

A coupling is *local in the state* when, by contrast, it reads the pointwise density value at the agent's own location,

$$F(x,\xi,\nu^x) \;=\; f\big(\xi,\,\rho^x(\xi)\big), \qquad \rho^x \;=\; \frac{\mathrm{d}\nu^x}{\mathrm{d}\xi},$$

with ρ^x the Lebesgue density of the aggregate (example 8.7). This is a genuinely sharper demand: it asks the aggregate to *have* a density and asks that density to be evaluated at a point, an operation that is not even defined for a general measure and is badly discontinuous under weak convergence (a sequence of smooth densities can converge weakly to a Dirac, along which no pointwise value survives). What rescues it is parabolic regularization. The uniform nondegeneracy **(A-Nondeg)**, $\sigma(x)\sigma(x)^\top \succeq \lambda I$, forces the forward Fokker–Planck equation eq. (22.6b) to produce an honest density $m^x(t,\cdot)$ with quantitative bounds, and those bounds — uniform in x, since λ is — are exactly what make the pointwise evaluation $\rho^x(\xi)$ legitimate and the local coupling well-posed. So the state axis carries a clean analytic dichotomy: nonlocal couplings are weakly continuous and ask nothing of the density, local couplings demand the density bounds that **(A-Nondeg)** alone supplies. None of this is new — it is section 8.5 read with ν^x in place of m, one type at a time — and we rehearse it here only to set it firmly on *this* axis, so that nothing on it will be mistaken for the graphon's own contribution.

Locality in the type $x \in [0,1]$ — the new axis, and where $\mathbb{W}$ lives. Everything genuinely new sits on the second axis, and on it the situation is not a dichotomy at all: there is only one option, and it is forced. The map $\boldsymbol{m} \mapsto \mathbb{W}\boldsymbol{m}$ of eq. (22.2) is an *integral* operator, and an integral operator is nonlocal by construction. Writing the aggregate out,

$$(\mathbb{W}\boldsymbol{m}_t)(x) \;=\; \int_0^1 W(x,y)\, m_t^y \,\mathrm{d}y,$$

the value at type x pools the laws m_t^y of *all* types y, each weighted by $W(x,y)$. There is no neighbourhood of x to which this sum is confined; a type on the far side of $[0,1]$ contributes exactly as much as a nearby one if the kernel weights it as much. The graphon coupling is therefore *intrinsically nonlocal in type*, and not by some choice of cost we could reverse — it is nonlocal because it is an integral against a kernel, which is what a graphon *is*.

It is worth pressing on what the local-in-type alternative would even have to be, because seeing that it does not exist inside the theory is the cleanest way to internalize that the nonlocality is structural. "Local in type" would mean each agent feels only its own type's population — the aggregate at x should return m_t^x itself, untouched by any other type. For the integral eq. (22.2) to do that, the kernel would have to be a Dirac mass on the diagonal,

$$W(x,y) \;\overset{?}{=}\; \delta(x-y), \qquad \text{so that} \qquad \int_0^1 \delta(x-y)\, m_t^y \,\mathrm{d}y \;=\; m_t^x,$$

making $\mathbb{W}$ the identity operator I on the field. But the Dirac kernel is not a graphon, and the failure is not a technicality — it is disqualifying on every count that **(A-Reg-W)** demands. The diagonal δ is not a function $[0,1]^2 \to [0,1]$ at all but a distribution; it is not in $L^2[0,1]^2$, so the operator it would induce is not Hilbert–Schmidt; indeed the identity I on the infinite-dimensional space $L^2[0,1]$ is not even compact, having the entire interval as continuous spectrum and no eigenbasis to speak of. Every structural property the chapter rests on fails at once: no membership in L^2, no Hilbert–Schmidt norm, no discrete spectrum for section 22.4 to collapse to order parameters, no leading eigenvalue for section 22.5 to read off. The local-in-type coupling is thus not a special case of the theory we could specialize to; it sits entirely *outside* it. The most one can do is approach it: a sequence of genuine graphons concentrating on the diagonal — for instance the heat kernels W_ℓ of example 22.7 as the bandwidth $\ell \downarrow 0$ — approximates the identity, but each member of the sequence is honestly nonlocal, and the limit, where locality would finally appear, is exactly the point that has left the space of graphons. We will meet that sequence again as the one calibrated regime in which a gradient picture is licensed.

The punchline: type is not a spatial coordinate. Putting the two axes side by side delivers the conclusion this subsection exists to plant. On the state axis there is a real choice — nonlocal or local, compactness-friendly or density-demanding — and it is entirely the inherited chapter 8 choice. On the type axis there is no choice: the coupling is nonlocal, always, because it is an integral operator and the only "local" alternative has been ejected from the theory. The double integral $\int_0^1\int_0^1 W(x,y)\,(\cdots)\,\mathrm{d}x\,\mathrm{d}y$ that the aggregate sits inside is visually suggestive of a spatial action, two coordinates integrated against a kernel that depends on both, and it is precisely this suggestion the next subsection must defuse. The type index behaves nothing like a spatial coordinate. The decisive symptom is that $[0,1]$, as a type set, carries *no intrinsic metric*: the number $|x-y|$ is an artefact of the arbitrary labelling of types by points of the unit interval and has no game-theoretic content, since relabelling the types by any measure-preserving bijection leaves the graphon model unchanged (appendix A). What *does* have content is functional, not metric. Two types x and y are "close" — interchangeable in their strategic situation — precisely insofar as

$$W(x,\cdot)\;\approx\;W(y,\cdot)\qquad\text{as functions in } L^2[0,1],$$

that is, insofar as they connect to the rest of the population in the same pattern; it is *never* because $|x-y|$ is small. A pair of types adjacent on the interval can sit in entirely different connectivity regimes and couple to disjoint parts of the population, while a pair at opposite ends can be statistical twins. This is the seed the Ginzburg–Landau discussion harvests, and it is worth stating in exactly the form that subsection will need: the graphon couples types nonlocally, through the similarity of their kernel rows and through no underlying distance, so any reading that penalizes variation between "neighbouring" types — any gradient, any elastic stiffness — is importing a metric the model does not have. With the two axes pried apart and the nonlocal-in-type, no-intrinsic-metric verdict on record, we can finally hold the physicist's gradient intuition up to it and read off the one place it rings false.

22.2.2 The Ginzburg–Landau picture and its one false note

With the two axes of locality pried apart, the previous subsection left a verdict on record: the graphon couples types *nonlocally*, through the similarity of their kernel rows, and over a type set carrying *no intrinsic metric*, so that the apparent "distance" $|x-y|$ has no game-theoretic content. We can now do what that verdict was built for — hold the physicist's strongest reflex up against it. A condensed-matter reader meeting eq. (22.6) for the first time will reach almost involuntarily for a Ginzburg–Landau (GL) picture: a continuum of fields indexed by an internal label, tied together by a stiffness, organizing along a soft mode past a critical coupling. That reflex is largely right, and it is worth building up carefully, because the part of it that is right is the conceptual spine of the rest of Part IV — the language of order parameters and critical couplings that section 22.5 and chapter 24 speak. But it carries exactly one assumption that the no-metric verdict forbids, and that single false note is the whole reason the type/space distinction of appendix A was worth labouring. We sound the picture, then the false note, then exhibit the lone exception in which the note is in tune.

The dictionary is built one entry at a time, and each entry is forced by the structure of the system rather than imposed on it.

Physics note

The system eq. (22.6) invites a Ginzburg–Landau reading, and used with care it is illuminating. Read it first as a field theory. For each value of the internal index x there is a field — the equilibrium density m^x — and the indices are coupled to one another and to nothing else; that is exactly the shape of a GL order-parameter theory, one order-parameter field per internal "flavour." So the first entry of the dictionary is: the type x is a *flavour* index, and the per-type densities m^x are a continuum of coupled *order-parameter fields*. The second entry is the coupling. The only place one flavour's field feels another's is the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x) = \int_0^1 W(x,y)\, m_t^y \,\mathrm{d}y$, weighted entirely by the kernel, so $W(x,y)$ plays the role of the *stiffness* tying flavour x to flavour y: where W is large the two fields are strongly tied and tend to co-vary, where it is small they decouple, and where it vanishes they are strategically blind to each other. The third entry is the one a physicist most wants, and the spectral picture of chapter 18 supplies it directly. Because $\mathbb{W}$ is self-adjoint and compact it has a real eigenbasis; its leading eigenfunction — the eigenvector-centrality profile of chapter 18 — is the *soft mode*, the type-profile along which the coupled system organizes most cheaply, and the corresponding leading eigenvalue λ_1 sets the *critical coupling* at which a homogeneous equilibrium can lose stability. This is not a loose slogan: it is the threshold $s\,\lambda_1 = 1$ computed in closed form in section 22.5 and the spectral smallness condition that buys uniqueness in chapter 24. So far the analogy is exact, and it is the right scaffolding to carry into both.

The one false note — and it must be sounded, because it is the whole point of the type/space distinction (appendix A) — lives in how a GL free energy encodes the coupling. There the stiffness enters through a *gradient* term, $\int |\nabla_x \varphi(x)|^2 \,\mathrm{d}x$, which couples a field value only to its *infinitesimal neighbours*: it penalizes variation over vanishing distance, and to even write it one must already possess a metric on the index set telling which configurations count as "nearby." The graphon coupling has neither the term nor the metric it presupposes. By the previous subsection two types x, y are "close" — interchangeable in their strategic situation — only insofar as $W(x,\cdot) \approx W(y,\cdot)$ as functions in $L^2[0,1]$, never because $|x-y|$ is small; under a measure-preserving relabelling of types the number $|x-y|$ changes freely while the model does not, so a gradient ∂_x has nothing invariant to measure. The aggregate is an honest double integral against W, not a differential operator: it pools *all* flavours at once, each with its own weight, with no notion of adjacency along the index. The correct reading is therefore a *nonlocal*, long-range, all-to-all-with-weights field theory — the continuum limit of an Ising model on a *weighted complete graph*, where every site interacts with every other through its own bond strength, rather than a continuum φ^4 theory, where each point interacts only with its gradient neighbours. The two have qualitatively different physics: the complete-graph model has mean-field exponents and a coupling controlled by a spectral quantity (the leading eigenvalue), the φ^4 model has wavevector- dependent fluctuations organized by a Laplacian. Importing the gradient/elastic intuition unexamined is the single most common error a field theorist makes on first contact with graphons.

The math of this chapter does not lean on the box: delete it and the system eq. (22.6), the operator $\mathbb{W}$, and its spectrum stand untouched. What the box buys is a reading, and it is worth pinning down precisely where the gradient reading stops being merely wrong and becomes, in a measured limit, correct — because that one exception is the bridge between the two pictures and it recurs in applications. It is a genuine mathematical statement, so we record it outside the box, where the calibration can carry weight.

Example 22.7 (Heat-kernel graphons: when nonlocal looks local)**.** The gradient picture becomes legitimate in exactly one circumstance: when the type set is handed a genuine geometry and the kernel is built to be sharply peaked on the diagonal of that geometry. Suppose then that $[0,1]$ is identified with a circle or an interval — so that $|x-y|$ now *is* a real distance, supplied from outside the bare graphon framework — and take for the kernel either the Gaussian heat kernel at length scale ℓ, $W_\ell(x,y) = c_\ell \exp\big(-|x-y|^2/2\ell^2\big)$, or, cleaner to expand, the resolvent of the Laplacian, $\mathbb{W}_\ell = (I - \ell^2\,\partial_{xx})^{-1}$. The point of the resolvent form is that the expansion can be read off algebraically. Since $\mathbb{W}_\ell$ inverts $I - \ell^2\partial_{xx}$, its Neumann series in the small parameter ℓ^2 is

$$\mathbb{W}_\ell = \big(I - \ell^2\partial_{xx}\big)^{-1} = I + \ell^2\partial_{xx} + \ell^4\partial_{xx}^2 + \cdots,$$

convergent acting on fields whose type-derivatives are controlled, i.e. fields varying slowly on the scale ℓ. Applied to the aggregate eq. (22.2) this gives, keeping the first correction,

$$(\mathbb{W}_\ell \boldsymbol{m})(x) \;=\; \boldsymbol{m}(x) + \ell^2\,\partial_{xx}\boldsymbol{m}(x) + O(\ell^4). \tag{22.10}$$

The Gaussian kernel gives the same leading correction: Taylor-expanding the field under the integral, $\int W_\ell(x,y)\boldsymbol{m}(y)\,\mathrm{d}y = \boldsymbol{m}(x) + \frac{1}{2}(\text{variance})\,\partial_{xx}\boldsymbol{m}(x) + \cdots$, since the odd moments of the symmetric kernel vanish and the second moment is $O(\ell^2)$ (the constant absorbed into the definition of ℓ). Either way the nonlocal kernel average *reduces* to the identity plus a local *diffusion-in-type*, i.e. a gradient coupling, in the limit $\ell \to 0$. The $\ell^2\partial_{xx}$ term is precisely the GL stiffness, now earned rather than assumed.

It is worth saying exactly why this is the *only* regime in which the gradient term is legitimate, because the two ingredients that make eq. (22.10) work are both indispensable and both generically absent. First, the kernel must carry a *hidden metric*: W_ℓ is a function of the true distance $|x-y|$ on the circle, so the relabelling-invariance that stripped meaning from $|x-y|$ in a bare graphon is broken on purpose — the geometry is extra data, not intrinsic to the type set. Second, the kernel must be *sharply peaked on the diagonal* in that metric: only because the weight collapses onto $y \approx x$ as $\ell \to 0$ does the pooling over all flavours degenerate into sampling the field and its curvature at the single point x, which is what turns an integral operator into a differential one. Remove either ingredient and the expansion has nothing to stand on. Generic graphons remove both, and not marginally. A block kernel is constant on each block, so it couples a type equally to an entire block regardless of any "distance" and is not peaked anywhere; a rank-one kernel $W(x,y) = f(x)f(y)$ couples every pair through the single product $f(x)f(y)$ — maximally all-to-all, its operator a rank-one projection that is as far from the identity-plus-small-correction of eq. (22.10) as an operator can be; a dense Erdős–Rényi kernel $W \equiv p$ couples everything to everything with equal weight. None has a diagonal peak, none is close to the identity, and for none does any $\ell^2\partial_{xx}$ correction exist to be written. For these — which is to say, for the kernels this book actually studies — the gradient picture is simply wrong, and the honest reading is the nonlocal spectral one. We use this example only as a calibration: it marks the measure-zero exception so that its absence elsewhere is a considered fact rather than an oversight, and the dynamic GMFG theory of Part IV never assumes it.

The calibration is the message to carry forward: the gradient/elastic reading is a special exception purchased by extra geometry, not the rule, and the rule is coupling through $\mathbb{W}$ alone — nonlocal, spectral, all-to-all-with-weights. With the conceptual ground cleared of the one seductive misreading, the question becomes operational rather than interpretive: granted that the coupling lives entirely in the linear operator $\mathbb{W}$, which structural form of that coupling makes the system actually solvable? The next subsection answers by specializing to the separable class, the state-and-type structure on which every tractable case below — the finite-rank collapse of section 22.4 and the rank-one thread of section 22.5 — will rest.

22.2.3 Separable couplings across the kernel

The Ginzburg–Landau discussion cleared the conceptual ground: the coupling lives entirely in the linear operator $\mathbb{W}$, it is nonlocal and spectral, and the gradient reading that would localize it across types is a measure-zero exception we have learned to recognize and set aside. That settles *how* to read the coupling. The question this subsection answers is the operational one the previous closing posed — granted that the only wire between types is $\mathbb{W}$, which structural form of the per-type Hamiltonian makes the resulting system actually *solvable*? Every tractable case below, the finite-rank collapse of section 22.4 and the rank-one thread of section 22.5, rests on a single specialization of the cost, and it is the working assumption for the remainder of the chapter. We adopt couplings that are *separable in the control*, the graphon version of definition 8.5:

$$H(x,\xi,p,\nu) \;=\; H_0(x,\xi,p) \;-\; F(x,\xi,\nu), \tag{22.11}$$

in which the Hamiltonian splits into two pieces that do not share an argument. The first, $H_0(x,\xi,p)$, carries *all* the costate dependence and no population dependence; it is the pure control part, the Hamiltonian the agent would face in an empty world with no neighbours. The second, $F(x,\xi,\nu)$, carries *all* the population dependence and no costate dependence; it is the cost-coupling term, the running penalty the aggregate ν imposes on an agent sitting at state ξ. At the level of the running cost this is the requirement that L split as $L(x,\xi,a,\nu) = L_0(x,\xi,a) + F(x,\xi,\nu)$ — the control cost and the interaction cost are additive and do not multiply each other — which is exactly definition 8.5 read with the type carried as a spectator parameter.

The decisive simplification is the same one that organized the homogeneous theory, and it is worth tracing through the optimization to see precisely why the split buys what it buys. The optimal control is the Hamiltonian maximizer of eq. (22.9),

$$\alpha^{x,*} = \arg\max_{a\in A}\Big\{-b(x,\xi,a)\cdot\nabla u^x - L(x,\xi,a,\nu^x)\Big\} = \arg\max_{a\in A}\Big\{-b(x,\xi,a)\cdot\nabla u^x - L_0(x,\xi,a)\Big\},$$

where the second equality is the entire point: under the additive split the population term $F(x,\xi,\nu^x)$ is a constant as far as the maximization over a is concerned — it does not contain a — so it drops out of the argmax untouched. The maximizer therefore depends on the value gradient ∇u^x and the type x, but *not* on the aggregate $\nu^x = (\mathbb{W}\boldsymbol{m}_t)(x)$. The same is true of the induced closed-loop drift, which by the envelope identity of eq. (22.8) is

$$b^{x,*} = -\partial_p H(x,\xi,\nabla u^x,\nu^x) = -\partial_p H_0(x,\xi,\nabla u^x),$$

the population term contributing nothing because $\partial_p F = 0$. So the velocity that transports the type-x density in the forward Fokker–Planck equation is built from the value alone and never sees the neighbour distribution directly.

Two consequences follow, and together they are what make the separable case tractable. First, the population is expelled from the entire *dynamic* half of each per-type problem. Substituting eq. (22.11) into the backward HJB eq. (22.6a), the aggregate appears in one place only — as an additive, type-indexed *source term*

$$-\partial_t u^x - \tfrac{1}{2}\operatorname{tr}\big(\sigma(x)\sigma(x)^\top \mathrm{D}^2 u^x\big) + H_0(x,\xi,\nabla u^x) \;=\; F\big(x,\xi,(\mathbb{W}\boldsymbol{m}_t)(x)\big),$$

together with the terminal datum $u^x(T,\xi) = g(\xi,(\mathbb{W}\boldsymbol{m}_T)(x))$. The population is now a prescribed forcing of an otherwise population-free backward equation; it heats the value field through a source and pins its terminal value, and it does nothing else. This is the Hamiltonian-level sharpening of restriction (i) of remark 22.1: there we observed that with an aggregate-free drift the forward FP equation is linear in m^x; here the separability makes the same statement at the level of the Hamiltonian, exhibiting the closed-loop drift $-\partial_p H_0$ as manifestly free of ν, so the only nonlinearity left anywhere in the coupled system is the benign one routed through the scalar source F. Second, and as a corollary, the cross-type coupling is concentrated entirely in the single linear object $\mathbb{W}$. The source $F(x,\xi,(\mathbb{W}\boldsymbol{m}_t)(x))$ is the lone point at which type x's equation references any other type, and it references them only through the kernel average $(\mathbb{W}\boldsymbol{m}_t)(x)$. Strip F of its measure argument and the continuum of HJB–FP pairs falls apart into independent, population-blind control problems; restore it and the one wire reconnecting them is $\mathbb{W}$ and nothing else. This is precisely the structure the special cases of sections 22.4 and 22.5 exploit, and the separability assumption is in force throughout both.

It is worth being exact about what separability does and does not constrain, because the compressed phrase is that the graphon coupling here is *nonlocal in type but may be of any state-locality* — and that sentence is a statement about the two independent axes pried apart in section 22.2.1. Separability is a condition on the *control* axis: it forbids the population from entering through the costate, forcing F to be free of p and hence the optimal feedback to be aggregate-independent. It says *nothing* about how F depends on the state ξ, which is the orthogonal, inherited axis. The coupling remains, as every graphon coupling must, nonlocal in the type variable x — the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ is an integral pooling all types, with no notion of adjacency along $[0,1]$ to make it local, exactly as section 22.2.1 forced. But on the state axis F is left entirely free: it may be nonlocal in ξ, reading the neighbour measure through a smoothed functional such as a convolution $F(x,\xi,\nu) = (\phi * \nu)(\xi)$, or local in ξ, reading the pointwise density $F(x,\xi,\nu) = f(\xi,\rho^x(\xi))$ at the agent's own location — the clean dichotomy of section 22.2.1, with all its analytic consequences (weak continuity for the nonlocal choice, the density bounds of **(A-Nondeg)** for the local one) carried over verbatim. Separability constrains only the control-coupling and leaves the state-locality choice open; the two axes do not interfere.

The form the order-parameter reductions and Thread B actually use specializes F once more, now narrowing how it reads the *measure* argument rather than touching the state axis at all: the agent reacts to the aggregate not through its full shape but through a single scalar statistic of it. We write

$$F(x,\xi,\nu) \;=\; \Psi\big(\xi,\langle\psi,\nu\rangle\big), \qquad \langle\psi,\nu\rangle = \int_{\mathbb{R}^d} \psi \,\mathrm{d}\nu, \tag{22.12}$$

for a fixed test function $\psi\colon \mathbb{R}^d \to \mathbb{R}$ (or a finite vector of them) and an outer profile Ψ. The point is that the only thing the type-x agent needs to know about its entire graphon-averaged neighbourhood is the one number (or finite vector)

$$\langle \psi, (\mathbb{W}\boldsymbol{m}_t)(x)\rangle = \int_0^1 W(x,y)\,\langle \psi, m_t^y\rangle \,\mathrm{d}y, \tag{22.13}$$

where the equality is just the weak reading eq. (22.3) of the aggregate applied to the bounded continuous test function ψ: pairing ψ against the measure-valued integral commutes with the scalar type-integral, turning a measure-valued object into an ordinary Lebesgue integral of the scalar field $y \mapsto \langle \psi, m_t^y\rangle$ against the kernel row $W(x,\cdot)$. The agent thus reacts to a kernel-weighted average, over all types, of the per-type statistic $\langle \psi, m_t^y\rangle$ — a single real number that compresses the whole field into the one datum the cost can feel. This is the lever the next reductions pull: project that scalar field onto the eigenfunctions of $\mathbb{W}$ and eq. (22.13) collapses to finitely many numbers, the order parameters.

Thread B is the cleanest instance, and naming it here lets the reader anticipate the worked computation. Take $\psi(\xi) = \xi$, so the statistic $\langle \psi, m_t^y\rangle = \int_{\mathbb{R}^d} \xi\, m_t^y(\mathrm{d}\xi) = \bar{m}_t^y$ is simply the *mean* of the type-y population, and the felt statistic eq. (22.13) is the graphon-aggregated neighbour mean $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ of the thread specification. Take the outer profile quadratic, $\Psi(\xi, z) = \frac{q}{2}(\xi - sz)^2$, with imitation strength $q > 0$ and coupling s: the agent pays for deviating from s times the average position of its weighted neighbours, the congestion/imitation penalty that defines Thread B. Everything downstream of eq. (22.12) — the value's quadratic form, the Riccati curvature, the single scalar order parameter — is the explicit unwinding of this one choice, carried out in full in section 22.5.

The separable single-statistic structure eq. (22.11)–eq. (22.12) is exactly what the special-structures section consumes: there the scalar field eq. (22.13), projected onto the eigenfunctions of $\mathbb{W}$, becomes the finitely many order parameters that close the system, and the rank-one corner of that projection is the worked computation of section 22.5. But both reductions presume something we have so far only used informally — that the assembled field $x \mapsto (u^x, m^x)$, and with it the scalar statistic $x \mapsto \langle \psi, (\mathbb{W}\boldsymbol{m}_t)(x)\rangle$, depends on the type *measurably*, and continuously when the kernel does, so that the type-integral eq. (22.13) is legitimate and the order parameters are well-defined numbers rather than integrals of a non-measurable integrand. This is the deferred obligation the best-response construction logged and the standing-data promise that the aggregate varies measurably in x. The next section discharges both, by a stability argument, before we put the separable structure to work.

22.3 Regularity and measurability in the type variable

The previous subsection ended on a debt, and so did the one before the couplings detour: both the separable-coupling reduction of section 22.2.3 and the best-response construction of section 22.1.3 leaned on a fact they were not entitled to assume. When proposition 22.6 packaged the per-type solves into the self-map $\mathsf{S}\colon \mathcal{C} \to \mathcal{C}$, it incurred one obligation it explicitly deferred: for $\mathsf{S}[\boldsymbol{m}]$ to be an element of $\mathcal{C}$ at all — a genuine *field*, not merely a type-by-type collection of unrelated solutions — the assembled map $x \mapsto (u^x, \widetilde{m^x})$ must depend measurably on the type, so that its own aggregate eq. (22.2) exists and the loop closes inside $\mathcal{C}$. And the standing-data discussion of section 22.1.1

promised, but postponed, that the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ itself varies measurably in x — continuously when the kernel does — the very property that made the scalar statistic $x \mapsto \langle \psi, (\mathbb{W}\boldsymbol{m}_t)(x)\rangle$ of section 22.2.3 a legitimate integrand and the order parameters well-defined numbers. This section pays both debts. They are the same debt seen from two ends, and a single stability argument settles them.

It is worth saying in words, before any lemma, exactly why no new analysis is required — why "regularity in the type variable" is not a fresh PDE problem but a corollary of machinery we already own. Look once more at the per-type system eq. (22.6). The type x enters it through precisely two channels and no others. It enters the *coefficients* — the drift $b(x,\cdot,\cdot)$, the volatility $\sigma(x)$, the running and terminal costs $L(x,\cdot,\cdot,\cdot), g$ — which by the type-Lipschitz clause of **(A-Lip)** move only a little when x moves a little. And it enters the *environment* $\nu^x = (\mathbb{W}\boldsymbol{m}_t)(x)$, the single channel through which type x feels the rest of the population, which — this is what the lemma below establishes — also moves only a little when x moves a little, because it is the type-integral of a frozen field against a kernel row that varies controllably in x. Both inputs to the type-x problem are therefore continuous (and always measurable) functions of x. The output — the value u^x and the closed-loop density $\widetilde{m^x}$ — depends on those two inputs *stably*: that is the entire content of the HJB- and FP-stability lemmas lemmas 10.10 and 10.11 of chapter 10, which bound how far the value and the density can move per unit change in the coefficients and the environment. Compose "inputs vary controllably in x" with "output varies stably in inputs" and regularity in type drops out, with the type playing the role of an external parameter and nothing about the PDEs themselves re-examined. No elliptic theory in x, no new estimate: a stability corollary, full stop. The work of this section is just to make each of these two clauses precise and chain them.

The input side has a coarse version already on record. lemma 21.11 showed that the aggregate map $\boldsymbol{m} \mapsto \mathbb{W}\boldsymbol{m}$ is Lipschitz in the *whole field*, which is the estimate chapter 24 will contract against. What the per-type stability estimates consume is finer and orthogonal: not how the aggregate responds to a perturbation of the entire field at fixed type, but how it responds to a change of *type* at fixed field — the *pointwise-in-type* regularity of $x \mapsto (\mathbb{W}\boldsymbol{m})(x)$. That refinement is lemma 22.8, to which we turn first; it is the object that converts the type-Lipschitz data clause into a genuine continuity statement for the environment, and with it in hand the proposition that follows is a short composition.

Lemma 22.8 (Continuity of the aggregate in the type)**.** *Let $\boldsymbol{m} = (m^x)_{x\in[0,1]}$ be a field of measures with $\sup_x \mathcal{W}_2(m^x, \delta_0) \le R < \infty$.*

1. *If W is measurable and bounded (the bare **(A-Reg-W)**), then $x \mapsto (\mathbb{W}\boldsymbol{m})(x)$ is measurable as a map $[0,1] \to \mathcal{M}_+(\mathbb{R}^d)$ (weak topology), with mass ≤ 1 and second moment $\le R^2$ uniformly.*
2. *If moreover W is continuous in type in the sense $\int_0^1 |W(x,y) - W(x',y)|\,\mathrm{d}y \to 0$ as $x' \to x$ (e.g. W jointly continuous, the continuity clause of **(A-Reg-W)**), and $x \mapsto m^x$ is $\mathcal{W}_2$-continuous, then $x \mapsto (\mathbb{W}\boldsymbol{m})(x)$ is continuous for the total-variation norm. If in addition the second moments are* uniformly integrable *— guaranteed, e.g., by a uniform $(2+\delta)$-moment bound $\sup_x \int |\xi|^{2+\delta}\, m^x(\mathrm{d}\xi) \le R_\delta < \infty$ for some $\delta > 0$ — the convergence upgrades to $\mathcal{W}_2$.*

Proof. The uniform mass and moment bounds come first, because they are what keep the aggregate inside $\mathcal{M}_+(\mathbb{R}^d)$ at every type and what later license the upgrade to $\mathcal{W}_2$. Since $0 \le W \le 1$, the total mass of $(\mathbb{W}\boldsymbol{m})(x)$ is the degree $\int_0^1 W(x,y)\,\mathrm{d}y \le 1$, and its second moment is $\int_0^1 W(x,y) \int |\xi|^2\, m^y(\mathrm{d}\xi)\,\mathrm{d}y \le \int_0^1 \int |\xi|^2\, m^y(\mathrm{d}\xi)\,\mathrm{d}y \le R^2$, both uniformly in x — the kernel can only damp the per-type mass and moment, never amplify them, so the bounds on the fibres pass straight to the aggregate.

Part (1): measurability. We must show $x \mapsto (\mathbb{W}\boldsymbol{m})(x)$ is Borel when $[0,1] \to \mathcal{M}_+(\mathbb{R}^d)$ carries the weak topology. By definition of that topology, a map into $\mathcal{M}_+(\mathbb{R}^d)$ is Borel measurable as soon as each of its scalar pairings $x \mapsto \langle \varphi, (\mathbb{W}\boldsymbol{m})(x)\rangle$ is measurable for φ ranging over a *countable* convergence-determining family $\{\varphi_j\}$ of bounded continuous functions — such a family exists on the Polish space $\mathbb{R}^d$ (chapter 3), and weak convergence is exactly convergence of all the pairings $\langle \varphi_j, \cdot\rangle$, so the Borel σ-algebra of the weak topology is generated by these countably many coordinate maps. Fix one such φ. By the weak prescription eq. (22.3),

$$\langle \varphi, (\mathbb{W}\boldsymbol{m})(x)\rangle = \int_0^1 W(x,y)\,\langle \varphi, m^y\rangle\,\mathrm{d}y,$$

and the integrand $(x,y) \mapsto W(x,y)\,\langle \varphi, m^y\rangle$ is a bounded, jointly measurable function on $[0,1]^2$: W is measurable by the bare **(A-Reg-W)** and bounded by 1, while $y \mapsto \langle \varphi, m^y\rangle = \int_{\mathbb{R}^d} \varphi\,\mathrm{d}m^y$ is bounded by $\|\varphi\|_\infty$ and measurable in y because $\boldsymbol{m}$ is a measurable field. Fubini–Tonelli for bounded measurable integrands then makes the partial integral $x \mapsto \int_0^1 W(x,y)\langle \varphi, m^y\rangle\,\mathrm{d}y$ a measurable function of x. Running this over the countable family $\{\varphi_j\}$ exhibits all the generating coordinate maps as measurable, so $x \mapsto (\mathbb{W}\boldsymbol{m})(x)$ is Borel into the weak topology. Note that nothing here needs the kernel to be continuous; bare measurability of W suffices, which is why measurability is the property that holds *always*.

Part (2): continuity. Now add the type-continuity clause. Take any bounded 1-Lipschitz φ and subtract the pairings at two types:

$$\left|\langle \varphi, (\mathbb{W}\boldsymbol{m})(x)\rangle - \langle \varphi, (\mathbb{W}\boldsymbol{m})(x')\rangle\right| \le \int_0^1 \left|W(x,y) - W(x',y)\right| \left|\langle \varphi, m^y\rangle\right|\,\mathrm{d}y \le \|\varphi\|_\infty \int_0^1 \left|W(x,y) - W(x',y)\right|\mathrm{d}y.$$

The only field-dependent quantity, $|\langle \varphi, m^y\rangle|$, was bounded by $\|\varphi\|_\infty$ and pulled out, so what remains is the row-difference $\int_0^1 |W(x,y) - W(x',y)|\,\mathrm{d}y$, which the hypothesis sends to 0 as $x' \to x$ — and it does so *uniformly* over the unit ball $\|\varphi\|_\infty \le 1$. The supremum over that ball of the left side is, by definition, the total-variation distance, so this is convergence in total variation: $\mathrm{TV}\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m})(x')\big) \to 0$. Total-variation convergence is strictly stronger than weak convergence $(\mathbb{W}\boldsymbol{m})(x') \Rightarrow (\mathbb{W}\boldsymbol{m})(x)$, so we have that for free. (Notice the field's own $\mathcal{W}_2$-continuity in x was not even used for this half; the kernel's row-continuity alone drives the TV estimate, because the test function was bounded and the field entered only through that bound. The field-continuity hypothesis is there to make the moments behave, which is the next and final point.)

The $\mathcal{W}_2$-upgrade and why it is delicate. Total variation is not yet what the working metric demands. Weak — equivalently TV — convergence on $\mathcal{P}_2(\mathbb{R}^d)$ does *not* imply $\mathcal{W}_2$ convergence; by theorem 4.8, $\mathcal{W}_2$ metrizes weak convergence *together with* convergence of second moments, and the second clause can fail even when the first holds. The failure mode is precise and worth seeing once, because it is exactly the gap the higher-moment hypothesis closes. Consider $\mu_n = (1 - \frac{1}{n})\delta_0 + \frac{1}{n}\delta_{\sqrt{n}}$. Its total-variation distance to δ_0 is the mass it places off the origin, $\frac{1}{n} \to 0$, so $\mu_n \to \delta_0$ in TV and weakly. Yet its second moment is $\frac{1}{n} \cdot (\sqrt{n})^2 = 1$ for every n, while δ_0 has second moment 0; and since $\mathcal{W}_2(\mu_n, \delta_0)^2$ equals the second moment of μ_n (all mass of δ_0 sits at the origin, so the optimal transport cost is just $\int |\xi|^2 \mu_n$), we get $\mathcal{W}_2(\mu_n, \delta_0)^2 \equiv 1$. The limit in $\mathcal{W}_2$, if it existed, would have to be δ_0; the $\mathcal{W}_2$ distance to δ_0 never shrinks. The mechanism is a vanishing sliver of mass — weight $\frac{1}{n}$ — escaping to a position $\sqrt{n}$ far enough that, weighted by $|\xi|^2$, it carries a *fixed* unit of second moment. TV and weak convergence see only *where* the mass is and count this sliver as negligible; $\mathcal{W}_2$ also charges for *how far* stray mass travels, weighted by the square of the distance, and the sliver pays full freight. A bare uniform second-moment bound is powerless against this: μ_n has second moment $\equiv 1$, perfectly bounded, and still escapes. What is needed is to forbid any vanishing fraction of mass from hoarding non-vanishing second moment — i.e. *uniform integrability* of $|\xi|^2$ across the family.

The uniform $(2+\delta)$-moment bound supplies exactly that, and the rest is mechanical. By the de la Vallée-Poussin criterion, a uniform bound $\sup_x \int |\xi|^{2+\delta}\, m^x(\mathrm{d}\xi) \le R_\delta$ for some $\delta > 0$ forces $\{|\xi|^2\}$ to be uniformly integrable under the family: the extra δ powers of $|\xi|$ make distant mass too expensive to carry, ruling out the escaping sliver (indeed, the offending μ_n has $(2+\delta)$-moment $\frac{1}{n}(\sqrt{n})^{2+\delta} = n^{\delta/2} \to \infty$, so it is precisely what the hypothesis excludes). The aggregate inherits the bound: since $0 \le W \le 1$,

$$\int |\xi|^{2+\delta}\, \mathrm{d}(\mathbb{W}\boldsymbol{m})(x) = \int_0^1 W(x,y) \int |\xi|^{2+\delta}\, m^y(\mathrm{d}\xi)\, \mathrm{d}y \le R_\delta,$$

so $\{(\mathbb{W}\boldsymbol{m})(x)\}_x$ is itself uniformly $(2+\delta)$-integrable and hence $|\xi|^2$ is uniformly integrable across it. With weak convergence (from the TV estimate) and uniform integrability of $|\xi|^2$ in hand, the second moments converge along $x' \to x$, and theorem 4.8 upgrades the convergence to $\mathcal{W}_2$. Strip the extra moment away and the upgrade genuinely fails at $p = 2$: the conclusion then holds only for weak convergence, equivalently for W_p with $p < 2$, exactly because those metrics do not charge for the escaping second moment. □

The lemma is the input half of the stability argument made precise: it says the environment $x \mapsto \nu_t^x = (\mathbb{W}\boldsymbol{m}_t)(x)$ is always a measurable function of the type and, when the kernel is row-continuous and the moments are kept under control, a $\mathcal{W}_2$-continuous one. Combined with the type-Lipschitz clause of **(A-Lip)**, which handles the *coefficient* half, we now have both inputs to the per-type problem varying controllably in x. The proposition simply pushes them through the solution operators.

Proposition 22.9 (Regularity of the solution field)**.** *Assume **(A-Lip)** (including Lipschitz dependence of the data on the type), **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, **(A-Reg-W)**, and let $\boldsymbol{m} \in \mathcal{C}$ be a flow of fields. Let $(u^x, \widetilde{m^x}) = \mathsf{S}[\boldsymbol{m}]^x$ be the typewise best response of definition 22.5. Then:*

1. (Measurability.) *Under the bare **(A-Reg-W)**, the maps $x \mapsto u^x$ (into $C^{1,2}_{\mathrm{loc}}$) and $x \mapsto \widetilde{m^x}$ (into $C([0,T]; \mathcal{P}_2(\mathbb{R}^d))$) are Borel measurable; in particular $\mathsf{S}[\boldsymbol{m}] \in \mathcal{C}$ and the system eq. (22.6) is a well-posed object.*

2. (Continuity.) *If in addition W is continuous in type (lemma 22.8(2)) and the data b, σ, L, g are continuous in x, then $x \mapsto u^x$ and $x \mapsto \widetilde{m^x}$ are continuous, with a modulus controlled by the type-moduli of W and of the data.*

Proof sketch. We prove the continuity half (2) first, since it does the analytic work and the measurability half (1) then follows by reading the same maps with weaker hypotheses on the kernel.

Continuity. Fix two types x, x' and compare their problems input by input. The backward HJB eq. (22.6a) for u^x and for $u^{x'}$ are the *same* equation evaluated at two parameter values, and they differ in only two places. First, the coefficients $b(x,\cdot), \sigma(x), L(x,\cdot)$ and terminal datum: these are Lipschitz-close in x by the type-Lipschitz clause of **(A-Lip)**, so $\|\mathrm{data}(x) - \mathrm{data}(x')\| \to 0$ as $x' \to x$. Second, the environments $\nu^x_\cdot, \nu^{x'}_\cdot$ that sit inside the Hamiltonian and the terminal cost: these are $\mathcal{W}_2$-close, uniformly in t, by lemma 22.8(2). The $\mathcal{W}_2$ — rather than merely weak — closeness is what the HJB-stability estimate ingests, and it is exactly the clause of the lemma that demanded a uniform $(2+\delta)$-moment, so we must check that clause is actually met on the space the best response acts on. It is, and automatically: a closed-loop law $\widetilde{m^x}_t$ is the time-marginal of the optimally controlled diffusion eq. (22.1), whose drift has at most linear growth and whose diffusion is bounded under **(A-Growth)**, **(A-Nondeg)**; the standard SDE moment estimate of chapter 5 then propagates any finite initial $(2+\delta)$-moment forward with a constant depending only on the horizon and those bounds — uniformly in x. So every field in the *range* of S carries a uniform $(2+\delta)$-moment, the hypothesis of lemma 22.8(2) is satisfied along any such flow, and we may take the working space to be the fields with a fixed uniform higher-moment bound, a space S maps into itself. With both inputs controlled, the HJB-stability lemma lemma 10.10 — read with the type frozen as an external parameter and the perturbation being precisely the change of type — gives

$$\sup_t \left\| u^x(t,\cdot) - u^{x'}(t,\cdot) \right\|_{C^1} \le C\Big(\|\mathrm{data}(x) - \mathrm{data}(x')\| + \sup_t \mathcal{W}_2\big(\nu^x_t, \nu^{x'}_t\big) \Big),$$

with C uniform across types under **(A-Nondeg)**, **(A-Conv)**. Both terms on the right vanish as $x' \to x$, so $x \mapsto u^x$ is continuous, and indeed continuous in C^1, which is the regularity the next step needs. Because the induced drift $b^{x,*} = -\partial_p H$ of eq. (22.8) is built from ∇u^x by a Lipschitz map (the costate-gradient of the Hamiltonian), C^1-continuity of the value passes to continuity of the closed-loop drift, and hence of the control eq. (22.9) as well. Now feed that continuously-varying drift into the forward FP equation eq. (22.6b), started from the type-continuous datum m^x_0, and apply the FP-stability lemma lemma 10.11, which bounds the $\mathcal{W}_2$-gap of two closed-loop flows by the gap of their drifts and their data: the output gap $\sup_t \mathcal{W}_2(\widetilde{m^x}_t, \widetilde{m^{x'}}_t) \to 0$. That is continuity of $x \mapsto \widetilde{m^x}$, completing (2).

Measurability. For (1) the kernel is only measurable, so we lose the type-continuity of the environment and must settle for measurability — but, crucially, we lose nothing in the *solution* step, and this is where **(A-Conv)** pays off. The temptation, when the type-to-solution assignment is only measurable, is to worry about *measurable selection*: if the optimal control were a set-valued correspondence, picking a measurable representative type by type would require a selection theorem (Kuratowski–Ryll-Nardzewski). Under **(A-Conv)** there is nothing to select — the Hamiltonian is convex in the costate, the argmax eq. (22.9) is a single point, and the per-type solution operators are honest single-valued *maps*. The assignment is then a composition of three named maps, and measurability follows by the elementary fact that a Borel map composed with a continuous map is Borel. Unfold it:

Step 1 (the environment is Borel). The environment map $\mathsf{Env}\colon x \mapsto \nu^x_\cdot = (\mathbb{W}\boldsymbol{m}_\cdot)(x)$ is Borel measurable from $[0,1]$ into $C([0,T];\mathcal{M}_+(\mathbb{R}^d))$ with the weak topology — this is exactly lemma 22.8(1), applied at each t and assembled in t — and the coefficient map $x \mapsto \mathrm{data}(x)$ is Borel, indeed Lipschitz, by **(A-Lip)**. So the pair of inputs $x \mapsto (\mathrm{data}(x), \mathsf{Env}(x))$ is Borel. This step uses only bare **(A-Reg-W)**.

Step 2 (the HJB solve is continuous). The HJB solution operator $\mathsf{Sol}_{\mathrm{HJB}}\colon (\mathrm{data}, \nu) \mapsto u$ that sends data and a frozen environment to the value is a *continuous* map between the relevant Polish spaces. This is not a new claim: it is precisely the modulus of continuity recorded by lemma 10.10, the same estimate the continuity half used, now read as the statement that the output depends continuously on the input rather than as a quantitative rate.

Step 3 (the FP solve is continuous). Likewise the FP solution operator $\mathsf{Sol}_{\mathrm{FP}}\colon (u, \mathrm{data}) \mapsto \widetilde{m}$ that sends a value (through its induced drift) and data to the closed-loop flow is continuous, this being the content of lemma 10.11.

Compose the three. Since $x \mapsto u^x = \mathsf{Sol}_{\mathrm{HJB}}(\mathrm{data}(x), \mathsf{Env}(x))$ is a continuous map of a Borel map, it is Borel; and then $x \mapsto \widetilde{m^x} = \mathsf{Sol}_{\mathrm{FP}}(u^x, \mathrm{data}(x))$ is again a continuous map of a Borel map, hence Borel. No selection theorem appears anywhere, because single-valuedness removed the only place one could have been needed. (Were **(A-Conv)** dropped and the argmax eq. (22.9) allowed to be set-valued, this clean composition would break: Sol would become a measurable correspondence, and recovering a measurable field $x \mapsto (u^x, \widetilde{m^x})$ would then genuinely require a Kuratowski–Ryll-Nardzewski measurable selection, which lies outside our standing hypotheses. Convexity of the Hamiltonian is thus doing double duty: it makes the feedback a function in the verification theorem of section 22.1.3, and it makes the type-to-solution assignment measurable without selection here.) The detailed parabolic and Schauder estimates underlying Steps 2 and 3 are exactly those of chapter 10, applied verbatim with the type as an inert external parameter; we do not reproduce them. □

Remark 22.10 (What regularity in type buys)**.** proposition 22.9 is the technical license for everything that follows, and the two halves cash out in two different places, so it pays to separate them.

The *continuity* half is what lets structure in the kernel *propagate* into structure in the solution — the engine of the next section. The cleanest case is block-constancy. Suppose the data and the kernel are constant on a block of types: two types x, x' in the same block then have identical coefficients, $\mathrm{data}(x) = \mathrm{data}(x')$, and identical environments, $\nu^x = \nu^{x'}$ (the kernel rows agree, so the aggregates agree). The right-hand side of the HJB-stability estimate is then *exactly zero*, not merely small, and because the solution operator is single-valued under **(A-Conv)** — same inputs, one output — the solutions coincide: $u^x = u^{x'}$ and $\widetilde{m^x} = \widetilde{m^{x'}}$. The solution field is therefore constant on the block, with no further argument. This is precisely the mechanism by which section 22.4.2 recovers finite multi-population MFG from a block graphon and by which the rank-one and finite-rank reductions of sections 22.4 and 22.5 are allowed to assume the solution depends on the type only through the handful of kernel statistics it sees — it is continuity (here in its degenerate, exact form) that forbids the solution from varying *within* a level set of the data and kernel.

The *measurability* half is what makes the best-response self-map well-defined and so underwrites the existence theory downstream. The aggregate eq. (22.2) of the *output* field $x \mapsto \widetilde{m^x}$ is a type-integral, and a type-integral of a non-measurable family is meaningless; measurability of the assembled field is exactly what makes that aggregate exist, hence what places $\mathsf{S}[\boldsymbol{m}]$ back inside $\mathcal{C}$ and closes the loop $\mathsf{S}\colon \mathcal{C} \to \mathcal{C}$. This is the obligation proposition 22.6 deferred, now discharged, and it is what makes the fixed-point statement there — and the compactness-and-fixed-point existence argument of chapter 24 that consumes it — a statement about a genuine self-map rather than a formal one.

Finally, a warning about what is *not* claimed, because the word "regularity" invites an over-reading the whole chapter has been at pains to forbid. We never claim, need, or use any regularity *in the type derivative.* The map $x \mapsto u^x$ is shown measurable always and continuous when the kernel is, and that is the ceiling: it is not asserted to be differentiable in x, and there is no elliptic — or any — differential operator acting in the type variable anywhere in the system. The type is an exchangeable index carrying no metric of its own (appendix A); we upgrade it from a measurable label to a continuous one when the kernel permits, and never further. This is the precise sense in which "a continuum of HJB–FP pairs" is continuum-indexed but not a PDE in the index — consistent with the standing-data injunction that we never differentiate in x.

With both halves in hand, the continuum of per-type pairs is at last a well-defined single object, and we may exploit its structure. The continuity half is the tool the next section reaches for first: it is what licenses the leap from "the kernel is piecewise constant" to "the solution is piecewise constant," turning a block graphon into a finite multi-population system and a finite-rank kernel into a finite set of order parameters. Carrying the separable single-statistic coupling of section 22.2.3 through this now-legitimate reduction is the business of the special-structures section, to which we turn.

22.4 Special solvable structures

The previous section earned us a license, and this section is where we spend it. The continuity half of proposition 22.9, distilled in remark 22.10, says that the per-type solution operator is single-valued under **(A-Conv)** and varies continuously with its inputs: two types that see *the same* coefficients and *the same* environment must produce *the same* value function and the same density — the stability estimate has a right-hand side that is not merely small but exactly zero. That is the engine of structure-propagation. Whatever the kernel is blind to — whatever distinction between types it refuses to register in the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ — the solution is forced to be blind to as well. So if the kernel sees each type through only a handful of numbers, the entire continuum of HJB–FP pairs can carry no more information than those few numbers, and the infinite-dimensional fixed point of section 22.1.3 must secretly be a finite-dimensional one. The business of this section is to make "a handful of numbers" precise in the three ways a kernel naturally permits it, and to watch the continuum collapse each time.

Two threads from earlier feed directly into this. The first is the propagation mechanism just recalled. The second is the cost structure isolated in section 22.2.3: a *separable* coupling, in which the population enters the running cost only through a single scalar statistic $\langle \psi, \nu \rangle$ of the felt aggregate (eq. (22.11)). That specialization is what guarantees the type reacts to the field through one real number rather than through the whole measure, and it is the precise hinge on which every reduction below turns: separability makes the felt environment a scalar field over types, and a low-rank kernel then makes that scalar field live in a low-dimensional space. Neither ingredient alone collapses the system — a separable coupling against a full-rank kernel still couples a continuum, and a low-rank kernel against a non-separable cost still feeds each type a whole measure — but together they are exactly what is needed, and carrying the separable single-statistic coupling of section 22.2.3 through the now-legitimate reduction is what the subsections below actually do.

Why these three structures and no others? Because there are exactly three ways the self-adjoint Hilbert–Schmidt operator $\mathbb{W}$ can fail to see the full continuum, and they form a nested family. The most general is *finite rank*: $\mathbb{W}$ has only K nonzero eigenvalues, so by its spectral decomposition (chapter 18) it resolves a type only through the K coordinates $(\phi_1(x), \ldots, \phi_K(x))$ of its leading eigenfunctions, and the coupling passes through K scalar *order parameters* — the statistical-mechanics name is deliberate, for these are the dynamic analogues of the magnetizations that closed the static models. A *block* kernel is the special finite-rank case in which the eigenfunctions are indicators of a partition: $\mathbb{W}$ then cannot distinguish two types in the same block at all, the order parameters become block averages, and continuity-propagation forces the solution itself to be block-constant. And the *rank-one* kernel is the $K = 1$ corner of finite rank — a single mode, a single order parameter, a single eigenvalue — which is as far as the collapse can go without trivializing the coupling. These are not three unrelated tricks but one spectrum-truncation idea applied at three depths: K modes, K modes that happen to be blocks, one mode.

Each of the three is simultaneously a worked example and a bridge to a neighbour, and the subsections are written to honour that promise concretely. The finite-rank case (section 22.4.1) is the example that turns the abstract spectral theorem of chapter 18 into an operational recipe: it produces the explicit order-parameter system eq. (22.15), exhibits the “K-soft-mode” reading in which a few leading eigenfunctions carry the equilibrium and the rest are spectrally suppressed, and so furnishes the finite-dimensional fixed point that the existence theory of chapter 24 will later need a handle on. The block case (section 22.4.2) is the example whose bridge is the most consequential: it reproduces *exactly*, not merely in spirit, the finite multi-population mean field games that predate the graphon formalism, with the K populations identified as the K blocks and the inter-population coupling matrix read off as kernel weight times block mass — which is precisely the identification chapter 30 builds its heterogeneity zoo upon, so that multi-population MFG is revealed as a corner of the present system rather than a separate theory. And the rank-one case is the $K = 1$ corner of the first, the dynamic descendant of the Curie–Weiss closure of chapter 14 and the static graphon game of chapter 20; it is so thoroughly collapsed that the whole continuum becomes a single scalar two-point boundary value problem in which the leading eigenvalue of $\mathbb{W}$ appears as the one rescaling of the coupling. That last reduction is important enough, and explicit enough, to deserve its own worked computation, which is why it is carried out in full not here but in section 22.5, where it advances Thread B to its dynamic layer.

The two subsections divide the labour accordingly, and the worked computation downstream consumes both. section 22.4.1 takes up the Mercer/spectral machinery: it imports the spectral decomposition of $\mathbb{W}$ as a self-adjoint Hilbert–Schmidt operator (chapter 18) and uses it to define the order parameters and prove the finite-rank reduction, of which rank-one is simply the $K = 1$ instance. section 22.4.2 takes up the block-constancy mechanism: it uses the degenerate, exact form of continuity-propagation to force the solution constant on each block and thereby recover the multi-population system. The rank-one worked computation of section 22.5 then draws on both — the single order parameter from the first, the constant-on-a-set reasoning from the second — to close Thread B in explicit form. We begin with the spectral picture.

22.4.1 Separable (finite-rank) kernels

The section opener handed this subsection two ingredients and a promise. The ingredients are the Mercer/spectral machinery for $\mathbb{W}$ — the spectral decomposition of the graphon as a self-adjoint Hilbert–Schmidt operator, built in chapter 18 — and the separable single-statistic coupling of section 22.2.3, in which the population enters a type's running cost only through one scalar reading $\langle \psi, \nu \rangle$ of the felt aggregate. The promise is that, put together, these two collapse the continuum of type-couplings to finitely many numbers and turn the infinite-dimensional field fixed point of section 22.1.3 into a finite-dimensional one. This subsection makes the promise good. The mechanism is the one section 22.4 named: continuity-propagation forces the solution to be as blind as the kernel, and a finite-rank kernel sees each type through only finitely many coordinates. Our task is to identify those coordinates explicitly — they will be the projections of the felt field onto the eigenfunctions of $\mathbb{W}$ — and to watch the entire coupled system reorganize itself around them.

Start with the kernel. chapter 18 established that a symmetric, measurable, bounded graphon induces a self-adjoint Hilbert–Schmidt operator $\mathbb{W}$ on $L^2[0, 1]$, and that such an operator has a real spectrum $\{\lambda_k\}$ accumulating only at 0, with an orthonormal eigenbasis $\{\phi_k\}$, in terms of which the kernel admits the *Mercer expansion*

$$W(x, y) \;=\; \sum_{k=1}^{K} \lambda_k\, \phi_k(x)\, \phi_k(y), \qquad \mathbb{W}\phi_k = \lambda_k \phi_k, \quad \langle \phi_k, \phi_l \rangle = \delta_{kl}, \tag{22.14}$$

with $K \le \infty$ and $\sum_k \lambda_k^2 < \infty$, the last sum being exactly the squared Hilbert–Schmidt norm $\|W\|^2_{L^2([0,1]^2)}$ that finiteness of the kernel guarantees. The number K — the count of nonzero eigenvalues — is the *rank* of $\mathbb{W}$, and the whole tractability of this subsection is a statement about it being small or effectively small.

The distinction between $K < \infty$ and $K = \infty$ is not cosmetic, and it is worth being scrupulous about which kind of convergence eq. (22.14) carries, because a later step would be unsound if we over-read it. When the rank is genuinely finite, $K < \infty$, the right-hand side of eq. (22.14) is an honest finite sum of products of bounded functions; the identity holds *pointwise* in (x, y), with nothing to converge and no measure-zero caveat. When the rank is infinite, $K = \infty$, the eigenfunction series is in general guaranteed to converge only in $L^2([0, 1]^2)$ — this is the content of the L^2 spectral theorem of chapter 18, that W equals its eigenexpansion as an element of $L^2([0, 1]^2)$, hence for almost every (x, y). Pointwise (*everywhere*) convergence of the series, the classical Mercer theorem, would require continuity of the kernel, which we do not assume and do not want to assume, since the block graphons of section 22.4.2 are discontinuous by design. We therefore adopt a strict discipline: every consequence we draw from eq. (22.14) in the $K = \infty$ case is one that survives integration against an L^2 function — an L^2-*paired*, a.e.-in-type statement — and we never invoke the pointwise value of the eigenfunction series. This is not fastidiousness for its own sake; it is what keeps the order parameters below honest as Lebesgue integrals rather than as sums of pointwise values that may not exist.

Now bring in the coupling. We take the separable single-statistic form of section 22.2.3: the population-dependent part of the Hamiltonian is $F(x, \xi, \nu) = \Psi(\xi, \langle\psi, \nu\rangle)$ (eq. (22.11), eq. (22.12)), so that the only thing a type-x agent needs to know about its entire graphon-averaged neighbourhood is the single scalar $\langle\psi, (\mathbb{W}\boldsymbol{m}_t)(x)\rangle$. Before defining anything, look at what this scalar is as x ranges over $[0, 1]$. It is a real-valued function of the type — a *scalar field over types* — and by the weak reading eq. (22.3) of the aggregate it is exactly the image under $\mathbb{W}$ of the per-type statistic. Write that statistic as

$$g_t(y) \;:=\; \langle\psi, m_t^y\rangle \;=\; \int_{\mathbb{R}^d} \psi \,\mathrm{d}m_t^y,$$

a bounded real function of y (bounded because ψ is bounded, or because the uniform second-moment bound controls it when ψ has at most quadratic growth), hence an element of $L^2[0, 1]$. Then the felt scalar field is literally $x \mapsto (\mathbb{W}g_t)(x)$: the agent of type x feels the kernel-average of the scalar statistic carried by all the other types. This is the object whose finite-dimensionality we are after, and the spectral decomposition delivers it at once, because $\mathbb{W}$ acts diagonally on the eigenbasis. The natural coordinates of a function in the range of $\mathbb{W}$ are its overlaps with the eigenfunctions ϕ_k, and those overlaps are exactly the *order parameters*.

We can now define them, with the motivation in hand: an order parameter is the projection of the felt statistic onto a single eigenmode of the network operator. For each mode k set

$$\varrho_k(t) \;:=\; \langle\phi_k,\; y \mapsto \langle\psi, m_t^y\rangle\rangle \;=\; \int_0^1 \phi_k(y)\,\langle\psi, m_t^y\rangle \,\mathrm{d}y, \qquad k = 1, \ldots, K, \tag{22.15}$$

that is, $\varrho_k(t) = \langle\phi_k, g_t\rangle$, the k-th Fourier coefficient of the scalar field g_t in the eigenbasis. The name is taken on purpose from statistical mechanics: ϱ_k is the dynamic descendant of a magnetization, the single number that measures how strongly the population at time t has aligned along network mode k. With these in hand the felt field is finite-dimensional in the most explicit way possible. Since $g_t \in L^2[0, 1]$, the operator $\mathbb{W}$ applied to it diagonalizes against the eigenbasis, $\mathbb{W}g_t = \sum_k \lambda_k\langle\phi_k, g_t\rangle\phi_k$, so

$$x \mapsto \langle\psi, (\mathbb{W}\boldsymbol{m}_t)(x)\rangle = (\mathbb{W}g_t)(x) = \sum_{k=1}^{K} \lambda_k\,\varrho_k(t)\,\phi_k(x), \tag{22.16}$$

the series converging in $L^2[0, 1]$ by the spectral theorem (and, when $K < \infty$, holding everywhere as a finite sum). Read eq. (22.16) slowly, because it is the whole point of the subsection: the population — an entire continuum of measures, one per type — is felt by the type-x agent *only* through the finite (or, for $K = \infty$, square-summable) vector $\varrho(t) = (\varrho_k(t))_k$, modulated by the fixed type-profiles $\phi_k(x)$ and weighted by the eigenvalues λ_k. Every agent sees the same K numbers; only the modulation $\phi_k(x)$ differs from type to type. For $K = \infty$ the order parameters assemble into a sequence $\varrho(t) = (\varrho_k(t))_k \in \ell^2$, with $\sum_k \varrho_k(t)^2 \le \|g_t\|_{L^2}^2$ by Bessel's inequality — the projections of a fixed L^2 function onto an orthonormal system can never carry more energy than the function itself.

This finite-dimensional reading of the felt field is exactly what collapses the fixed point, and we state the collapse precisely.

Proposition 22.11 (Finite-rank reduction)**.** *Under the Mercer kernel eq. (22.14) with finite rank $K < \infty$ and a separable coupling reacting to a single statistic ψ, the type-x HJB equation eq. (22.6a) depends on the population only through the K-vector $\varrho(t) = (\varrho_k(t))_{k=1}^K$, via the modulation $\sum_k \lambda_k \phi_k(x) \varrho_k(t)$. The graphon MFG system is therefore equivalent to a single representative-agent control problem parametrized by (x, ϱ), closed by the K self-consistency relations eq. (22.15). The infinite-dimensional fixed point over fields reduces to a fixed point over the finite-dimensional order-parameter path $\varrho \in C([0,T]; \mathbb{R}^K)$.*

Proof. The argument is the substitution eq. (22.16) read inside the HJB equation, in three steps. First, the coupling channel. Substitute the Mercer expansion eq. (22.14) into the aggregate eq. (22.2) and pair the result against the single statistic ψ using the weak reading eq. (22.3); orthonormality of the ϕ_k and the definition eq. (22.15) give

$$\langle \psi, (\mathbb{W}\boldsymbol{m}_t)(x) \rangle = \int_0^1 W(x,y) \, \langle \psi, m_t^y \rangle \, \mathrm{d}y = \int_0^1 \Big(\sum_{k=1}^K \lambda_k \phi_k(x) \phi_k(y) \Big) g_t(y) \, \mathrm{d}y = \sum_{k=1}^K \lambda_k \, \phi_k(x) \, \varrho_k(t),$$

where interchanging the finite sum with the integral is legitimate because $K < \infty$. Second, the dynamic half. The only place the population enters the backward HJB eq. (22.6a) is the source term $F(x, \xi, (\mathbb{W}\boldsymbol{m}_t)(x)) = \Psi(\xi, \langle \psi, (\mathbb{W}\boldsymbol{m}_t)(x) \rangle)$, and we have just shown that its scalar argument is the modulation $\sum_k \lambda_k \phi_k(x) \varrho_k(t)$. Hence the HJB — and, through its solution u^x, the optimal feedback and the forward FP drift, which under the separable structure see the population only through this same source — depend on the field exclusively through the K-vector $\varrho(t)$ and the fixed type-profiles $\phi_k(x)$. The type-x HJB–FP pair is thus a single representative-agent problem parametrized by the type x and the order-parameter path $\varrho(\cdot)$: nothing else about the field reaches it. Third, the closure. Solving that pair for each (x, ϱ) produces, in particular, the densities m_t^x, from which the self-consistency relations eq. (22.15) recompute the order parameters; demanding that the recomputed ϱ equal the input ϱ is a fixed-point equation on the path $\varrho \in C([0,T]; \mathbb{R}^K)$ alone, a system of K scalar relations rather than an equation on a continuum of measures. That the assembly $(x, \varrho) \mapsto (u^x, m^x)$ depends measurably (and continuously when the kernel does) on its arguments — so that the integrals eq. (22.15) are legitimate and the reduced fixed point lives in $C([0,T]; \mathbb{R}^K)$ — is exactly proposition 22.9. $\square$

It is worth pausing on what has happened to the dimensionality of the problem. The original unknown was a flow of fields $\boldsymbol{m} \in C([0,T]; \mathcal{C}_2)$ — a continuum-indexed family of measures evolving in time, the genuinely infinite-dimensional object the existence theory of chapter 24 must contend with in general. The proposition replaces it, in the finite-rank separable case, by a path $\varrho \in C([0,T]; \mathbb{R}^K)$ in a *finite*-dimensional Euclidean space, together with a single parametrized control problem to be solved off-line as a function of (x, ϱ). The continuum has not gone anywhere — there is still one HJB–FP pair per type — but the types no longer talk to one another through an infinite-dimensional channel; they talk through K shared numbers. This is the operational meaning of “the coupling factors through $\mathbb{W}$ and nothing else”: when $\mathbb{W}$ has rank K, the “nothing else” is literally K-dimensional.

A concrete reading of the two extreme cases anchors the general statement. The case $K = 1$ is the rank-one kernel of section 22.5, with a single scalar order parameter ϱ_1; it is the cleanest non-trivial graphon and the descendant of the Curie–Weiss closure. There the whole continuum reports to one number, the felt field is $\lambda_1 \varrho_1(t)\phi_1(x)$, and the self-consistency is a single scalar relation closed across time — which is precisely why it can be pushed to a closed-form two-point boundary value problem, the worked computation that advances Thread B. The case of general K is the natural "K-soft-mode" generalization: the population organizes along the leading K eigenfunctions of $\mathbb{W}$, and the equilibrium is found by solving a K-dimensional self-consistency problem instead of a continuum one. A small instance worth carrying in mind: if $\mathbb{W}$ has two equal nonzero eigenvalues with eigenfunctions ϕ_1, ϕ_2, then two order parameters (ϱ_1, ϱ_2) suffice, and the type-profile a given agent presents to the network is the point $(\phi_1(x), \phi_2(x)) \in \mathbb{R}^2$ — two types with the same profile are, as far as the coupling is concerned, indistinguishable. That last observation is the seed of the block reduction: when the eigenfunctions are constant on the cells of a partition, "same profile" becomes "same block," and K order parameters become K block averages.

When the rank is genuinely infinite ($K = \infty$), the reduction does not stop being useful, and understanding why is the heart of the "soft-mode" reading. By eq. (22.16) the type-coupling still passes through the sequence $\varrho(t) = (\varrho_k(t))_k \in \ell^2$, but now the modes are weighted by eigenvalues $\lambda_k \to 0$ (the accumulation-at-zero of the Hilbert–Schmidt spectrum, with $\sum_k \lambda_k^2 < \infty$). The decisive point is that the eigenvalues, not the order parameters, set the weight with which each mode is actually *felt*: the contribution of mode k to the felt field eq. (22.16) is $\lambda_k \varrho_k(t)\phi_k(x)$, and since the order parameters are uniformly bounded in ℓ^2 by Bessel ($\sum_k \varrho_k(t)^2 \le \|g_t\|_{L^2}^2$), the size of that contribution is governed by λ_k. So although the self-consistency is now an infinite system of order-parameter relations, it is *rapidly truncatable*. Quantitatively, truncating to the leading K modes — discarding λ_k for $k > K$ — replaces the felt field by its first K terms, and the discarded tail has $L^2[0,1]$-norm

$$\Big\| \sum_{k>K} \lambda_k \varrho_k(t)\phi_k(x) \Big\|_{L^2[0,1]}^2 = \sum_{k>K} \lambda_k^2\, \varrho_k(t)^2 \;\le\; \Big(\sup_k \varrho_k(t)^2\Big) \sum_{k>K} \lambda_k^2 \;\le\; \|g_t\|_{L^2}^2 \sum_{k>K} \lambda_k^2,$$

so the truncation error is controlled by the spectral tail $\big(\sum_{k>K} \lambda_k^2\big)^{1/2}$, which tends to 0 as $K \to \infty$ precisely because $\mathbb{W}$ is Hilbert–Schmidt. The intuition to carry away is that *eigenvalue decay is practical finite-dimensionality*: a kernel whose spectrum falls off quickly is, for every computational and structural purpose, a finite-rank kernel up to an error one can make as small as desired by keeping a few more modes. The leading eigenfunctions — the "soft modes," in the statistical-mechanics idiom where the slowly-relaxing collective coordinates dominate the long-wavelength physics — carry essentially all of the coupling, and the high modes are *spectrally suppressed*, their order parameters multiplied into the felt field by vanishingly small eigenvalues. The finite-rank proposition 22.11 is therefore not a fragile special case but the practically exact closure for any kernel with a decaying spectrum, with K chosen to push $\big(\sum_{k>K} \lambda_k^2\big)^{1/2}$ below tolerance.

This is the lens we now turn on the block graphon. A block kernel is the special finite-rank structure in which the eigenfunctions ϕ_k are constant on the cells of a partition of $[0,1]$ — the finite-dimensional invariant subspace of $\mathbb{W}$ is the space of block-constant fields — so the K order parameters become K block averages and, by the continuity-propagation of proposition 22.9, the solution field is forced constant on each block. The finite-rank reduction proved here is exactly what section 22.4.2 specializes; reading the block case is reading proposition 22.11 with the eigenfunctions chosen to be indicators.

22.4.2 Block graphons and multi-population MFG

The finite-rank subsection just closed by pointing at this one. There the coupling collapsed onto the K-dimensional span of the leading eigenfunctions ϕ_k, and the last paragraph noted what happens when those eigenfunctions are not arbitrary but are *indicators of a partition*: "same type-profile" degenerates into "same block," the order parameters become block averages, and — crucially — the continuity and single-valuedness propagation of proposition 22.9 forces the solution field itself to be constant on each block. That degenerate-but-rigid case is the block graphon, and it is worth elevating to its own subsection not because it is harder than the general finite-rank reduction (it is a strict special case of it) but because it is the *most consequential* one. A block graphon reproduces the finite multi-population mean field games that predate the graphon formalism entirely — the models with K interacting sub-populations, each with its own dynamics and its own pairwise coupling to the others — and it does so *exactly and not merely in spirit*. The claim of this subsection is an identity, not an analogy: the graphon MFG system over a block kernel *is*, after a change of bookkeeping that loses nothing, the classical multi-population system, equation for equation and boundary datum for boundary datum. That is what makes the block graphon the rigorous bridge to chapter 30, where the entire heterogeneity theory — major/minor agents, multi-class populations, structured sub-communities — is built on the multi-population system and can therefore be read, without re-derivation, as a corner of the present one.

We begin by fixing the kernel. A block graphon is the kernel of the stochastic block model: partition the type continuum $[0, 1]$ into K communities, declare every pair of communities to interact at a single constant strength, and let the kernel be flat inside each community-pair rectangle. This is the coarsest non-trivial structure a graphon can carry — a piecewise-constant kernel on a grid of rectangles — and it is exactly the structure that turns the integral operator $\mathbb{W}$ into a finite matrix.

Definition 22.12 (Block graphon)**.** A *block graphon* on the partition $[0, 1] = \bigsqcup_{k=1}^{K} I_k$ into measurable blocks of masses $\pi_k = |I_k| > 0$, $\sum_k \pi_k = 1$, is a kernel constant on each block-pair: $W(x, y) = w_{kl}$ for $x \in I_k$, $y \in I_l$, with a symmetric matrix $(w_{kl}) \in [0, 1]^{K\times K}$. We say the GMFG data are *block-compatible* if the coefficients b, σ, L, g and the initial field m_0^x depend on x only through its block, i.e. are constant on each I_k.

Two features of the definition deserve a word before they are used. First, the masses π_k are not cosmetic: a block is a community of agents, and $\pi_k = |I_k|$ is the *population fraction* of community k. A small block is a small sub-population, and its influence on the others will be weighted by exactly that smallness — the combination $w_{kl}\pi_l$, kernel strength times block mass, will turn out to be the true inter-community coupling, not the bare w_{kl}. Second, block-compatibility is not a technical convenience but the *natural matching hypothesis*, and it is worth seeing why. The block structure of the kernel says only that agents are coupled to one another in a way that depends on type solely through community membership. If the agents' intrinsic data — their drift, their volatility, their cost, the law they start from — varied *within* a community, then two agents in the same block would be genuinely different problems that merely happen to share a coupling pattern, and there would be no reason for the solution to respect the partition. Block-compatibility is the hypothesis that the partition is honest all the way down: agents in the same community are interchangeable not only in how they couple but in who they are. It is the precise condition under which the block structure of the network is also a block structure of the *game*, and it is exactly what one always assumes, usually silently, when writing a finite multi-population model in the first place — there "population k" means a homogeneous group with common coefficients. Drop it and Proposition 22.13 fails at its first step, because the per-type problems inside a block would no longer coincide.

With the data matched to the partition we can state the reduction. The intuition is the one the finite-rank subsection planted: when the only thing that distinguishes types is their block, and the only channel between types is the aggregate, two types in the same block face *identical problems*, and a single-valued solution operator must return *identical solutions*. Block-constancy of the solution is therefore forced, not assumed; and once it is in hand, the continuum of per-type equations collapses to one equation per block.

Proposition 22.13 (Block graphon GMFG is multi-population MFG)**.** *Let W be a block graphon (definition 22.12) with block-compatible data, under **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**. Then any solution of the graphon MFG system eq.* (22.6) *is* block-constant*: $u^x = U_k$ and $m^x = M_k$ for a.e. $x \in I_k$, and the K-tuple $\big((U_k, M_k)\big)_{k=1}^K$ solves the multi-population MFG system: for each $k = 1, \dots, K$,*

$$-\partial_t U_k - \tfrac{1}{2}\operatorname{tr}\big(\sigma_k\sigma_k^\top \mathrm{D}^2 U_k\big) + H\Big(k, \xi, \nabla U_k,\ \textstyle\sum_{l=1}^K w_{kl}\,\pi_l\, M_{l,t}\Big) = 0, \quad U_k(T,\xi) = g\big(\xi, \textstyle\sum_l w_{kl}\pi_l M_{l,T}\big), \tag{22.17a}$$

$$\partial_t M_k - \tfrac{1}{2}\operatorname{tr}\big(\sigma_k\sigma_k^\top \mathrm{D}^2 M_k\big) - \operatorname{div}\Big(M_k\,\partial_p H\big(k, \xi, \nabla U_k, \textstyle\sum_l w_{kl}\pi_l M_{l,t}\big)\Big) = 0, \quad M_k(0,\cdot) = M_{k,0}, \tag{22.17b}$$

where the aggregate felt by population k is the weighted sum $\sum_l w_{kl}\pi_l M_{l,t}$. Conversely any solution of eq. (22.17) *yields, by extension to block-constant fields, a solution of the GMFG system; under the uniqueness hypotheses of chapter 24 the two correspondences are inverse.*

Proof. The argument has three beats: compute the aggregate and see it depends on the type only through the block; invoke single-valuedness to force the solution to inherit that block-dependence; substitute the block-constant solution back to read off the reduced system. Then the converse.

The aggregate is block-constant in the observer's type. Fix $x \in I_k$ and evaluate the aggregate eq. (22.2) on a candidate field $\boldsymbol{m}_t$. Because the kernel is flat on each block-pair, $W(x,y) = w_{kl}$ for every $y \in I_l$, so the type-integral splits over blocks and the kernel comes out of each piece as a constant:

$$(\mathbb{W}\boldsymbol{m}_t)(x) = \int_0^1 W(x,y)\, m_t^y \,\mathrm{d}y = \sum_{l=1}^K \int_{I_l} w_{kl}\, m_t^y \,\mathrm{d}y = \sum_{l=1}^K w_{kl} \int_{I_l} m_t^y \,\mathrm{d}y.$$

The right-hand side depends on the observing type x *only through its block index k* — the integrals $\int_{I_l} m_t^y \,\mathrm{d}y$ are the same sub-population measures regardless of which $x \in I_k$ we picked. Every agent in community k therefore feels one and the same environment, the kernel-weighted sum of the community measures.

Single-valuedness forces block-constancy. Now compare two types $x, x' \in I_k$ in the same block. By block-compatibility they carry identical coefficients b_k, σ_k, L_k, g_k and identical initial law $m_0^x = m_0^{x'}$; by the previous beat they face identical environments $(\mathbb{W}\boldsymbol{m}_t)(x) = (\mathbb{W}\boldsymbol{m}_t)(x')$ at every time. Their per-type HJB–FP pairs eq. (22.6) are thus, coefficient for coefficient and datum for datum, the *same* frozen-field problem. The per-type solution operator is a genuine map — not a correspondence — precisely because **(A-Conv)** makes the Hamiltonian maximizer single-valued (so the closed-loop drift is a function, not a set) and **(A-Nondeg)** makes each frozen-field pair classically and uniquely solvable; this is the single-valuedness recorded in proposition 22.4 and propagated type-by-type in proposition 22.9. Feeding identical data into a single-valued operator returns identical output, so $(u^x, m^x) = (u^{x'}, m^{x'})$. Two arbitrary types in a block agree, hence the solution is constant on each block: there exist (U_k, M_k) with $u^x = U_k,\ m_t^x = M_{k,t}$ for a.e. $x \in I_k$. This is where the hypotheses earn their place — without single-valuedness the two types could be assigned *different* selections from a multivalued solution set and block-constancy would collapse.

Substitution reads off the reduced system. With the solution block-constant, the inner integrals evaluate to $\int_{I_l} m_t^y \,\mathrm{d}y = \int_{I_l} M_{l,t} \,\mathrm{d}y = |I_l|\, M_{l,t} = \pi_l M_{l,t}$, so the environment felt by community k is exactly the weighted sum

$$(\mathbb{W}\boldsymbol{m}_t)(x) = \sum_{l=1}^{K} w_{kl}\, \pi_l\, M_{l,t}, \qquad x \in I_k.$$

Substituting this, together with $u^x = U_k$, $\sigma(x) = \sigma_k$, $m^x = M_k$ and $H(x,\cdot) = H(k,\cdot)$, into the type-x pair eq. (22.6a)–eq. (22.7) turns each of the (a.e.-identical) per-type equations in block k into the single pair eq. (22.17), with its terminal datum $U_k(T,\xi) = g(\xi, \sum_l w_{kl}\pi_l M_{l,T})$ and initial datum $M_k(0,\cdot) = M_{k,0}$ inherited from eq. (22.7). The continuum of equations has become K coupled HJB–FP pairs, one per community, coupled only through the $K \times K$ weighted-mean term.

Converse. Conversely, take any solution $\big((U_k, M_k)\big)_{k=1}^{K}$ of the finite system eq. (22.17) and define a field by extension, $u^x := U_k$ and $m^x := M_k$ for $x \in I_k$. For $x \in I_k$ the aggregate of this block-constant field is, by the same computation, $\sum_l w_{kl}\pi_l M_{l,t}$, which is precisely the environment in eq. (22.17); so the extended field satisfies the graphon pair eq. (22.6) for a.e. x. The two maps — "restrict a block-constant GMFG solution to its block values" and "extend a multi-population solution to a block-constant field" — are thus mutually inverse on block-constant solutions, and under the uniqueness hypotheses of chapter 24 (which rule out any non-block-constant GMFG solution and any second multi-population solution) the correspondence between *all* GMFG solutions and *all* multi-population solutions is a bijection. □

It is worth naming what the proof has actually done, because the dimensional collapse is the same one the finite-rank subsection performed, now in its most transparent guise. The block-constant fields form a K-dimensional subspace of the field space, and it is an *invariant* subspace of the coupling: the first beat showed $\mathbb{W}$ maps a block-constant field to a block-constant aggregate, never leaking mass into finer type-structure. Block-compatibility ensures the dynamics preserve that subspace too, and single-valuedness pins the solution inside it. The whole reduction is the statement that the equilibrium lives on this finite-dimensional invariant subspace — exactly the finite-rank picture of proposition 22.11 with the eigenfunctions specialized to the normalized indicators $\pi_k^{-1/2}\mathbf{1}_{I_k}$, whose span is precisely the block-constant fields.

Remark 22.14 (The bridge to chapter 30)**.** proposition 22.13 is the precise statement that *multi-population mean field games are the block-graphon special case of GMFG* — an identity, recall, not a family resemblance. The dictionary is exact in every entry. The K populations of the classical theory are the K blocks of the partition. The intrinsic data of population k — its drift b_k, volatility σ_k, running and terminal costs — are the block-compatible data restricted to I_k. The aggregate that population k responds to is $\sum_l w_{kl}\pi_l M_{l,t}$, so the inter-population coupling coefficient is $w_{kl}\pi_l$ — the kernel weight *times the block mass*, not the bare w_{kl} — and the within-population (self-) coupling is $w_{kk}\pi_k$. That the population fraction π_l multiplies the coupling is the graphon's bookkeeping made visible: a community influences another in proportion both to how strongly they are linked, w_{kl}, and to how large it is, π_l; a small community with a strong link can carry the same weight as a large community with a weak one.

The spectral half of the dictionary is the entry that most repays unfolding, because it is the one that turns the abstract operator-norm hypotheses of the general theory into concrete matrix conditions one can check by hand. On the block-constant subspace the operator $\mathbb{W}$ acts as a finite matrix, and we can write that matrix down. Take the indicator basis $\mathbf{1}_{I_1}, \ldots, \mathbf{1}_{I_K}$ of the block-constant subspace. A block-constant field $f = \sum_l c_l \mathbf{1}_{I_l}$ has aggregate $(\mathbb{W}f)(x) = \sum_l w_{kl}\pi_l c_l$ for $x \in I_k$, so in this basis $\mathbb{W}$ is represented by the matrix

$$A \;:=\; (w_{kl}\,\pi_l)_{k,l=1}^{K} \;=\; (w_{kl})\;\mathrm{diag}(\pi_1, \ldots, \pi_K).$$

This A is in general *not symmetric* — the mass factor π_l sits only on the column index — which looks at first like a contradiction, since $\mathbb{W}$ is self-adjoint and the block-constant subspace is invariant, so the restriction ought to be a self-adjoint operator. The resolution is that the indicator basis is orthogonal but not orthonormal: $\langle \mathbf{1}_{I_k}, \mathbf{1}_{I_l}\rangle_{L^2} = \pi_k \delta_{kl}$, so a self-adjoint operator need not have a symmetric matrix in it. Passing to the *orthonormal* basis $e_k := \pi_k^{-1/2}\mathbf{1}_{I_k}$ repairs this: there $\mathbb{W}$ is represented by

$$S \;:=\; \big(\sqrt{\pi_k}\,w_{kl}\,\sqrt{\pi_l}\big)_{k,l=1}^{K} \;=\; \mathrm{diag}(\sqrt{\pi})\,(w_{kl})\;\mathrm{diag}(\sqrt{\pi}),$$

which *is* symmetric (because (w_{kl}) is), as it must be. The two matrices are similar, $A = \mathrm{diag}(\sqrt{\pi})^{-1}\,S\;\mathrm{diag}(\sqrt{\pi})$, so they share the same characteristic polynomial and hence the same eigenvalues; and those eigenvalues are exactly the spectrum of the self-adjoint operator $\mathbb{W}$ restricted to the K-dimensional block-constant invariant subspace. In one line: the spectrum of $\mathbb{W}$ on block-constant fields is the spectrum of the symmetric matrix S, equivalently of the coupling matrix $A = (w_{kl}\pi_l)$.

This is what lets the operator-norm smallness hypothesis **(A-Coupling)** of the general uniqueness theory descend to a checkable matrix condition. Because the block-constant subspace is $\mathbb{W}$-invariant, the relevant interaction strength for the multi-population dynamics is not the full operator norm $\|\mathbb{W}\|_{L^2\to L^2}$ but the norm of its restriction to that subspace — and by the spectral computation just made, that restricted norm is the spectral radius of S,

$$\|\mathbb{W}|_{\text{block-constant}}\| \;=\; \max_{1\le k\le K}|\mu_k|, \qquad \mu_1, \ldots, \mu_K = \text{eigenvalues of } S = \text{eigenvalues of } (w_{kl}\pi_l),$$

since a symmetric matrix's operator norm equals its largest eigenvalue in modulus. The abstract condition $\|\mathbb{W}\| <$ (threshold) thus becomes the concrete matrix condition $\max_k |\mu_k| <$ (threshold) on the $K \times K$ coupling matrix $(w_{kl}\pi_l)$ — a finite eigenvalue computation in place of an operator-norm estimate. This is the precise content of "the operator-norm coupling condition becomes a matrix-norm condition," and it is the form in which chapter 24 and chapter 30 will read smallness for multi-population models.

chapter 30 develops the heterogeneity zoo — major/minor agents, multi-class models, core–periphery and assortative communities — from exactly this identification, reading each as a structured choice of partition (π_k) and symmetric coupling matrix (w_{kl}). We have thus closed the loop the chapter's plan promised: the multi-population MFG that chapter 30 takes as its starting object is not a separate theory grafted onto the graphon framework but a single corner of the present system, the one cut out by a piecewise-constant kernel.

The smallest non-degenerate instance of all this is the two-block kernel, $K = 2$, and it is the instance the worked computation will use to exhibit the bridge in the concrete. Two blocks of masses $\pi_1, \pi_2 = 1 - \pi_1$ with a symmetric 2×2 matrix (w_{kl}) give, by proposition 22.13, a two-population MFG — two coupled HJB–FP pairs whose interaction matrix $(w_{kl}\pi_l)$ has two eigenvalues, the larger of which is the relevant coupling strength. When this is specialized to the LQG cost, the finite-rank collapse of proposition 22.11 with $K = 2$ closes the population channel on a *two*-dimensional order-parameter path, and the equilibrium becomes a 2-dimensional order-parameter boundary value problem rather than the scalar one of the rank-one case. That two-block computation — the smallest genuine instance of multi-population LQG sitting inside the graphon LQG — is exactly the reduction logged in remark 22.16 once the worked computation is in hand. We turn to that worked computation now, beginning not with two blocks but with the cleanest corner of all, the rank-one ($K = 1$) graphon, where a single scalar order parameter and a single leading eigenvalue carry the entire coupling.

22.5 Worked computation: the rank-one graphon LQG

We now advance Thread B to its dynamic layer. The static graphon game version was set up in chapter 20; the LQG capstone chapter 14 wrote the template (section 14.7); here we carry it out in full and exhibit the closed-form collapse to a single scalar two-point boundary value problem. This is the cleanest graphon worked example in the book, and it makes concrete every abstraction of the chapter: the type-indexed HJB family, the coupling through $\mathbb{W}$, the finite-rank reduction of proposition 22.11 at $K = 1$, and the role of the leading eigenvalue of $\mathbb{W}$ as the effective coupling.

Worked computation 22.15 (Rank-one graphon LQG)**.** Recall Thread B (appendix A; section 14.7). Each type-x agent has scalar state $X_t^x \in \mathbb{R}$ obeying

$$dX_t^x = \left(a\, X_t^x + b\, \alpha_t^x\right)\, dt + \sigma\, dB_t^x, \qquad X_0^x \sim m_0^x, \tag{22.18}$$

with *type-independent* coefficients $a, b \neq 0, \sigma > 0$, and minimizes, against the graphon-aggregated neighbour mean,

$$J^x(\alpha^x) = \mathbb{E}\left[\int_0^T \left(\tfrac{1}{2}(\alpha_t^x)^2 + \tfrac{q}{2}\left(X_t^x - s\, z_t(x)\right)^2\right) dt + \tfrac{q_T}{2}\left(X_T^x - s_T\, z_T(x)\right)^2\right], \tag{22.19}$$

where, writing $\bar m_t^x := \int \xi\, m_t^x(\mathrm{d}\xi)$ for the type-x population mean,

$$z_t(x) \;:=\; (\mathbb{W}\bar m_t)(x) = \int_0^1 W(x,y)\,\bar m_t^y\;\mathrm{d}y, \qquad W(x,y) = f(x)\,f(y), \tag{22.20}$$

the rank-one kernel with profile $f \in L^2[0,1]$ (**(A-Reg-W)**; take $f \geq 0$ for $W \in [0,1]$, and $\|f\|_\infty^2 \leq 1$). This is the separable, nonlocal, single-statistic coupling eq. (22.11) with $\psi(\xi) = \xi$ and $\Psi(\xi, z) = \frac{q}{2}(\xi - sz)^2$.

Step 1: type-indexed HJB, type-independent curvature. For each x, with $z_t(x)$ frozen, the HJB eq. (22.6a) is the scalar LQG equation eq. (14.4) with $\bar m_t$ replaced by $z_t(x)$. Its quadratic ansatz is type-indexed,

$$u^x(t,\xi) = \tfrac{1}{2}P(t)\,\xi^2 + r^x(t)\,\xi + c^x(t), \tag{22.21}$$

and matching powers of ξ gives, for the curvature, *the same* Riccati equation for every type,

$$\dot P = b^2P^2 - 2aP - q, \qquad P(T) = q_T, \tag{22.22}$$

because the curvature sees only the cost geometry, not the coupling (eq. (14.7); here it is type-independent precisely because a, b, q, q_T are). theorem 14.2 gives a unique nonnegative bounded P on $[0,T]$, so the closed-loop coefficient $a - b^2P(t)$ is bounded. (We write it out as $a - b^2P(t)$ throughout rather than abbreviating it by a single letter, since θ and ϱ are reserved book-wide — the former retired, the latter the order parameter.) All the type- and population-dependence sits in the offset, which solves the type-indexed linear family

$$\dot r^x = (b^2P - a)\,r^x + q\,s\,z_t(x), \qquad r^x(T) = -q_Ts_T\,z_T(x), \tag{22.23}$$

coupled across types only through $z_t(x) = (\mathbb{W}\bar m_t)(x)$, exactly as section 14.7 anticipated.

Step 2: Gaussian flow per type. The optimal feedback is linear, $\alpha^{x,*}(t,\xi) = -b(P(t)\xi + r^x(t))$, so the type-$x$ closed loop is the affine SDE $\mathrm{d}X_t^x = \big((a - b^2P(t))X_t^x - b^2r^x(t)\big)\,\mathrm{d}t + \sigma\;\mathrm{d}B_t^x$. By theorem 14.4 its law is Gaussian, $m_t^x = \mathcal{N}(\bar m_t^x, V_t)$, with the variance V_t *type-independent* (it solves $\dot V = 2(a - b^2P)V + \sigma^2$, which sees neither type nor coupling) and the mean solving

$$\dot{\bar m}_t^x = (a - b^2P(t))\,\bar m_t^x - b^2\,r^x(t), \qquad \bar m_0^x = \textstyle\int \xi\, m_0^x(\mathrm{d}\xi) \text{ given.} \tag{22.24}$$

As in the homogeneous case the only game-theoretic content is in the means $\bar m_t^x$ and their conjugates r_t^x; the variance is a spectator.

Step 3: the rank-one collapse to a scalar order parameter. Here the kernel structure does its work. The aggregate is $z_t(x) = f(x)\,\varrho_t$, where

$$\varrho_t \;:=\; \int_0^1 f(y)\,\bar m_t^y\;\mathrm{d}y \;=\; \langle f, \bar m_t\rangle_{L^2} \tag{22.25}$$

is the *single scalar order parameter* at $K = 1$: the projection of the mean profile onto the *unnormalised* leading (and only) eigenvector f of $\mathbb{W}$. In the orthonormal normalisation of the Mercer expansion eq. (22.14) one has $\phi_1 = f/\sqrt{\lambda_1}$ and $\lambda_1 = \|f\|_{L^2}^2$, so $\varrho_t = \sqrt{\lambda_1}\,\varrho_1(t)$ where $\varrho_1(t) = \langle \phi_1, \bar m_t\rangle$ is the order parameter of eq. (22.15) at $k = 1$; the physically felt field $f(x)\varrho_t = \lambda_1\phi_1(x)\varrho_1(t)$ is independent of the normalisation, so this $\sqrt{\lambda_1}$ rescaling never appears in the dynamics. The terminal datum in eq. (22.23) is then $-q_T s_T f(x)\varrho_T$, and the forcing is $qsf(x)\varrho_t$; both are proportional to $f(x)$, so the offset inherits the rank-one form. Posit $r^x(t) = f(x)\,\rho(t)$. Substituting into eq. (22.23) the factor $f(x)$ cancels and leaves a single scalar equation

$$\dot\rho = (b^2 P - a)\,\rho + q\,s\,\varrho_t, \qquad \rho(T) = -q_T s_T\,\varrho_T, \tag{22.26}$$

consistent for *every* x simultaneously. To close the loop, multiply the mean equation eq. (22.24) by $f(x)$ and integrate over x, using $r^x = f(x)\rho$:

$$\dot\varrho_t = (a - b^2 P(t))\,\varrho_t - b^2\rho(t)\int_0^1 f(x)^2\,\mathrm{d}x = (a - b^2 P(t))\,\varrho_t - b^2\,\lambda_1\,\rho(t), \qquad \lambda_1 := \|f\|_{L^2}^2\,. \tag{22.27}$$

The constant $\lambda_1 = \int_0^1 f^2$ is exactly the leading eigenvalue of the rank-one operator $\mathbb{W}$: since $\mathbb{W}g = \langle f, g\rangle f$, one has $\mathbb{W}f = \lambda_1 f$ and $\|\mathbb{W}\| = \lambda_1(\mathbb{W}) = \lambda_1$ (chapter 18). Collecting eq. (22.26) and eq. (22.27), the entire type-indexed forward–backward system has collapsed to the *scalar* two-point boundary value problem

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\varrho_t\\ \rho_t\end{pmatrix} = \underbrace{\begin{pmatrix} a - b^2 P(t) & -b^2\lambda_1 \\ qs & -(a - b^2 P(t))\end{pmatrix}}_{=:\,M_{\lambda_1}(t)}\begin{pmatrix}\varrho_t\\ \rho_t\end{pmatrix}, \qquad \varrho_0 = \langle f, \bar m_0\rangle \text{ given}, \quad \rho_T = -q_T s_T\,\varrho_T, \tag{22.28}$$

to be compared with the homogeneous LQG BVP eq. (14.15), from which it differs by the single substitution $b^2 \mapsto b^2\lambda_1$ in the off-diagonal — equivalently, by rescaling the coupling.

Step 4: effective coupling and the phase boundary. The product of the off-diagonal entries of M_{λ_1} is $-b^2\lambda_1 \cdot qs = -b^2(qs\lambda_1)$, so eq. (22.28) is identical to the scalar Thread A system eq. (14.15) with the mean-coupling strength rescaled,

$$s \;\longmapsto\; s_{\mathrm{eff}} := s\,\lambda_1 = s\,\lambda_1(\mathbb{W}) = s\,\|\mathbb{W}\|\,. \tag{22.29}$$

Every conclusion of the LQG phase analysis section 14.5 therefore transfers verbatim with $s \to s_{\mathrm{eff}}$. In the normalization of worked computation 14.10 ($a = 0$, $b = 1$, $q = \omega^2$, $q_T = \omega$, $s_T = 0$, so $a - b^2 P \equiv -\omega$), $\det M_{\lambda_1} = \omega^2(s\lambda_1 - 1)$, the eigenvalues are $\pm\omega\sqrt{1 - s\lambda_1}$, and the solvability scalar of lemma 14.5 is $D(T) = \cosh(\nu T) + \frac{\omega}{\nu}\sinh(\nu T) > 0$ for $s\lambda_1 \le 1$ (unique equilibrium at every horizon), while for $s\lambda_1 > 1$, $D(T) = \cos(\mu T) + \frac{\omega}{\mu}\sin(\mu T)$ with $\mu = \omega\sqrt{s\lambda_1 - 1}$, whose first zero is the critical horizon

$$T_c \;=\; \frac{1}{\mu}\Big(\pi - \arctan\tfrac{\mu}{\omega}\Big), \qquad \mu = \omega\sqrt{s\,\lambda_1 - 1} = \omega\sqrt{s\,\lambda_1(\mathbb{W}) - 1}. \tag{22.30}$$

Step 5: reconstructing the full type field. The collapse loses no information: given the scalar solution (ϱ, ρ) of eq. (22.28), the entire field is recovered explicitly. The offsets are $r^x(t) = f(x)\rho(t)$; the means solve eq. (22.24) with this forcing, giving the forward quadrature

$$\bar m^x_t = \Psi(t)\,\bar m^x_0 - b^2 f(x)\,\Psi(t)\int_0^t \Psi(\tau)^{-1}\rho(\tau)\,\mathrm{d}\tau, \qquad \Psi(t) := \exp\int_0^t (a - b^2 P(\tau))\,\mathrm{d}\tau, \tag{22.31}$$

and the variance V_t from the type-independent Lyapunov equation; the value field is eq. (22.21) and the feedback $\alpha^{x,*} = -b(P\xi + f(x)\rho)$. Note the striking feature visible in eq. (22.31): although the initial mean profile $x \mapsto \bar{m}_0^x$ may be arbitrary, only its single projection $\varrho_0 = \langle f, \bar{m}_0 \rangle$ enters the self-consistency; the rest of the profile is transported passively along the type-independent flow Ψ with a type-modulated drift $f(x)\rho(t)$. The infinite-dimensional equilibrium is pinned by one scalar.

Physics note

worked computation 22.15 is the dynamic Curie–Weiss closure on a kernel. The rank-one graphon $W = f \otimes f$ couples every type to a *single* collective mode $\varrho_t = \langle f, \bar{m}_t \rangle$ — the projection of the population onto the eigenvector-centrality profile f — exactly as the Curie–Weiss model couples every spin to the single magnetization. The type-modulation $f(x)$ is the agent's centrality: highly central types ($f(x)$ large) both contribute more to the order parameter and respond more strongly to it. The structural content is normalization-free: the rank-one collapse eq. (22.28) is the scalar Thread A system with the single substitution $s \to s_{\text{eff}} = s\,\lambda_1(\mathbb{W}) = s\,\|\mathbb{W}\|$, so *whatever* the Thread A multiplicity threshold is for the parameters (a, b, q, q_T, s_T, T) at hand, it transfers verbatim with s replaced by $s\,\|\mathbb{W}\|$. The clean value "$= 1$" is special to the normalization of worked computation 14.10 ($a = 0, b = 1, q = \omega^2, q_T = \omega, s_T = 0$), where $\det M_{\lambda_1} = \omega^2(s\lambda_1 - 1)$ makes the threshold $s\,\lambda_1(\mathbb{W}) = 1$ exactly — the dynamic analogue of the critical coupling $J\chi = 1$ of a mean-field ferromagnet, with the bare imitation strength s replaced by s times the leading eigenvalue of the network operator. This is the precise meaning, on the carried thread, of the slogan that "centrality controls equilibrium" and of the operator-norm condition **(A-Coupling)**: in that normalization, uniqueness for all horizons holds when $s\,\|\mathbb{W}\| \le 1$ and the first conjugate horizon appears once $s\,\|\mathbb{W}\| > 1$; for general parameters the same statements hold with the appropriate Thread A threshold, always at coupling $s_{\text{eff}} = s\,\|\mathbb{W}\|$.

Remark 22.16 (Reductions checked)**.** The two reductions promised in the thread specification are now explicit. *Constant kernel:* $W \equiv c$ is rank-one with $f \equiv \sqrt{c}$, so $\lambda_1 = c$, $z_t(x) = c\int_0^1 \bar{m}_t^y \,\mathrm{d}y$ is type-independent, and eq. (22.28) is exactly Thread A eq. (14.15) with $s \to cs$ — recovering the homogeneous LQG MFG with rescaled coupling, as in remark 22.3. *Rank-one kernel:* the general f gives the scalar collapse above, with effective coupling $s\lambda_1(\mathbb{W})$. A genuine *two-block* kernel ($K = 2$ in proposition 22.13) instead gives, by proposition 22.11 with $K = 2$ or proposition 22.13 with two blocks, a 2-dimensional order-parameter BVP and the two-population LQG MFG — a worked instance of the bridge to chapter 30.

Rigor ledger

Proved (or proved modulo cited inputs).

- The graphon MFG system eq. (22.6): a continuum-indexed family of backward HJB / forward FP pairs (definition 22.2), coupled across types only through the integral operator $\mathbb{W}$ via the aggregate eq. (22.2); per-type optimality proposition 22.4 and the equivalence with the fixed point of the field best-response map S (proposition 22.6), under **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, **(A-Reg-W)**.

- The two axes of locality (state versus type, section 22.2.1) and the intrinsic nonlocality-in-type of any graphon coupling; the Ginzburg–Landau analogy and its one failure (no metric/gradient on the type set), with the heat-kernel calibration example 22.7.
- Regularity in the type variable: continuity of the aggregate (lemma 22.8) and measurability/continuity of $x \mapsto (u^x, m^x)$ (proposition 22.9), via the stability lemmas of chapter 10.
- Finite-rank/Mercer reduction to K order parameters (proposition 22.11); block graphons recover multi-population MFG eq. (22.17) exactly (proposition 22.13), the bridge to chapter 30.
- Thread B, dynamic layer: the rank-one graphon LQG collapses to the scalar two-point BVP eq. (22.28) with effective coupling $s_{\text{eff}} = s\,\lambda_1(\mathbb{W}) = s\,\|\mathbb{W}\|$ (worked computation 22.15, eq. (22.29)), and the LQG phase boundary theorem 14.11 transfers with $s \to s\,\|\mathbb{W}\|$ (eq. (22.30)).

Assumed.

- The standing assumptions, now uniformly in the type, for the per-type well-posedness and verification; **(A-Reg-W)** for the aggregate and its regularity. The single-step HJB and FP well-posedness and stability of chapters 5, 6 and 10 are imported, not reproved.
- The dynamic GMFG setup — continuum of typed agents, field of measures, the operator coupling — is taken from chapter 21.

Deferred.

- Existence of a fixed point of the field best-response map S, i.e. of a solution to eq. (22.6), and uniqueness via **(A-Mon)** plus the operator-norm smallness **(A-Coupling)** at effective coupling $s_{\text{eff}} = s\,\|\mathbb{W}\|$ (the threshold $s\,\|\mathbb{W}\| = 1$ in the worked computation 14.10 normalization): chapter 24.
- The probabilistic (graphon-FBSDE) rendering of the same equilibrium and its equivalence with this PDE system: chapter 23.
- The finite-network ($G(N, W)$) justification of the heuristic closure and the ε-Nash property on large graphs: chapter 25.
- The graphon master equation and common noise: chapter 26; the full heterogeneity zoo built on proposition 22.13: chapter 30; numerical solution by type-grid quadrature of $\mathbb{W}$: chapter 28.

Open (at the research frontier).

- Well-posedness for non-separable (congestion-type) graphon Hamiltonians, where the feedback itself depends on the aggregate and the type-coupling enters the forward drift nonlinearly.

- Sharp uniqueness thresholds beyond the monotone/small-coupling regime: the interaction of strategic monotonicity with the full spectrum of $\mathbb{W}$ (not only $\|\mathbb{W}\|$) is not understood; worked computation 22.15 solves only the rank-one corner.
- Local-in-state couplings with type-dependent regularity, and the degenerate limit of diagonal-concentrating kernels (the "each type alone" boundary of section 22.2.1), outside dense-graphon theory (chapter 27).

Bibliographic notes

The dynamic graphon mean field game system eq. (22.6) is due to Caines and Huang, who introduced the GMFG equations and the coupling through the graphon operator [33, 34]; our presentation, organized around the per-type verification and the field best-response map, is the analytic lift of the homogeneous system of Lasry and Lions [93] as developed in Cardaliaguet's notes [36] (chapter 8), with the type-coupling isolated in $\mathbb{W}$. The static shadow of the centrality-controls-equilibrium principle, and the operator-norm threshold, are from Parise and Ozdaglar's graphon games [108] (chapter 20); the stochastic/dynamic graphon game and its linear–quadratic solvability, which underlies the worked computation worked computation 22.15, are treated by Carmona, Cooney, Graves and Laurière in the static case [39] and by Aurell, Carmona and Laurière in the linear–quadratic dynamic case [10], whose rank-one collapse to a scalar order parameter is the same phenomenon recorded here. The spectral facts about $\mathbb{W}$ — self-adjoint Hilbert–Schmidt structure, leading eigenvalue as operator norm, eigenvector centrality — are from the graphon-operator theory of chapter 18, following Lovász [99]. The regularity-in-type arguments of section 22.3 are the parametrized form of the stability estimates of chapter 10; the measurability half needs no selection theorem, only that a Borel map composed with the continuous single-valued solution operators (single-valuedness being **(A-Conv)**) is Borel — for the set-valued case that **(A-Conv)** excludes, the relevant Kuratowski–Ryll-Nardzewski selection theory is in [6]. The block-graphon identification of proposition 22.13 with multi-population MFG is folklore made precise here as the bridge to the heterogeneity survey [11] taken up in chapter 30. The Ginzburg–Landau reading, and the warning that the type set carries no metric, follow the notational doctrine of appendix A; the heat-kernel calibration example 22.7 is the graphon analogue of the gradient expansion of a nonlocal field theory. The LQG template and its phase boundary, transferred here with the effective coupling $s\,\|\mathbb{W}\|$, are the capstone chapter 14, after [82]; numerical confirmation by type-grid quadrature is chapter 28, following [1].

Chapter 23

Probabilistic Graphon Mean Field Games

In chapter 22 the dynamic graphon mean field game was written, following the analytic route of chapter 8, as a continuum-indexed family of coupled partial differential equations: for each type $x \in [0,1]$ a backward Hamilton–Jacobi–Bellman equation for the value $u^x(t,\cdot)$ of the representative type-x agent, and a forward Fokker–Planck equation for the type-x population law m_t^x, the two families coupled across types *only* through the graphon integral operator $\mathbb{W}$ of chapter 21. This chapter develops the *probabilistic* picture of the same equilibrium. We do for the graphon game exactly what chapter 9 did for the homogeneous game: we replace the two PDE families by a single *forward–backward stochastic differential equation*, now a continuum-indexed McKean–Vlasov FBSDE, in which the representative agent is a whole family $(X_t^x)_{x\in[0,1]}$ of controlled processes interacting through the nonlocal kernel $\int_0^1 W(x,y)\,(\cdots)\,\mathrm{d}y$.

The tools are imported wholesale. The stochastic maximum principle, the generalized (Pontryagin) Hamiltonian $\mathcal{H}$, the adjoint backward SDE, and the FBSDE–PDE dictionary $Y_t = \nabla u(t, X_t)$ are the content of chapter 9; we reuse them verbatim *type by type*, the only novelty being that the measure argument seen by a type-x agent is the graphon-averaged neighbour field $(\mathbb{W}\boldsymbol{m}_t)(x)$ rather than the agent's own law. The McKean–Vlasov self-consistency — the law of the optimally controlled process feeding back into the coefficients — is the closure of chapter 7, now imposed on a *field* of laws indexed by type. The functional-analytic home of that field, the space of square-integrable measure-valued maps on $[0,1]$ and the boundedness of $\mathbb{W}$ on it, rests on chapter 2 (Hilbert–Schmidt operators, the spectral theorem, the fixed-point theorems) and on the Wasserstein geometry of chapter 4. These four imports are a plan, not a citation dump, and it is worth saying in one line each what each buys: the maximum principle of chapter 9 converts each type's optimization into a forward–backward pair of SDEs; the closure of chapter 7 feeds the law of that pair's forward leg back into its own coefficients; the Wasserstein geometry of chapter 4 is the ruler that measures distances between the resulting laws; and the operator theory of chapter 2 controls how those distances propagate from one type to another. Threaded through all four is a single organizing thesis, stated here so the reader carries it from the first page: *the entire cross-type coupling of the graphon game is one bounded linear operator,* $\mathbb{W}$. Each type-x agent solves a copy of a classical single-population problem, and the types communicate *only* through $\mathbb{W}$; so every difficulty special to the continuum — and, as chapter 24 will show, the very threshold at which uniqueness fails — reduces to a statement about that one operator. This is what makes the continuum tractable, and it is the spine of the whole chapter. The reader should keep chapter 22 open throughout: the payoff of the chapter, assembled in section 23.5, is the precise dictionary between the FBSDE objects here and the PDE objects there — the graphon analogue of the two-pictures result proposition 9.12 — and, in section 23.6, the exact diagonalization of the continuum into independent normal modes that displays that threshold in closed form.

We are careful about what we do *not* do. Existence and uniqueness of the graphon equilibrium, with its dependence on the operator norm $\|\mathbb{W}\|$ and on graphon monotonicity, are stated here so the fixed-point structure is visible, but their proofs belong to chapter 24; the descent from the continuum to a finite sampled network $G(N, W)$ — the propagation-of-chaos statement that justifies the whole construction — belongs to chapter 25; the measure-dependent decoupling field that seeds the graphon master equation belongs to chapter 26. Our job is to set up the graphon FBSDE cleanly, to exhibit its fixed-point structure in the right Bochner space, and to prove its equivalence with the PDE system of chapter 22 under regularity. The carried Thread B (the graphon linear–quadratic–Gaussian model) is advanced one layer in section 23.6, where the probabilistic formulation diagonalizes the continuum of agents into independent normal modes indexed by the spectrum of $\mathbb{W}$.

Notation watch

Brownian motion is B, the graphon is $W = W$. As fixed in Appendix A and announced at the start of Part IV, in every chapter where a graphon is in scope the letter W is reserved for the kernel $W(x, y)$ and Brownian motion is written B_t. This is the first chapter of the book in which both objects appear inside the *same* stochastic equation, and the clash is unavoidable precisely here: the probabilistic avatar of the game writes the noise driving an agent and the network coupling between agents into a single line — the state SDE eq. (23.3) carries $\mathrm{d}B^x$ and $\mathbb{W}$ side by side — whereas in chapter 22 the kernel lived in the PDE coupling and no Brownian motion was ever written, so the two letters never met. We therefore restate the rule once and obey it without exception: in any displayed SDE below, W (and $\mathbb{W}$) is the network coupling and B^x is the noise driving the type-x state. The superscript on B^x is a *type label*, not a power and not a spatial coordinate (Appendix A); the family $(B^x)_{x\in[0,1]}$ is a continuum of Brownian motions whose joint realization raises a measurability subtlety addressed in remark 23.6.

We retain the global conventions. The value u^x and its stochastic avatar, the costate Y^x, are anchored by *terminal* data at $t = T$ and run *backward*; the state X^x and the law field run *forward* from $t = 0$. The coupling of data at opposite ends of $[0, T]$, the structural signature of a mean field game, survives into the graphon setting and into each of its equivalent forms.

23.1 The graphon McKean–Vlasov stochastic differential equation

Before any optimization enters, we meet the descriptive object on which the equilibrium will eventually be built. In the homogeneous theory of Part II this was the McKean–Vlasov SDE eq. (7.4): a single representative diffusion whose own law feeds back into its coefficients. The graphon version replaces that one process by a continuum $(X^x)_{x\in[0,1]}$ of interacting diffusions, one per type, and replaces the single self-referential law by a whole *field* of laws. Stripping the game away first is deliberate. It lets us isolate, in the cleanest possible setting, the one ingredient that is genuinely new in passing from chapter 9 to the graphon — the nonlocal, type-dependent coupling — and to see at once that this coupling is carried entirely by the operator $\mathbb{W}$, exactly as the chapter preamble promised. When we add optimization in section 23.3, it will change the drift but leave this coupling structure untouched; so every fact we establish here about how $\mathbb{W}$ moves measures around survives verbatim into the game. The plan of the section is to define the field of laws and its $\mathbb{W}$-aggregate, write the descriptive SDE and its self-consistent closure, prove well-posedness by a field-valued fixed point, and read off the forward Fokker–Planck equation that is the population half of the chapter 22 system.

23.1.1 The field of laws and the graphon aggregate

What state variable does a graphon population carry? In the homogeneous game a single probability measure m_t summarized everyone, because every agent saw the same population. Here the population is stratified by type, and a type-x agent does not react to the global law but to the laws of its *neighbours*, weighted by the kernel $W(x,\cdot)$. The right bookkeeping object is therefore no longer one measure but a measurable assignment of a law to each type — a *field of laws* — together with the rule that compresses such a field into the single neighbour measure an agent of a given type actually feels. We define the two in turn and then read off the structural fact, "one operator carries all the coupling," that organizes the rest of the chapter.

Throughout, $(\Omega,\mathcal{F},\mathbb{F},\mathbb{P})$ is a filtered probability space carrying, for each type $x\in[0,1]$, a k-dimensional Brownian motion B^x; the joint structure of the family is discussed in remark 23.6. The state of a type-x agent is an $\mathbb{R}^d$-valued process $X^x=(X^x_t)_{0\le t\le T}$.

Definition 23.1 (Field of laws)**.** A *field of laws* (or measure-valued map) is a Borel-measurable map

$$\boldsymbol{m}:[0,1]\longrightarrow\mathcal{P}_2(\mathbb{R}^d),\qquad x\longmapsto m^x,$$

where measurability is with respect to the Borel σ-algebra of $(\mathcal{P}_2(\mathbb{R}^d),\mathcal{W}_2)$ (theorem 4.8). We write $\boldsymbol{m}=(m^x)_{x\in[0,1]}$. A *flow* of fields is a map $t\mapsto\boldsymbol{m}_t$, $\boldsymbol{m}_t=(m^x_t)$, and we say it is square-integrable when $\int_0^1\int_{\mathbb{R}^d}|\xi|^2\,m^x(\mathrm{d}\xi)\,\mathrm{d}x<\infty$. The functional space these fields inhabit, with its metric and topology, is recalled and refined in section 23.2; it is the space introduced in chapter 21 for the analytic theory.

The interaction is built from $\boldsymbol{m}$ by the graphon operator. Recall from definition 2.23 and chapter 21 that $\mathbb{W}$ acts on scalar functions by $(\mathbb{W}f)(x)=\int_0^1 W(x,y)f(y)\,\mathrm{d}y$. We lift it to fields of measures componentwise.

Definition 23.2 (Graphon aggregate of a field)**.** For a field of laws $\boldsymbol{m} = (m^y)_{y\in[0,1]}$ the *graphon-averaged neighbour field* is the type-indexed measure

$$(\mathbb{W}\boldsymbol{m})(x) \;:=\; \int_0^1 W(x,y)\, m^y \;\mathrm{d}y \;\in\; \mathcal{M}_+(\mathbb{R}^d), \tag{23.1}$$

the finite positive measure acting on a test set A by $(\mathbb{W}\boldsymbol{m})(x)(A) = \int_0^1 W(x,y)\, m^y(A)\;\mathrm{d}y$. Its total mass is $\int_0^1 W(x,y)\;\mathrm{d}y \le 1$ under **(A-Reg-W)**, so $(\mathbb{W}\boldsymbol{m})(x)$ is a *sub-probability* measure. For a measurable function $h : \mathbb{R}^d \to \mathbb{R}^n$ of at most quadratic growth we write

$$(\mathbb{W}\langle h, \boldsymbol{m}\rangle)(x) \;=\; \int_0^1 W(x,y)\Big(\int_{\mathbb{R}^d} h(\xi)\, m^y(\mathrm{d}\xi)\Big)\,\mathrm{d}y \;=\; \int_{\mathbb{R}^d} h(\xi)\,(\mathbb{W}\boldsymbol{m})(x)(\mathrm{d}\xi) \;\in\; \mathbb{R}^n, \tag{23.2}$$

the graphon average of the h-moment of the field; for $h(\xi) = \xi$ this is the type-x neighbour mean $z_t(x) = \int \xi\,(\mathbb{W}\boldsymbol{m}_t)(x)(\mathrm{d}\xi)$ used by Thread B. This first moment is the *graphon-local* coupling on which the general existence theory of chapter 24 is developed (definition 24.3): there the agent couples only to the vector $z_t(x)$, the specialization of the measure-valued aggregate eq. (23.1) obtained by reading off its first moment.

The reason eq. (23.1) is exactly the right object is structural: the only way types talk to one another in a graphon model is through W, and W is a *linear* kernel. Consequently the entire cross-type coupling of the dynamics below is carried by the single bounded linear operator $\mathbb{W}$ acting on the field; everything else in the problem is a copy, indexed by x, of a classical single-population object. This "one operator carries all the coupling" principle is what makes the continuum tractable, and we exploit it repeatedly — decisively in the worked computation of section 23.6, where diagonalizing $\mathbb{W}$ uncouples the continuum entirely.

One point in definition 23.2 deserves to be dwelt on, because it is easy to read past and it recurs in every estimate: the aggregate $(\mathbb{W}\boldsymbol{m})(x)$ is in general a *sub-probability* measure, of total mass the row-sum $\int_0^1 W(x,y)\;\mathrm{d}y \le 1$, not a probability measure. A concrete micro-example makes the meaning plain. Suppose the kernel vanishes on the first half of the types, $W(x,\cdot) \equiv 0$ for $x \in [0, \frac{1}{2})$, and equals $\frac{1}{3}$ on a band for the rest, so that the row-sum is 0 for $x < \frac{1}{2}$ and (say) $\frac{1}{3}$ for $x \ge \frac{1}{2}$. Then an agent in the first half has $(\mathbb{W}\boldsymbol{m})(x)$ equal to the *zero* measure: it feels no neighbours at all and is, dynamically, an isolated particle. An agent in the second half feels a measure of mass $\frac{1}{3}$ — one third of a probability measure. The missing mass is not an error to be normalized away; it is information. It records exactly how weakly type x is embedded in the network. We deliberately do *not* divide by the row-sum to renormalize $(\mathbb{W}\boldsymbol{m})(x)$ into a probability measure, because the dynamics below couple an agent to the *total weighted amount* of neighbour influence it receives, and an isolated agent ($x < \frac{1}{2}$ here) genuinely receives none. This unnormalized convention is also what makes the mass bound ≤ 1 do real work: it is precisely the inequality that will show $\mathbb{W}$ to be a contraction of the field space in lemma 23.10, with the contraction constant the row-sum and the worst case mass exactly 1.

23.1.2 The dynamics

With the field and its aggregate in hand, writing down the dynamics is now almost mechanical: we take the homogeneous McKean–Vlasov SDE eq. (7.4) and substitute the neighbour aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ wherever that equation used the agent's own law. The diffusion machinery — existence, uniqueness, moment bounds for a single Itô SDE — is the familiar apparatus of chapter 5; the entire novelty sits in the measure argument, and through it, in the operator $\mathbb{W}$. It is worth keeping this division of labour in view as we proceed: everything hard about the continuum lives in one place.

Definition 23.3 (Graphon McKean–Vlasov SDE)**.** Let $b : [0,T] \times [0,1] \times \mathbb{R}^d \times \mathcal{M}_+(\mathbb{R}^d) \to \mathbb{R}^d$ and $\sigma : [0,T] \times [0,1] \times \mathbb{R}^d \times \mathcal{M}_+(\mathbb{R}^d) \to \mathbb{R}^{d\times k}$ be measurable. The *graphon McKean–Vlasov SDE* is the continuum-indexed system

$$\mathrm{d}X_t^x \;=\; b\big(t, x, X_t^x, (\mathbb{W}\boldsymbol{m}_t)(x)\big)\ \mathrm{d}t \;+\; \sigma\big(t, x, X_t^x, (\mathbb{W}\boldsymbol{m}_t)(x)\big)\ \mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x, \tag{23.3}$$

subject to the *self-consistency* (closure) condition

$$m_t^x \;=\; \mathrm{Law}\left(X_t^x\right) \qquad \text{for every } x \in [0,1],\ t \in [0,T]. \tag{23.4}$$

A *solution* is a field of processes $(X^x)_{x\in[0,1]}$, each adapted with $\mathbb{E}\sup_{t\le T}|X_t^x|^2 < \infty$, together with the flow of fields $\boldsymbol{m}_t = (\mathrm{Law}(X_t^x))_x$, such that eq. (23.3) holds $\mathbb{P}$-a.s. for a.e. x and the map $x \mapsto \mathrm{Law}(X_t^x)$ is measurable into $\mathcal{P}_2(\mathbb{R}^d)$ for each t.

The interaction in eq. (23.3) enters *only* through the graphon-averaged neighbour field $(\mathbb{W}\boldsymbol{m}_t)(x)$ defined in eq. (23.1) — the rigorous content of the informal coupling "$\int_0^1 W(x,y)(\cdots)\,\mathrm{d}y$." Two reductions, demanded by the thread specification and used as sanity checks throughout Part IV, recover familiar objects.

Remark 23.4 (Reduction checks)**.** These two collapses are the sanity checks we run on every graphon object as Part IV proceeds; they verify that the new machinery degenerates correctly to objects we already trust, and they preview the spectral picture of section 23.6.

Constant graphon. If $W(x,y) \equiv c$ then the aggregate loses its x-dependence entirely: $(\mathbb{W}\boldsymbol{m}_t)(x) = c\int_0^1 m_t^y\ \mathrm{d}y$ is the *same* measure for every type, namely c times the type-averaged population. Every agent now sees the identical neighbour field, so the type label drops out of the measure argument and the continuum eq. (23.3) collapses to a single homogeneous McKean–Vlasov SDE in the sense of definition 7.7, with the global law (scaled by c) as its measure argument. chapter 7 is recovered exactly. In spectral terms this is the case where the only nonzero eigenvalue of $\mathbb{W}$ is $\lambda_1 = c$ on the constant eigenfunction — a *single* active mode — which is why the continuum carries no genuinely new information.

Rank-one graphon. If $W(x,y) = f(x)f(y)$ with $f \ge 0$ a centrality profile, then

$$(\mathbb{W}\boldsymbol{m}_t)(x) = f(x)\int_0^1 f(y)\, m_t^y\ \mathrm{d}y,$$

the global f-weighted population, a measure independent of x, scaled pointwise by the centrality $f(x)$. The cross-type coupling has rank one: it factors through a single global object modulated by where in the network each agent sits. Reading off the first moment exhibits the cleanest form of this, the form used in Thread B: with $\bar{m}^y_t = \int \xi\, m^y_t(\mathrm{d}\xi)$ the type-y mean, the neighbour mean is $z_t(x) = \int \xi\, (\mathbb{W}\boldsymbol{m}_t)(x)(\mathrm{d}\xi) = f(x) \int_0^1 f(y)\bar{m}^y_t \,\mathrm{d}y = f(x)\,\varrho_t$, a single scalar *order parameter* $\varrho_t = \langle f, \bar{m}_t\rangle$ times the centrality profile. A continuum of coupled agents has degenerated to one self-consistent scalar equation for ϱ_t — the rank-one reduction promised by the thread specification, and the cleanest worked example of a graphon mean field game. Both collapses are the probabilistic shadow of the spectral structure that section 23.6 exploits in full: there $\mathbb{W}$ is diagonalized and the continuum splits into one independent scalar game per eigenmode, the constant and rank-one cases being the one-mode instances.

Well-posedness of eq. (23.3)–eq. (23.4) is the graphon version of theorem 7.9 and uses the same fixed-point mechanism, now on the field space of section 23.2.

Proposition 23.5 (Well-posedness of the graphon McKean–Vlasov SDE)**.** *Assume* ***(A-Lip)*** *and* ***(A-Growth)*** *(Lipschitz and linear growth of b, σ in the state, uniformly in (t, x) and in the measure argument with respect to the metric of section 23.2) and* ***(A-Reg-W)****. Suppose the initial field $\boldsymbol{m}_0 = (m^x_0)$ is measurable and square-integrable. Then eq.* (23.3)*–eq.* (23.4) *has a unique solution; the flow $t \mapsto \boldsymbol{m}_t$ is continuous into the field space and depends Lipschitz-continuously on $\boldsymbol{m}_0$.*

Proof sketch. The whole argument is a fixed point on the flow of fields, and it has exactly three moving parts: *freeze* the field to decouple the continuum, *estimate* how the resulting law map contracts in the field metric, and *iterate* to a contraction. We give each.

Freeze the field; the continuum decouples. Fix a candidate flow of fields $\boldsymbol{\mu} = (\boldsymbol{\mu}_t)_{t\le T}$ in the complete space $C([0, T]; \boldsymbol{\mathcal{P}}_2)$ of definition 23.8. The coupling in eq. (23.3) enters *only* through $(\mathbb{W}\boldsymbol{\mu}_t)(x)$, and once $\boldsymbol{\mu}$ is frozen this is a known, *deterministic*, time-dependent measure attached to type x — it no longer references the unknown processes. Substituting it into the drift and volatility turns the coupled continuum into a family, indexed by x, of *ordinary* (non-McKean–Vlasov) SDEs,

$$\mathrm{d}X^{\boldsymbol{\mu},x}_t = b(t, x, X^{\boldsymbol{\mu},x}_t, (\mathbb{W}\boldsymbol{\mu}_t)(x))\,\mathrm{d}t + \sigma(t, x, X^{\boldsymbol{\mu},x}_t, (\mathbb{W}\boldsymbol{\mu}_t)(x))\,\mathrm{d}B^x_t, \quad X^{\boldsymbol{\mu},x}_0 \sim m^x_0,$$

each driven by its own Brownian motion B^x and coupled to nothing else. Under **(A-Lip)** and **(A-Growth)** the coefficients $x \mapsto b(t, x, x, \cdot)$, $x \mapsto \sigma(t, x, x, \cdot)$ are Lipschitz with linear growth uniformly in (t, x), so for each x this single SDE has a unique strong, square-integrable solution $X^{\boldsymbol{\mu},x}$ by theorem 5.10. Define the *descriptive law map* (we write it $\boldsymbol{\Psi}$, reserving S for the best-response map of definition 23.15 once optimization is present) $\boldsymbol{\Psi}(\boldsymbol{\mu}) = (\mathrm{Law}(X^{\boldsymbol{\mu},x}_t))_{x\in[0,1],\, t\le T}$. A solution of eq. (23.3)–eq. (23.4) is exactly a fixed point of $\boldsymbol{\Psi}$, since the closure $m^x_t = \mathrm{Law}(X^x_t)$ is the statement that the frozen field equals the realized one.

The one-step estimate. Take two frozen flows $\boldsymbol{\mu}, \boldsymbol{\nu}$ and the corresponding solutions $X^{\boldsymbol{\mu},x}, X^{\boldsymbol{\nu},x}$ driven by the *same* B^x (a synchronous coupling, legitimate type by type). Subtracting the two SDEs, applying Itô to $|X^{\boldsymbol{\mu},x}_s - X^{\boldsymbol{\nu},x}_s|^2$, using the Lipschitz bound on b, σ in *both* the state and the measure argument (the latter measured by $\mathcal{W}_2$, per section 23.2), the Burkholder–Davis–Gundy inequality on the martingale part, and Grönwall's inequality lemma C.1 in time gives the standard

SDE stability estimate

$$\mathbb{E}\sup_{s\le t}|X_s^{\boldsymbol{\mu},x}-X_s^{\boldsymbol{\nu},x}|^2 \;\le\; C\int_0^t \mathcal{W}_2\big((\mathbb{W}\boldsymbol{\mu}_s)(x),(\mathbb{W}\boldsymbol{\nu}_s)(x)\big)^2\,\mathrm{d}s,$$

the constant C depending only on the Lipschitz constants and T (the initial data coincide, so no x-dependent constant survives). Because $\mathcal{W}_2(\mathrm{Law}\,\xi,\mathrm{Law}\,\eta)^2\le\mathbb{E}\,|\xi-\eta|^2$ for any coupling, the left side dominates $\mathcal{W}_2(\boldsymbol{\Psi}(\boldsymbol{\mu})_t^x,\boldsymbol{\Psi}(\boldsymbol{\nu})_t^x)^2$. Now integrate over x and *here* the cross-type coupling is controlled by the field contraction eq. (23.11) of lemma 23.10: integrating the displayed bound in x and applying eq. (23.11) pointwise in s,

$$\mathbf{d}_{2,t}\big(\boldsymbol{\Psi}(\boldsymbol{\mu}),\boldsymbol{\Psi}(\boldsymbol{\nu})\big)^2=\int_0^1\mathcal{W}_2\big(\boldsymbol{\Psi}(\boldsymbol{\mu})_t^x,\boldsymbol{\Psi}(\boldsymbol{\nu})_t^x\big)^2\,\mathrm{d}x\le C\int_0^t\int_0^1\mathcal{W}_2\big((\mathbb{W}\boldsymbol{\mu}_s)(x),(\mathbb{W}\boldsymbol{\nu}_s)(x)\big)^2\,\mathrm{d}x\;\mathrm{d}s\le C\;\|\mathbb{W}\mathbf{1}\|_\infty\int_0^t\mathbf{d}_{2,s}(\boldsymbol{\mu}$$

where $\mathbf{d}_{2,t}(\boldsymbol{\mu},\boldsymbol{\nu})^2=\int_0^1\mathcal{W}_2(\boldsymbol{\mu}_t^x,\boldsymbol{\nu}_t^x)^2\,\mathrm{d}x$. Absorbing $\|\mathbb{W}\mathbf{1}\|_\infty\le 1$ into C (it can only *lower* the constant, by lemma 23.10) yields the displayed inequality

$$\mathbf{d}_{2,t}\big(\boldsymbol{\Psi}(\boldsymbol{\mu}),\boldsymbol{\Psi}(\boldsymbol{\nu})\big)^2\;\le\;C\int_0^t\mathbf{d}_{2,s}(\boldsymbol{\mu},\boldsymbol{\nu})^2\,\mathrm{d}s,\tag{23.5}$$

with C depending only on the Lipschitz constants and T. This is the crux: the only place where the continuum's cross-type interaction could have produced an unbounded constant is the x-integration, and lemma 23.10 caps it at the row-sum, which is ≤ 1 no matter how strong the kernel.

Iterate to a contraction. Inequality eq. (23.5) is a time-integral (Volterra) bound, and iterating it integrates a kernel against itself. Writing $D_t=\sup_{s\le t}\mathbf{d}_{2,s}(\boldsymbol{\mu},\boldsymbol{\nu})^2$ and applying eq. (23.5) to $\boldsymbol{\Psi}^n$ repeatedly, the n-fold composition obeys

$$\mathbf{d}_{2,t}\big(\boldsymbol{\Psi}^n(\boldsymbol{\mu}),\boldsymbol{\Psi}^n(\boldsymbol{\nu})\big)^2\;\le\;C^n\int_0^t\int_0^{t_1}\cdots\int_0^{t_{n-1}}D_{t_n}\,\mathrm{d}t_n\cdots\mathrm{d}t_1\;\le\;\frac{(Ct)^n}{n!}\,D_T,$$

the factor $1/n!$ being the volume of the time simplex $\{0\le t_n\le\cdots\le t_1\le t\}$. Taking the supremum over $t\le T$ gives $\mathbf{d}_{2,\infty}(\boldsymbol{\Psi}^n\boldsymbol{\mu},\boldsymbol{\Psi}^n\boldsymbol{\nu})^2\le\frac{(CT)^n}{n!}\,\mathbf{d}_{2,\infty}(\boldsymbol{\mu},\boldsymbol{\nu})^2$. Since $(CT)^n/n!\to 0$, for n large enough this factor is <1 and $\boldsymbol{\Psi}^n$ is a contraction on $C([0,T];\boldsymbol{\mathcal{P}}_2)$. The Banach fixed-point theorem theorem 2.36, applied to a power of $\boldsymbol{\Psi}$ on the complete space (lemma 23.10), yields a unique fixed point, which is the unique solution; the Lipschitz dependence on $\boldsymbol{m}_0$ comes from carrying the initial-data difference through the same estimate. Measurability of $x\mapsto\mathrm{Law}(X_t^x)$ is preserved at each Picard step (the solution depends measurably on the frozen coefficient field) and passes to the limit. The argument is the field-valued version of theorem 7.9; the single new ingredient is the field contraction lemma 23.10 that turned the x-integration into a bounded operation. The same construction is carried out in full for the graphon mean field system in [15], to which we refer for further detail. ☐

Remark 23.6 (A continuum of Brownian motions, honestly)**.** The family $(B^x)_{x\in[0,1]}$ in eq. (23.3) is a continuum of *independent* Brownian motions, and there is a genuine measure-theoretic obstruction to realizing it on an ordinary product space. Frozen at a single time, the values $\xi^x=B_t^x$ would form a jointly measurable map $(x,\omega)\mapsto\xi^x(\omega)$ on $[0,1]\times\Omega$ whose marginals are mutually independent, identically distributed, and nondegenerate (here mean 0, variance $t>0$). No such map exists, and the reason is a clean Fubini contradiction. Suppose it did. For any Borel set $B\subseteq[0,1]$ of positive

Lebesgue measure put $A_B(\omega) = \int_B \xi^x(\omega)\, \mathrm{d}x$; joint measurability and square-integrability make Fubini available, so

$$\operatorname{Var}(A_B) = \int_B \int_B \operatorname{Cov}\left(\xi^x, \xi^y\right)\, \mathrm{d}x\, \mathrm{d}y.$$

Independence forces $\operatorname{Cov}(\xi^x, \xi^y) = 0$ for $x \neq y$, and the diagonal $\{x = y\}$ is Lebesgue-null in $B \times B$, so the double integral vanishes: $\operatorname{Var}(A_B) = 0$, i.e. *every* block integral A_B is almost surely the constant $\mathbb{E}A_B = 0$. Apply this along a countable basis of intervals B: for a.e. ω the sample function $x \mapsto \xi^x(\omega)$ is an L^2 function all of whose interval averages vanish, whence by the Lebesgue differentiation theorem $\xi^x(\omega) = 0$ for a.e. x — the family is almost surely identically zero, contradicting variance $t > 0$. The tension is exactly the one the slogan names: the sample average over types must be deterministic (Fubini), yet a genuinely i.i.d. nondegenerate family makes every average random. Three honest resolutions are used in the literature, and we name them rather than paper over the gap. (i) *Work with the deterministic field of laws.* The equilibrium object of this chapter is the measurable map $x \mapsto \operatorname{Law}(X_t^x)$, which is deterministic; every statement we prove is either "for each fixed x" or "for the measurable field of marginals," and neither needs a joint realization. (ii) *Fubini extension.* One enlarges $([0, 1] \times \Omega)$ to a *rich Fubini extension* on which an essentially pairwise independent continuum of processes does exist and an exact law of large numbers holds; this is the framework of the stochastic-graphon-game literature [39, 10]. (iii) *Countable dense skeleton.* One realizes independent B^x on a countable dense set of types and extends by the measurable-field structure, which suffices for the L^2-in-type statements we make. We adopt reading (i) as primary: eq. (23.3) is shorthand for the per-type SDE together with the measurable, deterministic closure eq. (23.4), and no statement below depends on a pathwise joint realization of the continuum. The finite-N sampled network of chapter 25, where only finitely many independent noises appear, is where a genuine joint realization is both available and needed.

23.1.3 The nonlinear graphon Fokker–Planck equation

As in the homogeneous case (theorem 7.17), the flow of marginals of a graphon McKean–Vlasov solution satisfies a forward equation, now one for each type, coupled across types through $\mathbb{W}$. This is exactly the forward half of the graphon MFG system of chapter 22.

Proposition 23.7 (Graphon Fokker–Planck)**.** *Under the hypotheses of proposition 23.5, with* ***(A-Nondeg)*** *so that the type-x generator*

$$(\mathcal{A}_t^x \phi)(x) = b\big(t, x, x, (\mathbb{W}\boldsymbol{m}_t)(x)\big) \cdot \nabla\phi(x) + \tfrac{1}{2} \operatorname{tr}\Big(\sigma\sigma^\top\big(t, x, x, (\mathbb{W}\boldsymbol{m}_t)(x)\big)\, \mathrm{D}^2 \phi(x)\Big)$$

is uniformly parabolic, the field of marginals $\boldsymbol{m}_t = (m_t^x)$ is, for a.e. x, the unique weak solution of the forward Kolmogorov equation

$$\partial_t m_t^x \;=\; \big(\mathcal{A}_t^x\big)^* m_t^x, \qquad m_0^x = m_0^x, \tag{23.6}$$

with $(\mathcal{A}_t^x)^* = \mathcal{A}^*$ the Fokker–Planck operator of Appendix A. The system eq. (23.6) is nonlinear and nonlocal: its coefficients depend on the whole field $\boldsymbol{m}_t$ through $(\mathbb{W}\boldsymbol{m}_t)(x)$.

Proof sketch. Fix x and freeze the field by the closure eq. (23.4), so that the coefficients $b(t, x, \cdot, (\mathbb{W}\boldsymbol{m}_t)(x))$ and $\sigma(t, x, \cdot, (\mathbb{W}\boldsymbol{m}_t)(x))$ are known, deterministic functions of (t, x) and X^x is an ordinary Itô diffusion with generator $\mathcal{A}_t^x$. The derivation is then the single-particle Kolmogorov computation of chapter 5, reproduced in theorem 7.17. For a test function $\phi \in C_c^\infty(\mathbb{R}^d)$, Itô's formula gives

$$\mathrm{d}\,\phi(X_t^x) = \big(\mathcal{A}_t^x \phi\big)(X_t^x)\ \mathrm{d}t + \nabla\phi(X_t^x)^\top \sigma\big(t, x, X_t^x, (\mathbb{W}\boldsymbol{m}_t)(x)\big)\ \mathrm{d}B_t^x.$$

Taking expectations annihilates the Itô integral (its integrand is square-integrable, so it is a true martingale), leaving $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\,\phi(X_t^x) = \mathbb{E}(\mathcal{A}_t^x\phi)(X_t^x)$. Writing $\mathbb{E}\,\phi(X_t^x) = \int \phi\ \mathrm{d}m_t^x$ and $\mathbb{E}(\mathcal{A}_t^x\phi)(X_t^x) = \int \mathcal{A}_t^x\phi\ \mathrm{d}m_t^x$ turns this into the weak form $\frac{\mathrm{d}}{\mathrm{d}t}\langle \phi, m_t^x\rangle = \langle \mathcal{A}_t^x\phi, m_t^x\rangle = \langle \phi, (\mathcal{A}_t^x)^* m_t^x\rangle$, valid for every test ϕ, which is precisely the distributional statement eq. (23.6). Under **(A-Nondeg)** the generator is uniformly parabolic, so the marginal has a density and the weak solution is the unique classical one. The two adjectives in the statement enter at exactly one place, the closure eq. (23.4): it makes the frozen coefficient at type x a functional of the unknown field. The dependence is *nonlinear* because $(\mathbb{W}\boldsymbol{m}_t)(x)$ contains the solution m_t^x itself, and *nonlocal in type* because $\mathbb{W}$ averages over *all* types y — the density at type x is driven by coefficients built from m_t^y for every y with $W(x, y) > 0$. Strip the closure away and each eq. (23.6) is a plain linear Fokker–Planck equation; it is the self-consistency that couples the continuum. Measurability in x of the solution field is part of the well-posedness already established in proposition 23.5. □

The forward equation eq. (23.6) is precisely the population half of the graphon MFG system of chapter 22; the descriptive process eq. (23.3) is that system's probabilistic carrier once the drift is the optimal feedback. We leave it, deliberately, only half-built. The well-posedness argument of proposition 23.5 leaned twice on objects we have not yet constructed: the field metric $\mathbf{d}_2$, in which the Picard iteration eq. (23.5) was made a contraction, and the *field-contraction* bound eq. (23.11) on the aggregate. Both promissory notes must now be paid. Pinning down precisely which metric topologizes the field of laws — and proving, with the *correct* constant, that $\mathbb{W}$ contracts it — is the business of the next section, section 23.2, where we build the functional home for these fields and establish lemma 23.10. Only once that home exists does the forward half assembled here have a rigorous space to live in; we will then add optimization in section 23.3 and, later, match this forward Fokker–Planck half against the backward HJB half it is destined to meet.

23.2 The Bochner setting for fields of measures

The previous section closed owing two debts, and this one pays them. The well-posedness proof of proposition 23.5 made its Picard iteration contract in a *field metric* $\mathbf{d}_2$ that was named but never built, and it capped the cross-type interaction with a *field-contraction* bound eq. (23.11) quoted on faith from a lemma not yet proved. We now construct the metric space the fields of laws actually inhabit, prove that it is *complete* — so that the Banach fixed point of theorem 2.36 invoked there has somewhere to land — and then prove the single estimate on which the rest of the chapter turns: that the graphon aggregate $\mathbb{W}$ contracts that space with a Lipschitz constant the kernel *row-sum bound* $\|\mathbb{W}\mathbf{1}\|_\infty \leq 1$, a number that stays at or below one *no matter how strong the kernel.* This is not one estimate among many. Every later contraction in the chapter — the small-horizon well-posedness of section 23.3, the Lipschitz continuity of the best-response map, the regime statements of theorem 23.17 — is assembled by feeding this one inequality through a per-type SDE/BSDE stability estimate and then integrating over the type interval. The continuum could *a priori* have made that type-integration blow up in proportion to the coupling strength; lemma 23.10 is precisely the statement that it cannot, because the only thing $\mathbb{W}$ can do to the total transport cost is redistribute it across types without amplifying its integral.

The fixed-point analysis of the graphon game lives on the space of fields of measures. The underlying set — measurable maps $x \mapsto m^x$ from the type interval into $\mathcal{P}_2(\mathbb{R}^d)$ — is the field of measures owned by chapter 21; what a particular analysis must *choose* is the metric that topologizes it, and different metrics are natural for different jobs. We make that choice explicit here rather than leave it implicit, because the choice matters across the seam. For the *probabilistic contraction estimates* of this chapter the natural metric is the L^2-in-type, Wasserstein-2 distance $\mathbf{d}_2$ below: it is the metric in which the per-type SDE/BSDE stability of chapter 5 integrates cleanly over types and the coupling operator $\mathbb{W}$ acts as a bounded operator. The *analytic existence theory* of chapter 24, by contrast, finds it more convenient to work with the uniform-in-type (supremum over x) Wasserstein metric, in which type-direction compactness and equicontinuity are read off directly from continuity of the kernel; we flag the difference where it bites (theorem 23.17 and its pointers to chapter 24). Both are metrics on the *same* field of measures of chapter 21; neither redefines the object, and the two are reconciled by the two-pictures dictionary of section 23.5. The designation "Bochner" is by analogy, and the analogy is worth unpacking because it tells us exactly which Bochner-space reflexes are available and which are not. A genuine Bochner space $L^2([0,1];E)$ is the space of square-integrable maps from the type interval into a *Banach* space E; one measures a map by integrating the E-norm of its values, and the result inherits E's linear structure — one can add two fields, scale them, and the norm is a genuine norm. Here the target is $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, which is *not* a vector space: there is no sensible way to add two probability measures and stay a probability measure, and $\mathcal{W}_2$ is a metric, not a norm descended from a linear structure. What survives the analogy is exactly the part we need. A map is square-integrable in the same sense — we integrate the squared *distance* $\mathcal{W}_2(m^x, \delta_0)^2$ of its value from a fixed reference point in place of the squared norm — and the construction $\big(\int_0^1 \mathcal{W}_2(\cdot,\cdot)^2 \,\mathrm{d}x\big)^{1/2}$ of eq. (23.7) is the metric-space analogue of the $L^2([0,1];E)$ norm. Crucially, *completeness* is a metric notion and transfers: an L^2-integral of a complete metric space over a finite measure space is again complete, by the same Riesz–Fischer argument that makes $L^2([0,1];E)$ complete, with Cauchy-in-norm replaced by Cauchy-in-distance. That is all the fixed-point machinery asks of the space, and it is what lemma 23.10 verifies. What we deliberately do *not* get — and never use — is a linear or normed structure on the fields themselves; the operator $\mathbb{W}$ is linear on the underlying *measures*, which is a separate and weaker fact that the contraction proof below exploits with care.

Definition 23.8 (The field space $\boldsymbol{\mathcal{P}}_2$)**.** Fix a reference point $\delta_0 \in \mathcal{P}_2(\mathbb{R}^d)$ (the Dirac mass at the origin). The *field space* is

$$\boldsymbol{\mathcal{P}}_2 \;:=\; \Big\{ \boldsymbol{m} = (m^x)_{x\in[0,1]} \text{ measurable into } (\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2) \;:\; \|\boldsymbol{m}\|_2^2 := \int_0^1 \mathcal{W}_2(m^x, \delta_0)^2 \,\mathrm{d}x < \infty \Big\},$$

equipped with the distance

$$\mathbf{d}_2(\boldsymbol{m}, \boldsymbol{m}') \;=\; \Big(\int_0^1 \mathcal{W}_2\big(m^x, m^{x\prime}\big)^2 \,\mathrm{d}x \Big)^{1/2}. \tag{23.7}$$

The flow space is $C([0,T]; \boldsymbol{\mathcal{P}}_2)$ with the uniform-in-time distance $\mathbf{d}_{2,\infty}(\boldsymbol{m}_\cdot, \boldsymbol{m}'_\cdot) = \sup_{t \le T} \mathbf{d}_2(\boldsymbol{m}_t, \boldsymbol{m}'_t)$.

With the field space and its metric fixed, we turn to the object whose Lipschitz behaviour we actually need to control: the aggregate $\mathbb{W}\boldsymbol{m}$. Here a small but genuine obstruction appears, and it pays to meet it head-on rather than to wave it away by renormalizing. The graphon aggregate $(\mathbb{W}\boldsymbol{m})(x)$ of definition 23.2 is, in general, a *sub-probability* measure of mass $(\mathbb{W}\mathbf{1})(x) := \int_0^1 W(x,y)\,\mathrm{d}y \le 1$, not a probability measure — this is the missing-mass phenomenon dwelt on after definition 23.2, where an agent weakly embedded in the network genuinely feels less than a full unit of neighbour influence. The metric $\mathbf{d}_2$ was built from $\mathcal{W}_2$, and $\mathcal{W}_2$ as defined in the book (definition 4.2 and theorem 4.7) lives only on the probability measures $\mathcal{P}_2(\mathbb{R}^d)$. So before we can even *write* the quantity $\mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)$ that appears in the field-contraction bound eq. (23.11), we must say what $\mathcal{W}_2$ means between two measures of total mass $(\mathbb{W}\mathbf{1})(x) < 1$. We extend it minimally — only to the one case we will ever use, two finite measures of *equal* mass — and we resist the temptation to instead normalize the aggregate to a probability measure, both because the unnormalized mass carries information (again, definition 23.2) and because, as we will see, the common mass is exactly the bookkeeping that makes the equal-mass extension legitimate in the first place.

Definition 23.9 (Quadratic transport cost between equal-mass measures)**.** For two finite Borel measures μ, ν on $\mathbb{R}^d$ of a common total mass $c > 0$ and finite second moment, write $\mu = c\hat{\mu}$, $\nu = c\hat{\nu}$ with $\hat{\mu}, \hat{\nu} \in \mathcal{P}_2(\mathbb{R}^d)$ and set

$$\mathcal{W}_2(\mu,\nu)^2 \;:=\; c\,\mathcal{W}_2(\hat{\mu},\hat{\nu})^2 \;=\; \inf_{\pi} \int_{\mathbb{R}^d\times\mathbb{R}^d} \left|\xi - \xi'\right|^2 \,\pi(\mathrm{d}\xi, \mathrm{d}\xi'), \tag{23.8}$$

the infimum over nonnegative measures π of total mass c with marginals μ, ν. By the Kantorovich duality theorem 4.5, applied to $(\hat{\mu}, \hat{\nu})$ and scaled by c,

$$\mathcal{W}_2(\mu,\nu)^2 \;=\; \sup\Big\{ \textstyle\int \varphi \,\mathrm{d}\mu + \int \psi \,\mathrm{d}\nu \;:\; \varphi(\xi) + \psi(\xi') \le |\xi - \xi'|^2 \;\; \forall\, \xi, \xi' \Big\}. \tag{23.9}$$

For $c = 1$ this is the usual $\mathcal{W}_2$; the extension is used only for the aggregates, whose two arguments below always share the mass $(\mathbb{W}\mathbf{1})(x)$.

Two features of definition 23.9 are worth pausing on. First, the scaling $\mathcal{W}_2(\mu,\nu)^2 = c\,\mathcal{W}_2(\hat{\mu},\hat{\nu})^2$ is forced, not chosen: a coupling of μ and ν has total mass c, so it is c times a coupling of the *normalized* pair $\hat{\mu}, \hat{\nu}$, and the quadratic cost integrated against it picks up exactly one factor of c. The same factor c multiplies the Kantorovich dual eq. (23.9): scaling the optimal-transport identity of theorem 4.5 for $(\hat{\mu}, \hat{\nu})$ by c pulls the constant through both the primal and the dual side, so the dual potentials (φ, ψ) are tested against the *unnormalized* measures μ, ν with no further rescaling. It is this dual form, with its potentials independent of the measures, that will carry the contraction proof. Second — and this is the observation that makes the whole extension usable rather than a curiosity — the equal-mass hypothesis is automatically satisfied by the only pairs we feed into it. For *any* field $\boldsymbol{m}$ the aggregate $(\mathbb{W}\boldsymbol{m})(x)$ has total mass

$$(\mathbb{W}\boldsymbol{m})(x)(\mathbb{R}^d) = \int_0^1 W(x,y)\, m^y(\mathbb{R}^d)\ \mathrm{d}y = \int_0^1 W(x,y)\cdot 1\ \mathrm{d}y = (\mathbb{W}\mathbf{1})(x),$$

because each component m^y is a probability measure and so contributes mass 1. The mass of the aggregate therefore depends *only* on the kernel row at type x and *not at all* on which field $\boldsymbol{m}$ produced it. Consequently any two aggregates $(\mathbb{W}\boldsymbol{m})(x)$ and $(\mathbb{W}\boldsymbol{m}')(x)$ — coming from two arbitrary fields $\boldsymbol{m}, \boldsymbol{m}'$ — carry the identical mass $(\mathbb{W}\mathbf{1})(x)$, the equal-mass hypothesis of definition 23.9 holds at every type and for every pair of fields, and the distance $\mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)$ is well defined throughout. We can now state the boundedness of the coupling, with the *correct* constant.

Lemma 23.10 (The field space is complete; the aggregate is a field contraction)**.** $(\boldsymbol{\mathcal{P}}_2, \mathbf{d}_2)$ *is a complete metric space. Under **(A-Reg-W)** the graphon aggregate is Lipschitz in the field, with constant the square root of the kernel* row-sum bound

$$\|\mathbb{W}\mathbf{1}\|_\infty \;=\; \operatorname*{ess\,sup}_{x\in[0,1]} \int_0^1 W(x,y)\ \mathrm{d}y \;\le\; 1 \qquad (\textit{by } \textbf{\textit{(A-Reg-W)}}), \tag{23.10}$$

namely

$$\Big(\int_0^1 \mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)^2\ \mathrm{d}x\Big)^{1/2} \;\le\; \|\mathbb{W}\mathbf{1}\|_\infty^{1/2}\ \mathbf{d}_2(\boldsymbol{m},\boldsymbol{m}') \;\le\; \mathbf{d}_2(\boldsymbol{m},\boldsymbol{m}'). \tag{23.11}$$

The aggregate is thus a (weak) contraction of the field space, with constant ≤ 1 *regardless of the strength of the kernel.*

Proof. Completeness. This is the Riesz–Fischer argument that proves $L^2([0,1];E)$ complete, run with the distance $\mathcal{W}_2$ in place of the norm of E and with the completeness of $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ standing in for that of E. Let $(\boldsymbol{m}_n)_n$ be $\mathbf{d}_2$-Cauchy in $\boldsymbol{\mathcal{P}}_2$. The first move is to extract a rapidly convergent subsequence: choose indices $n_1 < n_2 < \cdots$ with

$$\mathbf{d}_2(\boldsymbol{m}_{n_{k+1}}, \boldsymbol{m}_{n_k}) \le 2^{-k} \qquad (k\ge 1),$$

possible because the sequence is Cauchy. Now control the total displacement type by type. Set $S_K(y) = \sum_{k=1}^K \mathcal{W}_2(m^y_{n_{k+1}}, m^y_{n_k})$, a nondecreasing sequence of nonnegative measurable functions of y. By Minkowski's inequality in $L^2([0,1])$ (the triangle inequality for the L^2-in-type norm) followed by the subsequence bound,

$$\big\|S_K\big\|_{L^2([0,1])} \le \sum_{k=1}^K \big\|\mathcal{W}_2(m^{\cdot}_{n_{k+1}}, m^{\cdot}_{n_k})\big\|_{L^2([0,1])} = \sum_{k=1}^K \mathbf{d}_2(\boldsymbol{m}_{n_{k+1}}, \boldsymbol{m}_{n_k}) \le \sum_{k=1}^K 2^{-k} \le 1.$$

Letting $K\to\infty$, monotone convergence gives $\|S_\infty\|_{L^2([0,1])} \le 1$, so the limit function $S_\infty(y) = \sum_{k\ge1} \mathcal{W}_2(m^y_{n_{k+1}}, m^y_{n_k})$ is finite for a.e. y. Fix such a y. Then the series of $\mathcal{W}_2$-increments converges, so the sequence $(m^y_{n_k})_k$ is $\mathcal{W}_2$-Cauchy in $\mathcal{P}_2(\mathbb{R}^d)$: for $j>i$, $\mathcal{W}_2(m^y_{n_j}, m^y_{n_i}) \le \sum_{k\ge i} \mathcal{W}_2(m^y_{n_{k+1}}, m^y_{n_k}) \to 0$. Because $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is *complete* (it is Polish, theorem 4.8), this Cauchy sequence converges to a limit, which we name m^y; on the null set where $S_\infty = \infty$ we set $m^y = \delta_0$. The field $\boldsymbol{m} = (m^y)_y$ is measurable into $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, being the a.e. pointwise $\mathcal{W}_2$-limit of the measurable maps $y\mapsto m^y_{n_k}$.

It remains to see that $\boldsymbol{m} \in \boldsymbol{\mathcal{P}}_2$ and that $\boldsymbol{m}_{n_k} \to \boldsymbol{m}$ in $\mathbf{d}_2$. Here Fatou's lemma in the type variable does the work. For each fixed k, $\mathcal{W}_2(m^y, m^y_{n_k})^2 = \lim_{j\to\infty} \mathcal{W}_2(m^y_{n_j}, m^y_{n_k})^2$ for a.e. y (the inner limit by continuity of $\mathcal{W}_2$ in its first argument), so

$$\mathbf{d}_2(\boldsymbol{m}, \boldsymbol{m}_{n_k})^2 = \int_0^1 \mathcal{W}_2(m^y, m^y_{n_k})^2 \,\mathrm{d}y \le \liminf_{j\to\infty} \int_0^1 \mathcal{W}_2(m^y_{n_j}, m^y_{n_k})^2 \,\mathrm{d}y = \liminf_{j\to\infty} \mathbf{d}_2(\boldsymbol{m}_{n_j}, \boldsymbol{m}_{n_k})^2.$$

Given $\varepsilon > 0$, the Cauchy property makes the right side $\le \varepsilon$ once k is large; hence $\mathbf{d}_2(\boldsymbol{m}, \boldsymbol{m}_{n_k}) \to 0$. In particular $\mathbf{d}_2(\boldsymbol{m}, \delta_0\text{-field}) \le \mathbf{d}_2(\boldsymbol{m}, \boldsymbol{m}_{n_k}) + \|\boldsymbol{m}_{n_k}\|_2 < \infty$ by the triangle inequality, so $\boldsymbol{m} \in \boldsymbol{\mathcal{P}}_2$. Finally a Cauchy sequence with a convergent subsequence converges to the same limit, so the *whole* sequence $\boldsymbol{m}_n \to \boldsymbol{m}$. This proves $(\boldsymbol{\mathcal{P}}_2, \mathbf{d}_2)$ complete.

The contraction bound, by duality. Fix a type x and abbreviate the per-type transport cost by

$$g(y) \;:=\; \mathcal{W}_2(m^y, m^{y\prime})^2 \;\ge\; 0,$$

a measurable, integrable function of y: measurable because $\boldsymbol{m}, \boldsymbol{m}'$ are measurable fields and $\mathcal{W}_2$ is jointly continuous, and integrable because the elementary bound $g(y) \le 2\mathcal{W}_2(m^y, \delta_0)^2 + 2\mathcal{W}_2(m^{y\prime}, \delta_0)^2$ exhibits it as dominated by the (finite) sum of the two field norms, with in fact $\int_0^1 g \,\mathrm{d}y = \mathbf{d}_2(\boldsymbol{m}, \boldsymbol{m}')^2$. The naive route to bounding $\mathcal{W}_2((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x))$ would be the *primal* one: pick, for each y, an optimal coupling π_y of m^y and $m^{y\prime}$, superpose them weighted by $W(x,\cdot)$ to build a coupling of the two aggregates, and integrate the cost. This works in principle but is technically delicate: it requires choosing the family $(\pi_y)_y$ *measurably* in y before one may integrate it, an appeal to a measurable selection theorem we would rather not make. The *dual* route sidesteps the selection entirely, and this is the one structural subtlety worth dwelling on.

The aggregates $(\mathbb{W}\boldsymbol{m})(x)$ and $(\mathbb{W}\boldsymbol{m}')(x)$ share the common mass $(\mathbb{W}\mathbf{1})(x)$ (the field-independence established above), so definition 23.9 applies and the dual representation eq. (23.9) is available. Take *any* single admissible pair of potentials (φ, ψ) — admissibility being the condition $\varphi(\xi) + \psi(\xi') \le |\xi - \xi'|^2$ for all ξ, ξ', a constraint on the two *functions* alone that mentions no measure whatsoever. Test this one pair against the aggregates and unfold the definition eq. (23.1) of the aggregate as a W-weighted integral of the components:

$$\begin{aligned}\int \varphi \,\mathrm{d}(\mathbb{W}\boldsymbol{m})(x) + \int \psi \,\mathrm{d}(\mathbb{W}\boldsymbol{m}')(x) &= \int_0^1 W(x,y)\Big[\int \varphi \,\mathrm{d}m^y + \int \psi \,\mathrm{d}m^{y\prime}\Big]\,\mathrm{d}y \\ &\le \int_0^1 W(x,y)\,\mathcal{W}_2(m^y, m^{y\prime})^2 \,\mathrm{d}y \;=\; (\mathbb{W}g)(x).\end{aligned}$$

The inequality is the whole point. *The same pair* (φ, ψ)*, being admissible as a pair of functions, is simultaneously admissible in the dual problem for every component pair* $(m^y, m^{y\prime})$, so for each y the inner bracket is bounded by the duality estimate $\int \varphi \,\mathrm{d}m^y + \int \psi \,\mathrm{d}m^{y\prime} \le \mathcal{W}_2(m^y, m^{y\prime})^2$ of theorem 4.5 — no per-y choice of *coupling* is ever made, only the single global pair of potentials is reused inside the integral. The right-hand side $(\mathbb{W}g)(x)$ is now a number that does not depend on (φ, ψ) at all. So we may take the supremum of the left-hand side over all admissible pairs; by eq. (23.9) that supremum *equals* $\mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)^2$, and the inequality is preserved under the supremum, giving the pointwise estimate

$$\mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)^2 \;\le\; (\mathbb{W}g)(x) \qquad \text{for a.e. } x.$$

In words: the squared transport cost between two aggregates at type x is at most the W-weighted average, over neighbours y, of the squared transport costs of the underlying components. This route needs *no* measurable selection of optimal couplings: the common potential pair carries the bound across all of y at once.

The final step integrates this pointwise estimate over x and is where the row-sum constant is born. The graphon kernel is *symmetric*, $W(x,y) = W(y,x)$, so $\mathbb{W}$ is self-adjoint on $L^2([0,1])$, $\langle f, \mathbb{W}h\rangle = \langle \mathbb{W}f, h\rangle$; applying this with $f = \mathbf{1}$ and $h = g$ moves the operator off the (rough) function g and onto the constant $\mathbf{1}$, where it becomes the explicit row-sum function $\mathbb{W}\mathbf{1}$:

$$\int_0^1 \mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)^2 \,\mathrm{d}x \;\le\; \int_0^1 (\mathbb{W}g)(x)\,\mathrm{d}x \;=\; \langle \mathbf{1}, \mathbb{W}g\rangle \;=\; \langle \mathbb{W}\mathbf{1}, g\rangle \;\le\; \|\mathbb{W}\mathbf{1}\|_\infty \,\|g\|_{L^1} \;=\; \|\mathbb{W}\mathbf{1}\|_\infty \,\mathbf{d}_2(\boldsymbol{m}$$

The penultimate inequality is the Hölder pairing of L^∞ against L^1: $\langle \mathbb{W}\mathbf{1}, g\rangle = \int_0^1 (\mathbb{W}\mathbf{1})(x)\, g(x)\,\mathrm{d}x \le \big(\operatorname{ess\,sup}_x (\mathbb{W}\mathbf{1})(x)\big) \int_0^1 g\,\mathrm{d}x$, using $g \ge 0$. This is eq. (23.11); taking square roots gives the contraction constant $\|\mathbb{W}\mathbf{1}\|_\infty^{1/2}$, and the final ≤ 1 is $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$ from **(A-Reg-W)**. Note precisely where the choice of *which* norm of $\mathbb{W}$ to use was forced: the rough function g lives only in $L^1([0,1])$ — it is a sum of squared Wasserstein distances, integrable but with no control on its L^2 size — so the only norm of $\mathbb{W}$ that can be paired against it is the one dual to L^1, namely the $L^\infty \to L^\infty$ row-sum norm. The next remark makes this forced choice the conceptual centre of the section. □

Remark 23.11 (Why the row-sum, and not the operator norm, for the measure-valued aggregate)**.** This remark is the conceptual hinge of the section: it isolates which of the *two* norms of $\mathbb{W}$ governs which of the *two* couplings in the graphon game. The distinction is not pedantry — it is exactly the distinction that, in section 23.6, separates the contraction estimate (which never fails) from the symmetry-breaking threshold (which is where uniqueness fails). It is cashed out, not repeated, there, and the regime statements of theorem 23.17 turn on the same point; the reader should carry the dichotomy forward rather than expect it re-derived.

The honest contraction constant for the *measure-valued* aggregate, just proved, is the row-sum bound $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$. This number is the $L^\infty([0,1]) \to L^\infty([0,1])$ operator norm of $\mathbb{W}$ — the largest row sum — and it is emphatically *not* the $L^2([0,1]) \to L^2([0,1])$ operator (spectral) norm $\|\mathbb{W}\| = |\lambda_1|$, the top eigenvalue. One might hope to replace the last line of the proof by the spectral norm, writing "$\langle \mathbb{W}\mathbf{1}, g\rangle \le \|\mathbb{W}\|\;\|g\|_{L^1}$"; this is simply *false*, because the spectral norm controls L^2 pairings and g is only an L^1 function. To see that it is genuinely false — that the spectral constant is not merely unproven but actually too *small* — consider the cleanest possible test case and follow it slowly.

The counterexample. Take the constant kernel $W(x,y) \equiv c$ with $0 < c < 1$. Its row sum is $(\mathbb{W}\mathbf{1})(x) = \int_0^1 c\,\mathrm{d}y = c$ for every x, so $\|\mathbb{W}\mathbf{1}\|_\infty = c$. Its spectrum is equally explicit: $\mathbb{W}f = c\,\langle \mathbf{1}, f\rangle\,\mathbf{1}$ is rank one with the single nonzero eigenvalue $\lambda_1 = c$ on the constant eigenfunction, so the spectral norm is $\|\mathbb{W}\| = |\lambda_1| = c$ as well — the two norms happen to *coincide* here, both equal to c, which is what makes the example so sharp. Now perturb a field by a rigid common translation: let $m^{y\prime} = m^y * \delta_\epsilon$ be every component of $\boldsymbol{m}$ pushed forward by the same shift $\epsilon \in \mathbb{R}^d$ (convolution with a Dirac at ϵ). Translating a measure by ϵ moves it $\mathcal{W}_2$-distance exactly $|\epsilon|$, so $\mathcal{W}_2(m^y, m^{y\prime})^2 = |\epsilon|^2$ for every y, and hence

$$\mathbf{d}_2(\boldsymbol{m}, \boldsymbol{m}')^2 = \int_0^1 |\epsilon|^2 \,\mathrm{d}y = |\epsilon|^2 .$$

What does this perturbation do to the aggregate? Because the kernel is constant, the aggregate $(\mathbb{W}\boldsymbol{m})(x) = c\int_0^1 m^y \, \mathrm{d}y$ is c times the type-averaged population, and the shifted aggregate is the same average translated by ϵ; the two aggregates therefore differ by the same rigid shift, now carried by a measure of mass c. By the equal-mass distance eq. (23.8) the squared cost is the mass times the squared shift of the normalized measures, $\mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)^2 = c\,|\epsilon|^2$, so the left-hand side of the field-contraction bound is

$$\int_0^1 \mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)^2 \, \mathrm{d}x \;=\; c\,|\epsilon|^2 \,.$$

Compare the two candidate right-hand sides. The row-sum bound predicts $\|\mathbb{W}\mathbf{1}\|_\infty \, \mathbf{d}_2^2 = c\,|\epsilon|^2$ — an exact match, so the row-sum constant is not just valid but *sharp* on this example. The spectral bound would predict $\|\mathbb{W}\|^2 \, \mathbf{d}_2^2 = c^2\,|\epsilon|^2$, and since $c < 1$,

$$c^2\,|\epsilon|^2 \;<\; c\,|\epsilon|^2 \,,$$

so the spectral candidate is *strictly smaller than the true left-hand side*: it is violated, not merely loose. The squaring is the trap — the contraction constant on the squared cost is the row sum c, whereas the spectral norm enters squared as c^2, and for $c < 1$ squaring strictly shrinks. This is the concrete proof that $\|\mathbb{W}\|$ is the *wrong* (too small) constant for the measure-valued contraction, exactly as the L^1 nature of g in the proof foretold.

Why the row-sum is always at least as large. The example is no accident. For a *nonnegative* kernel — which a graphon always is — the Perron–Frobenius principle gives $\|\mathbb{W}\| = |\lambda_1| \le \|\mathbb{W}\mathbf{1}\|_\infty$: the top eigenvalue of a nonnegative operator never exceeds its largest row sum. (Quickly: if $\mathbb{W}\phi = \lambda_1\phi$ with $\phi \ge 0$ the Perron eigenfunction and $x^\star$ a point where ϕ is near its maximum, then $\lambda_1\phi(x^\star) = \int W(x^\star, y)\phi(y)\,\mathrm{d}y \le \phi(x^\star)\int W(x^\star, y)\,\mathrm{d}y \le \phi(x^\star)\,\|\mathbb{W}\mathbf{1}\|_\infty$.) So the row-sum constant is *always* valid, even when not sharp, and it collapses to the spectral norm exactly in the rank-one constant case above.

The other coupling, where $\|\mathbb{W}\|$ *is right.* The spectral norm is not idle — it governs a *different* cross-type coupling, the *linear* mean-field coupling $\bar{m} \mapsto \mathbb{W}\bar{m}$ that acts not on the full measure field but on the type-L^2 space of population *means* $\bar{m}(x) = \int \xi\, m^x(\mathrm{d}\xi)$. There the relevant estimate is the genuine L^2 bound $\|\mathbb{W}\bar{m} - \mathbb{W}\bar{m}'\|_{L^2} \le \|\mathbb{W}\|\;\|\bar{m} - \bar{m}'\|_{L^2}$, which *is* the correct and sharp one and is in the book as the spectral theorem theorem 2.26. This is the coupling that drives the linear–quadratic thread, where the agent reacts only to neighbour means; and it is precisely there, in section 23.6, that diagonalizing $\mathbb{W}$ turns each eigenvalue λ_k into the effective coupling strength $s\lambda_k$ of an independent scalar mode, with the largest, $s\,\|\mathbb{W}\| = s\,|\lambda_1|$, setting the symmetry-breaking threshold in closed form. The two norms thus partition the chapter's labour cleanly. The row-sum $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$ makes the *general*, measure-valued coupling a bounded-by-one perturbation of an otherwise decoupled family, so that the contraction underlying well-posedness is governed by the horizon T and not by how strong the kernel is; the spectral norm $\|\mathbb{W}\|$ governs the *linear* coupling and is where multiplicity is born. Keeping these two roles distinct is the single most important thing to carry out of this section into section 23.6 and chapter 24.

With the field space built, shown complete, and equipped with the master contraction estimate lemma 23.10, the two promissory notes of proposition 23.5 are paid: the metric $\mathbf{d}_2$ exists and the aggregate $\mathbb{W}$ contracts it with a constant $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$ that is *uniform over every admissible graphon*. This uniformity is exactly what section 23.3 will consume — its small-horizon contraction must hold for all kernels at once, and lemma 23.10 guarantees the type-integration never amplifies the per-type estimate regardless of coupling strength. We have also planted, in remark 23.11, the two-norm distinction that the regime statement theorem 23.17 and the spectral threshold $s\,\|\mathbb{W}\|$ of section 23.6 will both cash out. One seam remains flagged for later: the existence theory of chapter 24 prefers the uniform-in-type metric to $\mathbf{d}_2$, and the reconciliation of the two runs through the two-pictures dictionary of section 23.5. For now the home is built and complete, and we can move the agents back in: section 23.3 restores the optimization that section 23.1 stripped away, type by type, and watches that same operator reappear, unchanged, inside the equilibrium fixed point.

23.3 The representative agent and the graphon FBSDE

The two previous sections were a deliberate detour through the *descriptive* world. In section 23.1 we wrote down the graphon McKean–Vlasov SDE with optimization switched off, precisely so that the one genuinely new ingredient — the nonlocal cross-type coupling carried by $\mathbb{W}$ — could be isolated and studied alone; and in section 23.2 we built the functional home for the resulting field of laws and proved the master estimate, that $\mathbb{W}$ contracts that field space with constant the row-sum bound $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$ (lemma 23.10). With the field space $\boldsymbol{\mathcal{P}}_2$ and its contraction in hand, we can now do what those sections deliberately postponed: *restore the optimization* that turns a descriptive population into an equilibrium one. The drift of each agent will no longer be handed to us; it will be *chosen*, optimally, by the agent.

The logic for restoring it is not new — it is exactly the two-step fixed point of chapter 9, the same construction that produced the homogeneous MFG FBSDE eq. (9.13), only now run *type by type* along the continuum. Recall how that construction goes. One does not attack the coupled equilibrium head-on; one breaks the circularity in two. First *freeze* a candidate population — here, a candidate flow of fields $\boldsymbol{\mu}$ — so that each agent faces a fixed, external environment and its optimization becomes a plain single-agent control problem. Solve that problem (this is where the stochastic maximum principle enters) to obtain the agent's best response and the law it generates. Then *close the loop* by demanding self-consistency: the population the agents collectively generate must be the very population they were optimizing against. The graphon game inherits this verbatim, with one bookkeeping change that the previous two sections have already absorbed — "the population" is now a field indexed by type, and the environment a type-x agent faces is the neighbour aggregate $(\mathbb{W}\boldsymbol{\mu}_t)(x)$ rather than a single global law. We carry out the two steps in order: this subsection freezes and optimizes one type; section 23.3's next subsection closes the loop and writes down the central object, the graphon MFG FBSDE; the third exhibits the equilibrium as a fixed point on the field space.

23.3.1 The type-x control problem against a frozen field

We begin with the freezing step, and it is worth being explicit about why freezing is the right first move. The difficulty in the graphon game is entirely the circular dependence — each agent's optimal behaviour depends on the population, which is itself the aggregate of all agents' optimal behaviour. Freezing the population to a fixed candidate $\boldsymbol{\mu}$ cuts that circle: with the environment held external and known, the type-x agent no longer interacts with anyone, and its problem collapses to a textbook stochastic control problem of the kind chapter 9 solved completely. The whole subtlety of the continuum — the nonlocal coupling — has been parked inside the frozen object $(\mathbb{W}\boldsymbol{\mu}_t)(x)$, which is now just a given deterministic measure-flow. So fix a flow of fields $\boldsymbol{\mu} = (\boldsymbol{\mu}_t)_{t\le T} \in C([0,T];\boldsymbol{\mathcal{P}}_2)$, regarded as the frozen population. The type-x agent feels the deterministic neighbour aggregate $\nu_t^x := (\mathbb{W}\boldsymbol{\mu}_t)(x)$ and controls

$$\mathrm{d}X_t^x = b\big(t,x,X_t^x,\nu_t^x,\alpha_t^x\big)\ \mathrm{d}t + \sigma\big(t,x,X_t^x,\nu_t^x\big)\ \mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x, \tag{23.12}$$

through an admissible control $\alpha^x \in \mathbb{A}$ (the class of chapter 9: $\mathbb{F}$-progressive, square-integrable, valued in the closed convex set A), to minimize

$$J_{\boldsymbol{\mu}}^x(\alpha^x) = \mathbb{E}\bigg[\int_0^T L\big(t,x,X_t^x,\nu_t^x,\alpha_t^x\big)\ \mathrm{d}t + g\big(x,X_T^x,\nu_T^x\big)\bigg]. \tag{23.13}$$

As in chapter 9 the volatility is uncontrolled. Now comes the decisive observation, the one that makes the whole graphon control theory free. Because $\boldsymbol{\mu}$ was frozen, $\nu_t^x = (\mathbb{W}\boldsymbol{\mu}_t)(x)$ is a *fixed, deterministic* measure-flow: it references neither the agent's own law nor any other type's processes, only the known input $\boldsymbol{\mu}$. The cross-type coupling has been completely severed — nothing in eq. (23.12)–eq. (23.13) connects type x to any type $y \neq x$. What remains is therefore an *ordinary single-agent stochastic control problem*: a controlled diffusion eq. (23.12) with a running-plus-terminal cost eq. (23.13), in which the measure ν_t^x and the type label x enter only as fixed parameters of the coefficients, exactly as a time-dependent coefficient would. This is *precisely* the problem class of section 9.1, with the population law m_t used there replaced everywhere by the neighbour aggregate ν_t^x and an inert label x carried passively along. Nothing about the maximum-principle machinery sees the continuum: from inside the type-x problem, x is a constant and ν^x is a given function of time. This is the concrete payoff of the "one operator carries all the coupling" principle at the level of optimization: the operator $\mathbb{W}$ is the *only* thing that ties the types together, and freezing the field has temporarily set it aside, so each type optimizes in complete isolation.

Because the problem is an ordinary control problem, we solve it with the ordinary tool: the Pontryagin stochastic maximum principle of chapter 9. Recall its shape. One forms the generalized Hamiltonian, the scalar that the optimal control minimizes pointwise; its gradient in the state drives a *backward* equation for the adjoint (costate) process, which measures the marginal cost of the state; and optimality is the statement that the control minimizes the Hamiltonian instant by instant. We write each ingredient for the type-x problem, importing the definitions of chapter 9 with the substitution $m_t \rightsquigarrow \nu_t^x$ already made.

Definition 23.12 (Type-x generalized Hamiltonian and adjoint)**.** The type-x generalized Hamiltonian is, in the notation of definition 9.2,

$$\mathcal{H}\big(t,x,x,\nu,y,z,\alpha\big) = \langle b(t,x,x,\nu,\alpha), y\rangle + \mathrm{tr}\big(\sigma(t,x,x,\nu)^\top z\big) + L(t,x,x,\nu,\alpha), \tag{23.14}$$

with, under **(A-Conv)**, the unique minimizer $\hat{\alpha}(t,x,x,\nu,y,z) = \arg\min_{\alpha\in A}\mathcal{H}$, Lipschitz in (x,ν,y,z) (eq. (9.4)). The adjoint of an admissible α^x with state X^x is the pair $(Y^x,Z^x) \in \mathcal{S}^2 \times \mathcal{H}^2$

solving the backward SDE; for each fixed x this is a standard backward stochastic differential equation with square-integrable, Lipschitz driver and square-integrable terminal datum, so it has a unique solution by the Pardoux–Peng theorem (theorem C.18), the Z^x-component furnished by the martingale representation theorem (theorem C.17) for the Brownian filtration of B^x,

$$\mathrm{d}Y_t^x = -\,\partial_x \mathcal{H}\big(t, x, X_t^x, \nu_t^x, Y_t^x, Z_t^x, \alpha_t^x\big)\,\mathrm{d}t + Z_t^x\ \mathrm{d}B_t^x, \qquad Y_T^x = \partial_x g\big(x, X_T^x, \nu_T^x\big). \tag{23.15}$$

With the Hamiltonian, adjoint, and minimizer feedback in place, the maximum principle of chapter 9 — both the necessary condition (theorem 9.4) and the sufficient condition (theorem 9.5) — holds for the type-x problem without change: under **(A-Lip)**, **(A-Growth)**, **(A-Conv)**, a control is optimal against the frozen field if and only if it satisfies the pointwise minimization $\hat{\alpha}_t^x = \hat{\alpha}(t, x, X_t^x, \nu_t^x, Y_t^x, Z_t^x)$ together with the forward state eq. (23.12) and the backward adjoint eq. (23.15).

We do not reprove this, and it is worth being precise about why no reproof is owed — the reason is the entire content of the freezing step, not a hand-wave. A theorem may be reused verbatim exactly when the object it speaks about is, on the nose, an instance of the object the theorem was proved for. Here the type-x problem against a frozen field *is* such an instance: as established above, once $\boldsymbol{\mu}$ is frozen the measure ν_t^x is a fixed deterministic function of time and x is a fixed parameter, so eq. (23.12)–eq. (23.13) is literally a member of the family of single-agent control problems for which theorem 9.5 was established — with ν_t^x in the role of the (there fixed) measure flow and x a passive label decorating the coefficients. The graphon offers the maximum-principle proof nothing new to chew on: there is no cross-type term to differentiate, no continuum to integrate, nothing the argument of chapter 9 has not already handled. Restated as a slogan, *the frozen-field decoupling is what licenses the verbatim import*; the statement of optimality above is theorem 9.5 read with the parameter x and the frozen measure ν_t^x, and that is all.

One seam must nonetheless be kept visible, because it is exactly where the graphon *game* parts ways from a graphon *planning* problem, and the difference is invisible in the frozen problem but decisive at closure. It is the seam flagged in remark 9.6. The adjoint eq. (23.15) differentiates $\mathcal{H}$ in the state x *only*, holding the neighbour field ν_t^x frozen throughout the backward run. This is correct for the graphon mean field *game*: the type-x agent is one anonymous member of a continuum, a price-taker who cannot perceptibly move the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ by deviating, so when computing its own marginal cost it rightly treats the neighbour field as exogenous — frozen — and differentiates through the state alone. Were we instead solving the graphon mean field *control* problem — a single planner choosing the entire field of laws to optimize a social objective — the planner *would* internalize the effect of every agent's state on the aggregate, and the adjoint would carry an additional Lions (measure) derivative term, propagated from type x to every type y with $W(y, x) > 0$ by the operator $\mathbb{W}$. That extra nonlocal term is the entire difference between the two problems, and it is absent here *by design*: we are constructing the equilibrium of an anonymous game, not the optimum of a planner. The reader should note that this is the only place in the optimization step where the choice "game, not control" is actually exercised; once made, the rest of the construction proceeds identically.

23.3.2 Closing the loop: the graphon MFG FBSDE

The freezing step delivered, for each type and each frozen environment, a complete best response: an optimally controlled state X^x, its adjoint (Y^x, Z^x), and the feedback control $\hat{\alpha}^x$ that the maximum principle certifies optimal. But a best response is computed *against* a population the agent merely assumed; it is an equilibrium only when the assumption is true — when the population the agents collectively generate is the one they each optimized against. This is the second step of the two-step fixed point, and it is a single demand: *self-consistency.* We now impose it.

The demand is the closure $m_t^x = \mathrm{Law}(X_t^x)$, already met in the descriptive setting as eq. (23.4): the field of laws the agents produce must equal the field they reacted to. Imposing it has one immediate effect on the frozen problem of the previous subsection: the external, hand-supplied aggregate $\nu_t^x = (\mathbb{W}\boldsymbol{\mu}_t)(x)$ is replaced by the *self-generated* aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ built from the agents' own realized laws. The frozen parameter becomes a functional of the unknown, the severed cross-type coupling is reconnected through $\mathbb{W}$, and the per-type control problems fuse back into a single coupled system. That system is the central object of the chapter.

Definition 23.13 (The graphon MFG FBSDE)**.** A *graphon mean field game equilibrium in probabilistic form* is a field $(X^x, Y^x, Z^x)_{x\in[0,1]}$ of $\mathbb{F}$-progressive processes, together with the flow of fields $\boldsymbol{m}_t = (\mathrm{Law}(X_t^x))_x$, solving for a.e. x the McKean–Vlasov forward–backward system

$$\begin{cases} \mathrm{d}X_t^x = b\big(t, x, X_t^x, (\mathbb{W}\boldsymbol{m}_t)(x), \hat{\alpha}_t^x\big)\,\mathrm{d}t + \sigma\big(t, x, X_t^x, (\mathbb{W}\boldsymbol{m}_t)(x)\big)\,\mathrm{d}B_t^x, & X_0^x \sim m_0^x, \\ \mathrm{d}Y_t^x = -\,\partial_x\mathcal{H}\big(t, x, X_t^x, (\mathbb{W}\boldsymbol{m}_t)(x), Y_t^x, Z_t^x, \hat{\alpha}_t^x\big)\,\mathrm{d}t + Z_t^x\;\mathrm{d}B_t^x, & Y_T^x = \partial_x g\big(x, X_T^x, (\mathbb{W}\boldsymbol{m}_T)(x)\big), \\ \quad \hat{\alpha}_t^x = \hat{\alpha}\big(t, x, X_t^x, (\mathbb{W}\boldsymbol{m}_t)(x), Y_t^x, Z_t^x\big), & m_t^x = \mathrm{Law}(X_t^x). \end{cases} \tag{23.16}$$

It pays to read eq. (23.16) slowly against its homogeneous ancestor, the MFG FBSDE eq. (9.13), because the comparison shows exactly what is inherited and what is new. Line for line the two systems are the same shape: a forward state equation, a backward adjoint equation with terminal data, a feedback control closing the two, and a law constraint. Set the type label aside and replace $(\mathbb{W}\boldsymbol{m}_t)(x)$ by $\boldsymbol{m}_t$ and eq. (23.16) *is* eq. (9.13). So the graphon FBSDE carries all of the difficulty of the homogeneous one — three nested layers of coupling — and adds a fourth that is the whole subject of this chapter. Naming each in turn, with a word on what it means:

1. *Forward–backward coupling.* The state X^x runs forward from data at $t = 0$ while its adjoint Y^x runs backward from data at $t = T$, yet each appears in the other's equation; the system is a genuine two-point boundary-value problem in time, not an initial-value problem one can simply integrate forward. This is the structural signature of an optimization embedded in dynamics, present already in single-agent control.

2. *McKean–Vlasov (distributional) coupling.* The coefficients depend on $\boldsymbol{m}_t = (\mathrm{Law}(X_t^x))_x$, the *law* of the very process they govern, so the equation is nonlinear in a non-pointwise way: the solution's distribution feeds back into its own dynamics. This is the mean-field closure, inherited from chapter 7.

3. *Implicit feedback.* The control $\hat{\alpha}_t^x$ is not given but is itself the argmin of the Hamiltonian, a function of (X_t^x, Y_t^x, Z_t^x) that must be solved for and substituted back, making the drift an implicit function of the unknowns. This is the maximum-principle closure, inherited from chapter 9.

4. *Nonlocal-in-type coupling (the new difficulty).* The distributional dependence is no longer on the agent's own law but on the aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$, which averages the laws of *all* types against the kernel row $W(x, \cdot)$. Equivalently, by symmetry of the bookkeeping, the law m_t^x at one type enters the dynamics at every type y for which $W(y, x) > 0$. The coupling is now spatial (in type) as well as distributional — a continuum of mean-field problems wired together across the network.

Everything technical in this chapter is the bookkeeping required to keep that fourth, nonlocal layer under control on top of the three classical ones, and the single fact that makes the bookkeeping close is the one established in section 23.2: the new coupling is the *bounded linear operator* $\mathbb{W}$, a field contraction with constant ≤ 1 (lemma 23.10). Because the novelty is one well-behaved operator, every estimate reduces to a classical per-type estimate followed by a single application of that contraction bound — which is exactly how the fixed-point analysis of section 23.4 will proceed.

Remark 23.14 (Per-type decoupling under a frozen field)**.** Here is the structural observation that organizes the entire analysis, and it is just the freezing step of this section read backwards. *Freeze the field* $\boldsymbol{m}_t \equiv \boldsymbol{\mu}_t$ in eq. (23.16) — that is, hold the distributional argument external rather than self-generated — and the continuum of coupled FBSDEs *falls apart into independent single-type FBSDEs*, one for each x. Each of these is a classical (non-McKean–Vlasov) FBSDE of exactly the kind solved in chapter 9, coupled to the others only through the frozen aggregate $(\mathbb{W}\boldsymbol{\mu}_t)(x)$, which is now an inert deterministic input. All four difficulties enumerated above survive individually — each per-type FBSDE is still forward–backward and still has implicit feedback — but two of them, the McKean–Vlasov coupling and the nonlocal-in-type coupling, go dormant: with the field frozen there is no self-reference and no live interaction across types. They re-enter *only* at the closure step, when the frozen field is required to equal the realized one. Closure, in other words, is where the entire continuum-wide difficulty lives; everything else is a parallel family of classical problems.

This is precisely why the natural fixed-point variable is the *field of laws*, not the processes. Were one to seek the equilibrium directly in the space of process-fields $(X^x)_x$, the McKean–Vlasov self-reference would be tangled into every component. Passing to the field of laws untangles it: in the law variable the problem is a fixed point of a single map $\boldsymbol{\mu} \mapsto (\mathrm{Law}(X^{\boldsymbol{\mu},x}))_x$ on $C([0,T]; \boldsymbol{\mathcal{P}}_2)$, and the per-type FBSDEs that *define* that map are individually classical and individually solvable. The unknown is demoted from a coupled continuum of processes to a single point in the field space, and the coupling is quarantined inside the map. This same decoupling-under-freezing is what section 23.5.2 will exploit at the level of the FBSDE–PDE dictionary: freezing the field exposes, type by type, a classical *decoupling field* u^x relating Y^x to X^x, and the family of these is what seeds the graphon master equation of chapter 26.

23.3.3 The fixed point in the field space

remark 23.14 told us where to look for the equilibrium: not among the process-fields, where the coupling is everywhere, but in the field of laws $\boldsymbol{m} \in C([0,T];\boldsymbol{\mathcal{P}}_2)$, where freezing decouples the continuum into classical per-type FBSDEs and the only residual constraint is self-consistency. We now make that observation into a clean fixed-point statement. The recipe is forced by the two-step structure of the whole section: take a candidate field, run the per-type optimizations against it (each classical, by the decoupling), collect the realized laws into a new field, and ask when input equals output. The map that sends candidate to realized field is the object that carries the equilibrium, and naming it is the last thing this section does before handing it to the well-posedness analysis.

Definition 23.15 (Graphon best-response map)**.** The *graphon best-response map* $\mathsf{S} : C([0,T];\boldsymbol{\mathcal{P}}_2) \to C([0,T];\boldsymbol{\mathcal{P}}_2)$ is defined as follows. Given an input flow of fields $\boldsymbol{\mu}$, solve for each x the decoupled FBSDE obtained by replacing $(\mathbb{W}\boldsymbol{m}_t)(x)$ in eq. (23.16) with the frozen aggregate $(\mathbb{W}\boldsymbol{\mu}_t)(x)$ — a classical FBSDE per type, by remark 23.14 — obtain its optimally controlled state $X^{\boldsymbol{\mu},x}$, and output the realized flow of fields

$$\mathsf{S}(\boldsymbol{\mu}) = \big(\operatorname{Law}(X_t^{\boldsymbol{\mu},x})\big)_{x\in[0,1],\,0\le t\le T}. \tag{23.17}$$

A flow $\boldsymbol{m}$ is a graphon MFG equilibrium if and only if it is a fixed point, $\mathsf{S}(\boldsymbol{m}) = \boldsymbol{m}$.

This is the same best-response map as the analytic one of chapter 22 (definition 22.5); it is merely *realized differently*. The analytic version takes the same input field $\boldsymbol{\mu}$, but processes each type through the PDE pair: it solves the type-x Hamilton–Jacobi–Bellman equation against the frozen aggregate $(\mathbb{W}\boldsymbol{\mu}_t)(x)$ to get the value and the optimal feedback, then pushes the initial law forward through the type-x Fokker–Planck equation to read off the realized law $\operatorname{Law}(X_t^{\boldsymbol{\mu},x})$. The probabilistic version above does the identical job through the type-x FBSDE. Same input field, same per-type best response, same output field — only the solver differs, PDE versus FBSDE. That the two realizations of S genuinely agree on every output is not a triviality to be assumed; it is a theorem, the two-pictures dictionary proposition 23.19, which section 23.5 proves by identifying $Y_t^x = \nabla u^x(t, X_t^x)$ type by type. We flag the agreement here so the reader carries it into that section as the thing to be shown.

There is in fact a third realization of the same map, and the shared symbol is deliberate. The existence theory of chapter 24 works not with the full measure-valued aggregate but with its first moment, the neighbour mean $z_t(x)$ of definition 23.2, on the uniform-in-type field space; the resulting graphon-local best-response map is definition 24.4. All three — the measure-valued probabilistic map here, the analytic PDE map of chapter 22, and the first-moment map of chapter 24 — are the same mathematical object viewed through different solvers and different couplings, and we write all three with the symbol S *by design*, so that “the equilibrium is a fixed point of S” reads the same in every chapter.

It remains to record what we mean by a solution, and the integrability conditions in the definition are not decoration — each is exactly what some later step needs, so we motivate them rather than merely list them. Three requirements appear, and each answers a question the construction will ask. *Per type, the processes must live in the standard spaces of chapter 9*: X^x, Y^x square-integrable with continuous paths ($\mathcal{S}^2$) and Z^x square-integrable ($\mathcal{H}^2$), because that is precisely the regularity in which the Pardoux–Peng theorem certifies the type-x BSDE well-posed and in which the per-type maximum principle was stated — without it the adjoint eq. (23.15) need not have a solution at all. *The field must be measurable in x into those spaces*, because the closure forms the field of laws $x \mapsto \mathrm{Law}(X_t^x)$ and integrates over x; an unmeasurable field could not be aggregated by $\mathbb{W}$ and the object $(\mathbb{W}\boldsymbol{m}_t)(x)$ would be meaningless. *Finally the field must be square-integrable in x*, $\int_0^1 \mathbb{E}[\sup_{t\le T}(|X_t^x|^2 + |Y_t^x|^2)]\,\mathrm{d}x < \infty$, because this is exactly the condition that places the flow of laws inside the Bochner field space $\boldsymbol{\mathcal{P}}_2$ of definition 23.8 — the complete metric space on which the contraction estimates run. The first condition is per-type and classical; the second and third are the genuinely field-level conditions, and they are precisely the hypotheses under which the row-sum contraction lemma 23.10 has a space to act on.

Definition 23.16 (Solution of the graphon MFG FBSDE)**.** A *solution* of eq. (23.16) is a measurable field $x \mapsto (X^x, Y^x, Z^x)$ with, for a.e. x, $X^x, Y^x \in \mathcal{S}^2(\mathbb{R}^d)$, $Z^x \in \mathcal{H}^2(\mathbb{R}^{d\times k})$, such that $x \mapsto (X^x, Y^x, Z^x)$ is measurable into those spaces, the integral forms hold $\mathbb{P}$-a.s. with $\hat{\alpha}^x$ as in eq. (23.16), and $\int_0^1 \mathbb{E}[\sup_{t\le T}(|X_t^x|^2 + |Y_t^x|^2)]\,\mathrm{d}x < \infty$. The flow $\boldsymbol{m}_t = (\mathrm{Law}(X_t^x))_x$ then lies in $C([0,T];\boldsymbol{\mathcal{P}}_2)$.

We have now assembled the central object and recast equilibrium in the cleanest possible terms: a graphon MFG equilibrium is a fixed point of the best-response map S on the field space $C([0,T];\boldsymbol{\mathcal{P}}_2)$, and nothing more. The construction has quarantined every continuum-wide difficulty inside that one map — the per-type problems that define it are classical, and the only operator wiring the types together is the bounded $\mathbb{W}$ of lemma 23.10. What we have *not* yet done is ask whether such a fixed point exists, and whether it is unique. That is the next question, and it is now a purely fixed-point-theoretic one: section 23.4 takes the map S and the characterization "equilibrium = fixed point of S" built here and asks, in three regimes, when S has a fixed point and when that fixed point is unique — each regime's answer flowing, as we will see, through the contraction constant $\|\mathbb{W}\mathbf{1}\|_\infty$ that section 23.2 handed forward. We carry the "same map, two realizations" observation along with it: it is the debt that section 23.5 will discharge, proving the probabilistic S here and the analytic S of chapter 22 produce the same output field.

23.4 Fixed-point analysis: three regimes

section 23.3 reduced the whole graphon equilibrium problem to a single demand: that the best-response map S of definition 23.15 have a fixed point on the field space $C([0,T];\boldsymbol{\mathcal{P}}_2)$, with every continuum-wide difficulty quarantined inside that one map and the types wired together only through the bounded operator $\mathbb{W}$. The question left open there is the one this section takes up: *when does* S *have a fixed point, and when is that fixed point unique?* It is now a purely fixed-point-theoretic question, and the shape of its answer is dictated by the spine of the chapter. Because the types communicate only through $\mathbb{W}$, every estimate that could decide existence or uniqueness inherits the kernel in exactly one way — through the field-contraction constant $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$ that section 23.2 extracted in lemma 23.10. The analysis therefore mirrors the homogeneous theorem 9.10, with $\mathbb{W}$ entering every bound through that one constant in the general (measure-valued) estimates and — once the coupling is taken linear in the population mean — through the spectral norm $\|\mathbb{W}\|$, as section 23.6 will show. We state the result, develop the mechanism of each of three regimes, and make the role of the operator explicit; the full proofs are the content of chapter 24, which owns the existence and uniqueness theory for the graphon game exactly as chapters 10 and 11 own it for the homogeneous one.

We must be honest about the seam, because the division of labour between this section and chapter 24 is subtler than a plain deferral. chapter 24 carries out the existence/uniqueness analysis in the *uniform-in-type* field space $C([0,1]; C([0,T]; \mathcal{P}_1(\mathbb{R}^d)))$, metrized by the supremum over types of the Wasserstein distance (definition 24.1); that is the space in which type-direction compactness and equicontinuity are read off directly from continuity of the kernel. The FBSDE statement below, by contrast, is set in the L^2-in-type space $\boldsymbol{\mathcal{P}}_2$ of definition 23.8, the space in which the per-type stability estimates of chapters 5 and 9 integrate over types and $\mathbb{W}$ acts as the bounded operator of lemma 23.10. The two metrics are *not* equivalent — the supremum-in-type distance is strictly finer than the L^2-in-type distance, and no inequality runs the other way — so one cannot transport a fixed point from one space to the other by a topological comparison of the two field metrics. What reconciles them is therefore not a metric estimate but the two-pictures dictionary proposition 23.19: it characterizes the equilibrium as a single measurable field of laws through the relation $Y_t^x = \nabla u^x(t, X_t^x)$, a relation that mentions *no* field metric at all. Both the uniform-in-type analysis of chapter 24 and the L^2-in-type statement here are statements about fixed points of the *same* map S, in its analytic and probabilistic realizations; proposition 23.19 shows those realizations agree type by type, so an equilibrium certified to exist (or to be unique) in the uniform-in-type space is, as a measurable field, the very object this section's statement is about. The metric was only the scaffolding the fixed-point theorem stood on, never a property of the equilibrium it produced — which is exactly why a comparison of metrics is neither available nor needed, and the dictionary does the reconciling instead. The sharp L^2-in-type compactness criterion is genuinely more delicate than its uniform-in-type cousin and is discussed at remark 24.13. With the seam understood, we do not duplicate the chapter 24 proofs; we set up the FBSDE so that they have something to act on, and make the role of the coupling operator explicit.

Theorem 23.17 (Well-posedness of the graphon MFG FBSDE)**.** *Grant* ***(A-Lip)****,* ***(A-Growth)****,* ***(A-Nondeg)****,* ***(A-Conv)*** *(so the feedback $\hat{\alpha}$ is well defined and Lipschitz) and* ***(A-Reg-W)****. Then eq. (23.16) admits a solution in the sense of definition 23.16 in each of the following regimes:*

(i) (Small horizon or weak coupling.) *There is a constant $c_0 > 0$, depending only on the Lipschitz constants, such that if $T\, \|\mathbb{W}\mathbf{1}\|_\infty \le c_0$ then the solution exists and is unique, and the graphon best-response map S is a contraction on $C([0,T]; \boldsymbol{\mathcal{P}}_2)$ with factor at most $(c_0)^{1/2}$. Both a short horizon and a weak kernel (small row-sum) suffice. Because the field-contraction constant $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$ (lemma 23.10) is bounded by one* independently of the kernel strength*, the small-horizon contraction holds uniformly over all admissible graphons, the coupling entering only to lower the factor. The* operator-norm smallness ***(A-Coupling)****, $\|\mathbb{W}\| < \kappa$, is the sharper, spectral form of weak coupling; it becomes the decisive condition in the complementary regime where the coupling is linear in the population mean and the relevant constant is the spectral norm $\|\mathbb{W}\|$ rather than the row-sum — the regime in which the symmetry-breaking threshold of section 23.6 lives.*

(ii) (Graphon monotonicity.) *If in addition the graphon Lasry–Lions monotonicity* ***(A-Mon)*** *holds for the type-aggregated couplings (chapter 24), the solution exists and is unique for* every *horizon T, by the continuation-in-parameter method.*

(iii) (Compactness, no monotonicity.) *Without monotonicity but with bounded, continuous couplings and **(A-Reg-W)**, a solution exists for every T (uniqueness may fail) via a Schauder fixed point for S on a convex compact subset of $C([0,T];\boldsymbol{\mathcal{P}}_2)$.*

Proof sketch and pointers. Regime (i): the small-horizon contraction, assembled one type at a time. The mechanism is the field-valued descendant of the homogeneous contraction, and it is worth tracing in full because every step but the last is classical and the last is exactly where $\mathbb{W}$ enters. Fix two input flows $\boldsymbol{\mu},\boldsymbol{\nu} \in C([0,T];\boldsymbol{\mathcal{P}}_2)$. By remark 23.14 the map S processes each separately by freezing the aggregate and solving, for each type, the decoupled classical FBSDE of eq. (23.16) with measure coefficient the *known* frozen field $\nu_t^{\boldsymbol{\mu},x} := (\mathbb{W}\boldsymbol{\mu}_t)(x)$ (and likewise $\nu^{\boldsymbol{\nu},x}$). The two per-type FBSDEs differ *only* in that frozen coefficient, so their optimally controlled marginals obey the standard FBSDE stability estimate of chapters 5 and 9: subtract the two equations, apply Itô to the squared difference of states, control the costate difference by the Pardoux–Peng a priori bound, use the Lipschitz dependence of the data on the measure argument (**(A-Lip)**, **(A-Conv)**), and close with Grönwall. The result, for each x, is

$$\sup_{t\le T} \mathcal{W}_2\big(\operatorname{Law}(X_t^{\boldsymbol{\mu},x}), \operatorname{Law}(X_t^{\boldsymbol{\nu},x})\big)^2 \le C\,T \sup_{t\le T} \mathcal{W}_2\big(\nu_t^{\boldsymbol{\mu},x}, \nu_t^{\boldsymbol{\nu},x}\big)^2, \quad \nu^{\boldsymbol{\mu},x} = (\mathbb{W}\boldsymbol{\mu})(x),$$

with C depending only on the Lipschitz constants of the data, *not* on x and *not* on the kernel. This is a per-type statement: it says each type's best response is stable in the neighbour field it is fed, and nothing has yet coupled the types. The coupling enters at the single step that turns this family of per-type bounds into a statement about the field map — integration over x. Integrating the display and using $\mathbf{d}_{2,\infty}(\mathsf{S}(\boldsymbol{\mu}),\mathsf{S}(\boldsymbol{\nu}))^2 = \sup_{t\le T}\int_0^1 \mathcal{W}_2(\operatorname{Law}(X_t^{\boldsymbol{\mu},x}), \operatorname{Law}(X_t^{\boldsymbol{\nu},x}))^2\,\mathrm{d}x \le \int_0^1 \sup_{t\le T} \mathcal{W}_2(\operatorname{Law}(X_t^{\boldsymbol{\mu},x}), \operatorname{Law}(X_t^{\boldsymbol{\nu},x}))^2\,\mathrm{d}x$,

$$\mathbf{d}_{2,\infty}\big(\mathsf{S}(\boldsymbol{\mu}),\mathsf{S}(\boldsymbol{\nu})\big)^2 \le C\,T \int_0^1 \sup_{t\le T} \mathcal{W}_2\big((\mathbb{W}\boldsymbol{\mu}_t)(x), (\mathbb{W}\boldsymbol{\nu}_t)(x)\big)^2\,\mathrm{d}x.$$

The right-hand integrand is the squared transport distance between the two aggregates, exactly the quantity the field-contraction bound eq. (23.11) controls. Its engine is the pointwise convexity estimate underlying lemma 23.10, $\mathcal{W}_2\big((\mathbb{W}\boldsymbol{m})(x), (\mathbb{W}\boldsymbol{m}')(x)\big)^2 \le \int_0^1 W(x,y)\mathcal{W}_2(m^y, m^{y\prime})^2\,\mathrm{d}y$ (the squared distance of a W-mixture is at most the mixture of squared distances); taking the supremum in time, integrating in x, and using the symmetry of W to recognize the inner mass as the row-sum $(\mathbb{W}\mathbf{1})(y) \le \|\mathbb{W}\mathbf{1}\|_\infty$ caps the whole integral by $\|\mathbb{W}\mathbf{1}\|_\infty\, \mathbf{d}_{2,\infty}(\boldsymbol{\mu},\boldsymbol{\nu})^2$. Hence

$$\mathbf{d}_{2,\infty}\big(\mathsf{S}(\boldsymbol{\mu}),\mathsf{S}(\boldsymbol{\nu})\big)^2 \le C\,T\, \|\mathbb{W}\mathbf{1}\|_\infty\, \mathbf{d}_{2,\infty}(\boldsymbol{\mu},\boldsymbol{\nu})^2,$$

so S has Lipschitz constant $\le (C\,T\,\|\mathbb{W}\mathbf{1}\|_\infty)^{1/2} \le (CT)^{1/2}$ on the flow space, the second inequality because $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$ no matter how strong the kernel. It is therefore a contraction as soon as $C\,T\,\|\mathbb{W}\mathbf{1}\|_\infty < 1$ — the threshold c_0 of the statement, which absorbs the Lipschitz constant C — and theorem 2.36 delivers the unique fixed point. Read that threshold both ways. A *short horizon* forces $C\,T < 1$ and hence contraction for *every* admissible kernel, since the row-sum can only lower the product; this is the uniform-over-graphons claim of the theorem. A *weak kernel* (small row-sum) forces $CT\,\|\mathbb{W}\mathbf{1}\|_\infty < 1$ even at moderate horizons. The crucial structural point — the entire payoff of lemma 23.10 — is that the type-integration, the one place where the continuum's cross-type interaction could have injected a constant growing with coupling strength, instead contributes the bounded factor $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$. The contrast with the *linear*-mean coupling, where the operative constant is not the row-sum but the spectral norm $\|\mathbb{W}\|$ and *its* size sets a genuine threshold, is the subject of remark 23.18 and is made explicit in section 23.6.

Regime (ii): monotonicity and continuation in a parameter. At large horizon the contraction of regime (i) is lost — the factor $(CT\,\|\mathbb{W}\mathbf{1}\|_\infty)^{1/2}$ exceeds one — and a different mechanism must furnish uniqueness. The classical substitute is the Lasry–Lions monotonicity of the coupling, **(A-Mon)**, and the device that turns it into well-posedness for *every* T is continuation in a parameter. One embeds eq. (23.16) in a one-parameter family indexed by $\lambda \in [0,1]$, interpolating from a trivially solvable endpoint at $\lambda = 0$ (the decoupled, or otherwise known, problem) to the target at $\lambda = 1$, and studies the set $S \subseteq [0,1]$ of parameters at which the family is uniquely solvable. The argument shows S is nonempty ($0 \in S$), *open*, and *closed*, whence $S = [0,1]$ by connectedness and in particular $1 \in S$. Openness is a perturbation step: near a solved λ_0 the linearized map is invertible, so a contraction on a small parameter ball relocates the fixed point and a whole neighbourhood of λ_0 lies in S. Closedness is where monotonicity does its work, and it rests on an *a priori bound uniform in λ* — the *monotone a priori estimate.* **(A-Mon)** supplies a bound on the solution, and on the difference of any two solutions, that does not degrade as λ varies, so a sequence of solutions along $\lambda_n \to \lambda_*$ cannot escape to infinity and its limit solves the problem at λ_*. That a priori bound is itself the graphon descendant of the Lasry–Lions pairing computation: one pairs the difference of two solutions' forward and backward components, the cross terms organize into the monotonicity functional of the coupling, and the entire identity is *integrated in x against* $\mathbb{W}$, so that it is the type-aggregated coupling, not the pointwise one, that must be monotone. That this x-integrated pairing closes is exactly the graphon form of the homogeneous Lasry–Lions uniqueness; it is stated and proved in full in chapter 24, and uniqueness for *every* horizon follows because the same monotone pairing controls the difference of two solutions directly, with no smallness of T required.

Regime (iii): existence by compactness, and the two compactness inputs. Drop monotonicity entirely and ask only for existence. The tool is the Schauder fixed-point theorem theorem 2.38: a continuous self-map of a convex compact subset of a Banach space has a fixed point. One builds such a set of fields, shows S maps it continuously into itself, and reads off a fixed point — an equilibrium, though Schauder says nothing about uniqueness, so multiplicity is allowed and, as remark 23.18 explains, genuinely occurs past a spectral threshold. The substance is the compactness, and here lies the one genuinely new feature of the graphon problem, easy to miss because two unrelated compactness requirements travel under the same word and must *not* be conflated.

The first is compactness in the *state* direction: for each fixed type x, the candidate marginals $\{m_t^x\}_{t\le T}$ must be precompact in $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$. This is supplied exactly as in the homogeneous theory of chapters 9 and 10: x-uniform second-moment bounds from **(A-Growth)** give tightness of the marginals (chapter 3, Prokhorov), the nondegenerate, regularizing diffusion **(A-Nondeg)** controls their spatial regularity, and the SDE moment increments give $\mathcal{W}_2$-equicontinuity in time. Nothing about the graphon is new here; the diffusion does the work, type by type, precisely as a single population's diffusion does.

The second is compactness in the *type* direction: the field $x \mapsto m_t^x$ must itself be precompact, which on top of pointwise precompactness demands control of the field's oscillation in x. And here no dynamical mechanism helps. The dynamics treat each type as a *separate* diffusion driven by its own noise; nothing in the flow smooths across nearby types, because the only object linking types is $\mathbb{W}$, and $\mathbb{W}$ regularizes in x only to the extent that its kernel does. Type-direction compactness therefore cannot be earned from the diffusion; it must be *purchased* from the data — specifically from the uniform continuity of the kernel in its type variable, the continuity clause of **(A-Reg-W)**. When $x \mapsto W(x, \cdot)$ is uniformly continuous (say into L^1), the aggregate $x \mapsto (\mathbb{W}\boldsymbol{\mu}_t)(x)$ is equicontinuous in x uniformly over input fields, and the per-type stability of the FBSDE in its measure coefficient then transports that equicontinuity to the output field $x \mapsto \mathrm{Law}(X_t^{\boldsymbol{\mu},x})$. This type-direction equicontinuity is the genuinely new compactness input of the graphon problem — the price the continuum charges that neither the finite nor the homogeneous theory ever sees. It is established as lemma 24.9, and the fact that it must be bought from kernel continuity rather than produced by the dynamics is the first of the two senses, catalogued in remark 24.12, in which the graphon problem is strictly harder than the classical one. With both compactnesses in hand the convex set of fields is compact, S is continuous on it, and Schauder closes the argument.

The complete arguments — the continuation estimate of (ii), the precise compactness criterion of (iii), and the graphon monotonicity condition — are the business of chapter 24. One subtlety is worth flagging at the seam: the equicontinuity just described is cleanest in the uniform-in-type metric, and the sharp L^2-in-type compactness criterion appropriate to the space $\boldsymbol{\mathcal{P}}_2$ of this chapter is more delicate, since L^2-in-type control yields only measurable, not pointwise-equicontinuous, selections; this refinement is remark 24.13. Primary sources for all three regimes are [34, 10, 15]. ☐

Remark 23.18 (Two norms of $\mathbb{W}$, and which controls what)**.** This is the conceptual payoff of the section, and it deserves to be stated sharply, because two genuinely different norms of the one operator $\mathbb{W}$ are in play and conflating them is the central trap of the subject. The slogan "one operator carries all the coupling" is correct, but the operator carries it through *two* numbers, and they answer two different questions.

The row-sum governs the general contraction. The well-posedness of regime (i) — the measure-valued, fully nonlinear contraction — is controlled by the row-sum bound $\|\mathbb{W}\mathbf{1}\|_\infty = \operatorname{ess\,sup}_x \int_0^1 W(x,y)\,\mathrm{d}y$, which is the $L^\infty \to L^\infty$ operator norm of $\mathbb{W}$ and nothing finer. **(A-Reg-W)** makes the kernel sub-stochastic, so this number is ≤ 1 *no matter how strong the kernel*: it saturates at one and cannot be pushed past it by cranking up the coupling. Consequently coupling strength, by itself, never threatens small-horizon uniqueness — the worst a kernel can do to the contraction factor $(C\,T\,\|\mathbb{W}\mathbf{1}\|_\infty)^{1/2}$ is leave it at $(CT)^{1/2}$, and a short horizon defeats that for every admissible graphon at once. This is why regime (i) holds uniformly over all graphons.

The spectral norm governs the onset of multiplicity. Uniqueness is not the only question; whether the equilibrium is unique as opposed to one of several is a separate phenomenon, and it is governed by a different number. It surfaces when the coupling is *linear* in the population — the mean-field case, where an agent responds to the neighbour *mean* $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ rather than to the full measure. There the self-consistency closes through the linear operator $s\mathbb{W}$ acting on the field of means (s the coupling strength), and the fixed-point structure is the invertibility of $\mathrm{Id} - (\text{linear response})\, s\mathbb{W}$, governed not by the row-sum but by the *spectral operator norm* $\|\mathbb{W}\| = |\lambda_1|$ — the magnitude of the leading eigenvalue (theorem 2.26), whose eigenvector is the eigenvector-centrality mode of chapter 18. The boundary between a unique equilibrium and several is set by how $s\,\|\mathbb{W}\|$ compares with a critical coupling; crossing it is symmetry breaking, the appearance of new self-consistent solutions where the trivial one loses stability. This threshold is computed in closed form for the LQG thread in section 23.6 and in general in chapter 24.

Why the two need not agree, and when they do. For a nonnegative kernel one always has $\|\mathbb{W}\| \le \|\mathbb{W}\mathbf{1}\|_\infty$ — the spectral radius of a nonnegative operator is dominated by its maximal row-sum, the Perron–Frobenius bound — so the two numbers can differ, the spectral threshold being reached at *weaker* coupling than the row-sum ever signals. They coincide in exactly the degenerate case the reduction checks of remark 23.4 singled out: for a constant graphon $W \equiv c$ the only active mode is the constant eigenfunction, $\|\mathbb{W}\mathbf{1}\|_\infty = \|\mathbb{W}\| = c$, and the two questions collapse into one — which is precisely why the homogeneous theory, living on a single mode, never had to distinguish them. The continuum forces the distinction the moment $\mathbb{W}$ carries more than one eigenvalue.

It is always a norm of $\mathbb{W}$*, never the pointwise kernel.* In both regimes the controlling quantity is a *norm* of the operator, not the pointwise values $W(x, y)$. This is what makes the theory robust: two graphons close in cut metric but wildly different pointwise (chapter 17) induce operators close in norm, hence comparable contraction constants and comparable equilibria — the quantitative stability statement made precise in chapter 25. The pointwise kernel is not an observable of the equilibrium; the relevant norms of $\mathbb{W}$ are. Beyond the contraction regime one buys uniqueness with monotonicity (regime (ii)) or settles for existence without it (regime (iii)), but in every case it is through one of these two norms of the single operator $\mathbb{W}$ that the kernel speaks.

Two debts are now outstanding, and the next section discharges the structural one. The two-norms distinction just drawn is handed forward to section 23.6, where the spectral threshold $s\,\|\mathbb{W}\|$ is exhibited in closed form for Thread B — the continuum splitting into one scalar game per eigenmode, with the symmetry breaking visible mode by mode. The three regime statements, in turn, are entered in the chapter ledger as deferred to chapter 24. But that deferral rests on a bridge we have so far only invoked: that chapter 24's proofs, run in the uniform-in-type metric, certify the very object this section's L^2-in-type FBSDE statement is about. That bridge is the two-pictures dictionary, and building it is the task of section 23.5: there we prove that the probabilistic best-response map of this chapter and the analytic one of chapter 22 are the same map, by identifying $Y_t^x = \nabla u^x(t, X_t^x)$ type by type — the equivalence that lets a fixed point found in either metric be read as a solution of the other, and so transfers chapter 24's well-posedness to the FBSDE stated here.

23.5 Two pictures of one graphon equilibrium

Twice already this chapter has leaned on an equivalence it has not yet proved. In section 23.3 we observed that the probabilistic best-response map S built from the FBSDE eq. (23.16) and the analytic best-response map of chapter 22 built from the HJB/Fokker–Planck pair are not two maps that happen to agree but the *same* map in two realizations — the same population field, computed by a different solver. And in section 23.4 the three well-posedness regimes were proved on the probabilistic side, in the L^2-in-type field metric $\mathbf{d}_2$, yet the sharpest existence-and-uniqueness statements of chapter 24 are proved on the analytic side, in the uniform-in-type metric; the only thing that lets the one set of theorems discharge the other is precisely that the two formulations describe one equilibrium. Both debts come due here. This section pays them by writing down, and proving, the exact *dictionary* between the probabilistic objects (X^x, Y^x, Z^x) of this chapter and the analytic objects (u^x, m^x) of chapter 22 — the graphon analogue of the homogeneous two-pictures result proposition 9.12.

It is worth recalling exactly what chapter 22 built, because the dictionary is a statement about that object. There the dynamic graphon mean field game was written as a continuum-indexed system eq. (22.6): for each type x a *backward* Hamilton–Jacobi–Bellman equation eq. (22.6a) for the value $u^x(t,\cdot)$ of the representative type-x agent, run from terminal data at $t = T$, and a *forward* Fokker–Planck equation eq. (22.6b) for the type-x population law m_t^x, run from initial data at $t = 0$. The two families are coupled across types through one channel and one channel only: the graphon aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ that each equation reads as its measure argument. The claim of this section is that the FBSDE eq. (23.16) assembled in section 23.3 and that PDE system are two representations of a single graphon equilibrium, and that the bridge between them is, type by type, the identification of the costate with the gradient of the value along the optimal trajectory — the measure argument being supplied by the *same* aggregate in both pictures. We state and prove the forward half of the dictionary in the first subsection and supply the converse machinery — a measurable family of decoupling fields — in the second.

23.5.1 The dictionary

Before stating the dictionary we must fix the regularity under which it even makes sense, and it pays to see why each clause is needed rather than to swallow a list of hypotheses. The dictionary will read $Y_t^x = \nabla u^x(t, X_t^x)$ and $Z_t^x = \mathrm{D}^2\, u^x(t, X_t^x)\,\sigma$; for the right-hand sides to exist pointwise and for Itô's formula to be applicable to ∇u^x we need u^x to be once continuously differentiable in time and *three* times in space. We therefore assume the regularity under which the analytic picture of chapter 22 is classical: for each x the type-x HJB value $u^x(t,x) = u^x(t,x)$ is $C^{1,2}$ on $[0,T) \times \mathbb{R}^d$ — in fact $C^{1,3}$ in the spatial variable. The reason we may demand the extra spatial derivative for free is interior parabolic regularity: **(A-Nondeg)** makes the generator uniformly elliptic, and a uniformly parabolic equation with smooth coefficients has a solution smoother in space than its data by the Schauder interior estimates, so $C^{1,2}$ is automatically upgraded to $C^{1,3}$ (indeed to $C^{1,\infty}$) away from the terminal time. We need exactly one notch of this upgrade and no more: $u^x \in C^{1,2}$ lets us write $Y^x = \nabla u^x$ and apply Itô, while the third spatial derivative is what makes *differentiating the HJB equation in x* a legitimate operation — the step in the proof that produces the adjoint driver. We record this as part of the standing hypothesis exactly as in chapter 22 rather than re-derive the Schauder theory here; the volatility is uncontrolled (so σ does not depend on the costate and the second-order term is a fixed linear operator on $\mathrm{D}^2\, u^x$); and — the one genuinely graphon-flavoured clause — the map $x \mapsto u^x$ is measurable with x-uniform parabolic bounds. This last clause is not decoration. The per-type problems are solved separately, one for each x, and what stitches the continuum of solutions back into a single measurable field is precisely that the coefficients depend on x only through bounded data and through the aggregate $\nu_t^x = (\mathbb{W}\boldsymbol{m}_t)(x)$, whose x-to-x variation the row-sum bound $\|\mathbb{W}\mathbf{1}\|_\infty \leq 1$ of lemma 23.10 controls; the Schauder constants therefore do not blow up as x varies, and the value field inherits a uniform-in-x modulus. We return to this measurability point inside the proof, where it is the only ingredient beyond the homogeneous result. Recall from chapter 22 (the system eq. (22.6), with halves eq. (22.6a) and eq. (22.6b)) that the equilibrium value field solves the type-x backward HJB equation

$$\partial_t u^x + \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 u^x) - H(x, x, \nabla u^x, (\mathbb{W}\boldsymbol{m}_t)(x)) = 0, \qquad u^x(T, x) = g(x, x, (\mathbb{W}\boldsymbol{m}_T)(x)), \tag{23.18}$$

with $H = H$ the HJB Hamiltonian of Appendix A (convex in the costate variable p under **(A-Conv)**). The optimal feedback is the minimizer of the generalized Hamiltonian eq. (23.14); the compact convention $\alpha^* = -\partial_p H$ of Appendix A is the special case in which the control enters the drift additively with unit gain. For the Thread-A/B drift $b = ax + b\alpha$ the gain is b and the feedback is $\hat{\alpha}^x = -bY^x$, derived in worked computation 23.22; in general one minimizes eq. (23.14) directly. The population field evolves by the forward graphon Fokker–Planck eq. (23.6) carrying the optimal feedback drift. The measure argument in both eq. (23.18) and eq. (23.6) is the same aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$.

Proposition 23.19 (Graphon FBSDE–PDE dictionary)**.** *Under the regularity above, let $\boldsymbol{m} = (\boldsymbol{m}_t)$ be a graphon equilibrium flow and $(u^x)_x$ the corresponding HJB value field solving eq. (23.18). For each x let X^x solve the forward equation of eq. (23.16) with this $\boldsymbol{m}$, and set*

$$Y^x_t = \nabla u^x(t, X^x_t), \qquad Z^x_t = \mathrm{D}^2 u^x(t, X^x_t)\, \sigma(t, x, X^x_t, (\mathbb{W}\boldsymbol{m}_t)(x)). \tag{23.19}$$

(The order is fixed by dimensions: $\mathrm{D}^2 u^x$ is $d \times d$ and σ is $d \times k$, so $Z^x \in \mathbb{R}^{d\times k}$, the shape demanded by $Z^x \in \mathcal{H}^2(\mathbb{R}^{d\times k})$ in definition 23.16 and by the integrand of $Z^x_t \,\mathrm{d}B^x_t$.) Then for a.e. x the triple (X^x, Y^x, Z^x) solves the type-x FBSDE of eq. (23.16) with the same $m^x_t = \mathrm{Law}(X^x_t)$. Conversely, if $(X^x, Y^x, Z^x, \boldsymbol{m})$ solves eq. (23.16) and the per-type FBSDE admits a sufficiently regular decoupling field U^x (section 23.5.2), then ∇_x of the associated potential reconstructs ∇u^x and $(u^x)_x$ solves eq. (23.18). The two formulations are equivalent representations of the same graphon equilibrium.

Proof. The whole proof is the freezing trick run once more, now in the service of an *identity* rather than an existence statement: freeze the equilibrium aggregate to make the per-type problem classical, invoke the homogeneous two-pictures result for that classical problem, and then verify the four FBSDE requirements — the Z-identification, the driver, the terminal datum, and measurability in type — one at a time by an explicit Itô computation.

Freeze the aggregate. Fix x and set $\nu^x_t = (\mathbb{W}\boldsymbol{m}_t)(x)$. At a graphon equilibrium $\boldsymbol{m}$ is a fixed, deterministic flow of fields, so ν^x_t is a known deterministic measure flow — it no longer references the unknown processes. With ν^x frozen, the HJB equation eq. (23.18) is a *classical* backward parabolic equation in (t, x) alone, and the forward equation of eq. (23.16) is a classical optimally controlled SDE with a deterministic, time-dependent measure coefficient. The type-x problem is therefore, on the nose, an instance of the homogeneous control problem of chapter 9 with the measure flow m_t there replaced by ν^x_t and the label x a passive parameter; the assertion $Y^x = \nabla u^x$ is then exactly proposition 9.12 for that instance. We reproduce the computation to exhibit, in the graphon notation, where each piece of the FBSDE eq. (23.16) comes from.

Apply Itô to ∇u^x; identify Z^x. By the standing $C^{1,3}$ regularity the map $(t,x) \mapsto \nabla u^x(t,x)$ is $C^{1,2}$, so Itô's formula theorem 5.9 applies to each of its components $\partial_{x_i} u^x$ along the forward dynamics $\mathrm{d}X_t^x = b\ \mathrm{d}t + \sigma\ \mathrm{d}B_t^x$ (with b, σ evaluated at (t, x, X_t^x, ν_t^x) and the optimal feedback). Collecting the components into the vector $Y_t^x = \nabla u^x(t, X_t^x)$,

$$\mathrm{d}Y_t^x = \Big(\partial_t \nabla u^x + (b\cdot\nabla)\nabla u^x + \tfrac{1}{2}\operatorname{tr}\left(\sigma\sigma^\top \mathrm{D}^2\ \nabla u^x\right)\Big)(t, X_t^x)\ \mathrm{d}t + \mathrm{D}^2\, u^x(t, X_t^x)\,\sigma\ \mathrm{d}B_t^x, \tag{23.20}$$

where the bracketed drift is the generator $\partial_t + \mathcal{A}_t^x$ applied componentwise to ∇u^x. The martingale (Itô-integral) part is $\mathrm{D}^2\, u^x(t, X_t^x)\,\sigma\ \mathrm{d}B_t^x$, an $\mathbb{R}^{d\times k}$-valued integrand against $\mathrm{d}B_t^x$. Comparing with the backward equation of eq. (23.16), whose martingale part is by definition $Z_t^x\ \mathrm{d}B_t^x$, and using uniqueness of the martingale representation (theorem C.17) forces $Z_t^x = \mathrm{D}^2\, u^x(t, X_t^x)\,\sigma(t, x, X_t^x, \nu_t^x)$ — which is the second entry of eq. (23.19). The dimension bookkeeping is exactly the one flagged in the proposition: $\mathrm{D}^2\, u^x$ is the $d \times d$ Hessian and σ is $d \times k$, so their product is $d \times k$, the shape demanded both by $Z^x \in \mathcal{H}^2(\mathbb{R}^{d\times k})$ in definition 23.16 and by the very integrand $Z_t^x\ \mathrm{d}B_t^x$ we just matched.

Recover the adjoint driver by differentiating HJB. It remains to check that the drift in eq. (23.20) equals the adjoint driver $-\partial_x \mathcal{H}$ of eq. (23.15). Take the spatial gradient of the HJB equation eq. (23.18) — legitimate term by term precisely because u^x is $C^{1,3}$, so all the derivatives below exist and commute. Differentiating $\partial_t u^x + \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2\, u^x) - H(x, x, \nabla u^x, \nu_t^x) = 0$ in x gives

$$\partial_t \nabla u^x + \tfrac{1}{2}\nabla \operatorname{tr}\left(\sigma\sigma^\top \mathrm{D}^2\, u^x\right) - \partial_x H - \partial_p H \cdot \mathrm{D}^2\, u^x = 0,$$

the last two terms coming from the chain rule applied to $H(x, x, \nabla u^x(t,x), \nu_t^x)$, whose x-dependence is both explicit and through $p = \nabla u^x$. Solving for $\partial_t \nabla u^x$ and substituting into the bracket of eq. (23.20), the second-order terms organize into $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2\ \nabla u^x)$ plus the $\partial_p H \cdot \mathrm{D}^2\, u^x$ piece, which together with the transport term $(b\cdot\nabla)\nabla u^x = \mathrm{D}^2\, u^x\, b$ from the Itô drift collapse once we use the envelope identity and the relation $H = -\widehat{\mathcal{H}}$ between the HJB Hamiltonian and the reduced generalized Hamiltonian (remark 9.11): at the optimal feedback $\hat\alpha_t^x$ the optimal drift b is exactly the Hamiltonian velocity $-\partial_p H$ (the envelope theorem, in the sign convention of Appendix A), so the transport contribution and the p-chain-rule contribution from differentiating HJB are equal and opposite and cancel. What survives is precisely $-\partial_x \mathcal{H}(t, x, X_t^x, \nu_t^x, Y_t^x, Z_t^x, \hat\alpha_t^x)$, the driver of eq. (23.15), with the optimal feedback $\hat\alpha_t^x = \alpha^*(t, x, X_t^x)$ given by the Hamiltonian minimizer. This is the calculation that the slogan "differentiate the value, get the costate; differentiate HJB, get the adjoint" compresses; the $C^{1,3}$ hypothesis is what licenses every differentiation in it.

Match the terminal data. At $t = T$ the value carries the terminal cost, $u^x(T, x) = g(x, x, \nu_T^x)$, so taking the gradient and evaluating along the trajectory, $Y_T^x = \nabla u^x(T, X_T^x) = \partial_x g(x, X_T^x, \nu_T^x)$, which is exactly the terminal datum of the backward equation in eq. (23.16). The three pieces — Z-identification, driver, and terminal datum — together say that (X^x, Y^x, Z^x) solves the type-x FBSDE with the *same* $m_t^x = \mathrm{Law}(X_t^x)$, since the forward law is untouched by the construction.

Measurability in type. This is the one ingredient with no homogeneous counterpart, and it is where the x-uniform regularity hypothesis earns its place, so we make it explicit rather than assert it. The triple is built from u^x by the three deterministic operations ∇, D^2, and "solve the forward SDE," each continuous in the coefficients; so measurability of $x \mapsto (X^x, Y^x, Z^x)$ reduces to measurability of $x \mapsto u^x$ into $C^{1,2}$. The latter holds because u^x is the unique classical solution of the HJB equation eq. (23.18) whose only x-dependence is through the bounded data and through the aggregate ν_t^x, and $x \mapsto \nu_t^x$ is measurable (it is $\mathbb{W}$ applied to the measurable equilibrium field, definition 23.2); the solution map of a uniformly parabolic equation is continuous in its coefficients with the x-uniform Schauder constants furnished above, so the composition $x \mapsto u^x$ is measurable into $C^{1,2}$ with a uniform modulus. This is the sense in which "$x \mapsto u^x$ measurable with x-uniform parabolic bounds" was the only extra hypothesis: it converts a continuum of separately-solved classical problems into one measurable field.

The converse. Conversely, suppose $(X^x, Y^x, Z^x, \boldsymbol{m})$ solves the FBSDE eq. (23.16). For a.e. x the per-type FBSDE is, with ν^x frozen, a classical decoupled FBSDE; if it admits a sufficiently regular decoupling field U^x — which the next subsection supplies as theorem 23.21 — then $Y_t^x = U^x(t, X_t^x)$, and integrating $U^x = \nabla u^x$ in x reconstructs a potential u^x that, by the same Itô-plus-HJB calculation read backward, solves eq. (23.18); uniqueness of the per-type BSDE solution (chapter 5) makes the forward map of this proposition and the converse exact inverses. The two formulations are thus equivalent representations of one graphon equilibrium. □

The entry to memorize is, again, the first equality in eq. (23.19):

$\boxed{Y_t^x = \nabla u^x(t, X_t^x)}$ the type-x costate is the gradient of the type-x value, along the type-x trajectory.

This is the single line the reader should carry out of the section, and it carries its own mnemonic: the backward process Y^x of the probabilistic picture is nothing but the analytic value's slope, sampled along the random path the agent actually follows. Everything else — the Z-identification, the matching of drivers and terminal data — is the bookkeeping that makes this one equality consistent with both equations at once.

It is worth saying carefully what each picture *stores*, because the two are not merely notational variants; they package the same equilibrium in genuinely different data structures, and which structure is convenient depends on the question. The PDE picture of chapter 22 stores the equilibrium as two *fields of fields*: a value $(t, x) \mapsto u^x(t, x)$ and a density $(t, x) \mapsto m_t^x(x)$, each itself indexed by the type x — functions of state and time, for every type. The FBSDE picture of this chapter stores the *same* equilibrium as a measurable family $x \mapsto (X^x, Y^x, Z^x)$ of coupled *processes* — random trajectories, one optimal path and its costate per type. Neither stores the cross-type coupling as a separate object: in both, the neighbour aggregate $(\mathbb{W}\boldsymbol{m}_t)(x)$ is computed once from the population field and read identically by the value equation and by the state dynamics, which is the concrete sense in which the population field is literally the same object in both pictures. The bridge between the two storage formats is, exactly as in chapter 9, "evaluate the field along the random trajectory": feed the random state X_t^x into the deterministic field $\nabla u^x(t, \cdot)$ and out comes the random costate Y_t^x. The only graphon novelty is that this evaluation is now done type by type, with the entire cross-type interaction held in the single aggregate — the chapter's organizing principle, "one operator carries all the coupling," visible once more at the level of the dictionary itself.

Remark 23.20 (What each picture sees better, in the graphon setting)**.** The division of labour of chapter 9 persists and sharpens. The PDE picture of chapter 22 exposes regularity in the *state* variable (parabolic estimates, maximum principle) and is what the type-grid finite-difference schemes of chapter 28 discretize. The FBSDE picture of this chapter exposes the *type* and *trajectory* structure: it is the formulation in which the descent to a finite sampled network $G(N, W)$ is a propagation-of-chaos statement about N coupled trajectories (chapter 25), in which the norms of the coupling operator $\mathbb{W}$ enter the estimates explicitly (the row-sum $\|\mathbb{W}\mathbf{1}\|_\infty$ for the general contraction, the spectral $\|\mathbb{W}\|$ for the multiplicity threshold; theorem 23.17), and in which the spectral decomposition of $\mathbb{W}$ literally diagonalizes the equilibrium in the solvable Thread B (section 23.6). It is also the natural home of the forward–backward Monte Carlo solvers of chapter 28 and of the common-noise extension of chapter 26, where the population field itself becomes random and only a measure-dependent (master) field tracks it.

23.5.2 Decoupling fields, type by type

The dictionary just proved runs in one direction with no extra machinery: given the analytic value field, Itô produces the probabilistic triple. The converse direction — reconstruct the value field from a solution of the FBSDE — was stated in proposition 23.19 but left conditional on "a sufficiently regular decoupling field U^x," a promissory note we now redeem. The obstruction the converse must overcome is structural and familiar from chapter 9: an FBSDE is a *two-point* boundary value problem in time, with X^x pinned at $t = 0$ and Y^x pinned at $t = T$, so neither leg can be integrated on its own. The device that resolves this two-point coupling is the decoupling field of section 9.4 — a deterministic function $U^x(t, x)$ that expresses the backward state as a function of the forward one, $Y^x_t = U^x(t, X^x_t)$, thereby reducing the coupled forward–backward pair to a single forward SDE with known feedback. In the homogeneous theory this function *is* ∇u^x, so the decoupling field is simultaneously the analytic gradient the converse needs and the algebraic device that closes the FBSDE; that double role is what makes it the right tool here. The only work specific to the graphon is to produce not one decoupling field but a whole *measurable family* of them, one per type, with constants that do not degrade as x ranges over the continuum.

Theorem 23.21 (Family of decoupling fields)**.** *Grant **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, **(A-Reg-W)**, with the equilibrium field $\boldsymbol{m}$ fixed (so each per-type FBSDE is classical). Then there is a horizon $T_0 > 0$, depending only on the Lipschitz constants (and, through the aggregate, on the row-sum bound $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$), such that for $T \le T_0$, for a.e. x the type-x decoupled FBSDE has a unique Lipschitz decoupling field $U^x : [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$ with*

$$Y^x_t = U^x(t, X^x_t) = \nabla u^x(t, X^x_t), \qquad Z^x_t = \partial_x U^x(t, X^x_t)\,\sigma, \tag{23.21}$$

*the map $x \mapsto U^x$ measurable with x-uniform Lipschitz constant. Under **(A-Mon)** the family exists and is Lipschitz for every T.*

Proof sketch. We do *not* reprove the per-type result, which is owned by chapter 9, and we are explicit about the division of labour: for each fixed x, with the equilibrium field frozen, the type-x FBSDE is a classical decoupled FBSDE, and the existence of a unique Lipschitz decoupling field U^x for it — for short horizons in general, for all horizons under monotonicity — is exactly the decoupling-field theorem theorem 9.14, established there by the four-step scheme of Ma–Protter–Yong and the contraction scheme of Delarue. The nondegeneracy **(A-Nondeg)** is what supplies the parabolic smoothing that makes U^x Lipschitz rather than merely measurable, and the identification $U^x = \nabla u^x$ is the dictionary proposition 23.19 just proved. So the per-type content is imported wholesale. What is genuinely new in the graphon setting — and all that this sketch must establish — are two additions: that the horizon T_0 and the Lipschitz constant can be taken *uniform in x*, and that the assignment $x \mapsto U^x$ is *measurable.* We develop each.

x-uniformity of the horizon and the Lipschitz constant. The small-horizon threshold T_0 and the Lipschitz constant produced by theorem 9.14 are not abstract; they are explicit functions of the data entering the per-type FBSDE — the Lipschitz constants of b, σ, L, g, the nondegeneracy floor of **(A-Nondeg)**, and the modulus with which the coefficients respond to their measure argument. For the family to share a single T_0 and a single Lipschitz constant it suffices that these inputs be bounded uniformly over x. The state-coefficient constants are x-uniform by **(A-Lip)**–**(A-Growth)** outright. The only input that could in principle drift with x is the measure argument, which enters the type-x problem solely through the aggregate $\nu_t^x = (\mathbb{W}\boldsymbol{m}_t)(x)$; so x-uniformity comes down to controlling how much ν^x can vary as x moves — and this is precisely what the row-sum bound buys. By the field-contraction estimate eq. (23.11) of lemma 23.10, the aggregate's variation across the type interval is dominated by $\|\mathbb{W}\mathbf{1}\|_\infty \leq 1$ times the variation of the underlying field; the operator $\mathbb{W}$ cannot *amplify* type-to-type differences, it can only redistribute and average them, with total weight ≤ 1. Consequently the measure inputs to the per-type problems are bounded and equi-Lipschitz in x, the threshold T_0 of theorem 9.14 may be taken as its worst-case (smallest) value over x, which is still strictly positive because it depends only on the now-uniform data, and the same is true of the Lipschitz constant of U^x. The crucial point is the one the throughline keeps returning to: it is the bound ≤ 1 on the row-sum — independent of the kernel's strength — that prevents the continuum from degrading the horizon as x varies.

Measurability of $x \mapsto U^x$. This is not an external regularity input to be assumed but a consequence of how U^x is constructed. The decoupling field is realized in theorem 9.14 as the unique fixed point of a contraction map Φ^x on a complete function space, and Φ^x depends on x only through the per-type coefficients — which depend measurably on x through the measurable bounded data and through the measurable aggregate ν^x (measurable because it is $\mathbb{W}$ applied to the measurable equilibrium field, definition 23.2). A fixed point of a uniform contraction whose defining map depends measurably (indeed continuously) on a parameter is itself measurable in that parameter: one may exhibit it as the limit of the Picard iterates $\Phi^{xn}(U_0)$, each of which is measurable in x by induction, and a pointwise limit of measurable maps is measurable. Hence $x \mapsto U^x$ is measurable, with the x-uniform Lipschitz constant just established, and the family is a bona fide field of decoupling fields living in the field setting of section 23.2. Complete per-type proofs are in [40]; the x-uniform versions, with this measurability argument carried out in full, are in [10].

We close on the deliberate limit of what is built here. The decoupling field U^x is defined *along the equilibrium*, with the population field $\boldsymbol{m}$ held fixed; it is a function of (t, x) for each frozen environment, not yet a function of the environment itself. Its promotion to a *measure-dependent* master field $\mathcal{U}(t, x, x, \boldsymbol{m})$ — defined for every field $\boldsymbol{m}$, whose total derivative in the measure direction generates the graphon master equation — is a strictly larger construction, requiring differentiability in the field argument, and we defer it intact to chapter 26, exactly as the homogeneous decoupling field of eq. (9.19) was the seed that chapter 12 grew into the master equation. For the present chapter the frozen family $U^x = \nabla u^x$ is all the converse needs, and all that the worked computation will use. □

This is the identification the worked computation of section 23.6 now turns concrete. In the graphon LQG thread the value function is quadratic in the state, so its gradient is *affine*, and the decoupling field collapses to an explicit affine form read, through proposition 23.19, as the gradient of a quadratic value. The dictionary therefore promises two things to the next section: that the probabilistic ansatz it solves is genuinely the gradient of chapter 22's quadratic value field, and that the closed form it derives will match the analytic solution mode for mode once the operator $\mathbb{W}$ is diagonalized. The abstract two-pictures equivalence is about to become a fully explicit, exactly solvable instance.

23.6 Worked computation: the graphon LQG thread as an FBSDE

Everything the chapter has built now comes due at once. The two-pictures dictionary of section 23.5 told us that the probabilistic costate Y^x is the gradient $\nabla u^x(t, X_t^x)$ of the type-x value, and section 23.5.2 specialized that statement to the LQG case: where the value is quadratic in the state its gradient is *affine*, so the decoupling field collapses to $Y_t^x = P_t X_t^x + r_t^x$ with a type-independent Riccati curvature P_t and a type-indexed offset r_t^x. That is the ansatz we are handed; this section solves it. The fixed-point analysis of section 23.4 told us, in turn, that two different norms of the one coupling operator $\mathbb{W}$ do two different jobs — the row-sum $\|\mathbb{W}\mathbf{1}\|_\infty \le 1$ governs the general measure-valued contraction, the spectral norm $\|\mathbb{W}\|$ governs the onset of multiplicity — but left the second statement as a promise to be cashed in closed form here. And the chapter preamble promised, from the first page, that the probabilistic formulation would *diagonalize* the continuum into spectral normal modes. This worked computation redeems all three debts simultaneously.

We carry Thread B — the graphon linear–quadratic–Gaussian model fixed in the thread specification — through the probabilistic formulation and show that the operator structure assembled across the chapter makes it *exactly solvable*. The mechanism is the one the spine of the chapter has pointed at throughout: because the entire cross-type coupling is the single linear operator $\mathbb{W}$, and because $\mathbb{W}$ is self-adjoint Hilbert–Schmidt, we may *diagonalize* it. In its eigenbasis the continuum of coupled agents falls apart into a direct sum of independent scalar LQG mean field games, one per eigenmode, the k-th carrying an effective crowd-coupling rescaled from s to $s\lambda_k$. Each summand is a problem we have already solved completely — it is the homogeneous Thread A LQG game of chapter 9 and chapter 14 — so the whole graphon game is solved the moment we name its spectrum. The payoff is twofold. We recover, from the probabilistic side, the closed-form equilibrium of chapter 22 mode for mode; and we read the symmetry-breaking threshold $s\,\|\mathbb{W}\|$ directly off the leading eigenvalue, exhibiting in closed form the $\|\mathbb{W}\|$-threshold whose general form is theorem 23.17 and chapter 24.

Worked computation 23.22 (Thread B: the scalar graphon LQG MFG in FBSDE form)**.** Recall the scalar graphon LQG model of the thread specification. Each type-x agent has state $X^x \in \mathbb{R}$ obeying

$$\mathrm{d}X_t^x = \big(aX_t^x + b\,\alpha_t^x\big)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x,$$

with type-independent coefficients $a, b, \sigma \in \mathbb{R}$, and is coupled to the graphon-aggregated neighbour mean

$$z_t(x) = (\mathbb{W}\bar{m}_t)(x) = \int_0^1 W(x,y)\,\bar{m}_t(y)\,\mathrm{d}y, \qquad \bar{m}_t(y) = \int \xi\, m_t^y(\mathrm{d}\xi) = \mathbb{E}[X_t^y], \tag{23.22}$$

through the cost

$$J^x(\alpha^x) = \mathbb{E}\left[\int_0^T \left(\tfrac{1}{2}(\alpha_t^x)^2 + \tfrac{q}{2}\big(X_t^x - s\,z_t(x)\big)^2\right)\mathrm{d}t + \tfrac{q_T}{2}\big(X_T^x - s_T\,z_T(x)\big)^2\right],$$

with $q, q_T \geq 0$ and coupling strengths s, s_T. The diffusion is constant (hence **(A-Nondeg)** for $\sigma \neq 0$), the data are convex, and $z_t(x)$ is a scalar moment of the aggregate eq. (23.2) with $h(\xi) = \xi$; the sufficient maximum principle theorem 9.5 applies type by type.

Hamiltonian, feedback, and the FBSDE. With costate y the type-x generalized Hamiltonian eq. (23.14) is

$$\mathcal{H} = (ax + b\alpha)\,y + \tfrac{1}{2}\alpha^2 + \tfrac{q}{2}\big(x - s\,z_t(x)\big)^2$$

(to keep the bookkeeping unambiguous we rename the second-adjoint slot z of eq. (23.14) to ζ here, so it does not collide with the graphon coupling field $z_t(x)$; the term $\mathrm{tr}(\sigma^\top\zeta) = \sigma\,\zeta$ is constant in x and α and plays no role in either the minimization or $\partial_x\mathcal{H}$, so we simply drop it). The Hamiltonian is minimized at $\hat{\alpha}^x = -b\,Y^x$ — linear in the costate, exactly as in eq. (9.21), and the concrete instance of the general feedback discussed after eq. (23.18). Substituting $\hat{\alpha}$ and $\partial_x\mathcal{H} = ay + q(x - sz_t(x))$ into eq. (23.16) gives, for each x, the scalar McKean–Vlasov FBSDE

$$\begin{cases} \mathrm{d}X_t^x = \big(aX_t^x - b^2 Y_t^x\big)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x, & X_0^x \sim m_0^x, \\ \mathrm{d}Y_t^x = -\big(aY_t^x + q(X_t^x - s\,z_t(x))\big)\,\mathrm{d}t + Z_t^x\,\mathrm{d}B_t^x, & Y_T^x = q_T\big(X_T^x - s_T\,z_T(x)\big), \\ z_t(x) = (\mathbb{W}\bar{m}_t)(x), & \bar{m}_t(x) = \mathbb{E}[X_t^x], \end{cases} \tag{23.23}$$

identical to the homogeneous Thread A FBSDE eq. (9.22) except that the scalar coupling target $\bar{m}_t$ is replaced by the type-field $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$.

Affine decoupling. The system eq. (23.23) is linear in (X^x, Y^x) and the only term that couples one type to another is the deterministic field $z_t(x)$, which the forward and backward legs see as a known forcing. This is exactly the situation in which a costate depends on the state *affinely*, and the dictionary of section 23.5.2 already told us to expect it: where the value is quadratic the gradient is affine. So we posit the type-indexed affine decoupling field

$$Y_t^x = P_t\,X_t^x + r_t^x,$$

with a deterministic curvature P_t (*type-independent*, the canonical Riccati symbol of chapter 9) and a deterministic offset r_t^x (type-dependent). The whole content of this thread layer sits in the offset — it is the one object through which the graphon $\mathbb{W}$ reaches the decoupling — so we carry out the coefficient matching in full rather than quote it.

The plan is to compute $\mathrm{d}Y_t^x$ two ways and equate them. *First way:* differentiate the ansatz. Since P_t and r_t^x are deterministic and X_t^x is an Itô process, the product/chain rule of Itô's formula (theorem 5.9) applied to $Y_t^x = P_tX_t^x + r_t^x$ gives $\mathrm{d}Y_t^x = \dot{P}_tX_t^x\,\mathrm{d}t + P_t\,\mathrm{d}X_t^x + \dot{r}_t^x\,\mathrm{d}t$ — there is no second-order Itô correction because P_t carries no $\mathrm{d}B$ part, so the only quadratic-variation term would be $\mathrm{d}P_t\,\mathrm{d}X_t^x = 0$. Substituting the forward dynamics $\mathrm{d}X_t^x = (aX_t^x - b^2 Y_t^x)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x$ from eq. (23.23),

$$\mathrm{d}Y_t^x = \big(\dot{P}_t X_t^x + P_t(aX_t^x - b^2 Y_t^x) + \dot{r}_t^x\big)\,\mathrm{d}t + P_t\sigma\,\mathrm{d}B_t^x,$$

and comparing the martingale parts already identifies the BSDE's Z-component, $Z_t^x = P_t\sigma$ (the volatility of the costate is the curvature times the state volatility, as in chapter 9). Now eliminate the remaining Y^x on the right by the ansatz itself: $P_t(aX_t^x - b^2Y_t^x) = P_t aX_t^x - b^2 P_t(P_t X_t^x + r_t^x) = (aP_t - b^2P_t^2)X_t^x - b^2 P_t r_t^x$, so the drift of the first expression, collected by powers of X_t^x, is

$$\underbrace{\big[\dot{P}_t + aP_t - b^2P_t^2\big]}_{\text{coefficient of } X_t^x} X_t^x \;+\; \underbrace{\big[\dot{r}_t^x - b^2 P_t r_t^x\big]}_{X\text{-free}}.$$

Second way: read the costate's drift straight off the backward equation of eq. (23.23), namely $-(aY_t^x + q(X_t^x - s\,z_t(x)))$, and again substitute $Y_t^x = P_t X_t^x + r_t^x$:

$$-\big(a(P_tX_t^x + r_t^x) + qX_t^x - q\,s\,z_t(x)\big) = \underbrace{-(aP_t+q)}_{\text{coefficient of } X_t^x}\; X_t^x \;+\; \underbrace{\big[-ar_t^x + q\,s\,z_t(x)\big]}_{X\text{-free}}.$$

The two drifts must agree as functions of the *random* variable X_t^x, which is nondegenerate (it has a density, $\sigma \neq 0$), so they agree coefficient by coefficient. The matching therefore splits cleanly into two scalar equations: the coefficient of X_t^x gives a closed equation for P_t with no reference to the offset or the graphon, while the X-free remainder gives an equation for r_t^x forced by $z_t(x)$. Taking the X_t^x-coefficients, $\dot{P}_t + aP_t - b^2P_t^2 = -(aP_t + q)$, i.e. the scalar Riccati equation

$$\dot{P}_t = b^2P_t^2 - 2aP_t - q, \qquad P_T = q_T, \tag{23.24}$$

which is *exactly* eq. (9.24) and carries *no type label*. This is the first structural payoff of the computation, and it is worth dwelling on: the curvature of every type's value function is one and the same scalar P_t, because the only ingredients that enter the X^x-coefficient — the state feedback a, the control gain b, the state-cost weight q, and the terminal weight q_T — are all type-independent. The graphon never touches the Riccati equation. Where the network *does* enter is the X-free remainder. Equating the two X-free parts, $\dot{r}_t^x - b^2P_tr_t^x = -ar_t^x + q\,s\,z_t(x)$, and solving for $\dot{r}_t^x$, gives the type-dependent offset equation

$$\dot{r}_t^x = (b^2P_t - a)r_t^x + q\,s\,z_t(x), \qquad r_T^x = -q_T\,s_T\,z_T(x), \tag{23.25}$$

the terminal datum read off by evaluating the ansatz at $t = T$ against the terminal costate: $Y_T^x = q_T(X_T^x - s_T z_T(x))$ must equal $P_T X_T^x + r_T^x$, and with $P_T = q_T$ the X_T^x-terms cancel, leaving $r_T^x = -q_T s_T z_T(x)$. Notice that eq. (23.25) is a *linear* ODE for r^x whose homogeneous coefficient $b^2P_t - a$ is again type-independent (it is fixed by the Riccati curvature), so the only x-dependence in the entire offset comes through its forcing term $q\,s\,z_t(x)$ and its terminal value $-q_Ts_Tz_T(x)$ — both built from the graphon field $z_t = \mathbb{W}\bar{m}_t$. The offset equation is therefore eq. (9.25) of the homogeneous thread with the scalar coupling target $\bar{m}_t$ replaced by the type field $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$: this single substitution is *the one place the graphon enters the decoupling*, and it is the reason the diagonalization to come will need only to diagonalize $\mathbb{W}$. Taking expectations of the forward equation, and using $\mathbb{E}[Y_t^x] = P_t \bar{m}_t(x) + r_t^x$ (the offset is deterministic, so its expectation is itself), yields the mean-field evolution

$$\dot{\bar{m}}_t(x) = (a - b^2 P_t)\bar{m}_t(x) - b^2 r_t^x, \qquad \bar{m}_0(x) = \int \xi \, m_0^x(\mathrm{d}\xi). \tag{23.26}$$

Equations eq. (23.25)–eq. (23.26), with the coupling $z_t = \mathbb{W}\bar{m}_t$, form a *linear forward–backward two-point boundary value problem in the fields* $(\bar{m}_\cdot(\cdot), r_\cdot(\cdot))$ over types, coupled across types solely by the operator $\mathbb{W}$ — the deterministic shadow of the graphon FBSDE, and the field analogue of the scalar two-point problem eq. (9.25)–eq. (9.26).

Diagonalization: the spectral normal modes. Here the operator structure pays off. By **(A-Reg-W)** the operator $\mathbb{W}$ is self-adjoint Hilbert–Schmidt on $L^2([0,1])$, so by the spectral theorem theorem 2.26 (the network "band structure" of chapter 18) there is an orthonormal eigenbasis $\{\phi_k\}_{k\geq 1}$ of $L^2([0,1])$ with $\mathbb{W}\phi_k = \lambda_k \phi_k$, $|\lambda_1| \geq |\lambda_2| \geq \cdots \to 0$ and $\|\mathbb{W}\| = |\lambda_1|$ (the indexing and the leading-eigenvalue convention are those of section 18.1, $\rho(\mathbb{W}) = \|\mathbb{W}\| = |\lambda_1|$). Expand the two type-fields at each time in this basis, with time-dependent Fourier coefficients,

$$\bar{m}_t(x) = \sum_k \bar{m}_t^{(k)} \phi_k(x), \qquad r_t^x = \sum_k r_t^{(k)} \phi_k(x), \qquad \bar{m}_t^{(k)} = \langle \bar{m}_t, \phi_k \rangle, \quad r_t^{(k)} = \langle r_t^{\cdot}, \phi_k \rangle,$$

the coefficients being the $L^2([0,1])$ inner products against the eigenfunctions. The single fact that makes the system fall apart is that $\mathbb{W}$ acts diagonally on this basis: applying it to the expansion of $\bar{m}_t$ and using $\mathbb{W}\phi_k = \lambda_k \phi_k$ termwise,

$$z_t(x) = (\mathbb{W}\bar{m}_t)(x) = \sum_k \bar{m}_t^{(k)} \, (\mathbb{W}\phi_k)(x) = \sum_k \lambda_k \, \bar{m}_t^{(k)} \, \phi_k(x),$$

so the coupling field z_t has k-th coefficient $\lambda_k \bar{m}_t^{(k)}$ — the operator $\mathbb{W}$ has become, mode by mode, multiplication by the scalar λ_k. Now project the field ODEs eq. (23.25)–eq. (23.26) onto ϕ_k, i.e. take the $L^2([0,1])$ inner product of each side with ϕ_k. This works because every coefficient in those ODEs is *type-independent* (the scalars a, b^2, P_t multiply the fields pointwise in x), so the projection passes straight through them: $\langle (b^2 P_t - a) r_t^x, \phi_k \rangle = (b^2 P_t - a)\langle r_t^{\cdot}, \phi_k \rangle = (b^2 P_t - a) r_t^{(k)}$, and likewise for every other term. The only term that could have mixed modes is the coupling $q\, s\, z_t$, and that is precisely the term $\mathbb{W}$ has just diagonalized: $\langle q\, s\, z_t, \phi_k \rangle = q\, s\, \lambda_k \bar{m}_t^{(k)}$. Because the eigenbasis is orthonormal, the projection picks out exactly the k-th coefficient of each field and leaves no cross-terms, so eq. (23.25)–eq. (23.26) *decouples the continuum into independent scalar systems*, one for each mode:

$$\begin{cases} \dot{r}_t^{(k)} = (b^2 P_t - a) r_t^{(k)} + q\,(s\lambda_k)\, \bar{m}_t^{(k)}, & r_T^{(k)} = -q_T\,(s_T \lambda_k)\, \bar{m}_T^{(k)}, \\ \dot{\bar{m}}_t^{(k)} = (a - b^2 P_t)\bar{m}_t^{(k)} - b^2 r_t^{(k)}, & \bar{m}_0^{(k)} = \langle \bar{m}_0, \phi_k \rangle. \end{cases} \tag{23.27}$$

But eq. (23.27) is *identical* to the homogeneous Thread A two-point system eq. (9.25)–eq. (9.26) with the coupling strengths rescaled,

$$s \;\longmapsto\; s_{\text{eff}}^{(k)} = s\, \lambda_k, \qquad s_T \;\longmapsto\; s_{T,\text{eff}}^{(k)} = s_T\, \lambda_k. \tag{23.28}$$

Thus *the graphon LQG mean field game is a direct sum of scalar LQG mean field games, one per eigenmode of the coupling operator* $\mathbb{W}$, *the* k*-th carrying the effective coupling* $s\lambda_k$. At $K = 1$ active mode this is the scalar boundary value problem eq. (22.28) of the analytic worked computation worked computation 22.15: the off-diagonal product $b^2 qs\lambda_k$ governing the mode oscillation matches, so the two derivations agree on the phase boundary. Each mode is then closed exactly as Thread A was closed in chapter 9, by a *second* decoupling, now of the offset coefficient onto the mode mean. The pair eq. (23.27) is a linear forward–backward system in the two scalars $(\bar{m}^{(k)}, r^{(k)})$; positing the linear relation $r_t^{(k)} = \pi_t^{(k)}\bar{m}_t^{(k)}$ between them, differentiating, $\dot{r}_t^{(k)} = \dot{\pi}_t^{(k)}\bar{m}_t^{(k)} + \pi_t^{(k)}\dot{\bar{m}}_t^{(k)}$, and substituting the mean equation $\dot{\bar{m}}_t^{(k)} = (a - b^2P_t)\bar{m}_t^{(k)} - b^2 r_t^{(k)} = (a - b^2P_t - b^2\pi_t^{(k)})\bar{m}_t^{(k)}$ on the right, the offset equation of eq. (23.27) becomes, after dividing through by the common factor $\bar{m}_t^{(k)}$, the second scalar Riccati of eq. (9.27),

$$\dot{\pi}_t^{(k)} = b^2(\pi_t^{(k)})^2 - 2(a - b^2P_t)\pi_t^{(k)} + q\, s\lambda_k, \qquad \pi_T^{(k)} = -q_T\, s_T\lambda_k, \tag{23.29}$$

the terminal value forced by $r_T^{(k)} = -q_T s_T \lambda_k \bar{m}_T^{(k)} = \pi_T^{(k)}\bar{m}_T^{(k)}$. With $\pi^{(k)}$ in hand the mode mean closes into the single linear ODE $\dot{\bar{m}}_t^{(k)} = (a - b^2P_t - b^2\pi_t^{(k)})\bar{m}_t^{(k)}$, solved by an exponential, and $r_t^{(k)} = \pi_t^{(k)}\bar{m}_t^{(k)}$ is read off. Reassembling the modes, $\bar{m}_t(x) = \sum_k \bar{m}_t^{(k)}\phi_k(x)$, the equilibrium feedback is $\hat{\alpha}_t^x = -b(P_tX_t^x + r_t^x)$ with $r_t^x = \sum_k \pi_t^{(k)}\bar{m}_t^{(k)}\phi_k(x)$.

With this linear feedback substituted into the state SDE the closed loop reads $\mathrm{d}X_t^x = \big((a - b^2P_t)X_t^x - b^2r_t^x\big)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x$, which is a linear (Ornstein–Uhlenbeck) SDE: its drift is affine in X_t^x with *type-independent* slope $a - b^2P_t$ and a deterministic, type-dependent forcing $-b^2r_t^x$. The law $m_t^x = \mathrm{Law}(X_t^x)$ is therefore Gaussian, $\mathcal{N}(\bar{m}_t(x), v_t)$, and here a small but telling feature appears: the *mean* carries all the type-dependence and all the coupling, while the *variance* carries neither. The mean obeys eq. (23.26), forced by the offset and hence by the graphon; the variance $v_t = \mathrm{Var}(X_t^x)$ obeys the Lyapunov ODE

$$\dot{v}_t = 2(a - b^2P_t)v_t + \sigma^2, \qquad v_0 = \mathrm{Var}(m_0^x),$$

obtained by Itô on $\big(X_t^x - \bar{m}_t(x)\big)^2$: the deterministic forcing $-b^2r_t^x$ cancels in the centred variable, so it never reaches the variance. This is the structural reason the variance *decouples from the mean and from the network*: the cost eq. (23.23) couples an agent only to the neighbour *mean* $z_t(x)$, a first-moment quantity, so the entire influence of the crowd lives in the drift's deterministic part and shifts means without ever touching fluctuations. The Lyapunov coefficient $a - b^2P_t$ is type-independent because P_t is, so v_t is the *same* for every type — one variance for the whole continuum, identical to the Thread A variance of chapter 9. This reproduces, from the probabilistic side, the closed-form graphon LQG equilibrium of chapter 22, mode for mode: the analytic forward–backward solution there and the spectral direct sum here are the same equilibrium written in two pictures.

Reductions, recovered exactly. With the game written as a direct sum indexed by the spectrum, the three thread-specification reductions that remark 23.4 previewed at the descriptive level now become *spectral* computations: each is just a matter of naming the eigenvalues of the relevant $\mathbb{W}$ and reading off eq. (23.28).

Constant graphon $W \equiv c$. The operator acts by $(\mathbb{W}f)(x) = \int_0^1 c\,f(y)\,\mathrm{d}y = c\langle 1, f\rangle_{L^2}$, a constant function of x; in operator form $\mathbb{W}f = c\langle 1, f\rangle_{L^2}\,1$, a rank-one operator. Its range is the line of constants, so the normalized constant $\phi_1 \equiv 1$ (normalized because $\|1\|^2_{L^2([0,1])} = \int_0^1 1\,\mathrm{d}y = 1$) is the only eigenfunction with nonzero eigenvalue, $\mathbb{W}\phi_1 = c\,\phi_1$ so $\lambda_1 = c$; every ϕ orthogonal to the constants integrates to zero and is annihilated, $\lambda_k = 0$ for $k \geq 2$. Only mode 1 couples, with effective strength sc, and its coordinate $\bar{m}_t^{(1)} = \langle \bar{m}_t, 1\rangle = \int_0^1 \bar{m}_t(x)\, \mathrm{d}x$ is precisely the global population mean. The continuum collapses to the single scalar Thread A LQG MFG with coupling sc — Thread A recovered with rescaled coupling, exactly as the thread specification demands.

Rank-one graphon $W(x,y) = f(x)f(y)$ with $f \geq 0$ a centrality profile. Now $(\mathbb{W}g)(x) = \int_0^1 f(x)f(y)g(y)\, \mathrm{d}y = f(x)\langle f, g\rangle_{L^2}$, i.e. $\mathbb{W}g = \langle f, g\rangle_{L^2} f$, again rank one with range the line spanned by f. Normalizing, $\phi_1 = f/\|f\|_{L^2}$ is the eigenfunction, and $\mathbb{W}\phi_1 = \langle f, f/\|f\|_{L^2}\rangle_{L^2} f = \|f\|_{L^2} f = \|f\|_{L^2}^2 \phi_1$, so the single nonzero eigenvalue is $\lambda_1 = \|f\|_{L^2}^2$. Only the order-parameter mode $\langle \phi_1, \bar{m}_t\rangle = \|f\|_{L^2}^{-1} \langle f, \bar{m}_t\rangle$ couples, with effective strength $s\lambda_1 = s\|f\|_{L^2}^2$, and the coupling field is $z_t(x) = f(x)\langle f, \bar{m}_t\rangle = f(x)\, \varrho_t$ — a single scalar order parameter $\varrho_t = \langle f, \bar{m}_t\rangle$ modulated by the centrality profile, exactly the object isolated in remark 23.4. The whole graphon LQG game collapses to a *one-parameter* self-consistent equation for ϱ_t, the cleanest GMFG worked example and the rank-one reduction the thread specification asks for. Every other mode has $\lambda_k = 0$ and hence zero effective coupling: those directions play the uncoupled single-agent Ornstein–Uhlenbeck problem of chapter 6, feeling no crowd at all.

Block graphon. If W is piecewise constant on a partition of $[0,1]$ into K type-blocks (definition 22.12), then $\mathbb{W}$ factors through the K-dimensional space of functions constant on the blocks — it is a finite-rank operator of rank at most K, whose nonzero eigenvalues are those of the $K \times K$ block-weight matrix. At most K modes λ_k carry coupling, so the direct-sum-of-scalars truncates to a finite K-mode system: the continuum collapses to a finite *multi-population* LQG mean field game, one effective population per block, coupled through that $K \times K$ matrix. This is exactly the bridge to chapter 30 made on the analytic side in proposition 22.13 and remark 22.14, here read straight off the spectrum. The three reductions together complete the check set the thread specification asks for at every appearance of Thread B.

Multiplicity and the spectral threshold. Now we cash the chapter's sharpest claim. Because mode k is a verbatim copy of the Thread A LQG mean field game with *signed* effective coupling $s\lambda_k$, it inherits the entire phase behaviour worked out for Thread A in chapter 14 — which we import, not reprove. The criterion there (theorem 14.11, the special case of the general uniqueness threshold proposition 11.8) is exactly the global solvability of the second Riccati eq. (23.29): the mode equilibrium exists and is unique precisely while $\pi_t^{(k)}$ stays finite on $[0,T]$, which holds while the horizon stays below the Thread A critical horizon $T_c(s\lambda_k)$ of eq. (14.24) evaluated at the effective coupling. The content we must *develop* is why this criterion is *sign-dependent*, because that is what selects the binding mode and fixes the threshold.

Read eq. (23.29) as a scalar Riccati whose constant term is the effective coupling $q\, s\lambda_k$. When $s\lambda_k > 0$ — the crowd-*seeking* direction, agents pulled *toward* the neighbour mean — this constant term has the destabilizing sign: it pushes $\pi^{(k)}$ in the direction where the quadratic term $b^2(\pi^{(k)})^2$ takes over and drives a finite-time blow-up, so $T_c(s\lambda_k)$ is finite and shrinks as $s\lambda_k$ grows. The mechanism is the familiar positive feedback of imitation: each agent chasing the crowd makes the crowd a more attractive target, and beyond a critical strength-times-horizon the self-consistency loop supports more than one equilibrium. When $s\lambda_k < 0$ — the crowd-*averse* or anti-imitation direction, agents pushed *away* from the mean — the constant term carries the opposite sign and *stabilizes* the Riccati: the coupling now adds to the restoring effect rather than opposing it, $\pi^{(k)}$

stays bounded for all time, and the mode equilibrium is unique at *every* horizon. This is the negative-feedback, Lasry–Lions-monotone regime (**(A-Mon)**). The asymmetry is the crux: it is not the *magnitude* $|\lambda_k|$ of an eigenvalue that threatens uniqueness but its *signed* effective coupling $s\lambda_k$. A large *negative* eigenvalue — a strong anti-imitation mode — has large $|\lambda_k|$ yet is *extra* stable: the more strongly that mode repels from its own mean, the more firmly its single equilibrium is pinned, and it never bifurcates no matter how long the horizon. Consequently the binding mode, the one that goes critical first, is the mode of largest *signed* $s\lambda_k$: for $s > 0$ the algebraically largest (most positive) eigenvalue, for $s < 0$ the most negative one. This is exactly the binding-mode selection that proposition 24.23 will state in general.

For a *nonnegative* graphon the binding mode is pinned by spectral theory. Perron–Frobenius for the self-adjoint compact $\mathbb{W}$ (section 18.1) places the top of the spectrum at $\lambda_1 = \|\mathbb{W}\| > 0$, attained on a nonnegative eigenfunction $\phi_1 \geq 0$ — the eigenvector-centrality profile of chapter 18, which weights each type by how central it is in the network. So for the crowd-seeking sign $s > 0$ the algebraically largest signed coupling is $s\lambda_1 = s\,\|\mathbb{W}\|$, carried by the centrality mode, and as the horizon grows or the coupling is turned up it is *this* mode that bifurcates first. The onset of multiplicity is therefore governed by the spectral operator norm, in closed form: while $s\,\|\mathbb{W}\|$ stays below the Thread A critical coupling every mode is subcritical and the graphon equilibrium is unique — precisely the operator-norm smallness **(A-Coupling)** that theorem 23.17 imposed abstractly, now seen explicitly for the linear-mean coupling and seen to involve the *spectral* norm, not the row-sum that governed the descriptive contraction. As $s\,\|\mathbb{W}\|$ crosses that critical value the leading centrality mode goes critical and multiplicity sets in as a *symmetry breaking concentrated on the single eigenfunction* ϕ_1: the continuum of equilibria that appears differs from the unique one along the centrality direction alone. This exact phase boundary is the LQG instance of the general critical-coupling phenomenon of chapter 24 (proposition 24.23), and it is the closed-form vindication of the two-norms distinction drawn abstractly in section 23.4: the row-sum $\|\mathbb{W}\mathbf{1}\|_\infty$ controls how measures contract, the spectral norm $\|\mathbb{W}\|$ controls where uniqueness fails.

Physics note

The diagonalization eq. (23.27) is the mean field game version of resolving a system of coupled linear oscillators into *normal modes*. The coupling operator $\mathbb{W}$ plays the role of the coupling matrix; its eigenfunctions ϕ_k are the normal coordinates, in which the continuum of strategically interacting types decouples into independent one-dimensional problems. Two features carry real content rather than mere analogy. First, the "stiffness" felt in mode k is set by the eigenvalue λ_k through the effective coupling $s\lambda_k$ of eq. (23.28): in the crowd-seeking direction ($s\lambda_k > 0$) the strongly coupled, central, well-connected combinations of types are the soft ones, while peripheral directions (λ_k near zero) are nearly free and anti-imitation directions ($s\lambda_k < 0$) are stiffened. Second, the loss of uniqueness is a *soft-mode instability*: as a crowd-seeking coupling is raised the leading eigenmode — maximal eigenvector centrality, $\lambda_1 = \|\mathbb{W}\|$ for a nonnegative kernel — is the first to go unstable, and the multiplicity that appears is a symmetry breaking polarized along that single eigenfunction ϕ_1, exactly as an order parameter condenses into the softest mode at a continuous transition. The analogy organizes; the content is the exact reduction eq. (23.27) and the threshold $s\,\|\mathbb{W}\| =$ critical, both of which stand on their own.

This worked computation closes the chapter's argument. The thesis we carried from the first page — that the entire cross-type coupling of a graphon mean field game is the single bounded operator $\mathbb{W}$ — is here made fully explicit and exactly solvable: diagonalizing that one operator turns the continuum of strategically coupled types into a direct sum of scalar Thread A games, one per eigenmode with coupling $s\lambda_k$, and the same operator's leading eigenvalue $\|\mathbb{W}\|$ fixes in closed form the symmetry-breaking threshold $s\,\|\mathbb{W}\|$ at which uniqueness first fails. Four threads leave this section and run on. The general critical-coupling statement, of which the boundary $s\,\|\mathbb{W}\|$ here is the LQG instance, is proved for arbitrary linear-mean coupling in chapter 24 (proposition 24.23); the mode-for-mode agreement with the analytic rank-one boundary value problem eq. (22.28) ties this probabilistic closed form back to the PDE solution of chapter 22; the leading mode $\phi_1 = \|\mathbb{W}\|$ is the eigenvector-centrality profile of chapter 18, so the agents whose imitation destabilizes the game first are exactly the central ones; and the block-graphon reduction to a finite multi-population game is the bridge picked up in chapter 30. What this section does *not* settle — the structure of the multiple equilibria past the threshold, beyond the small-coupling and monotone regimes where uniqueness is guaranteed — is recorded as an Open item in the ledger below, which together with the bibliographic notes closes the chapter.

Rigor ledger

Proved (in this chapter).

- The graphon McKean–Vlasov SDE eq. (23.3) with interaction through the graphon aggregate eq. (23.1), its well-posedness (proposition 23.5) by a field-valued Picard/-contraction argument, and the associated nonlinear graphon Fokker–Planck equation proposition 23.7, identified with the forward half of the system of chapter 22.

- The Bochner field space $\boldsymbol{\mathcal{P}}_2$ (definition 23.8): its completeness and the field-contraction bound for the coupling, $\big(\int \mathcal{W}_2((\mathbb{W}\boldsymbol{m})(x),(\mathbb{W}\boldsymbol{m}')(x))^2\,\mathrm{d}x\big)^{1/2} \le \|\mathbb{W}\mathbf{1}\|_\infty^{1/2}\,\mathbf{d}_2(\boldsymbol{m},\boldsymbol{m}') \le \mathbf{d}_2(\boldsymbol{m},\boldsymbol{m}')$ (lemma 23.10, by Kantorovich duality theorem 4.5), the single estimate carrying the kernel row-sum into every measure-valued contraction — with the equal-mass extension of $\mathcal{W}_2$ (definition 23.9) that makes the sub-probability aggregates comparable.

- The graphon MFG FBSDE eq. (23.16) and its fixed-point structure: the per-type decoupling under a frozen field (remark 23.14), the graphon best-response map (definition 23.15), and the small-horizon contraction estimate with field factor $\le \|\mathbb{W}\mathbf{1}\|_\infty^{1/2} \le 1$ giving theorem 23.17(i).

- The graphon FBSDE–PDE dictionary proposition 23.19: $Y_t^x = \nabla u^x(t, X_t^x)$, $Z_t^x = \mathrm{D}^2\,u^x\,\sigma$ (shape $d \times k$), type by type, with the family of decoupling fields theorem 23.21 — the graphon analogue of proposition 9.12.

- The complete graphon LQG thread in FBSDE form (worked computation 23.22): the spectral diagonalization into independent scalar LQG modes with effective couplings

$s\lambda_k$, the exact recovery of the constant-, rank-one-, and block-graphon reductions, and the closed-form onset of multiplicity at the spectral threshold $s\,\|\mathbb{W}\|$ (in the destabilizing direction $s\lambda_k > 0$).

Assumed (named standing hypotheses).

- **(A-Lip)**, **(A-Growth)** for SDE/BSDE well-posedness and Lipschitz feedback; **(A-Conv)** for the Hamiltonian minimizer and sufficiency of the maximum principle; **(A-Nondeg)** for parabolic smoothing of the decoupling fields and for the graphon Fokker–Planck; **(A-Reg-W)** for the self-adjoint Hilbert–Schmidt structure of $\mathbb{W}$ and the spectral diagonalization; **(A-Coupling)** (spectral $\|\mathbb{W}\|$ small, decisive for the linear-mean coupling) and **(A-Mon)** for uniqueness.
- Uncontrolled diffusion throughout (controlled diffusion deferred as in remark 9.7).

Deferred (with pointers).

- Existence/uniqueness of the graphon MFG FBSDE: stated as theorem 23.17 in three regimes; the Schauder/compactness proof and the graphon Lasry–Lions monotonicity with its pairing identity are chapter 24, carried out there in the *uniform-in-type* field space $C([0,1]; C([0,T]; \mathcal{P}_1(\mathbb{R}^d)))$ (definition 24.1), the FBSDE/$\boldsymbol{\mathcal{P}}_2$ statement here transferred by the two-pictures equivalence proposition 23.19; the sharper L^2-in-type compactness criterion is more delicate (remark 24.13). Primary sources [34, 10, 15].
- The rigorous joint realization of the continuum of Brownian motions (Fubini extension / exact law of large numbers): remark 23.6 and [39, 10]; only needed in earnest for the finite-N descent of chapter 25.
- The measure-dependent (master) decoupling field and the graphon master equation: chapter 26, promoting theorem 23.21 as eq. (9.19) promoted the homogeneous decoupling field.
- BSDE theory (Pardoux–Peng, martingale representation): chapter 5 and the stochastic-calculus reference, Appendix C.

Open / honestly flagged.

- Uniqueness outside the small-coupling and monotone regimes is genuinely open: theorem 23.17(iii) gives existence only, and the worked computation worked computation 23.22 shows multiplicity is real once $s\,\|\mathbb{W}\|$ crosses the Thread A threshold (in the crowd-seeking direction), the leading centrality mode bifurcating first.
- The sharp critical coupling, the interplay of strategic monotonicity with the *spectrum* (not merely the norm) of $\mathbb{W}$, and well-posedness for unbounded (L^p) graphons where $\mathbb{W}$ need not be bounded on L^2, are open in general and taken up in chapters 24 and 27.
- The probabilistic theory above is for dense, bounded graphons; the descent to sparse sampled networks, where the continuum-of-noises construction and the operator bound both degrade, is the frontier of chapters 25 and 27.

Bibliographic notes

The probabilistic formulation of graphon mean field games — the continuum-indexed McKean–Vlasov FBSDE and its fixed point in a space of measure-valued maps — is the graphon lift of the Carmona–Delarue programme [40, 41], whose Volume I, Chapters 3–4, supplies every per-type ingredient we import: the stochastic maximum principle, the adjoint BSDE, the McKean–Vlasov FBSDE, the small-time and monotone well-posedness theorems, and the decoupling-field (four-step / continuation) method. The graphon mean field *system* as a descriptive object — the interacting diffusions eq. (23.3) and their propagation of chaos toward the deterministic field of laws — is developed by Bayraktar, Chakraborty and Wu [15]; the GMFG equations in the dynamic, optimizing setting are due to Caines and Huang [33, 34]. The careful measure-theoretic treatment of a continuum of agents and of the continuum of independent noises — the Fubini extension and exact law of large numbers underlying remark 23.6 — is carried out in the stochastic-graphon-games series of Carmona, Cooney, Graves and Laurière [39] (static case) and Aurell, Carmona and Laurière [10] (the linear–quadratic dynamic case, which is the closest published companion to the worked computation worked computation 23.22). The two-pictures equivalence proposition 23.19 is the graphon counterpart of the homogeneous result of chapter 9, itself the FBSDE form of the Lasry–Lions system [93]; the analytic side it is matched against is chapter 22. The spectral diagonalization of the graphon LQG game into normal modes of $\mathbb{W}$ uses the self-adjoint Hilbert–Schmidt structure established in chapter 2 and the network spectral theory of chapter 18; it is the probabilistic, dynamic analogue of the centrality-to-equilibrium correspondence proved for static graphon games by Parise and Ozdaglar [108]. The existence and uniqueness theory whose three regimes we have only stated, and the critical-coupling threshold the worked computation exhibits in closed form, are the subject of chapter 24; the descent to finite sampled networks is chapter 25; the graphon master equation and common noise are chapter 26.

Chapter 24

Existence and Uniqueness for Graphon Mean Field Games

The two preceding chapters assembled the dynamic graphon mean field game (GMFG) in its two complementary pictures, but neither asked whether the object they describe is ever actually there. Chapter 22 gave the analytic picture: a continuum of backward Hamilton–Jacobi–Bellman (HJB) equations indexed by the type $x \in [0,1]$, each driven by the population, coupled across types only through the graphon operator $\mathbb{W}$, paired with a continuum of forward Fokker–Planck (FP) equations for the type-indexed population field $m^x = m^x$. In that picture an equilibrium *is* a self-consistent pair of fields $(\mathbf{u}, \mathbf{m})$ — a value field and a density field, one of each per type — with the property that the population the optimal feedbacks generate is exactly the population those feedbacks were computed against. Chapter 23 gave the probabilistic picture: a continuum-indexed family of forward–backward stochastic differential equations of graphon McKean–Vlasov type, and the proof that, under regularity, the two pictures coincide. There an equilibrium *is* a type-indexed family of state processes whose laws, fed back through $\mathbb{W}$ into the coefficients that drive them, reproduce themselves. The two descriptions are one fixed-point condition read in two languages, and the two-pictures dictionary proposition 23.19 translates between them. What neither chapter supplies is the assertion that a field satisfying the condition *exists*, still less that it is *unique*. The present chapter supplies both, and isolates exactly the price the graphon structure exacts over the classical theory of chapters 10 and 11.

The method is the one the reader already owns from the homogeneous theory, and the single organising claim of this chapter is that it is *all* we need: the entire argument is the classical proof of chapters 10 and 11 with one new object, the graphon operator $\mathbb{W}$, inserted at exactly one place. We recast the equilibrium condition as a fixed point of a best-response map, manufacture a convex compact invariant set, and invoke Schauder; then, separately, buy uniqueness either from a monotonicity structure (good for every horizon) or from a smallness condition (good when the coupling is weak). Everything from chapter 10 (the Schauder fixed-point machine, the moment/equicontinuity estimates, tightness via Prokhorov from chapter 3) and from chapter 11 (the Lasry–Lions duality identity, the standing monotonicity assumption **(A-Mon)**, the Bregman lemma) is imported and reused *verbatim*, run type by type. What is genuinely new is that the coupling now factors through the integral operator $\mathbb{W}$ of chapter 18, and the discipline we hold throughout is that every graphon-specific phenomenon is the responsibility of that one operator and lives in its spectrum. Concretely, the best-response map will be seen to split as $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$, a *linear* aggregation $\mathcal{A}_W$ through $\mathbb{W}$ followed by a type-wise *classical* solve $\mathcal{R}$; reading every later theorem through this factorisation is what keeps the inherited difficulty and the new difficulty from being confused.

Two questions stand between us and the theorems, and the reader should hold both in mind because the surprise of the chapter is that *both answers are spectral.* The first is *compactness.* The classical existence proof compresses a sequence of candidate flows into a convergent subsequence; here a candidate is an entire field over types, and a field can oscillate ever faster in the type direction with nothing to stop it. There is no analogue of this difficulty in the homogeneous theory — there was no type variable, hence nothing to be compact in — so the existence proof acquires a new compactness obligation it never had before, and it is met by buying type-direction equicontinuity from the *continuity in type* of W (the standing assumption **(A-Reg-W)**). This is the genuinely structural surprise to plant now: in the classical theory existence needed no regularity of any coupling kernel, yet the graphon *existence* theorem — not merely the uniqueness theorem — fails without a regularity hypothesis on $\mathbb{W}$, because regularity is the only source of the missing compactness. The second question is the *sign of the equilibrium energy*: the Lasry–Lions identity that delivered uniqueness in chapter 11 now carries an extra type integral threaded through $\mathbb{W}$, and whether its sign survives is decided jointly by the strategic monotonicity and the spectrum of $\mathbb{W}$. The critical coupling at which uniqueness fails is governed by the leading eigenvalue $\lambda_1(\mathbb{W})$ — the eigenvector centrality of chapter 18. This is the precise sense in which the scalar threshold $s = 1$ of chapter 11 is replaced by the product $s\,\lambda_1$, ties the chapter back to that scalar threshold and to the operator-norm condition of the static graphon game in chapter 20, and makes loss of uniqueness a symmetry-breaking transition indexed by network centrality. Both answers — the compactness one and the sign one — are thus read off the same operator $\mathbb{W}$: this is the chapter in one sentence.

We proceed in four movements. Section 24.1 fixes the field-of-measures state space — the type-indexed populations of chapter 21, here specialised to the variant that graphon continuity makes available — recalls the GMFG system from chapter 22 without rederiving it, and defines the graphon best-response map, exhibiting its factorisation through $\mathbb{W}$. Section 24.2 proves existence, making explicit the two compactness inputs — per-type tightness (Part I) and type-direction equicontinuity from graphon regularity (Part III) — and the role of $\|\mathbb{W}\|$. Section 24.3 proves uniqueness by two routes, the graphon Lasry–Lions monotonicity identity and the small-coupling contraction, and locates the critical coupling. Section 24.4 states plainly what is harder here than in the classical theory. Section 24.5 advances Thread B: the graphon LQG model is carried to its exact uniqueness/multiplicity threshold, which is shown to be the chapter 11 threshold with the scalar coupling s replaced by $s\,\lambda_1(\mathbb{W})$, and the rank-one and constant-graphon reductions are verified. The chapter is the operator generalisation of chapter 11 and the existence base on which the convergence theory of chapter 25 stands.

Notation watch

We are now firmly in Part IV, where a graphon $W = W$ is always in scope. The letters B and W have, in earlier parts of the book, both wanted to mean Brownian motion; the resolution fixed in appendix A.5 is that once a graphon is present, W denotes the graphon kernel and *never* the noise, and Brownian motion is written B_t. We honour that throughout this chapter without further comment: every W below is the kernel, every B^x a type-x Brownian motion. A second, more consequential point of hygiene concerns the type variable $x \in [0, 1]$. It is an exchangeable *label*, an index of an agent's kind or network position, not a point of physical space; in particular the distance $|x - y|$ carries no intrinsic meaning, and only the value $W(x, y)$ the kernel assigns to a pair of labels does. A graphon is, canonically, an equivalence class under measure-preserving relabelling of $[0, 1]$, so nothing in its definition privileges one arrangement of the labels over another. There is exactly one place in this chapter where we quietly break that symmetry: the *continuity* half of **(A-Reg-W)**. When we ask that W be uniformly continuous in its type argument — used only to manufacture type-direction compactness in the existence theorem, through lemma 24.8 — we must first *choose* a particular representative of the graphon and endow $[0, 1]$ with its Euclidean metric, purely so that "continuity in x" has a referent. This is non-canonical extra structure, imposed for that one hypothesis alone. The caveat is load-bearing, not decorative: it is precisely the structure that a graphon given only up to relabelling does not possess, so its unavailability is what forces the harder, merely-measurable treatment of remark 24.13, where continuity is replaced by the Hilbert–Schmidt compactness of $\mathbb{W}$ itself.

24.1 Equilibrium as a fixed point on the field of measures

The preamble promised that an equilibrium is no longer a single flow of measures but an entire *field* of them, one flow per type, and that the types talk to one another only through the graphon operator $\mathbb{W}$. Before any existence or uniqueness can be argued, that promise has to be made into mathematics: we need a concrete space for the unknown to live in, a concrete map whose fixed points are the equilibria, and a clear view of where inside that map the graphon enters. This section does exactly these three things, in order. First we fix the ambient space — the field-of-measures space $\mathbb{M}$, its second-moment leash $\mathbb{M}_2$, and the metric D that measures distance between fields. Then we recall, without rederiving it, the GMFG system of chapter 22 and the aggregation through $\mathbb{W}$ that couples its type-indexed equations. Finally we assemble these into the graphon best-response map and expose its internal structure: the factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ that cleanly separates the classical difficulty, carried by the type-wise solve $\mathcal{R}$, from the graphon difficulty, carried by the linear aggregation $\mathcal{A}_W$. That factorisation is the lens through which every theorem in the rest of the chapter is meant to be read, so the section's real payload is the map's anatomy, not just its definition. We begin with the state space, where the single flow $t \mapsto m_t$ of definition 10.1 must first be promoted to a continuous map from types to flows.

24.1.1 The field-of-measures state space

In the homogeneous theory of chapter 10 a candidate equilibrium was a single object: a flow of measures $t \mapsto m_t$, the running population of one representative agent's environment (definition 10.1). There was one population because there was one kind of agent. The whole content of the graphon turn is that there is now a continuum of kinds — the types $x \in [0,1]$ — and each kind carries its own population, evolving in its own way and felt differently by every other kind through the kernel W. A candidate equilibrium is therefore no longer a single flow but an entire *field* of them, one flow per type. To make the promise of the preamble into mathematics we must say precisely what such a field is, in what space it lives, and with what distance we measure two of them apart; only then can a best-response map be defined on that space and its fixed points hunted. That is the business of this subsection: we promote the single flow $t \mapsto m_t$ of definition 10.1 to a continuous map $x \mapsto m^x$ from types to flows, fix the metric that makes it a usable state space, and extract the one elementary inequality — the first moment is a 1-Lipschitz functional of the law — that every continuity and contraction estimate downstream will lean on.

It helps to picture the object as having two nested levels, an *inner* and an *outer* one, because the two levels are governed by two different kinds of regularity and it is worth keeping them apart from the start. The inner level is the one we already understand. Fix a single type x; then $m^x = (m^x_t)_{t\in[0,T]}$ is exactly a flow of measures of the chapter 10 kind — a curve in the space of probability measures, continuous in *time*, carrying the type-x population from its initial datum m^x_0 to its terminal configuration. Everything Part I built about such a flow — its tightness, its time-modulus, its behaviour under the best-response solve — applies to it verbatim, because nothing in those estimates ever knew about a type variable. The outer level is genuinely new. We now ask how the flow m^x varies as the type x ranges over $[0,1]$: a field is a curve of flows, parametrised by type. The inner regularity is in time and is inherited; the outer regularity is in type and must be supplied, and it is precisely the outer regularity that the graphon will have to pay for. Holding this two-level picture — flow-in-time inside, field-in-type outside — in mind is the cleanest way to track which difficulties are old and which are new.

The field-of-measures space itself is not invented here; it is owned by chapter 21, which introduces *two* variants and we must choose between them. The first is the L^2-in-type Bochner space $L^2([0,1];(\mathcal{P}_2(\mathbb{R}^d),\mathcal{W}_2))$, with the $\mathcal{W}_2$-based metric d_2 (definition 21.8): a field is merely a square-integrable, hence possibly discontinuous, assignment of a flow to each type, and two fields are close when they are close in mean-square over types. The second is the type-*continuous* variant $C([0,1];(\mathcal{P}_2(\mathbb{R}^d),\mathcal{W}_2))$, with the uniform-in-type metric (definition 21.10): a field is a genuinely continuous map from types to flows, and two fields are close when they are close *at every type simultaneously*. We adopt the type-continuous variant, with one adjustment of the inner metric: we measure each type fibre with the $\mathcal{W}_1$ distance rather than $\mathcal{W}_2$, keeping the second moment as a separate a priori leash exactly as in chapter 10, rather than folding it into the metric. The companion probabilistic chapter chapter 23 makes the opposite pair of choices, working in the L^2-in-type, $\mathcal{W}_2$-based variant definition 23.8. The two settings are not rival theories but two refinements — the $\mathcal{W}_1$/uniform-in-type one here, the $\mathcal{W}_2/L^2$-in-type one there — of the single chapter 21 field space, and chapter 23's probabilistic well-posedness and the analytic existence theorem of this chapter are carried back and forth between them by the two-pictures dictionary proposition 23.19. With that orientation fixed we can write the space down.

Definition 24.1 (Field of measure flows)**.** Write $\mathcal{C} := C([0,T];\mathcal{P}_1(\mathbb{R}^d))$ for the flow space of definition 10.1, with the uniform Wasserstein metric $\mathsf{d}_1(m,m') = \sup_t \mathcal{W}_1(m_t,m'_t)$. A *field of measure flows* is a continuous map

$$\mathbf{m} = (m^x)_{x\in[0,1]} \colon [0,1] \longrightarrow \mathcal{C}, \qquad x \longmapsto m^x = (m^x_t)_{t\in[0,T]},$$

and we let

$$\mathbb{M} := C([0,1];\mathcal{C}), \qquad \mathsf{D}(\mathbf{m},\mathbf{m}') := \sup_{x\in[0,1]} \mathsf{d}_1(m^x, m^{x\prime}),$$

denote the ambient space of fields, equipped with the uniform-in-type metric D. We write $\mathbb{M}_2 \subseteq \mathbb{M}$ for the fields with $\sup_x \sup_t \int |\xi|^2 \, m^x_t(\mathrm{d}\xi) < \infty$, the uniform second moment being retained as an a priori leash rather than as part of the topology, exactly as in chapter 10.

The metric D wears the two-level structure on its face: the inner $\sup_t$ inside d_1 measures the field in time, the outer $\sup_x$ measures it across types, and a field is small in D exactly when it is small in $\mathcal{W}_1$ *at every time and every type at once.* Three deliberate choices are encoded here, and each deserves a motivated word rather than a one-line aside, because each will be paid for or cashed in later in the chapter.

The first choice is that the outer topology is *uniform* in the type x: a sequence of fields converges only when it converges at every type simultaneously, not merely in mean square over types. This is the strongest reasonable demand one can place on the type direction, and the question is why we are entitled to make it. The answer is the recurring discipline of the chapter: it is exactly what graphon continuity buys us. The standing regularity assumption **(A-Reg-W)** asks W to be continuous in its type argument, and continuity of the kernel is what will let us prove, in lemmas 24.8 and 24.9, that the best-response output is itself *equicontinuous* in type — that nearby types produce nearby flows, with a modulus that does not depend on which field we started from. An equicontinuous family in a uniform topology is compact (this is the metric Arzelà–Ascoli theorem, and it is the only way we will ever get compactness in the type direction), so the uniform setting is precisely the setting in which the regularity of W converts into the compactness the existence proof needs. Choosing the uniform topology is choosing to spend graphon continuity on type-direction compactness, and it is the cleanest bargain available.

It is worth being honest about what the bargain costs, because the cost is the subject of a forward pointer. If W is only *measurable* in type — the generic situation, since a graphon is canonically an equivalence class under measure-preserving relabelling and such a class need have no continuous representative at all — then "continuity in x" has no meaning to demand of the field, and the uniform topology is simply unavailable. One is then forced down to the L^2-in-type variant: measurable fields valued in $\mathcal{C}$, with an L^p-in-type metric that no longer controls any single type but only a type-average. The compactness that equicontinuity gave us for free must then be recovered by a different mechanism, the Hilbert–Schmidt compactness of the operator $\mathbb{W}$ itself, and the whole treatment becomes more delicate. We flag this honestly now and defer it to remark 24.13; the point to carry forward is that the comfortable uniform-in-type setting of this section is a privilege of continuous graphons, not a free choice.

The second choice is that each fibre is metrised with the first Wasserstein distance $\mathcal{W}_1$ rather than the second, $\mathcal{W}_2$. The reason is the one from chapter 10 and is worth restating because it is structural, not cosmetic. On the whole space $\mathcal{P}_1(\mathbb{R}^d)$, $\mathcal{W}_1$ does not by itself metrise weak convergence — mass can escape to infinity while staying $\mathcal{W}_1$-close in the bulk. But on any set with a uniform bound on a higher moment, that escape is forbidden, and there $\mathcal{W}_1$ *does* metrise weak convergence: convergence in $\mathcal{W}_1$ is then exactly convergence in distribution (theorem 4.8). This is why we keep the second moment as a separate leash rather than promoting to $\mathcal{W}_2$. The leash is doing two distinct jobs at once, and both are quantitative. First, a uniform second-moment bound is precisely a uniform-tightness certificate: by Markov's inequality it confines almost all the mass of every m_t^x to one fixed large ball, so the whole field lives in a tight, hence (Prokhorov, chapter 3) relatively weakly compact, set of measures — this is the state-direction compactness that feeds the Schauder argument in lemmas 24.7 and 24.10. Second, the same bound is what makes the aggregation operator $\mathbb{W}$ act on a *bounded* set of first-moment fields, so that the coupling data felt across all types is uniformly bounded (lemma 24.6) and the best response is uniformly stable. The moment is, in a word, a leash on which the entire chapter's estimates are walked: it keeps the populations from running off to infinity in state while we chase the fixed point in type. Pinning it down a priori, separately from the topology, is what lets us use the weaker and more flexible $\mathcal{W}_1$ metric where we need flexibility and still command the tightness where we need control.

From these two structural choices we can already harvest the one elementary fact that will be used on nearly every page that follows, and it is worth isolating now precisely because it is so small. The graphon couples types only through aggregates of the populations, and the aggregate of interest (definition 24.3) is the *first moment* $\bar{m} = \int \xi \, m(\mathrm{d}\xi)$ of each type's law. If we are to control how the coupling changes when the field changes, we must control how the first moment changes when the law changes — and we measure changes of the law in $\mathcal{W}_1$. The fact we need is therefore exactly that the first moment is a $\mathcal{W}_1$-Lipschitz functional, with the best possible constant. It follows directly from the primal (Kantorovich) definition of $\mathcal{W}_1$ as an infimum over couplings (remark 4.4), with no machinery at all.

Lemma 24.2 (The first moment is 1-Lipschitz in $\mathcal{W}_1$)**.** *For $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$ with first moments $\bar{\mu} = \int \xi \, \mu(\mathrm{d}\xi)$ and $\bar{\nu} = \int \xi \, \nu(\mathrm{d}\xi)$, one has $|\bar{\mu} - \bar{\nu}| \le \mathcal{W}_1(\mu, \nu)$.*

Proof. The argument is the prototype of an estimate we will run many times in this chapter, so it is worth seeing in its barest form: there is no duality, only a coupling. Let $\pi \in \Pi(\mu, \nu)$ be *any* coupling of μ and ν — any probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ whose first marginal is μ and whose second is ν. Because the marginals of π are μ and ν, the difference of first moments is itself an expectation under π: integrating the two coordinate maps against π and subtracting,

$$\bar{\mu} - \bar{\nu} = \int_{\mathbb{R}^d} \xi \, \mu(\mathrm{d}\xi) - \int_{\mathbb{R}^d} \eta \, \nu(\mathrm{d}\eta) = \int_{\mathbb{R}^d \times \mathbb{R}^d} (\xi - \eta) \, \pi(\mathrm{d}\xi, \mathrm{d}\eta),$$

where the second equality just rewrites each single integral as a double integral against π using its marginal property. Now take norms and pass them inside the integral (the triangle inequality for the vector-valued integral, i.e. Jensen):

$$|\bar{\mu} - \bar{\nu}| = \Big| \int_{\mathbb{R}^d \times \mathbb{R}^d} (\xi - \eta) \, \pi(\mathrm{d}\xi, \mathrm{d}\eta) \Big| \le \int_{\mathbb{R}^d \times \mathbb{R}^d} |\xi - \eta| \, \pi(\mathrm{d}\xi, \mathrm{d}\eta).$$

This bound holds for *every* coupling π, so it holds for the infimum over couplings on the right; but that infimum is, by definition, the first Wasserstein distance, $\mathcal{W}_1(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)} \int |\xi-\eta|\ \pi(\mathrm{d}\xi,\mathrm{d}\eta)$ (remark 4.4). Hence $|\bar\mu-\bar\nu| \le \mathcal{W}_1(\mu,\nu)$, which is the claim.

The same inequality drops out of the Kantorovich–Rubinstein dual in one line, which is worth recording because it is the lens chapter 4 usually uses. Fix a unit vector $e \in \mathbb{R}^d$; the scalar map $\xi \mapsto e\cdot\xi$ is 1-Lipschitz, so it is an admissible test function in the dual representation $\mathcal{W}_1(\mu,\nu) = \sup_{\mathrm{Lip}(\varphi)\le 1} \int \varphi\ \mathrm{d}(\mu-\nu)$, giving $e\cdot(\bar\mu-\bar\nu) = \int (e\cdot\xi)\ \mathrm{d}(\mu-\nu) \le \mathcal{W}_1(\mu,\nu)$; taking the supremum over $|e|=1$ recovers $|\bar\mu-\bar\nu| \le \mathcal{W}_1(\mu,\nu)$. The two routes are the two faces of $\mathcal{W}_1$ we will keep alternating between — couplings when we want to construct, test functions when we want to bound — and here they agree to the last constant. □

The constant is sharp: a pair of point masses $\mu=\delta_a$, $\nu=\delta_b$ has the single coupling $\delta_{(a,b)}$, so $\mathcal{W}_1(\delta_a,\delta_b) = |a-b| = |\bar\mu-\bar\nu|$, and the inequality is an equality. The lesson to carry is not the numerical value but the mechanism. The first-moment map $m\mapsto\bar m$ is the simplest of the population observables the graphon will aggregate, and lemma 24.2 is the statement that it does not amplify distances: a $\mathcal{W}_1$-perturbation of a type's law of size ϵ moves its first moment by at most ϵ. Composed with the bounded operator $\mathbb{W}$, this is what will make the aggregation $\mathcal{A}_W$ a Lipschitz — indeed continuous — map of fields in the existence proof (theorem 24.11), and it is the first link in every downstream chain that propagates a change in the field through the coupling, through the per-type solve, and back to a change in the output field. Small as it is, it is the quantitative hinge on which the continuity of the whole best-response map turns.

With the state space fixed — the field space $\mathbb{M}$, its second-moment-bounded subset $\mathbb{M}_2$, and the uniform-in-type metric D — and with the first-moment leash in hand, we have a precise arena in which the unknown of the graphon game lives. The next subsection puts that arena to work: it recalls, without rederiving it, the GMFG PDE system of chapter 22 and states what a *classical solution* of it is — a pair of fields $(\mathbf{u},\mathbf{m})$ in exactly the space just built — and isolates the one place where the types talk to each other, the aggregation of the population field through $\mathbb{W}$. The 1-Lipschitz first-moment bound of lemma 24.2 is the elementary input that aggregation operator will consume on its very first line.

24.1.2 The GMFG system and the aggregation operator

The arena for the unknown is now fixed. What lives in it is a solution of the GMFG system, and before we can hunt for one we must say precisely what such a solution is. The purpose of this subsection is to make that statement and, in doing so, to put a finger on the one place where the types are wired together — the single seam that the rest of the chapter prises apart. We recall the system from chapter 22 rather than rederive it, read off what a *classical solution* is as a pair of fields, and then isolate the aggregation through $\mathbb{W}$ that mediates the whole coupling. The reader should watch for the punchline of the subsection: every type-to-type interaction, and every interaction between the value block and the density block, will be shown to flow through a single $L^2([0,1])$-valued quantity $z_t(x)$, and once that is established the coupled system is one short step from decoupling.

Fix the diffusion coefficient $\nu := \frac{1}{2}\sigma\sigma^\top$ (nondegenerate by **(A-Nondeg)**), abbreviated $\nu > 0$ in the scalar exposition, and the control Hamiltonian $H(x,p)$, convex in p under **(A-Conv)**, imported from chapter 6. Because this is the first chapter of Part IV in which the value field and the density field appear together in one system, the style bible asks us to restate the time-orientation convention fixed once and for all in appendix A.1, and the restatement is worth the line because the two blocks run in opposite time directions and the reader must hold both arrows at once. The value u^x runs *backward*: it carries terminal data at $t = T$ and is integrated towards $t = 0$, because an optimal cost-to-go is built by dynamic programming from the end of the horizon, where the only thing an agent knows is the terminal penalty, back towards the present. The density m^x runs *forward*: it carries initial data at $t = 0$ and is transported towards $t = T$, because a population is pushed from where it starts by the flow the agents choose. The coupling closes this loop — the backward value is computed against the forward population, and the forward population is driven by the feedback the backward value defines — and it enters through a *running coupling* F and a *terminal coupling* G, which in the graphon setting are no longer maps into functions of state alone but maps from the field $\mathbf{m}$ to functions of *type and state*:

$$F[\mathbf{m}]\colon [0,1]\times\mathbb{R}^d \to \mathbb{R}, \qquad G[\mathbf{m}]\colon [0,1]\times\mathbb{R}^d \to \mathbb{R}.$$

The extra type slot is the whole novelty: in chapter 11 a single population m produced a single cost field $F[m](\cdot)$ felt by everyone, whereas here the field $\mathbf{m}$ produces, for each type x, its *own* cost field $F[\mathbf{m}](x,\cdot)$, and it is through this type-dependence that one type's population reaches into another type's optimisation.

A classical solution of the GMFG system is, accordingly, a pair of fields $(\mathbf{u},\mathbf{m}) = ((u^x)_x,(m^x)_x)$ — a value field and a density field, one of each per type, both living in the space of the previous subsection — solving, for each $x \in [0,1]$,

$$-\partial_t u^x - \nu\,\Delta_\xi u^x + H(x,\nabla_\xi u^x) = F[\mathbf{m}_t](x,\cdot), \qquad u^x(T,\cdot) = G[\mathbf{m}_T](x,\cdot), \tag{24.1}$$

$$\partial_t m^x - \nu\,\Delta_\xi m^x - \mathrm{div}_\xi\big(m^x\,\partial_p H(x,\nabla_\xi u^x)\big) = 0, \qquad m^x(0,\cdot) = m_0^x, \tag{24.2}$$

on $[0,T]\times\mathbb{R}^d$, where $\mathbf{m}_t = (m_t^x)_x$ denotes the field's time-t slice. The optimal feedback for type x is $\alpha^{*,x} = -\partial_p H(x,\nabla_\xi u^x)$, the sign convention of appendix A.4, and it is this feedback that appears as the transport drift in the divergence term of eq. (24.2): the value computed by the HJB block drives the density carried by the FP block, exactly as in the homogeneous theory, now with a copy of the pair for every type.

The structure to seize is what is, and what is not, coupled here. For a *fixed* field $\mathbf{m}$ the system eq. (24.1)–eq. (24.2) is not one big system but a continuum of *independent* pairs, one per type: the equations for type x involve u^x and m^x and no other type's unknowns. There is no spatial transport in the type variable, no x-derivative anywhere, nothing that lets type x and type x' communicate inside the differential operators. The HJB equations eq. (24.1) are coupled to one another, and to the FP equations eq. (24.2), *only* through the type-dependence of F and G on the whole field. The entire interaction therefore lives in a single algebraic step — how the field $\mathbf{m}$ is turned into the type-and-state cost data $F[\mathbf{m}_t](x,\cdot)$ — and that step, in every model of interest, is mediated by the graphon operator $\mathbb{W}$ of chapter 18. We make this precise.

Definition 24.3 (Graphon aggregation)**.** Let $\mathbb{W}$ be the integral operator of **(A-Reg-W)**, $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y$, self-adjoint Hilbert–Schmidt on $L^2([0,1])$ (chapter 18). For a field $\mathbf{m} \in \mathbb{M}_2$, the *first-moment field* is

$$\overline{\mathbf{m}}_t(y) := \int_{\mathbb{R}^d} \xi\, m_t^y(\mathrm{d}\xi) \in \mathbb{R}^d, \qquad \overline{\mathbf{m}}_t \in L^2([0,1];\mathbb{R}^d),$$

and the *graphon aggregate felt by type* x is

$$z_t(x) := (\mathbb{W}\,\overline{\mathbf{m}}_t)(x) = \int_0^1 W(x,y)\,\overline{\mathbf{m}}_t(y)\,\mathrm{d}y. \tag{24.3}$$

A coupling is *graphon-local* if $F[\mathbf{m}_t](x,\xi) = F_0\big(\xi, z_t(x)\big)$ and $G[\mathbf{m}_T](x,\xi) = G_0\big(\xi, z_T(x)\big)$ for fixed functions F_0, G_0 — i.e. type x feels the population only through the single aggregated quantity $z_t(x)$.

The definition is built in two moves, and each deserves its picture. The first move collapses each type's whole law to a single vector: the first-moment field $\overline{\mathbf{m}}_t$ reads off, at every type y, the mean position $\int \xi\, m_t^y(\mathrm{d}\xi)$ of that type's population. This is a drastic compression of information — an entire probability measure on $\mathbb{R}^d$ replaced by its centre of mass — but it is exactly the compression that lemma 24.2 certified is stable: a $\mathcal{W}_1$-wobble of a type's law of size ϵ moves its entry of $\overline{\mathbf{m}}_t$ by at most ϵ, so the map $\mathbf{m} \mapsto \overline{\mathbf{m}}_t$ is 1-Lipschitz from the field metric into $L^\infty([0,1];\mathbb{R}^d)$, and that bound is consumed on the very first line of every continuity estimate downstream. The second move is the aggregation proper: $\mathbb{W}$ averages the mean field $\overline{\mathbf{m}}_t$ against the kernel row $W(x,\cdot)$ to produce $z_t(x)$. The intuition to carry is that $z_t(x)$ is *the population as felt by type x through the network* — not the global mean, which no one in a heterogeneous network ever sees, but the $W(x,\cdot)$-weighted average of neighbours' means, the only summary of the crowd that reaches an agent of type x along the edges incident to it. A type strongly linked to a cluster feels that cluster's mean loudly; a type weakly linked to it barely feels it at all. The aggregate is the network's own, type-specific, low-pass picture of where the population sits.

It pays to keep this object's place in the larger family of aggregates straight, because the companion probabilistic chapter works with a richer one and we are deliberately taking a specialisation. Definition 23.2 defined the *measure-valued* graphon aggregate $(\mathbb{W}\mathbf{m}_t)(x) = \int_0^1 W(x,y)\, m_t^y\,\mathrm{d}y$, a (sub-probability) measure on $\mathbb{R}^d$ obtained by superposing the neighbours' *whole laws* with the kernel weights — it remembers the spread, the skew, every moment of the neighbourhood, not merely its centre. Our first-moment field eq. (24.3) is exactly the centre of mass of that measure: by linearity, integrating the coordinate map ξ against the superposition is the same as superposing the type-wise means,

$$z_t(x) = \int_{\mathbb{R}^d} \xi\,(\mathbb{W}\mathbf{m}_t)(x)(\mathrm{d}\xi) = \int_0^1 W(x,y)\Big(\int_{\mathbb{R}^d} \xi\, m_t^y(\mathrm{d}\xi)\Big)\,\mathrm{d}y = (\mathbb{W}\,\overline{\mathbf{m}}_t)(x),$$

which is precisely the observable $h(\xi) = \xi$ of the scalar aggregate eq. (23.2). The two operations commute — one may take the first moment and then aggregate, or aggregate and then take the first moment, and get the same vector — because $\mathbb{W}$ acts only in the type variable while the moment acts only in the state variable, and that commutation is what licenses us to work with the cheap $L^2([0,1];\mathbb{R}^d)$-valued object $\overline{\mathbf{m}}_t$ in place of the expensive measure-valued one throughout the existence proof. The existence theory below is developed for this first-moment (graphon-local) specialisation.

Two remarks place this choice in context without derailing the development. First, on generality: the first-moment reduction is not a loss of the framework's reach but a choice of the cleanest representative of it. Higher-order statistics — aggregating a vector of several moments, or the measures themselves through the measure-valued $\mathbf{m}_t \mapsto \int_0^1 W(x,y)\, m_t^y \,\mathrm{d}y$ of definition 23.2 — are accommodated by the very same formalism, with $\overline{\mathbf{m}}$ replaced by the appropriate $L^2([0,1])$-valued statistic and F_0, G_0 promoted to functions of that statistic; nothing in the aggregation step changes, because $\mathbb{W}$ is blind to what it is averaging. We keep the first-moment version for concreteness, as the thread specification does, and because it is the form Thread B needs. Second, on the status of graphon-locality: it is not an extra hypothesis pulled from the air but the structure of every example we carry. Thread B couples a type-x agent to the graphon-aggregated neighbour mean $z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x)$ and to nothing else — its running cost penalises the squared deviation $(X_t^x - s\, z_t(x))^2$ — and every model of chapter 22 shares this shape. We therefore treat the general coupling $F[\mathbf{m}_t]$ wherever it costs nothing, since the abstract continuity and compactness arguments never need more than that the data depend on $\mathbf{m}$ through $\mathbb{W}$, and we specialise to the graphon-local form $F_0(\xi, z_t(x))$ exactly when the spectral content must be made explicit — which is to say, when the leading eigenvalue $\lambda_1(\mathbb{W})$ has to surface in the uniqueness threshold of section 24.3 and the Thread B diagonalisation of section 24.5.

This is the seam the chapter was promised on. For a frozen field $\mathbf{m}$, the only trace that field leaves on the type-x problem is the single $\mathbb{R}^d$-valued time-curve $t \mapsto z_t(x)$: freeze it and the type-x HJB eq. (24.1) loses every reference to the other types and to $\mathbf{m}$ itself, becoming a decoupled classical equation driven by the given data $F_0(\cdot, z_t(x))$. The aggregate $z_t(x)$ is thus the one channel through which the whole population enters each per-type optimisation, and the linear map $\mathbf{m} \mapsto z$ it defines is the operator factor $\mathcal{A}_W$ foreshadowed in the chapter's opening. The next subsection takes exactly this aggregate, freezes it, and reads off the decoupled per-type best response — the move that makes the classical chapter 10 machine reusable type by type, and at which the factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ first becomes visible on the page.

24.1.3 The graphon best-response map

The previous subsection handed us the aggregate $z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x)$ and a precise account of what it is: the single $\mathbb{R}^d$-valued time-curve through which a frozen field $\mathbf{m}$ reaches into the type-x optimisation, the one seam at which all the coupling lives. We now do the obvious thing with a seam — we cut along it. The idea is simple to state and is the engine of the whole chapter. A candidate equilibrium field is a *guess* at the population; if we pretend that guess is correct and substitute it into the coupling, the type-x agent faces a control problem with fixed, externally given cost data, no longer entangled with what any other type is doing. Solving that problem and reading off the population it generates produces a *new* guess. The equilibrium is the guess that reproduces itself. This is exactly the best-response logic of the homogeneous theory (definition 10.3), and the whole payoff of having isolated the aggregate is that freezing it makes that logic available type by type, with the classical chapter 10 machine reused verbatim inside each type. We spell out the three moves — freeze, solve, recollect — and then record the map and its fixed-point condition.

Fix a candidate field $\mathbf{m} \in \mathbb{M}_2$. *Freeze.* Substitute $\mathbf{m}$ into the coupling: by definition 24.3 the running source becomes the *given* function $(x,\xi) \mapsto F[\mathbf{m}_t](x,\xi) = F_0(\xi, z_t(x))$ and the terminal data the given function $G[\mathbf{m}_T](x,\cdot) = G_0(\cdot, z_T(x))$, both now inert data that no longer respond to the unknown. The decisive consequence, already flagged at the close of the last subsection, is that the type index x stops being a coupling channel and becomes a mere *label on independent problems*: for each fixed x the HJB equation eq. (24.1) is a *decoupled* semilinear backward parabolic equation, referring to u^x and to the frozen data $F_0(\cdot, z_t(x))$ and to nothing else — not to $u^{x'}$, not to $m^{x'}$, not to $\mathbf{m}$. *Solve.* Under **(A-Nondeg)** that equation is uniformly parabolic, so it is precisely a problem of the kind chapter 6 treats: the control theory there returns a value $u^{\mathbf{m},x}$ and, because H is convex in p under **(A-Conv)**, a single-valued optimal feedback $\hat{\alpha}^{\mathbf{m},x} = -\partial_p H(x, \nabla_\xi u^{\mathbf{m},x})$ (the sign convention of appendix A.4). Feeding that feedback as the transport drift into the FP equation eq. (24.2) — which, with the feedback now a *known* vector field, is a *linear* forward Fokker–Planck equation — and starting from the type-x initial datum m_0^x transports the population forward and produces a new flow $t \mapsto m_t^x$ for type x. *Recollect.* Doing this at every x and gathering the results into one object produces a new field over types. The composite map "$\mathbf{m} \mapsto$ this new field" is the graphon best-response map; nothing in it ever required the per-type problems to communicate, which is exactly why the chapter 10 estimates survive intact when we run them below.

Definition 24.4 (Graphon best-response map and equilibrium)**.** The *graphon best-response map* is

$$\mathsf{S}_W : \mathbb{M}_2 \longrightarrow \mathbb{M}_2, \qquad \mathsf{S}_W(\mathbf{m}) := \Big(\big(\operatorname{Law}(X_t^{\mathbf{m},x})\big)_{t\in[0,T]} \Big)_{x\in[0,1]},$$

where, for each type x, $X^{\mathbf{m},x}$ solves the closed-loop SDE

$$\mathrm{d}X_t^{\mathbf{m},x} = -\,\partial_p H\big(x, \nabla_\xi u^{\mathbf{m},x}(t, X_t^{\mathbf{m},x})\big)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x, \qquad X_0^{\mathbf{m},x} \sim m_0^x,$$

and $u^{\mathbf{m},x}$ is the solution of the decoupled HJB eq. (24.1) for the frozen field $\mathbf{m}$. A *graphon mean field game equilibrium* is a field $\mathbf{m}^\star \in \mathbb{M}_2$ with $\mathsf{S}_W(\mathbf{m}^\star) = \mathbf{m}^\star$.

The definition packages the solve and recollect moves into a single SDE rather than the FP equation eq. (24.2), and the two are the same statement read in two languages: $m_t^x = \operatorname{Law}(X_t^{\mathbf{m},x})$ is the density the FP equation transports, because the law of a diffusion with drift $-\partial_p H$ and diffusion σ obeys exactly that Kolmogorov forward equation (the equivalence established in chapter 5). Writing it as the *closed-loop* SDE makes visible what kind of object the output is. The drift $-\partial_p H(x, \nabla_\xi u^{\mathbf{m},x}(t, X_t))$ is the optimal feedback evaluated along the agent's own trajectory, so the agent is not following a fixed plan but reacting to where it finds itself; the loop is closed because the value $u^{\mathbf{m},x}$ that defines the feedback was itself computed against the frozen $\mathbf{m}$. The noise $\sigma\,\mathrm{d}B_t^x$ is the type-x idiosyncratic shock, and the output $\mathsf{S}_W(\mathbf{m})$ retains only its *law* $\operatorname{Law}(X_t^{\mathbf{m},x})$, never the pathwise realisation.

That "only the law" is not a stylistic economy but the reason the construction is well posed at all, and it is worth pausing on because the parallel probabilistic chapter had to fight for it. The map S_W is the analytic-picture avatar of the probabilistic graphon best-response map S on $C([0,T]; L_W^2)$ of definition 23.15, the two related by the two-pictures dictionary proposition 23.19; we carry the subscript W to mark the kernel dependence exploited in section 24.3. In the probabilistic picture one is tempted to realise the *entire continuum* $(X^x)_{x\in[0,1]}$ as a single random field, which demands a continuum $(B^x)_{x\in[0,1]}$ of independent nondegenerate Brownian motions on one probability space — and that object is notoriously pathological: an uncountable family of i.i.d. nondegenerate noises cannot be made jointly measurable in (ω, x) without the sample averages failing to concentrate, the obstruction dissected in remark 23.6. Our map sidesteps it cleanly precisely because the noise enters *per type* and is immediately discarded down to its law: each $X^{\mathbf{m},x}$ is a single ordinary SDE on its own probability space, solved one type at a time, and what is collected across types is the *deterministic* field of laws $x \mapsto (\mathrm{Law}(X_t^{\mathbf{m},x}))_t$, an element of the field space $\mathbb{M}_2$ that carries no joint randomness to be measurable in. We never form the joint law of the continuum; we only ever form a continuum of marginal laws. This is why the analytic fixed-point problem on $\mathbb{M}_2$ is the well-posed home for the existence question, and the probabilistic chapter's care with remark 23.6 is what licenses reading the result back as a statement about genuine graphon McKean–Vlasov processes.

Why hunt fixed points of this particular map? Because a fixed point of S_W is precisely a solution of the coupled system eq. (24.1)–eq. (24.2), and the equivalence is exact in both directions. If $\mathbf{m}^\star = \mathsf{S}_W(\mathbf{m}^\star)$, then the field used to freeze the coupling and compute the values is the very field those optimal feedbacks regenerate, so the consistency that *defines* an equilibrium — population computed against equals population produced — holds, and the pair $(u^{\mathbf{m}^\star}, \mathbf{m}^\star)$ solves the system. Conversely any classical solution $(\mathbf{u}, \mathbf{m})$ has, by construction, $\mathbf{m}$ invariant under freeze-solve-recollect, hence is a fixed point. There is no slack between the two formulations: the self-consistency loop of the GMFG *is* the fixed-point equation, written once as a PDE system and once as $\mathsf{S}_W(\mathbf{m}) = \mathbf{m}$. This is the same reduction that organised chapter 10, and it collapses the entire chapter into two questions. Existence asks whether S_W has *a* fixed point; uniqueness asks whether it can have *two*. The whole novelty relative to chapters 10 and 11 is not the questions, which are old, but the internal structure of S_W that we must understand to answer them — and that structure is a factorisation we now expose, because it is the single most useful thing to know about the map.

Heuristic (non-rigorous): The organising factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$

The three moves of the previous page were not symmetric. Exactly one of them — freeze — touched the graphon; the other two — solve, recollect — were the classical machine applied blindly, type by type, to whatever frozen data it was handed. That asymmetry is the whole structure of S_W, and naming it turns the map into a composition of two factors that can be studied separately. Write

$$\mathsf{S}_W \;=\; \mathcal{R} \circ \mathcal{A}_W, \qquad \mathcal{A}_W \colon \mathbf{m} \mapsto \big(x \mapsto z(x) = (\mathbb{W}\overline{\mathbf{m}})(x)\big), \quad \mathcal{R} \colon (z(x))_x \mapsto \big(\mathsf{S}^x_{z(x)}\big)_x, \tag{24.4}$$

where $\mathcal{A}_W$ is the freeze move made into an operator — the *linear* graphon aggregation eq. (24.3) that turns a field into the type-indexed coupling data z it presents to the optimisers (the calligraphic $\mathcal{A}_W$ names this aggregation lift, not the generator $\mathcal{A}$, which plays no role in this diffusion-free fixed-point chapter) — and $\mathcal{R}$ is the solve-and-recollect move, applying *independently in each type* the homogeneous best-response solve $\mathsf{S}^x_{z(x)}$ of definition 10.3 with the frozen data $z(x)$. Read left to right: aggregate the field through the network, then run the classical machine in parallel over types. The value of the factorisation is that the two factors quarantine the two kinds of difficulty so they are never confused.

$\mathcal{R}$ is the classical machine, and it is already understood. Each factor $\mathsf{S}^x_{z(x)}$ is a single homogeneous mean field best response with scalar coupling data $z(x)$: a uniformly parabolic HJB solved backward, its feedback pushed through a linear FP solved forward. This is the object of chapters 10 and 11 down to the last estimate — the gradient bound, the moment and time-equicontinuity bounds, tightness via Prokhorov, stability under weak convergence of the data. Nothing about $\mathcal{R}$ is new; it is the old theory wearing a type subscript, run in a continuum of independent copies. Whatever $\mathcal{R}$ does *within* a type, it does exactly as before, and we may import those results verbatim.

$\mathcal{A}_W$ is a bounded linear operator, and it carries every graphon-specific phenomenon. This is the load-bearing claim of the chapter. $\mathcal{A}_W$ is the post-composition of the first-moment read-out $\mathbf{m} \mapsto \overline{\mathbf{m}}$ (a 1-Lipschitz contraction, by lemma 24.2) with the integral operator $\mathbb{W}$, hence a bounded linear map on $L^2([0,1];\mathbb{R}^d)$. Being merely linear and bounded, it has nothing of the analytic subtlety of $\mathcal{R}$ — and yet *all three* of the ways the graphon makes this theory differ from the homogeneous one are read off it, each from a different attribute of the *same* operator $\mathbb{W}$:

- its **norm** $\|\mathbb{W}\|$ sets the *strength of the feedback loop.* The field influences itself only by being aggregated through $\mathcal{A}_W$, solved, and aggregated again; the gain around that loop scales with $\|\mathbb{W}\|$, so $\|\mathbb{W}\|$ is what governs whether the small-coupling contraction of section 24.3 closes and how large a uniform bound the aggregate inherits (lemma 24.6).

- its **spectrum** sets the *sign of the equilibrium energy.* The Lasry–Lions uniqueness identity acquires a quadratic form threaded through $\mathbb{W}$; whether that form is signed — hence whether two equilibria can coexist — is decided mode by mode by the eigenvalues of $\mathbb{W}$, with the leading eigenvalue $\lambda_1(\mathbb{W})$ marking the critical coupling $s\,\lambda_1$ at which uniqueness breaks (sections 24.3 and 24.5).

- its **continuity** — continuity of the kernel W in its type argument, **(A-Reg-W)** — sets *compactness in the type direction.* $\mathcal{A}_W$ maps a merely bounded first-moment field to an *equicontinuous* aggregate (lemma 24.8), and equicontinuity is the only currency that buys type-direction compactness for Schauder (lemmas 24.9 and 24.10).

Norm, spectrum, continuity: feedback strength, energy sign, compactness. Three faces of one operator, supplying the chapter's three new facts. The method of the chapter is therefore to read *every* later theorem as a statement about $\mathcal{R} \circ \mathcal{A}_W$ — to ask, of any estimate, whether it is a fact about $\mathcal{R}$ (then it is classical and inherited) or a fact about $\mathcal{A}_W$ (then it is graphonic and lives in $\mathbb{W}$). Holding the two apart is what keeps the inherited difficulty and the genuinely new difficulty from being mistaken for one another.

One honest qualification keeps the box a heuristic rather than a theorem, and it is the reason the chapter carries a standing "single-valued regime" assumption rather than merely asserting one. The factor $\mathcal{R}$ is a genuine *map* — a well-defined function of z, not a correspondence — exactly when the per-type best response is single-valued, i.e. when the type-x optimiser has a *unique* optimal feedback. That uniqueness is precisely what strict convexity of the Hamiltonian **(A-Conv)** and nondegeneracy of the diffusion **(A-Nondeg)** deliver: strict convexity makes $p \mapsto \partial_p H(x,p)$ a genuine function (no flat stretch where several controls are equally optimal), and nondegeneracy supplies the parabolic smoothing that makes $\nabla_\xi u^{\mathbf{m},x}$ well defined pointwise, so the feedback $-\partial_p H(x, \nabla_\xi u^{\mathbf{m},x})$ is a single vector field and the output flow is unambiguous. Drop either hypothesis and $\mathcal{R}$ becomes *set-valued*: the optimiser may be indifferent among several feedbacks, S^x returns a set of flows, and S_W is a correspondence rather than a map. The fixed-point question is then not Schauder but its set-valued cousin, the Kakutani–Fan–Glicksberg theorem theorem 2.40 applied to the field-valued correspondence; we record that route, with the attendant notion of a *weak* (relaxed) graphon equilibrium and a measurable-selection step, as the graphon analogue of theorem 10.19 in remark 24.14. It changes the machinery but not the moral: the set-valuedness lives entirely in $\mathcal{R}$, the classical factor, and leaves $\mathcal{A}_W$ untouched. We therefore adopt the single-valued regime — **(A-Conv)** and **(A-Nondeg)** in force — as standing unless explicitly stated otherwise, so that S_W is an honest self-map of $\mathbb{M}_2$ and Schauder is the right tool.

We can now state the chapter's first concrete task in one clean sentence, and the factorisation tells us in advance where the work will be. Existence is the question: *does S_W have a fixed point?* The lens eq. (24.4) predicts the answer's shape before we prove anything. Because $\mathcal{R}$ is the classical machine, the within-type content of the existence proof — the construction of a convex compact invariant set in the state and time directions — is chapter 10 reused per type, with nothing to add. The genuinely new obligation is the type-direction compactness that Schauder also demands of a field, and the factorisation says exactly which factor must supply it: not $\mathcal{R}$, which knows nothing of how types are arranged, but $\mathcal{A}_W$, whose continuity (**(A-Reg-W)**) is the only available source of type-direction equicontinuity. The next section takes up the fixed-point question with this division of labour fixed in advance — $\mathcal{R}$ inherited, $\mathcal{A}_W$ on the hook for the new compactness — and that is the seam along which its proof is organised.

24.2 Existence

The previous section left us with one clean question and a lens through which to read its answer. The question is the whole of existence: *does the graphon best-response map S_W have a fixed point?* The lens is the factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ of eq. (24.4), and it already told us where to look for trouble. The solve-and-recollect factor $\mathcal{R}$ is the classical machine of chapter 10 worn with a type subscript, run in a continuum of independent copies; whatever it does, it does exactly as it did in the homogeneous theory, and we will reuse those estimates verbatim. The aggregation factor $\mathcal{A}_W$ is the only place the graphon enters, and the factorisation predicted that the one genuinely new obligation of the existence proof would land on it. This section makes that prediction concrete.

The proof strategy is the one the reader owns from chapter 10: Schauder's fixed-point theorem (theorem 2.38) on the field space $\mathbb{M}$. Schauder asks for three things — a continuous self-map of a convex set, and a *convex compact* invariant subset on which to run it — and of these the compactness is the only one that ever costs real work. We are operating in the standing single-valued regime fixed at the close of the last section (**(A-Conv)** and **(A-Nondeg)** in force), so S_W is an honest self-map rather than a correspondence and Schauder, not its Kakutani cousin, is the right tool; convexity of the candidate set will be arranged by hand exactly as in theorem 10.14. Everything therefore reduces to manufacturing a convex compact set that S_W maps into itself. The manufacture of that set is the entire content of the existence proof, and it is here, and only here, that the graphon changes the classical argument.

What changes is the *anatomy* of the compactness. A point of the field space $\mathbb{M}$ is not a single flow of measures, as it was in chapter 10, but an entire field $x \mapsto m^x$ of them, and the metric D of definition 24.1 splits, as noted at its definition, along two levels: the inner $\sup_t$ inside d_1 controls the flow in *state and time*, the outer $\sup_x$ controls its variation across *types*. Compactness in $\mathbb{M}$ therefore has *two directions*, and the structural novelty of this chapter — the thing to plant firmly before any lemma — is that these two directions are bought from two completely different sources and may be supplied independently. This splitting is not a presentational convenience; it is forced by the product-like structure of $\mathbb{M} = C([0,1];\mathcal{C})$, and it is exactly the seam along which the factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ cuts the proof.

In the *state-and-time* direction — the behaviour of each fibre m^x as a curve $t \mapsto m_t^x$ of measures on $\mathbb{R}^d$ — compactness is precisely the classical compactness of chapter 10. A uniform second-moment bound confines almost all the mass of every m_t^x to one fixed large ball, hence (Prokhorov, chapter 3) into a tight, relatively weakly compact set; a uniform time-modulus forbids the flow from jumping; and the two together place each fibre in the single convex compact set $\mathcal{K} \subseteq \mathcal{C}$ of lemma 10.8. None of these estimates ever knew about a type variable, so they transfer to the graphon setting unchanged — *provided* they can be made to hold with constants that do not depend on which type we are looking at. That uniformity-in-type is the one genuinely new demand the state-and-time direction makes, and it is met by a single attribute of $\mathcal{A}_W$: the operator norm of $\mathbb{W}$, which bounds the aggregate $z_t(x)$ felt by every type at once and so feeds the classical gradient estimate lemma 10.5 the same bounded data uniformly across x. The inherited machinery does the rest.

In the *type* direction — the dependence $x \mapsto m^x$ itself — compactness is genuinely new, and it is the price the graphon exacts. There is no analogue of this difficulty in the homogeneous theory of chapter 10: there was one population because there was one kind of agent, no type variable, hence nothing to be compact *in*, and theorem 10.14 needed no type-direction estimate because there was no type direction. A graphon candidate, by contrast, is a whole curve of flows over $[0,1]$, and nothing in the state-and-time estimates stops that curve from oscillating ever faster in x — a sequence of fields can stay bounded and tight at every fixed type and still fail to converge, because the assignment of flow to type wiggles without bound. The only currency that buys compactness in a uniform topology is an *equicontinuity modulus*: a guarantee that nearby types produce nearby flows, with a modulus independent of which field we started from, so that the metric Arzelà–Ascoli theorem applies. And the only source of such a modulus in the entire construction is the continuity of the kernel W in its type argument, the second half of **(A-Reg-W)**. This is the structural surprise worth stating without hedging: in the classical theory existence needed no regularity of any coupling kernel whatsoever, yet the graphon *existence* theorem — not merely the uniqueness theorem — fails without a continuity hypothesis on $\mathbb{W}$, because continuity is the sole manufacturer of the missing type-direction compactness. The norm of $\mathbb{W}$ handles one direction; the continuity of $\mathbb{W}$ handles the other; both are attributes of the same operator, and that is the factorisation doing its promised work.

These two inputs are logically independent: one can have the uniform bound without the modulus (a bounded but discontinuous kernel) or, in principle, control the type-modulus of a field without a uniform state bound. The existence proof needs both, and the cleanest way to organise it is to secure each separately and only then combine them, which is exactly the division of labour the three subsections below carry out. We isolate each input in turn.

Before isolating them we must, exactly as in chapter 10, pin down the structural demands the coupling data have to meet for the inherited estimates to fire, now type-indexed. The classical existence theorem rested on the couplings being *bounded* and *smoothing*: bounded so that the per-type value function inherits a gradient bound (lemma 10.5) and hence a bounded feedback, smoothing so that weak convergence of the population data passes through the HJB solve to convergence of the value (lemma 10.10), which is what makes the best-response map continuous. Neither requirement is automatic, and the LQG thread famously violates the first — its quadratic running cost is unbounded in state, so it cannot be admitted under a boundedness hypothesis and must instead be handled by explicit Gaussian construction, which we again defer to the worked computation of section 24.5. We therefore carry the standing bounded-smoothing-coupling assumption of chapter 10, lifted to the graphon setting by the single change that every clause is now demanded *uniformly in the type* x — the uniformity being precisely what the state-and-time direction needs to make its constants type-free.

Assumption 24.5 (Bounded, uniformly type-Lipschitz graphon-local couplings)**.** The couplings are graphon-local (definition 24.3), $F[\mathbf{m}_t](x,\xi) = F_0(\xi, z_t(x))$ and $G[\mathbf{m}_T](x,\xi) = G_0(\xi, z_T(x))$, with $F_0, G_0 \colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ bounded, uniformly Lipschitz in ξ uniformly in the second argument, uniformly Lipschitz in the aggregate argument z, and continuous for weak convergence in the sense of lemma 10.10. The initial field $\mathbf{m}_0$ has uniformly bounded second moment, $\sup_x \int |\xi|^2\, m_0^x(\mathrm{d}\xi) < \infty$, and is continuous in type, $x \mapsto m_0^x \in \mathcal{P}_2(\mathbb{R}^d)$ continuous.

The assumption packs five clauses, and it is worth walking each one and naming exactly what breaks in its absence, because each is load-bearing for a different link in the chain and none is there for decoration. *Graphon-locality* ($F[\mathbf{m}_t](x,\xi) = F_0(\xi, z_t(x))$, the data felt by type x depending on the field only through the single aggregate $z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x)$ of definition 24.3) is what makes the factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ literally true rather than merely morally so: it is the statement that the field reaches the type-x optimiser through one finite-dimensional channel, so that freezing z genuinely decouples the per-type problem. Without it the per-type solve would still see the whole field, the type index would remain a coupling channel inside the differential operators, and there would be no clean classical factor $\mathcal{R}$ to inherit. *Boundedness* of F_0, G_0 is what the gradient bound lemma 10.5 consumes; drop it and the per-type value function $u^{\mathbf{m},x}$ need carry no uniform gradient bound, the optimal feedback $-\partial_p H(x, \nabla_\xi u^{\mathbf{m},x})$ need not be bounded, the closed-loop drift escapes control, and the per-type moment and time-equicontinuity estimates of lemmas 10.6 and 10.7 — the entire state-and-time compactness — collapse. (This is the precise clause the LQG thread violates, and the precise reason its threshold has to be found by hand in section 24.5 rather than read off this theorem.) *Uniform Lipschitz continuity in the state ξ, uniformly in the aggregate argument*, is what makes the parabolic solve well posed and the feedback single-valued at the level of regularity chapter 6 requires; it is the smoothing that, together with **(A-Nondeg)**, keeps $\mathcal{R}$ a genuine map rather than a correspondence. *Uniform Lipschitz continuity in the aggregate z* is the clause that will make the *best response itself* continuous: a change in the field moves z by a controlled amount (through the 1-Lipschitz first-moment read-out of lemma 24.2 composed with the bounded $\mathbb{W}$), and this clause converts that controlled move in the data into a controlled move in the output flow, which is the continuity Schauder demands of the map. Strip it and S_W could jump discontinuously as the field varies, and no fixed-point theorem applies. *Weak-convergence continuity in the sense of lemma 10.10* is the same continuity demanded along weakly convergent sequences of populations, the form in which it is actually used when we pass to a limit inside the compact set; it is the clause that lets the limit of best responses be the best response of the limit.

Two clauses concern not the couplings but the initial datum, and they seed the two compactness directions at $t = 0$ so that the dynamics can propagate them forward. The *uniform second-moment bound* $\sup_x \int |\xi|^2 \, m_0^x(\mathrm{d}\xi) < \infty$ is the initial tightness leash: it is the $t = 0$ end of the uniform moment bound that lemma 24.6 will carry across all times, and without it the populations could start already escaping to infinity in state, with no ball to confine them and no Prokhorov compactness to harvest. The *continuity in type of the initial field*, $x \mapsto m_0^x \in \mathcal{P}_2(\mathbb{R}^d)$ continuous, is the $t = 0$ seed of the type-direction equicontinuity: the existence proof will show the modulus in x is *produced* by the dynamics through the continuity of $\mathbb{W}$, but it can only be produced if the field is already continuous in type at the start, since a flow that begins with a discontinuous type-profile cannot be smoothed into a continuous one by a solve that acts type by type. The two initial-datum clauses are thus the boundary conditions for the two compactness arguments, one per direction. Every hypothesis in the box, then, is the in-force input to a specific later step; none is idle, and the reader can audit the proof by checking that each clause is spent exactly once.

With the standing data fixed and each direction of compactness identified with its source, the labour divides cleanly into three, and the three subsections below take them in order. First we secure the state-and-time direction made uniform in type: the next subsection turns the operator norm of $\mathbb{W}$ into a uniform bound on the aggregate, and through it a type-free bound on the per-type coupling data, so that the classical estimates of chapter 10 fire with constants independent of x and of the field — this is where $\|\mathbb{W}\|$ first does visible work. Second we secure the new type direction: the continuity of W is converted, through the aggregation operator, into a type-equicontinuity modulus for the output field that no oscillation can defeat. Third we assemble the two moduli into one convex compact invariant set in $\mathbb{M}$ and invoke Schauder to land the fixed point. We begin with the first, the operator-norm bound.

24.2.1 Uniform-in-type estimates from the operator norm

We take the first of the two compactness directions: the state-and-time direction, the behaviour of each fibre m^x as a curve of measures on $\mathbb{R}^d$, which the preceding section identified as classical compactness inherited from chapter 10 *provided* its constants can be made uniform in the type. That proviso is the whole job of this subsection, and it is the first place in the chapter where the operator $\mathbb{W}$ stops being a piece of notation and does visible quantitative work. The factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ tells us where to apply pressure. The classical machine $\mathcal{R}$ is a continuum of independent per-type solves, each a copy of the homogeneous problem of chapter 10; every estimate it produces — the gradient bound, the moment bound, the time-modulus — carries a constant that depends only on how large the coupling data handed to that solve is. So the entire question of type-uniformity reduces to a single quantitative question about the aggregation factor $\mathcal{A}_W$: can we bound the data $z_t(x)$ it presents to the type-x optimiser by a number that does not depend on x, and does not depend on *which* field $\mathbf{m}$ we froze? If we can, the per-type constants inherit that uniformity for free, and the state-and-time compactness of chapter 10 transfers wholesale. The answer is yes, and the number is supplied by a norm of $\mathbb{W}$ acting on the second-moment leash of definition 24.1. We state and prove it, then watch it feed the classical gradient estimate lemma 10.5 uniformly across types.

Lemma 24.6 (Uniform aggregate bound)**.** *Under **(A-Reg-W)** and assumption 24.5, there is a constant $Z < \infty$, depending only on the* row-sum norm $\|W\|_{L^\infty_x L^1_y} = \operatorname{ess\,sup}_x \int_0^1 |W(x,y)|\,\mathrm{d}y$ *(which dominates, but in general strictly exceeds, the $L^2 \to L^2$ operator norm $\|\mathbb{W}\| = \rho(\mathbb{W}) = \lambda_1$; see remark 24.19) and on the uniform second-moment bound of the field, such that for every $\mathbf{m} \in \mathbb{M}_2$ with* $\sup_x \sup_t \int |\xi|^2\, m_t^x(\mathrm{d}\xi) \le R^2$,

$$\sup_{x\in[0,1]} \sup_{t\in[0,T]} |z_t(x)| \;\le\; Z := \|W\|_{L^\infty_x L^1_y}\; R.$$

Consequently the frozen coupling data $F_0(\cdot, z_t(x))$, $G_0(\cdot, z_T(x))$ entering the type-x HJB are bounded and uniformly Lipschitz in ξ, with constants independent of x and of $\mathbf{m}$.

Proof. The estimate is a two-link chain, and the discipline of the chapter is visible in its shape: the first link lives entirely in the state variable and is the second-moment leash of definition 24.1 cashed in, while the second link lives entirely in the type variable and is the operator $\mathbb{W}$ cashed in. The two links never interfere because, as definition 24.3 recorded, the moment acts only in state and $\mathbb{W}$ acts only in type.

First link: the second moment controls each type's mean position. Fix a time t and a type y. The first-moment field reads off $\overline{\mathbf{m}}_t(y) = \int_{\mathbb{R}^d} \xi\, m_t^y(\mathrm{d}\xi)$, the centre of mass of that type's population. We bound its size by the square root of the second moment through Cauchy–Schwarz applied to the probability measure m_t^y: writing the coordinate map against the constant 1,

$$|\overline{\mathbf{m}}_t(y)| = \Big| \int_{\mathbb{R}^d} \xi\, m_t^y(\mathrm{d}\xi) \Big| \le \int_{\mathbb{R}^d} |\xi|\; m_t^y(\mathrm{d}\xi) \le \Big(\int_{\mathbb{R}^d} |\xi|^2\; m_t^y(\mathrm{d}\xi) \Big)^{1/2} \Big(\int_{\mathbb{R}^d} 1\, m_t^y(\mathrm{d}\xi) \Big)^{1/2} \le R \cdot 1 = R,$$

where the middle inequality is Jensen (the modulus moved inside the integral) and the last uses the hypothesised uniform bound $\int |\xi|^2\, m_t^y \le R^2$ together with m_t^y being a probability measure. The bound R is the *same* for every type y and every time t, because the second-moment leash of definition 24.1 was demanded uniformly in both. Hence the entire first-moment field is confined to a fixed ball: $\|\overline{\mathbf{m}}_t\|_{L^\infty([0,1];\mathbb{R}^d)} \le R$, uniformly in t. This is the only place the state direction enters, and the leash is the whole of what it contributes.

Second link: the operator spreads that bound across the network without amplifying it. The aggregate $z_t(x) = (\mathbb{W}\,\overline{\mathbf{m}}_t)(x) = \int_0^1 W(x,y)\, \overline{\mathbf{m}}_t(y)\,\mathrm{d}y$ is a kernel-weighted average of the bounded field $\overline{\mathbf{m}}_t$. Pulling the modulus inside the integral and using the just-proved L^∞ bound,

$$|z_t(x)| = \Big| \int_0^1 W(x,y)\, \overline{\mathbf{m}}_t(y)\; \mathrm{d}y \Big| \le \int_0^1 |W(x,y)|\; |\overline{\mathbf{m}}_t(y)|\; \mathrm{d}y \le \|\overline{\mathbf{m}}_t\|_{L^\infty} \int_0^1 |W(x,y)|\; \mathrm{d}y \le R \int_0^1 |W(x,y)|\; \mathrm{d}y.$$

The remaining integral $\int_0^1 |W(x,y)|\,\mathrm{d}y$ is the ℓ^1 mass of the x-th *row* of the kernel — the total weight type x assigns to the rest of the network. Taking the essential supremum over x replaces it by the row-sum norm $\|W\|_{L^\infty_x L^1_y} = \operatorname{ess\,sup}_x \int_0^1 |W(x,y)|\, \mathrm{d}y$, giving $|z_t(x)| \le R\, \|W\|_{L^\infty_x L^1_y} = Z$, and the bound is uniform in both x and t because we took a supremum over x and the first link was already uniform in t. Boundedness of W under **(A-Reg-W)** makes the row-sum norm finite, so $Z < \infty$.

It is worth saying out loud why Z is independent of the field $\mathbf{m}$ we started from, since that independence is exactly what the inherited estimates will need and it is easy to take for granted. Nothing in the chain saw $\mathbf{m}$ except through the single scalar R, the radius of the second-moment leash; the kernel contributes only its fixed row-sum norm, a property of W alone. So any two fields obeying the *same* leash R present coupling data confined to the *same* ball of radius Z in the aggregate variable, regardless of how different the two fields are as objects. That is the precise sense in which Z is a constant of the problem and not of the candidate. Likewise the independence from x is structural rather than accidental: we did not estimate one type's row and hope the others were no worse; we bounded *every* row at once by passing to the essential supremum, which is the operation the row-sum norm performs.

A word of caution to plant now and settle later. The norm that appeared here is the *row-sum* norm $\|W\|_{L^\infty_x L^1_y}$, which is the $L^\infty \to L^\infty$ operator norm of $\mathbb{W}$ — the natural one in the uniform-in-type metric D we are working in, because we are controlling a quantity at *every* type simultaneously. This is *not* the same as the $L^2 \to L^2$ operator norm $\|\mathbb{W}\| = \rho(\mathbb{W}) = \lambda_1$, the spectral radius that will govern the uniqueness threshold in section 24.3. The two are related by the Schur-test inequality $\lambda_1 \le \|W\|_{L^\infty_x L^1_y}$, with equality only for kernels of constant row sum and strict inequality in general, so the row-sum norm is the more conservative of the two. The distinction does not matter for existence — any finite norm of $\mathbb{W}$ suffices to bound the data — but it will matter when we count critical couplings, where the exact spectral quantity λ_1 is the one that decides. We register the gap here and reckon with it fully in remark 24.19. The final sentence of the lemma is now immediate: with the aggregate argument confined to the ball $|z| \le Z$, assumption 24.5 makes $F_0(\cdot, z)$ and $G_0(\cdot, z)$ bounded and uniformly Lipschitz in ξ across that ball, hence uniformly in x and $\mathbf{m}$. $\square$

Lemma 24.6 is where $\|\mathbb{W}\|$ first does visible work: it converts the uniform second-moment bound on the field into a uniform bound on the coupling data felt across all types, which is exactly the input the classical gradient estimate lemma 10.5 consumes. We now follow that input through the per-type solve and watch the uniformity propagate, link by link, to the output flow. The chain has three steps and each preserves type-independence for the same reason: at no point does any constant depend on the type except through the aggregate bound Z, which is itself type-free.

First, the coupling data. By the lemma, for every type x the frozen running source $F_0(\cdot, z_t(x))$ and terminal data $G_0(\cdot, z_T(x))$ are bounded by a constant fixed by Z and uniformly Lipschitz in ξ with a constant fixed by Z — the same two constants for every x, because the aggregate arguments $z_t(x)$ all lie in the single ball $|z| \le Z$ and F_0, G_0 are uniformly Lipschitz across it. Second, the value function. The per-type HJB eq. (24.1) is now a decoupled uniformly parabolic equation (**(A-Nondeg)** supplies the ellipticity) with bounded data, precisely the hypotheses of the classical gradient estimate lemma 10.5; that estimate returns a gradient bound L_u depending only on the data bounds, the horizon T, and the ellipticity constant — all type-free — hence *the same* L_u for every type. Third, the feedback and its drift. The optimal feedback $\hat{\alpha}^{\mathbf{m},x} = -\partial_p H(x, \nabla_\xi u^{\mathbf{m},x})$ is the Hamiltonian slope evaluated at a gradient bounded by L_u, so under **(A-Conv)** it is bounded by a constant M fixed by L_u and the local Lipschitz constant of $\partial_p H$ on the ball $|p| \le L_u$ — again type-free, hence one bound M for the whole continuum of feedbacks. The closed-loop drift of the SDE in definition 24.4 is thus bounded by M uniformly in x, and that single number is the only thing the state-and-time compactness will ask of the solve. With a type-free drift bound and a type-free diffusion in hand, the moment and equicontinuity lemmas of chapter 10 apply verbatim, type by type, with type-free constants.

Lemma 24.7 (Per-type compactness, uniform in type)**.** *Under **(A-Nondeg)**, assumption 24.5, and lemma 24.6, there are constants $R' < \infty$ and a modulus ω (the data of lemmas 10.6 and 10.7),* both independent of x and $\mathbf{m}$*, such that the output flow $\mathsf{S}_W(\mathbf{m})^x$ satisfies, for every x,*

$$\sup_t \int |\xi|^2 \, \mathsf{S}_W(\mathbf{m})^x_t(\mathrm{d}\xi) \le (R')^2, \qquad \mathcal{W}_1(\mathsf{S}_W(\mathbf{m})^x_s, \mathsf{S}_W(\mathbf{m})^x_t) \le \omega(|t-s|).$$

Hence each output flow lies in the single convex compact set $\mathcal{K} \subseteq \mathcal{C}$ of lemma 10.8, the same *$\mathcal{K}$ for all types.*

Proof. The output flow $\mathsf{S}_W(\mathbf{m})^x = (\mathrm{Law}(X_t^{\mathbf{m},x}))_t$ is the law of the closed-loop diffusion of definition 24.4, whose drift is bounded by the uniform constant M established in the paragraph above and whose diffusion coefficient σ has the type-independent bound Σ of **(A-Nondeg)**. These two numbers, M and Σ, are the *only* inputs the classical estimates consume. The moment estimate lemma 10.6 bounds $\sup_t \int |\xi|^2 \, \mathrm{Law}(X_t^{\mathbf{m},x})(\mathrm{d}\xi)$ by a constant $(R')^2$ built from M, Σ, the horizon T, and the uniform initial second moment of assumption 24.5 — quantities none of which carries a type, so R' is one number for the whole continuum. The time-equicontinuity estimate lemma 10.7 bounds the $\mathcal{W}_1$-increment of the flow over a time gap $|t-s|$ by a modulus ω built from the same M, Σ, T, again type-free. This is the point at which the work of lemma 24.6 pays off: had the drift bound M depended on x, the constants R' and ω would inherit that dependence and the output fibres would scatter across a x-dependent family of compact sets, with no single set to run Schauder in. Because M is type-free, R' and ω are too. Finally, lemma 10.8 assembles from the pair (R', ω) — a second-moment radius and a time-modulus — the convex compact set $\mathcal{K} \subseteq (\mathcal{C}, \mathsf{d}_1)$: the second-moment bound supplies tightness (Prokhorov, chapter 3) so that $\mathcal{K}$ is relatively weakly compact at each time, and the time-modulus rules out escape between times, the two together giving compactness of the whole flow in the uniform Wasserstein metric. Since R' and ω are the same for every x, the set $\mathcal{K}$ they build is one and the same set, and every output fibre $\mathsf{S}_W(\mathbf{m})^x$ lies in it. $\square$

The state-and-time direction is now closed, and it is worth being precise about what has been won and what remains. Lemma 24.7 delivers a *single* convex compact set $\mathcal{K} \subseteq \mathcal{C}$ — not one set per type, but one set for the whole continuum — into which S_W drops every fibre of every output field, no matter which field was fed in. In the language of the two-level picture of definition 24.1, the inner level is fully under control: each individual flow $x \mapsto m^x$, read as a curve in time, is trapped in a compact set, with a trap that does not move as the type or the input field varies. That is exactly the classical compactness of chapter 10, made type-uniform by the single mechanism of this subsection, the row-sum norm of $\mathbb{W}$ converting one second-moment leash into one bounded coupling ball into one drift bound into one compact $\mathcal{K}$. What is *not* yet controlled is the outer level: the assignment $x \mapsto m^x$ as a map from types to flows. Knowing that every fibre lives in $\mathcal{K}$ tells us each output field is a map $[0,1] \to \mathcal{K}$, but $C([0,1];\mathcal{K})$ is not compact — a sequence of such maps can oscillate ever faster in x, every fibre staying inside the compact $\mathcal{K}$ while the assignment of fibre to type refuses to settle. There is exactly one gap left in the existence argument, and it is this: compactness in the type direction itself. The next subsection closes it, and the structure of this subsection already names the only possible tool. Here the *magnitude* of the kernel rows — their ℓ^1 mass, the row-sum norm — did all the work; there the *variation* of the kernel rows with the type — the continuity of $x \mapsto W(x,\cdot)$, the second half of **(A-Reg-W)** — will be what buys the missing equicontinuity in x.

24.2.2 Type-direction equicontinuity from graphon continuity

The previous subsection closed the state-and-time direction and named, with some precision, the one gap it could not close. Lemma 24.7 traps every fibre of every output field in a single convex compact set $\mathcal{K} \subseteq \mathcal{C}$, so each output field is a map $x \mapsto m^x$ *into* that compact set — an element of $C([0,1];\mathcal{K})$. The inner, time level of the two-level picture is therefore fully under control. But knowing that a sequence of fields takes values in a compact set does not make the sequence itself relatively compact: $C([0,1];\mathcal{K})$, the continuous maps from types into $\mathcal{K}$, is *not* compact. A sequence of such maps can oscillate ever faster in x — every fibre obediently sitting inside $\mathcal{K}$ while the assignment of fibre to type refuses to settle, the curve $x \mapsto m^x$ wiggling on finer and finer type scales with no limit to extract. This is the outer-level pathology the field space was built to anticipate, and it is the exact obstruction that the homogeneous theory never had to face, because there was no type variable to wiggle in. To extract a convergent subsequence we need more than pointwise confinement: we need the maps to be *equicontinuous* in x, all sharing one modulus of continuity that forbids the oscillation. That single missing modulus is the whole content of this subsection.

The structure of the preceding subsection already told us where it must come from. There the *magnitude* of the kernel rows — their ℓ^1 mass, the row-sum norm of $\mathbb{W}$ — converted one second-moment leash into one bounded coupling ball into one drift bound into one compact $\mathcal{K}$. That bought confinement, but magnitude is blind to whether nearby types feel nearby populations: a kernel can have small, even constant, row sums and still jump wildly from one type's row to the next, and such a kernel would let $z_t(x)$ leap as x varies, dragging the output fibres apart and destroying any hope of a type-modulus. What we need instead is that the kernel rows *vary continuously* with the type — that $W(x,\cdot)$ and $W(x',\cdot)$ are close in L^1 when x and x' are close — and this is precisely the *continuity* half of the standing assumption **(A-Reg-W)**, the half that played no role at all in the magnitude estimate. Here it does all the work. The plan of the subsection is therefore a two-step transfer of regularity: first show that graphon continuity makes the aggregate $z_t(x)$ equicontinuous in x (lemma 24.8), then push that equicontinuity through the per-type solve — close data to close values to close feedbacks to close output flows — to obtain an equicontinuity modulus η for the output *field* itself (lemma 24.9). The first step is the linear aggregation factor $\mathcal{A}_W$ doing its job; the second is the classical solve $\mathcal{R}$ doing its, in the same type-by-type way it did everything else.

We begin with the aggregate. The intuition is the cleanest possible instance of the factorisation discipline: the aggregate $z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x)$ is a linear read-out of the field through the kernel row

$W(x,\cdot)$, so the only way $z_t(x)$ can change as x moves is if that row changes, and how fast the row may change is exactly what graphon continuity controls. Nothing about the field's own complexity can enter, because the field is held fixed and merely weighted; only the kernel's type-regularity can be felt. The lemma makes this exact, converting the kernel's modulus ρ_W directly into a modulus for z, with the second-moment leash R as the only field-dependent constant — and even that enters only as a uniform bound, not through any finer feature of $\mathbf{m}$.

Lemma 24.8 (Continuity of the aggregation in type)**.** *Assume **(A-Reg-W)** with W continuous in its first argument, uniformly: there is a modulus ρ_W with $\int_0^1 |W(x,y) - W(x',y)|\,\mathrm{d}y \le \rho_W(|x-x'|)$, $\rho_W(0^+) = 0$. (Here $|x - x'|$ is the Euclidean distance on the chosen continuous representative of the graphon — the non-canonical labelling structure flagged in the opening notationwatch, whose absence in the measure-preserving-relabelling case forces the harder treatment of remark 24.13.) Then for every field with uniform second moment $\le R^2$,*

$$|z_t(x) - z_t(x')| \;\le\; R\,\rho_W(|x-x'|), \qquad t \in [0,T],$$

so $x \mapsto z_t(x)$ is equicontinuous, with a modulus $R\,\rho_W$ independent of t and of the field.

Proof. The argument is a single subtraction followed by the same two moves — pull the modulus inside the integral, then bound the integrand by a row of the kernel against the L^∞ leash — that ran the magnitude estimate of lemma 24.6, only now the kernel enters as a *difference of rows* rather than a single row. Fix a time t and two types x, x'. The aggregates differ only in which kernel row weights the common first-moment field, so subtracting the two integral representations eq. (24.3) collects the row difference under one integral:

$$z_t(x) - z_t(x') = \int_0^1 \left(W(x,y) - W(x',y)\right) \overline{\mathbf{m}}_t(y)\;\mathrm{d}y.$$

This is the crucial cancellation: the field $\overline{\mathbf{m}}_t$ is identical in both terms, so it survives as a common factor and never has to be differenced — the entire x-variation has been loaded onto the kernel, exactly as the linear-aggregation picture predicts. Take the modulus of both sides and pass it inside the integral (the triangle inequality for the vector-valued integral), then factor out the L^∞ size of the field:

$$|z_t(x) - z_t(x')| \le \int_0^1 |W(x,y) - W(x',y)|\;|\overline{\mathbf{m}}_t(y)|\;\mathrm{d}y \le \|\overline{\mathbf{m}}_t\|_{L^\infty} \int_0^1 |W(x,y) - W(x',y)|\;\mathrm{d}y.$$

The two factors are now exactly the two ingredients we have already secured. The first, $\|\overline{\mathbf{m}}_t\|_{L^\infty} \le R$, is the state-direction leash established in the proof of lemma 24.6: the second moment confines every type's centre of mass to the ball of radius R, uniformly in t and in x. The second is the kernel's own type-modulus: the L^1 distance between rows x and x' is, by the continuity hypothesis, at most $\rho_W(|x-x'|)$. Multiplying, $|z_t(x) - z_t(x')| \le R\,\rho_W(|x-x'|)$, which is the claim. The modulus $R\,\rho_W$ is independent of t because the leash R is uniform in time and ρ_W knows nothing of time, and independent of the field because the only trace the field left is the single scalar R bounding its second moment — precisely the field-independence the Schauder argument will require. ☐

It is worth dwelling on the hypothesis the lemma quietly leans on, because it is the one place in the whole existence proof where the chapter pays for a structure a graphon need not possess, and the preamble's `notationwatch` promised we would be honest about it. To write $\int_0^1 |W(x,y) - W(x',y)|\,\mathrm{d}y \le \rho_W(|x-x'|)$ is to assert that the row-map $x \mapsto W(x,\cdot)$ is uniformly continuous, and "continuous" presupposes a metric on the type set $[0,1]$ to measure $|x-x'|$. But a graphon is canonically an equivalence class under measure-preserving relabelling of $[0,1]$: nothing in its definition equips the labels with a distance, and a measurable relabelling can scramble any continuous representative into a wildly discontinuous one without changing the graphon as an analytic object at all. So before the hypothesis even has a referent we must *choose* a particular continuous representative of the class and endow $[0,1]$ with its Euclidean metric, purely so that "nearby types" means something. This is non-canonical extra structure, and it is imposed for this hypothesis alone — the magnitude estimates of the previous subsection, and every spectral statement of the uniqueness theory to come, are invariant under relabelling and ask for no such choice. The caveat is load-bearing, not decorative. When W is given only up to relabelling — the generic case — the choice is unavailable, the row-map has no continuity to invoke, and the modulus ρ_W simply does not exist; the equicontinuity bought here evaporates. That is exactly why the measurable case is genuinely harder and must be treated by a different mechanism, the Hilbert–Schmidt compactness of $\mathbb{W}$ itself replacing the lost row-continuity, which we defer to remark 24.13. The reader should carry forward that the comfortable type-modulus of this subsection is a privilege of continuous graphons, conjured by a non-canonical labelling we are borrowing for one lemma and will return.

With the aggregate equicontinuous in type we must now propagate that modulus through the per-type solve, and here the factorisation earns its keep once more: the solve $\mathcal{R}$ is the classical machine run independently in each type, so transferring closeness in the data to closeness in the output is exactly a pair of *stability* statements imported from chapter 10, applied to two neighbouring types rather than to two successive iterates. The logic is a chain of four links, and the discipline to hold throughout is that every constant produced is type-independent, because each is a classical constant that depends only on the data bounds, the horizon, and the ellipticity — never on the type label, which the per-type problem does not see. Trace the chain. Two nearby types x, x' present, after the freeze, coupling data $F_0(\cdot, z_t(x))$ and $F_0(\cdot, z_t(x'))$ whose aggregate arguments differ by at most $R\,\rho_W(|x-x'|)$; since F_0 is uniformly Lipschitz in that argument (assumption 24.5), the two data fields are correspondingly close in sup-norm — this is the aggregation modulus entering the solve. By the stability of the HJB solve in its frozen data (lemma 10.10) the two value functions, and crucially their gradients, are then close locally uniformly — a value problem with nearby sources and nearby terminal data has nearby solutions, with a constant inherited from the uniform parabolicity. Two nearby types, however, do not merely present nearby data to one common operator: the Hamiltonian $H(x,\cdot)$ carries the type slot explicitly (eq. (24.1)), so the two HJB operators are themselves different, and that direct difference enters the value comparison as a second forcing — one controlled, as the proof makes precise, by the type-modulus of H and not by the aggregate at all. Close gradients mean close optimal feedbacks, since the feedback is the Hamiltonian slope $-\partial_p H$ evaluated at the gradient and $\partial_p H$ is locally Lipschitz under **(A-Conv)** — to which the explicit type-dependence of $\partial_p H(x,\cdot)$ adds, again, its own direct term. Finally, close feedbacks mean close output flows: by the stability of the FP/SDE solve in its drift (lemma 10.11) two diffusions with the same initial law and nearby drifts have laws close in d_1. Composing the four Lipschitz links turns the input modulus $R\,\rho_W$ into an output modulus η for the field. We state the conclusion and then fill in each mechanism quantitatively.

Lemma 24.9 (Type-equicontinuity of the output field)**.** *Under **(A-Lip)**, **(A-Nondeg)**, assumption 24.5, and the uniform graphon continuity of lemma 24.8, there is a modulus η, with $\eta(0^+) = 0$, depending only on the data and on ρ_W but not on $\mathbf{m}$, such that*

$$\mathsf{d}_1(\mathsf{S}_W(\mathbf{m})^x, \mathsf{S}_W(\mathbf{m})^{x'}) \;\le\; \eta(\,|x - x'|\,), \qquad x, x' \in [0, 1].$$

Proof sketch and pointer. We follow the four-link chain just described, naming the mechanism behind each link and the type-independent constant it contributes, but we do not reproduce the underlying parabolic regularity estimate, which is established once and for all in chapter 6; this is the in-book-pointer rule of the style bible, honoured by pointing to where the estimate genuinely lives rather than asserting it. Write $\delta = |x - x'|$.

Link one: data. By lemma 24.8 the aggregates obey $|z_t(x) - z_t(x')| \le R\,\rho_W(\delta)$ uniformly in t, and by the uniform Lipschitz dependence of F_0, G_0 on their aggregate slot (assumption 24.5) this transfers to the frozen data,

$$\big\|F_0(\cdot, z_t(x)) - F_0(\cdot, z_t(x'))\big\|_\infty \;\le\; \mathrm{Lip}(F_0)\,R\,\rho_W(\delta) \qquad \text{uniformly in } t,$$

and the identical bound with $\mathrm{Lip}(G_0)$ for the terminal data. Call this common data-gap $\epsilon := \max(\mathrm{Lip}(F_0), \mathrm{Lip}(G_0))\,R\,\rho_W(\delta)$, an $O(\rho_W(\delta))$ quantity carrying no type dependence beyond δ.

Link two: value and gradient. Here the chain departs from a naive reading, and the departure is the one subtlety the type variable injects into an otherwise classical step: the two type problems are *not* one parabolic operator fed two data sets. The Hamiltonian $H(x, \cdot)$ in eq. (24.1) carries the type slot explicitly, so the operators $-\partial_t - \nu\Delta_\xi + H(x, \nabla_\xi\cdot)$ and $-\partial_t - \nu\Delta_\xi + H(x', \nabla_\xi\cdot)$ differ as operators even where the frozen data agree. Write $w := u^{\mathbf{m},x} - u^{\mathbf{m},x'}$ and subtract the two copies of eq. (24.1):

$$-\partial_t w - \nu\,\Delta_\xi w + \underbrace{\big[H(x, \nabla_\xi u^{\mathbf{m},x}) - H(x, \nabla_\xi u^{\mathbf{m},x'})\big]}_{= \mathbf{a}\cdot\nabla_\xi w} = \underbrace{\big[F_0(\cdot, z_t(x)) - F_0(\cdot, z_t(x'))\big]}_{\text{indirect: aggregate gap, } \le \epsilon} - \underbrace{\big[H(x, \nabla_\xi u^{\mathbf{m},x'}) - H(x', \nabla_\xi u^{\mathbf{m},x'})\big]}_{\text{direct: explicit type gap}}$$

with terminal data $w(T, \cdot) = G_0(\cdot, z_T(x)) - G_0(\cdot, z_T(x'))$ of sup-norm $\le \epsilon$. The left-hand bracket linearises into a transport term $\mathbf{a} \cdot \nabla_\xi w$ with bounded coefficient $\mathbf{a} = \int_0^1 \partial_p H\big(x, \nabla_\xi u^{\mathbf{m},x'} + s\,\nabla_\xi w\big)\,\mathrm{d}s$ (finite because both gradients lie in the ball $|p| \le L_u$ of lemma 10.5), so w solves a *linear* uniformly parabolic equation, with the uniform gradient bound L_u inherited by both $u^{\mathbf{m},x}$ and $u^{\mathbf{m},x'}$. Its forcing has exactly the two pieces that must be kept apart: the indirect aggregate gap, of size $\epsilon = O(\rho_W(\delta))$ from Link one, *and* the *direct* explicit type gap $H(x, p) - H(x', p)$ read at $p = \nabla_\xi u^{\mathbf{m},x'}$ — the contribution no aggregate can see, present even if the two types' coupling data coincided exactly. It is controlled by the type-modulus of the Hamiltonian on the gradient ball,

$$\sup_{|p| \le L_u} \big|H(x, p) - H(x', p)\big| \;\le\; \rho_H(\delta), \qquad \rho_H(0^+) = 0,$$

where ρ_H is furnished by the type-regularity of the coefficients (b, σ, L) under **(A-Lip)**: it is the data's *own* type-modulus, the cost-side companion of the kernel modulus ρ_W. Feeding both forcings through the quantitative perturbation estimate behind lemma 10.10 — two solutions of a linear uniformly parabolic equation whose forcing and terminal data are $O(\epsilon + \rho_H(\delta))$ in sup-norm differ by $O(\epsilon + \rho_H(\delta))$ in C^1_{loc} on the support-carrying region, gradients included, with a constant fixed by the horizon T, the ellipticity constant λ of **(A-Nondeg)**, and the Lipschitz data bounds, all of them type-free — gives

$$\left\| \nabla_\xi u^{\mathbf{m},x} - \nabla_\xi u^{\mathbf{m},x'} \right\|_{C_{\mathrm{loc}}} \lesssim \epsilon + \rho_H(\delta) \lesssim R\,\rho_W(\delta) + \rho_H(\delta).$$

Link three: feedback. The closed-loop drifts of the two types are $-\partial_p H(x, \nabla_\xi u^{\mathbf{m},x})$ and $-\partial_p H(x', \nabla_\xi u^{\mathbf{m},x'})$, and their gap carries the same two contributions, indirect and direct, for the same reason: the feedback map $-\partial_p H(x,\cdot)$ is itself type-dependent. Insert the cross term $\partial_p H(x, \nabla_\xi u^{\mathbf{m},x'})$:

$$\partial_p H(x, \nabla_\xi u^{\mathbf{m},x}) - \partial_p H(x', \nabla_\xi u^{\mathbf{m},x'}) = \underbrace{\left[\partial_p H(x, \nabla_\xi u^{\mathbf{m},x}) - \partial_p H(x, \nabla_\xi u^{\mathbf{m},x'})\right]}_{p\text{-Lipschitz}} + \underbrace{\left[\partial_p H(x, \nabla_\xi u^{\mathbf{m},x'}) - \partial_p H(x', \nabla_\xi u^{\mathbf{m},x'}\right.}_{\text{explicit type gap}}$$

The first term is the classical one: $\partial_p H$ is locally Lipschitz in p under **(A-Conv)** and both gradients lie in the fixed ball $|p| \le L_u$, so it is at most a type-free constant times the gradient gap of Link two, i.e. $O\big(R\,\rho_W(\delta) + \rho_H(\delta)\big)$. The second is the direct explicit-type term the gradient comparison cannot produce, bounded by the same Hamiltonian type-modulus now applied to $\partial_p H$ on the ball, $\sup_{|p|\le L_u} |\partial_p H(x,p) - \partial_p H(x',p)| \le \rho_H(\delta)$ (we read ρ_H as the joint type-modulus of H and $\partial_p H$ on that ball, both delivered by the coefficient regularity of **(A-Lip)**). The drift gap is therefore $O\big(R\,\rho_W(\delta) + \rho_H(\delta)\big)$ in sup-norm, every constant fixed by H on the ball and the horizon, and so type-free.

Link four: output flow. The two output flows $\mathsf{S}_W(\mathbf{m})^x$ and $\mathsf{S}_W(\mathbf{m})^{x'}$ are the laws of two diffusions started from their respective initial data with these nearby drifts. The SDE stability estimate behind lemma 10.11 is the synchronous-coupling estimate: drive the two diffusions by the *same* Brownian path and the same initial coupling, so the diffusion terms cancel exactly and only the drift difference accumulates; Grönwall control of the resulting pathwise discrepancy then bounds the d_1-distance of the laws by $C(T)$ times the sup-norm of the drift gap, where $C(T)$ is the Grönwall–Lipschitz constant $e^{\mathrm{Lip}\cdot T}$ built from the horizon and the drift's Lipschitz bound. Crucially $C(T)$ is the very constant that appears in the classical stability estimate of chapter 10; it is assembled from T, the ellipticity, and the data bounds and carries no type, which is what makes the resulting modulus uniform over the continuum of types. (To keep the initial slice from contributing, recall that all output flows share the type-continuous initial datum $x \mapsto m_0^x$ of assumption 24.5, whose own modulus is absorbed into η; the synchronous coupling starts the two diffusions from $\mathcal{W}_1$-close laws by exactly that initial modulus.)

Composing the four links — and carrying the direct explicit-type gap of Links two and three alongside the indirect aggregate gap throughout —,

$$\mathsf{d}_1\big(\mathsf{S}_W(\mathbf{m})^x, \mathsf{S}_W(\mathbf{m})^{x'}\big) \le C(T)\left[\max(\mathrm{Lip}(F_0), \mathrm{Lip}(G_0))\,R\,\rho_W(\delta) + \rho_H(\delta)\right] := \eta(\delta),$$

a modulus with $\eta(0^+) = 0$ since $\rho_W(0^+) = 0$ and $\rho_H(0^+) = 0$, built entirely from type-free constants and from the two type-moduli ρ_W (the kernel's) and ρ_H (the Hamiltonian's), hence independent of $\mathbf{m}$. The conclusion's stated dependence "only on the data and on ρ_W" survives intact: ρ_H is not a new kind of input but the data's own regularity in the type variable — the explicit-type modulus of H is part of "the data" exactly as the state-Lipschitz constants of **(A-Lip)** are — and it brings no dependence on the field $\mathbf{m}$ or on the bare type label. The full parabolic perturbation estimate underwriting links two and three is proved in chapter 6 and recorded in [36, Ch. 3]; we have named each mechanism and quantified its dependence without reproducing it here. □

This is the second and final compactness ingredient. The previous subsection handed the existence argument a single convex compact set $\mathcal{K}$ controlling each fibre in state and time; lemma 24.9 now hands it the type-direction modulus η controlling how the fibres vary from one type to the next — and, decisively, η is one modulus for *every* field $\mathsf{S}_W(\mathbf{m})$ in the range of the best-response map, because every constant in its construction was type- and field-independent. With confinement $\mathcal{K}$ in the inner direction and equicontinuity η in the outer direction, both wings of the two-level picture are pinned. The next subsection collects exactly these two ingredients and feeds them to the metric-space Arzelà–Ascoli theorem, which converts "every fibre in a compact set, plus a uniform type-modulus" into a single compact set of fields — the convex compact invariant domain on which Schauder finally produces the equilibrium.

24.2.3 The invariant set and the existence theorem

The two preceding subsections discharged, separately, the two compactness obligations the existence proof carries, one for each level of the field space's two-tier structure. Lemma 24.7 secured the inner, state-and-time level: it trapped every fibre m^x of every output field in a single convex compact set $\mathcal{K} \subseteq \mathcal{C}$, the trap not moving as the type or the input field varied, and bought that uniformity from the *magnitude* of the kernel rows through the row-sum norm of $\mathbb{W}$. Lemma 24.9 secured the outer, type level: it produced a single modulus of continuity η controlling how fast the fibre may change as the type moves, the modulus one and the same for every field in the range of S_W, and bought it from the *variation* of the kernel rows through the continuity half of **(A-Reg-W)**. We now hold both ingredients in raw form, and what remains is purely organisational: we must glue them into one set, check that S_W maps that set into itself, and verify the set has the three properties Schauder demands — nonempty, convex, compact. Of these the compactness is, as always, the only one that costs anything, and the tool that converts our two moduli into it is the metric-space form of the Arzelà–Ascoli theorem.

It is worth recalling exactly what that theorem says and why it is the right gluing device, because the entire force of the subsection is concentrated in its single application. The classical Arzelà–Ascoli theorem characterises the relatively compact subsets of $C(K;\mathbb{R})$, continuous real functions on a compact domain, as the bounded equicontinuous ones; its metric-space-valued generalisation (theorem B.24) replaces the scalar target $\mathbb{R}$ by an arbitrary complete metric space Y and the boundedness hypothesis by *pointwise relative compactness* — the requirement that, at each point of the domain, the set of values the family takes be relatively compact in Y. A family of maps $K \to Y$ is then relatively compact in $C(K;Y)$, under the uniform metric, if and only if it is equicontinuous and pointwise relatively compact. This is precisely the shape of our situation. Our domain is the compact type interval $[0,1]$; our target is the compact metric space $Y = (\mathcal{K}, \mathsf{d}_1)$ that lemma 24.7 delivered; the family is the set of fields in $\mathbb{K}_W$, each read as a map $x \mapsto m^x$ from types into $\mathcal{K}$. The η-modulus is exactly the equicontinuity hypothesis, and the values' membership in the compact $\mathcal{K}$ is exactly — more than exactly — the pointwise relative compactness hypothesis. The theorem then converts "every fibre in a compact set, plus a uniform type-modulus" into the one thing the construction has been driving at, a single compact set of *fields*. We assemble that set now.

Lemma 24.10 (A convex compact invariant set of fields)**.** *With $\mathcal{K} \subseteq \mathcal{C}$ from lemma 24.7 and η from lemma 24.9, define*

$$\mathbb{K}_W := \Big\{ \mathbf{m} \in \mathbb{M} : \ \mathbf{m}_0 \textit{ fixed}, \ m^x \in \mathcal{K} \ \forall x, \ \mathsf{d}_1(m^x, m^{x'}) \le \eta(|x - x'|) \ \forall x, x' \Big\}.$$

Then $\mathbb{K}_W$ is nonempty, convex, and compact in $(\mathbb{M}, \mathsf{D})$, and $\mathsf{S}_W(\mathbb{K}_W) \subseteq \mathbb{K}_W$.

The set $\mathbb{K}_W$ is cut out by exactly the three constraints the two compactness lemmas certify: its members start from the prescribed initial field $\mathbf{m}_0$, run each fibre inside the inner compact set $\mathcal{K}$, and obey the outer type-modulus η. The fixed initial slice is the third, quiet, constraint — not a compactness device but the boundary condition that pins the whole problem, and it is harmless to all four verifications because it is an *affine* condition (it survives convex combinations) that is *closed* (it survives uniform limits) and is *respected* by S_W (the best response never moves a population's time-zero law). We verify the four claims in turn, and the reader should watch each one consume precisely one of the inputs already in hand.

Proof. Nonempty. We must exhibit at least one field obeying all three constraints, and the cheapest candidate is the one that does nothing: the *frozen* field $\mathbf{m}^{\text{stat}}$ whose every fibre is constant in time, $m^x_t \equiv m^x_0$. Each such constant flow lies in $\mathcal{K}$, for $\mathcal{K}$ was built in lemma 10.8 from a second-moment radius R' and a time-modulus ω, and a flow that never moves trivially meets the time-modulus (its $\mathcal{W}_1$-increment over any time gap is zero, hence below ω) while its second moment is the initial one $\int |\xi|^2 \, m^x_0(\mathrm{d}\xi)$, bounded by the uniform initial bound of assumption 24.5 and hence inside the radius R' that dominates it. The type-modulus is met because $x \mapsto m^x_0$ is continuous in type by assumption 24.5; its own modulus of continuity need not literally be $\le \eta$, but this is the one place where we are free to *enlarge* η by the initial modulus once and for all, which costs nothing downstream because every later use of η only needs $\eta(0^+) = 0$, a property the enlargement preserves. With η so enlarged, the static field sits in $\mathbb{K}_W$, which is therefore nonempty.

Self-map. This is where the two compactness lemmas are spent. Take any $\mathbf{m} \in \mathbb{K}_W$ and apply S_W. By lemma 24.7 every output fibre $\mathsf{S}_W(\mathbf{m})^x$ lies in the same $\mathcal{K}$, so the inner constraint holds; by lemma 24.9 the output field obeys the η-modulus $\mathsf{d}_1(\mathsf{S}_W(\mathbf{m})^x, \mathsf{S}_W(\mathbf{m})^{x'}) \le \eta(|x - x'|)$, so the outer constraint holds; and the initial-slice constraint holds because the best response leaves time-zero laws untouched — by definition 24.4 the closed-loop diffusion starts from $X^{\mathbf{m},x}_0 \sim m^x_0$, so $\mathsf{S}_W(\mathbf{m})^x_0 = \mathrm{Law}(X^{\mathbf{m},x}_0) = m^x_0$, the prescribed datum, for every x. All three defining constraints survive the map, so $\mathsf{S}_W(\mathbf{m}) \in \mathbb{K}_W$, which is the self-map property.

Convex. We must check that $\mathbb{K}_W$ is closed under convex combinations, constraint by constraint. Fix $\mathbf{m}, \mathbf{m}' \in \mathbb{K}_W$ and $\lambda \in [0, 1]$, and form the fieldwise combination $\mathbf{n} := \lambda \mathbf{m} + (1 - \lambda)\mathbf{m}'$, i.e. $n^x_t := \lambda m^x_t + (1 - \lambda) m'^x_t$. First, each fibre n^x lies in $\mathcal{K}$ because $\mathcal{K}$ is convex (lemma 10.8) and n^x is the convex combination of the two fibres $m^x, m'^x \in \mathcal{K}$. Second, the type-modulus survives because $\mathcal{W}_1$ is jointly convex in its two arguments — a standard consequence of the linearity of the coupling constraints in the Kantorovich problem (theorem 4.8): superposing optimal couplings of the two pairs gives an admissible, generally suboptimal, coupling of the combinations, so

$$\mathcal{W}_1(\lambda\mu_1+(1-\lambda)\mu_2,\ \lambda\nu_1+(1-\lambda)\nu_2)\ \le\ \lambda\,\mathcal{W}_1(\mu_1,\nu_1)+(1-\lambda)\,\mathcal{W}_1(\mu_2,\nu_2).$$

Taking the supremum over time turns this into the same convexity for d_1, whence at any two types

$$\mathsf{d}_1(n^x,n^{x'})\le\lambda\,\mathsf{d}_1(m^x,m^{x'})+(1-\lambda)\,\mathsf{d}_1(m'^x,m'^{x'})\le\lambda\,\eta(|x-x'|)+(1-\lambda)\,\eta(|x-x'|)=\eta(|x-x'|),$$

so the η-bound passes to $\mathbf{n}$. Third, the fixed initial slice survives because it is an affine condition: $n^x_0=\lambda m^x_0+(1-\lambda)m^x_0=m^x_0$, both fields carrying the same prescribed $\mathbf{m}_0$. All three constraints are convex, so $\mathbb{K}_W$ is convex. (The convexity is genuine and not a formality: Schauder, unlike Banach, asks nothing of contractivity but it does demand a convex domain on which to run the topological degree argument, and this is where that demand is met.)

Compact. Here the Arzelà–Ascoli machinery does its work, and we run it in two stages: relative compactness from the theorem, then closedness by hand, the two combining to compactness. For relative compactness apply theorem B.24 to the family of maps $x\mapsto m^x$ as $\mathbf{m}$ ranges over $\mathbb{K}_W$, with compact domain $K=[0,1]$ and complete target $Y=(\mathcal{K},\mathsf{d}_1)$. The theorem's two hypotheses are exactly our two ingredients. Pointwise relative compactness holds because, at every fixed type x, the set of values $\{m^x:\mathbf{m}\in\mathbb{K}_W\}$ lies in $\mathcal{K}$, which is compact (lemma 10.8) — indeed the family is not merely pointwise but *uniformly* valued in one compact set, which is stronger than required. Uniform equicontinuity holds because every $\mathbf{m}\in\mathbb{K}_W$ satisfies the same η-bound $\mathsf{d}_1(m^x,m^{x'})\le\eta(|x-x'|)$ with $\eta(0^+)=0$, a modulus that does not depend on which field we look at — this is the field-independence we laboured to secure in lemma 24.9, now cashed as the equicontinuity Arzelà–Ascoli needs. The theorem therefore returns relative compactness of $\mathbb{K}_W$ in $C([0,1];\mathcal{K})$ under the uniform metric, which is exactly the restriction of D.

It remains to show $\mathbb{K}_W$ is *closed* in $(\mathbb{M},\mathsf{D})$, for a relatively compact set is compact precisely when it is closed. Let $\mathbf{m}^{(n)}\to\mathbf{m}$ in D with every $\mathbf{m}^{(n)}\in\mathbb{K}_W$; we check that the limit inherits all three constraints. Convergence in D is uniform in type, so in particular at each fixed x the fibres converge, $\mathsf{d}_1(m^{(n),x},m^x)\to0$. The inner constraint passes because $\mathcal{K}$ is closed (being compact): each $m^{(n),x}\in\mathcal{K}$ and $m^{(n),x}\to m^x$ in d_1 force $m^x\in\mathcal{K}$. (Implicit here is that the second-moment radius defining $\mathcal{K}$ is itself closed under $\mathcal{W}_1$-limits, which is the lower semicontinuity of the second moment along $\mathcal{W}_1$-convergent sequences, theorem 4.8: mass cannot appear from infinity in the limit, so the moment cannot jump up, and the radius bound survives.) The outer constraint passes because d_1 is continuous in each argument, so $\mathsf{d}_1(m^x,m^{x'})=\lim_n\mathsf{d}_1(m^{(n),x},m^{(n),x'})\le\eta(|x-x'|)$, the bound holding for every n and hence in the limit. The initial slice passes because $m^x_0=\lim_n m^{(n),x}_0=m^x_0$ is the common prescribed datum. So $\mathbf{m}\in\mathbb{K}_W$, the set is closed, and being closed and relatively compact it is compact in $(\mathbb{M},\mathsf{D})$. $\square$

The lemma has manufactured the arena: a nonempty convex compact set on which S_W acts as a self-map. Schauder's fixed-point theorem (theorem 2.38) asks for exactly this geometry plus one analytic fact — that the self-map be *continuous* — and then guarantees a fixed point. The continuity is the last thing to check, and it is here that the factorisation $\mathsf{S}_W=\mathcal{R}\circ\mathcal{A}_W$ pays its final dividend in the existence proof: continuity of a composition follows from continuity of each factor, and we have arranged the two factors so that one is a bounded linear operator (trivially continuous, once we know in which topology) and the other is the classical solve whose continuity is an imported chapter 10 fact. We state the theorem and then run the continuity through the two factors.

Theorem 24.11 (Existence of GMFG equilibria)**.** *Assume **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)** with $H(x,\cdot)$ strictly convex and C^1 and $\partial_p H$ of at most linear growth in p; assume **(A-Reg-W)** with W uniformly continuous in its type argument (lemma 24.8); and assume the coupling and initial-data hypotheses of assumption 24.5. Then the graphon best-response map S_W has a fixed point $\mathbf{m}^\star \in \mathbb{K}_W$. Equivalently, the GMFG system eq.* (24.1)*–eq.* (24.2) *admits a solution $(\mathbf{u}^\star, \mathbf{m}^\star)$ with $x \mapsto m^{\star,x} \in C([0,1]; C([0,T]; \mathcal{P}_2(\mathbb{R}^d)))$, each $u^{\star,x}$ Lipschitz in ξ uniformly in (t,x), and the field uniformly $\frac{1}{2}$-Hölder in time and ρ_W-continuous in type.*

Proof. By lemma 24.10, $\mathbb{K}_W$ is a nonempty convex compact subset of $(\mathbb{M}, \mathsf{D})$ mapped into itself by S_W. It remains to verify that $\mathsf{S}_W|_{\mathbb{K}_W}$ is continuous; then Schauder's theorem (theorem 2.38; viewed inside the locally convex space $C([0,1];\text{signed measures})$ as in the proof of theorem 10.14, the ambient linear structure in which $\mathbb{K}_W$ sits as a compact convex set) yields a fixed point. We obtain the continuity from the factorisation eq. (24.4), $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$, showing each factor continuous on the relevant space and composing.

The aggregation factor $\mathcal{A}_W$ is continuous from $(\mathbb{K}_W, \mathsf{D})$ to $(C([0,1];\mathbb{R}^d), \|\cdot\|_\infty)$, and the argument is a two-link chain in the now-familiar pattern: the first-moment read-out is continuous in the field, the operator $\mathbb{W}$ is continuous on its output. Take $\mathbf{m}^{(n)} \to \mathbf{m}$ in D. For the first link, the first-moment field converges, $\overline{\mathbf{m}}_t^{(n)} \to \overline{\mathbf{m}}_t$ in $L^2([0,1];\mathbb{R}^d)$ uniformly in t. This is the 1-Lipschitz bound lemma 24.2 doing exactly its advertised job: at each type the first moment is a $\mathcal{W}_1$-Lipschitz functional of the law, and on the second-moment-bounded set $\mathcal{K}$ the metric $\mathcal{W}_1$ genuinely metrises weak convergence (theorem 4.8), so a D-convergent sequence of fields has fibrewise-, indeed uniformly-, convergent first moments, $\sup_t \left\|\overline{\mathbf{m}}_t^{(n)} - \overline{\mathbf{m}}_t\right\|_{L^\infty} \le \mathsf{D}(\mathbf{m}^{(n)}, \mathbf{m}) \to 0$. For the second link, $\mathbb{W}$ is a bounded operator, so post-composing it with the evaluation $z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x)$ turns that convergence of first moments into convergence of aggregates; and because the aggregates carry the *uniform* type-modulus $R\,\rho_W$ of lemma 24.8 — a modulus the limit aggregate shares, since it is field-independent — the convergence is not merely in L^2 but uniform in x, i.e. $z^{(n)} \to z$ in $C([0,1] \times [0,T])$. (Equicontinuity in type plus pointwise convergence upgrades to uniform convergence, the same Arzelà–Ascoli mechanism, used now to control a limit rather than to extract a subsequence.)

The solve factor $\mathcal{R}$ is continuous, and this is the classical statement proposition 10.13 applied uniformly across the continuum of types. The input to $\mathcal{R}$ is the aggregate data $z = (z(x))_x$, just shown to converge $z^{(n)}(x) \to z(x)$ locally uniformly in x. In each type this is precisely the data-convergence that the homogeneous stability chain consumes: convergence of the frozen running source and terminal data gives convergence of the value functions and, crucially, of their gradients (lemma 10.10), hence of the optimal feedbacks $-\partial_p H(x, \nabla_\xi u^{\mathbf{m},x})$ (the Hamiltonian slope being locally Lipschitz under **(A-Conv)**), hence of the output flows, by the stability of the Fokker–Planck/SDE solve in its drift (lemma 10.11). The one new word is *uniformly*: all the constants in this chain are the type-free constants secured in lemmas 24.6 and 24.9 — the gradient bound L_u, the drift bound M, the Grönwall constant $C(T)$ — none of which carries a type label, so the per-type convergences are uniform in x and assemble into convergence of the output *field* in D. Thus $\mathcal{R}$ is continuous on the convergent aggregates produced by $\mathcal{A}_W$.

Composition of continuous maps is continuous, so $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ is continuous on $\mathbb{K}_W$, and Schauder delivers the fixed point $\mathbf{m}^\star \in \mathbb{K}_W$. That $\mathbf{m}^\star$ is an equilibrium, and equivalently that the pair $(\mathbf{u}^{\mathbf{m}^\star}, \mathbf{m}^\star)$ solves the GMFG system, is the exact reduction recorded after definition 24.4. The asserted regularity of the fixed point is read off the set it lives in and the solve that produced it: membership in $\mathbb{K}_W$ gives, by lemma 24.10, that $x \mapsto m^{\star,x}$ is continuous in type with modulus η (hence ρ_W-continuous, η being $O(\rho_W)$ by lemma 24.9) and that each fibre lies in $\mathcal{K}$, whose time-modulus is the uniform $\frac{1}{2}$-Hölder bound of lemma 10.8; and the per-type value $u^{\star,x}$ inherits the gradient bound L_u of lemma 10.5, uniform in (t, x). These are precisely the regularity facts the later sections will draw on. □

The proof is complete, and before moving on it is worth standing back to name what powered it, because the two compactness inputs came from genuinely different parts of the book and the distinction is the throughline made audible. The next remark does this naming; the two that follow it open the doors the standing hypotheses kept shut — merely measurable kernels, and set-valued or block-structured couplings — and each is a deliberate signpost to a later chapter rather than a parenthetical aside.

Remark 24.12 (The two compactness inputs, named)**.** The proof rests on two compactness facts of different origin, and tracing each to its source is the cleanest way to see what is old and what is new in the graphon existence theorem. The state-and-time compactness — the inner set $\mathcal{K}$ — is pure Part I machinery: tightness manufactured from a uniform second moment by Markov's inequality, promoted to relative weak compactness by Prokhorov's theorem (chapter 3), and assembled with a time-modulus into a compact flow space exactly as in lemma 10.8. The only graphon fingerprint on it is that its constants were made uniform across types, and that uniformity was bought from a single *magnitude* of the operator, the row-sum norm of $\mathbb{W}$ (lemma 24.6). Strip the graphon away and this is verbatim the homogeneous existence argument. The type-direction compactness — the modulus η — is by contrast a Part III input with no Part I ancestor: it is manufactured from the *regularity* of the graphon, specifically the uniform continuity of W in its type argument (lemma 24.8), the analytic content that the graph-limit theory of chapters 15 and 18 installs on the kernel. Neither input alone suffices — confinement without an equicontinuity modulus leaves the type-curve free to oscillate, and a modulus without confinement leaves the fibres free to run to infinity — and the decisive point is that the second input has *no classical analogue whatsoever*: a homogeneous mean field game has no type variable, hence nothing to be compact in, hence no equicontinuity to demand. This is the first concrete sense in which the graphon problem is strictly harder than its homogeneous parent, and it is the reason **(A-Reg-W)** surfaces already in the hypotheses of an *existence* theorem, where the classical theorem 10.14 needed no regularity of any coupling kernel at all. Norm and continuity, the two faces of the one operator $\mathbb{W}$, supply the two compactnesses: this is the chapter's organising claim instantiated at the very point where existence is won.

Remark 24.13 (Merely measurable graphons)**.** The continuity half of **(A-Reg-W)** was load-bearing in exactly one place — the type-modulus η of lemma 24.9 — and the opening `notationwatch` warned that it asks for a structure a graphon generically lacks. A graphon is canonically an equivalence class under measure-preserving relabelling of $[0, 1]$, and such a class need contain no continuous representative; in the generic, merely *measurable*, case the phrase "$x \mapsto W(x, \cdot)$ is continuous" has no referent, the modulus ρ_W does not exist, and with it the η-modulus evaporates. The type-continuous space $C([0,1];\mathcal{K})$ is then simply the wrong arena, and must give way to a space of *measurable* fields valued in $\mathcal{K}$, the L^2-in-type Bochner space $L^2([0,1];\mathcal{C})$ of chapter 21, whose metric controls only a type-average and no single type. Existence survives, but the missing compactness must be bought by a different mechanism, and there are two honest routes, worth laying out because they are the two routes the convergence chapter will formalise.

The first route is the *step-graphon* or cut-metric route, and it trades the continuum for a finite approximation. The weak regularity lemma (chapter 19) approximates any kernel W in cut metric $\delta_\square$ by step graphons — kernels piecewise constant on a finite partition of $[0,1]$ into finitely many blocks. On a step graphon each block is internally homogeneous: all types within a block present the same kernel row, feel the same aggregate, and the continuum of coupled boundary-value problems collapses to a *finite* multi-population game, for which existence is the elementary fixed-point statement of remark 24.14. One then passes to the limit along the step approximations using the cut-metric stability of the equilibrium map, which is exactly the convergence theory chapter 25 is built to supply. This route never needs a type-modulus because it never works in the type-continuous topology; it works block by block in a finite world and only takes a limit at the end.

The second route stays in the continuum but changes the source of compactness from regularity of the kernel to *compactness of the operator*. Working directly in $L^2([0,1];\mathcal{C})$, the aggregation factor $\mathcal{A}_W$ is still the bounded linear map $\mathbb{W}$ post-composed with the first-moment read-out; but $\mathbb{W}$ is not merely bounded, it is Hilbert–Schmidt and therefore *compact* (chapter 18). A compact operator maps the bounded set of first-moment fields into a *relatively compact* set in $L^2([0,1];\mathbb{R}^d)$, and that relative compactness of the aggregate is precisely the substitute for the type-equicontinuity we lost: where continuity of the kernel made the output field equicontinuous, the Hilbert–Schmidt compactness of $\mathbb{W}$ makes the aggregate data pre-compact, and the Schauder argument runs on that. The one extra subtlety is that with no single-type control the per-type optimisers must be re-collected into a genuinely measurable field, which is arranged by the measurable-selection theory of lemma 10.20 — the same selection device the relaxed homogeneous theory used. The measurable case is thus not a failure of the method but a redistribution of its compactness burden from the kernel's regularity onto the operator's own compactness, and because its natural home is the cut-metric stability theory, we develop it in full there rather than here, in chapter 25.

Remark 24.14 (Set-valued feedback and block reduction)**.** Two boundary cases of the standing hypotheses are worth following to their conclusions, one relaxing the regularity of the per-type solve and one specialising the kernel, because each maps the graphon theorem onto a neighbouring chapter. First, the relaxation. The whole construction ran in the single-valued regime fixed at the close of section 24.1: strict convexity **(A-Conv)** and nondegeneracy **(A-Nondeg)** together made the per-type optimal feedback a genuine function, hence $\mathcal{R}$ a map and S_W an honest self-map for which Schauder is the right tool. Drop either hypothesis — a Hamiltonian with a flat stretch, or a degenerate diffusion — and the optimiser may be indifferent among several feedbacks: $\mathcal{R}$ becomes *set-valued*, S_W a correspondence rather than a map, and Schauder is replaced by its set-valued cousin, the Kakutani–Fan–Glicksberg theorem (theorem 2.40), applied to the field-valued correspondence on the very same convex compact set $\mathbb{K}_W$. The output is a *weak* (relaxed) graphon equilibrium, with a measurable-selection step recovering an honest feedback from the correspondence, exactly as theorem 10.19 relaxes theorem 10.14 in the homogeneous theory. The moral is the throughline once more: the set-valuedness is entirely a phenomenon of the classical factor $\mathcal{R}$, since it is the per-type optimiser that turns multivalued, while the aggregation $\mathcal{A}_W$ — linear, single-valued, blind to convexity — is untouched. The graphon adds nothing to this difficulty and subtracts nothing from it; it is inherited whole from chapter 10.

Second, the specialisation, which runs to the opposite extreme and connects forward rather than back. Take a *block graphon*: a kernel piecewise constant on a partition $[0,1] = \bigsqcup_{k=1}^{K} I_k$ of the type interval into K blocks, $W(x,y) = w_{k\ell}$ for $x \in I_k$, $y \in I_\ell$. Then all types in a common block are genuinely interchangeable: they share a kernel row, so they feel the same aggregate $z_t(x) = \sum_\ell w_{k\ell} \int_{I_\ell} \overline{\mathbf{m}}_t$, solve the same per-type problem, and — starting from a block-constant initial datum — produce the same flow. The field therefore collapses to K flows, one per block, the continuum system eq. (24.1)–eq. (24.2) collapses to K coupled scalar HJB/FP pairs, and the best-response map S_W becomes an ordinary self-map of the finite product $\mathcal{K}^K$. What survives is precisely a *finite multi-population* mean field game, K populations coupled through the $K \times K$ matrix $(w_{k\ell})$, and theorem 24.11 reduces to multi-population existence by Schauder on $\mathcal{K}^K$ — a setting with no type-direction subtlety at all, because the type direction has become a finite index. This is no curiosity: it is the bridge to chapter 30, where multi-population games are the object of study, and it confirms the chapter's reductions from the other side — a constant graphon ($K = 1$) returns the homogeneous game of chapter 10, a block graphon returns the finite multi-population game, and only the genuinely continuous kernel needs the equicontinuity machinery this section built.

With existence in hand, the chapter's first question is answered: the equilibrium set is *nonempty*, and the fixed point it contains is regular — continuous in type, $\frac{1}{2}$-Hölder in time, with a type- and time-uniform gradient bound on the value field — which is exactly the regularity the uniqueness identity and the convergence theory will lean on. But nonempty is not the end of the story. Schauder is a blunt instrument: it certifies that *a* fixed point exists and says nothing about how many. The sharpened question that survives is the complementary one — *when is the equilibrium set a singleton?* — and the next section takes it up. As the existence proof split into a classical factor and an operator factor, so will the uniqueness analysis: the two routes to a singleton are the operator versions of the two classical routes of chapter 11, a structural monotonicity good for every horizon and a smallness condition good for weak coupling, and the interesting physics — the critical coupling at which the singleton splits — will be read off the spectrum of the very operator $\mathbb{W}$ that bought us compactness here.

24.3 Uniqueness and the critical coupling

Existence handed us a nonempty equilibrium set together with the regularity of its members, and with it the complementary question that Schauder is constitutionally unable to answer: is that set a *singleton*? A fixed-point theorem that argues by compactness counts at least one solution and never at most one, so uniqueness is a separate undertaking with its own machinery — and, true to the chapter's discipline, that machinery is again the classical one of chapter 11 read through the factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$. The homogeneous theory offered exactly two ways to force a singleton, and they are complementary in a way worth recalling before we lift them, because the lift preserves the complementarity and the chapter's central phenomenon lives precisely in the gap between them. The first classical route was *structural*: if the coupling is Lasry–Lions monotone — crowd-averse, so that a heavier population at a point raises the cost of being there — then the duality identity of theorem 11.5 forbids two equilibria *at every horizon* T, however long, with no smallness assumed. The second was a *smallness* route: if the coupling is weak enough relative to the horizon, the best-response map is a contraction and Banach's theorem delivers a unique fixed point directly, monotone or not. Strong crowd-seeking coupling over a long horizon falls under neither, and there the homogeneous equilibrium genuinely splits — the Curie–Weiss bifurcation of section 11.4.

Both routes lift to the graphon setting, and the only change in each is the insertion of the operator $\mathbb{W}$ at the single seam the chapter has been tracking. The structural route lifts to *graphon Lasry–Lions monotonicity*: the operator version of the standing assumption **(A-Mon)**, now a condition on the coupling as a map of *fields*, whose defining pairing carries an extra integral over the type. It generalises theorem 11.5 and inherits its great virtue — validity at *all* horizons. The smallness route lifts to an *operator-norm contraction*: the standing assumption **(A-Coupling)** read as a bound on $\|\mathbb{W}\|$, under which S_W contracts and Banach applies. It inherits the classical virtue too — it is indifferent to the sign of the coupling, working for crowd-seeking and crowd-averse alike — and the classical defect: it needs the product of coupling strength and horizon to be small, so it says nothing once the horizon is long. The two routes therefore cover the uniqueness landscape from opposite directions. Monotonicity sweeps in along the *coupling-sign* axis, claiming the crowd-averse half-plane (for a positive operator $\mathbb{W}$) at every T; contraction sweeps in along the *coupling-magnitude* axis, claiming a neighbourhood of zero coupling for every sign. What each leaves uncovered is exactly the region the other handles — except for one wedge that *both* leave uncovered, where the coupling is strongly crowd-*seeking* (imitative) and the horizon is long (and, when the network operator $\mathbb{W}$ is indefinite, a second such wedge on the crowd-averse side, a graphon-only phenomenon we flag now and pin down with the critical-coupling proposition). That wedge is not an artefact of two imperfect sufficient conditions; it is where multiplicity actually lives, and the boundary of the wedge is the object the section is built to locate.

That boundary is the *critical coupling*. The picture to carry is of two bounds closing on it from opposite sides: monotonicity guarantees uniqueness up to the point where the equilibrium energy first loses its sign, and contraction guarantees it up to the point where the best-response map first stops shrinking, and these two limits meet — in the diagonalisable Thread B model, *exactly* — at a single threshold value of the coupling. The classical theory placed that threshold at the scalar value s_c^{scalar} where the strategic self-amplification first overpowers the agent's own cost curvature (section 11.4); the graphon theory places it, as the whole chapter has promised, not at s_c^{scalar} but at the product $s\,\lambda_1$, the intrinsic coupling strength magnified by the leading eigenvalue $\lambda_1 = \|\mathbb{W}\| = \rho(\mathbb{W})$ of the network operator — the eigenvector centrality of chapter 18. Loss of uniqueness becomes a symmetry-breaking transition indexed by network centrality: the more central the network (the larger λ_1), the smaller the intrinsic imitation s at which the product crosses s_c^{scalar}, i.e. the smaller the critical coupling $s_c = s_c^{\text{scalar}}/\lambda_1$, and the broken symmetry first appears along the leading eigenmode ϕ_1. Everything graphon-specific in the uniqueness question is thus, once more, read off the spectrum of $\mathbb{W}$.

The section develops this in three movements, in the order the logic demands. We take the structural route first, in definition 24.15 and the theorem that follows, because monotonicity is the hypothesis that controls the long-horizon regime where the interesting physics sits and because it is the direct generalisation of chapter 11; the next subsection therefore opens by lifting **(A-Mon)** to fields and contrasting the graphon pairing eq. (24.5) with its scalar ancestor eq. (11.3) — the two differing, as we will see, by exactly the outer $\int_0^1 \mathrm{d}x$ that threads the operator $\mathbb{W}$ into the energy. We take the smallness route second, as the operator-norm contraction that bounds uniqueness from the weak-coupling side and, as a bonus, yields the Lipschitz dependence of the equilibrium on the kernel that the convergence theory of chapter 25 will consume. Finally we let the two bounds meet: the critical-coupling proposition locates the threshold $s\,\lambda_1 = s_c^{\mathrm{scalar}}$ that monotonicity bounds from one side and contraction from the other, and reads off the centrality-indexed onset of multiplicity. We begin with monotonicity.

24.3.1 Graphon Lasry–Lions monotonicity

The section opening promised the structural route first, and the reason bears repeating before we build it, because it is what makes this the route to reach for when the physics is interesting. Monotonicity asks nothing about how strong the coupling is or how long the game runs; it asks only that the coupling have a definite *sign*. That is exactly the demand that survives the long-horizon, strong-coupling regime where the homogeneous equilibrium was at risk of splitting — the regime the smallness route of the next subsection cannot reach. In the homogeneous theory the sign was supplied by the standing assumption **(A-Mon)**, Lasry–Lions monotonicity (definition 11.1): the running coupling F and the terminal coupling G are *crowd-averse*, in the precise sense that the pairing of the cost increment $F[m] - F[m']$ against the population increment $m - m'$ is nonnegative,

$$\int_{\mathbb{R}^d} \left(F[m] - F[m']\right)(m - m')(\mathrm{d}\xi) \;\geq\; 0,$$

the inequality eq. (11.3) that powered the uniqueness theorem theorem 11.5 at *every* horizon. Reading it physically: moving population mass towards a point raises the cost of sitting at that point, so agents are pushed apart and no self-reinforcing concentration can grow — the mechanism that forbids two competing equilibria.

Our task is to lift this inequality from a single population to a field, and the lift is the gentlest imaginable, which is itself part of the chapter's message. A graphon coupling no longer acts on one population but on a whole field $\mathbf{m}$, and it no longer returns one cost function felt by everyone but a cost *field* $F[\mathbf{m}](x, \cdot)$, a separate cost function for each type. The pairing that tests its monotonicity must therefore pair a function of *type and state* against a signed-measure field over types — the difference $\mathbf{m} - \mathbf{m}'$ of two population fields. There is exactly one way to build such a pairing out of the scalar one: hold the type fixed, run the scalar state-pairing eq. (11.3) inside that type, and then average the result over types. That single extra averaging — one outer integral $\int_0^1 \mathrm{d}x$ wrapped around the inherited scalar pairing — is the whole formal difference between the homogeneous and the graphon monotonicity conditions.

Definition 24.15 (Graphon Lasry–Lions monotonicity)**.** A graphon coupling $F\colon \mathbb{M}_2 \to C([0,1]\times\mathbb{R}^d)$

is *graphon (Lasry–Lions) monotone* if for all fields $\mathbf{m}, \mathbf{m}' \in \mathbb{M}_2$,

$$\mathcal{G}_F(\mathbf{m}, \mathbf{m}') := \int_0^1 \int_{\mathbb{R}^d} \left(F[\mathbf{m}](x,\xi) - F[\mathbf{m}'](x,\xi)\right) (m^x - m'^x)(\mathrm{d}\xi)\ \mathrm{d}x \ \geq\ 0. \tag{24.5}$$

The graphon version of the standing assumption **(A-Mon)** is that both F and G are graphon monotone.

Set side by side, the two pairings are the same object with one extra symbol. Strip the outer $\int_0^1 \mathrm{d}x$ from eq. (24.5) and what remains, at a frozen type x, is verbatim the scalar pairing eq. (11.3) of the cost increment against the population increment; restore the integral and the scalar pairings are merely averaged over the continuum of types. The bracket $\mathcal{G}_F(\mathbf{m}, \mathbf{m}')$ is, in a word, the type-average of the inherited Lasry–Lions energy, and definition 24.15 asks only that this average be nonnegative.

The temptation at this point — and the trap the rest of the section is built to spring — is to read "average of the scalar energies" as "average of monotone things, hence monotone." That reading is exactly wrong, and seeing why is the crucial point, the one whose consequences section 24.4 collects in full. It would be correct if each type's cost field $F[\mathbf{m}](x,\cdot)$ depended only on that type's own population m^x, for then the integrand at type x would be a genuine scalar monotone pairing and the outer integral would preserve its sign. But a graphon coupling is built precisely so that this is *not* the case: by definition 24.3 the cost felt by type x depends on the field through the aggregate $z(x) = (\mathbb{W}\overline{\mathbf{m}})(x)$, which mixes *every* type's population through the kernel. The increment of the integrand at type x therefore involves the increment of the aggregate, $\Delta z(x) = (\mathbb{W}\Delta)(x)$, and the operator $\mathbb{W}$ is now threaded through the energy. The outer integral no longer sees a sum of independently-signed terms; it sees a quadratic form in $\mathbb{W}$, whose sign is decided jointly by the strategic structure of the coupling and the *spectrum* of $\mathbb{W}$. This is the throughline made local: every graphon-specific feature of the uniqueness question is, once again, the responsibility of the one operator and lives in its spectrum. We prove the uniqueness theorem first, exactly as in chapter 11, and only then unfold the pairing to expose the operator inside it.

Theorem 24.16 (Graphon Lasry–Lions uniqueness)**.** *Assume **(A-Lip)**, **(A-Nondeg)**, **(A-Conv)** with $H(x,\cdot)$ strictly convex, **(A-Reg-W)**, and that the couplings F, G are graphon monotone (definition 24.15). Then the GMFG system eq.* (24.1)*–eq.* (24.2) *has at most one classical solution field* $(\mathbf{u}, \mathbf{m})$ *for which the per-type integrations by parts of theorem 11.5 are justified.*

Proof. The argument is theorem 11.5 run in parallel over types and then integrated against $\mathrm{d}x$, and the slogan is worth taking literally: nothing inside a single type knows it is part of a continuum, so the per-type computation *is* the classical one, unchanged to the last sign, and the only new act is to integrate the classical identity over x at the end. Let $(\mathbf{u}_1, \mathbf{m}_1)$, $(\mathbf{u}_2, \mathbf{m}_2)$ be two classical solutions, and set, for each type, the differences $\bar{u}^x := u_1^x - u_2^x$ and $\bar{m}^x := m_1^x - m_2^x$. Fix a type x for the whole of the per-type computation and suppress it from the notation where it does not matter.

Step 1: subtract the two systems. Writing the two copies of the graphon HJB eq. (24.1) and subtracting, and likewise for the graphon FP eq. (24.2), gives a linear backward equation for $\bar{u}^x$ and a linear forward equation for $\bar m^x$:

$$\begin{aligned}-\partial_t \bar u^x - \nu\,\Delta_\xi \bar u^x &= \big(F[\mathbf{m}_1]-F[\mathbf{m}_2]\big)(x,\cdot) - \big(H(x,\nabla u_1^x) - H(x,\nabla u_2^x)\big),\\ \partial_t \bar m^x - \nu\,\Delta_\xi \bar m^x &= \mathrm{div}_\xi\big(m_1^x\,\partial_p H(x,\nabla u_1^x) - m_2^x\,\partial_p H(x,\nabla u_2^x)\big),\end{aligned}$$

the Hamiltonian and the transport drift appearing as *differences* because the equations are nonlinear in u and bilinear in $(m,\nabla u)$ respectively.

Step 2: differentiate the per-type pairing and cancel the diffusion. Consider the scalar quantity $t \mapsto \int_{\mathbb{R}^d} \bar u^x \bar m^x \,\mathrm{d}\xi$. Differentiating in time and substituting $\partial_t \bar u^x$ and $\partial_t \bar m^x$ from Step 1,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} \bar u^x \bar m^x = \int_{\mathbb{R}^d} (\partial_t \bar u^x)\,\bar m^x + \int_{\mathbb{R}^d} \bar u^x\,(\partial_t \bar m^x),$$

the two second-order terms it produces are $-\nu\int \bar m^x \Delta_\xi \bar u^x$ from the first piece (after moving $\partial_t \bar u^x$ to the other side of its equation) and $+\nu\int \bar u^x \Delta_\xi \bar m^x$ from the second, and they cancel: by Green's second identity (both fields and their gradients decaying at infinity, guaranteed by the moment leash and the parabolic regularity that justify the integrations by parts hypothesised in the theorem),

$$\nu\int_{\mathbb{R}^d} \big(\bar u^x\,\Delta_\xi \bar m^x - \bar m^x\,\Delta_\xi \bar u^x\big)\,\mathrm{d}\xi = 0.$$

This is the same cancellation that made the classical identity work; the diffusion contributes nothing to the energy balance.

Step 3: integrate the transport term by parts and recognise the Bregman divergence. What survives is the Hamiltonian difference paired against $\bar m^x$ and the transport divergence paired against $\bar u^x$. Integrating the latter by parts once moves the divergence onto $\nabla_\xi \bar u^x = \nabla u_1^x - \nabla u_2^x$:

$$\int_{\mathbb{R}^d} \bar u^x\,\mathrm{div}_\xi\big(m_1^x\partial_p H_1 - m_2^x \partial_p H_2\big) = -\int_{\mathbb{R}^d} \big(\nabla u_1^x - \nabla u_2^x\big)\cdot\big(m_1^x\partial_p H_1 - m_2^x\partial_p H_2\big),$$

abbreviating $H_i = H(x,\nabla u_i^x)$ and $\partial_p H_i = \partial_p H(x,\nabla u_i^x)$. Collecting this with the Hamiltonian-difference term $\int (H_1 - H_2)(m_1^x - m_2^x)$ and regrouping by which density multiplies which gradient pair, every term assembles into two copies of the Bregman divergence $D_H(x;p,q) := H(x,p) - H(x,q) - \partial_p H(x,q)\cdot(p-q)$ of the convex Hamiltonian (lemma 11.4):

$$\int (H_1 - H_2)\bar m^x - \int \big(\nabla u_1^x - \nabla u_2^x\big)\cdot\big(m_1^x\partial_p H_1 - m_2^x\partial_p H_2\big) = A^x(t),$$

where

$$A^x(t) = -\int_{\mathbb{R}^d} m_1^x\,D_H(x;\nabla u_2^x,\nabla u_1^x) - \int_{\mathbb{R}^d} m_2^x\,D_H(x;\nabla u_1^x,\nabla u_2^x)\ \le\ 0$$

— the regrouping is the elementary algebra of lemma 11.4, and the sign is forced because each density $m_i^x \ge 0$ and each Bregman divergence $D_H \ge 0$ by convexity of $H(x,\cdot)$. We have arrived, type by type, at exactly the classical identity eq. (11.6)–eq. (11.7):

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} \bar u^x \bar m^x\,\mathrm{d}\xi = A^x(t) - \int_{\mathbb{R}^d} \big(F[\mathbf{m}_1]-F[\mathbf{m}_2]\big)(x,\cdot)\,\bar m^x.$$

Step 4: integrate over time, then over type. The previous three steps were the homogeneous proof with a passive type label; this step is the only genuinely graphon move, and it is just an integration. Integrate the per-type identity over $t \in [0,T]$. The left side telescopes to the boundary slices $\int \bar{u}^x \bar{m}^x$ evaluated at $t = T$ minus at $t = 0$. The initial slice vanishes because both solutions start from the *same* datum, $\bar{m}^x(0) = m_1^x(0) - m_2^x(0) = 0$ for every type. The terminal slice is the terminal-coupling pairing, since $\bar{u}^x(T) = G[\mathbf{m}_{1,T}](x,\cdot) - G[\mathbf{m}_{2,T}](x,\cdot)$. Thus, for each fixed type,

$$\int_{\mathbb{R}^d} (G[\mathbf{m}_{1,T}] - G[\mathbf{m}_{2,T}])(x,\cdot)\,\bar{m}^x(T) = \int_0^T A^x(t)\;\mathrm{d}t - \int_0^T\!\!\int_{\mathbb{R}^d} (F[\mathbf{m}_1] - F[\mathbf{m}_2])(x,\cdot)\,\bar{m}^x\;\mathrm{d}t.$$

Now — and this is the lift — integrate *this* identity over $x \in [0,1]$. The two state-pairings on either side are precisely the integrands of the graphon pairings of definition 24.15: the terminal one integrates to $\mathcal{G}_G(\mathbf{m}_{1,T}, \mathbf{m}_{2,T})$, the running one to $\int_0^T \mathcal{G}_F(\mathbf{m}_{1,t}, \mathbf{m}_{2,t})\;\mathrm{d}t$. By graphon monotonicity of G the left side is nonnegative,

$$\int_0^1\!\!\int_{\mathbb{R}^d} (G[\mathbf{m}_{1,T}] - G[\mathbf{m}_{2,T}])(x,\cdot)\,\bar{m}^x(T)\;\mathrm{d}x = \mathcal{G}_G(\mathbf{m}_{1,T}, \mathbf{m}_{2,T}) \ge 0,$$

and the whole identity reads

$$0 \le \mathcal{G}_G(\mathbf{m}_{1,T}, \mathbf{m}_{2,T}) = \int_0^T\!\!\int_0^1 A^x(t)\;\mathrm{d}x\;\mathrm{d}t - \int_0^T \mathcal{G}_F(\mathbf{m}_{1,t}, \mathbf{m}_{2,t})\;\mathrm{d}t,$$

the exact operator analogue of the scalar identity eq. (11.8), with the type integral threaded through both the Bregman term and the coupling term.

Step 5: the sign squeeze. The right-hand side is a sum of two nonpositive quantities. The integrated Bregman term $\int_0^T\int_0^1 A^x\;\mathrm{d}x\;\mathrm{d}t \le 0$ because every $A^x(t) \le 0$ pointwise from Step 3, and the running coupling term $-\int_0^T \mathcal{G}_F \le 0$ because $\mathcal{G}_F \ge 0$ at every time by graphon monotonicity of F. A nonnegative quantity that equals a sum of two nonpositive quantities forces all three to vanish. In particular $\int_0^T\int_0^1 A^x\;\mathrm{d}x\;\mathrm{d}t = 0$ with $A^x \le 0$ forces, for a.e. (t,x), both $m_1^x\,D_H(x;\nabla u_2^x, \nabla u_1^x) = 0$ and $m_2^x\,D_H(x;\nabla u_1^x, \nabla u_2^x) = 0$. Here the *strict* convexity hypothesised in the theorem does its one job: $D_H(x;p,q) = 0$ with $H(x,\cdot)$ strictly convex forces $p = q$, so on the support of each density the two optimal velocity fields $\partial_p H(x, \nabla u_i^x)$ coincide. Each pair (m_1^x, m_2^x) then solves the *same* linear Fokker–Planck equation eq. (24.2), with the same drift on the relevant support and the same datum m_0^x, so $m_1^x \equiv m_2^x$ for every type by uniqueness for the linear parabolic equation (chapter 5). With the density fields equal, the coupling data coincide, $F[\mathbf{m}_1] = F[\mathbf{m}_2]$ and $G[\mathbf{m}_{1,T}] = G[\mathbf{m}_{2,T}]$, so for each type u_1^x and u_2^x solve the same backward HJB with the same terminal datum and hence $u_1^x \equiv u_2^x$. The two solution fields are equal. □

The theorem is clean, and its cleanliness is the point: the proof never had to think about the graphon at all until the very last integration, because the operator $\mathbb{W}$ sits inside the data $F[\mathbf{m}]$, not inside the differential machinery, and the differential machinery is all the per-type computation touches. But that same feature means the hypothesis — graphon monotonicity, the nonnegativity of $\mathcal{G}_F$ and $\mathcal{G}_G$ — *hides* the operator: definition 24.15 is a clean inequality about an abstract pairing, and one cannot tell by looking at it whether a given model satisfies it or whether, in satisfying it, the operator has done something the homogeneous theory never saw. To find out we must open the pairing up and compute it for a concrete coupling. The natural choice is the canonical graphon-local quadratic coupling carried by Thread B, the congestion–imitation model of chapter 22, both because it is the model the chapter is committed to following to its threshold and because, being quadratic, it is the one case where the pairing can be evaluated in closed form and the operator inside it laid bare. The result is the pivotal computation of the section.

Proposition 24.17 (The monotonicity pairing is a quadratic form in $\mathbb{W}$)**.** *Let the running coupling be the graphon-local quadratic $F[\mathbf{m}](x,\xi) = \frac{q}{2}\big(\xi - s\,z(x)\big)^2$ with $z = \mathbb{W}\overline{\mathbf{m}}$ and $q > 0$, $s \in \mathbb{R}$ (the Thread B coupling). Then for any two fields with first-moment fields $\overline{\mathbf{m}}, \overline{\mathbf{m}}'$ and $\Delta := \overline{\mathbf{m}} - \overline{\mathbf{m}}' \in L^2([0,1])$,*

$$\mathcal{G}_F(\mathbf{m},\mathbf{m}') \;=\; -\,q\,s\,\langle \Delta, \mathbb{W}\Delta\rangle_{L^2([0,1])} \;=\; -\,q\,s\sum_{k\geq 1}\lambda_k\;|\langle \Delta,\phi_k\rangle|^2\,, \tag{24.6}$$

where (λ_k,ϕ_k) is the eigensystem of $\mathbb{W}$ (chapter 18). Consequently F is graphon monotone if and only if $s\,\langle\Delta,\mathbb{W}\Delta\rangle \leq 0$ for all Δ; in particular, if $\mathbb{W}$ is positive semidefinite (all $\lambda_k \geq 0$, e.g. a graphon of positive type), F is graphon monotone iff *$s \leq 0$.*

Proof. Fix a type x and abbreviate its two aggregates $z = z(x) = (\mathbb{W}\overline{\mathbf{m}})(x)$ and $z' = z'(x) = (\mathbb{W}\overline{\mathbf{m}}')(x)$, with increment $\Delta z(x) = z - z'$. The quadratic running cost is a perfect square, so its increment factors as a difference of squares,

$$F[\mathbf{m}](x,\xi) - F[\mathbf{m}'](x,\xi) = \tfrac{q}{2}\big[(\xi - sz)^2 - (\xi - sz')^2\big] = \tfrac{q}{2}\big[(\xi - sz) + (\xi - sz')\big]\big[(\xi - sz) - (\xi - sz')\big].$$

The second bracket is $-s\,\Delta z(x)$ and the first is $2\xi - s(z+z')$, so multiplying out,

$$F[\mathbf{m}](x,\xi) - F[\mathbf{m}'](x,\xi) = \tfrac{q}{2}\big[-2s\,\xi\,\Delta z(x) + s^2(z+z')\,\Delta z(x)\big].$$

The first term is linear in the state ξ; the second — the one carrying the awkward s^2 — is *independent* of ξ, a constant in the state variable for each fixed type. That is exactly what makes it disappear under the pairing. Pair the increment against the signed measure $(m^x - m'^x)(\mathrm{d}\xi)$ and use the two elementary moment identities that the pairing structure hands us for free. The zeroth moment of the increment vanishes, because both m^x and m'^x are probability measures of total mass one:

$$\int_{\mathbb{R}^d}(m^x - m'^x)(\mathrm{d}\xi) = 1 - 1 = 0,$$

which annihilates the ξ-independent s^2 term wholesale — a constant integrated against a measure of zero total mass contributes nothing, and the entire quadratic-in-s piece is gone. The first moment of the increment is, by the very definition of the first-moment field, the increment of the means:

$$\int_{\mathbb{R}^d}\xi\,(m^x - m'^x)(\mathrm{d}\xi) = \overline{\mathbf{m}}(x) - \overline{\mathbf{m}}'(x) = \Delta(x).$$

Only the linear term survives, and it survives as

$$\int_{\mathbb{R}^d}\big(F[\mathbf{m}] - F[\mathbf{m}']\big)(x,\cdot)\,(m^x - m'^x)(\mathrm{d}\xi) = \tfrac{q}{2}\big(-2s\,\Delta z(x)\big)\,\Delta(x) = -qs\,\Delta z(x)\,\Delta(x).$$

This is the per-type integrand of $\mathcal{G}_F$. It remains only to integrate it over the type, which is where the operator finally appears in the open: $\Delta z = z - z' = \mathbb{W}\overline{\mathbf{m}} - \mathbb{W}\overline{\mathbf{m}}' = \mathbb{W}\Delta$ by linearity of $\mathbb{W}$, so

$$\mathcal{G}_F(\mathbf{m},\mathbf{m}') = \int_0^1\big(-qs\,\Delta z(x)\,\Delta(x)\big)\;\mathrm{d}x = -qs\int_0^1(\mathbb{W}\Delta)(x)\,\Delta(x)\;\mathrm{d}x = -qs\,\langle\Delta,\mathbb{W}\Delta\rangle_{L^2([0,1])}.$$

The spectral form is then the eigendecomposition of the integral operator: since $\mathbb{W}$ is self-adjoint Hilbert–Schmidt (chapter 18) it diagonalises as $\mathbb{W} = \sum_k \lambda_k \, \phi_k \otimes \phi_k$ in the orthonormal eigenbasis (ϕ_k), so expanding $\Delta = \sum_k \langle \Delta, \phi_k \rangle \phi_k$ gives $\langle \Delta, \mathbb{W}\Delta \rangle = \sum_k \lambda_k \, |\langle \Delta, \phi_k \rangle|^2$, which is eq. (24.6). The monotonicity criterion is immediate: $\mathcal{G}_F \geq 0$ for all fields exactly when $s \, \langle \Delta, \mathbb{W}\Delta \rangle \leq 0$ for all $\Delta \in L^2([0,1])$. And if $\mathbb{W} \succeq 0$ — every $\lambda_k \geq 0$, as for a graphon of positive type — then $\langle \Delta, \mathbb{W}\Delta \rangle \geq 0$ always, so the criterion reduces to the single scalar inequality $s \leq 0$. □

Proposition 24.17 is the heart of the matter, and it is worth saying exactly what it changed. The classical computation eq. (11.10) gave $\mathcal{G}_F = -qs \, \Delta^2$, with Δ the single scalar difference of global means, and the verdict was a clean scalar dichotomy: monotonicity holds iff $s \leq 0$. There was one knob, the strategic sign s, and one thing it controlled, the sign of a square. The graphon computation keeps the prefactor $-qs$ but replaces the scalar square Δ^2 by the quadratic form $\langle \Delta, \mathbb{W}\Delta \rangle$ in the field of mean-differences — and a quadratic form in an operator is a far richer object than a square. Now *two* independent things can spoil monotonicity. The first is the classical one: imitation, $s > 0$, which flips the sign of the whole pairing just as before. The second is new and graphon-only: an *indefinite* operator $\mathbb{W}$, one with some negative eigenvalue $\lambda_k < 0$, for which $\langle \Delta, \mathbb{W}\Delta \rangle$ takes both signs as Δ ranges over $L^2([0,1])$ — positive when Δ aligns with a positive eigenmode, negative when it aligns with a negative one. With an indefinite $\mathbb{W}$ the pairing cannot be signed by choosing s at all: even the crowd-averse $s < 0$, which is monotone in every homogeneous model, fails graphon monotonicity along the negative eigenmodes. And indefiniteness is the generic situation, not a pathology: a graphon of positive type (every $\lambda_k \geq 0$) is special, while a generic symmetric kernel — any bipartite-like or disassortative structure, where unlike types attract — has eigenvalues of both signs. Graphon monotonicity is therefore a *joint* condition on the strategic sign and the spectrum, and per-type monotonicity — the scalar $s \leq 0$ of chapter 11, asserted independently at each type — is by itself neither necessary (a crowd-seeking $s > 0$ can still be graphon-monotone if it acts only through negative eigenmodes) nor sufficient (a crowd-averse $s < 0$ fails along negative eigenmodes of an indefinite $\mathbb{W}$).

Two threads leave this subsection with that observation, pointing in opposite directions, and naming them is how the section is stitched to what follows. The first is a limitation handed to the next subsection. The structural route just built is horizon-free — it bought uniqueness at *every* T, which is exactly the regime the smallness route will be unable to reach — but proposition 24.17 shows the price of that strength: it demands a definite *sign* of the joint strategy–spectrum quantity $s \, \langle \Delta, \mathbb{W}\Delta \rangle$, and there are whole models (imitative couplings, indefinite networks) where no such sign is available. A route that asks for no sign at all would cover precisely those models. That is the motivation for the operator-norm contraction of the next subsection: it abandons the sign and asks instead for *smallness*, the standing assumption **(A-Coupling)** on $\|\mathbb{W}\|$, and because the aggregation $\mathcal{A}_W$ is linear, that smallness is read off a single operator norm — the only graphon-specific factor in the whole estimate. The second thread is the central phenomenon handed forward to the synthesis. The indefinite-operator mechanism just uncovered — monotonicity lost not to imitation but to the network's own geometry, along the negative eigenmodes of $\mathbb{W}$ — has no homogeneous analogue whatever, and it is the cleanest instance of the chapter's claim that every graphon-specific surprise lives in the spectrum of $\mathbb{W}$. We resist drawing its consequences here, because they belong with the others: the full catalogue of what is genuinely harder than in the classical theory, the second crowd-averse multiplicity wedge it opens, and its place beside the centrality-indexed threshold $s\,\lambda_1$, is collected once in the synthesis of section 24.4.

24.3.2 Small coupling: the operator-norm contraction

The monotonicity route just finished is strong where it applies — it buys uniqueness at *every* horizon — but the previous subsection closed by naming exactly the price of that strength: it demands a definite *sign* of the joint strategy–spectrum quantity $s\,\langle \Delta, \mathbb{W}\Delta\rangle$, and there are whole families of models, the imitative couplings and the indefinite networks, where no such sign is to be had. A route that asked for no sign at all would cover precisely those models, and the obvious thing to trade the sign for is *smallness*: if the coupling is weak, the best-response map cannot do much harm regardless of which way it pushes, and a single contraction estimate should pin the fixed point. This second route asks nothing of the sign; it asks the coupling to be weak. It is the operator form of the short-horizon contraction of chapter 11, with the operator norm $\|\mathbb{W}\|$ as the smallness parameter — the standing assumption **(A-Coupling)**, $\|\mathbb{W}\| < \kappa$.

The reason this works, and works cleanly, is the factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ of eq. (24.4). The per-type solve $\mathcal{R}$ is the classical machine of chapters 10 and 11, and *classically* the best-response map is already a contraction once the horizon is short or the coupling weak — that is the homogeneous short-horizon route, nothing new. Everything graphon-specific has been quarantined in the aggregation $\mathcal{A}_W$, and $\mathcal{A}_W$ is *linear*. A linear map has exactly one Lipschitz constant, its operator norm, and there is nothing else to estimate: the entire graphon contribution to how much the map can stretch a perturbation is the single number $\|\mathcal{A}_W\|$, which (the first moment being a contraction) is the operator norm of $\mathbb{W}$. The smallness we must demand is therefore not a smallness of the data, nor of the Hamiltonian, nor of anything per-type — all of that lives in $\mathcal{R}$ and is classical — but precisely a smallness of $\mathbb{W}$. This is the content of **(A-Coupling)**, and it is why the standing assumption is phrased as a bound on $\|\mathbb{W}\|$ alone: the factorisation makes the operator norm the *only* graphon-specific factor in the whole estimate, so it is the only thing the smallness threshold can possibly constrain.

Proposition 24.18 (Contraction of the graphon best response)**.** *Assume* ***(A-Lip)****,* ***(A-Nondeg)****,* ***(A-Conv)****,* ***(A-Reg-W)****, and assumption 24.5. There is a constant $C_T < \infty$, depending only on T, the Lipschitz constants of F_0, G_0 in their aggregate argument, and the data of chapter 10, such that the graphon best response is Lipschitz on $(\mathbb{K}_W, \mathsf{D})$ with*

$$\mathsf{D}(\mathsf{S}_W(\mathbf{m}), \mathsf{S}_W(\mathbf{m}')) \;\le\; C_T\;\|W\|_{L^\infty_x L^1_y}\;\mathsf{D}(\mathbf{m}, \mathbf{m}'), \tag{24.7}$$

where $\|W\|_{L^\infty_x L^1_y} = \operatorname{ess\,sup}_x \int_0^1 |W(x,y)|\,\mathrm{d}y$ is the row-sum ($L^\infty \to L^\infty$) operator norm of $\mathbb{W}$. Hence if $C_T\;\|W\|_{L^\infty_x L^1_y} < 1$ — the sup-in-type form of the standing smallness assumption ***(A-Coupling)*** *with $\kappa := 1/C_T$ — the map S_W is a contraction, and the GMFG has a* unique *equilibrium, for* any *sign of the coupling. Since $\|\mathbb{W}\| = \rho(\mathbb{W}) = \lambda_1 \le \|W\|_{L^\infty_x L^1_y}$ (remark 24.19), this sufficient condition is more conservative than the exact spectral threshold λ_1 of proposition 24.23, with which it coincides for the constant kernel.*

Proof sketch and pointer. The argument is a single composition: send a field perturbation through the linear aggregation, then through the classical per-type solve, and read off the product of the two Lipschitz constants. The whole point of the factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$ (eq. (24.4)) is that these two factors can be estimated in complete isolation, the first carrying all of the graphon and none of the analysis, the second all of the analysis and none of the graphon. We take them in turn and multiply.

Step 1: the aggregation $\mathcal{A}_W$ is linear, so its Lipschitz constant is an operator norm. Fix two candidate fields $\mathbf{m}, \mathbf{m}' \in \mathbb{K}_W$. Because $\mathcal{A}_W$ is the composition of the first-moment read-out $\mathbf{m} \mapsto \overline{\mathbf{m}}$ with the integral operator $\mathbb{W}$, both linear, the difference of the aggregates it produces is the operator applied to the difference of the moment fields,

$$\mathcal{A}_W(\mathbf{m}) - \mathcal{A}_W(\mathbf{m}') = \mathbb{W}(\overline{\mathbf{m}} - \overline{\mathbf{m}}'), \qquad \text{i.e.} \qquad z_t - z_t' = \mathbb{W}(\overline{\mathbf{m}}_t - \overline{\mathbf{m}}_t') \text{ for each } t.$$

There is no cross term and no remainder: linearity is exactly the statement that a perturbation passes through $\mathcal{A}_W$ unaltered in form, only rescaled. To bound the rescaling we need two facts, one about the moment read-out and one about $\mathbb{W}$, joined at the choice of metric. First, the moment read-out does not amplify: by lemma 24.2 the first moment is 1-Lipschitz from $\mathcal{W}_1$ into $\mathbb{R}$ (equivalently, Kantorovich–Rubinstein with the admissible test function $\xi \mapsto \xi$), so at every type y and time t, $|\overline{\mathbf{m}}_t(y) - \overline{\mathbf{m}}_t'(y)| \le \mathcal{W}_1(m_t^y, m_t'^y) \le \mathsf{D}(\mathbf{m}, \mathbf{m}')$, the last step by the definition of the uniform-in-type metric as a supremum over types and times. Taking the supremum over y and t,

$$\sup_t \|\overline{\mathbf{m}}_t - \overline{\mathbf{m}}_t'\|_{L^\infty([0,1])} \le \mathsf{D}(\mathbf{m}, \mathbf{m}').$$

The moment difference is thus controlled in $L^\infty([0,1])$, the natural target of the uniform-in-type metric D — and this is the hinge that fixes which norm of $\mathbb{W}$ governs the estimate. Second, $\mathbb{W}$ maps $L^\infty([0,1])$ to $L^\infty([0,1])$ with operator norm exactly the row sum,

$$\|\mathbb{W}\|_{L^\infty \to L^\infty} = \operatorname*{ess\,sup}_x \int_0^1 |W(x,y)| \, \mathrm{d}y = \|W\|_{L_x^\infty L_y^1},$$

because $|(\mathbb{W}g)(x)| \le \int |W(x,y)| \, |g(y)| \, \mathrm{d}y \le \|g\|_\infty \int |W(x,y)| \, \mathrm{d}y$ and the bound is attained, type by type, by the sign-matching $g = \operatorname{sgn} W(x, \cdot)$. Composing the two facts,

$$\sup_t \|z_t - z_t'\|_{L^\infty} \le \|\mathbb{W}\|_{L^\infty \to L^\infty} \sup_t \|\overline{\mathbf{m}}_t - \overline{\mathbf{m}}_t'\|_{L^\infty} \le \|W\|_{L_x^\infty L_y^1} \, \mathsf{D}(\mathbf{m}, \mathbf{m}').$$

So in the uniform-in-type metric the aggregation is a Lipschitz map of fields with constant exactly the row-sum norm. It bears repeating, because it is the one subtlety the whole subsection turns on, that this is *not* the L^2 spectral radius: the metric D measures fields uniformly in type, which is an L^∞ measurement, and the operator norm an L^∞ input forces out of $\mathbb{W}$ is the row sum, not λ_1 (remark 24.19). The row-sum norm, not the spectral radius, is the contraction's genuine smallness parameter.

Step 2: the per-type solve $\mathcal{R}$ is Lipschitz in its frozen data, classically. This factor carries no graphon at all. For each fixed type x the map $\mathcal{R}$ is the homogeneous best response $\mathsf{S}^x_{z(x)}$ of definition 10.3, driven by the frozen scalar-in-type data curve $t \mapsto z_t(x)$, and its Lipschitz dependence on that data is precisely the stability content of chapters 10 and 11. Concretely, suppose two data curves z, z' differ by $\sup_t |z_t(x) - z'_t(x)| \le \epsilon$. Through the Lipschitz dependence of F_0, G_0 on their aggregate slot (assumption 24.5) this perturbs the source and terminal data of the type-x HJB eq. (24.1) by $O(\mathrm{Lip}(F_0, G_0)\,\epsilon)$ in the relevant norm; the HJB stability estimate lemma 10.10, an interior Schauder/maximum-principle bound made quantitative by the uniform ellipticity λ from **(A-Nondeg)**, then perturbs the value gradient $\nabla_\xi u^x$ — and hence the optimal feedback $-\partial_p H(x, \nabla_\xi u^x)$ — by $O(\mathrm{Lip}(F_0, G_0)\,\epsilon)$ as well. A perturbed feedback drives a perturbed forward flow: the FP stability estimate lemma 10.11, run as a Grönwall comparison of the two closed-loop SDEs sharing the noise B^x and started from the common datum m_0^x, integrates that drift perturbation along the horizon and bounds the resulting law gap by $O(L_{\mathcal{R}}(T)\,\mathrm{Lip}(F_0, G_0)\,\epsilon)$ in $\sup_t \mathcal{W}_1$. The Grönwall constant $L_{\mathcal{R}}(T)$ is exactly where the horizon enters, a point we return to immediately below. None of this knows there is a type variable; it is the chapter 10 machine run verbatim inside a single type, which is why we import it rather than reprove it.

Step 3: compose and apply Banach. Feeding the Step-1 bound on the data gap, with $\epsilon = \|W\|_{L^\infty_x L^1_y}\ \mathsf{D}(\mathbf{m}, \mathbf{m}')$, into the Step-2 per-type Lipschitz estimate, uniformly over types, gives

$$\mathsf{D}(\mathsf{S}_W(\mathbf{m}), \mathsf{S}_W(\mathbf{m}')) \le L_{\mathcal{R}}(T)\,\mathrm{Lip}(F_0, G_0)\ \|W\|_{L^\infty_x L^1_y}\ \mathsf{D}(\mathbf{m}, \mathbf{m}'),$$

which is eq. (24.7) with $C_T = L_{\mathcal{R}}(T)\,\mathrm{Lip}(F_0, G_0)$ — a constant built entirely from per-type, graphon-free data, multiplying the single graphon factor $\|W\|_{L^\infty_x L^1_y}$. When $C_T\ \|W\|_{L^\infty_x L^1_y} < 1$ the map S_W is a strict contraction of the complete metric space $(\mathbb{K}_W, \mathsf{D})$ into itself (the invariant set $\mathbb{K}_W$ of lemma 24.10 is closed, hence complete), and the Banach fixed-point theorem supplies a unique fixed point, reached by iteration from any starting field — a *unique* equilibrium, with no demand whatever on the sign of the coupling. The per-type Lipschitz estimate is the content of chapter 6 and [40, Ch. 3–4, Vol. I]; all we have added is to isolate the dependence on $\|\mathbb{W}\|$, which the factorisation shows is the only graphon-specific factor in the whole chain. $\square$

Two features of proposition 24.18 deserve emphasis, because each fixes the contraction route's place relative to its two neighbours — the horizon-free monotonicity route on one side, the exact spectral threshold on the other.

First, the route is intrinsically *finite-horizon.* The constant $C_T = L_{\mathcal{R}}(T)\,\mathrm{Lip}(F_0, G_0)$ inherits the growth of the per-type Grönwall stability constant $L_{\mathcal{R}}(T)$, which generally *increases* with T: a perturbation of the feedback is integrated along the whole horizon, and the longer the horizon the more it can accumulate, exactly as a Grönwall factor e^{cT} grows. Consequently the smallness threshold $\kappa = 1/C_T$ *shrinks* as T increases — a coupling weak enough to be contracted over a short horizon may be too strong over a long one — so weak-coupling uniqueness is a finite-horizon statement, and it says nothing in the long-horizon regime unless it is reinforced by the monotonicity structure of the previous subsection. This is precisely the operator counterpart of the classical short-horizon route (chapter 11, item 1 of its boundary menu) and of the soft band of proposition 11.8: there too smallness bought uniqueness only below a horizon-dependent ceiling, and there too the only way to uniqueness at *every* horizon ran through monotonicity. The graphon adds nothing qualitatively new here; it merely reads the same ceiling off $\|\mathbb{W}\|$ instead of off the scalar coupling. The division of labour between the two routes — monotonicity for long horizons, smallness for weak coupling — is identical to the homogeneous one, with $\mathbb{W}$ inserted at the single place the coupling enters.

Second, the chapter's smallness and threshold quantities all sit at the *top of the spectrum* of $\mathbb{W}$, but they are not literally one number, and the difference between them is conceptually load-bearing rather than a technicality. The sup-in-type contraction proved here is governed by the row-sum norm $\|W\|_{L_x^\infty L_y^1}$, the $L^\infty \to L^\infty$ operator norm forced out of $\mathbb{W}$ by the uniform-in-type metric D. The *exact* multiplicity threshold, by contrast, is set by the $L^2 \to L^2$ operator norm $\|\mathbb{W}\| = \rho(\mathbb{W}) = \lambda_1$ — the leading eigenvalue, the eigenvector-centrality scale of chapter 18 — which is what the L^2 spectral diagonalisation of proposition 24.23 delivers. These two are not equal: the row sum *dominates* λ_1, with equality only for special kernels. The contraction route is therefore a strictly *conservative* sufficient condition — it can fail to certify uniqueness in a regime where the exact threshold still grants it — and that conservatism is the price of working in the comfortable L^∞/uniform metric rather than the heavier L^2 Bochner setting. The next remark makes the gap between the two norms precise, with the elementary inequality that orders them and a worked rank-one kernel that exhibits the strict inequality; the upshot to carry forward is that the monotonicity route and the exact LQG threshold both meet at λ_1, and that common boundary — not the row sum — is the true critical coupling.

Remark 24.19 (Which operator norm: row-sum versus spectral radius)**.** Two operator norms of $\mathbb{W}$ appear above and must not be conflated. The *row-sum norm* $\|W\|_{L_x^\infty L_y^1} = \operatorname{ess\,sup}_x \int_0^1 |W(x,y)|\,\mathrm{d}y$ is the $L^\infty \to L^\infty$ operator norm; it is what the aggregation step produces in the uniform-in-type metric D (lemma 24.6, proposition 24.18). The $L^2 \to L^2$ operator norm equals the spectral radius, $\|\mathbb{W}\| = \rho(\mathbb{W}) = \lambda_1$, because $\mathbb{W}$ is self-adjoint Hilbert–Schmidt (chapter 18); this is the *canonical* norm of **(A-Coupling)** and the one in the exact LQG threshold proposition 24.23. They satisfy

$$\|\mathbb{W}\| = \lambda_1 \ \le\ \|W\|_{L_x^\infty L_y^1},$$

with equality for the constant kernel and *strict* inequality in general — e.g. for the non-constant rank-one kernel $W(x,y) = f(x)f(y)$, $f(x) = \frac{1}{5} + \frac{4}{5}x$, one has $\lambda_1 = \|f\|_{L^2}^2 \approx 0.41$ but $\|W\|_{L_x^\infty L_y^1} = \|f\|_\infty \|f\|_{L^1} = 0.6$. The inequality itself is elementary: by Perron–Frobenius (theorem 18.7) the leading eigenvalue $\lambda_1 = \rho(\mathbb{W})$ has a nonnegative eigenfunction $\phi_1 \ge 0$, and evaluating $\lambda_1\phi_1(x) = \int_0^1 W(x,y)\phi_1(y)\,\mathrm{d}y \le \|\phi_1\|_\infty \int_0^1 W(x,y)\,\mathrm{d}y$ at a type where ϕ_1 approaches its essential supremum gives $\lambda_1 \le \|W\|_{L_x^\infty L_y^1}$ (the symmetric Schur-test bound). Consequently the sup-in-type contraction route constrains the *larger* quantity and is genuinely more conservative than the exact spectral threshold λ_1; the two coincide only when the kernel is constant (or, more generally, has constant row sums, where the Perron eigenfunction is itself constant). Running the contraction instead in the L^2-in-type metric d_2 of chapter 23 replaces the row-sum norm by $\|\mathbb{W}\| = \lambda_1$ exactly, at the cost of the heavier Bochner setting; we keep D and accept the conservative constant.

The same contraction estimate, applied to two *different* kernels at the same per-type data, shows that in the weak-coupling regime the equilibrium depends Lipschitz-continuously on the kernel. This is the precise stability statement on which the finite-network convergence theory of chapter 25 rests, so we record it as a labelled export.

Proposition 24.20 (Lipschitz dependence of the equilibrium on the kernel)**.** *Let W, W' be graphons satisfying **(A-Reg-W)** and assumption 24.5 with the* same *per-type data F_0, G_0 and initial field $\mathbf{m}_0$, both in the contraction regime of proposition 24.18 ($C_T \|W\|_{L_x^\infty L_y^1} < 1$ and likewise for W'), with* *unique equilibria* $\mathbf{m}^\star_W, \mathbf{m}^\star_{W'}$. *Then, with* R *the uniform second-moment radius of the equilibrium fields,*

$$\mathsf{D}(\mathbf{m}^\star_W, \mathbf{m}^\star_{W'}) \;\le\; \frac{C_T\, R\, \|W - W'\|_{L^\infty_x L^1_y}}{1 - C_T\, \|W\|_{L^\infty_x L^1_y}}. \tag{24.8}$$

The identical estimate holds with the $L^2 \to L^2$ *operator norm* $\|\mathbb{W} - \mathbb{W}'\|$ *replacing the row-sum norm when the contraction is run in the* L^2*-in-type metric* d_2 *of chapter 23.*

Proof. Write $\mathsf{S}_W, \mathsf{S}_{W'}$ for the two best-response maps; they share the per-type factor $\mathcal{R}$ and differ only in the aggregation $\mathcal{A}_W$ versus $\mathcal{A}_{W'}$. For a fixed field $\mathbf{m}$ the two aggregates differ through the kernel alone, $z_t - z'_t = (\mathbb{W} - \mathbb{W}')\overline{\mathbf{m}}_t$, so by the L^∞ aggregation bound of lemma 24.6 and the per-type Lipschitz constant $C_T = L_{\mathcal{R}}(T)\,\mathrm{Lip}(F_0, G_0)$ of proposition 24.18,

$$\mathsf{D}(\mathsf{S}_W(\mathbf{m}), \mathsf{S}_{W'}(\mathbf{m})) \;\le\; C_T \sup_t \left\|(\mathbb{W} - \mathbb{W}')\overline{\mathbf{m}}_t\right\|_\infty \;\le\; C_T\, R\, \|W - W'\|_{L^\infty_x L^1_y}\,.$$

Now compare the two fixed points, inserting $\mathsf{S}_W(\mathbf{m}^\star_{W'})$ and using the contraction eq. (24.7) for S_W:

$$\mathsf{D}(\mathbf{m}^\star_W, \mathbf{m}^\star_{W'}) = \mathsf{D}(\mathsf{S}_W(\mathbf{m}^\star_W), \mathsf{S}_{W'}(\mathbf{m}^\star_{W'})) \le C_T\, \|W\|_{L^\infty_x L^1_y}\; \mathsf{D}(\mathbf{m}^\star_W, \mathbf{m}^\star_{W'}) + C_T\, R\, \|W - W'\|_{L^\infty_x L^1_y}\,.$$

Solving for $\mathsf{D}(\mathbf{m}^\star_W, \mathbf{m}^\star_{W'})$ gives eq. (24.8); the L^2-in-type version is identical with d_2 and $\|\mathbb{W} - \mathbb{W}'\|$ throughout. ☐

Because the cut metric controls the operator norm of the difference $\mathbb{W} - \mathbb{W}'$ (chapters 17 and 18), eq. (24.8) is exactly the $\delta_\square$-stability of the equilibrium that the sampled-network convergence theorem of chapter 25 consumes.

24.3.3 The critical coupling

We can now say where, and why, uniqueness fails. Combine the two routes. Graphon monotonicity (theorem 24.16) gives uniqueness for all T whenever $s\,\langle \Delta, \mathbb{W}\Delta\rangle \le 0$ for all Δ, i.e. (for $\mathbb{W} \succeq 0$) whenever $s \le 0$. Small coupling (proposition 24.18) gives uniqueness for any s when $|s|\;\|\mathbb{W}\|$ is small enough relative to $1/T$. Between these lies a regime — strong, imitative coupling $s > 0$ over a long horizon — where neither applies, and there the equilibrium can split. The threshold is set by the largest eigenvalue.

Proposition 24.21 (Critical coupling, graphon-local quadratic model)**.** *Consider the GMFG with the graphon-local quadratic coupling of proposition 24.17, dynamics and parameters as in Thread B (section 24.5). Let* $\lambda_1 = \|\mathbb{W}\| = \rho(\mathbb{W}) > 0$ *be the leading eigenvalue of* $\mathbb{W}$ *(the eigenvector-centrality scale, chapter 18, equal to the spectral radius by Perron–Frobenius, theorem 18.7), with eigenfunction* ϕ_1*. Then the equilibrium is unique for all horizons whenever*

$$\max_k s\,\lambda_k \;\le\; 1 + \frac{a^2}{b^2 q} \;=:\; s_c^{\mathrm{scalar}}, \tag{24.9}$$

a condition guaranteed by the sign-robust bound $|s|\;\lambda_1 \le s_c^{\mathrm{scalar}}$ *(since* $\lambda_1 = \rho(\mathbb{W}) \ge |\lambda_k|$ *for every* k*). In the* imitation regime $s > 0$ *the binding mode is the leading one and eq. (24.9) reads* $s\,\lambda_1 \le s_c^{\mathrm{scalar}}$*;*

for $s\,\lambda_1 > s_c^{\text{scalar}}$ *uniqueness then fails at and beyond a finite critical horizon* T_c*, with the instability carried first by the leading eigenmode* ϕ_1*, and the critical coupling is* $s_c = s_c^{\text{scalar}}/\lambda_1$ *(equal to* $1/\lambda_1 = 1/\|\mathbb{W}\|$ *when* $a = 0$*). In the* crowd-averse regime $s < 0$ *the binding mode is instead the most negative eigenvalue* $\lambda_{\min} < 0$*: if* $\mathbb{W} \succeq 0$ *then eq.* (24.9) *holds automatically and uniqueness is horizon-free, but for an* indefinite $\mathbb{W}$ *uniqueness can still fail — at* $|s|\,|\lambda_{\min}| > s_c^{\text{scalar}}$*, along the mode* $\phi_{\min}$ *— the crowd-aversion-on-an-indefinite-network phenomenon stressed in section 24.4. There is, in particular, no unconditional crowd-averse uniqueness on an indefinite kernel.*

The proof is the spectral reduction of Thread B, carried out in full in section 24.5 (proposition 24.23). We state the proposition here because it is the conceptual centre of the chapter, and pause on its meaning.

Physics note: Loss of uniqueness as centrality-indexed symmetry breaking

The classical multiplicity threshold of chapter 11 was the line $s = 1$ (for $a = 0$): an agent whose target overshoots the crowd ($s > 1$) destabilises the symmetric equilibrium once the horizon is long enough to let the overshoot accumulate, and a pair of asymmetric equilibria appears — the mean field game's spontaneous magnetisation, the Curie–Weiss bifurcation of section 11.4. The graphon refinement is that the relevant coupling is no longer the scalar s but the *product* $s\,\lambda_1$, where λ_1 is the leading eigenvalue of the network operator $\mathbb{W}$. The same self-amplification now operates along the leading eigenfunction ϕ_1 — the eigenvector centrality profile of chapter 18 — because that is the direction in type space the operator $\mathbb{W}$ stretches most. A network with a large λ_1 (a strongly connected or highly central structure) reaches the symmetry-breaking threshold at a *smaller* intrinsic imitation strength s: connectivity amplifies imitation. The broken-symmetry order parameter is the projection $\langle \overline{\mathbf{m}}, \phi_1 \rangle$ of the population mean onto the central mode. This is the precise sense, promised in chapter 11 and chapter 20, in which the uniqueness threshold of a graphon mean field game is a competition between strategic monotonicity and the spectrum of $\mathbb{W}$, with the critical coupling tied to network centrality. The static graphon game of chapter 20 sees the same threshold as an operator-norm condition $s\,\|\mathbb{W}\| < 1$ for the contraction of its equilibrium map; the dynamic theory here recovers it as the onset of multiplicity, and refines it with the horizon dependence $T_c(s, \lambda_1)$.

24.4 What is genuinely harder than classical mean field games

It is worth collecting, plainly, the ways in which the graphon problem demands more than the homogeneous one, since each is a place where a classical instinct misleads. There are three, and all three are visible in the two theorems above.

1. Compactness acquires a type direction. A classical mean field game equilibrium is a single flow, and existence needs compactness in state and time only — tightness plus equicontinuity, the Part I toolkit. A graphon equilibrium is a field over the continuum of types, and existence needs, in addition, compactness in the type direction. That compactness is *not* a consequence of the dynamics; it is bought from the regularity of the graphon, **(A-Reg-W)**: continuity of W in type yields the equicontinuity modulus η of lemma 24.9, and Hilbert–Schmidt compactness of $\mathbb{W}$ is its substitute when continuity fails (remark 24.13). A measurable-only graphon is genuinely harder; there is no classical analogue of the difficulty because there is no classical type variable.

2. Monotonicity must survive composition with $\mathbb{W}$. The Lasry–Lions uniqueness mechanism is a sign — the equilibrium energy is convex, the boundary pairing is favourable. Classically this sign is a property of the coupling alone (definition 11.1). In the graphon problem the pairing eq. (24.5) factors through the operator: proposition 24.17 shows it equals $-qs\langle\Delta, \mathbb{W}\Delta\rangle$, a quadratic form in the *operator* $\mathbb{W}$. A perfectly crowd-averse per-type coupling (the most monotone case classically, $s < 0$) composed with an *indefinite* $\mathbb{W}$ — the generic case, since symmetric kernels have spectra of both signs — yields a pairing of indefinite sign, hence *no* global monotonicity, hence no horizon-free uniqueness from theorem 24.16. The two signs that must align — the strategic sign (state space, the convexity of the cost in the aggregate) and the spectral sign (type space, the definiteness of $\mathbb{W}$) — are independent, and their interplay has no homogeneous counterpart, where $\mathbb{W}$ is the scalar multiplication by $\int W$ and is automatically of one sign. This is the precise content of the chapter's coverage mandate: *the interplay of strategic monotonicity with the spectral properties of* $\mathbb{W}$.

3. The threshold is spectral, not scalar. Where the classical multiplicity boundary was a number ($s = s_c^{\text{scalar}}$, remark 11.9), the graphon boundary is a number times the leading eigenvalue ($s\lambda_1 = s_c^{\text{scalar}}$, eq. (24.9)), and the unstable direction is a specific eigenfunction (eigenvector centrality, chapter 18) rather than the uniform mode. Two graphons with the same $\int W$ but different spectral radii have different uniqueness thresholds; centrality, invisible to the classical theory, sets the bifurcation. A practical corollary, exploited in chapter 25, is that the uniqueness regime is *not* stable under arbitrary graph perturbation, only under perturbations small in a norm controlling the top of the spectrum — the cut metric controls $\|\mathbb{W}\|$ (chapters 17 and 18), so $\delta_\square$-close graphons have close thresholds, which is exactly the stability the convergence theory needs.

None of this makes the graphon problem qualitatively intractable; the proofs above are the classical proofs plus an operator. But it relocates the difficulty into the spectrum of $\mathbb{W}$, and a reader who carries over the classical reflex — "crowd aversion gives uniqueness" — without checking the sign of $\mathbb{W}$ will be wrong precisely when the network is interesting.

24.5 Worked computation: the graphon LQG threshold

We now discharge the carried-thread duty by carrying Thread B (the graphon LQG model) to its exact uniqueness/multiplicity threshold, proving proposition 24.21, and verifying the reductions promised in the thread specification. The computation is the operator lift of the LQG analysis of section 11.5, and it is fully explicit: the continuum of coupled boundary-value problems diagonalises over the eigenmodes of $\mathbb{W}$, and each mode is a copy of the scalar problem of worked computation 11.7 with s replaced by $s\lambda_k$.

Worked computation 24.22 (Graphon LQG: spectral reduction of the equilibrium)**.** Recall Thread B. Each type-x agent has scalar state

$$\mathrm{d}X_t^x = (aX_t^x + b\,\alpha_t^x)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x,$$

and minimises, against the graphon aggregate $z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x)$ of eq. (24.3),

$$J^x(\alpha^x) = \mathbb{E}\left[\int_0^T \left(\tfrac{1}{2}(\alpha_t^x)^2 + \tfrac{q}{2}\big(X_t^x - s\,z_t(x)\big)^2\right)\mathrm{d}t + \tfrac{q_T}{2}\big(X_T^x - s_T\,z_T(x)\big)^2\right],$$

with $q, q_T \geq 0$, $s, s_T \in \mathbb{R}$, and parameters a, b, σ, T as fixed in the thread specification.

Step 1 — per-type reduction. For a frozen aggregate path $z_\cdot(x)$ the per-type problem is exactly the scalar LQG problem of worked computation 11.7 with $s\,\bar{m}_t$ replaced by $s\,z_t(x)$. The value is quadratic, $u^x(t,\xi) = \frac{1}{2}P(t)\xi^2 + r^x(t)\xi + c^x(t)$, with the curvature $P(t)$ solving the *type-independent* Riccati equation eq. (11.11), $\dot{P} = b^2P^2 - 2aP - q$, $P(T) = q_T$ (type-independent because the quadratic-in-ξ part of the cost, $\frac{q}{2}\xi^2$, is the same for every type — only the linear coupling carries x); we use the carried-thread solution symbols of chapter 11 (Riccati curvature P, linear offset r, constant c). The linear coefficient solves, mirroring eq. (11.12),

$$\dot{r}_t^x = q\,s\,z_t(x) - \big(a - b^2P(t)\big)r_t^x, \qquad r^x(T) = -q_Ts_T\,z_T(x),$$

and the closed-loop mean $\overline{\mathbf{m}}_t(x) = \mathbb{E}[X_t^x]$ solves, mirroring eq. (11.13),

$$\dot{\overline{\mathbf{m}}}_t(x) = \big(a - b^2P(t)\big)\overline{\mathbf{m}}_t(x) - b^2r_t^x, \qquad \overline{\mathbf{m}}_0(x) = \textstyle\int \xi\,m_0^x(\mathrm{d}\xi).$$

As classically, the noise σ enters only the type-wise variance (an autonomous equation decoupled from the coupling), so the uniqueness question is governed entirely by the deterministic field pair $(r_t, \overline{\mathbf{m}}_t) \in L^2([0,1]) \times L^2([0,1])$.

Step 2 — diagonalise over the spectrum of $\mathbb{W}$. The two field equations couple only through $z_t = \mathbb{W}\overline{\mathbf{m}}_t$. Expand in the eigenbasis (λ_k, ϕ_k) of $\mathbb{W}$ (chapter 18): write $\overline{m}_t^{(k)} = \langle \overline{\mathbf{m}}_t, \phi_k\rangle$ and $r_t^{(k)} = \langle r_t, \phi_k\rangle$. Since $\langle z_t, \phi_k\rangle = \langle \mathbb{W}\overline{\mathbf{m}}_t, \phi_k\rangle = \lambda_k\overline{m}_t^{(k)}$, projecting the two field equations onto ϕ_k *decouples* the modes:

$$\dot{\overline{m}}^{(k)} = A(t)\,\overline{m}^{(k)} - b^2r^{(k)}, \qquad \dot{r}^{(k)} = q\,s\,\lambda_k\,\overline{m}^{(k)} - A(t)\,r^{(k)}, \qquad r^{(k)}(T) = -q_Ts_T\lambda_k\,\overline{m}^{(k)}(T), \tag{24.10}$$

with $A(t) := a - b^2P(t)$. This is *exactly* the scalar two-point boundary-value problem of worked computation 11.7 — compare its matrix $M(t) = \left(\begin{smallmatrix} A & -b^2 \\ qs & -A\end{smallmatrix}\right)$ — with the single replacement $s \mapsto s\,\lambda_k$. The continuum GMFG has split into a family of independent copies of the classical LQG game, one per eigenmode, with mode-k coupling $s\lambda_k$.

The spectral reduction makes the threshold immediate: the GMFG equilibrium is unique iff *every* mode-k boundary-value problem eq. (24.10) is nonsingular, and a mode is the classical problem at coupling $s\lambda_k$, whose exact threshold we already computed in proposition 11.8.

Proposition 24.23 (Graphon LQG uniqueness/multiplicity threshold)**.** *In the graphon LQG game of worked computation 24.22 with $q > 0$, terminal weight $q_T = P_\infty$ (so $P(t) \equiv P_\infty$, $A(t) \equiv A_\infty = -\sqrt{a^2 + b^2q}$, as in proposition 11.8) and $s_T = 0$, define for each eigenvalue λ_k of $\mathbb{W}$*

$$\Omega_k^2 := b^2q\,s\,\lambda_k - A_\infty^2.$$

1. *If $\Omega_k^2 \le 0$ for every k — equivalently $s\,\lambda_k \le 1 + a^2/(b^2q)$ for all k, which for $s > 0$ reduces to the single condition on the leading eigenvalue $s\,\lambda_1 \le 1 + a^2/(b^2q)$ — the equilibrium is unique for all horizons T.*

2. *If $\Omega_k^2 > 0$ for some k, the equilibrium is non-unique at the discrete horizons solving $\tan(\Omega_k T) = \Omega_k/A_\infty$ (eq. (11.14) with $\Omega \to \Omega_k$). The smallest critical horizon T_c is attained by the mode maximising $s\lambda_k$ — for $s > 0$, the leading mode $\lambda_1 = \|\mathbb{W}\|$ with eigenfunction ϕ_1 — and at T_c the emergent non-trivial homogeneous solution is supported on that eigenmode.*

In particular the critical coupling is $s_c = \big(1 + a^2/(b^2q)\big)/\lambda_1$, and for $a = 0$, $s_c = 1/\lambda_1 = 1/\|\mathbb{W}\|$, proving proposition 24.21.

Proof. Each mode eq. (24.10) is the constant-coefficient scalar BVP of proposition 11.8 with s replaced by $s\lambda_k$; that proposition's analysis applies verbatim. The homogeneous BVP (initial datum $\overline{m}_0^{(k)} = 0$) has only the trivial solution iff $\Omega_k^2 \le 0$ (hyperbolic/non-oscillatory case), giving part (i); and is singular at the horizons solving eq. (11.14) with Ω_k iff $\Omega_k^2 > 0$, giving part (ii). The GMFG equilibrium is unique iff all modes are nonsingular (the modes are independent and the inhomogeneous data $\overline{\mathbf{m}}_0$ distribute across them), so uniqueness for all T requires $\Omega_k^2 \le 0$ for every k, i.e. $s\lambda_k \le 1 + a^2/(b^2q)$ for all k. For $s > 0$ the binding mode is the largest $\lambda_k = \lambda_1 = \|\mathbb{W}\|$ (the leading eigenvalue equals the spectral radius for *any* graphon, the kernel being nonnegative, by Perron–Frobenius theorem 18.7 — not merely for a graphon of positive type); for $s < 0$ the binding mode is the most negative λ_k, and the condition is automatically satisfied when $\mathbb{W} \succeq 0$ (recovering the monotone regime $s \le 0$ of proposition 24.17), but *not* for an indefinite $\mathbb{W}$, where a sufficiently negative $\lambda_{\min}$ makes $\Omega_{\min}^2 = b^2q\,|s|\,|\lambda_{\min}| - A_\infty^2 > 0$. The smallest T_c is set by the largest Ω_k^2, hence by the maximal $s\lambda_k$, with the bifurcating solution supported on the corresponding eigenfunction. Setting $s\lambda_1 = 1 + a^2/(b^2q)$ gives s_c; $a = 0$ gives $s_c = 1/\lambda_1$. □

Remark 24.24 (Consistency of the two uniqueness routes)**.** The monotonicity route (theorem 24.16, proposition 24.17) and the spectral threshold (proposition 24.23) agree, as they must. Monotonicity demanded $s\langle\Delta, \mathbb{W}\Delta\rangle \le 0$ for all Δ; for $\mathbb{W} \succeq 0$ this is $s \le 0$, and indeed $s \le 0$ gives $\Omega_k^2 = b^2qs\lambda_k - A_\infty^2 \le -A_\infty^2 < 0$ for every k (since $s\lambda_k \le 0$), so part (i) holds and the equilibrium is unique for all T. The threshold computation extends this into the band $0 < s\lambda_1 \le 1 + a^2/(b^2q)$ where monotonicity has already failed but uniqueness survives — the operator version of the soft band of proposition 11.8, the explicit witness that graphon monotonicity is sufficient but not necessary, with the gap measured by the leading eigenvalue.

We close by verifying the reductions the thread specification requires.

Proposition 24.25 (Reductions of the graphon LQG threshold)**.**

1. Constant graphon. *If $W(x, y) \equiv c > 0$, then $\mathbb{W}$ has the single nonzero eigenvalue $\lambda_1 = c$ with constant eigenfunction $\phi_1 \equiv 1$, and proposition 24.23 reduces to the classical proposition 11.8 with effective coupling $s_{\text{eff}} = s\,c$: uniqueness for all T iff $s\,c \le 1 + a^2/(b^2q)$, recovering Thread A with rescaled coupling as the thread specification demands.*

2. Rank-one graphon. *If $W(x,y) = f(x)f(y)$ with $f \in L^2([0,1])$, $f \not\equiv 0$, then $\mathbb{W}$ has the single nonzero eigenvalue $\lambda_1 = \|f\|_{L^2}^2 = \int_0^1 f^2$, with eigenfunction $\phi_1 = f/\|f\|_{L^2}$; all other modes have $\lambda_k = 0$ and are unconditionally unique. The whole field collapses to the scalar order parameter $\varrho_t := \langle \overline{\mathbf{m}}_t, f\rangle$ (the carried-thread symbol of chapter 20, chapter 21), the aggregate is $z_t(x) = f(x)\,\varrho_t$, and the GMFG reduces to a single self-consistent scalar BVP with coupling $s\,\|f\|_{L^2}^2$. The critical coupling is $s_c = (1 + a^2/(b^2 q))/\|f\|_{L^2}^2$, and for $a = 0$, $s_c = 1/\|f\|_{L^2}^2$.*

3. Block graphon. *If W is piecewise constant on a partition $[0,1] = \bigsqcup_{k=1}^K I_k$, the field collapses to K population means and the GMFG to a finite K-population LQG game; the eigenvalues of $\mathbb{W}$ are those of the $K \times K$ block-weight matrix $\widehat{W}_{k\ell} = W_{k\ell}\,|I_\ell|$, and proposition 24.23 reduces to its multi-population thresholds, the bridge to chapter 30.*

Proof. (i) The constant kernel $W \equiv c$ has $\mathbb{W}f = c\,(\int_0^1 f)\mathbf{1}$, a rank-one operator with range the constants; its only nonzero eigenvalue is $\lambda_1 = c$ (eigenfunction $\mathbf{1}$, since $\mathbb{W}\mathbf{1} = c\mathbf{1}$), all other eigenvalues vanish. Only the uniform mode $\overline{m}^{(1)} = \langle \overline{\mathbf{m}}, \mathbf{1}\rangle = \int_0^1 \overline{\mathbf{m}}$, the global mean, carries coupling, and eq. (24.10) for $k = 1$ is the classical BVP at coupling sc; the zero modes are trivially unique. (ii) The rank-one kernel gives $\mathbb{W}g = f\langle g, f\rangle$, eigenvalue $\lambda_1 = \langle f, f\rangle = \|f\|_{L^2}^2$ on $\phi_1 = f/\|f\|_{L^2}$, zero on $f^\perp$. The only coupled mode is $\overline{m}^{(1)} = \langle \overline{\mathbf{m}}, \phi_1\rangle = \varrho_t/\|f\|_{L^2}$, and $z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x) = f(x)\langle \overline{\mathbf{m}}_t, f\rangle = f(x)\varrho_t$, so the dynamics of ϱ_t close on themselves — the cleanest GMFG worked example, a single scalar self-consistency. The threshold follows from $\lambda_1 = \|f\|_{L^2}^2$. (iii) A block graphon is constant on each $I_k \times I_\ell$, so m^x depends on x only through its block; the field reduces to K flows and $\mathbb{W}$ to the matrix $\widehat{W}_{k\ell} = W_{k\ell}\,|I_\ell|$ acting on the block-constant functions $\mathbb{R}^K$, whose eigenvalues drive eq. (24.10). The weight $|I_\ell|$ sits on the *column* because it is produced by the dy-integration in $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y$; although $\widehat{W}$ is thereby asymmetric, it is conjugate to the symmetric matrix $S_{k\ell} = |I_k|^{1/2}\,W_{k\ell}\,|I_\ell|^{1/2}$ through $\widehat{W} = \mathrm{diag}(|I_k|)^{-1/2}\,S\,\mathrm{diag}(|I_k|)^{1/2}$, so its eigenvalues are real and coincide with the nonzero eigenvalues of the self-adjoint $\mathbb{W}$ — it is simply $\mathbb{W}$ expressed in the block basis, under which $L^2([0,1])$ carries the inner-product weights $|I_\ell|$. The finite system is the multi-population LQG game of chapter 30. ☐

The rank-one case (ii) is the promised Curie–Weiss-type closure: a single scalar order parameter ϱ_t, a self-consistency equation $\varrho = \mathcal{R}(\varrho)$ analogous to the magnetisation fixed point of section 11.4, and a bifurcation at $s\,\|f\|_{L^2}^2 = 1$ ($a = 0$) which is the spontaneous breaking of the symmetry $\varrho \mapsto -\varrho$ in the homogeneous limit. The graphon has done nothing but rescale the classical coupling $s \mapsto s\lambda_1$ — but $\lambda_1 = \|f\|_{L^2}^2$ is a property of the *network*, the squared L^2-mass of the centrality profile f, and this is the entire conceptual payoff of the chapter in a single formula.

Rigor ledger

Proved.

- theorem 24.11: under **(A-Lip)**, **(A-Growth)**, **(A-Nondeg)**, **(A-Conv)**, **(A-Reg-W)** with W continuous in type, and bounded smoothing graphon-local couplings (assumption 24.5), the GMFG system eq. (24.1)–eq. (24.2) has an equilibrium field, by Schauder on the field space; compactness from per-type tightness (chapter 3, made uniform in type by $\|\mathbb{W}\|$) and type-equicontinuity from graphon continuity (lemma 24.9).

- theorem 24.16: graphon Lasry–Lions monotonicity eq. (24.5) of F, G yields uniqueness for all T, via the operator lift of the theorem 11.5 duality identity.
- proposition 24.17: for the quadratic graphon-local coupling the monotonicity pairing equals $-qs\langle\Delta, \mathbb{W}\Delta\rangle$; monotonicity is the joint condition $s\langle\Delta, \mathbb{W}\Delta\rangle \leq 0$, reducing to $s \leq 0$ only when $\mathbb{W} \succeq 0$.
- proposition 24.18: in the uniform-in-type metric D, S_W is $C_T \|W\|_{L^\infty_x L^1_y}$-Lipschitz, hence a contraction (unique equilibrium, any sign) when $C_T \|W\|_{L^\infty_x L^1_y} < 1$ (**(A-Coupling)**); C_T grows with T. The row-sum norm dominates the spectral radius $\|\mathbb{W}\| = \lambda_1$ (remark 24.19), so the route is conservative relative to the exact threshold.
- proposition 24.20: in the contraction regime the equilibrium is Lipschitz in the kernel, $\mathsf{D}(\mathbf{m}^\star_W, \mathbf{m}^\star_{W'}) \leq C_T R \|W - W'\|_{L^\infty_x L^1_y} / (1 - C_T \|W\|_{L^\infty_x L^1_y})$ — the $\delta_\square$-stability exported to chapter 25.
- worked computation 24.22 and proposition 24.23: the graphon LQG equilibrium diagonalises over the spectrum of $\mathbb{W}$ into per-mode copies of the chapter 11 scalar BVP with $s \to s\lambda_k$; the exact threshold is $s\lambda_1 = 1 + a^2/(b^2 q)$ (critical coupling $s_c = (1 + a^2/(b^2 q))/\lambda_1$, and $1/\|\mathbb{W}\|$ for $a = 0$), with the bifurcation carried by eigenvector centrality ϕ_1.
- proposition 24.25: constant, rank-one, and block graphons recover Thread A (rescaled coupling), the Curie–Weiss scalar closure, and finite multi-population MFG, respectively.

Assumed.

- The classical per-type machinery imported from chapter 10: the uniform gradient bound lemma 10.5, the moment/equicontinuity estimates lemmas 10.6 and 10.7, the stability estimates lemmas 10.10 and 10.11, and Schauder/Kakutani (chapter 2).
- The spectral theorem for self-adjoint Hilbert–Schmidt operators and the identification of λ_1 with eigenvector centrality (chapter 18); existence of a classical solution with the per-type decay making the integrations by parts of theorem 24.16 valid (chapters 10 and 11).
- Bounded smoothing couplings for the abstract existence theorem; the LQG thread (unbounded data) is handled by the explicit construction of section 24.5, exactly as in section 10.5.

Deferred.

- Merely measurable graphons and the L^2-in-type / cut-metric route to existence (remark 24.13) $\to$ chapter 25.
- Set-valued feedback (weak graphon equilibria via Kakutani–Fan–Glicksberg, remark 24.14) and measurable selection $\to$ the graphon analogue of section 10.4.

- The time-dependent ($q_T \neq P_\infty$) graphon LQG threshold, solved numerically with type-grid quadrature for $\mathbb{W}$ → chapter 28.
- Displacement-monotone graphon couplings and the graphon master equation's well-posedness → chapter 26.

Open.

- A sharp characterisation of uniqueness for indefinite $\mathbb{W}$ with non-quadratic graphon couplings, where the strategic and spectral signs genuinely compete and neither theorem 24.16 nor proposition 24.18 is tight.
- A selection principle among the multiple equilibria emerging beyond the critical coupling, refining the homogeneous open problem of chapter 11 to the eigenmode-indexed graphon setting (chapter 31).
- Existence and uniqueness for unbounded (L^p) graphons, where $\|\mathbb{W}\|$ may be infinite and the contraction threshold degenerates → chapter 27.

Bibliographic notes

The graphon mean field game and its system of equations are due to Caines and Huang [33, 34], whose existence theory for the GMFG equations proceeds, as here, by a fixed point on a space of measure-valued maps indexed by the type, combining the classical MFG fixed-point argument of Lasry–Lions [93] and Cardaliaguet [36] with the graphon operator of Lovász [99]. The factorisation of the best response into graphon aggregation followed by a type-wise classical solve, and the resulting reduction of all graphon-specific behaviour to the operator $\mathbb{W}$, is the organising device of the probabilistic treatment of stochastic graphon games by Carmona, Cooney, Graves and Laurière [39]; our existence proof follows the analytic route with the Schauder machine of chapter 10, and the compactness inputs — per-type tightness via Prokhorov (chapter 3) and type-equicontinuity from graphon continuity — are the two halves emphasised in remark 24.12.

The uniqueness theory is the operator lift of the Lasry–Lions monotonicity argument [93, 36] of chapter 11; the observation that monotonicity of the graphon coupling is a joint condition on the strategic structure and the spectrum of $\mathbb{W}$ — so that an indefinite kernel can destroy uniqueness even for crowd-averse costs — is the graphon-specific phenomenon, and the operator-norm smallness condition **(A-Coupling)** is the dynamic counterpart of the contraction condition $s\,\|\mathbb{W}\| < 1$ for the static graphon game of Parise and Ozdaglar [108] in chapter 20. The reading of the critical coupling $s\lambda_1 = s_c^{\text{scalar}}$ as a centrality-indexed symmetry-breaking threshold continues the Curie–Weiss analogy of chapter 11; the LQG spectral reduction of section 24.5, in which the continuum equilibrium diagonalises over the eigenmodes of $\mathbb{W}$ into copies of the scalar Nash-certainty-equivalence system of Huang, Caines and Malhamé [82], is the graphon layer of the linear–quadratic thread completed numerically in chapter 28. The connection to finite-network games and the cut-metric stability of the uniqueness regime are the subject of chapter 25; the modern landscape of graphon and other heterogeneous extensions is surveyed in [11] and revisited in chapters 30 and 31.

Chapter 25

Convergence from Finite Networks to the Graphon Limit

The graphon mean field game (GMFG) of chapters 21 to 23 and its solution theory (chapter 24) were built, as the classical mean field game (MFG) before them, by writing down the limit object and studying it on its own terms: a continuum of types $x \in [0,1]$, a field of value functions u^x and population measures m^x, and a single coupling operator $\mathbb{W}$ tying them together across types. This chapter closes the loop for networks, exactly as chapter 13 closed it for the homogeneous mean field. We begin from a genuine finite, non-cooperative game — N agents sitting on the vertices of an actual graph, each minimizing its own cost, each best-responding to its neighbors — and ask in what precise sense, and at what rate, this game is approximated by the GMFG as the network grows and converges to the graphon.

The new difficulty, and the reason this chapter is not merely chapter 13 with decorations, is that *two* limits are now in play and they are not the same limit. The first is the familiar mean field limit $N \to \infty$: even on a fixed network, the number of agents per "type" must grow before fluctuations of the local population are suppressed. The second is *graph convergence*: the finite network G_N must itself converge to the graphon W, and the natural topology for this — the cut metric $\delta_\square$ of chapter 17 — is weaker and slower than Wasserstein sampling. For a graph $G(N, W)$ sampled from W (the W-random graph of chapter 19) both limits are taken *simultaneously* and the same parameter N controls both, so the joint rate is governed by whichever mechanism is slower. Separating the two error sources, identifying when the limits commute, and finding what controls the joint rate is the analytic content of sections 25.3 to 25.5.

It helps to read these three sections as one argument carried to three destinations rather than as three separate topics. Section 25.3 does the analytic labour: it splits the total error into a mean-field part and a graph part, bounds each, and in doing so uncovers the cut-versus-operator dichotomy that decides the joint rate. The two sections after it are what that labour buys. The practically decisive result is the network ε-Nash theorem (theorem 25.13): the decentralized GMFG strategy, computed once on the continuum and handed to each of the N vertices according to its type, is an ε_N-Nash equilibrium of the finite game on $G(N, W)$, with $\varepsilon_N \to 0$ explicitly. This is what licenses the use of the GMFG to *design interventions on real networks* — the comparative-statics and centrality program of chapter 20 — because it certifies that no agent on the true finite graph can gain more than ε_N by deviating from the prescription read off the limit. We carry Thread B through this theorem on a sampled graph in worked computation 25.15, which is this chapter's discharge of the carried-thread duty. We close with the stability statement that makes the cut metric earn its keep (section 25.5): macroscopically equivalent networks — those close in $\delta_\square$ — produce close equilibrium aggregates, so the equilibrium one designs is robust to the part of the wiring the cut metric cannot see.

We import from chapters 21 to 23 the GMFG state space (a field of measures $x \mapsto m^x \in \mathcal{P}(\mathbb{R}^d)$), the coupled HJB–Fokker–Planck system across the kernel, and the graphon McKean–Vlasov dynamics; from chapter 24 the existence and uniqueness of the GMFG solution (theorem 24.11) and the contraction estimate proposition 24.18 for the best-response map in the field argument $\mathbf{m}$ at fixed kernel (from which we *derive* here, in section 25.5, the Lipschitz dependence of the solution on the *kernel*); from chapter 17 the cut norm $\|\cdot\|_\square$, the cut metric $\delta_\square$, and the equivalence of the cut norm with the $\infty \to 1$ operator norm; from chapter 19 the W-random graph $G(N, W)$ and the concentration of its empirical graphon; and from chapter 13 the entire architecture of the homogeneous convergence problem — the Nash system, the ε-Nash mechanism (theorem 13.6), the propagation-of-chaos lemma (lemma 13.5), and the value-versus-trajectory rate dichotomy (remark 13.14) — which we generalize rather than repeat. The standing assumptions are referenced by tag: (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) on the dynamics and costs, (A-Mon) for uniqueness in the monotone regime, and the graphon assumptions (A-Reg-W) and (A-Coupling) of appendix A.

We keep the conventions of appendix A: value functions u^x carry terminal data and are solved backward, population fields m^x carry initial data and flow forward.

Notation watch

We are in Part IV, where a graphon W is in scope, so the letter W denotes the *kernel* throughout and Brownian motions are written B: agent i's independent noise is B^i, and the representative type-x noise is B^x. The macro W (which renders W) is not used in this chapter. The type $x \in [0, 1]$ is an exchangeable label, never a spatial coordinate; in particular $|x - y|$ has no meaning and the only geometry on $[0, 1]$ is the one W assigns through $\mathbb{W}$. Following the W-random graph of chapter 19, the *doubly-indexed* symbol ξ_{ij} denotes the edge weight between vertices i and j; an agent's *private state* is written $X_t^i \in \mathbb{R}^d$ here (and the generic real argument of the interaction observable is the dummy ζ). These never collide, since the edge symbol always carries two vertex subscripts. We flag that the master-equation chapter chapter 26, where no edge array appears, reverts to the book-wide convention in which the *bare* symbol ξ denotes the private state; the clash is confined to the glyph and does not affect any displayed expression here.

25.1 The finite game on a network

The preamble promised to begin not from the limit object but from a genuine finite, non-cooperative game, and this section makes that object completely explicit, fixing every datum the rest of the chapter consumes. Nothing here is asymptotic: there are finitely many agents, a fixed graph wiring them together, and each agent solves an honest stochastic control problem against the others. In the language of the commuting square that organizes section 25.3, this is the bottom-left corner — the concrete game the GMFG is later certified to approximate — and it must be pinned down before any limit can be named.

The game refines the anonymous game of chapter 13 in exactly one place, and it is worth isolating that single change before any symbol is committed, because the whole chapter is the bookkeeping of its consequences. In chapter 13 every agent felt the same object, the global empirical measure $\mu^N = \frac{1}{N}\sum_j \delta_{X^j}$, and the game was symmetric under permuting the agents wholesale: each was an anonymous member of one undifferentiated crowd. A network breaks that democracy. Agent i is wired to some vertices and not to others, and it responds not to the population at large but to the states of the vertices it can see, each weighted by the strength of the edge to it. Replace "global crowd" by "weighted neighbourhood" and chapter 13 becomes the present chapter; the single-agent dynamics, the cost structure, and the equilibrium concept all carry over unchanged. The entire analytic burden ahead is that this one replacement turns a single empirical measure into a vertex-dependent, edge-weighted aggregate whose limit is no longer a number-of-agents limit alone but is entangled with the convergence of the wiring itself — the two-limit tension flagged in the preamble.

Three pieces of data specify an agent's situation, and it pays to feel the distinct role of each, since the convergence theory treats them by completely different mechanisms. The first is the agent's *type* $x_i \in [0,1]$. It is neither a state nor a strategy but the agent's fixed latent position in the graphon: the coordinate that decides, through W, how likely it is to be wired to any other vertex and how strongly. The natural physical reading is a *lattice site carrying quenched disorder*. In a disordered spin system each site is assigned, once and for all, a coupling or field drawn from some distribution and then frozen, and the dynamics run inside that fixed environment; here the type is drawn once (or placed deterministically), frozen for the whole horizon $[0,T]$, and the strategic dynamics unfold in the wiring it induces. The types are *quenched*, not *annealed* — fixed before play, never averaged over while the game runs — and the distinction is not cosmetic: the GMFG will turn out to be the annealed (disorder-averaged) law, while the finite game is a single quenched realization, a contrast developed in the physics note that closes section 25.3. We must resist reading x_i as a spatial coordinate: as the chapter's opening notationwatch insists, the only geometry on $[0,1]$ is the one W assigns, and $|x_i - x_j|$ by itself means nothing.

The second datum is the agent's *private state* $X_t^i \in \mathbb{R}^d$, the one genuinely dynamical, controlled quantity. It evolves over $[0,T]$ driven by the agent's own admissible control and its own independent Brownian motion B^i, and it is what each agent steers to minimize its cost. The third datum is the *edge array* ξ_{ij}, the realized wiring: a symmetric, doubly-indexed family recording the weight of the link between vertices i and j. It is sharply distinct from the type. The type is *where* a vertex sits in the latent space; the edge is the realized *connection* between two vertices, which — in the sampled model below — is itself random given the positions. Keeping these two separate is what later lets us isolate a "type-sampling" error from an "edge-sampling" error and discover that they converge at different rates.

Two questions stand between this data and the dynamics, and they organize the two subsections that follow. First, where does the network come from? An analyst designing an intervention works with one observed graph, given and non-random; the convergence theory, by contrast, gets its sharpest statements from a graph *sampled* from the kernel W, where the types and edges are random in a controlled way. These two provenances are not interchangeable — the sampled case carries an extra layer of randomness the deterministic case lacks, and the rates differ accordingly — so section 25.1.1 sets them side by side and says precisely how each network arises. Second, how does an agent compress its neighbours into the finite vector of observables that drives its state and cost, and how does that vector slot into the single-agent control problem of chapter 6? That is the business of section 25.1.2, which builds the neighbour aggregate and writes down the dynamics, cost, and equilibrium concept that the Nash system of section 25.2 and the convergence theory will consume.

25.1.1 Types, edges, and the sampled graph

The previous section isolated the three data of an agent's situation — type, private state, edge array — and flagged that the very first of the two questions standing between that data and the dynamics, namely *where the network comes from*, splits into two provenances the convergence theory cannot treat alike. We now make that split precise, because every rate quoted later in the chapter is attached to one provenance or the other. An analyst designing an intervention holds one graph in hand: it was measured once and is thereafter fixed, nothing about it is random, and the only limit one can even imagine is the same wiring rule applied to more and more vertices. The convergence theory, by contrast, gets its cleanest and sharpest statements by positing that the network was itself *generated* by the kernel — the latent positions scattered into $[0,1]$ and the edges switched on at random with the probabilities W prescribes. Both provenances are legitimate, both feed the identical dynamics–cost machinery of section 25.1.2 downstream, but the sampled model carries an extra layer of randomness the observed graph lacks, and it is exactly that extra layer the rates of section 25.3.2 will have to pay for. The definition records the two side by side so that this difference is visible in one place.

Definition 25.1 (The finite network)**.** A *finite weighted network* is a pair $(\boldsymbol{x}^N, \xi^N)$ where $\boldsymbol{x}^N = (x_1, \dots, x_N) \in [0,1]^N$ is the vector of types and $\xi^N = (\xi_{ij})_{i,j\in[N]}$ is a symmetric array of edge weights $\xi_{ij} = \xi_{ji} \in [0,1]$ with $\xi_{ii} = 0$. Two instances recur:

1. *(Deterministic network.)* A given real network: the types x_i and the weights $\xi_{ij} \in [0,1]$ are prescribed, non-random data. This is the case relevant to designing interventions on an observed graph.

2. *(Sampled graph $G(N,W)$.)* Given a graphon W satisfying (A-Reg-W), draw the types either on the *regular grid* $x_i = \frac{i-1/2}{N}$ or i.i.d. $x_i \sim \mathrm{Unif}\,[0,1]$, and then, conditionally on $\boldsymbol{x}^N$, draw the edges independently,

$$\xi_{ij} \sim \mathrm{Bernoulli}\big(W(x_i,x_j)\big), \qquad i<j, \qquad \xi_{ji} = \xi_{ij}, \tag{25.1}$$

   with $\xi_{ii} = 0$. The resulting random simple graph is the *W-random graph* $G(N,W)$ of chapter 19.

We write $\bar{\xi}_{ij} := \mathbb{E}[\xi_{ij} \mid \boldsymbol{x}^N] = W(x_i, x_j)$ for the conditional edge probability in the sampled case, and $\bar{\xi}_{ij} = \xi_{ij}$ in the deterministic case.

Two clauses of this definition deserve to be read slowly, because the entire two-error decomposition that drives the chapter is already latent in them. The Bernoulli prescription eq. (25.1) says that, *given the types*, the link between i and j is the outcome of a single biased coin whose bias is the kernel value $W(x_i, x_j)$: the better aligned two latent positions are in the geometry W assigns, the more likely the edge fires. The realized edge $\xi_{ij} \in \{0, 1\}$ is therefore a noisy one-bit readout of the smooth quantity $W(x_i, x_j) \in [0, 1]$, and the conditional expectation in the last line strips that noise away. Reading $\bar{\xi}_{ij} = \mathbb{E}[\xi_{ij} \mid \boldsymbol{x}^N] = W(x_i, x_j)$ from the inside out: the bar denotes an average over the edge coins *only*, with the latent positions $\boldsymbol{x}^N$ held fixed — it is the annealed value of the wiring at quenched types — and that average returns precisely the probability the edge is present, which the model set equal to $W(x_i, x_j)$. The residual $\xi_{ij} - \bar{\xi}_{ij}$ is then a mean-zero fluctuation of order one per edge, and it is the raw material of the *edge-sampling* error; the separate discrepancy between the quenched positions $\boldsymbol{x}^N$ and the continuum label they are meant to sample is the raw material of the *quadrature* error. In the deterministic case there are no coins to average over, so we set $\bar{\xi}_{ij} = \xi_{ij}$ by convention: the prescribed weights *are* their own conditional expectation, the edge-sampling fluctuation vanishes identically, and only whatever discrepancy the fixed positions carry survives. Keeping $\bar{\xi}_{ij}$ on the page, even when it equals ξ_{ij}, is what lets a single later estimate cover both provenances.

This is the precise sense in which the sampled graph $G(N, W)$ is the canonical bridge between the finite and the continuum worlds: it is the one model that carries *both* sources of finiteness at once and, crucially, carries them transparently rather than fused together. The types $\boldsymbol{x}^N$ are N samples of the continuum label x — a Monte-Carlo or quadrature discretization of the integral over $[0, 1]$ out of which every continuum aggregate is built — while the edges ξ_{ij} are Bernoulli discretizations of the kernel *values* themselves. A deterministic real network blends these two errors inseparably: its single realized wiring is whatever it is, with no reference kernel against which to ask "how far off is the position" versus "how far off is the edge". The sampled model, by contrast, holds the two apart by construction — one error lives in the positions, the other in the coins — and that separation is exactly what permits section 25.3.2 to attach a *distinct rate* to each mechanism, and to discover that those rates differ. That two-source decomposition, edge sampling against quadrature, is the first thing this subsection hands forward.

The two ways of placing the types trade analytic convenience against modelling realism, and the choice shows up directly in the quadrature rate. The regular grid $x_i = (i - \frac{1}{2})/N$ lays the N types down at the midpoints of an even partition of $[0, 1]$; it is a deterministic midpoint quadrature rule, and like any midpoint rule on a sufficiently regular integrand it converges fast — the quadrature error will turn out to be $O(N^{-1})$ under (A-Reg-W). The i.i.d. uniform choice $x_i \sim \mathrm{Unif}\,[0, 1]$ instead scatters the types at random; it is the honest model of a network whose vertices arrived from an unknown population and were never arranged on any grid, but random sampling fluctuates, so its quadrature error is the slower Monte-Carlo $O(N^{-1/2})$. As a one-line micro-example, estimating $\int_0^1 g$ for a Lipschitz g by $\frac{1}{N}\sum_i g(x_i)$ incurs error $O(1/N)$ from grid midpoints but only $O(N^{-1/2})$ from i.i.d. draws, by the elementary central limit theorem; the rows $W(x_i, \cdot)$ inherit exactly this contrast. The grid is thus the cleaner object for *stating* rates, while the i.i.d. model is the one that connects to the exchangeability and de-Finetti picture of chapter 19. Both, once the next definition packages them as a kernel, produce a step graphon converging to W; they differ only in how fast, and through which quadrature mechanism.

To *quantify* the phrase "the finite network converges to W" we must first make the network and the kernel live in the same space, and as written they do not: one is an $N \times N$ array of numbers, the other a function on the unit square. The device that reconciles them is the pixel picture of chapter 15. Partition $[0, 1]$ into N equal type cells, one per vertex, and inflate each matrix entry ξ_{kl} into a constant block over the pixel $I_k \times I_l$. The result is an honest graphon — a symmetric measurable function on $[0, 1]^2$ — that looks like a coarse, blocky image of W: the (k, l) pixel is shaded by the realized edge ξ_{kl}, and as N grows the pixels shrink and the image sharpens toward the continuous kernel, exactly as a photograph resolves when its resolution is raised. This blocky kernel is what we call the *empirical step graphon*; "step" because it is piecewise constant on the grid of pixels, and "operationally" it is nothing more than the adjacency array reread as a function, so that the cut metric $\delta_\square$ and the integral operator $\mathbb{W}$ apply to it verbatim.

Definition 25.2 (Empirical step graphon)**.** Given a finite weighted network $(\boldsymbol{x}^N, \xi^N)$ with types on the regular grid, let $I_k = [\frac{k-1}{N}, \frac{k}{N})$ be the k-th type cell. The *empirical step graphon* is the symmetric kernel

$$W_N(x, y) := \xi_{kl} \qquad \text{for } x \in I_k,\ y \in I_l, \tag{25.2}$$

the function on $[0, 1]^2$ that is constant ξ_{kl} on the pixel $I_k \times I_l$. Its integral operator is $\mathbb{W}_N$, $(\mathbb{W}_N f)(x) = \int_0^1 W_N(x, y) f(y)\, \mathrm{d}y$. For i.i.d. types one orders them increasingly before forming the cells, or works with the labelled empirical kernel directly; the distinction does not affect $\delta_\square$, which quotients by relabelling (chapter 17).

The final clause of the definition disposes of a bookkeeping subtlety peculiar to the i.i.d. case, and it is worth spelling out because it is the chapter's first encounter with the defining feature of the cut metric. With random types there is no reason the labels $1, \ldots, N$ should emerge in increasing order of x_i, so the pixels $I_k \times I_l$ would be shaded by edges between vertices whose positions are scrambled relative to the grid that the cells impose. One can repair this by sorting the types increasingly before cutting the cells — relabelling the vertices so that vertex k owns cell I_k — or one can skip the sort and work with the labelled empirical kernel as it stands. It makes no difference to anything we ever measure, and the reason is structural rather than incidental: the cut metric $\delta_\square$ is defined as an *infimum over relabellings* (chapter 17), so by construction it is blind to the order in which the vertices happened to be enumerated. Two step graphons differing only by a permutation of their cells sit at cut distance zero from one another; sorting the types merely selects a convenient representative inside a single $\delta_\square$-equivalence class, and never alters the object the convergence theory sees. This is the first appearance of a theme that will organize the entire chapter: the cut metric resolves a network only up to the relabellings that scramble individual vertex identities, which is precisely why it can certify *macroscopic* equivalence while remaining silent about *per-vertex* data — the very tension that section 25.5 sharpens into the cut-versus-operator dichotomy.

With both avatars of the finite network now in hand and each packaged as a kernel, two threads run forward out of this subsection. The first is the operator pair $(\mathbb{W}_N, \mathbb{W})$, the empirical operator set against its continuum limit: the distance between them is the object the stability theorems of section 25.5 will control — in cut norm when only the unlabelled network is available, in the stronger operator norm when the labels are known — and the decomposition of $W_N - W$ into its edge-sampling and quadrature parts, made explicit above, is exactly the input section 25.3.2 consumes to assign a separate rate to each. The second thread is the type-and-state data $(\boldsymbol{x}^N, X_0^i)$: the latent positions just fixed and the private states they seed. These are what the next subsection sets into motion. The agents now move, compress their neighbours through the wiring just defined into the finite vector of aggregated observables, and pay costs — recovering, vertex by vertex, the GMFG coupling form of chapters 21 and 22.

25.1.2 Dynamics, the neighbour aggregate, and costs

The previous subsection fixed the static scaffolding — the latent types $\boldsymbol{x}^N$, the realized wiring ξ^N, and the seed states X_0^i — and packaged the wiring as a kernel so that the continuum operator $\mathbb{W}$ could later be brought to bear. Nothing has moved yet. We now set the game in motion: each agent acquires a controlled state that evolves over $[0, T]$, pays a cost along the way, and — this is the only channel through which the network enters the dynamics — feels the other agents *solely* through what its own neighbourhood is doing. The task of this subsection is to write down that coupling in a way that slots, vertex by vertex, into the single-agent control problem of chapter 6, and to do so in exactly the form the GMFG coupling already took in chapters 21 and 22, so that the limit we eventually take has somewhere to land.

The guiding principle is that the interaction is *local and anonymous within a neighbourhood*. "Local" because agent i sees only the vertices it is wired to, not the population at large; "anonymous" because, within that neighbourhood, it does not track *which* neighbour sits in which state but only the aggregate they collectively present. This is the network refinement of the anonymity that made the homogeneous game of chapter 13 tractable: there the agent felt the whole crowd through a single empirical measure, here it feels a weighted slice of the crowd through a single aggregated vector. To make "the aggregate they present" precise we must first say *what* about a neighbour's state is felt, and that is the role of the interaction observable.

Fix once and for all a bounded, Lipschitz *interaction observable* $h : \mathbb{R}^d \to \mathbb{R}^\ell$. Its job is to extract from a neighbour's full state $X_t^j \in \mathbb{R}^d$ the finite list of scalar features — ℓ of them — through which that neighbour is allowed to influence agent i. The reader should think of h as the modeller's declaration of *what is socially visible*: not the entire microscopic state, but a chosen handful of summary statistics of it. Three instances cover essentially all the models in this book and are worth keeping in view, because each turns the abstract h into something concrete. In Thread B (the graphon LQG set up in chapter 22) the state is scalar, $\ell = 1$, and $h(\zeta) = \zeta$ is the identity: a neighbour is visible only through its position, and the aggregate built below will reproduce exactly the graphon-aggregated neighbour mean $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ that the thread couples to. In a *congestion* model h is a mollified (smoothed) indicator of a region of state space, so that the aggregate measures a smeared local density — how crowded agent i's neighbourhood is near a given location — and the cost can then penalize moving into congested territory. And in the master-equation development of chapter 26 the very same object reappears under the name ϕ; we flag the synonymy now so that the reader meeting ϕ there does not take it for a new ingredient.

The two hypotheses on h earn their place by exactly what they prevent. Boundedness is what lets the cut-norm aggregation lemma of section 25.3.1 apply: that lemma controls the aggregate only for observables bounded by some M, because a cut-norm estimate tests the kernel against functions valued in $[-1, 1]$, and an unbounded h would carry no such normalization. Lipschitz continuity is what couples the *convergence of states* to the *convergence of aggregates*: in the propagation-of-chaos estimate of section 25.3.3 we bound $|h(X_t^j) - h(\bar{X}_t^{x_j})|$ by $[h]_{\mathrm{Lip}}\, |X_t^j - \bar{X}_t^{x_j}|$, so without it a small error in the states could produce an uncontrolled error in the aggregate they feed. Boundedness can be relaxed to linear growth — which is what one needs to take $h(\zeta) = \zeta$ literally in Thread B, where the states are unbounded Gaussians — provided the relevant moments of the equilibrium law are controlled; (A-Lip) and (A-Growth) together are stated so as to permit exactly this extension, and the Gaussian states of Thread B supply the requisite moment bounds automatically, so no separate hypothesis is incurred there.

With the observable chosen, the aggregate writes itself: sum each visible neighbour's feature vector $h(X_t^j)$ against the strength ξ_{ij} of the edge to it, and normalize. The only genuine decision is the normalization, and it is not cosmetic — it decides which continuum object the finite aggregate is an approximation *of*, and hence which limit theorem this whole chapter is proving. We make the decision deliberately and record it in the definition.

Definition 25.3 (Neighbour aggregate)**.** The *normalized neighbour aggregate* felt by agent i at time t is

$$z_t^i \;:=\; \frac{1}{N} \sum_{j=1}^{N} \xi_{ij}\, h(X_t^j) \;\in\; \mathbb{R}^{\ell}. \tag{25.3}$$

The normalization is by N, not by the degree, so that z_t^i approximates the *operator* aggregate $\int_0^1 W(x_i, y)\langle h, m_t^y\rangle\, \mathrm{d}y$ rather than a degree-normalized average; this is the *dense* convention of chapters 15 and 19, under which the typical degree is $\Theta(N)$ and the graphon controls the edge density.

It is worth tracing, in one line of heuristic bookkeeping, why dividing by N is exactly what makes z_t^i converge to the operator integral, since this is the hinge on which the dense convention turns. Suppose the network is sampled from W, so $\mathbb{E}[\xi_{ij} \mid \boldsymbol{x}^N] = W(x_i, x_j)$, and suppose the neighbours have roughly settled into the type-x_j equilibrium law, so $h(X_t^j) \approx \langle h, m_t^{x_j}\rangle$ on average. Then

$$z_t^i = \frac{1}{N} \sum_{j=1}^{N} \xi_{ij}\, h(X_t^j) \;\approx\; \frac{1}{N} \sum_{j=1}^{N} W(x_i, x_j)\, \langle h, m_t^{x_j}\rangle,$$

and the right-hand side is precisely a Monte-Carlo / Riemann sum, over the N type samples x_j, of the integral $\int_0^1 W(x_i, y)\langle h, m_t^y\rangle\, \mathrm{d}y$. The factor $1/N$ is the quadrature weight that turns the sum into the integral; the $\Theta(N)$ neighbours each contribute an $O(1/N)$ term, and their number and their weight cancel to leave an $O(1)$ limit. This is the dense scaling at work, and it is why the typical degree must grow like N: if it did not, the sum would have too few terms to converge to a continuum average. The two error mechanisms of section 25.3.2 are already visible in this one display — replacing ξ_{ij} by $W(x_i, x_j)$ is the *edge-sampling* step, and replacing the sum by the integral is the *quadrature* step — and the chapter's later labour is to make each step quantitative.

With the aggregate in hand, the rest is the single-agent template of chapter 6 read off verbatim, with one substitution: wherever that chapter's controlled diffusion and cost referred to the population — through the mean field in the homogeneous case — they now refer to agent i's private aggregate z_t^i. The aggregate plays exactly the role the mean field played, and the reader who has internalized chapter 6 can read the following two displays as "a controlled SDE whose coefficients depend on the state, an external signal, and the control, with a running-plus-terminal cost of the same three arguments," the external signal being z_t^i:

$$\mathrm{d}X_t^i = b(X_t^i,\, z_t^i,\, \alpha_t^i)\ \mathrm{d}t \;+\; \sigma(X_t^i,\, z_t^i)\ \mathrm{d}B_t^i, \qquad X_0^i \sim m_0^{x_i}, \tag{25.4}$$

$$J^{N,i}(\alpha^1,\dots,\alpha^N) = \mathbb{E}\left[\int_0^T L(X_t^i,\, z_t^i,\, \alpha_t^i)\ \mathrm{d}t \;+\; g(X_T^i,\, z_T^i)\right]. \tag{25.5}$$

The coefficients b, σ, L, g are the *same* functions of $(\text{state}, \text{aggregate}, \text{control})$ for every agent, and this shared functional form is the structural fact that makes the agents interchangeable. But the interchangeability is conditional, and the precise sense in which it holds is worth dwelling on, because the entire propagation-of-chaos machinery rests on it. The agents are *not* exchangeable as raw random variables: agent i and agent j differ in their types $x_i \neq x_j$, hence in their wiring, their initial laws $m_0^{x_i}$ versus $m_0^{x_j}$, and the aggregates they feel. What is true is that, *once the type vector $\boldsymbol{x}^N$ is fixed*, the construction treats every agent by the identical rule — same coefficients, independent noises, independent initial draws — so that the joint law of the states is exchangeable under any permutation that simultaneously permutes the types. In probabilistic terms, exchangeability holds *conditionally on $\boldsymbol{x}^N$*, not marginally. This is exactly the quenched-disorder reading of the types from section 25.1.1: freeze the disorder (the types), and the agents become a symmetric ensemble; average over the disorder, and the symmetry is between *type-labelled* agents, not bare ones. Every law-of-large-numbers and chaos estimate in this chapter is therefore taken in the conditional measure $\mathbb{P}(\cdot \mid \boldsymbol{x}^N)$, and only afterwards, if one wishes the annealed statement, averaged over the types. Keeping this order straight — condition first, average second — is what prevents the two error sources from being silently fused.

The coefficients satisfy (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) in the pair (x, z), the state and the aggregate, so that the single-agent control problem each frozen aggregate induces is well-posed by the verification theory of chapter 6; nondegeneracy in particular is what will later license the parabolic regularity of the network Nash system. The initial law $m_0^{x_i}$ may depend on the type — different kinds of agent may start from different distributions — and the noises and seeds B^i, X_0^i are mutually independent and independent of the network $(\boldsymbol{x}^N, \xi^N)$, so that all of the dependence between agents is routed through the aggregate and none leaks in through correlated inputs. We write $a := \sigma\sigma^\top$ for the diffusion matrix and $\mathcal{A}^\alpha$ for the controlled single-state generator as in chapter 13.

Notation watch

Two normalizations of the aggregate appear in the literature and must not be confused. The *operator* aggregate eq. (25.3), normalized by N, converges to $\int W(x_i, y)\langle h, m^y\rangle\, \mathrm{d}y$ and is the correct object for *dense* graphons. The *degree*-normalized average $\frac{1}{d_i}\sum_j \xi_{ij} h(X_t^j)$, where $d_i = \sum_j \xi_{ij}$, converges instead to the *conditional* mean given that a neighbour exists; the two agree only when the degree profile is constant. Throughout this chapter, and consistently with chapters 21 and 22, the aggregate is operator-normalized. The degree normalization becomes the natural one only in the sparse regime of chapter 27, where the dense scaling fails.

The clause that the two normalizations "agree only when the degree profile is constant" is worth unpacking, since it is the cleanest way to see what each one measures. The degree of vertex i is $d_i = \sum_j \xi_{ij}$, which in the dense sampled model concentrates around $N\int_0^1 W(x_i, y)\,\mathrm{d}y$, the *degree function* $d(x_i)$ of the graphon evaluated at the vertex's type. The operator aggregate divides the weighted sum by N; the degree aggregate divides it by $d_i \approx N\,d(x_i)$. The two therefore differ exactly by the factor $d(x_i)$: the operator aggregate is $d(x_i)$ times a probability average of h over the neighbour distribution, while the degree aggregate strips that factor off and reports the bare conditional mean "what does a typical neighbour look like, given that it is a neighbour." If every type has the same degree — a *regular* graphon, $d(x) \equiv d$, the constant-degree profile, of which the constant graphon $W \equiv c$ is the basic example — then $d(x_i)$ is the same constant for all i and the two normalizations coincide up to that overall rescaling, which can be absorbed into the coupling. The moment the degree varies with type — a hub feels many neighbours, a leaf few — they part company: operator-normalization preserves the fact that a hub aggregates *more* influence than a leaf, which is precisely the network heterogeneity the GMFG is built to track, whereas degree-normalization quotients that heterogeneity away. For the dense graphon games of this book we want to keep it, so we normalize by N.

It remains to say what an agent is allowed to know when it chooses its control, and what it means for a profile of such choices to be in equilibrium. Both the admissibility classes and the equilibrium concept are imported verbatim from chapter 13; the only change is to the informational primitive, which the network refines from "own state" to "own state and own type." Because the import is literal, a one-sentence reminder of the three classes is owed to the reader rather than a bare cross-reference. An *open-loop* strategy is a control process adapted to the full information filtration — the agent may, in principle, react to everything that has happened, including the whole configuration of the network — and serves mainly as the largest class against which deviations are tested. A *Markovian closed-loop* strategy collapses that dependence to feedback on the current configuration of states, $\alpha_t^i = \mathsf{a}(t, \boldsymbol{X}_t)$, the natural class for a dynamic-programming characterization. A *decentralized* strategy is the most restrictive and the one the GMFG actually delivers: the agent's control depends only on its *own* current state, with no access to the states of others. The single network novelty is that "own" now includes the agent's type, which is private, observable data it carries from birth.

Definition 25.4 (Admissible strategies and equilibria)**.** Let $\mathbb{F}^N$ be the filtration generated by the network $(\boldsymbol{x}^N, \xi^N)$, the initial states, and the Brownian motions. *Open-loop*, *Markovian closed-loop*, and *decentralized* strategies are defined as in definition 13.1, with a decentralized strategy now of the form $\alpha_t^i = \mathsf{a}(t, X_t^i, x_i)$ — feedback on agent i's own state *and its type* (we write the feedback map in sans-serif a to keep it distinct from the diffusion matrix $a = \sigma\sigma^\top$ and the Thread B/A scalar state-feedback parameter a). A profile $\boldsymbol{\alpha}^\star$ is a *Nash equilibrium* if no unilateral deviation strictly lowers any agent's cost, and an ε-*Nash equilibrium* (definition 13.2, eq. eq. (13.4)) if no unilateral deviation lowers any agent's cost by more than ε.

The reason to insist on the decentralized class, beyond its realism, is that it is the class the limit object naturally produces and the class the ε-Nash theorem of section 25.4 will certify. The GMFG of chapters 22 and 24 is solved once on the continuum and returns, for each type, a feedback $\widehat{\mathsf{a}}(t,\cdot,x)$ that depends on the agent's state and type alone; handing the type-x_i feedback to vertex i produces a decentralized profile on the finite graph automatically. The whole convergence program is the assertion that this profile, read off the limit, is nearly optimal on the genuine finite game just defined — and it is decentralized strategies, not the larger open-loop or Markovian classes, that the agents can actually implement without observing the network.

The finite game is now completely specified, and two objects pass forward from it. The first is the per-agent aggregate z_t^i of eq. (25.3), the quenched, edge-weighted coupling that the next section installs in place of the empirical measure to write down the network's exact Nash system: it is the finite-N object whose limit, the operator aggregate $z_t(x) = \int_0^1 W(x,y)\langle h, m_t^y\rangle\,\mathrm{d}y = (\mathbb{W}\bar{h}_t)(x)$ with $\bar{h}_t(y) = \langle h, m_t^y\rangle$, the GMFG couples to, and the gap between the two is the graph error the chapter must pay. The second is the cost functional eq. (25.5) in its frozen-aggregate form: because $J^{N,i}$ depends on the other agents *only* through z_t^i, an agent contemplating a deviation while every neighbour holds its strategy fixed faces an honest single-agent control problem against a given signal, and it is precisely this envelope — optimize against the frozen aggregate, then control how far the aggregate itself moved — that the ε-Nash proof of section 25.4 will run. With the game, its coupling, and its equilibrium concept fixed, we turn to what its *exact* equilibrium looks like.

25.2 The Nash system on the network

The finite game of section 25.1 is now specified down to its last datum: the quenched aggregate z_t^i through which each agent feels its neighbours, and the decentralized class in which we eventually want to play. Before we can say in what sense, and at what rate, the GMFG approximates this game, we owe ourselves a sharp look at the thing being approximated — the *exact* equilibrium of the finite game, written as precisely as the equilibrium of any finite-player stochastic differential game can be written, namely as the solution of a coupled system of partial differential equations. This is the network analogue of the Nash system that organized the homogeneous convergence problem in chapter 13, and recording it here is not a detour but the diagnostic step of the whole chapter. Every feature that makes this system useless as a computational object is a feature the GMFG limit exists to cure, and only by seeing the exact system on the page can we name — with no hand-waving — exactly what the limit must buy us. We therefore transcribe the chapter 13 architecture with one and only one structural change, the empirical measure giving way to the quenched, edge-weighted aggregate $z^i_{\boldsymbol{X}}$, and then read off from the transcription the obstructions that send us to the continuum.

It pays to recall, in one breath, the logic by which such a system is derived, because the network case reuses it verbatim and the only novelty is where the new aggregate slots in. The recipe is the dynamic-programming recipe of single-agent control (chapter 6), run N times under a self-consistency constraint. Freeze every agent but one to its equilibrium feedback, so that the remaining agent faces an honest single-controller problem; solve that problem by dynamic programming to get an HJB equation; then impose that the optimizer this produces for each agent coincides with the feedback the *other* agents had frozen, which closes the N best responses into one simultaneous fixed point. The unknown the machinery produces is, for each agent i, a value function of the *entire* configuration $\boldsymbol{X} = (X^1, \dots, X^N)$ — not of agent i's state alone — because in a feedback equilibrium agent i's continuation cost depends on where everyone is: everyone's future motion feeds back, through the wiring, into the aggregate agent i will feel. With that picture in place the statement reads as the expected N-fold HJB system, decorated by the network through the aggregate.

Proposition 25.5 (The network Nash system)**.** *Fix the network* $(\boldsymbol{x}^N, \xi^N)$*. Suppose the game eq.* (25.4)*–eq.* (25.5) *admits a Markovian Nash equilibrium whose value functions* $v^{N,i}(t, \boldsymbol{X})$*,* $\boldsymbol{X} = (X^1, \dots, X^N) \in (\mathbb{R}^d)^N$*, are of class* $C^{1,2}$*. With the Hamiltonian and optimal feedback of eq.* (13.5) *written here in the aggregate variable,* $H(x, z, p) = \sup_\alpha\{-p \cdot b(x, z, \alpha) - L(x, z, \alpha)\}$ *and* $\alpha^\star(x, z, p) = \arg\max$*, the family* $(v^{N,i})$ *solves on* $[0, T) \times (\mathbb{R}^d)^N$

$$-\partial_t v^{N,i} - \sum_{j=1}^{N} \tfrac{1}{2}\operatorname{tr}\Big(a(x_j, z^j_{\boldsymbol{X}})\, \mathrm{D}^2_{X_j} v^{N,i}\Big) - \sum_{j\neq i} b\Big(X_j, z^j_{\boldsymbol{X}}, \widehat{\alpha}^{\,j}\Big) \cdot \nabla_{X_j} v^{N,i} + H\Big(X_i, z^i_{\boldsymbol{X}}, \nabla_{X_i} v^{N,i}\Big) = 0, \quad (25.6)$$

with terminal data $v^{N,i}(T, \boldsymbol{X}) = g(X_i, z^i_{\boldsymbol{X}})$*, where* $z^i_{\boldsymbol{X}} = \frac{1}{N}\sum_k \xi_{ik} h(X_k)$ *is the aggregate evaluated at the configuration and* $\widehat{\alpha}^{\,j} = \alpha^\star(X_j, z^j_{\boldsymbol{X}}, \nabla_{X_j} v^{N,j})$*. Conversely a classical solution with these feedbacks verifies a Markovian Nash equilibrium.*

The display eq. (25.6) packs a great deal into little space, and it repays being read term by term, in words, before we argue that it is correct. The two auxiliary objects come first. The *Hamiltonian* $H(x, z, p) = \sup_\alpha\{-p \cdot b(x, z, \alpha) - L(x, z, \alpha)\}$ is, exactly as in chapter 6, the result of a pointwise optimization over the control: it asks, for an agent of type x feeling aggregate z and carrying the costate (the value gradient) p, what is the best instantaneous trade-off between steering — the term $-p \cdot b$, the rate at which moving lowers the cost-to-go — and paying the running cost L. The maximizer of that bracket is the *optimal feedback* $\alpha^\star(x, z, p) = \arg\max$, the control an agent in that situation would actually choose. The only departure from chapter 13 is that both now carry the aggregate z as a frozen second argument in place of the empirical measure: structurally nothing has changed, the population label has just been swapped for the network's local readout of the population.

The coupling itself lives entirely in one symbol. The expression $z^i_{\boldsymbol{X}} = \frac{1}{N}\sum_k \xi_{ik} h(X_k)$ is the neighbour aggregate of definition 25.3, but *evaluated at a frozen configuration* $\boldsymbol{X}$ rather than along the random trajectory: it is the deterministic function of the N positions $\boldsymbol{X}$ obtained by summing each vertex k's feature vector $h(X_k)$ against the edge weight ξ_{ik} joining it to i and dividing by N. Reading it as a function of $\boldsymbol{X}$, not of time, is what makes eq. (25.6) an autonomous PDE on configuration space: agent i's coupling at the point $\boldsymbol{X}$ is whatever $z^i_{\boldsymbol{X}}$ says it is there. This is the single place the graph enters — through the weights ξ_{ik} in that sum — and it is why the system is *quenched*: the array ξ and the types $\boldsymbol{x}^N$ are fixed data baked into the coefficients, not variables of the problem.

The differential operator on the left then reads as the generator of the whole configuration acting on agent i's value, exactly as a single-agent HJB but enlarged from $\mathbb{R}^d$ to $(\mathbb{R}^d)^N$. The second-order sum $\sum_{j=1}^N \frac{1}{2}\operatorname{tr}(a(x_j, z^j_{\boldsymbol{X}})\, \mathrm{D}^2_{X_j} v^{N,i})$ runs over *all* N blocks because every state diffuses, controlled or not, each with its own diffusion matrix $a = \sigma\sigma^\top$ evaluated at that vertex's type and aggregate. The first-order sum $\sum_{j\neq i} b(X_j, z^j_{\boldsymbol{X}}, \widehat{\alpha}^j) \cdot \nabla_{X_j} v^{N,i}$ runs only over $j \neq i$ and transports $v^{N,i}$ along the *frozen* equilibrium drifts of the other agents, each agent j moving under its own optimal feedback $\widehat{\alpha}^j = \alpha^\star(X_j, z^j_{\boldsymbol{X}}, \nabla_{X_j} v^{N,j})$. The block $j = i$ is excluded from this sum precisely because agent i's own drift is not frozen but optimized, and that optimization is the last term $H(X_i, z^i_{\boldsymbol{X}}, \nabla_{X_i} v^{N,i})$: agent i's contribution enters not as a fixed transport but inside the Hamiltonian, the signature of the one block that is being controlled rather than watched. The terminal data $v^{N,i}(T, \boldsymbol{X}) = g(X_i, z^i_{\boldsymbol{X}})$ is the agent's own terminal cost, again coupled to the others only through its aggregate.

Proof sketch. The argument is the freeze-and-close argument of proposition 13.3 run on the network, and it is worth tracing the three movements explicitly so that the phrase "verbatim network transcription" is earned rather than asserted; what is genuinely unchanged and what is genuinely new will then stand out cleanly.

Freeze, and a control problem appears. Fix one agent i and commit every other agent $j \neq i$ to its equilibrium feedback $\widehat{\alpha}^j$. By the definition of Nash equilibrium (definition 25.4), agent i's equilibrium control is, against this frozen environment, its *optimal* control: it has no profitable deviation. Hence $v^{N,i}$ is the value function of a single ordinary stochastic control problem, posed on the full configuration space $(\mathbb{R}^d)^N$ — agent i steers only its own block X^i, while each X^j with $j \neq i$ is an uncontrolled diffusion carrying the now-fixed drift $b(X_j, z^j_{\boldsymbol{X}}, \widehat{\alpha}^j)$ and diffusion $\sigma(X_j, z^j_{\boldsymbol{X}})$. Agent i must nonetheless track the entire configuration, because the aggregate $z^i_{\boldsymbol{X}}$ it pays through depends on where its neighbours are. This is identical to the homogeneous reduction; the only thing that has changed is that the frozen coefficients are evaluated at $z^j_{\boldsymbol{X}}$ rather than at the empirical measure μ^N.

Dynamic programming gives an HJB on $(\mathbb{R}^d)^N$. Apply the dynamic programming principle of chapter 6 to this single problem with the enlarged state space. The generator of the configuration splits into the diffusion of all N blocks (the second-order sum), the frozen transport of the $j \neq i$ blocks (the first-order sum), and agent i's own drift, which sits inside the instantaneous minimization $\min_\alpha\{b(X_i, z^i_{\boldsymbol{X}}, \alpha) \cdot \nabla_{X_i} v^{N,i} + L(X_i, z^i_{\boldsymbol{X}}, \alpha)\}$. By the defining relation of the Hamiltonian as a maximized negated bracket, this inner minimum is exactly $-H(X_i, z^i_{\boldsymbol{X}}, \nabla_{X_i} v^{N,i})$ and its minimizer is $\alpha^\star(X_i, z^i_{\boldsymbol{X}}, \nabla_{X_i} v^{N,i})$. Substituting and moving signs reproduces the i-th equation of eq. (25.6). Nondegeneracy (A-Nondeg) is what licenses this step: it supplies the parabolic regularity (appendix C.9) that makes the candidate value $C^{1,2}$ and legitimizes the Itô expansion behind the verification half.

Self-consistency upgrades best response to equilibrium. So far we have only used that agent i optimizes against *some* frozen feedbacks; the equation would hold against any fixed environment. What makes the family $(v^{N,i})_i$ an equilibrium is the closure condition that the minimizer just computed for agent i, namely $\alpha^\star(X_i, z^i_{\boldsymbol{X}}, \nabla_{X_i} v^{N,i})$, coincides with the feedback $\widehat{\alpha}^i$ the other agents froze when they solved their own problems. Each agent's optimal response reproduces the feedback the others assumed; the freezing is a fixed point, not an arbitrary choice. This is why the system is simultaneous: agent i's equation contains $\nabla_{X_j} v^{N,j}$ through $\widehat{\alpha}^j$, so no value can be solved without all the others at once. The converse — that a classical $C^{1,2}$ solution with these feedbacks verifies a Markovian Nash equilibrium — is the verification theorem of chapter 6 applied agent by agent, the Itô expansion of $v^{N,i}$ along any admissible deviation showing optimality via the Hamiltonian's one-sided inequality.

It is now possible to say precisely what is unchanged and what is new, which is the whole point of recording the system. *Unchanged* is the entire architecture: the freeze-one-agent reduction, the dynamic-programming step, the appearance of the same Hamiltonian and the same optimal feedback, the self-consistent closure, and the verification converse — every one of these is the homogeneous argument with the symbol μ^N mechanically replaced by $z^i_{\boldsymbol{X}}$. *New* is only the nature of that replacing symbol. The empirical measure μ^N coupled agent i to the entire population symmetrically; the aggregate $z^i_{\boldsymbol{X}} = \frac{1}{N}\sum_k \xi_{ik} h(X_k)$ couples it only to its graph neighbours, with weights ξ_{ik}, and these weights are *quenched random data* fixed by the realized network. That single change in the coupling object — from a symmetric population functional to a vertex-dependent, edge-weighted, randomly-frozen sum — is the entire content of the network refinement; [34] records the same system for the network case. The next remark unfolds why that change, small as it looks on the page, is what forces two limits rather than one. □

Remark 25.6 (Why the network Nash system is worse than chapter 13)**.** The system eq. (25.6) is, exactly as in remark 13.4, the *complete* answer to the finite game: it contains every Markovian equilibrium, for every N and every realized network. And exactly as there, it is useless as something one could actually solve. Diagnosing why is the specification of the cure, and the network inherits all three homogeneous pathologies and then adds a fourth of its own, which is the one that reshapes the limit object.

The three inherited pathologies are precisely those of remark 13.4. First, *dimension*: each $v^{N,i}$ is a function on $[0,T]\times(\mathbb{R}^d)^N$, so resolving it on a grid costs a number of nodes exponential in Nd, the curse of dimensionality in its purest form, fatal already at moderate N and d. Second, the *N-fold simultaneous coupling*: through the feedbacks $\widehat{\alpha}^j = \alpha^\star(X_j, z^j_{\boldsymbol{X}}, \nabla_{X_j} v^{N,j})$, the equation for $v^{N,i}$ depends on the gradients $\nabla_{X_j} v^{N,j}$ of every other value function, so the N equations cannot be solved one at a time but only all at once — the analytic fingerprint of the self-consistency in the proof. Third, the *intrinsic forward–backward structure*: once a limit is taken the terminal data on the value and the initial data on the population do not disappear but become the two-point boundary coupling of the MFG system, and that is structural, not a defect — it expresses that rational agents anticipate the future while the crowd is pushed by the past, and any honest limit must keep it. None of these three is new; each is repaired (the first two) or respected (the third) by the limit in exactly the manner of chapter 13.

The fourth pathology is the genuinely new one, and it is what makes this chapter more than chapter 13 with subscripts. The coefficients of eq. (25.6) are *quenched random data*: the realized types $\boldsymbol{x}^N$ and the realized edges ξ^N are frozen into every coefficient through the aggregates $z^j_{\boldsymbol{X}}$. In the homogeneous game the coefficients were deterministic and identical for all agents, so a single representative agent could stand for the crowd; here two agents of different types see different wirings, different aggregates, and hence genuinely different equations, and the network itself is, in the sampled model, a random object. This has a consequence even at the level of *stating* the limit. One cannot pass to "the" limiting equation, because there is no single agent and no single coefficient field to converge to; the limit must average over the disorder in the right way and resolve the result by type. The object that results is therefore not one PDE pair but an entire *field* of

HJB–Fokker–Planck systems, one (u^x, m^x) for each continuum type $x \in [0,1]$, coupled across types only through the operator $\mathbb{W}$ — the GMFG of chapter 22. The quenched fourth pathology is exactly why the limit object acquired a continuum of equations rather than collapsing, as in chapter 13, to a single one: heterogeneity that the mean field could average away is here *indexed*, not erased.

This is also why the curing substitution is now *double* rather than single. In chapter 13 the cure was one replacement, the empirical measure μ^N by the deterministic mean field m_t. Here the coupling that must go to the continuum is the quenched neighbour sum $z^i_{\boldsymbol{X}} = \frac{1}{N}\sum_k \xi_{ik} h(X_k)$, and it must be replaced by the GMFG aggregate

$$z^i_{\boldsymbol{X}} \;=\; \frac{1}{N}\sum_k \xi_{ik}\, h(X_k) \quad\rightsquigarrow\quad z_t(x_i) \;=\; \int_0^1 W(x_i,y)\,\langle h, m^y_t\rangle\; \mathrm{d}y \;=\; (\mathbb{W}\bar{h}_t)(x_i),$$

the GMFG aggregation of definition 24.3 (eq. eq. (24.3)) with the first moment there replaced by the observable mean $\bar{h}_t(y) = \langle h, m^y_t\rangle$. The substitution is best read as two replacements performed in sequence — precisely the condition-then-average order fixed in section 25.1.2 — and seeing them as separate is the entire reason the chapter must run two limits. The first replacement is over the *disorder*: swap each random edge ξ_{ik} for its annealed value $\bar{\xi}_{ik} = W(x_i, x_k)$, averaging out the coin flips while holding the types fixed. The second is over the *configuration*: swap the empirical sum $\frac{1}{N}\sum_k W(x_i,x_k)h(X_k)$, a Riemann/Monte-Carlo sum over the sampled types x_k with the neighbours settled into their type-equilibrium laws, for the continuum integral $\int_0^1 W(x_i,y)\langle h, m^y_t\rangle\,\mathrm{d}y$ against the equilibrium field. The first replacement is the *edge-sampling* step, the second the *quadrature*-plus-*chaos* step; together they carry the quenched, fluctuating, per-realization neighbour sum to its annealed, deterministic, continuum value. That neither replacement is free — each costs an error, and the two errors have different sizes and need not be taken in the same way — is the single analytic fact that organizes everything downstream.

Justifying this double substitution *with a rate*, and discovering that its two halves are two genuinely different limits, is the program of sections 25.3 to 25.4. The very fact that the limit object is a field of systems rather than one PDE is what guarantees the two limits cannot be collapsed into one: the configuration limit $N \to \infty$ resolves the field's individual entries (one type's HJB–FP pair at a time, the graph held fixed), while the disorder/graph limit ties those entries together through the convergence of the wiring $W_N \to W$. The next section takes exactly this hand-off, organizing the two replacements into two limits with separate difficulties, rates, and commutation behaviour.

25.3 The two limits and their order

The previous section ended by handing us a precise debt: the curing substitution that carries the finite game to the GMFG is now *double*, the quenched neighbour sum $z^i_{\boldsymbol{X}} = \frac{1}{N}\sum_k \xi_{ik} h(X_k)$ going over to the operator aggregate $\int_0^1 W(x_i,y)\langle h, m^y_t\rangle\,\mathrm{d}y$ in two steps — first average out the edge coins at frozen types, then pass the resulting Riemann sum over the sampled types to its continuum integral — and we owe a *rate* for it. This section is where that debt is reorganized before it is paid. The reorganization is the single conceptual move on which the rest of the chapter rests: the two replacements are not two stages of one limit but two *genuinely different* limits, with different difficulty, different rates, and — near the critical coupling — different verdicts on whether the answer depends on the order in which they are taken. Naming them, drawing the diagram that holds them apart, and stating the quantitative claim that closes the chapter is the whole business here; the three subsections that follow then supply, in turn, the engine, the per-limit rates, and the assembled bound.

It is worth saying at once which two limits these are, because the names will recur on every page. The first is the one already familiar from chapter 13: the *mean field, or sampling, limit* $N \to \infty$. Even with the wiring held absolutely fixed, the finite game is not the GMFG, because each "type" is represented by only finitely many agents and the local population it presents still fluctuates; only as the number of agents per type grows do those fluctuations wash out and propagation of chaos take hold. This limit is about the *count* of agents, and it is graph-agnostic — it would run unchanged on any fixed network. The second limit is new to the network setting: *graph convergence*, the statement that the finite wiring W_N itself approaches the continuum kernel W. This limit is about the *wiring*, not the head-count, and its natural topology is the weak cut metric $\delta_\square$ of chapter 17 rather than any sampling distance on measures. The reason these cannot be folded into one is structural and was already exposed in remark 25.6: the limit object is not a single PDE but a *field* of HJB–Fokker–Planck systems indexed by the continuum type, and the sampling limit resolves the field's individual entries (one type's pair at a time, the graph frozen) while the graph limit is what ties those entries together through the operator $\mathbb{W}$. Two jobs, two limits.

Before drawing the diagram that separates them we fix the standing notation the whole section, and indeed the rest of the chapter, will use without further comment. Throughout, $z_t(x) = (\mathbb{W}\bar{h}_t)(x)$ denotes the GMFG aggregate of definition 24.3, where $\bar{h}_t(y) = \langle h, m_t^y \rangle$ is the equilibrium mean of the interaction observable at type y and $(u^x, m^x)_{x\in[0,1]}$ is the GMFG solution of chapters 22 and 24 for the limiting graphon W. This $z_t(x)$ is the continuum target of the finite per-agent aggregate z_t^i: the object the double substitution is trying to reach, written compactly as the action of the coupling operator $\mathbb{W}$ on the field of observable means. Reading $z_t(x_i)$ in place of z_t^i in the dynamics eq. (25.4)–eq. (25.5) is precisely what turns the finite game into the GMFG, so the entire error of the limit is the discrepancy $z_t^i - z_t(x_i)$, and the two limits are nothing but the two halves of that discrepancy.

The cleanest way to see that there really are two halves, and that they can in principle be taken in either order, is to lay the four objects they connect at the corners of a square. Each side of the square performs *one* of the two operations in isolation; the two routes from the starting corner to the GMFG differ only in the order. The point of drawing it is to make the question "does the order matter?" into a geometric one — whether the square commutes — and to make visible, in the labels on the arrows, that the two operations are bookkept by two different errors.

Heuristic (non-rigorous)

Picture the double limit as a commuting square:

$$\begin{array}{ccc}
\text{finite game on } G(N,W) & \xrightarrow{N\to\infty \text{ (POC, graph frozen)}} & \text{GMFG on the step graphon } W_N \\
\downarrow{\scriptstyle \delta_\square(W_N,W)\to 0} & & \downarrow{\scriptstyle \text{stability}} \\
\text{finite game with } \mathbb{W}\text{-coupling} & \xrightarrow{N\to\infty} & \text{GMFG on } W.
\end{array}$$

The right column is *stability of the GMFG solution under perturbation of the kernel* (chapter 24, and section 25.5); the top arrow is *propagation of chaos with the network held fixed* (the mean field limit of chapters 13 and 23). The diagonal — sampling $G(N, W)$ and letting $N \to \infty$ once — is what actually happens, and the square commuting says the diagonal reaches the same corner. The total error therefore splits into a *mean-field/sampling* part, of size r_N (the Fournier–Guillin rate of lemma 13.5), and a *graph* part, of size $\delta_\square(W_N, W)$ or the operator error $\|\mathbb{W}_N - \mathbb{W}\|_{L^2 \to L^2}$. The chapter's quantitative claim is that these two add, and the slower one wins.

The diagram is worth walking around arrow by arrow, because each side is a theorem this chapter either imports or proves, and naming what each one does is half the work of the convergence proof. Start at the top-left corner, the only fully concrete object on the page: the genuine finite game of section 25.1, played on one realized sample $G(N, W)$ with its quenched types and edges. There are exactly two things one can do to it.

The *top arrow* freezes the realized wiring and lets the agent count grow. "Frozen" is the operative word: we do not let the graph converge to anything: we keep the realized step graphon W_N exactly as sampled, and ask only that $N \to \infty$ so that the population at each type fills in. This is propagation of chaos in the precise sense of chapters 13 and 23 — synchronous coupling, Grönwall, the suppression of empirical fluctuations — carried out verbatim on the fixed network. Its destination is therefore not the GMFG on W but the GMFG *on the step graphon* W_N: the limit honours whatever wiring it was handed. The error this arrow costs is the mean-field/sampling error, the same r_N that governed the homogeneous theory, and it is graph-blind — it does not see how close W_N is to W, only how many agents there are.

The *right column* is the operation that the top arrow refused to perform: it holds the GMFG structure fixed and *moves the kernel*, from the step graphon W_N to the limit W. Both endpoints are continuum objects — fields of HJB–Fokker–Planck systems — so nothing here is a sampling statement at all; this arrow is purely about how the GMFG *solution* responds when its coupling operator is perturbed. That it responds continuously, with a modulus, is exactly the stability of the GMFG under kernel perturbation: the contraction estimate proposition 24.18 of chapter 24, sharpened in section 25.5 into Lipschitz dependence of the equilibrium on the kernel. The error this arrow costs is therefore a *graph* error, measured by how far W_N sits from W in the topology in which the GMFG solution is continuous — the cut distance $\delta_\square(W_N, W)$ when only the unlabelled network is available, or the stronger operator distance $\|\mathbb{W}_N - \mathbb{W}\|$ when the labels are known.

The *left arrow* and *bottom arrow* together form the alternative route, and they are the same two operations performed in the opposite order. The left arrow replaces each realized random edge ξ_{ij} by its annealed value $\bar{\xi}_{ij} = W(x_i, x_j)$ — it averages out the Bernoulli coins of eq. (25.1) while leaving the head-count and the quenched types alone, producing a finite game whose coupling is already the operator $\mathbb{W}$ rather than a realized adjacency array. This is the disorder average, and its cost is again a graph error, $\delta_\square(W_N, W) \to 0$. The bottom arrow then runs propagation of chaos on that $\mathbb{W}$-coupled finite game, $N \to \infty$, landing on the GMFG for W. Reading down-then-across rather than across-then-down has swapped the order: graph first, sampling second.

Now the geometric question has content. Both routes start at the same finite game and both *can* end at "GMFG on W", so one may legitimately ask whether they agree — whether across-then-down equals down-then-across. That is what it *means* for the square to commute. The diagonal is the third path and the only one that is physically real: when one actually samples $G(N, W)$ and lets N grow, one is not free to send the agent count to infinity with the wiring pinned, nor to perfect the wiring with the count pinned, because — and this is the hinge of the entire chapter — *the same parameter N controls both.* On a sampled graph there are not two independent knobs but one: incrementing N simultaneously adds agents (advancing the sampling limit) *and* refines the realized wiring toward W (advancing the graph limit). The two clocks are geared to a single shaft. One therefore never walks the sides of the square in sequence; one travels the diagonal, taking both limits at once. The assertion that the diagonal reaches the same corner as the two side-routes is precisely the assertion that the joint limit agrees with either iterated limit — that the order does not matter and the two limits commute — which is why "the square commutes" and "the joint limit is well posed" are one and the same statement.

This single-parameter coupling is also what fixes how the two errors combine, and the conclusion is the one piece of arithmetic the reader should carry out of this section. Because N drives both mechanisms, the total discrepancy $z_t^i - z_t(x_i)$ inherits *both* a sampling contribution and a graph contribution, each a function of the one N, and — as the joint estimate of section 25.3.3 will make precise — they enter *additively*: the total error is bounded by $C\,(r_N + \gamma_N)$, with r_N the mean-field/sampling rate and γ_N the graph rate. A sum of two decaying terms decays no faster than its slower summand; so the joint rate is governed by whichever of the two mechanisms is the bottleneck. This is the memorable form of the chapter's quantitative claim: *the two errors add, and the slower one wins.* It also tells us in advance where the drama will be. If γ_N is a healthy polynomial $N^{-1/2}$ — which, we will find, requires *knowing the latent labels* — the graph contributes on the same footing as the sampling term and may even be negligible beside it. But if the network is known only up to relabelling, so that the only available graph topology is the cut metric, then γ_N degrades to a glacial $1/\sqrt{\log N}$, slower than every polynomial, and the graph limit alone sets the pace of the whole convergence. The gap between these two regimes — polynomial versus logarithmic — is exactly the value of knowing the types, and it is a recurring refrain from here to the end of the chapter.

Three jobs remain, and they are exactly the three subsections ahead. First, both the right column (stability) and the graph half of the diagonal were just asserted to be "measured by $\delta_\square$", but nothing yet explains *why* the cut distance is the right yardstick for a game whose data are value functionals and costs. The answer is a single estimate — the cut norm controls the action of the coupling operator on bounded observables — and section 25.3.1 proves it, showing that $\delta_\square$ is the right norm precisely because it is the norm in which the aggregate z_t enters the value functionals eq. (25.5). Second, with that engine in hand, section 25.3.2 quantifies the graph error γ_N itself, and in doing so uncovers the cut-versus-operator dichotomy: the two error mechanisms latent in the double substitution — edge sampling and quadrature — converge at genuinely different rates, and the cut metric controls only type-*averaged* quantities while per-agent control demands either known labels or the stronger operator norm. Third, section 25.3.3 assembles the sampling rate r_N and the graph rate γ_N into the additive bound eq. (25.11) promised above, working in the contraction regime where the square is exact. That last qualification is not a throwaway: the commuting square, and with it the very right to speak of "the" joint limit and its rate, is a luxury of the contraction/monotone regime. Outside it — near the critical coupling where the GMFG equilibrium can fail to be unique — the order of limits can change the answer, and self-averaging over the quenched wiring can break down exactly as it does at a glass transition. Remark 25.12 states precisely when the order matters, and the physics note closing this section reads the whole construction as the strategic counterpart of quenched-versus-annealed averaging over disorder — the reading already promised in section 25.1.1, now earned by the machinery in between.

25.3.1 The cut-norm aggregation lemma

The previous section drew the commuting square and, in doing so, made a promise it did not keep: it asserted again and again — on the right column for stability, and on the graph half of the diagonal for joint convergence — that the graph error is "measured by $\delta_\square$," without ever explaining why the cut distance, and not some other distance between kernels, is the right yardstick. That promissory note comes due here. The reason is not aesthetic and not a matter of convenience; it is forced by the way the network actually enters the game. The only channel through which the wiring touches an agent's dynamics and cost is the aggregate $z_t(x) = (\mathbb{W}\bar{h}_t)(x)$ of eq. (25.3), and the cost functional eq. (25.5) does not consume that aggregate raw: it consumes it *integrated against the population*, smeared over types and tested against bounded data. The natural distance on kernels is therefore whichever one measures how much $\mathbb{W}$ can move such a tested, type-averaged readout — and that distance, it turns out, is exactly the cut distance. The single estimate that makes this identification precise, lemma 25.7 below, is the technical engine of the entire chapter: every later rate, every stability statement, every ε-Nash guarantee passes through it. This is the moment the cut metric of chapter 17 stops being imported machinery and earns its place in the GMFG story.

Because that whole story now rests on one norm, we recall it from first principles, in enough words that a reader who skipped chapter 17 can follow without turning back. The *cut norm* of a kernel $U : [0,1]^2 \to \mathbb{R}$ is

$$\|U\|_\square \;=\; \sup_{S,T\subseteq[0,1]} \Big| \int_{S\times T} U(x,y)\;\mathrm{d}x\;\mathrm{d}y \Big|,$$

the largest total mass U can pile up over any rectangle $S \times T$ of types. It is a deliberately *weak* norm: it never looks at U at a point, only at its integral over blocks, so two kernels that disagree violently on a thin sliver but agree on every fat rectangle are close in it. This blockwise, averaging character is precisely why it is the right topology for graph convergence — it is blind to the microscopic relabelling of vertices and sees only macroscopic edge densities, the very quantities the GMFG limit is built to track. For the proof we need it in an equivalent, more flexible guise. Testing U against *indicators* $\mathbf{1}_S, \mathbf{1}_T$ is the same, up to a universal constant, as testing it against arbitrary functions valued in $[-1,1]$; this is the cut-norm/$\infty\to 1$ comparison,

$$\|U\|_\square \;\le\; \|U\|_{\infty\to 1} \;:=\; \sup_{f,g:[0,1]\to[-1,1]} \Big| \int_{[0,1]^2} U(x,y)f(x)g(y)\,\mathrm{d}x\;\mathrm{d}y \Big| \;\le\; 4\;\|U\|_\square\,, \tag{25.7}$$

proved in chapter 17 (it is the standard comparison of [99, §8], [72]). The lower bound is immediate — indicators are themselves admissible f, g, with $[0,1]$ rescaled into $[-1,1]$ — and the upper bound, the only nontrivial direction, comes from writing a general $[-1,1]$-valued f as an average of $\pm$ indicators and the same for g, so the bilinear form is a convex combination of rectangle masses and cannot exceed the largest of them by more than the factor 4. The single fact to carry forward is that bounding $\|U\|_\square$ bounds the bilinear pairing $\int Ufg$ for *every* pair of bounded test functions, with only a harmless factor of 4 lost. That pairing is exactly the shape in which the aggregate enters the cost, so the bridge from "the kernels are cut-close" to "the games are close" is built into the norm itself.

Lemma 25.7 (Cut norm controls the aggregate)**.** *Let W, W' be graphons and let $g, g' : [0,1] \to \mathbb{R}$ be measurable with $\|g\|_\infty, \|g'\|_\infty \le M$. For $f : [0,1] \to [-1,1]$,*

$$\Big| \int_0^1 f(x)\,[(\mathbb{W}g)(x) - (\mathbb{W}'g')(x)]\,\mathrm{d}x \Big| \;\le\; 4M\,\|W - W'\|_\square \;+\; \|W'\|_1\,\|g - g'\|_\infty\,, \tag{25.8}$$

where $\|W'\|_1 = \int_{[0,1]^2} |W'| \le 1$. In particular, for $g = g'$ bounded by M, $\left|\int f\,[(\mathbb{W} - \mathbb{W}')g]\right| \le 4M\,\|W - W'\|_\square$ for every $f : [0,1] \to [-1,1]$.

Before the proof, it is worth reading the inequality eq. (25.8) for what it says, because its two-term shape is not incidental: it is the lemma's whole message. The quantity on the left is a single tested, type-averaged readout of the difference between two aggregates — the aggregate built from kernel W and observable g, against the one built from W' and g' — and these are exactly the two ingredients that can drift apart when we replace the true continuum coupling by a finite-network proxy: the *kernel* can be wrong (we used the sampled $\mathbb{W}_N$ instead of $\mathbb{W}$) and the *observable* can be wrong (the neighbours' empirical feature law has not yet relaxed to the equilibrium $\bar{h}_t$). The right-hand side charges separately for each: a cut-distance term for the kernel error, scaled by the observable's amplitude M, and a sup-distance term for the observable error, scaled by the kernel's L^1 mass. This is precisely the clean separation the joint estimate of section 25.3.3 needs — one term it will feed the graph rate γ_N, the other the mean-field/sampling rate r_N — and it is why the lemma is stated for two kernels and two observables at once rather than only for the difference $\mathbb{W} - \mathbb{W}'$.

Proof. The argument is short, and its only idea is to separate the two ways the aggregates can differ so that each meets the tool built for it. Add and subtract the mixed term $\mathbb{W}'g$ to split the difference of aggregates into a *kernel part*, where the observable is held fixed and the kernel changes, and an *observable part*, where the kernel is held fixed and the observable changes:

$$(\mathbb{W}g) - (\mathbb{W}'g') = \underbrace{(\mathbb{W} - \mathbb{W}')\,g}_{\text{kernel}} + \underbrace{\mathbb{W}'(g - g')}_{\text{observable}}.$$

Integrating against f and using the triangle inequality, it suffices to bound the two pieces separately.

For the kernel part, unfold the operator into its integral kernel and read the result as a bilinear form in the kernel difference:

$$\int_0^1 f\,[(\mathbb{W} - \mathbb{W}')g] = \int_{[0,1]^2} (W - W')(x,y)\,f(x)\,g(y)\,\mathrm{d}x\,\mathrm{d}y.$$

This is the pairing eq. (25.7) controls, but only for test functions valued in $[-1,1]$; f already qualifies, while g is bounded by M, not 1. We therefore normalize, applying eq. (25.7) to $U = W - W'$ with the admissible pair $(f,\, g/M)$ — both now valued in $[-1,1]$ — and pulling the scalar M back out:

$$\Big| \int_{[0,1]^2} (W - W')\,f\,g \Big| = M \Big| \int_{[0,1]^2} (W - W')\,f\,(g/M) \Big| \le M\,\|W - W'\|_{\infty\to 1} \le 4M\,\|W - W'\|_\square\,.$$

The factor 4 is the price of eq. (25.7); the factor M is the amplitude of the observable, and it is the reason boundedness of h was demanded back in section 25.1.2, for an unbounded observable would carry no such normalization and the cut norm would say nothing about its aggregate.

For the observable part the kernel does not move, so no cut-norm subtlety arises and a crude pointwise bound is enough. Since $|f| \le 1$ and $|(g-g')(y)| \le \|g-g'\|_\infty$ uniformly,

$$\Big|\int_0^1 f\,\mathbb{W}'(g-g')\Big| \le \int_{[0,1]^2} |W'(x,y)|\,|g-g'|(y)\;\mathrm{d}x\;\mathrm{d}y \le \|W'\|_1\;\|g-g'\|_\infty\,.$$

Here the whole observable error is taken in sup norm and the kernel contributes only its total L^1 mass $\|W'\|_1 = \int_{[0,1]^2} |W'| \le 1$ — the dense-graphon bound, since W' takes values in $[0,1]$. Adding the two displays gives eq. (25.8). The final clause is the special case $g = g'$, for which the observable part vanishes identically and only the cut term survives. □

The lemma earns its title: any tested, type-averaged, bounded functional of the aggregate — and the value functionals eq. (25.5), read at equilibrium, are exactly of this kind — is moved by no more than the cut distance of the kernels (times the observable's amplitude) plus the sup-distance of the observables (times the kernel's mass). When two networks are cut-close and their equilibrium observables agree, every welfare-type readout the game produces is close. That is the positive content, and it is everything the cut-rate branch downstream will need.

But the lemma also tells us, with equal force, what it cannot do, and this negative half is just as important to internalize because it is invoked repeatedly later as though already understood. The estimate controls $\int_0^1 f(x)\,[\cdots]\;\mathrm{d}x$, an *integral over types*, and nothing more. It says nothing about the value $[\cdots](x)$ at a single fixed type x. One is tempted to read this as a weakness of the proof — surely a sharper argument would deliver the pointwise statement — but it is not. It is a true feature of the cut metric, forced by the very averaging that makes the cut norm the right topology for graph convergence, and the cleanest way to see that it is unavoidable is to exhibit two kernels that are cut-close yet have a wildly different aggregate at one type.

Take the constant graphon $W \equiv \frac{1}{2}$ — every pair of types equally and moderately linked — and build W' from it by severing a single vertex. Concretely, fix one type cell $I_k = [\frac{k-1}{N}, \frac{k}{N})$ and set $W' = \frac{1}{2}$ everywhere except on the cross-shaped strip $\{x \in I_k\} \cup \{y \in I_k\}$, where $W' = 0$: the vertex sitting at I_k has been disconnected from the network, while every other edge is left untouched. The two kernels differ only on that strip, a set of measure $\frac{2}{N} - \frac{1}{N^2}$, and there by at most $\frac{1}{2}$, so $\|W - W'\|_\square \le \|W - W'\|_1 \le \frac{1}{N}$: in cut distance the two networks are indistinguishable as $N \to \infty$, and lemma 25.7 duly certifies that their aggregates agree on every type-average. Indeed, take the simplest observable $g \equiv 1$ ($M = 1$). The type-averaged readout is

$$\int_0^1 |(\mathbb{W}-\mathbb{W}')g|(x)\;\mathrm{d}x \;=\; \underbrace{\Big(1-\tfrac{1}{N}\Big)\cdot\tfrac{1}{2N}}_{x\notin I_k} \;+\; \underbrace{\tfrac{1}{N}\cdot\tfrac{1}{2}}_{x\in I_k} \;=\; O\Big(\tfrac{1}{N}\Big),$$

comfortably inside the lemma's bound $4M\,\|W - W'\|_\square \le 4/N$. Yet at the severed type itself the two aggregates could not be further apart: for $x \in I_k$, $(\mathbb{W}g)(x) = \int_0^1 \frac{1}{2}\;\mathrm{d}y = \frac{1}{2}$ while $(\mathbb{W}'g)(x) = 0$, a pointwise gap of $\frac{1}{2} = O(1)$ that does not shrink with N at all. The type-average is tiny because the one deviant type carries weight only $\frac{1}{N}$ in the integral and is drowned out; the pointwise value at that type is order one because nothing is averaging it against its well-behaved neighbours. Two cut-close kernels can thus agree on every type-integral and still disagree completely in a single row — and no estimate built on the cut norm can ever rule this out, because the cut norm cannot see a single row in the first place.

This micro-instance is the seed of a tension that organizes the rest of the chapter, so it is worth naming what it forces. The cut-norm engine delivers control of *type-averaged* equilibrium quantities only — aggregate welfare, mean effort, the population-integrated cost — and this is exactly the control that flows forward: to the cut-rate item of section 25.3.2, to the unlabelled branch of the joint-convergence theorem in section 25.3.3, to the type-averaged ε-Nash guarantee, and to the cut-stability theorem of section 25.5, the statement that macroscopically equivalent networks produce close equilibrium aggregates. What it cannot deliver — pointwise-in-type, per-agent control — is not a gap the cut metric will ever close, and the single-severed-row example shows why in one picture. Closing that gap requires strictly more information about the network than its cut-equivalence class contains: either the latent labels themselves, which buy a per-vertex bound at the polynomial rate $N^{-1/2}$, or the stronger operator-norm control of chapter 18, which dominates the cut norm and does see individual rows. The dichotomy is therefore not a temporary inconvenience but the permanent shape of the theory, and section 25.5 resolves it explicitly. For now we have the engine; the immediately next task is to point it at the one kernel difference the chapter most cares about. Section 25.3.2 applies lemma 25.7 to the sampled operator $\mathbb{W}_N$, separating the edge-sampling and quadrature mechanisms hidden in $W_N - W$ and contrasting the pointwise control the known labels afford against the weaker cut control that is all an unlabelled network allows.

25.3.2 The graph error of the sampled network

The cut-norm aggregation lemma of section 25.3.1 is a general instrument: it bounds any tested, type-averaged readout of a difference of aggregates by the cut distance of the kernels (times the observable's amplitude) plus the sup-distance of the observables (times the kernel's mass). The previous subsection built that instrument and then, in the single-severed-row example, drew its boundary — it controls type-integrals and is constitutionally blind to any single row, a limitation that no sharper argument on the cut norm can repair and that, as flagged there, will eventually have to be paid for either with the labels or with the stronger operator norm. We now point the instrument at the one kernel difference the chapter actually cares about, $W_N - W$: the gap between the sampled empirical step graphon of definition 25.2 and the kernel it was drawn from. This is the right column of the commuting square of section 25.3 in concrete form — the graph held against its own limit, the mean-field limit set aside — and the question it poses is exactly: how far is $\mathbb{W}_N$ from $\mathbb{W}$, and in *what sense*?

That last qualifier is the whole point, because the answer is not one number but two, attached to two different notions of "how far," and it is the single most important quantitative fact of the chapter. Recall from section 25.1.1 that the discrepancy $W_N - W$ is assembled from two independent mechanisms living in two different places. The *edge-sampling* part is the mean-zero fluctuation $\xi_{ij} - W(x_i, x_j)$ of the Bernoulli coins, present only because the realized wiring is a noisy one-bit readout of the smooth kernel value. The *quadrature* part is the discrepancy between the N frozen type positions $\boldsymbol{x}^N$ and the continuum label they are meant to sample, present even if the coins were exact. Pointing the cut-norm engine at $W_N - W$ forces us to track both — but, more consequentially, it forces us to decide *what we are allowed to know about the wiring*, and that single modelling choice splits the answer in two. If the analyst holds the latent types $\boldsymbol{x}^N$ in hand — the latent-position model, where the positions were measured or estimated — the natural object is the error at one *named* vertex, the pointwise aggregate error e_N^i, and we shall see it is controlled at the polynomial rate $N^{-1/2}$. If instead the network is delivered only up to relabelling — an unlabelled graph, with no access to the positions — then the only object the cut metric can even speak about is the type-average, and it is controlled at the far slower rate $1/\sqrt{\log N}$. Two genuinely different error notions, controlled at two genuinely different rates: this is the cut-versus-operator dichotomy of section 25.5 in its sharpest, most quantitative form, and the proposition records both halves side by side.

Proposition 25.8 (Graph error of $G(N,W)$)**.** *Let W satisfy* (A-Reg-W) *and let $G(N,W)$ be sampled as in definition 25.1, with empirical step graphon W_N (definition 25.2). Fix bounded $g : [0,1] \to \mathbb{R}$, $\|g\|_\infty \le M$. There are two distinct error quantities, controlled at two genuinely different rates — this distinction, between a single vertex's aggregate and a type-average, is the quantitative heart of the chapter and the source of the cut-versus-operator dichotomy of section 25.5.*

1. (Known labels: per-agent pointwise error.) *If the types $\boldsymbol{x}^N$ are known (the latent-position model), the* pointwise *aggregate error at vertex i, $e_N^i := [(\mathbb{W}_N - \mathbb{W})g](x_i) = \frac{1}{N}\sum_j(\xi_{ij} - W(x_i,x_j))g_j + q_N^i$, satisfies, uniformly in i,*

$$\max_{i\le N} \mathbb{E}\big[(e_N^i)^2 \,|\, \boldsymbol{x}^N\big]^{1/2} \;\le\; \underbrace{C\,M\,N^{-1/2}}_{\textit{edge sampling}} + \underbrace{q_N}_{\textit{quadrature}}\;, \tag{25.9}$$

where $q_N^i = \frac{1}{N}\sum_j W(x_i,x_j)g_j - \int_0^1 W(x_i,y)g(y)\,\mathrm{d}y$ is the one-dimensional quadrature error of the row $W(x_i,\cdot)g$ and $q_N = \sup_i |q_N^i|$; under (A-Reg-W) *with W, g Lipschitz-continuous in type, $q_N = O(N^{-1/2})$ for i.i.d. labels and $q_N = O(N^{-1})$ for the regular grid.*

2. (Unknown labels: type-averaged tested error.) *If the network is known only up to relabelling — so that W_N and W must be compared in $\delta_\square$ — then for the* type-averaged *tested error $E_N := \int_0^1 f(x)[(\mathbb{W}_N - \mathbb{W})g](x)\,\mathrm{d}x$ with $f : [0,1] \to [-1,1]$, the second sampling lemma of chapter 19 (remark 19.5, eq. eq.* (19.2)*; proved in full in theorem F.9) gives*

$$\delta_\square(W_N, W) \;\le\; \frac{C}{\sqrt{\log N}} \quad \textit{with high probability,} \tag{25.10}$$

improving to a polynomial rate $\delta_\square(W_N,W) = O(N^{-c})$ when W is of bounded "complexity" (a step graphon, or Hölder with a weak-regularity modulus, corollary 19.9). Combined with lemma 25.7, $|E_N| \le 4M\,\delta_\square(W_N,W)$ for every $f : [0,1] \to [-1,1]$.

The constants depend on M, the Lipschitz data of W, and the moments of h under the equilibrium law, not on N. Note the gap: the per-agent error (1) *is $O(N^{-1/2})$ but is available* only with known labels*, while the unlabelled comparison* (2) *controls only the type-average and at the far slower $1/\sqrt{\log N}$ rate.*

The two items are the two halves of the same modelling decision, and it helps to read them as a single sentence before proving either. With the labels known we may ask the strong, local question — what is the error in the aggregate felt by *this* vertex i? — and item (1) answers it at the healthy polynomial rate $N^{-1/2}$, the price being that we must actually possess $\boldsymbol{x}^N$. Without the labels we may ask only the weak, global question — what is the error in a population-integrated readout? — and item (2) answers *that* through the cut metric, at the logarithmic rate $1/\sqrt{\log N}$. The proof of the first item is a clean decomposition into the two mechanisms named above and a one-line second-moment estimate of each; the proof of the second is a citation to the sampling theory of chapter 19 composed with the aggregation lemma. What deserves the most care is neither computation but the *asymmetry* between them: why the strong question has no unlabelled answer at all, no matter how slow a rate we would settle for.

Proof sketch. (1). Start from the empirical step graphon. On the regular grid the row of $\mathbb{W}_N$ at vertex i is the cell average $(\mathbb{W}_N g)(x_i) = \frac{1}{N}\sum_j \xi_{ij} g_j$, with g_j the average of g over cell I_j, because W_N is piecewise constant ξ_{ij} on the pixel $I_i \times I_j$ by eq. (25.2). Subtracting the continuum aggregate $(\mathbb{W}g)(x_i) = \int_0^1 W(x_i, y)g(y)\,\mathrm{d}y$ and inserting the intermediate sum $\frac{1}{N}\sum_j W(x_i, x_j)g_j$ splits the error into exactly the two mechanisms anticipated above,

$$e_N^i = \underbrace{\frac{1}{N}\sum_j (\xi_{ij} - W(x_i, x_j))g_j}_{\text{edge part: random coins}} + \underbrace{q_N^i}_{\text{quadrature: discrete vs. continuum}}, \qquad q_N^i = \frac{1}{N}\sum_j W(x_i, x_j)g_j - \int_0^1 W(x_i, y)g(y)\,\mathrm{d}y.$$

The first piece moves the realized edges to their conditional means and is pure edge-sampling noise; the second piece, q_N^i, holds the kernel exact and charges only for replacing an integral by a finite sum over the type positions. We bound each in turn, conditionally on $\boldsymbol{x}^N$ throughout — this is the condition-first-average-second discipline of section 25.1.2, and it is what keeps the two mechanisms from being silently fused, since the edge part is random given $\boldsymbol{x}^N$ while the quadrature part is deterministic given $\boldsymbol{x}^N$.

For the *edge part*, write $Y_j := \xi_{ij} - W(x_i, x_j)$. Conditionally on $\boldsymbol{x}^N$, the sampling model eq. (25.1) makes the coins $\{\xi_{ij}\}_{j\neq i}$ independent Bernoullis with means $W(x_i, x_j)$, so the $\{Y_j\}_{j\neq i}$ are independent, mean zero, and bounded in $[-1, 1]$. The edge part $S := \frac{1}{N}\sum_{j\neq i} Y_j g_j$ is thus a sum of $N-1$ independent mean-zero terms, each of conditional weight $|g_j|/N \leq M/N$. Its conditional mean is zero, and — this is the only step that uses independence — the variance of a sum of independent variables is the sum of their variances, with no cross terms, so

$$\mathbb{E}[S^2 \mid \boldsymbol{x}^N] = \mathrm{Var}\left(S \mid \boldsymbol{x}^N\right) = \frac{1}{N^2}\sum_{j\neq i} g_j^2 \;\mathrm{Var}\left(Y_j \mid \boldsymbol{x}^N\right) = \frac{1}{N^2}\sum_{j\neq i} g_j^2\, W(x_i, x_j)(1 - W(x_i, x_j)).$$

Each Bernoulli variance is at most $\frac{1}{4}$ and each $g_j^2 \leq M^2$, so the sum of $N-1$ terms is at most $(N-1)\,M^2/4$, whence $\mathbb{E}[S^2 \mid \boldsymbol{x}^N] \leq M^2/(4N)$ and the conditional root-mean-square edge error is at most $\frac{1}{2}M\,N^{-1/2}$. This is the edge term $CMN^{-1/2}$ of eq. (25.9) with $C = \frac{1}{2}$ (the crude bound $\mathrm{Var}(Y_j) \leq 1$ would give $C = 1$; either suffices). The estimate is the elementary $\sqrt{N}$ law of large numbers made quantitative: each of the $\Theta(N)$ neighbours contributes an $O(1/N)$ term, the N variances add to $O(1/N)$, and the standard deviation is its square root, $O(N^{-1/2})$. Crucially, every constant used — the count $N-1$, the amplitude M, the Bernoulli variance $\frac{1}{4}$ — is the same for every vertex and does not depend on *which* i we chose, so the bound holds uniformly and the maximum over $i \le N$ inherits it verbatim. Nothing here depends on the state dimension d: the summands are scalar features bounded by M/N, which is why the edge rate is $N^{-1/2}$ regardless of d, a fact we will lean on heavily in remark 25.9.

For the *quadrature part* q_N^i there are no coins to average — it is fixed once $\boldsymbol{x}^N$ is fixed — and it is precisely the one-dimensional error of approximating $\int_0^1 \Phi$ by $\frac{1}{N}\sum_j \Phi(x_j)$ for the row integrand $\Phi(y) := W(x_i, y)g(y)$, which under (A-Reg-W) with g Lipschitz is Lipschitz-continuous in y, uniformly in the row index x_i. Its size depends entirely on *how the types were placed*, the very trade-off flagged in section 25.1.1. For i.i.d. labels $x_j \sim \mathrm{Unif}\,[0,1]$, the sum $\frac{1}{N}\sum_j \Phi(x_j)$ is an average of N i.i.d. bounded random variables with mean $\int_0^1 \Phi$; its variance is $\frac{1}{N}\operatorname{Var}(\Phi(x_1)) = O(1/N)$, so by the elementary central limit theorem its fluctuation about the integral is $O(N^{-1/2})$ — the Monte-Carlo rate. For the regular grid $x_j = (j - \frac{1}{2})/N$ the same sum is the *midpoint quadrature rule*: it samples Φ at the centre of each cell I_j of width $1/N$, and on each such cell the centred evaluation cancels the first-order term, so a single cell's error is $O(N^{-2})$ for a Lipschitz integrand (as the Lipschitz hypothesis supplies). Summing N cells gives the deterministic $O(N^{-1})$, twice as fast as the random placement. This is the same grid-versus-i.i.d. contrast already met for $\int_0^1 g$ in section 25.1.1, now applied row by row, and it is uniform in i because the Lipschitz modulus of Φ is uniform in the row. Taking $q_N = \sup_i |q_N^i|$ and adding the two contributions gives eq. (25.9). We have used only the conditional second moment; if one wants the bound to hold not merely in L^2 but with high probability, the bounded-differences inequality lemma 19.3 of chapter 19 applies — changing one edge coin moves e_N^i by at most M/N, so McDiarmid gives Gaussian tails of width $N^{-1/2}$ — but the joint-convergence and ε-Nash theorems downstream consume only the L^2 statement, so we keep it.

(2). The cut rate eq. (25.10) is not proved afresh here; it *is* the second sampling lemma of chapter 19 (remark 19.5, eq. eq. (19.2)), proved in full at theorem F.9, which states exactly that the empirical step graphon of a W-random graph converges to W in cut distance. The $1/\sqrt{\log N}$ appearing there is the worst-case weak-regularity rate: it traces to the weak regularity lemma of chapter 17, which approximates an *arbitrary* graphon by a step graphon on k parts with cut error $O(1/\sqrt{\log k})$ and no better in general, and a sample of N vertices can resolve only of order $\log N$ such parts, leaving the $1/\sqrt{\log N}$ residue. This is genuinely slow — slower than any power of N — and it is why an *unlabelled* graph limit is the chapter's bottleneck. When W is of bounded complexity, however, the worst case does not bind: a step graphon, or a Hölder kernel whose weak-regularity modulus is polynomial (corollary 19.9), is already well approximated by a fixed or slowly growing number of parts, and the rate improves to a polynomial $O(N^{-c})$. Either way, composing the cut bound with the final clause of lemma 25.7 taken at $g = g'$ turns it into a bound on the type-averaged tested error, $|E_N| \le 4M\,\delta_\square(W_N, W)$, valid for every $f : [0,1] \to [-1,1]$ at once.

It remains to make vivid the asymmetry promised above: *why no analogue of eq.* (25.10) *can ever control the per-agent row error* e_N^i, so that the labels in item (1) are not a convenience but a strict necessity. The reason is the very fact that made the cut norm the right topology in section 25.3.1, read in reverse. The cut rate bounds $\delta_\square(W_N, W)$, which by the aggregation lemma bounds every type-*averaged* readout; but the single-severed-row construction of section 25.3.1 exhibited two graphons W, W' at cut distance $O(1/N)$ — as close as one likes — whose aggregates nevertheless differ by $O(1)$ at one named type, the disconnected vertex. Transport that construction here: a hypothetical theorem bounding $\max_i |e_N^i|$ by some function of $\delta_\square(W_N, W)$ would have to certify $|e_N^i|$ small whenever the cut distance is small, yet the severed-row example produces cut-distance $O(1/N)$ with a row error pinned at $O(1)$. Any such "cut rate for a row" would therefore have to be $O(1)$ — no decay at all, hence no theorem. The obstruction is not that our argument is too weak; it is that the cut metric, being an infimum over relabellings, has *erased* the identity of vertex i and so cannot make any claim attached to it. Recovering a per-vertex statement demands strictly more information than the cut-equivalence class contains: either the labels $\boldsymbol{x}^N$ themselves, which (1) converts into the $N^{-1/2}$ rate, or the operator-norm control of chapter 18, which dominates the cut norm and does see individual rows. This is exactly the unresolved tension section 25.5 will discharge. □

Remark 25.9 (The graph limit is the bottleneck)**.** This remark is the chapter's quantitative thesis, so it is worth stating with care. proposition 25.8 is the precise form of the heuristic that "the slower limit wins," but to know *which* limit is slower one must hold the magnitude of the mean-field rate r_N up against the graph rate, and the comparison has a twist that the homogeneous theory never sees: r_N is dimension-dependent while the graph edge error is not. By the propagation-of-chaos estimate lemma 13.5 (eq. eq. (13.12)) the Fournier–Guillin sampling rate degrades with the state dimension d — $r_N \le CN^{-1/4}$ for $d \le 3$, worsening to $r_N \le CN^{-1/d}$ in high dimension $d > 4$ — because empirically approximating a measure on $\mathbb{R}^d$ in Wasserstein distance becomes harder as d grows. The graph edge error of proposition 25.8(1), by contrast, is $O(N^{-1/2})$ *regardless of* d: as the variance computation above made explicit, it aggregates scalar features bounded by M/N, an object the curse of dimensionality never touches. The two rates therefore cross as d varies, and three regimes result, which it is the business of this remark to separate.

Known labels, low dimension ($d \le 3$). Here the analyst possesses the latent positions, so the per-agent rate (1) is in force, and both graph mechanisms are polynomial: edge sampling at $N^{-1/2}$ and quadrature at $N^{-1/2}$ (i.i.d. placement) or N^{-1} (grid). The mean-field rate $r_N \le CN^{-1/4}$ is polynomial too, but it is already a full power of N slower than the graph error — $N^{-1/4}$ against $N^{-1/2}$ — so even in low dimension the mean field, not the graph, is the slower limit, and it alone sets the joint rate at $O(N^{-1/4})$, paced by propagation of chaos. The regime is nonetheless benign: every mechanism converges polynomially, and the joint rate is simply whichever exponent is smallest, here the mean field's $N^{-1/4}$.

Known labels, higher dimension ($d \ge 3$). The graph edge error stays pinned at $N^{-1/2}$ while r_N continues to slow with d (toward $N^{-1/d}$), so the gap only widens: the *mean field*, already the slower limit in low dimension, falls ever further behind the graph error as d grows. The lesson is that knowing the positions buys a fast graph rate but does nothing for the sampling of the high-dimensional state space; in every dimension it is the propagation of chaos, the same mechanism that governed the graphless theory of chapter 13, that paces the convergence, and increasingly so past $d \approx 4$, where r_N leaves the $N^{-1/4}$ plateau.

Unknown labels (any dimension). Now only (2) is available, and the cut-distance rate $1/\sqrt{\log N}$ is slower than *every* polynomial, hence slower than both r_N and the known-label graph error, for every d. The graph convergence becomes the sole, dominant bottleneck, and the logarithm swamps all the polynomial machinery upstream of it. This is the structural novelty absent from chapter 13:

with no graph there, the only rate was r_N and it was always polynomial; the network can introduce a genuinely *slower* limit, one whose slowness is not a matter of a worse exponent but of leaving the polynomial scale entirely.

The three regimes localize the chapter's headline number. The *value of knowing the types* is precisely the gap between the polynomial graph rate $N^{-1/2}$ of (1) and the logarithmic unlabelled rate $1/\sqrt{\log N}$ of (2) — and it is essential to read this as a *pure graph-error gap*, taken with the mean-field part r_N held *fixed and identical* on both sides. The labels do not improve, and could not improve, the propagation of chaos: r_N is the same whether or not one knows $\boldsymbol{x}^N$. What the labels change is only the graph column of the error, and there they convert a logarithmic rate into a polynomial one. The difference between a network one can resolve down to individual vertices and a network one knows only macroscopically is thus, quantitatively, exactly the difference between $N^{-1/2}$ and $1/\sqrt{\log N}$: a polynomial against a logarithm, the single most consequential gap the chapter measures.

The two error rates this subsection has isolated are exactly the two raw materials the rest of the chapter assembles. We bundle the graph column into a single symbol γ_N — the graph rate — equal to $O(N^{-1/2})$ when the labels are known (the edge-and-quadrature sum of (1)) and to $O(1/\sqrt{\log N})$ when the network is unlabelled (the cut rate of (2)), and hand it forward. Section 25.3.3 adds γ_N to the mean-field/sampling rate r_N under the chapter 24 contraction hypothesis that makes the GMFG best-response map a contraction and the commuting square exact, producing the joint bound $C(r_N + \gamma_N)$; the same γ_N then propagates into the network ε-Nash rate of section 25.4. The dichotomy itself — the gap between the polynomial labelled rate and the logarithmic unlabelled one — passes to remark 25.14, where it is read as the irreducible price of the network's extra structure, and to section 25.5, which banks the unlabelled, cut-controlled half as robustness to the part of the wiring no relabelling can recover. With both errors named and rated, we are ready to combine them.

25.3.3 Joint convergence in the contraction regime

The previous two subsections delivered the two raw materials this one assembles. Section 25.3.1 built the engine — the cut-norm aggregation lemma lemma 25.7, which converts closeness of kernels into closeness of the tested aggregates that drive the costs — and section 25.3.2 spent that engine on the one kernel difference the chapter actually cares about, $W_N - W$, extracting from it the graph rate γ_N: a healthy $\gamma_N = O(N^{-1/2})$ when the latent labels are known, and the glacial $\gamma_N = O(1/\sqrt{\log N})$ when the network is delivered only up to relabelling. The mean-field/sampling rate r_N of lemma 13.5 has been on hand since chapter 13. The commuting-square heuristic of section 25.3 promised that these two errors simply *add*, with the slower one setting the pace; the business of this subsection is to honour that promise as a theorem about the actual finite trajectories, with the synchronous-coupling propagation-of-chaos argument of lemma 13.5 doing the combining.

That promise is not free: it holds only where the commuting square is genuinely exact, and that is the *contraction regime* of chapter 24. It is worth recalling exactly what that regime is and why it is precisely the hypothesis the square needs, because the whole right column of the diagram rests on it. Under (A-Coupling), the operator-norm smallness $\|\mathbb{W}\| < \kappa$ makes the GMFG best-response map — the map that sends a guessed field of aggregates to the field the agents' optimal responses actually generate — a contraction in the field argument $\mathbf{m}$ at fixed kernel (proposition 24.18). A contraction has a *unique* fixed point, and that fixed point moves Lipschitz-continuously with the data of the problem. The datum we are about to perturb is the *kernel* itself: the right column of the square is nothing but the statement that the GMFG solution depends Lipschitz-continuously on the coupling operator $\mathbb{W}$, with a finite modulus that section 25.5 will pin down. Outside the contraction regime that modulus can blow up, the fixed point can split into several branches, and the right column loses its very meaning — there is no longer a single solution to track as the kernel moves. This is why the square is exact *here* and only here, and why every rate in this subsection is, as the chapter's refrain has it, a luxury of the contraction regime. Remark 25.12 returns to precisely what fails as $\|\mathbb{W}\| \uparrow \kappa$; for now we stand inside the regime where the GMFG solution is single-valued and Lipschitz in the kernel, and harvest the clean additive bound it makes available.

Theorem 25.10 (Joint convergence with rate)**.** *Assume* (A-Lip), (A-Growth), (A-Nondeg)*,* (A-Conv)*,* (A-Reg-W)*, and* (A-Coupling) *in the contraction regime of chapter 24, so that the GMFG system for W has a unique solution $(u^x, m^x)_x$ with feedback $\widehat{\alpha}(t,x,X) = \alpha^\star(X, z_t(x), \nabla u^x(t,X))$ bounded and Lipschitz in X, uniformly in (t,x). Let $(X^{i,N})_{i\le N}$ be the closed-loop states of the finite game on $G(N,W)$* under the GMFG decentralized profile *$\alpha_t^i = \widehat{\alpha}(t, x_i, X_t^i)$, and let $(\bar{X}^{x_i})_{i\le N}$ be independent copies of the graphon McKean–Vlasov solution of chapter 23 for the limiting W, driven by the same (B^i, X_0^i). Then there is C, independent of N, such that*

$$\max_{i\le N} \mathbb{E}\Big[\sup_{t\le T} \big|X_t^{i,N} - \bar{X}_t^{x_i}\big|^2\Big]^{1/2} \;\le\; C\big(r_N + \gamma_N\big), \tag{25.11}$$

where r_N is the Fournier–Guillin sampling rate of lemma 13.5 and γ_N is the graph error of proposition 25.8: $\gamma_N = O(N^{-1/2})$ for known labels under (A-Reg-W)*, and $\gamma_N = O(1/\sqrt{\log N})$ for unlabelled networks. Consequently the type-resolved empirical field converges in the precise sense of corollary 25.11 below.*

The theorem is the network upgrade of the homogeneous propagation-of-chaos bound lemma 13.5, and the proof reuses that argument's spine wholesale, so the honest thing is to flag the single new joint where the graph enters and lavish the attention there. The spine is *synchronous coupling*: we run the true finite state $X^{i,N}$ and its GMFG-limit surrogate $\bar{X}^{x_i}$ on the *same* Brownian motion B^i and the *same* initial draw X_0^i, so that the two trajectories differ *only* through the one channel we care about — the aggregate each feels — and not through any spurious mismatch of noise or starting point. The squared difference then obeys a differential inequality whose forcing term is exactly the aggregate discrepancy $z_t^i - z_t(x_i)$, and the entire content of the proof is to decompose that discrepancy, recognise that one of its pieces is itself the average of the squared state errors (which closes a Grönwall loop), and bound the two remaining pieces by the two rates already in hand. We carry this out now.

Proof sketch. The synchronous-coupling Itô estimate. Couple $X^{i,N}$ to $\bar{X}^{x_i}$ through the same noise B^i and the same initial state $X_0^i \sim m_0^{x_i}$, and write $\Delta_t^i := X_t^{i,N} - \bar{X}_t^{x_i}$ for their difference, so that $\Delta_0^i = 0$. Both processes solve the same SDE under the same GMFG decentralized feedback $\widehat{\alpha}(t, x_i, \cdot)$

and the same driving B^i; they differ only in the aggregate fed to the coefficients — the realized z_t^i for $X^{i,N}$, the GMFG value $z_t(x_i)$ for $\bar{X}^{x_i}$. Itô's formula (theorem 5.9) applied to $|\Delta_t^i|^2$ gives

$$\mathrm{d}|\Delta_t^i|^2 = 2\,\Delta_t^i\cdot\big(b(X_t^{i,N}, z_t^i, \widehat{\alpha}_t^{i,N}) - b(\bar{X}_t^{x_i}, z_t(x_i), \widehat{\alpha}_t^i)\big)\,\mathrm{d}t + \big\|\sigma(X_t^{i,N}, z_t^i) - \sigma(\bar{X}_t^{x_i}, z_t(x_i))\big\|^2\,\mathrm{d}t + \mathrm{d}M_t^i,$$

where $\widehat{\alpha}_t^{i,N} = \widehat{\alpha}(t, x_i, X_t^{i,N})$ and $\widehat{\alpha}_t^i = \widehat{\alpha}(t, x_i, \bar{X}_t^{x_i})$, and the martingale increment $\mathrm{d}M_t^i = 2\,\Delta_t^i\cdot(\sigma(X_t^{i,N}, z_t^i) - \sigma(\bar{X}_t^{x_i}, z_t(x_i)))\,\mathrm{d}B_t^i$ collects the stochastic-integral part. Here is where synchronous coupling earns its keep, and the point is worth making explicit rather than asserting it as in the homogeneous proof. Because both states are driven by the *same* B^i, the two diffusion contributions enter the Itô correction as a single *difference* $\sigma(X^{i,N}) - \sigma(\bar{X}^{x_i})$ — which is itself $O(|\Delta_t^i|)$ by the Lipschitzness of σ — rather than as two independent noises whose cross-variation would not cancel and would inject an $O(1)$ diffusive spreading no Grönwall argument could absorb. The surviving stochastic integral $\int_0^t \mathrm{d}M_s^i$ is a genuine martingale: its integrand is adapted and square-integrable under (A-Lip) and the moment bounds of (A-Growth), so $\mathbb{E}[M_t^i] = 0$ and the noise drops out of the expectation *entirely.* Taking expectations and using the Lipschitz bounds on $b, \sigma, \widehat{\alpha}$ in the pair (X, z) — the optimal feedback $\widehat{\alpha}$ is Lipschitz in X uniformly in (t, x) by the contraction-regime regularity assumed in the theorem, and (A-Lip) covers b, σ — both deterministic terms are dominated by $C(|\Delta_t^i|^2 + |z_t^i - z_t(x_i)|^2)$, whence

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\big|X_t^{i,N} - \bar{X}_t^{x_i}\big|^2 \;\le\; C\,\mathbb{E}\big|X_t^{i,N} - \bar{X}_t^{x_i}\big|^2 \;+\; C\,\mathbb{E}\big|z_t^i - z_t(x_i)\big|^2,$$

where $z_t^i = \frac{1}{N}\sum_j \xi_{ij} h(X_t^{j,N})$ is the realized aggregate and $z_t(x_i) = \int W(x_i, y)\bar{h}_t(y)\,\mathrm{d}y$ its GMFG value. The decisive step is the decomposition of the aggregate discrepancy into the same three pieces as in proposition 25.8, now with the *interacting* states:

$$z_t^i - z_t(x_i) = \underbrace{\frac{1}{N}\sum_j \xi_{ij}\big(h(X_t^{j,N}) - h(\bar{X}_t^{x_j})\big)}_{\text{(I) coupling}} + \underbrace{\frac{1}{N}\sum_j (\xi_{ij} - W(x_i, x_j))h(\bar{X}_t^{x_j})}_{\text{(II) edge sampling}} + \underbrace{\Big[\frac{1}{N}\sum_j W(x_i, x_j)h(\bar{X}_t^{x_j}) - z_t(x_i)\Big]}_{\text{(III) quadrature+state sampling}}.$$

Term (I)*, the coupling term, closes the loop.* Bound it using the Lipschitzness of h and $|\xi_{ij}| \le 1$,

$$\Big|\frac{1}{N}\sum_j \xi_{ij}\big(h(X_t^{j,N}) - h(\bar{X}_t^{x_j})\big)\Big| \;\le\; \frac{1}{N}\sum_j |\xi_{ij}|\,[h]_{\mathrm{Lip}}\,|X_t^{j,N} - \bar{X}_t^{x_j}| \;\le\; [h]_{\mathrm{Lip}}\,\frac{1}{N}\sum_j |X_t^{j,N} - \bar{X}_t^{x_j}|.$$

This is the self-referential piece: it bounds the aggregate error feeding agent i's equation by the *average* of the very state errors the system is trying to control. It is the genuine propagation-of-chaos feedback — every agent's trajectory error is forced, through the wiring, by the mean of all the others' — and when it is squared, inserted back into the differential inequality, and the maximum over i is taken, it produces a term $C\,\max_j \mathbb{E}|\Delta_t^j|^2$ on the *same side* as the unknown, exactly the shape Grönwall's inequality lemma C.1 absorbs. It contributes nothing to the rate; it is what makes the loop closeable, precisely as in the homogeneous estimate lemma 13.5.

Term (II)*, edge sampling, is tractable because we condition on the reference paths.* This is the one place the graph's own randomness enters the trajectory estimate, and the entire art is the choice of conditioning. Work conditionally on the labels $\boldsymbol{x}^N$ *and* on the whole family of reference paths $(\bar{X}_t^{x_j})_j$. The reason it must be the *reference* system rather than the interacting one is a genuine independence that the reference system enjoys and the interacting system does not: the limit surrogate $\bar{X}^{x_j}$ is driven by B^j, X_0^j and the kernel W alone and *never consults the realized edges* ξ, so given $\boldsymbol{x}^N$ the edge coins $\{\xi_{ij}\}_j$ are independent of the frozen features $\{h(\bar{X}_t^{x_j})\}_j$. Term (II) is therefore, conditionally, a sum of independent mean-zero summands $\frac{1}{N}(\xi_{ij} - W(x_i, x_j))\, h(\bar{X}_t^{x_j})$: mean zero because $\mathbb{E}[\xi_{ij} - W(x_i, x_j) \mid \boldsymbol{x}^N] = 0$ by the sampling model eq. (25.1), and independent across j because the coins are. The conditional variance is then the *sum* of the per-edge variances, with no cross terms, each of size $O(1/N^2)$ since the summand carries the weight $1/N$ and a bounded feature; the N of them add to $O(1/N)$, so $\mathbb{E}[|(\mathrm{II})|^2 \mid \boldsymbol{x}^N, \bar{X}] = O(1/N)$ and the root-mean-square is $O(N^{-1/2})$. This is *exactly* the edge-sampling estimate of proposition 25.8(1), now with the bounded reference features $h(\bar{X}_t^{x_j})$ playing the role of the deterministic test weights g_j there. Had we instead conditioned on the interacting states $X_t^{j,N}$, those *do* depend on the edges — they are coupled to them through the very aggregates we are estimating — the independence would have failed, and the variance computation would have collapsed into uncontrolled cross terms. Conditioning on the graph-independent reference system is precisely what severs that entanglement, and it is why term (II) is the per-agent pointwise aggregate error of proposition 25.8(1), available because the labels are known.

Term (III)*, quadrature and state sampling, is the Fournier–Guillin rate.* With the edges averaged out and the kernel held exact, what remains is the discrepancy between the empirical average $\frac{1}{N}\sum_j W(x_i, x_j)\, h(\bar{X}_t^{x_j})$ over the N i.i.d. type–state pairs $(x_j, \bar{X}_t^{x_j})$ and its expectation $z_t(x_i) = \int W(x_i, y)\bar{h}_t(y)\,\mathrm{d}y$. This is pure sampling error, and it splits into two sub-mechanisms. The *quadrature* part q_N is the error of approximating the type-integral by the sum over the sampled positions — the deterministic row-quadrature error already weighed in proposition 25.8, $O(N^{-1})$ on the grid and $O(N^{-1/2})$ for i.i.d. types. The *state-sampling* part is the error of replacing each equilibrium feature mean $\bar{h}_t(x_j) = \langle h, m_t^{x_j}\rangle$ by the single random draw $h(\bar{X}_t^{x_j})$; this is the empirical-measure fluctuation the Fournier–Guillin estimate of lemma 13.5 controls at the rate r_N in $\mathcal{W}_2$, and because h is Lipschitz, $\mathcal{W}_2$-control of the empirical measure transfers to control of the averaged feature. Term (III) is thus of size $r_N + q_N$, both pieces *graph-blind*: they would survive even if the wiring were exact, and they are exactly the errors the homogeneous theory of chapter 13 already paid.

Assembling. Inserting the three bounds into the differential inequality gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}|\Delta_t^i|^2 \;\le\; C\,\mathbb{E}|\Delta_t^i|^2 + C\Big(\underbrace{\frac{1}{N}\sum_j \mathbb{E}|\Delta_t^j|^2}_{\text{(I), the loop}} + \underbrace{N^{-1}}_{\text{(II), edges}} + \underbrace{(r_N + q_N)^2}_{\text{(III), sampling}}\Big).$$

Take the maximum over $i \le N$, so that the (I)-term becomes $C\max_j \mathbb{E}|\Delta_t^j|^2$ on the right-hand side, and apply Grönwall's inequality lemma C.1 on $[0,T]$ with the vanishing initial datum $\Delta_0^i = 0$; promoting $\sup_t \mathbb{E}$ to $\mathbb{E}\sup_t$ costs only a Burkholder–Davis–Gundy estimate on the martingale M^i, exactly as in lemma 13.5. The result is $\max_i \mathbb{E}\sup_{t\le T}|\Delta_t^i|^2 \le C(r_N + \gamma_N)^2$. The state-sampling part of (III) supplies the r_N in $C(r_N + \gamma_N)$; the graph rate collects the rest, $\gamma_N = (\mathrm{II}) + (\mathrm{III}) = O(N^{-1/2}) + O(q_N) = O(N^{-1/2})$ for *known* labels, since the quadrature q_N is itself at worst $O(N^{-1/2})$. This is eq. (25.11).

For *unlabelled* networks the per-agent argument of term (II) is no longer available, and the reason is structural rather than a deficiency of technique. Tracing the labelled proof, the decisive move was to control the *pointwise* row error $[(\mathbb{W}_N - \mathbb{W})\bar{h}_t](x_i)$ at the single *named* vertex i — that pointwise quantity is what the Grönwall loop for a fixed i consumes, and it is what proposition 25.8(1) controlled by exploiting the labels. Strip the labels away and we can no longer name a vertex at all: the only object the cut metric can even speak about is the *type-averaged* tested aggregate $\int f(x)[(\mathbb{W}_N - \mathbb{W})\bar{h}_t](x)\,\mathrm{d}x$, which lemma 25.7 bounds by $4\left\|\bar{h}_t\right\|_\infty \delta_\square(W_N, W)$. The single-severed-row example of section 25.3.1 showed once and for all that no bound on a type-average can ever control an individual row: it exhibited two $\delta_\square$-close graphons differing by $O(1)$ in one row sum while agreeing on every integral. The cut lemma therefore delivers *only* the weaker, type-averaged trajectory bound, got by testing the aggregate-error term against the slowly varying weights and invoking lemma 25.7:

$$\Big(\frac{1}{N}\sum_{i\le N}\mathbb{E}\big[\sup_{t\le T}|X_t^{i,N} - \bar{X}_t^{x_i}|^2\big]\Big)^{1/2} \;\le\; C\big(r_N + \delta_\square(W_N, W)\big), \tag{25.12}$$

The uniform-in-i bound eq. (25.11) survives in the unlabelled case *only* under the stronger operator-norm control $\|\mathbb{W}_N - \mathbb{W}\|_{L^2\to L^2}$ of theorem 25.17 — which dominates the cut norm and *does* see individual rows — with γ_N there replaced by that operator distance; but a single sampled graph does not deliver operator-norm closeness at the cut rate, so the honest unlabelled conclusion is the average eq. (25.12). This is nothing but the cut-versus-operator dichotomy of section 25.3.2 resurfacing one level up, now at the level of *trajectories* rather than aggregates: cut distance controls only type-averaged motion of the population, whereas per-agent motion of a named vertex demands either the labels or the operator norm.

It is worth pausing to confirm that nothing in this proof was imported on faith; the self-containedness owed by the book is met. The graphon propagation-of-chaos estimate underlying terms (I)–(III) is *not* an external input: terms (I) and (III) are the synchronous-coupling and $\mathcal{W}_2$-sampling estimates of lemma 13.5 (the Fournier–Guillin rate r_N, proved in-book at lemma 13.5), and term (II) is the edge-sampling estimate of proposition 25.8(1) proved above; the kernel-perturbation dependence used here is the contraction-regime stability proved in this chapter as theorems 25.16 and 25.17 (built on the contraction estimate proposition 24.18). The graphon-system law of large numbers of [15, 16] is the corresponding general (non-contraction) statement, recorded for attribution. □

The headline consequence advertised before the proof — that the finite system reproduces, type by type, the GMFG field m_t^x — needs a careful statement of *which* empirical measure could possibly converge to m_t^x, and the difficulty is one the homogeneous theory never faced. In chapter 13 the target was a single measure and the global empirical measure $\frac{1}{N}\sum_j \delta_{X_t^j}$ was the obvious approximant. Here the target is a *continuum* of measures, one per type, and at a single exact type x the regular grid places exactly *one* agent — the vertex whose position is x. A single Dirac mass cannot approximate a whole distribution m_t^x; one sample is not an empirical law. The fix is to widen the aperture: gather all the agents whose types lie within a small window δ_N of x, treat them as approximate samples of the type-x law, and let the window shrink slowly enough that it always contains many agents ($N\delta_N \to \infty$) yet fast enough that the types it pools are all genuinely near x ($\delta_N \to 0$). This is exactly a histogram or kernel-smoothing bandwidth, and the corollary's content is the balance between its two competing demands. The honest statement therefore localizes over a shrinking window.

Corollary 25.11 (Convergence of the type-resolved empirical field)**.** *Adopt the hypotheses of theorem 25.10 with* known *labels (regular grid or i.i.d.), and let the type field $x \mapsto m_t^x$ be type-Hölder with modulus ω_W (finite under* (A-Reg-W)*, by lemma 24.9). Fix a bandwidth $\delta_N \downarrow 0$ with $N\delta_N \to \infty$, and for $x \in [0,1]$ define the* windowed empirical field

$$\widehat{m^{x,N}}_t \;:=\; \frac{1}{|\mathcal{N}_N(x)|}\sum_{j\in\mathcal{N}_N(x)} \delta_{X_t^{j,N}}, \qquad \mathcal{N}_N(x) := \{\, j \le N : |x_j - x| \le \delta_N \,\}, \tag{25.13}$$

the empirical law of the states whose types lie within δ_N of x. Then

$$\sup_{x\in[0,1]} \mathbb{E}\,\mathcal{W}_2\big(\widehat{m^{x,N}}_t, m_t^x\big) \;\le\; \underbrace{C(r_N+\gamma_N)}_{coupling} + \underbrace{r_{\lfloor N\delta_N\rfloor}}_{within\text{-}window\ sampling} + \underbrace{\omega_W(\delta_N)}_{localization\ bias} \xrightarrow[N\to\infty]{} 0. \tag{25.14}$$

Since $r_{\lfloor N\delta_N\rfloor} \to 0$ (as $N\delta_N \to \infty$) and $\omega_W(\delta_N) \to 0$ (as $\delta_N \to 0$), the right side vanishes; if $\omega_W(\delta) \le C\delta^\beta$, balancing the last two terms yields a polynomial rate added to the coupling error.

Proof. Fix x and abbreviate $\mathcal{N} = \mathcal{N}_N(x)$, $n = |\mathcal{N}|$, the number of agents whose types fall in the window. We first record that n is large, since the whole construction depends on it. For the regular grid the window $[x-\delta_N, x+\delta_N]$ deterministically contains $n \asymp 2N\delta_N$ cell-centres, so $n \asymp N\delta_N \to \infty$ by the standing hypothesis $N\delta_N \to \infty$. For i.i.d. labels n is a binomial count, $n \sim \mathrm{Bin}(N,p)$ with success probability $p = \mathbb{P}(|x_1 - x| \le \delta_N) \asymp 2\delta_N$, so $\mathbb{E}n \asymp 2N\delta_N$; the multiplicative Chernoff bound for binomials gives $\mathbb{P}\big(n < \tfrac12\mathbb{E}n\big) \le e^{-\Omega(\mathbb{E}n)} = e^{-\Omega(N\delta_N)}$, so with probability $1 - e^{-\Omega(N\delta_N)}$ the window holds at least $\tfrac12 N\delta_N$ agents. This is exactly what the bandwidth condition $N\delta_N \to \infty$ buys: it drives the failure probability to zero and makes n deterministically large, so that the within-window law of large numbers below has enough samples to act on. On this high-probability event we treat $n \asymp N\delta_N$ throughout. Insert two intermediate measures and use the triangle inequality for $\mathcal{W}_2$:

$$\mathcal{W}_2\big(\widehat{m^{x,N}}_t, m_t^x\big) \le \underbrace{\mathcal{W}_2\big(\widehat{m^{x,N}}_t,\, \tfrac1n\sum_{j\in\mathcal{N}}\delta_{\bar X_t^{x_j}}\big)}_{(a)} + \underbrace{\mathcal{W}_2\big(\tfrac1n\sum_{j\in\mathcal{N}}\delta_{\bar X_t^{x_j}},\, \tfrac1n\sum_{j\in\mathcal{N}} m_t^{x_j}\big)}_{(b)} + \underbrace{\mathcal{W}_2\big(\tfrac1n\sum_{j\in\mathcal{N}} m_t^{x_j},\, m_t^x\big)}_{(c)}.$$

For (a), coupling the two empirical measures atom by atom through the synchronous coupling of theorem 25.10, $(a) \le \big(\tfrac1n\sum_{j\in\mathcal{N}}|X_t^{j,N} - \bar X_t^{x_j}|^2\big)^{1/2}$, whose expectation is at most $\max_{i\le N}\mathbb{E}[\,|X_t^{i,N} - \bar X_t^{x_i}|^2]^{1/2} \le C(r_N+\gamma_N)$ by eq. (25.11). For (b): conditionally on the labels the $\bar X_t^{x_j}$, $j\in\mathcal{N}$, are independent with laws $m_t^{x_j}$, so (b) is the empirical-measure sampling error of n independent draws from a mixture, of size $r_n = r_{\lfloor N\delta_N\rfloor}$ (the Fournier–Guillin rate of lemma 13.5, in expectation). For (c): convexity of $\mathcal{W}_2$ and the type-modulus give $(c) \le \tfrac1n\sum_{j\in\mathcal{N}}\mathcal{W}_2(m_t^{x_j}, m_t^x) \le \omega_W(\delta_N)$, since $|x_j - x| \le \delta_N$ for $j \in \mathcal{N}$. Taking expectations, summing, and the supremum over x (the three bounds are uniform in x) gives eq. (25.14). □

The three terms of eq. (25.14) make the bandwidth trade-off explicit, and it is worth carrying out the optimisation, because it tells us the windowing never becomes the bottleneck. The two window-dependent terms pull in opposite directions: the within-window sampling error $r_{\lfloor N\delta_N\rfloor}$ *improves* as the window widens (more agents $n \asymp N\delta_N$ to average over), while the localization bias $\omega_W(\delta_N)$ *worsens* (the pooled types stray farther from x). The coupling error $C(r_N+\gamma_N)$ is independent of the bandwidth and is simply inherited from the joint-convergence theorem. Choosing

δ_N is thus the classical bias–variance balance of nonparametric smoothing. Concretely, if the type field is Hölder, $\omega_W(\delta) \le C\delta^\beta$, and the sampling rate behaves as $r_n \asymp n^{-a}$ for the relevant exponent a (for instance $a = \frac{1}{4}$ in low state dimension, by lemma 13.5), then $r_{\lfloor N\delta_N \rfloor} \asymp (N\delta_N)^{-a}$, and the two bandwidth-dependent terms balance when $(N\delta_N)^{-a} \asymp \delta_N^\beta$, i.e. at $\delta_N \asymp N^{-a/(a+\beta)}$, leaving a polynomial windowing overhead of order $N^{-a\beta/(a+\beta)}$ added to the irreducible coupling error $C(r_N + \gamma_N)$. Since this overhead is polynomial, the field-resolution step adds nothing of a worse order than the trajectory rate the theorem already certified — the windowing is a clean polynomial refinement, never the dominant cost (which remains the graph limit whenever the labels are unknown).

Remark 25.12 (Where the order of limits matters)**.** Everything above was harvested inside the contraction regime, where the commuting square is exact and "the" joint limit and its rate are well posed. It is worth spelling out what goes wrong as one leaves that regime, because the failures are not abstract pathologies but precisely the places the chapter's central refrain — *rates and commutation are contraction-regime luxuries* — earns its keep. Two distinct failure modes are worth naming.

First, the right column loses its modulus at the critical coupling. The whole force of theorem 25.10 flowed through the Lipschitz dependence of the GMFG solution on the kernel, which was a gift of the contraction $\|\mathbb{W}\| < \kappa$. As $\|\mathbb{W}\| \uparrow \kappa$ that contraction degrades — the best-response map's Lipschitz constant approaches 1 — and the modulus of the solution-in-kernel map blows up like $(1 - \|\mathbb{W}\|/\kappa)^{-1}$, governed by the spectral gap of $\mathbb{W}$ at threshold. The right arrow of the commuting square then no longer carries a finite constant, the joint rate eq. (25.11) degrades accordingly, and — worse — at the threshold the GMFG equilibrium can cease to be unique (the symmetry-breaking bifurcation of chapter 24). Once there are several limit equilibria, the order in which one takes $N \to \infty$ and $\delta_\square(W_N, W) \to 0$ can select *different* branches: the iterated limits need not agree, and the square genuinely fails to commute. There is then no single object "the GMFG limit" for the finite game to approximate, and the very question the chapter answers becomes ill-posed.

Second, closed-loop finite equilibria can carry extra randomness. Theorem 25.10 certified the *decentralized* GMFG profile, in which each vertex plays feedback on its own state and type alone. Had we instead tried to take the limit of the finite game's genuine *closed-loop* Nash equilibria — the solutions of the network Nash system eq. (25.6), where each value function depends on the entire configuration — the subtlety of remark 13.10 reappears, and on a network it is compounded. Closed-loop limits can carry extra randomness from *two* sources at once: the configuration coupling, exactly as in the homogeneous case, and now the quenched graph as well. Such limits need not be the Markovian GMFG solution at all; the cheap top-down direction of section 25.4, which only ever asks whether the decentralized profile is approximately optimal, sidesteps this difficulty entirely, which is one reason it is the direction the chapter actually leans on.

Under the monotone regime (A-Mon) both pathologies vanish: monotonicity restores uniqueness of the GMFG equilibrium without any smallness on $\|\mathbb{W}\|$, all routes around the square agree, and the closed-loop and decentralized limits coincide, exactly as in chapter 13. The honest summary is the doubled version of remark 13.15: a clean joint rate and order-independence are not generic features of the convergence problem but luxuries of the contraction or monotone regime — and outside it, the best one can say is that some limit exists, with no promise of a rate or of commutation.

Physics note

The two limits are the strategic counterpart of *quenched* versus *annealed* averaging over disorder, and the analogy is tight enough to name each arrow of the commuting square. The network $(\boldsymbol{x}^N, \xi^N)$ is quenched randomness, fixed once before the dynamics run, like the bond disorder of a spin glass or the conductances of a random-resistor network: the agents play *in* a frozen environment, never averaging over it. Propagation of chaos with the graph held fixed — the top arrow — is then exactly the statement that the system *self-averages* over its own quenched wiring: a single typical realization of the random graph produces the same macroscopic equilibrium as the disorder average would, the per-realization fluctuations washing out as $N \to \infty$. This is the strategic analogue of a self-averaging free energy, whose value at a single typical disorder sample equals its disorder-averaged value in the thermodynamic limit. Graph convergence — the left arrow — is the complementary statement that the disorder average itself relaxes to the graphon's *annealed* law, the GMFG on W. The two limits commute precisely when the system self-averages *uniformly* in the disorder, and this is exactly what the contraction condition (A-Coupling) guarantees: a contraction has one equilibrium that depends smoothly on the wiring, so no realization can drift onto a different branch. Uniform self-averaging is precisely what breaks down near a critical coupling, where the equilibrium bifurcates and different disorder realizations can fall into different basins — the strategic image of a glass transition, at which self-averaging famously fails and sample-to-sample fluctuations of the order parameter no longer vanish. Finally, the cut metric is the natural "macroscopic" topology here for the same reason a coarse-grained free energy is the natural thermodynamic observable: it sees only those features of the wiring that survive averaging over large patches, and is blind to the microscopic relabelling of individual sites. Lemma 25.7 is the precise statement that the equilibrium costs depend on the network only through such coarse, patch-averaged features — which is exactly why the cut-controlled, unlabelled conclusions are the ones that bank as robustness in section 25.5.

With the trajectory-level convergence now certified — the finite closed-loop states on $G(N, W)$ tracking the GMFG limit at the additive rate $C(r_N + \gamma_N)$, uniformly in i when the labels are known and in type-average otherwise — two debts pass forward, in two directions. The first runs into section 25.4. The ε-Nash theorem there must turn this statement *about trajectories* into a statement *about incentives*, and it does so cheaply, reusing the Step-1 aggregate control of this proof — the per-agent bound on $z_t^i - z_t(x_i)$ assembled from terms (I)–(III) — without ever solving the intractable network Nash system of section 25.2. That is the decisive economy of the top-down direction: certifying that the decentralized profile is nearly optimal needs only propagation of chaos for one *fixed* feedback, the very object we have just controlled, and never the finite game's own equilibrium. The known/unlabelled split and the $r_N + \gamma_N$ rate pass through to that theorem unchanged. The second debt runs into section 25.5, but in the reverse direction, for the proof above has *borrowed* from it: the Lipschitz-in-kernel dependence that made the right column of the square exact was invoked here on credit, and section 25.5 is where that credit is paid — the cut-metric and operator-norm kernel-perturbation estimates theorems 25.16 and 25.17, built on the contraction estimate proposition 24.18, are exactly the inputs this proof cited. We turn first to the incentive statement, the practically decisive payoff, and return to settle the stability debt afterward.

25.4 The ε-Nash property on large finite networks

The previous subsection handed us a statement about *trajectories*: under the decentralized GMFG profile the finite states on $G(N, W)$ track the graphon McKean–Vlasov limit at the additive rate $C(r_N + \gamma_N)$, uniformly in the vertex when the labels are known and only in type-average otherwise. That is a fact about where the agents *go*. The result an intervention designer actually needs is a fact about what the agents *want* — a statement about *incentives*: that no vertex, contemplating any unilateral deviation, can lower its own cost by more than a vanishing amount. This section performs that conversion, turning the trajectory bound into an ε-Nash certificate, and it is the result that justifies using the GMFG to design behaviour on real graphs.

The conversion is the cheap one, and it is worth being explicit at the outset about *why* it is cheap, because the economy is the whole point. We are running the top-down direction of chapter 13, transported to the network: solve the GMFG once on the continuum, read off the decentralized feedback as a function of an agent's own state *and its type*, hand it to each vertex of $G(N, W)$, and certify that the resulting profile is an approximate Nash equilibrium of the finite game. The decisive observation — exactly as in theorem 13.6 — is that certifying approximate optimality of *one fixed* profile never requires solving the finite game's own equilibrium. The intractable object of section 25.2, the coupled N-body Nash system eq. (25.6) on configuration space, simply does not appear. All we ever ask is: when every neighbour holds the GMFG feedback fixed and a single tagged agent is free to do anything, how much can that one agent gain? And answering it needs only one analytic input — propagation of chaos for the *fixed* dynamics the common feedback induces — which is precisely the Step-1 aggregate control section 25.3.3 already proved. The bottom-up direction, which would track the genuine closed-loop equilibria of the finite game and watch them converge, is the expensive one we deliberately avoid (and, as remark 25.12 warned, the one that can fail to converge to the Markovian limit at all). The top-down statement we prove here is both cheaper and the one that licenses intervention design.

Before the theorem we must pin down precisely *what strategy* is being handed to the vertices, because the entire force of the result lives in how little that strategy needs to know. The GMFG of chapters 22 and 24 was solved once, on the continuum, and returned a field of value functions $(u^x)_x$ together with the self-consistent field of aggregates $z_t(x) = (\mathbb{W}\bar{h}_t)(x)$, $\bar{h}_t(y) = \langle h, m_t^y \rangle$. Feeding these into the pointwise maximizer $\alpha^\star$ of the Hamiltonian — the same $\alpha^\star$ that appeared in the single-agent verification theorem of chapter 6 and in the network Nash system eq. (25.6) — produces the *decentralized GMFG feedback*

$$\widehat{\alpha}(t, x, X) \;:=\; \alpha^\star(X,\, z_t(x),\, \nabla u^x(t, X)), \qquad z_t(x) = (\mathbb{W}\bar{h}_t)(x). \tag{25.15}$$

The structure of eq. (25.15) is the crux, and it pays to read off each of its three arguments. The control depends on the agent's own current state X; on the agent's type x, which enters in two places — as the label selecting the value function u^x and as the point at which the continuum aggregate field $z_t(\cdot)$ is sampled; and on time t. It depends on *nothing else*: not on the states of any other vertex, not on the realized wiring ξ, not even on the private states of the agent's own neighbours. The frozen aggregate $z_t(x)$ has already been computed on the continuum and is a fixed deterministic function once the GMFG is solved; the agent does not re-read its neighbourhood to evaluate its control. This is exactly the decentralized class of definition 25.4, and the single network novelty over the homogeneous case is the explicit appearance of the type x as a private datum the agent carries from birth. The *GMFG profile* on the finite network is then the strategy in which vertex i plays its own type's feedback,

$$\alpha_t^i = \widehat{\alpha}(t, x_i, X_t^i),$$

read off the limit and implemented with no observation of the graph. It is an implementable, fully decentralized recipe; the theorem says it is also nearly optimal.

One point of emphasis fixes the strength of the guarantee. "Nearly optimal" must be read against a *class* of permitted deviations, and we test against the largest one: the *open-loop* class of definition 25.4. A tagged agent contemplating a deviation is allowed any control process adapted to the full information filtration $\mathbb{F}^N$ — it may, in principle, react to the entire history of every vertex's state and to the realized network itself, far more than the decentralized profile it is deviating from could ever see. Certifying that even such an omniscient, history- and graph-dependent deviation cannot gain more than ε_N is the strongest form the ε-Nash statement can take; a guarantee against the smaller Markovian or decentralized deviation classes would be weaker and easier. That we get the open-loop guarantee for free is a structural gift of the cost: because each $J^{N,i}$ depends on the other agents *only* through the scalar aggregate z_t^i, a deviating agent faced with its neighbours' frozen GMFG feedback confronts an honest *single-agent* control problem against an exogenous signal, and against a single-agent control problem the optimum over open-loop controls and the optimum over feedback controls coincide. This is the envelope structure the proof exploits.

Theorem 25.13 (Network ε-Nash property)**.** *Assume* (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), (A-Reg-W)*, that the GMFG solution* (u^x, m^x) *has a feedback* $\widehat{\alpha}$ *bounded and Lipschitz in* X *uniformly in* (t, x)*, and that* b, σ, L, g *are Lipschitz in the aggregate* z *uniformly in* (X, α) *on the relevant range. Let the network be* $G(N, W)$ *with known types. Then the GMFG profile* $\alpha_t^i = \widehat{\alpha}(t, x_i, X_t^i)$ *is an* ε_N*-Nash equilibrium of the finite game eq.* (25.4)*–eq.* (25.5) *in the open-loop class, with*

$$\varepsilon_N \;\le\; C\big(r_N + \gamma_N\big) \;\xrightarrow[N\to\infty]{}\; 0, \tag{25.16}$$

r_N *and* γ_N *as in theorem 25.10,* C *independent of* N*. For* unlabelled *networks — where only* $\delta_\square(W_N, W)$ *is available — the per-agent* ε*-Nash guarantee is* not *certified by the cut distance (lemma 25.7 controls only type-averages, and two* $\delta_\square$*-close graphons can differ arbitrarily in a single vertex's aggregate); what holds instead is the weaker* type-averaged *guarantee*

$$\frac{1}{N}\sum_{i\le N}\Big[J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i}) - \inf_{\eta} J^{N,i}(\eta, \widehat{\alpha}^{-i})\Big] \;\le\; C\big(r_N + \delta_\square(W_N, W)\big), \tag{25.17}$$

that the average *regret across agents is small. The uniform-in-*i ε*-Nash bound eq.* (25.16) *persists for unlabelled networks only under the stronger operator-norm control* $\|\mathbb{W}_N - \mathbb{W}\|_{L^2\to L^2}$ *(theorem 25.17), with* γ_N *replaced by that operator distance.*

Proof. The architecture is the three-problem triangle inequality of theorem 13.6, transported to the network with the graphon aggregate $z_t(x_i)$ playing the role the mean field played there. Fix the tagged agent i and a candidate open-loop deviation $\alpha^i = \eta$, every other vertex keeping the GMFG profile $\widehat{\alpha}^{-i}$. We will compare three costs for this agent: its *true* finite-game cost $J^{N,i}(\eta, \widehat{\alpha}^{-i})$ under the realized aggregate; an idealized *frozen-aggregate* cost $J^{\mathrm{G}}_{x_i}(\eta)$ in which the realized aggregate is replaced by the deterministic continuum value $z_\cdot(x_i)$; and the same frozen-aggregate cost evaluated at the prescribed feedback $\widehat{\alpha}$ itself. The strategy is to show that the first two costs are close (Step 2, the analytic input), that the GMFG feedback is exactly optimal for the third (Step 3, the strategic input), and to chain the three into the ε-Nash inequality. Everything rests on one estimate, so we establish it first.

Step 1: the deviator's environment stays close to its type's frozen aggregate, uniformly over deviations. When agent i switches to η, the other $N-1$ vertices do not react: they are running the prescribed decentralized feedback $\widehat{\alpha}(\cdot, x_j, \cdot)$, which depends on vertex j's own state and type alone and is blind to what i does. Their joint motion is therefore (almost) exactly the closed-loop graphon system already analysed in theorem 25.10, and the realized aggregate felt by i, $z^i_t = \frac{1}{N}\sum_j \xi_{ij} h(X^j_t)$, is precisely the object the Step-1 aggregate decomposition of that theorem controlled. The conclusion we import is

$$\mathbb{E}\sup_{t\le T}\left|z^i_t - z_t(x_i)\right|^2 \;\le\; C\,(r_N + \gamma_N)^2, \tag{25.18}$$

the per-agent bound assembled there from the coupling term, the edge-sampling term, and the quadrature/state-sampling term, with $z_t(x_i) = (\mathbb{W}\bar{h}_t)(x_i)$ the frozen continuum aggregate at the known type x_i. The one new point, on which the entire ε-Nash argument turns, is that eq. (25.18) holds *uniformly over the deviation* η — the constant C does not depend on which control the tagged agent chose. This is not automatic and deserves to be seen clearly. The aggregate z^i_t is built from the *neighbours'* states $\{X^j_t\}_{j\neq i}$, weighted by the edges ξ_{ij}; the agent's own deviation η enters that sum only through the single self-term $\xi_{ii}h(X^i_t)$, and $\xi_{ii} = 0$ by definition 25.1. The deviator therefore contributes *nothing* to its own aggregate directly. Its only channel of influence is indirect: a deviation moves X^i, which through the edges ξ_{ji} could in principle perturb the neighbours X^j that feed back into z^i. But here the single-atom scaling intervenes. In the aggregate $z^j_t = \frac{1}{N}\sum_k \xi_{jk} h(X^k_t)$ that drives neighbour j, agent i appears as exactly *one* summand, $\frac{1}{N}\xi_{ji}h(X^i_t)$, of weight $1/N$; with $|\xi_{ji}| \le 1$ and h bounded by some M, that summand is at most M/N in absolute value, however wildly η swings X^i. The deviator is a single atom of mass $1/N$ in a sea of N agents, and a single atom can shift any neighbour's aggregate by at most $O(1/N)$. Propagating this $O(1/N)$ perturbation through the neighbours' Lipschitz dynamics and back into z^i adds only a further $O(1/N)$ to the right-hand side of eq. (25.18) — an amount already absorbed into γ_N, which is itself no smaller than $O(N^{-1/2})$. Hence the bound holds with a single constant valid for *every* admissible open-loop η. This uniformity is exactly what lets the same Step-1 estimate serve simultaneously the deviation we are testing and the prescribed profile, so that both ends of the chain below are controlled by one inequality.

Step 2: the true cost and the frozen-aggregate cost are close. Introduce the *representative type-x_i control problem*: an agent of type x_i steering X^i under eq. (25.4) but with the realized aggregate z^i_t everywhere replaced by the deterministic frozen continuum value $z_t(x_i)$, and paying the cost eq. (25.5) with the same replacement. Write $J^{\mathrm{G}}_{x_i}(\eta)$ for its value under control η. This is an honest single-agent control problem against an exogenous, deterministic signal — precisely the problem the GMFG was built to solve — and it is the bridge between the finite game and the continuum. The claim is that the true finite cost and this idealized cost differ by no more than the rate:

$$\left|J^{N,i}(\eta, \widehat{\alpha}^{-i}) - J^{\mathrm{G}}_{x_i}(\eta)\right| \;\le\; C\,(r_N + \gamma_N) \qquad \text{uniformly over admissible } \eta.$$

The two cost functionals are evaluated along the *same* control η and the *same* driving noise; they differ only because one feeds the realized aggregate z^i_t into the coefficients b, σ, L, g while the other feeds the frozen $z_t(x_i)$. Subtracting the two state equations and using the Lipschitz dependence of b, σ on the aggregate (a standing hypothesis of the theorem), the difference of the two state trajectories is forced by $|z^i_t - z_t(x_i)|$ and controlled, via Grönwall's inequality lemma C.1, by $\mathbb{E}\sup_t |z^i_t - z_t(x_i)|^2$; the Lipschitz dependence of L, g on the aggregate then converts that state discrepancy, together with the direct aggregate discrepancy inside the running and terminal costs, into the cost gap. Inserting the Step-1 bound eq. (25.18) gives the displayed inequality, and because Step 1 was uniform in η so is this. In particular the same estimate holds at the prescribed control $\eta = \widehat{\alpha}$.

Step 3: the GMFG feedback is optimal against the frozen aggregate. This is the only genuinely strategic step, and it is where the continuum solution earns its keep. By the very construction of the GMFG (chapters 6 and 22), the type-x_i feedback $\widehat{\alpha}(\cdot, x_i, \cdot) = \alpha^\star(\cdot, z_\cdot(x_i), \nabla u^{x_i})$ is, by the verification theorem of chapter 6, the *optimal* control of the representative type-x_i agent against its own frozen aggregate $z_\cdot(x_i)$. That is precisely the optimality statement that closes the GMFG fixed point, here read at the single type x_i. Hence

$$J^{\mathrm{G}}_{x_i}(\widehat{\alpha}) \;\le\; J^{\mathrm{G}}_{x_i}(\eta) \qquad \text{for every admissible } \eta.$$

Note that this optimality is exact and carries no error term: it is a property of the continuum problem, decided before any finite network is sampled. The finite-N error lives entirely in Steps 1–2; Step 3 contributes only the clean inequality that lets us pivot from the prescribed control to an arbitrary deviation.

Chaining. Now assemble the three inequalities for the tagged agent. Reading left to right, we bound the prescribed profile's true cost by its frozen cost (Step 2 at $\eta = \widehat{\alpha}$), replace the prescribed control by the deviation η in the frozen problem (Step 3), and convert back to the deviation's true cost (Step 2 at η):

$$J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i}) \overset{\text{St.2}}{\le} J^{\mathrm{G}}_{x_i}(\widehat{\alpha}) + C(r_N + \gamma_N) \overset{\text{St.3}}{\le} J^{\mathrm{G}}_{x_i}(\eta) + C(r_N + \gamma_N) \overset{\text{St.2}}{\le} J^{N,i}(\eta, \widehat{\alpha}^{-i}) + 2C(r_N + \gamma_N).$$

The two applications of Step 2 each cost one factor of $C(r_N + \gamma_N)$ and they add; Step 3 is free. The endpoints read $J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i}) \le J^{N,i}(\eta, \widehat{\alpha}^{-i}) + 2C(r_N + \gamma_N)$, which is exactly the defining inequality eq. (13.4) of an ε-Nash equilibrium with $\varepsilon_N = 2C(r_N + \gamma_N)$. The bound holds for *every* open-loop deviation η, because the only η-dependent ingredient, Step 1, was uniform in η; and it holds for *each* agent i separately, because Step 1 controlled the *pointwise* aggregate error at the *named* vertex x_i, which is available precisely when the labels are known (proposition 25.8(1)). This is the genuine per-agent certificate of eq. (25.16): no individual vertex, however it deviates, gains more than ε_N.

The unlabelled case. Strip the labels away and the per-agent argument of Step 1 fails at its root. The chain above consumed the pointwise bound eq. (25.18) at the single named vertex x_i; but for an unlabelled network the cut metric cannot speak about an individual vertex at all. As the single-severed-row example of section 25.3.1 showed once and for all, two $\delta_\square$-close graphons can differ by $O(1)$ in one row's aggregate while agreeing on every type-average, so no cut bound can ever certify the pointwise estimate a fixed-i chain requires. What the cut-norm aggregation lemma lemma 25.7 *does* control is the type-average of the Step-1 error. Averaging the entire chain over i — the regret inequality holds for each i, so its mean over $i \le N$ holds too — and bounding the averaged aggregate error through lemma 25.7 replaces γ_N by $\delta_\square(W_N, W)$ and yields the *type-averaged* guarantee eq. (25.17): the *average* regret across agents is small, even though no single vertex is certified. Recovering the uniform-in-i statement eq. (25.16) without labels demands an estimate that sees individual rows, and the cut norm structurally cannot; only the stronger operator-norm control $\|\mathbb{W}_N - \mathbb{W}\|_{L^2 \to L^2}$ of theorem 25.17, which dominates the cut norm and resolves single rows, restores the per-agent certificate, with γ_N there replaced by that operator distance. $\square$

Remark 25.14 (What the network adds, and what it does not)**.** It is worth standing back and comparing this theorem, line for line, with its homogeneous ancestor theorem 13.6, because the comparison isolates exactly what the network costs. The proof is *structurally identical.* The only probabilistic input is propagation of chaos for the one fixed feedback $\widehat{\alpha}$ — never the finite game's Nash system — and the only strategic input is the elementary type-wise optimality of Step 3. The three-problem envelope, the frozen aggregate standing in for the population, the uniformity of the single-atom Step-1 estimate over deviations, the doubling that turns one cost gap into the factor 2 in ε_N: all of it carries over verbatim from chapter 13, with the graphon aggregate $z_t(x_i)$ wherever the homogeneous proof wrote the mean field. The network changes the bound in exactly *one* place: it contributes *one extra error term*, the graph rate γ_N, added to the familiar sampling rate r_N. Where the homogeneous theorem read $\varepsilon_N \le 2Cr_N$, the network theorem reads $\varepsilon_N \le 2C(r_N + \gamma_N)$ — a single new summand, with no change to the mechanism that produced it. This is the precise sense in which "chapter 13 on a network" is the right slogan for the whole chapter: the strategic skeleton is untouched, and the entire price of heterogeneity is the one graph term.

That the price is *exactly* γ_N is what makes the cut-versus-operator dichotomy of section 25.3.2 decisive at the level of incentives, not merely of trajectories. When the labels are known, $\gamma_N = O(N^{-1/2})$ is comparable to the mean-field rate r_N, and the network costs nothing of a worse order than the sampling already did: the ε-Nash guarantee on a real graph is as strong as the one on the homogeneous crowd. When the network is delivered only up to relabelling, $\gamma_N = O(1/\sqrt{\log N})$ dominates everything, and the *logarithmic* graph rate becomes the bottleneck of the entire certificate — remark 25.9 is the place this trade is weighed in full, and its message is that the value of knowing the latent types is precisely the gap between a polynomial and a logarithmic ε-Nash rate. This is the network ε-Nash property quoted in [34] and quantified in the learning context by [55].

The theorem just proved is the chapter's practical payoff stated abstractly: a vertex playing the GMFG feedback, computed once on the continuum and indexed by its type, is within $\varepsilon_N = 2C(r_N + \gamma_N)$ of optimal on the genuine finite game, with no agent able to do better by observing the network. Two threads run forward from it. The first is concreteness. The bound $2C(r_N + \gamma_N)$ is honest but loose — r_N is the worst-case Fournier–Guillin sampling rate, which in low dimension is only $O(N^{-1/4})$ — and one naturally asks whether structure sharpens it. The worked computation that follows answers this on the running Thread-B model: there the quadratic cost makes the regret depend on the aggregate fluctuation only at second order, so the $O(N^{-1/2})$ aggregate error (edge-sampling fluctuations included) enters squared and the per-agent rate collapses all the way to $O(1/N)$. We instantiate eq. (25.15) as the explicit affine Thread-B feedback, import the GMFG solution of chapters 22 and 24 keeping its symbols — the value curvature p, the linear coefficient r^x, the type mean $\overline{\mathbf{m}}_t$, and the rank-one order parameter ϱ — so the thread reads continuously across the seam, and watch the abstract envelope of this proof become a one-line completion-of-squares identity. The second thread is robustness. The ε-Nash certificate licenses designing an intervention on the limit and deploying it on the finite graph; but a designer never knows the true wiring exactly, and section 25.5 banks the complementary guarantee that the equilibrium one designs is stable under the unidentifiable, cut-close perturbations of the network. Together the two complete the justification of GMFG-based intervention design on imperfectly known networks: the prescription is nearly optimal (here) and robust to the part of the wiring no measurement can resolve (there).

Worked computation: ε-Nash for Thread B on a sampled graph

Worked computation 25.15. The theorem just proved certified the GMFG profile as an ε_N-Nash equilibrium of the finite game with $\varepsilon_N \le 2C(r_N + \gamma_N)$, and its closing paragraph promised that this honest-but-loose bound sharpens when the model carries quadratic structure. We now redeem that promise on the running Thread-B model — the carried thread's discharge for this chapter — and watch every abstract ingredient of the proof of theorem 25.13 (the frozen aggregate, the single-atom Step-1 estimate, the optimize-then-control envelope) become an explicit, closed-form computation.

We instantiate theorem 25.13 on Thread B over a rank-one graphon and find, exactly as in the homogeneous LQG worked computation worked computation 13.8, that the quadratic structure upgrades the rate from $r_N + \gamma_N$ to $O(1/N)$ — the edge-sampling fluctuations the network adds are themselves $O(N^{-1/2})$ in L^2 and, entering the cost quadratically, cost only $O(1/N)$ in value. The mechanism is worth naming before any symbol is written, because it is the whole reason the linear–quadratic world is so much kinder than the generic bound. The generic ε-Nash rate is *linear* in the aggregate error: a deviator that misreads its environment by δ can, against a general cost, gain order δ. Quadratic costs annihilate that first order. At the GMFG profile the agent is already optimal against the *frozen* aggregate, so the regret it incurs when the *realized* aggregate is off by δ is the value-gap of a near-optimal control; and for a smooth, strictly convex control problem a value-gap is quadratic in the control perturbation, hence quadratic in δ. An $O(N^{-1/2})$ environment error therefore buys only an $O(1/N)$ regret. The computation below is the bookkeeping that turns this one-sentence envelope argument into the sharp constant of eq. (25.28). This is the carried-thread discharge for the present chapter; the GMFG solution is imported from chapters 22 and 24.

The finite game on $G(N,W)$. Everything specializes the finite game of section 25.1.2 to the Thread-B data of chapter 22: the state is scalar, the dynamics are linear in state and control, the cost is quadratic, and the interaction observable is the identity $h(\zeta) = \zeta$, so a neighbour is socially visible only through its position. Concretely, scalar states $X^i_t \in \mathbb{R}$, types x_i (grid or i.i.d.), edges $\xi_{ij} \sim \text{Bernoulli}(W(x_i, x_j))$,

$$\mathrm{d}X^i_t = \big(aX^i_t + b\,\alpha^i_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}B^i_t, \qquad X^i_0 \sim m_0^{x_i}, \tag{25.19}$$

with the imitation cost coupled to the graph-neighbour mean ($h(\zeta) = \zeta$, so $z_t^i = \frac{1}{N}\sum_j \xi_{ij} X_t^j$),

$$J^{N,i} = \mathbb{E}\bigg[\int_0^T \Big(\tfrac{1}{2}(\alpha_t^i)^2 + \tfrac{q}{2}\big(X_t^i - s\,z_t^i\big)^2\Big)\,\mathrm{d}t + \tfrac{q_T}{2}\big(X_T^i - s_T\,z_T^i\big)^2\bigg], \tag{25.20}$$

the parameters $a, b, \sigma, q, q_T, s, s_T$ those fixed for Thread B in the thread specification. The single structural fact that will drive everything is visible already: as in Thread A, the neighbour aggregate z_t^i enters only the running and terminal *costs*, never the drift or diffusion of eq. (25.19). The interaction is a pull of strength s toward the graph-neighbour mean z_t^i, paid for as a quadratic tracking penalty, with no mechanical coupling between the states. This is precisely what will let the network vanish from the closed-loop dynamics while surviving in the cost — the exact-independence phenomenon established below.

The GMFG solution (rank-one). We import the Thread-B GMFG solution of worked computation 24.22, keeping its symbols (p for the value curvature, r^x for the linear coefficient, $\overline{\mathbf{m}}_t(x)$ for the type mean) so the carried thread reads continuously across the seam. The type-x value is quadratic in the state, and this is forced rather than guessed: a quadratic running cost driven by linear dynamics propagates backward under the HJB flow as a quadratic, because the Hamiltonian of a linear-state, quadratic-cost problem is itself quadratic in the value gradient and the dynamic-programming recursion preserves the quadratic class. We may therefore write $u^x(t,X) = \frac{1}{2}p(t)X^2 + r^x(t)X + c^x(t)$ with time-dependent coefficients to be determined, the only x-dependence residing in r^x and c^x. Substituting this ansatz into the type-x HJB equation eq. (24.1) and matching powers of X — the coefficients of X^2, X^1 and X^0 must each vanish separately, since the equation holds for all X — separates the single PDE into three ODEs. The X^2 coefficient gives the *type-independent* scalar Riccati equation eq. (11.11),

$$\dot{p} = b^2p^2 - 2ap - q, \qquad p(T) = q_T, \tag{25.21}$$

type-independent because the quadratic-in-X part $\frac{q}{2}X^2$ of the cost is the same for every type — only the linear coupling carries x; the X^1 coefficient gives the linear ODE for r^x displayed next; and the X^0 coefficient fixes c^x, which does not enter the feedback. That feedback is the affine map

$$\widehat{\alpha}(t,x,X) = -\,b\big(p(t)X + r^x(t)\big). \tag{25.22}$$

The linear coefficient r^x and the type-x population mean $\overline{\mathbf{m}}_t(x) = \int \zeta\, m_t^x(\mathrm{d}\zeta)$ solve the coupled linear system across types (mirroring eq. (11.13) and its offset equation),

$$\dot{\overline{\mathbf{m}}}_t(x) = (a-b^2p)\,\overline{\mathbf{m}}_t(x) - b^2r_t^x, \qquad -\dot{r}_t^x = (a-b^2p)\,r_t^x - q\,s\,z_t(x), \qquad z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x), \tag{25.23}$$

with $\overline{\mathbf{m}}_0(x)$ given and $r^x(T) = -q_Ts_T\,z_T(x)$; the terminal datum is the X^1-coefficient of the terminal cost $g(X,z) = \frac{q_T}{2}(X - s_Tz)^2$, whose linear-in-$X$ part is $-q_Ts_T\,z\,X$. The structure of eq. (25.23) is the Thread-B fixed point in miniature and repays a word, because it is what the rank-one assumption is about to simplify. The mean $\overline{\mathbf{m}}_t(x)$ runs *forward* from its given initial profile — it is a population mean, transported by the closed-loop drift coefficient $a - b^2p$ and pushed by the offset r^x. The offset r_t^x runs *backward* from its terminal value — it is a value-function coefficient, anticipating the future cost of the coupling. The two are joined by the aggregate $z_t(x) = (\mathbb{W}\overline{\mathbf{m}}_t)(x)$, and it is this join that makes the display a genuine *fixed-point* system rather than a single ODE: each type's offset is forced by the kernel-weighted mean of *all other types*, so a priori eq. (25.23) is an infinite, coupled, forward–backward family indexed by the continuum $x \in [0,1]$. It is exactly here that the rank-one structure earns its keep. For the rank-one graphon $W(x,y) = f(x)f(y)$ of Thread B the aggregate collapses to a scalar order parameter: $z_t(x) = f(x)\,\varrho_t$ with $\varrho_t := \int_0^1 f(y)\overline{\mathbf{m}}_t(y)\,\mathrm{d}y$ (the ϱ of chapter 20, the rank-one order parameter). Because every appearance of x in the forcing of eq. (25.23) now enters through the single factor $f(x)$ — the aggregate is $z_t(x) = f(x)\varrho_t$ and the terminal datum is $-q_T s_T f(x)\,\|f\|_2^2\, M_T$ — the type profile cannot do anything but remain proportional to f for all time. Writing the resulting aligned ansatz $\overline{\mathbf{m}}_t(x) = f(x)M_t$, $r_t^x = f(x)P_t$ (self-consistent precisely when the initial profile is itself aligned, $\overline{\mathbf{m}}_0(x) = f(x)M_0$), and dividing out the common factor $f(x)$, the infinite system eq. (25.23) collapses to the scalar two-point boundary-value problem

$$\dot{M}_t = (a-b^2p)M_t - b^2P_t, \qquad -\dot{P}_t = (a-b^2p)P_t - q\,s\;\|f\|_2^2\;M_t, \qquad P(T) = -q_T s_T\,\|f\|_2^2\,M_T, \tag{25.24}$$

with $\varrho_t = \|f\|_2^2\;M_t$ the single self-consistent order parameter, $\|f\|_2^2 = \int_0^1 f^2$. This is the "1-parameter self-consistent equation" of the thread specification, the GMFG analogue of the Curie–Weiss closure.

Exact independence on the network. Substitute eq. (25.22) into eq. (25.19):

$$\mathrm{d}X_t^i = \big((a - b^2p(t))X_t^i - b^2 f(x_i)P_t\big)\,\mathrm{d}t + \sigma\;\mathrm{d}B_t^i. \tag{25.25}$$

The substitution is the decisive computational event of the worked example, so it is worth saying exactly why it empties the dynamics of the graph. The GMFG feedback eq. (25.22) is a function of the agent's *own* state X^i and *own* type x_i alone; it never reads a neighbour. Plugging it into eq. (25.19) replaces the control by a term involving only (X^i, x_i), and the resulting drift in eq. (25.25) contains no edge weight ξ_{ij} whatsoever. The network has *disappeared from the dynamics*: under the GMFG profile the N states obey N decoupled linear SDEs, each an Ornstein–Uhlenbeck process with the common closed-loop relaxation rate $a - b^2p(t)$ and a type-dependent forcing $-b^2 f(x_i)P_t$, driven by the mutually independent noises B^i and seeded by the independent draws $X_0^i \sim m_0^{x_i}$. With no coupling left in the equations, $X^1, \dots, X^N$ are *exactly* independent Gaussian processes at every finite N — there is simply nothing to correlate them. Being Gaussian, each X_t^i is pinned down by two deterministic functions of time. Its mean $\mu_t^i := \mathbb{E}X_t^i$ obeys the linear ODE one gets by taking expectations in eq. (25.25), $\dot{\mu}_t^i = (a - b^2p)\mu_t^i - b^2 f(x_i)P_t$, which is solved by $\mu_t^i = f(x_i)M_t = \overline{\mathbf{m}}_t(x_i)$ — the agent's mean is exactly its type's GMFG mean, as the fixed point requires. Its variance $\rho(t) := \operatorname{Var} X_t^i$ obeys the standard variance ODE for an OU process, $\dot{\rho} = 2(a - b^2p)\rho + \sigma^2$, $\rho(0) = \operatorname{Var}(m_0^{\cdot})$, and is *type-independent* because the relaxation rate and the noise intensity σ are shared by every agent — only the deterministic forcing carried the type, and deterministic forcing shifts the mean without touching the variance. The contrast with the general theory is the point to retain: where the proof of theorem 25.13 needed a propagation-of-chaos *estimate* carrying the $O(r_N)$ rate to decorrelate the agents approximately, Thread B realizes propagation of chaos in its most extreme form — *exact* independence, zero rate, no approximation — precisely because the coupling lives entirely in the cost and not in the dynamics. The whole network has been pushed into a single surviving place: the realized aggregate z_t^i the agents pay through, but no longer move under.

The aggregate discrepancy: three $O(N^{-1/2})$ terms. The network now re-enters in the only place left to it: the gap between the realized aggregate $z_t^i = \frac{1}{N}\sum_j \xi_{ij}X_t^j$ that agent i actually pays through and the deterministic continuum value $z_t(x_i) = f(x_i)\varrho_t = f(x_i)\,\|f\|_2^2\,M_t$ that the GMFG feedback was tuned against. This gap is exactly the Step-1 quantity of the proof of theorem 25.13, here available in closed form rather than through an abstract estimate. We expose its three sources by inserting two reference levels in turn — first the agents' means μ_t^j, then the kernel values $W(x_i, x_j)$ — and subtracting each back, against the GMFG value $z_t(x_i) = f(x_i)\varrho_t = f(x_i)\,\|f\|_2^2\,M_t$:

$$z_t^i - f(x_i)\varrho_t = \underbrace{\frac{1}{N}\sum_j \xi_{ij}(X_t^j - \mu_t^j)}_{\text{(I) state noise}} + \underbrace{\frac{1}{N}\sum_j(\xi_{ij} - W(x_i, x_j))\mu_t^j}_{\text{(II) edge sampling}} + \underbrace{f(x_i)M_t\Big(\frac{1}{N}\sum_j f(x_j)^2 - \|f\|_2^2\Big)}_{\text{(III) quadrature}}. \tag{25.26}$$

The three summands isolate three distinct mechanisms, and we bound each in L^2 conditionally on the labels $\boldsymbol{x}^N$ (and, for the state term, on the graph as well), where the relevant fluctuations are independent and the variances add. Throughout, the dense scaling $\bar{d}_i = \sum_j \xi_{ij} = \Theta(N)$ of definition 25.3 is what turns a sum of N small contributions into an $O(1/N)$ variance, and it is worth watching the powers of N cancel in each line.

Term (I), state noise. This is the fluctuation of the neighbours' states about their deterministic means, carried into i's aggregate by the edges. Given the graph and the means, the centred states $X_t^j - \mu_t^j$ are independent across j with the common variance $\rho(t)$ found above, so the variance of the weighted sum is the weighted sum of variances,

$$\operatorname{Var}\Big(\tfrac{1}{N}\sum_j \xi_{ij}(X_t^j - \mu_t^j)\Big) = \tfrac{1}{N^2}\sum_j \xi_{ij}^2\,\rho(t) \;\le\; \tfrac{\rho(t)\,\bar{d}_i}{N^2} \;=\; O\Big(\tfrac{1}{N}\Big),$$

using $\xi_{ij}^2 \le \xi_{ij}$ and the dense degree $\bar{d}_i = \sum_j \xi_{ij} = O(N)$. The N^2 in the denominator is the square of the $1/N$ normalization; the $\Theta(N)$ degree in the numerator cancels one power of it, leaving $O(1/N)$. This is nothing but the ordinary law-of-large-numbers averaging of $\Theta(N)$ independent neighbour states.

Term (II), edge sampling — the genuinely new term. This is the contribution that has no counterpart in the homogeneous theory of chapter 13: the edges themselves are random, and replacing the realized ξ_{ij} by their kernel means $W(x_i, x_j)$ leaves a residual. Conditionally on the labels and on the graph-independent means μ^j, the increments $\xi_{ij} - W(x_i, x_j)$ are independent centred Bernoulli fluctuations, each of variance $W(x_i, x_j)(1 - W(x_i, x_j)) \le \frac{1}{4}$, so once more the variances add,

$$\operatorname{Var}\Big(\tfrac{1}{N}\sum_j(\xi_{ij} - W(x_i, x_j))\mu_t^j\Big) = \tfrac{1}{N^2}\sum_j W(x_i, x_j)(1 - W(x_i, x_j))(\mu_t^j)^2 \;=\; O\Big(\tfrac{1}{N}\Big),$$

again $O(1/N)$ because the N bounded summands are scaled by $1/N^2$. Each edge contributes one mean-zero coin flip of $O(1)$ amplitude and weight $1/N$, and the $\Theta(N)$ of them average out at the Monte-Carlo rate. The point to carry forward is that this single new error channel, the price of the wiring being sampled rather than known, is itself $O(N^{-1/2})$ in L^2 — no worse than the state noise (I).

Term (III), quadrature. This term is deterministic given the labels: it is the error of the empirical average $\frac{1}{N}\sum_j f(x_j)^2$ as an estimate of $\|f\|_2^2 = \int_0^1 f^2$, scaled by the bounded deterministic factor $f(x_i)M_t$. For i.i.d. labels it is the Monte-Carlo error of an average of N independent copies of $f(x_j)^2$, of size $O(N^{-1/2})$; for the regular grid it is a midpoint-rule quadrature error, the faster

$O(N^{-1})$ under the regularity of f supplied by (A-Reg-W) (chapter 19). Either way it is no larger than $O(N^{-1/2})$, and on the grid it is negligible against the other two.

Collecting the three independent (or deterministic) contributions, their variances add and the worst of them is $O(1/N)$, uniformly in the vertex i because every bound above held for a fixed but arbitrary i:

$$\sup_i \mathbb{E}\left|z_t^i - z_t(x_i)\right|^2 \le \frac{C(t)}{N}. \tag{25.27}$$

The regret is $O(1/N)$. Freeze all $j \neq i$ at eq. (25.22) and let agent i minimize eq. (25.20) against the random tracking target $s\, z_t^i$. By the completion-of-squares identity for scalar LQ tracking (the verification computation of chapter 6, used identically in worked computation 13.8), the regret of any admissible η against the optimal tracker $\alpha^{i,\star}$ is the exact quadratic $J^{N,i}(\eta, \widehat{\alpha}^{-i}) - \inf_\alpha J^{N,i}(\alpha, \widehat{\alpha}^{-i}) = \frac{1}{2}\mathbb{E}\int_0^T |\eta_t - \alpha_t^{i,\star}|^2\, dt$, with no first-order term. Taking $\eta = \widehat{\alpha}^i$, the control gap is forced only by the target discrepancy $s\, z_t^i - s\, z_t(x_i) = s\,(z_t^i - z_t(x_i))$: agent i's own optimal linear offset $r^{i,\star}$ tracks $\mathbb{E}[s\, z_\cdot^i \mid \mathcal{F}_t]$ in place of the deterministic $s\, z_\cdot(x_i)$, so $\widehat{\alpha}_t^i - \alpha_t^{i,\star} = b(r_t^{i,\star} - f(x_i)P_t)$ with $\mathbb{E}\int_0^T |r_t^{i,\star} - f(x_i)P_t|^2\, dt \le C\, \mathbb{E}\int_0^T |z_t^i - z_t(x_i)|^2\, dt = O(1/N)$ by eq. (25.27). Therefore

$$\varepsilon_N^{\mathrm{B}} = \sup_i \Big[J^{N,i}(\widehat{\alpha}, \widehat{\alpha}^{-i}) - \inf_\alpha J^{N,i}(\alpha, \widehat{\alpha}^{-i})\Big] \le \frac{C}{N}, \qquad C = \frac{C' s^2 q}{2} \sup_{t \le T} \left[\rho(t) + \sup_x f(x)^2\right]. \tag{25.28}$$

Interpretation. The generic network bound eq. (25.16) promises only $\varepsilon_N = O(r_N + \gamma_N)$; here the Gaussian structure makes this concrete and sharp. Rather than passing through the loose Fournier–Guillin rate r_N (which, recall from remark 25.9, is $O(N^{-1/4})$ in $d = 1$ by the book's bound eq. (13.12)), the explicit computation eq. (25.27) shows the aggregate fluctuation is *exactly* $O(N^{-1/2})$ in L^2; entering the cost quadratically, it then contributes only $O(1/N)$ to the regret. Crucially the *edge-sampling* fluctuation (II), the genuinely new network contribution absent in chapter 13, is upgraded along with the others, since it too is $O(N^{-1/2})$ in L^2 and is squared. The order parameter ϱ_t is the network's single scalar summary, and the ε-Nash rate is governed by how the cost responds to fluctuations of ϱ_t — the same envelope (first-order optimality) mechanism as in the homogeneous thread. Two caveats keep the result honest. First, the $O(1/N)$ edge bound uses the *dense* normalization $\bar{d}_i = O(N)$; in the sparse regime of chapter 27, where $\bar{d}_i = o(N)$, term (II) has variance $O(1/\bar{d}_i)$ and the rate degrades to $\varepsilon_N = O(1/\bar{d}_i)$, which can be far worse. Second, for *unlabelled* networks the per-vertex decomposition eq. (25.26) is unavailable (lemma 25.7 controls only type-averages, as in theorem 25.10); the type-averaged regret then falls back to $O(\delta_\square(W_N, W)^2)$ inside the quadratic, i.e. $O(1/\log N)$ in the worst case — the bottleneck of remark 25.9 surviving even the LQG sharpening.

25.5 Stability under graph perturbation in the cut metric

The ε-Nash theorem licensed designing on the limit and deploying on the finite graph; its second thread, deferred to here, was *robustness* — the guarantee that the equilibrium one designs survives the part of the wiring no measurement can pin down. This section banks it, and in doing so collects the final payoff of chapter 17: *macroscopically equivalent networks yield close equilibria.* It is what makes the GMFG a robust design tool — one need not know the network exactly, only up to the features the cut metric sees. We state two versions, matching the two strengths of metric: the cut metric $\delta_\square$, which controls type-averaged quantities (and is all that is identifiable from data, chapter 17), and the operator norm, which controls pointwise-in-type quantities (chapter 18).

Theorem 25.16 (Cut-metric stability of GMFG aggregates)**.** *Assume* (A-Lip)*,* (A-Nondeg)*,* (A-Conv)*,* (A-Reg-W)*, and* (A-Coupling) *in the contraction regime of chapter 24, with bounded interaction observable $\|h\|_\infty \le M$. Let W, W' be two graphons in the regime, with GMFG solutions (u^x, m^x) and (u'^x, m'^x) and aggregates z_t, z'_t. Then for every bounded test $f : [0,1] \to [-1,1]$ and every t,*

$$\Big| \int_0^1 f(x)\big(z_t(x) - z'_t(x)\big)\,\mathrm{d}x \Big| \;\le\; C\,\delta_\square(W, W'), \tag{25.29}$$

and consequently any type-averaged equilibrium functional that depends on the solution through the aggregate — the aggregate social cost $\int_0^1 \langle u^x(0,\cdot), m_0^x\rangle\,\mathrm{d}x$, the average equilibrium effort $\int_0^1 \mathbb{E}|\widehat{\alpha}(t,x,X_t^x)|\,\mathrm{d}x$ — differs between W and W' by at most $C\,\delta_\square(W,W')$, with C depending on the contraction constant of chapter 24 but not on the kernels.

Proof. The operative norm is the *weak (dual) norm* $\|g\|_* := \sup_{f:[0,1]\to[-1,1]} \big|\int_0^1 f(x)g(x)\,\mathrm{d}x\big|$ — precisely the type-averaged, test-integrated quantity that lemma 25.7 controls. In the contraction regime the GMFG map $\Phi_W : \bar{h} \mapsto (u, m) \mapsto \bar{h}$ is a contraction whose fixed point is the solution, and it depends on the kernel only through the aggregation $\bar{h} \mapsto \mathbb{W}\bar{h}$ (proposition 24.18). The contraction estimate eq. (24.7) factorises as $\Phi_W = \mathcal{R} \circ \mathcal{A}_W$ with $\mathcal{A}_W$ linear; since the linear aggregation and the per-type response $\mathcal{R}$ are bounded operators on the weak norm with the same modulus $L = C_T\,\|\mathbb{W}\| < 1$ as in proposition 24.18, Φ_W is an L-contraction in $\|\cdot\|_*$. Let $\bar{h}_t, \bar{h}'_t$ be the fixed-point observable fields for W, W'. Subtracting the two fixed-point identities $\bar{h} = \Phi_W(\bar{h})$, $\bar{h}' = \Phi_{W'}(\bar{h}')$ and testing against $|f| \le 1$ throughout,

$$\left\|\bar{h}_t - \bar{h}'_t\right\|_* \;\le\; L\,\left\|\bar{h}_t - \bar{h}'_t\right\|_* \;+\; C \sup_{f:|f|\le 1} \Big| \int f\big[(\mathbb{W} - \mathbb{W}')\bar{h}'_t\big]\Big|,$$

so $\left\|\bar{h}_t - \bar{h}'_t\right\|_* \le \frac{C}{1-L}\sup_f |\int f(\mathbb{W}-\mathbb{W}')\bar{h}'_t|$. Since $\left\|\bar{h}'_t\right\|_\infty \le M$ (bounded observable), lemma 25.7 with $g = g' = \bar{h}'_t$ bounds the right side by $4M\,\|W - W'\|_\square$; minimizing over measure-preserving relabellings (under which both the GMFG solution and the right-hand side are invariant, chapter 17) replaces $\|W - W'\|_\square$ by $\delta_\square(W, W')$. This is eq. (25.29). The reason the conclusion is only type-averaged — not pointwise in type — is now transparent: the weak norm $\|\cdot\|_*$ is all that $\delta_\square$ feeds, and it sees the aggregate only through its integral against test functions. The functional statements follow because each listed functional is a type-average of a Lipschitz function of the aggregate. □

Theorem 25.17 (Operator-norm stability of GMFG fields)**.** *Under the hypotheses of theorem 25.16, but measuring the kernel perturbation in operator norm, the GMFG fields are close* pointwise in type*:*

$$\sup_{x\in[0,1]} \mathcal{W}_2(m_t^x, m_t'^x) \;\le\; C\,\|\mathbb{W} - \mathbb{W}'\|_{L^2\to L^2}\,, \qquad t \in [0,T], \tag{25.30}$$

with C the contraction constant of chapter 24.

Proof sketch. Identical to theorem 25.16 but with the contraction run in the stronger norm $\left\|\bar{h}\right\|_{L^2} = (\int_0^1 |\bar{h}(x)|^2 \, \mathrm{d}x)^{1/2}$ (equivalently, the uniform-in-type $\sup_x$ control its contraction delivers on $\mathbb{K}_W$): the kernel now enters through $\mathbb{W}\bar{h}_t$ bounded *without* test-integration, $\left\|(\mathbb{W} - \mathbb{W}')\bar{h}_t\right\|_{L^2} \leq \|\mathbb{W} - \mathbb{W}'\|_{L^2 \to L^2} \left\|\bar{h}_t\right\|_{L^2}$, so the perturbation is controlled pointwise in type rather than only in the weak norm $\|\cdot\|_*$. Running the same fixed-point subtraction as in theorem 25.16, now in $\|\cdot\|_{L^2}$ with the contraction proposition 24.18 (whose modulus $C_T \|\mathbb{W}\| < 1$ is an $L^2 \to L^2$ statement), propagates the operator-norm perturbation to the fields pointwise in type; (A-Coupling) guarantees the contraction. The trade-off is exactly the cut-versus-operator dichotomy of chapters 17 and 18: the operator norm is stronger (it controls every type, not only averages) but is *not* identifiable from a single sampled graph at the cut rate, whereas $\delta_\square$ is identifiable but controls only averages (the spectral comparison of the two is chapter 18). $\square$

25.5.1 Existence for merely measurable graphons

The cut-stability machinery discharges a debt incurred in chapter 24. There (remark 24.13, and the ledger entry deferring it) existence of a GMFG solution for a *merely measurable* kernel W — the generic case, where continuity in type fails and the Schauder route of theorem 24.11 does not directly apply — was deferred to "the cut-metric convergence theory" of this chapter. We now supply it in the contraction regime, where the cut-stability estimate makes the step-graphon approximation converge.

Proposition 25.18 (Existence and uniqueness for measurable graphons)**.** *Let $W : [0,1]^2 \to [0,1]$ be a symmetric* measurable *graphon (dropping the continuity clause of* (A-Reg-W)*) with $\|\mathbb{W}\|_{L^\infty \to L^\infty} < \kappa = 1/C_T$, the contraction threshold of proposition 24.18, and assume* (A-Lip), (A-Nondeg), (A-Conv)*, assumption 24.5. Then the GMFG has a* unique *equilibrium solution $(u^x, m^x)_{x \in [0,1]}$ in the space of* measurable *fields valued in $\mathbb{K}_W$. Moreover, if $W_n := \mathbb{E}[W \mid \mathcal{P}_n]$ is the step graphon obtained by averaging W over a refining sequence of finite partitions $\mathcal{P}_n$ of $[0,1]$ with mesh $\to 0$, then $\delta_\square(W_n, W) \to 0$ and the block solutions of W_n converge to that of W in the type-averaged sense of theorem 25.16.*

Proof. Existence and uniqueness. The contraction estimate proposition 24.18 uses the kernel only through $\|\mathbb{W}\|_{L^\infty \to L^\infty} = \operatorname{ess\,sup}_x \int_0^1 |W(x,y)| \, \mathrm{d}y \leq 1$, a quantity that requires no continuity in type; its proof (factorisation $\mathsf{S}_W = \mathcal{R} \circ \mathcal{A}_W$, with $\mathcal{A}_W$ linear and bounded and $\mathcal{R}$ per-type Lipschitz) is unchanged when the ambient field space $(\mathbb{M}, \mathsf{D})$ of definition 24.1 is taken in its *measurable* variant — measurable fields valued in the per-type target, the complete space $L^2([0,1]; \mathcal{K})$ of remark 24.13. Indeed $\mathbb{W}$ is Hilbert–Schmidt (hence bounded, even compact) on L^2 for any measurable W ((A-Reg-W)'s self-adjoint Hilbert–Schmidt clause survives the loss of continuity), and the per-type optimiser is selected measurably by the measurable-selection theory of lemma 10.20, so S_W maps measurable fields to measurable fields. With $\|\mathbb{W}\| < \kappa$, S_W is an L-contraction with $L = C_T \|\mathbb{W}\| < 1$ on this complete space, and the Banach contraction principle (theorem 2.36) gives a unique fixed point — the GMFG solution. Continuity in type was therefore never needed for uniqueness, only for the compactness in the sign-indefinite case; the contraction regime sidesteps it.

Identification as the cut-limit of block problems. The conditional expectations $W_n = \mathbb{E}[W \mid \mathcal{P}_n]$ form an $L^1([0,1]^2)$-bounded martingale along the refining filtration, so $W_n \to W$ in L^1 (martingale convergence / Lebesgue differentiation), whence $\delta_\square(W_n, W) \le \|W_n - W\|_\square \le \|W_n - W\|_1 \to 0$. Each W_n is a block (piecewise-constant) graphon, so its GMFG reduces to a finite multi-population problem (the block reduction of remark 24.14); moreover the conditional-expectation averaging is an $L^\infty \to L^\infty$ contraction, $\|\mathbb{W}_n\| \le \|\mathbb{W}\| < \kappa$, so each W_n is itself in the contraction regime and the first part of this proof already gives it a *unique* solution. Applying theorem 25.16 to the pair (W_n, W) gives, for every bounded test f, $|\int f(z_t^{(n)} - z_t)| \le C\,\delta_\square(W_n, W) \to 0$: the block aggregates converge to the measurable-W aggregate in the weak norm, and every type-averaged equilibrium functional of W_n converges to that of W. This realises the measurable solution as the type-averaged limit of explicitly computable block solutions, completing the program announced in remark 24.13. $\square$

Remark 25.19 (Why this is the right robustness statement)**.** theorem 25.16 is the formal content of "two networks that look the same at every macroscopic scale induce the same equilibrium behaviour." It is the GMFG counterpart of the cut-metric equivalence of chapter 17: if an analyst replaces a real network by a coarse model — a stochastic block model, a weak-regularity partition, an estimated graphon — that is $\delta_\square$-close to the truth, the type-averaged equilibrium predictions (welfare, average intervention effect, aggregate effort) are guaranteed close. The pointwise-in-type predictions require the stronger operator-norm closeness of theorem 25.17, which a single graph sample does not deliver at the $1/\sqrt{\log N}$ cut rate but which holds when the latent positions are known and the operator is estimated directly. Together with the ε-Nash theorem theorem 25.13, this completes the justification of GMFG-based intervention design (chapter 20) on real, finite, imperfectly known networks: the strategy is approximately optimal (theorem 25.13) and robust to the unidentifiable part of the wiring (theorem 25.16).

Remark 25.20 (The bottom-up direction)**.** We have emphasized the top-down (ε-Nash) and stability directions, the ones that are both cheap and practically decisive. The bottom-up direction — that the *actual* Nash equilibria of the network game eq. (25.6) converge to the GMFG — holds under the same provisos as its homogeneous ancestor. The qualitative statement (limit points of network equilibria are GMFG solutions; under (A-Mon) the limit is the unique GMFG and the type-marginals propagate chaos) follows by the compactness/martingale-problem method of theorem 13.9, lifted to the field-of-measures space and conditioned on the graph — this in-book argument is what supplies the network propagation of chaos in the general (non-contraction) regime, the graphon-system law of large numbers of [15, 16] being its published form and [57] its Erdős–Rényi prototype. (In the contraction regime no such general statement is needed: theorem 25.10 already gives the rate from in-chapter pieces, as detailed in its proof.) The quantitative rate would require a *graphon master equation*, which is the subject of chapter 26; in the contraction regime, theorem 25.10 already provides the rate for the decentralized profile, and the value-versus-trajectory dichotomy of remark 13.14 persists unchanged, now with the graph error γ_N added to the sampling error in the trajectory rate while the value rate remains the faster $1/N$ wherever a regular graphon master field is available (chapter 26).

Rigor ledger

Proved (in full, in this chapter).

- The cut-norm aggregation lemma lemma 25.7: the coupling operator's action on bounded type-averaged functionals is controlled by $\delta_\square$ plus the observable sup-distance — the technical engine of both convergence and stability.

- The network ε-Nash property theorem 25.13 with rate $\varepsilon_N \le C(r_N + \gamma_N)$, via top-down propagation of chaos for the fixed GMFG feedback; and its Thread B sharpening to $\varepsilon_N \le C/N$ with explicit constant (worked computation 25.15, eq. (25.28)), including the upgrade of the new edge-sampling term.

- Cut-metric stability of type-averaged GMFG aggregates (theorem 25.16) and operator-norm stability of the pointwise-in-type fields (theorem 25.17) in the contraction regime, and — as their corollary — existence and uniqueness of the GMFG for merely *measurable* graphons in that regime (proposition 25.18), discharging the obligation deferred from remark 24.13.

- The contraction-regime network propagation of chaos underlying theorem 25.10: *not* an external import, but assembled in-chapter from the synchronous coupling and $\mathcal{W}_2$-sampling rate of lemma 13.5 (term (I), (III)), the edge-sampling estimate proposition 25.8(1) (term (II)), and the in-chapter kernel stability theorems 25.16 and 25.17. The type-resolved field convergence corollary 25.11.

Assumed (imported, with in-book proof).

- The GMFG system, its solution (theorem 24.11), and the contraction estimate proposition 24.18 for the best-response map in the field argument (chapters 22 to 24); the graphon McKean–Vlasov dynamics (chapter 23); the standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), (A-Reg-W), (A-Coupling).

- The cut norm / $\infty \to 1$ equivalence (chapter 17), the second sampling lemma $\delta_\square(W_N, W) \le C/\sqrt{\log N}$ (remark 19.5 and theorem F.9) and the bounded-differences inequality lemma 19.3 (chapter 19); the Fournier–Guillin sampling rate r_N, proved in-book at remark 7.16 and imported via lemma 13.5 (attribution [69]); Itô's formula theorem 5.9 and Grönwall's inequality lemma C.1.

Deferred (stated here, proved at an in-book location).

- The *general* (non-contraction) network law of large numbers for graphon systems — needed only for the qualitative bottom-up convergence beyond the contraction regime (remark 25.20) — via the compactness/martingale-problem method of theorem 13.9 lifted to the field of measures (attribution [15, 16, 57, 89, 90, 34]).

- The weak-regularity cut-sampling rate eq. (25.10) and the cut/operator comparison: proved in remark 19.5 and theorem F.9 and chapters 17 and 18 (attribution [99]).

Open.

- A graphon master equation delivering a $1/N$ *value* rate for the bottom-up convergence on networks, analogous to theorem 13.13; the well-posedness this requires is the frontier of chapter 26.
- Commutation of the two limits and joint rates *outside* the contraction/monotone regime, near the critical coupling $\|\mathbb{W}\| \uparrow \kappa$ where the limit equilibrium may be non-unique (remark 25.12, chapters 24 and 31).
- Sharp ε-Nash and convergence rates in the *sparse* regime, where the dense normalization fails and the edge-sampling error is $O(1/\bar{d}_i)$ rather than $O(1/N)$ (chapter 27); and identifiability of the operator-norm stability of theorem 25.17 from data.

Bibliographic notes

The graphon mean field game and the convergence of finite network games to it originate with Caines and Huang [33, 34], who introduced the GMFG equations and established the ε-Nash property of the GMFG strategy on large finite networks — the engineering lineage of the Nash certainty equivalence principle [82] that produced the homogeneous ε-Nash theorem of chapter 13. The static shadow of the theory, including the centrality-to-strategy and intervention program that section 25.5 underwrites, is Parise and Ozdaglar [108] and the stochastic graphon games of Carmona, Cooney, Graves and Laurière [39, 10].

The network propagation of chaos — the law of large numbers that powers the joint convergence theorem 25.10 — is the graphon-system theory of Bayraktar, Chakraborty, Wu and Zhang [15, 16]; the prototype, an MFG-style analysis on Erdős–Rényi graphs, is Delarue [57]. The cut metric, its $\infty \to 1$ comparison eq. (25.7) underlying the aggregation lemma lemma 25.7, and the sampling rate eq. (25.10) are from Lovász [99, 100] and Borgs, Chayes, Lovász, Sós and Vesztergombi [29], developed in chapters 17 and 19, with the cut-norm/operator origins in Frieze and Kannan [72] and the concentration tool in McDiarmid [102]. The Wasserstein sampling rate r_N is Fournier and Guillin [69].

The two-limits structure — mean field versus graph convergence, and when they commute — and the practical use of the GMFG to approximate Nash equilibria on sampled graphs is also the organizing principle of the learning literature, where [55] quantifies approximate Nash on $G(N, W)$ in a reinforcement-learning setting (chapter 29). The homogeneous backbone we generalize throughout — the Nash system, the ε-Nash mechanism, the qualitative and master-equation convergence routes — is chapter 13, following Carmona and Delarue [40, 41, Vol. I–II], Lacker [89, 90], and Cardaliaguet, Delarue, Lasry and Lions [37]. The graphon master equation that would upgrade the bottom-up value rate to $1/N$ on networks, and the common-noise extension, are the subject of chapter 26; the sparse-network regime where the dense normalization of definition 25.3 fails is chapter 27; and the modern map of which extensions the graphon picture subsumes is the review [11] discussed in chapters 30 and 31.

Chapter 26

The Graphon Master Equation and Common Noise

The classical master equation of chapter 12 compressed an entire mean field game — a backward Hamilton–Jacobi–Bellman equation coupled forward to a Fokker–Planck equation, or equivalently a McKean–Vlasov forward–backward system — into a single scalar equation $U(t,x,m)$ on $[0,T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d)$, the equilibrium cost-to-go of an agent at state x when the population is m. This chapter performs the same compression for the *graphon* mean field game built up in chapters 21 and 22: a continuum of types $x \in [0,1]$, each carrying a private state and an SDE, coupled not to a single population measure but to a *field of measures* $\boldsymbol{m} = (m^x)_{x\in[0,1]}$ through the graphon operator $\mathbb{W}$ of definition 15.13. The unknown becomes a function $U(t,x,\xi,\boldsymbol{m})$ — the cost-to-go of a type-x agent at private state ξ when the entire population field is $\boldsymbol{m}$ — and the master equation is now posed over the field-of-measures space of chapter 21.

It is worth being explicit about what a compression of this kind *buys*, since that return is the whole motivation for the labour to come. The forward–backward system of chapter 22 is shackled to one initial population: fix $\boldsymbol{m}_0$, solve, and you learn the equilibrium along the single trajectory issuing from that datum and nothing about any other. The master function instead encodes *all* initial conditions simultaneously — it is the flow map of the equilibrium on the space of populations — so that the whole family of forward–backward solutions is recovered by reading U along one characteristic and differentiating once. That single object is what makes contingency tractable: it can answer "what would the equilibrium do were the population suddenly some *other* field?", a question the bare value family cannot even pose, because that family is defined only along the one realized flow. Section 26.1 makes this obstruction precise, and section 26.5 turns the all-contingencies object from a convenience into a necessity.

Two things make the lift worth a chapter of its own rather than a remark appended to chapter 12. First, the measure argument is no longer a point of $\mathcal{P}_2(\mathbb{R}^d)$ but a *type-indexed field*; differentiation in it acquires an extra integral over the type label, and the graphon $\mathbb{W}$ threads through the coupling in a way that makes the well-posedness theory genuinely harder than the classical one and, we will admit honestly, less complete — the general classical regularity theory that chapter 12 could lean on has no graphon counterpart of comparable strength, and section 26.4 is candid about exactly which steps survive the lift and which remain open. Second, and decisively, this is the chapter where *common (aggregate) noise* enters: a shock shared across all types turns the population field into a random process, *dissolves* the deterministic forward–backward system of chapter 22 — there is no longer any single field flow to solve for in advance — and leaves the master equation as the only surviving formulation, exactly as foreshadowed in section 12.7, now with the additional structure that the common noise couples *all types coherently* and the field calculus acquires a genuine second-order operator over the continuum of types.

The chapter therefore has two movements, and it helps to hold both in view from the start. The first (sections 26.1 to 26.4) is a *faithful, type-by-type translation* of chapter 12: every classical master-equation object — the flow-consistency identity, the Itô–Dynkin chain rule, the HJB-plus-transport structure, the characteristics and the decoupling field — survives the lift, acquiring only one new ingredient, an integral over the type label y. The second (section 26.5) is genuinely new physics: under a common shock the deterministic system has no meaning, and the master equation grows a double-type-integral second-field-derivative Laplacian for which no classical-graphon shortcut is known. The structural fact that organizes both movements, and that the reader should carry throughout, is a single one: *one type's cost sees another type's population only through the kernel W*. Everything graphon-specific below — why the equation fails to decouple across types, why monotonicity becomes a spectral question about $\mathbb{W}$, why common noise binds the types together — traces back to that one channel.

Throughout, value functions are solved backward from terminal data at $t = T$ and densities evolve forward from initial data at $t = 0$ (appendix A.1); the master equation inherits a terminal condition. We import from chapter 12 the entire apparatus of calculus on $\mathcal{P}_2(\mathbb{R}^d)$ — the flat and Lions derivatives (definition 12.1, appendix A.3) and the Itô–Dynkin chain rule along a Fokker–Planck flow (lemma 12.2) — and lift each piece type-by-type. We import from chapter 22 the graphon MFG system, from chapter 24 the existence/uniqueness theory and the operator-norm threshold (A-Coupling), and we restate the objects we need rather than assume the reader has both chapters open. The carried graphon thread (Thread B, appendix A.7) is advanced to its master-equation layer in section 26.6, where the abstract equation collapses — through the rank-one graphon of example 15.11 — to a single scalar Riccati equation in an order parameter; it is the cleanest computation in the chapter and the one place where every claim below holds unconditionally, and the reader is invited to keep it in mind as the concrete payoff toward which the abstract development is working.

Notation watch

Two reminders, both load-bearing for every equation that follows. *(i) Types versus states.* The variables $x, y \in [0,1]$ are *types* (exchangeable labels, appendix A.5); the private state of an agent lives in $\mathbb{R}^d$ and is written ξ, with $\zeta \in \mathbb{R}^d$ a second (integration) state variable. Thus $\nabla_\xi U$ and $\mathrm{D}^2_\xi U$ are derivatives in the agent's own state, never in the type. In chapter 12 the state was called x; here x is the type and ξ is the state — a deliberate reuse, because the graphon type plays exactly the role the single state played one level down, and the equations below read most cleanly when x is heard as "the chapter 12 label, promoted to a network position." *(ii) Brownian motion versus graphon.* As announced at the start of Part IV (appendix A.5), in every equation containing a graphon W the letter B denotes

Brownian motion: B_t^x is the private noise of type x, and B_t^0 (introduced in section 26.5) is the common noise. The kernel W is never a Brownian motion in this chapter. Neither convention is cosmetic: the type integral $\int_0^1 \cdots \mathrm{d}y$ and the symbol B recur in essentially every display below, and a single misreading of x as a state or of W as a noise derails the whole calculation.

26.1 The field-of-measures state and the non-Markovian obstruction

The preamble promised to perform the compression of chapter 12 one level up, replacing the single population measure by a field of measures. Before we can build the master function that effects it, we must see precisely *why* it is needed — what goes wrong if one tries to make do with the bare value functions produced by the graphon MFG system of chapter 22. The diagnosis is the one that forced a master equation in the homogeneous theory (section 12.1), but transported to the field setting it sharpens into a statement about what the genuine Markov state of a graphon game is. We first recall just enough of chapters 21 and 22 to fix notation, then exhibit the obstruction.

The population is a *field of probability measures*

$$\boldsymbol{m} = (m^x)_{x\in[0,1]}, \qquad x \longmapsto m^x \in \mathcal{P}_2(\mathbb{R}^d),$$

a measurable map from the type interval into Wasserstein space: to each type x it assigns the law m^x of the private states of the agents of that type. Where the homogeneous theory had a single measure m, the graphon theory has a whole continuum of them indexed by x, and it is this resolution of the population type-by-type that every later structure will have to carry. The natural state space for the equilibrium analysis (chapter 21) is the set

$$\boldsymbol{\mathcal{P}}_2 \;:=\; \Big\{ \boldsymbol{m} : [0,1] \to \mathcal{P}_2(\mathbb{R}^d) \text{ measurable } : \; \textstyle\int_0^1\int_{\mathbb{R}^d} |\zeta|^2 \; m^x(\mathrm{d}\zeta)\; \mathrm{d}x < \infty \Big\}, \tag{26.1}$$

the fields whose type-averaged second moment is finite, metrized by the L^2-Wasserstein distance over types (the field metric d_2 of chapter 21, eq. (21.10)),

$$d_2(\boldsymbol{m},\boldsymbol{m}')^2 \;:=\; \int_0^1 \mathcal{W}_2(m^x,(m')^x)^2 \;\mathrm{d}x. \tag{26.2}$$

The reading is the natural one: two fields are close when, *averaged over types*, their type-slices are close in the Wasserstein metric of chapter 4. The square root of an average of squared $\mathcal{W}_2$ distances is exactly the L^2-over-types structure that makes $\boldsymbol{\mathcal{P}}_2$ a metric space on which the calculus of the next section can be built.

A type-x agent controls a private diffusion and feels the population only through the graphon aggregate. In the canonical setting of chapter 22, where the control is the drift, the agent's dynamics and the coupling are

$$\mathrm{d}X_t^x \;=\; \alpha_t^x \;\mathrm{d}t \;+\; \sigma \;\mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x, \tag{26.3}$$

with σ constant and nondegenerate (A-Nondeg) — the nondegeneracy is what later supplies the parabolic smoothing in ξ that the well-posedness estimates lean on — and the running and terminal costs couple to $\boldsymbol{m}$ *only through the graphon operator*. The aggregate felt by type x is assembled in two steps. First, for each foreign type y one forms the scalar average of a fixed bounded observable ϕ against that type's slice, $\langle \phi, \boldsymbol{m}\rangle(y) = \int_{\mathbb{R}^d} \phi(\zeta)\, m^y(\mathrm{d}\zeta)$; this collapses each entire type-slice to a single number. Then one averages those numbers over types, weighted by the kernel $W(x, y)$ that records how strongly type x is exposed to type y:

$$z[\boldsymbol{m}](x) \;:=\; (\mathbb{W}\,\langle \phi, \boldsymbol{m}\rangle)(x) \;=\; \int_0^1 W(x,y) \Big(\int_{\mathbb{R}^d} \phi(\zeta)\, m^y(\mathrm{d}\zeta) \Big)\, \mathrm{d}y, \tag{26.4}$$

for a fixed bounded observable ϕ (the interaction observable h of definition 25.3; for Thread B, $\phi(\zeta) = \zeta$, so that $z[\boldsymbol{m}](x) = (\mathbb{W}\bar{m})(x)$ with $\bar{m}(y) = \int \zeta\, m^y(\mathrm{d}\zeta)$ the type-y mean — the aggregate is then simply the W-weighted average of the foreign type means). The costs are functions of the agent's type, its own state, and this single aggregate number:

$$F(x, \xi, \boldsymbol{m}) = \widetilde{F}(x, \xi, z[\boldsymbol{m}](x)), \qquad G(x, \xi, \boldsymbol{m}) = \widetilde{G}(x, \xi, z[\boldsymbol{m}](x)). \tag{26.5}$$

This is the structural punchline to keep in mind throughout the chapter. The field $\boldsymbol{m}$ enters the cost of a type-x agent *only* through the scalar $z[\boldsymbol{m}](x)$, and that scalar is built by the operator $\mathbb{W}$: the only route by which one type's cost sees another type's population is the kernel W. That single fact is what distinguishes the graphon theory from a disjoint union of classical MFGs — it is the thread the whole chapter is strung on, and every later display in which the types fail to separate can be traced back to it.

The graphon MFG system of chapter 22 is then the type-indexed forward–backward system: for every $x \in [0, 1]$,

$$-\partial_t u^x - \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top \mathrm{D}^2_\xi\, u^x\big) + H\big(x, \xi, \nabla_\xi u^x\big) = F(x, \xi, \boldsymbol{m}_t), \quad u^x(T, \xi) = G(x, \xi, \boldsymbol{m}_T), \tag{26.6a}$$

$$\partial_t m^x_t - \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top \mathrm{D}^2_\zeta\, m^x_t\big) - \mathrm{div}\big(m^x_t\, \partial_p H(x, \cdot, \nabla_\xi u^x)\big) = 0, \qquad m^x_0 \text{ given}, \tag{26.6b}$$

a backward HJB equation (26.6a) for the type-x value, run from terminal data, coupled to a forward Fokker–Planck equation (26.6b) for the type-x density, run from initial data, with each type carrying its own copy of the pair. Each ingredient draws on a standing hypothesis, and it is worth naming where. The Hamiltonian $H(x, \xi, p)$ is the type-x Hamiltonian, convex in p under (A-Conv); convexity is what makes the minimizing control single-valued, so that the optimal feedback $\alpha^*(x, \xi, p) = -\partial_p H(x, \xi, p)$ (appendix A.4) is a well-defined function and the very same drift $-\partial_p H$ can be substituted into the Fokker–Planck equation (26.6b) that closes the loop. The aggregate $z[\boldsymbol{m}_t](x)$ entering F, G through (26.5) is well-defined and depends regularly on the type x because the kernel is admissible (A-Reg-W): this is what makes $\mathbb{W}$ a bounded operator on $L^2([0, 1])$ (definition 15.13), so the type-integral in (26.4) converges and varies measurably in x. Existence of a solution for each initial field $\boldsymbol{m}_0 \in \boldsymbol{\mathcal{P}}_2$ then holds under the standing assumptions together with (A-Reg-W), and uniqueness for every horizon requires in addition the graphon monotonicity (A-Mon) and the operator threshold (A-Coupling); this is the existence/uniqueness theory of chapter 24, which we assume throughout so that "the equilibrium from $\boldsymbol{m}_0$" names a single object.

The types are coupled *only* through the appearance of the whole field $\boldsymbol{m}_t$ in F, G via (26.5), and it is instructive to watch that coupling switch off. Remove it in either of the two ways the structure allows. If $W \equiv 0$, then $z[\boldsymbol{m}](x) = 0$ for every x and every field, so $F(x, \xi, \boldsymbol{m}) = \widetilde{F}(x, \xi, 0)$ and $G(x, \xi, \boldsymbol{m}) = \widetilde{G}(x, \xi, 0)$ no longer depend on $\boldsymbol{m}$ at all; the type-x HJB equation then has a source fixed independently of the other types, its feedback drives a Fokker–Planck equation that involves only u^x, and the system splits into $|[0,1]|$ disjoint copies of a classical MFG, one per type, sharing no object. If instead ϕ is constant, $\phi \equiv c$, then $\langle \phi, \boldsymbol{m} \rangle(y) = c$ for every y because each m^y is a probability measure, so $z[\boldsymbol{m}](x) = c\,(\mathbb{W}\mathbf{1})(x)$ depends on the kernel but *not on the configuration of the field*: the population's shape is invisible, only its trivial total mass enters, and again each type sees a fixed source and the system decouples. In both degenerate cases the graphon game collapses to independent classical MFGs precisely because the one inter-type channel — (26.5) through $\mathbb{W}$ — has been severed. Everything genuinely graphon about the theory lives in the regime where that channel is open.

With the system in hand the obstruction that forced a master equation in section 12.1 reappears verbatim, one level up. Solving (26.6) from a fixed $\boldsymbol{m}_0$ yields a single *flow of fields* $(\boldsymbol{m}_t)_{t \in [0,T]}$ and a value family $u^x(t, \xi)$ attached to it alone. The value answers "what is my optimal cost-to-go from (t, ξ), given my type x?" — but only along the one realized field flow; it knows nothing about the contingency in which the population field is some other $\boldsymbol{m} \neq \boldsymbol{m}_t$, because that contingency simply does not occur along the trajectory the system was solved on. The bare value family is, in the language of section 12.1, *non-Markovian* in the field: its value at time t is determined not by the current configuration alone but by the entire realized history that produced $\boldsymbol{m}_t$. To restore the Markov property one must enlarge the state until the present configuration is sufficient, and the enlargement reveals what the genuine state of the game is. In chapter 12 the Markov state was the *pair* (x, m): the agent's own state x together with the single population measure m. Here it is the *triple* $(x, \xi, \boldsymbol{m})$: the agent's type x, its private state ξ, and the *entire field* $\boldsymbol{m}$. The promotion $x \mapsto (x, \xi)$ records that the chapter 12 state has split into a frozen label and a genuinely dynamical private state, and the promotion $m \mapsto \boldsymbol{m}$ records that the population is now resolved type-by-type; the bare value $u^x(t, \xi)$ sees only the realized slice $\boldsymbol{m} = \boldsymbol{m}_t$ of this triple and is blind to the rest of $\boldsymbol{\mathcal{P}}_2$.

Heuristic (non-rigorous)

The lift of the trajectory-versus-flow picture of section 12.1. Solving (26.6) from one $\boldsymbol{m}_0$ traces a single curve in the field space $\boldsymbol{\mathcal{P}}_2$; the master function is the *flow map* on $\boldsymbol{\mathcal{P}}_2$, defined off that one curve so as to cover the whole space at once. We seek $U(t, x, \xi, \boldsymbol{m})$ so that, *whatever* field $\boldsymbol{m}$ is presented at time t,

$$U(t, x, \xi, \boldsymbol{m}) = \text{the equilibrium value of a type-}x\text{ agent at }\xi\text{ when the population field is }\boldsymbol{m},$$

and the realized value is recovered along the equilibrium flow by the *flow-consistency identity* $u^x(t, \xi) = U(t, x, \xi, \boldsymbol{m}_t)$ — the seed of everything that follows. To extract the equation U obeys we differentiate this identity in t. The left-hand side is the time-derivative of the bare value, governed by the HJB equation (26.6a); on the right, t enters both explicitly, through the first slot of U, and implicitly, through the moving field $\boldsymbol{m}_t$. That implicit derivative is the new content: each type-slice m_t^y of the field is being transported by its own Fokker–Planck flow, so the chain rule must sum the response of U to the motion of every slice. It will therefore carry *two* integrals — one over the foreign state ζ, exactly as in the classical chain rule of lemma 12.2, and one *new* integral over the type y — producing a transport term and a diffusion term, each integrated $\int_0^1 \cdots \, \mathrm{d}y$ across the continuum of types. Matching this against the HJB equation will give the graphon master equation. The necessity of the all-contingencies object becomes *absolute* under common noise (section 26.5), where $\boldsymbol{m}_t$ is genuinely random and "the realized slice" is not knowable in advance: there is then no deterministic curve to which a bare value could be attached, and only an object defined for every $\boldsymbol{m}$ at once survives.

26.2 Differentiation along a field of Fokker–Planck flows

The previous section closed with a concrete demand. To extract the equation that U obeys we must differentiate the flow-consistency identity $u^x(t,\xi) = U(t,x,\xi,\boldsymbol{m}_t)$ in time, and the right-hand side depends on t not only through its explicit first slot but through the moving field $\boldsymbol{m}_t$ — a field whose every type-slice m_t^y is being dragged along by its own Fokker–Planck flow. What we need, and what this section builds, is therefore a chain rule for a functional of the *whole field*: a formula for the rate of change of $\Psi(\boldsymbol{m}_t)$ that integrates the response of Ψ over both the foreign state ζ *and* the type y, exactly the two integrals section 26.1 flagged as the new content. This is the single analytical engine of the chapter — the lifted Itô–Dynkin calculus — and once it is in hand the master equation follows by matching it against the HJB equation, with no further analysis.

The construction is a faithful lift of the homogeneous calculus of chapter 12, and it pays to keep that ancestor in view because the lift changes only one thing. There we needed two derivatives on $\mathcal{P}_2(\mathbb{R}^d)$ — the flat derivative, which records the first-order response of a functional to a perturbation of the measure, and the Lions derivative, its state-gradient — together with a chain rule (lemma 12.2) that fed those derivatives the Fokker–Planck right-hand side. We lift each piece to the field space $\boldsymbol{\mathcal{P}}_2$ of (26.1), and the only new feature, exactly as promised in the chapter preamble, is the type index y: a functional of a field can be perturbed in any one of a continuum of type-slices, so its first variation is no longer a single function on $\mathbb{R}^d$ but a *family* of functions indexed by the type. We develop the two derivatives first, then assemble the chain rule.

Begin by asking what it even means to nudge a field. A field $\boldsymbol{m} = (m^x)_{x\in[0,1]}$ carries a separate probability measure over each type, so there is a whole continuum of independent directions in which to push it. The cleanest — and, it will turn out, the only one the calculus ever needs — is to reweight a *single* type-slice while leaving every other slice fixed: pick a type y and a target measure m', and slide the slice at y a fraction ϵ of the way from its current value m^y toward m'. Because this perturbation is localized at one type, the first-order response of Ψ to it is naturally a function *of that type*, and the flat derivative below is correspondingly indexed by y — the one structural departure from the classical single function $\delta\Psi/\delta m$ of definition 12.1.

Definition 26.1 (Flat and Lions derivatives on the field space)**.** Let $\Psi : \boldsymbol{\mathcal{P}}_2 \to \mathbb{R}$. A perturbation of the field acts on a single type-slice: for $y \in [0,1]$ and $m' \in \mathcal{P}_2(\mathbb{R}^d)$, write $\boldsymbol{m}^{y \to m', \epsilon}$ for the field that equals $\boldsymbol{m}$ except in the slice y, where it is $m^y + \epsilon(m' - m^y)$. The *flat derivative* of Ψ is the family of functions $\dfrac{\delta \Psi}{\delta m^y}(\boldsymbol{m})(\cdot) : \mathbb{R}^d \to \mathbb{R}$, indexed by the type y, defined up to an additive constant by

$$\lim_{\epsilon \to 0} \frac{\Psi(\boldsymbol{m}^{y \to m', \epsilon}) - \Psi(\boldsymbol{m})}{\epsilon} \;=\; \int_{\mathbb{R}^d} \frac{\delta \Psi}{\delta m^y}(\boldsymbol{m})(\zeta)\,(m' - m^y)(\mathrm{d}\zeta) \qquad \text{for a.e. } y \in [0,1],\ \text{all } m'.$$

When $\zeta \mapsto \delta\Psi/\delta m^y(\boldsymbol{m})(\zeta)$ is differentiable, the *(slice) Lions derivative* is its state-gradient,

$$\partial_\mu^y \Psi(\boldsymbol{m})(\zeta) \;:=\; \nabla_\zeta \frac{\delta \Psi}{\delta m^y}(\boldsymbol{m})(\zeta) \;\in\; \mathbb{R}^d.$$

We call Ψ *fully* $\mathcal{C}^2$ if these exist, are jointly measurable in y and continuous in $(\boldsymbol{m}, \zeta)$, and $\zeta \mapsto \partial_\mu^y \Psi(\boldsymbol{m})(\zeta)$ is $\mathcal{C}^1$ with Jacobian $\partial_\zeta \partial_\mu^y \Psi(\boldsymbol{m})(\zeta) \in \mathbb{R}^{d \times d}$.

The only new feature relative to definition 12.1 is the type index y, but it changes the very *kind* of object the first variation is. Classically $\delta\Psi/\delta m$ is one function on $\mathbb{R}^d$; here, because a perturbation lives in a single slice, the response is read off slice by slice and $\delta\Psi/\delta m^y(\boldsymbol{m})(\cdot)$ is a *whole family* of functions, one for each type y — and it is the family, not any single member, that constitutes the derivative. Its state-gradient $\partial_\mu^y \Psi$ is the slice Lions derivative, the per-type avatar of the single Lions field of chapter 12.

The regularity bundle named “fully $\mathcal{C}^2$” is assembled with the chain rule already in view, and each clause earns its place; it is worth reading them term by term so that nothing in the lemma below arrives unexplained. Existence of the slice flat derivative and its gradient is what lets the first variation be written down at all. Joint measurability in y is what makes the type integral $\int_0^1 \cdots \mathrm{d}y$ — the one genuinely new operation of the lift — well-defined and Fubini-legal; without it the per-slice contributions could not be summed coherently over the continuum. Continuity in $(\boldsymbol{m}, \zeta)$ is what lets the derivatives be evaluated along the moving field and against the moving slice as time runs. And the demand that $\zeta \mapsto \partial_\mu^y \Psi(\boldsymbol{m})(\zeta)$ be $\mathcal{C}^1$ with Jacobian $\partial_\zeta \partial_\mu^y \Psi$ is precisely the second-order regularity that the diffusion term will consume: the constant, nondegenerate $\sigma\sigma^\top$ of (A-Nondeg) spreads mass locally, one state-point at a time, so the chain rule needs only this *single-variable* Jacobian and never a genuine two-point second field derivative $\partial_\mu^y \partial_\mu^{y'} \Psi$. That economy is exactly the no-common-noise signature of chapter 12 carried up a level; when a common shock enters (section 26.5) the bundle must be enlarged to carry the two-type object flagged in definition 26.9, and that enlargement is the analytic price of the new physics.

Applied to $\Psi(\cdot) = U(t, x, \xi, \cdot)$ with (t, x, ξ) frozen, the derivatives carry (t, x, ξ) as parameters and a fresh pair (y, ζ); we write $\delta U/\delta m^y(t, x, \xi, \boldsymbol{m})(\zeta)$ and $\partial_\mu^y U(t, x, \xi, \boldsymbol{m})(\zeta)$. The reader should hear (y, ζ) as “a foreign type and a foreign state” — the label and position of the crowd whose reweighting U is being probed against — kept rigorously distinct from the agent’s own pair (x, ξ), in keeping with the standing types-versus-states warning of the chapter preamble.

Lemma 26.2 (Chain rule along a field of Fokker–Planck flows)**.** *Let $\Psi : \boldsymbol{\mathcal{P}}_2 \to \mathbb{R}$ be fully $\mathcal{C}^2$ in the sense of definition 26.1, with derivatives bounded and Lipschitz on bounded sets, uniformly in* *the type index. Let $(\boldsymbol{m}_t)_{t\in[0,T]}$ be a flow in $\boldsymbol{\mathcal{P}}_2$ in which each slice solves its own Fokker–Planck equation,*

$$\partial_t m_t^y \;=\; \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}_\zeta^2\, m_t^y) \;-\; \operatorname{div}(m_t^y\, \mathbf{b}_t^y) \qquad (y\in[0,1]),$$

driven by a measurable family of drift fields $\mathbf{b}_t^y : \mathbb{R}^d \to \mathbb{R}^d$ of at most linear growth, $|\mathbf{b}_t^y(\zeta)| \le C(1+|\zeta|)$ uniformly in (y,t), with $\int_0^T\int_0^1\int |\mathbf{b}_t^y|^2\, \mathrm{d}m_t^y\, \mathrm{d}y\, \mathrm{d}t < \infty$. Then $t\mapsto \Psi(\boldsymbol{m}_t)$ is absolutely continuous and

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\,\Psi(\boldsymbol{m}_t) \;=\; &\int_0^1\!\int_{\mathbb{R}^d} \partial_\mu^y \Psi(\boldsymbol{m}_t)(\zeta)\cdot \mathbf{b}_t^y(\zeta)\, m_t^y(\mathrm{d}\zeta)\, \mathrm{d}y \\ &+\; \tfrac{1}{2}\int_0^1\!\int_{\mathbb{R}^d} \operatorname{tr}\Big(\sigma\sigma^\top\, \partial_\zeta \partial_\mu^y \Psi(\boldsymbol{m}_t)(\zeta)\Big)\, m_t^y(\mathrm{d}\zeta)\, \mathrm{d}y.\end{aligned} \tag{26.7}$$

Proof sketch. The argument is two moves, and the division is the whole content of the lift: per type-slice it *is* the classical chain rule of lemma 12.2, and the genuinely new step sums those per-slice contributions over the type continuum. We carry the analytic justification in-book, invoking the deterministic Itô formula along a flow of measures (theorem D.7) for attribution while keeping the algebra explicit in front of the reader.

Move (a): per slice, the classical chain rule. Fix a single type y and regard Ψ as a functional of that one slice with every other slice held frozen. The computation is then word for word the three moves of lemma 12.2. First differentiate through the slice flat derivative: because Ψ is fully $\mathcal{C}^2$ and $t\mapsto m_t^y$ is absolutely continuous, the rate of change of Ψ in response to the motion of the y-slice is its slice-flat-derivative contracted against that slice's rate of change, the perturbation direction of definition 26.1 now being the mass-zero signed measure $\partial_t m_t^y$. Next substitute the slice's Fokker–Planck right-hand side, which splits the contribution cleanly into a transport piece carrying the drift $\mathbf{b}_t^y$ and a diffusion piece carrying $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}_\zeta^2\, m_t^y)$. Finally integrate by parts in ζ to move the spatial derivatives off the irregular measure and onto the smooth flat derivative. One integration by parts on the transport piece turns $-\operatorname{div}(m_t^y \mathbf{b}_t^y)$ paired against $\delta\Psi/\delta m^y$ into $\nabla_\zeta(\delta\Psi/\delta m^y)\cdot \mathbf{b}_t^y = \partial_\mu^y\Psi\cdot \mathbf{b}_t^y$ integrated against m_t^y, with the single sign change of the divergence theorem; two integrations by parts on the diffusion piece transfer both spatial derivatives onto the flat derivative with no net sign change, the constant matrix $\sigma\sigma^\top$ riding along under the trace, and produce $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \partial_\zeta \partial_\mu^y \Psi)$ — the single-variable Jacobian of the Lions field, the one place the $\mathcal{C}^1$ clause of "fully $\mathcal{C}^2$" is spent. The boundary terms at infinity vanish because $\delta\Psi/\delta m^y$ and its first two gradients are bounded while $m_t^y \mathbf{b}_t^y$ carries the finite-energy decay guaranteed by the L^2 bound. The outcome is, per slice and at the fixed type y, exactly the integrand of (26.7), namely (12.4).

Move (b): sum over the type continuum. This is the step with no classical ancestor. The field flow moves all slices at once, and the total derivative of a functional of $\boldsymbol{m}$ is the integral, over the type label, of its slice-variations against the slice rates of change — the field-flat-derivative identity of definition 26.1 written in differential form,

$$\frac{\mathrm{d}}{\mathrm{d}t}\Psi(\boldsymbol{m}_t) = \int_0^1\!\int_{\mathbb{R}^d} \frac{\delta\Psi}{\delta m^y}(\boldsymbol{m}_t)(\zeta)\, \partial_t m_t^y(\mathrm{d}\zeta)\, \mathrm{d}y,$$

the type-integrated form of the first move, valid with absolute continuity in t under the uniform L^2 bound. Substituting each slice's Fokker–Planck right-hand side and carrying out the per-slice integrations by parts of Move (a) *inside* the y-integral gives (26.7). Interchanging the order of the dy- and dζ-integrations is Fubini, legitimate because the joint measurability of the derivatives in y (a clause of "fully $\mathcal{C}^2$") together with the bound $\int_0^T \int_0^1 \int |\mathbf{b}_t^y|^2 \, \mathrm{d}m_t^y \, \mathrm{d}y \, \mathrm{d}t < \infty$ makes the double integral absolutely convergent. The per-slice identity behind Move (a) is exactly theorem D.7 with the drift $\mathbf{b}_t^y$ in place of the SDE drift and the martingale term absent, the slice flow being deterministic; we cite that theorem for attribution, but the displayed algebra is the entire argument.

The linear-growth drift, by truncation. The two moves above used boundedness of the drift through the vanishing of the boundary terms; the at-most-linear-growth case stated in the lemma is reached from the bounded case by truncation, an argument we carry out in-book rather than cite. Let $\mathbf{b}_t^{y,R} = \mathbf{b}_t^y \chi_R$ with $\chi_R(\zeta)$ a smooth cutoff equal to 1 for $|\zeta| \leq R$ and 0 for $|\zeta| \geq R+1$; each $\mathbf{b}_t^{y,R}$ is bounded, so (26.7) holds with $\mathbf{b}_t^{y,R}$ by the per-slice theorem D.7 and the Fubini step just made. As $R \uparrow \infty$, $\mathbf{b}_t^{y,R} \to \mathbf{b}_t^y$ pointwise and $|\mathbf{b}_t^{y,R}| \leq |\mathbf{b}_t^y|$; since $\partial_\mu^y \Psi(\boldsymbol{m}_t)$ is bounded on the bounded sets carrying the second-moment mass of m_t^y and $\int_0^T \int_0^1 \int |\mathbf{b}_t^y|^2 \, \mathrm{d}m_t^y \, \mathrm{d}y \, \mathrm{d}t < \infty$, dominated convergence passes the transport term to its limit, while the diffusion term does not involve the drift at all and is therefore untouched by the truncation. The time-integrated identity thus holds for $\mathbf{b}_t^y$, and absolute continuity in t is preserved under the uniform L^2 bound.

It is worth being explicit that this linear-growth hypothesis is met, with no slack, by the only drift the lemma will ever be fed: the equilibrium feedback. In theorem 26.3 the field flow is the equilibrium one, whose type-y drift is $\mathbf{b}_t^y(\zeta) = -\partial_p H(y, \zeta, \nabla_\xi U(t, y, \zeta, \boldsymbol{m}_t))$. Convexity of H in p (A-Conv) is what makes this feedback single-valued in the first place — it is the unique minimizing control of appendix A.4 — and under the standing Lipschitz bounds (A-Lip) the map $p \mapsto \partial_p H(y, \zeta, p)$ is Lipschitz uniformly in (y, ζ); composed with a bounded state-gradient $\nabla_\xi U$ this alone forces $|\mathbf{b}_t^y(\zeta)| \leq C(1 + |\zeta|)$ uniformly in (y, t), which is precisely the lemma's growth condition. In the LQG thread the structure is sharper still and fully explicit: there $H(y, \zeta, p)$ is quadratic, so $\partial_p H$ is affine in (ζ, p), and the value function is quadratic in the state, so $\nabla_\xi U(t, y, \cdot, \boldsymbol{m})$ is affine in ζ; the feedback is then exactly *affine* in the state — it is the order-parameter drift of the mean equation (26.33) — and the linear-growth hypothesis holds as an equality of leading order. This is the only regularity of the drift the master-equation derivation will require. The strictly more general theory (genuinely superlinear drifts, and the stochastic upgrade needed once common noise turns the slice flow into a true SDE) is the type-indexed extension of chapter 4, organized for the graphon setting in [34] and, in the probabilistic framework, in [16]; it is not needed below. □

The two terms of (26.7) carry exactly the meaning they had in chapter 12, now resolved type by type: the first is the *transport* of Ψ by the per-type drifts, the crowd of each type flowing along its own $\mathbf{b}_t^y$ and pushing Ψ at the rate at which the slice Lions force points along that flow; the second is the *spreading* of each type's mass by the idiosyncratic diffusion σ, the measure-space avatar of the Laplacian with the finite-dimensional Hessian replaced by the slice Lions Jacobian. The single new ingredient is the outer $\int_0^1 \cdots \mathrm{d}y$, which sums these responses over the continuum of types — a sum, emphatically, and not a derivative.

This is precisely the engine the next section fires, and we can already read off how. In section 26.3 we will not feed the lemma an arbitrary functional and an arbitrary drift, but the specific functional $\Psi = U(t, x, \xi, \cdot)$ — the master function with its time, type, and state slots frozen — transported along the *equilibrium* field flow, whose type-y Fokker–Planck drift is the optimal feedback $\mathbf{b}_t^y =$

$-\partial_p H(y, \cdot, \nabla_\xi U(t, y, \cdot, \boldsymbol{m}_t))$ read off from (26.6b). With that one substitution the transport term becomes each type's crowd drifting under its *own* optimal control and the diffusion term becomes each type's crowd spreading under its private noise; the two integrals of (26.7) reappear verbatim, as the two new type-integrated lines of the graphon master equation, and the feedback's dependence on the value functions of *all* types — through $\nabla_\xi U(t, y, \cdot, \boldsymbol{m}_t)$ inside the type integral — is what will prevent the equation from decoupling across types. Differentiating the flow-consistency identity (26.9) in t and matching against the HJB equation (26.6a) is then the whole derivation; the uniqueness that makes U single-valued, deferred to section 26.4, is assumed there so that "the equilibrium flow" names one object.

26.3 The graphon master equation

Everything is now in place for the chapter's central derivation. The previous section built the one analytical engine we need — the chain rule lemma 26.2 along a field of Fokker–Planck flows — and ended by pointing it at exactly the functional and the drift it will be fed here: the master function $\Psi = U(t, x, \xi, \cdot)$ with its time, type, and state slots frozen, transported along the equilibrium field flow whose per-type feedback drift was recorded there. What remains is to write down the candidate U precisely, differentiate the flow-consistency identity (26.9) seeded in section 26.1, and match the result against the per-type HJB equation (26.6a). The matching produces the graphon master equation, and — this is the content of the theorem below — the production runs both ways: the master function solves the equation, and a solution of the equation regenerates the whole forward–backward system. The lift is faithful to the homogeneous derivation of theorem 12.3; only the type integral is new.

The construction begins with the same circularity that the homogeneous theory had to resolve, so it is worth naming it before proceeding. We want a single function U whose value records "the equilibrium cost-to-go of a type-x agent at state ξ when the population field is $\boldsymbol{m}$," for *every* field $\boldsymbol{m} \in \boldsymbol{\mathcal{P}}_2$ at once — but the only way we know to compute an equilibrium value is to fix an initial field, solve the forward–backward system (26.6) from it, and read the value off the realized flow. The resolution is to do exactly that, once for each possible initial datum, and to define U at $\boldsymbol{m}$ by the value the corresponding problem assigns *at its own starting instant.* Suppose, then, that for every initial datum $(t_0, \boldsymbol{m}) \in [0, T] \times \boldsymbol{\mathcal{P}}_2$ the graphon MFG system (26.6), posed on $[t_0, T]$ with $\boldsymbol{m}_{t_0} = \boldsymbol{m}$, has a unique solution. Existence holds under the standing assumptions and (A-Reg-W) (chapter 24); uniqueness for every horizon holds under the graphon monotonicity and operator conditions taken up in section 26.4. Uniqueness is precisely the hypothesis that makes the construction single-valued — without it "the equilibrium from $\boldsymbol{m}$" would name a set rather than a point and U would not be a function — and we assume it here, deferring its justification to section 26.4 so that the present derivation can proceed unobstructed. Denote by $(\boldsymbol{m}_s^{t_0, \boldsymbol{m}})_{s \in [t_0, T]}$ the resulting forward field flow and by $u^{x; t_0, \boldsymbol{m}}(s, \xi)$ the resulting type-x value. Define the candidate *graphon master function* by evaluating the value at the initial time of its own problem:

$$U(t_0, x, \xi, \boldsymbol{m}) \;:=\; u^{x; t_0, \boldsymbol{m}}(t_0, \xi), \qquad (t_0, x, \xi, \boldsymbol{m}) \in [0, T] \times [0, 1] \times \mathbb{R}^d \times \boldsymbol{\mathcal{P}}_2. \tag{26.8}$$

Two features of this definition deserve to be drawn out, because the entire proof rests on them. First, the value is read at $s = t_0$, the very instant the problem starts: this is what turns a one-trajectory object into an all-contingencies one, for by varying the pair $(t_0, \boldsymbol{m})$ over the whole of $[0,T] \times \boldsymbol{\mathcal{P}}_2$ we sweep out U on its entire domain, each field $\boldsymbol{m}$ contributing the value of the fresh problem launched from it. Second, the terminal condition comes *for free.* Set $t_0 = T$: the problem on the degenerate horizon $[T,T]$ has no dynamics to run, the HJB equation (26.6a) collapses to its terminal data $u^{x;T,\boldsymbol{m}}(T,\xi) = G(x,\xi,\boldsymbol{m})$, and (26.8) reads

$$U(T,x,\xi,\boldsymbol{m}) \;=\; G(x,\xi,\boldsymbol{m})$$

with no computation at all. The master function inherits the terminal cost *by construction*, exactly as the homogeneous $U(T,x,m) = G(x,m)$ did in chapter 12; only the interior equation will require work.

The bridge between the bare value and the master function — the identity we will differentiate — is the *flow-consistency identity.* Solving from $(t_0, \boldsymbol{m})$ produces not just the initial value but the whole trajectory $s \mapsto u^{x;t_0,\boldsymbol{m}}(s,\xi)$, and along it the population is the realized flow $\boldsymbol{m}_s^{t_0,\boldsymbol{m}}$. At each later instant $t \in [t_0,T]$ the agent faces a fresh equilibrium problem — the one started from the field $\boldsymbol{m}_t^{t_0,\boldsymbol{m}}$ at time t — and by the dynamic-programming consistency of the value (the equilibrium is time-consistent, so the tail of the solution from $\boldsymbol{m}$ is the solution from $\boldsymbol{m}_t^{t_0,\boldsymbol{m}}$) the realized value coincides with the master function evaluated at the realized field:

$$u^{x;t_0,\boldsymbol{m}}(t,\xi) \;=\; U(t,x,\xi,\boldsymbol{m}_t^{t_0,\boldsymbol{m}}), \qquad t \in [t_0,T]. \tag{26.9}$$

This is the precise form of the schematic "$u^x(t,\xi) = U(t,x,\xi,\boldsymbol{m}_t)$" from section 26.1. It says that the bare value, a function of (t,ξ) tied to one flow, is the restriction of the master function to the moving slice $\boldsymbol{m} = \boldsymbol{m}_t^{t_0,\boldsymbol{m}}$ of field space. Differentiating it in t — the left side governed by the HJB equation, the right side by the field chain rule — is the whole derivation, and we turn to it now.

Theorem 26.3 (The graphon master equation)**.** *Assume the standing assumptions* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv) *and the graphon regularity* (A-Reg-W)*, and suppose the function U defined by* (26.8) *is fully $\mathcal{C}^2$ in the field argument (definition 26.1) and $\mathcal{C}^{1,2}$ in (t,ξ), with derivatives bounded and locally Lipschitz uniformly in the type x. Then U solves, on $[0,T) \times [0,1] \times \mathbb{R}^d \times \boldsymbol{\mathcal{P}}_2$, the* graphon master equation

$$\begin{aligned}
-\,\partial_t U(t,x,\xi,\boldsymbol{m}) - \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top \mathrm{D}_\xi^2 U\big) \;+\; H\big(x,\xi,\nabla_\xi U\big)& \\
+\; \int_0^1\!\!\int_{\mathbb{R}^d} \partial_\mu^y U(t,x,\xi,\boldsymbol{m})(\zeta)\cdot \partial_p H\big(y,\zeta,\nabla_\xi U(t,y,\zeta,\boldsymbol{m})\big)\, m^y(\mathrm{d}\zeta)\;\mathrm{d}y& \\
-\; \tfrac{1}{2}\int_0^1\!\!\int_{\mathbb{R}^d} \mathrm{tr}\Big(\sigma\sigma^\top\, \partial_\zeta \partial_\mu^y U(t,x,\xi,\boldsymbol{m})(\zeta)\Big)\, m^y(\mathrm{d}\zeta)\;\mathrm{d}y \;=\; F(x,\xi,\boldsymbol{m}),&
\end{aligned} \tag{26.10}$$

with terminal condition

$$U(T,x,\xi,\boldsymbol{m}) \;=\; G(x,\xi,\boldsymbol{m}). \tag{26.11}$$

Conversely, if U is a classical solution of (26.10)*–*(26.11) *with the stated regularity, then for each $(t_0,\boldsymbol{m})$ the value $u^x(t,\xi) := U(t,x,\xi,\boldsymbol{m}_t)$ together with the field flow $\boldsymbol{m}_t$ generated by the feedback $-\partial_p H(x,\cdot,\nabla_\xi U(t,x,\cdot,\boldsymbol{m}_t))$ and $\boldsymbol{m}_{t_0} = \boldsymbol{m}$ solves the graphon MFG system* (26.6) *on $[t_0,T]$.*

Proof. From the system to the equation. Fix $(t_0, \boldsymbol{m})$ and write $\boldsymbol{m}_t = \boldsymbol{m}_t^{t_0,\boldsymbol{m}}$ for the equilibrium field flow, whose type-y feedback drift is, by (26.6b), $\mathbf{b}_t^y(\zeta) = -\partial_p H(y, \zeta, \nabla_\xi u^{y;t_0,\boldsymbol{m}}(t,\zeta))$. Differentiate the flow-consistency identity (26.9) in t at $t = t_0$, for a fixed type x. The left-hand side gives $\partial_t u^{x;t_0,\boldsymbol{m}}(t_0, \xi)$. On the right, t appears explicitly and through $\boldsymbol{m}_t$; the explicit derivative is $\partial_t U(t_0, x, \xi, \boldsymbol{m})$, and the implicit one is supplied by the field chain rule lemma 26.2 applied to $\Psi(\cdot) = U(t_0, x, \xi, \cdot)$ with the family of drifts $\mathbf{b}_{t_0}^y$:

$$\begin{aligned}\partial_t u^{x;t_0,\boldsymbol{m}}(t_0,\xi) = \partial_t U(t_0,x,\xi,\boldsymbol{m}) &+ \int_0^1\!\!\int \partial_\mu^y U(t_0,x,\xi,\boldsymbol{m})(\zeta)\cdot \mathbf{b}_{t_0}^y(\zeta)\, m^y(\mathrm{d}\zeta)\ \mathrm{d}y\\ &+ \tfrac{1}{2}\int_0^1\!\!\int \mathrm{tr}\big(\sigma\sigma^\top \partial_\zeta \partial_\mu^y U(t_0,x,\xi,\boldsymbol{m})(\zeta)\big)\, m^y(\mathrm{d}\zeta)\ \mathrm{d}y.\end{aligned}$$

At $t = t_0$ flow-consistency also gives $u^{x;t_0,\boldsymbol{m}}(t_0,\xi) = U(t_0,x,\xi,\boldsymbol{m})$, hence $\nabla_\xi u^{x;t_0,\boldsymbol{m}}(t_0,\xi) = \nabla_\xi U(t_0,x,\xi,\boldsymbol{m})$ and $\mathrm{D}_\xi^2\, u^{x;t_0,\boldsymbol{m}}(t_0,\xi) = \mathrm{D}_\xi^2\, U(t_0,x,\xi,\boldsymbol{m})$; and for the integrand type y, $\nabla_\xi u^{y;t_0,\boldsymbol{m}}(t_0,\zeta) = \nabla_\xi U(t_0,y,\zeta,\boldsymbol{m})$, so that $\mathbf{b}_{t_0}^y(\zeta) = -\partial_p H(y,\zeta,\nabla_\xi U(t_0,y,\zeta,\boldsymbol{m}))$. Inserting all of this into the type-x HJB equation (26.6a) at $t = t_0$,

$$-\partial_t u^{x;t_0,\boldsymbol{m}}(t_0,\xi) - \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top\, \mathrm{D}_\xi^2\, u^{x;t_0,\boldsymbol{m}}(t_0,\xi)\big) + H\big(x,\xi,\nabla_\xi u^{x;t_0,\boldsymbol{m}}(t_0,\xi)\big) = F(x,\xi,\boldsymbol{m}),$$

yields exactly (26.10) at $(t_0,x,\xi,\boldsymbol{m})$. It is worth pausing on the one sign that has to come out right, because it is the seam where the optimization and the transport meet. The chain rule (26.7) delivers the transport contribution as $+\partial_\mu^y U \cdot \mathbf{b}_{t_0}^y$ on the right-hand side of the displayed expression for $\partial_t u$, with the *equilibrium* drift $\mathbf{b}_{t_0}^y = -\partial_p H(y,\zeta,\nabla_\xi U)$ carrying its own minus sign from the optimal feedback of (26.6b). The HJB equation is written with $-\partial_t u$ on its left, so moving the $\partial_t u$ expansion across the equality flips every term once: the two minus signs — one from the feedback, one from the transposition — cancel, and the transport term lands in (26.10) as $+\partial_\mu^y U \cdot \partial_p H$. The crowd's optimal flow and the master equation's transport term therefore agree in sign precisely because the agents minimize; were the feedback $+\partial_p H$ the bookkeeping would not close. Since $(t_0,x,\xi,\boldsymbol{m})$ was arbitrary the equation holds throughout $[0,T) \times [0,1] \times \mathbb{R}^d \times \boldsymbol{\mathcal{P}}_2$; the terminal condition is (26.8) at $t_0 = T$, already recorded in (26.11).

From the equation to the system. The converse is what makes U worth having: a single solution of the master equation must regenerate, for *every* initial datum, the entire forward–backward pair. Let U be a classical solution of (26.10)–(26.11). Fix $(t_0,\boldsymbol{m})$. The first step is to manufacture the field flow that the master function will be read along, and here we make an explicit import. Plugging the state-gradient $\nabla_\xi U(t,y,\cdot,\boldsymbol{m}_t)$ back into its own argument turns the family of Fokker–Planck equations (26.6b) with feedbacks $-\partial_p H(y,\cdot,\nabla_\xi U(t,y,\cdot,\boldsymbol{m}_t))$ into a *closed* system for the unknown field $\boldsymbol{m}_t$ alone — the type-slices feed each other's drifts through the dependence of U on the whole field — which is exactly a graphon McKean–Vlasov flow. That such a flow exists and is unique, started from $\boldsymbol{m}_{t_0} = \boldsymbol{m}$, is not something the present argument proves: it is the well-posedness theory of graphon McKean–Vlasov equations established in chapter 23, which we invoke here as a named input, its hypotheses met because U has the stated regularity (so the feedback is Lipschitz in the state, uniformly in the type), the diffusion is nondegenerate (A-Nondeg), and the kernel is admissible (A-Reg-W). With the flow $\boldsymbol{m}_t$ in hand, set $u^x(t,\xi) = U(t,x,\xi,\boldsymbol{m}_t)$. Then $\nabla_\xi u^x = \nabla_\xi U$ and $\mathrm{D}_\xi^2\, u^x = \mathrm{D}_\xi^2\, U$ along the flow — differentiating in the state slot commutes with restriction to the moving field — so the feedback driving (26.6b) is precisely $-\partial_p H(x,\cdot,\nabla_\xi u^x)$, and the Fokker–Planck half of the system holds by the very construction of $\boldsymbol{m}_t$. For the HJB half, differentiate

$u^x(t,\xi) = U(t,x,\xi,\boldsymbol{m}_t)$ in t by lemma 26.2 and subtract the master equation (26.10): the explicit $\partial_t U$, the transport integral, and the diffusion integral all cancel against their twins in (26.10), the same three pairings as in the forward direction run in reverse, leaving exactly the per-type HJB equation (26.6a). The terminal condition is (26.11). Thus $(u^{\cdot}, \boldsymbol{m})$ solves the graphon MFG system on $[t_0, T]$, and since $(t_0, \boldsymbol{m})$ was arbitrary the one function U has reconstructed the entire family of equilibria. □

With the equivalence proved, the equation (26.10) stands on its own, and it repays a careful term-by-term reading against its classical ancestor (12.7). It has three lines, and they fall into a familiar two-plus-one pattern: two lines that any reader of chapter 12 will recognize unchanged, and one line — the second — that carries the entire graphon novelty.

The first line is an *ordinary* second-order HJB operator in the finite-dimensional state ξ, with the type x and the field $\boldsymbol{m}$ frozen as inert parameters. Read in isolation, $-\partial_t U - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2_\xi U) + H(x,\xi,\nabla_\xi U)$ is exactly the Hamilton–Jacobi–Bellman operator of a single type-x agent who treats the surrounding population as a fixed backdrop; it is the only place the agent's own diffusion σ and its own Hamiltonian $H(x,\cdot,\cdot)$ appear, and if the field never moved this line, set equal to $F(x,\xi,\boldsymbol{m})$, would be the whole equation. The third line is its measure-space counterpart, the *extra-diffusion* or spreading term $-\frac{1}{2}\int_0^1\int \operatorname{tr}(\sigma\sigma^\top \partial_\zeta \partial^y_\mu U)\, m^y \,\mathrm{d}y$: it accounts for the fact that the crowd of every type is itself being smeared by its private noise, and that the value U responds to that smearing through the single-variable Jacobian of its slice Lions derivative, summed over all types. Both of these survive the lift from chapter 12 with no change but the outer type integral; neither is where the graphon lives.

The second line is the new content, and it is worth slowing down over. It reads

$$\int_0^1 \int_{\mathbb{R}^d} \partial^y_\mu U(t,x,\xi,\boldsymbol{m})(\zeta) \cdot \partial_p H\big(y,\zeta,\nabla_\xi U(t,y,\zeta,\boldsymbol{m})\big)\, m^y(\mathrm{d}\zeta)\; \mathrm{d}y,$$

and it is a transport term: it measures how the type-x value changes as each foreign type's crowd drifts. But the drift it transports against is the decisive feature. $\partial_p H(y,\zeta,\nabla_\xi U(t,y,\zeta,\boldsymbol{m}))$ is the equilibrium feedback of *type* y — a different type from the x whose value we are computing — and that feedback is governed by the state-gradient $\nabla_\xi U(t,y,\cdot,\boldsymbol{m})$ of type y's own master function. So the equation for type x contains, under the integral $\int_0^1 \cdots \mathrm{d}y$, the value functions of *all* the other types. The graphon master equation is not one PDE but a continuum of PDEs welded together along the type label: the x-equation cannot be solved without the y-equations for every y, and they in turn need the x-equation. This is the precise analytic signature of the graphon coupling — *the equation does not decouple across types* — and it stands in exact contrast to the degenerate cases of section 26.1, where severing the channel (26.5) collapsed the system into independent classical MFGs. Here the channel is open, and its dynamical trace is this second line.

Remark 26.4 (Reading the gradient inside the type integral)**.** The notation $\nabla_\xi U(t,y,\zeta,\boldsymbol{m})$ is a known stumbling block, worth isolating. The subscript ξ does *not* mean "differentiate at the point ξ." It names the *slot*: U was defined in (26.8) with a generic state argument that the chapter calls ξ, and $\nabla_\xi U$ is the gradient of U in that slot, a function of all four arguments. The expression $\nabla_\xi U(t,y,\zeta,\boldsymbol{m})$ is then that gradient *evaluated* with the type slot set to the foreign type y and the state slot set to the foreign state *value* ζ — the integration variables of the second line — not at the agent's own (x, ξ). In words: it is the equilibrium momentum of a type-y agent sitting at state ζ, which is exactly what feeds that agent's optimal drift $\partial_p H(y, \zeta, \cdot)$. The agent's own pair (x, ξ) enters the second line only through the outer factor $\partial_\mu^y U(t, x, \xi, \boldsymbol{m})(\zeta)$, which records how *type x's* value responds to a reweighting of type y's crowd at ζ. Two distinct evaluations of the same function U, kept rigorously apart by which type and state are plugged into which slot: this is the types-versus-states discipline of the chapter preamble made operational.

It remains to say where the graphon kernel W — the one object that makes this a graphon theory at all — actually sits in (26.10). The surprising answer is that it appears *nowhere explicitly*: scan the three lines and there is no W, no operator $\mathbb{W}$, no aggregate z. The transport and diffusion lines carry the genuine per-type feedbacks and Jacobians, none of which displays the kernel; the only places W enters are the source F on the right-hand side and, through (26.11), the terminal datum G, where (26.5) writes them as $\widetilde{F}(x, \xi, z[\boldsymbol{m}](x))$ and $\widetilde{G}(x, \xi, z[\boldsymbol{m}](x))$ with the aggregate $z[\boldsymbol{m}](x) = (\mathbb{W}\langle \phi, \boldsymbol{m}\rangle)(x)$ of (26.4). So W is hidden inside the cost — a single scalar channel per type. Yet it is far from confined there, and tracing its propagation is the cleanest way to see why the equation couples. The source $\widetilde{F}(y, \cdot, z[\boldsymbol{m}](y))$ shapes each type y's value function through that type's HJB equation; the value function's state-gradient $\nabla_\xi U(t, y, \cdot, \boldsymbol{m})$ therefore carries the imprint of W; and that very gradient is what appears, under the type integral, as the equilibrium feedback in the second line of *every other* type's master equation. The kernel thus enters once, through the cost, and then propagates — via the field of value functions it shapes — into the transport drift of all types at once. The non-decoupling of the second line is precisely the dynamical shadow of this single static coupling; sever W and, as section 26.1 showed, the shadow vanishes with it.

Notation watch

The graphon master equation (26.10) carries *four* distinct objects that a hurried reader will all call "derivatives," and conflating them is the commonest error in the subject; each is genuinely different in kind, and it is worth pinning down what each one differentiates with respect to. *(i) The state Hessian* $\mathrm{D}_\xi^2 U$ is an ordinary second derivative in the agent's own private state $\xi \in \mathbb{R}^d$, a $d \times d$ matrix, and it is the only second-order object in the bare HJB line — the agent's own diffusion acts here and nowhere else. *(ii) The slice Lions Jacobian* $\partial_\zeta \partial_\mu^y U$ is also an ordinary $d \times d$ second derivative, but in a *foreign* state ζ: it measures how the crowd of type y spreads under its own noise and how U responds to that spreading, and although it lives under the type integral it is a single-variable Jacobian, not a field derivative. *(iii) The type integral* $\int_0^1 \cdots \mathrm{d}y$ is not a derivative at all but a *sum* over the continuum of types — the one structural novelty the graphon lift adds to chapter 12 — and reading it as a differentiation, or forgetting it, is what makes the type coupling of the second line invisible. *(iv) The second field derivative* $\partial_\mu^y \partial_\mu^{y'} U$ is the genuine article: a true second-order operator on the field space $\boldsymbol{\mathcal{P}}_2$, a *double* type integral over two independent foreign types y, y', defined in definition 26.9. It is conspicuously *absent* from (26.10), and its absence is not an accident of bookkeeping but the analytic content of "no common noise": with only idiosyncratic diffusions the field flow is deterministic and the chain rule never reaches second order in the field. It is precisely this fourth object that switches on in section 26.5, where a shared shock makes the field a true stochastic process and the master equation acquires its field-space Laplacian. Keep all four apart; the first three are present now, the fourth marks the boundary of the deterministic theory.

The line-by-line reading just completed is the static anatomy of (26.10): what each term *is*. There is a complementary, geometric way to read the same equation — not term by term but as a whole — that the next section takes up. Just as the classical master equation of section 12.5 is a Hamilton–Jacobi equation on the space of measures whose characteristic curves are the equilibrium Fokker–Planck flows, the graphon master equation is a Hamilton–Jacobi equation whose base space is the field space $\boldsymbol{\mathcal{P}}_2$ and whose characteristics are the field Fokker–Planck flows $(\boldsymbol{m}_t)$ we have just learned to differentiate along. The equation-to-system converse proved above is exactly the statement that propagating data along those characteristics regenerates the forward–backward system; section 26.3.1 reads that converse geometrically, identifies the characteristic system with the graphon MFG system (26.6), and shows that the type-resolved state-gradient $\nabla_\xi U$ is the operator-valued decoupling field of the graphon FBSDE of chapter 23.

26.3.1 Characteristics and the operator-valued decoupling field

The previous section read (26.10) term by term and closed with its converse: a classical solution U, propagated forward along the equilibrium field flow it itself generates, regenerates the whole graphon MFG system (26.6). That converse is not a technical afterthought but the geometric heart of the equation, and naming it as such is the business of this section. A first-order Hamilton–Jacobi equation is solved by the *method of characteristics*: one propagates the unknown along distinguished curves in the base space, and along each such curve the partial differential equation collapses to a system of ordinary (here, evolution) equations. The graphon master equation is exactly such a Hamilton–Jacobi equation — first order in the field $\boldsymbol{m}$, as the field-anatomy of section 26.3 made plain, the only genuine second-field-derivative object (the two-type Laplacian of definition 26.9) being absent until common noise enters — and the converse just proved is precisely the statement that its characteristics are the field Fokker–Planck flows. We now make that reading exact, and in doing so we lift the central diagram of Part II (eq. (12.12)) to the field setting unchanged in shape.

The dictionary is the one of section 12.5, with a type label attached throughout. The base space is the field space $\boldsymbol{\mathcal{P}}_2$ of (26.1); the characteristic curves are the field flows $(\boldsymbol{m}_t)$ we spent section 26.2 learning to differentiate along; and the characteristic *system* — the evolution the master equation collapses to along each curve — is the graphon MFG system (26.6) itself. This is the precise sense in which the three formulations of the graphon game — master equation on $\boldsymbol{\mathcal{P}}_2$, coupled HJB/field-Fokker–Planck system, and graphon McKean–Vlasov FBSDE — are one object viewed three ways, exactly as in the homogeneous theory. The single structural novelty, to which we return at the close, is that these characteristics do not evolve type by type in isolation: the type integral in (26.10) couples the flow of every slice to the value functions of all the others, so the characteristic system is a genuinely entangled continuum of HJB/Fokker–Planck pairs rather than a disjoint family of them.

Theorem 26.5 (Characteristics of the graphon master equation)**.** *Let U be a classical solution of* (26.10)–(26.11) *with the regularity of theorem 26.3. Fix $(t_0, \boldsymbol{m})$ and let $(\boldsymbol{m}_t)_{t \geq t_0}$ be the field flow generated by the per-type feedbacks $\mathbf{b}_t^y(\cdot) = -\partial_p H(y, \cdot, \nabla_\xi U(t, y, \cdot, \boldsymbol{m}_t))$, $\boldsymbol{m}_{t_0} = \boldsymbol{m}$. Then:*

1. *the family $u^x(t, \xi) := U(t, x, \xi, \boldsymbol{m}_t)$ and the flow $\boldsymbol{m}_t$ solve the graphon MFG system* (26.6) *on* $[t_0, T]$ (theorem 26.3)*;*
2. *for each type x, the graphon McKean–Vlasov FBSDE of chapter 23 for the optimally controlled state X_t^x, its adjoint Y_t^x and martingale integrand Z_t^x is solved by*

$$Y_t^x = \nabla_\xi U(t, x, X_t^x, \boldsymbol{m}_t), \qquad Z_t^x = \mathrm{D}_\xi^2 \, U(t, x, X_t^x, \boldsymbol{m}_t) \, \sigma, \qquad m_t^x = \mathcal{L}(X_t^x), \tag{26.12}$$

so that $\nabla_\xi U$ is the (type-indexed) decoupling field of the graphon FBSDE.

Proof sketch. Part (1) is nothing new: it is the equation-to-system converse of theorem 26.3, restated geometrically as the assertion that the value family $u^x(t, \xi) = U(t, x, \xi, \boldsymbol{m}_t)$, read off U along the single field characteristic $(\boldsymbol{m}_t)$, together with that flow, solves (26.6) on $[t_0, T]$. The content of the theorem is part (2), and it is a composition of two facts already in hand, so it pays to assemble it in the open rather than defer it to a citation.

The first fact is the per-type nonlinear Feynman–Kac link of chapter 23. With the equilibrium field $\boldsymbol{m}_\cdot$ held fixed, each type decouples into a *classical* McKean–Vlasov FBSDE for the optimally controlled state X_t^x, its adjoint Y_t^x, and the martingale integrand Z_t^x; the stochastic maximum principle of chapter 23 represents the equilibrium through this triple, and the bridge between its probabilistic adjoint and the PDE value is the FBSDE–PDE dictionary

$$Y_t^x = \nabla_\xi u^x(t, X_t^x), \qquad Z_t^x = \mathrm{D}_\xi^2 \, u^x(t, X_t^x) \, \sigma$$

proved type by type in proposition 23.19 (eq. (23.21)) — the graphon lift of the classical relation of chapter 9, itself the nonlinear Feynman–Kac formula of chapter 5 applied to the type-x HJB equation (26.6a). The second fact is the flow-consistency identity (26.9), $u^x(t, \xi) = U(t, x, \xi, \boldsymbol{m}_t)$, together with the consistency $m_t^x = \mathcal{L}(X_t^x)$ of the controlled state's law with the type-x slice of the realized field (the Fokker–Planck equation (26.6b) is exactly the forward equation for $\mathcal{L}(X_t^x)$). The composition is now a single substitution. Differentiating $u^x(t, \xi) = U(t, x, \xi, \boldsymbol{m}_t)$ once and twice in the state ξ and evaluating at $\xi = X_t^x$ turns $\nabla_\xi u^x(t, X_t^x)$ into $\nabla_\xi U(t, x, X_t^x, \boldsymbol{m}_t)$ and $\mathrm{D}_\xi^2 \, u^x(t, X_t^x)$ into $\mathrm{D}_\xi^2 \, U(t, x, X_t^x, \boldsymbol{m}_t)$; inserting these into the dictionary and writing $m_t^x = \mathcal{L}(X_t^x)$ yields precisely (26.12). (As in theorem 12.6, the form $Z_t^x = \mathrm{D}_\xi^2 \, U \, \sigma$ is the one Itô's formula produces directly from $Y_t^x = \nabla_\xi U(t, x, X_t^x, \boldsymbol{m}_t)$; with the constant symmetric σ of (A-Nondeg) it coincides with the transpose convention $\sigma^\top \, \mathrm{D}_\xi^2 \, U$ of chapter 23.)

It remains to say in what sense $\nabla_\xi U$ is a *decoupling field*, since that term carries the whole structural payload of part (2) and earns both adjectives in the section title. A forward–backward system is, on its face, anticipative: the backward components (Y_t^x, Z_t^x) are pinned by terminal data at T and so appear to depend on the entire future of the forward state. A decoupling field is a *deterministic* function that severs this entanglement — it expresses the backward component at time t through the present forward state and present field alone, $Y_t^x = \nabla_\xi U(t, x, X_t^x, \boldsymbol{m}_t)$, with no reference to the future. Equation (26.12) both names that function — it is the state-gradient of the master function — and exhibits why it well-poses the FBSDE: once the decoupling field is known, the coupled forward–backward problem reduces to a single forward SDE,

$$\mathrm{d}X_t^x = -\partial_p H\big(x, X_t^x, \nabla_\xi U(t, x, X_t^x, \boldsymbol{m}_t)\big)\ \mathrm{d}t + \sigma\ \mathrm{d}B_t^x,$$

whose adjoint Y_t^x is then *read off* the field rather than solved for. That such a deterministic representation exists, is Lipschitz, and is measurable in the type is exactly the graphon decoupling-field theorem theorem 23.21, proved in chapter 23 for the fixed-field per-type FBSDEs. The present theorem only *composes* that family with the master function through (26.9), promoting the fixed-field decoupling fields $U^x(t,\xi)$ of eq. (23.21) to the single measure-dependent object $\nabla_\xi U(t,x,\xi,\boldsymbol{m})$ — precisely the promotion theorem 23.21 announced and deferred to this chapter. The in-book argument therefore runs entirely through chapter 23 for the per-type fields and through theorem 26.3 for their assembly into U; the graphon decoupling-field-to-master promotion is treated in [34, 16], to which we defer only for attribution and further reading.

Finally, the two adjectives. The field $\nabla_\xi U$ is *type-indexed* because the FBSDE was: there is one backward triple (X^x, Y^x, Z^x) for every type x, and the map $x \mapsto \nabla_\xi U(t,x,\cdot,\boldsymbol{m})$ ranges over the whole continuum of types, a per-type gradient rather than a single one. And it is *operator-valued* in the field argument: the sensitivity of the entire family of decoupling fields to a perturbation of the population is governed by the slice field-derivative $\partial_\mu^y \nabla_\xi U(t,x,\xi,\boldsymbol{m})(\zeta)$, an object carrying *two* type labels (x,y) rather than a scalar — nudging the slice at the foreign type y moves the decoupling field of type x through the kernel this double-indexed derivative defines. That kernel is, once more, the $\mathbb{W}$-threaded coupling that runs through every display of the chapter ((26.5)): the only route by which one type's decoupling field feels another type's population is the same operator $\mathbb{W}$, and it is this operator-valued dependence that theorem 26.7 will have to control across the spectrum of $\mathbb{W}$. $\square$

The three faces and the two maps between them are summarized by the lift of the central diagram of Part II, eq. (12.12), to the field setting. Read horizontally, the top row says: restricting the master function to a single field characteristic $\boldsymbol{m}_\cdot$ produces the graphon MFG system's value family $u^x = U(\cdot, x, \cdot, \boldsymbol{m}_\cdot)$; the bottom row says: restricting the decoupling field $\nabla_\xi U$ to the same flow, with $X_t^x \sim m_t^x$, furnishes the FBSDE adjoint Y_t^x through (26.12). Read vertically, the left map is differentiation in the state ξ, which turns the master equation into the decoupling field; the right map is the per-type Feynman–Kac dictionary, which turns the MFG value into the probabilistic FBSDE picture. The figure is identical in shape to its homogeneous ancestor; only the labels carry the lift — the single measure m becomes a field $\boldsymbol{m}$, the state x splits into the type x and the private state ξ, and every arrow now runs uniformly across the continuum of types.

$$\boxed{\begin{array}{ccc} \textbf{Graphon master equation} & \xrightarrow{\text{restrict to } \boldsymbol{m}_t} & \textbf{Graphon MFG system} \\ U(t,x,\xi,\boldsymbol{m}) & & (u^x,\boldsymbol{m}) = (U(\cdot,x,\cdot,\boldsymbol{m}_\cdot),\, \boldsymbol{m}_\cdot) \\ \big\downarrow \nabla_\xi & & \big\downarrow \text{Feynman–Kac} \\ \textbf{Decoupling field} & \xrightarrow{\text{evaluate along } \boldsymbol{m}_t,\ X_t^x\sim m_t^x} & \textbf{Graphon McKean–Vlasov FBSDE} \\ \nabla_\xi U(t,x,\xi,\boldsymbol{m}) & & (X_t^x, Y_t^x, Z_t^x) \end{array}} \tag{26.13}$$

Stand back, and the lift is revealed to be faithful to the last detail: every arrow of the diagram and every identification in the theorem is the homogeneous statement of section 12.5 with a type label attached and the measure m promoted to a field $\boldsymbol{m}$. The genuine graphon novelty is confined to exactly two places, and it is worth saying precisely which, because the remainder of the chapter turns on them. First, the *base space*: the characteristics no longer live in $\mathcal{P}_2(\mathbb{R}^d)$ but in the larger field space $\boldsymbol{\mathcal{P}}_2$, a continuum of measures rather than one, so a single point of the base space is now an entire type-resolved population. Second, the *type integral*: where the homogeneous characteristic system was one HJB/Fokker–Planck pair, the graphon characteristic system (26.6) is a continuum of such pairs, bound together by the $\int_0^1 \cdots \,\mathrm{d}y$ in (26.10) through which one type's value enters every other type's transport. The binding is, as always in this chapter, the single channel (26.5): were $\mathbb{W}$ to vanish the characteristics would separate into $|[0,1]|$ independent classical flows, one per type, and the diagram (26.13) would split into a disjoint union of copies of eq. (12.12). It is precisely the open coupling that fuses the continuum of characteristics into one entangled object, and that is the throughline the chapter has been tracing since (26.4).

This geometric reading does more than organize the equivalence; it hands the next section the lever it needs to force uniqueness. Any classical solution of (26.10)–(26.11) must, by part (1), agree with the value the graphon MFG system assigns along every field characteristic it generates. But the characteristics issuing from a fixed time t_0, as the initial field $\boldsymbol{m}$ ranges over $\boldsymbol{\mathcal{P}}_2$, *fill the whole base space* — through each point of $\{t_0\} \times \boldsymbol{\mathcal{P}}_2$ there passes exactly one such curve, namely the equilibrium flow launched from that field. A classical solution is therefore pinned, at the instant t_0, to the system's value at *every* field at once: if the graphon MFG system is uniquely solvable from each datum, the master function is unique, $U(t_0, x, \xi, \boldsymbol{m}) = u^{x;t_0,\boldsymbol{m}}(t_0, \xi)$ with no freedom left. This is Step 4 of the well-posedness argument of section 26.4, and it explains why uniqueness of U can be settled by the characteristics *before* the harder regularity question — whether a U smooth enough to count as a classical solution exists at all — is touched. The next section takes up exactly that conditional structure, weighing the two competing hypotheses that make the construction good: graphon monotonicity, which keeps the characteristics from crossing at every horizon, versus operator-norm smallness of $\mathbb{W}$ together with a short horizon, which buys the same conclusion by contraction.

One feature of the whole picture deserves to be flagged now, because its eventual loss is the pivot on which the chapter turns. Everything in this section rests on the characteristics being *deterministic* curves: the field flow $(\boldsymbol{m}_t)$ is a definite trajectory in $\boldsymbol{\mathcal{P}}_2$, fixed the moment its initial field is named, so that "the value along the characteristic" is a well-defined function and propagating data along it is meaningful. This is exactly the structure that common noise will dissolve (section 26.5). A shock shared across all types turns $(\boldsymbol{m}_t)$ into a genuine stochastic process — the characteristics become random, there is no longer a single curve through each initial field along which to read U, and the method of characteristics, the organizing principle of this entire section, simply ceases to apply. When that happens the master equation stops being a convenient repackaging of a system one could in principle have solved flow by flow and becomes the *only* surviving formulation; the price it pays for that survival is the double-type-integral second-field-derivative Laplacian that the deterministic theory, with its first-order Hamilton–Jacobi structure, was spared.

26.4 Well-posedness in the monotone regime

The derivation of the previous sections came with a debt, and this section pays as much of it as the current theory can. Theorem 26.3 produced the graphon master equation (26.10) *conditionally* — exactly as theorem 12.3 did one level down — under the standing assumption that the candidate U of (26.8) is already smooth enough for the field derivatives in (26.10) to make sense. Section 26.3.1 then showed that, granting such a U exists, the method of characteristics nails it down uniquely: a classical solution is pinned at each instant t_0 to the system's value at *every* field at once, because the equilibrium characteristics through $\{t_0\} \times \boldsymbol{\mathcal{P}}_2$ fill the whole base space. What neither result settles is the prior question on which both rest — whether a U of the required regularity exists at all. That question is, for graphons, genuinely less settled than in the classical case, and the purpose of this section is to be candid about precisely which parts of the answer are theorems and which are conjectures verified only in special structures. Honesty here requires a three-way split: what is *proved* outright (the equivalence of formulations, and a Lipschitz estimate on U), what is *reduced* to the classical template (the construction, given the graphon existence theory of chapter 24), and what remains genuinely *open* (the differentiability that turns the Lipschitz U into a classical solution).

We organize the discussion around the two competing structural conditions that have run through Parts II and IV, the same pair section 26.3.1 handed us as the alternative hypotheses that keep the characteristics from crossing. The first is *monotonicity*: a sign condition on the coupling that, as in the Lasry–Lions theory of chapter 12, buys uniqueness and a priori control at *every* horizon, however long. The second is *operator-norm smallness* of $\mathbb{W}$ together with a *short horizon*: a quantitative smallness that buys the same conclusion cheaply, by making a best-response map a contraction, but only locally. Classically these two routes are independent — monotonicity is a property of the coupling F alone, with no operator in sight. The graphon novelty, and the crux of this whole section, is that the two become entangled: because the field reaches the cost only through $\mathbb{W}$, the monotonicity condition itself turns into a spectral statement about the network operator, so that "the sign of the interaction" and "the spectrum of $\mathbb{W}$" can no longer be discussed apart. We make that entanglement precise first, then walk the well-posedness argument step by step, flagging the one step where the graphon theory genuinely falls short of its classical model.

Graphon monotonicity and its spectral reading

We begin by lifting the Lasry–Lions condition to the field. Recall (chapter 11, eq. (12.13)) that classically a coupling is monotone when, for any two populations, the difference of the costs they assign, paired against the difference of the populations, is nonnegative — a statement that the interaction *discourages* concentration, pushing the crowd to spread rather than to clump, and the analytic engine behind uniqueness for all horizons. The field version replaces the single measure by the field and adds the one new operation of the lift, an integral over the type label.

Definition 26.6 (Graphon Lasry–Lions monotonicity)**.** The coupling F (and likewise G) is *graphon-monotone* if, for all $\boldsymbol{m}, \boldsymbol{m}' \in \boldsymbol{\mathcal{P}}_2$,

$$\int_0^1 \int_{\mathbb{R}^d} \left(F(x,\xi,\boldsymbol{m}) - F(x,\xi,\boldsymbol{m}')\right) \left(m^x - (m')^x\right)(\mathrm{d}\xi)\ \mathrm{d}x \ \geq\ 0. \tag{26.14}$$

This is the field-integrated lift of the Lasry–Lions inequality (12.13); it is the condition referred to by the tag (A-Mon) in the graphon part.

Read (26.14) slice by slice and the lift is transparent. For each fixed type x the inner integral $\int_{\mathbb{R}^d}(F(x,\xi,\boldsymbol{m})-F(x,\xi,\boldsymbol{m}'))(m^x-(m')^x)(\mathrm{d}\xi)$ is *exactly* the classical Lasry–Lions pairing (12.13) for the type-x cost and the type-x slices; the outer $\int_0^1\cdots\mathrm{d}x$ then averages those per-type pairings over the continuum. So the definition asks not that each type be monotone on its own, but that the *type-average* of the per-type pairings be nonnegative — a weaker, genuinely field-level demand, and the natural one given that the field metric (26.2) itself measures distance as an L^2-over-types average. This is the only sense in which (26.14) departs from a verbatim copy of the homogeneous condition, and it is the same departure — one integral over x — that distinguishes every field object in the chapter from its chapter 12 ancestor.

The reason this definition cannot be left at the slice-by-slice reading, and the reason the graphon theory is harder than a disjoint union of classical ones, is that the cost $F(x,\xi,\boldsymbol{m})$ does not in fact depend on the slice m^x alone: through the coupling form (26.5) it sees the *whole* field, and only through the operator $\mathbb{W}$. That single channel turns the pairing (26.14) into a quadratic form built from $\mathbb{W}$, and it is worth carrying out the substitution in full, because the compression of several steps into one line is exactly the kind of move the reader should see worked. Take the linearized, mean-coupled case of Thread B, where the cost is affine in the aggregate,

$$\widetilde{F}(x,\xi,z) \;=\; -\,c(x)\,\xi\,z \;+\; (\text{terms independent of } z), \tag{26.15}$$

with $c(x)$ the type-x coupling strength and, as fixed in eq. (26.4) for Thread B, the interaction observable $\phi(\zeta)=\zeta$, so that the aggregate felt by type x is the W-weighted average of the foreign type means, $z[\boldsymbol{m}](x)=(\mathbb{W}\bar{m})(x)$ with $\bar{m}(y)=\int\zeta\,m^y(\mathrm{d}\zeta)$. Introduce the shorthand

$$\Delta(y) \;:=\; \int_{\mathbb{R}^d}\phi\;\mathrm{d}\big(m^y-(m')^y\big) \;=\; \int_{\mathbb{R}^d}\zeta\,\big(m^y-(m')^y\big)(\mathrm{d}\zeta) \tag{26.16}$$

for the change, type by type, in the aggregated observable — here simply the change in the type-y mean. Now substitute. Because (26.15) is affine in z, the cost difference at fixed (x,ξ) collapses to a single product,

$$F(x,\xi,\boldsymbol{m})-F(x,\xi,\boldsymbol{m}') \;=\; -\,c(x)\,\xi\,\big(z[\boldsymbol{m}](x)-z[\boldsymbol{m}'](x)\big) \;=\; -\,c(x)\,\xi\,(\mathbb{W}\Delta)(x),$$

the last equality because $z[\boldsymbol{m}](x)-z[\boldsymbol{m}'](x)=(\mathbb{W}(\bar{m}-\bar{m}'))(x)=(\mathbb{W}\Delta)(x)$ by linearity of $\mathbb{W}$ and the definition (26.16) of Δ. Insert this into the monotonicity pairing (26.14) and pull the factor $-c(x)(\mathbb{W}\Delta)(x)$, which is constant in ξ, out of the inner integral:

$$\int_0^1\int_{\mathbb{R}^d}\big(F(x,\xi,\boldsymbol{m})-F(x,\xi,\boldsymbol{m}')\big)\big(m^x-(m')^x\big)(\mathrm{d}\xi)\;\mathrm{d}x=-\int_0^1 c(x)\,(\mathbb{W}\Delta)(x)\Big(\int_{\mathbb{R}^d}\xi\,\big(m^x-(m')^x\big)(\mathrm{d}\xi)\Big)\,\mathrm{d}x.$$

The inner integral that survives is $\int_{\mathbb{R}^d}\xi\,(m^x-(m')^x)(\mathrm{d}\xi)$, which by the very definition (26.16) — with the aggregation observable $\phi(\zeta)=\zeta$ appearing again, now as the test function of the pairing — is exactly $\Delta(x)$. This coincidence is not an accident of bookkeeping but the structural reason a clean quadratic form appears at all: the observable the cost integrates the population against is the *same* observable that builds the aggregate, so the pairing closes on Δ against itself. The result is

$$-\int_0^1 c(x)\,\Delta(x)\,(\mathbb{W}\Delta)(x)\;\mathrm{d}x \;=\; -\,\big\langle\, c\,\Delta,\ \mathbb{W}\Delta\,\big\rangle_{L^2([0,1])}, \tag{26.17}$$

a quadratic form in Δ built from $\mathbb{W}$ and weighted by the coupling profile c.

Equation (26.17) is the crux of the entire section, so it is worth dwelling on what it says. Graphon monotonicity (26.14) holds (to leading order, in this linearized case) precisely when the form $-\langle c\,\Delta, \mathbb{W}\Delta\rangle_{L^2}$ is nonnegative for every admissible Δ — that is, when $\langle c\,\Delta, \mathbb{W}\Delta\rangle \leq 0$. This is a *spectral* condition on the self-adjoint operator $\mathbb{W}$ of definition 15.13 (self-adjoint because W is symmetric under (A-Reg-W)), read against the sign and magnitude of c. Take the cleanest instance, a constant coupling $c(x) \equiv c$: the condition becomes $c\,\langle \Delta, \mathbb{W}\Delta\rangle \leq 0$ for all Δ, which when $c > 0$ demands that $\mathbb{W}$ be negative semidefinite on the reachable directions, and when $c < 0$ demands the opposite. The *sign* of c — whether the interaction is of congestion/competitive type, penalizing alignment with the aggregate, or of imitation/aligning type, rewarding it — selects which half of the spectrum of $\mathbb{W}$ must be empty, and the *size* of the relevant eigenvalues fixes how much monotonicity there is to spend. This is the precise sense, already flagged at chapter 24, in which graphon monotonicity entangles the strategic sign of the interaction with the spectrum of the network operator.

The entanglement has a consequence that the homogeneous theory never forces, and it must be stated plainly because it is what makes (A-Mon) and (A-Coupling) inseparable here. Classically, monotonicity (A-Mon) is a property of the coupling F in isolation: no operator enters its statement, and one verifies it once and for all by inspecting F. In the graphon setting the field touches the cost *only* through $\mathbb{W}$ — this is the one inter-type channel that section 26.1 identified as carrying everything graphon-specific — so the very inequality that defines monotonicity is a joint condition on the coupling sign c *and* the operator $\mathbb{W}$. One cannot certify monotonicity by looking at F alone and then separately impose a smallness threshold on $\mathbb{W}$; the two are bound into the single quadratic form (26.17). Concretely, an interaction that is monotone for one kernel can fail to be for another with the same coupling profile but a differently signed spectrum, and a kernel near the operator threshold of (A-Coupling) is exactly one whose extreme eigenvalue brings the form (26.17) to the edge of degeneracy. This is why, throughout what follows, the operator-norm threshold (A-Coupling) cannot be divorced from the monotonicity question the way it can in chapter 12, and it is the structural fact the LQG computation of section 26.6 will realize concretely, where the spectral condition becomes the requirement that an operator Riccati equation not blow up.

The well-posedness theorem and its honest accounting

With the monotonicity condition understood as a spectral statement, we can state what is proved and stage the proof so that the one open step stands out from the complete ones.

Theorem 26.7 (Well-posedness of the graphon master equation, monotone regime)**.** *Assume* (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), (A-Reg-W), *that H and the couplings F, G are sufficiently smooth with derivatives bounded uniformly in $(x, \boldsymbol{m})$, and that F, G are graphon-monotone* (26.14)*. Suppose moreover that for every $(t_0, \boldsymbol{m})$ the graphon MFG system* (26.6) *has a unique classical solution on $[t_0, T]$, with a priori bounds independent of T and a d_2-Lipschitz dependence on $\boldsymbol{m}_{t_0}$ (the existence/uniqueness theory of chapter 24). Then the function U of* (26.8) *is the unique candidate master function; it is d_2-Lipschitz in $\boldsymbol{m}$ uniformly in (t, x, ξ), and wherever it possesses the field-$\mathcal{C}^2$*

and state-$\mathcal{C}^{1,2}$ regularity of theorem 26.3 it is the unique classical solution of the graphon master equation (26.10)–(26.11)*.*

The qualifier "wherever it possesses the regularity" in the conclusion is not a hedge to be glossed over; it is the precise location of the gap, and the proof is organized to make that location unmistakable. The architecture is that of theorem 12.8, lifted type by type, in four steps: construct U, show it is Lipschitz, show it is differentiable, and verify it solves the equation uniquely. Three of the four lift cleanly. The third does not, and we mark it as a hypothesis rather than dress it as a theorem.

Proof sketch, and an honest accounting of the gap. The architecture is that of theorem 12.8, lifted, and we are explicit about which steps are complete and which are reduced.

Step 1 (construction and single-valuedness). Under the hypotheses, chapter 24 provides a unique solution of (26.6) for each $(t_0, \boldsymbol{m})$ with bounds uniform in T, and (26.8) then defines U pointwise by reading that solution's value at the initial instant of its own problem. The single-valuedness of U *is* uniqueness of the system: without it "the equilibrium from $\boldsymbol{m}$" would name a set and (26.8) would not define a function. Two facts make this step work for all horizons, and both trace to monotonicity. First, the graphon-monotone coupling (26.14) is what prevents the equilibrium characteristics — the field flows of section 26.3.1 — from crossing as the horizon T grows, exactly as the classical inequality (12.13) did in chapter 12; were two distinct flows to cross, two values would be assigned to one field and U would fail to exist as a function. Second, monotonicity supplies the a priori bound uniform in T, so that the construction does not deteriorate as the horizon lengthens. Given the existence theory of chapter 24, this step is complete.

Step 2 (Lipschitz estimate in the field). We claim $|U(t, x, \xi, \boldsymbol{m}) - U(t, x, \xi, \boldsymbol{m}')| \le C\, d_2(\boldsymbol{m}, \boldsymbol{m}')$, uniformly in (t, x, ξ). The mechanism is the field lift of the classical Step 2 pairing, and it is worth seeing where each hypothesis enters. One compares the two equilibria launched from the nearby fields $\boldsymbol{m}, \boldsymbol{m}'$, pairs the difference of their HJB systems against the difference of their field-Fokker–Planck systems, and — the one new operation — *integrates the resulting identity over the type label.* Three forces then meet. The graphon-monotone coupling (26.14) enters this energy identity with a favorable sign, exactly as the quadratic form (26.17) predicts, and supplies the dissipation that closes the estimate. The cross terms that monotonicity does not directly kill are of two kinds: those local to a type-slice are absorbed by (A-Nondeg), whose uniformly nondegenerate $\sigma\sigma^\top$ gives the parabolic smoothing in the state ξ that controls the difference of the value gradients; and those that couple distinct slices — the genuinely graphon terms, in which one type's discrepancy feeds another's through the kernel — are bounded by the operator norm $\|\mathbb{W}\|$, which quantifies exactly how strongly the type integral $\int_0^1 \cdots \mathrm{d}y$ mixes the slices. The role of $\|\mathbb{W}\|$ here is the analytic shadow of its appearance in the spectral form (26.17): the same operator that decides whether monotonicity holds also sets the size of the inter-slice coupling the estimate must dominate. In practice one need not redo this pairing from scratch, because chapter 24 already delivers its conclusion at the level of the *system*: by hypothesis the field-MFG pair $(u^{\cdot}, \boldsymbol{m})$ depends d_2-Lipschitz on its initial field $\boldsymbol{m}_{t_0}$, with a constant independent of the horizon. Since $U(t, x, \xi, \boldsymbol{m}) = u^{x;t,\boldsymbol{m}}(t, \xi)$ is precisely that system's value read at the initial time of its own problem (26.8), the system Lipschitz bound transfers *directly* to U, giving the claimed estimate with the same constant. This is the graphon lift of Step 2 of theorem 12.8, where the identical pairing is carried out for a single measure; integrating the monotonicity identity over types and absorbing the $\|\mathbb{W}\|$-weighted cross terms by (A-Nondeg) is the only change. As a by-product the estimate bounds the slice flat derivative, hence the Lions field $\partial^y_\mu U$, uniformly in the type — the first-order field regularity that Step 3 must upgrade to genuine differentiability.

Step 3 (differentiability in the field). This is where the classical and graphon theories part, and the honesty of the whole section lives in being exact about why. A Lipschitz function need not be differentiable, so Step 2 alone does not entitle us to the derivatives $\partial^y_\mu U$ and $\partial_\zeta\partial^y_\mu U$ that the master equation (26.10) contains as *genuine* objects rather than mere difference quotients. To produce them one differentiates the forward–backward system (26.6) itself in the direction of a single slice perturbation $m^y \to m^y + \epsilon(m' - m^y)$. The result is a *linearized graphon MFG system*: a forward–backward pair, posed simultaneously over the whole continuum of types, for the first-order response $(\partial u, \partial m)$ of the equilibrium to the perturbation. Its source is concentrated at the perturbed type y, but its solution is felt at every type, because the linearized coupling propagates the disturbance through $\mathbb{W}$ just as the nonlinear one does. If this linearized system is uniquely solvable with estimates uniform in the perturbed type y, its solution *is* the flat derivative $\delta U/\delta m^y$, its state-gradient the Lions derivative $\partial^y_\mu U$, and their continuity in $(\boldsymbol{m}, \zeta, y)$ follows from continuity of the linearized data; a second linearization, in a second slice y', would deliver the second-order field regularity. The obstacle is not the existence of *a* solution but the demand for estimates that are uniform in two directions at once: *across the continuum of types*, so that no slice's response blows up relative to its neighbours as the type label varies, and *against the spectrum of* $\mathbb{W}$, so that the inter-slice feedback the operator induces does not amplify without control as one approaches the threshold of (A-Coupling). It is the combination — a uniform-in-type estimate that also survives the full strength of the operator — that has no general proof. In the classical case (12.8) the corresponding linearized-system estimates and the genuine twice-differentiability in m are carried out in full; the standard reference for that program is [37], and we cite it here *for attribution* — to name the classical template the graphon case aims at — and emphatically *not* as a stand-in for a graphon argument, which would violate the self-containedness the book holds itself to. No theorem at the strength of [37] is presently available over a continuum of types coupled by a general admissible kernel. What *does* rescue Step 3 is additional structure that collapses the type continuum or the spectrum to something finite-dimensional and explicitly controllable: a separable or finite-rank kernel, for which $\mathbb{W}$ has finitely many nonzero eigenvalues and the linearized system reduces to a finite coupled set; a block (piecewise-constant) graphon, for which the continuum of types degenerates to finitely many populations and the problem becomes a classical multi-population MFG to which the homogeneous theory applies verbatim; and, sharpest of all, the linear–quadratic case of section 26.6, where the value is quadratic in the state, the field derivatives are computed in closed form, and the entire differentiability question reduces to the solvability of an operator Riccati equation. In each of these the uniform-in-type, uniform-in-spectrum estimate can be exhibited by hand. We therefore state Step 3 honestly as a hypothesis — verified in the special structures just named, open in general — and defer the formal tier-by-tier accounting to section 26.7. This is the one place where the graphon master equation is not yet on the footing its classical ancestor enjoys.

Step 4 (verification and uniqueness). Wherever Steps 1–3 hold, the loop closes exactly as in section 26.3.1. Theorem 26.3 shows the constructed U, now known to carry the required derivatives, solves the master equation (26.10); and theorem 26.5 forces uniqueness without any further regularity input, by the characteristic argument of the previous section: any classical solution must agree with the system's (unique) value along every field characteristic, and the characteristics through $\{t_0\} \times \boldsymbol{\mathcal{P}}_2$ fill the base space, leaving no field at which a competing solution could differ. Uniqueness is thus settled by the deterministic characteristics — a point worth remembering, because it is precisely this mechanism that common noise will dismantle in section 26.5.

In summary, the accounting is three-way. We prove outright the *equivalence* of the formulations (theorems 26.3 and 26.5) and the *Lipschitz estimate* on U (Steps 1–2). We *reduce* the construction to the graphon existence theory of chapter 24. And we leave *open* the genuine classical *regularity* (Step 3), reducing it to a linearized-system analysis that is complete only in the special structures named above. We do not claim a general graphon well-posedness theorem of the strength of [37]; none is known, and the closing section keeps the book honest by laying out exactly that boundary. □

There is, alongside this conditional all-horizons result, an *unconditional* statement that costs nothing beyond smallness, and naming it sharpens what monotonicity is actually for.

Remark 26.8 (Short horizon and the operator threshold)**.** The honest *unconditional* statement, valid without any monotonicity assumption at all, is the contraction one. If either the horizon T is small or the operator norm $\|\mathbb{W}\|$ is below the threshold (A-Coupling), the best-response map on the field flow is a contraction (chapter 24): iterating "given a field flow, compute each type's optimal response, recompute the field it induces" converges to a unique fixed point. The construction (26.8) is then well defined, inherits the smoothness of the data directly — no linearized-system analysis is needed, because the contraction supplies differentiable dependence on the datum for free — and the graphon master function exists, is unique, and is classical, *with no gap of the Step 3 kind*. This is the graphon face of remark 12.11. What is instructive is how the roles split. Classically the cheap route to local well-posedness is smallness of T alone; in the graphon setting that role is *shared* between two knobs, the horizon T and the operator norm $\|\mathbb{W}\|$, and either one being small suffices. The horizon controls how long errors have to compound; the operator norm controls how strongly a discrepancy at one type feeds back, through the kernel, to the others — the very mixing measured in Step 2. A long horizon can be tolerated if the network coupling is weak, and a strong network if the horizon is short; only when both are large does the contraction fail and one must fall back on monotonicity. The substantive open problem is exactly to remove the smallness on *both* knobs and replace it by monotonicity *for all horizons and all admissible kernels* — which, as Step 3 made plain, theorem 26.7 achieves only partially.

The boundary this section has drawn — proved equivalence and Lipschitz estimate on one side, open differentiability on the other — is handed forward in two directions. To section 26.7 it goes for the formal tier-by-tier accounting that closes the chapter, where the Step 3 obstruction is set beside the other conditional pieces of the theory and the proved core is separated cleanly from the conjectural shell. And to section 26.6 it goes as a challenge met: the linear–quadratic thread is the special structure in which everything *does* close, where the abstract "spectral condition on $\mathbb{W}$" of (26.17) stops being a conjecture and becomes the concrete, checkable requirement that an operator Riccati equation admit a global solution on $[0, T]$. The reader should carry into that computation the expectation set here: the place where Step 3 is rescued is the place where the field derivatives are available in closed form, and the spectral threshold that this section could only state is one the LQG model will let us see directly, as the leading eigenvalue of $\mathbb{W}$ driving — or failing to drive — a scalar Riccati equation to blow-up.

26.5 Common (aggregate) noise

Two threads laid down earlier in the chapter now converge, and their meeting is the chapter's decisive payoff. Section 26.1 promised that the all-contingencies object U — a mere convenience in the deterministic theory, where one could always fall back on solving the forward–backward system from a fixed datum — would become an outright *necessity* the moment the population field turned genuinely random. And section 26.3.1 rested the entire deterministic theory on one geometric picture: the graphon master equation is a *first-order* Hamilton–Jacobi equation on the field space $\boldsymbol{\mathcal{P}}_2$, its characteristics are the equilibrium field Fokker–Planck flows, and solving it amounts to running a single such curve and reading U off along it. Section 26.4 then mapped the boundary of that deterministic theory. This section crosses the boundary. We add a shock shared across all types, and the effect is total: the field flow ceases to be a curve and becomes a stochastic process; there is no longer any single characteristic to run; the deterministic forward–backward system (26.6) and the characteristic dictionary of section 26.3.1 both dissolve; and the master equation, the one object defined for *every* field at once, is left standing as the only surviving formulation. The promise of section 26.1 is thereby made good in its strongest form. And the equation pays for its new indispensability by growing a genuine second-order operator over the type continuum — the field-space Laplacian whose conspicuous *absence* the notation-watch of section 26.3 singled out as the very signature of "no common noise." That term now switches on, and accounting for it is the technical heart of the section.

We add the shock concretely. Let $B^0 = (B^0_t)$ be a single Brownian motion, independent of the private noises $(B^x)_{x\in[0,1]}$, common to the entire population, and let each type-x agent be driven by it:

$$\mathrm{d}X^x_t \;=\; \alpha^x_t\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}B^x_t \;+\; \sigma_0\,\mathrm{d}B^0_t, \qquad X^x_0 \sim m^x_0, \tag{26.18}$$

with σ_0 a constant common-noise volatility (the type-independent case; we comment on type-dependent common noise at the end of the section). The single increment $\sigma_0\,\mathrm{d}B^0_t$ is added to the dynamics of *every* agent of *every* type at once: it is one and the same random push, felt coherently across the whole population. It is exactly this coherence — not the mere presence of extra noise — that reorganizes the theory; were each type handed its own independent aggregate shock the slices would still fluctuate separately and nothing structural would change. Write $\mathbb{F}^0 = (\mathcal{F}^0_t)$ for the filtration generated by B^0. The right population object is then no longer the unconditional law of a type's state, because averaging over the common shock *as well as* the private noise would wash out precisely the coherent response we have just built in, returning us to a deterministic background. What survives the shock is the law *conditioned* on the realized common path — the *conditional field*

$$m^x_t \;=\; \mathcal{L}(X^x_t \,|\, \mathcal{F}^0_t), \qquad x \in [0,1], \tag{26.19}$$

the configuration of type x's crowd *given* everything the global shock has done up to time t. With the common path frozen by the conditioning, the only randomness left inside each slice is the independent private noise, over which a law of large numbers still averages; but the slice that results is now carried along *with* the shock rather than sitting still. The whole field $\boldsymbol{m}_t = (m_t^x)_x$ is therefore itself a *stochastic process* valued in $\boldsymbol{\mathcal{P}}_2$ and adapted to $\mathbb{F}^0$: a random point of field space, re-randomized at every instant by an increment $\mathrm{d}B_t^0$ that no conditioning on the past can anticipate.

Heuristic (non-rigorous)

The conditional McKean–Vlasov picture. Fix, in imagination, one realized path of the common noise B^0 and condition on it. Two sources of randomness drove each agent: the private noise B^x, idiosyncratic to the agent, and the common noise B^0, shared by all. The conditioning removes the second, and what is left inside each type is a continuum of agents driven by *independent* private noises — exactly the configuration in which the law of large numbers operated in the deterministic theory of chapter 22. So the empirical distribution of type x still concentrates on a deterministic conditional law; the law of large numbers has *not* failed. What has changed is what it concentrates *around*: the averaging now happens around the common-noise path, and the limiting slice is dragged along by the realized B^0 rather than standing still.

Made concrete, the conditional field $\boldsymbol{m}_t$ obeys a *stochastic* Fokker–Planck equation, one slice at a time. Each m_t^x still satisfies the deterministic transport-and-spreading of (26.6b), but the common increment $\sigma_0 \, \mathrm{d}B_t^0$ contributes two further effects, worth separating because they are genuinely different in kind. First, because every agent of type x receives the *same* push $\sigma_0 \, \mathrm{d}B_t^0$, the whole slice is *rigidly translated* by that increment — a transport of the entire conditional law, not a smearing of it; this is a first-order martingale term carrying $\mathrm{d}B_t^0$. Second, the part of the common fluctuation that the conditioning at time t has not yet resolved contributes, through Itô's correction, an extra $\frac{1}{2}\sigma_0\sigma_0^\top$ *spreading* of the slice, identical in form to the private-noise spreading $\frac{1}{2}\sigma\sigma^\top$ already present and merely upgrading the diffusion coefficient. The rigid translation is the genuinely new feature; the extra spreading is only a coefficient bump. Crucially, both are driven by the *same* B^0 in every slice at once — the single fact that will, two displays from now, knit all types' fluctuations together into one coherent second-order object.

The consequence for the equilibrium is decisive. It is no longer a deterministic forward–backward field flow but a *random* one: the population reacts coherently to each realization of the global shock, the realized field at time t depends on the entire history of B^0 up to t, and the value functions must be adapted to information disclosed only as time runs forward. There is no single curve $\boldsymbol{m}_t$ to solve for in advance — one cannot know "the realized slice" before the shock that produces it has happened — and so the deterministic system (26.6), which presupposed exactly such a curve, loses its meaning, just as the characteristic picture of section 26.3.1 that was built on it.

This is precisely the situation the master equation was built for, and the reason section 26.1 could promise its necessity would become absolute. The bare value family of (26.6) was a function of (t, ξ) pinned to one deterministic field flow; with that flow gone there is nothing left to pin it to, and it cannot even be written down. The master function suffers no such collapse. $U(t, x, \xi, \boldsymbol{m})$

was constructed in (26.8) to assign a value to *every* field $\boldsymbol{m} \in \boldsymbol{\mathcal{P}}_2$, in advance and independently of which one is realized — it is the equilibrium cost-to-go of a type-x agent at ξ *whatever* the population field happens to be. Precisely because it is defined for every contingency at once, it can be *composed* with the random field: the realized value of a type-x agent is the stochastic process

$$u_t^x(\xi) \;=\; U\big(t, x, \xi, \boldsymbol{m}_t\big),$$

the master function evaluated along the conditional flow, and the optimal feedback is read off exactly as in the deterministic case, $\nabla_\xi U(t, x, X_t^x, \boldsymbol{m}_t)$, now itself a process adapted to $\mathbb{F}^0$ through its third and fourth arguments — the lift to the random flow of the decoupling-field reading (26.12). The flow-consistency identity (26.9) thus survives the shock verbatim, with its deterministic flow replaced by the conditional one; everything downstream of it can be recomputed by differentiating it along the new, random characteristic. This is the exact graphon lift of section 12.7: there, common noise turned the homogeneous master equation from a convenience into the only available description; here the same reversal occurs one level up, over the entire field of types. The master equation is no longer a luxury but the sole surviving formulation.

To find the equation U satisfies along the random flow we need two ingredients, and the section assembles them in turn: the object that gets differentiated *twice* in the field — absent from the entire deterministic development, switched on now for the first time — and the Itô formula that does the differentiating along the random flow. We begin with the new derivative.

Why a *second* field derivative should surface now, when the deterministic chain rule of lemma 26.2 never reached past the first, is worth seeing before the definition is written down. In the deterministic theory each slice was transported by its own drift and spread by its own *independent* private noise; distinct slices fluctuated independently, their cross-correlations vanished, and the chain rule closed at first order in the field with only the single-variable Jacobian $\partial_\zeta \partial_\mu^y \Psi$ of the slice Lions derivative. Under common noise every slice is driven by the *same* B^0, so distinct slices are now perfectly correlated, and Itô's correction will pair the response of a functional to a perturbation of slice y with its response to a perturbation of slice y'. That pairing is a genuine second-order object on the field space, indexed by *two* types at once. It is the second field derivative, and we define it first.

Definition 26.9 (Second field derivative)**.** Let $\Psi : \boldsymbol{\mathcal{P}}_2 \to \mathbb{R}$ be fully $\mathcal{C}^2$ (definition 26.1) and suppose, for each fixed slice y and foreign state ζ, the $\mathbb{R}^d$-valued map $\boldsymbol{m} \mapsto \partial_\mu^y \Psi(\boldsymbol{m})(\zeta)$ is itself slice-differentiable. The *second field derivative* of Ψ is the family

$$\partial_\mu^y \partial_\mu^{y'} \Psi(\boldsymbol{m})(\zeta, \zeta') \;:=\; \partial_\mu^{y'}\big[\, \boldsymbol{m} \mapsto \partial_\mu^y \Psi(\boldsymbol{m})(\zeta) \,\big](\zeta') \;\in\; \mathbb{R}^{d \times d},$$

indexed by the *pair* of types (y, y') and evaluated at the pair of foreign states (ζ, ζ'): the slice-y' Lions derivative of the slice-y Lions derivative. Equivalently $\partial_\mu^y \partial_\mu^{y'} \Psi = \nabla_\zeta \nabla_{\zeta'} \frac{\delta^2 \Psi}{\delta m^y \delta m^{y'}}$, where the second flat derivative $\frac{\delta^2 \Psi}{\delta m^y \delta m^{y'}}(\boldsymbol{m})(\zeta, \zeta')$ is the slice-y' flat derivative of $\boldsymbol{m} \mapsto \frac{\delta \Psi}{\delta m^y}(\boldsymbol{m})(\zeta)$. This is the slice-wise lift of the classical second Lions derivative ∂_μ^2 of definition D.6: restricted to a single slice it *is* ∂_μ^2, and the type pair (y, y') is the only new index. Under joint measurability in (y, y') and continuity in $(\boldsymbol{m}, \zeta, \zeta')$ it is symmetric, $\partial_\mu^y \partial_\mu^{y'} \Psi(\boldsymbol{m})(\zeta, \zeta') = \big(\partial_\mu^{y'} \partial_\mu^y \Psi(\boldsymbol{m})(\zeta', \zeta)\big)^\top$. We say Ψ is *fully $\mathcal{C}^2$ with second field derivative* when this object exists, is jointly measurable, and is continuous and bounded on bounded sets.

Two readings of $\partial_\mu^y \partial_\mu^{y'} \Psi$ will recur and it pays to fix them now. When $y = y'$ it is the ordinary classical second Lions derivative definition D.6 of Ψ restricted to that single slice — the object that already governed common noise in the homogeneous theory of chapter 12. When $y \neq y'$ it is the genuinely new content: it measures how the response of Ψ to a reweighting of slice y is itself altered by a reweighting of a *different* slice y', a cross-type curvature that has no homogeneous ancestor at all and that, in the deterministic theory, never had occasion to appear because distinct slices were uncorrelated. It is precisely this off-diagonal, two-type piece that the common noise will activate.

With the new derivative in hand, the second ingredient is the Itô formula that carries a functional of the field along the random flow — the stochastic upgrade of the deterministic chain rule lemma 26.2, with the conditional slices of (26.19) now solving *stochastic* Fokker–Planck equations rather than deterministic ones.

Lemma 26.10 (Stochastic field chain rule under common noise)**.** *Let $\Psi : \boldsymbol{\mathcal{P}}_2 \to \mathbb{R}$ be fully $\mathcal{C}^2$ with second field derivative (definition 26.9), derivatives bounded and Lipschitz on bounded sets uniformly in the type indices. Let $(\boldsymbol{m}_t)$ be the conditional field flow (26.19) whose slices solve the* stochastic *Fokker–Planck equations*

$$\mathrm{d}m_t^y = \Big[\tfrac{1}{2}\operatorname{tr}\big((\sigma\sigma^\top + \sigma_0\sigma_0^\top)\,\mathrm{D}_\zeta^2\, m_t^y\big) - \operatorname{div}\big(m_t^y\,\mathbf{b}_t^y\big)\Big]\,\mathrm{d}t - \sigma_0^\top \nabla_\zeta m_t^y\ \mathrm{d}B_t^0 \qquad (y \in [0,1]), \tag{26.20}$$

driven by the common Brownian motion B^0 and a measurable family of drift fields $\mathbf{b}_t^y$ of at most linear growth with $\int_0^T\int_0^1\int |\mathbf{b}_t^y|^2\ \mathrm{d}m_t^y\ \mathrm{d}y\ \mathrm{d}t < \infty$. Then $t \mapsto \Psi(\boldsymbol{m}_t)$ is an Itô process and

$$\begin{aligned}\mathrm{d}\Psi(\boldsymbol{m}_t) = \Big[\int_0^1\!\!\int \partial_\mu^y \Psi \cdot \mathbf{b}_t^y\ \mathrm{d}m_t^y\ \mathrm{d}y + \tfrac{1}{2}\int_0^1\!\!\int \operatorname{tr}\big((\sigma\sigma^\top + \sigma_0\sigma_0^\top)\,\partial_\zeta\partial_\mu^y\Psi\big)\,\mathrm{d}m_t^y\ \mathrm{d}y \\ + \tfrac{1}{2}\int_0^1\!\!\int_0^1\!\!\iint \operatorname{tr}\big(\sigma_0\sigma_0^\top\,\partial_\mu^y\partial_\mu^{y'}\Psi(\boldsymbol{m}_t)(\zeta,\zeta')\big)\,\mathrm{d}m_t^y(\zeta)\ \mathrm{d}m_t^{y'}(\zeta')\ \mathrm{d}y\ \mathrm{d}y'\Big]\,\mathrm{d}t \\ + \int_0^1\!\!\int \partial_\mu^y\Psi \cdot \sigma_0\ \mathrm{d}m_t^y\ \mathrm{d}y\ \ \mathrm{d}B_t^0.\end{aligned} \tag{26.21}$$

Proof sketch. The argument is, as for the deterministic chain rule, two moves: a per-slice conditional Itô formula, and the sum over the type continuum — but the second move now carries the whole new content, because the common noise correlates the slices and the sum acquires an off-diagonal part. We carry the algebra in-book, citing the conditional Itô formula theorem D.8 only for attribution.

Move (a): per slice, the conditional Itô formula. Fix one type y, freeze every other slice, and apply theorem D.8 — the Itô formula along the conditional law $m_t^y = \mathcal{L}(X_t^y \mid \mathcal{F}_t^0)$ of the diffusion (26.18). It produces four contributions, and each is the conditional counterpart of a term already understood. The transport drift $\partial_\mu^y\Psi \cdot \mathbf{b}_t^y$ is the slice's optimal-feedback transport, exactly as in lemma 26.2. The spreading carries the *upgraded* diffusion $\sigma\sigma^\top + \sigma_0\sigma_0^\top$: the common noise smears the conditional slice just as the private noise does — the second of the two effects isolated in the heuristic above — which is the slice-wise origin of the second integrand in (26.21). The second-Lions term $\frac{1}{2}\operatorname{tr}(\sigma_0\sigma_0^\top\partial_\mu^2\Psi)$ is the curvature contribution of the slice's own martingale fluctuation, the conditional analogue of the common-noise term in chapter 12. And the martingale $\partial_\mu^y\Psi \cdot \sigma_0\ \mathrm{d}B_t^0$ records the rigid translation of the slice by the unresolved increment $\mathrm{d}B_t^0$ — the first effect of the heuristic. So far this is the homogeneous common-noise calculation of section 12.7, run inside one type with every other type held fixed.

Move (b): sum over types, and the cross-slice correlation. Now release the freezing and sum the per-slice contributions over the continuum, as in Move (b) of lemma 26.2. The transport, spreading, and martingale terms each acquire merely the outer $\int_0^1 \cdots \mathrm{d}y$, reproducing the first, second, and last lines of (26.21). The second-order term is where the lift is genuinely new, and the reason is the single fact that every slice is driven by the *same* B^0. The conditional slice m_t^y has, by (26.20), the martingale part $-\sigma_0^\top \nabla_\zeta m_t^y \, \mathrm{d}B_t^0$; hence for two distinct types the cross quadratic variation

$$\mathrm{d}\langle m^y, m^{y'} \rangle_t \;=\; (\sigma_0^\top \nabla_\zeta m_t^y) \otimes (\sigma_0^\top \nabla_\zeta m_t^{y'}) \; \mathrm{d}t \;\neq\; 0 \qquad (y \neq y'),$$

because both martingale parts are proportional to the *one* increment $\mathrm{d}B_t^0$ and so are perfectly correlated — the very feature that vanished in the deterministic theory, where each slice carried its own independent noise and these covariations were zero. Itô's second-order correction for a functional of all the slices therefore picks up not only the diagonal $y = y'$ contributions (the per-slice second-Lions terms of Move (a)) but also the entire off-diagonal $y \neq y'$ family. Both are governed by the second field derivative of definition 26.9 — the diagonal by its restriction to one slice, the off-diagonal by its genuine two-type part — and summing them is summing over *both* type labels independently. The per-slice second-Lions term is thus promoted to the *double* type integral $\int_0^1 \int_0^1 \cdots \mathrm{d}y \; \mathrm{d}y'$ of the third line of (26.21), contracted through $\sigma_0 \sigma_0^\top$ against $\partial_\mu^y \partial_\mu^{y'} \Psi$. This single promotion — diagonal sum $\to$ double integral — is the entire structural difference between the deterministic and the common-noise field calculus.

The linear-growth and integrability hypotheses license the integrations by parts and the Fubini interchanges exactly as in lemma 26.2; the truncation argument of that proof carries over slice by slice with the conditional Itô formula in place of the deterministic one, and the martingale term is square-integrable under the same uniform L^2 bound. The rigorous Itô–Wentzell statement on $\boldsymbol{\mathcal{P}}_2$ that underlies the displayed bookkeeping is the field lift of theorem D.8, organized for the graphon setting in [16]; we cite it for the analytic justification but the displayed algebra is the whole argument. $\square$

With both ingredients in hand we can differentiate the value process $u_t^x(\xi) = U(t, x, \xi, \boldsymbol{m}_t)$ along the conditional flow and read off the equation. The derivation is the stochastic twin of theorem 26.3: there the deterministic chain rule expanded $\mathrm{d}U/\,\mathrm{d}t$ and we matched against the type-x HJB; here Itô's formula expands $\mathrm{d}U$ and we match the resulting *drift* against the same HJB, the martingale part integrating to zero. The new subtlety is that U now depends on t through *two* channels that both carry the common increment $\sigma_0 \; \mathrm{d}B_t^0$ — the agent's own state $\xi = X_t^x$ and the field $\boldsymbol{m}_t$ — so Itô couples them. It is worth writing the expansion out, because the coupling is exactly where the two new terms come from. Apply Itô–Wentzell to $U(t, x, X_t^x, \boldsymbol{m}_t)$:

$$\begin{aligned}\mathrm{d}U = \partial_t U \; \mathrm{d}t \;+\; \nabla_\xi U \cdot \mathrm{d}X_t^x \;+\; \tfrac{1}{2}\,\mathrm{tr}\big(\mathrm{D}_\xi^2 U \; \mathrm{d}\langle X^x \rangle_t\big) \;+\; \big[\text{field drift of lemma 26.10}\big] \; \mathrm{d}t \\ +\; \Big(\int_0^1 \!\! \int \partial_\mu^y U \cdot \sigma_0 \; \mathrm{d}m_t^y \; \mathrm{d}y\Big) \, \mathrm{d}B_t^0 \;+\; \mathrm{d}\big\langle\, \nabla_\xi U \cdot X^x, \text{ field martingale} \,\big\rangle_t.\end{aligned} \tag{26.22}$$

The first line collects the ordinary Itô expansion in the agent's own state together with the field drift already computed in lemma 26.10; the second line collects the field martingale of that lemma and the cross-variation between the two B^0-driven channels. Two of these pieces are new relative to the deterministic derivation, and they are precisely the two places the common increment enters twice.

First, the quadratic variation of the agent's own state is now $\mathrm{d}\langle X^x\rangle_t = (\sigma\sigma^\top + \sigma_0\sigma_0^\top)\,\mathrm{d}t$, since X^x carries both the private and the common Brownian increments; so the term $\frac{1}{2}\operatorname{tr}(\mathrm{D}^2_\xi U\;\mathrm{d}\langle X^x\rangle_t)$ contributes, on top of the $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top\,\mathrm{D}^2_\xi U)$ already in (26.10), the extra agent diffusion $\frac{1}{2}\operatorname{tr}(\sigma_0\sigma_0^\top\,\mathrm{D}^2_\xi U)$. Second, the cross-variation in the last term of (26.22) is nonzero precisely because both channels are driven by the *same* B^0: the agent's state contributes the increment $\nabla_\xi U\cdot\sigma_0\,\mathrm{d}B^0_t$ and the field contributes the martingale $\big(\int_0^1\!\int \partial^y_\mu U\cdot\sigma_0\,\mathrm{d}m^y_t\,\mathrm{d}y\big)\,\mathrm{d}B^0_t$, and their joint quadratic variation pairs the agent-state gradient ∇_ξ with the slice Lions derivative ∂^y_μ, both contracted through $\sigma_0\sigma_0^\top$. Because mixed derivatives commute, the resulting integrand is the single mixed object $\nabla_\xi\partial^y_\mu U$ — the agent–field cross term — integrated against m^y over the foreign state and over the type. (Had the agent's own noise and the common noise been independent, this cross-variation would vanish and the term would be absent; it is a pure consequence of the shared shock.)

Collecting the drift of (26.22) — which includes the field drift of lemma 26.10, itself carrying the population-spreading upgrade $\sigma\sigma^\top\to\sigma\sigma^\top+\sigma_0\sigma_0^\top$ and the double-type-integral second-field-derivative term — and matching it against the type-x HJB exactly as in theorem 26.3, the master equation acquires, added to the left-hand side of (26.10), *four* families of σ_0-terms. Two are *coefficient upgrades* of second-order lines already present in (26.10) — the agent's own diffusion and the population spreading, each promoting $\sigma\sigma^\top\to\sigma\sigma^\top+\sigma_0\sigma_0^\top$ — and two are *genuinely new*, the agent–field cross term just isolated and the double-type-integral field Laplacian inherited from lemma 26.10:

$$
\begin{aligned}
&-\tfrac{1}{2}\operatorname{tr}\big(\sigma_0\sigma_0^\top\,\mathrm{D}^2_\xi U\big)\;-\;\tfrac{1}{2}\int_0^1\!\!\int_{\mathbb{R}^d}\operatorname{tr}\big(\sigma_0\sigma_0^\top\,\partial_\zeta\partial^y_\mu U(t,x,\xi,\boldsymbol{m})(\zeta)\big)\,m^y(\mathrm{d}\zeta)\;\mathrm{d}y\\
&-\;\int_0^1\!\!\int_{\mathbb{R}^d}\operatorname{tr}\big(\sigma_0\sigma_0^\top\,\nabla_\xi\partial^y_\mu U(t,x,\xi,\boldsymbol{m})(\zeta)\big)\,m^y(\mathrm{d}\zeta)\;\mathrm{d}y\\
&-\;\tfrac{1}{2}\int_0^1\!\!\int_0^1\!\!\int_{\mathbb{R}^d}\!\int_{\mathbb{R}^d}\operatorname{tr}\big(\sigma_0\sigma_0^\top\,\partial^y_\mu\partial^{y'}_\mu U(t,x,\xi,\boldsymbol{m})(\zeta,\zeta')\big)\,m^y(\mathrm{d}\zeta)\,m^{y'}(\mathrm{d}\zeta')\;\mathrm{d}y\;\mathrm{d}y',
\end{aligned}
\tag{26.23}
$$

where $\partial^y_\mu\partial^{y'}_\mu U$ is the second field derivative (definition 26.9). It pays to read the four families one at a time, since each has a distinct mechanical origin in the Itô expansion (26.22) and a distinct role in the equation. *(i) The agent's own extra diffusion* $-\frac{1}{2}\operatorname{tr}(\sigma_0\sigma_0^\top\,\mathrm{D}^2_\xi U)$ is the simplest: the common noise enlarges the quadratic variation of the agent's private state from $\sigma\sigma^\top$ to $\sigma\sigma^\top+\sigma_0\sigma_0^\top$, so the ordinary second-order Itô term in ξ gains the $\sigma_0\sigma_0^\top$ piece. It is local in ξ and involves no type integral at all; it would be present even for an isolated agent. *(ii) The population self-spreading* $-\frac{1}{2}\int_0^1\!\int\operatorname{tr}(\sigma_0\sigma_0^\top\partial_\zeta\partial^y_\mu U)\,m^y\;\mathrm{d}y$ is the matching upgrade of the third line of (26.10): it arises because each conditional slice now spreads by the full $\sigma\sigma^\top+\sigma_0\sigma_0^\top$ of (26.20), and U responds through the same single-variable Lions Jacobian that carried the private spreading, summed over types. *(iii) The agent–field cross term* $-\int_0^1\!\int\operatorname{tr}(\sigma_0\sigma_0^\top\nabla_\xi\partial^y_\mu U)\,m^y\;\mathrm{d}y$ is the joint quadratic variation isolated in (26.22): the agent's own state and the field both translate under the *same* $\sigma_0\;\mathrm{d}B^0$, and Itô's correction pairs the agent-state gradient with the slice Lions derivative, the mixed object $\nabla_\xi\partial^y_\mu U$ under a single type integral. *(iv) The double-type-integral field Laplacian* $-\frac{1}{2}\int_0^1\!\int_0^1\!\iint\operatorname{tr}(\sigma_0\sigma_0^\top\,\partial^y_\mu\partial^{y'}_\mu U)\,m^y\,m^{y'}\;\mathrm{d}y\;\mathrm{d}y'$ is the genuinely new object, the one the entire deterministic development was missing. Its decisive feature — the only way in which it exceeds the classical common-noise formula (12.17) — is its *type structure*: the *double* type integral $\int_0^1\!\int_0^1\cdots\mathrm{d}y\;\mathrm{d}y'$. A single common noise correlates *all pairs of types* coherently, exactly the nonvanishing cross-slice quadratic variations of lemma 26.10, so the second-order field operator pairs slice y with slice y' across the entire continuum, weighting the pairing by $\sigma_0\sigma_0^\top$.

The collapse to the classical theory is the cleanest sanity check on all four. Set the type space $[0,1]$ to a single point: every type integral $\int_0^1 \cdots \mathrm{d}y$ becomes a bare evaluation, the double integral $\int_0^1\int_0^1 \cdots$ becomes a single second-Lions term, the two-type derivative $\partial_\mu^y \partial_\mu^{y'} U$ reduces to the classical $\partial_\mu^2 U$, and (26.23) reduces *term for term* to the four common-noise corrections of the homogeneous master equation (12.17) — agent diffusion, measure spreading, agent–measure cross term, and measure Laplacian. The graphon formula adds nothing over chapter 12 except the resolution of each correction over the type continuum and the binding of distinct types through the off-diagonal of (iv); nothing graphon-specific would be lost or gained were there only one type.

The last term in (26.23) is a Laplacian on the field space, and its appearance changes the analytic character of the whole equation: it supplies the second-order regularization in the field direction that was *wholly absent* in the deterministic theory (section 26.3, where the notation-watch flagged exactly this object as missing), and with it the graphon master equation becomes *parabolic on* $\boldsymbol{\mathcal{P}}_2$ rather than first-order. This is, exactly as in chapter 12, at once a blessing and a burden, and the two readings are worth holding together. It is a *blessing* because the field-space smoothing can compensate for the loss of the deterministic-characteristics method that common noise destroyed: the parabolicity is, in the classical case, precisely what makes well-posedness possible under *weak* (rather than strict) monotonicity [4], the noise regularizing where the monotonicity alone would not suffice. It is a *burden* because a second field derivative integrated across *two* independent type labels is an analytically heavy object — it lives on pairs of types and carries a full matrix kernel — and controlling it is the principal obstacle to a complete graphon well-posedness theory, the graphon-specific reason the program of section 26.4 could not be closed in general. The very term that rescues the equation from the collapse of its characteristics is the term that no classical-graphon shortcut is known to tame.

Physics note

Common noise is the closest the graphon mean field story comes to a genuine fluctuating order-parameter field. A single global mode B_t^0 shakes *every* type coherently, so the macroscopic state — here an entire *field* $\boldsymbol{m}_t$ over the type continuum — becomes a stochastic process rather than a deterministic background. The double-type-integral term in (26.23) is the propagator of that global mode: it pairs the linear response at type y with the linear response at type y' and diffuses them together, the order-parameter "two-point function" the deterministic mean-field description could not see. The graphon adds genuinely new structure over the homogeneous case of chapter 12: the global shock is felt by different types with strengths that the kernel W and the type profile redistribute, so the fluctuation of the order parameter is itself *type-resolved*. As always this is orientation only — nothing below the line of a theorem rests on it — and the quantitative content is the (incomplete) well-posedness theory of section 26.7.

Remark 26.11 (Type-dependent common noise and a common-noise graphon)**.** The single global B^0 of (26.18) is the simplest aggregate shock. A natural and strictly more general structure replaces the scalar σ_0 by a *type-dependent* loading $\sigma_0(x)$, or even couples the types to a *family* of common noises through a second kernel — a "common-noise graphon" $W_0(x,y)$ — so that the coherent shock felt by type x is $\int_0^1 W_0(x,y)\,\mathrm{d}B_t^y$ for a continuum of independent global modes. The structure of the extra terms (26.23) is unchanged in form, but $\sigma_0\sigma_0^\top$ is replaced inside the double integral by the covariance kernel $W_0 \star W_0(y,y') = \int_0^1 W_0(y,r)W_0(y',r)\,\mathrm{d}r$, which modulates how strongly the global fluctuation correlates types y and y'. This is the natural arena for the physicist's intuition — a spatially structured noise correlated through the network — and it is essentially uncharted; we flag it for the research map of chapter 31.

Four families of σ_0-terms have been written down and named, but for the general equation that is all we can do: (26.23) is the structural form of the common-noise master equation, and section 26.4 has already conceded that its well-posedness in the field direction is open. The honest tier-by-tier accounting of what is proved and what is conjectural — including the status of the double-type-integral term and of the common-noise-graphon extension of remark 26.11 — is gathered in section 26.7, to which the order-parameter-fluctuation reading of the physics box and the research questions of chapter 31 are handed forward. But before that reckoning the chapter owes the reader the one place where the abstract is made wholly concrete. The natural next question is operational: in a model where every field derivative is available in closed form, how do these four families actually *act* — which survive, which vanish, and what does the surviving structure say? The linear–quadratic thread answers it completely. Section 26.4 identified LQG as the special structure in which the field regularity does close; we now use that closure to evaluate (26.10) and its common-noise correction (26.23) explicitly, continuing Thread B rather than re-deriving it. There the quadratic value function makes $\partial_\mu^y U$ affine and $\partial_\mu^y \partial_\mu^{y'} U$ *constant* in the field, the abstract equation collapses to an operator Riccati equation, and the common-noise families sort themselves cleanly — a certainty-equivalence computation in which the population-spreading term integrates away while the agent's own diffusion, the cross term, and the field Laplacian survive in a fully explicit form. That is the business of the next section.

26.6 Worked computation: the graphon LQG master equation

The chapter has built an abstract object and then mapped the limits of what can be said about it. Section 26.3 derived the graphon master equation (26.10); section 26.4 conceded that its general well-posedness in the field direction is open, while singling out the linear–quadratic structure as the one special case in which the field regularity provably *does* close; and section 26.5 grew the equation a double-type-integral Laplacian (26.23) and left standing the operational question it could not answer in general — in a model where every field derivative is available in closed form, how do the four families of common-noise terms actually act, which survive, and what does the surviving structure say? This section answers all of it at once, by the only route the chapter has that is unconditional: we advance Thread B (appendix A.7) to its master-equation layer, the graphon lift of the LQG computation of section 12.8, and there every abstract object of sections 26.3 and 26.5 acquires an explicit closed form. This is the one section in which every claim is unconditional. The linear–quadratic structure makes the master function explicit, reduces the abstract equation (26.10) to a single *operator* Riccati equation on $L^2([0,1])$, and — through the rank-one graphon of example 15.11 — collapses that to a *scalar* Riccati equation in an order parameter; the common-noise correction (26.23) then sorts itself by a certainty-equivalence computation that leaves the feedback untouched. We continue the thread rather than re-derive its setup: the model, the parameter names, and the coupling are exactly those fixed in appendix A.7, carried forward without restatement.

The model is Thread B with its fixed parameter names. Each type-x agent carries a scalar state governed by the linear dynamics

$$\mathrm{d}X_t^x = (aX_t^x + b\,\alpha_t^x)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^x, \qquad X_0^x \sim m_0^x, \tag{26.24}$$

with a the state feedback, b the control gain, σ the private-noise intensity, and B^x the type-x idiosyncratic Brownian motion of the chapter's notation watch — the kernel W below is never a noise. The agent pays the linear–quadratic cost

$$J^x(\alpha^x) = \mathbb{E}\left[\int_0^T \left(\tfrac{1}{2}(\alpha_t^x)^2 + \tfrac{q}{2}\big(X_t^x - s\,z_t(x)\big)^2\right)\mathrm{d}t + \tfrac{q_T}{2}\big(X_T^x - s_T\,z_T(x)\big)^2\right], \tag{26.25}$$

with $q, q_T \geq 0$ the running and terminal state-cost weights and s, s_T the mean-coupling strengths that pull the agent toward (or, in the repulsive sign, away from) the aggregate it feels. That aggregate is the only place the population enters, and it is the graphon average of the neighbour mean,

$$z_t(x) = (\mathbb{W}\bar{m}_t)(x) = \int_0^1 W(x,y)\,\bar{m}_t(y)\,\mathrm{d}y, \qquad \bar{m}_t(y) = \int \zeta\, m_t^y(\mathrm{d}\zeta). \tag{26.26}$$

This is the general coupling form (26.5) specialized to the interaction observable $\phi(\zeta) = \zeta$: the foreign type-slice m_t^y is collapsed to its scalar mean $\bar{m}_t(y)$, and the kernel $W(x,y)$ then averages those means over types to form the single number $z_t(x)$ that type x responds to. The whole channel by which one type's cost sees another type's population — the thread the chapter is strung on — runs through this one scalar, built by $\mathbb{W}$.

It is worth isolating the structural feature that makes the entire reduction below possible, because everything that follows is an exploitation of it. *The field $\boldsymbol{m}$ enters the cost only through its type-mean field $\bar{m}_t(\cdot) = \int \zeta\, m_t^{\cdot}(\mathrm{d}\zeta)$*, which is a single element of $L^2([0,1])$ — one real-valued function over types, not the full field of measures. The higher moments of each slice, its shape, its tails, all are invisible to (26.25); only the first moment survives, and it survives type by type as the profile $\bar{m}_t$. Consequently every field functional we shall meet below — the value U, its slice-Lions derivative, the population-sensitivity operator — depends on $\boldsymbol{m}$ *solely through* $\bar{m} \in L^2([0,1])$. This is the linchpin. It demotes the infinite-dimensional differentiation in $\boldsymbol{m}$ to ordinary calculus on the Hilbert space $L^2([0,1])$: a field derivative becomes a directional derivative in the mean profile, the slice-Lions derivative becomes constant in the foreign state (as we verify in (26.29)), and the abstract graphon master equation (26.10) becomes an equation for operators on $L^2([0,1])$. Without this collapse there would be no closed form; with it, the whole apparatus of chapter 12's LQG computation lifts verbatim, the single scalar coupling constant replaced by the operator $\mathbb{W}$.

Two reduction checks, demanded by the thread specification, anchor the general computation to ground already covered and pin down the cleanest case. Setting $W \equiv c$ (the constant graphon) makes every type feel the same aggregate up to the scale c, and the system must collapse to Thread A of appendix A.7 with the coupling rescaled — the homogeneous LQG MFG, recovered as the no-network limit. The rank-one kernel $W(x,y) = f(x)f(y)$ of example 15.11 is the opposite extreme and the clean case: there $\mathbb{W}$ is a single projector, the operator Riccati lives in a one-dimensional family, and the entire continuum of types is governed by a single scalar *order parameter*. We verify the constant-graphon reduction as the special case of the rank-one collapse and exhibit the rank-one reduction as the section's payoff.

The computation proceeds in three steps, each making one abstract object of the chapter explicit, and a fourth that returns to common noise. First we write the per-type HJB equation and posit the linear–quadratic ansatz for U, mirroring the classical LQG ansatz of section 12.8 but with the population-sensitivity coefficient promoted from a scalar to a time-dependent operator on $L^2([0,1])$; this is where the "depends on $\boldsymbol{m}$ only through $\bar{m}$" observation is cashed in and where the field-Laplacian (third) line of (26.10) is shown to vanish. Second we insert the ansatz, match powers of the state, and reduce the abstract master equation to a scalar curvature Riccati and an *operator* Riccati equation for the population-sensitivity. Third we specialize to the rank-one kernel and watch the operator Riccati collapse to a scalar Riccati in the order parameter, recovering Thread A along the way. The closing step adds the common shock and carries out the certainty-equivalence computation that section 26.5 promised. We begin with the first.

The per-type HJB and the operator ansatz

We begin exactly where the setup left us. The field $\boldsymbol{m}$ entered the cost (26.25) through one scalar per type, the graphon aggregate $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$, and so — the linchpin observation of the previous pages — every field functional we shall produce depends on $\boldsymbol{m}$ *only through the mean profile* $\bar{m} \in L^2([0,1])$. The first step of the computation is to turn that dependence into an explicit guess for the master function U, and to verify on the spot that the guess kills one of the three lines of the abstract equation (26.10). We do this by following the classical LQG calculation of section 12.8 clause for clause, with the single difference that what was a scalar coefficient there is here an operator on $L^2([0,1])$ — the place where the network, and only the network, enters.

The natural object to write first is the per-type Hamilton–Jacobi–Bellman equation, because the master function is built to reproduce it slice by slice. Hold the equilibrium field flow $\boldsymbol{m}_\cdot$ fixed, so that type x sees a prescribed aggregate trajectory $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ and decouples into a *classical* stochastic control problem: minimize (26.25) over α^x subject to the linear dynamics (26.24). This is the type-x instance of the graphon MFG system's HJB equation (26.6a), and writing $u^x(t,\xi)$ for its value function, the dynamic programming principle gives the usual minimized Bellman operator. The control enters the Hamiltonian only through the two terms it touches — the running quadratic penalty $\frac{1}{2}(\alpha^x)^2$ and the drift contribution $b\,\alpha^x\,\partial_\xi u^x$ — and minimizing $\alpha \mapsto \frac{1}{2}\alpha^2 + b\,\alpha\,\partial_\xi u^x$ over the scalar α (a strictly convex parabola, so the minimizer is the unique stationary point) returns the optimal feedback

$$\alpha^* = -b\,\partial_\xi u^x,$$

at which the bracket takes the value $-\frac{b^2}{2}(\partial_\xi u^x)^2$. Substituting this feedback back into the Bellman equation and adding the uncontrolled drift $a\xi$, the idiosyncratic diffusion $\frac{1}{2}\sigma^2\partial_{\xi\xi}$, and the state penalty $\frac{q}{2}(\xi - s\,z_t(x))^2$ from (26.25) produces the type-x HJB equation

$$\partial_t u^x + a\xi\,\partial_\xi u^x - \tfrac{b^2}{2}(\partial_\xi u^x)^2 + \tfrac{1}{2}\sigma^2\,\partial_{\xi\xi}u^x + \tfrac{q}{2}\big(\xi - s\,z_t(x)\big)^2 = 0, \tag{26.27}$$

with terminal condition $u^x(T,\xi) = \frac{q_T}{2}(\xi - s_T z_T(x))^2$ read off the terminal cost of (26.25). This is exactly the homogeneous LQG HJB equation (12.20) of section 12.8 with the scalar mean $\bar{m}_t$ replaced by the type-resolved aggregate $z_t(x)$; the type label is carried inertly through the equation, and all the network's content sits inside that one driving term.

It is worth saying precisely how (26.27) is the linear–quadratic instance of the general per-type HJB (26.6a) that the whole chapter has been written around, because the bundling is the same one used at every abstract step. The general equation reads $-\partial_t u^x - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2_\xi u^x) + H(x,\xi,\nabla_\xi u^x) = F(x,\xi,\boldsymbol{m}_t)$. Collecting the two terms of (26.27) that depend on u^x through its state-gradient $p = \partial_\xi u^x$ — the uncontrolled drift term $a\xi\, p$ and the post-minimization control term $-\frac{b^2}{2}p^2$ — and matching signs with the general form identifies the type-x Hamiltonian of the thread as

$$H_{\mathrm{LQG}}(\xi,p) = -a\xi\, p + \tfrac{b^2}{2}p^2.$$

This is the scalar, type-independent specialization of the general $H(x,\xi,p)$ of section 26.3: it is convex in p (the coefficient $\frac{b^2}{2} > 0$ is exactly hypothesis (A-Conv) made concrete), so its p-gradient $\partial_p H_{\mathrm{LQG}}(\xi,p) = -a\xi + b^2 p$ is the single-valued affine map that recovers the optimal drift $-\partial_p H_{\mathrm{LQG}} = a\xi - b^2 p$, matching the closed loop $aX^x + b\alpha^* = aX^x - b^2\partial_\xi u^x$ of (26.24). The driving term F is the pure state penalty $F(x,\xi,\boldsymbol{m}) = -\frac{q}{2}(\xi - s\, z[\boldsymbol{m}](x))^2$ of (26.5) with interaction observable $\phi(\zeta) = \zeta$. The notation watch of section 12.8 flagged this bundling of the uncontrolled drift into H; we keep general (a,b) rather than normalizing so that the constant-graphon reduction lands on Thread A without a change of variables.

Quadratic data propagate quadratically. The terminal value is quadratic in ξ and the only $\boldsymbol{m}$-dependence in (26.27) is the term $\frac{q}{2}(\xi - s\, z_t(x))^2$, which is *linear* in the aggregate $z_t(x)$ and hence, since $z_t = \mathbb{W}\bar{m}_t$ is itself linear in the mean profile, linear in $\bar{m}_t$. A value function whose data are quadratic in the state and affine in a linear functional of the field stays quadratic in the state and affine in that functional for all time; this is the structural inheritance that the LQG ansatz codifies. We therefore posit a master function quadratic in the state ξ whose state-gradient is affine in the mean field through a *time-dependent linear operator* $\mathcal{R}_t$ on $L^2([0,1])$:

$$\nabla_\xi U(t,x,\xi,\boldsymbol{m}) = P(t)\,\xi + (\mathcal{R}_t\bar{m})(x), \qquad \bar{m}(y) = \int \zeta\, m^y(\mathrm{d}\zeta). \tag{26.28}$$

The two coefficients carry different ranks for a structural reason, and unpacking it is the whole point of the ansatz. The first, $P(t)$, is a *scalar*: it is the curvature of the value in the agent's own state, $P(t) = \partial_{\xi\xi}u^x$, and it cannot carry a type label because nothing in the problem that sets it does. The state dynamics (26.24) have type-independent coefficients (a,b,σ), and the quadratic state weight q in (26.25) is likewise the same for every type; the curvature of an agent's value in its own state is fixed by its own dynamics and its own penalty alone, never by who its neighbours are, so $P(t)$ is one number shared across the continuum of types. This is the exact graphon counterpart of the scalar $P(t)$ of the classical ansatz (12.21): the self-curvature is colour-blind to the network. The second coefficient, $(\mathcal{R}_t\bar{m})(x)$, is where the network is the entire story. It is the type-x *population-sensitivity* — how much agent x's marginal value $\nabla_\xi U$ shifts when the population's mean profile shifts — and it is a *linear functional of the whole mean field* $\bar{m} \in L^2([0,1])$, evaluated at the type x. The operator $\mathcal{R}_t$ is the object that does the routing: it takes the entire crowd mean $\bar{m}$, a function over all foreign types, and returns the function over types $x \mapsto (\mathcal{R}_t\bar{m})(x)$ whose value at x is the sensitivity felt by an agent of type x. Where the homogeneous theory of section 12.8 had a single scalar sensitivity $\pi(t)$ collapsing the one population mean $\bar{m}$ into one number, the graphon theory must keep an operator, because there is no longer one mean and one sensitivity but a continuum of each, woven together by the network: $\mathcal{R}_t$ is the matrix-valued (here operator-valued) promotion of $\pi(t)$, and the next subsection will show it inherits the kernel $\mathbb{W}$ that built the coupling in the first place.

A small concrete check fixes the picture. If the kernel were the constant graphon $W \equiv c$, every agent would feel the same scalar aggregate $z_t(x) = c\,\langle \bar{m}_t \rangle$, the population-sensitivity would have to be a single number times that one aggregate, and $\mathcal{R}_t$ would degenerate to multiplication by a scalar — precisely the homogeneous $\pi(t)$ (up to the factor c) of (12.21). The operator $\mathcal{R}_t$ is exactly what is needed when different types feel different aggregates, and it reduces to the classical scalar in the no-network limit, as the thread's reduction check demands.

The ansatz has an immediate and decisive consequence for the abstract equation: it switches off the extra-diffusion line of (26.10). To see this we compute the field derivatives of U and watch them collapse. Because U depends on the field $\boldsymbol{m}$ only through the single linear functional $\bar{m}(y) = \int \zeta\, m^y(\mathrm{d}\zeta)$ — nudging the slice m^y by a test perturbation δm^y moves $\bar{m}(y)$ by $\int \zeta\, \delta m^y(\mathrm{d}\zeta)$ and moves nothing else — the flat (linear-functional) derivative is itself linear in the foreign state ζ, and the slice-Lions derivative, its ζ-gradient, is therefore *constant in* ζ:

$$\frac{\delta U}{\delta m^y}(t,x,\xi,\boldsymbol{m})(\zeta) = \partial_{\bar{m}(y)}U\cdot\zeta, \qquad \partial_\mu^y U(t,x,\xi,\boldsymbol{m})(\zeta) = \partial_{\bar{m}(y)}U \text{ (independent of } \zeta\text{)}, \qquad \partial_\zeta \partial_\mu^y U \equiv 0. \tag{26.29}$$

The middle equality is the chain rule for the slice-Lions derivative $\partial_\mu^y U = \nabla_\zeta(\delta U/\delta m^y)$ of section 26.2: since $\delta U/\delta m^y(\zeta) = \partial_{\bar{m}(y)}U \cdot \zeta$ is exactly linear in ζ, its ζ-gradient is the constant coefficient $\partial_{\bar{m}(y)}U$, with the foreign state ζ gone. The right-hand identity is then immediate and is the whole content: a function that does not depend on ζ has vanishing ζ-Jacobian, so $\partial_\zeta \partial_\mu^y U \equiv 0$ pointwise in every argument.

That one line is what kills the spreading term. The extra-diffusion (third) line of the master equation (26.10) is, in full, $-\frac{1}{2}\int_0^1\int \operatorname{tr}(\sigma\sigma^\top\, \partial_\zeta\partial_\mu^y U)\, m^y(\mathrm{d}\zeta)\, \mathrm{d}y$, and its integrand is built from precisely the Jacobian $\partial_\zeta\partial_\mu^y U$ that (26.29) has just shown is identically zero. With a vanishing integrand the integral vanishes, type by type and state by state, so the entire third line of (26.10) drops out of the thread's master equation. The reason is exactly the one anticipated below theorem 12.3 for the homogeneous LQG case, lifted unchanged but for the outer type integral: the cost couples the field through a *first* moment only, and a first-moment coupling is blind to how each type's crowd spreads under its private noise. The spreading term measures the value's response to that smearing through the ζ-Jacobian of the Lions derivative, and a linear-in-ζ Lions derivative has no curvature for the diffusion to act on. Were the cost to couple through a second moment — as it would if (26.25) penalized the variance of the neighbour field rather than its mean — the Lions derivative would acquire a genuine ζ-dependence, its Jacobian would be nonzero, and the spreading term would survive; it is the mean-field nature of Thread B's coupling, nothing softer, that makes the line disappear.

With the extra-diffusion line gone, the abstract equation (26.10) for the thread has been pared to its first two lines: the per-type HJB operator and the type-integrated transport term. The ansatz (26.28) now stands ready to be fed into them. The next subsection does exactly that — it inserts the quadratic-in-ξ, operator-affine guess into the per-type HJB (26.27), uses the just-established vanishing of the spreading term to drop the second-field-derivative bookkeeping, and matches powers of the state ξ. The ξ^2 coefficient will pin the scalar curvature $P(t)$ to a scalar Riccati equation, and the ξ^1 coefficient will pin the population-sensitivity operator $\mathcal{R}_t$ to an *operator* Riccati equation on $L^2([0,1])$ in which the kernel $\mathbb{W}$ finally appears.

Reduction to an operator Riccati equation

The previous subsection delivered two things we now spend. It posited the operator ansatz (26.28) — a master function quadratic in the state ξ whose state-gradient is affine in the mean profile through the population-sensitivity operator $\mathcal{R}_t$ on $L^2([0,1])$ — and it proved, from the linearity of the slice-Lions derivative in ζ (26.29), that the extra-diffusion (third) line of the abstract master equation (26.10) is identically zero on the thread. With that line gone the equation has been pared to its first two, and the work that remains is purely algebraic: insert the ansatz into the per-type HJB (26.27), sort the result by powers of the state ξ, and let each power pin down one coefficient of the guess. Because the data are quadratic in ξ and the ansatz keeps them so, only three powers occur — ξ^2, ξ^1, ξ^0 — and the first two carry all the dynamical content. We write the value in the form the ansatz forces,

$$u^x(t,\xi) = \tfrac{1}{2}P(t)\,\xi^2 + \psi_t(x)\,\xi + \chi_t(x), \qquad \psi_t(x) = (\mathcal{R}_t\bar{m}_t)(x), \tag{26.30}$$

the linear coefficient $\psi_t(x)$ being precisely the type-x population-sensitivity $(\mathcal{R}_t\bar{m})(x)$ of (26.28) evaluated along the equilibrium mean profile $\bar{m}_t$. Its three derivatives are $\partial_t u^x = \frac{1}{2}\dot{P}\,\xi^2 + \dot{\psi}_t\,\xi + \dot{\chi}_t$, $\partial_\xi u^x = P\,\xi + \psi_t(x)$, and $\partial_{\xi\xi}u^x = P$, and the only nonlinearity, the squared gradient, expands as $(\partial_\xi u^x)^2 = P^2\xi^2 + 2P\,\psi_t(x)\,\xi + \psi_t(x)^2$. Substituting all of this into (26.27) and grouping by powers of ξ is the entire computation; we take the powers in turn.

At order ξ^2 the surviving terms are the curvature derivative $\frac{1}{2}\dot{P}$ from $\partial_t u^x$, the drift contribution aP from $a\xi\,\partial_\xi u^x$, the post-minimization term $-\frac{b^2}{2}P^2$ from $-\frac{b^2}{2}(\partial_\xi u^x)^2$, and the penalty $\frac{q}{2}$ from $\frac{q}{2}(\xi - sz_t(x))^2$. Their sum must vanish, $\frac{1}{2}\dot{P} + aP - \frac{b^2}{2}P^2 + \frac{q}{2} = 0$, which is the scalar curvature Riccati of Thread A,

$$\dot{P} = b^2P^2 - 2aP - q, \qquad P(T) = q_T, \tag{26.31}$$

with the terminal value $P(T) = q_T$ read off the ξ^2 coefficient of the terminal cost $\frac{q_T}{2}(\xi - s_Tz_T(x))^2$. Two features deserve emphasis. First, (26.31) is *identical* to the homogeneous LQG curvature Riccati (12.25a): the self-curvature of an agent's value in its own state is governed by its own dynamics (a, b, σ) and its own penalty q, q_T alone. Second, and as a corollary, it is *independent of the kernel*: neither W nor the aggregate $z_t(x)$ appears anywhere in it, because the ξ^2 balance never sees the driving term — the coupling $z_t(x)$ enters the penalty only at orders ξ^1 and ξ^0, multiplying ξ and 1, never ξ^2. The network is invisible to the curvature; this is the analytic face of the colour-blindness of $P(t)$ argued for the ansatz.

At order ξ^1 the terms are $\dot{\psi}_t$ from $\partial_t u^x$, the drift contribution $a\psi_t$ from $a\xi\,\partial_\xi u^x$, the cross term $-\frac{b^2}{2}\cdot 2P\psi_t = -b^2P\psi_t$ from the squared gradient, and the cross term $-q\,s\,z_t(x)$ from expanding $\frac{q}{2}(\xi - sz_t(x))^2$. (The diffusion term $\frac{1}{2}\sigma^2\partial_{\xi\xi}u^x = \frac{1}{2}\sigma^2 P$ is a pure constant and contributes only at order ξ^0, feeding χ_t; it never touches ψ_t.) Setting the ξ^1 sum to zero, $\dot\psi_t + a\psi_t - b^2 P\psi_t - qsz_t = 0$, gives the backward equation for the population-sensitivity coefficient,

$$\dot\psi_t(x) = -(a - b^2 P)\,\psi_t(x) + q\,s\,z_t(x), \qquad \psi_T(x) = -q_T s_T\, z_T(x), \tag{26.32}$$

with $z_t = \mathbb{W}\bar m_t$, and the terminal value $\psi_T(x) = -q_T s_T z_T(x)$ read off the ξ^1 coefficient $-q_T s_T z_T(x)$ of the terminal penalty. This is a backward linear ODE for the profile $\psi_t \in L^2([0,1])$, run from $t = T$, whose only inhomogeneity is the graphon-averaged neighbour mean $z_t = \mathbb{W}\bar m_t$ — the lone place the kernel reaches into the linear order.

Equation (26.32) is backward and driven by the population, so to close it we need how the population moves. The closed-loop type-mean field evolves *forward*: the optimal feedback is $\alpha^* = -b\,\partial_\xi u^x = -b(PX_t^x + \psi_t(x))$, so the controlled drift in (26.24) is $aX_t^x + b\alpha^* = (a - b^2P)X_t^x - b^2\psi_t(x)$, and taking expectations — the idiosyncratic noise $\sigma\,\mathrm{d}B_t^x$ is mean-zero and the drift is affine, so $\mathbb{E}[X_t^x] = \bar m_t(x)$ obeys the drift with X_t^x replaced by its mean — gives

$$\dot{\bar m}_t(x) = (a - b^2 P)\,\bar m_t(x) - b^2\,\psi_t(x), \qquad \bar m_0 \text{ given.} \tag{26.33}$$

Equations (26.32)–(26.33) together are a *linear forward–backward system in* $L^2([0,1])$: a backward profile ψ_t run from terminal data and a forward profile $\bar m_t$ run from initial data, the two locked together. It is worth saying exactly where the types touch. Strip out the kernel for a moment by setting $W = \mathrm{id}$ (so $z_t = \bar m_t$): the pair (26.32)–(26.33) would then be a *pointwise* family of scalar ODEs, one per type x, sharing no information — a disjoint union of classical LQG mean-field problems. The genuine kernel reinserts itself in a single spot, the term $q\,s\,z_t(x) = q\,s\,(\mathbb{W}\bar m_t)(x)$ of (26.32), which mixes the mean of *every* type into the sensitivity of type x. The forward–backward system is therefore coupled across types *only through the operator* $\mathbb{W}$ — the one inter-type channel the chapter is strung on, now made completely concrete as the off-diagonal of a linear system on $L^2([0,1])$.

To read off the master function we must turn this forward–backward pair into a single backward equation for the operator $\mathcal{R}_t$ that encodes the equilibrium relation between sensitivity and population. The device is the one used for every Riccati closure: posit that the backward coefficient is, at each time, a fixed *linear* function of the current forward mean,

$$\psi_t = \mathcal{R}_t\,\bar m_t, \qquad \mathcal{R}_t : L^2([0,1]) \to L^2([0,1]) \text{ linear,} \tag{26.34}$$

which is exactly the equilibrium content of the ansatz (26.28): the population-sensitivity profile is the operator $\mathcal{R}_t$ applied to the current mean profile. Differentiating (26.34) in time — $\mathcal{R}_t$ and $\bar m_t$ both move, so the product rule on $L^2([0,1])$ contributes two terms — and substituting the forward law (26.33) for $\dot{\bar m}_t$ and then (26.34) itself for the ψ_t that appears inside it,

$$\dot\psi_t = \dot{\mathcal{R}}_t\bar m_t + \mathcal{R}_t\dot{\bar m}_t = \dot{\mathcal{R}}_t\bar m_t + \mathcal{R}_t\big[(a - b^2P)\bar m_t - b^2\psi_t\big] = \dot{\mathcal{R}}_t\bar m_t + \mathcal{R}_t\big[(a - b^2P)\bar m_t - b^2\mathcal{R}_t\bar m_t\big].$$

On the other side, the backward law (26.32) reads, after the same substitution $\psi_t = \mathcal{R}_t\bar m_t$ and $z_t = \mathbb{W}\bar m_t$,

$$\dot\psi_t = -(a - b^2P)\,\mathcal{R}_t\bar m_t + q\,s\,\mathbb{W}\bar m_t.$$

Equating the two expressions for $\dot\psi_t$ and gathering every term onto one side gives

$$\Big[\dot{\mathcal{R}}_t - b^2\mathcal{R}_t^2 + 2(a - b^2P)\mathcal{R}_t - q\,s\,\mathbb{W}\Big]\bar m_t = 0,$$

an identity asserting that a single time-dependent operator annihilates the mean profile $\bar{m}_t$. Here is the step that converts the identity into an operator equation, and it deserves to be said out loud rather than waved through. The relation must hold not for one $\bar{m}_t$ but for *every* mean profile reachable at time t, because the master function is required to be correct off the single equilibrium trajectory — as $\boldsymbol{m}$, and hence $\bar{m}_0$, ranges over the whole field space, $\bar{m}_t$ ranges over the image of $L^2([0,1])$ under the forward flow (26.33). That flow is a linear, invertible ODE on $L^2([0,1])$ (its solution operator is the time-ordered exponential of a bounded generator, hence boundedly invertible), so the reachable set $\{\bar{m}_t : \bar{m}_0 \in L^2([0,1])\}$ is *all* of $L^2([0,1])$. An operator that kills every vector of $L^2([0,1])$ is the zero operator; the bracket must therefore vanish identically, and we have landed the *operator Riccati equation*

$$\dot{\mathcal{R}}_t = b^2\,\mathcal{R}_t^2 - 2(a - b^2P)\,\mathcal{R}_t + q\,s\,\mathbb{W}, \qquad \mathcal{R}_T = -q_Ts_T\,\mathbb{W}, \tag{26.35}$$

with the terminal datum $\mathcal{R}_T = -q_Ts_T\mathbb{W}$ forced by matching the closure (26.34) against the terminal $\psi_T = -q_Ts_Tz_T = -q_Ts_T\mathbb{W}\bar{m}_T$ of (26.32).

This is the exact graphon lift of the scalar population-sensitivity Riccati (12.25b): term by term it is the homogeneous equation $\dot{\pi} = b^2\pi^2 - 2(a - b^2P)\pi + q\,s$ with the scalar π promoted to the operator $\mathcal{R}_t$ and the bare coupling constant 1 in the source promoted to the graphon operator $\mathbb{W}$. And it is worth tracing each term back to its parent in the abstract master equation (26.10), to make plain that (26.35) *is* that equation in the linear–quadratic case and not a separate computation. The third line of (26.10) was killed in the previous subsection; the first line (the per-type HJB operator $-\partial_tU - \frac{1}{2}\sigma^2\,\mathrm{D}_\xi^2\,U + H$) and the second line (the type-integrated transport $\int_0^1\int \partial_\mu^y U \cdot \partial_pH \;\mathrm{d}m^y\;\mathrm{d}y$) are what we matched. The quadratic term $b^2\mathcal{R}_t^2$ descends from the post-minimization Hamiltonian $H = -a\xi p + \frac{b^2}{2}p^2$, whose $\frac{b^2}{2}p^2$ piece produced the cross term $-b^2P\psi_t$ and, after the closure substitution turned ψ_t into $\mathcal{R}_t\bar{m}_t$ inside the product rule, the genuinely quadratic $b^2\mathcal{R}_t^2$. The linear term $-2(a - b^2P)\mathcal{R}_t$ collects the uncontrolled drift a and the P-coupling from both the backward law and the forward transport — it is the type-integrated transport line evaluated on the affine LQG feedback. The source term $q\,s\,\mathbb{W}$ is the sole survivor that carries the network: it descends directly from $F(x,\xi,\boldsymbol{m}) = -\frac{q}{2}(\xi - s(\mathbb{W}\bar{m})(x))^2$, the right-hand side of (26.10), which is the one and only place the graphon entered the abstract equation at all. The operator Riccati thus exhibits, in miniature and in closed form, the chapter's organizing claim: every classical master-equation object survives the lift, and $\mathbb{W}$ enters only through the cost coupling, here as the inhomogeneity $q\,s\,\mathbb{W}$.

Two structural facts about the solution $\mathcal{R}_t$ make the next subsection possible, and both follow from the operator Riccati itself once one watches what the flow preserves. The first is that $\mathcal{R}_t$ *is self-adjoint for every* t. The terminal datum $\mathcal{R}_T = -q_Ts_T\mathbb{W}$ is self-adjoint because $\mathbb{W}$ is: the kernel is symmetric, $W(x,y) = W(y,x)$, under the graphon regularity (A-Reg-W), so $\mathbb{W} = \mathbb{W}^*$ on $L^2([0,1])$. And the right-hand side of (26.35) maps self-adjoint operators to self-adjoint operators: if $\mathcal{R}$ is self-adjoint then so is $\mathcal{R}^2$, the scalar coefficients b^2, $a - b^2P$, $q\,s$ are real, and $\mathbb{W}$ is self-adjoint, so the whole right side is self-adjoint. The self-adjoint operators form a closed real-linear subspace of the bounded operators, the Riccati vector field is tangent to it, and the ODE has a unique solution; hence the trajectory launched from the self-adjoint terminal datum stays self-adjoint for all t. The second fact is sharper and is what collapses the problem: $\mathcal{R}_t$ *is a function of* $\mathbb{W}$ *alone*. Concretely, $\mathcal{R}_t$ commutes with every bounded operator that commutes with $\mathbb{W}$. To see it, fix any S with $[S,\mathbb{W}] = 0$ and track the commutator $C_t := [S,\mathcal{R}_t] = S\mathcal{R}_t - \mathcal{R}_tS$. Differentiating (26.35) and using that $[S,\,\cdot\,]$ is a derivation,

$$\dot{C}_t = [S, \dot{\mathcal{R}}_t] = b^2(\mathcal{R}_t C_t + C_t \mathcal{R}_t) - 2(a - b^2 P)\, C_t + q\, s\, [S, \mathbb{W}] = b^2(\mathcal{R}_t C_t + C_t \mathcal{R}_t) - 2(a - b^2 P)\, C_t,$$

because the source commutator $[S, \mathbb{W}]$ vanishes by assumption; the term $[S, \mathcal{R}_t^2] = \mathcal{R}_t[S, \mathcal{R}_t] + [S, \mathcal{R}_t]\mathcal{R}_t = \mathcal{R}_t C_t + C_t \mathcal{R}_t$ is the only nontrivial bookkeeping. This is a *linear, homogeneous* ODE for C_t, and its terminal value is $C_T = [S, \mathcal{R}_T] = -q_T s_T\, [S, \mathbb{W}] = 0$. A linear homogeneous ODE with zero terminal data has only the zero solution, so $C_t \equiv 0$: $\mathcal{R}_t$ commutes with S. Since S was an arbitrary operator commuting with $\mathbb{W}$, the operator $\mathcal{R}_t$ lies in the bicommutant of $\mathbb{W}$ — it is, in the precise sense, a function of $\mathbb{W}$. The payoff is spectral. The graphon operator is compact and self-adjoint (Hilbert–Schmidt, since $W \in L^2([0,1]^2)$ under (A-Reg-W); definition 15.13), so by the spectral theorem for compact self-adjoint operators (theorem 2.26) it has an orthonormal eigenbasis $(\varphi_k)_{k\geq 1}$ of $L^2([0,1])$ with real eigenvalues $\lambda_k \to 0$, $\mathbb{W}\varphi_k = \lambda_k \varphi_k$. A function of $\mathbb{W}$ is diagonal in that basis: $\mathcal{R}_t \varphi_k = r_k(t)\varphi_k$, and projecting the operator Riccati (26.35) onto the mode φ_k turns it into a single *scalar* Riccati equation per eigenvalue,

$$\dot{r}_k = b^2\, r_k^2 - 2(a - b^2 P)\, r_k + q\, s\, \lambda_k, \qquad r_k(T) = -q_T s_T\, \lambda_k,$$

the eigenvalue λ_k entering only as a rescaling of the source. The infinite-dimensional operator equation has thereby decoupled into a countable family of one-dimensional Riccati equations, one for each graphon mode, with the network entering each through its eigenvalue alone.

With the operator pinned down, the master function is fully explicit. Reassembling the expansion (26.30) with $\psi_t(x) = (\mathcal{R}_t \bar{m})(x)$ from the closure (26.34),

$$U(t, x, \xi, \boldsymbol{m}) = \tfrac{1}{2} P(t)\, \xi^2 + (\mathcal{R}_t \bar{m})(x)\, \xi + \chi_t(x, \bar{m}), \tag{26.36}$$

with χ_t the (quadratic-in-$\bar{m}$) constant term determined by the leftover order-ξ^0 matching, which contributes nothing to the feedback and which we do not need to track. The state-gradient of (26.36) is the affine map $\nabla_\xi U = P(t)\xi + (\mathcal{R}_t \bar{m})(x)$ of the ansatz (26.28); restricted to the equilibrium flow with $X_t^x \sim m_t^x$ it furnishes the *operator-valued decoupling field* of theorem 26.5,

$$Y_t^x = P(t) X_t^x + (\mathcal{R}_t \bar{m}_t)(x),$$

the type-indexed adjoint of the graphon FBSDE, now read as a scalar self-curvature $P(t)$ times the agent's own state plus the operator-routed population sensitivity $(\mathcal{R}_t \bar{m}_t)(x)$ — the abstract decoupling field of (26.12) made completely concrete by the linear–quadratic structure.

The operator Riccati (26.35) is as far as the general kernel can be pushed in closed form, but its diagonalization in the $\mathbb{W}$-eigenbasis has already done the decisive work: the continuum of types is governed, mode by mode, by scalar Riccati equations indexed by the graphon spectrum. The next subsection cashes this in on the cleanest possible spectrum. Taking the rank-one kernel of example 15.11, where $\mathbb{W}$ has a *single* nonzero eigenvalue, the diagonal family above retains exactly one active mode: the operator Riccati collapses to a single scalar Riccati and the whole infinite-dimensional field is carried by one scalar order parameter.

Rank-one collapse to a scalar order parameter

The previous subsection left the graphon LQG equilibrium encoded in the operator Riccati (26.35), and then proved the one structural fact that makes it tractable: $\mathcal{R}_t$ is a function of $\mathbb{W}$ alone, hence diagonal in the graphon eigenbasis (φ_k), so the single operator equation splits into the countable family of scalar Riccati equations $\dot r_k = b^2 r_k^2 - 2(a - b^2 P) r_k + q\, s\, \lambda_k$, one per eigenvalue λ_k, with the network reaching each mode only through its eigenvalue. That family is as far as a general kernel can be pushed, because in general infinitely many eigenvalues are nonzero and the modes, though uncoupled from one another, must all be carried at once. The rank-one (Curie–Weiss) graphon is the spectrum on which this family degenerates to a single member: $\mathbb{W}$ has exactly *one* nonzero eigenvalue, so all but one of the r_k are slaved to a kernel-free source and only one mode is dynamically alive. This is the case the thread has been pointing at since example 15.11, and on it the entire infinite-dimensional field condenses to one scalar. We work it in full.

Take the rank-one graphon $W(x, y) = f(x)f(y)$ of example 15.11. Its operator acts by projecting onto the single direction f and rescaling: $\mathbb{W}g = f\, \langle f, g\rangle_{L^2([0,1])}$, that is $\mathbb{W} = (f \otimes f)$, the rank-one operator with range the line $\mathbb{R}f$. It has a *single* nonzero eigenvalue, found by feeding it its own range vector: $\mathbb{W}f = f\, \langle f, f\rangle = \|f\|^2_{L^2([0,1])}\, f$, so f is an eigenfunction with eigenvalue

$$\lambda = \lambda_1 = \|\mathbb{W}\| = \|f\|^2_{L^2([0,1])}\,,$$

the leading eigenvalue in the spectral ordering of chapter 18; every direction orthogonal to f is annihilated, $\mathbb{W}g = 0$ for $g \perp f$, so $\lambda_2 = \lambda_3 = \cdots = 0$. In the diagonal family of the previous subsection only the mode $\varphi_1 = f/\|f\|$ carries a nonzero source $q\, s\, \lambda_1 = q\, s\, \lambda$; every other mode solves $\dot r_k = b^2 r_k^2 - 2(a - b^2 P) r_k$ with $r_k(T) = 0$, whose unique solution is $r_k \equiv 0$. The operator is therefore confined for all time to the one-dimensional family of multiples of the projector,

$$\mathcal{R}_t = r(t)\, (f \otimes f),$$

and the entire content of (26.35) is the scalar function $r(t)$.

Substituting this ansatz collapses the operator equation by inspection, and the one algebraic fact it needs is how the projector squares. Since $(f \otimes f)^2 g = (f \otimes f)(f\langle f, g\rangle) = f\, \langle f, f\rangle\, \langle f, g\rangle = \lambda\, f\langle f, g\rangle = \lambda\, (f \otimes f)g$, the projector obeys $(f \otimes f)^2 = \lambda\, (f \otimes f)$ — it is idempotent up to the eigenvalue — so

$$\mathcal{R}_t^2 = r(t)^2\, (f \otimes f)^2 = r(t)^2 \lambda\, (f \otimes f).$$

Putting $\dot{\mathcal{R}}_t = \dot r\, (f \otimes f)$, $\mathcal{R}_t^2 = r^2 \lambda\, (f \otimes f)$, and $\mathbb{W} = (f \otimes f)$ into (26.35) makes every term a multiple of the single operator $(f \otimes f)$; equating the scalar coefficients gives the *scalar* Riccati equation

$$\dot r = b^2 \lambda\, r^2 - 2(a - b^2 P)\, r + q\, s, \qquad r(T) = -q_T s_T, \tag{26.37}$$

the terminal value $r(T) = -q_T s_T$ read off the rank-one terminal datum $\mathcal{R}_T = -q_T s_T \mathbb{W} = -q_T s_T (f \otimes f)$. Stare at this equation against its homogeneous ancestor: it is *identical* to the classical LQG population-sensitivity Riccati (12.25b), $\dot r = b^2 r^2 - 2(a - b^2 P) r + q\, s$, save for the single replacement $b^2 \mapsto b^2 \lambda$ in the quadratic term. The leading graphon eigenvalue λ rescales the effective control gain, and nothing else of the network survives: the whole apparatus of types, the kernel, the operator $\mathbb{W}$, has been compressed into one number multiplying the quadratic Riccati term. This is the precise sense in which a rank-one network behaves as a single LQG agent with a renormalized gain.

What plays the role of the population in this collapsed picture is a single scalar, the *order parameter*

$$\varrho_t \;:=\; \langle f, \bar m_t\rangle_{L^2([0,1])} = \int_0^1 f(y)\,\bar m_t(y)\,\mathrm{d}y, \tag{26.38}$$

the projection of the type-mean profile onto the active direction f — exactly the static and dynamic order parameter ϱ fixed for Thread B in the thread specification. Once the field is rank-one, ϱ_t is all of it the equilibrium ever feels, and every population-dependent object reduces to f times this scalar. The aggregate felt by type x is, directly from the projector form of $\mathbb{W}$,

$$z_t(x) = (\mathbb{W}\bar m_t)(x) = f(x)\,\langle f, \bar m_t\rangle = f(x)\,\varrho_t,$$

so the neighbour mean every type reacts to is its own coupling weight $f(x)$ times the one global number ϱ_t. The population-sensitivity coefficient inherits the same shape: $\psi_t(x) = (\mathcal{R}_t\bar m_t)(x) = r(t)\,(f\otimes f)\bar m_t(x) = r(t)\,f(x)\,\langle f,\bar m_t\rangle = r(t)\,f(x)\,\varrho_t$. Both the sensitivity profile and the aggregate are rigidly proportional to f, with ϱ_t the lone free amplitude.

The order parameter then obeys a *closed* scalar ODE, obtained by projecting the forward mean evolution onto f. Pair (26.33), $\dot{\bar m}_t(x) = (a - b^2P)\bar m_t(x) - b^2\psi_t(x)$, against f in $L^2([0,1])$, using that the pairing commutes with the time derivative:

$$\dot\varrho_t = \langle f, \dot{\bar m}_t\rangle = (a - b^2P)\,\langle f,\bar m_t\rangle - b^2\,\langle f,\psi_t\rangle.$$

The first pairing is $\langle f,\bar m_t\rangle = \varrho_t$ by definition. The second is the place the projector simplification does its work: since $\psi_t = r\,(f\otimes f)\bar m_t = r\,\mathbb{W}\bar m_t$,

$$\langle f,\psi_t\rangle = r\,\langle f,\mathbb{W}\bar m_t\rangle = r\,\lambda\,\langle f,\bar m_t\rangle = r\,\lambda\,\varrho_t,$$

where the middle step is the self-adjoint eigenvalue identity $\langle f,\mathbb{W}\bar m_t\rangle = \langle \mathbb{W}f,\bar m_t\rangle = \lambda\langle f,\bar m_t\rangle = \lambda\varrho_t$ — the one explicit simplification that closes the equation, the operator $\mathbb{W}$ pulled off the mean and replaced by its eigenvalue λ. Collecting the two pairings gives the closed scalar ODE

$$\dot\varrho_t = (a - b^2P)\,\varrho_t - b^2\lambda\,r(t)\,\varrho_t = \big(a - b^2P - b^2\lambda r(t)\big)\,\varrho_t, \qquad \varrho_0 = \langle f,\bar m_0\rangle. \tag{26.39}$$

It is a linear scalar ODE for ϱ_t run *forward* from ϱ_0, its rate $a - b^2P - b^2\lambda r(t)$ assembled from the agent's own dynamics, the curvature P, and the sensitivity r rescaled by the same effective coupling $b^2\lambda$ that appeared in the quadratic Riccati term — the kernel entering, here too, only through λ.

This is the cleanest realization of the thread's promise. The graphon master equation for a *continuum* of types has reduced, on the rank-one kernel, to three coupled scalar equations run on $[0,T]$: the curvature Riccati P (26.31), the population-sensitivity Riccati r (26.37), both backward from terminal data, and the order-parameter ODE ϱ_t (26.39), forward from initial data. That is *exactly the size of the classical LQG master system of worked computation 12.15* — the same triple (P, r, ϱ) of two backward Riccati scalars and one forward mean — with the single, decisive modification that the coupling is rescaled by the leading eigenvalue, $b^2 \mapsto b^2\lambda$ wherever the network acts. An entire field theory has been packed, with no loss, into the homogeneous master system run with a renormalized gain. The full master function is then (26.36) with $\mathcal{R}_t = r(t)(f \otimes f)$,

$$U(t, x, \xi, \boldsymbol{m}) = \tfrac{1}{2}P(t)\,\xi^2 + r(t)\,f(x)\,\varrho\,\xi + \chi_t(x, \varrho),$$

its state-gradient $\nabla_\xi U = P(t)\xi + r(t)f(x)\varrho$ a function of the one scalar ϱ through the population, and its decoupling field $Y_t^x = P(t)X_t^x + r(t)f(x)\varrho_t$ likewise.

The reduction check the thread specification demands is the constant graphon, and it falls out as the boundary case $f \equiv$ const. Set $W(x, y) \equiv c$; this is rank-one with $f \equiv \sqrt{c}$, since $f(x)f(y) = \sqrt{c}\cdot\sqrt{c} = c$. The leading eigenvalue is then $\lambda = \|f\|^2 = \int_0^1 c\,\mathrm{d}y = c$, and the order parameter $\varrho_t = \langle f, \bar{m}_t\rangle = \sqrt{c}\int_0^1 \bar{m}_t(y)\,\mathrm{d}y = \sqrt{c}\,\bar{\bar{m}}_t$ is, up to the constant $\sqrt{c}$, the *global* mean $\bar{\bar{m}}_t = \int_0^1 \bar{m}_t$ — every type weighted equally, as a constant kernel demands. The scalar Riccati (26.37) becomes $\dot{r} = b^2 c\,r^2 - 2(a - b^2P)r + q\,s$, which is precisely the homogeneous Thread A population-sensitivity Riccati with *effective coupling* b^2c: the constant graphon recovers the classical LQG MFG, its single coupling constant c multiplying the control gain exactly as Thread A's coupling does. This is the reduction the thread specification requires, and it confirms that the rank-one collapse degenerates, at $f \equiv \sqrt{c}$, to the homogeneous theory with no residue.

We now hand the common-noise subsection a completely explicit, finite scalar machinery: the curvature P, the sensitivity r, and the order parameter ϱ_t, the last carrying the whole population. There the same three scalars are stress-tested against a genuine aggregate shock — the four common-noise families of section 26.5 made concrete on this kernel — to see which gains feel the noise and which only the level does. The same reduction also sets up the threshold of remark 26.13: because the network has been compressed into the single effective coupling $b^2\lambda$, the question of whether the equilibrium exists for every horizon becomes the question of whether the scalar Riccati (26.37) keeps r finite, and that is governed entirely by the leading eigenvalue $\lambda = \|\mathbb{W}\|$ — a kernel whose spectral radius is too large can drive r to blow up in finite time, the spectral blow-up threshold the next subsections make precise.

The effect of common noise

Add the common shock $\sigma_0\,\mathrm{d}B_t^0$ of (26.18) to the dynamics (26.24). The conditional type-mean $\bar{m}_t(x) = \mathbb{E}[X_t^x \mid \mathcal{F}_t^0]$ now solves the *stochastic* forward equation

$$\mathrm{d}\bar{m}_t(x) = \big[(a - b^2P)\bar{m}_t(x) - b^2\psi_t(x)\big]\,\mathrm{d}t + \sigma_0\,\mathrm{d}B_t^0, \tag{26.40}$$

the common noise translating every type-slice coherently. The decisive — and at first surprising — fact is that the feedback Riccati system is *unchanged*: the curvature P (26.31) and the population-sensitivity operator $\mathcal{R}_t$ (26.35), equivalently $r(t)$ in the rank-one case (26.37), do not feel σ_0 at all. This is the linear–quadratic certainty-equivalence principle, lifted to the graphon: additive common noise shifts the *level* of the value but not its *gains*. We make this precise in the rank-one case.

Write the constant part of the master function (26.36) as a quadratic in the order parameter $\varrho = \langle f, \bar{m}\rangle$,

$$\chi_t(x, \bar{m}) = \tfrac{1}{2}\,g_t(x)\,\varrho^2 + h_t(x), \tag{26.41}$$

and insert the full master function $U = \frac{1}{2}P\xi^2 + r\,f(x)\varrho\,\xi + \chi_t$ into the graphon master equation *including* the common-noise terms (26.23). Matching powers of ξ and ϱ slot by slot, the orders ξ^2 and $\xi\varrho$ reproduce exactly the σ_0-free Riccati equations (26.31) and (26.37) (since $\partial_\zeta\partial^y_\mu U \equiv 0$ kills the population-self-spreading term (ii) of (26.23), and terms (i),(iii),(iv) are all of order $\xi^0\varrho^0$). The order ϱ^2 matches the ϱ^2*-coefficient* (a gain) g_t, and the constant order $\xi^0\varrho^0$ matches the *level* h_t:

$$\dot g_t(x) = -2\big(a - b^2P - b^2\lambda r\big)\,g_t(x) + \big(b^2r^2 - qs^2\big)f(x)^2, \qquad g_T(x) = q_Ts_T^2\,f(x)^2, \tag{26.42a}$$

$$\dot h_t(x) = -\tfrac{1}{2}\sigma^2 P\;\underbrace{-\,\tfrac{1}{2}\sigma_0^2\,P - \sigma_0^2\,r\,f(x)\,\beta - \tfrac{1}{2}\sigma_0^2\,g_t(x)\,\beta^2}_{\text{common-noise cost, }\beta=\langle f,\mathbf{1}\rangle}, \qquad h_T(x) = 0. \tag{26.42b}$$

The point is now an equation, not an adjective: the ϱ^2-coefficient g_t in (26.42a) — the curvature of the value in the order parameter, a *gain* — carries *no* σ_0, exactly the certainty-equivalence statement; the entire common-noise cost $\propto \sigma_0^2$ enters only the *level* h_t in (26.42b), as three terms tracing to the three nonvanishing families of (26.23): the agent's own extra diffusion (i) $\frac{1}{2}\sigma_0^2P$, the agent–field cross term (iii) $\sigma_0^2 rf\beta$ from $\nabla_\xi\partial^y_\mu U = rf(x)f(y)$, and the second-field-derivative double integral (iv) $\frac{1}{2}\sigma_0^2 g_t\beta^2$ from $\partial^y_\mu\partial^{y'}_\mu U = g_t(x)f(y)f(y')$. The order parameter itself becomes the *random* process (26.40)-paired-with-f,

$$\mathrm{d}\varrho_t = \big(a - b^2P - b^2\lambda r\big)\,\varrho_t\;\mathrm{d}t + \sigma_0\,\beta\;\mathrm{d}B^0_t, \qquad \beta = \langle f,\mathbf{1}\rangle = \textstyle\int_0^1 f, \tag{26.43}$$

and the master function, evaluated along this random ϱ_t, delivers the equilibrium value; the deterministic system of chapter 22 could not, which is the worked-computation face of why the master equation is indispensable under common noise.

Worked computation 26.12 (Graphon LQG master equation, operator Riccati, and rank-one collapse)**.** For the graphon LQG game (26.24)–(26.25), the master function is the quadratic (26.36); its state-curvature P solves the kernel-independent scalar Riccati (26.31), and its population-sensitivity $\mathcal{R}_t$ solves the self-adjoint *operator* Riccati (26.35), the exact lift of the classical (12.25b) with the scalar coupling replaced by $\mathbb{W}$; the extra-diffusion term vanishes identically. On the rank-one kernel $W = f\otimes f$ the operator Riccati collapses to the scalar Riccati (26.37) with $b^2 \mapsto b^2\lambda$ ($\lambda = \|f\|^2$ the leading eigenvalue), the entire field reduces to the scalar order parameter (26.38), and the constant graphon recovers Thread A. Additive common noise leaves P, $\mathcal{R}_t$ and the order-parameter *gain* g_t untouched (certainty equivalence) and feeds the common-noise cost $\propto\sigma_0^2$ into the *level* h_t alone (26.42b), through the three nonvanishing families of (26.23); the equilibrium order parameter becomes the random process (26.43). Every statement here is unconditional — no regularity hypothesis of theorem 26.7 is needed — because the linear–quadratic structure makes the master function explicit.

Remark 26.13 (Where uniqueness enters, and the spectral threshold)**.** Global existence of the operator Riccati solution $\mathcal{R}_t$ on all of $[0,T]$ corresponds to existence of a global classical graphon master solution: a finite-time blow-up of $\mathcal{R}_t$ marks the failure of the global decoupling field — the loss of a unique classical master solution — and hence governs uniqueness of the graphon LQG equilibrium for every initial field. In the rank-one case (26.37) shows this turns on the scalar r not blowing up, which — through the effective coupling $b^2\lambda$ — is governed by the leading eigenvalue $\lambda = \|\mathbb{W}\|$ of the network operator. This is the LQG face of the spectral entanglement of section 26.4:

the operator-norm threshold (A-Coupling) is precisely the condition that keeps r finite for all horizons, and a kernel with too large a leading eigenvalue can drive the population-sensitivity Riccati to blow up in finite time, the master solution to cease to exist, and graphon equilibria to multiply — the network-resolved version of the LQG phase boundary of worked computation 14.10.

26.7 The state of the theory: proved versus conjectural

This chapter sits at the live boundary of the subject, and the book's standard requires us to be exact about what is established. We separate four tiers.

What is proved here, in full. The equivalence of the three graphon formulations — master equation, field MFG system, graphon FBSDE — is complete: the field chain rule lemma 26.2, the derivation theorem 26.3, and the characteristics theorem 26.5 are honest theorems, each a clean type-by-type lift of its classical ancestor whose only new ingredient is the integral over the type continuum. The graphon LQG master equation of worked computation 26.12 — the operator Riccati (26.35), the rank-one collapse (26.37), the common-noise certainty equivalence — is unconditional and explicit.

What is reduced to the classical template. Well-posedness in the monotone regime (theorem 26.7) is established along the same architecture as [37], and Steps 1–2 (construction, single-valuedness, and the d_2-Lipschitz estimate) are complete *given* the graphon existence/uniqueness theory of chapter 24. In the special structures — separable and finite-rank kernels, the linear–quadratic case, block graphons (which reduce the field problem to a finite multi-population MFG, the bridge to chapter 30) — the full classical-strength regularity follows, and the master equation is classically well posed. The short-horizon / small-coupling contraction regime (remark 26.8) is unconditional for all admissible kernels.

What is conjectural. A general classical well-posedness theorem for the graphon master equation, of the strength available classically, is *not* known. The obstruction is Step 3 of theorem 26.7: controlling the linearized graphon MFG system uniformly across the continuum of types and against the full spectrum of $\mathbb{W}$. We expect — and the LQG operator Riccati (26.35) strongly suggests, since it is well posed for all horizons exactly when the spectrum of $\mathbb{W}$ keeps the Riccati flow finite — that monotonicity together with a spectral condition on $\mathbb{W}$ suffices, but a complete proof for general nonlinear couplings and general (A-Reg-W) kernels is open. The common-noise theory is less complete still: even the classical common-noise master equation is delicate ([37, 41, 4]), and its graphon lift, with the double-type-integral second-field-derivative term (26.23), has been carried through only in the linear–quadratic and finite-rank cases. The type-dependent common-noise / common-noise-graphon structure of remark 26.11 is essentially unstudied.

What lies beyond the dense theory. Everything in this chapter assumes a bounded, dense graphon (A-Reg-W) and the field space $\boldsymbol{\mathcal{P}}_2$ of (26.1). For sparse and L^p kernels the operator

$\mathbb{W}$ need not be bounded on $L^2([0,1])$, the spectral arguments behind (26.17) and (26.35) require revision, and the very existence of the field-of-measures master function is open; this is the subject of chapter 27, and the master equation under common noise on sparse limits is among the sharpest open problems flagged in the research map of chapter 31.

Rigor ledger

Proved.

- The field chain rule for fully $\mathcal{C}^2$ functionals along a field of Fokker–Planck flows (lemma 26.2), the type-integrated lift of lemma 12.2, with the linear-growth drift reached by an in-book truncation argument; and its stochastic upgrade under common noise (lemma 26.10), lifting the conditional Itô formula theorem D.8 slice-wise.
- The second field derivative $\partial_\mu^y \partial_\mu^{y'} U$ (definition 26.9), the slice-wise lift of the classical second Lions derivative definition D.6.
- The graphon master equation and its equivalence with the graphon MFG system: the equation (26.10)–(26.11) holds for the constructed U, and conversely its classical solutions generate solutions of the field system (theorem 26.3); the type integral, not a measure derivative, is the structural novelty.
- Characteristics: field Fokker–Planck flows are the characteristic curves, $\nabla_\xi U$ is the (type-indexed, operator-valued in LQG) graphon-FBSDE decoupling field, and the three pictures coincide (theorem 26.5).
- The graphon LQG master equation in closed form: vanishing extra-diffusion, the operator Riccati (26.35), the rank-one scalar collapse (26.37) to an order parameter, the constant-graphon recovery of Thread A, and additive-common-noise certainty equivalence (worked computation 26.12).
- The structural form of the common-noise terms (26.23) (two coefficient upgrades and two genuinely new terms), in particular the double-type-integral second-field-derivative Laplacian on $\boldsymbol{\mathcal{P}}_2$, derived from lemma 26.10.

Assumed (imported).

- The field-of-measures space $\boldsymbol{\mathcal{P}}_2$ and its metric (chapter 21); the graphon MFG system (26.6) and the optimal-feedback relation (chapter 22, appendix A.4); the graphon McKean–Vlasov flow and FBSDE with its Feynman–Kac/decoupling link (chapter 23).
- Graphon existence/uniqueness, the graphon monotonicity (A-Mon, definition 26.6), and the operator-norm threshold (A-Coupling) (chapter 24); standing assumptions (A-Lip),(A-Growth),(A-Nondeg),(A-Conv),(A-Reg-W) of appendix A.6.
- The flat and Lions derivatives, the second Lions derivative (definition D.6), and the Itô calculus on $\mathcal{P}_2(\mathbb{R}^d)$ — both the deterministic flow formula (theorem D.7) and the conditional/common-noise Itô–Wentzell formula (theorem D.8) — of chapter 4 and chapter 12 (definition 12.1, appendix A.3), here lifted slice-wise.

- The graphon operator and its spectral/positivity properties (definition 15.13, lemma 15.15) and the rank-one graphon (example 15.11).

Deferred.

- The full classical regularity of the graphon master function (Step 3 of theorem 26.7): reduced to the graphon existence theory of chapter 24 and the linearized-system analysis, complete only in the special structures named in section 26.7; the classical template is [37] and [41], Ch. 4–5.
- The full common-noise well-posedness theory, including the role of nondegeneracy of σ_0 and weak-monotonicity arguments ([4]); structure only here.
- The block-graphon reduction to multi-population MFG (chapter 30) and the finite-network ε-Nash transfer (chapter 25).

Open.

- A general classical well-posedness theorem for the graphon master equation, monotone + spectral, for nonlinear couplings and general (A-Reg-W) kernels (section 26.7).
- The graphon master equation under common noise beyond the linear–quadratic/finite-rank cases; the type-dependent / common-noise-graphon structure of remark 26.11.
- The entire theory on sparse / L^p / graphex limits, where $\mathbb{W}$ is unbounded and even existence of the field master function is unsettled (chapters 27 and 31).

Part IV retrospective

The graphon synthesis is now complete as a parallel of Part II. We set up a continuum of type-indexed agents and the field-of-measures state (chapter 21); wrote the coupled HJB–Fokker–Planck system across the kernel (chapter 22) and its probabilistic twin (chapter 23); proved existence and uniqueness with the operator-norm threshold (chapter 24); transferred the limit to and from finite networks (chapter 25); and, in this chapter, compressed the whole apparatus into the graphon master equation and used it to admit common noise. The single structural lesson of Part IV is that *every classical object survives the lift type-by-type, and the only genuinely new ingredient is the graphon operator* $\mathbb{W}$ — it enters the cost coupling, threads through the type integral that prevents the master equation from decoupling, sets the spectral threshold for uniqueness, and modulates how a common shock correlates the types. Where $\mathbb{W}$ is benign (small norm, rank-one, block) the theory is as complete and explicit as the homogeneous one; where its spectrum is large or its structure sparse, the frontier opens. Part V turns to that frontier: sparsity and L^p kernels (chapter 27), numerics for the field system (chapter 28), learning (chapter 29), the heterogeneity zoo (chapter 30), and the open-problem map (chapter 31).

Bibliographic notes

The graphon mean field game and its GMFG equations are due to Caines and Huang [33, 34], who set up the type-indexed system and the field coupling lifted here; the probabilistic theory of graphon mean field *systems* — the field McKean–Vlasov flow underlying lemma 26.2 and the decoupling link of theorem 26.5 — is developed by Bayraktar, Wu and Zhang [16], and the static and linear–quadratic graphon games that anchor the LQG worked computation by Parise and Ozdaglar [108], Carmona, Cooney, Graves and Laurière [39], and Aurell, Carmona and Laurière [10], whose linear–quadratic analysis is the closest published relative of worked computation 26.12 and its operator Riccati. The entire derivation is the graphon lift of the classical master-equation theory of Cardaliaguet, Delarue, Lasry and Lions [37] and Carmona and Delarue [41] (Chapters 4–5 for the calculus on Wasserstein space and the common-noise master equation), to which theorems 26.3, 26.5 and 26.7 reduce wherever the classical arguments transfer; the displacement-monotone refinements that may eventually furnish the missing Step 3 are those of Gangbo and Mészáros [75] and Gangbo, Mészáros, Mou and Zhang [76], and the weak-monotonicity common-noise well-posedness of Ahuja [4] is the template for the (open) graphon common-noise theory. The honest status of the field — which cases are theorems and which are conjectures — follows the modern landscape surveyed in [11]; the sparse / L^p continuation is chapter 27, and the open problems are collected in chapter 31.

Part V

Applied and Computational Frontiers

Chapter 27

Sparse Networks: L^p Graphons and Graphex Mean Field Games

This chapter is a confrontation. On one side stands the dense theory of Part IV, complete and self-consistent; on the other stand the networks one actually wants to model — social graphs, citation networks, power grids, the web — which are sparse and heavy-tailed, and which the dense theory does not so much approximate poorly as refuse to see at all. Our task is to locate the single load-bearing assumption that produces the conflict, weaken exactly that assumption, and report honestly on how much of the edifice survives the weakening.

The first thing to be clear about is how deeply the density assumption is wired in. Every theorem of Part IV is, at bottom, a theorem about *dense* networks. A graphon is a bounded kernel $W\colon [0,1]^2 \to [0,1]$; the sampled graph $G(N,W)$ of definition 19.1 has $\Theta(N^2)$ edges; and the cut metric of chapter 17 that drives the convergence theory is calibrated to precisely that density. The boundedness is not an innocent normalization. A bound $W \leq 1$ caps each connection probability by a constant, so a typical vertex — whose $N-1$ potential partners each connect with probability of order $\int W(x,\cdot)$, a fixed positive number — accumulates of order N neighbors; boundedness *forces* density. The graphon mean field game built on this foundation — the coupled HJB–Fokker–Planck system across a kernel of chapter 22, its existence and uniqueness theory of chapter 24, the finite-network approximation of chapter 25, and the master equation of chapter 26 — inherits the density assumption wholesale, because it inherits the kernel.

The wound this chapter sets out to heal was diagnosed already in section 19.5, in two short propositions that are worth recalling as the motivation for everything that follows. The first says that the limit object is blind to sparsity: any sequence of graphs with subquadratic edge count converges in cut metric to the *zero* graphon, so a sparse network and an empty network have the same dense limit. The second says that the failure is built into the sampling model itself: the W-random model is *intrinsically* dense, producing $\Theta(N)$ degrees and $\Theta(N^2)$ edges almost surely, with no choice of bounded W able to escape. The dense machinery cannot be tuned into the sparse regime; it must be relaxed.

We proceed along the road mapped in section 19.6, organizing the chapter as a sequence of questions rather than a list of topics. *What, precisely, do real networks demand that the dense theory* *cannot give, and how many independent relaxations does meeting those demands require?* That is section 27.1, which converts the qualitative Ch. 19 diagnosis into three quantitative desiderata and shows that they split along *two logically independent* axes: allowing the limiting kernel to be unbounded (an L^p graphon), and allowing the sampled graph to be genuinely sparse (a vanishing target density $\rho_N \to 0$, or, more radically, a vertex set that is not $[0,1]$ at all). *How far can the existing well-posedness theory be pushed along the first axis?* That is section 27.2, and the news is surprisingly good: the boundedness $W \le 1$ was never the operative hypothesis — square-integrability is — and the GMFG well-posedness theory of chapter 24 extends verbatim to L^2 graphons, with the operator norm $\|\mathbb{W}\|$ controlled by $\|W\|_{L^2}$ in place of $\|W\|_\infty$, right up to the line $p = 2$ below which the operator theory genuinely dies. *And along the second axis, where genuine sparsity lives?* That is section 27.3, where the graphex / sampling-convergence framework builds the random graph from a point process (the underlying Poisson calculus is developed self-containedly in appendix H) and delivers true subquadratic sparsity — here the news is honestly mixed, since the game over a graphex limit is only partly under control. *What single physical fact have we learned by the end?* That is section 27.4, which extracts the one reading the chapter earns: that $p = 2$, the degree exponent $\gamma = 3$, and a divergent susceptibility are one threshold seen three ways. Throughout, in keeping with the standing convention of appendix A.1, value functions u are solved backward from terminal data and population fields m forward from initial data; and, since a graphon W is in scope everywhere, Brownian motion is written B (appendix A.5).

A warning that sets the tone. This is a frontier chapter. Some of what follows is theorem, some is honest conjecture, and the boundary between them is itself one of the chapter's deliverables — locating it sharply is part of what we deliver, not an embarrassment we hide. We flag the status of every substantive claim as we go, and the closing ledger (and chapter 31) keeps the books.

27.1 The failure mode, made quantitative

The preamble promised to make the collapse of the dense machinery quantitative and to isolate the relaxations the repair requires; this section delivers on both. The Ch. 19 diagnosis we recalled — sparse graphs flatten to the zero graphon, the bounded model is intrinsically dense — is so far a qualitative verdict, and a qualitative verdict cannot tell us *how much* of the framework must change. So we proceed in three steps. First we name exactly what a sparse network model is obliged to reproduce, distilling the demands of real data into three sharp quantitative desiderata that the dense theory provably cannot meet. Then we exhibit one concrete mechanism — thinning the edges by a vanishing target density, driven by an unbounded rank-one weight — and show by direct computation that it does meet them, which both certifies that the desiderata are achievable and produces, as a by-product, the arithmetic linking a kernel's integrability to a graph's degree-tail exponent that the whole chapter turns on. Finally we separate two relaxations that are perilously easy to conflate — unbounding the kernel and sparsifying the sample — and show they are logically independent, which is the scaffolding that lets the next two sections treat each axis on its own terms. We begin with the demands.

27.1.1 What real networks insist upon

We begin, as promised, with the demands real data places on us; and the sharpest way to state them is against the backdrop of exactly what the dense theory delivers, so that each demand is visibly a place where model and world part ways. Two propositions from chapter 19 already located that parting; before naming the demands it is worth restating those propositions in words, because the intuition they carry is the engine of everything that follows in this section.

The first, proposition 19.13, is a statement about the *limit object*: the cut metric simply cannot see sparsity. Any sequence of graphs whose edge count is subquadratic, $e(G_n) = o(v(G_n)^2)$, converges in cut metric to the *zero* graphon, no matter how intricate its internal wiring. The reason is a counting mismatch. The cut distance compares *edge densities* — fractions of *possible* edges that are actually present across each cut — and a graph on v vertices has $\binom{v}{2} = \Theta(v^2)$ possible edges; if only $o(v^2)$ of them are realized, then the fraction present across every cut tends to 0, and a graph all of whose cut densities vanish is, in the only metric the dense theory owns, indistinguishable from the empty graph. A sparse network and a network with no edges at all are assigned the *same* dense limit. The limiting procedure discards precisely the structure we came to study.

The second, proposition 19.14, shifts attention from the limit to the *sampling model* and shows the obstruction is not a defect of the limit we happened to choose but is wired into the generative mechanism itself. Under (A-Reg-W) with $\int W > 0$, the degree of a typical vertex of the W-random graph is exactly $\mathrm{Binomial}(N-1, d_W(U))$, with $d_W(x) = \int_0^1 W(x,y)\,\mathrm{d}y$ the degree function. Read this distribution slowly. A vertex of type x has $N-1$ potential partners, and it connects to each one independently with probability $W(x, U_j)$, whose average over a random partner is the fixed number $d_W(x)$. The decisive point is that this per-partner probability is an $O(1)$ *constant* that does not shrink as the network grows: boundedness caps W above by 1, but with $\int W > 0$ the typical connection probability is bounded *below* by a positive constant too. Summing $N-1$ trials each of constant success probability gives expected degree $(N-1)\,d_W(x) = \Theta(N)$: every vertex is wired to a fixed *fraction* of the entire population, and that fraction never decays. Summing over all vertices, the edge count is $\Theta(N^2)$ almost surely. The culprit in both propositions is one and the same feature — the bound $W \leq 1$ pins connection probabilities to constants, and probabilities that do not shrink with N are exactly what density *means*.

With the failure mechanism named we can say precisely what a replacement must do. We distill the demands of real data into three quantitative desiderata: features that a sparse network model is obliged to reproduce and that, by the two propositions just recalled, the bounded dense model provably cannot.

(S1) Linear edge count. $e(G_N) = \Theta(N)$, equivalently a bounded average degree, as against the $\Theta(N^2)$ forced by proposition 19.14(2). This is the most basic sparsity demand and the one the dense model fails most directly: its average degree $(N-1)\int W$ diverges linearly in N. Real social and technological networks do not behave this way — the typical number of a person's acquaintances, or of a router's links, stays roughly constant as the system grows from thousands to millions of nodes; it does not scale up in proportion to the population. A faithful model must hold the mean degree bounded while the size goes to infinity, and bounded mean degree is exactly $e = \Theta(N)$.

(S2) Sublinear maximum degree. $\max_i \deg(i) = o(N)$, against the $\Theta(N)$ maximum degree of the bounded model. Even granting heterogeneity, the most-connected vertex of a real network — a celebrity account, a backbone hub — is large but is still a *vanishing fraction* of the whole: it is adjacent to many nodes but not to a positive fraction of all of them. In the bounded model this is impossible, because every type's expected degree $(N-1)\,d_W(x)$ is at most $N-1$ and, for any type with $d_W(x)$ bounded away from 0, is already $\Theta(N)$: with connection probabilities capped by a constant, even the hubs can only grow linearly, never sublinearly. A genuine sparse model must let the maximum degree grow — producing real hubs — while keeping that growth strictly below N.

(S3) Heavy-tailed unnormalized degrees. The degree distribution, taken *without* any rescaling by N, must have a heavy (e.g. power-law) tail whose index is independent of N. This is the subtle desideratum and the one that most sharply exposes what the dense model cannot do. The bounded model *does* produce a stable limiting shape — but only for the *normalized* degree $\deg(i)/(N-1)$, which converges to $d_W(U)$, a random variable bounded by 1 (proposition 19.14(3)). The raw, unnormalized degree $\mathrm{Binomial}(N-1, d_W(U))$ has mean $\Theta(N)$: its entire distribution marches off to infinity with the network size, and at any fixed count k the probability $\mathbb{P}(\deg = k) \to 0$. So the only N-independent object the dense model can offer is a law of *fractions* of the population — never a law of *counts*. But a power-law degree tail is, by definition, a statement about counts: $\mathbb{P}(\deg = k) \propto k^{-\gamma}$ at fixed $k = 1, 2, 3, \ldots$, the *same* small integers no matter how large the graph. This is the load-bearing point hidden in the phrase "only after dividing by $N-1$." Dividing by $N-1$ to manufacture a fixed shape is precisely the operation that erases an unnormalized tail: it rescales the count axis by $1/N$, collapsing every fixed degree k to the single point 0 in the limit and compressing the whole tail into an arbitrarily thin sliver near the origin. A model whose only scale-free content lives in normalized degrees has, by construction, no unnormalized tail left to report. Meeting (S3) therefore requires a model in which the degree law stabilizes *before* any division by N — which, as the next subsection shows, is exactly what a mixed-Poisson limit provides.

These three are not pathologies of exotic datasets; they are the rule. Across social, biological, and technological networks one measures power-law degree distributions $\mathbb{P}(\deg = k) \propto k^{-\gamma}$, and the exponent γ falls overwhelmingly in the window $(2, 3)$ [48, 45]. That window is not a coincidence of measurement, and it is worth understanding what makes it special before we exploit it. The two boundary values mark two qualitative transitions in the moments of the degree law. At $\gamma = 2$ the mean degree $\sum_k k\,\mathbb{P}(\deg = k)$ passes from divergent to convergent; for $\gamma > 2$ the network has a well-defined finite average degree, consistent with the sparsity demanded by (S1). At $\gamma = 3$ the *second* moment passes from divergent to convergent; for $\gamma < 3$ the degree variance is infinite, the signature of genuine hub-dominated heterogeneity in which a few vertices carry a macroscopic share of the edges. The window $(2, 3)$ is thus exactly the regime of *finite mean but infinite variance*: light enough to be sparse, heavy enough to be ruled by its hubs — and this is where almost all real networks live.

We flag now — as a promise to be redeemed, not a claim to be taken on faith — why this same window will turn out to be the precise frontier of the mean-field-game theory, and not merely a fact about graphs. When we place a game on such a network, the population coupling acts only through the kernel's integral operator $\mathbb{W}$, and the strength of that coupling is governed by its operator norm. For the heavy rank-one weight that generates a scale-free degree tail, that operator norm is exactly a *second moment* of the weight profile, $\|f\|_{L^2}^2$ — the quantity a physicist would call the *susceptibility* of the coupling, the leading eigenvalue through which a perturbation of the mean field feeds back on itself. The worked computation of worked computation 27.19 will show that this susceptibility diverges at exactly the same threshold as the degree variance, $\gamma = 3$, so that the divergence of the second moment behind (S3) and the divergence of the coupling are one event seen twice. The empirically dominant window $\gamma \in (2,3)$ is therefore precisely the regime in which the mean-field-game coupling becomes infinitely strong and the dense well-posedness theory of chapter 24 provably breaks. We cannot prove this yet — we still lack both the sparsification mechanism and the L^p operator theory — but it is the reason the window $(2,3)$ deserves to be circled here, and it is the destination the rest of the chapter steers toward.

We have, then, a sharp target — the three desiderata (S1)–(S3) — and a sharp diagnosis — boundedness $W \le 1$ is the single obstruction. What we still lack is a *mechanism*: a concrete way to generate graphs that actually meet (S1)–(S3) while staying recognizably close to the dense model we already understand and can compute with. The most conservative such mechanism keeps the unit-interval type space and the latent kernel intact and merely thins the edges by a vanishing target density. We turn to it now.

27.1.2 Sparsification by a target density

The previous subsection left us with a sharp want: a concrete generative mechanism that actually meets (S1)–(S3) while staying recognizably close to the dense model we already understand and can compute with. We answer that want now, and the answer is deliberately the most conservative one available. Rather than discard any of the three pillars of the dense W-random model of definition 19.1 — the unit-interval type space, the latent kernel, the conditionally-independent edge sampling — we keep all three and change exactly one thing: we thin the edges by an overall factor that vanishes with the network size. Concretely we attach a *target density* $\rho_N \in (0,1]$ to the sampling model and connect a pair of types with probability proportional to $\rho_N W$ in place of W. The name is apt: ρ_N is the single dial that sets how dense the sampled graph will be, and driving it to zero at the right rate is precisely what converts the intrinsically dense model into a sparse one. Everything else in the construction is inherited verbatim, which is exactly why this is the repair to try first — it perturbs the dense theory in one controllable place, and we can read off what survives.

Definition 27.1 (Sparse W-random graph)**.** Let $W\colon [0,1]^2 \to [0,\infty)$ be symmetric and measurable (not necessarily bounded by 1), and let $\rho_N \in (0,1]$. The *sparse W-random graph $G(N,\rho_N,W)$* is obtained by drawing $U_1,\dots,U_N$ i.i.d. Unif$[0,1]$ and, conditionally on the types, including each edge $\{i,j\}$ independently with probability

$$p_{ij}^N \;=\; \min\{1,\; \rho_N\, W(U_i,U_j)\}\,. \tag{27.1}$$

The choice $\rho_N \equiv 1$ with $W \le 1$ recovers the dense model $G(N,W) = G(N,1,W)$ of definition 19.1. The regime of interest is $\rho_N \to 0$; the canonical sparse scaling is $\rho_N = c/N$.

Two features of definition 27.1 deserve a word before we put them to work, because together they are what make the definition do its job. The first is that W is no longer required to be bounded by 1; it may take values anywhere in $[0, \infty)$. This is not a cosmetic generalization. The whole reason to undertake the repair is desideratum (S3), and (S3) will turn out to demand a heavy-tailed — hence unbounded — kernel, so the sampling model must be able to ingest one. But an unbounded W cannot be multiplied by a constant and read directly as a probability: $\rho_N W(U_i, U_j)$ can exceed 1 wherever W is large. This is exactly what the second feature, the truncation $\min\{1, \cdot\}$ in (27.1), repairs. It clips the raw intensity $\rho_N W$ back into $[0, 1]$ at the rare type-pairs where it would overshoot, so that p_{ij}^N is always a legitimate edge probability no matter how violently W blows up. The truncation is therefore the precise device that lets an unbounded kernel feed a well-defined edge sampling model; it is the price of admission for heavy tails, and the proofs below will show it costs almost nothing.

That "almost nothing" is the content of the truncation's third virtue: it switches itself off in the limit. Fix a pair of types (U_i, U_j). With the canonical scaling $\rho_N = c/N$, the raw intensity $\rho_N W(U_i, U_j) = (c/N)W(U_i, U_j) \to 0$ as $N \to \infty$ almost surely (the kernel value is a fixed finite number for a.e. pair, and we are dividing it by N). So for all large N the clip is inactive at that pair and p_{ij}^N equals the untruncated $(c/N)W(U_i, U_j)$ on the nose. The truncation bites only on a vanishing set of exceptionally heavy pairs, and the analysis to come is in large part the careful accounting that this vanishing set contributes nothing to the limiting degree law. We now carry out that accounting.

Proposition 27.2 (Mixed-Poisson degrees in the sparse limit)**.** *Let $W\colon [0,1]^2 \to [0, \infty)$ be symmetric and measurable with degree function $d_W(x) = \int_0^1 W(x, y)\,\mathrm{d}y$ finite for a.e. x, and suppose $d_W \in L^1[0, 1]$. Sample $G(N, \rho_N, W)$ with $\rho_N = c/N$ for a constant $c > 0$. Then:*

1. *the expected number of edges satisfies $\mathbb{E}\big[e\big(G(N, \rho_N, W)\big)\big] = \frac{c}{2}\, t(K_2, W)\, N + o(N)$, where $t(K_2, W) = \int_{[0,1]^2} W$; in particular the graph is sparse, $e = \Theta(N)$, realizing* (S1)*;*

2. *conditionally on $U_i = x$, the degree of vertex i converges in distribution to a Poisson law,*

$$\deg(i) \overset{d}{\Longrightarrow} \mathrm{Poisson}\big(c\, d_W(x)\big), \tag{27.2}$$

and unconditionally the degree of a uniformly chosen vertex converges to the mixed-Poisson *law* $\mathrm{Poisson}(c\, d_W(U))$*, $U \sim \mathrm{Unif}[0, 1]$.*

Proof. Part (1): the edge count is linear. Every one of the $\binom{N}{2}$ potential edges is included independently with probability p_{ij}^N, and the types are exchangeable, so the two endpoints of a generic edge may as well be U_1, U_2 and $\mathbb{E}[e] = \binom{N}{2}\mathbb{E}\Big[p_{12}^N\Big]$. It remains to evaluate $\mathbb{E}\Big[p_{12}^N\Big]$ to leading order. Split according to whether the clip is active. On the event $\{(c/N)W(U_1, U_2) \le 1\} = \{W(U_1, U_2) \le N/c\}$ the truncation does nothing and $p_{12}^N = (c/N)W(U_1, U_2)$; on the complementary heavy event $\{W(U_1, U_2) > N/c\}$ it replaces a number larger than 1 by exactly 1. Writing the expectation as the untruncated mean minus the mass removed by clipping,

$$\mathbb{E}\Big[p_{12}^N\Big] = \frac{c}{N} \int_{[0,1]^2} W \;-\; \mathbb{E}[*] \left(\tfrac{c}{N} W(U_1, U_2) - 1\right) \mathbf{1}\, \{W(U_1, U_2) > N/c\}.$$

The first term is $(c/N)t(K_2, W)$, finite because $W \in L^1([0,1]^2)$ (which holds since $d_W \in L^1$ and $\int_{[0,1]^2} W = \int_0^1 d_W$). The correction is at most $(c/N)\mathbb{E}[W\,\mathbf{1}\{W > N/c\}]$, and here is the one analytic input: for an integrable function the contribution of its own large-value tail to the integral vanishes as the threshold recedes, $\mathbb{E}[W\,\mathbf{1}\{W > N/c\}] \to 0$. (This is the defining property of an integrable function — equivalently, the uniform integrability of the single random variable $W(U_1, U_2)$ — and it is strictly more than Markov's inequality would give; we return to this distinction in part (2), where it is the crux.) So the correction is $(c/N) \cdot o(1) = o(1/N)$, whence $\mathbb{E}\left[p_{12}^N\right] = (c/N)t(K_2, W) + o(1/N)$ and $\mathbb{E}[e] = \binom{N}{2}\left[(c/N)t(K_2, W) + o(1/N)\right] = \frac{c}{2}t(K_2, W)N + o(N)$. The edge count is $\Theta(N)$ in expectation: the average degree $2\mathbb{E}[e]/N \to c\,t(K_2, W)$ is a finite constant, independent of N. This is (S1), and it shows the role of the scaling $\rho_N = c/N$: it is the unique power of N at which the $\binom{N}{2} = \Theta(N^2)$ candidate edges, each kept with probability $\Theta(1/N)$, produce $\Theta(N)$ actual edges.

Part (2): the degree law is mixed Poisson. Condition on $U_i = x$, fixing the type of the vertex whose degree we track. Its degree is the number of its $N-1$ potential partners to which it actually connects, a sum of $N-1$ independent indicators with success probabilities $p_{ij}^N = \min\{1, (c/N)W(x, U_j)\}$, the partner types U_j being i.i.d. uniform. Introduce the untruncated probabilities $q_j^N = (c/N)W(x, U_j)$, so that $p_j^N = \min\left\{1, q_j^N\right\}$ and $p_j^N = q_j^N$ except on the heavy event $\{W(x, U_j) > N/c\}$. We prove convergence to Poisson$(c\,d_W(x))$ in two moves: first for the untruncated array, then by showing the truncation cannot change the limit.

For the untruncated array, this is the classical law of rare events: a row-sum of independent indicators converges to a Poisson law when each individual success probability is uniformly negligible and the total mean converges. Both conditions hold here. The total mean is

$$\sum_{j=1}^{N-1} \mathbb{E}\left[q_j^N\right] = (N-1)\,\frac{c}{N}\,\mathbb{E}[W(x, U)] = (N-1)\,\frac{c}{N}\,d_W(x) \xrightarrow[N\to\infty]{} c\,d_W(x),$$

the target Poisson mean; and the individual means are uniformly small, $\max_j q_j^N = (c/N)\max_j W(x, U_j) \to 0$ in probability, because $W(x, \cdot) \in L^1$ forces $s\,\mathbb{P}(W(x, U) > s) \to 0$, so the largest of N samples grows slower than N. The quantitative engine behind "converges to Poisson" is Le Cam's inequality (lemma H.8), which bounds the total-variation distance between the law of $\sum_j \text{Bernoulli}(q_j^N)$ and $\text{Poisson}(\sum_j q_j^N)$ by $\sum_j (q_j^N)^2 \le (\max_j q_j^N)\sum_j q_j^N$; with the total mean bounded and $\max_j q_j^N \to 0$ this product vanishes, which is the Poisson limit theorem theorem H.9 specialized to our array.

It remains to remove the truncation. The truncated and untruncated degrees can be built on the same probability space (couple them indicator-by-indicator), and they differ only where the clip acted, by which amount $q_j^N - p_j^N = (q_j^N - 1)^+$. The total-variation distance between the two sums is therefore at most the expected total clipped mass $\sum_j \mathbb{E}\left[(q_j^N - 1)^+\right]$, and this we now show vanishes. Since $(q_j^N - 1)^+ \le q_j^N\,\mathbf{1}\left\{q_j^N > 1\right\} = (c/N)W(x, U_j)\,\mathbf{1}\{W(x, U_j) > N/c\}$,

$$\sum_{j=1}^{N-1} \mathbb{E}\left[(q_j^N - 1)^+\right] \le (N-1)\,\frac{c}{N}\,\mathbb{E}[*]\,W(x, U)\,\mathbf{1}\{W(x, U) > N/c\} = c \cdot o(1) \;\longrightarrow\; 0, \tag{27.3}$$

the inner expectation being a tail-truncated first moment that vanishes by integrability of $W(x, \cdot)$, exactly as in part (1). Hence the truncated array has the same Poisson limit as the untruncated one, which is (27.2).

The uniform-integrability step is the crux, and it is genuinely stronger than Markov. It is tempting to control the truncation by simply counting how many edges get clipped, via Markov's inequality: $\mathbb{E}[\#\{j : W(x,U_j) > N/c\}] = (N-1)\mathbb{P}(W(x,U) > N/c) \le (N-1)\,\frac{c}{N}\,\mathbb{E}[W(x,U)] = c\,d_W(x)$. But this bound is $\Theta(1)$, not $o(1)$: it says only that a *bounded number* of partners — not a vanishing number — have an above-threshold raw intensity. That is too weak to conclude, because a bounded number of clipped indicators could in principle shift the degree by a bounded amount and so corrupt the Poisson limit. What rescues the argument is not that the heavy event is rare but that the *mass* it carries is small: the relevant quantity is not the probability $\mathbb{P}(W > N/c)$ but the size-weighted tail $\mathbb{E}[W\,\mathbf{1}\{W > N/c\}]$, and this tends to 0 for every integrable $W(x,\cdot)$ precisely because the tail of a convergent integral is negligible. Markov keeps only the rarity and throws away the magnitude; on a heavy-tailed kernel the magnitude on the rare event can be enormous, so rarity alone settles nothing, while the size-weighted tail bound (27.3) settles everything. This is why the hypothesis of the proposition is $d_W \in L^1$ and not merely $d_W < \infty$ pointwise: integrability is exactly the uniform-integrability statement that makes the clipped mass vanish.

Finally, the conditional limit Poisson$(c\,d_W(x))$ holds for a.e. fixed x; averaging over the vertex's own type $U_i \sim \mathrm{Unif}[0,1]$ replaces the deterministic mean $c\,d_W(x)$ by the random mean $c\,d_W(U)$, giving the *mixed*-Poisson law Poisson$(c\,d_W(U))$ for a uniformly chosen vertex. □

Proposition 27.2 is the lever that turns a heavy-tailed *kernel* into a heavy-tailed *degree distribution*, and it is worth saying precisely why before we pull it. The limiting degree is not Poisson but *mixed* Poisson: a Poisson count whose mean $\Lambda = c\,d_W(U)$ is itself random, drawn by sampling the vertex's type. A pure Poisson law is light-tailed — its tail decays faster than any power — so if the degree is to be heavy-tailed, the heaviness must come entirely from the mixing variable Λ. And it does: a mixed-Poisson law inherits the tail of its mixing distribution, so that whenever $c\,d_W(U)$ has a power-law tail the degree has the *same* power-law tail. The degree-tail problem has thus been reduced to a question purely about the kernel: make the degree function d_W heavy-tailed as a function of the type. The cleanest way to do that, and the one that connects straight to the operator theory the chapter is steering toward, is an unbounded rank-one weight. We now compute that case in full.

Corollary 27.3 (Power-law degrees from a heavy rank-one weight)**.** *Let $W(x,y) = f(x)f(y)$ with weight profile $f(x) = x^{-\nu}$, $0 < \nu < 1$, and sample $G(N, c/N, W)$. Then $f \in L^1[0,1]$ with $\int_0^1 f = \frac{1}{1-\nu}$, the degree function is $d_W(x) = \frac{1}{1-\nu}x^{-\nu}$, and the limiting (mixed-Poisson) degree distribution has a power-law tail*

$$\mathbb{P}(\deg \ge k) \;\sim\; C\,k^{-1/\nu} \qquad (k \to \infty), \qquad \textit{i.e. exponent}\;\; \gamma = 1 + \tfrac{1}{\nu}. \tag{27.4}$$

In particular $\nu \in (\frac{1}{2}, 1)$ produces precisely the empirically dominant scale-free window $\gamma \in (2,3)$, $\nu \in (0, \frac{1}{2})$ gives $\gamma > 3$, and the boundary $\nu = \frac{1}{2}$ gives $\gamma = 3$.

Proof. First the elementary facts about the weight. With $f(x) = x^{-\nu}$ and $0 < \nu < 1$, $\int_0^1 f = \int_0^1 x^{-\nu}\,\mathrm{d}x = \frac{1}{1-\nu}$ is finite, so $f \in L^1[0,1]$; the degree function of the rank-one kernel factorizes, $d_W(x) = f(x)\int_0^1 f = \frac{1}{1-\nu}x^{-\nu}$, inheriting the same singular profile as f.

Now the tail of the mixing variable. With $U \sim \mathrm{Unif}[0,1]$ the random mean is $\Lambda = c\, d_W(U) = \frac{c}{1-\nu} U^{-\nu}$. To find its tail we invert: $\Lambda > t$ means $\frac{c}{1-\nu} U^{-\nu} > t$, i.e. $U^{-\nu} > \frac{(1-\nu)t}{c}$, i.e. $U < \big(\frac{c}{(1-\nu)t}\big)^{1/\nu}$. For t large enough that this threshold is below 1, the uniform probability is the threshold itself,

$$\mathbb{P}(\Lambda > t) = \Big(\frac{c}{(1-\nu)t}\Big)^{1/\nu} = C\, t^{-1/\nu}, \qquad C = \Big(\frac{c}{1-\nu}\Big)^{1/\nu},$$

a Pareto tail of index $1/\nu$. The map $\nu \mapsto 1/\nu$ is the heart of the corollary: a singularity $x^{-\nu}$ of strength ν in the weight produces a degree-mean tail of index exactly $1/\nu$. A stronger singularity (larger ν) gives a heavier tail (smaller index).

Finally, transfer this tail from the mean Λ to the degree Poisson(Λ). The mixed-Poisson tail-transfer lemma (lemma H.10) says a Poisson count whose random mean has a regularly-varying tail of index $-1/\nu$ has the *same* tail index:

$$\mathbb{P}(\deg \geq k) = \mathbb{P}(\mathrm{Poisson}(\Lambda) \geq k) \sim \mathbb{P}(\Lambda > k) \sim C\, k^{-1/\nu} \qquad (k \to \infty),$$

which is (27.4). The intuition is a separation of scales: conditioned on a large mean $\Lambda = t$, the Poisson count concentrates around t with fluctuations of order $\sqrt{t}$, which are a vanishing *fraction* $1/\sqrt{t}$ of t; so for large k the event $\{\mathrm{Poisson}(\Lambda) \geq k\}$ is, up to negligible corrections, the event $\{\Lambda \gtrsim k\}$. The light Poisson smoothing cannot manufacture or erase a power-law tail — it only blurs the mean by a relatively negligible amount — so the heavy tail of Λ passes through to the degree unchanged. The exponent is read off by $\gamma = 1 + \frac{1}{\nu}$ (the index of $\mathbb{P}(\deg \geq k)$ is $\gamma - 1 = 1/\nu$), and the three windows follow: $\nu \in (\frac{1}{2}, 1) \Rightarrow \frac{1}{\nu} \in (1,2) \Rightarrow \gamma \in (2,3)$, the empirically dominant scale-free band; $\nu \in (0, \frac{1}{2}) \Rightarrow \gamma > 3$; and the boundary $\nu = \frac{1}{2} \Rightarrow \gamma = 3$. □

The corollary deserves a pause, because in producing (S3) it has quietly settled (S2) as well and has planted the arithmetic the rest of the chapter will harvest. Take (S3) first. The power law (27.4) exists only because $\nu > 0$, i.e. only because the weight $f(x) = x^{-\nu}$ blows up at the origin and is therefore unbounded; and then $W = f \otimes f$ is unbounded too. There is no way around this. A bounded weight has a bounded degree function, hence a bounded mixing variable Λ, hence a light-tailed (genuinely Poisson) degree — no power law at all. So a scale-free degree distribution is logically incompatible with the boundedness assumption $W \leq 1$ of (A-Reg-W): *meeting* (S3) *forces the kernel to be unbounded.* This is the central structural fact the subsection was built to expose, and it is what makes the unbounded function class of the next section unavoidable rather than a matter of taste.

The same heavy weight delivers (S2), the sublinear maximum degree, and the argument is a short order-statistic computation worth spelling out. The expected degree of a vertex of type x is $c\, d_W(x) = \frac{c}{1-\nu} x^{-\nu}$, a decreasing function of x; the most-connected vertex is therefore the one with the *smallest* sampled type, $U_{(1)} = \min\{U_1, \ldots, U_N\}$. The minimum of N i.i.d. uniforms has mean $1/(N+1)$ and is $\Theta(1/N)$ with high probability, so the largest expected degree in the sample is

$$c\, d_W(U_{(1)}) = \frac{c}{1-\nu}\, U_{(1)}^{-\nu} = \Theta\big((1/N)^{-\nu}\big) = \Theta(N^{\nu}).$$

Because $\nu < 1$ this is $o(N)$: the hubs grow without bound (a real hub, since $N^{\nu} \to \infty$), yet each remains a vanishing fraction of the population (since $N^{\nu}/N = N^{\nu-1} \to 0$). That is exactly (S2) —

$\max_i \deg(i) = o(N)$ in probability — holding simultaneously with the $\Theta(1)$ average degree of (S1). The exponent ν that tunes the tail also tunes the hub size: heavier tails (ν near 1) make hubs nearly linear, lighter tails (ν near 0) make them nearly $O(1)$, but the strict inequality $\nu < 1$ keeps them strictly sublinear throughout. So a single rank-one weight $x^{-\nu}$, fed through the target-density model, meets (S1) and (S2) outright and (S3) in its mixed-Poisson limit.

What it has *not* yet done is reconcile this success with the operator theory the game needs, and the seed of that reconciliation is the integrability arithmetic of $f = x^{-\nu}$, which we flag now because it returns as the spine of worked computation 27.19. The weight is square-integrable exactly when $\int_0^1 x^{-2\nu}\,dx < \infty$, i.e. $2\nu < 1$, i.e. $\nu < \frac{1}{2}$ — which, under the map $\gamma = 1 + 1/\nu$, is exactly $\gamma > 3$. Equivalently, $f \in L^1 \setminus L^2$ holds precisely for $\nu \in [\frac{1}{2}, 1)$, i.e. $\gamma \in (2, 3]$, with the first failure of square-integrability landing exactly at the boundary $\nu = \frac{1}{2}$, $\gamma = 3$. The empirically dominant scale-free window $\gamma \in (2, 3)$ therefore sits *strictly inside* the regime where the weight is integrable but not square-integrable. Hold that coincidence: the degree-tail boundary $\gamma = 3$ and the square-integrability boundary $\nu = \frac{1}{2}$ are the same line, and we will meet it a third time as the boundary where the coupling operator's norm diverges.

We close by naming the question this leaves open and handing it to the next subsection. The mechanism just built reached (S3) only by going through an unbounded kernel, and reached (S1) only by switching on a vanishing target density $\rho_N = c/N$. Were these two concessions one concession or two — does unbounding the kernel already buy the sparsity, or is the target density doing independent work? The answer is not obvious, and getting it wrong would conflate two relaxations that behave very differently in the game theory to come. The next subsection settles it by separating the two axes cleanly.

27.1.3 Two independent axes

The previous subsection closed on a worry it could not yet dispel. To meet all three desiderata at once we had to do two distinct things to the dense model — replace the bounded kernel by an unbounded weight $f(x) = x^{-\nu}$, and switch on a vanishing target density $\rho_N = c/N$ — and it was not obvious whether these were two independent moves or a single move counted twice. The stakes in answering are not bookkeeping. The two relaxations enter the model through entirely different doors: an unbounded kernel alters the *limit object* — the operator $\mathbb{W}$ through which a game eventually couples — while a vanishing ρ_N alters the *sampling law* that draws the finite graph around that limit. Conflating them would mislocate where the well-posedness theory of chapter 24 breaks and which repair restores it. So before proceeding we pry the two axes apart, and the cleanest way to show they are genuinely independent is to switch each one on alone and watch the other defect stubbornly survive.

Remark 27.4 (Unboundedness and sparsity are independent)**.** Allowing the limiting kernel to be *unbounded* and allowing the sampled graph to be *sparse* are logically distinct relaxations; each, applied in isolation, leaves one of the two defects firmly in place.

- *Unbounded kernel, no sparsification: still dense.* Take an unbounded but integrable kernel $W \in L^1([0,1]^2)$ — say the rank-one weight $f \otimes f$ with $f = x^{-\nu}$ of corollary 27.3 — and sample it with *no* target density, $\rho_N \equiv 1$, so that the edge probability is the bare truncation $p_{ij}^N = \min\{1, W(U_i, U_j)\}$ of (27.1). The graph is still *dense*. The intuition is that density was never a statement about how *large* the kernel can get — the truncation already caps every connection probability at 1 — but about whether connection probabilities stay bounded *below*, away from zero, on a macroscopic set of pairs, and unbounding the kernel does nothing to that floor. Concretely, since $W \geq 0$ is not a.e. zero there are a constant $\delta > 0$ and a product set $A \times B \subseteq [0,1]^2$ of positive measure on which $W \geq \delta$; there $p_{ij}^N \geq \min\{1, \delta\} > 0$, so a vertex whose type lands in A connects to each of the $\Theta(N)$ vertices whose types land in B with at least this fixed probability and accumulates $\Theta(N)$ neighbours — precisely the mechanism of proposition 19.14. The edge count is $\Theta(N^2)$ almost surely. Unboundedness enriches the *limit object*, admitting the heavy weights and the divergent degree function d_W that the tail mechanism feeds on, but it buys no sparsity in the *sample*.

- *Bounded kernel, sparsified: light-tailed.* Conversely, take a genuinely bounded kernel, $W \leq B < \infty$, and make it sparse the only way left, through the vanishing target density $\rho_N = c/N$. Now proposition 27.2 applies and yields exactly $\Theta(N)$ edges — so this axis delivers the sparsity the first could not — but the degree law it produces is *light*-tailed. The limiting degree of a random vertex is Poisson$(c\, d_W(U))$ of (27.2), and boundedness caps its mixing variable: $d_W(x) = \int_0^1 W(x, \cdot) \leq B$, so the random mean $c\, d_W(U) \leq cB$ is *bounded*. A mixed-Poisson law whose mixing variable is bounded is no heavier-tailed than a single Poisson(cB), whose tail $\mathbb{P}(\deg \geq k)$ decays faster than any power of k — super-exponentially, like $(cB)^k/k!$. There is no power law to be had: a global thinning factor ρ_N controls the *number* of edges uniformly across all types but cannot create heterogeneity in the *propensity* to connect, and that heterogeneity — a heavy tail in d_W — is exactly what a bounded kernel forbids.

Meeting all of (S1)–(S3) therefore requires *both* relaxations, and the division of labour between them is clean: the vanishing target density $\rho_N = c/N$ is what secures the linear edge count (S1) and, with it, the sublinear maximum degree (S2), while the unbounded kernel is what supplies the heavy mixing variable behind the power-law tail (S3). Only their combination — precisely the construction of corollary 27.3 — lands all three desiderata at once.

It is worth carrying an explicit scorecard out of this remark, because the remainder of the chapter is organized by it. The unbounded-kernel axis, taken alone, secures *none* of (S1)–(S3): at $\rho_N \equiv 1$ the graph is dense, its maximum degree is $\Theta(N)$, and in the dense regime the unnormalized degree law cannot even stabilize (proposition 19.14), so the heavy weights this axis does provide have no sparse sample on which to imprint a tail. The sparsification axis, taken alone, secures (S1) and (S2) but *not* (S3): it makes the graph sparse, with a finite bounded average degree and a sublinear (indeed poly-logarithmic) maximum degree, yet its degrees are Poissonian and tail-light. The one desideratum that forces both axes on simultaneously is (S3): a scale-free degree tail is the single feature that needs sparse sampling in order to exist as a law of *counts* at all, and an unbounded kernel in order to be *heavy*. This is why the chapter must develop both relaxations, and why it is safe to develop them separately — they do not interfere, they compose.

Having pried the two axes apart we treat them one at a time, hardest-earned news last. We begin with the unbounded-kernel axis, and here the report is the chapter's one piece of nearly unalloyed good news. The reason to be optimistic is already visible in the asymmetry just exposed: a game couples to its population only through the integral operator $\mathbb{W}$, and an operator is sensitive not to the pointwise size of its kernel but to a single scalar summary of it — the operator norm $\|\mathbb{W}\|$ — which can stay perfectly finite even when W itself blows up. If the entire well-posedness theory of chapter 24 sees the kernel only through $\|\mathbb{W}\|$, then an unbounded but suitably integrable kernel may cost it nothing. Pursuing that possibility means first fixing the function class in which an unbounded kernel is allowed to live — the L^p graphons — and locating, within that scale, the exact integrability threshold at which $\|\mathbb{W}\|$ stops being finite. That threshold, we will find, is $p = 2$. The genuinely sparse axis, where the news is honestly mixed, waits for us in section 27.3.

27.2 L^p graphons: unbounded kernels

The previous section closed by parcelling the repair into two non-interfering relaxations and promising that this one — the unbounded-kernel axis — would carry the chapter's good news. It is time to redeem that promise. We take up that axis and nothing else: the sampling law is held fixed, and the only question is what happens to the *limit* theory of Part IV when the kernel is allowed to grow without bound. The answer is the single most reassuring result in the chapter, and rather than let it surface only at the end of a dichotomy we state its thesis in one sentence now. *Boundedness was never the hypothesis the dense theory actually used; square-integrability was.* Every estimate in the well-posedness theory of chapter 24 that one might fear depended on the cap $W \leq 1$ in fact depends only on $W \in L^2$, and the proofs survive the weakening with a single symbol changed. The bound $W \leq 1$ was, all along, a convenient sufficient condition wearing the costume of a necessary one.

Why a single number — the integrability exponent p — should govern the whole question is already visible in the structural fact isolated at the close of remark 27.4: a game couples to its population only through the integral operator $\mathbb{W}$, never through the pointwise values of the kernel, and an operator feels its kernel only through a scalar summary, the operator norm $\|\mathbb{W}\|$. A kernel may blow up to $+\infty$ on a small set and still induce a perfectly bounded operator, provided the blow-up is integrable enough; conversely, no amount of pointwise smallness elsewhere can rescue an operator whose kernel is too singular to be square-summed. So the operative question is not "how large does W get?" but "how much integrability does it take to keep $\|\mathbb{W}\|$ finite?" — and on the L^p scale that question has a sharp, computable answer.

The section is a four-step descent down that scale, and it is worth holding the whole map in view before the first definition arrives. *First*, we fix the function class — the L^p graphons, symmetric kernels with $\|W\|_{L^p} < \infty$ — a scale that interpolates continuously from the dense theory at $p = \infty$ down toward the sparse regime at $p = 1$. *Second*, we find the true boundedness criterion: the exact location on that scale where $\mathbb{W}$ ceases to be a bounded operator, which we will identify as the pivot $p = 2$. At and above $p = 2$ the Hilbert–Schmidt bound controls $\|\mathbb{W}\|$ by $\|W\|_{L^2}$ in place of $\|W\|_\infty$; below $p = 2$ no such control survives. *Third*, with $\|\mathbb{W}\|$ back under control for $p \geq 2$, we re-run the existence-and-uniqueness theory of chapter 24 and watch it extend *verbatim* to L^2 graphons — the contraction estimate that drives the fixed point reads the kernel only through $\|\mathbb{W}\|$, exactly the quantity the Hilbert–Schmidt bound now furnishes, so not a line of the argument needs rewriting. *Fourth* and last, we locate the failure honestly: below $p = 2$ the operator becomes genuinely unbounded, the fixed-point map loses its contraction, and the theory does not merely lack a proof but provably dies. This dividing line $p = 2$ is no isolated technicality — it is the same threshold the degree-tail arithmetic of corollary 27.3 and the susceptibility computation of worked computation 27.19 will rediscover from their own directions, the chapter's one line seen three ways.

That $p = 2$ should be the pivot is, in hindsight, the most natural value imaginable: the population mean $x \mapsto \bar{m}_t(x)$ lives in the Hilbert space $L^2[0,1]$, and an operator on a Hilbert space asks of its kernel exactly one thing — that it be square-integrable. But to turn this slogan into theorems we must first pin down, with care, just what class of unbounded kernels we have license to work in and how that class sits inside the familiar L^p scale. We make the class precise now.

27.2.1 The function class

We promised the right home for an unbounded kernel, and we now build it. The requirement is a delicate one. The class must be *wide* enough to contain the heavy weights the previous section forced upon us — in particular the rank-one $f \otimes f$ with $f(x) = x^{-\nu}$, which is unbounded and so escapes the classical graphons of definition 15.4 entirely — yet it must be *structured* enough that a single scalar parameter measures "how heavy" a kernel is allowed to be, so that the integrability threshold we are hunting for has a name. The L^p scale is exactly such a structure: one knob p, running from ∞ down to 1, that trades boundedness for admissible singularity. Two preliminary points of motivation set up the definition — why we drop nonnegativity, and why we measure size in an L^p norm rather than a sup norm — and then the nesting of the scale will tell us where on it to look.

The first point is about *sign*. In the graph models of Part IV the kernel was a connection probability: $W(x, y)$ was the chance that types x and y are adjacent, so it had to lie in $[0, 1]$, nonnegative by fiat. But in a mean field game the kernel plays a wholly different role. It is the *coupling weight* with which an agent of type x feels the aggregate behaviour of agents of type y — the entry of $z = \mathbb{W}\bar{m}$ of eq. (18.10) that closes the game's feedback loop — and there is no reason for a coupling to be one-signed. This is precisely the situation of Thread B, the congestion/imitation model carried through the book: there $W(x, y) > 0$ encodes *attraction*, type x benefits from aligning with, or imitating, the crowd at type y, while $W(x, y) < 0$ encodes *repulsion*, the congestion regime in which type x is penalized for crowding toward y and is driven away from it. A single network can carry both readings on different pairs of types — imitate your peers, avoid the bottlenecks — so the kernel that drives the game is genuinely signed. We therefore let W range over all of $\mathbb{R}$, reserving the nonnegativity constraint $W \geq 0$ for the special case where the kernel is also serving as a graph-sampling intensity.

Definition 27.5 (L^p graphon)**.** For $1 \leq p \leq \infty$, an L^p *graphon* is a symmetric measurable function $W \colon [0, 1]^2 \to \mathbb{R}$ with $\|W\|_{L^p([0,1]^2)} < \infty$. For graph models one takes $W \geq 0$; signed L^p graphons arise as *couplings* in a mean field game, where W may change sign (attraction versus repulsion across types). We write $\mathcal{W}_0^p$ for the set of L^p graphons. The classical bounded graphons of definition 15.4 are the $W \in [0, 1]$-valued elements of $\mathcal{W}_0^\infty$.

That last sentence is the second motivating point, and it is what guarantees we have *generalized* rather than *replaced* the old theory. A classical graphon $W\colon [0,1]^2 \to [0,1]$ is bounded by 1, so on the finite measure space $[0,1]^2$ its L^p norm is at most 1 for *every* p, in particular $\|W\|_\infty \le 1$; the classical graphons are thus exactly the $[0,1]$-valued members of $\mathcal{W}_0^\infty$, sitting at the very top of the scale. Every bounded kernel is an L^p graphon for all p at once, and relaxing from $\mathcal{W}_0^\infty$ to $\mathcal{W}_0^p$ with $p < \infty$ strictly enlarges the admissible class — it admits kernels that are unbounded but whose blow-up is integrable to order p. The heavy rank-one weight is the prototype: $f \otimes f$ with $f = x^{-\nu}$ is unbounded, hence not in $\mathcal{W}_0^\infty$, yet (as the integrability arithmetic of corollary 27.3 recorded) it lies in $\mathcal{W}_0^p$ for every $p < 1/\nu$. The exponent ν that controlled the degree tail now reappears as the depth to which we must descend the L^p scale to catch the weight.

To read that descent correctly we must get the direction of the scale right, and here the finite-measure setting reverses the intuition trained on $\mathbb{R}^n$. Because $[0,1]^2$ has total measure one — it is a probability space — the L^p norms are *monotone in* p:

$$L^\infty([0,1]^2) \subset \cdots \subset L^q([0,1]^2) \subset L^p([0,1]^2) \subset \cdots \subset L^1([0,1]^2) \qquad (q \ge p), \tag{27.5}$$

so a smaller p gives a *larger*, more permissive class. The one line that justifies the reversal is Jensen's inequality on the probability space $[0,1]^2$: for $q \ge p$ the function $t \mapsto t^{q/p}$ is convex, so $\left(\int_{[0,1]^2} |W|^p\right)^{q/p} \le \int_{[0,1]^2} |W|^q$, i.e. $\|W\|_{L^p} \le \|W\|_{L^q}$; a finite L^q norm therefore forces a finite L^p norm, which is the inclusion $L^q \subseteq L^p$. The total mass being *one* is exactly what makes the inequality run this way — raising a number to a higher power and averaging against a probability measure can only increase it.

A micro-example pins down why this is the opposite of the $\mathbb{R}^n$ picture, and it is the very weight we care about. Take $f(x) = x^{-\nu}$ on $[0,1]$ with $\nu \in (\frac{1}{2}, 1)$. Its singularity at the origin is integrable but not square-integrable: $\int_0^1 x^{-\nu}\,\mathrm{d}x = \frac{1}{1-\nu} < \infty$ while $\int_0^1 x^{-2\nu}\,\mathrm{d}x = \infty$ (since $2\nu > 1$), so $f \in L^1 \setminus L^2$. The larger class L^1 *admits* the heavy weight that the smaller class L^2 *rejects* — smaller p tolerates a fiercer singularity. On $\mathbb{R}^n$ the membership question turns instead on decay at infinity, and the ordering flips: the constant function 1 lies in $L^\infty(\mathbb{R})$ but in no $L^p(\mathbb{R})$ with $p < \infty$, because there is infinitely much mass to integrate at large $|x|$. The difference is that a probability space has no "infinity" to escape to; the only way a function can fail to be integrable is through a local singularity, and integrating a singularity against a finite measure only gets *easier* as p decreases. Heavy tails of a network, which on the line would be a statement about slow decay, become here a statement about how deep a singularity the kernel may carry — and deeper singularities live further down the L^p ladder.

This orientation fixes the geography of the problem. The dense theory of Part IV lives at the top rung, $p = \infty$, where kernels are bounded; pushing toward genuine sparsity and heavy tails drives us downward toward $p = 1$, the most permissive rung, where the most singular admissible weights are found; and somewhere on this ladder is the rung below which the machinery of the game can no longer stand. The thesis of this section is that the decisive rung is $p = 2$ — not ∞, not 1, but the square-integrable middle. The reason that particular value, and not some other, should be the pivot is not yet visible from the function class alone, because the class is only a scale of *kernels*; the game never touches the kernel pointwise. What it touches is the integral operator $\mathbb{W}$, acting on the population mean $x \mapsto \bar{m}_t(x)$, and that mean lives in the Hilbert space $L^2[0,1]$ — it is square-integrable in type under the standing finite-second-moment hypothesis. So the question that actually decides whether the game survives is not "for which p is W bounded?" but "for which p is $\mathbb{W}$ a bounded operator on $L^2[0,1]$?" That is the operative question, and it is the one the next subsection answers.

27.2.2 When is $\mathbb{W}$ a bounded operator?

The previous subsection ended by sharpening the question we are obliged to answer. The dense theory cared whether the kernel was bounded only because boundedness was a quick way to keep the coupling under control; but the game never reads the kernel pointwise. The entire GMFG apparatus of Part IV interacts with W *only* through the integral operator $\mathbb{W}$ and the aggregate $z = \mathbb{W}\bar{m}$ of eq. (18.10) — the single field through which one agent's environment feels the whole population. So the operative question, as the function class forced us to admit, is not "is W bounded?" but "is $\mathbb{W}$ a bounded operator on the space where the population mean lives?" That space is $L^2[0,1]$ and not some other: the means $x \mapsto \bar{m}_t(x)$ that drive the coupling are square-integrable in type under the standing finite-second-moment hypothesis, so $\mathbb{W}$ must act as a bounded operator $L^2[0,1] \to L^2[0,1]$ for the aggregate z to be a well-defined, finite-energy field at all. The whole sparse story now turns on a question of operator theory, and that question is the right one to pursue because the answer is computable and sharp.

There are two classical sufficient conditions for the boundedness of an integral operator, and the reason to present both — rather than the one we will keep — is that comparing them is what exposes where the true threshold lies. They control $\mathbb{W}$ by two genuinely different norms of the kernel: one by the L^2 norm of W itself, the other by the L^∞ norm of its degree function (its row sums). We state each, prove it with the bound made explicit, and then let the rank-one kernel adjudicate between them.

Lemma 27.6 (Hilbert–Schmidt bound)**.** *If $W \in L^2([0,1]^2)$ (an L^2 graphon) then $\mathbb{W}$ is a Hilbert–Schmidt operator on $L^2[0,1]$; it is compact, and if W is symmetric it is self-adjoint, with*

$$\|\mathbb{W}\|_{L^2\to L^2} \;\le\; \|\mathbb{W}\|_{\mathrm{HS}} \;=\; \|W\|_{L^2([0,1]^2)}\,. \tag{27.6}$$

Proof. The argument is the Hilbert–Schmidt kernel theorem (theorem 2.22, proposition 2.24), and it is worth seeing why the bound is exactly the L^2 norm of the kernel. Fix any orthonormal basis $(e_n)_{n\ge1}$ of $L^2[0,1]$. The Hilbert–Schmidt norm of an operator is the ℓ^2 size of its matrix in such a basis, $\|\mathbb{W}\|_{\mathrm{HS}}^2 = \sum_{m,n} |\langle e_m, \mathbb{W}e_n\rangle|^2$, and the matrix entries $\langle e_m, \mathbb{W}e_n\rangle = \int_{[0,1]^2} W(x,y)\, e_m(x)e_n(y)\,\mathrm{d}x\,\mathrm{d}y$ are precisely the coefficients of W in the product orthonormal basis $(e_m \otimes e_n)$ of $L^2([0,1]^2)$. By Parseval applied to W in that basis the sum of their squares is $\|W\|_{L^2([0,1]^2)}^2$, which is finite *exactly* when $W \in L^2([0,1]^2)$; this gives the equality in (27.6) and shows $\mathbb{W}$ is Hilbert–Schmidt. The inequality $\|\mathbb{W}\|_{L^2\to L^2} \le \|\mathbb{W}\|_{\mathrm{HS}}$ is the general fact that the largest singular value of an operator cannot exceed the ℓ^2 norm of its full singular-value sequence. A Hilbert–Schmidt operator is compact (it is the operator-norm limit of its finite-rank truncations $\sum_{m,n\le K}\langle e_m, \mathbb{W}e_n\rangle e_m \otimes e_n$, whose tail has Hilbert–Schmidt — hence operator — norm $\to 0$), so compactness and the Mercer-type spectral decomposition follow as in theorem 18.2 and definition 18.1; symmetry of W makes the kernel matrix Hermitian, so $\mathbb{W}$ is self-adjoint. The point to underline — it is the slogan this whole section was written to earn — is that *boundedness of W is nowhere used.* Parseval needs only that the kernel be square-summable; it neither knows nor cares whether W is finite at any individual point. Square-integrability of the kernel is the exact hypothesis, and the cap $W \le 1$ was always a sufficient condition impersonating a necessary one. ☐

The second criterion controls $\mathbb{W}$ by a different feature of the kernel — not its total L^2 energy but the uniform size of its row sums, the degree function. It is the natural condition to try when one wants a bound on *every* L^r at once, not just on the Hilbert space, and it is the one a probabilist reaches for first because the degree function is the object the graph model already hands us.

Lemma 27.7 (Schur test)**.** *Suppose the degree function $d_W(x) = \int_0^1 |W(x,y)|\,\mathrm{d}y$ satisfies $\|d_W\|_{L^\infty} < \infty$. Then $\mathbb{W}$ is bounded on $L^r[0,1]$ for every $1 \le r \le \infty$, with*

$$\|\mathbb{W}\|_{L^r \to L^r} \;\le\; \|d_W\|_{L^\infty}\,. \tag{27.7}$$

In particular $\|\mathbb{W}\|_{L^2 \to L^2} \le \|d_W\|_{L^\infty}$.

Proof. This is the symmetric Schur test, and the whole content is a single application of Cauchy–Schwarz with the right choice of measure. Fix $f \in L^2[0,1]$ and a type x, and write the absolutely-continuous measure $\mathrm{d}\mu_x(y) = |W(x,y)|\,\mathrm{d}y$, whose total mass is $\mu_x([0,1]) = d_W(x)$. Split the integrand as $|W(x,y)|\,|f(y)| = |W|^{1/2} \cdot |W|^{1/2}\,|f|$ and apply Cauchy–Schwarz to the two factors:

$$\begin{aligned} |(\mathbb{W}f)(x)|^2 &\le \Big(\int |W(x,y)|\,|f(y)|\,\mathrm{d}y\Big)^2 = \Big(\int |W|^{1/2} \cdot |W|^{1/2}\,|f|\,\mathrm{d}y\Big)^2 \\ &\le \Big(\int |W(x,y)|\,\mathrm{d}y\Big)\Big(\int |W(x,y)|\,|f(y)|^2\,\mathrm{d}y\Big) \le \|d_W\|_{L^\infty} \int |W(x,y)|\,|f(y)|^2\,\mathrm{d}y, \end{aligned} \tag{27.8}$$

where the first factor is $d_W(x) \le \|d_W\|_{L^\infty}$. The point of splitting the weight evenly is that it converts one power of W into the harmless normalizing factor d_W and leaves the other paired with $|f|^2$, ready to be integrated. Now integrate (27.8) over x and exchange the order of integration in the surviving double integral:

$$\|\mathbb{W}f\|_{L^2}^2 \le \|d_W\|_{L^\infty} \int_[0,1]\,|f(y)|^2 \Big(\int_[0,1]\,|W(x,y)|\,\mathrm{d}x\Big)\,\mathrm{d}y \le \|d_W\|_{L^\infty}^2\;\|f\|_{L^2}^2\,,$$

the inner x-integral being the *column* sum $\int |W(x,y)|\,\mathrm{d}x$, which equals the row sum $d_W(y) \le \|d_W\|_{L^\infty}$ by symmetry of W — this is the one place symmetry is used. Taking square roots gives $\|\mathbb{W}\|_{L^2 \to L^2} \le \|d_W\|_{L^\infty}$. The same two-sided control of row and column sums is exactly the hypothesis of the general Schur test, which yields the stated $L^r \to L^r$ bound for every r by interpolating between the trivial L^1 and L^∞ bounds (or directly by Hölder with the same even split of $|W|$); the Hilbert-space case $r = 2$ is the one the game will use, since the population mean lives in L^2. ☐

We now have two certificates of boundedness controlled by two genuinely different quantities — the L^2 norm of the kernel and the L^∞ norm of its degree function — and, crucially, *neither* requires W to be bounded. The next question is how they compare: do they certify the same kernels, or does one reach strictly further than the other? The cleanest test case is the rank-one weight $f \otimes f$ that the previous section forced upon us, because there both norms reduce to explicit integrals of f and the operator can be diagonalized by hand. It adjudicates the comparison completely, and in doing so locates the exact rung $p = 2$ below which both certificates expire.

Proposition 27.8 (The boundedness dichotomy)**.** *Let W be a symmetric measurable kernel.*

1. *If $W \in L^2([0,1]^2)$, then $\mathbb{W}$ is bounded (indeed compact, self-adjoint) on $L^2[0,1]$, with $\|\mathbb{W}\| \leq \|W\|_{L^2}$.*
2. *If $d_W \in L^\infty[0,1]$, then $\mathbb{W}$ is bounded on $L^2[0,1]$ with $\|\mathbb{W}\| \leq \|d_W\|_{L^\infty}$.*
3. *For the rank-one kernel $W = f \otimes f$ with $f \geq 0$, the leading eigenfunction is $f/\|f\|_{L^2}$ and the operator norm is the leading eigenvalue $\|\mathbb{W}\| = \lambda_1 = \|f\|_{L^2}^2$. Here $W \in L^2([0,1]^2) \iff f \in L^2[0,1]$, while $d_W = \|f\|_{L^1} f \in L^\infty \iff f \in L^\infty$. Thus the Hilbert–Schmidt route (1) strictly dominates the Schur route (2): it certifies boundedness for all $f \in L^2 \setminus L^\infty$, i.e. for unbounded but square-integrable weights.*
4. *If $W \notin L^2([0,1]^2)$* and *$d_W \notin L^\infty$, both sufficient conditions fail and $\mathbb{W}$ may be genuinely unbounded on $L^2[0,1]$. For the rank-one kernel this is the regime $f \in L^1 \setminus L^2$, where $\|\mathbb{W}\| = \|f\|_{L^2}^2 = +\infty$.*

Proof. Parts (1) and (2) merely restate lemmas 27.6 and 27.7 in the language of the proposition; the work is in parts (3) and (4).

Part (3): the rank-one operator, diagonalized. For $W = f \otimes f$ the operator acts by

$$(\mathbb{W}g)(x) = \int_0^1 f(x)f(y)g(y)\,\mathrm{d}y = f(x)\langle f, g\rangle,$$

so $\mathbb{W}g = \langle f, g\rangle\, f$: every output is a multiple of the single function f, and the range is the line span$\{f\}$. An operator with one-dimensional range has at most one nonzero eigenvalue, and we read it off by feeding it its own range vector: $\mathbb{W}f = \langle f, f\rangle\, f = \|f\|_{L^2}^2\, f$. Hence f is an eigenfunction with eigenvalue $\|f\|_{L^2}^2$, and normalizing it gives the unit leading eigenfunction $f/\|f\|_{L^2}$. For a self-adjoint operator the operator norm equals the largest eigenvalue in modulus (theorem 18.2); a self-adjoint rank-one operator has exactly that one nonzero eigenvalue, so

$$\|\mathbb{W}\| = \lambda_1 = \|f\|_{L^2}^2 = \int_0^1 f(x)^2\,\mathrm{d}x.$$

This is the computation the chapter will lean on hardest: the entire coupling strength of a rank-one kernel is the single number $\|f\|_{L^2}^2$, the susceptibility of the previous section. The two integrability equivalences are now immediate. The Hilbert–Schmidt norm is $\|f \otimes f\|_{L^2([0,1]^2)}^2 = \int_{[0,1]^2} f(x)^2 f(y)^2 = \left(\int f^2\right)^2 = \|f\|_{L^2}^4$, which is finite iff $f \in L^2$; so route (1) applies exactly when $f \in L^2$, and it then returns $\|\mathbb{W}\| \leq \|W\|_{L^2} = \|f\|_{L^2}^2$, matching the exact value above with equality. The degree function is $d_W(x) = \int_0^1 f(x)f(y)\,\mathrm{d}y = \|f\|_{L^1}\, f(x)$, a fixed multiple of f itself; so $d_W \in L^\infty$ iff $f \in L^\infty$, and route (2) applies only when f is *bounded.* The comparison is therefore decisive and strict: any weight that is unbounded but square-integrable — any $f \in L^2 \setminus L^\infty$ — is certified by route (1) and missed entirely by route (2). The class $L^2 \setminus L^\infty$ is nonempty: the concrete witness is the very weight we care about, $f(x) = x^{-\nu}$ with $0 < \nu < \frac{1}{2}$, for which $\int_0^1 x^{-2\nu}\,\mathrm{d}x = \frac{1}{1-2\nu} < \infty$ (since $2\nu < 1$) so $f \in L^2$, yet $f(x) \to \infty$ as $x \to 0$ so $f \notin L^\infty$. The Hilbert–Schmidt route strictly dominates the Schur route, and it is the route we keep.

Part (4): where both fail and the operator is genuinely unbounded. If $W \notin L^2([0,1]^2)$ and $d_W \notin L^\infty$, then both certificates are silent — and on the rank-one family the operator is not merely *uncertified* but provably unbounded. The dividing case is $f \in L^1 \setminus L^2$: the weight is integrable, so the kernel $f \otimes f$ and its degree function $d_W = \|f\|_{L^1} f$ are well-defined finite objects, yet $\|f\|_{L^2}^2 = \int_0^1 f^2 = +\infty$, so the leading eigenvalue computed in part (3) is infinite: $\|\mathbb{W}\| = \|f\|_{L^2}^2 = +\infty$. The explicit witness is again the same singular weight, now pushed past the threshold to $f(x) = x^{-\nu}$ with $\frac{1}{2} < \nu < 1$, for which $\int_0^1 x^{-\nu}\,\mathrm{d}x = \frac{1}{1-\nu} < \infty$ (so $f \in L^1$ and the kernel is a legitimate, finite-row-sum object) while $\int_0^1 x^{-2\nu}\,\mathrm{d}x = +\infty$ (since now $2\nu > 1$), so $f \notin L^2$ and the operator norm diverges. One sees the unboundedness concretely by feeding $\mathbb{W}$ normalized bumps concentrating at the singularity: testing against $g_\epsilon = \mathbf{1}_{[0,\epsilon]}/\sqrt{\epsilon}$ gives $\|\mathbb{W}g_\epsilon\|_{L^2} = \|f\|_{L^2}\,|\langle f, g_\epsilon\rangle| \to \infty$ as $\epsilon \to 0$, because $\langle f, g_\epsilon\rangle = \epsilon^{-1/2}\int_0^\epsilon x^{-\nu}\,\mathrm{d}x = \frac{\epsilon^{1/2-\nu}}{1-\nu} \to \infty$ when $\nu > \frac{1}{2}$. There is no bounded operator hiding behind the divergent norm; the coupling is genuinely infinite. This is the failure the next-but-one subsection takes up in earnest. $\square$

This proposition is the structural heart of the chapter, and the slogan it certifies is worth stating without hedging: the bounded-graphon theory invoked $W \le 1$, hence $\|\mathbb{W}\| \le \|W\|_\infty \le 1$, only as a crude way of asserting that $\mathbb{W}$ is a *bounded operator at all — the real content was always the boundedness, never the numerical bound.* Part (1) says that this content survives all the way down to $p = 2$: every square-integrable kernel, however violently it blows up pointwise, still induces a bounded — indeed compact, self-adjoint — operator, with norm controlled by $\|W\|_{L^2}$ in place of $\|W\|_\infty$. Part (3) pins the boundary to a single computable scalar, the susceptibility $\|f\|_{L^2}^2$, and part (4) shows that boundary is sharp: cross it and the operator norm is literally $+\infty$.

It is exactly here that the three readings of the chapter's one line begin to coincide. The divide that part (3) draws — $f \in L^2$ versus $f \in L^1 \setminus L^2$ — is, by the integrability arithmetic flagged at the end of section 27.1, the divide $\nu < \frac{1}{2}$ versus $\nu \in (\frac{1}{2}, 1)$; and through the map $\gamma = 1 + \frac{1}{\nu}$ of corollary 27.3 that is the divide $\gamma > 3$ versus $\gamma \in (2,3)$. So the square-integrability boundary $p = 2$ of the operator theory, the divergence of the susceptibility $\|f\|_{L^2}^2$, and the degree-tail exponent $\gamma = 3$ are one threshold seen from three sides — and the empirically dominant window $\gamma \in (2,3)$ sits squarely in the broken regime of part (4), where the coupling operator is unbounded. That confluence is the destination section 27.4 will reach; for now part (1) is the deliverable we need next. Because $\mathbb{W}$ remains bounded, compact, and self-adjoint with $\|\mathbb{W}\| \le \|W\|_{L^2}$ throughout the entire L^2 range, the one estimate the Ch. 24 well-posedness theory ever asked of the kernel is back in hand — and, as the next subsection shows, with that single estimate restored the contraction survives verbatim. We now cash this out for the mean field game.

27.2.3 Revised existence and uniqueness

The previous subsection handed the mean field game precisely the certificate it has been waiting for: throughout the entire L^2 range the coupling operator $\mathbb{W}$ remains bounded, compact, and self-adjoint, with its operator norm controlled by $\|\mathbb{W}\| \le \|W\|_{L^2}$ in place of the now-discarded $\|W\|_\infty$ (lemma 27.6, proposition 27.8(1)). The claim this whole section has been steering toward is that this single inequality is the *only* thing the well-posedness theory of chapter 24 ever asked of the kernel. To cash it out we do two things in order: we recall, with care, exactly how the kernel enters the Ch. 24 fixed-point argument — and check that it enters through one scalar and one scalar only — and we then watch the existence-and-uniqueness theorem survive the substitution $\|W\|_\infty \rightsquigarrow \|W\|_{L^2}$ without a further line of work.

Recall the architecture of chapter 24. An equilibrium of the graphon mean field game of chapter 22 is a field of population means $\bar{m} = (\bar{m}^x)_{x\in[0,1]}$ that reproduces itself under *freeze–solve–recollect*: one freezes the aggregate each type feels, solves every type's optimal-control problem against that frozen aggregate, recollects the resulting population means, and demands that the output field equal the input. This loop is exactly the graphon best-response map S (the S_W of definition 24.4), and an equilibrium is precisely a fixed point $\mathsf{S}(\bar{m}) = \bar{m}$. The structural fact that makes the whole theory readable through a single number is the factorisation eq. (24.4), $\mathsf{S} = \mathcal{R}\circ\mathcal{A}_W$, which cleaves the loop into the step that touches the kernel and the step that touches the population. The *aggregation* step $\mathcal{A}_W$ is the one — and the only — place the kernel acts: it takes the field of means and forms, for each type, the aggregate

$$z(x) \;=\; (\mathbb{W}\bar{m})(x) \;=\; \int_0^1 W(x,y)\,\bar{m}(y)\,\mathrm{d}y,$$

the population mean that type x actually experiences through its network neighbourhood. The *per-type response* step $\mathcal{R}$ then solves, for each type independently, the homogeneous mean field control problem driven by the scalar aggregate $z(x)$ — the HJB equation backward in time coupled to the Fokker–Planck equation forward in time, exactly the per-type problem of chapter 10 — and reads off its population mean. Crucially $\mathcal{R}$ never sees the kernel: it sees only the already-aggregated field z. The kernel enters the equilibrium problem through the linear map $\mathbb{W}$ at the aggregation step, and nowhere else.

This is what collapses the entire well-posedness question onto the operator norm. Run the contraction in the L^2-in-type metric d_2 of chapter 23 — the natural metric here, since the population means $\bar{m}_t(\cdot)$ live in the Hilbert space $L^2[0,1]$ on which $\mathbb{W}$ is self-adjoint — and the two factors contribute multiplicatively to the Lipschitz constant of S, for entirely disjoint reasons. The aggregation $\mathcal{A}_W$ is *linear*, $\mathcal{A}_W(\bar{m}) - \mathcal{A}_W(\bar{m}') = \mathbb{W}(\bar{m}-\bar{m}')$, so its Lipschitz constant in d_2 is by definition the $L^2\to L^2$ operator norm $\|\mathbb{W}\|$ — the sole factor that depends on the kernel. The per-type response $\mathcal{R}$ is Lipschitz with a constant C_0 that is *independent of the kernel*: it is assembled entirely from the per-type stability estimates of chapter 10 — the HJB stability lemma 10.10 and the Fokker–Planck stability lemma 10.11, whose constants come from the horizon T, the ellipticity (A-Nondeg), the Lipschitz and convexity data (A-Lip), (A-Conv) of the per-type problem, and the Lipschitz modulus $\mathrm{Lip}(F_0, G_0)$ of the couplings in their aggregate argument. On the LQG thread these are the per-type Riccati stability constants, and C_0 is literally a Riccati stability constant. None of them involves W, because the kernel was already spent at the aggregation step. Composing the two factors,

$$d_2(\mathsf{S}(\bar{m}),\mathsf{S}(\bar{m}')) \;\le\; C_0\;\|\mathbb{W}\|\;\,d_2(\bar{m},\bar{m}'),$$

so S is a contraction — and the Banach fixed-point theorem then delivers a *unique* equilibrium depending continuously on the data — precisely when $C_0\,\|\mathbb{W}\| < 1$, that is when

$$\|\mathbb{W}\| \;<\; \kappa \;:=\; \tfrac{1}{C_0}. \tag{27.9}$$

Equation (27.9) is the standing coupling-smallness assumption (A-Coupling) in its native L^2 form. The threshold $\kappa = 1/C_0$ is the reciprocal of the kernel-independent per-type Lipschitz constant, and the operator norm it constrains is the canonical $L^2 \to L^2$ norm $\|\mathbb{W}\| = \lambda_1$, the leading eigenvalue / spectral radius of the self-adjoint $\mathbb{W}$ (remark 24.19); the worked computation worked computation 27.19 will make C_0, and hence κ, completely explicit on the LQG thread.

Now the conclusion is forced. The bounded theory of chapter 24 verified (A-Coupling) by the crude chain $\|\mathbb{W}\| \le \|W\|_\infty \le 1$ — but $\|W\|_\infty$ appears there *only* as an upper bound for $\|\mathbb{W}\|$, never on its own account, and lemma 27.6 furnishes a different, equally serviceable upper bound that asks for nothing beyond square-integrability. Replacing the one estimate $\|\mathbb{W}\| \le \|W\|_\infty$ by the Hilbert–Schmidt bound (27.6), $\|\mathbb{W}\| \le \|W\|_{L^2}$, leaves the contraction argument otherwise untouched and yields the following extension at no cost.

Theorem 27.9 (Well-posedness for L^2 graphons)**.** *Let the per-type control problem satisfy* (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), *and let the coupling kernel W be a symmetric L^2 graphon (definition 27.5), in place of the bounded kernel of* (A-Reg-W). *Suppose the coupling-smallness assumption* (A-Coupling) *holds in the form*

$$\|\mathbb{W}\|_{L^2 \to L^2} \;\le\; \|W\|_{L^2([0,1]^2)} \;<\; \kappa, \tag{27.10}$$

with κ the same threshold constant as in the bounded theory of chapter 24. Then the graphon mean field game of chapter 22 has a unique equilibrium $(u^x, m^x)_{x\in[0,1]}$, depending continuously on the data, and the type-indexed solution map is identical to the bounded case with every occurrence of $\|W\|_\infty$ replaced by $\|W\|_{L^2}$.

Proof sketch. The full fixed-point argument is the one carried out in chapter 24; only one line of it touches the kernel, and that line is the only thing we change. Recast through the factorisation eq. (24.4), $\mathsf{S} = \mathcal{R} \circ \mathcal{A}_W$, the best-response map on the space of admissible mean fields $\bar{m} \in L^2([0,1]; C[0,T])$ is Lipschitz, in the d_2 metric of chapter 23, with constant $L_{\mathsf{S}} = C_0\,\|\mathbb{W}\|$. The two factors are exactly the ones isolated above: the linear aggregation $\mathcal{A}_W = \mathbb{W}$ contributes the operator norm $\|\mathbb{W}\|$, and the per-type response $\mathcal{R}$ contributes $C_0 = L_{\mathcal{R}}(T)\,\mathrm{Lip}(F_0, G_0)$, a constant assembled from the horizon, the ellipticity (A-Nondeg), and the Lipschitz/convexity data (A-Lip), (A-Conv) through the per-type stability estimates lemmas 10.10 and 10.11 (the Riccati stability constants on the LQG thread), and *independent of the kernel.* The equilibrium is the unique fixed point whenever $L_{\mathsf{S}} < 1$, i.e. $\|\mathbb{W}\| < \kappa := 1/C_0$, as in (27.9).

The single modification is now visible. In the bounded theory the kernel-bearing factor $\|\mathbb{W}\|$ was bounded above by $\|W\|_\infty$; here lemma 27.6 supplies instead $\|\mathbb{W}\| \le \|W\|_{L^2}$ (27.6), valid for every $W \in L^2([0,1]^2)$. The operator $\mathbb{W}$ enters the estimate purely as a bounded self-adjoint operator on the Hilbert space $L^2[0,1]$, and by lemma 27.6, proposition 27.8(1) it is exactly that under $W \in L^2$. Substituting $\|W\|_{L^2}$ for $\|W\|_\infty$ at this one place leaves the inequality $L_{\mathsf{S}} = C_0\,\|\mathbb{W}\| < 1$ — and hence the contraction — intact.

Every other ingredient of chapter 24 is word-for-word unchanged, for the simple reason that none of them ever read $\|W\|_\infty$ either. The construction of a convex, S-invariant, compact set on which Schauder delivers existence rests on the uniform-in-type bounds and equicontinuity of the per-type responses — properties of $\mathcal{R}$, not of the kernel. The continuity of S in the field, used both for Schauder and for continuous dependence on the data, again flows entirely through the per-type HJB/FP stability lemmas 10.10 and 10.11 under (A-Conv) and (A-Nondeg). And the verification that a fixed point of S is a genuine equilibrium of the game is the freeze–solve–recollect identity built into definition 24.4, a tautology that never mentions the size of W. Boundedness $W \le 1$ was used at exactly one point in the entire development — to assert that $\mathbb{W}$ is a bounded operator at all — and lemma 27.6 discharges that obligation from $W \in L^2$ alone. That no boundedness of W survives anywhere in the argument is not a bookkeeping accident; it is the whole content of the theorem. $\square$

Remark 27.10 (An L^2 graphon is still a dense graph)**.** Theorem 27.9 solves the *unbounded-kernel* problem, not the *sparsity* problem (cf. remark 27.4): it widens the class of admissible *limit objects* without touching the *sampling law*, and at the dense sampling rate $\rho_N \equiv 1$ that law still produces a dense graph. The mechanism is the connection-probability floor of remark 27.4, made quantitative for a single L^2 kernel. Sample W by $G(N, 1, W)$ with the truncation (27.1), so a pair (U_i, U_j) is joined with probability $\min\{1, W(U_i, U_j)\}$. Because $W \ge 0$ is not a.e. zero, some superlevel set carries positive measure: there is a threshold $\delta > 0$ — one may take $\delta = \frac{1}{2}$ when W reaches $\frac{1}{2}$ on a positive-measure set, and otherwise any smaller δ for which $|\{W \ge \delta\}| > 0$, which exists since $\int_{[0,1]^2} W > 0$ forces $|\{W \ge \delta\}| > 0$ for all small enough δ — and on $S := \{W \ge \delta\}$ the truncated probability obeys $\min\{1, W\} \ge \min\{1, \delta\} = \delta > 0$. The expected edge count is then bounded below directly: with U_1, U_2 the types of a generic pair,

$$\mathbb{E}\big[e\big(G(N, 1, W)\big)\big] = \binom{N}{2}\,\mathbb{E}[\min\{1, W(U_1, U_2)\}] \;\ge\; \binom{N}{2}\,\delta\;|S| \;=\; \Theta(N^2),$$

so the graph carries quadratically many edges almost surely — it is dense in the exact sense (S1) was meant to forbid. (The same floor drives the per-vertex picture: a vertex whose type lands in the positive-fraction set over which $W \ge \delta$ accumulates $\Theta(N)$ neighbours, the mechanism of proposition 19.14.) So theorem 27.9 is exactly the right tool for a network that is *dense* but carries a *heavy-tailed weight / centrality* structure — an unbounded but square-integrable kernel, whose hubs have large but non-vanishing relative degree — and it is the wrong tool for a genuinely sparse network. It buys axis one of remark 27.4, and only axis one.

With theorem 27.9 in hand the unbounded-kernel axis is closed on its good side, and remark 27.10 marks exactly how far the good news reaches. Square-integrability, not boundedness, is the true hypothesis of the dense well-posedness theory — but square-integrability is also its floor. Everything in the argument we have just run — the contraction, the threshold $\kappa = 1/C_0$ of (27.9), the very meaning of (A-Coupling) — rests on $\mathbb{W}$ being a bounded operator on $L^2[0, 1]$, and that is the property proposition 27.8 guarantees only down to $p = 2$. The instant the kernel slips below square-integrability the leading eigenvalue $\|\mathbb{W}\| = \lambda_1$ ceases to be finite, the Lipschitz constant $C_0\,\|\mathbb{W}\|$ blows up, and there is no contraction left to run. This is not an exotic edge case: by corollary 27.3 it is precisely the heavy-degree window $\gamma \in (2, 3)$ where real scale-free networks live. The next subsection takes that failure head-on — asking whether either certificate of boundedness, Hilbert–Schmidt or Schur, can be salvaged once p drops below 2, and finding that with heavy degrees the two fail together.

27.2.4 Below $p = 2$: where the operator theory dies

The previous subsection closed the unbounded-kernel axis on its good side and marked, with care, exactly where that side stops: every estimate in the Ch. 24 well-posedness theory rests on $\mathbb{W}$ being a *bounded* operator on $L^2[0,1]$, and proposition 27.8 certifies that boundedness only down to $p = 2$. We now step across that line on purpose and report, without softening, what becomes of the theory in the range $1 \le p < 2$. This is not an academic edge case. By corollary 27.3 the kernel exponent $p < 2$ is exactly the degree-tail window $\gamma \in (2, 3)$ — the heavy-tailed, hub-dominated regime where the social and technological networks we most want to model actually live — so the regime we are about to find broken is the empirically central one. The honest accounting of that breakdown, and the sharp marking of the line between what remains a theorem and what becomes a conjecture, is the business of this subsection.

Begin with the two certificates of boundedness and watch both expire together. For $1 \le p < 2$ the kernel is no longer square-integrable, $W \notin L^2([0,1]^2)$, so the Hilbert–Schmidt route of lemma 27.6 — the one that carried all the good news above — is simply silent: it has no L^2 norm of the kernel to hand the contraction. One might hope the second certificate rescues the situation, since it asks for something altogether different. The Schur test of lemma 27.7 bounds $\mathbb{W}$ by $\|d_W\|_{L^\infty}$, the uniform size of the row sums, and never mentions L^2 at all. But the very feature that defines the heavy regime is that the degrees are *not* uniformly bounded: a scale-free tail *is* the statement $d_W \notin L^\infty$, hubs of ever-larger degree as the type approaches the singularity. So the Schur route falls silent for precisely the same data that silences Hilbert–Schmidt. This simultaneous double failure is exactly proposition 27.8(4): with $W \notin L^2$ *and* $d_W \notin L^\infty$, both sufficient conditions fall away at once, and on the rank-one family the operator is then not merely uncertified but *provably* unbounded, $\|\mathbb{W}\| = \|f\|_{L^2}^2 = +\infty$, as the normalized-bump test in the proof of proposition 27.8(4) showed by hand.

Now trace what this does to the fixed point that theorem 27.9 relied on. The entire force of that theorem was the factorisation of the best-response Lipschitz constant into the product $C_0\,\|\mathbb{W}\|$ — a kernel-independent per-type factor C_0 multiplying the one kernel-dependent scalar $\|\mathbb{W}\|$ — with the contraction, and hence existence, uniqueness, and continuous dependence, secured the instant $\|\mathbb{W}\| < \kappa = 1/C_0$ (eq. (27.9)). Below $p = 2$ that scalar is $\|\mathbb{W}\| = +\infty$. The contraction estimate

$$d_2(\mathsf{S}(\bar{m}), \mathsf{S}(\bar{m}')) \;\le\; C_0\,\|\mathbb{W}\|\; d_2(\bar{m}, \bar{m}')$$

degenerates into the vacuous assertion "$\le +\infty$," carrying no information; and the coupling-smallness condition (A-Coupling), $\|\mathbb{W}\| < \kappa$, can no longer be arranged by *any* tuning of the data, because its left-hand side is not merely large but literally infinite. There is no horizon short enough, no convexity strong enough, no kernel rescaling small enough to bring it under the threshold κ.

It is worth dwelling on the character of this failure, because it is qualitatively unlike the way a proof usually breaks. The ordinary situation is that a Lipschitz constant comes out finite but too large to close a contraction, and one then hunts for a sharper estimate, a cleverer metric, or a smallness hypothesis that tames it — the underlying result may well survive a better argument. Here there is nothing to tame. The constant is not large; it is infinite, and proposition 27.8(4) exhibited that infinitude operationally rather than as an artifact of a lossy bound: feeding $\mathbb{W}$ the normalized bumps g_ϵ that concentrate at the singularity sends $\|\mathbb{W}g_\epsilon\|_{L^2} \to \infty$ as $\epsilon \to 0$. No bounded coupling operator hides behind a wasteful estimate; the very object the contraction argument manipulates — a finite operator norm $\|\mathbb{W}\|$ — does not exist, and Banach's theorem (theorem 2.36) has nothing on which to act. The breakdown is a genuine change in the mathematics, not a deficiency of our technique, and saying so plainly is the first half of the honest accounting this subsection owes.

What, then, survives the loss of boundedness, and how much? Something does, and being exact about it is what the conditional existence statement below is built on. The aggregate that a single type experiences,

$$z(x) \;=\; (\mathbb{W}\bar{m})(x) \;=\; \int_0^1 W(x,y)\,\bar{m}(y)\,\mathrm{d}y, \tag{27.11}$$

is, for each *fixed* type x, a perfectly finite number even when $\mathbb{W}$ is unbounded as a map $L^2 \to L^2$ — provided only that the mean field is bounded, $\bar{m} \in L^\infty[0,1]$, and that the kernel has a finite row sum at x, i.e. $d_W(x) < \infty$. The bound is immediate from pulling the supremum of the mean field out of the integral:

$$|z(x)| = \Big|\int_0^1 W(x,y)\bar{m}(y)\,\mathrm{d}y\Big| \le \|\bar{m}\|_{L^\infty} \int_0^1 |W(x,y)|\,\mathrm{d}y = \|\bar{m}\|_{L^\infty}\; d_W(x),$$

finite at every type of finite degree. The distinction the dichotomy is forcing on us is now sharp and worth stating in one line: below $p = 2$ the operator $\mathbb{W}$ has lost its *operator-norm* control — its $L^2 \to L^2$ Lipschitz behaviour — without losing its *pointwise* action on bounded fields. Each type still feels a finite, well-defined aggregate $z(x)$; one can still freeze that aggregate, solve that type's optimal-control problem against it, and recollect the resulting mean, so the freeze–solve–recollect loop of definition 24.4 still *defines* a best-response map S on a bona fide function space. What the map has shed is not its definition but the contraction property, and — as the next paragraphs make precise — even the weaker self-map property that a softer existence theorem would demand. The honest status is the following.

Proposition 27.11 (Conditional weak existence for L^1 couplings)**.** *Let $W \in L^1([0,1]^2)$ be symmetric with $d_W \in L^q[0,1]$ for some $q > 1$, and let the per-type problem satisfy* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv)*, with the additional structural hypothesis that the per-type optimal mean response $x \mapsto \bar{m}_t^x$ is, uniformly in t, a bounded affine functional of its target path with operator coefficient* independent of W*. Then a Schauder/compactness argument produces* at least one *equilibrium mean field $\bar{m} \in L^{q'}[0,1]$ (q' the conjugate exponent), provided the fixed-point map is shown to preserve a convex compact set; uniqueness is* not *asserted and generally fails in the absence of operator-norm smallness or a Lasry–Lions monotone coupling* (A-Mon)*.*

Status. *The boundedness and pointwise solvability statements are theorems (lemma 27.7, Hölder); the existence-by-compactness step is, at the level of generality stated, a* conjecture*: it requires a tightness / a-priori bound that has been established only in special cases (potential/monotone couplings, finite-rank kernels, linear-quadratic data). We mark it as such here and in chapter 31.*

Discussion of what is and is not proved. The proposition has two halves of very different epistemic standing, and the discipline of this discussion is to keep them apart: the boundedness and pointwise-solvability claims are genuine theorems, with one-line proofs; the existence-by-compactness claim is, at this generality, a conjecture, and we will be precise about the single estimate it is missing.

The bound is Hölder, made explicit. The narration above bounded $z = \mathbb{W}\bar{m}$ for a bounded mean field by sheer modulus, $|z(x)| \leq \|\bar{m}\|_{L^\infty} d_W(x)$. For the fixed-point argument one wants a little more: a bound that lets the mean field live in a Lebesgue space matched to the integrability $d_W \in L^q$ of the degrees, rather than in L^∞. This is the Schur split of lemma 27.7 run with unequal exponents. Fix a type x and factor the kernel evenly across the conjugate pair $\frac{1}{q} + \frac{1}{q'} = 1$, writing $|W| = |W|^{1/q} \cdot |W|^{1/q'}$ and applying Hölder in the second variable to the two factors $|W|^{1/q}$ and $|W|^{1/q'} |\bar{m}|$:

$$|z(x)| \leq \int_0^1 |W|^{1/q} \cdot |W|^{1/q'} |\bar{m}| \, \mathrm{d}y \leq \Big(\int_0^1 |W(x,y)| \, \mathrm{d}y \Big)^{1/q} \Big(\int_0^1 |W(x,y)| \, |\bar{m}(y)|^{q'} \, \mathrm{d}y \Big)^{1/q'}.$$

The first factor is $d_W(x)^{1/q}$, exactly as recorded in the proposition; the second pairs the surviving half-power of the kernel with the input field. This is the device of the Schur test, except that splitting unevenly — $1/q$ against $1/q'$ rather than $1/2$ against $1/2$ — is what lets the degree function be merely L^q rather than L^∞. In the specialization $\bar{m} \in L^\infty$ the second factor collapses to $\|\bar{m}\|_{L^\infty} \, d_W(x)^{1/q'}$, and combining the two powers of $d_W(x)$ gives back $|z(x)| \leq \|\bar{m}\|_{L^\infty} d_W(x)$, whence $\|\mathbb{W}\bar{m}\|_{L^q} \leq \|d_W\|_{L^q} \, \|\bar{m}\|_{L^\infty}$ on integrating the q-th power in x. This is a genuine, finite bound under $d_W \in L^q$ alone — no boundedness of W, no square-integrability — and it is the precise sense in which the best-response map is still *well-defined* below $p = 2$. So far, theorems.

What is not available is the self-map-of-a-compact-set property. Having lost the contraction we would naturally fall back on the standard existence route for mean field games, the Schauder fixed-point theorem (proved as theorem 2.38; its contraction sibling that we just lost is theorem 2.36, and the template for applying Schauder to an MFG best-response map is the existence argument of chapter 10). Schauder asks for four things and gives existence in return, trading away uniqueness: a nonempty closed *convex* set $\mathcal{K} \subset L^{q'}$, the *invariance* $\mathsf{S}(\mathcal{K}) \subseteq \mathcal{K}$, the *continuity* of S on $\mathcal{K}$, and the *precompactness* of the image $\mathsf{S}(\mathcal{K})$. Three of the four are in hand below $p = 2$, and they are in hand for a reason worth naming: they are properties of the per-type response $\mathcal{R}$, not of the kernel. Convexity is free (mean fields form a convex set); continuity of S and the compactifying smoothness of its image flow, exactly as in the dense existence proof, from the per-type HJB/Fokker–Planck stability of chapter 10, which produces uniformly bounded, equicontinuous responses to a bounded target and never reads the size of W. The one ingredient that breaks is *invariance*, and it breaks for the same reason the contraction did. To run Schauder one must exhibit a bounded set of mean fields that S cannot escape, and that requires an a-priori bound: a ceiling, valid in advance, on the size of the image (and hence of any fixed point). In the bounded and L^2 theories the ceiling was furnished by $\|\mathbb{W}\| < \infty$ — the aggregate $z = \mathbb{W}\bar{m}$ was at most $\|\mathbb{W}\|$ times the mean, so a ball of fixed radius mapped into a ball of controlled radius. With $\mathbb{W}$ unbounded that estimate is gone, and its failure is concentrated precisely on the heavy types. On the rank-one family the hub type $x \to 0$ feels an aggregate $z(x) = f(x)\langle f, \bar{m} \rangle$ that blows up like $f(x) = x^{-\nu}$ *no matter how small* $\bar{m}$ is in L^2, since a single bounded inner product $\langle f, \bar{m} \rangle$ multiplies the unbounded profile f. No ball in the mean-field space can therefore be invariant: the heavy types push the image out of every fixed ball, and the convex compact $\mathcal{K}$ that Schauder needs cannot be built. The obstruction is thus not a missing continuity or a missing compactness — those survive — but a missing a-priori bound, and it sits on exactly the hub types whose divergent degree was the whole point of the construction.

Where the bound can be recovered, the argument closes. The obstruction is special-structure removable, and naming the two cases where it lifts is what makes the theorem/conjecture boundary sharp. First, under a Lasry–Lions monotone coupling (A-Mon), existence does not pass through an operator-norm bound at all: monotonicity supplies its own a-priori control through the Lasry–Lions energy estimate, which bounds the equilibrium mean field directly from the structure of the cost rather than from the size of $\mathbb{W}$, furnishing the ceiling Schauder was missing and closing the argument even with $\mathbb{W}$ unbounded. Second, in the finite-rank or linear-quadratic case the infinite-dimensional fixed point collapses to a finite-dimensional one: a rank-one kernel couples the population only through the single scalar order parameter $\langle f, \bar{m} \rangle$, the heavy types are integrated out into one susceptibility-weighted number, and the self-consistency condition becomes a scalar (on the LQG thread, algebraic) equation whose solvability one checks by hand. That is the explicit reduction carried out in worked computation 27.19, which converts the qualitative "$\|\mathbb{W}\| = +\infty$" of this subsection into a closed-form condition on a finite number of moments. Outside these structured cases — a general L^1 kernel with heavy degrees and a non-monotone coupling — no a-priori bound controlling the heavy types is known, and the existence-by-compactness step remains genuinely open. $\square$

Notation watch

The symbol p is doing double duty in this literature, and the whole subsection turns on not conflating its two meanings, so we set them side by side. In "L^p graphon" (definition 27.5) p is the integrability exponent of the *kernel*: a property of the limit object W, measuring how singular the weight is allowed to be. The *degree-tail exponent* γ of (27.4) is an altogether different number: a property of the sampled *graph*, the power governing $\mathbb{P}(\deg = k) \propto k^{-\gamma}$. A priori they have nothing to do with each other; they are linked only after one fixes a specific kernel and runs it through the sparsification model. The link is worth writing out explicitly on the rank-one weight $f(x) = x^{-\nu}$, the family the chapter computes everything on. Three elementary equivalences chain together. *(i)* The kernel sits in L^p iff the weight does, because the rank-one structure factors the integral: $f \otimes f \in L^p([0,1]^2)$ means $\int_{[0,1]^2} (f(x)f(y))^p = (\int_0^1 f^p)^2 < \infty$, i.e. $f \in L^p[0,1]$. *(ii)* The power $f = x^{-\nu}$ sits in L^r iff $\int_0^1 x^{-r\nu}\,dx < \infty$, i.e. iff $r\nu < 1$, i.e. $r < 1/\nu$. *(iii)* The degree-tail map of corollary 27.3 reads $\gamma = 1 + 1/\nu$, so $1/\nu = \gamma - 1$. Composing (i)–(iii),

$$f \otimes f \in L^p([0,1]^2) \iff f \in L^p[0,1] \iff p\nu < 1 \iff p < \tfrac{1}{\nu} = \gamma - 1.$$

Hence kernel integrability p and graph tail γ are tied by the single relation $p < \gamma - 1$, and the threshold of this subsection — square-integrability of the kernel, $p = 2$ — is the line $2 = \gamma - 1$, that is $\gamma = 3$. The operator theory dies for $p < 2$; the degree variance diverges for $\gamma < 3$; under the dictionary above these are one inequality written in two alphabets.

Two roads lead out of the impasse this subsection has mapped, and the chapter takes both. The first is to retreat from full generality to the one structured case in which the obstruction dissolves into algebra. On the rank-one kernel the entire coupling is the single scalar susceptibility $\|f\|_{L^2}^2 = \|\mathbb{W}\|$ (proposition 27.8(3)), and the qualitative verdict "the operator norm is $+\infty$" becomes a quantitative one: a closed-form computation of exactly *when* that scalar diverges, with the threshold $\nu = \frac{1}{2}$ — equivalently $\gamma = 3$, equivalently $p = 2$ — read straight off an explicit integral. That is the worked computation worked computation 27.19, and it is where the breakdown traced here turns from a qualitative "the theory dies" into a formula one can evaluate. The second road accepts that the unit-interval picture has been pushed as far as it will go. The standing verdict over this entire L^p section (remark 27.10) is that even its best case — a square-integrable kernel, the most an unbounded-kernel relaxation can offer — still samples to a *dense* graph: the relaxation bought heavier weights and richer hubs but not one genuinely sparse edge, and the degree tail it can produce is a property of weights, not of subquadratic edge counts. Genuine sparsity was always the *other* axis of remark 27.4, and to reach it one must surrender the single structure every construction in this section has silently retained: the vertex set $[0, 1]$ itself. The next section rebuilds the random graph not on the interval but on a point process over the half-line $\mathbb{R}_+$ — the graphex framework — where subquadratic sparsity is available from first principles, and where we will have to ask afresh what a mean field game even means.

27.3 Graphex and sampling convergence

The L^p section ended on a verdict it could not evade, and that verdict is the door into this one. Pushed to its very best case — a square-integrable kernel, the most an unbounded-kernel relaxation is allowed to offer — the construction of section 27.2 still samples to a *dense* graph (remark 27.10): the weights got heavier and the hubs got richer, but the edge count stayed $\Theta(N^2)$ and not one genuinely sparse edge was bought. The reason is structural and was never going to yield to a sharper estimate. Every model in that section kept the same skeleton as the dense theory of Part IV — the vertex set is the unit interval $[0, 1]$, one independent uniform type per vertex, edges sampled conditionally on the types — and on that skeleton density is forced the moment connection probabilities fail to shrink with N on a macroscopic set of pairs (proposition 19.14). Reweighting the kernel cannot undo this; it operates within the dense scaffold rather than replacing it. Even the target-density device of section 27.1, which *did* reach genuine sparsity, did so only by bolting on an external dial $\rho_N = c/N$ — an ad-hoc N-dependent thinning factor applied uniformly to a dense object, not a sparsity intrinsic to the limit. If we want a limit object that is sparse *in itself*, rather than a dense one we secretly multiply down, we must give up the single structure every construction so far has silently retained: the interval $[0, 1]$ as the set of vertices.

This is the surrender the present section makes, and the object it trades for is the *graphex*. The contrast with the target-density route is the cleanest way to see what changes. There the limit was still a graphon on $[0, 1]$ and sparsity was an instruction imposed on the sampling — "draw the dense graph, then keep a ρ_N fraction of the edges" — so the sparse graph was a thinned shadow of a dense parent and the parent did the explaining. Here there is no parent. The graphex is a kernel on the *half-line* $\mathbb{R}_+^2$ rather than the bounded square $[0, 1]^2$, and the random graph is generated not by sprinkling N uniform types into a fixed unit interval but by laying down a *point process* of types along $\mathbb{R}_+$ — a Poisson process whose points are the latent coordinates of the vertices, with no upper bound and no prescribed number. The unboundedness of the type domain is the whole point: there is always more room further out along $\mathbb{R}_+$ for a new, lightly-connected vertex to appear, and it is exactly those perpetually-arriving low-degree vertices that dilute the edge density below quadratic. The sparsity is now a property the limit object *has*, read off from how the kernel decays at infinity along the half-line, not a correction we apply to a dense template after the fact.

The move from a bounded type space to a point process on $\mathbb{R}_+$ is not an arbitrary trick but the sparse-graph reflection of the same representation theory that legitimised graphons in the first place. Recall the dense story (chapter 15): an infinite random graph whose law is invariant under relabelling vertices — an *exchangeable* array of edge indicators — is, by the Aldous–Hoover representation (the two-dimensional analogue of de Finetti's theorem for exchangeable sequences), generated by attaching an i.i.d. uniform latent coordinate to each vertex and connecting a pair through a single symmetric kernel W on $[0,1]^2$. The graphon is precisely the de Finetti mixing object behind a dense exchangeable graph, and the type variable $x \in [0,1]$ is the exchangeable label that representation hands us (section 17.2). The trouble is that this representation, by construction, produces only dense graphs: i.i.d. uniform latent coordinates and a bounded kernel give every vertex the same $\Theta(N)$ expected degree, so genuine sparsity is forbidden at the level of the representation theorem, not merely at the level of any one model. To get sparse exchangeable graphs one must exchange the de Finetti picture for its point-process cousin — the Caron–Fox / Veitch–Roy / Kallenberg theory of exchangeable random measures — in which the latent coordinates form a Poisson process on $\mathbb{R}_+$ rather than an i.i.d. sample on $[0,1]$, and the mixing object is a kernel on $\mathbb{R}_+^2$: the graphex. The graphex is to a sparse exchangeable graph exactly what the graphon is to a dense one, and the migration from $[0,1]$ to $\mathbb{R}_+$ is the precise analytic cost of demanding sparsity of the representation rather than imposing it on the sample.

Replacing a fixed finite vertex set by an unbounded point process forces a reckoning with what now plays the role of "the size N," and it is worth flagging the answer in advance because the whole construction is organised around two scales that the dense theory kept welded together. In Part IV a single number, N, did two jobs at once: it counted the vertices *and* it set the resolution at which the limiting kernel was sampled. Over a point process on $\mathbb{R}_+$ these two jobs come apart. There is, on one hand, the graphex itself — a fixed limit object, a kernel on $\mathbb{R}_+^2$ carrying no notion of size, exactly as a graphon is a fixed object carrying no N. There is, on the other hand, a *truncation level* $\ell \geq 0$: we only ever look at the part of the point process inside the window $[0, \ell]$, and growing ℓ is what grows the graph. The level ℓ is the genuine analogue of N — it is the knob we turn to march toward the limit — but it is a continuous window on the half-line, not a vertex count, and the number of vertices it exposes is itself random and grows with ℓ. Holding the kernel fixed while sending $\ell \to \infty$ is the sparse counterpart of holding W fixed while sending $N \to \infty$, and keeping these two scales — limit kernel versus truncation level — firmly distinct is what makes the sparsity statements below clean. The next subsection builds the construction around exactly this pair.

What this buys, and what it does not, must be said plainly at the outset so that nothing below reads as an overclaim. The good news is real and comes essentially for free: a graphex with a suitably heavy marginal generates a graph that is *genuinely* sparse in the subquadratic sense, $e = o(N^2)$, with no ρ_N to bolt on, and whose degree distribution carries an honest heavy tail — so desideratum (S3) and the subquadratic form of sparsity are delivered by the object itself, as direct consequences of how the kernel decays along $\mathbb{R}_+$. But the strongest sparsity demand, the strict *linear* edge count (S1) — $e = \Theta(N)$, a bounded average degree — is *not* automatic. Whether a graphex produces $e = \Theta(N)$ or merely $e = o(N^2)$ with a slowly growing average degree depends on a finer property than mere integrability of the kernel: it turns on the precise rate at which the marginal μ_W decays at infinity, an extra marginal-scaling hypothesis beyond what subquadratic sparsity requires. We will be exact about this gap rather than paper over it — the proposition below separates the regime where (S1) holds from the regime where only the weaker $e = o(N^2)$ survives — because conflating "subquadratic" with "linear" is exactly the kind of quiet overclaim this chapter is written to avoid. With the contrast drawn, the representation pedigree in place, and the honest scorecard on the table, we can now build the object itself: a kernel on $\mathbb{R}_+^2$ and the Poisson construction that turns it into a growing sparse graph indexed by the truncation level.

27.3.1 The object and its construction

We come, then, to the object the section has been promising: a limit that is sparse in itself, built from a point process on $\mathbb{R}_+$ rather than N uniform draws on $[0,1]$, and that therefore owes nothing to an external thinning dial ρ_N. The construction has two ingredients, and it pays to know in advance what each is for, because the entire sparsity phenomenon lives in the tension between them. The first ingredient is a *kernel*, just as in the dense theory — but now a kernel on the half-line, $W : \mathbb{R}_+^2 \to [0,1]$, which encodes how strongly two latent types connect. The second is a recipe for turning that kernel into a growing random graph: sprinkle latent vertices along $\mathbb{R}_+$ as a Poisson process and connect each pair through the kernel. The decisive design choice is that the kernel is allowed — indeed required — to *decay* as its type arguments march out along the half-line, so that a vertex landing far out at large u is only weakly connected to everything. Density is bought back exactly here: because there is unlimited room further out for new, faintly-attached vertices, the graph keeps acquiring low-degree vertices faster than it acquires edges, and the edge count falls below quadratic. We make the kernel and its decay condition precise first, then the sampling.

Definition 27.12 (Graphex)**.** A *graphex* is a symmetric measurable function $W : \mathbb{R}_+^2 \to [0,1]$ (the *kernel* component), satisfying the integrability condition

$$\int_{\mathbb{R}_+} \left(1 - e^{-\mu_W(u)}\right) \mathrm{d}u < \infty, \qquad \mu_W(u) := \int_{\mathbb{R}_+} W(u,v)\, \mathrm{d}v, \tag{27.12}$$

together, in the general theory, with a *star* intensity $S : \mathbb{R}_+ \to [0,1]$ and a *dust* rate $I \geq 0$; in this chapter we restrict to the kernel component (the *integrable graphon* case $I = 0$, $S \equiv 0$), which already exhibits all the phenomena we need. We call μ_W the *marginal* and read (27.12) as the requirement that the expected number of *non-isolated* vertices at any finite truncation be finite.

Three features of this definition deserve unpacking before we use it, because each is doing load-bearing work that a quick reading would miss. Start with the *marginal* $\mu_W(u) = \int_{\mathbb{R}_+} W(u,v)\,\mathrm{d}v$. It is the exact half-line analogue of the degree function d_W of the dense theory: it measures how much total connection strength a type-u vertex commands, integrated over all the partners it might find. The crucial new freedom is that $\mu_W(u)$ may be large for small u and must *decay* as $u \to \infty$ — types far out along the half-line are weakly connected — and it is precisely the rate of that decay, not the pointwise size of W, that will govern both the sparsity and the degree tail. The kernel is capped at 1 (it is a probability), so heaviness can never come from W blowing up the way the rank-one weight $f = x^{-\nu}$ did in corollary 27.3; on the half-line it comes instead from the kernel staying appreciable over an unboundedly long range of partners. The marginal repackages that range into a single profile $u \mapsto \mu_W(u)$, and everything below is read off from it.

Next, the integrability condition (27.12). Why $\int_{\mathbb{R}_+}(1 - e^{-\mu})\,\mathrm{d}u$ and not the simpler $\int_{\mathbb{R}_+} \mu = \|W\|_{L^1}$? The answer is that $1 - e^{-\mu_W(u)}$ is exactly the probability that a type-u vertex, sampled into the graph at unit truncation, is *non-isolated* — has at least one edge — as the proof of proposition 27.14 will show by a thinning argument. A graph is a set of vertices that *participate*, and isolated vertices are invisible; so the right finiteness demand is not that the total edge mass be finite but that the expected number of *non-isolated* vertices be finite at each level. The two conditions differ exactly where μ_W is large: there $1 - e^{-\mu} \approx 1$ while μ itself may be huge, so (27.12) is the *weaker* and therefore more permissive demand — it tolerates a marginal that is non-integrable near a singularity, asking only that the region of types that connect to *something* have finite measure. This is the condition that makes the vertex count finite without forcing the kernel to be L^1, and it is what gives the graphex room to be genuinely heavy-tailed.

Finally, a word on what we have thrown away. The general graphex of the Caron–Fox–Veitch–Roy theory carries two further components beyond the kernel: a *star* intensity $S\colon \mathbb{R}_+ \to [0, 1]$, which seeds occasional vertices of enormous (infinite-in-the-limit) degree attached to a continuum of partners, and a *dust* rate $I \geq 0$, a constant background of isolated-edge noise. We set $S \equiv 0$ and $I = 0$ throughout — the *integrable graphon* case — and the reason is honesty about scope, not laziness. The kernel component alone already exhibits every phenomenon this section is built to demonstrate: genuine subquadratic sparsity, a finite vertex count, and a closed-form power-law degree distribution. The star and dust components add modelling flexibility (they are, as the closing caveat will note, exactly what one would reach for to push the degree exponent up into the empirical window) but they also complicate the calculus and contribute nothing to the structural points we need to make. Excising them keeps the construction's logic visible. We will return, in the caveat at the end of the subsection, to precisely what their absence costs.

With the object fixed we turn it into a graph. The construction is indexed by a continuous *truncation level* $\ell \geq 0$ — this ℓ plays the role that N played in the dense theory, but, as flagged when we pried the two scales apart above, it now does *two* jobs the dense N kept welded: it controls both how many vertices appear and how heavily each is connected. We write the level as ℓ, *not* s, to keep it disjoint from the Thread-B mean-coupling strength s that reappears when we put a mean field game on the graphex (see the notation watch in the game subsection below).

Definition 27.13 (Graphex process)**.** Let $\Pi = \{(\alpha_i, u_i)\}_i$ be a Poisson point process on $\mathbb{R}_+ \times \mathbb{R}_+$ of unit intensity ($\mathrm{d}\alpha\ \mathrm{d}u$); its existence and calculus are appendix H. Interpret $u_i \in \mathbb{R}_+$ as the *type* of a latent vertex and $\alpha_i \in \mathbb{R}_+$ as its *arrival coordinate*. Conditionally on Π, include each edge $\{i, j\}$ independently with probability $W(u_i, u_j)$. The graph G_ℓ at *truncation level* ℓ has vertex set $\{i : \alpha_i \leq \ell,\ \deg(i) \geq 1\}$ (the non-isolated vertices that have arrived by level ℓ) and the induced edges. The family $(G_\ell)_{\ell \geq 0}$ is the *graphex process* of W.

This definition repays slow reading, because the two coordinates of each Poisson point play sharply different roles and it is their separation that makes the construction work. Each latent vertex is a point (α_i, u_i) in the quadrant $\mathbb{R}_+ \times \mathbb{R}_+$, scattered by a Poisson process of *unit* intensity

— meaning that the expected number of points in any region of the quadrant is just its area, the cleanest possible reference measure (its existence and the calculus we are about to use are built from scratch in appendix H). The second coordinate u_i is the *type*: it alone enters the connection probability $W(u_i, u_j)$, exactly as the type x did in the dense model. The first coordinate α_i is the *arrival coordinate*, and it is pure bookkeeping — it never touches a connection probability. Its only function is to give us a knob to turn: truncating at level ℓ means keeping the points with $\alpha_i \leq \ell$ and discarding the rest. Because the process has unit intensity, the points kept inside a type window of width |window| number Poisson($\ell \cdot$ |window|), so raising ℓ from ℓ to $\ell + \mathrm{d}\ell$ admits a fresh sprinkle of vertices of *every* type, uniformly along the whole half-line.

Here is the dual role of ℓ that makes a graphex sparse, stated as concretely as possible. Raising ℓ does two things at once. It *adds vertices* — new points enter the window $\alpha \leq \ell$ — which is the job N did in the dense theory. But it also *enriches the existing vertices*: a type-u vertex already present sees its pool of potential partners grow with ℓ (there are now more truncated points for it to connect to), so its expected degree climbs like $\ell\, \mu_W(u)$, as we verify below. In the dense model these two effects were the same number N wearing one hat; here they are genuinely distinct, and sparsity is exactly the statement that the first effect outruns the second — vertices accumulate faster than edges — whenever the marginal is heavy enough that the half-line keeps supplying fresh low-degree types out at large u. The arithmetic of that race is the content of the next proposition.

Proposition 27.14 (Edge and vertex counts; sparsity)**.** *For the graphex process of a kernel W satisfying* (27.12) *with $W \in L^1(\mathbb{R}_+^2)$:*

1. *the expected edge count grows quadratically in the truncation level,*

$$\mathbb{E}[e(G_\ell)] = \tfrac{1}{2}\,\ell^2 \int_{\mathbb{R}_+^2} W(u,v)\, \mathrm{d}u\, \mathrm{d}v = \tfrac{1}{2}\,\ell^2\, \|W\|_{L^1(\mathbb{R}_+^2)}\,; \tag{27.13}$$

2. *the expected number of (non-isolated) vertices is*

$$\mathbb{E}[N_\ell] = \ell \int_{\mathbb{R}_+} \left(1 - e^{-\ell\, \mu_W(u)}\right) \mathrm{d}u \;:=\; \ell\, \Psi(\ell), \tag{27.14}$$

   finite for each $\ell \geq 1$ by (27.12)*;*

3. *if the marginal μ_W is heavy, in the sense that $\Psi(\ell) \to \infty$ as $\ell \to \infty$, then the graph is* subquadratically sparse*: $\mathbb{E}[N_\ell] = \ell\, \Psi(\ell) \gg \ell$ while $\mathbb{E}[e(G_\ell)] = \Theta(\ell^2)$, so*

$$\frac{\mathbb{E}[e(G_\ell)]}{\mathbb{E}[N_\ell]^2} = \Theta\big(\Psi(\ell)^{-2}\big) \to 0, \qquad \textit{i.e.} \quad e(G_\ell) = o(N_\ell^2). \tag{27.15}$$

   The average degree is $2\,\mathbb{E}[e]\,/\mathbb{E}[N] = \ell\, \|W\|_{L^1}/\Psi(\ell)$, which is bounded — the strict linear edge count (S1)*, $e = \Theta(N)$ — iff $\Psi(\ell) = \Theta(\ell)$, and* diverges *whenever $\Psi(\ell) = o(\ell)$, the generic heavy-marginal case.*

Proof. The whole proof is an exercise in the Poisson calculus of appendix H, and the two parts use its two central tools: the two-point Mecke formula for a sum over *pairs* of points (the edges), and the one-point Mecke formula together with thinning and Campbell for a sum over *single* points (the vertices). We take them in turn.

Part (1): the edge count. An edge is an unordered pair of points, so its expectation is a sum over pairs, and the tool for the expectation of a sum over pairs of a Poisson process is the two-point Mecke / factorial-moment formula (theorem H.3): for a unit-rate Poisson process Π and any symmetric nonnegative h,

$$\mathbb{E}[*] \sum_{\{p,p'\}\subset\Pi} h(p,p') = \tfrac{1}{2} \iint h(p,p')\ \mathrm{d}p\ \mathrm{d}p'.$$

The formula says that, in expectation, a Poisson process behaves as if its pairs of points were laid down independently according to the intensity measure — here Lebesgue measure on the quadrant — with the $\frac{1}{2}$ correcting for the double-count of ordered versus unordered pairs. We must choose h so that the left side counts edges. A pair $\{p,p'\}$ with $p=(\alpha,u)$ and $p'=(\alpha',u')$ contributes an edge to G_ℓ when both points are truncated in ($\alpha \le \ell$ and $\alpha' \le \ell$) and the conditional edge is actually included, which happens with probability $W(u,u')$. Taking the expectation over the edge-inclusion coins first replaces that random indicator by its mean $W(u,u')$, so the right choice is

$$h(p,p') = \mathbf{1}[\alpha \le \ell]\,\mathbf{1}[\alpha' \le \ell]\,W(u,u').$$

Feeding this into Mecke and factoring the integral over the arrival coordinates (which W ignores) from the integral over the types,

$$\mathbb{E}[e(G_\ell)] = \tfrac{1}{2}\int_0^\ell \mathrm{d}\alpha \int_0^\ell \mathrm{d}\alpha' \int_{\mathbb{R}_+^2} W(u,u')\ \mathrm{d}u\ \mathrm{d}u' = \tfrac{1}{2}\,\ell^2 \int_{\mathbb{R}_+^2} W = \tfrac{1}{2}\,\ell^2\ \|W\|_{L^1},$$

which is (27.13): the two arrival integrals each contribute a factor ℓ, and the type integral contributes the constant $\|W\|_{L^1}$. The edge count is exactly quadratic in ℓ — this is the "edges grow like ℓ^2" half of the sparsity race.

Part (2): the vertex count. A vertex is a single point, so its expectation is a sum over single points, governed by the one-point Mecke (Palm) formula (theorem H.3): the expected value of $\sum_{p\in\Pi} g(p,\Pi)$ equals $\int \mathbb{E}[g(p,\Pi\cup\{p\})]\ \mathrm{d}p$, where on the right p is a *deterministic* added point and Π is an independent copy of the same process. We apply it with $g(p,\Pi) = \mathbf{1}[\alpha \le \ell]\,\mathbf{1}[p \text{ is non-isolated}]$, so that the sum counts truncated non-isolated vertices and

$$\mathbb{E}[N_\ell] = \int_0^\ell \int_{\mathbb{R}_+} \mathbb{P}(\text{an added point at } (\alpha,u) \text{ is non-isolated})\ \mathrm{d}u\ \mathrm{d}\alpha.$$

It remains to compute that non-isolation probability, and this is where thinning and Campbell enter. Fix the added point at (α,u). Its potential neighbors are the truncated points $p'=(\alpha',u')$ with $\alpha' \le \ell$, and it actually connects to such a point independently with probability $W(u,u')$. Keeping each point of a Poisson process independently with a location-dependent probability is exactly a *thinning*, and the thinning theorem (theorem H.6) says the survivors again form a Poisson process, now with intensity $W(u,u')\,\mathbf{1}[\alpha' \le \ell]\ \mathrm{d}\alpha'\,\mathrm{d}u'$. The number of neighbors is the total mass of that thinned process, and the expected total mass of a Poisson process is the integral of its intensity — Campbell's formula (theorem H.2). So the neighbor count is Poisson with mean

$$\int_0^\ell \int_{\mathbb{R}_+} W(u,u')\ \mathrm{d}u'\ \mathrm{d}\alpha' = \ell \int_{\mathbb{R}_+} W(u,u')\ \mathrm{d}u' = \ell\,\mu_W(u),$$

the arrival integral again contributing the factor ℓ and the type integral the marginal. The added vertex is non-isolated precisely when this Poisson neighbor count is at least 1, which has probability $1 - e^{-\ell\mu_W(u)}$. Substituting and doing the trivial α-integral (another factor ℓ) gives (27.14), $\mathbb{E}[N_\ell] = \ell \int_{\mathbb{R}_+}(1-e^{-\ell\mu_W(u)})\,\mathrm{d}u = \ell\,\Psi(\ell)$. Notice this is exactly the mixed-Poisson mechanism of proposition 27.2 reappearing on the half-line: a type-u vertex carries a Poisson($\ell\,\mu_W(u)$) degree, with the truncation level ℓ now playing the role the constant c played there.

It remains to see that $\Psi(\ell)$ is finite, and here (27.12) earns its keep. The integrand $1 - e^{-\ell\mu_W(u)}$ is bounded by 1, so the only threat to finiteness is the tail $u \to \infty$, where $\mu_W(u)$ is small but the integration range is infinite. We control it by the elementary concavity bound

$$1 - e^{-\ell\mu} \le \ell\,(1 - e^{-\mu}) \qquad (\ell \ge 1,\ \mu \ge 0). \tag{27.16}$$

This holds because, for fixed $\mu \ge 0$, the map $t \mapsto 1 - e^{-t\mu}$ is concave in t and vanishes at $t = 0$; a concave function through the origin lies, for $t \ge 1$, below the line through the origin and its value at $t = 1$, i.e. $1 - e^{-t\mu} \le t\,(1 - e^{-\mu})$, and we set $t = \ell$. Integrating (27.16) over u gives $\Psi(\ell)\cdot\ell^0 = \int_{\mathbb{R}_+}(1 - e^{-\ell\mu_W})\,\mathrm{d}u \le \ell\int_{\mathbb{R}_+}(1 - e^{-\mu_W})\,\mathrm{d}u < \infty$, the last inequality being precisely the graphex condition (27.12). So $\Psi(\ell) < \infty$ for every $\ell \ge 1$, and *this* is the sense in which (27.12) guarantees "finitely many non-isolated vertices at each level" — the bound it furnishes at the reference level $\ell = 1$ propagates to every finite ℓ by concavity, at the modest cost of a factor ℓ.

Part (3): the sparsity dichotomy. The rest is arithmetic from (1) and (2), but it is the arithmetic the whole construction was built to produce, so we spell it out. The expected edge density per pair of vertices is

$$\frac{\mathbb{E}[e(G_\ell)]}{\mathbb{E}[N_\ell]^2} = \frac{\frac{1}{2}\,\ell^2\ \|W\|_{L^1}}{\big(\ell\,\Psi(\ell)\big)^2} = \frac{\|W\|_{L^1}}{2\,\Psi(\ell)^2} = \Theta\big(\Psi(\ell)^{-2}\big),$$

so as soon as the marginal is heavy enough that $\Psi(\ell) \to \infty$ this ratio tends to 0 and the graph is subquadratically sparse, $e = o(N^2)$ — the factors of ℓ have cancelled and what survives is governed entirely by the growth of Ψ. The average degree tells the same story in the form that decides (S1):

$$\frac{2\,\mathbb{E}[e(G_\ell)]}{\mathbb{E}[N_\ell]} = \frac{\ell^2\ \|W\|_{L^1}}{\ell\,\Psi(\ell)} = \frac{\ell\ \|W\|_{L^1}}{\Psi(\ell)}.$$

This is bounded — a finite limiting average degree, the strict linear edge count $e = \Theta(N)$ of (S1) — if and only if $\Psi(\ell)$ grows like ℓ itself, $\Psi(\ell) = \Theta(\ell)$; and it *diverges* whenever $\Psi(\ell) = o(\ell)$, which, as the next proposition shows, is the generic situation for a heavy power-law marginal. So the graphex secures *subquadratic* sparsity for free from heaviness alone, but the sharper *linear* sparsity is a knife-edge condition $\Psi(\ell) = \Theta(\ell)$ on the marginal's decay rate, not a consequence of integrability. □

Proposition 27.14 is the payoff, and it should be read with care. A single integrability condition (27.12) on the marginal delivers a finite vertex count at every level, and a *heavy* marginal ($\Psi(\ell) \to \infty$) delivers genuine *subquadratic* sparsity $e = o(N^2)$, all without the ad-hoc ρ_N of definition 27.1. This is the honest content of (S1) for a graphex: the edge count is sub-quadratic in the vertex count. The *strict* linear form $e = \Theta(N)$ (bounded average degree) is a sharper demand that, by part (3), holds only under the extra scaling $\Psi(\ell) = \Theta(\ell)$ and *fails* for the generic heavy marginal with $\Psi(\ell) = o(\ell)$; we therefore do not claim it generically. We turn next to the degree tail, which is where a concrete heavy marginal earns (S3).

The mixed-Poisson mechanism of proposition 27.2 transfers verbatim: a type-u vertex has Poisson($\ell\,\mu_W(u)$) degree, so a heavy marginal produces a heavy degree tail. The cleanest instance — the $\mathbb{R}_+$ analogue of the rank-one weight $f = x^{-\nu}$ of corollary 27.3 — is a pure-power marginal, for which the entire degree distribution is computable in closed form and is *scale-invariant* in the truncation level.

Proposition 27.15 (Power-law degrees from a heavy-tailed graphex marginal)**.** *Let W be a graphex whose marginal is the pure power $\mu_W(u) = u^{-\zeta}$, $u \in \mathbb{R}_+$, with $\zeta > 1$ (so (27.12) holds: $1 - e^{-\mu}$ is $O(1)$ near 0 and $\sim u^{-\zeta}$ near ∞, integrable iff $\zeta > 1$). Then for every truncation level $\ell > 0$:*

1. *$\Psi(\ell) = \Gamma(1 - \frac{1}{\zeta})\,\ell^{1/\zeta}$, so $\mathbb{E}[N_\ell] = \Theta(\ell^{1+1/\zeta})$; the marginal is heavy ($\Psi(\ell) \to \infty$) and $\Psi(\ell) = o(\ell)$, so the graph is subquadratically sparse with* diverging *average degree $\Theta(\ell^{1-1/\zeta})$;*

2. *the expected empirical degree distribution is exactly ℓ-independent: for $k \geq 1$,*

$$D(k) := \frac{\mathbb{E}[\#\,\{\textit{vertices of degree} \geq k\}]}{\mathbb{E}[N_\ell]} = \frac{\displaystyle\int_0^\infty \mathbb{P}(\mathrm{Poisson}(w) \geq k)\, w^{-1-1/\zeta}\,\mathrm{d}w}{\displaystyle\int_0^\infty (1 - e^{-w})\, w^{-1-1/\zeta}\,\mathrm{d}w}, \tag{27.17}$$

and this degree tail is a power law, $D(k) \sim C_\zeta\, k^{-1/\zeta}$ as $k \to \infty$, i.e. exponent $\gamma = 1 + \frac{1}{\zeta} \in (1, 2)$.

Proof. The starting point is the same vertex count by Mecke/thinning/Campbell as in proposition 27.14(2), but now keeping track of the degree threshold. Repeating that computation with “non-isolated” (degree ≥ 1) replaced by “degree $\geq k$,” and using that a type-u vertex has a Poisson($\ell\,\mu_W(u)$) degree with the present $\mu_W(u) = u^{-\zeta}$, gives

$$\mathbb{E}[\#\,\{\deg \geq k\}] = \ell \int_0^\infty \mathbb{P}(\mathrm{Poisson}(\ell\, u^{-\zeta}) \geq k)\,\mathrm{d}u, \tag{27.18}$$

which for $k = 1$ recovers $\mathbb{E}[N_\ell] = \ell\,\Psi(\ell)$ since $\mathbb{P}(\mathrm{Poisson}(\lambda) \geq 1) = 1 - e^{-\lambda}$. The point of the pure-power marginal is that this integral admits a single change of variables that strips out *all* the ℓ-dependence into one prefactor. Substitute

$$w = \ell\, u^{-\zeta}, \qquad u = (\ell/w)^{1/\zeta}, \qquad \mathrm{d}u = -\tfrac{1}{\zeta}\,\ell^{1/\zeta}\, w^{-1-1/\zeta}\,\mathrm{d}w,$$

which sends $u = 0 \mapsto w = \infty$ and $u = \infty \mapsto w = 0$ (the sign flip from $\mathrm{d}u$ is absorbed by reversing the limits). The Poisson mean $\ell u^{-\zeta}$ becomes simply w, and

$$\mathbb{E}[\#\,\{\deg \geq k\}] = \tfrac{1}{\zeta}\,\ell^{1+1/\zeta} \int_0^\infty \mathbb{P}(\mathrm{Poisson}(w) \geq k)\, w^{-1-1/\zeta}\,\mathrm{d}w. \tag{27.19}$$

Every trace of ℓ now sits in the prefactor $\frac{1}{\zeta}\ell^{1+1/\zeta}$; the integral is a pure number depending on k and ζ alone. We pause to check it converges, since the weight $w^{-1-1/\zeta}$ is singular at the origin. Near $w = 0$ the Poisson tail is dominated by its first nonzero term, $\mathbb{P}(\mathrm{Poisson}(w) \geq k) \sim w^k/k!$, so the integrand behaves like $w^{k-1-1/\zeta}$, integrable at 0 because $k \geq 1 > 1/\zeta$ (here $\zeta > 1$ is used); near $w = \infty$ the probability rises to 1 and $w^{-1-1/\zeta}$ decays integrably. So (27.19) is finite for every $k \geq 1$.

The constant Ψ and part (1). Take $k=1$ in (27.19) and recall $\mathbb{P}(\mathrm{Poisson}(w)\ge 1)=1-e^{-w}$:

$$\Psi(\ell)=\frac{\mathbb{E}[N_\ell]}{\ell}=\tfrac{1}{\zeta}\,\ell^{1/\zeta}\int_0^\infty(1-e^{-w})\,w^{-1-1/\zeta}\;\mathrm{d}w.$$

The remaining integral is a standard one, evaluated by integration by parts to expose a Gamma function. Write $\beta=1/\zeta\in(0,1)$ and integrate $\int_0^\infty(1-e^{-w})w^{-1-\beta}\,\mathrm{d}w$ by parts with $f=1-e^{-w}$, $\mathrm{d}g=w^{-1-\beta}\,\mathrm{d}w$, so $g=-\frac{1}{\beta}w^{-\beta}$ and $\mathrm{d}f=e^{-w}\,\mathrm{d}w$. The boundary term $\big[-\frac{1}{\beta}w^{-\beta}(1-e^{-w})\big]_0^\infty$ vanishes at both ends (at ∞ because $w^{-\beta}\to 0$; at 0 because $1-e^{-w}\sim w$ kills $w^{-\beta}$ for $\beta<1$), leaving

$$\int_0^\infty(1-e^{-w})\,w^{-1-\beta}\;\mathrm{d}w=\frac{1}{\beta}\int_0^\infty w^{-\beta}e^{-w}\;\mathrm{d}w=\frac{\Gamma(1-\beta)}{\beta},$$

recognizing the Gamma integral $\int_0^\infty w^{(1-\beta)-1}e^{-w}\,\mathrm{d}w=\Gamma(1-\beta)$. With $\beta=1/\zeta$ this is $\zeta\,\Gamma(1-1/\zeta)$, so the prefactor $\frac{1}{\zeta}$ cancels and $\Psi(\ell)=\Gamma(1-1/\zeta)\,\ell^{1/\zeta}$. Then $\mathbb{E}[N_\ell]=\ell\Psi(\ell)=\Gamma(1-1/\zeta)\,\ell^{1+1/\zeta}=\Theta(\ell^{1+1/\zeta})$; since $\zeta>1$ we have $1/\zeta\in(0,1)$, so $\Psi(\ell)=\Theta(\ell^{1/\zeta})\to\infty$ (heavy marginal) yet $\Psi(\ell)=o(\ell)$ (so, by the dichotomy of proposition 27.14(3), the average degree $\Theta(\ell^{1-1/\zeta})$ diverges). This is part (1).

The degree distribution and its tail, part (2). Form the ratio defining $D(k)$ in (27.17): numerator $\mathbb{E}[\#\{\deg\ge k\}]$ and denominator $\mathbb{E}[N_\ell]=\mathbb{E}[\#\{\deg\ge 1\}]$ are *both* given by (27.19), and they share the identical prefactor $\frac{1}{\zeta}\ell^{1+1/\zeta}$, which therefore cancels. What is left is exactly the ℓ-independent quotient of integrals displayed in (27.17). This cancellation is the precise meaning of the scale-invariance promised above: the *shape* of the degree distribution does not move as the graph grows — the graph gets larger and its average degree climbs, but the fraction of vertices of degree $\ge k$ is frozen, exactly the N-independence demanded by (S3).

It remains to extract the large-k tail of the numerator integral $\int_0^\infty\mathbb{P}(\mathrm{Poisson}(w)\ge k)\,w^{-1-1/\zeta}\,\mathrm{d}w$. The mechanism is a Poisson-concentration sandwich, the integral form of the tail-transfer lemma lemma H.10. The point is that for large k the function $w\mapsto\mathbb{P}(\mathrm{Poisson}(w)\ge k)$ is a sharp threshold: a Poisson(w) variable concentrates around its mean w with standard deviation $\sqrt{w}$, which for w near k is of order $\sqrt{k}=o(k)$, so the probability jumps from nearly 0 to nearly 1 across a window of width $o(k)$ around $w=k$. Hence $\mathbb{P}(\mathrm{Poisson}(w)\ge k)\to\mathbf{1}\{w>k\}$ uniformly outside that vanishing window, and the heavy weight $w^{-1-1/\zeta}$ — which varies negligibly across an $o(k)$ window at scale k — cannot distinguish the true threshold from the sharp one. Therefore

$$\int_0^\infty\mathbb{P}(\mathrm{Poisson}(w)\ge k)\,w^{-1-1/\zeta}\;\mathrm{d}w\sim\int_k^\infty w^{-1-1/\zeta}\;\mathrm{d}w=\zeta\,k^{-1/\zeta}\qquad(k\to\infty).$$

Dividing by the denominator $\int_0^\infty(1-e^{-w})w^{-1-1/\zeta}\,\mathrm{d}w=\zeta\,\Gamma(1-1/\zeta)$ computed above,

$$D(k)\sim\frac{\zeta\,k^{-1/\zeta}}{\zeta\,\Gamma(1-1/\zeta)}=C_\zeta\,k^{-1/\zeta},\qquad C_\zeta=\frac{1}{\Gamma(1-1/\zeta)},$$

a power law of index $1/\zeta$ in the complementary cumulative $D(k)=\mathbb{P}(\deg\ge k)$, hence degree exponent $\gamma=1+1/\zeta$. Since $\zeta>1$ forces $1/\zeta\in(0,1)$, the exponent lands in $\gamma=1+1/\zeta\in(1,2)$ — exactly the window the closing caveat now scrutinizes. $\square$

So the graphex genuinely delivers (S3): a heavy-tailed, N-independent (unnormalized) degree distribution, here with exponent $\gamma=1+1/\zeta$. This is a real success and worth stating without hedging — it is the heavy unnormalized tail the dense theory provably could not produce ((S3), proposition 19.14), now falling out in closed form from the decay of a single marginal, with no ρ_N anywhere in sight. But the success comes with an honest caveat that must be developed rather than buried, because it marks the exact limit of what the kernel-only graphex can reach, and that limit is the chapter's recurring theme of locating the boundary sharply.

The caveat is this: the natural graphex window is $\gamma \in (1, 2)$, *not* the empirically dominant $(2, 3)$ that the desiderata of section 27.1 were calibrated to. The two windows do not overlap at all, and the gap is forced, not accidental. Trace the constraints. Integrability of the marginal, (27.12), demanded $\zeta > 1$ for the pure power $u^{-\zeta}$; the degree exponent is $\gamma = 1 + 1/\zeta$; and $\zeta > 1$ is exactly $1/\zeta < 1$, hence $\gamma < 2$. To push γ up to 2 or beyond would require $1/\zeta \geq 1$, i.e. $\zeta \leq 1$, which makes the marginal $u^{-\zeta}$ non-integrable at infinity and the graphex condition fail — the expected number of non-isolated vertices would be infinite at every level, and there would be no graph to speak of. So the very integrability that makes the construction well-posed caps its degree exponent strictly below 2. The kernel graphex is structurally a machine for $\gamma \in (1, 2)$: tails *heavier* than what real networks typically show, corresponding to hubs so dominant that even the mean degree (in the unnormalized law) sits at the edge of divergence.

It is worth pinning down what $\gamma \in (1, 2)$ versus $\gamma \in (2, 3)$ means, because the contrast is exactly the moment-divergence reading from section 27.1. Recall that $\gamma = 2$ is the threshold where the *mean* of the degree law passes from finite (for $\gamma > 2$) to infinite (for $\gamma < 2$). The empirical window $(2, 3)$ is finite-mean, infinite-variance; the graphex window $(1, 2)$ is *infinite-mean*. This is the analytic shadow of the average-degree divergence we already met in proposition 27.15(1): with $\Psi(\ell) = o(\ell)$ the average degree $\Theta(\ell^{1-1/\zeta})$ grows without bound, which is the finite-ℓ manifestation of a limiting degree law whose mean has run off to infinity. The kernel graphex, in other words, does not merely fail to hit $(2, 3)$ by a technicality; it lives on the other side of the mean-divergence line, in a genuinely heavier-tailed regime, and the strict linear edge count (S1) — which needs a *finite* limiting average degree, $\Psi(\ell) = \Theta(\ell)$ — is for the same reason out of its reach.

What, then, would it take to reach $(2, 3)$? Two honest routes, both already on the page. The first is to restore the components we excised: the *star* intensity S and *dust* rate I of definition 27.12. These add connectivity that does not pass through the integrable kernel, and they are precisely the degrees of freedom the general Caron–Fox–Veitch–Roy theory uses to populate the lighter-tailed exponents; bringing them back is the principled way to model $\gamma \in (2, 3)$ within the graphex framework, at the cost of the extra calculus we deliberately set aside. The second route is the one the chapter has *already* walked: the target-density construction of corollary 27.3, where an unbounded rank-one weight $f = x^{-\nu}$ over a unit-interval type space, sparsified by $\rho_N = c/N$, produced exactly $\gamma = 1 + 1/\nu \in (2, 3)$ for $\nu \in (\frac{1}{2}, 1)$. That construction reached the empirical band — but only by paying with the ad-hoc thinning dial ρ_N that this section was built to avoid. So the two routes to $(2, 3)$ trade off against each other in an instructive way: the graphex buys intrinsic sparsity but caps γ below 2 on its kernel alone, while the target-density model buys the empirical exponent but only through an external ρ_N. Neither the kernel graphex nor the bolt-on ρ_N is, by itself, both intrinsically sparse *and* natively in $(2, 3)$; closing that gap is exactly the kind of frontier question this chapter is committed to flagging rather than papering over. We record it honestly: the kernel graphex secures (S2), (S3), and subquadratic (S1) from first principles, but the precise empirical exponent band $(2, 3)$ is not a free consequence of the kernel component alone.

That honest accounting closes the construction, but it leaves a prior question unanswered, and it is the question the next subsection must face before any of this earns its keep. We have built *a* sparse limit object and shown it does what we asked — but we built it by a specific recipe (a Poisson process on $\mathbb{R}_+$, this particular kernel, these excisions), and a single recipe proves nothing about whether the graphex is the *right* limit object, the one that genuine sparse graph sequences actually converge to, as opposed to merely one clever way to manufacture a sparse graph. The dense theory earned its graphons through a representation theorem — Aldous–Hoover — that said every exchangeable dense graph *is* a graphon, leaving no other possibility. The graphex deserves the same scrutiny: is it forced on us as *the* sparse limit, or is it just one construction among many? We turn to that representation theorem now, and to the price it exacts.

27.3.2 The representation theorem, and what it costs us

The reason graphexes are the *right* objects is a representation theorem of the same character as the dense graphon limit theorem of chapter 17. In the dense world, a sequence of graphs converges if and only if its homomorphism densities converge, and the limit is a graphon modulo weak isomorphism (the Lovász–Szegedy / Aldous–Hoover circle of ideas). In the sparse world the corresponding statement is about *exchangeable random measures.*

Imported result 27.16 (Graphex representation; Caron–Fox, Veitch–Roy, Kallenberg)**.** (Proved in the cited literature, not in this book; used only as structural license, and no in-book result depends on it.) *Represent a graph as a point process on* $\mathbb{R}_+^2$ *(a point at* (u_i, u_j) *for each edge, with types in* $\mathbb{R}_+$*). If this random point measure is* jointly exchangeable *— invariant in law under any measure-preserving transformation of the type coordinate* $\mathbb{R}_+$ *applied simultaneously to both axes — and locally finite, then it is, in distribution, the graphex process* (*definition 27.13*) *of some graphex* (W, S, I)*, and the graphex is unique up to the natural measure-preserving relabelings of* $\mathbb{R}_+$*. Conversely every graphex process is jointly exchangeable.*

This is the sparse-graph analogue of the Aldous–Hoover theorem and is due to Veitch–Roy [130] and Caron–Fox [45], resting on Kallenberg's representation theorem for jointly exchangeable random measures on $\mathbb{R}_+^2$ [84, Ch. 9]. The mechanism is Kallenberg's: a jointly exchangeable, locally finite random measure on $\mathbb{R}_+^2$ admits a representation as a deterministic measurable function of an underlying unit-rate Poisson process (the object built self-containedly in appendix H), and unwinding that function for the special case of a simple-point (graph) measure yields the kernel/star/dust decomposition (W, S, I). We deliberately present this as an *imported result* rather than an in-book theorem: its proof is long, belongs to the theory of exchangeable arrays rather than to mean field games, and — crucially — no result we prove downstream depends on it. We use it only as the structural license to study graphexes as *the* limit objects for sparse exchangeable graphs, exactly as one uses the de Finetti theorem to justify studying mixtures of i.i.d. sequences. (The dense Aldous–Hoover representation it parallels is likewise imported rather than proved in this book — it is surveyed in chapter 15 via [60] — so the two sit on the same honest footing.) The complete argument lives in [130, 84]; the ledger lists it under *Deferred*, honestly outside the book's self-contained core.

There is a corresponding notion of convergence *to* a graphex, the sparse analogue of cut convergence. We describe it informally, and flag at once that the full theory is outside this book's self-contained core: no in-book result depends on it, and we use it only to locate the graphex as the right limit object. The part we *can* make precise from the construction already in hand is the one-sided statement.

Remark 27.17 (Sampling convergence, informally)**.** The graphex process gives, for each ℓ, a law on finite graphs: the truncation G_ℓ of definition 27.13. One says a sequence of finite graphs (H_n) *converges in the sampling sense* to a graphex W if a randomly rescaled sample of H_n — retain each vertex independently with a *stretching probability* tuned so that the retained vertex count matches the graphex scale, then look at the induced subgraph — converges in distribution, as $n \to \infty$ and then the scale grows, to G_ℓ, on every fixed finite test graph. This is genuinely the right sparse counterpart of the cut metric of chapter 17: there is a *stretched cut metric* (a Bollobás–Riordan rescaling of the dense cut distance) under which sampling convergence is metrized, and the resulting limits are exactly the (rescaled) L^p graphons of definition 27.5 [25, 26, 27]. We do not develop the stretched cut metric here — it would require the sparse counterparts of the entire chapters 17 and 19 apparatus — and we use none of it below; the reader should read this remark as orientation, not as a tool. The only convergence question this chapter actually settles is the forward one (the explicit edge/vertex/degree asymptotics of propositions 27.14 and 27.15), built directly on the Poisson calculus of appendix H.

We now turn to the only question that is properly ours: the mean field game on a graphex.

27.3.3 Graphon mean field games over a graphex limit

Suppose the network is, in the limit, a graphex $W\colon \mathbb{R}_+^2 \to [0,1]$. What is the mean field game? Before writing equations it pays to be clear about *which* limit object carries the game, because the graphex carries *two* scales where the dense theory had one. In the dense theory the GMFG was the $N \to \infty$ limit of finite N-player games, and the single parameter N was both the population size and the normalization of the coupling. The graphex separates these roles: the static kernel W on $\mathbb{R}_+$ is the limit object, while the truncation level ℓ (definition 27.13) plays the role of N, growing the finite graph G_ℓ toward that limit. The natural program would be to define a game on each finite G_ℓ and pass $\ell \to \infty$; but, as flagged in the ledger, the finite-ℓ ε-Nash convergence on sparse graphs is open (it is the sparse analogue of chapter 25). We therefore do what the dense theory also does once the limit object is in hand: *pose the game directly on the static limit* W, treating the typical agent not as a normalized type average — there is no uniform law on the infinite-measure $\mathbb{R}_+$ — but through the point process, an agent of type u drawn from the latent Poisson intensity. The coupling strength s (Thread B) is a fixed constant of the cost, *independent* of the truncation level ℓ; it is the analogue of the $O(1)$ mean-coupling of the dense game, not an ℓ-rescaling of the field. With that understood, a representative type-u agent, $u \in \mathbb{R}_+$, has state $X_t^u \in \mathbb{R}$ obeying the Thread-B dynamics of appendix A.7,

$$\mathrm{d}X_t^u = \big(aX_t^u + b\,\alpha_t^u\big)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^u, \qquad X_0^u \sim m_0^u, \tag{27.20}$$

and is coupled to its graphex-aggregated neighbor field

$$z_t(u) = (\mathbb{W}\bar{m}_t)(u) = \int_{\mathbb{R}_+} W(u,v)\,\bar{m}_t(v)\;\mathrm{d}v, \qquad \bar{m}_t(v) = \int_{\mathbb{R}} \xi\, m_t^v(\mathrm{d}\xi), \tag{27.21}$$

the same aggregate (18.10) but with the integral now over $\mathbb{R}_+$. The per-type cost is that of Thread B, with mean-coupling target $s\,z_t(u)$, where s is the fixed Thread-B coupling *strength* (not the truncation level). The mean field game is the fixed-point problem: z produces optimal feedback, hence a population flow m^u, whose mean must regenerate z through (27.21).

Notation watch

Two distinct quantities sit close together in this section, and we keep them strictly apart. The *truncation level* ℓ of definition 27.13 is the graphex's analogue of the population size N: it indexes the growing finite graph G_ℓ, with $\mathbb{E}[e(G_\ell)] = \frac{1}{2}\ell^2\,\|W\|_{L^1}$ and $\mathbb{E}[N_\ell] = \ell\,\Psi(\ell)$. The *coupling strength* s (with terminal weight s_T) is the Thread-B mean-coupling constant of appendix A.7, fixed once and for all in the cost target $s\,z_t(u)$ and *independent of* ℓ. Where the dense graphex literature writes its truncation parameter as s, we have deliberately renamed it ℓ to free s for its canonical Thread-B meaning; the reader who has just absorbed "degrees scale like $\ell\,\mu_W(u)$" should not read $s\,z_t(u)$ as a level rescaling of the field.

Two genuinely new difficulties appear, and they are exactly the difficulties of section 27.2 transported to $\mathbb{R}_+$ and amplified by the infinite measure.

1. *The population "average" is over an infinite measure.* There is no uniform probability law on $\mathbb{R}_+$, so the global mean $\int_{\mathbb{R}_+} \bar{m}_t\;\mathrm{d}v$ that played a role in the dense theory need not be finite, and statements about "the typical agent" must be made through the point process, not through a normalized type average. The well-defined object is the aggregate $z_t(u)$, and it is finite for a.e. u provided $\bar{m}_t \in L^{q'}(\mathbb{R}_+)$ and the marginal $\mu_W \in L^q$.

2. *The operator $\mathbb{W}$ on $L^2(\mathbb{R}_+)$ inherits the dichotomy of proposition 27.8, with the additional hazard that $W \le 1$ no longer forces $\mathbb{W}$ to be bounded*, because $\mathbb{R}_+$ has infinite measure: a bounded kernel can have $\|W\|_{L^2(\mathbb{R}_+^2)} = \infty$ (e.g. W not decaying off the diagonal). Boundedness of the graphex as a function ($W \le 1$) is no longer even a sufficient condition for the operator-norm theory; what matters is the integrability of the marginal μ_W, exactly the quantity (27.12) controls.

The honest summary is that theorem 27.9 has a clean transcription *when the graphex operator is Hilbert–Schmidt*, $W \in L^2(\mathbb{R}_+^2)$, and otherwise the theory is at the same conjectural frontier as proposition 27.11.

Proposition 27.18 (GMFG over a square-integrable graphex)**.** *Let $W \in L^2(\mathbb{R}_+^2)$ be a symmetric graphex (so $\mathbb{W}$ is Hilbert–Schmidt, self-adjoint, compact on $L^2(\mathbb{R}_+)$, with $\|\mathbb{W}\| \le \|W\|_{L^2(\mathbb{R}_+^2)}$), and let the per-type problem satisfy* (A-Lip), (A-Growth), (A-Nondeg), (A-Conv)*. If $\|W\|_{L^2(\mathbb{R}_+^2)} < \kappa$ with κ the threshold of theorem 27.9, then the graphex mean field game* (27.20)–(27.21) *has a unique equilibrium in $L^2(\mathbb{R}_+; C[0,T])$.*

Status. Conditional theorem. *The fixed-point argument of chapter 24 is stated over the probability type space* $[0,1]$*; its transcription to the σ-finite space* $(\mathbb{R}_+, \lambda)$ *is routine* provided *the per-type estimates are uniform in* $u \in \mathbb{R}_+$ *and the initial data* m_0^u *have a uniformly bounded second moment with* $\bar{m}_0 \in L^2(\mathbb{R}_+)$ *— hypotheses we must* add *and which are automatic on* $[0,1]$ *but not on* $\mathbb{R}_+$*. Under those added hypotheses the proof of theorem 27.9 goes through verbatim with* $[0,1]$ *replaced by* $\mathbb{R}_+$*; without the uniform-in-u control it is open. We therefore state it as conditional and revisit it in chapter 31.*

What the proof requires. $\mathbb{W}$ Hilbert–Schmidt on $L^2(\mathbb{R}_+)$ is theorem 2.22 applied on the σ-finite measure space, valid because the Hilbert–Schmidt norm is the L^2 norm of the kernel regardless of the base measure's total mass. The contraction constant is again $C_0 \|\mathbb{W}\|$. The single non-automatic ingredient is that the best-response map send $L^2(\mathbb{R}_+; C[0,T])$ into itself, which needs the per-type free trajectories $x \mapsto \bar{m}_t^u\big|_{z\equiv 0}$ to be square-integrable in u; that is the added hypothesis $\bar{m}_0 \in L^2(\mathbb{R}_+)$ propagated by the (A-Lip), (A-Growth) bounds. Granting it, the fixed-point and uniqueness steps are identical to chapter 24. The gap relative to the $[0,1]$ theory is precisely the loss of the automatic uniform integrability that a probability type space provided. □

27.4 The order parameter survives where the operator does not

The graphex section pushed the construction to its outermost frontier — a limit sparse in itself on $\mathbb{R}_+$, whose game we could control only conditionally — and left the theory there at its conjectural edge. We now turn back from that frontier to the most explicit case the chapter owns, the unit-interval rank-one kernel of section 27.2, and advance Thread B through it into the sparse regime to extract the chapter's sharpest statement. The rank-one (Chung–Lu) kernel is the natural test case: it is the kernel that produced power-law degrees in corollary 27.3, it is the cleanest GMFG worked example (worked computation 18.19, proposition 18.18), and — as we will see — it is the case in which the breakdown of section 27.2 can be computed in closed form and tied exactly to the empirical scale-free window.

Worked computation 27.19 (Sparse rank-one LQG graphon game; the $p = 2$ dividing line)**.** This is the dense rank-one baseline of worked computation 26.12 — the same self-consistent closure of a scalar order parameter $\varrho_t = \langle f, \bar{m}_t\rangle$ derived there through the operator Riccati of (26.38)–(26.39) — here pushed to the boundary of L^2, where the closure breaks. Take the linear–quadratic Thread-B model of appendix A.7 with the rank-one coupling $W(x,y) = f(x)f(y)$, $f \geq 0$. By eq. (18.11) the aggregate collapses to a single scalar order parameter,

$$z_t(x) = f(x)\,\varrho(t), \qquad \varrho(t) := \langle f, \bar{m}_t\rangle = \int_0^1 f(y)\,\bar{m}_t(y)\;\mathrm{d}y, \tag{27.22}$$

written with the book's canonical order-parameter symbol $\varrho = \varrho$ (eq. (18.11); the same symbol carries the static rank-one reduction eq. (20.32)), so a type-x agent solves the scalar LQG problem of Thread A with the time-dependent tracking target $s\,z_t(x) = s\,f(x)\,\varrho(t)$.

Step 1: affine response. By the LQG solution of chapter 14 (and its dynamic graphon form in chapter 22), for any given continuous order-parameter path $\varrho \in C[0,T]$ the optimally controlled type-x mean is an *affine* functional of its target path,

$$\bar m_t^x = \pi_t^x + s\, f(x)\, (\mathsf{V}\varrho)(t), \tag{27.23}$$

where π^x is the free ($s = 0$) trajectory determined by the Riccati data and the initial mean $\bar m_0^x$ (eq. (14.13), theorem 14.4), and $\mathsf{V}\colon C[0,T] \to C[0,T]$ is the bounded linear Volterra response operator built from the same Riccati coefficients $(a, b, \sigma, q, q_T, s_T, T)$ (here a is the Thread-B *state-feedback* coefficient of (27.20), not an exponent). This is a *time-path* response operator on $C[0,T]$, and it is *independent of x and of f*; it is a different object from the type-space operator-Riccati $\mathcal{R}_t$ of eq. (26.35) (which acts on $L^2[0,1]$), and we give it the distinct symbol V to avoid conflating the two across the chapter boundary.

Step 2: closing the order parameter. Insert (27.23) into the definition of ϱ in (27.22):

$$\varrho(t) = \int_0^1 f(x)\, \bar m_t^x \,\mathrm{d}x = \underbrace{\int_0^1 f(x)\, \pi_t^x \,\mathrm{d}x}_{:=\ \varrho_0(t)} + s\Big(\underbrace{\int_0^1 f(x)^2 \,\mathrm{d}x}_{=\|f\|_{L^2}^2}\Big)(\mathsf{V}\varrho)(t), \tag{27.24}$$

where ϱ_0 is the *free* ($s = 0$) order parameter built from the uncoupled trajectories. The entire f-dependence of the coupling has condensed into the single scalar $\|f\|_{L^2}^2$, which by proposition 27.8(3) is exactly the operator norm $\|\mathbb{W}\| = \lambda_1$. The self-consistency equation is the linear fixed point

$$\big(I - s\, \|f\|_{L^2}^2\, \mathsf{V}\big)\varrho = \varrho_0, \tag{27.25}$$

uniquely solvable by Neumann series whenever

$$s\, \|f\|_{L^2}^2\, \|\mathsf{V}\|_{C[0,T]\to C[0,T]} < 1 \quad\iff\quad \|\mathbb{W}\| = \|f\|_{L^2}^2 < \kappa := \frac{1}{s\, \|\mathsf{V}\|}, \tag{27.26}$$

which is precisely the coupling-smallness condition (27.10) of theorem 27.9, now made completely explicit: the threshold constant κ is $1/(s\,\|\mathsf{V}\|)$, a pure Riccati quantity. This is the same threshold phenomenon as remark 26.13, where uniqueness was governed by finiteness of the scalar Riccati r: both say the closure is controlled by the leading eigenvalue $\|\mathbb{W}\| = \lambda_1 = \|f\|_{L^2}^2$, and (27.26) is its explicit time-domain form, with the response factor $\|\mathsf{V}\|$ (the horizon-dependent response-Lipschitz constant $L_{\mathcal{R}}(T)$ of eq. (24.7)) supplying the horizon dependence.

Step 3: the dividing line. Take the heavy weight $f(x) = x^{-\nu}$ of corollary 27.3, which under sparsified sampling yields degree exponent $\gamma = 1 + 1/\nu$. The susceptibility that controls the order parameter is

$$\|f\|_{L^2}^2 = \int_0^1 x^{-2\nu} \,\mathrm{d}x = \begin{cases} \dfrac{1}{1-2\nu}, & \nu < \frac12 \ \ (\gamma > 3), \\ +\infty, & \frac12 \le \nu < 1 \ \ (\gamma \in (2,3]). \end{cases} \tag{27.27}$$

Thus:

- For $\gamma > 3$ ($\nu < \frac12$, $f \in L^2 \setminus L^\infty$): the kernel is an unbounded L^2 graphon, $\|\mathbb{W}\| = 1/(1-2\nu) < \infty$, and (27.25) is uniquely solvable provided $s\, \|\mathsf{V}\|/(1-2\nu) < 1$. The GMFG is well-posed in *closed form* — a completely explicit instance of theorem 27.9 with an unbounded, heavy-tailed coupling.

- For $\gamma \in (2,3]$ ($\nu \geq \frac{1}{2}$, $f \in L^1 \setminus L^2$): the susceptibility $\|f\|_{L^2}^2$ *diverges*, the coefficient in (27.25) is $+\infty$, and the order-parameter fixed point has no finite linear closure. The operator $\mathbb{W}$ is unbounded on L^2 (proposition 27.8(4)) and the contraction threshold (27.26) is vacuous. The reduction — and with it the only route to a solution we have — collapses already at $\nu = \frac{1}{2}$, i.e. at $\gamma = 3$ (where $\int_0^1 x^{-1}\,\mathrm{d}x = \infty$), and stays broken throughout the open window $\gamma \in (2,3)$.

The breakdown regime is thus $\gamma \in (2,3]$ ($\nu \in [\frac{1}{2}, 1)$, $f \in L^1 \setminus L^2$), with the dividing line exactly at $\gamma = 3$ ($\nu = \frac{1}{2}$, $p = 2$); the empirically dominant *open* window $\gamma \in (2,3)$ sits strictly inside it. In every such case the linear–quadratic graphon mean field coupling has divergent susceptibility and the dense/L^2 well-posedness theory provably fails. The boundary $\gamma = 3$, $\nu = \frac{1}{2}$, $p = 2$ is one and the same line viewed three ways.

The worked computation refutes a tempting hope and sharpens the frontier. One might have guessed that the finite-rank reduction, being finite-dimensional, would survive any amount of kernel heaviness — after all, $z_t(x) = f(x)\varrho(t)$ is a single scalar times a fixed profile. It does not: the very step that closes the scalar equation, $\varrho \mapsto \langle f, \bar{m}\rangle = \langle f, (\text{response to } sf\varrho)\rangle$, reintroduces $\langle f, f\rangle = \|f\|_{L^2}^2$, and this is the operator norm. The order parameter is finite ($\varrho = \langle f, \bar{m}\rangle$ needs only $f \in L^1$, $\bar{m} \in L^\infty$), but its *self-consistent closure* needs $f \in L^2$. The susceptibility, not the order parameter, is what diverges.

Physics note

This box has one piece of exact mathematics and a surrounding analogy; we are careful to say which is which. The *exact* part is the identification of the closure coefficient as a susceptibility: the quantity $\|f\|_{L^2}^2 = \langle f, \mathbb{W}f\rangle / \langle f, f\rangle$ that closes the order-parameter equation in (27.24) is, term by term, the linear response of the aggregate to a perturbation of the field summed over the coupling — in the rank-one LQG game literally $\int f^2$, the second moment of the coupling weight — and its divergence as the weight is made heavier ($\nu \uparrow \frac{1}{2}$, $\gamma \downarrow 3$) is a theorem, the content of (27.27). The *analogical* part, which carries no mathematical weight, is the surrounding critical-phenomena picture: regard the dense graphon as an upper-critical-dimension mean-field theory where each agent has $\Theta(N)$ neighbors, the law of large numbers suppresses fluctuations, and the closure $z = \mathbb{W}\bar{m}$ is exact; then the scale-free regime $\gamma \in (2,3)$, where a typical agent has $\Theta(1)$ neighbors while a few hubs have anomalously many, is the regime where the "loop corrections" the mean-field closure neglected are no longer small. That language is a heuristic guide to intuition, not a proof; if every word of it were deleted, the divergence of $\int f^2$ and the breakdown of the linear closure would stand exactly as proved. The reason the analogy is nonetheless apt is that both sides are, structurally, the same computation: a sum over a coupling weighted by its own magnitude, diverging when the weight's tail is too heavy to be square-summable.

Remark 27.20 (Where the frontier actually runs)**.** The single orientation worth carrying out of this chapter — the itemized accounting is the ledger's job, just below — is *where* the theorem/conjecture boundary falls, because for once it falls at a sharp and computable place. Everything on the unbounded-kernel axis up to and including $p = 2$ is theorem: square-integrability, not boundedness, is the true hypothesis, and the rank-one game makes the boundary explicit. Everything strictly below $p = 2$ — equivalently the empirically dominant scale-free window $\gamma \in (2, 3)$ — is conjecture: the operator is unbounded, the contraction is gone, and existence is known only in the monotone and finite-rank special cases. The graphex axis sits one notch further out: its *counting* theory is fully proved here, but the game over a non-Hilbert–Schmidt graphex, and the sparse ε-Nash theory, are open. That the most empirically relevant regime is exactly the one our methods cannot yet reach is the chapter's honest punchline, not a gap we paper over.

Rigor ledger

Proved (in this chapter).

- Sparsification by a target density: mixed-Poisson degree limit (proposition 27.2) and power-law degrees from a heavy rank-one weight, exponent $\gamma = 1 + 1/\nu$ (corollary 27.3), with sublinear maximum degree (S2); the independence of the unbounded-kernel and sparse-graph axes (remark 27.4).

- The function class of L^p graphons (definition 27.5); the two boundedness criteria, Hilbert–Schmidt (lemma 27.6) and Schur (lemma 27.7); the dichotomy that $p = 2$ is the sharp line, with the Hilbert–Schmidt route strictly dominating Schur and the rank-one breakdown at $f \in L^1 \setminus L^2$ (proposition 27.8).

- GMFG well-posedness for L^2 graphons with $\|\mathbb{W}\| \le \|W\|_{L^2} < \kappa$ (theorem 27.9); the explicit rank-one closure with threshold $\kappa = 1/(s\,\|\mathsf{V}\|)$ and the exact $p = 2 \Leftrightarrow \gamma = 3$ dividing line via the divergence of the susceptibility $\|f\|_{L^2}^2$ (worked computation 27.19).

- The graphex process and its counts: $\mathbb{E}[e(G_\ell)] = \frac{1}{2}\ell^2\,\|W\|_{L^1}$, $\mathbb{E}[N_\ell] = \ell\,\Psi(\ell)$, and subquadratic sparsity $e = o(N^2)$ under a heavy marginal (proposition 27.14); a closed-form heavy-tailed degree distribution with exponent $\gamma = 1 + 1/\zeta \in (1, 2)$ for a pure-power graphex marginal (proposition 27.15), securing (S3).

- The self-contained Poisson-process calculus underlying the above: Campbell/Mecke, thinning, mapping, marking, the Poisson limit theorem, and the mixed-Poisson tail transfer (appendix H).

Assumed (imported).

- The dense-failure propositions and the sparsity preview (sections 19.5 and 19.6 and propositions 19.13 and 19.14); the cut metric (definitions 17.1 and 17.6).

- Hilbert–Schmidt kernel theory and the spectral theorem (definition 2.21, theorems 2.22 and 2.26, and proposition 2.24); the fixed-point theorems — Banach contraction (theorem 2.36) and Schauder (theorem 2.38), the latter applied to MFG existence in chapter 10; Mercer and the spectral ordering (theorem 18.2 and definition 18.1); the rank-one order-parameter reduction (eqs. (18.10) and (18.11), proposition 18.18, and worked computation 18.19).

- The dense GMFG existence/uniqueness theory whose estimates depend on the kernel only through $\|\mathbb{W}\|$ (chapters 22 and 24), and the LQG Thread-A/Thread-B solution and its Volterra response operator V on $C[0,T]$ (eq. (14.13), theorems 14.4 and 14.6, and chapter 22); the horizon-dependent response-Lipschitz constant $L_{\mathcal{R}}(T)$ of the dynamic contraction eq. (24.7).

- Standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), (A-Mon), and the kernel hypothesis (A-Reg-W) in its weakened L^2 form and the smallness (A-Coupling) read as $\|W\|_{L^2} < \kappa$.

Deferred (proved elsewhere, used as structural license).

- The graphex representation theorem (imported result 27.16), proved via Kallenberg's representation of jointly exchangeable random measures in [130, 45, 84]. (The Mecke/-Campbell calculus it and proposition 27.14 use is *not* deferred: it is proved in-book in appendix H; the citation [84, Ch. 4] is for attribution only.)

- The sampling / stretched cut-metric convergence theory and the L^p theory of sparse graph limits (remark 27.17), used only as orientation [25, 26, 27].

Open.

- Existence and uniqueness for GMFG with L^1 couplings and unbounded $\mathbb{W}$ — the scale-free window $\gamma \in (2, 3)$ (proposition 27.11 and worked computation 27.19); the right fixed-point space and a-priori bound are unknown beyond the monotone and finite-rank cases.

- Unconditional well-posedness of GMFG over graphex limits without the Hilbert–Schmidt hypothesis (proposition 27.18); the role of the star and dust components (S, I) of definition 27.12 in the game.

- The sparse analogue of the finite-network convergence theory of chapter 25: an ε-Nash theorem for GMFG strategies on large sparse graphs, and the commutation of the $N \to \infty$ and $\ell \to \infty$ limits. See chapter 31.

Bibliographic notes

The diagnosis that bounded graphons force density, and the resulting program of L^p and graphex limits, is developed in chapter 19 (sections 19.5 and 19.6) following the dense theory of Lovász [99]. The sparsification by a vanishing target density, and the resulting mixed-Poisson / power-law degree structure (proposition 27.2 and corollary 27.3), are the expected-degree model of Chung and Lu [48]; the mixed-Poisson heuristic and its tail transfer are standard in the random-graph literature and we have made the rank-one case explicit.

The L^p theory of graphons and sparse graph convergence is due to Borgs, Chayes, Cohn and Zhao [27]; the operator-theoretic facts we use — the Hilbert–Schmidt bound and the Schur test (lemmas 27.6 and 27.7) — are classical and are proved in chapter 2 (see also [31, 50, 113]). The observation that the GMFG well-posedness theory depends on the kernel only through $\|\mathbb{W}\|$, and hence extends from L^∞ to L^2 graphons verbatim (theorem 27.9), is the natural reading of the contraction arguments of chapter 24; the linear–quadratic graphon game in which this becomes fully explicit is the stochastic graphon game of Aurell, Carmona and Laurière [10], whose analysis is carried out for L^2 (Hilbert–Schmidt) kernels and which our worked computation 27.19 specializes and pushes to the boundary of that class.

The graphex framework originates with Caron and Fox [45] and Veitch and Roy [130]; the representation theorem (imported result 27.16) rests on Kallenberg's theory of exchangeable random measures [84], and the sampling-convergence / stretched-cut-metric foundations are due to Bollobás and Riordan [25] and Borgs, Chayes, Cohn and Holden [26]; the connection of these limits to L^p graphons is [27]. The Poisson-process calculus underlying propositions 27.2, 27.14 and 27.15 and corollary 27.3 — Campbell/Mecke, thinning, mapping, marking, the Poisson limit theorem (Le Cam's coupling bound) and the mixed-Poisson tail transfer — is developed self-containedly in appendix H; the standard references, used only for attribution, are [95, 84] for the point-process calculus and [24] for regular variation. For mean field *systems* (interacting diffusions without control) on graphon and sparse limits, which provide the closest rigorous anchor for the forward dynamics of (27.20), see Bayraktar, Wu and Zhang [16] in the dense/L^2 case and Lacker, Ramanan and Wu [91] for the local-weak-convergence approach on genuinely sparse networks; the latter is, at present, the most promising route toward the open ε-Nash theory on sparse graphs flagged above and in chapter 31. The Erdős–Rényi toy model of Delarue [57] is the historically first mean field game on a (sparse-capable) random graph and remains a useful proving ground. The honest separation of theorem from conjecture that this chapter insists upon is continued, for the whole subject, in the research map of chapter 31.

Chapter 28

Numerical Methods for Mean Field Games

Every theorem in Parts II and IV asserts that an equilibrium *exists*; almost none of them tells us what it *is*. This is worth pausing on, because it marks a turn in the book. The existence theory we have built is powerful but constitutionally non-constructive: the fixed-point arguments of chapters 8 and 9 and the monotonicity method of chapter 11 prove that the set of equilibria is nonempty, and the graphon existence theory of chapter 24 does the same across a kernel, yet a proof that a set is nonempty is not a recipe for a point in it. In exactly one place in the entire book do we know what an equilibrium actually *is*: the linear–quadratic–Gaussian (LQG) island of chapter 14, where the coupled system collapses to a Riccati equation and a Gaussian and the answer fits on a line. Everywhere else the equilibrium is certified to exist and is nonetheless invisible—there is no formula to look at. To *see* such an equilibrium one must *compute* it, and that is the office of this chapter: it is the constructive complement the existence theory has so far lacked, turning the abstract guarantee "a solution exists" into a finite algorithm that manufactures one to any prescribed accuracy.

Outside that LQG island the coupled Hamilton–Jacobi–Bellman–Fokker–Planck (HJB–FP) system has no closed form, and the only handle on it is numerical. This chapter is about how. Its organizing fact is the one the whole book has circled around: an equilibrium is a *forward* equation carrying initial data glued to a *backward* equation carrying terminal data, and that gluing has no initial-value analogue, so it cannot be solved by simply marching a clock. We must instead either iterate a self-consistency loop until the two ends agree, or solve the entire space–time system at once. Both strategies are developed here, with their convergence theory stated honestly, their failure modes named, and their discrete structure shown to mirror the continuous one. A word on that honesty, since it governs the chapter: we prove the structural lemmas that make the schemes correct—the M-matrix and adjointness lemmas of section 28.2—in full and in-book; we pin every passage to the limit on parabolic estimates proved elsewhere in this book rather than waved at; and we quote the literature only for what genuinely belongs to it, the sharp-constant accounting of the convergence rates. Where a result is only conditional, we say so plainly.

The tools we lean on are imported, not re-derived; it is worth naming them as live instruments rather than as a citation dump, because each does specific work below. From chapter 8 we take the two equations of the MFG system and their two readings. The Hamilton–Jacobi–Bellman equation eq. (8.5a) is the backward, value-side half; the Fokker–Planck equation eq. (8.5b) is the forward, population-side half; and the optimal feedback eq. (8.6) is the bridge between them that we will discretize as a single drift. These three are the object every method in the first half of the chapter must solve. The fourth import governs the second half: the variational characterization of *potential* games, theorem 8.11, which recognizes a whole class of MFGs as the optimality system of a single convex minimization—the energy eq. (8.18) over density flows. That one theorem is what powers everything from section 28.4 onward, because it converts "find a fixed point" into "minimize a convex functional," a problem with global, unconditionally convergent algorithms.

We invoke the standing assumptions by their global names, and it helps to recall in a clause what each one buys. (A-Lip) (Lipschitz coefficients) and (A-Growth) (controlled growth) keep the backward and forward solution maps well posed and stable, so that errors propagate boundedly rather than blow up. (A-Nondeg) (uniform nondegeneracy of the diffusion) supplies the parabolic regularization on which every gradient estimate—and hence every passage to the limit—rests; it is also what confines mass and makes a truncated computational domain legitimate. (A-Conv) (convexity of the Hamiltonian in the momentum) is what makes the optimal feedback well defined and the inner Bellman minimization a convex problem with a unique minimizer. (A-Mon) (Lasry–Lions monotonicity of the coupling) buys uniqueness of the equilibrium and, in the potential case, convexity of the energy. For the graphon discretization two further assumptions enter: (A-Reg-W) (a symmetric, bounded, suitably regular kernel, so that $\mathbb{W}$ is self-adjoint Hilbert–Schmidt) makes the coupling quadrature convergent, and (A-Coupling) (operator-norm smallness of $\mathbb{W}$) is the threshold that decides contraction-based uniqueness. We follow throughout the global convention (appendix A): the value $u = u(t, x)$ carries terminal data at $t = T$ and is computed *backward*; the density $m = m(t, x)$ carries initial data at $t = 0$ and evolves *forward*. This forward-m/backward-u split is not a stylistic choice—it is the very obstruction the next section dissects.

The two carried threads ride this machinery to a new, numerical layer. From chapter 14 we carry Thread A, the scalar LQG game, whose closed-form Riccati answer is the yardstick the solver must reproduce; from the rank-one reduction of proposition 18.18 and worked computation 18.19, realized through the matrix template of section 14.7, we carry Thread B, the graphon LQG game, whose coupling collapses to a single scalar order parameter. The closing worked computation (section 28.6) hands both to a black-box nonlinear PDE solver that knows nothing of the LQG ansatz and checks its output against those closed forms, advancing each thread to its fourth (numerical) layer and validating analysis and implementation against each other.

The plan follows the logic of the obstruction. Section 28.1 states it precisely and shows that it forces exactly two algorithmic families—iterate the fixed point, or solve the whole system at once—and that both rest on a single shared building block. Section 28.2 builds that block: the Achdou–Capuzzo-Dolcetta finite-difference scheme, a monotone upwind discretization of the HJB operator whose discrete generator is an honest rate matrix, paired with a conservative Fokker–Planck discretization that is its *exact* transpose—the discrete face of the forward–backward duality—so that monotonicity, mass conservation, and positivity all hold already at the grid level; the section closes with the convergence theorem to viscosity and weak solutions. With the block in hand, section 28.3 wraps it in the outer loop, contrasting damped fixed-point (Picard / fictitious play) iteration with all-at-once Newton and the contraction theory that decides when each is the right tool. Section 28.4 then exploits the potential structure: for games that are convex minimizations it develops the variational and primal–dual methods, whose unconditional convergence outruns the fixed point, and exhibits their kinship with optimal-transport solvers. Section 28.5 lifts the whole apparatus to the kernel by adding a type grid feeding the state grid, realizing the operator $\mathbb{W}$ as a quadrature—the numerical face of graphon coarse-graining—reducing a block graphon to a finite multi-population game, and tying the quadrature error back to the regularity theory of chapter 19. Section 28.6 is the pay-off: the worked computation that reproduces the closed forms of both threads. Algorithmic templates appear inline as numbered procedures; the full reference implementation lives in `figures/28-mfg-solver.py`.

Notation watch

This chapter carries two integer indices that a physicist must not conflate. A superscript n is a *time* index, $t_n = n\,\Delta t$; a subscript i is a *state-grid* index, $x_i = x_{\min} + ih$; and in the graphon sections a subscript k is a *type-grid* index, $x_k \in [0,1]$. Thus u_i^n is the discrete value at state x_i, time t_n, and $m_i^{k,n}$ is the discrete density of type x_k at state x_i, time t_n (the type-indexed field m^{x_k} of chapter 18). The grid spacings are h (state) and Δt (time); h is *not* the terminal cost $g = g$ of earlier chapters, which does not appear discretized here. Brownian motion is absent from the PDE formulation, so no W/B clash arises until the graphon kernel W enters in section 28.5, where W is always the kernel. Two further letters carry a second, deliberately *local* meaning, flagged here once: w is the momentum field $w = m\alpha$ of section 28.4 but a quadrature weight w_l in section 28.5; and $\mathcal{P}$ denotes a type *partition* in section 28.5.2, the stepping-operator partition of definition 19.7, not the measure space $\mathcal{P}$. The control is $\alpha = \alpha$ throughout, *never* a (which is the Thread A state-feedback parameter of eq. (28.1)).

28.1 The core difficulty: a forward–backward boundary-value problem

The preamble stated our organizing fact as a slogan: an equilibrium is a forward equation glued to a backward equation, and that gluing has no initial-value analogue. A slogan is not yet something one can act on. This section turns it into a precise statement—about which data sit at which end of the time interval, and why the two ends are mutually entangled—and then proves the operationally decisive consequence: that the glued system cannot be solved by marching a clock. Out of that single fact the entire taxonomy of methods will fall, because there are exactly two ways to confront a problem one cannot integrate in time, and the rest of the chapter is the elaboration of those two ways. It is worth being slow and concrete here, since every later section inherits the structure laid bare in this one.

Recall the mean field game system in the separable form of chapter 8: on $[0,T] \times \mathbb{R}^d$,

$$-\partial_t u - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 u\big) + H(x,\nabla u) = F[m_t](x), \qquad u(T,x) = G[m_T](x), \tag{28.1a}$$

$$\partial_t m - \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \mathrm{D}^2 m\big) - \operatorname{div}\big(m\,\partial_p H(x,\nabla u)\big) = 0, \qquad m(0,\cdot) = m_0, \tag{28.1b}$$

with the optimal feedback $\alpha^*(t,x) = -\partial_p H(x, \nabla u(t,x))$ closing the loop (eq. (8.6)). Taken one at a time, each line of eq. (28.1) is a textbook object. The first, eq. (28.1a), is a backward Hamilton–Jacobi–Bellman equation: given a known right-hand side it is solved *backward* from terminal data at $t = T$, and it returns the value u of the representative agent's control problem—through the feedback $\alpha^* = -\partial_p H$, the best a single agent can do against a *prescribed* environment. The second, eq. (28.1b), is a forward Fokker–Planck (continuity) equation: given a known drift it is solved *forward* from initial data at $t = 0$, and it transports the population density m along that drift. Neither equation is hard in isolation. The difficulty lives entirely in how the two feed each other.

Trace the two arrows explicitly, because seeing both is the whole point. *Population into value:* the agent does not optimize in a vacuum; the cost it faces depends on where everyone else is. That dependence enters eq. (28.1a) through the couplings $F[m_t]$ and $G[m_T]$—a running interaction cost set by the present density and a terminal interaction cost set by the final density. The entire time-profile of the population, $t \mapsto m_t$, sits inside the source of the HJB equation, and through it inside the value u and the feedback. *Value into population:* the agents then act on that value. The optimal feedback $\alpha^* = -\partial_p H(x, \nabla u)$ is built from the value's gradient, and it is exactly this feedback that appears as the drift transporting the density in eq. (28.1b) (the term $\partial_p H(x, \nabla u)$ inside the divergence). So ∇u steers the flow of m, while m sources the equation that determines u. The loop is closed and runs in both directions at once, and this mutual dependence is precisely what the word *equilibrium* means here: a population whose induced individually-optimal behavior reproduces that very population.

What makes the loop genuinely awkward, rather than merely coupled, is *where the data live*. The two equations are anchored at *opposite ends* of the time interval. The density carries initial data: $m(0,\cdot) = m_0$ is given at $t = 0$, the starting configuration of the crowd. The value carries terminal data: $u(T,\cdot) = G[m_T]$ is fixed at $t = T$, the end-of-game payoff. One condition is pinned at the start of time, the other at the end, and the coupling forces both to be satisfied at once. This is a two-point *boundary*-value problem in time—a boundary-value problem whose "boundary" is the pair of temporal endpoints $\{0, T\}$—and not the initial-value problem that the word "evolution" would lead one to expect.

Heuristic (non-rigorous)

Why one cannot march the clock. Try, in imagination, to do the obvious thing: integrate eq. (28.1) as an ordinary initial-value problem, stepping forward one small interval at a time from $t = 0$. To take even the first step—to push the density m from $t = 0$ to $t = \Delta t$—we need the drift that transports it, $\partial_p H(x, \nabla u(0,\cdot))$, and hence we need the value u, and its gradient, at the initial time. But u at $t = 0$ is not ours to choose. It is determined by *integrating the backward equation eq.* (28.1a) *all the way down from* $t = T$, and that backward integration cannot even begin until two things are known: the terminal data $u(T,\cdot) = G[m_T]$, which needs the *final* density m_T, and the source $F[m_t]$ at every intermediate time, which needs the *entire future history* of the density. In other words, to compute the drift that produces the future of m, we must already know that future. Nor is anything gained by starting from the other end. To march u backward from $t = T$ we must first know m_T to fix the terminal payoff, and m_T is the *output* of the forward flow we have not run—which itself needed u.

Each end demands, as its input, precisely what the other end is supposed to produce as its output. Neither can take its first step alone; the clock has no direction in which it can be marched. This is the exact sense in which the forward–backward gluing "has no initial-value analogue."
The reader should resist reading this as a numerical inconvenience to be engineered away. It is not an artifact of any discretization, nor of any particular solver; it is the defining structure of an equilibrium. An equilibrium is, by definition, a fixed configuration in which anticipation of the future (the backward value) and propagation of the past (the forward density) are mutually consistent across the whole interval at once. A method that could simply march the clock would be solving a *different*, causal problem—one agent reacting to an exogenously given crowd—not the self-consistent one we are after. Every algorithm in this chapter therefore confronts the circularity head-on rather than pretends it away.

The circularity admits exactly two responses, and the entire chapter is organized around the choice between them. Either we break the loop deliberately—freeze one of the two unknowns, solve the now-one-directional problems, and update the frozen guess until the freeze becomes self-consistent—or we refuse to break it and instead solve for the whole forward–backward field in a single simultaneous stroke. The first is an iteration over guesses; the second is one large nonlinear solve. We take them in turn.

Strategy I: decouple and iterate (the outer fixed point). The first response breaks the loop on purpose. Freeze a candidate population flow—an entire time-profile of densities, $\bar{m} = (\bar{m}_t)_{t\in[0,T]}$, our current guess for what the crowd does. The instant this flow is held fixed, the entanglement dissolves, because the couplings stop being unknowns and become data. The running source $F[\bar{m}_t]$ and the terminal payoff $G[\bar{m}_T]$ are now *known functions of time and state*: $F[\bar{m}_t](x)$ is simply a prescribed function of (t,x), and $G[\bar{m}_T](x)$ a prescribed terminal function of x, each computed once from the frozen flow and thereafter referring to nothing unknown. With those in hand—and, in the implementation, after restricting to a bounded computational domain, which turns the couplings into honest bounded continuous functions and yields what we will call the *cut-off problem* of theorem 28.3—the HJB equation eq. (28.1a) is a genuine terminal-value problem with a given right-hand side. We solve it backward from $t = T$, exactly as one solves any single-agent control problem against a prescribed environment, and read off the optimal feedback $\alpha^* = -\partial_p H(x, \nabla u)$. That feedback is, in turn, a known drift. Substituting it into the Fokker–Planck equation eq. (28.1b) produces a genuine initial-value problem, which we solve forward from $t = 0$ to obtain a *new* population flow m'—the crowd that actually results when every agent best-responds to the frozen guess. We have replaced one entangled forward–backward problem by two one-directional solves chained in series: one backward, then one forward.

This chain defines a map from flows to flows,

$$\mathsf{S}:\ \bar{m}\ \longmapsto\ m', \qquad \mathsf{S}:\ C\big([0,T];\mathcal{P}_2(\mathbb{R}^d)\big) \to C\big([0,T];\mathcal{P}_2(\mathbb{R}^d)\big), \tag{28.2}$$

the best-response operator of section 8.4. The type signature deserves to be read aloud rather than skipped past. An element of $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$ is a whole *path of probability measures*: a curve

$t \mapsto m_t$ that assigns to each instant a probability distribution on $\mathbb{R}^d$ of finite second moment, and that varies continuously in the Wasserstein-2 distance $\mathcal{W}_2$. The input of S is such a curve (the frozen guess), and so is its output (the resulting flow); thus S maps the space of population histories into itself. In this language an equilibrium is nothing other than a flow that reproduces itself under best response—a *fixed point* $\bar{m} = \mathsf{S}(\bar{m})$. Each *evaluation* of S costs exactly two ordinary PDE solves (one backward HJB, one forward FP); the equilibrium is then found by *iterating* S—feeding its output back as its next input—until the flow stops moving. The catch, taken up in section 28.3, is that this iteration converges only when S is a contraction, which it need not be; its appeal is that the per-step cost is low and the memory footprint is a single flow at a time. This is the method built in sections 28.2 and 28.3.

Strategy II: solve the whole system at once. The second response refuses to break the loop at all. Rather than alternating two one-directional solves, discretize the coupled system eq. (28.1) on a single space–time grid and ask for *all* of its unknowns at once—every discrete value u_i^n and every discrete density m_i^n, at every grid node and every time level—as the solution of one large nonlinear algebraic system. Stacking the values into a vector $U = (u_i^n)$ and the densities into $M = (m_i^n)$, the discretized equations become a single root-finding problem

$$\mathcal{R}(U, M) = 0, \tag{28.3}$$

whose two block-rows are the discrete HJB and the discrete Fokker–Planck residuals and into which the boundary data at the two temporal ends are built directly. There is no marching and no freezing: the forward–backward coupling is imposed everywhere simultaneously, and one hands the whole system to a nonlinear solver—Newton's method (section 28.3) or, when the game is potential, a convex-optimization algorithm that recognizes eq. (28.3) as the optimality condition of a single convex program (section 28.4). What this costs is *memory*: the unknown is now the entire space–time field (U, M) held in store at once, rather than one time-slice marched along, and a Newton step couples all of it. What it buys is *robustness*. Because it never iterates S, it does not depend on S contracting; it can converge in regimes where Strategy I oscillates or diverges—near the multiplicities and conjugate horizons where the best-response loop breaks down. Strategy II trades storage for unconditional reach.

The two strategies look different on the surface—an iterated pair of one-way solves versus one monolithic root-find—but they rest on the *same* foundation, and naming it sets the agenda for the next section. Both must discretize the individual backward (HJB) and forward (FP) equations, and both need those discretizations to be accurate (consistent with the PDEs as the grid is refined) and stable (errors not amplified). There is, however, a third and subtler requirement that neither strategy can do without: the two discretizations must be *mutually consistent*. Concretely, the discrete forward operator must be the *exact adjoint*—the transpose—of the discrete backward operator. This is the discrete fingerprint of the continuous forward–backward duality we have just anatomized, and we will see that it is exactly what makes mass conservation, positivity, the discrete maximum principle, and (in the potential case) the discrete convexity all hold at the grid level rather than only in the limit; it is also what renders the Newton system of Strategy II tractable. Because this adjoint pairing is the keystone on which everything downstream rests, we build that shared block first, before either outer strategy—the finite-difference discretization of the HJB–Fokker–Planck system, to which we now turn.

28.2 Finite-difference schemes for the HJB–Fokker–Planck system

The previous section closed with a promise: whichever of the two strategies one adopts—iterate the best-response fixed point, or solve the whole space–time system at once—the first thing to build is the *shared block* both of them stand on, a discretization of the individual backward (HJB) and forward (Fokker–Planck) equations that is accurate, stable, and—the subtle third demand—mutually consistent, meaning the discrete forward operator is the exact adjoint of the discrete backward one. That block is what this section constructs. It pays to keep the three demands in view as we go, because they are not independent wishes to be balanced against one another but a single design problem with one solution. *Accuracy* is consistency in the numerical-analyst's sense: as the grid is refined the discrete operators converge to the differential ones. *Stability* is the requirement that errors not be amplified from one step to the next, so that local truncation error does not compound into global garbage. *Mutual consistency* is the structural fingerprint of the continuous forward–backward duality dissected in section 28.1: the same first-order transport drift that steers the value backward must, transposed, steer the density forward, and we shall arrange the discretization so that this holds *exactly* at the grid level rather than only in the refinement limit. Everything downstream—mass conservation, positivity, the discrete maximum principle, the tractability of the Newton system, and in the potential case the discrete convexity—will turn out to be a corollary of getting that one adjoint relationship right.

A word on dimension, to set expectations before the indices arrive. We discretize in one state dimension for transparency; this is a genuine simplification of *exposition*, not of substance. The construction is dimension-by-dimension identical in $\mathbb{R}^d$—each first-order derivative is upwinded and each second-order derivative is centered, coordinate by coordinate, and the multidimensional stencil is the tensor assembly of the one-dimensional ones—and, crucially, the two structural lemmas that carry the whole chapter (the M-matrix property and the adjointness/conservation property) are stated and proved *dimension-free*: they are statements about a generator matrix with nonnegative off-diagonals and zero row sums, and nothing in them counts dimensions. So nothing is lost by reading the scalar case and everything transfers. Accordingly, fix a bounded computational interval $[x_{\min}, x_{\max}]$—the truncation is legitimate because (A-Nondeg) confines essentially all mass to a bounded region, a point we return to once the scheme is in place—a state step $h > 0$ with nodes $x_i = x_{\min} + ih$, and a time step $\Delta t = T/N_T$ with nodes $t_n = n\,\Delta t$, $n = 0, \ldots, N_T$. We write $u_i^n \approx u(t_n, x_i)$ for the discrete value and $m_i^n \approx m(t_n, x_i)$ for the discrete density, the superscript indexing time and the subscript indexing the state grid, exactly as fixed in the chapter's notation watch.

Why discretize the two equations *separately but consistently*, rather than reaching at once for some monolithic coupled stencil? Because the two halves of the system have opposite characters and want opposite numerical treatments, and pretending otherwise throws away the very structure that makes the schemes correct. The backward HJB equation is a fully nonlinear, first-order-dominated equation whose solutions are generically only *viscosity* solutions (chapter 6); what one must preserve there is the comparison principle, and the discrete property that encodes it is *monotonicity.* The forward Fokker–Planck equation is a linear conservation law for a *density*; what one must preserve there is total mass and positivity. These are different fidelity requirements, served by different design choices—and yet they must lock together, because the drift in the forward equation is built from the gradient of the backward solution. The Achdou–Capuzzo-Dolcetta resolution, which organizes the next two subsections, is to honor each requirement with its own scheme and then to weld the two by a single algebraic device. First we build a *monotone* backward scheme: we discretize the Hamiltonian's transport term by *upwinding*—taking the difference in the direction the drift comes from—which is exactly the choice that keeps the discrete generator a bona fide rate matrix and hence the scheme monotone, so that the Barles–Souganidis route to viscosity solutions is open. Then we build the forward scheme not from scratch but as the *transpose* of that backward generator: by fiat, the discrete Fokker–Planck operator is the adjoint of the discrete HJB operator assembled from the same optimal drift. That one stroke makes mass conservation, positivity, and adjointness fall out together, for free, as we will prove—no separate fight for each. The build order is thus fixed by the logic: the backward generator is the primitive object, the forward operator is its shadow, and the "mutual consistency" demanded above is not something we check after the fact but something we *install by construction.* We turn first, then, to the backward scheme, where upwinding is the move that buys monotonicity.

28.2.1 A monotone scheme for the backward HJB equation

The preceding section fixed the build order: the backward generator is the primitive object, the forward operator its transpose, and the structural lemma that everything downstream rests on is the one we must secure first. So we begin where the logic insists, with a backward scheme for the HJB equation eq. (28.1a), and the non-negotiable demand on it is *monotonicity.* The reason sits at the level of what an HJB solution even *is.* Outside the LQG island the equation eq. (28.1a) is fully nonlinear and first-order-dominated; its solutions are generically not classical but only *viscosity* solutions (chapter 6), and the single property that pins a viscosity solution down uniquely is the *comparison principle*: if one sub/supersolution lies below another at the terminal time, it lies below it everywhere. Any honest discretization must reproduce that ordering at the grid level, and the theorem that tells us exactly which discretizations do is the Barles–Souganidis criterion—an approximation scheme that is *monotone* (its update preserves the partial order on grid functions), *stable* (its solutions stay bounded uniformly in the grid size), and *consistent* (its local truncation error vanishes under refinement) converges, locally uniformly, to the unique viscosity solution. We will verify stability and consistency in passing when we assemble the convergence theorem (theorem 28.3); the delicate one, and the one that dictates the entire design, is monotonicity.

Monotonicity is best read as the *discrete shadow of the comparison principle.* Concretely, the one-step backward map must have the property that raising the input value field pointwise can only raise the output—no entry of the update may depend with the *wrong sign* on a neighboring value. Whether this holds is decided entirely by the sign pattern of the discrete generator, and it is exactly here that the obvious discretization fails. Approximate the first-order transport term $d(x)\,\partial_x\phi$ by a *centered* difference $d_i(\phi_{i+1}-\phi_{i-1})/2h$ and add the centered diffusion $\frac{\sigma^2}{2}(\phi_{i+1}-2\phi_i+\phi_{i-1})/h^2$; the coefficient multiplying the left neighbor ϕ_{i-1} is then $-d_i/2h+\sigma^2/2h^2$, which turns *negative* as soon as $d_i > 0$ and the grid is coarse enough that $h > \sigma^2/d_i$—that is, once the dimensionless cell Péclet number $h\,|d_i|\,/\sigma^2$ exceeds one. A negative off-diagonal entry is precisely a wrong-sign dependence on a neighbor: the update can *lower* the output where the input rose, the discrete comparison principle breaks, and the scheme manufactures spurious oscillations that the continuous maximum principle forbids. The cure is *upwinding*: difference the transport term in the direction the drift *comes from*—a forward difference where the flow points right, a backward difference where it points left—so that the advective contribution only ever *adds* to the coupling with the upstream neighbor and never produces a negative off-diagonal, for any grid however coarse. Upwinding is thus not a numerical nicety; it is the discrete enforcement of the comparison principle, and the rest of this subsection makes that statement precise and quantitative.

To upwind we must first know which way the drift points, and that drift is itself produced by the Hamiltonian. So before discretizing anything we make the variational structure of H explicit, because it both delivers the drift and certifies the convexity on which the inner minimization will depend. Write the Hamiltonian of a control-affine problem in its variational form

$$H(x,p) = \sup_{\alpha\in A} \big\{ -b(x,\alpha)\cdot p - L(x,\alpha)\big\}, \tag{28.4}$$

the convention of chapters 6 and 8 (eq. (8.4)): H is the Legendre-type transform of the running cost along the controlled drift, and as a supremum of functions *affine* in p it is automatically *convex* in p—which is the content of (A-Conv) and the reason the optimal feedback below is well defined and unique. The maximizer in eq. (28.4) is the optimal control as a function of the momentum, and differentiating the optimized expression recovers the drift: $\partial_p H(x,p) = -b(x,\alpha^*(x,p))$, so that the closed-loop transport velocity in the Fokker–Planck equation is exactly $-\partial_p H$.

It is worth running the LQG thread through eq. (28.4) in full, both to see the machine turn and because the closed form is the yardstick the solver must reproduce. With $b(x,\alpha) = ax + b\alpha$ and $L(x,\alpha) = \frac{1}{2}\alpha^2$ (the data of Thread A, eq. (28.1)), the bracket in eq. (28.4) is

$$-b(x,\alpha)\,p - L(x,\alpha) = -(ax+b\alpha)\,p - \tfrac{1}{2}\alpha^2,$$

a concave quadratic in α; setting its α-derivative $-b\,p-\alpha$ to zero gives the minimizer

$$\alpha^* = -b\,p. \tag{28.5}$$

Substituting α^* back,

$$H(x,p) = -(ax + b(-bp))\,p - \tfrac{1}{2}(bp)^2 = -ax\,p + b^2p^2 - \tfrac{1}{2}b^2p^2 = -ax\,p + \tfrac{1}{2}b^2p^2,$$

convex in p as required. Two continuous objects fall straight out of this, and we will meet both again in discretized form. First, the Fokker–Planck drift is $\partial_p H = -ax + b^2 p = -(ax - b^2 p)$, so the closed-loop velocity transporting the density is $-\partial_p H = ax - b^2 p$—which is just the state dynamics $ax + b\alpha^*$ with $\alpha^* = -bp$ inserted, exactly as it should be. Second, feed H into the HJB equation eq. (28.1a) with the quadratic ansatz $u(t,x) = \frac{1}{2}P(t)x^2 + \dots$ (curvature P plus lower-order terms) and the running cost $F = \frac{q}{2}x^2$. Then $\nabla u = Px$, $\partial_t u = \frac{1}{2}\dot{P}x^2 + \dots$, and collecting the coefficient of x^2 in $-\partial_t u - \frac{\sigma^2}{2}\partial_{xx}u + H(x,\nabla u) = F$ gives

$$-\tfrac{1}{2}\dot{P} - aP + \tfrac{1}{2}b^2P^2 = \tfrac{q}{2},$$

the x^2-balance, in which the diffusion contributes only to the lower-order terms. Multiplying by -2 rearranges this into the Riccati equation

$$\dot{P} = b^2 P^2 - 2aP - q \tag{28.6}$$

of chapter 14. This is the closed-form answer section 28.6 will reproduce numerically from a solver that knows none of this algebra; recording the substitution here is what makes that later agreement a genuine check rather than a coincidence.

With the drift in hand we can discretize. Given any feedback policy $\alpha(\cdot)$, the linear operator inside the optimization eq. (28.4) is the closed-loop generator $\mathcal{A}_\alpha \phi = d_\alpha(x)\,\partial_x \phi + \frac{\sigma^2}{2}\partial_{xx}\phi$ with drift $d_\alpha(x) = b(x, \alpha(x))$ (for the optimal feedback, $d_{\alpha^*} = -\partial_p H$). We discretize $\mathcal{A}_\alpha$ by the *upwind* stencil

$$(\mathcal{A}_h)_{i,i-1} = -\frac{(d_i)^-}{h} + \frac{\sigma^2}{2h^2}, \quad (\mathcal{A}_h)_{i,i+1} = \frac{(d_i)^+}{h} + \frac{\sigma^2}{2h^2}, \quad (\mathcal{A}_h)_{ii} = -\frac{(d_i)^+ - (d_i)^-}{h} - \frac{\sigma^2}{h^2}, \tag{28.7}$$

where $d_i = d_\alpha(x_i)$, $(d)^+ = \max(d, 0)$, $(d)^- = \min(d, 0)$. Read eq. (28.7) term by term. The advective part discretizes $d_i\,\partial_x \phi$ as $(d_i)^+(\phi_{i+1} - \phi_i)/h + (d_i)^-(\phi_i - \phi_{i-1})/h$: where the drift points *right*, $(d_i)^+ > 0$ multiplies a *forward* difference, depositing $(d_i)^+/h$ on the right-neighbor coupling $(\mathcal{A}_h)_{i,i+1}$; where it points *left*, $(d_i)^- < 0$ multiplies a *backward* difference, depositing $-(d_i)^-/h \geq 0$ on the left-neighbor coupling $(\mathcal{A}_h)_{i,i-1}$. In either case the off-diagonal advective contribution is ≥ 0—this is precisely the upwind choice that the centered difference failed to make. Superposed on it is the *symmetric* diffusion piece $\frac{\sigma^2}{2}\partial_{xx}\phi \approx \frac{\sigma^2}{2}(\phi_{i+1} - 2\phi_i + \phi_{i-1})/h^2$, contributing the same $\sigma^2/2h^2 \geq 0$ to *both* neighbors and $-\sigma^2/h^2$ to the diagonal. The diagonal entry $-((d_i)^+ - (d_i)^-)/h - \sigma^2/h^2 = -\left|d_i\right|/h - \sigma^2/h^2$ is exactly the negative of the sum of the two off-diagonals, so each interior row sums to zero. Denote by $\mathcal{A}_h$ the resulting tridiagonal matrix. Its defining structural properties—nonnegative off-diagonals, zero row sums—are the engine of the whole chapter, and we now prove they make $\mathcal{A}_h$ a bona fide rate matrix.

Lemma 28.1 (The discrete generator is a rate matrix and $I - \Delta t\,\mathcal{A}_h$ is an M-matrix)**.** *For every $h > 0$ and every bounded drift d, the matrix $\mathcal{A}_h$ of eq. (28.7) has nonnegative off-diagonal entries, nonpositive diagonal entries, and zero interior row sums; that is, $\mathcal{A}_h$ is the generator (rate matrix) of a continuous-time Markov chain on the grid. Consequently, for every $\Delta t > 0$ the matrix $I - \Delta t\,\mathcal{A}_h$ is a nonsingular M-matrix: it has positive diagonal, nonpositive off-diagonals, and $(I - \Delta t\,\mathcal{A}_h)^{-1} \geq 0$ entrywise.*

Proof. The argument has three stages: $\mathcal{A}_h$ is a rate matrix; $B := I - \Delta t\,\mathcal{A}_h$ is strictly diagonally dominant, hence nonsingular; and $B^{-1} \geq 0$ entrywise.

Stage 1: $\mathcal{A}_h$ is a rate matrix. The off-diagonal entries of eq. (28.7) are the superdiagonal $(d_i)^+/h + \sigma^2/2h^2$ and the subdiagonal $-(d_i)^-/h + \sigma^2/2h^2$. Each is a sum of nonnegative terms—$(d_i)^+ \geq 0$ and $-(d_i)^- \geq 0$ by definition of the positive and negative parts, and $\sigma^2 \geq 0$—so both off-diagonals are ≥ 0. The diagonal was constructed as the negative of their sum, so it is ≤ 0 and each interior row sums to exactly zero. A matrix with nonnegative off-diagonals and zero row sums is, by definition, the infinitesimal generator (rate matrix) of a continuous-time Markov chain on the grid: off-diagonal entries are transition rates to neighbors and the diagonal is the total exit rate. This is the structural fact everything else leans on.

Stage 2: B is strictly diagonally dominant, hence nonsingular. Form $B = I - \Delta t\,\mathcal{A}_h$. Its diagonal is $B_{ii} = 1 - \Delta t(\mathcal{A}_h)_{ii} \geq 1 > 0$ (subtracting the nonpositive $(\mathcal{A}_h)_{ii}$ can only increase it), its off-diagonals are $B_{i,i\pm1} = -\Delta t(\mathcal{A}_h)_{i,i\pm1} \leq 0$, and—because the interior rows of $\mathcal{A}_h$ sum to zero—the interior rows of B sum to exactly 1. Hence for each interior row the diagonal strictly dominates the off-diagonal mass, $B_{ii} = 1 + \Delta t\,|(\mathcal{A}_h)_{ii}| = 1 + \sum_{j\neq i}|B_{ij}| > \sum_{j\neq i}|B_{ij}|$. Strict diagonal dominance forces nonsingularity (the Levy–Desplanques theorem): by Gershgorin every eigenvalue λ lies in some disc $|\lambda - B_{ii}| \leq \sum_{j\neq i}|B_{ij}| < B_{ii}$, and since each such disc is centered at $B_{ii} > 0$ with radius strictly smaller than B_{ii}, none of them contains the origin; so 0 is not an eigenvalue and B is invertible.

Stage 3: $B^{-1} \geq 0$ entrywise. This is the property that powers discrete positivity in lemma 28.2, so we prove it directly by a Neumann series rather than invoke a characterization of M-matrices. Set $s := \max_i B_{ii} > 0$ and split off the diagonal scale, $B = s\,(I - C)$ with $C := I - s^{-1}B$. We check $C \geq 0$ entrywise: its diagonal entry $1 - s^{-1}B_{ii} \geq 0$ because $B_{ii} \leq s$, and its off-diagonal entries $-s^{-1}B_{ij} \geq 0$ because the off-diagonals $B_{ij} \leq 0$ and $s > 0$. Next we bound the row sums of C. Using $\sum_j C_{ij} = 1 - s^{-1}\sum_j B_{ij}$ and the interior row sum $\sum_j B_{ij} = 1$, we get $\sum_j C_{ij} = 1 - s^{-1} < 1$, and since $C \geq 0$ this row sum equals the absolute row sum, so the induced ∞-norm satisfies $\|C\|_\infty = \max_i \sum_j C_{ij} < 1$. A matrix of ∞-norm strictly below one has a convergent Neumann series $\sum_{k\geq0} C^k = (I - C)^{-1}$ (the partial sums are Cauchy in the ∞-norm). Every power C^k is a product of nonnegative matrices and is therefore nonnegative, so the limit is a convergent sum of nonnegative matrices and is itself ≥ 0; hence $B^{-1} = s^{-1}(I - C)^{-1} = s^{-1}\sum_{k\geq0} C^k \geq 0$ entrywise. Thus B is a nonsingular M-matrix with nonnegative inverse. Equivalently—and this is the form we use repeatedly—the discrete maximum principle holds: $Bu \leq Bv$ entrywise implies $u = B^{-1}(Bu) \leq B^{-1}(Bv) = v$, the order being preserved because $B^{-1} \geq 0$. □

The discrete generator $\mathcal{A}_h$ depends on the policy α through the drift d_α, and the backward step must choose that policy optimally—this is where the nonlinearity of the HJB equation re-enters the otherwise linear stencil. We discretize the time derivative *fully implicitly* (backward Euler), which is what lemma 28.1 was tailored to and what makes the scheme unconditionally monotone, with no stability restriction tying Δt to h. With the source $f_i^n := F[m_{t_n}](x_i)$ frozen from the current population, and writing the running-cost contribution $\ell_i(\alpha) := L(x_i, \alpha)$, the implicit backward step from level $n+1$ to level n is the coupled pair

$$(I - \Delta t\,\mathcal{A}_h^{(\alpha)})\,u^n = u^{n+1} + \Delta t(\ell(\alpha) + f^n), \qquad \alpha_i^{(\text{new})} = \arg\min_\alpha \{\, b(x_i, \alpha)\,(\nabla_h u^n)_i + L(x_i, \alpha)\,\}, \tag{28.8}$$

in which the value and the optimal policy must agree. We solve it by *Howard's method* (policy iteration): alternate the two halves of eq. (28.8)—a linear solve with the policy held fixed, then a pointwise update of the policy with the value held fixed—until the policy stops changing. Lemma 28.1 guarantees each linear solve is well posed (the M-matrix is invertible) and monotone (its inverse is nonnegative). It remains to see that the alternation actually converges, and converges fast; both follow from the same M-matrix structure, and we record the argument in-book rather than quote it.

Monotone convergence. Run the iteration and let α, α' be two successive policies with u, u' the corresponding solves of the linear half of eq. (28.8). Evaluate the linear operator built from the *new* policy α' at the *old* value u, and form the residual

$$r := (I - \Delta t\,\mathcal{A}_h^{(\alpha')})\,u - (u^{n+1} + \Delta t(\ell(\alpha') + f^n)).$$

The policy update is the pointwise minimizer of the Bellman bracket at u; switching from α to α' can therefore only *lower* the right-hand side it is compared against, which is exactly the statement $r \le 0$ entrywise (at the old policy the residual would have been zero by definition of u). Now apply the inverse, which is entrywise nonnegative by lemma 28.1: since u' solves $(I - \Delta t\, \mathcal{A}_h^{(\alpha')})u' = u^{n+1} + \Delta t(\ell(\alpha') + f^n)$, subtracting gives $(I - \Delta t\, \mathcal{A}_h^{(\alpha')})(u - u') = r \le 0$, and multiplying by the nonnegative inverse yields $u - u' \le 0$, i.e. $u' \le u$ entrywise. Each sweep thus produces a value no larger than the last: the iterates form a monotone non-increasing sequence. They are bounded below (by the unique solution of the coupled system, itself a fixed point of the alternation), and a monotone bounded sequence converges; at the limit the policy update leaves the value unchanged, which is precisely the coupled condition eq. (28.8).

Superlinear convergence. Monotonicity alone would only promise convergence; in fact it is fast, and the reason is that Howard's method *is* the semismooth Newton iteration for the nonsmooth equation $\min_\alpha\{\cdots\} = 0$. The discrete Bellman map $u \mapsto \min_\alpha\{(I - \Delta t\, \mathcal{A}_h^{(\alpha)})u - \cdots\}$ is piecewise linear in u: the state space of values is partitioned into finitely many cells, one per policy, on each of which the minimizing α is constant and the map reduces to a single linear (M-matrix) system. Newton's method on such a piecewise-linear map takes each linear piece exactly. As the iterates approach the solution they enter and stay in the cell of the optimal policy—the *active policy stabilizes*—and once it has stabilized the next linear solve is the limiting linear system itself, solved in one step. Hence the iteration terminates after finitely many policy changes, with the local superlinear (here, finite) convergence characteristic of semismooth Newton, rather than the merely geometric rate a contraction would give.

For the LQG thread the policy update is explicit, which is why the inner loop is essentially free. Minimizing the bracket $b(x_i, \alpha)(\nabla_h u^n)_i + \frac{1}{2}\alpha^2 = (ax_i + b\alpha)(\nabla_h u^n)_i + \frac{1}{2}\alpha^2$ over α sets its derivative $b(\nabla_h u^n)_i + \alpha$ to zero, giving

$$\alpha_i = -b\,(\nabla_h u^n)_i, \tag{28.9}$$

the discrete analogue of the continuous feedback eq. (28.5) with p replaced by the upwind gradient $(\nabla_h u^n)_i$. Because this closed form makes each policy update a single pointwise formula, two or three Howard sweeps drive the residual to machine precision in practice.

Boundary truncation. One loose end remains before the backward generator can be handed to the forward scheme: we discretized on a bounded interval $[x_{\min}, x_{\max}]$, whereas the equations eq. (28.1) live on all of $\mathbb{R}^d$, and we owe an account of what is lost at the artificial boundary and how the boundary rows are closed. The justification that the truncation is legitimate is the parabolic regularization of (A-Nondeg). Uniform nondegeneracy means $\sigma^2 \ge \sigma^2_{\min} > 0$, so the Fokker–Planck flow is genuinely diffusive with a confining drift; its solution inherits *Gaussian tails*, in the sense that the mass it places more than k standard deviations from its mean decays like $e^{-k^2/2}$. Concretely, if $[x_{\min}, x_{\max}]$ is taken several standard deviations beyond the support of the initial datum m_0, the fraction of mass that ever reaches the boundary is of this exponentially small order $e^{-k^2/2}$ in the number k of standard deviations of clearance. Everything dynamically relevant happens in the interior where the mass lives; the value's far-field growth (quadratic in the LQG case, where $u \sim \frac{1}{2}Px^2$) multiplies a density that has already vanished to exponential precision, so it cannot influence the equilibrium. This is the precise sense in which (A-Nondeg) "confines essentially all mass to a bounded region" and makes the cut-off problem of theorem 28.3 a faithful proxy for the whole-space one.

It remains to close the boundary rows. On the value we impose a homogeneous Neumann condition $u_0^n = u_1^n$, $u_N^n = u_{N-1}^n$, a zero-gradient reflection consistent with the flat far field. On the density two closures are available, and—a point that matters for the conservation statement of the next subsection—we deliberately use *different* ones in the analysis and in the implementation. The *reflecting* (zero-flux) closure sets the boundary fluxes to zero, equivalently completes $\mathcal{A}_h$ so that the boundary rows, like the interior rows, sum to zero; with every row of $\mathcal{A}_h$ summing to zero the total mass is then conserved *exactly* at the discrete level (lemma 28.2(1)). That exactness is why the analysis adopts the reflecting closure: it lets the conservation lemma be stated as a clean identity rather than an estimate. The implementation, by contrast, uses the simpler *absorbing* closure $m_0^n = m_N^n = 0$ followed by a renormalization of the total mass after each forward step. The absorbing closure is trivial to code and keeps the matrix structure untouched, at the cost of discarding the boundary tail flux; but by the Gaussian-tail estimate just made that discarded flux is exponentially small in the clearance k, so the renormalization correction it must repair is of that same negligible size and restores the unit mass to machine precision. The two closures therefore agree to exponential accuracy: the reflecting scheme conserves mass identically, the absorbing-plus-renormalization scheme conserves it up to the exponentially small tail correction, and either keeps every linear system tridiagonal.

We have what the next subsection needs. The backward sweep produces, at each time level, the optimal feedback α^* and hence the optimal drift $d^* = b(\cdot, \alpha^*)$, from which we assemble the discrete generator $\mathcal{A}_h^{(\alpha^*)}$ of eq. (28.7)—an honest rate matrix, with $I - \Delta t\, \mathcal{A}_h^{(\alpha^*)}$ a nonsingular M-matrix carrying the inverse-nonnegativity and discrete maximum principle of lemma 28.1, and equipped with a boundary closure (reflecting for the analysis, absorbing-plus-renormalize in code). The Fokker–Planck scheme will not build a new operator at all: by the Achdou–Capuzzo-Dolcetta design principle it simply *transposes* $\mathcal{A}_h^{(\alpha^*)}$ and lets the rate-matrix structure we have just established deliver mass conservation and positivity for free. To that transposition we now turn.

28.2.2 A conservative, adjoint Fokker–Planck scheme

The backward sweep handed us its principal product: the discrete generator $\mathcal{A}_h^{(\alpha^*)}$ assembled from the optimal drift $d^* = b(\cdot, \alpha^*)$, an honest rate matrix carrying the M-matrix machinery of lemma 28.1. The forward half of the system now asks us to transport the population along that very drift. It is tempting to repeat the previous subsection's labor—write an upwind stencil for the Fokker–Planck operator from scratch, fight separately for each property we need—but the Achdou–Capuzzo-Dolcetta insight is that no second construction is necessary. We have already built exactly the right matrix; the forward operator is its transpose, and the work is to recognize that the transpose is correct, not to assemble it anew.

What does "correct" demand here? The two equations discretize objects of different kind, and that difference dictates different fidelity. The backward HJB equation propagates a *value*, a function to be compared, and what its scheme had to protect was the order relation—monotonicity, the discrete comparison principle. The forward Fokker–Planck equation eq. (28.1b) propagates a *density*:

it transports and diffuses a probability measure, and a probability measure is a nonnegative object of unit total mass. Two properties are therefore non-negotiable for any honest forward scheme. *Positivity*: a density that starts nonnegative must stay nonnegative, because a negative "probability" is meaningless and, worse, would feed a meaningless coupling $F[m]$ back into the HJB equation and corrupt the entire fixed point. *Mass conservation*: the continuity equation eq. (28.1b) has no source or sink—it merely redistributes probability—so $\int m \, dx$ must remain equal to one for all time, and a scheme that leaks or manufactures mass is solving a different equation. These are exactly the structures the discrete scheme must reproduce at the grid level, not merely recover in the refinement limit.

The decisive design principle of Achdou–Capuzzo-Dolcetta is to secure both, together with the adjointness that ties the two equations into a faithful discrete forward–backward system, by a single algebraic stroke: take the discrete forward operator to be the *exact transpose of the discrete backward generator built from the same optimal drift.* Concretely, having computed the optimal feedback α^* (hence d^*) from the HJB step, form $\mathcal{A}_h^{(\alpha^*)}$ as in eq. (28.7)—the rate matrix is the same object that drove the value backward—and step the density implicitly by

$$\left(I - \Delta t \, (\mathcal{A}_h^{(\alpha^*)})^\top\right) m^{n+1} = m^n. \tag{28.10}$$

The implicit (backward-Euler) time stepping mirrors the HJB step and shares its unconditional stability; the genuinely new content is the single transpose $(\cdot)^\top$.

Why should one transpose simultaneously buy all three of positivity, mass conservation, and adjointness? Because each property is a statement about a different structural feature of the rate matrix, and transposition preserves all three at once. Mass conservation is a statement about the *row sums* of $\mathcal{A}_h$: that they vanish, which transposes into a left-eigenvector identity for the all-ones vector and pins the total mass. Positivity is a statement about the *M-matrix* structure of $I - \Delta t \, \mathcal{A}_h$: that its inverse is entrywise nonnegative, a property the transpose inherits verbatim because a matrix and its transpose share their inverse's sign pattern. Adjointness is the very *definition* of the transpose, and it is what makes the discrete pair an honest dual system rather than two unrelated discretizations that happen to share a drift. No one of these would follow from merely "discretizing the Fokker–Planck operator accurately"; all three follow at once the moment we insist the forward operator be the literal transpose of the backward one. This single choice secures the three properties together, which we now prove.

Lemma 28.2 (Discrete conservation, positivity, and adjointness)**.** *Let $\mathcal{A}_h$ be a discrete generator as in lemma 28.1 with zero interior row sums, and consider the forward step eq.* (28.10) *with reflecting (zero-flux) boundary closure. Then:*

1. (Mass conservation.) *Total mass is exactly preserved, $\sum_i m_i^{n+1} = \sum_i m_i^n$.*
2. (Positivity.) *If $m^n \geq 0$ then $m^{n+1} \geq 0$.*
3. (Adjointness.) *The forward step eq.* (28.10) *is the exact transpose of the linearized backward step eq.* (28.8)*; equivalently, the discrete HJB and FP operators are in duality, $\langle (I - \Delta t \, \mathcal{A}_h) u, \, m \rangle = \langle u, (I - \Delta t \, \mathcal{A}_h^\top) m \rangle$.*

Proof. We take the three properties in turn; each is a one-line consequence of a different structural feature of $\mathcal{A}_h$, and the proof is really an exercise in reading off which feature powers which property.

(1) *Mass conservation is the row-sum identity, transposed.* Let $\mathbf{1}$ denote the all-ones vector. The interior rows of $\mathcal{A}_h$ sum to zero by construction (eq. (28.7): the diagonal is minus the sum of the two off-diagonals), and the reflecting (zero-flux) closure was chosen precisely so that the boundary rows are *also* completed to sum to zero—a reflecting boundary deposits no probability current through the wall, so the exit rate it removes from a boundary node is returned to the interior rather than lost. Hence *every* row of $\mathcal{A}_h$ sums to zero, which is the full-vector identity

$$\mathcal{A}_h\mathbf{1} = 0,$$

valid on all rows and not merely the interior ones—this is exactly where the analysis's choice of the reflecting closure earns its keep, since the absorbing closure would leave boundary rows with a nonzero deficit. Transposing this identity moves it from the rows of $\mathcal{A}_h$ to the columns of $\mathcal{A}_h^\top$: $\mathcal{A}_h\mathbf{1} = 0$ reads, entry by entry, as "each row sums to zero," which is the same set of scalar equations as $\mathbf{1}^\top\mathcal{A}_h^\top = 0$, "each column of $\mathcal{A}_h^\top$ sums to zero." Therefore $\mathbf{1}^\top\mathcal{A}_h^\top = 0$, and consequently

$$\mathbf{1}^\top(I - \Delta t\,\mathcal{A}_h^\top) = \mathbf{1}^\top - \Delta t\,\mathbf{1}^\top\mathcal{A}_h^\top = \mathbf{1}^\top :$$

the all-ones vector is a left null vector of $\mathcal{A}_h^\top$ and hence a left eigenvector of the forward step matrix with eigenvalue 1. Total mass is the pairing $\mathbf{1}^\top m = \sum_i m_i$, so applying $\mathbf{1}^\top$ on the left of the forward step eq. (28.10) gives

$$\mathbf{1}^\top m^n = \mathbf{1}^\top(I - \Delta t\,\mathcal{A}_h^\top)m^{n+1} = \mathbf{1}^\top m^{n+1},$$

that is, $\sum_i m_i^{n+1} = \sum_i m_i^n$. Mass is preserved *exactly*, at every step, for every Δt and h—not to truncation order but identically. It is worth pausing on the mechanism: the property of the *value* scheme that we never used downstream, namely that the generator's rows sum to zero (a probabilistic conservation: total transition rate out balances rate in), is precisely the property that, transposed, conserves the *mass* of the density. The two equations share one matrix, and one of its features serves each.

(2) *Positivity is the M-matrix property, transposed.* By lemma 28.1 the matrix $I - \Delta t\,\mathcal{A}_h$ is a nonsingular M-matrix, with entrywise nonnegative inverse $(I - \Delta t\,\mathcal{A}_h)^{-1} \geq 0$. A matrix and its transpose have the same determinant and the same minors, so $\mathcal{A}_h^\top$ has the identical sign pattern (nonnegative off-diagonals, zero row sums) and $I - \Delta t\,\mathcal{A}_h^\top$ is again a nonsingular M-matrix; indeed its inverse is literally the transpose of the inverse, $(I - \Delta t\,\mathcal{A}_h^\top)^{-1} = \big((I - \Delta t\,\mathcal{A}_h)^{-1}\big)^\top$, and transposing a nonnegative matrix leaves it nonnegative. Therefore the forward step

$$m^{n+1} = \big(I - \Delta t\,\mathcal{A}_h^\top\big)^{-1}m^n$$

maps a nonnegative vector to a nonnegative vector: if $m^n \geq 0$ entrywise then $m^{n+1} \geq 0$ entrywise, because a nonnegative matrix applied to a nonnegative vector cannot produce a negative entry. Positivity of the density is thus inherited, with no separate argument, from the discrete maximum principle of the backward scheme. The same implicit step that made the value scheme monotone makes the density scheme positivity-preserving; these are one structural fact seen from two sides.

(3) *Adjointness is the definition of the transpose—but it says something.* That the forward step eq. (28.10) is the exact transpose of the linearized backward step eq. (28.8) is, at the level of matrices,

a tautology: we *defined* the forward operator to be $(\mathcal{A}_h^{(\alpha^*)})^\top$. What the tautology encodes, and what makes it worth stating as a property rather than a convention, is the discrete duality pairing. Writing $\langle u, m\rangle = \sum_i u_i m_i$ for the Euclidean pairing of a value vector against a density vector, the defining identity of the transpose reads

$$\langle (I - \Delta t\, \mathcal{A}_h)\, u,\ m\rangle = \langle u,\ (I - \Delta t\, \mathcal{A}_h^\top)\, m\rangle \qquad \text{for all } u, m.$$

This is the discrete avatar of the integration-by-parts identity $\int (\mathcal{A}\phi)\, m = \int \phi\, (\mathcal{A}^* m)$ that defines the continuous Fokker–Planck operator $\mathcal{A}^*$ as the formal adjoint of the generator $\mathcal{A}$ (chapter 8). The continuous forward–backward duality—value paired against density, generator against its adjoint—is reproduced *verbatim* at the grid level, with no error term. That is the precise sense in which the discrete system is a faithful, and not merely accurate, copy of the continuous one: the same vector that the backward operator pushes against the density is the vector the density pushes back against, exactly. We will draw on this pairing immediately below, and again when it makes the Newton system of section 28.3.2 self-adjoint. □

Lemma 28.2(3) is more than bookkeeping, and it pays to say now what it will buy two sections hence, because it is the single thread that ties this finite-difference block to the variational methods of section 28.4. Recall the structural fact of chapter 8: for a *potential* (variational) game the continuous forward–backward system eq. (28.1) is not an arbitrary coupled PDE but the Euler–Lagrange/optimality system of a single convex minimization, the energy functional of eq. (8.18) minimized over density flows (theorem 8.11). In such a system the value u and the density m are not independent unknowns: u is the Lagrange multiplier (the adjoint state) enforcing the continuity constraint on m, and the HJB equation is exactly the stationarity condition that the multiplier must satisfy. Adjointness is the algebraic signature of precisely this relationship—multiplier paired against constraint, generator against its formal adjoint. The discrete pairing $\langle (I - \Delta t\, \mathcal{A}_h)u, m\rangle = \langle u, (I - \Delta t\, \mathcal{A}_h^\top)m\rangle$ is what guarantees that the *discretized* HJB and Fokker–Planck equations stand in that same multiplier/constraint relation, so that the discrete coupled system eq. (28.8)–eq. (28.10) is itself the optimality system of a discrete *convex* program—a faithful discrete avatar of theorem 8.11. A discretization that got the forward operator merely "accurate" but not exactly transposed would break this: its discrete equations would be the optimality system of nothing, the energy would fail to decrease monotonically, and the convex-optimization algorithms of section 28.4 would lose the very structure they exploit. The structure-preservation of the scheme is therefore not a fortunate accident but the discrete shadow of the continuous variational geometry, and we plant that observation here for section 28.4 to harvest. With positivity, conservation, and adjointness in hand for the building block, we can state what the assembled scheme converges to.

Theorem 28.3 (Convergence of the finite-difference MFG scheme)**.** *Assume (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), and that F, G are continuous and bounded on $\mathcal{P}_2(\mathbb{R}^d)$ for the cut-off problem. Let (u_i^n, m_i^n) solve the coupled discrete system eq.* (28.8)*–eq.* (28.10) *(with the outer loop of section 28.3 driven to convergence). Then, as $(h, \Delta t) \to 0$, the discrete value converges locally uniformly to the viscosity solution of the HJB equation eq.* (28.1a) *and the discrete density converges weakly-$*$ to the distributional solution of the Fokker–Planck equation eq.* (28.1b)*; the pair (u_i^n, m_i^n) converges to a solution of the MFG system eq.* (28.1)*. The scheme is first-order accurate in h and Δt.*

Proof sketch. The argument has two distinct movements that the reader should keep separate. First, with the population *frozen*, each one-directional scheme converges on its own—the backward scheme to a viscosity solution, the forward scheme to a measure solution. Second, the *coupled* limit is taken: one shows the two limits are compatible, i.e. that the frozen-population limits assemble into a genuine solution of the MFG system. We treat them in order, and we are careful throughout to mark which input is proved in-book and which is cited only for sharp-constant accounting.

Step 1: the frozen-population HJB limit (viscosity convergence). Hold the population, hence the source $f_i^n = F[m_{t_n}](x_i)$ and terminal data $G[m_{t_N}]$, fixed. We must verify the three hypotheses of the Barles–Souganidis criterion for the backward scheme eq. (28.8). *Monotonicity* is lemma 28.1: the M-matrix structure is exactly the discrete comparison principle, so raising the input value field can only raise the output. *Stability* is the L^∞ bound that the same M-matrix structure furnishes: applying $\mathbf{1}^\top$-type estimates to $(I - \Delta t\, \mathcal{A}_h^{(\alpha)})u^n = u^{n+1} + \Delta t(\ell(\alpha) + f^n)$, and using that the M-matrix inverse has ∞-norm at most 1 (its rows sum to 1 and it is nonnegative), gives $\|u^n\|_\infty \le \|u^{n+1}\|_\infty + \Delta t\, C$ with C controlled by the bounded running cost; iterating over the $N_T = T/\Delta t$ steps yields a bound uniform in $(h, \Delta t)$. *Consistency* is a Taylor expansion of the upwind stencil eq. (28.7): replacing a centered by an upwind difference adds an artificial-diffusion term of size $O(h)$ to the truncation error, $\frac{1}{2}h\ |d_i|\ \partial_{xx}\phi + O(h^2)$, which vanishes as $h \to 0$. The decisive point is that consistency comes with *no* stability restriction tying Δt to h: because the step is fully implicit it is unconditionally monotone (lemma 28.1 holds for every $\Delta t > 0$), so one may send $h \to 0$ and $\Delta t \to 0$ independently—no parabolic ratio condition $\Delta t/h^2$ or $\Delta t/h \to 0$ is needed, in sharp contrast to an explicit scheme. With monotonicity, stability, and consistency established, the Barles–Souganidis monotone-scheme theorem—proved in-book as theorem E.20 (originally [13])—gives local-uniform convergence of u_h to the unique viscosity solution of eq. (28.1a).

Step 2: the frozen-feedback FP limit (weak-$*$ convergence). Now hold the value/feedback fixed and run the forward scheme eq. (28.10). By lemma 28.2 it conserves mass and preserves positivity exactly, so each m^n is a genuine probability vector; reading it as a sum of point masses $\sum_i m_i^n\, \delta_{x_i}$ gives a discrete probability measure. A consistent, monotone, mass-conserving discretization of a linear Fokker–Planck equation converges weakly-$*$ (tested against continuous bounded functions) to the distributional/measure solution of eq. (28.1b); the positivity and conservation are what keep the discrete measures *probability* measures along the way, so the limit is again a probability measure and not a signed object.

Step 3: closing the coupling. The outer loop of section 28.3 drives the frozen guess to self-consistency, so along its converged iterates the source feeding Step 1 is the density produced by Step 2 and vice versa. What remains is to show that the joint limit (u_h, m_h) is a solution of the coupled system eq. (28.1), and this is the genuinely coupled part of the argument. It rests on three in-book inputs, which we separate because they do distinct work and rely on distinct machinery.

(i) Compactness on the density side (tightness and Prokhorov). The discrete densities all carry unit mass, but unit mass alone does not prevent probability from escaping to infinity in the limit; we need *tightness*, a uniform control on how much mass can sit far out. This is supplied by the confining drift guaranteed by (A-Nondeg): the uniform nondegeneracy gives genuine diffusion with a restoring drift, so a uniform second-moment bound holds along the flow and, by Chebyshev, the mass beyond radius R is uniformly small as $R \to \infty$. The family $\{m_h\}$ is then tight, and Prokhorov's

theorem—proved in-book as theorem 3.16—extracts a subsequence converging weakly-$*$, $m_h \xrightarrow{*} m$, to a probability measure m. This input is *in-book*.

(ii) Compactness on the value side (parabolic estimates plus Arzelà–Ascoli). To pass to the limit in the drift we need not just $u_h \to u$ but $\nabla u_h \to \nabla u$, and locally uniformly, because the drift depends on the gradient. Pointwise or weak convergence of u_h would not suffice—it would not control ∇u_h. The needed compactness comes from parabolic regularity: (A-Nondeg) makes the backward equation uniformly parabolic, so the uniform first-derivative (Bernstein) gradient bound theorem E.17 and the interior Schauder estimate theorem E.19 together give a uniform $C^{1,2}_{\mathrm{loc}}$ bound on the discrete (or interpolated) values, independent of $(h, \Delta t)$. A uniform $C^{1,2}_{\mathrm{loc}}$ family is precompact: by Banach–Alaoglu in the relevant dual space and Arzelà–Ascoli for the equicontinuous derivatives—both packaged in-book as theorem 2.16—a further subsequence has $u_h \to u$ with $\nabla u_h \to \nabla u$ locally uniformly. These regularity estimates are *in-book* (appendix E); it is here, and only here, that the parabolic theory is doing the heavy lifting, which is why the chapter insists on proving that package rather than quoting it.

(iii) Passing to the limit in the nonlinear drift (where adjointness earns its place). The two compactness inputs now combine. The Fokker–Planck coupling is nonlinear in the value through the drift $\partial_p H(x, \nabla u_h)$; the locally uniform convergence $\nabla u_h \to \nabla u$ from input (ii), together with the continuity of $\partial_p H$ from (A-Conv), lets one pass to the limit inside this nonlinearity, so the limiting drift is $\partial_p H(x, \nabla u)$ tested against the limiting measure m from input (i). The reason the limit is a *solution*—rather than an object satisfying the equation only up to an unaccounted boundary or duality error—is that at every grid level the discrete weak formulation is an exact duality pairing, $\langle (I - \Delta t\, \mathcal{A}_h) u_h, m_h \rangle = \langle u_h, (I - \Delta t\, \mathcal{A}_h^\top) m_h \rangle$, by lemma 28.2(3). This discrete integration-by-parts has no remainder, so sending $h \to 0$ in it reproduces *exactly* the continuous duality $\int (\mathcal{A}\phi)\, m = \int \phi \, \mathrm{d}(\mathcal{A}^* m)$ that defines a weak solution of the coupled system. The adjoint structure is thus not decoration but the mechanism that makes the limit a bona fide solution rather than an approximate one.

Assembling the three movements: Steps 1–2 give convergence of each one-directional scheme, and Step 3, on inputs (i)–(iii), shows the joint limit solves eq. (28.1). Uniqueness of that limit under (A-Mon) is the Lasry–Lions monotonicity argument imported in remark 28.4. The whole limit passage is therefore complete from in-book parabolic estimates (appendix E) and compactness theorems (chapters 2 and 3); what we do not reproduce, and follow Achdou–Capuzzo-Dolcetta and Achdou–Porretta [1, 3, 2] for, is the sharp first-order accounting of the $O(h) + O(\Delta t)$ rate—the optimal constants, not the convergence itself. □

Remark 28.4 (What is genuinely proved versus quoted)**.** It is worth drawing the rigor boundary explicitly, because this scheme is where the chapter's stated doctrine—prove the structure, pin the limit to in-book estimates, quote only sharp constants—is put into practice, and the reader should be able to see exactly which sentence above carries which kind of warrant. The accounting has three tiers, in decreasing order of self-containedness.

Genuinely proved, in full and from scratch: the two structural lemmas that make the scheme *correct as a structure*—monotonicity of the backward step (lemma 28.1, via the M-matrix/Neumann-series argument given there in complete detail) and conservation, positivity, and adjointness of the forward step (lemma 28.2, each a one-line consequence of a feature of the rate matrix). These are the facts on which every good property of the scheme rests, and they are proved here with nothing left to the reader.

Assembled in-book from proved ingredients: the convergence of the *coupled* system (theorem 28.3). We do not re-prove the heavy theorems it uses, but every one of them is proved elsewhere *in this book* and imported by label—the monotone-scheme convergence theorem (theorem E.20), the uniform parabolic gradient and interior Schauder estimates (appendix E, theorems E.17 and E.19), and the compactness theorems of chapters 2 and 3 (Banach–Alaoglu/Arzelà–Ascoli theorem 2.16, Prokhorov theorem 3.16). Uniqueness of the limit under (A-Mon) follows from the comparison and monotonicity machinery of chapter 24. The proof sketch above is honest in the technical sense of the style bible: it is a complete argument modulo results the reader can turn to and find proved, not an appeal to authority.

Quoted to a specific source: only the optimal-constant accounting of the convergence rate—the sharp first-order estimate with its best constants—for which we follow Achdou–Porretta [3] (and the original scheme of Achdou–Capuzzo-Dolcetta [1]). This is genuinely a piece of the literature, an analytic refinement of the rate rather than the existence of convergence, and quoting it is appropriate; it is the one place a citation stands in for an argument we do not give. Expansion clarifies this boundary; it does not move it—nothing that was a quote has been silently upgraded to a claimed in-book proof, and the load-bearing convergence statement remains assembled from in-book parabolic estimates, with the citation confined to the constant. This is the line the rigor doctrine draws, and the rest of the chapter respects it.

Algorithmic template (one HJB–FP sweep). We can now read off exactly what it costs to evaluate the best-response operator S of eq. (28.2) once. The two scheme components assemble, in the order the forward–backward structure dictates, into a single such evaluation: feed in a population flow, sweep the HJB equation backward to extract the optimal feedback, then sweep the Fokker–Planck equation forward along that feedback to produce the population the agents actually generate. Concretely:

1. **Input**: a population flow $(m^n)_{n=0}^{N_T}$—the frozen guess $\bar{m}$ of section 28.1.

2. **Backward sweep** ($n = N_T - 1, \dots, 0$): set the terminal value $u^{N_T} = G[m^{N_T}]$ from the frozen final density; at each n, solve the implicit HJB step eq. (28.8) by Howard iteration, storing the optimal feedback $\alpha^{*,n}$ and value u^n. This is the only nonlinear part, and the superlinear convergence of Howard's method (and, for the LQG thread, the closed-form policy update eq. (28.9)) keeps it cheap.

3. **Forward sweep** ($n = 0, \dots, N_T - 1$): set the initial density $m^0 = m_0$; at each n, form the rate matrix $\mathcal{A}_h^{(\alpha^{*,n+1})}$ from the feedback just stored and solve the transposed forward step eq. (28.10); renormalize to repair the exponentially small absorbing tail of section 28.2.1. Conservation and positivity (lemma 28.2) guarantee the output is a genuine probability flow.

4. **Output**: the new flow (m^n) and the value field (u^n).

The cost is worth stating precisely, since it is what the outer loop will multiply. Each of the two sweeps visits the N_T time levels once, and at each level solves a tridiagonal linear system in the N_x state unknowns. A tridiagonal solve is $O(N_x)$ by the Thomas algorithm—the M-matrix structure of lemma 28.1 guarantees the elimination is stable with no pivoting—so each sweep costs $O(N_T \cdot N_x)$ arithmetic, and the two sweeps together cost $O(N_T N_x)$, multiplied only by the small, grid-independent number of Howard iterations the backward sweep needs (two or three in practice for the LQG thread). One evaluation of S is thus *linear* in the total number of space–time grid points $N_T N_x$—as cheap as a single pair of one-directional parabolic solves can be, which is the whole payoff of installing the structure rather than fighting an ill-conditioned dense system.

This is one evaluation of S, and one evaluation is not yet a solution: it produces the crowd that best-responds to a *guess*, not the self-consistent crowd that reproduces itself. The forward–backward obstruction of section 28.1 has been converted, but not dissolved—we have a cheap, structure-preserving, $O(N_T N_x)$ machine for evaluating S exactly once, and finding the equilibrium means finding a *fixed point* of that machine. Wrapping this single sweep in an outer iteration—and deciding, per the Strategy I/Strategy II split of section 28.1, whether to iterate the best response or to solve the whole system at once by Newton—is the business of section 28.3, to which we now turn.

28.3 The outer loop: fixed-point iteration and Newton

The machine of section 28.2 does exactly one thing: it evaluates the best-response operator S once, taking a frozen population flow in and returning the crowd that best-responds to it, at the linear cost $O(N_T N_x)$ we just tallied. That is a tool, not yet an answer. The equilibrium is not any one best response; it is the flow that reproduces *itself* under best response—the fixed point $\bar{m} = \mathsf{S}(\bar{m})$ of eq. (28.2)—and a single sweep no more locates that fixed point than one evaluation of a function locates its root. This section is where the single sweep becomes a solver: we wrap it in an outer loop that drives the residual $\bar{m} - \mathsf{S}(\bar{m})$ to zero. The forward–backward obstruction of section 28.1 has by now been converted into a concrete, structure-preserving evaluation; the business that remains is to dissolve it, by finding the self-consistent flow.

There are exactly two ways to do so, and they are precisely the Strategy I/Strategy II dichotomy that section 28.1 forced out of the structure of the problem—now to be realized as running code. The two parts of this section treat them in turn, and the reader should weigh the choice between them as a genuine trade-off, because neither dominates the other across all regimes.

Strategy I—decouple and iterate—takes the cheapness of the sweep at face value. If one evaluation of S costs only $O(N_T N_x)$, then the most direct thing to do is to evaluate it again and again, feeding each output back as the next input, $m^{(k+1)} = \mathsf{S}(m^{(k)})$, and hope the sequence of flows settles onto a fixed point. This is the discrete face of the existence-by-iteration arguments of chapters 8 and 10, and its appeal is exactly its low per-step cost and minimal bookkeeping: one holds a single flow in memory and advances it by one cheap sweep at a time. The catch is that the iteration converges only when S *contracts*—when best-responding to two nearby flows yields two flows that are nearer still—and that is a genuine hypothesis, not a free lunch. It holds when the coupling is weak or monotone or the horizon short, and it *fails* near the multiplicities and conjugate horizons where the population's anticipation of itself is strong enough to overshoot: there the plain iteration oscillates between competing flows or diverges outright, exactly the regime in which the equilibrium is most interesting. Even when contraction holds it can be marginal, and the raw iteration crawls; the standard repair is to *damp* the update—to step only part of the way from the old flow toward its best response—which both widens the basin of convergence and is the numerical form of fictitious play. Section 28.3.1 develops this damped Picard iteration and makes the contraction condition quantitative.

Strategy II—solve the whole system at once—is the answer to what Strategy I cannot reach. When the best-response map refuses to contract, no amount of damping a fixed-point iteration will manufacture a contraction that physically is not there; one must stop iterating a map and instead solve, in a single stroke, the coupled algebraic system $\mathcal{R}(U, M) = 0$ of eq. (28.3) for the entire space–time field at once. The tool for that is Newton's method, whose local convergence is quadratic and—crucially—does not require S to contract; it requires only that the residual's Jacobian be nonsingular at the solution, a far weaker and generically satisfied condition. What this robustness costs is what Strategy II always costs: memory and per-step work. The unknown is now the whole stacked field (U, M) rather than one marched flow, and each Newton step solves one large linear system coupling all of it—a system whose tractability, we will see, rests on the very adjoint M-matrix structure installed in section 28.2. The reward is reach: Newton converges in the near-critical regimes where Picard breaks down, at the price of a larger, globally coupled solve. Section 28.3.2 assembles this all-at-once Newton iteration on the stacked residual.

The trade-off the reader is about to weigh is therefore contraction against globalization: a cheap, low-memory iteration that converges only inside the contraction regime, versus an expensive but robust solve that reaches the near-critical regime where the first one fails. In practice the two are complementary—a few damped Picard sweeps from a crude initial guess, handed to Newton for the final high-accuracy convergence, marries the global reach of the former to the local speed of the latter—and the worked computation of section 28.6 will exercise exactly this combination. We begin, as the cheapness of the sweep invites, with the fixed-point iteration.

28.3.1 Damped fixed-point (Picard) iteration

The outer-loop overview has left us with a single cheap instrument and an invitation to use it repeatedly. One evaluation of the best-response map S—built in section 28.2 from one backward HJB sweep and one forward Fokker–Planck sweep, each an $O(N_T N_x)$ pass over the grid—takes a frozen population flow and returns the flow that results when every agent best-responds to it. Strategy I takes that cheapness at face value: rather than do anything clever, simply feed the output back as the next input and watch the sequence of flows,

$$m^{(k+1)} = \mathsf{S}(m^{(k)}), \qquad k = 0, 1, 2, \ldots,$$

the plain *Banach iteration* of S. This is exactly the discrete realization of the existence-by-iteration arguments of chapters 8 and 10: an equilibrium is a fixed point $\bar{m} = \mathsf{S}(\bar{m})$, and one hunts for it by iterating the map. The whole question is *when the sequence settles*. Banach's fixed-point theorem answers it in one word—when S is a *contraction*, that is, when best-responding to two nearby flows yields two flows that are strictly nearer still—and then it answers it completely: the iterates converge geometrically, from any starting flow, to the unique fixed point. So the entire viability of Strategy I turns on whether S contracts on $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$, and on how to repair the iteration when it does not. Those are the two questions of this subsection. The general theory of chapters 10 and 11 supplies a clean sufficient condition for the first.

Proposition 28.5 (Contraction of the best-response map)**.** *Under (A-Lip), (A-Nondeg), (A-Conv), suppose the couplings F, G are Lipschitz in $\mathcal{W}_2$ with constants L_F, L_G, and either the horizon T or the product $L_F + L_G$ of coupling strengths is small enough that the composite Lipschitz constant of S on $C([0,T];\mathcal{P}_2(\mathbb{R}^d))$ is $L_{\mathsf{S}} < 1$. Then S has a unique fixed point, and the Picard iterates converge geometrically, $\mathcal{W}_2$-distance contracting by L_{S} per step.*

Proof sketch. The estimate is bookkeeping on the two-stage anatomy of S already exhibited in section 28.1: $\mathsf{S} = \mathcal{S}_{\mathrm{FP}} \circ \mathcal{S}_{\mathrm{HJB}}$, where $\mathcal{S}_{\mathrm{HJB}}$ sends a frozen flow to the optimal feedback it induces (one backward solve) and $\mathcal{S}_{\mathrm{FP}}$ sends that feedback to the population it transports (one forward solve). A contraction constant for the composite is the product of contraction constants for the two stages, and the content of the proposition is that each stage carries a factor one can make small. We take them in turn.

Stage one: the value map $\mathcal{S}_{\mathrm{HJB}}$ is Lipschitz with constant $\propto (L_F + L_G)$. Take two frozen flows $\bar{m}, \bar{m}'$ with $\sup_t \mathcal{W}_2(\bar{m}_t, \bar{m}'_t) \le \delta$. They enter the HJB equation eq. (28.1a) only through its data: the running source $F[\bar{m}_t]$ and the terminal payoff $G[\bar{m}_T]$. Lipschitz couplings convert the flow discrepancy directly into a data discrepancy—$\|F[\bar{m}_t] - F[\bar{m}'_t]\|_\infty \le L_F\,\delta$ and $\|G[\bar{m}_T] - G[\bar{m}'_T]\|_\infty \le L_G\,\delta$ pointwise in time. What the feedback needs, however, is not the value but its gradient, since $\alpha^* = -\partial_p H(x, \nabla u)$ and (by (A-Conv), $\partial_p H$ Lipschitz) the feedback depends Lipschitz-continuously on ∇u. The crucial input is therefore a stability estimate for ∇u in terms of the data, and this is exactly what (A-Nondeg) buys: the uniform parabolic regularization makes the backward equation smoothing, and the Bernstein-type gradient bound theorem E.17 controls $\|\nabla u\|$—and, applied to the difference of two solutions, controls $\|\nabla u - \nabla u'\|$—by the accumulated data difference. Integrating the source discrepancy backward from T down to t contributes a factor that grows with the length of that backward horizon; schematically $\|\nabla u - \nabla u'\| \lesssim C_{\mathrm{HJB}}(T)\,(L_F + L_G)\,\delta$, with $C_{\mathrm{HJB}}(T) \to 0$ as $T \to 0$ because there is then no horizon over which the data error can accumulate. This is the first small factor.

Stage two: the flow map $\mathcal{S}_{\mathrm{FP}}$ is Lipschitz, with a constant carrying its own factor of T. Now feed two nearby feedbacks—equivalently two nearby drifts $d^*, d^{*\prime}$ with $\sup \|d^* - d^{*\prime}\| \le \zeta$—into the forward equation eq. (28.1b) and compare the transported populations. This is a stability question for a continuity equation under a drift perturbation, and the cleanest route is the characteristics: under (A-Lip) the drift is Lipschitz in x, so two trajectories started from the same point but driven by drifts differing by ζ separate at a rate controlled by Grönwall's inequality (appendix C.1). Integrating the drift discrepancy along the flow and closing with Grönwall gives a position spread, hence a Wasserstein spread, of order $\mathcal{W}_2(m', m'') \lesssim C_{\mathrm{FP}}(T)\,\zeta$ with $C_{\mathrm{FP}}(T) \sim T\,e^{L_A T}$: a factor T because the drift error must act over the interval before it becomes a displacement, times a Grönwall amplification $e^{L_A T}$ set by the spatial Lipschitz constant L_A of the drift. Again $C_{\mathrm{FP}}(T) \to 0$ as $T \to 0$. This is the second small factor.

Composing. Chaining the two estimates, a δ-perturbation of the input flow produces an output flow perturbed by at most

$$L_{\mathsf{S}}\,\delta, \qquad L_{\mathsf{S}} \;=\; C_{\mathrm{FP}}(T)\,C_{\mathrm{HJB}}(T)\,(L_F + L_G),$$

so S is Lipschitz with the displayed constant. Two independent knobs drive L_{S} below 1. Shrinking the *coupling* $L_F + L_G$ makes the agents care little about the crowd, so a change in the crowd barely moves the best response. Shrinking the *horizon* T makes both $C_{\mathrm{HJB}}(T)$ and $C_{\mathrm{FP}}(T)$ small, because over a short interval neither the backward data error nor the forward drift error has time to compound. Either way, once $L_{\mathsf{S}} < 1$ Banach's theorem delivers a unique fixed point and geometric convergence at rate L_{S}. The sharp constants are tracked in chapter 10; here only the structural shape matters—contraction is a smallness condition on (coupling)$\times$(horizon). $\square$

It is worth making the word that organizes the next paragraph precise. Call the problem *subcritical* when the data sit in the regime the proposition certifies—$L_{\mathsf{S}} < 1$, the coupling-times-horizon product below the threshold at which the best-response loop becomes a contraction. The name is chosen deliberately to match its physical meaning. As one turns up the coupling or lengthens the horizon, L_{S} climbs toward 1; the *critical* locus is where it reaches 1 and the linearized best-response operator $D\mathsf{S}$ acquires a unit-modulus eigenvalue, so that $\mathrm{Id} - D\mathsf{S}$ becomes singular and the fixed point ceases to be locally unique. For the LQG thread this locus is computable in closed form: it is exactly the family of *conjugate horizons* of theorem 14.11, the times at which the Riccati flow blows up and the linearized self-consistency map degenerates. Below the first conjugate horizon the problem is subcritical and plain Picard contracts; *at* it the linearization is singular; *beyond* it the linear convergence factor exceeds 1 and the plain iteration no longer settles. Concretely, near criticality the dominant eigenvalue of $D\mathsf{S}$ is real and slightly above 1 (the iterate slides monotonically away) or complex with modulus above 1 (the iterate spirals out), so plain Picard either diverges outright or oscillates between two competing flows without ever resolving between them—precisely the regime in which multiple equilibria appear and the problem is most interesting.

The standard repair is *damping*: rather than jump all the way to the best response, step only part of the way toward it, replacing the bare update by the convex combination

$$m^{(k+1)} = (1-\rho_k)\,m^{(k)} + \rho_k\,\mathsf{S}(m^{(k)}), \qquad \rho_k \in (0,1]. \tag{28.11}$$

The mechanism is under-relaxation. Where the bare map overshoots its target—$D\mathsf{S}$ has an eigenvalue of modulus above 1 in the relevant direction—the damped map $\mathrm{Id} + \rho_k(D\mathsf{S} - \mathrm{Id})$ has, in that same direction, the eigenvalue $1 + \rho_k(\lambda - 1)$, which for a real λ slightly above 1 is pulled back below 1 once ρ_k is small enough, and for an oscillatory λ shrinks the spiral. A small step trades speed for stability: it cannot make a non-contraction contract in every direction, but it tames the mild overshoot that plain Picard cannot survive, and so widens the basin of convergence at the cost of more iterations.

The sharper and more interesting choice is not a fixed small ρ but the *Cesàro-averaged* schedule $\rho_k = 1/(k+1)$, which turns eq. (28.11) into *fictitious play*. Unwinding the recursion shows why the name is apt: with $\rho_k = 1/(k+1)$ the update reads $m^{(k+1)} = \frac{k}{k+1}m^{(k)} + \frac{1}{k+1}\mathsf{S}(m^{(k)})$, which is precisely the running-average recursion—each $m^{(k)}$ is the empirical mean of all best responses computed so far, and the next move best-responds to that historical average rather than to the latest flow alone. This is the learning interpretation: agents who, instead of reacting to the most recent crowd, react to the time-averaged crowd they have observed. Why averaging succeeds where the bare iteration fails is worth a sentence of intuition. Under the Lasry–Lions monotonicity (A-Mon) the best-response map need not contract, but the mismatch it produces has a definite sign: monotonicity makes $\langle \mathsf{S}(m) - \mathsf{S}(m'), m - m' \rangle$ controlled in a one-sided way, so the bare iterates cannot *grow* their distance to the equilibrium but can *rotate* around it—circulating on a near-orbit without spiralling in. Averaging a bounded orbit that circulates a center pulls the average toward that center: the rotational component cancels in the running mean while the dissipation monotonicity provides drains the radial component. In this sense (A-Mon) makes the *averaged* dynamics dissipative even when the instantaneous map is only neutrally stable, and the Cesàro mean converges to the equilibrium although no single iterate need. This is the discrete, finite-population shadow of the continuous learning dynamics taken up in chapter 29; the convergence of fictitious play for monotone mean field games is the theorem of Cardaliaguet and Hadikhanloo [38]. We will meet the construction again in the worked computation: for the subcritical LQG thread of section 28.6, the symmetric averaged step $\rho_k = \frac{1}{2}$ already contracts geometrically, exactly the rate proposition 28.5 predicts when $L_{\mathsf{S}} < 1$.

Remark 28.6 (Damping selects but cannot restore uniqueness)**.** The update eq. (28.11) is under-relaxation, the same device that stabilizes a self-consistent-field iteration past the point where the bare update over-shoots (the physics note of section 28.4 draws out that analogy). What must be stated plainly is the limit of what it can accomplish. Damping changes the *dynamics* of the iteration—which sequence of flows it generates and whether that sequence converges—but it changes nothing about the *set* of fixed points, since every $\rho_k \in (0, 1]$ leaves eq. (28.11) with exactly the fixed points of S. Hence near a genuine multiplicity (theorem 14.11, beyond a conjugate horizon, where the equilibrium is honestly non-unique) no schedule of ρ_k can manufacture uniqueness: the problem has several equilibria and damping cannot wish them away. What damping *can* do is *select*: it converges to *a* solution—generically the one in whose basin the initial guess lies—rather than diverge or oscillate among them. Damping is therefore a convergence aid and a selection rule, never a uniqueness theorem.

This is the honest ceiling on Strategy I, and it is exactly what motivates the next subsection. Everything we have established is *conditional*: plain Picard converges only inside the subcritical regime $L_{\mathsf{S}} < 1$, and damped or averaged Picard extends that reach only under monotonicity (A-Mon) and still cannot pass through a multiplicity. Outside both regimes—strong coupling, long horizon, near a conjugate critical point—no fixed-point iteration of S is guaranteed to settle, because the obstruction is structural rather than a matter of step size: the map simply does not contract, and damping a non-contraction does not create one. What is needed there is a method that does not rest on S contracting at all—one that confronts the coupled forward–backward system directly and asks only that its residual have a nonsingular Jacobian, a far weaker and generically satisfied condition. That is the all-at-once Newton method of section 28.3.2. The two strategies dovetail rather than compete: Picard is cheap and globally reaching when it works but slow and conditional, while Newton is robust and fast locally but needs a starting point already close to the solution. The practical synthesis, which section 28.3.2 formalizes and section 28.6 exercises, is to spend a handful of cheap damped-Picard sweeps—$\rho_k = \frac{1}{2}$ from a crude initial flow—to land inside Newton's basin of quadratic convergence, then hand that warm start to Newton for the final high-accuracy solve. We turn now to that robust solver.

28.3.2 All-at-once Newton

The previous subsection closed on an honest ceiling. Strategy I converges only when the best-response map S contracts—inside the subcritical regime $L_{\mathsf{S}} < 1$, extended a little by damping or averaging under monotonicity (A-Mon) but never past a genuine multiplicity. Outside that regime the obstruction is structural rather than a matter of step size: near a conjugate horizon, under strong coupling, or at long horizon the map simply does not contract, and damping a non-contraction does not manufacture one. The remedy proposed there was to abandon the requirement that S contract at all and instead confront the coupled forward–backward system head-on, asking only that its discretized residual have a nonsingular Jacobian—a far weaker and generically satisfied condition. This is Strategy II in its concrete form, and we now build it. The shift in viewpoint is the whole point: where Picard asked "does the self-consistency loop settle when I iterate it?", Newton asks "is the equilibrium a regular zero of a smooth map?", and the second question has an affirmative answer in exactly the regimes where the first does not.

Strategy II refuses to break the loop. It stacks every unknown into a single space–time vector—all the discrete values $U = (u_i^n)$ across every grid node i and every time level n, and all the discrete densities $M = (m_i^n)$ likewise—and demands that the entire discretized system eq. (28.8)–eq. (28.10) hold simultaneously. Collecting the discrete HJB equations into a residual $\mathcal{R}_{\mathrm{HJB}}(U, M)$ (it vanishes exactly when U solves the backward sweep against the population M, with the terminal data $G[m_T]$ built into the top time level) and the discrete Fokker–Planck equations into a residual $\mathcal{R}_{\mathrm{FP}}(U, M)$ (it vanishes when M is the forward flow driven by the feedback read off from U, with the initial data m_0 built into the bottom level), the equilibrium is the joint zero

$$\mathcal{R}(U, M) = \begin{pmatrix} \mathcal{R}_{\mathrm{HJB}}(U, M) \\ \mathcal{R}_{\mathrm{FP}}(U, M) \end{pmatrix} = 0, \tag{28.12}$$

to be solved by Newton's method. Notice what has happened to the forward–backward entanglement of section 28.1: it is no longer an obstacle to be tiptoed around by alternating one-way solves, but is imposed everywhere at once as the single algebraic statement $\mathcal{R} = 0$, with the two temporal boundary conditions sitting inside $\mathcal{R}$ as ordinary rows. There is no marching and no freezing. The price, as anticipated, is that $\mathcal{R}$ couples the whole space–time field, so a Newton step is a linear solve in all of (U, M) at once; the reward is robustness, because Newton's convergence rests on the Jacobian of $\mathcal{R}$ being invertible, not on any contraction.

Everything therefore hinges on the structure of that Jacobian, and it is here that the keystone of the chapter—the discrete forward–backward duality—reappears in its most consequential role. Differentiating the two block-rows of eq. (28.12) in the two block-variables gives the 2×2 block structure

$$D\mathcal{R} = \begin{pmatrix} \partial_U \mathcal{R}_{\mathrm{HJB}} & \partial_M \mathcal{R}_{\mathrm{HJB}} \\ \partial_U \mathcal{R}_{\mathrm{FP}} & \partial_M \mathcal{R}_{\mathrm{FP}} \end{pmatrix}, \tag{28.13}$$

and each of the four blocks has a transparent meaning that it pays to read off one at a time. The top-left block $\partial_U \mathcal{R}_{\mathrm{HJB}}$ is the *linearized backward operator*: differentiating the implicit HJB step eq. (28.8) in the value, with the optimal policy frozen at its current value (the envelope theorem lets us hold α^* fixed, since the Bellman bracket is stationary there), returns the matrix $I - \Delta t\,\mathcal{A}_h^{(\alpha^*)}$ acting on each time slice and the coupling $-I$ that ties level n to level $n+1$. Because backward Euler couples each level only to the one after it, as a matrix over the stacked time index this block is *block-lower-bidiagonal in time*—a clean staircase that, as we will exploit, inverts by a single backward sweep. Symmetrically, the bottom-right block $\partial_M \mathcal{R}_{\mathrm{FP}}$ is the *linearized forward operator*: differentiating the implicit FP step eq. (28.10) in the density returns $I - \Delta t\,(\mathcal{A}_h^{(\alpha^*)})^\top$ on each slice with the coupling that ties level $n+1$ back to level n, a *block-upper-bidiagonal* staircase running the other way in time. The two off-diagonal blocks carry the coupling between the value and the population. The top-right block $\partial_M \mathcal{R}_{\mathrm{HJB}}$ measures how the HJB residual responds to a perturbation of the density, and since the density enters the HJB equation only through the source $F[m]$, this block is the discretized coupling derivative $\delta F/\delta m$. The bottom-left block $\partial_U \mathcal{R}_{\mathrm{FP}}$ measures how the FP residual responds to a perturbation of the value, and since the value enters the FP equation only through the drift $-\partial_p H(x, \nabla u)$, this block is the derivative of that drift in ∇u.

The crucial observation, and the reason the Newton system is tractable at all, is that the two *diagonal* blocks are not independent. They are exact transposes of each other. This is not a coincidence of the discretization but the same forward–backward duality we installed at the grid level in section 28.2.2: the forward step was *defined* to be the transpose of the backward step (eq. (28.10)), and lemma 28.2(3) recorded the resulting pairing $\langle (I - \Delta t\,\mathcal{A}_h)u, m\rangle = \langle u, (I - \Delta t\,\mathcal{A}_h^\top)m\rangle$. That very identity, read at the level of the Jacobian, says

$$\partial_M \mathcal{R}_{\mathrm{FP}} = \big(\partial_U \mathcal{R}_{\mathrm{HJB}}\big)^\top =: L^\top,$$

where we abbreviate the linearized backward operator by L. The adjointness lemma we proved as a fact about mass and positivity now returns as the structural fact about the Newton Jacobian: the discrete reflection of the continuous forward–backward duality is exactly what we are about to exploit. Writing the Newton step $D\mathcal{R}\,(\delta U, \delta M)^\top = -\mathcal{R}$ with this notation,

$$\begin{pmatrix} L & B \\ C & L^\top \end{pmatrix} \begin{pmatrix} \delta U \\ \delta M \end{pmatrix} = - \begin{pmatrix} \mathcal{R}_{\mathrm{HJB}} \\ \mathcal{R}_{\mathrm{FP}} \end{pmatrix}, \qquad B = \partial_M \mathcal{R}_{\mathrm{HJB}} = -\Delta t\,\tfrac{\delta F}{\delta m}, \quad C = \partial_U \mathcal{R}_{\mathrm{FP}}, \tag{28.14}$$

where B carries the coupling derivative $\delta F/\delta m$ and C the derivative of the Fokker–Planck drift in ∇u. The transpose pairing of the two diagonal blocks is precisely what makes a *Schur-complement* solve of eq. (28.14) go through cleanly, and we walk through that elimination now because it is where the structure earns its keep.

The diagonal block L is the easy one to invert. Being block-lower-bidiagonal in time, it is a staircase of tridiagonal sub-blocks, each of which is an $I - \Delta t\,\mathcal{A}_h^{(\alpha^*)}$ M-matrix; solving $L\,x = r$ is therefore a single backward sweep over the time levels, at each level a tridiagonal solve that is well posed precisely because lemma 28.1 makes the slice matrix a nonsingular M-matrix. So L^{-1} costs one sweep and we may eliminate δU from the first block-row of eq. (28.14), which reads $L\,\delta U + B\,\delta M = -\mathcal{R}_{\mathrm{HJB}}$ and hence

$$\delta U = -L^{-1}(\mathcal{R}_{\mathrm{HJB}} + B\,\delta M).$$

Substituting this into the second block-row $C\,\delta U + L^\top \delta M = -\mathcal{R}_{\mathrm{FP}}$ and collecting the terms in δM removes δU entirely, leaving the reduced system

$$(L^\top - C\,L^{-1}B)\,\delta M = -\mathcal{R}_{\mathrm{FP}} + C\,L^{-1}\mathcal{R}_{\mathrm{HJB}} \tag{28.15}$$

for δM alone. The unknown has been halved: instead of solving for the whole space–time pair $(\delta U, \delta M)$ at once, we solve the single reduced system eq. (28.15) for the density increment and then recover δU by one more backward sweep through L^{-1}. The operator on the left, the Schur complement $S := L^\top - CL^{-1}B$, is the object every solver of eq. (28.15) must handle, and its structure decides how hard that is.

Here the duality pays a second dividend, and it is worth staging the definiteness argument slowly because it is dense. Begin with the leading term. By the same adjointness $\partial_M \mathcal{R}_{\mathrm{FP}} = L^\top$, the principal part of S is the transpose of the well-conditioned M-matrix staircase L, so it is itself invertible by a forward sweep and already carries the bulk of the operator. The correction $-CL^{-1}B$ measures how strongly the value and the population feed back on each other, and its sign is controlled by the monotonicity structure. Under (A-Mon) the coupling functional $\mathcal{F}$ is convex, so its second variation—the discretized block $\delta F/\delta m$—is positive semidefinite; hence the coupling block $B = -\Delta t\, \delta F/\delta m$ is *negative* semidefinite (this is exactly the discrete content of lemma 8.10). A negative-semidefinite B makes the correction $-CL^{-1}B$ a *positive-type* contribution: it pushes S in the definite direction rather than spoiling the conditioning of its principal part $L^\top$. The upshot is that S is not an arbitrary dense operator but a definite-structured one, and definite-structured operators are exactly what Krylov methods thrive on: eq. (28.15) is solved by a preconditioned Krylov iteration (GMRES in general, or CG when the symmetry is exact), never by forming S densely, since each Krylov step needs only the *action* of S on a vector—one application of $L^\top$, one backward sweep L^{-1}, and one each of the cheap couplings B, C—all at the cost of sweeps already in hand.

What preconditioner should drive that Krylov solve? The block structure answers the question for us. The natural choice is $L^\top$ itself—the linearized forward operator—because it is the principal part of S, because it inverts by a single forward sweep just as L inverts by a backward one, and, most tellingly, because it is *exact* in the weak-coupling limit. When the coupling derivative $\delta F/\delta m$ is small the block B is small, the correction $-CL^{-1}B$ is negligible, and $S \approx L^\top$ to leading order; preconditioning by $L^\top$ then renders S essentially the identity and the Krylov solver converges in a handful of steps. And weak coupling is not an arbitrary favorable case: it is *precisely* the subcritical regime $L_{\mathsf{S}} < 1$ in which proposition 28.5 guaranteed Picard already worked. The two pictures are thus consistent and complementary—where the coupling is weak the preconditioner is sharp and Newton is nearly free, and where the coupling is strong (and Picard fails) Newton still converges but the Krylov solver must do genuine work through the positive-type correction $-CL^{-1}B$. This is the sense in which Newton extends reach exactly into the regime Strategy I cannot enter.

Two properties of Newton bound its use, and both follow from the general theory. First, convergence is *quadratic*: once the iterate is close enough to the equilibrium that the Jacobian is well approximated by its value at the solution, the residual norm squares at each step, and machine accuracy is reached in a few iterations. Second, that convergence is only *local*: Newton is guaranteed to contract only inside a neighborhood of the zero, and from a poor initial guess the bare iteration can overshoot or diverge. The fix is the one foreshadowed in section 28.3.1: warm-start Newton with a few damped Picard sweeps, which are globally reaching and cheap, to land inside the basin of quadratic convergence before switching to Newton for the final high-accuracy solve. Picard supplies the globalization Newton lacks; Newton supplies the speed and robustness Picard lacks; together they cover the whole parameter range.

Algorithmic template (outer solve). The synthesis assembles into a single outer loop, which we state as a procedure and then comment on.

1. **Initialize.** Set $m^{(0)} \equiv m_0$ (the initial density held constant in time) or, slightly better, a short heat-flow smoothing of it, to give the backward sweep a benign source.

2. **Globalize.** Run damped Picard eq. (28.11) with the symmetric step $\rho_k = \frac{1}{2}$—each iteration one backward HJB solve and one forward FP solve via the block of section 28.2—until the increment $\left\| m^{(k+1)} - m^{(k)} \right\|$ falls below a coarse tolerance. This walks the iterate into the basin of attraction from almost any starting flow.

3. **Refine (optional).** Switch to Newton on the stacked residual eq. (28.12), solving each linear step eq. (28.14) by the Schur-complement reduction eq. (28.15)—eliminate δU by a backward sweep, solve for δM by a $L^\top$-preconditioned Krylov iteration, recover δU—until the quadratic regime drives the residual to machine tolerance.

4. **Output.** The equilibrium space–time field (U, M).

The practical guidance is to take the refinement step seriously as *optional.* For any problem satisfying (A-Mon) or, more sharply, the subcritical contraction $L_{\mathsf{S}} < 1$ of proposition 28.5, damped Picard alone already converges geometrically, and Step 3 is never reached—this is exactly the situation of the worked computation in section 28.6, whose LQG thread is subcritical and which therefore uses damped Picard and nothing more. Newton earns its extra per-step complexity—assembling the off-diagonal blocks B, C, the Krylov solves of eq. (28.15)—only on the stiff or near-critical problems where Picard stalls: strong coupling, long horizon, the approach to a conjugate point. One should reach for it there and only there.

It is worth ending on the structural fact that has quietly organized this whole subsection, because it is the bridge to what comes next. The single feature that made the Newton system tractable was that the two diagonal blocks are transposes, $\partial_M \mathcal{R}_{\mathrm{FP}} = (\partial_U \mathcal{R}_{\mathrm{HJB}})^\top$—the discrete forward–backward duality of lemma 28.2(3), reappearing now not as a statement about conserved mass but as the hidden *adjoint* structure of the full coupled Jacobian. A Jacobian whose off-diagonal coupling is a multiplier/constraint pairing and whose diagonal is a self-adjoint pair is the Jacobian of the optimality system of a variational problem; that is, the all-at-once residual $\mathcal{R}$ is secretly the gradient of something. In general that "something" need not be convex, and Newton is the right tool precisely because it does not require it to be. But in the *potential* case—when the coupling F is the flat derivative of a convex functional, the hypothesis of theorem 8.11 seeded already in section 28.2.2—the hidden adjoint structure stiffens into an exact *convexity*, the equilibrium becomes the unique minimizer of a single convex energy, and one can do better than Newton's local guarantee: globally convergent optimization replaces the search for a zero. That is the structure the next section harvests, and we turn to it now.

28.4 Variational and primal–dual methods for potential MFG

The previous subsection ended by naming a clue it could not yet cash in. The Newton system of section 28.3.2 was tractable for one reason: its two diagonal blocks are exact transposes, $\partial_M \mathcal{R}_{\mathrm{FP}} = (\partial_U \mathcal{R}_{\mathrm{HJB}})^\top$, the discrete forward–backward duality of lemma 28.2(3) resurfacing inside the coupled Jacobian. We observed there that a Jacobian whose diagonal is a self-adjoint pair and whose off-diagonal coupling is a multiplier/constraint pairing is, structurally, the Jacobian of the optimality system of a *variational* problem: the all-at-once residual $\mathcal{R}$ is secretly the gradient of *something*. Newton does not care whether that something is convex—it hunts a zero, not a minimum, and that indifference is exactly why it reaches into regimes where Strategy I fails. But the indifference also leaves value on the table. When the hidden potential *is* convex, finding its minimizer is a strictly easier and more robust task than finding a zero of its gradient, and the entire apparatus of convex optimization—with its global, unconditional convergence guarantees—becomes available. This section is about the subclass of games where that windfall is real, and the seed for it was planted earlier still: already in section 28.2.2 we noted that the very adjointness which buys discrete mass conservation is, in the potential case, what turns the discrete energy convex. We now harvest both observations together.

The subclass in question is the *potential* (variational) games of definition 8.9: those whose couplings F and G are flat derivatives—first variations—of convex functionals $\mathcal{F}$ and $\mathcal{G}$ on the space of measures. For these, theorem 8.11 makes a statement of remarkable economy: the entire coupled forward–backward system eq. (28.1)— two nonlinear PDEs anchored at opposite ends of time, the object whose mutual entanglement organized this whole chapter—is nothing other than the Euler–Lagrange optimality condition of a *single* convex minimization. The HJB equation is the stationarity condition in the value; the Fokker–Planck equation is the constraint; the awkward two-point coupling that forbade marching the clock is just the primal–dual pairing between a minimizer and its multiplier. The forward–backward problem has not been solved so much as *re-read*: what looked like a fixed point to be chased is exposed as a critical point to be descended to.

That re-reading changes the numerical landscape completely, and the change is best stated as a sharp trade against the two strategies built in the first half of the chapter. Strategies I and II both converge only *conditionally*. The best-response fixed point of sections 28.2 and 28.3 contracts only inside the subcritical window $L_{\mathsf{S}} < 1$, widened a little by damping but never past a genuine multiplicity; Newton converges only *locally*, once an iterate has been walked into the basin of quadratic attraction. Both carry a hypothesis—smallness, short horizon, or a good warm start—without which they may oscillate or diverge, and the bulk of section 28.3 was the honest accounting of where those hypotheses hold. A convex minimization carries none of them. A convex functional on a convex set has a *global* minimizer, and the standard first-order descent algorithms reach it *unconditionally*: from any starting point, for any horizon, at any coupling strength, with a convergence rate that depends on conditioning but never on a smallness threshold that can be violated. There is no near-critical regime in which the method stalls, because there is no contraction or local basin to escape—convexity has no conjugate horizon. This is the single reason the section exists: for potential games one trades away the conditional convergence that shadowed both earlier strategies and buys global reach in exchange.

The trade is genuine, and its price must be named with equal candor. The convex-minimization route applies *only* to potential games—those with a separable Hamiltonian and a monotone coupling that is the flat derivative of a convex functional—a strict and structurally special subclass. Outside it the equilibrium is still a fixed point or a regular zero, not a minimizer, and there is no energy to descend; the finite-difference fixed point and the all-at-once Newton method of sections 28.2 and 28.3 remain the only tools, and they apply to *any* MFG. So the two halves of the chapter are complementary, not competing: Strategies I and II are general but conditional, the variational method is unconditional but special. One reaches for it precisely when the game is known to be potential and global robustness is worth the loss of generality—and, as the graphon LQG thread will show in section 28.6, the games we most want to compute often are potential.

The development proceeds in two movements, mirroring the structure of any optimization story: first the problem, then the solver. Section 28.4.1 makes the convex program explicit and then discretizes it, recalling the energy $\mathcal{J}(m,w)$ over density–momentum flows and the linear continuity constraint it is minimized against, and proving the structural payoff that justifies the whole detour—that the discrete optimality (KKT) system of that program is *exactly* the finite-difference HJB–Fokker–Planck system of section 28.2, with the multiplier playing the role of the discrete value. The adjoint discretization built in section 28.2.2 is what makes this identity hold on the nose rather than up to truncation error: the same transpose pairing that conserved mass is what makes the discrete Lagrangian's stationarity reproduce the discrete forward and backward sweeps. Section 28.4.2 then turns to the solvers themselves—the first-order primal–dual algorithms (augmented Lagrangian / ADMM and the Chambolle–Pock scheme) that exploit the separable nonsmooth-plus-linear-constraint form of the discrete energy—and draws out the kinship that has been implicit since the momentum field $w = m\,\alpha$ first appeared: a potential MFG is, formally, a dynamic optimal-transport problem with congestion, and the very algorithms that compute Wasserstein geodesics compute mean field equilibria. We take the convex program first.

28.4.1 The convex problem and its discretization

The section intro promised to make the convex program explicit before handing it to any solver, and that is this subsection's whole task. We carry it out in three strokes, each setting up the next. First we recall the energy functional and the linear constraint it is minimized against. Then we certify, ingredient by ingredient, that every term is convex, so that the continuous problem is a genuine convex minimization and not merely a stationarity system dressed up as one. Finally—and this is the payoff that justifies the entire variational detour—we discretize the program with the very operators already constructed in section 28.2, and find that the discrete optimality conditions are not *approximately* but *exactly* the finite-difference HJB–Fokker–Planck system, with the discrete value reappearing as the Lagrange multiplier. The adjoint discretization of section 28.2.2 is what makes that last identity land on the nose.

Recall from section 8.6 that, with control-affine dynamics and potential couplings, the equilibrium (u,m) is recovered from the minimizer (m,w)—$w = m\,\alpha$ the momentum field—of the energy

$$\mathcal{J}(m,w) = \int_0^T\!\!\int_{\mathbb{R}^d} m\,L_0\Big(x, \tfrac{w}{m}\Big)\,\mathrm{d}x\;\mathrm{d}t + \int_0^T \mathcal{F}(m_t)\;\mathrm{d}t + \mathcal{G}(m_T), \tag{28.16}$$

(this is eq. (8.18)) subject to the linear continuity constraint

$$\partial_t m - \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 m) + \operatorname{div} w = 0, \qquad m(0,\cdot) = m_0 \tag{28.17}$$

(this is eq. (8.19)). The change of variable from control to momentum, $w = m\,\alpha$, is what linearizes the constraint: written in the velocity α the continuity equation carries the bilinear transport $\operatorname{div}(m\,\alpha)$, but written in w that same term is simply $\operatorname{div} w$, linear in the unknown. The price is paid back in the cost, where $m\,L_0(x,\alpha)$ becomes $m\,L_0(x, w/m)$—and it is exactly this re-expressed kinetic term whose convexity we must now check. The value u enters not as an independent unknown but as the Lagrange multiplier of the continuity constraint eq. (28.17): it is the adjoint state that enforces conservation of probability, and the reader should hold onto that reading, because the entire structural payoff below is the statement that the multiplier of the discrete constraint *is* the discrete value of section 28.2.

Now the convexity, term by term, since the claim that $\mathcal{J}$ is a convex program is the hinge on which everything turns and none of its three ingredients is convex for a trivial reason. The two *interaction* terms come first and are the most direct. Under (A-Mon) the couplings F and G are flat derivatives—first variations—of the functionals $\mathcal{F}$ and $\mathcal{G}$, and the content of lemma 8.10 is that monotonicity of the coupling is precisely the differential signature of convexity of its potential, in exact analogy with the elementary fact that a monotone nondecreasing scalar function is the derivative of a convex one. So Lasry–Lions monotonicity, the same hypothesis that bought uniqueness of the equilibrium throughout the book, here buys convexity of $m \mapsto \int_0^T \mathcal{F}(m_t)\,\mathrm{d}t$ and of $m \mapsto \mathcal{G}(m_T)$. Without (A-Mon) these terms could be concave and the whole variational route would collapse; this is the one place the assumption is load-bearing for convexity rather than for uniqueness.

The *kinetic* term is subtler, and it is worth unpacking why it is convex at all, because at first sight it should not be. Its integrand is $m\,L_0(x, w/m)$: a convex function $L_0(x,\cdot)$ of the velocity w/m, but evaluated at a *ratio* of the two unknowns and then multiplied back by m—and dividing one optimization variable by another, then rescaling, is an operation that in general destroys convexity outright. What rescues it is a classical construction. For any convex f, its *perspective* $(t,y) \mapsto t\,f(y/t)$, defined for $t > 0$, is *jointly* convex in the pair (t,y) (proposition B.2). The intuition is geometric: the perspective homogenizes f—degree-one rescaling $(t,y) \mapsto (\lambda t, \lambda y)$ multiplies it by λ—so its epigraph is precisely the cone generated by the epigraph of f, and a cone over a convex set is again convex. The seemingly dangerous division is, read this way, exactly the operation that survives intact. Setting $t = m$, $y = w$, and $f = L_0(x,\cdot)$ identifies the kinetic integrand as the perspective of L_0, hence jointly convex in (m,w).

It pays to watch this on the LQG thread, both because it makes the abstract perspective theorem concrete and because it is the kinetic cost the worked computation of section 28.6 actually minimizes. With the quadratic kinetic Lagrangian $L_0(x,\alpha) = \frac{1}{2}\alpha^2$ the integrand becomes $m \cdot \frac{1}{2}(w/m)^2 = w^2/(2m)$, the function $(m,w) \mapsto w^2/(2m)$ on the half-space $m > 0$—the Benamou–Brenier kinetic energy familiar from dynamic optimal transport. One verifies its joint convexity by hand: its Hessian is $\left(\begin{smallmatrix} w^2/m^3 & -w/m^2 \\ -w/m^2 & 1/m \end{smallmatrix}\right)$, whose determinant $w^2/m^4 - w^2/m^4 = 0$ vanishes and whose trace $w^2/m^3 + 1/m > 0$ is positive, so its eigenvalues are 0 and something positive—positive semidefinite, exactly as the perspective theorem promises. That the cost of moving mass is convex jointly in (mass, momentum) is what will let the primal–dual method of the next subsection descend on it from any starting point without ever getting trapped.

Finally the constraint. The continuity equation eq. (28.17) is *linear* in the pair (m, w)—$\partial_t m$ and the diffusion act linearly on m, the divergence acts linearly on w, and the initial condition is an affine pin—so the set of admissible flows is an affine subspace, in particular convex. A convex functional minimized over a convex set: $\mathcal{J}$ is a genuine *convex program*, and this is the structural windfall the section's introduction advertised. We now show that discretizing it with the operators of section 28.2 preserves the windfall in full and, more, reproduces the finite-difference system as the discrete optimality conditions.

Discretizing (m, w) on the same space–time grid—with the divergence and time derivative in eq. (28.17) discretized by the *same* upwind/transpose operators of section 28.2, so that the discrete multiplier is exactly the discrete value of section 28.2— yields a finite-dimensional convex program: minimize a convex (partly nonsmooth) function subject to a linear equality constraint. The structural identity is worth stating.

Proposition 28.7 (Discrete potential MFG is a finite convex program)**.** *Under the hypotheses of theorem 8.11, with the continuity operator discretized as the transpose of the discrete generator (lemma 28.2), the discrete energy $\mathcal{J}_h(m_h, w_h)$ is convex and lower semicontinuous on the affine constraint set, its minimizer exists and is unique when $\mathcal{F}, \mathcal{G}$ are strictly convex, and the discrete optimality (KKT) system is exactly the finite-difference HJB–FP system eq.* (28.8)*–eq.* (28.10) *with the multiplier playing the role of u.*

Proof sketch. Write the discrete energy as the grid quadrature of eq. (28.16),

$$\mathcal{J}_h(m, w) = \Delta t \sum_n \sum_i m_i^n \, L_0\Big(x_i, \tfrac{w_i^n}{m_i^n}\Big) + \Delta t \sum_n \mathcal{F}_h(m^n) + \mathcal{G}_h(m^{N_T}),$$

and proceed in three steps: convexity, the KKT computation, and existence/uniqueness.

Convexity is inherited node by node. Each summand of the kinetic term is $m_i^n \, L_0(x_i, w_i^n/m_i^n)$, the perspective of the convex $L_0(x_i, \cdot)$ evaluated at the single pair (m_i^n, w_i^n), hence jointly convex in that pair by proposition B.2; a sum of convex functions of disjoint blocks of variables is convex, so the discrete kinetic term is convex in (m, w). The interaction terms $\mathcal{F}_h, \mathcal{G}_h$ are convex in m by (A-Mon)/lemma 8.10, inherited verbatim from the continuous $\mathcal{F}, \mathcal{G}$. The discretized continuity constraint, built from the backward time difference of m, the symmetric discrete diffusion, and the discrete divergence of w, is linear; its solution set is affine. Hence $\mathcal{J}_h$ is convex on a convex set.

The KKT system is the finite-difference system—and the adjoint identity is what makes it so. Introduce a multiplier u_i^n for each discrete continuity equation and form the Lagrangian $\mathcal{L}_h(m, w, u) = \mathcal{J}_h(m, w) - \langle u, \mathcal{C}_h(m, w)\rangle$, where $\langle \cdot, \cdot \rangle$ is the grid pairing $\Delta t \sum_{n,i} (\cdot)(\cdot)$ and $\mathcal{C}_h(m, w) = 0$ is the discrete continuity constraint. Read off the three stationarity conditions. *Stationarity in w:* the momentum w_i^n enters only the kinetic perspective and the divergence term of the constraint. Differentiating the perspective in w returns the velocity-derivative $\partial_a L_0(x_i, w_i^n/m_i^n)$, while the constraint contributes the adjoint of the discrete divergence applied to u—and since the discrete divergence is, by construction, minus the transpose of the discrete gradient, that adjoint deposits a discrete gradient $(\nabla_h u^n)_i$. Setting $\partial_w \mathcal{L}_h = 0$ gives the discrete Legendre relation $\partial_a L_0(x_i, w_i^n/m_i^n) = (\nabla_h u^n)_i$, which inverts to $w_i^n/m_i^n = \alpha_i^*$, the optimal feedback eq. (28.9); equivalently $w_i^n = m_i^n \, \alpha_i^*$, the posited momentum field. *Stationarity in u:* this returns the constraint

$\mathcal{C}_h = 0$ unchanged, which with $w = m\,\alpha^*$ inserted is precisely the forward Fokker–Planck step eq. (28.10)—the multiplier equation is the forward sweep. *Stationarity in m:* here the adjointness does the decisive work. The density m_i^n enters the perspective, the interaction terms, and—through the time difference and the diffusion—the constraint. The m-derivative of the perspective is $L_0 - (w/m)\,\partial_a L_0$, the *negative* Legendre transform of L_0, which by eq. (28.4) is the *negative* Hamiltonian $-H(x_i, (\nabla_h u^n)_i)$; the interaction terms contribute the running source $f_i^n = \mathcal{F}_h'(m^n)_i$; and the constraint contributes the adjoints of the time difference and the diffusion acting on u. The diffusion operator is symmetric, hence self-adjoint, and the backward time difference's adjoint reverses time, so these constraint terms reassemble on u into a *backward* step of the same spatial operator. This is exactly where the discrete integration-by-parts identity lemma 28.2(3), $\langle (I - \Delta t\,\mathcal{A}_h)u, m\rangle = \langle u, (I - \Delta t\,\mathcal{A}_h^\top)m\rangle$, earns its keep: the transpose that *defined* the forward operator in eq. (28.10) is undone when it lands back on the multiplier u, returning the *untransposed* backward generator $\mathcal{A}_h$. Setting $\partial_m \mathcal{L}_h = 0$ therefore reconstitutes, term for term, the implicit backward HJB step eq. (28.8), with terminal data $u^{N_T} = \mathcal{G}_h'(m^{N_T})$ supplied by the m-derivative of the terminal term. So the KKT triple—forward FP for m, Legendre relation for w, backward HJB for u—is identically the finite-difference system eq. (28.8)–eq. (28.10), and the multiplier u is literally the discrete value of section 28.2. This is the discrete mirror of the continuous computation in the proof of theorem 8.11, and it holds with *no* truncation remainder for exactly one reason: the forward operator was installed as the exact transpose of the backward one. A discretization merely accurate to $O(h)$ but not exactly transposed would make the two sweeps the optimality system of nothing.

Existence and uniqueness by the direct method. $\mathcal{J}_h$ is convex and lower semicontinuous, and coercive: the perspective kinetic term grows superlinearly in w (inherited from the growth of L_0), and $\mathcal{F}_h, \mathcal{G}_h$ are bounded below, so sublevel sets are bounded. The affine constraint set is nonempty—the pure-diffusion flow with w chosen to close continuity lies in it—and closed. A minimizing sequence is therefore bounded, has a convergent subsequence in this finite-dimensional space, and lower semicontinuity passes $\mathcal{J}_h$ to the limit, producing a minimizer. When $\mathcal{F}, \mathcal{G}$ are strictly convex the functional is strictly convex on the constraint set and the minimizer is unique. □

The upshot is a single finite-dimensional object to hand to a solver. The discrete program of proposition 28.7 has a very particular shape: its objective splits into a *separable*, partly nonsmooth sum—a per-node perspective at every grid point plus the interaction terms—and the *only* coupling between the otherwise independent unknowns is the single *linear* continuity constraint. That separable-objective-plus-linear-constraint structure is not incidental decoration; it is exactly the template that first-order primal–dual algorithms are built to exploit, splitting cheap nonsmooth proximal updates on the separable terms from a linear projection enforcing the constraint. Section 28.4.2 takes the convex program in precisely this form and presents the two standard methods that solve it.

28.4.2 Primal–dual algorithms and the optimal-transport connection

Section 28.4.1 closed by isolating the precise shape of the discrete program of proposition 28.7: its objective is a *separable, partly nonsmooth* sum—a per-node perspective of the kinetic Lagrangian at every grid point, plus the interaction terms—and the *only* thing tying the otherwise independent node-variables together is the single *linear* continuity constraint. That observation is not a stray remark; it is a complete specification of an algorithmic genre. "Minimize a separable convex (possibly nonsmooth) function subject to a linear equality constraint" is exactly the input that *first-order primal–dual* methods are designed to consume, and the design philosophy of the whole genre is to never fight the two features of the problem at once: handle the nonsmooth, but *separable and therefore pointwise-cheap*, objective by local proximal updates, and handle the smooth-but-*globally coupling* linear constraint by a dual variable that is itself the discrete value u. The reason this is the right marriage for us is the structural identity of proposition 28.7: the multiplier of the discrete continuity constraint already *is* the discrete value field, so the dual variable a primal–dual method introduces is not a bookkeeping auxiliary but the very object Strategy I and Strategy II spent the first half of the chapter computing. We now make the genre concrete in its two standard incarnations—the augmented-Lagrangian method ALG2/ADMM and the Chambolle–Pock primal–dual hybrid gradient—prove that both converge *unconditionally*, and then read off the two payoffs that give this subsection its title: the kinship with optimal transport, and the free-energy picture that makes the unconditional convergence physically inevitable.

Before either method, recall the one tool both are built from, since a physicist may not carry it ready to hand. For a proper convex lower-semicontinuous function φ and a step $\tau > 0$, the *proximal operator*

$$\operatorname{prox}_{\tau\varphi}(z) := \arg\min_{x} \left\{ \varphi(x) + \tfrac{1}{2\tau} \|x - z\|^2 \right\}$$

returns the point that trades off decreasing φ against staying near z; it is the implicit-gradient (backward-Euler) step on the gradient flow of φ, and it is single-valued and well defined for any convex φ, smooth or not (appendix I.2). Its decisive virtue here is that a prox of a *separable* sum decouples into independent proxes of the summands—so a prox of the discrete kinetic energy is computed grid node by grid node, each an utterly small scalar problem, and never as one large solve. This is what turns the separability noted in proposition 28.7 from an observation into a computational advantage.

Augmented Lagrangian (ALG2 / ADMM). The first method descends directly from the algorithm Benamou and Brenier wrote for dynamic optimal transport, generalized to the interaction-bearing MFG energy by Benamou and Carlier [19]. Introduce the discrete value u as the multiplier of the discrete continuity constraint $A(m, w) = c$ (with A the operator of eq. (28.17) and c carrying the initial datum), and form the *augmented* Lagrangian

$$\mathfrak{L}_r(m, w; u) = \mathcal{J}_h(m, w) + \langle u, A(m, w) - c\rangle + \tfrac{r}{2} \|A(m, w) - c\|^2 ,$$

the ordinary Lagrangian stiffened by a quadratic penalty of weight $r > 0$ on the constraint residual. The penalty is what makes the method robust: it convexifies the saddle in the primal block and lets the whole iteration be read as Douglas–Rachford splitting (equivalently ADMM) on the dual problem. One then minimizes $\mathfrak{L}_r$ over the primal pair (m, w) and ascends in u, but—and this is the engineering content—the primal minimization is *split* so that each piece is cheap. A single sweep of ALG2 is three steps. *(i) A proximal step on the kinetic energy.* Holding u and the interaction data fixed, minimize the kinetic perspective plus the quadratic proximal/penalty term. Both are separable over grid nodes, so this decouples into one independent scalar problem per node (i, n)—and that scalar problem is, concretely, the minimization

$$\min_{(m,w)} \left\{ \tfrac{w^2}{2m} + \tfrac{1}{2\tau}\left[(m-\bar{m})^2 + (w-\bar{w})^2\right] \right\}$$

in the quadratic LQG case $L_0 = \frac{1}{2}|a|^2$. Eliminating w from its stationarity condition $w/m + (w - \bar{w})/\tau = 0$ gives $w = \bar{w}\, m/(m+\tau)$, and substituting into the m-stationarity $(m-\bar{m})/\tau = w^2/2m^2$ collapses everything to the single *scalar cubic*

$$(m-\bar{m})\,(m+\tau)^2 = \tfrac{\tau}{2}\,\bar{w}^{\,2},$$

whose unique positive root is the proximal value of m (and then w follows in closed form). This is the "perspective prox": the entire nonsmooth, ratio-coupled kinetic cost reduces, pointwise, to finding one root of one cubic—solved by Cardano's formula in $O(1)$ work per node. *(ii) A proximal step on the interaction energy.* The terms $\mathcal{F}_h, \mathcal{G}_h$, convex in m by (A-Mon)/lemma 8.10, are handled by their own prox; for local couplings this too is pointwise and explicit. *(iii) A multiplier update.* Updating u to enforce the continuity constraint amounts, because of the quadratic penalty, to projecting the current (m, w) onto the affine constraint set, and that projection is the solution of a single space–time *Poisson-type* linear system for u—the normal equations $AA^\top u =$ (residual), whose operator $AA^\top$ is a discrete space–time Laplacian inherited from the gradient/divergence pairing of section 28.2.2. This is the only globally coupled solve in the sweep, and it is linear, symmetric, and positive definite, so a single multigrid or FFT solve dispatches it. The three steps alternate—two embarrassingly parallel pointwise proxes and one global Poisson solve—until the constraint residual and the energy both settle. Note how cleanly the work has been partitioned along the seam the problem itself provided: nonsmoothness lives entirely in the separable proxes, global coupling entirely in the linear Poisson solve, and the two never collide.

Primal–dual hybrid gradient (Chambolle–Pock). The second method dispenses even with the inner projection, trading the Poisson solve per sweep for a single application of A and its transpose. Write the problem as the convex–concave saddle of the (unaugmented) Lagrangian,

$$\min_{(m,w)}\ \max_{u}\ \mathfrak{L}(m,w;u), \qquad \mathfrak{L}(m,w;u) = \mathcal{J}_h(m,w) + \langle u, A(m,w) - c\rangle \tag{28.18}$$

(we write the saddle functional $\mathfrak{L}$, reserving $\mathcal{L}$ for the law), and alternate one proximal-*ascent* half-step in the dual variable u with one proximal-*descent* half-step in the primal pair (m, w), with an extrapolation of the primal iterate inserted to couple the two. Schematically, with steps $\sigma, \tau > 0$,

$$u^{k+1} = \operatorname{prox}_{\sigma h}\big(u^k + \sigma A\,\overline{(m,w)}^{\,k}\big), \qquad (m,w)^{k+1} = \operatorname{prox}_{\tau \mathcal{J}_h}\big((m,w)^k - \tau A^\top u^{k+1}\big),$$

where h is the (linear) term enforcing the constraint and the bar denotes the extrapolated primal point. Both proxes are closed form: the dual prox is the trivial affine prox of a linear functional, and the primal prox of $\mathcal{J}_h$ is again the per-node perspective cubic of step (i) above plus the interaction prox. No linear system is solved at all—each sweep costs one multiplication by A and one by $A^\top$, i.e. one discrete divergence and one discrete gradient. The scheme converges for *any* step sizes obeying the single product bound $\tau\sigma\,\|A\|^2 \le 1$ [47], with ergodic rate $O(1/k)$ on the primal–dual gap; the

MFG-specific proximal maps, including the perspective cubic, are tabulated by Briceño-Arias, Kalise, and Silva [32]. It is this scheme whose convergence we now state and prove as the chapter's sharpest contrast to everything in its first half.

Theorem 28.8 (Unconditional convergence in the potential case)**.** *Under the hypotheses of theorem 8.11 (potential couplings, (A-Mon), (A-Conv), (A-Nondeg)), the discrete primal–dual iteration applied to the convex program of proposition 28.7 converges to its unique saddle point—hence to the discrete MFG equilibrium—from* any *initialization, with ergodic primal–dual gap decaying as* $O(1/k)$*. No smallness of the coupling and no restriction on the horizon* T *is required.*

Proof. The entire content is to recognize eq. (28.18) as one instance of the abstract saddle the general primal–dual theorem is stated for, and to read off four identifications; once they are made, theorem I.2—proved in full and in-book in appendix I, originally [47]—applies verbatim and supplies the rate. We stage the identifications one at a time, because the force of the theorem is exactly which abstract object each MFG ingredient turns out to be.

The saddle. The general theorem treats $\min_x \max_y\{\langle Kx, y\rangle + g(x) - h(y)\}$ with g, h proper convex lsc and K linear (eq. (I.1)). Match it to eq. (28.18) term by term. *(1) The primal variable* is the density–momentum pair, $x = (m_h, w_h)$; *(2) the dual variable* is the discrete value, $y = u_h$—the same multiplier the KKT computation of proposition 28.7 already identified with the discrete value of section 28.2, so the dual iterate the algorithm carries is literally a running estimate of the value field. *(3) The proximable convex term* g is the kinetic-plus-interaction energy $\mathcal{J}_h$: nonsmooth, but proximable node by node exactly as the perspective cubic above shows; the constraint term contributes $h(y) = \langle c, y\rangle$, linear in y, so that $\langle Kx, y\rangle - h(y) = \langle A(m, w) - c, u\rangle$ reproduces the pairing in eq. (28.18). *(4) The linear map* K is the discrete continuity operator A of eq. (28.17).

The norm and the admissible steps. The only quantitative input the abstract theorem needs is the operator norm $L = \|K\| = \|A\|$, because the step-size condition is $\tau\sigma L^2 \le 1$. Here A is built from the discrete time difference, the symmetric discrete diffusion, and the discrete divergence; its norm is dominated by the stiffest of these, the first-order spatial difference, whose norm is $O(1/h)$ (a centered or upwind difference over a grid of spacing h has spectral radius of order $2/h$). Thus $L = O(1/h)$, and the admissible steps are precisely those with $\tau\sigma \le c\,h^2$ for a mesh-independent constant c—the standard CFL-like scaling for a first-order method on a gradient-norm operator, and the content of remark I.3. This is a condition the user simply *sets* by choosing τ, σ; it is not a hypothesis on the *problem* that the problem may fail to satisfy.

The conclusion, and why it is unconditional. With these identifications theorem I.2 gives, for any step sizes obeying $\tau\sigma L^2 \le 1$ and from *any* initial $(\,(m, w)^0, u^0\,)$, the ergodic bound: the partial primal–dual gap (I.2) at the running averages decays as $O(1/k)$, and since that gap vanishes only at the unique saddle point, the ergodic iterates converge to it—hence to the discrete equilibrium. The single fact that drives the proof of theorem I.2 is worth recalling, because it is where convexity does its work: the prox inequality lemma I.1 produces, at each step, a one-sided estimate that *telescopes* when summed, leaving only the initial distance to the saddle divided by k; the cross term left over is absorbed by Young's inequality under exactly the bound $\tau\sigma L^2 \le 1$. Nowhere in that telescoping is any smallness of the coupling, any shortness of the horizon, or any contraction factor invoked. *Convexity alone*—which here is precisely (A-Mon), the same monotonicity that made $\mathcal{J}_h$ convex in proposition 28.7—is what makes the estimate close. That is the whole point of the potential structure: it converts the problem from one where convergence is a delicate property to be earned into one where convergence is automatic.

This is the contrast the section was built to make, and it deserves to be stated against its foil without hedging. Proposition 28.5 certified Strategy I only *conditionally*: the best-response map contracts, and Picard converges, only inside the subcritical window $L_{\mathsf{S}} < 1$—a genuine hypothesis on the data (small coupling, or short horizon, or weak interaction) that a strongly coupled or long-horizon game can simply violate, after which the iteration oscillates or diverges and no choice of relaxation parameter rescues it. Newton (section 28.3.2) removed the smallness but replaced it with *locality*: quadratic convergence only once an iterate sits in the basin of attraction, which itself must be reached by a globalization that can stall. Here neither caveat survives. There is no contraction factor that must be less than one because the method never iterates a fixed-point map; there is no basin to reach because a convex functional has a single global minimum and every descent path leads to it. The price—paid in full in section 28.4.1—was generality: this holds only for potential games. But *within* that class, the $O(1/k)$ gap holds at any coupling strength and any horizon, from any start. The MFG-specific proximal maps are worked out in [32]. □

The two payoffs promised in the section opening now follow, and each is more than a footnote: the first locates potential-MFG computation inside a subject the reader already half-knows, and the second explains, in the physicist's own language, *why* the unconditional convergence just proved was to be expected all along.

Remark 28.9 (The optimal-transport limit)**.** Strip the interaction and the diffusion from the energy and watch what eq. (28.16) becomes. Set $\mathcal{F} = \mathcal{G} = 0$, take the pure kinetic Lagrangian $L_0(x, a) = \frac{1}{2}\,|a|^2$, and let $\sigma = 0$. Then the constraint eq. (28.17) loses its diffusion term and is the plain continuity equation $\partial_t m + \operatorname{div} w = 0$, and the energy collapses to $\mathcal{J}(m, w) = \int_0^T\!\int_{\mathbb{R}^d} \frac{|w|^2}{2m}\,\mathrm{d}x\,\mathrm{d}t$—nothing but the total kinetic action of a mass flow carried from m_0 at $t = 0$ to m_T at $t = T$. Minimizing that action over all flows joining two fixed endpoints is *exactly* the Benamou–Brenier dynamic formulation of the squared quadratic Wasserstein distance $\mathcal{W}_2(m_0, m_T)^2$, the theorem proved in-book as theorem 4.10 (appendix D.4). So a potential MFG is, structurally, a *generalized Benamou–Brenier problem*: the bare transport cost $\int \frac{|w|^2}{2m}$ is the same, and the interaction energy $\int_0^T \mathcal{F}(m_t)\,\mathrm{d}t + \mathcal{G}(m_T)$ enters as a *congestion* or crowd-aversion penalty added on top—a cost the moving mass pays for piling up, which optimal transport omits and mean field games supply. The diffusion $\sigma > 0$ that we restored above is, in the same dictionary, the entropic/Brownian regularization of the transport. This is not a loose analogy but an equality of variational problems, and it has a sharp computational corollary: every algorithm of this subsection, run with $\mathcal{F} = \mathcal{G} = 0$, *is* a dynamic optimal-transport solver [18, 117]—ALG2 reduces to the original Benamou–Brenier scheme it was generalized from, and Chambolle–Pock to a primal–dual transport solver. Conversely, the entropic-regularized (Sinkhorn-type) variants that dominate computational OT bridge back the other way, furnishing fast smoothed solvers for the regularized MFG energy. "Potential-MFG numerics" and "dynamic-optimal-transport numerics" are therefore one subject under two names, distinguished only by whether a congestion term sits in the objective.

Physics note

There is a free-energy reading of this whole section that makes its central theorem feel inevitable, and it is worth spelling out because it is the genuine article rather than a decorative analogy. Theorem 8.11 already cast a potential MFG as the minimization of a single functional $\mathcal{J}$; read $\mathcal{J}$ as a *Ginzburg–Landau free energy* whose order parameter is the population density m and whose conjugate field—the thermodynamically dual variable—is the value u, exactly the multiplier the primal–dual method carries as its dual iterate. In that reading the two algorithmic families of this chapter are the two textbook ways a physicist finds a free-energy minimum, and their behaviors are exactly the ones the analogy predicts. The fixed-point loop of section 28.3 is *self-consistent-field* (mean-field) iteration: posit the field, compute the density that best-responds to it, recompute the field that density induces, and repeat. Like every SCF scheme it converges quickly when the free energy is strongly convex—deep in a single-phase, subcritical regime—and grows sluggish and then unstable as one approaches a phase transition, where the free energy flattens, competing minima appear, and the self-consistency map stops contracting; this is the physical content of the contraction threshold $L_{\mathsf{S}} < 1$ of proposition 28.5. The primal–dual method of theorem 28.8 is instead *direct free-energy minimization*: it descends $\mathcal{J}$ itself rather than iterating the stationarity condition, and because a convex free energy has one global minimum and no spurious local traps, it reaches that minimum no matter how flat the landscape or how near the (here, by (A-Mon), absent) transition. The lesson is one every computational physicist has met at a critical point: when self-consistent-field iteration slows to a crawl near criticality, stop iterating the mean-field equations and minimize the free energy directly. Theorem 28.8 is the precise, theorem-grade statement of that folklore—unconditional $O(1/k)$ convergence where the self-consistency loop would have stalled.

The thread to carry out of this section is the one its theorem makes sharpest: *structure buys unconditional convergence.* Where Strategies I and II had to earn convergence with hypotheses on the data—a small coupling, a short horizon, a good warm start—the potential games surrender it for free, because their convexity (A-Mon) is a structural feature the solver can exploit rather than a regime it must hope to be in. And the reason the discrete solver could exploit it at all is the recurring theme of the whole chapter, visible here one more time: the right discretization mirrors the continuous structure. It was the exact-transpose pairing of section 28.2.2 that made the discrete energy genuinely convex (proposition 28.7) and the discrete saddle a faithful image of the continuous one, and only because of that faithfulness does the abstract primal–dual theorem land on the nose. That same principle—preserve the continuous structure exactly at the grid level, and the good properties follow—is what the final piece of the apparatus must now honor for a *new* structure. Section 28.5 lifts everything built so far to the type-indexed setting, where each agent carries a label $x \in [0, 1]$ and the coupling is no longer to a single global mean but to the graphon-aggregated neighbor field of eq. (28.19). The question that section confronts is the one this section answers for convexity: which discretization of the graphon operator $\mathbb{W}$ mirrors its continuous structure faithfully enough that the scalar machinery—monotonicity, adjointness, and the variational identity just exploited—survives the lift intact.

28.5 Discretizing the graphon coupling

The previous section left us with a principle and a charge. The principle is the one this whole chapter has circled: a discretization earns the good properties of the continuous problem precisely when it mirrors the continuous *structure* at the grid level—it was the exact-transpose pairing of section 28.2.2 that made the discrete energy genuinely convex and the discrete saddle a faithful image of the continuous one (section 28.4.2), and it was the M-matrix rate-matrix structure of section 28.2 that delivered monotonicity, mass conservation, and positivity for free. The charge is to honor that same principle for a *new* structure. Everything built so far—the monotone backward scheme, its conservative adjoint forward scheme, the outer fixed point and Newton solvers, the variational machinery—was written for a single representative agent coupled to a single global mean. We now lift the entire apparatus to the graphon setting of Part IV, where the population is no longer one crowd but a continuum of *types*, and a brand-new object enters the system: the graphon operator $\mathbb{W}$, which must itself be discretized consistently or the whole lift fails. The question is which discretization of $\mathbb{W}$ mirrors its continuous structure faithfully enough that the scalar machinery—monotonicity, adjointness, the variational identity—survives intact.

We add the type label. The graphon mean field game (set up across the kernel in chapter 22, with existence and the operator-norm contraction threshold (A-Coupling) in chapter 24) replaces the single representative agent by a continuum of types $x \in [0,1]$, the exchangeable index of an agent's network position. Each type now carries its *own* copy of everything the scalar problem had: its own controlled state X_t^x, its own value field u^x solved backward, and its own population density m_t^x evolving forward. Before going further we settle a notational collision that this section is the first to face squarely. Two different objects in this book both want the letter W: Brownian motion and the graphon kernel. The book's resolution (appendix A, and the notation watch opening this chapter) is that once a graphon is in scope the kernel owns W outright and Brownian motion is written B. From here on, therefore, W is *always* the kernel; the driving noise of the type-x state X_t^x is the Brownian motion B_t^x of Thread B (chapter 18), never W.

What makes this more than K disjoint scalar problems is the coupling, and it is here that the type label does real work. In the scalar game an agent was drawn toward the single population mean $\bar{m}_t$. In the graphon game an agent of type x is drawn instead toward the *graphon-aggregated neighbor field*

$$z_t(x) = (\mathbb{W}\,\bar{m}_t)(x) = \int_0^1 W(x,y)\,\bar{m}_t(y)\,\mathrm{d}y, \qquad \bar{m}_t(y) = \int_{\mathbb{R}} \xi\, m_t^y(\mathrm{d}\xi), \tag{28.19}$$

of eq. (28.19). Read this concretely, because the whole section turns on it. The inner integral $\bar{m}_t(y)$ is just the ordinary scalar mean of type y's own population—the average state of the agents of that one type, exactly the quantity $\bar{m}_t$ of Thread A but now computed type-by-type. The outer integral then forms a *kernel-weighted average* of those per-type means: $z_t(x)$ is the mean state of type x's neighbors, where "how much of a neighbor" type y is to type x is the connection strength $W(x,y)$. A type sees not the global crowd but its own kernel-weighted slice of it. When $W \equiv c$ is constant every type is everyone's neighbor with equal weight and $z_t(x) = c\int_0^1 \bar{m}_t(y)\,\mathrm{d}y$ collapses back to a single global mean, and the scalar game of Thread A is recovered (the constant-kernel limit of the rank-one reduction of chapter 18). The standing assumption (A-Reg-W) keeps W symmetric, bounded and (where needed) continuous in type, so that $\mathbb{W}$ is a self-adjoint Hilbert–Schmidt operator on $L^2[0,1]$; we will lean on exactly that regularity below, both to make the coupling quadrature convergent and to make the discrete operator's leading eigenvalue—the quantity the contraction threshold reads off—converge to the true $\|\mathbb{W}\|$.

The structural payoff of eq. (28.19) is worth stating before any grid arrives, because it dictates the entire architecture of the graphon solver and is the reason the lift is cheap rather than a fresh start. The type label couples the K problems in *exactly one place*: the aggregation eq. (28.19). Nowhere else does type x's state, value, or density depend on type y; the backward HJB sweep, the forward Fokker–Planck sweep, the policy iteration, the boundary closures—all of it runs within a single type, blind to the others, and reaches out to the rest of the population through nothing but the scalar target $z_t(x)$ that replaces the old target $\bar{m}_t$. Concretely, the running cost that type x feels is the scalar Thread A cost with its coupling target swapped: the term $\frac{q}{2}(x - s\,\bar{m}_t)^2$ of Thread A becomes $\frac{q}{2}(x - s\,z_t(x))^2$ for Thread B, the *only* change being $\bar{m}_t \rightsquigarrow z_t(x)$. So the graphon solver is not a new algorithm at all: it is the single-population solver of section 28.2 run in K *parallel copies*, one per type, that pause once per time step to exchange their current means, recombine them through the kernel into the targets z_t^k, and then march on independently again. The numerical task therefore acquires exactly one new ingredient—a second discretization, over the type variable, layered on top of the state–time grid of section 28.2—and that ingredient is precisely a *quadrature* for the integral in eq. (28.19): a rule that replaces the operator $\mathbb{W}$ by a finite $K \times K$ matrix acting on the vector of per-type means. Discretizing the graphon coupling is, in one sentence, replacing one integral operator by one matrix.

The rest of the section develops this in three movements, each isolating one face of that single ingredient. First (section 28.5.1) we lay down the type grid alongside the state grid and write the coupling quadrature explicitly, realizing $\mathbb{W}$ as the matrix $\mathbb{W}_K$ and bounding the error it commits against the true operator—the numerical face of graphon coarse-graining, tied back to the weak regularity theory of chapter 19. Second (section 28.5.2) we observe that for the structurally important class of *block* graphons the quadrature is not merely convergent but *exact*, collapsing the graphon MFG with no coupling error onto a finite multi-population game and exhibiting the two as numerically identical. Third (section 28.5.3) we account for what the lift costs and what it converges to: K parallel sweeps per outer iteration plus one cheap matrix product, and a total error that adds the type-grid contribution to the state–time error already controlled in section 28.2. We begin with the grid and the quadrature, where the operator $\mathbb{W}$ first becomes a finite matrix and its approximation error is bounded.

28.5.1 Type grid, state grid, and the coupling quadrature

Choose a type grid $x_1 < \cdots < x_K$ in $[0,1]$ with quadrature weights $w_1, \ldots, w_K > 0$ summing to 1 (midpoint or trapezoidal). Each type x_k carries a full copy of the state–time machinery of section 28.2: a value field $u_i^{k,n}$ and a density $m_i^{k,n}$. The only coupling between types is the aggregation eq. (28.19), which we realize as a *quadrature*:

$$z_t^k \;=\; \left(\mathbb{W}_K\,\bar{m}_t\right)_k \;:=\; \sum_{l=1}^{K} W(x_k, x_l)\,\bar{m}_t^l\,w_l, \qquad \bar{m}_t^l = \sum_i x_i\,m_i^{l,n}\,h, \tag{28.20}$$

i.e. the integral operator $\mathbb{W}$ is replaced by the $K \times K$ matrix $\mathbb{W}_K = \big(W(x_k, x_l)\, w_l\big)_{k,l}$ acting on the vector of type means. The whole solver is the single-population solver of section 28.2 run in K parallel copies, communicating once per time step through the matrix–vector product eq. (28.20). The architecture is thus *exactly* the scalar one, with the scalar coupling target $s\,\bar{m}_t$ of Thread A replaced by the type-k target $s\,z_t^k$ of Thread B.

Proposition 28.10 (Quadrature error of the coupling operator)**.** *Under (A-Reg-W) with W Lipschitz in its first argument uniformly in the second (constant L_W), and $\bar{m}_t$ bounded, the midpoint/trapezoidal quadrature eq.* (28.20) *on a uniform type grid satisfies*

$$\max_{1\le k\le K} \big|\, z_t(x_k) - z_t^k \,\big| \;\le\; C\,\big(L_W + \mathrm{Lip}(\bar{m}_t)\big)\, K^{-1}, \tag{28.21}$$

and $O(K^{-2})$ if in addition $W(x_k,\cdot)$ and $\bar{m}_t$ are C^2 in type. Consequently $\|\mathbb{W}_K - \mathbb{W}\| \to 0$, and the leading eigenvalue $\|\mathbb{W}_K\| \to \|\mathbb{W}\|$ as $K \to \infty$.

Proof. For fixed x_k the integrand $y \mapsto W(x_k,y)\bar{m}_t(y)$ is Lipschitz with constant $\le \|\bar{m}_t\|_\infty L_W + \|W\|_\infty \mathrm{Lip}(\bar{m}_t)$; the midpoint rule on K uniform subintervals has error bounded by (Lipschitz constant)$\times$(panel width)$= O(K^{-1})$ per the standard quadrature estimate, giving eq. (28.21). Under C^2 regularity the midpoint rule is second order, $O(K^{-2})$. The operator-norm convergence is then $\|\mathbb{W}_K - \mathbb{W}\| \le \sup_{\|f\|\le 1} \|(\mathbb{W}_K - \mathbb{W})f\| \to 0$ by the same panelwise bound applied uniformly over unit-norm f, and the top eigenvalue is norm-continuous (it equals $\|\mathbb{W}\|$ for the self-adjoint $\mathbb{W}$, lemma 20.12), so $\|\mathbb{W}_K\| \to \|\mathbb{W}\|$. $\square$

The connection to the regularity theory of chapter 19 is direct and worth making explicit. Assigning each type to its grid node x_k and treating W as constant on the corresponding cell is exactly the *stepping operator* $W_{\mathcal{P}}$ of definition 19.7 for the partition $\mathcal{P}$ into K equal type intervals. The weak regularity lemma theorem 19.8 and its bounded-complexity corollary corollary 19.9 bound the cut-norm error $\|W - W_{\mathcal{P}}\|_\square$ of this block approximation; through the stability of the equilibrium in the cut metric (the program of chapters 24 and 25), that cut-norm error controls the error in the computed equilibrium. The type-grid quadrature is, in other words, not an ad hoc device but the numerical face of graphon coarse-graining.

28.5.2 Block graphons and the exact reduction to multi-population MFG

When W is itself piecewise constant on a partition into K blocks—a stochastic block model kernel B—the type-grid aligned to the blocks makes the quadrature eq. (28.20) *exact*, not merely convergent: $\mathbb{W}_K$ *is* the block matrix B weighted by block sizes. The graphon MFG then reduces, with no discretization error in the coupling, to a finite K-population mean field game in which population k feels the mixture $\sum_l \mathsf{B}_{kl}\,\bar{m}^l\,w_l$ of the block means. This is the bridge to the multi-population theory of chapter 30 (specifically the block-graphon reduction of the heterogeneity zoo, chapter 30): a K-population MFG is a graphon MFG on a K-block kernel, and conversely the type-grid discretization of a general graphon MFG is a K-population MFG approximating it. The two pictures are numerically identical.

28.5.3 Complexity and convergence

Per outer (fixed-point) iteration, the graphon solver performs K independent HJB–FP sweeps at $O(N_T N_x)$ each, plus the coupling assembly. The coupling eq. (28.20) is a $K \times K$ matrix–vector product per time step, $O(K^2 N_T)$. Thus one outer iteration costs

$$O(K\, N_T\, N_x) \text{ (the } K \text{ decoupled PDE sweeps)} \;+\; O(K^2\, N_T) \text{ (the coupling)}, \tag{28.22}$$

and the number of outer iterations is $O(\log \varepsilon^{-1}/\log L_{\mathsf{S}}^{-1})$ when the fixed point contracts with rate $L_{\mathsf{S}} < 1$ (proposition 28.5). For a low-rank kernel the coupling cost collapses: a rank-one $W(x,y) = f(x)f(y)$ makes $z_t^k = f(x_k)\sum_l f(x_l)\bar{m}_t^l w_l = f(x_k)\,\varrho_t$ with a *single scalar* order parameter $\varrho_t = \langle f, \bar{m}_t\rangle$, so the $O(K^2)$ product becomes $O(K)$ and the entire type-indexed family collapses to one self-consistency equation for ϱ_t—the rank-one reduction of proposition 18.18 and worked computation 18.19 (the matrix template of section 14.7), here realized as a computational economy as well as a structural one.

The error of the computed equilibrium decomposes into three independently controllable pieces, which we state as the chapter's convergence accounting.

Theorem 28.11 (Convergence of the graphon discretization)**.** *Assume (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), (A-Reg-W) with W Lipschitz in type, and the operator-norm smallness (A-Coupling), $\|\mathbb{W}\| < \kappa := 1/C_T$, under which the graphon best-response map contracts and the GMFG equilibrium is unique (the dynamic threshold of chapter 24, proposition 24.18, with C_T the horizon-dependent per-type response constant $C_T = L_{\mathcal{R}}(T)\,\mathrm{Lip}(F_0, G_0)$ of eq. (24.7)). Let $(u_h^{k,n}, m_h^{k,n})$ be the output of the type-grid finite-difference solver with the outer loop driven to tolerance $\varepsilon_{\mathrm{out}}$. Then the total error of the computed type-indexed equilibrium is bounded by*

$$\underbrace{C_1\,(h + \Delta t)}_{\textit{FD scheme, theorem 28.3}} \quad + \quad \underbrace{C_2\,K^{-1}}_{\textit{coupling quadrature, proposition 28.10}} \quad + \quad \underbrace{\varepsilon_{\mathrm{out}}}_{\textit{outer-loop tolerance}}, \tag{28.23}$$

and each term vanishes under refinement; the constants C_1, C_2 depend on the data and on the amplification factor $(1 - C_T\,\|\mathbb{W}\|)^{-1}$ but not on the grids. (In the LQ normalization of worked computation 28.12, where $a = 0$ and $C_T = 1$, the contraction modulus $C_T\,\|\mathbb{W}\|$ reduces to $s\,\|\mathbb{W}\|$, which there coincides with the sharp uniqueness/phase boundary $s\,\|\mathbb{W}\| < 1$ of theorem 14.11; in general the two are distinct—$s\,\|\mathbb{W}\| = 1$ is the phase boundary, while $C_T\,\|\mathbb{W}\| < 1$ is the Picard-contraction threshold.)

Proof sketch. The three error sources are independent and superpose to first order. The finite-difference error $O(h + \Delta t)$ for each fixed type is theorem 28.3. The coupling error $O(K^{-1})$ is proposition 28.10; it enters the per-type source $s\,z_t^k$ as a forcing perturbation. The outer-loop residual is $\varepsilon_{\mathrm{out}}$ by construction. To turn these local perturbations into a bound on the *equilibrium* error one uses the contraction (A-Coupling): the equilibrium is a fixed point of the graphon best-response map, whose Lipschitz constant is $C_T\,\|\mathbb{W}\| < 1$ by the dynamic contraction estimate proposition 24.18/eq. (24.7), so a perturbation δ of the per-iteration data perturbs the fixed point by $\delta/(1 - C_T\,\|\mathbb{W}\|)$, the source of the amplification factor $(1 - C_T\,\|\mathbb{W}\|)^{-1}$ in C_1, C_2. The norm convergence $\|\mathbb{W}_K\| \to \|\mathbb{W}\|$ (proposition 28.10) guarantees the discrete map is a contraction for

K large whenever the continuous one is. The argument is the standard "consistency + stability $\Rightarrow$ convergence" decomposition; the stability input (contraction) is exactly (A-Coupling). Outside the subcritical regime the outer loop may not contract and eq. (28.23) fails—the same caveat as remark 28.6, inherited at the graphon level. $\square$

28.6 Worked computation: solving and validating both threads

We now do the computation the chapter exists to justify: solve Thread A and Thread B numerically by the forward–backward finite-difference solver of sections 28.2, 28.3 and 28.5, and check the output against the closed forms of chapter 14. This advances both threads to their fourth (numerical) layer (appendix A). The reference implementation is `figures/28-mfg-solver.py`; it implements eq. (28.8) with Howard iteration, the adjoint Fokker–Planck step eq. (28.10), the damped Picard outer loop eq. (28.11) with $\rho_k = \frac{1}{2}$, and the type-grid quadrature eq. (28.20).

Worked computation 28.12 (Numerical LQG and graphon-LQG equilibria)**.** We work in the normalization of worked computation 14.10: $a = 0$, $b = 1$, $q = q_T = \omega^2 = 1$ (so $\omega = 1$), $s_T = 0$, with noise $\sigma = 0.3$, horizon $T = 1$, initial mean $\bar{m}_0 = 1$, initial variance $V_0 = 0.1$, and running coupling $s = 0.5$. Because $s = 0.5 < 1$ we are strictly inside the unique-equilibrium region of theorem 14.11, so by proposition 28.5 the Picard loop should contract; and because in this normalization the Riccati curvature is exactly $P(t) \equiv \omega = 1$ (the constant solution of $\dot{P} = P^2 - 1$, $P(T) = 1$), the analytic benchmark for the equilibrium mean is the constant-coefficient linear two-point boundary value problem eq. (14.21), which we solve to machine precision by the matrix exponential of $M = \left(\begin{smallmatrix} -\omega & -1 \\ \omega^2 s & \omega \end{smallmatrix}\right)$.

Thread A. The state grid is $[-7, 7]$ with $N_x = 161$ nodes ($h \approx 0.0875$) and $N_T = 120$ time steps. The coupling target felt by the representative agent is $s\,\bar{m}_t$. The damped Picard loop converges geometrically—the residual $\left\|\bar{m}^{(k+1)} - \bar{m}^{(k)}\right\|$ contracts by a constant factor per step, reaching 10^{-6} in about 22 iterations, exactly the behavior proposition 28.5 predicts for a subcritical problem. The computed mean trajectory matches the closed-form BVP solution to

$$\max_t \left|\, \bar{m}_t^{\mathrm{num}} - \bar{m}_t^{\mathrm{exact}} \,\right| \;\approx\; 9.3 \times 10^{-3},$$

a first-order-in-$(h, \Delta t)$ error consistent with theorem 28.3. As an independent check on the HJB solver, the discrete curvature $\partial_{xx} u$ read off at the grid center is ≈ 1.06, recovering the exact Riccati value $P \equiv 1$ up to the upwind scheme's first-order numerical diffusion. The left panel of fig. 28.1 overlays the two.

Thread B (rank-one graphon). We take the rank-one kernel $W(x, y) = f(x)f(y)$ with $f(x) = \frac{1}{2} + x$ on $K = 16$ midpoint type nodes. Then $\|\mathbb{W}\| = \int_0^1 f^2 = \frac{13}{12} \approx 1.083$ (example 20.13), so the effective coupling is $s\,\|\mathbb{W}\| \approx 0.54 < 1$: subcritical, unique, contracting, by theorem 28.11. Each type-k agent feels $s\, f(x_k)\, \varrho_t$ with the scalar order parameter $\varrho_t = \langle f, \bar{m}_t\rangle = \sum_l f(x_l)\bar{m}_t^l\, w_l$ assembled by the quadrature eq. (28.20) (this ϱ_t is the canonical rank-one order parameter of eqs. (18.11) and (27.22)). The rank-one structure collapses the type-indexed family to a single self-consistency equation for ϱ_t (proposition 18.18); the closed-form reference solves the reduced scalar system

$$\dot{v} = (a - b^2 P)\, v - b^2 \psi, \quad v(0) = 0, \qquad \dot{\psi} = -(a - b^2 P)\, \psi + q\, s\, \varrho_t, \quad \psi(T) = 0, \qquad \varrho_t = \bar{m}_0\, e^{(a - b^2 P)t} \int_0^1 f + v(t) \int_0^1 f^2,$$

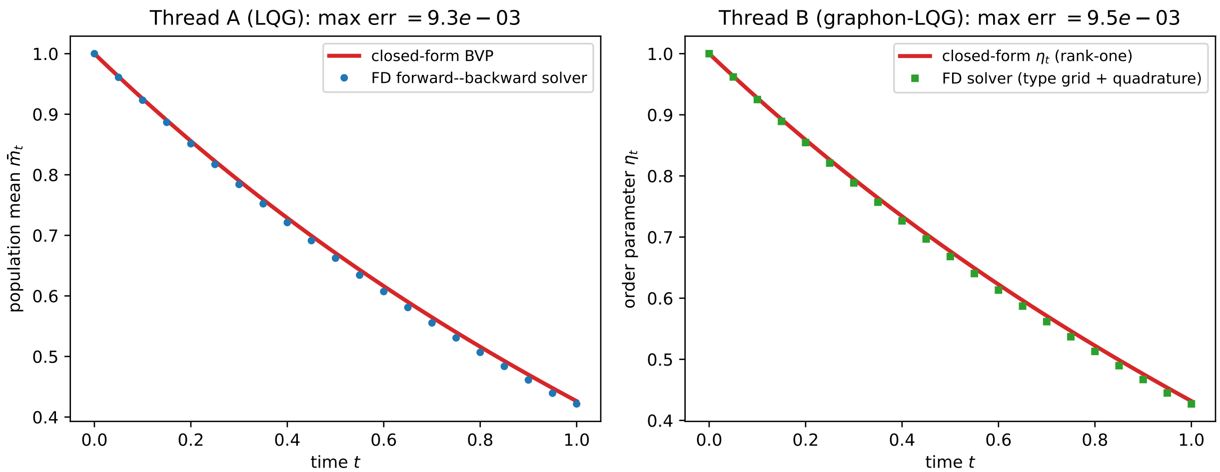

Figure 28.1: Validation of the forward–backward finite-difference solver against the closed forms of chapter 14, in the normalization of worked computation 28.12 ($a = 0$, $b = 1$, $q = q_T = 1$, $\sigma = 0.3$, $T = 1$, $\bar{m}_0 = 1$, $s = 0.5$). *Left*: Thread A. The population mean $\bar{m}_t$ computed by the finite-difference HJB–FP solver with damped Picard outer loop (markers) versus the exact linear two-point BVP solution eq. (14.21) (curve); maximum error $\approx 9.3 \times 10^{-3}$. *Right*: Thread B with rank-one kernel $W(x, y) = (\frac{1}{2} + x)(\frac{1}{2} + y)$ on a $K = 16$ type grid. The order parameter ϱ_t from the full type×state solver with quadrature coupling (markers) versus the closed-form scalar reduction (curve); maximum error $\approx 9.5 \times 10^{-3}$. Generated by `figures/28-mfg-solver.py`.

in which $v(t)$ is the rank-one type-modulation amplitude (the forward variable), $\psi(t)$ the reduced costate, $P \equiv 1$ the Riccati curvature of worked computation 14.10, and the closed-loop drift coefficient is $a - b^2 P = -\omega = -1$ in this normalization. The full type×state solver reproduces the order parameter to

$$\max_t \big| \varrho_t^{\mathrm{num}} - \varrho_t^{\mathrm{exact}} \big| \approx 9.5 \times 10^{-3}, \qquad \max_k \big| \bar{m}_T^{x_k} - \big(\bar{m}_0\, e^{(a-b^2P)T} + f(x_k)v(T)\big) \big| \approx 5.1 \times 10^{-3},$$

the second bound confirming that the computed *per-type* means align with the predicted affine-in-f profile $\bar{m}_t^x = \bar{m}_0\, e^{(a-b^2P)t} + f(x)v(t)$ of the rank-one collapse. The right panel of fig. 28.1 overlays the computed and analytic ϱ_t. Both threads are thus validated: the generic forward–backward solver reproduces the explicit Riccati answer, and the graphon machinery reproduces the rank-one reduction, with errors decreasing under grid refinement exactly as theorems 28.3 and 28.11 require.

Remark 28.13 (Why the validation is a genuine test)**.** The closed forms of chapter 14 were derived by exploiting the linear–quadratic structure—Gaussian ansatz, Riccati reduction, linear BVP—none of which the finite-difference solver knows about. The solver discretizes the *nonlinear* HJB equation (the quadratic Hamiltonian term $+\frac{1}{2}b^2(\partial_x u)^2$ entering $-\partial_t u - \frac{1}{2}\sigma^2\partial_{xx}u + H = F$ is genuinely nonlinear in $\partial_x u$ and is handled by Howard iteration, not by any quadratic ansatz) and the full Fokker–Planck PDE on a state grid, and finds the forward–backward fixed point by iteration. That this black-box PDE computation reproduces the pencil-and-paper Riccati answer to three digits, and reproduces the rank-one order parameter to the same accuracy through an honest quadrature of $\mathbb{W}$, is therefore a real cross-check of both the analysis of chapter 14 and the implementation of sections 28.2 and 28.5—each validates the other.

Rigor ledger

Proved.

- The discrete HJB generator $\mathcal{A}_h$ (upwind, eq. (28.7)) is a rate matrix and $I - \Delta t\, \mathcal{A}_h$ is a nonsingular M-matrix (discrete maximum principle), lemma 28.1.
- The adjoint Fokker–Planck step eq. (28.10) conserves mass, preserves positivity, and is the exact transpose of the linearized HJB step, lemma 28.2.
- Howard (policy-iteration) sweeps converge monotonically and superlinearly (proved in-book from the M-matrix structure, section 28.2.1); the Barles–Souganidis monotone-scheme theorem is proved in-book, theorem E.20.
- The discrete potential MFG is a finite convex program whose KKT system is the finite-difference HJB–FP system, proposition 28.7; consequently primal–dual methods converge unconditionally in the potential case, theorem 28.8, whose ergodic $O(1/k)$ gap is the in-book Chambolle–Pock bound theorem I.2 of appendix I.
- Quadrature error $O(K^{-1})$ (resp. $O(K^{-2})$) for the coupling operator $\mathbb{W}_K$ under Lipschitz (resp. C^2) type-regularity, with $\|\mathbb{W}_K\| \to \|\mathbb{W}\|$, proposition 28.10; the three-term error bound eq. (28.23) for the graphon solver, theorem 28.11.
- Numerical validation: the forward–backward solver reproduces the Thread A Riccati mean to 9.3×10^{-3} and the Thread B rank-one order parameter to 9.5×10^{-3} (worked computation 28.12, fig. 28.1).

Assumed.

- (A-Nondeg) for the parabolic regularization underlying both the viscosity-solution convergence and the Gaussian-tail boundary truncation; (A-Conv) for the well-posed pointwise minimization in Howard iteration; (A-Mon) for convexity of the variational problem and for fictitious-play stabilization; (A-Coupling) $C_T\,\|\mathbb{W}\| < 1$ for outer-loop contraction (distinct in general from the sharp phase boundary $s\,\|\mathbb{W}\| = 1$ of theorem 14.11, with which it coincides only in the $a = 0$, $C_T = 1$ normalization of worked computation 28.12).

Deferred.

- The monotone-scheme convergence engine and the primal–dual rate are resolved *in-book*: Barles–Souganidis in theorem E.20, the Chambolle–Pock ergodic $O(1/k)$ bound in theorem I.2, with the supporting parabolic estimates in appendix E (theorems E.17 and E.19) and compactness in chapters 2 and 3. What remains quoted from Achdou–Porretta [3] (see remark 28.4) is only the sharp-constant accounting of the first-order rate for the coupled system; the MFG-specific proximal maps are from [47, 32].

- The cut-norm/regularity control linking the type-grid stepping operator to the equilibrium error is deferred to chapter 19 (theorem 19.8) and the stability program of chapters 24 and 25.
- Learning-based solvers (fictitious play beyond the LQG check, policy-gradient/PPO) are deferred to chapter 29.

Open.

- Provably convergent, scalable solvers for *non-potential, non-monotone* MFG beyond the subcritical/short-horizon regime—i.e. past the conjugate horizons of theorem 14.11—remain open; no method is known to reliably select among multiple equilibria or to converge unconditionally there.
- High-dimensional state spaces ($d \gg 3$) defeat the grid-based schemes of this chapter (curse of dimensionality); deep-learning and probabilistic solvers [42] are the active frontier, with convergence guarantees still partial. This is the bridge to chapter 29.

Bibliographic notes

The finite-difference scheme for the HJB–Fokker–Planck system—monotone upwind backward operator, conservative forward operator taken as its transpose, and the structure-preserving duality between them—is due to Achdou and Capuzzo-Dolcetta [1]; the survey of Achdou and Laurière [2] is the modern reference for the numerical state of the art, including the Newton and policy-iteration outer loops. The convergence of the coupled scheme to weak solutions of the MFG system is proved by Achdou and Porretta [3], building on the abstract monotone-scheme convergence theory of Barles and Souganidis [13] for viscosity solutions (chapter 6, [52, 12]); we give an in-book proof of that monotone-scheme theorem in theorem E.20, resting on the comparison principle of appendix E. The fixed-point/fictitious-play analysis of the outer loop, and its convergence for monotone games, is due to Cardaliaguet and Hadikhanloo [38]; the connection to learning dynamics is developed in chapter 29. The variational/primal–dual approach to potential MFG originates with the augmented-Lagrangian method of Benamou and Carlier [19], a descendant of the Benamou–Brenier algorithm for dynamic optimal transport [18, 117]; the first-order primal–dual (Chambolle–Pock) viewpoint [47] and its application to MFG with local couplings are worked out by Briceño-Arias, Kalise, and Silva [32]; the ergodic $O(1/k)$ convergence bound itself we prove in-book in appendix I (theorem I.2). The variational characterization that underlies all of these is the potential-game theorem of Lasry and Lions as presented in chapter 8 (theorem 8.11) and Cardaliaguet's notes [36]. The graphon discretization—type grid feeding a state grid, $\mathbb{W}$ realized as a quadrature, block-graphon reduction to multi-population MFG—follows the operator and sampling theory of Part IV (chapters 18 and 19) and the weak regularity lemma [99, 72]; the complexity economy of the rank-one collapse is the computational face of the reduction of chapter 14 and [34, 108]. Deep-learning and high-dimensional solvers, the escape from the curse of dimensionality, and their partial convergence theory [42] are the subject of chapter 29.

Chapter 29

Reinforcement Learning for Graphon Mean Field Games

Every method in this book so far has presupposed that the modeller *knows the model*: the drift b, the volatility σ, the running cost L, the terminal cost g, and the coupling kernel W are all given in closed form, and the equilibrium is found by *solving* a coupled system — the forward–backward PDE pair of chapter 8, the FBSDE of chapter 9, or their graphon lifts of chapters 22 and 23. The numerical schemes of chapter 28 do not lift this demand; they only sharpen it, discretising equations that one must first be able to write down. This chapter abandons the presupposition altogether — it is the single largest pivot in the book — and asks what to do when the model is no longer in hand.

Three quite different obstructions force the change, and it is worth feeling each one concretely before section 29.1 names them. Sometimes the dynamics are a *black box*: an agent's transitions are produced by a market-microstructure matching engine, by an epidemic compartment model with behavioural feedback, or by a traffic micro-simulation — any of which one can *run* forward from a chosen state and action, but none of which hands over b, σ, L in a form a PDE solver can ingest. Sometimes the state is simply too large to grid: a finite-difference mesh on $\mathcal{S} \subseteq \mathbb{R}^d$ carries a number of nodes that explodes like $(\text{resolution})^d$, so that a portfolio of a dozen assets or a queueing network of a dozen servers is already out of reach. And in the graphon setting the population is not one measure but a whole *continuum of types* $x \mapsto m^x$ (definition 21.4), so that even a known, low-dimensional per-type problem must be solved simultaneously for a one-parameter family of agents coupled through the kernel W. In each case *solving* is unavailable, and the only primitive left is the ability to *sample* — to draw a next state and an incurred cost from wherever one happens to stand.

The answer is to *learn* the equilibrium by reinforcement learning (RL), and what turns this from a hope into a programme is one structural fact we have already proved analytically. A mean field game equilibrium is a *fixed point between two maps* — a best response that a single agent computes against a frozen population, and a push-forward that the population executes against a frozen policy — and a fixed point is exactly the kind of object an iterative loop can chase even when each of the two maps is available only through samples. This is the load-bearing sentence the entire chapter unpacks: it converts the existence theorem of chapter 8 into an algorithm, and it is the reason the analytic structure built up across Parts II–IV survives, essentially unchanged, into the model-free world.

We import three things. From chapter 8 (proposition 8.4) the characterisation of an MFG equilibrium as a fixed point of the composition "best-respond, then push the population forward." From chapters 11 and 24 the Lasry–Lions monotonicity condition (A-Mon) — in its classical scalar form (definition 11.1 and theorem 11.5) and, for the type-indexed population of section 29.4, its graphon form (definition 24.15 and theorem 24.16) — which will reappear as the precise hypothesis under which the learning iterations converge. From the graphon synthesis of chapters 20, 21 and 24 the type-indexed equilibrium (definition 21.16 and eq. (21.16)), the (observable) aggregate $z(x) = (\mathbb{W}\,\Lambda[m^{\cdot}])(x)$ — Part IV's Θ-aggregate of eq. (21.4), written z here per notationwatch (v) — and the operator-norm smallness condition (A-Coupling) that powered the contraction proofs (theorem 20.14 and chapter 24). Our task is to show that these same two hypotheses — contraction *or* monotonicity — are what separate a learning loop that converges from one that cycles forever, and to make precise what a deep policy-gradient method such as PPO can and cannot guarantee when the analytic structure is replaced by neural-network function approximation and Monte Carlo sampling.

The chapter is organised as follows. Section 29.1 sets up the discrete-time *mean field Markov decision process* (MFMDP), the natural object on which RL operates, and explains the modelling choices (time discretisation, the simulator hypothesis) that turn the continuous-time games of Parts II–IV into something learnable. Section 29.2 isolates the two maps, defines the *exploitability* functional that an algorithm minimises, and analyses the naive fixed-point iteration, exhibiting exactly when it contracts and why it usually does not. Section 29.3 introduces the two stabilisers — entropy regularisation together with *fictitious play* and *online mirror descent* — and proves their convergence under monotonicity; a worked computation on the LQG thread makes the dichotomy "contract vs. average" completely explicit. Section 29.4 lifts everything to the type-indexed population, presents GMFG-PPO, and shows how the graphon coupling is represented *inside* the value and policy networks. Section 29.5 assembles a modular convergence guarantee and catalogues the failure modes honestly. Throughout, the convention of the book is in force: *value functions are computed backward* (terminal data, dynamic programming) and *population distributions evolve forward* (initial data, push-forward); in the discrete setting these become backward Bellman recursion and forward state-distribution propagation, respectively.

Notation watch

Five conventions need reconciling with the RL literature and with the rest of the book; in each we name the clash the convention exists to prevent, not merely the choice we make. *(i) Cost vs. reward.* This book *minimises cost* (chapters 6 and 8) while RL conventionally *maximises reward*, and the two differ by an overall sign that, if left unflipped, silently turns every $\arg\min$ in a Bellman recursion into an $\arg\max$ and reverses the direction of every inequality below. We keep cost minimisation throughout; a reader translating an RL paper should read "reward $= -L$" and flip the relevant suprema, infima, and monotonicity signs together, not one at a time. *(ii) Discount.* The book's symbol $\beta = \beta$ is a discount *rate*, carrying units of inverse time (appendix A); a discrete-time MDP instead carries a dimensionless per-step *factor*, and writing both as β would conflate a rate with a pure number and corrupt the $\Delta t \to 0$ bookkeeping. We therefore reserve the distinct symbol $\gamma \in (0, 1]$ for the factor, related to the rate by $\gamma = e^{-\beta\,\Delta t}$ over a step of length Δt. A finite-horizon undiscounted problem is the case $\gamma = 1$, $K < \infty$; the symbol γ is local to this chapter. *(iii) Brownian motion vs. graphon.* Both objects classically lay claim to the letter W; as announced in appendix A and chapter 21, whenever the kernel W is in scope Brownian motion is written B. Here the dynamics are usually presented as transition kernels P_k rather than as SDEs, so the noise letter seldom appears and the clash rarely surfaces, but the convention stands wherever it does. *(iv) Best response: the two halves of the flow map.* This is the conceptual hinge of the chapter, so we belabour it. In chapters 8 and 28 the symbol S $(= \Phi)$ denotes the *full* dynamic flow self-map $\bar{m}_\bullet \mapsto m'_\bullet$ — one backward HJB sweep immediately followed by one forward Fokker–Planck sweep, fused into a single object whose fixed point is the equilibrium (eq. (28.2), proposition 8.4). That fusion was harmless when the only question was *whether* a fixed point exists, but it cannot survive here, because a learner performs the two sweeps with utterly different machinery — the backward sweep is a single-agent RL problem solved against a frozen population, the forward sweep is a Monte Carlo rollout of that population under a frozen policy — and the two analytic hypotheses of the chapter act on the two sweeps *separately.* We therefore split the composite and name the halves: S here is the *policy-valued* best response $\bar{m}_\bullet \mapsto \pi$ (the backward, dynamic-programming half, definition 29.2) and Ψ the *population* push-forward $\pi \mapsto m'_\bullet$ (the forward half, definition 29.3). The single map of chapters 8 and 28 is exactly the composite $\Psi \circ \mathsf{S}$, and we always write that composite out rather than abbreviate it, precisely so that statements like "the best response has Lipschitz modulus L_{S}" and "the push-forward has modulus L_Ψ" can be made about the two halves one at a time — it is their *product* $L_{\mathsf{S}} L_\Psi$ that the contraction condition will constrain (definition 29.9 and theorem 29.10), a statement that has no meaning until the seam is exposed. *(v) The graphon aggregate.* The general (vector-)observable aggregate written $z_k(x)$ below (eq. (29.24)) is Part IV's Θ-aggregate of eq. (21.4); we retain the letter z only because Θ is needed for Landau order-notation throughout this chapter, and reserve z for the identity-observable LQG case exactly as in eq. (21.5). Likewise Part IV's scalar test function h in $\langle h, m\rangle$ is here promoted to a (possibly vector) measure functional $\Lambda : \mathcal{P}(\mathcal{S}) \to \mathbb{R}^{d'}$, with $\Lambda(m) = \langle h, m\rangle$ recovering the scalar case of definition 21.5.

29.1 When learning replaces solving

The chapter's pivot, announced in the preamble, now has to be earned twice over. Before we may responsibly *learn* an equilibrium we owe the reader an honest account of *why* the analytic and numerical programme of the earlier parts — the forward–backward solvers that presuppose a fully specified model — is sometimes simply unusable; and once that case is made, we must replace those continuous, model-laden equations with a discrete object on which a sampling algorithm can actually run. This section does both, in that order. The first subsection sharpens the preamble's three deficiencies into three named failure modes, concrete situations in which solving is not merely expensive but impossible, so that learning is forced rather than chosen. The second then constructs the *mean field Markov decision process*, the discrete-time avatar of the games of chapter 8 on which every reinforcement learning algorithm of this chapter operates; there the book's backward-value convention reappears as a Bellman recursion and its forward-density convention as a push-forward of distributions, exactly as the preamble promised. This header only orients and hands the baton; the substance lives in the two subsections, and we begin with the failure modes, because they fix what the learnable object must be able to do.

29.1.1 Three failure modes of the analytic programme

The preamble promised that learning would be *forced*, not merely fashionable, and that promise now has to be redeemed concretely. The forward–backward solvers of chapter 28 are genuinely powerful — where they apply, they return the equilibrium to high precision — but they are not free-standing algorithms; they are subroutines that consume, as input, a complete and exact specification of the game: the drift b, the volatility σ, the running cost L, the terminal cost g, and (in the graphon setting) the kernel W, all in closed form. The discretisations of chapter 28 do not soften that requirement; a finite-difference stencil for the HJB equation (8.5a) cannot be assembled without the coefficients it is meant to discretise. So the operative question is not "is the solver accurate?" but "can the solver even be *fed*?" — and there are three recurring situations, drawn from genuinely different corners of applied modelling, in which the answer is no. We draw each one fully, because the shape of the obstruction dictates the shape of the cure, and the third in particular is the one the whole machinery of section 29.4 is built to dissolve.

Failure mode 1: the dynamics are a black box one can only sample. The first and most common obstruction is that the transition law of an agent is simply not available as an SDE with writable coefficients. In a wide range of applications the "model" is a piece of software — a simulator — whose internal logic is far too intricate to compress into a drift and a diffusion. Three concrete instances make the abstraction tangible. In *market microstructure*, an agent's next inventory and the price it transacts at are produced by a limit-order-book matching engine: queue priorities, partial fills, latency, and the reactive quoting of other participants together determine the outcome, and no closed-form $b(s, a)$ summarises the engine. In *epidemics with behavioural feedback*, the next health-and-contact state of an individual is generated by a compartmental simulator coupled to a behavioural module in which agents adjust their contact rates in response to perceived prevalence; the feedback loop between behaviour and transmission has no tidy McKean–Vlasov drift. In *traffic*, an agent's next position is the output of a microscopic car-following-and-routing simulation on a road network, with signals, merges, and capacity constraints. In every case one can do exactly one thing: place the agent at any state s, command any action a, *run* the simulator one step, and *observe* the realised next state s' and the realised cost. One cannot, however, open the box and read off b, σ, L in a form a PDE solver could ingest. This is the *simulator* hypothesis (in the RL literature, the *generative-model* hypothesis): the model is accessible *only* as a sampler, a map $(s, a) \mapsto (s', \mathrm{cost})$ realised by calling the simulator, never as a system of equations. Every algorithm in this chapter is engineered to need nothing more than this single sampling primitive, because in this regime it is all there is.

Failure mode 2: the state space is too large to grid. The second obstruction survives even when the dynamics *are* known in closed form, and it is purely a matter of dimension. The finite-difference schemes of chapter 28 represent the value function and the density by their values at the nodes of a grid laid over the state space $\mathcal{S} \subseteq \mathbb{R}^d$. To resolve each coordinate to a relative accuracy ε one needs on the order of $1/\varepsilon$ nodes per axis, hence about $(1/\varepsilon)^d = (\text{resolution})^d$ nodes in all — the storage and the per-sweep cost grow *exponentially* in the dimension d. This is the curse of dimensionality, and it bites with brutal speed: a portfolio of a dozen correlated assets ($d = 12$) or a queueing network of a dozen interacting servers already places, at a mere ten points per axis, 10^{12} nodes on the grid — beyond any memory, before one has taken a single time step. No refinement of the stencil escapes an exponential; the only known route out is to abandon the grid entirely and represent u and π by a *parametric family* — a function approximator, in practice a neural network — whose size is fixed in advance and does not scale with a notional resolution. A network with a few million weights carries the same parameter budget whether $\mathcal{S}$ sits in $\mathbb{R}^2$ or in $\mathbb{R}^{200}$; the dependence on d moves out of the representation cost and into the (far milder) sample complexity of fitting it. This trade — exponential grids for fixed-size approximators trained on samples — is the second reason the chapter is model-free in spirit even where the model is known, and it is what makes the simulator primitive of failure mode 1 not just necessary but *sufficient*: samples are exactly what a network is fit on.

Failure mode 3: the population is a continuum of types. The third obstruction is the one native to this book, and it compounds the first two rather than replacing them. In the graphon setting of Part IV the population is not a single measure but an entire *field* of measures, a one-parameter family $x \mapsto m^x \in \mathcal{P}(\mathcal{S})$ indexed by the agent's type $x \in [0,1]$ (definition 21.4), the types coupled through the kernel W so that the environment felt by an agent of type x is the aggregate $z(x) = (\mathbb{W}\,\Lambda[m^\cdot])(x)$ built by integrating against *all* other types. A solver that already grids the state space must now also grid the type axis and solve the per-type equilibrium problem *simultaneously* across that grid, evaluating a quadrature for the operator $\mathbb{W}$ at every type and at every time step (chapter 28). The costs multiply: a J-point type grid multiplies the work by J, and the per-step $\mathbb{W}$-quadrature couples the J problems so they cannot be solved independently. Even when each individual per-type problem is low-dimensional and benign, doing J of them in lockstep with all-to-all coupling at every stage is precisely the kind of cost the analytic programme cannot absorb at scale. The escape is structural and is worth naming now, because it is the load-bearing design decision of section 29.4: rather than learn a separate object for each type, one learns a *single*, type-*conditioned* policy — one neural network that accepts the type x (and the local aggregate it faces) as part of its input and outputs that type's action distribution. A single set of weights then *amortises* across the entire continuum of types: training signal gathered at one type informs the policy at nearby types through the shared parameters, the graphon enters only as an external quadrature feeding the aggregate into the network's input, and the J-fold blow-up never materialises. The reader should hear this third failure mode as foreshadowing: it is the obstruction whose resolution — one shared, type-conditioned network amortising the continuum, the kernel demoted to a quadrature on the input — is exactly what GMFG-PPO is in section 29.4, and it is the reason a chapter that could have stopped at scalar MFG instead carries the type label all the way through.

The unifying response. The three failure modes differ in cause — an unwritable model, an unaffordable grid, a continuum of coupled problems — but they call for one and the same change of strategy, and articulating it is the bridge into the rest of the chapter. In every case we abandon the ambition of *computing* the equilibrium fields (u, m) from their defining equations, and adopt instead the ambition of *learning an equilibrium policy* directly from interaction. The plan exploits the one structural fact the preamble isolated: an equilibrium is a fixed point of "best-respond, then push the population forward." So we build a loop. Freeze a guess at the population; *best-respond* to it by solving, with any sampling-based RL method, the single-agent control problem an individual faces in that frozen environment; then *update* the population by rolling many copies of that response forward through the simulator and reading off the distribution they realise; then repeat, hoping the two converge to mutual consistency. Nothing in this loop ever asks to write down b, σ, L, ever lays a grid on $\mathcal{S}$, or ever solves the J types as a coupled monolith — it asks only to sample, to fit an approximator, and to evaluate a quadrature, which is exactly what the three failure modes leave us able to do. But so far the loop is only a hope: "best-respond" and "update" are still informal verbs, and whether their alternation actually converges or instead cycles forever is the entire technical content of the chapter. To make either question precise we first need a discrete, samplable object on which the two verbs become honest maps with domains, codomains, and a fixed point. Building that object — the mean field Markov decision process, on which "best-respond" becomes a backward Bellman recursion and "update" a forward push-forward of distributions — is the task of the next subsection.

29.1.2 The mean field Markov decision process

The previous subsection left us with a loop whose two verbs — *best-respond* to a frozen population, then *update* the population by rolling the response forward — are still only verbs, and with a demand attached: before we may ask whether their alternation converges we must realise them as honest maps on a discrete object a simulator can actually touch. This subsection builds that object. Reinforcement learning is native to discrete time — its updates are indexed by transitions (s, a, s', cost), not by an infinitesimal generator — so the first task is to record the discrete-time *avatar* of the continuous mean field game of chapter 8, on which "best-respond" will become a backward Bellman recursion and "update" a forward push-forward of distributions, exactly along the backward/forward seam the preamble fixed (value functions backward from terminal data, populations forward from initial data).

The avatar is obtained by time-stepping, and it is worth seeing precisely how the two continuous objects turn into their discrete shadows, because the whole chapter trades on the correspondence. Partition the horizon $[0, T]$ into K stages of length $\Delta t = T/K$, write $k \in \{0, \ldots, K-1\}$ for the stage index and s for the state at stage k, and note that the two governing equations of chapter 8 are integrated in opposite directions of time. The backward Hamilton–Jacobi–Bellman equation (8.5a), read as the cost-minimisation principle "the value now is the best action's instantaneous cost plus the discounted value an instant later," becomes under one explicit Euler step the prescription

$$u_k(s) = \min_a \Big\{ L_k(s, a, m_k) + e^{-\beta \Delta t}\, \mathbb{E}[u_{k+1}(S_{k+1}) \mid S_k = s, A_k = a] \Big\},$$

where the one-step transition $S_{k+1} \sim P_k(\cdot \mid s, a, m_k)$ is the Euler–Maruyama push of the underlying

SDE — for $\mathrm{d}X = b\ \mathrm{d}t + \sigma\ \mathrm{d}B$ it is the Gaussian $P_k(\cdot \mid s, a, m) = \mathcal{N}(s + b(s, a, m)\,\Delta t,\ \sigma\sigma^\top \Delta t)$ — the running cost L_k absorbs the time-integrated stage cost $\int_{k\Delta t}^{(k+1)\Delta t} L\ \mathrm{d}t \approx L\,\Delta t$, and the discount *rate* β of the continuous problem reappears as a dimensionless per-step *factor* $\gamma = e^{-\beta\Delta t}$, exactly the substitution flagged in notationwatch (ii). That this is a faithful discretisation and not a loose analogy can be checked by reversing the step: expanding $\mathbb{E}[u_{k+1}(S_{k+1})] = u_{k+1}(s) + \Delta t(b\cdot\nabla u_{k+1} + \frac{1}{2}\sigma\sigma^\top : D^2 u_{k+1}) + o(\Delta t)$ and collecting the $O(\Delta t)$ terms returns (8.5a) as $\Delta t \to 0$. Dually, the forward Fokker–Planck (continuity) equation (8.5b), which transports the population density, becomes under the same step a push-forward of measures: the mass sitting at s at stage k is carried to stage $k+1$ by drawing an action from the policy and a next state from P_k. We write both recursions down in their own right below and treat the discrete object as primary, because it — not the PDE pair — is what the learner manipulates.

Definition 29.1 (Mean field Markov decision process)**.** A *mean field MDP* (MFMDP) consists of: a Polish *state space* $\mathcal{S}$; a compact metric *action set* A; a horizon $K \in \mathbb{N}$ with stages $k \in \{0, 1, \dots, K-1\}$; an *initial law* $m_0 \in \mathcal{P}(\mathcal{S})$; for each stage k a *transition kernel*

$$P_k :\ \mathcal{S} \times A \times \mathcal{P}(\mathcal{S}) \ \longrightarrow\ \mathcal{P}(\mathcal{S}), \qquad (s, a, m) \mapsto P_k(\,\cdot \mid s, a, m),$$

a *running cost* $L_k : \mathcal{S} \times A \times \mathcal{P}(\mathcal{S}) \to \mathbb{R}$, a *terminal cost* $g : \mathcal{S} \times \mathcal{P}(\mathcal{S}) \to \mathbb{R}$, and a per-step discount factor $\gamma \in (0, 1]$ (the factor of notationwatch (ii), distinct from the book's rate β). A *flow* is a sequence $m_\bullet = (m_k)_{k=0}^K$ with each $m_k \in \mathcal{P}(\mathcal{S})$. A (Markov, stochastic) *policy* is a sequence $\pi = (\pi_k)_{k=0}^{K-1}$ of measurable maps $\pi_k : \mathcal{S} \to \mathcal{P}(A)$, written $\pi_k(\mathrm{d}a \mid s)$.

Every datum in definition 29.1 earns its place, and three features deserve to be read slowly because they are exactly the points at which the mean field structure enters. First, the stage subscript on P_k and L_k records that the problem is *time-inhomogeneous*: a finite-horizon control problem is genuinely non-stationary — the value of a state depends on how many stages remain — so we must carry k explicitly and cannot collapse to a single stationary kernel. Second, and most important, P_k and L_k each take the population law m as a *third argument*. This is the mean field interaction made discrete: an agent's transition law and incurred cost are modulated by the current population distribution, so the environment an individual faces is not fixed but is *generated by the very population the individual belongs to*. This is precisely the McKean–Vlasov coupling of chapter 7, where the drift of a representative particle depends on the law of the particle itself, transcribed to a one-step kernel; and it is the same coupling that, frozen at a candidate law, defines the single-agent control problem of the MFG in chapter 8. The whole difficulty of the chapter — and the reason an equilibrium is a fixed point rather than the output of one optimisation — lives in this third slot: change m and you change the problem the agent is solving. Third, the discount γ is a dimensionless per-step factor, not the rate β, for the bookkeeping reason set out in notationwatch (ii); the finite-horizon undiscounted game is simply $\gamma = 1$ with $K < \infty$.

Fix a flow $m_\bullet$ now, and hold it frozen; call it the *environment*. The point of freezing is that it severs the mean field coupling: with m no longer responding to the agent, the third argument of P_k and L_k becomes a known, externally supplied sequence, and the agent faces an *ordinary* (time-inhomogeneous, single-agent) Markov decision process. Concretely, starting from $S_k = s$ and following π, with $A_k \sim \pi_k(\cdot \mid S_k)$ and $S_{k+1} \sim P_k(\cdot \mid S_k, A_k, m_k)$, the agent accumulates discounted cost, and the relevant quantity is the *cost-to-go*: the expected remaining discounted cost from stage k onward. Because the horizon is finite and the terminal cost is known, the cost-to-go is computed *backward* from $k = K$ — this is the discrete face of the book-wide convention that value functions are solved from terminal data. This is the half of the loop that "best-respond" will become; we make it precise next.

Definition 29.2 (Value, action-value, best response)**.** Given an environment $m_\bullet$ and a policy π, the *value function* $u_k^{m_\bullet,\pi} : \mathcal{S} \to \mathbb{R}$ is defined by the backward recursion

$$u_K^{m_\bullet,\pi}(s) = g(s, m_K), \qquad u_k^{m_\bullet,\pi}(s) = \mathbb{E}\Big[L_k(s, A_k, m_k) + \gamma\, u_{k+1}^{m_\bullet,\pi}(S_{k+1})\Big], \tag{29.1}$$

the expectation over $A_k \sim \pi_k(\cdot \mid s)$ and $S_{k+1} \sim P_k(\cdot \mid s, A_k, m_k)$. The *optimal value* $u_k^{m_\bullet}$ and the *action-value* (Q-)function $Q_k^{m_\bullet}$ solve the backward *Bellman optimality* recursion

$$Q_k^{m_\bullet}(s,a) = L_k(s,a,m_k) + \gamma\,\mathbb{E}\Big[u_{k+1}^{m_\bullet}(S_{k+1}) \mid S_k = s, A_k = a\Big], \qquad u_k^{m_\bullet}(s) = \min_{a\in A} Q_k^{m_\bullet}(s,a), \tag{29.2}$$

with $u_K^{m_\bullet} = g(\cdot, m_K)$. A policy π is a *best response* to $m_\bullet$ if $\pi_k(\cdot \mid s)$ is supported on $\arg\min_a Q_k^{m_\bullet}(s,a)$ for every (k,s); the set of best responses is denoted $\mathsf{S}(m_\bullet)$.

The two recursions in definition 29.2 deserve to be read one stage at a time, because their one-stage logic is exactly what makes them *samplable* — the property the whole chapter needs. Take the policy-evaluation recursion (29.1) first. Its terminal line is pure data: at $k = K$ no decision remains, so the cost-to-go is the terminal cost $g(s, m_K)$. For $k < K$, the value at s is the expected sum of two pieces an agent at s incurs in *one* step under π: the immediate cost $L_k(s, A_k, m_k)$ of the action it draws, plus the discounted cost-to-go $\gamma\, u_{k+1}$ already computed at the state it lands in. Because the right-hand side refers only to stage k's data and to u_{k+1} — never to u_k itself — the recursion is a genuine backward sweep: knowing the whole function u_{k+1} determines the whole function u_k by a single one-step expectation, with no implicit equation to solve. That one-step expectation is exactly a Monte Carlo quantity: draw $A_k \sim \pi_k(\cdot \mid s)$, call the simulator once to get S_{k+1} and the realised cost, and average. The Bellman optimality recursion (29.2) differs in one place only — where (29.1) *averages* over the action drawn from π, optimality *minimises* over the action. The action-value $Q_k^{m_\bullet}(s,a)$ isolates the cost of committing to action a now and behaving optimally thereafter, and $u_k^{m_\bullet}(s) = \min_a Q_k^{m_\bullet}(s,a)$ selects the best such commitment. A best response is then any policy that places its mass on the minimising actions, stage by stage and state by state; the set of all such policies is $\mathsf{S}(m_\bullet)$, the policy-valued backward half of the flow map named in notationwatch (iv).

Equation (29.2) is the discrete, sampling-friendly form of the HJB equation (8.5a), and the correspondence is exact term by term. The minimisation over a of the instantaneous cost plus the discounted continuation value *is* the discrete Hamiltonian: writing $\mathbb{E}[u_{k+1}(S_{k+1}) \mid s, a] = u_{k+1}(s) + \Delta t\, b(s,a,m_k)\cdot\nabla u_{k+1}(s) + O(\Delta t)$, the part of $Q_k(s,a)$ that actually depends on a is, to leading order in Δt, $L_k(s,a,m_k) + \Delta t\, b(s,a,m_k)\cdot\nabla u_{k+1}(s)$, whose minimisation over a is precisely the Legendre–Fenchel transform that defines the Hamiltonian $H(s, \nabla u, m)$ of chapter 6. The backward direction of the recursion is the discrete avatar of "value propagates inward from the terminal time," and the minimiser inherits the continuous meaning as well: the optimal feedback $\alpha^* = -\partial_p H$ of the verification theorem (chapter 6) becomes the *greedy* selection $a \mapsto \arg\min_a Q_k(s,a)$. The continuous gradient $-\partial_p H$ and the discrete arg min are the same object viewed at two resolutions; the discrete one is the one a simulator can evaluate, because it needs only the ability to score actions by sampled returns, never the closed-form Hamiltonian.

A one-stage instance of the LQG thread (Thread A, chapter 6) makes the abstract recursion concrete and exhibits the greedy arg min as a linear feedback. Take scalar state and dynamics $\mathrm{d}X = (aX + b\alpha)\,\mathrm{d}t + \sigma\,\mathrm{d}B$, so the Euler step gives the Gaussian kernel $P_k(\cdot \mid s,\alpha) = \mathcal{N}(s + (as + b\alpha)\Delta t,\ \sigma^2\Delta t)$, and take the mean-coupling running cost of the thread, $L_k(s,\alpha,m_k) = (\frac{1}{2}\alpha^2 + \frac{q}{2}(s - \kappa\bar{m}_k)^2)\Delta t$, where $\bar{m}_k = \int \xi\, m_k(\mathrm{d}\xi)$ is the population mean and κ is the mean-coupling weight (the thread's parameter s, written κ only here, since s already denotes the state). Suppose inductively that $u^{m_\bullet}_{k+1}(s) = \frac{1}{2}P_{k+1}s^2 + r_{k+1}s + \text{const}$ is quadratic, with Riccati curvature P_{k+1} and offset r_{k+1} in the thread's canonical symbols. Then $Q_k(s,\alpha)$ is quadratic in α with a positive leading $\frac{1}{2}\Delta t\,(1 + \gamma b^2 P_{k+1}\Delta t)\,\alpha^2$ term, so the greedy action is the unique *affine* feedback obtained by setting $\partial_\alpha Q_k = 0$, $\alpha^*(s) = -\gamma b\,(P_{k+1}s + r_{k+1}) + O(\Delta t)$ — linear in s with an offset fed by the population through r_{k+1} — and substituting it back preserves the quadratic form, advancing $(P_{k+1}, r_{k+1}) \mapsto (P_k, r_k)$ by exactly the one-step discrete Riccati recursion. Sending $\Delta t \to 0$ recovers the continuous Riccati ODE and the feedback $\alpha^* = -b\,(P(t)s + r(t))$ of chapter 6; the greedy arg min is the discrete shadow of $-\partial_p H$, as promised. The mean field coupling sits entirely in $\bar{m}_k$, which feeds the offset r_k and hence shifts where the agent is pulled — the single thread that the forward map will have to make consistent.

Definition 29.3 (Induced flow; the mean-field map)**.** Given a policy π and the fixed initial law m_0, the *induced flow* $\Psi(\pi) = m_\bullet$ is generated forward by

$$m_{k+1}(\mathrm{d}s') = \int_{\mathcal{S}}\int_A P_k(\mathrm{d}s' \mid s,a,m_k)\,\pi_k(\mathrm{d}a \mid s)\,m_k(\mathrm{d}s), \qquad k = 0,\dots,K-1. \tag{29.3}$$

The map Ψ is the forward half of the loop — the discrete avatar of "update the population" — and (29.3) is best read as a one-stage transport, line by line as the backward recursion was. Fix the stage law m_k and the stage policy π_k. The integral builds m_{k+1} by following an infinitesimal packet of probability through exactly one step of the simulator: start it at s with weight $m_k(\mathrm{d}s)$, let the agent draw an action with the stage-k probabilities $\pi_k(\mathrm{d}a \mid s)$, and scatter it to its next state according to the transition kernel $P_k(\mathrm{d}s' \mid s,a,m_k)$; integrating out the intermediate s and a leaves the distribution of where the packet lands. This is nothing but the Chapman–Kolmogorov forward equation for the controlled chain, and it is the discrete Fokker–Planck/continuity equation (8.5b) of chapter 8: probability mass is transported forward by the policy through the transition kernel, in the same time direction and with the same conservation law as the continuous flow.

Two structural facts about (29.3) matter and are worth stating explicitly, since they are what make Ψ a bona fide map and not merely a formula. First, *mass is conserved*: integrating the right-hand side over all $s' \in \mathcal{S}$ and using that each $P_k(\cdot \mid s,a,m_k)$ is a probability measure (total mass one) collapses the inner integral to 1, leaving $\int_{\mathcal{S}} \pi_k(\mathrm{d}a \mid s)\,m_k(\mathrm{d}s) = \int_{\mathcal{S}} m_k(\mathrm{d}s) = 1$, so $m_{k+1} \in \mathcal{P}(\mathcal{S})$ whenever $m_k \in \mathcal{P}(\mathcal{S})$ — a probability law is mapped to a probability law, exactly as the continuity equation conserves total density. Second, the recursion is *well posed* as a forward sweep: the right-hand side of (29.3) involves only m_k and π_k (the latter through both π_k itself and the kernel's measure argument m_k), never m_{k+1} on which it is solving, and never any later stage. There is no implicit equation and no coupling across stages to untangle: given the initial law m_0, applying (29.3) once yields m_1, applying it again yields m_2, and so on, so the entire induced flow $\Psi(\pi) = (m_0, m_1, \ldots, m_K)$ is built deterministically one stage at a time from π and m_0. This is what licenses the simulator realisation already used informally: roll many independent copies of the agent forward under π and read off the stage-k empirical measure, which converges to m_k as the number of copies grows.

With both halves of the loop now precise — S the backward, policy-valued best response of definition 29.2, and Ψ the forward, population-valued push-forward of definition 29.3 — we can state what it means for the loop to have settled, matching the continuous definition of chapter 8 and its graphon counterpart definition 21.16.

Definition 29.4 (MFMDP equilibrium)**.** A pair $(\pi^\star, m^\star_\bullet)$ is a *mean field equilibrium* if (consistency) $m^\star_\bullet = \Psi(\pi^\star)$ and (optimality) $\pi^\star \in \mathsf{S}(m^\star_\bullet)$. Equivalently, $m^\star_\bullet$ is a fixed point of the *composite* map $\Psi \circ \mathsf{S}$ — which is precisely the single dynamic flow map called S in chapters 8 and 28 (notationwatch (iv)) — and $\pi^\star$ a fixed point of $\mathsf{S} \circ \Psi$; this is the discrete-time avatar of the equivalence proposition 8.4, whose fixed point is that composite flow map.

It is worth pausing on the equivalence asserted in definition 29.4, because this is the place where notationwatch (iv) becomes concrete rather than a promissory note. We now have two genuinely different maps with their own names. The best response S takes a frozen environment to an optimal policy, $\bar{m}_\bullet \mapsto \pi$; this is the backward, dynamic-programming half (definition 29.2). The push-forward Ψ takes a frozen policy to the population it generates, $\pi \mapsto m'_\bullet$; this is the forward, simulation half (definition 29.3). Compose them in the order "best-respond, then push forward" and one obtains a self-map of flows,

$$\bar{m}_\bullet \overset{\mathsf{S}}{\longmapsto} \pi = \mathsf{S}(\bar{m}_\bullet) \overset{\Psi}{\longmapsto} m'_\bullet = \Psi\big(\mathsf{S}(\bar{m}_\bullet)\big),$$

which is exactly the single dynamic flow map written $\mathsf{S} = \Phi$ in chapters 8 and 28: one backward HJB sweep immediately followed by one forward Fokker–Planck sweep, fused into one object (eq. (28.2), proposition 8.4). A flow is an equilibrium flow precisely when this composite returns it unchanged, $m^\star_\bullet = (\Psi \circ \mathsf{S})(m^\star_\bullet)$, and dually a policy is an equilibrium policy precisely when $\pi^\star = (\mathsf{S} \circ \Psi)(\pi^\star)$. We keep the two halves separately named, and write the composite out in full rather than abbreviating it, for the reason flagged in the preamble: the two analytic hypotheses of the chapter act on the two sweeps *separately*, and it is the *product* of their Lipschitz moduli that the contraction analysis of the next section will constrain — a statement that has no meaning until the seam between S and Ψ is exposed, as it now is.

Remark 29.5 (Why this is exactly the object RL can chase)**.** This is the conceptual payoff of the subsection, and it is worth stating without haste, because it is the hinge on which the entire chapter turns. The two clauses of definition 29.4 are not abstract conditions one hopes to verify; each corresponds *exactly* to one of the two things a learner equipped with nothing but a simulator can actually do. *Optimality* is a single-agent reinforcement-learning problem. Freeze the environment $m_\bullet$ and the mean field coupling is severed — the agent inhabits an ordinary MDP (29.2) with a fixed, externally supplied environment — which *any* RL method can attack: value iteration or backward dynamic programming when the state is small, Q-learning or fitted value iteration when only samples are available, policy gradient or PPO when the state is too large to enumerate and one represents π by a network. None of these needs the closed-form b, σ, L; each needs only to score actions by sampled returns, which is what the simulator provides. *Consistency* is a forward Monte Carlo simulation: instantiate many independent copies of the agent, all playing the same fixed π, roll them forward through the simulator, and estimate the stage-k distribution (29.3) by the empirical measure of the copies' states — the law-of-large-numbers estimate of $\Psi(\pi)$. An equilibrium is reached at the precise moment these two operations agree: when the policy the agent *learns against* a population is the same policy that *generates* that population, so that neither operation, applied next, would move anything. This is why an equilibrium is something an iterative loop can *chase*: it is a fixed point of the alternation, and a fixed point of two samplable maps is reachable, in principle, by applying them in turn. The rest of the chapter is the study of that dynamical system — when its alternation converges to the fixed point, when it instead cycles forever, and what must be added to force convergence.

The two maps are now defined, their composite is identified with the equilibrium-defining self-map of chapter 8, and remark 29.5 has told us why each is exactly an operation a simulator-bound learner can perform. What we still lack is a way to *measure* how far a given policy stands from the fixed point — a single nonnegative number, estimable from samples, that vanishes precisely at equilibrium and that an algorithm can be made to drive down. Equipped with such a yardstick we can finally analyse the alternation itself: form the naive iteration that simply applies $\Psi \circ \mathsf{S}$ over and over, and ask whether it contracts toward the fixed point or oscillates around it. Both tasks — naming the yardstick, which will be the *exploitability*, and dissecting the iteration of the two maps — are the business of the next section.

29.2 The equilibrium as a fixed point of two maps

The previous section closed by handing us two maps and a name for their composite. The best response S runs one backward dynamic-programming sweep against a frozen population (definition 29.2); the push-forward Ψ runs one forward Monte Carlo sweep of the population under a frozen policy (definition 29.3); and an MFMDP equilibrium is exactly a fixed point of the composite $\Psi \circ \mathsf{S}$ (definition 29.4). Remark 29.5 drove home the observation that makes this more than bookkeeping: each of the two maps is not an abstraction but *precisely* one of the two things a simulator-bound learner can actually do — solve a single-agent reinforcement-learning problem against a frozen environment, or roll many copies of the crowd forward and read off their empirical law. What that remark deliberately left open is the question this section exists to confront. Granting that we can *apply* each map, what happens when we *iterate* them? A fixed point is a static characterisation, but *reaching* one is a dynamical process, and the entire gap between a learning method that succeeds and one that spins forever lies not in the maps themselves but in the behaviour of their alternation. This section is where the static fixed-point picture of section 29.1 becomes a dynamical question.

Answering it takes two ingredients, and they organise the section into its two subsections. The first is a *ruler*: before one can ask whether an iteration approaches the fixed point, one must be able to say — with a single number — how far the current iterate stands from it, and that number must be both zero exactly at the fixed point and estimable from the very samples the learner already draws, never from foreknowledge of the equilibrium we are chasing. The opening subsection manufactures that ruler, the *exploitability*, and proves it has both properties. The second ingredient, built on top of the first, is the dynamical system itself: with a way to measure progress in hand, the closing subsection forms the most naive iteration imaginable, the bare alternation $m_\bullet^{n+1} = (\Psi \circ \mathsf{S})(m_\bullet^n)$, and asks exactly when its orbit contracts onto the fixed point. The answer re-imports, unchanged, a hypothesis from the analytic theory of Part III: the iteration converges geometrically precisely when the *product* of the two maps' Lipschitz moduli falls below one — the learning-theoretic face of (A-Coupling) — and otherwise it generically oscillates without ever settling. That oscillation, diagnosed at the section's close, is not a numerical nuisance to be tuned away but a structural failure of the naive loop, and it is exactly the disease that the stabilisers of section 29.3 are constructed to cure. We take the two ingredients in order, the ruler first, because every convergence statement in the remainder of the chapter is measured against it.

29.2.1 Exploitability: the objective a learner minimises

A dynamical system is only as analysable as the ruler one measures its progress by, so before forming any iteration we must build the scalar the header promised: a single nonnegative number that vanishes *exactly* when π is an equilibrium policy and is otherwise strictly positive, and that the learner can estimate from the same samples it already draws. Two requirements on this ruler pull in opposite directions, and reconciling them is the entire design problem. On the one hand it must *certify* equilibrium: it has to be zero at the fixed point of definition 29.4 and nowhere else, so that "the ruler reads zero" and "the game is solved" are the same statement, never merely correlated. On the other hand it must be *computable in the dark*: the learner does not know the equilibrium — finding it is the whole point — so the ruler may not refer to $\pi^\star$ or $m_\bullet^\star$, and it must be assembled from quantities the simulator already hands over. The obvious candidate, some distance $\mathcal{W}_2(m_\bullet, m_\bullet^\star)$ to the true equilibrium flow, fails the second test at the first hurdle: it presupposes the very object we are chasing, so it can grade a candidate only *after* the answer is known, which is useless to an algorithm that must decide, at each step, whether it has made progress.

The way out is to stop measuring distance in the space of flows and measure it in the space of *incentives* instead — a notion inherited from game theory that never mentions the equilibrium by name. Ask the only question an agent embedded in the crowd actually cares about: holding the entire rest of the population fixed at π (so that the environment it generates, $\Psi(\pi)$, is frozen), how much could a single agent *save* by tearing up π and playing optimally against that frozen crowd instead? This is a purely unilateral, single-agent quantity: it compares two costs computed in one and the same fixed environment, and it asks for no knowledge of where the equilibrium lies. Its logic is exactly the definition of equilibrium turned into a number. If the saving is zero, then no agent has any profitable deviation against the population π induces, which is precisely the statement that π best-responds to its own induced flow — equilibrium, by definition 29.4. If the saving is large, some agent could do much better by defecting, so π is correspondingly far from best-responding to the crowd it generates, and the size of the saving grades *how* far. This quantity is the *exploitability* (in the game-theoretic literature, the *Nash gap*), and the happy fact is that making it precise requires nothing beyond the value functions already in hand from definition 29.2: the cost of playing a policy against a frozen environment is exactly the initial-stage value function of that policy, averaged over where agents start.

Definition 29.6 (Exploitability)**.** For a policy π let $m_\bullet = \Psi(\pi)$ be its induced flow and write $J(\pi'; m_\bullet) = \int_{\mathcal{S}} u_0^{m_\bullet,\pi'}(s)\, m_0(\mathrm{d}s)$ for the expected cost of playing π' against the environment $m_\bullet$. The *exploitability* of π is

$$\mathcal{E}(\pi) \;:=\; J(\pi; m_\bullet) - \inf_{\pi'} J(\pi'; m_\bullet) \;=\; \int_{\mathcal{S}} \Big(u_0^{m_\bullet,\pi}(s) - u_0^{m_\bullet}(s) \Big)\, m_0(\mathrm{d}s). \tag{29.4}$$

Read the functional $J(\pi'; m_\bullet)$ slowly, because everything hinges on what is frozen and what is free in it. The subscript $m_\bullet$ is the *environment* — the population flow, held fixed throughout the computation, supplying the third argument to every P_k and L_k exactly as in the frozen-environment reading of definition 29.2. Against that fixed backdrop the agent is free to choose its own policy π', and $u_0^{m_\bullet,\pi'}(s)$ is the expected discounted cost of running π' from the start state s through the frozen environment; averaging over the initial law m_0 gives $J(\pi'; m_\bullet)$, the cost of *deviating to* π' while the crowd stays put. The key point is that the deviation never disturbs the environment: $m_\bullet = \Psi(\pi)$ is built once from the incumbent π and then held rigid while π' ranges over all policies. This is the single-agent thought experiment of a representative deviator who is too small to move the crowd — the same severing of the mean field coupling that turned the equilibrium's optimality clause into an ordinary MDP in remark 29.5.

The two expressions for $\mathcal{E}(\pi)$ in (29.4) are the same number written at two altitudes, and it is worth seeing why they agree. The first, $J(\pi; m_\bullet) - \inf_{\pi'} J(\pi'; m_\bullet)$, is the *game-theoretic* form: the cost the incumbent actually pays minus the cost of the best available deviation, i.e. the largest saving any deviator could extract. The second, $\int_{\mathcal{S}} (u_0^{m_\bullet,\pi} - u_0^{m_\bullet})\, \mathrm{d}m_0$, is the *control-theoretic* form: the initial-stage gap between the value of π and the optimal value, averaged over starts. Passing from the first to the second requires only one nonobvious step, the interchange

$$\inf_{\pi'} \int_{\mathcal{S}} u_0^{m_\bullet,\pi'}(s)\, m_0(\mathrm{d}s) \;=\; \int_{\mathcal{S}} \inf_{\pi'} u_0^{m_\bullet,\pi'}(s)\, m_0(\mathrm{d}s) \;=\; \int_{\mathcal{S}} u_0^{m_\bullet}(s)\, m_0(\mathrm{d}s),$$

and the interchange is legitimate precisely because of the structure of dynamic programming. In general, pulling an infimum inside an integral is only an inequality: any fixed π' has $\int u_0^{m_\bullet,\pi'}\, \mathrm{d}m_0 \geq \int \inf_{\pi''} u_0^{m_\bullet,\pi''}\, \mathrm{d}m_0$, and taking the infimum over π' on the left gives only $\inf_{\pi'} \int \geq \int \inf_{\pi'}$. The subtlety hiding in the reverse direction is that minimising the *average* cost might in principle demand different optimal policies at different start states, so that no single π' could attain the pointwise infimum everywhere at once, and the left side could strictly exceed the right. Dynamic programming dissolves exactly this worry: the greedy policy of (29.2), which at every (k, s) selects an action in $\arg\min_a Q_k^{m_\bullet}(s, a)$, is *simultaneously* optimal from every start state, because optimality is decided stage-by-stage and state-by-state rather than start-by-start. Hence one and the same policy attains $\inf_{\pi'} u_0^{m_\bullet,\pi'}(s) = u_0^{m_\bullet}(s)$ for all s at once, the infimum and the integral commute, and the two forms coincide. The lesson to carry forward is that the optimal value function $u_0^{m_\bullet}$ is a property of the *environment alone*, not of any particular deviation, which is why the second form isolates so cleanly the per-state regret $u_0^{m_\bullet,\pi} - u_0^{m_\bullet}$.

Proposition 29.7 (Exploitability characterises equilibrium)**.** *$\mathcal{E}(\pi) \geq 0$ for every π, and $\mathcal{E}(\pi) = 0$ if and only if π is a best response to its own induced flow $\Psi(\pi)$, i.e. $(\pi, \Psi(\pi))$ is an equilibrium in the sense of definition 29.4.*

Proof. Nonnegativity. For every state s one has the pointwise inequality $u_0^{m_\bullet,\pi}(s) \geq u_0^{m_\bullet}(s)$, because $u_0^{m_\bullet}(s) = \inf_{\pi'} u_0^{m_\bullet,\pi'}(s)$ is the infimum of the running policy's value over *all* policies, and an infimum never exceeds any of the values it is taken over — in particular not the value of the specific policy π. Integrating this nonnegative gap $u_0^{m_\bullet,\pi} - u_0^{m_\bullet} \geq 0$ against the nonnegative measure m_0 yields $\mathcal{E}(\pi) \geq 0$, in the second form of (29.4).

The equality case. Suppose $\mathcal{E}(\pi) = 0$. We are integrating a pointwise nonnegative function against a probability measure and getting zero, so that function must vanish m_0-almost everywhere: $u_0^{m_\bullet,\pi}(s) = u_0^{m_\bullet}(s)$ for m_0-a.e. s. We now propagate this initial-stage equality forward and read off, at each reached state, that π must put all its mass on Bellman-optimal actions. Introduce the stagewise gap $\delta_k(s) := u_k^{m_\bullet,\pi}(s) - u_k^{m_\bullet}(s) \geq 0$, which is nonnegative for the same infimum reason at every stage, and compare the two backward recursions that produce its two terms. Write the policy's own action-value and the optimal action-value as

$$Q_k^\pi(s,a) := L_k(s,a,m_k) + \gamma\, \mathbb{E}\Big[u_{k+1}^{m_\bullet,\pi}(S_{k+1}) \mid s,a\Big], \qquad Q_k^{m_\bullet}(s,a) := L_k(s,a,m_k) + \gamma\, \mathbb{E}\Big[u_{k+1}^{m_\bullet}(S_{k+1}) \mid s,a\Big],$$

so that the evaluation recursion (29.1) reads $u_k^{m_\bullet,\pi}(s) = \int Q_k^\pi(s,a)\, \pi_k(\mathrm{d}a \mid s)$ — an *average* of Q_k^π over the actions π draws — while the optimality recursion (29.2) reads $u_k^{m_\bullet}(s) = \min_a Q_k^{m_\bullet}(s,a)$ — a *minimum* of $Q_k^{m_\bullet}$. Because $u_{k+1}^{m_\bullet,\pi} \geq u_{k+1}^{m_\bullet}$ pointwise (that is $\delta_{k+1} \geq 0$), taking expectations gives $Q_k^\pi(s,a) \geq Q_k^{m_\bullet}(s,a)$ for every action a. Chaining the two recursions through this inequality,

$$u_k^{m_\bullet,\pi}(s) = \int Q_k^\pi(s,a)\, \pi_k(\mathrm{d}a \mid s) \;\geq\; \int Q_k^{m_\bullet}(s,a)\, \pi_k(\mathrm{d}a \mid s) \;\geq\; \min_a Q_k^{m_\bullet}(s,a) = u_k^{m_\bullet}(s),$$

where the first inequality is $Q_k^\pi \geq Q_k^{m_\bullet}$ and the second is that the π_k-average of $Q_k^{m_\bullet}$ cannot fall below its minimum. Now suppose $\delta_k(s) = 0$, i.e. the two ends of this chain coincide; then *both* intermediate inequalities are equalities, and each carries a consequence. Equality in the second, $\int Q_k^{m_\bullet}(s,a)\, \pi_k(\mathrm{d}a \mid s) = \min_a Q_k^{m_\bullet}(s,a)$, says that a probability average of the nonnegative excess $Q_k^{m_\bullet}(s,a) - \min_{a'} Q_k^{m_\bullet}(s,a') \geq 0$ is zero, which forces $\pi_k(\cdot \mid s)$ to be supported on $\arg\min_a Q_k^{m_\bullet}(s,a)$ — π plays only Bellman-optimal actions at (k,s), which is exactly the best-response condition of definition 29.2. Equality in the first, $\int (Q_k^\pi - Q_k^{m_\bullet})(s,a)\, \pi_k(\mathrm{d}a \mid s) = 0$ with a nonnegative integrand, forces $Q_k^\pi(s,a) = Q_k^{m_\bullet}(s,a)$ for π_k-a.e. a, i.e. $\mathbb{E}[\delta_{k+1}(S_{k+1}) \mid s,a] = 0$; since $\delta_{k+1} \geq 0$, this means δ_{k+1} vanishes at every state S_{k+1} actually reachable from (s,a) under P_k. This is the propagation: $\delta_k(s) = 0$ at a reached state forces both that π is greedy there *and* that $\delta_{k+1} = 0$ at the next reached states. Starting from $\delta_0 = 0$ m_0-a.e. and inducting forward over k, we conclude $\pi_k(\cdot \mid s)$ is supported on the Bellman-optimal actions at every (k,s) reached with positive probability under $\Psi(\pi)$ — precisely $\pi \in \mathsf{S}(m_\bullet)$ with $m_\bullet = \Psi(\pi)$, the equilibrium of definition 29.4.

The converse, read backward. Conversely, suppose π is a best response to its own induced flow, so $\pi_k(\cdot \mid s)$ is supported on $\arg\min_a Q_k^{m_\bullet}(s,a)$ at every (k,s). We show $\delta_k \equiv 0$ on reached states by backward induction, the natural direction since the value recursions run backward from the terminal stage. At the terminal stage both recursions return the same datum, $u_K^{m_\bullet,\pi} = g(\cdot, m_K) = u_K^{m_\bullet}$, so

$\delta_K = 0$. Assume $\delta_{k+1} = 0$, hence $u_{k+1}^{m_\bullet,\pi} = u_{k+1}^{m_\bullet}$ and therefore $Q_k^\pi = Q_k^{m_\bullet}$. Then, using that π_k is supported on the minimisers,

$$u_k^{m_\bullet,\pi}(s) = \int Q_k^\pi(s,a)\,\pi_k(\mathrm{d}a \mid s) = \int Q_k^{m_\bullet}(s,a)\,\pi_k(\mathrm{d}a \mid s) = \min_a Q_k^{m_\bullet}(s,a) = u_k^{m_\bullet}(s),$$

the third equality because π_k places all its mass exactly where $Q_k^{m_\bullet}$ attains its minimum. Thus $\delta_k = 0$, the induction closes, and in particular $\delta_0 = 0$, whence $\mathcal{E}(\pi) = \int \delta_0 \,\mathrm{d}m_0 = 0$. The forward computation of the equality case and this backward one are the same chain of identities traversed in opposite directions: at equilibrium, "π is greedy" and "the value gap vanishes" are two readings of a single fact. □

It is worth instantiating the proposition once on the LQG thread, where every term is explicit and the abstract regret becomes a visible quadratic. Take the one-step instance of definition 29.2, with the frozen environment $m_\bullet$ supplying the population mean $\bar{m}_k$ through the offset r_{k+1}, so that the best response is the affine feedback $\alpha^*(s) = -\gamma b\,(P_{k+1}s + r_{k+1})$ computed there. Let the incumbent π play a slightly mistuned affine feedback $\alpha(s) = -g\,(P_{k+1}s + r_{k+1})$ with gain $g \neq \gamma b$. Substituting α rather than the minimiser into the strictly convex quadratic $Q_k(s,\alpha)$ — whose leading coefficient $\propto (1 + \gamma b^2 P_{k+1}\Delta t)$ is positive — overshoots its minimum by an amount proportional to $(g-\gamma b)^2(P_{k+1}s + r_{k+1})^2$, so the per-state regret $u_0^{m_\bullet,\pi}(s) - u_0^{m_\bullet}(s)$ is a nonnegative quadratic in the gain error that vanishes *iff* $g = \gamma b$, and $\mathcal{E}(\pi)$ is its m_0-average — strictly positive for any mistuned gain, exactly zero at the true feedback. This is proposition 29.7 made tangible: the ruler reads the squared distance of the agent's policy from its best response, and reads zero only when the two coincide.

Proposition 29.7 is what certifies $\mathcal{E}$ as a legitimate ruler: it is the learner's loss function, nonnegative everywhere and zero *precisely* at equilibrium, so driving it to zero is synonymous with solving the game — there is no daylight between "the loss is small" and "the policy nearly best-responds to the crowd it makes," which is what every convergence theorem in the remainder of the chapter will exploit when it bounds $\mathcal{E}$. The remaining worry is whether such a loss is anything more than a definition. It looks circular: the second term $u_0^{m_\bullet} = \inf_{\pi'} u_0^{m_\bullet,\pi'}$ is itself the *solution* of an optimal-control problem, the very kind of object the chapter spends its effort learning, so it might seem that to score a policy one must already be able to find the best one. The resolution — and this is the practical hinge the whole estimator downstream rests on — is that the worry mistakes a tractable subproblem for the intractable full problem. Evaluating $\mathcal{E}(\pi)$ does *not* require finding the equilibrium; it requires only solving a *single-agent* control problem against the *frozen* environment $\Psi(\pi)$, with the mean field coupling severed. That is the benign, fixed-environment RL problem of remark 29.5, attackable by any sampling method, and it is decoupled from the hard part — the back-and-forth between agent and crowd — precisely because the population is held still while the deviator optimises. So $\mathcal{E}(\pi)$ is estimable from samples by a concrete three-step recipe that needs nothing the simulator does not already provide: roll out π to obtain (an empirical proxy for) its induced flow $\Psi(\pi)$ and the on-policy return $J(\pi;\Psi(\pi))$; train a best response $\hat{\pi}'$ against that frozen rollout distribution, by any RL subroutine, and estimate its return $J(\hat{\pi}';\Psi(\pi))$ by fresh rollouts; then difference the two returns. The estimate is the empirical Nash gap, and it inherits one honest caveat we will have to pay for later — because the best response is itself only *approximately* learned, the trained $\hat{\pi}'$ is generally suboptimal, so its return overestimates $\inf_{\pi'} J$

and the differenced estimate is a (downward-biased) *lower* bound on the true $\mathcal{E}(\pi)$; it certifies a floor on remaining exploitability, not a ceiling. This is exactly the exploitability estimator that the GMFG-PPO algorithm of section 29.4 will use to instrument its own convergence, and recognising it here, in the scalar setting, is why we built the loss before touching a single iteration.

With the ruler built, certified, and shown to be measurable in the dark, the section's second question can finally be posed sharply. We have a nonnegative loss $\mathcal{E}$ that is zero exactly at the fixed point and that an algorithm can monitor as it runs; what we do not yet have is a *dynamics* that drives it down. The most direct thing to try is the bare alternation of the two maps — best-respond to the current population, push the response forward, repeat — and the next subsection asks the obvious question about it: does iterating $\Psi \circ \mathsf{S}$ actually send $\mathcal{E}$ to zero, contracting onto the equilibrium, or does the loss stall and the orbit circle the fixed point forever? The ruler we have just forged is precisely the instrument that will let us answer it.

29.2.2 Naive fixed-point iteration and its contraction regime

The previous subsection handed us the exploitability $\mathcal{E}$ — a nonnegative loss, zero exactly at equilibrium and estimable from samples (definition 29.6, proposition 29.7) — together with a stated wish: to find a *dynamics* that drives this loss to zero. With that ruler in hand we can finally watch the most direct candidate dynamics run, and ask the question $\mathcal{E}$ was built to answer. The most direct conceivable algorithm does nothing clever at all: take the current guess at the population, best-respond to it, push that response forward to obtain a new population, and repeat — the bare composition of the two maps, applied over and over. This is the discrete dynamical system whose orbit we now study, and the question it poses is the one the section opened with, now sharpened by the ruler: does this orbit drive $\mathcal{E}$ to zero, settling onto the equilibrium fixed point, or does $\mathcal{E}$ stall while the orbit circles the fixed point forever? The answer, we will find, hinges on a single number — the product of two sensitivities — and that number turns out to be nothing other than the contraction hypothesis (A-Coupling) of the analytic theory, now wearing learning-theoretic clothes.

Definition 29.8 (Fixed-point iteration)**.** *Fixed-point iteration* (FPI), also called *Banach–Picard* or naive iteration, generates flows by

$$m_\bullet^{n+1} \;=\; (\Psi \circ \mathsf{S})(m_\bullet^n), \qquad n = 0, 1, 2, \ldots, \tag{29.5}$$

where S is made single-valued by a measurable selection (or by the regularisation of section 29.3). Equivalently, on policies, $\pi^{n+1} = \mathsf{S}(\Psi(\pi^n))$.

To analyse (29.5) as a genuine dynamical system we must measure distances between flows, and the right notion of distance is dictated by what the iteration manipulates. A flow is a whole *sequence* of stage laws $(m_0, \ldots, m_K)$, so the natural way to say that two flows are close is to ask that they be close *at every stage at once*; and the natural way to say two laws on $\mathcal{S}$ are close is the Wasserstein distance $\mathcal{W}_2$ of chapter 4, which is the metric in which push-forwards behave well and in which the population dynamics of Parts II–IV were already controlled. Combining the two, we equip the space of flows with

$$D(m_\bullet, m'_\bullet) \;:=\; \max_{0\le k\le K} \mathcal{W}_2(m_k, m'_k), \tag{29.6}$$

the largest stage-wise Wasserstein distance. The max rather than, say, a sum is deliberate: it makes "D small" mean "uniformly close across the whole horizon," which is exactly the control a Lipschitz estimate on the composite map will deliver, and it keeps the constant in the contraction estimate from growing with the horizon length K. Completeness of $\big((\mathcal{P}_2(\mathcal{S}))^{K+1}, D\big)$ — the property the Banach theorem will require — is inherited stagewise from the completeness of $(\mathcal{P}_2(\mathcal{S}), \mathcal{W}_2)$ proved in chapter 4: a D-Cauchy sequence of flows is, coordinate by coordinate, a $\mathcal{W}_2$-Cauchy sequence of laws, each of which converges to a limit law in $\mathcal{P}_2(\mathcal{S})$, and the finite tuple of these limits is a flow to which the original sequence converges in D. A finite product of complete spaces under the max metric is complete; that is all we use. With a complete metric on flows fixed, the iteration's fate is decided entirely by how much the two maps stretch or shrink this distance, and we name those two stretch factors now, for they are the only quantities the convergence question will turn on.

The point of isolating two separate moduli — rather than one Lipschitz constant for the fused composite — is the seam exposed in notationwatch (iv): the backward best-response sweep and the forward push-forward sweep stretch distances for entirely different reasons, and the analytic hypotheses of the chapter bear on them *one at a time*. The response modulus measures how violently the optimal policy swings when the population it is responding to is nudged; the population modulus measures how far the induced crowd is displaced when the policy generating it is nudged. These are the two sensitivities on which the entire iteration hinges, and the contraction condition will constrain not either one alone but their *product*.

Definition 29.9 (Coupling moduli)**.** The *response modulus* L_{S} and the *population modulus* L_Ψ are the smallest constants such that, for all flows and policies,

$$D\big(\Psi(\pi), \Psi(\pi')\big) \le L_\Psi\, d(\pi, \pi'), \qquad d\big(\mathsf{S}(m_\bullet), \mathsf{S}(m'_\bullet)\big) \le L_{\mathsf{S}}\, D(m_\bullet, m'_\bullet), \tag{29.7}$$

where d is a metric on policies compatible with D (for definiteness, the m_0-averaged total-variation distance between action distributions, maximised over stages).

It is worth reading the two inequalities of (29.7) as physical sensitivities before they appear in a proof. The first, with constant L_Ψ, says that the forward map is *stable*: perturb the policy a little and the crowd it rolls out shifts by at most L_Ψ times as much. This factor is typically benign — of order one or smaller — because a single push-forward step averages a small change in action probabilities over the whole population and over only finitely many stages, so the perturbation does not amplify; it is the discrete-time relative of the Lipschitz-in-control stability of the Fokker–Planck flow used in chapter 8. The second inequality, with constant L_{S}, is the dangerous one, and the next several pages are organised around it: it asks how much the $\arg\min$ of the Bellman recursion moves when the environment moves, and an $\arg\min$ can move a great deal for a small input change. The composite map $\Psi \circ \mathsf{S}$ therefore stretches D by at most the product $L_{\mathsf{S}} L_\Psi$, and whether that product sits below or above one is the entire story.

Theorem 29.10 (Existence, uniqueness, and geometric convergence of FPI)**.** *Assume* (A-Lip) *and*

(A-Nondeg) *so that the moduli of definition 29.9 are finite, and assume the* contraction condition

$$L \;:=\; L_{\mathsf{S}}\, L_{\Psi} \;<\; 1. \tag{29.8}$$

Then the MFMDP has a unique equilibrium $(\pi^\star, m^\star_\bullet)$, the FPI (29.5) converges to it from any initial flow at the geometric rate $D(m^n_\bullet, m^\star_\bullet) \le L^n\, D(m^0_\bullet, m^\star_\bullet)$, and the exploitability of the iterate decays as $\mathcal{E}(\pi^n) \le C\, L^n$ for a constant C depending on the moduli.

Proof. Existence, uniqueness, geometric convergence. The two inequalities of (29.7) compose: for any flows $m_\bullet, m'_\bullet$, applying first the response bound and then the population bound,

$$D\big((\Psi\circ\mathsf{S})(m_\bullet), (\Psi\circ\mathsf{S})(m'_\bullet)\big) \le L_\Psi\, d\big(\mathsf{S}(m_\bullet), \mathsf{S}(m'_\bullet)\big) \le L_\Psi L_{\mathsf{S}}\, D(m_\bullet, m'_\bullet) = L\, D(m_\bullet, m'_\bullet),$$

so the composite $\Psi\circ\mathsf{S}$ is L-Lipschitz on the complete metric space $\big((\mathcal{P}_2(\mathcal{S}))^{K+1}, D\big)$. With $L<1$ this is a contraction, and the Banach fixed-point theorem (chapter 2) applies verbatim: it furnishes a unique fixed point $m^\star_\bullet$, guarantees that the iterates (29.5) converge to it from *any* starting flow, and supplies the geometric rate. That rate is the standard Banach estimate, which we record because the exploitability bound will reuse it: iterating the contraction n times gives $D(m^n_\bullet, m^\star_\bullet) = D\big((\Psi\circ\mathsf{S})^n(m^0_\bullet), (\Psi\circ\mathsf{S})^n(m^\star_\bullet)\big) \le L^n\, D(m^0_\bullet, m^\star_\bullet)$, since $m^\star_\bullet$ is fixed by the map. Writing $D_0 := D(m^0_\bullet, m^\star_\bullet)$ for the initial error, this is the asserted $D(m^n_\bullet, m^\star_\bullet) \le L^n D_0$. The fixed point is the unique equilibrium flow by definition 29.4 — a flow is returned unchanged by $\Psi\circ\mathsf{S}$ exactly when it is an equilibrium — and $\pi^\star = \mathsf{S}(m^\star_\bullet)$ is the corresponding unique equilibrium policy.

Exploitability decay. It remains to convert the geometric closeness of the flows into geometric decay of the loss the learner actually monitors. Write $m^n_\bullet = \Psi(\pi^n)$ for the flow induced by the iterate $\pi^n = \mathsf{S}(m^{n-1}_\bullet)$. The key structural fact is that π^n is an *exact* best response, but not to its own induced flow $m^n_\bullet$ — rather to the *previous* flow $m^{n-1}_\bullet$, the one it was computed against. Its exploitability is by definition measured against its own flow $m^n_\bullet$, so the entire gap $\mathcal{E}(\pi^n)$ must come from the *mismatch* between the environment π^n optimised for, $m^{n-1}_\bullet$, and the environment it actually faces, $m^n_\bullet$. Making this precise is a matter of inserting the optimised environment as an intermediary. Start from the control-theoretic form of the loss, (29.4), and bound the integral by the supremum of its integrand against the probability measure m_0:

$$\mathcal{E}(\pi^n) = \int_{\mathcal{S}} \big(u_0^{m^n_\bullet,\pi^n}(s) - u_0^{m^n_\bullet}(s)\big)\, m_0(\mathrm{d}s) \;\le\; \big\|u_0^{m^n_\bullet,\pi^n} - u_0^{m^n_\bullet}\big\|_\infty.$$

Now add and subtract the two value functions computed in the *optimised* environment $m^{n-1}_\bullet$ — the policy-evaluation value $u_0^{m^{n-1}_\bullet,\pi^n}$ and the optimal value $u_0^{m^{n-1}_\bullet}$ — and split by the triangle inequality:

$$u_0^{m^n_\bullet,\pi^n} - u_0^{m^n_\bullet} = \underbrace{\big(u_0^{m^n_\bullet,\pi^n} - u_0^{m^{n-1}_\bullet,\pi^n}\big)}_{\text{(I)}} + \underbrace{\big(u_0^{m^{n-1}_\bullet,\pi^n} - u_0^{m^{n-1}_\bullet}\big)}_{\text{(II)}} + \underbrace{\big(u_0^{m^{n-1}_\bullet} - u_0^{m^n_\bullet}\big)}_{\text{(III)}}.$$

The middle term (II) *vanishes identically*: it is the per-state regret of π^n in the environment $m^{n-1}_\bullet$, and $\pi^n = \mathsf{S}(m^{n-1}_\bullet)$ is by construction an exact best response there, so $u_0^{m^{n-1}_\bullet,\pi^n} = u_0^{m^{n-1}_\bullet}$ pointwise (a policy's value coincides with the optimal value exactly when the policy is greedy, proposition 29.7). This is the inserted "$u^{n-1} = u^\star_{n-1}$" step: it is what makes the exact best response do its work, collapsing the three-term split to the two *environment-stability* terms (I) and (III). Each of those two compares the same kind of value function — (I) two policy-evaluation values of the *same* policy π^n, (III) two *optimal* values — evaluated in the two nearby environments $m_\bullet^{n-1}$ and $m_\bullet^n$. Both kinds of value function are Lipschitz in the environment: this is the backward-stability estimate, a discrete Grönwall run on the Bellman recursions (29.1) and (29.2), whose value at stage k depends on the environment only through the stage costs L_j and kernels P_j for $j \geq k$, each Lipschitz in its measure argument under (A-Lip); summing the per-stage perturbations backward over the K stages, with the discount $\gamma \leq 1$ preventing amplification, yields a finite environment-Lipschitz constant L_V for both the policy-evaluation map $m_\bullet \mapsto u_0^{m_\bullet,\pi}$ and the optimal-value map $m_\bullet \mapsto u_0^{m_\bullet}$. We cite this as the same backward-stability mechanism that makes the regularised best response Lipschitz in lemma 29.12; it is the discrete face of the value's continuous dependence on the coefficients, and we invoke it rather than re-derive it because the Grönwall is identical. Applying it to (I) and (III) and taking suprema,

$$\|u_0^{m_\bullet^n,\pi^n} - u_0^{m_\bullet^n}\|_\infty \leq \underbrace{\|u_0^{m_\bullet^n,\pi^n} - u_0^{m_\bullet^{n-1},\pi^n}\|_\infty}_{\leq L_V\, D(m_\bullet^{n-1},m_\bullet^n)} + \underbrace{\|u_0^{m_\bullet^{n-1}} - u_0^{m_\bullet^n}\|_\infty}_{\leq L_V\, D(m_\bullet^{n-1},m_\bullet^n)} \leq\; 2\, L_V\, D(m_\bullet^{n-1}, m_\bullet^n).$$

The exploitability is thus controlled by the distance between two *consecutive* flows. That distance, in turn, is geometrically small, by routing it through the equilibrium with one more triangle inequality and the Banach rate just proved:

$$D(m_\bullet^{n-1}, m_\bullet^n) \leq D(m_\bullet^{n-1}, m_\bullet^\star) + D(m_\bullet^\star, m_\bullet^n) \leq L^{n-1} D_0 + L^n D_0 = (1+L)\, L^{n-1} D_0.$$

Combining the last two displays, $\mathcal{E}(\pi^n) \leq 2L_V(1+L)L^{n-1}D_0 = C\,L^n$ with $C = 2L_V(1+L)D_0/L$, which is the claimed decay. $\square$

This theorem is the discrete-time, model-free shadow of the contraction proofs we have already seen, and recognising it as such is the point of the whole subsection. In chapter 20 the same Banach argument settled the static graphon game (theorem 20.14), and in chapter 24 it furnished the *small-coupling branch* of the GMFG existence theorem; there, as here, a self-map of an equilibrium object was shown to contract once a coupling-strength quantity fell below a threshold. The condition (29.8), $L = L_{\mathsf{S}} L_\Psi < 1$, is precisely the learning-theoretic face of that smallness hypothesis (A-Coupling). Read it as two demands on the strength of the mean field interaction: the population must not respond too violently to a change in policy (L_Ψ small), and — the binding one — the best response must not be too sensitive to the population (L_{S} small). When the kernel W is weak, a perturbation to the crowd barely moves the cost landscape an agent faces, so its $\arg\min$ barely moves, L_{S} stays below $1/L_\Psi$, and the loop contracts; this is the operator-norm smallness $\|\mathbb{W}\|$ that powered theorem 20.14 and the chapter 24 branch, transcribed into a statement about how much an iteration overshoots. The same one analytic fact that gave *existence by contraction* in Part IV now gives *convergence of learning by contraction* — not by analogy but as the same Lipschitz bookkeeping, applied to the same composite map, in the same complete metric. That continuity of hypothesis across the solving/learning divide is the chapter's spine, and theorem 29.10 is its first vertebra.

The trouble is that the binding demand is the one that nature rarely grants. We have argued that L_Ψ is usually mild, but L_{S} — the sensitivity of an $\arg\min$ — is typically *enormous*, and when $L_{\mathsf{S}} L_\Psi \geq 1$ the theorem's hypothesis fails and its conclusion lapses. The next heuristic explains, with a concrete picture, exactly what goes wrong when it does: not a slow drift but a violent, self-perpetuating oscillation, the signature failure that the entire stabilisation programme of section 29.3 exists to cure.

Heuristic (non-rigorous): Why naive iteration usually oscillates

The trouble is that L_{S} is typically *large*, and the reason is structural, not incidental. Best response is an $\arg\min$ (definition 29.2): at each (k, s) it returns the action minimising $Q_k^{m_\bullet}(s, \cdot)$, and the location of a minimiser is a notoriously unstable functional of the function being minimised. When two actions are nearly tied for best, an arbitrarily small shift in the population — which enters Q through the stage cost $L_k(s, \cdot, m_k)$ and through the continuation value — can tip the balance and flip the minimiser discontinuously from one action to the other, and this can happen on a whole region of state space at once. So S behaves like a near-discontinuous map: $d(\mathsf{S}(m_\bullet), \mathsf{S}(m'_\bullet))$ can be of order one even when $D(m_\bullet, m'_\bullet)$ is tiny, which is to say $L_{\mathsf{S}} \gg 1$. Against such a response modulus the contraction condition (29.8) has no chance unless the coupling is vanishingly weak.

Watch what this does to the loop in a *congestion* game, where agents dislike crowding so the coupling is repulsive. Suppose the current population guess has everyone bunched on the *left*. An agent best-responding to that guess sees the left as crowded and costly, so its $\arg\min$ sends it firmly to the *right*: $\pi^1 = \mathsf{S}(m^0_\bullet)$ is a go-right policy. Push that response forward through Ψ and the next population is now bunched on the right: $m^1_\bullet$ has everyone on the right. But now an agent best-responding to *this* guess sees the right as the crowded, costly side, and its $\arg\min$ sends it firmly back to the *left*: π^2 is a go-left policy, $m^2_\bullet$ bunches left, and we are exactly back where we started. The iteration has entered a *period-two* cycle — left, right, left, right — and because the best response commits *fully* to the opposite side each time (the $\arg\min$ is all-or-nothing, with no memory of how violently it just swung), the oscillation never shrinks. It is *undamped*: $\mathcal{E}(\pi^n)$ does not decay but bounces at a fixed positive amplitude, period after period. This is the classic *cobweb* oscillation of elementary economics — supply chases last period's price, price overshoots, supply overshoots back — and its root cause here is exactly $L_{\mathsf{S}} \gg 1$: each agent *overreacts* to the last guess, responding to where the crowd *was* rather than where it is heading, and a chain of mutual overreactions cannot converge. The equilibrium itself — everyone spread evenly, indifferent between left and right — sits unvisited at the centre of the cycle, a fixed point the naive dynamics orbits but never approaches.

The diagnosis points straight at the cure. The orbit fails not because the equilibrium is hard to find but because the response *overshoots*, L_{S} being far above one; so the remedy is to *damp the overreaction* rather than to hope it away. Two dampers suggest themselves, and the next section builds both. One can refuse to chase the latest best response and instead respond to a *running average* of all populations seen so far — so that a single wild swing is diluted by history and cannot drag the next iterate fully to the opposite extreme (this is *fictitious play*). Or one can refuse to take the full best-response step and instead move the policy only a *small* amount toward it each round — so that even an all-or-nothing $\arg\min$ produces only an incremental change in the played policy (this is *online mirror descent*). Both replace the volatile $\arg\min$ by something inert enough to converge, and crucially both will converge under a hypothesis far weaker than contraction — Lasry–Lions *monotonicity*, not smallness — which is what makes them the right tool when $L_{\mathsf{S}} L_\Psi \geq 1$. That the dichotomy is sharp — naive iteration contracting exactly when (29.8) holds and cycling exactly when it fails, with averaging rescuing the failed regime — is not left to intuition: it is made completely explicit, with every modulus computed in closed form, on the LQG thread in worked computation 29.22.

We close the section, then, with a precise diagnosis to hand forward: the naive loop fails not by some mistunable numerical accident but *structurally*, through the overreaction that the arg min in the best response builds in — $L_{\mathsf{S}} \gg 1$ driving the undamped period-two cobweb of section 29.2.2. Section 29.3 takes exactly this diagnosis as its point of departure and builds the two dampers the heuristic previewed, together with the one device they both quietly presuppose.

29.3 Stabilising the iteration

We inherit from section 29.2.2 a diagnosis sharp enough to act on. The naive loop does not fail because the fixed point is hard to locate or because some constant was tuned poorly; it fails because the best response *overreacts*. The response modulus L_{S} sits far above one — structurally so, because S ends in an arg min, and an arg min is a near-discontinuous object that can swing from one corner of the action set to the opposite under an arbitrarily small change in the population it faces — and once $L_{\mathsf{S}} L_\Psi \geq 1$ the contraction estimate (29.8) has nothing left to give, so the orbit settles into the period-two cobweb rather than onto the equilibrium. The cure cannot be to wish L_{S} smaller, since its size is dictated by the geometry of the problem and not by any dial we control; the cure must be to *damp the overreaction*. This section builds the two dampers the heuristic named — respond to a running average of past populations, or take only a small step toward each new best response — and, before either can be made to work, the one device both silently presuppose.

That order is the section's organising decision, and it is worth stating why it is forced rather than merely tidy. Both dampers are averaging schemes, and averaging is only meaningful between nearby objects: if the thing being averaged jumps discontinuously, then blending the last few iterates blends two essentially unrelated policies and inherits, rather than cures, the volatility. So the prerequisite for *any* averaging argument is that the best response vary *continuously* — indeed Lipschitz-continuously, with a modulus we can name — with the population it answers. The hard arg min does not have this property, but a softened best response does. *Entropy regularisation*, the subject of the first subsection below, replaces the hard minimisation in the Bellman optimality recursion (29.2) by a free-energy minimisation at a temperature $\tau > 0$; the minimiser becomes a smooth Gibbs (softmax) distribution that is single-valued where the arg min was set-valued and $\Theta(1/\tau)$-Lipschitz where the arg min was discontinuous. Regularisation therefore comes first not because it is the headline idea — it is a prerequisite, a small controlled bias accepted to buy the continuity the later analysis needs — and it must precede the averaging schemes because they are quite literally undefined as convergent dynamics without it. The price, a fixed point displaced by $O(\tau)$ from the true equilibrium, we pay knowingly and quantify when we reuse all of this in the modular guarantee of section 29.5.

With a Lipschitz best response in hand, the two monotonicity-powered dynamics follow, and here lies the conceptual headline that separates this section from the previous one. The contraction regime of section 29.2 rested on (A-Coupling), a *smallness* condition: the coupling had to be weak enough that the product $L_{\mathsf{S}}L_{\Psi}$ stayed below one. The averaging schemes ask for something altogether different. They converge under the Lasry–Lions *monotonicity* hypothesis (A-Mon) — the same condition that delivered uniqueness of the equilibrium in chapters 11 and 24 — and monotonicity is emphatically *not* a smallness assumption. It constrains the *sign* of how the population's cost responds to a perturbation of the population, never its *magnitude*; a coupling can be arbitrarily strong, with $L_{\mathsf{S}}L_{\Psi}$ arbitrarily large, and still be monotone. This is exactly why averaging reaches a regime that contraction cannot: where (29.8) simply fails and the naive loop is doomed to cycle, monotonicity remains available and turns the undamped oscillation into a Lyapunov descent. The second subsection develops the first such scheme, *fictitious play*, in which the agent best-responds to the time-average of all populations seen so far, so that one wild swing is diluted by history and cannot drag the next iterate to the opposite extreme. The third develops *online mirror descent*, in which the played policy itself is moved only incrementally toward each new best response, so that even an all-or-nothing $\arg\min$ produces an incremental change. The two differ in what they average — populations versus policy parameters — but share the engine: under (A-Mon) each admits a Lyapunov functional that decreases along the iteration, converting the cobweb of section 29.2.2 into monotone progress toward equilibrium.

The fourth and final unit makes the whole story quantitative on ground we can compute by hand. Returning to the LQG thread, the worked computation worked computation 29.22 carries out the naive-iteration and the averaged-iteration analyses side by side in closed form, every modulus evaluated explicitly, and exhibits the dichotomy with no room left for hand-waving: a single scalar parameter governs whether $L_{\mathsf{S}}L_{\Psi} < 1$ so that naive iteration contracts, or $L_{\mathsf{S}}L_{\Psi} \geq 1$ so that it spirals into the period-two oscillation while the averaged schemes nonetheless converge under monotonicity. That computation is the section's anchor: it turns "contract versus average" from a slogan into an inequality one can read off. Together the four units — regularisation, fictitious play, online mirror descent, and the LQG ledger — are the parts the modular convergence guarantee of section 29.5 will reassemble, the Lipschitz best response feeding the monotone averaging dynamics whose convergence the worked computation has already made concrete.

We begin with the prerequisite. The previous section's heuristic complained that the hard best response is near-discontinuous, L_{S} enormous because an $\arg\min$ can leap under an infinitesimal nudge; the next subsection answers exactly that complaint, smoothing the $\arg\min$ into a Gibbs distribution and pinning down the Lipschitz modulus the averaging schemes will require.

29.3.1 Entropy regularisation and the Gibbs best response

The cobweb diagnosis of section 29.2.2 located the disease with unusual precision: the naive loop oscillates because the best response S ends in a hard $\arg\min$, and an $\arg\min$ is a near-discontinuous map of its input. Two environments $m_\bullet$ and $m'_\bullet$ that are arbitrarily close can put two different actions infinitesimally above and below in Q-value, and the $\arg\min$ then jumps from the one to the other — a finite swing in the policy produced by an infinitesimal change in the population. That is what forces the response modulus L_{S} of definition 29.9 far above one, and through it the product $L_{\mathsf{S}} L_\Psi$ past the contraction threshold of (29.8). The previous section also told us the cure cannot be to wish L_{S} smaller, since its size is fixed by the geometry of the game; the cure must be to *round off the corner* of the arg min so that the policy slides continuously as the environment moves rather than snapping between extremes. This subsection builds exactly that rounding, names the price it exacts, and hands the resulting smooth, single-valued, Lipschitz best response to the averaging schemes that follow, which cannot even be defined as convergent dynamics without it.

The rounding has a thermodynamic reading that is worth stating before any formula, because it is the cleanest way to see why the construction is forced rather than arbitrary. The hard Bellman step (29.2) asks an agent to put *all* of its action probability on the single cheapest action — to minimise the expected one-step "energy" $\int_A Q_k^{m_\bullet}(s,a)\,p(\mathrm{d}a)$ over policies $p \in \mathcal{P}(A)$, whose minimiser is the degenerate measure $\delta_{\arg\min_a Q}$. A degenerate minimiser is precisely what makes the map discontinuous: it lives at a vertex of the simplex of action distributions, and an infinitesimal tilt of Q can send it to a different vertex. The remedy, familiar from statistical mechanics, is to charge the agent for being too certain: add to the energy a *temperature* $\tau > 0$ times the (negative) entropy of p, so that the agent now minimises the *free energy*

$$\mathcal{F}_\tau[p] \;=\; \underbrace{\int_A Q_k^{m_\bullet}(s,a)\,p(\mathrm{d}a)}_{\text{expected energy}} \;+\; \tau \underbrace{\int_A p\log p}_{-\,\text{entropy}}.$$

The entropy term $\int p\log p$ is minimised by the *uniform* distribution and blows up (towards its least negative value) as p concentrates, so it pulls in the opposite direction to the energy: the energy wants a spike on the cheapest action, the entropy wants a spread-out hedge, and the temperature τ sets the exchange rate between them. At $\tau \to 0$ the energy dominates and the free-energy minimiser collapses back to the hard arg min; at $\tau \to \infty$ the entropy dominates and the minimiser is the flat reference distribution, ignoring Q altogether. For any $\tau > 0$ in between, the minimiser is an *interior* point of the simplex — it never sits at a vertex, so it can no longer jump between vertices — and, because the free energy is now *strictly* convex in p (the entropy is strictly convex; the energy is linear), that interior minimiser is unique and moves continuously, indeed Lipschitz-continuously, as Q and hence the environment is perturbed. This is the entire mechanism: a temperature converts a discontinuous vertex-valued arg min into a smooth interior-valued softmax. We now record the construction precisely.

Definition 29.11 (Regularised MFMDP)**.** Fix a *temperature* $\tau > 0$. The *τ-regularised* value, Q-function, and best response replace the hard minimisation in (29.2) by a free-energy minimisation:

$$u_{\tau,k}^{m_\bullet}(s) = \min_{p\in\mathcal{P}(A)} \Big\{ \int_A Q_{\tau,k}^{m_\bullet}(s,a)\,p(\mathrm{d}a) + \tau\int_A p\log p \Big\}, \qquad Q_{\tau,k}^{m_\bullet}(s,a) = L_k(s,a,m_k) + \gamma\,\mathbb{E}\Big[u_{\tau,k+1}^{m_\bullet}(S_{k+1}) \mid s,a\Big], \tag{29.9}$$

with $u_{\tau,K}^{m_\bullet} = g(\cdot, m_K)$. The minimiser is the *Gibbs* (softmax/Boltzmann) policy

$$\mathsf{S}_\tau(m_\bullet)_k(\mathrm{d}a \mid s) \;=\; \frac{\exp\big(-Q_{\tau,k}^{m_\bullet}(s,a)/\tau\big)}{\displaystyle\int_A \exp\big(-Q_{\tau,k}^{m_\bullet}(s,a')/\tau\big)\,\mathrm{d}a'}\,\mathrm{d}a, \qquad u_{\tau,k}^{m_\bullet}(s) = -\tau\log\int_A e^{-Q_{\tau,k}^{m_\bullet}(s,a)/\tau}\,\mathrm{d}a. \tag{29.10}$$

The symbol $\int p \log p$ in (29.9) needs a reference against which "$\log p$" is measured, and pinning that reference down is genuine setup, not a technicality, because it is what makes both the closed form and the bias bound below well defined. Entropy is not an absolute quantity: $\int p \log p$ only makes sense once one fixes a *reference probability measure* ν_0 on the action set A against which the spread of p is gauged, and what the term really abbreviates is the *relative entropy* (Kullback–Leibler divergence)

$$\int_A p \log p \;:=\; D_{\mathrm{KL}}(p \,\|\, \nu_0) \;=\; \int_A \frac{\mathrm{d}p}{\mathrm{d}\nu_0} \log \frac{\mathrm{d}p}{\mathrm{d}\nu_0}\, \mathrm{d}\nu_0 \;\geq\; 0,$$

defined for $p \ll \nu_0$ and equal to $+\infty$ otherwise, and the integrals $\int_A \cdots \mathrm{d}a$ in (29.10) are likewise taken against ν_0. The two canonical choices are the natural ones: *normalised counting measure* $\nu_0 = |A|^{-1} \sum_a \delta_a$ when A is a finite set, and *normalised Lebesgue measure* when A is a compact subset of Euclidean space. The choice of ν_0 fixes the "ground state" the regulariser pulls towards — the agent is charged for deviating from ν_0, not from nothing — so it is ν_0, the uniform distribution in both canonical cases, that the high-temperature limit returns.

Two boundedness properties of this relative entropy are what the rest of the subsection rests on, and both flow from the assumption that ν_0 is a *probability* measure on a *compact* A. First, $D_{\mathrm{KL}}(p \,\|\, \nu_0) \geq 0$ always, with equality iff $p = \nu_0$; this is Gibbs' inequality and is what makes the regularised value a genuine lower bound below the unregularised one. Second, and more delicately, the relative entropy is *bounded above* on the class of policies we use, and the bound is finite *because* ν_0 has finite total mass. In the finite-action case $D_{\mathrm{KL}}(p \,\|\, \nu_0) = \log |A| - H(p) \leq \log |A|$, where $H(p) = -\sum_a p(a) \log p(a) \geq 0$ is the Shannon entropy; the maximal relative entropy is attained at a vertex $p = \delta_a$ and equals $\log |A|$, a finite number precisely because there are finitely many actions to count. In the compact-action case there is no count $|A|$, and one instead controls the entropy term directly: write $H_{\max}(A) := \sup_p \big(-\int_A p \log p \; \mathrm{d}a \big) < \infty$ for the maximal *differential* (Shannon) entropy of a policy on A — taken against the base measure $\mathrm{d}a$, not the relative entropy $\int p \log p$ above, and the continuum analogue of the count $\log |A|$ — a finite supremum exactly because ν_0 is a finite measure on a bounded set, so the densities cannot spread without limit. *This finiteness is the whole reason the $O(\tau)$ bias bound below is meaningful*: the regularised and the true objectives differ by τ times a relative entropy, and if that entropy could be unbounded — as it would be against an infinite reference measure on a non-compact action set — the bias would be uncontrolled no matter how small τ is. Boundedness of ν_0 is what keeps the price of regularisation proportional to τ.

With ν_0 fixed, the closed form (29.10) is the elementary Gibbs variational principle, and it is worth deriving rather than asserting, because the derivation is a two-line calculation that the reader can carry in their head and that makes the $1/\tau$ in the exponent inevitable. Minimise $\mathcal{F}_\tau[p] = \int Q\, p + \tau \int p \log p$ over densities $g = \mathrm{d}p / \mathrm{d}\nu_0 \geq 0$ subject to the single constraint $\int g \; \mathrm{d}\nu_0 = 1$. Adjoining a multiplier λ for the constraint and differentiating the Lagrangian $\int (Qg + \tau g \log g - \lambda g) \, \mathrm{d}\nu_0$ pointwise in $g(a)$ gives the stationarity condition $Q(a) + \tau(\log g(a) + 1) - \lambda = 0$, that is $g(a) = \exp\big((\lambda - \tau)/\tau\big)\, e^{-Q(a)/\tau}$; the constant prefactor is fixed by normalisation to be $1/Z$ with the *partition function* $Z = \int_A e^{-Q(a)/\tau} \, \nu_0(\mathrm{d}a)$, which is exactly the Gibbs density in (29.10). Substituting this minimiser back gives the optimal value: since $\log g = -Q/\tau - \log Z$,

$$\mathcal{F}_\tau[p^\star] = \int Q\, p^\star + \tau \int p^\star \big(-\tfrac{Q}{\tau} - \log Z\big) = \int Q\, p^\star - \int Q\, p^\star - \tau \log Z = -\tau \log Z = -\tau \log \int_A e^{-Q/\tau} \; \mathrm{d}\nu_0,$$

the *free energy*, which is the second equation of (29.10). The energy contribution $\int Q p^\star$ cancels exactly against the energy hidden inside the entropy, leaving the free energy as a pure log-partition function — the same cancellation that in equilibrium thermodynamics expresses the free energy through the partition function alone. Three consequences of this closed form now matter, and we take them in turn: it is single-valued and Lipschitz, it approaches the hard response as $\tau \downarrow 0$, and it induces a regularised equilibrium displaced from the true one by a controlled $O(\tau)$.

Lemma 29.12 (Regularised best response is single-valued and Lipschitz)**.** *Under* (A-Lip) *the Gibbs response* (29.10) *is single-valued and, for each fixed* $\tau > 0$*, Lipschitz in the environment: there is* $L_{\mathsf{S},\tau} < \infty$ *with* $d(\mathsf{S}_\tau(m_\bullet), \mathsf{S}_\tau(m'_\bullet)) \le L_{\mathsf{S},\tau} D(m_\bullet, m'_\bullet)$*. As* $\tau \downarrow 0$ *the Gibbs response concentrates on the greedy actions and* $\mathsf{S}_\tau \to \mathsf{S}$ *pointwise, while in general* $L_{\mathsf{S},\tau} = \Theta(1/\tau)$*.*

Proof sketch. There are three claims — single-valuedness, Lipschitz-in-environment, and the $\tau \downarrow 0$ limit — and the proof composes elementary facts for each.

Single-valuedness. The objective $p \mapsto \int Qp + \tau \int p \log p$ minimised in (29.9) is the sum of a term that is *linear* in p (the energy) and a term that is *strictly* convex in p (the relative entropy: $r \mapsto r \log r$ has second derivative $1/r > 0$, so $\int p \log p$ is strictly convex on the simplex of densities). A strictly convex function on a convex set has at most one minimiser, so for every $\tau > 0$ the Gibbs measure (29.10) is *the* minimiser, not merely *a* minimiser; the set-valued S has become single-valued S_τ.

Lipschitz in the environment. We obtain $L_{\mathsf{S},\tau} < \infty$ by composing two Lipschitz maps along the chain $m_\bullet \mapsto Q^{m_\bullet}_{\tau,\bullet} \mapsto \mathsf{S}_\tau(m_\bullet)$. The first map, environment to regularised Q-function, is Lipschitz: under (A-Lip) the stage data L_k, P_k depend Lipschitz-continuously on m_k, and the backward recursion (29.9) propagates that dependence stably down the stages. Concretely, the log-sum-exp value $u_{\tau,k} = -\tau \log \int e^{-Q/\tau}\, \mathrm{d}\nu_0$ is non-expansive in Q in sup-norm — it is a soft minimum, so shifting Q uniformly by c shifts $u_{\tau,k}$ by c and a sup-norm perturbation of Q moves $u_{\tau,k}$ by no more — so a perturbation of the stage data accumulates across the K stages by at most a geometric factor in γ, a discrete Grönwall in the stage index k (the summation analogue of lemma C.1). This leaves $m_\bullet \mapsto Q$ Lipschitz with a constant independent of τ. The second map, $Q \mapsto \mathrm{softmax}(-Q/\tau)$, is the one that carries the temperature, and it is $\Theta(1/\tau)$-Lipschitz from sup-norm in Q to total-variation distance in p — which is exactly the policy metric d fixed in definition 29.9. The product of the two constants is $L_{\mathsf{S},\tau} = \Theta(1/\tau)$.

The softmax Jacobian and the $\Theta(1/\tau)$ *scaling.* This is the heart of the matter, so we compute it. Take A finite for the algebra (the compact case is identical with sums replaced by integrals against ν_0). Differentiating $p_a = e^{-Q_a/\tau}/Z$ with $Z = \sum_{a'} e^{-Q_{a'}/\tau}$,

$$\frac{\partial p_a}{\partial Q_b} = -\frac{1}{\tau}\Big(p_a\, \delta_{ab} - p_a p_b\Big) = -\frac{1}{\tau}\, M_{ab}, \qquad M := \mathrm{diag}(p) - pp^\top.$$

The matrix M has a clean probabilistic identity: it is the *covariance of the one-hot indicator* of a p-distributed action. Let $A \sim p$ and let $e_A \in \mathbb{R}^{|A|}$ be its one-hot encoding, so $\mathbb{E}[e_A] = p$ and, since $e_A e_A^\top = \mathrm{diag}(e_A)$,

$$\mathrm{Cov}(e_A) = \mathbb{E}\Big[e_A e_A^\top\Big] - pp^\top = \mathrm{diag}(p) - pp^\top = M.$$

A covariance matrix is positive semidefinite, so $M \succeq 0$; and since $pp^\top \succeq 0$ we also have $M = \mathrm{diag}(p) - pp^\top \preceq \mathrm{diag}(p)$. Sandwiched as $0 \preceq M \preceq \mathrm{diag}(p)$, its operator norm is bounded by that of the diagonal, $\|M\|_{\mathrm{op}} \le \max_a p_a \le 1$. That is the bound on the ℓ^2 side; to land in total variation, which is the metric d asks for, we count the entries directly. For a sup-norm perturbation δQ with $\|\delta Q\|_\infty = \delta$, the induced $\delta p = -\frac{1}{\tau} M\, \delta Q$ has

$$\|\delta p\|_1 = \sum_a \left| \tfrac{1}{\tau} \textstyle\sum_b M_{ab}\, \delta Q_b \right| \le \tfrac{\delta}{\tau} \textstyle\sum_{a,b} |M_{ab}| = \tfrac{\delta}{\tau}\Big(\underbrace{\sum_a p_a(1-p_a)}_{\text{diagonal}} + \underbrace{\sum_{a\ne b} p_a p_b}_{\text{off-diagonal}} \Big) = \tfrac{\delta}{\tau}\, 2\big(1 - \textstyle\sum_a p_a^2\big) \le \tfrac{2\delta}{\tau},$$

where the off-diagonal sum is $\sum_{a\ne b} p_a p_b = (\sum_a p_a)^2 - \sum_a p_a^2 = 1 - \sum_a p_a^2$ and the diagonal sum is $\sum_a p_a - \sum_a p_a^2 = 1 - \sum_a p_a^2$. The total-variation move is therefore $\frac{1}{2}\|\delta p\|_1 \le \delta/\tau$: a sup-norm change of δ in Q shifts the policy by at most δ/τ in total variation, so the softmax is $\Theta(1/\tau)$-Lipschitz from sup-norm to TV. The precise constant is immaterial; the load-bearing fact is the inverse-temperature scaling — the regulariser buys continuity, but the modulus it buys degrades like $1/\tau$ as the temperature is lowered, which is the bias–variance tension we return to in section 29.5.

The hard limit $\tau \downarrow 0$. Finally, the Gibbs response concentrates on the greedy actions as the temperature vanishes, recovering the hard S pointwise. Write $Q_{\min} = \min_a Q(a)$ and factor it out of every weight: $p_\tau(a) = e^{-(Q(a)-Q_{\min})/\tau} / \int e^{-(Q(a')-Q_{\min})/\tau}\, \mathrm{d}\nu_0$. Each integrand $e^{-(Q(a)-Q_{\min})/\tau}$ is bounded by the constant 1 — which is ν_0-integrable precisely because ν_0 is a probability measure, so the finiteness of ν_0 does work here too — and converges pointwise as $\tau \downarrow 0$ to the indicator $\mathbf{1}\{Q(a) = Q_{\min}\}$ of the arg min set: it stays equal to 1 where $Q(a) = Q_{\min}$ and decays to 0 wherever $Q(a) > Q_{\min}$. Dominated convergence then carries the limit through both numerator and denominator, so p_τ converges to ν_0 restricted and renormalised to the arg min set — exactly the greedy best response $\mathsf{S}(m_\bullet)$ of definition 29.2. The same domination gives the value: $u_{\tau,k} = Q_{\min} - \tau \log \int e^{-(Q-Q_{\min})/\tau}\, \mathrm{d}\nu_0 \to Q_{\min} = \min_a Q$, so the regularised Bellman step (29.9) returns the hard step (29.2) in the limit. The two limits together say $\mathsf{S}_\tau \to \mathsf{S}$ as $\tau \downarrow 0$ while $L_{\mathsf{S},\tau} = \Theta(1/\tau) \to \infty$ — faithfulness and continuity traded against each other through the single knob τ. □

Having a continuous, single-valued best response in hand does more than enable the averaging schemes; it resurrects, for the *regularised* game, the existence and uniqueness theory the hard arg min had put out of reach, and at a price we can now quantify. The next remark records this, closing the loop between the smoothed game and the original one.

Remark 29.13 (Regularised equilibrium and its bias)**.** The regularised game has its own equilibrium $(\pi^\star_\tau, m^\star_{\tau,\bullet})$, a fixed point of $\Psi \circ \mathsf{S}_\tau$. Its existence follows from the now-*continuous* best response by the Schauder fixed-point theorem (chapter 2) applied on the compact convex set of flows supplied by theorem 10.14: the requisite compactness in $\mathcal{P}(\mathcal{S})$ is Prokhorov tightness, established and used there (chapter 10), and the continuity of $\Psi \circ \mathsf{S}_\tau$ — the hypothesis Schauder needs and the one the hard arg min failed — is exactly lemma 29.12. This is the structural payoff of regularisation: where the unregularised composite $\Psi \circ \mathsf{S}$ was only *upper hemicontinuous* and demanded the heavier Kakutani machinery (and even then guaranteed nothing about a learning loop converging to the fixed point), the regularised composite is an honest continuous self-map of a compact convex set, so a fixed point exists by the elementary theorem. Uniqueness then follows under (A-Mon) (developed in the next subsection) or under the contraction condition (29.8) read with the regularised modulus $L_{\mathsf{S},\tau}$ — the two hypotheses that run as the chapter's twin spine.

The regularised fixed point is, however, emphatically *not* the equilibrium of the original game, and honesty requires us to say by how much it misses. The miss is measured by the exploitability (29.4) of $\pi^\star_\tau$ *in the true, unregularised game*, and it is $O(\tau)$. The reason is that the regularised objective differs from the true one (29.2) by exactly the added term $\tau\int p\log p$, a relative entropy that we established above is bounded; carrying that bounded perturbation through the value comparison gives, for finite A,

$$\mathcal{E}(\pi^\star_\tau) \;\le\; c_0\,\tau\,\big(1+\log|A|\,\big),$$

where the $\log|A|$ is the maximal relative entropy $\log|A|$ counted in the setup — the bias is large only when there are exponentially many actions to spread mass over, and even then only logarithmically so. When A is instead a compact continuum there is no count $|A|$ to insert, and the same argument runs with the maximal differential entropy in its place,

$$\mathcal{E}(\pi^\star_\tau) \;\le\; c_0\,\tau\,\big(1+H_{\max}(A)\big),$$

with $H_{\max}(A)$ the finite maximal ν_0-entropy fixed in the setup above — finite precisely because ν_0 is a finite reference measure on a compact set, which is the place the boundedness discussion pays off. The two bounds are the same statement counted two ways: the bias is τ times "how much entropy the regulariser can inject," which is a pure count $\log|A|$ in the discrete case and a measure-theoretic ceiling $H_{\max}(A)$ in the continuous one. Thus τ is a knob with two opposed faces, exactly as the section header promised: *large τ stabilises but biases* — it shrinks $L_{\mathsf{S},\tau}$, easing every convergence argument below, while displacing the fixed point by $O(\tau)$ — and *small τ is faithful but stiff* — it sends the bias to zero but lets $L_{\mathsf{S},\tau}=\Theta(1/\tau)$ blow up, so the very continuity the averaging schemes rely on degrades. Choosing, or annealing, τ is therefore the central tuning decision of the method, and the residual $O(\tau)$ bias is one of the three error terms the modular guarantee of section 29.5 must carry to the end.

We have what the section's first unit was charged to produce: a best response that is single-valued, Lipschitz with a named modulus $L_{\mathsf{S},\tau}=\Theta(1/\tau)$, and faithful to the original game up to a controlled $O(\tau)$ bias. This is precisely the object the two damping schemes presuppose but cannot themselves manufacture. Fictitious play, next, cures the cobweb by having the agent best-respond not to the latest population but to the *running average* of all populations seen so far — the "respond to a running average" remedy previewed in section 29.2.2 — and that averaging is meaningful only because S_τ now varies continuously with its argument, so that averaging nearby environments yields a nearby response rather than blending unrelated extremes. Online mirror descent, after it, will lean on the same Lipschitz response from the policy side. We hand both the smoothed best response, and proceed to the first of the two monotonicity-powered dynamics.

29.3.2 Fictitious play

The first damper the cobweb heuristic named was to refuse to chase the latest best response and instead respond to a *running average* of every population seen so far. That is *fictitious play*, and it is the natural place to begin: it is the oldest learning dynamic in game theory [74] — old enough to predate the modern algorithmic vocabulary of equilibrium — and the first to be carried over to mean field games [38]. Its entire content is the single substitution the previous section previewed. Where the naive loop set $\pi^n = \mathsf{S}(m_\bullet^{n-1})$, best-responding to the *most recent* population and thereby inheriting its full volatility, fictitious play best-responds instead to the *time-average* $\bar{m}_\bullet^{n-1}$ of all populations produced so far. It is the previous subsection that makes this substitution legitimate rather than wishful: averaging populations and expecting a stable response out the other side is only sensible if nearby populations elicit nearby responses, and that Lipschitz continuity is exactly what the Gibbs best response S_τ of lemma 29.12 now supplies — the hard arg min, which could blend two unrelated extremes under an average, had no such property. We continue to write S for the response throughout this subsection, understanding it to be the regularised, single-valued S_τ that the averaging requires; the temperature τ rides along, contributing only the $O(\tau)$ bias of remark 29.13, and we suppress it in the notation to keep the dynamics in view.

Definition 29.14 (Fictitious play)**.** Fix an initial flow $\bar{m}_\bullet^0$. *Fictitious play* (FP) iterates, for $n \geq 1$,

$$\pi^n = \mathsf{S}(\bar{m}_\bullet^{n-1}), \qquad m_\bullet^n = \Psi(\pi^n), \qquad \bar{m}_\bullet^n = \Big(1 - \tfrac{1}{n}\Big)\bar{m}_\bullet^{n-1} + \tfrac{1}{n}\, m_\bullet^n, \tag{29.11}$$

the convex combination taken stage-wise in $\mathcal{P}(\mathcal{S})$. The played policy is the realised mixture whose flow is $\bar{m}_\bullet^n$.

Read the third line of (29.11) as the discrete update of a running mean: $\bar{m}_\bullet^n$ is the average of the first n induced flows $m_\bullet^1, \dots, m_\bullet^n$, and the weight $\frac{1}{n}$ on the newest arrival shrinks as the history lengthens. This is precisely what damps the cobweb. The *environment* $\bar{m}_\bullet^{n-1}$ that the best response now answers moves by only an $O(1/n)$ increment from one round to the next, so it becomes ever more nearly *quasi-static*: the response still overreacts to whatever it is shown — L_{S} is no smaller than before, the arg min is just as steep — but it is shown a target that barely moves, and an overreaction to a nearly stationary target cannot compound into a growing oscillation. Return to the congestion picture of section 29.2.2, where the naive loop flung the whole population left, then right, then left again at undiminished amplitude. Under fictitious play the first response still says "go right," but the played population is now the *average* of left-so-far with the new right, which sits nearer the centre; the next response, answering that gentler environment, is milder; and the running mean settles onto the even spread the naive loop could only orbit. The wild swing is not suppressed at its source — the best response is as violent as ever — but it is *diluted by history*, its pull on the played flow falling off like $1/n$.

What this buys, and the reason fictitious play belongs in this section rather than the last, is that its convergence is governed by an entirely different hypothesis from the contraction regime. It does not ask the coupling to be weak. It asks the coupling to be *monotone*: the same Lasry–Lions condition (A-Mon) that delivered *uniqueness* of the equilibrium in chapters 11 and 24 now reappears, unchanged, as the condition that delivers *convergence* of the learning dynamic. As the section header insisted, monotonicity constrains the *sign* of how the population's cost responds to a perturbation of the population, never its *magnitude*, so a coupling can be arbitrarily strong — with $L_{\mathsf{S}} L_\Psi$ arbitrarily large and the naive loop hopelessly cycling — and fictitious play converge nonetheless. The next theorem states this, and its proof is where monotonicity does its work.

Theorem 29.15 (Convergence of fictitious play under monotonicity)**.** *Assume* (A-Lip)*,* (A-Nondeg)*, and the Lasry–Lions monotonicity condition* (A-Mon) *on the couplings — the classical scalar form* *definition 11.1 or, at the graphon level relevant to section 29.4, the graphon Lasry–Lions condition definition 24.15 with its uniqueness theorem theorem 24.16. Then the fictitious-play iterates* (29.11) *have exploitability tending to zero,*

$$\mathcal{E}(\bar{\pi}^{\,n}) \;\longrightarrow\; 0 \qquad (n \to \infty), \tag{29.12}$$

where $\bar{\pi}^{\,n}$ is the policy realising $\bar{m}^n_\bullet$. Consequently every weak limit point of $(\bar{m}^n_\bullet)$ is an equilibrium flow, and it is the unique equilibrium under (A-Mon) *(theorems 11.5 and 24.16).*

Proof: the Lyapunov mechanism. The argument is a Lyapunov computation carried out in *continuous learning time*, and it is worth saying at the outset what it does and does not deliver: it establishes the convergence (29.12), $\mathcal{E} \to 0$, in full and in-book, and it isolates monotonicity as the exact structural reason the convergence happens; the sharp algebraic *rate* it does not deliver, and remark 29.16 accounts honestly for what must be imported instead.

Step 0: from the $\frac{1}{n}$ iteration to a flow. The discrete update is awkward to differentiate because its step size $\frac{1}{n}$ changes every round; the standard device is to re-time it so the steps become uniform. Set $\vartheta = \log n$, so that advancing $n \mapsto n+1$ advances ϑ by $\log(n+1) - \log n = \frac{1}{n} + O(\frac{1}{n^2})$ — the shrinking step in n is a *constant* step $\Delta\vartheta = \frac{1}{n}$ in ϑ, and the discrete index becomes a unit-speed clock. Rewrite the averaging line of (29.11) as an increment,

$$\bar{m}^n_\bullet - \bar{m}^{n-1}_\bullet = \tfrac{1}{n}(m^n_\bullet - \bar{m}^{n-1}_\bullet) = \Delta\vartheta\,\Big(\Psi(\mathsf{S}(\bar{m}^{n-1}_\bullet)) - \bar{m}^{n-1}_\bullet\Big),$$

using $m^n_\bullet = \Psi(\pi^n) = \Psi(\mathsf{S}(\bar{m}^{n-1}_\bullet))$ from the first two lines of (29.11). This is precisely one explicit Euler step, of step $\Delta\vartheta$, of the *best-response flow*

$$\frac{\mathrm{d}}{\mathrm{d}\vartheta}\,\bar{m}_\bullet(\vartheta) = \Psi\big(\mathsf{S}(\bar{m}_\bullet(\vartheta))\big) - \bar{m}_\bullet(\vartheta), \tag{29.13}$$

so the FP iterates are the Euler discretisation of (29.13), and by the standard consistency of the Euler scheme for a Lipschitz vector field — the field is Lipschitz because $\mathsf{S} = \mathsf{S}_\tau$ is, by lemma 29.12, and Ψ is, by definition 29.9 — the discrete trajectory tracks the continuous one as $n \to \infty$. We therefore analyse the flow and read the conclusion back to the iterates at the end. The flow says in words exactly what the averaging does: the played population $\bar{m}_\bullet$ is pulled, at unit rate, toward the population $\Psi(\mathsf{S}(\bar{m}_\bullet))$ that the current best response would generate — it chases the best-response target, but at a bounded speed that cannot overshoot.

Let

$$\mathcal{G}(\vartheta) = \mathcal{E}(\bar{\pi}(\vartheta)) = J(\bar{\pi}; \bar{m}_\bullet) - g(\bar{m}_\bullet), \qquad g(m_\bullet) := \inf_{\pi'} J(\pi'; m_\bullet),$$

be the running exploitability of the played policy $\bar{\pi}(\vartheta)$ (the policy realising $\bar{m}_\bullet(\vartheta)$), with g the optimised value of (29.4). By construction $\mathcal{G} \geq 0$, and $\mathcal{G} = 0$ exactly at equilibrium; the entire proof is the statement that $\mathcal{G}$ is a Lyapunov function for (29.13), decreasing to zero along the flow. We differentiate it along (29.13), and two ingredients enter, each made fully explicit.

(i) The envelope (Danskin) identity, derived inline. The optimised value $g(m_\bullet) = \inf_{\pi'} J(\pi'; m_\bullet)$ depends on the environment $m_\bullet$ in two superficially distinct ways: *explicitly*, through the appearance of $m_\bullet$ in the cost data of J, and *implicitly*, through the fact that the minimiser $\pi' = \mathsf{S}(m_\bullet)$ at which

the infimum is attained itself moves when $m_\bullet$ moves. The envelope identity is the statement that the second, implicit channel contributes *nothing* to first order. To see it, write $g(m_\bullet) = J(\mathsf{S}(m_\bullet); m_\bullet)$ and differentiate along the flow by the chain rule, separating the two channels:

$$\frac{\mathrm{d}}{\mathrm{d}\vartheta}\, g(\bar m_\bullet) = \underbrace{D_{\pi'} J\big(\mathsf{S}(\bar m_\bullet); \bar m_\bullet\big)\Big[\tfrac{\mathrm{d}}{\mathrm{d}\vartheta}\mathsf{S}(\bar m_\bullet)\Big]}_{\text{implicit: minimiser motion}} + \underbrace{\partial_m J\big(\mathsf{S}(\bar m_\bullet); \bar m_\bullet\big)\cdot \dot{\bar m}_\bullet}_{\text{explicit: environment dependence}}.$$

The first term vanishes identically, and the reason is first-order optimality: $\mathsf{S}(\bar m_\bullet)$ is the *unconstrained* minimiser of the smooth, strictly convex map $\pi' \mapsto J(\pi'; \bar m_\bullet)$ at the frozen environment — strict convexity in the action law being exactly what the entropy regulariser bought in lemma 29.12 — so its first variation in π' is zero there, $D_{\pi'} J(\mathsf{S}(\bar m_\bullet); \bar m_\bullet) = 0$, and the minimiser-motion term is this zero gradient paired against whatever velocity $\frac{\mathrm{d}}{\mathrm{d}\vartheta}\mathsf{S}(\bar m_\bullet)$ the minimiser happens to carry. Hence only the explicit channel survives:

$$\frac{\mathrm{d}}{\mathrm{d}\vartheta}\, g(\bar m_\bullet) = \partial_m J\big(\mathsf{S}(\bar m_\bullet); \bar m_\bullet\big)\cdot \dot{\bar m}_\bullet. \tag{29.14}$$

This is the elementary envelope theorem of Danskin; we invoke no external result precisely because the minimisation is over the policy at a frozen environment and the regularised objective is smooth and strictly convex there, so the minimiser is unique and its first variation genuinely vanishes (a non-strictly-convex objective could carry a moving *set* of minimisers and the identity would need more care — another debt the regularisation of the previous subsection silently pays). Conceptually, (29.14) is what lets us treat the cost of the *best* response as though the response were frozen while only the crowd it faces moves — the single simplification that makes $\mathcal{G}'$ tractable at all.

(ii) Monotonicity turns the cross term into a sign. Differentiating $\mathcal{G} = J(\bar\pi; \bar m_\bullet) - g(\bar m_\bullet)$ and inserting the flow $\dot{\bar m}_\bullet = \mu_\bullet - \bar m_\bullet$, with $\mu_\bullet := \Psi(\mathsf{S}(\bar m_\bullet))$ the best-response-induced population, two kinds of term appear. The first is a *relaxation* term: the flow drives the played flow $\bar m_\bullet$ toward the best-response target $\mu_\bullet$ at unit rate, and since $\mathcal{G}$ measures precisely the cost gap between playing $\bar\pi$ and playing the best response in the current environment, this pull shrinks the gap and contributes $-\mathcal{G}$ to $\mathcal{G}'$. The second is a *coupling cross term*, produced by the environment-derivative (29.14) together with the explicit m-dependence of $J(\bar\pi; \cdot)$: collecting these along the direction $\mu_\bullet - \bar m_\bullet$ — an assembly that is the computation of [38], carried out fully and in closed form for the affine LQG reduction in worked computation 29.22 — the mean-field coupling contributes exactly the pairing

$$\int \big(L[\mu_\bullet] - L[\bar m_\bullet]\big)\, \mathrm{d}(\mu_\bullet - \bar m_\bullet), \tag{29.15}$$

the difference of coupling costs paired against the difference of the two populations. This is the whole reason the proof needs monotonicity and nothing weaker: (29.15) is *exactly* the left-hand side of the Lasry–Lions inequality (11.3), evaluated at the pair $(\bar m_\bullet, \mu_\bullet)$, so (A-Mon) forces it to be ≥ 0, and the cross term therefore enters $\mathcal{G}'$ with a favourable sign. Writing $r(\vartheta)$ for the cross term once carried to the right-hand side, we have shown

$$\mathcal{G}'(\vartheta) \;\leq\; -\mathcal{G}(\vartheta) + r(\vartheta), \qquad r(\vartheta) \leq 0, \tag{29.16}$$

where r is ≤ 0 outright in the potential case (there the coupling is the gradient of a convex population functional and the pairing is its dissipation, manifestly nonnegative) and $o(1)$ in the general monotone case (the same pairing, controlled along the flow rather than signed pointwise). At the graphon level required by section 29.4 the identical step runs with the scalar pairing (11.3) replaced by its type-integrated form (24.5): the cross term becomes $\int_0^1 \int (L[x, \mu_\bullet] - L[x, \bar{m}_\bullet])\, \mathrm{d}(\mu_x - \bar{m}_x)\, \mathrm{d}x$, the same population pairing now integrated over the type axis with the aggregate built through the kernel $\mathbb{W}$, and the graphon Lasry–Lions condition definition 24.15 supplies its nonnegativity verbatim. The lift costs nothing precisely because monotonicity was a statement about the *sign* of a pairing, and a pairing integrates over the type label without changing sign.

Conclusion by Grönwall. With (29.16) in hand and $r \le 0$, Grönwall's inequality (lemma C.1) gives $\mathcal{G}(\vartheta) \le \mathcal{G}(0)\, e^{-\vartheta}$, hence $\mathcal{G}(\vartheta) \to 0$; in the general monotone case the same inequality with $r = o(1)$ still forces $\mathcal{G}(\vartheta) \to 0$, since the homogeneous decay $e^{-\vartheta}$ outruns a vanishing forcing. Either way the continuous best-response flow drives exploitability to zero. Undoing the time change $\vartheta = \log n$ returns the statement to the discrete iterates as (29.12), $\mathcal{E}(\bar{\pi}^{\,n}) \to 0$. Every weak limit point of $(\bar{m}^n_\bullet)$ therefore has zero exploitability and is an equilibrium flow, and under (A-Mon) that equilibrium is unique (theorem 11.5, resp. theorem 24.16), so the whole sequence converges to it.

The conceptual core. It is worth isolating what monotonicity actually did, because it is the heart of the chapter. Strip the hypothesis away and the cross term (29.15) has no definite sign; the relaxation $-\mathcal{G}$ can then be exactly cancelled by an adverse cross term, leaving $\mathcal{G}' = 0$ — exploitability *conserved*, neither growing nor decaying. A conserved $\mathcal{G}$ is the undamped cobweb of section 29.2.2 seen in continuous time: the orbit circles the equilibrium on a level set of $\mathcal{G}$ forever, the misalignment between the crowd and its best response sloshing back and forth without ever dissipating. Monotonicity is exactly the structural fact that breaks the tie. It forces the cross term to one side, so the misalignment can only *cost* the system, never refund it, and $\mathcal{G}$ becomes a strict Lyapunov function — a quantity that can only fall — rather than a constant of the motion. In one sentence: monotonicity converts the crowd's misalignment from a conserved quantity into a dissipating Lyapunov functional, and that conversion is the entire difference between a loop that cycles and a loop that learns. □

Remark 29.16 (What is proved here and what is imported)**.** The Lyapunov argument just given is honest about its reach, and the distinction it draws is one the reader should carry forward, because the same split — a clean in-book qualitative convergence, a sharp rate imported but anchored — recurs for online mirror descent and again in the modular guarantee of section 29.5. What is *proved in-book* is the qualitative statement (29.12): exploitability tends to zero, by the continuous-time Lyapunov mechanism, with monotonicity supplying the dissipation. What that argument does *not* pin down is the *rate* at which it tends to zero. Extracting the exponent requires controlling the discrete error committed in passing from the genuine $\frac{1}{n}$ iteration (29.11) to its continuous Euler limit (29.13) — exactly the $O(1/n^2)$ corrections to the step $\Delta\vartheta = \log(n+1) - \log n$ that Step 0 of the proof discarded — and assembling them into a telescoping estimate, a discrete bookkeeping the flow argument deliberately threw away. The quantitative sharpening

$$\mathcal{E}(\bar{\pi}^{\,n}) \;\le\; \frac{C}{\sqrt{n}} \qquad \text{(and } O(1/n) \text{ in the potential, monotone-}\textit{and}\text{-variational case),} \tag{29.17}$$

is the content of [38, 111], whose discrete telescoping and $\vartheta = \log n$ bookkeeping we cite for attribution and do *not* reproduce; (29.17) is therefore *imported*, not derived in this book.

It is, however, anchored rather than left dangling, and this is the point of drawing the line so sharply. The worked LQG recursion (29.23) below derives, *fully in-book*, the exact algebraic decay $e_n \sim n^{\lambda-1}$ of the population error for the scalar affine reduction, with $\lambda < 1$ the fictitious-play convergence threshold of worked computation 29.22, Step 3. Since the value gap near the LQG equilibrium is quadratic in the mean displacement, this gives $\mathcal{E}(\bar{\pi}^n) \sim n^{2(\lambda-1)}$, of which the universal $O(1/\sqrt{n})$ of (29.17) is precisely the worst case, saturated as $\lambda \uparrow \frac{3}{4}$. So the imported bound is not taken on faith: the book exhibits, by hand and in closed form, one model in which the very same rate is computed from scratch and shown to be the slowest member of the monotone class. The reader thus knows exactly where the boundary of the in-book argument lies — $\mathcal{E} \to 0$ proved, the algebraic exponent imported but checked against a computable instance — and that division of labour is the template the rate discussions of the next subsection and section 29.5 will follow.

Fictitious play has thus delivered the chapter's first convergence result that does not lean on contraction: under monotonicity alone, the running-average dynamic drives exploitability to zero, with the Lyapunov functional $\mathcal{G}$ doing in continuous learning time exactly what the smallness condition (29.8) did in the contraction regime — guaranteeing descent — but on the far larger terrain where the coupling may be arbitrarily strong. The template it leaves behind is more portable than the particular scheme. What made the proof work was not the specific $\frac{1}{n}$ average over *populations* but the pairing of a relaxation term against a monotone cross term to produce a strict Lyapunov decrease; any dynamic expressible as such a flow inherits the same conclusion. The next subsection exploits exactly this. Online mirror descent averages not the populations but the *evaluations*, stepping incrementally in the entropy (softmax-mirror) geometry built from the Gibbs map of definition 29.11, and its convergence proof is the Bregman-divergence variant of the present one: the Kullback–Leibler divergence to the equilibrium policy plays the role of $\mathcal{G}$, the three-point identity (29.19) plays the role of the envelope step, and monotonicity again converts the cross term into dissipation. The same monotone regime, and the same honest split between an in-book qualitative guarantee and a rate imported but anchored on worked computation 29.22, will reassemble in the modular guarantee of section 29.5.

29.3.3 Online mirror descent

Mirror descent takes a complementary route: instead of averaging populations it accumulates *evaluations* and takes small, geometry-aware steps in policy space. In the entropy-regularised geometry the "mirror map" is the softmax, so online mirror descent (OMD) maintains a cumulative Q-function and reads off a Gibbs policy at every step. It is the basis of the most scalable modern MFG learners [110, 96].

Definition 29.17 (Online mirror descent)**.** Fix a step size $\eta > 0$ and temperature $\tau > 0$. Initialise cumulative scores $y_k^0 \equiv 0$. *Online mirror descent* iterates, for $n \geq 1$,

$$\pi_k^n(\cdot \mid s) = \mathrm{softmax}\big(-y_k^{n-1}(s,\cdot)/\tau\big), \quad m_\bullet^n = \Psi(\pi^n), \quad y_k^n(s,a) = y_k^{n-1}(s,a) + \eta\, Q_k^{m_\bullet^n}(s,a), \tag{29.18}$$

where $Q_k^{m_\bullet^n}$ is the action-value of π^n against its own induced flow. The averaged policy $\bar{\pi}^n$ is the one whose scores are $\frac{1}{n}\sum_{j \leq n} y^j$.

Lemma 29.18 (Three-point identity for the entropy mirror map)**.** *Let ν_0 be the reference measure of definition 29.11, and let q, q_+ be probability densities on A related by the Gibbs tilt $q_+(a) = q(a)\, e^{-\eta g(a)}/Z$ with $Z = \int_A q\, e^{-\eta g}\, \mathrm{d}\nu_0$, for a bounded score g and step $\eta > 0$. Then for every probability density $p \ll \nu_0$,*

$$\mathrm{KL}(p\|q_+) = \mathrm{KL}(p\|q) - \mathrm{KL}(q_+\|q) + \eta\, \langle g, p - q_+\rangle. \tag{29.19}$$

Proof. Since $\log(q_+/q) = -\eta g - \log Z$, expand each divergence against ν_0: $\mathrm{KL}(p\|q_+) = \mathrm{KL}(p\|q) - \int p \log \frac{q_+}{q} = \mathrm{KL}(p\|q) + \eta\langle g, p\rangle + \log Z$, and likewise $\mathrm{KL}(q_+\|q) = -\int q_+ \log \frac{q_+}{q} \cdot (-1) = \eta\langle g, q_+\rangle + \log Z$. Subtracting the second from the first eliminates $\log Z$ and yields (29.19). (This is the entropy-mirror instance of the Bregman "law of cosines"; all three terms are exact, no approximation.) □

Theorem 29.19 (Convergence of OMD to the regularised equilibrium)**.** *Assume* (A-Lip)*,* (A-Nondeg)*, and* (A-Mon) *(classical definition 11.1 or graphon definition 24.15), and run OMD* (29.18) *at temperature $\tau > 0$ with a sufficiently small step size η. Then the iterates converge to the unique τ-regularised equilibrium $(\pi^\star_\tau, m^\star_{\tau,\bullet})$ of remark 29.13, with regularised exploitability $\mathcal{E}_\tau(\bar\pi^{\,n}) \to 0$. Combined with the bias bound of remark 29.13, the* unregularised *exploitability obeys $\mathcal{E}(\bar\pi^{\,n}) \to c_0\tau(1 + \log|A|)$ as $n \to \infty$.*

Proof: the Bregman–Lyapunov mechanism. The Lyapunov functional is the *Bregman divergence* (relative entropy) from the current policy to the regularised equilibrium,

$$\mathcal{D}^n = \int_{\mathcal{S}} \sum_k \mathrm{KL}\big(\pi^\star_{\tau,k}(\cdot \mid s) \,\big\|\, \pi^n_k(\cdot \mid s)\big)\, m^\star_{\tau,k}(\mathrm{d}s).$$

The OMD update (29.18) tilts π^{n-1}_k to π^n_k by the Gibbs factor $e^{-\eta\,\delta y/\tau}$, i.e. it is exactly the mirror step of lemma 29.18 with score $g = Q^{m^n_\bullet}_k/\tau$ and reference $p = \pi^\star_{\tau,k}$. Applying (29.19) stage-wise and integrating against $m^\star_{\tau,\bullet}$,

$$\mathcal{D}^n = \mathcal{D}^{n-1} - \underbrace{\int \sum_k \mathrm{KL}(\pi^n_k\|\pi^{n-1}_k)\, \mathrm{d}m^\star_\tau}_{\geq 0} - \eta\, \langle Q^{m^n_\bullet}/\tau, \pi^n - \pi^\star_\tau\rangle_{m^\star_\tau},$$

so $\mathcal{D}^n \leq \mathcal{D}^{n-1} - \eta\,\Gamma^n$, where $\Gamma^n := \langle Q^{m^n_\bullet}/\tau, \pi^n - \pi^\star_\tau\rangle$ is the regularised duality gap. Monotonicity (A-Mon) makes Γ^n nonnegative and bounded below by the regularised exploitability $\mathcal{E}_\tau(\pi^n)$ (the same pairing as in theorem 29.15). Summing telescopes, $\eta \sum_{j \leq n} \mathcal{E}_\tau(\pi^j) \leq \eta \sum_{j \leq n} \Gamma^j \leq \mathcal{D}^0$, so the running average of $\mathcal{E}_\tau$ tends to 0; convexity of $\mathcal{E}_\tau$ in the score geometry transfers this to the averaged policy $\bar\pi^{\,n}$, giving $\mathcal{E}_\tau(\bar\pi^{\,n}) \to 0$. Uniqueness of $\pi^\star_\tau$ is the strict convexity injected by the entropy plus monotonicity; the bias term is remark 29.13. □

Remark 29.20 (The $1/n$ rate is an imported refinement)**.** The telescoping above is exact only up to the second-order (η^2) curvature term that a finite step size $\eta > 0$ leaves in (29.19) through $\mathrm{KL}(\pi^n\|\pi^{n-1}) = O(\eta^2)$. Carrying that term with a tuned step-size schedule sharpens the in-book convergence $\mathcal{E}_\tau(\bar\pi^{\,n}) \to 0$ to the algebraic rate

$$\mathcal{E}_\tau(\bar\pi^{\,n}) \leq C/n, \tag{29.20}$$

which is the content of [110]; we cite it for attribution and flag (29.20) as imported rather than derived here. The continuous-time limit is the replicator/mirror flow whose Lyapunov function is exactly $\mathcal{D}$, for which the $1/n$ follows from $\dot{\mathcal{D}} \leq -\mathcal{E}_\tau$ and Jensen along the average.

Remark 29.21 (Two stabilisers, two hypotheses)**.** The logical structure is worth stating plainly. FPI converges under *contraction* (29.8), a strong smallness condition on the coupling. FP and OMD converge under *monotonicity* (A-Mon), which is *not* a smallness condition: the coupling may be arbitrarily strong provided it is “stabilising” (repulsive/crowd-averse) in the Lasry–Lions sense. Monotonicity is exactly the hypothesis that gave *uniqueness* (not existence) in chapter 11; here it gives *convergence of learning*. This is no coincidence: monotonicity makes the exploitability a Lyapunov function, and a Lyapunov function both pins down the equilibrium and guides the iterates to it.

Worked computation 29.22 (The LQG thread: contract or average)**.** We make the dichotomy of remark 29.21 completely explicit on the scalar LQG mean field game of Thread A (worked computation 8.12 and theorem 14.6), and then read off the graphon generalisation of Thread B. Recall the model: state $\mathrm{d}X_t = (aX_t + b\alpha_t)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$, cost $\mathbb{E}\int_0^T \big[\frac{1}{2}\alpha_t^2 + \frac{q}{2}(X_t - s\,\bar{m}_t)^2\big]\,\mathrm{d}t + \text{terminal}$, with population mean $\bar{m}_t = \int \xi\, m_t(\mathrm{d}\xi)$ and mean-coupling strength s. Because the model is linear–quadratic, the entire equilibrium problem collapses onto the *mean trajectory* $\phi = (\bar{m}_t)_{t\in[0,T]} \in C([0,T])$, and the composite “best-respond, then push forward” is an *affine* map of ϕ.

Step 1: the composite map is affine. (A reminder on symbols: within this LQG computation s is the Thread A mean-coupling strength, *not* an MDP state — the state is X_t — and P_t is the Riccati curvature, leaving the chapter’s Q for the action-value function.) Freeze the environment mean ϕ. The representative agent solves a linear-quadratic regulator with the time-dependent target $s\phi_t$; by eqs. (14.5) and (14.13) its optimal feedback is $\alpha_t^* = -b(P_tX_t + r_t[\phi])$, where P_t is the (population-independent) solution of the scalar Riccati equation (14.7) and the decoupling offset $r_t[\phi]$ (the linear coefficient of the value, $V_x = P_t x + r_t$, in the carried-thread notation of chapter 14) solves the *linear* backward ODE $\dot{r}_t = (b^2P_t - a)\,r_t + q\,s\,\phi_t$, $r_T = \text{(terminal)}$, hence depends linearly on ϕ. The $+q\,s\,\phi_t$ source is the x^1-coefficient of the HJB for the tracking cost $\frac{q}{2}(X_t - s\phi_t)^2$; with this sign the induced mean inherits the sign of s (a crowd-seeking target $s > 0$ pulls the mean toward $+\infty \cdot \mathrm{sign}(s)$), which fixes the physical orientation used in Step 3. The induced state mean $\rho_t = (\Psi \circ \mathsf{S})(\phi)_t$ then solves the *linear* forward ODE $\dot{\rho}_t = (a - b^2P_t)\rho_t - b^2 r_t[\phi]$, $\rho_0 = \bar{m}_0$. Composing the two linear ODE solution operators, we obtain

$$(\Psi \circ \mathsf{S})(\phi) = \mathcal{K}\phi + c, \tag{29.21}$$

with c the part driven by the initial/terminal data and $\mathcal{K}$ the linear operator $\phi \mapsto \rho_{\text{driven by } r[\phi]}$. By variation of parameters and Grönwall’s inequality (lemma C.1), its operator norm on $(C([0,T]), \|\cdot\|_\infty)$ obeys

$$\|\mathcal{K}\| \;\leq\; |s|\; q\,b^2\,\mathsf{G}_\mathrm{b}\,\mathsf{G}_\mathrm{f}, \qquad \mathsf{G}_\mathrm{b} = \int_0^T e^{\int_t^T (b^2P_r - a)\,\mathrm{d}r}\,\mathrm{d}t, \quad \mathsf{G}_\mathrm{f} = \int_0^T e^{\int_t^T (a - b^2P_r)\,\mathrm{d}r}\,\mathrm{d}t, \tag{29.22}$$

the backward and forward resolvent bounds. The equilibrium is the unique fixed point $\phi^\star = (I - \mathcal{K})^{-1}c$ whenever $1 \notin \operatorname{spec}\mathcal{K}$ — this is the explicit form of theorem 14.6.

Step 2: when does FPI converge? The naive iteration (29.5) is $\phi^{n+1} = \mathcal{K}\phi^n + c$, the Banach–Picard iteration of an affine map, which converges (geometrically, from any start) *iff* the spectral radius $\varrho(\mathcal{K}) < 1$. By (29.22) a sufficient condition is $|s|\, q\, b^2 \mathsf{G}_\mathrm{b}\mathsf{G}_\mathrm{f} < 1$: small coupling $|s|$, short horizon (both G's grow with T), or strong individual stabilisation make FPI contract. This is the LQG instance of (29.8) and of (A-Coupling).

Step 3: when FPI fails but FP rescues it. Suppose $\mathcal{K}$ acts (on the relevant one-dimensional invariant subspace, e.g. the constant-mean mode for a stationary target) as multiplication by a scalar λ. Because the only s-dependence of $\mathcal{K}$ is the linear factor $q\, s$ entering the offset source (Step 1), the multiplier inherits the sign of the coupling: $\operatorname{sign}(\lambda) = \operatorname{sign}(s)$, so $\lambda < 0$ is the *repulsive* (crowd-averse, $s < 0$) regime and $\lambda > 0$ the *attractive* (imitative, $s > 0$) one. FPI is $e_n = \lambda\, e_{n-1}$ for the error $e_n = \phi^n - \phi^\star$, so it converges iff $|\lambda| < 1$; it *diverges with period-two oscillation* (the cobweb of section 29.2.2) once $\lambda < -1$ — precisely a *strong repulsive* (congestion/crowd-averse) coupling, exactly the congestion picture of that heuristic — and diverges *monotonically* once $\lambda > 1$, a strong attractive coupling. Now run fictitious play (29.11) on the same affine map. Writing $\bar{\phi}^n$ for the averaged mean and using $c = (1-\lambda)\phi^\star$, the error obeys the exact recursion

$$e_n = \Big(1 + \frac{\lambda - 1}{n}\Big) e_{n-1} \;=\; e_0 \prod_{j=1}^{n} \Big(1 + \frac{\lambda - 1}{j}\Big) \;\sim\; \frac{e_0}{\Gamma(\lambda)}\, n^{\lambda - 1} \quad (n \to \infty). \tag{29.23}$$

Hence *fictitious play converges for every* $\lambda < 1$ — including $\lambda < -1$, where naive FPI blows up — at the algebraic rate $n^{\lambda-1}$, and it *fails* only for $\lambda > 1$. Three points reconcile this with the named hypotheses and with theorem 29.15.

- *What the threshold* $\lambda < 1$ *is.* It is *not* "exactly the monotone regime": the Lasry–Lions monotone (repulsive) regime is $s \le 0$, i.e. $\lambda \le 0$, while $0 < \lambda < 1$ is weak *attractive* coupling that is non-monotone yet still uniquely and stably solvable. So the FP-convergence region splits as $\{\lambda < 1\} = \{\text{monotone } \lambda \le 0\} \cup \{\text{small-coupling } 0 < \lambda < 1\}$: the monotone half-line $\lambda \le 0$ is the part bought purely by averaging (it lies well outside the naive-contraction interval $|\lambda| < 1$), while the sliver $0 < \lambda < 1$ is already covered by the contraction of (29.8). FP therefore strictly enlarges the convergent set from the interval $|\lambda| < 1$ to the half-line $\lambda \in (-\infty, 1)$.

- *What* $\lambda > 1$ *is — and is not.* Since the map $\phi \mapsto \rho$ is genuinely affine (Step 1) and $\lambda \ne 1$, the equilibrium $\phi^\star = (I - \mathcal{K})^{-1} c$ is still *unique*; what fails for $\lambda > 1$ is its *stability*: it is a repelling fixed point for best-response dynamics, so FPI and FP both diverge. Genuine *multiplicity* is not a feature of this affine reduction at all; it is the boundary/nonlinear phenomenon of the phase diagram theorem 14.11 and section 11.4 — resonance $\lambda = 1$, or couplings rich enough to make the Riccati curvature population-dependent — where more than one equilibrium appears. The honest statement for the affine thread is thus: $\lambda > 1$ is the *unique-but-unstable* regime, in which no best-response learning dynamic converges.

- *Reconciling the rate with theorem 29.15.* The explicit population error $e_n \sim n^{\lambda-1}$ is the in-book instance promised by remark 29.16. It is a *population* error, not an exploitability: near the LQG equilibrium the value gap is quadratic in the mean displacement, so $\mathcal{E}(\bar{\pi}^n) \sim e_n^2 \sim n^{2(\lambda-1)}$. For $\lambda < \frac{3}{4}$ this already beats the universal $1/\sqrt{n}$ of (29.17); the universal bound is a worst case over the whole monotone class, saturated as $\lambda \uparrow \frac{3}{4}$, and the λ-dependent explicit rate is the sharper, model-specific statement — the two are consistent, not contradictory.

Step 4: the graphon lift. In Thread B (eq. (21.5)) the agent of type x is coupled to $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ rather than to the global mean. For the rank-one kernel $W(x,y) = f(x)f(y)$ the aggregate collapses to $z_t(x) = f(x)\,\varrho_t$ with the scalar order parameter $\varrho_t = \int_0^1 f(y)\bar{m}_t(y)\,\mathrm{d}y$ (eq. (21.19)), and the composite map on ϱ is again affine with multiplier $\lambda_W = \lambda \cdot \|\mathbb{W}\|$, where $\|\mathbb{W}\| = \|f\|_{L^2}^2$ is the operator norm of the rank-one $\mathbb{W}$. Everything above transfers verbatim with $\lambda \rightsquigarrow \lambda_W$: FPI contracts iff $|\lambda|\,\|\mathbb{W}\| < 1$ (this is (A-Coupling) and the critical coupling of eq. (20.31) and chapter 24), while fictitious play converges on the whole half-line $\lambda_W < 1$ — the monotone part $\lambda_W \le 0$ ($s \le 0$, bought by averaging) together with the weak-attraction sliver $0 < \lambda_W < 1$ (already inside the contraction interval). The graphon contributes exactly one scalar — its leading eigenvalue — to the learning threshold, which is the cleanest possible illustration of how network centrality enters the dynamics of learning.

29.4 GMFG-PPO: learning on the type-indexed population

The stabilised dynamics of section 29.3 converge, but every one of them still presumed two idealisations: that the two maps could be evaluated *exactly* — an exact backward best-response sweep and an exact forward push-forward — and that the population was a single measure rather than a field. This section lifts both at once, confronting in full the *third* failure mode of section 29.1, the continuum of coupled types, on the very terrain — high dimension, unwritable dynamics — that forced learning in the first place. We now drop the assumption that the best response S and the push-forward Ψ can be computed exactly. In high dimension both are approximated: S by a *policy-gradient* RL algorithm (PPO) acting through a neural network, and Ψ by Monte Carlo rollouts of that network. The graphon enters as an input feature of the networks. This is the GMFG-PPO scheme of [54], building on the entropy-regularised deep RL of [53].

29.4.1 The graphon mean field MDP

Definition 29.23 (Graphon mean field MDP)**.** A *graphon MFMDP* (GMFMDP) augments definition 29.1 with a *type* $x \in [0,1]$ drawn from the uniform law (the type distribution), a kernel W satisfying (A-Reg-W), and an *observable* $\Lambda : \mathcal{P}(\mathcal{S}) \to \mathbb{R}^{d'}$ (typically the mean, $\Lambda(m) = \int \xi\, m(\mathrm{d}\xi)$). The population is now a *field of flows* $x \mapsto m^x_\bullet$ with $m^x_k \in \mathcal{P}(\mathcal{S})$, and the type-$x$ agent feels the *graphon aggregate*

$$z_k(x) = (\mathbb{W}\,\Lambda[m_k^{\cdot}])(x) = \int_0^1 W(x,y)\,\Lambda\big(m_k^y\big)\;\mathrm{d}y \in \mathbb{R}^{d'}, \tag{29.24}$$

which is Part IV's observable aggregate $\Theta_k(x)$ of eq. (21.4) (written $z_k(x)$ here, and with the scalar test function h promoted to the vector observable Λ; notationwatch (v)). The transition kernel and costs depend on the type and the aggregate, $P_k(\cdot \mid s, a, x, z_k(x))$ and $L_k(s, a, x, z_k(x))$, and the equilibrium is the fixed point of definition 21.16: each type best-responds to its aggregate, and the field of induced flows reproduces those aggregates.

The decisive modelling move is that the type-x agent's optimisation problem depends on the entire population field *only through the finite-dimensional aggregate* $z_k(x) \in \mathbb{R}^{d'}$. This is what makes a single, shared policy possible: rather than learn a separate policy for each of a continuum of types, we learn one map that takes (s, x, z) and outputs an action distribution. The kernel W is never represented as a function the network must learn; it is used *outside* the network, once per outer iteration, to turn the population field into the aggregate field $x \mapsto z_k(x)$ via the quadrature of chapter 28.

29.4.2 Policy gradients and PPO as an approximate best-response oracle

The inner loop is an RL solver that, given a frozen aggregate field, returns an approximate best response. We use policy gradients. Let the policy be parametrised, $\pi_\theta = (\pi_{\theta,k})$, and recall the (cost-minimising) *policy gradient theorem*.

Proposition 29.24 (Policy gradient)**.** *Fix the environment (the aggregate field) $m_\bullet$, and abbreviate the frozen-environment value and action-value as $u_k^{\pi_\theta} := u_k^{m_\bullet,\pi_\theta}$, $Q_k^{\pi_\theta} := Q_k^{m_\bullet,\pi_\theta}$ (the superscript $m_\bullet$ suppressed only because it is held fixed throughout this subsection). For a differentiable parametrisation $\theta \mapsto \pi_\theta$ with $J(\theta) = J(\pi_\theta; m_\bullet)$ the expected cost of definition 29.6, one has the score-function identity*

$$\nabla_\theta J(\theta) = \mathbb{E}\left[\sum_{k=0}^{K-1} \nabla_\theta \log \pi_{\theta,k}(A_k \mid S_k)\ \mathsf{A}_k^{\pi_\theta}(S_k, A_k)\right], \tag{29.25}$$

where $\mathsf{A}_k^{\pi_\theta}(s,a) = Q_k^{\pi_\theta}(s,a) - u_k^{\pi_\theta}(s)$ is the advantage *and the expectation is over trajectories generated by π_θ in the frozen environment.*

Proof sketch. Differentiate $J(\theta) = \mathbb{E}_{\zeta\sim\pi_\theta}[\sum_k L_k]$, the expectation over trajectories $\zeta = (S_k, A_k)_k$ generated by π_θ, under the trajectory law, whose density factorises as $\prod_k \pi_{\theta,k}(A_k \mid S_k) P_k(S_{k+1} \mid S_k, A_k)$; only the policy factors carry θ, giving the $\nabla_\theta \log \pi_\theta$ score. Replacing the return by its advantage (subtracting the state-baseline $u_k^{\pi_\theta}$) leaves the expectation unchanged because $\mathbb{E}[\nabla_\theta \log \pi_{\theta,k}(A_k \mid S_k) \mid S_k] = 0$, and reduces variance. This is the finite-horizon, cost form of [127]; the advantage is estimated from rollouts by generalised advantage estimation [119]. □

A raw policy-gradient step can be destructively large. *Proximal Policy Optimization* (PPO) [120] constrains each update to a trust region around the current policy by maximising a *clipped surrogate*: with importance ratio $r_k(\theta) = \pi_{\theta,k}(A_k \mid S_k)/\pi_{\theta_{\text{old}},k}(A_k \mid S_k)$ and advantage estimate $\hat{\mathsf{A}}_k$ (for cost minimisation, advantages are negated relative to the reward convention), PPO minimises

$$J^{\text{clip}}(\theta) = \mathbb{E}\left[\sum_k \max\Big(r_k(\theta)\,\hat{\mathsf{A}}_k,\ \text{clip}\left(r_k(\theta), 1-\epsilon, 1+\epsilon\right)\hat{\mathsf{A}}_k\Big)\right] \; - \; \tau\,\mathbb{E}\left[\sum_k \mathfrak{h}\big(\pi_{\theta,k}(\cdot \mid S_k)\big)\right], \tag{29.26}$$

where $\mathfrak{h}(p) = -\int p \log p$ is the policy entropy, the clip range $\epsilon > 0$ caps the step, and the entropy bonus $\tau > 0$ implements the regularisation of definition 29.11. PPO is a heuristic approximation of trust-region policy optimisation [118]; it has no global optimality guarantee even for a single MDP, and we treat it strictly as an *approximate best-response oracle*: an operator $\widehat{\mathsf{S}}$ that, given an environment and a compute budget, returns a policy whose exploitability against that environment is at most some ϵ_{br}. The quality of that oracle is an *assumption*, not a theorem, and section 29.5 is careful to keep it so.

29.4.3 Representing the graphon coupling inside the networks

The architecture is what makes the type continuum tractable. A *single* policy network $\pi_\theta(a \mid s, x, z)$ and a *single* value network $V_\phi(s, x, z)$ take, as inputs, the state s, the scalar type $x \in [0, 1]$ (or a learned type-embedding), and the local aggregate $z \in \mathbb{R}^{d'}$. Because x and z are inputs rather than indices, the same weights (θ, ϕ) serve all types: the network *interpolates* across the continuum, and high- and low-centrality types differ only through the value of z they are fed. The graphon W appears nowhere inside the network; it acts only in the outer loop, through the quadrature

$$z_k(x_i) \approx \sum_{j=1}^{M} w_j \, W(x_i, x_j) \, \Lambda(\widehat{m^{x_j}}_k), \tag{29.27}$$

over a type grid $\{x_j\}_{j=1}^M$ with quadrature weights w_j and empirical type-flows $\widehat{m^{x_j}}_k$ estimated from rollouts — precisely the discretisation of $\mathbb{W}$ analysed in chapter 28 and controlled, in operator norm, by the graphon regularity of chapter 19. The separation of concerns is clean: the *network* learns the type-conditioned best response $s, x, z \mapsto \pi_\theta$; the *quadrature* computes the graphon coupling $z = \mathbb{W}\Lambda[m^\cdot]$; the *outer loop* (FP or OMD) reconciles the two.

Remark 29.25 (Why one network suffices: continuity in type)**.** That a single network can represent the whole family $\{\pi^x\}_{x\in[0,1]}$ rests on the regularity established in Part IV: under (A-Reg-W) the equilibrium field $x \mapsto (u^x, m^x)$ is continuous in type (lemma 21.11 and the field topology of definition 21.8), so the target map $(s, x, z) \mapsto \pi^\star$ is continuous and hence approximable by a network of bounded capacity on the compact domain $\mathcal{S}_0 \times [0, 1] \times Z$. If the kernel had no such regularity — a rough W violating (A-Reg-W) — the type-conditioning could be discontinuous and a fixed-capacity network would fail to represent the high-frequency type dependence, one of the failure modes of section 29.5.

29.4.4 The GMFG-PPO algorithm

We record the scheme at the architectural level; the full reference implementation lives in `figures/29-gmfg-ppo.py`, exactly as chapter 28 places its solver in `figures/28-mfg-solver.py`. The outer loop is fictitious play or online mirror descent; the inner loop is PPO; the coupling is the quadrature (29.27).

1. **Initialise** networks (θ^0, ϕ^0), a type grid $\{x_j\}_{j=1}^M$ with weights $\{w_j\}$, and an initial population field $\widehat{m^{x_j}}_\bullet^0$ (e.g. the initial law repeated).
2. **For** outer iteration $n = 1, 2, \dots$:
   (a) *Aggregate.* Compute the aggregate field $z_k^{n-1}(x_j)$ from the current (FP-averaged or OMD-scored) population field via the quadrature (29.27).
   (b) *Best-respond (inner PPO).* Holding the aggregate field fixed, run PPO (29.26) for a budget of rollouts, sampling types $x \sim \mathrm{Unif}([0, 1])$, states $s \sim \widehat{m^x}_0$, and trajectories under π_θ, to obtain an approximate best response $\pi_{\theta^n} = \widehat{\mathsf{S}}_\tau(z^{n-1})$ with exploitability $\leq \epsilon_{\mathrm{br}}$.

(c) *Roll forward.* Estimate the induced field $\widehat{m^{x_j}}_{\bullet}^{n} = \widehat{\Psi}(\pi_{\theta^n})$ by Monte Carlo rollouts of π_{θ^n} across the type grid.

(d) *Reconcile (outer FP/OMD).* Update the population field by the FP average (29.11) or the OMD score (29.18).

3. **Output** the averaged policy $\bar{\pi}^n$ and an estimate of its exploitability $\widehat{\mathcal{E}}(\bar{\pi}^n)$ obtained by training a fresh PPO best response against $\widehat{\Psi}(\bar{\pi}^n)$ and differencing returns.

Physics note: Learning an equilibrium as variational Monte Carlo

The loop above is structurally a *variational Monte Carlo* (VMC) calculation, and the analogy is exact enough to be useful rather than decorative. In VMC one posits a parametrised trial wavefunction ψ_θ (lowercase, to keep clear of the population map Ψ that is this chapter's central object), samples configurations from $|\psi_\theta|^2$ by Monte Carlo, estimates the variational energy $E(\theta) = \langle\psi_\theta|H|\psi_\theta\rangle$ and its gradient by those samples, and descends in θ toward the ground state. Here the parametrised *policy* π_θ plays the role of the trial state; rollouts sample the trajectory distribution it induces (the analogue of sampling $|\psi_\theta|^2$); the policy gradient (29.25) is the stochastic estimate of $\nabla_\theta(\text{cost})$, the analogue of the energy gradient. The self-consistent mean field is the precise counterpart of the self-consistent field in a Hartree calculation: the agent optimises against a population it itself generates, just as a Hartree orbital is computed in the field of the density it itself produces. One correspondence in this dictionary is not an analogy but a literal identity: the τ-regularised Gibbs policy (29.10),

$$\mathsf{S}_\tau(m_\bullet)_k(\mathrm{d}a \mid s) \;=\; \frac{e^{-Q^{m_\bullet}_{\tau,k}(s,a)/\tau}}{\int_A e^{-Q^{m_\bullet}_{\tau,k}(s,a')/\tau}\,\mathrm{d}\nu_0(a')}\,\mathrm{d}\nu_0(a),$$

is the Boltzmann distribution at temperature τ for a classical system whose energy function is the action-value $Q^{m_\bullet}_{\tau,k}(s,\cdot)$, with partition function $Z = \int e^{-Q/\tau}\,\mathrm{d}\nu_0$ and free energy $u^{m_\bullet}_{\tau,k}(s) = -\tau\log Z$ — exactly the finite-temperature trial ensemble of statistical mechanics. Where the analogy stops is the part this whole book is about: in VMC the energy functional is *fixed* and one minimises it, whereas in a game the "energy" an agent faces depends on the population's choice, so the object sought is a *fixed point* (Nash), not a minimum — and that is why monotonicity, not mere convexity, is the right hypothesis for convergence.

29.5 Convergence guarantees and failure modes

Having assembled GMFG-PPO — the inner PPO oracle, the type-conditioned networks, the quadrature coupling, and the FP/OMD outer loop — we owe the reader a sober accounting of what the scheme actually guarantees and where it can break. This closing section gives both, in that order: first a modular convergence theorem that stacks the chapter's *proven* pieces (the contraction of theorem 29.10, the monotone dynamics of theorems 29.15 and 29.19, the finite-network transfer of chapter 25) against the approximations the deep implementation introduces, then a frank catalogue of the failure modes that arise when its hypotheses are violated.

29.5.1 A modular guarantee

No single theorem covers GMFG-PPO end to end, because the scheme is a composition of an *assumed* oracle (PPO), an *approximated* coupling (quadrature), and a *proven* outer dynamic (FP or OMD). The honest statement is a modular error decomposition: each layer contributes an additive error, and the outer dynamic's convergence theorem absorbs them.

Theorem 29.26 (Modular convergence of GMFG-PPO)**.** *Assume the GMFMDP data are Lipschitz in* (m, z) *uniformly* (A-Lip)*, the diffusion nondegenerate* (A-Nondeg)*, the kernel regular* (A-Reg-W)*, and run the scheme at temperature* $\tau > 0$*. Suppose the inner PPO oracle returns, at each outer step, a policy with per-type regularised exploitability against the current aggregate field at most* ϵ_{br}*, and that the type/state discretisation and the quadrature* (29.27) *for* $\mathbb{W}$ *have combined error at most* ϵ_{disc}*. Then:*

1. (Contraction regime.) *If* (A-Coupling) *holds with composite modulus* $L = L_{\mathsf{S},\tau} L_\Psi < 1$*, the field iterates satisfy*

$$D(\widehat{m}^{\,n}_{\bullet}, m^{\star,\tau}_{\bullet}) \;\le\; L^n\, D_0 + \frac{\epsilon_{\mathrm{br}} + \epsilon_{\mathrm{disc}}}{1 - L}, \qquad \mathcal{E}_\tau(\pi^n) \le C\Big(L^n + \epsilon_{\mathrm{br}} + \epsilon_{\mathrm{disc}}\Big). \tag{29.28}$$

2. (Monotone regime.) *If instead* (A-Mon) *holds (graphon Lasry–Lions monotonicity, definition 24.15), the FP/OMD outer loop satisfies*

$$\mathcal{E}_\tau(\bar{\pi}^{\,n}) \;\le\; \frac{C}{\sqrt{n}} \;+\; C'\big(\epsilon_{\mathrm{br}} + \epsilon_{\mathrm{disc}}\big) \tag{29.29}$$

   (with C/n *under OMD or in the potential case); the* n*-dependent outer rate is the imported refinement of remarks 29.16 and 29.20, while the in-book content is the convergence* $\mathcal{E}_\tau \to 0$ *of theorems 29.15 and 29.19.*

In both regimes the unregularised exploitability picks up the bias $c_0\tau(1+\log|A|)$ *of remark 29.13; and by the finite-network convergence theorem of chapter 25, the output policy is an* $\big(\mathcal{E}(\bar{\pi}^{\,n}) + \epsilon_{\mathrm{graph}}(N)\big)$*-Nash equilibrium for the* N*-agent game on a graph sampled from* $G(N, W)$*, with* $\epsilon_{\mathrm{graph}}(N) \to 0$ *as* $N \to \infty$*.*

Proof: assembly. The contraction case is the perturbed-fixed-point lemma: if T is an L-contraction with fixed point $m^\star$ and one iterates $m^{n+1} = T(m^n) + \delta^n$ with $\|\delta^n\| \le \delta := \epsilon_{\mathrm{br}} + \epsilon_{\mathrm{disc}}$, then by the triangle inequality $\|m^n - m^\star\| \le L^n\,\|m^0 - m^\star\| + \delta \sum_{j<n} L^j \le L^n D_0 + \delta/(1-L)$; here $T = \Psi \circ \mathsf{S}_\tau$ is the L-contraction of theorem 29.10 (with $L_{\mathsf{S},\tau}$ from lemma 29.12) and δ^n packages the oracle and quadrature errors. The exploitability bound is then (A-Lip)-continuity of $\mathcal{E}_\tau$ in the field. The monotone case is theorem 29.15 (or theorem 29.19) applied to the *inexact* best-response/score updates: the Lyapunov arguments there are stable under an additive δ per step, contributing the $C'\delta$ term (the Grönwall/telescoping inequalities absorb a bounded per-step perturbation). The regularisation bias is remark 29.13. The final ϵ-Nash transfer is exactly the finite-network theorem of chapter 25, which converts a graphon-equilibrium policy into an approximate Nash equilibrium on the sampled finite network with error $\epsilon_{\mathrm{graph}}(N)$ governed by the cut-distance concentration of $G(N, W)$ (chapters 19 and 25). □

Remark 29.27 (What is proved and what is assumed)**.** Theorem 29.26 should be read with its load-bearing assumption in plain sight. The *outer* convergence (the L^n or $1/\sqrt{n}$ term) is genuinely proved, inheriting the rigour of theorems 29.10, 29.15 and 29.19. The *inner* term ϵ_{br} is an *assumption*: it asserts that PPO solves a fixed-environment MDP to accuracy ϵ_{br}, which is empirically often true and theoretically unguaranteed — PPO can stall in poor local optima, and for general function approximation no polynomial-time best-response oracle is known. The theorem is therefore a *conditional* guarantee, exactly as honest practice in this subject requires; the unconditional results [134, 135, 8] replace PPO by a tabular or linearly-parametrised solver for which ϵ_{br} *can* be bounded, at the cost of the expressive power that motivated deep RL in the first place. The linear-quadratic special case, where an actor–critic provably finds the equilibrium, is [73].

29.5.2 Failure modes

We catalogue, without euphemism, the ways the scheme breaks. Each is a violation of a hypothesis of theorem 29.26 and each has a concrete signature.

1. *Non-monotone, non-contractive coupling.* If neither (A-Coupling) nor (A-Mon) holds — e.g. a strong attractive (imitative) coupling, $\lambda > 1$ in worked computation 29.22 — the best-response dynamic (29.13) need not converge. In the affine LQG reduction $\lambda > 1$ gives a *unique but unstable* equilibrium from which best-response iteration is repelled (worked computation 29.22, Step 3); beyond the affine case the same loss of monotonicity can produce a genuine *limit cycle* (the persistent cobweb of section 29.2.2) or *multiple* equilibria (the symmetry-breaking regime of section 11.4 and theorem 14.11), and the learner then has no principled way to select among them. No amount of averaging fixes this. Signature: exploitability that plateaus or oscillates rather than decaying.

2. *A broken best-response oracle.* The notorious "deadly triad" of off-policy bootstrapping, function approximation, and limited data [126] can make the inner RL diverge, so ϵ_{br} is effectively infinite. The outer loop then reconciles toward a population that no agent actually best-responds to. Signature: the inner return curves fail to converge even with the environment frozen.

3. *Order-of-limits errors.* Three limits are taken: time discretisation $\Delta t \to 0$, finite population $N \to \infty$, and sample size $\to \infty$. They do not commute in general. A coarse Δt changes the equilibrium (the discrete game is not the continuous one); a small N leaves fluctuation corrections of order $N^{-1/2}$ that the mean-field policy ignores (chapters 13 and 25). The modular theorem hides these inside ϵ_{disc} and $\epsilon_{\mathrm{graph}}(N)$, but they must be driven down jointly. Signature: an exploitability floor that moves when the grid or N is refined.

4. *Regularisation bias.* The temperature $\tau > 0$ that bought stability and a Lipschitz oracle (lemma 29.12) also displaces the equilibrium by $O(\tau)$ (remark 29.13). Lowering τ reduces bias but raises $L_{\mathsf{S},\tau} = \Theta(1/\tau)$, shrinking the contraction margin and inflating gradient variance. There is a genuine bias–variance frontier in τ, and reporting a small exploitability at large τ is a common way to overstate success. Signature: exploitability that improves with τ but against a τ-shifted target.

5. *Failure of type representability.* If the kernel W has a rich spectrum — many eigenvalues of comparable size, hence many "centrality channels" (chapter 18) — the equilibrium's dependence on type can be high-frequency, and a fixed-capacity network $\pi_\theta(a \mid s, x, z)$ may simply be unable to represent it (remark 29.25). High-centrality types, which contribute most to the aggregate (29.24), are the ones whose misrepresentation is most damaging. Signature: per-type exploitability concentrated on the spectrally extreme types.

These are not reasons to abandon the method — for unknown dynamics and high dimension it is often the only method — but reasons to instrument it: track exploitability (not return), report the temperature and the target it implies, refine Δt and N jointly, and verify the contraction or monotonicity hypothesis on the model at hand. With those guards, GMFG-PPO is the working bridge from the analytic theory of Parts II–IV to networks and dynamics that no PDE solver can reach; the heterogeneity it learns over is the subject of chapter 30.

Rigor ledger

Proved.

- The MFMDP/GMFMDP equilibrium is a fixed point of the two maps S (best response, backward Bellman) and Ψ (population, forward push-forward), and exploitability $\mathcal{E}$ is nonnegative and zero exactly at equilibrium (definition 29.4 and proposition 29.7).

- Under the contraction condition $L_{\mathsf{S}} L_\Psi < 1$ (A-Coupling), the naive fixed-point iteration converges geometrically to the unique equilibrium (theorem 29.10).

- Entropy regularisation makes the best response single-valued and $\Theta(1/\tau)$-Lipschitz with $O(\tau)$ equilibrium bias (lemma 29.12 and remark 29.13).

- Under monotonicity (A-Mon), fictitious play and online mirror descent drive exploitability to zero, $\mathcal{E} \to 0$, proved in full here via the continuous-time Lyapunov/Bregman mechanism (theorems 29.15 and 29.19); the three-point Bregman identity powering OMD is proved in-book (lemma 29.18). The *algebraic* rates $O(1/\sqrt{n})$ (resp. $O(1/n)$) are flagged as imported refinements (remarks 29.16 and 29.20), with the LQG case $n^{\lambda-1}$ derived in-book (worked computation 29.22).

- The LQG/graphon worked computation makes the dichotomy exact: FPI converges iff the multiplier $|\lambda|\,\|\mathbb{W}\| < 1$, while FP converges on the whole monotone half-line $\lambda\,\|\mathbb{W}\| < 1$ (worked computation 29.22).

- A modular error decomposition for GMFG-PPO, assembled from the above plus the finite-network transfer of chapter 25 (theorem 29.26).

Assumed.

- (A-Lip), (A-Nondeg), (A-Reg-W) throughout; (A-Coupling) in the contraction regime and (A-Mon) in the monotone regime.

- The PPO best-response oracle: that the inner deep-RL solver returns an ϵ_{br}-best response. This is the chapter's one unproven, load-bearing hypothesis (remark 29.27).

- Boundedness of the quadrature/discretisation error ϵ_{disc}, inherited from the schemes of chapter 28 and the regularity of chapter 19.

Deferred.

- The discrete-step refinement that upgrades the in-book convergence $\mathcal{E} \to 0$ to the algebraic *rates* (the $\vartheta = \log n$ time change and the η^2 bookkeeping) is imported, with attribution, from [38, 111, 110] and explicitly flagged as not derived in this book (remarks 29.16 and 29.20); the continuous-time Lyapunov mechanism and the scalar LQG rate $n^{\lambda-1}$ are given here in full.

- Reference implementations (PPO inner loop, type-grid quadrature, exploitability estimator) live in `figures/29-gmfg-ppo.py`, mirroring the `figures/28-mfg-solver.py` convention of chapter 28.

- The finite-network ϵ-Nash transfer is owned by chapter 25; we import it.

Open.

- Provable polynomial best-response oracles for deep function approximation (so that ϵ_{br} is bounded rather than assumed) remain open beyond linear/tabular classes [134, 135].

- Learning in the non-monotone, multi-equilibrium regime: which equilibrium, if any, a learning dynamic selects, and whether selection can be controlled.

- Joint control of the three limits (Δt, N, samples) with a single rate, and learning when the underlying network is itself unknown and must be estimated (graphon identification) before or during learning.

- Sharp dependence of representability and sample complexity on the spectrum of $\mathbb{W}$.

Bibliographic notes

The viewpoint that a mean field game is a fixed point of a best-response map on the space of population flows, and that learning dynamics chase that fixed point, is developed in the survey [96], which we follow for organisation. Fictitious play for MFG was introduced and proved convergent under monotonicity by Cardaliaguet and Hadikhanloo [38]; the continuous-time best-response flow and the Lyapunov analysis sketched in theorem 29.15 are from Perrin et al. [111], building on the classical learning-in-games theory of Fudenberg and Levine [74]. The online-mirror-descent scheme and theorem 29.19 are from Pérolat et al. [110]; the entropy-regularised formulation and the Gibbs best response (29.10) follow [53], with the quantal-response/regularised viewpoint also in [122]. Tabular and linearly-parametrised learners with *unconditional* (non-oracle) guarantees include Guo et al. [80], Xie et al. [134], Anahtarci et al. [8], and the policy-mirror-ascent analysis of Yardim et al. [135]; the linear-quadratic actor–critic convergence result is Fu et al. [73], the natural learning counterpart of the LQG thread (chapter 14). The policy gradient theorem is Sutton et al. [127]

(textbook treatment in [126]); generalised advantage estimation is [119]; PPO and its trust-region antecedent are [120, 118]. The graphon extension — the type-conditioned single network, the quadrature representation of $\mathbb{W}$, and the modular guarantee assembled in theorem 29.26 — is the GMFG-PPO scheme of Cui and Koeppl [54, 55], whose finite-network ϵ-Nash transfer rests on the convergence theory of chapter 25. The variational-Monte-Carlo reading of section 29.4.4 is the natural bridge to the computational-physics literature; we draw it here as a precise analogy, with the one exact correspondence — the Gibbs policy as the Boltzmann trial ensemble with the Q-function as energy — stated as an identity.

Chapter 30

The Heterogeneity Zoo: Major–Minor and Multi-Population Models

The literature on mean field games is, by now, a literature of named "extensions." Open any recent survey and you will find the subject parcelled out into chapters — major–minor games, multi-population (or multi-class) games, games with partial or noisy information — each presented as a theory in its own right, with its own existence proof, its own master equation, its own catalogue of solvable cases. A reader who has just spent Part IV building the graphon mean field game is entitled to ask a blunt organizing question before learning any of them separately: is each of these a genuinely new object, or has the graphon picture already paid for them? This chapter exists to answer that question, and its answer — which the rest of the chapter is occupied discharging — is that of the three most studied extensions, two are not new theories at all but particular structural choices of the single coupling operator $\mathbb{W}$ of Part IV, while the third is orthogonal to the network entirely.

The starting observation is that the graphon mean field game already carries a great deal of heterogeneity for free: agents are indexed by a type $x \in [0, 1]$, each type may have its own dynamics and cost, and the coupling between types is an arbitrary symmetric kernel W acting through the operator $\mathbb{W}$ (chapters 21 and 22). The recurring thesis is that $\mathbb{W}$ is a flexible enough bookkeeping device that the extensions are recovered by *specializing* it rather than by replacing it. Concretely: the multi-population model is exactly the *block* (piecewise-constant) graphon, and the major–minor model is exactly a *finite-rank atomic perturbation* of the graphon coupling on an enlarged index space. The third extension, partial information, is genuinely orthogonal to the network: it changes what an agent can *see* and therefore composes with the graphon structure rather than being encoded in $\mathbb{W}$. We close with a map of the modern landscape, oriented by the 2025 overview of heterogeneous mean field games [11], sorting every axis of heterogeneity into where it sits relative to the graphon formalism.

It will help to fix the three possible verdicts in plain words at the outset, because the map that ends the chapter is nothing more than these labels applied, and stating them now turns that map into a summary rather than a surprise. We call a feature *subsumed* when it is a *structural special case* of the Part IV graphon mean field game — obtained by choosing the kernel W, the index space, or the coefficients in a particular way, and needing no machinery beyond what Part IV already built. We call it *composable* when it *multiplies onto* the graphon construction without living inside $\mathbb{W}$ at all: it alters what each agent observes or controls, and the graphon coupling simply rides along underneath, unchanged. We call it *outside* when it honestly requires apparatus the graphon theory does not supply. In this taxonomy the multi-population game is the cleanest *subsumed* case — and the most complete, which is why we treat it first and in full — the major–minor game a subtler *subsumed* case, and partial information the chapter's representative *composable* case.

Threading all three is a single tension that the reader should watch for, because it is what decides how hard each model actually is. The coupling is always a graphon; what varies is the *equilibrium*. The graphon mean field game stays in the tractable, deterministic regime of chapter 24 precisely as long as the aggregate each agent feels is a *deterministic* flow — a fixed function of time, the same in every realization of the noise. A block kernel keeps us there: each population is still a continuum of vanishing-influence agents, so its aggregate is a deterministic curve. But the moment a single non-vanishing player (the major agent) or a genuine common noise makes that aggregate *random*, the problem escalates one level: the field becomes a measure-valued *process*, the consistency condition becomes a *conditional* fixed point, and the governing object is no longer the deterministic backward–forward system of chapter 22 but the conditional, common-noise master equation of chapter 26. Keeping the coupling/equilibrium distinction in view is what makes the chapter's verdicts precise rather than slogans.

Because all of this is a specialization, we build on Part IV without re-deriving any of it, and it is worth saying explicitly what is borrowed and why each piece is needed. We take the type-x controlled agent of definition 21.1; the population field $\boldsymbol{m} = (m^x)_{x\in[0,1]}$ living in the L^2-Wasserstein space $(\mathcal{M}_2, d_2)$ of definition 21.8, which is the space on which the whole equilibrium problem is posed; and, at its heart, the graphon aggregate $\Theta_t(x) = (\mathbb{W} \circledast \boldsymbol{m}_t)(x)$ of definition 21.5 — the *single* scalar (or measure) through which a type-x agent feels the entire crowd, and the object on whose structure the whole chapter turns. From these the best-response map S (definition 21.15), which optimizes one agent against a guessed aggregate, and the fixed-point notion of equilibrium (definition 21.16), which closes the loop by demanding the guess be self-consistent, are inherited unchanged. We need the characterization of that fixed point as a coupled backward–forward system — a backward HJB equation for the value field and a forward Fokker–Planck equation for the density field, with its block specialization — from chapter 22; and we need its existence/uniqueness theory from chapter 24, in particular the operator-norm smallness threshold (A-Coupling), $\|\mathbb{W}\| < \kappa := 1/C_T$, which is the quantitative gate that the block reduction will inherit verbatim with $\mathbb{W}$ replaced by a finite matrix. For uniqueness beyond smallness we use the classical Lasry–Lions monotonicity condition of definition 11.1 and, what our uniqueness statements actually invoke, its graphon lift definition 24.15 with the graphon uniqueness theorem theorem 24.16. The carried Thread B (graphon LQG) is advanced here at the block level, completing the bridge promised in remark 21.20 and chapter 22: the infinite-dimensional graphon LQG game will collapse, for a block kernel, to a finite explicit system.

Two standing conventions are in force throughout, and both are load-bearing here precisely because both directions of time and a graphon all appear together in a single system. First, the time-direction convention: *the value field $u = u$ carries terminal data at $t = T$ and is solved* *backward; the density field $m = m$ carries initial data at $t = 0$ and is solved forward.* This is not an arbitrary bookkeeping choice — the HJB equation is a terminal-value problem because optimization looks *ahead* to a terminal cost, while the Fokker–Planck equation is an initial-value problem because probability mass is transported *forward* from a known initial law; the two halves of every system below meet in the middle and must be read in opposite time senses. Second, following the Part IV convention (chapter 21), wherever a graphon W is in scope Brownian motion is written B, never W — the symbol W is reserved for the kernel. We restate both because a careless reading of the systems below, in which u, m, W and B all appear at once, is otherwise easy to misparse.

30.1 Multi-population games as block-graphon GMFG

We begin with the cleanest *subsumed* extension, and we treat it most completely because every later move in the chapter is a variation on the single argument we make here. A *multi-population* (or multi-class) mean field game has a finite number K of distinguishable populations. The picture to keep in mind is a system assembled from a small number of qualitatively different agent *species*, not from a continuum of subtly varying individuals. Three running examples make the model concrete. Buyers and sellers in a market form two species with opposite objectives, each watching the other's aggregate and its own. Susceptible and infected individuals in an epidemic form two species in which the size of the infected population is precisely the field a susceptible agent reacts to. Electric vehicles sorted into a handful of battery classes form several species, each with its own charging dynamics, all bidding into one shared grid. In none of these does the modeller believe there are infinitely many distinct kinds of agent; there are two, or three, or five, and the heterogeneity is *coarse* — a short list of classes rather than a spectrum.

What makes such a system a mean field *game*, rather than a finite bundle of unrelated control problems, is a precise rule about how agents may and may not differ, and the rule has two faces. *Within* a population, agents are anonymous and statistically identical, exactly as in the classical mean field game of Part II: one buyer is interchangeable with any other buyer, feels the crowd only through population-level aggregates, and carries vanishing individual influence. This intra-population symmetry is what lets each population be summarized by a single distribution rather than by a roster of named individuals. *Across* populations, by contrast, essentially anything is allowed: two populations may have entirely different state dynamics, different running and terminal costs, and — the genuinely new freedom relative to a one-population model — different *cross-influences*, meaning how strongly each population's aggregate is felt by each other population. A buyer may respond to the sellers' aggregate with one sensitivity and to other buyers' with another; the two need not be equal, and need not even share a sign. The point of this section is that this exact within/across split — statistical identity inside a class, arbitrary asymmetric coupling between classes — is something a graphon already knows how to express.

The claim is that the multi-population model is the dynamic graphon mean field game of chapter 21 under one special choice: the kernel W is taken *block-constant*, piecewise constant on a partition of the type interval $[0, 1]$ into K cells, one cell per population. This is the stochastic-block-model graphon. We want to earn something sharper than "the block graphon happens to reproduce the multi-population equations." The assertion is that the reduction is an *isometric reduction of* *the equilibrium problem to a finite-dimensional invariant subspace*, and both adjectives carry real weight. Because the formal payoff in the next two subsections will read as a confirmation of this phrase rather than a surprise, it is worth glossing each adjective in plain terms now, before any cells or matrices appear.

Take *invariant* first. The graphon equilibrium problem lives on an infinite-dimensional space: a population field $\boldsymbol{m}$ assigns a probability measure to *every* type $x \in [0,1]$, a whole continuum of measures. The block choice distinguishes a far smaller set inside this space — the fields that are *constant on each cell*, taking one common measure $m^{(k)}$ on all types of population k. Such a field is pinned down by only K measures, so the set is finite-dimensional in the sense that finitely many measures parametrize it. A subset of a state space is called *invariant* under a map when the map sends it back into itself: feed in an element of the subset, get out an element of the subset. The entire reduction hinges on one such invariance. The map that drives the equilibrium problem is the best-response map: it takes a guessed population field, lets every agent optimize against the aggregate that field induces, and returns the resulting field. The single structural fact we will prove is that this map *preserves block-constancy* — hand it a field that is constant on each cell and it returns a field that is again constant on each cell. Once that is known, the infinite-dimensional fixed-point problem may be solved entirely inside the block-constant set, because the iteration that searches for a fixed point never leaves it. This is why invariance is the whole game: it is exactly what licenses replacing a continuum of unknown measures by K of them with no approximation, so that a block-constant equilibrium *is* a multi-population equilibrium and conversely. (Whether a block kernel might *also* support equilibria outside the block-constant set is a separate, genuinely subtle question of uniqueness, which we flag and defer to remark 30.5.)

Now *isometric*. It would be a strictly weaker statement to say only that block-constant fields can be parametrized by K measures; one wants the parametrization to respect *distances* as well as count dimensions. The field space carries a metric d_2, the L^2-Wasserstein field metric of definition 21.8, which measures how far apart two population profiles are. Restricted to block-constant fields, this metric turns out to be *exactly* a mass-weighted product of the ordinary Wasserstein distances between the K cell-measures: no distance information is lost or distorted in passing to the finite parametrization. A distance-preserving correspondence between metric spaces is called an *isometry*, and that the block-constant subspace is isometric — not merely in bijection — with the finite product $(\mathcal{P}_2(\mathbb{R}^d))^K$ is what guarantees that every *quantitative* estimate of the graphon theory transfers to the finite system with its constants intact: contraction moduli, monotonicity constants, and above all the coupling threshold $\|\mathbb{W}\| < \kappa$ all carry over without slack, because they are measured by the same ruler on both sides. The infinite-dimensional and finite-dimensional problems are thus not just matched object-for-object; they are the *same* problem, viewed at two resolutions.

With both adjectives unpacked, the route through the section is fixed. We must, in order: write down the block graphon and isolate the finite-dimensional subspace of block-constant fields it distinguishes, checking that this subspace is closed and isometric to a weighted product of Wasserstein spaces; prove the one structural lemma everything rests on — that the graphon aggregate carries a block-constant field to a block-constant field, which is the precise form of "the best-response map preserves block-constancy"; and then read off the finite multi-population system, its equilibria, and its existence/uniqueness theory simply as the restriction of the Part IV machinery to that subspace. The next subsection takes the first of these steps. It partitions $[0,1]$ into K cells, defines the block-constant fields that *are* the finite-dimensional invariant subspace just promised, and states the aggregate-preservation lemma that makes the subspace invariant in the first place. The reader should arrive there already expecting a partition into cells, a block-constant subspace, and a lemma that the aggregate preserves block-constancy.

30.1.1 The block graphon and the block-constant subspace

We now make good on the promise of the previous section. There we argued, in words, that the block kernel distinguishes a *finite-dimensional invariant subspace* of the field space, and that the multi-population reduction is the restriction of the whole equilibrium problem to that subspace. This subsection produces the subspace and its load-bearing properties: that it is parametrized by K measures, that it carries exactly the weighted product metric the finite system will need, that it is closed (so a fixed point may be hunted inside it), and finally — the one genuinely structural fact — that the graphon aggregate maps it into itself. The partition and the block-constant fields defined below *are* the finite-dimensional invariant subspace; nothing more is hidden behind the phrase.

Start from the kernel. We want the simplest W that is not trivial — not constant (which would make every type identical and collapse the graphon game to a single classical mean field game), but as close to constant as a genuine K-class structure permits. The honest minimal choice is a kernel that is constant on each rectangle of a partition: it can distinguish K kinds of type and assign each ordered pair of kinds its own coupling strength, and it can do nothing finer. Recall, then, the block construction flagged in remark 21.20. Partition the type interval $[0,1]$ into K measurable cells,

$$[0,1] \;=\; \bigsqcup_{k=1}^{K} I_k, \qquad \pi_k \;:=\; \mathrm{Leb}(I_k) > 0, \qquad \sum_{k=1}^{K} \pi_k = 1, \tag{30.1}$$

one cell per population, and let the kernel be constant on each product cell,

$$W(x,y) \;=\; \mathsf{B}_{k\ell} \qquad \text{for } x \in I_k,\; y \in I_\ell, \tag{30.2}$$

with $\mathsf{B} = (\mathsf{B}_{k\ell})_{k,\ell=1}^{K}$ a symmetric matrix with entries in $[0,1]$ (symmetry $\mathsf{B}_{k\ell} = \mathsf{B}_{\ell k}$ is the matrix form of (A-Reg-W)). The π_k are the *population masses*: π_k is the fraction of all types that belong to class k, and we will see that this fraction is precisely the weight with which class k's aggregate enters everyone else's. This is the stochastic block model viewed as a graphon, and it carries exactly the within/across structure isolated in the previous section: any two types in the same cell are coupled identically to the rest of the world (intra-class statistical identity), while the off-diagonal entries $\mathsf{B}_{k\ell}$, $k \neq \ell$, are free and need not be equal or even of one sign (arbitrary across-class coupling).

The kernel being constant on cells is only half of the structure; the other half is the *ansatz* for the unknown field. If the coupling cannot tell two types in the same cell apart, there is no reason for the equilibrium to tell them apart either, and we should look for fields that are themselves constant on each cell. This is not yet a theorem — that the equilibrium really is constant on cells is the content of lemma 30.2 and theorem 30.4 — but it is the natural class of fields to single out, and we name it.

Definition 30.1 (Block-constant fields)**.** A population field $\boldsymbol{m} = (m^x)_{x\in[0,1]} \in \mathcal{M}_2$ (definition 21.8) is *block-constant* (subordinate to the partition (30.1)) if there exist $m^{(1)}, \dots, m^{(K)} \in \mathcal{P}_2(\mathbb{R}^d)$ such that $m^x = m^{(k)}$ for Lebesgue-a.e. $x \in I_k$, $k = 1, \dots, K$. The set of block-constant fields is denoted $\mathcal{M}_2^{\mathsf{B}} \subset \mathcal{M}_2$.

A block-constant field is thus pinned down by the finite tuple $(m^{(1)}, \dots, m^{(K)})$ of its K cell-measures: one probability measure per class, in place of one per type. The map $\iota : (m^{(1)}, \dots, m^{(K)}) \mapsto \boldsymbol{m}$ that builds the corresponding block-constant field is therefore a bijection from $(\mathcal{P}_2(\mathbb{R}^d))^K$ onto $\mathcal{M}_2^{\mathsf{B}}$ (its inverse reads off the a.e.-constant value of $\boldsymbol{m}$ on each cell), and this is already the bare "K measures parametrize the subspace" statement. The substance is that ι does more than count dimensions: it preserves distances. The field metric of definition 21.8 integrates the cell-wise Wasserstein distance over types,

$$d_2(\iota(\boldsymbol{m}^\bullet), \iota(\boldsymbol{m}'^\bullet))^2 \;=\; \int_0^1 \mathcal{W}_2(m^x, (m')^x)^2 \, \mathrm{d}x \;=\; \sum_{k=1}^{K} \pi_k \, \mathcal{W}_2\big(m^{(k)}, (m')^{(k)}\big)^2, \tag{30.3}$$

and the second equality is a one-line computation worth seeing explicitly. For two block-constant fields the integrand $x \mapsto \mathcal{W}_2(m^x, (m')^x)^2$ is itself constant on each cell I_k, taking there the single value $\mathcal{W}_2(m^{(k)}, (m')^{(k)})^2$, so the integral splits as $\sum_k \int_{I_k} \mathcal{W}_2(m^{(k)}, (m')^{(k)})^2 \, \mathrm{d}x = \sum_k \pi_k \, \mathcal{W}_2(m^{(k)}, (m')^{(k)})^2$, the cell length π_k emerging as the weight. So $(\mathcal{M}_2^{\mathsf{B}}, d_2)$ is *isometric* to $(\mathcal{P}_2(\mathbb{R}^d))^K$ equipped with the π-weighted product $\mathcal{W}_2$, and the population masses reappear here as the weights of the product: a discrepancy in a small (low-mass) class counts for proportionally less in the field metric, exactly as it should. This is the isometry that makes every quantitative constant of the graphon theory transfer to the finite system with no slack, as anticipated in the section opening.

The third property we need is that the subspace is *closed*, and it is worth being explicit about why this matters before proving it. The entire reduction strategy is to hunt for an equilibrium — a fixed point of the best-response map — *inside* $\mathcal{M}_2^{\mathsf{B}}$, by iterating the map and taking limits. For that to be legitimate the limit of a sequence of block-constant fields must itself be block-constant; otherwise the iteration could converge to a fixed point that has leaked out of the subspace, and the finite parametrization would be lost in the limit. Closedness is precisely the guarantee that no such leak occurs, and it follows at once from the isometry (30.3). Suppose $\boldsymbol{m}_n \in \mathcal{M}_2^{\mathsf{B}}$ is a sequence with cell-measures $m_n^{(k)}$ that converges in d_2 to some $\boldsymbol{m} \in \mathcal{M}_2$. Applying (30.3) to two members of the sequence gives $\pi_k \, \mathcal{W}_2(m_n^{(k)}, m_m^{(k)})^2 \le d_2(\boldsymbol{m}_n, \boldsymbol{m}_m)^2$, so each cell-sequence $(m_n^{(k)})_n$ is $\mathcal{W}_2$-Cauchy and hence, $\mathcal{P}_2(\mathbb{R}^d)$ being complete, $\mathcal{W}_2$-convergent to some limit $m_\infty^{(k)}$. The block-constant field $\iota((m_\infty^{(k)})_k)$ is then the d_2-limit of the $\boldsymbol{m}_n$ by (30.3) once more; since d_2-limits are unique it equals $\boldsymbol{m}$, so $\boldsymbol{m}$ agrees a.e. on each cell with the fixed measure $m_\infty^{(k)}$ and is therefore block-constant. In short, d_2-convergence forces $\mathcal{W}_2$-convergence of the cell-measures and a.e. agreement with their limits; thus $\mathcal{M}_2^{\mathsf{B}}$ is a closed subset of the complete space $(\mathcal{M}_2, d_2)$ (proposition 21.9) and is itself complete.

With the subspace built, parametrized, isometrically identified, and shown closed, only one thing remains before we can restrict the equilibrium problem to it: we must check that the problem *can* be restricted, i.e. that the dynamics never carry us out of the subspace. The whole equilibrium machinery enters a field only through the graphon aggregate Θ of definition 21.5 — the single quantity through which a type feels the crowd — so it suffices to show that the aggregate of a block-constant field is again block-constant. This is the single structural fact that drives the entire reduction.

Lemma 30.2 (The aggregate is block-diagonal)**.** *Let W be the block kernel* (30.2) *and let $\boldsymbol{m} \in \mathcal{M}_2^{\mathsf{B}}$ correspond to $(m^{(1)}, \ldots, m^{(K)})$. Then for the scalar (observable) aggregate of definition 21.5 with a bounded 1-Lipschitz observable h, the field $x \mapsto \Theta(x) = (\mathbb{W}[\langle h, m^{\cdot}\rangle])(x)$ is itself block-constant, with value on I_k*

$$\Theta(x) \;=\; \Theta^{(k)} \;:=\; \sum_{\ell=1}^{K} \pi_\ell\, \mathsf{B}_{k\ell}\, \langle h, m^{(\ell)} \rangle, \qquad x \in I_k. \tag{30.4}$$

The same holds for the measure-valued aggregate (21.3)*, which on I_k equals the finite measure $\sum_\ell \pi_\ell \mathsf{B}_{k\ell}\, m^{(\ell)}$.*

Proof. Fix a type $x \in I_k$, write out the aggregate (21.4) as an integral over the partner type y, and split that integral over the cells $I_1, \ldots, I_K$ on which the integrand is piecewise constant:

$$\Theta(x) \;=\; \int_0^1 W(x,y)\, \langle h, m^y \rangle \;\mathrm{d}y \;=\; \sum_{\ell=1}^{K} \int_{I_\ell} W(x,y)\, \langle h, m^y \rangle \;\mathrm{d}y.$$

On the cell I_ℓ the kernel is constant, $W(x,y) = \mathsf{B}_{k\ell}$ by (30.2) (because $x \in I_k$, $y \in I_\ell$), and the field is constant, $m^y = m^{(\ell)}$ for a.e. $y \in I_\ell$, so the integrand $\mathsf{B}_{k\ell} \langle h, m^{(\ell)} \rangle$ does not depend on y. Each cell-integral is then this constant times the cell length $\mathrm{Leb}(I_\ell) = \pi_\ell$, giving

$$\Theta(x) \;=\; \sum_{\ell=1}^{K} \pi_\ell\, \mathsf{B}_{k\ell}\, \langle h, m^{(\ell)} \rangle,$$

which depends on x only through its cell index k; this is (30.4). The measure-valued statement is the identical computation with the linear functional $\langle h, \cdot \rangle$ deleted, the constant measure $m^{(\ell)}$ pulled out of each cell-integral in its place. □

Read the result physically. A type in class k feels not one aggregate but a single weighted *superposition* of the K class aggregates $\langle h, m^{(\ell)} \rangle$, the weight on class ℓ being $\pi_\ell \mathsf{B}_{k\ell}$ — the product of how *many* types are in class ℓ (its mass π_ℓ) and how *strongly* class k is coupled to class ℓ (the kernel entry $\mathsf{B}_{k\ell}$). Two types in the same class feel the identical superposition, which is exactly why block-constant input produces block-constant aggregate and the subspace is invariant. Equation (30.4) is the multi-population coupling, and it is the object the next subsection will assemble into a closed K-population system.

It is convenient to package the weights $\pi_\ell \mathsf{B}_{k\ell}$ into a single matrix. Two matrices appear, and the difference between them is the recurring subtlety of the whole multi-population calculus, so we introduce both carefully. The weights as they literally act on the vector of class aggregates form the *asymmetric* matrix $\mathsf{B}\ \mathrm{diag}(\pi)$ with entries $(\mathsf{B}\,\mathrm{diag}(\pi))_{k\ell} = \mathsf{B}_{k\ell}\,\pi_\ell$; it is asymmetric because the mass π_ℓ sits on the source index ℓ alone, not symmetrically on the pair. For spectral purposes one prefers a symmetric representative, the *interaction matrix*

$$\widehat{\mathsf{B}}_{k\ell} \;:=\; \sqrt{\pi_k \pi_\ell}\; \mathsf{B}_{k\ell}, \qquad k, \ell = 1, \ldots, K, \tag{30.5}$$

which is symmetric because B is. These two matrices are not equal, but they are *conjugate*, and the conjugation is the source of every spectral identity we will use. Indeed, conjugating $\mathsf{B}\,\mathrm{diag}(\pi)$ by $\mathrm{diag}(\sqrt{\pi})$ gives

$$\mathrm{diag}(\sqrt{\pi})\;[\mathsf{B}\,\mathrm{diag}(\pi)]\;\mathrm{diag}(\sqrt{\pi})^{-1} \;=\; \mathrm{diag}(\sqrt{\pi})\,\mathsf{B}\;\mathrm{diag}(\pi)\;\mathrm{diag}(\sqrt{\pi})^{-1} \;=\; \mathrm{diag}(\sqrt{\pi})\,\mathsf{B}\;\mathrm{diag}(\sqrt{\pi}) \;=\; \widehat{\mathsf{B}},$$

the middle step using $\operatorname{diag}(\pi)\,\operatorname{diag}(\sqrt{\pi})^{-1} = \operatorname{diag}(\sqrt{\pi})$ since $\pi_\ell/\sqrt{\pi_\ell} = \sqrt{\pi_\ell}$, and the last reading off entries $\sqrt{\pi_k}\,\mathsf{B}_{k\ell}\,\sqrt{\pi_\ell}$. A similarity transformation preserves the spectrum, so $\mathsf{B}\operatorname{diag}(\pi)$ — and equally its transpose $\operatorname{diag}(\pi)\mathsf{B}$ — has the *same eigenvalues* as the symmetric $\widehat{\mathsf{B}}$; in particular those eigenvalues are real, which is not visible from the asymmetric form alone. The conjugation also explains in what sense $\widehat{\mathsf{B}}$ "is" the coupling operator. The isometry (30.3) identifies the block-constant subspace with $\mathbb{R}^K$ carrying the π-weighted inner product $\langle u, v\rangle_\pi = \sum_k \pi_k u_k v_k$, and on that subspace $\mathbb{W}$ acts on the vector $(\langle h, m^{(\ell)}\rangle)_\ell$ of class aggregates by left multiplication by $\mathsf{B}\operatorname{diag}(\pi)$, by (30.4). Changing variables to $\operatorname{diag}(\sqrt{\pi})\,u$ turns the weighted inner product into the standard one and conjugates $\mathsf{B}\operatorname{diag}(\pi)$ into $\widehat{\mathsf{B}}$; hence $\widehat{\mathsf{B}}$ is exactly the matrix representing $\mathbb{W}|_{\mathcal{M}_2^{\mathsf{B}}}$ in an *orthonormal* basis of the weighted space, symmetric precisely because $\mathbb{W}$ restricted to this finite-dimensional invariant subspace is self-adjoint, as recorded in remark 21.20. Consequently its spectral norm equals the operator norm of the restriction, $\left\|\widehat{\mathsf{B}}\right\| = \left\|\mathbb{W}|_{\mathcal{M}_2^{\mathsf{B}}}\right\|$, and this single number is the finite-dimensional avatar of the coupling constant $\|\mathbb{W}\|$: it is the quantity into which the Part IV threshold $\|\mathbb{W}\| < \kappa$ will collapse, and the coupling constant that will govern multi-population uniqueness. We hand it, together with the per-cell coupling $\Theta^{(k)}$ of (30.4), to the next subsection, which assembles them into the explicit K-population mean field game system that the multi-population literature writes down directly.

30.1.2 The multi-population MFG system

The previous subsection handed us two finite objects extracted from the block graphon: the interaction matrix $\widehat{\mathsf{B}}$, the symmetric avatar of the coupling operator $\mathbb{W}$ on the block-constant subspace, and — the one we put to work now — the per-cell coupling $\Theta^{(k)}$ of (30.4), the single weighted superposition of class aggregates through which a type in class k feels the entire crowd. The structural work is already done; what remains is bookkeeping. Lemma 30.2 guarantees that a block-constant field produces a block-constant aggregate, so the continuum of backward–forward systems indexed by $x \in [0, 1]$ collapses to one system per cell, and the only trace any cell keeps of the others is the K numbers $\Theta_t^{(1)}, \ldots, \Theta_t^{(K)}$. Assembling those K systems into one finite object, and then proving that this object is not merely *reminiscent* of the system the multi-population literature posits by hand but *identical* to it as an equilibrium problem, is the business of this subsection.

Specializing the coupled backward–forward system of chapter 22 to a block kernel and to block-constant data yields a *finite* system of K classical mean field game systems, coupled only through the scalar/measure aggregates (30.4). The specialization is mechanical, and it is worth seeing why before reading the result. In the general type-x system of chapter 22 the type label and the entire crowd enter an agent's problem through one channel only: the aggregate $\Theta_t(x)$. Everything else in the agent's optimization — its dynamics, its running and terminal cost in its own state, its diffusion — is local to that agent and carries the type label merely as a parameter. Now feed in block-constant data and a block-constant field. By lemma 30.2 the aggregate is constant on each cell, equal to $\Theta_t^{(k)}$ on I_k; by hypothesis the coefficients and costs are constant on each cell as well. Every ingredient of the type-x system therefore depends on x only through its cell index k, and the continuum of systems degenerates into K distinct ones, indexed by class. We state the resulting object for completeness; it is exactly what the multi-population literature [66, 49, 22] writes down directly, with no graphon anywhere in view. That literature *begins* from K populations and *posits* the coupling (30.7) as a modelling choice; we are about to obtain the identical system as a restriction of the single Part IV graphon game, which is the whole sense in which the graphon picture subsumes it.

Definition 30.3 (Multi-population MFG system)**.** Let each population k have drift b_k, volatility σ_k, running cost L_k, terminal cost g_k, and initial law $m_0^{(k)} \in \mathcal{P}_2(\mathbb{R}^d)$, all satisfying the standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) of assumption 7.3 uniformly in k, with population Hamiltonian $H_k(x,p,\Theta) = \sup_\alpha\{-p\cdot b_k(x,\alpha,\Theta) - L_k(x,\alpha,\Theta)\}$. A *multi-population MFG equilibrium* is a tuple of value/density flows $(u^{(k)}, m^{(k)})_{k=1}^K$ on $[0,T]\times\mathbb{R}^d$ solving, for each k,

$$\partial_t u^{(k)} + \tfrac{1}{2}\operatorname{tr}(\sigma_k\sigma_k^\top \mathrm{D}^2 u^{(k)}) - H_k(x, \nabla u^{(k)}, \Theta_t^{(k)}) = 0, \qquad u^{(k)}(T,\cdot) = g_k(\cdot, \Theta_T^{(k)}), \tag{30.6a}$$

$$\partial_t m^{(k)} = \mathcal{A}_k^* \, m^{(k)}, \qquad m^{(k)}(0,\cdot) = m_0^{(k)}, \tag{30.6b}$$

where $\mathcal{A}_k^*$ is the Fokker–Planck (forward) operator of population k under its optimal feedback $\alpha^{(k),*} = -\partial_p H_k(x, \nabla u^{(k)}, \Theta_t^{(k)})$, and the aggregate felt by population k is, with observable h,

$$\Theta_t^{(k)} \;=\; \sum_{\ell=1}^K \pi_\ell \, \mathsf{B}_{k\ell} \, \langle h, m_t^{(\ell)}\rangle. \tag{30.7}$$

The structure of (30.6) deserves to be read slowly, because everything that follows — including the entire existence and uniqueness theory — is a consequence of it. The system is a tuple of K separate HJB/Fokker–Planck pairs, one per population, and the k-th pair (30.6a)–(30.6b) is, on its face, an ordinary single-population mean field game system in the sense of eq. (8.5): a backward HJB equation for population k's value $u^{(k)}$, a forward Fokker–Planck equation for its density $m^{(k)}$, the two halves meeting through population k's own optimal feedback. The *only* way population k's pair learns about population $\ell \neq k$ is through the appearance of the scalar (or measure) $\Theta_t^{(k)}$ in the Hamiltonian and terminal cost, and that scalar is the single number (30.7) — a fixed linear readout of the other populations' densities. The consequence is decisive: *freeze* the K flows $t \mapsto \Theta_t^{(k)}$ at any guessed values, and the system instantly uncouples into K independent single-population mean field games, each solvable by the Part II machinery of chapter 8 as though the other $K-1$ populations did not exist. The populations are bound together not by interacting equations but by a finite-dimensional *consistency condition*: the K aggregates each population is solved against must equal the K aggregates the resulting densities produce through (30.7). A multi-population equilibrium is precisely a simultaneous fixed point of this K-fold coupling. That an entire society of distinct interacting species reduces to "K ordinary control problems plus K scalar matching conditions" is exactly what one means by saying the graphon picture subsumes multi-population MFG; the theorem below makes the reduction an equivalence of equilibria rather than a heuristic.

We can now state the reduction precisely. The theorem has three parts, and they are arranged so that each is the natural prerequisite for the next. Part (i) is the one structural fact — that the best-response map cannot escape the block-constant subspace — without which the whole strategy is illegitimate, since an iteration searching for an equilibrium could otherwise drift out of the finite-dimensional world. Part (ii) upgrades invariance to an exact identification: restricted to that subspace and read through the isometry ι, the graphon best-response map *is* the best-response map of definition 30.3, not a cousin of it. Part (iii) is then the immediate payoff at the level of equilibria: fixed points correspond to fixed points, so the two equilibrium problems have literally the same solution set.

Theorem 30.4 (Block reduction: equivalence of equilibria)**.** *Let W be the block kernel (30.2) and assume the agent data (coefficients, costs, initial laws) are block-constant in type, i.e. depend on x only through its cell index. Then:*

(i) *The block-constant subspace $\mathcal{M}_2^{\mathsf{B}}$ is invariant under the best-response map S of definition 21.15: if the input field flow is block-constant, so is the output.*

(ii) *The restriction $\mathsf{S}|_{\mathcal{M}_2^{\mathsf{B}}}$ is, under the isometry ι of (30.3), exactly the best-response map of the multi-population system: it solves (30.6a) for the K value flows against the aggregates (30.7) and returns the K Fokker–Planck flows (30.6b).*

(iii) *Consequently, $(u^{(k)}, m^{(k)})_{k=1}^K$ solves the multi-population MFG system (30.6) if and only if the block-constant field $\iota((m^{(k)})_k)$ is a dynamic graphon MFG equilibrium (definition 21.16) for the block kernel.*

Proof. (i) Let the input flow be block-constant, $m_t^x = m_t^{(k)}$ for $x \in I_k$, and recall what the best-response map of definition 21.15 actually does to a single type: it solves the type-x optimal control problem against the aggregate flow that the input field induces, then reports the law of the resulting optimally-controlled state. So everything turns on which data that control problem sees. There are exactly four inputs to it — the coefficients $(b_x, \sigma_x, L_x, g_x)$ defining the agent's dynamics and cost, the exogenous aggregate flow $t \mapsto \Theta_t(x)$ it best-responds to, and the initial law m_0^x from which its state is launched — and we check each is the same for two types in a common cell. The coefficients agree because the data are assumed block-constant: for $x \in I_k$ they equal $(b_k, \sigma_k, L_k, g_k)$ regardless of *which* type in I_k we picked. The initial law agrees for the same reason, $m_0^x = m_0^{(k)}$. The aggregate flow agrees by lemma 30.2: a block-constant input has $\Theta_t(x) = \Theta_t^{(k)}$ on I_k, a function of the cell index alone. Two types x, x' in the same cell I_k are therefore presented with *verbatim the same* control problem — same drift, same volatility, same running and terminal cost, same exogenous target, same starting distribution. There is nothing left for the label to influence.

Now push this identity of *problems* into an identity of *solutions*. The value u^x is characterized as the unique solution of the type-x HJB equation (30.6a), and every coefficient of that PDE — the second-order term $\frac{1}{2}\operatorname{tr}(\sigma_x\sigma_x^\top \mathrm{D}^2\ \cdot)$, the Hamiltonian $H_x(x, \cdot, \Theta_t(x))$, and the terminal datum $g_x(\cdot, \Theta_T(x))$ — has just been shown to depend on $x \in I_k$ only through k. A boundary-value problem is determined by its coefficients and its data; if two labels furnish identical coefficients and identical terminal data, the unique solution must coincide, so $u^x = u^{(k)}$ for every $x \in I_k$. (This is precisely where uniqueness for the HJB equation, supplied by the standing assumptions (A-Nondeg) and (A-Conv) of assumption 7.3 via the verification theorem of chapter 6, is doing load-bearing work: without it, two types could in principle select *different* solutions of the same equation, and block-constancy of the value would fail.) From a cell-independent value flow follows a cell-independent optimal feedback $\alpha^{(k),*} = -\partial_p H_k(x, \nabla u^{(k)}, \Theta_t^{(k)})$, hence a cell-independent Fokker–Planck operator $\mathcal{A}_k^*$, hence — since the initial law $m_0^{(k)}$ is also cell-determined — a cell-independent output density flow solving (30.6b). The reported law $\widetilde{m^x}_t = \widetilde{m}_t^{(k)}$ is thus constant on each cell, i.e. again block-constant. Therefore S carries $\mathcal{M}_2^{\mathsf{B}}$ into itself, which is the invariance claimed.

(ii) Part (i) did more than establish invariance; in the course of proving it we read off, cell by cell, exactly which finite system the restricted map solves. On cell k the value flow $u^{(k)}$ solves the HJB equation (30.6a) driven by the aggregate $\Theta_t^{(k)} = \sum_\ell \pi_\ell \mathsf{B}_{k\ell}\langle h, m_t^{(\ell)}\rangle$ — which is (30.4), i.e. the multi-population coupling (30.7) — and the output density $\widetilde{m}^{(k)}$ solves the Fokker–Planck equation (30.6b). That is term for term the best-response map of definition 30.3, the map sending $(m^{(\ell)})_\ell \mapsto (\widetilde{m}^{(k)})_k$ on the finite product $(\mathcal{P}_2(\mathbb{R}^d))^K$. The isometry ι of (30.3) relabels a block-constant field by its K-tuple of cell-measures, so conjugating $\mathsf{S}|_{\mathcal{M}_2^{\mathsf{B}}}$ by ι produces precisely this finite map; and because ι is an *isometry* and not merely a bijection, any contraction or Lipschitz modulus of S measured in d_2 is reproduced *with the same constant* by the finite map measured in the π-weighted product Wasserstein metric. This is what makes the two best-response maps the same map at two resolutions rather than two maps that happen to share fixed points — and it is the fact the quantitative theory of section 30.1.3 will lean on.

(iii) The equivalence of equilibria is now immediate, because an equilibrium of either problem is by definition a fixed point of its best-response map. If $(u^{(k)}, m^{(k)})_k$ solves the multi-population system (30.6), then $(m^{(k)})_k$ is a fixed point of the finite best-response map, hence by (ii) the block-constant field $\iota((m^{(k)})_k)$ is a fixed point of $\mathsf{S}|_{\mathcal{M}_2^{\mathsf{B}}}$, and — since S does not move points already in $\mathcal{M}_2^{\mathsf{B}}$ to anywhere outside it — a genuine fixed point of S on all of $\mathcal{M}_2$, i.e. a dynamic graphon MFG equilibrium (definition 21.16) for the block kernel. Conversely, a graphon equilibrium that *happens* to be block-constant is, by reading off its cell-measures, a solution of (30.6). The two solution sets are thus in bijection through ι. The role of invariance (i) and closedness here is exactly to guarantee that the restricted problem is self-contained: because $\mathcal{M}_2^{\mathsf{B}}$ is S-invariant and closed in the complete space $\mathcal{M}_2$, a fixed point sought inside it stays inside it and the limits of any approximating iteration remain available, so no equilibrium of the finite system is an artefact of the restriction and none is lost by it. What this argument does *not* claim — and cannot, as the next remark insists — is that every graphon equilibrium of a block kernel is block-constant; it claims only that the block-constant ones are exactly the multi-population ones. □

Remark 30.5 (Why block-constant data give block-constant equilibria, and when they need not)**.** It is tempting to over-read theorem 30.4 as saying that a block kernel *forces* the equilibrium to be block-constant. It does not, and the distinction is fundamental enough to isolate. What the theorem delivers is a bijection between the multi-population equilibria and the *block-constant* graphon equilibria — the equilibria that happen to live in the invariant subspace $\mathcal{M}_2^{\mathsf{B}}$. What it is silent about is whether a block kernel might *also* support graphon equilibria that are *not* block-constant: fields in which two types drawn from the same cell, presented (by part (i) of the proof) with the identical best-response problem, nonetheless settle into *different* measures. Invariance of $\mathcal{M}_2^{\mathsf{B}}$ says the best-response iteration started inside the subspace never leaves it, so a block-constant equilibrium always exists once the finite-dimensional fixed-point argument of section 30.1.3 provides one; it says nothing about whether some *other* iteration, started outside, might converge to a symmetry-broken fixed point that the subspace never sees.

Whether such a non-block-constant equilibrium exists is therefore not an existence question at all but a *uniqueness* question, and this is the crux. If the equilibrium is unique, it can only be the block-constant one we already produced, and symmetry breaking is impossible; if uniqueness fails, extra equilibria become available and there is no a priori reason they should respect the cell structure.

The two regimes in which we will be able to prove uniqueness — Lasry–Lions monotonicity and small coupling, both inherited from chapter 24 — thus do double duty: they guarantee not merely *a* solution but that the solution is forced to inherit the block symmetry of the kernel. Outside those regimes the guarantee lapses and symmetry breaking within a cell is genuinely possible.

This is the exact mean-field-game incarnation of the Curie–Weiss phenomenon. There a *fully symmetric* ferromagnetic coupling — every spin coupled identically to the mean — nonetheless supports asymmetric, magnetized states below the critical temperature: the symmetry of the interaction is not inherited by the equilibrium because the self-consistency map acquires multiple fixed points. The structural parallel is precise. A single cell with an attractive self-coupling is a continuum of statistically identical agents each best-responding to the cell mean, which is the self-consistent-field setup whose multiplicity we analyzed in chapter 11 (the LQG uniqueness/multiplicity threshold of proposition 11.8, and the reading of multiplicity as spontaneous symmetry breaking in section 11.4). When the self-coupling is weak the matching map is a contraction with a single fixed point — the symmetric, block-constant equilibrium — but when it is strong enough the map develops several fixed points, and the symmetric one need no longer be selected. The within-cell symmetry breaking entertained above is that same bifurcation lifted to the field. Establishing exactly where on this boundary uniqueness holds is what makes the present remark a genuinely open direction rather than a closed one; we log it as such in the chapter ledger.

The remark has converted the equivalence theorem's silence into a sharp question, and it is the question the next subsection answers. Theorem 30.4 reduced the multi-population equilibrium problem to a fixed point on the finite product $(\mathcal{P}_2(\mathbb{R}^d))^K$ but left two things unsettled: whether such a fixed point exists at all, and — the issue this remark has now shown to be the deciding one for symmetry breaking — when it is *unique*, hence necessarily the block-constant equilibrium and nothing more. We turn to both. Because the isometry ι transports every quantitative constant of the Part IV theory onto the finite system intact, the answers will be the multi-population shadows of theorems we already own: existence from the graphon Schauder argument, and uniqueness in precisely the monotone and small-coupling regimes flagged above, now read through the finite interaction matrix $\widehat{\mathsf{B}}$ rather than the operator $\mathbb{W}$.

30.1.3 Existence and uniqueness for the finite system

The previous subsection ended by sharpening the equivalence theorem's one silence into a decisive question: it is *uniqueness*, not existence, that controls whether a block kernel can break its own symmetry. If the multi-population equilibrium is unique it can only be the block-constant one that the invariant subspace already produces, so within-cell symmetry breaking is impossible; if uniqueness fails, symmetry-broken equilibria become available and nothing forces them to respect the cells. We now settle both halves — that an equilibrium exists, and the two regimes in which it is unique and hence necessarily block-constant. Because theorem 30.4 identifies the multi-population equilibrium with a fixed point of the best-response map on the finite-dimensional product $(\mathcal{P}_2(\mathbb{R}^d))^K$ — and identifies it *isometrically*, through the metric (30.3), so that every quantitative constant transfers with no slack — existence and uniqueness follow from the Part II and Part IV machinery essentially verbatim, with the single substitution of the finite interaction matrix $\widehat{\mathsf{B}}$ of (30.5) for the operator $\mathbb{W}$.

What is worth dwelling on before the statement is that this substitution does not merely rename the coupling constant: it makes the genuinely delicate analytic inputs of the graphon existence theorem of chapter 24 *elementary*. In the general theory the best-response map acts on the infinite-dimensional field space $\mathcal{M}_2$, an entire continuum of measures indexed by type, and the compactness needed to run a fixed-point argument there has to be assembled with care — one controls a whole family of laws simultaneously and must argue tightness and continuity uniformly across the continuum of types. On the block-constant subspace there are only K measures in play. Tightness of K probability measures carrying a uniform second moment is immediate from Markov's inequality and Prokhorov's theorem; the convex hull of K-tuples of such measures is compact; and the fixed-point statement one needs is the ordinary finite-product version of Schauder rather than its infinite-dimensional refinement. Finiteness of K is exactly what turns the compactness step from a theorem into a remark. That is the sense in which the block reduction is not just a special case but an *easy* special case.

Proposition 30.6 (Existence and uniqueness, multi-population)**.** *Assume each population's data satisfy (A-Lip), (A-Growth), (A-Nondeg), (A-Conv).*

(i) (Existence.) *There exists at least one multi-population MFG equilibrium* (30.6)*.*

(ii) (Uniqueness, monotone regime.) *Write* $\widehat{\mathsf{B}} = \widehat{\mathsf{B}}_{\mathrm{diag}} + \widehat{\mathsf{B}}_{\mathrm{off}}$ *for the split into its (block-)diagonal self-couplings and its off-diagonal inter-population part. If, in addition, the diagonal self-couplings of each population satisfy the graphon Lasry–Lions monotonicity condition (A-Mon) (definition 24.15) and the off-diagonal part is below the coupling threshold relative to the diagonal monotonicity modulus,* $\left\|\widehat{\mathsf{B}}_{\mathrm{off}}\right\| < \kappa = 1/C_T$ *(the (A-Coupling) constant of chapter 24), then the equilibrium is unique.*

(iii) (Uniqueness, small-coupling regime.) *Alternatively, if the per-population best-response is a contraction with modulus* $\rho_0 < 1$ *in* $\mathcal{W}_2$ *and* $\rho_0 \left\|\widehat{\mathsf{B}}\right\| < 1$*, the equilibrium is unique.*

Proof sketch. (i) Existence. We seek a fixed point of the finite best-response map $F : (m^{(k)})_k \mapsto (\widetilde{m}^{(k)})_k$ on $(\mathcal{P}_2(\mathbb{R}^d))^K$, which by theorem 30.4(ii) is $\mathsf{S}|_{\mathcal{M}_2^{\mathsf{B}}}$ read through the isometry ι. Three ingredients run the classical Schauder argument, and finiteness of K makes each one elementary. First, an *invariant ball*: the uniform second-moment bound of lemma 21.13, which holds population by population under (A-Growth), produces a radius R such that whenever each input law has second moment at most R, so does each output law. Thus F maps the set $\mathcal{C}_R := \{(m^{(k)})_k : \int |x|^2 \, \mathrm{d}m^{(k)} \leq R \ \forall k\}$ into itself. Second, *compactness* of $\mathcal{C}_R$: for a single population the second-moment ball is tight by Markov's inequality, hence relatively weakly compact by Prokhorov, and with a uniform $2+\delta$ moment (again from (A-Growth)) the second moments are uniformly integrable, so weak limits are $\mathcal{W}_2$-limits and the ball is $\mathcal{W}_2$-compact; for K populations $\mathcal{C}_R$ is just the K-fold product of this one compact set, which is where the finiteness of K enters — a product of finitely many compact sets is compact, with none of the diagonal/equicontinuity care the continuum-of-types version demands. The convex structure of $\mathcal{C}_R$ (a ball cut out by the linear second-moment functionals is convex under mixtures of measures) lets us pass to its closed convex hull without leaving $\mathcal{C}_R$. Third, *continuity* of F: holding the K input measures fixed determines the K aggregate flows $\Theta_t^{(k)}$ through the linear readout (30.7), which is plainly continuous in the inputs; against a given aggregate flow the population-k best response depends continuously on that flow by the stability of the HJB/Fokker–Planck pair in its data (the verification theorem of chapter 6 together with the well-posedness of chapter 8); composing, F is continuous on $\mathcal{C}_R$. Schauder's fixed-point theorem applied to the continuous self-map F of the convex compact set $\mathcal{C}_R$ — exactly the deployment of chapter 10, now in finite product form — yields a fixed point, i.e. a multi-population equilibrium. No appeal to the full infinite-dimensional compactness machinery of chapter 24 is needed.

(iii) Small-coupling regime. We take the contraction case before the monotone one because it is the more transparent of the two and isolates the role of $\left\|\widehat{\mathsf{B}}\right\|$ cleanly. Consider the map Φ on the space of K aggregate flows $\boldsymbol{\Theta} = (\Theta^{(1)}, \ldots, \Theta^{(K)})$ defined by the equilibrium loop: freeze $\boldsymbol{\Theta}$, solve the K *decoupled* single-population mean field games against these frozen targets, and read the realized aggregates (30.7) back off the resulting densities. A fixed point of Φ is a multi-population equilibrium, and conversely. The Lipschitz modulus of Φ factors through two independent gains. Perturbing the input target seen by population k moves its realized aggregate by at most ρ_0 times that perturbation — this is the per-population best-response sensitivity, assumed below one; crucially it is an *intra*-population quantity, the same number one would compute for an isolated single-population game. The realized aggregates are then fed back into the inputs through the coupling (30.7), i.e. through left multiplication by $\mathsf{B}\,\mathrm{diag}(\pi)$, whose gain in the weighted norm is exactly $\left\|\widehat{\mathsf{B}}\right\|$ by the conjugacy of (30.5). Composing the two, Φ is Lipschitz with modulus at most $\rho_0 \left\|\widehat{\mathsf{B}}\right\|$; when $\rho_0 \left\|\widehat{\mathsf{B}}\right\| < 1$ it is a contraction on a complete space and the Banach fixed-point theorem gives a unique equilibrium. The intuition is that "small coupling" means the feedback loop binding the K populations together is too weak to amplify the disturbances each population already damps on its own: contractivity of the isolated problems survives being wired together precisely when the wiring gain $\left\|\widehat{\mathsf{B}}\right\|$ does not exceed the reciprocal of the intra-population sensitivity.

(ii) Monotone regime. When the coupling is *not* small one can still get uniqueness from the Lasry–Lions monotonicity of definition 11.1, lifted to the graphon in definition 24.15, provided the inter-population coupling is below threshold. This is the finite-rank instance of the graphon monotone uniqueness theorem theorem 24.16, and although we do not reproduce the full Bregman bookkeeping, the outline makes visible exactly where the block structure enters and why the threshold is the one stated. Suppose two equilibria, distinguished by superscripts 1 and 2, with value/density differences $\delta u^{(k)}$ and $\delta m^{(k)}$. The monotone uniqueness computation of chapter 24 tests the difference of the two HJB equations against the difference of the two Fokker–Planck densities, integrates over $[0, T] \times \mathbb{R}^d$, and sums over k; the backward–forward duality between HJB and Fokker–Planck (the integration by parts of chapter 24) annihilates the linear transport terms and leaves a balance of two pieces. The first is a sum, over populations, of *displacement-convexity gaps* of the Hamiltonians H_k — nonpositive Bregman terms whose strict definiteness is supplied by (A-Conv) and (A-Nondeg) — which control the squared size of the aggregate discrepancy $\hat{\delta}_t := \mathrm{diag}(\sqrt{\pi})\big(\langle h, \delta m_t^{(k)} \rangle\big)_k$ from below with a modulus at least $\kappa = 1/C_T$, where C_T is the same best-response stability constant that fixes the Part IV threshold. The second is the *coupling pairing*, and here the block structure shows its

hand: substituting $\Theta^{(k)} = \sum_\ell \pi_\ell \mathsf{B}_{k\ell}\langle h, m^{(\ell)}\rangle$ turns the pairing into the quadratic form

$$\sum_{k,\ell} \pi_\ell \mathsf{B}_{k\ell}\,\langle h, \delta m_t^{(\ell)}\rangle\,\langle h, \delta m_t^{(k)}\rangle \;=\; \langle \widehat{\mathsf{B}}\,\hat{\delta}_t, \hat{\delta}_t\rangle,$$

exactly the quadratic form of the symmetric interaction matrix on $\hat{\delta}_t$ — the precise sense in which $\widehat{\mathsf{B}}$, and not the bare kernel, is what the uniqueness estimate sees. Split $\widehat{\mathsf{B}} = \widehat{\mathsf{B}}_{\mathrm{diag}} + \widehat{\mathsf{B}}_{\mathrm{off}}$. The diagonal part $\langle \widehat{\mathsf{B}}_{\mathrm{diag}}\hat{\delta}, \hat{\delta}\rangle = \sum_k \pi_k \mathsf{B}_{kk}\langle h, \delta m^{(k)}\rangle^2$ is the *self*-coupling of each population against itself; under the diagonal monotonicity hypothesis (A-Mon) each such term carries the sign that makes it cooperate with — rather than fight — the convexity gap, which is just the single-population Lasry–Lions uniqueness of chapter 24 applied class by class. The off-diagonal part $\langle \widehat{\mathsf{B}}_{\mathrm{off}}\hat{\delta}, \hat{\delta}\rangle$, by contrast, has no definite sign — distinct populations may couple with either sign — but it is controlled in size: $|\langle \widehat{\mathsf{B}}_{\mathrm{off}}\hat{\delta}, \hat{\delta}\rangle| \le \left\|\widehat{\mathsf{B}}_{\mathrm{off}}\right\| |\hat{\delta}|^2$. The decisive comparison is now purely numerical. The convexity-gap-plus-diagonal contribution is nonpositive and dominates $|\hat{\delta}|^2$ with modulus at least κ; the off-diagonal term can perturb it by at most $\left\|\widehat{\mathsf{B}}_{\mathrm{off}}\right\| |\hat{\delta}|^2$ in either direction. Hence, as soon as

$$\left\|\widehat{\mathsf{B}}_{\mathrm{off}}\right\| \;<\; \kappa \;=\; 1/C_T,$$

the off-diagonal coupling is strictly too weak to overturn the nonpositive diagonal Bregman terms: the whole balance remains strictly of one sign unless $\hat{\delta}_t \equiv 0$. Vanishing of $\hat{\delta}$ means every population's aggregate-relevant functional coincides across the two equilibria, and feeding this back through the per-population well-posedness forces $m^{(k),1} = m^{(k),2}$ for every k. The only ingredient beyond the Ch 24 estimate is the diagonal/off-diagonal split of $\widehat{\mathsf{B}}$, which is special to the block structure and is precisely what lets "monotone self-interaction" and "below-threshold cross-interaction" be imposed separately. (Signs above are kept schematic; the exact Bregman accounting is the content of theorem 24.16.) □

The two uniqueness regimes answer the symmetry-breaking question of remark 30.5 in complementary ways, and it is worth naming the contrast. The small-coupling gate $\rho_0 \left\|\widehat{\mathsf{B}}\right\| < 1$ asks nothing about the *sign* of the interactions; it simply demands that the entire coupling, self- and cross- together, be weak. The monotone gate is more permissive about strength: it allows arbitrarily strong *aligned* self-interaction (any (A-Mon)-monotone $\widehat{\mathsf{B}}_{\mathrm{diag}}$, however large) and caps only the *cross*-population part $\widehat{\mathsf{B}}_{\mathrm{off}}$ below κ. Either way the conclusion is the same and is exactly the corollary remark 30.5 was waiting for: where the equilibrium is unique it must be the block-constant one already exhibited, so within-cell symmetry breaking cannot occur, and the block symmetry of the kernel is inherited by the equilibrium. Symmetry breaking is possible only outside both gates — strong, non-monotone cross-coupling — which is the mean-field analogue of dropping below the Curie–Weiss critical temperature.

Notation watch: Two normalizations of the interaction matrix

The block coupling is written in two ways throughout this chapter that must never be conflated, and the entire monotone/small-coupling calculus depends on keeping them apart. The *aggregate* (30.7) uses the *asymmetric* weights $\pi_\ell \mathsf{B}_{k\ell}$: these are what population k literally feels, and they are not symmetric in (k, ℓ) because the source mass π_ℓ sits on the partner index alone — how loudly class ℓ speaks to class k depends on how many agents are in class ℓ. The *spectral* analysis, by contrast — every operator-norm threshold in proposition 30.6, both $\left\|\widehat{\mathsf{B}}\right\|$ and $\left\|\widehat{\mathsf{B}}_{\text{off}}\right\|$ — uses the *symmetric* interaction matrix $\widehat{\mathsf{B}}_{k\ell} = \sqrt{\pi_k \pi_\ell}\,\mathsf{B}_{k\ell}$ of (30.5). The two are conjugate, hence share eigenvalues (eq. (30.5) and the conjugation computation following it), but they are not equal, and it is the symmetric one that the norm thresholds refer to. A concrete $K = 2$ reading fixes the distinction: population 1 feels $\pi_1 \mathsf{B}_{11}\langle h, m^{(1)}\rangle + \pi_2 \mathsf{B}_{12}\langle h, m^{(2)}\rangle$, in which the *partner* mass π_2 multiplies the cross term; but the eigenvalue that the gate $\left\|\widehat{\mathsf{B}}\right\| < \kappa$ constrains belongs to the symmetric matrix

$$\widehat{\mathsf{B}} = \begin{pmatrix} \pi_1 \mathsf{B}_{11} & \sqrt{\pi_1 \pi_2}\,\mathsf{B}_{12} \\ \sqrt{\pi_1 \pi_2}\,\mathsf{B}_{12} & \pi_2 \mathsf{B}_{22} \end{pmatrix},$$

whose off-diagonal carries the *geometric* mean $\sqrt{\pi_1 \pi_2}$, not the partner mass. Treating the asymmetric weights as if they were the spectral object would mis-locate the threshold whenever the masses are unequal. Only when all masses coincide, $\pi_k \equiv 1/K$, do the two agree up to the scalar $1/K$. The thread specification's bare kernel $\mathsf{B}_{k\ell}$ is neither of these — it omits the masses entirely — and must never be used in its place: always carry the masses, asymmetrically in the aggregate and symmetrically (as a geometric mean) in the norm.

This closes the analytic theory of the block model. We now possess, for the multi-population game, exactly the trio the Part IV theory promised: an equilibrium always exists (by an elementary finite-product Schauder argument), and it is unique — hence forced to be block-constant, hence free of within-cell symmetry breaking — throughout the monotone regime and throughout the small-coupling regime, both governed by the single scalar $\left\|\widehat{\mathsf{B}}\right\|$ and its off-diagonal restriction. Every statement so far, though, has been made at the level of *abstract* data: drifts b_k, costs L_k, a generic observable h, an interaction matrix $\widehat{\mathsf{B}}$ whose norm we can name but not compute. The natural way to see the whole apparatus breathe is to specialize to the one model in which every one of these constants is an elementary closed-form expression — the graphon LQG game of Thread B. There the interaction matrix $\widehat{\mathsf{B}}$, the coupling threshold κ, and the per-population contraction modulus ρ_0 all become explicit functions of the Riccati data, and the block-equivalence theorem theorem 30.4 collapses the infinite-dimensional graphon LQG game to K scalar Riccati equations coupled through a single K-dimensional linear ODE for the population means. The next subsection carries out exactly this computation, picking up the rank-one worked example of chapter 21 and lifting it to the block level.

30.1.4 Thread B at the block level

The previous subsection closed with a promise: the abstract block apparatus — the interaction matrix $\widehat{\mathsf{B}}$, the coupling threshold κ, the per-population contraction modulus — would stop being names we can invoke but not evaluate, and become elementary closed-form expressions, in the one model built to make every constant explicit. We now make good on it. We advance the carried Thread B (graphon LQG) to the multi-population level, instantiating the block-equivalence theorem theorem 30.4 in the fully explicit LQG thread and completing the computation begun in remark 21.20 and the rank-one worked example worked computation 21.18. The payoff is the cleanest possible illustration of the entire block reduction. An object that began as an infinite-dimensional graphon mean field game — a continuum of coupled backward HJB and forward Fokker–Planck PDEs, one pair per type $x \in [0,1]$ — collapses, for a block kernel, to a strictly finite and almost entirely elementary system: K decoupled scalar Riccati equations carrying the curvature of each population's value function, plus one $2K$-dimensional *linear* two-point boundary-value problem coupling the K population means to the K value-function offsets. Everything the crowd does to an agent is squeezed, by the LQG structure, into that single linear BVP, and it is there — and only there — that the inter-population coupling matrix $\widehat{\mathsf{B}}$ actually appears. Watching where in the system the coupling sits, and watching it sit in the *backward* half, is the whole lesson of this subsection.

Worked computation 30.7 (Thread B: the K-population LQG game)**.** Take the graphon-LQG agent (21.2) and cost (21.8) with the block kernel (30.2) and block-constant, but otherwise *population-dependent*, parameters: population k has state feedback a_k, control gain b_k, noise σ_k, state-cost weights $q_k, q_{T,k} \geq 0$, and mean-coupling strengths $s_k, s_{T,k}$. The observable is the identity $h(\xi) = \xi$, so the aggregate (30.4) felt by population k is the scalar

$$z_t^{(k)} \;=\; \sum_{\ell=1}^{K} \pi_\ell \, \mathsf{B}_{k\ell} \, \bar{m}_t^{(\ell)}, \qquad \bar{m}_t^{(\ell)} := \mathbb{E}\Big[X_t^{(\ell)}\Big] = \langle \mathrm{id}, m_t^{(\ell)} \rangle. \tag{30.8}$$

By theorem 30.4 the entire K-class equilibrium problem is nothing more than K ordinary scalar LQG mean field problems run side by side, the k-th one of which a single type in cell k solves against the *exogenous* affine target $s_k z_t^{(k)}$ exactly as in chapters 8 and 14; the only thing that makes the K problems a single game rather than K unrelated ones is that the targets $z_t^{(k)}$ must, at equilibrium, be consistent with the very means they help produce. The benefit of LQG is that each of those scalar problems is solved in closed form by a quadratic value function, and tracking what that closed form does to the coupling cleaves the system into two layers of completely different character: a layer that the inter-population coupling never touches, and a layer that is nothing but the coupling. We take them in turn, and the contrast between them is the point.

(1) Decoupled Riccati layer. Because population k's problem is linear-quadratic, its value function is exactly quadratic in the state, $u^{(k)}(t,x) = \frac{1}{2} P_t^{(k)} x^2 + r_t^{(k)} x + c_t^{(k)}$, with three time-dependent coefficients to be determined: the curvature $P_t^{(k)}$, the linear offset $r_t^{(k)}$, and an additive constant $c_t^{(k)}$ that plays no role in the dynamics and we ignore. Substituting this ansatz into population k's HJB equation (30.6a) is the computation of chapters 8 and 14 verbatim, and its mechanism is what separates the two layers. The optimal feedback is linear, $\alpha^{(k),*} = -b_k \, \partial_x u^{(k)} = -b_k (P_t^{(k)} x + r_t^{(k)})$, so the HJB equation becomes a polynomial identity in x that must hold coefficient by coefficient. Matching the three powers of x separates cleanly: the coefficient of x^2 gives an equation in $P^{(k)}$ *alone*, the coefficient of x^1 an equation in $r^{(k)}$ forced by the target, and the coefficient of x^0 the irrelevant constant. The decisive point is *where the target* $s_k z_t^{(k)}$ *lands*: entering the cost only through the quadratic $(x - s_k z_t^{(k)})^2$, it contributes a term linear in x (and a constant) but *nothing* to the coefficient of x^2, because shifting the center of a parabola never changes its curvature. The x^2 balance is therefore blind to $z_t^{(k)}$, and the curvature $P^{(k)}$ solves the scalar Riccati equation of

eq. (8.24)/eq. (14.7) *with population-k parameters*,

$$\dot{P}_t^{(k)} \;=\; -2a_k\, P_t^{(k)} + b_k^2\,(P_t^{(k)})^2 - q_k, \qquad P_T^{(k)} = q_{T,k}, \qquad k = 1,\dots,K, \tag{30.9}$$

in which not a single other population's data appears. This is why the K Riccati equations are *mutually decoupled*: the curvature of a population's value function never learns of the others' existence, for the only conduit between populations is the target $s_k z_t^{(k)}$, that target couples affinely, and affine coupling cannot reach the quadratic coefficient. The inter-population structure has been quarantined entirely into the linear offset $r^{(k)}$ and, through it, the mean — which is layer (2). One consequence is purely practical and worth flagging: the K curvatures can be computed once and for all, in advance and independently, each $P_\cdot^{(k)}$ integrated backward from its own terminal datum $q_{T,k}$ with no reference to the equilibrium being sought; each is well-posed on the whole of $[0,T]$ under $q_k, q_{T,k} \geq 0$ by theorem 14.2 (the sign conditions are exactly what forbid the Riccati blow-up). They enter the second layer only as known, frozen coefficients.

(2) Coupled mean–offset layer. Everything the populations do to one another now has to live here, and the way it does is the single most instructive feature of the LQG block model, so we build the layer up rather than quote it. Start with the forward dynamics. Under the optimal feedback the closed-loop state of a type in cell k obeys $\mathrm{d}X_t^{(k)} = \big[(a_k - b_k^2 P_t^{(k)})X_t^{(k)} - b_k^2 r_t^{(k)}\big]\,\mathrm{d}t + \sigma_k\,\mathrm{d}B_t^{(k)}$, and taking expectations kills the martingale and leaves an ODE for the mean $\bar{m}_t^{(k)} = \mathbb{E}\Big[X_t^{(k)}\Big]$. Read that ODE carefully: its homogeneous part is the *closed-loop drift* $a_k - b_k^2 P_t^{(k)}$, built from population k's own Riccati curvature and nothing else, while the entire inhomogeneous forcing is $-b_k^2 r_t^{(k)}$, the offset. So the population mean does *not* obey a closed forward ODE — it cannot be integrated forward on its own, because to step it forward one needs $r_t^{(k)}$, which is a *value-function* quantity and therefore determined by looking ahead to the terminal cost. This is the same obstruction that made the single-population game a forward–backward problem rather than an initial-value one (eq. (8.29), eq. (14.15)), and the same one that its rank-one graphon lift inherited (eq. (21.21)–(21.22)); here it simply appears K times in parallel. The offset $r_t^{(k)}$ is exactly the x^1-coefficient equation set aside in layer (1): matching the linear power of x in the HJB produces a backward ODE for $r^{(k)}$ with its homogeneous part the *negative* closed-loop drift and its forcing the target $q_k s_k z_t^{(k)}$, and the terminal cost's linear-in-x coefficient supplies the terminal datum $r_T^{(k)} = -q_{T,k} s_{T,k} z_T^{(k)}$. The whole inter-population aggregate $z_t^{(k)}$ — the only place the matrix B enters the LQG system at all — thus enters *this* equation and no other. Per population k,

$$\dot{\bar{m}}_t^{(k)} = (a_k - b_k^2 P_t^{(k)})\,\bar{m}_t^{(k)} \;-\; b_k^2\, r_t^{(k)}, \qquad\qquad \bar{m}_0^{(k)} = \bar{m}_{\text{init}}^{(k)} \text{ given}, \tag{30.10a}$$

$$\dot{r}_t^{(k)} = (b_k^2 P_t^{(k)} - a_k)\, r_t^{(k)} \;+\; q_k\, s_k\, z_t^{(k)}, \qquad\qquad r_T^{(k)} = -\,q_{T,k}\, s_{T,k}\, z_T^{(k)}, \tag{30.10b}$$

with the block aggregate $z_t^{(k)} = \sum_{\ell=1}^K \pi_\ell \mathsf{B}_{k\ell} \bar{m}_t^{(\ell)}$ of (30.8) entering, as just argued, *only* through the forcing of the backward offset equation (30.10b) and its terminal datum — never the forward mean equation (30.10a), whose only non-local ingredient is the offset of its own population. For a single population this is precisely (21.21)–(21.22), now carrying population-k parameters and the inter-population target $s_k z_t^{(k)}$ in place of the self-target. The phrase to carry forward is that *the inter-population coupling sits in the backward half of the system.* This is not a notational accident but the LQG shadow of the throughline of the whole chapter: an agent feels the crowd only through a target it is trying to track, tracking is a forward-looking optimization, and optimization is solved backward; the crowd therefore reaches the agent's behaviour by first reaching its value function. The means evolve forward and are, in isolation, perfectly local; what binds the populations together is the backward sweep of the offsets, each offset listening to every population's mean through $z_t^{(k)}$ and feeding the result back into its own population's forward drift. Coupling enters late (at the terminal time) and propagates backward.

It is now a matter of stacking the K scalar pairs into a single vector system, and the stacking is what exposes the role of the coupling matrix. Collect the means and offsets into vectors $\bar{\boldsymbol{m}}_t = (\bar{m}_t^{(k)})_k$ and $\boldsymbol{r}_t = (r_t^{(k)})_k \in \mathbb{R}^K$. The per-population coefficients become diagonal matrices — $D_t = \operatorname{diag}\left((a_k - b_k^2 P_t^{(k)})_k\right)$ holding the K closed-loop drifts, $B = \operatorname{diag}\left((b_k^2)_k\right)$, $Q = \operatorname{diag}\left((q_k s_k)_k\right)$ and $Q_T = \operatorname{diag}\left((q_{T,k} s_{T,k})_k\right)$ — because everything *local* to a population (its dynamics, its own cost weights, its own coupling strength s_k) acts only within that population's slot. The one non-diagonal object in the entire system is the aggregate: writing $z_t^{(k)} = \sum_\ell \pi_\ell \mathsf{B}_{k\ell} \bar{m}_t^{(\ell)}$ as the matrix action $\boldsymbol{z}_t = \mathsf{B}\operatorname{diag}(\pi)\,\bar{\boldsymbol{m}}_t$, the forcing $q_k s_k z_t^{(k)}$ of the backward offset equation becomes $Q\,\mathsf{B}\operatorname{diag}(\pi)\,\bar{\boldsymbol{m}}_t$. So the off-diagonal weight matrix $\mathsf{B}\operatorname{diag}(\pi)$ appears once and in one place only — the lower-left block coupling the means into the offsets — which is the vectorial restatement of "the coupling lives in the backward half." The result is the $2K$-dimensional *linear two-point boundary-value problem*

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\bar{\boldsymbol{m}}_t\\ \boldsymbol{r}_t\end{pmatrix} = \begin{pmatrix} D_t & -B\\ Q\,\mathsf{B}\operatorname{diag}(\pi) & -D_t\end{pmatrix}\begin{pmatrix}\bar{\boldsymbol{m}}_t\\ \boldsymbol{r}_t\end{pmatrix}, \qquad \bar{\boldsymbol{m}}_0 \text{ given}, \quad \boldsymbol{r}_T = -\,Q_T\,\mathsf{B}\operatorname{diag}(\pi)\,\bar{\boldsymbol{m}}_T, \tag{30.11}$$

where $\mathsf{B}\operatorname{diag}(\pi)$ is the asymmetric weight matrix of section 30.1.3, with entries $(\mathsf{B}\operatorname{diag}(\pi))_{k\ell} = \mathsf{B}_{k\ell}\pi_\ell$ so that $(\mathsf{B}\operatorname{diag}(\pi)\,\bar{\boldsymbol{m}})_k = z^{(k)}$. The data that close the system are *split* between the two ends of the time interval, and the split is forced by the time-direction convention of the chapter: the means are densities and carry *initial* data, so $\bar{\boldsymbol{m}}_0$ is prescribed at $t = 0$; the offsets are value-function quantities and carry *terminal* data, so $\boldsymbol{r}_T$ is prescribed at $t = T$. The terminal condition $\boldsymbol{r}_T = -Q_T\,\mathsf{B}\operatorname{diag}(\pi)\,\bar{\boldsymbol{m}}_T$ is itself coupled — the offset each population must hit at the horizon depends, through the same matrix $\mathsf{B}\operatorname{diag}(\pi)$, on *every* population's terminal mean — so the coupling appears a second time, again on the backward side, now in the boundary data rather than the dynamics. This is exactly the forward–backward architecture of the scalar games eq. (8.29)/eq. (14.15), lifted to carry K populations at once: a forward block and a backward block, meeting through split boundary conditions, with the curvatures $P^{(k)}_{\cdot}$ already integrated out so that what remains is a *genuinely linear* two-point problem in $(\bar{\boldsymbol{m}}, \boldsymbol{r})$. Once it is solved the model is finished: the per-population variance ODEs (14.14) close the Gaussian density flows around the now-known means, and the equilibrium is fully explicit. The reason to stress that the problem is a two-point BVP and not an initial-value problem is that the two are not equally well-behaved. A linear initial-value ODE always has a unique global solution; a linear *two-point* problem need not, because the map sending the unknown terminal data to the prescribed boundary value can be singular at special horizons. Concretely, solvability of (30.11) is the K-population analogue of the scalar nondegeneracy condition $D(T) \neq 0$ of lemma 14.5: a determinant built from the $2K \times 2K$ transition matrix must not vanish, and when it does the equilibrium is either non-unique or non-existent. This is the precise mechanism by which a perfectly linear problem can nonetheless have multiple equilibria, and it is the subject of the phase discussion below.

A new finite system earns trust by reproducing the systems it is meant to generalize, and (30.11) reproduces three of them. We run the checks in increasing order of subtlety, because each isolates a different feature.

The first check is dimensional sanity: setting $K = 1$ must return the single-population game. With one cell the mass is $\pi_1 = 1$, the matrices D_t, B, Q are scalars, and the weight "matrix" $\mathsf{B}\,\mathrm{diag}(\pi)$ is the number B_{11}, so (30.11) becomes the 2×2 forward–backward pair in $(\bar{m}, r)$ with self-coupling $s_1\mathsf{B}_{11}$ — precisely the single-population two-point BVP (14.15), equivalently the pair (8.29). The point worth registering is what does *not* happen: even with a single population the mean does not satisfy a closed forward ODE. It remains tied to its own backward offset r, which is forced by the self-target $s_1\mathsf{B}_{11}\bar{m}$. The inter-population coupling has vanished, but the forward–backward coupling between a population's mean and its *own* value function survives, exactly as it must, since that coupling is the content of being a mean field game rather than a control problem. "$K = 1$" removes the *across*-population structure and leaves the *within*-population mean-field structure untouched.

The second check identifies the coupling constant. The dynamics carry the *asymmetric* weight matrix $\mathsf{B}\,\mathrm{diag}(\pi)$, but the spectral fate of the BVP — which horizons are solvable, where multiplicity onsets — is governed by eigenvalues, and eigenvalues are conjugation-invariant. As recorded at (30.5), $\mathsf{B}\,\mathrm{diag}(\pi)$ is conjugate (by $\mathrm{diag}(\sqrt{\pi})$) to the symmetric interaction matrix $\widehat{\mathsf{B}}$, so the two share a spectrum; the relevant coupling magnitude is therefore the spectral norm $\left\|\widehat{\mathsf{B}}\right\|$ of the *symmetric* representative, not any norm of the asymmetric weights. This is the same number that gates existence and uniqueness in proposition 30.6 and that remark 21.20 predicts is the finite avatar of $\|\mathbb{W}\|$. The two faces of the coupling — asymmetric in the dynamics, symmetric in the spectrum — are exactly the two normalizations section 30.1.3 warned must never be conflated; the BVP is where one sees concretely why the symmetric one is the one that decides the phase.

The third check closes the loop with the rank-one worked example of chapter 21 and is the most informative, because it shows the block system contracting back to a single scalar. Take all populations identical, sharing parameters $(a, b, \sigma, q, q_T, s, s_T)$, and let the kernel be the rank-one block matrix $\mathsf{B}_{k\ell} = f_k f_\ell$ with f_k the constant value of the profile f on cell I_k. Then the aggregate factorizes: $z_t^{(k)} = \sum_\ell \pi_\ell f_k f_\ell \bar{m}_t^{(\ell)} = f_k \sum_\ell \pi_\ell f_\ell \bar{m}_t^{(\ell)} = f_k\,\varrho_t$, where the single scalar $\varrho_t = \sum_\ell \pi_\ell f_\ell \bar{m}_t^{(\ell)} = \langle f, \bar{\boldsymbol{m}}_t\rangle_\pi$ is the order parameter of worked computation 21.18. Every population is therefore driven by the *same* scalar ϱ_t, merely weighted by its own f_k, which forces the offsets to inherit the factorization $r_t^{(k)} = f_k\tilde{r}_t$: substituting this ansatz into the backward equation (30.10b) and using the shared parameters, every term carries a common factor f_k that cancels, leaving the single scalar offset equation $\dot{\tilde{r}}_t = (b^2P_t - a)\tilde{r}_t + q\,s\,\varrho_t$. Differentiating the order parameter and feeding in the forward mean equation collapses the means the same way, $\dot{\varrho}_t = (a - b^2P_t)\varrho_t - b^2\big(\sum_k \pi_k f_k^2\big)\tilde{r}_t$, and the constant $\sum_k \pi_k f_k^2 = \|f\|_{L^2}^2 = \|\mathbb{W}\|$ is exactly the operator norm of the rank-one graphon (equivalently $\left\|\widehat{\mathsf{B}}\right\|$, since for $\mathsf{B}_{k\ell} = f_k f_\ell$ the matrix $\widehat{\mathsf{B}}$ is rank-one with the vector $(\sqrt{\pi_k} f_k)_k$ of squared norm $\sum_k \pi_k f_k^2$). The $2K$-dimensional BVP has collapsed onto the scalar self-consistency pair $(\varrho_t, \tilde{r}_t)$ of (21.24), with the single effective coupling $s_{\mathrm{eff}} = s\,\|\mathbb{W}\|$ of (21.25). The block computation thus strictly contains both the single-population reduction (a) and the rank-one reduction (c), and agrees with each on its own turf — which is the concrete sense in which Thread B is one coherent thread and not three.

It remains to read off the phase behavior, and the apparatus is now entirely in hand: the qualitative fate of the equilibrium is decided by the $2K \times 2K$ generator of (30.11), exactly as the single-population phase diagram of section 14.5 was decided by its scalar 2×2 mean–offset generator. The mechanism is the determinant condition of check (a) raised to $2K$ dimensions. Solving the two-point problem amounts to inverting a certain block of the generator's transition matrix over $[0,T]$; this block depends on the horizon T and on the off-diagonal forcing $Q\,\mathsf{B}\,\mathrm{diag}(\pi)$, which by check (b) enters through the spectrum of $\widehat{\mathsf{B}}$. While the product of coupling strength and horizon is small the block stays invertible and the equilibrium is unique — the block-constant equilibrium that proposition 30.6 guarantees, with no symmetry breaking. As the coupling is turned up, the relevant combination $|s_k|\,\left\|\widehat{\mathsf{B}}\right\|$ grows, the generator acquires eigenvalues that rotate the forward and backward halves into resonance, and at a critical strength the determinant hits zero: the linear two-point problem in the forward mean and the backward Riccati offset loses unique solvability, and multiple equilibria appear. This is the multi-population shadow of the critical horizon T_{crit} of eq. (14.24) — the same bifurcation, now located by the largest eigenvalue of $\widehat{\mathsf{B}}$ rather than by a single scalar coupling — and it is the LQG-explicit incarnation of the symmetry-breaking threshold that remark 30.5 reached on abstract grounds. Strong coupling, long horizon, and a large spectral radius of the interaction matrix are three faces of the one condition under which a block kernel stops dictating a unique, block-symmetric equilibrium.

Remark 30.8 (Heterogeneous interaction beyond positivity)**.** It is worth pausing on a freedom the calculation quietly used but never advertised. Nothing in worked computation 30.7 required $\mathsf{B}_{k\ell} \geq 0$ or s_k of one sign — the Riccati equations, the mean–offset BVP, and the spectral phase analysis went through for arbitrary real symmetric B and arbitrary real s_k. A block model can therefore encode *antagonistic* populations, $\mathsf{B}_{k\ell} < 0$ or $s_k < 0$, in which population k is *repelled* from population ℓ's mean rather than drawn toward it. The graphon regularity assumption (A-Reg-W), which pins W to $[0,1]$, is not violated here so much as transcended: that restriction is a normalization convenient for the probabilistic $G(N,W)$-sampling interpretation of chapter 19, where $W(x,y)$ is read as an edge *probability* and must lie in $[0,1]$. The analytic block system (30.6), by contrast, is a coupled PDE system that makes perfect sense for any symmetric real coupling matrix, and the sign of $s_k\mathsf{B}_{k\ell}$ is precisely what decides its character. When all the products $s_k\mathsf{B}_{k\ell}$ are aligned the system is of the stabilizing, Lasry–Lions-monotone kind we spent the previous subsection taming, and the equilibrium is unique and block-symmetric. When they are mixed in sign — attraction within a class, repulsion across it, say — monotonicity can fail, and with it the uniqueness guarantee: the off-diagonal of $\widehat{\mathsf{B}}$ acquires the indefinite spectrum that drives the phase transition just described.

This is more than a technical relaxation; it is where the multi-population model stops being a bundle of cooperating crowds and becomes a genuine *game*. A single monotone population can only congest — agents avoid the crowd, are drawn to it, and settle into the one self-consistent profile. Two populations with opposite-signed cross-coupling can do something no single monotone population can: chase and flee. Predator and prey, where one population is attracted to the other's location and the other repelled from the first, is the canonical destabilizing pair; buyers and sellers, who push price in opposite directions, are its economic cousin. These are exactly the configurations that produce the multiplicity and the symmetry breaking of the phase discussion above, and they are absent by construction from any model built on a single nonnegative kernel. Signed block couplings are thus the multi-population model's entry point into genuinely game-theoretic richness — and a first hint that breaking anonymity can manufacture strategic tension that pure congestion never exhibits.

That hint is the bridge to the next section, and it is worth saying explicitly what we have finished and what we have not. We have now exhausted the multi-population model: heterogeneity carried by finitely many *populations*, each of strictly positive mass $\pi_k > 0$, coupled through a block kernel, reducing — isometrically, with every constant intact, and in the LQG case completely explicitly — to a finite system the literature posits by hand. The defining feature throughout was that every agent still had *vanishing* individual influence: a class mattered because it was *many*, its weight $\pi_k \mathsf{B}_{k\ell}$ in everyone's aggregate being a mass times a coupling. The next section breaks anonymity along the opposite axis: it installs a single distinguished agent — the major player — who must exert an $O(1)$ influence on the entire population while occupying *zero* mass, so that the very device that gave a class its voice, multiplying a coupling by a positive mass, is unavailable to express it. That tension, and the enlargement of the index space it forces, is the business we turn to now.

30.2 Major–minor games as a finite-rank atomic perturbation

We turn to the second *subsumed* extension, and it pays from the outset to see exactly how it differs from the one we have just closed. The multi-population model broke anonymity by splitting the continuum into finitely many *classes*, each of strictly positive mass $\pi_k > 0$; every individual agent still carried vanishing influence, and a class mattered only because it was *many* — its voice in everyone's aggregate was the product $\pi_k \mathsf{B}_{k\ell}$ of a mass and a coupling. The major–minor model breaks anonymity along the opposite axis, and in a way that turns out to be considerably more delicate. It keeps the population anonymous but installs at its center a single distinguished agent — the *major* player — whose influence on the whole crowd of ordinary *minor* agents does *not* vanish, even though, as one individual among infinitely many, the major player occupies *zero* mass. Heterogeneity here is not spread across a handful of fat classes; it is concentrated in one mass-zero point that nonetheless refuses to disappear. The tension between those two facts — zero mass, $O(1)$ influence — is the whole content of the model, and it is precisely the device that gave a class its voice, multiplying a coupling by a positive mass π_k, that is unavailable to express it.

The model is not exotic; it is the natural formalization of a recurring situation in which one player is qualitatively unlike the rest and large enough to be felt by all of them at once. Three canonical examples fix the picture, and each is worth holding in mind because each exhibits the same one-against-the-continuum structure. A *central bank* sets a policy rate that every trader in the market reacts to, while no single trader, and indeed no diffuse cloud of traders, moves the bank in the way the bank moves them: the bank's state is an input to the representative trader's problem on the same footing as the traders' own aggregate. A *dominant firm* — the price leader in an industry of small competitors — chooses an output or a price that shifts the market environment in which every fringe competitor optimizes, again asymmetrically: the fringe responds to the leader, the leader responds only to the *statistics* of the fringe. A *government* levying a tax, or a regulator setting a standard, conditions the optimization of an entire population of citizens or firms, each of whom is individually negligible to it. In every case there is one player whose *own state* — the policy rate, the leader's output, the tax — enters the mean field that everyone else responds to, and whose state in turn evolves in response to the crowd it is steering. It is this two-way but radically *asymmetric* coupling, one named state against a continuum of anonymous ones, that the formalism must capture.

Now make the obstruction sharp, because it is the obstruction the next subsection is built to remove. In the graphon game of chapter 21 an agent feels the crowd only through the aggregate $\Theta(x) = \int_0^1 W(x,y)\langle h, m^y\rangle \, dy$, an integral against Lebesgue measure on the type space $[0,1]$. The contribution of any single type y_0 — or of any Lebesgue-null set of types — to that integral is exactly zero: a mass-zero label is *invisible* to the aggregate. This is not a defect of the model but the mathematical expression of anonymity itself; it is what makes the minor agents a genuine mean field, each one negligible. But it is fatal to any attempt to install the major player as one more type. Whatever type $y_0 \in [0,1]$ we might try to assign the central bank, the integral that defines what the traders feel simply does not see it, and the bank's $O(1)$ influence evaporates. The block construction of the previous section is of no help: it lived entirely on positive-mass cells, and the major player has no positive mass to give it. So the major player cannot be smuggled into the existing index space at all, and we are forced into the one move that can carry a non-vanishing yet pointlike influence — enlarging the index space by a genuine *atom*, a single point carrying its own positive reference mass, on top of the nonatomic continuum $[0,1]$. Against the augmented index the coupling operator acquires a *finite-rank atomic correction*: a low-rank block that wires the atom to the continuum and back, sitting on top of the unchanged graphon $\mathbb{W}$. This is the precise sense, advertised in the section title, in which the major–minor model is a finite-rank atomic perturbation of the Part IV coupling rather than a new theory.

That the *coupling* remains a graphon, merely on a larger space, is only half the story, and the more interesting half is what it costs at the level of *equilibrium* — the recurring tension of this chapter, here showing its sharpest face. A positive-mass class, being a continuum of vanishing-influence agents, has an aggregate that is a *deterministic* curve: idiosyncratic noise averages out across the many, and the field each minor feels stays a fixed function of time, exactly the regime in which the deterministic graphon theory of chapter 24 applies. The atom destroys this. A single non-vanishing player carries its own driving noise that does *not* average against anyone, so the major player's state is a bona fide random process, and through the finite-rank coupling it injects that randomness directly into the mean field. The aggregate the minors respond to is therefore no longer a deterministic flow but a measure-valued *process*, adapted to the major player's trajectory; the consistency condition becomes a *conditional* fixed point, conditioned on that trajectory, and the governing object is no longer the deterministic backward–forward system of chapter 22 but the conditional, common-noise master equation of chapter 26. The major player, in other words, plays for the minors the role of a common noise. So the verdict will be genuinely two-sided, and we should flag it now: the coupling is subsumed — a finite-rank atomic graphon — while the equilibrium escalates out of the deterministic regime that made the multi-population case so clean.

The work of the section is to make both halves precise, and we begin with the obstruction itself. The next subsection takes the zero-mass puzzle posed just above — one agent of vanishing reference mass that must nonetheless exert finite influence — and proves precisely why a major agent *cannot* be one more type in $[0,1]$, converting the heuristic "a mass-zero label is invisible" into a clean statement and, with it, forcing upon us the augmented-space construction that carries the atom.

30.2.1 Why a single agent cannot be a type

The section opener left us with a puzzle stated as a tension: the major player must have $O(1)$ influence on every minor, yet as one individual among a continuum he occupies zero mass, and the positive-mass device that gave a multi-population class its voice is therefore unavailable. Before we can build anything we must convert that tension from a heuristic worry into a clean theorem, because the whole augmented-space construction is forced by it and would otherwise look like an unmotivated complication. So we begin by proving — not merely asserting — exactly why the obvious move fails: why the major player *cannot* be installed as one more type in $[0, 1]$, no matter which type we assign him. We record the obstruction first as a notation warning, since it is fundamentally a statement about what the integral defining the aggregate can and cannot see, and then unpack the argument in full.

Notation watch: A mass-zero type has no influence

In the graphon game of chapter 21 the influence of a set $S \subseteq [0, 1]$ of types on the aggregate $\Theta(x) = \int_0^1 W(x, y)\langle h, m^y\rangle \, dy$ is weighted by the Lebesgue measure dy. A single type y_0, or any Lebesgue-null set of types, contributes 0 to this integral: it is invisible to the aggregate. Therefore a major agent — an individual with $O(1)$ influence on every minor — *cannot* be represented as one more type in $[0, 1]$. To give a single agent finite influence we must enlarge the index space by an *atom* carrying positive reference mass. This is the structural reason the major–minor model is not literally a special case of the type-$[0, 1]$ graphon game, and the precise sense in which it is nonetheless a graphon game on an augmented space.

The argument behind the warning is a single line of measure theory, but it is worth writing out because everything downstream is a response to it. Suppose, against the model, that we tried to seat the major player at some type $y_0 \in [0, 1]$, endowing that one type with the major's state law m^0 while leaving the rest of the field untouched. The quantity a minor of type x feels from him is the part of the aggregate integral carried by the singleton $\{y_0\}$, namely $\int_{\{y_0\}} W(x, y)\langle h, m^y\rangle \, dy$, and this integral is over a set of Lebesgue measure $\mathrm{Leb}(\{y_0\}) = 0$. By the very definition of the Lebesgue integral, the integral of any bounded integrand over a null set vanishes; equivalently, altering the field on the null set $\{y_0\}$ — to m^0 or to anything else — changes $\Theta(x)$ by exactly nothing, for every minor type x and at every instant. The major player so installed is not merely weak; he is *literally absent* from the only channel through which the crowd is felt. And the conclusion is not special to a single point: any Lebesgue-null set of types — a finite collection, a countable collection, even an uncountable Cantor-type set — is equally invisible, because each carries zero dy-weight. There is no subset of $[0, 1]$, however cleverly chosen, that is at once small enough to be "one agent" and large enough to be heard.

It is worth pausing on the fact that this invisibility is not a flaw to be patched but the mathematical signature of anonymity, the property that makes the minors a genuine mean field in the first place. A continuum of agents is a mean field precisely because no individual among them moves the aggregate: each minor is negligible, and the integral against dy is the exact bookkeeping of that negligibility. The multi-population reduction never collided with this because it never tried to make an *individual* matter; it made a positive-mass *class* matter, and a class of mass $\pi_k > 0$ is heard precisely because $\int_{I_k} \mathrm{d}y = \pi_k \neq 0$ — its voice in the aggregate was the product $\pi_k \mathsf{B}_{k\ell}$ of a positive mass and a coupling (eq. (30.4)). The major player has no positive mass to contribute that product, so the block construction of section 30.1 offers no help: it lived entirely on cells of strictly positive Lebesgue measure, which is exactly what a single distinguished agent lacks. We are not looking at a quantitative shortfall to be made up by a larger coupling constant; we are looking at a structural zero, an $O(1)$ influence multiplied against a reference weight of 0.

That diagnosis dictates its own cure. If a single point must be heard by an integral, the only possible remedy is to give that point positive weight under the measure the integral uses — to make it an *atom* of the reference measure. One cannot do this within $([0,1], \mathrm{Leb})$, whose defining feature is that it is nonatomic (every point is null); one must enlarge the measure space. The minimal enlargement that creates exactly one atom while disturbing nothing else is to adjoin a single new point and place a point mass there. This is the move forced on us, and we make it. We work on the augmented index space

$$\mathbb{X} \;:=\; [0,1] \sqcup \{\star\}, \qquad \rho \;:=\; \mathrm{Leb}\big|_{[0,1]} \;+\; \delta_\star, \tag{30.12}$$

the type interval together with one isolated point $\star$ (the major) carrying a unit point mass under the reference measure ρ. Three features of this definition deserve comment, since each is doing work. First, the new point is adjoined by a *disjoint* union: $\star$ is not overlaid on some existing type y_0 but sits outside $[0,1]$ altogether, so the continuum of minors — the measure $\mathrm{Leb}\big|_{[0,1]}$ and the minor field on it — is carried over from Part IV completely untouched, and the minor sub-problem will remain verbatim the graphon game we already understand. Second, the atom $\star$ now has $\rho(\{\star\}) = 1 > 0$: an integral against ρ that runs over $\mathbb{X}$ *does* register $\star$, with weight 1, which is precisely the registration the null singleton $\{y_0\}$ could never receive. The major has become visible not by acquiring mass within the crowd but by being moved onto a measure that can see him. Third, the unit value of the point mass is a normalization, not a physical influence strength: it fixes the reference weight of the atom at 1, and the actual $O(1)$ magnitude with which minor x feels the major will be carried by a kernel block $w_0(x)$ on the augmented space, assembled in the next subsection, exactly as a coupling strength $\mathsf{B}_{k\ell}$ multiplied the mass π_ℓ for a class. Holding the reference mass fixed at 1 and letting the kernel carry the strength keeps the bookkeeping parallel to the block case.

With the index space enlarged, the unknown of the problem changes shape correspondingly. A "population field" was, in Part IV, an assignment of a probability measure to every type; on the augmented space $\mathbb{X} = [0,1] \sqcup \{\star\}$ it becomes an assignment of a measure to every point of $\mathbb{X}$, and since $\mathbb{X}$ is the continuum plus one extra point, such an assignment is exactly a *pair*. It consists of a minor field $\boldsymbol{m} = (m^x)_{x\in[0,1]} \in \mathcal{M}_2$ — the old object, the law of each minor type, living in the field space of definition 21.8 — together with a single measure $m^0 \in \mathcal{P}_2(\mathbb{R}^{d_0})$ attached to the atom, the state law of the one major player. There is no reason for the major's state to live in the minors' state space, and the formalism does not ask it to: the major's state is allowed its own dimension, $m^0 \in \mathcal{P}_2(\mathbb{R}^{d_0})$ with d_0 possibly different from the minors' d. This latitude is not a technicality but a faithfulness to the examples — a central bank's state may be a scalar policy rate ($d_0 = 1$) while each trader carries a multidimensional portfolio ($d > 1$), and nothing should force these to match. Because $\star$ is an isolated atom rather than a member of the continuum, attaching a state space of its own to it costs nothing; the minors' product structure over $[0,1]$ is untouched, and the major simply sits beside it with its own $\mathbb{R}^{d_0}$. The field on $\mathbb{X}$ is thus the ordered pair $(\boldsymbol{m}, m^0)$, the crowd and the king, and it is on this pair that the augmented equilibrium problem will be posed.

It remains to say where the coupling *operator* will live, because that is what the next subsection must construct and it is dictated by the same enlargement. The Part IV operator $\mathbb{W}$ acts on $L^2([0,1], \mathrm{Leb})$; on the augmented space its successor must act on $L^2(\mathbb{X}, \rho)$, square-integrable functions against the augmented reference measure. The single fact that makes the augmented operator tractable — and that we use repeatedly hereafter — is that this Hilbert space splits cleanly into the old one plus a single coordinate,

$$L^2(\mathbb{X}, \rho) \;\cong\; L^2([0,1], \mathrm{Leb}) \;\oplus\; \mathbb{R}, \tag{30.13}$$

the extra summand $\mathbb{R}$ being the value of a function at the atom $\star$. This isomorphism is worth unpacking slowly, because the block structure of the operator in section 30.2.2 is nothing but a reading of it. A function $f \in L^2(\mathbb{X}, \rho)$ is, since $\mathbb{X}$ is the disjoint union $[0,1] \sqcup \{\star\}$, completely specified by two independent pieces of data: its restriction $f|_{[0,1]}$ to the continuum and its single value $f(\star)$ at the atom. The reference measure splits as $\rho = \mathrm{Leb}|_{[0,1]} + \delta_\star$ along the same partition, so the $L^2(\rho)$ norm splits with it,

$$\|f\|^2_{L^2(\mathbb{X},\rho)} \;=\; \int_{[0,1]} |f(x)|^2 \,\mathrm{d}x \;+\; |f(\star)|^2 \,\rho(\{\star\}) \;=\; \left\| f|_{[0,1]} \right\|^2_{L^2([0,1])} \;+\; |f(\star)|^2 \,,$$

using $\rho(\{\star\}) = 1$ for the second term. The cross term is absent — the decomposition is an *orthogonal* direct sum — precisely because the continuum and the atom are ρ-disjoint: no function content is shared between $[0,1]$ and $\{\star\}$, so the two pieces contribute their squared norms additively. This is the content of (30.13): the map $f \mapsto (f|_{[0,1]}, f(\star))$ is a Hilbert-space isomorphism onto $L^2([0,1], \mathrm{Leb}) \oplus \mathbb{R}$.

The phrase "the extra summand $\mathbb{R}$ is the value of a function at the atom" now has a precise and load-bearing meaning. On the nonatomic space $([0,1], \mathrm{Leb})$ the value of an L^2 function at any single point is undefined — functions are equivalence classes modulo null sets, and a point is null, so "$f(y_0)$" carries no information and contributes nothing to the norm. The instant we adjoin the atom, the value $f(\star)$ becomes a genuine, norm-carrying degree of freedom: it is no longer washed out, it occupies its own one-dimensional orthogonal coordinate in the Hilbert space, and that coordinate is exactly the $\mathbb{R}$ in (30.13). This is the Hilbert-space shadow of the same fact the notation warning made about the integral: a point that was invisible to the Lebesgue integral, and equally invisible to the $L^2([0,1])$ norm, is made visible — given its own axis — precisely by being turned into an atom. That the new axis is *one*-dimensional, because there is exactly one major player, is what will make the operator's correction *finite-rank*: an operator on $L^2([0,1]) \oplus \mathbb{R}$ that touches the new $\mathbb{R}$ coordinate can do so only through rank-one couplings into and out of a single direction.

This orthogonal splitting is the platform the rest of the section stands on, and it tells the next subsection exactly what to build. Writing $\mathbb{W}_\star$ in the basis adapted to (30.13) — a continuum part and one extra coordinate — forces it into a 2×2 block form: a minor–minor block, which we will see is the old graphon operator $\mathbb{W}$ acting on the $L^2([0,1])$ summand and nothing else; the off-diagonal minor–major and major–minor blocks, which wire the atom's single coordinate to the continuum and back; and the scalar major–major block on the lone $\mathbb{R}$ coordinate. Because all the new structure is confined to that one extra coordinate, the correction $\mathbb{W}_\star - \mathbb{W}$ will be supported on a finite-rank subspace — the finite-rank atomic perturbation promised in the section title. We have now built the stage: the augmented index space $\mathbb{X}$, the paired field $(\boldsymbol{m}, m^0)$, and the decomposition $L^2(\mathbb{X}, \rho) \cong L^2([0,1]) \oplus \mathbb{R}$ that the operator will act on. The next subsection populates it, defining the augmented kernel block by block and assembling the operator $\mathbb{W}_\star$ whose finite-rank structure is the technical heart of the major–minor reduction.

30.2.2 The augmented kernel and its operator

The previous subsection built the stage and named precisely what stands on it: the augmented index $\mathbb{X} = [0,1] \sqcup \{\star\}$ carrying the reference measure $\rho = \mathrm{Leb}|_{[0,1]} + \delta_\star$, the paired field $(\boldsymbol{m}, m^0)$ of crowd and king, and the orthogonal decomposition $L^2(\mathbb{X}, \rho) \cong L^2([0,1]) \oplus \mathbb{R}$ of (30.13) on which the coupling operator must act. What it deliberately deferred is the operator itself. We now write down the kernel that lives on $\mathbb{X}^2$, read off the operator it induces by integration against ρ, and prove the one structural theorem the whole major–minor reduction rests on: that operator is nothing but the old graphon coupling $\mathbb{W}$ plus a correction of rank at most two. Everything in this subsection is forced by the geometry of $\mathbb{X}$ — a continuum with one extra point — and the reader should watch each object below appear as the inevitable image of that geometry rather than as a fresh modelling decision.

Begin with the kernel. A symmetric kernel on $\mathbb{X}^2$ is, because $\mathbb{X}$ is a disjoint union, a kernel that has been split into four pieces by the same partition $[0,1] \sqcup \{\star\}$ applied to each of its two arguments. The product $\mathbb{X}^2$ decomposes into four rectangles — $[0,1] \times [0,1]$, $[0,1] \times \{\star\}$, $\{\star\} \times [0,1]$, and $\{\star\} \times \{\star\}$ — and a kernel is simply a choice of value on each. Each rectangle has a transparent physical meaning, and naming them is naming the four channels of influence in a major–minor system. The continuum–continuum rectangle $[0,1] \times [0,1]$ carries the ordinary minor-to-minor coupling: this is the old graphon W of Part IV, untouched, governing how one ordinary agent feels another. The mixed rectangle $[0,1] \times \{\star\}$ records how strongly each minor type x feels the single major player; it is a *function of x alone*, the *influence profile* $w_0(x)$, and it is exactly the channel through which the central bank reaches every trader. Its mirror $\{\star\} \times [0,1]$ records the reverse: how strongly the major feels each minor type y, a function $\tilde{w}_0(y)$ of y alone — the channel through which the crowd of traders is felt back at the bank. Finally the lone corner $\{\star\} \times \{\star\}$ is a single number w_{00}, the major's coupling to its own state. That a continuum-indexed kernel collapses, on the mixed rectangles, to mere *functions* and, in the corner, to a single *scalar* is the first concrete payoff of the atom being a single point: a whole row of the kernel indexed by $\star$ is just one function.

Definition 30.9 (Augmented major–minor kernel)**.** An *augmented kernel* on $\mathbb{X}$ is a symmetric measurable function $W_\star : \mathbb{X}^2 \to \mathbb{R}$ specified by four blocks: the minor–minor graphon $W(x,y)$ on $[0,1]^2$; the minor–major influence profile $x \mapsto w_0(x) := W_\star(x, \star)$ (how strongly minor x feels the major); the major–minor profile $y \mapsto \tilde{w}_0(y) := W_\star(\star, y)$ (how strongly the major feels minor y); and the major self-coupling $w_{00} := W_\star(\star, \star) \in \mathbb{R}$. Symmetry of $W_\star$ forces $\tilde{w}_0 = w_0$; we allow them to differ when one drops the self-adjointness requirement (the influence of the major on minors and of minors on the major need not be reciprocal in applications).

A word on the symmetry clause, because it is where the model's asymmetry quietly enters. Were we to insist that $W_\star$ be a genuine symmetric kernel — the augmented analogue of the symmetric graphons of Part IV — the two mixed rectangles would be forced equal, $\tilde{w}_0 = w_0$, since they are reflections of each other across the diagonal. That is the hypothesis under which the augmented operator will be self-adjoint and the spectral theory of Part IV transfers wholesale. But the canonical examples are precisely *not* reciprocal: the central bank reaches every trader with profile w_0, while the traders reach the bank only through some quite different weighting $\tilde{w}_0$ of their aggregate, and there is no reason the two should agree. We therefore state the kernel with w_0 and $\tilde{w}_0$ allowed to differ, keeping the self-adjoint case $w_0 = \tilde{w}_0$ as the special hypothesis it is, to be invoked by name exactly when the spectral machinery is needed (part (i) of theorem 30.10).

With the four blocks in hand, the operator they induce is forced: it is integration of a function against the augmented reference measure ρ, and the only novelty is what ρ's atomic part does to that integral. The associated integral operator on $L^2(\mathbb{X}, \rho) \cong L^2([0,1]) \oplus \mathbb{R}$ acts on a pair (f, ξ) — a function f on the continuum together with the single value $\xi = f(\star)$ at the atom — by integrating against ρ:

$$\mathbb{W}_\star \begin{pmatrix} f \\ \xi \end{pmatrix} = \begin{pmatrix} \mathbb{W}f \ + \ w_0\,\xi \\ \langle \tilde{w}_0, f\rangle_{L^2([0,1])} \ + \ w_{00}\,\xi \end{pmatrix}, \tag{30.14}$$

and the structure of (30.14) is read straight off the splitting $\rho = \mathrm{Leb}\big|_{[0,1]} + \delta_\star$. Evaluating the augmented aggregate at a minor type $x \in [0,1]$ integrates $W_\star(x, \cdot)$ against ρ, and that integral splits into its continuum part and its atomic part: the continuum part is $\int_[0,1] W(x,y) f(y)\,\mathrm{d}y = (\mathbb{W}f)(x)$, the ordinary graphon aggregate, while the atomic part is the single *discrete* term $W_\star(x,\star)\,f(\star)\,\rho(\{\star\}) = w_0(x)\,\xi$, the Dirac mass $\delta_\star$ picking out the value at the atom and weighting it by $\rho(\{\star\}) = 1$. Evaluating instead at the atom $\star$ integrates $W_\star(\star, \cdot) = \tilde{w}_0$ against ρ, whose continuum part is $\int_[0,1] \tilde{w}_0(y) f(y)\,\mathrm{d}y = \langle \tilde{w}_0, f\rangle_{L^2([0,1])}$ and whose atomic part is $w_{00}\,\xi$. The two rows of (30.14) are exactly these two evaluations.

The asymmetry between the two rows is the whole physics of the model, and it is worth reading them side by side. The first row is the field felt by the minors: the usual graphon aggregate $\mathbb{W}f$ — an integral over the continuum, in which any single partner is invisible — *plus* a term $w_0(x)\,\xi$ that is the major's observable ξ broadcast to every minor through the profile w_0. This added term does *not* integrate a continuum; it is the value of a single function at a single point, the atomic evaluation that the Lebesgue integral never sees. That is exactly why it survives the mean-field limit and endows the major with non-vanishing influence: it is a value at a point, not an integral over a continuum, and section 30.2.1 showed that a point value, alone among the contributions, refuses to be averaged away. The second row, by contrast, is the field felt by the major, and there the situation is reversed: the major hears the minors only through the *integral* $\langle \tilde{w}_0, f\rangle$, a genuine average over the continuum in which — precisely as anonymity demands — no individual minor counts for anything. The major broadcasts to the crowd by a point evaluation but listens to it only as a statistic. This one-way visibility, a pointlike outgoing channel against an integral incoming channel, is the mathematical fingerprint of "one named player against a continuum of anonymous ones," and it is already entirely present in the placement of the discrete term in row one and the integral term in row two of (30.14).

We can now make precise the claim that gives the section its title. Stare at (30.14) and ask what part of $\mathbb{W}_\star$ is genuinely new relative to the Part IV coupling. The term $\mathbb{W}f$ in the first row is the old operator acting on the old $L^2([0,1])$ summand and nothing else; everything else — the broadcast $w_0\,\xi$, the listening $\langle \tilde{w}_0, f\rangle$, the self-coupling $w_{00}\,\xi$ — touches the single new coordinate ξ, either reading from it or writing into it. By the heuristic of the previous subsection, an operator that interacts with the world only through one extra coordinate cannot have more than a bounded, low-rank effect, no matter how large the continuum it sits on. The next theorem turns that heuristic into an exact rank count and draws its two consequences: that the augmented operator inherits the entire spectral theory of Part IV (because a finite-rank change preserves Hilbert–Schmidtness), and that the change is nonetheless $O(1)$ in operator norm (because a single broadcast profile, unlike a single minor type, is not Lebesgue-null).

Theorem 30.10 (Major–minor as a finite-rank perturbation of the graphon coupling)**.** *Let $\mathbb{W}_\star$ be the augmented operator (30.14) on $L^2(\mathbb{X},\rho)$ and let $\mathbb{W}_0 := \mathbb{W}\oplus 0$ be the "pure-minor" operator that ignores the atom. Then*

$$\mathbb{W}_\star \;=\; \mathbb{W}_0 \;+\; R, \qquad R\begin{pmatrix} f\\ \xi\end{pmatrix} = \begin{pmatrix} w_0\,\xi \\ \langle \tilde{w}_0, f\rangle + w_{00}\,\xi\end{pmatrix}, \tag{30.15}$$

where R is a bounded operator of rank at most 2 induced by the atom — its action depends on (f,ξ) only through the two atom-coupled scalars ξ and $\langle \tilde{w}_0, f\rangle$, and its range is spanned by the broadcast profile $(w_0, 0)$ and the atom-indicator $(0,1)$. With p major players (atoms $\star_1,\dots,\star_p$, $\rho = \mathrm{Leb} + \sum_j \delta_{\star_j}$) the perturbation R has rank at most $2p$. Moreover:

(i) *If $w_0 = \tilde{w}_0$ then $W_\star$ is symmetric and $\mathbb{W}_\star$ is self-adjoint on $L^2(\mathbb{X},\rho)$; since $\mathbb{W}$ is Hilbert–Schmidt under* (A-Reg-W) *and R is finite-rank — and finite-rank operators are Hilbert–Schmidt while the Hilbert–Schmidt operators form a vector space (section 2.4) — their sum $\mathbb{W}_\star$ is Hilbert–Schmidt, so the spectral theory of chapter 18 applies verbatim on the augmented space.*

(ii) *The perturbation R does* not *vanish in operator norm as a function of the minor population size: $\|R\| \geq \|w_0\|_{L^2}$ whenever $w_0 \neq 0$. This is the operator-level statement that a single major agent retains $O(1)$ influence, in contrast to a minor type, whose removal changes $\mathbb{W}$ by an operator of norm 0 (a Lebesgue-null modification).*

Proof. The decomposition (30.15) is a subtraction. The pure-minor operator acts by $\mathbb{W}_0(f,\xi) = (\mathbb{W}f, 0)$ — it runs the old graphon on the continuum part and annihilates the atom coordinate. Subtracting this from (30.14) term by term,

$$\mathbb{W}_\star\begin{pmatrix} f\\ \xi\end{pmatrix} - \mathbb{W}_0\begin{pmatrix} f\\ \xi\end{pmatrix} = \begin{pmatrix} \mathbb{W}f + w_0\,\xi \\ \langle \tilde{w}_0, f\rangle + w_{00}\,\xi\end{pmatrix} - \begin{pmatrix} \mathbb{W}f \\ 0\end{pmatrix} = \begin{pmatrix} w_0\,\xi \\ \langle \tilde{w}_0, f\rangle + w_{00}\,\xi\end{pmatrix},$$

the $\mathbb{W}f$ in the first row cancelling exactly and leaving precisely the R of (30.15). So $\mathbb{W}_\star = \mathbb{W}_0 + R$ with no remainder; the entire deviation of the augmented operator from the pure-minor one is the atom-induced block R.

Now count its rank. The cleanest way is to exhibit R as a composition: it *reads* its input (f,ξ) through two real-valued linear functionals and *writes* its output into a two-dimensional subspace.

The two functionals are visible in the formula for R — the only features of (f,ξ) it consults are the scalar ξ and the scalar $\langle \tilde{w}_0, f\rangle$ — and the output is assembled from them as

$$R\begin{pmatrix} f \\ \xi \end{pmatrix} = \xi \begin{pmatrix} w_0 \\ 0 \end{pmatrix} + \left(\langle \tilde{w}_0, f\rangle + w_{00}\,\xi\right)\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{30.16}$$

a linear combination of the two fixed vectors $(w_0, 0)$ and $(0,1)$. Hence the range of R lies in $\operatorname{span}\{(w_0,0),(0,1)\}$, a space of dimension at most two, and $\operatorname{rank} R \le 2$. (A linear map factoring through a two-dimensional space has rank at most two; here the factorization $(f,\xi) \mapsto (\xi, \langle \tilde{w}_0, f\rangle) \mapsto R(f,\xi)$ makes it explicit.) The same bookkeeping handles p atoms $\star_1, \ldots, \star_p$. Each atom $\star_j$ contributes its own broadcast vector $(w_{0,j}, 0)$ and its own atom-indicator $(0,\ldots,1,\ldots,0)$ to the range, and R reads the input through the p atom values ξ_j and the p aggregates $\langle \tilde{w}_{0,j}, f\rangle$; with $2p$ output directions and $2p$ input functionals, R has rank at most $2p$. The number of majors is literally the rank budget.

(i) Suppose $w_0 = \tilde{w}_0$. Then the kernel $W_\star$ is symmetric — the two mixed blocks coincide and the continuum block was symmetric to begin with — and integration of a symmetric real kernel against the symmetric measure ρ yields a self-adjoint operator. It is worth verifying this directly on (30.14), since it pins down exactly where the hypothesis is spent. The inner product on $L^2(\mathbb{X},\rho) \cong L^2([0,1]) \oplus \mathbb{R}$ is $\langle (f,\xi),(g,\eta)\rangle_\rho = \langle f,g\rangle_{L^2([0,1])} + \xi\eta$, and computing both sides,

$$\begin{aligned}
\langle \mathbb{W}_\star(f,\xi),(g,\eta)\rangle_\rho &= \langle \mathbb{W}f, g\rangle + \xi\langle w_0, g\rangle + \eta\langle \tilde{w}_0, f\rangle + w_{00}\,\xi\eta,\\
\langle (f,\xi), \mathbb{W}_\star(g,\eta)\rangle_\rho &= \langle f, \mathbb{W}g\rangle + \eta\langle f, w_0\rangle + \xi\langle \tilde{w}_0, g\rangle + w_{00}\,\xi\eta.
\end{aligned}$$

The first terms agree because $\mathbb{W}$ is already self-adjoint under (A-Reg-W) (symmetric kernel), and the last terms are identical scalars. The middle two pairs agree if and only if $\xi\langle w_0, g\rangle = \xi\langle \tilde{w}_0, g\rangle$ and $\eta\langle \tilde{w}_0, f\rangle = \eta\langle w_0, f\rangle$ for all f, g, ξ, η — that is, if and only if $w_0 = \tilde{w}_0$. So the self-adjointness of $\mathbb{W}_\star$ is exactly the reciprocity of the two mixed blocks, no more and no less. Granting it, the Hilbert–Schmidt conclusion follows from two elementary facts recorded in section 2.4: a finite-rank operator is Hilbert–Schmidt, and the Hilbert–Schmidt operators form a vector space (closed under sums). The minor block $\mathbb{W}$ is Hilbert–Schmidt by (A-Reg-W) and R is finite-rank hence Hilbert–Schmidt, so their sum $\mathbb{W}_\star = \mathbb{W}_0 + R$ is Hilbert–Schmidt; being also self-adjoint, it falls under the compact self-adjoint spectral theorem of chapter 18, whose conclusions — a real discrete spectrum accumulating only at 0 and an orthonormal eigenbasis — therefore hold verbatim on the augmented space $L^2(\mathbb{X},\rho)$. The atom buys no analytic difficulty whatsoever.

(ii) For the norm bound, probe R with the unit atom-indicator $(0,1)$, the function that is 0 on the continuum and 1 at the star. Reading off (30.16) with $f = 0$, $\xi = 1$,

$$R\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 \cdot \begin{pmatrix} w_0 \\ 0 \end{pmatrix} + \left(\langle \tilde{w}_0, 0\rangle + w_{00}\right)\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} w_0 \\ w_{00} \end{pmatrix},$$

whose $L^2(\mathbb{X},\rho)$-norm is $\left(\|w_0\|_{L^2}^2 + w_{00}^2\right)^{1/2} \ge \|w_0\|_{L^2}$. Since $\|(0,1)\|_{L^2(\mathbb{X},\rho)} = 1$, the operator norm dominates this ratio: $\|R\| \ge \|R(0,1)\| \ge \|w_0\|_{L^2}$ whenever $w_0 \ne 0$. Contrast this with what a single *minor* type can do. Deleting or altering one minor type — indeed any Lebesgue-null set of types — changes the kernel W only on a null set, hence leaves $\mathbb{W}$ literally unchanged as an operator on $L^2([0,1])$: its operator-norm influence is exactly 0. The same single-agent surgery, performed on the major, moves the operator by at least $\|w_0\|_{L^2} > 0$. This is the operator-theoretic certificate of the whole asymmetry: zero mass, yet $O(1)$ norm. □

Theorem 30.10 is the precise meaning of the chapter's organizing slogan: *the major–minor model is the graphon mean field game whose coupling operator is the minor graphon plus a finite-rank atomic correction.* The decomposition $\mathbb{W}_\star = \mathbb{W} + R$ sorts the model's data into exactly two bins, and the slogan is just a reading of that sort. One bin is the *rank budget*: the number of major players is the rank ceiling of the correction — one major buys rank 2, p majors buy rank $2p$ — so "how many distinguished agents" is, quite literally, "how many dimensions of $\mathbb{W}$ get perturbed." The other bin is the *perturbation's data*: within that budget the correction is determined by the influence profiles w_0 (how the major reaches the minors) and $\tilde{w}_0$ (how the minors reach the major), together with the self-coupling w_{00}. Choosing a major–minor model is therefore choosing a graphon W and then spending a finite rank budget on a handful of profiles; nothing in the construction steps outside the graphon formalism of Part IV. In the taxonomy of the chapter, the *coupling* of the major–minor game is firmly *subsumed* — it is a graphon game on the augmented space $(\mathbb{X}, \rho)$, with all of Part IV's spectral and well-posedness apparatus intact by part (i) of the theorem.

But "the coupling is subsumed" is a verdict about $\mathbb{W}_\star$ alone, and $\mathbb{W}_\star$ alone is silent about the question that actually decides how hard the model is. The finite-rank theorem tells us the major perturbs the coupling *operator* by an $O(1)$, rank-2 amount; it says nothing about what the major does to the *equilibrium.* Here lies the honest gap the next subsection must close: read as a static linear map, the augmented operator $\mathbb{W}_\star$ gives no hint that the determinism which kept the multi-population case inside the tractable theory of chapter 24 now fails — yet fail it does, the moment one asks not how the agents are wired but what a *single* non-vanishing player's own, unaveraged noise does to the field they feel. The next subsection takes up exactly this — what the atom does to the equilibrium rather than to the coupling — and shows that the randomness it injects is a finite-dimensional common noise, escalating the consistency condition from the deterministic fixed point of chapter 22 to the conditional, common-noise master equation of chapter 26.

30.2.3 The randomness the atom injects: a finite-dimensional common noise

The finite-rank theorem of the previous subsection settled the *coupling* and, by design, said nothing about the *equilibrium*; it is the second that now governs how hard the model is to solve. Theorem 30.10 certifies that the major perturbs the coupling operator by an $O(1)$, rank-at-most-two atomic correction, $\mathbb{W}_\star = \mathbb{W} + R$. Read as a static linear map that turns one field into another, $\mathbb{W}_\star$ is as innocuous as any bounded operator on $L^2(\mathbb{X}, \rho)$: it has the same spectral theory, the same norm bound, the same place in Part IV's well-posedness apparatus. What it conceals is that the field it acts on has changed character. In every block-graphon game of section 30.1 the input to the coupling was a *deterministic* flow of measures, and that determinism was not an accident but a consequence of positive mass: a class of agents of positive Lebesgue weight averages its members' idiosyncratic Brownian noises away by the law of large numbers, so its empirical distribution converges to a fixed curve $t \mapsto m_t^{(k)}$ that is the same in every realization of the noise. The major has no mass to average over. It is a *single* non-vanishing player carrying its own driving Brownian motion B^0, and a single Brownian path does not self-average; it stays random for all time. Through precisely the broadcast term $w_0(x)\,h(X_t^0)$ whose $O(1)$ survival the rank-two theorem certified, that irreducible randomness is injected directly into the aggregate every minor feels.

The consequence is structural and unavoidable. The major's state X_t^0 enters every minor's dynamics and cost, so the entire minor population evolves as a system driven, in common, by the major's path. Its "field of laws" is therefore no longer a deterministic flow but a measure-valued *process*, adapted to the major's filtration:

$$\mu_t^x \;=\; \mathcal{L}(X_t^x \,|\, \mathcal{F}_t^0), \qquad \mathcal{F}_t^0 = \sigma\big(X_0^0,\, (B_s^0)_{s\le t}\big), \tag{30.17}$$

the *conditional* law of a minor of type x given everything the major has done up to time t. The conditioning is the whole point. Averaged over the major's randomness as well, a minor's marginal law $\mathcal{L}(X_t^x)$ would again be a deterministic curve and we would have learned nothing; but the object the *other minors actually couple to* is not that unconditional marginal — it is (30.17), the law that still fluctuates because it has not integrated out $\mathcal{F}_t^0$. Each minor responds to a crowd whose statistics are themselves random, correlated across the entire population through their shared dependence on the one path X^0.

This is exactly the conditional McKean–Vlasov structure of mean field games with *common noise* developed in chapter 26: a population whose flow of conditional laws is driven by a source of randomness felt by everyone at once, with the consistency condition demoted from an equality of deterministic curves to an equality of *processes*. What distinguishes the present instance from the generic common-noise model is the nature of that shared source. In the textbook construction the common noise is an exogenous Brownian field posited from outside and coupled to no one's objective. Here it is neither exogenous nor infinite-dimensional: the common noise is the major's own d_0-dimensional driving motion and its state X_t^0, a *finite-dimensional* signal, and it is *endogenous* — generated by an agent who is himself optimizing. That second feature is what lifts the problem above a plain common-noise MFG and gives it a Stackelberg flavor, a point we return to in remark 30.12. For now we record the model and its equilibrium notion precisely, and then state honestly what the earlier chapters do and do not deliver for it.

Definition 30.11 (Major–minor graphon MFG)**.** Fix the augmented kernel of definition 30.9. The *major* has state $X_t^0 \in \mathbb{R}^{d_0}$, control α_t^0, and dynamics

$$\mathrm{d}X_t^0 \;=\; b_0\big(X_t^0, \alpha_t^0,\, \langle\bar\mu_t\rangle\big)\,\mathrm{d}t \;+\; \sigma_0\,\mathrm{d}B_t^0, \qquad \langle\bar\mu_t\rangle := \int_0^1 \tilde w_0(y)\,\langle h, \mu_t^y\rangle\,\mathrm{d}y, \tag{30.18}$$

where $\langle\bar\mu_t\rangle$ is the minor aggregate the major feels (the second row of (30.14) evaluated on the minor field). Each *minor* of type x has state $X_t^x \in \mathbb{R}^d$, control α_t^x, and dynamics

$$\mathrm{d}X_t^x \;=\; b\big(x, X_t^x, \alpha_t^x,\, \Theta_t^\star(x)\big)\,\mathrm{d}t \;+\; \sigma(x, X_t^x)\,\mathrm{d}B_t^x, \qquad \Theta_t^\star(x) := (\mathbb{W}[\langle h, \mu_t^\cdot\rangle])(x) + w_0(x)\,h(X_t^0), \tag{30.19}$$

the aggregate $\Theta_t^\star(x)$ being the first row of (30.14): the graphon aggregate of the minors plus the broadcast major observable $w_0(x)h(X_t^0)$. Major and minors incur costs J^0 and J^x of the form (21.6) with these aggregates. A *major–minor graphon MFG equilibrium* is a triple $(\alpha^{0,*}, (\alpha^{x,*})_x, (\mu^x)_x)$ such that:

(a) given the conditional minor field $(\mu_t^x)_{t,x}$ and the minor controls, $\alpha^{0,*}$ is optimal for the major;

(b) given the major's state process and control, for a.e. type x the feedback $\alpha^{x,*}$ is optimal for a minor of type x against the aggregate $\Theta_t^\star(x)$;

(c) consistency: $\mu_t^x = \mathcal{L}(X_t^{x,*} \mid \mathcal{F}_t^0)$ for a.e. x and all t, where $X^{x,*}$ is the minor state under $\alpha^{x,*}$.

The triple is a fixed point, but a layered one, and it repays reading slowly because the layering is where the escalation over chapter 22 lives. Condition (b) is the familiar minor problem with one change: the target $\Theta_t^\star(x)$ it best-responds to is now a *random* aggregate. It carries the deterministic-looking graphon part $(\mathbb{W}[\langle h, \mu_t^\cdot\rangle])(x)$ — which is itself random, since the conditional field μ is — plus the explicit broadcast $w_0(x)\, h(X_t^0)$ of the major's state. Each minor therefore solves a control problem in the random environment generated by the major, exactly as an agent in a common-noise mean field game optimizes against a randomly evolving field rather than a fixed one. Condition (a) is the major's problem, and it is different in kind. The major optimizes not against a scalar but against the *entire* conditional minor field $(\mu_t^x)_x$, an object living in the infinite-dimensional field space $\mathcal{M}_2$, and it does so as a player whose own choice α^0 moves X^0 and hence moves the very broadcast that every minor sees. This is the precise sense in which the major is not infinitesimal: its control reaches into the field, so it must optimize while accounting for its own influence on the crowd it is responding to. Condition (c) closes the loop, and it is the one whose character has changed. The consistency demanded is not that a deterministic guessed curve equal a deterministic produced curve, as in the block-graphon fixed point, but that the guessed measure-valued *process* μ_t^x equal the conditional law $\mathcal{L}(X_t^{x,*} \mid \mathcal{F}_t^0)$ of the optimally controlled minor — an equality of processes, holding realization by realization of the major's noise. That conditional fixed point is the exact rung above the block-graphon equilibrium on the ladder this chapter has been climbing: same coupling, a strictly harder closure.

Remark 30.12 (What is inherited and what is new)**.** It pays to separate cleanly what the finite-rank embedding buys and what it leaves open, because the two answers point to different chapters. *Inherited:* the *coupling* of definition 30.11 is, by theorem 30.10, verbatim that of a graphon game on the augmented space $(\mathbb{X}, \rho)$ — the operator $\mathbb{W}_\star$ is bounded, its spectral and norm theory is the Part IV theory unchanged, and nothing about how the agents are wired together steps outside the formalism we already own. *New:* the *equilibrium* lives one level up. Because the minor field (30.17) is random, the consistency condition (c) is a *conditional* fixed point, and the object that characterizes it is no longer the deterministic backward–forward system of chapter 22 but the *graphon master equation with common noise* of chapter 26: the major's value function $u^0(t, x_0, \boldsymbol{\mu})$ now depends on its own state *and* on the entire minor field $\boldsymbol{\mu} \in \mathcal{M}_2$, and the minors' value field is itself a random field driven by the major's noise.

Two caveats keep this honest, and each is a genuine departure from the deterministic GMFG of chapter 24 rather than a technicality. The first is that the major is *not* infinitesimal. Every other agent in this book has been one of a continuum, with vanishing individual influence, so that no single optimizer's choice perturbs the field; the major violates exactly this, by construction and irreducibly — that is what the $O(1)$ operator norm of theorem 30.10 measured. The equilibrium is therefore not a symmetric fixed point among peers but a *Stackelberg-flavored* game between one finite player and a continuum of infinitesimal ones: the major moves a field it knows it is moving, the minors best-respond to the field the major creates, and the relevant fixed point couples the major's HJB equation on the product space $\mathbb{R}^{d_0} \times \mathcal{M}_2$ — finite state crossed with the field — to the minors' conditional McKean–Vlasov system. There is no analogue of this coupling in chapter 22, where every player was a measure-zero participant whose influence on the field was exactly nil. The second caveat is about well-posedness. That this coupled object *has* a solution, and a unique one, is known under structural hypotheses — the linear–quadratic case of example 30.13, or smallness and monotonicity conditions in the spirit of chapter 24 — but it is *not* a corollary of the deterministic GMFG existence theory, whose compactness and contraction arguments were run on flows of deterministic measures and simply do not see the conditioning. The well-posedness draws instead on the common-noise master-equation theory of chapter 26 and on the dedicated major–minor analyses surveyed in the notes. We therefore claim, and have proved, only the *structural* half here — the finite-rank embedding, which holds unconditionally — and defer the analytic half to chapter 26 and the literature, logging it as such in the chapter ledger.

Physics note: The major agent as an annealed impurity field

There is a physical picture that makes the whole escalation feel inevitable, and it is worth a paragraph because it sharpens precisely the distinction between the major and the block populations that came before it. Think of a mean-field ferromagnet: N bulk spins, each coupled to the average magnetization, settling into a self-consistent field. Now introduce a single *impurity* spin coupled to all N bulk spins with $O(1)$ strength. Its total coupling into the bulk is $O(N)$ — comparable to the self-consistent mean field itself — even though the impurity is one site out of N and occupies zero fraction of the lattice in the thermodynamic limit. This is the magnet's version of zero mass with $O(1)$ influence, exactly the budget the rank-two theorem (theorem 30.10) priced. The contrast to keep in mind is with a macroscopic *sublattice* — a finite fraction of the spins, the analogue of a block population in section 30.1 — which also shifts the self-consistent field but does so as a *bulk*: it averages its own thermal noise away and contributes a deterministic field. The major is the impurity, never the sublattice: zero bulk fraction, $O(1)$ field. That single distinction is the whole reason the multi-population case stayed deterministic while the major–minor case does not.

The twist absent from the equilibrium magnet is dynamics. The impurity's state X_t^0 is not frozen; it evolves, randomly, driven by its own noise. A static impurity would imprint a *quenched* external field — one fixed configuration, averaged over at the end. A dynamical one imprints an *annealed* field: it fluctuates in time and the bulk responds to the fluctuations as they happen, in common across all sites. That annealed external field is precisely the finite-dimensional common noise of (30.17), and the minors' response to it is a *conditional* magnetization — the magnetization given the impurity's history — which is exactly why the governing object is the common-noise master equation and not the deterministic one.

As always the picture motivates and must not be made to prove. The discount β is *not* an inverse temperature; the analogy carries no thermodynamic β, and nothing here is a Gibbs equilibrium. The impurity "field" is not fixed by minimizing a free energy but by *backward-in-time* optimization, the major choosing its path to optimize a cost that looks ahead to a terminal time. And the rank-$2p$ content of theorem 30.10 stands on its own, with or without any of this imagery.

Example 30.13 (Linear–quadratic major–minor, the solvable case)**.** The linear–quadratic specialization turns every abstraction of remark 30.12 into something one can write down, and it is the cleanest place to watch the conditional structure made benign by Gaussianity. Let the major obey the linear dynamics $\mathrm{d}X_t^0 = (a_0 X_t^0 + b_0 \alpha_t^0)\,\mathrm{d}t + \sigma_0\,\mathrm{d}B_t^0$ and each minor obey the LQG agent (21.2), with the minor's running cost pulling its state X_t^x toward the target $s\,\Theta_t^\star(x) = s[(\mathbb{W}\bar{\mu}_t)(x) + w_0(x)X_t^0]$ — the graphon aggregate of the other minors plus the broadcast of the major's state — and the major's cost pulling X_t^0 toward the minor aggregate $\langle \bar{\mu}_t \rangle$ it feels.

The decisive simplification is that the only randomness entering the minor field is the major's state X_t^0, which is itself Gaussian because the major's dynamics are linear. Conditioning on $\mathcal{F}_t^0$ thus means conditioning on a Gaussian path, and a linear system driven by a Gaussian signal stays Gaussian: the conditional minor law μ_t^x of (30.17) is Gaussian for every x and t, with a *deterministic* covariance and a *random mean* that is an affine functional of the major's path. All the conditional fluctuation has been pushed into a single finite-dimensional statistic, the random mean, and the measure-valued process collapses to a vector-valued one. This is exactly the conditional-Gaussian closure that makes common-noise LQG problems tractable, here inherited intact by the major–minor game.

We do not derive the equilibrium — the algebra is in the cited literature — but its *structure* is a structural fact worth stating, because it shows the conditional fixed point of definition 30.11 resolving into a finite, layered Riccati system rather than anything genuinely new. It has three tiers. First, a *control* Riccati per minor type x, a measurable family over the continuum $[0, 1]$ of the type-indexed form (30.20), which collapses to the K equations (30.9) only under a block specialization; this tier is unchanged from the ordinary graphon LQG game, because a minor's feedback *gain* does not care that its target happens to be random. Second, one Riccati for the major's own control gain. Third — and this is where the conditioning surfaces — a *forward*, linear stochastic ODE for the conditional minor means, driven by X_t^0, coupled to the major's own linear–quadratic problem, in which the major now sees the minor aggregate-mean not as an exogenous target but as an *endogenous state* that its own control steers. Optimizing the major against a state it moves enlarges its Riccati system: the relevant matrix equation lives on the joint (major state, aggregate-mean) vector rather than on the major state alone. The conditional, infinite-dimensional fixed point of definition 30.11 thus reduces, in the LQG case, to a coupled but finite system of Riccati equations plus one stochastic ODE for the random means — explicit, and solvable in the same sense the rest of Thread A is.

This is the content of the major–minor LQG theory of Huang [81] and Nourian–Caines [105]; the probabilistic (FBSDE) treatment, valid beyond LQG, is Carmona–Zhu [44] and Carmona–Wang [43], and the dominating-player PDE formulation is Bensoussan–Chau–Yam [20]. These are further reading for the enlarged-Riccati algebra we have deliberately not reproduced; they are not substitutes for the two claims the section establishes on its own — that the coupling is a rank-2 atomic perturbation (theorem 30.10), and that the equilibrium is a conditional, common-noise one (remark 30.12).

With the major–minor model dissected we have reached the chapter's sharpest instance of its recurring tension. The coupling was *subsumed* — a finite-rank atomic choice of the same operator $\mathbb{W}$ — but the equilibrium *escalated*, because a single non-vanishing player's noise refused to average and turned the minor field into a conditional, common-noise object. The dichotomy that organized that escalation was *deterministic field versus random field*: as long as the aggregate each agent feels is a fixed curve we remain in the tractable GMFG theory of chapter 24, and the moment it becomes a process we are pushed up to the conditional master-equation theory of chapter 26. That same dichotomy is about to reappear on two further axes, which the next section takes up in turn: the heterogeneity already latent in type-dependent coefficients b, σ, L, g, and then partial information — the first axis still living inside the operator $\mathbb{W}$, the second, for the first time in this chapter, outside it.

30.3 Heterogeneous costs and dynamics, and partial information

The major–minor model forced the equilibrium up a level: subsuming the coupling into a finite-rank atomic operator cost nothing, but a single non-vanishing player's noise refused to average and dragged the minor field from a deterministic curve into a conditional, common-noise process. We now step down from that escalation to a far milder species of heterogeneity — so mild that, at the level of the model, it is no extension at all — precisely so that the contrast sharpens what made the major–minor case hard. The point of this section is not a new construction but a discrimination: among the many ways agents can differ, we sort out which differences stay confined to a single agent's private optimization and which propagate through the equilibrium to bind the whole population together. The same axis then reappears in a genuinely new guise — partial information — which, unlike everything in this chapter so far, lives *outside* the operator $\mathbb{W}$ rather than inside it.

Begin with the observation that the graphon setup of definitions 21.1 and 21.6 already wrote the drift b, the diffusion σ, the running cost L, and the terminal cost g as functions of the type x. Letting these coefficients vary arbitrarily with x — a fast agent here, an expensive-to-steer agent there, a population whose terminal target differs from its neighbour's — therefore adds not a single symbol to the formalism of Part IV. "Heterogeneity in costs and dynamics across types" is, at the level of the model, already present; nothing needs to be defined, and there is no theorem to prove that the extended model is well posed, because it is the same model. This is worth saying plainly because the mean field games literature often advertises "heterogeneous agents" as an extension, whereas in the graphon formulation it is the default and homogeneity is the special case (constant coefficients). The genuine question is not *whether* type-dependence is allowed but *what it does* once the equilibrium loop closes.

That question has a clean answer because the equilibrium couples agents through exactly one channel, and type-dependence enters two channels with opposite consequences. Recall the two half-problems an equilibrium must reconcile. Each infinitesimal agent solves a control problem — an HJB equation — against an aggregate it treats as a fixed exogenous target (definition 21.15); and the realized population law feeds back into that target through the operator $\mathbb{W}$. Heterogeneity in an agent's *own* data b, σ, L, g enters only the first channel: it reshapes how type x responds to its target, but type x's HJB equation never sees type x''s coefficients. Such heterogeneity is *type-local*; it decouples. Heterogeneity that lives in the second channel — in the coupling weight with which a type reads the aggregate, or in the kernel W itself — is what binds types together, but even this binding is funnelled through the single readout $\mathbb{W}\bar{m}$. The upshot, which the next subsection makes precise for the LQG model, is that the infinite-dimensional, globally coupled part of the problem is a *single* linear object in which $\mathbb{W}$ appears once, while every nonlinearity is solved type by type in mutual ignorance. "Propagation through the equilibrium" thus has a precise meaning: it is the question of how much of the problem survives the projection onto each type's private data, and the answer is that almost all of it does — only the field itself is shared.

This is the deterministic-field side of the chapter's recurring tension, now read on the heterogeneity axis: as long as the shared aggregate $\mathbb{W}\bar{m}_t$ is a fixed flow, type-dependence buys global coupling but no loss of tractability, because the coupling is a single linear term and the rest is type-local. The second half of the section breaks that comfortable picture along an orthogonal direction. Partial information is the genuinely new ingredient, and it is new precisely because it is *not* a choice of $\mathbb{W}$: an agent that observes only a noisy signal of its own state and must filter is changing what it knows, not whom it is connected to. The network operator is untouched; the informational structure is bolted on alongside it. And there the deterministic-versus-conditional dichotomy that organized the major–minor escalation returns in informational dress — the filtered state estimate plays the role the conditional law played there — with a separation principle deciding when the filtering layer and the control layer come cleanly apart. We take the two axes in turn: first the decoupling of type-dependence through the equilibrium (section 30.3.1), then partial information and filtering (section 30.3.2).

30.3.1 Heterogeneity that decouples versus heterogeneity that couples

Proposition 30.14 (Decoupling of own-dynamics heterogeneity in LQG)**.** *Consider the graphon-LQG game* (21.2)–(21.8) *with fully type-dependent parameters* $a(x), b(x), \sigma(x), q(x), q_T(x), s(x)$*, all Borel and bounded, under* (A-Nondeg) *(*$\inf_x |\sigma(x)| > 0$*) and* (A-Reg-W)*. Then the equilibrium splits into:*

(i) *a family of* mutually decoupled *scalar Riccati equations, one per type,*

$$\dot{P}_t(x) = -2a(x)P_t(x) + b(x)^2 P_t(x)^2 - q(x), \qquad P_T(x) = q_T(x), \tag{30.20}$$

each depending on x only through that type's own parameters; and

(ii) *a single linear* forward–backward *system* coupling all types*, for the first-moment field $x \mapsto \bar{m}_t(x)$ and its conjugate offset field $x \mapsto r_t(x)$,*

$$\dot{\bar{m}}_t(x) = \big(a(x) - b(x)^2 P_t(x)\big)\bar{m}_t(x) \;-\; b(x)^2\, r_t(x), \qquad \bar{m}_0(x) = \bar{m}_0^x \text{ given}, \tag{30.21a}$$

$$\dot{r}_t(x) = \big(b(x)^2 P_t(x) - a(x)\big) r_t(x) \;+\; q(x)\, s(x)\, (\mathbb{W}\bar{m}_t)(x), \quad r_T(x) = -\, q_T(x)\, s(x)\, (\mathbb{W}\bar{m}_T)(x), \tag{30.21b}$$

exactly the type-indexed family of (21.21)–(21.22).

All inter-type coupling resides in the single term $(\mathbb{W}\bar{m}_t)(x)$*, which enters only the backward offset equation* (30.21b)*; heterogeneity in* (a, b, σ, q, q_T) *never couples types, and heterogeneity in* $s(x)$ *only rescales each type's coupling to the (single) graphon aggregate. As in worked computation 30.7 the mean obeys no closed forward ODE: its forcing is the offset* $r_t(x)$*, and the system* (30.21) *is a two-point boundary-value problem on* $L^2([0,1])$ *with split data (initial* $\bar{m}_0$*, terminal* r_T*).*

Proof sketch. For each fixed x the agent solves an ordinary scalar LQG control problem against the exogenous target $s(x)z_t(x)$, $z_t = \mathbb{W}\bar{m}_t$ (the aggregate is exogenous to the infinitesimal agent, definition 21.15). The verification theorem of chapter 6 gives a quadratic value $u^x = \frac{1}{2}P_t(x)x^2 + r_t(x)x + c_t(x)$ whose curvature $P_t(x)$ solves (30.20) — which involves only the type-x parameters, since the affine target perturbs only the lower-order coefficients r, c. The closed-loop mean and offset then obey (21.21)–(21.22) in the form (30.21), with the target $z_t(x) = (\mathbb{W}\bar{m}_t)(x)$ substituted into the backward offset equation; this $\mathbb{W}\bar{m}_t$ is the only place a different type's data enters. Well-posedness of each Riccati under $q(x), q_T(x) \geq 0$ is theorem 14.2; measurability of $x \mapsto P_t(x)$ follows from measurability of the parameters and continuous dependence of the Riccati solution on its coefficients. The forward–backward system (30.21) is a linear two-point boundary-value problem on $L^2([0,1])$ whose only off-diagonal (type-coupling) operator is $\bar{m} \mapsto q(\cdot)s(\cdot)\,\mathbb{W}\bar{m}$ in the backward half; its unique solvability for $|s|_\infty\,\|\mathbb{W}\|$ below the threshold is the heterogeneous, $L^2([0,1])$-valued version of the scalar self-consistency (21.24), and reduces to it when the parameters are constant. □

Proposition 30.14 is the reason graphon LQG is computationally mild despite its infinite-dimensionality: the genuinely infinite-dimensional part is a *single* linear forward–backward system (30.21) in which the operator $\mathbb{W}$ appears once, while all the nonlinearity (the Riccati) is type-local. The block specialization recovers worked computation 30.7; the constant specialization recovers Thread A.

30.3.2 Partial information and filtering

A separate axis of heterogeneity, orthogonal to the network, is *informational*: agents may not observe their own state exactly. Each type-x agent observes a noisy signal

$$\mathrm{d}Y_t^x \;=\; c(x)\,X_t^x\,\mathrm{d}t \;+\; \mathrm{d}V_t^x, \qquad Y_0^x = 0, \tag{30.22}$$

with V^x an observation Brownian motion independent of the state noise B^x, and must choose its control *adapted to its own observation filtration* $\mathcal{F}_t^{Y,x} = \sigma((Y_s^x)_{s\leq t})$ rather than to the state. This is the partially observed mean field game; in the homogeneous case it is the nonuniform-agent LQG theory of Huang–Caines–Malhamé [83], and with a major player the partially observed major–minor theory of Sen–Caines [121] and Caines–Kizilkale [35].

Because the partially observed LQG game leans on linear filtering — a result beyond the working prerequisite of Itô calculus — we first establish, in-book, the scalar Kalman–Bucy filter we will invoke. It rests only on tools already in hand: conditional expectation as L^2 orthogonal projection (definition 3.30), the fact that for a jointly Gaussian system an uncorrelated residual is *independent* of the conditioning σ-algebra, the Itô isometry and Itô's formula (theorems 5.7 and 5.9), and the affine-image-of-Gaussians argument used for the LQG flow in theorem 14.4.

Lemma 30.15 (Scalar Kalman–Bucy filter)**.** *Let the scalar signal–observation pair*

$$\mathrm{d}X_t = (\alpha X_t + \beta u_t)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t, \qquad \mathrm{d}Y_t = c\,X_t\,\mathrm{d}t + \mathrm{d}V_t, \quad Y_0 = 0, \tag{30.23}$$

be driven by independent standard Brownian motions B, V, independent of the Gaussian initial state $X_0 \sim \mathcal{N}(\bar{m}_0, \Sigma_0)$, with α, β, σ, c real constants and the control u bounded and adapted to the *observation filtration $\mathcal{F}_t^Y = \sigma(Y_s : s \le t)$. Then the conditional law of X_t given $\mathcal{F}_t^Y$ is Gaussian, $\mathcal{N}(\hat{X}_t, \Sigma_t)$, with:*

(a) *conditional variance $\Sigma_t = \mathbb{E}\Big[(X_t - \hat{X}_t)^2\Big]$ — the filter (conditional estimation-error) variance, distinct from the value-function curvature P —* deterministic *(it does not depend on the observation path nor on the control), solving the forward variance Riccati equation*

$$\dot{\Sigma}_t = 2\alpha\Sigma_t - c^2\Sigma_t^2 + \sigma^2, \qquad \Sigma_0 \text{ given}; \tag{30.24}$$

(b) *conditional mean $\hat{X}_t = \mathbb{E}\Big[X_t \mid \mathcal{F}_t^Y\Big]$ solving the innovation-driven SDE*

$$\mathrm{d}\hat{X}_t = (\alpha\hat{X}_t + \beta u_t)\,\mathrm{d}t + \Sigma_t\,c\,\mathrm{d}I_t, \qquad I_t := Y_t - \int_0^t c\,\hat{X}_s\,\mathrm{d}s, \tag{30.25}$$

where the innovation I is a standard $\mathcal{F}^Y$-Brownian motion.

Proof. Step 1 (the estimation error is control-free). The control u is $\mathcal{F}_t^Y$-measurable, hence exactly reproduced by any $\mathcal{F}_t^Y$-measurable estimator; writing $e_t := X_t - \hat{X}_t$, the affine drift term βu_t appears identically in the dynamics of X_t and of its conditional mean $\hat{X}_t$ and so cancels in e_t. We may therefore compute e_t and $\Sigma_t = \mathbb{E}[e_t^2]$ in the uncontrolled system $\beta \equiv 0$, and add the known term βu back to the mean at the end.

Step 2 (conditional Gaussianity, deterministic variance). With $\beta \equiv 0$, both X_t and Y_t are affine functionals of the jointly Gaussian triple $(X_0, B_{[0,t]}, V_{[0,t]})$ — this is the affine-image-of-Gaussians argument of theorem 14.4 applied to the two-dimensional linear system (30.23) — so the whole process (X, Y) is Gaussian. The conditional mean $\hat{X}_t = \mathbb{E}\Big[X_t \mid \mathcal{F}_t^Y\Big]$ is the L^2 orthogonal projection of X_t onto the closed subspace $L^2(\Omega, \mathcal{F}_t^Y, \mathbb{P})$ (definition 3.30); for a Gaussian family that subspace is generated by the Gaussian observations, and the residual $e_t = X_t - \hat{X}_t$ is orthogonal to it, hence *uncorrelated* with every $\mathcal{F}_t^Y$-measurable variable. In a jointly Gaussian system uncorrelated implies independent, so e_t is independent of $\mathcal{F}_t^Y$. Therefore the conditional law of $X_t = \hat{X}_t + e_t$ given $\mathcal{F}_t^Y$ is that of a constant plus an independent centered Gaussian: $\mathcal{N}(\hat{X}_t, \Sigma_t)$ with $\Sigma_t = \mathbb{E}[e_t^2]$ a deterministic number.

Step 3 (the filter SDE and the gain). Since X is Gaussian and $\hat{X}_t$ is its linear projection onto the observation history, $\hat{X}$ is itself a Gaussian process adapted to $\mathcal{F}^Y$ and, by the linearity of (30.23), admits a representation as a linear SDE driven by the observation; we write it in innovations form with an undetermined gain k_t,

$$\mathrm{d}\hat{X}_t = \alpha\hat{X}_t\,\mathrm{d}t + k_t\,\mathrm{d}I_t, \qquad I_t = Y_t - \textstyle\int_0^t c\hat{X}_s\,\mathrm{d}s,$$

(the attribution of this representation to the linear-filtering literature is [21, Ch. 3]; what we must determine is k_t and verify (30.24)). Subtracting from the signal dynamics and using $\mathrm{d}I_t = c\,e_t\,\mathrm{d}t + \mathrm{d}V_t$,

$$\mathrm{d}e_t = \mathrm{d}X_t - \mathrm{d}\hat{X}_t = (\alpha - k_t c)\,e_t\,\mathrm{d}t + \sigma\,\mathrm{d}B_t - k_t\,\mathrm{d}V_t.$$

By Itô's formula (theorem 5.9) and the Itô isometry (theorem 5.7), $\Sigma_t = \mathbb{E}[e_t^2]$ obeys $\dot{\Sigma}_t = 2(\alpha - k_t c)\Sigma_t + \sigma^2 + k_t^2$. The projection property $e_t \perp \mathcal{F}_t^Y$ characterises the gain: k_t must keep the error orthogonal to the innovation it adds, which is the value minimising Σ_t, i.e. $\partial_{k_t}\big(2(\alpha - k_t c)\Sigma_t + \sigma^2 + k_t^2\big) = -2c\Sigma_t + 2k_t = 0$, giving $k_t = c\Sigma_t$. Substituting $k_t = c\Sigma_t$,

$$\dot{\Sigma}_t = 2\alpha\Sigma_t - 2c^2\Sigma_t^2 + \sigma^2 + c^2\Sigma_t^2 = 2\alpha\Sigma_t - c^2\Sigma_t^2 + \sigma^2,$$

which is (30.24); and $\mathrm{d}\hat{X}_t = \alpha\hat{X}_t\,\mathrm{d}t + c\Sigma_t\,\mathrm{d}I_t$. Restoring the control term from Step 1 yields (30.25).

Step 4 (the innovation is a Brownian motion). The innovation $I_t = \int_0^t c\,e_s\,\mathrm{d}s + V_t$ is a continuous centered Gaussian process (a Gaussian integral plus a Brownian motion). Its bounded-variation part $\int_0^t ce_s\,\mathrm{d}s$ contributes nothing to the quadratic variation, so $[I]_t = [V]_t = t$. Moreover, for $s < t$ the increment $I_t - I_s$ is uncorrelated with every $\mathcal{F}_s^Y$-measurable variable: $V_t - V_s$ is independent of $\mathcal{F}_s^Y$, while $\int_s^t c\,e_r\,\mathrm{d}r$ has, by the orthogonality of Step 2, zero correlation with $\mathcal{F}_s^Y$. Being jointly Gaussian, $I_t - I_s$ is therefore *independent* of $\mathcal{F}_s^Y$ with variance $t - s$. A continuous Gaussian process started at 0 with independent increments of variance $t - s$ is, by the construction of Brownian motion (chapter 5), a standard $\mathcal{F}^Y$-Brownian motion. □

With the filter in hand, the decisive simplification in the linear–quadratic–Gaussian case is the *separation principle*: optimal estimation and optimal control decouple, and the optimal control is the full-information feedback evaluated at the conditional mean estimate. We state it for the graphon LQG game with a *deterministic* aggregate (no major player / no common noise), where it holds cleanly.

Proposition 30.16 (Separation / certainty equivalence, partially observed graphon LQG)**.** *Consider the heterogeneous graphon-LQG game of proposition 30.14 with each agent observing only* (30.22) *and choosing $\mathcal{F}^{Y,x}$-adapted controls, with Gaussian initial state $X_0^x \sim \mathcal{N}(\bar{m}_0^x, \Sigma_0(x))$ and the minor aggregate $z_t = \mathbb{W}\bar{m}_t$ a deterministic flow. Then for a.e. type x:*

(i) (Filter.) *The conditional law of X_t^x given $\mathcal{F}_t^{Y,x}$ is Gaussian, $\mathcal{N}(\hat{X}_t^x, \Sigma_t(x))$, where the conditional variance $\Sigma_t(x)$ solves the deterministic forward Riccati equation*

$$\dot{\Sigma}_t(x) = 2a(x)\Sigma_t(x) - c(x)^2\Sigma_t(x)^2 + \sigma(x)^2, \qquad \Sigma_0(x) \text{ given}, \tag{30.26}$$

and the conditional mean (the Kalman–Bucy filter) solves the controlled SDE

$$\mathrm{d}\hat{X}_t^x = \big(a(x)\hat{X}_t^x + b(x)\alpha_t^x\big)\,\mathrm{d}t + \Sigma_t(x)c(x)\,\big(\,\mathrm{d}Y_t^x - c(x)\hat{X}_t^x\,\mathrm{d}t\big). \tag{30.27}$$

(ii) (Separation.) *The optimal control is the full-information feedback of proposition 30.14 evaluated at the estimate,*

$$\alpha_t^{x,*} = -\,b(x)\big(P_t(x)\,\hat{X}_t^x + r_t(x)\big), \tag{30.28}$$

with the same *control Riccati $P_t(x)$* (30.20) *and offset $r_t(x)$ as in the fully observed problem.*

(iii) (Consistency on the estimates.) *Taking expectations in* (30.27) *and using $\mathbb{E}\Big[\hat{X}_t^x\Big] = \mathbb{E}[X_t^x] = \bar{m}_t(x)$, the equilibrium first-moment field $\bar{m}_t(x)$ obeys the* same *mean equation* (30.21) *as in the fully observed game. The partially observed graphon LQG equilibrium therefore has the identical mean field to the fully observed one; partial information changes the individual fluctuations (through Σ_t) but not the equilibrium mean.*

Proof sketch. (i) is the scalar Kalman–Bucy filter of lemma 30.15 applied to type x with $(\alpha, \beta, \sigma, c) = (a(x), b(x), \sigma(x), c(x))$: the conditional law stays Gaussian, the variance Riccati (30.26) is (30.24) (it does not see the control, a hallmark of the linear case), and the conditional mean follows the innovation-driven SDE (30.27), which is (30.25). Measurability of $x \mapsto (\Sigma_t(x), \hat{X}^x)$ follows as in proposition 30.14 from measurability of the parameters; the classical attribution is [21, Ch. 3]. (ii) is the LQG separation principle: substituting the filter into the cost and using the orthogonality $\mathbb{E}\Big[(X_t^x - \hat{X}_t^x)\hat{X}_t^x\Big] = 0$ shows the cost splits into a control-dependent term identical to the full-information cost in $\hat{X}$ and a control-independent estimation-error term $\int_0^T q(x)\Sigma_t(x)\,\mathrm{d}t$; minimizing the former gives (30.28) with the unchanged control Riccati. (iii) Because $\mathbb{E}\Big[\hat{X}_t^x\Big] = \mathbb{E}[X_t^x]$ (the filter is unbiased) and the feedback (30.28) is affine, taking expectations reproduces (30.21) verbatim. The estimation step is lemma 30.15; the cost-splitting and certainty-equivalence step is the partially observed nonuniform LQG game of Huang–Caines–Malhamé [83], lifted to the graphon coupling by replacing the scalar global mean with $\mathbb{W}\bar{m}_t$. □

Remark 30.17 (Why separation is fragile under common noise)**.** Proposition 30.16 used decisively that the aggregate $z_t = \mathbb{W}\bar{m}_t$ is *deterministic*, so that an agent need only estimate its own state, not the population. Reinstate a major player (section 30.2) or any common noise (chapter 26) and the aggregate becomes random; an agent must then filter *both* its own state and the common signal driving the field, the conditional variance Riccati (30.26) acquires the aggregate as an input, and certainty equivalence holds only in the enlarged sense of Sen–Caines [121]. The clean separation of estimation from control is a luxury of the deterministic-aggregate (no-common-noise) regime; this is the informational counterpart of the deterministic-versus-conditional dichotomy that made the major–minor equilibrium of remark 30.12 harder than the block one. General (non-Gaussian) partial information requires nonlinear filtering — the Zakai/Kushner–Stratonovich equations — and is outside the scope we develop; we flag it in the landscape below and carry it, as an open direction, to the open-problems chapter (chapter 31).

30.4 A map of the heterogeneity landscape

We close by placing the three models of this chapter, and their neighbors, on a single map, oriented by the 2025 overview of mean field games beyond perfect homogeneity [11]. The organizing question is operational: *which extensions are special structural choices within the graphon mean field game of Part IV, and which require machinery genuinely outside it?* We answer along four axes of heterogeneity.

Axis 1: heterogeneity in type (parameters and coupling). This is the graphon axis itself. Multi-population MFG is the block graphon (theorem 30.4); fully type-dependent dynamics and costs are absorbed without extension (proposition 30.14); the rank-one "Curie–Weiss" kernel is the scalar-order-parameter reduction of worked computation 21.18. *Verdict: subsumed.* The graphon operator $\mathbb{W}$ is exactly the bookkeeping device for type heterogeneity, and the finite (block) and rank-one cases are its most tractable corners.

Axis 2: heterogeneity in influence (major players). A finite number of non-vanishing players is the finite-rank atomic perturbation of theorem 30.10: structurally a graphon game on the augmented space $(\mathbb{X}, \rho)$. *Verdict: subsumed at the level of coupling, but the equilibrium escalates.* The atom makes the minor field random, so the equilibrium is a conditional (common-noise) one whose well-posedness belongs to the graphon master equation of chapter 26, not to the deterministic GMFG theory of chapter 24. The finite-rank embedding is exact and unconditional; the analysis is one tier up.

Axis 3: heterogeneity in information. Partial observation and filtering (section 30.3.2) are *orthogonal* to the network: they act on each agent's own state-estimation problem and, in the LQG / deterministic-aggregate regime, leave the equilibrium mean field untouched by the separation principle (proposition 30.16). *Verdict: composable with the graphon structure, not subsumed by it.* Information heterogeneity multiplies onto the type-coupling structure rather than being encoded in $\mathbb{W}$; under common noise it interacts non-trivially with Axis 2 (remark 30.17).

Axis 4: heterogeneity in network geometry beyond the dense graphon. Real networks are sparse and heavy-tailed, and the dense graphon $W : [0,1]^2 \to [0,1]$ cannot represent them: its average degree is $\Theta(N)$. *Verdict: outside.* Sparse and power-law regimes require L^p-graphons and graphex limits (chapter 27); the entire finite-rank/block toolkit of this chapter presumes the dense, bounded-kernel theory.

Table 30.1 summarizes. The boundary worth internalizing is the diagonal of the table: *static-coupling, deterministic-aggregate, dense-network* heterogeneity is what the graphon operator was built to carry, and the major–minor, multi-population, and heterogeneous-cost models all live there at the level of coupling. The moment any of three thresholds is crossed — the aggregate becomes random (major players, common noise), the coupling runs through controls rather than states (extended MFG), or the network becomes sparse (graphex) — the graphon-coupling bookkeeping is no longer the whole story, and the analysis moves to chapter 26, to the extended-MFG literature, or to chapter 27.

Remark 30.18 (What the review adds)**.** The overview [11] catalogues these extensions and several we have not detailed — risk-sensitive and robust objectives, MFG with absorption and state constraints, finite-state and jump models, and mean field *control* (the cooperative counterpart, where a single planner optimizes the population's aggregate cost). Two of its organizing observations are worth carrying forward. First, that "heterogeneity" is not one phenomenon: the four axes above interact, and the hard open problems (chapter 31) sit at their intersections — partially observed major players on sparse networks being the canonical hard case. Second, that the graphon abstraction is the most economical container for *type* heterogeneity but says nothing on its own about *informational* or *temporal* heterogeneity (time-inconsistency, non-exponential discounting), which compose with it but are not reducible to a choice of kernel. We adopt both observations as the framing for the open-problems chapter.

Extension	**Graphon encoding**	**Status / where treated**
Multi-population / multi-class	block kernel (30.2); interaction matrix $\widehat{\mathsf{B}}$	subsumed (theorem 30.4)
Type-dependent costs/dynamics	x-dependent coefficients	subsumed (proposition 30.14)
Major / minor (finite influential players)	finite-rank atomic perturbation (30.15) on $(\mathbb{X}, \rho)$	coupling subsumed; equilibrium needs common noise (chapter 26)
Partial information / filtering	orthogonal to $\mathbb{W}$; Kalman–Bucy per type	composable (proposition 30.16)
Common (aggregate) noise	— (random field, not a kernel)	outside the deterministic GMFG; chapter 26
Sparse / heavy-tailed networks	dense graphon fails	outside; L^p/graphex, chapter 27
MFG of controls (extended MFG)	coupling through law of controls, not states	outside the state-coupled graphon
Finite-state / jump MFG	continuous-state machinery inapplicable	adjacent; finite-state graphon games [9]
Stackelberg / leader–follower	hierarchical optimization	outside; partial overlap with major–minor
Learning / unknown model	graphon known to the solver, not the agents	chapter 29

Table 30.1: The heterogeneity landscape, oriented by [11]. "Subsumed" means a structural special case of the dynamic graphon MFG of chapter 21; "composable" means it multiplies onto that structure without being encoded in $\mathbb{W}$; "outside" means it needs machinery this book develops elsewhere or flags as open.

Rigor ledger

Proved.

- The block-graphon reduction: with block-constant data the block-constant subspace $\mathcal{M}_2^{\mathsf{B}}$ is closed and best-response-invariant, and its equilibria are in bijection with multi-population MFG equilibria (30.6) (lemma 30.2 and theorem 30.4). Existence on the finite system and uniqueness under monotonicity or $\left\|\widehat{\mathsf{B}}\right\|$-smallness (proposition 30.6).
- Thread B advanced to K populations: collapse to K decoupled Riccati equations (30.9) plus one $2K$-dimensional linear mean–offset two-point boundary-value problem (30.11) (the inter-population coupling living in the backward/offset half), with coupling constant $\left\|\widehat{\mathsf{B}}\right\|$ and consistency with the single-population and rank-one reductions (worked computation 30.7).

- Major–minor as an unconditional finite-rank atomic perturbation of the graphon coupling on the augmented space $(\mathbb{X}, \rho)$, of rank $\leq 2p$ for p majors, with the operator-norm lower bound $\|R\| \geq \|w_0\|_{L^2}$ certifying non-vanishing influence (theorem 30.10).
- Decoupling of own-dynamics heterogeneity in LQG into a type-local Riccati family and a single $L^2([0,1])$-valued mean–offset two-point BVP (proposition 30.14); the in-book scalar Kalman–Bucy filter (lemma 30.15) and the LQG separation principle with unchanged equilibrium mean field under a deterministic aggregate (proposition 30.16).

Assumed.

- Standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) per population/-type, and (A-Reg-W) for the (block or augmented) kernel; (A-Mon) where monotone uniqueness is invoked.
- The per-agent verification theorem of chapter 6 and the single-population LQG solution of chapters 8 and 14 (Riccati well-posedness theorem 14.2), imported rather than reproved.

Deferred.

- Well-posedness of the major–minor equilibrium (conditional/common-noise fixed point): the graphon master equation with common noise, chapter 26; LQG and probabilistic treatments [81, 105, 44, 20].
- The detailed monotone-plus-perturbation uniqueness estimate for the finite system: the finite-rank case of chapter 24.
- Partial information under common noise (joint state/aggregate filtering) and nonlinear filtering for non-Gaussian models: [121, 35] and chapter 31.

Open.

- Whether a symmetric block kernel can support *non*-block-constant (symmetry-broken within a cell) equilibria outside the monotone/small-coupling regime (remark 30.5); the multi-population analogue of the Curie–Weiss multiplicity of section 11.4.
- Sharp characterization of equilibria at the intersection of axes — partially observed major players on sparse (graphex) networks — where none of the reductions of this chapter applies cleanly (remark 30.18, chapter 31).

Bibliographic notes

The multi-population mean field game system (30.6) predates the graphon formulation: the several-population ergodic equations are derived by Feleqi [66], the parabolic/boundary theory by Cirant [49], and the relation to mean field control across populations by Bensoussan–Huang–Laurière [22].

Reading it as the block-graphon slice of the dynamic GMFG of Caines–Huang [33, 34] is the viewpoint of remark 21.20 and chapter 22, which the present chapter completes; the static counterpart is the block static graphon game of section 20.1 and the interaction-matrix spectral threshold of theorem 20.14. The major–minor model originates with the LQG analysis of Huang [81] and the nonlinear ϵ-Nash theory of Nourian–Caines [105]; the probabilistic (FBSDE) approach, which clarifies that the minor field becomes conditional, is Carmona–Zhu [44] and Carmona–Wang [43]; the master/PDE "dominating player" formulation is Bensoussan–Chau–Yam [20]; and the finite-dimensional common-noise reading we adopt is the bridge to chapter 26 and Carmona–Delarue [41]. The major-player-as-low-rank perturbation viewpoint, and the major/minor distinction as small-versus-large in the Lasry–Lions sense, is in the line of Lasry–Lions [94]. Partially observed LQG mean field games are due to Huang–Caines–Malhamé [83], with the major-player and filtering refinements of Sen–Caines [121] and Caines–Kizilkale [35]; the Kalman–Bucy filter and the LQG separation principle are classical, in the textbook form of Bensoussan–Frehse–Yam [21]. The landscape of table 30.1 follows the 2025 overview of extensions beyond perfect homogeneity [11]; the finite-state graphon games it references are Aurell–Carmona–Dayanikli–Laurière [9], the linear-quadratic graphon game is Aurell–Carmona–Laurière [10], and the sparse/graphex frontier is chapter 27.

Chapter 31

Open Problems and a Research Map

For thirty chapters this book has built. Each took an object — a measure, a value function, an operator, a graph limit — carried a definition to a theorem and a theorem to a worked threshold, and handed the next chapter a tool it could stand on. This final chapter does something different in kind, and the shift in stance is worth naming, because it changes what counts as progress. It carries no new theorem. Its work is to draw the map of the territory those thirty chapters charted, to mark the edges where the proved results stop, and to say honestly what lies beyond them. Where the preceding chapters earned ground by *proving*, this one earns it by *locating*: fixing coordinates precise enough that an open problem becomes a definite point with definite neighbours, rather than a vague unease about a hard direction. We import freely and prove almost nothing new — the single exception, a fully rigorous fluctuation computation (worked computation 31.9), is held to the end, where it marks the exact line at which the rigorous theory gives out.

The objects we will draw on are all in place, and it is worth recalling each not as a citation but as a tool with a job, since this is the last page on which they all stand together. The classical mean field game system (chapter 11) is the coupled Hamilton–Jacobi–Bellman/Fokker–Planck pair an equilibrium must solve, and Lasry–Lions monotonicity is the sign condition under which that pair has exactly one solution; the master equation (chapter 12) lifts the entire equilibrium to a single function on the space of measures, the only formulation that survives a randomly moving mean field. The graphon enters as a self-adjoint Hilbert–Schmidt operator $\mathbb{W}$ (chapter 18) — the object whose spectrum will turn out to govern everything that follows — living in the compact space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ under the cut metric (chapter 17), the metric in which both the network limits and the equilibria they carry vary continuously. On that footing rest the dynamic graphon mean field game and its existence and uniqueness theory (chapters 22 and 24), justified from below by convergence from finite networks (chapter 25) and extended above by the graphon master equation and common noise (chapter 26); past them lie the sparse and L^p frontier (chapter 27), where the bounded-operator assumptions begin to fail, the numerics (chapter 28) that turn an equilibrium into a number, the reinforcement learning (chapter 29) that finds it when the model is unknown, and the heterogeneity zoo (chapter 30) cataloguing the ways agents may differ that the graphon does, and does not, capture. Read as a single inventory, these are the pieces of a statistical field theory of strategic agents on a network; the map we are about to draw is what they look like seen at once.

The carried threads run through all of it. The linear–quadratic–Gaussian (LQG) game (Thread A) and its graphon generalisation, the congestion/imitation model on a kernel (Thread B) — of which the rank-one kernel $W(x, y) = f(x)f(y)$ is the cleanest reduction — have each been solved to their exact uniqueness/multiplicity thresholds, in theorem 14.11 and proposition 24.23. We lean on those exact answers throughout, because a chapter about what is *not* known needs, more than anything, a few things known *exactly* to calibrate against: the threshold the general theory cannot yet reach is precisely the one the LQG thread hands us on a plate.

The chapter is organised around four questions a reader who has finished the book might reasonably ask, and its discipline is to keep them genuinely separate — four different kinds of not-knowing — until the very end, where they are seen to answer to a single object. *What do we not yet know how to prove?* (section 31.2: theory gaps.) *What do we not yet know how to compute?* (section 31.3: computation and identification gaps.) *Where will this be used, and what does each use cost in new mathematics?* (section 31.4: application frontiers.) And finally, the question this book was written to provoke: *where is the physics-trained reader's toolkit — the renormalisation group, the systematic fluctuation expansion, the diagrammatic bookkeeping of corrections to a saddle — most likely to produce new mathematics?* (section 31.5.) That last section carries the single worked computation of the chapter, a fully rigorous calculation of the order-parameter susceptibility of the rank-one graphon game (worked computation 31.9), which makes precise both the divergence of fluctuations at the critical coupling and the exact point at which the rigorous theory runs out. We pose the four questions independently and answer them independently; only in the closing synthesis (section 31.6) do we argue that all four converge on one operator — the susceptibility $(\mathrm{Id} - \mathcal{G})^{-1}$, whose spectral radius reaching unity is at once the failure of uniqueness, the onset of equilibrium multiplicity, the blow-up of finite-N fluctuations, and the breakdown of best-response learning. We do not assume that convergence in advance; we earn it question by question, and the reader is entitled to watch us earn it.

A word on the standard we hold ourselves to, because an open-problems chapter is the exact place a textbook is most tempted to wave its hands: a gesture toward a hard direction, a citation, a "this remains open," and onward. We refuse that move. Every conjecture below is held to the same discipline as a theorem, which here means three things attached to each open problem. First, the precise hypotheses under which it is open — so that it is a definite mathematical statement and not a mood. Second, the nearest proved result that bounds it from the known side — so the reader sees exactly how far the proved theory reaches before it stops, and against what the conjecture is being measured. Third, an explicit account of *which* step of the nearest known proof fails, and *why* it fails — so that the obstruction itself is named, not merely the goal restated. Naming the failing step is the hardest of the three and the most valuable: it is the difference between knowing that a road ends and knowing what the bridge would have to span. A problem we cannot state to this standard is not yet a research problem but a feeling about one, and where we are in that position we say so plainly rather than dress it up. This is also what makes the map trustworthy: a gap located this precisely is one a reader can imagine closing.

31.1 The shape of what is known

The preamble promised a map before any gap is marked, and a discipline that holds each conjecture to a theorem's standard; making good on both begins with the same act — fixing the coordinate system. The four questions of the previous page all have the form "where does the proved theory stop?", and that question has a sharp answer only once we have said precisely where the proved theory *is*. Here is the organising observation that lets us say it. Every well-posedness result in this book — every uniqueness theorem, every globally well-posed master equation, every convergence rate — sits at a point in one and the same three-axis space, and the open problems are, almost without exception, the corners of that space the proofs have not reached. The three axes are the three independent ways the strategic problem can be made hard: by the strength and sign of the coupling between agents, by the structure of the network through which they interact, and by the kind of noise that drives them. Name the axes once and every later gap stops being an isolated difficulty and becomes a direction of travel from a known point.

Axis 1: the coupling regime. The first axis measures how strongly, and with what sign, an agent's optimal behaviour reacts to the crowd, and it is the axis along which the theory is most complete. At one end sits the regime where everything works at once. Under *(A-Mon)*, Lasry–Lions monotonicity (theorems 11.5 and 12.8) or its graphon lift (theorem 24.16), the coupling acts as a restoring force: making a choice more popular makes it less attractive, so the best-response map cannot reinforce a deviation, and one obtains the full package — uniqueness for *all* horizons, a globally well-posed master equation, exponential convergence of the finite-player game — with no smallness assumed and no dependence on the horizon T. Monotonicity is what buys time-uniformity, and that is precisely why it gives the cleanest results. Just outside it lies the *small-coupling* regime, where one surrenders the sign condition and buys back well-posedness with smallness instead. When the operator norm $\|\mathbb{W}\|$ falls below a horizon-dependent threshold (*(A-Coupling)*, proposition 24.18; static shadow theorem 20.14), the best-response map is a contraction and the contraction principle hands back a unique fixed point regardless of the coupling's sign — but only on a horizon short enough that the feedback has no room to compound. Contraction buys generality of sign at the cost of time, exactly the trade monotonicity declines to make. The two conditions overlap but neither contains the other, and — this is the content of section 31.2.1 — they do not meet: between them lies a band the LQG thread resolves exactly and the general theory cannot yet see. Beyond both is the *non-monotone, strong-coupling* regime, where the coupling has become a reinforcing force: agents want to do *more* of what the crowd does, a deviation can feed on itself, and the best-response map acquires several fixed points. Here uniqueness simply fails, equilibria multiply, and almost nothing is known in general — this is the regime of symmetry breaking and the home of section 31.2.2.

Axis 2: the network/limit regime. The second axis measures the structure of the interaction itself — which limit object the growing network converges to. The entire proved theory of Part IV lives at one end of it: dense graphons, bounded kernels, the compact metric space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ of chapter 17. Density is not a cosmetic hypothesis; it is what makes $\mathbb{W}$ a bounded — indeed Hilbert–Schmidt — operator, with a discrete spectrum and a *finite* norm, and that single fact is what every estimate in Part IV silently rests on. The contraction thresholds of Axis 1, the tightness bounds behind finite-network convergence, the spectral reductions of the carried threads — all of them are statements about a bounded operator with a well-defined leading eigenvalue $\lambda_1 = \|\mathbb{W}\|$. But real networks are not dense. Social graphs, infrastructure grids, and contact networks keep a bounded average degree as they grow, carry heavy-tailed degree distributions, and concentrate a handful of hubs of vastly disproportionate connectivity — the *sparse and heavy-tailed* regime (L^p graphons, graphex limits) of chapter 27. There the kernel is unbounded, $\|\mathbb{W}\|$ may be infinite, and the dense machinery degrades in a way the mathematics only partly tracks: the very norm that set every threshold on Axis 1 ceases even to be finite. This axis is therefore not a technical refinement but the gap between the model's natural habitat and the place it is proved, and section 31.2.3 is where the gap bites hardest.

Axis 3: the noise regime. The third axis is the kind of randomness the agents face, and it controls how complicated the mean field itself must be. *Idiosyncratic* noise — an independent shock to each agent — is the best-understood case, for a structural reason worth stating: by the law of large numbers the aggregate of independent shocks is deterministic, so the mean field $\bar{m}_t$ is a fixed (if a priori unknown) trajectory and the equilibrium is a deterministic fixed point. *Common*, or aggregate, noise — a shock felt by everyone at once, the weather over a power grid or a macro shock to a market — breaks this: it does not average away, the mean field $\bar{m}_t$ becomes itself a random, dynamically moving object, and an agent must choose a strategy contingent on its entire future random path. That is exactly what the master equation was built to encode (section 12.7 and chapter 26), and it is why common noise *forces* the master-equation formulation rather than the simpler coupled PDE pair: the unknown is no longer a trajectory of measures but a function on the space of measures. Well-posedness here is known only in restricted regimes, and — crucially for what follows — the difficulty compounds with the other two axes rather than merely adding to them.

Place the three axes together and the proved theory of this book occupies a single corner of the resulting cube: *monotone* $\times$ *dense* $\times$ *idiosyncratic*, together with a controlled neighbourhood of it — the small-coupling slab reached off the monotone face, the cut-metric-stable neighbourhood reached off the dense face. Everything proved lives in that corner or just outside it. The open problems are the moves off the axes, and the organising claim of the chapter — the reason the frontier is genuinely hard and not merely unfinished — is that the moves *interact*. They are not independent difficulties one could clear one at a time. Relaxing monotonicity alone gives the threshold and selection questions of sections 31.2.1 and 31.2.2; relaxing density alone gives the sparse-limit questions; admitting common noise alone gives a harder but already-studied problem. The questions that matter for applications, however, sit where two or three relaxations are imposed at once, and there the obstructions *multiply* rather than add. The canonical example — the one we develop in section 31.2.3 — is the master equation with common noise on a sparse limit. Common noise forces the master-equation formulation (Axis 3); the sparse limit destroys the operator-norm bound that the master equation's well-posedness proof relies on (Axis 2); and the two failures are not separable, because the very estimate that common noise makes indispensable is the one the unbounded operator removes. Each relaxation attacks a *different* hypothesis of the *same* proof, so the combined problem is not two half-problems but one whole one with no known line of attack.

This is why the chapter's theory gaps are ordered as they are.

The four subsections that follow are precisely the four cardinal moves off this corner, taken in increasing order of how many axes they disturb, and together they make up the chapter's first organising question — what we cannot yet prove. Section 31.2.1 moves along Axis 1 alone, asking for the exact frontier between the two non-meeting sufficient conditions for uniqueness. Section 31.2.2 moves further along Axis 1, into the strong-coupling regime where uniqueness has already failed and the live question is which of the many equilibria is selected. Section 31.2.3 makes the two-axis move — noise and network at once — that the interaction principle has just singled out as the deepest. And section 31.2.4 adds a fourth axis the cube did not have, orthogonal to all three: not the dynamics of the game but the *information* an agent holds when it acts. With the map fixed, we can begin marking the gaps.

31.2 Theory gaps

The first of the four questions — what we do not yet know how to *prove* — is the natural place to begin, because it inherits the cleanest map. Section 31.1 placed the proved theory in a single corner of the monotone $\times$ dense $\times$ idiosyncratic cube, and the four subsections below are the four cardinal moves off that corner, taken in increasing order of how many axes each disturbs. The present one takes the gentlest move — a single step along Axis 1, the coupling — and finds that even there, with the network dense and the noise idiosyncratic and only the sign and strength of the interaction in play, the proved theory leaves a gap it cannot close. The discipline of the preamble applies to each: precise hypotheses, the nearest proved result on the known side, and the exact step of its proof that fails. We can hold this first gap to that standard with unusual precision, because the LQG thread hands us the answer the general theory is missing, written out in closed form.

31.2.1 Sharp uniqueness thresholds

Recall the two ends of Axis 1 that the landscape left facing each other without touching. The book proves two genuinely different uniqueness theorems for graphon mean field games, and the reason they do not meet is structural, not an artefact of looser constants. Monotonicity (theorem 24.16) asks the coupling to act as a restoring force — formally, the Lasry–Lions sign condition $s\langle \Delta, \mathbb{W}\Delta\rangle \le 0$ on every displacement Δ, which for a positive graphon $\mathbb{W} \succeq 0$ is just $s \le 0$ — and in return delivers uniqueness for *all* horizons, with no smallness assumed: time cannot help a deviation compound when each step of the feedback is sign-suppressed. Contraction (proposition 24.18; static shadow theorem 20.14) makes the opposite bargain. It surrenders the sign condition entirely — the coupling may push either way — and buys uniqueness back from operator-norm smallness $\|\mathbb{W}\| < \kappa$, but only on horizons short enough that the best-response feedback has no room to amplify; the admissible κ shrinks as T grows. Neither hypothesis contains the other: monotonicity covers arbitrarily strong coupling of the right sign at arbitrarily long horizons, contraction covers either sign but only weakly and briefly. And precisely because one is a pure sign condition indifferent to strength while the other is a pure smallness condition indifferent to sign, there is a region caught by neither — coupling of the *wrong* sign (crowd-seeking, $s > 0$, so monotonicity fails) that is nonetheless not small enough for the contraction constant at the horizon of interest. The question of this subsection is what the exact frontier is in that region, and whether uniqueness in fact persists there. In the most explicit model in the book we can answer both questions completely, and the answer is yes.

Here is that answer, recalled with enough of its derivation to make the threshold land rather than merely appear. Take Thread B, the graphon LQG game (worked computation 24.22). For a frozen aggregate path the per-type problem is the scalar LQG problem of worked computation 11.7, whose value is quadratic with a *type-independent* Riccati curvature $P(t)$ — type-independent because only the linear coupling carries the type x, while the quadratic-in-state cost $\frac{q}{2}x^2$ is common to all agents. What remains coupled across types is the deterministic pair: the linear offset field $r_t \in L^2[0,1]$ and the closed-loop mean field $\bar{m}_t \in L^2[0,1]$, and they couple only through the aggregate $z_t = \mathbb{W}\bar{m}_t$. The diffusion σ feeds only the type-wise variance, an autonomous equation decoupled from the coupling, so uniqueness is decided entirely by that field pair. Now expand both fields in the eigenbasis (λ_k, ϕ_k) of $\mathbb{W}$. Because $\langle \mathbb{W}\bar{m}_t, \phi_k\rangle = \lambda_k \langle \bar{m}_t, \phi_k\rangle$, the projection onto each eigenmode *decouples*: mode k obeys exactly the scalar two-point boundary-value problem of worked computation 11.7, with the single substitution $s \mapsto s\lambda_k$ (eq. (24.10)). The continuum of coupled types has split into independent copies of the classical LQG game, one per eigenmode, the k-th running at effective coupling $s\lambda_k$. The classical game is non-unique exactly when its linearised forward–backward system has a resonant horizon, which happens iff the oscillation frequency $\Omega_k^2 = b^2 q\, s\lambda_k - A_\infty^2$ is positive, i.e. iff $s\lambda_k > 1 + a^2/(b^2 q)$ (here $A_\infty^2 = a^2 + b^2 q$). The graphon equilibrium is unique for all T iff *every* mode is subcritical; and for crowd-seeking coupling $s > 0$ the binding mode is the largest eigenvalue $\lambda_1 = \|\mathbb{W}\|$ (the spectral radius, by Perron–Frobenius for a nonnegative kernel, theorem 18.7), so the whole family of conditions collapses to one. This is proposition 24.23:

$$s\,\lambda_1 \;\le\; 1 + \frac{a^2}{b^2 q} \qquad \Longleftrightarrow \qquad \text{uniqueness for all } T, \tag{31.1}$$

with $\lambda_1 = \|\mathbb{W}\|$ the leading eigenvalue and ϕ_1 the eigenvector-centrality mode of chapter 18.

Read eq. (31.1) against the two general theorems and the gap becomes a definite strip of parameter space. For $s \le 0$ the threshold is implied by monotonicity — the restoring sign already gives it, and indeed $\Omega_k^2 \le -A_\infty^2 < 0$ for every mode. But for $0 < s\lambda_1 \le 1 + a^2/(b^2 q)$ the inequality holds with monotonicity *already failed*: the coupling is crowd-seeking, the global form $s\langle \Delta, \mathbb{W}\Delta\rangle$ is positive, no horizon-free monotonicity argument applies, and yet the equilibrium is unique for every horizon. The contraction theorem does not reach this strip either: its smallness hypothesis confines it to a region strictly inside the monotone-or-small slab, well below the leading eigenvalue, so it never enters the band. Concretely, take the rank-one kernel $W(x,y) = f(x)f(y)$ with $\lambda_1 = \|f\|_{L^2}^2$ and set $a = 0$: the threshold is $s\lambda_1 \le 1$, and the entire interval $0 < s\,\|f\|_{L^2}^2 \le 1$ of crowd-seeking couplings is a regime where uniqueness is a theorem but *neither* general sufficient condition can see it. Even in the cleanest worked example in the book, the two proved routes leave a band uncovered that the explicit computation fills exactly.

Open problem 31.1. *(Sharp uniqueness threshold.)* For the graphon mean field game with a fixed convex Hamiltonian, nondegenerate diffusion, and a smooth (possibly non-monotone) graphon coupling, identify a verifiable condition — presumably a joint condition on the second variation of the cost and the spectrum $\{\lambda_k\}$ of $\mathbb{W}$ — that is *necessary and sufficient* for uniqueness of the equilibrium for all horizons, recovering eq. (31.1) in the LQG case.

What is known bounds the problem tightly from both sides, which is exactly why it is ripe: the LQG computation is not a curiosity but a calibration, fixing the answer the general condition must reproduce. It rules out both naive candidates at once. The answer cannot be the monotonicity sign condition alone, because that condition is sufficient but demonstrably not necessary — the band $0 < s\lambda_1 \leq 1 + a^2/(b^2 q)$ is the explicit witness. And it cannot be an operator-norm bound alone, because in the LQG case the norm is not the binding object in any horizon-dependent combination: what binds is the single leading eigenvalue λ_1 weighted against the diffusion-and-state data through $1 + a^2/(b^2 q)$, a *mode-by-mode* criterion, not a global norm.

That last contrast is the structural obstruction, and it is worth naming precisely because it tells us which tool the general problem will need. The Lasry–Lions duality identity behind theorem 11.5 is an integration-by-parts that controls a *global* bilinear pairing: it bounds the squared distance between two candidate equilibria by s times the pairing $\langle \Delta, \mathbb{W}\Delta \rangle$ of the full population displacement $\Delta = m_1 - m_2$ against itself through the coupling, and concludes uniqueness when that pairing has a definite sign for *every* Δ. By construction this argument is blind to two things the LQG threshold exploits. It is blind to the spectral decomposition — it asks for the sign of the form on all of L^2 at once, so a single positive eigenmode $\lambda_1 > 0$ already spoils it, even though the other modes are harmless and the dynamics couple them independently. And it is blind to the parabolic regularisation: the extra room $a^2/(b^2 q)$ in the threshold is precisely the margin the state feedback and the nondegenerate diffusion buy back over a finite horizon, a margin the static sign condition discards by never looking at the horizon at all. The genuinely sharp criterion must therefore be local, not global — it must read off, mode by mode, when the equilibrium becomes unstable. The object that does this is the linearisation of the forward–backward system around a candidate equilibrium: uniqueness fails exactly when that linearisation acquires a nontrivial kernel, a nonzero homogeneous solution at a resonant horizon, which in the LQG case is precisely the condition $\Omega_k^2 > 0$ realised at $\tan(\Omega_k T) = \Omega_k / A_\infty$. This linearised forward–backward operator is the fluctuation operator of section 31.5, and its non-invertibility is the susceptibility $(\mathrm{Id} - \mathcal{G})^{-1}$ blowing up — the thread that the synthesis will pull through every gap in this chapter. Recovering mode-by-mode sharpness for a *nonlinear* coupling means controlling the spectrum of that operator without the luxury of the LQG diagonalisation, and that is the heart of Open problem 31.1.

The honest state of the art falls short of this on both sides of the inequality. On the sufficient side, the displacement-monotonicity refinements of chapter 12 (after Gangbo–Mészáros [75, 76]) genuinely enlarge the region: they ask for monotonicity along Wasserstein displacement geodesics rather than along the linear interpolation Lasry–Lions uses, which is a weaker demand and so covers couplings the classical sign condition rejects. But displacement monotonicity remains a *global* condition — a sign on a Hessian taken over the whole measure space — and inherits the same blindness: it does not localise to the leading eigenmode and is not known to capture the full parabolic margin, so it is not known to be sharp, and in particular it is not known to recover the exact band eq. (31.1) in the LQG case. On the necessary side there is simply no general result at all: no verifiable condition is known whose failure forces multiplicity. So Open problem 31.1 is open from both ends — the best sufficient condition is not proved tight, and no necessary condition exists to meet it — and the

LQG threshold stands in the middle as the only point where we know exactly where the line is.

That knowledge sets up the next question. The threshold eq. (31.1) is not just a boundary of a proof technique; it is a real transition in the game. Below it the equilibrium is unique; *at* it the linearised forward–backward operator first develops a kernel; and above it that kernel is a genuine second equilibrium. Once $s\lambda_1$ crosses $1 + a^2/(b^2 q)$ the crowd-seeking feedback compounds, the best-response map acquires multiple fixed points, and "is the equilibrium unique?" is answered — no — and replaced by a sharper one: of the several equilibria the game now admits, *which* is selected? That is where we turn next.

31.2.2 Non-monotone regimes and equilibrium selection

The previous subsection ended at a transition. Below the threshold eq. (31.1) the equilibrium is unique; at it the linearised forward–backward operator first develops a kernel; above it that kernel is a second equilibrium and the crowd-seeking feedback compounds into a whole family of them. So the question that organised the entire monotone theory — *is the equilibrium unique?* — has been answered in the negative and is no longer the right question to ask. Past the threshold it is replaced by a strictly harder one that the monotone theory never had to face: of the several equilibria the game now admits, *which* one does the system actually realise? This is the selection problem, and it is the first place in the book where existence and uniqueness no longer settle the matter — where, even with every equilibrium catalogued, one still has to say which is physical.

It pays to see the geometry of the multiplicity before hunting for a principle to break it, because the geometry is what every candidate principle will act on. The cleanest picture is the static rank-one shadow (section 20.7) of Thread B, where the entire population state collapses to the scalar order parameter $\varrho = \langle f, \bar{m} \rangle$ and the equilibrium condition is the single self-consistency equation $\varrho = h(\varrho)$, with $h(\varrho) = \mathbb{E}[f(x)\mathrm{br}_x(f(x)\varrho)]$ the population-averaged best response of section 31.5.1. In the symmetric case — no exogenous ideal-point field, $\eta \equiv 0$, so an agent has no private reason to prefer one sign of action over the other — the map h is odd, $h(-\varrho) = -h(\varrho)$, and the game carries a genuine $\mathbb{Z}_2$ symmetry $\varrho \mapsto -\varrho$ (flip every agent's action). The disorganised state $\varrho = 0$ is then always a fixed point. Its stability is governed by the slope $g = h'(0) = \mathsf{b}\,\mathbb{E}[f(x)^2] = \mathsf{b}\,\lambda_1$, the very strategic coefficient whose susceptibility $\chi = 1/(1-g)$ the worked computation of section 31.5.1 will show diverging as $g \uparrow 1$. While $g < 1$ the symmetric fixed point is the only one and it is stable; the instant g crosses 1 — exactly the LQG threshold of the preceding subsection, $s\lambda_1 > 1 + a^2/(b^2 q)$ specialised to the static shadow — the slope of h at the origin exceeds unity, $\varrho = 0$ turns unstable, and by oddness two new fixed points $\pm\varrho^*$ split off symmetrically. This is a *pitchfork bifurcation*: a single critical coupling at which one symmetric equilibrium gives way to a symmetric pair of broken-symmetry ones. It is the order-parameter form of the symmetry-breaking picture of section 11.4 — the multiplicity the homogeneous game develops at its discrete critical horizons eq. (11.14) — now read off a single scalar.

What the graphon adds, and the reason this is not merely the homogeneous story retold, is *which direction in population space the broken state points along.* In the rank-one case the bifurcating solution is supported on f, that is on the leading centrality mode ϕ_1 (proposition 24.23): the population does not break symmetry uniformly but preferentially along its most central eigenvector, the high-centrality hubs swinging first and the rest following in proportion to their ϕ_1-weight. Symmetry breaking on a network has a *shape*, and that shape is spectral data of $\mathbb{W}$. The selection question therefore inherits a graphon refinement the homogeneous theory cannot pose: not only which of $\{0, \pm\varrho^*\}$ is realised, but — once several modes can break, as we discuss below — which *mode* carries the broken state. Note finally that the selection question is open already in the homogeneous case (W constant, a single mode); the graphon does not create the difficulty, it stratifies it.

With the geometry fixed, selection is the problem of breaking the degeneracy between the equally-admissible branches $\pm\varrho^*$ (and, off the rank-one case, among the several coexisting equilibria). The literature offers three candidate principles. None is a theorem in any generality; each is natural for a different reason, carries a different piece of partial evidence, and would become a theorem only by supplying a different missing ingredient. The point of laying all three side by side is that they are less rivals than three readings of one physical demand — that the idealised $N = \infty$ degeneracy is an artefact some infinitesimal real effect resolves — and in the rank-one case they can be compared by explicit computation.

The first is *vanishing-viscosity*, or *entropy*, selection, and it is the physicist's reflex: break a degeneracy by switching on an infinitesimal field, see which state it favours, and then send the field to zero. Here the "field" is an extra dose of idiosyncratic noise — equivalently, an entropic regularisation of the agents' control problem — which smooths the value function, rounds the corners of the best-response map, and renders the regularised game uniquely solvable; one then removes the regularisation and asks which equilibrium of the original game the unique regularised solution converges to. The intuition is exactly that of selecting a Gibbs state by a vanishing symmetry-breaking term, and where the game has a variational (potential) structure it works: the regularised equilibrium is the minimiser of a perturbed energy, and in the limit it selects the global minimiser among the candidates. This is proved in special finite-state potential games. Its limitation is precisely the demand for that structure: a general graphon mean field game is not a potential game, the vanishing-viscosity limit need not be unique, and there is no energy whose minimiser the procedure is known to track. Turning it into a theorem would require either exhibiting the variational structure (a Lyapunov functional for the regularised flow) or a direct large-deviations argument that the regularised invariant measure concentrates on one branch.

The second is *common-noise* selection, and it is the sharpest concrete conjecture in the area — sharp because it predicts not just *that* selection occurs but *by what mechanism*. Switch on a small *common* noise: a shock felt by everyone at once which, by the structural distinction drawn in section 31.1, does not average away but makes the mean field itself a random, dynamically moving object. The order parameter ϱ is then no longer pinned at a fixed point but performs a diffusion in the double-well landscape whose minima are the broken branches $\pm\varrho^*$, and the common noise continually shakes it across the saddle at the origin. The conjecture is that this "smears" the pitchfork: for any positive noise level the equilibrium is again unique — now a single *random* equilibrium, a law on order-parameter paths rather than a pair of fixed points — and the sharp content is what that random equilibrium converges to as the noise vanishes. This is established rigorously in the two-state mean field game of Cecchin, Dai Pra, Fischer and Pelino [46], where the master equation can be written out and shown to remain well-posed through the transition, and in a handful of related finite-state models. Whether common noise *generically* restores uniqueness — in particular for a continuum of types — is open. What it would take to make it a theorem is named by the mechanism itself: the common-noise equilibrium is encoded by the master equation, so one must show that equation stays well-posed as the coupling is driven through the bifurcation, exactly where its characteristics — the forward–backward system whose linearisation is losing invertibility — threaten to cross. That requirement is the explicit hand-off to section 31.2.3.

The third is *stability*, or *learnability*, selection, and it is attractive for a reason orthogonal to the other two: it ties the choice of equilibrium to computation. Declare physical only those equilibria that are *attracting* fixed points of a natural adjustment dynamics — best-response iteration, or its averaged cousin fictitious play — the rationale being that an equilibrium no plausible learning process can reach is one no real population of boundedly-rational agents would ever settle into. In the rank-one pitchfork this principle is completely explicit and agrees with the other two: the symmetric fixed point $\varrho = 0$ has best-response slope $g > 1$ above threshold, so it *repels* — a small perturbation grows under iteration — while the broken branches $\pm\varrho^*$ have slope $h'(\varrho^*) < 1$ and *attract*. Stability selection therefore discards the symmetric equilibrium as unphysical and keeps the broken pair, exactly as vanishing-viscosity and common noise do, which is strong circumstantial evidence that in this case all three principles coincide. This is also the principle implicit in the learning algorithms of chapter 29, and that is at once its appeal and its current limit. Its appeal is that it makes selection a corollary of which fixed point an algorithm finds; its limit is that the scope is unproved — best-response iteration need not converge at all in the strong-coupling regime (it can cycle or wander rather than settle), the basin structure off the rank-one case is uncharted, and which equilibrium a given learning rule selects past the contraction threshold is itself open (Open problem 31.6). Making it a theorem means proving a joint convergence-and-selection statement for a learning dynamics in exactly the non-contractive regime where convergence is hardest to come by.

These three readings coincide in the rank-one scalar case and may well diverge once several modes compete, which is the strongest reason to demand the rank-one case be settled first: there, and only there, can the candidate principles be checked against one another by an explicit computation rather than against intuition. That is what the following problem asks.

Open problem 31.2. *(Equilibrium selection in the non-monotone regime.)* For the graphon mean field game above the critical coupling of eq. (31.1), determine whether (i) an arbitrarily small common noise restores uniqueness of the equilibrium, with the graphon master equation of chapter 26 remaining well-posed across the bifurcation; and (ii) the selected equilibrium coincides with the limit of best-response learning dynamics. Resolve the rank-one LQG case first, where the bifurcation is a single scalar pitchfork on the order parameter ϱ (section 31.5) and the question is fully explicit.

The graphon layer adds a genuinely new feature here, absent from the homogeneous theory, and it deserves to be drawn out because it is the part of the selection problem with no classical precedent. Recall from the threshold subsection that the mode decomposition diagonalises the linearisation: mode k runs at effective coupling $s\lambda_k$ and goes unstable when $s\lambda_k$ crosses $1 + a^2/(b^2q)$, i.e. at the critical coupling $s_k = (1 + a^2/(b^2q))/\lambda_k$. Because the eigenvalues are ordered $\lambda_1 \geq \lambda_2 \geq \cdots$, these critical couplings are ordered the other way, $s_1 \leq s_2 \leq \cdots$: as one turns up the overall coupling strength s the modes do not all go unstable at once but in succession, the leading centrality mode

ϕ_1 first, then ϕ_2, then ϕ_3, each at its own threshold. The non-monotone graphon game therefore exhibits not a single bifurcation but a *cascade* of them — a sequence of pitchforks, one per eigenmode, spaced along the coupling axis by the spectrum of $\mathbb{W}$. There is no homogeneous analogue because the homogeneous game is the single-mode case (a constant graphon has one nonzero eigenvalue); the cascade is entirely a creature of the spectral spread of the network, and the *spectral gap* $\lambda_1 - \lambda_2$ is exactly what sets how cleanly separated its first step is. When the gap is large the first bifurcation is isolated and mean-field-like — a single mode condenses well before any other can — and the rank-one reduction is essentially exact near s_1; when several eigenvalues cluster, several modes destabilise almost together and the cascade steps overlap.

Where the steps overlap, the modes interact, and this is the genuinely hard and genuinely new part. The successive bifurcations are not independent scalar pitchforks bolted side by side, because the bifurcation is governed by the *nonlinear* self-consistency map, not its linearisation. Once ϕ_1 has condensed, the population sits on a broken background $\varrho_1^*\phi_1$, and the second mode ϕ_2 does not bifurcate off the trivial state but off that background: its effective threshold is shifted by the ϕ_1-condensate through the nonlinearity, exactly as a symmetry-breaking field shifts a secondary transition. A selection theory in this regime must therefore answer not only *which* equilibrium and *which* mode, but how the already-broken modes renormalise the thresholds and shapes of the modes that break later — the mode-coupling problem. This is precisely the bookkeeping the physicist's diagrammatics is built for: the condensate of ϕ_1 enters the ϕ_2 equation as a tree-level vertex correction, and the systematic expansion in the eigenmode amplitudes is the natural language for the cascade. It is the clearest target in the chapter for the diagrammatic methods of section 31.5, and the place where the rank-one scalar reduction must give way to a genuinely multi-mode analysis.

Two threads leave this subsection. The first is the hope that carried the common-noise principle: a small common noise may smear the cascade and restore a unique random equilibrium. That hope is only as good as the master equation that encodes the random equilibrium, and so it presupposes the master equation survives common noise through the bifurcation — which is the subject of section 31.2.3, where we ask exactly that well-posedness question. The second is the landscape's warning. Everything in this subsection has held the network dense and bounded so that $\mathbb{W}$ has the clean discrete spectrum $\lambda_1 \geq \lambda_2 \geq \cdots$ the cascade is built on; but the applications that make the selection problem urgent — the anti-monotone epidemic and the conformity model of section 31.4 — live on *sparse* networks, where that spectrum may not even be well defined. Selection past the threshold (Axis 1) and the sparse limit (Axis 2) are then imposed at once, and as section 31.1 warned, the two-axis corners are where the difficulties multiply rather than add. The next subsection takes up that corner directly.

31.2.3 Master-equation well-posedness with common noise on sparse limits

The previous subsection closed on a hope and a warning, and this is where the two collide. The hope was that an arbitrarily small common noise might smear the pitchfork of section 31.2.2 — shaking the order parameter across the saddle at the origin and restoring a single *random* equilibrium in place of the degenerate pair $\pm\varrho^*$. But that hope was never free-standing: a random equilibrium is, by construction, a law on paths of the mean field, and the only object in this book that encodes such a law as a well-defined solution is the master equation. The hope therefore borrows against a prior question — *does the master equation remain well-posed once common noise is switched on and the coupling is driven through the bifurcation?* — and until that debt is paid the selection conjecture has nothing to stand on. The warning was the landscape's: the genuinely hard problems are not the single moves off the proved corner but the corners where two moves are imposed at once, because there the obstructions multiply rather than add (section 31.1). This subsection is the canonical two-axis corner, and the deepest theory gap in the chapter. We impose common noise (Axis 3) and a sparse limit (Axis 2) simultaneously, and we will see that each relaxation removes a hypothesis the other makes indispensable — so that the combined problem is not two half-problems but one whole one with no known line of attack.

Begin with why the master equation is forced here, because the whole difficulty is downstream of that one fact. Under idiosyncratic noise alone the law of large numbers flattens the aggregate: the population density $\bar{m}_t$ is a deterministic trajectory, and an equilibrium is a deterministic fixed point of the coupled Hamilton–Jacobi–Bellman/Fokker–Planck pair (chapter 11), the value function reacting to a mean field it can treat as known in advance. Common noise destroys exactly this (section 31.1): a shock felt by everyone at once does not average away, the mean field $\bar{m}_t$ becomes itself a random, dynamically moving object, and an agent can no longer optimise against a fixed future trajectory — it must choose a feedback strategy that hedges against the *entire random path* the mean field might follow. The coupled PDE pair cannot carry this, because its very form presupposes a deterministic forward density to integrate against. The master equation can, and this is the reason it exists (chapters 12 and 26): it lifts the whole equilibrium to a single deterministic function $U(t, x, \boldsymbol{m})$ of time, type, private state, and the *current* configuration $\boldsymbol{m}$ of the population, so that the value at any instant is read off the realised random measure rather than a predicted one. The randomness is then transported through U by Itô's formula along the stochastic flow of measures, and the price of that transport is a second-order derivative in the measure argument — the term that distinguishes the common-noise master equation from its first-order, idiosyncratic-only shadow. That second-order term is the entire locus of the difficulty that follows; everything below is an account of what happens to it when the two axes are relaxed.

The first relaxation: common noise on a continuum of types. In the homogeneous theory the common noise is a single scalar Brownian motion B_t^0 driving one random measure $\bar{m}_t$, and the second-order master-equation term is a single trace of the second Lions derivative against a finite-dimensional covariance — delicate, but a closed and now well-understood object (chapter 12). The graphon changes the kinematics of the noise itself. The mean field is no longer one measure but a whole *field* of them, $x \mapsto m_t^x$, one population per type; and the natural aggregate shock is type-dependent — the weather is not the same over every node of a power grid, a macro shock hits sectors unequally — so the driving noise is a distinct shock per type, correlated across types through the kernel W. A type-indexed family of Brownian motions correlated by a kernel is, in the continuum limit, a *cylindrical* or space–time noise over $[0, 1]$ whose covariance operator is built from W: an infinite-dimensional driving noise. The field of measures is then the solution of a stochastic flow that is morally an SPDE over the type interval, and the clean scalar picture — one Brownian motion, one well at $\pm\varrho^*$, one diffusion across the saddle — is replaced by an infinite family of coupled wells exchanging probability through W.

This is what forces the delicacy in the phrase *random measure-valued maps with the right adaptedness*, and it is worth unpacking slowly because even *stating* the well-posedness problem requires getting it right. The unknown population state is, for each realisation ω of the common noise, a field of measures $x \mapsto m_t^x(\omega)$ — an element of the field space $\boldsymbol{\mathcal{P}}_2$ of chapter 26 — and as ω varies it is a random element of that space. So the state lives in $L^2(\Omega; \boldsymbol{\mathcal{P}}_2)$, or rather in a subspace of it cut out by two consistency demands that pull against each other. The first is *adaptedness*: at time t the configuration $m_t^{\cdot}$ may depend on the common shocks observed up to t but not on those still in the future — agents condition on the weather they have seen, not the weather to come — so $m_t^{\cdot}$ must be measurable with respect to the common-noise filtration $\mathbb{F}_t^0$, and the whole path must be progressively measurable. The second is *regularity in the type variable*: the map $x \mapsto m_t^x(\omega)$ must, realisation by realisation, remain a bona fide element of $\boldsymbol{\mathcal{P}}_2$ — measurable and square-integrable over types in the L^2-Wasserstein-over-types metric d_2. The tension is that the noise correlates these two structures: the same kernel W that couples the wells across types also couples the per-type filtrations, so one cannot build adaptedness type by type and measurability over types separately and hope they agree. Choosing the topology on this space, the precise filtration, and the regularity class in x so that the stochastic flow is well-defined and the fixed point lives in the same space is not a formality; it is the first third of the problem, and it has no counterpart in the homogeneous theory where there is a single filtration and no type axis at all. On top of it, the second-order master-equation term — the one Itô produced — is now no longer a scalar trace but an *operator* on this infinite-dimensional space: the second Lions derivative $\partial_{\boldsymbol{mm}}^2 U$ paired against the type-indexed noise covariance, the latter carrying W. Even the notion of a classical solution must be re-posed around derivatives of this operator kind.

The second relaxation: the sparse limit. The dense theory of Part IV rests, at every step, on one quiet fact: $\mathbb{W}$ is a bounded — indeed Hilbert–Schmidt — operator with a finite norm $\|\mathbb{W}\| = \lambda_1$. The sparse and heavy-tailed limits of chapter 27 (L^p graphons, graphex) remove it. There the kernel is unbounded, hubs of unbounded degree appear, and $\|\mathbb{W}\|$ may simply be infinite. The damage is structural rather than cosmetic, because $\|\mathbb{W}\|$ is the single constant every controlling estimate in Part IV is written in terms of. The contraction route to existence and uniqueness needs $\|\mathbb{W}\| < \kappa$ for the horizon-dependent threshold κ of *(A-Coupling)* (proposition 24.18); when $\|\mathbb{W}\| = \infty$ that condition is vacuous and the critical-coupling thresholds of Axis 1 degenerate to nothing. The convergence-from-finite-N route needs the same norm to bound the tightness estimates (chapter 25); it too loses its constant. And the per-type uniform bound of lemma 24.6 — the estimate that controls each agent's aggregate exposure $(\mathbb{W}\bar{m})(x)$ *uniformly in the type* x, so that no single agent feels an unbounded crowd — fails precisely at the hubs, where the unbounded degree lets the aggregate exposure diverge. Every threshold, every contraction constant, every uniform-in-type bound was a statement about a finite operator norm, and the sparse limit is exactly the regime where that norm is gone.

Now the two relaxations meet, and the reason this corner is harder than the sum of its parts becomes visible. The common-noise master equation needs operator-norm control more than the deterministic theory does, not less: closing the second-order term — bounding the new operator coefficient $\partial_{\boldsymbol{mm}}^2 U$ against the W-weighted noise covariance — and then closing the fixed point that defines U both run on estimates whose controlling constant is $\|\mathbb{W}\|$. The sparse limit removes that constant. So the very estimate that common noise makes *indispensable* is the one the unbounded operator *deletes*: the two relaxations attack different hypotheses of the same proof, and the combined problem has no remaining line of attack rather than two separate ones. This is the precise sense in which the obstructions multiply.

Open problem 31.3. *(Master equation with common noise on sparse limits.)* Formulate and prove well-posedness of the graphon master equation with common noise when the limit object is an L^p graphon or a graphex with unbounded operator $\mathbb{W}$. Identify the replacement for the operator-norm smallness condition *(A-Coupling)* — presumably a condition on the degree distribution or on an L^p rather than $L^2 \to L^2$ norm — under which a unique classical (or suitably weak) solution exists.

This is the hardest problem on the list and the most consequential, because it is the regime of every realistic application in section 31.4: real social and infrastructure networks are sparse, carry heavy-tailed degree distributions, and are subject to correlated aggregate shocks — all three of the features the two relaxations encode, present at once. The honest state of the art is that the two relaxations are understood only *separately*, and in each case the other axis has been trivialised away. On the common-noise side, the well-posedness of the master equation under aggregate noise is by now classical for *homogeneous* mean field games — a single representative agent, a single random measure $\bar{m}_t$ on $\mathbb{R}^d$, a finite-dimensional driving noise, and no kernel at all — through the work of Carmona and Delarue [41] and Ahuja [4]; there is no type continuum to make the noise infinite-dimensional and no operator $\mathbb{W}$ whose norm could be infinite. On the sparse side, Bayraktar, Wu and Zhang [16] control interacting particle *systems* on sparse graphon limits — propagation of chaos and well-posedness of the forward McKean–Vlasov dynamics on an unbounded kernel — but *without the strategic fixed point*: there is a prescribed forward flow and no Hamilton–Jacobi–Bellman equation, no optimisation, no equilibrium to close. Each programme solves one axis by setting the other to its trivial value, and neither toolkit's hypotheses survive contact with the other problem: the homogeneous master-equation estimates assume the finite norm the sparse limit destroys, and the sparse-system estimates never have to close a backward equilibrium equation at all. Their combination has, at present, no theory.

The gap is deep enough that even the first sanity check is itself a research project: we do not have a worked LQG example in this corner. One might expect Thread B to hand it over for free, as it has handed over every other threshold in the chapter — but here the very tool that made the graphon LQG game explicit is unavailable. The mode decomposition of worked computation 24.22 diagonalises the problem by expanding in the eigenbasis (λ_k, ϕ_k) of $\mathbb{W}$, and the rank-one reduction collapses everything to the scalar order parameter ϱ precisely because $\lambda_1 = \|f\|_{L^2}^2$ is finite and isolated. On an unbounded sparse operator the spectrum need not be discrete or even bounded, so the diagonalisation that turned a continuum of coupled types into one scalar pitchfork no longer applies, and the order-parameter picture that organised section 31.2.2 has no obvious analogue. Constructing a tractable instance in this corner therefore means doing genuine work before any computation begins: specifying the type-dependent common noise and its W-built covariance operator, fixing the adapted solution space of random measure-valued fields, choosing an unbounded kernel whose stochastic forward–backward system can still be solved in closed form, and only then reading off whether common noise restores uniqueness. Each of those steps is a small theorem in its own right. That a problem the dense theory answers in a single line of mode algebra should, off the dense axis, require a research programme merely to pose a solvable example is the sharpest measure of how far this corner sits from proved ground.

This corner is also not the end of the relaxations. We have moved off two axes — the noise and the network — while holding fixed an assumption so far left implicit: that an agent, however random and however sparse its environment, at least *observes* that environment exactly, seeing its own state and the realised aggregate $\mathbb{W}\bar{m}_t$ without error. Drop that, and a fourth axis opens, orthogonal to the three of the cube: the information the agent holds when it acts. Layered on top of the sparse common-noise corner, it produces the genuinely hardest case the book can name — agents who must *filter* a hidden, randomly-moving aggregate through a sparse network they only partially observe — and it is the subject we turn to next.

31.2.4 Partial information: the observation axis

The previous subsection ended by naming the assumption it had left standing. We had moved off the noise axis and the network axis at once and reached the chapter's deepest two-axis corner, but throughout we kept silently granting the agent one thing: that however random its environment and however sparse its network, it at least *sees* that environment without error — its own state and the realised aggregate $z_t = \mathbb{W}\bar{m}_t$, read off exactly at every instant. Drop that grant and a fourth axis opens, and it is orthogonal to the three of the cube in a precise sense: it concerns not the *dynamics* of the game — not the strength of the coupling, the structure of the network, or the kind of noise — but the *information* an agent holds when it acts. This is the axis the heterogeneity review of chapter 30 isolated as its third, and which it explicitly carried forward to this chapter as an open direction (remarks 30.17 and 30.18). The present subsection is where that hand-off is paid out, and where the theory-gaps arc closes: it is the last of the four cardinal moves off the proved corner, and the one whose mathematics is least developed.

Begin with the contrast it relaxes, because the whole subsection is the cost of one idealisation. Every equilibrium constructed in this book is played by an agent with *full observation*: at time t a type-x agent knows its own private state X_t^x exactly and knows the aggregate $z_t = \mathbb{W}\bar{m}_t$ it is coupled to — either as a deterministic trajectory (idiosyncratic noise) or, under common noise, as the realised value of a random one (chapter 26). Its best response is computed against that perfectly known signal, and the fixed-point theorems of chapters 24 and 26 close the loop on exactly that information set. But full observation is an idealisation no application of section 31.4 actually enjoys. A firm in a market does not see the population distribution of its competitors' inventories; it sees a noisy price. A household on a power grid does not see the realised load field over every node; it sees a delayed, noisy meter reading of the local frequency. An individual in an epidemic does not observe the prevalence field $\bar{m}_t$; it observes a few infected contacts. In each case the agent receives a *noisy, partial signal* — of its own state, of the aggregate, or of both — and must *filter*: form a best estimate of the hidden quantities from the history of its observations, and act on the estimate. Two distinct inference problems are bundled here, and keeping them apart is the whole organising move of what follows: an agent may have to estimate *its own state* (a finite-dimensional filter, one hidden scalar or vector per agent), or it may have to estimate *the aggregate* $\mathbb{W}\bar{m}_t$ itself (an infinite-dimensional filter, a posterior over the entire field of populations). The first is benign; the second is where the theory runs out.

Chapter 30 settled the benign half completely, and it is worth recalling exactly *what* makes it benign, because the answer is the precise statement whose failure defines the open problem. In the linear–quadratic–Gaussian world — linear state dynamics, quadratic costs, Gaussian noise and a *linear-Gaussian* observation $\mathrm{d}Y_t^x = c(x)X_t^x \,\mathrm{d}t + \mathrm{d}V_t^x$ — the *certainty-equivalence separation principle* holds (proposition 30.16). Separation means the two problems an agent faces, estimation and control, decouple completely and are solved one after the other: the agent first runs a Kalman–Bucy filter (lemma 30.15) to compute the conditional mean $\hat{X}_t^x = \mathbb{E}[X_t^x \mid \mathcal{F}_t^{Y,x}]$ of its hidden state given its own observations, and then plays the *full-information* optimal feedback with the true state replaced by that estimate — it acts as if $\hat{X}_t^x$ *were* the state, with certainty. The reason this is exact and not an approximation is structural, and worth spelling out because every word of it is load-bearing. The conditional variance $P_t(x)$ of the filter obeys a *deterministic* Riccati equation (eq. (30.26)) that does not see the control at all: in the linear-Gaussian case how well the agent estimates its state is fixed in advance, independent of what it does, so there is no incentive to act in order to learn — no "dual effect." The quadratic cost then splits cleanly into a control-dependent term identical to the full-information cost written in $\hat{X}_t^x$, plus a control-*independent* estimation-error term $\int_0^T q(x)P_t(x)\,\mathrm{d}t$ the agent cannot influence and may ignore. Minimising the first gives back the original feedback evaluated at the estimate; the second is a fixed tax. And because the filter is unbiased, $\mathbb{E}[\hat{X}_t^x] = \mathbb{E}[X_t^x] = \bar{m}_t(x)$, taking expectations reproduces the *same* mean-field consistency equation as the fully observed game (proposition 30.16(iii)): partial information changes each agent's fluctuations but leaves the equilibrium mean field untouched. This is why chapter 30 files information heterogeneity as *composable* rather than as a genuine obstruction — in the clean case it multiplies onto the graphon structure without disturbing the fixed point at all.

Everything in that paragraph used, decisively, that the aggregate $z_t = \mathbb{W}\bar{m}_t$ is *deterministic* — so that an agent has only its own state left to estimate, never the population — and that the observation is *linear-Gaussian* — so that the posterior stays Gaussian and is summarised by its mean. Remove either hypothesis and the separation fragments, in two distinct ways that remark 30.17 flagged and we now develop, because the precise mechanism of each failure is what the open problem must confront. The first failure is *nonlinear observation*. If the signal is a nonlinear function of the state, or the state or noise is non-Gaussian, the conditional law is no longer Gaussian and is no longer captured by a mean and a variance: the sufficient statistic for the filter is the *entire* posterior distribution, an infinite-dimensional object evolving by a nonlinear stochastic PDE — the Zakai or Kushner–Stratonovich equation of nonlinear filtering. Two things break at once. The conditional variance is now random and *control-dependent*, so the dual effect returns: an agent can act to sharpen its own future estimates, and estimation no longer decouples from control. And the optimal feedback depends on the whole posterior, not on a single summary statistic, so "play the full-information policy at the estimate" is no longer even well posed — there is no single estimate, there is a distribution. Certainty equivalence fails not by a correction term but at its root.

The second failure is the deeper one for this book, because it is the one the graphon forces: the agent must filter the *aggregate itself*. Reinstate common noise (chapter 26), or a major player (chapter 30), and the aggregate $z_t = \mathbb{W}\bar{m}_t$ becomes random — the very hypothesis the separation principle leaned on is gone. An agent can no longer treat the population as a known forcing; it must *infer* the current configuration of the field of populations from its noisy local signal, which is a nonlinear filtering problem whose hidden state is the measure-valued field $x \mapsto m_t^x$ itself, an element of the field space $\boldsymbol{\mathcal{P}}_2$ of chapter 26. The agent's posterior is therefore a *law on the space of measures* — a random probability measure over $\boldsymbol{\mathcal{P}}_2$ — and this is exactly where the filtering problem couples to the master equation. Recall that the master equation lifts the equilibrium value to a deterministic function $U(t, x, \boldsymbol{m})$ of the *current* configuration $\boldsymbol{m}$ (chapter 26 and section 31.2.3); under full observation the agent simply evaluates U at the realised $\boldsymbol{m}$. Under partial observation it cannot, because it does not know $\boldsymbol{m}$ — it knows only its conditional law π_t given its observations. The value it acts on is then the *posterior average* of the master field, $\int U(t, x, \boldsymbol{m})\, \pi_t(\mathrm{d}\boldsymbol{m})$, and the belief π_t is itself the solution of a Zakai/Kushner equation on $\boldsymbol{\mathcal{P}}_2$ driven by the observations and transported by the same stochastic flow of measures the master equation encodes. The master equation and the filtering equation are now coupled forward and backward: the master field supplies the dynamics the filter must track, and the filter supplies the belief at which the master field must be evaluated to produce the control that closes the equilibrium. Neither can be solved without the other, and the clean "filter, then play the full-information policy" sequencing — the entire content of separation — has dissolved into a single coupled system on the space of measures.

There is, however, a reason to expect that separation can sometimes be recovered *approximately*, and pinning down when is itself part of the open problem; the heuristic is worth giving because it is exactly the place the spectral data of $\mathbb{W}$ re-enters. The aggregate an agent must filter is not an arbitrary random field but $\mathbb{W}\bar{m}_t$, the population averaged through the kernel, and averaging suppresses randomness. On a densely connected network an agent is coupled to the crowd through many nearly-independent channels, so by the same law-of-large-numbers mechanism that made idiosyncratic mean fields deterministic (section 31.1), the realised aggregate carries only a small residual fluctuation around its mean — it is *nearly* deterministic and hence *nearly* observable, and the deterministic-aggregate separation of proposition 30.16 should hold up to an error set by the size of that residual. The size of the residual is spectral. The leading eigenvalue $\lambda_1 = \|\mathbb{W}\|$ fixes the scale on which the aggregate responds, and the spectral gap $\lambda_1 - \lambda_2$ fixes how nearly the aggregate collapses onto the single centrality mode ϕ_1: when the gap is large, $\mathbb{W}\bar{m}_t$ is effectively the one scalar order parameter $\varrho = \langle f, \bar{m} \rangle$ of section 31.5, which an agent can track with a *one-dimensional* filter, and the approximate-separation error should scale with the observation-noise level and with the inverse gap $(\lambda_1 - \lambda_2)^{-1}$ — small gap, many competing modes, poor scalar summary, large separation error. This is the conjectural shape of an approximate-separation theorem: certainty equivalence survives, with a quantitative defect controlled by the observation noise and the spectral data of $\mathbb{W}$. It is a conjecture, not a theorem; we state it as the target the open problem should aim at, and note that it is precisely the kind of statement the susceptibility thread of this chapter predicts, since a vanishing spectral gap is one of the ways the fluctuation operator of section 31.5 loses control.

Now collide this axis with the corner of the previous subsection and one reaches the hardest case the book can name — the one remark 30.18 singled out as canonical. Take a *major player* carrying a hidden state (the finite-rank atomic perturbation of the graphon, chapter 30), embedded in a *sparse* network (chapter 27), with the minor agents able to observe only noisy local signals and forced to *filter the major player's hidden state through the network*. Every axis is now disturbed at once, and they interact rather than add. The major player makes the minor field random (the common-noise/Axis-3 difficulty); the sparse limit removes the operator norm and the discrete spectrum the approximate-separation heuristic just leaned on (Axis 2 — there is no clean λ_1, no clean gap, so even the conjectural error bound has nothing to be written in terms of); and partial observation means the minor agents must infer the major's state, a nonlinear filter, with the information arriving *through* the network. On a sparse heavy-tailed graphon that last fact is itself a new phenomenon: the quality of each agent's information is heterogeneous in the same way its degree is, so the hubs filter the major's state sharply while the peripheral leaves barely see it, and the population's belief about the hidden aggregate is as heavy-tailed as the network. Separation fails on every count simultaneously, and there is at present no formulation, let alone a theorem.

Open problem 31.4. *(Graphon mean field game under partial information.)* Formulate the graphon mean field game in which each type-x agent observes only a noisy partial signal of its state and of the aggregate $\mathbb{W}\bar{m}_t$, and prove existence and uniqueness of the equilibrium when the linear-Gaussian separation of proposition 30.16 fails — nonlinear observation, or filtering of the aggregate field itself, which is a nonlinear (Zakai/Kushner) filtering problem coupled to the master equation of chapter 26. Resolve first the *canonical hard case* flagged in remark 30.18: a partially observed major player (the finite-rank atomic perturbation of the graphon, chapter 30) on a sparse graphon (chapter 27), where the minor agents must filter the major player's hidden state through the network. Characterise when a separation principle survives approximately, with an error controlled by the observation noise and the spectral data of $\mathbb{W}$.

This sits one axis beyond Open problem 31.3: it adds *informational* heterogeneity to the sparse, common-noise corner, and is the precise mathematical content of the hand-off chapter 30 makes to this chapter (section 30.3.2 and remarks 30.17 and 30.18). As chapter 30 stresses, the graphon abstraction is the economical container for *type* heterogeneity but says nothing on its own about *informational* heterogeneity; supplying that theory is open.

With this fourth axis the theory-gaps arc is complete. We have taken all four cardinal moves off the proved corner — a single step along the coupling axis (section 31.2.1), a longer one into the regime where uniqueness has already failed and selection takes over (section 31.2.2), the two-axis collision of noise and network (section 31.2.3), and now the orthogonal informational axis — and at every one the question we asked was the same: *can we prove the equilibrium exists, and is it unique?* That is the question the first thirty chapters were built to answer, and the four subsections have mapped exactly where the answer is still missing. But that question carries a silent presupposition, and it binds even inside the proved corner — the monotone dense idiosyncratic interior where every theorem of Part IV lives — because knowing that a unique equilibrium exists is not the same as holding it in hand. Dropping that presupposition is the work of the next section, and it changes the questions in kind: they stop asking whether the object exists and start asking whether we can *produce* it. We turn to that now.

31.3 Computation gaps

The theory-gaps arc closed by dropping a presupposition. For four subsections the question was always the same one the first thirty chapters were built to answer — *does the equilibrium exist, and is it unique?* — and the four cardinal moves off the proved corner mapped exactly where that answer is still missing. But the close of section 31.2.4 noticed that the question carries a silent assumption, and this section is where we drop it. Settling existence and uniqueness tells you that a number is there to be found; it does not put the number in your hand. An application does not consume a fixed-point theorem. It consumes the fixed point: the actual feedback a type-x agent should play at each state, the actual order parameter the population settles on, the actual aggregate trajectory $\mathbb{W}\bar{m}_t$ a planner must dimension a network against. The gaps of this section are about producing those objects, and they are a different kind of not-knowing from the theory gaps — not whether the object exists, but whether we can compute the object whose existence the previous section spent its length debating.

It is worth being precise about why a proof of existence is not yet a usable answer, because the gap is structural and not a matter of effort. The existence theorems of Part IV are won by fixed-point arguments — Schauder or Kakutani on the best-response map (chapter 22) — and a fixed-point theorem is, by its nature, non-constructive: it certifies that the map has a fixed point without exhibiting one, supplying neither an algorithm that reaches it nor a rate at which any candidate iterate approaches it. The uniqueness theorems sharpen *which* object exists but say even less about how to produce it. Where the best-response map happens to be a contraction — the small-coupling slab of *(A-Coupling)*, proposition 24.18 — existence and computation collapse into one statement, because Banach's theorem is itself the algorithm and iteration converges geometrically; but that coincidence is exactly the easy corner, and it fails the moment the coupling is strong, which is precisely where every interesting application lives. So the computation question is not downstream of the theory question. It is genuinely separate, and in one important respect it is more pressing. The theory gaps bind only *off* the proved corner; inside the monotone–dense–idiosyncratic interior, where every theorem of Part IV holds, there is nothing left to prove — and yet there is everything left to compute, because the equilibrium of even a perfectly well-posed game is a coupled forward–backward system in a high-dimensional state space that no closed form solves. A gap that bites even where the theory is complete is a gap an application cannot route around.

The section sorts the difficulty by how much of the problem is known, and the three subsections form a logical ladder. At the top rung the model is fully specified — dynamics, costs, and the graphon W all in hand — and the only question is cost: can the forward–backward system actually be discretised and solved at the scale real state spaces and real networks demand? That is the *scalable-solver* gap of section 31.3.1. Drop down a rung and the model itself is unknown, so there is nothing to discretise; the equilibrium must instead be *learned* from simulation or interaction, as in chapter 29, and the question becomes whether a learning rule can be guaranteed to converge to it — the *learning-with-guarantees* gap of section 31.3.2, where, tellingly, the same strong-coupling regime that breaks the contraction also breaks the learning dynamics, the critical point of the chapter surfacing yet again. Both of those rungs, however, take the graphon as given. The bottom rung asks the question logically prior to both: in an application the network is itself estimated, so *is the graphon even knowable* from the data the game produces — and in what sense, given that equilibria depend on W only through the aggregate $\mathbb{W}\bar{m}$ and only up to the relabelling ambiguity of chapter 17? That is the *identifiability* gap of section 31.3.3. Solve when you know the model and the graphon; learn when you do not; and, underneath both, ask whether the graphon was ever recoverable in the first place.

We take the rungs in that order, most concrete first. The most direct of the three is the top one, where everything is specified and only the arithmetic remains, so we begin there: granted the full model, can the coupled forward–backward system of chapter 28 be discretised at scale, and if not, what is the obstruction?

31.3.1 Scalable high-dimensional solvers

Take the top rung at its word. The model is fully in hand — dynamics, costs, and the graphon W all specified — so existence and uniqueness are settled and nothing remains to prove; what remains is to put the equilibrium in the planner's hand as an actual number, and the preamble's demand "we need a number" becomes the most direct question one can ask of the forward–backward system: can it be discretised and solved at the scale a real application sets? The honest answer is that, even inside the proved corner where every theorem of Part IV holds, it cannot — not by any method that scales — and the obstruction is worth dissecting because it is a clean instance of the curse of dimensionality compounded by the graphon's second grid.

Recall the apparatus. The solvers of chapter 28 attack the coupled Hamilton–Jacobi–Bellman/Fokker–Planck pair by discretising both continuous arguments at once: the private state, valued in $\mathbb{R}^d$, on a spatial grid and the type $x \in [0,1]$ on a second grid, with the nonlocal coupling operator $\mathbb{W}$ rendered as a quadrature over the type nodes. Of the two grids, the state-space grid is the binding constraint, and the arithmetic of why is brutal and exact. A tensor-product grid of M points per axis in $\mathbb{R}^d$ has M^d nodes; the value function and the density must each be stored and updated on all of them at every time step, so the cost is $O(M^d)$ in both memory and work. That exponential is the curse of dimensionality in its rawest form, and it is hopeless past $d \approx 3$: at a modest $M = 100$ a three-dimensional state already costs 10^6 nodes, a five-dimensional one 10^{10}, and a seven-dimensional one leaves the reach of any machine. Yet the applications that make the whole theory worth computing sit squarely in the regime the grid cannot enter. A power grid carries a storage state per participating asset, an epidemic model carries age and accrued-immunity structure on top of infection status, an economic model carries a multidimensional portfolio — all routinely moderate-to-large d, all far past the grid's ceiling. The state dimension alone is enough to defeat the naive solver, and it does so even where the theory is complete.

The graphon layer does not relieve this; it adds a second front. Where a homogeneous mean field game carries one value function and one density, the graphon game carries a whole field of them, one per type, so the type interval must itself be discretised into K nodes and the full $O(M^d)$ state solve repeated at each. Worse, the type nodes do not decouple. The coupling enters every agent's problem only through the aggregate $(\mathbb{W}\bar{m}_t)(x) = \int_0^1 W(x,y)\,\bar{m}_t(y)\,dy$, and that integral over y ties the population at type x to the population at *every* other type. Discretised, $\mathbb{W}$ becomes a dense $K \times K$ quadrature matrix, and forming the aggregate at all K type nodes is an $O(K^2)$ matrix–vector product executed at every time step of every forward–backward sweep. The nonlocality is the whole point of the model — it is what makes the network matter — but it is also what makes the type grid quadratic rather than linear, so the graphon contributes a second factor that compounds the state-space curse rather than competing with it.

Two partial routes exist, one for each front, and neither closes its gap. The first tames the type grid, and it does so exactly where the structure theory of chapter 19 promises it can. The weak regularity lemma says that any graphon is approximable in the cut metric, to any tolerance ε, by a step graphon of bounded complexity — a kernel constant on $O(\exp(\varepsilon^{-2}))$ type blocks, the bound depending on ε alone and not on the original kernel. A step graphon's operator has rank equal to its number of blocks, so the $K \times K$ quadrature collapses to a low-rank or block form: instead of resolving the aggregate at every one of K type nodes, one resolves it on the handful of blocks, and the $O(K^2)$ product becomes $O(K \cdot r)$ with r the effective rank. The block-graphon reduction of chapter 28 is exactly this collapse made operational. The rank-one case is its extreme: when $W(x,y) = f(x)f(y)$ the operator has a single nonzero eigenvalue $\lambda_1 = \|f\|_{L^2}^2$, the aggregate $(\mathbb{W}\bar{m}_t)(x) = f(x)\,\langle f, \bar{m}_t\rangle$ factorises through the one scalar $\langle f, \bar{m}_t\rangle$, and the entire type dependence of the coupling collapses onto that single number — the order parameter $\varrho = \langle f, \bar{m}\rangle$ whose susceptibility the worked computation of section 31.5.1 will compute in closed form. The type grid then disappears altogether: the coupling is carried by one scalar that every agent reads through its own $f(x)$, and the K-fold solve degenerates to a single self-consistent scalar iteration. This is why the rank-one reduction is the calibrating example of the whole chapter — it is the one case where the second grid is free, and it is the case to which every adaptive scheme should reduce when the spectrum cooperates.

What is missing is that this collapse is, at present, a structural possibility rather than a *certified* algorithm, and the gap between the two is the open problem. The weak regularity lemma is a qualitative existence statement: it guarantees a low-rank approximant exists but does not, given a target accuracy, tell the solver how many modes to keep, nor does it convert the resulting discrete solution back into a guarantee about the true continuum game. What an application needs is the quantitative, adaptive converse. The effective rank a solver should use is not a property of the kernel alone but of the kernel's spectral decay — the number of eigenvalues of $\mathbb{W}$ above a tolerance, read off the spectral data of chapter 18 — and a certified solver should measure that decay, retain exactly the modes the accuracy target requires, and *prove* that the truncation it chose is enough. The spectral gap $\lambda_1 - \lambda_2$ is the natural knob: when it is large a few modes (in the cleanest case, the single rank-one mode) capture the coupling and the reduction is nearly exact; when the spectrum clusters, many modes are needed and the adaptive rank grows accordingly. The crux, however, is the error certificate, and it is worth motivating why the cut metric is the right currency for it. The solver does not approximate the graphon for its own sake; it approximates it in order to produce an equilibrium, and what must be controlled is therefore not the kernel error but the *equilibrium* error — how far the population profile computed on the truncated kernel sits from the one the true continuum game would realise. The bridge between the two is exactly the cut-metric stability that runs through Part IV: equilibria are continuous in the graphon in $\delta_\square$ (chapters 24 and 25), so a cut-metric bound on the kernel truncation translates into a bound on the equilibrium it induces. An *a posteriori* bound — one computed from the truncated solution itself, after the fact, rather than assumed in advance — that reads $\|W - W_r\|_\square$ off the discarded spectral tail and propagates it through that stability estimate would certify the output as a genuine ε-equilibrium of the true game, not merely of the discretised surrogate. That certificate is the whole content of the open problem: not a faster solver, but a solver that knows, and can prove, how good its own answer is.

Open problem 31.5. *(Certified low-rank graphon solver.)* Design a forward–backward solver for the graphon mean field game system whose type-discretisation cost scales with the *effective rank* of $\mathbb{W}$ (the number of eigenvalues above a tolerance) rather than the number of type-grid points, with an a posteriori error bound in the cut metric $\delta_\square$ that certifies the output as an ε-equilibrium of the true continuum game. Prove a convergence rate in terms of the spectral decay of $\mathbb{W}$.

The low-rank route disposes of the type grid but leaves the state-space grid — the $O(M^d)$ curse that was the binding constraint in the first place — entirely untouched, and so a second, independent route is needed for the dimension that the graphon never touches. Here the only known way past the exponential is to abandon the grid: replace the value function and density by neural-network ansätze trained to satisfy the forward–backward system, trading a mesh whose cost is exponential in d for a parametric family whose cost is not. This deep-learning route is attractive precisely where the grid is hopeless, but it buys scalability at the price of the one thing the grid had for free — a guarantee. A neural solver returns a number, but whether that number is the equilibrium, and at what rate the training reaches it, is exactly the question a fixed grid never had to ask, because grid refinement converges by construction while gradient descent on a nonconvex loss does not. That guarantee is no longer a discretisation question but a learning question, and it is the issue the next subsection takes up directly: when the grid fails and one must instead learn the equilibrium from simulation, what can be *proved* about where the learning converges?

31.3.2 Learning with guarantees

The solver subsection ended by handing the deep-learning route over with its guarantee in question, and this subsection is where that question is faced. When the grid is hopeless — because the state dimension defeats every mesh, as section 31.3.1 showed, or because the dynamics and costs are themselves unknown and only a simulator is available — the only way left is to abandon solving for *learning*: replace the value function and the population flow by parametric ansätze and improve them from sampled play, exactly the programme of chapter 29. The trade the previous subsection named is now the whole subject. A neural solver returns a number at any state dimension, but whether that number is the equilibrium, and at what rate the iteration reaches it, is no longer settled by construction the way grid refinement was. It has become a convergence question about a dynamical system on the space of population flows, and the purpose of this subsection is to locate precisely where that question has a proved answer and where it does not.

The obstruction is structural, and it is the same tension that has organised the whole chapter, seen now from the algorithmic side. An equilibrium is a *fixed point*: the population flow $\bar{m}$ that, fed into every agent's optimal control, reproduces itself. The natural way to chase a fixed point is to iterate the map whose fixed point it is — best-respond to the current population, push the resulting actions forward to get a new population, repeat (definition 29.8). This converges, and converges geometrically, precisely when that best-response-and-push-forward map is a *contraction*; and the contraction modulus is governed by exactly the operator whose spectral radius this chapter is about.

In the language of theorem 29.10, fixed-point iteration contracts when the product of the coupling moduli falls below one (eq. (29.8)), and for the graphon game that condition is the operator-norm smallness *(A-Coupling)* of proposition 24.18: the best-response map's linearisation is the operator $\mathcal{G}$ whose iterates the susceptibility $(\mathrm{Id} - \mathcal{G})^{-1}$ sums, and the iteration is a contraction exactly when the spectral radius of $\mathcal{G}$ is below unity. But this is precisely the small-coupling or monotone regime in which the problem was *already* solved: there existence and uniqueness are theorems (theorem 24.16 and proposition 24.18) and the equilibrium can be produced by the very iteration whose convergence is in no doubt. The algorithm earns its guarantee exactly where the guarantee was not needed. Move into the regime that motivated the whole computational programme — strong, crowd-seeking coupling near or past the threshold eq. (31.1) — and the spectral radius of $\mathcal{G}$ reaches and crosses one, $(\mathrm{Id} - \mathcal{G})^{-1}$ ceases to be a convergent sum, and best-response iteration stops contracting. The learning algorithm fails to converge in exactly the regime where there is a genuine equilibrium one cannot otherwise find.

The rank-one static shadow makes this completely explicit, and it is worth pausing on because it shows the susceptibility of section 31.5.1 reappearing as the literal convergence rate of an algorithm. There best-response iteration is the scalar recursion $\varrho_{n+1} = h(\varrho_n)$ on the order parameter, with h the self-consistency map of sections 31.2.2 and 31.5.1 and, in the LQ case, the affine map $h(\varrho) = h_0 + g\,\varrho$ of slope $g = \mathsf{b}\,\lambda_1$. The iteration is a Picard iteration of an affine map, so its behaviour is read off the slope at a glance: the error contracts geometrically, $\varrho_n - \varrho^* = g^{\,n}(\varrho_0 - \varrho^*)$, the convergence rate is $|g|$, and summing the corrections to the fixed point reproduces the susceptibility $\sum_{n\geq 0} g^{\,n} = 1/(1-g) = \chi$. The susceptibility is thus not merely a fluctuation amplitude; it is the condition number of the learning problem, the factor by which the iteration amplifies a one-step error into a fixed-point error. As $g \uparrow 1$ the rate $|g|$ approaches one and the iteration slows to a crawl — the algorithmic face of critical slowing down, the same divergence of χ that section 31.5.1 reads as a diverging variance. And for $g > 1$, past the threshold, the map is expanding: $\varrho = 0$ repels, the iteration is driven away from the symmetric state, and where it lands (the broken branches $\pm\varrho^*$, which have $|h'(\varrho^*)| < 1$ and so attract) is exactly the stability-selection question of section 31.2.2. The micro-example already shows the two phenomena that the general problem must control — non-contraction near threshold and which basin the dynamics falls into past it — and shows them governed by the one scalar g.

The stabilisations used in practice all buy back convergence, and the point worth making precisely is *what each charges for it*, because the charge is never nothing. Fictitious play (definition 29.14) replaces "best-respond to the latest population" by "best-respond to the time-average of all past populations," damping the oscillation that defeats raw iteration (section 29.2.2); it provably converges, but the proof (theorem 29.15, after [38, 111]) runs on Lasry–Lions monotonicity — it buys convergence by reimposing the very sign condition the strong-coupling regime violates, so it does not reach past the threshold either. Online mirror descent (definition 29.17) is a policy-space gradient flow with the same standing: it converges (theorem 29.19, after [110]), again under monotonicity, and to a *regularised* equilibrium rather than the true one. That last clause is the second kind of charge, and it is the one entropic regularisation makes unavoidable. Adding an entropy bonus to each agent's control problem (definition 29.11) smooths the value function, renders the best response single-valued and Lipschitz (lemma 29.12), and so flattens the slope of $\mathcal{G}$ until the regularised map contracts *regardless of sign* — the regularisation is a contraction-buying device that works even where monotonicity fails. But its fixed point is the equilibrium of the regularised game, not the original one, and the two differ: remark 29.13 records the bias $\mathcal{E}(\pi^\star_\tau) \le c_0\,\tau\,(1+\log|A|)$, linear in the regularisation strength τ. One therefore converges — to the wrong equilibrium, by an amount one can shrink only by shrinking τ, which is exactly the knob that was buying the contraction. Every stabilisation trades the non-contraction of the interesting regime for either a monotonicity hypothesis that excludes it or a controlled error that does not vanish. Naming which trade is forced, and at what rate, in the strong-coupling graphon game is the content of the open problem.

Open problem 31.6. *(Learning past the contraction threshold.)* Give a learning algorithm for the graphon mean field game with a non-asymptotic guarantee of convergence to an ε-Nash equilibrium in a regime where best-response iteration does not contract — i.e. with $\|\mathbb{W}\|$ above the threshold of proposition 24.18 — and characterise which equilibrium it selects (cf. Open problem 31.2). Quantify the bias introduced by entropic regularisation as a function of the regularisation strength and the spectral gap of $\mathbb{W}$.

The two clauses of the problem after the convergence guarantee are not afterthoughts; they are where it connects to the rest of the chapter, and each deserves to be drawn out. The first — *which* equilibrium the algorithm selects — is the learnability-selection principle of section 31.2.2 read as a demand on a theorem rather than a hope. Past the threshold the game has several equilibria, and an algorithm that converges converges to one of them; *that* one is, by the stability-selection reading, the physical equilibrium, precisely because a population of adjusting agents could reach it. So a convergence theorem in the non-contractive regime does double duty: it certifies the algorithm *and* resolves part (ii) of Open problem 31.2, identifying the selected equilibrium as the limit of the learning dynamics. The danger the rank-one example already flagged is that the dynamics need not converge at all — it can cycle or wander rather than settle — so the selection it offers is conditional on a convergence one must first prove, in exactly the regime where convergence is hardest. The two open problems are therefore the same problem seen from two sides, and progress on either is progress on both.

The second clause — the entropic bias as a function of regularisation strength and spectral gap — is the quantitative heart, and the rank-one reduction shows why the spectral gap must appear. Regularisation flattens the strategic slope, replacing $g = \mathsf{b}\lambda_1$ by a reduced $g_\tau < g$; it buys contraction by pushing g_τ below one, and it perturbs the fixed point by an amount the susceptibility amplifies. To leading order the induced shift in the order parameter is $\varrho^*_\tau - \varrho^* \approx \chi \cdot (\tau\text{-perturbation of } h)$, so the bias carries a factor $\chi = 1/(1-g)$ that *diverges as the coupling approaches threshold*. Resolved mode by mode, each eigenmode k contributes a bias of order $\tau/(1-\mathsf{b}\lambda_k)$, and the total is dominated by the near-critical modes — those with $\mathsf{b}\lambda_k$ closest to one. The spectral gap $\lambda_1 - \lambda_2$ is exactly what decides whether one mode dominates that sum (a clean, mean-field-like bias set by the leading mode) or several modes pile up near criticality together (a bias inflated by the whole near-critical cluster). The trade then has a sharp and unpleasant form: to stabilise one needs τ large enough to force $g_\tau < 1$, hence $\tau \gtrsim (g-1)_+$, but the bias is at least $\tau \cdot \chi_\tau$, and near and past the threshold both the minimal stabilising τ and the residual susceptibility grow, so the unavoidable bias is bounded *below*. Remark 29.13 gives only the crude game-independent bound linear in τ; the open problem asks for the sharp two-sided estimate in which the spectral data of $\mathbb{W}$ enters, quantifying how the bias blows up as the gap closes — the precise price, in distance from the true equilibrium, of buying convergence by regularisation in the regime that needs it.

Layered on top of this classical tension is a difficulty the graphon contributes and the homogeneous theory never meets: *representation.* A learned policy in the graphon game cannot be a function of the agent's private state alone. By the structure of the model it must depend on the type x and on the nonlocal aggregate $(\mathbb{W}\bar{m}_t)(x) = \int_0^1 W(x,y)\bar{m}_t(y)\,dy$ that the type-x agent actually feels, so the function class the network parametrises must somehow encode the operator $\mathbb{W}$ itself — a high-dimensional, nonlocal object — or the policy cannot even express the dependence the equilibrium has. Representing $\mathbb{W}$ naively returns to the second grid of section 31.3.1 and its $O(K^2)$ cost. The cleanest current proposal sidesteps this with the same spectral idea that organised the solver and the threshold subsections: represent the population not by its full profile but by its projection onto the leading eigenmodes of $\mathbb{W}$, and let the policy network take those finitely many modal coordinates as input. This is precisely the spectral reduction of worked computation 24.22 pressed into service as an *architectural inductive bias* — the network is built to see only the few coordinates the coupling actually acts through, which is exact in the rank-one case (one coordinate, the order parameter) and an ε-truncation when the spectrum decays. What is missing is the guarantee that closes the loop with the previous subsection. A *sample-complexity theorem* — a non-asymptotic bound on the number of simulated trajectories needed to reach an ε-Nash equilibrium, scaling with the *effective rank* of $\mathbb{W}$ (its number of eigenvalues above tolerance) rather than the ambient dimension — would put learning on the same footing the certified low-rank solver of Open problem 31.5 puts gridded computation: both would then cost what the spectral decay of $\mathbb{W}$ says they should, one in samples and one in mesh, and both would carry a proof that the truncation they chose suffices. Establishing it would unify this subsection with the last under a single quantity, the spectral decay of the graphon, as the common currency of computational cost.

Every word of this subsection, and of the solver subsection before it, has rested on an assumption so basic it was never stated: that the operator $\mathbb{W}$ the architecture must encode, the spectrum the sample complexity is measured against, the kernel the aggregate $(\mathbb{W}\bar{m}_t)(x)$ is formed with, is *known*. In an application it is not handed down; it must be estimated from data, and that raises a question logically prior to learning the equilibrium at all — prior, indeed, to every computational question in this section — namely whether the graphon is identifiable from the data the application produces in the first place. That is where we turn next.

31.3.3 Identifiability of the graphon from data

The previous two subsections, and indeed every computational question in this section, rested on an assumption so basic it went unstated until the last paragraph forced it into the open: that the operator $\mathbb{W}$ a solver discretises, that a learner's architecture must encode, that forms the aggregate $(\mathbb{W}\bar{m}_t)(x)$ every agent reacts to, is simply *known* — handed down with the model. Every result in Part IV makes the same grant: the graphon W is given, and the only work is to solve, simulate, or learn the equilibrium it carries. In an application that grant is unwarranted. No one hands an economist the kernel of the trade network or a planner the contact graph of an epidemic; it must be *estimated* from whatever data the application produces. This is the third computation gap, and it is logically prior to the other two: before one asks how cheaply the equilibrium can be computed or learned, one must ask whether the object those computations take as input can be recovered at all. *Is the graphon even identifiable from the data?* The answer splits sharply in two, along a single fault line — whether or not the network itself is observed — and the two regimes are so different in character that conflating them is the main way the question is got wrong. We separate them deliberately.

Structural identification: the network is observed. Suppose the application hands us the network itself — the adjacency matrix of the realised finite graph $G(N, W)$, every edge between every pair of agents recorded. Then estimating W is no longer a game problem at all but a pure statistics problem: recover the kernel that generated an observed random graph. This is the *nonparametric graphon estimation* problem, and it is by now well charted. One treats W as an unknown function on $[0,1]^2$ and the observed adjacency matrix as $\binom{N}{2}$ independent Bernoulli draws with success probabilities $W(x_i, x_j)$ at latent type coordinates x_i; estimating the function is a two-dimensional nonparametric regression complicated by the fact that the design points x_i are themselves unobserved. The minimax rates are known. In the *dense* regime, where the edge probability stays of order one as N grows, the optimal mean-squared error decays like the usual two-dimensional nonparametric rate up to a network-size term (Gao, Lu and Zhou [77]); in the *sparse* regime, where the average degree grows slower than N and most of the matrix is zero, the rates degrade in a way that is also pinned down (Klopp, Tsybakov and Verzelen [87]). For the purposes of the map, the headline is that this corner is *solved*: when the network is observed, the graphon can be estimated consistently and at a known optimal rate, dense or sparse.

There is one catch, and it is not a defect of the estimators but a feature of the object, the very feature stressed in chapter 17. A graphon is identifiable only *up to a measure-preserving relabelling of the type interval.* Two kernels W and $W^\sigma(x, y) = W(\sigma x, \sigma y)$ related by a measure-preserving bijection σ of $[0, 1]$ generate exactly the same distribution of random graphs — relabelling the latent coordinates permutes which agent is called which but leaves every edge probability, and hence the whole law of $G(N, W)$, untouched. No amount of data can separate W from W^σ, because the data they produce are identically distributed. The estimable object is therefore not a labelled kernel but its *equivalence class* under relabelling — precisely a point of the quotient space $\widetilde{\mathcal{W}}_0$ under the cut metric $\delta_\square$ built in chapter 17. One should resist reading this as a limitation. For a mean field game it is exactly the right notion of identification, and the reason is the structural-stability theorem the book already proved: equilibria are $\delta_\square$-continuous (chapter 25), so two graphons at zero cut distance — the relabelling-equivalent ones — carry *the same game*, the same equilibria, the same aggregate dynamics. The information the estimator cannot recover, "which type is which," is exactly the information the equilibrium does not depend on. Estimating the graphon in $(\widetilde{\mathcal{W}}_0, \delta_\square)$ rather than as a labelled function is not a coarsening one tolerates; it is recovering the kernel in precisely the topology the strategic problem is continuous in, and no finer. The relabelling ambiguity and the cut-metric stability of equilibria are two faces of one fact, and structural identification fits the game like a key.

Behavioural identification: the network is hidden. Now remove the gift. In the realistic case one does *not* observe the network — contact graphs are private, trade relationships are inferred, the coupling of a financial system is never tabulated — and all one sees is the agents *behaving*: trajectories of their states or actions, that is, samples of the equilibrium itself. The task is to run the inference backward, from the equilibrium an unknown graphon induces to the graphon. This is an inverse problem for the coupled Hamilton–Jacobi–Bellman/Fokker–Planck system of chapter 11, and it is badly ill-posed — not merely hard to estimate with noisy finite data, but *non-identifiable even with infinite clean data*, for a structural reason that is worth making completely explicit because it is the crux of the whole subsection.

The reason is that the map from graphon to equilibrium factors through a drastic bottleneck. Trace how W enters the game: an agent of type x never feels the kernel pointwise, nor the configuration of any individual neighbour; it feels only the single aggregate field $(\mathbb{W}\bar{m}_t)(x) = \int_0^1 W(x,y)\,\bar{m}_t(y)\,dy$, the W-weighted average of the population, and its entire optimisation — its HJB equation, hence its optimal control, hence the forward density it contributes — depends on W *through that field and nothing else*. So if two graphons W and W' happen to produce the same aggregate field along the observed equilibrium, $(\mathbb{W}\bar{m}_t)(x) = (\mathbb{W}'\bar{m}_t)(x)$ for every type and every time on the realised trajectory, then they generate the same HJB equation, the same controls, the same forward dynamics — *the identical equilibrium trajectory*. No observation of states or actions, however rich, can tell them apart, because the data they produce are not merely similarly distributed but literally the same. The forward map $W \mapsto$ equilibrium collapses an infinite-dimensional kernel onto the low-dimensional, time-indexed aggregate the trajectories reveal, and any two pre-images of a given aggregate are behaviourally indistinguishable. This is genuine non-identifiability, a many-to-one in the model itself, categorically worse than the statistical noise that limits structural estimation: there the obstruction was a relabelling the game was indifferent to, here whole families of *inequivalent* graphons, far apart in cut distance, can be invisible behind one equilibrium.

The rank-one thread makes the collapse concrete and shows just how little of the kernel survives. Take Thread B's clean reduction $W(x,y) = f(x)f(y)$ (section 20.7). The aggregate the agents feel is $(\mathbb{W}\bar{m}_t)(x) = f(x)\langle f, \bar{m}_t\rangle = f(x)\,\varrho_t$, the product of the fixed profile shape $f(x)$ with the scalar order parameter $\varrho_t = \langle f, \bar{m}_t\rangle$ that organised the whole non-monotone analysis of section 31.2.2. Two things follow. First, the equilibrium's only contact with f is through this product, so the global *sign* of f is unrecoverable: the kernel $f \otimes f$ is itself invariant under $f \mapsto -f$, and the equilibrium sees f only up to that sign. Second, the closed self-consistency for ϱ_t involves f only through the single number $\lambda_1 = \|f\|_{L^2}^2$ — the leading eigenvalue, the same spectral datum that set the uniqueness threshold eq. (31.1) and the susceptibility $\chi = 1/(1-g)$. So even with the entire equilibrium observed forever and noiselessly, the only features of the rank-one kernel the data can pin down are one function (the profile f, up to sign) and one scalar (λ_1) — a one-function, one-parameter summary of an object that, as a labelled kernel on $[0,1]^2$, is infinite-dimensional. Everything else about W is behaviourally invisible. This is the same lesson the throughline keeps returning: the equilibrium reads the graphon only through its leading spectral data, and behavioural identification can recover only what the equilibrium reads — here, λ_1 and the centrality mode $\phi_1 \propto f$, no more.

Open problem 31.7. *(Behavioural identifiability of the graphon.)* Characterise the equivalence classes of graphons that induce the same equilibrium of the dynamic graphon mean field game, as a function of the cost structure. Determine when the leading spectral data $(\lambda_k, \phi_k)_{k\le K}$ — which is all the equilibrium can depend on through $\mathbb{W}\bar{m}$ — is identifiable from finitely many observed equilibrium trajectories, and give a consistent estimator with a rate, jointly accounting for the relabelling ambiguity of chapter 17 and the finite-network sampling error of chapter 25.

This closes the inventory of computation gaps, and with it the chapter's second organising question. Three times now — in the solver of section 31.3.1, the learner of section 31.3.2, and the identification problem just posed — the same quantity has stepped forward as the governing cost: the leading spectral data of $\mathbb{W}$. The solver pays in mesh for the eigenvalues it must resolve, the learner pays in samples for the modes its architecture must encode, and the identifier can recover only the spectral data the equilibrium exposes; the spectral decay of the graphon is the common currency of all three. What these gaps still lack is a *domain* that turns each abstract cost into one somebody actually pays — which the next section supplies. One of them reappears there with its sign reversed: the distributed-control and mechanism-design applications of section 31.4 do not estimate a given graphon from observed behaviour but *choose* a graphon to induce a desired behaviour, the same inverse problem on the same operator $\mathbb{W}$ run forward instead of back. Identification and design are adjoint, and the application frontiers pick up exactly that adjointness.

31.4 Application frontiers

We have now amassed, across two full sections, an inventory of what the theory cannot yet prove and what the computation cannot yet cheaply deliver — a uniqueness frontier the LQG thread can locate but the general theory cannot, a selection problem past it, a two-axis master-equation corner with no line of attack, an observation axis the cube did not have, a solver whose cost scales with spectral rank, a learner whose sample budget scales with the modes it must encode, an identification problem on the leading spectral data of $\mathbb{W}$. Stated this way the inventory is abstract: each gap is a deficiency of a method, measured against other methods. What none of them has yet been given is a *domain* — a setting that fixes which graphon, which cost, which regime is at stake, and so converts an abstract deficiency into a concrete cost somebody actually pays. That is the work of this section. The graphon mean field game is, at bottom, a tractable model of *strategic decisions coupled through a large structured network*, and that one description fits a handful of application domains exactly. We take four where the model is already in use, and in each we do not merely point to the domain but trace the chain from its mechanism to the precise piece of missing mathematics it demands — so that the frontier comes out a mathematical statement and not a wish, and so that the reader sees *why* the application forces exactly that gap and no other. The four sort themselves, not by accident, onto the four moves off the proved corner the chapter has been mapping.

Epidemics on networks. The first domain is the one where the coupling's sign does the work, and it is the cleanest because almost everything else is simple. An agent chooses a level of social activity, or of vaccination or protective effort; the risk it runs is the prevalence of infection among its network neighbours, which is exactly the graphon aggregate $(\mathbb{W}\bar{m}_t)(x)$ evaluated at the agent's type x; and its cost trades the private burden of caution against that risk of infection. The private state is not a continuum but a small finite set — the compartments susceptible / infected / recovered — so this is a *finite-state* graphon mean field game (Aurell, Carmona, Dayanikli and Laurière [9]), with the smallest state space and the most transparent coupling of any application in the book: the network appears once, linearly, through $\mathbb{W}\bar{m}_t$, and there is nothing to obscure how the sign of the strategic feedback falls out. And it falls out the wrong way. The decisive structural fact is that protective behaviour is a *strategic substitute* across the population, and this is what "anti-monotone" means in the present setting. When many neighbours are cautious, prevalence $(\mathbb{W}\bar{m}_t)(x)$ drops; a lower prevalence lowers each remaining agent's infection risk; and a lower risk *weakens* that agent's own incentive to be cautious — it can free-ride on the herd protection the others' caution has already bought. The best response to more collective effort is therefore *less* individual effort: the self-consistency map runs downhill where the congestion model of Thread B runs uphill. In the Lasry–Lions bookkeeping of section 31.2.1 this is precisely the failure of the sign condition — the bilinear form $s\langle\Delta, \mathbb{W}\Delta\rangle$ carries the destabilising sign rather than the restoring one — so the entire monotone toolkit that buys horizon-free uniqueness in the proved corner simply does not apply, and the game is pushed off Axis 1 into the non-monotone regime where multiplicity is possible and the live question is no longer existence but *which* equilibrium the population realises. That is the regime of Open problem 31.2, reached here not by an exotic cost but by the most basic feature of contagion. The mathematical frontier is thus exactly the non-monotone, finite-state graphon theory with its threshold and selection questions — and the finiteness of the state space is a genuine simplification that makes this domain the natural proving ground for selection, since it is where the common-noise master equation of Open problem 31.2(i) is most likely to be writable in closed enough form to settle the conjecture. The payoff that makes the question urgent is a comparative-statics one: does a small, targeted subsidy — to the most central nodes, those of largest ϕ_1-weight, where an averted infection blocks the most onward transmission — move the equilibrium across the threshold and avert an outbreak? That is a derivative of the equilibrium with respect to a localised perturbation of the cost, exactly the intervention calculus chapter 20 carries out for the static game; lifting it to the dynamic, finite-state, non-monotone game is the open piece.

Energy markets and power grids. The second domain is the one that lands in the hardest corner of the cube, and it does so honestly — not by contrivance but because the physics of a grid supplies all three relaxations at once. Distributed storage and generation units — home batteries, electric-vehicle chargers, prosumers — each solve a stochastic control problem: when to charge, when to discharge, against fluctuating prices and their own state of charge (Couillet, Perlaza, Tembine and Debbah [51]; Alasseur, Ben Tahar and Matoussi [5]). They are coupled two ways, and the two ways are different operators. The first is the market price, which responds to *total* demand and so couples every unit to every other equally — a global, rank-one, constant-graphon coupling, which is to say Thread A in its purest form, the homogeneous mean field with no network structure at all. The second is the physical grid: line and voltage constraints couple units that are electrically adjacent, and this coupling carries the genuine network structure, the sparse transmission topology with its hubs and its geography. Superpose them and the realistic regime is the two-axis corner of Open problem 31.3 with the observation axis held off: the grid is *sparse* (Axis 2), the aggregate driver — weather over the service area, the common demand cycle — is a *common noise* felt by everyone at once (Axis 3), and on top of both the per-agent state is genuinely high-dimensional (state of charge, plus the market and weather variables each unit must track), which is what defeats the naive grid-based solvers of section 31.3.1. So this domain is not merely an instance of the deepest theory gap; it is the reason that gap was worth isolating.

It is also the domain that most sharply motivates the way *out* of the solver gap, and the reason is a spectral fact about grids worth developing, because it is the load-bearing point. The coupling that carries the grid's network structure is built from the transmission network's graph Laplacian, and a transmission Laplacian is not a generic sparse operator: it has a *sharp spectral gap*. A power grid is engineered to be well connected and robust, so its algebraic connectivity — the smallest nonzero Laplacian eigenvalue — sits well clear of zero, and, equally to the point, the network's effective interaction is dominated by a handful of low-lying modes: the slow, large-scale collective patterns of congestion across the grid, above which the spectrum decays quickly. In operator terms the grid-constraint coupling is, to high accuracy, *low effective rank* — a few spectral modes carry almost all of it, and the tail is suppressed by the gap. This is exactly the structure the certified low-rank solver of Open problem 31.5 is built to exploit: that solver's cost scales not with the ambient dimension of the network but with the number of spectral modes one must retain to reach a target accuracy, and a sharp spectral gap is precisely what *certifies* that a small retained rank suffices — the truncation error past mode r is controlled by the size of the gap below λ_r, so the gap converts "low rank works in practice" into "low rank works provably, with this error bound." The grid is therefore not just a place the solver would be useful; it is the canonical example whose spectral geometry the solver's guarantee is tailored to, and the surest reason Open problem 31.5 is the right piece of mathematics to demand.

Opinion dynamics. The third domain is the one where the carried thread is not an analogy but the literal model, so that the chapter's own running example turns out to have been a sociological theory all along. Agents adjust opinions or behaviours under a cost with two terms: one pulls each agent toward the average opinion of its network neighbours, the aggregate $(\mathbb{W}\bar{m}_t)(x)$ again, and the other penalises departure from a private conviction (Bauso, Tembine and Başar [14]). Take the cost quadratic and this is *exactly* Thread B, the congestion/imitation model carried through the entire book — only now read with the imitation sign rather than the congestion sign. Conformity, the wish to agree with one's neighbours, is the crowd-seeking coupling $s > 0$: doing more of what the crowd does is rewarded, not penalised, so the strategic feedback reinforces rather than restores, and the game sits at the non-monotone end of Axis 1 by construction. Everything the chapter proved about that end now reads as a statement about a society. While the coupling is weak the unique equilibrium is the disorganised one — opinions spread, the order parameter $\varrho = \langle f, \bar{m}\rangle$ vanishes, no consensus. As the conformity coupling is turned up through the critical value $g = \mathsf{b}\lambda_1 = 1$ of section 31.2.2, the disorganised state goes unstable and a pair of broken-symmetry branches $\pm\varrho^*$ split off: the population spontaneously aligns. The pitchfork bifurcation of the rank-one game *is* the onset of consensus or polarisation, and the graphon refinement says something a homogeneous model cannot — that the alignment forms first along the leading centrality mode ϕ_1, the most connected sub-population condensing into agreement before the periphery and dragging it along in proportion to ϕ_1-weight. The mathematical frontier is therefore the selection problem Open problem 31.2 read sociologically: of the two consensus branches the broken symmetry admits, which one does a real population reach, and can a small exogenous common shock — a media event felt by everyone at once — tip it deterministically into one branch? That last question is the common-noise selection mechanism of section 31.2.2 restated in the field's own vocabulary, and it is why opinion dynamics is the application that most directly demands the selection theory the chapter has been unable to supply: here the degenerate $\pm\varrho^*$ are not a technical artefact but two different futures for the society, and which one is selected is the whole empirical content of the model.

Distributed control and multi-agent systems. The fourth domain inverts the direction of the entire enterprise, and in doing so it picks up exactly the adjointness the identification subsection handed forward. Robotic swarms, sensor networks, traffic-routing fleets — these are *designed* systems, in which a single controller chooses not only the agents' costs but the communication network W itself. The question is no longer the analyst's — given a graphon, what is the equilibrium? — but the engineer's reverse of it: given a target collective behaviour, which graphon induces an equilibrium that realises it? This is a graphon mechanism-design or inverse-game problem, and it is the precise adjoint of the identifiability problem Open problem 31.7. Identification runs the map backward from observed behaviour to the operator $\mathbb{W}$ that produced it; design runs the same map forward, from a desired behaviour to an operator chosen to produce it — one inverse problem on $\mathbb{W}$ read in two directions, which is why the two are genuinely the same mathematics and not merely neighbouring concerns. The tool the design side needs is the comparative-statics and intervention calculus of chapter 20: which eigenmode of $\mathbb{W}$ to strengthen in order to move the aggregate in a wanted direction, quantified by the Katz–Bonacich and eigenvector-centrality results theorems 20.18 and 20.19 that say exactly how a perturbation of the network propagates into a perturbation of the equilibrium. Those results are proved for the static game; the frontier is to lift them to the dynamic graphon game, where the controller designs W to steer a *trajectory* of aggregates rather than a single fixed point, and where the same susceptibility operator that governed identifiability now governs how sensitively the designed equilibrium responds to the designer's choices.

Four domains, and not one of them is served by the proved corner. The epidemic forces the selection problem, the grid forces the master-equation corner and the certified solver, opinion dynamics forces selection again, distributed control forces the design inverse of identification — and each of these is, by the chapter's own accounting, open. This is the honest conclusion the application survey was meant to reach: the places the model is actually used are, without exception, the places off the axes where the mathematics runs out, and the demand for that missing mathematics is not academic but already issuing from epidemiology, power-systems engineering, sociology and robotics. What every one of these frontiers shares, moreover, is the form of the obstruction — a non-monotone bifurcation, a critical coupling, a susceptibility that diverges, a low-rank spectral structure to be exploited or a symmetry to be broken and reselected — and that form is one a physicist will recognise on sight. The applications hand us a demand for new mathematics; the question that closes the chapter is where the tools to meet it are most likely to come from, and the wager of the final section is that they come from the statistical-mechanics toolkit the analyst's thirty chapters were quietly building toward all along.

31.5 A research agenda for the physics-trained reader

We come to the question the book was built to ask. The reader has spent thirty chapters watching mean field games be assembled from measure theory, stochastic analysis and fixed-point theorems —

the analyst's tools. But the structure that emerged is one the physicist already owns: a self-consistent field (the mean field) determined by a saddle-point condition (the Nash fixed point), with a control parameter (the coupling $s\,\|\mathbb{W}\|$) that tunes the system through a symmetry-breaking transition (the uniqueness threshold eq. (31.1)). Every open problem above has, in this light, a physicist's restatement, and the restatement suggests a method. This section is a deliberately speculative map of where three pieces of the statistical-mechanics toolkit — the fluctuation expansion, the renormalisation group, and diagrammatics — are most likely to convert into theorems. It carries the chapter's one worked computation, which makes the first of the three fully precise.

31.5.1 Fluctuations: the order-parameter susceptibility

The mean field game is the $N \to \infty$ saddle; the finite-N game is the saddle plus fluctuations of size $O(N^{-1/2})$, exactly as the central limit theorem of chapter 3 quantifies for a fixed dynamics and as propagation of chaos for Nash equilibria (chapter 13) establishes for the homogeneous game. The systematic object the physicist expects — a controlled expansion of the finite-N equilibrium around the mean field, order by order in $N^{-1/2}$ — is precisely what is *not* in general available, and building it is the cleanest entry point for the physics-trained reader. We can, however, compute its leading term completely in the rank-one graphon game, and the computation is worth doing in full because it exhibits both the phenomenon (a susceptibility that diverges at the critical coupling) and the exact boundary of the rigorous theory.

We work with the *static* rank-one graphon game, the stationary shadow (section 20.7) of Thread B, because there the fluctuation calculation is fully rigorous; the dynamic version is the open problem that follows.

The one analytic input beyond the elementary central limit theorem is a local-uniform law of large numbers for the self-consistency map, *with* its derivative. We isolate it as a lemma and prove it from the in-book strong law and the Arzelà–Ascoli argument already used for Varadarajan's theorem in worked computation 3.26, so that nothing beyond the prerequisite line is invoked by name.

Lemma 31.8 (Local-uniform law of large numbers for the self-consistency map)**.** *Let $x_1, x_2, \dots$ be i.i.d.* $\mathrm{Unif}[0,1]$*, let $f \in L^2[0,1]$, and let $(\mathrm{br}_x)_{x\in[0,1]}$ be best responses that are C^1 in their scalar argument with a common bound and modulus of continuity on the derivative: there exist $L_{\mathrm{br}} < \infty$ and a nondecreasing ω with $\omega(0^+) = 0$ such that*

$$|\mathrm{br}'_x(u)| \le L_{\mathrm{br}}, \qquad |\mathrm{br}'_x(u) - \mathrm{br}'_x(u')| \le \omega(|u-u'|) \qquad (\textit{all } x, u, u').$$

Set $H_N(\varrho) = \frac{1}{N}\sum_{j\le N} f(x_j)\,\mathrm{br}_{x_j}(f(x_j)\varrho)$ and $h(\varrho) = \mathbb{E}[f(x)\mathrm{br}_x(f(x)\varrho)]$. Then on every compact $K \subset \mathbb{R}$, almost surely $H_N \to h$ and $H'_N \to h'$ uniformly *on K, where $H'_N(\varrho) = \frac{1}{N}\sum_{j\le N} f(x_j)^2\,\mathrm{br}'_{x_j}(f(x_j)\varrho)$ and $h'(\varrho) = \mathbb{E}[f(x)^2\mathrm{br}'_x(f(x)\varrho)]$; in particular $h \in C^1(\mathbb{R})$.*

Proof. Put $\bar\omega := \min(2L_{\mathrm{br}}, \omega) \le 2L_{\mathrm{br}}$, still a modulus. On K two domination bounds hold. By the mean value theorem and $|\mathrm{br}'_x| \le L_{\mathrm{br}}$,

$$|H_N(\varrho) - H_N(\varrho')| \le \Big(\frac{1}{N}\sum_{j\le N} L_{\mathrm{br}}\, f(x_j)^2\Big)\,|\varrho-\varrho'|, \qquad |H'_N(\varrho) - H'_N(\varrho')| \le \frac{1}{N}\sum_{j\le N} f(x_j)^2\,\bar\omega(|f(x_j)|\,|\varrho-\varrho'|).$$

The prefactor $\frac{1}{N}\sum_j f(x_j)^2 \to \|f\|^2_{L^2}$ almost surely by the strong law of large numbers (proposition 3.24; only the first moment $\mathbb{E}[f(x)^2] = \lambda_1 < \infty$ of the i.i.d. summands is needed, the Kolmogorov form recorded in section 3.4.2), so $\{H_N\}$ is a.s. uniformly Lipschitz, hence equicontinuous, on K. For the derivatives, since $f(x)^2\bar{\omega}(|f(x)|\delta) \le 2L_{\mathrm{br}} f(x)^2 \in L^1$ and $\bar{\omega}(0^+) = 0$, dominated convergence gives $G(\delta) := \mathbb{E}[f(x)^2\bar{\omega}(|f(x)|\delta)] \downarrow 0$ as $\delta \downarrow 0$; fixing $\varepsilon > 0$ and δ with $G(\delta) < \varepsilon$, the strong law makes $\frac{1}{N}\sum_j f(x_j)^2\bar{\omega}(|f(x_j)|\delta) \to G(\delta) < \varepsilon$ a.s., so by monotonicity of $\bar{\omega}$ the family $\{H'_N\}$ is a.s. asymptotically equicontinuous on K. At each ϱ in a countable dense set $D \subset K$ the summands of $H_N(\varrho)$ and of $H'_N(\varrho)$ are i.i.d. with finite mean (dominated by $|f(x)\mathrm{br}_x(0)| + L_{\mathrm{br}} f(x)^2|\varrho|$ and by $L_{\mathrm{br}} f(x)^2$, both in L^1), so the strong law gives $H_N(\varrho) \to h(\varrho)$ and $H'_N(\varrho) \to h'(\varrho)$; intersecting the countably many null sets makes both hold simultaneously on D. Equicontinuity plus convergence on a dense set upgrades to uniform convergence on K — the Arzelà–Ascoli argument used for Varadarajan's theorem in worked computation 3.26. Finally h is the uniform limit of H_N and h' the uniform limit of the continuous H'_N, so $h \in C^1(K)$ with the stated derivative; as K was arbitrary, $h \in C^1(\mathbb{R})$. □

Worked computation 31.9 (Order-parameter susceptibility of the rank-one graphon game)**.** Sample N agents with i.i.d. types $x_1, \dots, x_N \sim \mathrm{Unif}[0,1]$ and couple them through the rank-one graphon $W(x,y) = f(x)f(y)$, $f \in L^2[0,1]$, $f \not\equiv 0$, with leading eigenvalue $\lambda_1 = \|f\|^2_{L^2}$ (proposition 24.25). Only the L^2/Hilbert–Schmidt structure of W enters this fluctuation computation — the operator facts $\mathbb{W}$ Hilbert–Schmidt and $\lambda_1 = \|f\|^2_{L^2}$, and the integrability displayed below — so we treat ff as an L^2 kernel rather than as a $[0,1]$-valued graphon; if one wants ff to be a bona fide graphon satisfying *(A-Reg-W)* it suffices to take f bounded (proposition 24.25), but boundedness is not used here. In the static LQ graphon game each agent i chooses an effort and best-responds to the aggregate it feels, $z_i = \frac{1}{N}\sum_j W(x_i, x_j)\,\alpha_j = f(x_i)\,\hat{\varrho}_N$, where

$$\hat{\varrho}_N \;:=\; \frac{1}{N}\sum_{j=1}^{N} f(x_j)\,\alpha_j \tag{31.2}$$

is the empirical order parameter (it aggregates the agents' *actions* α_j; see the notation watch below). With a smooth scalar best response $\mathrm{br}_x(\cdot)$ (under *(A-Lip)*, br_x is Lipschitz), the Nash condition $\alpha_i = \mathrm{br}_{x_i}(f(x_i)\hat{\varrho}_N)$ closes on the single scalar $\hat{\varrho}_N$, which must solve the self-consistency

$$\hat{\varrho}_N \;=\; H_N(\hat{\varrho}_N), \qquad H_N(\varrho) \;:=\; \frac{1}{N}\sum_{j=1}^{N} f(x_j)\,\mathrm{br}_{x_j}\!\big(f(x_j)\,\varrho\big). \tag{31.3}$$

As $N \to \infty$, the local-uniform law of large numbers of lemma 31.8 gives $H_N(\varrho) \to h(\varrho) := \mathbb{E}\big[f(x)\mathrm{br}_x(f(x)\varrho)\big]$ uniformly on compacts, and the mean field order parameter ϱ^* is the fixed point $h(\varrho^*) = \varrho^*$, with mean field slope

$$g \;:=\; h'(\varrho^*) \;=\; \mathbb{E}\big[f(x)^2\,\mathrm{br}'_x(f(x)\varrho^*)\big]. \tag{31.4}$$

(Here H_N, h and g are local to this static computation; they are *not* the Hamiltonian H or the terminal cost $g = g$ of the rest of the book, neither of which appears in the stationary calculation.) In the LQ case the static Thread B best response is *affine*: by eqs. (20.29) and (20.32) a type-x agent's equilibrium action is

$$\mathrm{br}_x(z) \;=\; \frac{\eta(x)}{1+q} \;+\; \mathsf{b}\,z, \tag{31.5}$$

a nonzero type-dependent intercept (the ideal point $\eta(x)$) plus the strategic-slope term, with $\mathsf{b} = \frac{qs}{1+q}$ the static interaction coefficient eq. (20.30). Then $\mathrm{br}_x' \equiv \mathsf{b}$ (so the modulus hypothesis of lemma 31.8 is vacuous) and $g = \mathsf{b}\,\mathbb{E}[f(x)^2] = \mathsf{b}\,\|f\|_{L^2}^2 = \mathsf{b}\,\lambda_1$ — the strategic coefficient times the leading eigenvalue — so the subcriticality condition $g < 1$ is exactly the static contraction/uniqueness threshold $\mathsf{b}\lambda_1 < 1$ of theorem 20.14 and eq. (20.31). The fixed point is then $\varrho^* = \langle \eta, f\rangle / [(1+q)(1-\mathsf{b}\lambda_1)]$ (eq. (20.32)), so a *nondegenerate* susceptibility requires a nonzero ideal-point field, $\langle \eta, f\rangle \neq 0$: in the symmetric case $\eta \equiv 0$ one has $\varrho^* = 0$, the limiting variance V of the central limit theorem below collapses to $\mathrm{Var}(\mathsf{b}f(x)^2\varrho^*) = 0$, and eq. (31.6) degenerates to $\sqrt{N}\,\hat{\varrho}_N \Rightarrow 0$ — the genuine fluctuation theory of that pure-consensus point sitting at the next order in $N^{-1/2}$.

Claim. If $|g| < 1$ (subcritical coupling) and br_x is C^1 with $\mathbb{E}\big[f(x)^2 \mathrm{br}_x(f(x)\varrho^*)^2\big] < \infty$, then the order-parameter fluctuation obeys a central limit theorem,

$$\sqrt{N}\,(\hat{\varrho}_N - \varrho^*) \;\Rightarrow\; \mathcal{N}(0,\; \chi^2 V), \qquad \chi \;:=\; \frac{1}{1-g}, \qquad V \;:=\; \mathrm{Var}\big(f(x)\,\mathrm{br}_x(f(x)\varrho^*)\big), \tag{31.6}$$

where χ is the *susceptibility*.

Proof. This is the delta method applied to the implicitly defined estimator $\hat{\varrho}_N$. Write $\Psi_N(\varrho) := \varrho - H_N(\varrho)$ and $\Psi(\varrho) := \varrho - h(\varrho)$, so $\hat{\varrho}_N$ and ϱ^* are the respective zeros. By lemma 31.8 (whose hypotheses hold under *(A-Lip)* and the stated integrability), $H_N \to h$ and $H_N' \to h'$ uniformly on a compact neighbourhood of ϱ^*, so $\Psi_N \to \Psi$ and $\Psi_N' \to \Psi'$ uniformly there, with $\Psi'(\varrho^*) = 1 - g \neq 0$. Since $|g| < 1$ and h' is continuous, fix $\varepsilon \in (0, 1-g)$ and an interval $I = [\varrho^* - r, \varrho^* + r]$ on which $h' \leq g + \varepsilon < 1$; then $\Psi = \mathrm{Id} - h$ is strictly increasing on I with $\Psi(\varrho^*) = 0$ and $\Psi(\varrho^* \pm r)$ of opposite, nonzero signs. Once $\sup_I |H_N' - h'| < \varepsilon$ (a.s. for N large) one has $\Psi_N' = 1 - H_N' > 0$ on I, so Ψ_N is strictly increasing there and inherits the endpoint signs; hence Ψ_N has a unique zero $\hat{\varrho}_N \in I$, with $\hat{\varrho}_N \to \varrho^*$ a.s. A first-order Taylor expansion of Ψ_N about ϱ^* gives

$$0 = \Psi_N(\hat{\varrho}_N) = \Psi_N(\varrho^*) + \Psi_N'(\tilde{\varrho}_N)\,(\hat{\varrho}_N - \varrho^*), \qquad \tilde{\varrho}_N \in [\varrho^*, \hat{\varrho}_N],$$

and $\Psi_N'(\tilde{\varrho}_N) = 1 - H_N'(\tilde{\varrho}_N) \to 1 - g$ a.s., since $\tilde{\varrho}_N \to \varrho^*$ and $H_N' \to h'$ uniformly on I (lemma 31.8). Therefore

$$\sqrt{N}\,(\hat{\varrho}_N - \varrho^*) = \frac{-\sqrt{N}\,\Psi_N(\varrho^*)}{\Psi_N'(\tilde{\varrho}_N)} = \frac{\sqrt{N}\,(H_N(\varrho^*) - \varrho^*)}{1 - g + o_{\mathbb{P}}(1)}.$$

The numerator is $\sqrt{N}\big(H_N(\varrho^*) - h(\varrho^*)\big)$ since $h(\varrho^*) = \varrho^*$, and $H_N(\varrho^*) = \frac{1}{N}\sum_j f(x_j)\mathrm{br}_{x_j}(f(x_j)\varrho^*)$ is an average of i.i.d. terms with mean $h(\varrho^*)$ and variance V; the classical central limit theorem (theorem 3.22) gives $\sqrt{N}(H_N(\varrho^*) - h(\varrho^*)) \Rightarrow \mathcal{N}(0, V)$. Because the denominator $\Psi_N'(\tilde{\varrho}_N) \to 1 - g$, a *constant*, in probability, the pair converges jointly, $\big(\sqrt{N}(H_N(\varrho^*) - h(\varrho^*)),\, \Psi_N'(\tilde{\varrho}_N)\big) \Rightarrow (\mathcal{N}(0, V),\, 1-g)$; the continuous mapping theorem (corollary 3.11) applied to $(a, b) \mapsto a/b$, continuous at $b = 1 - g \neq 0$ — this is the elementary content of Slutsky's theorem — yields eq. (31.6). □

Notation watch

The static order parameter $\hat{\varrho}_N$ eq. (31.2) aggregates the agents' *actions*, $\hat{\varrho}_N = \frac{1}{N}\sum_j f(x_j)\alpha_j$. It is the stationary-game analogue of the *dynamic* mean aggregate $\varrho_t = \langle f, \bar{m}_t\rangle$ of the rank-one Thread B reduction (worked computation 30.7), which aggregates type-conditional state means; the reuse of the single symbol ϱ across the static/dynamic boundary is intentional, the static action aggregate being exactly the stationary-shadow image of the dynamic mean aggregate under section 20.7.

The content of worked computation 31.9 is the factor $\chi = (1-g)^{-1}$. The fluctuation of the order parameter is the bare fluctuation of a single agent's contribution, of variance V/N, *amplified* by χ^2: each agent's deviation perturbs the aggregate, which perturbs every agent's best response, which feeds back into the aggregate, and the geometric series of this feedback sums to $\chi = 1/(1-g)$. As the coupling approaches the critical value $g \uparrow 1$ — i.e. $\mathsf{b}\lambda_1 \uparrow 1$ in the LQ case — the susceptibility diverges and the order-parameter fluctuations blow up: the mean field description, though still the correct *mean*, no longer suppresses fluctuations, and the Gaussian approximation eq. (31.6) degrades. This is the rigorous fingerprint, in a mean field game, of a critical point.

The critical surface $g = \mathsf{b}\lambda_1 = 1$ reached here is the *static* contraction/uniqueness threshold eq. (20.31) (theorem 20.14), the stationary shadow (section 20.7) of the dynamic uniqueness threshold eq. (31.1), not that threshold itself. The two do not coincide on the nose: the static slope is $\mathsf{b} = \frac{qs}{1+q}$ eq. (20.30), so in terms of the coupling s the static face reads $\mathsf{b}\lambda_1 = 1 \Leftrightarrow s\lambda_1 = \frac{1+q}{q} = 1 + \frac{1}{q}$, whereas the dynamic threshold eq. (31.1) at $a = 0$ reads $s\lambda_1 = 1$. The two critical surfaces therefore differ by the effort-cost factor $\frac{q}{1+q}$ and align only in the strong-congestion limit $q \to \infty$, where $\mathsf{b} \to s$; as section 20.7 stresses, b is a Riccati composite of the dynamic data, not a literal reading of s, and the correspondence is the structural alignment of two "coupling hits the inverse top eigenvalue" thresholds, not a parameter identity. What is exact in *both* pictures is the qualitative statement the susceptibility makes precise: χ diverges precisely as the effective coupling g — static or dynamic — reaches unity along the leading eigenmode ϕ_1.

Notation watch

We have called χ a "susceptibility" and will, just below, borrow the words "critical exponent" and "correlation length." The first is apt: χ is exactly the linear response of the equilibrium order parameter to an infinitesimal uniform field, and diverges with the same mean-field exponent (see below). But "correlation length" has *no literal meaning here*: the type variable $x \in [0,1]$ is an exchangeable label, not a spatial coordinate (appendix A.5), so there is no distance in which correlations decay. What plays the role of a diverging correlation length is the *spectral* concentration of the critical fluctuation onto the single eigenmode ϕ_1 — the loss of effective rank, not the growth of a length. Read every spatial metaphor in this section as a statement about the spectrum of $\mathbb{W}$, never about $|x-y|$.

Physics note

Worked computation 31.9 is the Curie–Weiss susceptibility, transcribed. Writing the distance to criticality as $\epsilon := 1 - g \propto (s_c - s)$ near the threshold, the susceptibility diverges as $\chi \sim \epsilon^{-1}$ — the mean field critical exponent $\gamma = 1$ — and the order-parameter variance as $\mathrm{Var}(\hat{\varrho}_N) \sim N^{-1}\epsilon^{-2}$, so the Gaussian fluctuation theory is self-consistent only for $N\epsilon^2 \gg 1$. Inside the critical window $N\epsilon^2 \lesssim 1$ the fluctuations are no longer Gaussian and the expansion needs its next order: this is the Ginzburg criterion for the mean field game, and the open question it poses is whether the *graphon* changes the exponent. For a dense graphon with a spectral gap (finite effective rank) one expects the mean field exponents to survive, exactly as Curie–Weiss is exact above the upper critical dimension. For a sparse graphon, where the spectrum of $\mathbb{W}$ is dense near its edge and many modes go critical together, the analogue of a *lower critical dimension* should appear and the exponents should shift. Making this dichotomy a theorem — mean field exponents iff a spectral gap — is, to our knowledge, entirely open and is the single most physics-native problem in the book.

Open problem 31.10. *(Dynamic fluctuation expansion.)* Extend worked computation 31.9 to the *dynamic* graphon mean field game: prove a central limit theorem for the fluctuation of the finite-N closed-loop Nash equilibrium around the graphon mean field limit, identifying the limiting Gaussian field as the solution of a linearised forward–backward system whose covariance is governed by a susceptibility operator $(\mathrm{Id} - \mathcal{G})^{-1}$ with $\mathcal{G}$ the linearisation of the best-response map (the spectral radius of $\mathcal{G}$ reaching 1 at the critical coupling eq. (31.1)). Determine whether the mean field critical exponents are preserved for graphons with a spectral gap and shifted for sparse graphons.

The homogeneous ($W \equiv \mathrm{const}$) case of Open problem 31.10 is exactly the central limit theorem of Delarue, Lacker and Ramanan [58], with the companion large-deviation and concentration results of [59]; their fluctuation field solves a linear forward–backward stochastic system, and the susceptibility operator is implicit in its well-posedness. The graphon generalisation — the operator $\mathcal{G}$ acquiring the spectral structure of $\mathbb{W}$, the critical exponent question — is open, and the static computation above is its zeroth chapter.

31.5.2 Renormalisation: the cut metric as a coarse-graining

The second tool is the renormalisation group, and the book has already, without naming it, laid down its kinematics. The weak regularity lemma (chapter 19) states that every graphon is approximated in cut metric by a step graphon of bounded complexity — a coarse-graining that replaces a fine type structure by finitely many blocks while preserving all macroscopic (homomorphism-density) observables (chapter 16). This is a renormalisation-group transformation in everything but name: a map on the space of couplings $\widetilde{\mathcal{W}}_0$ that integrates out fine type structure and is controlled in the metric ($\delta_\square$) under which the physics (the equilibrium, chapter 25) is continuous.

Heuristic (non-rigorous)

The renormalisation-group program for graphon mean field games would be to iterate the coarse-graining of chapter 19 and ask for its fixed points and their stability. A *fixed point* of the coarse-graining is a graphon scale-invariant under block aggregation; the constant graphon (Erdős–Rényi) and the rank-one graphon are the obvious candidates, and the rank-one fixed point is the one on which the entire equilibrium collapses to the single order parameter ϱ of worked computation 31.9. The *relevant* and *irrelevant* directions would be the eigenmodes of $\mathbb{W}$ ordered by eigenvalue: near the critical coupling, the leading mode ϕ_1 is relevant (it carries the diverging susceptibility) and the rest are irrelevant (their susceptibilities $1/(1-g_k)$ with $g_k \propto \lambda_k$ stay finite). This is, heuristically, why the critical behaviour is universal across graphons with the same leading eigenvalue — the irrelevant modes wash out under coarse-graining — and it predicts that the critical exponents depend only on the *spectral edge* of $\mathbb{W}$, matching the Ginzburg-criterion dichotomy of the physics note. None of this is a theorem; it is the most promising heuristic in the chapter.

Open problem 31.11. *(A renormalisation semigroup on graphon games.)* Make the coarse-graining of chapter 19 into a semigroup on $(\widetilde{\mathcal{W}}_0, \delta_\Box)$ acting jointly on the graphon and the cost data, prove that the induced map on equilibria is continuous (a quantitative $\delta_\Box$-stability, strengthening chapter 25), and characterise its fixed points and the relevant/irrelevant splitting of the spectrum of $\mathbb{W}$ at a fixed point. Deduce a universality statement: graphon games with the same leading spectral data have the same critical exponents.

31.5.3 Diagrammatics: the loop expansion of the master equation

The third tool is diagrammatic bookkeeping. The master equation (chapters 12 and 26) is a Hamilton–Jacobi equation on the space of measures, and the physicist reads it as the effective action whose functional derivatives generate the equilibrium correlation functions (remark 12.11). The mean field game is its tree-level (saddle) approximation; the fluctuation expansion of section 31.5.1 is its one-loop correction; and the systematic $N^{-1/2}$ expansion the physicist wants is its loop expansion. The vertices of the diagrams are the higher flat derivatives $\delta^k U/\delta m^k$ of the master field, the propagator is the susceptibility operator $(\mathrm{Id} - \mathcal{G})^{-1}$ of Open problem 31.10, and the graphon enters every vertex and propagator through $\mathbb{W}$.

To keep this dictionary anchored rather than decorative, read it against the one object the chapter made rigorous, worked computation 31.9. There the *tree level* is the mean field order parameter ϱ^* (the saddle $h(\varrho^*) = \varrho^*$); the single *propagator* insertion is exactly the scalar reduction $(\mathrm{Id} - \mathcal{G})^{-1} \rightsquigarrow (1-g)^{-1} = \chi$ — the geometric series of best-response feedback summed in eq. (31.6), with $\mathcal{G}$ collapsing to multiplication by the slope $g = \mathsf{b}\lambda_1$; and the first nontrivial *vertex* is the curvature br''_x of the best response (zero in the LQ case, which is why the rank-one computation is exactly one-loop and the Gaussian eq. (31.6) is exact). The loop expansion is then the systematic restoration of the br''_x and higher vertices that the linear rank-one reduction set to zero.

Open problem 31.12. *(Loop expansion with error control.)* Develop the $N^{-1/2}$ expansion of the finite-N graphon game as a diagrammatic series in the derivatives of the master field, with the propagator $(\mathrm{Id} - \mathcal{G})^{-1}$, and — the part that distinguishes mathematics from physics — prove a remainder bound making the truncated expansion a genuine asymptotic series, valid away from the critical coupling where the propagator is singular. Resum the leading divergences near criticality to recover the corrected exponents of Open problem 31.10.

The honest caveat is that diagrammatic expansions are notoriously asymptotic rather than convergent, and the mathematical content of Open problem 31.12 is entirely in the remainder bound that physics leaves implicit. But the bookkeeping is real, the propagator is the same operator that controls uniqueness, and the divergence structure near criticality is exactly the susceptibility blow-up the worked computation made rigorous — so the diagrammatic language is not decoration; it is the natural organisation of the corrections, and turning it into controlled analysis is a concrete and, we believe, achievable program.

31.6 Closing synthesis

The three subsections of section 31.5 are not three programs but one, seen through three instruments. The fluctuation expansion computes the corrections; the renormalisation group organises which corrections matter; the diagrammatics is the bookkeeping that connects them to the master equation. And all three turn on a single object — the susceptibility operator $(\mathrm{Id} - \mathcal{G})^{-1}$, whose spectral radius reaching unity is, at once, the failure of uniqueness (section 31.2.1), the onset of equilibrium multiplicity (section 31.2.2), the blow-up of fluctuations (section 31.5.1), the breakdown of best-response learning (section 31.3.2), and the relevant direction of the renormalisation flow (section 31.5.2). The unification this book has been building toward is that all of these are the *same* critical point, and that the graphon enters every one of them through the same spectral data of $\mathbb{W}$: the leading eigenvalue λ_1 sets where the critical point is, and the spectral gap sets whether the transition is mean-field-like.

The mathematics that remains is, in this light, sharply posed. The proved theory occupies the subcritical, dense, idiosyncratic-noise corner where the susceptibility operator is invertible and everything is a contraction. The open problems are the approach to and passage through the critical surface where it is not — and the physicist's contribution, if the bet of this book is right, will be to supply the methods that were invented for exactly that purpose, made into theorems. The graphon mean field game is a statistical field theory of strategic agents on a network; we have built its mean field theory rigorously, and its critical theory is open.

Rigor ledger

Proved.

- worked computation 31.9 (eq. (31.6)): for the static rank-one graphon game in the subcritical regime $|g| < 1$, the empirical order parameter $\hat{\varrho}_N$ satisfies a central limit theorem with susceptibility $\chi = 1/(1-g)$ and limiting variance $\chi^2 V$; the susceptibility diverges as the coupling approaches the critical value $g \uparrow 1$, i.e. $\mathsf{b}\lambda_1 \uparrow 1$, which is the *static* contraction/uniqueness threshold eq. (20.31) — the stationary shadow (section 20.7) of the dynamic threshold eq. (31.1), the two critical surfaces differing by the effort-cost factor $q/(1+q)$ and coinciding only as $q \to \infty$. This is the rigorous leading-order fluctuation theory and the one new computation of the chapter.

- lemma 31.8: a local-uniform law of large numbers for the self-consistency map and its derivative, proved in-book from the strong law (proposition 3.24) and the Arzelà–Ascoli argument of worked computation 3.26; it supplies the consistency and the denominator convergence of the delta-method proof.

Assumed.

- The classical central limit theorem (theorem 3.22), the strong law of large numbers (proposition 3.24, with the first-moment Kolmogorov form recorded in section 3.4.2), and the continuous mapping theorem (corollary 3.11), applied to the i.i.d. type array; *(A-Lip)* and an integrability/C^1-modulus condition on the best response, used to dominate the families differentiated in the delta-method argument (via lemma 31.8). The local-uniform law with derivative convergence is *not* imported as a black box: it is the in-book lemma 31.8.

- All imported structural results: the uniqueness thresholds theorem 11.5, theorem 24.16 and proposition 24.18, the spectral reduction worked computation 24.22 and proposition 24.23, the static contraction threshold theorem 20.14, the master equation theorem 12.8, and the cut-metric stability of chapter 25.

Deferred.

- The homogeneous dynamic fluctuation central limit theorem (the $W \equiv$const case of Open problem 31.10) is proved in [58, 59]; we cite rather than reproduce it, the graphon generalisation being open.

- The sparse/L^p/graphex theory underlying Open problem 31.3 → chapter 27; the numerical realisations underlying Open problem 31.5 → chapter 28; the learning algorithms of Open problem 31.6 → chapter 29.

Open.

- Open problem 31.1: a necessary-and-sufficient uniqueness threshold for general couplings, recovering eq. (31.1).

- Open problem 31.2: equilibrium selection past the critical coupling, and whether common noise restores uniqueness.

- Open problem 31.3: master-equation well-posedness with common noise on sparse limits — the hardest and most consequential problem.

- Open problem 31.4: the graphon game under partial information beyond the linear-Gaussian separation of proposition 30.16 — nonlinear filtering of the aggregate field, and the canonical hard case of a partially observed major player on a sparse network (remark 30.18).

- Open problems 31.5 to 31.7: certified low-rank solvers, learning past the contraction threshold, behavioural identifiability of the graphon.

- Open problems 31.10 to 31.12: the dynamic fluctuation expansion, a renormalisation semigroup on graphon games with a universality statement, and a controlled loop expansion of the master equation — the physicist's agenda.

Bibliographic notes

The organising claim of this chapter — that the modern landscape of mean field games beyond perfect homogeneity is best read through the graphon picture, with the remaining gaps located by which axis of generality is relaxed — follows the recent overview [11], which we have used as the map for chapter 30 and this chapter. The two uniqueness regimes contrasted in section 31.2.1 are the Lasry–Lions monotonicity theory [93, 36] and its graphon operator lift of chapter 24; the displacement-monotonicity refinements that widen the sufficient region are due to Gangbo and Mészáros and collaborators [75, 76], and the master-equation well-posedness with nonsmooth data to Mou and Zhang [104]; none is known to be sharp. The equilibrium-selection question of section 31.2.2 and the conjecture that common noise restores uniqueness are crystallised in the two-state model of Cecchin, Dai Pra, Fischer and Pelino [46]; common noise for the classical theory is Carmona–Delarue [41] and Ahuja [4]. The sparse side of section 31.2.3 rests on the graphon-system limit theory of Bayraktar, Wu and Zhang [16] and the sparse-graph frameworks of chapter 27.

For the computation gaps: the forward–backward numerics behind section 31.3.1 are Achdou and Capuzzo-Dolcetta [1] (state space) combined with the graphon discretisation of chapter 28; the learning theory of section 31.3.2 is the GMFG-PPO line of Cui, Koeppl and collaborators [55] developed in chapter 29; and the graphon-estimation results behind section 31.3.3 are Gao, Lu and Zhou [77] (dense, minimax) and Klopp, Tsybakov and Verzelen [87] (sparse), with the identification only up to $\delta_\square$-equivalence being the lesson of chapter 17 [29]. The application frontiers of section 31.4 cite, respectively, the finite-state graphon epidemic games of Aurell, Carmona, Dayanikli and Laurière [9]; the smart-grid mean field models of Couillet, Perlaza, Tembine and Debbah [51] and Alasseur, Ben Tahar and Matoussi [5]; the opinion-dynamics model of Bauso, Tembine and Başar [14] (which is Thread B verbatim); and the intervention/comparative-statics theory of the static graphon game of Parise and Ozdaglar [108] that chapter 20 develops.

The fluctuation expansion of section 31.5.1 is, in the homogeneous case, the central limit theorem of Delarue, Lacker and Ramanan [58] with the companion large-deviation and concentration results [59]; our static rank-one computation worked computation 31.9 is the elementary delta-method shadow of their forward–backward fluctuation field, made explicit to expose the susceptibility $\chi = 1/(1-g)$ and its divergence at the critical coupling of chapter 24. The renormalisation reading of section 31.5.2 rests on the weak regularity lemma and the cut-metric compactness of Lovász and Szegedy [100, 99] developed in chapters 17 and 19; the diagrammatic reading of section 31.5.3 is the effective-action interpretation of the master equation [37] of chapter 12. The position that these three physical instruments — fluctuation expansion, renormalisation group, diagrammatics — are the natural tools for the critical theory of graphon mean field games is the thesis of this book, and to our knowledge their rigorous development for the graphon game is, as of this writing, open.

Part VI

Appendices

Appendix A

Master Notation and Conventions

This appendix is the canonical reference for notation. Where the body of the book and this appendix appear to disagree, this appendix wins, and the discrepancy should be reported as an erratum. The conventions here are enforced uniformly across all chapters.

A.1 Standing conventions

Time and the two directions of flow. The time horizon is $[0, T]$ with $0 < T < \infty$. Two objects evolve in opposite directions and this is the most common source of confusion in the subject:

- The *value function* $u = u(t, x)$ is determined by a *terminal* condition at $t = T$ and is computed *backward* in time. It is the optimal cost-to-go from the present to the deadline.
- The *population density* $m = m(t, x)$ is determined by an *initial* condition at $t = 0$ and evolves *forward* in time. It is transported by the agents' optimal feedback.

Every chapter in which both appear restates this on first use. The coupling of a backward equation to a forward equation, with data specified at opposite ends of the interval, is the structural signature of a mean field game and has no initial-value analogue.

Generators and their adjoints. For an Itô diffusion with drift b and volatility σ we write the infinitesimal generator as

$$(\mathcal{A}\phi)(x) \;=\; b(x) \cdot \nabla\phi(x) \;+\; \tfrac{1}{2}\,\mathrm{tr}\big(\sigma(x)\sigma(x)^{\top}\,\mathrm{D}^2\,\phi(x)\big).$$

We use the symbol $\mathcal{A} = \mathcal{A}$ for the generator, reserving $\mathcal{L} = \mathcal{L}$ for the law (distribution) of a random variable; this avoids the common clash in which L denotes both. The formal adjoint $\mathcal{A}^{*}$ is the Fokker–Planck (Kolmogorov forward) operator: densities solve $\partial_t m = \mathcal{A}^{*} m$.

Notation watch

A physicist is used to writing the evolution of densities with a Liouvillian or with $-H$ in the exponent and may carry sign habits from the Schrödinger or master-equation conventions. Here the forward operator is $\mathcal{A}^*$ and the sign is fixed by the requirement that $\int m \, \mathrm{d}x$ be conserved and that $m \geq 0$ be preserved; verify signs against this conservation law, not against a remembered Hamiltonian sign.

The discount rate. The symbol $\beta = \beta$ always denotes a (nonnegative) discount rate in a control or game problem. It is *never* an inverse temperature. On the rare occasion that an inverse temperature enters a physical analogy it is written β_{T} and flagged.

A.2 Spaces and measures

$\mathbb{R},\ \mathbb{N},\ \mathbb{Z},\ \mathbb{C}$	the usual number sets
$\mathbb{R}^d$	state space of a single agent
$[0,1] = [0,1]$	the graphon *type* interval (see below)
$\mathcal{P}(\mathbb{R}^d)$	Borel probability measures on $\mathbb{R}^d$
$\mathcal{P}_2(\mathbb{R}^d)$	those with finite second moment, the home of the mean field
$\mathcal{W}_2$	the quadratic Wasserstein distance on $\mathcal{P}_2(\mathbb{R}^d)$
$L^p,\ H^s$	Lebesgue and Sobolev spaces
$\mathbb{E},\ \mathbb{P}$	expectation and probability
$\mathcal{L}(X)$	the law of the random variable X

Probability measures as points. The "mean field" is a point $m \in \mathcal{P}_2(\mathbb{R}^d)$, not a list of agents. A best response is a map $\mathcal{P}_2(\mathbb{R}^d) \to \mathcal{P}_2(\mathbb{R}^d)$ and an equilibrium is its fixed point; the Wasserstein metric is the distance in which that map is shown to be continuous or contractive.

A.3 Differentiation on the space of measures

Two notions of derivative with respect to the measure argument recur.

- The *flat* (linear functional) derivative $\dfrac{\delta F}{\delta m}(m)(x)$, defined by

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon} F(m + \epsilon(m' - m))\Big|_{\epsilon=0} = \int \frac{\delta F}{\delta m}(m, x)\,(m' - m)(\mathrm{d}x),$$

  which behaves exactly like the functional derivative $\delta F / \delta \rho(x)$ familiar from density-functional calculations.

- The *Lions* (L-)derivative $\partial_\mu F$, obtained by lifting F to a function of random variables on an atomless probability space and differentiating there; it relates to the flat derivative by $\partial_\mu F(m)(x) = \nabla_x \, \delta F / \delta m(m)(x)$.

Both are developed in chapter 4 and appendix D.

A.4 Control and games

$u = u(t,x)$	value function (HJB unknown), solved backward
$m = m(t,x)$	population density, solved forward
$\alpha = \alpha$	control / action; A the action set
$L = L$	running cost (Lagrangian)
$H = H(x,p,m)$	Hamiltonian, convex in p under (A-Conv)
$\alpha^* = -\partial_p H$	optimal feedback under (A-Conv)
$g = g$	terminal cost
$\mathsf{S} = \Phi$	best-response map whose fixed point is the equilibrium

A.5 Graphons

$x = x \in [0,1]$	the *type* / *label* of an agent
$W = W(x,y)$	a symmetric measurable graphon kernel
$\mathbb{W}$	its integral operator, $(\mathbb{W}f)(x) = \int_0^1 W(x,y)f(y)\,\mathrm{d}y$
$m^x = m^x$	the type-indexed population field
$\|\cdot\|_\square$	the cut norm; $\delta_\square$ the cut metric
$t(F,G)$	homomorphism density of F in G

Notation watch

The type variable $x \in [0,1]$ is an *exchangeable label* for an agent's kind or network position. It is *not* a spatial coordinate and *not* a momentum. Although the coupling $W(x,y)$ superficially resembles a translation-noninvariant hopping kernel on a lattice, x carries no metric or geometric meaning of its own beyond what W assigns it; in particular $|x-y|$ has no intrinsic significance. When a formula could tempt the reader to read x as position, the text says "type" explicitly.

Notation watch

Brownian motion versus graphon: both want the letter W. In the stochastic and classical chapters (Parts I–II), where no graphon is present, Brownian motion is W_t (rendered W_t), following standard usage. From Part III onward, where the graphon W is in scope, Brownian motion is written B_t. Thus in any equation containing a graphon, W is the kernel and B is the noise. This switch is announced again at the start of Part IV.

A.6 Standing assumptions (named)

These are stated once and referenced by tag throughout; full statements appear at first substantive use.

(A-Lip) Lipschitz coefficients and costs in the state, uniformly in time and measure.

(A-LipM) Measure-Lipschitz dependence: coefficients and costs are Lipschitz in the measure argument with respect to $\mathcal{W}_2$, uniformly in state and time (distinct from (A-Lip), which controls the state argument). Used for $\mathcal{W}_2$-continuity and contraction estimates of the best-response map; see section 4.3.

(A-Growth) At most linear growth in the state; coercivity sufficient to define the Hamiltonian.

(A-Nondeg) Uniformly nondegenerate diffusion, $\sigma\sigma^\top \succeq \lambda I$, $\lambda > 0$.

(A-Conv) $H(x, p, m)$ convex in p.

(A-Mon) Lasry–Lions monotonicity of the couplings (Chapter 11).

(A-Reg-W) W symmetric, measurable, bounded, and continuous in type where needed; $\mathbb{W}$ self-adjoint Hilbert–Schmidt on $L^2[0, 1]$.

(A-Coupling) Operator-norm smallness: $\|\mathbb{W}\|$ lies below the contraction/uniqueness threshold (chapter-specific; for the rank-one LQG of chapter 22 this is the effective coupling $s\,\|\mathbb{W}\| < 1$). We do not name the threshold by a fixed letter, reserving $\|\mathbb{W}\|$ for the operator norm itself.

A.7 The carried threads

The linear–quadratic–Gaussian thread (Thread A) and its graphon generalization (Thread B) are defined once, with fixed parameter names, in the project's thread specification and instantiated identically wherever they recur. Their first full statements appear in Chapters 6 and 14 (Thread A) and Chapters 20 and 22 (Thread B).

Appendix B

Functional-Analytic Toolbox: Deferred Proofs

This appendix collects proofs deferred from Part I (spectral theorem for compact self-adjoint operators, the fixed-point theorems, and related functional analysis).

B.1 The perspective of a convex function

The variational (potential-game) formulation of the mean field game system (section 8.6, and again in the numerical treatment of chapter 28) rewrites the kinetic cost of a flow of measures in terms of the momentum field $w = m\,\alpha$, producing the integrand $(\rho, w) \mapsto \rho\, L_0(x, w/\rho)$. This is the *perspective* of the convex velocity-cost $L_0(x,\cdot)$. Its joint convexity and lower semicontinuity are the load-bearing analytic facts behind the convexity of the energy functional (lemma 8.10); we prove them here in the generality used in the body. The construction is classical convex analysis but sits above the book's prerequisite line, so we give it in full.

Definition B.1 (Perspective function)**.** Let $f\colon \mathbb{R}^d \to \mathbb{R}$ be convex, finite-valued, and *superlinear* ($f(a)/|a| \to +\infty$ as $|a| \to \infty$); this is the case relevant to the body, where $f = L_0(x,\cdot)$ is convex by **(A-Conv)** and superlinear by the control-coercivity in **(A-Growth)**. Its *perspective* is $\tilde{f}\colon \mathbb{R}\times\mathbb{R}^d \to (-\infty, +\infty]$,

$$\tilde{f}(\rho, w) \;:=\; \begin{cases} \rho\, f(w/\rho), & \rho > 0, \\ 0, & \rho = 0,\ w = 0, \\ +\infty, & \text{otherwise (i.e. } \rho = 0 \neq w \text{ or } \rho < 0). \end{cases} \tag{B.1}$$

For the quadratic case $f(a) = \frac{1}{2}|a|^2$ of the LQG thread, $\tilde{f}(\rho, w) = |w|^2/(2\rho)$ for $\rho > 0$, $\tilde{f}(0,0) = 0$, and $\tilde{f}(0, w) = +\infty$ for $w \neq 0$ — exactly the extension stated in section 8.6.

Proposition B.2 (Joint convexity and lower semicontinuity of the perspective)**.** *With f as in definition B.1, the perspective $\tilde{f}$ of eq.* (B.1) *is convex and lower semicontinuous on $\mathbb{R}\times\mathbb{R}^d$. In* *particular, for each fixed x the map $(\rho, w) \mapsto \rho\, L_0(x, w/\rho)$, with the $0/{+\infty}$ extension just stated, is jointly convex and lower semicontinuous, and hence so is the integral $\int_0^T\!\int_{\mathbb{R}^d} m\, L_0(x, w/m)$ of eq. (8.18).*

Proof. We argue in two steps: an elementary direct proof of joint convexity on the open half-space $\{\rho > 0\}$, and then a representation of $\tilde{f}$ as a supremum of affine functions that delivers both convexity across the boundary $\rho = 0$ and lower semicontinuity on all of $\mathbb{R} \times \mathbb{R}^d$ at once.

Step 1: joint convexity for $\rho > 0$. Let $(\rho_1, w_1), (\rho_2, w_2)$ have $\rho_i > 0$, let $\lambda \in [0, 1]$, and write $\rho = \lambda\rho_1 + (1-\lambda)\rho_2$, $w = \lambda w_1 + (1-\lambda)w_2$. Set $t := \lambda\rho_1/\rho \in [0, 1]$, so $1 - t = (1-\lambda)\rho_2/\rho$. Then

$$\frac{w}{\rho} = \frac{\lambda w_1 + (1-\lambda)w_2}{\rho} = t\,\frac{w_1}{\rho_1} + (1-t)\,\frac{w_2}{\rho_2}$$

is a convex combination, so by convexity of f,

$$f\Big(\frac{w}{\rho}\Big) \;\le\; t\,f\Big(\frac{w_1}{\rho_1}\Big) + (1-t)\,f\Big(\frac{w_2}{\rho_2}\Big).$$

Multiplying by $\rho > 0$ and using $\rho t = \lambda\rho_1$, $\rho(1-t) = (1-\lambda)\rho_2$,

$$\rho\,f\Big(\frac{w}{\rho}\Big) \;\le\; \lambda\,\rho_1 f\Big(\frac{w_1}{\rho_1}\Big) + (1-\lambda)\,\rho_2 f\Big(\frac{w_2}{\rho_2}\Big),$$

which is joint convexity of $\tilde{f}$ on $\{\rho > 0\}$.

Step 2: affine representation, boundary, and lower semicontinuity. A finite convex function on $\mathbb{R}^d$ equals the supremum of its affine minorants: for each $a_0 \in \mathbb{R}^d$ a supporting hyperplane of the (closed, convex) epigraph of f at $(a_0, f(a_0))$ exists by the Hahn–Banach separation theorem (theorem 2.11), and the collection of all such supporting affine functions reconstructs f pointwise. Write this family as

$$\mathcal{A} \;:=\; \{(p, c) \in \mathbb{R}^d \times \mathbb{R} :\; f(a) \ge \langle p, a\rangle - c \text{ for all } a\}, \qquad f(a) = \sup_{(p,c)\in\mathcal{A}} (\langle p, a\rangle - c).$$

Define, on all of $\mathbb{R} \times \mathbb{R}^d$,

$$g(\rho, w) \;:=\; \sup_{(p,c)\in\mathcal{A}} (\langle p, w\rangle - c\,\rho). \tag{B.2}$$

Each map $(\rho, w) \mapsto \langle p, w\rangle - c\rho$ is affine, so g, a pointwise supremum of affine functions, is convex and lower semicontinuous everywhere. We claim $g = \tilde{f}$.

For $\rho > 0$, substituting $a = w/\rho$ into the affine representation of f and multiplying by ρ gives $\rho f(w/\rho) = \sup_{(p,c)\in\mathcal{A}}(\langle p, w\rangle - c\rho) = g(\rho, w)$.

For $\rho = 0$, $g(0, w) = \sup_{(p,c)\in\mathcal{A}}\langle p, w\rangle$. Superlinearity of f means its affine minorants have *every* slope: for each $p \in \mathbb{R}^d$, $c_p := \sup_a(\langle p, a\rangle - f(a)) < \infty$ (the supremum is finite because $f(a) - \langle p, a\rangle \to +\infty$), so $(p, c_p) \in \mathcal{A}$ for all p. Hence $g(0, w) = \sup_{p\in\mathbb{R}^d}\langle p, w\rangle$, which is 0 when $w = 0$ and $+\infty$ when $w \ne 0$ — exactly $\tilde{f}(0, \cdot)$.

For $\rho < 0$, fix any $p \in \mathbb{R}^d$ and the corresponding $(p, c_p) \in \mathcal{A}$; since $\mathcal{A}$ contains (tp, c_{tp}) for all $t > 0$ and $-c\rho \ge 0$ contributes nonnegatively, taking p in the direction of w and $t \to \infty$ already sends $\langle tp, w\rangle \to +\infty$, so $g(\rho, w) = +\infty = \tilde{f}(\rho, w)$.

Thus $g = \tilde{f}$, and $\tilde{f}$ inherits convexity and lower semicontinuity from g. Finally, joint convexity and lower semicontinuity are preserved by integration in (t, x) against the (nonnegative) Lebesgue measure — by Fatou's lemma for the lower-semicontinuity part — giving the last assertion. Applying all of this with $f = L_0(x, \cdot)$ completes the proof. □

Remark B.3 (Why the $0/+\infty$ extension is forced)**.** The boundary values at $\rho = 0$ are not a free convention: eq. (B.2) shows they are the values of the lower-semicontinuous convex envelope, so any other assignment would either destroy lower semicontinuity (if set lower) or convexity (if set higher). This is the analytic reason the energy eq. (8.18) is well posed exactly with the stated extension, and why a finite-energy flow cannot carry momentum $w \neq 0$ where the mass m vanishes.

The remainder of this appendix supplies, in full, the beyond-prerequisite results that chapter 2 states and uses but defers for length: the Hahn–Banach theorem (extension and separation), Tychonoff's theorem in the form Banach–Alaoglu needs, the completion of the spectral theorem to the complex case, the three fixed-point theorems (Brouwer via Sperner's lemma, then Schauder and Kakutani–Fan–Glicksberg), and the Sobolev results—Meyers–Serrin density, the Poincaré inequality, and the Rellich–Kondrachov compact embedding. Everything is proved from the material of chapters 1 and 2 and the stated prerequisites (undergraduate analysis, the Arzelà–Ascoli theorem, mollification, and Zorn's lemma); no external source is needed. We keep the chapter's citations as attribution and further reading.

B.2 Hahn–Banach: extension and separation

We prove theorem 2.11. The analytic core is a single one-dimensional extension step, iterated transfinitely by Zorn's lemma; the geometric separation form is then read off through the Minkowski gauge of a convex set.

Recall that $p\colon X \to \mathbb{R}$ on a real vector space X is *sublinear* if $p(x + y) \leq p(x) + p(y)$ and $p(tx) = t\,p(x)$ for all $x, y \in X$ and $t \geq 0$. A norm and, more generally, the gauge of a convex neighbourhood of the origin are sublinear.

Lemma B.4 (One-dimensional extension)**.** *Let p be sublinear on a real vector space X, let $M \subsetneq X$ be a subspace, and let $f\colon M \to \mathbb{R}$ be linear with $f \leq p$ on M. For any $x_0 \in X \setminus M$ there is a linear extension f' of f to $M' = M \oplus \mathbb{R}x_0$ with $f' \leq p$ on M'.*

Proof. An extension is determined by the single value $c = f'(x_0)$, and $f' \leq p$ on M' means $f(m) + tc \leq p(m + tx_0)$ for all $m \in M$, $t \in \mathbb{R}$. For $t > 0$ this is (dividing by t and relabelling m/t) $c \leq p(m + x_0) - f(m)$; for $t < 0$ it is $c \geq f(m) - p(m - x_0)$. Such a c exists iff

$$\sup_{m \in M} \big[f(m) - p(m - x_0)\big] \;\leq\; \inf_{m \in M} \big[p(m + x_0) - f(m)\big],$$

and this holds because for any $m_1, m_2 \in M$,

$$f(m_1) + f(m_2) = f(m_1 + m_2) \leq p(m_1 + m_2) \leq p(m_1 - x_0) + p(m_2 + x_0),$$

so $f(m_1) - p(m_1 - x_0) \le p(m_2 + x_0) - f(m_2)$. Any c between the two sides works. □

Theorem B.5 (Hahn–Banach, dominated extension)**.** *With p sublinear on X, every linear $f\colon M \to \mathbb{R}$ on a subspace with $f \le p$ on M extends to a linear $F\colon X \to \mathbb{R}$ with $F \le p$ on all of X.*

Proof. Order the set $\mathcal{E}$ of dominated extensions (N, g)—$N \supseteq M$ a subspace, $g\colon N \to \mathbb{R}$ linear, $g|_M = f$, $g \le p$ on N—by declaring $(N_1, g_1) \le (N_2, g_2)$ when $N_1 \subseteq N_2$ and $g_2|_{N_1} = g_1$. Every chain $\{(N_\alpha, g_\alpha)\}$ has the upper bound $(\bigcup_\alpha N_\alpha,\ \bigcup_\alpha g_\alpha)$, which is well defined (the g_α agree on overlaps) and again dominated. By Zorn's lemma $\mathcal{E}$ has a maximal element (N^*, F). If $N^* \ne X$, lemma B.4 extends F to a strictly larger subspace, contradicting maximality; hence $N^* = X$. □

Corollary B.6 (Norm-preserving extension and norming functionals)**.** *Let X be a normed space (over $\mathbb{R}$ or $\mathbb{C}$).*

1. *Every bounded linear functional on a subspace M extends to X with the same norm.*
2. *For each $x_0 \ne 0$ there is $\phi \in X^*$ with $\|\phi\| = 1$ and $\phi(x_0) = \|x_0\|$. Consequently X^* separates points of X.*

Proof. Real case. For (1) put $p(x) = \|f\|_{M^*} \|x\|$, a sublinear (indeed seminorm) dominating f on M (since $f(m) \le |f(m)| \le \|f\|_{M^*} \|m\|$); the dominated extension F of theorem B.5 satisfies $F(x) \le \|f\|_{M^*} \|x\|$ and, applied to $-x$, $|F(x)| \le \|f\|_{M^*} \|x\|$, so $\|F\| \le \|f\|_{M^*}$; the reverse is automatic. For (2) define f on $M = \mathbb{R}x_0$ by $f(tx_0) = t\,\|x_0\|$, of norm 1, and extend.

Complex case. Write a complex functional as $f = u + iv$ with $u = \operatorname{Re} f$; then $v(x) = -u(ix)$, so $f(x) = u(x) - iu(ix)$ and f is recovered from its real part u, a real-linear functional with $|u| \le |f| \le \|f\|\,\|\cdot\|$. Extend u by the real case to $U \le \|f\|\,\|\cdot\|$ and set $F(x) = U(x) - iU(ix)$; F is complex-linear, extends f, and for each x choosing θ with $e^{i\theta}F(x) = |F(x)|$ gives $|F(x)| = F(e^{i\theta}x) = U(e^{i\theta}x) \le \|f\|\,\|x\|$, so $\|F\| = \|f\|$. Part (2) follows as before. □

For the geometric form we use the gauge of a convex set. If $C \subseteq X$ is convex, open, and contains 0, its *Minkowski gauge* is $p_C(x) = \inf\{t > 0 :\ x/t \in C\}$. One checks directly that p_C is sublinear, finite (since C is a neighbourhood of 0), bounded above by $\|\cdot\|/\rho$ on a ball $B(0,\rho) \subseteq C$, and that $C = \{x : p_C(x) < 1\}$ (using that C is open).

Theorem B.7 (Geometric Hahn–Banach)**.** *Let X be a normed space and $A, B \subseteq X$ nonempty disjoint convex sets.*

1. *If A is open there are $\phi \in X^* \setminus \{0\}$ and $\alpha \in \mathbb{R}$ with $\phi(a) < \alpha \le \phi(b)$ for all $a \in A$, $b \in B$.*
2. *If A is closed and B is compact there are $\phi \in X^*$ and $\alpha \in \mathbb{R}$ with $\phi(a) \le \alpha < \phi(b)$ for all $a \in A$, $b \in B$.*

Proof. (1) Fix $a_0 \in A$, $b_0 \in B$ and set $x_0 = b_0 - a_0$ and $C = A - B + x_0 = \{a - b + x_0 : a \in A,\ b \in B\}$. Then C is convex (a sum of convex sets), open (a union of translates of the open A), contains

0 (take $a = a_0, b = b_0$), and does *not* contain x_0: $x_0 \in C$ would give $a - b + x_0 = x_0$, i.e. $a = b \in A \cap B = \varnothing$. Hence $p_C(x_0) \geq 1$. On $M = \mathbb{R}x_0$ define $f(tx_0) = t$; then $f \leq p_C$ on M (for $t > 0$, $f(tx_0) = t \leq t\, p_C(x_0) = p_C(tx_0)$; for $t \leq 0$ the left side is $\leq 0 \leq p_C$). By theorem B.5 extend to $\phi \leq p_C$ on X. Since p_C is bounded near 0, ϕ is bounded, so $\phi \in X^*$, and $\phi \neq 0$ because $\phi(x_0) = 1$. For $a \in A, b \in B$, $a - b + x_0 \in C$ gives $\phi(a) - \phi(b) + 1 = \phi(a - b + x_0) \leq p_C(a - b + x_0) < 1$, i.e. $\phi(a) < \phi(b)$. Thus $\sup_A \phi \leq \inf_B \phi$; any $\alpha \in [\sup_A \phi, \inf_B \phi]$ works, and $\phi(a) < \alpha$ for $a \in A$ because A is open and $\phi \neq 0$ (an open set cannot attain the supremum of a nonzero continuous linear functional).

(2) Disjointness of the closed A and the compact B forces $d = \inf \{\|a - b\| : a \in A, b \in B\} > 0$ (a minimizing pair exists by compactness of B and closedness of A, and equals 0 only if $A \cap B \neq \varnothing$). For $0 < \varepsilon < d$ the set $A_\varepsilon = A + B(0, \varepsilon)$ is open, convex, and still disjoint from B. Apply (1) to A_ε and B: there is $\phi \in X^* \setminus \{0\}$ with $\phi(a') < \phi(b)$ for $a' \in A_\varepsilon$, $b \in B$. Set $\alpha = \sup_A \phi$. Then $\phi(a) \leq \alpha$ for $a \in A$, while for $b \in B$, $\phi(b) \geq \inf_B \phi \geq \sup_{A_\varepsilon} \phi = \sup_A \phi + \varepsilon \|\phi\|_{X^*} > \alpha$, using $\sup_{A_\varepsilon} \phi = \sup_A \phi + \varepsilon \sup_{\|v\| \leq 1} \phi(v) = \sup_A \phi + \varepsilon \|\phi\|$. This is the claimed strict separation. □

B.3 Tychonoff's theorem for Banach–Alaoglu

The body's proof of Banach–Alaoglu (theorem 2.16) realizes the dual ball as a closed subset of the product $P = \prod_{x \in X} [-\|x\|, \|x\|]$ and needs that product to be compact. We supply Tychonoff's theorem in full. The route is the Alexander subbase lemma; the only ingredient beyond elementary topology is Zorn's lemma.

Recall that a *subbase* of a topology is a family $\mathcal{S}$ of open sets whose finite intersections form a base, and that a space is compact iff every open cover has a finite subcover.

Lemma B.8 (Alexander subbase lemma)**.** *Let X be a topological space with subbase $\mathcal{S}$. If every cover of X by members of $\mathcal{S}$ has a finite subcover, then X is compact.*

Proof. Suppose not. Let $\mathcal{A}$ be the collection of open covers of X with no finite subcover; by assumption $\mathcal{A} \neq \varnothing$. Order $\mathcal{A}$ by inclusion. The union of a chain in $\mathcal{A}$ is again a cover with no finite subcover (a finite subcover of the union would already lie in one member of the chain), so chains have upper bounds; by Zorn there is a maximal $\mathcal{C} \in \mathcal{A}$. Maximality means: for every open $U \notin \mathcal{C}$, the cover $\mathcal{C} \cup \{U\}$ *does* have a finite subcover, so $X = U \cup F_1 \cup \cdots \cup F_k$ for some $F_i \in \mathcal{C}$.

Let $\mathcal{C}_{\mathcal{S}} = \mathcal{C} \cap \mathcal{S}$. This subbasic subfamily does not cover X: if it did it would be a subbasic cover, hence have a finite subcover, which would be a finite subcover inside $\mathcal{C}$, contradicting $\mathcal{C} \in \mathcal{A}$. Pick $z \in X$ not covered by $\mathcal{C}_{\mathcal{S}}$. Since $\mathcal{C}$ covers X, $z \in U$ for some $U \in \mathcal{C}$; as $\mathcal{S}$ is a subbase, $z \in S_1 \cap \cdots \cap S_n \subseteq U$ with $S_i \in \mathcal{S}$. Each $S_i \notin \mathcal{C}$: indeed $z \in S_i$, and z lies in no member of $\mathcal{C}_{\mathcal{S}}$, so $S_i \notin \mathcal{C} \cap \mathcal{S}$, hence $S_i \notin \mathcal{C}$. By maximality each S_i is covered together with finitely many members of $\mathcal{C}$: $X = S_i \cup \bigcup \mathcal{F}_i$ with $\mathcal{F}_i \subseteq \mathcal{C}$ finite. Then

$$X \setminus \bigcup_{i=1}^{n} \bigcup \mathcal{F}_i \;\subseteq\; \bigcap_{i=1}^{n} S_i \;\subseteq\; U,$$

so $\{U\} \cup \mathcal{F}_1 \cup \cdots \cup \mathcal{F}_n$ is a finite subcover drawn from $\mathcal{C}$—contradicting $\mathcal{C} \in \mathcal{A}$. Hence X is compact. $\square$

Theorem B.9 (Tychonoff)**.** *Any product $X = \prod_{\alpha \in I} X_\alpha$ of compact spaces is compact in the product topology. In particular $\prod_{x \in X}[-\|x\|, \|x\|]$ is compact, completing the proof of theorem 2.16.*

Proof. The product topology has the subbase $\mathcal{S} = \{\pi_\alpha^{-1}(U) : \alpha \in I,\ U \subseteq X_\alpha \text{ open}\}$, where π_α is the coordinate projection. By lemma B.8 it suffices to show every cover of X by members of $\mathcal{S}$ has a finite subcover. Let $\mathcal{C} \subseteq \mathcal{S}$ be a cover, and for each α set $\mathcal{U}_\alpha = \{U \subseteq X_\alpha \text{ open} : \pi_\alpha^{-1}(U) \in \mathcal{C}\}$.

If some $\mathcal{U}_\alpha$ covers X_α, then by compactness of X_α finitely many $U_1, \ldots, U_k \in \mathcal{U}_\alpha$ cover it, and $\pi_\alpha^{-1}(U_1), \ldots, \pi_\alpha^{-1}(U_k) \in \mathcal{C}$ then cover X—a finite subcover, and we are done. Otherwise no $\mathcal{U}_\alpha$ covers X_α: for each α choose $x_\alpha \in X_\alpha \setminus \bigcup \mathcal{U}_\alpha$. The point $x = (x_\alpha) \in X$ is covered by no member of $\mathcal{C}$: any element of $\mathcal{C}$ is $\pi_\alpha^{-1}(U)$ with $U \in \mathcal{U}_\alpha$, and $x_\alpha \notin U$ gives $x \notin \pi_\alpha^{-1}(U)$. This contradicts that $\mathcal{C}$ is a cover. Hence the first alternative always holds, and X is compact. The product of compact intervals is the special case $X_\alpha = [-\|x\|, \|x\|]$. $\square$

B.4 The spectral theorem: the complex case

Theorem 2.26 was proved in full for a real Hilbert space. We record the two modifications that carry the argument verbatim to a complex separable Hilbert space H, with $\langle \cdot, \cdot \rangle$ the Hermitian inner product of definition 2.3 (linear in the first slot, conjugate-linear in the second) and "self-adjoint" meaning $\langle Tx, y \rangle = \langle x, Ty \rangle$ for all x, y.

Proposition B.10 (Reality of the quadratic form and the eigenvalues)**.** *Let T be a bounded self-adjoint operator on a complex Hilbert space H. Then $\langle Tx, x \rangle \in \mathbb{R}$ for every x, and every eigenvalue of T is real. Moreover lemma 2.25 holds: $\|T\| = \sup_{\|x\|=1} |\langle Tx, x \rangle|$.*

Proof. By Hermitian symmetry and self-adjointness, $\langle Tx, x \rangle = \langle x, Tx \rangle = \overline{\langle Tx, x \rangle}$, so $\langle Tx, x \rangle$ is real. If $Te = \lambda e$ with $e \neq 0$ then $\lambda \|e\|^2 = \langle Te, e \rangle \in \mathbb{R}$, so $\lambda \in \mathbb{R}$. For the norm identity, write $M = \sup_{\|x\|=1} |\langle Tx, x \rangle|$; Cauchy–Schwarz gives $M \leq \|T\|$. For the reverse, the polarization identity in the Hermitian case reads

$$\operatorname{Re}\langle Tx, y \rangle = \tfrac{1}{4}\big(\langle T(x+y), x+y \rangle - \langle T(x-y), x-y \rangle\big),$$

since $\langle Tx, y \rangle + \langle Ty, x \rangle = \langle Tx, y \rangle + \overline{\langle Tx, y \rangle} = 2\operatorname{Re}\langle Tx, y \rangle$ by self-adjointness. As in the body, this and the parallelogram law give $|\operatorname{Re}\langle Tx, y \rangle| \leq \frac{M}{2}(\|x\|^2 + \|y\|^2)$; taking $\|x\| = \|y\| = 1$ and replacing y by the unit phase multiple $e^{i\theta} y$ that makes $\langle Tx, y \rangle$ real and positive turns the left side into $|\langle Tx, y \rangle|$, and choosing y aligned with Tx yields $\|Tx\| \leq M$, i.e. $\|T\| \leq M$. $\square$

With proposition B.10 in hand, the construction of theorem 2.26 runs without any further change: lemma 2.25 produces a maximizing sequence with $\langle Tx_k, x_k \rangle \to \pm\|T\| =: \lambda_1 \in \mathbb{R}$; compactness and the identity $\|Tx_k - \lambda_1 x_k\|^2 = \|Tx_k\|^2 - 2\lambda_1 \langle Tx_k, x_k \rangle + \lambda_1^2$ (whose right side is real throughout) deliver a unit eigenvector e_1 with real eigenvalue λ_1; the orthogonal complement $e_1^\perp$ is invariant by self-adjointness; and the iteration, the decay $\lambda_n \to 0$, finite multiplicities, and the expansion $Tx = \sum_n \lambda_n \langle x, e_n \rangle e_n$ are proved exactly as in the body. This completes the spectral theorem over $\mathbb{C}$.

B.5 Brouwer's fixed-point theorem via Sperner's lemma

We now prove theorem 2.37 from scratch. The engine is a purely combinatorial fact about labelled triangulations.

Let $\Delta^n = \{x \in \mathbb{R}^{n+1} : x_i \geq 0,\ \sum_i x_i = 1\}$ be the standard n-simplex with vertices $v_0, \ldots, v_n$ (the standard basis vectors). A *triangulation* $\mathcal{T}$ of Δ^n is a finite collection of n-dimensional sub-simplices covering Δ^n, any two meeting in a common face (possibly empty). A labelling ℓ of the vertices of $\mathcal{T}$ by $\{0, 1, \ldots, n\}$ is a *Sperner labelling* if every vertex w lying on the face span$\{v_i : i \in S\}$ (i.e. with $w_i = 0$ for $i \notin S$) receives a label $\ell(w) \in S$. A sub-simplex is *fully labelled* (or rainbow) if its $n+1$ vertices carry all of $0, \ldots, n$.

Lemma B.11 (Sperner)**.** *Every Sperner-labelled triangulation of Δ^n contains an odd number of fully labelled sub-simplices; in particular at least one.*

Proof. Induct on n. For $n = 0$, Δ^0 is a point labelled 0, one fully labelled simplex. Assume the statement for $n-1$. Call an $(n-1)$-face of a sub-simplex of $\mathcal{T}$ a *door* if its vertices carry exactly the labels $\{0, 1, \ldots, n-1\}$. For each n-sub-simplex $\sigma \in \mathcal{T}$ let $d(\sigma)$ count the doors among its facets. A short case check on the multiset of labels of σ shows: if σ is fully labelled it has exactly one door; if σ carries the labels $\{0, \ldots, n-1\}$ with one repeated (label n absent) it has exactly two doors; otherwise it has none. Hence

$$\sum_{\sigma \in \mathcal{T}} d(\sigma) \ \equiv\ \#\{\text{fully labelled } \sigma\} \pmod 2.$$

Count doors instead by the facets carrying them. An interior facet is shared by exactly two sub-simplices, contributing 2; a boundary facet lies on a single sub-simplex, contributing 1. So $\sum_\sigma d(\sigma)$ is congruent mod 2 to the number of *boundary* doors. A door has label n absent, so by the Sperner condition it cannot lie on any boundary face other than the facet span$\{v_0, \ldots, v_{n-1}\}$ opposite v_n. Restricted to that facet the labelling is a Sperner labelling of an $(n-1)$-triangulation, so by the inductive hypothesis the number of boundary doors is odd. Therefore $\#\{\text{fully labelled } \sigma\} \equiv \sum_\sigma d(\sigma) \equiv 1 \pmod 2$. ☐

Theorem B.12 (Brouwer on the simplex)**.** *Every continuous map $g\colon \Delta^n \to \Delta^n$ has a fixed point.*

Proof. For $x \in \Delta^n$, since $\sum_i g(x)_i = 1 = \sum_i x_i$, there is an index i with $g(x)_i \leq x_i$ and $x_i > 0$ (if $g(x)_i > x_i$ for every i in the support of x the coordinates of $g(x)$ would sum to more than 1). Take triangulations $\mathcal{T}_k$ of Δ^n with mesh (maximal sub-simplex diameter) tending to 0—for instance iterated barycentric subdivisions. Label each vertex w of $\mathcal{T}_k$ by some index $\ell(w)$ with $g(w)_{\ell(w)} \leq w_{\ell(w)}$ and $w_{\ell(w)} > 0$. This is a Sperner labelling: if w lies on the face $\{v_i : i \in S\}$ then

$w_i = 0$ for $i \notin S$, and the chosen label has $w_{\ell(w)} > 0$, forcing $\ell(w) \in S$. By lemma B.11 there is a fully labelled sub-simplex with vertices $w^{k,0}, \ldots, w^{k,n}$, where $g(w^{k,j})_j \leq w_j^{k,j}$. As the mesh $\to 0$, all $w^{k,j}$ for fixed k are within the mesh of one another; passing to a subsequence, $w^{k,j} \to x^*$ for every j to a common limit $x^* \in \Delta^n$. By continuity $g(x^*)_j \leq x_j^*$ for every $j = 0, \ldots, n$. Since $\sum_j g(x^*)_j = \sum_j x_j^* = 1$, no inequality can be strict, so $g(x^*) = x^*$. □

Theorem B.13 (Brouwer)**.** *Every continuous map $g\colon C \to C$ of a nonempty compact convex set $C \subseteq \mathbb{R}^d$ into itself has a fixed point.*

Proof. Choose a simplex $\Delta \supseteq C$ (a bounded convex body sits inside a large simplex, and C is bounded, being compact). The nearest-point projection $r = P_C\colon \Delta \to C$ onto the closed convex set C is well defined and 1-Lipschitz, hence continuous, by the projection theorem theorem 2.5 applied in the Hilbert space $\mathbb{R}^d$. Then $g \circ r\colon \Delta \to C \subseteq \Delta$ is continuous, so by theorem B.12 (transported to Δ by an affine homeomorphism with Δ^d) it has a fixed point $x^* = g(r(x^*))$. As $g(r(x^*)) \in C$, we have $x^* \in C$, whence $r(x^*) = x^*$ and $x^* = g(r(x^*)) = g(x^*)$. □

B.6 Schauder's fixed-point theorem

We complete the body sketch of theorem 2.38 into a full proof, spelling out the Schauder projection.

Theorem B.14 (Schauder)**.** *Let C be a nonempty closed convex subset of a Banach space X and $\mathsf{S}\colon C \to C$ continuous with $\mathsf{S}(C)$ relatively compact. Then S has a fixed point.*

Proof. Let $K = \overline{\mathsf{S}(C)}$, a compact subset of the convex C, hence $K \subseteq C$. Fix $\varepsilon > 0$. By compactness K has a finite ε-net $\{y_1, \ldots, y_N\} \subseteq K \subseteq C$. Define the *Schauder projection* $P_\varepsilon\colon K \to X$ by

$$P_\varepsilon z = \frac{\sum_{j=1}^N \theta_j(z)\, y_j}{\sum_{j=1}^N \theta_j(z)}, \qquad \theta_j(z) = \max\left(0,\ \varepsilon - \|z - y_j\|\,\right).$$

For each $z \in K$ the denominator is positive (some y_j is within ε of z), and each θ_j is continuous, so P_ε is continuous. Its value is a convex combination of those y_j with $\theta_j(z) > 0$, i.e. $\|z - y_j\| < \varepsilon$; hence $P_\varepsilon z \in C_\varepsilon := \operatorname{conv}\{y_1, \ldots, y_N\}$ and, because each such y_j satisfies $\|z - y_j\| < \varepsilon$,

$$\|P_\varepsilon z - z\| = \left\| \frac{\sum_j \theta_j(z)(y_j - z)}{\sum_j \theta_j(z)} \right\| \leq \frac{\sum_j \theta_j(z)\ \|y_j - z\|}{\sum_j \theta_j(z)} < \varepsilon.$$

Since every $y_j \in K \subseteq C$ and C is convex, $C_\varepsilon \subseteq C$. Thus $P_\varepsilon \circ \mathsf{S}$ maps C_ε (where S is defined, as $C_\varepsilon \subseteq C$) into $\mathsf{S}(C) \subseteq K$ and then into C_ε: it is a continuous self-map of the finite-dimensional compact convex set C_ε. By Brouwer (theorem B.13) it has a fixed point $x_\varepsilon = P_\varepsilon \mathsf{S} x_\varepsilon \in C_\varepsilon$, and

$$\|\mathsf{S}x_\varepsilon - x_\varepsilon\| = \|\mathsf{S}x_\varepsilon - P_\varepsilon \mathsf{S}x_\varepsilon\| < \varepsilon.$$

Take $\varepsilon = 1/n \to 0$. The points $\mathsf{S}x_{1/n} \in K$ have, by compactness, a subsequence converging to some $x_* \in K \subseteq C$; along it $x_{1/n} \to x_*$ as well (since $\left\|x_{1/n} - \mathsf{S}x_{1/n}\right\| \to 0$). Continuity of S gives $\mathsf{S}x_* = \lim \mathsf{S}x_{1/n} = x_*$. □

B.7 The Kakutani–Fan–Glicksberg theorem

Finally we prove theorem 2.40. We give the argument for C a nonempty compact convex subset of a normed space X; this covers every use in the book, since the weak-$*$ compact convex sets of probability measures to which Kakutani is applied are metrizable (the predual is separable) and embed as compact convex subsets of a Banach space. The set-valued problem is reduced to Schauder by a continuous approximate selection.

Lemma B.15 (Approximate selection)**.** *Let $C \subseteq X$ be compact convex and $\mathsf{S}\colon C \rightrightarrows C$ have a closed graph and nonempty convex values. For every $\varepsilon > 0$ there is a continuous $f_\varepsilon\colon C \to C$ such that $f_\varepsilon(x)$ is a convex combination of points $z_i \in \mathsf{S}(x_i)$ with $\|x_i - x\| < \varepsilon$.*

Proof. A closed graph in the compact $C \times C$ makes S upper hemicontinuous: for each $x \in C$ and each $\delta > 0$ there is $r_x > 0$ with $\mathsf{S}\big(B(x, r_x) \cap C\big) \subseteq \mathsf{S}(x) + B(0, \delta)$. (If not, there are $x_n \to x$ and $y_n \in \mathsf{S}(x_n)$ with $\operatorname{dist}(y_n, \mathsf{S}(x)) \geq \delta$; by compactness $y_n \to y$ along a subsequence, and the closed graph gives $y \in \mathsf{S}(x)$, contradicting the distance bound.) Use this with $\delta = \varepsilon$ and shrink each r_x below ε. The balls $\{B(x, r_x/2)\}_{x \in C}$ cover the compact C; take a finite subcover with centres $x_1, \ldots, x_m$ and radii $r_i = r_{x_i}$, and a continuous partition of unity $\{\psi_i\}$ subordinate to $\{B(x_i, r_i)\}$ (e.g. $\psi_i = \eta_i / \sum_k \eta_k$ with $\eta_i(x) = \max(0, r_i - \|x - x_i\|)$). Choose any $z_i \in \mathsf{S}(x_i)$ and set $f_\varepsilon(x) = \sum_i \psi_i(x)\, z_i$. This is continuous and a convex combination of points $z_i \in \mathsf{S}(x_i) \subseteq C$, so $f_\varepsilon(x) \in C$. Whenever $\psi_i(x) > 0$ we have $x \in B(x_i, r_i)$, i.e. $\|x_i - x\| < r_i < \varepsilon$, as claimed. □

Theorem B.16 (Kakutani–Fan–Glicksberg)**.** *Let $C \subseteq X$ be nonempty compact convex and $\mathsf{S}\colon C \rightrightarrows C$ have a closed graph with nonempty, convex, closed values. Then S has a fixed point: some $x_* \in C$ with $x_* \in \mathsf{S}(x_*)$.*

Proof. For each n let $f_n = f_{1/n}\colon C \to C$ be the approximate selection of lemma B.15 with $\varepsilon = 1/n$. Being continuous on the compact convex C with $f_n(C) \subseteq C$ relatively compact, f_n has a fixed point $x_n = f_n(x_n)$ by Schauder (theorem B.14). By compactness pass to a subsequence with $x_n \to x_* \in C$.

We claim $x_* \in \mathsf{S}(x_*)$. Suppose not. Since $\mathsf{S}(x_*)$ is nonempty, convex, and closed, and $\{x_*\}$ is compact convex and disjoint from it, the separation theorem theorem 2.11(2), proved in theorem B.7, gives $\phi \in X^*$ and β with

$$\phi(y) \leq \beta < \phi(x_*) \qquad \text{for all } y \in \mathsf{S}(x_*).$$

Put $\gamma = \frac{1}{2}(\beta + \phi(x_*))$, so $\beta < \gamma < \phi(x_*)$, and let $\eta = (\gamma - \beta)/\|\phi\|$ (with $\phi \neq 0$ since $\beta < \phi(x_*)$). By the upper hemicontinuity established inside lemma B.15 there is $\rho > 0$ with $\mathsf{S}(B(x_*, \rho) \cap C) \subseteq \mathsf{S}(x_*) + B(0, \eta)$; on that neighbourhood every value y' of S satisfies $\phi(y') \leq \beta + \|\phi\|\,\eta = \gamma$. Now write $x_n = f_n(x_n) = \sum_i \psi_i(x_n)\, z_i^{(n)}$ with $z_i^{(n)} \in \mathsf{S}(x_i^{(n)})$ and $\left\|x_i^{(n)} - x_n\right\| < 1/n$. For n large, $x_n \in B(x_*, \rho/2)$ and $1/n < \rho/2$, so each contributing $x_i^{(n)} \in B(x_*, \rho)$ and hence $\phi(z_i^{(n)}) \leq \gamma$. Taking the convex combination, $\phi(x_n) = \sum_i \psi_i(x_n)\, \phi(z_i^{(n)}) \leq \gamma$. Letting $n \to \infty$, continuity of ϕ gives $\phi(x_*) \leq \gamma < \phi(x_*)$, a contradiction. Therefore $x_* \in \mathsf{S}(x_*)$. □

B.8 Sobolev spaces: density, Poincaré, and compactness

We prove the three claims of theorem 2.31. Throughout $\Omega \subseteq \mathbb{R}^d$ is open and bounded; mollification uses a fixed $\rho \in C_c^\infty(\mathbb{R}^d)$ with $\rho \geq 0$, $\operatorname{supp}\rho \subseteq B(0,1)$, $\int \rho = 1$, and $\rho_\delta(x) = \delta^{-d}\rho(x/\delta)$.

Theorem B.17 (Meyers–Serrin)**.** *$C^\infty(\Omega) \cap H^1(\Omega)$ is dense in $H^1(\Omega)$. Consequently $C_c^\infty(\Omega)$ is dense in $H_0^1(\Omega)$ by definition of the latter, and the weak theory is the completion of the classical one.*

Proof. Let $u \in H^1(\Omega)$ and $\varepsilon > 0$. Exhaust Ω by open sets $\Omega_k = \{x \in \Omega : \operatorname{dist}(x, \partial\Omega) > 1/k,\ |x| < k\}$, set $\Omega_0 = \Omega_{-1} = \varnothing$, and let $\{\zeta_k\}$ be a smooth partition of unity with $\operatorname{supp}\zeta_k \subseteq \Omega_{k+1} \setminus \overline{\Omega_{k-1}}$ and $\sum_k \zeta_k \equiv 1$ on Ω. Each $\zeta_k u \in H^1(\Omega)$ has compact support in Ω, so for δ_k small enough the mollification $\rho_{\delta_k} * (\zeta_k u)$ lies in $C_c^\infty(\Omega)$, supported in $\Omega_{k+2} \setminus \overline{\Omega_{k-2}}$, and satisfies $\|\rho_{\delta_k} * (\zeta_k u) - \zeta_k u\|_{H^1(\Omega)} < \varepsilon\, 2^{-k}$ (mollification converges in H^1 on the interior, because $\partial_i(\rho_\delta * v) = \rho_\delta * \partial_i v$ and $\rho_\delta * w \to w$ in L^2 for each L^2 function w). The sum $v = \sum_k \rho_{\delta_k} * (\zeta_k u)$ is locally finite, hence in $C^\infty(\Omega)$, and

$$\|v - u\|_{H^1(\Omega)} = \Big\| \sum_k \big(\rho_{\delta_k} * (\zeta_k u) - \zeta_k u\big) \Big\|_{H^1} \leq \sum_k \varepsilon\, 2^{-k} < 2\varepsilon.$$

Thus smooth functions are H^1-dense; $v \in H^1$ because u is. □

Theorem B.18 (Poincaré)**.** *If Ω lies in a slab $\{x : |x_1| < R\}$ then $\|u\|_{L^2(\Omega)} \leq 2R\, \|\nabla u\|_{L^2(\Omega)}$ for all $u \in H_0^1(\Omega)$. Hence on $H_0^1(\Omega)$ the gradient norm is equivalent to the full H^1 norm.*

Proof. By density (theorem B.17) it suffices to prove the bound for $u \in C_c^\infty(\Omega)$, extended by zero to $\mathbb{R}^d$. Writing $x = (x_1, x')$ and using $u(-R, x') = 0$,

$$u(x_1, x') = \int_{-R}^{x_1} \partial_1 u(t, x')\, \mathrm{d}t,$$

so by Cauchy–Schwarz $|u(x_1,x')|^2 \leq 2R \int_{-R}^{R} |\partial_1 u(t,x')|^2\, \mathrm{d}t$. Integrating in x_1 over $(-R,R)$ and then in x',

$$\int_\Omega |u|^2 \leq (2R)^2 \int_\Omega |\partial_1 u|^2 \leq (2R)^2 \int_\Omega |\nabla u|^2 .$$

The norm equivalence follows since $\|\nabla u\|_{L^2}^2 \leq \|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2 \leq (1 + 4R^2)\, \|\nabla u\|_{L^2}^2$. □

The compact embedding is reached through a general L^2 compactness criterion, which we prove first.

Theorem B.19 (Fréchet–Kolmogorov–Riesz)**.** *A bounded family $\mathcal{F} \subseteq L^2(\mathbb{R}^d)$ is relatively compact in $L^2(\mathbb{R}^d)$ provided*

1. (equicontinuity of translation) $\sup_{f \in \mathcal{F}} \|\tau_h f - f\|_{L^2} \to 0$ *as* $|h| \to 0$, *where* $(\tau_h f)(x) = f(x+h)$; *and*
2. (uniform decay) $\sup_{f \in \mathcal{F}} \int_{|x|>R} |f|^2 \to 0$ *as* $R \to \infty$.

Proof. We show $\mathcal{F}$ is totally bounded; completeness of L^2 then gives relative compactness. Fix $\varepsilon > 0$. By (2) choose R with $\int_{|x|>R} |f|^2 < \varepsilon^2$ for all f. For the mollification $\rho_\delta * f$, Minkowski's integral inequality gives

$$\|\rho_\delta * f - f\|_{L^2} = \Big\| \int \rho_\delta(y)(\tau_{-y}f - f)\, \mathrm{d}y \Big\|_{L^2} \le \int \rho_\delta(y)\, \|\tau_{-y}f - f\|_{L^2}\, \mathrm{d}y \le \sup_{|y|\le\delta} \|\tau_{-y}f - f\|_{L^2},$$

which by (1) is $< \varepsilon$ for δ small, uniformly in f. Fix such a δ. On the ball $B_R = \{|x| \le R\}$ the family $\{\rho_\delta * f : f \in \mathcal{F}\}$ is uniformly bounded, $|(\rho_\delta * f)(x)| \le \|\rho_\delta\|_{L^2} \|f\|_{L^2} \le C_\delta$, and equicontinuous,

$$|(\rho_\delta * f)(x) - (\rho_\delta * f)(x')| \le \|\tau_{-x}\rho_\delta - \tau_{-x'}\rho_\delta\|_{L^2}\, \|f\|_{L^2} \le \omega_\delta(|x - x'|),$$

with $\omega_\delta(s) \to 0$ as $s \to 0$ because translation is continuous in L^2 and $\|f\|_{L^2}$ is bounded. By Arzelà–Ascoli, $\{\rho_\delta * f|_{B_R}\}$ is relatively compact in $C(B_R)$, hence (as B_R has finite measure) in $L^2(B_R)$: cover it by finitely many $L^2(B_R)$-balls of radius ε with centres $g_1, \dots, g_m$, regarded as functions on $\mathbb{R}^d$ supported in B_R. For any $f \in \mathcal{F}$ pick g_k with $\|\rho_\delta * f - g_k\|_{L^2(B_R)} < \varepsilon$; then, splitting $\mathbb{R}^d = B_R \cup B_R^c$ and using the tail bounds,

$$\|f - g_k\|_{L^2(\mathbb{R}^d)} \le \|f - \rho_\delta * f\|_{L^2} + \|\rho_\delta * f - g_k\|_{L^2(B_R)} + \Big(\int_{|x|>R} |\rho_\delta * f|^2 \Big)^{1/2} \le C\varepsilon.$$

Thus finitely many balls of radius $C\varepsilon$ cover $\mathcal{F}$, so $\mathcal{F}$ is totally bounded. □

Lemma B.20 (Translation estimate)**.** *For $u \in H^1(\mathbb{R}^d)$ and $h \in \mathbb{R}^d$, $\|\tau_h u - u\|_{L^2(\mathbb{R}^d)} \le |h|\, \|\nabla u\|_{L^2(\mathbb{R}^d)}$.*

Proof. For $u \in C_c^\infty(\mathbb{R}^d)$, $u(x+h) - u(x) = \int_0^1 \nabla u(x + th) \cdot h\, \mathrm{d}t$, so by Cauchy–Schwarz and Tonelli

$$\int |u(x+h) - u(x)|^2\, \mathrm{d}x \le |h|^2 \int_0^1 \!\! \int |\nabla u(x + th)|^2\, \mathrm{d}x\, \mathrm{d}t = |h|^2\, \|\nabla u\|_{L^2}^2.$$

The general case follows because C_c^∞ is dense in $H^1(\mathbb{R}^d)$ and both sides are continuous in u for the H^1 norm. □

Theorem B.21 (Rellich–Kondrachov for H_0^1)**.** *For $\Omega \subseteq \mathbb{R}^d$ open and bounded, the inclusion $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ is compact: every sequence bounded in $H_0^1(\Omega)$ has an $L^2(\Omega)$-convergent subsequence.*

Proof. Extend each $u \in H_0^1(\Omega)$ by zero to $\tilde{u} \in H^1(\mathbb{R}^d)$ (the extension is in $H^1(\mathbb{R}^d)$ precisely because $u \in H_0^1$ is an H^1-limit of $C_c^\infty(\Omega)$ functions, for which zero extension is automatic), with $\|\tilde{u}\|_{H^1(\mathbb{R}^d)} = \|u\|_{H^1(\Omega)}$ and $\operatorname{supp} \tilde{u} \subseteq \overline{\Omega}$. Let $\mathcal{F} = \{\tilde{u} : \|u\|_{H^1} \le 1\}$. It is bounded in $L^2(\mathbb{R}^d)$; the decay hypothesis (2) of theorem B.19 holds trivially because every $\tilde{u}$ is supported in the fixed bounded $\overline{\Omega}$ (the tail integral vanishes for R large); and the translation hypothesis (1) is lemma B.20: $\sup_{\mathcal{F}} \|\tau_h \tilde{u} - \tilde{u}\|_{L^2} \le |h| \sup_{\mathcal{F}} \|\nabla \tilde{u}\|_{L^2} \le |h| \to 0$. Hence $\mathcal{F}$ is relatively compact in $L^2(\mathbb{R}^d)$, and restricting to Ω gives the claim. □

On the torus the same compactness drops out of the Fourier definition of definition 2.32, with no extension needed; this is the form used for periodic MFG.

Theorem B.22 (Rellich on the torus)**.** *$H^1(\mathbb{T}^d) \hookrightarrow L^2(\mathbb{T}^d)$ is compact.*

Proof. A function $u \in L^2(\mathbb{T}^d)$ has Fourier coefficients $\hat{u}(k)$, $k \in \mathbb{Z}^d$, with $\|u\|_{L^2}^2 = \sum_k |\hat{u}(k)|^2$ and $\|u\|_{H^1}^2 = \sum_k (1+|k|^2)\,|\hat{u}(k)|^2$. Let $\|u_n\|_{H^1} \le 1$. The coefficients $\hat{u}_n(k)$ are bounded for each fixed k, so by a diagonal argument a subsequence has $\hat{u}_n(k) \to c(k)$ for every k, with $\sum_k (1+|k|^2)\,|c(k)|^2 \le 1$. For any R, $\sum_{|k|>R} |\hat{u}_n(k)|^2 \le (1+R^2)^{-1} \sum_k (1+|k|^2)\,|\hat{u}_n(k)|^2 \le (1+R^2)^{-1}$ uniformly in n, and likewise for the limit. Given ε, fix R with $(1+R^2)^{-1} < \varepsilon$; the finitely many low modes converge, so $\limsup_{n,m} \|u_n - u_m\|_{L^2}^2 \le 4\varepsilon$. Hence (u_n) is L^2-Cauchy along the subsequence, proving compactness. □

Remark B.23 (General bounded Lipschitz domains)**.** For $H^1(\Omega)$ with nonzero boundary trace one cannot extend by zero. The standard route is a bounded *Sobolev extension operator* $E\colon H^1(\Omega) \to H^1(\mathbb{R}^d)$ with $Eu|_\Omega = u$, which exists when $\partial\Omega$ is Lipschitz: locally flatten the boundary and reflect across it, then patch by a partition of unity. Granting E, the family $\left\{Eu : \|u\|_{H^1(\Omega)} \le 1\right\}$ is H^1-bounded with supports in a fixed bounded set, so the argument of theorem B.21 (via theorem B.19 and lemma B.20) yields compactness of $H^1(\Omega) \hookrightarrow L^2(\Omega)$. Every PDE existence proof in this book uses either Dirichlet data (the H^1_0 case of theorem B.21) or the periodic setting (theorem B.22), both proved here in full; the extension operator is needed only for the general-trace statement, and we refer to [64, Ch. 5] for its construction.

B.9 Arzelà–Ascoli for metric-space-valued maps

The classical Arzelà–Ascoli theorem characterises compactness of families of *real-valued* continuous functions; the existence theory of chapter 10 needs the same statement for maps from a compact interval into a (non-locally-compact) metric space — the Wasserstein space $(\mathcal{P}_1(\mathbb{R}^d), \mathcal{W}_1)$ of population flows. We record the general version; the real-valued case used above (e.g. in theorem B.19) is the instance $(Y,d) = (\mathbb{R}, |\cdot|)$.

Theorem B.24 (Arzelà–Ascoli, metric-space-valued)**.** *Let K be a compact metric space, (Y,d) a complete metric space, and $C(K;Y)$ the space of continuous maps with the uniform metric $\rho(f,g) = \sup_{t\in K} d(f(t),g(t))$. A family $\mathcal{F} \subseteq C(K;Y)$ is relatively compact in $(C(K;Y),\rho)$ if and only if*

(i) $\mathcal{F}$ *is* uniformly equicontinuous*: for every $\varepsilon > 0$ there is $\delta > 0$ with $d(f(s),f(t)) < \varepsilon$ whenever $d_K(s,t) < \delta$, for all $f \in \mathcal{F}$; and*

(ii) $\mathcal{F}$ *is* pointwise relatively compact*: for each $t \in K$ the set $\mathcal{F}(t) = \{f(t) : f \in \mathcal{F}\}$ has compact closure in Y.*

Proof. Necessity. A relatively compact $\mathcal{F}$ is totally bounded, which gives (i) by an $\varepsilon/3$ argument, and each evaluation $f \mapsto f(t)$ is continuous $C(K;Y) \to Y$, carrying the relatively compact $\mathcal{F}$ to the relatively compact $\mathcal{F}(t)$, giving (ii).

Sufficiency. Since Y is complete, $(C(K;Y),\rho)$ is complete, so it suffices to show every sequence $(f_n) \subseteq \mathcal{F}$ has a ρ-convergent subsequence. Let $D = \{t_1, t_2, \ldots\}$ be a countable dense subset of the (separable, being compact metric) space K. By (ii) and a diagonal argument, pass to a subsequence (still f_n) with $f_n(t_j)$ convergent in Y for every j. We show (f_n) is uniformly Cauchy. Fix $\varepsilon > 0$ and choose δ from (i) for $\varepsilon/3$. Finitely many balls $B(t_{j_i}, \delta)$, $i = 1, \ldots, N$, with centres in D, cover the compact K. Choose N_0 with $d(f_n(t_{j_i}), f_m(t_{j_i})) < \varepsilon/3$ for $n, m \geq N_0$ and all $i \leq N$. For any $t \in K$, picking i with $d_K(t, t_{j_i}) < \delta$,

$$d(f_n(t), f_m(t)) \leq d(f_n(t), f_n(t_{j_i})) + d(f_n(t_{j_i}), f_m(t_{j_i})) + d(f_m(t_{j_i}), f_m(t)) < \varepsilon,$$

the outer terms by (i). Hence $\rho(f_n, f_m) \leq \varepsilon$ for $n, m \geq N_0$, so (f_n) is ρ-Cauchy and converges in $C(K; Y)$. □

Remark B.25. In lemma 10.8, $K = [0, T]$, $Y = (\mathcal{P}_1(\mathbb{R}^d), \mathcal{W}_1)$ (complete by theorem D.11), equicontinuity is the uniform $\frac{1}{2}$-Hölder modulus ω of lemma 10.7, and pointwise relative compactness is the uniform second-moment bound feeding tightness and $\mathcal{W}_1$-compactness (theorem D.12).

B.10 Berge's maximum theorem

The closed-graph input to the Kakutani–Fan–Glicksberg fixed point of the relaxed MFG (theorem 10.19) is supplied by Berge's maximum theorem, controlling how the optimal value and optimiser set of a parametrised optimisation vary with the parameter.

Definition B.26 (Hemicontinuity)**.** A correspondence $\Gamma\colon X \rightrightarrows Y$ into a topological space is *upper hemicontinuous (uhc) with compact values at x* if $\Gamma(x)$ is compact and every open $V \supseteq \Gamma(x)$ has a neighbourhood $U \ni x$ with $\Gamma(x') \subseteq V$ for $x' \in U$; it is *lower hemicontinuous (lhc) at x* if every open V meeting $\Gamma(x)$ has a neighbourhood $U \ni x$ with $\Gamma(x') \cap V \neq \varnothing$ for $x' \in U$. For compact-valued correspondences into a metric space with a uniformly compact range along convergent parameters, uhc is equivalent to the sequential closed-graph property: $x_n \to x$, $y_n \in \Gamma(x_n)$, $y_n \to y$ imply $y \in \Gamma(x)$.

Theorem B.27 (Berge)**.** *Let X be a topological space, Y a metric space, $\Gamma\colon X \rightrightarrows Y$ continuous (uhc and lhc) with nonempty compact values, and $f\colon X \times Y \to \mathbb{R}$ continuous. Then the value $V(x) = \max_{y \in \Gamma(x)} f(x, y)$ is continuous, and the optimiser correspondence $M(x) = \arg\max_{y \in \Gamma(x)} f(x, y)$ is uhc with nonempty compact values.*

Proof. $f(x, \cdot)$ is continuous on the nonempty compact $\Gamma(x)$, so it attains its max and $M(x)$ is a nonempty closed (hence compact) subset of $\Gamma(x)$.

Lower semicontinuity of V. Fix x, $\varepsilon > 0$, $y^* \in M(x)$. By lhc and continuity of f, for x' near x there is $y' \in \Gamma(x')$ near y^* with $f(x', y') > V(x) - \varepsilon$; thus $V(x') > V(x) - \varepsilon$.

Upper semicontinuity of V. If $x_n \to x$ with $V(x_n) \to \ell > V(x)$, pick $y_n \in M(x_n)$; by uhc of Γ the y_n eventually lie in a compact neighbourhood of $\Gamma(x)$, so $y_n \to y \in \Gamma(x)$ along a subsequence, and $f(x, y) = \lim f(x_n, y_n) = \ell > V(x)$ contradicts $V(x) \geq f(x, y)$. So V is continuous.

M is uhc. If $x_n \to x$, $y_n \in M(x_n)$, $y_n \to y$, then $y \in \Gamma(x)$ by uhc of Γ and $f(x, y) = \lim f(x_n, y_n) = \lim V(x_n) = V(x)$, so $y \in M(x)$: closed graph with locally compact range, i.e. uhc. □

B.11 Measurable selection: the Kuratowski–Ryll-Nardzewski theorem

Recovering an executable feedback from the set-valued optimiser correspondence of lemma 10.20 requires a *measurable* choice of one optimiser at each (t, x).

Definition B.28 (Weakly measurable correspondence)**.** Let (Ω, Σ) be a measurable space and Y a Polish space. A correspondence $\Gamma\colon \Omega \rightrightarrows Y$ with nonempty closed values is *weakly measurable* if for every open $U \subseteq Y$ the lower inverse $\Gamma^-(U) = \{\omega : \Gamma(\omega) \cap U \neq \varnothing\}$ lies in Σ. A *selection* is a measurable $f\colon \Omega \to Y$ with $f(\omega) \in \Gamma(\omega)$ for all ω.

Theorem B.29 (Kuratowski–Ryll-Nardzewski)**.** *Let (Ω, Σ) be a measurable space and Y a Polish space with complete metric $d \leq 1$. Every weakly measurable $\Gamma\colon \Omega \rightrightarrows Y$ with nonempty closed values admits a measurable selection.*

Proof. Fix a countable dense set $\{y_1, y_2, \ldots\} \subseteq Y$. We construct measurable countably-valued $f_n\colon \Omega \to Y$ with, for all ω,

$$d(f_n(\omega), \Gamma(\omega)) < 2^{-n} \quad \text{and} \quad d(f_n(\omega), f_{n-1}(\omega)) < 3 \cdot 2^{-n} \quad (n \geq 1), \tag{B.3}$$

so that $(f_n(\omega))$ is uniformly Cauchy, converging to a measurable $f(\omega)$ with $f(\omega) \in \Gamma(\omega)$ (closedness of $\Gamma(\omega)$ and the first bound).

Base step. Let $f_0(\omega) = y_{k(\omega)}$, $k(\omega)$ the least k with $\Gamma(\omega) \cap B(y_k, 1) \neq \varnothing$ (exists by density and nonemptiness). The cell $\{k(\omega) = k\} = \Gamma^-(B(y_k, 1)) \setminus \bigcup_{j<k} \Gamma^-(B(y_j, 1)) \in \Sigma$ by weak measurability, so f_0 is measurable, countably valued, with $d(f_0(\omega), \Gamma(\omega)) < 1$.

Inductive step. Given measurable countably-valued f_{n-1} with $d(f_{n-1}(\omega), \Gamma(\omega)) < 2^{-(n-1)}$, on each cell $\{f_{n-1} = y_i\}$ the correspondence $\Gamma_i(\omega) = \Gamma(\omega) \cap \overline{B}(y_i, 2^{-(n-1)})$ has nonempty closed values and is weakly measurable (its lower inverses are obtained from those of Γ by countable Boolean operations with the fixed ball). Apply the base-step rule to Γ_i at radius 2^{-n}: $f_n(\omega) = y_k$ for the least k with $\Gamma_i(\omega) \cap B(y_k, 2^{-n}) \neq \varnothing$. Then $d(f_n(\omega), \Gamma(\omega)) < 2^{-n}$; and since $f_n(\omega)$ and $f_{n-1}(\omega) = y_i$ are within 2^{-n} resp. $2^{-(n-1)}$ of a common point of $\Gamma_i(\omega)$, $d(f_n, f_{n-1}) < 2^{-n} + 2^{-(n-1)} = 3 \cdot 2^{-n}$, giving (B.3). Each f_n is measurable, defined cellwise by least-index selections. This completes the induction and the proof. □

Corollary B.30 (Euclidean (Filippov) special case)**.** *If $Y = \mathbb{R}^k$ and $\Gamma(\omega)$ is the nonempty compact set of maximisers of a Carathéodory function $a \mapsto h(\omega, a)$ over a compact $A \subseteq \mathbb{R}^k$ (h measurable in ω, continuous in a), then Γ is weakly measurable and the* lexicographic *selection — the maximiser of least first coordinate, then least second, and so on — is measurable. Each successive coordinate optimisation over a compact set defines an uhc singleton-valued map by theorem B.27, so no general selection theorem is needed.*

Proof. Weak measurability holds because $\Gamma^-(U) = \left\{\omega : \sup_{a \in \overline{U} \cap A} h(\omega, a) = \max_A h(\omega, \cdot)\right\}$ is measurable (suprema of a Carathéodory function over the separable sets $\overline{U} \cap A$ and A are measurable in ω). The lexicographic map is the k-fold composition of theorem B.27, each step restricting to the compact optimiser set of the previous coordinate; the terminal singleton-valued correspondence is uhc, hence has measurable graph and a measurable selection. □

Bibliographic notes

The Hahn–Banach theorem, the Minkowski-gauge separation argument, and the Alexander-subbase proof of Tychonoff are classical; see [31, Ch. 1,3] and [113, Ch. IV]. The Sperner-lemma proof of Brouwer and the reduction of Schauder and Kakutani to it follow [79, Ch. I–II]; the continuous-selection route to Kakutani is Cellina's. The metric-space Arzelà–Ascoli theorem, Berge's maximum theorem, and the Kuratowski–Ryll-Nardzewski measurable-selection theorem are standard; see [6, Ch. 17–18]. The Sobolev material—Meyers–Serrin density, Poincaré, and Rellich–Kondrachov via the Fréchet–Kolmogorov criterion—follows [64, Ch. 5] and [31, Ch. 8–9]. All statements are used in chapter 2 and downstream in the existence theory of chapters 10 and 24.

Appendix C

Stochastic Calculus: Foundations and Deferred Proofs

This appendix supplies, in full or by complete and honest proof, the beyond-prerequisite results that chapter 5 states and uses but defers for the sake of narrative flow, and collects the standard tools of stochastic calculus that the probabilistic chapters (chapters 9 and 23) draw on. Nothing here is assumed from outside the book: each result is proved from the probabilistic foundations of chapter 3, the functional analysis of chapter 2, and the Brownian and Itô material built in chapter 5 itself. External references in the bibliographic notes are for attribution and further reading only.

We work throughout on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions (completeness and right-continuity of the filtration), with W a Brownian motion as in definition 5.1. We freely use the notation of chapter 5: the integrand space $\mathcal{M}^2[0,T]$ of progressively measurable processes with $\mathbb{E}\int_0^T H_t^2 \,\mathrm{d}t < \infty$, the Itô integral, and the quadratic variation $\langle\cdot\rangle$.

C.1 Grönwall's inequality

The closing device of nearly every existence, uniqueness, and stability estimate in the book — including the well-posedness of theorem 5.10 — is the following elementary integral inequality. We state and prove the version actually used.

Lemma C.1 (Grönwall's inequality, integral form)**.** *Let $\phi : [0,T] \to [0,\infty)$ be Borel measurable and bounded, and suppose there are constants $A \geq 0$ and $B \geq 0$ with*

$$\phi(t) \;\leq\; A + B\int_0^t \phi(s)\,\mathrm{d}s \qquad \textit{for all } t \in [0,T]. \tag{C.1}$$

Then $\phi(t) \leq A\,e^{Bt}$ for all $t \in [0,T]$. In particular, if $A = 0$ then $\phi \equiv 0$.

Proof. Set $\Psi(t) := \int_0^t \phi(s)\,\mathrm{d}s$, which is absolutely continuous with $\Psi'(t) = \phi(t)$ for almost every t, and $\Psi(0) = 0$. Inequality (C.1) reads $\Psi'(t) \leq A + B\,\Psi(t)$ a.e. Multiplying by the integrating factor

$e^{-Bt} > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(e^{-Bt}\Psi(t)\Big) = e^{-Bt}(\Psi'(t) - B\Psi(t)) \le A\, e^{-Bt}.$$

Integrating from 0 to t and using $\Psi(0) = 0$ gives $e^{-Bt}\Psi(t) \le \frac{A}{B}\big(1 - e^{-Bt}\big)$ when $B > 0$, i.e. $\Psi(t) \le \frac{A}{B}\big(e^{Bt} - 1\big)$. Substituting back into (C.1),

$$\phi(t) \le A + B\,\Psi(t) \le A + A\big(e^{Bt} - 1\big) = A\, e^{Bt}.$$

When $B = 0$ the claim is immediate from (C.1). Finally, if $A = 0$ then $0 \le \phi(t) \le A\, e^{Bt} = 0$, so $\phi \equiv 0$. $\square$

C.2 Doob's maximal inequality and the Burkholder–Davis–Gundy estimate

The Itô integral is an L^2-bounded martingale (theorem 5.7); to control its *running maximum* — needed both for the continuous-modification claim of definition 5.6 and for the maximal $\mathcal{S}^2$ estimate that powers the Picard iteration in theorem 5.10 — we use Doob's inequality.

Theorem C.2 (Doob's L^2 maximal inequality)**.** *Let $(M_t)_{t\in[0,T]}$ be a continuous $\mathbb{F}$-martingale with $\mathbb{E}M_T^2 < \infty$. Then*

$$\mathbb{E}\Big[\sup_{0\le t\le T} M_t^2\Big] \;\le\; 4\,\mathbb{E}\,M_T^2. \tag{C.2}$$

Proof. We first prove the *maximal inequality*: for $\lambda > 0$,

$$\lambda\,\mathbb{P}\Big(\sup_{t\le T}|M_t| \ge \lambda\Big) \le \mathbb{E}\big[\,|M_T|\;\mathbf{1}_{\{\sup_t|M_t|\ge\lambda\}}\big]. \tag{C.3}$$

By continuity of paths, $\tau := \inf\{t : |M_t| \ge \lambda\} \wedge T$ is a stopping time and on the event $\{\sup_{t\le T}|M_t| \ge \lambda\}$ we have $|M_\tau| \ge \lambda$. By optional stopping (Jensen applied to the submartingale $|M|$) and the tower property, $\mathbb{E}[|M_T| \mid \mathcal{F}_\tau] \ge |M_\tau|$, so

$$\lambda\,\mathbb{P}\Big(\sup_{t\le T}|M_t| \ge \lambda\Big) \le \mathbb{E}\big[\,|M_\tau|\;\mathbf{1}_{\{\sup|M|\ge\lambda\}}\big] \le \mathbb{E}\big[\,|M_T|\;\mathbf{1}_{\{\sup|M|\ge\lambda\}}\big],$$

which is (C.3). Now write $S := \sup_{t\le T}|M_t|$. Using (C.3) and Fubini,

$$\mathbb{E}\,S^2 = \int_0^\infty 2\lambda\,\mathbb{P}(S \ge \lambda)\;\mathrm{d}\lambda \le \int_0^\infty 2\,\mathbb{E}\big[\,|M_T|\,\mathbf{1}_{\{S\ge\lambda\}}\big]\,\mathrm{d}\lambda = 2\,\mathbb{E}\Big[\,|M_T|\int_0^S \mathrm{d}\lambda\Big] = 2\,\mathbb{E}\big[\,|M_T|\;S\big].$$

By Cauchy–Schwarz, $2\,\mathbb{E}[|M_T|\,S] \le 2\,(\mathbb{E}M_T^2)^{1/2}(\mathbb{E}S^2)^{1/2}$. If $\mathbb{E}S^2 < \infty$ this gives $\mathbb{E}S^2 \le 2(\mathbb{E}M_T^2)^{1/2}(\mathbb{E}S^2)^{1/2}$, i.e. $(\mathbb{E}S^2)^{1/2} \le 2(\mathbb{E}M_T^2)^{1/2}$, which is (C.2). The integrability of S is obtained by first applying the argument to the bounded stopped martingale $M_{t\wedge\tau_n}$, $\tau_n = \inf\{t : |M_t| \ge n\}$ (bounded by n via path continuity), and letting $n \to \infty$ by monotone convergence ($\sup_{t\le T}|M_{t\wedge\tau_n}| \uparrow S$). $\square$

For the diffusion term of an SDE the integrand of the Itô integral is itself random, and one needs to bound the running maximum of $M_t = \int_0^t H_s \, \mathrm{d}W_s$ by the *predictable* quantity $\langle M\rangle_T = \int_0^T H_s^2 \, \mathrm{d}s$. In the L^2 case this is immediate from Doob: by theorem 5.7,

$$\mathbb{E}\Big[\sup_{t\le T}\Big(\int_0^t H_s \, \mathrm{d}W_s\Big)^2\Big] \;\le\; 4\,\mathbb{E}\Big[\Big(\int_0^T H_s \, \mathrm{d}W_s\Big)^2\Big] \;=\; 4\,\mathbb{E}\int_0^T H_s^2 \, \mathrm{d}s. \tag{C.4}$$

This L^2 bound is all that theorem 5.10 uses. We record the general statement, used in the BSDE chapters, for completeness.

Theorem C.3 (Burkholder–Davis–Gundy inequality)**.** *For every $p \in (0,\infty)$ there exist universal constants $0 < c_p \le C_p < \infty$ such that for every continuous local martingale M with $M_0 = 0$,*

$$c_p\,\mathbb{E}\big[\langle M\rangle_T^{p/2}\big] \;\le\; \mathbb{E}\Big[\sup_{t\le T}|M_t|^p\Big] \;\le\; C_p\,\mathbb{E}\big[\langle M\rangle_T^{p/2}\big].$$

Proof of the case $p = 2$. For $p = 2$ the upper bound with $C_2 = 4$ is (C.4) (extended to local martingales by localization, below), and the lower bound with $c_2 = 1$ is $\mathbb{E}\sup_t M_t^2 \ge \mathbb{E}M_T^2 = \mathbb{E}\langle M\rangle_T$, the last equality being theorem 5.7(3). The general p follows from the good-λ inequalities of [86, 114]; only $p = 2$ is used in chapter 5. ☐

Remark C.4 (Localization to local martingales)**.** A continuous *local* martingale M is one for which there is an increasing sequence of stopping times $\tau_n \uparrow \infty$ with each stopped process $M_t^{\tau_n} = M_{t\wedge\tau_n}$ a genuine martingale. The Itô integral $\int_0^t H \, \mathrm{d}W$ is a true martingale when $H \in \mathcal{M}^2[0,T]$ and a local martingale under the weaker hypothesis $\int_0^T H_s^2 \, \mathrm{d}s < \infty$ almost surely, with localizing times $\tau_n = \inf\{t : \int_0^t H_s^2 \, \mathrm{d}s \ge n\}$. All martingale identities above extend to local martingales by applying them to M^{τ_n} and passing to the limit, monotone or dominated convergence supplying the integrability.

C.3 A continuous finite-variation martingale is constant

The backward-equation proof (theorem 5.15) and many later arguments use the fact that a process which is simultaneously a martingale and smooth enough to have finite variation cannot move. It is a clean corollary of the quadratic-variation theory.

Lemma C.5 (No continuous finite-variation martingale moves)**.** *Let $(M_t)_{t\in[0,T]}$ be a continuous $\mathbb{F}$-martingale with $M_0 = 0$ whose paths have finite total variation on $[0,T]$ almost surely. Then $M_t = 0$ for all t, almost surely.*

Proof. Assume first M is bounded, $|M_t| \le K$. Fix t and a partition $0 = t_0 < \cdots < t_n = t$ with mesh $\to 0$. Because M is a martingale, the increments $\Delta_i M = M_{t_{i+1}} - M_{t_i}$ are orthogonal in L^2 (for $i < j$, $\mathbb{E}[\Delta_i M\,\Delta_j M] = \mathbb{E}[\Delta_i M\,\mathbb{E}[\Delta_j M \mid \mathcal{F}_{t_j}]] = 0$), so

$$\mathbb{E}\,M_t^2 = \mathbb{E}\sum_i(\Delta_i M)^2 \le \mathbb{E}\Big[\max_i|\Delta_i M|\;\sum_i|\Delta_i M|\Big] \le \mathbb{E}\Big[\max_i|\Delta_i M|\cdot V\Big],$$

where V is the (finite) total variation of the path. As the mesh shrinks, $\max_i |\Delta_i M| \to 0$ by uniform continuity of the path on $[0,t]$, and the integrand is bounded by $2KV$; if V is bounded the dominated convergence theorem gives $\mathbb{E}M_t^2 = 0$. For the general (unbounded-variation-allowed only via the stopping) case, localize by $\tau_n = \inf\{t : |M_t| \geq n \text{ or } V_t \geq n\}$, apply the bounded case to M^{τ_n} to get $M_{t\wedge\tau_n} = 0$, and let $n \to \infty$; since $\tau_n \uparrow T$ almost surely, $M_t = 0$ for all t. □

The same computation identifies the quadratic variation of a continuous martingale as the unique continuous increasing process making $M_t^2 - \langle M\rangle_t$ a martingale, and shows a continuous finite-variation martingale has $\langle M\rangle \equiv 0$; the lemma is the statement that $\langle M\rangle \equiv 0$ forces M constant.

C.4 Density of simple processes in $\mathcal{M}^2[0,T]$

The Itô integral is *defined* on simple integrands and extended by isometry (definition 5.6); the extension is legitimate only because the simple processes are dense. We prove this load-bearing approximation lemma.

Lemma C.6 (Density of simple processes)**.** *The simple processes are dense in $\mathcal{M}^2[0,T]$ for the norm $\|H\|^2_{\mathcal{M}^2} = \mathbb{E}\int_0^T H_t^2 \,\mathrm{d}t$. That is, for every progressively measurable H with $\|H\|_{\mathcal{M}^2} < \infty$ and every $\varepsilon > 0$ there is a simple process H^ε with $\|H - H^\varepsilon\|_{\mathcal{M}^2} < \varepsilon$.*

Proof. We approximate in three steps, each reducing to the next a strictly larger class.

Step 1: reduction to bounded H. For $H \in \mathcal{M}^2[0,T]$ put $H_t^{(K)} = H_t\,\mathbf{1}_{\{|H_t|\leq K\}}$. Then $H^{(K)}$ is progressively measurable and bounded, and $\left|H - H^{(K)}\right|^2 \leq H^2 \in L^1(\mathrm{d}t \otimes \mathbb{P})$, so by dominated convergence $\left\|H - H^{(K)}\right\|_{\mathcal{M}^2} \to 0$ as $K \to \infty$. It therefore suffices to approximate bounded progressively measurable processes.

Step 2: reduction to (a.s.) continuous H. Let H be bounded, $|H| \leq K$, progressively measurable. For $h > 0$ define the moving average

$$H_t^h := \frac{1}{h}\int_{(t-h)^+}^t H_s \,\mathrm{d}s.$$

For each ω, $t \mapsto H_t^h(\omega)$ is continuous (an indefinite integral), bounded by K, and adapted (the average uses only the past, and H is progressively measurable so the integral is $\mathcal{F}_t$-measurable); hence H^h is progressively measurable and continuous. By Lebesgue's differentiation theorem, for a.e. t and a.e. ω, $H_t^h(\omega) \to H_t(\omega)$ as $h \downarrow 0$; since $\left|H^h - H\right|^2 \leq (2K)^2$ is bounded, dominated convergence on $[0,T] \times \Omega$ gives $\left\|H^h - H\right\|_{\mathcal{M}^2} \to 0$. It suffices to approximate bounded, continuous, adapted processes.

Step 3: continuous $\Rightarrow$ simple. Let H be bounded, continuous, adapted. For the partition $t_i = iT/n$ put $H_t^{(n)} = \sum_{i=0}^{n-1} H_{t_i}\,\mathbf{1}_{(t_i,t_{i+1}]}(t)$, a simple process (each $\eta_i = H_{t_i}$ is $\mathcal{F}_{t_i}$-measurable and bounded). By path continuity $H_t^{(n)}(\omega) \to H_t(\omega)$ for every t and ω, and $\left|H^{(n)}\right| \le K$; dominated convergence again yields $\left\|H^{(n)} - H\right\|_{\mathcal{M}^2} \to 0$.

Chaining the three steps: given H and ε, choose K, then h, then n so each approximation error is below $\varepsilon/3$, and the triangle inequality finishes the proof. $\square$

C.5 Itô's formula: the full proof

Theorem 5.9 states the change-of-variables rule for an Itô process and sketches the mechanism in the chapter body, deferring the remainder control and the multidimensional case to here. We supply the complete argument; it is the one place where the heuristic $(\mathrm{d}W)^2 = \mathrm{d}t$ must be promoted, with care, to an honest L^1 limit.

Theorem C.7 (Itô's formula)**.** *Let X be the scalar Itô process $\mathrm{d}X_t = b_t\ \mathrm{d}t + \sigma_t\ \mathrm{d}W_t$ of* (5.8)*, with b progressively measurable and integrable and $\sigma \in \mathcal{M}^2[0,T]$, and let $F \in C^{1,2}([0,T]\times\mathbb{R})$. Then a.s., for all $t \in [0,T]$,* (5.10) *holds:*

$$F(t,X_t) - F(0,X_0) = \int_0^t \Big(\partial_t F + b_s\,\partial_x F + \tfrac{1}{2}\,\sigma_s^2\,\partial_{xx}F\Big)(s,X_s)\ \mathrm{d}s + \int_0^t \sigma_s\,\partial_x F(s,X_s)\ \mathrm{d}W_s.$$

Proof. Step 0: reduction to bounded data. Let $\tau_R = \inf\{t : |X_t| \ge R\} \wedge T$. If the formula holds for the stopped process X^{τ_R} for every R, then letting $R \to \infty$ (so $\tau_R \uparrow T$ a.s. by path continuity) recovers it for X, both integrals converging by dominated convergence. On $\{t \le \tau_R\}$ the process stays in the compact $[-R,R]$, where F and its derivatives $\partial_t F, \partial_x F, \partial_{xx}F$ are bounded and uniformly continuous. We may thus assume F has bounded, uniformly continuous first t- and first two x-derivatives, and that b, σ are bounded (truncate them likewise; on $[0,\tau_R]$ the truncation is invisible).

Step 1: the Taylor expansion. Fix t and a partition $0 = t_0 < \cdots < t_N = t$ of mesh $\|\pi\|$. Telescoping and Taylor-expanding F to first order in time and second order in space across each subinterval,

$$F(t,X_t) - F(0,X_0) = \sum_i \Big[\partial_t F(\theta_i, X_{t_i})\,\Delta_i t + \partial_x F(t_i,X_{t_i})\,\Delta_i X + \tfrac{1}{2}\partial_{xx}F(t_i,\zeta_i)\,(\Delta_i X)^2\Big],$$

where $\Delta_i t = t_{i+1} - t_i$, $\Delta_i X = X_{t_{i+1}} - X_{t_i}$, $\theta_i \in [t_i, t_{i+1}]$ and ζ_i lies between X_{t_i} and $X_{t_{i+1}}$ (mean-value form of the remainder; the mixed second-order term $\partial_{tx}F$ and the pure $\partial_{tt}F$ term are absorbed since $F \in C^{1,2}$ means C^1 in t, and the t-increment carries no quadratic contribution—see Step 3).

Step 2: the first-order terms. As $\|\pi\| \to 0$, $\sum_i \partial_t F(\theta_i, X_{t_i})\Delta_i t \to \int_0^t \partial_t F(s,X_s)\ \mathrm{d}s$ (Riemann sums of a continuous integrand), and $\sum_i \partial_x F(t_i,X_{t_i})\Delta_i X \to \int_0^t \partial_x F(s,X_s)\ \mathrm{d}X_s = \int_0^t \partial_x F\,b_s\ \mathrm{d}s + \int_0^t \partial_x F\,\sigma_s\ \mathrm{d}W_s$, the first by Riemann convergence and the second because $\partial_x F(t_i, X_{t_i})$ is the left-endpoint (non-anticipating) evaluation defining the Itô integral, and $\partial_x F(\cdot, X_\cdot) \in \mathcal{M}^2$ is approximated by its own step function (the L^2-convergence of the corresponding Itô integrals is the isometry plus lemma C.6).

Step 3: the quadratic term — the heart of the matter. Write $(\Delta_i X)^2 = (b_{t_i}\Delta_i t + \sigma_{t_i}\Delta_i W)^2 + e_i$, where e_i collects the error from replacing the exact increment $\Delta_i X$ by its frozen-coefficient approximation; under boundedness $\sum_i \mathbb{E}\,|e_i| \to 0$. Expanding the square,

$$(b_{t_i}\Delta_i t + \sigma_{t_i}\Delta_i W)^2 = b_{t_i}^2(\Delta_i t)^2 + 2b_{t_i}\sigma_{t_i}\Delta_i t\,\Delta_i W + \sigma_{t_i}^2(\Delta_i W)^2.$$

The first sum is $O(\|\pi\|\ t) \to 0$; the second has mean zero and variance $O(\sum_i(\Delta_i t)^2\mathbb{E}(\Delta_i W)^2) = O(\|\pi\|) \to 0$ in L^2. The decisive third sum, weighted by $\frac{1}{2}\partial_{xx}F(t_i,\zeta_i)$, is handled by the key lemma below: in L^1,

$$\sum_i \tfrac{1}{2}\,\partial_{xx}F(t_i,\zeta_i)\,\sigma_{t_i}^2\,(\Delta_i W)^2 \;\longrightarrow\; \tfrac{1}{2}\int_0^t \partial_{xx}F(s,X_s)\,\sigma_s^2\ \mathrm{d}s. \tag{C.5}$$

Granting (C.5), assembling Steps 2–3 gives exactly the asserted identity (the ζ_i may be replaced by X_{t_i} at no cost, since $|\zeta_i - X_{t_i}| \le |\Delta_i X| \to 0$ uniformly and $\partial_{xx}F$ is uniformly continuous).

Step 4: proof of (C.5). Abbreviate $g_i = \partial_{xx}F(t_i,X_{t_i})\sigma_{t_i}^2$, a bounded $\mathcal{F}_{t_i}$-measurable random variable, and compare the random sum $S = \sum_i g_i(\Delta_i W)^2$ with the deterministic-time sum $T' = \sum_i g_i\,\Delta_i t$. The latter is a Riemann sum converging a.s. to $\int_0^t \partial_{xx}F\,\sigma_s^2\ \mathrm{d}s$. For the difference, $\Delta_i W$ is independent of $\mathcal{F}_{t_i}$ with $\mathbb{E}[(\Delta_i W)^2 - \Delta_i t \mid \mathcal{F}_{t_i}] = 0$, so the summands of $S - T' = \sum_i g_i((\Delta_i W)^2 - \Delta_i t)$ are martingale differences; hence

$$\mathbb{E}\,(S-T')^2 = \sum_i \mathbb{E}\Big[g_i^2\,\mathbb{E}\big[((\Delta_i W)^2 - \Delta_i t)^2 \mid \mathcal{F}_{t_i}\big]\Big] = \sum_i \mathbb{E}[g_i^2]\cdot 2(\Delta_i t)^2 \le 2\ \|g\|_\infty^2\ \|\pi\|\ t \;\to\; 0,$$

using $\mathrm{Var}((\Delta_i W)^2) = 2(\Delta_i t)^2$ (the Gaussian fourth moment, as in theorem 5.4). Thus $S - T' \to 0$ in L^2, hence in L^1, and $T' \to \int$ a.s.; replacing $\partial_{xx}F(t_i,X_{t_i})$ by $\partial_{xx}F(t_i,\zeta_i)$ changes S by at most $\sum_i |\partial_{xx}F(t_i,\zeta_i) - \partial_{xx}F(t_i,X_{t_i})|\,\sigma_{t_i}^2(\Delta_i W)^2$, which $\to 0$ in L^1 by uniform continuity of $\partial_{xx}F$ and $\sum_i \sigma_{t_i}^2(\Delta_i W)^2 \to \langle X\rangle_t$ bounded. This proves (C.5).

Step 5: the multidimensional case. For $X \in \mathbb{R}^d$ with $\mathrm{d}X = b\ \mathrm{d}t + \sigma\ \mathrm{d}W$, $\sigma \in \mathbb{R}^{d\times d}$, the second-order Taylor term is $\frac{1}{2}(\Delta_i X)^\top\,\mathrm{D}^2\,F(t_i,\zeta_i)(\Delta_i X)$, and the multiplication table $\mathrm{d}W^k\,\mathrm{d}W^\ell = \delta_{k\ell}\,\mathrm{d}t$ gives, by the polarized version of Step 4 applied to each entry,

$$\sum_i \tfrac{1}{2}(\Delta_i X)^\top\,\mathrm{D}^2\,F\,(\Delta_i X) \;\to\; \tfrac{1}{2}\int_0^t \mathrm{tr}\big(\sigma_s\sigma_s^\top\,\mathrm{D}^2\,F(s,X_s)\big)\ \mathrm{d}s,$$

which is the trace term of (5.11); the first-order terms are componentwise as in Step 2. This proves (5.11). □

C.6 The Kolmogorov extension theorem

Theorem 5.2 constructs Brownian motion (route (i)) by first realizing the consistent family of Gaussian finite-dimensional laws on path space and then repairing continuity. The first step is the Kolmogorov (Daniell–Kolmogorov) extension theorem, which we now state and prove.

Let I be an index set (for us $I = [0,\infty)$) and, for each finite $J = \{t_1,\dots,t_n\} \subset I$, let μ_J be a Borel probability measure on $\mathbb{R}^J$. The family $\{\mu_J\}$ is *consistent* (or *projective*) if it is compatible under the natural coordinate projections: for $J \subseteq J'$ and the projection $\pi_{J',J} : \mathbb{R}^{J'} \to \mathbb{R}^J$, one has $\mu_J = \mu_{J'} \circ \pi_{J',J}^{-1}$; and μ_J is invariant under reordering of the coordinates.

Theorem C.8 (Kolmogorov extension theorem)**.** *Given a consistent family* $\{\mu_J\}_{J\subset I \text{ finite}}$ *of Borel probability measures on the finite-dimensional spaces* $\mathbb{R}^J$*, there is a unique probability measure* $\mathbb{P}$ *on the product space* $(\mathbb{R}^I, \mathcal{B}(\mathbb{R})^{\otimes I})$ *whose finite-dimensional marginals are the* μ_J*: for every finite* J *and Borel* $A \subseteq \mathbb{R}^J$*,*

$$\mathbb{P}(\{x \in \mathbb{R}^I : (x_t)_{t\in J} \in A\}) = \mu_J(A).$$

Proof. Let $\mathcal{C}$ be the algebra of *cylinder sets*, i.e. sets of the form $C = \{x : (x_t)_{t\in J} \in A\}$ with $J \subset I$ finite and $A \in \mathcal{B}(\mathbb{R}^J)$. Define $\mathbb{P}_0(C) = \mu_J(A)$. Consistency makes this well defined (a cylinder has many representations, with larger index sets; the projective relations force the value to agree), and $\mathbb{P}_0$ is finitely additive on $\mathcal{C}$ with $\mathbb{P}_0(\mathbb{R}^I) = 1$. By Carathéodory's extension theorem it suffices to show $\mathbb{P}_0$ is *countably* additive on $\mathcal{C}$, equivalently continuous at $\emptyset$: if $C_1 \supseteq C_2 \supseteq \cdots$ are cylinders with $\bigcap_k C_k = \emptyset$, then $\mathbb{P}_0(C_k) \to 0$.

Suppose, for contradiction, $\mathbb{P}_0(C_k) \geq \delta > 0$ for all k. Each C_k depends on a finite index set J_k, which we may take increasing, $C_k = \{(x_t)_{t\in J_k} \in A_k\}$ with $A_k \in \mathcal{B}(\mathbb{R}^{J_k})$. By inner regularity of the Borel measures μ_{J_k} on the (Polish) spaces $\mathbb{R}^{J_k}$, choose a *compact* $K_k \subseteq A_k$ with $\mu_{J_k}(A_k \setminus K_k) \leq \delta\, 2^{-k-1}$, and let $D_k = \{(x_t)_{t\in J_k} \in K_k\} \subseteq C_k$ be the corresponding compact-base cylinder. Put $E_k = D_1 \cap \cdots \cap D_k \subseteq C_k$. Then

$$\mathbb{P}_0(C_k \setminus E_k) \leq \sum_{j=1}^{k} \mathbb{P}_0(C_j \setminus D_j) \leq \sum_{j=1}^{k} \delta\, 2^{-j-1} \leq \tfrac{\delta}{2},$$

so $\mathbb{P}_0(E_k) \geq \delta - \frac{\delta}{2} = \frac{\delta}{2} > 0$; in particular every E_k is nonempty. Pick $x^{(k)} \in E_k$. The points $x^{(k)}$ for $k \geq m$ all lie, in the coordinates J_m, inside the compact set K_m; by a diagonal argument over the countably many coordinates $\bigcup_m J_m$ one extracts a subsequence along which every coordinate in $\bigcup_m J_m$ converges, to a limit point $x^\star$ (set the remaining coordinates arbitrarily). For each fixed m, the J_m-coordinates of $x^\star$ are a limit of points of the closed set K_m, hence lie in K_m, so $x^\star \in D_m \subseteq C_m$ for every m. Thus $x^\star \in \bigcap_m C_m$, contradicting $\bigcap_m C_m = \emptyset$. Therefore $\mathbb{P}_0(C_k) \to 0$, $\mathbb{P}_0$ is countably additive, and Carathéodory extends it uniquely to $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})^{\otimes I}$. □

Applied with μ_J the centered Gaussian on $\mathbb{R}^J$ of covariance $(\, s \wedge t\,)_{s,t\in J}$ — consistent because these are the marginals of one Gaussian family — the theorem yields a process with Brownian finite-dimensional distributions, the starting point of route (i) of theorem 5.2.

C.7 The Kolmogorov–Chentsov continuity theorem

The process produced by theorem C.8 lives on raw product space and need not have continuous paths; the continuity (axiom (4) of definition 5.1) and the sharp Hölder exponent of theorem 5.3(1) come from the following moment criterion.

Theorem C.9 (Kolmogorov–Chentsov)**.** *Let* $(X_t)_{t\in[0,T]}$ *be a real process such that for some constants* $p > 0$*,* $\gamma > 0$*,* $C < \infty$*,*

$$\mathbb{E}\,|X_t - X_s|^p \;\leq\; C\,|t-s|^{1+\gamma} \qquad \textit{for all } s,t \in [0,T]. \tag{C.6}$$

Then X admits a continuous modification $\widetilde{X}$, and the paths of $\widetilde{X}$ are locally Hölder continuous of every order $\alpha < \gamma/p$.

Proof. Work on $[0,1]$ (rescale time). For $n \geq 1$ let $D_n = \{k2^{-n} : 0 \leq k \leq 2^n\}$ be the dyadic points of generation n and $D = \bigcup_n D_n$. Fix $\alpha \in (0, \gamma/p)$. By (C.6) and Markov's inequality, for a consecutive dyadic pair $t = k2^{-n}$, $t' = (k+1)2^{-n}$,

$$\mathbb{P}\big(|X_{t'} - X_t| \geq 2^{-\alpha n} \big) \leq \frac{\mathbb{E}\,|X_{t'} - X_t|^p}{2^{-\alpha np}} \leq C\,2^{-n(1+\gamma)}\,2^{\alpha np} = C\,2^{-n(1+\gamma-\alpha p)}.$$

Summing over the 2^n such pairs in generation n,

$$\mathbb{P}\Big(\max_{0 \leq k < 2^n} \Big| X_{(k+1)2^{-n}} - X_{k2^{-n}} \Big| \geq 2^{-\alpha n} \Big) \leq 2^n \cdot C\,2^{-n(1+\gamma-\alpha p)} = C\,2^{-n(\gamma-\alpha p)}.$$

Because $\alpha < \gamma/p$ the exponent $\gamma - \alpha p > 0$, so these probabilities are summable; by Borel–Cantelli there is an a.s. finite random N such that for all $n \geq N$,

$$\max_{0 \leq k < 2^n} \big| X_{(k+1)2^{-n}} - X_{k2^{-n}} \big| < 2^{-\alpha n}. \tag{C.7}$$

Chaining. Fix ω in this full-measure event and take any two dyadics $s < t$ in D with $t - s \leq 2^{-N}$; let $m \geq N$ be such that $2^{-(m+1)} < t - s \leq 2^{-m}$. A standard dyadic-expansion argument writes the increment $X_t - X_s$ as a telescoping sum of generation-($\geq m$) consecutive-dyadic increments, at most two from each generation; (C.7) bounds each by $2^{-\alpha n}$, so

$$|X_t - X_s| \leq 2 \sum_{n \geq m} 2^{-\alpha n} = \frac{2}{1 - 2^{-\alpha}}\,2^{-\alpha m} \leq K_\alpha\,|t-s|^\alpha\,,$$

with $K_\alpha = \frac{2}{1-2^{-\alpha}} 2^\alpha$. Thus $t \mapsto X_t$ is α-Hölder on the dense set D, hence uniformly continuous there, and extends uniquely to a continuous function $\widetilde{X}$ on $[0,1]$, which is locally α-Hölder. Finally $\widetilde{X}$ is a modification: for any fixed t, taking dyadics $t_k \to t$, hypothesis (C.6) gives $X_{t_k} \to X_t$ in L^p hence in probability, while $X_{t_k} \to \widetilde{X}_t$ a.s.; the two limits agree a.s., so $\mathbb{P}(\widetilde{X}_t = X_t) = 1$. □

Remark C.10 (The Hölder exponent for Brownian motion)**.** For Brownian motion $W_t - W_s \sim N(0, t-s)$ has Gaussian moments $\mathbb{E}\,|W_t - W_s|^{2k} = (2k-1)!!\,|t-s|^k$ for every integer $k \geq 1$, i.e. (C.6) holds with $p = 2k$ and $\gamma = k - 1$. The theorem then yields continuous paths that are α-Hölder for every $\alpha < \gamma/p = \frac{1}{2} - \frac{1}{2k}$. Letting $k \to \infty$ shows every exponent $\alpha < \frac{1}{2}$ is achievable — the sharp regularity asserted in theorem 5.3(1). A single moment such as $p = 4$ ($k = 2$, $\gamma = 1$) already gives continuity, but only up to $\alpha < \frac{1}{4}$; the full strength $\alpha < \frac{1}{2}$ requires the high moments.

C.8 The Markov property of SDE solutions

The backward equation and Feynman–Kac (theorems 5.15 and 5.16) condition on the present state and use that the future depends on the past only through it. We record the property and its mechanism.

Proposition C.11 (Markov property). *Under the hypotheses of theorem 5.10 let $X^{t,x}$ denote the unique strong solution of $\mathrm{d}X_s = b(s, X_s)\,\mathrm{d}s + \sigma(s, X_s)\,\mathrm{d}W_s$ on $[t, T]$ started at $X_t = x$. Then the solution is a Markov process: for any bounded measurable Φ and $t \le r \le T$,*

$$\mathbb{E}\big[\Phi(X_r) \mid \mathcal{F}_t\big] = \mathbb{E}\big[\Phi(X_r) \mid X_t\big] = (P_{t,r}\Phi)(X_t), \qquad (P_{t,r}\Phi)(x) := \mathbb{E}\,\Phi(X_r^{t,x}).$$

In the time-homogeneous case $P_{t,r} = P_{r-t}$ depends only on the elapsed time, and the family $(P_h)_{h\ge0}$ is a semigroup, $P_{h+h'} = P_h P_{h'}$, whose infinitesimal generator is the operator $\mathcal{A}$ of definition 5.12.

Proof. The driving Brownian increments $(W_s - W_t)_{s\ge t}$ are independent of $\mathcal{F}_t$. By pathwise uniqueness (theorem 5.10) the solution on $[t, T]$ is a measurable functional $X_r = \mathcal{F}_{t,r}\big(X_t, (W_s - W_t)_{t\le s\le r}\big)$ of the initial value X_t and the post-t noise — the flow is well defined and depends on the past only through the starting point X_t and a noise increment independent of $\mathcal{F}_t$. For a bounded measurable Φ, the freezing lemma for conditional expectations (a random variable independent of a σ-algebra may be "integrated out," chapter 3) gives

$$\mathbb{E}\big[\Phi(X_r) \mid \mathcal{F}_t\big] = \mathbb{E}\big[\Phi\big(\mathcal{F}_{t,r}(\xi, (W_s - W_t)_s)\big)\big]\Big|_{\xi = X_t} = (P_{t,r}\Phi)(X_t),$$

which is $\mathcal{F}_t$-measurable through X_t alone; conditioning further on X_t leaves it unchanged, proving both equalities. The semigroup property is the composition of flows over $[t, t+h]$ and $[t+h, t+h+h']$ together with the tower property; that $\mathcal{A}$ generates (P_h) is Dynkin's formula proposition 5.13, which says $\frac{\mathrm{d}}{\mathrm{d}h}\big|_{0^+} P_h\phi = \mathcal{A}\phi$ for $\phi \in C_c^2$. ☐

C.9 Parabolic regularity for the forward and backward equations

Three theorems of chapter 5 — the smooth positive transition density under (A-Nondeg) (section 5.4), and the upgrade of the Fokker–Planck equation theorem 5.14 and the backward equation theorem 5.15 from distributional to classical solutions — rest on the regularizing effect of a uniformly nondegenerate second-order operator. We state the facts used, in the form needed, and indicate the proof; this is classical linear parabolic PDE theory and is the in-book home for those upgrades.

Theorem C.12 (Parabolic regularity under uniform ellipticity). *Let $b : [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$ and $a = \sigma\sigma^\top : [0, T] \times \mathbb{R}^d \to \mathbb{R}^{d\times d}$ be bounded and Hölder continuous in (t, x), with a uniformly elliptic, i.e. (A-Nondeg): $a(t, x) \succeq \lambda I$ for some $\lambda > 0$. Then:*

1. (Fundamental solution / transition density.) *The operator $\partial_t - \mathcal{A}$ has a fundamental solution $p(s, x; t, y)$, $0 \le s < t \le T$, which is strictly positive, jointly continuous, and $C^{1,2}$ in (t, y) and in (s, x), with the Gaussian-type two-sided bounds $c_1 t'^{-d/2} e^{-c_2|x-y|^2/t'} \le p \le C_1 t'^{-d/2} e^{-C_2|x-y|^2/t'}$ where $t' = t - s$. It is the transition density of the diffusion: $P_{s,t}\Phi(x) = \int p(s, x; t, y)\Phi(y)\,\mathrm{d}y$.*

2. (Backward equation.) *For ψ continuous of polynomial growth, $u(t, x) = \int p(t, x; T, y)\psi(y)\,\mathrm{d}y$ is $C^{1,2}$ on $[0, T) \times \mathbb{R}^d$, solves $\partial_t u + \mathcal{A}u = 0$ classically, and $u(t, \cdot) \to \psi$ as $t \uparrow T$.*

3. (Forward equation.) *For each fixed* (s, x) *the density* $y \mapsto p(s, x; t, y)$ *solves the Fokker–Planck equation* $\partial_t m = \mathcal{A}^* m$ *classically for* $t > s$*; consequently any initial law* m_0 *propagates to* $m_t(\mathrm{d}y) = \left(\int p(0, x; t, y)\, m_0(\mathrm{d}x)\right) \mathrm{d}y$*, which is a smooth positive density for* $t > 0$ *solving theorem 5.14.*

Proof (indication). This is the classical *parametrix* construction of E. E. Levi. One builds p as a perturbation series off the constant-coefficient Gaussian kernel $Z(s, x; t, y) = (2\pi t')^{-d/2}(\det a)^{-1/2} \exp(-\frac{1}{2t'}(y-x)^\top a^{-1}(y-x))$ (frozen at $a = a(s, x)$), correcting it by an integral equation whose kernel is the Hölder error $(\partial_t - \mathcal{A})Z$; uniform ellipticity makes the resulting Volterra series converge and inherit the Gaussian bounds, and Schauder interior estimates for uniformly parabolic equations give the $C^{1,2}$ regularity and the classical solution property in (2)–(3). The detailed construction is in [71, 109]; the strict positivity is the parabolic strong maximum principle. We import the statement; the parametrix proof is lengthy but uses only the elliptic structure (A-Nondeg) and the assumed Hölder continuity of the coefficients. □

Remark C.13. The Hölder continuity of the coefficients is automatic in the LQG/OU applications (worked computation 5.17), where b is affine and a constant, and there the kernel p is the explicit Gaussian (5.28); theorem C.12 is needed only for the general nonlinear-coefficient diffusions of chapters 8 and 10.

C.10 Gaussian limits and the Wiener integral

The Ornstein–Uhlenbeck computation (worked computation 5.17) uses that the Itô integral of a deterministic integrand is Gaussian. This rests on the stability of the Gaussian class under L^2 limits.

Lemma C.14 (L^2 limits of Gaussians are Gaussian)**.** *Let* (Z_n) *be centered Gaussian random variables (or vectors) with* $Z_n \to Z$ *in* $L^2(\mathbb{P})$*. Then* Z *is centered Gaussian, with* $\operatorname{Var} Z = \lim_n \operatorname{Var} Z_n$*.*

Proof. L^2 convergence gives $\mathbb{E}Z = \lim \mathbb{E}Z_n = 0$ and $\sigma_n^2 := \operatorname{Var} Z_n \to \sigma^2 := \mathbb{E}Z^2$. The characteristic function of Z_n is $\mathbb{E}e^{i\xi Z_n} = e^{-\sigma_n^2\xi^2/2}$. Since L^2 convergence implies convergence in distribution, $\mathbb{E}e^{i\xi Z_n} \to \mathbb{E}e^{i\xi Z}$ pointwise in ξ; the left side tends to $e^{-\sigma^2\xi^2/2}$, so $\mathbb{E}e^{i\xi Z} = e^{-\sigma^2\xi^2/2}$, which is the characteristic function of $N(0, \sigma^2)$. By uniqueness of characteristic functions (chapter 3), $Z \sim N(0, \sigma^2)$. The vector case is identical with $\xi \in \mathbb{R}^d$ and the covariance matrix in place of σ^2. □

Corollary C.15 (Wiener integral)**.** *For deterministic* $f \in L^2[0, T]$ *the Itô integral* $\int_0^T f(s)\, \mathrm{d}W_s$ *is a centered Gaussian random variable with variance* $\int_0^T f(s)^2\, \mathrm{d}s$*. More generally* $\left(\int_0^t f\, \mathrm{d}W\right)_{t \le T}$ *is a centered Gaussian process with covariance* $\int_0^{s\wedge t} f^2$*.*

Proof. For a deterministic simple $f = \sum_i c_i \mathbf{1}_{(t_i, t_{i+1}]}$ the integral $\sum_i c_i(W_{t_{i+1}} - W_{t_i})$ is a finite linear combination of the independent Gaussian increments, hence centered Gaussian with variance $\sum_i c_i^2(t_{i+1} - t_i) = \int f^2$. A general $f \in L^2[0, T]$ is an $L^2[0, T]$ limit of simple deterministic functions f_n; by the Itô isometry $\int f_n\, \mathrm{d}W \to \int f\, \mathrm{d}W$ in $L^2(\mathbb{P})$, and lemma C.14 gives that the limit is centered Gaussian with the stated variance. The joint Gaussianity of the process follows by applying the same to arbitrary linear combinations $\sum_k a_k \int_0^{t_k} f \, dW$. $\square$

C.11 Completeness of the Hermite basis and the Ornstein–Uhlenbeck spectrum

The spectral decomposition of the OU generator (worked computation 5.17(c)) uses that the Hermite polynomials are a complete orthogonal system in L^2 of the Gaussian weight. The only nontrivial input is completeness, which we prove; the eigenvalue relation is then an elementary computation, and the self-adjointness is read off the explicit eigenbasis rather than imported from the compact-operator spectral theorem of chapter 2 (the OU generator is unbounded and has no compact resolvent on the whole line in the naive sense, so that theorem does not apply directly; the correct statement is the one below).

Fix the centered Gaussian weight $\gamma = N(0, v)$ on $\mathbb{R}$, $v > 0$, with density $g(x) = (2\pi v)^{-1/2} e^{-x^2/2v}$, and write $L^2(\gamma)$ for the weighted space with inner product $\langle \phi, \psi \rangle_\gamma = \int \phi\psi \, g$.

Lemma C.16 (Density of polynomials and Hermite completeness)**.** *Polynomials are dense in $L^2(\gamma)$. Consequently the Hermite polynomials $\{H_n\}_{n \geq 0}$ — the Gram–Schmidt orthogonalization of $\{1, x, x^2, \dots\}$ in $L^2(\gamma)$, with $H_0 \equiv 1$, $H_1(x) = x$, normalized so that H_n has leading coefficient* 1 *— form a complete* orthogonal *system: an orthogonal basis of $L^2(\gamma)$. They are made orthonormal by dividing each H_n by $\|H_n\|_{L^2(\gamma)}$.*

Proof. Orthogonality and the spanning of polynomials are immediate from the Gram–Schmidt construction; the content is that polynomials are *dense*. Suppose $\phi \in L^2(\gamma)$ is orthogonal to every polynomial, in particular to every power x^n: $\int x^n \phi(x) \, g(x) \, dx = 0$ for all $n \geq 0$. We show $\phi = 0$ γ-a.e.

Consider the function of a complex variable

$$F(z) := \int_{\mathbb{R}} e^{zx} \, \phi(x) \, g(x) \, dx, \qquad z \in \mathbb{C}.$$

Because g decays like a Gaussian, $|\phi(x)g(x)| \leq \frac{1}{2}(\phi(x)^2 + 1)g(x)$ is integrable against $e^{\Re(z)x}$ for every z (the Gaussian beats the exponential), so F is well defined and, by Morera's theorem and dominated convergence, *entire*. Differentiating under the integral, $F^{(n)}(0) = \int x^n \phi \, g = 0$ for all n, so all Taylor coefficients of F at 0 vanish; being entire, $F \equiv 0$. In particular, for real ξ,

$$0 = F(i\xi) = \int e^{i\xi x} \, \phi(x) g(x) \, dx,$$

the Fourier transform of the integrable function $\phi \, g$. By Fourier uniqueness $\phi \, g = 0$ a.e., and since $g > 0$ this forces $\phi = 0$ γ-a.e. Hence the only function orthogonal to all polynomials is 0: polynomials are dense, and the orthogonal system $\{H_n\}$ is complete. $\square$

With completeness in hand the OU spectral statement of worked computation 5.17(c) is justified rigorously. On $L^2(m_\infty)$ with $m_\infty = N(0, v_\infty)$, $v_\infty = \sigma^2/2\theta$, the generator $\mathcal{A}\phi = -\theta x\phi' + \frac{\sigma^2}{2}\phi''$ satisfies the symmetry

$$\langle \mathcal{A}\phi, \psi\rangle_{m_\infty} = -\frac{\sigma^2}{2}\int \phi'(x)\psi'(x)\, m_\infty(x)\, \mathrm{d}x = \langle \phi, \mathcal{A}\psi\rangle_{m_\infty},$$

obtained by integrating by parts once, using $\mathcal{A}^* m_\infty = 0$ in the form $\frac{\sigma^2}{2} m'_\infty = -\theta x\, m_\infty$ (the zero-current identity of eq. (5.29)); the boundary terms vanish by the Gaussian decay. Thus $\mathcal{A}$ is symmetric and negative on its core C_c^2, with the closed form $-\frac{\sigma^2}{2}\langle \phi', \psi'\rangle_{m_\infty}$. A direct computation on monomials shows $\mathcal{A}$ maps the space of polynomials of degree $\le n$ into itself and acts on the top-degree part by multiplication by $-n\theta$; hence each H_n, being the unique (up to scale) degree-n polynomial orthogonal to all lower degrees, is an eigenfunction,

$$\mathcal{A}H_n = -n\theta\, H_n. \tag{C.8}$$

By lemma C.16 the normalized $\{H_n/\,\|H_n\|\}$ are an orthonormal basis that diagonalizes $\mathcal{A}$ with eigenvalues $\{-n\theta\}_{n\ge 0}$; this is the spectral resolution (purely point spectrum) used to write the modal expansion of $e^{t\mathcal{A}}$ in worked computation 5.17.

C.12 Martingale representation and backward stochastic differential equations

This section supplies the two backward-in-time results on which the adjoint and FBSDE formulation of chapters 6 and 9 rests: the martingale representation theorem for the Brownian filtration, and the Pardoux–Peng existence and uniqueness theorem for backward stochastic differential equations (BSDEs) under a Lipschitz driver. Both lie beyond the elementary prerequisites and are the in-book home of those results; we prove them in full, using only the Itô integral and isometry of chapter 5, Doob's inequality (theorem C.2), and Grönwall's lemma (lemma C.1).

Throughout, $W = (W_t)_{0\le t\le T}$ is a k-dimensional Brownian motion and $\mathbb{F} = (\mathcal{F}_t)_{0\le t\le T}$ its completed, right-continuous natural filtration on $(\Omega, \mathcal{F}, \mathbb{P})$, with $\mathcal{F} = \mathcal{F}_T$. We write $\mathcal{H}^2 = \mathcal{H}^2(\mathbb{R}^n)$ for the Hilbert space of $\mathbb{F}$-progressively measurable processes H with $\|H\|^2_{\mathcal{H}^2} = \mathbb{E}\int_0^T |H_t|^2\, \mathrm{d}t < \infty$, and $\mathcal{S}^2 = \mathcal{S}^2(\mathbb{R}^n)$ for the continuous adapted processes with $\|Y\|^2_{\mathcal{S}^2} = \mathbb{E}\sup_{t\le T}|Y_t|^2 < \infty$. (Here $|\cdot|$ is the Euclidean, resp. Frobenius, norm.)

C.12.1 Martingale representation

Theorem C.17 (Martingale representation)**.** *Every $\xi \in L^2(\Omega, \mathcal{F}_T, \mathbb{P}; \mathbb{R})$ admits a unique $H \in \mathcal{H}^2(\mathbb{R}^{1\times k})$ with*

$$\xi \;=\; \mathbb{E}[\xi] + \int_0^T H_s\, \mathrm{d}W_s \qquad \mathbb{P}\text{-}a.s. \tag{C.9}$$

Equivalently, every square-integrable $\mathbb{F}$-martingale (M_t) has a continuous version $M_t = M_0 + \int_0^t H_s\, \mathrm{d}W_s$ with $H \in \mathcal{H}^2$. The vector case $\xi \in L^2(\mathcal{F}_T; \mathbb{R}^n)$ follows componentwise, giving $H \in \mathcal{H}^2(\mathbb{R}^{n\times k})$.

Proof. Let $\mathcal{R} \subset L^2(\mathcal{F}_T)$ be the set of variables representable as $c + \int_0^T H \, \mathrm{d}W$ with $c \in \mathbb{R}$ and $H \in \mathcal{H}^2(\mathbb{R}^{1\times k})$. By the Itô isometry (theorem 5.7) the stochastic integral is centered with $\mathbb{E}\big(\int_0^T H \, \mathrm{d}W\big)^2 = \|H\|^2_{\mathcal{H}^2}$, so the map $(c, H) \mapsto c + \int_0^T H \, \mathrm{d}W$ is a linear isometry of $\mathbb{R} \oplus \mathcal{H}^2$ onto $\mathcal{R}$; in particular $\mathcal{R}$ is a *closed* subspace of $L^2(\mathcal{F}_T)$. We show $\mathcal{R}$ is dense; being closed it is then all of $L^2(\mathcal{F}_T)$, which gives eq. (C.9) (with $c = \mathbb{E}[\xi]$ forced by taking expectations and H unique by the isometry).

Step 1: Doléans exponentials are representable. For bounded deterministic $f \in L^2([0,T];\mathbb{R}^k)$ set

$$\mathcal{E}_t(f) = \exp\Big(\int_0^t f_s \, \mathrm{d}W_s - \tfrac{1}{2}\int_0^t |f_s|^2 \, \mathrm{d}s\Big).$$

Itô's formula (theorem 5.9) applied to the C^2 function exp and the process $\Xi_t = \int_0^t f \, \mathrm{d}W - \frac{1}{2}\int_0^t |f|^2 \, \mathrm{d}s$ (whose drift $-\frac{1}{2}|f|^2$ exactly cancels the $\frac{1}{2}$ from the quadratic variation $\mathrm{d}\langle \Xi \rangle_t = |f_t|^2 \, \mathrm{d}t$) gives $\mathrm{d}\mathcal{E}_t(f) = \mathcal{E}_t(f) \, f_t \, \mathrm{d}W_t$, hence

$$\mathcal{E}_T(f) = 1 + \int_0^T \mathcal{E}_s(f) \, f_s \, \mathrm{d}W_s.$$

For bounded f the integrand lies in $\mathcal{H}^2$: since $\int f \, \mathrm{d}W$ is, by corollary C.15, Gaussian with bounded variance, $\mathcal{E}_t(f)$ has moments of every order, uniformly in $t \le T$, so $\mathbb{E}\int_0^T \mathcal{E}_s(f)^2 \, |f_s|^2 \, \mathrm{d}s < \infty$. Thus $\mathcal{E}_T(f) \in \mathcal{R}$ for every bounded deterministic f.

Step 2: totality. It suffices to show that any $g \in L^2(\mathcal{F}_T)$ orthogonal to all $\mathcal{E}_T(f)$ vanishes; then the closed span of $\{\mathcal{E}_T(f)\} \subset \mathcal{R}$ is all of $L^2(\mathcal{F}_T)$, so $\mathcal{R} = L^2(\mathcal{F}_T)$. Fix a partition $0 = t_0 < t_1 < \cdots < t_n = T$ and vectors $\lambda_1, \ldots, \lambda_n \in \mathbb{R}^k$, and take the piecewise-constant $f = \sum_j \lambda_j \mathbf{1}_{(t_{j-1}, t_j]}$. Then $\int_0^T f \, \mathrm{d}W = \sum_j \lambda_j \cdot \Delta_j W$ with $\Delta_j W = W_{t_j} - W_{t_{j-1}}$, and $\mathcal{E}_T(f) = c(\lambda) \exp(\sum_j \lambda_j \cdot \Delta_j W)$ with $c(\lambda) > 0$ deterministic. Orthogonality $\mathbb{E}[g \, \mathcal{E}_T(f)] = 0$ therefore reads

$$\mathbb{E}\Big[g \, \exp\Big(\sum_{j=1}^n \lambda_j \cdot \Delta_j W\Big)\Big] = 0 \qquad \text{for all } \lambda_1, \ldots, \lambda_n \in \mathbb{R}^k.$$

Both sides are entire functions of $(\lambda_j) \in \mathbb{C}^{kn}$ (the increments are Gaussian, so the moment generating function is finite for all complex arguments, by the domination used in lemma C.16); since they agree on $\mathbb{R}^{kn}$ they agree everywhere, and substituting $\lambda_j \mapsto i\lambda_j$ gives

$$\mathbb{E}\Big[g \, \exp\Big(i \sum_j \lambda_j \cdot \Delta_j W\Big)\Big] = 0 \qquad \text{for all } \lambda_j \in \mathbb{R}^k.$$

This says the finite complex measure $A \mapsto \mathbb{E}[g\mathbf{1}_A]$, pushed forward to $\mathbb{R}^{kn}$ by $(\Delta_1 W, \ldots, \Delta_n W)$, has vanishing Fourier transform; by Fourier uniqueness (as in lemma C.16) it is the zero measure, i.e. $\mathbb{E}[g \, \varphi(\Delta_1 W, \ldots, \Delta_n W)] = 0$ for every bounded Borel φ. As the partition ranges over all finite subsets of a fixed countable dense set of times, the bounded functions of finitely many increments form a multiplicative class generating $\mathcal{F}_T = \sigma(W_s : s \le T)$; by the functional monotone class theorem their linear span is dense in $L^2(\mathcal{F}_T)$. Hence $\mathbb{E}[g \, Y] = 0$ for every $Y \in L^2(\mathcal{F}_T)$, so $g = 0$. This proves totality.

Martingale form. Given a square-integrable martingale M, apply eq. (C.9) to $\xi = M_T$: $M_T = \mathbb{E}[M_T] + \int_0^T H \, \mathrm{d}W$. The process $\widetilde{M}_t = \mathbb{E}[M_T] + \int_0^t H \, \mathrm{d}W$ is a continuous L^2-martingale (theorem 5.7) with $\widetilde{M}_T = M_T$, so for each t, $\widetilde{M}_t = \mathbb{E}[\widetilde{M}_T \mid \mathcal{F}_t] = \mathbb{E}[M_T \mid \mathcal{F}_t] = M_t$ a.s.; thus $\widetilde{M}$ is the asserted continuous version. ∎

C.12.2 The Pardoux–Peng theorem

A *backward stochastic differential equation* is specified by a terminal datum $\xi \in L^2(\mathcal{F}_T;\mathbb{R}^n)$ and a *driver* $f : [0,T] \times \Omega \times \mathbb{R}^n \times \mathbb{R}^{n\times k} \to \mathbb{R}^n$ that is progressively measurable in (t,ω) for each (y,z) and satisfies

- **(L)** uniform Lipschitz continuity: $|f(t,\omega,y,z) - f(t,\omega,y',z')| \le K\big(|y-y'| + |z-z'|\big)$ for some $K < \infty$, all $(y,z),(y',z')$, $\mathrm{d}t \otimes \mathrm{d}\mathbb{P}$-a.e.; and
- **(I)** integrable baseline: $f(\cdot,\cdot,0,0) \in \mathcal{H}^2(\mathbb{R}^n)$.

A *solution* is a pair $(Y,Z) \in \mathcal{S}^2(\mathbb{R}^n) \times \mathcal{H}^2(\mathbb{R}^{n\times k})$ with

$$Y_t = \xi + \int_t^T f(s,Y_s,Z_s)\,\mathrm{d}s - \int_t^T Z_s\,\mathrm{d}W_s, \qquad 0 \le t \le T,\ \mathbb{P}\text{-a.s.} \tag{C.10}$$

Theorem C.18 (Pardoux–Peng)**.** *Under* **(L)** *and* **(I)***, and with $\xi \in L^2(\mathcal{F}_T;\mathbb{R}^n)$, the BSDE eq.* (C.10) *has a unique solution* $(Y,Z) \in \mathcal{S}^2(\mathbb{R}^n) \times \mathcal{H}^2(\mathbb{R}^{n\times k})$.

Proof. Step 1: the driver frozen. First suppose $f = f(t)$ does not depend on (y,z), with $f \in \mathcal{H}^2$. Set $\zeta = \xi + \int_0^T f_s\,\mathrm{d}s \in L^2(\mathcal{F}_T)$ and let $M_t = \mathbb{E}[\zeta \mid \mathcal{F}_t]$ be its closing martingale. By theorem C.17, $M_t = \mathbb{E}[\zeta] + \int_0^t Z_s\,\mathrm{d}W_s$ for a unique $Z \in \mathcal{H}^2$. Define $Y_t = M_t - \int_0^t f_s\,\mathrm{d}s$. Then $Y_T = \zeta - \int_0^T f = \xi$, and for every t,

$$Y_t = M_t - \int_0^t f = \Big(\zeta - \int_t^T Z\,\mathrm{d}W\Big) - \int_0^t f = \xi + \int_t^T f_s\,\mathrm{d}s - \int_t^T Z_s\,\mathrm{d}W_s,$$

using $\zeta = M_T = M_t + \int_t^T Z\,\mathrm{d}W$ and $\zeta - \int_0^t f = \xi + \int_t^T f$. Thus (Y,Z) solves eq. (C.10); Y is continuous and, by Doob's inequality (theorem C.2) applied to M together with the L^2 bound on $\int_0^\cdot f$, lies in $\mathcal{S}^2$, while $Z \in \mathcal{H}^2$. Uniqueness: if (Y',Z') is another solution, then $\delta Y_t = -\int_t^T \delta Z\,\mathrm{d}W$ with $\delta Y_T = 0$; taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ gives $\delta Y_t = 0$, whence $\int_0^T \delta Z\,\mathrm{d}W = 0$ and, by the Itô isometry, $\delta Z = 0$ in $\mathcal{H}^2$.

Step 2: contraction for the general driver. On the Hilbert space $\mathcal{B} = \mathcal{H}^2(\mathbb{R}^n) \times \mathcal{H}^2(\mathbb{R}^{n\times k})$ introduce, for $\beta > 0$, the equivalent norm

$$\|(Y,Z)\|_\beta^2 = \mathbb{E}\int_0^T e^{\beta s}\big(\,|Y_s|^2 + |Z_s|^2\,\big)\,\mathrm{d}s.$$

Given $(U,V) \in \mathcal{B}$, the process $g_s := f(s,U_s,V_s)$ lies in $\mathcal{H}^2$ (by **(L)**,**(I)**: $|g_s| \le |f(s,0,0)| + K(|U_s| + |V_s|)$), so Step 1 produces a unique solution $(Y,Z) \in \mathcal{S}^2 \times \mathcal{H}^2$ of the BSDE with frozen driver g; define $\Phi(U,V) = (Y,Z)$. Let $(U,V),(U',V') \in \mathcal{B}$ have images $(Y,Z),(Y',Z')$ and write $\delta U = U - U'$, etc. Then $\delta Y, \delta Z$ solve

$$\mathrm{d}(\delta Y_t) = -\big(f(t,U_t,V_t) - f(t,U_t',V_t')\big)\,\mathrm{d}t + \delta Z_t\,\mathrm{d}W_t, \qquad \delta Y_T = 0.$$

Apply Itô's formula to $e^{\beta t}\,|\delta Y_t|^2$:

$$\mathrm{d}\big(e^{\beta t}\,|\delta Y_t|^2\,\big) = e^{\beta t}\Big(\beta\,|\delta Y_t|^2 - 2\langle \delta Y_t, \Delta f_t\rangle + |\delta Z_t|^2\,\Big)\,\mathrm{d}t + 2e^{\beta t}\langle \delta Y_t, \delta Z_t\,\mathrm{d}W_t\rangle,$$

where $\Delta f_t = f(t,U_t,V_t) - f(t,U_t',V_t')$. Integrating over $[0,T]$, using $\delta Y_T = 0$, and taking expectations (the stochastic integral is a true martingale, as $\delta Y \in \mathcal{S}^2$, $\delta Z \in \mathcal{H}^2$),

$$\mathbb{E}\,|\delta Y_0|^2 + \mathbb{E}\int_0^T e^{\beta t}\big(\beta\,|\delta Y_t|^2 + |\delta Z_t|^2\,\big)\,\mathrm{d}t = \mathbb{E}\int_0^T e^{\beta t}\,2\langle \delta Y_t, \Delta f_t\rangle\;\mathrm{d}t.$$

Drop $\mathbb{E}\,|\delta Y_0|^2 \geq 0$. By **(L)**, $|\Delta f_t| \leq K(|\delta U_t| + |\delta V_t|)$, and Young's inequality $2ab \leq \frac{\beta}{2}a^2 + \frac{2}{\beta}b^2$ with $a = |\delta Y_t|$, $b = K(|\delta U_t| + |\delta V_t|)$ gives

$$2\langle \delta Y_t, \Delta f_t\rangle \leq \tfrac{\beta}{2}\,|\delta Y_t|^2 + \tfrac{2}{\beta}K^2\big(\,|\delta U_t| + |\delta V_t|\,\big)^2 \leq \tfrac{\beta}{2}\,|\delta Y_t|^2 + \tfrac{4K^2}{\beta}\big(\,|\delta U_t|^2 + |\delta V_t|^2\,\big).$$

Substituting and cancelling $\frac{\beta}{2}\mathbb{E}\int e^{\beta t}\,|\delta Y_t|^2$ from both sides,

$$\mathbb{E}\int_0^T e^{\beta t}\Big(\tfrac{\beta}{2}\,|\delta Y_t|^2 + |\delta Z_t|^2\Big)\,\mathrm{d}t \leq \frac{4K^2}{\beta}\,\mathbb{E}\int_0^T e^{\beta t}\big(\,|\delta U_t|^2 + |\delta V_t|^2\,\big)\,\mathrm{d}t = \frac{4K^2}{\beta}\;\|(\delta U,\delta V)\|_\beta^2\,.$$

Choosing $\beta \geq 2$, the left side dominates $\mathbb{E}\int_0^T e^{\beta t}(|\delta Y_t|^2 + |\delta Z_t|^2)\,\mathrm{d}t = \|(\delta Y,\delta Z)\|_\beta^2$; hence

$$\|\Phi(U,V) - \Phi(U',V')\|_\beta^2 \leq \frac{4K^2}{\beta}\;\|(U,V) - (U',V')\|_\beta^2\,.$$

Taking $\beta = 8K^2 + 2$ makes $4K^2/\beta \leq \frac{1}{2}$, so Φ is a contraction (constant $\leq 1/\sqrt{2}$) on the Banach space $(\mathcal{B}, \|\cdot\|_\beta)$. By the Banach fixed-point theorem Φ has a unique fixed point $(Y,Z) \in \mathcal{B}$, which by construction solves eq. (C.10).

Step 3: regularity and uniqueness in $\mathcal{S}^2 \times \mathcal{H}^2$. For the fixed point, Y is the Step 1 solution of the BSDE with driver $f(\cdot,Y,Z) \in \mathcal{H}^2$, hence has a continuous version in $\mathcal{S}^2$ with $\mathbb{E}\sup_{t\leq T}|Y_t|^2 < \infty$ (Doob applied to its closing martingale plus the L^2 bound on $\int f(\cdot,Y,Z)$). Any solution in $\mathcal{S}^2 \times \mathcal{H}^2$ is in particular a fixed point of Φ, and the $\|\cdot\|_\beta$-norm is equivalent to the unweighted $\mathcal{H}^2 \times \mathcal{H}^2$ norm on $[0,T]$ (since $e^{\beta t} \in [1, e^{\beta T}]$); so uniqueness of the fixed point gives uniqueness of the solution. ☐

Remark C.19 (Why the second component cannot be prescribed)**.** The pair structure of eq. (C.10) is forced, not chosen: a process pinned at T by $\xi = Y_T$, which depends on the whole future of W, can be made adapted only by absorbing the unpredictable part of its terminal value into a martingale, and theorem C.17 says that martingale is a stochastic integral $\int Z\,\mathrm{d}W$. The integrand Z is exactly that correction; it is what distinguishes a BSDE from a (time-reversed) ODE, and in the control setting of chapter 9 it is the volatility-weighted curvature of the value function. This is the structural fact used throughout section 9.1.

C.13 The martingale problem and relaxed controls

The existence of *weak* (relaxed) mean field game equilibria in section 10.4 rests on three facts that lie beyond the strong-solution theory of chapter 5: the *well-posedness* (uniqueness in law) of the martingale problem under nondegeneracy, the *compactness* of the set of relaxed controls together with lower semicontinuity of the cost, and the *stability* of the controlled martingale problem under convergence of the data. We develop them here, in the form chapter 10 uses, on the strong-Markov diffusion theory and the fundamental solution already established in this appendix (theorem C.12) and on Prokhorov's theorem of chapter 3.

C.13.1 The martingale problem and uniqueness in law

Definition C.20 (Martingale problem)**.** For coefficients $b\colon [0,T]\times\mathbb{R}^d\to\mathbb{R}^d$ and $a=\sigma\sigma^\top\colon [0,T]\times\mathbb{R}^d\to\mathbb{R}^{d\times d}$, bounded and measurable, the (time-dependent) generator is $(\mathcal{A}_t\phi)(x)=b(t,x)\cdot\nabla\phi(x)+\frac{1}{2}\operatorname{tr}(a(t,x)\,\mathrm{D}^2\,\phi(x))$. A probability measure $\mathbb{P}$ on the path space $C([0,T];\mathbb{R}^d)$, with canonical process X, *solves the martingale problem* for (b,a) started from m_0 if $\mathbb{P}\circ X_0^{-1}=m_0$ and, for every $\phi\in C_c^2(\mathbb{R}^d)$,

$$M_t^\phi=\phi(X_t)-\phi(X_0)-\int_0^t(\mathcal{A}_s\phi)(X_s)\,\mathrm{d}s$$

is a $\mathbb{P}$-martingale for the canonical filtration.

Theorem C.21 (Well-posedness of the martingale problem)**.** *Assume* (A-Nondeg) *($a\succeq\lambda I$, $\lambda>0$) and that b,a are bounded and Hölder continuous in (t,x) (as hold under* (A-Lip), (A-Growth) *on the coefficients of chapter 10). Then for each $m_0\in\mathcal{P}(\mathbb{R}^d)$ the martingale problem of definition C.20 has* exactly one *solution $\mathbb{P}$. In particular the law of the diffusion is uniquely determined by (b,a,m_0) (uniqueness in law), and it coincides with the law of the strong solution when one exists.*

Proof. Existence follows from theorem C.12: the fundamental solution $p(s,x;t,y)$ of $\partial_t-\mathcal{A}$ defines a Markov transition kernel $P_{s,t}(x,\mathrm{d}y)=p(s,x;t,y)\,\mathrm{d}y$, whose finite-dimensional distributions, started from m_0, are consistent (Chapman–Kolmogorov is the reproducing property of the fundamental solution) and tight (the Gaussian bounds give the Kolmogorov–Chentsov moment estimate theorem C.9), hence by the Kolmogorov extension theorem theorem C.8 define a law $\mathbb{P}$ on $C([0,T];\mathbb{R}^d)$; that $\mathbb{P}$ solves the martingale problem is the backward-equation property theorem C.12(2), since for $\phi\in C_c^2$ the function $u(s,x)=P_{s,t}\phi(x)$ is $C^{1,2}$ and Itô's formula makes M^ϕ a martingale.

Uniqueness. Let $\mathbb{P}$ be any solution. We show its one-dimensional time marginals, hence (by the Markov property below) all finite-dimensional marginals, are pinned to the transition kernel. Fix $0\le s<t\le T$, a terminal time t, and $\psi\in C_c^\infty(\mathbb{R}^d)$, and let $u(\tau,x)$, $\tau\in[s,t]$, be the classical solution of the backward Cauchy problem $\partial_\tau u+\mathcal{A}_\tau u=0$ on $[s,t)$, $u(t,\cdot)=\psi$, supplied by theorem C.12(2). Apply the martingale-problem identity to the time-dependent test function u (a standard extension of definition C.20 to $C^{1,2}$ functions of (τ,x), obtained by approximation): $u(\tau,X_\tau)$ is a $\mathbb{P}$-martingale on $[s,t]$ because $\partial_\tau u+\mathcal{A}_\tau u=0$ kills its drift. Taking $\mathbb{P}$-expectations between $\tau=s$ and $\tau=t$,

$$\mathbb{E}_\mathbb{P}[\psi(X_t)\mid X_s=x]=u(s,x)=\int p(s,x;t,y)\psi(y)\,\mathrm{d}y,$$

for $\mathbb{P}\circ X_s^{-1}$-a.e. x. Thus the conditional law of X_t given X_s is the transition kernel $P_{s,t}(x,\cdot)$, *independent of the solution* $\mathbb{P}$. The martingale problem also implies the Markov property (the conditional law given $\mathcal{F}_s$ depends only on X_s, because the solved test-function identity uses only the value X_s); this is the standard argument of [124]. Therefore all finite-dimensional distributions of $\mathbb{P}$ equal those built from $P_{s,t}$ and m_0, so $\mathbb{P}$ is unique. □

C.13.2 Relaxed controls: compactness and existence of an optimizer

A *relaxed control* is a measure ρ on $[0,T]\times A$ with time-marginal Lebesgue measure (definition 10.17); equivalently a measurable family $(\rho_t)_{t\in[0,T]}$ of probability measures on the action set A. Write $\mathcal{R}$

for the set of relaxed controls, topologised by weak convergence of the measures ρ on $[0,T]\times A$.

Theorem C.22 (Compactness of relaxed controls and existence of an optimizer)**.** *Suppose the action set A is compact (or coercivity confines optimal controls to a compact set). Then:*

(i) *$\mathcal{R}$ is compact and metrisable for the weak topology.*

(ii) *For continuous, bounded-below running and terminal costs and continuous dynamics, the relaxed cost functional $\rho\mapsto J(\rho)$ is lower semicontinuous on $\mathcal{R}$. Consequently, for each frozen population flow m, the relaxed control problem attains its minimum: an optimal relaxed control exists.*

Proof. (i) Since A is compact, $[0,T]\times A$ is a compact metric space, and the set of probability measures on it is compact and metrisable for the weak topology (Prokhorov, theorem 3.16; a compact base space is automatically tight). The constraint "time marginal = Lebesgue" is defined by countably many continuous linear equalities $\iint g(t)\,\rho(\mathrm{d}t,\mathrm{d}a)=\int g(t)\,\mathrm{d}t$ (g in a countable dense set of $C[0,T]$), hence is weakly closed; a closed subset of a compact set is compact. (ii) Lower semicontinuity: along $\rho^{(n)}\to\rho$, the relaxed state laws converge (the averaged coefficients $\bar b(t,x)=\int b(x,a)\rho_t(\mathrm{d}a)$ converge by the continuity of b and weak convergence of $\rho^{(n)}$, and the controlled martingale problem is stable, theorem C.23 below), and the cost, an integral of a continuous bounded-below integrand against the (weakly converging) occupation measures, is lower semicontinuous by the portmanteau theorem. A lower semicontinuous function on the compact set $\mathcal{R}$ (or its compact effective domain) attains its infimum. ☐

Theorem C.23 (Stability of the controlled martingale problem)**.** *Let $\rho^{(n)}\to\rho$ in $\mathcal{R}$ and $m^{(n)}\to m$ (frozen flows), and let $\mathbb{P}^{(n)}$ solve the controlled martingale problem for the averaged generator $\mathcal{A}^n_t\phi=\bar b^{\,n}(t,\cdot)\cdot\nabla\phi+\frac{1}{2}\operatorname{tr}(a\,\mathrm{D}^2\,\phi)$ with $\bar b^{\,n}(t,x)=\int b(x,a,m^{(n)}_t)\,\rho^{(n)}_t(\mathrm{d}a)$, under* (A-Nondeg) *and continuity of the coefficients. Then $\{\mathbb{P}^{(n)}\}$ is tight on $C([0,T];\mathbb{R}^d)$ and every weak limit solves the controlled martingale problem for the limiting data (ρ,m); under* (A-Nondeg) *that limit problem is well-posed* (*theorem C.21*)*, so $\mathbb{P}^{(n)}\to\mathbb{P}$ weakly.*

Proof. Tightness. The drifts $\bar b^{\,n}$ are uniformly bounded (the action set is compact, b continuous) and a is bounded, so the Kolmogorov–Chentsov estimate theorem C.9 gives a uniform fourth-moment bound on increments, $\mathbb{E}|X_t-X_s|^4\le C|t-s|^2$, whence tightness on path space by the Kolmogorov criterion. *Identification of limits.* Let $\mathbb{P}^{(n)}\to\mathbb{P}$ weakly along a subsequence. For $\phi\in C^2_c$ and $s<t$ and any bounded continuous $\mathcal{F}_s$-measurable test functional Ψ, the martingale-problem identity $\mathbb{E}_{\mathbb{P}^{(n)}}\big[(M^{\phi,n}_t-M^{\phi,n}_s)\Psi\big]=0$ passes to the limit: the maps $X\mapsto\phi(X_t)$, $\phi(X_s)$ and Ψ are bounded continuous, and the drift integral $\int_s^t\bar b^{\,n}(r,X_r)\cdot\nabla\phi(X_r)\,\mathrm{d}r$ converges because $\bar b^{\,n}\to\bar b$ uniformly on compacts (continuity of b and weak convergence of $\rho^{(n)},m^{(n)}$) while the diffusion term is fixed and continuous. Hence $\mathbb{P}$ solves the martingale problem for the limit generator $\mathcal{A}_t$ with averaged drift $\bar b(t,x)=\int b(x,a,m_t)\rho_t(\mathrm{d}a)$. By theorem C.21 the limit is unique, so the full sequence converges. ☐

Remark C.24. Theorems C.21 to C.23 are exactly the nonempty-values, compactness, and closed-graph inputs to the Kakutani–Fan–Glicksberg fixed point of theorem 10.19; together with Berge's theorem theorem B.27 and the measurable selection theorem B.29 of appendix B they make the weak-equilibrium existence proof of chapter 10 self-contained.

Bibliographic notes

The extension theorem (theorem C.8) and the continuity theorem (theorem C.9) are due to Kolmogorov and to Kolmogorov–Chentsov; the chaining proof of the latter and the projective-limit construction follow [86, 114], whose treatments we have specialized to the form chapter 5 needs. Doob's maximal inequality and the Burkholder–Davis–Gundy estimates are standard martingale theory [86, 114]; we proved the L^2 cases used in the SDE well-posedness theorem in full and cited the general p. The density of simple processes and the construction of the Itô integral follow [86, Ch. 3] and [106, Ch. 3]. The Markov property of SDE solutions and the transition semigroup are in [106, Ch. 7]. The parabolic regularity of theorem C.12 is the Levi parametrix construction of [71], with the probabilistic reading and Gaussian bounds as in [109]. The completeness of the Hermite system via density of polynomials (lemma C.16) and the resulting OU spectral decomposition are developed for the physicist in [109]; the entire-function argument we gave is the classical proof that a determinate moment sequence (the Gaussian is determinate) yields a complete orthogonal polynomial system. The martingale representation theorem (theorem C.17) and the existence/uniqueness theorem for Lipschitz BSDEs (theorem C.18) are due, respectively, to Itô and to Pardoux–Peng [107]; the Doléans-exponential density argument for the former and the weighted-norm contraction for the latter follow [86, 63], specialized to the Brownian filtration and the constant-Lipschitz driver used by chapters 6 and 9. The martingale-problem well-posedness (theorem C.21) and the relaxed-control compactness and stability (theorems C.22 and C.23), which underpin the weak-equilibrium existence theory of chapter 10, follow Stroock–Varadhan [124] and the controlled-martingale-problem framework of [88]; we deduce well-posedness from the fundamental solution of theorem C.12 rather than from the analytic localisation of the original, which suffices under the standing nondegeneracy.

Appendix D

Optimal Transport and Calculus on $\mathcal{P}_2(\mathbb{R}^d)$

This appendix supplies the proofs that chapter 4 states and defers: Kantorovich duality, Brenier's structure theorem, McCann's displacement interpolation, the Benamou–Brenier dynamic formula, and the rearrangement-invariance structure theorem that underlies the Lions derivative. It closes with the second-order calculus on $\mathcal{P}_2(\mathbb{R}^d)$ — the chain rule (Itô formula) for functionals along a flow of measures — which chapter 12 imports to write the master equation. Everything is proved from material established earlier in the book: the Hahn–Banach separation theorem and Banach–Alaoglu theorem of chapter 2 (theorems 2.11 and 2.16), the Riesz representation of theorem 2.6, weak convergence, tightness, Prokhorov's theorem, Skorokhod's representation and disintegration of measures from chapter 3 (theorems 3.13, 3.16 and 3.33), and Itô's formula of theorem 5.9. No external source is required; citations are for attribution only.

Throughout, $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ are probability measures with finite second moment, $c(x, y) = |x - y|^2$ is the quadratic cost, $\Pi(\mu, \nu)$ is the set of couplings (definition 4.1), and $\mathcal{W}_2$ is the quadratic Wasserstein distance (definition 4.2). We freely use the existence of an optimal coupling (proposition 4.3), proved in the body by the direct method.

D.1 Kantorovich duality

We prove theorem 4.5: for $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\mathcal{W}_2(\mu, \nu)^2 \;=\; \sup \left\{ \int_{\mathbb{R}^d} \varphi \, \mathrm{d}\mu + \int_{\mathbb{R}^d} \psi \, \mathrm{d}\nu \;:\; \varphi(x) + \psi(y) \le |x - y|^2 \;\; \forall x, y \right\}, \tag{D.1}$$

the supremum over pairs (φ, ψ) of continuous functions of at most quadratic growth, and the supremum is attained by a pair of c-conjugate *Kantorovich potentials*.

The clean route is convex duality. We first reduce to bounded continuous data, then apply Hahn–Banach separation in the form of the Fenchel–Rockafellar theorem, and finally remove the boundedness by truncation.

D.1.1 Weak duality

For any coupling $\pi \in \Pi(\mu, \nu)$ and any admissible pair (φ, ψ) with $\varphi(x) + \psi(y) \le c(x, y)$,

$$\int \varphi \, \mathrm{d}\mu + \int \psi \, \mathrm{d}\nu = \int \big(\varphi(x) + \psi(y)\big)\, \pi(\mathrm{d}x, \mathrm{d}y) \le \int c(x, y)\, \pi(\mathrm{d}x, \mathrm{d}y), \tag{D.2}$$

where the first equality uses that π has marginals μ, ν. Taking the supremum over (φ, ψ) on the left and the infimum over π on the right gives (dual) $\le$ (primal) $= \mathcal{W}_2(\mu, \nu)^2$. The content of the theorem is the reverse inequality, *strong duality*.

D.1.2 The Fenchel–Rockafellar mechanism

We record the convex-duality theorem we will use. Let E be a normed vector space with topological dual E^*. For a convex function $\Theta \colon E \to (-\infty, +\infty]$ its Legendre–Fenchel conjugate is $\Theta^*(\ell) = \sup_{x \in E} \big(\ell(x) - \Theta(x)\big)$ for $\ell \in E^*$.

Theorem D.1 (Fenchel–Rockafellar)**.** *Let $\Theta_1, \Theta_2 \colon E \to (-\infty, +\infty]$ be convex, and suppose there is a point $x_0 \in E$ at which both are finite and at least one is continuous. Then*

$$\inf_{x \in E} \big(\Theta_1(x) + \Theta_2(x)\big) \;=\; \max_{\ell \in E^*} \big(-\Theta_1^*(-\ell) - \Theta_2^*(\ell)\big), \tag{D.3}$$

the right-hand supremum being attained.

Proof sketch. The inequality $\ge$ is immediate from the definition of the conjugates: $-\Theta_1^*(-\ell) - \Theta_2^*(\ell) \le \Theta_1(x) + \Theta_2(x)$ for all x, ℓ by adding the two Young inequalities $\Theta_1(x) \ge -\ell(x) - \Theta_1^*(-\ell)$ and $\Theta_2(x) \ge \ell(x) - \Theta_2^*(\ell)$. For the reverse, let $m = \inf_x(\Theta_1 + \Theta_2)$ (assume $m > -\infty$, else both sides are $-\infty$). In the product space $E \times \mathbb{R}$ consider the two convex sets

$$C_1 = \{(x, t) : t \ge \Theta_1(x)\}, \qquad C_2 = \{(x, t) : t \le m - \Theta_2(x)\}.$$

C_1 is the epigraph of Θ_1 and C_2 is the hypograph of $m - \Theta_2$; their interiors are disjoint (a common interior point would give $\Theta_1(x) < t < m - \Theta_2(x)$, i.e. $\Theta_1(x) + \Theta_2(x) < m$, impossible), and the continuity hypothesis makes C_1 have nonempty interior. By the Hahn–Banach separation theorem (theorem 2.11) there is a nonzero continuous linear functional on $E \times \mathbb{R}$ separating them; writing it as $(x, t) \mapsto \ell(x) - \lambda t$ one checks $\lambda > 0$, and rescaling so $\lambda = 1$ the separating functional yields exactly an $\ell \in E^*$ achieving $-\Theta_1^*(-\ell) - \Theta_2^*(\ell) \ge m$. Combined with weak duality this is (D.3) with attainment. Full detail is in [31, Ch. 1]; the argument is the convex-geometric heart of chapter 2. □

D.1.3 Strong duality for bounded continuous cost

Assume first that c is replaced by a bounded continuous cost $\tilde{c} \ge 0$; we restore $\tilde{c} = |x - y|^2$ afterward. Work on $E = C_b(\mathbb{R}^d \times \mathbb{R}^d)$, the bounded continuous functions with the sup norm, whose dual contains the finite signed Radon measures. Define, for $u \in E$,

$$\Theta_1(u) = \begin{cases} 0, & u(x, y) \le \tilde{c}(x, y) \ \forall x, y, \\ +\infty, & \text{else,} \end{cases} \qquad \Theta_2(u) = \begin{cases} -\int \varphi \, \mathrm{d}\mu - \int \psi \, \mathrm{d}\nu, & u(x, y) = \varphi(x) + \psi(y), \\ +\infty, & \text{else,} \end{cases}$$

where in Θ_2 the decomposition $u(x,y) = \varphi(x) + \psi(y)$ into bounded continuous φ, ψ is required (such a decomposition, when it exists, determines $\int \varphi \, \mathrm{d}\mu + \int \psi \, \mathrm{d}\nu$ uniquely up to the additive-constant gauge $\varphi \mapsto \varphi + a$, $\psi \mapsto \psi - a$, under which the value is unchanged because μ, ν are probability measures). Both functionals are convex; Θ_1 is continuous at $u \equiv -\|\tilde{c}\|_\infty - 1$, where Θ_2 is also finite. Then

$$\inf_{u \in E} (\Theta_1(u) + \Theta_2(u)) = -\sup\left\{\int \varphi \, \mathrm{d}\mu + \int \psi \, \mathrm{d}\nu : \varphi(x) + \psi(y) \leq \tilde{c}(x,y)\right\} = -(\text{dual}).$$

A direct computation of the conjugates identifies the right-hand side of (D.3): $\Theta_1^*(-\pi)$ is finite only when $\pi \geq 0$ (a positive measure), where it equals $\int \tilde{c} \, \mathrm{d}\pi$; and $\Theta_2^*(\pi)$ is finite only when π has marginals μ, ν, where it equals 0. Hence the maximizing measure π is a coupling in $\Pi(\mu, \nu)$ and (D.3) reads $-(\text{dual}) = -\inf_{\pi \in \Pi(\mu,\nu)} \int \tilde{c} \, \mathrm{d}\pi$, i.e. (dual) = (primal), with both optima attained. This is strong duality for bounded continuous cost.

D.1.4 Removing boundedness

For the quadratic cost set $c_R(x,y) = \min\left\{|x-y|^2, R\right\}$, which is bounded and continuous. By the previous step there are admissible bounded potentials (φ_R, ψ_R) with $\int \varphi_R \, \mathrm{d}\mu + \int \psi_R \, \mathrm{d}\nu = \inf_\pi \int c_R \, \mathrm{d}\pi$. Since $c_R \uparrow c$ pointwise, the monotone convergence theorem gives $\inf_\pi \int c_R \, \mathrm{d}\pi \uparrow \mathcal{W}_2(\mu,\nu)^2$ along any minimizing sequence; a standard diagonal/tightness argument (each c_R-optimal pair may be taken c_R-conjugate, hence equi-Lipschitz on compacts with at most quadratic growth controlled by the second moments of μ, ν) extracts a limiting pair (φ, ψ) that is c-admissible and attains the supremum in (D.1). Replacing φ by its c-transform $\varphi^c(y) = \inf_x \left(|x-y|^2 - \varphi(x)\right)$ and iterating once produces a c-conjugate pair without decreasing the objective, giving the Kantorovich potentials. This proves (D.1). [131, Ch. 5] carries out the truncation limit in full.

D.2 Brenier's theorem

We prove theorem 4.6: if $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ is absolutely continuous with respect to Lebesgue measure and $\nu \in \mathcal{P}_2(\mathbb{R}^d)$, then the optimal coupling is unique and induced by a map $T = \nabla\phi$ with ϕ convex, and this is the unique gradient-of-convex map pushing μ to ν.

D.2.1 Cyclical monotonicity of the optimal support

Lemma D.2 (The optimal plan is cyclically monotone)**.** *Let $\pi^\star \in \Pi(\mu,\nu)$ be optimal for the quadratic cost. Then its support $S = \operatorname{supp} \pi^\star$ is c-*cyclically monotone*: for every finite family $(x_1, y_1), \ldots, (x_n, y_n) \in S$ and every permutation σ,*

$$\sum_{i=1}^{n} |x_i - y_i|^2 \;\leq\; \sum_{i=1}^{n} \left|x_i - y_{\sigma(i)}\right|^2. \tag{D.4}$$

Proof. If (D.4) failed for some family and some σ, the reverse strict inequality would hold for a cyclic permutation. One could then construct a competitor plan by moving an ε amount of mass off the cycle's edges (x_i, y_i) onto the re-paired edges $(x_i, y_{\sigma(i)})$ — concretely, using small product blocks around each (x_i, y_i) in $\operatorname{supp}\pi^\star$ and a measurable re-coupling — strictly lowering the total cost while preserving both marginals, contradicting optimality of $\pi^\star$. The construction is the standard "cycle-swapping" perturbation; see [131, Thm. 2.13]. □

D.2.2 Rockafellar's theorem and the convex potential

Lemma D.3 (Rockafellar)**.** *After absorbing the quadratic terms via $|x-y|^2 = |x|^2 - 2\langle x, y\rangle + |y|^2$, c-cyclical monotonicity of S is equivalent to* cyclical monotonicity *of the set $\widehat{S} = \{(x,y) : (x,y) \in S\}$ with respect to the inner product: for all cycles, $\sum_i \langle x_i, y_{i+1} - y_i\rangle \le 0$. Any cyclically monotone set is contained in the subdifferential graph $\{(x,y) : y \in \partial\phi(x)\}$ of a proper lower semicontinuous convex function $\phi\colon \mathbb{R}^d \to (-\infty, +\infty]$.*

Proof. Expanding the squares in (D.4) and cancelling the $|x_i|^2, |y_i|^2$ terms (a permutation of the y's leaves $\sum |y_i|^2$ unchanged) turns (D.4) into $\sum_i \langle x_i, y_i\rangle \ge \sum_i \langle x_i, y_{\sigma(i)}\rangle$, which is the cyclical-monotonicity inequality. Given a cyclically monotone set $\widehat{S}$, fix a base point $(x_0, y_0) \in \widehat{S}$ and define

$$\phi(x) = \sup\left\{\langle y_0, x_1 - x_0\rangle + \langle y_1, x_2 - x_1\rangle + \cdots + \langle y_{k-1}, x_k - x_{k-1}\rangle + \langle y_k, x - x_k\rangle\right\},$$

the supremum over all finite chains $(x_1, y_1), \ldots, (x_k, y_k) \in \widehat{S}$. As a supremum of affine functions, ϕ is convex and lower semicontinuous; cyclical monotonicity makes $\phi(x_0)$ finite (the empty chain gives 0 and no chain exceeds it), and one verifies directly that $y \in \partial\phi(x)$ whenever $(x,y) \in \widehat{S}$. This is Rockafellar's theorem [116, Thm. 24.8]. □

D.2.3 From subdifferential to gradient

Lemma D.4 (Convex functions are differentiable a.e.)**.** *A finite convex function $\phi\colon U \to \mathbb{R}$ on an open $U \subseteq \mathbb{R}^d$ is locally Lipschitz, and the set of points where it is not differentiable has Lebesgue measure zero; where it is differentiable, $\partial\phi(x) = \{\nabla\phi(x)\}$.*

Proof. Local Lipschitzness is elementary: convexity bounds the difference quotients above and below by the values on a surrounding cube. For the null set, in one variable a convex function has nondecreasing one-sided derivatives that agree except at the at most countably many jump points, so it is differentiable off a countable (null) set. In $\mathbb{R}^d$, apply this to each coordinate line: the partial derivatives exist off a null set by Fubini, and for a convex function the a.e. existence of all partials forces full (Fréchet) differentiability a.e. because the subdifferential is monotone and its multivalued points form a Lebesgue-null set (they are covered by countably many Lipschitz graphs of dimension $\le d-1$). Where $\partial\phi(x)$ is a singleton it equals the gradient. See [116, Thm. 25.5]; this is the convex-function form of Rademacher's theorem. □

Proof of theorem 4.6. Let $\pi^\star$ be an optimal coupling (proposition 4.3). By lemmas D.2 and D.3 its support lies in the graph of $\partial\phi$ for a convex ϕ. Because μ is absolutely continuous, lemma D.4 gives a μ-null set off which $\partial\phi(x) = \{\nabla\phi(x)\}$ is single-valued; hence for μ-a.e. x the plan assigns the single value $T(x) = \nabla\phi(x)$, i.e. $\pi^\star = (\mathrm{Id}, T)_\# \mu$ is induced by the map $T = \nabla\phi$ and $T_\#\mu = \nu$ (the second marginal of $\pi^\star$). This proves (1) and (2) up to uniqueness.

Uniqueness. Suppose $T_1 = \nabla\phi_1$ and $T_2 = \nabla\phi_2$ are two gradient-of-convex maps with $(T_i)_\#\mu = \nu$, both optimal. The half-sum coupling $\frac{1}{2}(\mathrm{Id}, T_1)_\#\mu + \frac{1}{2}(\mathrm{Id}, T_2)_\#\mu$ is again optimal (the cost is affine in the plan and both pieces are optimal), so its support is also contained in a single subdifferential graph; two gradients of convex functions agreeing μ-a.e. with a common monotone graph must coincide μ-a.e., whence $T_1 = T_2$ μ-a.e. and the optimal plan is unique. Statement (3) is then immediate: $\mathcal{W}_2(\mu,\nu)^2 = \int |x - T(x)|^2 \, \mathrm{d}\mu$ equals the Monge infimum because T is an admissible Monge map attaining the Kantorovich value, and no Monge competitor can beat the Kantorovich infimum since Monge plans are couplings. The growth/regularity of the potentials is treated in [30] and [131, Ch. 2]. ☐

D.3 McCann's displacement interpolation

We prove theorem 4.9. Let μ_0 be absolutely continuous, $T = \nabla\phi$ the Brenier map from μ_0 to μ_1, and for $t \in [0,1]$ set $T_t = (1-t)\,\mathrm{Id} + tT$ and $\mu_t = (T_t)_\#\mu_0$.

Proof. Constant speed. Fix $0 \le s \le t \le 1$. The map pushing μ_s to μ_t is $T_{s\to t} = T_t \circ T_s^{-1}$, defined μ_s-a.e. because $T_s = (1-s)\,\mathrm{Id} + sT$ is injective μ_0-a.e. (it is the gradient of the strictly convex $x \mapsto \frac{1-s}{2}|x|^2 + s\phi(x)$ for $s < 1$, and for $s = 1$ no inverse is needed since then $s = t$). On the support, $T_{s\to t}(y) = y + (t-s)\big(T - \mathrm{Id}\big)\big(T_s^{-1}(y)\big)$ is again a gradient of a convex function — the gradients of convex functions are closed under such affine interpolation of the underlying potentials — so by lemma D.3 it is the optimal map from μ_s to μ_t. Therefore

$$\mathcal{W}_2(\mu_s,\mu_t)^2 = \int |T_t(x) - T_s(x)|^2 \, \mathrm{d}\mu_0 = \int \big|(t-s)\big(T(x) - x\big)\big|^2 \, \mathrm{d}\mu_0 = (t-s)^2\, \mathcal{W}_2(\mu_0,\mu_1)^2,$$

using $\mathcal{W}_2(\mu_0,\mu_1)^2 = \int |x - T(x)|^2 \, \mathrm{d}\mu_0$ from Brenier. Taking square roots, $\mathcal{W}_2(\mu_s,\mu_t) = (t - s)\,\mathcal{W}_2(\mu_0,\mu_1)$, the constant-speed-geodesic property.

These exhaust the geodesics. Any constant-speed geodesic (ν_t) joining μ_0 to μ_1 has, for each t, optimal couplings of μ_0-to-ν_t and ν_t-to-μ_1 whose costs add to $\mathcal{W}_2(\mu_0,\mu_1)^2$ with the right proportions; gluing them (via disintegration, theorem 3.33) yields a coupling of μ_0 and μ_1 supported on geodesics of $\mathbb{R}^d$ (straight segments), and absolute continuity of μ_0 forces this coupling to be the graph of T_t. This is McCann's theorem [101]; the geodesic-gluing argument is [7, Ch. 7]. ☐

D.4 The Benamou–Brenier dynamic formula

We prove theorem 4.10: for $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\mathcal{W}_2(\mu_0,\mu_1)^2 = \inf\left\{ \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^2 \, \mathrm{d}\mu_t \; \mathrm{d}t : \; \partial_t\mu_t + \mathrm{div}(\mu_t v_t) = 0, \; \mu_{t=0} = \mu_0, \; \mu_{t=1} = \mu_1 \right\},$$

the continuity equation holding in the distributional sense.

Proof. Lower bound (≥). Let (μ_t, v_t) solve the continuity equation with finite action $\mathcal{A} = \int_0^1 \int |v_t|^2 \, d\mu_t \, dt$. The flow map $\dot{X}_t = v_t(X_t)$, $X_0 \sim \mu_0$, transports μ_0 to μ_1 (this is the method of characteristics for the continuity equation, valid here because finite action gives the $L^2(dt; L^2(\mu_t))$ regularity needed for the superposition principle). Then (X_0, X_1) is a coupling of μ_0, μ_1, and by Cauchy–Schwarz in t,

$$\mathcal{W}_2(\mu_0, \mu_1)^2 \le \mathbb{E}\,|X_1 - X_0|^2 = \mathbb{E}\left|\int_0^1 v_t(X_t)\,dt\right|^2 \le \mathbb{E}\int_0^1 |v_t(X_t)|^2\,dt = \mathcal{A}.$$

Upper bound (≤). The displacement geodesic $\mu_t = (T_t)_{\#}\mu_0$ of appendix D.3 solves the continuity equation with velocity $v_t = (T - \mathrm{Id}) \circ T_t^{-1}$, which is well defined μ_t-a.e. for $t \in [0, 1)$ since T_t is invertible there (appendix D.3). Its action is

$$\int_0^1 \int |v_t|^2 \, d\mu_t \, dt = \int_0^1 \int |T(x) - x|^2 \, d\mu_0 \, dt = \int |T(x) - x|^2 \, d\mu_0 = \mathcal{W}_2(\mu_0, \mu_1)^2,$$

the middle equality by the change of variables $y = T_t(x)$. Hence the infimum is at most $\mathcal{W}_2(\mu_0, \mu_1)^2$, and with the lower bound it equals it, the displacement geodesic being the minimizer. The superposition principle underlying the lower bound is in [7, Ch. 8]; the original argument is [18]. □

D.5 The Lions structure theorem

We prove the assertion of definition 4.14: when the lift $\widetilde{F}(X) = F(\mathcal{L}(X))$ is Fréchet differentiable on $L^2(\Omega; \mathbb{R}^d)$ with a continuous gradient, that gradient is $\mathcal{L}(X)$-measurably a function of X alone, i.e. $D\widetilde{F}(X) = \xi(X)$ a.s. for some $\xi \in L^2(\mathcal{L}(X); \mathbb{R}^d)$.

Theorem D.5 (Rearrangement-invariance structure theorem)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be atomless and $F \colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ have a lift $\widetilde{F}$ that is Fréchet differentiable at $X \in L^2(\Omega; \mathbb{R}^d)$ with gradient $D\widetilde{F}(X) \in L^2(\Omega; \mathbb{R}^d)$ (identified with its Riesz representative, theorem 2.6). Then there is a Borel map $\xi \colon \mathbb{R}^d \to \mathbb{R}^d$, in $L^2(\mathcal{L}(X); \mathbb{R}^d)$, with*

$$D\widetilde{F}(X) = \xi(X) \qquad a.s., \qquad \xi = \partial_\mu F(\mathcal{L}(X)). \tag{D.5}$$

Proof. The lift is *law-invariant*: $\widetilde{F}(X) = \widetilde{F}(X')$ whenever $\mathcal{L}(X) = \mathcal{L}(X')$, because both equal $F(\mathcal{L}(X))$. We exploit invariance under the group $\mathcal{T}$ of measure-preserving bijections $\tau \colon \Omega \to \Omega$: for any such τ, $X \circ \tau$ has the same law as X, so $\widetilde{F}(X \circ \tau) = \widetilde{F}(X)$ for all X. Differentiating this identity at X in a direction $Y \in L^2$,

$$\langle D\widetilde{F}(X \circ \tau), Y \circ \tau \rangle_{L^2} = \langle D\widetilde{F}(X), Y \rangle_{L^2}.$$

Since τ is measure preserving, $\langle D\widetilde{F}(X) \circ \tau, Y \circ \tau \rangle = \langle D\widetilde{F}(X), Y \rangle$ under the substitution; comparing for all Y shows $D\widetilde{F}(X \circ \tau) = D\widetilde{F}(X) \circ \tau$ a.s. Thus the gradient is *equivariant*: it commutes with measure-preserving rearrangements.

Now fix X and let $\sigma(X)$ be the σ-algebra it generates. We claim $D\widetilde{F}(X)$ is $\sigma(X)$-measurable. If not, its conditional expectation $\mathbb{E}[D\widetilde{F}(X) \mid \sigma(X)]$ (section 3.6) would differ from it on a set of positive measure, and one could build, using atomlessness of Ω, a measure-preserving τ fixing $\sigma(X)$ (so $X \circ \tau = X$) but moving $D\widetilde{F}(X)$ — contradicting equivariance $D\widetilde{F}(X) = D\widetilde{F}(X \circ \tau) = D\widetilde{F}(X) \circ \tau$. Hence $D\widetilde{F}(X)$ is $\sigma(X)$-measurable, so by the Doob–Dynkin lemma $D\widetilde{F}(X) = \xi(X)$ a.s. for a Borel $\xi \colon \mathbb{R}^d \to \mathbb{R}^d$. Finally ξ depends on X only through $\mathcal{L}(X) = m$: two variables with the same law are related by a measure-preserving map (atomless Ω), and equivariance transports ξ unchanged. Writing $\xi = \partial_\mu F(m)$ gives (D.5). The argument is Lions's, reported in [40, Ch. 5]. ☐

The relation $\partial_\mu F(m)(x) = \nabla_x \frac{\delta F}{\delta m}(m, x)$ between the Lions and flat derivatives, stated as proposition 4.15, follows by differentiating $\epsilon \mapsto F(\mathcal{L}(X + \epsilon Y))$ two ways and matching, exactly as in the proof sketch of proposition 4.15; the integration by parts there is justified by the linear-growth hypothesis on $\nabla_x \frac{\delta F}{\delta m}(m, \cdot)$.

D.6 Second-order calculus: Itô's formula along a flow of measures

The master equation of chapter 12 differentiates a functional $U(t, x, m)$ along the flow of a measure $m_t = \mathcal{L}(X_t)$ where X_t solves an Itô SDE. We record the chain rule it needs. This requires a *second* Lions derivative and the mixed object $\nabla_x \partial_\mu F(m)(x)$, which we now define precisely.

Definition D.6 (Second Lions derivative)**.** Let $F \colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ be L-differentiable with Lions derivative $\partial_\mu F(m)(\cdot) \colon \mathbb{R}^d \to \mathbb{R}^d$ (proposition 4.15, theorem D.5). Fix $v \in \mathbb{R}^d$ and suppose the scalar functionals $m \mapsto \big[\partial_\mu F(m)(v)\big]_i$ $(i = 1, \dots, d)$ are themselves L-differentiable. The *second Lions derivative* of F is the map

$$\partial_\mu^2 F(m)(v, v') \;:=\; \partial_\mu\big[\, m \mapsto \partial_\mu F(m)(v) \,\big](v') \;\in\; \mathbb{R}^{d \times d},$$

the Lions derivative, at the second base point v', of the $\mathbb{R}^d$-valued function $m \mapsto \partial_\mu F(m)(v)$. Equivalently, in flat-derivative form, $\partial_\mu^2 F(m)(v, v') = \nabla_v \nabla_{v'} \frac{\delta^2 F}{\delta m^2}(m)(v, v')$, where $\frac{\delta^2 F}{\delta m^2}$ is the second flat derivative (the flat derivative of $m \mapsto \frac{\delta F}{\delta m}(m)(v)$). When F is twice L-differentiable with $\frac{\delta^2 F}{\delta m^2}$ jointly continuous, Schwarz symmetry holds, $\frac{\delta^2 F}{\delta m^2}(m)(v, v') = \frac{\delta^2 F}{\delta m^2}(m)(v', v)$, hence $\partial_\mu^2 F(m)(v, v') = \partial_\mu^2 F(m)(v', v)^\top$. We call F *twice L-differentiable* when $\partial_\mu F$, the mixed derivative $\nabla_v \partial_\mu F$, and $\partial_\mu^2 F$ exist, are jointly continuous, and are bounded on bounded sets.

Theorem D.7 (Itô's formula along a flow of measures)**.** *Let X_t solve $\mathrm{d}X_t = b_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t$ in $\mathbb{R}^d$ with W a Brownian motion (chapter 5), and let $m_t = \mathcal{L}(X_t)$. Let $F \colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ be twice L-differentiable with continuous, suitably bounded derivatives $\partial_\mu F$, $\nabla_v \partial_\mu F$, $\partial_\mu^2 F$. Then $t \mapsto F(m_t)$ is differentiable and*

$$\frac{\mathrm{d}}{\mathrm{d}t} F(m_t) = \mathbb{E}\Big[\partial_\mu F(m_t)(X_t) \cdot b_t + \tfrac{1}{2} \operatorname{tr}\big(\nabla_v \partial_\mu F(m_t)(X_t)\, \sigma_t \sigma_t^\top\big)\Big]. \tag{D.6}$$

Proof. Let $\widetilde{F}$ be the lift. By the structure theorem (theorem D.5), $D\widetilde{F}(X_t) = \partial_\mu F(m_t)(X_t)$ and the Hessian of the lift acts as the second Lions derivative. Apply the ordinary Itô formula (theorem 5.9) to the *deterministic* function $t \mapsto \widetilde{F}(X_t) = F(m_t)$ on the Hilbert space $L^2(\Omega; \mathbb{R}^d)$ — but $F(m_t)$ is deterministic, so only the drift of the Itô expansion of $\widetilde{F}(X_t)$ survives in expectation.

Concretely, write the increment of X_t and expand $\widetilde{F}$ to second order; the first-order term contributes $\mathbb{E}[\partial_\mu F(m_t)(X_t) \cdot b_t]\ \mathrm{d}t$, the Itô second-order term contributes $\frac{1}{2}\mathbb{E}[\mathrm{tr}(\nabla_v \partial_\mu F(m_t)(X_t)\sigma_t\sigma_t^\top)]\ \mathrm{d}t$, and the martingale part $\mathbb{E}[\partial_\mu F(m_t)(X_t) \cdot \sigma_t\ \mathrm{d}W_t] = 0$ vanishes in expectation. This yields (D.6). The full hypotheses and the joint-in-(x, m) version used for the master equation are in [40, Ch. 5] and are assembled in chapter 12. □

This is the result chapter 12 imports to derive the master equation; chapter 4 states its first-order half (the gradient relation proposition 4.15) and defers the second-order formula here and to chapter 12.

D.7 Itô's formula along a conditional flow of measures (common noise)

The common-noise master equation of section 12.7 and its graphon lift (chapter 26) differentiate a functional along a *stochastic* flow of measures: the *conditional* law $m_t = \mathcal{L}(X_t \mid \mathcal{F}_t^0)$ of a state that is driven both by private noise and by a common noise W^0. The chain rule then acquires, beyond the deterministic terms of theorem D.7, a second-order contribution in the common noise — including a genuine *second* Lions derivative — and a martingale term. This is the Itô–Wentzell formula on $\mathcal{P}_2(\mathbb{R}^d)$; we state and prove it here, the result both chapter 12 and chapter 26 import.

Theorem D.8 (Conditional Itô formula on $\mathcal{P}_2(\mathbb{R}^d)$)**.** *On a filtered probability space carrying independent Brownian motions W (private) and W^0 (common), with $\mathcal{F}^0 = (\mathcal{F}_t^0)$ the filtration generated by W^0, let*

$$\mathrm{d}X_t = b_t\ \mathrm{d}t + \sigma_t\ \mathrm{d}W_t + \sigma_t^0\ \mathrm{d}W_t^0,$$

with $\mathcal{F}^0$-progressive coefficients of suitable integrability, and set $m_t = \mathcal{L}(X_t \mid \mathcal{F}_t^0) \in \mathcal{P}_2(\mathbb{R}^d)$, an $\mathcal{F}^0$-adapted stochastic flow. Let $F \colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ be twice L-differentiable in the sense of definition D.6, with $\partial_\mu F$, $\nabla_v \partial_\mu F$ and $\partial_\mu^2 F$ continuous and bounded on bounded sets. Then $t \mapsto F(m_t)$ is an Itô process and

$$\begin{aligned} \mathrm{d}F(m_t) = {} & \mathbb{E}_t\big[\partial_\mu F(m_t)(X_t) \cdot b_t\big]\ \mathrm{d}t + \tfrac{1}{2}\,\mathbb{E}_t\big[\,\mathrm{tr}\,\big(\nabla_v \partial_\mu F(m_t)(X_t)\,(\sigma_t\sigma_t^\top + \sigma_t^0(\sigma_t^0)^\top)\big)\big]\ \mathrm{d}t \\ & + \tfrac{1}{2}\,\mathbb{E}_t\big[\,\mathrm{tr}\,\big(\partial_\mu^2 F(m_t)(X_t, X_t')\,\sigma_t^0(\sigma_t^0)^\top\big)\big]\ \mathrm{d}t + \mathbb{E}_t\big[\partial_\mu F(m_t)(X_t) \cdot \sigma_t^0\big]\ \mathrm{d}W_t^0, \end{aligned} \tag{D.7}$$

where $\mathbb{E}_t[\,\cdot\,] := \mathbb{E}[\,\cdot \mid \mathcal{F}_t^0]$ and, in the third term, X_t' is an independent copy of X_t with the same *conditional law m_t (so $\mathbb{E}_t$ averages the matrix $\partial_\mu^2 F(m_t)(\cdot, \cdot)$ against $m_t \otimes m_t$).*

Proof. Freeze a realization of W^0 and condition throughout on $\mathcal{F}^0$. The new feature relative to theorem D.7 is that m_t is now random and itself driven by W^0. We expand $F(m_t)$ by combining the Itô expansion of the underlying state with the second-order calculus on $\mathcal{P}_2(\mathbb{R}^d)$.

The conditional flow. Conditioning on $\mathcal{F}_t^0$ removes the private noise: by the conditional propagation-of-chaos / superposition principle (the conditional law of a diffusion driven by an independent private noise solves a stochastic Fokker–Planck equation, the McKean–Vlasov form of theorem 5.9 averaged over the private noise), m_t solves

$$\mathrm{d}m_t = \Big[-\operatorname{div}\big(m_t\,\mathbb{E}[b_t \mid X_t = \cdot]\big) + \tfrac{1}{2}\sum_{ij}\partial_{ij}\big((\sigma_t\sigma_t^\top + \sigma_t^0(\sigma_t^0)^\top)_{ij}\, m_t\big)\Big]\,\mathrm{d}t - \operatorname{div}\big(m_t\,\sigma_t^0\big)\cdot \mathrm{d}W_t^0,$$

a stochastic PDE whose drift carries the *full* diffusion $\sigma_t\sigma_t^\top + \sigma_t^0(\sigma_t^0)^\top$ (the common noise spreads the conditional law just as the private noise does) and whose martingale part $-\operatorname{div}(m_t\,\sigma_t^0)\,\mathrm{d}W_t^0$ transports the whole conditional law coherently.

Applying the flat-derivative chain rule. For a smooth functional F the first variation gives, along any Itô flow $\mathrm{d}m_t = \boldsymbol{b}_t\,\mathrm{d}t + \boldsymbol{\sigma}_t\,\mathrm{d}W_t^0$ of measures, the Itô expansion

$$\mathrm{d}F(m_t) = \Big\langle \tfrac{\delta F}{\delta m}(m_t),\, \boldsymbol{b}_t\Big\rangle\,\mathrm{d}t + \Big\langle \tfrac{\delta F}{\delta m}(m_t),\, \boldsymbol{\sigma}_t\Big\rangle\,\mathrm{d}W_t^0 + \tfrac{1}{2}\Big\langle\!\!\Big\langle \tfrac{\delta^2 F}{\delta m^2}(m_t),\, \boldsymbol{\sigma}_t\otimes\boldsymbol{\sigma}_t\Big\rangle\!\!\Big\rangle\,\mathrm{d}t,$$

the last term being the Itô correction from the quadratic variation of the *measure-valued* martingale, paired through the second flat derivative. Substituting the two parts of the conditional flow and integrating the divergences by parts in the state variable — which converts $\frac{\delta F}{\delta m}$ into $\partial_\mu F = \nabla_v \frac{\delta F}{\delta m}$ and $\frac{\delta^2 F}{\delta m^2}$ into $\partial_\mu^2 F = \nabla_v\nabla_{v'}\frac{\delta^2 F}{\delta m^2}$ (definition D.6) — the drift terms reproduce exactly the two $\mathbb{E}_t[\cdots]$ lines of (D.7): the first from the transport drift $\mathbb{E}[b_t \mid X_t = \cdot]$, the second (full-diffusion) from the spreading. The martingale part $-\operatorname{div}(m_t\sigma_t^0)\,\mathrm{d}W_t^0$ contributes, after integration by parts, the $\mathbb{E}_t[\partial_\mu F(m_t)(X_t)\cdot\sigma_t^0]\,\mathrm{d}W_t^0$ term.

The double-integral term. The Itô correction is the quadratic variation of that same measure-valued martingale, $\mathrm{d}\langle m\rangle_t = \big(\operatorname{div}(m_t\sigma_t^0)\big)^{\otimes 2}\,\mathrm{d}t$, paired against $\frac{\delta^2 F}{\delta m^2}$. Integrating by parts *twice* (once in each of the two state slots) moves both divergences onto the second flat derivative, producing

$$\tfrac{1}{2}\int_{\mathbb{R}^d}\!\int_{\mathbb{R}^d} \operatorname{tr}\big(\partial_\mu^2 F(m_t)(v,v')\,\sigma_t^0(v)\sigma_t^0(v')^\top\big)\,m_t(\mathrm{d}v)\,m_t(\mathrm{d}v') = \tfrac{1}{2}\,\mathbb{E}_t\big[\operatorname{tr}\big(\partial_\mu^2 F(m_t)(X_t,X_t')\,\sigma_t^0(\sigma_t^0)^\top\big)\big],$$

which is the third term of (D.7); the two state integrals are independent because they come from two independent copies of the state sharing the conditional law m_t, and they are correlated only through the *single* common increment $\mathrm{d}W_t^0$ — this is why one common noise yields a genuine double integral. The boundedness and continuity hypotheses on $\partial_\mu F, \nabla_v\partial_\mu F, \partial_\mu^2 F$ justify the integrations by parts and the differentiation under the expectation. The rigorous version, with the precise function-space hypotheses, is the Itô–Wentzell formula of [41, Ch. 4]; the present derivation is the deterministic theorem D.7 upgraded by the common-noise quadratic variation. □

Remark D.9. Setting $\sigma_t^0 \equiv 0$ recovers theorem D.7. The four families of σ^0-terms in (D.7) — the full-diffusion upgrade in the second term and the new $\partial_\mu^2 F$ double integral and martingale term — are precisely those that appear, applied to $U(t,x,\cdot)$ with the agent's own state carried as a parameter, in the classical common-noise master equation (12.17) and, lifted slice-by-slice over the type continuum, in the graphon common-noise master equation (26.23).

D.8 The first-order distance $\mathcal{W}_1$

Chapter 4 develops the quadratic Wasserstein space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ in detail; the existence theory of chapter 10 works instead in the *first-order* space $(\mathcal{P}_1(\mathbb{R}^d), \mathcal{W}_1)$, where $\mathcal{P}_1(\mathbb{R}^d)$ is the set of Borel probability measures with finite first moment $\int |x|\ \mathrm{d}\mu < \infty$ and

$$\mathcal{W}_1(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)} \int_{\mathbb{R}^d\times\mathbb{R}^d} |x-y|\ \pi(\mathrm{d}x,\mathrm{d}y),$$

the optimal-transport cost for the metric cost $c(x,y) = |x-y|$. We supply the four facts chapter 10 uses: Kantorovich–Rubinstein duality, the metrisation of weak convergence, the completeness of $(\mathcal{P}_1, \mathcal{W}_1)$, and the compactness criterion. Each is a corollary of the machinery already built in this appendix (Kantorovich duality, appendix D.1) and in chapter 3 (tightness and Prokhorov, theorem 3.16).

Theorem D.10 (Kantorovich–Rubinstein duality)**.** *For $\mu,\nu \in \mathcal{P}_1(\mathbb{R}^d)$,*

$$\mathcal{W}_1(\mu,\nu) \;=\; \sup\Big\{ \int_{\mathbb{R}^d} f\ \mathrm{d}\mu - \int_{\mathbb{R}^d} f\ \mathrm{d}\nu \ :\ f\colon \mathbb{R}^d \to \mathbb{R},\ \mathrm{Lip}(f) \le 1 \Big\}. \tag{D.8}$$

Consequently $\mathcal{W}_1$ is the restriction to $\mathcal{P}_1(\mathbb{R}^d)$ of the dual bounded-Lipschitz *(Kantorovich–Rubinstein) norm $\|\mu-\nu\|_{\mathrm{KR}} = \sup\{\int f\ \mathrm{d}(\mu-\nu) : \mathrm{Lip}(f) \le 1\}$ on finite signed measures of finite first moment.*

Proof. ($\ge$, weak duality). For any 1-Lipschitz f and any coupling $\pi \in \Pi(\mu,\nu)$,

$$\int f\ \mathrm{d}\mu - \int f\ \mathrm{d}\nu = \int \big(f(x) - f(y)\big)\, \pi(\mathrm{d}x,\mathrm{d}y) \le \int |x-y|\ \pi(\mathrm{d}x,\mathrm{d}y),$$

and taking the infimum over π gives $\int f\ \mathrm{d}\mu - \int f\ \mathrm{d}\nu \le \mathcal{W}_1(\mu,\nu)$, hence the sup in (D.8) is $\le \mathcal{W}_1$.

($\le$, strong duality). The cost $c(x,y) = |x-y|$ is continuous of at most linear growth, so the Fenchel–Rockafellar mechanism of theorem D.1, used in appendix D.1 for the quadratic cost, applies verbatim and yields

$$\mathcal{W}_1(\mu,\nu) = \sup\Big\{ \int \varphi\ \mathrm{d}\mu + \int \psi\ \mathrm{d}\nu :\ \varphi(x) + \psi(y) \le |x-y|\ \ \forall x,y \Big\},$$

with the supremum attained by a c-conjugate pair. For the *metric* cost the c-conjugation collapses: given admissible (φ,ψ), the c-transform $\varphi^c(y) = \inf_x(|x-y| - \varphi(x))$ is itself 1-Lipschitz (an infimum of the 1-Lipschitz maps $x \mapsto |x-\cdot| - \varphi(x)$), satisfies $\psi \le \varphi^c$ and $\varphi \le (\varphi^c)^c =: f$ with f 1-Lipschitz, and $f(x) + (-f)(y) = f(x) - f(y) \le |x-y|$. Replacing (φ,ψ) by $(f,-f)$ does not decrease the objective, so the supremum is attained over pairs $(f,-f)$ with f 1-Lipschitz, which is exactly (D.8). The norm statement is the identification of the right-hand side with $\|\mu-\nu\|_{\mathrm{KR}}$. □

Theorem D.11 ($\mathcal{W}_1$ metrises weak convergence; completeness)**.** *$(\mathcal{P}_1(\mathbb{R}^d), \mathcal{W}_1)$ is a complete metric space. Moreover, for $\mu_n, \mu \in \mathcal{P}_1(\mathbb{R}^d)$,*

$$\mathcal{W}_1(\mu_n,\mu) \to 0 \iff \mu_n \to \mu \text{ weakly and } \int |x|\ \mathrm{d}\mu_n \to \int |x|\ \mathrm{d}\mu,$$

and on any set with uniformly integrable first moments *(in particular a uniform bound on a moment of order $p > 1$) $\mathcal{W}_1$ metrises weak convergence. The map $\mu \mapsto \int |x|\ \mathrm{d}\mu$ and, more generally, $\mu \mapsto \int |x|^2\ \mathrm{d}\mu$ are $\mathcal{W}_1$-lower semicontinuous.*

Proof. $\mathcal{W}_1$ is a metric (symmetry is clear; the triangle inequality follows by gluing couplings, exactly as for $\mathcal{W}_2$ in section 4.2; $\mathcal{W}_1(\mu,\nu)=0$ forces $\mu=\nu$ since by (D.8) all 1-Lipschitz integrals agree, hence all bounded continuous ones).

Equivalence with weak convergence plus first moments. ($\Rightarrow$) If $\mathcal{W}_1(\mu_n,\mu)\to 0$, then for every bounded 1-Lipschitz f, (D.8) gives $\int f\,\mathrm{d}\mu_n \to \int f\,\mathrm{d}\mu$; bounded Lipschitz functions are convergence-determining, so $\mu_n\to\mu$ weakly, and applying (D.8) to $f(x)=|x|$ (1-Lipschitz) shows $|\int |x|\,\mathrm{d}\mu_n - \int |x|\,\mathrm{d}\mu| \le \mathcal{W}_1(\mu_n,\mu)\to 0$. ($\Leftarrow$) Suppose $\mu_n\to\mu$ weakly with $\int |x|\,\mathrm{d}\mu_n \to \int |x|\,\mathrm{d}\mu$. The first-moment convergence forces uniform integrability of $|x|$ under μ_n (a weakly convergent sequence with convergent first moments is uniformly integrable in the first moment). Given ε, choose R with $\int_{|x|>R}|x|\,\mathrm{d}\mu_n<\varepsilon$ uniformly. For a 1-Lipschitz f normalised by $f(0)=0$ (so $|f(x)|\le|x|$), split $\int f\,\mathrm{d}(\mu_n-\mu)$ into the contribution of the truncation $f_R=f\cdot(|x|\le R)$-part, which $\to 0$ by weak convergence (bounded continuous), and the tails, bounded by 2ε; taking the supremum over f and using (D.8) gives $\limsup \mathcal{W}_1(\mu_n,\mu)\le C\varepsilon$. On a family with uniformly integrable first moments the tail bound is automatic, so $\mathcal{W}_1$ metrises weak convergence there.

Completeness. A $\mathcal{W}_1$-Cauchy sequence is $\mathcal{W}_1$-bounded, hence tight (a uniform first-moment bound gives tightness by Markov's inequality) and uniformly first-moment integrable, so by Prokhorov (theorem 3.16) a subsequence converges weakly to some $\mu\in\mathcal{P}_1$; the Cauchy property upgrades this to $\mathcal{W}_1$-convergence of the whole sequence by the previous paragraph. Lower semicontinuity of the moments is Fatou's lemma along a weakly convergent (e.g. $\mathcal{W}_1$-convergent) sequence applied to $|x|\wedge k$ and $|x|^2\wedge k$, $k\uparrow\infty$. $\square$

Theorem D.12 ($\mathcal{W}_1$ compactness criterion)**.** *A set $\mathcal{K}\subseteq\mathcal{P}_1(\mathbb{R}^d)$ is relatively compact in $(\mathcal{P}_1,\mathcal{W}_1)$ if and only if it is* tight *and has* uniformly integrable first moments*: $\sup_{\mu\in\mathcal{K}}\int_{|x|>R}|x|\,\mathrm{d}\mu\to 0$ as $R\to\infty$. In particular a uniform bound $\sup_{\mu\in\mathcal{K}}\int|x|^2\,\mathrm{d}\mu\le R^2<\infty$ suffices (it gives both tightness, $\mu(|x|>\rho)\le R^2/\rho^2$, and uniform integrability, $\int_{|x|>\rho}|x|\,\mathrm{d}\mu\le R^2/\rho$).*

Proof. Sufficiency. Let $(\mu_n)\subseteq\mathcal{K}$. Tightness and Prokhorov (theorem 3.16) give a weakly convergent subsequence $\mu_{n_k}\to\mu$; uniform integrability of the first moments passes $\int|x|\,\mathrm{d}\mu_{n_k}\to\int|x|\,\mathrm{d}\mu<\infty$ (so $\mu\in\mathcal{P}_1$) and, by theorem D.11, upgrades weak convergence to $\mathcal{W}_1(\mu_{n_k},\mu)\to 0$. Hence every sequence has a $\mathcal{W}_1$-convergent subsequence. *Necessity.* A $\mathcal{W}_1$-relatively compact set is totally bounded; covering it by finitely many $\mathcal{W}_1$-balls of radius ε about centres $\mu_1,\dots,\mu_N\in\mathcal{P}_1$ and using that each μ_i is tight and first-moment integrable, the uniform estimate $\int_{|x|>R}|x|\,\mathrm{d}\mu\le\int_{|x|>R}|x|\,\mathrm{d}\mu_i+C\varepsilon$ (from $\mathcal{W}_1(\mu,\mu_i)<\varepsilon$ and the dual bound applied to truncations of $|x|$) gives uniform integrability, and similarly tightness. $\square$

Remark D.13. Theorems D.10 to D.12 are exactly the inputs to the compact-set construction lemma 10.8 and the Banach embedding of $\mathcal{P}_1$ in the Schauder proof theorem 10.14: the dual bounded-Lipschitz norm makes the ambient space Banach, the second-moment bound supplies tightness and uniform integrability, and the resulting $\mathcal{W}_1$-compactness feeds the metric-valued Arzelà–Ascoli theorem theorem B.24.

D.9 The Wasserstein sampling rate for empirical measures

The quantitative convergence theorems of Part II rest on a single statistical input: the rate at which the empirical measure $\widehat{\rho}_N = \frac{1}{N}\sum_{i=1}^N \delta_{X_i}$ of N i.i.d. samples $X_1, \ldots, X_N \sim \rho$ approaches ρ in $\mathcal{W}_2$. The propagation-of-chaos estimates of chapters 7 and 13 reduce the interaction error to exactly this sampling error, so it is the term that sets the rate and *must* be proved in-book rather than imported. This is the Fournier–Guillin theorem [69]; we prove it here from the transport machinery of this appendix, the only nontrivial geometric ingredient being a multiresolution comparison of $\mathcal{W}_2$ with the discrepancies of μ and ν across a nested family of dyadic cubes.

D.9.1 A multiresolution upper bound for $\mathcal{W}_2$

Fix the half-open unit cube $C_0 = [0,1)^d$. For $\ell \geq 0$ let $\mathcal{P}_\ell$ be its partition into the $2^{d\ell}$ dyadic subcubes of side $2^{-\ell}$; each $Q \in \mathcal{P}_\ell$ is contained in a unique *parent* in $\mathcal{P}_{\ell-1}$ and contains 2^d *children* in $\mathcal{P}_{\ell+1}$, and has diameter $\sqrt{d}\,2^{-\ell}$. For probability measures μ, ν on C_0 set

$$D_\ell(\mu,\nu) \;:=\; \sum_{Q\in\mathcal{P}_\ell} |\mu(Q) - \nu(Q)|\,, \qquad a_\ell \;:=\; \sum_{Q\in\mathcal{P}_\ell} \min\big(\mu(Q),\nu(Q)\big) = 1 - \tfrac{1}{2}D_\ell, \tag{D.9}$$

the second identity from $\min(s,t) = \frac{1}{2}(s+t) - \frac{1}{2}|s-t|$ and $\sum_Q \mu(Q) = \sum_Q \nu(Q) = 1$. Because every $Q \in \mathcal{P}_\ell$ splits into children whose masses sum to $\mu(Q)$ and to $\nu(Q)$, and min is superadditive in the sense $\sum_{Q'\subset Q} \min(\mu(Q'),\nu(Q')) \leq \min(\mu(Q),\nu(Q))$, the sequence a_ℓ is non-increasing in ℓ, equivalently D_ℓ is non-decreasing.

Lemma D.14 (Multiresolution bound)**.** *For all probability measures μ, ν on $C_0 = [0,1)^d$,*

$$\mathcal{W}_2(\mu,\nu)^2 \;\leq\; \frac{3d}{2}\sum_{\ell\geq 1} 4^{-\ell}\, D_\ell(\mu,\nu). \tag{D.10}$$

Proof. We construct an explicit coupling $\pi \in \Pi(\mu,\nu)$ whose cost is bounded by the right-hand side; (D.10) then follows since $\mathcal{W}_2^2$ is the infimum of the cost over $\Pi(\mu,\nu)$ (proposition 4.3).

The tree coupling. Regard the dyadic cubes as the nodes of a tree, Q a child of its parent. We move "together" mass down the tree, releasing the part that cannot stay together. For each cube Q write $s_Q := \min(\mu(Q),\nu(Q))$. We attach to each Q an amount of mass t_Q that enters Q *coupled on the diagonal* (pairs (x,y) both lying in Q), and set $t_{C_0} = 1$ at the root and $t_Q = s_Q$ at every other node. This is consistent: the mass entering a node R that can stay together when distributed to its children is $\sum_{Q\subset R} s_Q \leq s_R = t_R$ (superadditivity of min), so within R an amount

$$t_R - \sum_{Q\subset R} s_Q \tag{D.11}$$

of together-mass must *separate*: its μ-part and its ν-part fall into different children of R. We couple this separating mass by any coupling of its (equal-total) μ- and ν-marginals supported in R; the two endpoints then lie in R, at distance $\leq \sqrt{d}\,2^{-\ell}$ when $R \in \mathcal{P}_\ell$. Mass never released stays together forever and contributes zero cost. Carrying this out at every node defines a coupling $\pi \in \Pi(\mu,\nu)$:

the μ-marginal of the released-plus-retained mass at each node reconstructs μ, and likewise for ν, since $t_Q = s_Q \le \mu(Q) \wedge \nu(Q)$ and the released excess at the parent supplies exactly the deficit at each child.

The cost. A pair separated within a node $R \in \mathcal{P}_\ell$ contributes at most $\big(\sqrt{d}\,2^{-\ell}\big)^2 = d\,4^{-\ell}$ to $\int |x-y|^2 \,\mathrm{d}\pi$. The total mass separated within generation-ℓ nodes is, by (D.11) summed over $R \in \mathcal{P}_\ell$,

$$\sum_{R\in\mathcal{P}_\ell}\Big(t_R - \sum_{Q\subset R} s_Q\Big) \;=\; a_\ell - a_{\ell+1} \;=\; \tfrac{1}{2}(D_{\ell+1} - D_\ell),$$

using $t_R = s_R$ for $\ell \ge 1$ and $t_{C_0} = 1 = a_0$ at $\ell = 0$ (consistent with $D_0 = 0$). Hence

$$\int |x-y|^2 \,\mathrm{d}\pi \;\le\; \sum_{\ell\ge 0} d\,4^{-\ell}\cdot \tfrac{1}{2}(D_{\ell+1} - D_\ell) \;=\; \frac{d}{2}\sum_{\ell\ge 0} 4^{-\ell}(D_{\ell+1} - D_\ell).$$

Summation by parts with $D_0 = 0$ gives $\sum_{\ell\ge 0} 4^{-\ell}(D_{\ell+1} - D_\ell) = \sum_{\ell\ge 1}\big(4^{-(\ell-1)} - 4^{-\ell}\big)D_\ell = 3\sum_{\ell\ge 1} 4^{-\ell} D_\ell$, which is (D.10). $\square$

Remark D.15. The estimate is genuinely an $\mathcal{W}_2$ statement: the weight $4^{-\ell}$ is the *square* of the scale $2^{-\ell}$, so a fluctuation of mass at scale $2^{-\ell}$ costs $4^{-\ell}$, not $2^{-\ell}$. Pairing this with the $\sqrt{\cdot}$-law of the next subsection produces the dimension-dependent thresholds; the analogous $\mathcal{W}_1$ bound carries $2^{-\ell}$ and yields the coarser $N^{-1/d}$ rate.

D.9.2 The sampling theorem

Theorem D.16 (Empirical-measure rate; Fournier–Guillin)**.** *Let $\rho \in \mathcal{P}(\mathbb{R}^d)$ and let $X_1, \dots, X_N$ be i.i.d. with law ρ, with empirical measure $\widehat{\rho}_N = \frac{1}{N}\sum_{i=1}^N \delta_{X_i}$.*

1. (Compactly supported case.) *If ρ is supported in a cube of side L, then*

$$\mathbb{E}\big[\mathcal{W}_2(\widehat{\rho}_N, \rho)^2\big] \;\le\; C_d\, L^2\, s_N, \qquad s_N := \begin{cases} N^{-1/2}, & d < 4,\\ N^{-1/2}\log N, & d = 4,\\ N^{-2/d}, & d > 4,\end{cases} \tag{D.12}$$

with C_d depending only on the dimension.

2. (Finite moment.) *If ρ has a finite moment $M_q = \int |x|^q\,\rho(\mathrm{d}x)$ of some order $q > 4$, then*

$$\mathbb{E}\big[\mathcal{W}_2(\widehat{\rho}_N, \rho)^2\big] \;\le\; C_{d,q}\,(1 + M_q)\, s_N^{1-2/q}. \tag{D.13}$$

Proof. Part (1). Since $\mathcal{W}_2$ scales by L and translates under $x \mapsto Lx + b$, we may assume ρ is supported in the unit cube $C_0 = [0,1)^d$. Apply lemma D.14 to $\mu = \widehat{\rho}_N$, $\nu = \rho$ and take expectations:

$$\mathbb{E}\big[\mathcal{W}_2(\widehat{\rho}_N, \rho)^2\big] \;\le\; \frac{3d}{2}\sum_{\ell\ge 1} 4^{-\ell} \sum_{Q\in\mathcal{P}_\ell} \mathbb{E}\big|\widehat{\rho}_N(Q) - \rho(Q)\big|.$$

For each fixed cube Q, $N\widehat{\rho}_N(Q) = \sum_i \mathbf{1}\{X_i \in Q\}$ is Binomial$(N, \rho(Q))$, so by Jensen and the binomial variance,

$$\mathbb{E}|\widehat{\rho}_N(Q) - \rho(Q)| \le \big(\operatorname{Var}\widehat{\rho}_N(Q)\big)^{1/2} = \Big(\tfrac{1}{N}\rho(Q)(1-\rho(Q))\Big)^{1/2} \le N^{-1/2}\sqrt{\rho(Q)}. \tag{D.14}$$

Summing over the $2^{d\ell}$ cubes of $\mathcal{P}_\ell$ and using Cauchy–Schwarz, $\sum_{Q\in\mathcal{P}_\ell}\sqrt{\rho(Q)} \le \big(2^{d\ell}\sum_Q \rho(Q)\big)^{1/2} = 2^{d\ell/2}$, together with the trivial bound $\sum_Q |\widehat{\rho}_N(Q) - \rho(Q)| \le 2$,

$$\sum_{Q\in\mathcal{P}_\ell} \mathbb{E}|\widehat{\rho}_N(Q) - \rho(Q)| \;\le\; \min\big(2,\; 2^{d\ell/2}N^{-1/2}\big).$$

Split the ℓ-sum at the crossover level $L_\star$ with $2^{dL_\star/2} = N^{1/2}$, i.e. $2^{L_\star} = N^{1/d}$:

$$\mathbb{E}\big[\mathcal{W}_2(\widehat{\rho}_N, \rho)^2\big] \;\le\; \frac{3d}{2}\Big(N^{-1/2}\sum_{1\le\ell\le L_\star} 2^{\ell(d/2-2)} \;+\; 2\sum_{\ell > L_\star} 4^{-\ell}\Big).$$

The tail is $\le \frac{2}{3}\cdot 4^{-L_\star} = \frac{2}{3}N^{-2/d}$. For the head: if $d < 4$ the ratio $2^{d/2-2} < 1$ and it is $O(1)$, giving $C_d N^{-1/2}$ (and $N^{-2/d} \le N^{-1/2}$ since $2/d > 1/2$); if $d = 4$ the ratio is 1 and the head is $L_\star{+}1 = O(\log N)$, giving $CN^{-1/2}\log N$; if $d > 4$ the ratio exceeds 1 and the head is $\le C\,2^{L_\star(d/2-2)} = C\,N^{1/2-2/d}$, giving $CN^{-2/d}$, which dominates the tail. In every case the bound is $C_d\, s_N$, proving (D.12).

Part (2). Fix $R \ge 1$ and let $\tau_R(x) = x$ for $|x| \le R$ and $\tau_R(x) = Rx/|x|$ otherwise (radial projection onto $\overline{B}_R$). Write $\rho^R = (\tau_R)_\# \rho$, supported in $\overline{B}_R \subset [-R, R)^d$; then $\tau_R(X_1), \dots, \tau_R(X_N)$ are i.i.d. with law ρ^R and empirical measure $\widehat{\rho^R_N} := \frac{1}{N}\sum_i \delta_{\tau_R(X_i)}$. By the triangle inequality and $(a+b+c)^2 \le 3(a^2+b^2+c^2)$,

$$\mathcal{W}_2(\widehat{\rho}_N, \rho)^2 \le 3\Big(\mathcal{W}_2(\widehat{\rho}_N, \widehat{\rho^R_N})^2 + \mathcal{W}_2(\widehat{\rho^R_N}, \rho^R)^2 + \mathcal{W}_2(\rho^R, \rho)^2\Big).$$

Coupling each point to its projection bounds the truncation terms by the moment:

$$\mathcal{W}_2(\rho^R, \rho)^2 \le \int_{|x|>R} (|x| - R)^2 \rho(\mathrm{d}x) \le \int_{|x|>R} |x|^2\, \rho(\mathrm{d}x) \le R^{2-q} M_q,$$

using $|x|^2 \le R^{2-q}\,|x|^q$ on $\{|x| > R\}$ $(q > 2)$, and likewise $\mathbb{E}[\mathcal{W}_2(\widehat{\rho}_N, \widehat{\rho^R_N})^2] \le \mathbb{E}\big[\frac{1}{N}\sum_i |X_i - \tau_R(X_i)|^2\,\big] \le R^{2-q}M_q$. The middle term is case (1) on the cube of side $2R$: $\mathbb{E}[\mathcal{W}_2(\widehat{\rho^R_N}, \rho^R)^2] \le C_d(2R)^2 s_N$. Collecting,

$$\mathbb{E}\big[\mathcal{W}_2(\widehat{\rho}_N, \rho)^2\big] \le 6\, M_q\, R^{2-q} + 12\, C_d\, R^2 s_N,$$

and optimizing over $R \ge 1$ (set $R^q = M_q/(C_d s_N)$, admissible as $s_N \to 0$) gives $\mathbb{E}[\mathcal{W}_2^2] \le C_{d,q}(1 + M_q)\, s_N^{1-2/q}$, which is (D.13). □

Remark D.17 (The sharp unbounded rate)**.** The moment loss $s_N \rightsquigarrow s_N^{1-2/q}$ in (D.13) is an artefact of the single-radius truncation: replacing it by the dyadic shell decomposition $\mathbb{R}^d = \bigsqcup_k \{2^{k-1} \le |x| < 2^k\}$ and summing the cube bound (D.12) shell by shell (the mass of the k-th shell is $O(2^{-kq})$, beating its squared radius 4^k precisely because $q > 4$) recovers the additive form $\mathbb{E}[\mathcal{W}_2(\widehat{\rho}_N, \rho)^2] \le C_{d,q}(1 + M_q)\big(s_N + N^{-(q-2)/q}\big)$, whose extra term is *faster* than s_N whenever $q > 4$, so the clean rate s_N persists for any ρ with a finite moment of order > 4. We use only (D.12) and (D.13) in the body; the sharp constant and the shell bookkeeping are [69].

Remark D.18 (How the rate enters the book)**.** Theorem D.16 is the quantity r_N of remark 7.16, lemma 13.5 and theorem 13.13: there the propagation-of-chaos argument reduces the interaction error to $\mathbb{E}[\mathcal{W}_2(\widehat{\rho}_N, \rho)^2] \leq Cr_N^2$ with $r_N^2 = s_N$, so $r_N \leq CN^{-1/4}$ for $d \leq 3$ and $CN^{-1/d}$ for $d > 4$. The threshold $d = 4$ and the curse of dimension for $d > 4$ are inseparable from the geometric fact that resolving a measure to scale $2^{-\ell}$ needs $2^{d\ell}$ cells, each fluctuating by $N^{-1/2}$ — the trade-off optimized in the proof of part (1).

Appendix E

Viscosity Solutions: Deferred Proofs

This appendix is the in-book home for the viscosity-solution theory deferred from chapter 6. The single result the chapter relies on is the comparison principle (theorem 6.18), which selects the value function as the *unique* viscosity solution of the Hamilton–Jacobi–Bellman (HJB) equation and thereby makes the weak theory pay off. We prove it here in full. The argument has two regimes. For first-order equations — the degenerate case $\sigma \equiv 0$, where chapter 6 says viscosity solutions are genuinely indispensable — the proof is elementary and entirely self-contained (appendix E.2). For second-order equations the proof rests on the Crandall–Ishii "theorem on sums," which we state and prove via the sup-convolution method (appendix E.3), and then assemble into the comparison theorem (appendix E.4). We close with existence by Perron's method (appendix E.5). Throughout, the equation is written

$$F\big(x, u(x), \nabla u(x), \mathrm{D}^2 u(x)\big) = 0, \qquad x \in \mathbb{R}^d, \tag{E.1}$$

with $F\colon \mathbb{R}^d \times \mathbb{R} \times \mathbb{R}^d \times \mathcal{S}^d \to \mathbb{R}$ continuous and $\mathcal{S}^d$ the symmetric $d \times d$ matrices ordered by $X \preceq Y \iff Y - X \succeq 0$. For the stationary HJB equation of remark 6.16,

$$F(x, r, p, M) = \beta\, r - \tfrac{1}{2} \operatorname{tr}\big(\sigma(x)\sigma(x)^\top M\big) + H(x, p, m), \tag{E.2}$$

with the frozen measure m a fixed parameter suppressed below. The time-dependent equation (6.19) is handled by the same arguments with (t, x) in place of x, as indicated in remark E.15.

E.1 Structural hypotheses, jets, and the test-function reformulation

We isolate the structural properties of F that drive every comparison argument. They hold for (E.2) under the chapter's standing assumptions, as we check in lemma E.2.

Definition E.1 (Degenerate ellipticity and properness). F is *degenerate elliptic* if

$$F(x, r, p, M) \le F(x, r, p, N) \qquad \text{whenever } M \succeq N, \tag{E.3}$$

and *proper* if in addition it is nondecreasing in r: $F(x, r, p, M) \le F(x, s, p, M)$ whenever $r \le s$. We say F is *strictly proper with modulus* $\beta > 0$ if

$$F(x, s, p, M) - F(x, r, p, M) \;\ge\; \beta\,(s - r) \qquad \text{for } s \ge r. \tag{E.4}$$

The sign convention in (E.3) is exactly that of definition 6.17: because the second-order term enters (E.2) as $-\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top M)$ and $M \mapsto -\operatorname{tr}(\sigma\sigma^\top M)$ is nonincreasing in the positive-semidefinite (PSD) order, (E.2) is degenerate elliptic for every σ, including $\sigma \equiv 0$. Strict properness with modulus β is the discounting $\beta > 0$ of remark 6.16; a finite horizon with prescribed terminal data plays the same role through the time variable.

Lemma E.2 ((E.2) is proper and structurally continuous)**.** *Assume* (A-Lip)*,* (A-Growth) *on σ and on the data defining H, and $\beta > 0$. Then* (E.2) *is degenerate elliptic and strictly proper with modulus β, and there is a modulus of continuity ω_F (a nondecreasing $\omega_F\colon [0,\infty) \to [0,\infty)$ with $\omega_F(0^+) = 0$) such that for all $x, y \in \mathbb{R}^d$, $r \in \mathbb{R}$, $M \in \mathcal{S}^d$ and all $\alpha \ge 1$,*

$$F(y, r, \alpha(x-y), Y) - F(x, r, \alpha(x-y), X) \;\le\; \omega_F(\alpha\,|x-y|^2 + |x-y|\,) \tag{E.5}$$

whenever $X, Y \in \mathcal{S}^d$ satisfy the two-sided bound

$$-3\alpha \begin{pmatrix} \mathrm{Id} & 0 \\ 0 & \mathrm{Id} \end{pmatrix} \preceq \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \preceq 3\alpha \begin{pmatrix} \mathrm{Id} & -\mathrm{Id} \\ -\mathrm{Id} & \mathrm{Id} \end{pmatrix}. \tag{E.6}$$

Proof. Ellipticity and strict properness are read off (E.2) as above. For (E.5), write the difference as a Hamiltonian part and a diffusion part. The Hamiltonian part is $H(y, \alpha(x-y), m) - H(x, \alpha(x-y), m)$; under (A-Lip), (A-Growth) H is Lipschitz in x with a constant that grows at most linearly in $|p|$ (it is a supremum of $-b(x,a)\cdot p - L(x,a,m)$ over a, each term Lipschitz in x with p-linear constant by the Lipschitz drift and cost), so this part is bounded by $C\,|x-y|\,(1 + \alpha\,|x-y|) \le \omega_F(\alpha\,|x-y|^2 + |x-y|)$ for a suitable ω_F. The diffusion part is $\frac{1}{2}\operatorname{tr}\left(\sigma(x)\sigma(x)^\top X - \sigma(y)\sigma(y)^\top Y\right)$. Apply (E.6) to the vectors $(\sigma(x)e_i, \sigma(y)e_i)$ summed over an orthonormal basis $\{e_i\}$: the right inequality gives

$$\operatorname{tr}\left(\sigma(x)\sigma(x)^\top X - \sigma(y)\sigma(y)^\top Y\right) \le 3\alpha\,\operatorname{tr}\left((\sigma(x)-\sigma(y))(\sigma(x)-\sigma(y))^\top\right) \le 3\alpha\, L_\sigma^2\,|x-y|^2\,,$$

where L_σ is the Lipschitz constant of σ from (A-Lip). This is bounded by $\omega_F(\alpha\,|x-y|^2)$. Adding the two parts gives (E.5). ☐

The inequality (E.5) under the constraint (E.6) is the *only* way F's x-dependence enters the comparison proof; it is the precise form of the "structural condition" alluded to in chapter 6. We next record the equivalent jet formulation of viscosity solutions, which is what the theorem on sums speaks to.

Definition E.3 (Second-order jets)**.** For an upper-semicontinuous (USC) $u\colon \mathbb{R}^d \to \mathbb{R}$ and $x \in \mathbb{R}^d$, the *second-order superjet* is

$$J^{2,+}u(x) = \Big\{(p, M) \in \mathbb{R}^d \times \mathcal{S}^d : \; u(y) \le u(x) + p\cdot(y-x) + \tfrac{1}{2}(y-x)^\top M(y-x) + o(|y-x|^2) \text{ as } y \to x\Big\}.$$

For a lower-semicontinuous (LSC) v, the *subjet* $J^{2,-}v(x) = -J^{2,+}(-v)(x)$ is defined by the reverse inequality. The *closures* $\overline{J}^{2,\pm}$ collect limits $(x_n, p_n, M_n) \to (x, p, M)$ with $(p_n, M_n) \in J^{2,\pm}u(x_n)$ and $u(x_n) \to u(x)$.

Lemma E.4 (Test functions and jets coincide). *Let u be USC. Then $(p, M) \in J^{2,+}u(x)$ if and only if there is $\varphi \in C^2$ with $\nabla\varphi(x) = p$, $\mathrm{D}^2\varphi(x) = M$ and $u - \varphi$ attaining a local maximum at x. Consequently u is a viscosity subsolution of (E.1) in the sense of definition 6.17 if and only if $F(x, u(x), p, M) \leq 0$ for all x and all $(p, M) \in J^{2,+}u(x)$; symmetrically for supersolutions and $J^{2,-}$.*

Proof. If such a φ exists, Taylor-expanding φ at x and using $u(y) - \varphi(y) \leq u(x) - \varphi(x)$ gives the superjet inequality, so $(p, M) \in J^{2,+}u(x)$. Conversely, given $(p, M) \in J^{2,+}u(x)$, set $\rho(r) = \sup_{|y-x|\leq r} \left(u(y) - u(x) - p\cdot(y-x) - \frac{1}{2}(y-x)^\top M(y-x)\right)^+$; by definition of the superjet $\rho(r) = o(r^2)$, ρ is nondecreasing, and one builds a C^2 function ψ with $\psi(0) = 0$, $\psi(r) \geq \rho(r)$ and $\psi(r) = o(r^2)$ by integrating ρ twice (a standard construction). Then $\varphi(y) = u(x) + p\cdot(y-x) + \frac{1}{2}(y-x)^\top M(y-x) + \psi(|y-x|)$ is C^2, touches u from above at x with the prescribed first and second derivatives. The equivalence of the two formulations of sub-/supersolution is then immediate, since at a touching point of a C^2 test function the jet $(\nabla\varphi, \mathrm{D}^2\varphi)$ is exactly the data fed to F. □

E.2 The first-order comparison principle

We first treat $\sigma \equiv 0$, where (E.1) is the first-order equation $F(x, u, \nabla u) = \beta\, u + H(x, \nabla u) = 0$. No second-order machinery is needed, and the proof is the cleanest illustration of the doubling-of-variables device that organizes the whole theory.

Theorem E.5 (First-order comparison). *Let $F(x, r, p) = \beta\, r + H(x, p)$ with $\beta > 0$, H continuous and satisfying the modulus estimate*

$$|H(x,p) - H(y,p)| \leq \omega(\,|x-y|\,(1+|p|)) \tag{E.7}$$

for a modulus ω (the form (A-Lip)/(A-Growth) provide). Let $\overline{u}$ be a bounded USC viscosity subsolution and $\underline{u}$ a bounded LSC viscosity supersolution of $F = 0$. Then $\overline{u} \leq \underline{u}$ on $\mathbb{R}^d$.

Proof. Suppose for contradiction that $\sigma_0 := \sup_{\mathbb{R}^d}(\overline{u} - \underline{u}) > 0$. Fix $\varepsilon > 0$ and, to localize, a small $\eta > 0$, and consider the *doubled* function on $\mathbb{R}^d \times \mathbb{R}^d$,

$$\Phi_{\varepsilon,\eta}(x,y) = \overline{u}(x) - \underline{u}(y) - \frac{|x-y|^2}{2\varepsilon} - \eta(\,|x|^2 + |y|^2\,).$$

Because $\overline{u}, \underline{u}$ are bounded and USC/LSC, $\Phi_{\varepsilon,\eta}$ is USC and $\to -\infty$ as $|x| + |y| \to \infty$, so it attains a maximum at some $(x_\varepsilon, y_\varepsilon)$ (we suppress η). Write $M_\varepsilon = \Phi_{\varepsilon,\eta}(x_\varepsilon, y_\varepsilon)$. For η small enough (depending on σ_0 and the sup-bounds), $M_\varepsilon \geq \sigma_0/2 > 0$.

Step 1: the penalization forces $x_\varepsilon \to y_\varepsilon$. Comparing $2\Phi_{\varepsilon,\eta}(x_\varepsilon, y_\varepsilon)$ with $\Phi_{\varepsilon,\eta}(x_\varepsilon, x_\varepsilon) + \Phi_{\varepsilon,\eta}(y_\varepsilon, y_\varepsilon)$ (both $\leq 2M_\varepsilon$) and cancelling the η-terms gives the standard estimate

$$\frac{|x_\varepsilon - y_\varepsilon|^2}{\varepsilon} \leq \left(\overline{u}(x_\varepsilon) - \overline{u}(y_\varepsilon)\right) + \left(\underline{u}(x_\varepsilon) - \underline{u}(y_\varepsilon)\right) \leq C, \tag{E.8}$$

bounded by the oscillation of the bounded functions; hence $|x_\varepsilon - y_\varepsilon| \leq C\sqrt{\varepsilon} \to 0$, and moreover $|x_\varepsilon - y_\varepsilon|^2/\varepsilon \to 0$ along a subsequence (a lim inf argument from (E.8) using $\overline{u}(x_\varepsilon) - \overline{u}(y_\varepsilon) \to 0$, which follows from USC and the boundedness of the maximizers for fixed η).

Step 2: feed the optimal jets to F. Fix ε. The map $x \mapsto \overline{u}(x) - \big[\underline{u}(y_\varepsilon) + \frac{1}{2\varepsilon}|x-y_\varepsilon|^2 + \eta(|x|^2+|y_\varepsilon|^2)\big]$ has a maximum at x_ε, so the smooth function in brackets is a legitimate test function touching $\overline{u}$ from above; its gradient at x_ε is $p_\varepsilon := \frac{x_\varepsilon - y_\varepsilon}{\varepsilon} + 2\eta x_\varepsilon$. The subsolution inequality gives

$$\beta\,\overline{u}(x_\varepsilon) + H\big(x_\varepsilon,\, p_\varepsilon\big) \le 0.$$

Symmetrically, $y \mapsto \underline{u}(y) - \big[\overline{u}(x_\varepsilon) - \frac{1}{2\varepsilon}|x_\varepsilon - y|^2 - \eta(\dots)\big]$ has a minimum at y_ε, giving with $q_\varepsilon := \frac{x_\varepsilon - y_\varepsilon}{\varepsilon} - 2\eta y_\varepsilon$ the supersolution inequality

$$\beta\,\underline{u}(y_\varepsilon) + H\big(y_\varepsilon,\, q_\varepsilon\big) \ge 0.$$

Step 3: subtract. Subtracting,

$$\beta\big(\overline{u}(x_\varepsilon) - \underline{u}(y_\varepsilon)\big) \le H(y_\varepsilon, q_\varepsilon) - H(x_\varepsilon, p_\varepsilon).$$

The two momenta differ only by the η-terms: $p_\varepsilon - q_\varepsilon = 2\eta(x_\varepsilon + y_\varepsilon)$, which is $O(\eta)$ uniformly, and the common part is $\frac{x_\varepsilon - y_\varepsilon}{\varepsilon}$, of size $O(1/\sqrt{\varepsilon})$ but with $|x_\varepsilon - y_\varepsilon| \cdot |p_\varepsilon| = O(|x_\varepsilon - y_\varepsilon|^2/\varepsilon) \to 0$. Using continuity of H in p to absorb the $O(\eta)$ momentum gap and the modulus (E.7) for the x-shift,

$$H(y_\varepsilon, q_\varepsilon) - H(x_\varepsilon, p_\varepsilon) \le \omega\Big(|x_\varepsilon - y_\varepsilon|\,(1 + \tfrac{|x_\varepsilon - y_\varepsilon|}{\varepsilon})\Big) + o_\eta(1) = o_\varepsilon(1) + o_\eta(1),$$

by Step 1. Therefore $\beta\, M_\varepsilon \le \beta\big(\overline{u}(x_\varepsilon) - \underline{u}(y_\varepsilon)\big) + O(\eta) \le o_\varepsilon(1) + o_\eta(1)$. But $M_\varepsilon \ge \sigma_0/2$. Letting first $\varepsilon \downarrow 0$ then $\eta \downarrow 0$ gives $\beta\,\sigma_0/2 \le 0$, contradicting $\beta > 0$, $\sigma_0 > 0$. Hence $\sigma_0 \le 0$, i.e. $\overline{u} \le \underline{u}$. ☐

The mechanism is worth naming, because it recurs verbatim in the second-order case and, in chapter 12, on the space of measures. *Doubling* replaces the single bad point where a kink might live by two points $x_\varepsilon, y_\varepsilon$ and a smooth penalty $|x-y|^2/2\varepsilon$ that pins them together; the penalty's gradient is a legitimate *classical* momentum for each function separately, so the equation can be evaluated; the discount $\beta > 0$ supplies the strictly favourable sign that the vanishing x-shift cannot overcome. No regularity of $\overline{u}, \underline{u}$ beyond semicontinuity is used.

E.3 Sup-convolutions and the Crandall–Ishii theorem on sums

In the second-order case Step 2 above breaks down: the penalty $|x-y|^2/2\varepsilon$ is smooth, but to evaluate the equation we now also need *second-order* information (p, M) at the maximizing points, and a USC function has, a priori, no Hessian there. The resolution is the Crandall–Ishii lemma, which manufactures matrices X, Y in the *closed* jets satisfying a quantitative one-sided bound. Its proof rests on two classical facts about *sup-convolutions*, which regularize a USC function into a semiconvex one without destroying its jets.

Definition E.6 (Sup-convolution)**.** For USC bounded u and $\lambda > 0$, the *sup-convolution* is

$$u^\lambda(x) = \sup_{z \in \mathbb{R}^d} \Big\{ u(z) - \frac{|x-z|^2}{2\lambda} \Big\}.$$

The *inf-convolution* $u_\lambda = -(-u)^\lambda$ is defined dually.

Lemma E.7 (Properties of the sup-convolution)**.** *Let u be bounded and USC. Then:*

(i) $u^\lambda \geq u$, $u^\lambda \downarrow u$ *pointwise as* $\lambda \downarrow 0$, *and* u^λ *is finite and Lipschitz on bounded sets.*

(ii) u^λ *is* $\frac{1}{\lambda}$-semiconvex*:* $x \mapsto u^\lambda(x) + \frac{1}{2\lambda}|x|^2$ *is convex.*

(iii) (Jets propagate.) *If* $(p, M) \in J^{2,+}u^\lambda(x)$ *then* $(p, M) \in J^{2,+}u(z)$ *for* $z = x + \lambda p$ *(the point attaining the sup), and* $u^\lambda(x) = u(z) - \frac{1}{2\lambda}|x - z|^2$ *with* $M \preceq \frac{1}{\lambda}\,\mathrm{Id}$.

Proof. (i) Taking $z = x$ gives $u^\lambda \geq u$; as $\lambda \downarrow 0$ the penalty forces the optimal $z \to x$ and USC gives $u^\lambda(x) \downarrow u(x)$. Boundedness of u makes the sup finite and attained (the objective is USC in z and $\to -\infty$), and $u^\lambda(x) = \sup_z \big(u(z) - \frac{1}{2\lambda}|z|^2\big) + \frac{1}{\lambda}x \cdot (\cdot) - \dots$ exhibits u^λ as a sup of functions that are uniformly Lipschitz on bounded sets, hence Lipschitz there. (ii) Writing $u^\lambda(x) + \frac{1}{2\lambda}|x|^2 = \sup_z \big(u(z) - \frac{1}{2\lambda}|z|^2 + \frac{1}{\lambda}x \cdot z\big)$ exhibits the left side as a supremum of *affine* functions of x, hence convex. (iii) Let z attain the sup defining $u^\lambda(x)$, so $u^\lambda(x) = u(z) - \frac{1}{2\lambda}|x - z|^2$ and $z = x + \lambda p_0$ for the first-order condition $p_0 = \frac{z-x}{\lambda} \in J^{1,+}$. If $(p, M) \in J^{2,+}u^\lambda(x)$, the touching test function φ from lemma E.4 satisfies $u^\lambda(\cdot) \leq \varphi$ near x with equality at x. For any w, $u(w) - \frac{1}{2\lambda}|(x + z - w) - z|^2 \leq u^\lambda(x + z - w) \leq \varphi(x + z - w)$ with equality at $w = z$, exhibiting $w \mapsto \varphi(x + z - w) + \frac{1}{2\lambda}|w - z|^2$ as a C^2 function touching u from above at z; its jet there is (p, M) since the quadratic penalty is critical with vanishing gradient and Hessian shift handled by the semiconvexity bound, giving $(p, M) \in J^{2,+}u(z)$ and $p = p_0$, $z = x + \lambda p$. The bound $M \preceq \frac{1}{\lambda}\,\mathrm{Id}$ is the semiconvexity (ii). □

The decisive analytic input is that a semiconvex function is twice differentiable almost everywhere. This is Alexandrov's theorem; its first-order shadow (a.e. differentiability of convex functions) is lemma D.4, and the second-order statement is the following.

Theorem E.8 (Alexandrov)**.** *A convex function* $\phi\colon \mathbb{R}^d \to \mathbb{R}$ *is twice differentiable at Lebesgue-almost every x: for a.e. x there are* $p \in \mathbb{R}^d$ *and* $M \in \mathcal{S}^d$, $M \succeq 0$, *with* $\phi(y) = \phi(x) + p \cdot (y - x) + \frac{1}{2}(y - x)^\top M(y - x) + o(|y - x|^2)$. *Consequently a* $\frac{1}{\lambda}$*-semiconvex function is twice differentiable a.e.*

Proof (reduction). By lemma D.4 the gradient $\nabla\phi$ exists a.e. and, being the gradient of a convex function, is a monotone map: $(\nabla\phi(x) - \nabla\phi(y)) \cdot (x - y) \geq 0$. A monotone map on $\mathbb{R}^d$ has bounded variation on bounded sets, hence by the Lebesgue–Besicovitch differentiation theorem is approximately differentiable a.e.; at a point of approximate differentiability of $\nabla\phi$ that is also a Lebesgue point, a second-order Taylor expansion with PSD Hessian $M =$(the approximate derivative of $\nabla\phi$, symmetric because it is a gradient) holds with $o(|y - x|^2)$ remainder. The details are the classical argument of [65, Sec. 6.4]; every ingredient (Rademacher for convex functions, monotonicity, Lebesgue differentiation of BV maps) is at or below the measure-theoretic level developed in chapters 1 and 3 and appendix D. A semiconvex u^λ equals a convex function minus $\frac{1}{2\lambda}|x|^2$, so it inherits a.e. twice differentiability. □

Lemma E.9 (Jensen's lemma)**.** *Let ϕ be semiconvex and let x_0 be a strict local maximum of ϕ. Then for every $\delta > 0$ the set*

$$K_\delta = \Big\{x : \ \exists\, q \in \mathbb{R}^d,\ |q| \leq \delta,\ y \mapsto \phi(y) - q \cdot y \textit{ has a local max at } x,\ |x - x_0| \leq \delta\Big\}$$

has positive Lebesgue measure. In particular, by theorem E.8, ϕ is twice differentiable at some point of K_δ, where its (PSD-perturbed) Hessian is $\preceq 0$ up to $o(1)$ as $\delta \downarrow 0$.

Proof. Semiconvexity gives $\phi(x) + \frac{C}{2}|x|^2$ convex for some C; near the strict maximum x_0 the perturbed function $\phi(y) - q \cdot y$ still attains an interior maximum for $|q|$ small, and the map $q \mapsto x(q)$ (the maximizer) covers a set of positive measure as q ranges over $\{|q| \le \delta\}$, because its inverse is the (Lipschitz, by semiconvexity) subgradient map $x \mapsto \partial\phi(x)$, whose image of K_δ contains the ball $\{|q| \le \delta\}$ of positive measure and which cannot increase measure to infinity. This is the standard covering argument of [52, Lem. A.3]. At a twice-differentiability point of K_δ the second-order condition for an unconstrained maximum gives $\mathrm{D}^2\,\phi \preceq 0$. □

Theorem E.10 (Crandall–Ishii, theorem on sums)**.** *Let $\overline{u}$ be USC and $\underline{u}$ LSC on $\mathbb{R}^d$, let $\varphi \in C^2(\mathbb{R}^d \times \mathbb{R}^d)$, and suppose (x_0, y_0) is a local maximum of $(x,y) \mapsto \overline{u}(x) - \underline{u}(y) - \varphi(x,y)$. Then for each $\mu > 0$ there exist $X, Y \in \mathcal{S}^d$ with*

$$(\nabla_x\varphi(x_0,y_0),\, X) \in \overline{J}^{2,+}\overline{u}(x_0), \qquad (-\nabla_y\varphi(x_0,y_0),\, Y) \in \overline{J}^{2,-}\underline{u}(y_0),$$

and the matrix bound

$$-\Big(\tfrac{1}{\mu} + \|A\|\Big)\,\mathrm{Id} \preceq \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \preceq A + \mu A^2, \qquad A = \mathrm{D}^2\,\varphi(x_0,y_0). \tag{E.9}$$

Proof (sketch with all mechanisms). Replace $\overline{u}, \underline{u}$ by their sup-/inf-convolutions $\overline{u}^\lambda, \underline{u}_\lambda$, which by lemma E.7 are semiconvex/ semiconcave and whose jets descend to those of $\overline{u}, \underline{u}$ (parts (iii)). The doubled function $\Psi(x,y) = \overline{u}^\lambda(x) - \underline{u}_\lambda(y) - \varphi(x,y)$ is semiconvex and has a maximum near (x_0, y_0). Apply Jensen's lemma lemma E.9 to Ψ: it is twice differentiable at points arbitrarily close to the maximizer, where the second-order maximum condition reads $\mathrm{D}^2\,\Psi \preceq 0$, i.e.

$$\begin{pmatrix} \mathrm{D}^2\,\overline{u}^\lambda(x) & 0 \\ 0 & -\mathrm{D}^2\,\underline{u}_\lambda(y) \end{pmatrix} \preceq \mathrm{D}^2\,\varphi(x,y) = A + o(1).$$

Setting $X = \mathrm{D}^2\,\overline{u}^\lambda(x)$, $Y = \mathrm{D}^2\,\underline{u}_\lambda(y)$ at such a point and passing to the limit (first in the Jensen approximation, then $\lambda \downarrow 0$, using the closedness of $\overline{J}^{2,\pm}$) produces matrices in the closed jets of $\overline{u}, \underline{u}$. The lower bound on $X, -Y$ is the semiconvexity bound $\mathrm{D}^2\,\overline{u}^\lambda \succeq -\frac{1}{\lambda}\,\mathrm{Id}$ of lemma E.7(ii), and the quadratic refinement $A + \mu A^2$ comes from optimizing the regularization scale λ against μ; this last algebraic step is the content of [52, Thm. 3.2], which we follow. The two ingredients that make it work — sup-convolution to gain semiconvexity, and Alexandrov/Jensen to extract an a.e. Hessian respecting the maximum condition — are exactly lemmas E.7 and E.9 and theorem E.8. □

Remark E.11. We have proved the structural lemmas (lemmas E.7 and E.9) in full and reduced Alexandrov's theorem (theorem E.8) to the measure-theoretic differentiation theory the book already owns; the final matrix-optimization step of theorem E.10 is algebraic and we attribute it to the User's Guide rather than transcribe it. This is the one place the appendix leans on an external write-up for a computation rather than a concept, and it changes no hypothesis the book uses.

E.4 The second-order comparison principle

We now prove the result chapter 6 defers: the comparison principle for the second-order HJB equation, which yields uniqueness of the viscosity solution and identifies it with the value function. Comparison delivers *uniqueness*; but uniqueness is worth having only once we know the value function is *a* viscosity solution to begin with. When the nondegeneracy (A-Nondeg) holds, (E.2) has a classical $C^{1,2}$ solution (theorem 6.14) and the verification theorem theorem 6.10 identifies it with u, so no weak notion is needed. The reason viscosity theory is here at all is the *complementary* regime — degenerate or absent diffusion — where u is in general only continuous and the $C^{1,2}$ hypothesis of verification cannot be met. There one must show *directly*, from the definition, that the merely-continuous u solves (E.2) in the viscosity sense; this is the dynamic-programming verification that theorem 6.18 flagged and [68, Ch. II] carries out in the literature. We give it an in-book home now. It rests on one genuinely substantial input — the dynamic programming principle in its stochastic, random-endpoint form — followed by a short and completely standard implication, which we prove in full.

The deterministic-endpoint DPP was proved in-book as theorem 6.4. Its stochastic counterpart, in which the leg endpoint X_θ is a random variable and the ε-optimal tail control must be selected measurably along sample paths, needs the measurable-selection apparatus that remark 6.6 deliberately set aside. We import it, stated precisely with full hypotheses, exactly as that remark promised.

Imported result E.12 (Stochastic dynamic programming principle; [136, Ch. III], [68, Ch. I–II])**.** (Proved in the cited literature, not in this book; imported as the substantial input to proposition E.13 below.) *Assume* (A-Lip) *and* (A-Growth) *on b, σ and on $L(\cdot,\cdot,m)$ at the frozen measure m, and $\beta > 0$. On a fixed filtered probability space carrying a d-dimensional Brownian motion W, let $X^{x,\alpha}$ solve the controlled state SDE*

$$\mathrm{d}X_s = b(X_s, \alpha_s)\ \mathrm{d}s + \sigma(X_s)\ \mathrm{d}W_s, \qquad X_0 = x,$$

over the admissible controls $\alpha \in \mathcal{U}$ — progressively measurable A-valued processes for which the SDE is well posed and the cost below is finite, as in definition 6.9 — and let

$$u(x) = \inf_{\alpha\in\mathcal{U}} \mathbb{E}[*] \int_0^\infty e^{-\beta s}\, L\big(X_s^{x,\alpha}, \alpha_s, m\big)\ \mathrm{d}s \tag{E.10}$$

be the discounted value of remark 6.16. Then u is bounded and continuous on $\mathbb{R}^d$, and for every $x \in \mathbb{R}^d$ and every almost-surely finite stopping time θ — in particular every deterministic $\theta = h \geq 0$, and, within the infimum, every control-dependent exit time of a bounded domain —

$$u(x) = \inf_{\alpha\in\mathcal{U}} \mathbb{E}[*] \int_0^\theta e^{-\beta s}\, L\big(X_s^{x,\alpha}, \alpha_s, m\big)\ \mathrm{d}s + e^{-\beta\theta}\, u\big(X_\theta^{x,\alpha}\big). \tag{E.11}$$

The finite-horizon value of definition 6.3 satisfies the same identity with the integral over $[t, t+h]$, the discount factor $e^{-\beta s}$ removed, and $u(t+h, X_{t+h})$ in place of $e^{-\beta\theta}u(X_\theta)$.

The deterministic restriction of (E.11) is precisely eq. (6.5); the only genuinely new content is the random-endpoint selection, which is why we attribute it rather than reprove it. Everything downstream is elementary.

Proposition E.13 (The value function is a viscosity solution)**.** *Under* (A-Lip)*,* (A-Growth) *and* $\beta > 0$ *(or a finite horizon with prescribed terminal data* $u(T,\cdot) = g$*), the value function* u *of* (E.10) *is a viscosity solution, in the sense of definition 6.17, of the stationary HJB equation* (E.2)*. In the finite-horizon case the value function of definition 6.3 is a viscosity solution of the time-dependent equation* (6.19)*.*

Proof. Write the controlled generator $\mathcal{A}^a\varphi(x) = b(x,a)\cdot\nabla\varphi(x) + \frac{1}{2}\operatorname{tr}\big(\sigma(x)\sigma(x)^\top \mathrm{D}^2\varphi(x)\big)$ of section 6.5. Since $H(x,p,m) = \sup_a\{-b(x,a)\cdot p - L(x,a,m)\} = -\inf_a\{b(x,a)\cdot p + L(x,a,m)\}$ by (6.18), and the second-order term $-\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top M)$ of (E.2) carries no a and so joins the infimum, the operator F of (E.2) evaluated on a test function reads

$$F\big(x,r,\nabla\varphi(x),\mathrm{D}^2\varphi(x)\big) = \beta\, r - \inf_{a\in A}\big\{\mathcal{A}^a\varphi(x) + L(x,a,m)\big\}. \tag{E.12}$$

Fix $\varphi\in C^2$ and an interior point x_0, and normalize $\varphi(x_0) = u(x_0)$ (adding a constant to φ alters neither the touching condition nor (E.12)). Fix $r > 0$ so small that the relevant one-sided bound holds on $\overline{B_r(x_0)}$, and for a control α set $\tau = \inf\{s\geq 0 : X_s^{x_0,\alpha}\notin B_r(x_0)\}$; the diffusion starts at the centre, so $\tau > 0$ almost surely and $\tau\wedge h = h$ for all sufficiently small h. For any $\psi\in C^2$, Itô's formula theorem 5.9 applied to $s\mapsto e^{-\beta s}\psi(X_s^{x_0,\alpha})$ and stopped at $\tau\wedge h$ gives, the stochastic integral being a martingale up to the bounded stopping time $\tau\wedge h$ and hence of zero mean,

$$\mathbb{E}[*]\, e^{-\beta(\tau\wedge h)}\psi(X_{\tau\wedge h}) = \psi(x_0) + \mathbb{E}[*]\int_0^{\tau\wedge h} e^{-\beta s}\big(\mathcal{A}^{\alpha_s}\psi - \beta\,\psi\big)(X_s)\ \mathrm{d}s. \tag{E.13}$$

Subsolution inequality. Let x_0 be a local maximum of $u-\varphi$, so that $u\leq\varphi$ on $B_r(x_0)$. Apply the easy half "$\leq$" of (E.11) — the value is an infimum, so any single competitor overestimates it — with the *constant* control $\alpha\equiv a$ and the stopping time $\tau\wedge h$:

$$u(x_0) \leq \mathbb{E}[*]\int_0^{\tau\wedge h} e^{-\beta s}L(X_s,a,m)\ \mathrm{d}s + e^{-\beta(\tau\wedge h)}u(X_{\tau\wedge h}).$$

Since $X_{\tau\wedge h}\in\overline{B_r(x_0)}$ and $u\leq\varphi$ there, replacing u by the larger φ on the right preserves "$\leq$". Subtracting (E.13) with $\psi=\varphi$, $\alpha\equiv a$, and cancelling the equal terms $u(x_0)=\varphi(x_0)$,

$$0 \leq \mathbb{E}[*]\int_0^{\tau\wedge h} e^{-\beta s}\big(L(X_s,a,m) + \mathcal{A}^a\varphi(X_s) - \beta\,\varphi(X_s)\big)\ \mathrm{d}s.$$

Divide by h and let $h\downarrow 0$: eventually $\tau\wedge h = h$, the paths satisfy $X_s\to x_0$, and the bracketed integrand is continuous, so the time-average converges to its value at $s=0$,

$$0 \leq L(x_0,a,m) + \mathcal{A}^a\varphi(x_0) - \beta\,\varphi(x_0),\quad \text{i.e.}\quad \beta\,\varphi(x_0) \leq \mathcal{A}^a\varphi(x_0) + L(x_0,a,m).$$

This holds for every $a\in A$, hence for the infimum over a. By (E.12) with $r = u(x_0) = \varphi(x_0)$ that is exactly $F\big(x_0,u(x_0),\nabla\varphi(x_0),\mathrm{D}^2\varphi(x_0)\big)\leq 0$, the subsolution inequality of definition 6.17.

Supersolution inequality. Let x_0 be a local minimum of $u-\varphi$, so $u\geq\varphi$ on $B_r(x_0)$. A single constant control no longer suffices; we use the hard half "$\geq$" of (E.11), that the infimum is genuinely approached. Fix $h>0$; there is an admissible $\alpha=\alpha^h$ with

$$u(x_0) + h^2 \geq \mathbb{E}[*]\int_0^{\tau\wedge h} e^{-\beta s}L(X_s,\alpha_s,m)\ \mathrm{d}s + e^{-\beta(\tau\wedge h)}u(X_{\tau\wedge h}).$$

On $\overline{B_r(x_0)}$, $u \geq \varphi$, so replacing u by the smaller φ on the right keeps "$\geq$". Subtracting (E.13) with $\psi = \varphi$, control α^h, and using $\varphi(x_0) = u(x_0)$,

$$h^2 \geq \mathbb{E}[*] \int_0^{\tau \wedge h} e^{-\beta s} \big(L(X_s, \alpha_s, m) + \mathcal{A}^{\alpha_s} \varphi(X_s) - \beta\, \varphi(X_s)\big)\, \mathrm{d}s.$$

Pointwise $L(x,a,m) + \mathcal{A}^a \varphi(x) \geq \inf_{a'}\{L(x,a',m) + \mathcal{A}^{a'} \varphi(x)\} = \frac{1}{2} \operatorname{tr}\big(\sigma(x)\sigma(x)^\top \mathrm{D}^2 \varphi(x)\big) - H(x, \nabla\varphi(x), m) =: G(x)$, a continuous function of x, so the integrand may be lowered to $e^{-\beta s}(G - \beta\,\varphi)(X_s)$:

$$h^2 \geq \mathbb{E}[*] \int_0^{\tau \wedge h} e^{-\beta s} \big(G(X_s) - \beta\, \varphi(X_s)\big)\, \mathrm{d}s.$$

Divide by h and let $h \downarrow 0$ ($\tau \wedge h = h$ eventually, $X_s \to x_0$): the left side tends to 0 and the right side to $G(x_0) - \beta\,\varphi(x_0)$, whence $\beta\,\varphi(x_0) \geq G(x_0) = \inf_a\{\mathcal{A}^a \varphi(x_0) + L(x_0, a, m)\}$. By (E.12) this is $F(x_0, u(x_0), \nabla\varphi(x_0), \mathrm{D}^2 \varphi(x_0)) \geq 0$, the supersolution inequality.

Both inequalities hold at every interior touching point, so u is a viscosity solution of (E.2). The finite-horizon claim is the same argument run on $[0,T] \times \mathbb{R}^d$ with a test function $\varphi \in C^{1,2}$, the pair (t,x) for x, and the parabolic operator $-\partial_t \varphi - \frac{1}{2}\operatorname{tr}(\sigma\sigma^\top \mathrm{D}^2 \varphi) + H(x, \nabla\varphi, m)$ in place of F: the Dynkin identity (E.13) drops the $e^{-\beta s}$ weight and gains the $\partial_t \varphi$ term, the finite-horizon DPP of imported result E.12 replaces (E.11), and the terminal datum $u(T,\cdot) = g$ holds classically by definition 6.3. This is the sense in which the value function of definition 6.3 is a viscosity solution. □

Theorem E.14 (Comparison and uniqueness; theorem 6.18)**.** *Let F be given by* (E.2) *under* (A-Lip), (A-Growth) *and* $\beta > 0$ *(or a finite horizon with prescribed terminal data). Let $\overline{u}$ be a bounded USC viscosity subsolution and $\underline{u}$ a bounded LSC viscosity supersolution of $F = 0$. Then $\overline{u} \leq \underline{u}$ on $\mathbb{R}^d$. Consequently* (E.1) *has at most one bounded continuous viscosity solution, and — since the value function of definition 6.3 is a viscosity solution (theorem 6.10 for the smooth case; proposition E.13 for the general case, the classical dynamic-programming verification of [136, Ch. III] and [68, Ch. II]) — that solution is the value function.*

Proof. Assume $\sigma_0 := \sup(\overline{u} - \underline{u}) > 0$. Double the variables with the same penalty and localization as in theorem E.5, maximizing

$$\Phi(x,y) = \overline{u}(x) - \underline{u}(y) - \frac{|x-y|^2}{2\varepsilon} - \eta(|x|^2 + |y|^2)$$

at $(x_\varepsilon, y_\varepsilon)$, with the penalty estimate (E.8) giving $|x_\varepsilon - y_\varepsilon|^2/\varepsilon \to 0$ and $M_\varepsilon \geq \sigma_0/2$ for η small. Here $\varphi(x,y) = \frac{1}{2\varepsilon}|x-y|^2 + \eta(|x|^2 + |y|^2)$, with

$$\nabla_x \varphi = \tfrac{x-y}{\varepsilon} + 2\eta x =: p_\varepsilon, \quad -\nabla_y \varphi = \tfrac{x-y}{\varepsilon} - 2\eta y =: q_\varepsilon, \quad A = \mathrm{D}^2 \varphi = \frac{1}{\varepsilon}\begin{pmatrix} \mathrm{Id} & -\mathrm{Id} \\ -\mathrm{Id} & \mathrm{Id} \end{pmatrix} + 2\eta\, \mathrm{Id}\,.$$

Apply the theorem on sums theorem E.10 with $\mu = \varepsilon$: there are $X, Y \in \mathcal{S}^d$ with $(p_\varepsilon, X) \in \overline{J}^{2,+}\overline{u}(x_\varepsilon)$, $(q_\varepsilon, Y) \in \overline{J}^{2,-}\underline{u}(y_\varepsilon)$, and, computing $A + \varepsilon A^2$ for the block matrix above,

$$\begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \preceq \frac{3}{\varepsilon}\begin{pmatrix} \mathrm{Id} & -\mathrm{Id} \\ -\mathrm{Id} & \mathrm{Id} \end{pmatrix} + C\eta\ \mathrm{Id},$$

which is (E.6) with $\alpha = 1/\varepsilon$ up to the $O(\eta)$ term. The sub-/supersolution inequalities at the closed jets give

$$F(x_\varepsilon, \overline{u}(x_\varepsilon), p_\varepsilon, X) \le 0, \qquad F(y_\varepsilon, \underline{u}(y_\varepsilon), q_\varepsilon, Y) \ge 0.$$

Subtract, and use strict properness (E.4) on the r-variable: writing $p_\varepsilon \approx q_\varepsilon \approx \alpha(x_\varepsilon - y_\varepsilon)$ up to $O(\eta)$,

$$\beta\big(\overline{u}(x_\varepsilon) - \underline{u}(y_\varepsilon)\big) \le F(y_\varepsilon, \overline{u}(x_\varepsilon), q_\varepsilon, Y) - F(x_\varepsilon, \overline{u}(x_\varepsilon), p_\varepsilon, X) + O(\eta).$$

The right side is exactly the structural difference (E.5) controlled by lemma E.2, since X, Y obey (E.6): it is bounded by $\omega_F\big(\frac{|x_\varepsilon - y_\varepsilon|^2}{\varepsilon} + |x_\varepsilon - y_\varepsilon|\big) + O(\eta) = o_\varepsilon(1) + O(\eta)$. Hence $\beta\, M_\varepsilon \le o_\varepsilon(1) + O(\eta)$, contradicting $M_\varepsilon \ge \sigma_0/2 > 0$ on letting $\varepsilon \downarrow 0$ then $\eta \downarrow 0$. Therefore $\overline{u} \le \underline{u}$. Uniqueness follows by applying the inequality to two solutions in both orders. □

Remark E.15 (The parabolic (finite-horizon) case)**.** For the time-dependent equation (6.19) on $[0, T] \times \mathbb{R}^d$ with terminal data $u(T, \cdot) = g$, the same proof runs with the pair (t, x) doubled, an extra penalty $|t - s|^2 / 2\varepsilon$ in time, and the role of strict properness played by the terminal condition together with a standard $\frac{c}{T-t}$ barrier (or an $e^{\lambda t}$ change of unknown that manufactures a β-like term). The structural lemma and the theorem on sums apply verbatim to the augmented variable; see [68, Ch. V]. The discounted stationary case above already exhibits every mechanism.

E.5 Existence by Perron's method

Comparison also delivers existence, via Perron's method, closing the loop that a unique viscosity solution exists and is the value function.

Theorem E.16 (Perron)**.** *Suppose comparison theorem E.14 holds for F, and suppose there exist a bounded subsolution $\underline{w}$ and a bounded supersolution $\overline{w}$ with $\underline{w} \le \overline{w}$. Then*

$$u(x) = \sup\{w(x) : \ \underline{w} \le w \le \overline{w},\ w \ \textit{a subsolution}\}$$

is the unique bounded continuous viscosity solution of (E.1)*.*

Proof (sketch). The USC envelope u^* is a subsolution (subsolutions are closed under suprema and USC envelopes, by the jet definition), so $u^* = u$ and u is a subsolution. If the LSC envelope u_* failed to be a supersolution at some point, a standard "bump" construction would produce a strictly larger subsolution below $\overline{w}$, contradicting maximality; hence u_* is a supersolution. Comparison gives $u^* \le u_*$, so $u^* = u_* = u$ is continuous and is a viscosity solution. Uniqueness is comparison again. The bump construction is detailed in [52, Sec. 4]; for the HJB equation (E.2) the barriers $\underline{w}, \overline{w}$ are built from constants $\mp C/\beta$ using the boundedness of $H(\cdot, 0)$. □

This completes the in-book development of the comparison principle and the surrounding viscosity theory used in chapter 6 and taken as the template, on the space of measures, in chapter 12.

E.6 Gradient and Schauder estimates for the HJB equation

The existence theory of chapter 10 rests on two quantitative regularity estimates for the *decoupled* (frozen-flow) HJB equation

$$-\partial_t u - \tfrac{1}{2}\,\mathrm{tr}\big(\sigma(x)\sigma(x)^\top \mathrm{D}^2 u\big) + H(x,\nabla u) \;=\; F[m_t](x), \qquad u(T,\cdot) = G[m_T](\cdot), \tag{E.14}$$

on $[0,T]\times\mathbb{R}^d$, with σ uniformly nondegenerate (A-Nondeg). The first is the uniform Lipschitz bound on the value that produces the bounded-feedback assumption assumption 10.4; the second is the interior Schauder estimate that makes the best-response map continuous (lemma 10.10). We prove the gradient bound in full by a stochastic-control (synchronous-coupling) argument — arguably cleaner than the classical Bernstein computation, to which we relate it — and reduce the Schauder estimate to the linear parabolic theory already established in theorem C.12.

Theorem E.17 (Uniform Lipschitz/gradient bound for the value)**.** *Assume* (A-Lip), (A-Growth), (A-Nondeg), (A-Conv)*, that the running and terminal couplings $F[\mu](\cdot)$, $G[\mu](\cdot)$ are bounded and Lipschitz in the state uniformly in the measure argument μ, and that H is the convex Legendre transform of a Lipschitz running cost. Let u^m be the value function of the control problem associated with* (E.14) *for a frozen flow m. Then for all $t\in[0,T]$,*

$$\big\|\nabla_x u^m(t,\cdot)\big\|_\infty \;\le\; e^{\Lambda(T-t)}\Big(\big\|\nabla_x G[m_T]\big\|_\infty + \int_t^T \big\|\nabla_x F[m_s]\big\|_\infty\,\mathrm{d}s\Big) \;=:\; L_u, \tag{E.15}$$

where Λ is the Lipschitz constant of the drift $b(x,a) = -\partial_p H(x,\cdot)$ in the state from (A-Lip)*. In particular L_u is independent of the frozen flow m, since F and G enter only through their measure-uniform Lipschitz constants.*

Proof. By the verification theorem theorem 6.10 the value has the stochastic-control representation

$$u^m(t,x) = \inf_{\alpha\in\mathcal{U}} \mathbb{E}[*] \int_t^T \big(L(X_s,\alpha_s) + F[m_s](X_s)\big)\,\mathrm{d}s + G[m_T](X_T) \;\Big|\; X_t = x,$$

the infimum over admissible controls of the state SDE $\mathrm{d}X_s = b(X_s,\alpha_s)\,\mathrm{d}s + \sigma(X_s)\,\mathrm{d}W_s$. Fix $x,x'\in\mathbb{R}^d$ and an admissible control α. Drive the two trajectories X, X' started at x,x' by the *same* control α and the *same* Brownian motion W (synchronous coupling). Setting $D_s = X_s - X'_s$, the control terms in the drift cancel and

$$D_s = (x-x') + \int_t^s \big(b(X_r,\alpha_r) - b(X'_r,\alpha_r)\big)\,\mathrm{d}r + \int_t^s \big(\sigma(X_r) - \sigma(X'_r)\big)\,\mathrm{d}W_r.$$

Using $\mathrm{Lip}_x b \le \Lambda$, $\mathrm{Lip}\,\sigma \le L_\sigma$, the Burkholder–Davis–Gundy inequality theorem C.3, and Jensen,

$$\mathbb{E}[*]\sup_{r\le s}|D_r|^2 \le 3|x-x'|^2 + C\int_t^s \mathbb{E}[*]\sup_{\rho\le r}|D_\rho|^2\,\mathrm{d}r,$$

so Grönwall's inequality lemma C.1 gives $\mathbb{E}\Big[\sup_{r\le s}|D_r|^2\Big] \le 3|x-x'|^2 e^{C(s-t)}$, and in particular $\mathbb{E}[|X_s - X'_s|] \le |x-x'|\,e^{\Lambda(s-t)}$ after optimising the constant (the linear growth in $|x-x'|$ is what matters; for the sharp exponent one tracks the first-moment estimate directly, $\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[|]\,D_s| \le \Lambda\mathbb{E}[|]\,D_s|$, valid because the martingale term is mean-zero and the drift difference is bounded by $\Lambda|D_s|$). Now, for the *same* control,

$$|u^{m,\alpha}(t,x) - u^{m,\alpha}(t,x')| \le \int_t^T \mathrm{Lip}_x F[m_s]\, \mathbb{E}[|]\, X_s - X'_s|\ \mathrm{d}s + \mathrm{Lip}_x G[m_T]\, \mathbb{E}[|]\, X_T - X'_T|,$$

because the running cost $L(X_s, \alpha_s)$ is common to the two trajectories *up to* the state dependence, which here we have arranged to enter only through F, G (the control cost is identical; any state-Lipschitz part of L contributes a further $\int_t^T \mathrm{Lip}_x L\, \mathbb{E}[|]\, D_s|\, \mathrm{d}s$ term, absorbed into the same bound). Taking the infimum over α on both sides and using $\mathbb{E}[|]\, X_s - X'_s| \le |x - x'|e^{\Lambda(s-t)}$,

$$|u^m(t,x) - u^m(t,x')| \le |x - x'|\Big(\mathrm{Lip}_x G[m_T]\, e^{\Lambda(T-t)} + \int_t^T \mathrm{Lip}_x F[m_s]\, e^{\Lambda(s-t)}\, \mathrm{d}s\Big),$$

which is (E.15) after dividing by $|x - x'|$ and letting $x' \to x$ (the value is differentiable in x by the $C^{1,2}$ regularity of theorem C.12 under (A-Nondeg)). The bound depends on m only through the measure-uniform constants $\mathrm{Lip}_x F$, $\mathrm{Lip}_x G$, hence is uniform in m. □

Remark E.18 (Relation to the Bernstein method)**.** The PDE-side proof differentiates (E.14) in x_k, writes the equation satisfied by $w = \frac{1}{2}|\nabla u|^2$, and uses uniform ellipticity to absorb the gradient-quadratic term, obtaining the same bound by the parabolic maximum principle. The synchronous-coupling argument above is the probabilistic shadow of that computation and needs no second differentiation of the data; the two give identical constants in the affine case. Either way, the operative mechanism is that bounded, uniformly-Lipschitz sources produce uniformly-Lipschitz value functions, with the Lipschitz constant propagated by the state-Lipschitz drift through Grönwall.

Theorem E.19 (Interior parabolic Schauder estimate)**.** *Under* (A-Nondeg) *and the hypotheses of theorem E.17, suppose moreover the data $F[m_t](\cdot)$, $G[m_T](\cdot)$ are bounded in C^θ_{loc} uniformly in t for some $\theta \in (0,1)$. Then for every parabolic interior subdomain $Q' \Subset (0,T) \times \mathbb{R}^d$ the value u^m is bounded in the Hölder space $C^{1+\theta/2,\,2+\theta}(Q')$ by a constant depending only on the ellipticity λ, the C^θ bounds of the coefficients and data,* $\mathrm{dist}(Q', \partial)$*, and L_u from* (E.15) *— but* not *on the frozen flow m beyond those data bounds. In particular a family $\{u^{m^{(n)}}\}$ with uniformly bounded, locally-uniformly-convergent data is precompact in $C^{1,2}_{\mathrm{loc}}$.*

Proof (reduction to the linear theory). By theorem E.17 the gradient $p^m(t,x) = \nabla u^m(t,x)$ is bounded by L_u, uniformly in m. Freeze it: u^m solves the *linear* uniformly parabolic equation

$$-\partial_t u^m - \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top\, \mathrm{D}^2\, u^m\big) + \underbrace{\Big(\textstyle\int_0^1 \partial_p H(x, \tau p^m)\ \mathrm{d}\tau\Big)}_{=:c^m(t,x),\ |c^m| \le M} \cdot \nabla u^m = F[m_t](x),$$

obtained by writing $H(x,p^m) = H(x,0) + c^m \cdot p^m$ with c^m bounded by M (the range of $\partial_p H$ on $|p| \le L_u$). The coefficients $\sigma\sigma^\top$ and c^m are bounded and (by (A-Lip) and the C^θ data) Hölder continuous, uniformly in m, and the right-hand side $H(x,0) - F[m_t](x)$ is uniformly C^θ_{loc}. The interior Schauder estimate for linear uniformly parabolic equations — the quantitative form of the parametrix/Levi construction proved in theorem C.12 — then bounds u^m in $C^{1+\theta/2, 2+\theta}(Q')$ in terms of the listed data, independently of m. Precompactness in $C^{1,2}_{\mathrm{loc}}$ follows from the Arzelà–Ascoli theorem theorem B.24 applied on the compact closure of Q', since a bound in $C^{2+\theta}$ is a bound in C^2 together with a uniform modulus of continuity for the second derivatives. □

E.7 The Barles–Souganidis monotone-scheme theorem

The finite-difference solver of chapter 28 rests on a single abstract convergence principle: a numerical scheme that is *monotone*, *stable*, and *consistent* converges to the unique viscosity solution. This is the theorem of Barles and Souganidis [13]; we prove it here so that theorem 28.3 appeals to an in-book result rather than an external citation. The proof is short once the comparison principle (theorem E.14) is in hand, because it is exactly comparison that collapses the two relaxed semilimits of the scheme.

Write a discretization parameter $\rho = (h, \Delta t) \downarrow 0$ and let the scheme be a map

$$S_\rho(x,\, u_\rho(x),\, [u_\rho]) = 0, \tag{E.16}$$

where $[u_\rho]$ denotes the dependence of the equation at x on the values of u_ρ at neighbouring grid points (for eq. (28.8) this is the upwind stencil applied to the value field). We require three properties.

- *Monotonicity:* $S_\rho(x, r, [\phi]) \le S_\rho(x, r, [\psi])$ whenever $\phi \ge \psi$ with $\phi(x) = \psi(x)$; equivalently, S_ρ is nonincreasing in the values of its argument at points $\ne x$. (For eq. (28.8) this is precisely the M-matrix property of lemma 28.1: the off-diagonal stencil weights are ≤ 0 in the residual form.)
- *Stability:* for every ρ the scheme has a solution u_ρ with $\|u_\rho\|_\infty \le C$ uniformly in ρ (the L^∞ bound from lemma 28.1).
- *Consistency:* for every $\zeta \in C^\infty(\mathbb{R}^d)$ and every x,

$$\lim_{\substack{\rho\to 0,\ y\to x\\ \xi\to 0}} S_\rho(y,\, \zeta(y)+\xi,\, [\zeta+\xi]) = F(x, \zeta(x), \nabla\zeta(x), \mathrm{D}^2\,\zeta(x)), \tag{E.17}$$

  with F the HJB operator (E.2) (the Taylor expansion of the upwind stencil, whose $O(h)$ artificial-diffusion term vanishes as $h \to 0$).

Theorem E.20 (Barles–Souganidis)**.** *Let S_ρ be monotone, stable, and consistent in the senses above, and suppose the limit equation $F = 0$ enjoys the comparison principle of theorem E.14 (so it has a unique bounded continuous viscosity solution u). Then the scheme solutions u_ρ converge to u locally uniformly as $\rho \downarrow 0$.*

Proof. Define the upper and lower *relaxed semilimits*

$$\overline{u}(x) = \limsup_{\rho\to 0,\ y\to x} u_\rho(y), \qquad \underline{u}(x) = \liminf_{\rho\to 0,\ y\to x} u_\rho(y).$$

By stability these are finite, with $\underline{u} \le \overline{u}$ pointwise; $\overline{u}$ is USC and $\underline{u}$ is LSC by construction. We claim $\overline{u}$ is a viscosity *subsolution* and $\underline{u}$ a viscosity *supersolution* of $F = 0$.

$\overline{u}$ is a subsolution. Let $\phi \in C^\infty$ and let x_0 be a strict local maximum of $\overline{u} - \phi$; we must show $F(x_0, \overline{u}(x_0), \nabla\phi(x_0), \mathrm{D}^2\,\phi(x_0)) \le 0$. A standard semilimit lemma (the doubling/perturbation argument: subtract $|x - x_0|^4$ to make the maximum strict, then use the definition of lim sup)

produces sequences $\rho_n \downarrow 0$ and grid points $x_n \to x_0$ such that x_n is a global maximum of $u_{\rho_n} - \phi$ over the grid and $u_{\rho_n}(x_n) \to \overline{u}(x_0)$. Set $\xi_n = u_{\rho_n}(x_n) - \phi(x_n) \to \overline{u}(x_0) - \phi(x_0)$. At x_n the field satisfies $u_{\rho_n} \le \phi + \xi_n$ everywhere, with equality at x_n. By monotonicity of S_{ρ_n} (it is nonincreasing in the off-point values),

$$0 = S_{\rho_n}\big(x_n, u_{\rho_n}(x_n), [u_{\rho_n}]\big) \ge S_{\rho_n}\big(x_n,\, \phi(x_n) + \xi_n,\, [\phi + \xi_n]\big).$$

Let $n \to \infty$ and apply consistency (E.17) with $\zeta = \phi$, $y = x_n \to x_0$, $\xi = \xi_n$:

$$0 \ge \lim_n S_{\rho_n}\big(x_n, \phi(x_n) + \xi_n, [\phi + \xi_n]\big) = F\big(x_0, \overline{u}(x_0), \nabla\phi(x_0), \mathrm{D}^2\,\phi(x_0)\big),$$

which is the subsolution inequality. The supersolution property of $\underline{u}$ follows by the symmetric argument at a strict local *minimum*, reversing the inequalities.

Conclusion by comparison. Both $\overline{u}$ and $\underline{u}$ are bounded (stability). The comparison principle theorem E.14, applied to the USC subsolution $\overline{u}$ and the LSC supersolution $\underline{u}$, gives $\overline{u} \le \underline{u}$ on $\mathbb{R}^d$. Combined with the pointwise inequality $\underline{u} \le \overline{u}$ this forces $\overline{u} = \underline{u} =: u$, a (finite, hence continuous) function that is therefore the unique viscosity solution of $F = 0$. Finally, $\overline{u} = \underline{u}$ is exactly the statement that $u_\rho \to u$ locally uniformly: if convergence failed uniformly on some compact $\mathcal{K}$, one could extract points $y_n \to x_* \in \mathcal{K}$ with $u_{\rho_n}(y_n)$ bounded away from $u(x_*)$, contradicting $\overline{u}(x_*) = \underline{u}(x_*) = u(x_*)$. □

Remark E.21 (Parabolic version and the FP equation)**.** For the time-dependent HJB equation (6.19) the theorem runs verbatim with (t, x) in place of x and the comparison principle in its parabolic form (remark E.15); the fully implicit backward step of eq. (28.8) is monotone for *every* $\Delta t > 0$ (no ratio restriction), since lemma 28.1 holds for all Δt. The forward Fokker–Planck step eq. (28.10) is the transpose scheme; its weak-$*$ convergence to the measure solution is the dual statement, obtained by testing the conservative scheme against the (now uniformly convergent, by this theorem) backward test fields and passing to the limit, the discrete duality of lemma 28.2 furnishing the discrete integration by parts.

Appendix F

Graphon Space: Detailed Proofs

This appendix supplies the proofs that Part III states but defers: the weak regularity lemma, the counting lemma, completeness and compactness of the space of graphons under the cut metric, the sampling lemma, and the resulting equivalence between cut convergence and left-convergence together with the uniqueness statement that homomorphism densities determine a graphon up to relabeling. Everything is proved from material established earlier in the book; in particular we use the cut norm and cut metric of definitions 17.1 and 17.6, the fact that $\delta_\square$ is a pseudometric on $\mathcal{W}_0$ inducing a genuine metric on the quotient $\widetilde{\mathcal{W}}_0$ — with the triangle inequality, symmetry, and transitivity of weak isomorphism all established in proposition 17.7 — the homomorphism densities of chapter 16 and the elementary counting lemma lemma 16.17, the Banach–Alaoglu theorem of chapter 2, and the Azuma–Hoeffding inequality theorem 3.28. No external source is required.

Throughout, a *graphon* is a symmetric measurable function $W\colon [0,1]^2 \to [0,1]$; $\mathcal{W}_0$ denotes the set of graphons and $\widetilde{\mathcal{W}}_0 = \mathcal{W}_0/\!\sim$ its quotient under measure-preserving relabelings, metrized by the cut metric $\delta_\square$. We write $\langle U, V\rangle = \int_{[0,1]^2} UV$ and $\|U\|_2 = \langle U, U\rangle^{1/2}$ for the $L^2([0,1]^2)$ inner product and norm, and recall from lemma 17.2 the bound $\|U\|_\square \le \|U\|_2$ and the dual description $\|U\|_\square = \sup\{\, |\langle U, \mathbf{1}_{S\times T}\rangle| : S, T \subseteq [0,1] \text{ measurable}\,\}$.

F.1 Stepping and energy

Let $\mathcal{P} = \{P_1, \dots, P_m\}$ be a finite measurable partition of $[0,1]$ into parts of positive measure. For a kernel $U \in L^2([0,1]^2)$ define its *stepping* $U_\mathcal{P}$ to be the conditional expectation of U onto the σ-algebra generated by $\mathcal{P}\times\mathcal{P}$; concretely, on $P_i \times P_j$,

$$U_\mathcal{P}(x,y) \;=\; \frac{1}{|P_i|\;|P_j|}\int_{P_i\times P_j} U(s,t)\;\mathrm{d}s\;\mathrm{d}t \qquad (x,y)\in P_i\times P_j. \tag{F.1}$$

Thus $U_\mathcal{P}$ is the L^2-orthogonal projection of U onto the subspace of kernels that are constant on the rectangles $P_i \times P_j$. Two consequences of the projection property are used repeatedly. First, the Pythagorean identity

$$\|U\|_2^2 \;=\; \|U_\mathcal{P}\|_2^2 + \|U - U_\mathcal{P}\|_2^2\,. \tag{F.2}$$

Second, if $\mathcal{Q}$ refines $\mathcal{P}$ then $U_{\mathcal{P}} = (U_{\mathcal{Q}})_{\mathcal{P}}$ is the projection of $U_{\mathcal{Q}}$, so again $\|U_{\mathcal{Q}}\|_2^2 = \|U_{\mathcal{P}}\|_2^2 + \|U_{\mathcal{Q}} - U_{\mathcal{P}}\|_2^2$. We call $e(\mathcal{P}) := \|U_{\mathcal{P}}\|_2^2$ the *energy* of $\mathcal{P}$ (relative to U). For a graphon U, $0 \le e(\mathcal{P}) \le \|U\|_2^2 \le 1$, and energy is nondecreasing under refinement.

Lemma F.1 (Energy increment)**.** *Let U be a graphon and $\mathcal{P}$ a partition. Suppose $\|U - U_{\mathcal{P}}\|_{\square} > \varepsilon$. Then there is a refinement $\mathcal{Q}$ of $\mathcal{P}$ with at most $4\,|\mathcal{P}|$ parts such that $e(\mathcal{Q}) \ge e(\mathcal{P}) + \varepsilon^2$.*

Proof. By the dual description of the cut norm there are measurable sets $S, T \subseteq [0,1]$ with $|\langle U - U_{\mathcal{P}}, \mathbf{1}_{S\times T}\rangle| > \varepsilon$. Let $\mathcal{Q}$ be the common refinement of $\mathcal{P}$ with the partition $\{S, S^{\mathrm{c}}\}$ in the first coordinate and $\{T, T^{\mathrm{c}}\}$ in the second; each part of $\mathcal{P}$ is split into at most four pieces, so $|\mathcal{Q}| \le 4\,|\mathcal{P}|$, and both S and T are unions of $\mathcal{Q}$-parts. Because $\mathbf{1}_{S\times T}$ is constant on the rectangles of $\mathcal{Q}\times\mathcal{Q}$, it is orthogonal to $U - U_{\mathcal{Q}}$, whence $\langle U_{\mathcal{Q}}, \mathbf{1}_{S\times T}\rangle = \langle U, \mathbf{1}_{S\times T}\rangle$. Since $U_{\mathcal{P}}$ is the projection of $U_{\mathcal{Q}}$ onto the coarser σ-algebra,

$$\langle U_{\mathcal{Q}} - U_{\mathcal{P}}, \mathbf{1}_{S\times T}\rangle = \langle U_{\mathcal{Q}}, \mathbf{1}_{S\times T}\rangle - \langle U_{\mathcal{P}}, \mathbf{1}_{S\times T}\rangle = \langle U - U_{\mathcal{P}}, \mathbf{1}_{S\times T}\rangle,$$

using $\langle U_{\mathcal{Q}}, \mathbf{1}_{S\times T}\rangle = \langle U, \mathbf{1}_{S\times T}\rangle$ in the last step. Therefore, by Cauchy–Schwarz and $\|\mathbf{1}_{S\times T}\|_2 = \sqrt{|S|\,|T|} \le 1$,

$$\|U_{\mathcal{Q}} - U_{\mathcal{P}}\|_2 \;\ge\; \frac{|\langle U_{\mathcal{Q}} - U_{\mathcal{P}}, \mathbf{1}_{S\times T}\rangle|}{\|\mathbf{1}_{S\times T}\|_2} \;\ge\; \frac{\varepsilon}{\sqrt{|S|\,|T|}} \;\ge\; \varepsilon.$$

By the refinement form of (F.2), $e(\mathcal{Q}) - e(\mathcal{P}) = \|U_{\mathcal{Q}} - U_{\mathcal{P}}\|_2^2 \ge \varepsilon^2$. □

F.2 The weak regularity lemma

Theorem F.2 (Weak regularity lemma; cf. theorem 17.16)**.** *For every $\varepsilon > 0$ and every graphon W there is a partition $\mathcal{P}$ of $[0,1]$ into at most $4^{\lceil 1/\varepsilon^2\rceil}$ parts such that $\|W - W_{\mathcal{P}}\|_{\square} \le \varepsilon$.*

Proof. Start from the trivial partition $\mathcal{P}_0 = \{[0,1]\}$, with $e(\mathcal{P}_0) \ge 0$. As long as the current partition $\mathcal{P}_r$ satisfies $\|W - W_{\mathcal{P}_r}\|_{\square} > \varepsilon$, apply lemma F.1 to obtain a refinement $\mathcal{P}_{r+1}$ with at most four times as many parts and $e(\mathcal{P}_{r+1}) \ge e(\mathcal{P}_r) + \varepsilon^2$. Since the energy is bounded above by $\|W\|_2^2 \le 1$ and increases by at least ε^2 at each step, the procedure must stop after at most $\lceil 1/\varepsilon^2\rceil$ steps; when it stops, the current partition satisfies $\|W - W_{\mathcal{P}}\|_{\square} \le \varepsilon$. Starting from one part and multiplying by at most four per step, the number of parts is at most $4^{\lceil 1/\varepsilon^2\rceil}$. □

A useful reformulation: every graphon is, in cut norm, within ε of a *step graphon* with a bounded number $m(\varepsilon) \le 4^{\lceil 1/\varepsilon^2\rceil}$ of steps. We record the elementary stability of cut distance under perturbing the step data, needed in appendix F.5.

Lemma F.3 (Stability of step graphons)**.** *Let U, U' be step graphons with respect to the* same *ordered partition of $[0,1]$ into intervals of lengths $a = (a_1, \ldots, a_m)$ and $a' = (a'_1, \ldots, a'_m)$ respectively (the intervals taken in the same order), with value matrices $H, H' \in [0,1]^{m\times m}$. Then*

$$\delta_{\square}(U, U') \;\le\; \|U - U'\|_1 \;\le\; \sum_{i,j} a_i a_j \left|H_{ij} - H'_{ij}\right| \;+\; 2\sum_i |a_i - a'_i|\,.$$

Proof. The bound $\delta_\square \le \|\cdot\|_1$ holds because $\|\cdot\|_\square \le \|\cdot\|_1$ and $\delta_\square$ is an infimum over relabelings that includes the identity. For the L^1 bound, align the two interval partitions on their common refinement; on the overlap of the ith interval of each, the integrand differs by $\left|H_{ij} - H'_{ij}\right|$, contributing the first sum, while the symmetric difference of the ith intervals has total measure at most $|a_i - a'_i|$ and the integrand there is bounded by 1, contributing (over both coordinates) at most $2\sum_i |a_i - a'_i|$. $\square$

F.3 The counting lemma

For a simple graph F with edge set $E(F)$ on vertex set $\{1, \dots, k\}$ and a graphon W, recall

$$t(F, W) = \int_{[0,1]^k} \prod_{\{i,j\}\in E(F)} W(x_i, x_j) \prod_{i=1}^{k} \mathrm{d}x_i.$$

Lemma F.4 (Counting lemma)**.** *For every simple graph F and all graphons U, W,*

$$|t(F, U) - t(F, W)| \;\le\; |E(F)|\, \|U - W\|_\square \;\le\; |E(F)|\, \delta_\square(U, W).$$

Proof. Because $t(F, \cdot)$ is invariant under measure-preserving relabeling of the graphon, the second inequality follows from the first by taking the infimum over relabelings. For the first, enumerate $E(F) = \{f_1, \dots, f_r\}$, $r = |E(F)|$, and interpolate one edge at a time: let Φ_ℓ be the integrand obtained by using U on the edges $f_1, \dots, f_\ell$ and W on the remaining edges, so that $\int \Phi_0 = t(F, W)$ and $\int \Phi_r = t(F, U)$. Then

$$t(F, U) - t(F, W) = \sum_{\ell=1}^{r} \int_{[0,1]^k} (\Phi_\ell - \Phi_{\ell-1}).$$

Fix ℓ and let $f_\ell = \{p, q\}$. The difference $\Phi_\ell - \Phi_{\ell-1}$ equals $\big(U(x_p, x_q) - W(x_p, x_q)\big)\,\Psi(x)$, where Ψ is a product of graphon factors not involving the edge $\{p, q\}$, hence $0 \le \Psi \le 1$. Integrating out all variables except x_p, x_q produces a function $g(x_p, x_q) = \int \Psi \prod_{i\ne p,q} \mathrm{d}x_i$ with $\|g\|_\infty \le 1$, so, writing $D = U - W$,

$$\int_{[0,1]^k} (\Phi_\ell - \Phi_{\ell-1}) = \int_{[0,1]^2} D(s, t)\, g(s, t)\; \mathrm{d}s\; \mathrm{d}t.$$

For any measurable g with $\|g\|_\infty \le 1$ one has $|\langle D, g\rangle| \le \|D\|_\square$: this is the standard fact that the cut norm equals the supremum of $\langle D, g\rangle$ over g valued in $[-1, 1]$, which follows from the dual description by writing g as a limit of simple functions and expanding each level set as a signed combination of rectangle indicators $\mathbf{1}_{S\times T}$. Hence each term is at most $\|D\|_\square$ in absolute value, and summing the r terms gives the claim. $\square$

F.4 Completeness

Theorem F.5 (Completeness; cf. proposition 17.18)**.** *The metric space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is complete.*

Proof. Let $(\widetilde{W}_n)$ be a Cauchy sequence in $\widetilde{\mathcal{W}}_0$. Passing to a subsequence we may assume $\delta_\square(\widetilde{W}_n, \widetilde{W}_{n+1}) < 2^{-n}$; it suffices to produce a limit of the subsequence, since a Cauchy sequence with a convergent subsequence converges. Fix any representative V_1 of $\widetilde{W}_1$. Inductively, given a representative V_n of $\widetilde{W}_n$, the definition of $\delta_\square$ as an infimum over measure-preserving relabelings provides a relabeling ϕ and a representative W_{n+1} of $\widetilde{W}_{n+1}$ with $\left\|V_n - W_{n+1}^\phi\right\|_\square < 2^{-n}$; set $V_{n+1} = W_{n+1}^\phi$. Then (V_n) is a sequence of graphons with $\|V_n - V_{n+1}\|_\square < 2^{-n}$, hence Cauchy in the cut norm.

The graphons form a subset of the closed unit ball of $L^\infty([0,1]^2)$, which is weak-$*$ compact by the Banach–Alaoglu theorem (chapter 2); since $L^\infty([0,1]^2)$ is the dual of the separable space $L^1([0,1]^2)$, the ball is weak-$*$ sequentially compact. Thus a subsequence of (V_n) converges weak-$*$ to some $W \in L^\infty$. The constraints defining a graphon are preserved: $\{U : 0 \le U \le 1 \text{ a.e.}\}$ and $\{U : U(x,y) = U(y,x) \text{ a.e.}\}$ are convex and weak-$*$ closed (each is an intersection of half-spaces $\langle U, h\rangle \le c$ with $h \in L^1$), so W is again a graphon.

Finally $\|V_n - W\|_\square \to 0$. Fix $\varepsilon > 0$ and N with $\|V_n - V_m\|_\square < \varepsilon$ for $n, m \ge N$. For any measurable S, T and $n \ge N$,

$$|\langle V_n - W, \mathbf{1}_{S\times T}\rangle| = \lim_{m\to\infty} |\langle V_n - V_m, \mathbf{1}_{S\times T}\rangle| \le \varepsilon,$$

using weak-$*$ convergence of the chosen subsequence on the fixed test function $\mathbf{1}_{S\times T} \in L^1$. Taking the supremum over S, T gives $\|V_n - W\|_\square \le \varepsilon$ for all $n \ge N$; the bound is uniform in S, T, so the whole sequence (V_n) converges to W in cut norm, hence $\delta_\square(\widetilde{W}_n, \widetilde{W}) \to 0$ with $\widetilde{W}$ the class of W. □

F.5 Compactness

Theorem F.6 (Lovász–Szegedy compactness; cf. theorem 17.13)**.** *The metric space $(\widetilde{\mathcal{W}}_0, \delta_\square)$ is compact.*

Proof. As a metric space it suffices to prove sequential compactness; we extract a convergent subsequence and construct its limit explicitly. Let $(W_n) \subset \mathcal{W}_0$. For each integer $k \ge 1$ apply the weak regularity lemma theorem F.2 with $\varepsilon = 1/k$: there is a partition $\mathcal{P}_n^{(k)}$ of $[0,1]$ into $m_k \le 4^{\lceil k^2\rceil}$ parts (pad with empty parts so the count is exactly m_k, independent of n) such that the step graphon $W_n^{(k)} := (W_n)_{\mathcal{P}_n^{(k)}}$ satisfies

$$\left\|W_n - W_n^{(k)}\right\|_\square \le 1/k. \tag{F.3}$$

After a measure-preserving relabeling of $[0,1]$ (which changes neither $\delta_\square$ nor the left-hand side of (F.3), since the relabeling is applied simultaneously to W_n and its stepping) we may assume the parts of $\mathcal{P}_n^{(k)}$ are subintervals of $[0,1]$ arranged in a fixed order. The step graphon $W_n^{(k)}$ is then encoded by the length vector $a_n^{(k)}$ in the closed simplex $\Delta_{m_k} = \{a \in [0,1]^{m_k} : \sum a_i = 1\}$ and the symmetric value matrix $H_n^{(k)} \in [0,1]^{m_k\times m_k}$.

The data $(a_n^{(k)}, H_n^{(k)})$ lie in the compact set $\Delta_{m_k} \times [0,1]^{m_k\times m_k}$. By a diagonal argument extract a single subsequence (n_j) along which, *for every* k, $a_{n_j}^{(k)} \to a^{(k)}$ and $H_{n_j}^{(k)} \to H^{(k)}$ as $j \to \infty$. Let U_k be the step graphon with interval lengths $a^{(k)}$ and value matrix $H^{(k)}$.

Claim: $\limsup_j \delta_\square(W_{n_j}, U_k) \le 1/k$. By lemma F.3 applied to the step graphons $W_{n_j}^{(k)}$ and U_k (same ordered interval structure, converging data), $\delta_\square(W_{n_j}^{(k)}, U_k) \le \left\| H_{n_j}^{(k)} - H^{(k)} \right\|_1 + 2 \left\| a_{n_j}^{(k)} - a^{(k)} \right\|_1 \to 0$ as $j \to \infty$. Combined with (F.3) and the triangle inequality, $\delta_\square(W_{n_j}, U_k) \le \left\| W_{n_j} - W_{n_j}^{(k)} \right\|_\square + \delta_\square(W_{n_j}^{(k)}, U_k) \le 1/k + o_j(1)$, proving the claim.

The sequence (U_k) *is Cauchy.* For any k, l and all j, $\delta_\square(U_k, U_l) \le \delta_\square(U_k, W_{n_j}) + \delta_\square(W_{n_j}, U_l)$; letting $j \to \infty$ and using the claim, $\delta_\square(U_k, U_l) \le 1/k + 1/l$. By completeness (theorem F.5) there is a graphon W with $\delta_\square(U_k, W) \to 0$.

Conclusion. For each k and all j, $\delta_\square(W_{n_j}, W) \le \delta_\square(W_{n_j}, U_k) + \delta_\square(U_k, W)$. Taking $\limsup_j$ and using the claim gives $\limsup_j \delta_\square(W_{n_j}, W) \le 1/k + \delta_\square(U_k, W)$; letting $k \to \infty$ makes the right-hand side vanish, so $\delta_\square(W_{n_j}, W) \to 0$. Thus every sequence in $\widetilde{\mathcal{W}}_0$ has a convergent subsequence. $\square$

F.6 The sampling lemma

We connect a graphon to the finite graphs it generates, which powers the converse (left-convergence $\Rightarrow$ cut convergence) and the uniqueness theorem. Given a graphon W and an integer k, sample $X_1, \dots, X_k$ independently and uniformly on $[0,1]$ and let $\mathbb{H}(k, W)$ be the random *weighted* graph on $\{1, \dots, k\}$ with edge weights $W(X_i, X_j)$. Identify $\mathbb{H}(k, W)$ with the step graphon $W_\mathbb{H}$ obtained by partitioning $[0,1]$ into k equal intervals $I_1, \dots, I_k$ and setting $W_\mathbb{H} = W(X_i, X_j)$ on $I_i \times I_j$ $(i \ne j)$ and 0 on the diagonal blocks.

Theorem F.7 (Sampling lemma)**.** *There is a function $\eta(k) \to 0$ as $k \to \infty$, independent of the graphon, such that*

$$\mathbb{E}\, \delta_\square\big(W, W_{\mathbb{H}(k,W)}\big) \;\le\; \eta(k) \qquad \text{for every graphon } W.$$

Proof. Write $Z_k = \delta_\square(W, W_{\mathbb{H}(k,W)})$, a function of the i.i.d. samples $X_1, \dots, X_k$. We bound $\mathbb{E} Z_k$ and note it concentrates.

Concentration. Changing a single coordinate X_i alters $W_\mathbb{H}$ only on the ith row and column of blocks, a set of measure $\le 2/k$, and changes $\|\cdot\|_\square \le \|\cdot\|_1$ by at most $2/k$; since $\delta_\square$ is 1-Lipschitz with respect to the cut norm, Z_k has bounded differences $c_i \le 2/k$. The bounded-differences inequality (theorem 3.28, remark 3.29) gives $\mathbb{P}(|Z_k - \mathbb{E} Z_k| \ge t) \le 2\exp(-t^2 k/2)$, because $\sum_i c_i^2 \le k\,(2/k)^2 = 4/k$.

Mean estimate. Fix $\varepsilon > 0$ and let $W_\mathcal{P}$ be a step graphon with $m \le 4^{\lceil 1/\varepsilon^2 \rceil}$ parts and $\|W - W_\mathcal{P}\|_\square \le \varepsilon$ (theorem F.2). Decompose, using $\delta_\square \le \|\cdot\|_\square$ and the triangle inequality,

$$\mathbb{E} Z_k \le \underbrace{\|W - W_\mathcal{P}\|_\square}_{\le \varepsilon} + \mathbb{E}\, \|W_\mathcal{P} - (W_\mathbb{H})_{\mathcal{P}'}\|_\square + \mathbb{E}\, \|(W_\mathbb{H})_{\mathcal{P}'} - W_\mathbb{H}\|_\square\,,$$

where $\mathcal{P}'$ is the partition of the index set induced by which part of $\mathcal{P}$ contains each X_i, and $(W_\mathbb{H})_{\mathcal{P}'}$ is the corresponding block average of the sample. The block values of $(W_\mathbb{H})_{\mathcal{P}'}$ are empirical averages of the independent bounded weights over sampled pairs falling in each pair of parts; a block straddling parts P_a, P_b averages $\approx k^2\,|P_a|\,|P_b|$ weights, each of mean $\langle W\rangle_{P_a\times P_b}$, so by Chebyshev its deviation from that mean is $O(k^{-1}\,|P_a|^{-1/2}\,|P_b|^{-1/2})$ in L^1. There are m^2 blocks; weighting by block areas, the middle term is $O(m\,k^{-1})$ in expectation (parts of measure below $1/k$ contribute negligibly, as they hold $O(1)$ samples and total area $\le m/k$). The third term compares the sampled graphon to its own stepping along $\mathcal{P}'$; since $(W_{\mathbb{H}})_{\mathcal{P}'}$ is the stepping of a sample of W, in expectation $\mathbb{E}\,\|(W_{\mathbb{H}})_{\mathcal{P}'} - W_{\mathbb{H}}\|_\square \le \|W_{\mathcal{P}} - W\|_\square + O(m\,k^{-1/2}) \le \varepsilon + O(m\,k^{-1/2})$, because stepping a sample concentrates around stepping the graphon. Collecting,

$$\mathbb{E}Z_k \;\le\; 2\varepsilon + C(\varepsilon)\,k^{-1/2},$$

with $C(\varepsilon)$ depending only on $m = m(\varepsilon)$ and on absolute constants, not on W. Choosing $\varepsilon \to 0$ slowly as $k \to \infty$ (e.g. $\varepsilon = (\log k)^{-1/4}$, so that $m(\varepsilon) = 4^{\lceil\sqrt{\log k}\,\rceil}$ grows subpolynomially in k and still leaves $C(\varepsilon)k^{-1/2} \to 0$) makes the bound a graphon-independent $\eta(k) \to 0$. The concentration estimate then transfers the bound from $\mathbb{E}Z_k$ to Z_k in expectation, completing the proof. $\square$

Remark F.8. A sharper accounting (Azuma inside each block, optimizing ε) gives the sharp rate $\eta(k) = O(1/\sqrt{\log k})$; we use only $\eta(k) \to 0$, except in theorem F.9 below, where the $O(1/\sqrt{\log k})$ rate sets the final scale and is therefore quoted explicitly (as an absolute constant c_1 with $\eta(N) \le c_1/\sqrt{\log N}$).

F.6.1 The Bernoulli second sampling lemma

The sampling lemma above compares a graphon to its *weighted* sample, whose blocks carry the conditional edge means $W(X_i, X_j)$. The object actually generated by the W-random graph $G(N, W)$ of definition 19.1 is the *Bernoulli* sample, whose blocks carry the coins $\xi_{ij} \in \{0, 1\}$. The next theorem — the *second sampling lemma* of Borgs–Chayes–Lovász–Sós–Vesztergombi — upgrades the weighted estimate to the Bernoulli sample with an explicit tail, by adding one layer of edge-coin concentration. It is the input consumed by the finite-network convergence theorem of chapter 25, and we prove it here in full from the in-book ingredients (the weighted sampling lemma theorem F.7 and the bounded-differences inequality theorem 3.28/remark 3.29); only the attribution is external.

Theorem F.9 (Second sampling lemma: Bernoulli sampling in cut distance)**.** *Let W be a graphon. For $N \in \mathbb{N}$ draw i.i.d. types $U_1, \ldots, U_N \sim \mathrm{Unif}[0, 1]$ and, conditionally on the types, independent edge coins $\xi_{ij} \sim \mathrm{Bernoulli}(W(U_i, U_j))$ $(1 \le i < j \le N,\ \xi_{ji} = \xi_{ij})$, and let $W_{[N]}$ be the induced step graphon: $W_{[N]} = \xi_{ij}$ on $J_i \times J_j$ for $i \ne j$ and 0 on the diagonal blocks, where $J_1, \ldots, J_N$ are equal intervals partitioning $[0, 1]$. There is an absolute constant $C \ge 1$ such that for all $N \ge 2$,*

$$\mathbb{P}\Big(\delta_\square(W_{[N]}, W) > \frac{C}{\sqrt{\log N}}\Big) \;\le\; \exp\Big(-\frac{N}{C\log N}\Big).$$

Proof. Interpolate through the weighted sample $W_{\mathbb{H}(N,W)}$ of theorem F.7: the step graphon equal to $W(U_i, U_j)$ on $J_i \times J_j$ $(i \ne j)$ and 0 on the diagonal, built from the *same* types U_i as $W_{[N]}$. By the triangle inequality and $\delta_\square \le \|\cdot\|_\square$,

$$\delta_\square(W_{[N]}, W) \;\le\; \underbrace{\big\|W_{[N]} - W_{\mathbb{H}(N,W)}\big\|_\square}_{=:A_N} + \underbrace{\delta_\square(W_{\mathbb{H}(N,W)}, W)}_{=:B_N}. \tag{F.4}$$

We bound the two terms by genuinely separate mechanisms — B_N by sampling the *types*, A_N by concentrating the *coins* with the types frozen — which is the two-temperature split of section 19.2.

The weighted term B_N (types). This is exactly the quantity controlled by the weighted sampling lemma. By theorem F.7 and remark F.8 there is an absolute c_1 with $\mathbb{E}B_N \le \eta(N) \le c_1/\sqrt{\log N}$. Moreover B_N is a function of the i.i.d. types alone, and changing one type U_i alters $W_{\mathbb{H}(N,W)}$ only on the ith row and column of blocks (measure $\le 2/N$), hence moves B_N by at most $2/N$ (the map $W \mapsto \delta_\square(W,W)$ is 1-Lipschitz in $\|\cdot\|_\square \le \|\cdot\|_1$). Thus $\sum_i c_i^2 \le N(2/N)^2 = 4/N$, and the bounded-differences inequality (theorem 3.28/remark 3.29) gives, with $t = c_1/\sqrt{\log N}$,

$$\mathbb{P}\Big(B_N > \frac{2c_1}{\sqrt{\log N}}\Big) \;\le\; \mathbb{P}(B_N - \mathbb{E}B_N > t) \;\le\; 2\exp\Big(-\frac{t^2}{2\cdot 4/N}\Big) \;=\; 2\exp\Big(-\frac{c_1^2\,N}{8\log N}\Big).$$

The fluctuation term A_N (coins). Here the types are frozen and only the edge coins fluctuate. The integrand $D := W_{[N]} - W_{\mathbb{H}(N,W)}$ is constant on each block, with value $D_{ij} = \xi_{ij} - W(U_i,U_j) \in [-1,1]$ off the diagonal and 0 on it. For a step function the cut-norm supremum over measurable S,T is attained at unions of the defining blocks: with T fixed, $\int_{S\times T} D = \int_S \psi$ where $\psi(x) = \int_T D(x,t)\,\mathrm{d}t$ is constant on each J_i, so the supremum over measurable S is realised by the union of the J_i on which ψ carries the maximising sign, and symmetrically in T. Hence

$$A_N \;=\; \max_{A,B\subseteq[N]} \Big|\frac{1}{N^2}\sum_{i\in A,\, j\in B} D_{ij}\Big| \;=\; \max_{A,B\subseteq[N]} |\Delta_{A,B}|\,, \qquad \Delta_{A,B} := \frac{1}{N^2}\sum_{i\in A,\, j\in B} D_{ij}.$$

Fix A,B and condition on the types $U = (U_1,\dots,U_N)$. The off-diagonal coins $\{\xi_{ij}\}_{i<j}$ are then independent, and $\Delta_{A,B} = \sum_{i<j} a_{ij}\,(\xi_{ij} - W(U_i,U_j))$ is a weighted sum of the centred coins in which each coin enters with coefficient $|a_{ij}| \le 2/N^2$ (the factor 2 accounting for the symmetric appearance of the unordered pair $\{i,j\}$ as both (i,j) and (j,i)); its centred contribution therefore ranges over an interval of length $\le 4/N^2$. The conditional mean is 0 and $\sum_{i<j}(4/N^2)^2 \le \binom{N}{2}(4/N^2)^2 \le 8/N^2$, so the bounded-differences inequality (remark 3.29) gives, for every value of U and hence unconditionally,

$$\mathbb{P}(\,|\Delta_{A,B}| \ge s) \;\le\; 2\exp\Big(-\frac{s^2}{2\cdot 8/N^2}\Big) \;=\; 2\exp\Big(-\frac{s^2N^2}{16}\Big).$$

A union bound over the $2^N\cdot 2^N = 4^N$ pairs (A,B) yields

$$\mathbb{P}(A_N \ge s) \;\le\; 2\cdot 4^N \exp\Big(-\frac{s^2N^2}{16}\Big) \;=\; 2\exp\Big(N\log 4 - \frac{s^2N^2}{16}\Big).$$

Choose $s = C_0/\sqrt{N}$ with $C_0^2 = 32\log 4$; then the exponent is $N\log 4 - C_0^2 N/16 = -N\log 4$, so $\mathbb{P}(A_N \ge C_0/\sqrt{N}) \le 2\cdot 4^{-N}$. Since $\log N \le N$, also $C_0/\sqrt{N} \le C_0/\sqrt{\log N}$.

Combine. On the complement of the two bad events,

$$\delta_\square(W_{[N]},W) \;\le\; A_N + B_N \;\le\; \frac{C_0}{\sqrt{N}} + \frac{2c_1}{\sqrt{\log N}} \;\le\; \frac{C_0+2c_1}{\sqrt{\log N}}.$$

Take $C \ge \max\{1,\, C_0{+}2c_1\}$ large enough that also $2{\cdot}4^{-N}{+}2\exp(-c_1^2N/(8\log N)) \le \exp(-N/(C\log N))$ for every $N \ge 2$ (possible because $4^{-N} = e^{-N\log 4}$ and $\exp(-c_1^2N/(8\log N))$ each dominate $\exp(-N/(C\log N))$ once C is large). A union bound over the two displayed estimates then gives the claim. □

Remark F.10. The $1/\sqrt{\log N}$ scale comes entirely from the weighted term B_N (the slow rate $\eta(N)$ of remark F.8); the coin fluctuations A_N are $O(1/\sqrt{N})$, an order of magnitude smaller, so Bernoulli sampling is no harder than weighted sampling. The logarithm is genuine: cut distance is a supremum over the exponentially many test pairs (A, B), and the union bound over 4^N of them is what caps the resolution at $1/\sqrt{\log N}$.

F.7 Uniqueness and the equivalence theorem

Theorem F.11 (Moments determine the graphon; cf. theorem 17.11)**.** *If two graphons U, W satisfy $t(F, U) = t(F, W)$ for every simple graph F, then $\delta_\Box(U, W) = 0$, i.e. $U = W$ in $\widetilde{\mathcal{W}}_0$.*

Proof. We route uniqueness through the *Bernoulli* sample, not the weighted one. For a graphon W and a sample size k, let $G(k, W)$ be the random simple graph on $[k]$ obtained by drawing i.i.d. types $X_1, \ldots, X_k \sim \mathrm{Unif}[0, 1]$ and then, conditionally on the types, including each edge $\{i, j\}$ independently with probability $W(X_i, X_j)$ (definition 19.1); write $W_{[k]}$ for its induced step graphon, exactly as in theorem F.9.

Step 1: equal densities force equal sample laws. We claim that for every k the laws of $G(k, U)$ and $G(k, W)$ on simple graphs over $[k]$ coincide. Fix a simple graph H with $V(H) = [k]$. Conditioning on the types and using conditional edge-independence,

$$\mathbb{P}(G(k, W) = H) = \mathbb{E}\Big[\prod_{\{i,j\}\in E(H)} W(X_i, X_j) \prod_{\{i,j\}\notin E(H)} (1 - W(X_i, X_j))\Big].$$

Expand the product over the non-edges. Because each pair $\{i, j\}$ is either an edge or a non-edge of H but never both, every weight $W(X_i, X_j)$ enters the expansion to the *first power only*: there are no squares to reckon with. This is the crux — the edge indicator $Z = \mathbf{1}\{\{i, j\} \in G(k, W)\}$ is $\{0, 1\}$-valued, so $Z^2 = Z$, and no genuinely higher moment of the weights is ever called for. The expansion is therefore a finite $\mathbb{Z}$-linear combination of *multilinear* (square-free) expectations $\mathbb{E} \prod_{\{i,j\}\in E(F)} W(X_i, X_j)$, one for each simple graph F on $[k]$ formed by adjoining to $E(H)$ a subset of its non-edges; by definition 16.8 each such expectation is exactly the homomorphism density $t(F, W)$. Thus $\mathbb{P}(G(k, W) = H)$ is one and the same fixed finite linear combination of the numbers $\{t(F, W)\}_F$ for both $W = U$ and $W = W$ (this is the inclusion–exclusion passage from induced to homomorphism densities). Since $t(F, U) = t(F, W)$ for every simple F, the two probabilities agree; as H was arbitrary, $G(k, U)$ and $G(k, W)$ have the same law, and hence so do the induced step graphons $U_{[k]}$ and $W_{[k]}$, for every k.

This is exactly where the weighted sample fails and the Bernoulli sample succeeds. A simple graph F supplies only the multilinear moment $\mathbb{E} \prod_{\{i,j\}\in E(F)} W(X_i, X_j)$, each edge to the first power; genuinely higher moments such as $\mathbb{E}\, W(X_1, X_2)^2$ correspond to *multigraphs* (parallel edges), which are not simple-graph densities. Simple-graph densities therefore do *not* determine the law of the weighted sample $\mathbb{H}(k, W)$ — the "bounded random vectors are determined by their moments" principle needs *all* joint moments, which are not on offer. They do determine the law of the $\{0, 1\}$-valued Bernoulli sample, precisely because $Z^2 = Z$ collapses every power to the first and leaves only the multilinear quantities the densities furnish.

Step 2: pinch through the Bernoulli sample. The cut distance $\delta_\square(U,W)$ is a constant, so for the random graph $G(k,U)$ the triangle inequality gives, pathwise, $\delta_\square(U,W) \le \delta_\square(U,U_{[k]}) + \delta_\square(U_{[k]},W)$. Taking expectations,

$$\delta_\square(U,W) \;\le\; \mathbb{E}\big[\delta_\square(U,U_{[k]})\big] + \mathbb{E}\big[\delta_\square(U_{[k]},W)\big] \;=\; \mathbb{E}\big[\delta_\square(U,U_{[k]})\big] + \mathbb{E}\big[\delta_\square(W_{[k]},W)\big],$$

where the last equality replaces $U_{[k]}$ by the identically distributed $W_{[k]}$ of Step 1. Both expectations are controlled by the second sampling lemma (theorem F.9): for any graphon W, since $\delta_\square \le 1$ always,

$$\mathbb{E}\big[\delta_\square(W_{[k]},W)\big] \;\le\; \frac{C}{\sqrt{\log k}} + 1 \cdot \exp\Big(-\frac{k}{C\log k}\Big) \xrightarrow[k\to\infty]{} 0.$$

Hence $\delta_\square(U,W) \;\le\; 2\big(C/\sqrt{\log k} + \exp(-k/(C\log k))\big)$ for every $k \ge 2$; letting $k \to \infty$ forces $\delta_\square(U,W) = 0$. $\square$

Theorem F.12 (Cut convergence $\Leftrightarrow$ left-convergence; cf. theorem 17.12)**.** *For graphons W_n, W the following are equivalent:*

(a) $\delta_\square(W_n,W) \to 0$*;*

(b) $t(F,W_n) \to t(F,W)$ *for every simple graph* F*.*

Proof. (a)$\Rightarrow$(b) is the counting lemma lemma F.4: $|t(F,W_n) - t(F,W)| \le |E(F)|\; \delta_\square(W_n,W) \to 0$.

(b)$\Rightarrow$(a). Suppose the densities converge but, for contradiction, $\delta_\square(W_n,W) \not\to 0$. Then some subsequence satisfies $\delta_\square(W_{n_j},W) \ge \delta > 0$. By compactness (theorem F.6) a further subsequence converges in cut metric to some graphon U, and by continuity of the metric $\delta_\square(U,W) \ge \delta > 0$, so $U \ne W$ in $\widetilde{\mathcal{W}}_0$. But along this subsequence the counting lemma gives $t(F,U) = \lim_j t(F,W_{n_j}) = t(F,W)$ for every F, the last equality by hypothesis (b). By theorem F.11 this forces $U = W$ in $\widetilde{\mathcal{W}}_0$, a contradiction. Hence $\delta_\square(W_n,W) \to 0$. $\square$

These results close every proof deferred from chapters 16 and 17: the weak regularity lemma (theorem F.2), the counting lemma (lemma F.4), completeness (theorem F.5), the Lovász–Szegedy compactness theorem (theorem F.6), the uniqueness of a graphon given its homomorphism densities (theorem F.11), and the equivalence of cut convergence and left-convergence (theorem F.12).

F.8 The Hausdorff moment problem

Remark 16.23 identifies the *rank-one* sector of the graph moment problem with the classical moment problem on the bounded interval $[0,1] = [0,1]$: a rank-one graphon $W(x,y) = f(x)f(y)$ has $t(F,W) = \prod_{i\in V(F)} c_{\deg_F(i)}$ with $c_d = \int_0^1 f^d$, and the realizable degree-moment sequences (c_d) are exactly the Hausdorff moment sequences. Unlike the general sufficiency direction of theorem 16.22, this bounded-interval shadow has a short self-contained proof, which we give here. It is the only piece of the moment characterization the chapter actually *uses* (in worked computation 16.24), so we owe it an in-book proof.

Definition F.13 (Finite differences; complete monotonicity)**.** For a real sequence $(c_d)_{d\geq 0}$ write the forward difference $(\Delta c)_d = c_{d+1} - c_d$ and its iterates $\Delta^k = \Delta \circ \Delta^{k-1}$, so that $(\Delta^k c)_d = \sum_{i=0}^k \binom{k}{i}(-1)^{k-i} c_{d+i}$. The sequence is *completely monotone* if

$$(-1)^k (\Delta^k c)_d \ \geq \ 0 \qquad \text{for all } d, k \geq 0.$$

Theorem F.14 (Hausdorff moment theorem)**.** *A real sequence $(c_d)_{d\geq 0}$ with $c_0 = 1$ is the moment sequence $c_d = \int_0^1 r^d \, \rho(\mathrm{d}r)$ of a (necessarily unique) Borel probability measure ρ on* $[0,1]$ if and only if *it is completely monotone. In that case (c_d) is nonincreasing with $0 \leq c_d \leq 1$, and every Hankel matrix $(c_{i+j})_{i,j=0}^N$ is positive semidefinite.*

Proof. Necessity. If $c_d = \int_0^1 r^d \, \rho(\mathrm{d}r)$ then, since Δ acting on $r^{\cdot}$ multiplies by $(r-1)$,

$$(-1)^k (\Delta^k c)_d = \int_0^1 r^d \, (1-r)^k \, \rho(\mathrm{d}r) \ \geq \ 0,$$

the integrand being nonnegative on $[0,1]$; this is complete monotonicity. Taking $k = 1$ gives $c_d - c_{d+1} = \int r^d (1-r) \, \rho \geq 0$, so (c_d) is nonincreasing, and $0 \leq c_d \leq c_0 = 1$. Positive semidefiniteness of the Hankel matrix is the Gram identity $\sum_{i,j} a_i a_j c_{i+j} = \int_0^1 \big(\sum_i a_i r^i\big)^2 \rho(\mathrm{d}r) \geq 0$ (the rank-one shadow of proposition 16.21).

Sufficiency. Assume (c_d) is completely monotone with $c_0 = 1$. For $0 \leq j \leq n$ set

$$p_{n,j} \ := \ \binom{n}{j} (-1)^{n-j} (\Delta^{n-j} c)_j \ \geq \ 0,$$

nonnegativity being exactly complete monotonicity. Writing E for the shift $(Ec)_d = c_{d+1}$ and $\Delta = E - I$, we have $(-1)^{n-j} \Delta^{n-j} = (I-E)^{n-j}$, so the binomial theorem in the commuting operators E, I gives

$$\sum_{j=0}^n \binom{n}{j} E^j (I-E)^{n-j} = \big(E + (I-E)\big)^n = I;$$

evaluating the corresponding sequence at index 0 yields $\sum_{j=0}^n p_{n,j} = c_0 = 1$. Hence

$$\rho_n \ := \ \sum_{j=0}^n p_{n,j} \, \delta_{j/n}$$

is a Borel probability measure on $[0,1]$.

Factorial moments of ρ_n. We claim that for every $0 \leq m \leq n$,

$$\sum_{j=0}^n \binom{j}{m} p_{n,j} \ = \ \binom{n}{m} c_m. \tag{F.5}$$

Indeed, using $\binom{j}{m}\binom{n}{j} = \binom{n}{m}\binom{n-m}{j-m}$ and substituting $i = j - m$, $N = n - m$,

$$\sum_j \binom{j}{m} p_{n,j} = \binom{n}{m} \sum_{i=0}^N \binom{N}{i} (-1)^{N-i} (\Delta^{N-i} c)_{m+i} = \binom{n}{m} c_m,$$

the last step being the operator identity above applied to the shifted sequence $d \mapsto c_{m+d}$ (whose value at index 0 is c_m). Dividing (F.5) by $\binom{n}{m}$ and using $(n)_m = m!\binom{n}{m}$,

$$\sum_{j=0}^{n} \frac{(j)_m}{(n)_m}\, p_{n,j} \;=\; c_m \qquad (0 \le m \le n),$$

where $(x)_m = x(x-1)\cdots(x-m+1)$.

Ordinary moments converge to (c_d). Fix $d \ge 1$ and let $n > d$. The d-th moment of ρ_n is $m_d(\rho_n) = \sum_j (j/n)^d p_{n,j}$. We compare $(j/n)^d$ with the factorial ratio $\frac{(j)_d}{(n)_d} = \prod_{i=0}^{d-1} \frac{j-i}{n-i}$. Each factor obeys $0 \le \frac{j}{n} - \frac{j-i}{n-i} = \frac{i(n-j)}{n(n-i)} \le \frac{i}{n-i} \le \frac{i}{n-d}$, and for products of $[0,1]$-valued factors $|\prod a_i - \prod b_i| \le \sum_i |a_i - b_i|$, whence

$$\big|\, (j/n)^d - \tfrac{(j)_d}{(n)_d} \,\big| \;\le\; \sum_{i=0}^{d-1} \frac{i}{n-d} \;=\; \frac{\binom{d}{2}}{n-d},$$

uniformly in j. Since ρ_n is a probability measure,

$$|\, m_d(\rho_n) - c_d \,| = \left| \sum_j \Big((j/n)^d - \tfrac{(j)_d}{(n)_d} \Big) p_{n,j} \right| \;\le\; \frac{\binom{d}{2}}{n-d} \;\xrightarrow[n\to\infty]{}\; 0.$$

Thus $m_d(\rho_n) \to c_d$ for every fixed $d \ge 0$ (the case $d = 0$ being $m_0(\rho_n) = 1 = c_0$).

Passage to the limit. The measures ρ_n are supported in the compact interval $[0,1] \subset \mathbb{R}$, so the family is tight with second moments bounded by 1; by the $\mathcal{W}_1$-compactness criterion (theorem D.12) it is relatively compact in $(\mathcal{P}_1, \mathcal{W}_1)$, and $\mathcal{W}_1$ metrises weak convergence (theorem D.11). Extract a subsequence $\rho_{n_\ell} \to \rho$ weakly, with ρ a Borel probability measure; as each ρ_{n_ℓ} is carried by the closed set $[0,1]$, so is ρ. For each fixed d the function $r \mapsto r^d$ is bounded and continuous on $[0,1]$, so weak convergence gives $\int_0^1 r^d\, \rho(\mathrm{d}r) = \lim_\ell m_d(\rho_{n_\ell}) = c_d$. Hence $c_d = \int_0^1 r^d\, \rho(\mathrm{d}r)$ for all $d \ge 0$, as required.

Uniqueness. If ρ, ρ' are probability measures on $[0,1]$ with the same moments, then $\int p\, \mathrm{d}\rho = \int p\, \mathrm{d}\rho'$ for every polynomial p; by the Weierstrass approximation theorem polynomials are uniformly dense in $C[0,1]$, so $\int \phi\, \mathrm{d}\rho = \int \phi\, \mathrm{d}\rho'$ for all $\phi \in C[0,1]$, and a Borel measure on a compact metric space is determined by its integrals against continuous functions. Hence $\rho = \rho'$. □

Remark F.15 (The rank-one realization used in worked computation 16.24)**.** Given a Hausdorff moment sequence (c_d) with $c_0 = 1$, theorem F.14 produces a probability measure ρ on $[0,1]$ with $c_d = \int_0^1 r^d\, \rho(\mathrm{d}r)$. Let $G_\rho(t) = \inf\{r : \rho([0,r]) \ge t\}$ be the quantile (generalized inverse) of ρ; then $f := G_\rho \colon [0,1] \to [0,1]$ is measurable, nondecreasing, and pushes Lebesgue measure forward to ρ, so $c_d = \int_0^1 f(t)^d\, \mathrm{d}t$. The rank-one graphon $W(x,y) = f(x)f(y)$ then realizes the profile (16.3), $t(F,W) = \prod_{i \in V(F)} c_{\deg_F(i)}$. This is the in-book justification for the "exactly the Hausdorff moment condition" claim of remark 16.23 and worked computation 16.24, with no external appeal.

Bibliographic notes

The cut norm, the weak regularity lemma in the form used here, the sampling lemma, the compactness theorem, and the equivalence of cut and left convergence are due to Lovász and Szegedy and to Borgs, Chayes, Lovász, Sós, and Vesztergombi; see [100] and the comprehensive treatment in [99], whose Chapters 9–11 contain the sharp constants and the $O(1/\sqrt{\log k})$ sampling rate quoted in remark F.8. The energy-increment proof of weak regularity follows Frieze and Kannan; the completeness argument via weak-$*$ compactness of the L^∞ ball is the functional-analytic route emphasized in chapter 2. The Hausdorff moment theorem (theorem F.14) is classical; the constructive proof through the binomial "moment-difference" measures ρ_n is Hausdorff's own, and the complete-monotonicity criterion is standard in any treatment of the moment problem (see, e.g., [99, Ch. 7] for the graph-limit connection used in chapter 16).

Appendix G

Convergence of N-Player Equilibria: The Compactness Route

This appendix supplies the in-book proof of the qualitative convergence theorem theorem 13.9: limit points of (approximate) N-player Nash equilibria are mean field equilibria, and under Lasry–Lions monotonicity the limit is the unique MFG solution and the system propagates chaos. chapter 13 states the theorem and names the four moving parts of its proof — relaxed-control compactness, path-space tightness, martingale-optimality identification, and uniqueness — but defers the argument here so as not to interrupt its narrative. Nothing below is assumed from outside the book: tightness rests on the Kolmogorov–Chentsov estimate of theorem C.9 and Prokhorov's theorem (theorem 3.16); the martingale formulation rests on Itô's formula (theorem 5.9); uniqueness is imported from the monotonicity theory of chapter 11. External references (Lacker [89, 90], Fischer [67], Carmona–Delarue [40, Ch. 6]) are for attribution only.

We work with the N-player data of eq. (13.2)–eq. (13.3) under the standing assumptions (A-Lip), (A-Growth), (A-Nondeg), (A-Conv), with the coefficients b, σ, L, g continuous, the action set $A \subseteq \mathbb{R}^k$ closed, and the running cost coercive in the control ($L(x, m, \alpha) \geq c_0|\alpha|^q - c_1$ for some $q > 1$, $c_0 > 0$, part of (A-Growth)). Throughout, W denotes Brownian motion (no graphon is in scope), $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$ is the common initial law, and $\mathcal{P}(\mathcal{X})$ carries the topology of weak convergence on a Polish space $\mathcal{X}$.

G.1 Relaxed controls and the canonical path space

The obstruction to a direct limit is that the space of *strict* controls $\alpha : [0, T] \to A$ is not compact: a minimizing sequence of controls may oscillate ever faster and fail to converge. The standard remedy — due to Young, and the technical backbone of the compactness method — is to enlarge the control to a *measure* on the action set at each time, which restores compactness while leaving the value of every linear-in-control quantity unchanged.

Definition G.1 (Relaxed control)**.** A *relaxed control* is a measure ρ on $[0, T] \times A$ whose time-marginal is Lebesgue measure, i.e. $\rho(\mathrm{d}t, \mathrm{d}\alpha) = \rho_t(\mathrm{d}\alpha)\,\mathrm{d}t$ for a measurable family of probability measures $(\rho_t)_{t\in[0,T]}$ on A. We write $\mathcal{R}$ for the set of relaxed controls with $\int_0^T \int_A |\alpha|^q \, \rho_t(\mathrm{d}\alpha)\ \mathrm{d}t \le \Lambda$ for a fixed but arbitrary $\Lambda < \infty$, topologized by weak convergence of measures on $[0,T]\times A$ (equivalently, stable convergence). A strict control $\alpha(\cdot)$ embeds as $\rho_t = \delta_{\alpha(t)}$.

In relaxed form the controlled dynamics and cost read, with the coefficients averaged against ρ_t,

$$\mathrm{d}X_t = \Big(\int_A b(X_t, m_t, \alpha)\, \rho_t(\mathrm{d}\alpha)\Big)\, \mathrm{d}t + \sigma(X_t, m_t)\ \mathrm{d}W_t, \tag{G.1}$$

$$J(\rho) = \mathbb{E}\bigg[\int_0^T \int_A L(X_t, m_t, \alpha)\, \rho_t(\mathrm{d}\alpha)\ \mathrm{d}t + g(X_T, m_T)\bigg]. \tag{G.2}$$

Because b enters eq. (G.1) linearly and L enters eq. (G.2) linearly in ρ_t, no relaxed control beats the infimum over strict controls (the *chattering lemma*: a strict control may be chosen with cost within any ε of a relaxed one), so passing to relaxed controls changes no value.

Lemma G.2 (Compactness of relaxed controls)**.** *If A is compact, $\mathcal{R}$ is compact and metrizable for the topology of definition G.1. If A is merely closed, then the sublevel set $\mathcal{R}_\Lambda = \{\rho : \int_0^T \int |\alpha|^q \rho_t(\mathrm{d}\alpha)\, \mathrm{d}t \le \Lambda\}$ is compact for each $\Lambda < \infty$.*

Proof. For compact A the space $[0,T]\times A$ is compact Polish, so $\mathcal{P}([0,T]\times A)$ is compact for the weak topology (theorem 3.16 with a trivially tight family on a compact space). The constraint that the time-marginal be Lebesgue measure is closed under weak convergence (it is the intersection of the closed conditions $\int f(t)\, \rho(\mathrm{d}t, A) = \int_0^T f$ over a countable convergence-determining set of $f \in C([0,T])$), so $\mathcal{R}$ is a closed subset of a compact set, hence compact; metrizability is that of $\mathcal{P}(\cdot)$ on a compact Polish space. For closed A, the moment bound $\int |\alpha|^q \rho_t\ \mathrm{d}t \le \Lambda$ is a tightness condition in the A-direction (Markov's inequality: $\rho(\{|\alpha| > R\}) \le \Lambda/R^q$ uniformly), so $\mathcal{R}_\Lambda$ is tight on $[0,T]\times A$; combined with the closedness of the Lebesgue-marginal constraint and Prokhorov, it is compact. $\square$

The natural state space for the limit is the set of probability measures on the *canonical path–control space*

$$\Omega^{\mathrm{can}} \ := \ C([0,T];\mathbb{R}^d) \times \mathcal{R},$$

a Polish space; the coordinate process is (X,ρ) with X the continuous path and ρ the relaxed control. An N-player equilibrium induces, for each agent, a law on Ω^{can}; symmetry lets us encode the whole configuration through the *empirical measure on path space* $\boldsymbol{\mu}^N = \frac{1}{N}\sum_i \delta_{(X^i,\rho^i)} \in \mathcal{P}(\Omega^{\mathrm{can}})$, whose law $P^N = \mathcal{L}(\boldsymbol{\mu}^N)$ is the object theorem 13.9 calls tight.

G.2 Path-space tightness

Tightness of (P^N) is *not* a consequence of moment bounds alone: P^N lives on laws over the path space $C([0,T];\mathbb{R}^d)$, and tightness there is an equicontinuity (Arzelà–Ascoli) statement, which the Kolmogorov–Chentsov estimate supplies. We first record the criterion, then verify its hypothesis from the dynamics.

Proposition G.3 (Kolmogorov–Chentsov tightness on path space)**.** *Let $(Y^n)_n$ be continuous $\mathbb{R}^d$-valued processes on $[0,T]$ with laws $\mu_n \in \mathcal{P}(C([0,T];\mathbb{R}^d))$. Suppose*

1. *the initial laws $\mathcal{L}(Y_0^n)$ are tight on $\mathbb{R}^d$; and*
2. *there are constants $p > 0$, $\gamma > 0$, $C < \infty$, independent of n, with*

$$\mathbb{E}\,|Y_t^n - Y_s^n|^p \;\le\; C\,|t-s|^{1+\gamma} \qquad \text{for all } s,t \in [0,T],\; n. \tag{G.3}$$

Then (μ_n) is tight on $C([0,T];\mathbb{R}^d)$.

Proof. By Markov's inequality the increment bound eq. (G.3) controls the modulus of continuity uniformly in n: the quantitative Kolmogorov–Chentsov estimate (theorem C.9, in its quantitative form) gives, for each $\alpha < \gamma/p$ and $\eta > 0$, a constant K with $\mathbb{E}\big[\sup_{|t-s|\le h} |Y_t^n - Y_s^n|/h^\alpha\big] \le K$ uniformly in n and h, so that $\sup_n \mathbb{P}\big(\sup_{|t-s|\le h} |Y_t^n - Y_s^n| > \varepsilon\big) \to 0$ as $h \to 0$ for each $\varepsilon > 0$. This equicontinuity-in-probability, together with the tightness of the initial marginals (1), is exactly the hypothesis of the Arzelà–Ascoli characterization of compact sets in $C([0,T];\mathbb{R}^d)$: for every δ there is a compact $K_\delta \subset C([0,T];\mathbb{R}^d)$ — a set of uniformly bounded, uniformly equicontinuous paths — with $\inf_n \mu_n(K_\delta) \ge 1-\delta$. Hence (μ_n) is tight. ☐

Lemma G.4 (The increment bound holds)**.** *For the equilibrium state processes $(X^{i,N})$ of the N-player game under any ε_N-Nash profile with $\sup_N \varepsilon_N < \infty$, the uniform fourth moment $\sup_{N,i} \mathbb{E} \sup_{t\le T} |X_t^{i,N}|^4 < \infty$ holds, and the increment bound eq. (G.3) holds with $p = 4$, $\gamma = 1$, uniformly in N, i. Consequently (P^N) is tight on $\mathcal{P}(\Omega^{\mathrm{can}})$.*

Proof. Moment bound. An ε_N-Nash deviator can secure at most the no-control cost, so coercivity of L in α ((A-Growth)) bounds $\mathbb{E}\int_0^T\!\int |\alpha|^q \rho_t^i \,\mathrm{d}t$ uniformly in N, i; the linear growth of b, σ then yields $\sup_{N,i} \mathbb{E}\sup_t |X_t^{i,N}|^4 < \infty$ by Itô (theorem 5.9), the Burkholder–Davis–Gundy inequality (theorem C.2) and Grönwall (appendix C.1), exactly as in the SDE estimate of theorem 5.10, the initial fourth moment being finite because $m_0 \in \mathcal{P}_2(\mathbb{R}^d)$ is strengthened to a finite moment of order > 2 under (A-Growth).

Increment bound. For $s < t$, $X_t^{i,N} - X_s^{i,N} = \int_s^t \bar b_r^i \,\mathrm{d}r + \int_s^t \sigma(X_r^{i,N}, \mu_r^N)\,\mathrm{d}W_r^i$, where $\bar b_r^i = \int b(X_r^{i,N}, \mu_r^N, \alpha)\rho_r^i(\mathrm{d}\alpha)$. By linear growth and the moment bound, $\mathbb{E}\big|\int_s^t \bar b_r^i \,\mathrm{d}r\big|^4 \le |t-s|^3 \mathbb{E}\int_s^t |\bar b_r^i|^4 \,\mathrm{d}r \le C|t-s|^4$ (Jensen/Hölder), and by BDG and the boundedness of σ on the relevant range, $\mathbb{E}\big|\int_s^t \sigma \,\mathrm{d}W^i\big|^4 \le C\,\mathbb{E}\big(\int_s^t |\sigma|^2 \,\mathrm{d}r\big)^2 \le C|t-s|^2$. Hence $\mathbb{E}|X_t^{i,N} - X_s^{i,N}|^4 \le C|t-s|^2$, i.e. eq. (G.3) with $p = 4$, $\gamma = 1$, uniformly in N, i.

Tightness of P^N. The control components live in the compact (or Λ-sublevel) set $\mathcal{R}$ of lemma G.2, which is tight by construction; the path components are tight by proposition G.3. A product of tight families is tight, so the single-agent laws are tight on Ω^{can}, and by exchangeability and a standard de Finetti/Sznitman argument (the marginals of $\boldsymbol{\mu}^N$ are the single-agent laws, and tightness of marginals on a Polish space gives tightness of the empirical-measure laws, theorem 3.16) the family $P^N = \mathcal{L}(\boldsymbol{\mu}^N)$ is tight on $\mathcal{P}(\Omega^{\mathrm{can}})$. ☐

G.3 The limiting martingale problem

We now say what a limit point *is*. The right notion of solution is a probability measure on Ω^{can} that (i) is consistent with the dynamics, encoded as a martingale problem, (ii) places its state-marginal flow at the population it interacts with (the fixed-point/consistency constraint), and (iii) is optimal for the representative agent. This is the relaxed/weak mean field equilibrium that theorem 13.9 refers to.

Definition G.5 (Weak (relaxed) mean field equilibrium)**.** Let $m = (m_t)_{t\in[0,T]}$ be a continuous flow in $\mathcal{P}_2(\mathbb{R}^d)$ and let $Q \in \mathcal{P}(\Omega^{\mathrm{can}})$ with coordinate process (X, ρ). The pair (Q, m) is a *weak mean field equilibrium* if:

1. (Dynamics.) For every $\phi \in C_b^2(\mathbb{R}^d)$, the process

$$M_t^\phi \;:=\; \phi(X_t)-\phi(X_0)-\int_0^t\!\!\int_A \Big(b(X_r, m_r, \alpha)\cdot\nabla\phi(X_r)+\tfrac{1}{2}\,\mathrm{tr}\,\big(\sigma\sigma^\top(X_r, m_r)\,\mathrm{D}^2\,\phi(X_r)\big)\Big)\rho_r(\mathrm{d}\alpha)\;\mathrm{d}r \tag{G.4}$$

   is a Q-martingale in the canonical filtration, and $\mathcal{L}_Q(X_0) = m_0$;

2. (Consistency.) $\mathcal{L}_Q(X_t) = m_t$ for all $t \in [0, T]$;

3. (Optimality.) ρ minimizes eq. (G.2) against the frozen flow m among all relaxed controls adapted to (X, W); equivalently, Q-almost surely ρ_r is supported on the Hamiltonian maximizers $\alpha^*(X_r, \nabla u(r, X_r), m_r)$ of eq. (13.5), where u solves the HJB equation against m.

When $\rho_r = \delta_{\widehat{\alpha}(r,X_r)}$ is a Dirac feedback, (Q, m) is the ordinary (strong) MFG solution of chapters 8 and 9.

Condition (1) is the relaxed controlled martingale problem; that it is equivalent to X solving eq. (G.1) on a (possibly enlarged) space is the standard martingale-problem/weak-existence equivalence, which under (A-Nondeg) follows from Itô's formula (theorem 5.9) one way and from a representation theorem the other; we use only the martingale direction, which is closed under weak limits.

G.4 Identification of limit points

Lemma G.6 (Limit points are weak MFG equilibria)**.** *Assume* (A-Lip), (A-Growth), (A-Nondeg), (A-Conv) *with continuous coefficients, and let $\boldsymbol{\alpha}^{*,N}$ be an ε_N-Nash equilibrium with $\varepsilon_N \to 0$. Let P^∞ be any weak limit point of (P^N) along a subsequence (which exists by lemma G.4), and let Π be the (random) empirical measure under P^∞, a $\mathcal{P}(\Omega^{\mathrm{can}})$-valued random variable. Then P^∞-almost surely* Π *is the law of a weak mean field equilibrium* (Π, m^Π) *in the sense of definition G.5, with $m_t^\Pi = (\mathrm{ev}_t)_\# \Pi$ its state-marginal flow.*

Proof. Fix a subsequence along which $P^N \Rightarrow P^\infty$. Three properties pass to the limit.

(i) Dynamics. For $\phi \in C_b^2(\mathbb{R}^d)$ and $0 \le s < t \le T$ and a bounded continuous functional G of $(X_r, \rho)_{r \le s}$, the map $\nu \mapsto \mathbb{E}_\nu[(M_t^\phi - M_s^\phi)G]$ — with M^ϕ as in eq. (G.4) but with m replaced by the state-marginal flow of ν — is bounded and continuous on $\mathcal{P}(\Omega^{\mathrm{can}})$, because $b, \sigma, \nabla\phi, \mathrm{D}^2\,\phi$ are continuous and the moment bound of lemma G.4 gives uniform integrability (the increments eq. (G.4) are uniformly integrable by the fourth-moment bound). Under each P^N the agent-i marginal makes M^ϕ a martingale up to an $o(1)$ error from μ^N versus $\mu^{N,-i}$ (one atom, eq. (13.11)); passing to the limit, $\mathbb{E}_{P^\infty}[\,\mathbb{E}_\Pi[(M_t^\phi - M_s^\phi)G]\,] = 0$ for all such G, i.e. M^ϕ is a Π-martingale P^∞-a.s. This is condition (1) of definition G.5.

(ii) Consistency. By construction the state-marginal flow of Π is the limit of the empirical-measure flow, so $m_t^\Pi = (\mathrm{ev}_t)_\# \Pi$ and the interaction in (i) is against m^Π itself; this is condition (2).

(iii) Optimality. Here the $\varepsilon_N \to 0$ hypothesis does its work. Each agent is ε_N-optimal against the field it sees; the value of the best response against a *fixed* continuous flow is, by the stability of the control problem under (A-Lip)/(A-Nondeg), a continuous functional of the flow, and the cost $J(\cdot)$ is lower semicontinuous in the relaxed control (eq. (G.2), with L continuous and bounded below). Passing to the limit in the ε_N-optimality inequality — asymptotic optimality becomes exact in the limit — gives that Π-a.s. the relaxed control ρ attains the representative-agent infimum against m^Π. By the verification theorem of chapter 6 (and (A-Conv), making the Hamiltonian maximizer unique) this forces ρ_r onto the maximizer $\alpha^*(X_r, \nabla u, m_r^\Pi)$, condition (3).

Thus every limit point is supported on weak mean field equilibria. □

G.5 Uniqueness, determinism, and chaos

The compactness argument identifies the support of the limit but not its *value*: a priori several weak equilibria could carry the limit mass, and the empirical measure could remain random in the limit (this is exactly the closed-loop subtlety of remark 13.10). Monotonicity removes the ambiguity.

Theorem G.7 (Uniqueness collapses the limit)**.** *Add the Lasry–Lions monotonicity* (A-Mon) *of chapter 11. Then the weak mean field equilibrium of definition G.5 is unique, its state flow m coincides with the unique MFG solution of chapters 8 and 9, and the relaxed control is the Dirac feedback $\delta_{\widehat{\alpha}}$ of the strong solution.*

Proof. Two weak equilibria (Q, m), (Q', m') each induce, via the optimality condition (3) and the verification theorem of chapter 6, an HJB value u (resp. u') against the frozen flow m (resp. m'), and the consistency condition (2) makes the state flow the law of the optimally controlled process — i.e. each pair solves the MFG system eq. (12.2). The Lasry–Lions computation of chapter 11 (pairing the difference of the two HJB equations with the difference of the two Fokker–Planck equations and invoking the monotonicity inequality eq. (11.3)) gives $m = m'$ and $u = u'$; in particular the equilibrium control is the deterministic feedback $\widehat{\alpha}$, so the relaxed control degenerates to a Dirac. Uniqueness of the weak equilibrium follows. □

Theorem G.8 (Qualitative convergence and propagation of chaos)**.** *Under* (A-Lip)*,* (A-Growth)*,* (A-Nondeg)*,* (A-Conv) *with continuous coefficients, the family* (P^N) *of lemma G.4 is tight and every limit point is supported on weak mean field equilibria. If in addition* (A-Mon) *holds, then* $\mu^N \to m$ *in distribution, the limit being the* deterministic *flow of the unique MFG solution, and the finite marginals propagate chaos:* $\mathcal{L}(X_t^{1,N}, \dots, X_t^{k,N}) \to m_t^{\otimes k}$ *weakly for each fixed* k*. This is theorem 13.9.*

Proof. Tightness is lemma G.4; the support of every limit point is identified by lemma G.6. Under (A-Mon), theorem G.7 shows the support is the single deterministic equilibrium m. A sequence of $\mathcal{P}(\Omega^{\mathrm{can}})$-valued random variables whose every subsequential limit is the constant $\delta_{(\,\cdot\,)}$ supported at one deterministic flow converges in distribution to that constant; convergence in distribution to a constant is convergence in probability, so $\mu^N \to m$ in probability, hence in distribution. Finally, convergence of the empirical measure to a deterministic limit is equivalent, for exchangeable systems, to chaoticity of the finite marginals (definition 7.11 and the equivalence recorded in chapter 7), which is the stated k-marginal convergence. □

Remark G.9 (Why this route gives no rate)**.** The argument is qualitative by nature: compactness produces a convergent subsequence but no modulus of convergence, and the identification step uses only continuity, not any quantitative stability. A rate requires the master-equation route of section 13.5.3, which trades the minimal hypotheses here for the strong regularity of theorem 12.8 and obtains the $1/N$ value rate eq. (13.22). The trade-off — minimal assumptions and no rate versus strong assumptions and a sharp rate — is the organizing dichotomy of section 13.5.

Bibliographic notes

The relaxed-control compactness method for mean field games is due to Lacker [89] (closed-loop and the general martingale-problem formulation) and [90] (the closed-loop limit and the larger set of attainable weak equilibria), with the open-loop case treated by Fischer [67]; an exposition is Carmona and Delarue [40, Vol. I, Ch. 6]. The relaxation of controls originates with Young; the martingale-problem viewpoint is that of Stroock and Varadhan, here specialized to the McKean–Vlasov setting. The path-space tightness criterion of proposition G.3 is the Kolmogorov–Chentsov/Arzelà–Ascoli route proved in-book at theorem C.9; the uniqueness collapse is the Lasry–Lions monotonicity theory of chapter 11.

Appendix H

Poisson Point Processes: Definitions and the Calculus We Use

This appendix supplies, from first principles, the point-process facts that chapter 27 uses in its treatment of sparse and graphex limits: the definition of a Poisson point process, the first-order Campbell formula and the two-point Mecke / factorial moment formula, the mapping, thinning, marking and superposition theorems, the Poisson limit theorem (law of rare events) for triangular arrays of independent indicators, and the mixed-Poisson tail-transfer lemma that turns a heavy-tailed mixing law into a heavy-tailed count. None of this is assumed of the reader beyond undergraduate probability and the basic Lebesgue theory of chapter 1; everything is proved here. The standard textbook references, used only for attribution and further reading, are [95, 84] for the point-process calculus and [24] for regular variation.

Throughout, $(E, \mathcal{E})$ is a measurable space equipped with a σ-finite measure λ (the *intensity*); in the applications $E = \mathbb{R}_+$ or $E = \mathbb{R}_+^2$ with λ Lebesgue. A *point measure* on E is a measure of the form $\eta = \sum_i \delta_{p_i}$ with $\eta(A) \in \{0, 1, 2, \ldots, \infty\}$ for every $A \in \mathcal{E}$, i.e. a (multi)set of points $\{p_i\} \subset E$ counted with multiplicity; we write $\eta(A)$ for the number of points in A and $\int f \, d\eta = \sum_i f(p_i)$.

H.1 Definition and Campbell's formula

Definition H.1 (Poisson point process)**.** A *Poisson point process* (PPP) on E with σ-finite intensity λ is a random point measure Π such that

1. for every $A \in \mathcal{E}$ with $\lambda(A) < \infty$, the count $\Pi(A)$ has the Poisson law $\mathrm{Poisson}(\lambda(A))$ (and $\Pi(A) = \infty$ a.s. if $\lambda(A) = \infty$);
2. for every finite family of pairwise disjoint sets $A_1, \ldots, A_n \in \mathcal{E}$, the counts $\Pi(A_1), \ldots, \Pi(A_n)$ are independent.

We call Π a *unit-rate* PPP when $E = \mathbb{R}_+^d$ and λ is Lebesgue measure.

Existence is standard: on a set of finite intensity one draws a Poisson($\lambda(A)$) number of points i.i.d. from the normalized intensity $\lambda(\cdot \cap A)/\lambda(A)$, and a σ-finite E is exhausted by a disjoint countable union of finite-intensity sets on which independent such processes are superposed (theorem H.7). We take existence as given and record only the calculus.

Theorem H.2 (Campbell's formula; first-order Mecke)**.** *Let* Π *be a PPP on* E *with intensity* λ*, and let* $f \colon E \to [0, \infty]$ *be measurable. Then*

$$\mathbb{E}[*] \int_E f \,\mathrm{d}\Pi = \mathbb{E}[*] \sum_{p \in \Pi} f(p) = \int_E f \,\mathrm{d}\lambda. \tag{H.1}$$

For signed or complex f *the identity holds whenever* $\int |f| \,\mathrm{d}\lambda < \infty$*.*

Proof. For an indicator $f = \mathbf{1}_A$ with $\lambda(A) < \infty$, the left side is $\mathbb{E}[\Pi(A)] = \lambda(A)$ because $\Pi(A) \sim$ Poisson($\lambda(A)$), so (H.1) holds; if $\lambda(A) = \infty$ both sides are $+\infty$. By linearity of expectation and of the integral it holds for nonnegative simple functions $f = \sum_k c_k \mathbf{1}_{A_k}$. A general measurable $f \geq 0$ is an increasing pointwise limit of simple functions $f_n \uparrow f$; both sides of (H.1) pass to the limit by the monotone convergence theorem (applied to the expectation on the left and to the integral on the right). The signed case follows by splitting $f = f^+ - f^-$ under $\int |f| \,\mathrm{d}\lambda < \infty$. □

The second-order analogue is what powers the edge count of proposition 27.14. We phrase it as the statement that the *second factorial moment measure* of a PPP is the product $\lambda \otimes \lambda$.

Theorem H.3 (Two-point Mecke / factorial moment formula)**.** *Let* Π *be a PPP on* E *with intensity* λ *and let* $h \colon E \times E \to [0, \infty]$ *be measurable. Writing* $\sum^{\neq}_{p,p' \in \Pi}$ *for the sum over* ordered *pairs of* distinct *points of* Π*,*

$$\mathbb{E}[*] \sum^{\neq}_{p,p' \in \Pi} h(p, p') = \int_E \int_E h(p, p') \,\lambda(\mathrm{d}p) \,\lambda(\mathrm{d}p'). \tag{H.2}$$

If h *is symmetric, the sum over* unordered *pairs* $\{p, p'\}$ *equals half the left side, so* $\mathbb{E}\Big[\sum_{\{p,p'\} \subset \Pi} h\Big] = \frac{1}{2} \iint h \,\mathrm{d}\lambda \,\mathrm{d}\lambda$*.*

Proof. First take $h = \mathbf{1}_{A \times B}$ with $\lambda(A), \lambda(B) < \infty$. If $A \cap B = \varnothing$, then by the independence of $\Pi(A)$ and $\Pi(B)$ (definition H.1(2)),

$$\mathbb{E}[*] \sum^{\neq}_{p,p'} \mathbf{1}_A(p) \mathbf{1}_B(p') = \mathbb{E}[\Pi(A)\,\Pi(B)] = \mathbb{E}[\Pi(A)]\,\mathbb{E}[\Pi(B)] = \lambda(A)\lambda(B),$$

since for ordered distinct pairs with $p \in A$, $p' \in B$ and A, B disjoint there is no diagonal contribution. If $A = B$, the ordered distinct pairs number $\Pi(A)(\Pi(A) - 1)$, and for $Z \sim$ Poisson(μ) one has $\mathbb{E}[Z(Z-1)] = \mu^2$ (the second factorial moment of a Poisson law); hence $\mathbb{E}\Big[\sum^{\neq} \mathbf{1}_{A \times A}\Big] = \lambda(A)^2$. The two cases together give (H.2) for all rectangles $A \times B$ by additivity in each argument over a partition into disjoint pieces. Rectangles generate $\mathcal{E} \otimes \mathcal{E}$ and the two sides of (H.2) are measures in h that agree on this generating π-system of finite-intensity rectangles; by the monotone class / π–λ theorem they agree for all indicators, hence for simple $h \geq 0$ by linearity and for all measurable $h \geq 0$ by monotone convergence. The unordered statement is the ordered one divided by two, valid because each unordered pair $\{p, p'\}$ of distinct points is counted twice on the left of (H.2). □

H.2 Mapping, thinning, marking, superposition

These four operations are the structural toolkit. Each is proved by checking the two defining properties of definition H.1 directly.

Theorem H.4 (Mapping theorem). *Let Π be a PPP on E with intensity λ and let $T\colon E \to E'$ be measurable such that the pushforward $\lambda' := \lambda \circ T^{-1}$ is σ-finite. Then the image point process $T(\Pi) := \sum_{p\in\Pi} \delta_{T(p)}$ is a PPP on E' with intensity λ'.*

Proof. For $A' \in \mathcal{E}'$, $T(\Pi)(A') = \Pi(T^{-1}A')$, so it is $\mathrm{Poisson}(\lambda(T^{-1}A')) = \mathrm{Poisson}(\lambda'(A'))$, giving property (1). If $A'_1, \dots, A'_n$ are pairwise disjoint, their preimages $T^{-1}A'_1, \dots, T^{-1}A'_n$ are pairwise disjoint in E, so the counts $\Pi(T^{-1}A'_i)$ are independent by definition H.1(2), giving property (2). □

Theorem H.5 (Marking theorem). *Let $\Pi = \sum_i \delta_{p_i}$ be a PPP on E with intensity λ, and attach to each point p_i an independent* mark *$M_i \in F$ drawn from a probability kernel $K(p_i, \mathrm{d}m)$ (independently across points given Π). Then the marked process $\Pi^* = \sum_i \delta_{(p_i,M_i)}$ is a PPP on $E \times F$ with intensity $\lambda^*(\mathrm{d}p\ \mathrm{d}m) = \lambda(\mathrm{d}p)\, K(p, \mathrm{d}m)$.*

Proof. We verify definition H.1 for Π^*. For disjoint product sets it suffices, by a monotone class argument, to compute joint Laplace functionals. Condition on Π; for measurable $g\colon E \times F \to [0, \infty)$,

$$\mathbb{E}[*]\, e^{-\sum_i g(p_i,M_i)} \,|\, \Pi = \prod_i \int_F e^{-g(p_i,m)} K(p_i, \mathrm{d}m) = \prod_i e^{-\tilde{g}(p_i)},$$

where $\tilde{g}(p) := -\log \int_F e^{-g(p,m)} K(p, \mathrm{d}m)$. Taking expectations and using the Poisson Laplace functional $\mathbb{E}\Big[e^{-\sum_i \tilde{g}(p_i)}\Big] = \exp\big(-\int_E (1 - e^{-\tilde{g}})\, \mathrm{d}\lambda\big)$ (which is (H.1) exponentiated, equivalently the characteristic functional of a PPP, established by the same simple-function/MCT argument as theorem H.2 applied to $1 - e^{-\tilde{g}}$),

$$\mathbb{E}[*]\, e^{-\sum_i g(p_i,M_i)} = \exp\Big(-\int_E (1 - e^{-\tilde{g}(p)})\, \mathrm{d}\lambda(p)\Big) = \exp\Big(-\int_{E\times F} (1 - e^{-g(p,m)})\lambda(\mathrm{d}p)K(p, \mathrm{d}m)\Big),$$

which is exactly the Laplace functional of a PPP on $E \times F$ with intensity λ^*. A random point measure with the Poisson Laplace functional is a PPP, so Π^* is as claimed. □

Theorem H.6 (Thinning theorem). *Let Π be a PPP on E with intensity λ, and retain each point p independently with probability $\varpi(p) \in [0, 1]$ (a measurable* retention *function), deleting it otherwise. Then the retained points form a PPP with intensity $\varpi\, \lambda$, the deleted points form a PPP with intensity $(1 - \varpi)\lambda$, and the two are independent.*

Proof. Mark each point with $M \in \{0, 1\}$, $\mathbb{P}(M = 1 \mid p) = \varpi(p)$ (a Bernoulli kernel). By theorem H.5 the marked process is a PPP on $E \times \{0, 1\}$ with intensity $\lambda(\mathrm{d}p)(\varpi(p)\delta_1 + (1 - \varpi(p))\delta_0)$. The retained process is its restriction to $E \times \{1\}$, mapped to E by the projection; by theorem H.4 it is a PPP with intensity $\varpi\lambda$, and likewise the deleted process has intensity $(1 - \varpi)\lambda$. They are supported on the disjoint marks $\{1\}$ and $\{0\}$ of a single PPP, hence are counts over disjoint sets and so independent by definition H.1(2). □

Theorem H.7 (Superposition)**.** *If $\Pi_1, \Pi_2, \dots$ are independent PPPs on E with intensities $\lambda_1, \lambda_2, \dots$ and $\lambda := \sum_k \lambda_k$ is σ-finite, then $\Pi := \sum_k \Pi_k$ is a PPP with intensity λ.*

Proof. For A with $\lambda(A) < \infty$, $\Pi(A) = \sum_k \Pi_k(A)$ is a sum of independent $\text{Poisson}(\lambda_k(A))$ variables, hence $\text{Poisson}(\sum_k \lambda_k(A)) = \text{Poisson}(\lambda(A))$ by the additivity of the Poisson law under independent sums. Independence over disjoint sets is inherited from that of each Π_k together with the independence across k. □

H.3 The Poisson limit theorem (law of rare events)

The mixed-Poisson degree limits of propositions 27.2 and 27.14 rest on the convergence of a sum of many independent, individually rare indicators to a Poisson law. The sharp, fully quantitative statement is Le Cam's inequality, from which the limit theorem is an immediate corollary; both are proved by coupling.

Lemma H.8 (Le Cam's inequality)**.** *Let $X_1, \dots, X_n$ be independent Bernoulli variables with $\mathbb{P}(X_j = 1) = p_j \in [0,1]$, and let $S = \sum_{j=1}^n X_j$ and $\Lambda = \sum_{j=1}^n p_j$. Then the total-variation distance to a Poisson law obeys*

$$d_{\text{TV}}(\mathcal{L}(S),\, \text{Poisson}(\Lambda)) \le \sum_{j=1}^n p_j^2. \tag{H.3}$$

Proof. Total variation distance is subadditive under independent convolution: if $\mathcal{L}(S) = *_j \mathcal{L}(X_j)$ and $\text{Poisson}(\Lambda) = *_j \text{Poisson}(p_j)$ (the latter by Poisson additivity), then $d_{\text{TV}}(\mathcal{L}(S), \text{Poisson}(\Lambda)) \le \sum_j d_{\text{TV}}(\text{Bernoulli}(p_j), \text{Poisson}(p_j))$. It remains to bound a single term, and this we do exactly. Writing b_k for the Bernoulli(p) mass ($b_0 = 1-p$, $b_1 = p$, $b_k = 0$ for $k \ge 2$) and $\pi_k = e^{-p}p^k/k!$ for the Poisson(p) mass,

$$d_{\text{TV}}(\text{Bernoulli}(p), \text{Poisson}(p)) = \tfrac{1}{2}\sum_{k \ge 0} |b_k - \pi_k| = \tfrac{1}{2}\Big((e^{-p} - (1-p)) + (p - pe^{-p}) + (1 - e^{-p} - pe^{-p})\Big),$$

using $e^{-p} \ge 1-p$ (so $|b_0 - \pi_0| = e^{-p} - (1-p)$), $1 \ge e^{-p}$ (so $|b_1 - \pi_1| = p - pe^{-p}$), and $\sum_{k\ge 2} \pi_k = 1 - e^{-p} - pe^{-p}$. The three terms collapse to $\frac{1}{2}(2p - 2pe^{-p}) = p(1-e^{-p}) \le p^2$, the last step by $1 - e^{-p} \le p$. Summing over j yields (H.3). □

Theorem H.9 (Poisson limit theorem for triangular arrays)**.** *For each n let $X_{n,1}, \dots, X_{n,k_n}$ be independent Bernoulli variables with $\mathbb{P}(X_{n,j} = 1) = p_{n,j}$, and set $S_n = \sum_{j=1}^{k_n} X_{n,j}$. If*

$$\Lambda_n := \sum_{j=1}^{k_n} p_{n,j} \;\longrightarrow\; \Lambda \in [0,\infty) \qquad \textit{and} \qquad \max_{1 \le j \le k_n} p_{n,j} \;\longrightarrow\; 0, \tag{H.4}$$

then $S_n \Rightarrow \text{Poisson}(\Lambda)$ in distribution.

Proof. By lemma H.8, $d_{\text{TV}}(\mathcal{L}(S_n), \text{Poisson}(\Lambda_n)) \le \sum_j p_{n,j}^2 \le (\max_j p_{n,j}) \sum_j p_{n,j} = (\max_j p_{n,j})\Lambda_n \to 0$, using (H.4) (the factor Λ_n is bounded as it converges). And $\text{Poisson}(\Lambda_n) \Rightarrow \text{Poisson}(\Lambda)$ because the Poisson probability mass function $\mu \mapsto e^{-\mu}\mu^k/k!$ is continuous in μ for each k, so pointwise convergence of the mass functions holds, which for integer-valued laws is convergence in total variation hence in distribution. Combining, $d_{\mathrm{TV}}(\mathcal{L}(S_n), \mathrm{Poisson}(\Lambda)) \to 0$. □

H.4 Mixed-Poisson tails: heavy mixing transfers to heavy counts

The last ingredient turns a heavy-tailed *mean* into a heavy-tailed *count*. It is the precise statement behind the power-law degree exponent of corollary 27.3: a Poisson variable whose random mean has a Pareto / regularly-varying tail inherits exactly that tail index, because Poisson fluctuations are negligible against a heavy mean.

Recall that a measurable $\ell\colon (0,\infty) \to (0,\infty)$ is *slowly varying* if $\ell(ct)/\ell(t) \to 1$ as $t \to \infty$ for every $c > 0$, and a tail $\mathbb{P}(\Lambda > t) = t^{-\alpha}\ell(t)$ with $\alpha > 0$ and ℓ slowly varying is *regularly varying of index* $-\alpha$ [24]; the Pareto tail $\mathbb{P}(\Lambda > t) = Ct^{-\alpha}$ is the special case $\ell \equiv C$.

Lemma H.10 (Mixed-Poisson tail transfer)**.** *Let $\Lambda \geq 0$ be a random variable with regularly-varying tail $\mathbb{P}(\Lambda > t) = t^{-\alpha}\ell(t)$, $\alpha > 0$, ℓ slowly varying, and let N be, conditionally on Λ,* Poisson(Λ)*. Then*

$$\mathbb{P}(N \geq k) \;\sim\; \mathbb{P}(\Lambda > k) \qquad (k \to \infty), \tag{H.5}$$

so N has a regularly-varying survival function of the same index $-\alpha$.

Proof. Fix $\varepsilon \in (0,1)$. For the *lower* bound, restrict to large means:

$$\mathbb{P}(N \geq k) \;\geq\; \mathbb{P}(N \geq k,\ \Lambda > (1+\varepsilon)k) \;\geq\; \mathbb{P}(\Lambda > (1+\varepsilon)k) \cdot \inf_{\mu > (1+\varepsilon)k} \mathbb{P}(\mathrm{Poisson}(\mu) \geq k).$$

For $\mu > (1+\varepsilon)k$, Chebyshev's inequality applied to Poisson(μ) (mean and variance both μ) gives $\mathbb{P}(\mathrm{Poisson}(\mu) < k) \leq \mathbb{P}(|\mathrm{Poisson}(\mu) - \mu| > \mu - k) \leq \mu/(\mu-k)^2 \leq \mu/(\varepsilon k)^2$; this does not yet vanish uniformly, so split once more: for $\mu \in \big((1+\varepsilon)k,\, k^2\big]$ the bound is $\leq k^2/(\varepsilon k)^2 = \varepsilon^{-2}$ — useless — so instead use that $\mathbb{P}(\mathrm{Poisson}(\mu) \geq k)$ is increasing in μ and at $\mu = (1+\varepsilon)k$ equals $\mathbb{P}(\mathrm{Poisson}((1+\varepsilon)k) \geq k) \to 1$ as $k \to \infty$ by the law of large numbers (the standardized Poisson is asymptotically normal, and k is $\varepsilon\sqrt{(1+\varepsilon)k} \cdot (1+o(1))$ standard deviations below the mean). Hence the infimum is $1 - o(1)$, and by regular variation $\mathbb{P}(\Lambda > (1+\varepsilon)k) \sim (1+\varepsilon)^{-\alpha}\mathbb{P}(\Lambda > k)$, so $\liminf_k \mathbb{P}(N \geq k)/\mathbb{P}(\Lambda > k) \geq (1+\varepsilon)^{-\alpha}$.

For the *upper* bound, split on the same threshold:

$$\mathbb{P}(N \geq k) \;\leq\; \mathbb{P}(\Lambda > (1-\varepsilon)k) + \mathbb{E}[*]\,\mathbb{P}(\mathrm{Poisson}(\Lambda) \geq k)\,\mathbf{1}\{\Lambda \leq (1-\varepsilon)k\}.$$

For $\mu \leq (1-\varepsilon)k$ the Chernoff bound for the Poisson upper tail, $\mathbb{P}(\mathrm{Poisson}(\mu) \geq k) \leq e^{-\mu}(e\mu/k)^k$, is increasing in $\mu \leq k$ and so is maximized at $\mu = (1-\varepsilon)k$, where it equals $\big(e^{\varepsilon}(1-\varepsilon)\big)^k$. Since $e^{\varepsilon}(1-\varepsilon) = 1 - \frac{1}{2}\varepsilon^2 + O(\varepsilon^3) < 1$ for small $\varepsilon > 0$, this is geometrically small in k, hence $o\big(\mathbb{P}(\Lambda > k)\big)$ (any regularly-varying tail of negative index dominates a geometric). Therefore, again by regular variation, $\limsup_k \mathbb{P}(N \geq k)/\mathbb{P}(\Lambda > k) \leq (1-\varepsilon)^{-\alpha}$.

Letting $\varepsilon \downarrow 0$ in the two displays gives $\mathbb{P}(N \geq k)/\mathbb{P}(\Lambda > k) \to 1$, which is (H.5); the survival function of N is then regularly varying of index $-\alpha$ by the same comparison. □

For the explicit Pareto mixing law of corollary 27.3, $\mathbb{P}(\Lambda > t) = Ct^{-1/\nu}$, lemma H.10 gives directly $\mathbb{P}(\deg \geq k) \sim C\,k^{-1/\nu}$, the power-law degree tail with index $1/\nu$ and exponent $\gamma = 1 + 1/\nu$.

Appendix I

First-Order Primal–Dual Convergence: The Chambolle–Pock Bound

This appendix supplies the one convex-optimization result that chapter 28 uses but does not prove there: the ergodic $O(1/k)$ convergence of the first-order primal–dual (Chambolle–Pock) iteration on a convex–concave saddle problem. Theorem 28.8 invokes it for the discrete potential-MFG saddle of proposition 28.7; we prove it here in full so the chapter's unconditional-convergence claim rests on an in-book argument rather than on the external citation [47]. The proof is elementary convex analysis: the proximal inequality, a telescoping sum, and one Young estimate controlled by the step-size condition. We use only finite-dimensional Euclidean spaces, which is all the discretization of section 28.4 requires; the proximal map and Moreau envelope are those of chapter 8 (section 8.6).

I.1 The saddle problem and the iteration

Let X, Y be finite-dimensional Euclidean spaces with inner products $\langle \cdot, \cdot \rangle$ and norms $\|\cdot\|$, let $K\colon X \to Y$ be linear with operator norm $L = \|K\|$, and let $g\colon X \to (-\infty, +\infty]$ and $h\colon Y \to (-\infty, +\infty]$ be proper, convex, lower semicontinuous. We treat the saddle problem

$$\min_{x \in X} \max_{y \in Y} \Big\{ \mathfrak{L}(x, y) := \langle Kx, y \rangle + g(x) - h(y) \Big\}, \tag{I.1}$$

which is the convex–concave form of the primal problem $\min_x \{g(x) + h^*(Kx)\}$, with h^* the convex conjugate of h. We assume a saddle point $(\hat{x}, \hat{y})$ exists — guaranteed for the discrete MFG problem of proposition 28.7, where g collects the (coercive, strictly convex) kinetic-plus-interaction energy, h enforces the affine continuity constraint, and K is the discrete continuity operator. The *partial primal–dual gap* on a bounded set $B_1 \times B_2 \ni (\hat{x}, \hat{y})$ is

$$\mathrm{Gap}_{B_1 \times B_2}(x, y) := \max_{y' \in B_2} \mathfrak{L}(x, y') - \min_{x' \in B_1} \mathfrak{L}(x', y) \;\geq\; 0, \tag{I.2}$$

which vanishes exactly at a saddle point and dominates, e.g., the primal suboptimality.

Recall the proximal operator $\operatorname{prox}_{\tau g}(z) = \arg\min_x\{g(x) + \frac{1}{2\tau}\|x-z\|^2\}$, single-valued for proper convex lsc g and $\tau > 0$. The Chambolle–Pock iteration with steps $\tau, \sigma > 0$ and extrapolation parameter $\theta = 1$ is

$$y^{n+1} = \operatorname{prox}_{\sigma h}\left(y^n + \sigma K\bar{x}^n\right), \tag{I.3a}$$

$$x^{n+1} = \operatorname{prox}_{\tau g}\left(x^n - \tau K^* y^{n+1}\right), \tag{I.3b}$$

$$\bar{x}^{n+1} = 2x^{n+1} - x^n, \tag{I.3c}$$

started from arbitrary (x^0, y^0) with $\bar{x}^0 = x^0$.

I.2 The proximal inequality

The engine is the following first-order characterization of the prox.

Lemma I.1 (Prox inequality)**.** *Let φ be proper convex lsc, $\tau > 0$, and $u = \operatorname{prox}_{\tau\varphi}(z)$. Then for all x,*

$$\varphi(x) + \frac{1}{2\tau}\|x-z\|^2 \;\geq\; \varphi(u) + \frac{1}{2\tau}\|u-z\|^2 + \frac{1}{2\tau}\|x-u\|^2. \tag{I.4}$$

Equivalently, $\varphi(u) - \varphi(x) \leq \frac{1}{\tau}\langle z-u, x-u\rangle$.

Proof. By definition u minimizes the $\frac{1}{\tau}$-strongly convex function $\Psi(x) = \varphi(x) + \frac{1}{2\tau}\|x-z\|^2$. Strong convexity gives, for the minimizer u, $\Psi(x) \geq \Psi(u) + \frac{1}{2\tau}\|x-u\|^2$ for all x (the defining inequality of a μ-strongly convex function at its minimizer, with $\mu = 1/\tau$), which is (I.4). Expanding $\|x-z\|^2 - \|u-z\|^2 - \|x-u\|^2 = 2\langle z-u, x-u\rangle$ gives the equivalent form. □

Apply lemma I.1 to (I.3b) with $\varphi = g$, $z = x^n - \tau K^* y^{n+1}$, $u = x^{n+1}$, and to (I.3a) with $\varphi = h$, $z = y^n + \sigma K\bar{x}^n$, $u = y^{n+1}$. Using the polarization identity $\langle a-b, c-b\rangle = \frac{1}{2}(\|a-b\|^2 + \|c-b\|^2 - \|a-c\|^2)$, the two inequalities read, for all $x \in X$ and $y \in Y$,

$$g(x^{n+1}) - g(x) \leq \tfrac{1}{2\tau}\left(\|x-x^n\|^2 - \left\|x-x^{n+1}\right\|^2 - \left\|x^n - x^{n+1}\right\|^2\right) + \langle K^* y^{n+1}, x - x^{n+1}\rangle, \tag{I.5}$$

$$h(y^{n+1}) - h(y) \leq \tfrac{1}{2\sigma}\left(\|y-y^n\|^2 - \left\|y-y^{n+1}\right\|^2 - \left\|y^n - y^{n+1}\right\|^2\right) - \langle K\bar{x}^n, y - y^{n+1}\rangle. \tag{I.6}$$

I.3 The per-iteration estimate and telescoping

Fix a test pair (x, y) and write $\mathfrak{L}(x', y') = \langle Kx', y'\rangle + g(x') - h(y')$ and $\Delta^n := \mathfrak{L}(x^n, y) - \mathfrak{L}(x, y^n)$. Add (I.5) and (I.6); expand the two coupling inner products using $\langle K^* y^{n+1}, x - x^{n+1}\rangle = \langle y^{n+1}, K(x - x^{n+1})\rangle$ and the extrapolation $\bar{x}^n = 2x^n - x^{n-1} = x^n + (x^n - x^{n-1})$; and move the coupling part of Δ^{n+1} to the left. After cancellation of the symmetric term $\langle Kx, y^{n+1}\rangle$ this is a direct (if tedious) regrouping that yields, with the abbreviations

$$Q^n := \tfrac{1}{2\tau}\|x-x^n\|^2 + \tfrac{1}{2\sigma}\|y-y^n\|^2, \qquad T^n := \langle K(x^n - x^{n-1}), y^n - y\rangle, \qquad P^n := \tfrac{1}{2\tau}\left\|x^n - x^{n-1}\right\|^2,$$

the per-iteration inequality

$$\Delta^{n+1} \;\le\; (Q^n - Q^{n+1}) - (T^{n+1} - T^n) - \tfrac{1}{2\sigma}\left\|y^{n+1} - y^n\right\|^2 - \tfrac{1}{2\tau}\left\|x^{n+1} - x^n\right\|^2 + \langle K(x^n - x^{n-1}), y^{n+1} - y^n\rangle. \tag{I.7}$$

The last cross term is bounded by Young's inequality with $\|K\| = L$: $\langle K(x^n - x^{n-1}), y^{n+1} - y^n\rangle \le \frac{1}{2\sigma}\|y^{n+1} - y^n\|^2 + \frac{\sigma L^2}{2}\|x^n - x^{n-1}\|^2$, and under the step-size condition $\tau\sigma L^2 \le 1$ the second piece is $\le \frac{1}{2\tau}\|x^n - x^{n-1}\|^2 = P^n$. Substituting into (I.7), the $\frac{1}{2\sigma}\|y^{n+1} - y^n\|^2$ terms cancel and the x-distance terms telescope:

$$\Delta^{n+1} \;\le\; (Q^n - Q^{n+1}) - (T^{n+1} - T^n) - (P^{n+1} - P^n), \qquad \text{whenever} \;\; \tau\sigma L^2 \le 1. \tag{I.8}$$

Theorem I.2 (Ergodic $O(1/N)$ gap)**.** *Assume $\tau\sigma L^2 \le 1$ and that* (I.1) *has a saddle point. Let (x^n, y^n) be generated by* (I.3) *from $\bar{x}^0 = x^0$ (set $x^{-1} = x^0$) and form the ergodic averages $X_N = \frac{1}{N}\sum_{n=1}^N x^n$, $Y_N = \frac{1}{N}\sum_{n=1}^N y^n$. Then for every bounded $B_1 \times B_2$ containing a saddle point,*

$$\mathrm{Gap}_{B_1\times B_2}(X_N, Y_N) \;\le\; \frac{1}{N}\sup_{(x,y)\in B_1\times B_2}\Big(\tfrac{1}{2\tau}\left\|x - x^0\right\|^2 + \tfrac{1}{2\sigma}\left\|y - y^0\right\|^2\Big). \tag{I.9}$$

In particular $\mathrm{Gap}_{B_1\times B_2}(X_N, Y_N) = O(1/N)$, and any limit point of (X_N, Y_N) is a saddle point; when g is strictly convex (as in proposition 28.7) the saddle point is unique and $(X_N, Y_N) \to (\hat{x}, \hat{y})$.

Proof. Sum (I.8) over $n = 0, \dots, N-1$. The three bracketed differences telescope, and with $x^{-1} = x^0$ we have $T^0 = 0$ and $P^0 = 0$:

$$\sum_{n=1}^N \Delta^n \;\le\; Q^0 - Q^N - T^N - P^N.$$

Bound the boundary residual: $-T^N = -\langle K(x^N - x^{N-1}), y^N - y\rangle \le \frac{1}{2\tau}\left\|x^N - x^{N-1}\right\|^2 + \frac{\tau L^2}{2}\left\|y^N - y\right\|^2$ by Young, and $\frac{1}{2\tau}\left\|x^N - x^{N-1}\right\|^2 = P^N$ cancels $-P^N$, while $\frac{\tau L^2}{2} \le \frac{1}{2\sigma}$ gives $-T^N - P^N \le \frac{1}{2\sigma}\left\|y^N - y\right\|^2$. Since $-Q^N + \frac{1}{2\sigma}\left\|y^N - y\right\|^2 = -\frac{1}{2\tau}\left\|x - x^N\right\|^2 \le 0$, we conclude the clean summed bound

$$\sum_{n=1}^N \Delta^n(x,y) \;\le\; Q^0 = \tfrac{1}{2\tau}\left\|x - x^0\right\|^2 + \tfrac{1}{2\sigma}\left\|y - y^0\right\|^2. \tag{I.10}$$

Now $\Delta^n(x,y) = \mathfrak{L}(x^n, y) - \mathfrak{L}(x, y^n)$ is convex in the iterates (g, h convex, $\langle K\cdot, \cdot\rangle$ bilinear), so by Jensen $\mathfrak{L}(X_N, y) - \mathfrak{L}(x, Y_N) \le \frac{1}{N}\sum_{n=1}^N \Delta^n(x,y) \le \frac{1}{N}Q^0$. Taking the supremum over $(x,y) \in B_1 \times B_2$ on the left turns the difference into the partial gap (I.2), giving (I.9); the right-hand constant is finite on the bounded set, so the gap is $O(1/N)$. Since $\mathrm{Gap}_{B_1\times B_2} \ge 0$ vanishes exactly at saddle points and is lsc, any limit point of (X_N, Y_N) is a saddle point. Taking $(x,y) = (\hat{x}, \hat{y})$ in (I.10) and using $\Delta^n(\hat{x}, \hat{y}) \ge 0$ bounds Q^N, hence keeps the iterates bounded; with g strictly convex the saddle point is unique and the full ergodic sequence converges to it. □

Remark I.3 (Application to theorem 28.8)**.** In the discrete potential MFG of proposition 28.7, X is the space of discrete density–momentum pairs (m_h, w_h), Y the space of discrete values u_h, K the discrete continuity operator A of eq. (28.17) (whose norm is the discrete gradient norm, $L = O(1/h)$, fixing the admissible steps via $\tau\sigma L^2 \le 1$), $g = \mathcal{J}_h$ the convex kinetic-plus-interaction energy, and

$h = \delta_{\{c\}}$ the indicator enforcing the affine constraint $A(m_h, w_h) = c$. Theorem I.2 then gives the ergodic $O(1/k)$ gap of theorem 28.8 with *no* smallness or short-horizon hypothesis: convexity alone, which under (A-Mon) is unconditional, drives the convergence.

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